

EUCLID'S ELEMENTS OF GEOMETRY

The Greek text of J.L. Heiberg (1883–1885)

from *Euclidis Elementa*, edidit et Latine interpretatus est I.L. Heiberg, in aedibus
B.G. Teubneri, 1883–1885

edited, and provided with a modern English translation, by

Richard Fitzpatrick

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Introduction

Euclid's *Elements* is by far the most famous mathematical work of classical antiquity, and also has the distinction of being the world's oldest continuously used mathematical textbook. Little is known about the author, beyond the fact that he lived in Alexandria around 300 BCE. The main subjects of the work are geometry, proportion, and number theory.

Most of the theorems appearing in the *Elements* were not discovered by Euclid himself, but were the work of earlier Greek mathematicians such as Pythagoras (and his school), Hippocrates of Chios, Theaetetus of Athens, and Eudoxus of Cnidos. However, Euclid is generally credited with arranging these theorems in a logical manner, so as to demonstrate (admittedly, not always with the rigour demanded by modern mathematics) that they necessarily follow from five simple axioms. Euclid is also credited with devising a number of particularly ingenious proofs of previously discovered theorems: e.g., Theorem 48 in Book 1.

The geometrical constructions employed in the *Elements* are restricted to those which can be achieved using a straight-rule and a compass. Furthermore, empirical proofs by means of measurement are strictly forbidden: i.e., any comparison of two magnitudes is restricted to saying that the magnitudes are either equal, or that one is greater than the other.

The *Elements* consists of thirteen books. Book 1 outlines the fundamental propositions of plane geometry, including the three cases in which triangles are congruent, various theorems involving parallel lines, the theorem regarding the sum of the angles in a triangle, and the Pythagorean theorem. Book 2 is commonly said to deal with “geometric algebra”, since most of the theorems contained within it have simple algebraic interpretations. Book 3 investigates circles and their properties, and includes theorems on tangents and inscribed angles. Book 4 is concerned with regular polygons inscribed in, and circumscribed around, circles. Book 5 develops the arithmetic theory of proportion. Book 6 applies the theory of proportion to plane geometry, and contains theorems on similar figures. Book 7 deals with elementary number theory: e.g., prime numbers, greatest common denominators, etc. Book 8 is concerned with geometric series. Book 9 contains various applications of results in the previous two books, and includes theorems on the infinitude of prime numbers, as well as the sum of a geometric series. Book 10 attempts to classify incommensurable (i.e., irrational) magnitudes using the so-called “method of exhaustion”, an ancient precursor to integration. Book 11 deals with the fundamental propositions of three-dimensional geometry. Book 12 calculates the relative volumes of cones, pyramids, cylinders, and spheres using the method of exhaustion. Finally, Book 13 investigates the five so-called Platonic solids.

This edition of Euclid's *Elements* presents the definitive Greek text—i.e., that edited by J.L. Heiberg (1883–1885)—accompanied by a modern English translation, as well as a Greek-English lexicon. Neither the spurious books 14 and 15, nor the extensive scholia which have been added to the *Elements* over the centuries, are included. The aim of the translation is to make the mathematical argument as clear and unambiguous as possible, whilst still adhering closely to the meaning of the original Greek. Text within square parenthesis (in both Greek and English) indicates material identified by Heiberg as being later interpolations to the original text (some particularly obvious or unhelpful interpolations have been omitted altogether). Text within round parenthesis (in English) indicates material which is implied, but not actually present, in the Greek text.

My thanks to Mariusz Wodzicki (Berkeley) for typesetting advice, and to Sam Watson & Jonathan Fenno (U. Mississippi), and Gregory Wong (UCSD) for pointing out a number of errors in Book 1.

In the second edition of this work I have opted for more compact (and nicer looking) Greek and English fonts. I have also made some minor improvements and corrections to my translation.

ELEMENTS BOOK 1

*Fundamentals of Plane Geometry Involving
Straight-Lines*

Ὅροι.

1. Σημεῖόν ἐστιν, ὃ ὅ μέρους οὐθέν.
2. Γραμμὴ δὲ μήκος ἀπλατέξ.
3. Γραμμῆς δὲ πέρατα σημεία.
4. Εὐθεῖα γραμμὴ ἐστίν, ἥτις ἐχ' ἴσον τοῖς ἐφ' ἑαυτῆς σημείοις κεῖται.
5. Ἐπιφάνεια δὲ ἐστίν, ὃ μήκος καὶ πλάτος μόνον ἔθει.
6. Ἐπιφανείας δὲ πέρατα γραμμαί.
7. Ἐπίπεδος ἐπιφάνειά ἐστίν, ἥτις ἐχ' ἴσον ταῖς ἐφ' ἑαυτῆς εὐθείαις κεῖται.
8. Ἐπίπεδος δὲ γωνία ἐστίν ἡ ἐν ἐπιπέδῳ δύο γραμμῶν ἀπτομένων ἀλλήλων καὶ μὴ ἐπ' εὐθείας κειμένων πρὸς ἀλλήλας τῶν γραμμῶν κλίσις.
9. Ὅταν δὲ αἱ περιέθουσαι τὴν γωνίαν γραμμαὶ εὐθεῖαι ὦσιν, εὐθύγραμμος καλεῖται ἡ γωνία.
10. Ὅταν δὲ εὐθεῖα ἐπ' εὐθείαν σταθεῖσα τὰς ἐφεχῆς γωνίας ἴσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκάτερα τῶν ἴσων γωνιῶν ἐστί, καὶ ἡ ἐφεστηκυῖα εὐθεῖα κάθετος καλεῖται, ἐφ' ἣν ἐφέστηκεν.
11. Ἀμβλεία γωνία ἐστίν ἡ μείζων ὀρθῆς.
12. Ὀξεῖα δὲ ἡ ἐλάσσων ὀρθῆς.
13. Ὅρος ἐστίν, ὃ τινὸς ἐστὶ πέραξ.
14. Σθῆμά ἐστι τὸ ὑπὸ τινὸς ἢ τινων ὁρων περιεχόμενον.
15. Κύκλος ἐστὶ σθῆμα ἐπίπεδον ὑπὸ μιᾷς γραμμῆς περιεχόμενον [ἡ καλεῖται περιφέρεια], πρὸς ἣν ἀφ' ἐνὸς σημείου τῶν ἐντὸς τοῦ σθῆματος κειμένων πᾶσαι αἱ προσπίπτουσαι εὐθεῖαι [πρὸς τὴν τοῦ κύκλου περιφέρειαν] ἴσαι ἀλλήλαις εἰσίν.
16. Κέντρον δὲ τοῦ κύκλου τὸ σημεῖον καλεῖται.
17. Διάμετρος δὲ τοῦ κύκλου ἐστίν εὐθεῖα τις διὰ τοῦ κέντρον ἡγμένη καὶ περατομένη ἐφ' ἑκάτερα τὰ μέρη ὑπὸ τῆς τοῦ κύκλου περιφέρειας, ἥτις καὶ δίθια τέμνει τὸν κύκλον.
18. Ἡμικύκλιον δὲ ἐστὶ τὸ περιεχόμενον σθῆμα ὑπὸ τε τῆς διαμέτρου καὶ τῆς ἀπολαμβανομένης ὑπ' αὐτῆς περιφέρειας. κέντρον δὲ τοῦ ἡμικυκλίου τὸ αὐτό, ὃ καὶ τοῦ κύκλου ἐστίν.
19. Σθῆματα εὐθύγραμμά ἐστι τὰ ὑπὸ εὐθειῶν περιεθόμενα, τρίπλευρα μὲν τὰ ὑπὸ τριῶν, τετράπλευρα δὲ τὰ ὑπὸ τεσσάρων, πολύπλευρα δὲ τὰ ὑπὸ πλείονων ἢ τεσσάρων εὐθειῶν περιεθόμενα.
20. Τῶν δὲ τριπλεύρων σθημάτων ἰσοπλευρον μὲν τρίγωνόν ἐστι τὸ τὰς τρεῖς ἴσας ἔθον πλευράς, ἰσοσκελὲς δὲ τὸ τὰς δύο μόνας ἴσας ἔθον πλευράς, σκαληνόν δὲ τὸ τὰς τρεῖς ἀνίσους ἔθον πλευράς.
21. Ἐτι δὲ τῶν τριπλεύρων σθημάτων ὀρθογώνιον μὲν τρίγωνόν ἐστι τὸ ἔθον ὀρθὴν γωνίαν, ἀμβλυγώνιον δὲ τὸ ἔθον ἀμβλείαν γωνίαν, ὀξυγώνιον δὲ τὸ τὰς τρεῖς ὀξείας ἔθον γωνίας.
22. Τῶν δὲ τετραπλεύρων σθημάτων τετράγωνον μὲν ἐστίν, ὃ ἰσοπλευρόν τε ἐστὶ καὶ ὀρθογώνιον, ἑτερόμηκες δὲ, ὃ ὀρθογώνιον μὲν, οὐκ ἰσοπλευρον δὲ, ῥόμβος δὲ, ὃ ἰσοπλευρον μὲν, οὐκ ὀρθογώνιον δὲ, ῥομβοειδὲς δὲ τὸ τὰς ἀπεναντίον πλευράς

Definitions

1. A point is that of which there is no part.
2. And a line is a length without breadth.
3. And the extremities of a line are points.
4. A straight-line is (any) one which lies evenly with points on itself.
5. And a surface is that which has length and breadth only.
6. And the extremities of a surface are lines.
7. A plane surface is (any) one which lies evenly with the straight-lines on itself.
8. And a plane angle is the inclination of the lines to one another, when two lines in a plane meet one another, and are not lying in a straight-line.
9. And when the lines containing the angle are straight then the angle is called rectilinear.
10. And when a straight-line stood upon (another) straight-line makes adjacent angles (which are) equal to one another, each of the equal angles is a right-angle, and the former straight-line is called a perpendicular to that upon which it stands.
11. An obtuse angle is one greater than a right-angle.
12. And an acute angle (is) one less than a right-angle.
13. A boundary is that which is the extremity of something.
14. A figure is that which is contained by some boundary or boundaries.
15. A circle is a plane figure contained by a single line [which is called a circumference], (such that) all of the straight-lines radiating towards [the circumference] from one point amongst those lying inside the figure are equal to one another.
16. And the point is called the center of the circle.
17. And a diameter of the circle is any straight-line, being drawn through the center, and terminated in each direction by the circumference of the circle. (And) any such (straight-line) also cuts the circle in half.[†]
18. And a semi-circle is the figure contained by the diameter and the circumference cut off by it. And the center of the semi-circle is the same (point) as (the center of) the circle.
19. Rectilinear figures are those (figures) contained by straight-lines: trilateral figures being those contained by three straight-lines, quadrilateral by four, and multilateral by more than four.
20. And of the trilateral figures: an equilateral triangle is that having three equal sides, an isosceles (triangle) that having only two equal sides, and a scalene (triangle) that having three unequal sides.
21. And further of the trilateral figures: a right-angled triangle is that having a right-angle, an obtuse-angled (triangle) that having an obtuse angle, and an acute-angled (triangle) that having three acute angles.

τε καὶ γωνίαξ ἴσαξ ἀλλήλαιξ ἔθον, ὃ οὕτε ἰσόπλευρόν ἐστιν οὕτε ὀρθογώνιον· τὰ δὲ παρὰ ταῦτα τετράπλευρα τραπέζια καλεῖσθω.

23. Παράλληλοι εἰσιν εὐθεῖαι, αἵτινες ἐν τῷ αὐτῷ ἐπιπέδῳ οὔσαι καὶ ἐκβαλλόμεναι εἰς ἄπειρον ἐφ' ἑκάτερα τὰ μέρη ἐπὶ μηδέτερα συμπίπτουσιν ἀλλήλαιξ.

22. And of the quadrilateral figures: a square is that which is right-angled and equilateral, a rectangle that which is right-angled but not equilateral, a rhombus that which is equilateral but not right-angled, and a rhomboid that having opposite sides and angles equal to one another which is neither right-angled nor equilateral. And let quadrilateral figures besides these be called trapezia.

23. Parallel lines are straight-lines which, being in the same plane, and being produced to infinity in each direction, meet with one another in neither (of these directions).

† This should really be counted as a postulate, rather than as part of a definition.

Αἰτήματα.

1. Ἡτῆσθω ἀπὸ παντός σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.

2. Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνθεῖξ ἐπ' εὐθείας ἐκβαλεῖν.

3. Καὶ παντὶ κέντρῳ καὶ διαστήματι κύκλον γράφεσθαι.

4. Καὶ πάσαξ τὰξ ὀρθὰξ γωνίαξ ἴσαξ ἀλλήλαιξ εἶναι.

5. Καὶ ἔαν εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὰξ ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη γωνίαξ δύο ὀρθῶν ἐλάσσοναξ ποιῇ, ἐκβαλλόμεναξ τὰξ δύο εὐθείας ἐπ' ἄπειρον συμπίπτειν, ἐφ' ἃ μέρη εἰσὶν αἱ τῶν δύο ὀρθῶν ἐλάσσονες.

Postulates

1. Let it be postulated[†] (that it is possible) to draw a straight-line from any point to any point;

2. And to produce a finite straight-line continuously in a straight-line;

3. And to draw a circle with any center and radius.

4. And that all right-angles are equal to one another.

5. And that if a straight-line falling across two (other) straight-lines makes internal angles on the same side (of itself whose sum is) less than two right-angles, (then) the two (other) straight-lines, being produced to infinity, meet on that side (of the original straight-line) that the (sum of the internal angles) is less than two right-angles (and do not meet on the other side).[‡]

† Literally, "let it have been postulated". The Greek present perfect tense indicates a past action with present significance. Hence, the 3rd-person present perfect imperative *ἡτῆσθω* means "let it be postulated" in the sense "let it stand as postulated", but not "let the postulate be now brought forward". This peculiar Greek idiom is used throughout the Elements.

‡ This postulate effectively specifies that we are dealing with the geometry of flat, rather than curved, space.

Κοιναὶ ἔννοιαι.

1. Τὰ τῷ αὐτῷ ἴσα καὶ ἀλλήλοιξ ἐστὶν ἴσα.

2. Καὶ ἔαν ἴσοιξ ἴσα προστεθῇ, τὰ ὅλα ἐστὶν ἴσα.

3. Καὶ ἔαν ἀπὸ ἴσων ἴσα ἀφαιρεθῇ, τὰ καταλειπόμενά ἐστιν ἴσα.

4. Καὶ τὰ ἐφαρμόζοντα ἐπ' ἀλλήλα ἴσα ἀλλήλοιξ ἐστὶν.

5. Καὶ τὸ ὅλον τοῦ μέρους μείζον [ἐστίν].

Common Notions

1. Things equal to the same thing are also equal to one another.

2. And if equal things are added to equal things (then) the wholes are equal.

3. And if equal things are subtracted from equal things (then) the remainders are equal.[†]

4. And things coinciding with one another are equal to one another.

5. And the whole [is] greater than the part.

† As an obvious extension of C.N.s 2 & 3—if equal things are added or subtracted from the two sides of an inequality then the inequality remains an inequality of the same type.

1.

Proposition 1

Ἐπὶ τῇξ δοθείσης εὐθείας πεπερασμένης τρίγωνον ἰσόπλευ-

To construct an equilateral triangle on a given finite straight-

ρον συστήσασθαι.

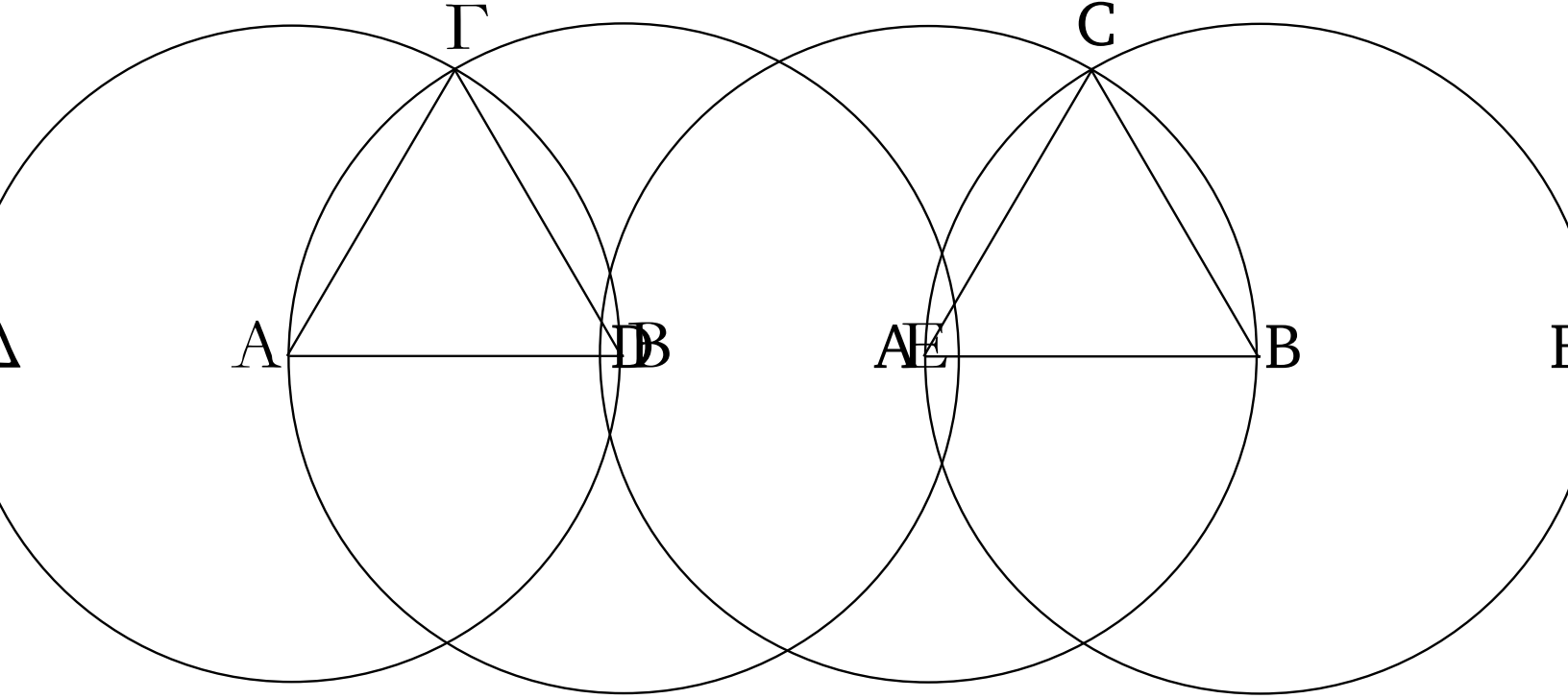
Ἐστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB .

Δεῖ δὴ ἐπὶ τῆς AB εὐθείας τρίγωνον ἰσόπλευρον συστήσασθαι.

line.

Let AB be the given finite straight-line.

So it is required to construct an equilateral triangle on the straight-line AB .



Κέντρῳ μὲν τῷ A διαστήματι δὲ τῷ AB κύκλος γεγράφθω ὁ $BΓΔ$, καὶ πάλιν κέντρῳ μὲν τῷ B διαστήματι δὲ τῷ BA κύκλος γεγράφθω ὁ $ΑΓΕ$, καὶ ἀπὸ τοῦ $Γ$ σημείου, καθ' ὃ τέμνουσιν ἀλλήλους οἱ κύκλοι, ἐπὶ τὰ A , B σημεία ἐπεζεύθωσαν εὐθεῖαι αἱ $ΓΑ$, $ΓΒ$.

Καὶ ἐπεὶ τὸ A σημεῖον κέντρον ἐστὶ τοῦ $ΓΔΒ$ κύκλου, ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$. πάλιν, ἐπεὶ τὸ B σημεῖον κέντρον ἐστὶ τοῦ $ΓΑΕ$ κύκλου, ἴση ἐστὶν ἡ $ΒΓ$ τῇ $ΒΑ$. ἐδείκθη δὲ καὶ ἡ $ΓΑ$ τῇ $ΑΒ$ ἴση· ἐκατέρα ἄρα τῶν $ΓΑ$, $ΓΒ$ τῇ $ΑΒ$ ἐστὶν ἴση. τὰ δὲ τῷ αὐτῷ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα· καὶ ἡ $ΓΑ$ ἄρα τῇ $ΓΒ$ ἐστὶν ἴση· αἱ τρεῖς ἄρα αἱ $ΓΑ$, $ΑΒ$, $ΒΓ$ ἴσαι ἀλλήλαις εἰσὶν.

Ἰσόπλευρον ἄρα ἐστὶ τὸ $ΑΒΓ$ τρίγωνον. καὶ συνέσταται ἐπὶ τῆς δοθείσης εὐθείας πεπερασμένης τῆς $ΑΒ$. ὅπερ ἔδει ποιῆσαι.

Let the circle BCD with center A and radius AB be drawn[†] [Post. 3], and again let the circle ACE with center B and radius BA be drawn [Post. 3]. And let the straight-lines CA and CB be joined from the point C , where the circles cut one another,[‡] to the points A and B (respectively) [Post. 1].

And since the point A is the center of the circle CDB , AC is equal to AB [Def. 1.15]. Again, since the point B is the center of the circle CAE , BC is equal to BA [Def. 1.15]. But CA was also shown (to be) equal to AB . Thus, CA and CB are each equal to AB . But things equal to the same thing are also equal to one another [C.N. 1]. Thus, CA is also equal to CB . Thus, the three (straight-lines) CA , AB , and BC are equal to one another.

Thus, the triangle ABC is equilateral, and has been constructed on the given finite straight-line AB . (Which is) the very thing it was required to do.

[†] Literally, "have been drawn". The use of this peculiar Greek idiom will pass without further comment.

[‡] The assumption that the circles do indeed cut one another should be counted as an additional postulate. There is also an implicit assumption that two straight-lines cannot share a common segment.

2.

Proposition 2[†]

Πρὸς τῷ δοθέντι σημείῳ τῇ δοθείσῃ εὐθείᾳ ἴσην εὐθεῖαν

To place a straight-line equal to a given straight-line at a

θέσθαι.

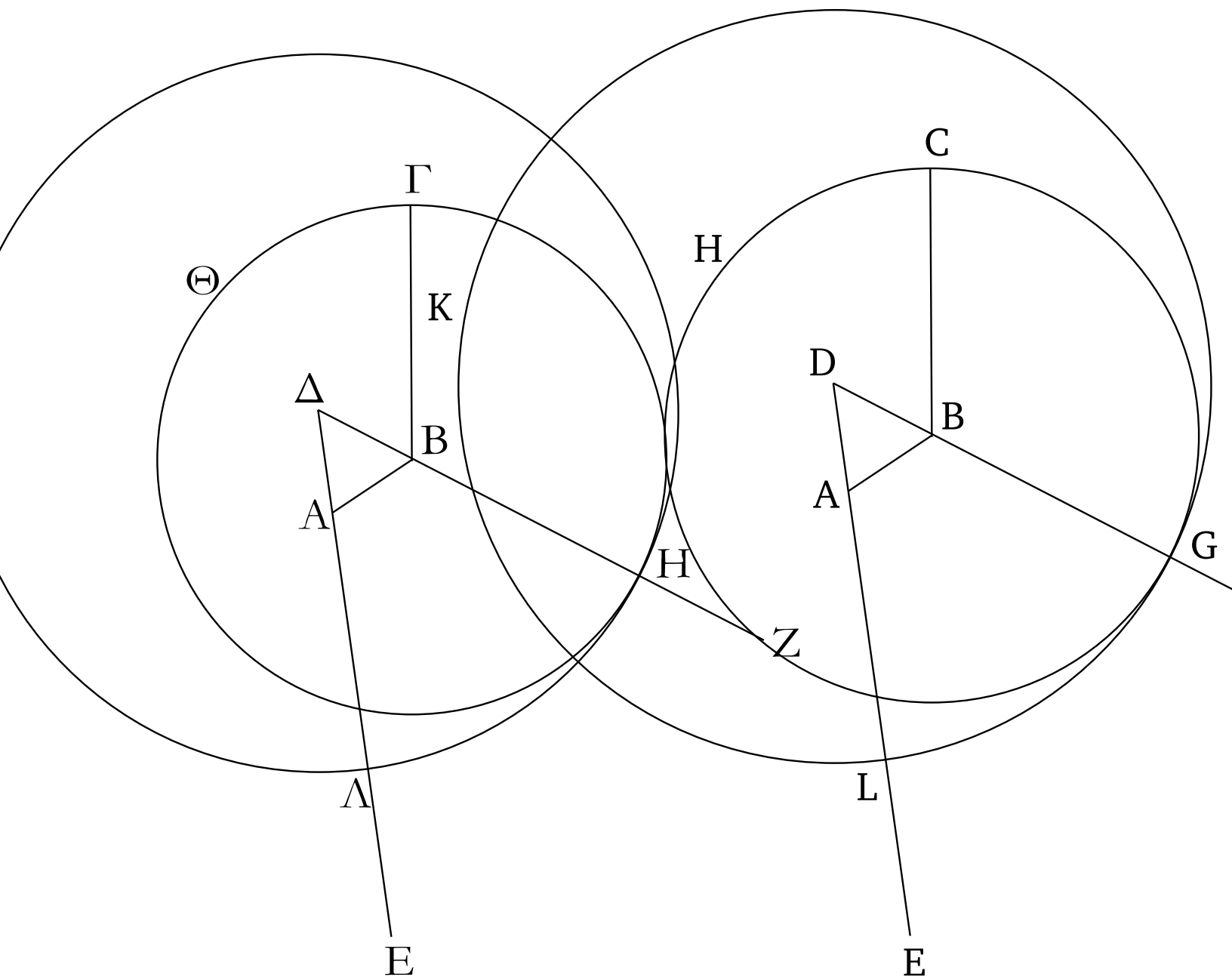
Ἐστω τὸ μὲν δοθὲν σημεῖον τὸ A , ἡ δὲ δοθεῖσα εὐθεῖα ἡ $ΒΓ$. δεῖ δὴ πρὸς τῷ A σημείῳ τῇ δοθείσῃ εὐθείᾳ τῇ $ΒΓ$ ἴσην εὐθεῖαν θέσθαι.

Ἐπεξεύθθω γὰρ ἀπὸ τοῦ A σημείου ἐπὶ τὸ B σημεῖον εὐθεῖα ἡ AB , καὶ συνεστάτω ἐπ' αὐτῇς τρίγωνον ἰσόπλευρον τὸ $ΔAB$, καὶ ἐκβεβλήσθωσαν ἐπ' εὐθείας ταῖς $ΔA$, $ΔB$ εὐθεῖαι αἱ AE , BZ , καὶ κέντρῳ μὲν τῷ B διαστήματι δὲ τῷ $ΒΓ$ κύκλος γεγράφθω ὁ $ΓΗΘ$, καὶ πάλιν κέντρῳ τῷ $Δ$ καὶ διαστήματι τῷ $ΔΗ$ κύκλος γεγράφθω ὁ $ΗΚΛ$.

given point (as an extremity).

Let A be the given point, and BC the given straight-line. So it is required to place a straight-line at point A equal to the given straight-line BC .

For let the straight-line AB be joined from point A to point B [Post. 1], and let the equilateral triangle DAB be constructed upon it [Prop. 1.1]. And let the straight-lines AE and BF be produced in a straight-line with DA and DB (respectively) [Post. 2]. And let the circle CGH with center B and radius BC be drawn [Post. 3], and again let the circle GKL with center D and radius DG be drawn [Post. 3].



Ἐπειὶ οὖν τὸ B σημεῖον κέντρον ἐστὶ τοῦ $\Gamma H \Theta$, ἴση ἐστὶν ἡ $B\Gamma$ τῇ BH . πάλιν, ἐπεὶ τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ $H\kappa\Lambda$ κύκλου, ἴση ἐστὶν ἡ $\Delta\Lambda$ τῇ ΔH , ὥς ἡ $\Delta\Lambda$ τῇ ΔB ἴση ἐστίν. λοιπὴ ἄρα ἡ $A\Lambda$ λοιπῇ τῇ BH ἐστὶν ἴση. ἐδείχθη δὲ καὶ ἡ $B\Gamma$ τῇ BH ἴση· ἐκάτερά ἄρα τῶν $A\Lambda$, $B\Gamma$ τῇ BH ἐστὶν ἴση. τὰ δὲ τῷ αὐτῷ ἴσα καὶ ἀλλήλοισι ἐστὶν ἴσα· καὶ ἡ $A\Lambda$ ἄρα τῇ $B\Gamma$ ἐστὶν ἴση.

Πρὸς ἅρα τῷ δοθέντι σημείῳ τῷ A τῇ δοθείσῃ εὐθείᾳ τῇ $B\Gamma$ ἴση εὐθείᾳ κεῖται ἡ $ΑΛ$ · ὅπερ ἔδει ποιῆσαι.

Therefore, since the point B is the center of (the circle) CGH , BC is equal to BG [Def. 1.15]. Again, since the point D is the center of the circle GKL , DL is equal to DG [Def. 1.15]. And within these, DA is equal to DB . Thus, the remainder AL is equal to the remainder BG [C.N. 3]. But BC was also shown (to be) equal to BG . Thus, AL and BC are each equal to BG . But things equal to the same thing are also equal to one another [C.N. 1]. Thus, AL is also equal to BC .

Thus, the straight-line AL , equal to the given straight-line BC , has been placed at the given point A . (Which is) the very

thing it was required to do.

† This proposition admits of a number of different cases, depending on the relative positions of the point A and the line BC . In such situations, Euclid invariably only considers one particular case—usually, the most difficult—and leaves the remaining cases as exercises for the reader.

3.

Proposition 3

Δύο δοθεισῶν εὐθειῶν ἀνίσων ἀπὸ τῆς μείζονος τῇ ἐλάσσονι ἴσην εὐθεῖαν ἀφελεῖν.

Ἐστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι ἴσους αἱ AB , Γ , ὧν μείζων ἔστω ἡ AB . δεῖ δὴ ἀπὸ τῆς μείζονος τῆς AB τῇ ἐλάσσονι τῇ Γ ἴσην εὐθεῖαν ἀφελεῖν.

Κείσθω πρὸς τῷ A σημείῳ τῇ Γ εὐθείᾳ ἴση ἡ $A\Delta$. καὶ κέντρῳ μὲν τῷ A διαστήματι δὲ τῷ $A\Delta$ κύκλος γεγράφθω ὁ ΔEZ .

Καὶ ἐπεὶ τὸ A σημεῖον κέντρον ἐστὶ τοῦ ΔEZ κύκλου, ἴση ἐστὶν ἡ AE τῇ $A\Delta$. ἀλλὰ καὶ ἡ Γ τῇ $A\Delta$ ἐστὶν ἴση. ἑκατέρα ἄρα τῶν AE , Γ τῇ $A\Delta$ ἐστὶν ἴση· ὥστε καὶ ἡ AE τῇ Γ ἐστὶν ἴση.

For two given unequal straight-lines, to cut off from the greater a straight-line equal to the lesser.

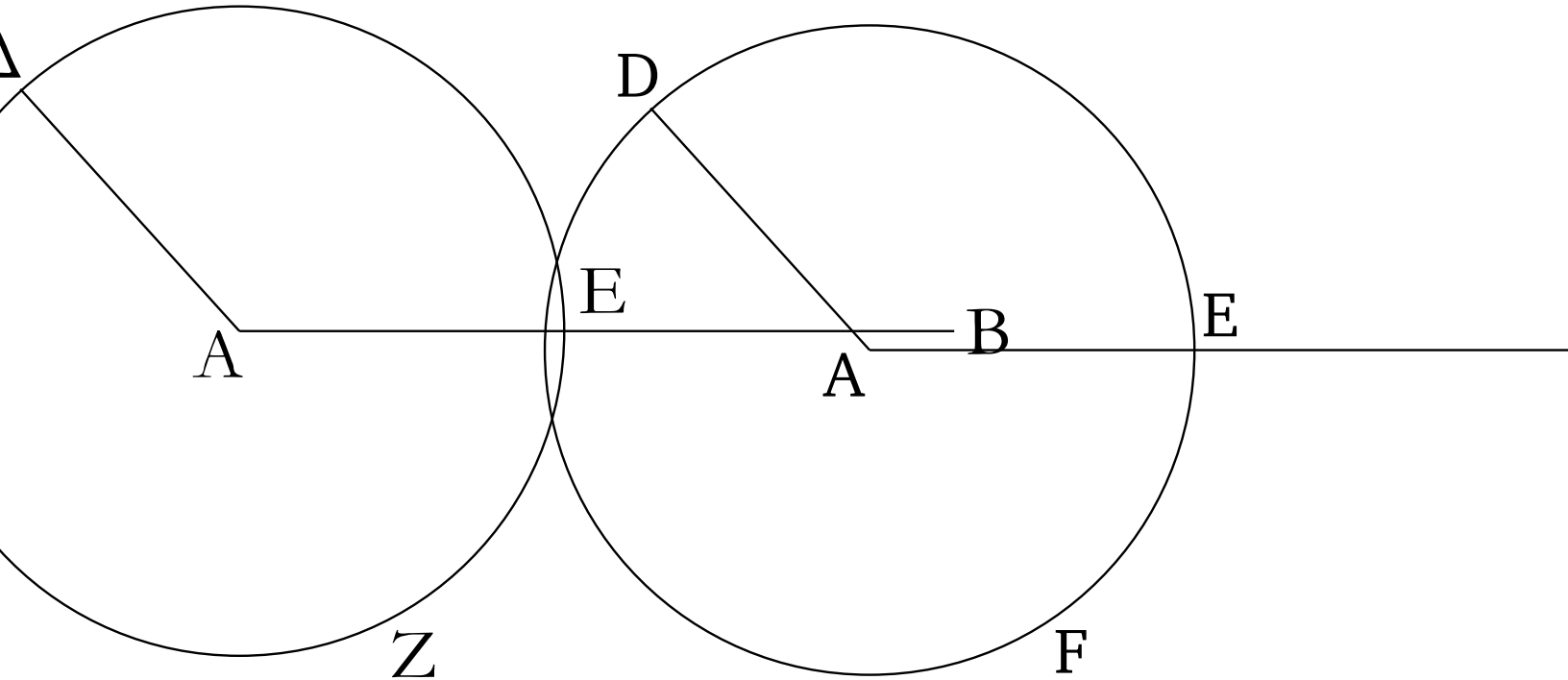
Let AB and C be the two given unequal straight-lines, of which let the greater be AB . So it is required to cut off a straight-line equal to the lesser C from the greater AB .

Let the line AD , equal to the straight-line C , be placed at point A [Prop. 1.2]. And let the circle DEF be drawn with center A and radius AD [Post. 3].

And since point A is the center of circle DEF , AE is equal to AD [Def. 1.15]. But, C is also equal to AD . Thus, AE and C are each equal to AD . So AE is also equal to C [C.N. 1].

Γ

C



Δύο ἄρα δοθεισῶν εὐθειῶν ἀνίσων τῶν AB , Γ ἀπὸ τῆς μείζονος τῆς AB τῇ ἐλάσσονι τῇ Γ ἴση ἀφίηται ἡ AE . ὅπερ ἔδει ποιῆσαι.

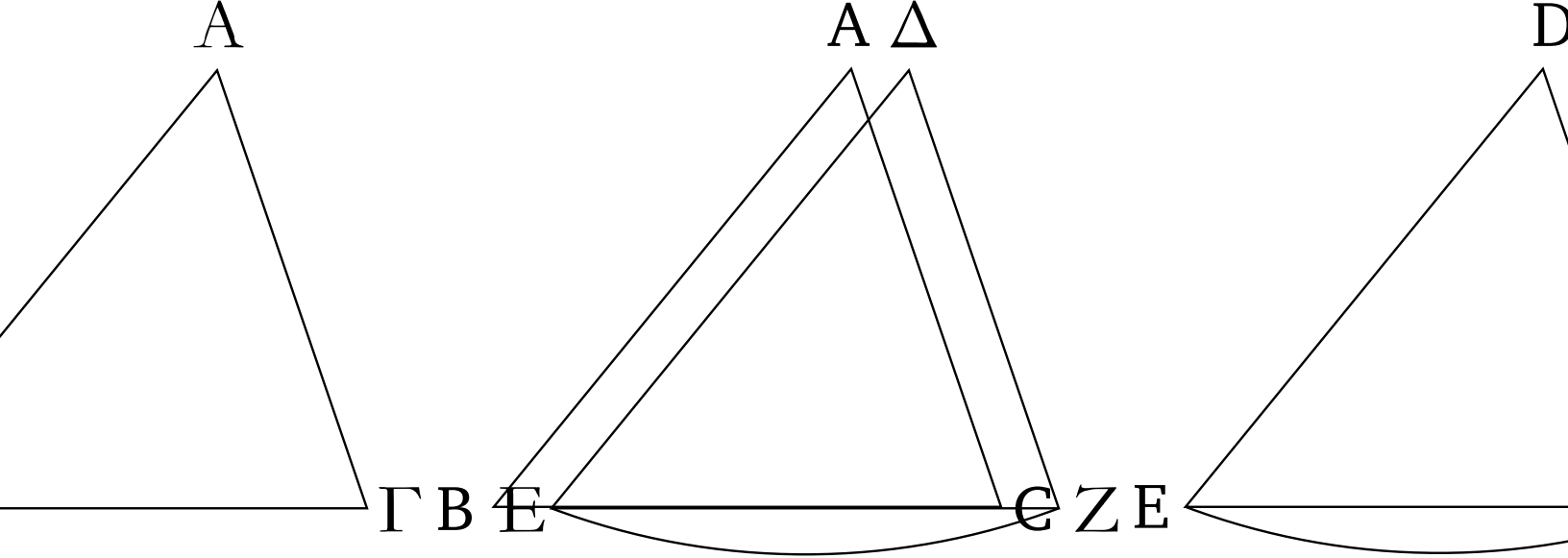
Thus, for two given unequal straight-lines, AB and C , the (straight-line) AE , equal to the lesser C , has been cut off from the greater AB . (Which is) the very thing it was required to do.

4.

Proposition 4

Ἐάν δύο τρίγωνα τὰς δύο πλευράς [ταῖς] δυοῖ πλευραῖς ἴσαξ ἔθῃ ἑκατέραν ἑκατέρᾳ καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔθῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῇ βάσει ἴσην ἔχει, καὶ τὸ τρίγωνον τῷ τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρα ἑκατέρᾳ, ὅφ' ἂξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν.

If two triangles have two sides equal to two sides, respectively, and have the angle(s) enclosed by the equal straight-lines equal, (then) they will also have the base equal to the base, and the triangle will be equal to the triangle, and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles.



Ἐστω δύο τρίγωνα τὰ ABG , $ΔEZ$ τὰς δύο πλευράς τὰς AB , AG ταῖς δυοῖς πλευραῖς ταῖς $ΔE$, $ΔZ$ ἴσαξ ἔθοντα ἑκατέραν ἑκατέρᾳ τὴν μὲν AB τῇ $ΔE$ τὴν δὲ AG τῇ $ΔZ$ καὶ γωνίαν τὴν ὑπὸ BAG γωνίᾳ τῇ ὑπὸ $EΔZ$ ἴσην. λέγω, ὅτι καὶ βάσει ἡ BG βάσει τῇ EZ ἴση ἔστί, καὶ τὸ ABG τρίγωνον τῷ $ΔEZ$ τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρα ἑκατέρᾳ, ὅφ' ἂξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ ABG τῇ ὑπὸ $ΔEZ$, ἡ δὲ ὑπὸ AGB τῇ ὑπὸ $ΔZE$.

Ἐφαρμοζόμενον γὰρ τοῦ ABG τριγώνου ἐπὶ τὸ $ΔEZ$ τρίγωνον καὶ τιθεμένον τοῦ μὲν A σημείου ἐπὶ τὸ $Δ$ σημεῖον τῆς δὲ AB εὐθείας ἐπὶ τὴν $ΔE$, ἐφαρμόσει καὶ τὸ B σημεῖον ἐπὶ τὸ E διὰ τὸ ἴσην εἶναι τὴν AB τῇ $ΔE$. ἐφαρμοσάσης δὲ τῆς AB ἐπὶ τὴν $ΔE$ ἐφαρμόσει καὶ ἡ AG εὐθεῖα ἐπὶ τὴν $ΔZ$ διὰ τὸ ἴσην εἶναι τὴν ὑπὸ BAG γωνίαν τῇ ὑπὸ $EΔZ$. ὥστε καὶ τὸ $Γ$ σημεῖον ἐπὶ τὸ Z σημεῖον ἐφαρμόσει διὰ τὸ ἴσην πάλιν εἶναι τὴν AG τῇ $ΔZ$. ἀλλὰ μὴν καὶ τὸ B ἐπὶ τὸ E ἐφηρμόκει· ὥστε βάσει ἡ BG ἐπὶ βάσιν τὴν EZ ἐφαρμόσει. εἰ γὰρ τοῦ μὲν B ἐπὶ τὸ E ἐφαρμόσαντος τοῦ δὲ $Γ$ ἐπὶ τὸ Z ἡ BG βάσις ἐπὶ τὴν EZ οὐκ ἐφαρμόσει, δύο εὐθεῖαι θωρίον περιέχουσιν· ὅπερ ἐστὶν ἀδύνατον. ἐφαρμόσει ἄρα ἡ BG βάσις ἐπὶ τὴν EZ καὶ ἴση αὐτῇ ἔσται· ὥστε καὶ ὅλον τὸ ABG τρίγωνον ἐπὶ ὅλον τὸ $ΔEZ$ τρίγωνον ἐφαρμόσει καὶ ἴσον αὐτῷ ἔσται, καὶ αἱ λοιπαὶ γωνίαι ἐπὶ τὰς λοιπὰς γωνίας ἐφαρμόσουσι καὶ ἴσαι

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is) AB to DE , and AC to DF . And (let) the angle BAC (be) equal to the angle EDF . I say that the base BC is also equal to the base EF , and triangle ABC will be equal to triangle DEF , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles. (That is) ABC to DEF , and ACB to DFE .

For if triangle ABC is applied to triangle DEF ,[†] the point A being placed on the point D , and the straight-line AB on DE , (then) the point B will also coincide with E , on account of AB being equal to DE . So (because of) AB coinciding with DE , the straight-line AC will also coincide with DF , on account of the angle BAC being equal to EDF . So the point C will also coincide with the point F , again on account of AC being equal to DF . But, point B certainly also coincided with point E , so that the base BC will coincide with the base EF . For if B coincides with E , and C with F , and the base BC does not coincide with EF , (then) two straight-lines will encompass an area. The very thing is impossible [Post. 1].[‡] Thus, the base BC will coincide with EF , and will be equal to it [C.N. 4]. So the whole triangle ABC will coincide with the whole triangle DEF , and will be equal to it [C.N. 4]. And the remaining angles will coincide with the remaining angles, and will be

αὐταῖς ἴσονται, ἡ μὲν ὑπὸ $ABΓ$ τῇ ὑπὸ $ΔΕΖ$ ἢ δὲ ὑπὸ $ΑΓΒ$ τῇ ὑπὸ $ΔΖΕ$.

Ἐὰν ἄρα δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρα καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔθῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῇ βάσει ἴσην ἔχει, καὶ τὸ τρίγωνον τῷ τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρα ἑκατέρα, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ὅπερ ἔδει δεῖξαι.

† The application of one figure to another should be counted as an additional postulate.

‡ Since Post. 1 implicitly assumes that the straight-line joining two given points is unique.

5.

Τῶν ἰσοσκελῶν τριγώνων αἱ πρὸς τῇ βάσει γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ προσεκβληθεῖσιν τῶν ἴσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι ἴσαι ἀλλήλαις ἔσονται.

Ἔστω τριγώνον ἰσοσκελὲς τὸ $ABΓ$ ἴσην ἔθον τὴν AB πλευρὰν τῇ $ΑΓ$ πλευρᾷ, καὶ προσεκβεβλήσθωσαν ἐπ' εὐθείας ταῖς AB , $ΑΓ$ εὐθεῖαι αἱ $BΔ$, $ΓΕ$. λέγω, ὅτι ἡ μὲν ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΑΓΒ$ ἴση ἐστίν, ἡ δὲ ὑπὸ $ΓΒΔ$ τῇ ὑπὸ $ΒΓΕ$.

Εἰλήφθω γάρ ἐπὶ τῇς $BΔ$ τυθὸν σημεῖον τὸ Z , καὶ ἀφηρήσθω ἀπὸ τῇς μείζονος τῇς AE τῇ ἐλάσσονι τῇ AZ ἴση ἡ AH , καὶ ἐπεζεύθωσαν αἱ $ZΓ$, HB εὐθεῖαι.

equal to them [C.N. 4]. (That is) ABC to DEF , and ACB to DFE [C.N. 4].

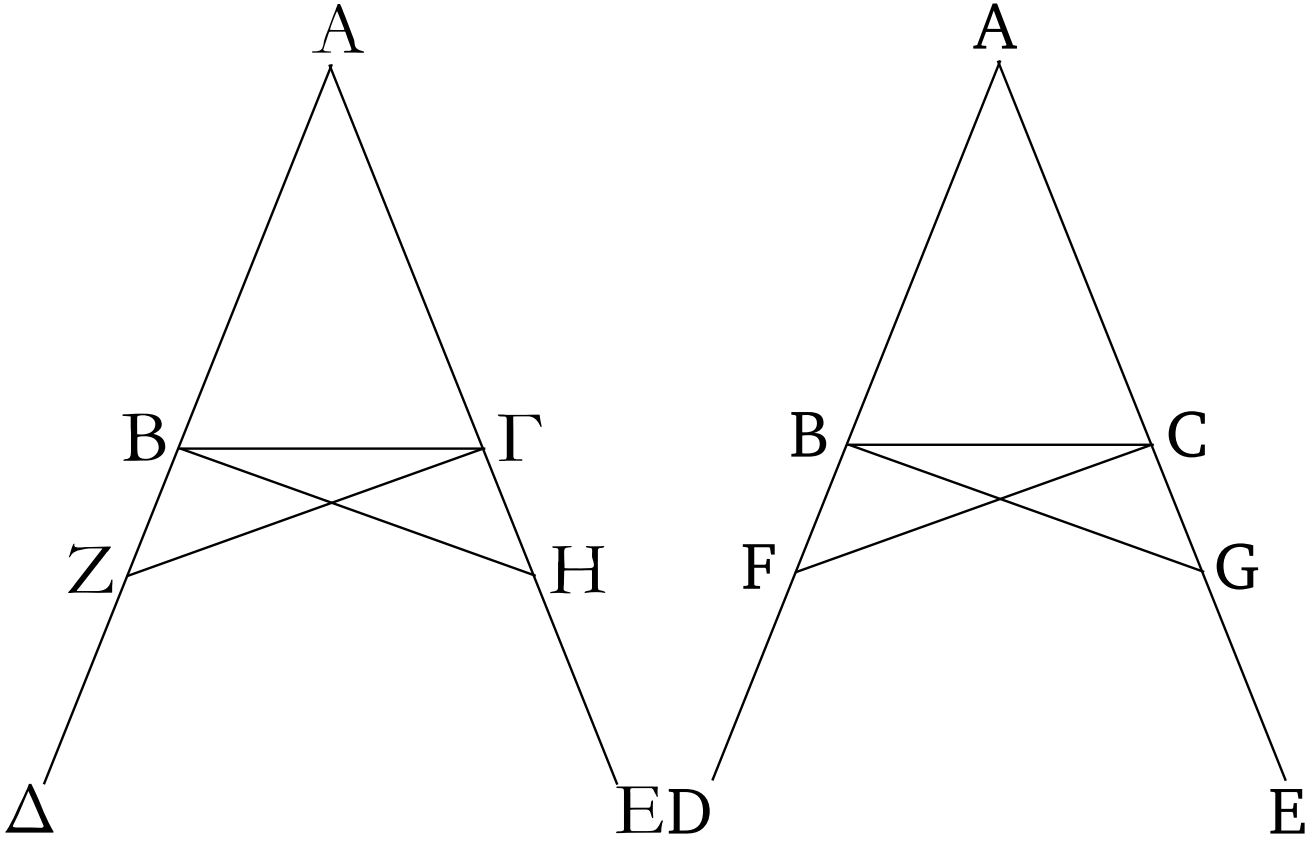
Thus, if two triangles have two sides equal to two sides, respectively, and have the angle(s) enclosed by the equal straight-lines equal, (then) they will also have the base equal to the base, and the triangle will be equal to the triangle, and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles. (Which is) the very thing it was required to show.

Proposition 5

For isosceles triangles, the angles at the base are equal to one another, and if the equal sides are produced (then) the angles under the base will be equal to one another.

Let ABC be an isosceles triangle having the side AB equal to the side AC , and let the straight-lines BD and CE be produced in a straight-line with AB and AC (respectively) [Post. 2]. I say that the angle ABC is equal to ACB , and (angle) CBD to BCE .

For let the point F be taken at random on BD , and let AG be cut off from the greater AE , equal to the lesser AF [Prop. 1.3]. Also, let the straight-lines FC and GB be joined [Post. 1].



Ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν AZ τῇ AH ἢ δὲ AB τῇ AG , δύο δὲ αἱ ZA , AG δυοὶ ταῖς HA , AB ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνίαν κοινὴν περιέθουσι τὴν ὑπὸ ZAH · βάσις ἄρα ἡ ZG βάσει τῇ HB ἴση ἐστίν, καὶ τὸ AZG τρίγωνον τῷ AHB τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρω ἑκατέρω, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ AGZ τῇ ὑπὸ ABH , ἡ δὲ ὑπὸ AZG τῇ ὑπὸ AHB . καὶ ἐπεὶ ὅλη ἡ AZ ὅλη τῇ AH ἐστὶν ἴση, ὧν ἡ AB τῇ AG ἐστὶν ἴση, λοιπὴ ἄρα ἡ BZ λοιπῇ τῇ GH ἐστὶν ἴση. ἐδείκθη δὲ καὶ ἡ ZG τῇ HB ἴση· δύο δὲ αἱ BZ , ZG δυοὶ ταῖς GH , HB ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνία ἡ ὑπὸ BZG γωνία τῇ ὑπὸ GHB ἴση, καὶ βάσις αὐτῶν κοινὴ ἡ BG · καὶ τὸ BZG ἄρα τρίγωνον τῷ GHB τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρω ἑκατέρω, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἐστὶν ἡ μὲν ὑπὸ ZBG τῇ ὑπὸ HGB ἢ δὲ ὑπὸ BGZ τῇ ὑπὸ GBH . ἐπεὶ οὖν ὅλη ἡ ὑπὸ ABH γωνία ὅλη τῇ ὑπὸ AGZ γωνίᾳ ἐδείκθη ἴση, ὧν ἡ ὑπὸ GBH τῇ ὑπὸ BGZ ἴση, λοιπὴ ἄρα ἡ ὑπὸ ABG λοιπῇ τῇ ὑπὸ AGB ἐστὶν ἴση· καὶ εἰσι πρὸς τῇ βάσει τοῦ ABG τριγώνου. ἐδείκθη δὲ καὶ ἡ ὑπὸ ZBG τῇ ὑπὸ HGB ἴση· καὶ εἰσιν ὑπὸ τὴν βάσιν.

Τῶν ἄρα ἰσοσκελῶν τριγώνων αἱ τρὸς τῇ βάσει γωνίαι ἴσαι ἀλλήλαις εἰσὶν, καὶ προσεκβληθεισῶν τῶν ἴσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι ἴσαι ἀλλήλαις ἔσονται· ὅπερ ἔδει

In fact, since AF is equal to AG , and AB to AC , the two (straight-lines) FA , AC are equal to the two (straight-lines) GA , AB , respectively. They also encompass a common angle, FAG . Thus, the base FC is equal to the base GB , and the triangle AFC will be equal to the triangle AGB , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles [Prop. 1.4]. (That is) ACF to ABG , and AFC to AGB . And since the whole of AF is equal to the whole of AG , within which AB is equal to AC , the remainder BF is thus equal to the remainder CG [C.N. 3]. But FC was also shown (to be) equal to GB . So the two (straight-lines) BF , FC are equal to the two (straight-lines) CG , GB , respectively, and the angle BFC (is) equal to the angle CGB , and the base BC is common to them. Thus, the triangle BFC will be equal to the triangle CGB , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles [Prop. 1.4]. Thus, FBC is equal to GCB , and BCF to CBG . Therefore, since the whole angle ABG was shown (to be) equal to the whole angle ACF , within which CBG is equal to BCF , the remainder ABC is thus equal to the remainder ACB [C.N. 3]. And they are at the base of triangle ABC . And FBC was also shown (to be) equal to GCB . And they are under the base.

Thus, for isosceles triangles, the angles at the base are

δειχαι.

equal to one another, and if the equal sides are produced (then) the angles under the base will be equal to one another. (Which is) the very thing it was required to show.

6.

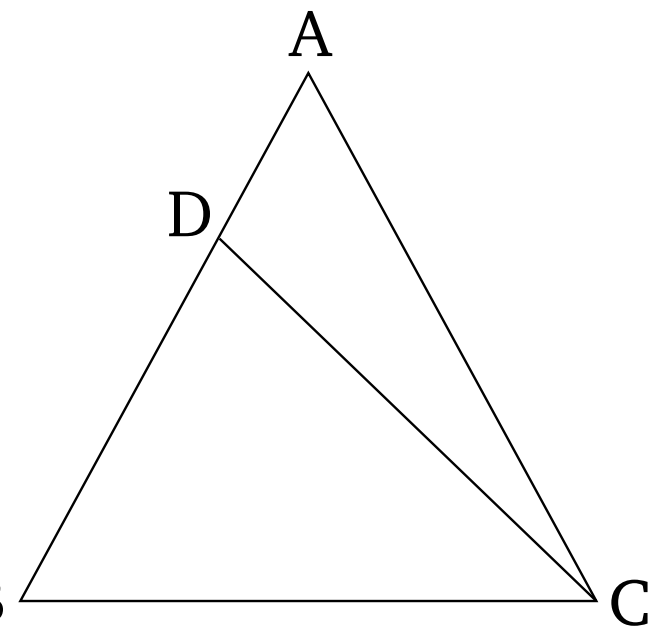
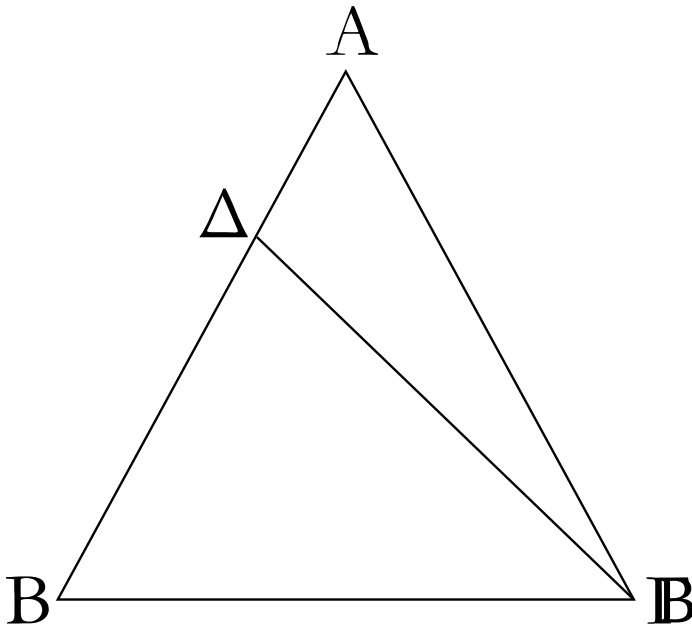
Proposition 6

Ἐάν τριγώνον αἱ δύο γωνίαι ἴσαι ἀλλήλαις ᾤσιν, καὶ α-
ῖ ὑπὸ τὰς ἴσας γωνίας ὑποτείνουσαι πλευραὶ ἴσαι ἀλλήλαις
έσονται.

If a triangle has two angles equal to one another (then) the
sides subtending the equal angles will also be equal to one
another.

Ἐστω τρίγωνον τὸ $ABΓ$ ἴσην ἔθον τὴν ὑπὸ $ABΓ$ γωνίαν
τῇ ὑπὸ $ΑΓΒ$ γωνίᾳ· λέγω, ὅτι καὶ πλευρὰ ἡ AB πλευρᾷ τῇ
 $ΑΓ$ ἔστιν ἴση.

Let ABC be a triangle having the angle ABC equal to the
angle ACB . I say that side AB is also equal to side AC .



Εἰ γὰρ ἄνισός ἐστιν ἡ AB τῇ $ΑΓ$, ἡ ἑτέρα αὐτῶν μείζων
ἐστίν. ἔστω μείζων ἡ AB , καὶ ἀφρηθήσθω ἀπὸ τῆς μείζονος
τῆς AB τῇ ἐλάττωι τῇ $ΑΓ$ ἴση ἡ $ΔB$, καὶ ἐπεξεύθθω ἡ $ΔΓ$.

For if AB is unequal to AC (then) one of them is greater.
Let AB be greater. And let DB , equal to the lesser AC , be
cut off from the greater AB [Prop. 1.3]. And let DC be joined
[Post. 1].

Ἐπεὶ οὖν ἴση ἐστὶν ἡ $ΔB$ τῇ $ΑΓ$ κοινῇ δὲ ἡ $BΓ$, δύο δὲ
αἱ $ΔB$, $BΓ$ δύο ταῖς $ΑΓ$, $ΓB$ ἴσαι εἰσὶν ἑκατέρω ἑκατέρω, καὶ
γωνία ἡ ὑπὸ $ΔBΓ$ γωνία τῇ ὑπὸ $ΑΓΒ$ ἔστιν ἴση· βάσει ἄρα
ἡ $ΔΓ$ βάσει τῇ AB ἴση ἐστίν, καὶ τὸ $ΔBΓ$ τρίγωνον τῷ $ΑΓΒ$
τρίγωνῳ ἴσον ἐσται, τὸ ἔλασσον τῷ μείζονι· ὅπερ ἄτοπον·
οὐκ ἄρα ἄνισός ἐστιν ἡ AB τῇ $ΑΓ$ · ἴση ἄρα.

Therefore, since DB is equal to AC , and BC (is) common,
the two sides DB , BC are equal to the two sides AC , CB , re-
spectively, and the angle DBC is equal to the angle ACB . Thus,
the base DC is equal to the base AB , and the triangle DBC
will be equal to the triangle ACB [Prop. 1.4], the lesser to the
greater. The very notion (is) absurd [C.N. 5]. Thus, AB is not
unequal to AC . Thus, (it is) equal.[†]

Ἐάν ἄρα τριγώνον αἱ δύο γωνίαι ἴσαι ἀλλήλαις ᾤσιν, καὶ
αἱ ὑπὸ τὰς ἴσας γωνίας ὑποτείνουσαι πλευραὶ ἴσαι ἀλλήλαις
έσονται· ὅπερ ἔδει δεῖχαι.

Thus, if a triangle has two angles equal to one another
(then) the sides subtending the equal angles will also be equal
to one another. (Which is) the very thing it was required to
show.

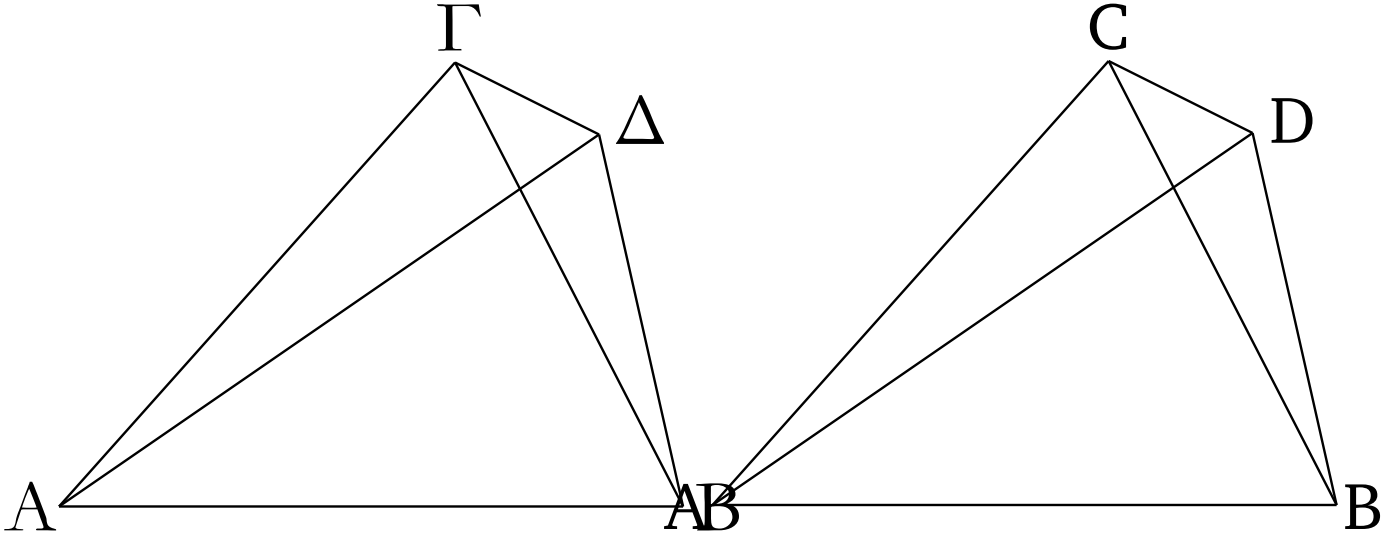
[†] Here, use is made of the previously unmentioned common notion that if two quantities are not unequal then they must be equal. Later on, use is made of the closely related common notion that if two quantities are not greater than or less than one another, respectively, then they must be equal to one another.

7.

Ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις ἄλλαι δύο εὐθεῖαι ἴσαι ἑκατέρα ἑκατέρῃ οὐ συνσταθήσονται πρὸς ἄλλῳ καὶ ἄλλῳ σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσαι ταῖς ἐξ ἀρθῆς εὐθείαις.

Proposition 7

On the same straight-line, two other straight-lines equal, respectively, to two (given) straight-lines (which meet) cannot be constructed (meeting) at a different point on the same side (of the straight-line), but having the same ends as the given straight-lines.



Εἰ γὰρ δυνατόν, ἐπὶ τῆς αὐτῆς εὐθείας τῆς AB δύο ταῖς αὐταῖς εὐθείαις ταῖς AG , GB ἄλλαι δύο εὐθεῖαι αἱ AD , DB ἴσαι ἑκατέρα ἑκατέρῃ συνεστάτωσαν πρὸς ἄλλῳ καὶ ἄλλῳ σημείῳ τῷ Γ καὶ Δ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσαι, ὥστε ἴσην εἶναι τὴν μὲν GA τῇ DA τὸ αὐτὸ πέραξ ἔθουσας αὐτῇ τὸ A , τὴν δὲ GB τῇ DB τὸ αὐτὸ πέραξ ἔθουσας αὐτῇ τὸ B , καὶ ἐπεξεύθθω ἡ GD .

Ἐπεὶ οὖν ἴση ἐστὶν ἡ AG τῇ AD , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ AGD τῇ ὑπὸ ADG . μείζων ἄρα ἡ ὑπὸ ADG τῆς ὑπὸ AGB . πολλῶν ἄρα ἡ ὑπὸ ADB μείζων ἐστὶ τῆς ὑπὸ AGB . πάλιν ἐπεὶ ἴση ἐστὶν ἡ GB τῇ DB , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ ADB γωνία τῇ ὑπὸ AGB . ἐδείχθη δὲ αὐτῆς καὶ πολλῶν μείζων· ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις ἄλλαι δύο εὐθεῖαι ἴσαι ἑκατέρα ἑκατέρῃ συνσταθήσονται πρὸς ἄλλῳ καὶ ἄλλῳ σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσαι ταῖς ἐξ ἀρθῆς εὐθείαις· ὅπερ ἔδει δεῖχαι.

For, if possible, let the two straight-lines AC , CB , equal to two other straight-lines AD , DB , respectively, be constructed on the same straight-line AB , meeting at different points, C and D , on the same side (of AB), and having the same ends (on AB). So CA is equal to DA , having the same end A as it, and CB is equal to DB , having the same end B as it. And let CD be joined [Post. 1].

Therefore, since AC is equal to AD , the angle ACD is also equal to angle ADC [Prop. 1.5]. Thus, ADC (is) greater than DCB [C.N. 5]. Thus, CDB is much greater than DCB [C.N. 5]. Again, since CB is equal to DB , the angle CDB is also equal to angle DCB [Prop. 1.5]. But it was shown that the former (angle) is also much greater (than the latter). The very thing is impossible.

Thus, on the same straight-line, two other straight-lines equal, respectively, to two (given) straight-lines (which meet) cannot be constructed (meeting) at a different point on the same side (of the straight-line), but having the same ends as the given straight-lines. (Which is) the very thing it was required to show.

8.

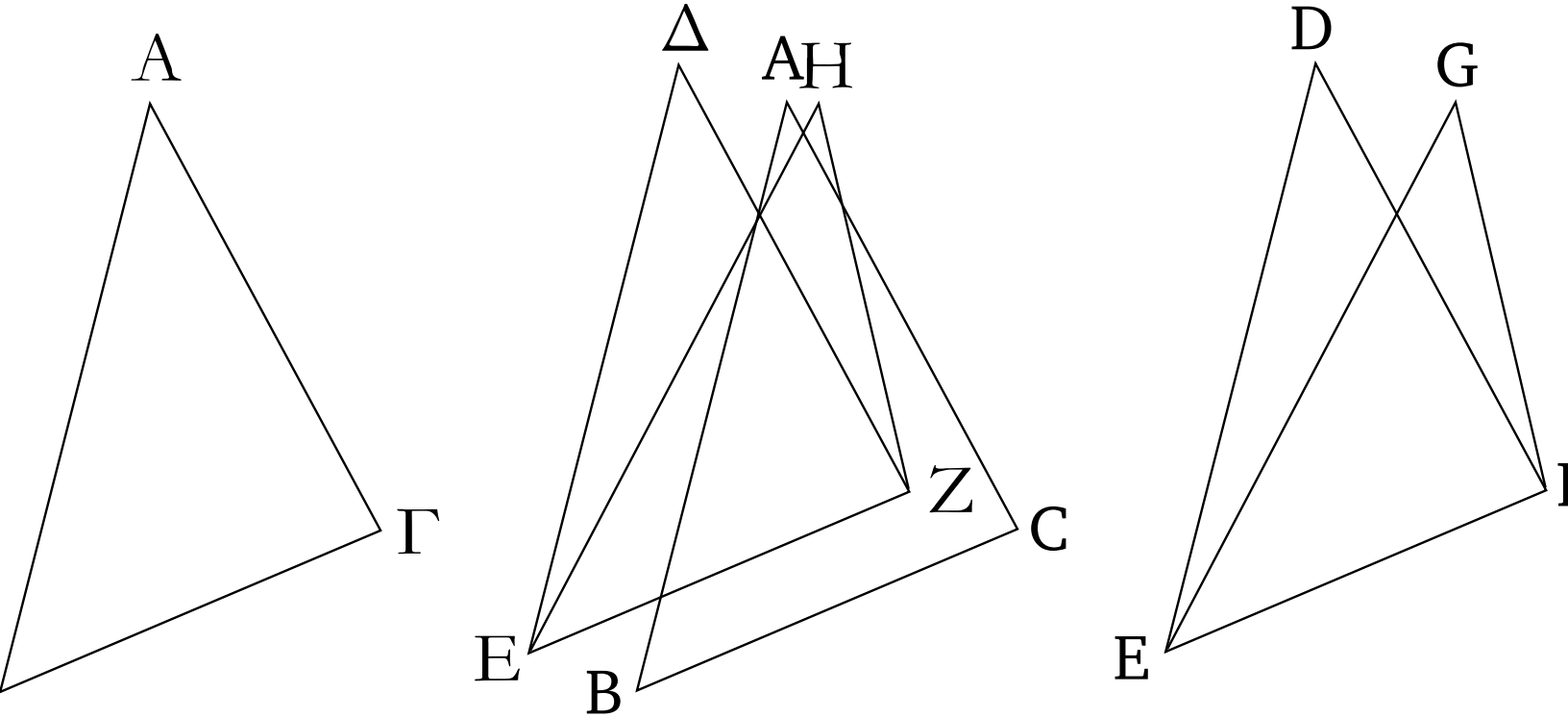
Ἐάν δύο τρίγωνα τὰς δύο πλευράς [ταῖς] δύο πλευραῖς ἴσαξ ἔθῃ ἑκατέραν ἑκατέρῃ, ἔθῃ δὲ καὶ τὴν βάσιν τῇ βάσει ἴσην, καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην.

Proposition 8

If two triangles have two sides equal to two sides, respectively, and also have the base equal to the base, (then) they will also have equal the angles encompassed by the equal straight-lines.

Ἐστω δύο τρίγωνα τὰ $ABΓ$, $ΔΕΖ$ τὰς δύο πλευρὰς τὰς AB , $ΑΓ$ ταῖς δύο πλευραῖς ταῖς $ΔΕ$, $ΔΖ$ ἴσας ἔχοντα ἑκατέραν ἑκατέρα, τὴν μὲν AB τῇ $ΔΕ$ τὴν δὲ $ΑΓ$ τῇ $ΔΖ$ · ἐθέτω δὲ καὶ βάσιν τὴν $ΒΓ$ βάσει τῇ $ΕΖ$ ἴσην· λέγω, ὅτι καὶ γωνία ἡ ὑπὸ $ΒΑΓ$ γωνία τῇ ὑπὸ $ΕΔΖ$ ἔστω ἴση.

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is) AB to DE , and AC to DF . Let them also have the base BC equal to the base EF . I say that the angle BAC is also equal to the angle EDF .



Ἐφαρμοζόμενον γάρ τοῦ $ABΓ$ τριγώνου ἐπὶ τὸ $ΔΕΖ$ τριγώνον καὶ τιθεμένον τοῦ μὲν B σημείου ἐπὶ τὸ $Ε$ σημῖον τῆς δὲ $ΒΓ$ εὐθείας ἐπὶ τὴν $ΕΖ$ ἐφαρμόσει καὶ τὸ $Γ$ σημῖον ἐπὶ τὸ $Ζ$ διὰ τὸ ἴσην εἶναι τὴν $ΒΓ$ τῇ $ΕΖ$ · ἐφαρμοσάσης δὲ τῆς $ΒΓ$ ἐπὶ τὴν $ΕΖ$ ἐφαρμόσουσι καὶ αἱ BA , $ΓΑ$ ἐπὶ τὰς $ΕΔ$, $ΔΖ$. εἰ γὰρ βάσις μὲν ἡ $ΒΓ$ ἐπὶ βάσιν τὴν $ΕΖ$ ἐφαρμόσει, αἱ δὲ BA , $ΑΓ$ πλευραὶ ἐπὶ τὰς $ΕΔ$, $ΔΖ$ οὐκ ἐφαρμόσουσιν ἀλλὰ παραλλάχουσιν ὥς αἱ $ΕΗ$, $ΗΖ$, συσταθήσονται ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις ἄλλαι δύο εὐθεῖαι ἴσαι ἑκατέρα ἑκατέρῳ πρόξ ἄλλῳ καὶ ἄλλῳ σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσαι. οὐ συνίστανται δέ· οὐκ ἔστι ἐφαρμοζομένης τῆς $ΒΓ$ βάσεως ἐπὶ τὴν $ΕΖ$ βάσιν οὐκ ἐφαρμόσουσι καὶ αἱ BA , $ΑΓ$ πλευραὶ ἐπὶ τὰς $ΕΔ$, $ΔΖ$. ἐφαρμόσουσιν ἄρα· ὥστε καὶ γωνία ἡ ὑπὸ $ΒΑΓ$ ἐπὶ γωνίαν τὴν ὑπὸ $ΕΔΖ$ ἐφαρμόσει καὶ ἴση αὐτῇ ἔσται.

Ἐὰν ἄρα δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρῳ καὶ τὴν βάσιν τῇ βάσει ἴσην ἔθῃ, καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην· ὅπερ ἔδει δεῖχαι.

For if triangle ABC is applied to triangle DEF , the point B being placed on point E , and the straight-line BC on EF , (then) point C will also coincide with F , on account of BC being equal to EF . So (because of) BC coinciding with EF , (the sides) BA and CA will also coincide with ED and DF (respectively). For if base BC coincides with base EF , but the sides AB and AC do not coincide with ED and DF (respectively), but miss like EG and GF (in the above figure), then we will have constructed upon the same straight-line two other straight-lines equal, respectively, to two (given) straight-lines, and (meeting) at a different point on the same side (of the straight-line), but having the same ends. But (such straight-lines) cannot be constructed [Prop. 1.7]. Thus, the base BC being applied to the base EF , the sides BA and AC cannot not coincide with ED and DF (respectively). Thus, they will coincide. So the angle BAC will also coincide with angle EDF , and will be equal to it [C.N. 4].

Thus, if two triangles have two sides equal to two side, respectively, and have the base equal to the base, (then) they will also have equal the angles encompassed by the equal straight-lines. (Which is) the very thing it was required to show.

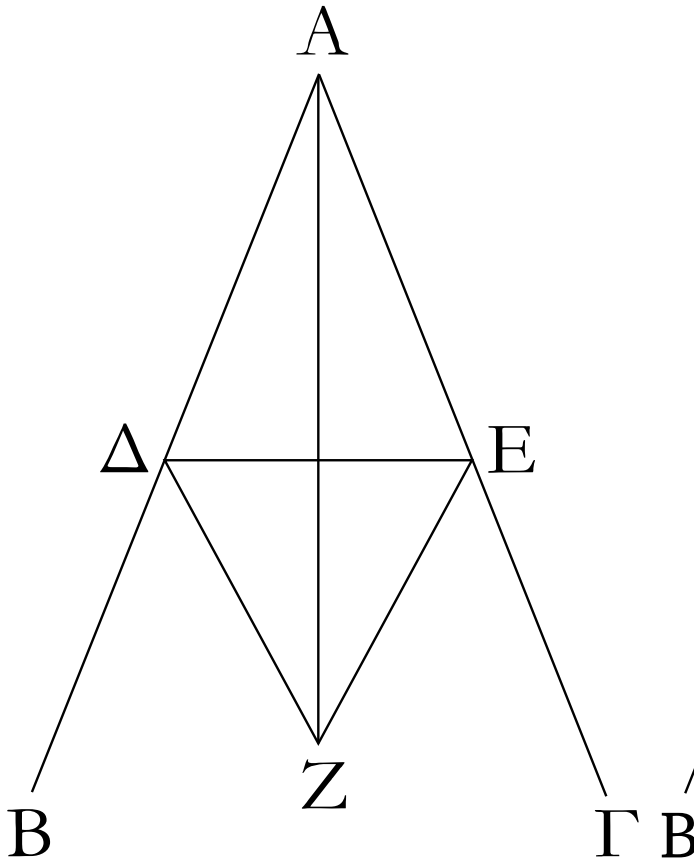
9.

Τὴν δοθεῖσαν γωνίαν εὐθύγραμμον δίῃ τεμεῖν.

Ἐστω ἡ δοθεῖσα γωνία εὐθύγραμμος ἡ ὑπὸ BAG . δεῖ δὴ αὐτὴν δίῃ τεμεῖν.

Εἰλήφθω ἐπὶ τῆς AB τυθὸν σημεῖον τὸ Δ , καὶ ἀφηρήσθω ἀπὸ τῆς AG τῇ AD ἴση ἡ AE , καὶ ἐπεζεύθθω ἡ DE , καὶ συνεστάτω ἐπὶ τῆς DE τρίγωνον ἰσόπλευρον τὸ DEZ , καὶ ἐπεζεύθθω ἡ AZ . λέγω, ὅτι ἡ ὑπὸ BAG γωνία δίῃ τέτμηται ὑπὸ τῆς AZ εὐθείας.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ AD τῇ AE , κοινὴ δὲ ἡ AZ , δύο δὴ αἱ DA , AZ δυοὶ ταῖς EA , AZ ἴσαι εἰσὶν ἑκατέρω ἑκατέρω. καὶ βάσει ἡ DZ βάσει τῇ EZ ἴση ἐστὶν γωνία ἄρα ἡ ὑπὸ DAZ γωνία τῇ ὑπὸ EAZ ἴση ἐστίν.



Ἡ ἄρα δοθεῖσα γωνία εὐθύγραμμος ἡ ὑπὸ BAG δίῃ τέτμηται ὑπὸ τῆς AZ εὐθείας· ὅπερ ἔδει ποιῆσαι.

10.

Τὴν δοθεῖσαν εὐθεῖαν πεπερασμένην δίῃ τεμεῖν.

Ἐστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB . δεῖ δὴ τὴν AB εὐθεῖαν πεπερασμένην δίῃ τεμεῖν.

Συνεστάτω ἐπ' αὐτῆς τρίγωνον ἰσόπλευρον τὸ $AB\Gamma$, καὶ

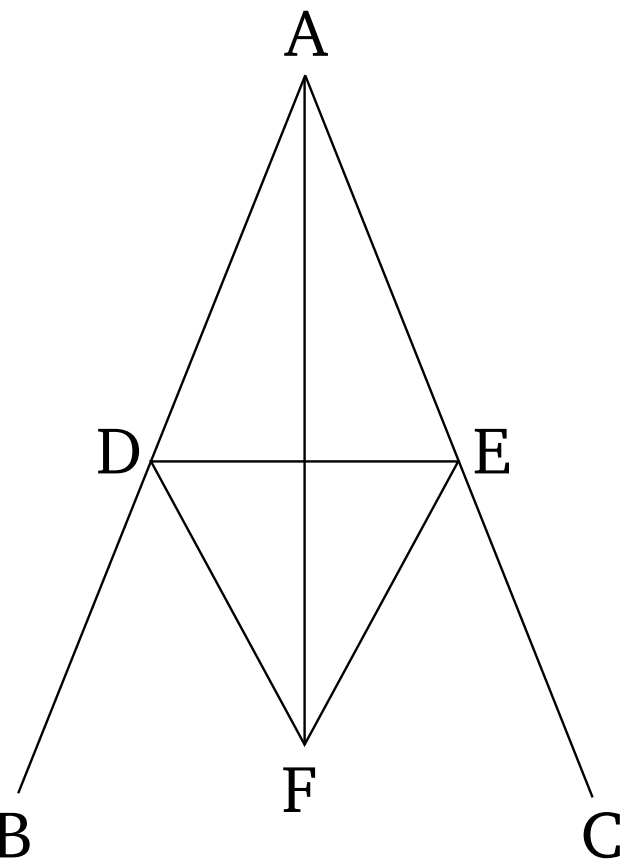
Proposition 9

To cut a given rectilinear angle in half.

Let BAC be the given rectilinear angle. So it is required to cut it in half.

Let the point D be taken at random on AB , and let AE , equal to AD , be cut off from AC [Prop. 1.3], and let DE be joined. And let the equilateral triangle DEF be constructed upon DE [Prop. 1.1], and let AF be joined. I say that the angle BAC has been cut in half by the straight-line AF .

For since AD is equal to AE , and AF is common, the two (straight-lines) DA , AF are equal to the two (straight-lines) EA , AF , respectively. And the base DF is equal to the base EF . Thus, angle DAF is equal to angle EAF [Prop. 1.8].



Thus, the given rectilinear angle BAC has been cut in half by the straight-line AF . (Which is) the very thing it was required to do.

Proposition 10

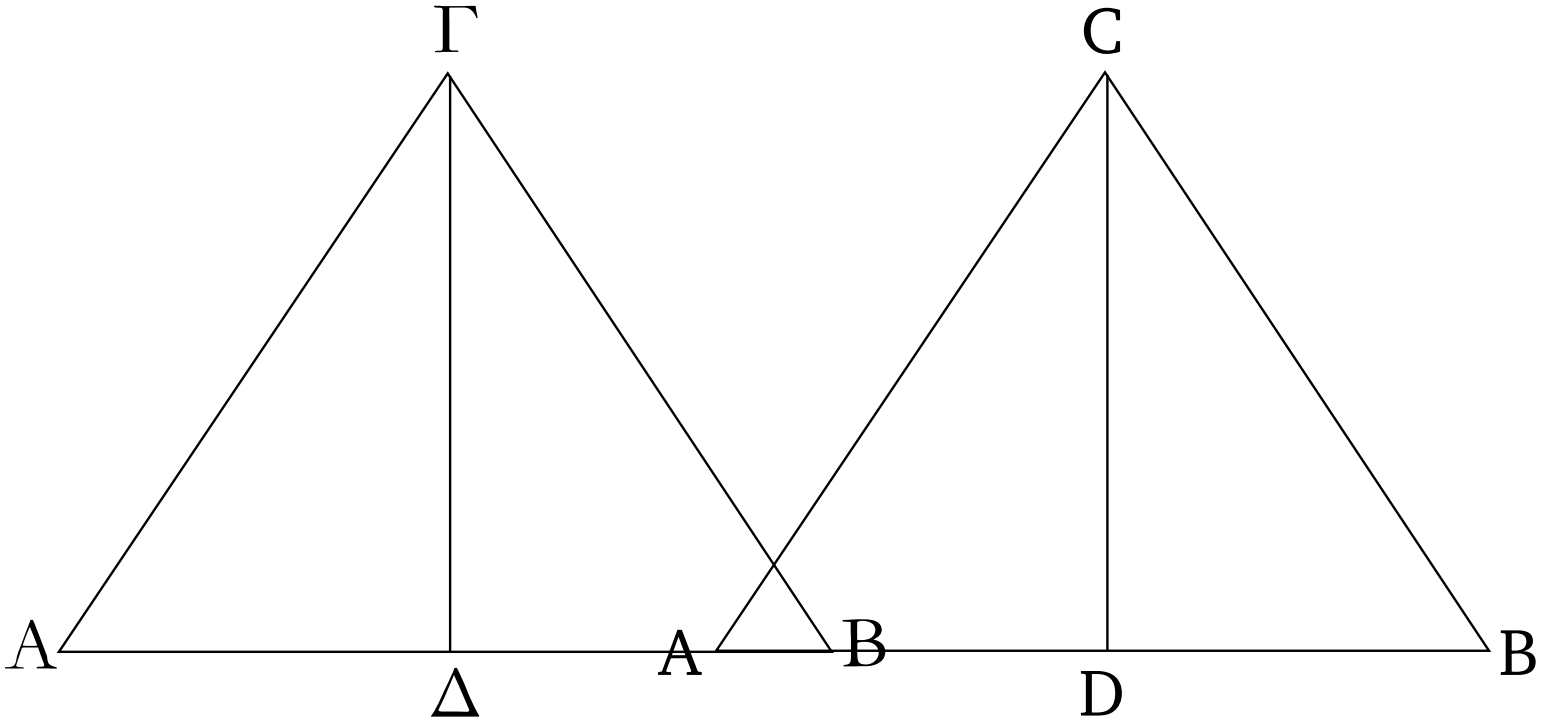
To cut a given finite straight-line in half.

Let AB be the given finite straight-line. So it is required to cut the finite straight-line AB in half.

Let the equilateral triangle ABC be constructed upon (AB)

τετμήσθω ἡ ὑπὸ AGB γωνία διὰ τῇ $ΓΔ$ εὐθείᾳ· λέγω, ὅτι ἡ AB εὐθεῖα διὰ τέμνεται κατὰ τὸ $Δ$ σημεῖον.

[Prop. 1.1], and let the angle ACB be cut in half by the straight-line CD [Prop. 1.9]. I say that the straight-line AB has been cut in half at point D .



Ἐπεὶ γάρ ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΓΒ$, κοινὴ δὲ ἡ $ΓΔ$, δύο δὴ αἱ $ΑΓ$, $ΓΔ$ δύο ταῖς $ΒΓ$, $ΓΔ$ ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνία ἡ ὑπὸ $ΑΓΔ$ γωνία τῇ ὑπὸ $ΒΓΔ$ ἴση ἐστὶν· βάσεις ἄρα ἡ $ΑΔ$ βάσει τῇ $ΒΔ$ ἴση ἐστὶν.

Ἦ ἄρα δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB διὰ τέμνεται κατὰ τὸ $Δ$ · ὅπερ ἔδει ποιῆσαι.

For since AC is equal to CB , and CD (is) common, the two (straight-lines) AC , CD are equal to the two (straight-lines) BC , CD , respectively. And the angle ACD is equal to the angle BCD . Thus, the base AD is equal to the base BD [Prop. 1.4].

Thus, the given finite straight-line AB has been cut in half at (point) D . (Which is) the very thing it was required to do.

11.

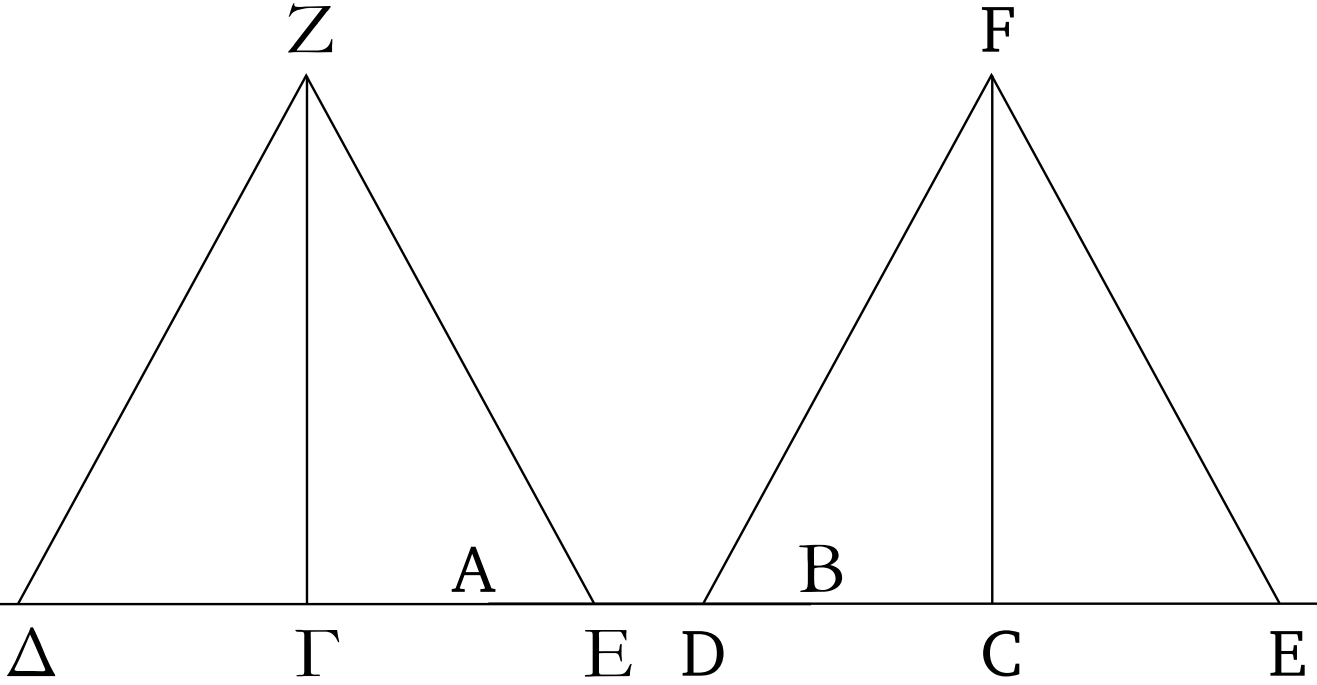
Proposition 11

Τῇ δοθείσῃ εὐθείᾳ ἀπὸ τοῦ πρὸς αὐτῇ δοθέντος σημείου πρὸς ὀρθὰς γωνία εὐθεῖαν γραμμὴν ἀγαγεῖν.

Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB τὸ δὲ δοθέν σημεῖον ἐπ' αὐτῇ τὸ $Γ$ · δεῖ δὲ ἀπὸ τοῦ $Γ$ σημείου τῇ AB εὐθείᾳ πρὸς ὀρθὰς γωνία εὐθεῖαν γραμμὴν ἀγαγεῖν.

To draw a straight-line at right-angles to a given straight-line from a given point on it.

Let AB be the given straight-line, and C the given point on it. So it is required to draw a straight-line from the point C at right-angles to the straight-line AB .



Εἰλήφθω ἐπὶ τῆς $ΑΓ$ τυθὸν σημείον τὸ Δ , καὶ κείσθω τῇ $\Gamma\Delta$ ἴση ἡ ΓE , καὶ συνεστώτω ἐπὶ τῆς ΔE τρίγωνον ἰσόπλευρον τὸ $Z\Delta E$, καὶ ἐπεξεύθθω ἡ $Z\Gamma$. λέγω, ὅτι τῇ δοθείσῃ εὐθείᾳ τῇ $ΑΒ$ ἀπὸ τοῦ πρὸς αὐτῇ δοθέντος σημείου τοῦ Γ πρὸς ὀρθὰς γωνίαξ εὐθεῖα γραμμὴ ἥκται ἡ $Z\Gamma$.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ $\Delta\Gamma$ τῇ ΓE , κοινὴ δὲ ἡ ΓZ , δύο δὴ αἱ $\Delta\Gamma$, ΓZ δυοὶ ταῖς $E\Gamma$, ΓZ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ βάσεις ἡ ΔZ βάσει τῇ ZE ἴση ἐστὶν· γωνία ἄρα ἡ ὑπὸ $\Delta\Gamma Z$ γωνία τῇ ὑπὸ $E\Gamma Z$ ἴση ἐστὶν· καὶ εἰσὶν ἐφεχῆς. ὅταν δὲ εὐθεῖα ἐπ' εὐθείαν σταθεῖσα τὰς ἐφεχῆς γωνίας ἴσαξ ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν ἴσων γωνιῶν ἐστὶν· ὀρθὴ ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ $\Delta\Gamma Z$, $Z\Gamma E$.

Τῇ ἄρα δοθείσῃ εὐθείᾳ τῇ $ΑΒ$ ἀπὸ τοῦ πρὸς αὐτῇ δοθέντος σημείου τοῦ Γ πρὸς ὀρθὰς γωνίαξ εὐθεῖα γραμμὴ ἥκται ἡ ΓZ . ὅπερ ἔδει ποιῆσαι.

Let the point D be taken at random on AC , and let CE be made equal to CD [Prop. 1.3], and let the equilateral triangle FDE be constructed on DE [Prop. 1.1], and let FC be joined. I say that the straight-line FC has been drawn at right-angles to the given straight-line AB from the given point C on it.

For since DC is equal to CE , and CF is common, the two (straight-lines) DC , CF are equal to the two (straight-lines), EC , CF , respectively. And the base DF is equal to the base FE . Thus, the angle DCF is equal to the angle ECF [Prop. 1.8], and they are adjacent. But when a straight-line stood on a(nother) straight-line makes the adjacent angles equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, each of the (angles) DCF and FCE is a right-angle.

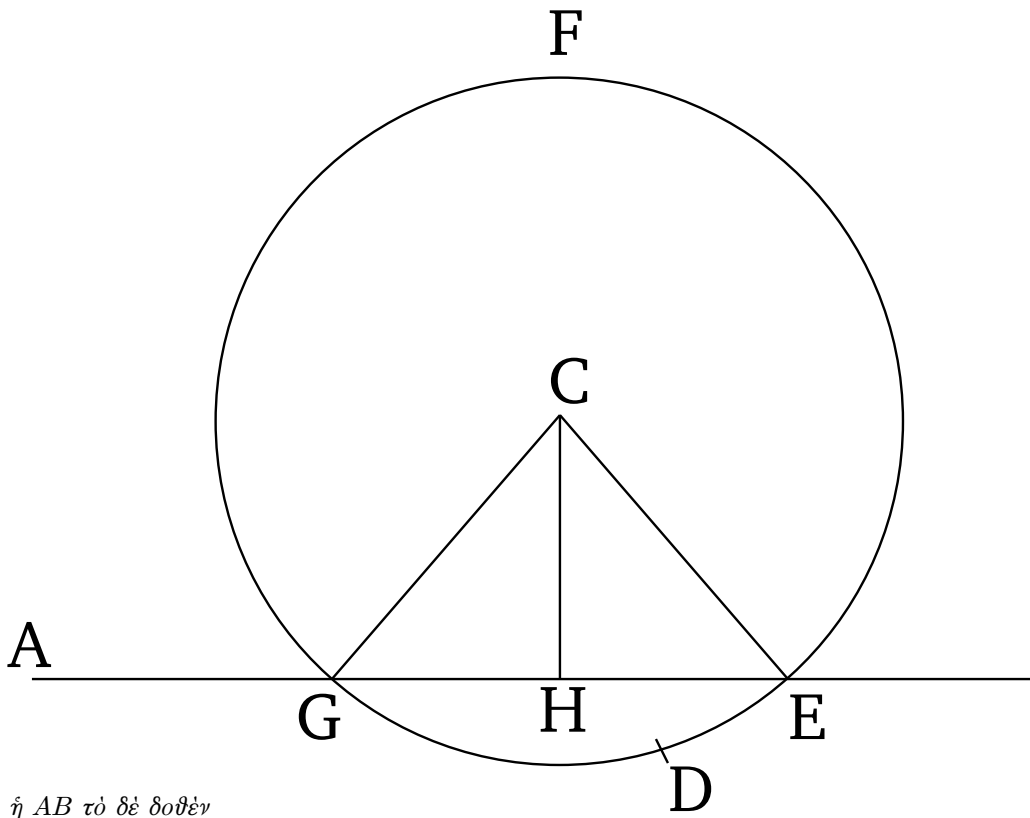
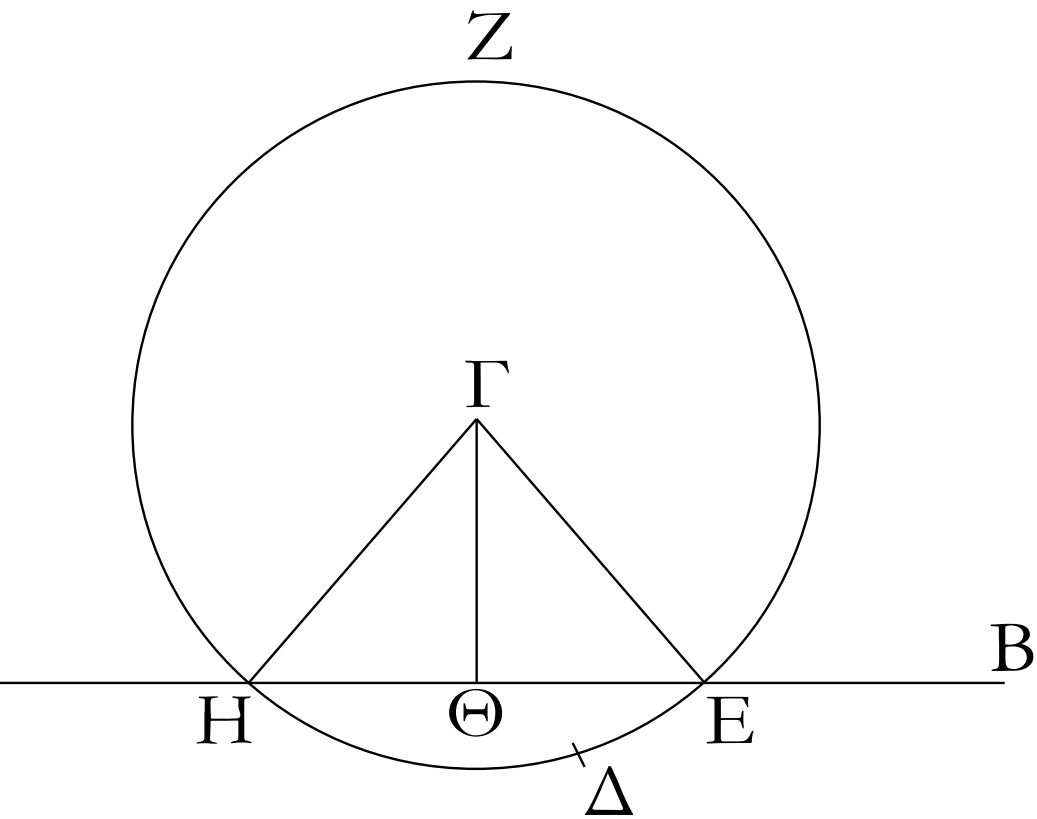
Thus, the straight-line CF has been drawn at right-angles to the given straight-line AB from the given point C on it. (Which is) the very thing it was required to do.

12.

Proposition 12

Ἐπὶ τὴν δοθεῖσαν εὐθεῖαν ἄπειρον ἀπὸ τοῦ δοθέντος σημείου, ὃ μὴ ἐστὶν ἐπ' αὐτῆς, κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

To draw a straight-line perpendicular to a given infinite straight-line from a given point which is not on it.



Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἀπειροξ ἡ AB τὸ δὲ δοθέν

σημείον, ὃ μὴ ἔστιν ἐπ' αὐτῆς, τὸ Γ . δεῖ δὲ ἐπὶ τὴν δοθεῖσαν εὐθεῖαν ἄπειρον τὴν AB ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ , ὃ μὴ ἔστιν ἐπ' αὐτῆς, κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Εἰλήφθω γὰρ ἐπὶ τὰ ἑτέρα μέρη τῆς AB εὐθείας τυθὸν σημείον τὸ Δ , καὶ κέντρῳ μὲν τῷ Γ διαστήματι δὲ τῷ $\Gamma\Delta$ κύκλος γεγράφθω ὁ EZH , καὶ τετμήσθω ἡ EH εὐθεῖα διὰ κατὰ τὸ Θ , καὶ ἐπεξεύθθωσαν αἱ GH , $\Gamma\Theta$, ΓE εὐθεῖαι· λέγω, ὅτι ἐπὶ τὴν δοθεῖσαν εὐθεῖαν ἄπειρον τὴν AB ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ , ὃ μὴ ἔστιν ἐπ' αὐτῆς, κάθετος ἥκται ἡ $\Gamma\Theta$.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ $H\Theta$ τῇ ΘE , κοινὴ δὲ ἡ $\Theta\Gamma$, δύο δὲ αἱ $H\Theta$, $\Theta\Gamma$ δύο ταῖς $E\Theta$, $\Theta\Gamma$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρῃ· καὶ βάσει ἡ GH βάσει τῇ ΓE ἔστιν ἴση· γωνία ἄρα ἡ ὑπὸ $\Gamma\Theta H$ γωνία τῇ ὑπὸ $E\Theta\Gamma$ ἔστιν ἴση. καὶ εἰσὶν ἐφεχῆς. ὅταν δὲ εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα τὰς ἐφεχῆς γωνίας ἴσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν ἴσων γωνιῶν ἐστὶν, καὶ ἡ ἐφεστηκυῖα εὐθεῖα κάθετος καλεῖται ἐφ' ἣν ἐφέστηκεν.

Ἐπὶ τὴν δοθεῖσαν ἄρα εὐθεῖαν ἄπειρον τὴν AB ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ , ὃ μὴ ἔστιν ἐπ' αὐτῆς, κάθετος ἥκται ἡ $\Gamma\Theta$. ὅπερ ἔδει ποιῆσαι.

Let AB be the given infinite straight-line and C the given point, which is not on it. So it is required to draw a straight-line perpendicular to the given infinite straight-line AB from the given point C , which is not on it.

For let point D be taken at random on the other side (to C) of the straight-line AB , and let the circle EFG be drawn with center C and radius CD [Post. 3], and let the straight-line EG be cut in half at (point) H [Prop. 1.10], and let the straight-lines CG , CH , and CE be joined. I say that the (straight-line) CH has been drawn perpendicular to the given infinite straight-line AB from the given point C , which is not on it.

For since GH is equal to HE , and HC (is) common, the two (straight-lines) GH , HC are equal to the two (straight-lines) EH , HC , respectively, and the base CG is equal to the base CE . Thus, the angle CHG is equal to the angle EHC [Prop. 1.8], and they are adjacent. But when a straight-line stood on a(nother) straight-line makes the adjacent angles equal to one another, each of the equal angles is a right-angle, and the former straight-line is called a perpendicular to that upon which it stands [Def. 1.10].

Thus, the (straight-line) CH has been drawn perpendicular to the given infinite straight-line AB from the given point C , which is not on it. (Which is) the very thing it was required to do.

13.

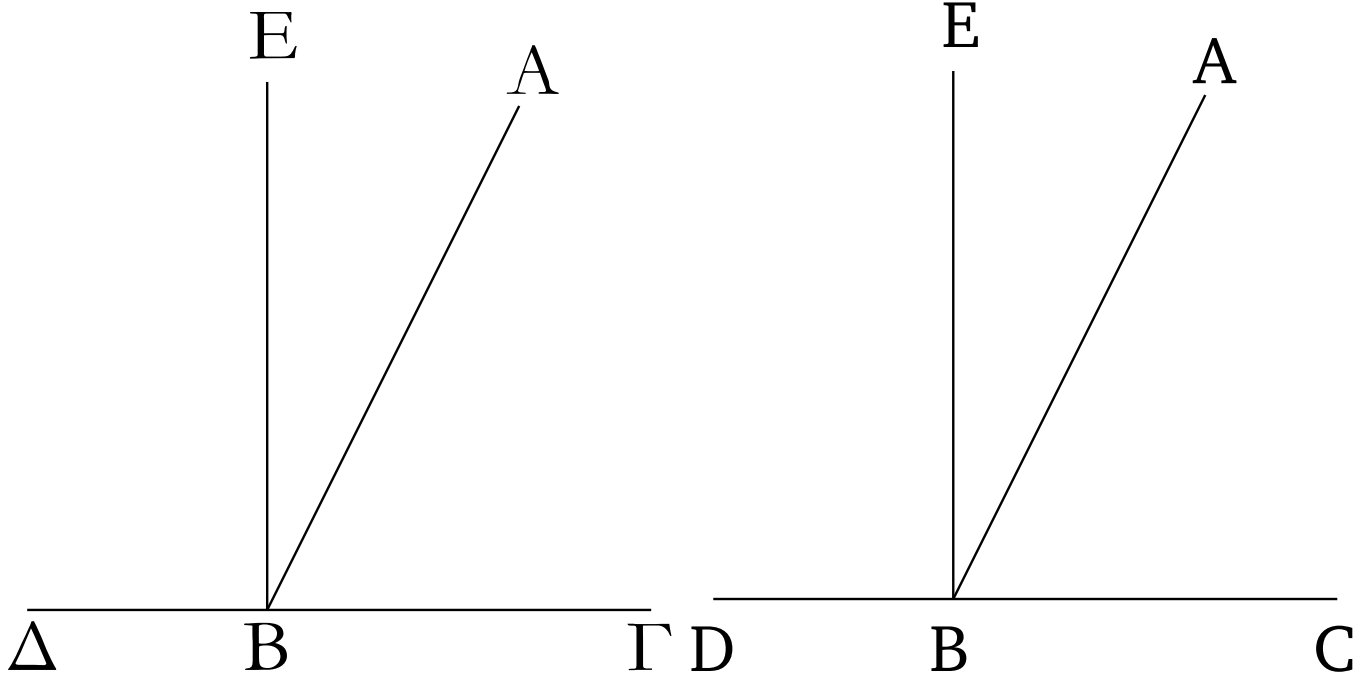
Proposition 13

Ἐὰν εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα γωνίας ποιῇ, ἥτοι δύο ὀρθὰς ἢ δυὸν ὀρθαῖς ἴσας ποιήσῃ.

Εὐθεῖα γὰρ τις ἡ AB ἐπ' εὐθεῖαν τὴν $\Gamma\Delta$ σταθεῖσα γωνίας ποιείτω τὰς ὑπὸ $\Gamma B A$, $A B \Delta$ · λέγω, ὅτι αἱ ὑπὸ $\Gamma B A$, $A B \Delta$ γωνίαι ἥτοι δύο ὀρθαὶ εἰσὶν ἢ δυὸν ὀρθαῖς ἴσαι.

If a straight-line stood on a(nother) straight-line makes angles, it will certainly either make two right-angles, or (angles whose sum is) equal to two right-angles.

For let some straight-line AB stood on the straight-line CD make the angles CBA and ABD . I say that the angles CBA and ABD are certainly either two right-angles, or (have a sum) equal to two right-angles.



Εἰ μὲν οὖν ἴση ἐστὶν ἡ ὑπὸ ΓΒΑ τῇ ὑπὸ ΑΒΔ, δύο ὀρθαὶ εἰσιν. εἰ δὲ οὐκ, ἤθθω ἀπὸ τοῦ Β σημείου τῇ ΓΔ [εὐθείᾳ] πρὸς ὀρθὰς ἡ ΒΕ· αἱ ἄρα ὑπὸ ΓΒΕ, ΕΒΔ δύο ὀρθαὶ εἰσιν· καὶ ἐπεὶ ἡ ὑπὸ ΓΒΕ δυοὶ ταῖς ὑπὸ ΓΒΑ, ΑΒΕ ἴση ἐστίν, κοινὴ προσκείσθω ἡ ὑπὸ ΕΒΔ· αἱ ἄρα ὑπὸ ΓΒΕ, ΕΒΔ τρισὶ ταῖς ὑπὸ ΓΒΑ, ΑΒΕ, ΕΒΔ ἴσαι εἰσιν. πάλιν, ἐπεὶ ἡ ὑπὸ ΔΒΑ δυοὶ ταῖς ὑπὸ ΔΒΕ, ΕΒΑ ἴση ἐστίν, κοινὴ προσκείσθω ἡ ὑπὸ ΑΒΓ· αἱ ἄρα ὑπὸ ΔΒΑ, ΑΒΓ τρισὶ ταῖς ὑπὸ ΔΒΕ, ΕΒΑ, ΑΒΓ ἴσαι εἰσιν. ἐδείκθησαν δὲ καὶ αἱ ὑπὸ ΓΒΕ, ΕΒΔ τρισὶ ταῖς αὐταῖς ἴσαι· τὰ δὲ τῶ αὐτῶ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα· καὶ αἱ ὑπὸ ΓΒΕ, ΕΒΔ ἄρα ταῖς ὑπὸ ΔΒΑ, ΑΒΓ ἴσαι εἰσιν· ἀλλὰ αἱ ὑπὸ ΓΒΕ, ΕΒΔ δύο ὀρθαὶ εἰσιν· καὶ αἱ ὑπὸ ΔΒΑ, ΑΒΓ ἄρα δυοὶν ὀρθαῖς ἴσαι εἰσιν.

Ἐάν ἄρα εὐθεῖα ἐπ' εὐθείαν σταθεῖσα γωνίας ποιῇ, ἥτοι δύο ὀρθὰς ἢ δυοὶν ὀρθαῖς ἴσας ποιήσῃ· ὅπερ ἔδει δεῖξαι.

In fact, if CBA is equal to ABD (then) they are two right-angles [Def. 1.10]. But, if not, let BE be drawn from the point B at right-angles to [the straight-line] CD [Prop. 1.11]. Thus, CBE and EBD are two right-angles. And since CBE is equal to the two (angles) CBA and ABE , let EBD be added to both. Thus, the (sum of the angles) CBE and EBD is equal to the (sum of the) three (angles) CBA , ABE , and EBD [C.N. 2]. Again, since DBA is equal to the two (angles) DBE and EBA , let ABC be added to both. Thus, the (sum of the angles) DBA and ABC is equal to the (sum of the) three (angles) DBE , EBA , and ABC [C.N. 2]. But (the sum of) CBE and EBD was also shown (to be) equal to the (sum of the) same three (angles). And things equal to the same thing are also equal to one another [C.N. 1]. Therefore, (the sum of) CBE and EBD is also equal to (the sum of) DBA and ABC . But, (the sum of) CBE and EBD is two right-angles. Thus, (the sum of) ABD and ABC is also equal to two right-angles.

Thus, if a straight-line stood on a(nother) straight-line makes angles, it will certainly either make two right-angles, or (angles whose sum is) equal to two right-angles. (Which is) the very thing it was required to show.

14.

Ἐάν πρὸς τινι εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ δύο εὐθεῖαι μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δυοὶν ὀρθαῖς ἴσας ποιῶσιν, ἐπ' εὐθείας ἔσσονται ἀλλήλαις αἱ εὐθεῖαι.

Πρὸς γάρ τινι εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ B δύο εὐθεῖαι αἱ BF , BD μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας τὰς ὑπὸ ABF , ABD δύο ὀρθαῖς ἴσας

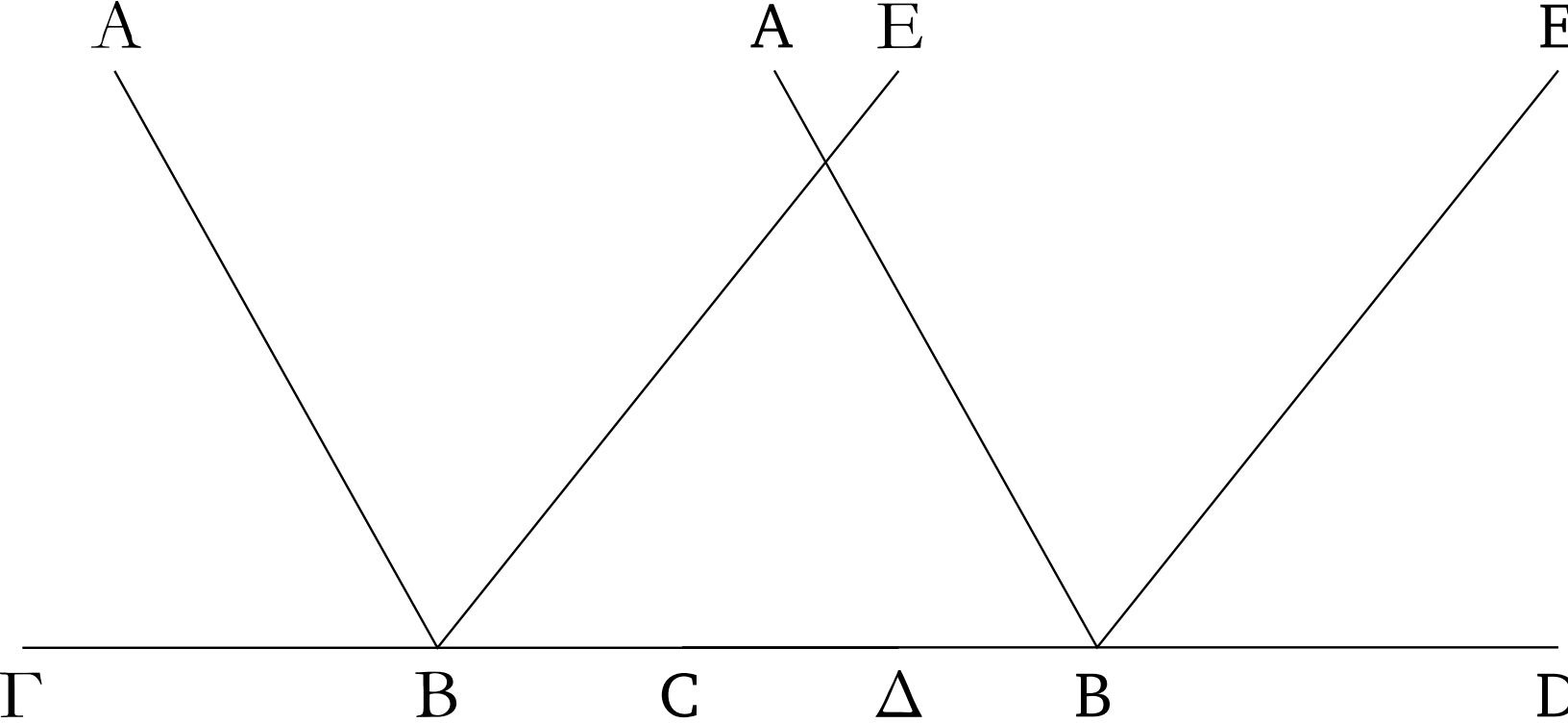
Proposition 14

If two straight-lines, not lying on the same side, make adjacent angles (whose sum is) equal to two right-angles with some straight-line, at a point on it, (then) the two straight-lines will be straight-on (with respect) to one another.

For let two straight-lines BC and BD , not lying on the same side, make adjacent angles ABC and ABD (whose sum

ποιείτωσαν· λέγω, ὅτι ἐπ' εὐθείας ἐστὶ τῇ $ΓΒ$ ἢ $ΒΔ$.

is) equal to two right-angles with some straight-line AB , at the point B on it. I say that BD is straight-on with respect to CB .



Εἰ γὰρ μὴ ἐστὶ τῇ $ΒΓ$ ἐπ' εὐθείας ἡ $ΒΔ$, ἔστω τῇ $ΓΒ$ ἐπ' εὐθείας ἡ $ΒΕ$.

Ἐπεὶ οὖν εὐθεῖα ἡ $ΑΒ$ ἐπ' εὐθείαν τὴν $ΓΒΕ$ ἐφέστηκεν, αἱ ἄρα ὑπὸ $ΑΒΓ$, $ΑΒΕ$ γωνίαι δύο ὀρθαῖς ἴσαι εἰσὶν· εἰσὶ δὲ καὶ αἱ ὑπὸ $ΑΒΓ$, $ΑΒΔ$ δύο ὀρθαῖς ἴσαι· αἱ ἄρα ὑπὸ $ΓΒΑ$, $ΑΒΕ$ ταῖς ὑπὸ $ΓΒΑ$, $ΑΒΔ$ ἴσαι εἰσὶν. κοινὴ ἀφηρήσθω ἡ ὑπὸ $ΓΒΑ$ · λοιπὴ ἄρα ἡ ὑπὸ $ΑΒΕ$ λοιπὴ τῇ ὑπὸ $ΑΒΔ$ ἐστὶν ἴση, ἡ ἐλάσσων τῇ μείζονι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐπ' εὐθείας ἐστὶν ἡ $ΒΕ$ τῇ $ΓΒ$. ὁμοίως δὲ δείχουμεν, ὅτι οὐδὲ ἄλλη τις πλὴν τῆς $ΒΔ$ ἐπ' εὐθείας ἄρα ἐστὶν ἡ $ΓΒ$ τῇ $ΒΔ$.

Ἐάν ἄρα πρόξ τινη εὐθείᾳ καὶ τῷ πρόξ αὐτῇ σημείῳ δύο εὐθεῖαι μὴ ἐπὶ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δυσὶν ὀρθαῖς ἴσας ποιῶσιν, ἐπ' εὐθείας ἔσσονται ἀλλήλαις αἱ εὐθεῖαι· ὅπερ ἔδει δεῖχαι.

For if BD is not straight-on to BC (then) let BE be straight-on to CB .

Therefore, since the straight-line AB stands on the straight-line CBE , the (sum of the) angles ABC and ABE is thus equal to two right-angles [Prop. 1.13]. But (the sum of) ABC and ABD is also equal to two right-angles. Thus, (the sum of angles) CBA and ABE is equal to (the sum of angles) CBA and ABD [C.N. 1]. Let (angle) CBA be subtracted from both. Thus, the remainder ABE is equal to the remainder ABD [C.N. 3], the lesser to the greater. The very thing is impossible. Thus, BE is not straight-on with respect to CB . Similarly, we can show that neither (is) any other (straight-line) than BD . Thus, CB is straight-on with respect to BD .

Thus, if two straight-lines, not lying on the same side, make adjacent angles (whose sum is) equal to two right-angles with some straight-line, at a point on it, (then) the two straight-lines will be straight-on (with respect) to one another. (Which is) the very thing it was required to show.

15.

Proposition 15

Ἐάν δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὰς κατὰ κορυφὴν γωνίας ἴσας ἀλλήλαις ποιῶσιν.

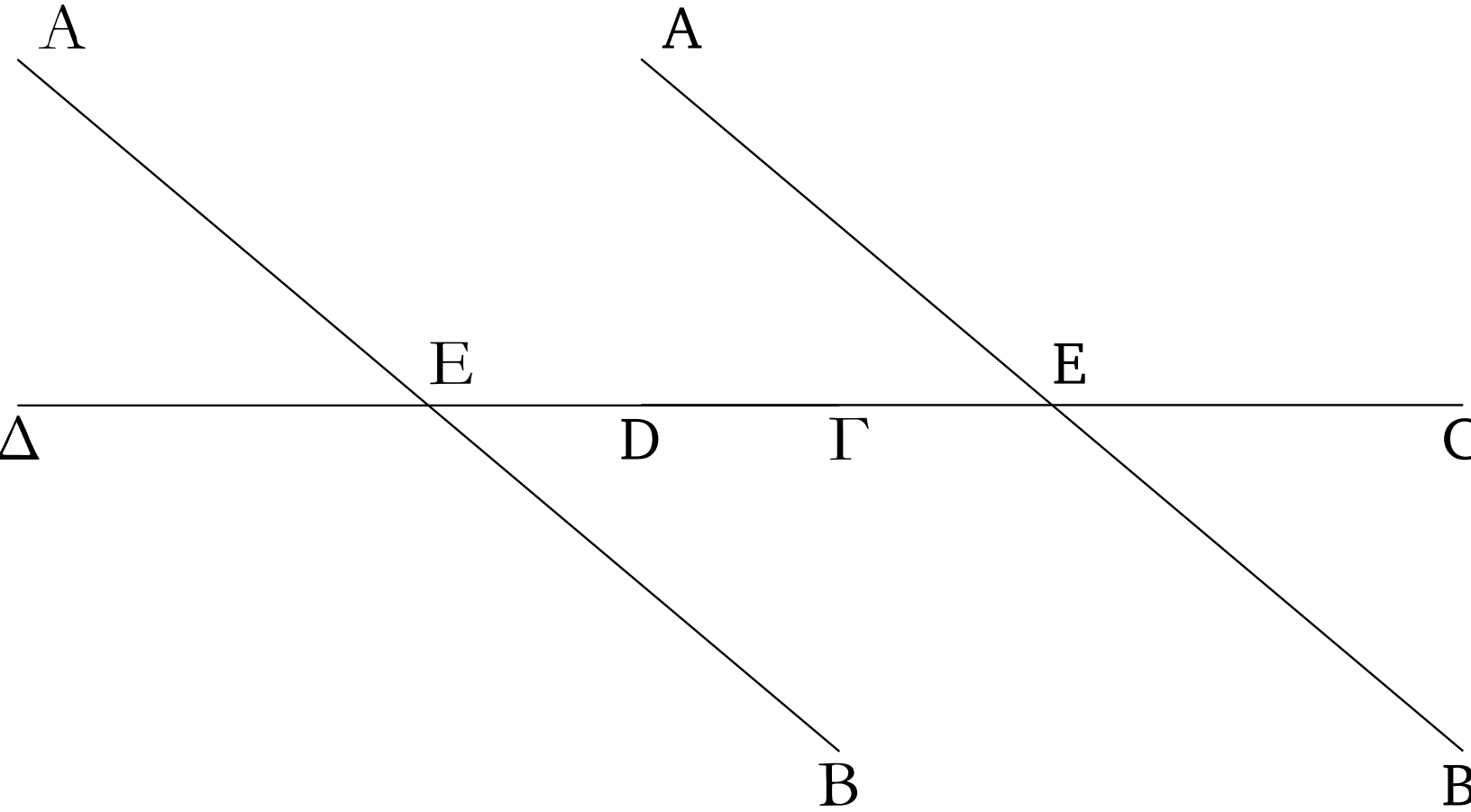
Δύο γὰρ εὐθεῖαι αἱ $ΑΒ$, $ΓΔ$ τεμνέτωσαν ἀλλήλας κατὰ τὸ $Ε$ σημεῖον· λέγω, ὅτι ἴση ἐστὶν ἡ μὲν ὑπὸ $ΑΕΓ$ γωνία τῇ ὑπὸ

If two straight-lines cut one another (then) they make the vertically opposite angles equal to one another.

For let the two straight-lines AB and CD cut one another at the point E . I say that angle AEC is equal to (angle) DEB ,

ΔEB , ἡ δὲ ὑπὸ ΓEB τῇ ὑπὸ $AE\Delta$.

and (angle) CEB to (angle) AED .



Ἐπεὶ γὰρ εὐθεῖα ἡ AE ἐπ' εὐθείαν τὴν $\Gamma\Delta$ ἐφέστηκε γωνίαξ ποιοῦσα τὰς ὑπὸ ΓEA , $AE\Delta$, αἱ ἄρα ὑπὸ ΓEA , $AE\Delta$ γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν. πάλιν, ἐπεὶ εὐθεῖα ἡ DE ἐπ' εὐθείαν τὴν AB ἐφέστηκε γωνίαξ ποιοῦσα τὰς ὑπὸ $AE\Delta$, ΔEB , αἱ ἄρα ὑπὸ $AE\Delta$, ΔEB γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν. ἐδείκθησαν δὲ καὶ αἱ ὑπὸ ΓEA , $AE\Delta$ δυοῖν ὀρθαῖς ἴσαι· αἱ ἄρα ὑπὸ ΓEA , $AE\Delta$ ταῖς ὑπὸ $AE\Delta$, ΔEB ἴσαι εἰσὶν. κοινὴ ἀφηρήσθω ἡ ὑπὸ $AE\Delta$ · λοιπὴ ἄρα ἡ ὑπὸ ΓEA λοιπῇ τῇ ὑπὸ $BE\Delta$ ἴση ἐστίν· ὁμοίως δὲ δευτέρησεται, ὅτι καὶ αἱ ὑπὸ ΓEB , ΔEA ἴσαι εἰσὶν.

Ἐὰν ἄρα δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὰς κατὰ κορυφὴν γωνίαξ ἴσας ἀλλήλαις ποιοῦσιν· ὅπερ ἔδει δεῖχαι.

For since the straight-line AE stands on the straight-line CD , making the angles CEA and AED , the (sum of the) angles CEA and AED is thus equal to two right-angles [Prop. 1.13]. Again, since the straight-line DE stands on the straight-line AB , making the angles AED and DEB , the (sum of the) angles AED and DEB is thus equal to two right-angles [Prop. 1.13]. But (the sum of) CEA and AED was also shown (to be) equal to two right-angles. Thus, (the sum of) CEA and AED is equal to (the sum of) AED and DEB [C.N. 1]. Let AED be subtracted from both. Thus, the remainder CEA is equal to the remainder DEB [C.N. 3]. Similarly, it can be shown that CEB and DEA are also equal.

Thus, if two straight-lines cut one another (then) they make the vertically opposite angles equal to one another. (Which is) the very thing it was required to show.

16.

Proposition 16

Παντὸς τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία ἐκατέρωξ τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν.

Ἐστω τριγώνου τὸ $AB\Gamma$, καὶ προσεκβεβλήσθω αὐτοῦ μία

For any triangle, when one of the sides is produced, the external angle is greater than each of the internal and opposite angles.

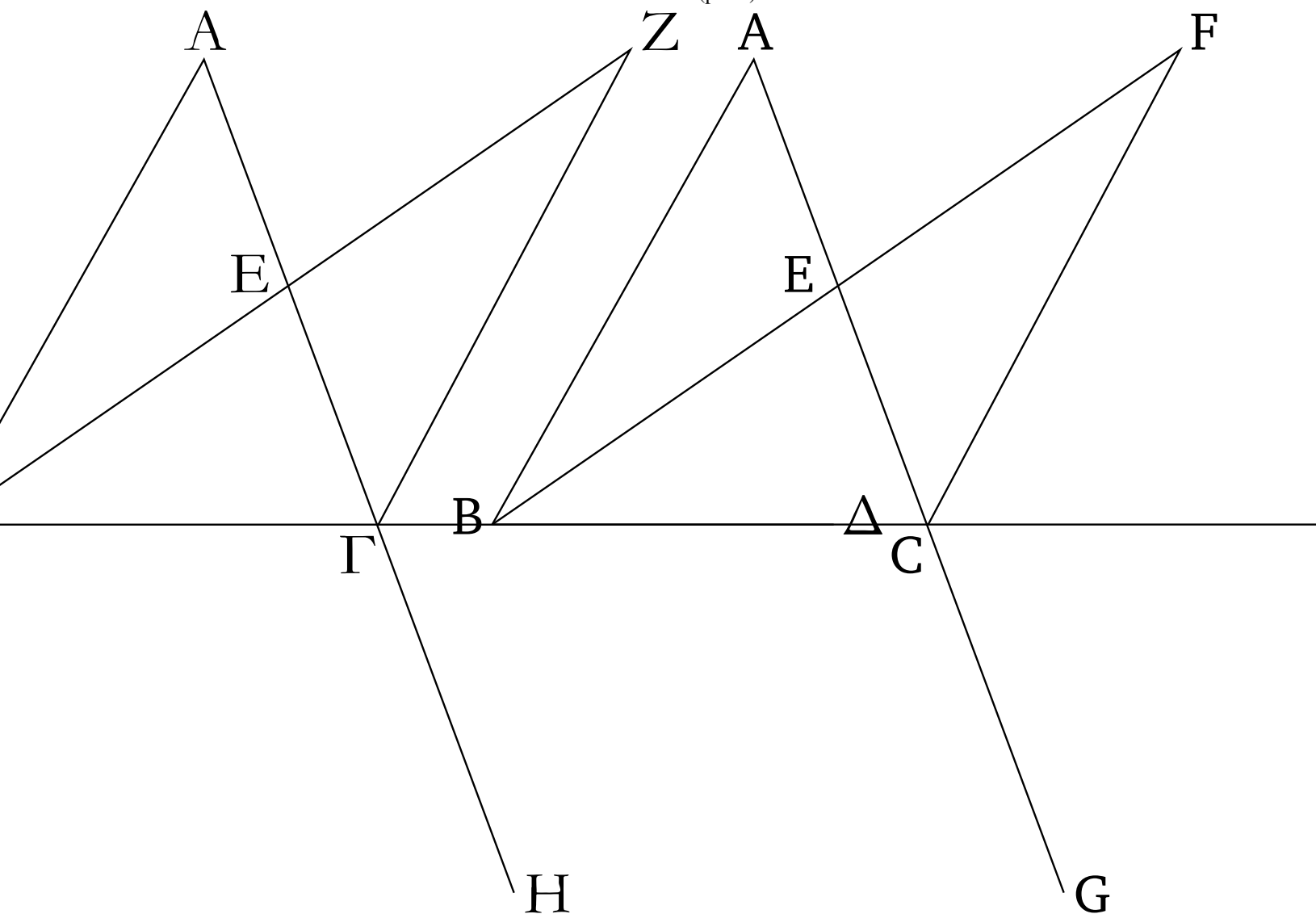
Let ABC be a triangle, and let one of its sides BC be pro-

πλευρὰ ἡ ΒΓ ἐπὶ τὸ Δ· λέγω, ὅτι ἡ ἐκτὸς γωνία ἡ ὑπὸ ΑΓΔ μείζων ἐστὶν ἑκατέρᾳ τῶν ἐντὸς καὶ ἀπεναντίων τῶν ὑπὸ ΓΒΑ, ΒΑΓ γωνιῶν.

Τετμήσθω ἡ ΑΓ διὰ κατὰ τὸ Ε, καὶ ἐπιξενυθθεῖσα ἡ ΒΕ ἐκβεβλήσθω ἐπ' εὐθείᾳς ἐπὶ τὸ Ζ, καὶ κείσθω τῇ ΒΕ ἴση ἡ ΕΖ, καὶ ἐπεξεύθθω ἡ ΖΓ, καὶ διήθθω ἡ ΑΓ ἐπὶ τὸ Η.

duced to D . I say that the external angle ACD is greater than each of the internal and opposite angles, CBA and BAC .

Let the (straight-line) AC be cut in half at (point) E [Prop. 1.10]. And BE being joined, let it be produced in a straight-line to (point) F .[†] And let EF be made equal to BE [Prop. 1.3], and let FC be joined, and let AC be drawn through to (point) G .



Ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν AE τῇ EF , ἡ δὲ BE τῇ EZ , δύο δὲ αἱ AE , EB δυοὶ ταῖς FE , EZ ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνία ἡ ὑπὸ AEB γωνία τῇ ὑπὸ ZEF ἴση ἐστίν· κατὰ κορυφὴν γάρ· βάσις ἄρα ἡ AB βάσει τῇ ZF ἴση ἐστίν, καὶ τὸ ABE τρίγωνον τῷ ZEF τριγώνῳ ἐστὶν ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι εἰσὶν ἑκατέρω ἑκατέρω, ὅφ' ἂς αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἐστὶν ἡ ὑπὸ BAE τῇ ὑπὸ EFZ . μείζων δὲ ἐστὶν ἡ ὑπὸ EBD τῇς ὑπὸ EFZ .

Therefore, since AE is equal to EC , and BE to EF , the two (straight-lines) AE, EB are equal to the two (straight-lines) CE, EF , respectively. Also, angle AEB is equal to angle FEC , for (they are) vertically opposite [Prop. 1.15]. Thus, the base AB is equal to the base FC , and the triangle ABE is equal to the triangle FEC , and the remaining angles subtended by the equal sides are equal to the corresponding remaining angles [Prop. 1.4]. Thus, BAE is equal to ECF . But ECD is greater

μείζων ἄρα ἢ ὑπὸ $ΑΓΔ$ τῆς ὑπὸ $ΒΑΕ$. Ὁμοίως δὴ τῆς $ΒΓ$ τετμημένης διῖθα δειθθήσεται καὶ ἢ ὑπὸ $ΒΓΗ$, τοντέστιν ἢ ὑπὸ $ΑΓΔ$, μείζων καὶ τῆς ὑπὸ $ΑΒΓ$.

Παντὸς ἄρα τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἢ ἐκτὸς γωνία ἐκατέρωξ τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν· ὅπερ ἔδει δεῖχαι.

† The implicit assumption that the point F lies in the interior of the angle ABC should be counted as an additional postulate.

17.

Παντὸν τριγώνον αἱ δύο γωνίαι δύο ὀρθῶν ἐλάσσονές εἰσι πάντῃ μεταλαμβανόμεναι.

Ἐστω τρίγωνον τὸ $ΑΒΓ$ · λέγω, ὅτι τοῦ $ΑΒΓ$ τριγώνου αἱ δύο γωνίαι δύο ὀρθῶν ἐλάττονές εἰσι πάντῃ μεταλαμβανόμεναι.

Ἐκβεβλήσθω γάρ ἢ $ΒΓ$ ἐπὶ τὸ $Δ$.

Καὶ ἐπεὶ τριγώνον τοῦ $ΑΒΓ$ ἐκτὸς ἐστὶ γωνία ἢ ὑπὸ $ΑΓΔ$, μείζων ἐστὶ τῆς ἐντὸς καὶ ἀπεναντίον τῆς ὑπὸ $ΑΒΓ$. κοινὴ προσκείσθω ἢ ὑπὸ $ΑΓΒ$ · αἱ ἄρα ὑπὸ $ΑΓΔ$, $ΑΓΒ$ τῶν ὑπὸ $ΑΒΓ$, $ΒΓΑ$ μείζονές εἰσιν. ἀλλ' αἱ ὑπὸ $ΑΓΔ$, $ΑΓΒ$ δύο ὀρθαῖς ἴσαι εἰσὶν· αἱ ἄρα ὑπὸ $ΑΒΓ$, $ΒΓΑ$ δύο ὀρθῶν ἐλάσσονές εἰσιν. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ ὑπὸ $ΒΑΓ$, $ΑΓΒ$ δύο ὀρθῶν ἐλάσσονές εἰσι καὶ ἔτι αἱ ὑπὸ $ΓΑΒ$, $ΑΒΓ$.

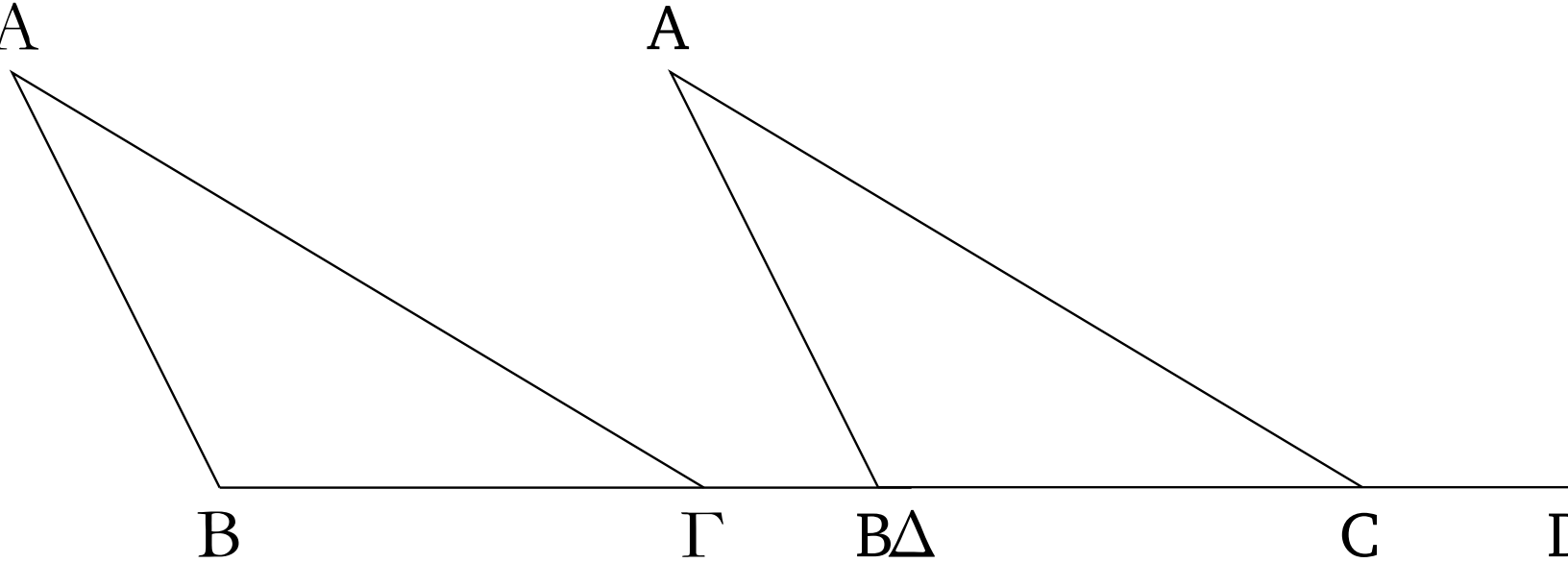
Proposition 17

For any triangle, (the sum of) two angles taken together in any (possible way) is less than two right-angles.

Let ABC be a triangle. I say that (the sum of) two angles of triangle ABC taken together in any (possible way) is less than two right-angles.

For let BC be produced to D .

And since the angle ACD is external to triangle ABC , it is greater than the internal and opposite angle ABC [Prop. 1.16]. Let ACB be added to both. Thus, the (sum of the angles) ACD and ACB is greater than the (sum of the angles) ABC and BCA . But, (the sum of) ACD and ACB is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) ABC and BCA is less than two right-angles. Similarly, we can show that (the sum of) BAC and ACB is also less than two right-angles, and further (that the sum of) CAB and ABC (is less than two right-angles).



Παντὸν ἄρα τριγώνον αἱ δύο γωνίαι δύο ὀρθῶν ἐλάσσονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖχαι.

Thus, for any triangle, (the sum of) two angles taken together in any (possible way) is less than two right-angles. (Which is) the very thing it was required to show.

18.

Proposition 18

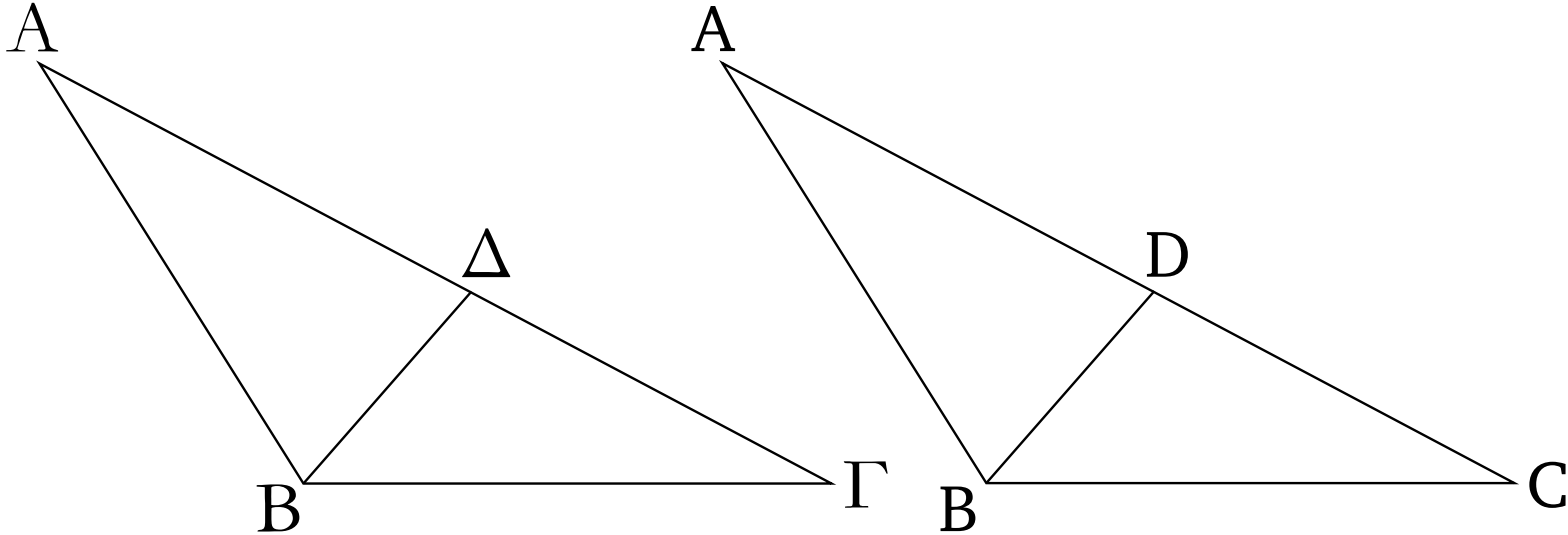
Παντὸς τριγώνου ἡ μείζων πλευρὰ τὴν μείζονα γωνίαν ὑπο-

In any triangle, the greater side subtends the greater angle.

τείνει.

Ἐστω γάρ τριγώνον τὸ $AB\Gamma$ μείζονα ἔθον τήν $ΑΓ$ πλευρὰν τῆς AB · λέγω, ὅτι καὶ γωνία ἡ ὑπὸ $AB\Gamma$ μείζων ἐστὶ τῆς ὑπὸ $B\Gamma A$.

For let ABC be a triangle having side AC greater than AB . I say that angle ABC is also greater than BCA .



Ἐπεὶ γὰρ μείζων ἐστὶν ἡ $ΑΓ$ τῆς AB , κείσθω τῇ AB ἴση ἡ $ΑΔ$, καὶ ἐπεζεύθθω ἡ $BΔ$.

Καὶ ἐπεὶ τριγώνον τοῦ $B\Gamma Δ$ ἐκτός ἐστι γωνία ἡ ὑπὸ $ΑΔB$, μείζων ἐστὶ τῆς ἐντός καὶ ἀπεναντίον τῆς ὑπὸ $ΔΓB$ · ἴση δὲ ἡ ὑπὸ $ΑΔB$ τῇ ὑπὸ $ABΔ$, ἐπεὶ καὶ πλευρὰ ἡ AB τῇ $ΑΔ$ ἐστὶν ἴση· μείζων ἄρα καὶ ἡ ὑπὸ $ABΔ$ τῆς ὑπὸ $ΑΓB$ · πολλῶν ἄρα ἡ ὑπὸ $AB\Gamma$ μείζων ἐστὶ τῆς ὑπὸ $ΑΓB$.

Παντὸς ἄρα τριγώνου ἡ μείζων πλευρὰ τὴν μείζονα γωνίαν ὑποτείνει· ὅπερ ἔδει δεῖξαι.

For since AC is greater than AB , let AD be made equal to AB [Prop. 1.3], and let BD be joined.

And since angle ADB is external to triangle BCD , it is greater than the internal and opposite (angle) DCB [Prop. 1.16]. But ADB (is) equal to ABD , since side AB is also equal to side AD [Prop. 1.5]. Thus, ABD is also greater than ACB . Thus, ABC is much greater than ACB .

Thus, in any triangle, the greater side subtends the greater angle. (Which is) the very thing it was required to show.

19.

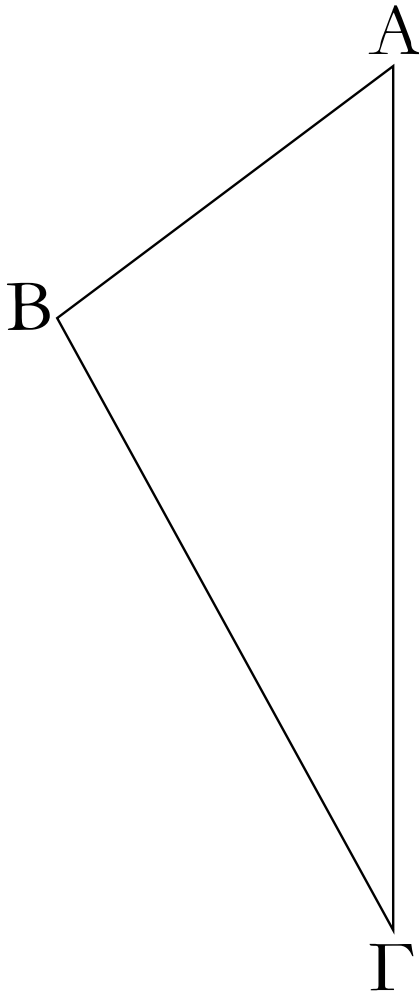
Proposition 19

Παντὸς τριγώνου ὑπὸ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει.

Ἐστω τριγώνον τὸ $AB\Gamma$ μείζονα ἔθον τὴν ὑπὸ $AB\Gamma$ γωνίαν τῆς ὑπὸ $B\Gamma A$ · λέγω, ὅτι καὶ πλευρὰ ἡ $ΑΓ$ πλευρᾷ τῆς AB μείζων ἐστίν.

In any triangle, the greater angle is subtended by the greater side.

Let ABC be a triangle having the angle ABC greater than BCA . I say that side AC is also greater than side AB .

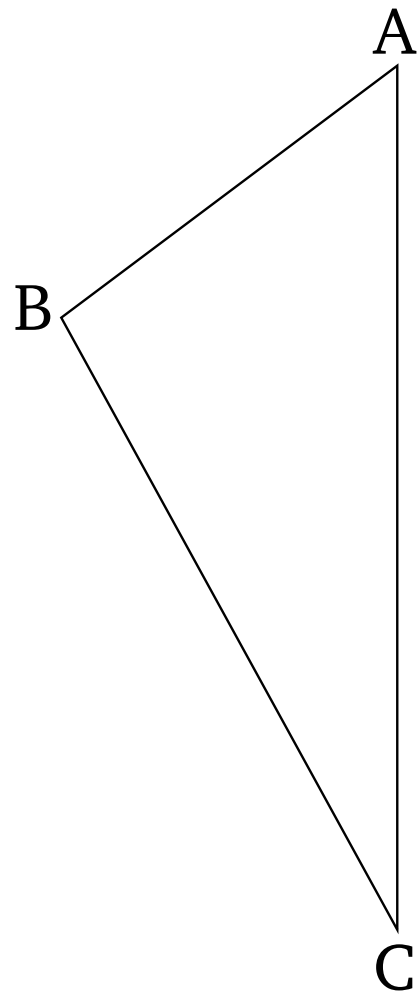


Εἰ γὰρ μή, ἦτοι ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$ ἢ ἐλάσσων· ἴση μὲν οὖν οὐκ ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$ · ἴση γὰρ ἂν ἦν καὶ γωνία ἡ ὑπὸ $ΑΒΓ$ τῇ ὑπὸ $ΑΓΒ$ · οὐκ ἐστὶ δέ· οὐκ ἄρα ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$. οὐδὲ μὴν ἐλάσσων ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$ · ἐλάσσων γὰρ ἂν ἦν καὶ γωνία ἡ ὑπὸ $ΑΒΓ$ τῇ ὑπὸ $ΑΓΒ$ · οὐκ ἐστὶ δέ· οὐκ ἄρα ἐλάσσων ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$. ἐδείκθη δέ, ὅτι οὐδὲ ἴση ἐστὶν. μείζων ἄρα ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$.

Παντὸς ἄρα τριγώνου ὑπὸ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει· ὅπερ ἔδει δεῖξαι.

20.

Παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονός εἰσι πάντῃ μεταλαμβάνόμεναι.

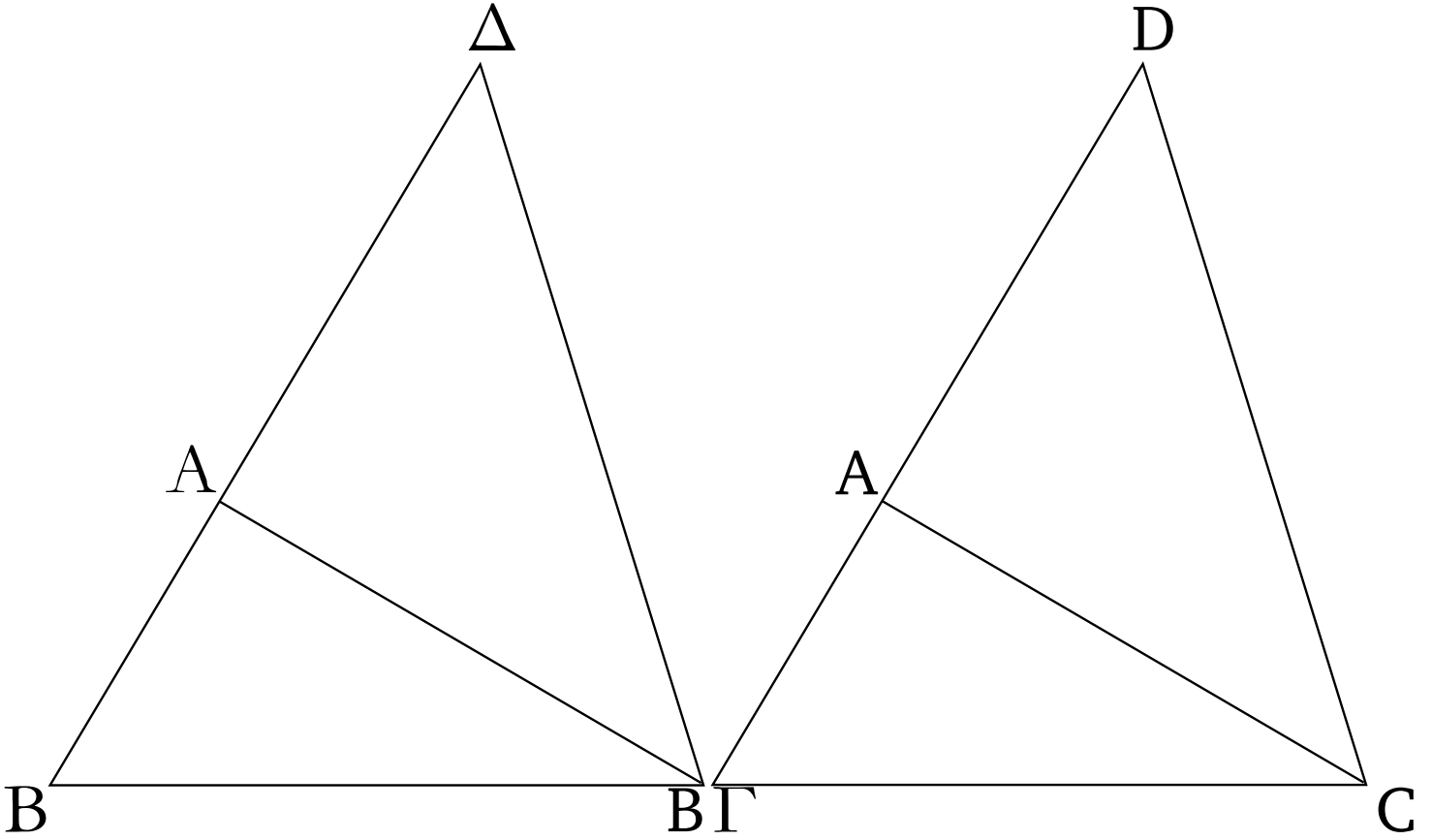


For if not, AC is certainly either equal to, or less than, AB . In fact, AC is not equal to AB . For then angle ABC would also be equal to ACB [Prop. 1.5]. But it is not. Thus, AC is not equal to AB . Neither, indeed, is AC less than AB . For then angle ABC would also be less than ACB [Prop. 1.18]. But it is not. Thus, AC is not less than AB . But it was shown that (AC) is not equal (to AB) either. Thus, AC is greater than AB .

Thus, in any triangle, the greater angle is subtended by the greater side. (Which is) the very thing it was required to show.

Proposition 20

In any triangle, (the sum of) two sides taken together in any (possible way) is greater than the remaining (side).



Ἐστω γάρ τριγώνον τὸ $AB\Gamma$. λέγω, ὅτι τοῦ $AB\Gamma$ τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν BA , $A\Gamma$ τῆς $B\Gamma$, αἱ δὲ AB , $B\Gamma$ τῆς $A\Gamma$, αἱ δὲ $B\Gamma$, ΓA τῆς AB .

Διήθηθω γάρ ἡ BA ἐπὶ τὸ Δ σημεῖον, καὶ κείσθω τῇ ΓA ἴση ἡ $A\Delta$, καὶ ἐπεζεύθηθω ἡ $\Delta\Gamma$.

Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΔA τῇ $A\Gamma$, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $A\Delta\Gamma$ τῇ ὑπὸ $A\Gamma\Delta$. μείζων ἄρα ἡ ὑπὸ $B\Gamma\Delta$ τῆς ὑπὸ $A\Delta\Gamma$. καὶ ἐπεὶ τριγώνον ἐστὶ τὸ $\Delta\Gamma B$ μείζονα ἔθον τὴν ὑπὸ $B\Gamma\Delta$ γωνίαν τῆς ὑπὸ $B\Delta\Gamma$, ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει, ἡ ΔB ἄρα τῆς $B\Gamma$ ἐστὶ μείζων. ἴση δὲ ἡ ΔA τῇ $A\Gamma$. μείζονες ἄρα αἱ BA , $A\Gamma$ τῆς $B\Gamma$. ὁμοίως δὲ δείξομεν, ὅτι καὶ αἱ μὲν AB , $B\Gamma$ τῆς ΓA μείζονές εἰσιν, αἱ δὲ $B\Gamma$, ΓA τῆς AB .

Παντὸς ἄρα τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι. ὅπερ ἔδει δείχαι.

For let ABC be a triangle. I say that in triangle ABC (the sum of) two sides taken together in any (possible way) is greater than the remaining (side). (So), (the sum of) BA and AC (is greater) than BC , (the sum of) AB and BC than AC , and (the sum of) BC and CA than AB .

For let BA be drawn through to point D , and let AD be made equal to CA [Prop. 1.3], and let DC be joined.

Therefore, since DA is equal to AC , the angle ADC is also equal to ACD [Prop. 1.5]. Thus, BCD is greater than ADC . And since DCB is a triangle having the angle BCD greater than BDC , and the greater angle subtends the greater side [Prop. 1.19], DB is thus greater than BC . But DA is equal to AC . Thus, (the sum of) BA and AC is greater than BC . Similarly, we can show that (the sum of) AB and BC is also greater than CA , and (the sum of) BC and CA than AB .

Thus, in any triangle, (the sum of) two sides taken together in any (possible way) is greater than the remaining (side). (Which is) the very thing it was required to show.

21.

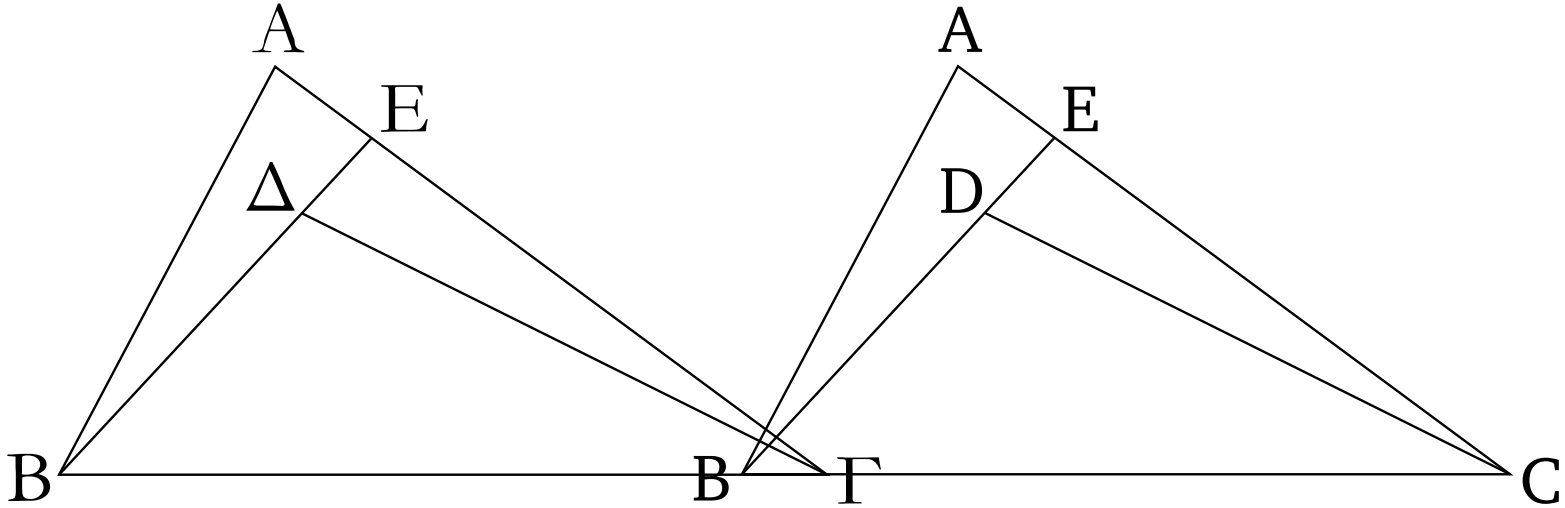
Proposition 21

Ἐάν τριγώνον ἐπὶ μιᾷ τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττωες μὲν ἔσονται, μείζονα δὲ

If two internal straight-lines are constructed on one of the sides of a triangle, from its ends, the constructed (straight-lines) will be less than the two remaining sides of the triangle,

γωνίαν περιέχουσιν.

Τριγώνον γάρ τοῦ $ABΓ$ ἐπὶ μιᾷ τῶν πλευρῶν τῆς $ΒΓ$ ἀπὸ τῶν περάτων τῶν $B, Γ$ δύο εὐθεῖαι ἐντὸς συνεστάτωσαν αἱ $ΒΔ, ΔΓ$. λέγω, ὅτι αἱ $ΒΔ, ΔΓ$ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τῶν $ΒΑ, ΑΓ$ ἐλάσσονες μὲν εἰσιν, μείζονα δὲ γωνίαν περιέθουσι τὴν ὑπὸ $ΒΔΓ$ τῆς ὑπὸ $ΒΑΓ$.



Διήθθω γάρ ἡ $ΒΔ$ ἐπὶ τὸ E . καὶ ἐπεὶ παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονες εἰσιν, τοῦ ABE ἄρα τριγώνου αἱ δύο πλευραὶ αἱ AB, AE τῆς BE μείζονες εἰσιν· κοινὴ προσκείσθω ἡ $ΕΓ$. αἱ ἄρα $BA, ΑΓ$ τῶν $BE, ΕΓ$ μείζονες εἰσιν. πάλιν, ἐπεὶ τοῦ $ΓΕΔ$ τριγώνου αἱ δύο πλευραὶ αἱ $ΓΕ, ΕΔ$ τῆς $ΓΔ$ μείζονες εἰσιν, κοινὴ προσκείσθω ἡ $ΔΒ$. αἱ $ΓΕ, ΕΒ$ ἄρα τῶν $ΓΔ, ΔΒ$ μείζονες εἰσιν. ἀλλὰ τῶν $BE, ΕΓ$ μείζονες ἐδείχθησαν αἱ $BA, ΑΓ$. πολλῶν ἄρα αἱ $BA, ΑΓ$ τῶν $ΒΔ, ΔΓ$ μείζονες εἰσιν.

Πάλιν, ἐπεὶ παντὸς τριγώνου ἡ ἐκτὸς γωνία τῆς ἐντὸς καὶ ἀπεναντίον μείζων ἐστίν, τοῦ $ΓΔΕ$ ἄρα τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ $ΒΔΓ$ μείζων ἐστὶ τῆς ὑπὸ $ΓΕΔ$. διὰ ταῦτά τοῖνυν καὶ τοῦ ABE τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ $ΓΕΒ$ μείζων ἐστὶ τῆς ὑπὸ $ΒΑΓ$. ἀλλὰ τῆς ὑπὸ $ΓΕΒ$ μείζων ἐδείχθη ἡ ὑπὸ $ΒΔΓ$. πολλῶν ἄρα ἡ ὑπὸ $ΒΔΓ$ μείζων ἐστὶ τῆς ὑπὸ $ΒΑΓ$.

Ἐὰν ἄρα τριγώνον ἐπὶ μιᾷ τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθείσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μὲν εἰσιν, μείζονα δὲ γωνίαν περιέθουσιν· ὅπερ ἔδει δεῖχαι.

but will encompass a greater angle.

For let the two internal straight-lines BD and DC be constructed on one of the sides BC of the triangle ABC , from its ends B and C (respectively). I say that BD and DC are less than the (sum of the) two remaining sides of the triangle BA and AC , but encompass an angle BDC greater than BAC .

For let BD be drawn through to E . And since in any triangle (the sum of any) two sides is greater than the remaining (side) [Prop. 1.20], in triangle ABE the (sum of the) two sides AB and AE is thus greater than BE . Let EC be added to both. Thus, (the sum of) BA and AC is greater than (the sum of) BE and EC . Again, since in triangle CED the (sum of the) two sides CE and ED is greater than CD , let DB be added to both. Thus, (the sum of) CE and EB is greater than (the sum of) CD and DB . But, (the sum of) BA and AC was shown (to be) greater than (the sum of) BE and EC . Thus, (the sum of) BA and AC is much greater than (the sum of) BD and DC .

Again, since in any triangle the external angle is greater than the internal and opposite (angles) [Prop. 1.16], in triangle CDE the external angle BDC is thus greater than CED . Accordingly, for the same (reason), the external angle CEB of the triangle ABE is also greater than BAC . But, BDC was shown (to be) greater than CEB . Thus, BDC is much greater than BAC .

Thus, if two internal straight-lines are constructed on one of the sides of a triangle, from its ends, the constructed (straight-lines) are less than the two remaining sides of the triangle, but encompass a greater angle. (Which is) the very thing it was required to show.

22.

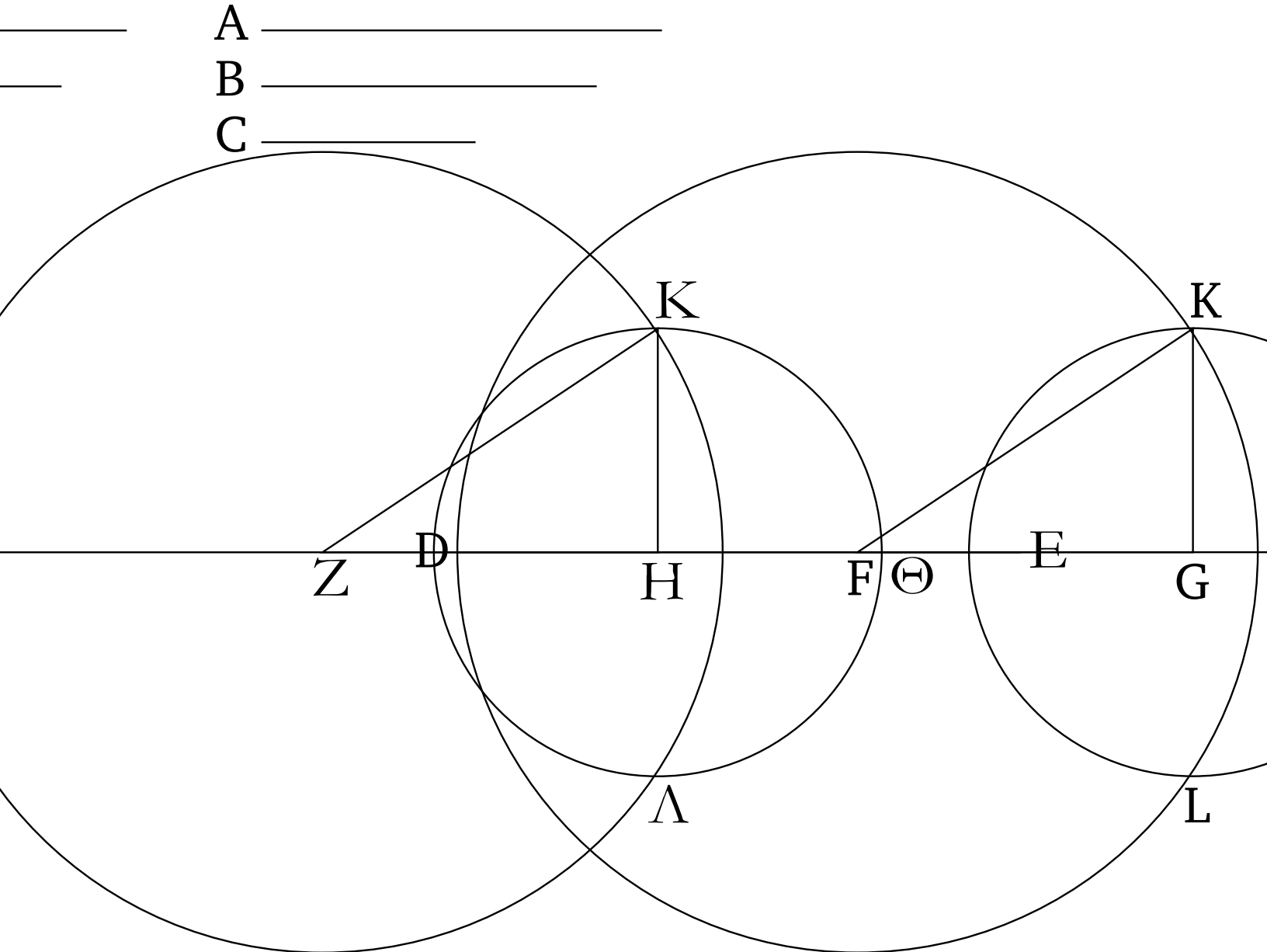
Proposition 22

Ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις [εὐθείαις], τριγώνον συστήσασθαι· δεῖ δὲ τὰς δύο τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας [διὰ τὸ καὶ παντὸς

To construct a triangle from three straight-lines which are equal to three given [straight-lines]. It is necessary for (the sum of) two (of the straight-lines) taken together in any (pos-

τριγώνον τὰς δύο πλευρὰς τῆς λοιπῆς μείζονα εἶναι πάντῃ μεταλαμβανομένας].

sible way) to be greater than the remaining (one), [on account of the (fact that) in any triangle (the sum of) two sides taken together in any (possible way) is greater than the remaining (one) [Prop. 1.20]].



Ἐστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ A, B, Γ , ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, αἱ μὲν A, B τῆς Γ , αἱ δὲ A, Γ τῆς B , καὶ ἔτι αἱ B, Γ τῆς A . δεῖ δὴ ἐκ τῶν ἴσων ταῖς A, B, Γ τριγώνον συστήσασθαι.

Ἐκκείσθω τις εὐθεῖα ἡ DE πεπερασμένη μὲν κατὰ τὸ Δ ἄπειροξ δὲ κατὰ τὸ E , καὶ κείσθω τῇ μὲν A ἴση ἡ ΔZ , τῇ δὲ B ἴση ἡ ZH , τῇ δὲ Γ ἴση ἡ $H\Theta$. καὶ κέντρῳ μὲν τῷ Z , διαστήματι δὲ τῷ $Z\Delta$ κύκλος γεγράφθω ὁ $\Delta K\Lambda$. πάλιν

Let A, B , and C be the three given straight-lines, of which let (the sum of) two taken together in any (possible way) be greater than the remaining (one). (Thus), (the sum of) A and B (is greater) than C , (the sum of) A and C than B , and also (the sum of) B and C than A . So it is required to construct a triangle from (straight-lines) equal to A, B , and C .

Let some straight-line DE be set out, terminated at D , and infinite in the direction of E . And let DF made equal to A , and

κέντρον μὲν τῷ H , διαστήματι δὲ τῷ $H\Theta$ κύκλος γεγράφθω ὁ $ΚΛ\Theta$, καὶ ἐπεξέυθθωσαν αἱ KZ , KH . λέγω, ὅτι ἐκ τριῶν εὐθειῶν τῶν ἴσων ταῖς A , B , Γ τρίγωνον συνέσταται τὸ KZH .

Ἐπεὶ γὰρ τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $\Delta K\Lambda$ κύκλου, ἴση ἐστὶν ἡ $Z\Delta$ τῇ ZK . ἀλλὰ ἡ $Z\Delta$ τῇ A ἐστὶν ἴση. καὶ ἡ KZ ἄρα τῇ A ἐστὶν ἴση. πάλιν, ἐπεὶ τὸ H σημεῖον κέντρον ἐστὶ τοῦ $\Delta K\Theta$ κύκλου, ἴση ἐστὶν ἡ $H\Theta$ τῇ HK . ἀλλὰ ἡ $H\Theta$ τῇ Γ ἐστὶν ἴση· καὶ ἡ KH ἄρα τῇ Γ ἐστὶν ἴση. ἐστὶ δὲ καὶ ἡ ZH τῇ B ἴση· αἱ τρεῖς ἄρα εὐθεῖαι αἱ KZ , ZH , HK τρισὶ ταῖς A , B , Γ ἴσαι εἰσίν.

Ἐκ τριῶν ἄρα εὐθειῶν τῶν KZ , ZH , HK , αἱ εἰσὶν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις ταῖς A , B , Γ , τρίγωνον συνέσταται τὸ KZH . ὅπερ ἔδει ποιῆσαι.

23.

Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ γωνίᾳ εὐθύγραμμω ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.

Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ πρὸς αὐτῇ σημεῖον τὸ A , ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἡ ὑπὸ $\Delta Γ Ε$. δεῖ δὴ πρὸς τῇ δοθείσῃ εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ δοθείσῃ γωνίᾳ εὐθύγραμμω τῇ ὑπὸ $\Delta Γ Ε$ ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.

FG equal to B , and GH equal to C [Prop. 1.3]. And let the circle DKL be drawn with center F and radius FD . Again, let the circle KLH be drawn with center G and radius GH . And let KF and KG be joined. I say that the triangle KFG has been constructed from three straight-lines equal to A , B , and C .

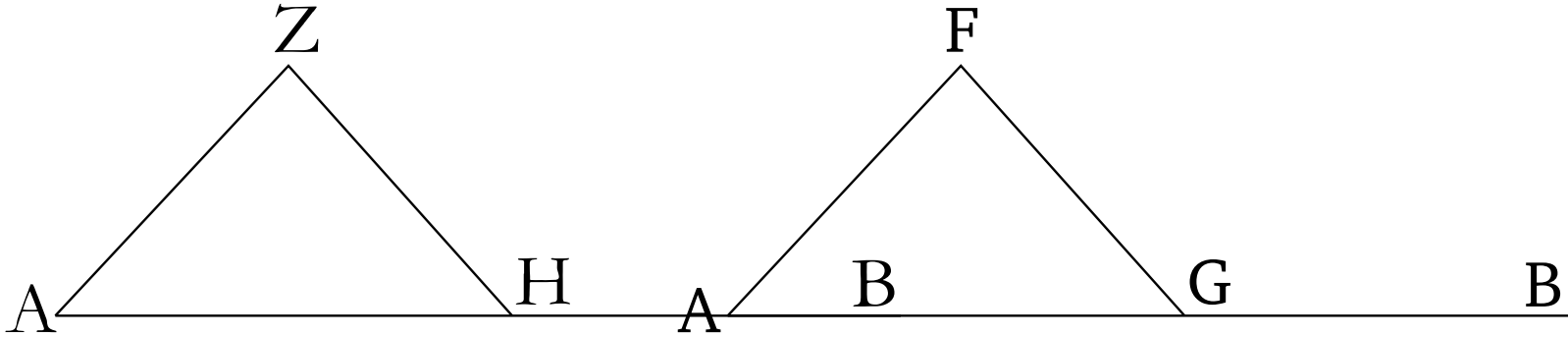
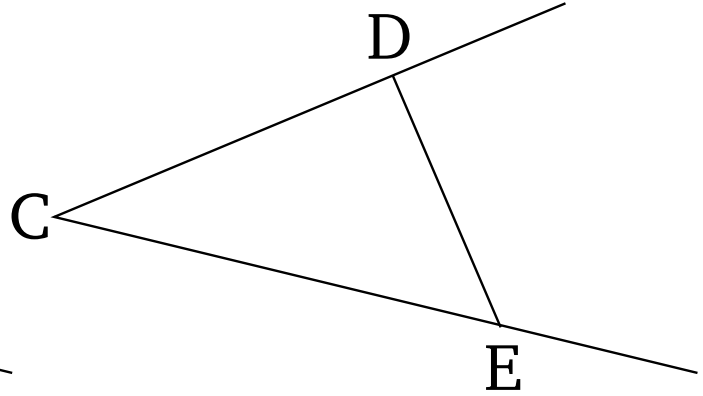
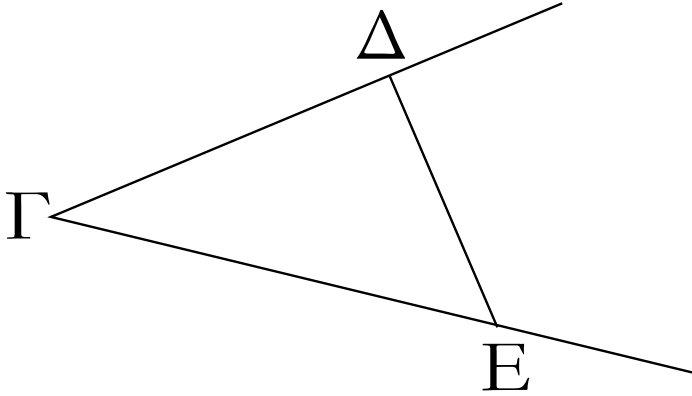
For since point F is the center of the circle DKL , FD is equal to FK . But, FD is equal to A . Thus, KF is also equal to A . Again, since point G is the center of the circle LKH , GH is equal to GK . But, GH is equal to C . Thus, KG is also equal to C . And FG is also equal to B . Thus, the three straight-lines KF , FG , and GK are equal to A , B , and C (respectively).

Thus, the triangle KFG has been constructed from the three straight-lines KF , FG , and GK , which are equal to the three given straight-lines A , B , and C (respectively). (Which is) the very thing it was required to do.

Proposition 23

To construct a rectilinear angle equal to a given rectilinear angle at a (given) point on a given straight-line.

Let AB be the given straight-line, A the (given) point on it, and DCE the given rectilinear angle. So it is required to construct a rectilinear angle equal to the given rectilinear angle DCE at the (given) point A on the given straight-line AB .



Εἰλήφθω ἐφ' ἑκατέρᾳ τῶν ΓΔ, ΓΕ τυθόντα σημεία τὰ Δ, Ε, καὶ ἐπεζεύθθω ἡ ΔΕ· καὶ ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς ΓΔ, ΔΕ, ΓΕ, τρίγωνον συνεστήτω τὸ ΑΖΗ, ὥστε ἴσῃν εἶναι τὴν μὲν ΓΔ τῇ ΑΖ, τὴν δὲ ΓΕ τῇ ΑΗ, καὶ ἔτι τὴν ΔΕ τῇ ΖΗ.

Ἐπεὶ οὖν δύο αἱ ΔΓ, ΓΕ δύο ταῖς ΖΑ, ΑΗ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ βάσις ἡ ΔΕ βάσει τῇ ΖΗ ἴση, γωνία ἄρα ἡ ὑπὸ ΔΓΕ γωνία τῇ ὑπὸ ΖΑΗ ἔστιν ἴση.

Πρὸς ἄρα τῇ δοθείσῃ εὐθείᾳ τῇ ΑΒ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ ὑπὸ ΔΓΕ ἴση γωνία εὐθύγραμμοῦ συνέσταται ἡ ὑπὸ ΖΑΗ· ὅπερ ἔδει ποιῆσαι.

Let the points D and E be taken at random on each of the (straight-lines) CD and CE (respectively), and let DE be joined. And let the triangle AFG be constructed from three straight-lines which are equal to CD , DE , and CE , such that CD is equal to AF , CE to AG , and further DE to FG [Prop. 1.22].

Therefore, since the two (straight-lines) DC , CE are equal to the two (straight-lines) FA , AG , respectively, and the base DE is equal to the base FG , the angle DCE is thus equal to the angle FAG [Prop. 1.8].

Thus, the rectilinear angle FAG , equal to the given rectilinear angle DCE , has been constructed at the (given) point A on the given straight-line AB . (Which is) the very thing it was required to do.

24.

Ἐὰν δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς ἴσαξ ἔθῃ ἑκατέραν ἑκατέρᾳ, τὴν δὲ γωνίαν τῆς γωνίας μείζονα ἔθῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῆς βάσεως μείζονα ἔχει.

Ἐστω δύο τρίγωνα τὰ ΑΒΓ, ΔΕΖ τὰς δύο πλευρὰς τὰς ΑΒ, ΑΓ ταῖς δύο πλευραῖς ταῖς ΔΕ, ΔΖ ἴσαξ ἔθοντα ἑκα-

Proposition 24

If two triangles have two sides equal to two sides, respectively, but (one) has the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the other), (then the former triangle) will also have a base greater than the base (of the latter).

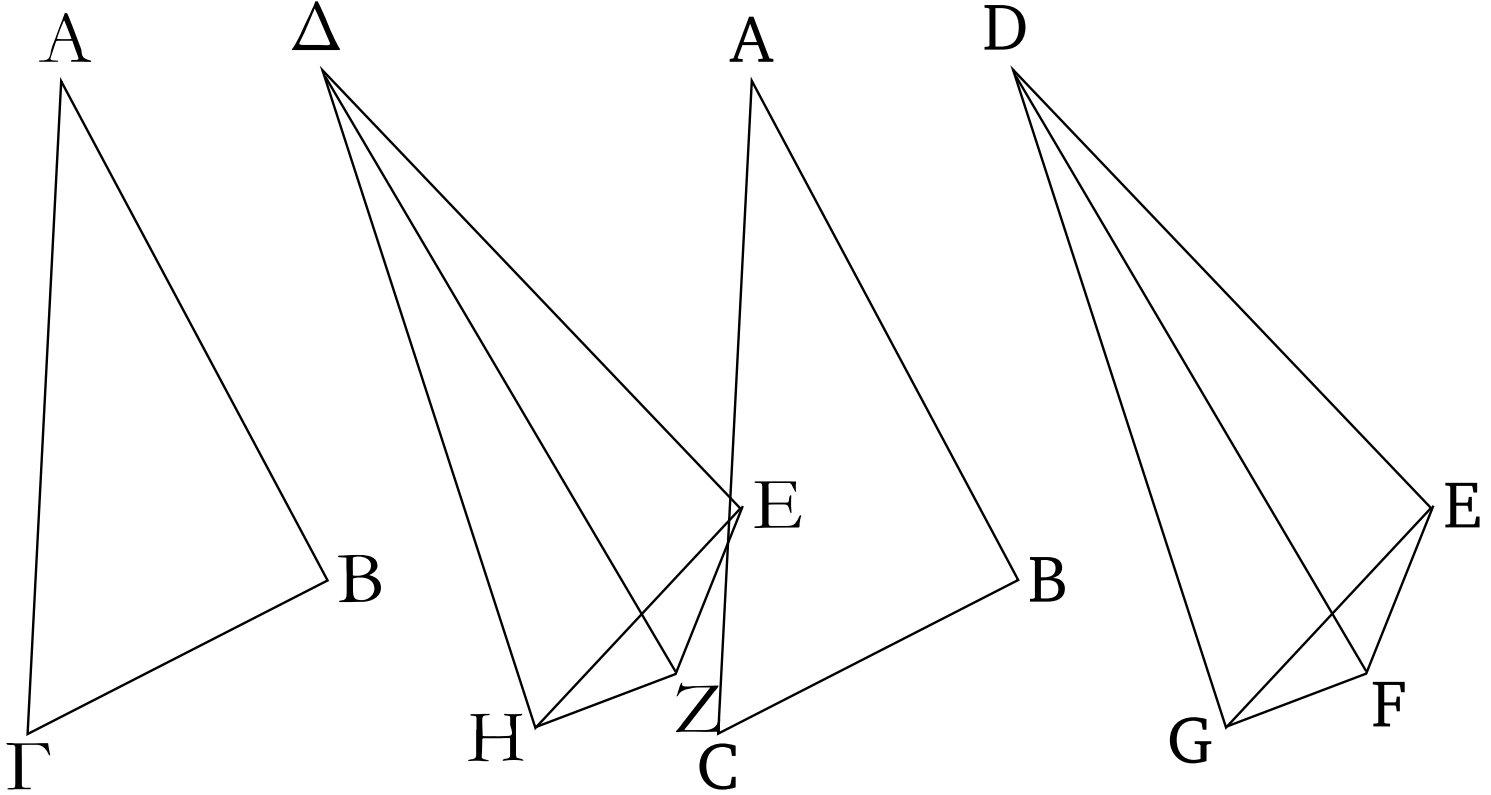
Let ABC and DEF be two triangles having the two sides

τέρων ἑκατέρω, τὴν μὲν AB τῇ ΔE τὴν δὲ AG τῇ ΔZ , ἥ δὲ πρὸς τῷ A γωνία τῇ πρὸς τῷ Δ γωνίᾳ μείζων ἔστω· λέγω, ὅτι καὶ βάσις ἡ BG βάσεως τῇς EZ μείζων ἔστί.

Ἐπεὶ γὰρ μείζων ἡ ὑπὸ BAG γωνία τῇς ὑπὸ $E\Delta Z$ γωνιάς, σννεσάτω πρὸς τῇ ΔE εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Δ τῇ ὑπὸ BAG γωνίᾳ ἴση ἡ ὑπὸ $E\Delta H$, καὶ κείσθω ὁποτέρω τῶν AG , ΔZ ἴση ἡ ΔH , καὶ ἐπεξέυθυσαν αἱ EH , ZH .

AB and AC equal to the two sides DE and DF , respectively. (That is), AB (equal) to DE , and AC to DF . Let them also have the angle at A greater than the angle at D . I say that the base BC is also greater than the base EF .

For since angle BAC is greater than angle EDF , let (angle) EDG , equal to angle BAC , be constructed at the point D on the straight-line DE [Prop. 1.23]. And let DG be made equal to either of AC or DF [Prop. 1.3], and let EG and FG be joined.



Ἐπεὶ οὖν ἴση ἔστί ἡ μὲν AB τῇ ΔE , ἡ δὲ AG τῇ ΔH , δύο δὴ αἱ BA , AG δυοὶ ταῖς $E\Delta$, ΔH ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνία ἡ ὑπὸ BAG γωνία τῇ ὑπὸ $E\Delta H$ ἴση· βάσις ἄρα ἡ BG βάσει τῇ EH ἔστιν ἴση. πάλιν, ἐπεὶ ἴση ἔστί ἡ ΔZ τῇ ΔH , ἴση ἐστὶ καὶ ἡ ὑπὸ ΔHZ γωνία τῇ ὑπὸ ΔZH · μείζων ἄρα ἡ ὑπὸ ΔZH τῇς ὑπὸ EZH · πολλῶν ἄρα μείζων ἔστί ἡ ὑπὸ EZH τῇς ὑπὸ EHZ . καὶ ἐπεὶ τρίγωνόν ἐστι τὸ EZH μείζονα ἔθον τὴν ὑπὸ EZH γωνίαν τῇς ὑπὸ EHZ , ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει, μείζων ἄρα καὶ πλευρὰ ἡ EH τῇς EZ . ἴση δὲ ἡ EH τῇ BG · μείζων ἄρα καὶ ἡ BG τῇς EZ .

Ἐάν ἄρα δύο τρίγωνα τὰς δύο πλευράς δυοὶ πλευραῖς ἴσας ἔθῃ ἑκατέρω ἑκατέρω, τὴν δὲ γωνίαν τῇς γωνιάς μείζονα ἔθῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῇς βάσεως μείζονα ἔχει· ὅπερ ἔδει δεῖξαι.

Therefore, since AB is equal to DE and AC to DG , the two (straight-lines) BA , AC are equal to the two (straight-lines) ED , DG , respectively. Also the angle BAC is equal to the angle EDG . Thus, the base BC is equal to the base EG [Prop. 1.4]. Again, since DF is equal to DG , angle DGF is also equal to angle DFG [Prop. 1.5]. Thus, DFG (is) greater than EGF . Thus, EFG is much greater than EGF . And since triangle EFG has angle EFG greater than EGF , and the greater angle is subtended by the greater side [Prop. 1.19], side EG (is) thus also greater than EF . But EG (is) equal to BC . Thus, BC (is) also greater than EF .

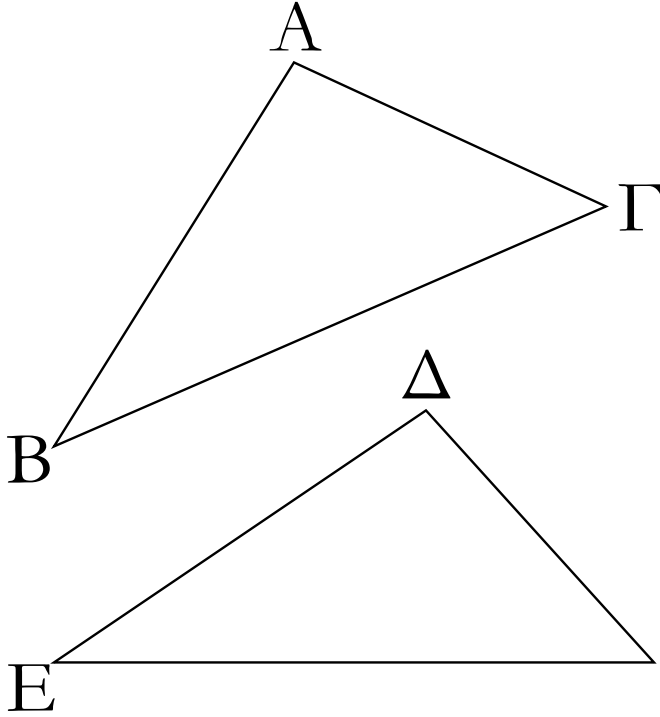
Thus, if two triangles have two sides equal to two sides, respectively, but (one) has the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the other), (then the former triangle) will also have a base greater than the base (of the latter). (Which is) the very thing it was required to show.

25.

Ἐάν δύο τρίγωνα τὰς δύο πλευράς δυοὶ πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρα, τὴν δὲ βάσιν τῆς βάσεως μείζονα ἔθῃ, καὶ τὴν γωνίαν τῆς γωνίας μείζονα ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην.

Ἐστω δύο τρίγωνα τὰ $ABΓ$, $ΔΕΖ$ τὰς δύο πλευράς τὰς AB , $ΑΓ$ ταῖς δύο πλευραῖς ταῖς $ΔΕ$, $ΔΖ$ ἴσας ἔθοντα ἑκατέραν ἑκατέρα, τὴν μὲν AB τῇ $ΔΕ$, τὴν δὲ $ΑΓ$ τῇ $ΔΖ$ · βάσις δὲ ἡ $ΒΓ$ βάσεως τῆς $ΕΖ$ μείζων ἔστω· λέγω, ὅτι καὶ γωνία ἡ ὑπὸ $ΒΑΓ$ γωνία τῆς ὑπὸ $ΕΔΖ$ μείζων ἔστί.

Εἰ γὰρ μή, ἦτοι ἴση ἔστιν αὐτῇ ἢ ἐλάσσων· ἴση μὲν οὐκ ἔστιν ἡ ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΕΔΖ$ · ἴση γὰρ ἂν ἦν καὶ βάσις ἡ $ΒΓ$ βάσει τῇ $ΕΖ$ · οὐκ ἔστι δέ. οὐκ ἄρα ἴση ἔστι γωνία ἡ ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΕΔΖ$ · οὐδὲ μὴν ἐλάσσων ἔστιν ἡ ὑπὸ $ΒΑΓ$ τῆς ὑπὸ $ΕΔΖ$ · ἐλάσσων γὰρ ἂν ἦν καὶ βάσις ἡ $ΒΓ$ βάσεως τῆς $ΕΖ$ · οὐκ ἔστι δέ· οὐκ ἄρα ἐλάσσων ἔστιν ἡ ὑπὸ $ΒΑΓ$ γωνία τῆς ὑπὸ $ΕΔΖ$. ἐδείχθη δέ, ὅτι οὐδὲ ἴση μείζων ἄρα ἔστιν ἡ ὑπὸ $ΒΑΓ$ τῆς ὑπὸ $ΕΔΖ$.



Ἐάν ἄρα δύο τρίγωνα τὰς δύο πλευράς δυοὶ πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρα, τὴν δὲ βάσιν τῆς βάσεως μείζονα ἔθῃ, καὶ τὴν γωνίαν τῆς γωνίας μείζονα ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην· ὅπερ ἔδει δεῖξαι.

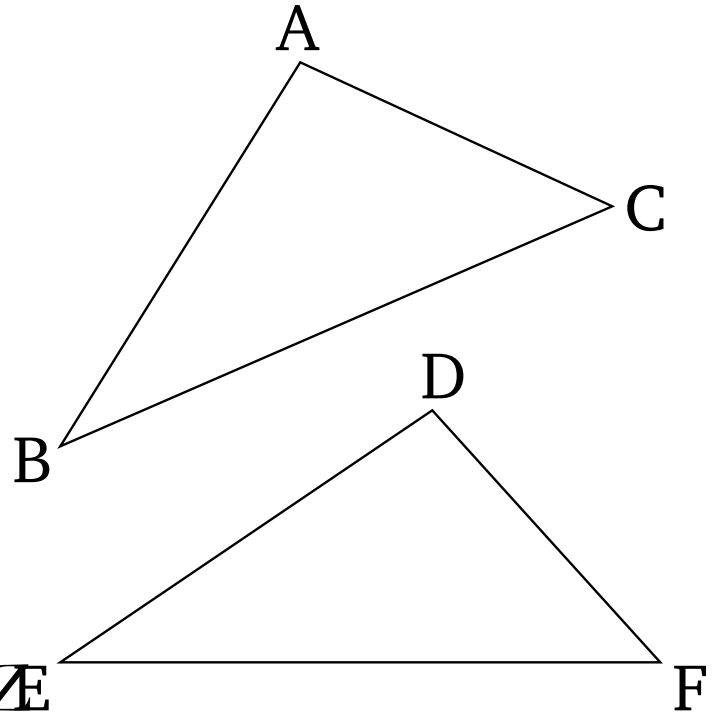
26.

Proposition 25

If two triangles have two sides equal to two sides, respectively, but (one) has a base greater than the base (of the other), (then the former triangle) will also have the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the latter).

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively (That is), AB (equal) to DE , and AC to DF . And let the base BC be greater than the base EF . I say that angle BAC is also greater than EDF .

For if not, (BAC) is certainly either equal to, or less than, (EDF). In fact, BAC is not equal to EDF . For then the base BC would also be equal to the base EF [Prop. 1.4]. But it is not. Thus, angle BAC is not equal to EDF . Neither, indeed, is BAC less than EDF . For then the base BC would also be less than the base EF [Prop. 1.24]. But it is not. Thus, angle BAC is not less than EDF . But it was shown that (BAC) is not equal (to EDF) either. Thus, BAC is greater than EDF .



Thus, if two triangles have two sides equal to two sides, respectively, but (one) has a base greater than the base (of the other), (then the former triangle) will also have the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the latter). (Which is) the very thing it was required to show.

Proposition 26

Ἐάν δύο τρίγωνα τὰς δύο γωνίας δυοὶ γωνίαις ἴσαις ἔθῃ ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην ᾗτοι τὴν πρὸς ταῖς ἴσαις γωνίαις ἢ τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν, καὶ τὰς λοιπὰς πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαις ἔχει [ἑκατέραν ἑκατέρᾳ] καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ.

Ἐστω δύο τρίγωνα τὰ $ABΓ$, $ΔΕΖ$ τὰς δύο γωνίας τὰς ὑπὸ $ABΓ$, $ΒΓΑ$ δυοὶ ταῖς ὑπὸ $ΔΕΖ$, $ΕΖΔ$ ἴσαις ἔθοντα ἑκατέραν ἑκατέρᾳ, τὴν μὲν ὑπὸ $ABΓ$ τῇ ὑπὸ $ΔΕΖ$, τὴν δὲ ὑπὸ $ΒΓΑ$ τῇ ὑπὸ $ΕΖΔ$. ἐθέτω δὲ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην, πρότερον τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν $ΒΓ$ τῇ $ΕΖ$. λέγω, ὅτι καὶ τὰς λοιπὰς πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαις ἔχει ἑκατέραν ἑκατέρᾳ, τὴν μὲν AB τῇ $ΔΕ$ τὴν δὲ $ΑΓ$ τῇ $ΔΖ$, καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ, τὴν ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΕΔΖ$.

Εἰ γὰρ ἄνισός ἐστιν ἡ AB τῇ $ΔΕ$, μία αὐτῶν μείζων ἐστίν. ἔστω μείζων ἡ AB , καὶ κείσθω τῇ $ΔΕ$ ἴση ἡ BH , καὶ ἐπεξεύθθω ἡ $ΗΓ$.

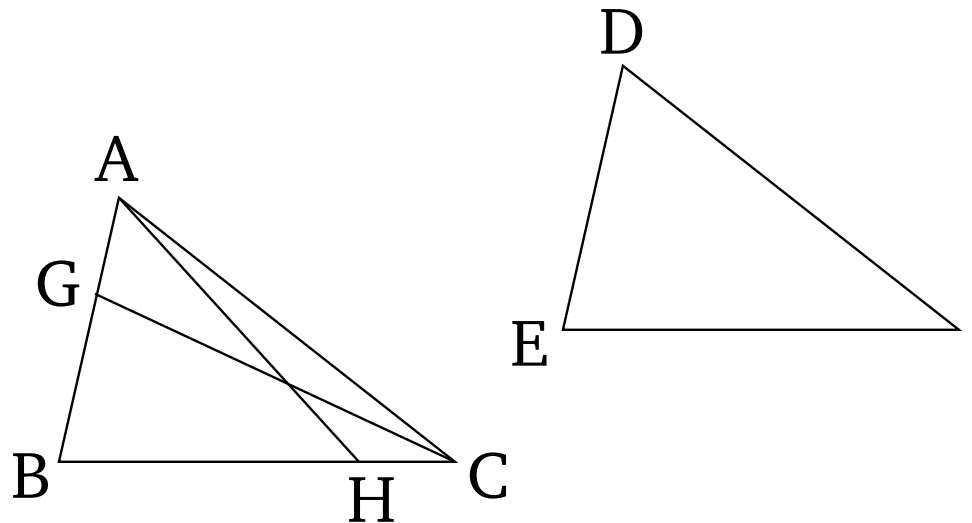
Ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν BH τῇ $ΔΕ$, ἡ δὲ $ΒΓ$ τῇ $ΕΖ$, δύο δὴ αἱ BH , $ΒΓ$ δυοὶ ταῖς $ΔΕ$, $ΕΖ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ $ΗΒΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ ἴση ἐστίν· βάσει ἄρα ἡ $ΗΓ$ βάσει τῇ $ΔΖ$ ἴση ἐστίν, καὶ τὸ $ΗΒΓ$ τρίγωνον τῷ $ΔΕΖ$ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἡ ὑπὸ $ΗΓΒ$ γωνία τῇ ὑπὸ $ΔΖΕ$. ἀλλὰ ἡ ὑπὸ $ΔΖΕ$ τῇ ὑπὸ $ΒΓΑ$ ὑπόκειται ἴση· καὶ ἡ ὑπὸ $ΒΓΗ$ ἄρα τῇ ὑπὸ $ΒΓΑ$ ἴση ἐστίν, ἡ ἐλάσσων τῇ μείζονι· ὅπερ ἀδύνατον. οὐκ ἄρα ἄνισός ἐστιν ἡ AB τῇ $ΔΕ$. ἴση ἄρα. ἔστι δὲ καὶ ἡ $ΒΓ$ τῇ $ΕΖ$ ἴση· δύο δὴ αἱ AB , $ΒΓ$ δυοὶ ταῖς $ΔΕ$, $ΕΖ$ ἴσαι εἰσὶν ἑκάτερα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ ἐστὶν ἴση· βάσει ἄρα ἡ $ΑΓ$ βάσει τῇ $ΔΖ$ ἴση ἐστίν, καὶ λοιπὴ γωνία ἡ ὑπὸ $ΒΑΓ$ τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ $ΕΔΖ$ ἴση ἐστίν.

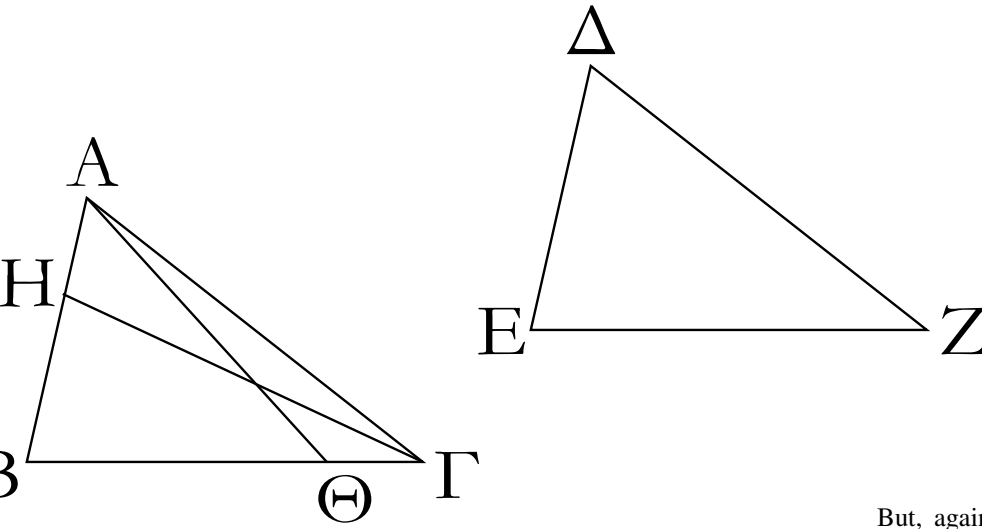
If two triangles have two angles equal to two angles, respectively, and one side equal to one side—in fact, either that by the equal angles, or that subtending one of the equal angles—(then the triangles) will also have the remaining sides equal to the [corresponding] remaining sides, and the remaining angle (equal) to the remaining angle.

Let ABC and DEF be two triangles having the two angles ABC and BCA equal to the two (angles) DEF and EFD , respectively. (That is) ABC (equal) to DEF , and BCA to EFD . And let them also have one side equal to one side. First of all, the (side) by the equal angles. (That is) BC (equal) to EF . I say that they will have the remaining sides equal to the corresponding remaining sides. (That is) AB (equal) to DE , and AC to DF . And (they will have) the remaining angle (equal) to the remaining angle. (That is) BAC (equal) to EDF .

For if AB is unequal to DE (then) one of them is greater. Let AB be greater, and let BG be made equal to DE [Prop. 1.3], and let GC be joined.

Therefore, since BG is equal to DE , and BC to EF , the two (straight-lines) GB , BC † are equal to the two (straight-lines) DE , EF , respectively. And angle GBC is equal to angle DEF . Thus, the base GC is equal to the base DF , and triangle GBC is equal to triangle DEF , and the remaining angles subtended by the equal sides will be equal to the (corresponding) remaining angles [Prop. 1.4]. Thus, GCB (is equal) to DFE . But, DFE was assumed (to be) equal to BCA . Thus, BCG is also equal to BCA , the lesser to the greater. The very thing (is) impossible. Thus, AB is not unequal to DE . Thus, (it is) equal. And BC is also equal to EF . So the two (straight-lines) AB , BC are equal to the two (straight-lines) DE , EF , respectively. And angle ABC is equal to angle DEF . Thus, the base AC is equal to the base DF , and the remaining angle BAC is equal to the remaining angle EDF [Prop. 1.4].





Ἀλλὰ δὴ πάλιν ἔστωσαν αἱ ὑπὸ τὰς ἴσαξ γωνίαξ πλευραὶ ὑποτείνουσαι ἴσαι, ὡς ἡ AB τῇ DE . λέγω πάλιν, ὅτι καὶ αἱ λοιπαὶ πλευραὶ ταῖς λοιπαῖς πλευραῖς ἴσαι ἔσονται, ἡ μὲν AG τῇ DZ , ἡ δὲ BG τῇ EZ καὶ ἔτι ἡ λοιπὴ γωνία ἡ ὑπὸ BAG τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ EDZ ἴση ἐστίν.

Εἰ γὰρ ἀνισὸς ἐστὶν ἡ BG τῇ EZ , μία αὐτῶν μείζων ἐστίν. ἔστω μείζων, εἰ δυνατόν, ἡ BG , καὶ κείσθω τῇ EZ ἴση ἡ $B\Theta$, καὶ ἐπεζεύθῃ ἡ $A\Theta$. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν $B\Theta$ τῇ EZ ἡ δὲ AB τῇ DE , δύο δὴ αἱ AB , $B\Theta$ δυοὶ ταῖς DE , EZ ἴσαι εἰσὶν ἑκατέρα ἑκατέρῃ· καὶ γωνίαξ ἴσαξ περιέθουσιν· βάσεις ἄρα ἡ $A\Theta$ βάσει τῇ DZ ἴση ἐστίν, καὶ τὸ $AB\Theta$ τρίγωνον τῷ DEZ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται, ὅφ' ἂς αἱ ἴσαξ πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἐστὶν ἡ ὑπὸ $B\Theta A$ γωνία τῇ ὑπὸ EZD . ἀλλὰ ἡ ὑπὸ EZD τῇ ὑπὸ BGA ἐστὶν ἴση· τριγώνον δὴ τοῦ $A\Theta G$ ἡ ἑκτὸς γωνία ἡ ὑπὸ $B\Theta A$ ἴση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ BGA . ὅπερ ἀδύνατον. οὐκ ἄρα ἀνισὸς ἐστὶν ἡ BG τῇ EZ ἴση ἄρα. ἐστὶ δὲ καὶ ἡ AB τῇ DE ἴση. δύο δὴ αἱ AB , BG δύο ταῖς DE , EZ ἴσαι εἰσὶν ἑκατέρα ἑκατέρῃ· καὶ γωνίαξ ἴσαξ περιέθουσι· βάσεις ἄρα ἡ AG βάσει τῇ DZ ἴση ἐστίν, καὶ τὸ ABG τρίγωνον τῷ DEZ τριγώνῳ ἴσον καὶ λοιπὴ γωνία ἡ ὑπὸ BAG τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ EDZ ἴση.

Ἐὰν ἄρα δύο τρίγωνα τὰς δύο γωνίαξ δυοὶ γωνίαις ἴσαξ ἔθῃ ἑκατέραν ἑκατέρῃ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην ἢ τοὶ τὴν πρὸς ταῖς ἴσαις γωνίαις, ἢ τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν, καὶ τὰς λοιπὰς πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαξ ἔχει καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ· ὅπερ ἔδει δεῖχαι.

But, again, let the sides subtending the equal angles be equal: for instance, (let) AB (be equal) to DE . Again, I say that the remaining sides will be equal to the remaining sides. (That is) AC (equal) to DF , and BC to EF . Furthermore, the remaining angle BAC is equal to the remaining angle EDF .

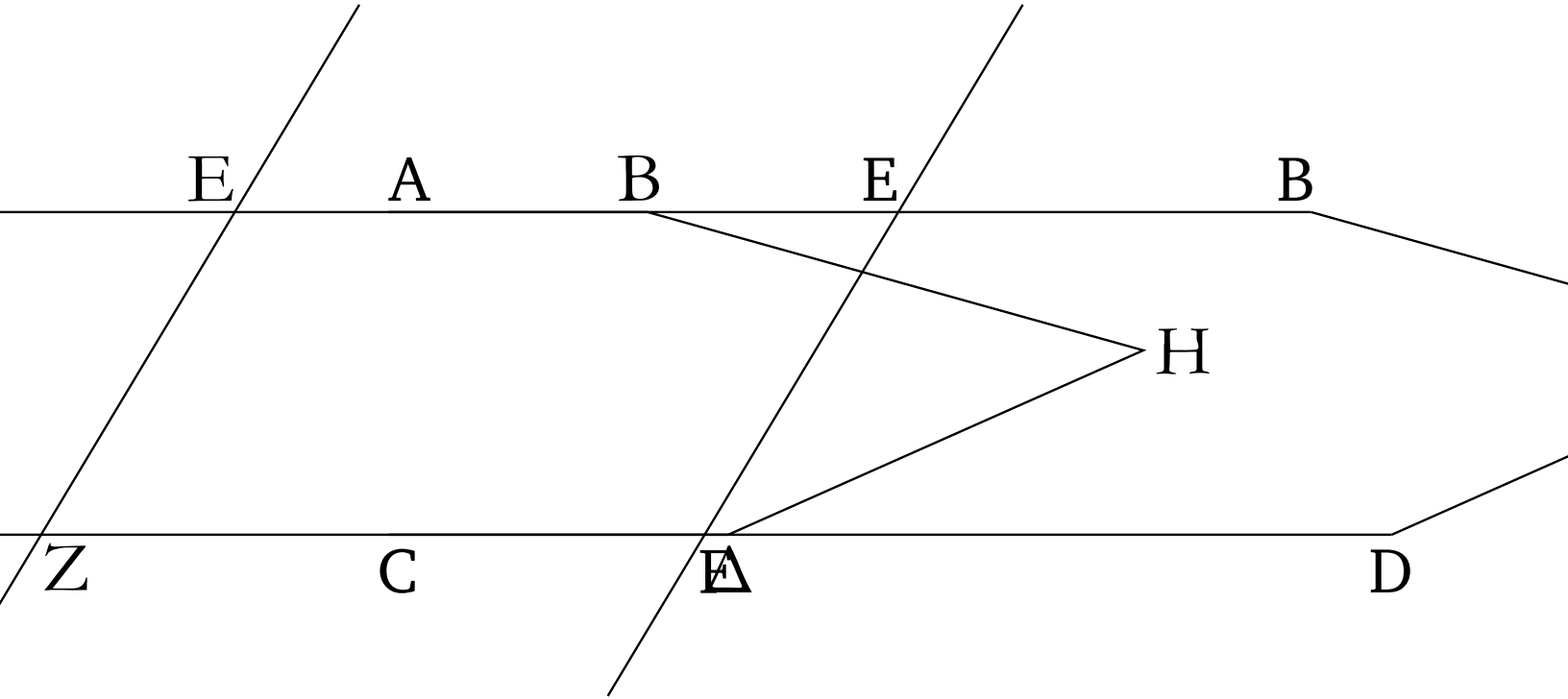
For if BC is unequal to EF (then) one of them is greater. If possible, let BC be greater. And let BH be made equal to EF [Prop. 1.3], and let AH be joined. And since BH is equal to EF , and AB to DE , the two (straight-lines) AB , BH are equal to the two (straight-lines) DE , EF , respectively. And the angles they encompass (are also equal). Thus, the base AH is equal to the base DF , and the triangle ABH is equal to the triangle DEF , and the remaining angles subtended by the equal sides will be equal to the (corresponding) remaining angles [Prop. 1.4]. Thus, angle BHA is equal to EFD . But, EFD is equal to BCA . So, in triangle AHC , the external angle BHA is equal to the internal and opposite angle BCA . The very thing (is) impossible [Prop. 1.16]. Thus, BC is not unequal to EF . Thus, (it is) equal. And AB is also equal to DE . So the two (straight-lines) AB , BC are equal to the two (straight-lines) DE , EF , respectively. And they encompass equal angles. Thus, the base AC is equal to the base DF , and triangle ABC (is) equal to triangle DEF , and the remaining angle BAC (is) equal to the remaining angle EDF [Prop. 1.4].

Thus, if two triangles have two angles equal to two angles, respectively, and one side equal to one side—in fact, either that by the equal angles, or that subtending one of the equal angles—(then the triangles) will also have the remaining sides equal to the (corresponding) remaining sides, and the remaining angle (equal) to the remaining angle. (Which is) the very thing it was required to show.

† The Greek text has “ BG , BC ”, which is obviously a mistake.

Ἐάν εἷς δύο εὐθείας εὐθεῖα ἐμπίπτονσα τὰς ἐναλλάχ γωνίας ἴσας ἀλλήλαις ποιῇ, παράλληλοι ἔσονται ἀλλήλαις αἱ εὐθεῖαι.

If a straight-line falling across two straight-lines makes the alternate angles equal to one another (then) the (two) straight-lines will be parallel to one another.



Εἷς γὰρ δύο εὐθείας τὰς AB , $ΓΔ$ εὐθεῖα ἐμπίπτονσα ἡ EZ τὰς ἐναλλάχ γωνίας τὰς ὑπὸ AEZ , $EZΔ$ ἴσας ἀλλήλαις ποιεῖτω· λέγω, ὅτι παράλληλός ἐστιν ἡ AB τῇ $ΓΔ$.

Εἰ γὰρ μή, ἐκβαλλόμεναι αἱ AB , $ΓΔ$ συμπεσοῦνται ἥτοι ἐπὶ τὰ B , $Δ$ μέρη ἢ ἐπὶ τὰ A , $Γ$. ἐκβεβλήσθωσαν καὶ συμπίπτέτωσαν ἐπὶ τὰ B , $Δ$ μέρη κατὰ τὸ H . τριγώνου δὴ τοῦ HEZ ἡ ἐκτὸς γωνία ἡ ὑπὸ AEZ ἴση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ EZH · ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα αἱ AB , $ΔΓ$ ἐκβαλλόμεναι συμπεσοῦνται ἐπὶ τὰ B , $Δ$ μέρη. ὁμοίως δὴ δευθθήσεται, ὅτι οὐδὲ ἐπὶ τὰ A , $Γ$ · αἱ δὲ ἐπὶ μηδέτερον τὰ μέρη συμπίπτονσαι παράλληλοί εἰσιν· παράλληλός ἄρα ἐστὶν ἡ AB τῇ $ΓΔ$.

Ἐάν ἄρα εἷς δύο εὐθείας εὐθεῖα ἐμπίπτονσα τὰς ἐναλλάχ γωνίας ἴσας ἀλλήλαις ποιῇ, παράλληλοι ἔσονται αἱ εὐθεῖαι· ὅπερ ἔδει δεῖξαι.

For let the straight-line EF , falling across the two straight-lines AB and CD , make the alternate angles AEF and EFD equal to one another. I say that AB and CD are parallel.

For if not, being produced, AB and CD will certainly meet together: either in the direction of B and D , or (in the direction) of A and C [Def. 1.23]. Let them be produced, and let them meet together in the direction of B and D at (point) G . So, for the triangle GEF , the external angle AEF is equal to the interior and opposite (angle) EFG . The very thing is impossible [Prop. 1.16]. Thus, being produced, AB and CD will not meet together in the direction of B and D . Similarly, it can be shown that neither (will they meet together) in (the direction of) A and C . But (straight-lines) meeting in neither direction are parallel [Def. 1.23]. Thus, AB and CD are parallel.

Thus, if a straight-line falling across two straight-lines makes the alternate angles equal to one another (then) the (two) straight-lines will be parallel (to one another). (Which is) the very thing it was required to show.

28.

Proposition 28

Ἐάν εἷς δύο εὐθείας εὐθεῖα ἐμπίπτονσα τὴν ἐκτὸς γωνίαν τῇ ἐντὸς καὶ ἀπεναντίον καὶ ἐπὶ τὰ αὐτὰ μέρη ἴσην ποιῇ ἢ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυσὶν ὀρθαῖς ἴσας, παράλληλοι

If a straight-line falling across two straight-lines makes the external angle equal to the internal and opposite angle on the same side, or (makes) the (sum of the) internal (angles) on the

ἔσονται ἀλλήλαις αἱ εὐθεῖαι.

Εἰς γὰρ δύο εὐθείας τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπίπτουσα ἡ EZ τὴν ἐκτὸς γωνίαν τὴν ὑπὸ EHB τῇ ἐντὸς καὶ ἀπεναντίον γωνίᾳ τῇ ὑπὸ $H\Theta\Delta$ ἴσην ποιεῖτω ἢ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη τὰς ὑπὸ $BH\Theta$, $H\Theta\Delta$ δυοῖν ὀρθαῖς ἴσας· λέγω, ὅτι παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ ὑπὸ EHB τῇ ὑπὸ $H\Theta\Delta$, ἀλλὰ ἡ ὑπὸ EHB τῇ ὑπὸ $AH\Theta$ ἐστὶν ἴση, καὶ ἡ ὑπὸ $AH\Theta$ ἄρα τῇ ὑπὸ $H\Theta\Delta$ ἐστὶν ἴση· καὶ εἰσὶν ἐναλλάχ· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

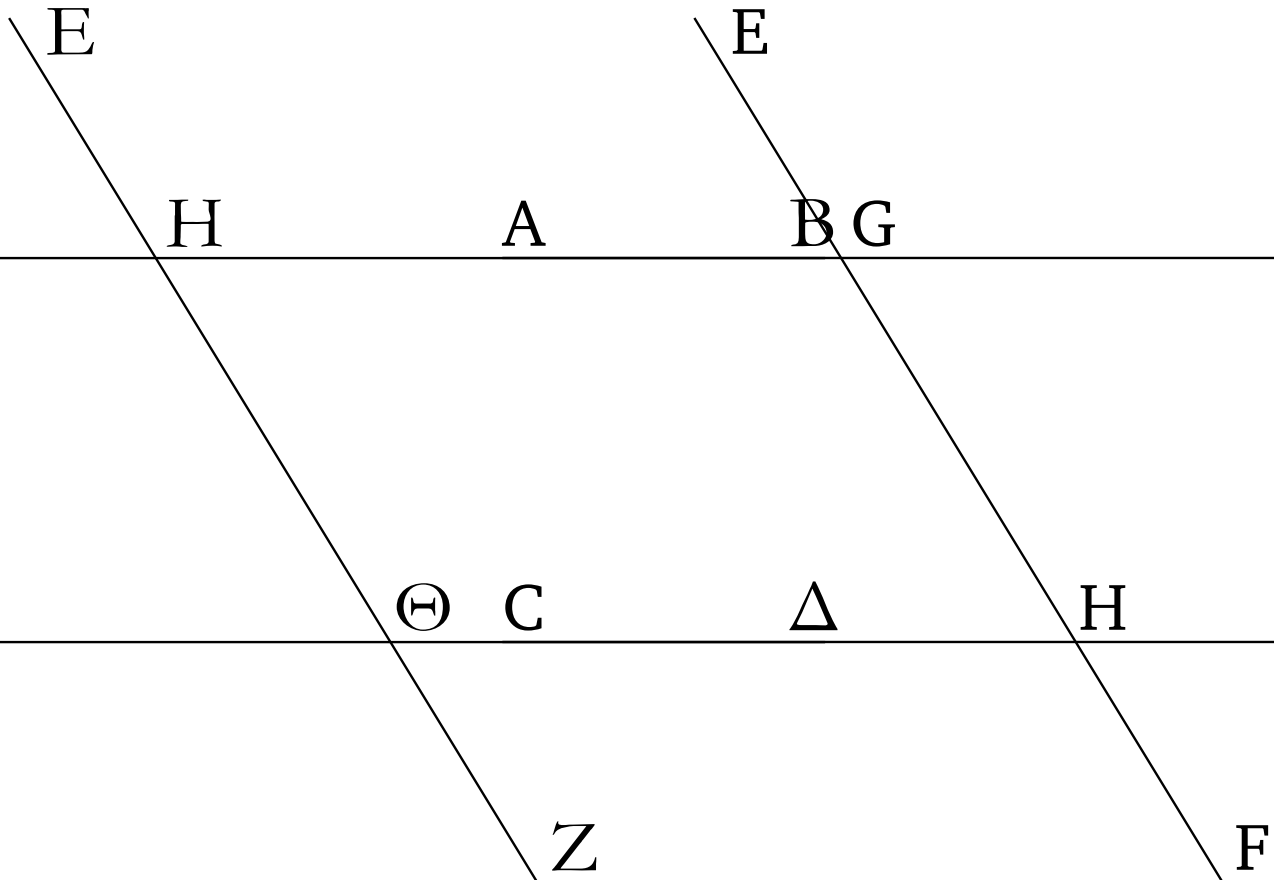
Πάλιν, ἐπεὶ αἱ ὑπὸ $BH\Theta$, $H\Theta\Delta$ δύο ὀρθαῖς ἴσαι εἰσὶν, εἰσὶ δὲ καὶ αἱ ὑπὸ $AH\Theta$, $BH\Theta$ δυοῖν ὀρθαῖς ἴσαι, αἱ ἄρα ὑπὸ $AH\Theta$, $BH\Theta$ ταῖς ὑπὸ $BH\Theta$, $H\Theta\Delta$ ἴσαι εἰσὶν· κοινὴ ἀφηρέσθω ἡ ὑπὸ $BH\Theta$ · λοιπὴ ἄρα ἡ ὑπὸ $AH\Theta$ λοιπῇ τῇ ὑπὸ $H\Theta\Delta$ ἐστὶν ἴση· καὶ εἰσὶν ἐναλλάχ· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

same side equal to two right-angles, (then) the (two) straight-lines will be parallel to one another.

For let EF , falling across the two straight-lines AB and CD , make the external angle EGB equal to the internal and opposite angle GHD , or the (sum of the) internal (angles) on the same side, BGH and GHD , equal to two right-angles. I say that AB is parallel to CD .

For since (in the first case) EGB is equal to GHD , but EGB is equal to AGH [Prop. 1.15], AGH is thus also equal to GHD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].

Again, since (in the second case, the sum of) BGH and GHD is equal to two right-angles, and (the sum of) AGH and BGH is also equal to two right-angles [Prop. 1.13], (the sum of) AGH and BGH is thus equal to (the sum of) BGH and GHD . Let BGH be subtracted from both. Thus, the remainder AGH is equal to the remainder GHD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].



Ἐάν ἄρα εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὴν ἐκτὸς γωνίαν τῇ ἐντὸς καὶ ἀπεναντίον καὶ ἐπὶ τὰ αὐτὰ μέρη ἴσην ποιῇ ἢ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυοῖν ὀρθαῖς ἴσας, παράλληλοι ἔσονται αἱ εὐθεῖαι· ὅπερ ἔδει δεῖξαι.

Thus, if a straight-line falling across two straight-lines makes the external angle equal to the internal and opposite angle on the same side, or (makes) the (sum of the) internal (angles) on the same side equal to two right-angles, (then) the (two) straight-lines will be parallel (to one another). (Which

29.

Ἡ εἷς τὰς παραλλήλων εὐθείας εὐθεῖα ἐμπίπτουσα τὰς τε ἐναλλάχ γωνίας ἴσας ἀλλήλαις ποιεῖ καὶ τὴν ἐκτὸς τῇ ἐντὸς καὶ ἀπεναντίον ἴσην καὶ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυοῖν ὀρθαῖς ἴσας.

Εἷς γὰρ παραλλήλων εὐθείας τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπίπτει ἡ EZ · λέγω, ὅτι τὰς ἐναλλάχ γωνίας τὰς ὑπὸ $AH\Theta$, $H\Theta\Delta$ ἴσας ποιεῖ καὶ τὴν ἐκτὸς γωνίαν τὴν ὑπὸ EHB τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ $H\Theta\Delta$ ἴσην καὶ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη τὰς ὑπὸ $BH\Theta$, $H\Theta\Delta$ δυοῖν ὀρθαῖς ἴσας.

Εἰ γὰρ ἄνισός ἐστιν ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ $H\Theta\Delta$, μία αὐτῶν μείζων ἐστίν. ἔστω μείζων ἡ ὑπὸ $AH\Theta$ · κοινὴ προσκείσθω ἡ ὑπὸ $BH\Theta$ · αἱ ἄρα ὑπὸ $AH\Theta$, $BH\Theta$ τῶν ὑπὸ $BH\Theta$, $H\Theta\Delta$ μείζονές εἰσιν. ἀλλὰ αἱ ὑπὸ $AH\Theta$, $BH\Theta$ δυοῖν ὀρθαῖς ἴσαι εἰσίν. [καὶ] αἱ ἄρα ὑπὸ $BH\Theta$, $H\Theta\Delta$ δύο ὀρθῶν ἐλάσσονές εἰσιν. αἱ δὲ ἀπ' ἐλασσόνων ἢ δύο ὀρθῶν ἐκβαλλόμεναι εἷς ἄπειρον συμπέπτουσιν· αἱ ἄρα AB , $\Gamma\Delta$ ἐκβαλλόμεναι εἷς ἄπειρον συμπεσοῦνται· οὐ συμπέπτουσι δὲ διὰ τὸ παραλλήλων αὐτὰς ὑποκεῖσθαι· οὐκ ἄρα ἄνισός ἐστιν ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ $H\Theta\Delta$ · ἴση ἄρα. ἀλλὰ ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ EHB ἐστὶν ἴση· καὶ ἡ ὑπὸ EHB ἄρα τῇ ὑπὸ $H\Theta\Delta$ ἐστὶν ἴση· κοινὴ προσκείσθω ἡ ὑπὸ $BH\Theta$ · αἱ ἄρα ὑπὸ EHB , $BH\Theta$ ταῖς ὑπὸ $BH\Theta$, $H\Theta\Delta$ ἴσαι εἰσίν. ἀλλὰ αἱ ὑπὸ EHB , $BH\Theta$ δύο ὀρθαῖς ἴσαι εἰσίν· καὶ αἱ ὑπὸ $BH\Theta$, $H\Theta\Delta$ ἄρα δύο ὀρθαῖς ἴσαι εἰσίν.

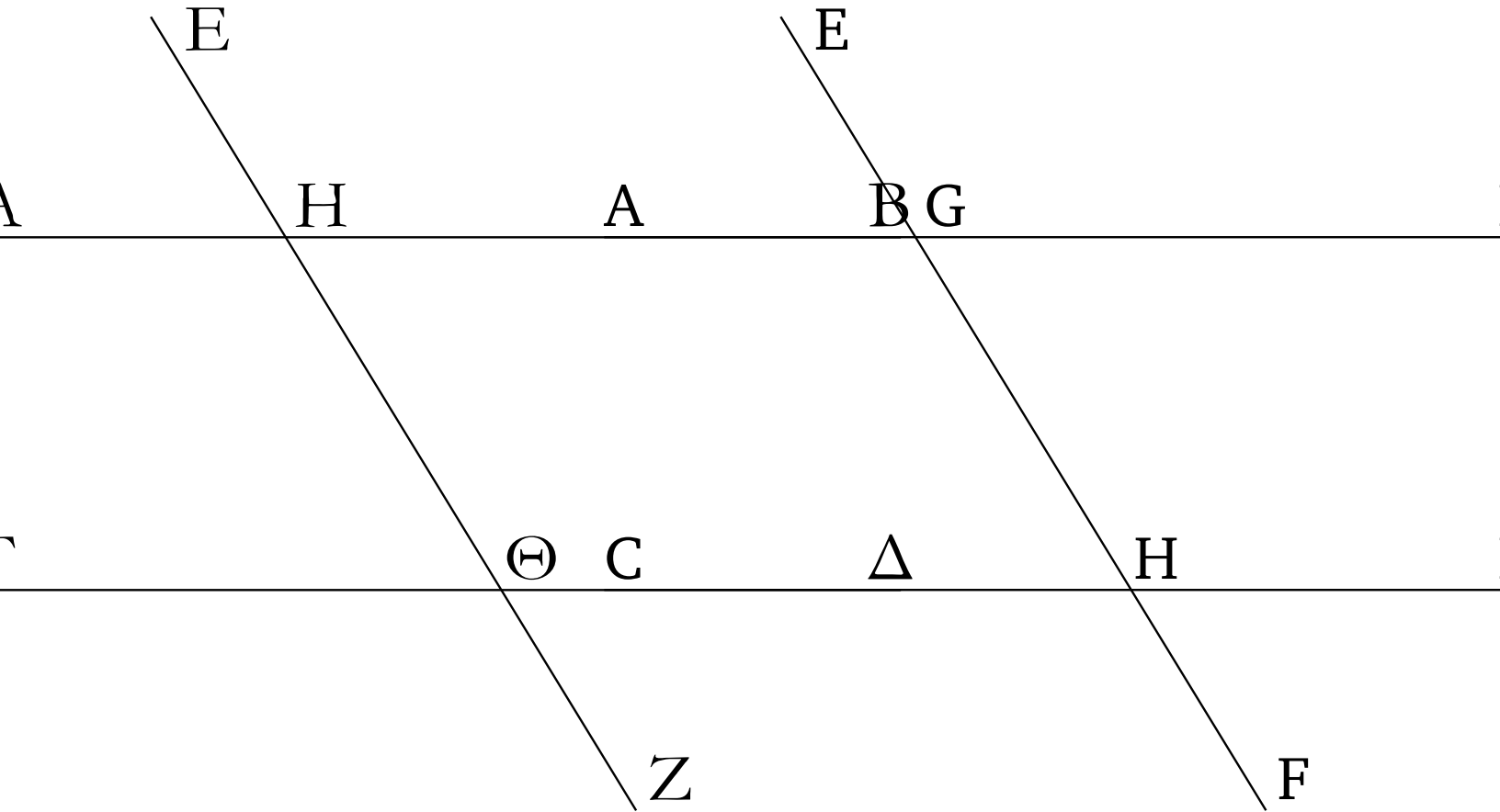
is) the very thing it was required to show.

Proposition 29

A straight-line falling across parallel straight-lines makes the alternate angles equal to one another, the external (angle) equal to the internal and opposite (angle), and the (sum of the) internal (angles) on the same side equal to two right-angles.

For let the straight-line EF fall across the parallel straight-lines AB and CD . I say that it makes the alternate angles, AGH and GHD , equal, the external angle EGB equal to the internal and opposite (angle) GHD , and the (sum of the) internal (angles) on the same side, BGH and GHD , equal to two right-angles.

For if AGH is unequal to GHD (then) one of them is greater. Let AGH be greater. Let BGH be added to both. Thus, (the sum of) AGH and BGH is greater than (the sum of) BGH and GHD . But, (the sum of) AGH and BGH is equal to two right-angles [Prop 1.13]. Thus, (the sum of) BGH and GHD is [also] less than two right-angles. But (straight-lines) being produced to infinity from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, AB and CD , being produced to infinity, will meet together. But they do not meet, on account of them (initially) being assumed parallel (to one another) [Def. 1.23]. Thus, AGH is not unequal to GHD . Thus, (it is) equal. But, AGH is equal to EGB [Prop. 1.15]. And EGB is thus also equal to GHD . Let BGH be added to both. Thus, (the sum of) EGB and BGH is equal to (the sum of) BGH and GHD . But, (the sum of) EGB and BGH is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) BGH and GHD is also equal to two right-angles.



Ἡ ἄρα εἷς τὰς παραλλήλους εὐθείας εὐθεῖα ἐμπίπτουσα
τάς τε ἐναλλάξ γωνίας ἴσας ἀλλήλαις ποιεῖ καὶ τὴν ἐκτὸς τῇ
ἐντὸς καὶ ἀπεναντίον ἴσην καὶ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη
δυσὶν ὀρθαῖς ἴσας· ὅπερ ἔδει δεῖξαι.

Thus, a straight-line falling across parallel straight-lines
makes the alternate angles equal to one another, the external
(angle) equal to the internal and opposite (angle), and the (sum
of the) internal (angles) on the same side equal to two right-
angles. (Which is) the very thing it was required to show.

30.

Proposition 30

Αἱ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ ἀλλήλαις εἰσὶ παράλλη-
λοι.

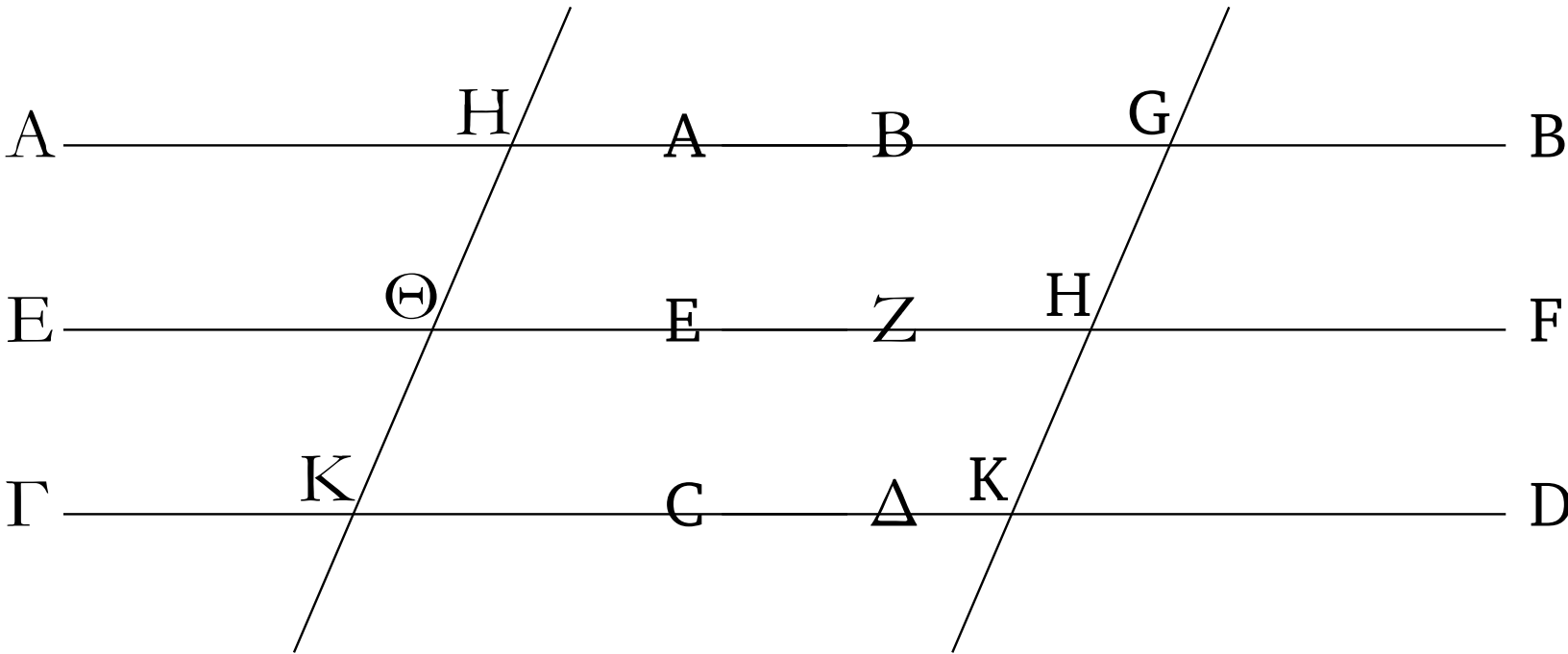
(Straight-lines) parallel to the same straight-line are also
parallel to one another.

Ἐστω ἑκατέρα τῶν AB , $ΓΔ$ τῇ EZ παράλληλος· λέγω,
ὅτι καὶ ἡ AB τῇ $ΓΔ$ ἐστὶ παράλληλος.

Let each of the (straight-lines) AB and CD be parallel to
 EF . I say that AB is also parallel to CD .

Ἐμπιπτέτω γάρ τις αὐτάς εὐθεῖα ἡ HK .

For let the straight-line GK fall across (AB , CD , and EF).



Καὶ ἐπεὶ εἷς παραλλήλων εὐθείας τὰς AB , EZ εὐθεία ἐμπέπτωκεν ἡ HK , ἴση ἄρα ἡ ὑπὸ AHK τῇ ὑπὸ $HΘZ$. πάλιν, ἐπεὶ εἷς παραλλήλων εὐθείας τὰς EZ , $ΓΔ$ εὐθεία ἐμπέπτωκεν ἡ HK , ἴση ἐστὶν ἡ ὑπὸ $HΘZ$ τῇ ὑπὸ $HKΔ$. ἐδείκθη δὲ καὶ ἡ ὑπὸ AHK τῇ ὑπὸ $HΘZ$ ἴση. καὶ ἡ ὑπὸ AHK ἄρα τῇ ὑπὸ $HKΔ$ ἐστὶν ἴση· καὶ εἰσιν ἐναλλάχ. παραλλήλοξ ἄρα ἐστὶν ἡ AB τῇ $ΓΔ$.

[Αἱ ἄρα τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ ἀλλήλαις εἰσὶ παράλληλοι.] ὅπερ ἔδει δεῖξαι.

31.

Διὰ τοῦ δοθέντος σημείου τῇ δοθείσῃ εὐθείᾳ παράλληλον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Ἐστω τὸ μὲν δοθὲν σημεῖον τὸ A , ἡ δὲ δοθεῖσα εὐθεῖα ἡ $BΓ$. δεῖ δὴ διὰ τοῦ A σημείου τῇ $BΓ$ εὐθείᾳ παράλληλον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Εἰλήφθω ἐπὶ τῇς $BΓ$ τυθὸν σημεῖον τὸ $Δ$, καὶ ἐπέξευθθω ἡ $AΔ$. καὶ συνεστάτω πρόξ τῇ $ΔA$ εὐθείᾳ καὶ τῷ πρόξ αὐτῇ σημείῳ τῷ A τῇ ὑπὸ $AΔΓ$ γωνίᾳ ἴση ἡ ὑπὸ $ΔAE$. καὶ ἐκβεβλήσθω ἐπ' εὐθείας τῇ EA εὐθείᾳ ἡ AZ .

And since the straight-line GK has fallen across the parallel straight-lines AB and EF , (angle) AGK (is) thus equal to GHF [Prop. 1.29]. Again, since the straight-line GK has fallen across the parallel straight-lines EF and CD , (angle) GHF is equal to GKD [Prop. 1.29]. But AGK was also shown (to be) equal to GKF . Thus, AGK is also equal to GKD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].

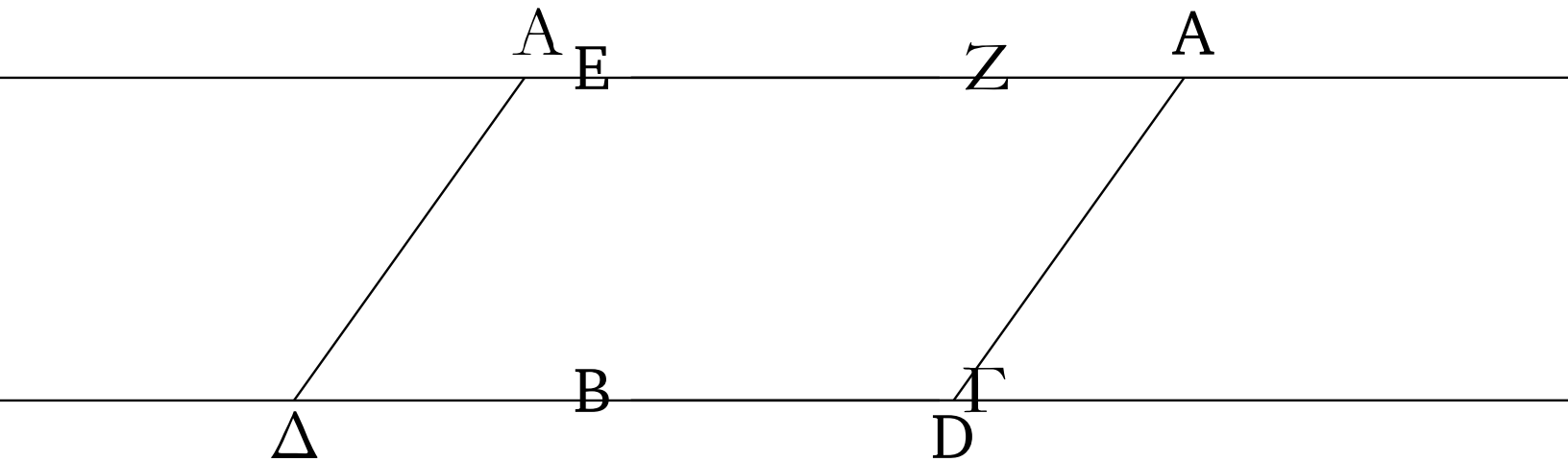
[Thus, (straight-lines) parallel to the same straight-line are also parallel to one another.] (Which is) the very thing it was required to show.

Proposition 31

To draw a straight-line parallel to a given straight-line, through a given point.

Let A be the given point, and BC the given straight-line. So it is required to draw a straight-line parallel to the straight-line BC , through the point A .

Let the point D be taken a random on BC , and let AD be joined. And let (angle) DAE , equal to angle ADC , be constructed on the straight-line DA at the point A on it [Prop. 1.23]. And let the straight-line AF be produced in a straight-line with EA .



Καὶ ἐπεὶ εἰς δύο εὐθείας τὰς BF , EZ εὐθεῖα ἐμπίπτουσα ἡ AD τὰς ἐναλλάχ γωνίας τὰς ὑπὸ EAD , ADF ἴσας ἀλλήλαις πεποίηκεν, παράλληλος ἄρα ἐστὶν ἡ EAF τῇ BF .

Διὰ τοῦ δοθέντος ἄρα σημείου τοῦ A τῇ δοθείσῃ εὐθείᾳ τῇ BF παράλληλος εὐθεῖα γραμμὴ ἤκται ἡ EAF · ὅπερ ἔδει ποιῆσαι.

And since the straight-line AD , (in) falling across the two straight-lines BC and EF , has made the alternate angles EAD and ADC equal to one another, EAF is thus parallel to BC [Prop. 1.27].

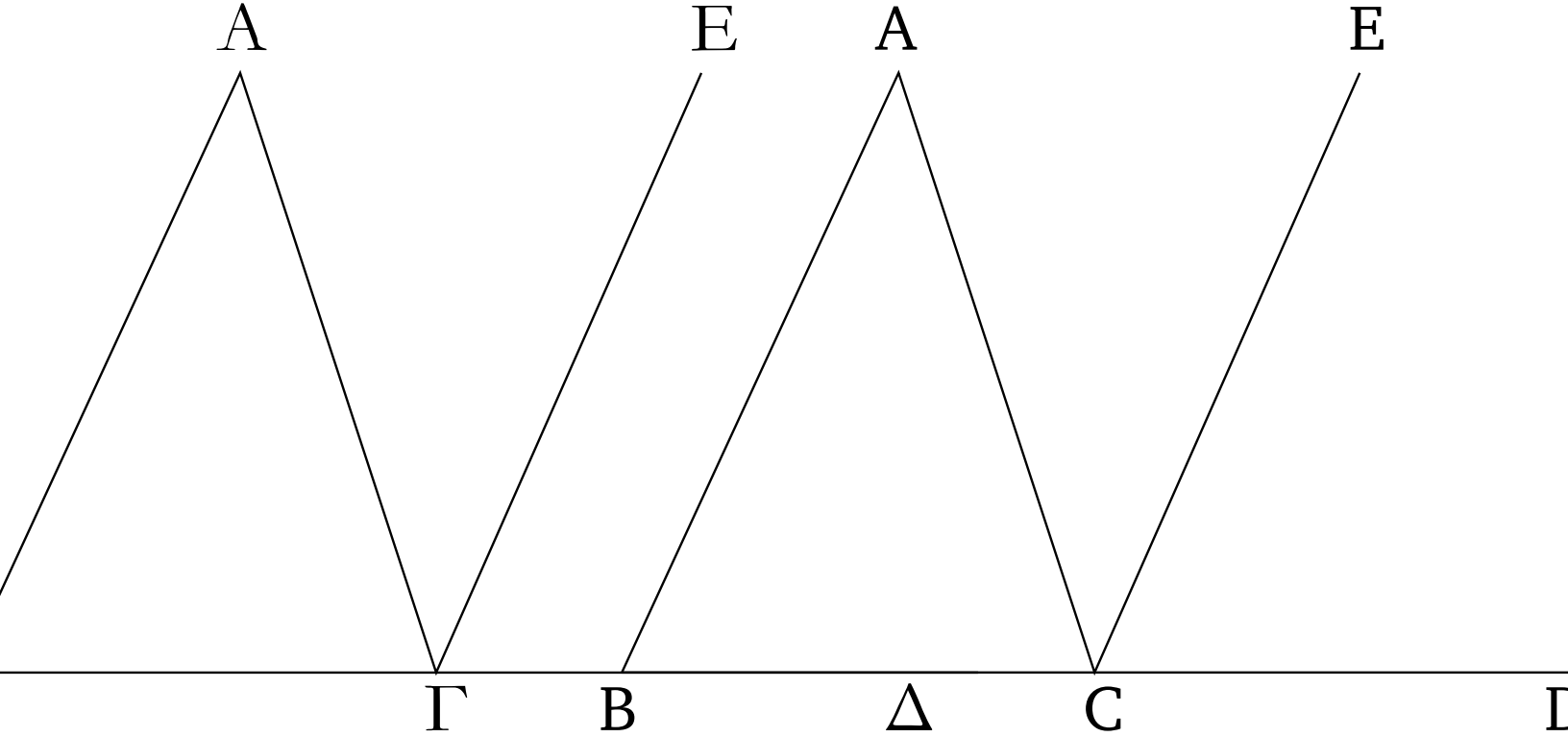
Thus, the straight-line EAF has been drawn parallel to the given straight-line BC , through the given point A . (Which is) the very thing it was required to do.

32.

Παντὸς τριγώνου μᾶξ τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον ἴση ἐστίν, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι δυοὶν ὀρθαῖς ἴσαι εἰσίν.

Proposition 32

In any triangle, (if) one of the sides (is) produced (then) the external angle is equal to the (sum of the) two internal and opposite (angles), and the (sum of the) three internal angles of the triangle is equal to two right-angles.



Ἐστω τρίγωνον τὸ $ABΓ$, καὶ προσεκβεβλήσθω αὐτοῦ μία πλευρὰ ἢ $ΒΓ$ ἐπὶ τὸ $Δ$ · λέγω, ὅτι ἡ ἐκτὸς γωνία ἢ ὑπὸ $ΑΓΔ$ ἴση ἐστὶ δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον ταῖς ὑπὸ $ΓΑΒ$, $ΑΒΓ$, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι αἱ ὑπὸ $ΑΒΓ$, $ΒΓΑ$, $ΓΑΒ$ δυοῖν ὀρθαῖς ἴσαι εἰσὶν.

Ἦθθω γάρ διὰ τοῦ $Γ$ σημείου τῇ $ΑΒ$ εὐθείᾳ παράλληλος ἢ $ΓΕ$.

Καὶ ἐπεὶ παράλληλός ἐστιν ἡ $ΑΒ$ τῇ $ΓΕ$, καὶ εἰς αὐτάς ἐμπίπτωκεν ἡ $ΑΓ$, αἱ ἐναλλάξ γωνίαι αἱ ὑπὸ $ΒΑΓ$, $ΑΓΕ$ ἴσαι ἀλλήλαις εἰσὶν. πάλιν, ἐπεὶ παράλληλός ἐστιν ἡ $ΑΒ$ τῇ $ΓΕ$, καὶ εἰς αὐτάς ἐμπίπτωκεν εὐθεῖα ἢ $ΒΔ$, ἡ ἐκτὸς γωνία ἢ ὑπὸ $ΕΓΔ$ ἴση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ $ΑΒΓ$. ἐδείχθη δὲ καὶ ἡ ὑπὸ $ΑΓΕ$ τῇ ὑπὸ $ΒΑΓ$ ἴση· ὅλη ἄρα ἢ ὑπὸ $ΑΓΔ$ γωνία ἴση ἐστὶ δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον ταῖς ὑπὸ $ΒΑΓ$, $ΑΒΓ$.

Κοινὴ προσκείσθω ἡ ὑπὸ $ΑΓΒ$ · αἱ ἄρα ὑπὸ $ΑΓΔ$, $ΑΓΒ$ τρισὶ ταῖς ὑπὸ $ΑΒΓ$, $ΒΓΑ$, $ΓΑΒ$ ἴσαι εἰσὶν. ἀλλ' αἱ ὑπὸ $ΑΓΔ$, $ΑΓΒ$ δυοῖν ὀρθαῖς ἴσαι εἰσὶν· καὶ αἱ ὑπὸ $ΑΓΒ$, $ΓΒΑ$, $ΓΑΒ$ ἄρα δυοῖν ὀρθαῖς ἴσαι εἰσὶν.

Παντὸς ἄρα τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον ἴση ἐστίν, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν· ὅπερ ἔδει δεῖξαι.

Let ABC be a triangle, and let one of its sides BC be produced to D . I say that the external angle ACD is equal to the (sum of the) two internal and opposite angles CAB and ABC , and the (sum of the) three internal angles of the triangle— ABC , BCA , and CAB —is equal to two right-angles.

For let CE be drawn through point C parallel to the straight-line AB [Prop. 1.31].

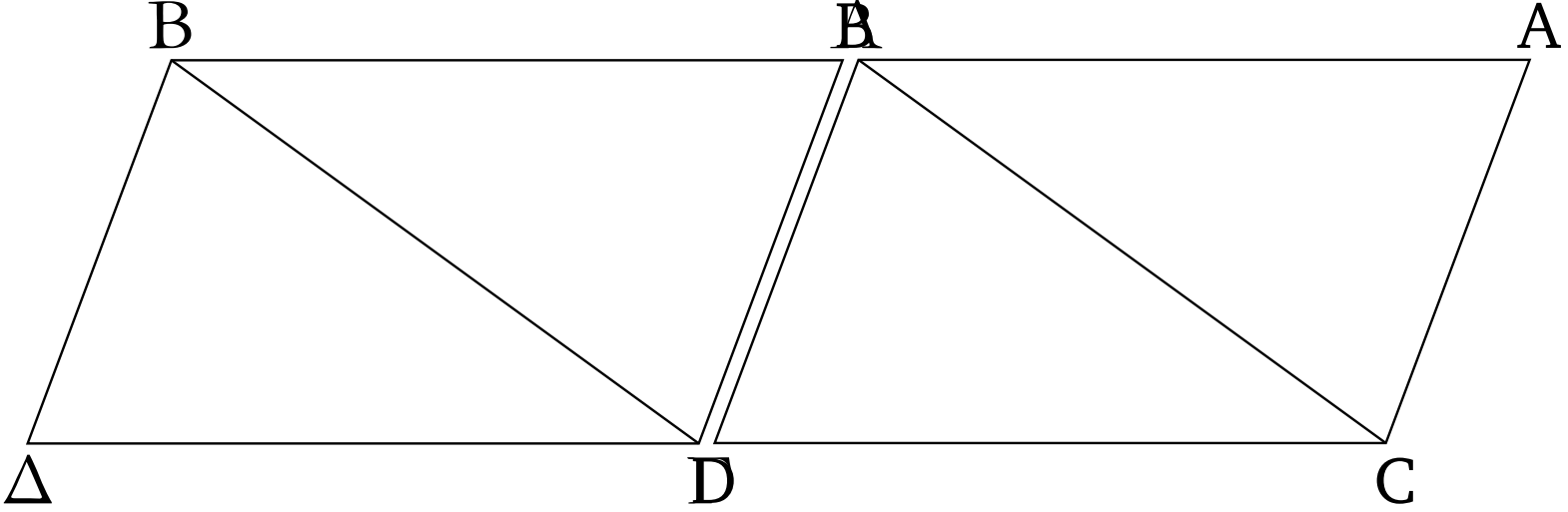
And since AB is parallel to CE , and AC has fallen across them, the alternate angles BAC and ACE are equal to one another [Prop. 1.29]. Again, since AB is parallel to CE , and the straight-line BD has fallen across them, the external angle ECD is equal to the internal and opposite (angle) ABC [Prop. 1.29]. But ACE was also shown (to be) equal to BAC . Thus, the whole angle ACD is equal to the (sum of the) two internal and opposite (angles) BAC and ABC .

Let ACB be added to both. Thus, (the sum of) ACD and ACB is equal to the (sum of the) three (angles) ABC , BCA , and CAB . But, (the sum of) ACD and ACB is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) ACB , CBA , and CAB is also equal to two right-angles.

Thus, in any triangle, (if) one of the sides (is) produced (then) the external angle is equal to the (sum of the) two internal and opposite (angles), and the (sum of the) three internal angles of the triangle is equal to two right-angles. (Which is) the very thing it was required to show.

Αἱ τὰς ἴσας τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη ἐπι-
ξενυγνύουσαι εὐθεῖαι καὶ αὐταὶ ἴσαι τε καὶ παράλληλοι εἰσιν.

Ἐστωσαν ἴσαι τε καὶ παράλληλοι αἱ AB , $ΓΔ$, καὶ ἐπιξεν-
γνύωσαν αὐτὰς ἐπὶ τὰ αὐτὰ μέρη εὐθεῖαι αἱ $ΑΓ$, $ΒΔ$. λέγω,
ὅτι καὶ αἱ $ΑΓ$, $ΒΔ$ ἴσαι τε καὶ παράλληλοι εἰσιν.



Ἐπεξεύθθω ἡ $ΒΓ$. καὶ ἐπεὶ παράλληλός ἐστιν ἡ AB τῇ
 $ΓΔ$, καὶ εἰς αὐτὰς ἐμπίπτωκεν ἡ $ΒΓ$, αἱ ἐναλλάξ γωνίαι αἱ
ὑπὸ $ΑΒΓ$, $ΒΓΔ$ ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ
 AB τῇ $ΓΔ$ κοινὴ δὲ ἡ $ΒΓ$, δύο δὴ αἱ AB , $ΒΓ$ δύο ταῖς $ΒΓ$,
 $ΓΔ$ ἴσαι εἰσὶν καὶ γωνία ἡ ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΒΓΔ$
ἴση. βάσει ἄρα ἡ $ΑΓ$ βάσει τῇ $ΒΔ$ ἐστὶν ἴση, καὶ τὸ $ΑΒΓ$
τρίγωνον τῷ $ΒΓΔ$ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι
ταῖς λοιπαῖς γωνίαις ἴσαι ἔσσονται ἑκατέρω ἑκατέρω, ὅφ' ἂν
αἱ ἴσαι πλευραὶ ὑποτείνουσιν. ἴση ἄρα ἡ ὑπὸ $ΑΓΒ$ γωνία
τῇ ὑπὸ $ΓΒΔ$. καὶ ἐπεὶ εἰς δύο εὐθείας τὰς $ΑΓ$, $ΒΔ$ εὐθεῖα
ἐμπίπτουσα ἡ $ΒΓ$ τὰς ἐναλλάξ γωνίας ἴσας ἀλλήλαις πεπο-
ίηκεν, παράλληλός ἄρα ἐστὶν ἡ $ΑΓ$ τῇ $ΒΔ$. ἐδείκθη δὲ αὐτῇ
καὶ ἴση.

Αἱ ἄρα τὰς ἴσας τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη
ἐπιξενυγνύουσαι εὐθεῖαι καὶ αὐταὶ ἴσαι τε καὶ παράλληλοι ε-
ἰσιν· ὅπερ ἔδει δεῖχαι.

Straight-lines joining equal and parallel (straight-lines) on
the same sides are themselves also equal and parallel.

Let AB and CD be equal and parallel (straight-lines), and
let the straight-lines AC and BD join them on the same sides.
I say that AC and BD are also equal and parallel.

Let BC be joined. And since AB is parallel to CD , and BC
has fallen across them, the alternate angles ABC and BCD are
equal to one another [Prop. 1.29]. And since AB is equal to
 CD , and BC is common, the two (straight-lines) AB , BC are
equal to the two (straight-lines) DC , CB .[†] And the angle ABC
is equal to the angle BCD . Thus, the base AC is equal to the
base BD , and triangle ABC is equal to triangle DCB ,[‡] and the
remaining angles will be equal to the corresponding remaining
angles subtended by the equal sides [Prop. 1.4]. Thus, angle
 ACB is equal to CBD . Also, since the straight-line BC , (in)
falling across the two straight-lines AC and BD , has made the
alternate angles (ACB and CBD) equal to one another, AC is
thus parallel to BD [Prop. 1.27]. And (AC) was also shown (to
be) equal to (BD).

Thus, straight-lines joining equal and parallel (straight-
lines) on the same sides are themselves also equal and parallel.
(Which is) the very thing it was required to show.

[†] The Greek text has “ BC , CD ”, which is obviously a mistake.

[‡] The Greek text has “ DCB ”, which is obviously a mistake.

34.

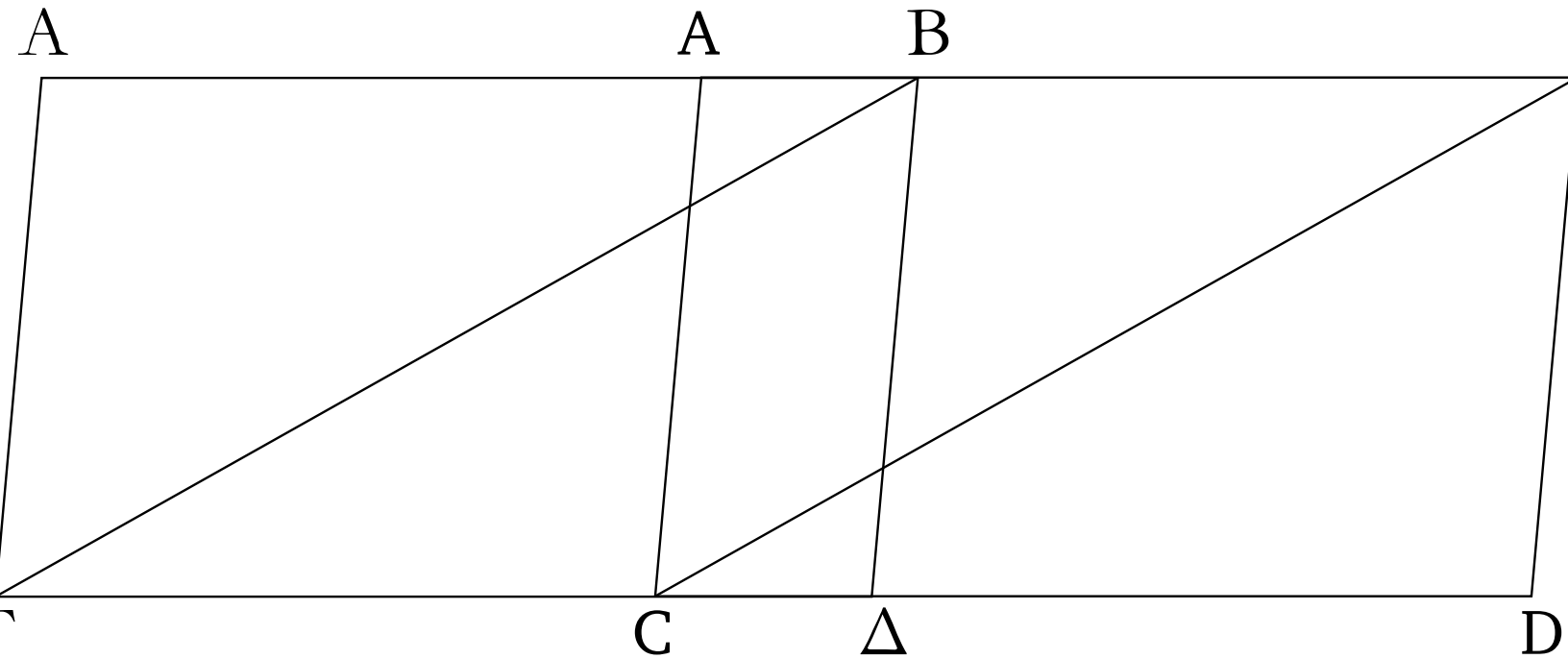
Proposition 34

Τῶν παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραὶ
τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ ἡ διάμετρος αὐτὰ διῖθα
τέμνει.

Ἐστω παραλληλόγραμμον θωρίον τὸ $ΑΓΔΒ$, διάμετρος
δὲ αὐτοῦ ἡ $ΒΓ$. λέγω, ὅτι τοῦ $ΑΓΔΒ$ παραλληλογράμμου αἱ
ἀπεναντίον πλευραὶ τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ ἡ
 $ΒΓ$ διάμετρος αὐτὸ διῖθα τέμνει.

In parallelogramic figures, the opposite sides and angles are
equal to one another, and a diagonal cuts them in half.

Let $ACDB$ be a parallelogramic figure, and BC its diagon-
al. I say that for parallelogram $ACDB$, the opposite sides and
angles are equal to one another, and the diagonal BC cuts it in
half.



Ἐπεὶ γὰρ παράλληλός ἐστιν ἡ AB τῇ $ΓΔ$, καὶ εἰς αὐτάς ἐμπεπτωκεν εὐθεῖα ἡ $ΒΓ$, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $ΑΒΓ$, $ΒΓΔ$ ἴσαι ἀλλήλαις εἰσίν. πάλιν ἐπεὶ παράλληλός ἐστιν ἡ $ΑΓ$ τῇ $ΒΔ$, καὶ εἰς αὐτάς ἐμπεπτωκεν ἡ $ΒΓ$, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $ΑΓΒ$, $ΓΒΔ$ ἴσαι ἀλλήλαις εἰσίν. δύο δὲ τρίγωνα ἐστί τὰ $ΑΒΓ$, $ΒΓΔ$ τὰς δύο γωνίας τὰς ὑπὸ $ΑΒΓ$, $ΒΓΔ$ δυοὶ ταῖς ὑπὸ $ΓΒΔ$, $ΓΒΔ$ ἴσαξ ἔθοντα ἑκατέραν ἑκατέρῳ καὶ μίαν πλευρὰν μὲν $ΒΓ$ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις κοινὴν αὐτῶν τὴν $ΒΓ$ · καὶ τὰς λοιπὰς ἄρα πλευρὰς ταῖς λοιπαῖς ἴσαξ ἔχει ἑκατέραν ἑκατέρῳ καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ· ἴση ἄρα ἡ μὲν $ΑΒ$ πλευρὰ τῇ $ΓΔ$, ἡ δὲ $ΑΓ$ τῇ $ΒΔ$, καὶ ἔτι ἴση ἐστὶν ἡ ὑπὸ $ΒΑΓ$ γωνία τῇ ὑπὸ $ΓΔΒ$. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΒΓΔ$, ἡ δὲ ὑπὸ $ΓΒΔ$ τῇ ὑπὸ $ΑΓΒ$, ὅλη ἄρα ἡ ὑπὸ $ΑΒΔ$ ὅλη τῇ ὑπὸ $ΑΓΔ$ ἐστὶν ἴση. ἐδείκθη δὲ καὶ ἡ ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΓΔΒ$ ἴση.

Τῶν ἄρα παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραὶ τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσίν.

Λέγω δὴ, ὅτι καὶ ἡ διάμετρος αὐτὰ διῖθα τέμνει. ἐπεὶ γὰρ ἴση ἐστὶν ἡ $ΑΒ$ τῇ $ΓΔ$, κοινὴ δὲ ἡ $ΒΓ$, δύο δὲ αἱ $ΑΒ$, $ΒΓ$ δυοὶ ταῖς $ΓΔ$, $ΒΓ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρῳ· καὶ γωνία ἡ ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΒΓΔ$ ἴση. καὶ βάσις ἄρα ἡ $ΑΓ$ τῇ $ΔΒ$ ἴση. καὶ τὸ $ΑΒΓ$ [ἄρα] τρίγωνον τῷ $ΒΓΔ$ τριγώνῳ ἴσον ἐστίν.

Ἡ ἄρα $ΒΓ$ διάμετρος διῖθα τέμνει τὸ $ΑΒΓΔ$ παραλληλόγραμμον· ὅπερ ἔδει δεῖξαι.

For since AB is parallel to CD , and the straight-line BC has fallen across them, the alternate angles ABC and BCD are equal to one another [Prop. 1.29]. Again, since AC is parallel to BD , and BC has fallen across them, the alternate angles ACB and CBD are equal to one another [Prop. 1.29]. So ABC and BCD are two triangles having the two angles ABC and BCA equal to the two (angles) BCD and CBD , respectively, and one side equal to one side—the (one) by the equal angles and common to them, (namely) BC . Thus, they will also have the remaining sides equal to the corresponding remaining (sides), and the remaining angle (equal) to the remaining angle [Prop. 1.26]. Thus, side AB is equal to CD , and AC to BD . Furthermore, angle BAC is equal to CDB . And since angle ABC is equal to BCD , and CBD to ACB , the whole (angle) ABD is thus equal to the whole (angle) ACD . And BAC was also shown (to be) equal to CDB .

Thus, in parallelogramic figures, the opposite sides and angles are equal to one another.

And, I also say that a diagonal cuts them in half. For since AB is equal to CD , and BC (is) common, the two (straight-lines) AB , BC are equal to the two (straight-lines) DC , CB [†], respectively. And angle ABC is equal to angle BCD . Thus, the base AC (is) also equal to DB , and triangle ABC is equal to triangle BCD [Prop. 1.4].

Thus, the diagonal BC cuts the parallelogram $ACDB$ [‡] in half. (Which is) the very thing it was required to show.

[†] The Greek text has “ CD , BC ”, which is obviously a mistake.

[‡] The Greek text has “ $ABCD$ ”, which is obviously a mistake.

35.

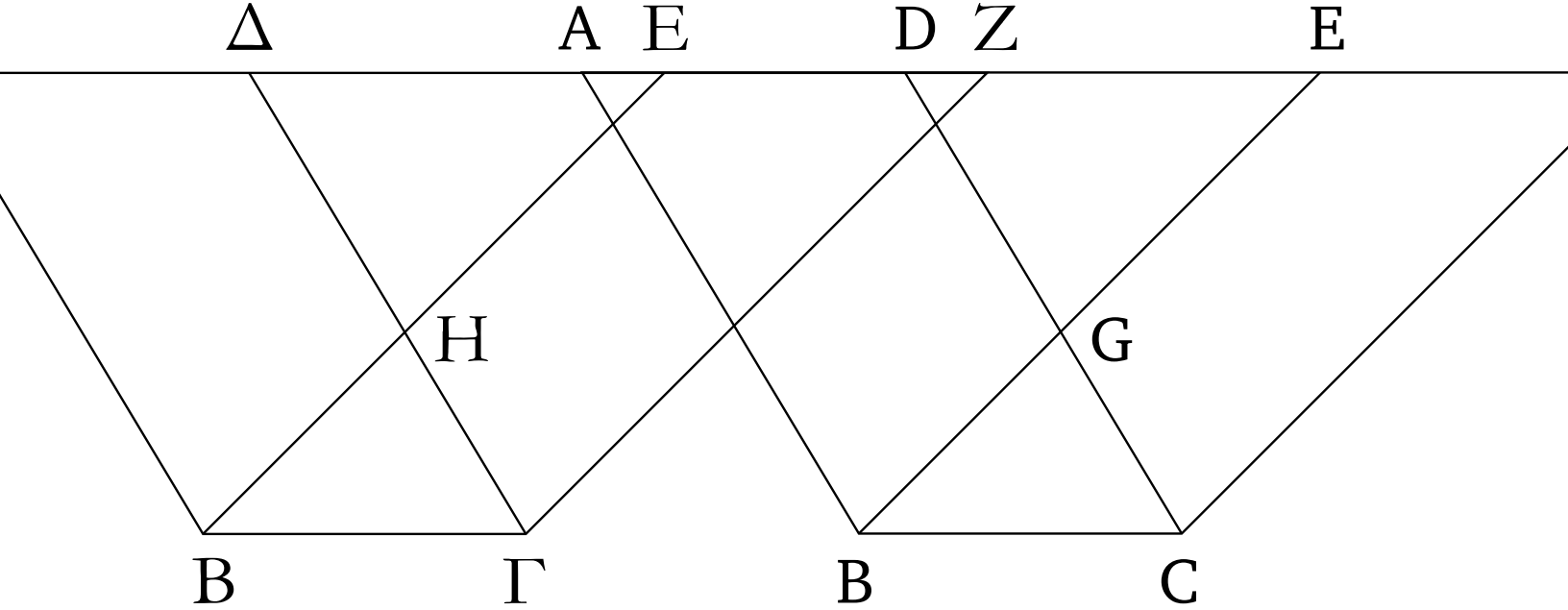
Proposition 35

Τὰ παραλληλόγραμμα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν.

Ἐστω παραλληλόγραμμα τὰ $ABΓΔ$, $EBΓΖ$ ἐπὶ τῆς αὐτῆς βάσεως τῆς $ΒΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $ΑΖ$, $ΒΓ$. λέγω, ὅτι ἴσον ἐστὶ τὸ $ABΓΔ$ τῷ $EBΓΖ$ παραλληλογράμῳ.

Parallelograms which are on the same base and between the same parallels are equal[†] to one another.

Let $ABCD$ and $EBCF$ be parallelograms on the same base BC , and between the same parallels AF and BC . I say that $ABCD$ is equal to parallelogram $EBCF$.



Ἐπεὶ γὰρ παραλληλόγραμμὸν ἐστὶ τὸ $ABΓΔ$, ἴση ἐστὶν ἡ $ΑΔ$ τῇ $ΒΓ$. διὰ τὰ αὐτὰ δὴ καὶ ἡ $ΕΖ$ τῇ $ΒΓ$ ἐστὶν ἴση· ὥστε καὶ ἡ $ΑΔ$ τῇ $ΕΖ$ ἐστὶν ἴση· καὶ κοινὴ ἡ $ΔΕ$ · ὅλη ἄρα ἡ $ΑΕ$ ὅλη τῇ $ΔΖ$ ἐστὶν ἴση. ἐστὶ δὲ καὶ ἡ $ΑΒ$ τῇ $ΔΓ$ ἴση· δύο δὴ αἱ $ΕΑ$, $ΑΒ$ δύο ταῖς $ΖΔ$, $ΔΓ$ ἴσαι εἰσὶν ἐκατέρω ἐκατέρω· καὶ γωνία ἡ ὑπὸ $ΖΔΓ$ γωνία τῇ ὑπὸ $ΕΑΒ$ ἐστὶν ἴση ἡ ἐκτὸς τῇ ἐντὸς· βάσει ἄρα ἡ $ΕΒ$ βάσει τῇ $ΖΓ$ ἴση ἐστίν, καὶ τὸ $ΕΑΒ$ τρίγωνον τῷ $ΔΖΓ$ τριγώνῳ ἴσον ἐστὶ· κοινὸν ἀφρηθήσθω τὸ $ΔΗΕ$ · λοιπὸν ἄρα τὸ $ΑΒΗΔ$ τραπέζιον λοιπῷ τῷ $ΕΗΓΖ$ τραπέζιῳ ἐστὶν ἴσον· κοινὸν προσκείσθω τὸ $ΗΒΓ$ τρίγωνον· ὅλον ἄρα τὸ $ΑΒΓΔ$ παραλληλόγραμμον ὅλῳ τῷ $ΕΒΓΖ$ παραλληλογράμῳ ἴσον ἐστίν.

Τὰ ἄρα παραλληλόγραμμα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν· ὅπερ εἶδει δεῖχαι.

For since $ABCD$ is a parallelogram, AD is equal to BC [Prop. 1.34]. So, for the same (reasons), EF is also equal to BC . So AD is also equal to EF . And DE is common. Thus, the whole (straight-line) AE is equal to the whole (straight-line) DF . And AB is also equal to DC . So the two (straight-lines) EA , AB are equal to the two (straight-lines) FD , DC , respectively. And angle FDC is equal to angle EAB , the external to the internal [Prop. 1.29]. Thus, the base EB is equal to the base FC , and triangle EAB will be equal to triangle DFC [Prop. 1.4]. Let DGE be taken away from both. Thus, the remaining trapezium $ABGD$ is equal to the remaining trapezium $EGCF$. Let triangle GBC be added to both. Thus, the whole parallelogram $ABCD$ is equal to the whole parallelogram $EBCF$.

Thus, parallelograms which are on the same base and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

[†] Here, for the first time, “equal” means “equal in area”, rather than “congruent”.

36.

Proposition 36

Τὰ παραλληλόγραμμα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν.

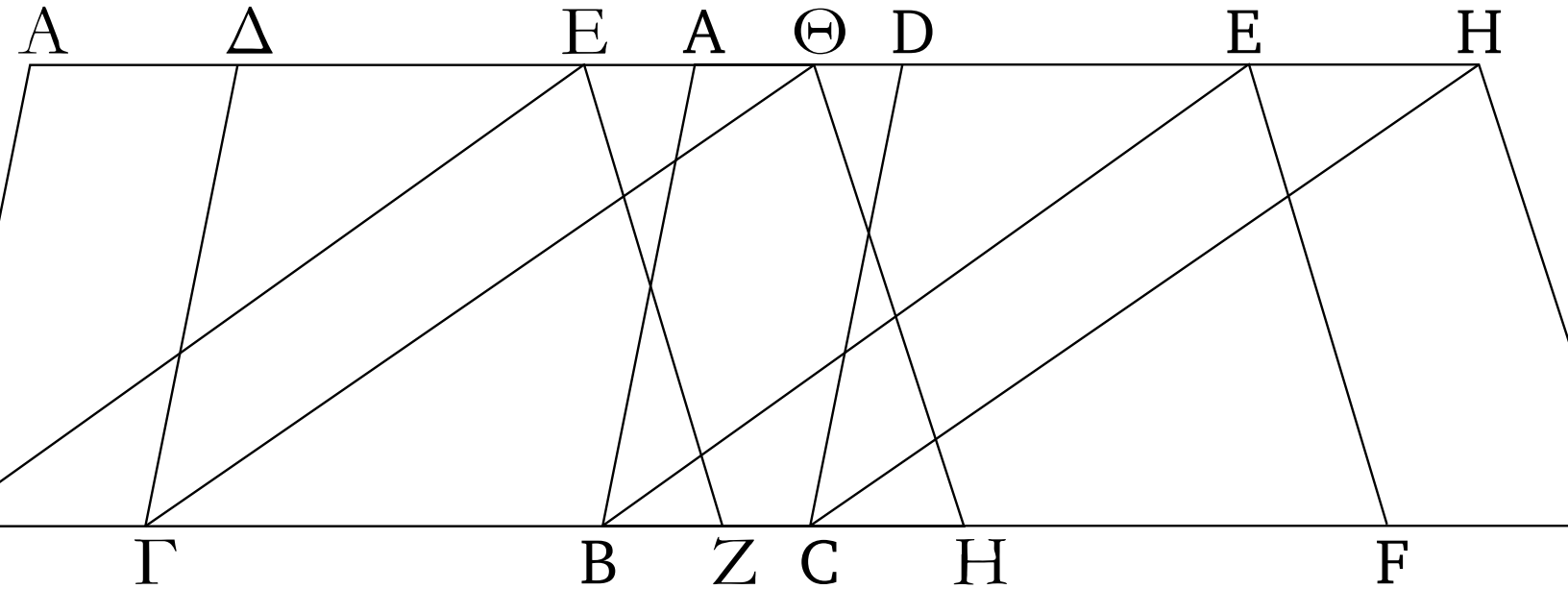
Ἐστω παραλληλόγραμμα τὰ $ABΓΔ$, $EΖΗΘ$ ἐπὶ ἴσων

Parallelograms which are on equal bases and between the same parallels are equal to one another.

Let $ABCD$ and $EFGH$ be parallelograms which are on the

βάσεων ὄντα τῶν $BΓ$, ZH καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $AΘ$, BH · λέγω, ὅτι ἴσον ἐστὶ τὸ $ABΓΔ$ παραλληλόγραμμον τῷ $EZHΘ$.

equal bases BC and FG , and (are) between the same parallels AH and BG . I say that the parallelogram $ABCD$ is equal to $EFGH$.



Ἐπεξεύθωσαν γὰρ αἱ BE , $ΓΘ$. καὶ ἐπεὶ ἴση ἐστὶν ἡ $BΓ$ τῇ ZH , ἀλλὰ ἡ ZH τῇ $EΘ$ ἐστὶν ἴση, καὶ ἡ $BΓ$ ἄρα τῇ $EΘ$ ἐστὶν ἴση. εἰσὶ δὲ καὶ παράλληλοι. καὶ ἐπιξεννύουσιν αὐτάς αἱ EB , $ΘΓ$ · αἱ δὲ τὰς ἴσας τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη ἐπιξεννύουσιν ἴσαι τε καὶ παράλληλοί εἰσι [καὶ αἱ EB , $ΘΓ$ ἄρα ἴσαι τέ εἰσι καὶ παράλληλοι]. παραλληλόγραμμον ἄρα ἐστὶ τὸ $EBΓΘ$. καὶ ἐστὶν ἴσον τῷ $ABΓΔ$ · βάσιν τε γὰρ αὐτῷ τὴν αὐτὴν ἔθει τὴν $BΓ$, καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστὶν αὐτῷ ταῖς $BΓ$, $AΘ$. διὰ τὰ αὐτὰ δὴ καὶ τὸ $EZHΘ$ τῷ αὐτῷ τῷ $EBΓΘ$ ἐστὶν ἴσον· ὥστε καὶ τὸ $ABΓΔ$ παραλληλόγραμμον τῷ $EZHΘ$ ἐστὶν ἴσον.

Τὰ ἄρα παραλληλόγραμμα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

For let BE and CH be joined. And since BC is equal to FG , but FG is equal to EH [Prop. 1.34], BC is thus equal to EH . And they are also parallel, and EB and HC join them. But (straight-lines) joining equal and parallel (straight-lines) on the same sides are (themselves) equal and parallel [Prop. 1.33] [thus, EB and HC are also equal and parallel]. Thus, $EBCH$ is a parallelogram [Prop. 1.34], and is equal to $ABCD$. For it has the same base, BC , (as $ABCD$), and is between the same parallels, BC and AH , (as $ABCD$) [Prop. 1.35]. So, for the same (reasons), $EFGH$ is also equal to the same (parallelogram) $EBCH$ [Prop. 1.34]. So that the parallelogram $ABCD$ is also equal to $EFGH$.

Thus, parallelograms which are on equal bases and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

37.

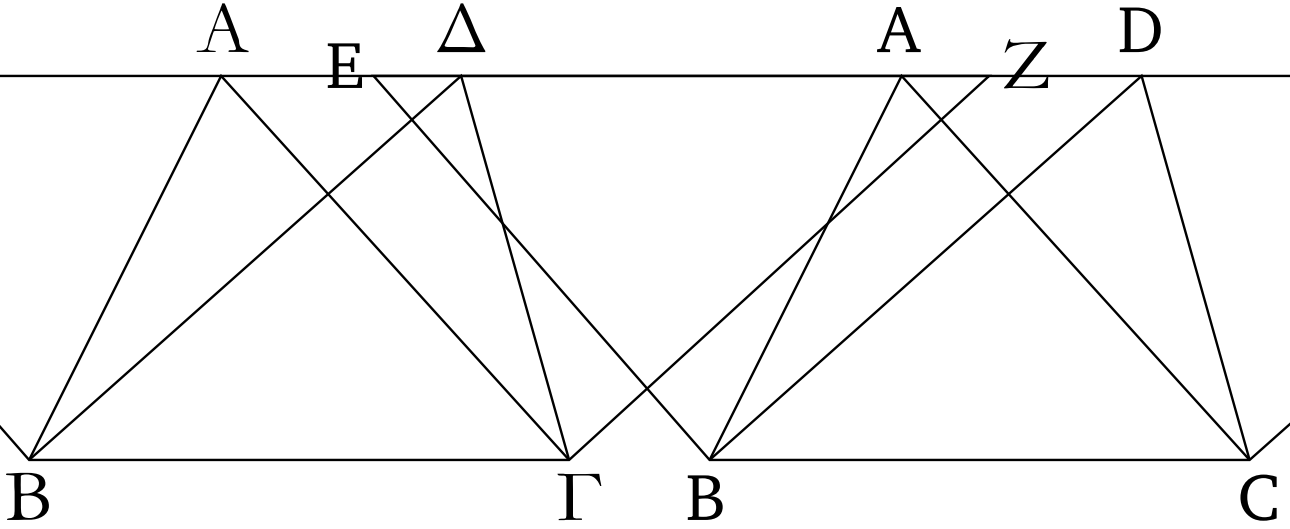
Proposition 37

Τὰ τρίγωνα τὰ ἐπὶ τῇ αὐτῇ βάσει ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν.

Ἐστω τρίγωνα τὰ $ABΓ$, $ΔBΓ$ ἐπὶ τῇ αὐτῇ βάσει τῇ $BΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς AD , $BΓ$ · λέγω, ὅτι ἴσον ἐστὶ τὸ $ABΓ$ τρίγωνον τῷ $ΔBΓ$ τριγώνῳ.

Triangles which are on the same base and between the same parallels are equal to one another.

Let ABC and DBC be triangles on the same base BC , and between the same parallels AD and BC . I say that triangle ABC is equal to triangle DBC .



Ἐκβεβλήσθω ἡ $ΑΔ$ ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ $Ε, Ζ$, καὶ διὰ μὲν τοῦ $Β$ τῇ $ΓΑ$ παράλληλος ἡθθω ἡ $ΒΕ$, διὰ δὲ τοῦ $Γ$ τῇ $ΒΔ$ παράλληλος ἡθθω ἡ $ΓΖ$. παραλληλόγραμμον ἄρα ἐστὶν ἑκάτερον τῶν $ΕΒΓΑ$, $ΔΒΓΖ$ · καὶ εἰσιν ἴσα· ἐπὶ τε γὰρ τῆς αὐτῆς βάσεώς εἰσι τῆς $ΒΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $ΒΓ$, $ΕΖ$ · καὶ ἐστὶ τοῦ μὲν $ΕΒΓΑ$ παραλληλογράμμου ἡμισὺν τὸ $ΑΒΓ$ τρίγωνον· ἡ γὰρ $ΑΒ$ διάμετρος αὐτὸ διίθα τέμνει· τοῦ δὲ $ΔΒΓΖ$ παραλληλογράμμου ἡμισὺν τὸ $ΔΒΓ$ τρίγωνον· ἡ γὰρ $ΔΓ$ διάμετρος αὐτὸ διίθα τέμνει. [τὰ δὲ τῶν ἴσων ἡμίση ἴσα ἀλλήλοις ἐστίν]. ἴσον ἄρα ἐστὶ τὸ $ΑΒΓ$ τρίγωνον τῷ $ΔΒΓ$ τριγώνῳ.

Τὰ ἄρα τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

Let AD be produced both directions to E and F , and let the (straight-line) BE be drawn through B parallel to CA [Prop. 1.31], and let the (straight-line) CF be drawn through C parallel to BD [Prop. 1.31]. Thus, $EBCA$ and $DBCF$ are both parallelograms, and are equal. For they are on the same base BC , and between the same parallels BC and EF [Prop. 1.35]. And the triangle ABC is half of the parallelogram $EBCA$. For the diagonal AB cuts the latter in half [Prop. 1.34]. And the triangle DBC (is) half of the parallelogram $DBCF$. For the diagonal DC cuts the latter in half [Prop. 1.34]. [And the halves of equal things are equal to one another.][†] Thus, triangle ABC is equal to triangle DBC .

Thus, triangles which are on the same base and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

[†] This is an additional common notion.

38.

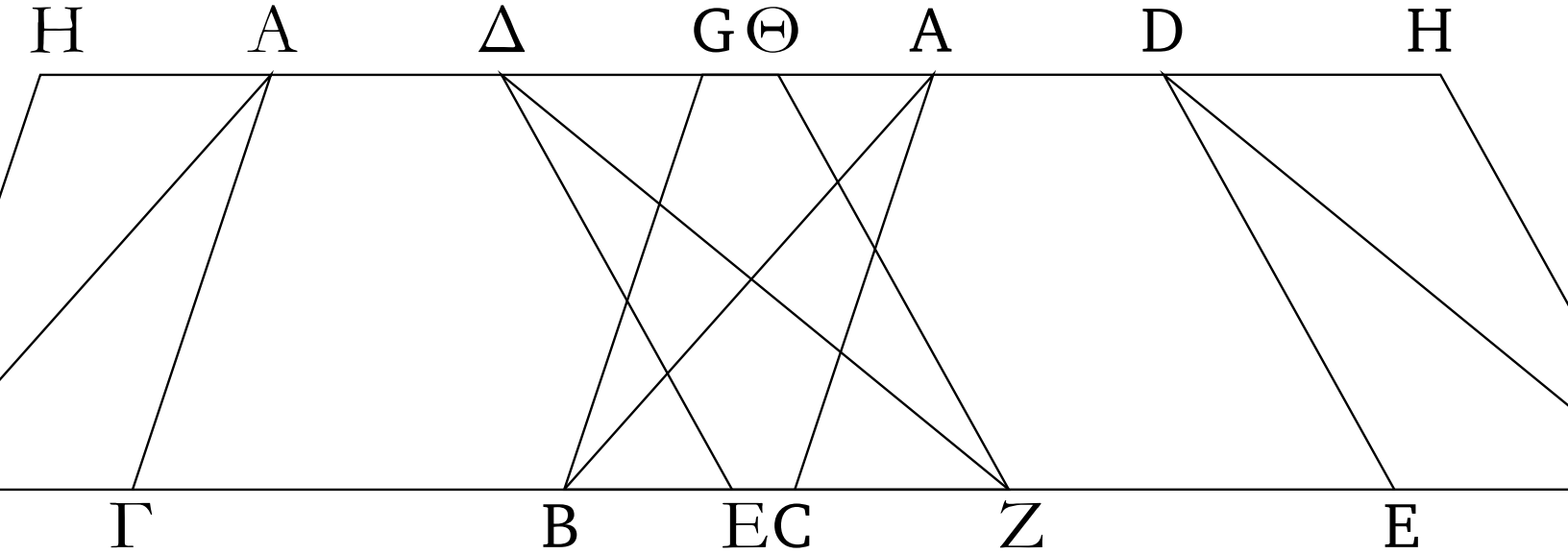
Τὰ τρίγωνα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν.

Ἐστω τρίγωνα τὰ $ΑΒΓ$, $ΔΕΖ$ ἐπὶ ἴσων βάσεων τῶν $ΒΓ$, $ΕΖ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $ΒΖ$, $ΑΔ$ · λέγω, ὅτι ἴσον ἐστὶ τὸ $ΑΒΓ$ τρίγωνον τῷ $ΔΕΖ$ τριγώνῳ.

Proposition 38

Triangles which are on equal bases and between the same parallels are equal to one another.

Let ABC and DEF be triangles on the equal bases BC and EF , and between the same parallels BF and AD . I say that triangle ABC is equal to triangle DEF .



Ἐκβεβλήσθω γὰρ ἡ AD ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ H , Θ , καὶ διὰ μὲν τοῦ B τῇ GA παράλληλος ἡθθω ἡ BH , διὰ δὲ τοῦ Z τῇ DE παράλληλος ἡθθω ἡ $Z\Theta$. παραλληλόγραμμον ἄρα ἐστὶν ἑκάτερον τῶν $HBGA$, $\Delta EZ\Theta$ · καὶ ἴσον τὸ $HBGA$ τῷ $\Delta EZ\Theta$ · ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν BG , EZ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς BZ , $H\Theta$ · καὶ ἐστὶ τοῦ μὲν $HBGA$ παραλληλογράμμου ἡμισὺν τὸ ABG τρίγωνον. ἡ γὰρ AB διάμετρος αὐτὸ διῖτα τέμνει· τοῦ δὲ $\Delta EZ\Theta$ παραλληλογράμμου ἡμισὺν τὸ $ZE\Delta$ τρίγωνον· ἡ γὰρ ΔZ διάμετρος αὐτὸ διῖτα τέμνει [τὰ δὲ τῶν ἴσων ἡμίση ἴσα ἀλλήλοις ἐστίν]. ἴσον ἄρα ἐστὶ τὸ ABG τρίγωνον τῷ ΔEZ τριγώνῳ.

Τὰ ἄρα τρίγωνα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖχαι.

For let AD be produced in both directions to G and H , and let the (straight-line) BG be drawn through B parallel to CA [Prop. 1.31], and let the (straight-line) FH be drawn through F parallel to DE [Prop. 1.31]. Thus, $GBCA$ and $DEFH$ are each parallelograms. And $GBCA$ is equal to $DEFH$. For they are on the equal bases BC and EF , and between the same parallels BF and GH [Prop. 1.36]. And triangle ABC is half of the parallelogram $GBCA$. For the diagonal AB cuts the latter in half [Prop. 1.34]. And triangle FED (is) half of parallelogram $DEFH$. For the diagonal DF cuts the latter in half. [And the halves of equal things are equal to one another.] Thus, triangle ABC is equal to triangle DEF .

Thus, triangles which are on equal bases and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

39.

Proposition 39

Τὰ ἴσα τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Ἔστω ἴσα τρίγωνα τὰ ABG , ΔBG ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη τῆς BG · λέγω, ὅτι καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Ἐπεξεύθθω γὰρ ἡ AD · λέγω, ὅτι παράλληλός ἐστιν ἡ AD τῇ BG .

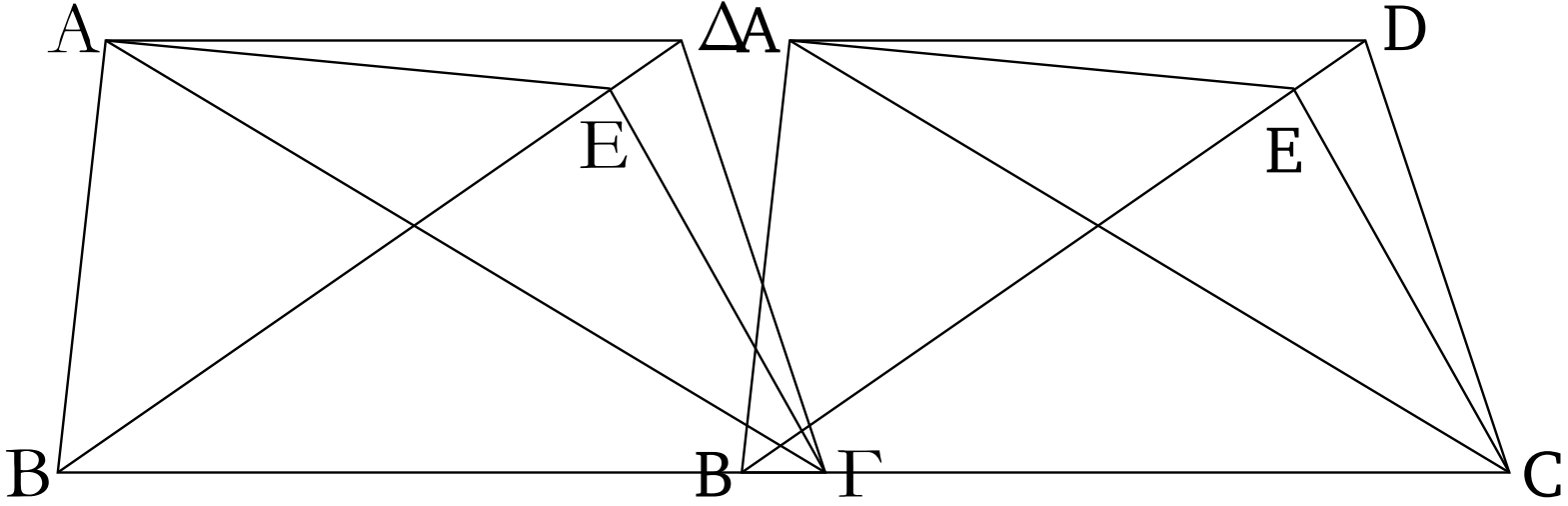
Εἰ γὰρ μή, ἡθθω διὰ τοῦ A σημείου τῇ BG εὐθείᾳ παράλληλος ἡ AE , καὶ ἐπεξεύθθω ἡ EF . ἴσον ἄρα ἐστὶ τὸ ABG τρίγωνον τῷ EBG τριγώνῳ· ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως ἐστὶν αὐτῷ τῇ BG καὶ ἐν ταῖς αὐταῖς παραλλήλοις. ἀλλὰ τὸ ABG τῷ ΔBG ἐστὶν ἴσον· καὶ τὸ ΔBG ἄρα τῷ EBG ἴσον ἐστὶ τὸ μείζον τῷ ἐλάσσονι· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα παράλληλός ἐστιν ἡ AE τῇ BG . ὁμοίως δὲ δείχμεν, ὅτι οὐδ' ἄλλη τις πλὴν τῆς AD · ἡ AD ἄρα τῇ BG ἐστὶ παράλληλος.

Equal triangles which are on the same base, and on the same side, are also between the same parallels.

Let ABC and DBC be equal triangles which are on the same base BC , and on the same side (of it). I say that they are also between the same parallels.

For let AD be joined. I say that AD and BC are parallel.

For, if not, let AE be drawn through point A parallel to the straight-line BC [Prop. 1.31], and let EC be joined. Thus, triangle ABC is equal to triangle EBC . For it is on the same base as it, BC , and between the same parallels [Prop. 1.37]. But ABC is equal to DBC . Thus, DBC is also equal to EBC , the greater to the lesser. The very thing is impossible. Thus, AE is not parallel to BC . Similarly, we can show that neither (is) any other (straight-line) than AD . Thus, AD is parallel to BC .



Τὰ ἄρα ἴσα τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

Thus, equal triangles which are on the same base, and on the same side, are also between the same parallels. (Which is) the very thing it was required to show.

40.

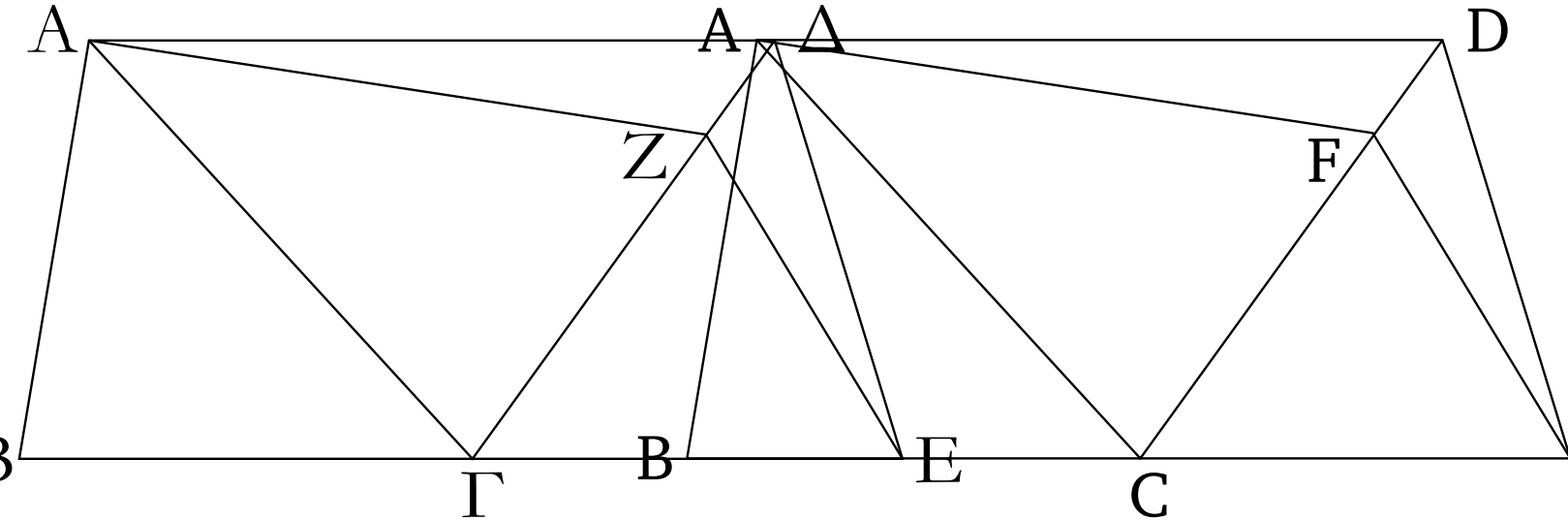
Proposition 40[†]

Τὰ ἴσα τρίγωνα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Equal triangles which are on equal bases, and on the same side, are also between the same parallels.

Ἐστω ἴσα τρίγωνα τὰ $AB\Gamma$, $\Gamma\Delta E$ ἐπὶ ἴσων βάσεων τῶν $B\Gamma$, ΓE καὶ ἐπὶ τὰ αὐτὰ μέρη. λέγω, ὅτι καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Let ABC and CDE be equal triangles on the equal bases BC and CE (respectively), and on the same side (of BE). I say that they are also between the same parallels.



Ἐπεξεύθθω γὰρ ἡ $A\Delta$. λέγω, ὅτι παραλλήλῳς ἐστὶν ἡ $A\Delta$ τῇ BE .

For let AD be joined. I say that AD is parallel to BE .

Εἰ γὰρ μή, ἤθθω διὰ τοῦ A τῇ BE παράλληλῳς ἡ AZ , καὶ ἐπεξεύθθω ἡ ZE . ἴσον ἄρα ἐστὶ τὸ $AB\Gamma$ τριγώνον τῷ $Z\Gamma E$ τριγώνῳ· ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν $B\Gamma$, ΓE καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς BE , AZ . ἀλλὰ τὸ $AB\Gamma$ τριγώνον ἴσον ἐστὶ τῷ $\Delta\Gamma E$ [τριγώνῳ]· καὶ τὸ $\Delta\Gamma E$ ἄρα

For if not, let AF be drawn through A parallel to BE [Prop. 1.31], and let FE be joined. Thus, triangle ABC is equal to triangle FCE . For they are on equal bases, BC and CE , and between the same parallels, BE and AF [Prop. 1.38]. But, triangle ABC is equal to [triangle] DCE . Thus, [triangle] DCE is also equal to triangle FCE , the greater to the lesser. The very

[τριγώνων] ἴσον ἐστὶ τῷ $ZΓΕ$ τριγώνῳ τὸ μείζον τῷ ἐλάσσονι· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα παράλληλος ἡ AZ τῇ BE . ὁμοίως δὴ δείχομεν, ὅτι οὐδ' ἄλλη τις πλὴν τῆς AD · ἡ AD ἄρα τῇ BE ἐστι παράλληλος.

Τὰ ἄρα ἴσα τριγώνων τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν· ὅπερ ἔδει δεῖχαι.

thing is impossible. Thus, AF is not parallel to BE . Similarly, we can show that neither (is) any other (straight-line) than AD . Thus, AD is parallel to BE .

Thus, equal triangles which are on equal bases, and on the same side, are also between the same parallels. (Which is) the very thing it was required to show.

† This whole proposition is regarded by Heiberg as a relatively early interpolation to the original text.

41.

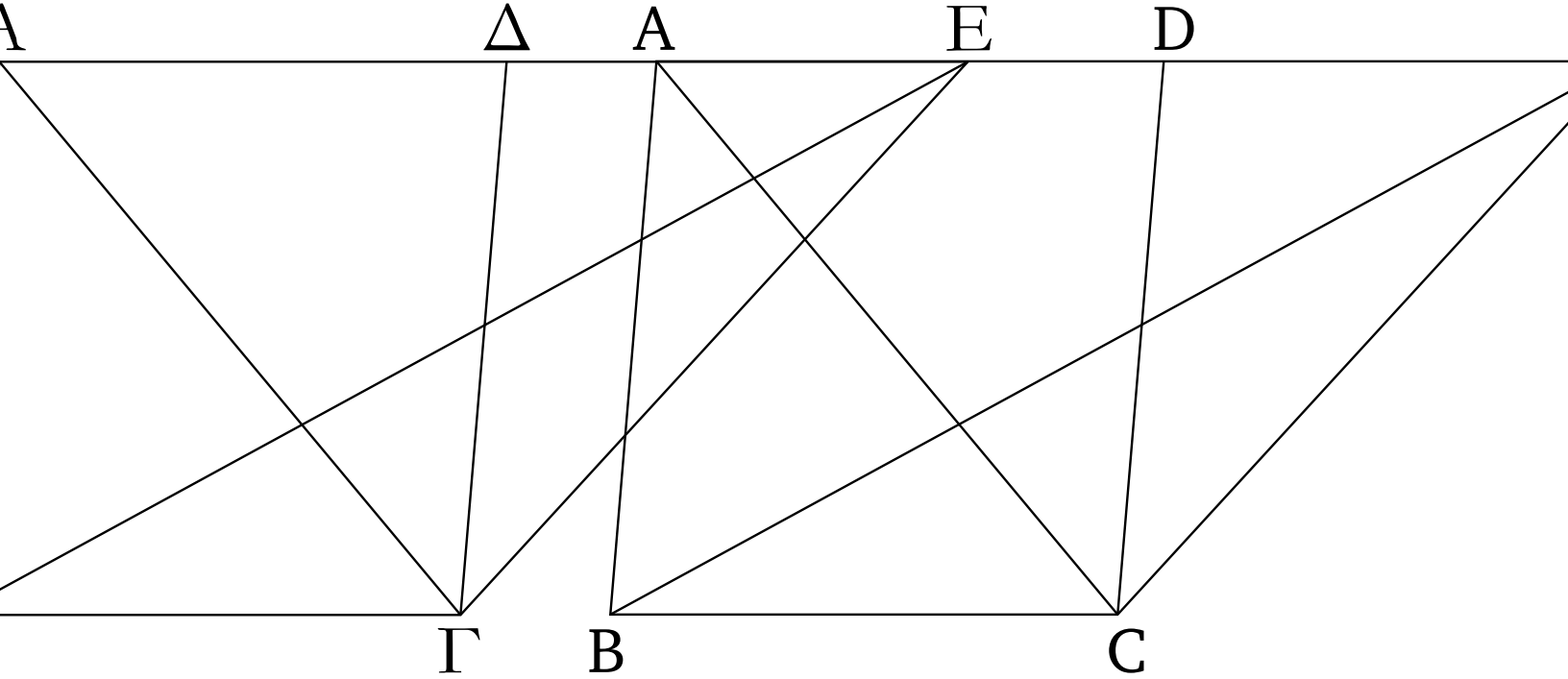
Ἐάν παραλληλόγραμμον τριγώνῳ βάσιν τε ἔσθῃ τὴν αὐτὴν καὶ ἐν ταῖς αὐταῖς παραλλήλοις ᾗ, διπλάσιόν ἐστὶ τὸ παραλληλόγραμμον τοῦ τριγώνου.

Παραλληλόγραμμον γάρ τὸ $ABΓΔ$ τριγώνῳ τῷ $EBΓ$ βάσιν τε ἔσθῃ τὴν αὐτὴν τὴν $ΒΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἔστω ταῖς $ΒΓ$, AE · λέγω, ὅτι διπλάσιόν ἐστι τὸ $ABΓΔ$ παραλληλόγραμμον τοῦ $EBΓ$ τριγώνου.

Proposition 41

If a parallelogram has the same base as a triangle, and is between the same parallels, (then) the parallelogram is double (the area) of the triangle.

For let parallelogram $ABCD$ have the same base BC as triangle EBC , and let it be between the same parallels, BC and AE . I say that parallelogram $ABCD$ is double (the area) of triangle BEC .



Ἐπεξεύσθω γάρ ἡ $ΑΓ$. ἴσον δὴ ἐστὶ τὸ $ABΓ$ τριγώνον τῷ $EBΓ$ τριγώνῳ· ἐπὶ τε γάρ τῆς αὐτῆς βάσεώς ἐστιν αὐτῷ τῆς $ΒΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $ΒΓ$, AE . ἀλλὰ τὸ $ABΓΔ$ παραλληλόγραμμον διπλάσιόν ἐστι τοῦ $ABΓ$ τριγώνου· ἡ γάρ $ΑΓ$ διάμετρος αὐτὸ διῖθα τέμνει· ὥστε τὸ $ABΓΔ$ παραλληλόγραμμον καὶ τοῦ $EBΓ$ τριγώνου ἐστὶ διπλάσιον.

Ἐάν ἄρα παραλληλόγραμμον τριγώνῳ βάσιν τε ἔσθῃ τὴν αὐτὴν καὶ ἐν ταῖς αὐταῖς παραλλήλοις ᾗ, διπλάσιόν ἐστὶ τὸ παραλληλόγραμμον τοῦ τριγώνου· ὅπερ ἔδει δεῖχαι.

For let AC be joined. So triangle ABC is equal to triangle EBC . For it is on the same base, BC , (as EBC), and between the same parallels, BC and AE [Prop. 1.37]. But, parallelogram $ABCD$ is double (the area) of triangle ABC . For the diagonal AC cuts the former in half [Prop. 1.34]. So parallelogram $ABCD$ is also double (the area) of triangle EBC .

Thus, if a parallelogram has the same base as a triangle, and is between the same parallels, (then) the parallelogram is double (the area) of the triangle. (Which is) the very thing it

42.

Τῷ δοθέντι τριγώνῳ ἴσον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ.

Ἐστω τὸ μὲν δοθέν τρίγωνον τὸ $ABΓ$, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμοξ ἡ Δ . δεῖ δὴ τῷ $ABΓ$ τριγώνῳ ἴσον παραλληλόγραμμον συστήσασθαι ἐν τῇ Δ γωνίᾳ εὐθυγράμμῳ.

Τετμήσθω ἡ $BΓ$ δίῃα κατὰ τὸ E , καὶ ἐπεζεύθθω ἡ AE , καὶ συνεστάτω πρὸς τῇ EF εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ E τῇ Δ γωνίᾳ ἴση ἡ ὑπὸ $ΓΕΖ$, καὶ διὰ μὲν τοῦ A τῇ EF παράλληλοξ ἤθθω ἡ AH , διὰ δὲ τοῦ $Γ$ τῇ EZ παράλληλοξ ἤθθω ἡ $ΓH$. παραλληλόγραμμον ἄρα ἐστὶ τὸ $ZΕΓH$. καὶ ἐπεὶ ἴση ἐστὶν ἡ BE τῇ EF , ἴσον ἐστὶ καὶ τὸ ABE τρίγωνον τῷ $AΕΓ$ τριγώνῳ. ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν BE , EF καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $BΓ$, AH . διπλάσιον ἄρα ἐστὶ τὸ $ABΓ$ τρίγωνον τοῦ $AΕΓ$ τριγώνου. ἐστὶ δὲ καὶ τὸ $ZΕΓH$ παραλληλόγραμμον διπλάσιον τοῦ $AΕΓ$ τριγώνου. βάσιν τε γὰρ αὐτῷ τὴν αὐτὴν ἔχει καὶ ἐν ταῖς αὐταῖς ἐστὶν αὐτῷ παραλλήλοις. ἴσον ἄρα ἐστὶ τὸ $ZΕΓH$ παραλληλόγραμμον τῷ $ABΓ$ τριγώνῳ. καὶ ἔθει τὴν ὑπὸ $ΓΕΖ$ γωνίαν ἴσην τῇ δοθείσῃ τῇ Δ .

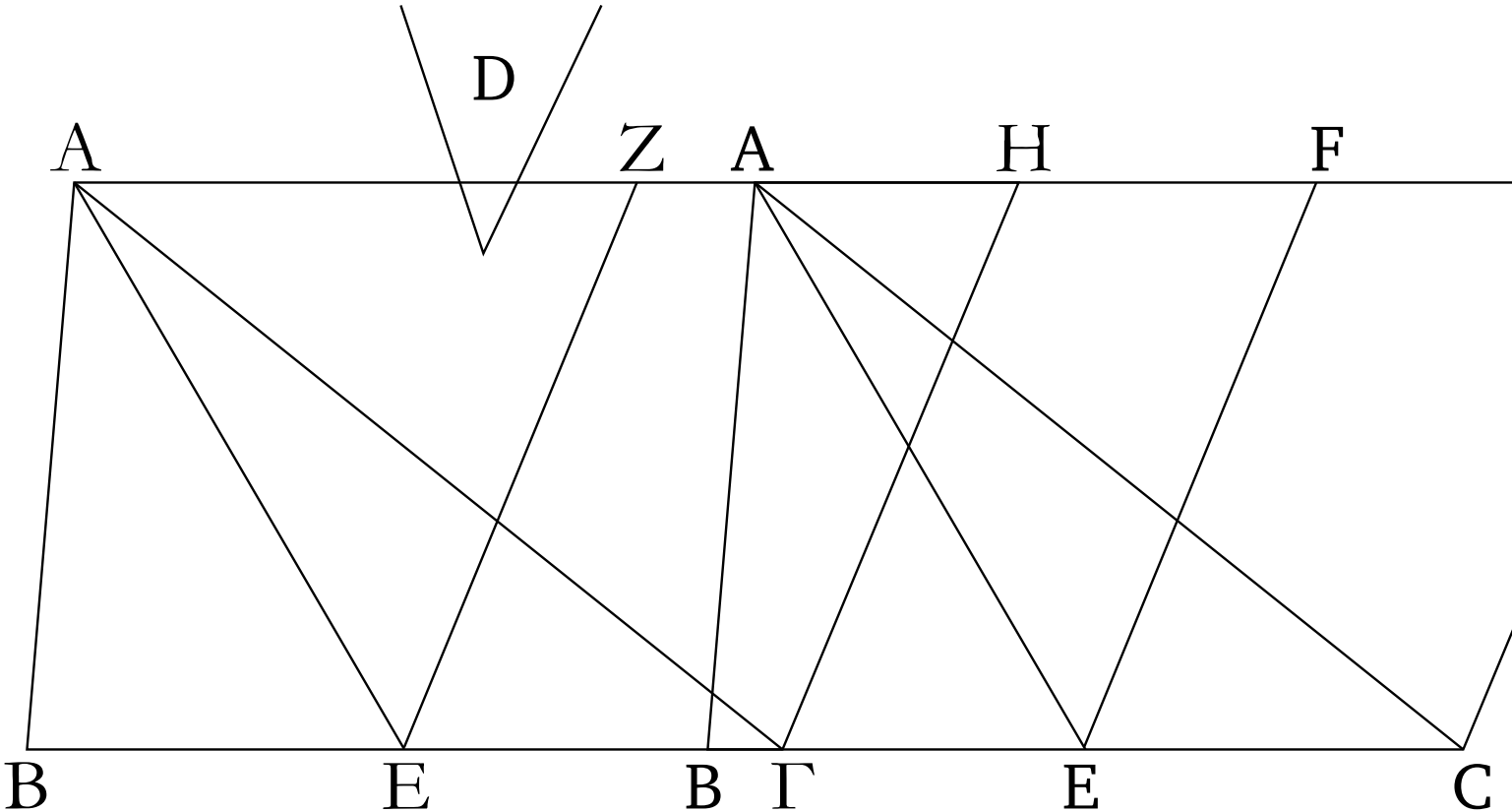
was required to show.

Proposition 42

To construct a parallelogram equal to a given triangle in a given rectilinear angle.

Let ABC be the given triangle, and D the given rectilinear angle. So it is required to construct a parallelogram equal to triangle ABC in the rectilinear angle D .

Let BC be cut in half at E [Prop. 1.10], and let AE be joined. And let (angle) CEF , equal to angle D , be constructed at the point E on the straight-line EC [Prop. 1.23]. And let AG be drawn through A parallel to EC [Prop. 1.31], and let CG be drawn through C parallel to EF [Prop. 1.31]. Thus, $FECG$ is a parallelogram. And since BE is equal to EC , triangle ABE is also equal to triangle AEC . For they are on the equal bases, BE and EC , and between the same parallels, BC and AG [Prop. 1.38]. Thus, triangle ABC is double (the area) of triangle AEC . And parallelogram $FECG$ is also double (the area) of triangle AEC . For it has the same base (as AEC), and is between the same parallels (as AEC) [Prop. 1.41]. Thus, parallelogram $FECG$ is equal to triangle ABC . ($FECG$) also has the angle CEF equal to the given (angle) D .



Τῷ ἄρα δοθέντι τριγώνῳ τῷ $ABΓ$ ἴσον παραλληλόγραμμον συνέσταιται τὸ $ZΕΓH$ ἐν γωνίᾳ τῇ ὑπὸ $ΓΕΖ$, ἥτις ἐστὶν

Thus, parallelogram $FECG$, equal to the given triangle ABC , has been constructed in the angle CEF , which is equal

ἴση τῇ Δ · ὅπερ ἔδει ποιῆσαι.

to D . (Which is) the very thing it was required to do.

43.

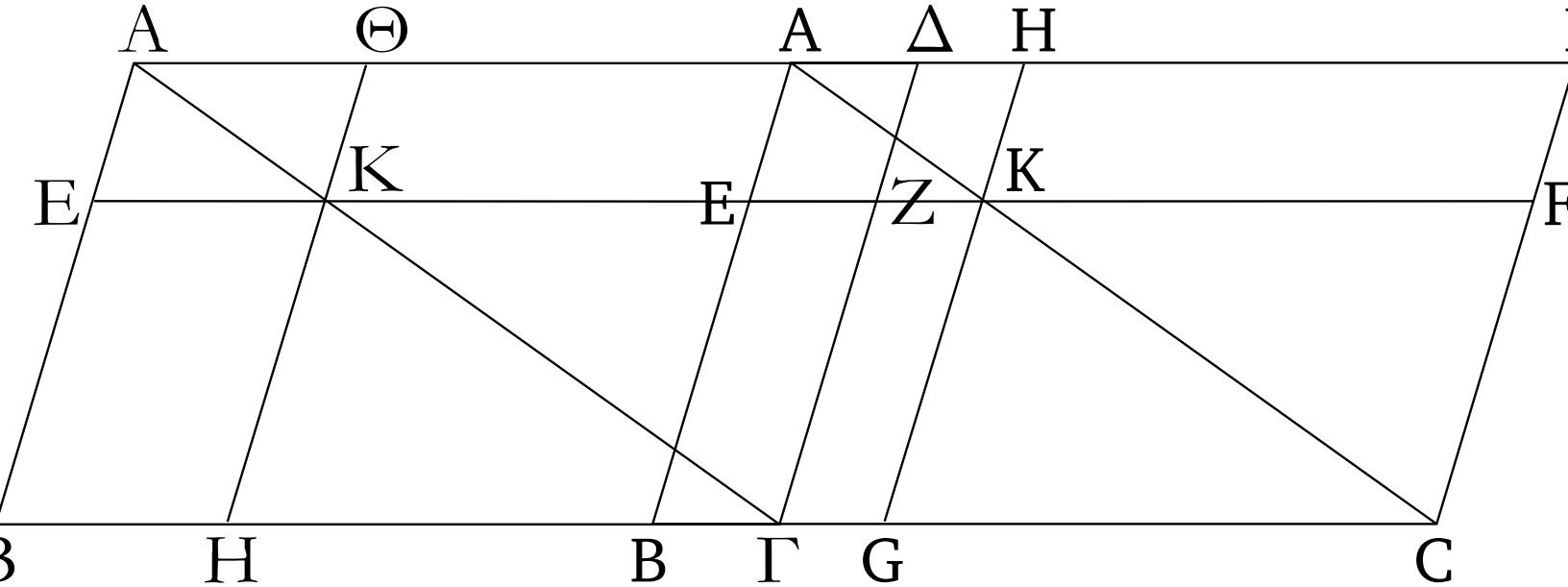
Proposition 43

Παντὸς παραλληλογράμμου τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα ἴσα ἀλλήλοις ἐστίν.

For any parallelogram, the complements of the parallelograms about the diagonal are equal to one another.

Ἐστω παραλληλόγραμμον τὸ $AB\Gamma\Delta$, διάμετρος δὲ αὐτοῦ ἡ AG , περὶ δὲ τὴν AG παραλληλόγραμμα μὲν ἔστω τὰ $E\Theta$, ZH , τὰ δὲ λεγόμενα παραπληρώματα τὰ BK , $K\Delta$ · λέγω, ὅτι ἴσον ἐστὶ τὸ BK παραπλήρωμα τῷ $K\Delta$ παραπληρώματι.

Let $ABCD$ be a parallelogram, and AC its diagonal. And let EH and FG be the parallelograms about AC , and BK and KD the so-called complements (about AC). I say that the complement BK is equal to the complement KD .



Ἐπεὶ γὰρ παραλληλόγραμμον ἐστὶ τὸ $AB\Gamma\Delta$, διάμετρος δὲ αὐτοῦ ἡ AG , ἴσον ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ $AG\Delta$ τριγώνῳ. πάλιν, ἐπεὶ παραλληλόγραμμον ἐστὶ τὸ $E\Theta$, διάμετρος δὲ αὐτοῦ ἐστὶν ἡ AK , ἴσον ἐστὶ τὸ AEK τρίγωνον τῷ $A\Theta K$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $KZ\Gamma$ τρίγωνον τῷ $K\Theta H$ ἐστὶν ἴσον. ἐπεὶ οὖν τὸ μὲν AEK τρίγωνον τῷ $A\Theta K$ τριγώνῳ ἐστὶν ἴσον, τὸ δὲ $KZ\Gamma$ τῷ $K\Theta H$, τὸ AEK τρίγωνον μετὰ τοῦ $K\Theta H$ ἴσον ἐστὶ τῷ $A\Theta K$ τριγώνῳ μετὰ τοῦ $KZ\Gamma$ · ἐστὶ δὲ καὶ ὅλον τὸ $AB\Gamma$ τρίγωνον ὅλῳ τῷ $AG\Delta$ ἴσον· λοιπὸν ἄρα τὸ BK παραπλήρωμα λοιπῷ τῷ $K\Delta$ παραπληρώματι ἐστὶν ἴσον.

For since $ABCD$ is a parallelogram, and AC its diagonal, triangle ABC is equal to triangle ACD [Prop. 1.34]. Again, since EH is a parallelogram, and AK is its diagonal, triangle AEK is equal to triangle AHK [Prop. 1.34]. So, for the same (reasons), triangle KFC is also equal to (triangle) KGC . Therefore, since triangle AEK is equal to triangle AHK , and KFC to KGC , triangle AEK plus KGC is equal to triangle AHK plus KFC . And the whole triangle ABC is also equal to the whole (triangle) ADC . Thus, the remaining complement BK is equal to the remaining complement KD .

Παντὸς ἄρα παραλληλογράμμου θεωρίον τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

Thus, for any parallelogramic figure, the complements of the parallelograms about the diagonal are equal to one another. (Which is) the very thing it was required to show.

44.

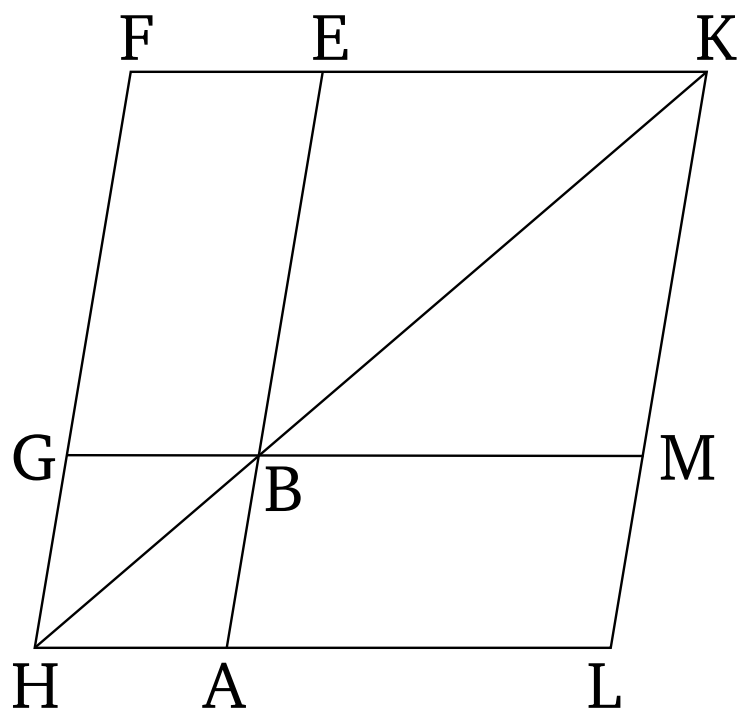
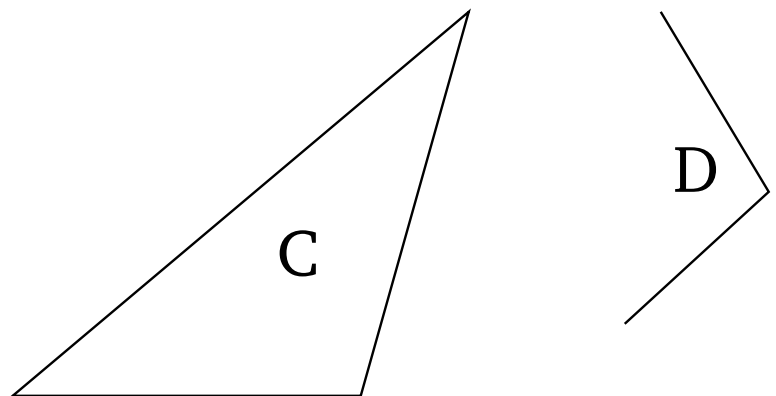
Proposition 44

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι τριγώνῳ ἴσον παραλληλόγραμμον παραβαλεῖν ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμω.

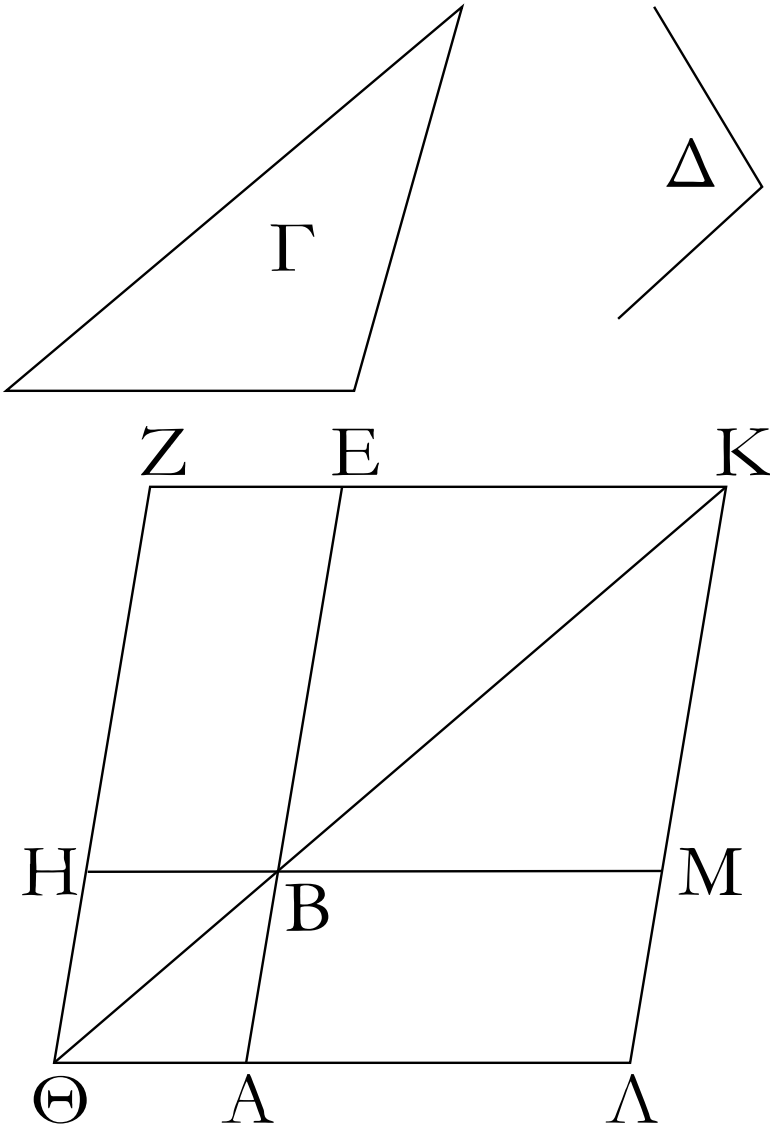
To apply a parallelogram equal to a given triangle to a given straight-line in a given rectilinear angle.

Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθέν τρίγωνον τὸ Γ , ἡ δὲ δοθεῖσα γωνία εὐθύγραμμοξ ἡ Δ · δεῖ δὴ παρὰ τὴν δοθεῖσαν εὐθεῖαν τὴν AB τῷ δοθέντι τριγώνῳ τῷ Γ ἴσον

Let AB be the given straight-line, C the given triangle, and D the given rectilinear angle. So it is required to apply a parallelogram equal to the given triangle C to the given straight-line AB in an angle equal to (angle) D .



παράλληλογράμμον παραβαλεῖν ἐν ἴσῃ τῇ Δ γωνίᾳ.



Συνεστάτω τῷ Γ τριγώνῳ ἴσον παραλληλόγραμμον τὸ $BEZH$ ἐν γωνίᾳ τῇ ὑπὸ EBH , ἥ ἐστίν ἴση τῇ Δ · καὶ κείσθω ὥστε ἐπ' εὐθείᾳ εἶναι τὴν BE τῇ AB , καὶ διήθθω ἡ ZH ἐπὶ τὸ Θ , καὶ διὰ τοῦ A ὁποτέρῳ τῶν BH , EZ παράλληλοξ ἡθθω ἡ $A\Theta$, καὶ ἐπεξεύθθω ἡ ΘB . καὶ ἐπεὶ εἰς παραλλήλων τὰς $A\Theta$, EZ εὐθεῖα ἐνέπεσεν ἡ ΘZ , αἱ ἄρα ὑπὸ $A\Theta Z$, ΘZE γωνίαι δυσὶν ὀρθαῖς εἰσιν ἴσαι. αἱ ἄρα ὑπὸ $B\Theta H$, HZE δύο ὀρθῶν ἐλάσσονές εἰσιν· αἱ δὲ ἀπὸ ἐλασσόνων ἢ δύο ὀρθῶν εἰς ἄπειρον ἐκβαλλόμεναι συμπίπτουσιν· αἱ ΘB , ZE ἄρα ἐκβαλλόμεναι συμπεσοῦνται. ἐκβεβλήσθωσαν καὶ συμμιπτέωσαν κατὰ τὸ K , καὶ διὰ τοῦ K σημείου ὁποτέρῳ τῶν EA , $Z\Theta$ παράλληλοξ ἡθθω ἡ KL , καὶ ἐκβεβλήσθωσαν αἱ ΘA , HB ἐπὶ τὰ Λ , M σημεία. παραλληλόγραμμον ἄρα ἐστὶ τὸ ΘAKZ , διάμετροξ δὲ αὐτοῦ ἡ ΘK , περὶ δὲ τὴν ΘK παραλληλόγραμμά μὲν τὰ AH , ME , τὰ δὲ λεγόμενα παραπληρώματα τὰ AB ,

Let the parallelogram $BEFG$, equal to the triangle C , be constructed in the angle EBG , which is equal to D [Prop. 1.42]. And let it be placed so that BE is straight-on to AB .[†] And let FG be drawn through to H , and let AH be drawn through A parallel to either of BG or EF [Prop. 1.31], and let HB be joined. And since the straight-line HF falls across the parallels AH and EF , the (sum of the) angles AHF and HFE is thus equal to two right-angles [Prop. 1.29]. Thus, (the sum of) BHG and GFE is less than two right-angles. And (straight-lines) produced to infinity from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, being produced, HB and FE will meet together. Let them be produced, and let them meet together at K . And let KL be drawn through point K parallel to either of EA or FH [Prop. 1.31]. And let HA and GB be produced to points L and M (respectively). Thus, $HLKF$ is a parallelogram, and HK its diagon-

BZ ἴσον ἄρα ἐστὶ τὸ AB τῷ BZ . ἀλλὰ τὸ BZ τῷ Γ τριγώνῳ ἐστὶν ἴσον· καὶ τὸ AB ἄρα τῷ Γ ἐστὶν ἴσον. καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ HBE γωνία τῇ ὑπὸ ABM , ἀλλὰ ἡ ὑπὸ HBE τῇ Δ ἐστὶν ἴση, καὶ ἡ ὑπὸ ABM ἄρα τῇ Δ γωνία ἐστὶν ἴση.

Παρὰ τὴν δοθεῖσαν ἄρα εὐθεῖαν τὴν AB τῷ δοθέντι τριγώνῳ τῷ Γ ἴσον παραλληλόγραμμον παραβέβληται τὸ AB ἐν γωνίᾳ τῇ ὑπὸ ABM , ἥ ἐστὶν ἴση τῇ Δ · ὅπερ ἔδει ποιῆσαι.

nal. And AG and ME (are) parallelograms, and LB and BF the so-called complements, about HK . Thus, LB is equal to BF [Prop. 1.43]. But, BF is equal to triangle C . Thus, LB is also equal to C . Also, since angle GBE is equal to ABM [Prop. 1.15], but GBE is equal to D , ABM is thus also equal to angle D .

Thus, the parallelogram LB , equal to the given triangle C , has been applied to the given straight-line AB in the angle ABM , which is equal to D . (Which is) the very thing it was required to do.

[†] This can be achieved using Props. 1.3, 1.23, and 1.31.

45.

Τῷ δοθέντι εὐθυγράμμῳ ἴσον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ.

Ἐστω τὸ μὲν δοθέν εὐθύγραμμον τὸ $AB\Gamma\Delta$, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἡ E . δεῖ δὴ τῷ $AB\Gamma\Delta$ εὐθυγράμμῳ ἴσον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ τῇ E .

Proposition 45

To construct a parallelogram equal to a given rectilinear figure in a given rectilinear angle.

Let $ABCD$ be the given rectilinear figure,[†] and E the given rectilinear angle. So it is required to construct a parallelogram equal to the rectilinear figure $ABCD$ in the given angle E .

$\Theta\text{ΗΛ}$ ταῖς ὑπὸ $\Theta\text{ΗΖ}$, $\Theta\text{ΗΛ}$ ἴσαι εἰσιν. ἀλλ' αἱ ὑπὸ ΜΘΗ , $\Theta\text{ΗΛ}$ δύο ὀρθαῖς ἴσαι εἰσιν· καὶ αἱ ὑπὸ $\Theta\text{ΗΖ}$, $\Theta\text{ΗΛ}$ ἄρα δύο ὀρθαῖς ἴσαι εἰσιν· ἐπ' εὐθείας ἄρα ἐστὶν ἡ ΖΗ τῇ ΗΛ . καὶ ἐπεὶ ἡ ΖΚ τῇ $\Theta\text{Η}$ ἴση τε καὶ παράλληλός ἐστιν, ἀλλὰ καὶ ἡ $\Theta\text{Η}$ τῇ ΜΛ , καὶ ἡ ΚΖ ἄρα τῇ ΜΛ ἴση τε καὶ παράλληλός ἐστιν· καὶ ἐπιξενγνύουσιν αὐτάς εὐθεῖαι αἱ ΚΜ , ΖΛ · καὶ αἱ ΚΜ , ΖΛ ἄρα ἴσαι τε καὶ παράλληλοί εἰσιν· παραλληλόγραμμον ἄρα ἐστὶ τὸ ΚΖΑΜ . καὶ ἐπεὶ ἴσον ἐστὶ τὸ μὲν ΑΒΔ τριγώνων τῷ ΖΘ παραλληλογράμμῳ, τὸ δὲ ΔΒΓ τῷ ΗΜ , ὅλον ἄρα τὸ ΑΒΓΔ εὐθύγραμμον ὅλῳ τῷ ΚΖΑΜ παραλληλογράμμῳ ἐστὶν ἴσον.

Τῷ ἄρα δοθέντι εὐθυγράμμῳ τῷ ΑΒΓΔ ἴσον παραλληλόγραμμον συνέσταται τὸ ΚΖΑΜ ἐν γωνίᾳ τῇ ὑπὸ ΖΚΜ , ἣ ἐστὶν ἴση τῇ δοθείσῃ τῇ Ε · ὅπερ ἔδει ποιῆσαι.

equal to one another [Prop. 1.29]. Let HGL have been added to both. Thus, (the sum of) mhG and HGL is equal to (the sum of) HGF and HGL . But, (the sum of) mhG and HGL is equal to two right-angles [Prop. 1.29]. Thus, (the sum of) HGF and HGL is also equal to two right-angles. Thus, FG is straight-on to GL [Prop. 1.14]. And since FK is equal and parallel to HG [Prop. 1.34], but also HG to ML [Prop. 1.34], KF is thus also equal and parallel to ML [Prop. 1.30]. And the straight-lines KM and FL join them. Thus, KM and FL are equal and parallel as well [Prop. 1.33]. Thus, KFLM is a parallelogram. And since triangle ABD is equal to parallelogram FH , and DBC to GM , the whole rectilinear figure ABCD is thus equal to the whole parallelogram KFLM .

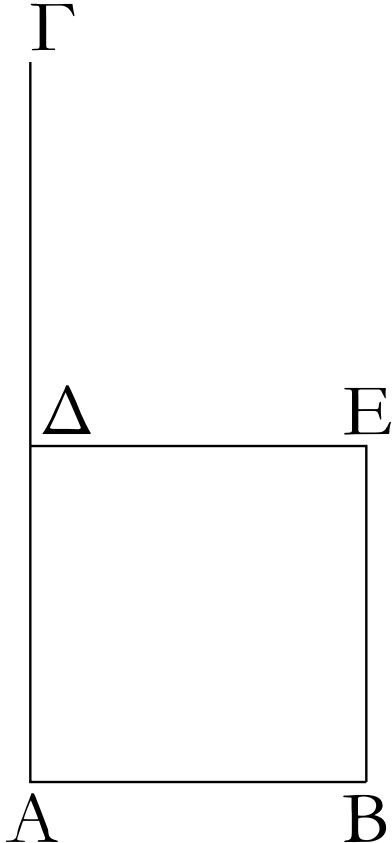
Thus, the parallelogram KFLM , equal to the given rectilinear figure ABCD , has been constructed in the angle FKM , which is equal to the given (angle) E . (Which is) the very thing it was required to do.

† The proof is only given for a four-sided figure. However, the extension to many-sided figures is trivial.

46.

Ἀπὸ τῆς δοθείσης εὐθείας τετράγωνον ἀναγράψαι.

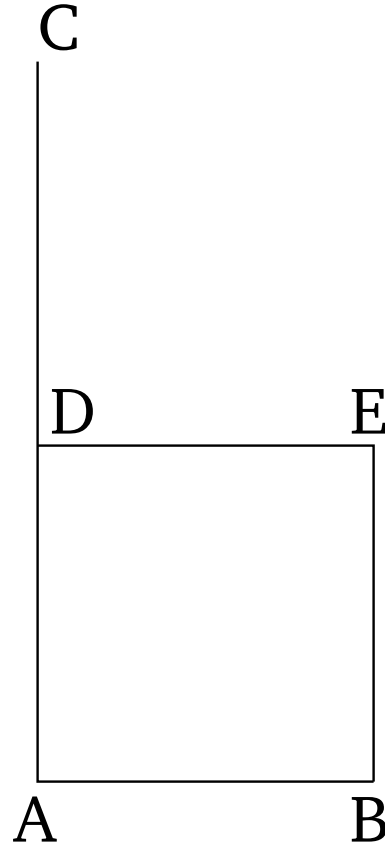
Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ ΑΒ · δεῖ δὴ ἀπὸ τῆς ΑΒ εὐθείας τετράγωνον ἀναγράψαι.



Proposition 46

To describe a square on a given straight-line.

Let AB be the given straight-line. So it is required to describe a square on the straight-line AB .



Ἦθθω τῇ AB εὐθείᾳ ἀπὸ τοῦ πρὸς αὐτῇ σημείου τοῦ A πρὸς ὀρθὰς ἡ AG , καὶ κείσθω τῇ AB ἴση ἡ AD . καὶ διὰ μὲν τοῦ Δ σημείου τῇ AB παράλληλος ἡθθω ἡ DE , διὰ δὲ τοῦ B σημείου τῇ AD παράλληλος ἡθθω ἡ BE . παραλληλόγραμμον ἄρα ἐστὶ τὸ $ADEB$. ἴση ἄρα ἐστὶν ἡ μὲν AB τῇ DE , ἡ δὲ AD τῇ BE . ἀλλὰ ἡ AB τῇ AD ἐστὶν ἴση· αἱ τέσσαρες ἄρα αἱ BA , AD , DE , EB ἴσαι ἀλλήλαις εἰσὶν· ἰσόπλευρον ἄρα ἐστὶ τὸ $ADEB$ παραλληλόγραμμον. λέγω δὴ, ὅτι καὶ ὀρθογώνιον. ἐπεὶ γὰρ εἷς παραλλήλων τάξ AB , DE εὐθεῖα ἐνέπεσεν ἡ AD , αἱ ἄρα ὑπὸ BAD , ADE γωνίαι δύο ὀρθαῖς ἴσαι εἰσὶν. ὀρθὴ δὲ ἡ ὑπὸ BAD . ὀρθὴ ἄρα καὶ ἡ ὑπὸ ADE . τῶν δὲ παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραὶ τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσὶν· ὀρθὴ ἄρα καὶ ἑκάτερα τῶν ἀπεναντίον τῶν ὑπὸ ABE , BED γωνιῶν· ὀρθογώνιον ἄρα ἐστὶ τὸ $ADEB$. ἐδείχθη δὲ καὶ ἰσόπλευρον.

Τετράγωνον ἄρα ἐστὶν καὶ ἐστὶν ἀπὸ τῆς AB εὐθείας ἀναγεγραμμένον· ὅπερ ἔδει ποιῆσαι.

47.

Ἐν τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσῃς πλευρᾶς τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν γωνίαν περιεθουσῶν πλευρῶν τετραγώνοις.

Ἔστω τρίγωνον ὀρθογώνιον τὸ ABG ὀρθὴν ἔθον τὴν ὑπὸ BAG γωνίαν· λέγω, ὅτι τὸ ἀπὸ τῆς BG τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν BA , AG τετραγώνοις.

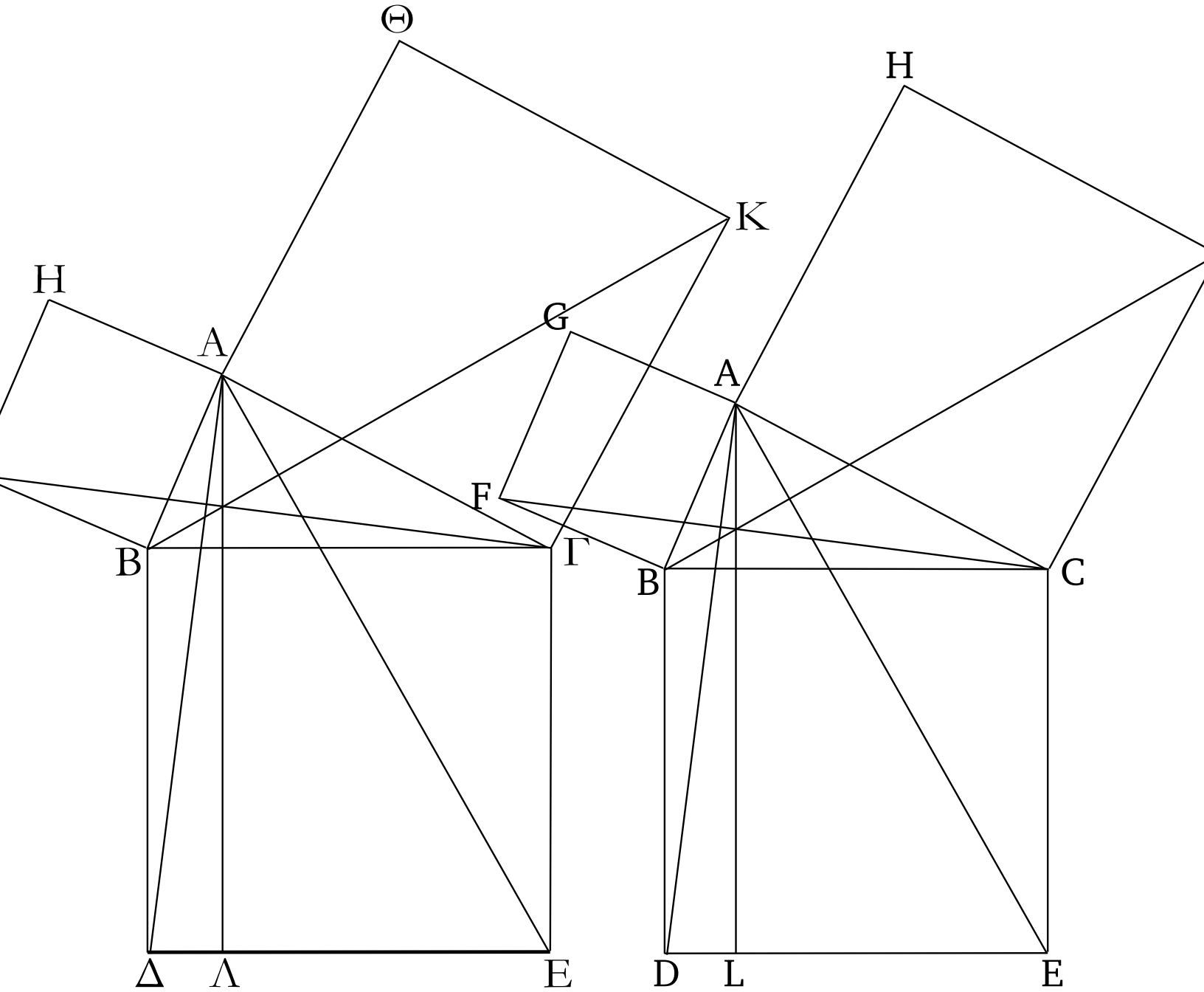
Let AC be drawn at right-angles to the straight-line AB from the point A on it [Prop. 1.11], and let AD be made equal to AB [Prop. 1.3]. And let DE be drawn through point D parallel to AB [Prop. 1.31], and let BE be drawn through point B parallel to AD [Prop. 1.31]. Thus, $ADEB$ is a parallelogram. Therefore, AB is equal to DE , and AD to BE [Prop. 1.34]. But, AB is equal to AD . Thus, the four (sides) BA , AD , DE , and EB are equal to one another. Thus, the parallelogram $ADEB$ is equilateral. So I say that (it is) also right-angled. For since the straight-line AD falls across the parallels AB and DE , the (sum of the) angles BAD and ADE is equal to two right-angles [Prop. 1.29]. But BAD (is a) right-angle. Thus, ADE (is) also a right-angle. And for parallelogramic figures, the opposite sides and angles are equal to one another [Prop. 1.34]. Thus, each of the opposite angles ABE and BED (are) also right-angles. Thus, $ADEB$ is right-angled. And it was also shown (to be) equilateral.

Thus, $(ADEB)$ is a square [Def. 1.22]. And it is described on the straight-line AB . (Which is) the very thing it was required to do.

Proposition 47

In right-angled triangles, the square on the side subtending the right-angle is equal to the (sum of the) squares on the sides containing the right-angle.

Let ABC be a right-angled triangle having the angle BAC a right-angle. I say that the square on BC is equal to the (sum of the) squares on BA and AC .



Ἀναγεγράφθω γάρ ἀπὸ μὲν τῆς $BΓ$ τετραγώνον τὸ $BΔΕΓ$, ἀπὸ δὲ τῶν BA , AC τὰ HB , $ΘΓ$, καὶ διὰ τοῦ A ὁποτέρῳ τῶν BD , $ΓΕ$ παράλληλος ἦθθω ἡ $ΑΛ$ · καὶ ἐπεξενέθθωσαν αἱ AD , $ZΓ$. καὶ ἐπεὶ ὀρθή ἐστὶν ἑκάτερα τῶν ὑπὸ $BAΓ$, BAH γωνιῶν, πρὸς δὴ τινι εὐθείᾳ τῇ BA καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A δύο εὐθεῖαι αἱ AG , AH μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δυοῖν ὀρθαῖς ἴσας ποιοῦσιν· ἐπ' εὐθείᾳ ἄρα ἐστὶν ἡ $ΓA$ τῇ AH . διὰ τὰ αὐτὰ δὴ καὶ ἡ BA τῇ $AΘ$ ἐστὶν ἐπ' εὐθείᾳ. καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ $ΔBΓ$ γωνία τῇ

For let the square $BDEC$ be described on BC , and (the squares) GB and HC on AB and AC (respectively) [Prop. 1.46]. And let AL be drawn through point A parallel to either of BD or CE [Prop. 1.31]. And let AD and FC be joined. And since angles BAC and BAG are each right-angles, then two straight-lines AC and AG , not lying on the same side, make the adjacent angles with some straight-line BA , at the point A on it, (whose sum is) equal to two right-angles. Thus, CA is straight-on to AG [Prop. 1.14]. So, for the same (rea-

ὑπὸ ZBA · ὀρθή γάρ ἑκατέρα· κοινή προσκείσθω ἡ ὑπὸ $ABΓ$ · ὅλη ἄρα ἡ ὑπὸ ΔBA ὅλη τῇ ὑπὸ $ZBΓ$ ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν ΔB τῇ $BΓ$, ἡ δὲ ZB τῇ BA , δύο δὲ αἱ ΔB , BA δύο ταῖς ZB , $BΓ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρα· καὶ γωνία ἡ ὑπὸ ΔBA γωνία τῇ ὑπὸ $ZBΓ$ ἴση· βάσεις ἄρα ἡ AD βάσει τῇ $ZΓ$ [ἐστὶν] ἴση, καὶ τὸ $AB\Delta$ τρίγωνον τῷ $ZBΓ$ τριγώνῳ ἐστὶν ἴσον· καὶ [ἐστὶ] τοῦ μὲν $AB\Delta$ τριγώνου διπλάσιον τὸ BA παραλληλόγραμμον· βάσιν τε γὰρ τὴν αὐτὴν ἔθουσι τὴν $B\Delta$ καὶ ἐν ταῖς αὐταῖς εἰσι παραλλήλοις ταῖς $B\Delta$, AL · τοῦ δὲ $ZBΓ$ τριγώνου διπλάσιον τὸ HB τετράγωνον· βάσιν τε γὰρ πάλιν τὴν αὐτὴν ἔθουσι τὴν ZB καὶ ἐν ταῖς αὐταῖς εἰσι παραλλήλοις ταῖς ZB , $HΓ$. [τὰ δὲ τῶν ἴσων διπλάσια ἴσα ἀλλήλοις ἐστὶν] ἴσον ἄρα ἐστὶ καὶ τὸ BA παραλληλόγραμμον τῷ HB τετραγώνῳ. ὁμοίως δὲ ἐπιξενυγνυμένων τῶν AE , BK δευτέρῃσιν καὶ τὸ $ΓA$ παραλληλόγραμμον ἴσον τῷ $\ThetaΓ$ τετραγώνῳ· ὅλον ἄρα τὸ $B\Delta EΓ$ τετράγωνον δυοῖ τοῖς HB , $\ThetaΓ$ τετραγώνοις ἴσον ἐστὶν. καὶ ἐστὶ τὸ μὲν $B\Delta EΓ$ τετράγωνον ἀπὸ τῆς $BΓ$ ἀναγραφέν, τὰ δὲ HB , $\ThetaΓ$ ἀπὸ τῶν BA , AG . τὸ ἄρα ἀπὸ τῆς $BΓ$ πλευρᾶς τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν BA , AG πλευρῶν τετραγώνοις.

Ἐν ἄρα τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσας πλευρᾶς τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν [γωνίαν] περιεθουσῶν πλευρῶν τετραγώνοις· ὅπερ ἔδει δεῖξαι.

sons), BA is also straight-on to AH . And since angle DBC is equal to FBA , for (they are) both right-angles, let ABC be added to both. Thus, the whole (angle) DBA is equal to the whole (angle) FBC . And since DB is equal to BC , and FB to BA , the two (straight-lines) DB , BA are equal to the two (straight-lines) CB , BF ,[†] respectively. And angle DBA (is) equal to angle FBC . Thus, the base AD [is] equal to the base FC , and the triangle ABD is equal to the triangle FBC [Prop. 1.4]. And parallelogram BL [is] double (the area) of triangle ABD . For they have the same base, BD , and are between the same parallels, BD and AL [Prop. 1.41]. And square GB is double (the area) of triangle FBC . For again they have the same base, FB , and are between the same parallels, FB and GC [Prop. 1.41]. [And the doubles of equal things are equal to one another.][‡] Thus, the parallelogram BL is also equal to the square GB . So, similarly, AE and BK being joined, the parallelogram CL can be shown (to be) equal to the square HC . Thus, the whole square $BDEC$ is equal to the (sum of the) two squares GB and HC . And the square $BDEC$ is described on BC , and the (squares) GB and HC on BA and AC (respectively). Thus, the square on the side BC is equal to the (sum of the) squares on the sides BA and AC .

Thus, in right-angled triangles, the square on the side subtending the right-angle is equal to the (sum of the) squares on the sides surrounding the right-[angle]. (Which is) the very thing it was required to show.

[†] The Greek text has “ FB , BC ”, which is obviously a mistake.

[‡] This is an additional common notion.

48.

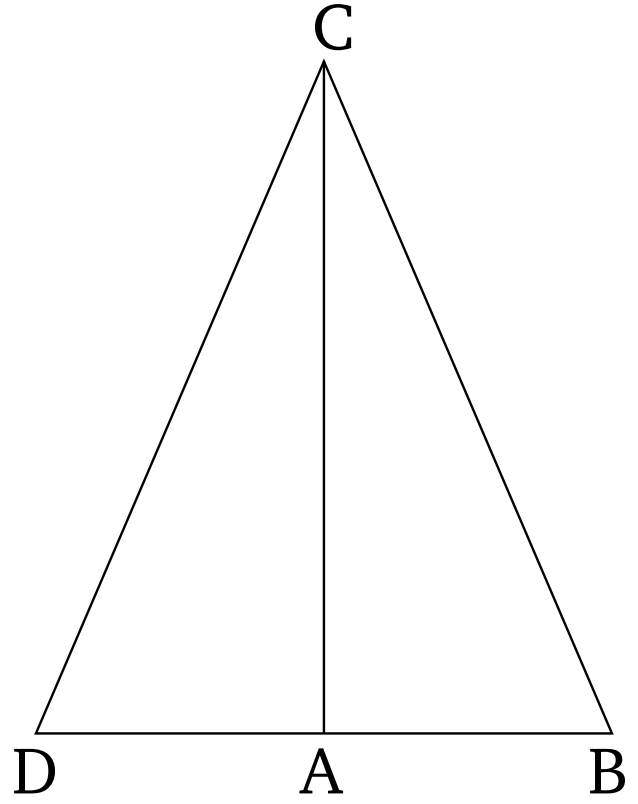
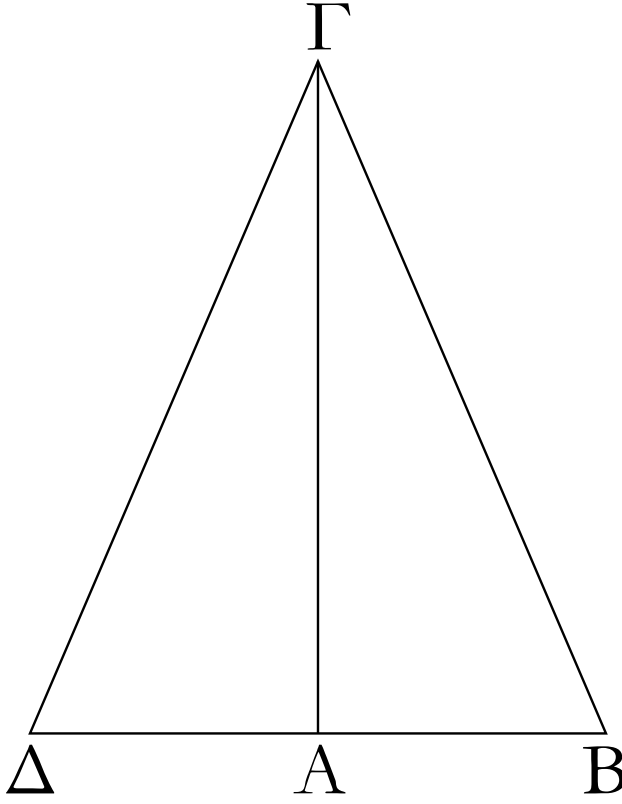
Ἐὰν τριγώνου τὸ ἀπὸ μιᾶς τῶν πλευρῶν τετράγωνον ἴσον ᾖ τοῖς ἀπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τετραγώνοις, ἡ περιεθομένη γωνία ὑπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ὀρθή ἐστὶν.

Τριγώνου γάρ τοῦ $ABΓ$ τὸ ἀπὸ μιᾶς τῆς $BΓ$ πλευρᾶς τετράγωνον ἴσον ἔστω τοῖς ἀπὸ τῶν BA , AG πλευρῶν τετραγώνοις· λέγω, ὅτι ὀρθή ἐστὶν ἡ ὑπὸ BAG γωνία.

Proposition 48

If the square on one of the sides of a triangle is equal to the (sum of the) squares on the two remaining sides of the triangle (then) the angle contained by the two remaining sides of the triangle is a right-angle.

For let the square on one of the sides, BC , of triangle ABC be equal to the (sum of the) squares on the sides BA and AC . I say that angle BAC is a right-angle.



Ἦθθω γάρ ἀπὸ τοῦ Α σημείον τῇ ΑΓ εὐθείᾳ πρὸς ὀρθὰς ἡ ΑΔ καὶ κείσθω τῇ ΒΑ ἴση ἡ ΑΔ, καὶ ἐπεξεύθθω ἡ ΔΓ. ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΑΒ, ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς ΔΑ τετραγώνον τῷ ἀπὸ τῆς ΑΒ τετραγώνῳ. κοινὸν προσκείσθω τὸ ἀπὸ τῆς ΑΓ τετραγώνον· τὰ ἄρα ἀπὸ τῶν ΔΑ, ΑΓ τετραγώνων ἴσα ἐστὶ τοῖς ἀπὸ τῶν ΒΑ, ΑΓ τετραγώνοις. ἀλλὰ τοῖς μὲν ἀπὸ τῶν ΔΑ, ΑΓ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΔΓ· ὀρθὴ γάρ ἐστιν ἡ ὑπὸ ΔΑΓ γωνία· τοῖς δὲ ἀπὸ τῶν ΒΑ, ΑΓ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΒΓ· ὑπόκειται γάρ· τὸ ἄρα ἀπὸ τῆς ΔΓ τετραγώνον ἴσον ἐστὶ τῷ ἀπὸ τῆς ΒΓ τετραγώνῳ· ὥστε καὶ πλευρὰ ἡ ΔΓ τῇ ΒΓ ἐστὶν ἴση· καὶ ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΑΒ, κοινὴ δὲ ἡ ΑΓ, δύο δὲ αἱ ΔΑ, ΑΓ δύο ταῖς ΒΑ, ΑΓ ἴσαι εἰσὶν καὶ βάσις ἡ ΔΓ βάσει τῇ ΒΓ ἴση· γωνία ἄρα ἡ ὑπὸ ΔΑΓ γωνία τῇ ὑπὸ ΒΑΓ [ἐστὶν] ἴση. ὀρθὴ δὲ ἡ ὑπὸ ΔΑΓ· ὀρθὴ ἄρα καὶ ἡ ὑπὸ ΒΑΓ.

Ἐὰν ἀρὰ τριγώνον τὸ ἀπὸ μιᾶς τῶν πλευρῶν τετραγώνον ἴσον ᾗ τοῖς ἀπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τετραγώνοις, ἡ περιεθρομένη γωνία ὑπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ὀρθὴ ἐστὶν· ὅπερ ἔδει δεῖχαι.

For let AD be drawn from point A at right-angles to the straight-line AC [Prop. 1.11], and let AD be made equal to BA [Prop. 1.3], and let DC be joined. Since DA is equal to AB , the square on DA is thus also equal to the square on AB .[†] Let the square on AC be added to both. Thus, the (sum of the) squares on DA and AC is equal to the (sum of the) squares on BA and AC . But, the (square) on DC is equal to the (sum of the squares) on DA and AC . For angle DAC is a right-angle [Prop. 1.47]. But, the (square) on BC is equal to (sum of the squares) on BA and AC . For (that) was assumed. Thus, the square on DC is equal to the square on BC . So side DC is also equal to (side) BC . And since DA is equal to AB , and AC (is) common, the two (straight-lines) DA, AC are equal to the two (straight-lines) BA, AC . And the base DC is equal to the base BC . Thus, angle DAC [is] equal to angle BAC [Prop. 1.8]. But DAC is a right-angle. Thus, BAC is also a right-angle.

Thus, if the square on one of the sides of a triangle is equal to the (sum of the) squares on the remaining two sides of the triangle (then) the angle contained by the remaining two sides of the triangle is a right-angle. (Which is) the very thing it was required to show.

[†] Here, use is made of the additional common notion that the squares of equal things are themselves equal. Later on, the inverse notion is used.

ELEMENTS BOOK 1

*Fundamentals of Plane Geometry Involving
Straight-Lines*

Ὅροι.

1. Σημεῖόν ἐστιν, ὃ ὅ μέρους οὐθέν.
2. Γραμμὴ δὲ μήκος ἀπλατέξ.
3. Γραμμῆς δὲ πέρατα σημεία.
4. Εὐθεῖα γραμμὴ ἐστίν, ἥτις ἐχ' ἴσον τοῖς ἐφ' ἑαυτῆς σημείοις κεῖται.
5. Ἐπιφάνεια δὲ ἐστίν, ὃ μήκος καὶ πλάτος μόνον ἔθει.
6. Ἐπιφανείας δὲ πέρατα γραμμαί.
7. Ἐπίπεδος ἐπιφάνειά ἐστίν, ἥτις ἐχ' ἴσον ταῖς ἐφ' ἑαυτῆς εὐθείαις κεῖται.
8. Ἐπίπεδος δὲ γωνία ἐστίν ἡ ἐν ἐπιπέδῳ δύο γραμμῶν ἀπτομένων ἀλλήλων καὶ μὴ ἐπ' εὐθείας κειμένων πρὸς ἀλλήλας τῶν γραμμῶν κλίσις.
9. Ὅταν δὲ αἱ περιέθουσαι τὴν γωνίαν γραμμαὶ εὐθεῖαι ὦσιν, εὐθύγραμμος καλεῖται ἡ γωνία.
10. Ὅταν δὲ εὐθεῖα ἐπ' εὐθείαν σταθεῖσα τὰς ἐφεχῆς γωνίας ἴσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκάτερα τῶν ἴσων γωνιῶν ἐστί, καὶ ἡ ἐφεστηκυῖα εὐθεῖα κάθετος καλεῖται, ἐφ' ἣν ἐφέστηκεν.
11. Ἀμβλεία γωνία ἐστίν ἡ μείζων ὀρθῆς.
12. Ὀξεῖα δὲ ἡ ἐλάσσων ὀρθῆς.
13. Ὅρος ἐστίν, ὃ τινὸς ἐστὶ πέραξ.
14. Σθῆμά ἐστι τὸ ὑπὸ τινὸς ἢ τινων ὁρων περιεχόμενον.
15. Κύκλος ἐστὶ σθῆμα ἐπίπεδον ὑπὸ μιᾷς γραμμῆς περιεχόμενον [ἡ καλεῖται περιφέρεια], πρὸς ἣν ἀφ' ἐνὸς σημείου τῶν ἐντὸς τοῦ σθήματος κειμένων πᾶσαι αἱ προσπίπτουσαι εὐθεῖαι [πρὸς τὴν τοῦ κύκλου περιφέρειαν] ἴσαι ἀλλήλαις εἰσίν.
16. Κέντρον δὲ τοῦ κύκλου τὸ σημεῖον καλεῖται.
17. Διάμετρος δὲ τοῦ κύκλου ἐστίν εὐθεῖα τις διὰ τοῦ κέντρον ἡγμένη καὶ περατομένη ἐφ' ἑκάτερα τὰ μέρη ὑπὸ τῆς τοῦ κύκλου περιφέρειας, ἥτις καὶ δίθια τέμνει τὸν κύκλον.
18. Ἡμικύκλιον δὲ ἐστὶ τὸ περιεχόμενον σθῆμα ὑπὸ τε τῆς διαμέτρου καὶ τῆς ἀπολαμβανομένης ὑπ' αὐτῆς περιφέρειας. κέντρον δὲ τοῦ ἡμικυκλίου τὸ αὐτό, ὃ καὶ τοῦ κύκλου ἐστίν.
19. Σθῆματα εὐθύγραμμά ἐστι τὰ ὑπὸ εὐθειῶν περιεθόμενα, τρίπλευρα μὲν τὰ ὑπὸ τριῶν, τετράπλευρα δὲ τὰ ὑπὸ τεσσάρων, πολὺπλευρα δὲ τὰ ὑπὸ πλείονων ἢ τεσσάρων εὐθειῶν περιεθόμενα.
20. Τῶν δὲ τριπλεύρων σθημάτων ἰσοπλευρον μὲν τρίγωνόν ἐστι τὸ τὰς τρεῖς ἴσας ἔθον πλευράς, ἰσοσκελὲς δὲ τὸ τὰς δύο μόνας ἴσας ἔθον πλευράς, σκαληνὸν δὲ τὸ τὰς τρεῖς ἀνίσους ἔθον πλευράς.
21. Ἐτι δὲ τῶν τριπλεύρων σθημάτων ὀρθογώνιον μὲν τρίγωνόν ἐστι τὸ ἔθον ὀρθὴν γωνίαν, ἀμβλυγώνιον δὲ τὸ ἔθον ἀμβλείαν γωνίαν, ὀξυγώνιον δὲ τὸ τὰς τρεῖς ὀξείας ἔθον γωνίας.
22. Τῶν δὲ τετραπλεύρων σθημάτων τετράγωνον μὲν ἐστίν, ὃ ἰσοπλευρόν τε ἐστὶ καὶ ὀρθογώνιον, ἑτερόμηκες δὲ, ὃ ὀρθογώνιον μὲν, οὐκ ἰσοπλευρον δὲ, ῥόμβος δὲ, ὃ ἰσοπλευρον μὲν, οὐκ ὀρθογώνιον δὲ, ῥομβοειδὲς δὲ τὸ τὰς ἀπεναντίον πλευράς

Definitions

1. A point is that of which there is no part.
2. And a line is a length without breadth.
3. And the extremities of a line are points.
4. A straight-line is (any) one which lies evenly with points on itself.
5. And a surface is that which has length and breadth only.
6. And the extremities of a surface are lines.
7. A plane surface is (any) one which lies evenly with the straight-lines on itself.
8. And a plane angle is the inclination of the lines to one another, when two lines in a plane meet one another, and are not lying in a straight-line.
9. And when the lines containing the angle are straight then the angle is called rectilinear.
10. And when a straight-line stood upon (another) straight-line makes adjacent angles (which are) equal to one another, each of the equal angles is a right-angle, and the former straight-line is called a perpendicular to that upon which it stands.
11. An obtuse angle is one greater than a right-angle.
12. And an acute angle (is) one less than a right-angle.
13. A boundary is that which is the extremity of something.
14. A figure is that which is contained by some boundary or boundaries.
15. A circle is a plane figure contained by a single line [which is called a circumference], (such that) all of the straight-lines radiating towards [the circumference] from one point amongst those lying inside the figure are equal to one another.
16. And the point is called the center of the circle.
17. And a diameter of the circle is any straight-line, being drawn through the center, and terminated in each direction by the circumference of the circle. (And) any such (straight-line) also cuts the circle in half.[†]
18. And a semi-circle is the figure contained by the diameter and the circumference cut off by it. And the center of the semi-circle is the same (point) as (the center of) the circle.
19. Rectilinear figures are those (figures) contained by straight-lines: trilateral figures being those contained by three straight-lines, quadrilateral by four, and multilateral by more than four.
20. And of the trilateral figures: an equilateral triangle is that having three equal sides, an isosceles (triangle) that having only two equal sides, and a scalene (triangle) that having three unequal sides.
21. And further of the trilateral figures: a right-angled triangle is that having a right-angle, an obtuse-angled (triangle) that having an obtuse angle, and an acute-angled (triangle) that having three acute angles.

τε καὶ γωνίαξ ἴσαξ ἀλλήλαιξ ἔθον, ὃ οὕτε ἰσόπλευρόν ἐστιν οὕτε ὀρθογώνιον· τὰ δὲ παρὰ ταῦτα τετράπλευρα τραπέζια καλεῖσθω.

23. Παράλληλοι εἰσιν εὐθεῖαι, αἵτινες ἐν τῷ αὐτῷ ἐπιπέδῳ οὔσαι καὶ ἐκβαλλόμεναι εἰς ἄπειρον ἐφ' ἑκάτερα τὰ μέρη ἐπὶ μηδέτερα συμπίπτουσιν ἀλλήλαιξ.

22. And of the quadrilateral figures: a square is that which is right-angled and equilateral, a rectangle that which is right-angled but not equilateral, a rhombus that which is equilateral but not right-angled, and a rhomboid that having opposite sides and angles equal to one another which is neither right-angled nor equilateral. And let quadrilateral figures besides these be called trapezia.

23. Parallel lines are straight-lines which, being in the same plane, and being produced to infinity in each direction, meet with one another in neither (of these directions).

† This should really be counted as a postulate, rather than as part of a definition.

Αἰτήματα.

1. Ἡτῆσθω ἀπὸ παντός σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.

2. Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνθεῖξ ἐπ' εὐθείας ἐκβαλεῖν.

3. Καὶ παντὶ κέντρῳ καὶ διαστήματι κύκλον γράφεσθαι.

4. Καὶ πάσαξ τὰξ ὀρθὰξ γωνίαξ ἴσαξ ἀλλήλαιξ εἶναι.

5. Καὶ ἔαν εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὰξ ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη γωνίαξ δύο ὀρθῶν ἐλάσσοναξ ποιῇ, ἐκβαλλόμεναξ τὰξ δύο εὐθείας ἐπ' ἄπειρον συμπίπτειν, ἐφ' ἃ μέρη εἰσὶν αἱ τῶν δύο ὀρθῶν ἐλάσσονες.

Postulates

1. Let it be postulated[†] (that it is possible) to draw a straight-line from any point to any point;

2. And to produce a finite straight-line continuously in a straight-line;

3. And to draw a circle with any center and radius.

4. And that all right-angles are equal to one another.

5. And that if a straight-line falling across two (other) straight-lines makes internal angles on the same side (of itself whose sum is) less than two right-angles, (then) the two (other) straight-lines, being produced to infinity, meet on that side (of the original straight-line) that the (sum of the internal angles) is less than two right-angles (and do not meet on the other side).[‡]

† Literally, "let it have been postulated". The Greek present perfect tense indicates a past action with present significance. Hence, the 3rd-person present perfect imperative *ἡτῆσθω* means "let it be postulated" in the sense "let it stand as postulated", but not "let the postulate be now brought forward". This peculiar Greek idiom is used throughout the Elements.

‡ This postulate effectively specifies that we are dealing with the geometry of flat, rather than curved, space.

Κοιναὶ ἔννοιαι.

1. Τὰ τῷ αὐτῷ ἴσα καὶ ἀλλήλοιξ ἐστὶν ἴσα.

2. Καὶ ἔαν ἴσοιξ ἴσα προστεθῇ, τὰ ὅλα ἐστὶν ἴσα.

3. Καὶ ἔαν ἀπὸ ἴσων ἴσα ἀφαιρεθῇ, τὰ καταλειπόμενά ἐστιν ἴσα.

4. Καὶ τὰ ἐφαρμόζοντα ἐπ' ἀλλήλα ἴσα ἀλλήλοιξ ἐστὶν.

5. Καὶ τὸ ὅλον τοῦ μέρους μείζον [ἐστίν].

Common Notions

1. Things equal to the same thing are also equal to one another.

2. And if equal things are added to equal things (then) the wholes are equal.

3. And if equal things are subtracted from equal things (then) the remainders are equal.[†]

4. And things coinciding with one another are equal to one another.

5. And the whole [is] greater than the part.

† As an obvious extension of C.N.s 2 & 3—if equal things are added or subtracted from the two sides of an inequality then the inequality remains an inequality of the same type.

1.

Proposition 1

Ἐπὶ τῇξ δοθείσης εὐθείας πεπερασμένης τρίγωνον ἰσόπλευ-

To construct an equilateral triangle on a given finite straight-

ρον συστήσασθαι.

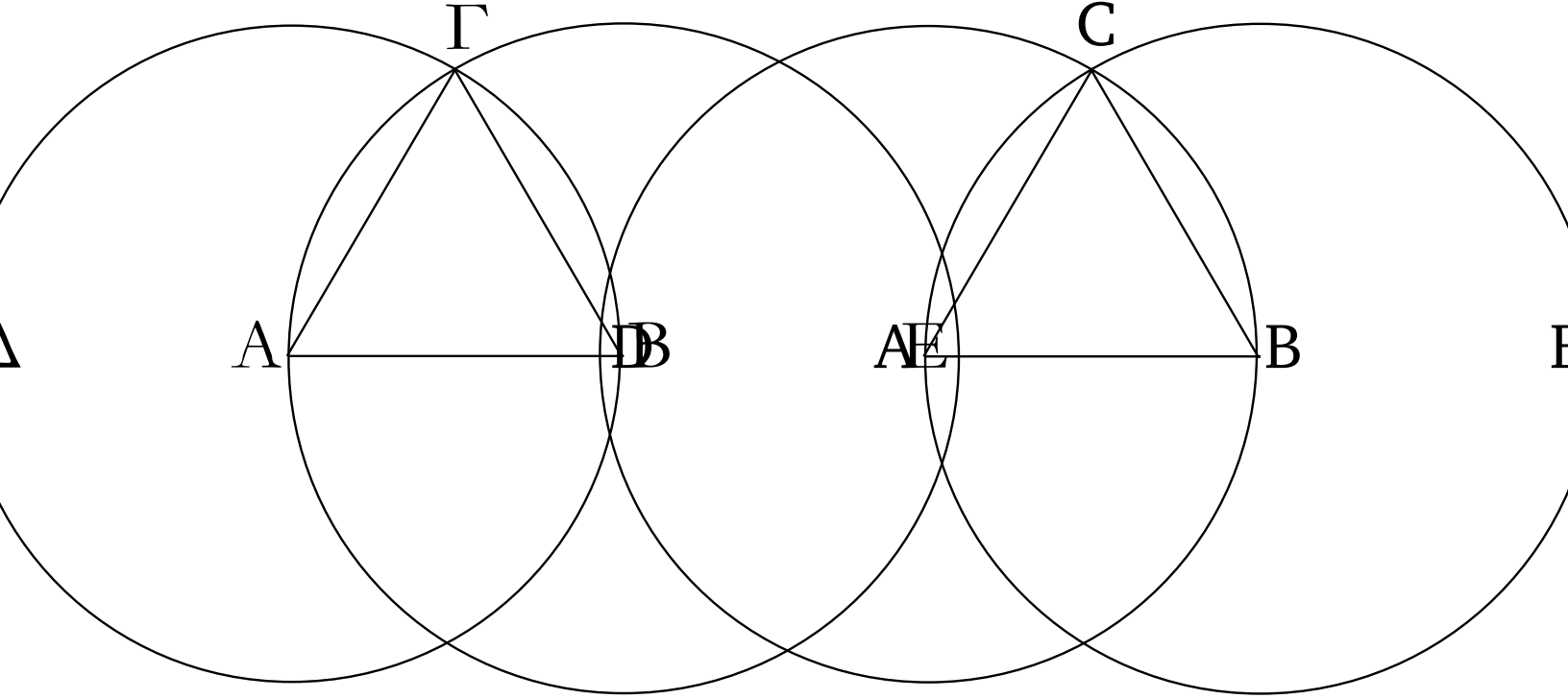
Ἐστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB .

Δεῖ δὴ ἐπὶ τῆς AB εὐθείας τρίγωνον ἰσόπλευρον συστήσασθαι.

line.

Let AB be the given finite straight-line.

So it is required to construct an equilateral triangle on the straight-line AB .



Κέντρῳ μὲν τῷ A διαστήματι δὲ τῷ AB κύκλος γεγράφθω ὁ $BΓΔ$, καὶ πάλιν κέντρῳ μὲν τῷ B διαστήματι δὲ τῷ BA κύκλος γεγράφθω ὁ $ΑΓΕ$, καὶ ἀπὸ τοῦ $Γ$ σημείου, καθ' ὃ τέμνουσιν ἀλλήλους οἱ κύκλοι, ἐπὶ τὰ A , B σημεία ἐπεζεύθωσαν εὐθεῖαι αἱ $ΓΑ$, $ΓΒ$.

Καὶ ἐπεὶ τὸ A σημεῖον κέντρον ἐστὶ τοῦ $ΓΔΒ$ κύκλου, ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$. πάλιν, ἐπεὶ τὸ B σημεῖον κέντρον ἐστὶ τοῦ $ΓΑΕ$ κύκλου, ἴση ἐστὶν ἡ $ΒΓ$ τῇ $ΒΑ$. ἐδείκθη δὲ καὶ ἡ $ΓΑ$ τῇ $ΑΒ$ ἴση· ἐκατέρα ἄρα τῶν $ΓΑ$, $ΓΒ$ τῇ $ΑΒ$ ἐστὶν ἴση. τὰ δὲ τῷ αὐτῷ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα· καὶ ἡ $ΓΑ$ ἄρα τῇ $ΓΒ$ ἐστὶν ἴση· αἱ τρεῖς ἄρα αἱ $ΓΑ$, $ΑΒ$, $ΒΓ$ ἴσαι ἀλλήλαις εἰσὶν.

Ἰσόπλευρον ἄρα ἐστὶ τὸ $ΑΒΓ$ τρίγωνον. καὶ συνέσταται ἐπὶ τῆς δοθείσης εὐθείας πεπερασμένης τῆς $ΑΒ$. ὅπερ ἔδει ποιῆσαι.

Let the circle BCD with center A and radius AB be drawn[†] [Post. 3], and again let the circle ACE with center B and radius BA be drawn [Post. 3]. And let the straight-lines CA and CB be joined from the point C , where the circles cut one another,[‡] to the points A and B (respectively) [Post. 1].

And since the point A is the center of the circle CDB , AC is equal to AB [Def. 1.15]. Again, since the point B is the center of the circle CAE , BC is equal to BA [Def. 1.15]. But CA was also shown (to be) equal to AB . Thus, CA and CB are each equal to AB . But things equal to the same thing are also equal to one another [C.N. 1]. Thus, CA is also equal to CB . Thus, the three (straight-lines) CA , AB , and BC are equal to one another.

Thus, the triangle ABC is equilateral, and has been constructed on the given finite straight-line AB . (Which is) the very thing it was required to do.

[†] Literally, "have been drawn". The use of this peculiar Greek idiom will pass without further comment.

[‡] The assumption that the circles do indeed cut one another should be counted as an additional postulate. There is also an implicit assumption that two straight-lines cannot share a common segment.

2.

Proposition 2[†]

Πρὸς τῷ δοθέντι σημείῳ τῇ δοθείσῃ εὐθείᾳ ἴσην εὐθεῖαν

To place a straight-line equal to a given straight-line at a

θέσθαι.

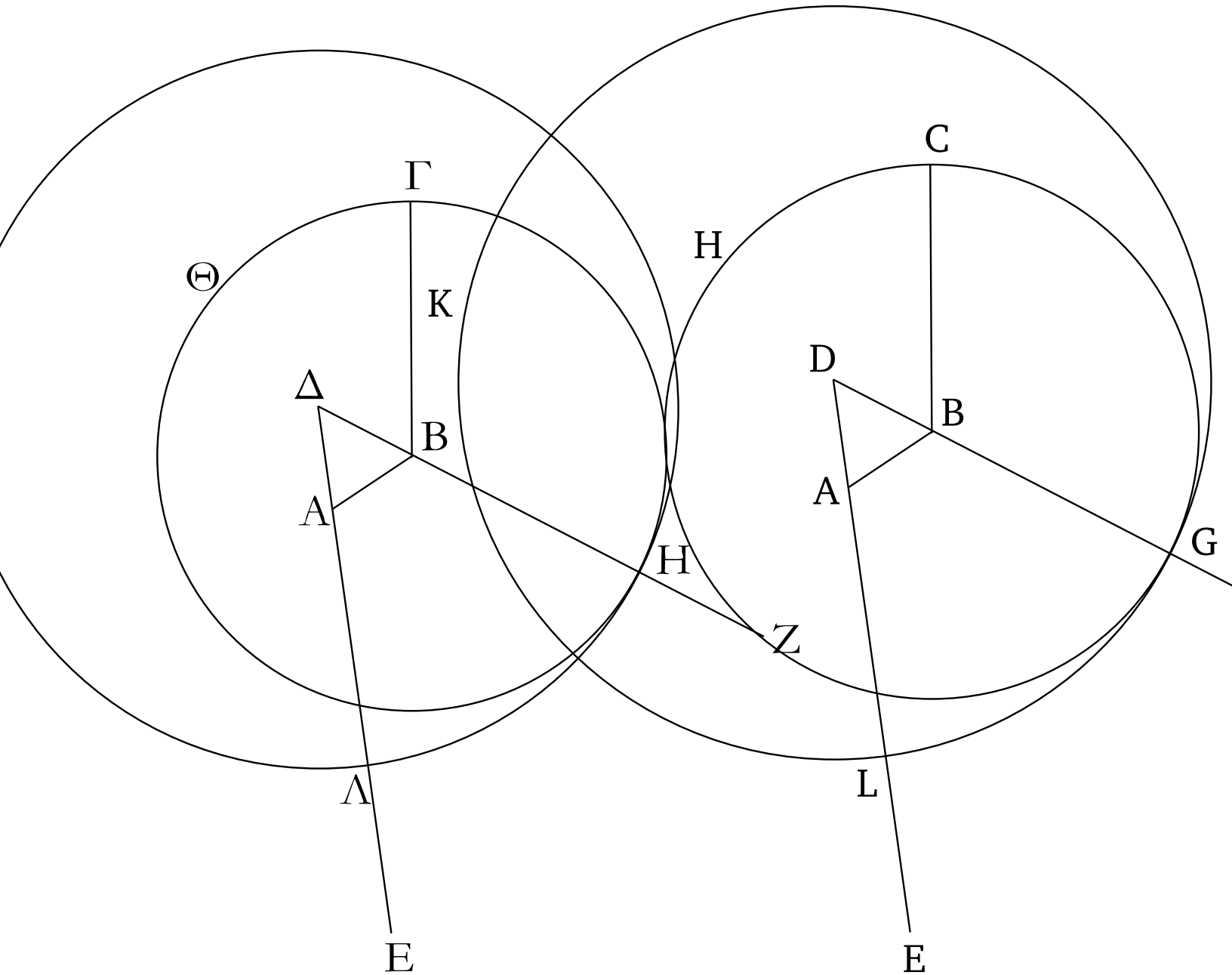
Ἐστω τὸ μὲν δοθὲν σημεῖον τὸ A , ἡ δὲ δοθεῖσα εὐθεῖα ἡ $BΓ$. δεῖ δὴ πρὸς τῷ A σημείῳ τῇ δοθείσῃ εὐθείᾳ τῇ $BΓ$ ἴσην εὐθεῖαν θέσθαι.

Ἐπεξεύθθω γὰρ ἀπὸ τοῦ A σημείου ἐπὶ τὸ B σημεῖον εὐθεῖα ἡ AB , καὶ συνεστάτω ἐπ' αὐτῆς τρίγωνον ἰσόπλευρον τὸ $ΔAB$, καὶ ἐκβεβλήσθωσαν ἐπ' εὐθείας ταῖς $ΔA$, $ΔB$ εὐθεῖαι αἱ AE , BZ , καὶ κέντρῳ μὲν τῷ B διαστήματι δὲ τῷ $BΓ$ κύκλος γεγράφθω ὁ $ΓΗΘ$, καὶ πάλιν κέντρῳ τῷ $Δ$ καὶ διαστήματι τῷ $ΔΗ$ κύκλος γεγράφθω ὁ $ΗΚΛ$.

given point (as an extremity).

Let A be the given point, and BC the given straight-line. So it is required to place a straight-line at point A equal to the given straight-line BC .

For let the straight-line AB be joined from point A to point B [Post. 1], and let the equilateral triangle DAB be constructed upon it [Prop. 1.1]. And let the straight-lines AE and BF be produced in a straight-line with DA and DB (respectively) [Post. 2]. And let the circle CGH with center B and radius BC be drawn [Post. 3], and again let the circle GKL with center D and radius DG be drawn [Post. 3].



Ἐπεὶ οὖν τὸ B σημεῖον κέντρον ἐστὶ τοῦ $\Gamma\Theta$, ἴση ἐστὶν ἡ $B\Gamma$ τῇ $B\Theta$. πάλιν, ἐπεὶ τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ HKL κύκλου, ἴση ἐστὶν ἡ $\Delta\Lambda$ τῇ $\Delta\mathrm{H}$, ὥν ἡ $\Delta\Lambda$ τῇ $\Delta\mathrm{B}$ ἴση ἐστὶν. λοιπὴ ἄρα ἡ $\Lambda\mathrm{A}$ λοιπῇ τῇ $B\Theta$ ἐστὶν ἴση. ἐδείκθη δὲ καὶ ἡ $B\Gamma$ τῇ $B\Theta$ ἴση· ἑκατέρα ἄρα τῶν $\Lambda\mathrm{A}$, $B\Gamma$ τῇ $B\Theta$ ἐστὶν ἴση. τὰ δὲ τῶν αὐτῶν ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα· καὶ ἡ $\Lambda\mathrm{A}$ ἄρα τῇ $B\Gamma$ ἐστὶν ἴση.

Πρὸς ἄρα τῶν δοθέντι σημείῳ τῶν A τῇ δοθείσῃ εὐθείᾳ τῇ $B\Gamma$ ἴση εὐθεῖα κείται ἡ $\Lambda\mathrm{A}$ · ὅπερ ἔδει ποιῆσαι.

Therefore, since the point B is the center of (the circle) CGH , BC is equal to BG [Def. 1.15]. Again, since the point D is the center of the circle GKL , DL is equal to DG [Def. 1.15]. And within these, DA is equal to DB . Thus, the remainder AL is equal to the remainder BG [C.N. 3]. But BC was also shown (to be) equal to BG . Thus, AL and BC are each equal to BG . But things equal to the same thing are also equal to one another [C.N. 1]. Thus, AL is also equal to BC .

Thus, the straight-line AL , equal to the given straight-line BC , has been placed at the given point A . (Which is) the very

thing it was required to do.

† This proposition admits of a number of different cases, depending on the relative positions of the point A and the line BC . In such situations, Euclid invariably only considers one particular case—usually, the most difficult—and leaves the remaining cases as exercises for the reader.

3.

Proposition 3

Δύο δοθεισῶν εὐθειῶν ἀνίσων ἀπὸ τῆς μείζονος τῇ ἐλάσσονι ἴσην εὐθεῖαν ἀφελεῖν.

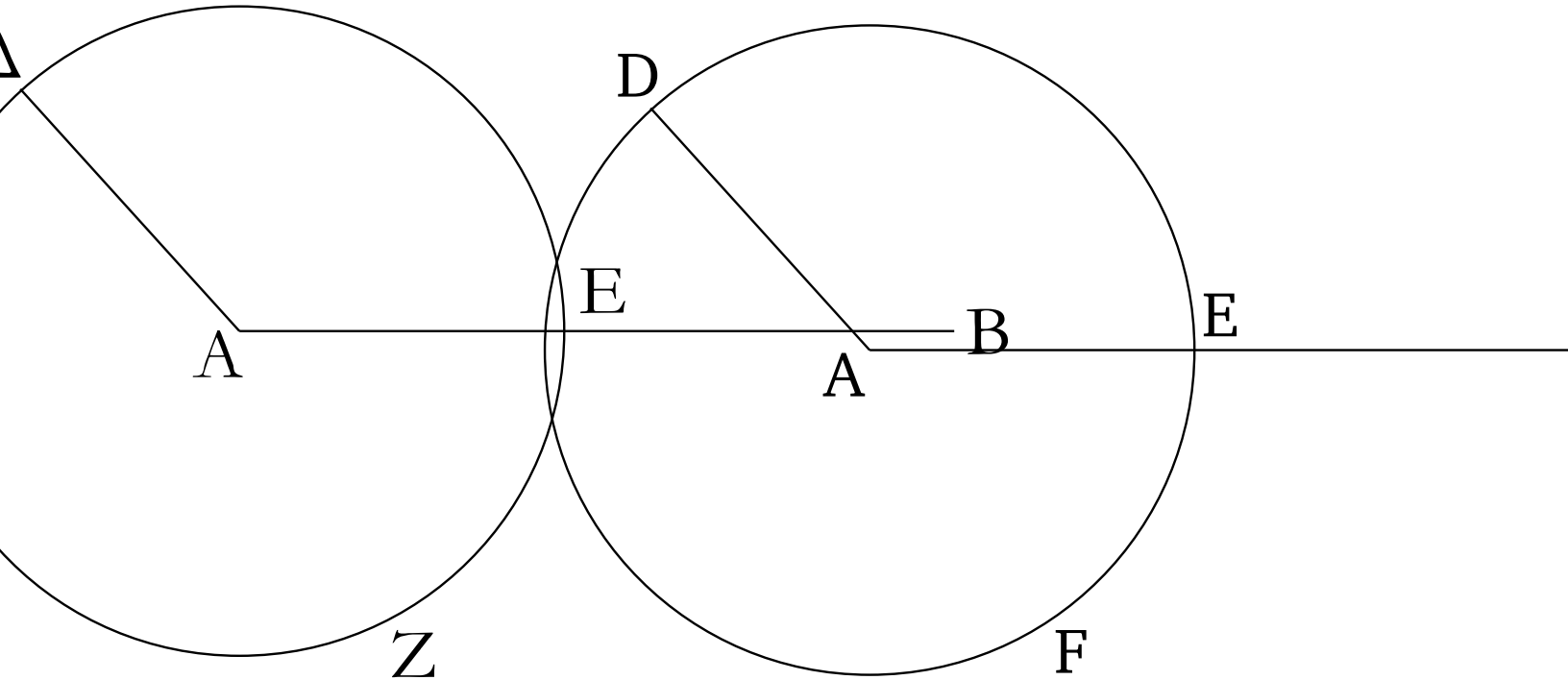
Ἐστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι ἴσους αἱ AB , Γ , ὧν μείζων ἔστω ἡ AB . δεῖ δὴ ἀπὸ τῆς μείζονος τῆς AB τῇ ἐλάσσονι τῇ Γ ἴσην εὐθεῖαν ἀφελεῖν.

Κείσθω πρὸς τῷ A σημείῳ τῇ Γ εὐθείᾳ ἴση ἡ $A\Delta$. καὶ κέντρῳ μὲν τῷ A διαστήματι δὲ τῷ $A\Delta$ κύκλος γεγράφθω ὁ ΔEZ .

Καὶ ἐπεὶ τὸ A σημεῖον κέντρον ἐστὶ τοῦ ΔEZ κύκλου, ἴση ἐστὶν ἡ AE τῇ $A\Delta$. ἀλλὰ καὶ ἡ Γ τῇ $A\Delta$ ἐστὶν ἴση. ἑκατέρω ἄρα τῶν AE , Γ τῇ $A\Delta$ ἐστὶν ἴση· ὥστε καὶ ἡ AE τῇ Γ ἐστὶν ἴση.

Γ

C



Δύο ἄρα δοθεισῶν εὐθειῶν ἀνίσων τῶν AB , Γ ἀπὸ τῆς μείζονος τῆς AB τῇ ἐλάσσονι τῇ Γ ἴση ἀφίηται ἡ AE . ὅπερ ἔδει ποιῆσαι.

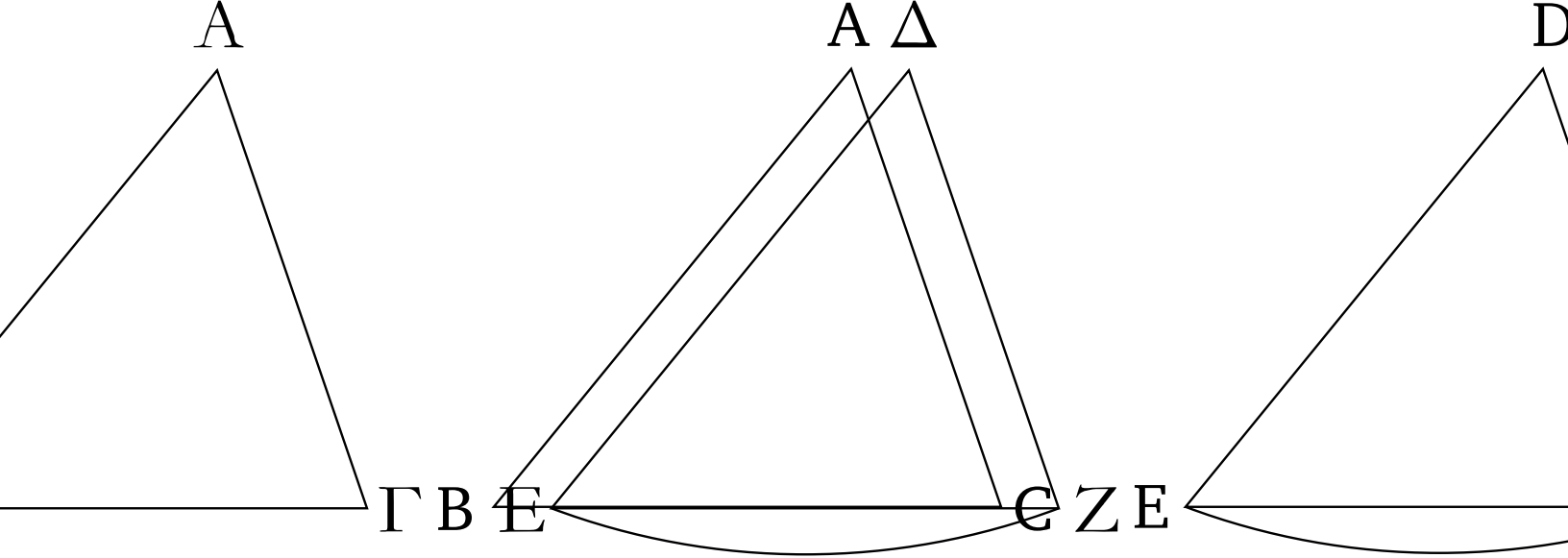
Thus, for two given unequal straight-lines, AB and C , the (straight-line) AE , equal to the lesser C , has been cut off from the greater AB . (Which is) the very thing it was required to do.

4.

Proposition 4

Ἐάν δύο τρίγωνα τὰς δύο πλευράς [ταῖς] δυοῖ πλευραῖς ἴσαξ ἔθῃ ἑκατέραν ἑκατέρᾳ καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔθῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῇ βάσει ἴσην ἔχει, καὶ τὸ τρίγωνον τῷ τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρα ἑκατέρᾳ, ὅφ' ἂξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν.

If two triangles have two sides equal to two sides, respectively, and have the angle(s) enclosed by the equal straight-lines equal, (then) they will also have the base equal to the base, and the triangle will be equal to the triangle, and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles.



Ἐστω δύο τρίγωνα τὰ ABG , $ΔEZ$ τὰς δύο πλευράς τὰς AB , AG ταῖς δυοῖς πλευραῖς ταῖς $ΔE$, $ΔZ$ ἴσαξ ἔθοντα ἑκατέραν ἑκατέρᾳ τὴν μὲν AB τῇ $ΔE$ τὴν δὲ AG τῇ $ΔZ$ καὶ γωνίαν τὴν ὑπὸ BAG γωνίᾳ τῇ ὑπὸ $EΔZ$ ἴσην. λέγω, ὅτι καὶ βάσει ἡ BG βάσει τῇ EZ ἴση ἔστί, καὶ τὸ ABG τρίγωνον τῷ $ΔEZ$ τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρα ἑκατέρᾳ, ὅφ' ἂξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ ABG τῇ ὑπὸ $ΔEZ$, ἡ δὲ ὑπὸ AGB τῇ ὑπὸ $ΔZE$.

Ἐφαρμοζόμενον γὰρ τοῦ ABG τριγώνου ἐπὶ τὸ $ΔEZ$ τρίγωνον καὶ τιθεμένον τοῦ μὲν A σημείου ἐπὶ τὸ $Δ$ σημεῖον τῆς δὲ AB εὐθείας ἐπὶ τὴν $ΔE$, ἐφαρμόσει καὶ τὸ B σημεῖον ἐπὶ τὸ E διὰ τὸ ἴσην εἶναι τὴν AB τῇ $ΔE$. ἐφαρμοσάσης δὲ τῆς AB ἐπὶ τὴν $ΔE$ ἐφαρμόσει καὶ ἡ AG εὐθεῖα ἐπὶ τὴν $ΔZ$ διὰ τὸ ἴσην εἶναι τὴν ὑπὸ BAG γωνίαν τῇ ὑπὸ $EΔZ$. ὥστε καὶ τὸ G σημεῖον ἐπὶ τὸ Z σημεῖον ἐφαρμόσει διὰ τὸ ἴσην πάλιν εἶναι τὴν AG τῇ $ΔZ$. ἀλλὰ μὴν καὶ τὸ B ἐπὶ τὸ E ἐφηρμόκει· ὥστε βάσει ἡ BG ἐπὶ βάσιν τὴν EZ ἐφαρμόσει. εἰ γὰρ τοῦ μὲν B ἐπὶ τὸ E ἐφαρμόσαντος τοῦ δὲ G ἐπὶ τὸ Z ἡ BG βάσις ἐπὶ τὴν EZ οὐκ ἐφαρμόσει, δύο εὐθεῖαι θωρίον περιέχουσιν· ὅπερ ἐστὶν ἀδύνατον. ἐφαρμόσει ἄρα ἡ BG βάσις ἐπὶ τὴν EZ καὶ ἴση αὐτῇ ἔσται· ὥστε καὶ ὅλον τὸ ABG τρίγωνον ἐπὶ ὅλον τὸ $ΔEZ$ τρίγωνον ἐφαρμόσει καὶ ἴσον αὐτῷ ἔσται, καὶ αἱ λοιπαὶ γωνίαι ἐπὶ τὰς λοιπὰς γωνίας ἐφαρμόσουσι καὶ ἴσαι

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is) AB to DE , and AC to DF . And (let) the angle BAC (be) equal to the angle EDF . I say that the base BC is also equal to the base EF , and triangle ABC will be equal to triangle DEF , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles. (That is) ABC to DEF , and ACB to DFE .

For if triangle ABC is applied to triangle DEF ,[†] the point A being placed on the point D , and the straight-line AB on DE , (then) the point B will also coincide with E , on account of AB being equal to DE . So (because of) AB coinciding with DE , the straight-line AC will also coincide with DF , on account of the angle BAC being equal to EDF . So the point C will also coincide with the point F , again on account of AC being equal to DF . But, point B certainly also coincided with point E , so that the base BC will coincide with the base EF . For if B coincides with E , and C with F , and the base BC does not coincide with EF , (then) two straight-lines will encompass an area. The very thing is impossible [Post. 1].[‡] Thus, the base BC will coincide with EF , and will be equal to it [C.N. 4]. So the whole triangle ABC will coincide with the whole triangle DEF , and will be equal to it [C.N. 4]. And the remaining angles will coincide with the remaining angles, and will be

αὐταῖς ἴσονται, ἡ μὲν ὑπὸ $ABΓ$ τῇ ὑπὸ $ΔΕΖ$ ἢ δὲ ὑπὸ $ΑΓΒ$ τῇ ὑπὸ $ΔΖΕ$.

Ἐὰν ἄρα δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρα καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔθῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῇ βάσει ἴσην ἔχει, καὶ τὸ τρίγωνον τῷ τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρα ἑκατέρα, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ὅπερ ἔδει δεῖχαι.

† The application of one figure to another should be counted as an additional postulate.

‡ Since Post. 1 implicitly assumes that the straight-line joining two given points is unique.

5.

Τῶν ἰσοσκελῶν τριγώνων αἱ πρὸς τῇ βάσει γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ προσεκβληθεισῶν τῶν ἴσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι ἴσαι ἀλλήλαις ἔσονται.

Ἔστω τριγώνον ἰσοσκελὲς τὸ $ABΓ$ ἴσην ἔθον τὴν AB πλευρὰν τῇ $ΑΓ$ πλευρᾷ, καὶ προσεκβεβλήσθωσαν ἐπ' εὐθείας ταῖς AB , $ΑΓ$ εὐθεῖαι αἱ $BΔ$, $ΓΕ$. λέγω, ὅτι ἡ μὲν ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΑΓΒ$ ἴση ἐστίν, ἡ δὲ ὑπὸ $ΓΒΔ$ τῇ ὑπὸ $ΒΓΕ$.

Εἰλήφθω γάρ ἐπὶ τῇς $BΔ$ τυθὸν σημεῖον τὸ Z , καὶ ἀφηρήσθω ἀπὸ τῇς μείζονος τῇς AE τῇ ἐλάσσονι τῇ AZ ἴση ἡ AH , καὶ ἐπεζεύθωσαν αἱ $ZΓ$, HB εὐθεῖαι.

equal to them [C.N. 4]. (That is) ABC to DEF , and ACB to DFE [C.N. 4].

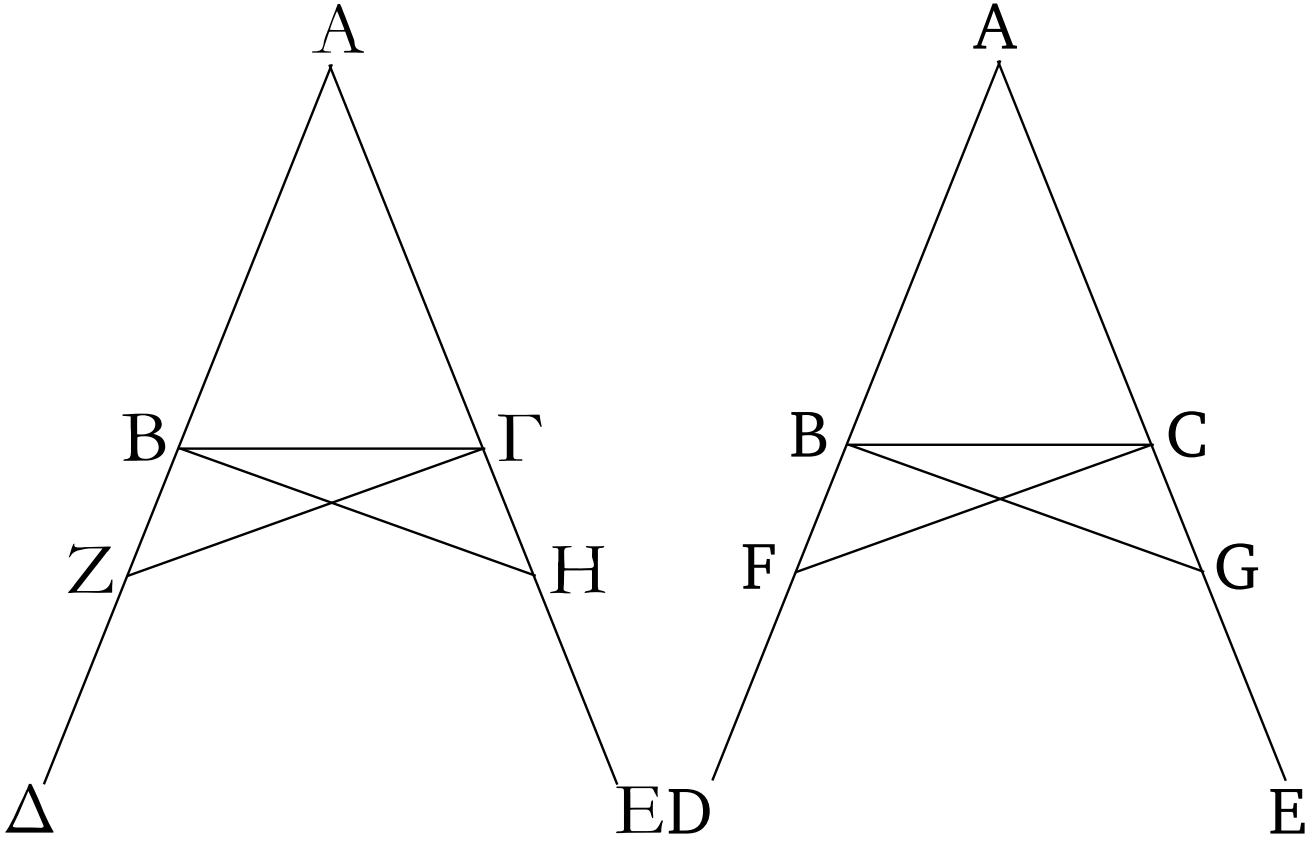
Thus, if two triangles have two sides equal to two sides, respectively, and have the angle(s) enclosed by the equal straight-lines equal, (then) they will also have the base equal to the base, and the triangle will be equal to the triangle, and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles. (Which is) the very thing it was required to show.

Proposition 5

For isosceles triangles, the angles at the base are equal to one another, and if the equal sides are produced (then) the angles under the base will be equal to one another.

Let ABC be an isosceles triangle having the side AB equal to the side AC , and let the straight-lines BD and CE be produced in a straight-line with AB and AC (respectively) [Post. 2]. I say that the angle ABC is equal to ACB , and (angle) CBD to BCE .

For let the point F be taken at random on BD , and let AG be cut off from the greater AE , equal to the lesser AF [Prop. 1.3]. Also, let the straight-lines FC and GB be joined [Post. 1].



Ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν AZ τῇ AH ἢ δὲ AB τῇ AG , δύο δὲ αἱ ZA , AG δυοὶ ταῖς HA , AB ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνίαν κοινὴν περιέθουσι τὴν ὑπὸ ZAH · βάσεις ἄρα ἢ ZG βάσει τῇ HB ἴση ἐστίν, καὶ τὸ AZG τριγώνον τῷ AHB τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρω ἑκατέρω, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ AGZ τῇ ὑπὸ ABH , ἡ δὲ ὑπὸ AZG τῇ ὑπὸ AHB . καὶ ἐπεὶ ὅλη ἡ AZ ὅλη τῇ AH ἐστὶν ἴση, ὧν ἡ AB τῇ AG ἐστὶν ἴση, λοιπὴ ἄρα ἡ BZ λοιπῇ τῇ GH ἐστὶν ἴση. ἐδείκθη δὲ καὶ ἡ ZG τῇ HB ἴση· δύο δὲ αἱ BZ , ZG δυοὶ ταῖς GH , HB ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνία ἡ ὑπὸ BZG γωνία τῇ ὑπὸ GHB ἴση, καὶ βάσεις αὐτῶν κοινὴ ἡ BG . καὶ τὸ BZG ἄρα τριγώνον τῷ GHB τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρω ἑκατέρω, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἐστὶν ἡ μὲν ὑπὸ ZBG τῇ ὑπὸ HGB ἢ δὲ ὑπὸ BGZ τῇ ὑπὸ GBH . ἐπεὶ οὖν ὅλη ἡ ὑπὸ ABH γωνία ὅλη τῇ ὑπὸ AGZ γωνίᾳ ἐδείκθη ἴση, ὧν ἡ ὑπὸ GBH τῇ ὑπὸ BGZ ἴση, λοιπὴ ἄρα ἡ ὑπὸ ABG λοιπῇ τῇ ὑπὸ AGB ἐστὶν ἴση· καὶ εἰσι πρὸς τῇ βάσει τοῦ ABG τριγώνου. ἐδείκθη δὲ καὶ ἡ ὑπὸ ZBG τῇ ὑπὸ HGB ἴση· καὶ εἰσιν ὑπὸ τὴν βάσιν.

Τῶν ἄρα ἰσοσκελῶν τριγώνων αἱ τρὸς τῇ βάσει γωνίαι ἴσαι ἀλλήλαις εἰσὶν, καὶ προσεκβληθεισῶν τῶν ἴσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι ἴσαι ἀλλήλαις ἔσονται· ὅπερ ἔδει

In fact, since AF is equal to AG , and AB to AC , the two (straight-lines) FA , AC are equal to the two (straight-lines) GA , AB , respectively. They also encompass a common angle, FAG . Thus, the base FC is equal to the base GB , and the triangle AFC will be equal to the triangle AGB , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles [Prop. 1.4]. (That is) ACF to ABG , and AFC to AGB . And since the whole of AF is equal to the whole of AG , within which AB is equal to AC , the remainder BF is thus equal to the remainder CG [C.N. 3]. But FC was also shown (to be) equal to GB . So the two (straight-lines) BF , FC are equal to the two (straight-lines) CG , GB , respectively, and the angle BFC (is) equal to the angle CGB , and the base BC is common to them. Thus, the triangle BFC will be equal to the triangle CGB , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles [Prop. 1.4]. Thus, FBC is equal to GCB , and BCF to CBG . Therefore, since the whole angle ABG was shown (to be) equal to the whole angle ACF , within which CBG is equal to BCF , the remainder ABC is thus equal to the remainder ACB [C.N. 3]. And they are at the base of triangle ABC . And FBC was also shown (to be) equal to GCB . And they are under the base.

Thus, for isosceles triangles, the angles at the base are

δειχαι.

equal to one another, and if the equal sides are produced (then) the angles under the base will be equal to one another. (Which is) the very thing it was required to show.

6.

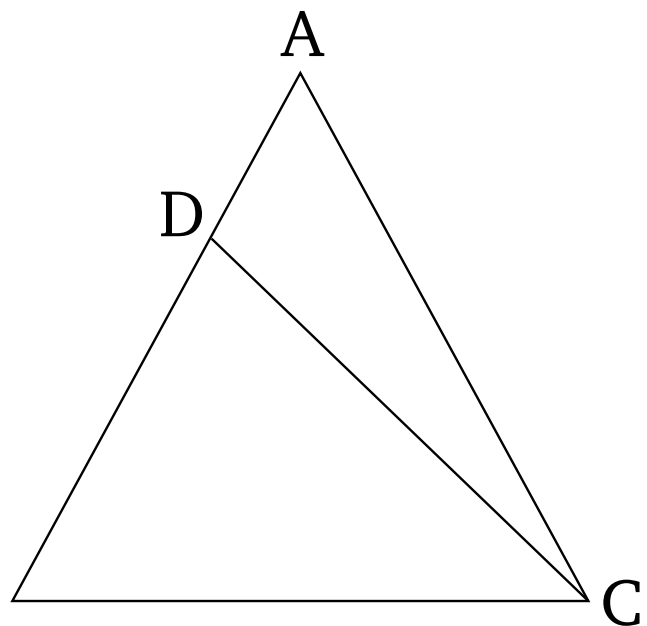
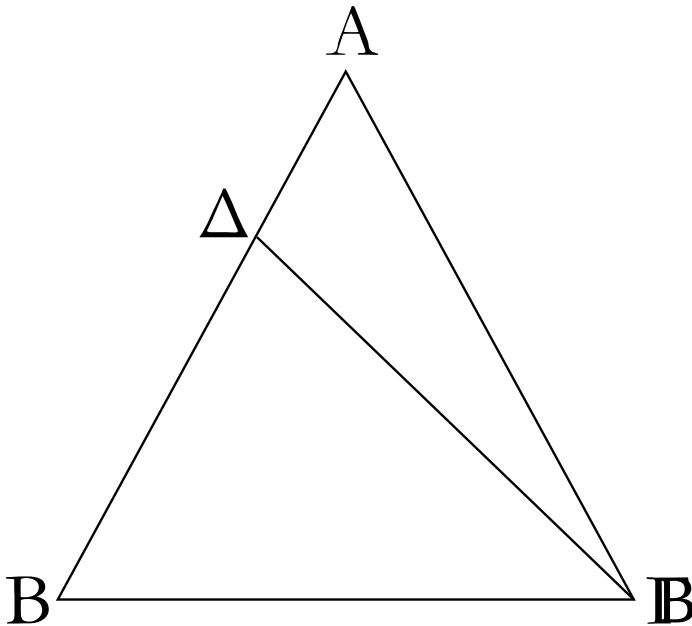
Proposition 6

Ἐάν τριγώνον αἱ δύο γωνίαι ἴσαι ἀλλήλαις ᾤσιν, καὶ α-
ῖ ὑπὸ τὰς ἴσαξ γωνίαξ ὑποτείνονσαι πλευραὶ ἴσαι ἀλλήλαιξ
έσονται.

If a triangle has two angles equal to one another (then) the
sides subtending the equal angles will also be equal to one
another.

Ἐστω τρίγωνον τὸ $ABΓ$ ἴσην ἔθον τὴν ὑπὸ $ABΓ$ γωνίαν
τῇ ὑπὸ $ΑΓΒ$ γωνίᾳ· λέγω, ὅτι καὶ πλευρὰ ἡ AB πλευρᾷ τῇ
 $ΑΓ$ ἔστιν ἴση.

Let ABC be a triangle having the angle ABC equal to the
angle ACB . I say that side AB is also equal to side AC .



Εἰ γὰρ ἄνισός ἐστιν ἡ AB τῇ $ΑΓ$, ἡ ἑτέρα αὐτῶν μείζων
ἐστίν. ἔστω μείζων ἡ AB , καὶ ἀφρηθήσθω ἀπὸ τῆς μείζονος
τῆς AB τῇ ἐλάττω τῇ $ΑΓ$ ἴση ἡ $ΔB$, καὶ ἐπεξεύθθω ἡ $ΔΓ$.

For if AB is unequal to AC (then) one of them is greater.
Let AB be greater. And let DB , equal to the lesser AC , be
cut off from the greater AB [Prop. 1.3]. And let DC be joined
[Post. 1].

Ἐπεὶ οὖν ἴση ἐστὶν ἡ $ΔB$ τῇ $ΑΓ$ κοινῇ δὲ ἡ $BΓ$, δύο δὲ
αἱ $ΔB$, $BΓ$ δύο ταῖς $ΑΓ$, $ΓB$ ἴσαι εἰσὶν ἑκατέρᾳ ἑκατέρᾳ, καὶ
γωνία ἡ ὑπὸ $ΔBΓ$ γωνία τῇ ὑπὸ $ΑΓB$ ἔστιν ἴση· βάσει ἄρα
ἡ $ΔΓ$ βάσει τῇ AB ἴση ἐστίν, καὶ τὸ $ΔBΓ$ τρίγωνον τῷ $ΑΓB$
τρίγωνῳ ἴσον ἐσται, τὸ ἔλασσον τῷ μείζονι· ὅπερ ἄτοπον·
οὐκ ἄρα ἄνισός ἐστιν ἡ AB τῇ $ΑΓ$ · ἴση ἄρα.

Therefore, since DB is equal to AC , and BC (is) common,
the two sides DB , BC are equal to the two sides AC , CB , re-
spectively, and the angle DBC is equal to the angle ACB . Thus,
the base DC is equal to the base AB , and the triangle DBC
will be equal to the triangle ACB [Prop. 1.4], the lesser to the
greater. The very notion (is) absurd [C.N. 5]. Thus, AB is not
unequal to AC . Thus, (it is) equal.[†]

Ἐάν ἄρα τριγώνον αἱ δύο γωνίαι ἴσαι ἀλλήλαιξ ᾤσιν, καὶ
αἱ ὑπὸ τὰς ἴσαξ γωνίαξ ὑποτείνονσαι πλευραὶ ἴσαι ἀλλήλαιξ
έσονται· ὅπερ ἔδει δεῖχαι.

Thus, if a triangle has two angles equal to one another
(then) the sides subtending the equal angles will also be equal
to one another. (Which is) the very thing it was required to
show.

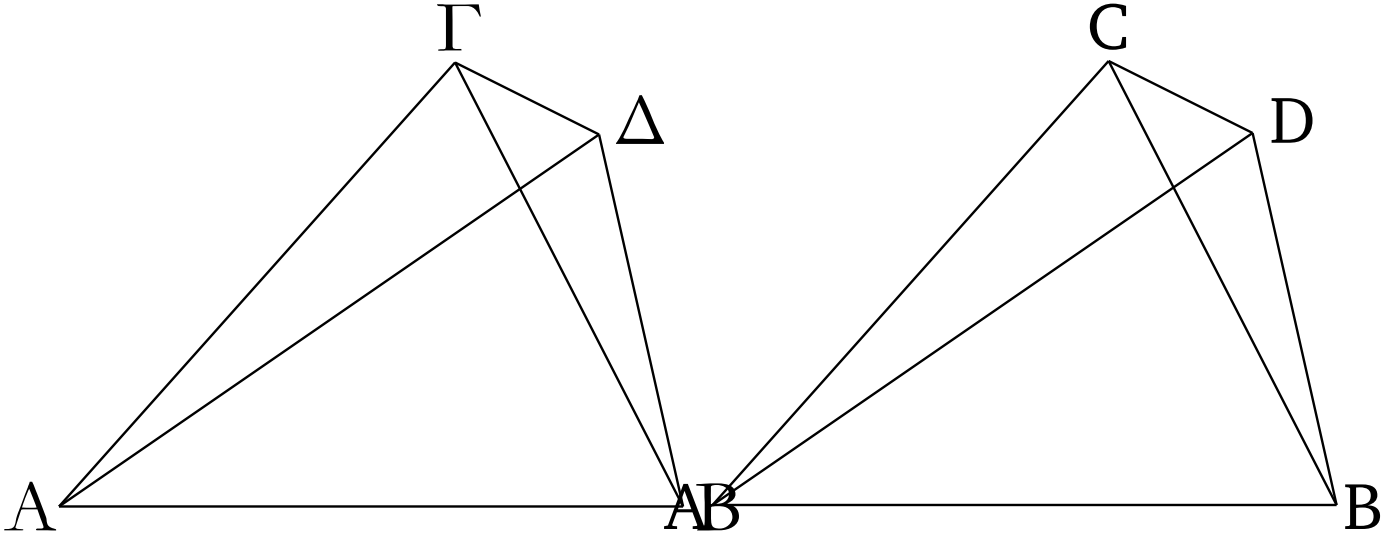
[†] Here, use is made of the previously unmentioned common notion that if two quantities are not unequal then they must be equal. Later on, use is made of the closely related common notion that if two quantities are not greater than or less than one another, respectively, then they must be equal to one another.

7.

Ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις ἄλλαι δύο εὐθεῖαι ἴσαι ἑκατέρα ἑκατέρῃ οὐ συνσταθήσονται πρὸς ἄλλῳ καὶ ἄλλῳ σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσαι ταῖς ἐξ ἀρθῆς εὐθείαις.

Proposition 7

On the same straight-line, two other straight-lines equal, respectively, to two (given) straight-lines (which meet) cannot be constructed (meeting) at a different point on the same side (of the straight-line), but having the same ends as the given straight-lines.



Εἰ γὰρ δυνατόν, ἐπὶ τῆς αὐτῆς εὐθείας τῆς AB δύο ταῖς αὐταῖς εὐθείαις ταῖς AG , GB ἄλλαι δύο εὐθεῖαι αἱ AD , DB ἴσαι ἑκατέρα ἑκατέρῃ συνεστάτωσαν πρὸς ἄλλῳ καὶ ἄλλῳ σημείῳ τῷ Γ καὶ Δ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσαι, ὥστε ἴσην εἶναι τὴν μὲν GA τῇ ΔA τὸ αὐτὸ πέραξ ἔθουσας αὐτῇ τὸ A , τὴν δὲ GB τῇ DB τὸ αὐτὸ πέραξ ἔθουσας αὐτῇ τὸ B , καὶ ἐπεξεύθθω ἡ GD .

Ἐπεὶ οὖν ἴση ἐστὶν ἡ AG τῇ AD , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ AGD τῇ ὑπὸ ADG . μείζων ἄρα ἡ ὑπὸ ADG τῆς ὑπὸ AGB . πολλῶν ἄρα ἡ ὑπὸ ADB μείζων ἐστὶ τῆς ὑπὸ AGB . πάλιν ἐπεὶ ἴση ἐστὶν ἡ GB τῇ DB , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ ADB γωνία τῇ ὑπὸ AGB . ἐδείχθη δὲ αὐτῆς καὶ πολλῶν μείζων· ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις ἄλλαι δύο εὐθεῖαι ἴσαι ἑκατέρα ἑκατέρῃ συνσταθήσονται πρὸς ἄλλῳ καὶ ἄλλῳ σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσαι ταῖς ἐξ ἀρθῆς εὐθείαις· ὅπερ ἔδει δεῖχαι.

For, if possible, let the two straight-lines AC , CB , equal to two other straight-lines AD , DB , respectively, be constructed on the same straight-line AB , meeting at different points, C and D , on the same side (of AB), and having the same ends (on AB). So CA is equal to DA , having the same end A as it, and CB is equal to DB , having the same end B as it. And let CD be joined [Post. 1].

Therefore, since AC is equal to AD , the angle ACD is also equal to angle ADC [Prop. 1.5]. Thus, ADC (is) greater than DCB [C.N. 5]. Thus, CDB is much greater than DCB [C.N. 5]. Again, since CB is equal to DB , the angle CDB is also equal to angle DCB [Prop. 1.5]. But it was shown that the former (angle) is also much greater (than the latter). The very thing is impossible.

Thus, on the same straight-line, two other straight-lines equal, respectively, to two (given) straight-lines (which meet) cannot be constructed (meeting) at a different point on the same side (of the straight-line), but having the same ends as the given straight-lines. (Which is) the very thing it was required to show.

8.

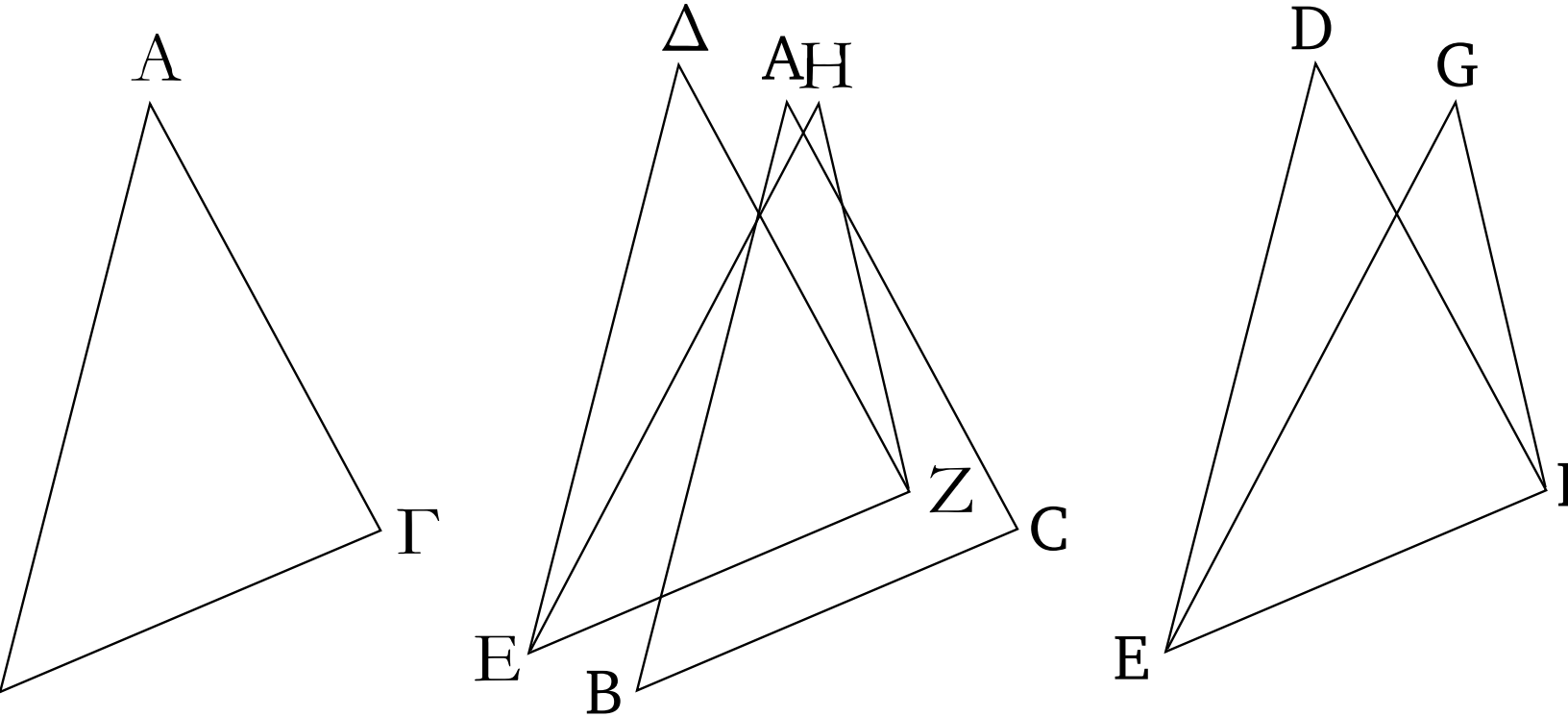
Ἐάν δύο τρίγωνα τὰς δύο πλευράς [ταῖς] δύο πλευραῖς ἴσαξ ἔθῃ ἑκατέραν ἑκατέρῃ, ἔθῃ δὲ καὶ τὴν βάσιν τῇ βάσει ἴσην, καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην.

Proposition 8

If two triangles have two sides equal to two sides, respectively, and also have the base equal to the base, (then) they will also have equal the angles encompassed by the equal straight-lines.

Ἐστω δύο τρίγωνα τὰ $ABΓ$, $ΔΕΖ$ τὰς δύο πλευρὰς τὰς AB , $ΑΓ$ ταῖς δύο πλευραῖς ταῖς $ΔΕ$, $ΔΖ$ ἴσας ἔχοντα ἑκατέραν ἑκατέρα, τὴν μὲν AB τῇ $ΔΕ$ τὴν δὲ $ΑΓ$ τῇ $ΔΖ$ · ἐθέτω δὲ καὶ βάσιν τὴν $ΒΓ$ βάσει τῇ $ΕΖ$ ἴσην· λέγω, ὅτι καὶ γωνία ἡ ὑπὸ $ΒΑΓ$ γωνία τῇ ὑπὸ $ΕΔΖ$ ἔστω ἴση.

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is) AB to DE , and AC to DF . Let them also have the base BC equal to the base EF . I say that the angle BAC is also equal to the angle EDF .



Ἐφαρμοζόμενον γάρ τοῦ $ABΓ$ τριγώνου ἐπὶ τὸ $ΔΕΖ$ τρίγωνον καὶ τιθεμένον τοῦ μὲν B σημείου ἐπὶ τὸ $Ε$ σημῖον τῆς δὲ $ΒΓ$ εὐθείας ἐπὶ τὴν $ΕΖ$ ἐφαρμόσει καὶ τὸ $Γ$ σημῖον ἐπὶ τὸ $Ζ$ διὰ τὸ ἴσην εἶναι τὴν $ΒΓ$ τῇ $ΕΖ$ · ἐφαρμοσάσης δὲ τῆς $ΒΓ$ ἐπὶ τὴν $ΕΖ$ ἐφαρμόσουσι καὶ αἱ BA , $ΓΑ$ ἐπὶ τὰς $ΕΔ$, $ΔΖ$. εἰ γὰρ βάσις μὲν ἡ $ΒΓ$ ἐπὶ βάσιν τὴν $ΕΖ$ ἐφαρμόσει, αἱ δὲ BA , $ΑΓ$ πλευραὶ ἐπὶ τὰς $ΕΔ$, $ΔΖ$ οὐκ ἐφαρμόσουσιν ἀλλὰ παραλλάχουσιν ὥς αἱ $ΕΗ$, $ΗΖ$, συσταθήσονται ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις ἄλλαι δύο εὐθεῖαι ἴσαι ἑκατέρα ἑκατέρῳ πρόξ ἄλλῳ καὶ ἄλλῳ σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔθουσαι. οὐ συνίστανται δέ· οὐκ ἴρα ἐφαρμοζομένης τῆς $ΒΓ$ βάσεως ἐπὶ τὴν $ΕΖ$ βάσιν οὐκ ἐφαρμόσουσι καὶ αἱ BA , $ΑΓ$ πλευραὶ ἐπὶ τὰς $ΕΔ$, $ΔΖ$. ἐφαρμόσουσιν ἴρα· ὥστε καὶ γωνία ἡ ὑπὸ $ΒΑΓ$ ἐπὶ γωνίαν τὴν ὑπὸ $ΕΔΖ$ ἐφαρμόσει καὶ ἴση αὐτῇ ἔσται.

Ἐὰν ἴρα δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρῳ καὶ τὴν βάσιν τῇ βάσει ἴσην ἔθῃ, καὶ τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην· ὅπερ ἔδει δεῖχαι.

For if triangle ABC is applied to triangle DEF , the point B being placed on point E , and the straight-line BC on EF , (then) point C will also coincide with F , on account of BC being equal to EF . So (because of) BC coinciding with EF , (the sides) BA and CA will also coincide with ED and DF (respectively). For if base BC coincides with base EF , but the sides AB and AC do not coincide with ED and DF (respectively), but miss like EG and GF (in the above figure), then we will have constructed upon the same straight-line two other straight-lines equal, respectively, to two (given) straight-lines, and (meeting) at a different point on the same side (of the straight-line), but having the same ends. But (such straight-lines) cannot be constructed [Prop. 1.7]. Thus, the base BC being applied to the base EF , the sides BA and AC cannot not coincide with ED and DF (respectively). Thus, they will coincide. So the angle BAC will also coincide with angle EDF , and will be equal to it [C.N. 4].

Thus, if two triangles have two sides equal to two side, respectively, and have the base equal to the base, (then) they will also have equal the angles encompassed by the equal straight-lines. (Which is) the very thing it was required to show.

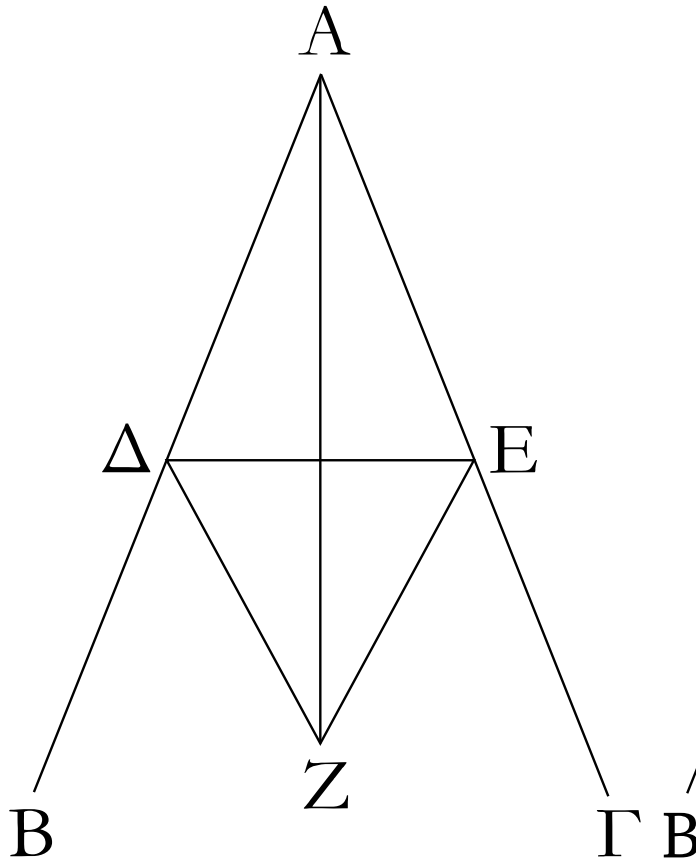
9.

Τὴν δοθεῖσαν γωνίαν εὐθύγραμμον δίῃ τεμεῖν.

Ἐστω ἡ δοθεῖσα γωνία εὐθύγραμμος ἡ ὑπὸ BAG . δεῖ δὴ αὐτὴν δίῃ τεμεῖν.

Εἰλήφθω ἐπὶ τῆς AB τυθὸν σημεῖον τὸ Δ , καὶ ἀφηρήσθω ἀπὸ τῆς AG τῇ AD ἴση ἡ AE , καὶ ἐπεζεύθθω ἡ DE , καὶ συνεστάτω ἐπὶ τῆς DE τρίγωνον ἰσόπλευρον τὸ DEZ , καὶ ἐπεζεύθθω ἡ AZ . λέγω, ὅτι ἡ ὑπὸ BAG γωνία δίῃ τέμνεται ὑπὸ τῆς AZ εὐθείας.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ AD τῇ AE , κοινὴ δὲ ἡ AZ , δύο δὴ αἱ DA , AZ δυοὶ ταῖς EA , AZ ἴσαι εἰσὶν ἑκατέρω ἑκατέρω. καὶ βάσει ἡ DZ βάσει τῇ EZ ἴση ἐστὶν γωνία ἄρα ἡ ὑπὸ DAZ γωνία τῇ ὑπὸ EAZ ἴση ἐστίν.



Ἡ ἄρα δοθεῖσα γωνία εὐθύγραμμος ἡ ὑπὸ BAG δίῃ τέμνεται ὑπὸ τῆς AZ εὐθείας· ὅπερ ἔδει ποιῆσαι.

10.

Τὴν δοθεῖσαν εὐθεῖαν πεπερασμένην δίῃ τεμεῖν.

Ἐστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB . δεῖ δὴ τὴν AB εὐθεῖαν πεπερασμένην δίῃ τεμεῖν.

Συνεστάτω ἐπ' αὐτῆς τρίγωνον ἰσόπλευρον τὸ $AB\Gamma$, καὶ

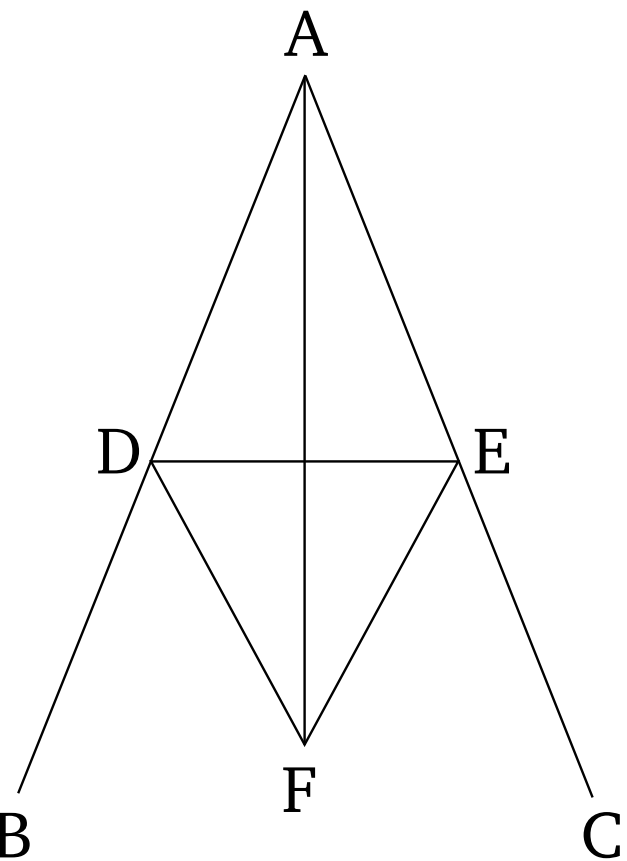
Proposition 9

To cut a given rectilinear angle in half.

Let BAC be the given rectilinear angle. So it is required to cut it in half.

Let the point D be taken at random on AB , and let AE , equal to AD , be cut off from AC [Prop. 1.3], and let DE be joined. And let the equilateral triangle DEF be constructed upon DE [Prop. 1.1], and let AF be joined. I say that the angle BAC has been cut in half by the straight-line AF .

For since AD is equal to AE , and AF is common, the two (straight-lines) DA , AF are equal to the two (straight-lines) EA , AF , respectively. And the base DF is equal to the base EF . Thus, angle DAF is equal to angle EAF [Prop. 1.8].



Thus, the given rectilinear angle BAC has been cut in half by the straight-line AF . (Which is) the very thing it was required to do.

Proposition 10

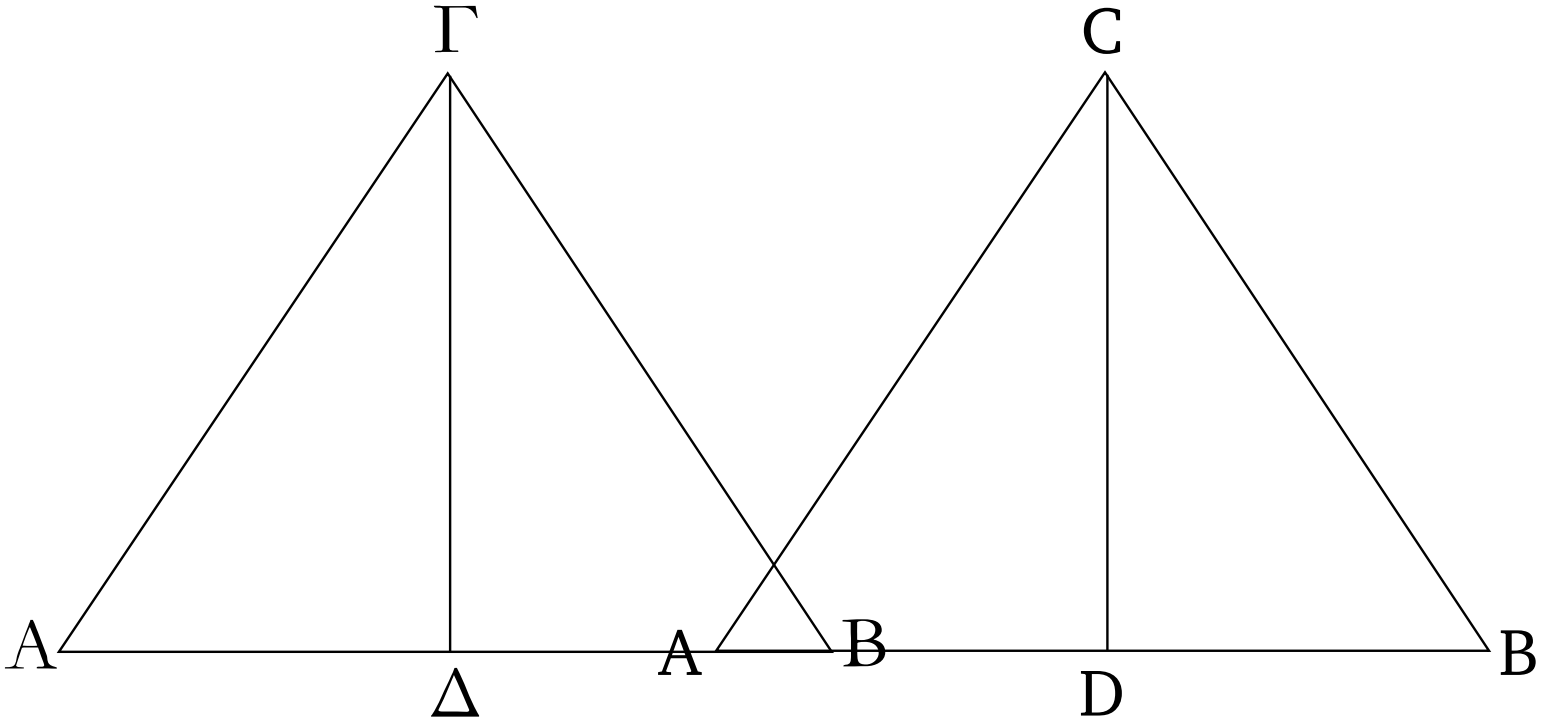
To cut a given finite straight-line in half.

Let AB be the given finite straight-line. So it is required to cut the finite straight-line AB in half.

Let the equilateral triangle ABC be constructed upon (AB)

τετμήσθω ἡ ὑπὸ AGB γωνία διὰ τῇ $ΓΔ$ εὐθείᾳ· λέγω, ὅτι ἡ AB εὐθεῖα διὰ τέτμηται κατὰ τὸ $Δ$ σημεῖον.

[Prop. 1.1], and let the angle ACB be cut in half by the straight-line CD [Prop. 1.9]. I say that the straight-line AB has been cut in half at point D .



Ἐπεὶ γάρ ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΓΒ$, κοινὴ δὲ ἡ $ΓΔ$, δύο δὴ αἱ $ΑΓ$, $ΓΔ$ δύο ταῖς $ΒΓ$, $ΓΔ$ ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνία ἡ ὑπὸ $ΑΓΔ$ γωνία τῇ ὑπὸ $ΒΓΔ$ ἴση ἐστὶν· βάσεις ἄρα ἡ $ΑΔ$ βάσει τῇ $ΒΔ$ ἴση ἐστὶν.

Ἦ ἄρα δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB διὰ τέτμηται κατὰ τὸ $Δ$ · ὅπερ ἔδει ποιῆσαι.

For since AC is equal to CB , and CD (is) common, the two (straight-lines) AC , CD are equal to the two (straight-lines) BC , CD , respectively. And the angle ACD is equal to the angle BCD . Thus, the base AD is equal to the base BD [Prop. 1.4].

Thus, the given finite straight-line AB has been cut in half at (point) D . (Which is) the very thing it was required to do.

11.

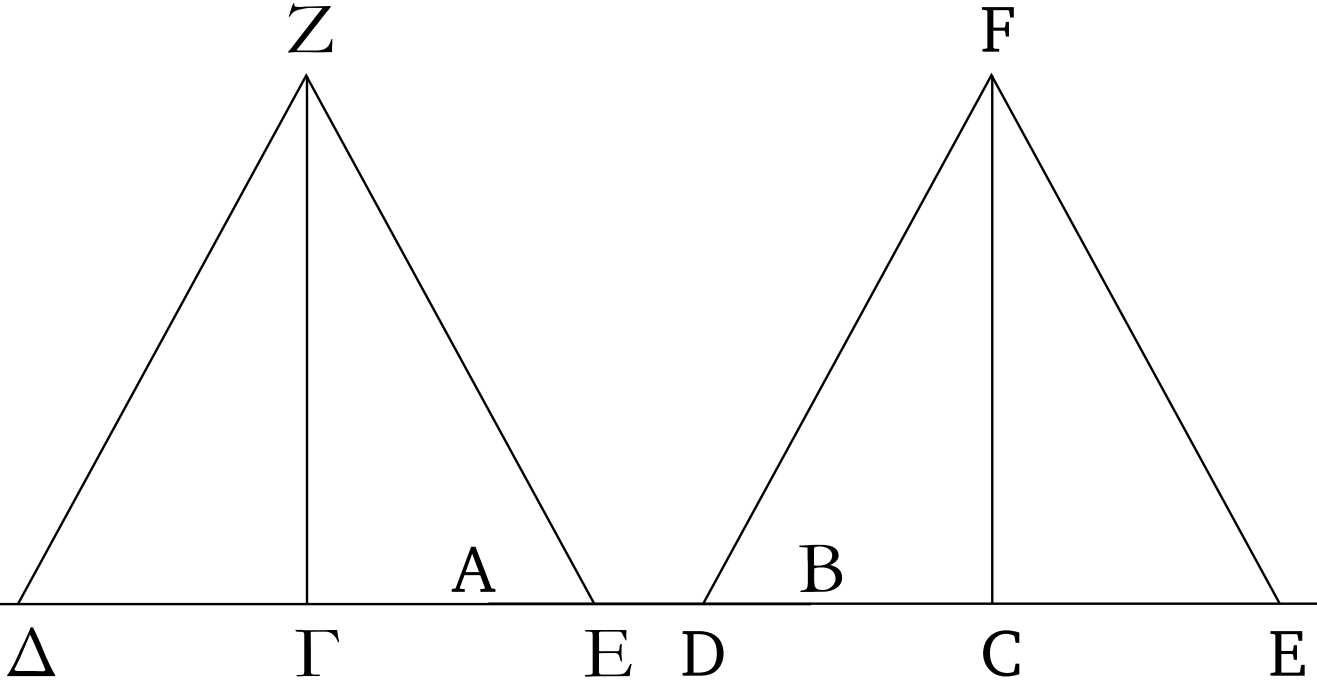
Proposition 11

Τῇ δοθείσῃ εὐθείᾳ ἀπὸ τοῦ πρὸς αὐτῇ δοθέντος σημείου πρὸς ὀρθὰς γωνία εὐθεῖαν γραμμὴν ἀγαγεῖν.

Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB τὸ δὲ δοθέν σημεῖον ἐπ' αὐτῇ τὸ $Γ$ · δεῖ δὲ ἀπὸ τοῦ $Γ$ σημείου τῇ AB εὐθείᾳ πρὸς ὀρθὰς γωνία εὐθεῖαν γραμμὴν ἀγαγεῖν.

To draw a straight-line at right-angles to a given straight-line from a given point on it.

Let AB be the given straight-line, and C the given point on it. So it is required to draw a straight-line from the point C at right-angles to the straight-line AB .



Εἰλήφθω ἐπὶ τῆς $ΑΓ$ τυθὸν σημείον τὸ Δ , καὶ κείσθω τῇ $Γ\Delta$ ἴση ἡ $ΓΕ$, καὶ συνεστάτω ἐπὶ τῆς $\Delta Ε$ τρίγωνον ἰσόπλευρον τὸ $Ζ\Delta Ε$, καὶ ἐπεζεύθθω ἡ $ΖΓ$. λέγω, ὅτι τῇ δοθείσῃ εὐθείᾳ τῇ $ΑΒ$ ἀπὸ τοῦ πρὸς αὐτῇ δοθέντος σημείου τοῦ $Γ$ πρὸς ὀρθὰς γωνίαξ εὐθεῖα γραμμὴ ἥκται ἡ $ΖΓ$.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ $\Delta Γ$ τῇ $ΓΕ$, κοινὴ δὲ ἡ $ΓΖ$, δύο δὴ αἱ $\Delta Γ$, $ΓΖ$ δυοὶ ταῖς $ΕΓ$, $ΓΖ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ βάσεις ἡ $\Delta Ζ$ βάσει τῇ $ΖΕ$ ἴση ἐστὶν· γωνία ἄρα ἡ ὑπὸ $\Delta Γ Ζ$ γωνία τῇ ὑπὸ $Ε Γ Ζ$ ἴση ἐστὶν· καὶ εἰσὶν ἐφεχῆς. ὅταν δὲ εὐθεῖα ἐπ' εὐθείαν σταθεῖσα τὰς ἐφεχῆς γωνίαξ ἴσαξ ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν ἴσων γωνιῶν ἐστὶν· ὀρθὴ ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ $\Delta Γ Ζ$, $Ζ Γ Ε$.

Τῇ ἄρα δοθείσῃ εὐθείᾳ τῇ $ΑΒ$ ἀπὸ τοῦ πρὸς αὐτῇ δοθέντος σημείου τοῦ $Γ$ πρὸς ὀρθὰς γωνίαξ εὐθεῖα γραμμὴ ἥκται ἡ $ΖΓ$. ὅπερ ἔδει ποιῆσαι.

Let the point D be taken at random on AC , and let CE be made equal to CD [Prop. 1.3], and let the equilateral triangle FDE be constructed on DE [Prop. 1.1], and let FC be joined. I say that the straight-line FC has been drawn at right-angles to the given straight-line AB from the given point C on it.

For since DC is equal to CE , and CF is common, the two (straight-lines) DC , CF are equal to the two (straight-lines), EC , CF , respectively. And the base DF is equal to the base FE . Thus, the angle DCF is equal to the angle ECF [Prop. 1.8], and they are adjacent. But when a straight-line stood on a(nother) straight-line makes the adjacent angles equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, each of the (angles) DCF and FCE is a right-angle.

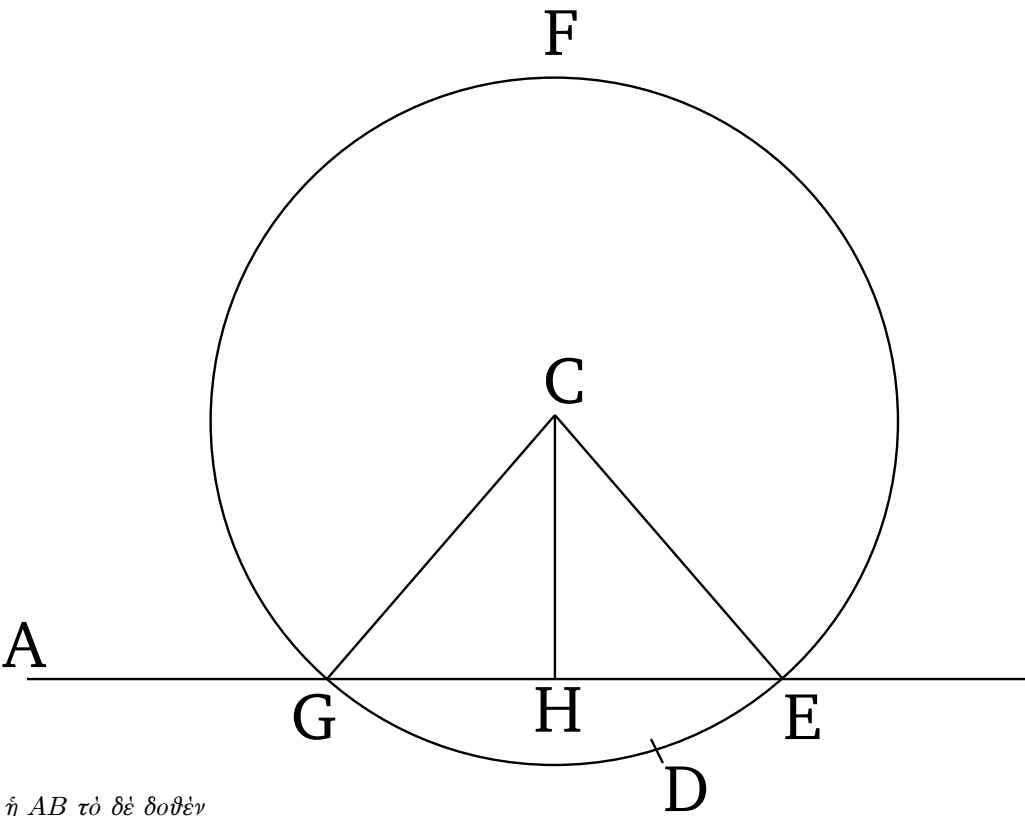
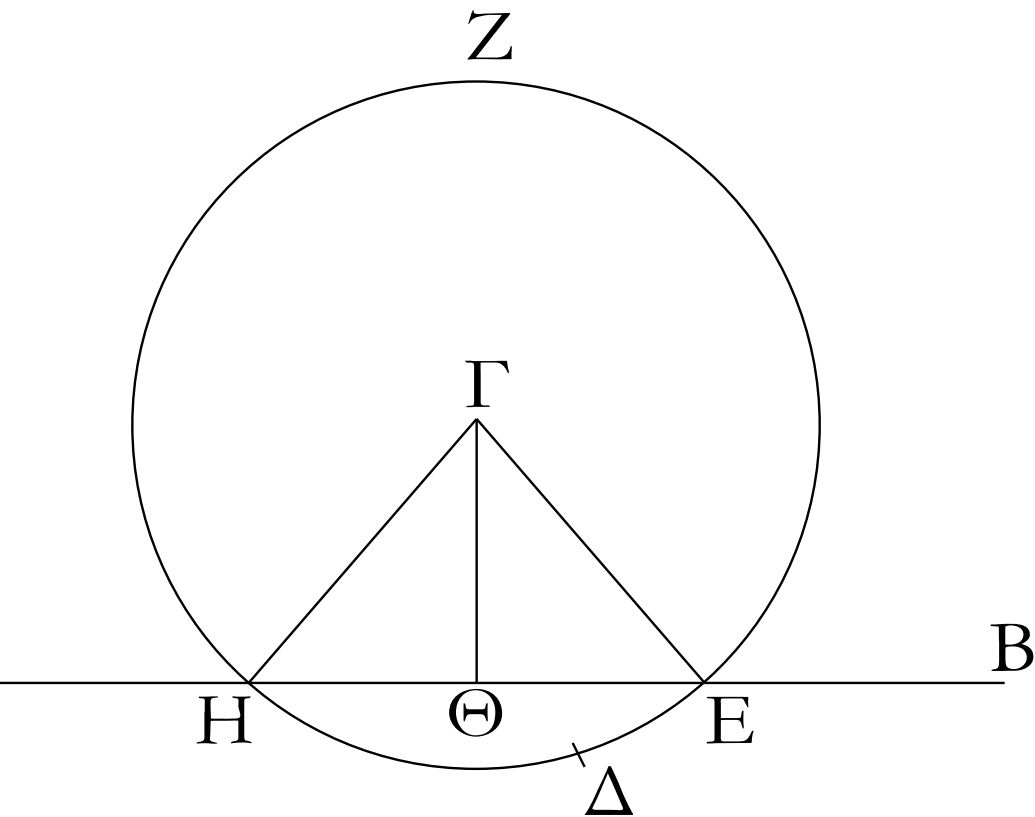
Thus, the straight-line CF has been drawn at right-angles to the given straight-line AB from the given point C on it. (Which is) the very thing it was required to do.

12.

Proposition 12

Ἐπὶ τὴν δοθεῖσαν εὐθεῖαν ἄπειρον ἀπὸ τοῦ δοθέντος σημείου, ὃ μὴ ἐστὶν ἐπ' αὐτῆς, κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

To draw a straight-line perpendicular to a given infinite straight-line from a given point which is not on it.



Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἀπειροξ ἡ AB τὸ δὲ δοθέν

σημείον, ὃ μὴ ἔστιν ἐπ' αὐτῆς, τὸ Γ . δεῖ δὲ ἐπὶ τὴν δοθεῖσαν εὐθεῖαν ἄπειρον τὴν AB ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ , ὃ μὴ ἔστιν ἐπ' αὐτῆς, κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Εἰλήφθω γάρ ἐπὶ τὰ ἑτέρα μέρη τῆς AB εὐθείας τυθὸν σημείον τὸ Δ , καὶ κέντρῳ μὲν τῷ Γ διαστήματι δὲ τῷ $\Gamma\Delta$ κύκλος γεγράφθω ὁ EZH , καὶ τετμήσθω ἡ EH εὐθεῖα διὰ κατὰ τὸ Θ , καὶ ἐπεξεύθθωσαν αἱ GH , $\Gamma\Theta$, ΓE εὐθεῖαι· λέγω, ὅτι ἐπὶ τὴν δοθεῖσαν εὐθεῖαν ἄπειρον τὴν AB ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ , ὃ μὴ ἔστιν ἐπ' αὐτῆς, κάθετος ἦκται ἡ $\Gamma\Theta$.

Ἐπεὶ γάρ ἴση ἐστὶν ἡ $H\Theta$ τῇ ΘE , κοινὴ δὲ ἡ $\Theta\Gamma$, δύο δὲ αἱ $H\Theta$, $\Theta\Gamma$ δύο ταῖς $E\Theta$, $\Theta\Gamma$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρῃ· καὶ βάσει ἡ GH βάσει τῇ ΓE ἔστιν ἴση· γωνία ἄρα ἡ ὑπὸ $\Gamma\Theta H$ γωνία τῇ ὑπὸ $E\Theta\Gamma$ ἔστιν ἴση. καὶ εἰσὶν ἐφεχῆς. ὅταν δὲ εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα τὰς ἐφεχῆς γωνίας ἴσαξ ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν ἴσων γωνιῶν ἔστιν, καὶ ἡ ἐφεστηκυῖα εὐθεῖα κάθετος καλεῖται ἐφ' ἣν ἐφέστηκεν.

Ἐπὶ τὴν δοθεῖσαν ἄρα εὐθεῖαν ἄπειρον τὴν AB ἀπὸ τοῦ δοθέντος σημείου τοῦ Γ , ὃ μὴ ἔστιν ἐπ' αὐτῆς, κάθετος ἦκται ἡ $\Gamma\Theta$. ὅπερ ἔδει ποιῆσαι.

Let AB be the given infinite straight-line and C the given point, which is not on it. So it is required to draw a straight-line perpendicular to the given infinite straight-line AB from the given point C , which is not on it.

For let point D be taken at random on the other side (to C) of the straight-line AB , and let the circle EFG be drawn with center C and radius CD [Post. 3], and let the straight-line EG be cut in half at (point) H [Prop. 1.10], and let the straight-lines CG , CH , and CE be joined. I say that the (straight-line) CH has been drawn perpendicular to the given infinite straight-line AB from the given point C , which is not on it.

For since GH is equal to HE , and HC (is) common, the two (straight-lines) GH , HC are equal to the two (straight-lines) EH , HC , respectively, and the base CG is equal to the base CE . Thus, the angle CHG is equal to the angle EHC [Prop. 1.8], and they are adjacent. But when a straight-line stood on a(nother) straight-line makes the adjacent angles equal to one another, each of the equal angles is a right-angle, and the former straight-line is called a perpendicular to that upon which it stands [Def. 1.10].

Thus, the (straight-line) CH has been drawn perpendicular to the given infinite straight-line AB from the given point C , which is not on it. (Which is) the very thing it was required to do.

13.

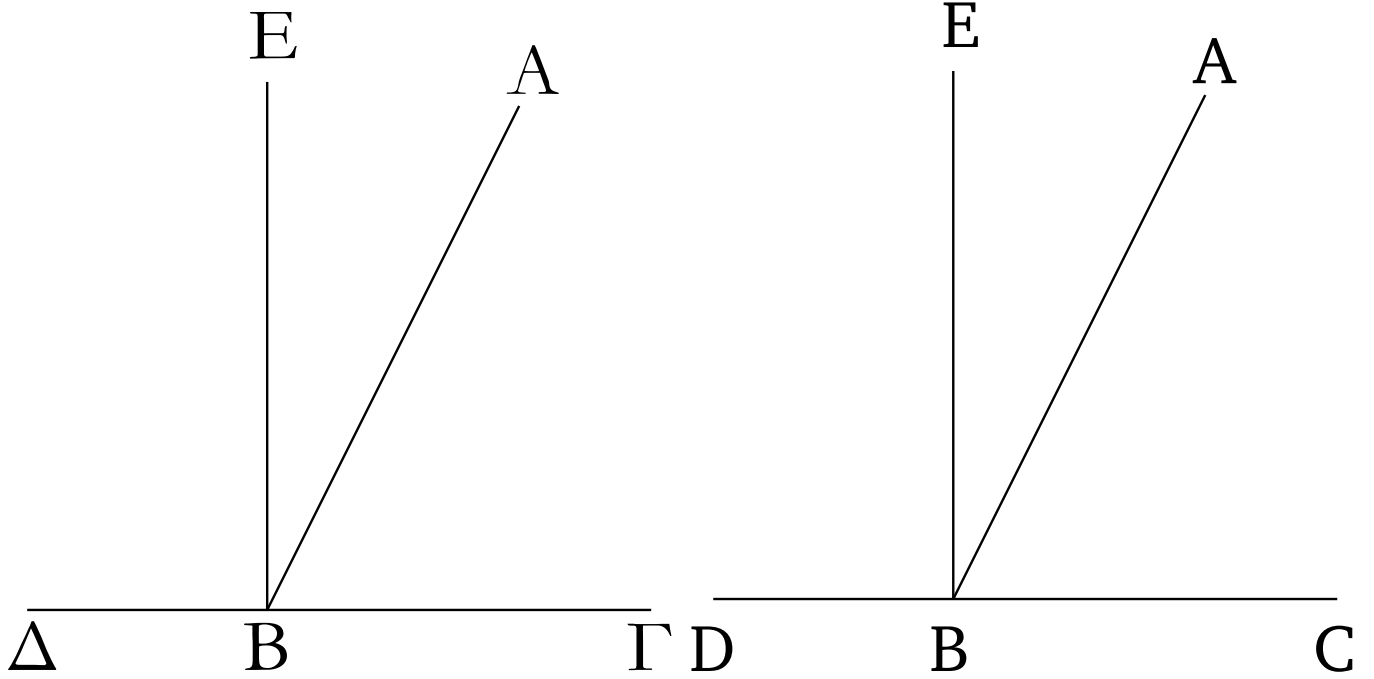
Proposition 13

Ἐὰν εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα γωνίας ποιῇ, ἦτοι δύο ὀρθὰς ἢ δυσὶν ὀρθαῖς ἴσαξ ποιήσῃ.

Εὐθεῖα γάρ τις ἡ AB ἐπ' εὐθεῖαν τὴν $\Gamma\Delta$ σταθεῖσα γωνίας ποιείτω τὰς ὑπὸ $\Gamma B A$, $A B \Delta$ · λέγω, ὅτι αἱ ὑπὸ $\Gamma B A$, $A B \Delta$ γωνίαι ἦτοι δύο ὀρθαὶ εἰσὶν ἢ δυσὶν ὀρθαῖς ἴσαι.

If a straight-line stood on a(nother) straight-line makes angles, it will certainly either make two right-angles, or (angles whose sum is) equal to two right-angles.

For let some straight-line AB stood on the straight-line CD make the angles CBA and ABD . I say that the angles CBA and ABD are certainly either two right-angles, or (have a sum) equal to two right-angles.



Εἰ μὲν οὖν ἴση ἐστὶν ἡ ὑπὸ ΓΒΑ τῇ ὑπὸ ΑΒΔ, δύο ὀρθαὶ εἰσιν. εἰ δὲ οὐκ, ἤθθω ἀπὸ τοῦ Β σημείου τῇ ΓΔ [εὐθείᾳ] πρὸς ὀρθὰς ἡ ΒΕ· αἱ ἄρα ὑπὸ ΓΒΕ, ΕΒΔ δύο ὀρθαὶ εἰσιν· καὶ ἐπεὶ ἡ ὑπὸ ΓΒΕ δυοὶ ταῖς ὑπὸ ΓΒΑ, ΑΒΕ ἴση ἐστίν, κοινὴ προσκείσθω ἡ ὑπὸ ΕΒΔ· αἱ ἄρα ὑπὸ ΓΒΕ, ΕΒΔ τρισὶ ταῖς ὑπὸ ΓΒΑ, ΑΒΕ, ΕΒΔ ἴσαι εἰσιν. πάλιν, ἐπεὶ ἡ ὑπὸ ΔΒΑ δυοὶ ταῖς ὑπὸ ΔΒΕ, ΕΒΑ ἴση ἐστίν, κοινὴ προσκείσθω ἡ ὑπὸ ΑΒΓ· αἱ ἄρα ὑπὸ ΔΒΑ, ΑΒΓ τρισὶ ταῖς ὑπὸ ΔΒΕ, ΕΒΑ, ΑΒΓ ἴσαι εἰσιν. ἐδείκθησαν δὲ καὶ αἱ ὑπὸ ΓΒΕ, ΕΒΔ τρισὶ ταῖς ἀνταῖς ἴσαι· τὰ δὲ τῶ ἀντῶ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα· καὶ αἱ ὑπὸ ΓΒΕ, ΕΒΔ ἄρα ταῖς ὑπὸ ΔΒΑ, ΑΒΓ ἴσαι εἰσιν· ἀλλὰ αἱ ὑπὸ ΓΒΕ, ΕΒΔ δύο ὀρθαὶ εἰσιν· καὶ αἱ ὑπὸ ΔΒΑ, ΑΒΓ ἄρα δυοὶν ὀρθαῖς ἴσαι εἰσιν.

Ἐὰν ἄρα εὐθεῖα ἐπ' εὐθείαν σταθεῖσα γωνίας ποιῇ, ἥτοι δύο ὀρθὰς ἢ δυοὶν ὀρθαῖς ἴσας ποιήσῃ· ὅπερ ἔδει δεῖξαι.

In fact, if CBA is equal to ABD (then) they are two right-angles [Def. 1.10]. But, if not, let BE be drawn from the point B at right-angles to [the straight-line] CD [Prop. 1.11]. Thus, CBE and EBD are two right-angles. And since CBE is equal to the two (angles) CBA and ABE , let EBD be added to both. Thus, the (sum of the angles) CBE and EBD is equal to the (sum of the) three (angles) CBA , ABE , and EBD [C.N. 2]. Again, since DBA is equal to the two (angles) DBE and EBA , let ABC be added to both. Thus, the (sum of the angles) DBA and ABC is equal to the (sum of the) three (angles) DBE , EBA , and ABC [C.N. 2]. But (the sum of) CBE and EBD was also shown (to be) equal to the (sum of the) same three (angles). And things equal to the same thing are also equal to one another [C.N. 1]. Therefore, (the sum of) CBE and EBD is also equal to (the sum of) DBA and ABC . But, (the sum of) CBE and EBD is two right-angles. Thus, (the sum of) ABD and ABC is also equal to two right-angles.

Thus, if a straight-line stood on a(nother) straight-line makes angles, it will certainly either make two right-angles, or (angles whose sum is) equal to two right-angles. (Which is) the very thing it was required to show.

14.

Ἐὰν πρὸς τινι εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ δύο εὐθεῖαι μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δυοὶν ὀρθαῖς ἴσας ποιῶσιν, ἐπ' εὐθείας ἔσσονται ἀλλήλαις αἱ εὐθεῖαι.

Πρὸς γάρ τινι εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ B δύο εὐθεῖαι αἱ BF , BD μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας τὰς ὑπὸ ABF , ABD δύο ὀρθαῖς ἴσας

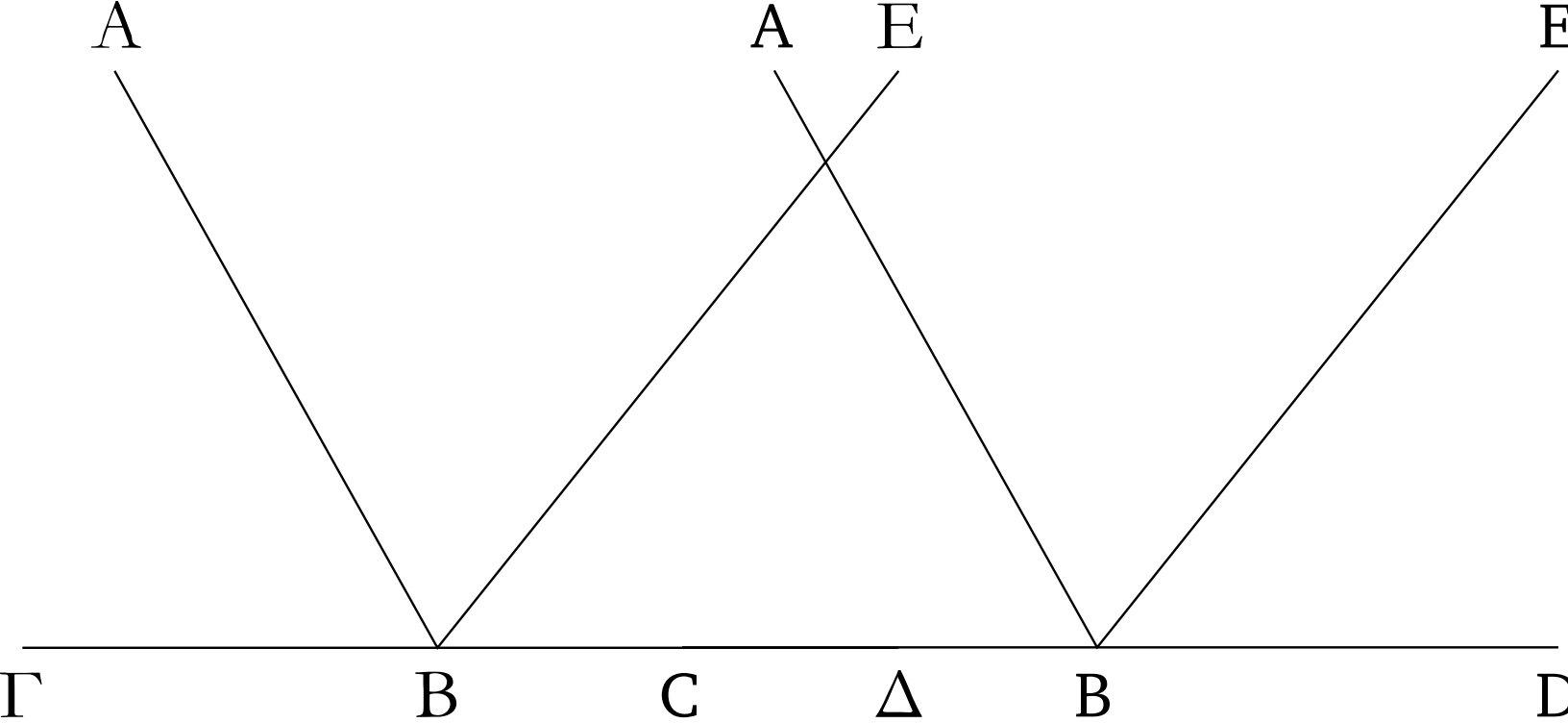
Proposition 14

If two straight-lines, not lying on the same side, make adjacent angles (whose sum is) equal to two right-angles with some straight-line, at a point on it, (then) the two straight-lines will be straight-on (with respect) to one another.

For let two straight-lines BC and BD , not lying on the same side, make adjacent angles ABC and ABD (whose sum

ποιείτωσαν· λέγω, ὅτι ἐπ' εὐθείας ἐστὶ τῇ $ΓΒ$ ἢ $ΒΔ$.

is) equal to two right-angles with some straight-line AB , at the point B on it. I say that BD is straight-on with respect to CB .



Εἰ γὰρ μὴ ἐστὶ τῇ $ΒΓ$ ἐπ' εὐθείας ἡ $ΒΔ$, ἔστω τῇ $ΓΒ$ ἐπ' εὐθείας ἡ $ΒΕ$.

Ἐπεὶ οὖν εὐθεῖα ἡ $ΑΒ$ ἐπ' εὐθείαν τὴν $ΓΒΕ$ ἐφέστηκεν, αἱ ἄρα ὑπὸ $ΑΒΓ$, $ΑΒΕ$ γωνίαι δύο ὀρθαῖς ἴσαι εἰσὶν· εἰσὶ δὲ καὶ αἱ ὑπὸ $ΑΒΓ$, $ΑΒΔ$ δύο ὀρθαῖς ἴσαι· αἱ ἄρα ὑπὸ $ΓΒΑ$, $ΑΒΕ$ ταῖς ὑπὸ $ΓΒΑ$, $ΑΒΔ$ ἴσαι εἰσὶν. κοινὴ ἀφηρήσθω ἡ ὑπὸ $ΓΒΑ$ · λοιπὴ ἄρα ἡ ὑπὸ $ΑΒΕ$ λοιπὴ τῇ ὑπὸ $ΑΒΔ$ ἐστὶν ἴση, ἡ ἐλάσσων τῇ μείζονι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐπ' εὐθείας ἐστὶν ἡ $ΒΕ$ τῇ $ΓΒ$. ὁμοίως δὲ δείχμεν, ὅτι οὐδὲ ἄλλη τις πλὴν τῆς $ΒΔ$ ἐπ' εὐθείας ἄρα ἐστὶν ἡ $ΓΒ$ τῇ $ΒΔ$.

Ἐάν ἄρα πρόξ τινι εὐθείᾳ καὶ τῷ πρόξ αὐτῇ σημείῳ δύο εὐθεῖαι μὴ ἐπὶ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δυσὶν ὀρθαῖς ἴσας ποιῶσιν, ἐπ' εὐθείας ἔσσονται ἀλλήλαις αἱ εὐθεῖαι· ὅπερ ἔδει δεῖχαι.

For if BD is not straight-on to BC (then) let BE be straight-on to CB .

Therefore, since the straight-line AB stands on the straight-line CBE , the (sum of the) angles ABC and ABE is thus equal to two right-angles [Prop. 1.13]. But (the sum of) ABC and ABD is also equal to two right-angles. Thus, (the sum of angles) CBA and ABE is equal to (the sum of angles) CBA and ABD [C.N. 1]. Let (angle) CBA be subtracted from both. Thus, the remainder ABE is equal to the remainder ABD [C.N. 3], the lesser to the greater. The very thing is impossible. Thus, BE is not straight-on with respect to CB . Similarly, we can show that neither (is) any other (straight-line) than BD . Thus, CB is straight-on with respect to BD .

Thus, if two straight-lines, not lying on the same side, make adjacent angles (whose sum is) equal to two right-angles with some straight-line, at a point on it, (then) the two straight-lines will be straight-on (with respect) to one another. (Which is) the very thing it was required to show.

15.

Proposition 15

Ἐάν δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὰς κατὰ κορυφὴν γωνίας ἴσας ἀλλήλαις ποιῶσιν.

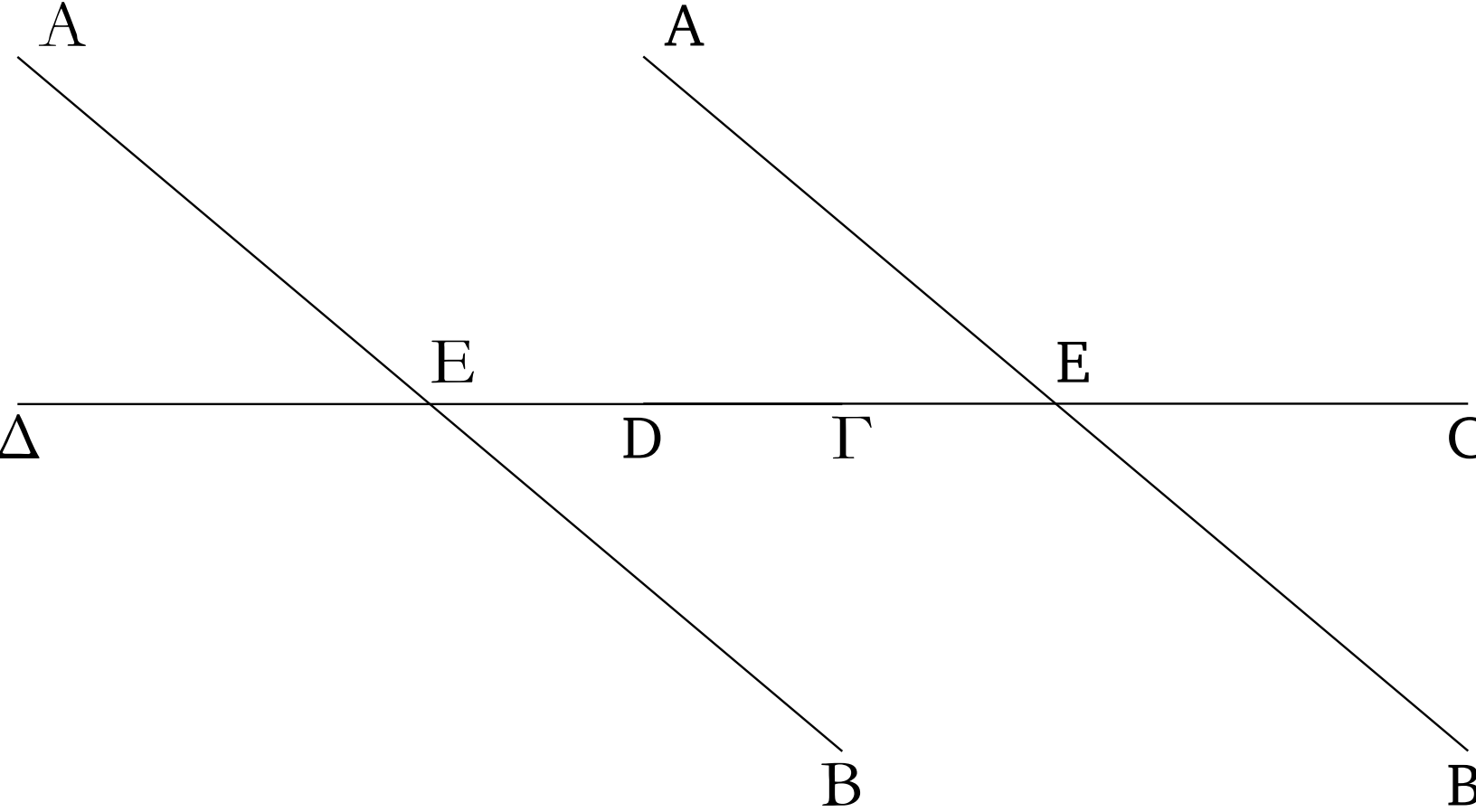
Δύο γὰρ εὐθεῖαι αἱ $ΑΒ$, $ΓΔ$ τεμνέτωσαν ἀλλήλας κατὰ τὸ $Ε$ σημεῖον· λέγω, ὅτι ἴση ἐστὶν ἡ μὲν ὑπὸ $ΑΕΓ$ γωνία τῇ ὑπὸ

If two straight-lines cut one another (then) they make the vertically opposite angles equal to one another.

For let the two straight-lines AB and CD cut one another at the point E . I say that angle AEC is equal to (angle) DEB ,

ΔEB , ἡ δὲ ὑπὸ ΓEB τῇ ὑπὸ $AE\Delta$.

and (angle) CEB to (angle) AED .



Ἐπεὶ γὰρ εὐθεῖα ἡ AE ἐπ' εὐθείαν τὴν $\Gamma\Delta$ ἐφέστηκε γωνίαξ ποιοῦσα τὰς ὑπὸ ΓEA , $AE\Delta$, αἱ ἄρα ὑπὸ ΓEA , $AE\Delta$ γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν. πάλιν, ἐπεὶ εὐθεῖα ἡ DE ἐπ' εὐθείαν τὴν AB ἐφέστηκε γωνίαξ ποιοῦσα τὰς ὑπὸ $AE\Delta$, ΔEB , αἱ ἄρα ὑπὸ $AE\Delta$, ΔEB γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν. ἐδείκθησαν δὲ καὶ αἱ ὑπὸ ΓEA , $AE\Delta$ δυοῖν ὀρθαῖς ἴσαι· αἱ ἄρα ὑπὸ ΓEA , $AE\Delta$ ταῖς ὑπὸ $AE\Delta$, ΔEB ἴσαι εἰσὶν. κοινὴ ἀφηρήσθω ἡ ὑπὸ $AE\Delta$ · λοιπὴ ἄρα ἡ ὑπὸ ΓEA λοιπῇ τῇ ὑπὸ $BE\Delta$ ἴση ἐστίν· ὁμοίως δὲ δευτέρησεται, ὅτι καὶ αἱ ὑπὸ ΓEB , ΔEA ἴσαι εἰσὶν.

Ἐὰν ἄρα δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὰς κατὰ κορυφὴν γωνίαξ ἴσας ἀλλήλαις ποιοῦσιν· ὅπερ ἔδει δεῖχαι.

For since the straight-line AE stands on the straight-line CD , making the angles CEA and AED , the (sum of the) angles CEA and AED is thus equal to two right-angles [Prop. 1.13]. Again, since the straight-line DE stands on the straight-line AB , making the angles AED and DEB , the (sum of the) angles AED and DEB is thus equal to two right-angles [Prop. 1.13]. But (the sum of) CEA and AED was also shown (to be) equal to two right-angles. Thus, (the sum of) CEA and AED is equal to (the sum of) AED and DEB [C.N. 1]. Let AED be subtracted from both. Thus, the remainder CEA is equal to the remainder DEB [C.N. 3]. Similarly, it can be shown that CEB and DEA are also equal.

Thus, if two straight-lines cut one another (then) they make the vertically opposite angles equal to one another. (Which is) the very thing it was required to show.

16.

Proposition 16

Παντὸς τριγώνου μιᾷ τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία ἐκατέραξ τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν.

Ἐστω τριγώνου τὸ $AB\Gamma$, καὶ προσεκβεβλήσθω αὐτοῦ μία

For any triangle, when one of the sides is produced, the external angle is greater than each of the internal and opposite angles.

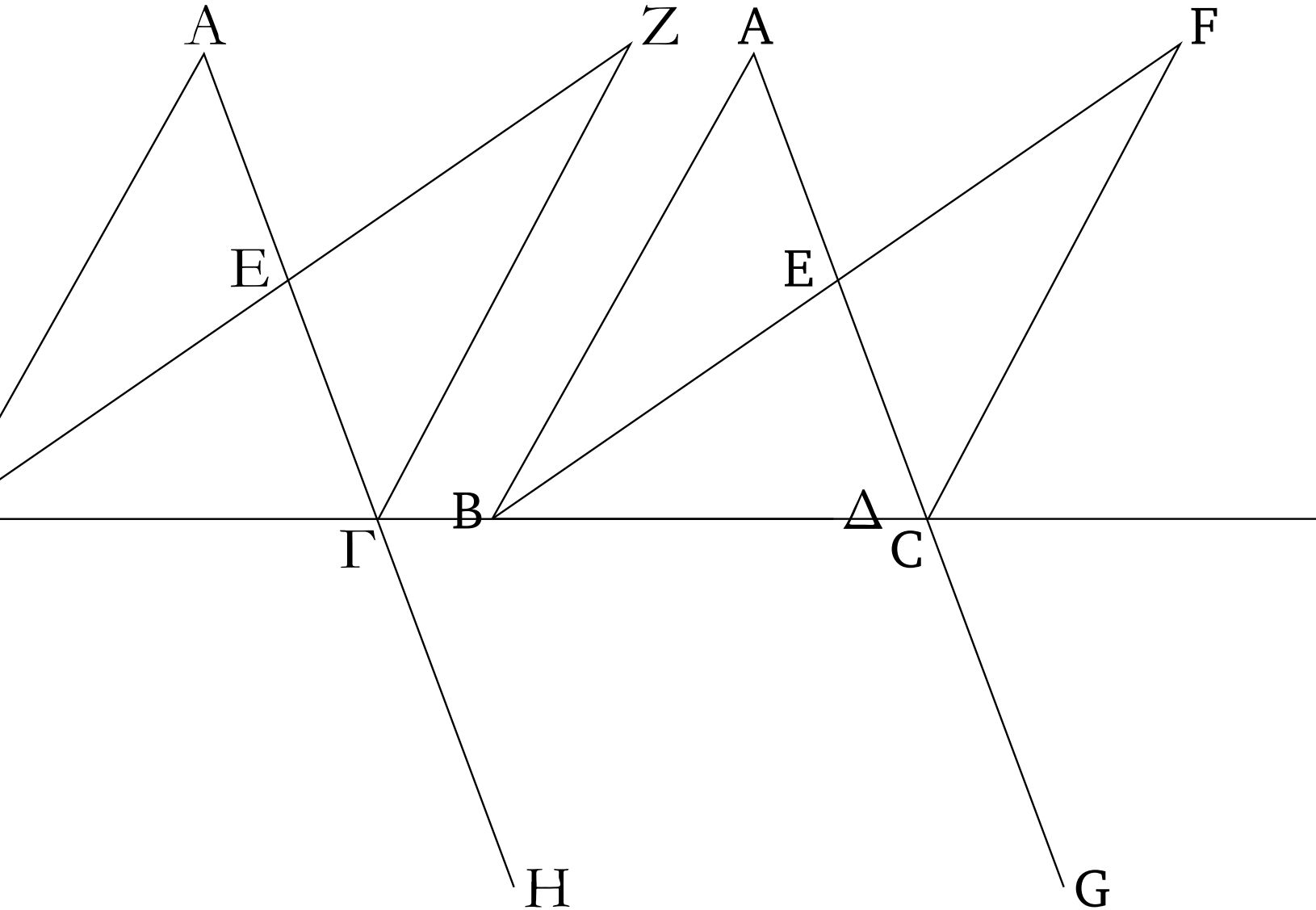
Let ABC be a triangle, and let one of its sides BC be pro-

πλευρά ἡ BF ἐπὶ τὸ Δ · λέγω, ὅτι ἡ ἐκτόξ γωνία ἡ ὑπὸ $ΑΓΔ$ μείζων ἐστὶν ἑκατέρᾳ τῶν ἐντὸς καὶ ἀπεναντίον τῶν ὑπὸ $ΓΒΑ$, $ΒΑΓ$ γωνιῶν.

Τετμήσθω ἡ $ΑΓ$ δίῃα κατὰ τὸ E , καὶ ἐπιζευθθεῖσα ἡ BE ἐκβεβλήσθω ἐπ' εὐθείᾳ ἐπὶ τὸ Z , καὶ κείσθω τῇ BE ἴση ἡ EZ , καὶ ἐπεξεύθθω ἡ $ZΓ$, καὶ διήθθω ἡ $ΑΓ$ ἐπὶ τὸ H .

duced to D . I say that the external angle ACD is greater than each of the internal and opposite angles, CBA and BAC .

Let the (straight-line) AC be cut in half at (point) E [Prop. 1.10]. And BE being joined, let it be produced in a straight-line to (point) F .[†] And let EF be made equal to BE [Prop. 1.3], and let FC be joined, and let AC be drawn through to (point) G .



Ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν AE τῇ EF , ἡ δὲ BE τῇ EZ , δύο δὴ αἱ AE , EB δυοὶ ταῖς FE , EZ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ AEB γωνία τῇ ὑπὸ ZEF ἴση ἐστὶν· κατὰ κορυφὴν γάρ· βάσεις ἄρα ἡ AB βάσει τῇ $ZΓ$ ἴση ἐστὶν, καὶ τὸ ABE τρίγωνον τῷ ZEF τριγώνῳ ἐστὶν ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἐστὶν ἡ ὑπὸ BAE τῇ ὑπὸ $EΓZ$. μείζων δὲ ἐστὶν ἡ ὑπὸ $ΕΓΔ$ τῇς ὑπὸ $ΕΓΖ$.

Therefore, since AE is equal to EC , and BE to EF , the two (straight-lines) AE , EB are equal to the two (straight-lines) CE , EF , respectively. Also, angle AEB is equal to angle FEC , for (they are) vertically opposite [Prop. 1.15]. Thus, the base AB is equal to the base FC , and the triangle ABE is equal to the triangle FEC , and the remaining angles subtended by the equal sides are equal to the corresponding remaining angles [Prop. 1.4]. Thus, BAE is equal to ECF . But ECD is greater

μείζων ἄρα ἢ ὑπὸ $ΑΓΔ$ τῆς ὑπὸ $ΒΑΕ$. Ὁμοίως δὴ τῆς $ΒΓ$ τετμημένης διῖθα δεινθήσεται καὶ ἡ ὑπὸ $ΒΓΗ$, τοντέστιν ἡ ὑπὸ $ΑΓΔ$, μείζων καὶ τῆς ὑπὸ $ΑΒΓ$.

Παντὸς ἄρα τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἢ ἐκτὸς γωνία ἐκατέρωξ τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν· ὅπερ ἔδει δεῖχαι.

† The implicit assumption that the point F lies in the interior of the angle ABC should be counted as an additional postulate.

17.

Παντὸν τριγώνον αἱ δύο γωνίαι δύο ὀρθῶν ἐλάσσονές εἰσι πάντῃ μεταλαμβανόμεναι.

Ἐστω τρίγωνον τὸ $ΑΒΓ$ · λέγω, ὅτι τοῦ $ΑΒΓ$ τριγώνου αἱ δύο γωνίαι δύο ὀρθῶν ἐλάττονές εἰσι πάντῃ μεταλαμβανόμεναι.

Ἐκβεβλήσθω γὰρ ἡ $ΒΓ$ ἐπὶ τὸ $Δ$.

Καὶ ἐπεὶ τριγώνον τοῦ $ΑΒΓ$ ἐκτὸς ἐστὶ γωνία ἢ ὑπὸ $ΑΓΔ$, μείζων ἐστὶ τῆς ἐντὸς καὶ ἀπεναντίον τῆς ὑπὸ $ΑΒΓ$. κοινὴ προσκείσθω ἡ ὑπὸ $ΑΓΒ$ · αἱ ἄρα ὑπὸ $ΑΓΔ$, $ΑΓΒ$ τῶν ὑπὸ $ΑΒΓ$, $ΒΓΑ$ μείζονές εἰσιν. ἀλλ' αἱ ὑπὸ $ΑΓΔ$, $ΑΓΒ$ δύο ὀρθαῖς ἴσαι εἰσὶν· αἱ ἄρα ὑπὸ $ΑΒΓ$, $ΒΓΑ$ δύο ὀρθῶν ἐλάσσονές εἰσιν. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ ὑπὸ $ΒΑΓ$, $ΑΓΒ$ δύο ὀρθῶν ἐλάσσονές εἰσι καὶ ἔτι αἱ ὑπὸ $ΓΑΒ$, $ΑΒΓ$.

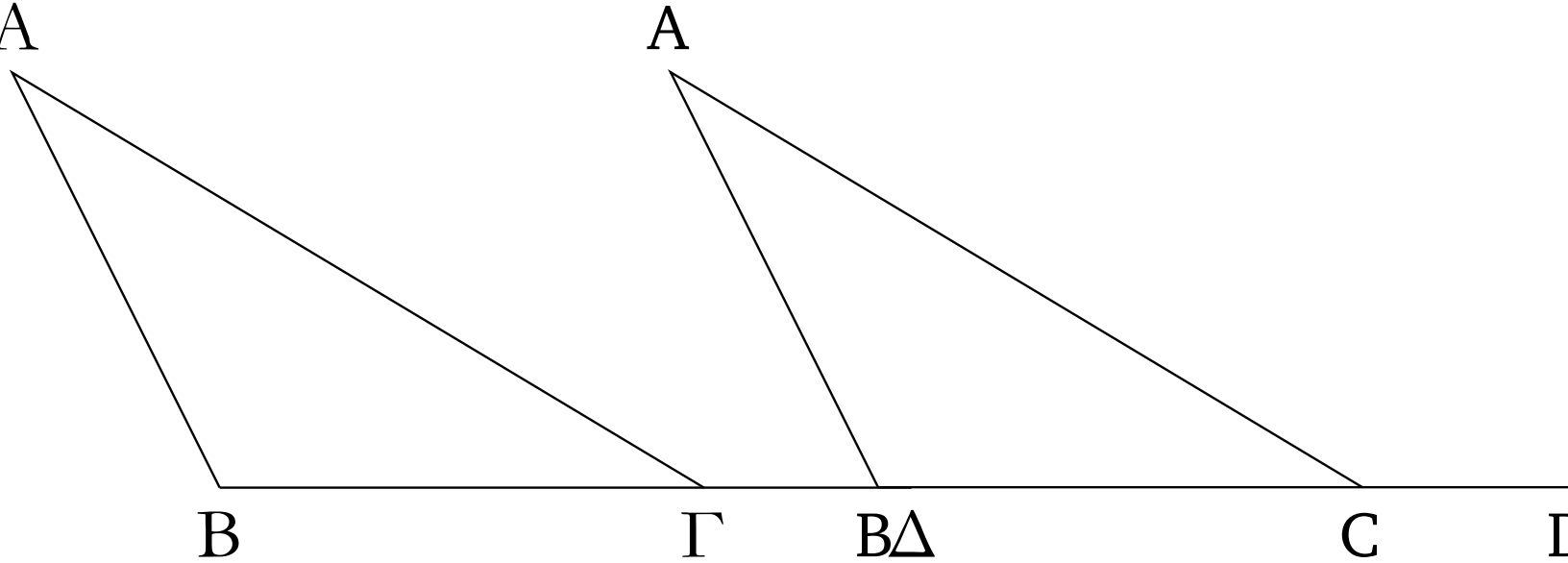
Proposition 17

For any triangle, (the sum of) two angles taken together in any (possible way) is less than two right-angles.

Let ABC be a triangle. I say that (the sum of) two angles of triangle ABC taken together in any (possible way) is less than two right-angles.

For let BC be produced to D .

And since the angle ACD is external to triangle ABC , it is greater than the internal and opposite angle ABC [Prop. 1.16]. Let ACB be added to both. Thus, the (sum of the angles) ACD and ACB is greater than the (sum of the angles) ABC and BCA . But, (the sum of) ACD and ACB is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) ABC and BCA is less than two right-angles. Similarly, we can show that (the sum of) BAC and ACB is also less than two right-angles, and further (that the sum of) CAB and ABC (is less than two right-angles).



Παντὸν ἄρα τριγώνον αἱ δύο γωνίαι δύο ὀρθῶν ἐλάσσονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖχαι.

Thus, for any triangle, (the sum of) two angles taken together in any (possible way) is less than two right-angles. (Which is) the very thing it was required to show.

18.

Proposition 18

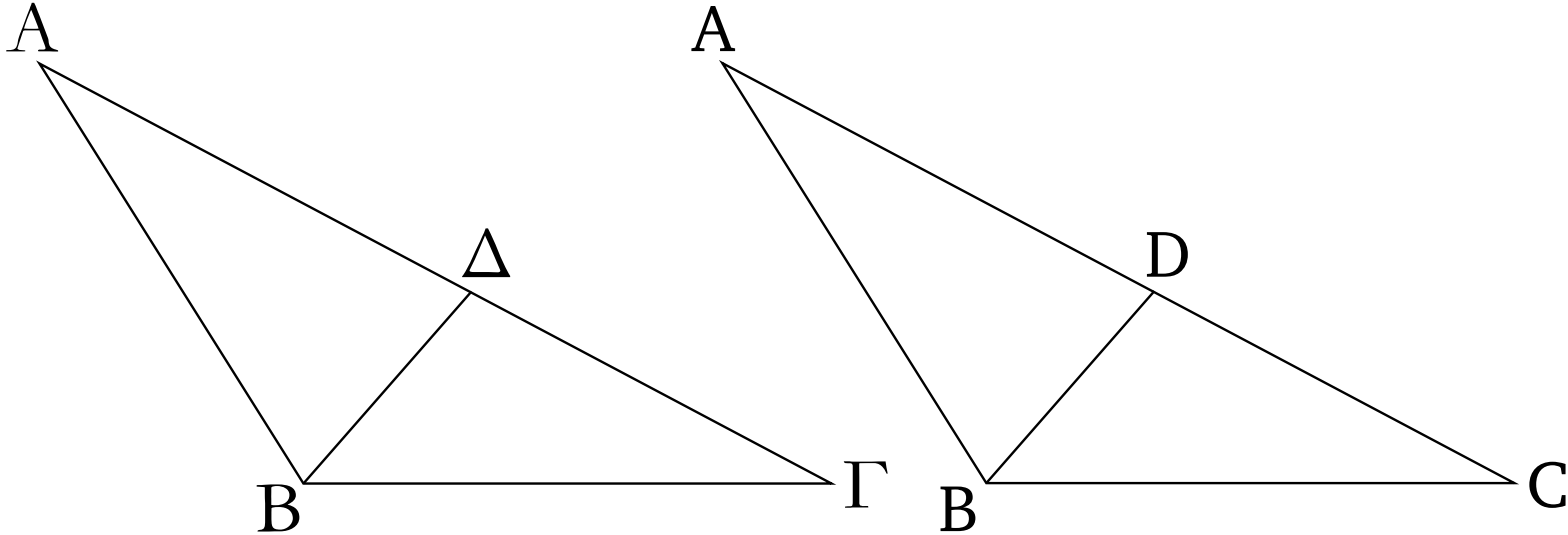
Παντὸς τριγώνου ἡ μείζων πλευρὰ τὴν μείζονα γωνίαν ὑπο-

In any triangle, the greater side subtends the greater angle.

τείνει.

Ἐστω γάρ τριγώνον τὸ $AB\Gamma$ μείζονα ἔθον τήν $ΑΓ$ πλευρὰν τῆς AB · λέγω, ὅτι καὶ γωνία ἡ ὑπὸ $AB\Gamma$ μείζων ἐστὶ τῆς ὑπὸ $B\Gamma A$.

For let ABC be a triangle having side AC greater than AB . I say that angle ABC is also greater than BCA .



Ἐπεὶ γὰρ μείζων ἐστὶν ἡ $ΑΓ$ τῆς AB , κείσθω τῇ AB ἴση ἡ $ΑΔ$, καὶ ἐπεζεύθθω ἡ $BΔ$.

Καὶ ἐπεὶ τριγώνον τοῦ $B\Gamma Δ$ ἐκτός ἐστι γωνία ἡ ὑπὸ $ΑΔΒ$, μείζων ἐστὶ τῆς ἐντός καὶ ἀπεναντίον τῆς ὑπὸ $ΔΓΒ$ · ἴση δὲ ἡ ὑπὸ $ΑΔΒ$ τῇ ὑπὸ $ABΔ$, ἐπεὶ καὶ πλευρὰ ἡ AB τῇ $ΑΔ$ ἐστὶν ἴση· μείζων ἄρα καὶ ἡ ὑπὸ $ABΔ$ τῆς ὑπὸ $ΑΓΒ$ · πολλῶν ἄρα ἡ ὑπὸ $AB\Gamma$ μείζων ἐστὶ τῆς ὑπὸ $ΑΓΒ$.

Παντὸς ἄρα τριγώνου ἡ μείζων πλευρὰ τὴν μείζονα γωνίαν ὑποτείνει· ὅπερ ἔδει δεῖξαι.

For since AC is greater than AB , let AD be made equal to AB [Prop. 1.3], and let BD be joined.

And since angle ADB is external to triangle BCD , it is greater than the internal and opposite (angle) DCB [Prop. 1.16]. But ADB (is) equal to ABD , since side AB is also equal to side AD [Prop. 1.5]. Thus, ABD is also greater than ACB . Thus, ABC is much greater than ACB .

Thus, in any triangle, the greater side subtends the greater angle. (Which is) the very thing it was required to show.

19.

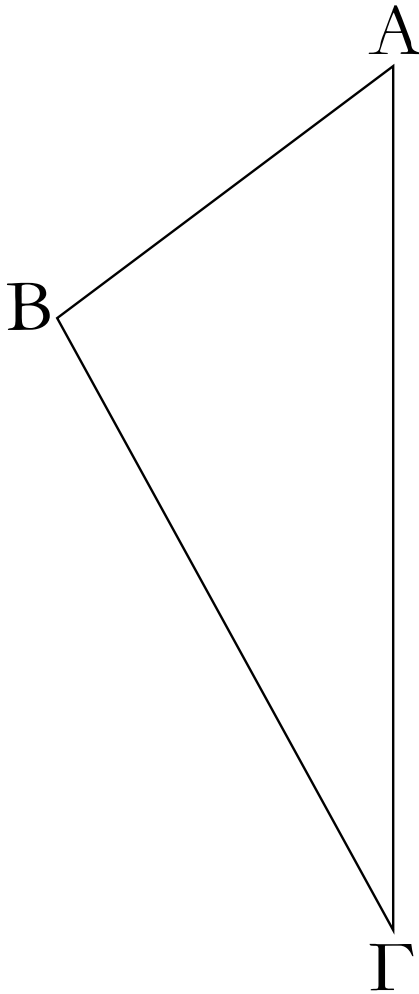
Proposition 19

Παντὸς τριγώνου ὑπὸ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει.

Ἐστω τριγώνον τὸ $AB\Gamma$ μείζονα ἔθον τὴν ὑπὸ $AB\Gamma$ γωνίαν τῆς ὑπὸ $B\Gamma A$ · λέγω, ὅτι καὶ πλευρὰ ἡ $ΑΓ$ πλευρᾷ τῆς AB μείζων ἐστίν.

In any triangle, the greater angle is subtended by the greater side.

Let ABC be a triangle having the angle ABC greater than BCA . I say that side AC is also greater than side AB .

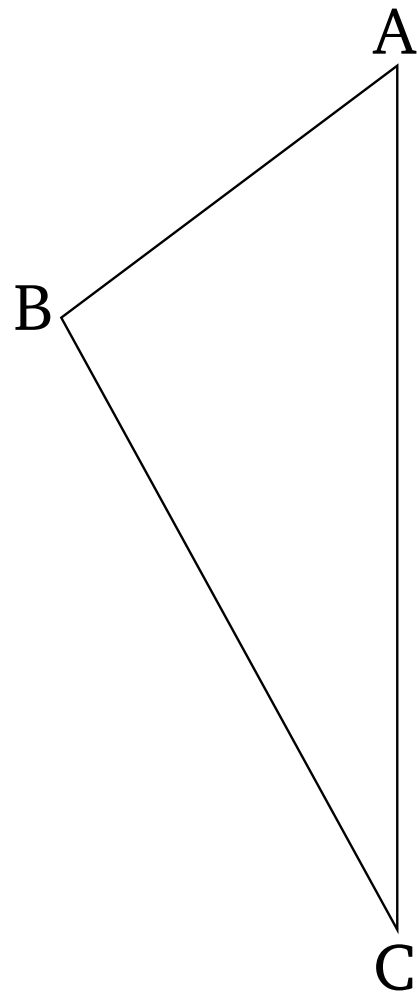


Εἰ γὰρ μή, ἤτοι ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$ ἢ ἐλάσσων· ἴση μὲν οὖν οὐκ ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$ · ἴση γὰρ ἂν ἦν καὶ γωνία ἡ ὑπὸ $ΑΒΓ$ τῇ ὑπὸ $ΑΓΒ$ · οὐκ ἐστὶ δέ· οὐκ ἄρα ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$. οὐδὲ μὴν ἐλάσσων ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$ · ἐλάσσων γὰρ ἂν ἦν καὶ γωνία ἡ ὑπὸ $ΑΒΓ$ τῇ ὑπὸ $ΑΓΒ$ · οὐκ ἐστὶ δέ· οὐκ ἄρα ἐλάσσων ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$. ἐδείκθη δέ, ὅτι οὐδὲ ἴση ἐστὶν. μείζων ἄρα ἐστὶν ἡ $ΑΓ$ τῇ $ΑΒ$.

Παντὸς ἄρα τριγώνου ὑπὸ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει· ὅπερ ἔδει δεῖξαι.

20.

Παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονός εἰσι πάντῃ μεταλαμβάνόμεναι.

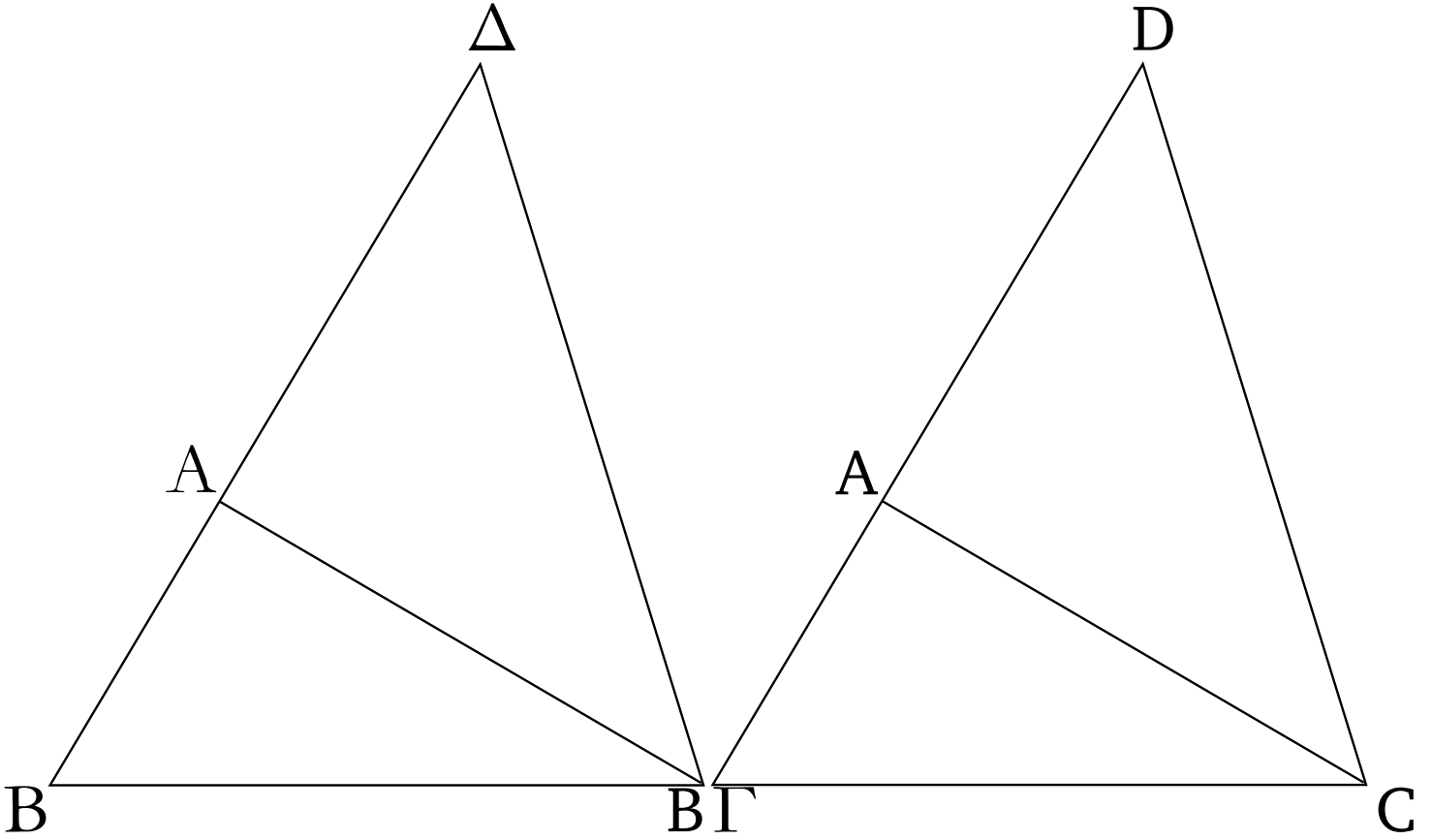


For if not, AC is certainly either equal to, or less than, AB . In fact, AC is not equal to AB . For then angle ABC would also be equal to ACB [Prop. 1.5]. But it is not. Thus, AC is not equal to AB . Neither, indeed, is AC less than AB . For then angle ABC would also be less than ACB [Prop. 1.18]. But it is not. Thus, AC is not less than AB . But it was shown that (AC) is not equal (to AB) either. Thus, AC is greater than AB .

Thus, in any triangle, the greater angle is subtended by the greater side. (Which is) the very thing it was required to show.

Proposition 20

In any triangle, (the sum of) two sides taken together in any (possible way) is greater than the remaining (side).



Ἐστω γάρ τριγώνον τὸ $AB\Gamma$. λέγω, ὅτι τοῦ $AB\Gamma$ τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν BA , $A\Gamma$ τῆς $B\Gamma$, αἱ δὲ AB , $B\Gamma$ τῆς $A\Gamma$, αἱ δὲ $B\Gamma$, ΓA τῆς AB .

Διήθηθω γάρ ἡ BA ἐπὶ τὸ Δ σημεῖον, καὶ κείσθω τῇ ΓA ἴση ἡ $A\Delta$, καὶ ἐπέξευθθω ἡ $\Delta\Gamma$.

Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΔA τῇ $A\Gamma$, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $A\Delta\Gamma$ τῇ ὑπὸ $A\Gamma\Delta$. μείζων ἄρα ἡ ὑπὸ $B\Gamma\Delta$ τῆς ὑπὸ $A\Delta\Gamma$. καὶ ἐπεὶ τριγώνον ἐστὶ τὸ $\Delta\Gamma B$ μείζονα ἔθον τὴν ὑπὸ $B\Gamma\Delta$ γωνίαν τῆς ὑπὸ $B\Delta\Gamma$, ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει, ἡ ΔB ἄρα τῆς $B\Gamma$ ἐστὶ μείζων. ἴση δὲ ἡ ΔA τῇ $A\Gamma$. μείζονες ἄρα αἱ BA , $A\Gamma$ τῆς $B\Gamma$. ὁμοίως δὲ δείχομεν, ὅτι καὶ αἱ μὲν AB , $B\Gamma$ τῆς ΓA μείζονές εἰσιν, αἱ δὲ $B\Gamma$, ΓA τῆς AB .

Παντὸς ἄρα τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι. ὅπερ ἔδει δείχαι.

For let ABC be a triangle. I say that in triangle ABC (the sum of) two sides taken together in any (possible way) is greater than the remaining (side). (So), (the sum of) BA and AC (is greater) than BC , (the sum of) AB and BC than AC , and (the sum of) BC and CA than AB .

For let BA be drawn through to point D , and let AD be made equal to CA [Prop. 1.3], and let DC be joined.

Therefore, since DA is equal to AC , the angle ADC is also equal to ACD [Prop. 1.5]. Thus, BCD is greater than ADC . And since DCB is a triangle having the angle BCD greater than BDC , and the greater angle subtends the greater side [Prop. 1.19], DB is thus greater than BC . But DA is equal to AC . Thus, (the sum of) BA and AC is greater than BC . Similarly, we can show that (the sum of) AB and BC is also greater than CA , and (the sum of) BC and CA than AB .

Thus, in any triangle, (the sum of) two sides taken together in any (possible way) is greater than the remaining (side). (Which is) the very thing it was required to show.

21.

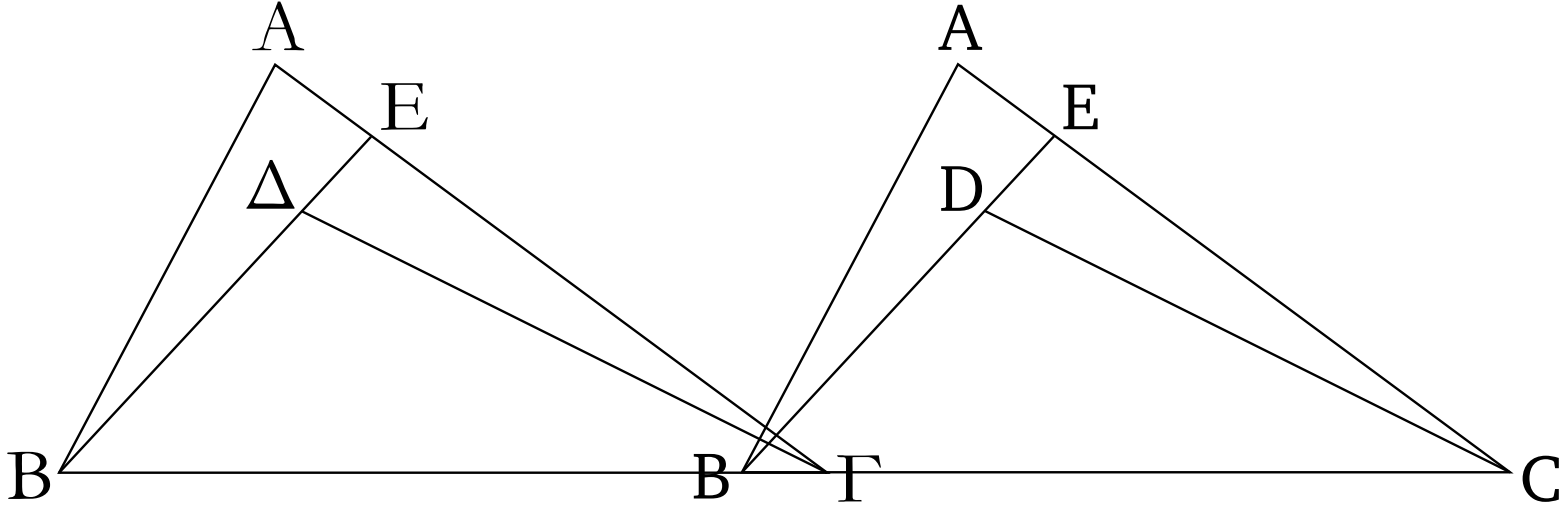
Proposition 21

Ἐάν τριγώνον ἐπὶ μᾶξ τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μὲν ἔσονται, μείζονα δὲ

If two internal straight-lines are constructed on one of the sides of a triangle, from its ends, the constructed (straight-lines) will be less than the two remaining sides of the triangle,

γωνίαν περιέχουσιν.

Τριγώνον γάρ τοῦ $ABΓ$ ἐπὶ μιᾷ τῶν πλευρῶν τῆς $ΒΓ$ ἀπὸ τῶν περάτων τῶν $B, Γ$ δύο εὐθεῖαι ἐντὸς συνεστάτωσαν αἱ $ΒΔ, ΔΓ$. λέγω, ὅτι αἱ $ΒΔ, ΔΓ$ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τῶν $ΒΑ, ΑΓ$ ἐλάσσονες μὲν εἰσιν, μείζονα δὲ γωνίαν περιέθουσι τὴν ὑπὸ $ΒΔΓ$ τῆς ὑπὸ $ΒΑΓ$.



Διήθηθω γάρ ἡ $ΒΔ$ ἐπὶ τὸ E . καὶ ἐπεὶ παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονες εἰσιν, τοῦ ABE ἄρα τριγώνου αἱ δύο πλευραὶ αἱ AB, AE τῆς BE μείζονες εἰσιν· κοινὴ προσκείσθω ἡ $ΕΓ$. αἱ ἄρα $BA, ΑΓ$ τῶν $BE, ΕΓ$ μείζονες εἰσιν. πάλιν, ἐπεὶ τοῦ $ΓΕΔ$ τριγώνου αἱ δύο πλευραὶ αἱ $ΓΕ, ΕΔ$ τῆς $ΓΔ$ μείζονες εἰσιν, κοινὴ προσκείσθω ἡ $ΔΒ$. αἱ $ΓΕ, ΕΒ$ ἄρα τῶν $ΓΔ, ΔΒ$ μείζονες εἰσιν. ἀλλὰ τῶν $BE, ΕΓ$ μείζονες ἐδείκθησαν αἱ $BA, ΑΓ$. πολλῶν ἄρα αἱ $BA, ΑΓ$ τῶν $ΒΔ, ΔΓ$ μείζονες εἰσιν.

Πάλιν, ἐπεὶ παντὸς τριγώνου ἡ ἐκτὸς γωνία τῆς ἐντὸς καὶ ἀπεναντίον μείζων ἐστίν, τοῦ $ΓΔΕ$ ἄρα τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ $ΒΔΓ$ μείζων ἐστὶ τῆς ὑπὸ $ΓΕΔ$. διὰ ταῦτά τοῖνυν καὶ τοῦ ABE τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ $ΓΕΒ$ μείζων ἐστὶ τῆς ὑπὸ $ΒΑΓ$. ἀλλὰ τῆς ὑπὸ $ΓΕΒ$ μείζων ἐδείκθη ἡ ὑπὸ $ΒΔΓ$. πολλῶν ἄρα ἡ ὑπὸ $ΒΔΓ$ μείζων ἐστὶ τῆς ὑπὸ $ΒΑΓ$.

Ἐὰν ἄρα τριγώνον ἐπὶ μιᾷ τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθείσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μὲν εἰσιν, μείζονα δὲ γωνίαν περιέθουσιν· ὅπερ ἔδει δεῖχαι.

but will encompass a greater angle.

For let the two internal straight-lines BD and DC be constructed on one of the sides BC of the triangle ABC , from its ends B and C (respectively). I say that BD and DC are less than the (sum of the) two remaining sides of the triangle BA and AC , but encompass an angle BDC greater than BAC .

For let BD be drawn through to E . And since in any triangle (the sum of any) two sides is greater than the remaining (side) [Prop. 1.20], in triangle ABE the (sum of the) two sides AB and AE is thus greater than BE . Let EC be added to both. Thus, (the sum of) BA and AC is greater than (the sum of) BE and EC . Again, since in triangle CED the (sum of the) two sides CE and ED is greater than CD , let DB be added to both. Thus, (the sum of) CE and EB is greater than (the sum of) CD and DB . But, (the sum of) BA and AC was shown (to be) greater than (the sum of) BE and EC . Thus, (the sum of) BA and AC is much greater than (the sum of) BD and DC .

Again, since in any triangle the external angle is greater than the internal and opposite (angles) [Prop. 1.16], in triangle CDE the external angle BDC is thus greater than CED . Accordingly, for the same (reason), the external angle CEB of the triangle ABE is also greater than BAC . But, BDC was shown (to be) greater than CEB . Thus, BDC is much greater than BAC .

Thus, if two internal straight-lines are constructed on one of the sides of a triangle, from its ends, the constructed (straight-lines) are less than the two remaining sides of the triangle, but encompass a greater angle. (Which is) the very thing it was required to show.

22.

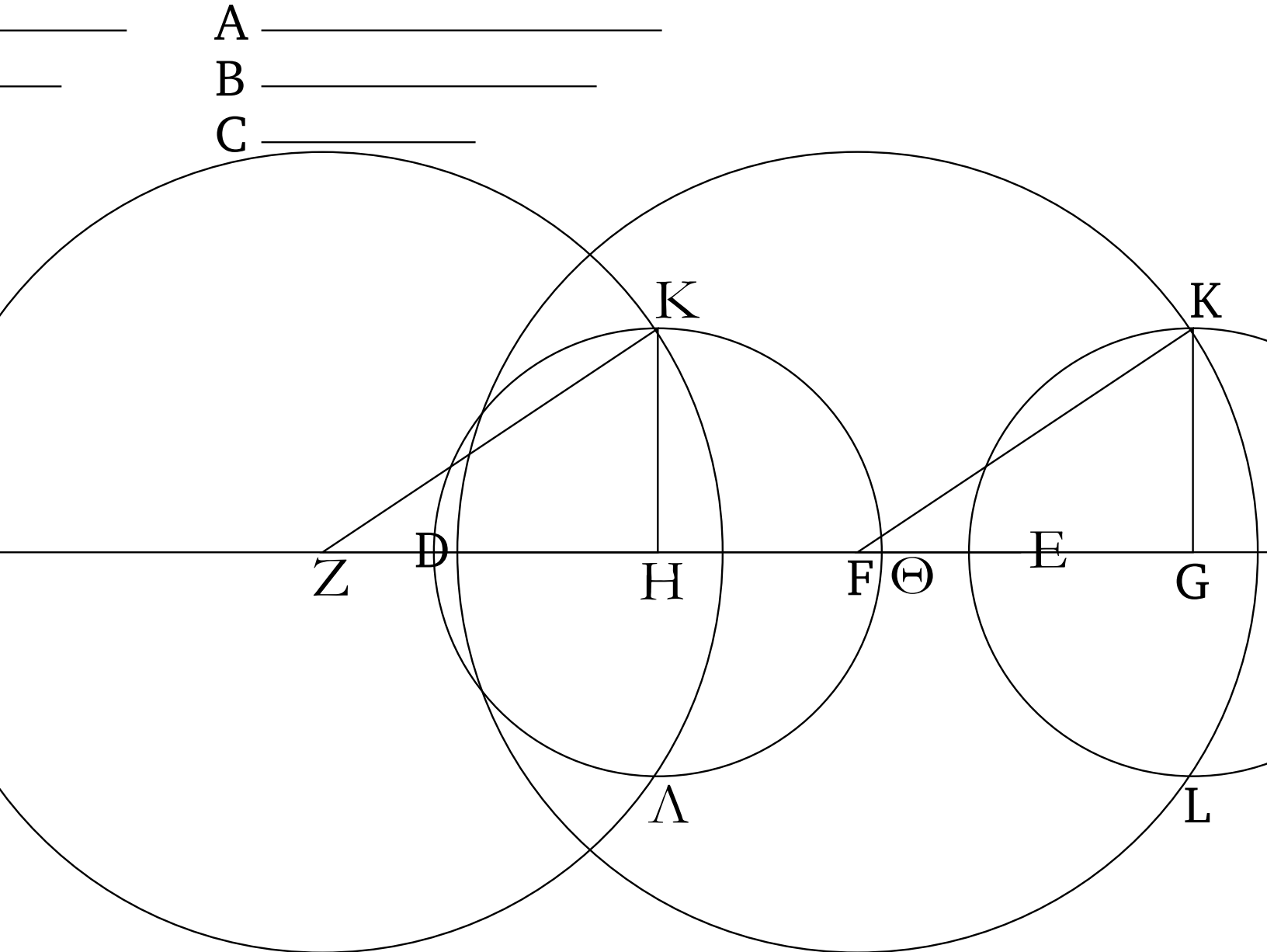
Proposition 22

Ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις [εὐθείαις], τριγώνον συστήσασθαι· δεῖ δὲ τὰς δύο τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας [διὰ τὸ καὶ παντὸς

To construct a triangle from three straight-lines which are equal to three given [straight-lines]. It is necessary for (the sum of) two (of the straight-lines) taken together in any (pos-

τριγώνον τὰς δύο πλευρὰς τῆς λοιπῆς μείζονα εἶναι πάντῃ μεταλαμβανομένας].

sible way) to be greater than the remaining (one), [on account of the (fact that) in any triangle (the sum of) two sides taken together in any (possible way) is greater than the remaining (one) [Prop. 1.20]].



Ἐστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ A, B, Γ , ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, αἱ μὲν A, B τῆς Γ , αἱ δὲ A, Γ τῆς B , καὶ ἔτι αἱ B, Γ τῆς A . δεῖ δὴ ἐκ τῶν ἴσων ταῖς A, B, Γ τριγώνον συστήσασθαι.

Ἐκκείσθω τις εὐθεῖα ἡ DE πεπερασμένη μὲν κατὰ τὸ Δ ἄπειροξ δὲ κατὰ τὸ E , καὶ κείσθω τῇ μὲν A ἴση ἡ ΔZ , τῇ δὲ B ἴση ἡ ZH , τῇ δὲ Γ ἴση ἡ $H\Theta$. καὶ κέντρῳ μὲν τῷ Z , διαστήματι δὲ τῷ $Z\Delta$ κύκλος γεγράφθω ὁ $\Delta K\Lambda$. πάλιν

Let A, B , and C be the three given straight-lines, of which let (the sum of) two taken together in any (possible way) be greater than the remaining (one). (Thus), (the sum of) A and B (is greater) than C , (the sum of) A and C than B , and also (the sum of) B and C than A . So it is required to construct a triangle from (straight-lines) equal to A, B , and C .

Let some straight-line DE be set out, terminated at D , and infinite in the direction of E . And let DF made equal to A , and

κέντρον μὲν τῷ H , διαστήματι δὲ τῷ $H\Theta$ κύκλος γεγράφθω ὁ $ΚΛ\Theta$, καὶ ἐπεξέυθθωσαν αἱ KZ , KH . λέγω, ὅτι ἐκ τριῶν εὐθειῶν τῶν ἴσων ταῖς A , B , Γ τρίγωνον συνέσταται τὸ KZH .

Ἐπεὶ γὰρ τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $\Delta K\Lambda$ κύκλου, ἴση ἐστὶν ἡ $Z\Delta$ τῇ ZK . ἀλλὰ ἡ $Z\Delta$ τῇ A ἐστὶν ἴση. καὶ ἡ KZ ἄρα τῇ A ἐστὶν ἴση. πάλιν, ἐπεὶ τὸ H σημεῖον κέντρον ἐστὶ τοῦ $\Lambda K\Theta$ κύκλου, ἴση ἐστὶν ἡ $H\Theta$ τῇ HK . ἀλλὰ ἡ $H\Theta$ τῇ Γ ἐστὶν ἴση· καὶ ἡ KH ἄρα τῇ Γ ἐστὶν ἴση. ἐστὶ δὲ καὶ ἡ ZH τῇ B ἴση· αἱ τρεῖς ἄρα εὐθεῖαι αἱ KZ , ZH , HK τρισὶ ταῖς A , B , Γ ἴσαι εἰσίν.

Ἐκ τριῶν ἄρα εὐθειῶν τῶν KZ , ZH , HK , αἱ εἰσὶν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις ταῖς A , B , Γ , τρίγωνον συνέσταται τὸ KZH . ὅπερ ἔδει ποιῆσαι.

23.

Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ γωνίᾳ εὐθύγραμμω ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.

Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ πρὸς αὐτῇ σημεῖον τὸ A , ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἡ ὑπὸ $\Delta Γ Ε$. δεῖ δὴ πρὸς τῇ δοθείσῃ εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ δοθείσῃ γωνίᾳ εὐθύγραμμω τῇ ὑπὸ $\Delta Γ Ε$ ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.

FG equal to B , and GH equal to C [Prop. 1.3]. And let the circle DKL be drawn with center F and radius FD . Again, let the circle KLH be drawn with center G and radius GH . And let KF and KG be joined. I say that the triangle KFG has been constructed from three straight-lines equal to A , B , and C .

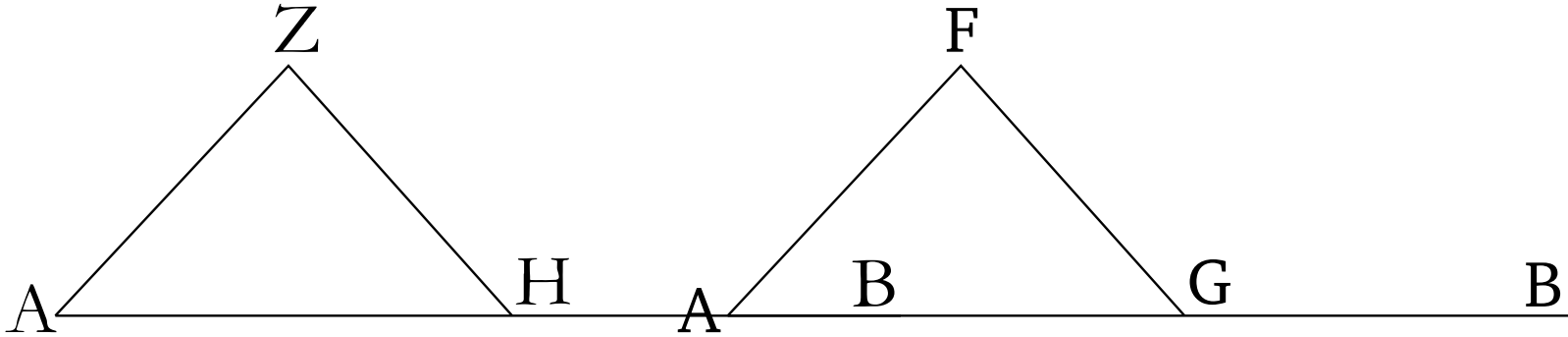
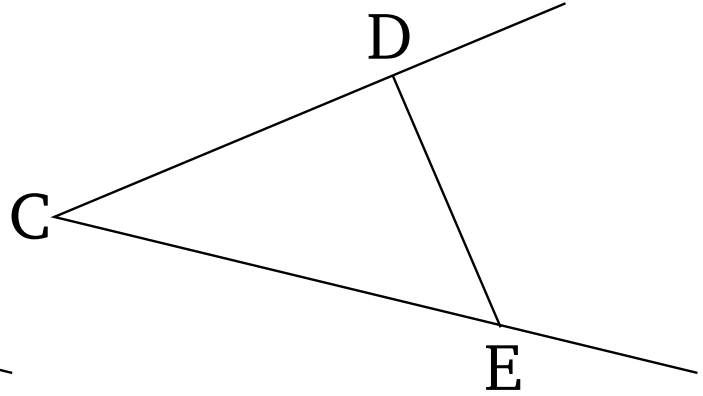
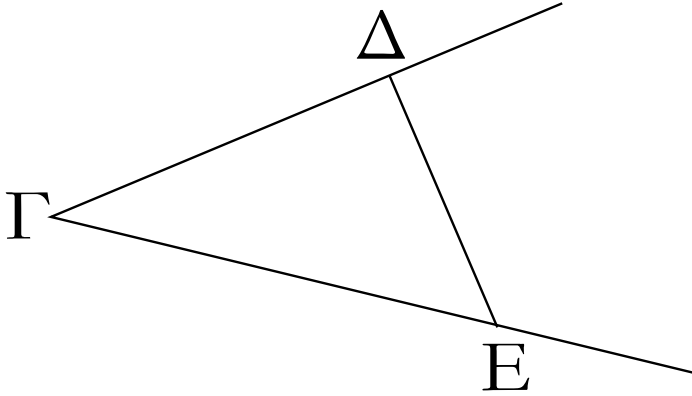
For since point F is the center of the circle DKL , FD is equal to FK . But, FD is equal to A . Thus, KF is also equal to A . Again, since point G is the center of the circle LKH , GH is equal to GK . But, GH is equal to C . Thus, KG is also equal to C . And FG is also equal to B . Thus, the three straight-lines KF , FG , and GK are equal to A , B , and C (respectively).

Thus, the triangle KFG has been constructed from the three straight-lines KF , FG , and GK , which are equal to the three given straight-lines A , B , and C (respectively). (Which is) the very thing it was required to do.

Proposition 23

To construct a rectilinear angle equal to a given rectilinear angle at a (given) point on a given straight-line.

Let AB be the given straight-line, A the (given) point on it, and DCE the given rectilinear angle. So it is required to construct a rectilinear angle equal to the given rectilinear angle DCE at the (given) point A on the given straight-line AB .



Εἰλήφθω ἐφ' ἑκατέρᾳ τῶν ΓΔ, ΓΕ τυθόντα σημεία τὰ Δ, Ε, καὶ ἐπεζεύθθω ἡ ΔΕ· καὶ ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς ΓΔ, ΔΕ, ΓΕ, τρίγωνον συνεστήτω τὸ ΑΖΗ, ὥστε ἴσῃν εἶναι τὴν μὲν ΓΔ τῇ ΑΖ, τὴν δὲ ΓΕ τῇ ΑΗ, καὶ ἔτι τὴν ΔΕ τῇ ΖΗ.

Ἐπεὶ οὖν δύο αἱ ΔΓ, ΓΕ δύο ταῖς ΖΑ, ΑΗ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ βάσις ἡ ΔΕ βάσει τῇ ΖΗ ἴση, γωνία ἄρα ἡ ὑπὸ ΔΓΕ γωνία τῇ ὑπὸ ΖΑΗ ἔστιν ἴση.

Πρὸς ἄρα τῇ δοθείσῃ εὐθείᾳ τῇ ΑΒ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ ὑπὸ ΔΓΕ ἴση γωνία εὐθύγραμμοῦς συνέσσεται ἡ ὑπὸ ΖΑΗ· ὅπερ ἔδει ποιῆσαι.

Let the points D and E be taken at random on each of the (straight-lines) CD and CE (respectively), and let DE be joined. And let the triangle AFG be constructed from three straight-lines which are equal to CD , DE , and CE , such that CD is equal to AF , CE to AG , and further DE to FG [Prop. 1.22].

Therefore, since the two (straight-lines) DC , CE are equal to the two (straight-lines) FA , AG , respectively, and the base DE is equal to the base FG , the angle DCE is thus equal to the angle FAG [Prop. 1.8].

Thus, the rectilinear angle FAG , equal to the given rectilinear angle DCE , has been constructed at the (given) point A on the given straight-line AB . (Which is) the very thing it was required to do.

24.

Ἐὰν δύο τρίγωνα τὰς δύο πλευρὰς [ταῖς] δύο πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρᾳ, τὴν δὲ γωνίαν τῆς γωνίας μείζονα ἔθῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῆς βάσεως μείζονα ἔχει.

Ἐστω δύο τρίγωνα τὰ ΑΒΓ, ΔΕΖ τὰς δύο πλευρὰς τὰς ΑΒ, ΑΓ ταῖς δύο πλευραῖς ταῖς ΔΕ, ΔΖ ἴσας ἔθοντα ἑκα-

Proposition 24

If two triangles have two sides equal to two sides, respectively, but (one) has the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the other), (then the former triangle) will also have a base greater than the base (of the latter).

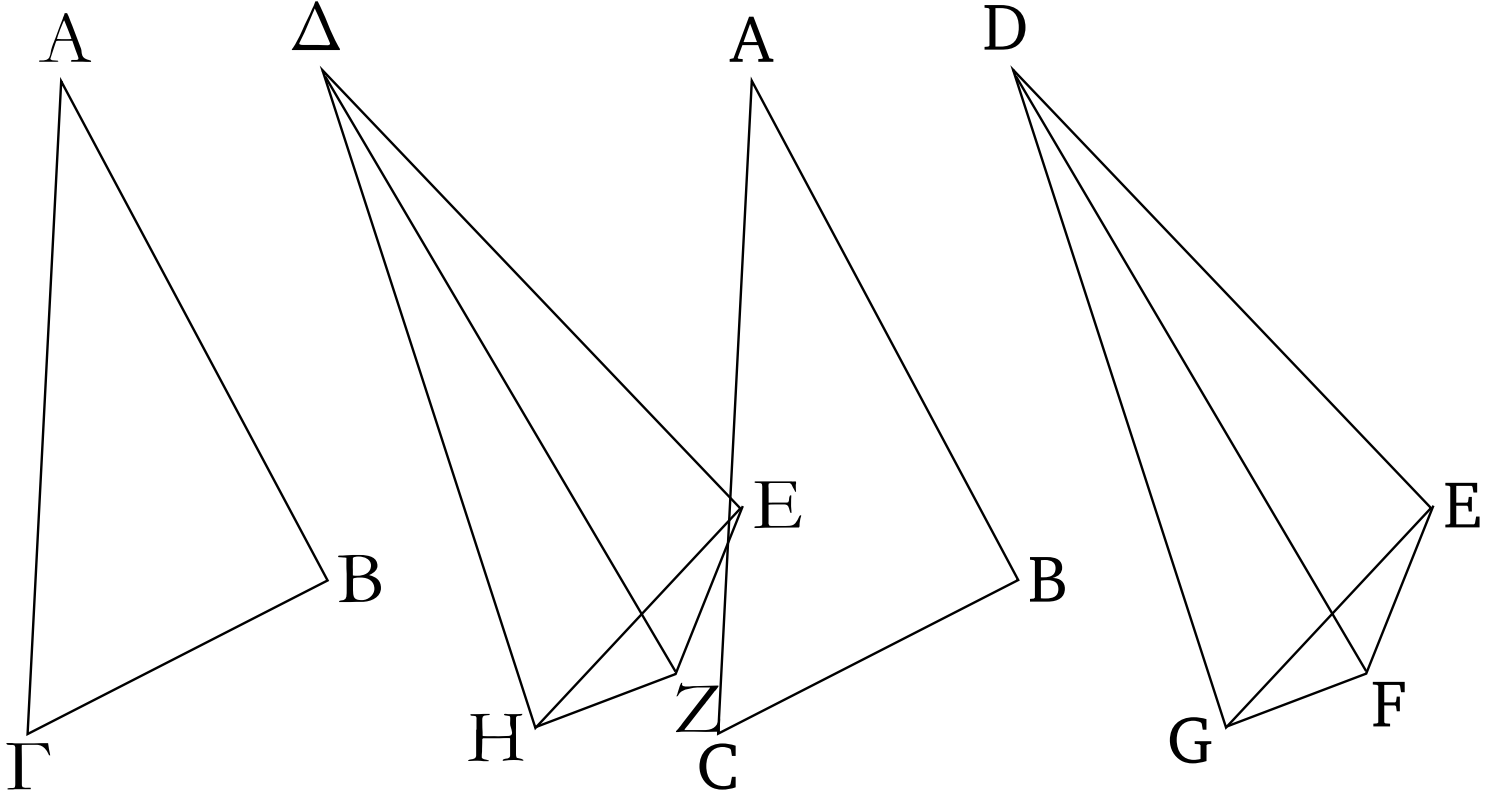
Let ABC and DEF be two triangles having the two sides

τέρων ἑκατέρω, τὴν μὲν AB τῇ ΔE τὴν δὲ AG τῇ ΔZ , ἥ δὲ πρὸς τῷ A γωνία τῆς πρὸς τῷ Δ γωνίας μείζων ἔστω· λέγω, ὅτι καὶ βάσις ἡ BG βάσεως τῆς EZ μείζων ἔστί.

Ἐπεὶ γὰρ μείζων ἡ ὑπὸ BAG γωνία τῆς ὑπὸ $E\Delta Z$ γωνίας, σννεσάτω πρὸς τῇ ΔE εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Δ τῇ ὑπὸ BAG γωνίᾳ ἴση ἡ ὑπὸ $E\Delta H$, καὶ κείσθω ὁποτέρω τῶν AG , ΔZ ἴση ἡ ΔH , καὶ ἐπεξέυθυσαν αἱ EH , ZH .

AB and AC equal to the two sides DE and DF , respectively. (That is), AB (equal) to DE , and AC to DF . Let them also have the angle at A greater than the angle at D . I say that the base BC is also greater than the base EF .

For since angle BAC is greater than angle EDF , let (angle) EDG , equal to angle BAC , be constructed at the point D on the straight-line DE [Prop. 1.23]. And let DG be made equal to either of AC or DF [Prop. 1.3], and let EG and FG be joined.



Ἐπεὶ οὖν ἴση ἔστί ἡ μὲν AB τῇ ΔE , ἥ δὲ AG τῇ ΔH , δύο δὴ αἱ BA , AG δυοὶ ταῖς $E\Delta$, ΔH ἴσαι εἰσὶν ἑκατέρω ἑκατέρω· καὶ γωνία ἡ ὑπὸ BAG γωνία τῇ ὑπὸ $E\Delta H$ ἴση· βάσις ἄρα ἡ BG βάσει τῇ EH ἔστιν ἴση. πάλιν, ἐπεὶ ἴση ἔστί ἡ ΔZ τῇ ΔH , ἴση ἐστὶ καὶ ἡ ὑπὸ ΔHZ γωνία τῇ ὑπὸ ΔZH · μείζων ἄρα ἡ ὑπὸ ΔZH τῆς ὑπὸ EZH · πολλῶν ἄρα μείζων ἔστί ἡ ὑπὸ EZH τῆς ὑπὸ EHZ . καὶ ἐπεὶ τρίγωνόν ἐστι τὸ EZH μείζονα ἔθον τὴν ὑπὸ EZH γωνίαν τῆς ὑπὸ EHZ , ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει, μείζων ἄρα καὶ πλευρὰ ἡ EH τῆς EZ . ἴση δὲ ἡ EH τῇ BG · μείζων ἄρα καὶ ἡ BG τῆς EZ .

Ἐάν ἄρα δύο τρίγωνα τὰς δύο πλευράς δυοὶ πλευραῖς ἴσας ἔθῃ ἑκατέρω ἑκατέρω, τὴν δὲ γωνίαν τῆς γωνίας μείζονα ἔθῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην, καὶ τὴν βάσιν τῆς βάσεως μείζονα ἔχει· ὅπερ ἔδει δεῖξαι.

Therefore, since AB is equal to DE and AC to DG , the two (straight-lines) BA , AC are equal to the two (straight-lines) ED , DG , respectively. Also the angle BAC is equal to the angle EDG . Thus, the base BC is equal to the base EG [Prop. 1.4]. Again, since DF is equal to DG , angle DGF is also equal to angle DFG [Prop. 1.5]. Thus, DFG (is) greater than EGF . Thus, EFG is much greater than EGF . And since triangle EFG has angle EFG greater than EGF , and the greater angle is subtended by the greater side [Prop. 1.19], side EG (is) thus also greater than EF . But EG (is) equal to BC . Thus, BC (is) also greater than EF .

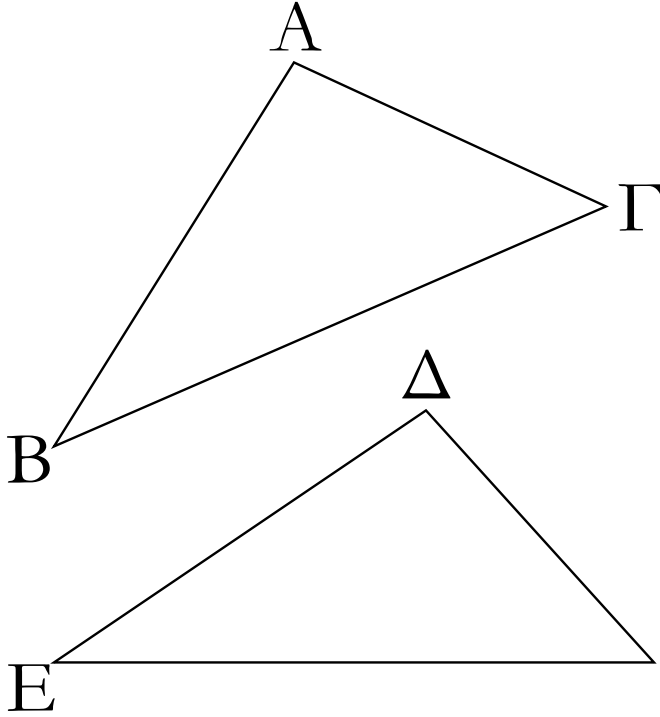
Thus, if two triangles have two sides equal to two sides, respectively, but (one) has the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the other), (then the former triangle) will also have a base greater than the base (of the latter). (Which is) the very thing it was required to show.

25.

Ἐάν δύο τρίγωνα τὰς δύο πλευράς δυοὶ πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρα, τὴν δὲ βάσιν τῆς βάσεως μείζονα ἔθῃ, καὶ τὴν γωνίαν τῆς γωνίας μείζονα ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην.

Ἐστω δύο τρίγωνα τὰ $ABΓ$, $ΔΕΖ$ τὰς δύο πλευράς τὰς AB , $ΑΓ$ ταῖς δύο πλευραῖς ταῖς $ΔΕ$, $ΔΖ$ ἴσας ἔθοντα ἑκατέραν ἑκατέρα, τὴν μὲν AB τῇ $ΔΕ$, τὴν δὲ $ΑΓ$ τῇ $ΔΖ$ · βάσις δὲ ἡ $ΒΓ$ βάσεως τῆς $ΕΖ$ μείζων ἔστω· λέγω, ὅτι καὶ γωνία ἡ ὑπὸ $ΒΑΓ$ γωνία τῆς ὑπὸ $ΕΔΖ$ μείζων ἔστί.

Εἰ γὰρ μή, ἦτοι ἴση ἔστιν αὐτῇ ἢ ἐλάσσων· ἴση μὲν οὐκ ἔστιν ἡ ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΕΔΖ$ · ἴση γὰρ ἂν ἦν καὶ βάσις ἡ $ΒΓ$ βάσει τῇ $ΕΖ$ · οὐκ ἔστι δέ. οὐκ ἄρα ἴση ἔστι γωνία ἡ ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΕΔΖ$ · οὐδὲ μὴν ἐλάσσων ἔστιν ἡ ὑπὸ $ΒΑΓ$ τῆς ὑπὸ $ΕΔΖ$ · ἐλάσσων γὰρ ἂν ἦν καὶ βάσις ἡ $ΒΓ$ βάσεως τῆς $ΕΖ$ · οὐκ ἔστι δέ· οὐκ ἄρα ἐλάσσων ἔστιν ἡ ὑπὸ $ΒΑΓ$ γωνία τῆς ὑπὸ $ΕΔΖ$. ἐδείχθη δέ, ὅτι οὐδὲ ἴση μείζων ἄρα ἔστιν ἡ ὑπὸ $ΒΑΓ$ τῆς ὑπὸ $ΕΔΖ$.



Ἐάν ἄρα δύο τρίγωνα τὰς δύο πλευράς δυοὶ πλευραῖς ἴσας ἔθῃ ἑκατέραν ἑκατέρα, τὴν δὲ βάσιν τῆς βάσεως μείζονα ἔθῃ, καὶ τὴν γωνίαν τῆς γωνίας μείζονα ἔχει τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεθρομένην· ὅπερ ἔδει δεῖξαι.

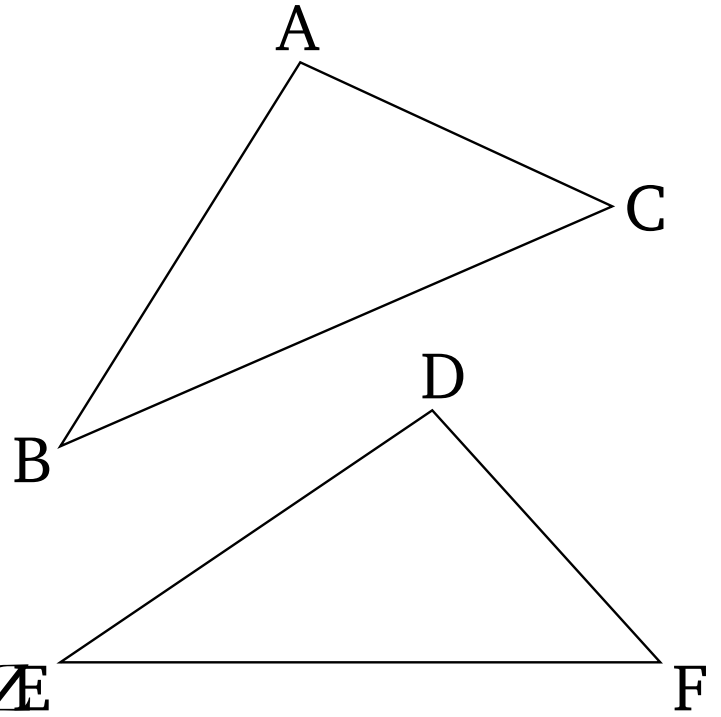
26.

Proposition 25

If two triangles have two sides equal to two sides, respectively, but (one) has a base greater than the base (of the other), (then the former triangle) will also have the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the latter).

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively (That is), AB (equal) to DE , and AC to DF . And let the base BC be greater than the base EF . I say that angle BAC is also greater than EDF .

For if not, (BAC) is certainly either equal to, or less than, (EDF). In fact, BAC is not equal to EDF . For then the base BC would also be equal to the base EF [Prop. 1.4]. But it is not. Thus, angle BAC is not equal to EDF . Neither, indeed, is BAC less than EDF . For then the base BC would also be less than the base EF [Prop. 1.24]. But it is not. Thus, angle BAC is not less than EDF . But it was shown that (BAC) is not equal (to EDF) either. Thus, BAC is greater than EDF .



Thus, if two triangles have two sides equal to two sides, respectively, but (one) has a base greater than the base (of the other), (then the former triangle) will also have the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the latter). (Which is) the very thing it was required to show.

Proposition 26

Ἐάν δύο τρίγωνα τὰς δύο γωνίας δυοὶ γωνίαις ἴσαις ἔθῃ ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην ᾗτοι τὴν πρὸς ταῖς ἴσαις γωνίαις ἢ τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν, καὶ τὰς λοιπὰς πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαις ἔχει [ἑκατέραν ἑκατέρᾳ] καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ.

Ἐστω δύο τρίγωνα τὰ $ABΓ$, $ΔΕΖ$ τὰς δύο γωνίας τὰς ὑπὸ $ABΓ$, $ΒΓΑ$ δυοὶ ταῖς ὑπὸ $ΔΕΖ$, $ΕΖΔ$ ἴσαις ἔθοντα ἑκατέραν ἑκατέρᾳ, τὴν μὲν ὑπὸ $ABΓ$ τῇ ὑπὸ $ΔΕΖ$, τὴν δὲ ὑπὸ $ΒΓΑ$ τῇ ὑπὸ $ΕΖΔ$. ἐθέτω δὲ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην, πρότερον τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν $ΒΓ$ τῇ $ΕΖ$. λέγω, ὅτι καὶ τὰς λοιπὰς πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαις ἔχει ἑκατέραν ἑκατέρᾳ, τὴν μὲν AB τῇ $ΔΕ$ τὴν δὲ $ΑΓ$ τῇ $ΔΖ$, καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ, τὴν ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΕΔΖ$.

Εἰ γὰρ ἄνισός ἐστιν ἡ AB τῇ $ΔΕ$, μία αὐτῶν μείζων ἐστίν. ἔστω μείζων ἡ AB , καὶ κείσθω τῇ $ΔΕ$ ἴση ἡ BH , καὶ ἐπεξεύθθω ἡ $ΗΓ$.

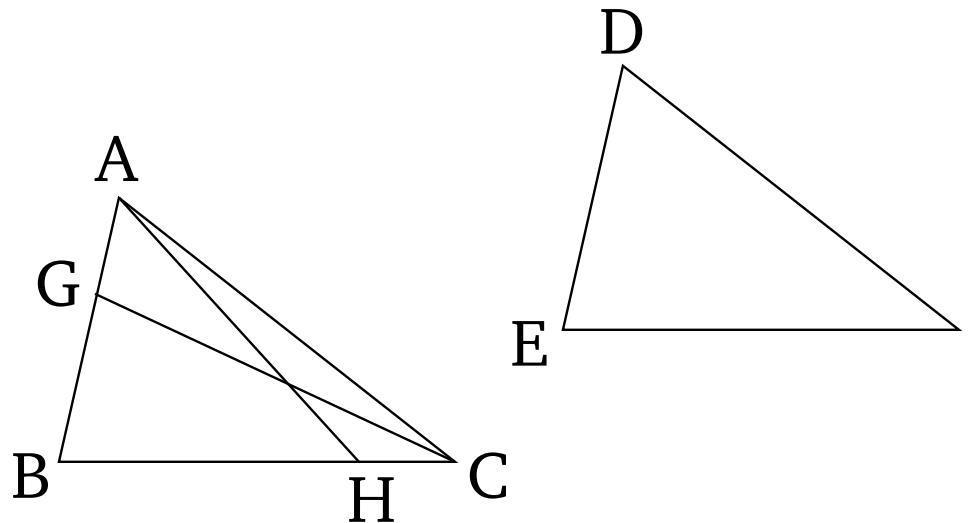
Ἐπεὶ οὖν ἴση ἐστίν ἡ μὲν BH τῇ $ΔΕ$, ἡ δὲ $ΒΓ$ τῇ $ΕΖ$, δύο δὴ αἱ BH , $ΒΓ$ δυοὶ ταῖς $ΔΕ$, $ΕΖ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ $ΗΒΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ ἴση ἐστίν· βάσει ἄρα ἡ $ΗΓ$ βάσει τῇ $ΔΖ$ ἴση ἐστίν, καὶ τὸ $ΗΒΓ$ τρίγωνον τῷ $ΔΕΖ$ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἡ ὑπὸ $ΗΓΒ$ γωνία τῇ ὑπὸ $ΔΖΕ$. ἀλλὰ ἡ ὑπὸ $ΔΖΕ$ τῇ ὑπὸ $ΒΓΑ$ ὑπόκειται ἴση· καὶ ἡ ὑπὸ $ΒΓΗ$ ἄρα τῇ ὑπὸ $ΒΓΑ$ ἴση ἐστίν, ἡ ἐλάσσων τῇ μείζονι· ὅπερ ἀδύνατον. οὐκ ἄρα ἄνισός ἐστιν ἡ AB τῇ $ΔΕ$. ἴση ἄρα. ἔστι δὲ καὶ ἡ $ΒΓ$ τῇ $ΕΖ$ ἴση· δύο δὴ αἱ AB , $ΒΓ$ δυοὶ ταῖς $ΔΕ$, $ΕΖ$ ἴσαι εἰσὶν ἑκάτερα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ ἐστίν ἴση· βάσει ἄρα ἡ $ΑΓ$ βάσει τῇ $ΔΖ$ ἴση ἐστίν, καὶ λοιπὴ γωνία ἡ ὑπὸ $ΒΑΓ$ τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ $ΕΔΖ$ ἴση ἐστίν.

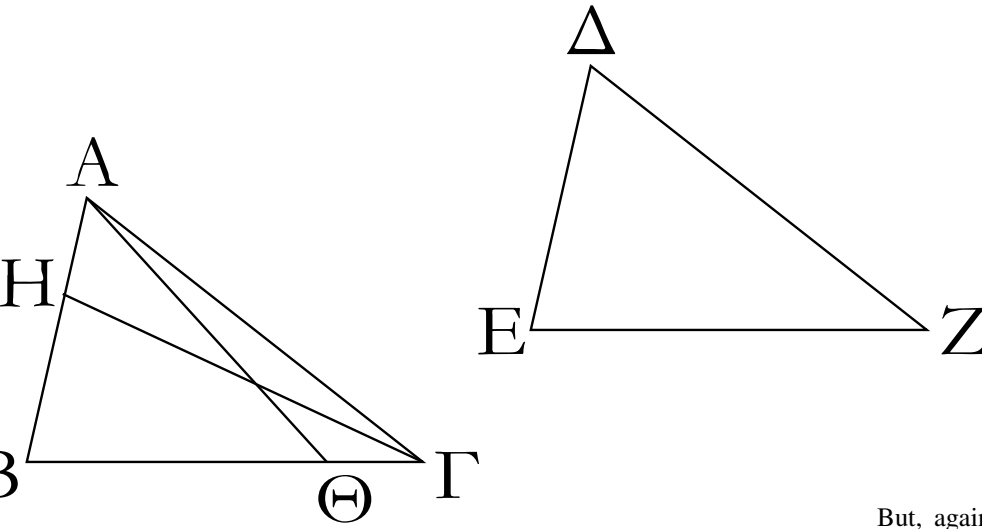
If two triangles have two angles equal to two angles, respectively, and one side equal to one side—in fact, either that by the equal angles, or that subtending one of the equal angles—(then the triangles) will also have the remaining sides equal to the [corresponding] remaining sides, and the remaining angle (equal) to the remaining angle.

Let ABC and DEF be two triangles having the two angles ABC and BCA equal to the two (angles) DEF and EFD , respectively. (That is) ABC (equal) to DEF , and BCA to EFD . And let them also have one side equal to one side. First of all, the (side) by the equal angles. (That is) BC (equal) to EF . I say that they will have the remaining sides equal to the corresponding remaining sides. (That is) AB (equal) to DE , and AC to DF . And (they will have) the remaining angle (equal) to the remaining angle. (That is) BAC (equal) to EDF .

For if AB is unequal to DE (then) one of them is greater. Let AB be greater, and let BG be made equal to DE [Prop. 1.3], and let GC be joined.

Therefore, since BG is equal to DE , and BC to EF , the two (straight-lines) GB , BC † are equal to the two (straight-lines) DE , EF , respectively. And angle GBC is equal to angle DEF . Thus, the base GC is equal to the base DF , and triangle GBC is equal to triangle DEF , and the remaining angles subtended by the equal sides will be equal to the (corresponding) remaining angles [Prop. 1.4]. Thus, GCB (is equal) to DFE . But, DFE was assumed (to be) equal to BCA . Thus, BCG is also equal to BCA , the lesser to the greater. The very thing (is) impossible. Thus, AB is not unequal to DE . Thus, (it is) equal. And BC is also equal to EF . So the two (straight-lines) AB , BC are equal to the two (straight-lines) DE , EF , respectively. And angle ABC is equal to angle DEF . Thus, the base AC is equal to the base DF , and the remaining angle BAC is equal to the remaining angle EDF [Prop. 1.4].





Ἀλλὰ δὴ πάλιν ἔστωσαν αἱ ὑπὸ τὰς ἴσας γωνίας πλευραὶ ὑποτείνουσαι ἴσαι, ὡς ἡ AB τῇ DE . λέγω πάλιν, ὅτι καὶ αἱ λοιπαὶ πλευραὶ ταῖς λοιπαῖς πλευραῖς ἴσαι ἔσονται, ἡ μὲν AG τῇ DZ , ἡ δὲ BG τῇ EZ καὶ ἔτι ἡ λοιπὴ γωνία ἡ ὑπὸ BAG τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ EDZ ἴση ἐστίν.

Εἰ γὰρ ἀνισὸς ἐστὶν ἡ BG τῇ EZ , μία αὐτῶν μείζων ἐστίν. ἔστω μείζων, εἰ δυνατόν, ἡ BG , καὶ κείσθω τῇ EZ ἴση ἡ $B\Theta$, καὶ ἐπεζεύθω ἡ $A\Theta$. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν $B\Theta$ τῇ EZ ἡ δὲ AB τῇ DE , δύο δὴ αἱ AB , $B\Theta$ δυοὶ ταῖς DE , EZ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνίαξ ἴσας περιέθουσιν· βάσειξ ἄρα ἡ $A\Theta$ βάσει τῇ DZ ἴση ἐστίν, καὶ τὸ $AB\Theta$ τρίγωνον τῷ ΔEZ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται, ὅφ' ἂς αἱ ἴσας πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἐστὶν ἡ ὑπὸ $B\Theta A$ γωνία τῇ ὑπὸ $EZ\Delta$. ἀλλὰ ἡ ὑπὸ $EZ\Delta$ τῇ ὑπὸ BGA ἐστὶν ἴση· τριγώνον δὴ τοῦ $A\Theta G$ ἡ ἐκτὸς γωνία ἡ ὑπὸ $B\Theta A$ ἴση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ BGA . ὅπερ ἀδύνατον. οὐκ ἄρα ἀνισὸς ἐστὶν ἡ BG τῇ EZ · ἴση ἄρα. ἐστὶ δὲ καὶ ἡ AB τῇ DE ἴση. δύο δὴ αἱ AB , BG δύο ταῖς DE , EZ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνίαξ ἴσας περιέθουσι· βάσειξ ἄρα ἡ AG βάσει τῇ DZ ἴση ἐστίν, καὶ τὸ ABG τρίγωνον τῷ ΔEZ τριγώνῳ ἴσον καὶ λοιπὴ γωνία ἡ ὑπὸ BAG τῇ λοιπῇ γωνίᾳ τῇ ὑπὸ EDZ ἴση.

Ἐὰν ἄρα δύο τρίγωνα τὰς δύο γωνίαξ δυοὶ γωνίαξ ἴσας ἔθῃ ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην ἢ τοὶ τὴν πρὸς ταῖς ἴσαις γωνίαις, ἢ τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν, καὶ τὰς λοιπὰς πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσας ἔχει καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ· ὅπερ ἔδει δεῖχαι.

But, again, let the sides subtending the equal angles be equal: for instance, (let) AB (be equal) to DE . Again, I say that the remaining sides will be equal to the remaining sides. (That is) AC (equal) to DF , and BC to EF . Furthermore, the remaining angle BAC is equal to the remaining angle EDF .

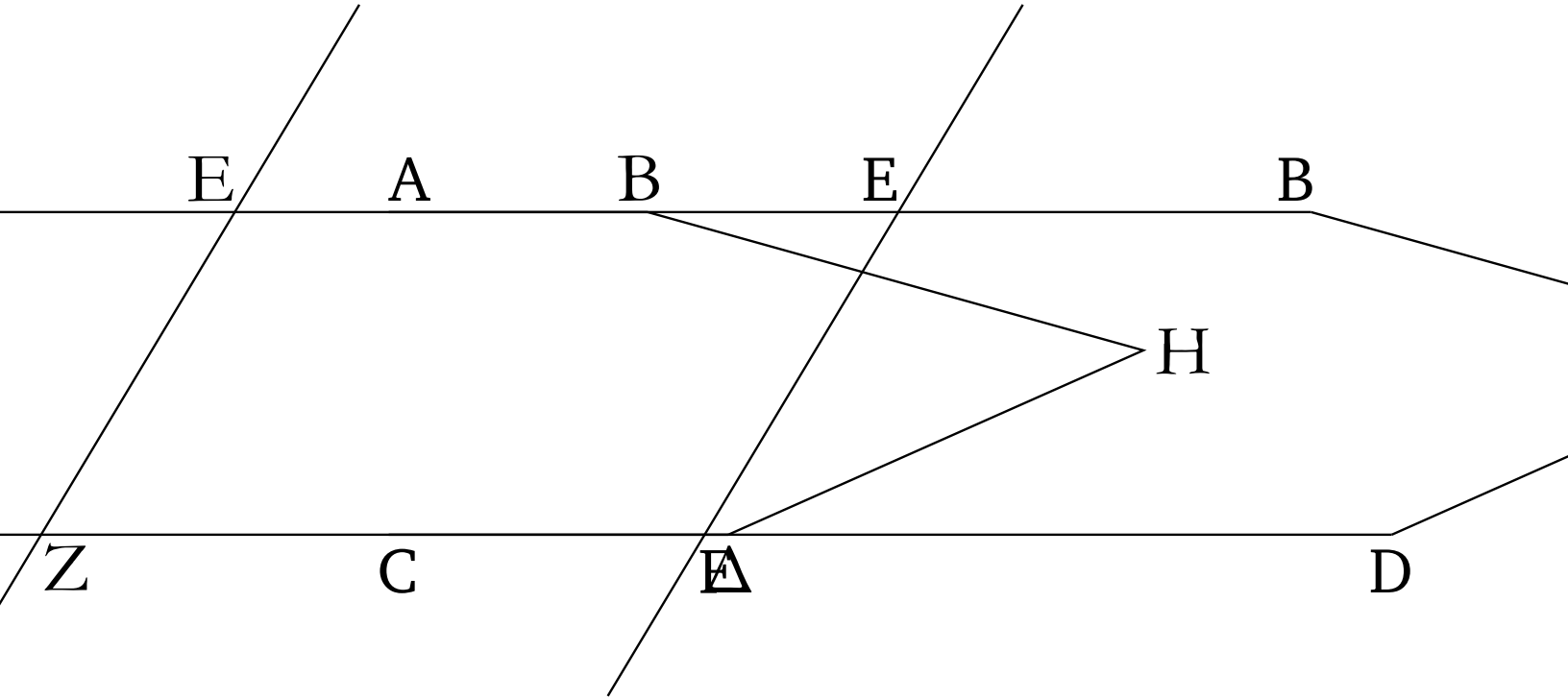
For if BC is unequal to EF (then) one of them is greater. If possible, let BC be greater. And let BH be made equal to EF [Prop. 1.3], and let AH be joined. And since BH is equal to EF , and AB to DE , the two (straight-lines) AB , BH are equal to the two (straight-lines) DE , EF , respectively. And the angles they encompass (are also equal). Thus, the base AH is equal to the base DF , and the triangle ABH is equal to the triangle DEF , and the remaining angles subtended by the equal sides will be equal to the (corresponding) remaining angles [Prop. 1.4]. Thus, angle BHA is equal to EFD . But, EFD is equal to BCA . So, in triangle AHC , the external angle BHA is equal to the internal and opposite angle BCA . The very thing (is) impossible [Prop. 1.16]. Thus, BC is not unequal to EF . Thus, (it is) equal. And AB is also equal to DE . So the two (straight-lines) AB , BC are equal to the two (straight-lines) DE , EF , respectively. And they encompass equal angles. Thus, the base AC is equal to the base DF , and triangle ABC (is) equal to triangle DEF , and the remaining angle BAC (is) equal to the remaining angle EDF [Prop. 1.4].

Thus, if two triangles have two angles equal to two angles, respectively, and one side equal to one side—in fact, either that by the equal angles, or that subtending one of the equal angles—(then the triangles) will also have the remaining sides equal to the (corresponding) remaining sides, and the remaining angle (equal) to the remaining angle. (Which is) the very thing it was required to show.

† The Greek text has “ BG , BC ”, which is obviously a mistake.

Ἐάν εἷς δύο εὐθείας εὐθεῖα ἐμπίπτονσα τὰς ἐναλλάχ γωνίας ἴσας ἀλλήλαις ποιῇ, παράλληλοι ἔσονται ἀλλήλαις αἱ εὐθεῖαι.

If a straight-line falling across two straight-lines makes the alternate angles equal to one another (then) the (two) straight-lines will be parallel to one another.



Εἷς γὰρ δύο εὐθείας τὰς AB , $ΓΔ$ εὐθεῖα ἐμπίπτονσα ἡ EZ τὰς ἐναλλάχ γωνίας τὰς ὑπὸ AEZ , $EZΔ$ ἴσας ἀλλήλαις ποιεῖτω· λέγω, ὅτι παράλληλός ἐστιν ἡ AB τῇ $ΓΔ$.

Εἰ γὰρ μή, ἐκβαλλόμεναι αἱ AB , $ΓΔ$ συμπεσοῦνται ἥτοι ἐπὶ τὰ B , $Δ$ μέρη ἢ ἐπὶ τὰ A , $Γ$. ἐκβεβλήσθωσαν καὶ συμπίπτέτωσαν ἐπὶ τὰ B , $Δ$ μέρη κατὰ τὸ H . τριγώνου δὴ τοῦ HEZ ἡ ἐκτὸς γωνία ἡ ὑπὸ AEZ ἴση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ EZH · ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα αἱ AB , $ΔΓ$ ἐκβαλλόμεναι συμπεσοῦνται ἐπὶ τὰ B , $Δ$ μέρη. ὁμοίως δὴ δευθθήσεται, ὅτι οὐδὲ ἐπὶ τὰ A , $Γ$ · αἱ δὲ ἐπὶ μηδέτερα τὰ μέρη συμπίπτονσαι παράλληλοί εἰσιν· παράλληλός ἄρα ἐστὶν ἡ AB τῇ $ΓΔ$.

Ἐάν ἄρα εἷς δύο εὐθείας εὐθεῖα ἐμπίπτονσα τὰς ἐναλλάχ γωνίας ἴσας ἀλλήλαις ποιῇ, παράλληλοι ἔσονται αἱ εὐθεῖαι· ὅπερ ἔδει δεῖξαι.

For let the straight-line EF , falling across the two straight-lines AB and CD , make the alternate angles AEF and EFD equal to one another. I say that AB and CD are parallel.

For if not, being produced, AB and CD will certainly meet together: either in the direction of B and D , or (in the direction) of A and C [Def. 1.23]. Let them be produced, and let them meet together in the direction of B and D at (point) G . So, for the triangle GEF , the external angle AEF is equal to the interior and opposite (angle) EFG . The very thing is impossible [Prop. 1.16]. Thus, being produced, AB and CD will not meet together in the direction of B and D . Similarly, it can be shown that neither (will they meet together) in (the direction of) A and C . But (straight-lines) meeting in neither direction are parallel [Def. 1.23]. Thus, AB and CD are parallel.

Thus, if a straight-line falling across two straight-lines makes the alternate angles equal to one another (then) the (two) straight-lines will be parallel (to one another). (Which is) the very thing it was required to show.

28.

Proposition 28

Ἐάν εἷς δύο εὐθείας εὐθεῖα ἐμπίπτονσα τὴν ἐκτὸς γωνίαν τῇ ἐντὸς καὶ ἀπεναντίον καὶ ἐπὶ τὰ αὐτὰ μέρη ἴσην ποιῇ ἢ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυσὶν ὀρθαῖς ἴσας, παράλληλοι

If a straight-line falling across two straight-lines makes the external angle equal to the internal and opposite angle on the same side, or (makes) the (sum of the) internal (angles) on the

ἔσονται ἀλλήλαις αἱ εὐθεῖαι.

Εἰς γὰρ δύο εὐθείας τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπίπτουσα ἡ EZ τὴν ἐκτὸς γωνίαν τὴν ὑπὸ EHB τῇ ἐντὸς καὶ ἀπεναντίον γωνίᾳ τῇ ὑπὸ $H\Theta\Delta$ ἴσην ποιεῖτω ἢ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη τὰς ὑπὸ $BH\Theta$, $H\Theta\Delta$ δυοῖν ὀρθαῖς ἴσας· λέγω, ὅτι παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ ὑπὸ EHB τῇ ὑπὸ $H\Theta\Delta$, ἀλλὰ ἡ ὑπὸ EHB τῇ ὑπὸ $AH\Theta$ ἐστὶν ἴση, καὶ ἡ ὑπὸ $AH\Theta$ ἄρα τῇ ὑπὸ $H\Theta\Delta$ ἐστὶν ἴση· καὶ εἰσὶν ἐναλλάχ· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

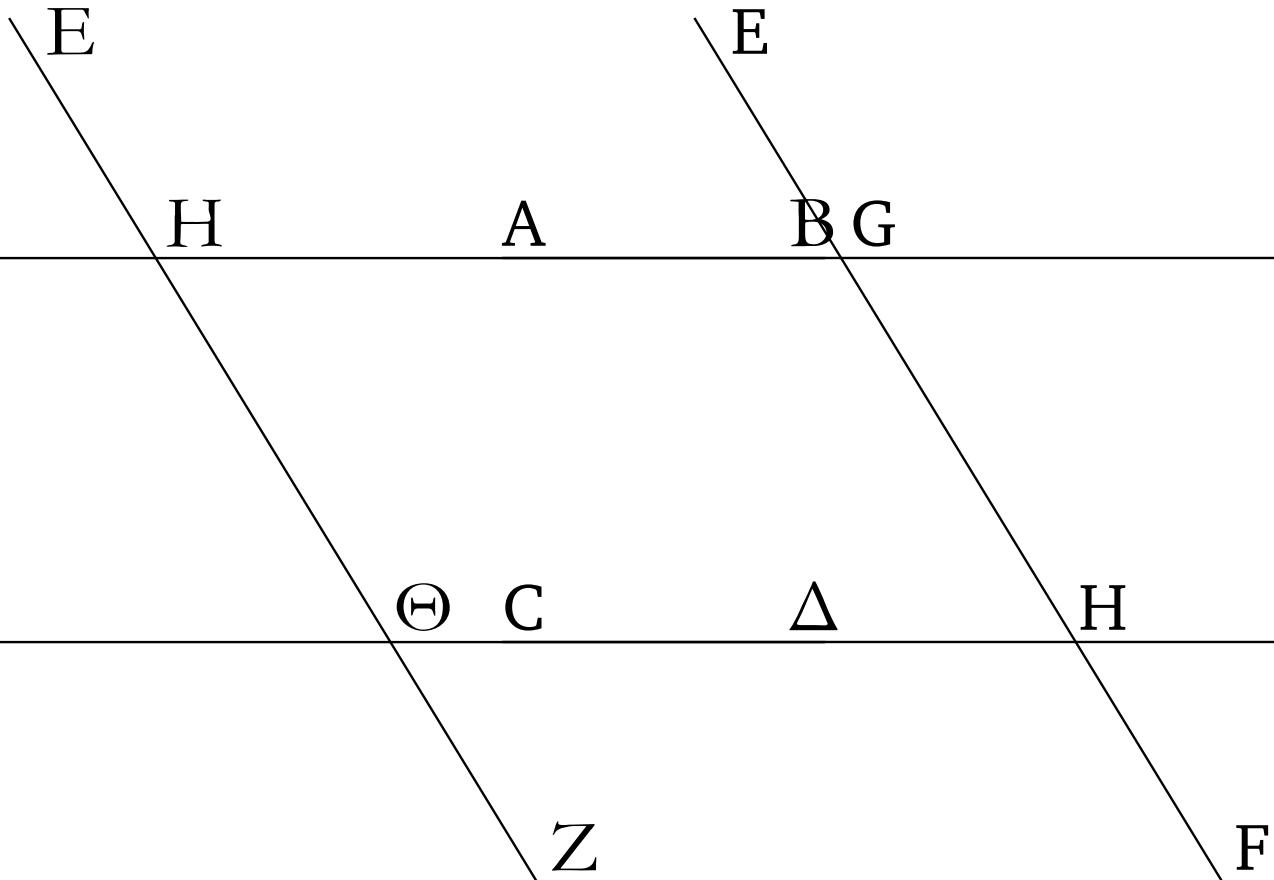
Πάλιν, ἐπεὶ αἱ ὑπὸ $BH\Theta$, $H\Theta\Delta$ δύο ὀρθαῖς ἴσαι εἰσὶν, εἰσὶ δὲ καὶ αἱ ὑπὸ $AH\Theta$, $BH\Theta$ δυοῖν ὀρθαῖς ἴσαι, αἱ ἄρα ὑπὸ $AH\Theta$, $BH\Theta$ ταῖς ὑπὸ $BH\Theta$, $H\Theta\Delta$ ἴσαι εἰσὶν· κοινὴ ἀφηρέσθω ἡ ὑπὸ $BH\Theta$ · λοιπὴ ἄρα ἡ ὑπὸ $AH\Theta$ λοιπῇ τῇ ὑπὸ $H\Theta\Delta$ ἐστὶν ἴση· καὶ εἰσὶν ἐναλλάχ· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

same side equal to two right-angles, (then) the (two) straight-lines will be parallel to one another.

For let EF , falling across the two straight-lines AB and CD , make the external angle EGB equal to the internal and opposite angle GHD , or the (sum of the) internal (angles) on the same side, BGH and GHD , equal to two right-angles. I say that AB is parallel to CD .

For since (in the first case) EGB is equal to GHD , but EGB is equal to AGH [Prop. 1.15], AGH is thus also equal to GHD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].

Again, since (in the second case, the sum of) BGH and GHD is equal to two right-angles, and (the sum of) AGH and BGH is also equal to two right-angles [Prop. 1.13], (the sum of) AGH and BGH is thus equal to (the sum of) BGH and GHD . Let BGH be subtracted from both. Thus, the remainder AGH is equal to the remainder GHD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].



Ἐάν ἄρα εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὴν ἐκτὸς γωνίαν τῇ ἐντὸς καὶ ἀπεναντίον καὶ ἐπὶ τὰ αὐτὰ μέρη ἴσην ποιῇ ἢ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυοῖν ὀρθαῖς ἴσας, παράλληλοι ἔσονται αἱ εὐθεῖαι· ὅπερ ἔδει δεῖξαι.

Thus, if a straight-line falling across two straight-lines makes the external angle equal to the internal and opposite angle on the same side, or (makes) the (sum of the) internal (angles) on the same side equal to two right-angles, (then) the (two) straight-lines will be parallel (to one another). (Which

29.

Ἡ εἷς τὰς παραλλήλους εὐθείας εὐθεῖα ἐπίπτουσα τὰς τε ἐναλλάχ γωνίας ἴσας ἀλλήλαις ποιεῖ καὶ τὴν ἐκτὸς τῇ ἐντὸς καὶ ἀπεναντίον ἴσην καὶ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη δυοῖν ὀρθαῖς ἴσας.

Εἷς γὰρ παραλλήλους εὐθείας τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπέτω ἢ EZ · λέγω, ὅτι τὰς ἐναλλάχ γωνίας τὰς ὑπὸ $AH\Theta$, $H\Theta\Delta$ ἴσας ποιεῖ καὶ τὴν ἐκτὸς γωνίαν τὴν ὑπὸ EHB τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ $H\Theta\Delta$ ἴσην καὶ τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη τὰς ὑπὸ $BH\Theta$, $H\Theta\Delta$ δυοῖν ὀρθαῖς ἴσας.

Εἰ γὰρ ἄνισός ἐστιν ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ $H\Theta\Delta$, μία αὐτῶν μείζων ἐστίν. ἔστω μείζων ἡ ὑπὸ $AH\Theta$ · κοινὴ προσκείσθω ἡ ὑπὸ $BH\Theta$ · αἱ ἄρα ὑπὸ $AH\Theta$, $BH\Theta$ τῶν ὑπὸ $BH\Theta$, $H\Theta\Delta$ μείζονές εἰσιν. ἀλλὰ αἱ ὑπὸ $AH\Theta$, $BH\Theta$ δυοῖν ὀρθαῖς ἴσαι εἰσίν. [καὶ] αἱ ἄρα ὑπὸ $BH\Theta$, $H\Theta\Delta$ δύο ὀρθῶν ἐλάσσονές εἰσιν. αἱ δὲ ἀπ' ἐλασσόνων ἢ δύο ὀρθῶν ἐκβαλλόμεναι εἷς ἄπειρον συμπέπτουσιν· αἱ ἄρα AB , $\Gamma\Delta$ ἐκβαλλόμεναι εἷς ἄπειρον συμπεσοῦνται· οὐ συμπέπτουσι δὲ διὰ τὸ παραλλήλους αὐτὰς ὑποκεῖσθαι· οὐκ ἄρα ἄνισός ἐστιν ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ $H\Theta\Delta$ · ἴση ἄρα. ἀλλὰ ἡ ὑπὸ $AH\Theta$ τῇ ὑπὸ EHB ἐστὶν ἴση· καὶ ἡ ὑπὸ EHB ἄρα τῇ ὑπὸ $H\Theta\Delta$ ἐστὶν ἴση· κοινὴ προσκείσθω ἡ ὑπὸ $BH\Theta$ · αἱ ἄρα ὑπὸ EHB , $BH\Theta$ ταῖς ὑπὸ $BH\Theta$, $H\Theta\Delta$ ἴσαι εἰσίν. ἀλλὰ αἱ ὑπὸ EHB , $BH\Theta$ δύο ὀρθαῖς ἴσαι εἰσίν· καὶ αἱ ὑπὸ $BH\Theta$, $H\Theta\Delta$ ἄρα δύο ὀρθαῖς ἴσαι εἰσίν.

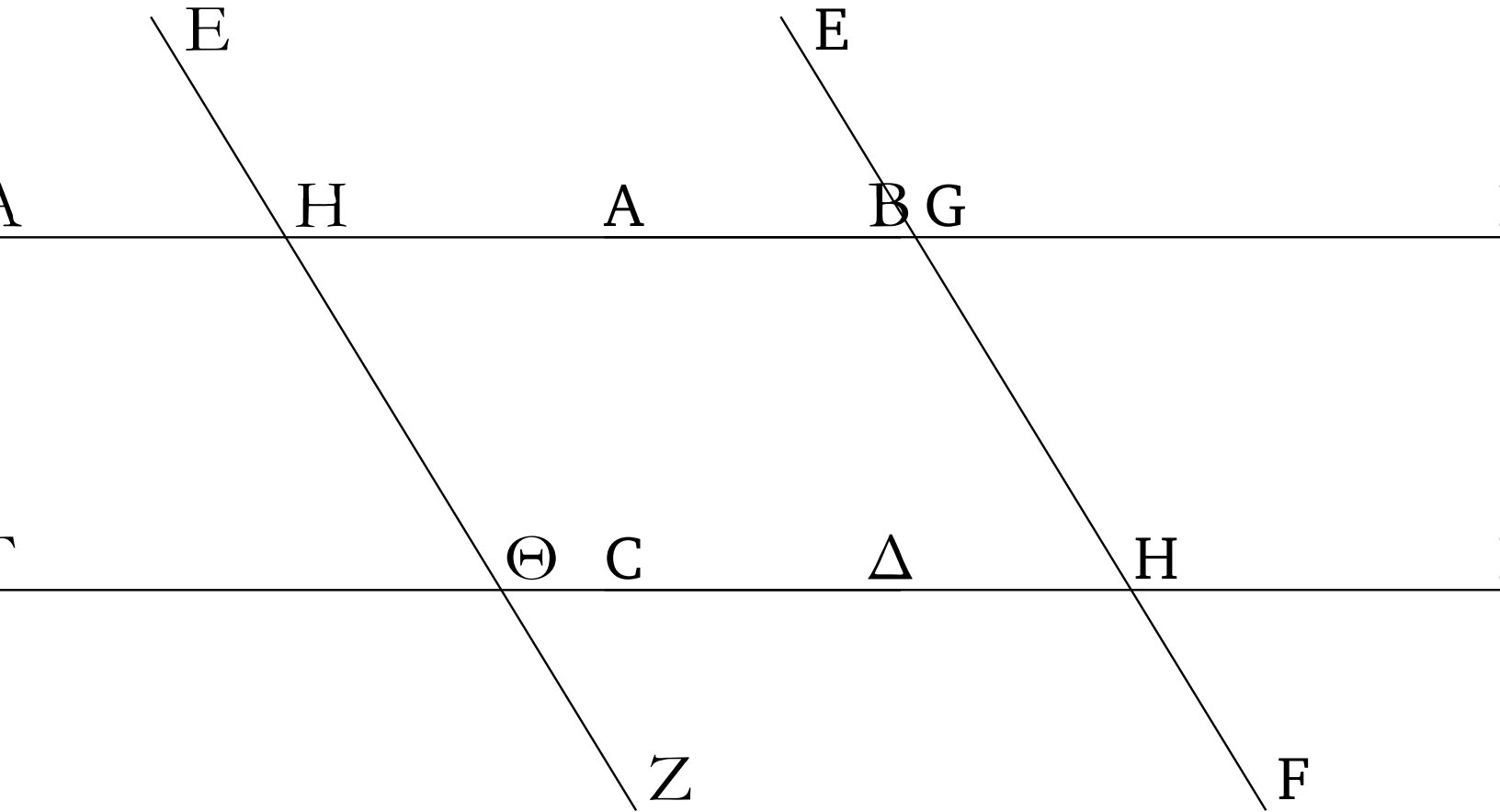
is) the very thing it was required to show.

Proposition 29

A straight-line falling across parallel straight-lines makes the alternate angles equal to one another, the external (angle) equal to the internal and opposite (angle), and the (sum of the) internal (angles) on the same side equal to two right-angles.

For let the straight-line EF fall across the parallel straight-lines AB and CD . I say that it makes the alternate angles, AGH and GHD , equal, the external angle EGB equal to the internal and opposite (angle) GHD , and the (sum of the) internal (angles) on the same side, BGH and GHD , equal to two right-angles.

For if AGH is unequal to GHD (then) one of them is greater. Let AGH be greater. Let BGH be added to both. Thus, (the sum of) AGH and BGH is greater than (the sum of) BGH and GHD . But, (the sum of) AGH and BGH is equal to two right-angles [Prop 1.13]. Thus, (the sum of) BGH and GHD is [also] less than two right-angles. But (straight-lines) being produced to infinity from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, AB and CD , being produced to infinity, will meet together. But they do not meet, on account of them (initially) being assumed parallel (to one another) [Def. 1.23]. Thus, AGH is not unequal to GHD . Thus, (it is) equal. But, AGH is equal to EGB [Prop. 1.15]. And EGB is thus also equal to GHD . Let BGH be added to both. Thus, (the sum of) EGB and BGH is equal to (the sum of) BGH and GHD . But, (the sum of) EGB and BGH is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) BGH and GHD is also equal to two right-angles.



Ἡ ἄρα εἷς τὰς παραλλήλους εὐθείας εὐθεῖα ἐμπίπτουσα
τάς τε ἐναλλάξ γωνίας ἴσας ἀλλήλαις ποιεῖ καὶ τὴν ἐκτός τῇ
ἐντός καὶ ἀπεναντίον ἴσην καὶ τὰς ἐντός καὶ ἐπὶ τὰ αὐτὰ μέρη
δυσὶν ὀρθαῖς ἴσας· ὅπερ ἔδει δεῖξαι.

Thus, a straight-line falling across parallel straight-lines
makes the alternate angles equal to one another, the external
(angle) equal to the internal and opposite (angle), and the (sum
of the) internal (angles) on the same side equal to two right-
angles. (Which is) the very thing it was required to show.

30.

Proposition 30

Αἱ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ ἀλλήλαις εἰσὶ παράλλη-
λοι.

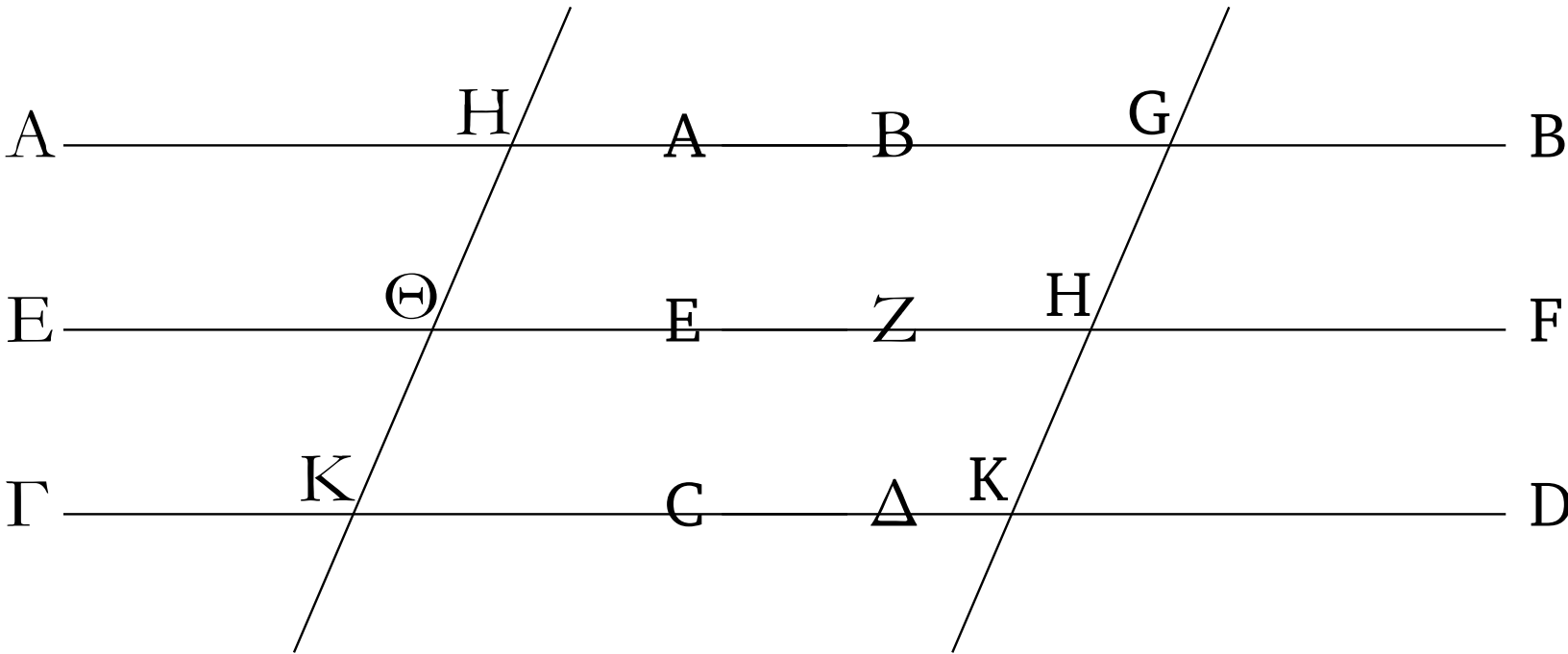
Ἐστω ἑκατέρα τῶν AB , $ΓΔ$ τῇ EZ παράλληλος· λέγω,
ὅτι καὶ ἡ AB τῇ $ΓΔ$ ἐστὶ παράλληλος.

Ἐμπιπτέτω γάρ εἷς αὐτάς εὐθεῖα ἡ HK .

(Straight-lines) parallel to the same straight-line are also
parallel to one another.

Let each of the (straight-lines) AB and CD be parallel to
 EF . I say that AB is also parallel to CD .

For let the straight-line GK fall across (AB , CD , and EF).



Καὶ ἐπεὶ εἷς παραλλήλων εὐθείας τὰς AB , EZ εὐθεία ἐμπίπτωκεν ἡ HK , ἴση ἄρα ἡ ὑπὸ AHK τῇ ὑπὸ $HΘZ$. πάλιν, ἐπεὶ εἷς παραλλήλων εὐθείας τὰς EZ , $ΓΔ$ εὐθεία ἐμπίπτωκεν ἡ HK , ἴση ἐστὶν ἡ ὑπὸ $HΘZ$ τῇ ὑπὸ $HKΔ$. ἐδείκθη δὲ καὶ ἡ ὑπὸ AHK τῇ ὑπὸ $HΘZ$ ἴση. καὶ ἡ ὑπὸ AHK ἄρα τῇ ὑπὸ $HKΔ$ ἐστὶν ἴση· καὶ εἰσιν ἐναλλάχ. παραλλήλοξ ἄρα ἐστὶν ἡ AB τῇ $ΓΔ$.

[Αἱ ἄρα τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ ἀλλήλαιξ εἰσὶ παράλληλοι.] ὅπερ ἔδει δεῖξαι.

31.

Διὰ τοῦ δοθέντος σημείου τῇ δοθείσῃ εὐθείᾳ παράλληλον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Ἐστω τὸ μὲν δοθὲν σημεῖον τὸ A , ἡ δὲ δοθεῖσα εὐθεῖα ἡ $BΓ$. δεῖ δὴ διὰ τοῦ A σημείου τῇ $BΓ$ εὐθείᾳ παράλληλον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Εἰλήφθω ἐπὶ τῇς $BΓ$ τυθὸν σημεῖον τὸ $Δ$, καὶ ἐπέξευθθω ἡ $AΔ$. καὶ συνεστάτω πρόξ τῇ $ΔA$ εὐθείᾳ καὶ τῷ πρόξ αὐτῇ σημείῳ τῷ A τῇ ὑπὸ $AΔΓ$ γωνίᾳ ἴση ἡ ὑπὸ $ΔAE$. καὶ ἐκβεβλήσθω ἐπ' εὐθείας τῇ EA εὐθείᾳ ἡ AZ .

And since the straight-line GK has fallen across the parallel straight-lines AB and EF , (angle) AGK (is) thus equal to GHF [Prop. 1.29]. Again, since the straight-line GK has fallen across the parallel straight-lines EF and CD , (angle) GHF is equal to GKD [Prop. 1.29]. But AGK was also shown (to be) equal to GKF . Thus, AGK is also equal to GKD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].

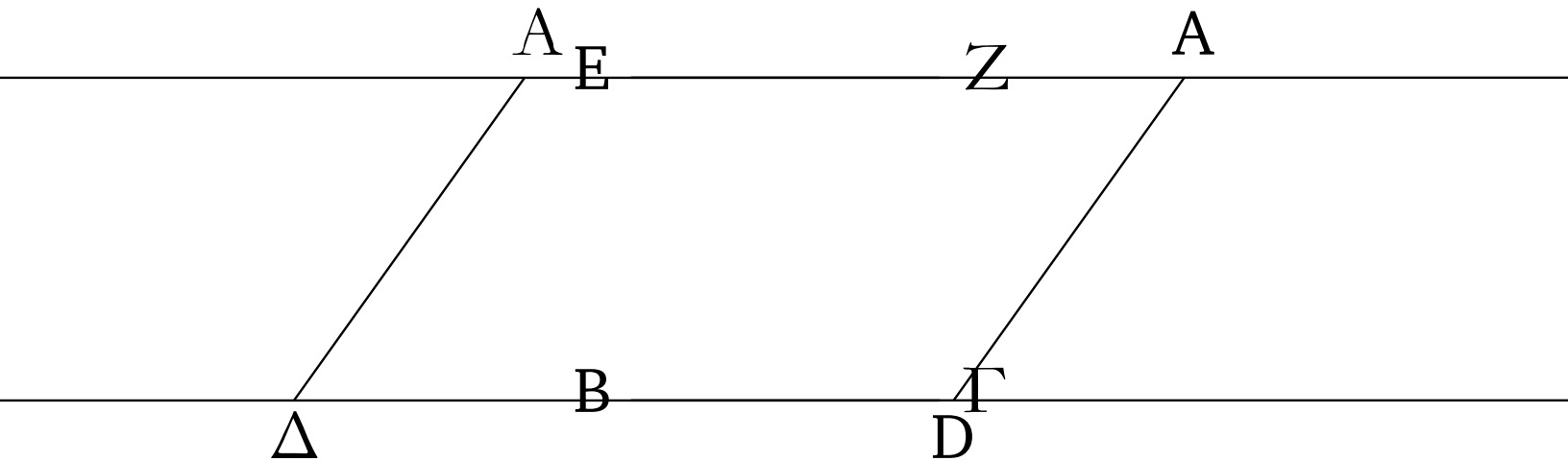
[Thus, (straight-lines) parallel to the same straight-line are also parallel to one another.] (Which is) the very thing it was required to show.

Proposition 31

To draw a straight-line parallel to a given straight-line, through a given point.

Let A be the given point, and BC the given straight-line. So it is required to draw a straight-line parallel to the straight-line BC , through the point A .

Let the point D be taken a random on BC , and let AD be joined. And let (angle) DAE , equal to angle ADC , be constructed on the straight-line DA at the point A on it [Prop. 1.23]. And let the straight-line AF be produced in a straight-line with EA .



Καὶ ἐπεὶ εἰς δύο εὐθείας τὰς BF , EZ εὐθεῖα ἐμπίπτουσα ἡ AD τὰς ἐναλλάχ γωνίας τὰς ὑπὸ EAD , ADF ἴσας ἀλλήλαις πεποίηκεν, παράλληλος ἄρα ἐστὶν ἡ EAF τῇ BF .

Διὰ τοῦ δοθέντος ἄρα σημείου τοῦ A τῇ δοθείσῃ εὐθείᾳ τῇ BF παράλληλος εὐθεῖα γραμμὴ ἤκται ἡ EAF · ὅπερ ἔδει ποιῆσαι.

And since the straight-line AD , (in) falling across the two straight-lines BC and EF , has made the alternate angles EAD and ADC equal to one another, EAF is thus parallel to BC [Prop. 1.27].

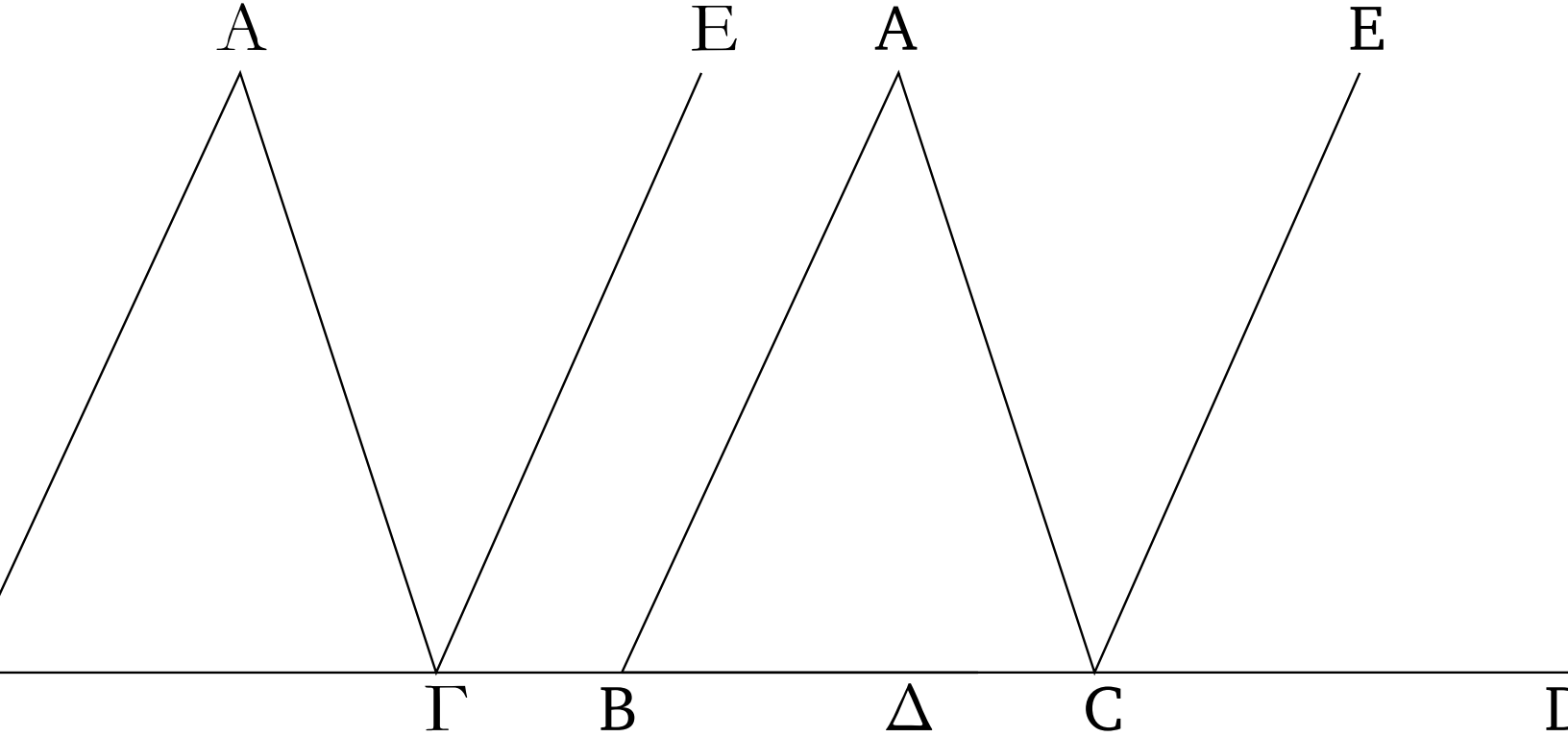
Thus, the straight-line EAF has been drawn parallel to the given straight-line BC , through the given point A . (Which is) the very thing it was required to do.

32.

Παντὸς τριγώνου μᾶξ τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία δυοὶ ταῖς ἐντὸς καὶ ἀπεναντίον ἴση ἐστίν, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι δυοὶν ὀρθαῖς ἴσαι εἰσίν.

Proposition 32

In any triangle, (if) one of the sides (is) produced (then) the external angle is equal to the (sum of the) two internal and opposite (angles), and the (sum of the) three internal angles of the triangle is equal to two right-angles.



Ἐστω τρίγωνον τὸ $AB\Gamma$, καὶ προσεκβεβλήσθω αὐτοῦ μία πλευρὰ ἢ $B\Gamma$ ἐπὶ τὸ Δ · λέγω, ὅτι ἡ ἐκτὸς γωνία ἢ ὑπὸ $AG\Delta$ ἴση ἐστὶ δυοῖν ταῖς ἐντὸς καὶ ἀπεναντίον ταῖς ὑπὸ ΓAB , $AB\Gamma$, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι αἱ ὑπὸ $AB\Gamma$, $B\Gamma A$, ΓAB δυοῖν ὀρθαῖς ἴσαι εἰσίν.

Ἦθθω γάρ διὰ τοῦ Γ σημείου τῇ AB εὐθείᾳ παράλληλος ἢ GE .

Καὶ ἐπεὶ παράλληλός ἐστιν ἡ AB τῇ GE , καὶ εἰς αὐτὰς ἐμπίπτωκεν ἡ AG , αἱ ἐναλλάξ γωνίαι αἱ ὑπὸ BAG , AGE ἴσαι ἀλλήλαις εἰσίν. πάλιν, ἐπεὶ παράλληλός ἐστιν ἡ AB τῇ GE , καὶ εἰς αὐτὰς ἐμπίπτωκεν εὐθεῖα ἢ BD , ἡ ἐκτὸς γωνία ἢ ὑπὸ $EG\Delta$ ἴση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ $AB\Gamma$. ἐδείκθη δὲ καὶ ἡ ὑπὸ AGE τῇ ὑπὸ BAG ἴση· ὅλη ἄρα ἢ ὑπὸ $AG\Delta$ γωνία ἴση ἐστὶ δυοῖν ταῖς ἐντὸς καὶ ἀπεναντίον ταῖς ὑπὸ BAG , $AB\Gamma$.

Κοινὴ προσκείσθω ἡ ὑπὸ AGB · αἱ ἄρα ὑπὸ $AG\Delta$, AGB τρισὶ ταῖς ὑπὸ $AB\Gamma$, $B\Gamma A$, ΓAB ἴσαι εἰσίν. ἀλλ' αἱ ὑπὸ $AG\Delta$, AGB δυοῖν ὀρθαῖς ἴσαι εἰσίν· καὶ αἱ ὑπὸ AGB , ΓBA , ΓAB ἄρα δυοῖν ὀρθαῖς ἴσαι εἰσίν.

Παντὸς ἄρα τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία δυοῖν ταῖς ἐντὸς καὶ ἀπεναντίον ἴση ἐστίν, καὶ αἱ ἐντὸς τοῦ τριγώνου τρεῖς γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσίν· ὅπερ ἔδει δεῖξαι.

Let ABC be a triangle, and let one of its sides BC be produced to D . I say that the external angle ACD is equal to the (sum of the) two internal and opposite angles CAB and ABC , and the (sum of the) three internal angles of the triangle— ABC , BCA , and CAB —is equal to two right-angles.

For let CE be drawn through point C parallel to the straight-line AB [Prop. 1.31].

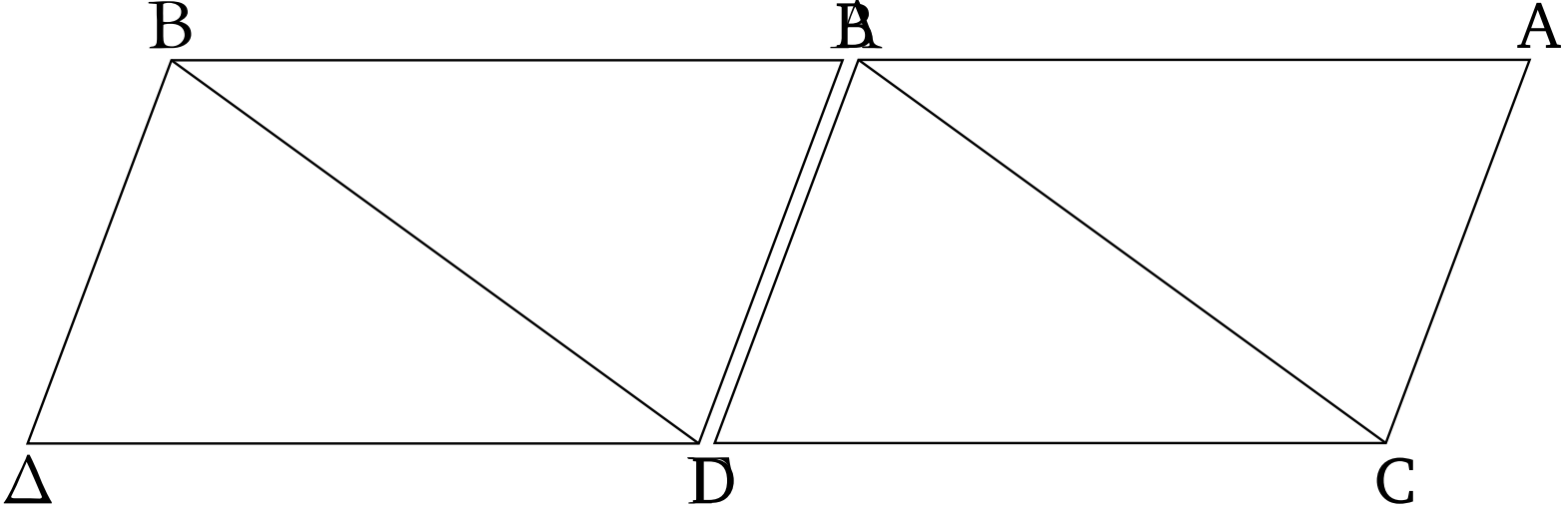
And since AB is parallel to CE , and AC has fallen across them, the alternate angles BAC and ACE are equal to one another [Prop. 1.29]. Again, since AB is parallel to CE , and the straight-line BD has fallen across them, the external angle ECD is equal to the internal and opposite (angle) ABC [Prop. 1.29]. But ACE was also shown (to be) equal to BAC . Thus, the whole angle ACD is equal to the (sum of the) two internal and opposite (angles) BAC and ABC .

Let ACB be added to both. Thus, (the sum of) ACD and ACB is equal to the (sum of the) three (angles) ABC , BCA , and CAB . But, (the sum of) ACD and ACB is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) ACB , CBA , and CAB is also equal to two right-angles.

Thus, in any triangle, (if) one of the sides (is) produced (then) the external angle is equal to the (sum of the) two internal and opposite (angles), and the (sum of the) three internal angles of the triangle is equal to two right-angles. (Which is) the very thing it was required to show.

Αἱ τὰς ἴσας τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη ἐπι-
ξενυγνύουσαι εὐθεῖαι καὶ αὐταὶ ἴσαι τε καὶ παράλληλοι εἰσιν.

Ἐστωσαν ἴσαι τε καὶ παράλληλοι αἱ AB , $\Gamma\Delta$, καὶ ἐπιξεν-
γνύωσαν αὐτὰς ἐπὶ τὰ αὐτὰ μέρη εὐθεῖαι αἱ AG , $B\Delta$. λέγω,
ὅτι καὶ αἱ AG , $B\Delta$ ἴσαι τε καὶ παράλληλοι εἰσιν.



Ἐπεξεύθθω ἡ $B\Gamma$. καὶ ἐπεὶ παράλληλός ἐστιν ἡ AB τῇ
 $\Gamma\Delta$, καὶ εἰς αὐτὰς ἐμπίπτωκεν ἡ $B\Gamma$, αἱ ἐναλλάξ γωνίαι αἱ
ὑπὸ $AB\Gamma$, $B\Gamma\Delta$ ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ
 AB τῇ $\Gamma\Delta$ κοινὴ δὲ ἡ $B\Gamma$, δύο δὴ αἱ AB , $B\Gamma$ δύο ταῖς $B\Gamma$,
 $\Gamma\Delta$ ἴσαι εἰσὶν καὶ γωνία ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ $B\Gamma\Delta$
ἴση. βάσεις ἄρα ἡ AG βάσει τῇ $B\Delta$ ἐστὶν ἴση, καὶ τὸ $AB\Gamma$
τρίγωνον τῷ $B\Gamma\Delta$ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι
ταῖς λοιπαῖς γωνίαις ἴσαι ἔσσονται ἑκατέρα ἑκατέρῃ, ὅφ' ἂν
αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἡ ὑπὸ AGB γωνία
τῇ ὑπὸ $GB\Delta$. καὶ ἐπεὶ εἰς δύο εὐθείας τὰς AG , $B\Delta$ εὐθεῖα
ἐμπίπτουσα ἡ $B\Gamma$ τὰς ἐναλλάξ γωνίας ἴσας ἀλλήλαις πεπο-
ίηκεν, παράλληλός ἄρα ἐστὶν ἡ AG τῇ $B\Delta$. ἐδείκθη δὲ αὐτῇ
καὶ ἴση.

Αἱ ἄρα τὰς ἴσας τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη
ἐπιξενυγνύουσαι εὐθεῖαι καὶ αὐταὶ ἴσαι τε καὶ παράλληλοι ε-
ἰσιν· ὅπερ ἔδει δεῖχαι.

Straight-lines joining equal and parallel (straight-lines) on
the same sides are themselves also equal and parallel.

Let AB and CD be equal and parallel (straight-lines), and
let the straight-lines AC and BD join them on the same sides.
I say that AC and BD are also equal and parallel.

Let BC be joined. And since AB is parallel to CD , and BC
has fallen across them, the alternate angles ABC and BCD are
equal to one another [Prop. 1.29]. And since AB is equal to
 CD , and BC is common, the two (straight-lines) AB , BC are
equal to the two (straight-lines) DC , CB .[†] And the angle ABC
is equal to the angle BCD . Thus, the base AC is equal to the
base BD , and triangle ABC is equal to triangle DCB ,[‡] and the
remaining angles will be equal to the corresponding remaining
angles subtended by the equal sides [Prop. 1.4]. Thus, angle
 ACB is equal to CBD . Also, since the straight-line BC , (in)
falling across the two straight-lines AC and BD , has made the
alternate angles (ACB and CBD) equal to one another, AC is
thus parallel to BD [Prop. 1.27]. And (AC) was also shown (to
be) equal to (BD).

Thus, straight-lines joining equal and parallel (straight-
lines) on the same sides are themselves also equal and parallel.
(Which is) the very thing it was required to show.

[†] The Greek text has “ BC , CD ”, which is obviously a mistake.

[‡] The Greek text has “ DCB ”, which is obviously a mistake.

34.

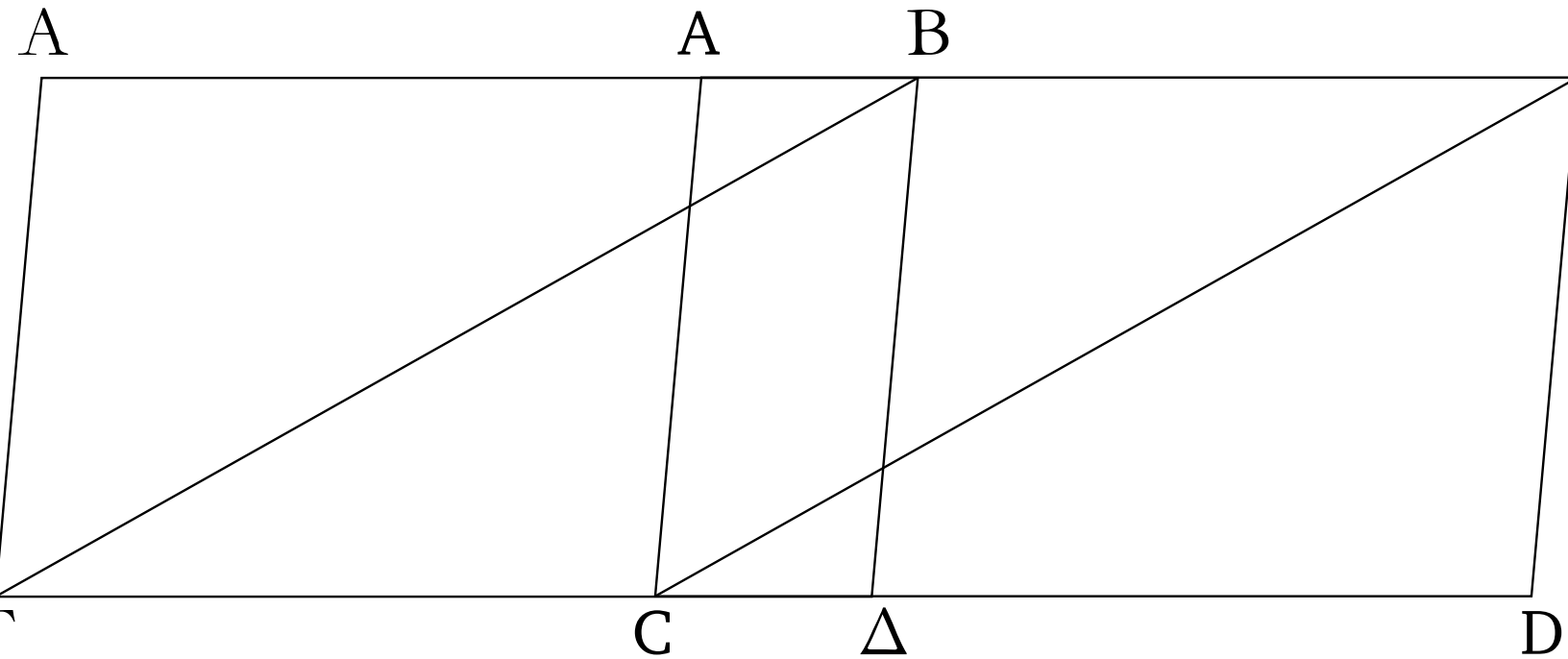
Proposition 34

Τῶν παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραὶ
τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ ἡ διάμετρος αὐτὰ διῖθα
τέμνει.

Ἐστω παραλληλόγραμμον θωρίον τὸ $AG\Delta B$, διάμετρος
δὲ αὐτοῦ ἡ $B\Gamma$. λέγω, ὅτι τοῦ $AG\Delta B$ παραλληλογράμμου αἱ
ἀπεναντίον πλευραὶ τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ ἡ
 $B\Gamma$ διάμετρος αὐτὸ διῖθα τέμνει.

In parallelogramic figures, the opposite sides and angles are
equal to one another, and a diagonal cuts them in half.

Let $ACDB$ be a parallelogramic figure, and BC its diagon-
al. I say that for parallelogram $ACDB$, the opposite sides and
angles are equal to one another, and the diagonal BC cuts it in
half.



Ἐπεὶ γὰρ παράλληλός ἐστιν ἡ AB τῇ $ΓΔ$, καὶ εἰς αὐτάς ἐμπεπτωκεν εὐθεῖα ἡ $ΒΓ$, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $ΑΒΓ$, $ΒΓΔ$ ἴσαι ἀλλήλαις εἰσίν. πάλιν ἐπεὶ παράλληλός ἐστιν ἡ $ΑΓ$ τῇ $ΒΔ$, καὶ εἰς αὐτάς ἐμπεπτωκεν ἡ $ΒΓ$, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $ΑΓΒ$, $ΓΒΔ$ ἴσαι ἀλλήλαις εἰσίν. δύο δὲ τρίγωνα ἐστί τὰ $ΑΒΓ$, $ΒΓΔ$ τὰς δύο γωνίας τὰς ὑπὸ $ΑΒΓ$, $ΒΓΔ$ δυοὶ ταῖς ὑπὸ $ΓΒΔ$, $ΓΒΔ$ ἴσαξ ἔθοντα ἑκατέραν ἑκατέρῳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις κοινὴν αὐτῶν τὴν $ΒΓ$. καὶ τὰς λοιπὰς ἄρα πλευρὰς ταῖς λοιπαῖς ἴσαξ ἔχει ἑκατέραν ἑκατέρῳ καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ. ἴση ἄρα ἡ μὲν $ΑΒ$ πλευρὰ τῇ $ΓΔ$, ἡ δὲ $ΑΓ$ τῇ $ΒΔ$, καὶ ἔτι ἴση ἐστὶν ἡ ὑπὸ $ΒΑΓ$ γωνία τῇ ὑπὸ $ΓΔΒ$. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΒΓΔ$, ἡ δὲ ὑπὸ $ΓΒΔ$ τῇ ὑπὸ $ΑΓΒ$, ὅλη ἄρα ἡ ὑπὸ $ΑΒΔ$ ὅλη τῇ ὑπὸ $ΑΓΔ$ ἐστὶν ἴση. ἐδείκθη δὲ καὶ ἡ ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΓΔΒ$ ἴση.

Τῶν ἄρα παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραί τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσίν.

Λέγω δὴ, ὅτι καὶ ἡ διάμετρος αὐτὰ διῖθα τέμνει. ἐπεὶ γὰρ ἴση ἐστὶν ἡ $ΑΒ$ τῇ $ΓΔ$, κοινὴ δὲ ἡ $ΒΓ$, δύο δὲ αἱ $ΑΒ$, $ΒΓ$ δυοὶ ταῖς $ΓΔ$, $ΒΓ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρῳ. καὶ γωνία ἡ ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΒΓΔ$ ἴση. καὶ βάσις ἄρα ἡ $ΑΓ$ τῇ $ΔΒ$ ἴση. καὶ τὸ $ΑΒΓ$ [ἄρα] τρίγωνον τῷ $ΒΓΔ$ τριγώνῳ ἴσον ἐστίν.

Ἡ ἄρα $ΒΓ$ διάμετρος διῖθα τέμνει τὸ $ΑΒΓΔ$ παραλληλόγραμμον· ὅπερ ἔδει δεῖξαι.

For since AB is parallel to CD , and the straight-line BC has fallen across them, the alternate angles ABC and BCD are equal to one another [Prop. 1.29]. Again, since AC is parallel to BD , and BC has fallen across them, the alternate angles ACB and CBD are equal to one another [Prop. 1.29]. So ABC and BCD are two triangles having the two angles ABC and BCA equal to the two (angles) BCD and CBD , respectively, and one side equal to one side—the (one) by the equal angles and common to them, (namely) BC . Thus, they will also have the remaining sides equal to the corresponding remaining (sides), and the remaining angle (equal) to the remaining angle [Prop. 1.26]. Thus, side AB is equal to CD , and AC to BD . Furthermore, angle BAC is equal to CDB . And since angle ABC is equal to BCD , and CBD to ACB , the whole (angle) ABD is thus equal to the whole (angle) ACD . And BAC was also shown (to be) equal to CDB .

Thus, in parallelogramic figures, the opposite sides and angles are equal to one another.

And, I also say that a diagonal cuts them in half. For since AB is equal to CD , and BC (is) common, the two (straight-lines) AB , BC are equal to the two (straight-lines) DC , CB [†], respectively. And angle ABC is equal to angle BCD . Thus, the base AC (is) also equal to DB , and triangle ABC is equal to triangle BCD [Prop. 1.4].

Thus, the diagonal BC cuts the parallelogram $ACDB$ [‡] in half. (Which is) the very thing it was required to show.

[†] The Greek text has “ CD , BC ”, which is obviously a mistake.

[‡] The Greek text has “ $ABCD$ ”, which is obviously a mistake.

35.

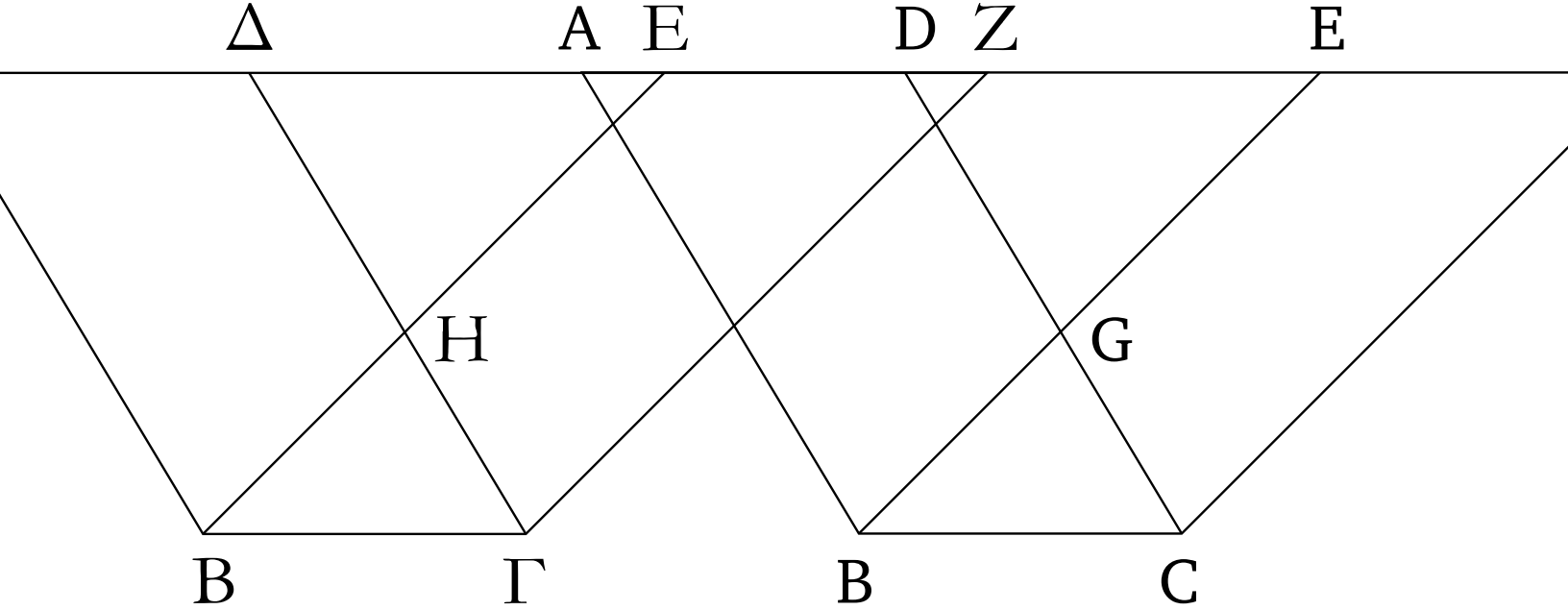
Proposition 35

Τὰ παραλληλόγραμμα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν.

Ἐστω παραλληλόγραμμα τὰ $ABΓΔ$, $EBΓΖ$ ἐπὶ τῆς αὐτῆς βάσεως τῆς $ΒΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $ΑΖ$, $ΒΓ$. λέγω, ὅτι ἴσον ἐστὶ τὸ $ABΓΔ$ τῷ $EBΓΖ$ παραλληλογράμμῳ.

Parallelograms which are on the same base and between the same parallels are equal[†] to one another.

Let $ABCD$ and $EBCF$ be parallelograms on the same base BC , and between the same parallels AF and BC . I say that $ABCD$ is equal to parallelogram $EBCF$.



Ἐπεὶ γὰρ παραλληλόγραμμὸν ἐστὶ τὸ $ABΓΔ$, ἴση ἐστὶν ἡ $ΑΔ$ τῇ $ΒΓ$. διὰ τὰ αὐτὰ δὴ καὶ ἡ $ΕΖ$ τῇ $ΒΓ$ ἐστὶν ἴση· ὥστε καὶ ἡ $ΑΔ$ τῇ $ΕΖ$ ἐστὶν ἴση· καὶ κοινὴ ἡ $ΔΕ$ · ὅλη ἄρα ἡ $ΑΕ$ ὅλη τῇ $ΔΖ$ ἐστὶν ἴση. ἐστὶ δὲ καὶ ἡ $ΑΒ$ τῇ $ΔΓ$ ἴση· δύο δὴ αἱ $ΕΑ$, $ΑΒ$ δύο ταῖς $ΖΔ$, $ΔΓ$ ἴσαι εἰσὶν ἐκατέρω ἐκατέρω· καὶ γωνία ἡ ὑπὸ $ΖΔΓ$ γωνία τῇ ὑπὸ $ΕΑΒ$ ἐστὶν ἴση ἡ ἐκτὸς τῇ ἐντὸς· βάσει ἄρα ἡ $ΕΒ$ βάσει τῇ $ΖΓ$ ἴση ἐστίν, καὶ τὸ $ΕΑΒ$ τρίγωνον τῷ $ΔΖΓ$ τριγώνῳ ἴσον ἐστὶ· κοινὸν ἀφρηθήσθω τὸ $ΔΗΕ$ · λοιπὸν ἄρα τὸ $ΑΒΗΔ$ τραπέζιον λοιπῷ τῷ $ΕΗΓΖ$ τραπέζιῳ ἐστὶν ἴσον· κοινὸν προσκείσθω τὸ $ΗΒΓ$ τρίγωνον· ὅλον ἄρα τὸ $ΑΒΓΔ$ παραλληλόγραμμον ὅλῳ τῷ $ΕΒΓΖ$ παραλληλογράμμῳ ἴσον ἐστίν.

Τὰ ἄρα παραλληλόγραμμα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν· ὅπερ εἶδει δεῖχαι.

For since $ABCD$ is a parallelogram, AD is equal to BC [Prop. 1.34]. So, for the same (reasons), EF is also equal to BC . So AD is also equal to EF . And DE is common. Thus, the whole (straight-line) AE is equal to the whole (straight-line) DF . And AB is also equal to DC . So the two (straight-lines) EA , AB are equal to the two (straight-lines) FD , DC , respectively. And angle FDC is equal to angle EAB , the external to the internal [Prop. 1.29]. Thus, the base EB is equal to the base FC , and triangle EAB will be equal to triangle DFC [Prop. 1.4]. Let DGE be taken away from both. Thus, the remaining trapezium $ABGD$ is equal to the remaining trapezium $EGCF$. Let triangle GBC be added to both. Thus, the whole parallelogram $ABCD$ is equal to the whole parallelogram $EBCF$.

Thus, parallelograms which are on the same base and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

[†] Here, for the first time, “equal” means “equal in area”, rather than “congruent”.

36.

Proposition 36

Τὰ παραλληλόγραμμα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν.

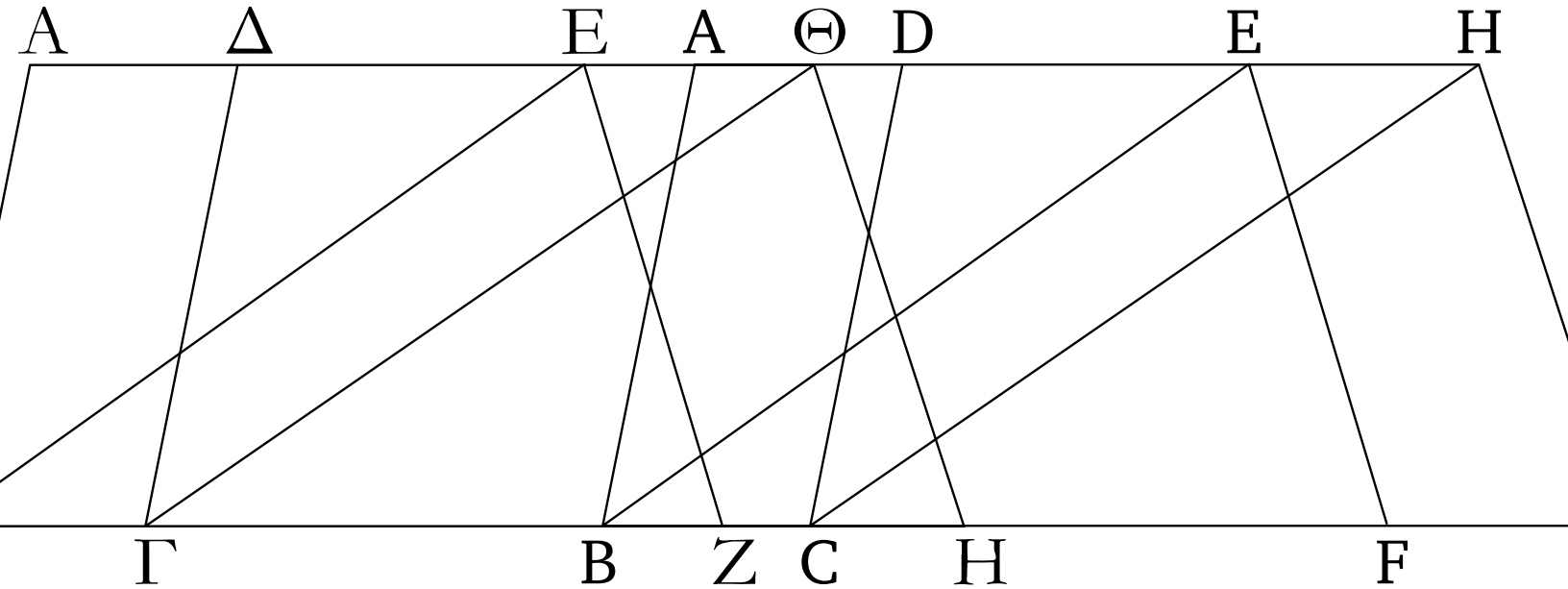
Ἐστω παραλληλόγραμμα τὰ $ABΓΔ$, $ΕΖΗΘ$ ἐπὶ ἴσων

Parallelograms which are on equal bases and between the same parallels are equal to one another.

Let $ABCD$ and $EFGH$ be parallelograms which are on the

βάσεων ὄντα τῶν $BΓ$, ZH καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $AΘ$, BH · λέγω, ὅτι ἴσον ἐστὶ τὸ $ABΓΔ$ παραλληλόγραμμον τῷ $EZHΘ$.

equal bases BC and FG , and (are) between the same parallels AH and BG . I say that the parallelogram $ABCD$ is equal to $EFGH$.



Ἐπεξεύθωσαν γὰρ αἱ BE , $ΓΘ$. καὶ ἐπεὶ ἴση ἐστὶν ἡ $BΓ$ τῇ ZH , ἀλλὰ ἡ ZH τῇ $EΘ$ ἐστὶν ἴση, καὶ ἡ $BΓ$ ἄρα τῇ $EΘ$ ἐστὶν ἴση. εἰσὶ δὲ καὶ παράλληλοι. καὶ ἐπιξεννύουσιν αὐτάς αἱ EB , $ΘΓ$ · αἱ δὲ τὰς ἴσας τε καὶ παραλλήλους ἐπὶ τὰ αὐτὰ μέρη ἐπιξεννύουσαι ἴσαι τε καὶ παράλληλοί εἰσι [καὶ αἱ EB , $ΘΓ$ ἄρα ἴσαι τέ εἰσι καὶ παράλληλοι]. παραλληλόγραμμον ἄρα ἐστὶ τὸ $EBΓΘ$. καὶ ἐστὶν ἴσον τῷ $ABΓΔ$ · βάσιν τε γὰρ αὐτῷ τὴν αὐτὴν ἔθει τὴν $BΓ$, καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστὶν αὐτῷ ταῖς $BΓ$, $AΘ$. διὰ τὰ αὐτὰ δὴ καὶ τὸ $EZHΘ$ τῷ αὐτῷ τῷ $EBΓΘ$ ἐστὶν ἴσον· ὥστε καὶ τὸ $ABΓΔ$ παραλληλόγραμμον τῷ $EZHΘ$ ἐστὶν ἴσον.

Τὰ ἄρα παραλληλόγραμμα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

For let BE and CH be joined. And since BC is equal to FG , but FG is equal to EH [Prop. 1.34], BC is thus equal to EH . And they are also parallel, and EB and HC join them. But (straight-lines) joining equal and parallel (straight-lines) on the same sides are (themselves) equal and parallel [Prop. 1.33] [thus, EB and HC are also equal and parallel]. Thus, $EBCH$ is a parallelogram [Prop. 1.34], and is equal to $ABCD$. For it has the same base, BC , (as $ABCD$), and is between the same parallels, BC and AH , (as $ABCD$) [Prop. 1.35]. So, for the same (reasons), $EFGH$ is also equal to the same (parallelogram) $EBCH$ [Prop. 1.34]. So that the parallelogram $ABCD$ is also equal to $EFGH$.

Thus, parallelograms which are on equal bases and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

37.

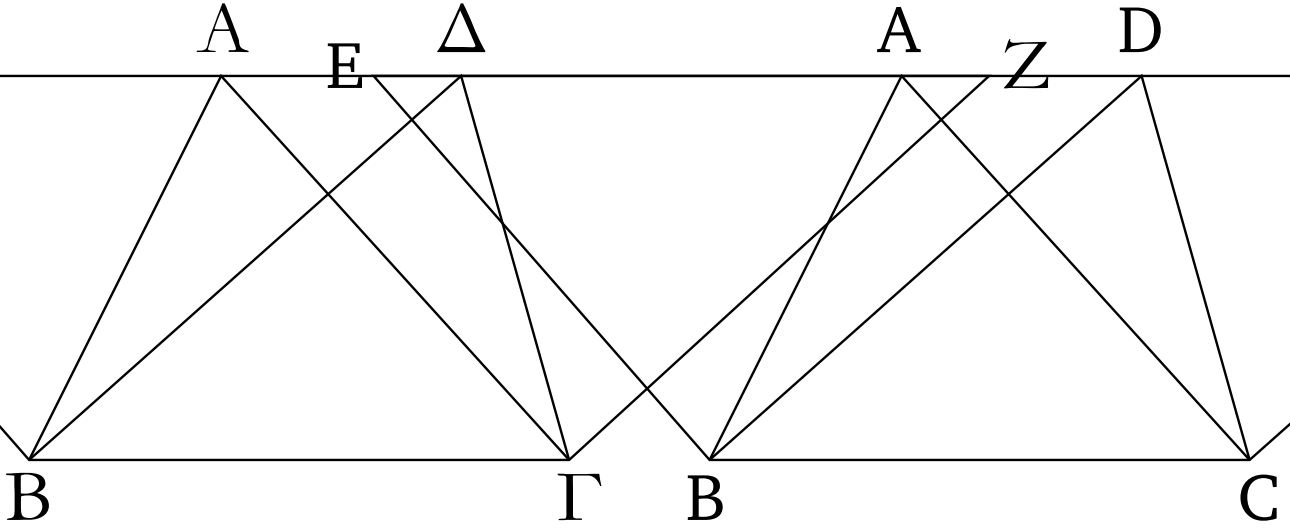
Proposition 37

Τὰ τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν.

Ἐστω τρίγωνα τὰ $ABΓ$, $ΔBΓ$ ἐπὶ τῆς αὐτῆς βάσεως τῆς $BΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς AD , $BΓ$ · λέγω, ὅτι ἴσον ἐστὶ τὸ $ABΓ$ τρίγωνον τῷ $ΔBΓ$ τριγώνῳ.

Triangles which are on the same base and between the same parallels are equal to one another.

Let ABC and DBC be triangles on the same base BC , and between the same parallels AD and BC . I say that triangle ABC is equal to triangle DBC .



Ἐκβεβλήσθω ἡ AD ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ E , Z , καὶ διὰ μὲν τοῦ B τῇ GA παράλληλος ἡθθω ἡ BE , διὰ δὲ τοῦ Γ τῇ BD παράλληλος ἡθθω ἡ ΓZ . παραλληλόγραμμον ἄρα ἐστὶν ἑκάτερον τῶν $EBGA$, $\Delta B\Gamma Z$ · καὶ εἰσιν ἴσα· ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς $B\Gamma$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $B\Gamma$, EZ · καὶ ἐστὶ τοῦ μὲν $EBGA$ παραλληλογράμμου ἡμισὺν τὸ $AB\Gamma$ τρίγωνον· ἡ γὰρ AB διάμετρος αὐτὸ διίτα τέμνει· τοῦ δὲ $\Delta B\Gamma Z$ παραλληλογράμμου ἡμισὺν τὸ $\Delta B\Gamma$ τρίγωνον· ἡ γὰρ $\Delta\Gamma$ διάμετρος αὐτὸ διίτα τέμνει. [τὰ δὲ τῶν ἴσων ἡμίση ἴσα ἀλλήλοις ἐστίν]. ἴσον ἄρα ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ $\Delta B\Gamma$ τριγώνῳ.

Τὰ ἄρα τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

Let AD be produced both directions to E and F , and let the (straight-line) BE be drawn through B parallel to CA [Prop. 1.31], and let the (straight-line) CF be drawn through C parallel to BD [Prop. 1.31]. Thus, $EBCA$ and $DBCF$ are both parallelograms, and are equal. For they are on the same base BC , and between the same parallels BC and EF [Prop. 1.35]. And the triangle ABC is half of the parallelogram $EBCA$. For the diagonal AB cuts the latter in half [Prop. 1.34]. And the triangle DBC (is) half of the parallelogram $DBCF$. For the diagonal DC cuts the latter in half [Prop. 1.34]. [And the halves of equal things are equal to one another.][†] Thus, triangle ABC is equal to triangle DBC .

Thus, triangles which are on the same base and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

[†] This is an additional common notion.

38.

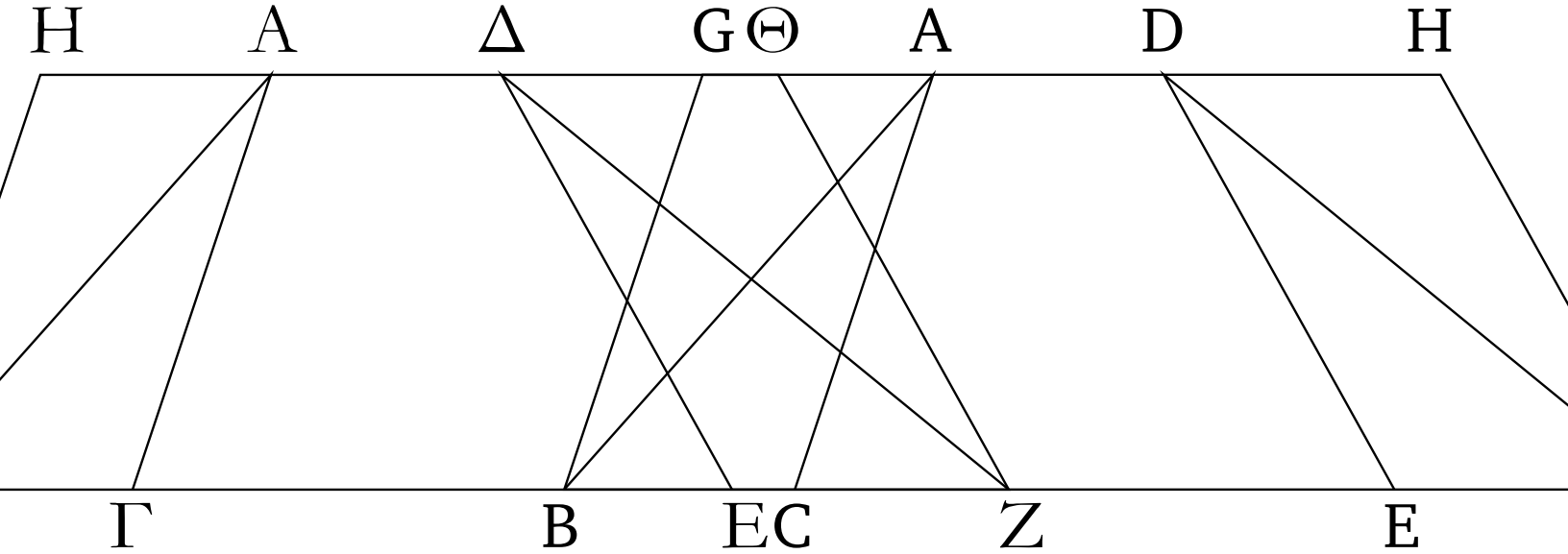
Τὰ τρίγωνα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν.

Ἐστω τρίγωνα τὰ $AB\Gamma$, ΔEZ ἐπὶ ἴσων βάσεων τῶν $B\Gamma$, EZ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς BZ , AD · λέγω, ὅτι ἴσον ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ ΔEZ τριγώνῳ.

Proposition 38

Triangles which are on equal bases and between the same parallels are equal to one another.

Let ABC and DEF be triangles on the equal bases BC and EF , and between the same parallels BF and AD . I say that triangle ABC is equal to triangle DEF .



Ἐκβεβλήσθω γὰρ ἡ AD ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ H , Θ , καὶ διὰ μὲν τοῦ B τῇ GA παράλληλος ἡθθω ἡ BH , διὰ δὲ τοῦ Z τῇ DE παράλληλος ἡθθω ἡ $Z\Theta$. παραλληλόγραμμον ἄρα ἐστὶν ἑκάτερον τῶν $HBGA$, $\Delta EZ\Theta$ · καὶ ἴσον τὸ $HBGA$ τῷ $\Delta EZ\Theta$ · ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν BG , EZ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς BZ , $H\Theta$ · καὶ ἐστὶ τοῦ μὲν $HBGA$ παραλληλογράμμου ἡμισυ τὸ ABG τρίγωνον. ἡ γὰρ AB διάμετρος αὐτὸ διῖτα τέμνει· τοῦ δὲ $\Delta EZ\Theta$ παραλληλογράμμου ἡμισυ τὸ $ZE\Delta$ τρίγωνον· ἡ γὰρ ΔZ διάμετρος αὐτὸ διῖτα τέμνει [τὰ δὲ τῶν ἴσων ἡμίση ἴσα ἀλλήλοις ἐστίν]. ἴσον ἄρα ἐστὶ τὸ ABG τρίγωνον τῷ ΔEZ τριγώνῳ.

Τὰ ἄρα τρίγωνα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

For let AD be produced in both directions to G and H , and let the (straight-line) BG be drawn through B parallel to CA [Prop. 1.31], and let the (straight-line) FH be drawn through F parallel to DE [Prop. 1.31]. Thus, $GBCA$ and $DEFH$ are each parallelograms. And $GBCA$ is equal to $DEFH$. For they are on the equal bases BC and EF , and between the same parallels BF and GH [Prop. 1.36]. And triangle ABC is half of the parallelogram $GBCA$. For the diagonal AB cuts the latter in half [Prop. 1.34]. And triangle FED (is) half of parallelogram $DEFH$. For the diagonal DF cuts the latter in half. [And the halves of equal things are equal to one another.] Thus, triangle ABC is equal to triangle DEF .

Thus, triangles which are on equal bases and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

39.

Proposition 39

Τὰ ἴσα τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Ἔστω ἴσα τρίγωνα τὰ ABG , ΔBG ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη τῆς BG · λέγω, ὅτι καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Ἐπεξέυθθω γὰρ ἡ AD · λέγω, ὅτι παράλληλός ἐστιν ἡ AD τῇ BG .

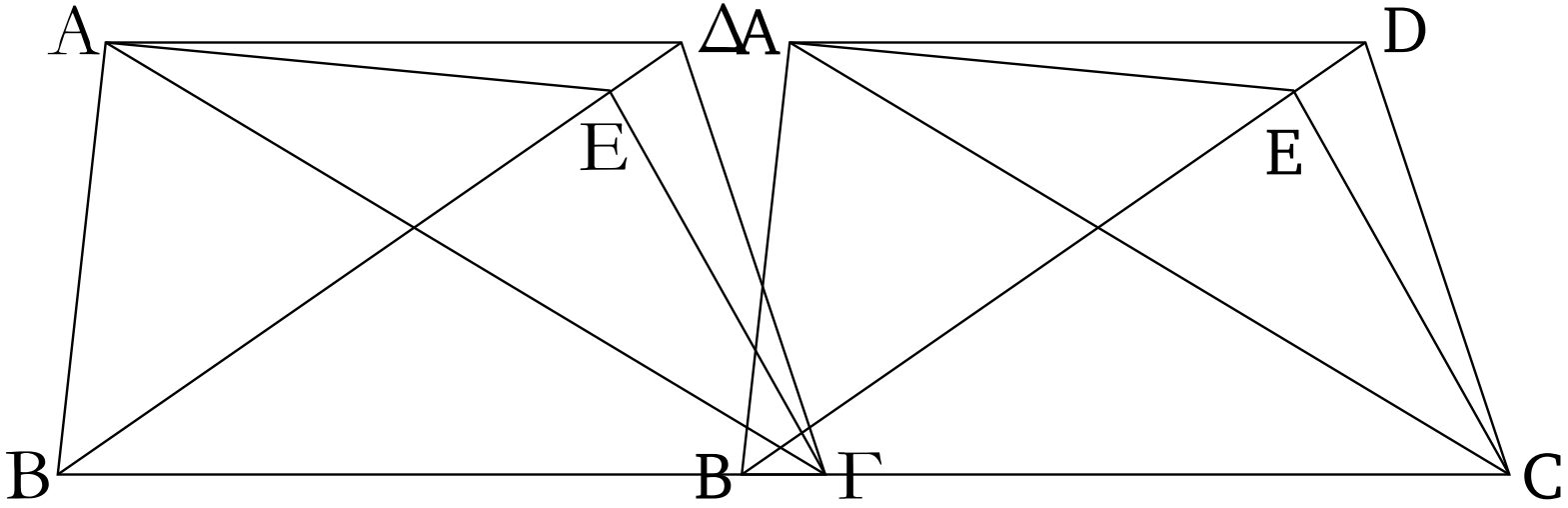
Εἰ γὰρ μή, ἡθθω διὰ τοῦ A σημείου τῇ BG εὐθείᾳ παράλληλος ἡ AE , καὶ ἐπεξέυθθω ἡ EF . ἴσον ἄρα ἐστὶ τὸ ABG τρίγωνον τῷ EBG τριγώνῳ· ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως ἐστὶν αὐτῷ τῇ BG καὶ ἐν ταῖς αὐταῖς παραλλήλοις. ἀλλὰ τὸ ABG τῷ ΔBG ἐστὶν ἴσον· καὶ τὸ ΔBG ἄρα τῷ EBG ἴσον ἐστὶ τὸ μείζον τῷ ἐλάσσονι· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα παράλληλός ἐστιν ἡ AE τῇ BG . ὁμοίως δὲ δείχομεν, ὅτι οὐδ' ἄλλη τις πλὴν τῆς AD · ἡ AD ἄρα τῇ BG ἐστὶ παράλληλος.

Equal triangles which are on the same base, and on the same side, are also between the same parallels.

Let ABC and DBC be equal triangles which are on the same base BC , and on the same side (of it). I say that they are also between the same parallels.

For let AD be joined. I say that AD and BC are parallel.

For, if not, let AE be drawn through point A parallel to the straight-line BC [Prop. 1.31], and let EC be joined. Thus, triangle ABC is equal to triangle EBC . For it is on the same base as it, BC , and between the same parallels [Prop. 1.37]. But ABC is equal to DBC . Thus, DBC is also equal to EBC , the greater to the lesser. The very thing is impossible. Thus, AE is not parallel to BC . Similarly, we can show that neither (is) any other (straight-line) than AD . Thus, AD is parallel to BC .



Τὰ ἄρα ἴσα τρίγωνα τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν· ὅπερ ἔδει δεῖχαι.

Thus, equal triangles which are on the same base, and on the same side, are also between the same parallels. (Which is) the very thing it was required to show.

40.

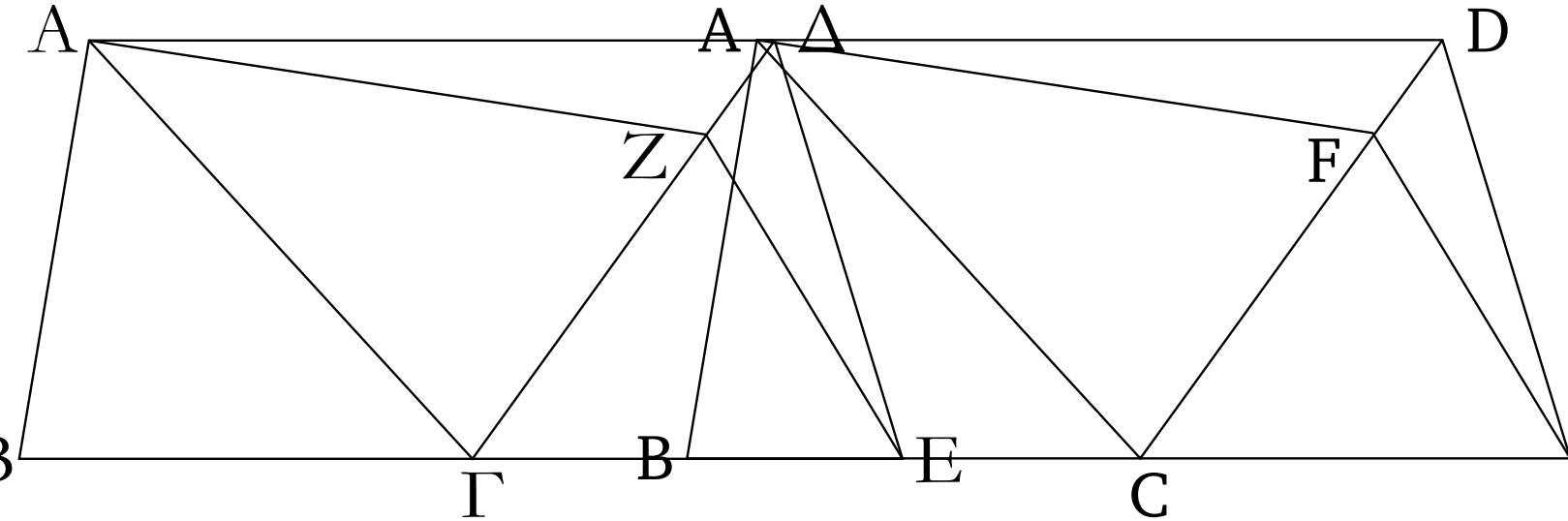
Proposition 40[†]

Τὰ ἴσα τρίγωνα τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Equal triangles which are on equal bases, and on the same side, are also between the same parallels.

Ἐστω ἴσα τρίγωνα τὰ $AB\Gamma$, $\Gamma\Delta E$ ἐπὶ ἴσων βάσεων τῶν $B\Gamma$, ΓE καὶ ἐπὶ τὰ αὐτὰ μέρη. λέγω, ὅτι καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν.

Let ABC and CDE be equal triangles on the equal bases BC and CE (respectively), and on the same side (of BE). I say that they are also between the same parallels.



Ἐπεξεύθθω γὰρ ἡ $A\Delta$. λέγω, ὅτι παραλλήλός ἐστιν ἡ $A\Delta$ τῇ BE .

For let AD be joined. I say that AD is parallel to BE .

Εἰ γὰρ μή, ῥηθθω διὰ τοῦ A τῇ BE παράλληλος ἡ AZ , καὶ ἐπεξεύθθω ἡ ZE . ἴσον ἄρα ἐστὶ τὸ $AB\Gamma$ τριγώνον τῷ $Z\Gamma E$ τριγώνῳ· ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν $B\Gamma$, ΓE καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς BE , AZ . ἀλλὰ τὸ $AB\Gamma$ τριγώνον ἴσον ἐστὶ τῷ $\Delta\Gamma E$ [τριγώνῳ]· καὶ τὸ $\Delta\Gamma E$ ἄρα

For if not, let AF be drawn through A parallel to BE [Prop. 1.31], and let FE be joined. Thus, triangle ABC is equal to triangle FCE . For they are on equal bases, BC and CE , and between the same parallels, BE and AF [Prop. 1.38]. But, triangle ABC is equal to [triangle] DCE . Thus, [triangle] DCE is also equal to triangle FCE , the greater to the lesser. The very

[τριγώνων] ἴσον ἐστὶ τῷ $ZΓΕ$ τριγώνῳ τὸ μείζον τῷ ἐλάσσονι· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα παράλληλος ἡ AZ τῇ BE . ὁμοίως δὴ δείχομεν, ὅτι οὐδ' ἄλλη τις πλὴν τῆς AD · ἡ AD ἄρα τῇ BE ἐστι παράλληλος.

Τὰ ἄρα ἴσα τριγώνων τὰ ἐπὶ ἴσων βάσεων ὄντα καὶ ἐπὶ τὰ αὐτὰ μέρη καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν· ὅπερ ἔδει δεῖχαι.

thing is impossible. Thus, AF is not parallel to BE . Similarly, we can show that neither (is) any other (straight-line) than AD . Thus, AD is parallel to BE .

Thus, equal triangles which are on equal bases, and on the same side, are also between the same parallels. (Which is) the very thing it was required to show.

† This whole proposition is regarded by Heiberg as a relatively early interpolation to the original text.

41.

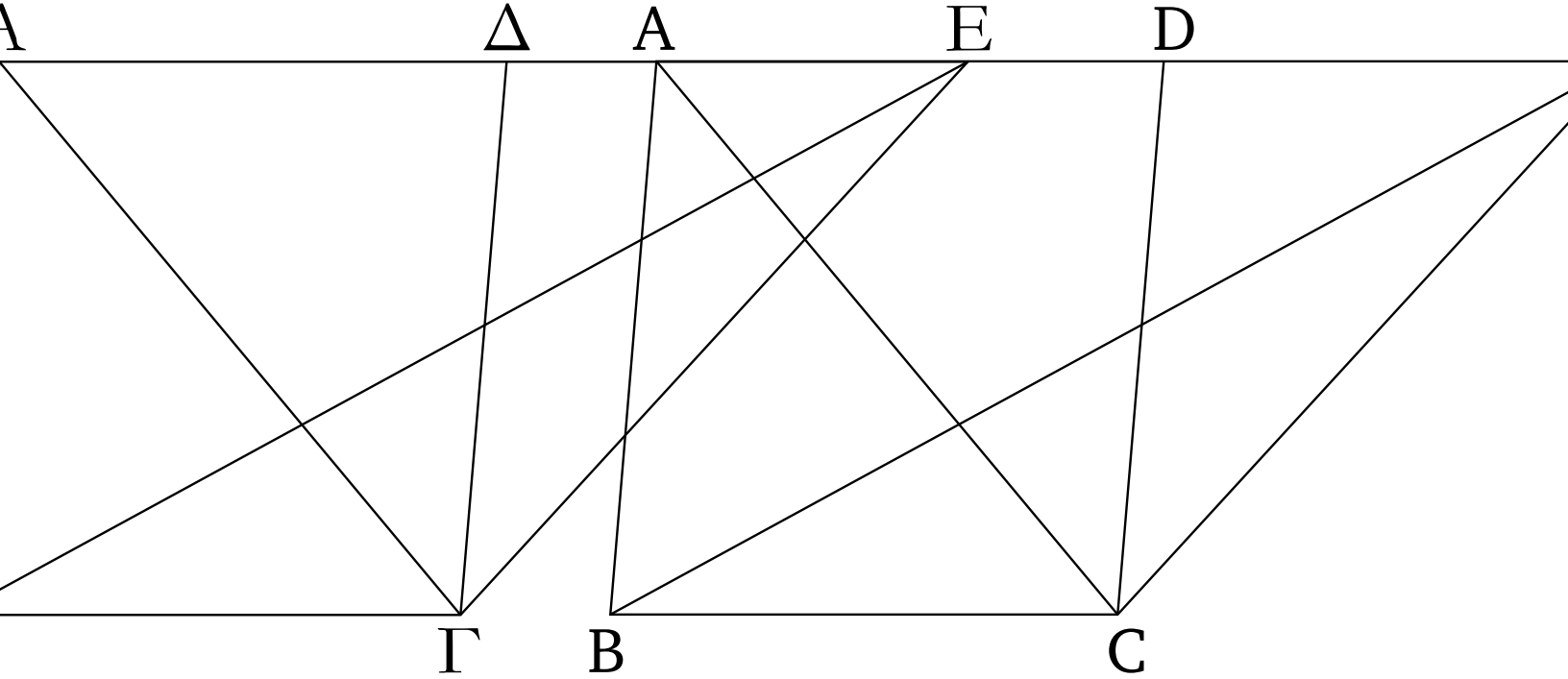
Ἐάν παραλληλόγραμμον τριγώνῳ βάσιν τε ἔσθῃ τὴν αὐτὴν καὶ ἐν ταῖς αὐταῖς παραλλήλοις ᾗ, διπλάσιόν ἐστὶ τὸ παραλληλόγραμμον τοῦ τριγώνου.

Παραλληλόγραμμον γάρ τὸ $ABΓΔ$ τριγώνῳ τῷ $EBΓ$ βάσιν τε ἔσθῃ τὴν αὐτὴν τὴν $ΒΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἔστω ταῖς $ΒΓ$, AE · λέγω, ὅτι διπλάσιόν ἐστι τὸ $ABΓΔ$ παραλληλόγραμμον τοῦ $EBΓ$ τριγώνου.

Proposition 41

If a parallelogram has the same base as a triangle, and is between the same parallels, (then) the parallelogram is double (the area) of the triangle.

For let parallelogram $ABCD$ have the same base BC as triangle EBC , and let it be between the same parallels, BC and AE . I say that parallelogram $ABCD$ is double (the area) of triangle BEC .



Ἐπεξεύσθω γάρ ἡ $ΑΓ$. ἴσον δὴ ἐστὶ τὸ $ABΓ$ τριγώνον τῷ $EBΓ$ τριγώνῳ· ἐπὶ τε γάρ τῆς αὐτῆς βάσεώς ἐστιν αὐτῷ τῆς $ΒΓ$ καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $ΒΓ$, AE . ἀλλὰ τὸ $ABΓΔ$ παραλληλόγραμμον διπλάσιόν ἐστι τοῦ $ABΓ$ τριγώνου· ἡ γάρ $ΑΓ$ διάμετρος αὐτὸ διῖθα τέμνει· ὥστε τὸ $ABΓΔ$ παραλληλόγραμμον καὶ τοῦ $EBΓ$ τριγώνου ἐστὶ διπλάσιον.

Ἐάν ἄρα παραλληλόγραμμον τριγώνῳ βάσιν τε ἔσθῃ τὴν αὐτὴν καὶ ἐν ταῖς αὐταῖς παραλλήλοις ᾗ, διπλάσιόν ἐστὶ τὸ παραλληλόγραμμον τοῦ τριγώνου· ὅπερ ἔδει δεῖχαι.

For let AC be joined. So triangle ABC is equal to triangle EBC . For it is on the same base, BC , (as EBC), and between the same parallels, BC and AE [Prop. 1.37]. But, parallelogram $ABCD$ is double (the area) of triangle ABC . For the diagonal AC cuts the former in half [Prop. 1.34]. So parallelogram $ABCD$ is also double (the area) of triangle EBC .

Thus, if a parallelogram has the same base as a triangle, and is between the same parallels, (then) the parallelogram is double (the area) of the triangle. (Which is) the very thing it

42.

Τῷ δοθέντι τριγώνῳ ἴσον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ.

Ἐστω τὸ μὲν δοθέν τρίγωνον τὸ $AB\Gamma$, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμοξ ἡ Δ . δεῖ δὴ τῷ $AB\Gamma$ τριγώνῳ ἴσον παραλληλόγραμμον συστήσασθαι ἐν τῇ Δ γωνίᾳ εὐθυγράμμῳ.

Τετμήσθω ἡ $B\Gamma$ δίῃα κατὰ τὸ E , καὶ ἐπεζεύθθω ἡ AE , καὶ συνεστάτω πρὸς τῇ EF εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ E τῇ Δ γωνία ἴση ἡ ὑπὸ $ΓΕΖ$, καὶ διὰ μὲν τοῦ A τῇ EF παράλληλοξ ἡθθῶ ἡ AH , διὰ δὲ τοῦ Γ τῇ EZ παράλληλοξ ἡθθῶ ἡ ΓH . παραλληλόγραμμον ἄρα ἐστὶ τὸ $ZEGH$. καὶ ἐπεὶ ἴση ἐστὶν ἡ BE τῇ EF , ἴσον ἐστὶ καὶ τὸ ABE τρίγωνον τῷ AEG τριγώνῳ. ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν BE , EF καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς $B\Gamma$, AH . διπλάσιον ἄρα ἐστὶ τὸ $AB\Gamma$ τρίγωνον τοῦ AEG τριγώνου. ἐστὶ δὲ καὶ τὸ $ZEGH$ παραλληλόγραμμον διπλάσιον τοῦ AEG τριγώνου. βάσιν τε γὰρ αὐτῷ τὴν αὐτὴν ἔχει καὶ ἐν ταῖς αὐταῖς ἐστὶν αὐτῷ παραλλήλοις. ἴσον ἄρα ἐστὶ τὸ $ZEGH$ παραλληλόγραμμον τῷ $AB\Gamma$ τριγώνῳ. καὶ ἔχει τὴν ὑπὸ $ΓΕΖ$ γωνίαν ἴσην τῇ δοθείσῃ τῇ Δ .

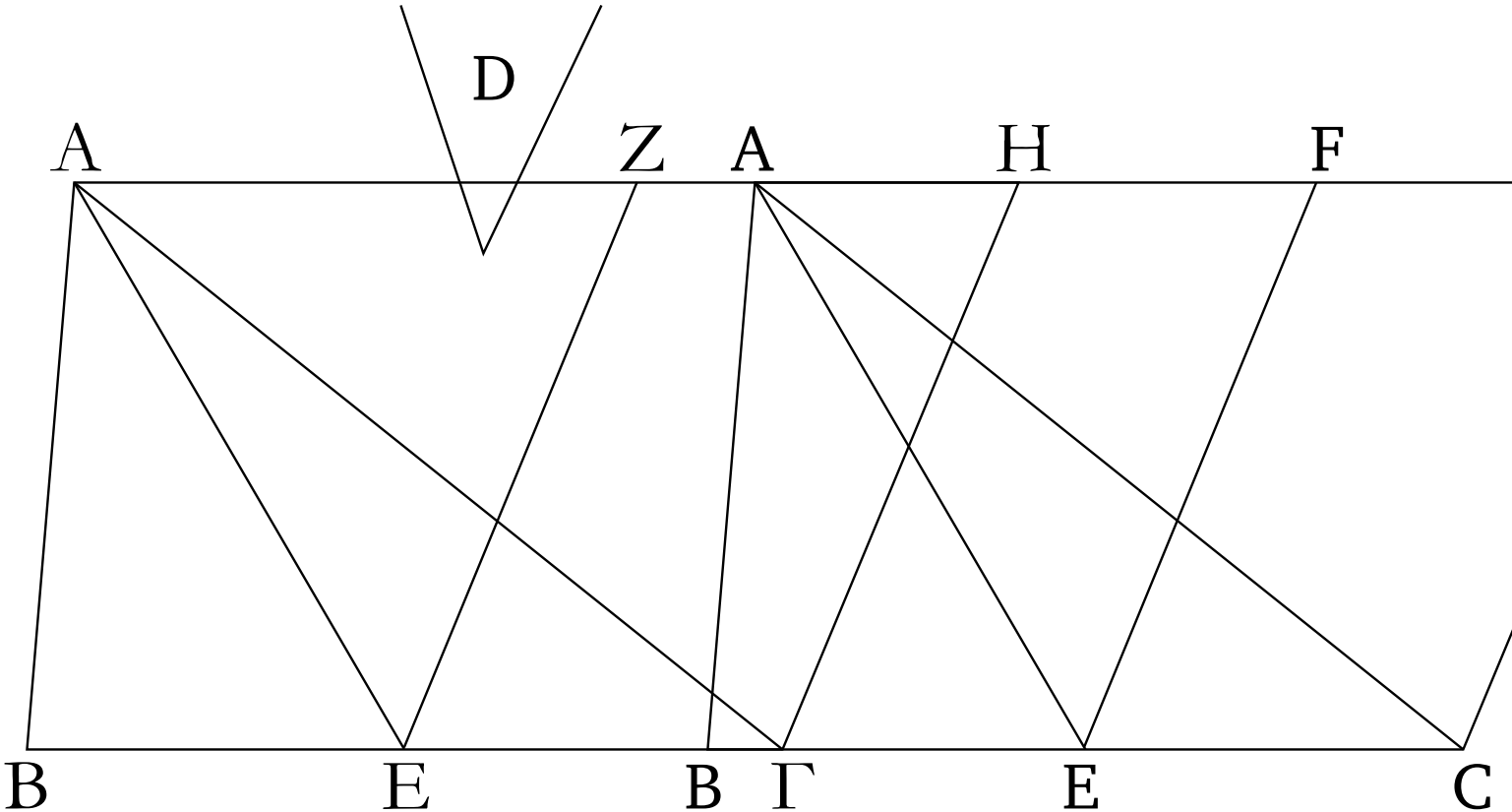
was required to show.

Proposition 42

To construct a parallelogram equal to a given triangle in a given rectilinear angle.

Let ABC be the given triangle, and D the given rectilinear angle. So it is required to construct a parallelogram equal to triangle ABC in the rectilinear angle D .

Let BC be cut in half at E [Prop. 1.10], and let AE be joined. And let (angle) CEF , equal to angle D , be constructed at the point E on the straight-line EC [Prop. 1.23]. And let AG be drawn through A parallel to EC [Prop. 1.31], and let CG be drawn through C parallel to EF [Prop. 1.31]. Thus, $FECG$ is a parallelogram. And since BE is equal to EC , triangle ABE is also equal to triangle AEC . For they are on the equal bases, BE and EC , and between the same parallels, BC and AG [Prop. 1.38]. Thus, triangle ABC is double (the area) of triangle AEC . And parallelogram $FECG$ is also double (the area) of triangle AEC . For it has the same base (as AEC), and is between the same parallels (as AEC) [Prop. 1.41]. Thus, parallelogram $FECG$ is equal to triangle ABC . ($FECG$) also has the angle CEF equal to the given (angle) D .



Τῷ ἄρα δοθέντι τριγώνῳ τῷ $AB\Gamma$ ἴσον παραλληλόγραμμον συνέσταται τὸ $ZEGH$ ἐν γωνίᾳ τῇ ὑπὸ $ΓΕΖ$, ἥτις ἐστὶν

Thus, parallelogram $FECG$, equal to the given triangle ABC , has been constructed in the angle CEF , which is equal

ἴση τῇ Δ · ὅπερ ἔδει ποιῆσαι.

to D . (Which is) the very thing it was required to do.

43.

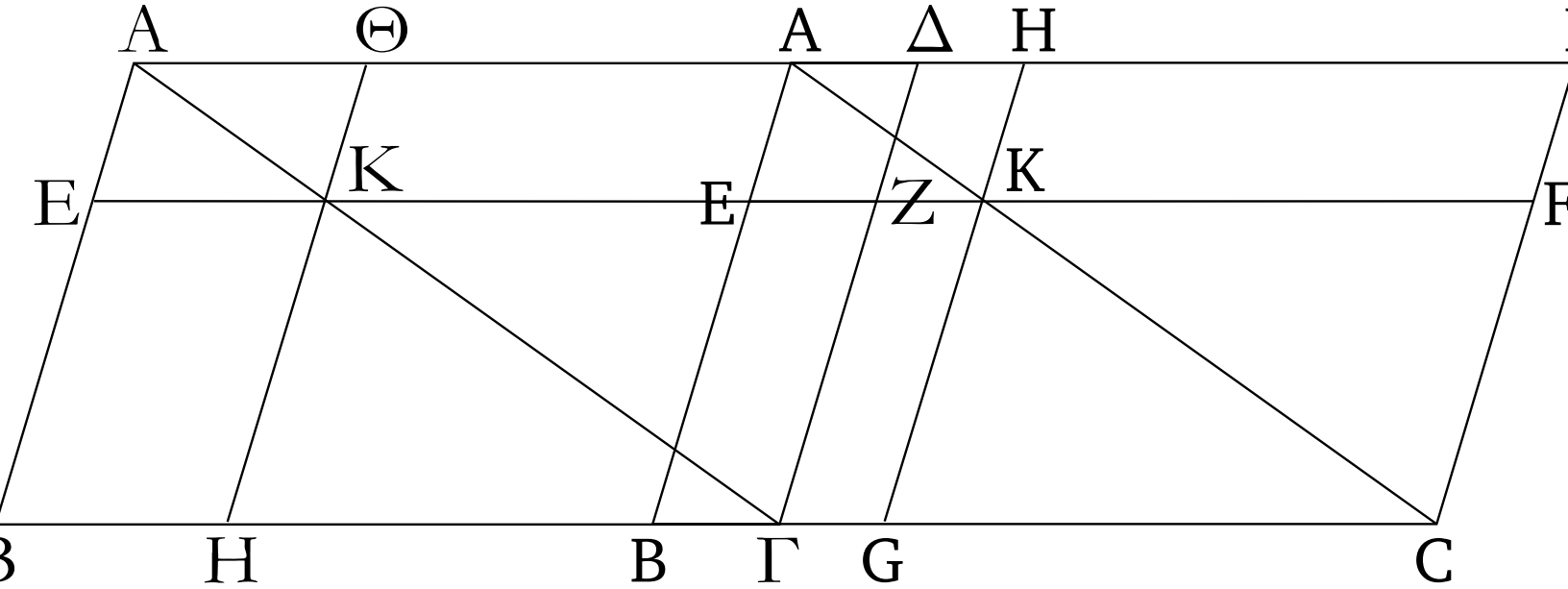
Proposition 43

Παντὸς παραλληλογράμμου τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα ἴσα ἀλλήλοις ἐστίν.

For any parallelogram, the complements of the parallelograms about the diagonal are equal to one another.

Ἐστω παραλληλόγραμμον τὸ $AB\Gamma\Delta$, διάμετρος δὲ αὐτοῦ ἡ $ΑΓ$, περὶ δὲ τὴν $ΑΓ$ παραλληλόγραμμα μὲν ἔστω τὰ $ΕΘ$, $ΖΗ$, τὰ δὲ λεγόμενα παραπληρώματα τὰ $ΒΚ$, $ΚΔ$ · λέγω, ὅτι ἴσον ἐστὶ τὸ $ΒΚ$ παραπλήρωμα τῷ $ΚΔ$ παραπληρώματι.

Let $ABCD$ be a parallelogram, and AC its diagonal. And let EH and FG be the parallelograms about AC , and BK and KD the so-called complements (about AC). I say that the complement BK is equal to the complement KD .



Ἐπεὶ γὰρ παραλληλόγραμμον ἐστὶ τὸ $AB\Gamma\Delta$, διάμετρος δὲ αὐτοῦ ἡ $ΑΓ$, ἴσον ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ $ΑΓΔ$ τριγώνῳ. πάλιν, ἐπεὶ παραλληλόγραμμον ἐστὶ τὸ $ΕΘ$, διάμετρος δὲ αὐτοῦ ἐστὶν ἡ $ΑΚ$, ἴσον ἐστὶ τὸ $ΑΕΚ$ τρίγωνον τῷ $ΑΘΚ$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $ΚΖΓ$ τρίγωνον τῷ $ΚΗΓ$ ἐστὶν ἴσον. ἐπεὶ οὖν τὸ μὲν $ΑΕΚ$ τρίγωνον τῷ $ΑΘΚ$ τριγώνῳ ἐστὶν ἴσον, τὸ δὲ $ΚΖΓ$ τῷ $ΚΗΓ$, τὸ $ΑΕΚ$ τρίγωνον μετὰ τοῦ $ΚΗΓ$ ἴσον ἐστὶ τῷ $ΑΘΚ$ τριγώνῳ μετὰ τοῦ $ΚΖΓ$ · ἐστὶ δὲ καὶ ὅλον τὸ $AB\Gamma$ τρίγωνον ὅλῳ τῷ $ΑΔΓ$ ἴσον· λοιπὸν ἄρα τὸ $ΒΚ$ παραπλήρωμα λοιπῷ τῷ $ΚΔ$ παραπληρώματι ἐστὶν ἴσον.

For since $ABCD$ is a parallelogram, and AC its diagonal, triangle ABC is equal to triangle ACD [Prop. 1.34]. Again, since EH is a parallelogram, and AK is its diagonal, triangle AEK is equal to triangle AHK [Prop. 1.34]. So, for the same (reasons), triangle KFC is also equal to (triangle) KGC . Therefore, since triangle AEK is equal to triangle AHK , and KFC to KGC , triangle AEK plus KGC is equal to triangle AHK plus KFC . And the whole triangle ABC is also equal to the whole (triangle) ADC . Thus, the remaining complement BK is equal to the remaining complement KD .

Παντὸς ἄρα παραλληλογράμμου θεωρίον τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

Thus, for any parallelogramic figure, the complements of the parallelograms about the diagonal are equal to one another. (Which is) the very thing it was required to show.

44.

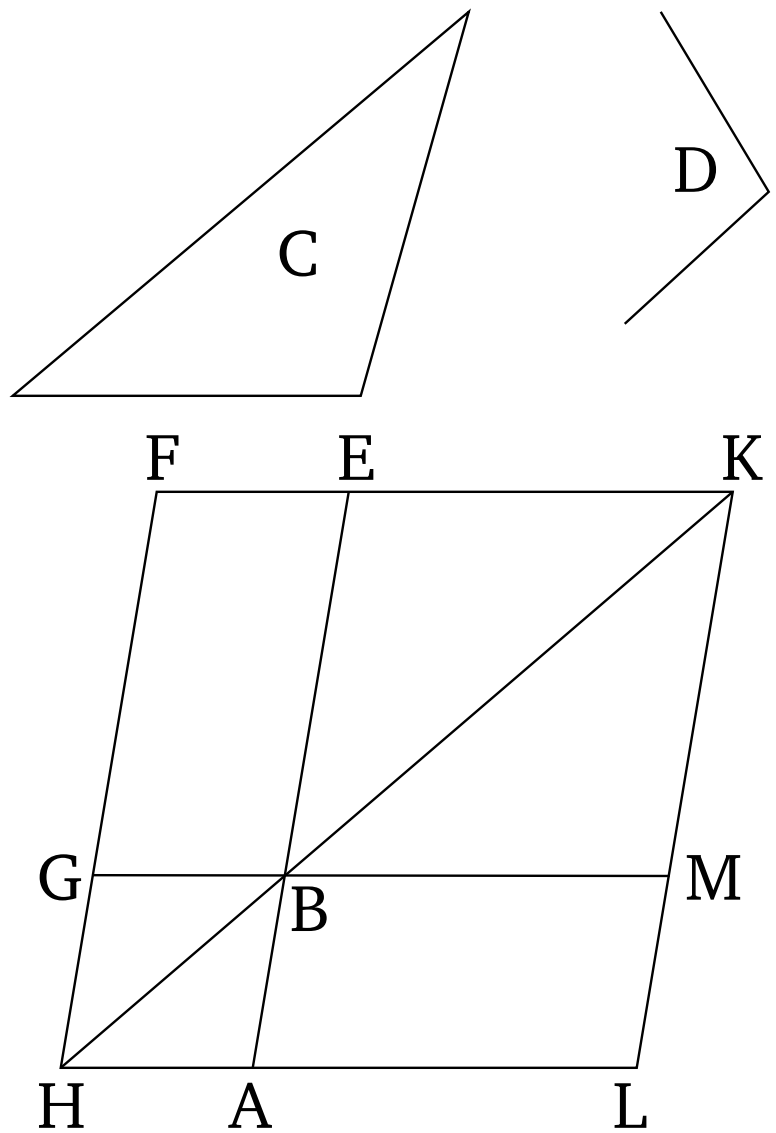
Proposition 44

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι τριγώνῳ ἴσον παραλληλόγραμμον παραβαλεῖν ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμω.

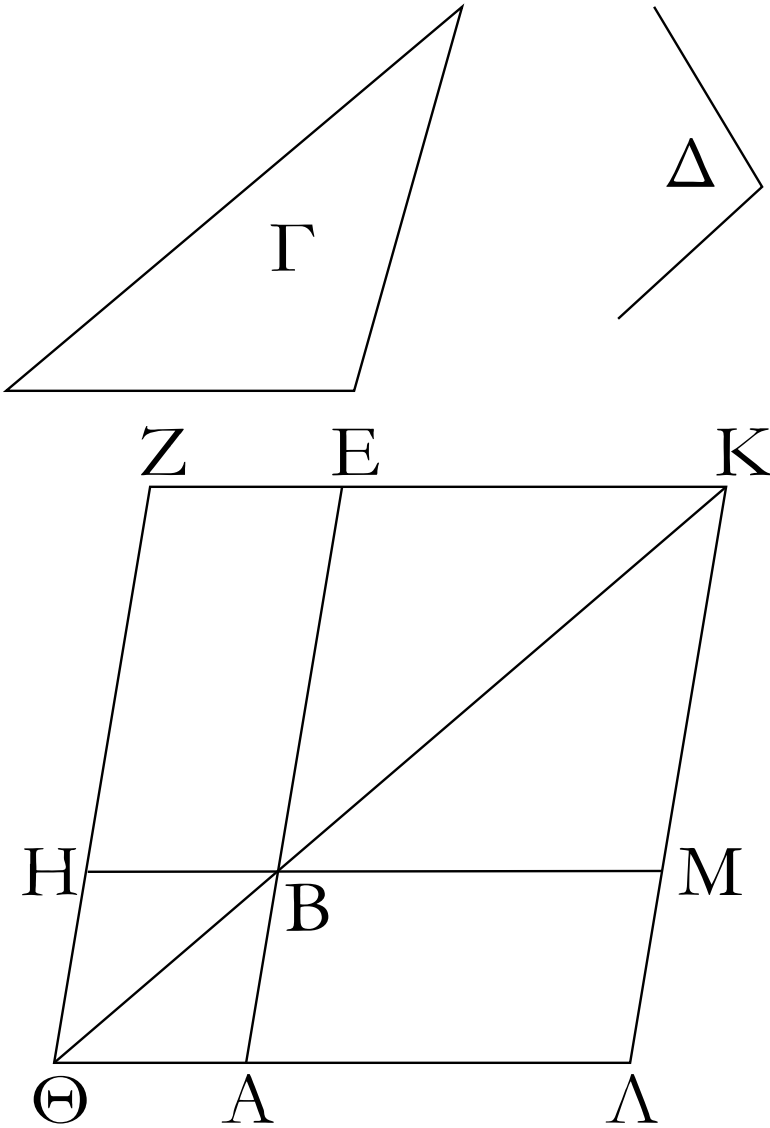
To apply a parallelogram equal to a given triangle to a given straight-line in a given rectilinear angle.

Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ $ΑΒ$, τὸ δὲ δοθὲν τρίγωνον τὸ Γ , ἡ δὲ δοθεῖσα γωνία εὐθύγραμμοξ ἡ Δ · δεῖ δὴ παρὰ τὴν δοθεῖσαν εὐθεῖαν τὴν $ΑΒ$ τῷ δοθέντι τριγώνῳ τῷ Γ ἴσον

Let AB be the given straight-line, C the given triangle, and D the given rectilinear angle. So it is required to apply a parallelogram equal to the given triangle C to the given straight-line AB in an angle equal to (angle) D .



παράλληλογράμμον παραβαλεῖν ἐν ἴσῃ τῇ Δ γωνίᾳ.



Συνεστάτω τῷ Γ τριγώνῳ ἴσον παραλληλόγραμμον τὸ $BEZH$ ἐν γωνίᾳ τῇ ὑπὸ EBH , ἥ ἐστίν ἴση τῇ Δ · καὶ κείσθω ὥστε ἐπ' εὐθείᾳ εἶναι τὴν BE τῇ AB , καὶ διήθθω ἡ ZH ἐπὶ τὸ Θ , καὶ διὰ τοῦ A ὁποτέρᾳ τῶν BH , EZ παράλληλοξ ἡθθω ἡ $A\Theta$, καὶ ἐπεζεύθθω ἡ ΘB . καὶ ἐπεὶ εἰς παραλλήλων τὰς $A\Theta$, EZ εὐθεῖα ἐνέπεσεν ἡ ΘZ , αἱ ἄρα ὑπὸ $A\Theta Z$, ΘZE γωνίαι δυσὶν ὀρθαῖς εἰσιν ἴσαι. αἱ ἄρα ὑπὸ $B\Theta H$, HZE δύο ὀρθῶν ἐλάσσονές εἰσιν· αἱ δὲ ἀπὸ ἐλασσόνων ἢ δύο ὀρθῶν εἰς ἄπειρον ἐκβαλλόμεναι συμπίπτουσιν· αἱ ΘB , ZE ἄρα ἐκβαλλόμεναι συμπεσοῦνται. ἐκβεβλήσθωσαν καὶ συμμιπτέωσαν κατὰ τὸ K , καὶ διὰ τοῦ K σημείου ὁποτέρᾳ τῶν EA , $Z\Theta$ παράλληλοξ ἡθθω ἡ KL , καὶ ἐκβεβλήσθωσαν αἱ ΘA , HB ἐπὶ τὰ Λ , M σημεία. παραλληλόγραμμον ἄρα ἐστὶ τὸ ΘAKZ , διάμετροξ δὲ αὐτοῦ ἡ ΘK , περὶ δὲ τὴν ΘK παραλληλόγραμμά μὲν τὰ AH , ME , τὰ δὲ λεγόμενα παραπληρώματα τὰ AB ,

Let the parallelogram $BEFG$, equal to the triangle C , be constructed in the angle EBG , which is equal to D [Prop. 1.42]. And let it be placed so that BE is straight-on to AB .[†] And let FG be drawn through to H , and let AH be drawn through A parallel to either of BG or EF [Prop. 1.31], and let HB be joined. And since the straight-line HF falls across the parallels AH and EF , the (sum of the) angles AHF and HFE is thus equal to two right-angles [Prop. 1.29]. Thus, (the sum of) BHG and GFE is less than two right-angles. And (straight-lines) produced to infinity from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, being produced, HB and FE will meet together. Let them be produced, and let them meet together at K . And let KL be drawn through point K parallel to either of EA or FH [Prop. 1.31]. And let HA and GB be produced to points L and M (respectively). Thus, $HLKF$ is a parallelogram, and HK its diagon-

BZ ἴσον ἄρα ἐστὶ τὸ AB τῷ BZ . ἀλλὰ τὸ BZ τῷ Γ τριγώνῳ ἐστὶν ἴσον· καὶ τὸ AB ἄρα τῷ Γ ἐστὶν ἴσον. καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ HBE γωνία τῇ ὑπὸ ABM , ἀλλὰ ἡ ὑπὸ HBE τῇ Δ ἐστὶν ἴση, καὶ ἡ ὑπὸ ABM ἄρα τῇ Δ γωνία ἐστὶν ἴση.

Παρὰ τὴν δοθεῖσαν ἄρα εὐθεῖαν τὴν AB τῷ δοθέντι τριγώνῳ τῷ Γ ἴσον παραλληλόγραμμον παραβέβληται τὸ AB ἐν γωνίᾳ τῇ ὑπὸ ABM , ἥ ἐστὶν ἴση τῇ Δ · ὅπερ ἔδει ποιῆσαι.

nal. And AG and ME (are) parallelograms, and LB and BF the so-called complements, about HK . Thus, LB is equal to BF [Prop. 1.43]. But, BF is equal to triangle C . Thus, LB is also equal to C . Also, since angle GBE is equal to ABM [Prop. 1.15], but GBE is equal to D , ABM is thus also equal to angle D .

Thus, the parallelogram LB , equal to the given triangle C , has been applied to the given straight-line AB in the angle ABM , which is equal to D . (Which is) the very thing it was required to do.

[†] This can be achieved using Props. 1.3, 1.23, and 1.31.

45.

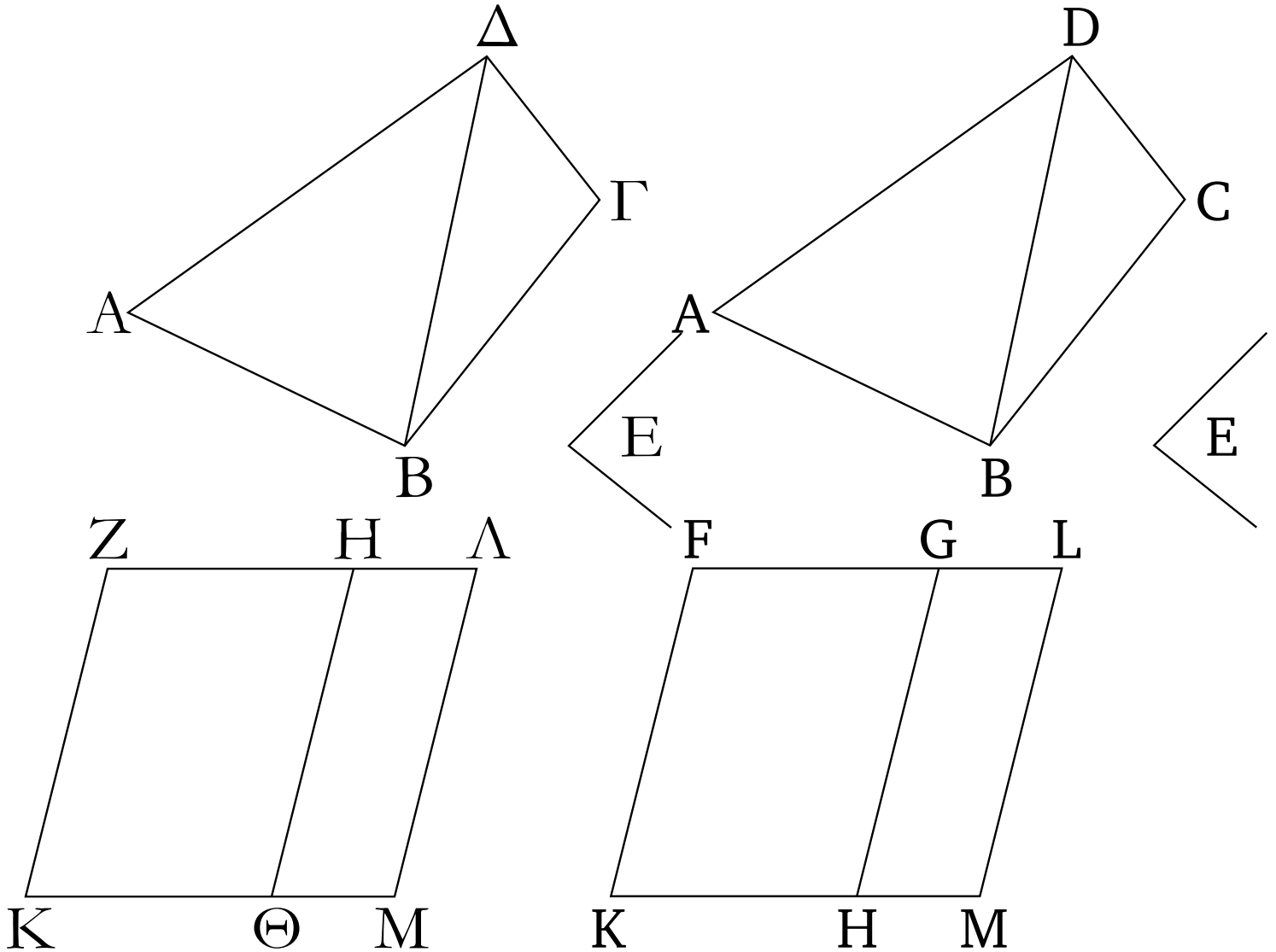
Τῷ δοθέντι εὐθυγράμμῳ ἴσον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ.

Ἐστω τὸ μὲν δοθέν εὐθύγραμμον τὸ $AB\Gamma\Delta$, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἡ E . δεῖ δὴ τῷ $AB\Gamma\Delta$ εὐθυγράμμῳ ἴσον παραλληλόγραμμον συστήσασθαι ἐν τῇ δοθείσῃ γωνίᾳ τῇ E .

Proposition 45

To construct a parallelogram equal to a given rectilinear figure in a given rectilinear angle.

Let $ABCD$ be the given rectilinear figure,[†] and E the given rectilinear angle. So it is required to construct a parallelogram equal to the rectilinear figure $ABCD$ in the given angle E .



Ἐπεξεύθθω ἡ ΔB , καὶ συνεστιάτω τῷ $AB\Delta$ τριγώνῳ ἴσον παραλληλόγραμμον τὸ $Z\Theta$ ἐν τῇ ὑπὸ ΘKZ γωνίᾳ, ἥ ἐστιν ἴση τῇ E : καὶ παραβεβλήσθω παρὰ τὴν $H\Theta$ εὐθεΐαν τῷ $\Delta B\Gamma$ τριγώνῳ ἴσον παραλληλόγραμμον τὸ HM ἐν τῇ ὑπὸ $H\Theta M$ γωνίᾳ, ἥ ἐστιν ἴση τῇ E . καὶ ἐπεὶ ἡ E γωνία ἑκατέρω τῶν ὑπὸ ΘKZ , $H\Theta M$ ἐστὶν ἴση, καὶ ἡ ὑπὸ ΘKZ ἄρα τῇ ὑπὸ $H\Theta M$ ἐστὶν ἴση. κοινὴ προσκείσθω ἡ ὑπὸ $K\Theta H$: αἱ ἄρα ὑπὸ $ZK\Theta$, $K\Theta H$ ταῖς ὑπὸ $K\Theta H$, $H\Theta M$ ἴσαι εἰσὶν. ἀλλ' αἱ ὑπὸ $ZK\Theta$, $K\Theta H$ δυοῖν ὀρθαῖς ἴσαι εἰσὶν: καὶ αἱ ὑπὸ $K\Theta H$, $H\Theta M$ ἄρα δύο ὀρθαῖς ἴσαι εἰσὶν. πρὸς δὲ τινι εὐθεΐᾳ τῇ $H\Theta$ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Θ δύο εὐθεΐαι αἱ $K\Theta$, ΘM μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δύο ὀρθαῖς ἴσας ποιοῦσιν· ἐπ' εὐθείᾳς ἄρα ἐστὶν ἡ $K\Theta$ τῇ ΘM : καὶ ἐπεὶ εἰς παραλλήλων τὰς KM , ZH εὐθεΐα ἐνέπεσεν ἡ ΘH , αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $M\Theta H$, ΘHZ ἴσαι ἀλλήλαις εἰσὶν. κοινὴ προσκείσθω ἡ ὑπὸ $\Theta H\Lambda$: αἱ ἄρα ὑπὸ $M\Theta H$,

Let DB be joined, and let the parallelogram FH , equal to the triangle ABD , be constructed in the angle HKF , which is equal to E [Prop. 1.42]. And let the parallelogram GM , equal to the triangle DBC , be applied to the straight-line GH in the angle GHM , which is equal to E [Prop. 1.44]. And since angle E is equal to each of (angles) HKF and GHM , (angle) HKF is thus also equal to GHM . Let KHG be added to both. Thus, (the sum of) FKH and KHG is equal to (the sum of) KHG and GHM . But, (the sum of) FKH and KHG is equal to two right-angles [Prop. 1.29]. Thus, (the sum of) KHG and GHM is also equal to two right-angles. So two straight-lines, KH and HM , not lying on the same side, make adjacent angles with some straight-line GH , at the point H on it, (whose sum is) equal to two right-angles. Thus, KH is straight-on to HM [Prop. 1.14]. And since the straight-line HG falls across the parallels KM and FG , the alternate angles mhG and HGF are

$\Theta\text{ΗΛ}$ ταῖς ὑπὸ $\Theta\text{ΗΖ}$, $\Theta\text{ΗΛ}$ ἴσαι εἰσιν. ἀλλ' αἱ ὑπὸ ΜΘΗ , $\Theta\text{ΗΛ}$ δύο ὀρθαῖς ἴσαι εἰσίν· καὶ αἱ ὑπὸ $\Theta\text{ΗΖ}$, $\Theta\text{ΗΛ}$ ἄρα δύο ὀρθαῖς ἴσαι εἰσίν· ἐπ' εὐθείας ἄρα ἐστὶν ἡ ΖΗ τῇ ΗΛ . καὶ ἐπεὶ ἡ ΖΚ τῇ $\Theta\text{Η}$ ἴση τε καὶ παράλληλός ἐστιν, ἀλλὰ καὶ ἡ $\Theta\text{Η}$ τῇ ΜΛ , καὶ ἡ ΚΖ ἄρα τῇ ΜΛ ἴση τε καὶ παράλληλός ἐστιν· καὶ ἐπιξενγνύουσιν αὐτάς εὐθεῖαι αἱ ΚΜ , ΖΛ · καὶ αἱ ΚΜ , ΖΛ ἄρα ἴσαι τε καὶ παράλληλοί εἰσιν· παραλληλόγραμμον ἄρα ἐστὶ τὸ ΚΖΑΜ . καὶ ἐπεὶ ἴσον ἐστὶ τὸ μὲν ΑΒΔ τριγώνων τῷ ΖΘ παραλληλογράμμῳ, τὸ δὲ ΔΒΓ τῷ ΗΜ , ὅλον ἄρα τὸ ΑΒΓΔ εὐθύγραμμον ὅλῳ τῷ ΚΖΑΜ παραλληλογράμμῳ ἐστὶν ἴσον.

Τῷ ἄρα δοθέντι εὐθυγράμμῳ τῷ ΑΒΓΔ ἴσον παραλληλόγραμμον συνέσταται τὸ ΚΖΑΜ ἐν γωνίᾳ τῇ ὑπὸ ΖΚΜ , ἥ ἐστὶν ἴση τῇ δοθείσῃ τῇ Ε · ὅπερ ἔδει ποιῆσαι.

equal to one another [Prop. 1.29]. Let HGL have been added to both. Thus, (the sum of) mhG and HGL is equal to (the sum of) HGF and HGL . But, (the sum of) mhG and HGL is equal to two right-angles [Prop. 1.29]. Thus, (the sum of) HGF and HGL is also equal to two right-angles. Thus, FG is straight-on to GL [Prop. 1.14]. And since FK is equal and parallel to HG [Prop. 1.34], but also HG to ML [Prop. 1.34], KF is thus also equal and parallel to ML [Prop. 1.30]. And the straight-lines KM and FL join them. Thus, KM and FL are equal and parallel as well [Prop. 1.33]. Thus, KFLM is a parallelogram. And since triangle ABD is equal to parallelogram FH , and DBC to GM , the whole rectilinear figure ABCD is thus equal to the whole parallelogram KFLM .

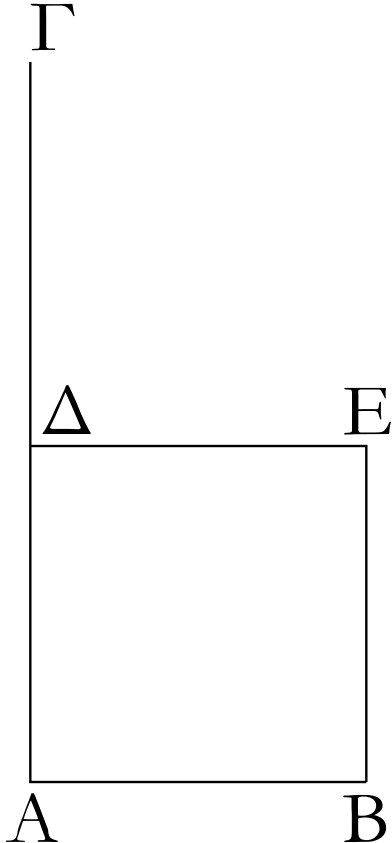
Thus, the parallelogram KFLM , equal to the given rectilinear figure ABCD , has been constructed in the angle FKM , which is equal to the given (angle) E . (Which is) the very thing it was required to do.

† The proof is only given for a four-sided figure. However, the extension to many-sided figures is trivial.

46.

Ἀπὸ τῆς δοθείσης εὐθείας τετράγωνον ἀναγράψαι.

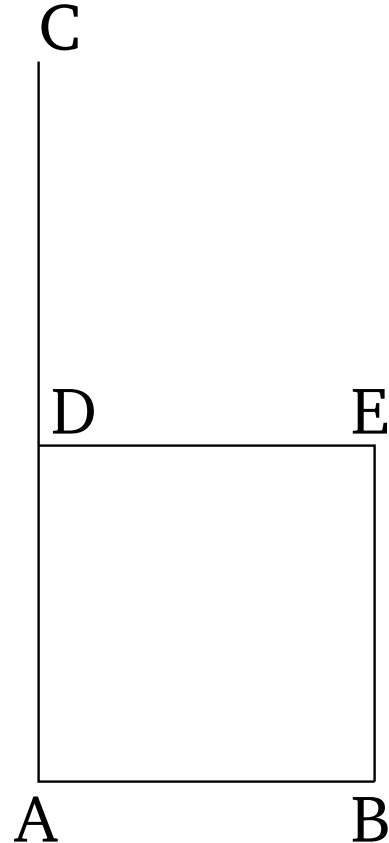
Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ ΑΒ · δεῖ δὴ ἀπὸ τῆς ΑΒ εὐθείας τετράγωνον ἀναγράψαι.



Proposition 46

To describe a square on a given straight-line.

Let AB be the given straight-line. So it is required to describe a square on the straight-line AB .



Ἦθθω τῇ AB εὐθείᾳ ἀπὸ τοῦ πρὸς αὐτῇ σημείου τοῦ A πρὸς ὀρθὰς ἡ AG , καὶ κείσθω τῇ AB ἴση ἡ AD . καὶ διὰ μὲν τοῦ Δ σημείου τῇ AB παράλληλος ἡθθω ἡ DE , διὰ δὲ τοῦ B σημείου τῇ AD παράλληλος ἡθθω ἡ BE . παραλληλόγραμμον ἄρα ἐστὶ τὸ $ADEB$. ἴση ἄρα ἐστὶν ἡ μὲν AB τῇ DE , ἡ δὲ AD τῇ BE . ἀλλὰ ἡ AB τῇ AD ἐστὶν ἴση· αἱ τέσσαρες ἄρα αἱ BA , AD , DE , EB ἴσαι ἀλλήλαις εἰσὶν· ἰσόπλευρον ἄρα ἐστὶ τὸ $ADEB$ παραλληλόγραμμον. λέγω δὴ, ὅτι καὶ ὀρθογώνιον. ἐπεὶ γὰρ εἷς παραλλήλων τάς AB , DE εὐθεῖα ἐνέπεσεν ἡ AD , αἱ ἄρα ὑπὸ BAD , ADE γωνίαι δύο ὀρθαῖς ἴσαι εἰσὶν. ὀρθὴ δὲ ἡ ὑπὸ BAD . ὀρθὴ ἄρα καὶ ἡ ὑπὸ ADE . τῶν δὲ παραλληλογράμμων θωρίων αἱ ἀπεναντίον πλευραὶ τε καὶ γωνίαι ἴσαι ἀλλήλαις εἰσὶν· ὀρθὴ ἄρα καὶ ἑκάτερα τῶν ἀπεναντίον τῶν ὑπὸ ABE , BED γωνιῶν· ὀρθογώνιον ἄρα ἐστὶ τὸ $ADEB$. ἐδείχθη δὲ καὶ ἰσόπλευρον.

Τετράγωνον ἄρα ἐστὶν καὶ ἐστὶν ἀπὸ τῆς AB εὐθείας ἀναγεγραμμένον· ὅπερ ἔδει ποιῆσαι.

47.

Ἐν τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσῃς πλευρᾶς τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν γωνίαν περιεθουσῶν πλευρῶν τετραγώνοις.

Ἔστω τρίγωνον ὀρθογώνιον τὸ ABG ὀρθὴν ἔχον τὴν ὑπὸ BAG γωνίαν· λέγω, ὅτι τὸ ἀπὸ τῆς BG τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν BA , AG τετραγώνοις.

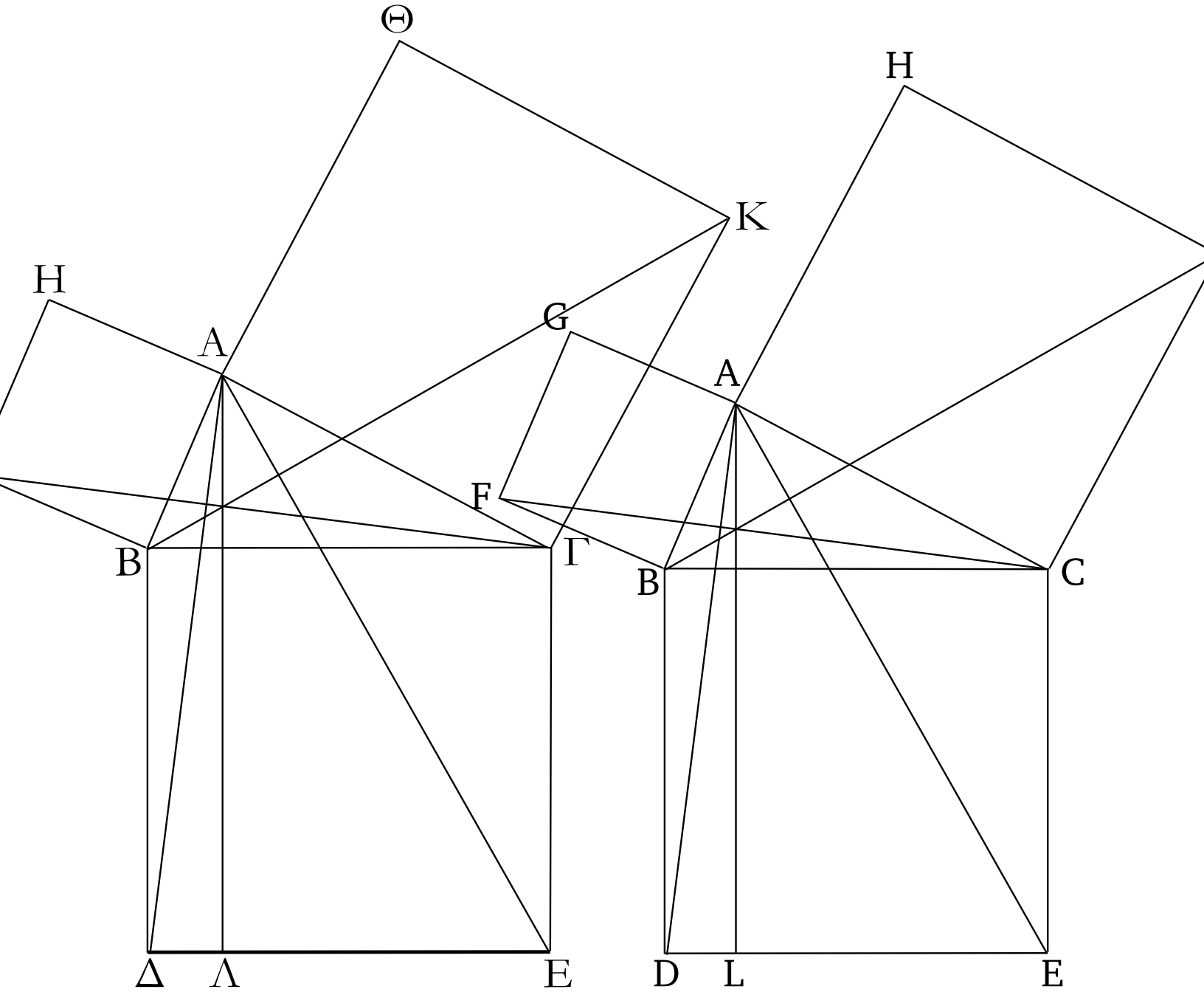
Let AC be drawn at right-angles to the straight-line AB from the point A on it [Prop. 1.11], and let AD be made equal to AB [Prop. 1.3]. And let DE be drawn through point D parallel to AB [Prop. 1.31], and let BE be drawn through point B parallel to AD [Prop. 1.31]. Thus, $ADEB$ is a parallelogram. Therefore, AB is equal to DE , and AD to BE [Prop. 1.34]. But, AB is equal to AD . Thus, the four (sides) BA , AD , DE , and EB are equal to one another. Thus, the parallelogram $ADEB$ is equilateral. So I say that (it is) also right-angled. For since the straight-line AD falls across the parallels AB and DE , the (sum of the) angles BAD and ADE is equal to two right-angles [Prop. 1.29]. But BAD (is a) right-angle. Thus, ADE (is) also a right-angle. And for parallelogramic figures, the opposite sides and angles are equal to one another [Prop. 1.34]. Thus, each of the opposite angles ABE and BED (are) also right-angles. Thus, $ADEB$ is right-angled. And it was also shown (to be) equilateral.

Thus, $(ADEB)$ is a square [Def. 1.22]. And it is described on the straight-line AB . (Which is) the very thing it was required to do.

Proposition 47

In right-angled triangles, the square on the side subtending the right-angle is equal to the (sum of the) squares on the sides containing the right-angle.

Let ABC be a right-angled triangle having the angle BAC a right-angle. I say that the square on BC is equal to the (sum of the) squares on BA and AC .



Ἀναγεγράφθω γάρ ἀπὸ μὲν τῆς $BΓ$ τετραγώνον τὸ $BΔΕΓ$, ἀπὸ δὲ τῶν BA , AC τὰ HB , $ΘΓ$, καὶ διὰ τοῦ A ὁποτέρῳ τῶν BD , $ΓΕ$ παράλληλος ἦθθω ἡ $ΑΛ$ · καὶ ἐπεξενέθωσαν αἱ $ΑΔ$, $ΖΓ$. καὶ ἐπεὶ ὀρθή ἐστὶν ἑκάτερα τῶν ὑπὸ $BAΓ$, BAH γωνιῶν, πρὸς δὴ τινι εὐθείᾳ τῇ BA καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A δύο εὐθεῖαι αἱ $ΑΓ$, AH μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας δυοῖν ὀρθαῖς ἴσας ποιοῦσιν· ἐπ’ εὐθείᾳς ἄρα ἐστὶν ἡ $ΓΑ$ τῇ AH . διὰ τὰ αὐτὰ δὴ καὶ ἡ BA τῇ $AΘ$ ἐστὶν ἐπ’ εὐθείᾳς. καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ $ΔΒΓ$ γωνία τῇ

For let the square $BDEC$ be described on BC , and (the squares) GB and HC on AB and AC (respectively) [Prop. 1.46]. And let AL be drawn through point A parallel to either of BD or CE [Prop. 1.31]. And let AD and FC be joined. And since angles BAC and BAG are each right-angles, then two straight-lines AC and AG , not lying on the same side, make the adjacent angles with some straight-line BA , at the point A on it, (whose sum is) equal to two right-angles. Thus, CA is straight-on to AG [Prop. 1.14]. So, for the same (rea-

ὑπὸ ZBA · ὀρθή γάρ ἑκατέρα· κοινὴ προσκείσθω ἡ ὑπὸ $ABΓ$ · ὅλη ἄρα ἡ ὑπὸ ΔBA ὅλη τῇ ὑπὸ $ZBΓ$ ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν ΔB τῇ $BΓ$, ἡ δὲ ZB τῇ BA , δύο δὲ αἱ ΔB , BA δύο ταῖς ZB , $BΓ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρα· καὶ γωνία ἡ ὑπὸ ΔBA γωνία τῇ ὑπὸ $ZBΓ$ ἴση· βάσις ἄρα ἡ AD βάσει τῇ $ZΓ$ [ἐστὶν] ἴση, καὶ τὸ $AB\Delta$ τρίγωνον τῷ $ZBΓ$ τριγώνῳ ἐστὶν ἴσον· καὶ [ἐστὶ] τοῦ μὲν $AB\Delta$ τριγώνου διπλάσιον τὸ BA παραλληλόγραμμον· βάσιν τε γὰρ τὴν αὐτὴν ἔθουσι τὴν $B\Delta$ καὶ ἐν ταῖς αὐταῖς εἰσι παραλλήλοις ταῖς $B\Delta$, AL · τοῦ δὲ $ZBΓ$ τριγώνου διπλάσιον τὸ HB τετράγωνον· βάσιν τε γὰρ πάλιν τὴν αὐτὴν ἔθουσι τὴν ZB καὶ ἐν ταῖς αὐταῖς εἰσι παραλλήλοις ταῖς ZB , $HΓ$. [τὰ δὲ τῶν ἴσων διπλάσια ἴσα ἀλλήλοις ἐστὶν] ἴσον ἄρα ἐστὶ καὶ τὸ BA παραλληλόγραμμον τῷ HB τετραγώνῳ. ὁμοίως δὲ ἐπιξενυγνυμένων τῶν AE , BK δευτέρῃσιν καὶ τὸ $ΓA$ παραλληλόγραμμον ἴσον τῷ $\ThetaΓ$ τετραγώνῳ· ὅλον ἄρα τὸ $B\Delta EΓ$ τετράγωνον δυοῖ τοῖς HB , $\ThetaΓ$ τετραγώνοις ἴσον ἐστὶν. καὶ ἐστὶ τὸ μὲν $B\Delta EΓ$ τετράγωνον ἀπὸ τῆς $BΓ$ ἀναγραφέν, τὰ δὲ HB , $\ThetaΓ$ ἀπὸ τῶν BA , AG . τὸ ἄρα ἀπὸ τῆς $BΓ$ πλευρᾶς τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν BA , AG πλευρῶν τετραγώνοις.

Ἐν ἄρα τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσας πλευρᾶς τετράγωνον ἴσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν [γωνίαν] περιεθουσῶν πλευρῶν τετραγώνοις· ὅπερ ἔδει δεῖξαι.

sons), BA is also straight-on to AH . And since angle DBC is equal to FBA , for (they are) both right-angles, let ABC be added to both. Thus, the whole (angle) DBA is equal to the whole (angle) FBC . And since DB is equal to BC , and FB to BA , the two (straight-lines) DB , BA are equal to the two (straight-lines) CB , BF ,[†] respectively. And angle DBA (is) equal to angle FBC . Thus, the base AD [is] equal to the base FC , and the triangle ABD is equal to the triangle FBC [Prop. 1.4]. And parallelogram BL [is] double (the area) of triangle ABD . For they have the same base, BD , and are between the same parallels, BD and AL [Prop. 1.41]. And square GB is double (the area) of triangle FBC . For again they have the same base, FB , and are between the same parallels, FB and GC [Prop. 1.41]. [And the doubles of equal things are equal to one another.][‡] Thus, the parallelogram BL is also equal to the square GB . So, similarly, AE and BK being joined, the parallelogram CL can be shown (to be) equal to the square HC . Thus, the whole square $BDEC$ is equal to the (sum of the) two squares GB and HC . And the square $BDEC$ is described on BC , and the (squares) GB and HC on BA and AC (respectively). Thus, the square on the side BC is equal to the (sum of the) squares on the sides BA and AC .

Thus, in right-angled triangles, the square on the side subtending the right-angle is equal to the (sum of the) squares on the sides surrounding the right-[angle]. (Which is) the very thing it was required to show.

[†] The Greek text has “ FB , BC ”, which is obviously a mistake.

[‡] This is an additional common notion.

48.

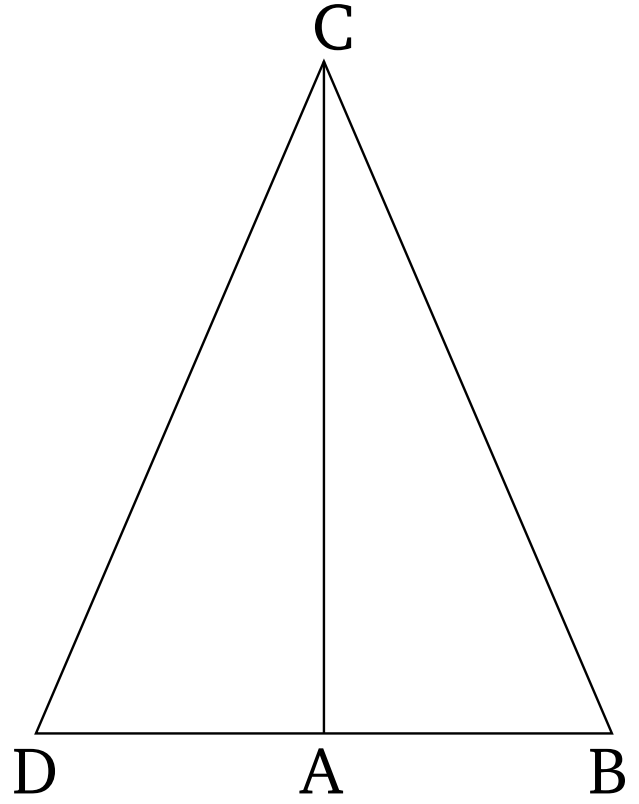
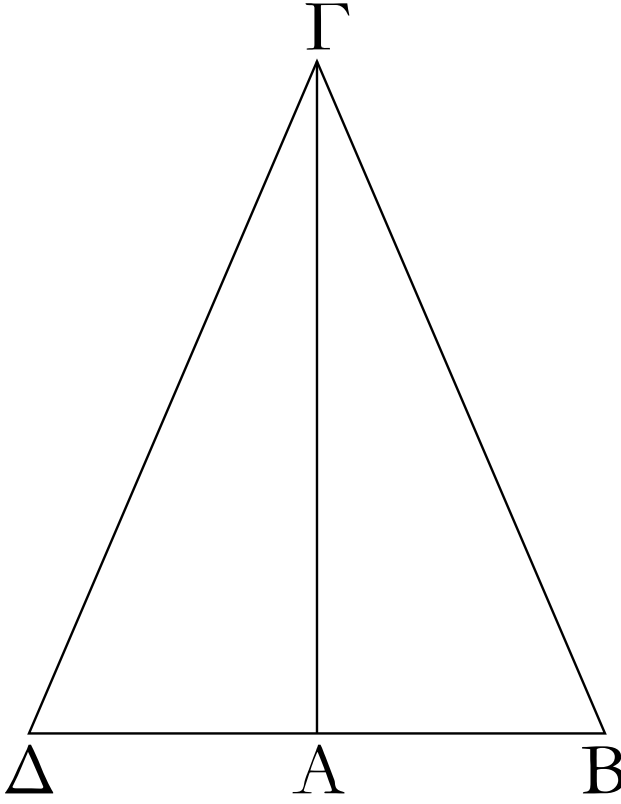
Ἐὰν τριγώνου τὸ ἀπὸ μιᾶς τῶν πλευρῶν τετράγωνον ἴσον ᾖ τοῖς ἀπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τετραγώνοις, ἡ περιεθομένη γωνία ὑπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ὀρθή ἐστὶν.

Τριγώνου γάρ τοῦ $ABΓ$ τὸ ἀπὸ μιᾶς τῆς $BΓ$ πλευρᾶς τετράγωνον ἴσον ἔστω τοῖς ἀπὸ τῶν BA , AG πλευρῶν τετραγώνοις· λέγω, ὅτι ὀρθή ἐστὶν ἡ ὑπὸ BAG γωνία.

Proposition 48

If the square on one of the sides of a triangle is equal to the (sum of the) squares on the two remaining sides of the triangle (then) the angle contained by the two remaining sides of the triangle is a right-angle.

For let the square on one of the sides, BC , of triangle ABC be equal to the (sum of the) squares on the sides BA and AC . I say that angle BAC is a right-angle.



Ἦθθω γάρ ἀπὸ τοῦ Α σημείον τῇ ΑΓ εὐθείᾳ πρὸς ὀρθὰς ἡ ΑΔ καὶ κείσθω τῇ ΒΑ ἴση ἡ ΑΔ, καὶ ἐπεξεύθθω ἡ ΔΓ. ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΑΒ, ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς ΔΑ τετραγώνον τῷ ἀπὸ τῆς ΑΒ τετραγώνῳ. κοινὸν προσκείσθω τὸ ἀπὸ τῆς ΑΓ τετραγώνον· τὰ ἄρα ἀπὸ τῶν ΔΑ, ΑΓ τετραγώνων ἴσα ἐστὶ τοῖς ἀπὸ τῶν ΒΑ, ΑΓ τετραγώνοις. ἀλλὰ τοῖς μὲν ἀπὸ τῶν ΔΑ, ΑΓ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΔΓ· ὀρθὴ γάρ ἐστιν ἡ ὑπὸ ΔΑΓ γωνία· τοῖς δὲ ἀπὸ τῶν ΒΑ, ΑΓ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΒΓ· ὑπόκειται γάρ· τὸ ἄρα ἀπὸ τῆς ΔΓ τετραγώνον ἴσον ἐστὶ τῷ ἀπὸ τῆς ΒΓ τετραγώνῳ· ὥστε καὶ πλευρὰ ἡ ΔΓ τῇ ΒΓ ἐστὶν ἴση· καὶ ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΑΒ, κοινὴ δὲ ἡ ΑΓ, δύο δὲ αἱ ΔΑ, ΑΓ δύο ταῖς ΒΑ, ΑΓ ἴσαι εἰσὶν καὶ βάσις ἡ ΔΓ βάσει τῇ ΒΓ ἴση· γωνία ἄρα ἡ ὑπὸ ΔΑΓ γωνία τῇ ὑπὸ ΒΑΓ [ἐστὶν] ἴση. ὀρθὴ δὲ ἡ ὑπὸ ΔΑΓ· ὀρθὴ ἄρα καὶ ἡ ὑπὸ ΒΑΓ.

Ἐάν ἄρὰ τριγώνον τὸ ἀπὸ μιᾶς τῶν πλευρῶν τετραγώνον ἴσον ᾗ τοῖς ἀπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τετραγώνοις, ἡ περιεθρομένη γωνία ὑπὸ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ὀρθὴ ἐστὶν· ὅπερ ἔδει δεῖχαι.

For let AD be drawn from point A at right-angles to the straight-line AC [Prop. 1.11], and let AD be made equal to BA [Prop. 1.3], and let DC be joined. Since DA is equal to AB , the square on DA is thus also equal to the square on AB .[†] Let the square on AC be added to both. Thus, the (sum of the) squares on DA and AC is equal to the (sum of the) squares on BA and AC . But, the (square) on DC is equal to the (sum of the squares) on DA and AC . For angle DAC is a right-angle [Prop. 1.47]. But, the (square) on BC is equal to (sum of the squares) on BA and AC . For (that) was assumed. Thus, the square on DC is equal to the square on BC . So side DC is also equal to (side) BC . And since DA is equal to AB , and AC (is) common, the two (straight-lines) DA, AC are equal to the two (straight-lines) BA, AC . And the base DC is equal to the base BC . Thus, angle DAC [is] equal to angle BAC [Prop. 1.8]. But DAC is a right-angle. Thus, BAC is also a right-angle.

Thus, if the square on one of the sides of a triangle is equal to the (sum of the) squares on the remaining two sides of the triangle (then) the angle contained by the remaining two sides of the triangle is a right-angle. (Which is) the very thing it was required to show.

[†] Here, use is made of the additional common notion that the squares of equal things are themselves equal. Later on, the inverse notion is used.

ELEMENTS BOOK 2

Fundamentals of Geometric Algebra

Ὅροι.

1. Πᾶν παραλληλόγραμμον ὀρθογώνιον περιέθεσθαι λέγεται ὑπὸ δύο τῶν τῇν ὀρθὴν γωνίαν περιεθουσῶν εὐθειῶν.
2. Παντὸς δὲ παραλληλογράμμου θωρίον τῶν περὶ τῇν διὰ μέτρον αὐτοῦ παραλληλογράμμου ἐν ὁποιοῦν σὺν τοῖς δυοῖ παραπληρώμασι γνῶμων καλεῖσθω.

1.

Ἐάν ᾧσι δύο εὐθεῖαι, τμηθῇ δὲ ἡ ἑτέρα αὐτῶν εἰς ὅσα-
δηποτοῦν τμήματα, τὸ περιεχόμενον ὀρθογώνιον ὑπὸ τῶν δύο
εὐθειῶν ἴσον ἐστὶ τοῖς ὑπὸ τε τῆς ἀτμήτου καὶ ἐκάστου τῶν
τμημάτων περιεθόμενοις ὀρθογωνίοις.

A —————

B

Δ

Ε Β

Γ

D

E

C

H

Z

K

G

F

Θ

K

L

H

Ἐστωσαν δύο εὐθεῖαι αἱ $A, B\Gamma$, καὶ τετμήσθω ἡ $B\Gamma$, ὥς
ἔνυθεν, κατὰ τὰ Δ, E σημεία· λέγω, ὅτι τὸ ὑπὸ τῶν $A, B\Gamma$
περιεθόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν $A, B\Delta$
περιεθόμενῳ ὀρθογωνίῳ καὶ τῷ ὑπὸ τῶν $A, \Delta E$ καὶ ἔτι τῷ
ὑπὸ τῶν $A, E\Gamma$.

Ἦθθω γάρ ἀπὸ τοῦ B τῇ $B\Gamma$ πρὸς ὀρθὰς ἡ BZ , καὶ
κείσθω τῇ A ἴση ἡ BH , καὶ διὰ μέν τοῦ H τῇ $B\Gamma$ παράλληλος
ἦθθω ἡ $H\Theta$, διὰ δὲ τῶν Δ, E, Γ τῇ BH παράλληλοι ἦθθωσαν
αἱ $\Delta K, EL, \Gamma\Theta$.

Ἴσον δὲ ἐστὶ τὸ $B\Theta$ τοῖς $BK, \Delta\Lambda, E\Theta$. καὶ ἐστὶ τὸ μὲν
 $B\Theta$ τὸ ὑπὸ τῶν $A, B\Gamma$ περιέθεται μὲν γὰρ ὑπὸ τῶν $HB, B\Gamma$,
ἴση δὲ ἡ BH τῇ A · τὸ δὲ BK τὸ ὑπὸ τῶν $A, B\Delta$ περιέθεται

Definitions

1. Any rectangular parallelogram is said to be contained by the two straight-lines containing the right-angle.
2. And in any parallelogrammic figure, let any one whatsoever of the parallelograms about its diagonal, (taken) with its two complements, be called a gnomon.

Proposition 1[†]

If there are two straight-lines, and one of them is cut into any number of pieces whatsoever, (then) the rectangle contained by the two straight-lines is equal to the (sum of the) rectangles contained by the uncut (straight-line), and every one of the pieces (of the cut straight-line).

A —————

Let A and BC be the two straight-lines, and let BC be cut, at random, at points D and E . I say that the rectangle contained by A and BC is equal to the rectangle(s) contained by A and BD , by A and DE , and, finally, by A and EC .

For let BF be drawn from point B , at right-angles to BC [Prop. 1.11], and let BG be made equal to A [Prop. 1.3], and let GH be drawn through (point) G , parallel to BC [Prop. 1.31], and let DK, EL , and CH be drawn through (points) D, E , and C (respectively), parallel to BG [Prop. 1.31].

So the (rectangle) BH is equal to the (rectangles) BK, DL , and EH . And BH is the (rectangle contained) by A and BC . For it is contained by GB and BC , and BG (is) equal to A .

μὲν γὰρ ὑπὸ τῶν HB , $B\Delta$, ἴση δὲ ἡ BH τῇ A . τὸ δὲ ΔA τὸ ὑπὸ τῶν A , ΔE . ἴση γὰρ ἡ ΔK , τοντέστιν ἡ BH , τῇ A . καὶ ἔτι ὁμοίως τὸ $E\Theta$ τὸ ὑπὸ τῶν A , $E\Gamma$. τὸ ἄρα ὑπὸ τῶν A , $B\Gamma$ ἴσον ἐστὶ τῷ τε ὑπὸ A , $B\Delta$ καὶ τῷ ὑπὸ A , ΔE καὶ ἔτι τῷ ὑπὸ A , $E\Gamma$.

Ἐάν ἄρα ᾧσι δύο εὐθεῖαι, τμηθῇ δὲ ἡ ἑτέρα αὐτῶν εἰς ὁσαδηποῦν τμήματα, τὸ περιεχόμενον ὀρθογώνιον ὑπὸ τῶν δύο εὐθειῶν ἴσον ἐστὶ τοῖς ὑπὸ τε τῆς ἀτμήτου καὶ ἐκάστου τῶν τμημάτων περιεθρομένοις ὀρθογωνίοις. ὅπερ ἔδει δεῖξαι.

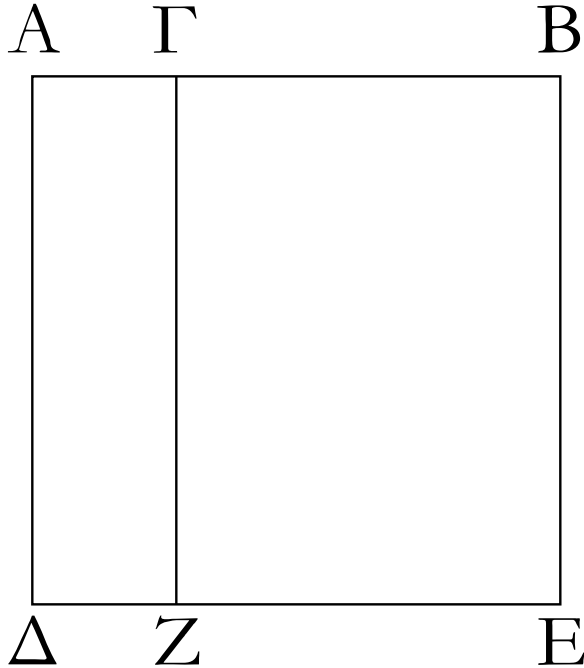
And BK (is) the (rectangle contained) by A and BD . For it is contained by GB and BD , and BG (is) equal to A . And DL (is) the (rectangle contained) by A and DE . For DK , that is to say BG [Prop. 1.34], (is) equal to A . Similarly, EH (is) also the (rectangle contained) by A and EC . Thus, the (rectangle contained) by A and BC is equal to the (rectangles contained) by A and BD , by A and DE , and, finally, by A and EC .

Thus, if there are two straight-lines, and one of them is cut into any number of pieces whatsoever, (then) the rectangle contained by the two straight-lines is equal to the (sum of the) rectangles contained by the uncut (straight-line), and every one of the pieces (of the cut straight-line). (Which is) the very thing it was required to show.

† This proposition is a geometric version of the algebraic identity: $a(b + c + d + \dots) = ab + ac + ad + \dots$.

2.

Ἐάν εὐθεῖα γραμμὴ τμηθῇ, ὡς ἔτυθεν, τὸ ὑπὸ τῆς ὅλης καὶ ἐκατέρου τῶν τμημάτων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς ὅλης τετραγώνῳ.



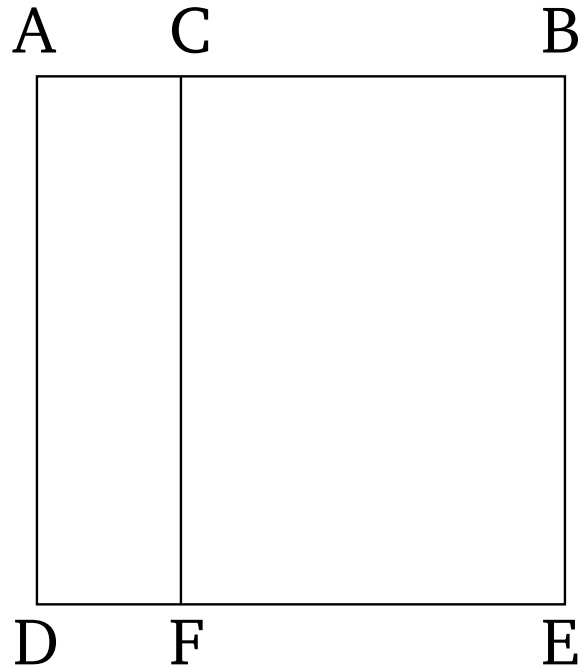
Εὐθεῖα γὰρ ἡ AB τετμήσθω, ὡς ἔτυθεν, κατὰ τὸ Γ σημειον. λέγω, ὅτι τὸ ὑπὸ τῶν AB , $B\Gamma$ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ὑπὸ BA , $A\Gamma$ περιεθρομένου ὀρθογωνίου ἴσον ἐστὶ τῷ ἀπὸ τῆς AB τετραγώνῳ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $ADEB$, καὶ ἦθθω διὰ τοῦ Γ ὁποτέρου τῶν AD , BE παράλληλος ἡ ΓZ .

Ἰσον δὲ ἐστὶ τὸ AE τοῖς AZ , ΓE . καὶ ἐστὶ τὸ μὲν AE τὸ ἀπὸ τῆς AB τετράγωνον, τὸ δὲ AZ τὸ ὑπὸ τῶν BA , $A\Gamma$

Proposition 2†

If a straight-line is cut at random, (then) the (sum of the) rectangle(s) contained by the whole (straight-line), and each of the pieces (of the straight-line), is equal to the square on the whole.



For let the straight-line AB be cut, at random, at point C . I say that the rectangle contained by AB and BC , plus the rectangle contained by BA and AC , is equal to the square on AB .

For let the square $ADEB$ be described on AB [Prop. 1.46], and let CF be drawn through C , parallel to either of AD or BE [Prop. 1.31].

So the (square) AE is equal to the (rectangles) AF and CE . And AE is the square on AB . And AF (is) the rectangle con-

περιεχόμενον ὀρθογώνιον· περιέθεται μὲν γὰρ ὑπὸ τῶν ΔA , $A\Gamma$, ἴση δὲ ἡ $A\Delta$ τῇ AB · τὸ δὲ ΓE τὸ ὑπὸ τῶν AB , $B\Gamma$ · ἴση γὰρ ἡ BE τῇ AB . τὸ ἄρα ὑπὸ τῶν BA , $A\Gamma$ μετὰ τοῦ ὑπὸ τῶν AB , $B\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB τετραγώνῳ.

Ἐὰν ἄρα εὐθεῖα γραμμὴ τμηθῇ, ὥς ἔτυχεν, τὸ ὑπὸ τῆς ὅλης καὶ ἑκατέρου τῶν τμημάτων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς ὅλης τετραγώνῳ· ὅπερ ἔδει δεῖχαι.

tained by the (straight-lines) BA and AC . For it is contained by DA and AC , and AD (is) equal to AB . And CE (is) the (rectangle contained) by AB and BC . For BE (is) equal to AB . Thus, the (rectangle contained) by BA and AC , plus the (rectangle contained) by AB and BC , is equal to the square on AB .

Thus, if a straight-line is cut at random, (then) the (sum of the) rectangle(s) contained by the whole (straight-line), and each of the pieces (of the straight-line), is equal to the square on the whole. (Which is) the very thing it was required to show.

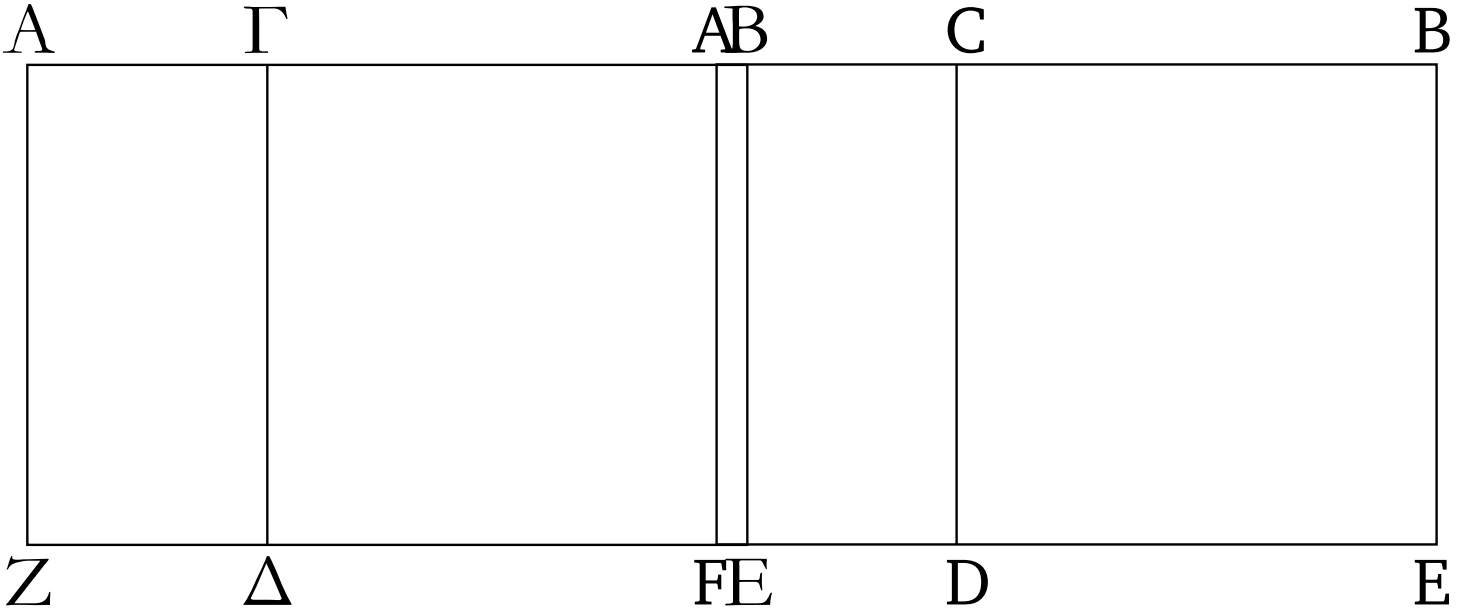
† This proposition is a geometric version of the algebraic identity: $ab + ac = a^2$ if $a = b + c$.

3.

Proposition 3[†]

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ, ὥς ἔτυχεν, τὸ ὑπὸ τῆς ὅλης καὶ ἑνὸς τῶν τμημάτων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ τε ὑπὸ τῶν τμημάτων περιεθομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τοῦ προειρημένου τμήματος τετραγώνῳ.

If a straight-line is cut at random, (then) the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), is equal to the rectangle contained by (both of) the pieces, and the square on the aforementioned piece.



Εὐθεῖα γὰρ ἡ AB τεμήσθω, ὥς ἔτυχεν, κατὰ τὸ Γ · λέγω, ὅτι τὸ ὑπὸ τῶν AB , $B\Gamma$ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ τε ὑπὸ τῶν $A\Gamma$, ΓB περιεθομένῳ ὀρθογωνίῳ μετὰ τοῦ ἀπὸ τῆς $B\Gamma$ τετραγώνῳ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΓB τετραγώνων τὸ $\Gamma\Delta EB$, καὶ διήθθω ἡ $E\Delta$ ἐπὶ τὸ Z , καὶ διὰ τοῦ A ὁποτέρῳ τῶν $\Gamma\Delta$, BE παράλληλος ἡθθῶ ἡ AZ . ἴσον δὲ ἐστὶ τὸ AE τοῖς $A\Delta$, ΓE · καὶ ἐστὶ τὸ μὲν AE τὸ ὑπὸ τῶν AB , $B\Gamma$ περιεχόμενον ὀρθογώνιον· περιέθεται μὲν γὰρ ὑπὸ τῶν AB , BE , ἴση δὲ ἡ BE τῇ $B\Gamma$ · τὸ δὲ $A\Delta$ τὸ ὑπὸ τῶν $A\Gamma$, ΓB · ἴση γὰρ ἡ $\Delta\Gamma$ τῇ ΓB · τὸ δὲ ΔB τὸ ἀπὸ τῆς ΓB τετραγώνων· τὸ ἄρα ὑπὸ τῶν AB , $B\Gamma$ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν $A\Gamma$, ΓB περιεθομένῳ ὀρθογωνίῳ μετὰ τοῦ ἀπὸ τῆς $B\Gamma$

For let the straight-line AB be cut, at random, at (point) C . I say that the rectangle contained by AB and BC is equal to the rectangle contained by AC and CB , plus the square on BC .

For let the square $CDEB$ be described on CB [Prop. 1.46], and let ED be drawn through to F , and let AF be drawn through A , parallel to either of CD or BE [Prop. 1.31]. So the (rectangle) AE is equal to the (rectangle) AD and the (square) CE . And AE is the rectangle contained by AB and BC . For it is contained by AB and BE , and BE (is) equal to BC . And AD (is) the (rectangle contained) by AC and CB . For DC (is) equal to CB . And DB (is) the square on CB . Thus, the rectangle contained by AB and BC is equal to the rectangle contained by AC and CB , plus the square on BC .

τετραγώνου.

Ἐάν ἄρα εὐθεία γραμμὴ τμηθῇ, ὥς ἔτυθεν, τὸ ὑπὸ τῆς ὅλης καὶ ἐνὸς τῶν τμημάτων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ τε ὑπὸ τῶν τμημάτων περιεχομένῳ ὀρθογώνιῳ καὶ τῷ ἀπὸ τοῦ προειρημένου τμήματος τετραγώνῳ· ὅπερ ἔδει δεῖχαι.

† This proposition is a geometric version of the algebraic identity: $(a + b)a = ab + a^2$.

4.

Ἐάν εὐθεία γραμμὴ τμηθῇ, ὥς ἔτυθεν, τὸ ἀπὸ τῆς ὅλης τετράγωνον ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν τμημάτων τετραγώνοις καὶ τῷ δις ὑπὸ τῶν τμημάτων περιεχομένῳ ὀρθογώνιῳ.

Εὐθεία γάρ γραμμὴ ἡ AB τετμήσθω, ὥς ἔτυθεν, κατὰ τὸ Γ . λέγω, ὅτι τὸ ἀπὸ τῆς AB τετράγωνον ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν AG , GB τετραγώνοις καὶ τῷ δις ὑπὸ τῶν AG , GB περιεχομένῳ ὀρθογώνιῳ.

Ἀναγεγράφθω γάρ ἀπὸ τῆς AB τετράγωνον τὸ $ADEB$, καὶ ἐπεξεύθθω ἡ BD , καὶ διὰ μὲν τοῦ Γ ὁποτέρῳ τῶν AD , EB παράλληλος ᾗθθω ἡ ΓZ , διὰ δὲ τοῦ H ὁποτέρῳ τῶν AB , DE παράλληλος ᾗθθω ἡ ΘK . καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΓZ τῇ AD , καὶ εἰς αὐτὰς ἐμπέπτωκεν ἡ BD , ἡ ἐκτὸς γωνία ἡ ὑπὸ ΓHB ἴση ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ ADB . ἀλλ' ἡ ὑπὸ ADB τῇ ὑπὸ ABD ἐστὶν ἴση, ἐπεὶ καὶ πλευρὰ ἡ BA τῇ AD ἐστὶν ἴση· καὶ ἡ ὑπὸ ΓHB ἄρα γωνία τῇ ὑπὸ HBG ἐστὶν ἴση· ὥστε καὶ πλευρὰ ἡ BG πλευρᾷ τῇ GH ἐστὶν ἴση· ἀλλ' ἡ μὲν GB τῇ HK ἐστὶν ἴση, ἡ δὲ GH τῇ KB · καὶ ἡ HK ἄρα τῇ KB ἐστὶν ἴση· ἰσόπλευρον ἄρα ἐστὶ τὸ ΓHKB . λέγω δὲ, ὅτι καὶ ὀρθογώνιον. ἐπεὶ γάρ παράλληλός ἐστιν ἡ GH τῇ BK [καὶ εἰς αὐτὰς ἐμπέπτωκεν εὐθεία ἡ ΓB], αἱ ἄρα ὑπὸ KBF , HGB γωνίαι δύο ὀρθαῖς εἰσιν ἴσαι. ὀρθὴ δὲ ἡ ὑπὸ KBF · ὀρθὴ ἄρα καὶ ἡ ὑπὸ BGH · ὥστε καὶ αἱ ἀπεναντίον αἱ ὑπὸ ΓHK , HKB ὀρθαὶ εἰσιν. ὀρθογώνιον ἄρα ἐστὶ τὸ ΓHKB · ἐδείχθη δὲ καὶ ἰσόπλευρον· τετράγωνον ἄρα ἐστὶν· καὶ ἐστὶν ἀπὸ τῆς GB . διὰ τὰ αὐτὰ δὲ καὶ τὸ ΘZ τετράγωνόν ἐστιν· καὶ ἐστὶν ἀπὸ τῆς ΘH , τουτέστιν [ἀπὸ] τῆς AG · τὰ ἄρα ΘZ , $K\Gamma$ τετράγωνα ἀπὸ τῶν AG , GB εἰσιν. καὶ ἐπεὶ ἴσον ἐστὶ τὸ AH τῷ HE , καὶ ἐστὶ τὸ AH τὸ ὑπὸ τῶν AG , GB · ἴση γάρ ἡ $H\Gamma$ τῇ GB · καὶ τὸ HE ἄρα ἴσον ἐστὶ τῷ ὑπὸ AG , GB · τὰ ἄρα AH , HE ἴσα ἐστὶ τῷ δις ὑπὸ τῶν AG , GB . ἐστὶ δὲ καὶ τὰ ΘZ , ΓK τετράγωνα ἀπὸ τῶν AG , GB · τὰ ἄρα τέσσαρα τὰ ΘZ , ΓK , AH , HE ἴσα ἐστὶ τοῖς τε ἀπὸ τῶν AG , GB τετραγώνοις καὶ τῷ δις ὑπὸ τῶν AG , GB περιεχομένῳ ὀρθογώνιῳ. ἀλλὰ τὰ ΘZ , ΓK , AH , HE ὅλον ἐστὶ τὸ $ADEB$, ὅ ἐστιν ἀπὸ τῆς AB τετράγωνον· τὸ ἄρα ἀπὸ τῆς AB τετράγωνον ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν AG , GB τετραγώνοις καὶ τῷ δις ὑπὸ τῶν AG , GB περιεχομένῳ ὀρθογώνιῳ.

Thus, if a straight-line is cut at random, (then) the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), is equal to the rectangle contained by (both of) the pieces, and the square on the aforementioned piece. (Which is) the very thing it was required to show.

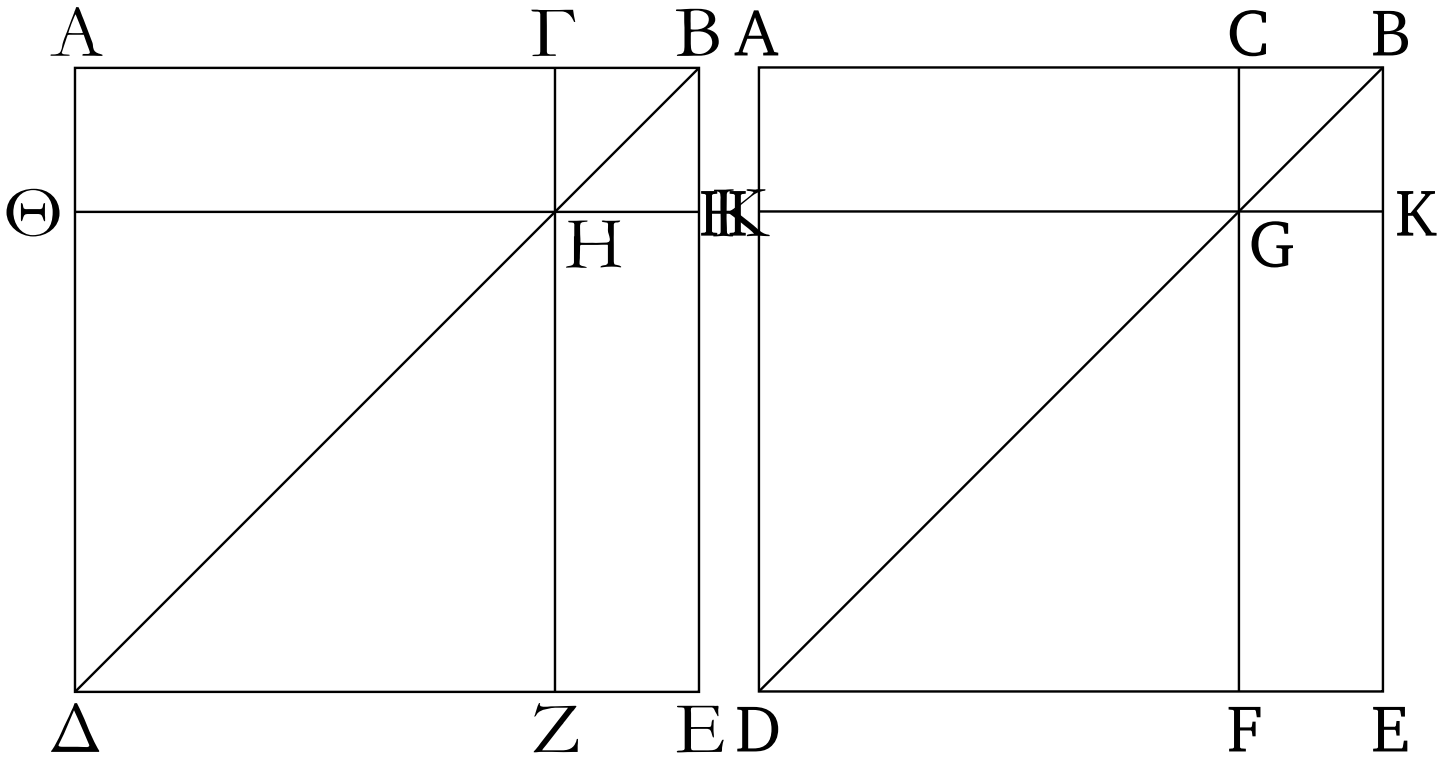
Proposition 4†

If a straight-line is cut at random, (then) the square on the whole (straight-line) is equal to the (sum of the) squares on the pieces (of the straight-line), and twice the rectangle contained by the pieces.

For let the straight-line AB be cut, at random, at (point) C . I say that the square on AB is equal to the (sum of the) squares on AC and CB , and twice the rectangle contained by AC and CB .

For let the square $ADEB$ be described on AB [Prop. 1.46], and let BD be joined, and let CF be drawn through C , parallel to either of AD or EB [Prop. 1.31], and let HK be drawn through G , parallel to either of AB or DE [Prop. 1.31]. And since CF is parallel to AD , and BD has fallen across them, the external angle CGB is equal to the internal and opposite (angle) ADB [Prop. 1.29]. But, ADB is equal to ABD , since the side BA is also equal to AD [Prop. 1.5]. Thus, angle CGB is also equal to GBC . So the side BC is equal to the side CG [Prop. 1.6]. But, CB is equal to GK , and CG to KB [Prop. 1.34]. Thus, GK is also equal to KB . Thus, $CGKB$ is equilateral. So I say that (it is) also right-angled. For since CG is parallel to BK [and the straight-line CB has fallen across them], the angles KBC and GCB are thus equal to two right-angles [Prop. 1.29]. But KBC (is) a right-angle. Thus, BCG (is) also a right-angle. So the opposite (angles) CGK and GKB are also right-angles [Prop. 1.34]. Thus, $CGKB$ is right-angled. And it was also shown (to be) equilateral. Thus, it is a square. And it is on CB . So, for the same (reasons), HF is also a square. And it is on HG , that is to say [on] AC [Prop. 1.34]. Thus, the squares HF and KC are on AC and CB (respectively). And the (rectangle) AG is equal to the (rectangle) GE [Prop. 1.43]. And AG is the (rectangle contained) by AC and CB . For GC (is) equal to CB . Thus, GE is also equal to the (rectangle contained) by AC and CB . Thus, the (rectangles) AG and GE are equal to twice the (rectangle contained) by AC and CB . And HF and CK are the squares on AC and CB (respectively). Thus, the four (figures) HF , CK , AG , and GE are equal to the (sum of the) squares on AC and BC , and twice the rectangle contained by AC and CB . But, the (figures) HF , CK , AG , and GE are (equivalent to) the whole of $ADEB$, which is the square on AB . Thus, the square on AB is equal to the (sum of the) squares on AC and CB , and twice

the rectangle contained by AC and CB .



Ἐάν ἄρα εὐθεῖα γραμμὴ τμηθῇ, ὥς ἔτυθεν, τὸ ἀπὸ τῆς ὅλης τετραγώνου ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν τμημάτων τετραγώνοις καὶ τῷ διπλῷ ὑπὸ τῶν τμημάτων περιεχομένῳ ὀρθογώνῳ· ὅπερ ἔδει δεῖξαι.

Thus, if a straight-line is cut at random, (then) the square on the whole (straight-line) is equal to the (sum of the) squares on the pieces (of the straight-line), and twice the rectangle contained by the pieces. (Which is) the very thing it was required to show.

† This proposition is a geometric version of the algebraic identity: $(a + b)^2 = a^2 + b^2 + 2ab$.

5.

Proposition 5[‡]

Ἐάν εὐθεῖα γραμμὴ τμηθῇ εἰς ἴσα καὶ ἄνισα, τὸ ὑπὸ τῶν ἀνίσων τῆς ὅλης τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς μεταχὺ τῶν τομῶν τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ἡμισείας τετραγώνῳ.

If a straight-line is cut into equal and unequal (pieces, then) the rectangle contained by the unequal pieces of the whole (straight-line), plus the square on the (difference) between the (equal and unequal) pieces, is equal to the square on half (of the straight-line).

[†] Note the (presumably mistaken) double use of the label *M* in the Greek text.

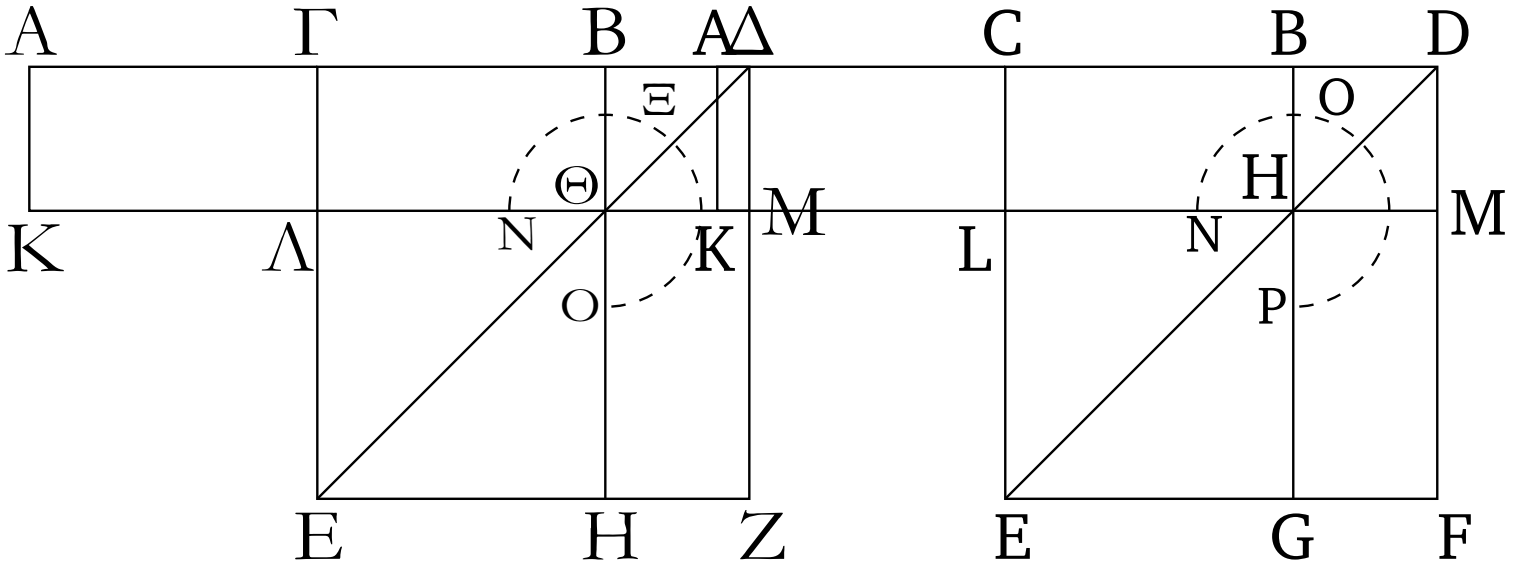
[‡] This proposition is a geometric version of the algebraic identity: $ab + [(a+b)/2 - b]^2 = [(a+b)/2]^2$.

6.

Proposition 6[†]

Ἐάν εὐθεῖα γραμμὴ τμηθῇ δίῃ, προστεθῇ δὲ τις αὐτῇ εὐθεῖα ἐπ' εὐθείας, τὸ ὑπὸ τῆς ὅλης σὺν τῇ προσκειμένῃ καὶ τῆς προσκειμένης περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ἡμισείας τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς συγκειμένης ἕκ τε τῆς ἡμισείας καὶ τῆς προσκειμένης τετραγώνῳ.

If a straight-line is cut in half, and any straight-line added to it straight-on, (then) the rectangle contained by the whole (straight-line) with the (straight-line) having being added, and the (straight-line) having being added, plus the square on half (of the original straight-line), is equal to the square on the sum of half (of the original straight-line) and the (straight-line) having been added.



Εὐθεῖα γάρ τις ἡ *AB* τετμήσθω δίῃ κατὰ τὸ *Γ* σημείον, προσκείσθω δὲ τις αὐτῇ εὐθεῖα ἐπ' εὐθείας ἡ *BD*. λέγω, ὅτι τὸ ὑπὸ τῶν *AD*, *DB* περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς *GB* τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς *ΓΔ* τετραγώνῳ.

Ἀναγεγράφθω γάρ ἀπὸ τῆς *ΓΔ* τετράγωνον τὸ *ΓΕΖΔ*, καὶ ἐπεξεύθθω ἡ *DE*, καὶ διὰ μὲν τοῦ *B* σημείου ὁποτέρῳ τῶν *ΕΓ*, *ΔΖ* παράλληλος ἦθθω ἡ *BH*, διὰ δὲ τοῦ *Θ* σημείου ὁποτέρῳ τῶν *AB*, *ΕΖ* παράλληλος ἦθθω ἡ *KM*, καὶ ἔτι διὰ τοῦ *A* ὁποτέρῳ τῶν *ΓΛ*, *ΔΜ* παράλληλος ἦθθω ἡ *AK*.

Ἐπεὶ οὖν ἴση ἐστὶν ἡ *ΑΓ* τῇ *ΓΒ*, ἴσον ἐστὶ καὶ τὸ *ΑΛ* τῷ *ΓΘ*. ἀλλὰ τὸ *ΓΘ* τῷ *ΘΖ* ἴσον ἐστίν. καὶ τὸ *ΑΛ* ἴσα τῷ *ΘΖ* ἐστὶν ἴσον. κοινὸν προσκείσθω τὸ *ΓΜ*. ὅλον ἴσα τὸ *ΑΜ* τῷ *ΝΧΟ* γνῶμονί ἐστιν ἴσον. ἀλλὰ τὸ *ΑΜ* ἐστὶ τὸ ὑπὸ τῶν *ΑΔ*, *ΔΒ*. ἴση γάρ ἐστὶν ἡ *ΔΜ* τῇ *ΔΒ*. καὶ ὁ *ΝΧΟ* ἴσα γνῶμων ἴσος ἐστὶ τῷ ὑπὸ τῶν *ΑΔ*, *ΔΒ* [περιεχομένῳ ὀρθογωνίῳ]. κοινὸν προσκείσθω τὸ *ΛΗ*, ὅ ἐστιν ἴσον τῷ ἀπὸ τῆς *ΒΓ* τετραγώνῳ. τὸ ἴσα ὑπὸ τῶν *ΑΔ*, *ΔΒ* περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς *ΓΒ* τετραγώνου ἴσον ἐστὶ τῷ *ΝΧΟ* γνῶμονι καὶ τῷ *ΛΗ*. ἀλλὰ ὁ *ΝΧΟ* γνῶμων καὶ τὸ *ΛΗ* ὅλον ἐστὶ τὸ *ΓΕΖΔ* τετράγωνον, ὅ ἐστιν ἀπὸ τῆς *ΓΔ*. τὸ ἴσα ὑπὸ τῶν *ΑΔ*, *ΔΒ* περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς

For let any straight-line *AB* be cut in half at point *C*, and let any straight-line *BD* be added to it straight-on. I say that the rectangle contained by *AD* and *DB*, plus the square on *CB*, is equal to the square on *CD*.

For let the square *CEFD* be described on *CD* [Prop. 1.46], and let *DE* be joined, and let *BG* be drawn through point *B*, parallel to either of *EC* or *DF* [Prop. 1.31], and let *KM* be drawn through point *H*, parallel to either of *AB* or *EF* [Prop. 1.31], and finally let *AK* be drawn through *A*, parallel to either of *CL* or *DM* [Prop. 1.31].

Therefore, since *AC* is equal to *CB*, (rectangle) *AL* is also equal to (rectangle) *CH* [Prop. 1.36]. But, (rectangle) *CH* is equal to (rectangle) *HF* [Prop. 1.43]. Thus, (rectangle) *AL* is also equal to (rectangle) *HF*. Let (rectangle) *CM* be added to both. Thus, the whole (rectangle) *AM* is equal to the gnomon *NOP*. But, *AM* is the (rectangle contained) by *AD* and *DB*. For *DM* is equal to *DB*. Thus, gnomon *NOP* is also equal to the [rectangle contained] by *AD* and *DB*. Let *LG*, which is equal to the square on *BC*, be added to both. Thus, the rectangle contained by *AD* and *DB*, plus the square on *CB*, is equal to the gnomon *NOP* and the (square) *LG*. But the

ΓΒ τετραγώνον ἴσον ἐστὶ τῷ ἀπὸ τῆς ΓΔ τετραγώνῳ.

Ἐάν ἄρα εὐθεῖα γραμμὴ τμηθῇ δίῃα, προστεθῇ δέ τις αὐτῇ εὐθεῖα ἐπ' εὐθείας, τὸ ὑπὸ τῆς ὅλης σὺν τῇ προσκειμένῃ καὶ τῆς προσκειμένης περιεχόμενον ὀρθόγωνιον μετὰ τοῦ ἀπὸ τῆς ἡμισείας τετραγώνον ἴσον ἐστὶ τῷ ἀπὸ τῆς συγκεκλιμένης ἕκ τε τῆς ἡμισείας καὶ τῆς προσκειμένης τετραγώνῳ· ὅπερ ἔδει δεῖξαι.

gnomon NOP and the (square) LG is (equivalent to) the whole square $CEFD$, which is on CD . Thus, the rectangle contained by AD and DB , plus the square on CB , is equal to the square on CD .

Thus, if a straight-line is cut in half, and any straight-line added to it straight-on, (then) the rectangle contained by the whole (straight-line) with the (straight-line) having being added, and the (straight-line) having being added, plus the square on half (of the original straight-line), is equal to the square on the sum of half (of the original straight-line) and the (straight-line) having been added. (Which is) the very thing it was required to show.

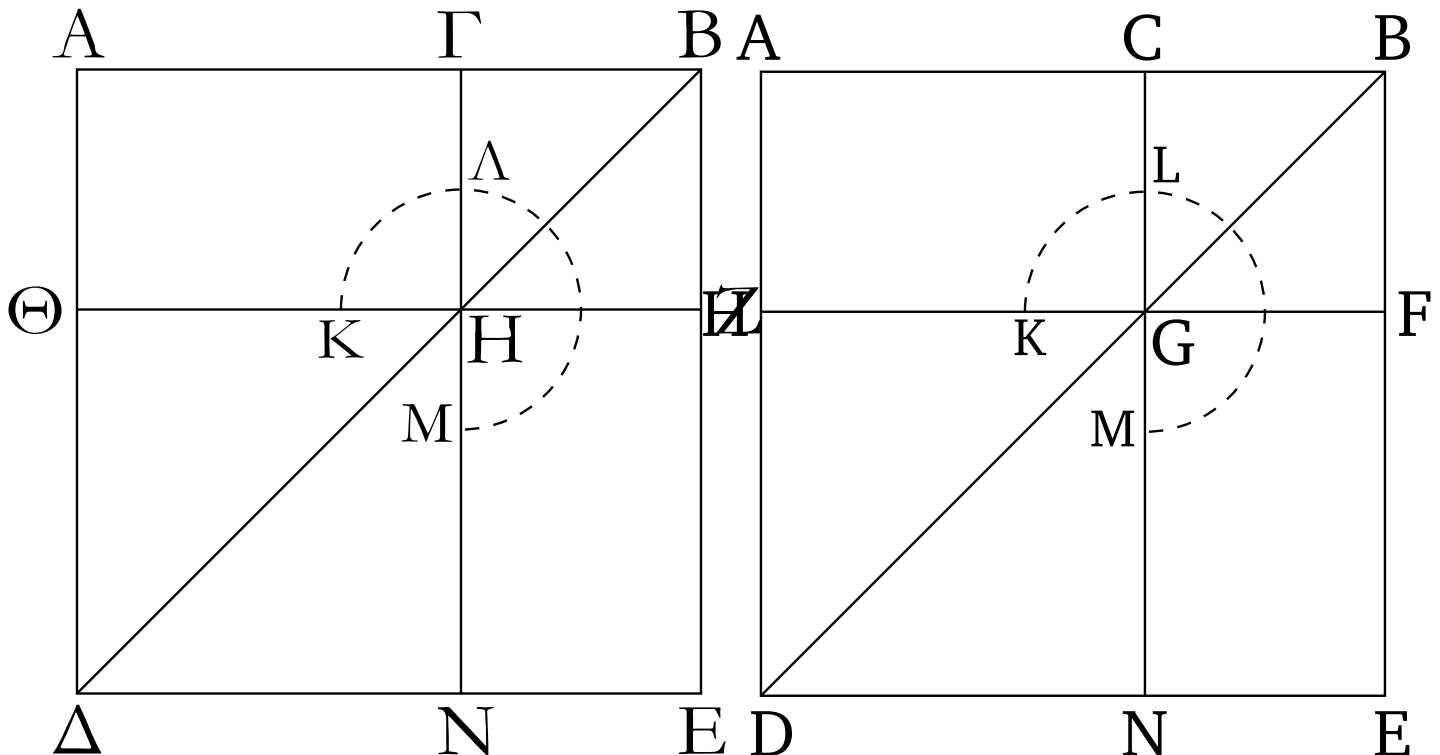
† This proposition is a geometric version of the algebraic identity: $(2a + b)b + a^2 = (a + b)^2$.

7.

Proposition 7†

Ἐάν εὐθεῖα γραμμὴ τμηθῇ, ὡς ἔτυθεν, τὸ ἀπὸ τῆς ὅλης καὶ τὸ ἀφ' ἐνὸς τῶν τμημάτων τὰ συναμφότερα τετράγωνα ἴσα ἐστὶ τῷ δι' ὑπὸ τῆς ὅλης καὶ τοῦ εἰρημένου τμήματος περιεθόμενῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνῳ.

If a straight-line is cut at random, (then) the sum of the squares on the whole (straight-line), and one of the pieces (of the straight-line), is equal to twice the rectangle contained by the whole, and the said piece, and the square on the remaining piece.



Εὐθεῖα γάρ τις ἡ AB τετμήσθω, ὡς ἔτυθεν, κατὰ τὸ Γ σημείον· λέγω, ὅτι τὰ ἀπὸ τῶν AB , $B\Gamma$ τετράγωνα ἴσα ἐστὶ τῷ δι' ὑπὸ τῶν AB , $B\Gamma$ περιεθόμενῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τῆς ΓA τετραγώνῳ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $ADEB$.

For let any straight-line AB be cut, at random, at point C . I say that the (sum of the) squares on AB and BC is equal to twice the rectangle contained by AB and BC , and the square on CA .

For let the square $ADEB$ be described on AB [Prop. 1.46],

καὶ καταγεγράφθω τὸ σθῆμα.

Ἐπεὶ οὖν ἴσον ἐστὶ τὸ AH τῷ HE , κοινὸν προσκείσθω τὸ GZ . ὅλον ἄρα τὸ AZ ὅλῳ τῷ GE ἴσον ἐστίν· τὰ ἄρα AZ , GE διπλάσιά ἐστι τοῦ AZ . ἀλλὰ τὰ AZ , GE ὁ KAM ἐστὶ γνῶμων καὶ τὸ GZ τετράγωνον· ὁ KAM ἄρα γνῶμων καὶ τὸ GZ διπλάσιά ἐστι τοῦ AZ . ἐστὶ δὲ τοῦ AZ διπλάσιον καὶ τὸ διζὺ ὑπὸ τῶν AB , $BΓ$. ἴση γὰρ ἡ BZ τῇ $BΓ$. ὁ ἄρα KAM γνῶμων καὶ τὸ GZ τετράγωνον ἴσον ἐστὶ τῷ διζὺ ὑπὸ τῶν AB , $BΓ$. κοινὸν προσκείσθω τὸ ΔH , ὅ ἐστιν ἀπὸ τῆς AG τετράγωνον· ὁ ἄρα KAM γνῶμων καὶ τὰ BH , $H\Delta$ τετράγωνα ἴσα ἐστὶ τῷ τε διζὺ ὑπὸ τῶν AB , $BΓ$ περιεθρομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τῆς AG τετραγώνῳ. ἀλλὰ ὁ KAM γνῶμων καὶ τὰ BH , $H\Delta$ τετράγωνα ὅλον ἐστὶ τὸ $A\Delta EB$ καὶ τὸ GZ , ὅ ἐστιν ἀπὸ τῶν AB , $BΓ$ τετράγωνα· τὰ ἄρα ἀπὸ τῶν AB , $BΓ$ τετράγωνα ἴσα ἐστὶ τῷ [τε] διζὺ ὑπὸ τῶν AB , $BΓ$ περιεθρομένῳ ὀρθογωνίῳ μετὰ τοῦ ἀπὸ τῆς AG τετραγώνου.

Ἐάν ἄρα εὐθεῖα γραμμὴ τμηθῇ, ὥς ἔτυθεν, τὸ ἀπὸ τῆς ὅλης καὶ τὸ ἀφ' ἐνόξ τῶν τμημάτων τὰ συνναμφότερα τετράγωνα ἴσα ἐστὶ τῷ τε διζὺ ὑπὸ τῆς ὅλης καὶ τοῦ εἰρημένον τμήματος περιεθρομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνῳ· ὅπερ ἔδει δεῖλαι.

and let the (rest of) the figure be drawn.

Therefore, since (rectangle) AG is equal to (rectangle) GE [Prop. 1.43], let the (square) CF be added to both. Thus, the whole (rectangle) AF is equal to the whole (rectangle) CE . Thus, (rectangle) AF plus (rectangle) CE is double (rectangle) AF . But, (rectangle) AF plus (rectangle) CE is the gnomon KLM , and the square CF . Thus, the gnomon KLM , and the square CF , is double the (rectangle) AF . But double the (rectangle) AF is also twice the (rectangle contained) by AB and BC . For BF (is) equal to BC . Thus, the gnomon KLM , and the square CF , are equal to twice the (rectangle contained) by AB and BC . Let DG , which is the square on AC , be added to both. Thus, the gnomon KLM , and the squares BG and GD , are equal to twice the rectangle contained by AB and BC , and the square on AC . But, the gnomon KLM and the squares BG and GD is (equivalent to) the whole of $ADEB$ and CF , which are the squares on AB and BC (respectively). Thus, the (sum of the) squares on AB and BC is equal to twice the rectangle contained by AB and BC , and the square on AC .

Thus, if a straight-line is cut at random, (then) the sum of the squares on the whole (straight-line), and one of the pieces (of the straight-line), is equal to twice the rectangle contained by the whole, and the said piece, and the square on the remaining piece. (Which is) the very thing it was required to show.

† This proposition is a geometric version of the algebraic identity: $(a + b)^2 + a^2 = 2(a + b)a + b^2$.

8.

Proposition 8[†]

Ἐάν εὐθεῖα γραμμὴ τμηθῇ, ὥς ἔτυθεν, τὸ τετράκις ὑπὸ τῆς ὅλης καὶ ἐνόξ τῶν τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τε τῆς ὅλης καὶ τοῦ εἰρημένον τμήματος ὥς ἀπὸ μιᾶς ἀναγραφέντι τετραγώνῳ.

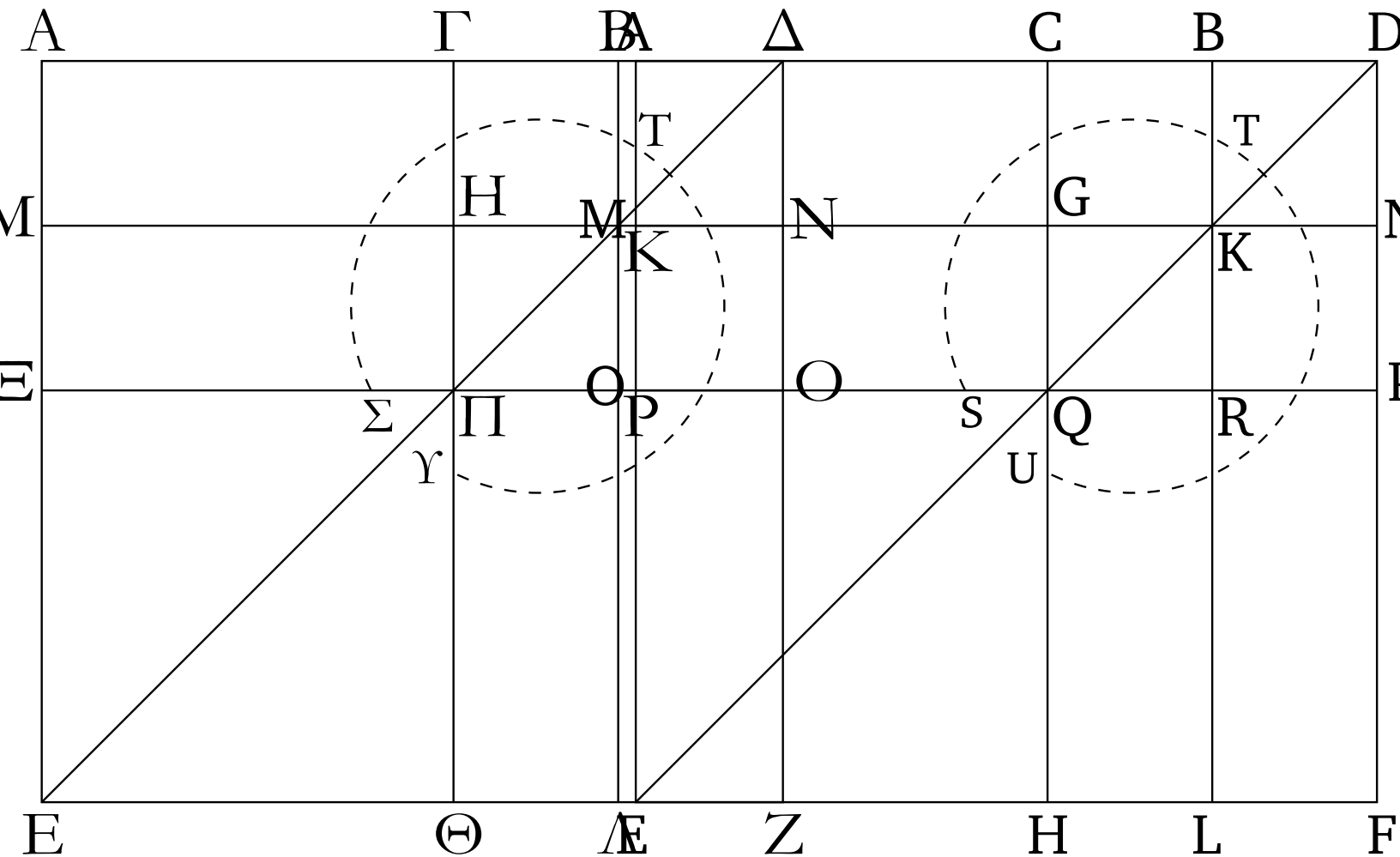
Εὐθεῖα γάρ τις ἡ AB τετμήσθω, ὥς ἔτυθεν, κατὰ τὸ Γ σημείον· λέγω, ὅτι τὸ τετράκις ὑπὸ τῶν AB , $BΓ$ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς AG τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς AB , $BΓ$ ὥς ἀπὸ μιᾶς ἀναγραφέντι τετραγώνῳ.

Ἐκβεβλήσθω γὰρ ἐπ' εὐθείας [τῇ AB εὐθείᾳ] ἡ $B\Delta$, καὶ κείσθω τῇ GB ἴση ἡ $B\Delta$, καὶ ἀναγεγράφθω ἀπὸ τῆς $A\Delta$ τετράγωνον τὸ $AEZ\Delta$, καὶ καταγεγράφθω διπλοῦν τὸ σθῆμα.

If a straight-line is cut at random, (then) four times the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), plus the square on the remaining piece, is equal to the square described on the whole and the former piece, as on one (complete straight-line).

For let any straight-line AB be cut, at random, at point C . I say that four times the rectangle contained by AB and BC , plus the square on AC , is equal to the square described on AB and BC , as on one (complete straight-line).

For let BD be produced in a straight-line [with the straight-line AB], and let BD be made equal to CB [Prop. 1.3], and let the square $AEFD$ be described on AD [Prop. 1.46], and let the (rest of the) figure be drawn double.



Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΓΒ τῇ ΒΔ, ἀλλὰ ἡ μὲν ΓΒ τῇ ΗΚ ἐστὶν ἴση, ἡ δὲ ΒΔ τῇ ΚΝ, καὶ ἡ ΗΚ ἄρα τῇ ΚΝ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΠΡ τῇ ΡΟ ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ ΒΓ τῇ ΒΔ, ἡ δὲ ΗΚ τῇ ΚΝ, ἴσον ἄρα ἐστὶ καὶ τὸ μὲν ΓΚ τῷ ΚΔ, τὸ δὲ ΗΡ τῷ ΡΝ. ἀλλὰ τὸ ΓΚ τῷ ΡΝ ἐστὶν ἴσον· παραπληρώματα γάρ τοῦ ΓΟ παραλληλογράμμον· καὶ τὸ ΚΔ ἄρα τῷ ΗΡ ἴσον ἐστὶν· τὰ τέσσαρα ἄρα τὰ ΔΚ, ΓΚ, ΗΡ, ΡΝ ἴσα ἀλλήλοις ἐστίν· τὰ τέσσαρα ἄρα τετραπλάσιά ἐστι τοῦ ΓΚ. πάλιν ἐπεὶ ἴση ἐστὶν ἡ ΓΒ τῇ ΒΔ, ἀλλὰ ἡ μὲν ΒΔ τῇ ΒΚ, τοντέστι τῇ ΓΗ ἴση, ἡ δὲ ΓΒ τῇ ΗΚ, τοντέστι τῇ ΗΠ, ἐστὶν ἴση, καὶ ἡ ΓΗ ἄρα τῇ ΗΠ ἴση ἐστὶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν ΓΗ τῇ ΗΠ, ἡ δὲ ΠΡ τῇ ΡΟ, ἴσον ἐστὶ καὶ τὸ μὲν ΑΗ τῷ ΜΠ, τὸ δὲ ΠΛ τῷ ΡΖ. ἀλλὰ τὸ ΜΠ τῷ ΠΛ ἐστὶν ἴσον· παραπληρώματα γάρ τοῦ ΜΛ παραλληλογράμμον· καὶ τὸ ΑΗ ἄρα τῷ ΡΖ ἴσον ἐστὶν· τὰ τέσσαρα ἄρα τὰ ΑΗ, ΜΠ, ΠΛ, ΡΖ ἴσα ἀλλήλοις ἐστίν· τὰ τέσσαρα ἄρα τοῦ ΑΗ ἐστὶ τετραπλάσια. ἐδείκθη δὲ καὶ τὰ τέσσαρα τὰ ΓΚ, ΚΔ, ΗΡ, ΡΝ τοῦ ΓΚ τετραπλάσια· τὰ ἄρα ὅκτω, ἃ περιέχει τὸν ΣΤΥ γνῶμονα, τετραπλάσιά ἐστι τοῦ ΑΚ. καὶ ἐπεὶ τὸ ΑΚ τὸ ὑπὸ τῶν ΑΒ, ΒΔ ἐστὶν ἴση γὰρ ἡ ΒΚ τῇ ΒΔ· τὸ

Therefore, since CB is equal to BD , but CB is equal to GK [Prop. 1.34], and BD to KN [Prop. 1.34], GK is thus also equal to KN . So, for the same (reasons), QR is equal to RP . And since BC is equal to BD , and GK to KN , (square) CK is thus also equal to (square) KD , and (square) GR to (square) RN [Prop. 1.36]. But, (square) CK is equal to (square) RN . For (they are) complements in the parallelogram CP [Prop. 1.43]. Thus, (square) KD is also equal to (square) GR . Thus, the four (squares) DK , CK , GR , and RN are equal to one another. Thus, the four (taken together) are quadruple (square) CK . Again, since CB is equal to BD , but BD (is) equal to BK —that is to say, CG —and CB is equal to GK —that is to say, GQ — CG is thus also equal to GQ . And since CG is equal to GQ , and QR to RP , (rectangle) AG is also equal to (rectangle) MQ , and (rectangle) QL to (rectangle) RF [Prop. 1.36]. But, (rectangle) MQ is equal to (rectangle) QL . For (they are) complements in the parallelogram ML [Prop. 1.43]. Thus, (rectangle) AG is also equal to (rectangle) RF . Thus, the four (rectangles) AG , MQ , QL , and RF are equal to one another. Thus, the four (taken together) are quadruple (rectangle) AG . And it

ἄρα τετράκις ὑπὸ τῶν AB, BD τετραπλάσιόν ἐστι τοῦ AK . ἐδείκθη δὲ τοῦ AK τετραπλάσιος καὶ ὁ ΣTY γνῶμων· τὸ ἄρα τετράκις ὑπὸ τῶν AB, BD ἴσον ἐστὶ τῷ ΣTY γνῶμονι. κοινὸν προσκείσθω τὸ $X\Theta$, ὅ ἐστιν ἴσον τῷ ἀπὸ τῆς AG τετραγώνου· τὸ ἄρα τετράκις ὑπὸ τῶν AB, BD περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ AG τετραγώνου ἴσον ἐστὶ τῷ ΣTY γνῶμονι καὶ τῷ $X\Theta$. ἀλλὰ ὁ ΣTY γνῶμων καὶ τὸ $X\Theta$ ὅλον ἐστὶ τὸ $AEZ\Delta$ τετράγωνον, ὅ ἐστιν ἀπὸ τῆς AD . τὸ ἄρα τετράκις ὑπὸ τῶν AB, BD μετὰ τοῦ ἀπὸ AG ἴσον ἐστὶ τῷ ἀπὸ AD τετραγώνου· ἴση δὲ ἡ BD τῇ BF . τὸ ἄρα τετράκις ὑπὸ τῶν AB, BF περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ AG τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς AD , τούτεστι τῷ ἀπὸ τῆς AB καὶ BF ὥς ἀπὸ μιᾶς ἀναγραφέντι τετραγώνου.

Ἐάν ἄρα εὐθεῖα γραμμὴ τμηθῇ, ὥς ἔτυθεν, τὸ τετράκις ὑπὸ τῆς ὅλης καὶ ἐνὸς τῶν τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τε τῆς ὅλης καὶ τοῦ εἰρημένου τμήματος ὥς ἀπὸ μιᾶς ἀναγραφέντι τετραγώνου· ὅπερ ἔδει δεῖχαι.

was also shown that the four (squares) CK, KD, GR , and RN (taken together are) quadruple (square) CK . Thus, the eight (figures taken together), which comprise the gnomon STU , are quadruple (rectangle) AK . And since AK is the (rectangle contained) by AB and BD , for BK (is) equal to BD , four times the (rectangle contained) by AB and BD is quadruple (rectangle) AK . But the gnomon STU was also shown (to be equal to) quadruple (rectangle) AK . Thus, four times the (rectangle contained) by AB and BD is equal to the gnomon STU . Let OH , which is equal to the square on AC , be added to both. Thus, four times the rectangle contained by AB and BD , plus the square on AC , is equal to the gnomon STU , and the (square) OH . But, the gnomon STU and the (square) OH is (equivalent to) the whole square $AEFD$, which is on AD . Thus, four times the (rectangle contained) by AB and BD , plus the (square) on AC , is equal to the square on AD . And BD (is) equal to BC . Thus, four times the rectangle contained by AB and BC , plus the square on AC , is equal to the (square) on AD , that is to say the square described on AB and BC , as on one (complete straight-line).

Thus, if a straight-line is cut at random, (then) four times the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), plus the square on the remaining piece, is equal to the square described on the whole and the former piece, as on one (complete straight-line). (Which is) the very thing it was required to show.

[†] This proposition is a geometric version of the algebraic identity: $4(a+b)a + b^2 = [(a+b) + a]^2$.

9.

Proposition 9[†]

Ἐάν εὐθεῖα γραμμὴ τμηθῇ εἰς ἴσα καὶ ἄνισα, τὰ ἀπὸ τῶν ἀνίσων τῆς ὅλης τμημάτων τετράγωνα διπλάσιά ἐστι τοῦ τε ἀπὸ τῆς ἡμισείας καὶ τοῦ ἀπὸ τῆς μεταχρὶ τῶν τομῶν τετραγώνου.

Εὐθεῖα γάρ τις ἡ AB τετμήσθω εἰς μὲν ἴσα κατὰ τὸ Γ , εἰς δὲ ἄνισα κατὰ τὸ Δ . λέγω, ὅτι τὰ ἀπὸ τῶν $AD, \Delta B$ τετράγωνα διπλάσιά ἐστι τῶν ἀπὸ τῶν $AG, \Gamma\Delta$ τετραγώνων.

Ἦθθω γάρ ἀπὸ τοῦ Γ τῇ AB πρὸς ὀρθὰς ἡ GE , καὶ κείσθω ἴση ἑκατέρω τῶν $AG, \Gamma B$, καὶ ἐπεξεύθωσαν αἱ EA, EB , καὶ διὰ μὲν τοῦ Δ τῇ EG παράλληλος ἡθθω ἡ ΔZ , διὰ δὲ τοῦ Z τῇ AB ἡ ZH , καὶ ἐπεξεύθω ἡ AZ . καὶ ἐπεὶ ἴση ἐστὶν ἡ AG τῇ GE , ἴση ἐστὶ καὶ ἡ ὑπὸ EAG γωνία τῇ ὑπὸ AEG . καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ πρὸς τῷ Γ , λοιπαὶ ἄρα αἱ ὑπὸ EAG, AEG μιᾶ ὀρθῇ ἴσαι εἰσὶν καὶ εἰσιν ἴσαι· ἡμίσεια ἄρα ὀρθῆς ἐστὶν ἑκατέρω τῶν ὑπὸ GEA, GAE . διὰ τὰ αὐτὰ δὲ καὶ ἑκατέρω τῶν ὑπὸ GEB, EBG ἡμίσειά ἐστὶν ὀρθῆς· ὅλη ἄρα ἡ ὑπὸ AEB ὀρθὴ ἐστὶν. καὶ ἐπεὶ ἡ ὑπὸ HEZ ἡμίσειά ἐστὶν ὀρθῆς, ὀρθὴ δὲ ἡ ὑπὸ EZH . ἴση γάρ ἐστι τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ $EG\Gamma$. λοιπὴ ἄρα ἡ ὑπὸ EZH ἡμίσειά ἐστὶν ὀρθῆς· ἴση ἄρα [ἐστὶν] ἡ ὑπὸ HEZ γωνία τῇ ὑπὸ EZH . ὥστε

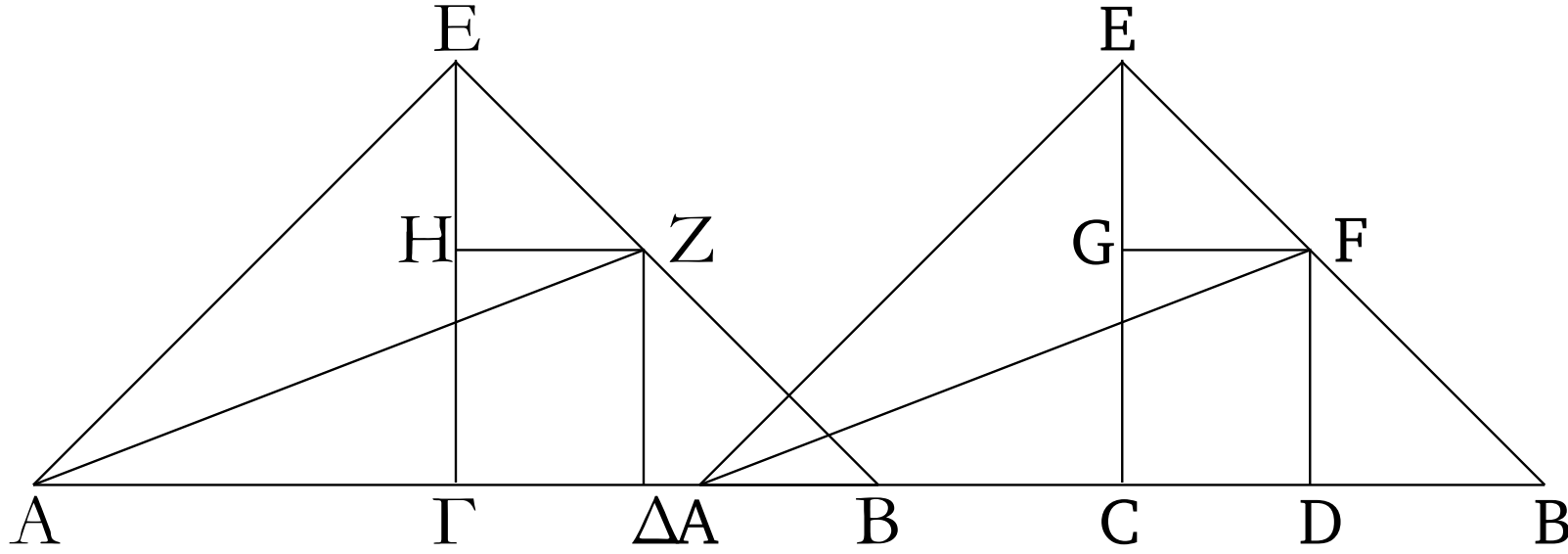
If a straight-line is cut into equal and unequal (pieces, then) the (sum of the) squares on the unequal pieces of the whole (straight-line) is double the (sum of the) square on half (the straight-line) and (the square) on the (difference) between the (equal and unequal) pieces.

For let any straight-line AB be cut—equally at C , and unequally at D . I say that the (sum of the) squares on AD and DB is double the (sum of the squares) on AC and CD .

For let CE be drawn from (point) C , at right-angles to AB [Prop. 1.11], and let it be made equal to each of AC and CB [Prop. 1.3], and let EA and EB be joined. And let DF be drawn through (point) D , parallel to EC [Prop. 1.31], and (let) FG (be drawn) through (point) F , (parallel) to AB [Prop. 1.31]. And let AF be joined. And since AC is equal to CE , the angle EAC is also equal to the (angle) AEC [Prop. 1.5]. And since the (angle) at C is a right-angle, the (sum of the) remaining angles (of triangle AEC), EAC and AEC , is thus equal to one right-angle [Prop. 1.32]. And they are equal. Thus, (angles) CEA and CAE are each half a right-angle. So, for the same (reasons), (angles) CEB and EBC are also each half a right-angle.

καὶ πλευρὰ ἡ EH τῇ HZ ἔστιν ἴση. πάλιν ἐπεὶ ἡ $πρὸς τῷ B$ γωνία ἡμίσειά ἐστιν ὀρθῆς, ὀρθὴ δὲ ἡ ὑπὸ ZAB . ἴση γὰρ πάλιν ἐστὶ τῇ ἐντὸς καὶ ἀπεναντίον τῇ ὑπὸ ETB . λοιπὴ ἄρα ἡ ὑπὸ BZD ἡμίσειά ἐστιν ὀρθῆς. ἴση ἄρα ἡ $πρὸς τῷ B$ γωνία τῇ ὑπὸ ΔZB . ὥστε καὶ πλευρὰ ἡ ZD πλευρᾷ τῇ ΔB ἔστιν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ AG τῇ GE , ἴσον ἐστὶ καὶ τὸ ἀπὸ AG τῷ ἀπὸ GE . τὰ ἄρα ἀπὸ τῶν AG , GE τετράγωνα διπλάσιά ἐστι τοῦ ἀπὸ AG . τοῖς δὲ ἀπὸ τῶν AG , GE ἴσον ἐστὶ τὸ ἀπὸ τῆς EA τετράγωνον. ὀρθὴ γὰρ ἡ ὑπὸ AGE γωνία. τὸ ἄρα ἀπὸ τῆς EA διπλάσιόν ἐστι τοῦ ἀπὸ τῆς AG . πάλιν, ἐπεὶ ἴση ἐστὶν ἡ EH τῇ HZ , ἴσον καὶ τὸ ἀπὸ τῆς EH τῷ ἀπὸ τῆς HZ . τὰ ἄρα ἀπὸ τῶν EH , HZ τετράγωνα διπλάσιά ἐστι τοῦ ἀπὸ τῆς HZ τετράγωνον. τοῖς δὲ ἀπὸ τῶν EH , HZ τετραγώνοις ἴσον ἐστὶ τὸ ἀπὸ τῆς EZ τετράγωνον. τὸ ἄρα ἀπὸ τῆς EZ τετράγωνον διπλάσιόν ἐστι τοῦ ἀπὸ τῆς HZ . ἴση δὲ ἡ HZ τῇ $\Gamma\Delta$. τὸ ἄρα ἀπὸ τῆς EZ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς $\Gamma\Delta$. ἔστι δὲ καὶ τὸ ἀπὸ τῆς EA διπλάσιον τοῦ ἀπὸ τῆς AG . τὰ ἄρα ἀπὸ τῶν AE , EZ τετράγωνα διπλάσιά ἐστι τῶν ἀπὸ τῶν AG , $\Gamma\Delta$ τετραγώνων. τοῖς δὲ ἀπὸ τῶν AE , EZ ἴσον ἐστὶ τὸ ἀπὸ τῆς AZ τετράγωνον. ὀρθὴ γὰρ ἐστὶν ἡ ὑπὸ $A EZ$ γωνία. τὸ ἄρα ἀπὸ τῆς AZ τετράγωνον διπλάσιόν ἐστι τῶν ἀπὸ τῶν AG , $\Gamma\Delta$. τῷ δὲ ἀπὸ τῆς AZ ἴσα τὰ ἀπὸ τῶν $A\Delta$, ΔZ . ὀρθὴ γὰρ ἡ $πρὸς τῷ \Delta$ γωνία. τὰ ἄρα ἀπὸ τῶν $A\Delta$, ΔZ διπλάσιά ἐστι τῶν ἀπὸ τῶν AG , $\Gamma\Delta$ τετραγώνων. ἴση δὲ ἡ ΔZ τῇ ΔB . τὰ ἄρα ἀπὸ τῶν $A\Delta$, ΔB τετράγωνα διπλάσιά ἐστι τῶν ἀπὸ τῶν AG , $\Gamma\Delta$ τετραγώνων.

Thus, the whole (angle) AEB is a right-angle. And since GEF is half a right-angle, and EGF (is) a right-angle—for it is equal to the internal and opposite (angle) ECB [Prop. 1.29]—the remaining (angle) EFG is thus half a right-angle [Prop. 1.32]. Thus, angle GEF [is] equal to EFG . So the side EG is also equal to the (side) GF [Prop. 1.6]. Again, since the angle at B is half a right-angle, and (angle) FDB (is) a right-angle—for again it is equal to the internal and opposite (angle) ECB [Prop. 1.29]—the remaining (angle) BFD is half a right-angle [Prop. 1.32]. Thus, the angle at B (is) equal to DFB . So the side FD is also equal to the side DB [Prop. 1.6]. And since AC is equal to CE , the (square) on AC (is) also equal to the (square) on CE . Thus, the (sum of the) squares on AC and CE is double the (square) on AC . And the square on EA is equal to the (sum of the) squares on AC and CE . For angle ACE (is) a right-angle [Prop. 1.47]. Thus, the (square) on EA is double the (square) on AC . Again, since EG is equal to GF , the (square) on EG (is) also equal to the (square) on GF . Thus, the (sum of the squares) on EG and GF is double the square on GF . And the square on EF is equal to the (sum of the) squares on EG and GF [Prop. 1.47]. Thus, the square on EF is double the (square) on GF . And GF (is) equal to CD [Prop. 1.34]. Thus, the (square) on EF is double the (square) on CD . And the (square) on EA is also double the (square) on AC . Thus, the (sum of the) squares on AE and EF is double the (sum of the) squares on AC and CD . And the square on AF is equal to the (sum of the squares) on AE and EF . For the angle AEF is a right-angle [Prop. 1.47]. Thus, the square on AF is double the (sum of the squares) on AC and CD . And the (sum of the squares) on AD and DF (is) equal to the (square) on AF . For the angle at D is a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AD and DF is double the (sum of the) squares on AC and CD . And DF (is) equal to DB . Thus, the (sum of the) squares on AD and DB is double the (sum of the) squares on AC and CD .



Ἐάν ἄρα εὐθεῖα γραμμὴ τμηθῇ εἰς ἴσα καὶ ἄνισα, τὰ ἀπὸ τῶν ἀνίσων τῆς ὅλης τμημάτων τετράγωνα διπλάσιά ἐστι τοῦ τε ἀπὸ τῆς ἡμισείας καὶ τοῦ ἀπὸ τῆς μεταχὺ τῶν τομῶν τετραγώνου· ὅπερ ἔδει δεῖχαι.

Thus, if a straight-line is cut into equal and unequal (pieces, then) the (sum of the) squares on the unequal pieces of the whole (straight-line) is double the (sum of the) square on half (the straight-line) and (the square) on the (difference) between the (equal and unequal) pieces. (Which is) the very thing it was required to show.

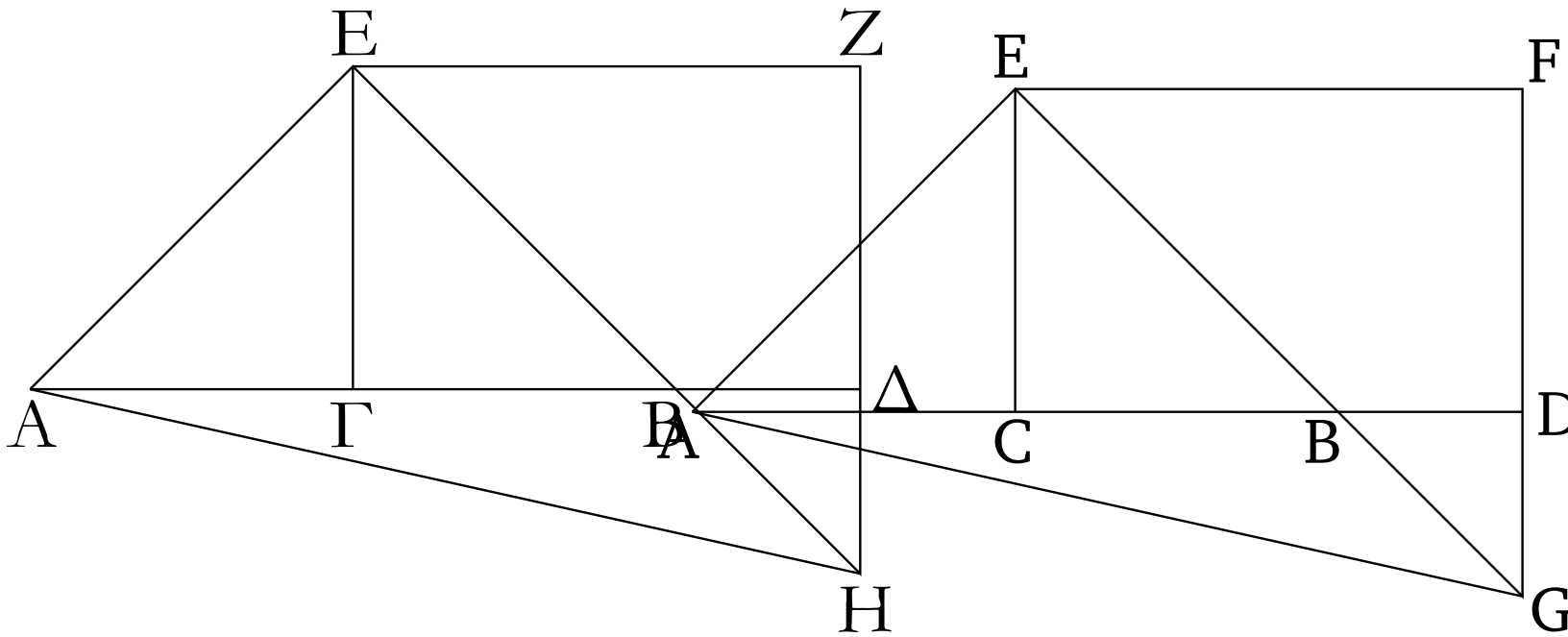
† This proposition is a geometric version of the algebraic identity: $a^2 + b^2 = 2 \left[\left(\frac{a+b}{2} \right)^2 + \left(\frac{a+b}{2} - b \right)^2 \right]$.

10.

Proposition 10[†]

Ἐάν εὐθεῖα γραμμὴ τμηθῇ δίθα, προστεθῇ δέ τις αὐτῇ εὐθεῖα ἐπ' εὐθείας, τὸ ἀπὸ τῆς ὅλης σὺν τῇ προσκειμένῃ καὶ τὸ ἀπὸ τῆς προσκειμένης τὰ συναμφότερα τετράγωνα διπλάσιά ἐστι τοῦ τε ἀπὸ τῆς ἡμισείας καὶ τοῦ ἀπὸ τῆς συγκειμένης ἕκ τε τῆς ἡμισείας καὶ τῆς προσκειμένης ὥς ἀπὸ μιᾶς ἀναγραφέντος τετραγώνου.

If a straight-line is cut in half, and any straight-line added to it straight-on, (then) the sum of the square on the whole (straight-line) with the (straight-line) having been added, and the (square) on the (straight-line) having been added, is double the (sum of the square) on half (the straight-line), and the square described on the sum of half (the straight-line) and (straight-line) having been added, as on one (complete straight-line).



Εὐθεία γάρ τις ἡ AB τετμήσθω δίῃα κατὰ τὸ C , προσκε-
 ῖσθω δὲ τις αὐτῇ εὐθείᾳ ἐπ' εὐθείᾳς ἡ BD · λέγω, ὅτι τὰ ἀπὸ
 τῶν AD , DB τετράγωνα διπλάσιά ἐστι τῶν ἀπὸ τῶν AC , CD
 τετραγώνων.

Ἦθθω γὰρ ἀπὸ τοῦ C σημείου τῇ AB πρὸς ὀρθάς ἡ CE ,
 καὶ κείσθω ἴση ἑκατέρᾳ τῶν AC , CB , καὶ ἐπεξεύθθωσαν αἱ
 EA , EB · καὶ διὰ μὲν τοῦ E τῇ AD παράλληλος ἦθθω ἡ EZ ,
 διὰ δὲ τοῦ D τῇ CE παράλληλος ἦθθω ἡ ZD . καὶ ἐπεὶ εἰς
 παραλλήλους εὐθείας τὰς EG , ZD εὐθείᾳ τις ἐνέπεσεν ἡ EZ ,
 αἱ ὑπὸ GEZ , EZD ἄρα δυοῖν ὀρθαῖς ἴσαι εἰσὶν· αἱ ἄρα ὑπὸ
 ZEB , EZD δύο ὀρθῶν ἐλάσσονές εἰσιν· αἱ δὲ ἀπ' ἐλασσόνων
 ἡ δύο ὀρθῶν ἐκβαλλόμεναι συμπίπτουσιν· αἱ ἄρα EB , ZD
 ἐκβαλλόμεναι ἐπὶ τὰ B , D μέρη συμπεσοῦνται. ἐκβεβλήσθω-
 σαν καὶ συμπιπτεύωσαν κατὰ τὸ H , καὶ ἐπεξεύθθω ἡ AH . καὶ
 ἐπεὶ ἴση ἐστὶν ἡ AG τῇ GE , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ EAG τῇ
 ὑπὸ AEG · καὶ ὀρθή ἡ πρὸς τῷ C · ἡμίσεια ἄρα ὀρθῆς [ἐστίν]
 ἑκατέρα τῶν ὑπὸ EAG , AEG . διὰ τὰ αὐτὰ δὲ καὶ ἑκατέρᾳ
 τῶν ὑπὸ GEB , EBG ἡμίσεια ἐστὶν ὀρθῆς· ὀρθή ἄρα ἐστὶν ἡ
 ὑπὸ AEB . καὶ ἐπεὶ ἡμίσεια ὀρθῆς ἐστὶν ἡ ὑπὸ EBG , ἡμίσεια
 ἄρα ὀρθῆς καὶ ἡ ὑπὸ DBH . ἐστὶ δὲ καὶ ἡ ὑπὸ $BΔH$ ὀρθή·
 ἴση γὰρ ἐστὶ τῇ ὑπὸ $ΔGE$ · ἐναλλάξ γάρ· λοιπὴ ἄρα ἡ ὑπὸ
 $ΔHB$ ἡμίσεια ἐστὶν ὀρθῆς· ἡ ἄρα ὑπὸ $ΔHB$ τῇ ὑπὸ $ΔBH$
 ἐστὶν ἴση· ὥστε καὶ πλευρὰ ἡ $BΔ$ πλευρᾷ τῇ $HΔ$ ἐστὶν ἴση.
 πάλιν, ἐπεὶ ἡ ὑπὸ EHZ ἡμίσεια ἐστὶν ὀρθῆς, ὀρθή δὲ ἡ πρὸς
 τῷ Z · ἴση γὰρ ἐστὶ τῇ ἀπεναντίον τῇ πρὸς τῷ C · λοιπὴ ἄρα
 ἡ ὑπὸ ZEH ἡμίσεια ἐστὶν ὀρθῆς· ἴση ἄρα ἡ ὑπὸ EHZ γωνία
 τῇ ὑπὸ ZEH · ὥστε καὶ πλευρὰ ἡ HZ πλευρᾷ τῇ EZ ἐστὶν
 ἴση. καὶ ἐπεὶ [ἴση ἐστὶν ἡ EG τῇ GA], ἴσον ἐστὶ [καὶ] τὸ
 ἀπὸ τῆς EG τετράγωνον τῷ ἀπὸ τῆς GA τετραγώνῳ· τὰ ἄρα
 ἀπὸ τῶν EG , GA τετράγωνα διπλάσιά ἐστι τοῦ ἀπὸ τῆς GA

For let any straight-line AB be cut in half at (point) C , and
 let any straight-line BD be added to it straight-on. I say that
 the (sum of the) squares on AD and DB is double the (sum of
 the) squares on AC and CD .

For let CE be drawn from point C , at right-angles to AB
 [Prop. 1.11], and let it be made equal to each of AC and CB
 [Prop. 1.3], and let EA and EB be joined. And let EF be
 drawn through E , parallel to AD [Prop. 1.31], and let FD be
 drawn through D , parallel to CE [Prop. 1.31]. And since some
 straight-line EF falls across the parallel straight-lines EC and
 FD , the (internal angles) CEF and EFD are thus equal to two
 right-angles [Prop. 1.29]. Thus, FEB and EFD are less than
 two right-angles. And (straight-lines) produced from (internal
 angles whose sum is) less than two right-angles meet together
 [Post. 5]. Thus, being produced in the direction of B and D ,
 the (straight-lines) EB and FD will meet. Let them be pro-
 duced, and let them meet together at G , and let AG be joined.
 And since AC is equal to CE , angle EAC is also equal to (an-
 gle) AEC [Prop. 1.5]. And the (angle) at C (is) a right-angle.
 Thus, EAC and AEC [are] each half a right-angle [Prop. 1.32].
 So, for the same (reasons), CEB and EBC are also each half
 a right-angle. Thus, (angle) AEB is a right-angle. And since
 EBC is half a right-angle, DBG (is) thus also half a right-angle
 [Prop. 1.15]. And BDG is also a right-angle. For it is equal
 to DCE . For (they are) alternate (angles) [Prop. 1.29]. Thus,
 the remaining (angle) DGB is half a right-angle. Thus, DGB is
 equal to DBG . So side BD is also equal to side GD [Prop. 1.6].
 Again, since EGF is half a right-angle, and the (angle) at F
 (is) a right-angle, for it is equal to the opposite (angle) at C
 [Prop. 1.34], the remaining (angle) FEG is thus half a right-

τετραγώνων. τοῖς δὲ ἀπὸ τῶν EF , GA ἴσον ἐστὶ τὸ ἀπὸ τῆς EA · τὸ ἄρα ἀπὸ τῆς EA τετράγωνον διπλάσιόν ἐστι τοῦ ἀπὸ τῆς AF τετραγώνου. πάλιν, ἐπεὶ ἴση ἐστὶν ἡ ZH τῇ EZ , ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς ZH τῷ ἀπὸ τῆς ZE · τὰ ἄρα ἀπὸ τῶν HZ , ZE διπλάσιά ἐστι τοῦ ἀπὸ τῆς EZ . τοῖς δὲ ἀπὸ τῶν HZ , ZE ἴσον ἐστὶ τὸ ἀπὸ τῆς EH · τὸ ἄρα ἀπὸ τῆς EH διπλάσιόν ἐστι τοῦ ἀπὸ τῆς EZ . ἴση δὲ ἡ EZ τῇ $ΓΔ$ · τὸ ἄρα ἀπὸ τῆς EH τετράγωνον διπλάσιόν ἐστι τοῦ ἀπὸ τῆς $ΓΔ$. ἐδείκθη δὲ καὶ τὸ ἀπὸ τῆς EA διπλάσιον τοῦ ἀπὸ τῆς AF · τὰ ἄρα ἀπὸ τῶν AE , EH τετράγωνα διπλάσιά ἐστι τῶν ἀπὸ τῶν AF , $ΓΔ$ τετραγώνων. τοῖς δὲ ἀπὸ τῶν AE , EH τετραγώνοις ἴσον ἐστὶ τὸ ἀπὸ τῆς AH τετράγωνον· τὸ ἄρα ἀπὸ τῆς AH διπλάσιόν ἐστι τῶν ἀπὸ τῶν AF , $ΓΔ$. τῷ δὲ ἀπὸ τῆς AH ἴσα ἐστὶ τὰ ἀπὸ τῶν AD , $ΔΗ$ · τὰ ἄρα ἀπὸ τῶν AD , $ΔΗ$ [τετράγωνα] διπλάσιά ἐστι τῶν ἀπὸ τῶν AF , $ΓΔ$ [τετραγώνων]. ἴση δὲ ἡ $ΔΗ$ τῇ $ΔΒ$ · τὰ ἄρα ἀπὸ τῶν AD , $ΔΒ$ [τετράγωνα] διπλάσιά ἐστι τῶν ἀπὸ τῶν AF , $ΓΔ$ τετραγώνων.

Ἐάν ἄρα εὐθεῖα γραμμὴ τμηθῇ δίῃ, προστεθῇ δὲ τις αὐτῇ εὐθεῖα ἐπ' εὐθείας, τὸ ἀπὸ τῆς ὅλης σὺν τῇ προσκειμένῃ καὶ τὸ ἀπὸ τῆς προσκειμένης τὰ συναμφότερα τετράγωνα διπλάσιά ἐστι τοῦ τε ἀπὸ τῆς ἡμισείας καὶ τοῦ ἀπὸ τῆς συγκειμένης ἕκ τε τῆς ἡμισείας καὶ τῆς προσκειμένης ὥς ἀπὸ μιᾶς ἀναγραφέντος τετραγώνου· ὅπερ ἔδει δεῖξαι.

angle. Thus, angle EGF (is) equal to FEG . So the side GF is also equal to the side EF [Prop. 1.6]. And since [EC is equal to CA] the square on EC is [also] equal to the square on CA . Thus, the (sum of the) squares on EC and CA is double the square on CA . And the (square) on EA is equal to the (sum of the squares) on EC and CA [Prop. 1.47]. Thus, the square on EA is double the square on AC . Again, since FG is equal to EF , the (square) on FG is also equal to the (square) on FE . Thus, the (sum of the squares) on GF and FE is double the (square) on EF . And the (square) on EG is equal to the (sum of the squares) on GF and FE [Prop. 1.47]. Thus, the (square) on EG is double the (square) on EF . And EF (is) equal to CD [Prop. 1.34]. Thus, the square on EG is double the (square) on CD . But it was also shown that the (square) on EA (is) double the (square) on AC . Thus, the (sum of the) squares on AE and EG is double the (sum of the) squares on AC and CD . And the square on AG is equal to the (sum of the) squares on AE and EG [Prop. 1.47]. Thus, the (square) on AG is double the (sum of the squares) on AC and CD . And the (sum of the squares) on AD and DG is equal to the (square) on AG [Prop. 1.47]. Thus, the (sum of the) [squares] on AD and DG is double the (sum of the) [squares] on AC and CD . And DG (is) equal to DB . Thus, the (sum of the) [squares] on AD and DB is double the (sum of the) squares on AC and CD .

Thus, if a straight-line is cut in half, and any straight-line added to it straight-on, (then) the sum of the square on the whole (straight-line) with the (straight-line) having been added, and the (square) on the (straight-line) having been added, is double the (sum of the square) on half (the straight-line), and the square described on the sum of half (the straight-line) and (straight-line) having been added, as on one (complete straight-line). (Which is) the very thing it was required to show.

[†] This proposition is a geometric version of the algebraic identity: $(2a + b)^2 + b^2 = 2[a^2 + (a + b)^2]$.

11.

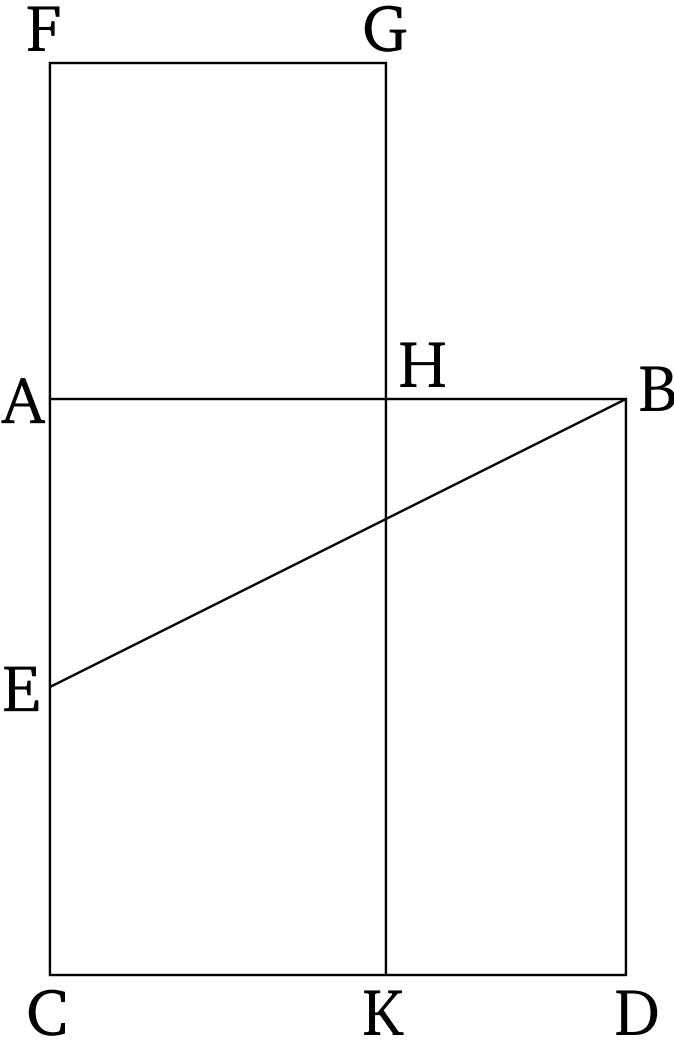
Τὴν δοθεῖσαν εὐθεῖαν τεμεῖν ὥστε τὸ ὑπὸ τῆς ὅλης καὶ τοῦ ἑτέρου τῶν τμημάτων περιεχόμενον ὀρθογώνιον ἴσον εἶναι τῷ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνῳ.

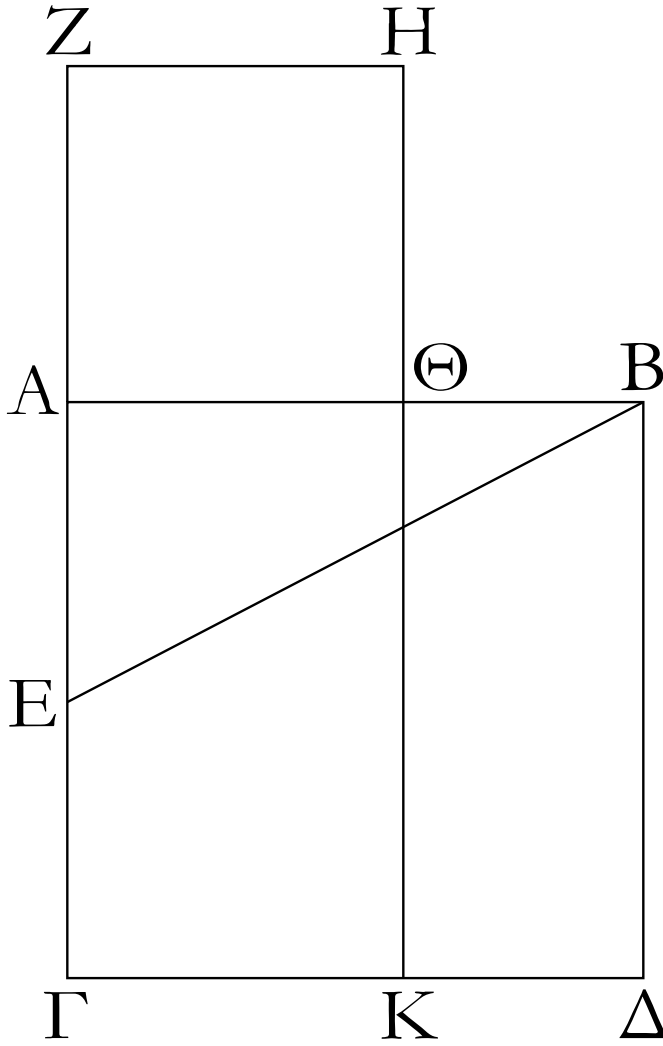
Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ AB · δεῖ δὴ τὴν AB τεμεῖν ὥστε τὸ ὑπὸ τῆς ὅλης καὶ τοῦ ἑτέρου τῶν τμημάτων περιεχόμενον ὀρθογώνιον ἴσον εἶναι τῷ ἀπὸ τοῦ λοιποῦ τμήματος τετραγώνῳ.

Proposition 11[†]

To cut a given straight-line such that the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), is equal to the square on the remaining piece.

Let AB be the given straight-line. So it is required to cut AB such that the rectangle contained by the whole (straight-line), and one of the pieces (of the straight-line), is equal to the square on the remaining piece.





Ἀναγεγράφθω γάρ ἀπὸ τῆς AB τετράγωνον τὸ $AB\Delta\Gamma$, καὶ τεμήσθω ἡ AG διῖθα κατὰ τὸ E σημείον, καὶ ἐπεξεύθθω ἡ BE , καὶ διήθθω ἡ GA ἐπὶ τὸ Z , καὶ κείσθω τῇ BE ἴση ἡ EZ , καὶ ἀναγεγράφθω ἀπὸ τῆς AZ τετράγωνον τὸ $Z\Theta$, καὶ διήθθω ἡ $H\Theta$ ἐπὶ τὸ K · λέγω, ὅτι ἡ AB τέμνεται κατὰ τὸ Θ , ὥστε τὸ ὑπὸ τῶν AB , $B\Theta$ περιεχόμενον ὀρθογώνιον ἴσον ποιεῖν τῷ ἀπὸ τῆς $A\Theta$ τετραγώνῳ.

Ἐπεὶ γάρ εὐθεία ἡ AG τέμνεται διῖθα κατὰ τὸ E , πρόσκει-
ται δὲ αὐτῇ ἡ ZA , τὸ ἄρα ὑπὸ τῶν ΓZ , ZA περιεχόμενον ὀρ-
θογώνιον μετὰ τοῦ ἀπὸ τῆς AE τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ
τῆς EZ τετραγώνῳ. ἴση δὲ ἡ EZ τῇ EB · τὸ ἄρα ὑπὸ τῶν
 ΓZ , ZA μετὰ τοῦ ἀπὸ τῆς AE ἴσον ἐστὶ τῷ ἀπὸ EB . ἀλλὰ
τῷ ἀπὸ EB ἴσα ἐστὶ τὰ ἀπὸ τῶν BA , AE · ὀρθὴ γάρ ἡ πρὸς
τῷ A γωνία· τὸ ἄρα ὑπὸ τῶν ΓZ , ZA μετὰ τοῦ ἀπὸ τῆς AE
ἴσον ἐστὶ τοῖς ἀπὸ τῶν BA , AE . κοινὸν ἀφηρήσθω τὸ ἀπὸ τῆς
 AE · λοιπὸν ἄρα τὸ ὑπὸ τῶν ΓZ , ZA περιεχόμενον ὀρθογώνιον
ἴσον ἐστὶ τῷ ἀπὸ τῆς AB τετραγώνῳ. καὶ ἐστὶ τὸ μὲν ὑπὸ
τῶν ΓZ , ZA τὸ ZK · ἴση γάρ ἡ AZ τῇ ZH · τὸ δὲ ἀπὸ τῆς

For let the square $ABDC$ be described on AB [Prop. 1.46], and let AC be cut in half at point E [Prop. 1.10], and let BE be joined. And let CA be drawn through to (point) F , and let EF be made equal to BE [Prop. 1.3]. And let the square FH be described on AF [Prop. 1.46], and let GH be drawn through to (point) K . I say that AB has been cut at H such as to make the rectangle contained by AB and BH equal to the square on AH .

For since the straight-line AC has been cut in half at E , and FA has been added to it, the rectangle contained by CF and FA , plus the square on AE , is thus equal to the square on EF [Prop. 2.6]. And EF (is) equal to EB . Thus, the (rectangle contained) by CF and FA , plus the (square) on AE , is equal to the (square) on EB . But, the (sum of the squares) on BA and AE is equal to the (square) on EB . For the angle at A (is) a right-angle [Prop. 1.47]. Thus, the (rectangle contained) by CF and FA , plus the (square) on AE , is equal to the (sum of the squares) on BA and AE . Let the square on AE have been subtracted from both. Thus, the remaining rectangle contained by CF and FA is equal to the square on AB . And FK is the

AB τὸ AD · τὸ ἄρα ZK ἴσον ἐστὶ τῷ AD . κοινὸν ἀρρηρήσθω τὸ AK · λοιπὸν ἄρα τὸ $Z\Theta$ τῷ $\Theta\Delta$ ἴσον ἐστίν. καὶ ἐστὶ τὸ μὲν $\Theta\Delta$ τὸ ὑπὸ τῶν AB , $B\Theta$ · ἴση γὰρ ἡ AB τῇ $B\Delta$ · τὸ δὲ $Z\Theta$ τὸ ἀπὸ τῆς $A\Theta$ · τὸ ἄρα ὑπὸ τῶν AB , $B\Theta$ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ ΘA τετραγώνῳ.

Ἡ ἄρα δοθεῖσα εὐθεῖα ἡ AB τέμνεται κατὰ τὸ Θ ὥστε τὸ ὑπὸ τῶν AB , $B\Theta$ περιεχόμενον ὀρθογώνιον ἴσον ποιεῖν τῷ ἀπὸ τῆς ΘA τετραγώνῳ· ὅπερ ἔδει ποιῆσαι.

(rectangle contained) by CF and FA . For AF (is) equal to FG . And AD (is) the (square) on AB . Thus, the (rectangle) FK is equal to the (square) AD . Let (rectangle) AK be subtracted from both. Thus, the remaining (square) FH is equal to the (rectangle) HD . And HD is the (rectangle contained) by AB and BH . For AB (is) equal to BD . And FH (is) the (square) on AH . Thus, the rectangle contained by AB and BH is equal to the square on HA .

Thus, the given straight-line AB has been cut at (point) H such as to make the rectangle contained by AB and BH equal to the square on HA . (Which is) the very thing it was required to do.

† This manner of cutting a straight-line—so that the ratio of the whole to the larger piece is equal to the ratio of the larger to the smaller piece—is sometimes called the “Golden Section”.

12.

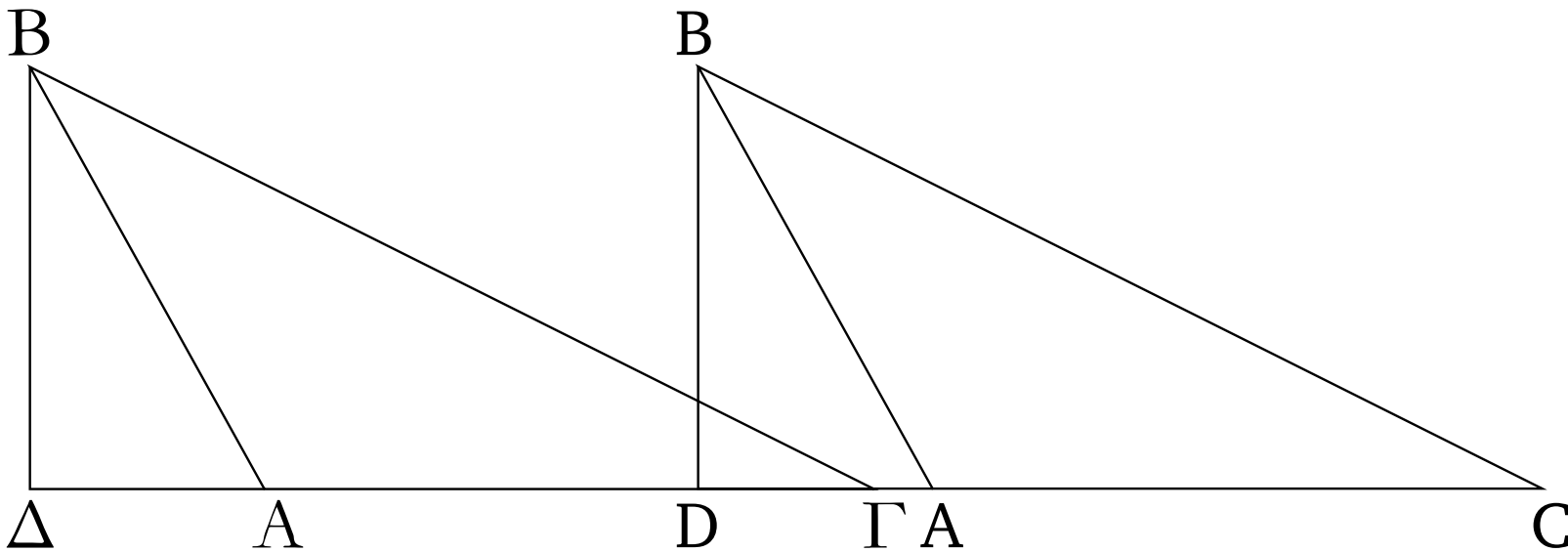
Proposition 12[†]

Ἐν τοῖς ἀμβλυγωνίοις τριγώνοις τὸ ἀπὸ τῆς τῆν ἀμβλείαν γωνίαν ὑποτείνουσας πλευρᾶς τετράγωνον μείζον ἐστὶ τῶν ἀπὸ τῶν τῆν ἀμβλείαν γωνίαν περιεθουσῶν πλευρῶν τετραγώνων τῷ περιεθομένῳ διξ ὑπὸ τε μιᾶς τῶν περὶ τῆν ἀμβλείαν γωνίαν, ἐφ' ἣν ἡ κάθετος πίπτει, καὶ τῆς ἀπολαμβανομένης ἐκτὸς ὑπὸ τῆς καθέτου προδὲς τῇ ἀμβλείᾳ γωνίᾳ.

Ἐστω ἀμβλυγώνιον τρίγωνον τὸ $AB\Gamma$ ἀμβλείαν ἔθον τῆν ὑπὸ BAG , καὶ ἤθθω ἀπὸ τοῦ B σημείου ἐπὶ τῆν GA ἐκβληθεῖσαν κάθετος ἡ BD . λέγω, ὅτι τὸ ἀπὸ τῆς $B\Gamma$ τετράγωνον μείζον ἐστὶ τῶν ἀπὸ τῶν BA , AG τετραγώνων τῷ διξ ὑπὸ τῶν GA , AD περιεθομένῳ ὀρθογώνῳ.

In obtuse-angled triangles, the square on the side subtending the obtuse angle is greater than the (sum of the) squares on the sides containing the obtuse angle by twice the (rectangle) contained by one of the sides around the obtuse angle, to which a perpendicular (straight-line) falls, and the (straight-line) cut off outside (the triangle) by the perpendicular (straight-line) towards the obtuse angle.

Let ABC be an obtuse-angled triangle, having the angle BAC obtuse. And let BD be drawn from point B , perpendicular to CA produced [Prop. 1.12]. I say that the square on BC is greater than the (sum of the) squares on BA and AC by twice the rectangle contained by CA and AD .



Ἐπεὶ γὰρ εὐθεῖα ἡ GA τέμνεται, ὡς ἔτυθεν, κατὰ τὸ A σημεῖον, τὸ ἄρα ἀπὸ τῆς $\Delta\Gamma$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν GA , AD τετραγώνοις καὶ τῷ διξ ὑπὸ τῶν GA , AD περιεθομένῳ ὀρθο-

For since the straight-line CD has been cut, at random, at point A , the (square) on DC is thus equal to the (sum of the) squares on CA and AD , and twice the rectangle contained by

γωνίᾳ. κοινὸν προσκείσθω τὸ ἀπὸ τῆς ΔB · τὰ ἄρα ἀπὸ τῶν $\Gamma\Delta$, ΔB ἴσα ἐστὶ τοῖς τε ἀπὸ τῶν ΓA , $A\Delta$, ΔB τετραγώνοις καὶ τῷ διξ ὑπὸ τῶν ΓA , $A\Delta$ [περιεθομένῳ ὀρθογωνίῳ]. ἀλλὰ τοῖς μὲν ἀπὸ τῶν $\Gamma\Delta$, ΔB ἴσον ἐστὶ τὸ ἀπὸ τῆς ΓB · ὀρθή γάρ ἡ πρὸς τῷ Δ γωνία· τοῖς δὲ ἀπὸ τῶν $A\Delta$, ΔB ἴσον τὸ ἀπὸ τῆς AB · τὸ ἄρα ἀπὸ τῆς ΓB τετράγωνον ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν ΓA , AB τετραγώνοις καὶ τῷ διξ ὑπὸ τῶν ΓA , $A\Delta$ περιεθομένῳ ὀρθογωνίῳ· ὥστε τὸ ἀπὸ τῆς ΓB τετράγωνον τῶν ἀπὸ τῶν ΓA , AB τετραγώνων μείζον ἐστὶ τῷ διξ ὑπὸ τῶν ΓA , $A\Delta$ περιεθομένῳ ὀρθογωνίῳ.

Ἐν ἄρα τοῖς ἀμβλυγωνίοις τριγώνοις τὸ ἀπὸ τῆς τήν ἀμβλείαν γωνίαν ὑποτείνουσας πλευρᾶς τετράγωνον μείζον ἐστὶ τῶν ἀπὸ τῶν τήν ἀμβλείαν γωνίαν περιεθουσῶν πλευρῶν τετραγώνων τῷ περιεθομένῳ διξ ὑπὸ τε μιᾶς τῶν περὶ τήν ἀμβλείαν γωνίαν, ἐφ' ἣν ἡ κάθετος πίπτει, καὶ τῆς ἀπολαμβανομένης ἐκτὸς ὑπὸ τῆς καθέτου πρὸς τῇ ἀμβλείᾳ γωνίᾳ· ὅπερ ἔδει δεῖχαι.

CA and AD [Prop. 2.4]. Let the (square) on DB be added to both. Thus, the (sum of the squares) on CD and DB is equal to the (sum of the) squares on CA , AD , and DB , and twice the [rectangle contained] by CA and AD . But, the (square) on CB is equal to the (sum of the squares) on CD and DB . For the angle at D (is) a right-angle [Prop. 1.47]. And the (square) on AB (is) equal to the (sum of the squares) on AD and DB [Prop. 1.47]. Thus, the square on CB is equal to the (sum of the) squares on CA and AB , and twice the rectangle contained by CA and AD . So the square on CB is greater than the (sum of the) squares on CA and AB by twice the rectangle contained by CA and AD .

Thus, in obtuse-angled triangles, the square on the side subtending the obtuse angle is greater than the (sum of the) squares on the sides containing the obtuse angle by twice the (rectangle) contained by one of the sides around the obtuse angle, to which a perpendicular (straight-line) falls, and the (straight-line) cut off outside (the triangle) by the perpendicular (straight-line) towards the obtuse angle. (Which is) the very thing it was required to show.

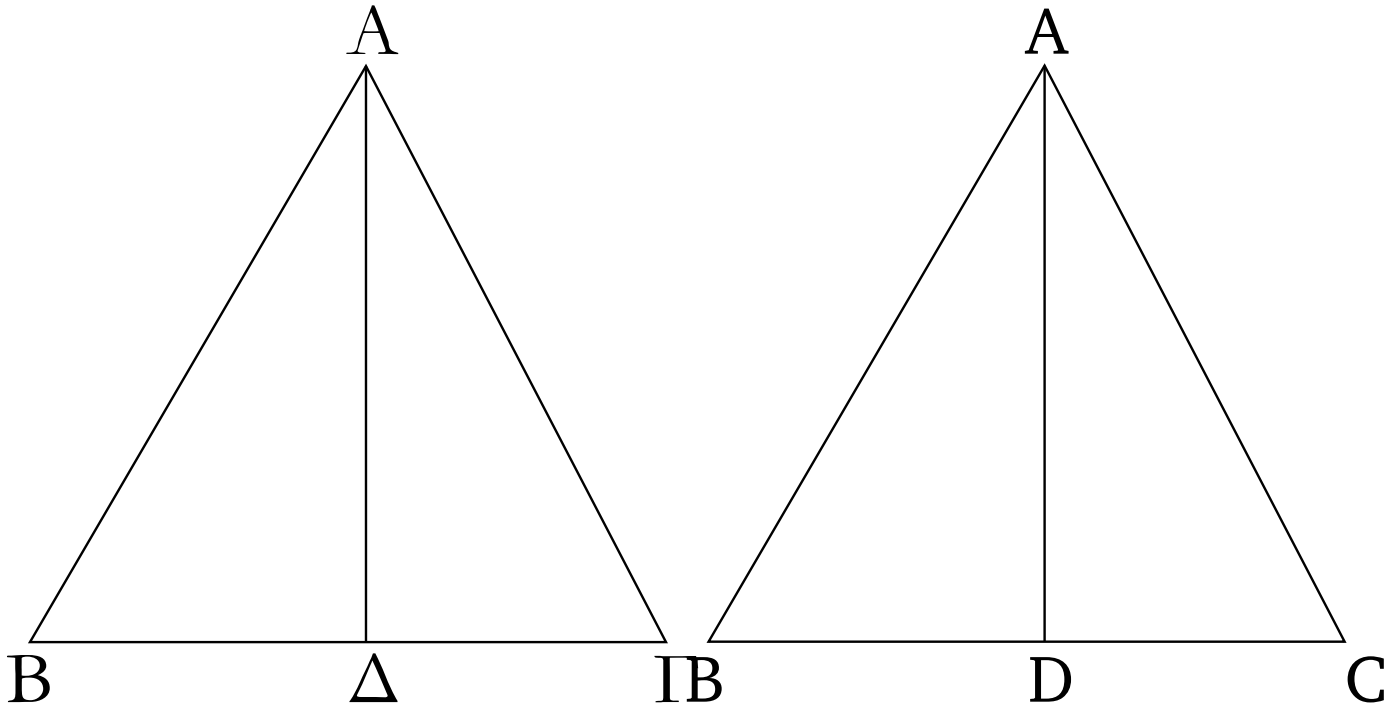
[†] This proposition is equivalent to the well-known cosine formula: $BC^2 = AB^2 + AC^2 - 2ABAC \cos BAC$, since $\cos BAC = -AD/AB$.

13.

Ἐν τοῖς ὀξυγωνίοις τριγώνοις τὸ ἀπὸ τῆς τήν ὀχείαν γωνίαν ὑποτείνουσας πλευρᾶς τετράγωνον ἑλαττόν ἐστὶ τῶν ἀπὸ τῶν τήν ὀχείαν γωνίαν περιεθουσῶν πλευρῶν τετραγώνων τῷ περιεθομένῳ διξ ὑπὸ τε μιᾶς τῶν περὶ τήν ὀχείαν γωνίαν, ἐφ' ἣν ἡ κάθετος πίπτει, καὶ τῆς ἀπολαμβανομένης ἐντὸς ὑπὸ τῆς καθέτου πρὸς τῇ ὀχείᾳ γωνίᾳ.

Proposition 13[†]

In acute-angled triangles, the square on the side subtending the acute angle is less than the (sum of the) squares on the sides containing the acute angle by twice the (rectangle) contained by one of the sides around the acute angle, to which a perpendicular (straight-line) falls, and the (straight-line) cut off inside (the triangle) by the perpendicular (straight-line) towards the acute angle.



Ἐστω ὀξυγώνιον τρίγωνον τὸ $AB\Gamma$ ὀξείαν ἔχον τὴν πρὸς τῷ B γωνίαν, καὶ ἦρθω ἀπὸ τοῦ A σημείου ἐπὶ τὴν $B\Gamma$ κάθετος ἡ AD . λέγω, ὅτι τὸ ἀπὸ τῆς AG τετραγώνον ἔλαττόν ἐστι τῶν ἀπὸ τῶν GB , BA τετραγώνων τῷ δις ὑπὸ τῶν GB , BD περιεθομένῳ ὀρθογωνίῳ.

Ἐπεὶ γὰρ εὐθεῖα ἡ GB τέμνεται, ὡς ἔτυθεν, κατὰ τὸ Δ , τὰ ἄρα ἀπὸ τῶν GB , BD τετράγωνα ἴσα ἐστί τῷ τε δις ὑπὸ τῶν GB , BD περιεθομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τῆς AD τετραγώνῳ. κοινὸν προσκείσθω τὸ ἀπὸ τῆς AD τετράγώνον· τὰ ἄρα ἀπὸ τῶν GB , BD , AD τετράγωνα ἴσα ἐστί τῷ τε δις ὑπὸ τῶν GB , BD περιεθομένῳ ὀρθογωνίῳ καὶ τοῖς ἀπὸ τῶν AD , AG τετραγώνοις. ἀλλὰ τοῖς μὲν ἀπὸ τῶν BD , AD ἴσον τὸ ἀπὸ τῆς AB · ὀρθή γάρ ἡ πρὸς τῷ Δ γωνία· τοῖς δὲ ἀπὸ τῶν AD , AG ἴσον τὸ ἀπὸ τῆς AG . τὰ ἄρα ἀπὸ τῶν GB , BA ἴσα ἐστί τῷ τε ἀπὸ τῆς AG καὶ τῷ δις ὑπὸ τῶν GB , BD · ὥστε μόνον τὸ ἀπὸ τῆς AG ἔλαττόν ἐστι τῶν ἀπὸ τῶν GB , BA τετραγώνων τῷ δις ὑπὸ τῶν GB , BD περιεθομένῳ ὀρθογωνίῳ.

Ἐν ἄρα τοῖς ὀξυγωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀξείαν γωνίαν ὑποτείνουσῃς πλευρᾶς τετραγώνον ἔλαττόν ἐστι τῶν ἀπὸ τῶν τὴν ὀξείαν γωνίαν περιεθουσῶν πλευρῶν τετραγώνων τῷ περιεθομένῳ δις ὑπὸ τε μιᾶς τῶν περὶ τὴν ὀξείαν γωνίαν, ἐφ' ἣν ἡ κάθετος πίπτει, καὶ τῆς ἀπολαμβανομένης ἐντὸς ὑπὸ τῆς καθέτου πρὸς τῇ ὀξείᾳ γωνίᾳ· ὅπερ ἔδει δεῖξαι.

Let ABC be an acute-angled triangle, having the angle at (point) B acute. And let AD be drawn from point A , perpendicular to BC [Prop. 1.12]. I say that the square on AC is less than the (sum of the) squares on CB and BA by twice the rectangle contained by CB and BD .

For since the straight-line CB has been cut, at random, at (point) D , the (sum of the) squares on CB and BD is thus equal to twice the rectangle contained by CB and BD , and the square on DC [Prop. 2.7]. Let the square on DA be added to both. Thus, the (sum of the) squares on CB , BD , and DA is equal to twice the rectangle contained by CB and BD , and the (sum of the) squares on AD and DC . But, the (square) on AB (is) equal to the (sum of the squares) on BD and DA . For the angle at (point) D is a right-angle [Prop. 1.47]. And the (square) on AC (is) equal to the (sum of the squares) on AD and DC [Prop. 1.47]. Thus, the (sum of the squares) on CB and BA is equal to the (square) on AC , and twice the (rectangle contained) by CB and BD . So the (square) on AC alone is less than the (sum of the) squares on CB and BA by twice the rectangle contained by CB and BD .

Thus, in acute-angled triangles, the square on the side subtending the acute angle is less than the (sum of the) squares on the sides containing the acute angle by twice the (rectangle) contained by one of the sides around the acute angle, to which a perpendicular (straight-line) falls, and the (straight-line) cut off inside (the triangle) by the perpendicular (straight-line) towards the acute angle. (Which is) the very thing it was required to show.

εὐθυγράμμω. καὶ τὸ A ἄρα εὐθύγραμμον ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Theta$ ἀναγραφισμένῳ τετραγώνῳ.

Τῷ ἄρα δοθέντι εὐθυγράμμῳ τῷ A ἴσον τετράγωνον συνέσταται τὸ ἀπὸ τῆς $E\Theta$ ἀναγραφισόμενον· ὅπερ ἔδει ποιῆσαι.

rectangle contained by BE and EF is equal to the square on EH . But, BD is the (rectangle contained) by BE and EF . For EF (is) equal to ED . Thus, the parallelogram BD is equal to the square on HE . And BD (is) equal to the rectilinear figure A . Thus, the rectilinear figure A is also equal to the square (which) can be described on EH .

Thus, a square—(namely), that (which) can be described on EH —has been constructed, equal to the given rectilinear figure A . (Which is) the very thing it was required to do.

ELEMENTS BOOK 3

*Fundamentals of Plane Geometry Involving
Circles*

Ὅροι.

1. Ἰσοὶ κύκλοι εἰσὶν, ὧν αἱ διάμετροι ἴσαι εἰσὶν, ἢ ὧν αἱ ἐκ τῶν κέντρων ἴσαι εἰσὶν.
2. Εὐθεῖα κύκλον ἐφάπτεσθαι λέγεται, ἥτις ἀπτομένη τοῦ κύκλου καὶ ἐκβαλλομένη οὐ τέμνει τὸν κύκλον.
3. Κύκλοι ἐφάπτεσθαι ἀλλήλων λέγονται ὅττινες ἀπτόμενοι ἀλλήλων οὐ τέμνουσιν ἀλλήλους.
4. Ἐν κύκλῳ ἴσον ἀπέθειν ἀπὸ τοῦ κέντρου εὐθεῖαι λέγονται, ὅταν αἱ ἀπὸ τοῦ κέντρου ἐπ' αὐτάς κάθεται ἀγόμεναι ἴσαι ᾧσιν.
5. Μείζον δὲ ἀπέθειν λέγεται, ἐφ' ἣν ἡ μείζων κάθετος πίπτει.
6. Τμήμα κύκλον ἐστὶ τὸ περιεχόμενον σθῆμα ὑπὸ τε εὐθείας καὶ κύκλου περιφερείας.
7. Τμήματοξ δὲ γωνία ἐστὶν ἡ περιεθρομένη ὑπὸ τε εὐθείας καὶ κύκλου περιφερείας.
8. Ἐν τμήματι δὲ γωνία ἐστίν, ὅταν ἐπὶ τῆς περιφερείας τοῦ τμήματος ληφθῇ τι σημεῖον καὶ ἀπ' αὐτοῦ ἐπὶ τὰ πέρατα τῆς εὐθείας, ἢ ἐστὶ βάσει τοῦ τμήματος, ἐπιξενθθῶσιν εὐθεῖαι, ἢ περιεθρομένη γωνία ὑπὸ τῶν ἐπιξενθθειςῶν εὐθειῶν.
9. Ὅταν δὲ αἱ περιέθουσαι τὴν γωνίαν εὐθεῖαι ἀπολαμβάνωσι τινα περιφέρειαν, ἐπ' ἐκείνης λέγεται βεβηκέναι ἡ γωνία.
10. Τομεὺς δὲ κύκλον ἐστίν, ὅταν πρὸς τῷ κέντρῳ τοῦ κύκλου συσταθῇ γωνία, τὸ περιεχόμενον σθῆμα ὑπὸ τε τῶν τὴν γωνίαν περιέθουσῶν εὐθειῶν καὶ τῆς ἀπολαμβανομένης ὑπ' αὐτῶν περιφερείας.
11. Ὅμοια τμήματα κύκλων ἐστὶ τὰ δεθόμενα γωνίας ἴσας, ἢ ἐν οἷς αἱ γωναὶ ἴσαι ἀλλήλαις εἰσὶν.

1.

Τοῦ δοθέντος κύκλου τὸ κέντρον εὐρεῖν.

Ἐστω ὁ δοθεὶς κύκλος ὁ $ABΓ$. δεῖ δὴ τοῦ $ABΓ$ κύκλου τὸ κέντρον εὐρεῖν.

Διήθθω τις εἷς αὐτόν, ὡς ἔτυθεν, εὐθεῖα ἡ AB , καὶ τετμήσθω δίῃ κατὰ τὸ Δ σημεῖον, καὶ ἀπὸ τοῦ Δ τῇ AB πρὸς ὀρθὰς ἡθθω ἡ $\Delta Γ$ καὶ διήθθω ἐπὶ τὸ E , καὶ τετμήσθω ἡ $ΓΕ$ δίῃ κατὰ τὸ Z . λέγω, ὅτι τὸ Z κέντρον ἐστὶ τοῦ $ABΓ$ [κύκλου].

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἔστω τὸ H , καὶ ἐπεξενθθωσαν αἱ HA , HD , HB . καὶ ἐπεὶ ἴση ἐστὶν ἡ AD τῇ ΔB , κοινὴ δὲ ἡ ΔH , δύο δὲ αἱ AD , ΔH δύο ταῖς HD , ΔB ἴσαι εἰσὶν ἑκατέρα ἑκατέρῃ· καὶ βάσει ἡ HA βάσει τῇ HB ἐστὶν ἴση· ἐκ κέντρου γάρ· γωνία ἄρα ἡ ὑπὸ $A\Delta H$ γωνία τῇ ὑπὸ $H\Delta B$ ἴση ἐστίν. ὅταν δὲ εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα τὰς ἐφεχθῇς γωνίας ἴσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν ἴσων γωνιῶν

Definitions

1. Equal circles are (circles) whose diameters are equal, or whose (distances) from the centers (to the circumferences) are equal (i.e., whose radii are equal).
2. A straight-line said to touch a circle is any (straight-line) which, meeting the circle and being produced, does not cut the circle.
3. Circles said to touch one another are any (circles) which, meeting one another, do not cut one another.
4. In a circle, straight-lines are said to be equally far from the center when the perpendiculars drawn to them from the center are equal.
5. And (that straight-line) is said to be further (from the center) on which the greater perpendicular falls (from the center).
6. A segment of a circle is the figure contained by a straight-line and a circumference of a circle.
7. And the angle of a segment is that contained by a straight-line and a circumference of a circle.
8. And the angle in a segment is the angle contained by the joined straight-lines, when any point is taken on the circumference of a segment, and straight-lines are joined from it to the ends of the straight-line which is the base of the segment.
9. And when the straight-lines containing an angle cut off some circumference, the angle is said to stand upon that (circumference).
10. And a sector of a circle is the figure contained by the straight-lines surrounding an angle, and the circumference cut off by them, when the angle is constructed at the center of a circle.
11. Similar segments of circles are those accepting equal angles, or in which the angles are equal to one another.

Proposition 1

To find the center of a given circle.

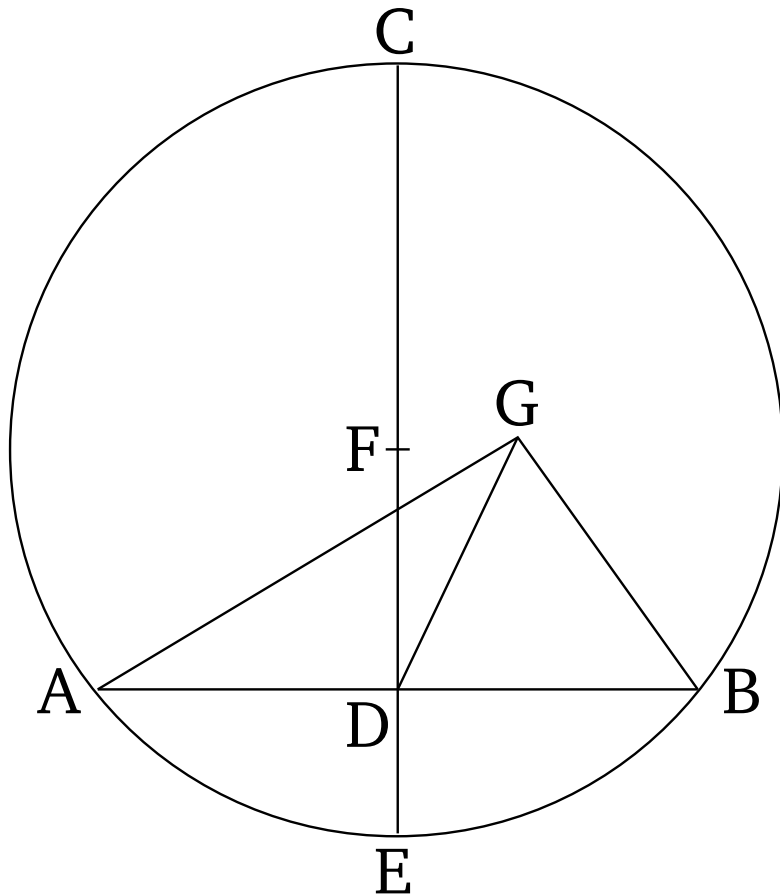
Let ABC be the given circle. So it is required to find the center of circle ABC .

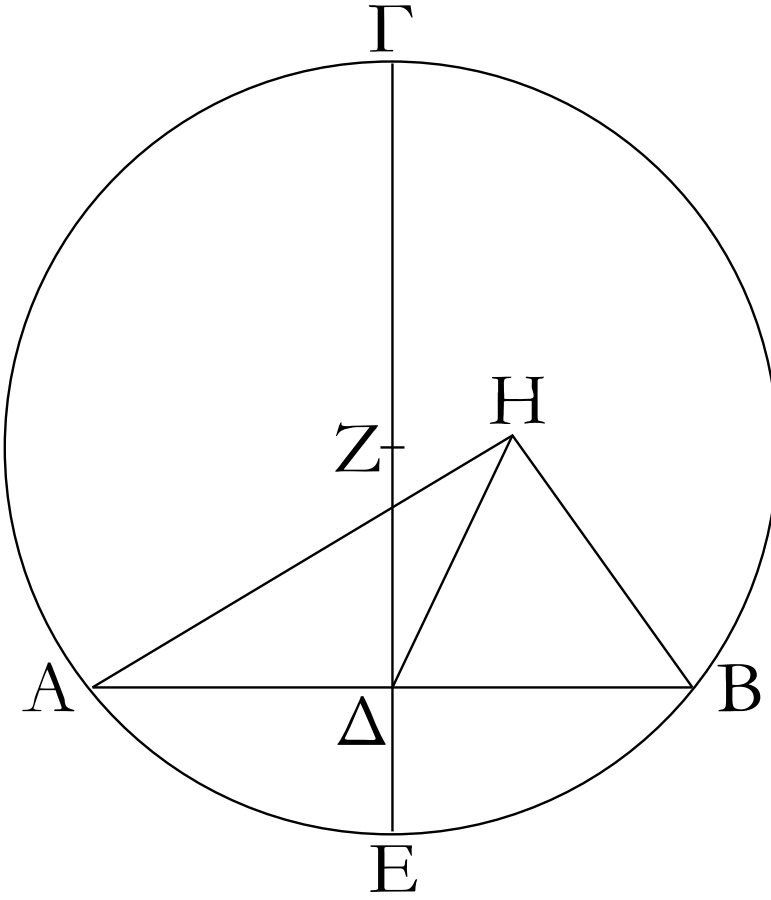
Let some straight-line AB be drawn through (ABC), at random, and let (AB) be cut in half at point D [Prop. 1.9]. And let DC be drawn from D , at right-angles to AB [Prop. 1.11]. And let (CD) be drawn through to E . And let CE be cut in half at F [Prop. 1.9]. I say that (point) F is the center of the [circle] ABC .

For (if) not (then), if possible, let G (be the center of the circle), and let GA , GD , and GB be joined. And since AD is equal to DB , and DG (is) common, the two (straight-lines) AD , DG are equal to the two (straight-lines) BD , DG ,[†] respectively. And the base GA is equal to the base GB . For (they are both) radii. Thus, angle ADG is equal to angle GDB

ἔστιν ὀρθή ᾗ ἄρα ἔστιν ἡ ὑπὸ $H\Delta B$. ἔστι δὲ καὶ ἡ ὑπὸ $Z\Delta B$ ὀρθή· ἴση ᾗ ἄρα ἡ ὑπὸ $Z\Delta B$ τῇ ὑπὸ $H\Delta B$, ἡ μείζων τῇ ἐλάττω· ὅπερ ἔστιν ἀδύνατον. οὐκ ᾗ ἄρα τὸ H κέντρον ἔστι τοῦ $AB\Gamma$ κύκλου. ὁμοίως δὲ δείχμεν, ὅτι οὐδ' ἄλλο τι πλὴν τοῦ Z .

[Prop. 1.8]. And when a straight-line stood upon (another) straight-line make adjacent angles (which are) equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, GDB is a right-angle. And FDB is also a right-angle. Thus, FDB (is) equal to GDB , the greater to the lesser. The very thing is impossible. Thus, (point) G is not the center of the circle ABC . So, similarly, we can show that neither is any other (point) except F .





Τὸ Z ἄρα σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ [κύκλου].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν ἐν κύκλῳ εὐθεῖα τις εὐθεῖαν τινα διῇ καὶ πρὸς ὀρθὰς τέμνῃ, ἐπὶ τῇς τεμονούσης ἐστὶ τὸ κέντρον τοῦ κύκλου. — ὅπερ ἔδει ποιῆσαι.

[†] The Greek text has “ GD, DB ”, which is obviously a mistake.

2.

Ἐάν κύκλον ἐπὶ τῇς περιφερείας ληφθῇ δύο τυθόντα σημεία, ἢ ἐπὶ τὰ σημεία ἐπιξεννυμένη εὐθεῖα ἐντὸς πεσεῖται τοῦ κύκλου.

Ἐστω κύκλος ὁ $AB\Gamma$, καὶ ἐπὶ τῇς περιφερείας αὐτοῦ εἰλήφθω δύο τυθόντα σημεία τὰ A, B · λέγω, ὅτι ἡ ἀπὸ τοῦ A ἐπὶ τὸ B ἐπιξεννυμένη εὐθεῖα ἐντὸς πεσεῖται τοῦ κύκλου.

μὴ γάρ, ἀλλ’ εἰ δυνατόν, πιπτέτω ἐκτὸς ὥς ἡ AEB , καὶ εἰλήφθω τὸ κέντρον τοῦ $AB\Gamma$ κύκλου, καὶ ἔστω τὸ Δ , καὶ ἐπεξεύθθωσαν αἱ $\Delta A, \Delta B$, καὶ διήθθω ἡ ΔZE .

Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΔA τῇ ΔB , ἴση ἄρα καὶ γωνία ἡ ὑπὸ ΔAE τῇ ὑπὸ ΔBE · καὶ ἐπεὶ τριγώνον τοῦ ΔAE μία πλευρὰ προσεκβέλλεται ἡ AEB , μείζων ἄρα ἡ ὑπὸ ΔEB γωνία

Thus, point F is the center of the [circle] ABC .

Corollary

So, from this, (it is) manifest that if any straight-line in a circle cuts any (other) straight-line in half, and at right-angles, (then) the center of the circle is on the former (straight-line). — (Which is) the very thing it was required to do.

Proposition 2

If two points are taken at random on the circumference of a circle, (then) the straight-line joining the points will fall inside the circle.

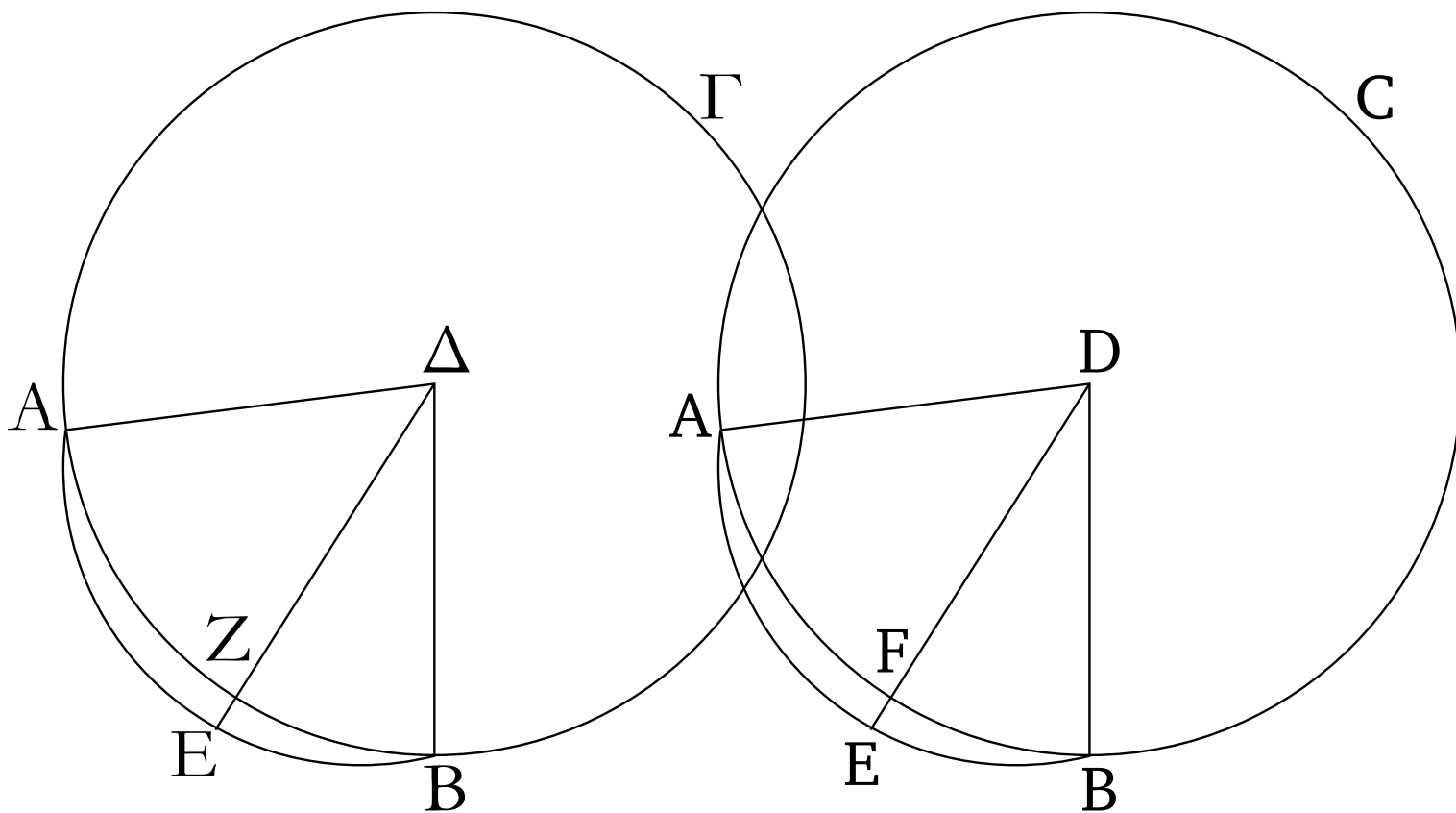
Let ABC be a circle, and let two points A and B be taken at random on its circumference. I say that the straight-line joining A to B will fall inside the circle.

For (if) not (then), if possible, let it fall outside (the circle), like AEB (in the figure). And let the center of the circle ABC be found [Prop. 3.1], and let it be (at point) D . And let DA and DB be joined, and let DFE be drawn through.

Therefore, since DA is equal to DB , the angle DAE (is) thus also equal to DBE [Prop. 1.5]. And since in triangle DAE

νία τῆς ὑπὸ ΔAE . ἴση δὲ ἡ ὑπὸ ΔAE τῇ ὑπὸ ΔBE · μείζων
 ἄρα ἡ ὑπὸ ΔEB τῆς ὑπὸ ΔBE . ὑπὸ δὲ τὴν μείζονα γωνίαν
 ἡ μείζων πλευρὰ ὑποτείνει· μείζων ἄρα ἡ ΔB τῆς ΔE . ἴση
 δὲ ἡ ΔB τῇ ΔZ . μείζων ἄρα ἡ ΔZ τῆς ΔE ἡ ἐλάττωον τῆς
 μείζονος· ὅπερ εἶναι ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ A ἐπὶ τὸ
 B ἐπιξεννυμένη εὐθεῖα ἐκτὸς πεσεῖται τοῦ κύκλου. ὁμοίως
 δὴ δείξομεν, ὅτι οὐδὲ ἐπ' αὐτῆς τῆς περιφερείας· ἐντὸς ἄρα.

the one side, AEB , has been produced, angle DEB (is) thus greater than DAE [Prop. 1.16]. And DAE (is) equal to DBE [Prop. 1.5]. Thus, DEB (is) greater than DBE . And the greater angle is subtended by the greater side [Prop. 1.19]. Thus, DB (is) greater than DE . And DB (is) equal to DF . Thus, DF (is) greater than DE , the lesser than the greater. The very thing is impossible. Thus, the straight-line joining A to B will not fall outside the circle. So, similarly, we can show that neither (will it fall) on the circumference itself. Thus, (it will fall) inside (the circle).



Ἐὰν ἄρα κύκλον ἐπὶ τῆς περιφερείας ληφθῇ δύο τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐντὸς πεσεῖται τοῦ κύκλου· ὅπερ ἔδει δεῖξαι.

Thus, if two points are taken at random on the circumference of a circle, (then) the straight-line joining the points will fall inside the circle. (Which is) the very thing it was required to show.

3.

Ἐάν ἐν κύκλῳ εὐθεΐά τις διὰ τοῦ κέντρου εὐθεΐαν τινα μὴ διὰ τοῦ κέντρου διῇα τέμνη, καὶ πρὸς ὁρθὰς αὐτὴν τέμνῃ· καὶ ἕαν πρὸς ὁρθὰς αὐτὴν τέμνη, καὶ διῇα αὐτὴν τέμνῃ.

Ἔστω κύκλος ὁ $ΑΒΓ$, καὶ ἐν αὐτῷ εὐθείᾳ τις διὰ τοῦ κέντρου ἡ $ΓΔ$ εὐθείαν τινα μὴ διὰ τοῦ κέντρου τὴν $ΑΒ$ διχα τεμνέτω κατὰ τὸ $Ζ$ σημεῖον· λέγω, ὅτι καὶ πρὸς ὀρθὰς αὐτὴν τέμνει.

Εἰλήφθω γὰρ τὸ κέντρον τοῦ $AB\Gamma$ κύκλου, καὶ ἔστω τὸ

Proposition 3

In a circle, if any straight-line through the center cuts in half any straight-line not through the center, (then) it also cuts it at right-angles. And (conversely) if it cuts it at right-angles, (then) it also cuts it in half.

Let ABC be a circle, and, within it, let some straight-line through the center, CD , cut in half some straight-line not through the center, AB , at the point F . I say that (CD) also cuts (AB) at right-angles.

E , καὶ ἐπεξεύθωσαν αἱ EA , EB .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ AZ τῇ ZB , κοινὴ δὲ ἡ ZE , δύο δυνὸν ἴσαι [εἰσὶν]· καὶ βάσεις ἡ EA βάσει τῇ EB ἴση· γωνία ἄρα ἡ ὑπὸ AZE γωνία τῇ ὑπὸ BZE ἴση ἐστίν. ὅταν δὲ εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα τὰς ἐφεχθῆς γωνίας ἴσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρω τῶν ἴσων γωνιῶν ἐστίν· ἑκατέρω ἄρα τῶν ὑπὸ AZE , BZE ὀρθή ἐστίν. ἡ $\Gamma\Delta$ ἄρα διὰ τοῦ κέντρου οὔσα τὴν AB μὴ διὰ τοῦ κέντρου οὔσαν δίχα τέμνουσα καὶ πρὸς ὀρθὰς τέμνει.

Ἀλλὰ δὴ ἡ $\Gamma\Delta$ τὴν AB πρὸς ὀρθὰς τεμνέτω· λέγω, ὅτι καὶ δίχα αὐτὴν τέμνει, τοντέστιν, ὅτι ἴση ἐστὶν ἡ AZ τῇ ZB .

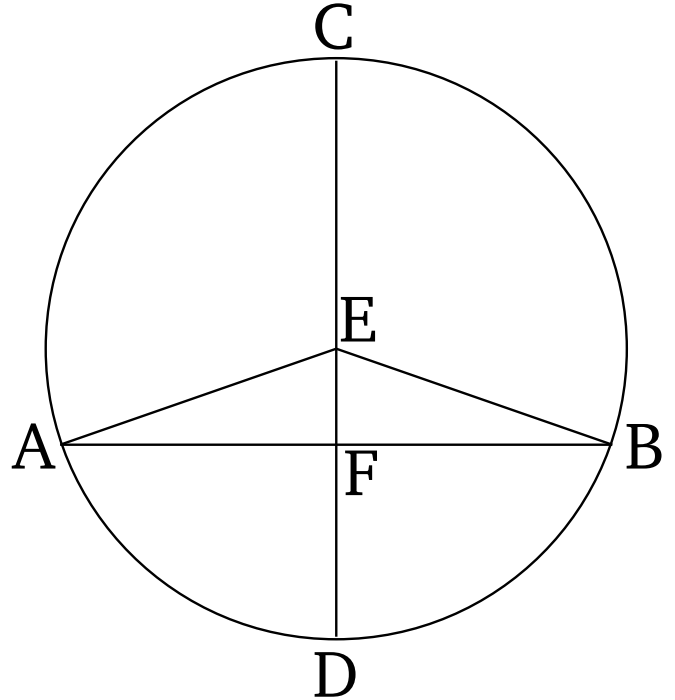
Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἴση ἐστὶν ἡ EA τῇ EB , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ EAZ τῇ ὑπὸ EBZ . ἐστὶ δὲ καὶ ὀρθή ἡ ὑπὸ AZE ὀρθὴ τῇ ὑπὸ BZE ἴση· δύο ἄρα τρίγωνά ἐστι EAZ , EZB τὰς δύο γωνίας δύο γωνίαις ἴσας ἔχοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην κοινήν αὐτῶν τὴν EZ ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν· καὶ τὰς λοιπὰς ἄρα πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσας ἔχει· ἴση ἄρα ἡ AZ τῇ ZB .

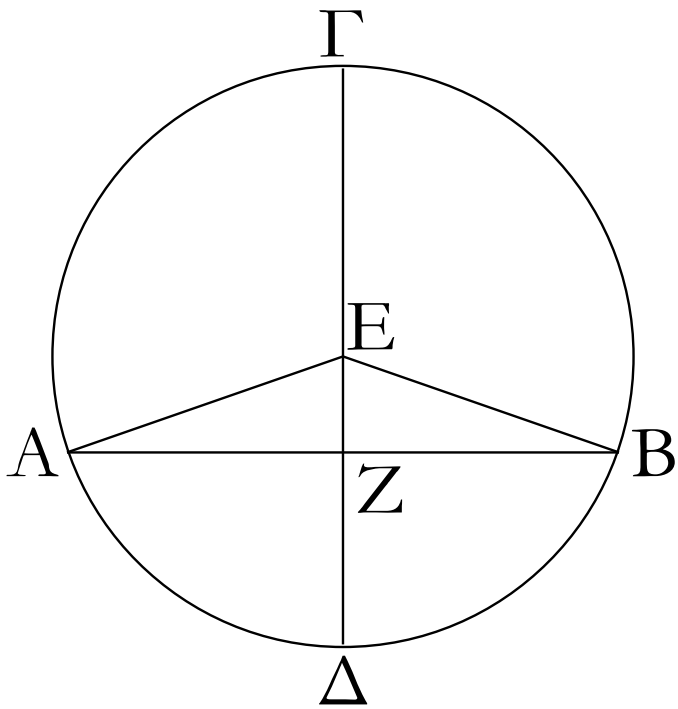
For let the center of the circle ABC be found [Prop. 3.1], and let it be (at point) E , and let EA and EB be joined.

And since AF is equal to FB , and FE (is) common, two (sides of triangle AFE) [are] equal to two (sides of triangle BFE). And the base EA (is) equal to the base EB . Thus, angle AFE is equal to angle BFE [Prop. 1.8]. And when a straight-line stood upon (another) straight-line makes adjacent angles (which are) equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, AFE and BFE are each right-angles. Thus, the (straight-line) CD , which is through the center and cuts in half the (straight-line) AB , which is not through the center, also cuts (AB) at right-angles.

And so let CD cut AB at right-angles. I say that it also cuts (AB) in half. That is to say, that AF is equal to FB .

For, with the same construction, since EA is equal to EB , angle EAF is also equal to EBF [Prop. 1.5]. And the right-angle AFE is also equal to the right-angle BFE . Thus, EAF and EBF are two triangles having two angles equal to two angles, and one side equal to one side—(namely), their common (side) EF , subtending one of the equal angles. Thus, they will also have the remaining sides equal to the (corresponding) remaining sides [Prop. 1.26]. Thus, AF (is) equal to FB .





Ἐάν ἄρα ἐν κύκλῳ εὐθεῖά τις διὰ τοῦ κέντρου εὐθεϊάν
τινα μὴ διὰ τοῦ κέντρου δίχα τέμνη, καὶ πρὸς ὀρθὰς αὐτὴν
τέμνει· καὶ ἔαν πρὸς ὀρθὰς αὐτὴν τέμνη, καὶ δίχα αὐτὴν τέμνει·
ὅπερ ἔδει δεῖχαι.

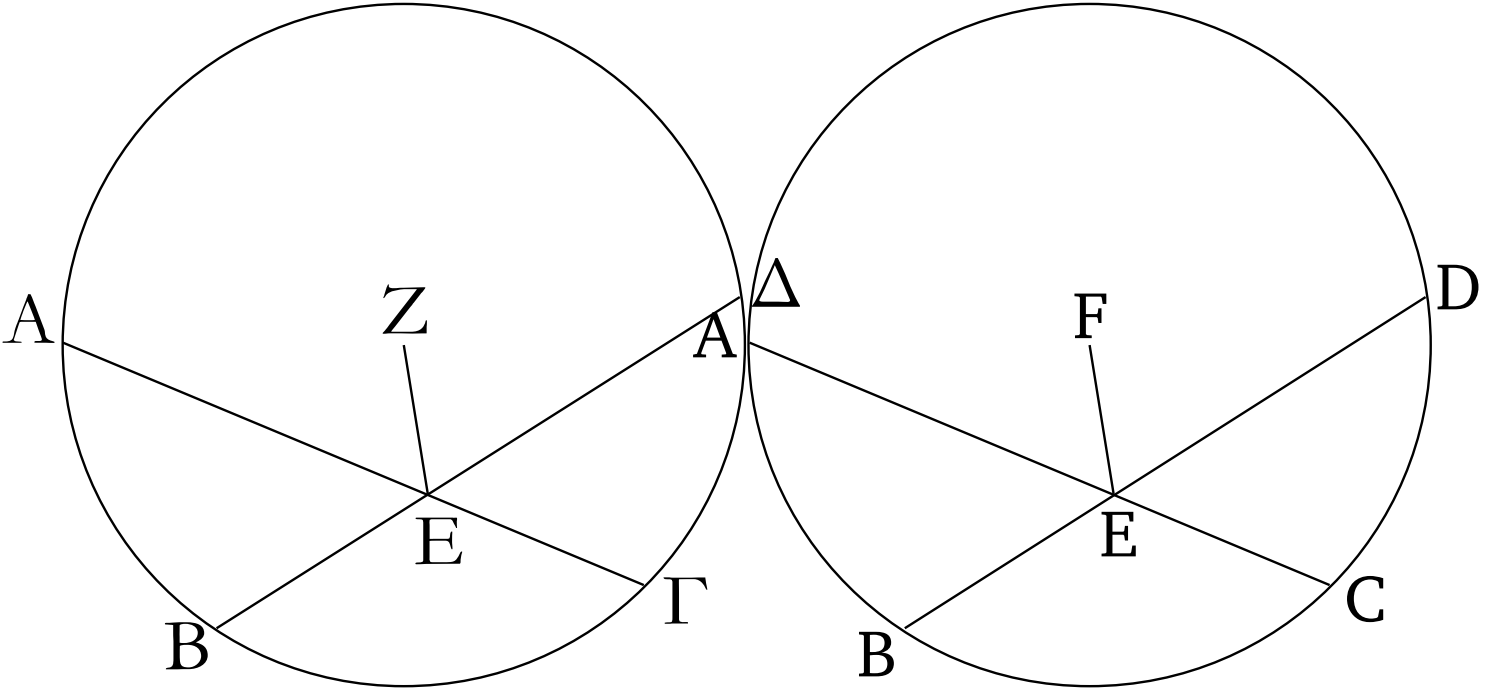
Thus, in a circle, if any straight-line through the center cuts
in half any straight-line not through the center, (then) it also
cuts it at right-angles. And (conversely) if it cuts it at right-
angles, (then) it also cuts it in half. (Which is) the very thing
it was required to show.

4.

Proposition 4

Ἐάν ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας μὴ διὰ τοῦ
κέντρου οὔσαι, οὐ τέμνουσιν ἀλλήλας δίχα.

In a circle, if two straight-lines, which are not through the
center, cut one another, (then) they do not cut one another in
half.



Ἐστω κύκλος ὁ $AB\Gamma\Delta$, καὶ ἐν αὐτῷ δύο εὐθεῖαι αἱ AG , $B\Delta$ τεμνέτωσαν ἀλλήλας κατὰ τὸ E μὴ διὰ τοῦ κέντρου οὔσαι· λέγω, ὅτι οὐ τέμνουσιν ἀλλήλας δίθλα.

Εἰ γὰρ δυνατόν, τεμνέτωσαν ἀλλήλας δίθλα ὥστε ἴσην εἶναι τὴν μὲν AE τῇ EG , τὴν δὲ BE τῇ ED · καὶ εἰληφθῶ τὸ κέντρον τοῦ $AB\Gamma\Delta$ κύκλου, καὶ ἔστω τὸ Z , καὶ ἐπεζεύθω ἡ ZE .

Ἐπεὶ οὖν εὐθεῖα τις διὰ τοῦ κέντρου ἡ ZE εὐθεϊάν τινα μὴ διὰ τοῦ κέντρου τὴν AG δίθλα τέμνει, καὶ πρὸς ὀρθὰς αὐτὴν τέμνει· ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ ZEA · πάλιν, ἐπεὶ εὐθεῖα τις ἡ ZE εὐθεϊάν τινα τὴν $B\Delta$ δίθλα τέμνει, καὶ πρὸς ὀρθὰς αὐτὴν τέμνει· ὀρθὴ ἄρα ἡ ὑπὸ ZEB . ἐδείκθη δὲ καὶ ἡ ὑπὸ ZEA ὀρθή· ἴση ἄρα ἡ ὑπὸ ZEA τῇ ὑπὸ ZEB ἢ ἐλάττων τῇ μείζον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα αἱ AG , $B\Delta$ τέμνουσιν ἀλλήλας δίθλα.

Ἐάν ἄρα ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας μὴ διὰ τοῦ κέντρου οὔσαι, οὐ τέμνουσιν ἀλλήλας δίθλα· ὅπερ ἔδει δεῖξαι.

Let $ABCD$ be a circle, and within it, let two straight-lines, AC and BD , which are not through the center, cut one another at (point) E . I say that they do not cut one another in half.

For, if possible, let them cut one another in half, such that AE is equal to EC , and BE to ED . And let the center of the circle $ABCD$ be found [Prop. 3.1], and let it be (at point) F , and let FE be joined.

Therefore, since some straight-line through the center, FE , cuts in half some straight-line not through the center, AC , it also cuts it at right-angles [Prop. 3.3]. Thus, FEA is a right-angle. Again, since some straight-line FE cuts in half some straight-line BD , it also cuts it at right-angles [Prop. 3.3]. Thus, FEB (is) a right-angle. But FEA was also shown (to be) a right-angle. Thus, FEA (is) equal to FEB , the lesser to the greater. The very thing is impossible. Thus, AC and BD do not cut one another in half.

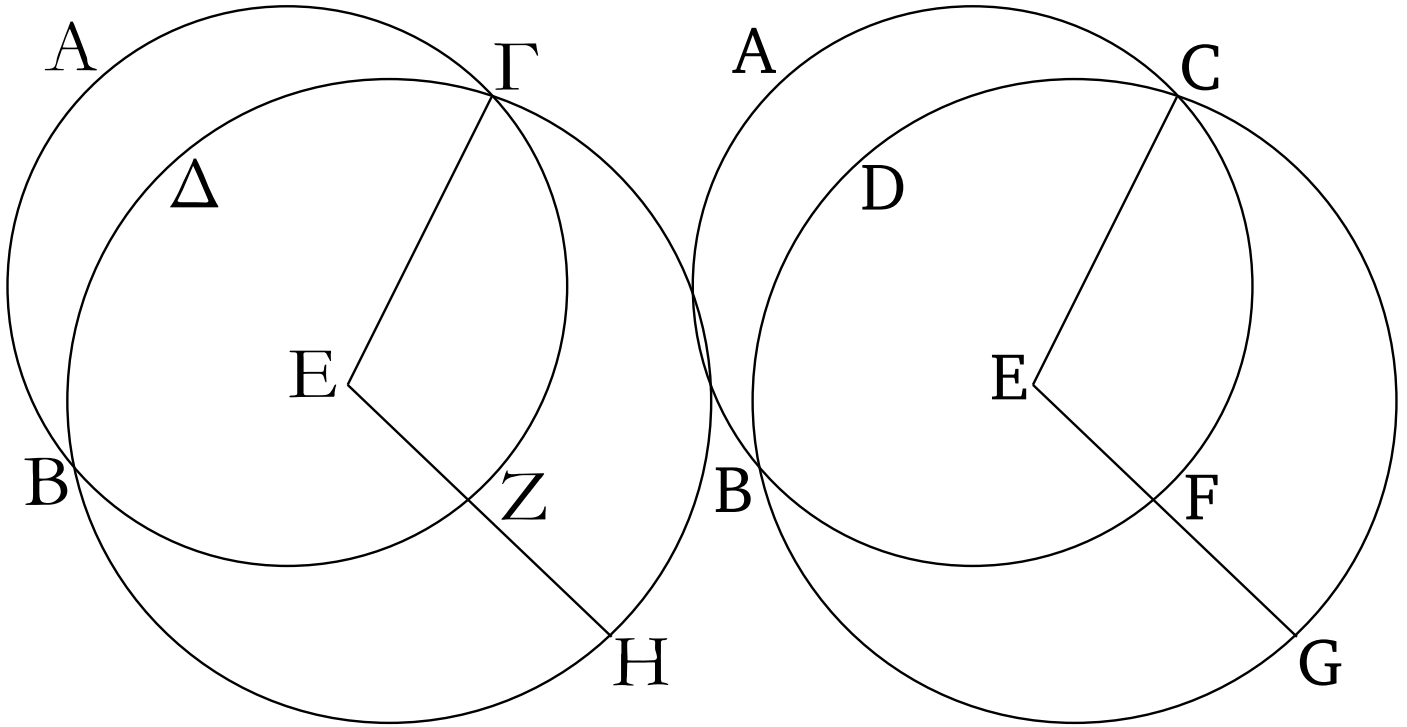
Thus, in a circle, if two straight-lines, which are not through the center, cut one another, (then) they do not cut one another in half. (Which is) the very thing it was required to show.

5.

Proposition 5

Ἐάν δύο κύκλοι τέμνωσιν ἀλλήλων, οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον.

If two circles cut one another, (then) they will not have the same center.



Δύο γὰρ κύκλοι οἱ $AB\Gamma$, $\Gamma\Delta\text{H}$ τεμνέτωσαν ἀλλήλων κατὰ τὰ B , Γ σημεία. λέγω, ὅτι οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον.

Εἰ γὰρ δυνατόν, ἔστω τὸ E , καὶ ἐπεζεύθω ἡ EG , καὶ διήθω ἡ EZH , ὥς ἔτυθεν. καὶ ἐπεὶ τὸ E σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου, ἴση ἐστὶν ἡ EG τῇ EZ . πάλιν, ἐπεὶ τὸ E σημεῖον κέντρον ἐστὶ τοῦ $\Gamma\Delta\text{H}$ κύκλου, ἴση ἐστὶν ἡ EG τῇ EH . ἐδείχθη δὲ ἡ EG καὶ τῇ EZ ἴση· καὶ ἡ EZ ἄρα τῇ EH ἐστὶν ἴση ἢ ἐλάσσων τῇ μείζον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὸ E σημεῖον κέντρον ἐστὶ τῶν $AB\Gamma$, $\Gamma\Delta\text{H}$ κύκλων.

Ἐὰν ἄρα δύο κύκλοι τέμνωσιν ἀλλήλους, οὐκ ἔστιν αὐτῶν τὸ αὐτὸ κέντρον· ὅπερ ἔδει δεῖξαι.

For let the two circles ABC and CDG cut one another at points B and C . I say that they will not have the same center.

For, if possible, let E be (the common center), and let EC be joined, and let EFG be drawn through (the two circles), at random. And since point E is the center of the circle ABC , EC is equal to EF . Again, since point E is the center of the circle CDG , EC is equal to EG . But EC was also shown (to be) equal to EF . Thus, EF is also equal to EG , the lesser to the greater. The very thing is impossible. Thus, point E is not the (common) center of the circles ABC and CDG .

Thus, if two circles cut one another, (then) they will not have the same center. (Which is) the very thing it was required to show.

6.

Proposition 6

Ἐὰν δύο κύκλοι ἐφάπτονται ἀλλήλων, οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον.

Δύο γὰρ κύκλοι οἱ $AB\Gamma$, $\Gamma\Delta E$ ἐφαπτέσθωσαν ἀλλήλων κατὰ τὸ Γ σημεῖον· λέγω, ὅτι οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον.

Εἰ γὰρ δυνατόν, ἔστω τὸ Z , καὶ ἐπεζεύθω ἡ $Z\Gamma$, καὶ διήθω, ὥς ἔτυθεν, ἡ ZEB .

Ἐπεὶ οὖν τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου, ἴση ἐστὶν ἡ $Z\Gamma$ τῇ ZB . πάλιν, ἐπεὶ τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $\Gamma\Delta E$ κύκλου, ἴση ἐστὶν ἡ $Z\Gamma$ τῇ ZE . ἐδείχθη δὲ ἡ $Z\Gamma$ τῇ ZB ἴση· καὶ ἡ ZE ἄρα τῇ ZB ἐστὶν ἴση, ἢ ἐλάττων τῇ μείζον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὸ Z σημεῖον κέντρον ἐστὶ τῶν

If two circles touch one another, (then) they will not have the same center.

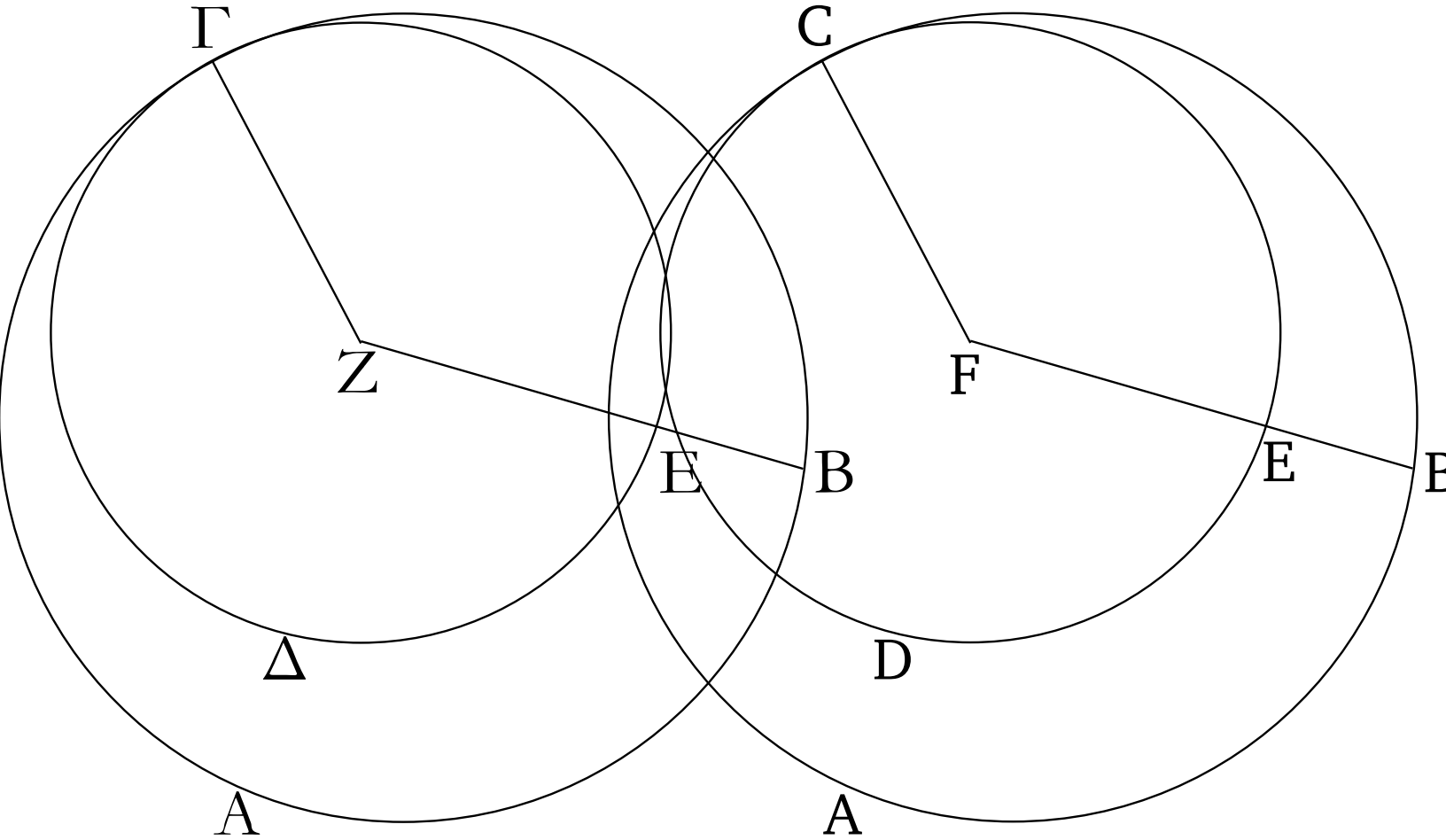
For let the two circles ABC and CDE touch one another at point C . I say that they will not have the same center.

For, if possible, let F be (the common center), and let FC be joined, and let FEB be drawn through (the two circles), at random.

Therefore, since point F is the center of the circle ABC , FC is equal to FB . Again, since point F is the center of the circle CDE , FC is equal to FE . But FC was shown (to be) equal to FB . Thus, FE is also equal to FB , the lesser to the greater. The very thing is impossible. Thus, point F is not the

$AB\Gamma$, $\Gamma\Delta E$ κύκλων.

(common) center of the circles ABC and CDE .



Ἐάν ἄρα δύο κύκλοι ἐφάπτονται ἀλλήλων, οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον· ὅπερ ἔδει δεῖχαι.

Thus, if two circles touch one another, (then) they will not have the same center. (Which is) the very thing it was required to show.

7.

Proposition 7

Ἐάν κύκλον ἐπὶ τῇ διαμέτρῳ ληφθῇ τι σημεῖον, ὃ μὴ ἔστι κέντρον τοῦ κύκλου, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσιν εὐθεῖαι τινεῖς, μεγίστη μὲν ἔσται, ἐφ' ἣς τὸ κέντρον, ἐλαθίστη δὲ ἡ λοιπή, τῶν δὲ ἄλλων αἰεὶ ἡ ἔγγιον τῇς διὰ τοῦ κέντρον τῇς ἀπώτερον μείζων ἔστί, δύο δὲ μόνον ἴσαι ἀπὸ τοῦ σημείου προσπεσοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῇς ἐλαθίστης.

Ἐστω κύκλος ὁ $AB\Gamma\Delta$, διάμετρος δὲ αὐτοῦ ἔστω ἡ $A\Delta$, καὶ ἐπὶ τῇς $A\Delta$ εἰληφθῶ τι σημεῖον τὸ Z , ὃ μὴ ἔστι κέντρον τοῦ κύκλου, κέντρον δὲ τοῦ κύκλου ἔστω τὸ E , καὶ ἀπὸ τοῦ Z πρὸς τὸν $AB\Gamma\Delta$ κύκλον προσπιπτέτωσαν εὐθεῖαι τινεῖς αἱ ZB , $Z\Gamma$, ZH · λέγω, ὅτι μεγίστη μὲν ἔστιν ἡ ZA , ἐλαθίστη δὲ ἡ $Z\Delta$, τῶν δὲ ἄλλων ἡ μὲν ZB τῇς $Z\Gamma$ μείζων, ἡ δὲ $Z\Gamma$ τῇς ZH .

If some point, which is not the center of the circle, is taken on the diameter of a circle, and some straight-lines radiate from the point towards the (circumference of the) circle, (then) the greatest (straight-line) will be that on which the center (lies), and the least the remainder (of the same diameter). And for the others, a (straight-line) nearer[†] to the (straight-line) through the center is always greater than a (straight-line) further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line).

Let $ABCD$ be a circle, and let AD be its diameter, and let some point F , which is not the center of the circle, be taken on AD . Let E be the center of the circle. And let some straight-lines, FB , FC , and FG , radiate from F towards (the circum-

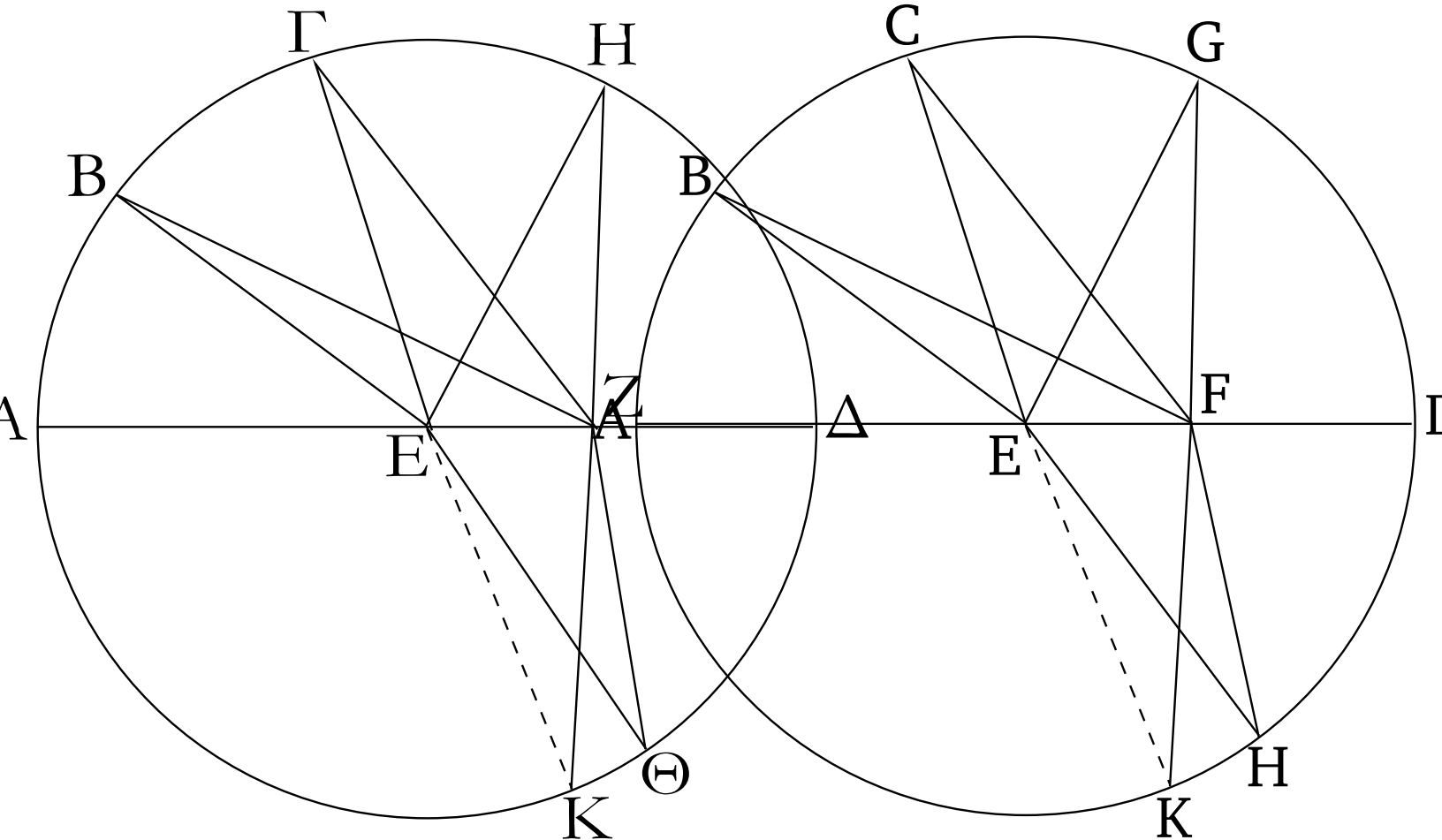
Ἐπεξεύχθησαν γάρ αἱ BE , $ΓE$, HE . καὶ ἐπεὶ παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσιν, αἱ ἄρα EB , EZ τῆς BZ μείζονές εἰσιν. ἴση δὲ ἡ AE τῇ BE [αἱ ἄρα BE , EZ ἴσαι εἰσὶ τῇ AZ]. μείζων ἄρα ἡ AZ τῆς BZ . πάλιν, ἐπεὶ ἴση ἐστὶν ἡ BE τῇ $ΓE$, κοινὴ δὲ ἡ ZE , δύο δὲ αἱ BE , EZ δυοὶ ταῖς $ΓE$, EZ ἴσαι εἰσίν. ἀλλὰ καὶ γωνία ἡ ὑπὸ BEZ γωνία τῆς ὑπὸ $ΓEZ$ μείζων· βάσει ἄρα ἡ BZ βάσεως τῆς $ΓZ$ μείζων ἐστίν. διὰ τὰ αὐτὰ δὲ καὶ ἡ $ΓZ$ τῆς ZH μείζων ἐστίν.

Πάλιν, ἐπεὶ αἱ HZ , ZE τῆς EH μείζονές εἰσιν, ἴση δὲ ἡ EH τῇ $EΔ$, αἱ ἄρα HZ , ZE τῆς $EΔ$ μείζονές εἰσιν. κοινὴ ἀφηρέσθω ἡ EZ · λοιπὴ ἄρα ἡ HZ λοιπῆς τῆς $ZΔ$ μείζων ἐστίν. μεγίστη μὲν ἄρα ἡ ZA , ἐλαθίστη δὲ ἡ $ZΔ$, μείζων δὲ ἡ μὲν ZB τῆς $ZΓ$, ἡ δὲ $ZΓ$ τῆς ZH .

ference of) circle $ABCD$. I say that FA is the greatest (straight-line), FD the least, and of the others, FB (is) greater than FC , and FC than FG .

For let BE , CE , and GE be joined. And since for every triangle (any) two sides are greater than the remaining (side) [Prop. 1.20], EB and EF is thus greater than BF . And AE (is) equal to BE [thus, BE and EF is equal to AF]. Thus, AF (is) greater than BF . Again, since BE is equal to CE , and FE (is) common, the two (straight-lines) BE , EF are equal to the two (straight-lines) CE , EF (respectively). But, angle BEF (is) also greater than angle CEF .[‡] Thus, the base BF is greater than the base CF . Thus, the base BF is greater than the base CF [Prop. 1.24]. So, for the same (reasons), CF is also greater than FG .

Again, since GF and FE are greater than EG [Prop. 1.20], and EG (is) equal to ED , GF and FE are thus greater than ED . Let EF be taken from both. Thus, the remainder GF is greater than the remainder FD . Thus, FA (is) the greatest (straight-line), FD the least, and FB (is) greater than FC , and FC than FG .



Λέγω, ὅτι καὶ ἀπὸ τοῦ Z σημείου δύο μόνον ἴσαι προσ-

I also say that from point F only two equal (straight-lines)

πεσοῦνται πρὸς τὸν $AB\Gamma\Delta$ κύκλον ἐφ' ἑκάτερα τῆς $Z\Delta$ ἐλαθίστης. συνεστάτω γὰρ πρὸς τῇ EZ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ E τῇ ὑπὸ HEZ γωνίᾳ ἴση ἢ ὑπὸ $ZE\Theta$, καὶ ἐπεξεύθθω ἡ $Z\Theta$. ἐπεὶ οὖν ἴση ἐστὶν ἡ HE τῇ $E\Theta$, κοινὴ δὲ ἡ EZ , δύο δὲ αἱ HE , EZ δυοὶ ταῖς ΘE , EZ ἴσαι εἰσὶν· καὶ γωνία ἢ ὑπὸ HEZ γωνία τῇ ὑπὸ ΘEZ ἴση· βάσεις ἄρα ἡ ZH βάσει τῇ $Z\Theta$ ἴση ἐστίν. λέγω δὴ, ὅτι τῇ ZH ἄλλη ἴση οὐ προσπeseῖται πρὸς τὸν κύκλον ἀπὸ τοῦ Z σημείου. εἰ γὰρ δυνατόν, προσπιπτέτω ἡ ZK . καὶ ἐπεὶ ἡ ZK τῇ ZH ἴση ἐστίν, ἀλλὰ ἡ $Z\Theta$ τῇ ZH [ἴση ἐστίν], καὶ ἡ ZK ἄρα τῇ $Z\Theta$ ἐστὶν ἴση, ἢ ἔργον τῆς διὰ τοῦ κέντρου τῇ ἀπώτερον ἴση· ὅπερ ἀδύνατον. οὐκ ἄρα ἀπὸ τοῦ Z σημείου ἑτέρα τις προσπeseῖται πρὸς τὸν κύκλον ἴση τῇ HZ · μία ἄρα μόνη.

Ἐάν ἄρα κύκλον ἐπὶ τῆς διαμέτρου ληφθῇ τι σημεῖον, ὃ μὴ ἐστὶ κέντρον τοῦ κύκλου, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτουσιν εὐθεῖαι τινεῖς, μεγίστη μὲν ἔσται, ἐφ' ἣς τὸ κέντρον, ἐλαθίστη δὲ ἡ λοιπή, τῶν δὲ ἄλλων αἰεὶ ἢ ἔργον τῆς διὰ τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν, δύο δὲ μόνον ἴσαι ἀπὸ τοῦ αὐτοῦ σημείου προσπeseοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης· ὅπερ ἔδει δεῖξαι.

will radiate towards (the circumference of) circle $ABCD$, (one) on each (side) of the least (straight-line) FD . For let the (angle) FEH , equal to angle GEF , be constructed on the straight-line EF , at the point E on it [Prop. 1.23], and let FH be joined. Therefore, since GE is equal to EH , and EF (is) common, the two (straight-lines) GE , EF are equal to the two (straight-lines) HE , EF (respectively). And angle GEF (is) equal to angle HEF . Thus, the base FG is equal to the base FH [Prop. 1.4]. So I say that another (straight-line) equal to FG will not radiate towards (the circumference of) the circle from point F . For, if possible, let FK (so) radiate. And since FK is equal to FG , but FH [is equal] to FG , FK is thus also equal to FH , the nearer to the (straight-line) through the center equal to the further away. The very thing (is) impossible. Thus, another (straight-line) equal to GF will not radiate from the point F towards (the circumference of) the circle. Thus, (there is) only one (such straight-line).

Thus, if some point, which is not the center of the circle, is taken on the diameter of a circle, and some straight-lines radiate from the point towards the (circumference of the) circle, (then) the greatest (straight-line) will be that on which the center (lies), and the least the remainder (of the same diameter). And for the others, a (straight-line) nearer to the (straight-line) through the center is always greater than a (straight-line) further away. And only two equal (straight-lines) will radiate from the same point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line). (Which is) the very thing it was required to show.

[†] Presumably, in an angular sense.

[‡] This is not proved, except by reference to the figure.

8.

Proposition 8

Ἐάν κύκλον ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον διαθῶσιν εὐθεῖαι τινεῖς, ὧν μία μὲν διὰ τοῦ κέντρου, αἱ δὲ λοιπαί, ὡς ἔτυθεν, τῶν μὲν πρὸς τὴν κοίλην περιφέρειαν προσπιπτουσῶν εὐθειῶν μεγίστη μὲν ἐστὶν ἢ διὰ τοῦ κέντρου, τῶν δὲ ἄλλων αἰεὶ ἢ ἔργον τῆς διὰ τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν, τῶν δὲ πρὸς τὴν κυρτὴν περιφέρειαν προσπιπτουσῶν εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἢ μεταχὺ τοῦ τε σημείου καὶ τῆς διαμέτρου, τῶν δὲ ἄλλων αἰεὶ ἢ ἔργον τῆς ἐλαθίστης τῆς ἀπώτερόν ἐστιν ἐλάττων, δύο δὲ μόνον ἴσαι ἀπὸ τοῦ σημείου προσπeseοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης.

Ἐστω κύκλος ὁ $AB\Gamma$, καὶ τοῦ $AB\Gamma$ εἰλήφθω τι σημεῖον ἐκτός τὸ Δ , καὶ ἀπ' αὐτοῦ διήθθωσαν εὐθεῖαι τινεῖς αἱ ΔA , ΔE , ΔZ , $\Delta \Gamma$, ἔστω δὲ ἡ ΔA διὰ τοῦ κέντρου. λέγω, ὅτι τῶν μὲν πρὸς τὴν $AEZ\Gamma$ κοίλην περιφέρειαν προσπιπτουσῶν εὐθειῶν μεγίστη μὲν ἐστὶν ἢ διὰ τοῦ κέντρου ἡ ΔA , μείζων δὲ ἢ μὲν ΔE τῆς ΔZ ἢ δὲ ΔZ τῆς $\Delta \Gamma$, τῶν δὲ πρὸς τὴν

If some point is taken outside a circle, and some straight-lines are drawn from the point to the (circumference of the) circle, one of which (passes) through the center, the remainder (being) random, (then) for the straight-lines radiating towards the concave (part of the) circumference, the greatest is that (passing) through the center. For the others, a (straight-line) nearer[†] to the (straight-line) through the center is always greater than one further away. For the straight-lines radiating towards the convex (part of the) circumference, the least is that between the point and the diameter. For the others, a (straight-line) nearer to the least (straight-line) is always less than one further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line).

Let ABC be a circle, and let some point D be taken outside ABC , and from it let some straight-lines, DA , DE , DF , and DC , be drawn through (the circle), and let DA be through

ΘΑΚΗ κυρτήν περιφέρειαν προσιπτονσών εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἡ ΔΗ ἢ μεταχὺ τοῦ σημείου καὶ τῆς διαμέτρου τῆς ΑΗ, αἰ δὲ ἡ ἔγγιον τῆς ΔΗ ἐλαθίστης ἐλάττων ἐστὶ τῆς ἀπώτερον, ἡ μὲν ΔΚ τῆς ΔΑ, ἡ δὲ ΔΑ τῆς ΔΘ.

Εἰλήφθω γάρ τὸ κέντρον τοῦ ΑΒΓ κύκλου καὶ ἔστω τὸ Μ· καὶ ἐπεζεύθωσαν αἱ ΜΕ, ΜΖ, ΜΓ, ΜΚ, ΜΑ, ΜΘ.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ ΑΜ τῇ ΕΜ, κοινὴ προσκείσθω ἡ ΜΔ· ἡ ἄρα ΑΔ ἴση ἐστὶ ταῖς ΕΜ, ΜΔ. ἀλλ' αἱ ΕΜ, ΜΔ τῆς ΕΔ μείζονές εἰσιν· καὶ ἡ ΑΔ ἄρα τῆς ΕΔ μείζων ἐστίν. πάλιν, ἐπεὶ ἴση ἐστὶν ἡ ΜΕ τῇ ΜΖ, κοινὴ δὲ ἡ ΜΔ, αἱ ΕΜ, ΜΔ ἄρα ταῖς ΖΜ, ΜΔ ἴσαι εἰσίν· καὶ γωνία ἡ ὑπὸ ΕΜΔ γωνία τῆς ὑπὸ ΖΜΔ μείζων ἐστίν. βάσις ἄρα ἡ ΕΔ βάσεως τῆς ΖΔ μείζων ἐστίν. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΖΔ τῆς ΓΔ μείζων ἐστίν· μεγίστη μὲν ἄρα ἡ ΔΑ, μείζων δὲ ἡ μὲν ΔΕ τῆς ΔΖ, ἡ δὲ ΔΖ τῆς ΔΓ.

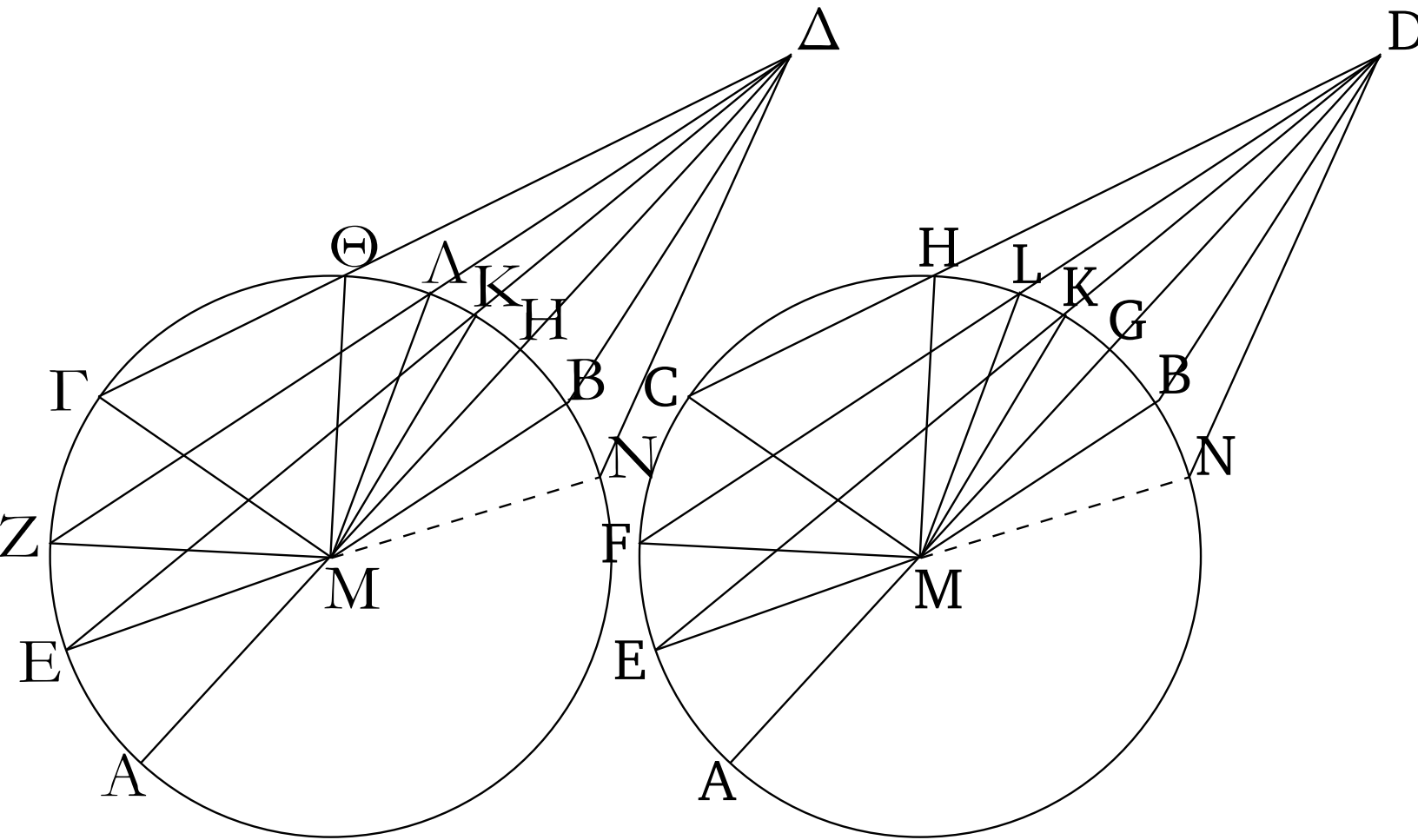
Καὶ ἐπεὶ αἱ ΜΚ, ΚΔ τῆς ΜΔ μείζονές εἰσιν, ἴση δὲ ἡ μὴ τῇ ΜΚ, λοιπὴ ἄρα ἡ ΚΔ λοιπῆς τῆς ΗΔ μείζων ἐστίν· ὥστε ἡ ΗΔ τῆς ΚΔ ἐλάττων ἐστίν· καὶ ἐπεὶ τριγώνον τοῦ ΜΑΔ ἐπὶ μιᾷ τῶν πλευρῶν τῆς ΜΔ δύο εὐθεῖαι ἐντὸς συνεστάθησαν αἱ ΜΚ, ΚΔ, αἱ ἄρα ΜΚ, ΚΔ τῶν ΜΑ, ΑΔ ἐλάττονές εἰσιν· ἴση δὲ ἡ ΜΚ τῇ ΜΑ· λοιπὴ ἄρα ἡ ΔΚ λοιπῆς τῆς ΔΑ ἐλάττων ἐστίν. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΔΑ τῆς ΔΘ ἐλάττων ἐστίν· ἐλαθίστη μὲν ἄρα ἡ ΔΗ, ἐλάττων δὲ ἡ μὲν ΔΚ τῆς ΔΑ ἡ δὲ ΔΑ τῆς ΔΘ.

the center. I say that for the straight-lines radiating towards the concave (part of the) circumference, $AEFC$, the greatest is the one (passing) through the center, (namely) AD , and (that) DE (is) greater than DF , and DF than DC . For the straight-lines radiating towards the convex (part of the) circumference, $HLKG$, the least is the one between the point and the diameter AG , (namely) DG , and a (straight-line) nearer to the least (straight-line) DG is always less than one farther away, (so that) DK (is less) than DL , and DL than DH .

For let the center of the circle be found [Prop. 3.1], and let it be (at point) M [Prop. 3.1]. And let ME , MF , MC , MK , ML , and mh be joined.

And since AM is equal to EM , let MD be added to both. Thus, AD is equal to EM and MD . But, EM and MD is greater than ED [Prop. 1.20]. Thus, AD is also greater than ED . Again, since ME is equal to MF , and MD (is) common, the (straight-lines) EM , MD are thus equal to FM , MD . And angle EMD is greater than angle FMD .[‡] Thus, the base ED is greater than the base FD [Prop. 1.24]. So, similarly, we can show that FD is also greater than CD . Thus, AD (is) the greatest (straight-line), and DE (is) greater than DF , and DF than DC .

And since MK and KD is greater than MD [Prop. 1.20], and MG (is) equal to MK , the remainder KD is thus greater than the remainder GD . So GD is less than KD . And since in triangle MLD , the two internal straight-lines MK and KD were constructed on one of the sides, MD , (then) MK and KD are thus less than ML and LD [Prop. 1.21]. And MK (is) equal to ML . Thus, the remainder DK is less than the remainder DL . So, similarly, we can show that DL is also less than DH . Thus, DG (is) the least (straight-line), and DK (is) less than DL , and DL than DH .



Λέγω, ὅτι καὶ δύο μόνον ἴσαι ἀπὸ τοῦ Δ σημείου προσ-
 πεσοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ΔΗ ἐλαθίστης·
 συνεστώτω πρὸς τῇ ΜΔ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ
 Μ τῇ ὑπὸ ΚΜΔ γωνίᾳ ἴση γωνία ἡ ὑπὸ ΔΜΒ, καὶ ἐπεζεύ-
 θω ἡ ΔΒ. καὶ ἐπεὶ ἴση ἐστὶν ἡ ΜΚ τῇ ΜΒ, κοινὴ δὲ ἡ ΜΔ,
 δύο δὴ αἱ ΚΜ, ΜΔ δύο ταῖς ΒΜ, ΜΔ ἴσαι εἰσὶν ἑκατέρω
 ἑκατέρω· καὶ γωνία ἡ ὑπὸ ΚΜΔ γωνία τῇ ὑπὸ ΒΜΔ ἴση·
 βάσει ἄρα ἡ ΔΚ βάσει τῇ ΔΒ ἴση ἐστίν. λέγω [δὴ], ὅτι
 τῇ ΔΚ εὐθείᾳ ἄλλη ἴση οὐ προσπεσεῖται πρὸς τὸν κύκλον
 ἀπὸ τοῦ Δ σημείου. εἰ γὰρ δυνατόν, προσπιπτέτω καὶ ἔστω
 ἡ ΔΝ. ἐπεὶ οὖν ἡ ΔΚ τῇ ΔΝ ἐστὶν ἴση, ἀλλ' ἡ ΔΚ τῇ ΔΒ
 ἐστὶν ἴση, καὶ ἡ ΔΒ ἄρα τῇ ΔΝ ἐστὶν ἴση, ἡ ἔγγιον τῆς ΔΗ
 ἐλαθίστης τῇ ἀπώτερων [ἐστίν] ἴση· ὅπερ ἀδύνατον ἐδείχθη.
 οὐκ ἄρα πλείους ἢ δύο ἴσαι πρὸς τὸν ΑΒΓ κύκλον ἀπὸ τοῦ
 Δ σημείου ἐφ' ἑκάτερα τῆς ΔΗ ἐλαθίστης προσπεσοῦνται.

Ἐάν ἄρα κύκλον ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ ση-
 μείου πρὸς τὸν κύκλον διαθῶσιν εὐθεῖαι τιναί, ὧν μία μὲν
 διὰ τοῦ κέντρου αἱ δὲ λοιπαί, ὥς ἔτυθεν, τῶν μὲν πρὸς τὴν
 κοίλην περιφέρειαν προσπιπτονσῶν εὐθειῶν μεγίστη μὲν ἐστὶν
 ἡ διὰ τοῦ κέντρου, τῶν δὲ ἄλλων αἰεὶ ἡ ἔγγιον τῆς διὰ τοῦ

I also say that only two equal (straight-lines) will radiate
 from point D towards (the circumference of) the circle, (one)
 on each (side) on the least (straight-line), DG . Let the angle
 DMB , equal to angle KMD , be constructed on the straight-line
 MD , at the point M on it [Prop. 1.23], and let DB be joined.
 And since MK is equal to MB , and MD (is) common, the two
 (straight-lines) KM , MD are equal to the two (straight-lines)
 BM , MD , respectively. And angle KMD (is) equal to angle
 BMD . Thus, the base DK is equal to the base DB [Prop. 1.4].
 [So] I say that another (straight-line) equal to DK will not radi-
 ate towards the (circumference of) the circle from point D . For,
 if possible, let (such a straight-line) radiate, and let it be DN .
 Therefore, since DK is equal to DN , but DK is equal to DB ,
 (then) DB is thus also equal to DN , (so that) a (straight-line)
 nearer to the least (straight-line) DG [is] equal to one further
 away. The very thing was shown (to be) impossible. Thus, not
 more than two equal (straight-lines) will radiate towards (the
 circumference of) circle ABC from point D , (one) on each side
 of the least (straight-line) DG .

Thus, if some point is taken outside a circle, and some

κέντρον τῆς ἀπώτερον μείζων ἐστίν, τῶν δὲ πρὸς τὴν κυρτὴν περιφέρειαν προσπιπτουσῶν εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἡ μεταχὺ τοῦ τε σημείου καὶ τῆς διαμέτρου, τῶν δὲ ἄλλων αἰεὶ ἢ ἔγγιον τῆς ἐλαθίστης τῆς ἀπώτερόν ἐστιν ἐλάττων, δύο δὲ μόνον ἴσαι ἀπὸ τοῦ σημείου προσπεσοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης· ὅπερ ἔδει δεῖχαι.

straight-lines are drawn from the point to the (circumference of the) circle, one of which (passes) through the center, the remainder (being) random, (then) for the straight-lines radiating towards the concave (part of the) circumference, the greatest is that (passing) through the center. For the others, a (straight-line) nearer to the (straight-line) through the center is always greater than one further away. For the straight-lines radiating towards the convex (part of the) circumference, the least is that between the point and the diameter. For the others, a (straight-line) nearer to the least (straight-line) is always less than one further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line). (Which is) the very thing it was required to show.

† Presumably, in an angular sense.

‡ This is not proved, except by reference to the figure.

9.

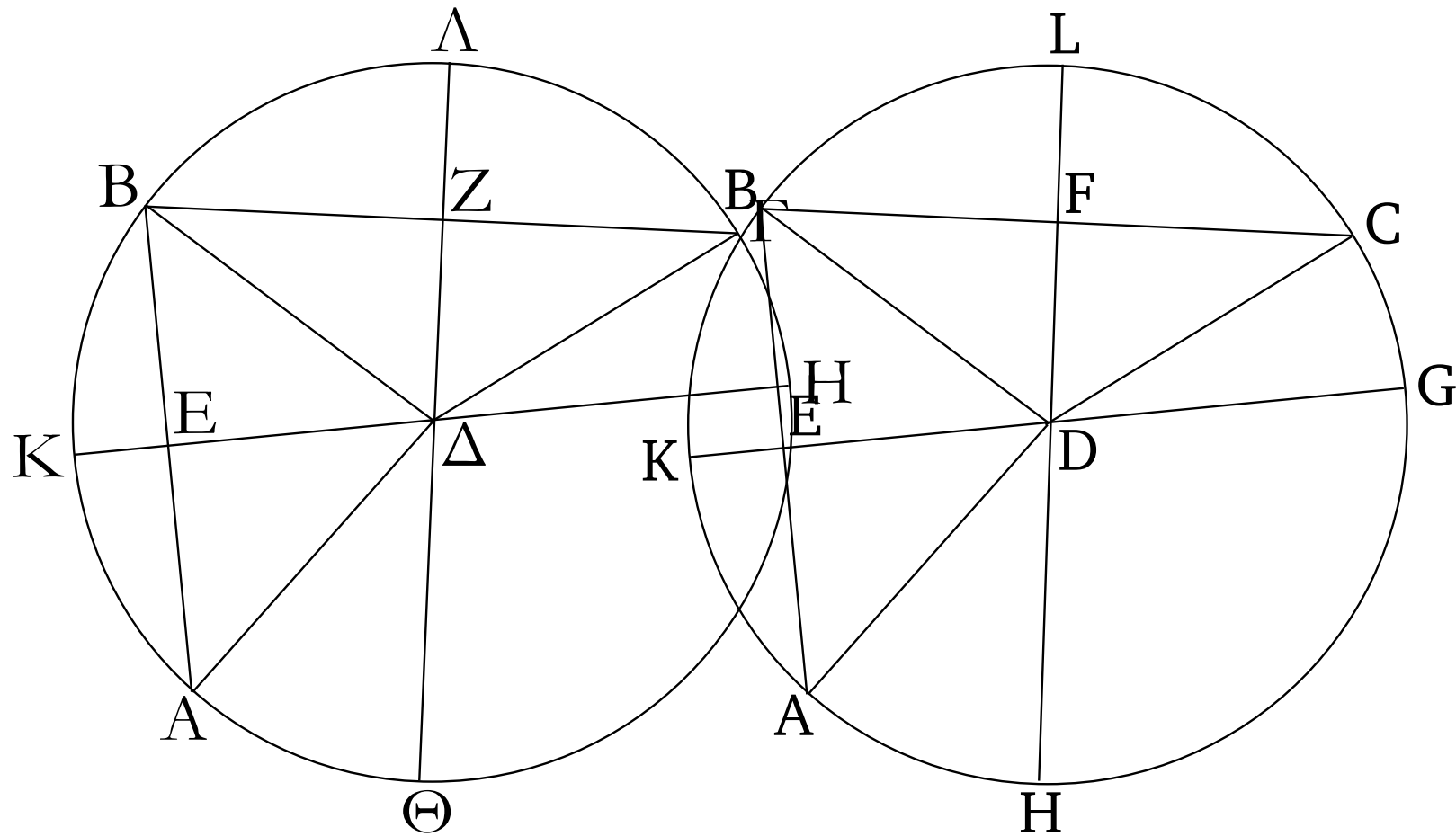
Ἐάν κύκλον ληφθῇ τι σημεῖον ἐντός, ἀπο δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι πλείονες ἢ δύο ἴσαι εὐθεῖαι, τὸ ληφθὲν σημεῖον κέντρον ἐστὶ τοῦ κύκλου.

Ἐστω κύκλος ὁ $ABΓ$, ἐντός δὲ αὐτοῦ σημεῖον τὸ Δ , καὶ ἀπὸ τοῦ Δ πρὸς τὸν $ABΓ$ κύκλον προσπιπτέτωσαν πλείονες ἢ δύο ἴσαι εὐθεῖαι αἱ ΔA , ΔB , $\Delta Γ$. λέγω, ὅτι τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ $ABΓ$ κύκλου.

Proposition 9

If some point is taken inside a circle, and more than two equal straight-lines radiate from the point towards the (circumference of the) circle, (then) the point taken is the center of the circle.

Let ABC be a circle, and D a point inside it, and let more than two equal straight-lines, DA , DB , and DC , radiate from D towards (the circumference of) circle ABC . I say that point D is the center of circle ABC .



Ἐπεξενθῶσαν γὰρ αἱ AB, BF καὶ τετμήσθωσαν διῖθα κατὰ τὰ E, Z σημεία, καὶ ἐπιζενθῆσαι αἱ ED, ZD διήθθωσαν ἐπὶ τὰ H, K, Θ, Λ σημεία.

Ἐπεὶ οὖν ἴση ἐστὶν ἡ AE τῇ EB , κοινὴ δὲ ἡ ED , δύο δὴ αἱ AE, ED δύο ταῖς BE, ED ἴσαι εἰσὶν καὶ βάσεις ἡ DA βάσει τῇ DB ἴση· γωνία ἄρα ἡ ὑπὸ $AE\Delta$ γωνία τῇ ὑπὸ $BE\Delta$ ἴση ἐστὶν· ὀρθὴ ἄρα ἑκατέρω τῶν ὑπὸ $AE\Delta, BE\Delta$ γωνιῶν· ἡ HK ἄρα τὴν AB τέμνει διῖθα καὶ πρὸς ὀρθάς. καὶ ἐπεὶ, ἔαν ἐν κύκλῳ εὐθεῖά τις εὐθεϊάν τινα διῖθα τε καὶ πρὸς ὀρθὰς τέμνη, ἐπὶ τῆς τεμνουσῆς ἐστὶ τὸ κέντρον τοῦ κύκλου, ἐπὶ τῆς HK ἄρα ἐστὶ τὸ κέντρον τοῦ κύκλου. διὰ τὰ αὐτὰ δὴ καὶ ἐπὶ τῆς $\Theta\Lambda$ ἐστὶ τὸ κέντρον τοῦ ABF κύκλου. καὶ οὐδέν ἕτερον κοινὸν ἔθουσιν αἱ $HK, \Theta\Lambda$ εὐθεῖαι ἢ τὸ Δ σημεῖον· τὸ Δ ἄρα σημεῖον κέντρον ἐστὶ τοῦ ABF κύκλου.

Ἐάν ἄρα κύκλον ληφθῇ τι σημεῖον ἐντός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι πλείονες ἢ δύο ἴσαι εὐθεῖαι, τὸ ληφθὲν σημεῖον κέντρον ἐστὶ τοῦ κύκλου· ὅπερ ἔδει δεῖχαι.

For let AB and BC be joined, and (then) be cut in half at points E and F (respectively) [Prop. 1.10]. And ED and FD being joined, let them be drawn through to points G, K, H , and L .

Therefore, since AE is equal to EB , and ED (is) common, the two (straight-lines) AE, ED are equal to the two (straight-lines) BE, ED (respectively). And the base DA (is) equal to the base DB . Thus, angle AED is equal to angle BED [Prop. 1.8]. Thus, angles AED and BED (are) each right-angles [Def. 1.10]. Thus, GK cuts AB in half, and at right-angles. And since, if some straight-line in a circle cuts some (other) straight-line in half, and at right-angles, (then) the center of the circle is on the former (straight-line) [Prop. 3.1 corr.], the center of the circle is thus on GK . So, for the same (reasons), the center of circle ABC is also on HL . And the straight-lines GK and HL have no common (point) other than point D . Thus, point D is the center of circle ABC .

Thus, if some point is taken inside a circle, and more than two equal straight-lines radiate from the point towards the (circumference of the) circle, (then) the point taken is the center of the circle. (Which is) the very thing it was required to show.

10.

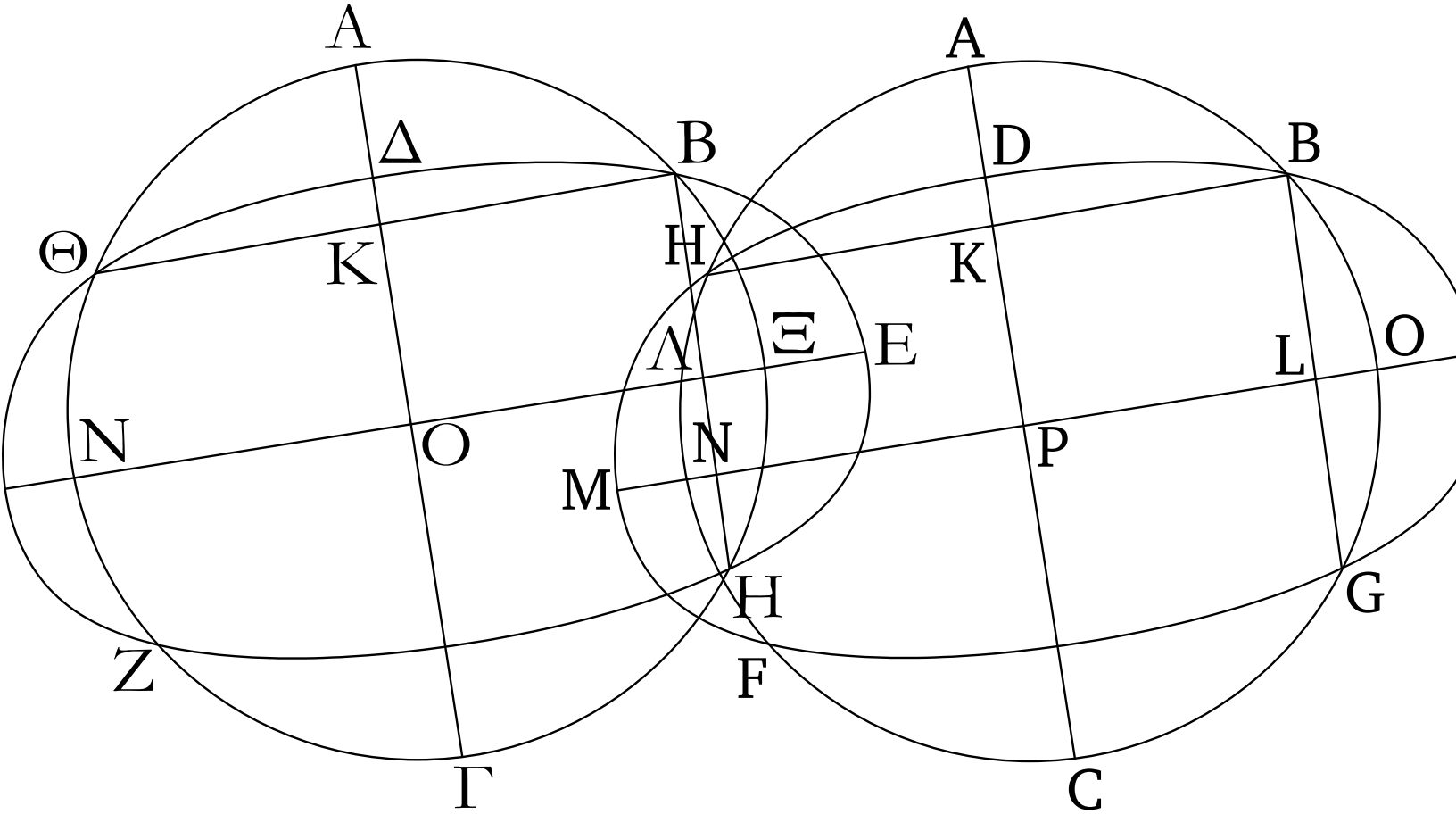
Proposition 10

Κύκλος κύκλον οὐ τέμνει κατὰ πλείονα σημεία ἢ δύο.

Εἰ γὰρ δυνατόν, κύκλος ὁ $ABΓ$ κύκλον τὸν $ΔΕΖ$ τεμνέτω κατὰ πλείονα σημεία ἢ δύο τὰ $B, H, Z, Θ$, καὶ ἐπιξενθθεῖσαι αἱ $BΘ, BH$ δίθλα τεμνέσθωσαν κατὰ τὰ $K, Λ$ σημεία· καὶ ἀπὸ τῶν $K, Λ$ ταῖς $BΘ, BH$ πρὸς ὀρθὰς ἀθθεῖσαι αἱ $KΓ, ΛΜ$ διήθθωσαν ἐπὶ τὰ A, E σημεία.

A circle does not cut a(nother) circle at more than two points.

For, if possible, let the circle ABC cut the circle DEF at more than two points, B, G, F , and H . And BH and BG being joined, let them (then) be cut in half at points K and L (respectively). And KC and LM being drawn at right-angles to BH and BG from K and L (respectively) [Prop. 1.11], let them (then) be drawn through to points A and E (respectively).



Ἐπεὶ οὖν ἐν κύκλῳ τῷ $ABΓ$ εὐθεῖά τις ἡ $ΑΓ$ εὐθεῖάν τινα τὴν $BΘ$ δίθλα καὶ πρὸς ὀρθὰς τέμνει, ἐπὶ τῇς $ΑΓ$ ἄρα ἐστὶ τὸ κέντρον τοῦ $ABΓ$ κύκλου. πάλιν, ἐπεὶ ἐν κύκλῳ τῷ αὐτῷ τῷ $ABΓ$ εὐθεῖά τις ἡ $ΝΧ$ εὐθεῖάν τινα τὴν BH δίθλα καὶ πρὸς ὀρθὰς τέμνει, ἐπὶ τῇς $ΝΧ$ ἄρα ἐστὶ τὸ κέντρον τοῦ $ABΓ$ κύκλου. ἐδείχθη δὲ καὶ ἐπὶ τῇς $ΑΓ$, καὶ κατ' οὐδέν συμβάλλουσιν αἱ $ΑΓ, ΝΧ$ εὐθεῖαι ἢ κατὰ τὸ $Ο$ · τὸ $Ο$ ἄρα σημείον κέντρον ἐστὶ τοῦ $ABΓ$ κύκλου. ὁμοίως δὲ δείχμεν, ὅτι καὶ τοῦ $ΔΕΖ$ κύκλου κέντρον ἐστὶ τὸ $Ο$ · δύο ἄρα κύκλων τεμνόντων ἀλλήλους τῶν $ABΓ, ΔΕΖ$ τὸ αὐτὸ ἐστὶ κέντρον τὸ $Ο$ · ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα κύκλος κύκλον τέμνει κατὰ πλείονα σημεία ἢ

Therefore, since in circle ABC some straight-line AC cuts some (other) straight-line BH in half, and at right-angles, the center of circle ABC is thus on AC [Prop. 3.1 corr.]. Again, since in the same circle ABC some straight-line NO cuts some (other straight-line) BG in half, and at right-angles, the center of circle ABC is thus on NO [Prop. 3.1 corr.]. And it was also shown (to be) on AC . And the straight-lines AC and NO meet at no other (point) than P . Thus, point P is the center of circle ABC . So, similarly, we can show that P is also the center of circle DEF . Thus, two circles cutting one another, ABC and DEF , have the same center P . The very thing is impossible [Prop. 3.5].

δύο· ὅπερ ἔδει δεῖχαι.

11.

Ἐάν δύο κύκλοι ἐφάπτονται ἀλλήλων ἐντός, καί ληφθῇ αὐτῶν τὰ κέντρα, ἡ ἐπὶ τὰ κέντρα αὐτῶν ἐπιζευγνυμένη εὐθεῖα καὶ ἐκβαλλομένη ἐπὶ τὴν συναφὴν πεσεῖται τῶν κύκλων.

Δύο γάρ κύκλοι οἱ $ABΓ$, $ΑΔΕ$ ἐφαπτόσθωσαν ἀλλήλων ἐντός κατὰ τὸ A σημείον, καὶ εἰλήφθω τοῦ μὲν $ABΓ$ κύκλου κέντρον τὸ Z , τοῦ δὲ $ΑΔΕ$ τὸ H · λέγω, ὅτι ἡ ἀπὸ τοῦ H ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖα ἐκβαλλομένη ἐπὶ τὸ A πεσεῖται.

μὴ γάρ, ἀλλ' εἰ δυνατόν, πιπτέτω ὡς ἡ $ZHΘ$, καὶ ἐπεζεύθωσαν αἱ AZ , AH .

Ἐπεὶ οὖν αἱ AH , HZ τῆς ZA , τοντέστι τῆς $ZΘ$, μείζονές εἰσιν, κοινὴ ἀφηρεῖσθω ἡ ZH · λοιπὴ ἄρα ἡ AH λοιπῆς τῆς $HΘ$ μείζων ἐστίν. ἴση δὲ ἡ AH τῇ $HΔ$ · καὶ ἡ $HΔ$ ἄρα τῆς $HΘ$ μείζων ἐστίν ἢ ἐλάττων τῆς μείζονος· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα ἡ ἀπὸ τοῦ Z ἐπὶ τὸ H ἐπιζευγνυμένη εὐθεῖα ἐκτὸς πεσεῖται· κατὰ τὸ A ἄρα ἐπὶ τῆς συναφῆς πεσεῖται.

Thus, a circle does not cut a(nother) circle at more than two points. (Which is) the very thing it was required to show.

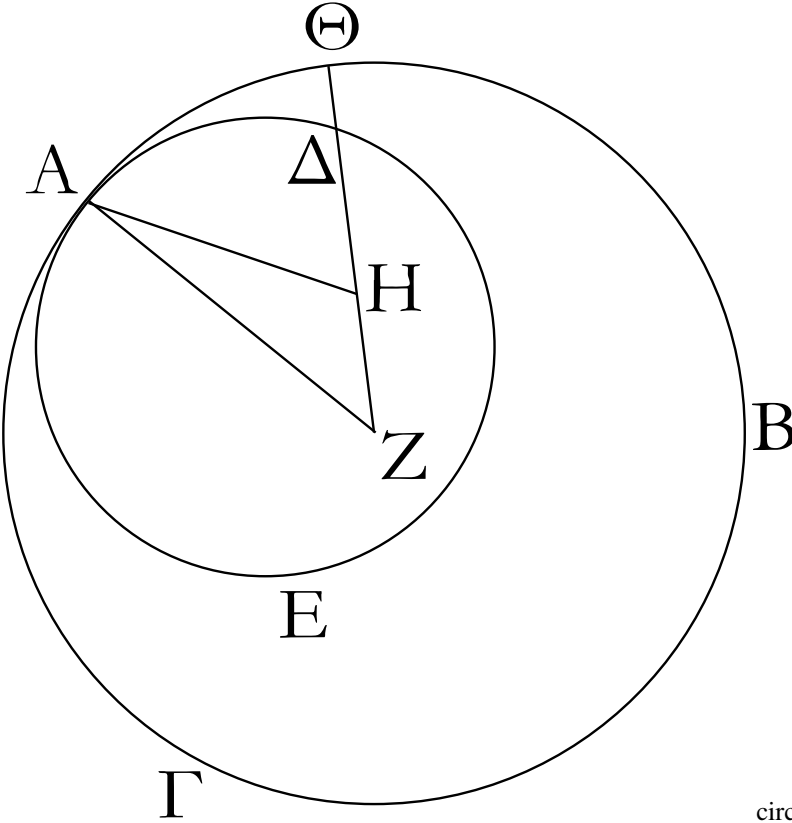
Proposition 11

If two circles touch one another internally, and their centers are found, (then) the straight-line joining their centers, being produced, will fall upon the point of union of the circles.

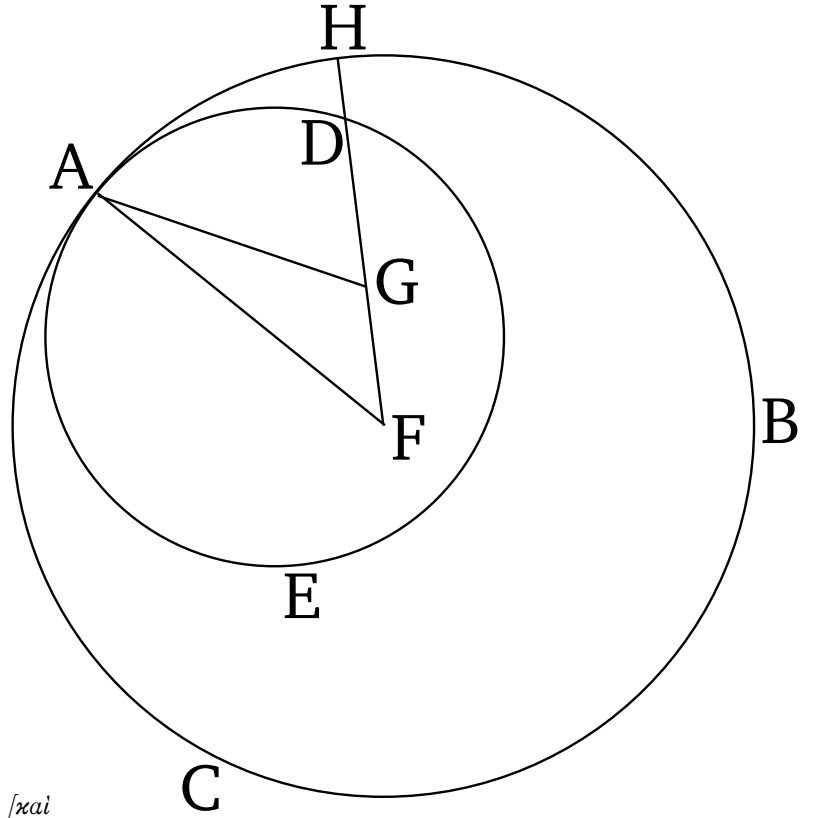
For let two circles, ABC and ADE , touch one another internally at point A , and let the center F of circle ABC be found [Prop. 3.1], and (the center) G of (circle) ADE [Prop. 3.1]. I say that the straight-line joining G to F , being produced, will fall on A .

For (if) not (then), if possible, let it fall like FGH (in the figure), and let AF and AG be joined.

Therefore, since AG and GF is greater than FA , that is to say FH [Prop. 1.20], let FG be taken from both. Thus, the remainder AG is greater than the remainder GH . And AG (is) equal to GD . Thus, GD is also greater than GH , the lesser than the greater. The very thing is impossible. Thus, the straight-line joining F to G will not fall outside (one circle but inside the other). Thus, it will fall upon the point of union (of the



circles) at point A .



Ἐάν ἄρα δύο κύκλοι ἐφάπτονται ἀλλήλων ἐντός, [καὶ
ληφθῇ αὐτῶν τὰ κέντρα], ἡ ἐπὶ τὰ κέντρα αὐτῶν ἐπιζευγνυ-
μένη εὐθεῖα [καὶ ἐκβαλλομένη] ἐπὶ τὴν συναφὴν πεσεῖται τῶν
κύκλων· ὅπερ ἔδει δεῖχαι.

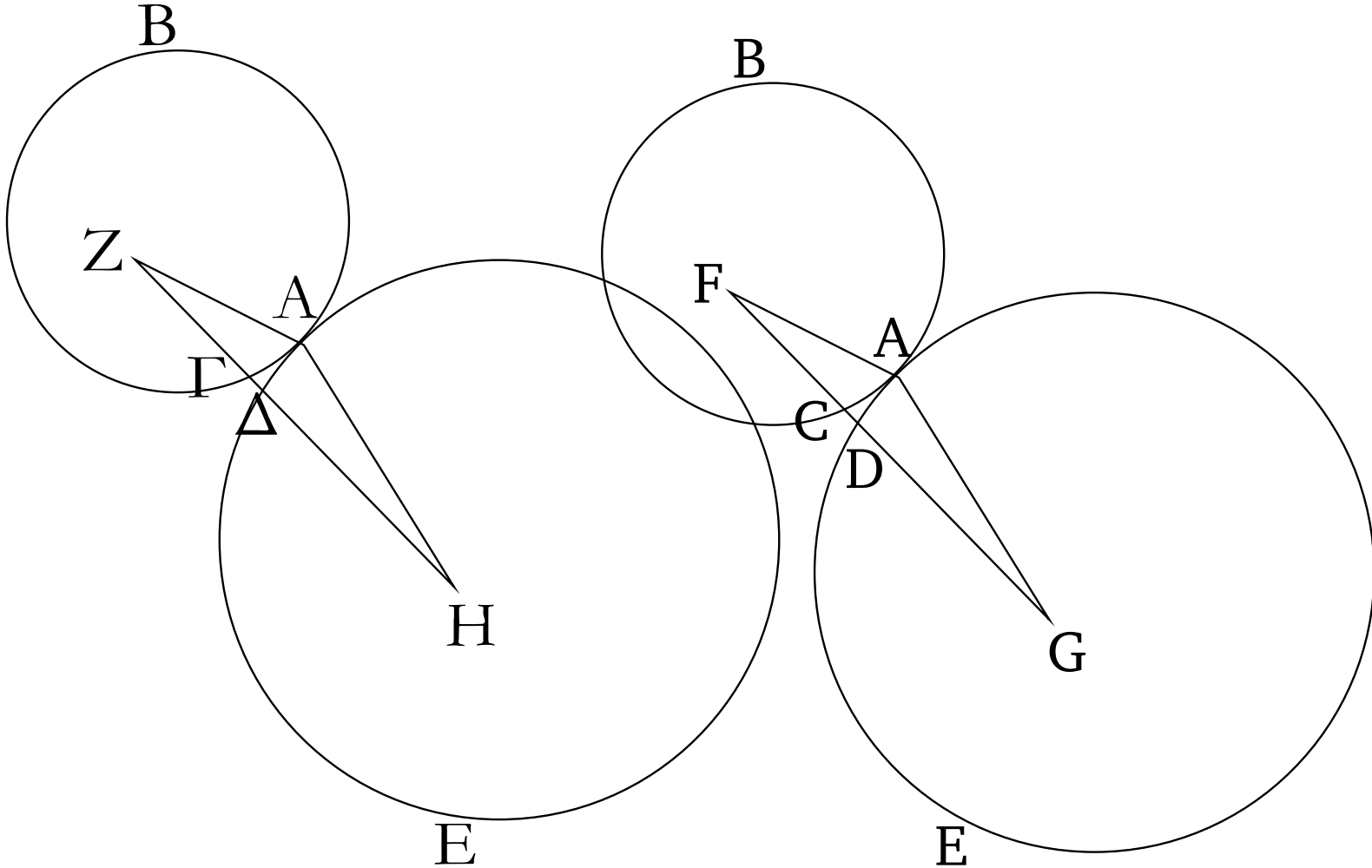
Thus, if two circles touch one another internally, [and their
centers are found], (then) the straight-line joining their cen-
ters, [being produced], will fall upon the point of union of the
circles. (Which is) the very thing it was required to show.

12.

Proposition 12

Ἐάν δύο κύκλοι ἐφάπτονται ἀλλήλων ἐκτός, ἡ ἐπὶ τὰ
κέντρα αὐτῶν ἐπιζευγνυμένη διὰ τῆς ἐπαφῆς ἐλεύσεται.

If two circles touch one another externally, (then) the
(straight-line) joining their centers will go through the point
of union.



Δύο γάρ κύκλοι οἱ $ABΓ$, $AΔΕ$ ἐφαπτέσθωσαν ἀλλήλων ἐκτός κατὰ τὸ A σημεῖον, καὶ εἰλήφθω τοῦ μὲν $ABΓ$ κέντρον τὸ Z , τοῦ δὲ $AΔΕ$ τὸ H . λέγω, ὅτι ἡ ἀπὸ τοῦ Z ἐπὶ τὸ H ἐπιξεννυμένη εὐθεῖα διὰ τῆς κατὰ τὸ A ἐπαφῆς ἐλεύσεται.

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἐρθέσθω ὡς ἡ $ZΓΔH$, καὶ ἐπεξεύθθωσαν αἱ AZ , AH .

Ἐπεὶ οὖν τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $ABΓ$ κύκλου, ἴση ἐστὶν ἡ ZA τῇ $ZΓ$. πάλιν, ἐπεὶ τὸ H σημεῖον κέντρον ἐστὶ τοῦ $AΔΕ$ κύκλου, ἴση ἐστὶν ἡ HA τῇ $HΔ$. ἐδείχθη δὲ καὶ ἡ ZA τῇ $ZΓ$ ἴση· αἱ ἄρα ZA , AH ταῖς $ZΓ$, $HΔ$ ἴσαι εἰσὶν· ὥστε ὅλη ἡ ZH τῶν ZA , AH μείζων ἐστίν· ἀλλὰ καὶ ἐλάττων· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ Z ἐπὶ τὸ H ἐπιξεννυμένη εὐθεῖα διὰ τῆς κατὰ τὸ A ἐπαφῆς οὐκ ἐλεύσεται· δι' αὐτῆς ἄρα.

Ἐάν ἄρα δύο κύκλοι ἐφάπτονται ἀλλήλων ἐκτός, ἡ ἐπὶ τὰ κέντρα αὐτῶν ἐπιξεννυμένη [εὐθεῖα] διὰ τῆς ἐπαφῆς ἐλεύσεται· ὅπερ ἔδει δεῖξαι.

For let two circles, ABC and ADE , touch one another externally at point A , and let the center F of ABC be found [Prop. 3.1], and (the center) G of ADE [Prop. 3.1]. I say that the straight-line joining F to G will go through the point of union at A .

For (if) not (then), if possible, let it go like $FCDG$ (in the figure), and let AF and AG be joined.

Therefore, since point F is the center of circle ABC , FA is equal to FC . Again, since point G is the center of circle ADE , GA is equal to GD . And FA was also shown (to be) equal to FC . Thus, the (straight-lines) FA and AG are equal to the (straight-lines) FC and GD . So the whole of FG is greater than FA and AG . But, (it is) also less [Prop. 1.20]. The very thing is impossible. Thus, the straight-line joining F to G cannot not go through the point of union at A . Thus, (it will go) through it.

Thus, if two circles touch one another externally, (then) the [straight-line] joining their centers will go through the point of

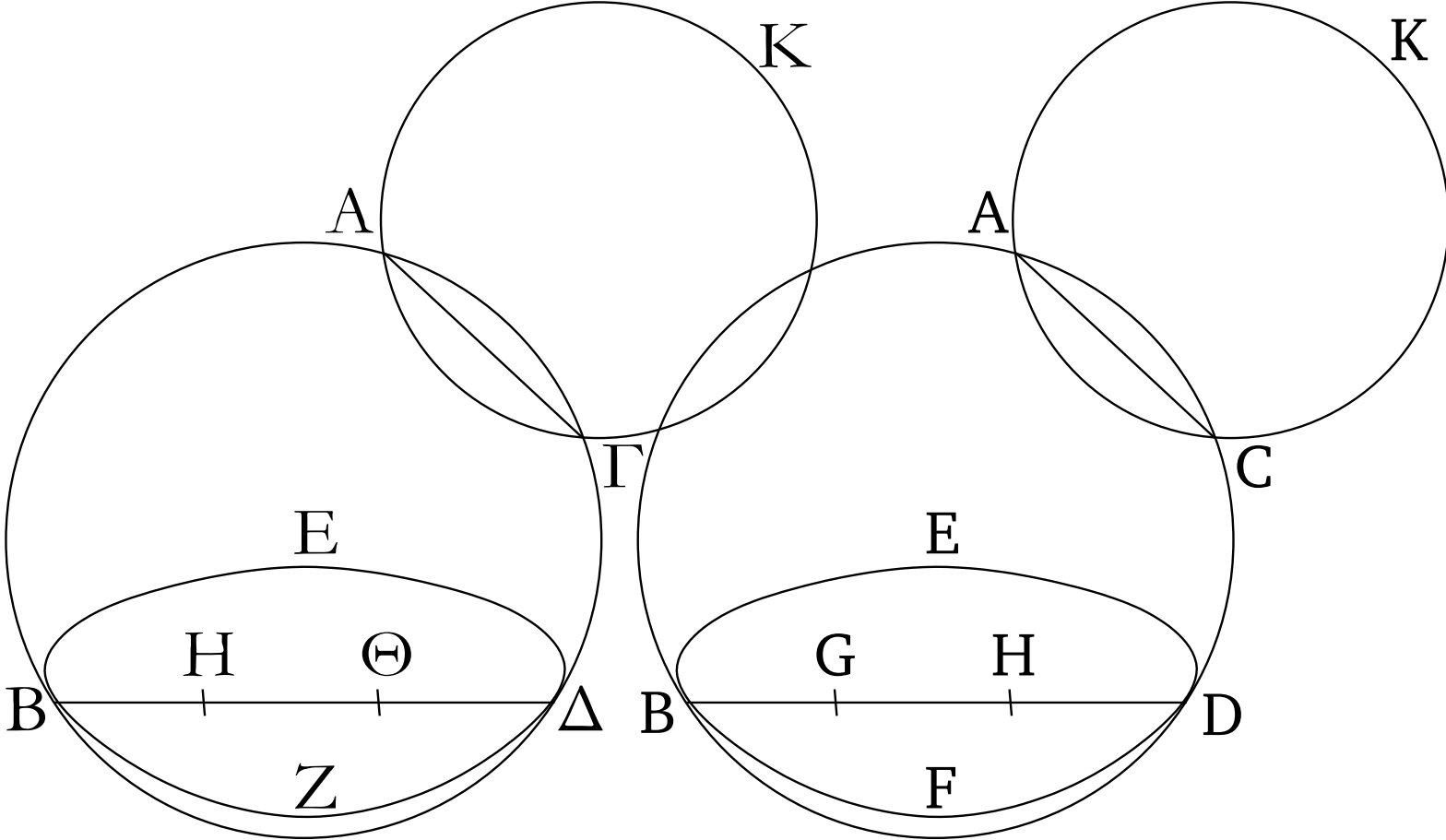
union. (Which is) the very thing it was required to show.

13.

Proposition 13

Κύκλος κύκλον οὐκ ἐφάπτεται κατὰ πλείονα σημεία ἢ καθ' ἓν, ἔαν τε ἐντὸς ἔαν τε ἐκτὸς ἐφάπτηται.

A circle does not touch a(nother) circle at more than one point, whether they touch internally or externally.



Εἰ γὰρ δυνατόν, κύκλος ὁ ABΓΔ κύκλον τοῦ EBZΔ ἐφάπτεσθω πρότερον ἐντὸς κατὰ πλείονα σημεία ἢ ἐν τὰ Δ, Β.

Καὶ εἰλήφθω τοῦ μὲν ABΓΔ κύκλου κέντρον τὸ Η, τοῦ δὲ EBZΔ τὸ Θ.

Ἡ ἄρα ἀπὸ τοῦ Η ἐπὶ τὸ Θ ἐπιξεννυμένη ἐπὶ τὰ Β, Δ πεσεῖται. πιπτέτω ὡς ἡ ΒΗΘΔ. καὶ ἐπεὶ τὸ Η σημεῖον κέντρον ἐστὶ τοῦ ABΓΔ κύκλου, ἴση ἐστὶν ἡ ΒΗ τῇ ΗΔ· μείζων ἄρα ἡ ΒΗ τῇς ΘΔ· πολλῶν ἄρα μείζων ἡ ΒΘ τῇς ΘΔ. πάλιν, ἐπεὶ τὸ Θ σημεῖον κέντρον ἐστὶ τοῦ EBZΔ κύκλου, ἴση ἐστὶν ἡ ΒΘ τῇ ΘΔ· ἐδείχθη δὲ αὐτῇς καὶ πολλῶν μείζων· ὅπερ ἀδύνατον· οὐκ ἄρα κύκλος κύκλον ἐφάπτεται ἐντὸς κατὰ πλείονα σημεία ἢ ἓν.

Λέγω δὴ, ὅτι οὐδὲ ἐκτὸς.

Εἰ γὰρ δυνατόν, κύκλος ὁ AΓΚ κύκλον τοῦ ABΓΔ ἐφάπτεσθω ἐκτὸς κατὰ πλείονα σημεία ἢ ἐν τὰ Α, Γ, καὶ ἐπεξεύθθω ἡ AΓ.

For, if possible, let circle $ABDC$ [†] touch circle $EBFD$ —first of all, internally—at more than one point, D and B .

And let the center G of circle $ABDC$ be found [Prop. 3.1], and (the center) H of $EBFD$ [Prop. 3.1].

Thus, the (straight-line) joining G and H will fall on B and D [Prop. 3.11]. Let it fall like $BGHD$ (in the figure). And since point G is the center of circle $ABDC$, BG is equal to GD . Thus, BG (is) greater than HD . Thus, BH (is) much greater than HD . Again, since point H is the center of circle $EBFD$, BH is equal to HD . But it was also shown (to be) much greater than it. The very thing (is) impossible. Thus, a circle does not touch a(nother) circle internally at more than one point.

So, I say that neither (does it touch) externally (at more than one point).

For, if possible, let circle ACK touch circle $ABDC$ externally at more than one point, A and C . And let AC be joined.

Ἐπεὶ οὖν κύκλων τῶν $AB\Gamma\Delta$, $ΑΓΚ$ ἔλληπται ἐπὶ τῇς περιφερείας ἑκατέρον δύο τυθόντα σημεῖα τὰ A , Γ , ἢ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐντὸς ἑκατέρον πεσεῖται· ἀλλὰ τοῦ μὲν $AB\Gamma\Delta$ ἐντὸς ἔπεσεν, τοῦ δὲ $ΑΓΚ$ ἐκτός· ὅπερ ἄτοπον· οὐκ ἄρα κύκλος κύκλον ἐφάπτεται ἐκτός κατὰ πλείονα σημεῖα ἢ ἓν. ἐδείκθη δέ, ὅτι οὐδὲ ἐντός.

Κύκλος ἄρα κύκλον οὐκ ἐφάπτεται κατὰ πλείονα σημεῖα ἢ [καθ'] ἓν, ἔαν τε ἐντός ἔαν τε ἐκτός ἐφάπτηται· ὅπερ ἔδει δεῖχαι.

Therefore, since two points, A and C , be taken at random on the circumference of each of the circles $ABDC$ and ACK , the straight-line joining the points will fall inside each (circle) [Prop. 3.2]. But, it fell inside $ABDC$, and outside ACK [Def. 3.3]. The very thing (is) absurd. Thus, a circle does not touch a(nother) circle externally at more than one point. And it was shown that neither (does it) internally.

Thus, a circle does not touch a(nother) circle at more than one point, whether they touch internally or externally. (Which is) the very thing it was required to show.

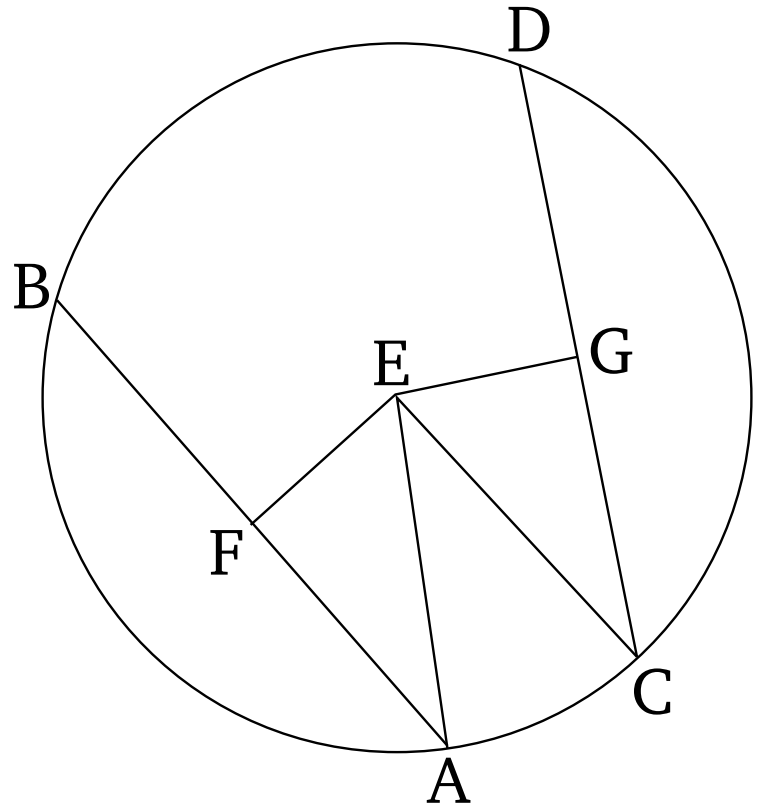
† The Greek text has “ $ABCD$ ”, which is obviously a mistake.

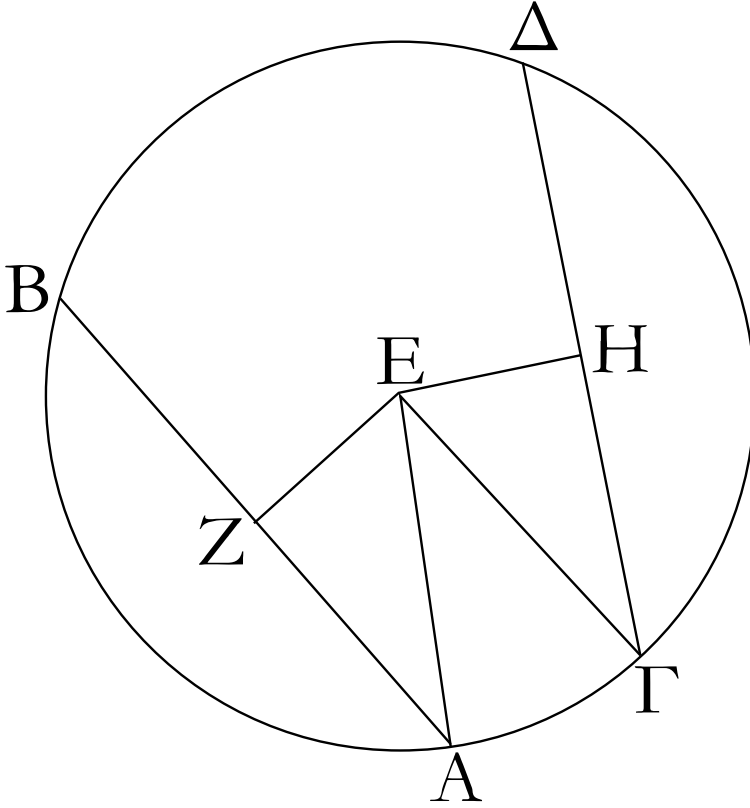
14.

Ἐν κύκλῳ αἱ ἴσαι εὐθεῖαι ἴσον ἀπέθουσιν ἀπὸ τοῦ κέντρου, καὶ αἱ ἴσον ἀπέθουσαι ἀπὸ τοῦ κέντρου ἴσαι ἀλλήλαις εἶσιν.

Proposition 14

In a circle, equal straight-lines are equally far from the center, and (straight-lines) which are equally far from the center are equal to one another.





Ἐστω κύκλος ὁ $AB\Gamma\Delta$, καὶ ἐν αὐτῷ ἴσαι εὐθεῖαι ἔστωσαν αἱ AB , $\Gamma\Delta$ · λέγω, ὅτι αἱ AB , $\Gamma\Delta$ ἴσον ἀπέθουσιν ἀπὸ τοῦ κέντρου.

Εἰλήφθω γάρ τὸ κέντρον τοῦ $AB\Gamma\Delta$ κύκλου καὶ ἔστω τὸ E , καὶ ἀπὸ τοῦ E ἐπὶ τὰς AB , $\Gamma\Delta$ κάθετοι ἡθθωσαν αἱ EZ , EH , καὶ ἐπεξεύθθωσαν αἱ AE , EG .

Ἐπεὶ οὖν εὐθεῖα τις δια τοῦ κέντρου ἡ EZ εὐθεῖάν τινα μὴ διὰ τοῦ κέντρου τὴν AB πρὸς ὀρθὰς τέμνει, καὶ δίχα αὐτὴν τέμνει. ἴση ἄρα ἡ AZ τῇ ZB · διπλὴ ἄρα ἡ AB τῆς AZ . διὰ τὰ αὐτὰ δὴ καὶ ἡ $\Gamma\Delta$ τῆς GH ἐστι διπλὴ· καὶ ἐστὶν ἴση ἡ AB τῇ $\Gamma\Delta$ · ἴση ἄρα καὶ ἡ AZ τῇ GH . καὶ ἐπεὶ ἴση ἐστὶν ἡ AE τῇ EG , ἴσον καὶ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς EG . ἀλλὰ τῷ μὲν ἀπὸ τῆς AE ἴσα τὰ ἀπὸ τῶν AZ , EZ · ὀρθὴ γάρ ἡ πρὸς τῷ Z γωνία· τῷ δὲ ἀπὸ τῆς EG ἴσα τὰ ἀπὸ τῶν EH , HG · ὀρθὴ γάρ ἡ πρὸς τῷ H γωνία· τὰ ἄρα ἀπὸ τῶν AZ , ZE ἴσα ἐστὶ τοῖς ἀπὸ τῶν GH , HE , ὧν τὸ ἀπὸ τῆς AZ ἴσον ἐστὶ τῷ ἀπὸ τῆς GH · ἴση γάρ ἐστιν ἡ AZ τῇ GH · λοιπὸν ἄρα τὸ ἀπὸ τῆς ZE τῷ ἀπὸ τῆς EH ἴσον ἐστὶν· ἴση ἄρα ἡ EZ τῇ EH . ἐν δὲ κύκλῳ ἴσον ἀπέθειν ἀπὸ τοῦ κέντρου εὐθεῖαι λέγονται, ὅταν αἱ ἀπὸ τοῦ κέντρου ἐπ' αὐτὰς κάθετοι ἀγόμεναι ἴσαι ᾤσιν· αἱ ἄρα AB , $\Gamma\Delta$ ἴσον ἀπέθουσιν ἀπὸ τοῦ κέντρου.

Ἀλλὰ δὴ αἱ AB , $\Gamma\Delta$ εὐθεῖαι ἴσον ἀπεθέτωσαν ἀπὸ τοῦ κέντρου, τουτέστιν ἴση ἔστω ἡ EZ τῇ EH . λέγω, ὅτι ἴση ἐστὶ καὶ ἡ AB τῇ $\Gamma\Delta$.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχνομεν, ὅτι

Let $ABDC^{\dagger}$ be a circle, and let AB and CD be equal straight-lines within it. I say that AB and CD are equally far from the center.

For let the center of circle $ABDC$ be found [Prop. 3.1], and let it be (at) E . And let EF and EG be drawn from (point) E , perpendicular to AB and CD (respectively) [Prop. 1.12]. And let AE and EC be joined.

Therefore, since some straight-line, EF , through the center (of the circle), cuts some (other) straight-line, AB , not through the center, at right-angles, it also cuts it in half [Prop. 3.3]. Thus, AF (is) equal to FB . Thus, AB (is) double AF . So, for the same (reasons), CD is also double CG . And AB is equal to CD . Thus, AF (is) also equal to CG . And since AE is equal to EC , the (square) on AE (is) also equal to the (square) on EC . But, the (sum of the squares) on AF and EF (is) equal to the (square) on AE . For the angle at F (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on EG and GC (is) equal to the (square) on EC . For the angle at G (is) a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AF and FE is equal to the (sum of the squares) on CG and GE , of which the (square) on AF is equal to the (square) on CG . For AF is equal to CG . Thus, the remaining (square) on FE is equal to the (remaining square) on EG . Thus, EF (is) equal to EG . And straight-lines in a circle are said to be equally far from the center when perpendicular (straight-lines) which are drawn to them from the center are equal [Def. 3.4]. Thus, AB

διπλὴ ἐστὶν ἡ μὲν AB τῆς AZ , ἡ δὲ $\Gamma\Delta$ τῆς ΓH · καὶ ἐπεὶ ἴση ἐστὶν ἡ AE τῇ ΓE , ἴσον ἐστὶ τὸ ἀπὸ τῆς AE τῶ ἀπὸ τῆς ΓE · ἀλλὰ τῶ μὲν ἀπὸ τῆς AE ἴσα ἐστὶ τὰ ἀπὸ τῶν EZ , ZA , τῶ δὲ ἀπὸ τῆς ΓE ἴσα τὰ ἀπὸ τῶν EH , $H\Gamma$. τὰ ἄρα ἀπὸ τῶν EZ , ZA ἴσα ἐστὶ τοῖς ἀπὸ τῶν EH , $H\Gamma$. ὦν τὸ ἀπὸ τῆς EZ τῶ ἀπὸ τῆς EH ἐστὶν ἴσον· ἴση γάρ ἡ EZ τῇ EH . λοιπὸν ἄρα τὸ ἀπὸ τῆς AZ ἴσον ἐστὶ τῶ ἀπὸ τῆς ΓH . ἴση ἄρα ἡ AZ τῇ ΓH · καὶ ἐστὶ τῆς μὲν AZ διπλὴ ἡ AB , τῆς δὲ ΓH διπλὴ ἡ $\Gamma\Delta$. ἴση ἄρα ἡ AB τῇ $\Gamma\Delta$.

Ἐν κύκλῳ ἄρα αἱ ἴσαι εὐθεῖαι ἴσον ἀπέθουσι ἀπὸ τοῦ κέντρου, καὶ αἱ ἴσον ἀπέθουσαι ἀπὸ τοῦ κέντρου ἴσαι ἀλλήλαις εἰσὶν· ὅπερ ἔδει δεῖξαι.

and CD are equally far from the center.

So, let the straight-lines AB and CD be equally far from the center. That is to say, let EF be equal to EG . I say that AB is also equal to CD .

For, with the same construction, we can, similarly, show that AB is double AF , and CD (double) CG . And since AE is equal to CE , the (square) on AE is equal to the (square) on CE . But, the (sum of the squares) on EF and FA is equal to the (square) on AE [Prop. 1.47]. And the (sum of the squares) on EG and GC (is) equal to the (square) on CE [Prop. 1.47]. Thus, the (sum of the squares) on EF and FA is equal to the (sum of the squares) on EG and GC , of which the (square) on EF is equal to the (square) on EG . For EF (is) equal to EG . Thus, the remaining (square) on AF is equal to the (remaining square) on CG . Thus, AF (is) equal to CG . And AB is double AF , and CD double CG . Thus, AB (is) equal to CD .

Thus, in a circle, equal straight-lines are equally far from the center, and (straight-lines) which are equally far from the center are equal to one another. (Which is) the very thing it was required to show.

† The Greek text has “ $ABCD$ ”, which is obviously a mistake.

15.

Ἐν κύκλῳ μεγίστη μὲν ἡ διάμετρος, τῶν δὲ ἄλλων αἰεὶ ἡ ἕγγιον τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν.

Ἐστω κύκλος ὁ $AB\Gamma\Delta$, διάμετρος δὲ αὐτοῦ ἔστω ἡ $A\Delta$, κέντρον δὲ τὸ E , καὶ ἕγγιον μὲν τῆς $A\Delta$ διαμέτρου ἔστω ἡ $B\Gamma$, ἀπώτερον δὲ ἡ ZH . λέγω, ὅτι μεγίστη μὲν ἐστὶν ἡ $A\Delta$, μείζων δὲ ἡ $B\Gamma$ τῆς ZH .

Ἦθθωσαν γάρ ἀπὸ τοῦ E κέντρου ἐπὶ τὰς $B\Gamma$, ZH κάθετοι αἱ $E\Theta$, EK . καὶ ἐπεὶ ἕγγιον μὲν τοῦ κέντρου ἐστὶν ἡ $B\Gamma$, ἀπώτερον δὲ ἡ ZH , μείζων ἄρα ἡ EK τῆς $E\Theta$. κείσθω τῇ $E\Theta$ ἴση ἡ $ΕΛ$, καὶ διὰ τοῦ Λ τῇ EK πρὸς ὀρθὰς ἀθθεῖσα ἡ ΛM διήθθω ἐπὶ τὸ N , καὶ ἐπεξέυθθωσαν αἱ ME , EN , ZE , EH .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ $E\Theta$ τῇ $ΕΛ$, ἴση ἐστὶ καὶ ἡ $B\Gamma$ τῇ MN . πάλιν, ἐπεὶ ἴση ἐστὶν ἡ μὲν AE τῇ EM , ἡ δὲ $ΕΔ$ τῇ EN , ἡ ἄρα $A\Delta$ ταῖς ME , EN ἴση ἐστίν. ἀλλ’ αἱ μὲν ME , EN τῆς MN μείζονές εἰσιν [καὶ ἡ $A\Delta$ τῆς MN μείζων ἐστίν], ἴση δὲ ἡ MN τῇ $B\Gamma$. ἡ $A\Delta$ ἄρα τῆς $B\Gamma$ μείζων ἐστίν. καὶ ἐπεὶ δύο αἱ ME , EN δύο ταῖς ZE , EH ἴσαι εἰσὶν, καὶ γωνία ἡ ὑπὸ MEN γωνία τῆς ὑπὸ ZEH μείζων [ἐστίν], βάσις ἄρα ἡ MN βάσεως τῆς ZH μείζων ἐστίν. ἀλλὰ ἡ MN τῇ $B\Gamma$ ἐδείχθη ἴση [καὶ ἡ $B\Gamma$ τῆς ZH μείζων ἐστίν]. μεγίστη μὲν ἄρα ἡ $A\Delta$ διάμετρος, μείζων δὲ ἡ $B\Gamma$ τῆς ZH .

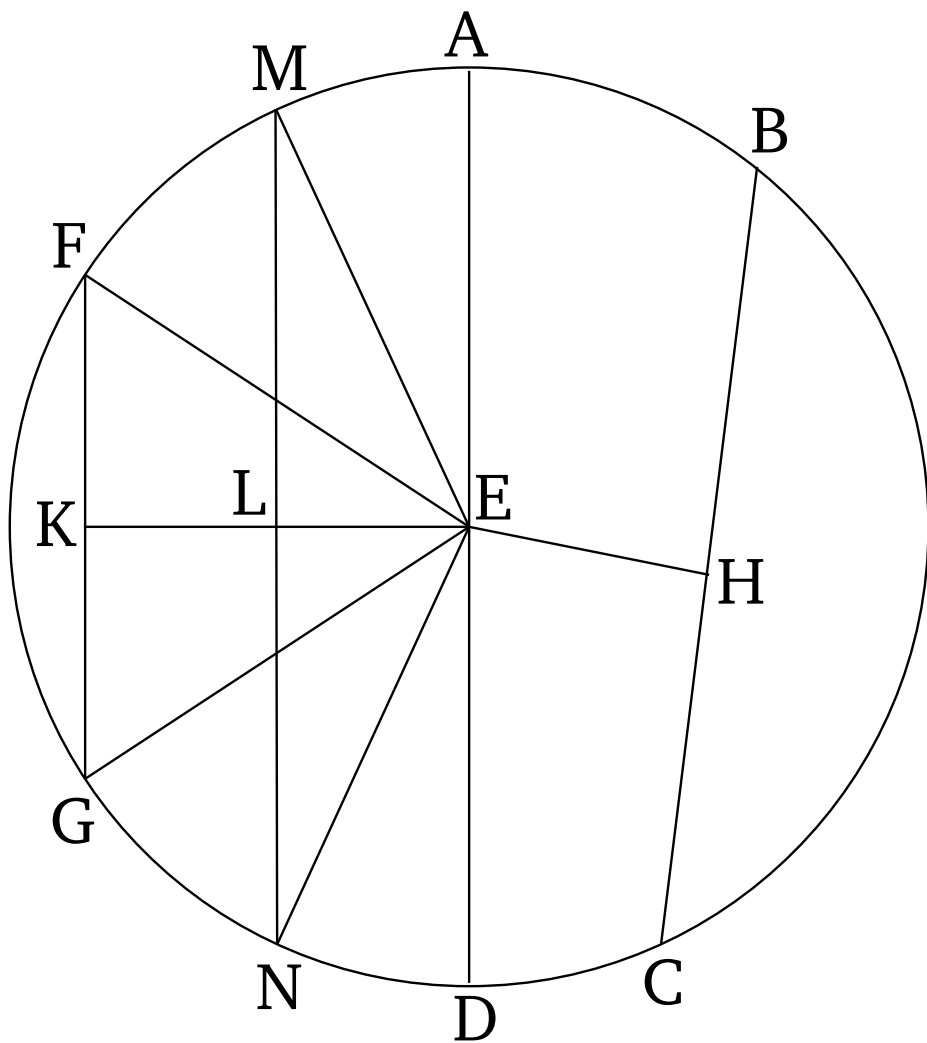
Proposition 15

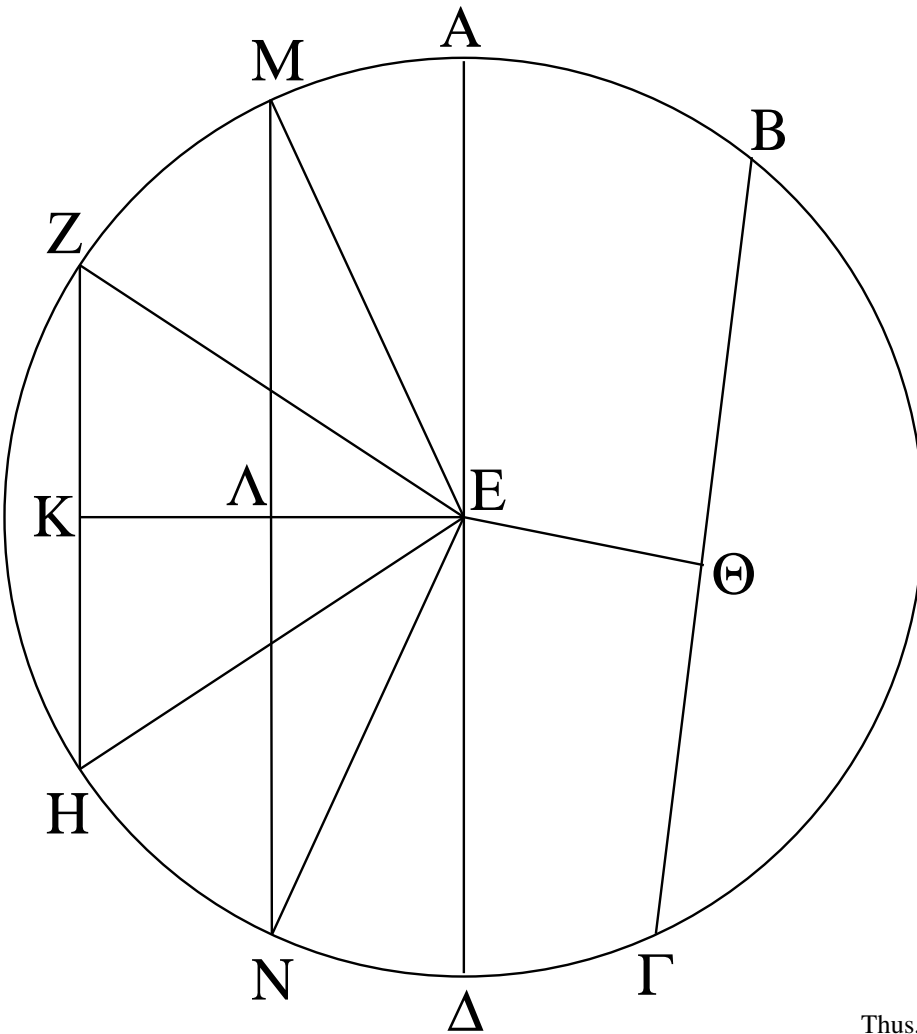
In a circle, a diameter (is) the greatest (straight-line), and for the others, a (straight-line) nearer to the center is always greater than one further away.

Let $ABCD$ be a circle, and let AD be its diameter, and E (its) center. And let BC be nearer to the diameter AD ,[†] and FG further away. I say that AD is the greatest (straight-line), and BC (is) greater than FG .

For let EH and EK be drawn from the center E , at right-angles to BC and FG (respectively) [Prop. 1.12]. And since BC is nearer to the center, and FG further away, EK (is) thus greater than EH [Def. 3.5]. Let EL be made equal to EH [Prop. 1.3]. And LM being drawn through L , at right-angles to EK [Prop. 1.11], let it be drawn through to N . And let ME , EN , FE , and EG be joined.

And since EH is equal to EL , BC is also equal to MN [Prop. 3.14]. Again, since AE is equal to EM , and ED to EN , AD is thus equal to ME and EN . But, ME and EN is greater than MN [Prop. 1.20] [also AD is greater than MN], and MN (is) equal to BC . Thus, AD is greater than BC . And since the two (straight-lines) ME , EN are equal to the two (straight-lines) FE , EG (respectively), and angle MEN [is] greater than angle FEG ,[‡] the base MN is thus greater than the base FG [Prop. 1.24]. But, MN was shown (to be) equal to BC [(so) BC is also greater than FG]. Thus, the diameter AD (is) the greatest (straight-line), and BC (is) greater than FG .





Ἐν κύκλῳ ἄρα μέγιστη μὲν ἐστὶν ἡ διάμετρος, τῶν δὲ ἄλλων αἰεὶ ἡ ἔγγιον τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν· ὅπερ ἔδει δεῖξαι.

Thus, in a circle, a diameter (is) the greatest (straight-line), and for the others, a (straight-line) nearer to the center is always greater than one further away. (Which is) the very thing it was required to show.

† Euclid should have said “to the center”, rather than “to the diameter AD”, since BC, AD and FG are not necessarily parallel.

‡ This is not proved, except by reference to the figure.

16.

Ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ’ ἀκρᾶς ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου, καὶ εἷς τὸν μεταχὺ τόπον τῆς τε εὐθείας καὶ τῆς περιφερείας ἑτέρα εὐθεΐα οὐ παρεμπεσεῖται, καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἀπάσης γωνίας ὀχείας εὐθυγράμμων μείζων ἐστίν, ἡ δὲ λοιπὴ ἐλάττων.

Ἐστω κύκλος ὁ ABΓ περὶ κέντρον τὸ Δ καὶ διάμετρον τὴν AB· λέγω, ὅτι ἡ ἀπὸ τοῦ A τῇ AB πρὸς ὀρθὰς ἀπ’ ἀκρᾶς ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου.

μὴ γάρ, ἀλλ’ εἰ δυνατόν, πιπτέτω ἐντὸς ὡς ἡ ΓΑ, καὶ ἐπεξεύθθω ἡ ΔΓ.

Proposition 16

A (straight-line) drawn at right-angles to the diameter of a circle, from its end, will fall outside the circle. And another straight-line cannot be inserted into the space between the (aforementioned) straight-line and the circumference. And the angle of the semi-circle is greater than any acute rectilinear angle whatsoever, and the remaining (angle is) less (than any acute rectilinear angle).

Let ABC be a circle around the center D and the diameter AB. I say that the (straight-line) drawn from A, at right-angles to AB [Prop 1.11], from its end, will fall outside the circle.

Ἐπεὶ ἴση ἐστὶν ἡ ΔA τῇ $\Delta \Gamma$, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $\Delta A \Gamma$ γωνία τῇ ὑπὸ $A \Gamma \Delta$. ὀρθὴ δὲ ἡ ὑπὸ $\Delta A \Gamma$. ὀρθὴ ἄρα καὶ ἡ ὑπὸ $A \Gamma \Delta$. τριγώνου δὲ τοῦ $A \Gamma \Delta$ αἱ δύο γωνίαι αἱ ὑπὸ $\Delta A \Gamma$, $A \Gamma \Delta$ δύο ὀρθαῖς ἴσαι εἰσὶν· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ A σημείου τῇ BA πρὸς ὀρθὰς ἀγομένη ἐντὸς πεσεῖται τοῦ κύκλου. ὁμοίως δὲ δεῖχμεν, ὅτι οὐδ' ἐπὶ τῇς περιφερείας· ἐκτὸς ἄρα.

Πιπτέτω ὡς ἡ AE . λέγω δὲ, ὅτι εἰς τὸν μεταχὺν τόπον τῇς τε AE εὐθείας καὶ τῇς $\Gamma\Theta A$ περιφερείας ἑτέρα εὐθεῖα οὐ παρεμπεσεῖται.

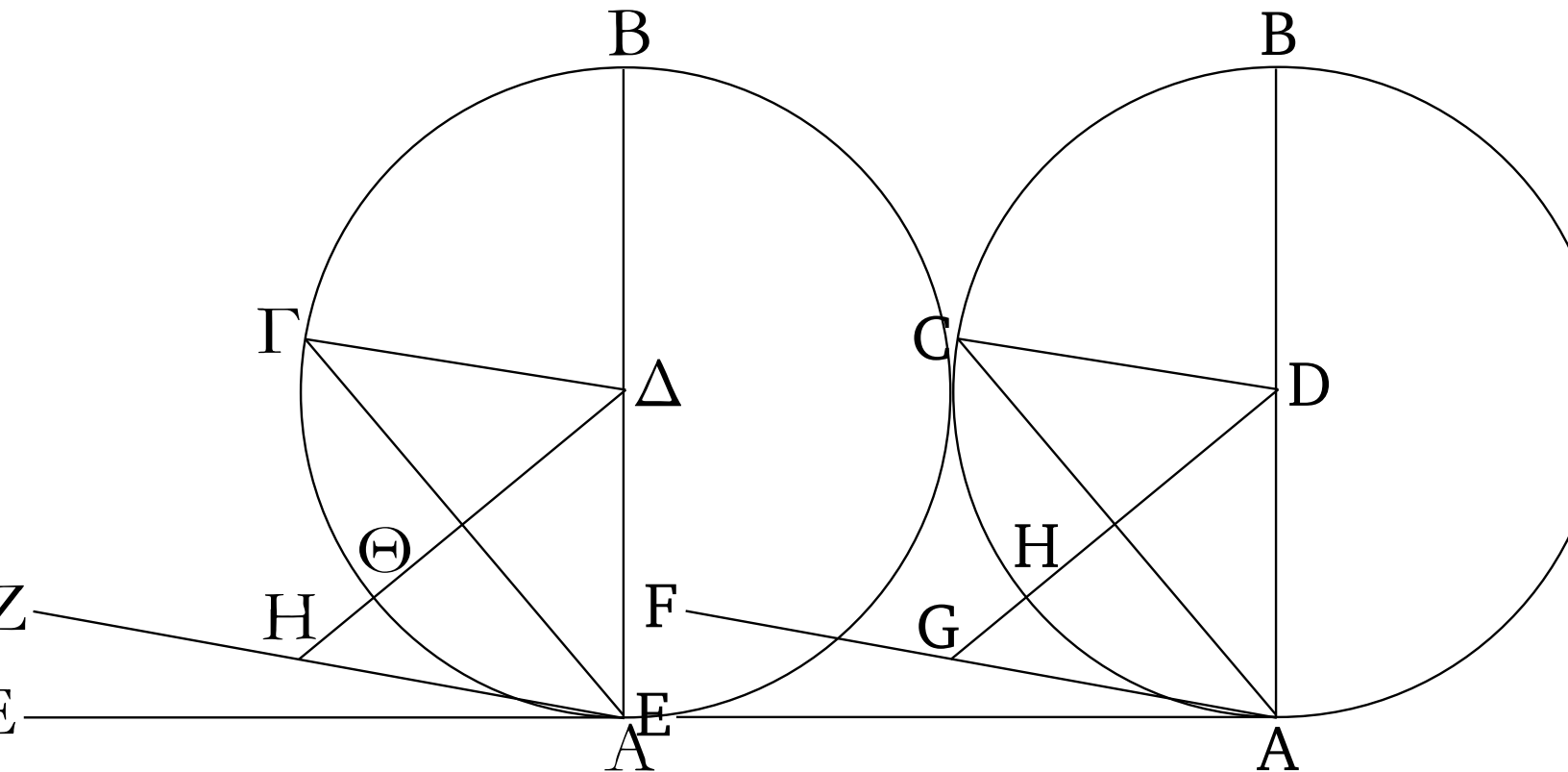
Εἰ γὰρ δυνατόν, παρεμπιπτέτω ὡς ἡ ZA , καὶ ἤθθω ἀπὸ τοῦ Δ σημείου ἐπὶ τὴν ZA κάθετος ἡ ΔH . καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ ὑπὸ $AH\Delta$, ἐλάττων δὲ ὀρθῆς ἡ ὑπὸ $\Delta A H$, μείζων ἄρα ἡ $A\Delta$ τῇς ΔH . ἴση δὲ ἡ ΔA τῇ $\Delta\Theta$. μείζων ἄρα ἡ $\Delta\Theta$ τῇς ΔH , ἢ ἐλάττων τῇς μείζονος· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα εἰς τὸν μεταχὺν τόπον τῇς τε εὐθείας καὶ τῇς περιφερείας ἑτέρα εὐθεῖα παρεμπεσεῖται.

For (if) not (then), if possible, let it fall inside, like CA (in the figure), and let DC be joined.

Since DA is equal to DC , angle DAC is also equal to angle ACD [Prop. 1.5]. And DAC (is) a right-angle. Thus, ACD (is) also a right-angle. So, in triangle ACD , the two angles DAC and ACD are equal to two right-angles. The very thing is impossible [Prop. 1.17]. Thus, the (straight-line) drawn from point A , at right-angles to BA , will not fall inside the circle. So, similarly, we can show that neither (will it fall) on the circumference. Thus, (it will fall) outside (the circle).

Let it fall like AE (in the figure). So, I say that another straight-line cannot be inserted into the space between the straight-line AE and the circumference CHA .

For, if possible, let it be inserted like FA (in the figure), and let DG be drawn from point D , perpendicular to FA [Prop. 1.12]. And since AGD is a right-angle, and DAG (is) less than a right-angle, AD (is) thus greater than DG [Prop. 1.19]. And DA (is) equal to DH . Thus, DH (is) greater than DG , the lesser than the greater. The very thing is impossible. Thus, another straight-line cannot be inserted into the space between the straight-line (AE) and the circumference.



Λέγω, ὅτι καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἡ περιεθρομένη ὑπὸ τε τῇς BA εὐθείας καὶ τῇς $\Gamma\Theta A$ περιφερείας ἀπόσης γωνίαξ ὀχλείας εὐθυγράμμου μείζων ἐστίν, ἢ δὲ λοιπὴ ἡ περιεθρομένη ὑπὸ τε τῇς $\Gamma\Theta A$ περιφερείας καὶ τῇς AE εὐθείας

And I also say that the semi-circular angle contained by the straight-line BA and the circumference CHA is greater than any acute rectilinear angle whatsoever, and the remaining (angle) contained by the circumference CHA and the straight-line

ἀπάσης γωνίας ὀξείας εὐθυγράμμων ἐλάττων ἐστίν.

Εἰ γὰρ ἐστί τις γωνία εὐθύγραμμος μείζων μὲν τῆς περιεθρομένης ὑπὸ τε τῆς BA εὐθείας καὶ τῆς $ΓΘΑ$ περιφερείας, ἐλάττων δὲ τῆς περιεθρομένης ὑπὸ τε τῆς $ΓΘΑ$ περιφερείας καὶ τῆς $ΑΕ$ εὐθείας, εἰς τὸν μεταχὺ τόπον τῆς τε $ΓΘΑ$ περιφερείας καὶ τῆς $ΑΕ$ εὐθείας εὐθεῖα παρεμπεσέται, ἥτις ποιήσει μείζονα μὲν τῆς περιεθρομένης ὑπὸ τε τῆς BA εὐθείας καὶ τῆς $ΓΘΑ$ περιφερείας ὑπὸ εὐθειῶν περιεθρομένην, ἐλάττωνα δὲ τῆς περιεθρομένης ὑπὸ τε τῆς $ΓΘΑ$ περιφερείας καὶ τῆς $ΑΕ$ εὐθείας. οὐ παρεμπίπτει δέ· οὐκ ἄρα τῆς περιεθρομένης γωνίας ὑπὸ τε τῆς BA εὐθείας καὶ τῆς $ΓΘΑ$ περιφερείας ἔσται μείζων ὀξεία ὑπὸ εὐθειῶν περιεθρομένη, οὐδὲ μὴν ἐλάττων τῆς περιεθρομένης ὑπὸ τε τῆς $ΓΘΑ$ περιφερείας καὶ τῆς $ΑΕ$ εὐθείας.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένη ἐφάπτεται τοῦ κύκλου [καὶ ὅτι εὐθεῖα κύκλου καθ' ἓν μόνον ἐφάπτεται σημεῖον, ἐπειδήπερ καὶ ἡ κατὰ δύο αὐτῶν συμβάλλονσα ἐντὸς αὐτοῦ πίπτουσα ἐδείχθη]· ὅπερ ἔδει δεῖχαι.

17.

Ἀπὸ τοῦ δοθέντος σημείου τοῦ δοθέντος κύκλου ἐφαπτομένην εὐθεῖαν γραμμὴν ἀγαγεῖν.

Ἔστω τὸ μὲν δοθὲν σημεῖον τὸ A , ὃ δὲ δοθεὶς κύκλος ὁ $ΒΓΔ$ · δεῖ δὴ ἀπὸ τοῦ A σημείου τοῦ $ΒΓΔ$ κύκλου ἐφαπτομένην εὐθεῖαν γραμμὴν ἀγαγεῖν.

AE is less than any acute rectilinear angle whatsoever.

For if any rectilinear angle is greater than the (angle) contained by the straight-line BA and the circumference CHA , or less than the (angle) contained by the circumference CHA and the straight-line AE , (then) a straight-line can be inserted into the space between the circumference CHA and the straight-line AE —anything which will make (an angle) contained by straight-lines greater than the angle contained by the straight-line BA and the circumference CHA , or less than the (angle) contained by the circumference CHA and the straight-line AE . But (such a straight-line) cannot be inserted. Thus, an acute (angle) contained by straight-lines cannot be greater than the angle contained by the straight-line BA and the circumference CHA , neither (can it be) less than the (angle) contained by the circumference CHA and the straight-line AE .

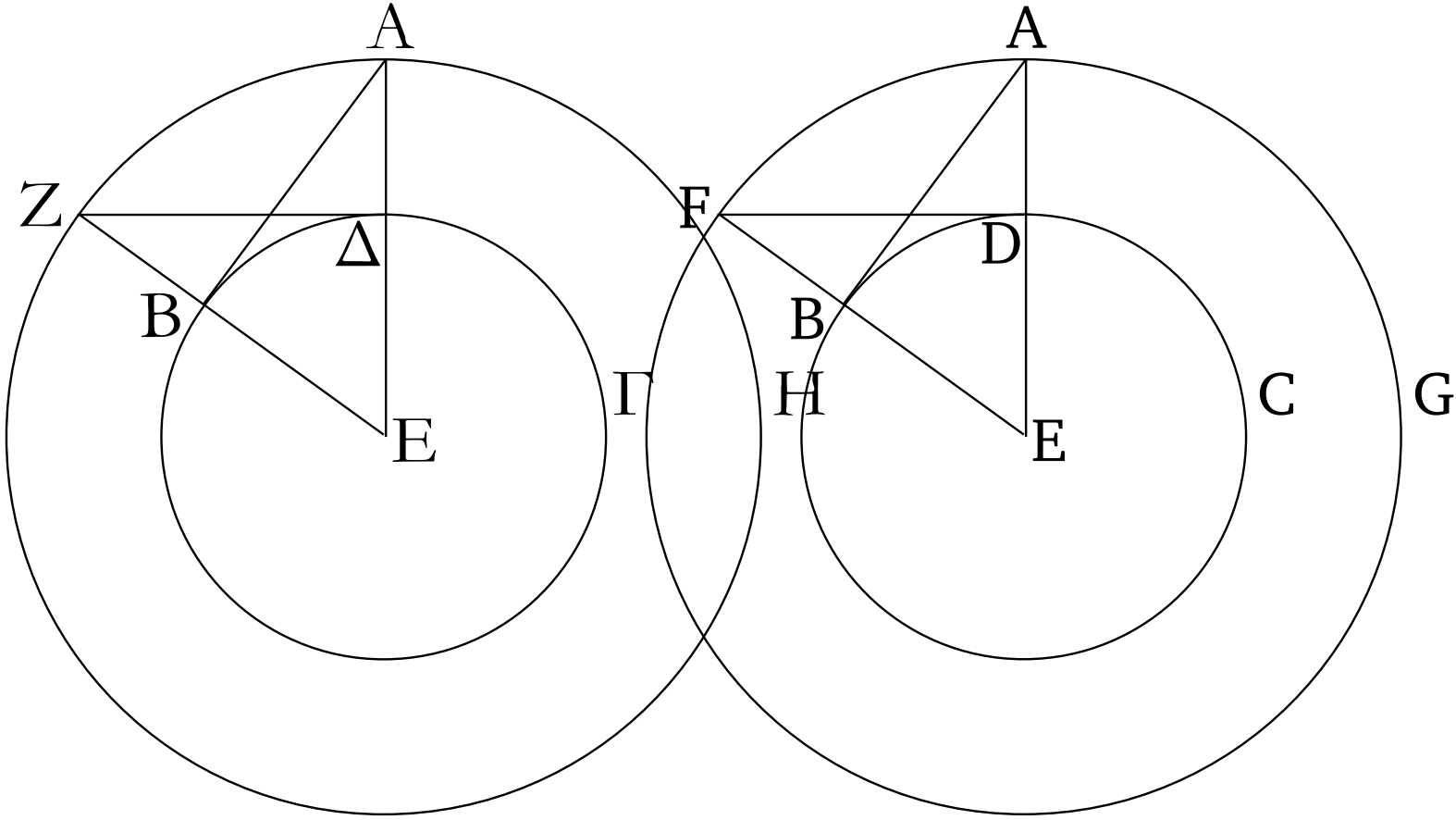
Corollary

So, from this, (it is) manifest that a (straight-line) drawn at right-angles to the diameter of a circle, from its extremity, touches the circle [and that the straight-line touches the circle at a single point, inasmuch as it was also shown that a (straight-line) meeting (the circle) at two (points) falls inside it [Prop. 3.2]]. (Which is) the very thing it was required to show.

Proposition 17

To draw a straight-line touching a given circle from a given point.

Let A be the given point, and BCD the given circle. So it is required to draw a straight-line touching circle BCD from point A .



Εἰλήφθω γάρ τὸ κέντρον τοῦ κύκλου τὸ E , καὶ ἐπεξεύθθω ἡ AE , καὶ κέντρῳ μὲν τῷ E διαστήματι δὲ τῷ EA κύκλῳ γεγράφθω ὁ AZH , καὶ ἀπὸ τοῦ Δ τῇ EA πρὸς ὀρθὰς ῥηθῇ ἡ ΔZ , καὶ ἐπεξεύθθωσαν αἱ EZ , AB . λέγω, ὅτι ἀπὸ τοῦ A σημείου τοῦ $B\Gamma\Delta$ κύκλου ἐφαπτομένη ἦκται ἡ AB .

Ἐπεὶ γάρ τὸ E κέντρον ἐστὶ τῶν $B\Gamma\Delta$, AZH κύκλων, ἴση ἄρα ἐστὶν ἡ μὲν EA τῇ EZ , ἡ δὲ $E\Delta$ τῇ EB . δύο δὲ αἱ AE , EB δύο ταῖς ZE , $E\Delta$ ἴσαι εἰσὶν· καὶ γωνίαν κοινὴν περιέθουσι τὴν πρὸς τῷ E . βάσει ἄρα ἡ ΔZ βάσει τῇ AB ἴση ἐστίν, καὶ τὸ ΔEZ τρίγωνον τῷ EBA τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις· ἴση ἄρα ἡ ὑπὸ $E\Delta Z$ τῇ ὑπὸ EBA . ὀρθὴ δὲ ἡ ὑπὸ $E\Delta Z$. ὀρθὴ ἄρα καὶ ἡ ὑπὸ EBA . καὶ ἐστὶν ἡ EB ἐκ τοῦ κέντρον· ἡ δὲ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἀκραξ ἀγομένη ἐφάπτεται τοῦ κύκλου· ἡ AB ἄρα ἐφάπτεται τοῦ $B\Gamma\Delta$ κύκλου.

Ἀπὸ τοῦ ἄρα δοθέντος σημείου τοῦ A τοῦ δοθέντος κύκλου τοῦ $B\Gamma\Delta$ ἐφαπτομένη εὐθεῖα γραμμὴ ἦκται ἡ AB . ὅπερ ἔδει ποιῆσαι.

For let the center E of the circle be found [Prop. 3.1], and let AE be joined. And let (the circle) AFG be drawn with center E and radius EA . And let DF be drawn from (point) D , at right-angles to EA [Prop. 1.11]. And let EF and AB be joined. I say that the (straight-line) AB has been drawn from point A touching circle BCD .

For since E is the center of circles BCD and AFG , EA is thus equal to EF , and ED to EB . So the two (straight-lines) AE , EB are equal to the two (straight-lines) FE , ED (respectively). And they contain a common angle at E . Thus, the base DF is equal to the base AB , and triangle DEF is equal to triangle EBA , and the remaining angles (are equal) to the (corresponding) remaining angles [Prop. 1.4]. Thus, (angle) EDF (is) equal to EBA . And EDF (is) a right-angle. Thus, EBA (is) also a right-angle. And EB is a radius. And a (straight-line) drawn at right-angles to the diameter of a circle, from its extremity, touches the circle [Prop. 3.16 corr.]. Thus, AB touches circle BCD .

Thus, the straight-line AB has been drawn touching the given circle BCD from the given point A . (Which is) the very thing it was required to do.

Ἐάν κύκλον ἐφάπτηται τις εὐθεΐα, ἀπό δὲ τοῦ κέντρου ἐπὶ τὴν ἀφὴν ἐπιξενθῇ τις εὐθεΐα, ἡ ἐπιξενθθεῖσα κάθετος ἔσται ἐπὶ τὴν ἐφαπτομένην.

Κύκλον γάρ τοῦ $ABΓ$ ἐφαπτέσθω τις εὐθεΐα ἡ $ΔΕ$ κατὰ τὸ $Γ$ σημεῖον, καὶ εἰλήφθω τὸ κέντρον τοῦ $ABΓ$ κύκλου τὸ Z , καὶ ἀπὸ τοῦ Z ἐπὶ τὸ $Γ$ ἐπεξενθῇ ἡ $ZΓ$. λέγω, ὅτι ἡ $ZΓ$ κάθετός ἐστιν ἐπὶ τὴν $ΔΕ$.

Εἰ γὰρ μή, ἦθθω ἀπὸ τοῦ Z ἐπὶ τὴν $ΔΕ$ κάθετος ἡ $ZΗ$.

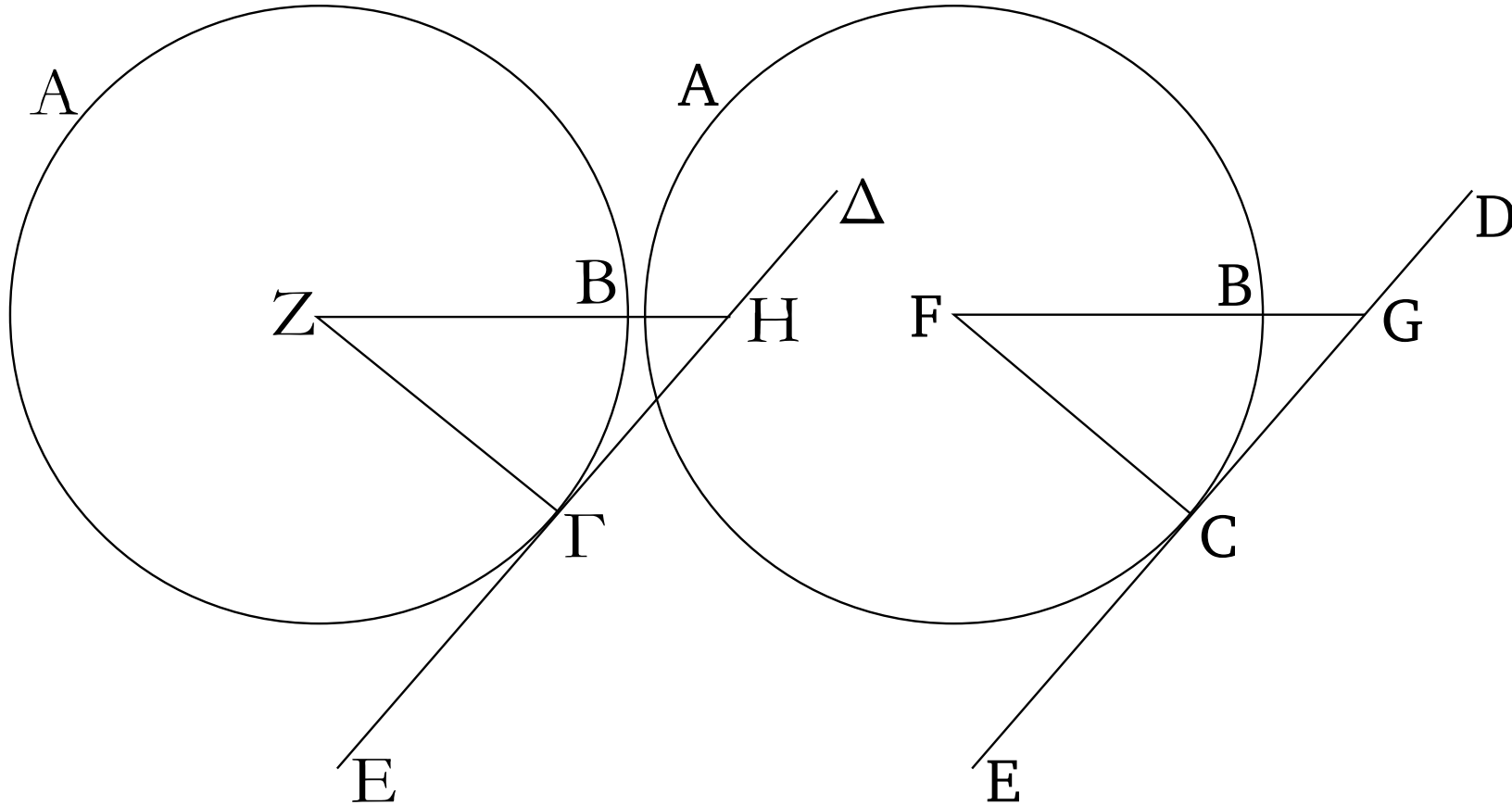
Ἐπεὶ οὖν ἡ ὑπὸ $ZΗΓ$ γωνία ὀρθή ἐστιν, ὁχεῖα ἄρα ἐστὶν ἡ ὑπὸ $ZΓΗ$. ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρά ὑποτείνει· μείζων ἄρα ἡ $ZΓ$ τῆς $ZΗ$. ἴση δὲ ἡ $ZΓ$ τῇ ZB . μείζων ἄρα καὶ ἡ ZB τῆς $ZΗ$ ἢ ἐλάττων τῆς μείζονος· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ $ZΗ$ κάθετός ἐστιν ἐπὶ τὴν $ΔΕ$. ὁμοίως δὲ δεῖχμεν, ὅτι οὐδ' ἄλλη τις πλὴν τῆς $ZΓ$. ἡ $ZΓ$ ἄρα κάθετός ἐστιν ἐπὶ τὴν $ΔΕ$.

If some straight-line touches a circle, and some (other) straight-line is joined from the center (of the circle) to the point of contact, (then) the (straight-line) so joined will be perpendicular to the tangent.

For let some straight-line DE touch the circle ABC at point C , and let the center F of circle ABC be found [Prop. 3.1], and let FC be joined from F to C . I say that FC is perpendicular to DE .

For if not, let FG be drawn from F , perpendicular to DE [Prop. 1.12].

Therefore, since angle FGC is a right-angle, (angle) FCG is thus acute [Prop. 1.17]. And the greater angle is subtended by the greater side [Prop. 1.19]. Thus, FC (is) greater than FG . And FC (is) equal to FB . Thus, FB (is) also greater than FG , the lesser than the greater. The very thing is impossible. Thus, FG is not perpendicular to DE . So, similarly, we can show that neither (is) any other (straight-line) except FC . Thus, FC is perpendicular to DE .



Ἐάν ἄρα κύκλον ἐφάπτηται τις εὐθεΐα, ἀπό δὲ τοῦ κέντρου ἐπὶ τὴν ἀφὴν ἐπιξενθῇ τις εὐθεΐα, ἡ ἐπιξενθθεῖσα κάθετος ἔσται ἐπὶ τὴν ἐφαπτομένην· ὅπερ ἔδει δεῖχαι.

Thus, if some straight-line touches a circle, and some (other) straight-line is joined from the center (of the circle) to the point of contact, (then) the (straight-line) so joined will be perpendicular to the tangent. (Which is) the very thing it

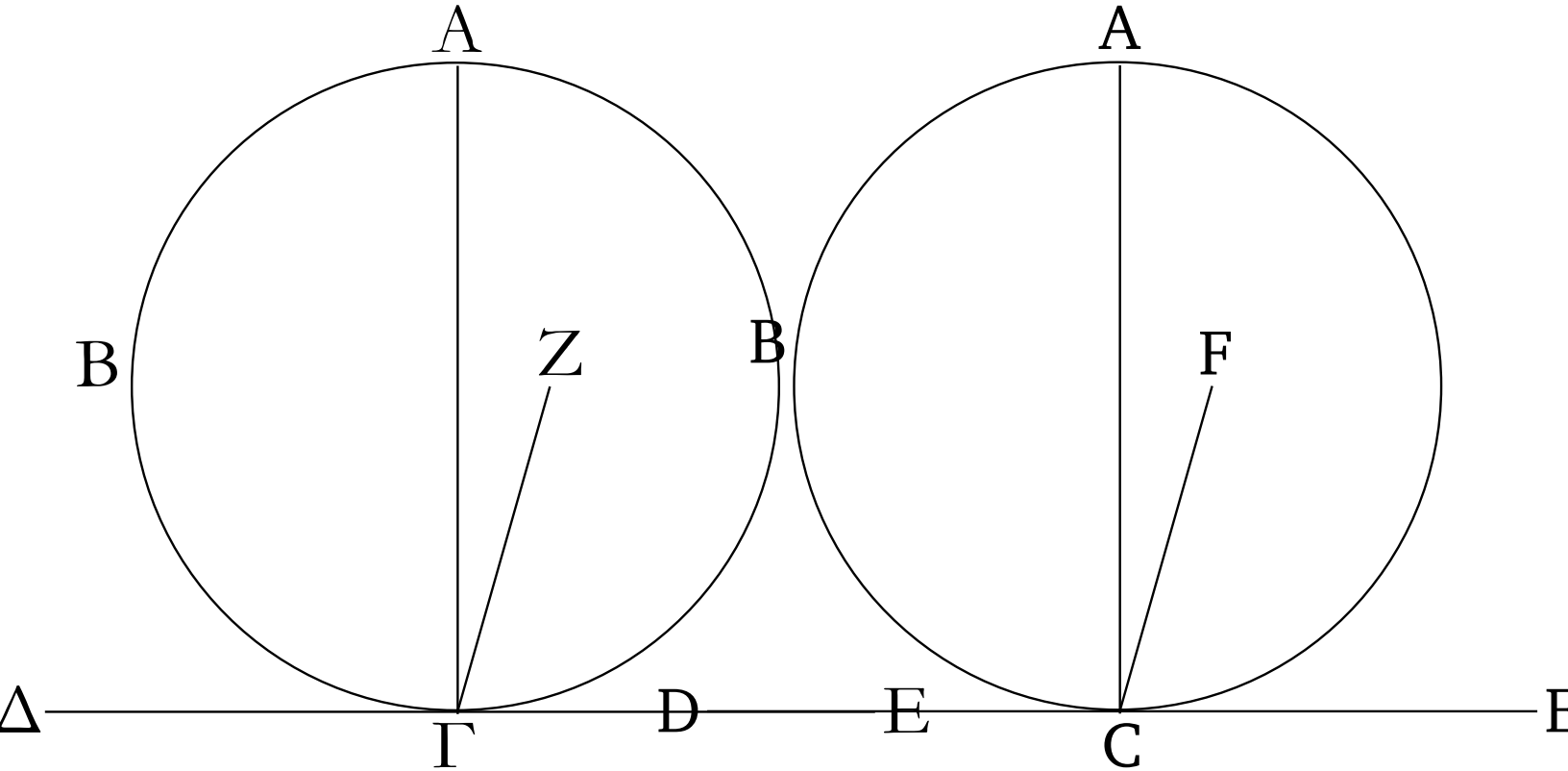
was required to show.

19.

Proposition 19

Ἐάν κύκλον ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τῆς ἀφ᾽ ἧς τῇ ἐφαπτομένῃ πρὸς ὀρθὰς [γωνίαξ] εὐθεΐα γραμμὴ ἀθθῇ, ἐπὶ τῆς ἀθθείσης ἔσται τὸ κέντρον τοῦ κύκλου.

If some straight-line touches a circle, and a straight-line is drawn from the point of contact, at right-[angles] to the tangent, (then) the center (of the circle) will be on the (straight-line) so drawn.



Κύκλον γάρ τοῦ ΑΒΓ ἐφαπτέσθω τις εὐθεΐα ἡ ΔΕ κατὰ τὸ Γ σημεῖον, καὶ ἀπὸ τοῦ Γ τῇ ΔΕ πρὸς ὀρθὰς ἡθθω ἡ ΓΑ· λέγω, ὅτι ἐπὶ τῆς ΑΓ ἔστι τὸ κέντρον τοῦ κύκλου.

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἔστω τὸ Ζ, καὶ ἐπέξενθω ἡ ΓΖ.

Ἐπεὶ [οὔν] κύκλον τοῦ ΑΒΓ ἐφάπτεται τις εὐθεΐα ἡ ΔΕ, ἀπὸ δὲ τοῦ κέντρον ἐπὶ τὴν ἀφῆν ἐπέξενεται ἡ ΖΓ, ἡ ΖΓ ἄρα κάθετόξ ἐστιν ἐπὶ τὴν ΔΕ· ὀρθή ἄρα ἐστὶν ἡ ὑπὸ ΖΓΕ. ἐστὶ δὲ καὶ ἡ ὑπὸ ΑΓΕ ὀρθή· ἴση ἄρα ἐστὶν ἡ ὑπὸ ΖΓΕ τῇ ὑπὸ ΑΓΕ ἡ ἐλάττων τῇ μείζονι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὸ Ζ κέντρον ἐστὶ τοῦ ΑΒΓ κύκλου. ὁμοίωξ δὲ δείχομεν, ὅτι οὐδ' ἄλλο τι πλὴν ἐπὶ τῆς ΑΓ.

Ἐάν ἄρα κύκλον ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τῆς ἀφ᾽ ἧς τῇ ἐφαπτομένῃ πρὸς ὀρθὰς εὐθεΐα γραμμὴ ἀθθῇ, ἐπὶ τῆς ἀθθείσης ἔσται τὸ κέντρον τοῦ κύκλου· ὅπερ ἔδει δείχαι.

For let some straight-line DE touch the circle ABC at point C . And let CA be drawn from C , at right-angles to DE [Prop. 1.11]. I say that the center of the circle is on AC .

For (if) not, if possible, let F be (the center of the circle), and let CF be joined.

[Therefore], since some straight-line DE touches the circle ABC , and FC has been joined from the center to the point of contact, FC is thus perpendicular to DE [Prop. 3.18]. Thus, FCE is a right-angle. And ACE is also a right-angle. Thus, FCE is equal to ACE , the lesser to the greater. The very thing is impossible. Thus, F is not the center of circle ABC . So, similarly, we can show that neither is any (point) other (than one) on AC .

Thus, if some straight-line touches a circle, and a straight-line is drawn from the point of contact, at right-angles to the tangent, (then) the center (of the circle) will be on the (straight-line) so drawn. (Which is) the very thing it was required to

show.

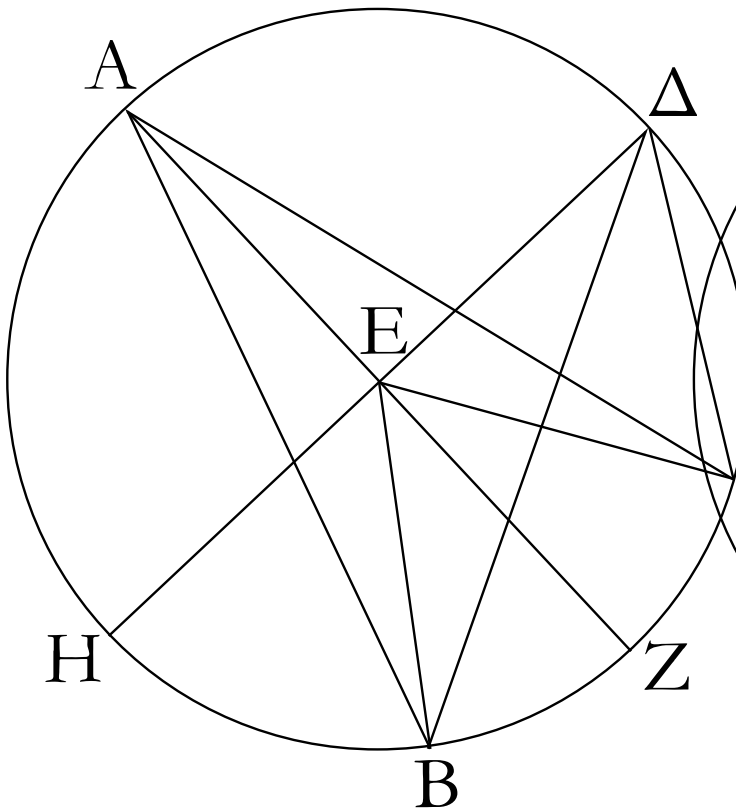
20.

Ἐν κύκλῳ ἡ πρὸς τῷ κέντρῳ γωνία διπλασίῳ ἐστὶ τῆς πρὸς τῇ περιφερείᾳ, ὅταν τὴν αὐτὴν περιφέρειαν βάσιν ἔθωσιν αἱ γωνίαι.

Ἐστω κύκλος ὁ $ABΓ$, καὶ πρὸς μὲν τῷ κέντρῳ αὐτοῦ γωνία ἔστω ἡ ὑπὸ BEG , πρὸς δὲ τῇ περιφερείᾳ ἡ ὑπὸ BAG , ἐθέτωσαν δὲ τὴν αὐτὴν περιφέρειαν βάσιν τὴν $BΓ$. λέγω, ὅτι διπλασίῳ ἐστὶν ἡ ὑπὸ BEG γωνία τῆς ὑπὸ BAG .

Ἐπιξευθθεῖσα γὰρ ἡ AE διήθθη ἐπὶ τὸ Z .

Ἐπεὶ οὖν ἴση ἐστὶν ἡ EA τῇ EB , ἴση καὶ γωνία ἡ ὑπὸ EAB τῇ ὑπὸ EBA . αἱ ἄρα ὑπὸ EAB , EBA γωνίαι τῆς ὑπὸ EAB διπλασίονες εἰσιν. ἴση δὲ ἡ ὑπὸ BEZ ταῖς ὑπὸ EAB , EBA . καὶ ἡ ὑπὸ BEZ ἄρα τῆς ὑπὸ EAB ἐστὶ διπλῇ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ZEG τῆς ὑπὸ EAG ἐστὶ διπλῇ. ὅλη ἄρα ἡ ὑπὸ BEG ὅλης τῆς ὑπὸ BAG ἐστὶ διπλῇ.



Κεκλάσθω δὴ πάλιν, καὶ ἔστω ἑτέρα γωνία ἡ ὑπὸ $BΔΓ$, καὶ ἐπιξευθθεῖσα ἡ $ΔE$ ἐκβεβλήσθω ἐπὶ τὸ H . ὁμοίως δὲ δεῖχμεν, ὅτι διπλῇ ἐστὶν ἡ ὑπὸ $HEΓ$ γωνία τῆς ὑπὸ $EΔΓ$, ὣν ἡ ὑπὸ HEB διπλῇ ἐστὶ τῆς ὑπὸ $EΔB$. λοιπὴ ἄρα ἡ ὑπὸ BEG διπλῇ ἐστὶ τῆς ὑπὸ $BΔΓ$.

Ἐν κύκλῳ ἄρα ἡ πρὸς τῷ κέντρῳ γωνία διπλασίῳ ἐστὶ τῆς πρὸς τῇ περιφερείᾳ, ὅταν τὴν αὐτὴν περιφέρειαν βάσιν ἔθωσιν [αἱ γωνίαι]. ὅπερ ἔδει δεῖξαι.

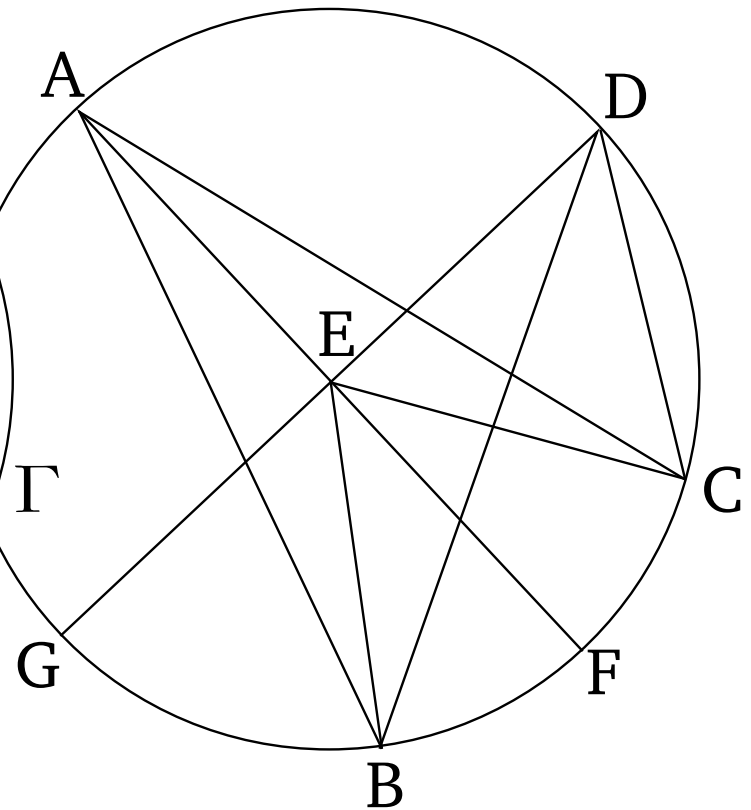
Proposition 20

In a circle, the angle at the center is double that at the circumference, when the angles have the same circumference base.

Let ABC be a circle, and let BEC be an angle at its center, and BAC (one) at (its) circumference. And let them have the same circumference base BC . I say that angle BEC is double (angle) BAC .

For being joined, let AE be drawn through to F .

Therefore, since EA is equal to EB , angle EAB (is) also equal to EBA [Prop. 1.5]. Thus, angle EAB and EBA is double (angle) EAB . And BEF (is) equal to EAB and EBA [Prop. 1.32]. Thus, BEF is also double EAB . So, for the same (reasons), FEC is also double EAC . Thus, the whole (angle) BEC is double the whole (angle) BAC .

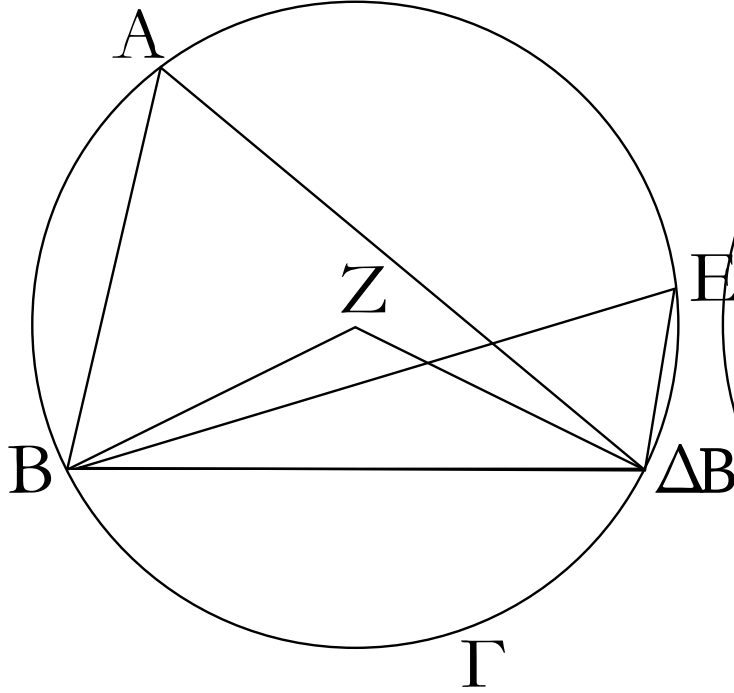


So let another (straight-line) be inflected, and let there be another angle, BDC . And DE being joined, let it be produced to G . So, similarly, we can show that angle GEC is double EDC , of which GEB is double EDB . Thus, the remaining (angle) BEC is double the (remaining angle) BDC .

Thus, in a circle, the angle at the center is double that at the circumference, when [the angles] have the same circumference base. (Which is) the very thing it was required to show.

21.

Ἐν κύκλῳ αἱ ἐν τῷ αὐτῷ τμήματι γωνίαι ἴσαι ἀλλήλαις εἰσίν.



Ἐστω κύκλος ὁ ΑΒΓΔ, καὶ ἐν τῷ αὐτῷ τμήματι τῷ ΒΑ-ΕΔ γωνίαι ἕστωσαν αἱ ὑπὸ ΒΑΔ, ΒΕΔ· λέγω, ὅτι αἱ ὑπὸ ΒΑΔ, ΒΕΔ γωνίαι ἴσαι ἀλλήλαις εἰσίν.

Εἰλήφθω γάρ τοῦ ΑΒΓΔ κύκλου τὸ κέντρον, καὶ ἕστω τὸ Ζ, καὶ ἐπεζεύθωσαν αἱ ΒΖ, ΖΔ.

Καὶ ἐπεὶ ἡ μὲν ὑπὸ ΒΖΔ γωνία πρὸς τῷ κέντρῳ ἐστίν, ἡ δὲ ὑπὸ ΒΑΔ πρὸς τῇ περιφερείᾳ, καὶ ἔθουσι τὴν αὐτὴν περιφέρειαν βάσιν τὴν ΒΓΔ, ἡ ἄρα ὑπὸ ΒΖΔ γωνία διπλασίον ἐστὶ τῆς ὑπὸ ΒΑΔ. διὰ τὰ αὐτὰ δὲ ἡ ὑπὸ ΒΖΔ καὶ τῆς ὑπὸ ΒΕΔ ἐστὶ διπλασίον· ἴση ἄρα ἡ ὑπὸ ΒΑΔ τῇ ὑπὸ ΒΕΔ.

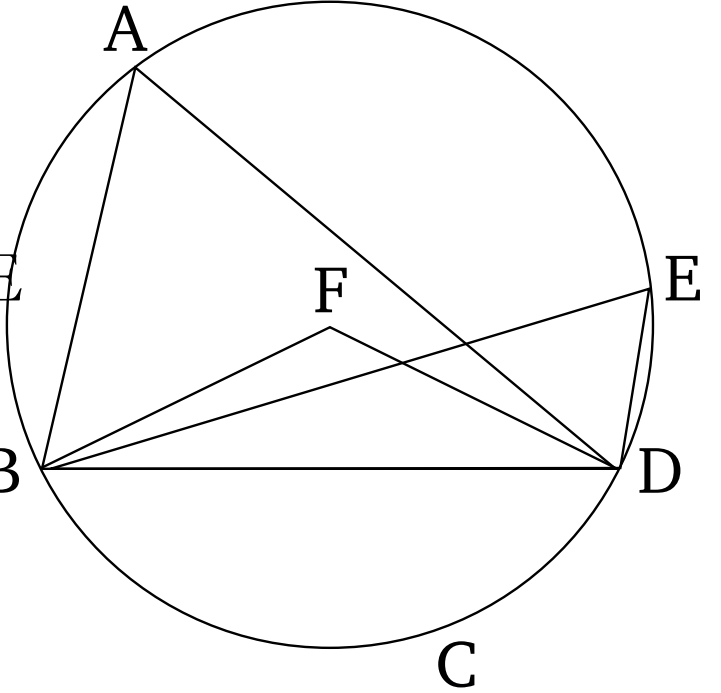
Ἐν κύκλῳ ἄρα αἱ ἐν τῷ αὐτῷ τμήματι γωνίαι ἴσαι ἀλλήλαις εἰσίν· ὅπερ ἔδει δεῖξαι.

22.

Τῶν ἐν τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυοὶν ὀρθαῖς ἴσαι εἰσίν.

Proposition 21

In a circle, angles in the same segment are equal to one another.



Let $ABCD$ be a circle, and let BAD and BED be angles in the same segment $BAED$. I say that angles BAD and BED are equal to one another.

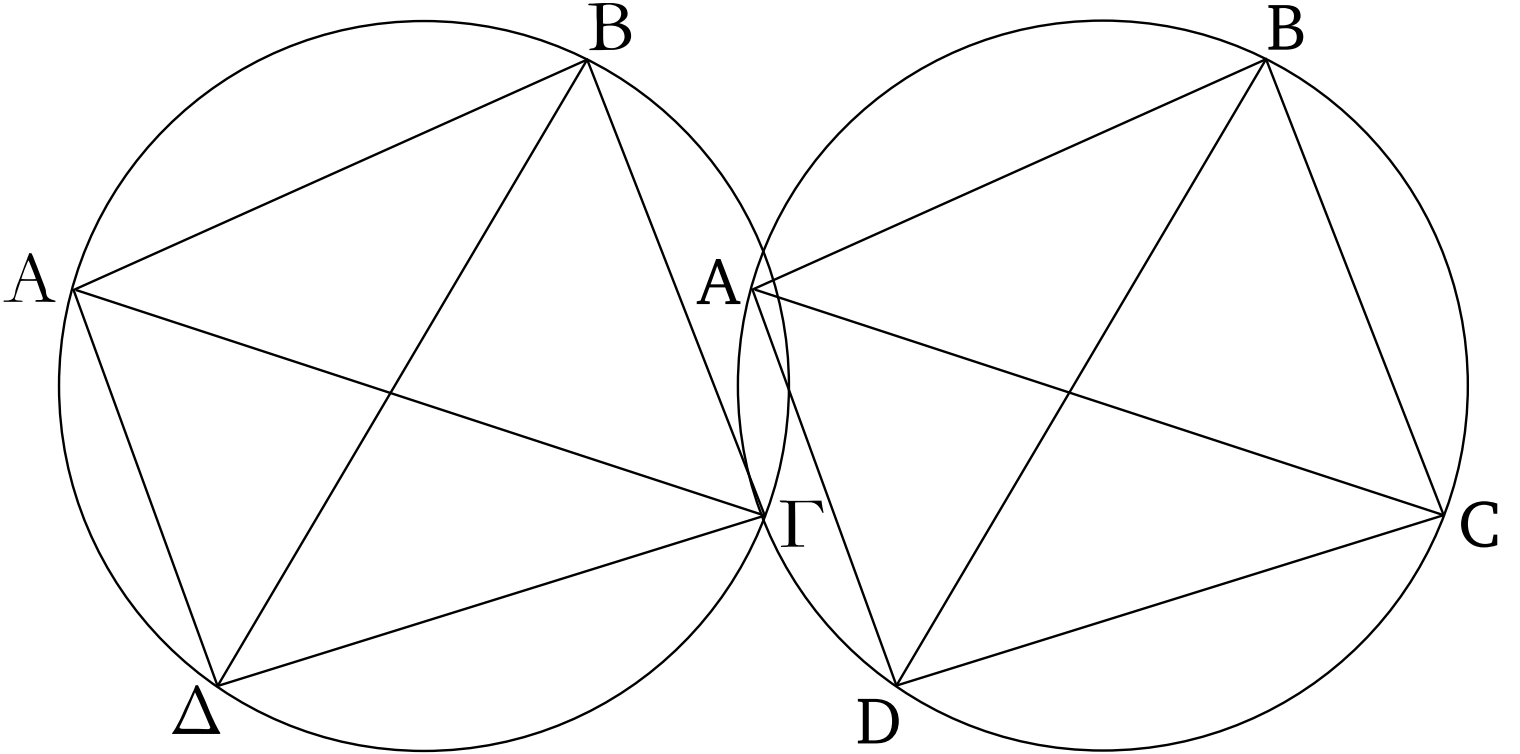
For let the center of circle $ABCD$ be found [Prop. 3.1], and let it be (at point) F . And let BF and FD be joined.

And since angle BFD is at the center, and BAD at the circumference, and they have the same circumference base BCD , angle BFD is thus double BAD [Prop. 3.20]. So, for the same (reasons), BFD is also double BED . Thus, BAD (is) equal to BED .

Thus, in a circle, angles in the same segment are equal to one another. (Which is) the very thing it was required to show.

Proposition 22

For quadrilaterals within circles, the (sum of the) opposite angles is equal to two right-angles.



Ἐστω κύκλος ὁ $AB\Gamma\Delta$, καὶ ἐν αὐτῷ τετράπλευρον ἔστω τὸ $AB\Gamma\Delta$. λέγω, ὅτι αἱ ἀπεναντίον γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν.

Ἐπεξεύϑθωσαν αἱ $A\Gamma$, $B\Delta$.

Ἐπεὶ οὖν παντὸς τριγώνου αἱ τρεῖς γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν, τοῦ $AB\Gamma$ ἄρα τριγώνου αἱ τρεῖς γωνίαι αἱ ὑπὸ ΓAB , $AB\Gamma$, $B\Gamma A$ δυοῖν ὀρθαῖς ἴσαι εἰσὶν. ἴση δὲ ἡ μὲν ὑπὸ ΓAB τῇ ὑπὸ $B\Delta\Gamma$. ἐν γὰρ τῷ αὐτῷ τμήματι εἰσι τῷ $BA\Delta\Gamma$. ἡ δὲ ὑπὸ $A\Gamma B$ τῇ ὑπὸ $A\Delta B$. ἐν γὰρ τῷ αὐτῷ τμήματι εἰσι τῷ $A\Delta\Gamma B$. ὅλη ἄρα ἡ ὑπὸ $A\Delta\Gamma$ ταῖς ὑπὸ $BA\Gamma$, $A\Gamma B$ ἴση ἐστίν. κοινὴ προσκείσθω ἡ ὑπὸ $AB\Gamma$. αἱ ἄρα ὑπὸ $AB\Gamma$, $BA\Gamma$, $A\Gamma B$ ταῖς ὑπὸ $AB\Gamma$, $A\Delta\Gamma$ ἴσαι εἰσὶν. ἀλλ' αἱ ὑπὸ $AB\Gamma$, $BA\Gamma$, $A\Gamma B$ δυοῖν ὀρθαῖς ἴσαι εἰσὶν. καὶ αἱ ὑπὸ $AB\Gamma$, $A\Delta\Gamma$ ἄρα δυοῖν ὀρθαῖς ἴσαι εἰσὶν. ὁμοίως δὲ δείχομεν, ὅτι καὶ αἱ ὑπὸ $BA\Delta$, $\Delta\Gamma B$ γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν.

Τῶν ἄρα ἐν τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν. ὅπερ ἔδει δεῖξαι.

Let $ABCD$ be a circle, and let $ABCD$ be a quadrilateral within it. I say that the (sum of the) opposite angles is equal to two right-angles.

Let AC and BD be joined.

Therefore, since the three angles of any triangle are equal to two right-angles [Prop. 1.32], the three angles CAB , ABC , and BCA of triangle ABC are thus equal to two right-angles. And CAB (is) equal to BDC . For they are in the same segment $BADC$ [Prop. 3.21]. And ACB (is equal) to ADB . For they are in the same segment $ADCB$ [Prop. 3.21]. Thus, the whole of ADC is equal to BAC and ACB . Let ABC be added to both. Thus, ABC , BAC , and ACB are equal to ABC and ADC . But, ABC , BAC , and ACB are equal to two right-angles. Thus, ABC and ADC are also equal to two right-angles. Similarly, we can show that angles BAD and DCB are also equal to two right-angles.

Thus, for quadrilaterals within circles, the (sum of the) opposite angles is equal to two right-angles. (Which is) the very thing it was required to show.

23.

Proposition 23

Ἐπὶ τῇ αὐτῇ εὐθείᾳ δύο τμήματα κύκλων ὅμοια καὶ ἄνισα οὐ συνσταθήσεται ἐπὶ τὰ αὐτὰ μέρος.

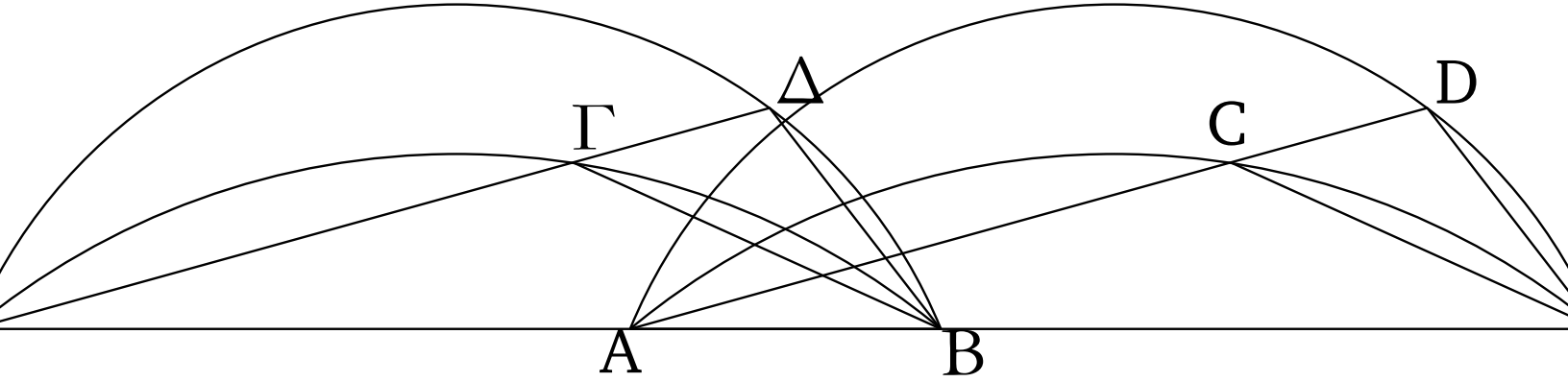
Εἰ γὰρ δυνατόν, ἐπὶ τῇ αὐτῇ εὐθείᾳ τῇ AB δύο τμήματα κύκλων ὅμοια καὶ ἄνισα συνεστιάτω ἐπὶ τὰ αὐτὰ μέρος τὰ $A\Gamma B$, $A\Delta B$, καὶ διήθθω ἡ $A\Gamma\Delta$, καὶ ἐπεξεύϑθωσαν

Two similar and unequal segments of circles cannot be constructed on the same side of the same straight-line.

For, if possible, let the two similar and unequal segments of circles, ACB and ADB , be constructed on the same side of the same straight-line AB . And let ACD be drawn through (the

αἱ GB , ΔB .

segments), and let CB and DB be joined.



Ἐπεὶ οὖν ὁμοίον ἐστὶ τὸ AGB τμήμα τῷ $A\Delta B$ τμήματι, ὅμοια δὲ τμήματα κύκλων ἐστὶ τὰ δευόμενα γωνίαξ ἴσαξ, ἴση ἄρα ἐστὶν ἡ ὑπὸ AGB γωνία τῇ ὑπὸ $A\Delta B$ ἢ ἐκτόξ τῇ ἐντόξ· ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἐπὶ τῆς αὐτῆς εὐθείας δύο τμήματα κύκλων ὅμοια καὶ ἄνισα συσταθήσεται ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἔδει δεῖξαι.

Therefore, since segment ACB is similar to segment ADB , and similar segments of circles are those accepting equal angles [Def. 3.11], angle ACB is thus equal to ADB , the external to the internal. The very thing is impossible [Prop. 1.16].

Thus, two similar and unequal segments of circles cannot be constructed on the same side of the same straight-line.

24.

Proposition 24

Τὰ ἐπὶ ἴσων εὐθειῶν ὅμοια τμήματα κύκλων ἴσα ἀλλήλοις ἐστίν.

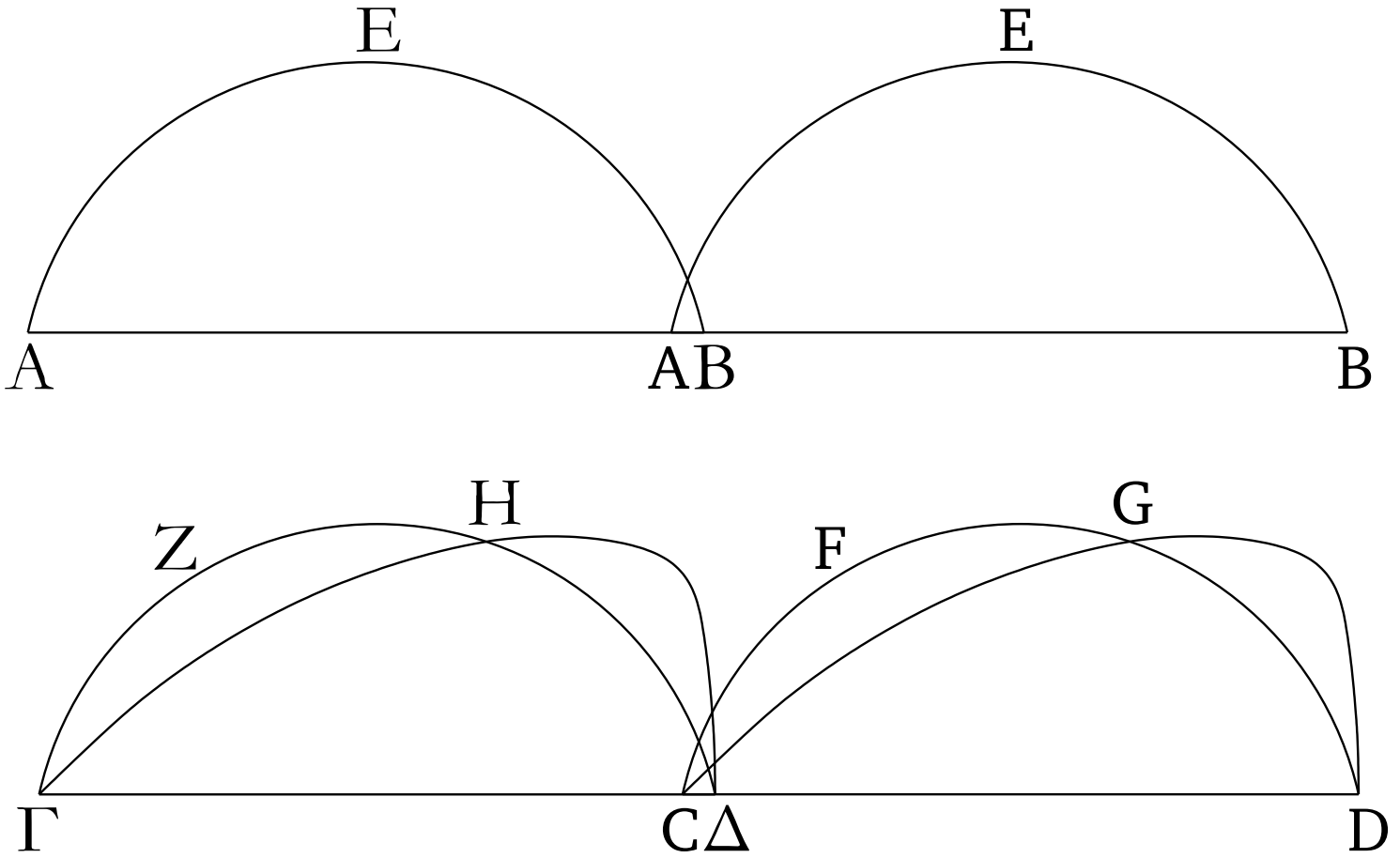
Ἐστωσαν γὰρ ἐπὶ ἴσων εὐθειῶν τῶν AB , $ΓΔ$ ὅμοια τμήματα κύκλων τὰ AEB , $ΓΖΔ$. λέγω, ὅτι ἴσον ἐστὶ τὸ AEB τμήμα τῷ $ΓΖΔ$ τμήματι.

Ἐφαρμοζόμενον γὰρ τοῦ AEB τμήματος ἐπὶ τὸ $ΓΖΔ$ καὶ τιθεμένον τοῦ μὲν A σημείου ἐπὶ τὸ $Γ$ τῆς δὲ AB εὐθείας ἐπὶ τὴν $ΓΔ$, ἐφαρμόσει καὶ τὸ B σημεῖον ἐπὶ τὸ $Δ$ σημεῖον διὰ τὸ ἴσων εἶναι τὴν AB τῇ $ΓΔ$. τῆς δὲ AB ἐπὶ τὴν $ΓΔ$ ἐφαρμοσάσης ἐφαρμόσει καὶ τὸ AEB τμήμα ἐπὶ τὸ $ΓΖΔ$. εἰ γὰρ ἡ AB εὐθεῖα ἐπὶ τὴν $ΓΔ$ ἐφαρμόσει, τὸ δὲ AEB τμήμα ἐπὶ τὸ $ΓΖΔ$ μὴ ἐφαρμόσει, ἥτοι ἐντόξ αὐτοῦ πεσεῖται ἢ ἐκτόξ ἢ παραλλάξει, ὥς τὸ $ΓΗΔ$, καὶ κύκλος κύκλον τέμνει κατὰ πλείονα σημεῖα ἢ δύο· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐφαρμολομένης τῆς AB εὐθείας ἐπὶ τὴν $ΓΔ$ οὐκ ἐφαρμόσει καὶ τὸ AEB τμήμα ἐπὶ τὸ $ΓΖΔ$ · ἐφαρμόσει ἄρα, καὶ ἴσον αὐτῷ ἔσται.

Similar segments of circles on equal straight-lines are equal to one another.

For let AEB and CFD be similar segments of circles on the equal straight-lines AB and CD (respectively). I say that segment AEB is equal to segment CFD .

For if the segment AEB is applied to the segment CFD , and point A is placed on (point) C , and the straight-line AB on CD , (then) point B will also coincide with point D , on account of AB being equal to CD . And if AB coincides with CD , (then) the segment AEB will also coincide with CFD . For if the straight-line AB coincides with CD , and the segment AEB does not coincide with CFD , (then) it will surely either fall inside it, outside (it),[†] or it will miss like CGD (in the figure), and a circle (will) cut (another) circle at more than two points. The very thing is impossible [Prop. 3.10]. Thus, if the straight-line AB is applied to CD , the segment AEB cannot not also coincide with CFD . Thus, it will coincide, and will be equal to it [C.N. 4].



Τὰ ἄρα ἐπὶ ἴσων εὐθειῶν ὅμοια τμήματα κύκλων ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖχαι.

Thus, similar segments of circles on equal straight-lines are equal to one another. (Which is) the very thing it was required to show.

† Both this possibility, and the previous one, are precluded by Prop. 3.23.

25.

Κύκλον τμήματος δοθέντος προσαναγράψαι τὸν κύκλον, ὃν ὑπὲρ ἐστὶ τμήμα.

Ἐστω τὸ δοθέν τμήμα κύκλου τὸ $AB\Gamma$. δεῖ δὴ τοῦ $AB\Gamma$ τμήματος προσαναγράψαι τὸν κύκλον, ὃν ὑπὲρ ἐστὶ τμήμα.

Τετμήσθω γάρ ἡ AG δίῃα κατὰ τὸ Δ , καὶ ᾗθθω ἀπὸ τοῦ Δ σημείου τῇ AG πρὸς ὀρθὰς ἡ ΔB , καὶ ἐπεξεύθθω ἡ AB . ἡ ὑπὸ $AB\Delta$ γωνία ἄρα τῆς ὑπὸ $BA\Delta$ ἥτοι μείζων ἐστὶν ἢ ἴση ἢ ἐλάττω.

Ἐστω πρότερον μείζων, καὶ συνεστάτω πρὸς τῇ BA εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ ὑπὸ $AB\Delta$ γωνίᾳ ἴση ἢ ὑπὸ BAE , καὶ διήθθω ἡ ΔB ἐπὶ τὸ E , καὶ ἐπεξεύθθω ἡ EF . ἐπεὶ οὖν ἴση ἐστὶν ἡ ὑπὸ ABE γωνία τῇ ὑπὸ BAE , ἴση ἄρα ἐστὶ καὶ ἡ EB εὐθεῖα τῇ EA . καὶ ἐπεὶ ἴση ἐστὶν ἡ $A\Delta$ τῇ $\Delta\Gamma$, κοινὴ δὲ ἡ ΔE , δύο δὲ αἱ $A\Delta$, ΔE δύο ταῖς $\Gamma\Delta$, ΔE ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ $A\Delta E$

Proposition 25

For a given segment of a circle, to complete the circle, the very one of which it is a segment.

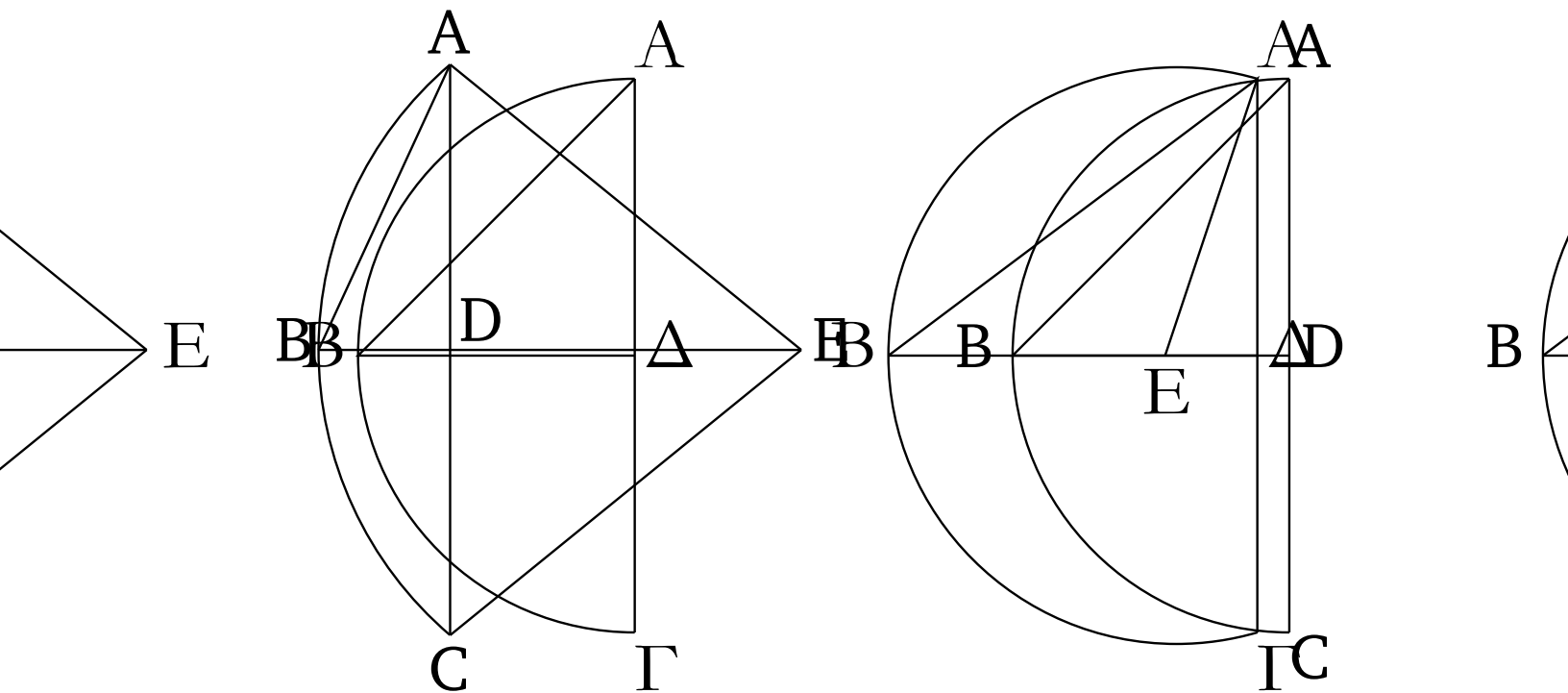
Let ABC be the given segment of a circle. So it is required to complete the circle for segment ABC , the very one of which it is a segment.

For let AC be cut in half at (point) D [Prop. 1.10], and let DB be drawn from point D , at right-angles to AC [Prop. 1.11]. And let AB be joined. Thus, angle ABD is surely either greater than, equal to, or less than (angle) BAD .

First of all, let it be greater. And let (angle) BAE , equal to angle ABD , be constructed on the straight-line BA , at the point A on it [Prop. 1.23]. And let DB be drawn through to E , and let EC be joined. Therefore, since angle ABE is equal to BAE , the straight-line EB is thus also equal to EA [Prop. 1.6]. And since AD is equal to DC , and DE (is) common, the two

γωνία τῇ ὑπὸ $\Gamma\Delta E$ ἔστιν ἴση· ὁρθή γάρ ἐκατέρα· βάσει ἄρα ἡ AE βάσει τῇ ΓE ἔστιν ἴση. ἀλλὰ ἡ AE τῇ BE ἐδείχθη ἴση· καὶ ἡ BE ἄρα τῇ ΓE ἔστιν ἴση· αἱ τρεῖς ἄρα αἱ AE , EB , ΓE ἴσαι ἀλλήλαις εἰσίν· ὁ ἄρα κέντρον τῷ E διαστήματι δὲ ἐνὶ τῶν AE , EB , ΓE κύκλος γραφόμενος ἔχει καὶ διὰ τῶν λοιπῶν σημείων καὶ ἔσται προσαναγεγραμμένος· κύκλον ἄρα τμήματος δοθέντος προσαναγράφεται ὁ κύκλος· καὶ δῆλον, ὥς τὸ $AB\Gamma$ τμήμα ἑλάττων ἔστιν ἡμικυκλίον διὰ τὸ τὸ E κέντρον ἐκτὸς αὐτοῦ τυγθάνειν.

(straight-lines) AD , DE are equal to the two (straight-lines) CD , DE , respectively. And angle ADE is equal to angle CDE . For each (is) a right-angle. Thus, the base AE is equal to the base CE [Prop. 1.4]. But, AE was shown (to be) equal to BE . Thus, BE is also equal to CE . Thus, the three (straight-lines) AE , EB , and EC are equal to one another. Thus, if a circle is drawn with center E , and radius one of AE , EB , or EC , it will also go through the remaining points (of the segment), and the (associated circle) will be completed [Prop. 3.9]. Thus, a circle has been completed from the given segment of a circle. And (it is) clear that the segment ABC is less than a semi-circle, because the center E happens to lie outside it.



Ὅμοίως [δὲ] κ' ἂν ἡ ὑπὸ $AB\Delta$ γωνία ἴση τῇ ὑπὸ $BA\Delta$, τῇς $A\Delta$ ἴσης γενομένης ἐκατέρω τῶν $B\Delta$, $\Delta\Gamma$ αἱ τρεῖς αἱ ΔA , ΔB , $\Delta\Gamma$ ἴσαι ἀλλήλαις ἔσονται, καὶ ἔσται τὸ Δ κέντρον τοῦ προσαναπεπληρωμένου κύκλου, καὶ δηλαδὴ ἔσται τὸ $AB\Gamma$ ἡμικύκλιον.

Ἐάν δὲ ἡ ὑπὸ $AB\Delta$ ἐλάττων ἢ τῇς ὑπὸ $BA\Delta$, καὶ συστησώμεθα πρὸς τῇ BA εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ ὑπὸ $AB\Delta$ γωνίᾳ ἴσην, ἐντὸς τοῦ $AB\Gamma$ τμήματος πεσεῖται τὸ κέντρον ἐπὶ τῇς ΔB , καὶ ἔσται δηλαδὴ τὸ $AB\Gamma$ τμήμα μείζον ἡμικυκλίον.

Κύκλον ἄρα τμήματος δοθέντος προσαναγράφεται ὁ κύκλος· ὅπερ ἔδει ποιῆσαι.

[And], similarly, even if angle ABD is equal to BAD , (since) AD becomes equal to each of BD [Prop. 1.6] and DC , the three (straight-lines) DA , DB , and DC will be equal to one another. And point D will be the center of the completed circle. And ABC will manifestly be a semi-circle.

And if ABD is less than BAD , and we construct (angle BAE), equal to angle ABD , on the straight-line BA , at the point A on it [Prop. 1.23], then the center will fall on DB , inside the segment ABC . And segment ABC will manifestly be greater than a semi-circle.

Thus, a circle has been completed from the given segment of a circle. (Which is) the very thing it was required to do.

ρειῶν βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὥσι βεβηκνῶται.

Ἐστωσαν ἴσοι κύκλοι οἱ $ABΓ$, $ΔΕΖ$ καὶ ἐν αὐτοῖς ἴσαι γωνίαι ἔστωσαν πρὸς μὲν τοῖς κέντροις αἱ ὑπὸ $BΗΓ$, $ΕΘΖ$, πρὸς δὲ ταῖς περιφερείαις αἱ ὑπὸ $ΒΑΓ$, $ΕΔΖ$. λέγω, ὅτι ἴση ἐστὶν ἡ $ΒΚΓ$ περιφέρεια τῇ $ΕΛΖ$ περιφερείᾳ.

Ἐπεξένθθωσαν γάρ αἱ $ΒΓ$, $ΕΖ$.

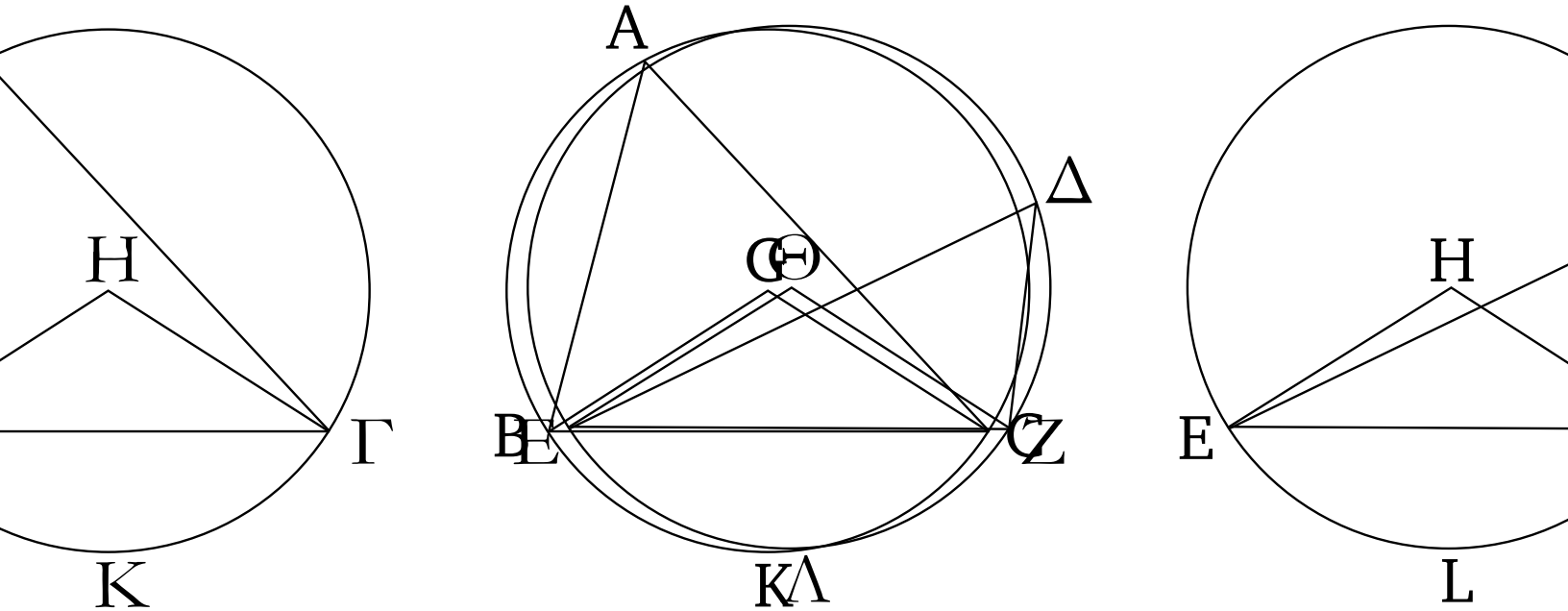
Καὶ ἐπεὶ ἴσοι εἰσὶν οἱ $ABΓ$, $ΔΕΖ$ κύκλοι, ἴσαι εἰσὶν αἱ ἐκ τῶν κέντρων δύο δὴ αἱ $BΗ$, $ΗΓ$ δύο ταῖς $ΕΘ$, $ΘΖ$ ἴσαι· καὶ γωνία ἡ πρὸς τῷ H γωνία τῇ πρὸς τῷ $Θ$ ἴση· βάσει ἄρα ἡ $ΒΓ$ βάσει τῇ $ΕΖ$ ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ πρὸς τῷ A γωνία τῇ πρὸς τῷ $Δ$, ὅμοιον ἄρα ἐστὶ τὸ $ΒΑΓ$ τμήμα τῷ $ΕΔΖ$ τμήματι· καὶ εἰσὶν ἐπὶ ἴσων εὐθειῶν [τῶν $ΒΓ$, $ΕΖ$]· τὰ δὲ ἐπὶ ἴσων εὐθειῶν ὅμοια τμήματα κύκλων ἴσα ἀλλήλοις ἐστὶν· ἴσον ἄρα τὸ $ΒΑΓ$ τμήμα τῷ $ΕΔΖ$. ἔστι δὲ καὶ ὅλος ὁ $ABΓ$ κύκλος ὅλῳ τῷ $ΔΕΖ$ κύκλῳ ἴσος· λοιπὴ ἄρα ἡ $ΒΚΓ$ περιφέρεια τῇ $ΕΛΖ$ περιφερείᾳ ἐστὶν ἴση.

ences whether they are standing at the center or at the circumference.

Let ABC and DEF be equal circles, and within them let BGC and EHF be equal angles at the center, and BAC and EDF (equal angles) at the circumference. I say that circumference BKC is equal to circumference ELF .

For let BC and EF be joined.

And since circles ABC and DEF are equal, their radii are equal. So the two (straight-lines) BG , GC (are) equal to the two (straight-lines) EH , HF (respectively). And the angle at G (is) equal to the angle at H . Thus, the base BC is equal to the base EF [Prop. 1.4]. And since the angle at A is equal to the (angle) at D , the segment BAC is thus similar to the segment EDF [Def. 3.11]. And they are on equal straight-lines [BC and EF]. And similar segments of circles on equal straight-lines are equal to one another [Prop. 3.24]. Thus, segment BAC is equal to (segment) EDF . And the whole circle ABC is also equal to the whole circle DEF . Thus, the remaining circumference BKC is equal to the (remaining) circumference ELF .



Ἐν ἄρα τοῖς ἴσοις κύκλοις αἱ ἴσαι γωνίαι ἐπὶ ἴσων περιφερειῶν βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὥσι βεβηκνῶται· ὅπερ ἔδει δεῖξαι.

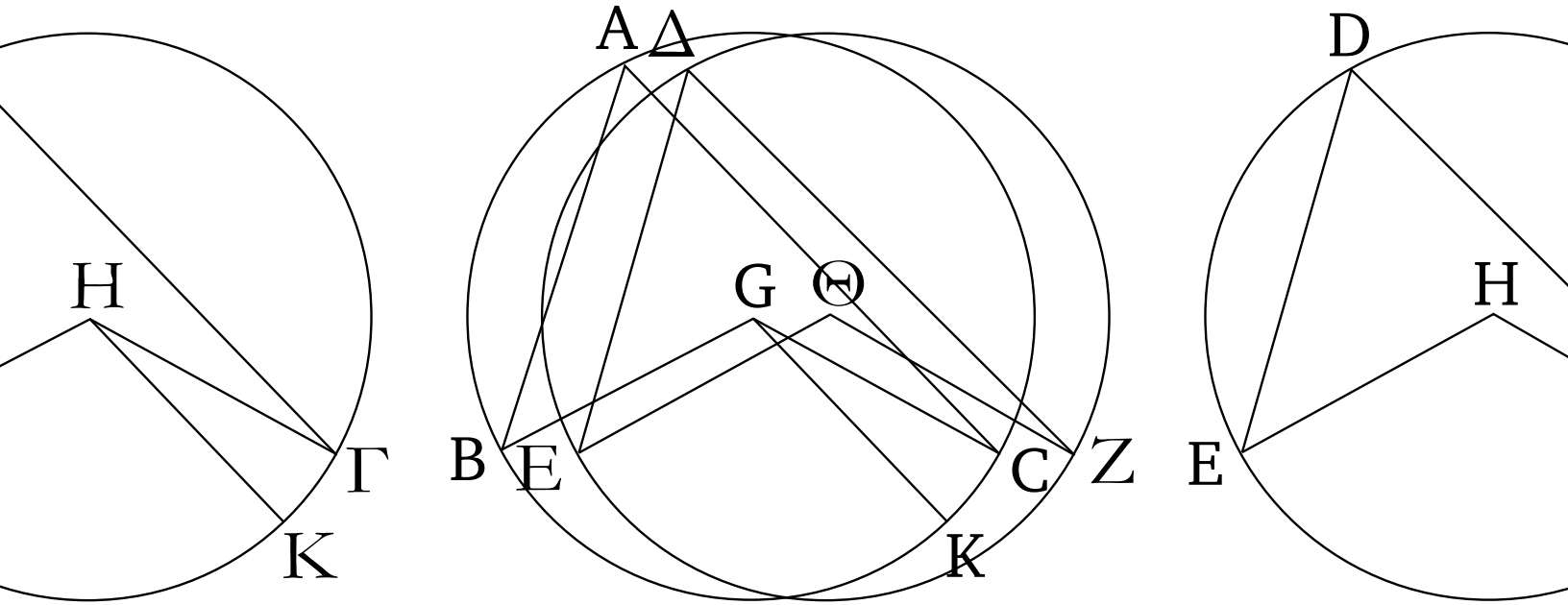
Thus, in equal circles, equal angles stand upon equal circumferences, whether they are standing at the center or at the circumference. (Which is) the very thing which it was required to show.

27.

Proposition 27

Ἐν τοῖς ἴσοις κύκλοις αἱ ἐπὶ ἴσων περιφερειῶν βεβηκνῶται γωνίαι ἴσαι ἀλλήλαις εἰσὶν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὥσι βεβηκνῶται.

In equal circles, angles standing upon equal circumferences are equal to one another, whether they are standing at the center or at the circumference.



Ἐν γάρ ἴσοις κύκλοις τοῖς $ABΓ$, $ΔEZ$ ἐπὶ ἴσων περιφερειῶν τῶν $BΓ$, EZ πρὸς μὲν τοῖς H , $Θ$ κέντροις γωνίαι βεβηκέτωσαν αἱ ὑπὸ $BΗΓ$, $EΘZ$, πρὸς δὲ ταῖς περιφερείαις αἱ ὑπὸ $BAΓ$, $EΔZ$ · λέγω, ὅτι ἡ μὲν ὑπὸ $BΗΓ$ γωνία τῇ ὑπὸ $EΘZ$ ἐστὶν ἴση, ἡ δὲ ὑπὸ $BAΓ$ τῇ ὑπὸ $EΔZ$ ἐστὶν ἴση.

Εἰ γὰρ ἄνισός ἐστιν ἡ ὑπὸ $BΗΓ$ τῇ ὑπὸ $EΘZ$, μία αὐτῶν μείζων ἐστίν. ἔστω μείζων ἡ ὑπὸ $BΗΓ$, καὶ συνεστάτω πρὸς τῇ $BΗ$ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ H τῇ ὑπὸ $EΘZ$ γωνίᾳ ἴση ἡ ὑπὸ $BΗΚ$ · αἱ δὲ ἴσαι γωνίαι ἐπὶ ἴσων περιφερειῶν βεβήκασι, ὅταν πρὸς τοῖς κέντροις ᾖσιν· ἴση ἄρα ἡ $BΚ$ περιφέρεια τῇ EZ περιφερείᾳ. ἀλλὰ ἡ EZ τῇ $BΓ$ ἐστὶν ἴση· καὶ ἡ $BΚ$ ἄρα τῇ $BΓ$ ἐστὶν ἴση ἢ ἐλάττω τῇ μείζον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἄνισός ἐστιν ἡ ὑπὸ $BΗΓ$ γωνία τῇ ὑπὸ $EΘZ$ · ἴση ἄρα. καὶ ἐστὶ τῆς μὲν ὑπὸ $BΗΓ$ ἡμίσεια ἡ πρὸς τῷ A , τῆς δὲ ὑπὸ $EΘZ$ ἡμίσεια ἡ πρὸς τῷ $Δ$ · ἴση ἄρα καὶ ἡ πρὸς τῷ A γωνία τῇ πρὸς τῷ $Δ$.

Ἐν ἄρα τοῖς ἴσοις κύκλοις αἱ ἐπὶ ἴσων περιφερειῶν βεβηκνύει γωνίαι ἴσαι ἀλλήλαις εἰσίν, ἥν τε πρὸς τοῖς κέντροις ἥν τε πρὸς ταῖς περιφερείαις ᾧσι βεβηκνύει· ὅπερ ἔδει δεῖξαι.

28.

Ἐν τοῖς ἴσοις κύκλοις αἱ ἴσαι εὐθεῖαι ἴσαξ περιφερείας ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττωνα τῇ ἐλάττω.

Ἐστῶσαν ἴσοι κύκλοι οἱ $ABΓ$, $ΔEZ$, καὶ ἐν τοῖς κύκλοις ἴσαι εὐθεῖαι ἔστωσαν αἱ AB , $ΔE$ τὰς μὲν $ΑΓB$, $ΑZE$ περιφερείας μείζονας ἀφαιροῦσαι τὰς δὲ AHB , $ΔΘE$ ἐλάττωνας·

For let the angles BGC and EHF at the centers G and H , and the (angles) BAC and EDF at the circumferences, stand upon the equal circumferences BC and EF , in the equal circles ABC and DEF (respectively). I say that angle BGC is equal to (angle) EHF , and BAC is equal to EDF .

For if BGC is unequal to EHF , one of them is greater. Let BGC be greater, and let the (angle) BGK , equal to angle EHF , be constructed on the straight-line BG , at the point G on it [Prop. 1.23]. But equal angles (in equal circles) stand upon equal circumferences, when they are at the centers [Prop. 3.26]. Thus, circumference BK (is) equal to circumference EF . But, EF is equal to BC . Thus, BK is also equal to BC , the lesser to the greater. The very thing is impossible. Thus, angle BGC is not unequal to EHF . Thus, (it is) equal. And the (angle) at A is half BGC , and the (angle) at D half EHF [Prop. 3.20]. Thus, the angle at A (is) also equal to the (angle) at D .

Thus, in equal circles, angles standing upon equal circumferences are equal to one another, whether they are standing at the center or at the circumference. (Which is) the very thing it was required to show.

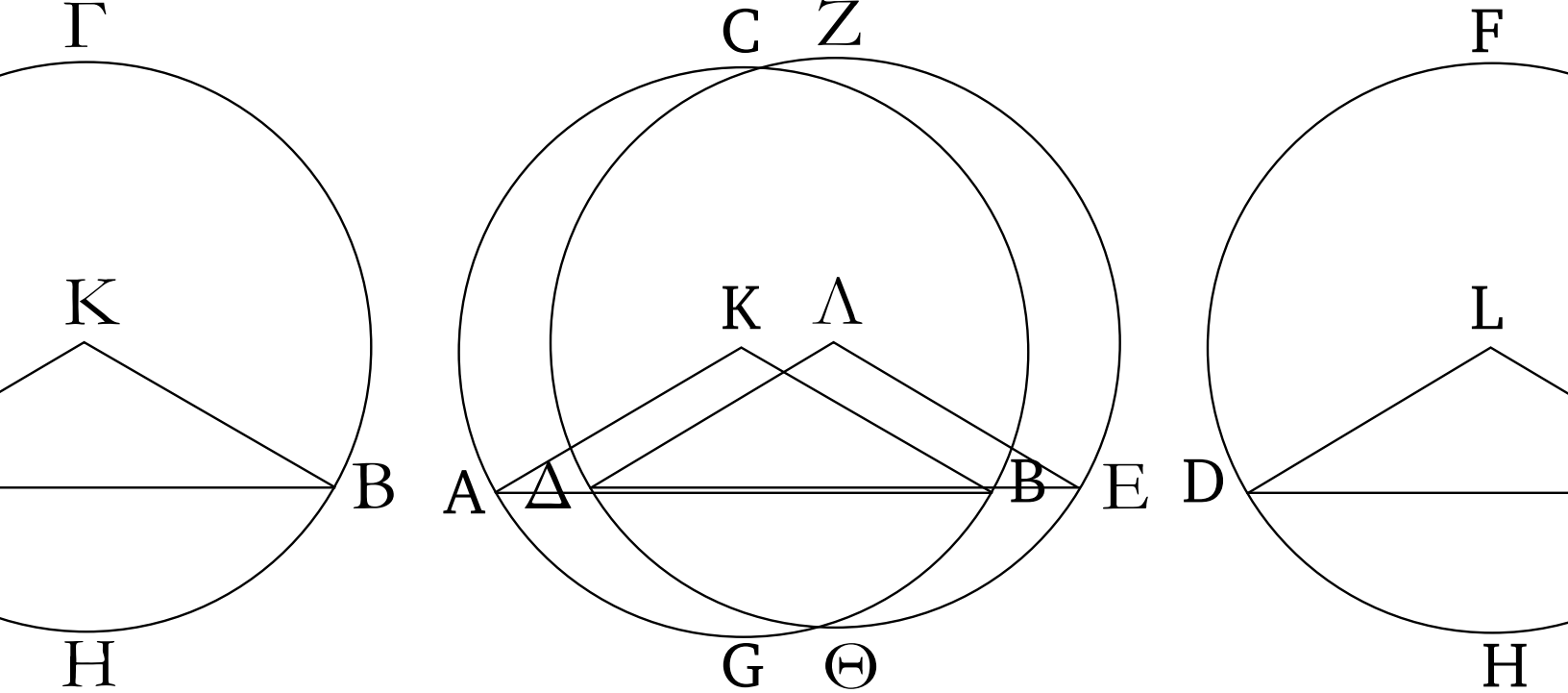
Proposition 28

In equal circles, equal straight-lines cut off equal circumferences, the greater (circumference being equal) to the greater, and the lesser to the lesser.

Let ABC and DEF be equal circles, and let AB and DE be equal straight-lines in these circles, cutting off the greater circumferences ACB and DFE , and the lesser (circumferences)

λέγω, ὅτι ἡ μὲν $ΑΓΒ$ μείζων περιφέρεια ἴση ἐστὶ τῇ $ΔΖΕ$ μείζονι περιφερείᾳ ἡ δὲ $ΑΗΒ$ ἐλάττων περιφέρεια τῇ $ΔΘΕ$.

$ΑΓΒ$ and $ΔΗΕ$ (respectively). I say that the greater circumference $ΑΒΓ$ is equal to the greater circumference $ΔΕΖ$, and the lesser circumference $ΑΒΗ$ to (the lesser) $ΔΗΕ$.



Εἰλήφθω γὰρ τὰ κέντρα τῶν κύκλων τὰ K , $Λ$, καὶ ἐπεξ-
ὑθῆωσαν αἱ AK , KB , $ΔΛ$, $ΛΕ$.

Καὶ ἐπεὶ ἴσοι κύκλοι εἰσὶν, ἴσαι εἰσὶ καὶ αἱ ἐκ τῶν κέντρων
δύο δὴ αἱ AK , KB δυοὶ ταῖς $ΔΛ$, $ΛΕ$ ἴσαι εἰσὶν καὶ βάσει
ἡ AB βάσει τῇ $ΔΕ$ ἴση· γωνία ἄρα ἡ ὑπὸ AKB γωνία τῇ
ὑπὸ $ΔΛΕ$ ἴση ἐστίν. αἱ δὲ ἴσαι γωνίαι ἐπὶ ἴσων περιφερειῶν
βεβήκασιν, ὅταν πρὸς τοῖς κέντροις ᾗσιν ἴση ἄρα ἡ $ΑΗΒ$
περιφέρεια τῇ $ΔΘΕ$. ἐστὶ δὲ καὶ ὅλος ὁ $ΑΒΓ$ κύκλος ὅλος τῷ
 $ΔΕΖ$ κύκλῳ ἴσος· καὶ λοιπὴ ἄρα ἡ $ΑΓΒ$ περιφέρεια λοιπῇ
τῇ $ΔΖΕ$ περιφερείᾳ ἴση ἐστίν.

Ἐν ἄρα τοῖς ἴσοις κύκλοις αἱ ἴσαι εὐθεῖαι ἴσαξ περιφε-
ρεῖαξ ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττονα
τῇ ἐλάττονι· ὅπερ εἶδει δεῖχαι.

For let the centers of the circles, K and L , be found
[Prop. 3.1], and let AK , KB , DL , and LE be joined.

And since $(ABC$ and $DEF)$ are equal circles, their radii
are also equal [Def. 3.1]. So the two (straight-lines) AK , KB
are equal to the two (straight-lines) DL , LE (respectively).
And the base AB (is) equal to the base DE . Thus, angle
 AKB is equal to angle DLE [Prop. 1.8]. And equal angles
stand upon equal circumferences, when they are at the centers
[Prop. 3.26]. Thus, circumference $ΑΒΓ$ (is) equal to $ΔΗΕ$.
And the whole circle ABC is also equal to the whole circle
 DEF . Thus, the remaining circumference ACB is also equal
to the remaining circumference DFE .

Thus, in equal circles, equal straight-lines cut off equal cir-
cumferences, the greater (circumference being equal) to the
greater, and the lesser to the lesser. (Which is) the very thing
it was required to show.

29.

Proposition 29

Ἐν τοῖς ἴσοις κύκλοις τὰς ἴσαξ περιφερεῖαξ ἴσαι εὐθεῖαι
ὑποτείνουσιν.

Ἐστωσαν ἴσοι κύκλοι οἱ $ΑΒΓ$, $ΔΕΖ$, καὶ ἐν αὐτοῖς ἴσαι
περιφέρειαι ἀπειλήφθωσαν αἱ $ΒΗΓ$, $ΕΘΖ$, καὶ ἐπεξῆνθωσαν
αἱ $ΒΓ$, $ΕΖ$ εὐθεῖαι· λέγω, ὅτι ἴση ἐστὶν ἡ $ΒΓ$ τῇ $ΕΖ$.

Εἰλήφθω γὰρ τὰ κέντρα τῶν κύκλων, καὶ ἔστω τὰ K , $Λ$,

In equal circles, equal straight-lines subtend equal cir-
cumferences.

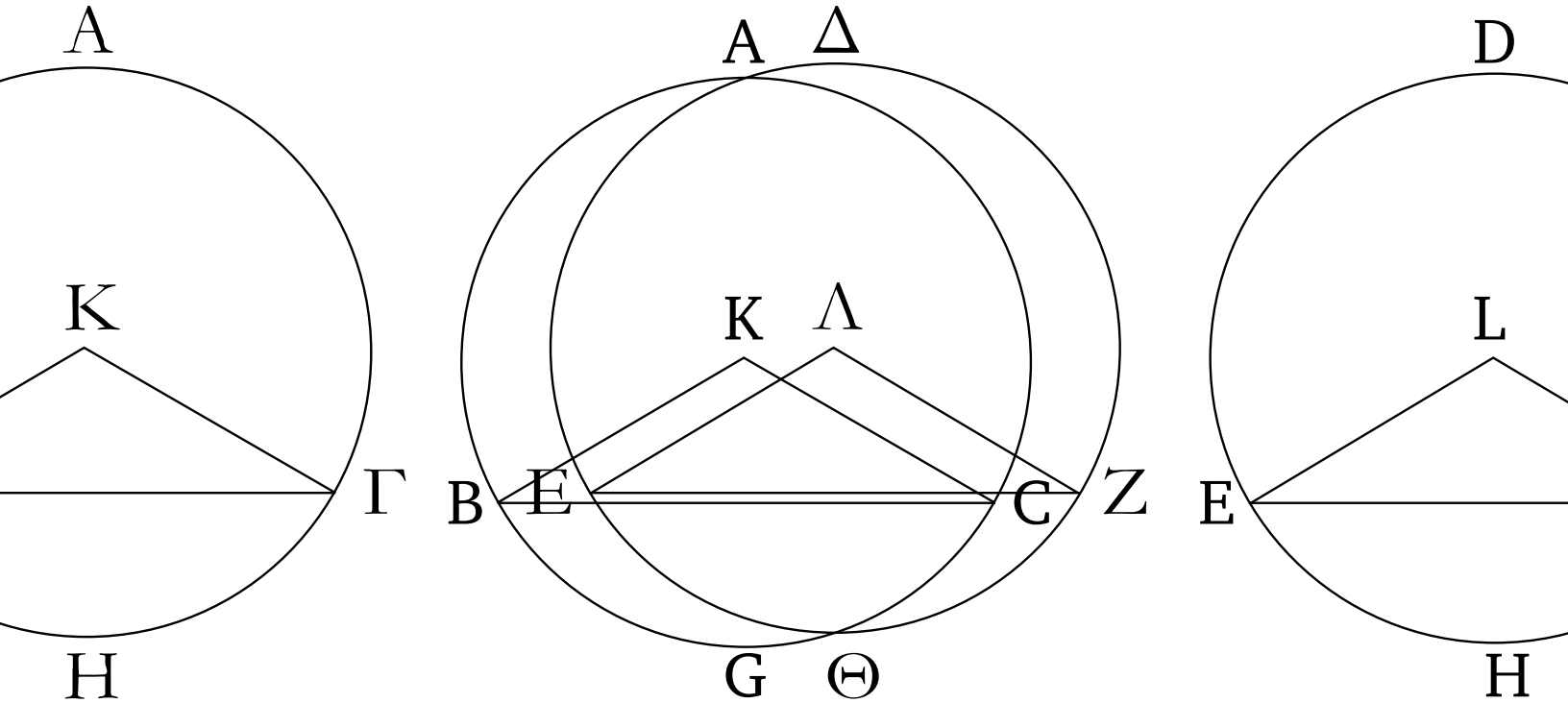
Let ABC and DEF be equal circles, and within them let
the equal circumferences BGC and EHF be cut off. And let
the straight-lines BC and EF be joined. I say that BC is equal
to EF .

καὶ ἐπεξεύχθησαν αἱ BK , $K\Gamma$, $ΕΛ$, $ΛΖ$.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ $BH\Gamma$ περιφέρεια τῇ $ΕΘΖ$ περιφερείᾳ, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $BK\Gamma$ τῇ ὑπὸ $ΕΛΖ$. καὶ ἐπεὶ ἴσοι εἰσὶν οἱ $AB\Gamma$, $\Delta ΕΖ$ κύκλοι, ἴσαι εἰσὶ καὶ αἱ ἐκ τῶν κέντρων δύο δὴ αἱ BK , $K\Gamma$ δυοὶ ταῖς $ΕΛ$, $ΛΖ$ ἴσαι εἰσὶν καὶ γωνίαξ ἴσαξ περιέθουσιν βάσει ἡ $B\Gamma$ βάσει τῇ $ΕΖ$ ἴση ἐστίν.

For let the centers of the circles be found [Prop. 3.1], and let them be (at) K and L . And let BK , KC , EL , and LF be joined.

And since the circumference BGC is equal to the circumference EHF , the angle BKC is also equal to (angle) ELF [Prop. 3.27]. And since the circles ABC and DEF are equal, their radii are also equal [Def. 3.1]. So the two (straight-lines) BK , KC are equal to the two (straight-lines) EL , LF (respectively). And they contain equal angles. Thus, the base BC is equal to the base EF [Prop. 1.4].



Ἐν ἄρα τοῖς ἴσοις κύκλοις τὰς ἴσαξ περιφερείας ἴσαι εὐθείαι ὑποτείνουσιν ὅπερ ἔδει δεῖχαι.

Thus, in equal circles, equal straight-lines subtend equal circumferences. (Which is) the very thing it was required to show.

30.

Τὴν δοθεῖσαν περιφέρειαν διῖτα τεμεῖν.

Ἐστω ἡ δοθεῖσα περιφέρεια ἡ $A\Delta B$. δεῖ δὴ τὴν $A\Delta B$ περιφέρειαν διῖτα τεμεῖν.

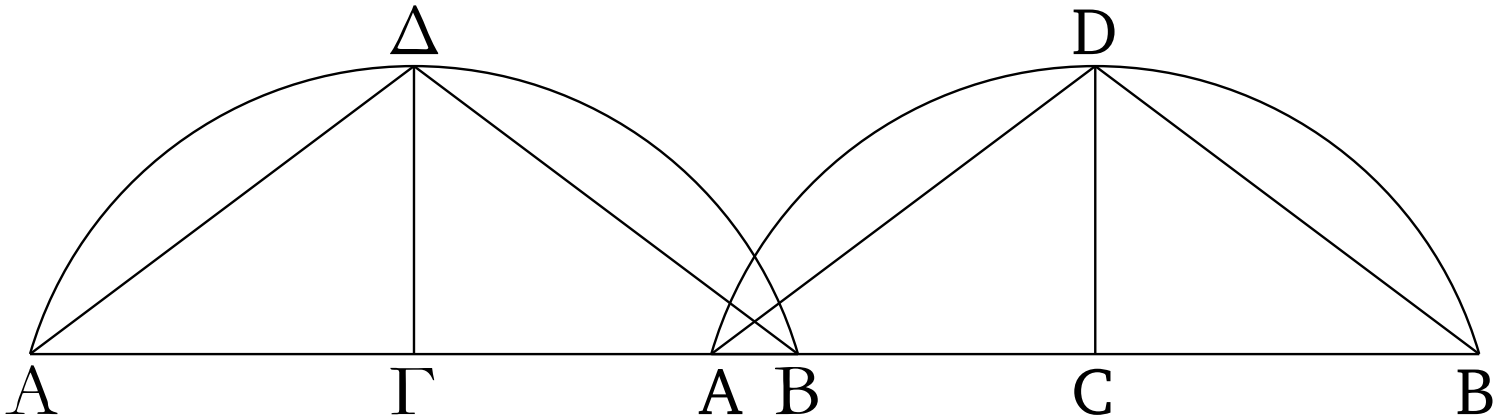
Ἐπεξεύχθω ἡ AB , καὶ τετμήσθω διῖτα κατὰ τὸ Γ , καὶ ἀπὸ τοῦ Γ σημείου τῇ AB εὐθείᾳ πρὸς ὀρθὰς ῥηθῶ ἡ $\Gamma\Delta$, καὶ ἐπεξεύχθησαν αἱ $A\Delta$, ΔB .

Proposition 30

To cut a given circumference in half.

Let ADB be the given circumference. So it is required to cut circumference ADB in half.

Let AB be joined, and let it be cut in half at (point) C [Prop. 1.10]. And let CD be drawn from point C , at right-angles to AB [Prop. 1.11]. And let AD , and DB be joined.



Καὶ ἐπεὶ ἴση ἐστὶν ἡ AG τῇ GB , κοινὴ δὲ ἡ GD , δύο δὲ αἱ AG , GD δυοὶ ταῖς BG , GD ἴσαι εἰσὶν· καὶ γωνία ἡ ὑπὸ AGD γωνία τῇ ὑπὸ BGD ἴση· ὀρθή γάρ ἐκατέρα· βάσις ἄρα ἡ AD βάσει τῇ DB ἴση ἐστίν. αἱ δὲ ἴσαι εὐθεῖαι ἴσαξ περιφερεῖαξ ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττωνα τῇ ἐλάττω· καὶ ἐστὶν ἐκατέρα τῶν AD , DB περιφερειῶν ἐλάττων ἡμικυκλίον· ἴση ἄρα ἡ AD περιφέρεια τῇ DB περιφερείᾳ.

Ἡ ἄρα δοθεῖσα περιφέρεια δίθῃ τέτμηται κατὰ τὸ D σημείον· ὅπερ ἔδει ποιῆσαι.

And since AC is equal to CB , and CD (is) common, the two (straight-lines) AC , CD are equal to the two (straight-lines) BC , CD (respectively). And angle ACD (is) equal to angle BCD . For (they are) each right-angles. Thus, the base AD is equal to the base DB [Prop. 1.4]. And equal straight-lines cut off equal circumferences, the greater (circumference being equal) to the greater, and the lesser to the lesser [Prop. 1.28]. And the circumferences AD and DB are each less than a semi-circle. Thus, circumference AD (is) equal to circumference DB .

Thus, the given circumference has been cut in half at point D . (Which is) the very thing it was required to do.

31.

Ἐν κύκλῳ ἡ μὲν ἐν τῷ ἡμικυκλίῳ γωνία ὀρθή ἐστίν, ἡ δὲ ἐν τῷ μείζονι τμήματι ἐλάττων ὀρθῆς, ἡ δὲ ἐν τῷ ἐλάττω τμήματι μείζων ὀρθῆς· καὶ ἐπὶ ἡ μὲν τοῦ μείζονος τμήματος γωνία μείζων ἐστὶν ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος γωνία ἐλάττων ὀρθῆς.

Proposition 31

In a circle, the angle in a semi-circle is a right-angle, and that in a greater segment (is) less than a right-angle, and that in a lesser segment (is) greater than a right-angle. And, further, the angle of a segment greater (than a semi-circle) is greater than a right-angle, and the angle of a segment less (than a semi-circle) is less than a right-angle.

τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν [αἱ ἄρα ὑπὸ $AB\Gamma$, $A\Delta\Gamma$ γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν], καὶ ἔστιν ἡ ὑπὸ $AB\Gamma$ ἐλάττων ὀρθῆς· λοιπὴ ἄρα ἡ ὑπὸ $A\Delta\Gamma$ γωνία μείζων ὀρθῆς ἔστιν· καὶ ἔστιν ἐν τῷ $A\Delta\Gamma$ ἐλάττονι τοῦ ἡμικυκλίου τμήματι.

Λέγω, ὅτι καὶ ἡ μὲν τοῦ μείζονος τμήματος γωνία ἡ περιεθρομένη ὑπὸ [τε] τῆς $AB\Gamma$ περιφερείας καὶ τῆς AF εὐθείας μείζων ἔστιν ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος γωνία ἡ περιεθρομένη ὑπὸ [τε] τῆς $A\Delta\Gamma$ περιφερείας καὶ τῆς AF εὐθείας ἐλάττων ἔστιν ὀρθῆς. καὶ ἔστιν αὐτόθεν φανερόν. ἐπεὶ γάρ ἡ ὑπὸ τῶν BA , AF εὐθειῶν ὀρθή ἐστιν, ἡ ἄρα ὑπὸ τῆς $AB\Gamma$ περιφερείας καὶ τῆς AF εὐθείας περιεθρομένη μείζων ἔστιν ὀρθῆς. πάλιν, ἐπεὶ ἡ ὑπὸ τῶν AF , AZ εὐθειῶν ὀρθή ἐστιν, ἡ ἄρα ὑπὸ τῆς FA εὐθείας καὶ τῆς $A\Delta\Gamma$ περιφερείας περιεθρομένη ἐλάττων ἔστιν ὀρθῆς.

Ἐν κύκλῳ ἄρα ἡ μὲν ἐν τῷ ἡμικυκλίῳ γωνία ὀρθή ἐστιν, ἡ δὲ ἐν τῷ μείζονι τμήματι ἐλάττων ὀρθῆς, ἡ δὲ ἐν τῷ ἐλάττονι [τμήματι] μείζων ὀρθῆς· καὶ ἔπι ἡ μὲν τοῦ μείζονος τμήματος [γωνία] μείζων [ἔστιν] ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος [γωνία] ἐλάττων ὀρθῆς· ὅπερ ἔδει δεῖχαι.

quadrilaterals within circles the (sum of the) opposite angles is equal to two right-angles [Prop. 3.22] [angles ABC and ADC are thus equal to two right-angles], and (angle) ABC is less than a right-angle. The remaining angle ADC is thus greater than a right-angle. And it is in segment ADC , (which is) less than a semi-circle.

I also say that the angle of the greater segment, (namely) that contained by the circumference ABC and the straight-line AC , is greater than a right-angle. And the angle of the lesser segment, (namely) that contained by the circumference $AD[C]$ and the straight-line AC , is less than a right-angle. And this is immediately apparent. For since the (angle contained by) the two straight-lines BA and AC is a right-angle, the (angle) contained by the circumference ABC and the straight-line AC is thus greater than a right-angle. Again, since the (angle contained by) the straight-lines AC and AF is a right-angle, the (angle) contained by the circumference $AD[C]$ and the straight-line CA is thus less than a right-angle.

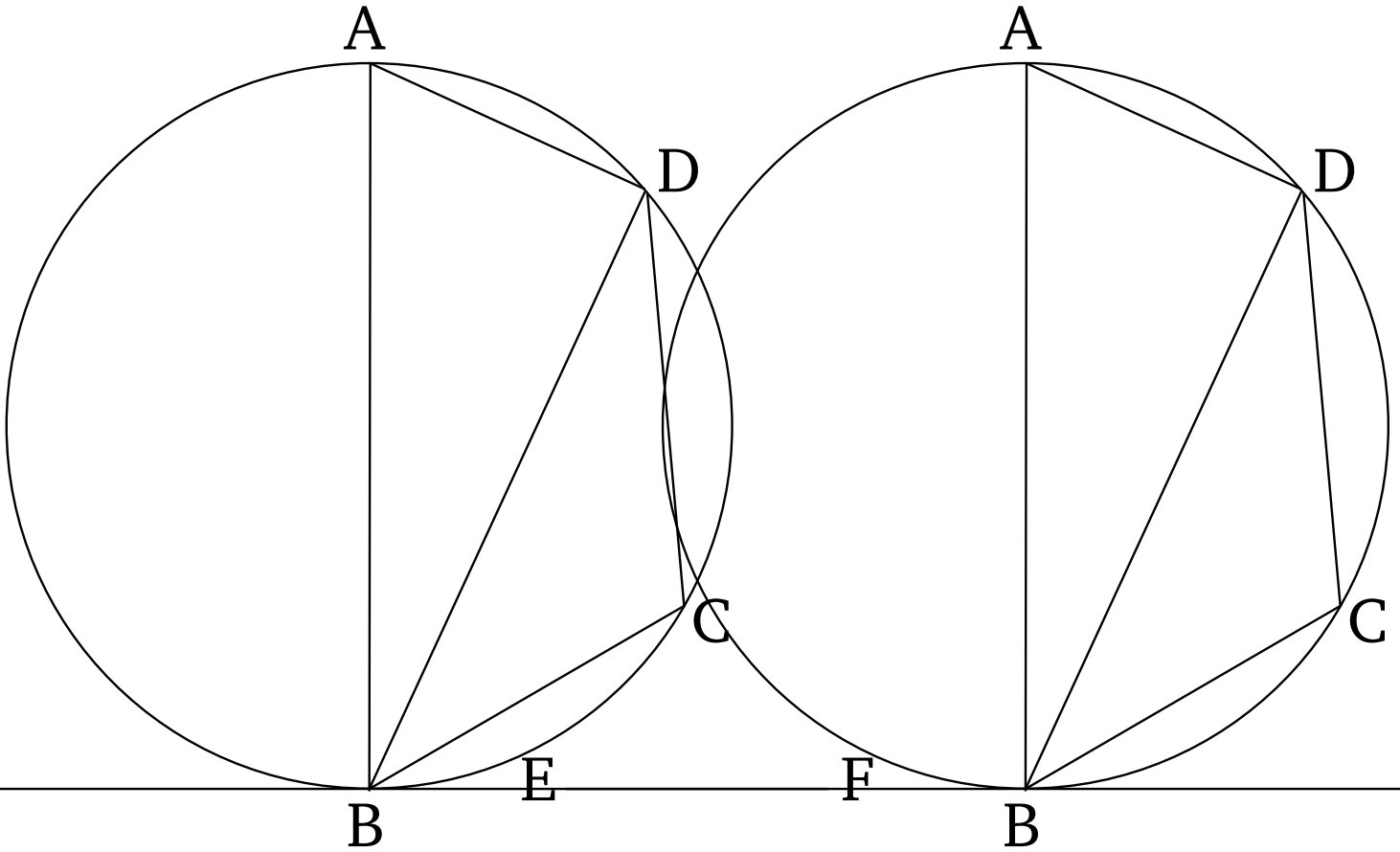
Thus, in a circle, the angle in a semi-circle is a right-angle, and that in a greater segment (is) less than a right-angle, and that in a lesser [segment] (is) greater than a right-angle. And, further, the [angle] of a segment greater (than a semi-circle) [is] greater than a right-angle, and the [angle] of a segment less (than a semi-circle) is less than a right-angle. (Which is) the very thing it was required to show.

32.

Proposition 32

Ἐὰν κύκλον ἐφάπτηται τις εὐθεῖα, ἀπὸ δὲ τῆς ἀφ᾽ ἧς εἰς τὸν κύκλον διαθῇ τις εὐθεῖα τέμνονσα τὸν κύκλον, ἃς ποιῇ γωνίας πρὸς τῇ ἐφαπτομένῃ, ἴσαι ἔσονται ταῖς ἐν τοῖς ἐναλλάχ τοῦ κύκλου τμήμασι γωνίαις.

If some straight-line touches a circle, and some (other) straight-line is drawn across, from the point of contact into the circle, cutting the circle (in two), (then) those angles the (straight-line) makes with the tangent will be equal to the angles in the alternate segments of the circle.



Κύκλον γάρ τοῦ $AB\Gamma\Delta$ ἐφαπτέσθω τις εὐθεΐα ἡ EZ κατὰ τὸ B σημεῖον, καὶ ἀπὸ τοῦ B σημείου διήθθω τις εὐθεΐα εἰς τὸν $AB\Gamma\Delta$ κύκλον τέμνουσα αὐτὸν ἡ $B\Delta$. λέγω, ὅτι ἅξ ποιεῖ γωνίαξ ἡ $B\Delta$ μετὰ τῆξ EZ ἐφαπτομένηξ, ἴσαξ ἔσσονται ταῖξ ἐν τοῖξ ἐναλλάξ τμήμασι τοῦ κύκλου γωνίαιξ, τουτέστιν, ὅτι ἡ μὲν ὑπὸ $ZB\Delta$ γωνία ἴση ἐστὶ τῇ ἐν τῷ $BA\Delta$ τμήματι συνισταμένη γωνία, ἡ δὲ ὑπὸ $EB\Delta$ γωνία ἴση ἐστὶ τῇ ἐν τῷ $\Delta\Gamma B$ τμήματι συνισταμένη γωνία.

Ἦθθω γάρ ἀπὸ τοῦ B τῇ EZ πρὸξ ὀρθάξ ἡ BA , καὶ εἰλήφθω ἐπὶ τῆξ $B\Delta$ περιφερείαξ τυθὸν σημεῖον τὸ Γ , καὶ ἐπεξεύθθωσαν αἱ $A\Delta$, $\Delta\Gamma$, ΓB .

Καὶ ἐπεὶ κύκλον τοῦ $AB\Gamma\Delta$ ἐφάπτεται τις εὐθεΐα ἡ EZ κατὰ τὸ B , καὶ ἀπὸ τῆξ ἀφῆξ ἦκται τῇ ἐφαπτομένη πρὸξ ὀρθάξ ἡ BA , ἐπὶ τῆξ BA ἄρα τὸ κέντρον ἐστὶ τοῦ $AB\Gamma\Delta$ κύκλου. ἡ BA ἄρα διάμετόξ ἐστὶ τοῦ $AB\Gamma\Delta$ κύκλου· ἡ ἄρα ὑπὸ $A\Delta B$ γωνία ἐν ἡμικυκλίῳ οὔσα ὀρθή ἐστιν. λοιπαὶ ἄρα αἱ ὑπὸ $BA\Delta$, $AB\Delta$ μᾶ ὀρθῇ ἴσαι εἰσίν. ἐστὶ δὲ καὶ ἡ ὑπὸ ABZ ὀρθή· ἡ ἄρα ὑπὸ ABZ ἴση ἐστὶ ταῖξ ὑπὸ $BA\Delta$, $AB\Delta$. κοινῇ ἀφηρήσθω ἡ ὑπὸ $AB\Delta$ · λοιπὴ ἄρα ἡ ὑπὸ ΔBZ γωνία ἴση ἐστὶ τῇ ἐν τῷ ἐναλλάξ τμήματι τοῦ κύκλου γωνία τῇ ὑπὸ $BA\Delta$. καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρόν ἐστι τὸ $AB\Gamma\Delta$, αἱ ἀπεναντίον αὐτοῦ γωνίαι δυοὶν ὀρθαῖξ ἴσαι εἰσίν. εἰσὶ δὲ καὶ αἱ ὑπὸ ΔBZ , ΔBE δυοὶν ὀρθαῖξ ἴσαι· αἱ ἄρα ὑπὸ ΔBZ ,

For let some straight-line EF touch the circle $ABCD$ at the point B , and let some (other) straight-line BD be drawn from point B into the circle $ABCD$, cutting it (in two). I say that the angles BD makes with the tangent EF will be equal to the angles in the alternate segments of the circle. That is to say, that angle FBD is equal to the angle constructed in segment BAD , and angle EBD is equal to the angle constructed in segment DCB .

For let BA be drawn from B , at right-angles to EF [Prop. 1.11]. And let the point C be taken at random on the circumference BD . And let AD , DC , and CB be joined.

And since some straight-line EF touches the circle $ABCD$ at point B , and BA has been drawn from the point of contact, at right-angles to the tangent, the center of circle $ABCD$ is thus on BA [Prop. 3.19]. Thus, BA is a diameter of circle $ABCD$. Thus, angle ADB , being in a semi-circle, is a right-angle [Prop. 3.31]. Thus, the remaining angles (of triangle ADB) BAD and ABD are equal to one right-angle [Prop. 1.32]. And ABF is also a right-angle. Thus, ABF is equal to BAD and ABD . Let ABD be subtracted from both. Thus, the remaining angle DBF is equal to the angle BAD in the alternate segment of the circle. And since $ABCD$ is a quadrilateral in a circle, (the sum of) its opposite angles is equal to two right-angles

ΔBE ταῖς ὑπὸ $BA\Delta$, $B\Gamma\Delta$ ἴσαι εἰσίν, ἢ ὑπὸ $BA\Delta$ τῇ ὑπὸ ΔBZ ἐδεῖσθῃ ἴση· λοιπὴ ἄρα ἡ ὑπὸ ΔBE τῇ ἐν τῷ ἐναλλάξ τοῦ κύκλου τμήματι τῷ $\Delta\Gamma B$ τῇ ὑπὸ $\Delta\Gamma B$ γωνίᾳ ἐστὶν ἴση.

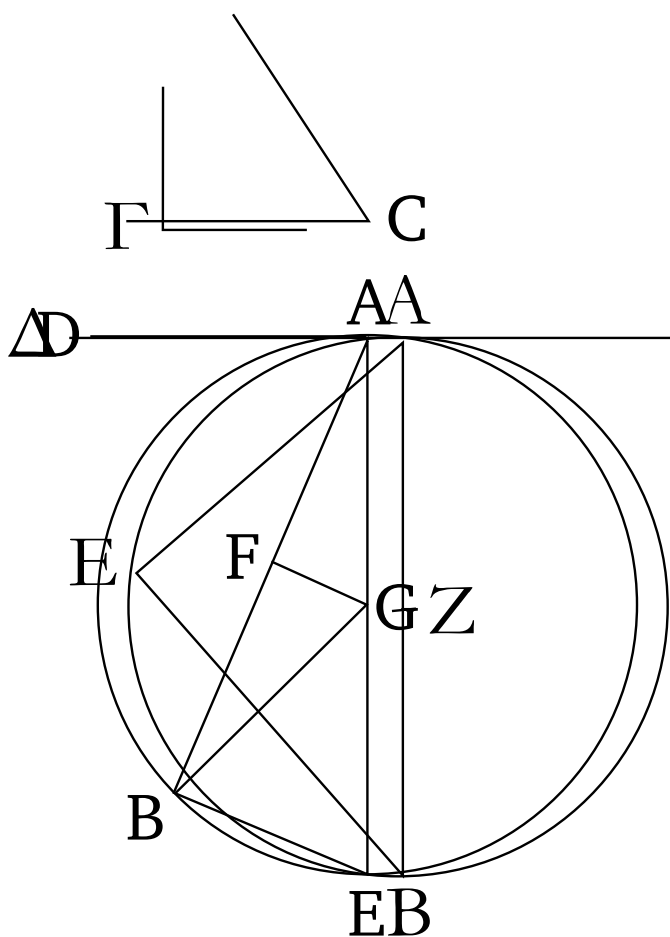
Ἐὰν ἄρα κύκλον ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τῆς ἀφ᾽ ἧς εἰς τὸν κύκλον διασθῇ τις εὐθεΐα τέμνουσα τὸν κύκλον, ἅς ποιεῖ γωνίας πρὸς τῇ ἐφαπτομένῃ, ἴσαι ἔσονται ταῖς ἐν τοῖς ἐναλλάξ τοῦ κύκλου τμήμασι γωνίαις· ὅπερ ἔδει δεῖξαι.

[Prop. 3.22]. And DBF and DBE is also equal to two right-angles [Prop. 1.13]. Thus, DBF and DBE is equal to BAD and BCD , of which BAD was shown (to be) equal to DBF . Thus, the remaining (angle) DBE is equal to the angle DCB in the alternate segment DCB of the circle.

Thus, if some straight-line touches a circle, and some (other) straight-line is drawn across, from the point of contact into the circle, cutting the circle (in two), (then) those angles the (straight-line) makes with the tangent will be equal to the angles in the alternate segments of the circle. (Which is) the very thing it was required to show.

33.

Ἐπὶ τῆς δοθείσης εὐθείας γράψαι τμήμα κύκλον δευτέρου γωνίαν ἴσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμω.

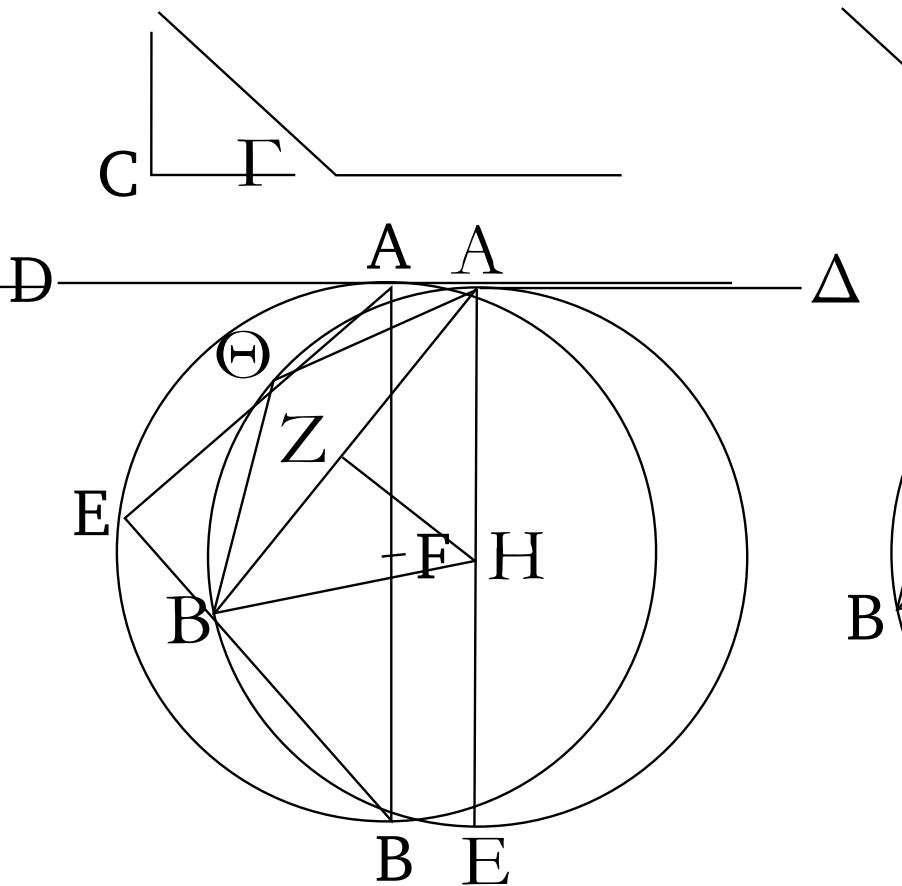


Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ AB , ἡ δὲ δοθεῖσα γωνία ἐκ-
 τὸν A ἡ Γ . δεῖ δὴ ἐπὶ τῆς δοθεῖσης εὐθείας
 τῆς AB γράψαι τμήμα κύκλου δεσθόμενον γωνίαν ἴσην τῇ
 ἐκ τὸν A ἡ Γ .

Ἡ δὲ πρὸς τῷ Γ [γωνία] ἦτοι ὀξεῖα ἐστὶν ἢ ὀρθή ἢ ἀμβλεῖα· ἔστω πρότερον ὀξεῖα, καὶ ὥς ἐπὶ τῇς πρώτης κατα-

Proposition 33

To draw a segment of a circle, accepting an angle equal to a given rectilinear angle, on a given straight-line.



Let AB be the given straight-line, and C the given rectilinear angle. So it is required to draw a segment of a circle, accepting an angle equal to C , on the given straight-line AB .

So the [angle] C is surely either acute, a right-angle, or obtuse. First of all, let it be acute. And, as in the first diagram (from the left), let (angle) BAD , equal to angle C ,

γραφήσιν συνεστάτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ A σημείῳ τῇ πρὸς τῷ Γ γωνία ἴση ἢ ὑπὸ $BA\Delta$. ὅχεια ἄρα ἐστὶ καὶ ἡ ὑπὸ $BA\Delta$. ἤθθω τῇ ΔA πρὸς ὀρθὰς ἡ AE , καὶ τετμήσθω ἡ AB δίθω κατὰ τὸ Z , καὶ ἤθθω ἀπὸ τοῦ Z σημείου τῇ AB πρὸς ὀρθὰς ἡ ZH , καὶ ἐπεξεύθθω ἡ HB .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ AZ τῇ ZB , κοινὴ δὲ ἡ ZH , δύο δὲ αἱ AZ , ZH δύο ταῖς BZ , ZH ἴσαι εἰσὶν καὶ γωνία ἡ ὑπὸ AZH [γωνία] τῇ ὑπὸ BZH ἴση· βάσει ἄρα ἡ AH βάσει τῇ BH ἴση ἐστίν. ὁ ἄρα κέντρον μὲν τῷ H διαστήματι δὲ τῷ HA κύκλος γραφόμενος ἔχει καὶ διὰ τοῦ B . γεγράφθω καὶ ἔστω ὁ ABE , καὶ ἐπεξεύθθω ἡ EB . ἐπεὶ οὖν ἀπ' ἀκρᾶς τῆς AE διαμέτρον ἀπὸ τοῦ A τῇ AE πρὸς ὀρθὰς ἐστὶν ἡ $A\Delta$, ἡ $A\Delta$ ἄρα ἐφάπτεται τοῦ ABE κύκλου· ἐπεὶ οὖν κύκλον τοῦ ABE ἐφάπτεται τις εὐθεῖα ἡ $A\Delta$, καὶ ἀπὸ τῆς κατὰ τὸ A ἀφῆς εἰς τὸν ABE κύκλον διηκταὶ τις εὐθεῖα ἡ AB , ἡ ἄρα ὑπὸ ΔAB γωνία ἴση ἐστὶ τῇ ἐν τῷ ἐναλλάξ τοῦ κύκλου τμήματι γωνία τῇ ὑπὸ AEB . ἀλλ' ἡ ὑπὸ ΔAB τῇ πρὸς τῷ Γ ἐστὶν ἴση· καὶ ἡ πρὸς τῷ Γ ἄρα γωνία ἴση ἐστὶ τῇ ὑπὸ AEB .

Ἐπὶ τῆς δοθείσης ἄρα εὐθείας τῆς AB τμήμα κύκλου γέγραπται τὸ ABE δεθόμενον γωνίαν τὴν ὑπὸ AEB ἴσην τῇ δοθείσῃ τῇ πρὸς τῷ Γ .

Ἀλλὰ δὴ ὀρθὴ ἔστω ἡ πρὸς τῷ Γ · καὶ δεόν πάλιν ἔστω ἐπὶ τῆς AB γράψαι τμήμα κύκλου δεθόμενον γωνίαν ἴσην τῇ πρὸς τῷ Γ ὀρθῇ [γωνία]. συνεστάτω [πάλιν] τῇ πρὸς τῷ Γ ὀρθῇ γωνία ἴση ἢ ὑπὸ $BA\Delta$, ὡς ἔθει ἐπὶ τῆς δευτέρας καταγραφῆς, καὶ τετμήσθω ἡ AB δίθω κατὰ τὸ Z , καὶ κέντρον τῷ Z , διαστήματι δὲ ὅποτέρω τῶν ZA , ZB , κύκλος γεγράφθω ὁ ABE .

Ἐφάπτεται ἄρα ἡ $A\Delta$ εὐθεῖα τοῦ ABE κύκλου διὰ τὸ ὀρθὴν εἶναι τὴν πρὸς τῷ A γωνίαν. καὶ ἴση ἐστὶν ἡ ὑπὸ $BA\Delta$ γωνία τῇ ἐν τῷ AEB τμήματι· ὀρθὴ γάρ καὶ αὐτὴ ἐν ἡμικυκλίῳ οὔσα. ἀλλὰ καὶ ἡ ὑπὸ $BA\Delta$ τῇ πρὸς τῷ Γ ἴση ἐστίν. καὶ ἡ ἐν τῷ AEB ἄρα ἴση ἐστὶ τῇ πρὸς τῷ Γ .

Γέγραπται ἄρα πάλιν ἐπὶ τῆς AB τμήμα κύκλου τὸ ABE δεθόμενον γωνίαν ἴσην τῇ πρὸς τῷ Γ .

Ἀλλὰ δὴ ἡ πρὸς τῷ Γ ἀμβλεία ἔστω· καὶ συνεστάτω αὐτῇ ἴση πρὸς τῇ AB εὐθείᾳ καὶ τῷ A σημείῳ ἡ ὑπὸ $BA\Delta$, ὡς ἔθει ἐπὶ τῆς τρίτης καταγραφῆς, καὶ τῇ $A\Delta$ πρὸς ὀρθὰς ἤθθω ἡ AE , καὶ τετμήσθω πάλιν ἡ AB δίθω κατὰ τὸ Z , καὶ τῇ AB πρὸς ὀρθὰς ἤθθω ἡ ZH , καὶ ἐπεξεύθθω ἡ HB .

Καὶ ἐπεὶ πάλιν ἴση ἐστὶν ἡ AZ τῇ ZB , καὶ κοινὴ ἡ ZH , δύο δὲ αἱ AZ , ZH δύο ταῖς BZ , ZH ἴσαι εἰσὶν καὶ γωνία ἡ ὑπὸ AZH γωνία τῇ ὑπὸ BZH ἴση· βάσει ἄρα ἡ AH βάσει τῇ BH ἴση ἐστίν. ὁ ἄρα κέντρον μὲν τῷ H διαστήματι δὲ τῷ HA κύκλος γραφόμενος ἔχει καὶ διὰ τοῦ B . ἐρθέσθω ὡς ὁ ABE . καὶ ἐπεὶ τῇ AE διαμέτρῳ ἀπ' ἀκρᾶς πρὸς ὀρθὰς ἐστὶν ἡ $A\Delta$, ἡ $A\Delta$ ἄρα ἐφάπτεται τοῦ ABE κύκλου. καὶ ἀπὸ τῆς κατὰ τὸ A ἐπαφῆς διηκταὶ ἡ AB · ἡ ἄρα ὑπὸ $BA\Delta$ γωνία ἴση ἐστὶ τῇ ἐν τῷ ἐναλλάξ τοῦ κύκλου τμήματι τῷ $A\Theta B$ συνισταμένη γωνία. ἀλλ' ἡ ὑπὸ $BA\Delta$ γωνία τῇ πρὸς τῷ Γ ἴση ἐστίν. καὶ ἡ ἐν τῷ $A\Theta B$ ἄρα τμήματι γωνία ἴση ἐστὶ τῇ πρὸς τῷ Γ .

be constructed on the straight-line AB , at the point A (on it) [Prop. 1.23]. Thus, BAD is also acute. Let AE be drawn, at right-angles to DA [Prop. 1.11]. And let AB be cut in half at F [Prop. 1.10]. And let FG be drawn from point F , at right-angles to AB [Prop. 1.11]. And let GB be joined.

And since AF is equal to FB , and FG (is) common, the two (straight-lines) AF , FG are equal to the two (straight-lines) BF , FG (respectively). And angle AFG (is) equal to [angle] BFG . Thus, the base AG is equal to the base BG [Prop. 1.4]. Thus, the circle drawn with center G , and radius GA , will also go through B (as well as A). Let it be drawn, and let it be (denoted) ABE . And let EB be joined. Therefore, since AD is at the extremity of diameter AE , (namely, point) A , at right-angles to AE , the (straight-line) AD thus touches the circle ABE [Prop. 3.16 corr.]. Therefore, since some straight-line AD touches the circle ABE , and some (other) straight-line AB has been drawn across from the point of contact A into circle ABE , angle DAB is thus equal to the angle AEB in the alternate segment of the circle [Prop. 3.32]. But, DAB is equal to C . Thus, angle C is also equal to AEB .

Thus, a segment AEB of a circle, accepting the angle AEB (which is) equal to the given (angle) C , has been drawn on the given straight-line AB .

And so let C be a right-angle. And let it again be necessary to draw a segment of a circle on AB , accepting an angle equal to the right-[angle] C . Let the (angle) BAD [again] be constructed, equal to the right-angle C [Prop. 1.23], as in the second diagram (from the left). And let AB be cut in half at F [Prop. 1.10]. And let the circle ABE be drawn with center F , and radius either FA or FB .

Thus, the straight-line AD touches the circle ABE , on account of the angle at A being a right-angle [Prop. 3.16 corr.]. And angle BAD is equal to the angle in segment AEB . For (the latter angle), being in a semi-circle, is also a right-angle [Prop. 3.31]. But, BAD is also equal to C . Thus, the (angle) in (segment) AEB is also equal to C .

Thus, a segment AEB of a circle, accepting an angle equal to C , has again been drawn on AB .

And so let (angle) C be obtuse. And let (angle) BAD , equal to (C), be constructed on the straight-line AB , at the point A (on it) [Prop. 1.23], as in the third diagram (from the left). And let AE be drawn, at right-angles to AD [Prop. 1.11]. And let AB again be cut in half at F [Prop. 1.10]. And let FG be drawn, at right-angles to AB [Prop. 1.10]. And let GB be joined.

And again, since AF is equal to FB , and FG (is) common, the two (straight-lines) AF , FG are equal to the two (straight-lines) BF , FG (respectively). And angle AFG (is) equal to angle BFG . Thus, the base AG is equal to the base BG [Prop. 1.4]. Thus, a circle of center G , and radius GA , being drawn, will also go through B (as well as A). Let it go like ABE (in the third diagram from the left). And since AD

Ἐπὶ τῆς ἄρα δοθείσης εὐθείας τῆς AB γέγραπται τμήμα κύκλον τὸ $A\Theta B$ δεθόμενον γωνίαν ἴσην τῇ πρὸς τῷ Γ . ὅπερ ἔδει ποιῆσαι.

is at right-angles to the diameter AE , at its extremity, AD thus touches circle AEB [Prop. 3.16 corr.]. And AB has been drawn across (the circle) from the point of contact A . Thus, angle BAD is equal to the angle constructed in the alternate segment AHB of the circle [Prop. 3.32]. But, angle BAD is equal to C . Thus, the angle in segment AHB is also equal to C .

Thus, a segment AHB of a circle, accepting an angle equal to C , has been drawn on the given straight-line AB . (Which is) the very thing it was required to do.

34.

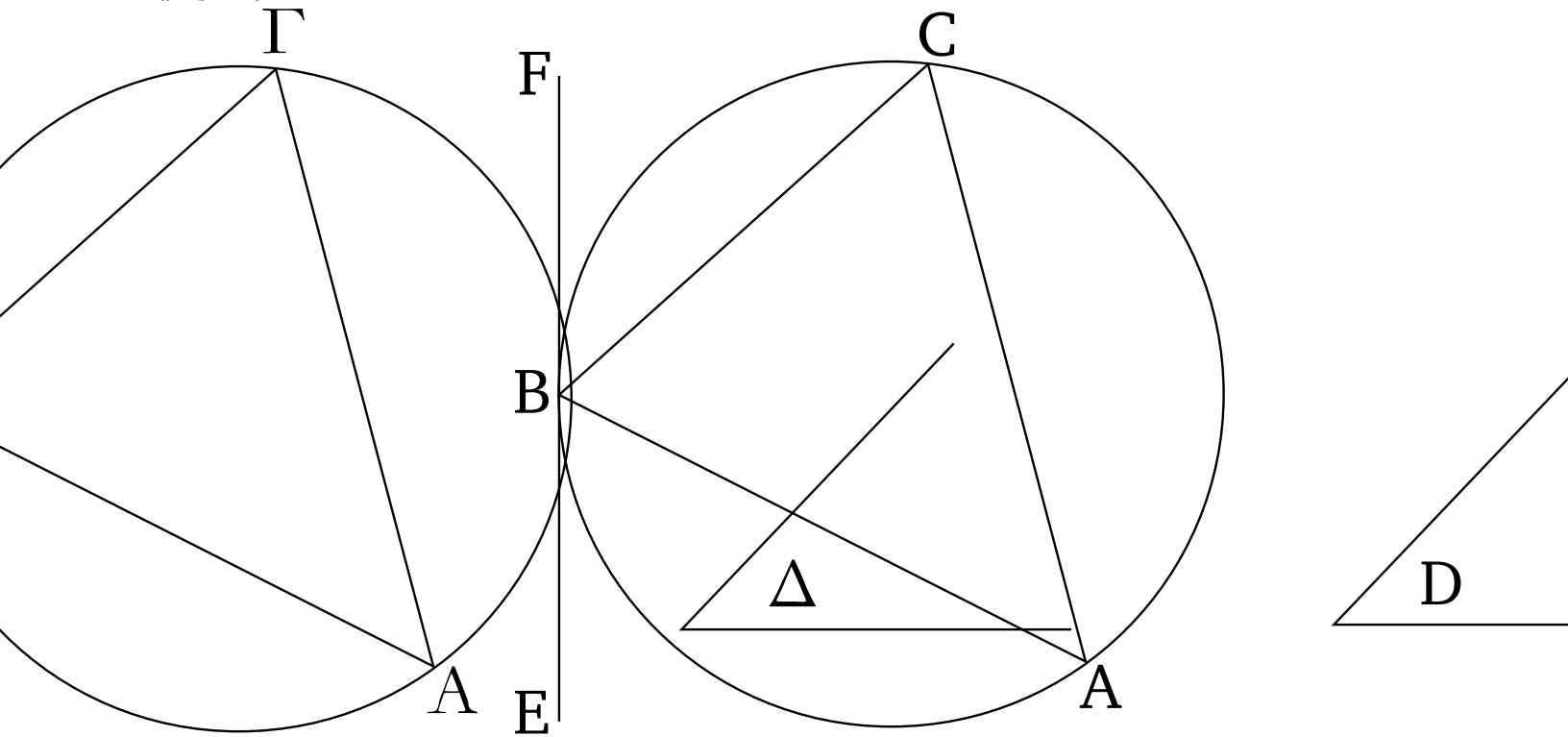
Ἀπὸ τοῦ δοθέντος κύκλου τμήμα ἀφελεῖν δεθόμενον γωνίαν ἴσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ.

Ἐστω ὁ δοθείς κύκλος ὁ $AB\Gamma$, ἡ δὲ δοθεῖσα γωνία εὐθυγράμμος ἡ πρὸς τῷ Δ . δεῖ δὲ ἀπὸ τοῦ $AB\Gamma$ κύκλου τμήμα ἀφελεῖν δεθόμενον γωνίαν ἴσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ πρὸς τῷ Δ .

Proposition 34

To cut off a segment, accepting an angle equal to a given rectilinear angle, from a given circle.

Let ABC be the given circle, and D the given rectilinear angle. So it is required to cut off a segment, accepting an angle equal to the given rectilinear angle D , from the given circle ABC .



Ἦθθω τοῦ $AB\Gamma$ ἐφαπτομένη ἡ EZ κατὰ τὸ B σημεῖον, καὶ συνεστώτω πρὸς τῇ ZB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ B τῇ πρὸς τῷ Δ γωνίᾳ ἴση ἡ ὑπὸ $ZB\Gamma$.

Ἐπεὶ οὖν κύκλον τοῦ $AB\Gamma$ ἐφάπτεται τις εὐθεία ἡ EZ , καὶ ἀπὸ τῆς κατὰ τὸ B ἐπαφῆς διήκται ἡ $B\Gamma$, ἡ ὑπὸ $ZB\Gamma$ ἄρα γωνία ἴση ἐστὶ τῇ ἐν τῷ BAG ἐναλλάξ τμήματι συνισταμένη γωνίᾳ. ἀλλ' ἡ ὑπὸ $ZB\Gamma$ τῇ πρὸς τῷ Δ ἐστὶν ἴση· καὶ ἡ ἐν τῷ BAG ἄρα τμήματι ἴση ἐστὶ τῇ πρὸς τῷ Δ [γωνίᾳ].

Ἀπὸ τοῦ δοθέντος ἄρα κύκλου τοῦ $AB\Gamma$ τμήμα ἀφήρηται

Let EF be drawn touching ABC at point B .[†] And let (angle) FBC , equal to angle D , be constructed on the straight-line FB , at the point B on it [Prop. 1.23].

Therefore, since some straight-line EF touches the circle ABC , and BC has been drawn across (the circle) from the point of contact B , angle FBC is thus equal to the angle constructed in the alternate segment BAC [Prop. 1.32]. But, FBC is equal to D . Thus, the (angle) in the segment BAC is also equal to [angle] D .

τὸ BAG δεθόμενον γωνίαν ἴσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμω τῇ πρὸς τῷ Δ · ὅπερ ἔδει ποιῆσαι.

Thus, the segment BAC , accepting an angle equal to the given rectilinear angle D , has been cut off from the given circle ABC . (Which is) the very thing it was required to do.

† Presumably, by finding the center of ABC [Prop. 3.1], drawing a straight-line between the center and point B , and then drawing EF through point B , at right-angles to the aforementioned straight-line [Prop. 1.11].

35.

Ἐάν ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὸ ὑπὸ τῶν τῆς μιᾶς τμημάτων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν τῆς ἑτέρας τμημάτων περιεχομένῳ ὀρθογώνιῳ.

Ἐν γάρ κύκλῳ τῷ $AB\Gamma\Delta$ δύο εὐθεῖαι αἱ AG , BD τέμνεται ἀλλήλας κατὰ τὸ E σημεῖον· λέγω, ὅτι τὸ ὑπὸ τῶν AE , EG περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν DE , EB περιεχομένῳ ὀρθογώνιῳ.

Εἰ μὲν οὖν αἱ AG , BD διὰ τοῦ κέντρου εἰσὶν ὥστε τὸ E κέντρον εἶναι τοῦ $AB\Gamma\Delta$ κύκλου, φανερόν, ὅτι ἴσων οὐσῶν τῶν AE , EG , DE , EB καὶ τὸ ὑπὸ τῶν AE , EG περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν DE , EB περιεχομένῳ ὀρθογώνιῳ.

μὴ ἔστωσαν δὴ αἱ AG , BD διὰ τοῦ κέντρου, καὶ εἰλήφθω τὸ κέντρον τοῦ $AB\Gamma\Delta$, καὶ ἔστω τὸ Z , καὶ ἀπὸ τοῦ Z ἐπὶ τὰς AG , BD εὐθείας κάθετοι ἡθῶσαν αἱ ZH , $Z\Theta$, καὶ ἐπεξέυθῃωσαν αἱ ZB , ZI , ZE .

Καὶ ἐπεὶ εὐθεῖα τις διὰ τοῦ κέντρου ἡ HZ εὐθειάν τινα μὴ διὰ τοῦ κέντρου τὴν AG πρὸς ὀρθὰς τέμνει, καὶ δίθῃ αὐτὴν τέμνει· ἴση ἄρα ἡ AH τῇ HG . ἐπεὶ οὖν εὐθεῖα ἡ AG τέμνεται εἰς μὲν ἴσα κατὰ τὸ H , εἰς δὲ ἄνισα κατὰ τὸ E , τὸ ἄρα ὑπὸ τῶν AE , EG περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς EH τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς HG · [κοινόν] προσκείσθω τὸ ἀπὸ τῆς HZ · τὸ ἄρα ὑπὸ τῶν AE , EG μετὰ τῶν ἀπὸ τῶν HE , HZ ἴσον ἐστὶ τοῖς ἀπὸ τῶν GH , HZ . ἀλλὰ τοῖς μὲν ἀπὸ τῶν EH , HZ ἴσον ἐστὶ τὸ ἀπὸ τῆς ZE , τοῖς δὲ ἀπὸ τῶν GH , HZ ἴσον ἐστὶ τὸ ἀπὸ τῆς ZI · τὸ ἄρα ὑπὸ τῶν AE , EG μετὰ τοῦ ἀπὸ τῆς ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς ZI . ἴση δὲ ἡ ZI τῇ ZB · τὸ ἄρα ὑπὸ τῶν AE , EG μετὰ τοῦ ἀπὸ τῆς ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς ZB . διὰ τὰ αὐτὰ δὴ καὶ τὸ ὑπὸ τῶν DE , EB μετὰ τοῦ ἀπὸ τῆς ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς ZB . ἐδείκθη δὲ καὶ τὸ ὑπὸ τῶν AE , EG μετὰ τοῦ ἀπὸ τῆς ZE ἴσον τῷ ἀπὸ τῆς ZB · τὸ ἄρα ὑπὸ τῶν AE , EG μετὰ τοῦ ἀπὸ τῆς ZE ἴσον ἐστὶ τῷ ὑπὸ τῶν DE , EB μετὰ τοῦ ἀπὸ τῆς ZE . κοινὸν ἀφῆρησθῶ τὸ ἀπὸ τῆς ZE · λοιπὸν ἄρα τὸ ὑπὸ τῶν AE , EG περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν DE , EB περιεχομένῳ ὀρθογώνιῳ.

Proposition 35

If two straight-lines in a circle cut one another, (then) the rectangle contained by the pieces of one is equal to the rectangle contained by the pieces of the other.

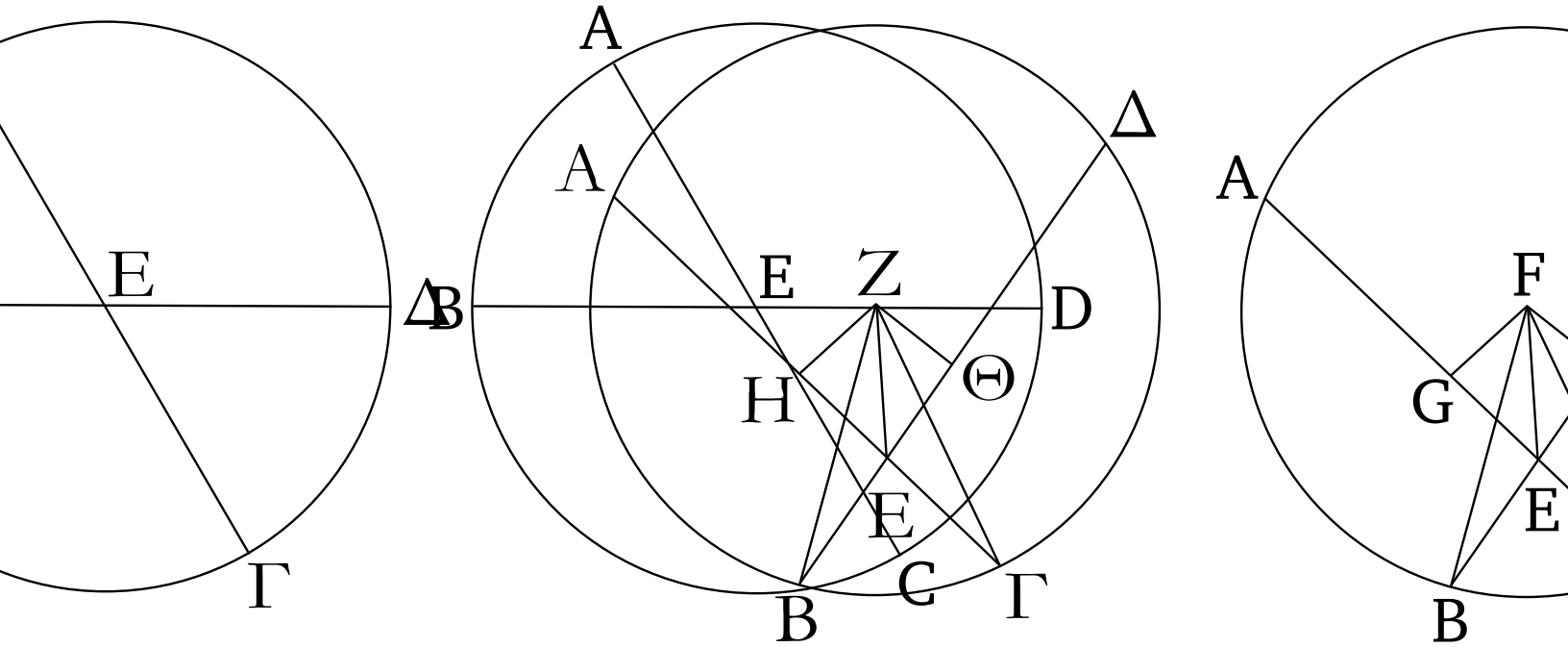
For let the two straight-lines AC and BD , in the circle $ABCD$, cut one another at point E . I say that the rectangle contained by AE and EC is equal to the rectangle contained by DE and EB .

In fact, if AC and BD are through the center (as in the first diagram from the left), so that E is the center of circle $ABCD$, (then it is) clear that, AE , EC , DE , and EB being equal, the rectangle contained by AE and EC is also equal to the rectangle contained by DE and EB .

So let AC and BD not be through the center (as in the second diagram from the left), and let the center of $ABCD$ be found [Prop. 3.1], and let it be (at) F . And let FG and FH be drawn from F , perpendicular to the straight-lines AC and BD (respectively) [Prop. 1.12]. And let FB , FC , and FE be joined.

And since some straight-line, GF , through the center, cuts at right-angles some (other) straight-line, AC , not through the center, (then) it also cuts it in half [Prop. 3.3]. Thus, AG (is) equal to GC . Therefore, since the straight-line AC is cut equally at G , and unequally at E , the rectangle contained by AE and EC plus the square on EG is thus equal to the (square) on GC [Prop. 2.5]. Let the (square) on GF be added [to both]. Thus, the (rectangle contained) by AE and EC plus the (sum of the squares) on GE and GF is equal to the (sum of the squares) on CG and GF . But, the (square) on FE is equal to the (sum of the squares) on EG and GF [Prop. 1.47], and the (square) on FC is equal to the (sum of the squares) on CG and GF [Prop. 1.47]. Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (square) on FC . And FC (is) equal to FB . Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (square) on FB . So, for the same (reasons), the (rectangle contained) by DE and EB plus the (square) on FE is equal to the (square) on FB . And the (rectangle contained) by AE and EC plus the (square) on FE was also shown (to be) equal to the (square) on FB . Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (rectangle contained) by DE and EB plus the (square) on FE . Let the (square) on FE be taken from both. Thus, the remaining rectangle contained by

AE and EC is equal to the rectangle contained by DE and EB .



Ἐάν ἄρα ἐν κύκλῳ εὐθεῖαι δύο τέμνωσιν ἀλλήλας, τὸ ὑπὸ τῶν τῆς μᾶξ τιμημάτων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν τῆς ἑτέρας τιμημάτων περιεχομένῳ ὀρθογώνιῳ· ὅπερ ἔδει δεῖξαι.

Thus, if two straight-lines in a circle cut one another, (then) the rectangle contained by the pieces of one is equal to the rectangle contained by the pieces of the other. (Which is) the very thing it was required to show.

36.

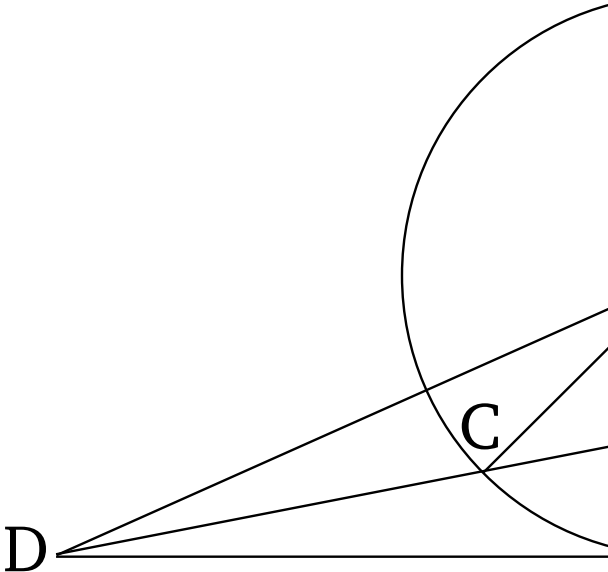
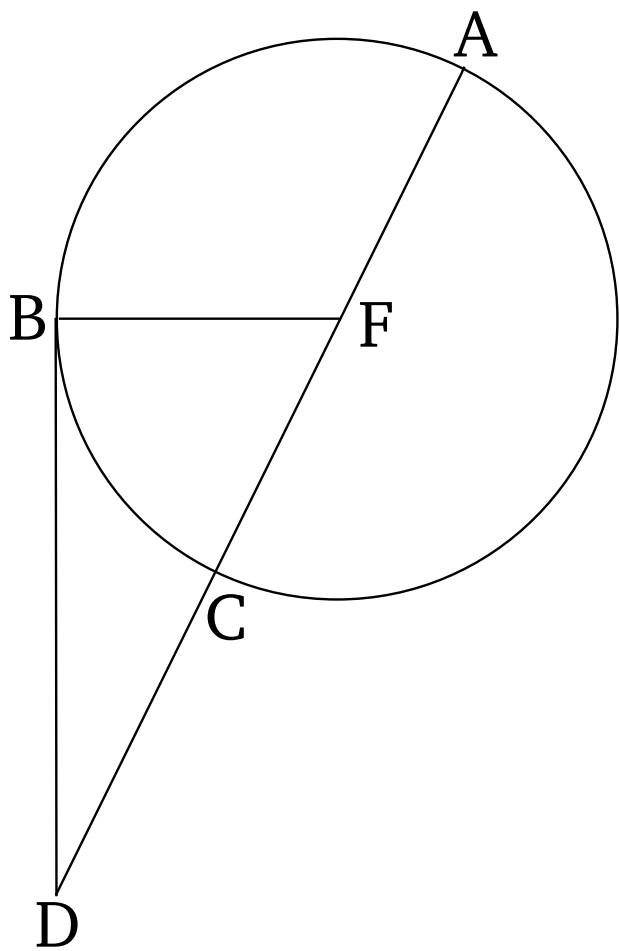
Proposition 36

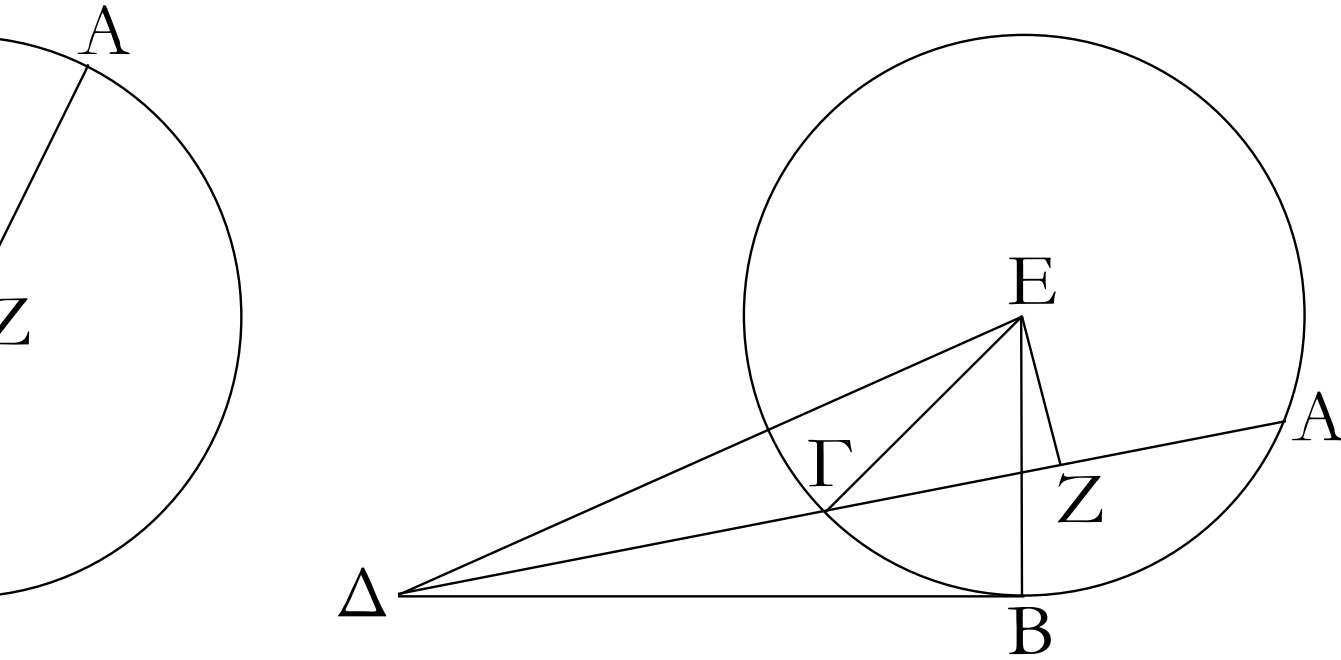
Ἐάν κύκλον ληφθῇ τι σημεῖον ἐκτός, καὶ ἀπ' αὐτοῦ πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ ἐφάπτεται, ἔσται τὸ ὑπὸ ὅλης τῆς τεμνοῦσης καὶ τῆς ἐκτός ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ἴσον τῷ ἀπὸ τῆς ἐφαπτομένης τετραγώνῳ.

Κύκλον γάρ τοῦ $AB\Gamma$ εἰλήφθω τι σημεῖον ἐκτός τὸ Δ , καὶ ἀπὸ τοῦ Δ πρὸς τὸν $AB\Gamma$ κύκλον προσπιπτέτωσαν δύο εὐθεῖαι αἱ $\Delta\Gamma[A]$, ΔB · καὶ ἡ μὲν $\Delta\Gamma A$ τεμνέτω τὸν $AB\Gamma$ κύκλον, ἡ δὲ $B\Delta$ ἐφαπτέσθω· λέγω, ὅτι τὸ ὑπὸ τῶν $A\Delta$, $\Delta\Gamma$ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔB τετραγώνῳ.

If some point is taken outside a circle, and two straight-lines radiate from it towards the circle, and (one) of them cuts the circle, and the (other) touches (it), (then) the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, will be equal to the square on the tangent (line).

For let some point D be taken outside circle ABC , and let two straight-lines, $DC[A]$ and DB , radiate from D towards circle ABC . And let DCA cut circle ABC , and let BD touch (it). I say that the rectangle contained by AD and DC is equal to the square on DB .





Ἡ ἄρα $[\Delta]ΓΑ$ ἤτοι διὰ τοῦ κέντρου ἐστὶν ἡ οὐ. ἔστω πρότερον διὰ τοῦ κέντρου, καὶ ἔστω τὸ Z κέντρον τοῦ $ABΓ$ κύκλου, καὶ ἐπεζεύθῃ ἡ ZB . ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ $ZBΔ$. καὶ ἐπεὶ εὐθεῖα ἡ $ΑΓ$ δίῃα τέτμηται κατὰ τὸ Z , πρόσκειται δὲ αὐτῇ ἡ $ΓΔ$, τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τοῦ ἀπὸ τῆς $ZΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ZΔ$. ἴση δὲ ἡ $ZΓ$ τῇ ZB . τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τοῦ ἀπὸ τῆς ZB ἴσον ἐστὶ τῷ ἀπὸ τῆς $ZΔ$. τῷ δὲ ἀπὸ τῆς $ZΔ$ ἴσα ἐστὶ τὰ ἀπὸ τῶν ZB , $BΔ$. τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τοῦ ἀπὸ τῆς ZB ἴσον ἐστὶ τοῖς ἀπὸ τῶν ZB , $BΔ$. κοινὸν ἀφηγήσθω τὸ ἀπὸ τῆς ZB . λοιπὸν ἄρα τὸ ὑπὸ τῶν $ΑΔ$, $ΔΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΔB$ ἐφαπτομένης.

Ἀλλὰ δὴ ἡ $ΔΓΑ$ μὴ ἔστω διὰ τοῦ κέντρου τοῦ $ABΓ$ κύκλου, καὶ εἰλήφθω τὸ κέντρον τὸ E , καὶ ἀπὸ τοῦ E ἐπὶ τὴν $ΑΓ$ κάθετος ἡθῇ ἡ EZ , καὶ ἐπεζεύθωσαν αἱ EB , $ΕΓ$, $ΕΔ$. ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ $EBΔ$. καὶ ἐπεὶ εὐθεῖα τις διὰ τοῦ κέντρου ἡ EZ εὐθεϊάν τινα μὴ διὰ τοῦ κέντρου τὴν $ΑΓ$ πρὸς ὀρθὰς τέμνει, καὶ δίῃα αὐτὴν τέμνει· ἡ AZ ἄρα τῇ $ZΓ$ ἐστὶν ἴση. καὶ ἐπεὶ εὐθεῖα ἡ $ΑΓ$ τέτμηται δίῃα κατὰ τὸ Z σημείον, πρόσκειται δὲ αὐτῇ ἡ $ΓΔ$, τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τοῦ ἀπὸ τῆς $ZΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ZΔ$. κοινὸν προσκείσθω τὸ ἀπὸ τῆς ZE . τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τῶν ἀπὸ τῶν

$[D]CA$ is surely either through the center, or not. Let it first of all be through the center, and let F be the center of circle ABC , and let FB be joined. Thus, (angle) FBD is a right-angle [Prop. 3.18]. And since straight-line AC is cut in half at F , let CD be added to it. Thus, the (rectangle contained) by AD and DC plus the (square) on FC is equal to the (square) on FD [Prop. 2.6]. And FC (is) equal to FB . Thus, the (rectangle contained) by AD and DC plus the (square) on FB is equal to the (square) on FD . And the (square) on FD is equal to the (sum of the squares) on FB and BD [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on FB is equal to the (sum of the squares) on FB and BD . Let the (square) on FB be subtracted from both. Thus, the remaining (rectangle contained) by AD and DC is equal to the (square) on the tangent DB .

And so let DCA not be through the center of circle ABC , and let the center E be found, and let EF be drawn from E , perpendicular to AC [Prop. 1.12]. And let EB , EC , and ED be joined. (Angle) EBD (is) thus a right-angle [Prop. 3.18]. And since some straight-line, EF , through the center, cuts some (other) straight-line, AC , not through the center, at right-angles, it also cuts it in half [Prop. 3.3]. Thus, AF is equal to

FZ , ZE ἴσον ἐστὶ τοῖς ἀπὸ τῶν $Z\Delta$, ZE . τοῖς δὲ ἀπὸ τῶν FZ , ZE ἴσον ἐστὶ τὸ ἀπὸ τῆς EF . ὁρθὴ γὰρ [ἐστίν] ἡ ὑπὸ EZF [γωνία]. τοῖς δὲ ἀπὸ τῶν ΔZ , ZE ἴσον ἐστὶ τὸ ἀπὸ τῆς $E\Delta$. τὸ ἄρα ὑπὸ τῶν $A\Delta$, ΔF μετὰ τοῦ ἀπὸ τῆς EF ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Delta$. ἴση δὲ ἡ EF τῇ EB . τὸ ἄρα ὑπὸ τῶν $A\Delta$, ΔF μετὰ τοῦ ἀπὸ τῆς EB ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Delta$. τῷ δὲ ἀπὸ τῆς $E\Delta$ ἴσα ἐστὶ τὰ ἀπὸ τῶν EB , $B\Delta$. ὁρθὴ γὰρ ἡ ὑπὸ $EB\Delta$ γωνία. τὸ ἄρα ὑπὸ τῶν $A\Delta$, ΔF μετὰ τοῦ ἀπὸ τῆς EB ἴσον ἐστὶ τοῖς ἀπὸ τῶν EB , $B\Delta$. κοινὸν ἀφαιρήσθω τὸ ἀπὸ τῆς EB . λοιπὸν ἄρα τὸ ὑπὸ τῶν $A\Delta$, ΔF ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔB .

Ἐὰν ἄρα κύκλον ληφθῇ τι σημεῖον ἐκτός, καὶ ἀπ' αὐτοῦ πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ ἐφάπτεται, ἔσται τὸ ὑπὸ ὅλης τῆς τεμνούσης καὶ τῆς ἐκτός ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ἴσον τῷ ἀπὸ τῆς ἐφαπτομένης τετραγώνῳ· ὅπερ ἔδει δεῖξαι.

FC . And since the straight-line AC is cut in half at point F , let CD be added to it. Thus, the (rectangle contained) by AD and DC plus the (square) on FC is equal to the (square) on FD [Prop. 2.6]. Let the (square) on FE be added to both. Thus, the (rectangle contained) by AD and DC plus the (sum of the squares) on CF and FE is equal to the (sum of the squares) on FD and FE . But the (square) on EC is equal to the (sum of the squares) on CF and FE . For [angle] EFC [is] a right-angle [Prop. 1.47]. And the (square) on ED is equal to the (sum of the squares) on DF and FE [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on EC is equal to the (square) on ED . And EC (is) equal to EB . Thus, the (rectangle contained) by AD and DC plus the (square) on EB is equal to the (square) on ED . And the (sum of the squares) on EB and BD is equal to the (square) on ED . For EBD (is) a right-angle [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on EB is equal to the (sum of the squares) on EB and BD . Let the (square) on EB be subtracted from both. Thus, the remaining (rectangle contained) by AD and DC is equal to the (square) on BD .

Thus, if some point is taken outside a circle, and two straight-lines radiate from it towards the circle, and (one) of them cuts the circle, and (the other) touches (it), (then) the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, will be equal to the square on the tangent (line). (Which is) the very thing it was required to show.

37.

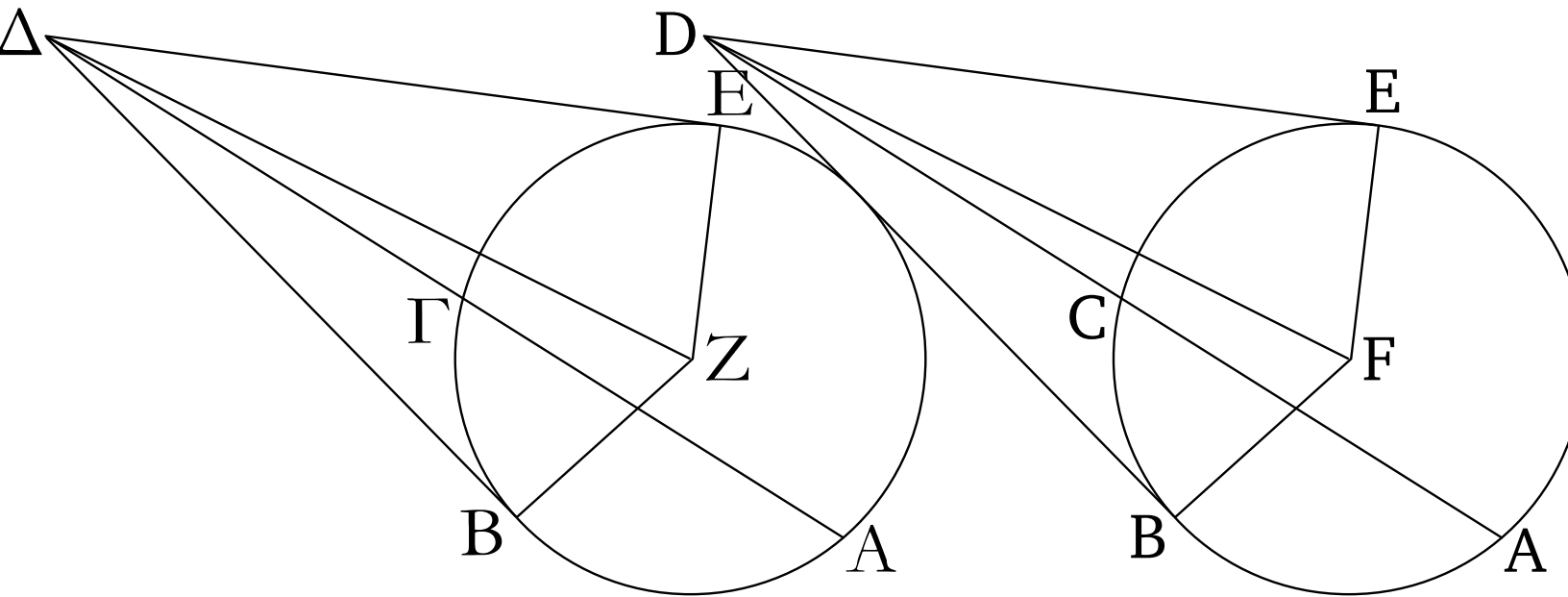
Proposition 37

Ἐὰν κύκλον ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ προσπίπτῃ, ἢ δὲ τὸ ὑπὸ [τῆς] ὅλης τῆς τεμνούσης καὶ τῆς ἐκτός ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ἴσον τῷ ἀπὸ τῆς προσπίπτουσας, ἡ προσπίπτουσα ἐφάπτεται τοῦ κύκλου.

Κύκλον γάρ τοῦ $AB\Gamma$ εἰλήφθω τι σημεῖον ἐκτός τὸ Δ , καὶ ἀπὸ τοῦ Δ πρὸς τὸν $AB\Gamma$ κύκλον προσπίπτέτωσαν δύο εὐθεῖαι αἱ $\Delta\Gamma A$, ΔB , καὶ ἡ μὲν $\Delta\Gamma A$ τεμνέτω τὸν κύκλον, ἡ δὲ ΔB προσπίπτέτω, ἔστω δὲ τὸ ὑπὸ τῶν $A\Delta$, $\Delta\Gamma$ ἴσον τῷ ἀπὸ τῆς ΔB . λέγω, ὅτι ἡ ΔB ἐφάπτεται τοῦ $AB\Gamma$ κύκλου.

If some point is taken outside a circle, and two straight-lines radiate from the point towards the circle, and one of them cuts the circle, and the (other) meets (it), and the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, is equal to the (square) on the (straight-line) meeting (the circle), (then) the (straight-line) meeting (the circle) will touch the circle.

For let some point D be taken outside circle ABC , and let two straight-lines, DCA and DB , radiate from D towards circle ABC , and let DCA cut the circle, and let DB meet (the circle). And let the (rectangle contained) by AD and DC be equal to the (square) on DB . I say that DB touches circle ABC .



Ἦθθω γάρ τοῦ $AB\Gamma$ ἐφαπτομένη ἡ ΔE , καὶ εἰλήφθω τὸ κέντρον τοῦ $AB\Gamma$ κύκλου, καὶ ἔστω τὸ Z , καὶ ἐπεζεύθωσαν αἱ ZE , ZB , $Z\Delta$. ἡ ἄρα ὑπὸ $ZE\Delta$ ὀρθή ἐστιν. καὶ ἐπεὶ ἡ ΔE ἐφάπτεται τοῦ $AB\Gamma$ κύκλου, τέμνει δὲ ἡ $\Delta\Gamma A$, τὸ ἄρα ὑπὸ τῶν $A\Delta$, $\Delta\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔE . ἦν δὲ καὶ τὸ ὑπὸ τῶν $A\Delta$, $\Delta\Gamma$ ἴσον τῷ ἀπὸ τῆς ΔB . τὸ ἄρα ἀπὸ τῆς ΔE ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔB . ἴση ἄρα ἡ ΔE τῇ ΔB . ἐστὶ δὲ καὶ ἡ ZE τῇ ZB ἴση· δύο δὲ αἱ ΔE , EZ δύο ταῖς ΔB , BZ ἴσαι εἰσὶν καὶ βάσεις αὐτῶν κοινὴ ἡ $Z\Delta$. γωνία ἄρα ἡ ὑπὸ ΔEZ γωνία τῇ ὑπὸ ΔBZ ἐστὶν ἴση. ὀρθὴ δὲ ἡ ὑπὸ ΔEZ . ὀρθὴ ἄρα καὶ ἡ ὑπὸ ΔBZ . καὶ ἐστὶν ἡ ZB ἐκβαλλομένη διάμετρος· ἡ δὲ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένη ἐφάπτεται τοῦ κύκλου· ἡ ΔB ἄρα ἐφάπτεται τοῦ $AB\Gamma$ κύκλου. ὁμοίως δὲ δειθθήσεται, κ' ἂν τὸ κέντρον ἐπὶ τῆς $A\Gamma$ τυγθάνῃ.

Ἐάν ἄρα κύκλον ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσιπύωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ προσιπύτῃ, ἥ δὲ τὸ ὑπὸ ὅλης τῆς τεμνούσης καὶ τῆς ἐκτός ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ἴσον τῷ ἀπὸ τῆς προσιπυτούσης, ἡ προσιπύτουσα ἐφάπτεται τοῦ κύκλου· ὅπερ ἔδει δεῖξαι.

For let DE be drawn touching ABC [Prop. 3.17], and let the center of the circle ABC be found, and let it be (at) F . And let FE , FB , and FD be joined. (Angle) FED is thus a right-angle [Prop. 3.18]. And since DE touches circle ABC , and DCA cuts (it), the (rectangle contained) by AD and DC is thus equal to the (square) on DE [Prop. 3.36]. And the (rectangle contained) by AD and DC was also equal to the (square) on DB . Thus, the (square) on DE is equal to the (square) on DB . Thus, DE (is) equal to DB . And FE is also equal to FB . So the two (straight-lines) DE , EF are equal to the two (straight-lines) DB , BF (respectively). And their base, FD , is common. Thus, angle DEF is equal to angle DBF [Prop. 1.8]. And DEF (is) a right-angle. Thus, DBF (is) also a right-angle. And FB produced is a diameter, And a (straight-line) drawn at right-angles to a diameter of a circle, at its extremity, touches the circle [Prop. 3.16 corr.]. Thus, DB touches circle ABC . Similarly, (the same thing) can be shown, even if the center happens to be on AC .

Thus, if some point is taken outside a circle, and two straight-lines radiate from the point towards the circle, and one of them cuts the circle, and the (other) meets (it), and the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, is equal to the (square) on the (straight-line) meeting (the circle), (then) the (straight-line) meeting (the circle) will touch the circle. (Which is) the very thing it was required to show.

ELEMENTS BOOK 3

*Fundamentals of Plane Geometry Involving
Circles*

Ὅροι.

1. Ἰσοὶ κύκλοι εἰσὶν, ὧν αἱ διάμετροι ἴσαι εἰσὶν, ἢ ὧν αἱ ἐκ τῶν κέντρων ἴσαι εἰσὶν.
2. Εὐθεῖα κύκλον ἐφάπτεσθαι λέγεται, ἥτις ἀπτομένη τοῦ κύκλου καὶ ἐκβαλλομένη οὐ τέμνει τὸν κύκλον.
3. Κύκλοι ἐφάπτεσθαι ἀλλήλων λέγονται ὅττινες ἀπτόμενοι ἀλλήλων οὐ τέμνουσιν ἀλλήλους.
4. Ἐν κύκλῳ ἴσον ἀπέθειν ἀπὸ τοῦ κέντρου εὐθεῖαι λέγονται, ὅταν αἱ ἀπὸ τοῦ κέντρου ἐπ' αὐτάς κάθεται ἀγόμεναι ἴσαι ᾧσιν.
5. Μείζον δὲ ἀπέθειν λέγεται, ἐφ' ἣν ἡ μείζων κάθετοξ πίπτει.
6. Τμήμα κύκλον ἐστὶ τὸ περιεχόμενον σθῆμα ὑπὸ τε εὐθείας καὶ κύκλου περιφερείας.
7. Τμήματοξ δὲ γωνία ἐστὶν ἡ περιεθρομένη ὑπὸ τε εὐθείας καὶ κύκλου περιφερείας.
8. Ἐν τμήματι δὲ γωνία ἐστὶν, ὅταν ἐπὶ τῆς περιφερείας τοῦ τμήματος ληφθῇ τι σημεῖον καὶ ἀπ' αὐτοῦ ἐπὶ τὰ πέρατα τῆς εὐθείας, ἢ ἐστὶ βάσει τοῦ τμήματος, ἐπιξενθθῶσιν εὐθεῖαι, ἢ περιεθρομένη γωνία ὑπὸ τῶν ἐπιξενθθειςῶν εὐθειῶν.
9. Ὅταν δὲ αἱ περιέθουσαι τὴν γωνίαν εὐθεῖαι ἀπολαμβάνωσι τινα περιφέρειαν, ἐπ' ἐκείνης λέγεται βεβηκέναι ἡ γωνία.
10. Τομεὺς δὲ κύκλον ἐστὶν, ὅταν πρὸς τῷ κέντρῳ τοῦ κύκλου συσταθῇ γωνία, τὸ περιεχόμενον σθῆμα ὑπὸ τε τῶν τὴν γωνίαν περιέθουσῶν εὐθειῶν καὶ τῆς ἀπολαμβανομένης ὑπ' αὐτῶν περιφερείας.
11. Ὅμοια τμήματα κύκλων ἐστὶ τὰ δεθόμενα γωνίας ἴσας, ἢ ἐν οἷς αἱ γωνίαι ἴσαι ἀλλήλαις εἰσὶν.

1.

Τοῦ δοθέντος κύκλου τὸ κέντρον εὐρεῖν.

Ἐστω ὁ δοθεὶς κύκλος ὁ $ABΓ$. δεῖ δὴ τοῦ $ABΓ$ κύκλου τὸ κέντρον εὐρεῖν.

Διήθθω τις εἷς αὐτόν, ὥς ἔτυθεν, εὐθεῖα ἡ AB , καὶ τετμήσθω δίθω κατὰ τὸ Δ σημεῖον, καὶ ἀπὸ τοῦ Δ τῇ AB πρὸς ὀρθὰς ἡθθω ἡ $\Delta Γ$ καὶ διήθθω ἐπὶ τὸ E , καὶ τετμήσθω ἡ $ΓΕ$ δίθω κατὰ τὸ Z . λέγω, ὅτι τὸ Z κέντρον ἐστὶ τοῦ $ABΓ$ [κύκλου].

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἔστω τὸ H , καὶ ἐπεξενθθωσαν αἱ HA , HD , HB . καὶ ἐπεὶ ἴση ἐστὶν ἡ AD τῇ ΔB , κοινὴ δὲ ἡ ΔH , δύο δὲ αἱ AD , ΔH δύο ταῖς HD , ΔB ἴσαι εἰσὶν ἑκατέρα ἑκατέρῃ· καὶ βάσει ἡ HA βάσει τῇ HB ἐστὶν ἴση· ἐκ κέντρου γάρ· γωνία ἄρα ἡ ὑπὸ $A\Delta H$ γωνία τῇ ὑπὸ $H\Delta B$ ἴση ἐστὶν. ὅταν δὲ εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα τὰς ἐφεχθῇς γωνίας ἴσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρα τῶν ἴσων γωνιῶν

Definitions

1. Equal circles are (circles) whose diameters are equal, or whose (distances) from the centers (to the circumferences) are equal (i.e., whose radii are equal).
2. A straight-line said to touch a circle is any (straight-line) which, meeting the circle and being produced, does not cut the circle.
3. Circles said to touch one another are any (circles) which, meeting one another, do not cut one another.
4. In a circle, straight-lines are said to be equally far from the center when the perpendiculars drawn to them from the center are equal.
5. And (that straight-line) is said to be further (from the center) on which the greater perpendicular falls (from the center).
6. A segment of a circle is the figure contained by a straight-line and a circumference of a circle.
7. And the angle of a segment is that contained by a straight-line and a circumference of a circle.
8. And the angle in a segment is the angle contained by the joined straight-lines, when any point is taken on the circumference of a segment, and straight-lines are joined from it to the ends of the straight-line which is the base of the segment.
9. And when the straight-lines containing an angle cut off some circumference, the angle is said to stand upon that (circumference).
10. And a sector of a circle is the figure contained by the straight-lines surrounding an angle, and the circumference cut off by them, when the angle is constructed at the center of a circle.
11. Similar segments of circles are those accepting equal angles, or in which the angles are equal to one another.

Proposition 1

To find the center of a given circle.

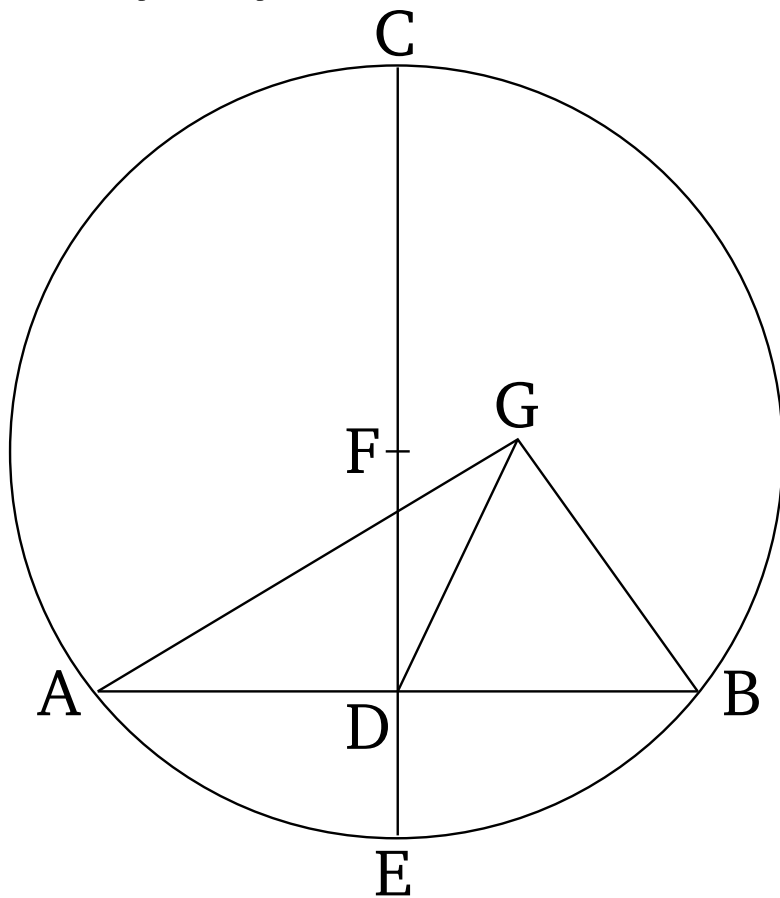
Let ABC be the given circle. So it is required to find the center of circle ABC .

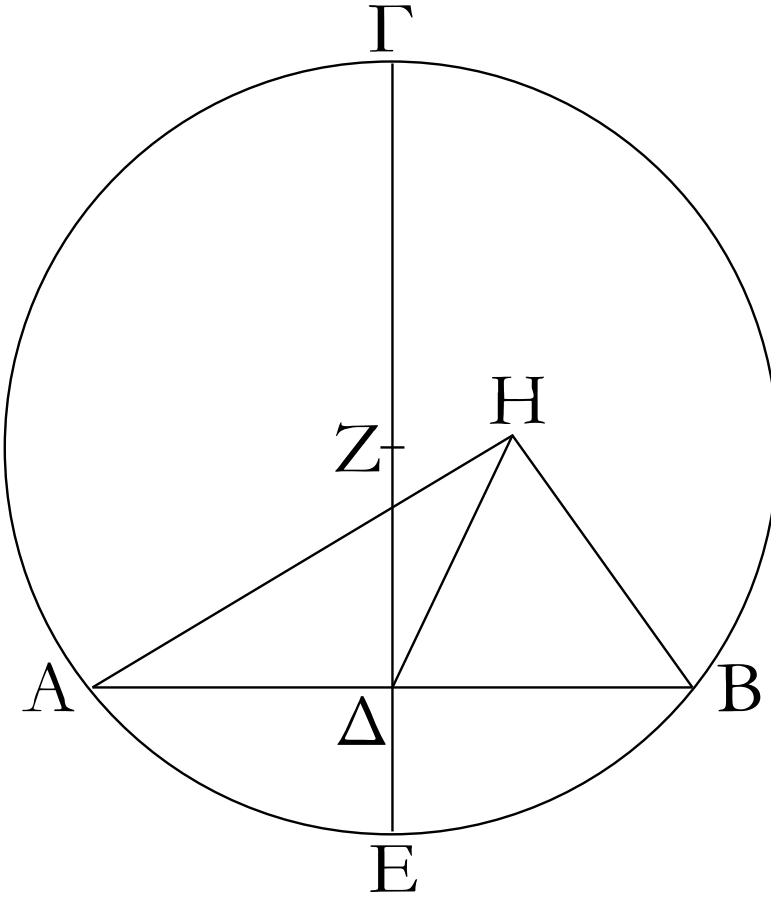
Let some straight-line AB be drawn through (ABC), at random, and let (AB) be cut in half at point D [Prop. 1.9]. And let DC be drawn from D , at right-angles to AB [Prop. 1.11]. And let (CD) be drawn through to E . And let CE be cut in half at F [Prop. 1.9]. I say that (point) F is the center of the [circle] ABC .

For (if) not (then), if possible, let G (be the center of the circle), and let GA , GD , and GB be joined. And since AD is equal to DB , and DG (is) common, the two (straight-lines) AD , DG are equal to the two (straight-lines) BD , DG ,[†] respectively. And the base GA is equal to the base GB . For (they are both) radii. Thus, angle ADG is equal to angle GDB

ἐστίν· ὀρθή ἄρα ἐστίν ἡ ὑπὸ $H\Delta B$. ἐστὶ δὲ καὶ ἡ ὑπὸ $Z\Delta B$ ὀρθή· ἴση ἄρα ἡ ὑπὸ $Z\Delta B$ τῇ ὑπὸ $H\Delta B$, ἡ μείζων τῇ ἐλάττω· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὸ H κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου. ὁμοίως δὲ δείχμεν, ὅτι οὐδ' ἄλλο τι πλὴν τοῦ Z .

[Prop. 1.8]. And when a straight-line stood upon (another) straight-line make adjacent angles (which are) equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, GDB is a right-angle. And FDB is also a right-angle. Thus, FDB (is) equal to GDB , the greater to the lesser. The very thing is impossible. Thus, (point) G is not the center of the circle ABC . So, similarly, we can show that neither is any other (point) except F .





Τὸ Z ἄρα σημεῖον κέντρον ἐστὶ τοῦ $ABΓ$ [κύκλου].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν ἐν κύκλῳ εὐθεῖα τις εὐθεῖαν τινα διῇ καὶ πρὸς ὀρθὰς τέμνῃ, ἐπὶ τῇς τεμονούσης ἐστὶ τὸ κέντρον τοῦ κύκλου. — ὅπερ ἔδει ποιῆσαι.

[†] The Greek text has “ GD, DB ”, which is obviously a mistake.

2.

Ἐάν κύκλον ἐπὶ τῇς περιφερείας ληφθῇ δύο τυθόντα σημεία, ἢ ἐπὶ τὰ σημεία ἐπιξεννυμένη εὐθεῖα ἐντὸς πεσεῖται τοῦ κύκλου.

Ἐστω κύκλος ὁ $ABΓ$, καὶ ἐπὶ τῇς περιφερείας αὐτοῦ εἰλήφθω δύο τυθόντα σημεία τὰ A, B · λέγω, ὅτι ἡ ἀπὸ τοῦ A ἐπὶ τὸ B ἐπιξεννυμένη εὐθεῖα ἐντὸς πεσεῖται τοῦ κύκλου.

μὴ γάρ, ἀλλ’ εἰ δυνατόν, πιπτέτω ἐκτὸς ὥς ἡ AEB , καὶ εἰλήφθω τὸ κέντρον τοῦ $ABΓ$ κύκλου, καὶ ἔστω τὸ Δ , καὶ ἐπεξεύθθωσαν αἱ $\Delta A, \Delta B$, καὶ διήθθω ἡ ΔZE .

Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΔA τῇ ΔB , ἴση ἄρα καὶ γωνία ἡ ὑπὸ ΔAE τῇ ὑπὸ ΔBE · καὶ ἐπεὶ τριγώνον τοῦ ΔAE μία πλευρὰ προσεκβέλλεται ἡ AEB , μείζων ἄρα ἡ ὑπὸ ΔEB γωνία

Thus, point F is the center of the [circle] ABC .

Corollary

So, from this, (it is) manifest that if any straight-line in a circle cuts any (other) straight-line in half, and at right-angles, (then) the center of the circle is on the former (straight-line). — (Which is) the very thing it was required to do.

Proposition 2

If two points are taken at random on the circumference of a circle, (then) the straight-line joining the points will fall inside the circle.

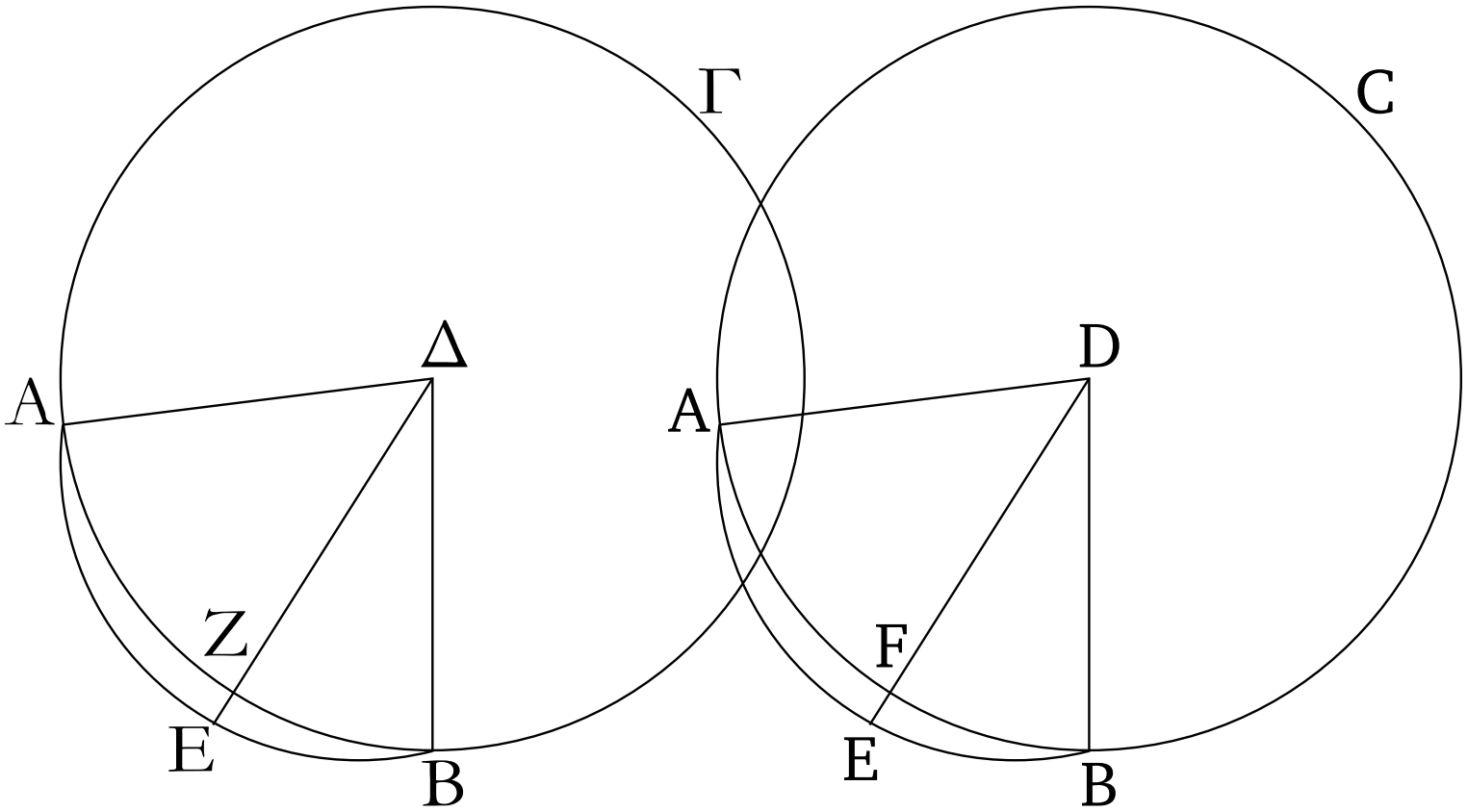
Let ABC be a circle, and let two points A and B be taken at random on its circumference. I say that the straight-line joining A to B will fall inside the circle.

For (if) not (then), if possible, let it fall outside (the circle), like AEB (in the figure). And let the center of the circle ABC be found [Prop. 3.1], and let it be (at point) D . And let DA and DB be joined, and let DFE be drawn through.

Therefore, since DA is equal to DB , the angle DAE (is) thus also equal to DBE [Prop. 1.5]. And since in triangle DAE

νία τῆς ὑπὸ ΔAE . ἴση δὲ ἡ ὑπὸ ΔAE τῇ ὑπὸ ΔBE · μείζων ἄρα ἡ ὑπὸ ΔEB τῆς ὑπὸ ΔBE . ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει· μείζων ἄρα ἡ ΔB τῆς ΔE . ἴση δὲ ἡ ΔB τῇ ΔZ . μείζων ἄρα ἡ ΔZ τῆς ΔE ἢ ἐλάττων τῆς μείζονος· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ A ἐπὶ τὸ B ἐπιζευγνυμένη εὐθεῖα ἐκτὸς πεσεῖται τοῦ κύκλου. ὁμοίως δὲ δείχομεν, ὅτι οὐδὲ ἐπ' αὐτῆς τῆς περιφερείας ἐντὸς ἄρα.

the one side, AEB , has been produced, angle DEB (is) thus greater than DAE [Prop. 1.16]. And DAE (is) equal to DBE [Prop. 1.5]. Thus, DEB (is) greater than DBE . And the greater angle is subtended by the greater side [Prop. 1.19]. Thus, DB (is) greater than DE . And DB (is) equal to DF . Thus, DF (is) greater than DE , the lesser than the greater. The very thing is impossible. Thus, the straight-line joining A to B will not fall outside the circle. So, similarly, we can show that neither (will it fall) on the circumference itself. Thus, (it will fall) inside (the circle).



Ἐὰν ἄρα κύκλον ἐπὶ τῆς περιφερείας ληφθῇ δύο τυθόντα σημεία, ἡ ἐπὶ τὰ σημεία ἐπιζευγνυμένη εὐθεῖα ἐντὸς πεσεῖται τοῦ κύκλου· ὅπερ ἔδει δεῖξαι.

Thus, if two points are taken at random on the circumference of a circle, (then) the straight-line joining the points will fall inside the circle. (Which is) the very thing it was required to show.

3.

Proposition 3

Ἐὰν ἐν κύκλῳ εὐθεῖα τις διὰ τοῦ κέντρου εὐθειάν τινα μὴ διὰ τοῦ κέντρου δίχα τέμνῃ, καὶ πρὸς ὀρθὰς αὐτὴν τέμνῃ· καὶ ἔαν πρὸς ὀρθὰς αὐτὴν τέμνῃ, καὶ δίχα αὐτὴν τέμνῃ.

Ἔστω κύκλος ὁ $ABΓ$, καὶ ἐν αὐτῷ εὐθεῖα τις διὰ τοῦ κέντρου ἡ $ΓΔ$ εὐθειάν τινα μὴ διὰ τοῦ κέντρου τὴν AB δίχα τεμνέτω κατὰ τὸ Z σημεῖον· λέγω, ὅτι καὶ πρὸς ὀρθὰς αὐτὴν τέμνει.

Εἰλήφθω γάρ τὸ κέντρον τοῦ $ABΓ$ κύκλου, καὶ ἔστω τὸ

In a circle, if any straight-line through the center cuts in half any straight-line not through the center, (then) it also cuts it at right-angles. And (conversely) if it cuts it at right-angles, (then) it also cuts it in half.

Let ABC be a circle, and, within it, let some straight-line through the center, CD , cut in half some straight-line not through the center, AB , at the point F . I say that (CD) also cuts (AB) at right-angles.

E , καὶ ἐπεξεύθωσαν αἱ EA , EB .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ AZ τῇ ZB , κοινὴ δὲ ἡ ZE , δύο δυνὸν ἴσαι [εἰσὶν]· καὶ βάσεις ἡ EA βάσει τῇ EB ἴση· γωνία ἄρα ἡ ὑπὸ AZE γωνία τῇ ὑπὸ BZE ἴση ἐστίν. ὅταν δὲ εὐθεῖα ἐπ' εὐθεῖαν σταθεῖσα τὰς ἐφεχθῆς γωνίας ἴσας ἀλλήλαις ποιῇ, ὀρθὴ ἑκατέρω τῶν ἴσων γωνιῶν ἐστίν· ἑκατέρω ἄρα τῶν ὑπὸ AZE , BZE ὀρθή ἐστίν. ἡ $\Gamma\Delta$ ἄρα διὰ τοῦ κέντρου οὔσα τὴν AB μὴ διὰ τοῦ κέντρου οὔσαν δίχα τέμνουσα καὶ πρὸς ὀρθὰς τέμνει.

Ἀλλὰ δὴ ἡ $\Gamma\Delta$ τὴν AB πρὸς ὀρθὰς τεμνέτω· λέγω, ὅτι καὶ δίχα αὐτὴν τέμνει, τοντέστιν, ὅτι ἴση ἐστὶν ἡ AZ τῇ ZB .

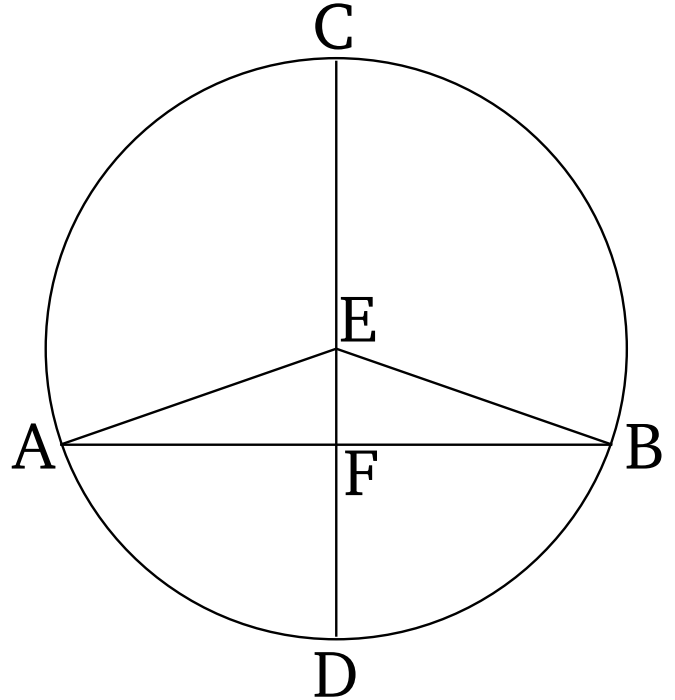
Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἴση ἐστὶν ἡ EA τῇ EB , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ EAZ τῇ ὑπὸ EBZ . ἐστὶ δὲ καὶ ὀρθὴ ἡ ὑπὸ AZE ὀρθὴ τῇ ὑπὸ BZE ἴση· δύο ἄρα τριγωνα ἐστὶ EAZ , EZB τὰς δύο γωνίας δυοὶ γωνίαις ἴσας ἔχοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην κοινήν αὐτῶν τὴν EZ ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν· καὶ τὰς λοιπὰς ἄρα πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσας ἔχει· ἴση ἄρα ἡ AZ τῇ ZB .

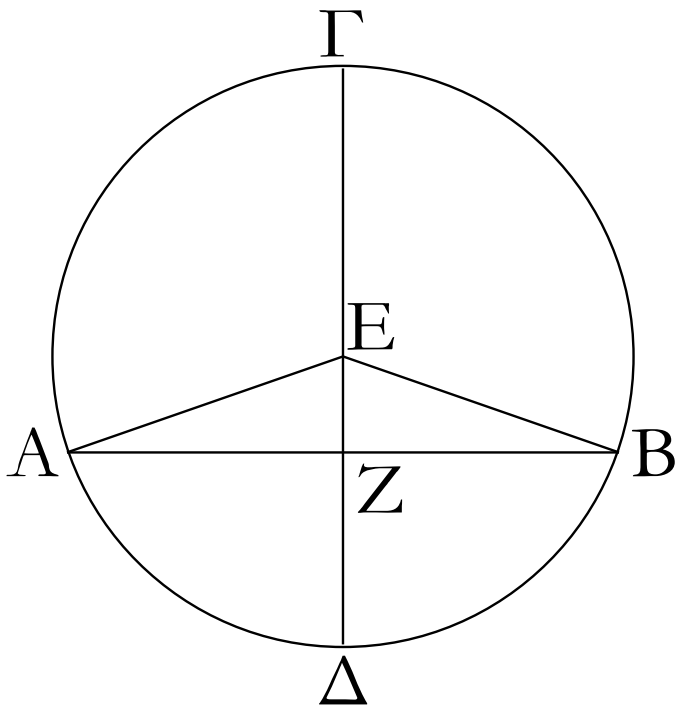
For let the center of the circle ABC be found [Prop. 3.1], and let it be (at point) E , and let EA and EB be joined.

And since AF is equal to FB , and FE (is) common, two (sides of triangle AFE) [are] equal to two (sides of triangle BFE). And the base EA (is) equal to the base EB . Thus, angle AFE is equal to angle BFE [Prop. 1.8]. And when a straight-line stood upon (another) straight-line makes adjacent angles (which are) equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, AFE and BFE are each right-angles. Thus, the (straight-line) CD , which is through the center and cuts in half the (straight-line) AB , which is not through the center, also cuts (AB) at right-angles.

And so let CD cut AB at right-angles. I say that it also cuts (AB) in half. That is to say, that AF is equal to FB .

For, with the same construction, since EA is equal to EB , angle EAF is also equal to EBF [Prop. 1.5]. And the right-angle AFE is also equal to the right-angle BFE . Thus, EAF and EBF are two triangles having two angles equal to two angles, and one side equal to one side—(namely), their common (side) EF , subtending one of the equal angles. Thus, they will also have the remaining sides equal to the (corresponding) remaining sides [Prop. 1.26]. Thus, AF (is) equal to FB .





Ἐάν ἄρα ἐν κύκλῳ εὐθεῖά τις διὰ τοῦ κέντρου εὐθεῖάν τινα μὴ διὰ τοῦ κέντρου δίχα τέμνη, καὶ πρὸς ὀρθὰς αὐτὴν τέμνει· καὶ ἔαν πρὸς ὀρθὰς αὐτὴν τέμνη, καὶ δίχα αὐτὴν τέμνει· ὅπερ ἔδει δεῖχαι.

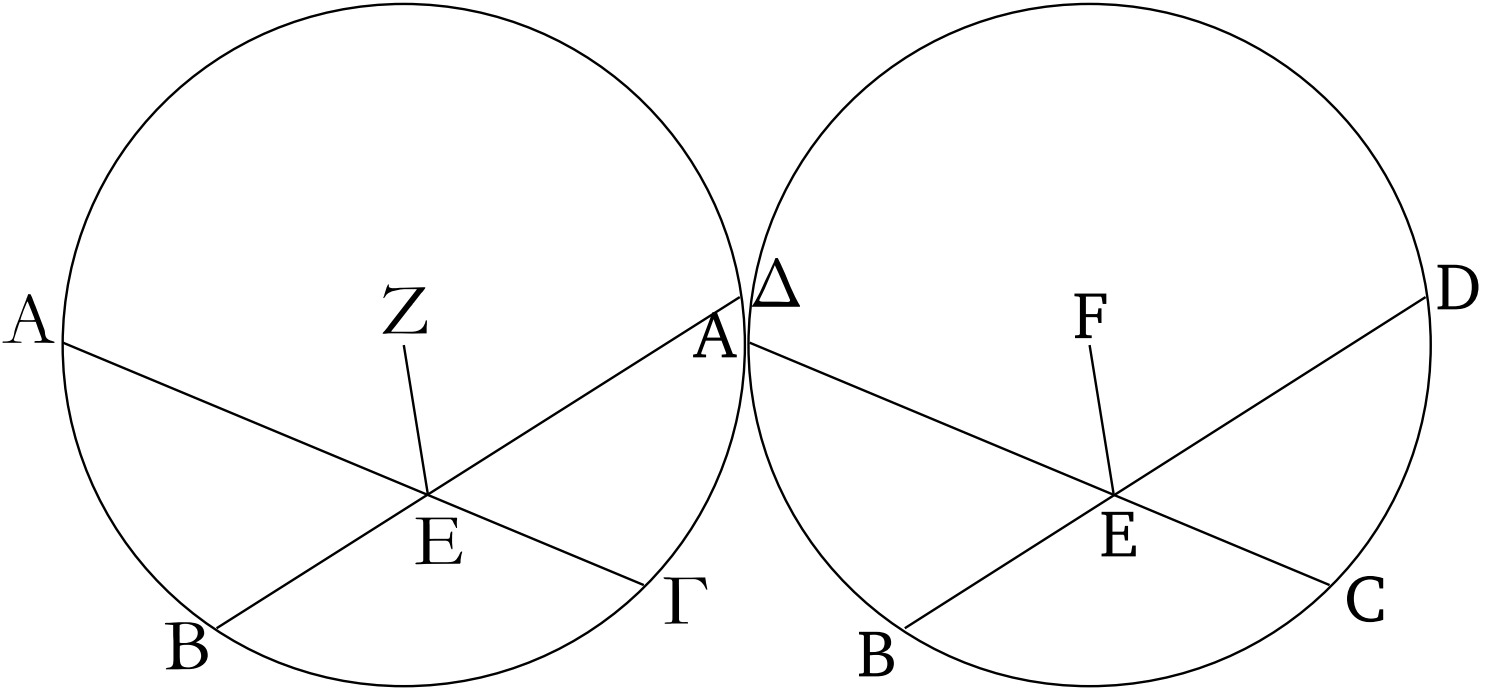
Thus, in a circle, if any straight-line through the center cuts in half any straight-line not through the center, (then) it also cuts it at right-angles. And (conversely) if it cuts it at right-angles, (then) it also cuts it in half. (Which is) the very thing it was required to show.

4.

Proposition 4

Ἐάν ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας μὴ δια τοῦ κέντρου οὔσαι, οὐ τέμνουσιν ἀλλήλας δίχα.

In a circle, if two straight-lines, which are not through the center, cut one another, (then) they do not cut one another in half.



Ἐστω κύκλος ὁ $AB\Gamma\Delta$, καὶ ἐν αὐτῷ δύο εὐθεῖαι αἱ AG , $B\Delta$ τεμνέτωσαν ἀλλήλας κατὰ τὸ E μὴ διὰ τοῦ κέντρου οὔσαι· λέγω, ὅτι οὐ τέμνουσιν ἀλλήλας δίθα.

Εἰ γὰρ δυνατόν, τεμνέτωσαν ἀλλήλας δίθα ὥστε ἴσην εἶναι τὴν μὲν AE τῇ EG , τὴν δὲ BE τῇ ED · καὶ εἰληφθῶ τὸ κέντρον τοῦ $AB\Gamma\Delta$ κύκλου, καὶ ἔστω τὸ Z , καὶ ἐπεζεύθω ἡ ZE .

Ἐπεὶ οὖν εὐθεῖα τις διὰ τοῦ κέντρου ἡ ZE εὐθεϊάν τινα μὴ διὰ τοῦ κέντρου τὴν AG δίθα τέμνει, καὶ πρὸς ὀρθὰς αὐτὴν τέμνει· ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ ZEA · πάλιν, ἐπεὶ εὐθεῖα τις ἡ ZE εὐθεϊάν τινα τὴν $B\Delta$ δίθα τέμνει, καὶ πρὸς ὀρθὰς αὐτὴν τέμνει· ὀρθὴ ἄρα ἡ ὑπὸ ZEB . ἐδείκθη δὲ καὶ ἡ ὑπὸ ZEA ὀρθή· ἴση ἄρα ἡ ὑπὸ ZEA τῇ ὑπὸ ZEB ἢ ἐλάττων τῇ μείζον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα αἱ AG , $B\Delta$ τέμνουσιν ἀλλήλας δίθα.

Ἐάν ἄρα ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας μὴ διὰ τοῦ κέντρου οὔσαι, οὐ τέμνουσιν ἀλλήλας δίθα· ὅπερ ἔδει δεῖξαι.

Let $ABCD$ be a circle, and within it, let two straight-lines, AC and BD , which are not through the center, cut one another at (point) E . I say that they do not cut one another in half.

For, if possible, let them cut one another in half, such that AE is equal to EC , and BE to ED . And let the center of the circle $ABCD$ be found [Prop. 3.1], and let it be (at point) F , and let FE be joined.

Therefore, since some straight-line through the center, FE , cuts in half some straight-line not through the center, AC , it also cuts it at right-angles [Prop. 3.3]. Thus, FEA is a right-angle. Again, since some straight-line FE cuts in half some straight-line BD , it also cuts it at right-angles [Prop. 3.3]. Thus, FEB (is) a right-angle. But FEA was also shown (to be) a right-angle. Thus, FEA (is) equal to FEB , the lesser to the greater. The very thing is impossible. Thus, AC and BD do not cut one another in half.

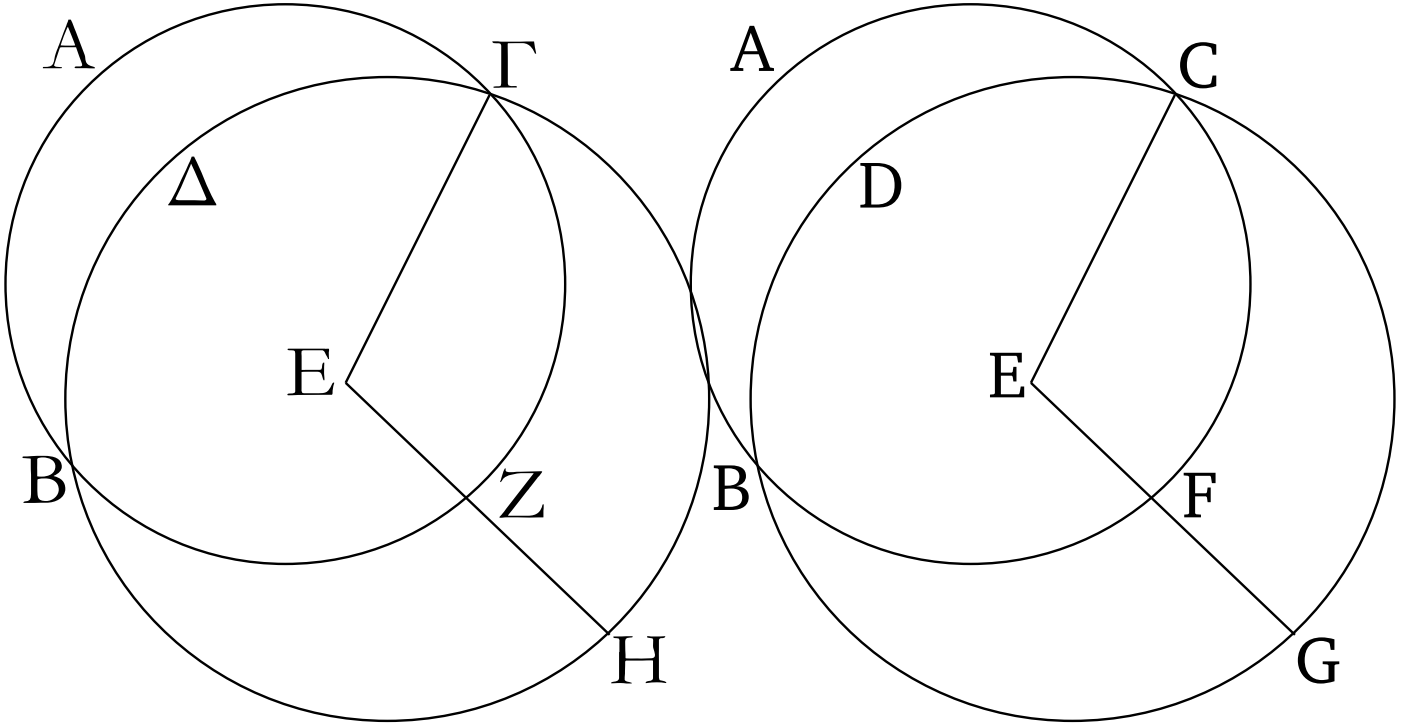
Thus, in a circle, if two straight-lines, which are not through the center, cut one another, (then) they do not cut one another in half. (Which is) the very thing it was required to show.

5.

Proposition 5

Ἐάν δύο κύκλοι τέμνωσιν ἀλλήλων, οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον.

If two circles cut one another, (then) they will not have the same center.



Δύο γὰρ κύκλοι οἱ $AB\Gamma$, $\Gamma\Delta\text{H}$ τεμνέτωσαν ἀλλήλων κατὰ τὰ B , Γ σημεία. λέγω, ὅτι οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον.

Εἰ γὰρ δυνατόν, ἔστω τὸ E , καὶ ἐπεζεύθω ἡ EG , καὶ διήθω ἡ EZH , ὥς ἔτυθεν. καὶ ἐπεὶ τὸ E σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου, ἴση ἐστὶν ἡ EG τῇ EZ . πάλιν, ἐπεὶ τὸ E σημεῖον κέντρον ἐστὶ τοῦ $\Gamma\Delta\text{H}$ κύκλου, ἴση ἐστὶν ἡ EG τῇ EH . ἐδείχθη δὲ ἡ EG καὶ τῇ EZ ἴση· καὶ ἡ EZ ἄρα τῇ EH ἐστὶν ἴση ἢ ἐλάσσων τῇ μείζον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὸ E σημεῖον κέντρον ἐστὶ τῶν $AB\Gamma$, $\Gamma\Delta\text{H}$ κύκλων.

Ἐὰν ἄρα δύο κύκλοι τέμνωσιν ἀλλήλους, οὐκ ἔστιν αὐτῶν τὸ αὐτὸ κέντρον· ὅπερ ἔδει δεῖξαι.

For let the two circles ABC and CDG cut one another at points B and C . I say that they will not have the same center.

For, if possible, let E be (the common center), and let EC be joined, and let EFG be drawn through (the two circles), at random. And since point E is the center of the circle ABC , EC is equal to EF . Again, since point E is the center of the circle CDG , EC is equal to EG . But EC was also shown (to be) equal to EF . Thus, EF is also equal to EG , the lesser to the greater. The very thing is impossible. Thus, point E is not the (common) center of the circles ABC and CDG .

Thus, if two circles cut one another, (then) they will not have the same center. (Which is) the very thing it was required to show.

6.

Proposition 6

Ἐὰν δύο κύκλοι ἐφάπτονται ἀλλήλων, οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον.

Δύο γὰρ κύκλοι οἱ $AB\Gamma$, $\Gamma\Delta E$ ἐφαπτέσθωσαν ἀλλήλων κατὰ τὸ Γ σημεῖον· λέγω, ὅτι οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον.

Εἰ γὰρ δυνατόν, ἔστω τὸ Z , καὶ ἐπεζεύθω ἡ $Z\Gamma$, καὶ διήθω, ὥς ἔτυθεν, ἡ ZEB .

Ἐπεὶ οὖν τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου, ἴση ἐστὶν ἡ $Z\Gamma$ τῇ ZB . πάλιν, ἐπεὶ τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $\Gamma\Delta E$ κύκλου, ἴση ἐστὶν ἡ $Z\Gamma$ τῇ ZE . ἐδείχθη δὲ ἡ $Z\Gamma$ τῇ ZB ἴση· καὶ ἡ ZE ἄρα τῇ ZB ἐστὶν ἴση, ἢ ἐλάττων τῇ μείζον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὸ Z σημεῖον κέντρον ἐστὶ τῶν

If two circles touch one another, (then) they will not have the same center.

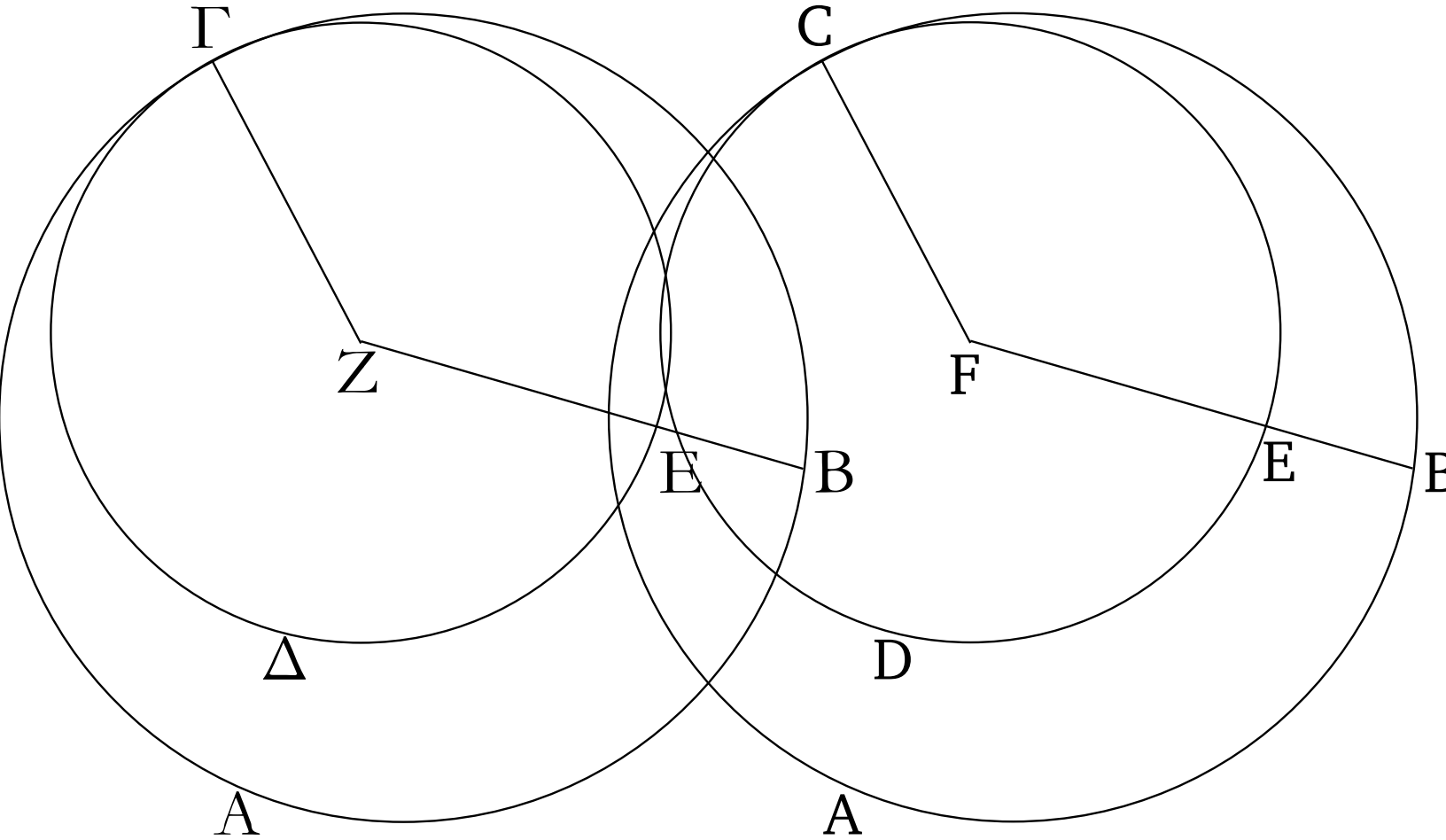
For let the two circles ABC and CDE touch one another at point C . I say that they will not have the same center.

For, if possible, let F be (the common center), and let FC be joined, and let FEB be drawn through (the two circles), at random.

Therefore, since point F is the center of the circle ABC , FC is equal to FB . Again, since point F is the center of the circle CDE , FC is equal to FE . But FC was shown (to be) equal to FB . Thus, FE is also equal to FB , the lesser to the greater. The very thing is impossible. Thus, point F is not the

ABΓ, ΓΔΕ κύκλων.

(common) center of the circles *ABC* and *CDE*.



Ἐάν ἄρα δύο κύκλοι ἐφάπτονται ἀλλήλων, οὐκ ἔσται αὐτῶν τὸ αὐτὸ κέντρον· ὅπερ ἔδει δεῖχαι.

Thus, if two circles touch one another, (then) they will not have the same center. (Which is) the very thing it was required to show.

7.

Proposition 7

Ἐάν κύκλον ἐπὶ τῇ διαμέτρῳ ληφθῇ τι σημεῖον, ὃ μὴ ἔστι κέντρον τοῦ κύκλου, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσιν εὐθεῖαι τινεῖς, μεγίστη μὲν ἔσται, ἐφ' ἣς τὸ κέντρον, ἐλαθίστη δὲ ἡ λοιπή, τῶν δὲ ἄλλων αἰεὶ ἡ ἔγγιον τῇς διὰ τοῦ κέντρον τῇς ἀπώτερον μείζων ἔστί, δύο δὲ μόνον ἴσαι ἀπὸ τοῦ σημείου προσπεσοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῇς ἐλαθίστης.

Ἐστω κύκλος ὁ *ABΓΔ*, διάμετρος δὲ αὐτοῦ ἔστω ἡ *ΑΔ*, καὶ ἐπὶ τῇς *ΑΔ* εἰληφθῶ τι σημεῖον τὸ *Ζ*, ὃ μὴ ἔστι κέντρον τοῦ κύκλου, κέντρον δὲ τοῦ κύκλου ἔστω τὸ *Ε*, καὶ ἀπὸ τοῦ *Ζ* πρὸς τὸν *ABΓΔ* κύκλον προσπιπτέτωσαν εὐθεῖαι τινεῖς αἱ *ΖΒ*, *ΖΓ*, *ΖΗ*· λέγω, ὅτι μεγίστη μὲν ἔστιν ἡ *ΖΑ*, ἐλαθίστη δὲ ἡ *ΖΔ*, τῶν δὲ ἄλλων ἡ μὲν *ΖΒ* τῇς *ΖΓ* μείζων, ἡ δὲ *ΖΓ* τῇς *ΖΗ*.

If some point, which is not the center of the circle, is taken on the diameter of a circle, and some straight-lines radiate from the point towards the (circumference of the) circle, (then) the greatest (straight-line) will be that on which the center (lies), and the least the remainder (of the same diameter). And for the others, a (straight-line) nearer[†] to the (straight-line) through the center is always greater than a (straight-line) further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line).

Let *ABCD* be a circle, and let *AD* be its diameter, and let some point *F*, which is not the center of the circle, be taken on *AD*. Let *E* be the center of the circle. And let some straight-lines, *FB*, *FC*, and *FG*, radiate from *F* towards (the circum-

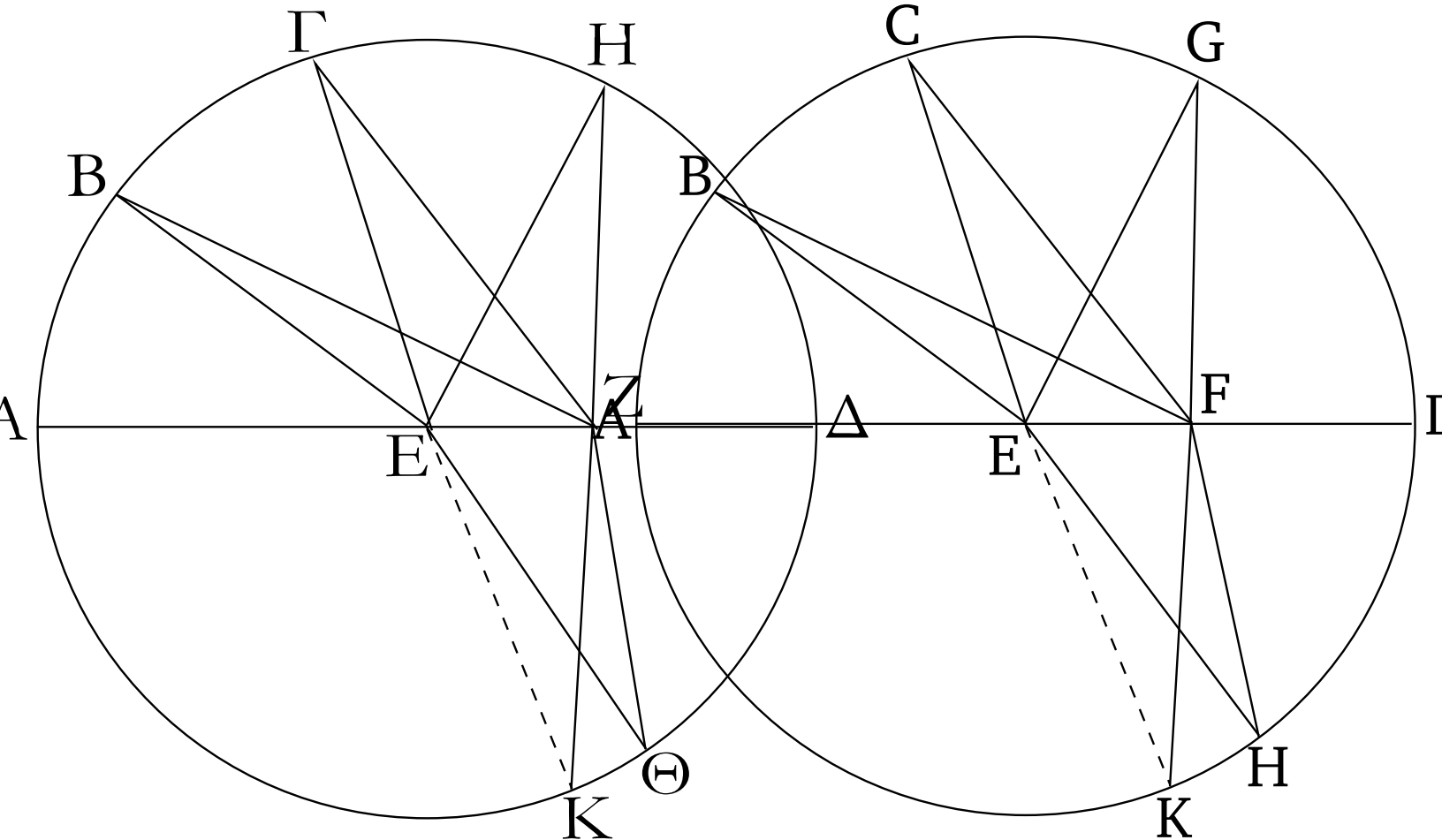
Ἐπεξεύχθησαν γάρ αἱ BE , $ΓE$, HE . καὶ ἐπεὶ παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσιν, αἱ ἄρα EB , EZ τῆς BZ μείζονές εἰσιν. ἴση δὲ ἡ AE τῇ BE [αἱ ἄρα BE , EZ ἴσαι εἰσὶ τῇ AZ]. μείζων ἄρα ἡ AZ τῆς BZ . πάλιν, ἐπεὶ ἴση ἐστὶν ἡ BE τῇ $ΓE$, κοινὴ δὲ ἡ ZE , δύο δὲ αἱ BE , EZ δυοὶ ταῖς $ΓE$, EZ ἴσαι εἰσὶν. ἀλλὰ καὶ γωνία ἡ ὑπὸ BEZ γωνία τῆς ὑπὸ $ΓEZ$ μείζων· βάσει ἄρα ἡ BZ βάσεως τῆς $ΓZ$ μείζων ἐστίν. διὰ τὰ αὐτὰ δὲ καὶ ἡ $ΓZ$ τῆς ZH μείζων ἐστίν.

Πάλιν, ἐπεὶ αἱ HZ , ZE τῆς EH μείζονές εἰσιν, ἴση δὲ ἡ EH τῇ $EΔ$, αἱ ἄρα HZ , ZE τῆς $EΔ$ μείζονές εἰσιν. κοινὴ ἀφηρέσθω ἡ EZ · λοιπὴ ἄρα ἡ HZ λοιπῆς τῆς $ZΔ$ μείζων ἐστίν. μεγίστη μὲν ἄρα ἡ ZA , ἐλαθίστη δὲ ἡ $ZΔ$, μείζων δὲ ἡ μὲν ZB τῆς $ZΓ$, ἡ δὲ $ZΓ$ τῆς ZH .

ference of) circle $ABCD$. I say that FA is the greatest (straight-line), FD the least, and of the others, FB (is) greater than FC , and FC than FG .

For let BE , CE , and GE be joined. And since for every triangle (any) two sides are greater than the remaining (side) [Prop. 1.20], EB and EF is thus greater than BF . And AE (is) equal to BE [thus, BE and EF is equal to AF]. Thus, AF (is) greater than BF . Again, since BE is equal to CE , and FE (is) common, the two (straight-lines) BE , EF are equal to the two (straight-lines) CE , EF (respectively). But, angle BEF (is) also greater than angle CEF .[‡] Thus, the base BF is greater than the base CF . Thus, the base BF is greater than the base CF [Prop. 1.24]. So, for the same (reasons), CF is also greater than FG .

Again, since GF and FE are greater than EG [Prop. 1.20], and EG (is) equal to ED , GF and FE are thus greater than ED . Let EF be taken from both. Thus, the remainder GF is greater than the remainder FD . Thus, FA (is) the greatest (straight-line), FD the least, and FB (is) greater than FC , and FC than FG .



Λέγω, ὅτι καὶ ἀπὸ τοῦ Z σημείου δύο μόνον ἴσαι προσ-

I also say that from point F only two equal (straight-lines)

πεσοῦνται πρὸς τὸν $ABΓΔ$ κύκλον ἐφ' ἑκάτερα τῆς $ZΔ$ ἐλαθίστης. συνεστάτω γὰρ πρὸς τῇ EZ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ E τῇ ὑπὸ HEZ γωνίᾳ ἴση ἢ ὑπὸ $ZEΘ$, καὶ ἐπεξεύθθω ἡ $ZΘ$. ἐπεὶ οὖν ἴση ἐστὶν ἡ HE τῇ $EΘ$, κοινὴ δὲ ἡ EZ , δύο δὲ αἱ HE , EZ δυοὶ ταῖς $ΘE$, EZ ἴσαι εἰσὶν· καὶ γωνία ἢ ὑπὸ HEZ γωνία τῇ ὑπὸ $ΘEZ$ ἴση· βάσεις ἄρα ἡ ZH βάσει τῇ $ZΘ$ ἴση ἐστίν. λέγω δὴ, ὅτι τῇ ZH ἄλλη ἴση οὐ προσπασσεται πρὸς τὸν κύκλον ἀπὸ τοῦ Z σημείου. εἰ γὰρ δυνατόν, προσπιπτέτω ἡ ZK . καὶ ἐπεὶ ἡ ZK τῇ ZH ἴση ἐστίν, ἀλλὰ ἡ $ZΘ$ τῇ ZH [ἴση ἐστίν], καὶ ἡ ZK ἄρα τῇ $ZΘ$ ἐστὶν ἴση, ἢ ἔργον τῆς διὰ τοῦ κέντρου τῇ ἀπώτερον ἴση· ὅπερ ἀδύνατον. οὐκ ἄρα ἀπὸ τοῦ Z σημείου ἕτερα τις προσπασσεται πρὸς τὸν κύκλον ἴση τῇ HZ · μία ἄρα μόνη.

Ἐάν ἄρα κύκλον ἐπὶ τῆς διαμέτρου ληφθῇ τι σημεῖον, ὃ μὴ ἐστὶ κέντρον τοῦ κύκλου, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσιν εὐθεῖαι τινεῖς, μεγίστη μὲν ἔσται, ἐφ' ἣς τὸ κέντρον, ἐλαθίστη δὲ ἡ λοιπή, τῶν δὲ ἄλλων αἰεὶ ἢ ἔργον τῆς διὰ τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν, δύο δὲ μόνον ἴσαι ἀπὸ τοῦ αὐτοῦ σημείου προσπασσονται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης· ὅπερ ἔδει δεῖξαι.

will radiate towards (the circumference of) circle $ABCD$, (one) on each (side) of the least (straight-line) FD . For let the (angle) FEH , equal to angle GEF , be constructed on the straight-line EF , at the point E on it [Prop. 1.23], and let FH be joined. Therefore, since GE is equal to EH , and EF (is) common, the two (straight-lines) GE , EF are equal to the two (straight-lines) HE , EF (respectively). And angle GEF (is) equal to angle HEF . Thus, the base FG is equal to the base FH [Prop. 1.4]. So I say that another (straight-line) equal to FG will not radiate towards (the circumference of) the circle from point F . For, if possible, let FK (so) radiate. And since FK is equal to FG , but FH [is equal] to FG , FK is thus also equal to FH , the nearer to the (straight-line) through the center equal to the further away. The very thing (is) impossible. Thus, another (straight-line) equal to GF will not radiate from the point F towards (the circumference of) the circle. Thus, (there is) only one (such straight-line).

Thus, if some point, which is not the center of the circle, is taken on the diameter of a circle, and some straight-lines radiate from the point towards the (circumference of the) circle, (then) the greatest (straight-line) will be that on which the center (lies), and the least the remainder (of the same diameter). And for the others, a (straight-line) nearer to the (straight-line) through the center is always greater than a (straight-line) further away. And only two equal (straight-lines) will radiate from the same point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line). (Which is) the very thing it was required to show.

[†] Presumably, in an angular sense.

[‡] This is not proved, except by reference to the figure.

8.

Proposition 8

Ἐάν κύκλον ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον διαθῶσιν εὐθεῖαι τινεῖς, ὧν μία μὲν διὰ τοῦ κέντρου, αἱ δὲ λοιπαί, ὡς ἔτυθεν, τῶν μὲν πρὸς τὴν κοίλην περιφέρειαν προσπιπτουσῶν εὐθειῶν μεγίστη μὲν ἐστὶν ἢ διὰ τοῦ κέντρου, τῶν δὲ ἄλλων αἰεὶ ἢ ἔργον τῆς διὰ τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν, τῶν δὲ πρὸς τὴν κυρτὴν περιφέρειαν προσπιπτουσῶν εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἢ μεταχὺ τοῦ τε σημείου καὶ τῆς διαμέτρου, τῶν δὲ ἄλλων αἰεὶ ἢ ἔργον τῆς ἐλαθίστης τῆς ἀπώτερόν ἐστιν ἐλάττων, δύο δὲ μόνον ἴσαι ἀπὸ τοῦ σημείου προσπασσονται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης.

Ἐστω κύκλος ὁ $ABΓ$, καὶ τοῦ $ABΓ$ εἰλήφθω τι σημεῖον ἐκτὸς τὸ $Δ$, καὶ ἀπ' αὐτοῦ διήθθωσαν εὐθεῖαι τινεῖς αἱ $ΔΑ$, $ΔΕ$, $ΔΖ$, $ΔΓ$, ἔστω δὲ ἡ $ΔΑ$ διὰ τοῦ κέντρου. λέγω, ὅτι τῶν μὲν πρὸς τὴν $AEZΓ$ κοίλην περιφέρειαν προσπιπτουσῶν εὐθειῶν μεγίστη μὲν ἐστὶν ἢ διὰ τοῦ κέντρου ἡ $ΔΑ$, μείζων δὲ ἢ μὲν $ΔΕ$ τῆς $ΔΖ$ ἢ δὲ $ΔΖ$ τῆς $ΔΓ$, τῶν δὲ πρὸς τὴν

If some point is taken outside a circle, and some straight-lines are drawn from the point to the (circumference of the) circle, one of which (passes) through the center, the remainder (being) random, (then) for the straight-lines radiating towards the concave (part of the) circumference, the greatest is that (passing) through the center. For the others, a (straight-line) nearer[†] to the (straight-line) through the center is always greater than one further away. For the straight-lines radiating towards the convex (part of the) circumference, the least is that between the point and the diameter. For the others, a (straight-line) nearer to the least (straight-line) is always less than one further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line).

Let ABC be a circle, and let some point D be taken outside ABC , and from it let some straight-lines, DA , DE , DF , and DC , be drawn through (the circle), and let DA be through

ΘΑΚΗ κυρτήν περιφέρειαν προσιπτονσών εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἡ ΔΗ ἢ μεταχὺ τοῦ σημείου καὶ τῆς διαμέτρου τῆς ΑΗ, αἰ δὲ ἡ ἔγγιον τῆς ΔΗ ἐλαθίστης ἐλάττων ἐστὶ τῆς ἀπώτερον, ἡ μὲν ΔΚ τῆς ΔΑ, ἡ δὲ ΔΑ τῆς ΔΘ.

Εἰλήφθω γάρ τὸ κέντρον τοῦ ΑΒΓ κύκλου καὶ ἔστω τὸ Μ· καὶ ἐπεζεύθωσαν αἱ ΜΕ, ΜΖ, ΜΓ, ΜΚ, ΜΑ, ΜΘ.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ ΑΜ τῇ ΕΜ, κοινὴ προσκείσθω ἡ ΜΔ· ἡ ἄρα ΑΔ ἴση ἐστὶ ταῖς ΕΜ, ΜΔ. ἀλλ' αἱ ΕΜ, ΜΔ τῆς ΕΔ μείζονές εἰσιν· καὶ ἡ ΑΔ ἄρα τῆς ΕΔ μείζων ἐστίν. πάλιν, ἐπεὶ ἴση ἐστὶν ἡ ΜΕ τῇ ΜΖ, κοινὴ δὲ ἡ ΜΔ, αἱ ΕΜ, ΜΔ ἄρα ταῖς ΖΜ, ΜΔ ἴσαι εἰσίν· καὶ γωνία ἡ ὑπὸ ΕΜΔ γωνία τῆς ὑπὸ ΖΜΔ μείζων ἐστίν. βάσις ἄρα ἡ ΕΔ βάσεως τῆς ΖΔ μείζων ἐστίν. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΖΔ τῆς ΓΔ μείζων ἐστίν· μεγίστη μὲν ἄρα ἡ ΔΑ, μείζων δὲ ἡ μὲν ΔΕ τῆς ΔΖ, ἡ δὲ ΔΖ τῆς ΔΓ.

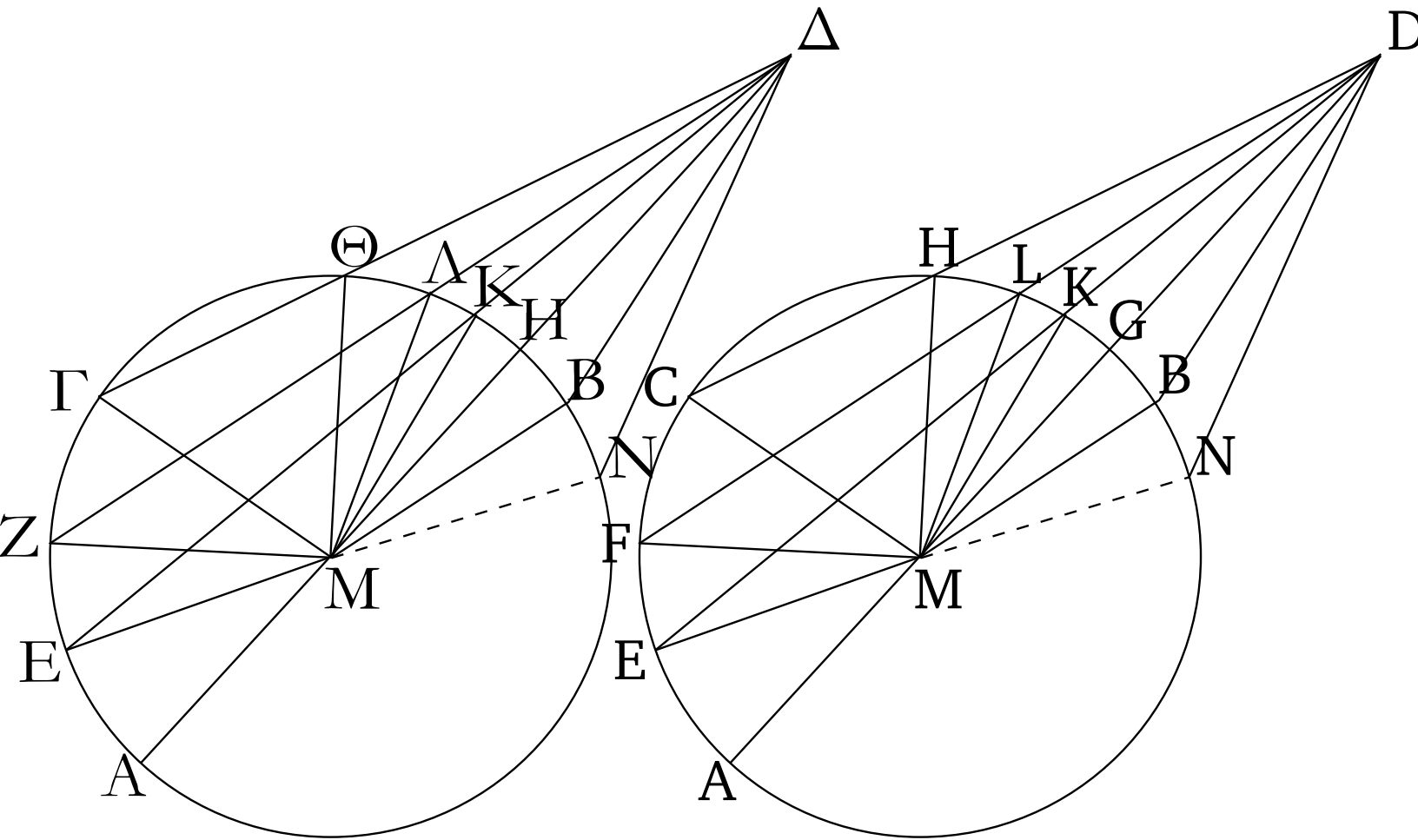
Καὶ ἐπεὶ αἱ ΜΚ, ΚΔ τῆς ΜΔ μείζονές εἰσιν, ἴση δὲ ἡ μὴ τῇ ΜΚ, λοιπὴ ἄρα ἡ ΚΔ λοιπῆς τῆς ΗΔ μείζων ἐστίν· ὥστε ἡ ΗΔ τῆς ΚΔ ἐλάττων ἐστίν· καὶ ἐπεὶ τριγώνον τοῦ ΜΑΔ ἐπὶ μιᾷ τῶν πλευρῶν τῆς ΜΔ δύο εὐθεῖαι ἐντὸς συνεστάθησαν αἱ ΜΚ, ΚΔ, αἱ ἄρα ΜΚ, ΚΔ τῶν ΜΑ, ΑΔ ἐλάττονές εἰσιν· ἴση δὲ ἡ ΜΚ τῇ ΜΑ· λοιπὴ ἄρα ἡ ΔΚ λοιπῆς τῆς ΔΑ ἐλάττων ἐστίν. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΔΑ τῆς ΔΘ ἐλάττων ἐστίν· ἐλαθίστη μὲν ἄρα ἡ ΔΗ, ἐλάττων δὲ ἡ μὲν ΔΚ τῆς ΔΑ ἡ δὲ ΔΑ τῆς ΔΘ.

the center. I say that for the straight-lines radiating towards the concave (part of the) circumference, $AEFC$, the greatest is the one (passing) through the center, (namely) AD , and (that) DE (is) greater than DF , and DF than DC . For the straight-lines radiating towards the convex (part of the) circumference, $HLKG$, the least is the one between the point and the diameter AG , (namely) DG , and a (straight-line) nearer to the least (straight-line) DG is always less than one farther away, (so that) DK (is less) than DL , and DL than DH .

For let the center of the circle be found [Prop. 3.1], and let it be (at point) M [Prop. 3.1]. And let ME , MF , MC , MK , ML , and mh be joined.

And since AM is equal to EM , let MD be added to both. Thus, AD is equal to EM and MD . But, EM and MD is greater than ED [Prop. 1.20]. Thus, AD is also greater than ED . Again, since ME is equal to MF , and MD (is) common, the (straight-lines) EM , MD are thus equal to FM , MD . And angle EMD is greater than angle FMD .[‡] Thus, the base ED is greater than the base FD [Prop. 1.24]. So, similarly, we can show that FD is also greater than CD . Thus, AD (is) the greatest (straight-line), and DE (is) greater than DF , and DF than DC .

And since MK and KD is greater than MD [Prop. 1.20], and MG (is) equal to MK , the remainder KD is thus greater than the remainder GD . So GD is less than KD . And since in triangle MLD , the two internal straight-lines MK and KD were constructed on one of the sides, MD , (then) MK and KD are thus less than ML and LD [Prop. 1.21]. And MK (is) equal to ML . Thus, the remainder DK is less than the remainder DL . So, similarly, we can show that DL is also less than DH . Thus, DG (is) the least (straight-line), and DK (is) less than DL , and DL than DH .



Λέγω, ὅτι καὶ δύο μόνον ἴσαι ἀπὸ τοῦ Δ σημείου προσ-
 πεσοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ΔΗ ἐλαθίστης·
 συννεσάτω πρὸς τῇ ΜΔ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ
 Μ τῇ ὑπὸ ΚΜΔ γωνίᾳ ἴση γωνία ἡ ὑπὸ ΔΜΒ, καὶ ἐπεζεύ-
 θω ἡ ΔΒ. καὶ ἐπεὶ ἴση ἐστὶν ἡ ΜΚ τῇ ΜΒ, κοινὴ δὲ ἡ ΜΔ,
 δύο δὴ αἱ ΚΜ, ΜΔ δύο ταῖς ΒΜ, ΜΔ ἴσαι εἰσὶν ἑκατέρω
 ἑκατέρω· καὶ γωνία ἡ ὑπὸ ΚΜΔ γωνία τῇ ὑπὸ ΒΜΔ ἴση·
 βάσει ἄρα ἡ ΔΚ βάσει τῇ ΔΒ ἴση ἐστίν. λέγω [δὴ], ὅτι
 τῇ ΔΚ εὐθείᾳ ἄλλη ἴση οὐ προσπεσεῖται πρὸς τὸν κύκλον
 ἀπὸ τοῦ Δ σημείου. εἰ γὰρ δυνατόν, προσπιπτέτω καὶ ἔστω
 ἡ ΔΝ. ἐπεὶ οὖν ἡ ΔΚ τῇ ΔΝ ἐστὶν ἴση, ἀλλ' ἡ ΔΚ τῇ ΔΒ
 ἐστὶν ἴση, καὶ ἡ ΔΒ ἄρα τῇ ΔΝ ἐστὶν ἴση, ἡ ἔγγιον τῆς ΔΗ
 ἐλαθίστης τῇ ἀπώτερων [ἐστίν] ἴση· ὅπερ ἀδύνατον ἐδείχθη.
 οὐκ ἄρα πλείους ἢ δύο ἴσαι πρὸς τὸν ΑΒΓ κύκλον ἀπὸ τοῦ
 Δ σημείου ἐφ' ἑκάτερα τῆς ΔΗ ἐλαθίστης προσπεσοῦνται.

Ἐάν ἄρα κύκλον ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ ση-
 μέιου πρὸς τὸν κύκλον διαθῶσιν εὐθεῖαι τιναί, ὧν μία μὲν
 διὰ τοῦ κέντρου αἱ δὲ λοιπαί, ὥς ἔτυθεν, τῶν μὲν πρὸς τὴν
 κοίλην περιφέρειαν προσπιπτονσῶν εὐθειῶν μεγίστη μὲν ἐστὶν
 ἡ διὰ τοῦ κέντρου, τῶν δὲ ἄλλων αἰεὶ ἡ ἔγγιον τῆς διὰ τοῦ

I also say that only two equal (straight-lines) will radiate
 from point D towards (the circumference of) the circle, (one)
 on each (side) on the least (straight-line), DG . Let the angle
 DMB , equal to angle KMD , be constructed on the straight-line
 MD , at the point M on it [Prop. 1.23], and let DB be joined.
 And since MK is equal to MB , and MD (is) common, the two
 (straight-lines) KM , MD are equal to the two (straight-lines)
 BM , MD , respectively. And angle KMD (is) equal to angle
 BMD . Thus, the base DK is equal to the base DB [Prop. 1.4].
 [So] I say that another (straight-line) equal to DK will not radi-
 ate towards the (circumference of) the circle from point D . For,
 if possible, let (such a straight-line) radiate, and let it be DN .
 Therefore, since DK is equal to DN , but DK is equal to DB ,
 (then) DB is thus also equal to DN , (so that) a (straight-line)
 nearer to the least (straight-line) DG [is] equal to one further
 away. The very thing was shown (to be) impossible. Thus, not
 more than two equal (straight-lines) will radiate towards (the
 circumference of) circle ABC from point D , (one) on each side
 of the least (straight-line) DG .

Thus, if some point is taken outside a circle, and some

κέντρον τῆς ἀπώτερον μείζων ἐστίν, τῶν δὲ πρὸς τὴν κυρτὴν περιφέρειαν προσπιπτουσῶν εὐθειῶν ἐλαθίστη μὲν ἐστὶν ἡ μεταχὺ τοῦ τε σημείου καὶ τῆς διαμέτρου, τῶν δὲ ἄλλων αἰεὶ ἢ ἔγγιον τῆς ἐλαθίστης τῆς ἀπώτερόν ἐστιν ἐλάττων, δύο δὲ μόνον ἴσαι ἀπὸ τοῦ σημείου προσπεσοῦνται πρὸς τὸν κύκλον ἐφ' ἑκάτερα τῆς ἐλαθίστης· ὅπερ ἔδει δεῖχαι.

straight-lines are drawn from the point to the (circumference of the) circle, one of which (passes) through the center, the remainder (being) random, (then) for the straight-lines radiating towards the concave (part of the) circumference, the greatest is that (passing) through the center. For the others, a (straight-line) nearer to the (straight-line) through the center is always greater than one further away. For the straight-lines radiating towards the convex (part of the) circumference, the least is that between the point and the diameter. For the others, a (straight-line) nearer to the least (straight-line) is always less than one further away. And only two equal (straight-lines) will radiate from the point towards the (circumference of the) circle, (one) on each (side) of the least (straight-line). (Which is) the very thing it was required to show.

† Presumably, in an angular sense.

‡ This is not proved, except by reference to the figure.

9.

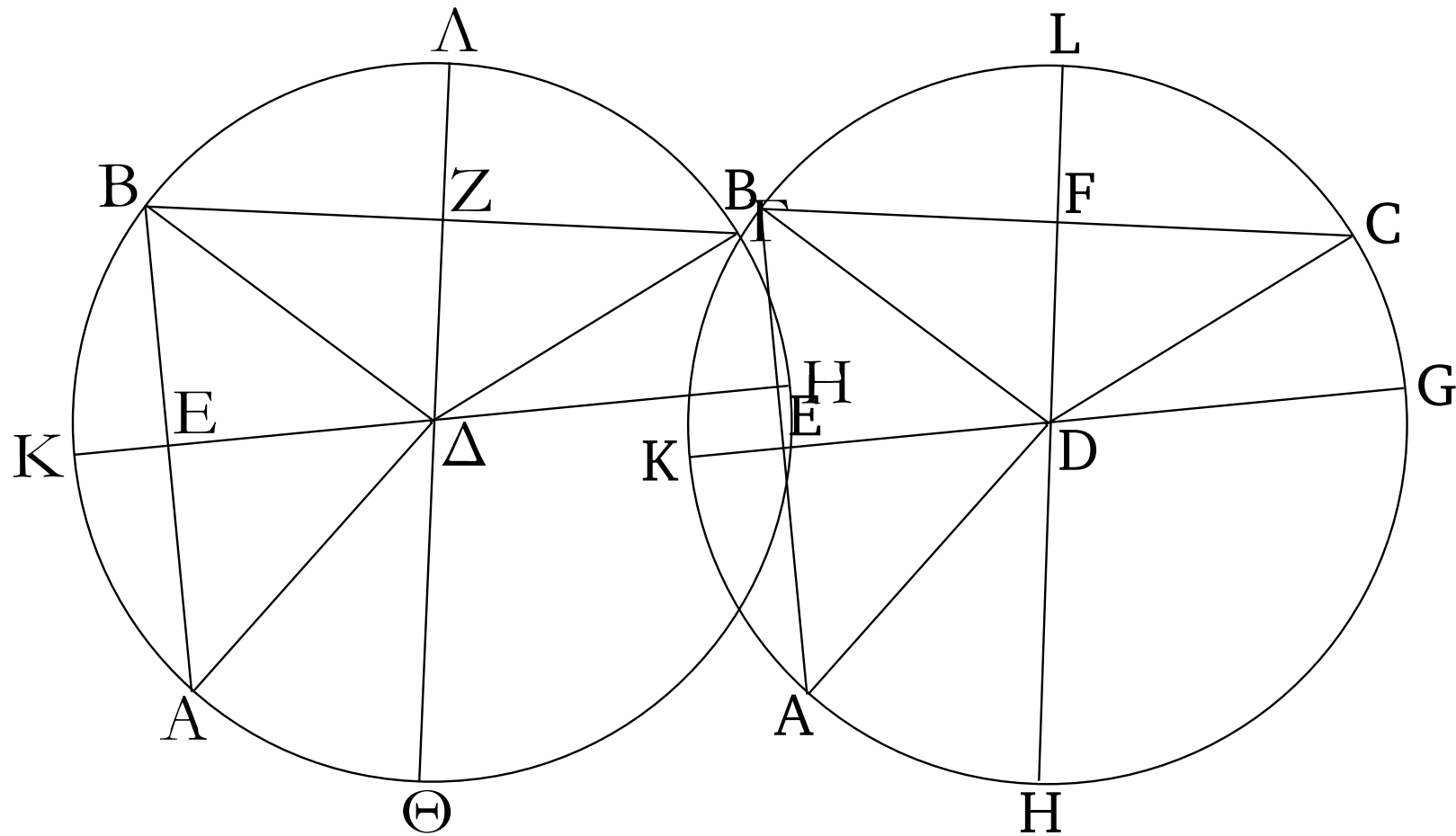
Ἐάν κύκλον ληφθῇ τι σημεῖον ἐντόξ, ἀπο δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι πλείονξ ἢ δύο ἴσαι εὐθεῖαι, τὸ ληφθὲν σημεῖον κέντρον ἐστὶ τοῦ κύκλου.

Ἐστω κύκλος ὁ $ABΓ$, ἐντόξ δὲ αὐτοῦ σημεῖον τὸ Δ , καὶ ἀπὸ τοῦ Δ πρὸς τὸν $ABΓ$ κύκλον προσπιπτέτωσαν πλείονξ ἢ δύο ἴσαι εὐθεῖαι αἱ ΔA , ΔB , $\Delta Γ$. λέγω, ὅτι τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ $ABΓ$ κύκλου.

Proposition 9

If some point is taken inside a circle, and more than two equal straight-lines radiate from the point towards the (circumference of the) circle, (then) the point taken is the center of the circle.

Let ABC be a circle, and D a point inside it, and let more than two equal straight-lines, DA , DB , and DC , radiate from D towards (the circumference of) circle ABC . I say that point D is the center of circle ABC .



Ἐπεξευθύνθωσαν γὰρ αἱ AB , BF καὶ τετμήσθωσαν δίῃ κατὰ τὰ E , Z σημεία, καὶ ἐπιζευθθεῖσαι αἱ ED , ZD διήθθωσαν ἐπὶ τὰ H , K , Θ , Λ σημεία.

Ἐπεὶ οὖν ἴση ἐστὶν ἡ AE τῇ EB , κοινὴ δὲ ἡ ED , δύο δὴ αἱ AE , ED δύο ταῖς BE , ED ἴσαι εἰσὶν καὶ βάσει ἡ DA βάσει τῇ DB ἴση· γωνία ἄρα ἡ ὑπὸ $AE\Delta$ γωνία τῇ ὑπὸ $BE\Delta$ ἴση ἐστίν· ὀρθὴ ἄρα ἑκατέρω τῶν ὑπὸ $AE\Delta$, $BE\Delta$ γωνιῶν· ἡ HK ἄρα τὴν AB τέμνει δίῃ καὶ πρὸς ὀρθὰς. καὶ ἐπεὶ, ἔαν ἐν κύκλῳ εὐθεῖά τις εὐθεϊάν τινα δίῃ τε καὶ πρὸς ὀρθὰς τέμνῃ, ἐπὶ τῆς τεμνουσῆς ἐστὶ τὸ κέντρον τοῦ κύκλου, ἐπὶ τῆς HK ἄρα ἐστὶ τὸ κέντρον τοῦ κύκλου. διὰ τὰ αὐτὰ δὴ καὶ ἐπὶ τῆς $\Theta\Lambda$ ἐστὶ τὸ κέντρον τοῦ $AB\Gamma$ κύκλου. καὶ οὐδέν ἕτερον κοινὸν ἔθουσιν αἱ HK , $\Theta\Lambda$ εὐθεῖαι ἢ τὸ Δ σημεῖον· τὸ Δ ἄρα σημεῖον κέντρον ἐστὶ τοῦ $AB\Gamma$ κύκλου.

Ἐάν ἄρα κύκλον ληφθῇ τι σημεῖον ἐντός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι πλείονες ἢ δύο ἴσαι εὐθεῖαι, τὸ ληφθὲν σημεῖον κέντρον ἐστὶ τοῦ κύκλου· ὅπερ ἔδει δεῖξαι.

For let AB and BC be joined, and (then) be cut in half at points E and F (respectively) [Prop. 1.10]. And ED and FD being joined, let them be drawn through to points G , K , H , and L .

Therefore, since AE is equal to EB , and ED (is) common, the two (straight-lines) AE , ED are equal to the two (straight-lines) BE , ED (respectively). And the base DA (is) equal to the base DB . Thus, angle AED is equal to angle BED [Prop. 1.8]. Thus, angles AED and BED (are) each right-angles [Def. 1.10]. Thus, GK cuts AB in half, and at right-angles. And since, if some straight-line in a circle cuts some (other) straight-line in half, and at right-angles, (then) the center of the circle is on the former (straight-line) [Prop. 3.1 corr.], the center of the circle is thus on GK . So, for the same (reasons), the center of circle ABC is also on HL . And the straight-lines GK and HL have no common (point) other than point D . Thus, point D is the center of circle ABC .

Thus, if some point is taken inside a circle, and more than two equal straight-lines radiate from the point towards the (circumference of the) circle, (then) the point taken is the center of the circle. (Which is) the very thing it was required to show.

10.

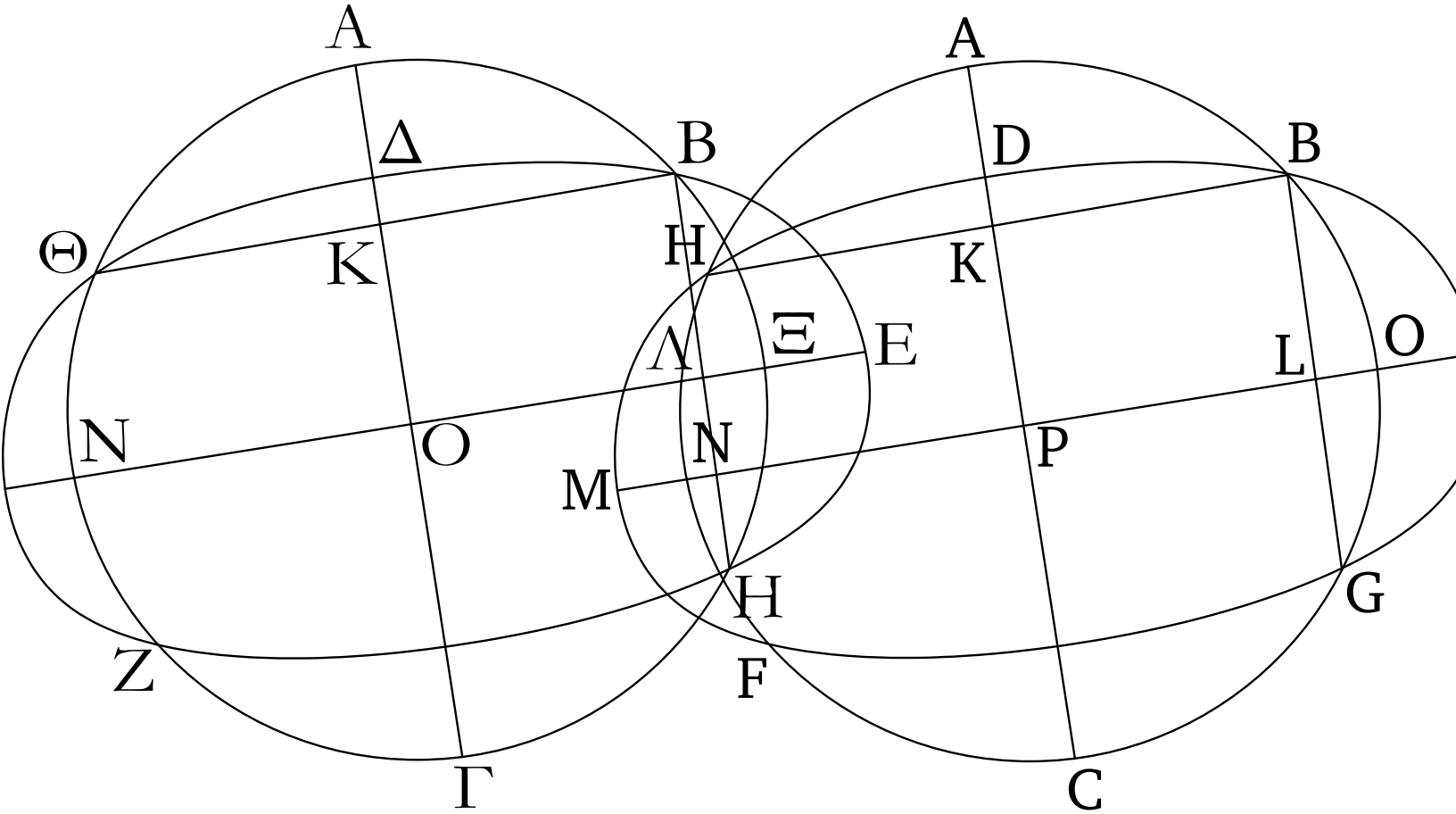
Proposition 10

Κύκλος κύκλον οὐ τέμνει κατὰ πλείονα σημεία ἢ δύο.

Εἰ γὰρ δυνατόν, κύκλος ὁ $ABΓ$ κύκλον τὸν $ΔΕΖ$ τεμνέτω κατὰ πλείονα σημεία ἢ δύο τὰ $B, H, Z, Θ$, καὶ ἐπιξενθθεῖσαι αἱ $BΘ$, BH δίθλα τεμνέσθωσαν κατὰ τὰ $K, Λ$ σημεία· καὶ ἀπὸ τῶν $K, Λ$ ταῖς $BΘ$, BH πρὸς ὀρθὰς ἀθθεῖσαι αἱ $KΓ$, $ΛΜ$ διήθθωσαν ἐπὶ τὰ A, E σημεία.

A circle does not cut a(nother) circle at more than two points.

For, if possible, let the circle ABC cut the circle DEF at more than two points, B, G, F , and H . And BH and BG being joined, let them (then) be cut in half at points K and L (respectively). And KC and LM being drawn at right-angles to BH and BG from K and L (respectively) [Prop. 1.11], let them (then) be drawn through to points A and E (respectively).



Ἐπεὶ οὖν ἐν κύκλῳ τῷ $ABΓ$ εὐθεῖά τις ἡ $ΑΓ$ εὐθεῖάν τινα τὴν $BΘ$ δίθλα καὶ πρὸς ὀρθὰς τέμνει, ἐπὶ τῇς $ΑΓ$ ἄρα ἐστὶ τὸ κέντρον τοῦ $ABΓ$ κύκλου. πάλιν, ἐπεὶ ἐν κύκλῳ τῷ αὐτῷ τῷ $ABΓ$ εὐθεῖά τις ἡ NX εὐθεῖάν τινα τὴν BH δίθλα καὶ πρὸς ὀρθὰς τέμνει, ἐπὶ τῇς NX ἄρα ἐστὶ τὸ κέντρον τοῦ $ABΓ$ κύκλου. ἐδείχθη δὲ καὶ ἐπὶ τῇς $ΑΓ$, καὶ κατ' οὐδέν συμβάλλουσιν αἱ $ΑΓ$, NX εὐθεῖαι ἢ κατὰ τὸ O · τὸ O ἄρα σημείον κέντρον ἐστὶ τοῦ $ABΓ$ κύκλου. ὁμοίως δὲ δείχμεν, ὅτι καὶ τοῦ $ΔΕΖ$ κύκλου κέντρον ἐστὶ τὸ O · δύο ἄρα κύκλων τεμνόντων ἀλλήλους τῶν $ABΓ$, $ΔΕΖ$ τὸ αὐτὸ ἐστὶ κέντρον τὸ O · ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα κύκλος κύκλον τέμνει κατὰ πλείονα σημεία ἢ

Therefore, since in circle ABC some straight-line AC cuts some (other) straight-line BH in half, and at right-angles, the center of circle ABC is thus on AC [Prop. 3.1 corr.]. Again, since in the same circle ABC some straight-line NO cuts some (other straight-line) BG in half, and at right-angles, the center of circle ABC is thus on NO [Prop. 3.1 corr.]. And it was also shown (to be) on AC . And the straight-lines AC and NO meet at no other (point) than P . Thus, point P is the center of circle ABC . So, similarly, we can show that P is also the center of circle DEF . Thus, two circles cutting one another, ABC and DEF , have the same center P . The very thing is impossible [Prop. 3.5].

δύο· ὅπερ ἔδει δεῖχαι.

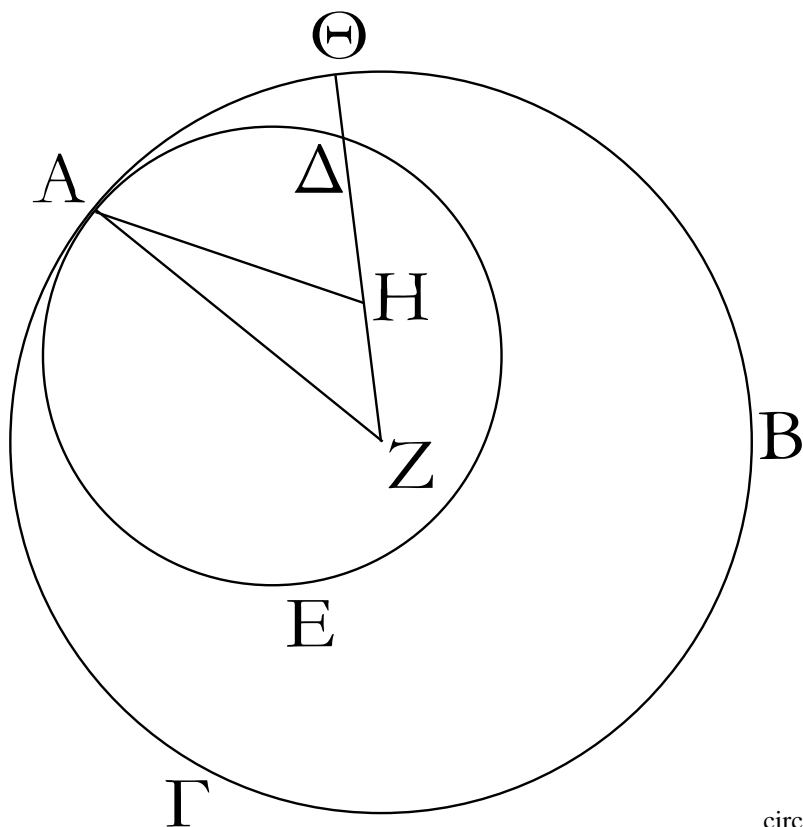
11.

Ἐὰν δύο κύκλοι ἐφάπτονται ἀλλήλων ἐντόξ, καὶ ληφθῇ αὐτῶν τὰ κέντρα, ἡ ἐπὶ τὰ κέντρα αὐτῶν ἐπιξηρνημένη εὐθεῖα καὶ ἐκβαλλομένη ἐπὶ τὴν συναφὴν πεσεῖται τῶν κύκλων.

Δύο γάρ κύκλοι οἱ $AB\Gamma$, $A\Delta E$ εφαπτέσθωσαν ἀλλήλων ἐντὸς κατὰ τὸ A σημεῖον, καὶ εἰλήθω τοῦ μὲν $AB\Gamma$ κύκλου κέντρον τὸ Z , τοῦ δὲ $A\Delta E$ τὸ H . λέγω, ὅτι ἡ ἀπὸ τοῦ H ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖα ἐκβαλλομένη ἐπὶ τὸ A περσεῖται.

μή γάρ, ἀλλ' εἰ δυνατόν, πιπτέτω ὡς ἡ ΖΗΘ, καὶ ἐπεξε-
ύθθωσαν αἱ ΑΖ, ΑΗ.

Ἐπειὶ οὖν αἱ AH, HZ τῆς ZA , τουτέστι τῆς $Z\Theta$, μείζονές εἰσιν, κοινὴ ἀφηρέσθω ἡ ZH . λοιπὴ ἄρα ἡ AH λοιπῆς τῆς $H\Theta$ μείζων ἐστίν. ἴση δὲ ἡ AH τῇ $H\Delta$. καὶ ἡ $H\Delta$ ἄρα τῆς $H\Theta$ μείζων ἐστίν ἡ ἐλάττων τῆς μείζονος· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα ἡ ἀπὸ τοῦ Z ἐπὶ τὸ H ἐπιγευγνυμένη εὐθεῖα ἐκτὸς πεσεῖται· κατὰ τὸ A ἄρα ἐπὶ τῆς συναφῆς πεσεῖται.



Thus, a circle does not cut a(nother) circle at more than two points. (Which is) the very thing it was required to show.

Proposition 11

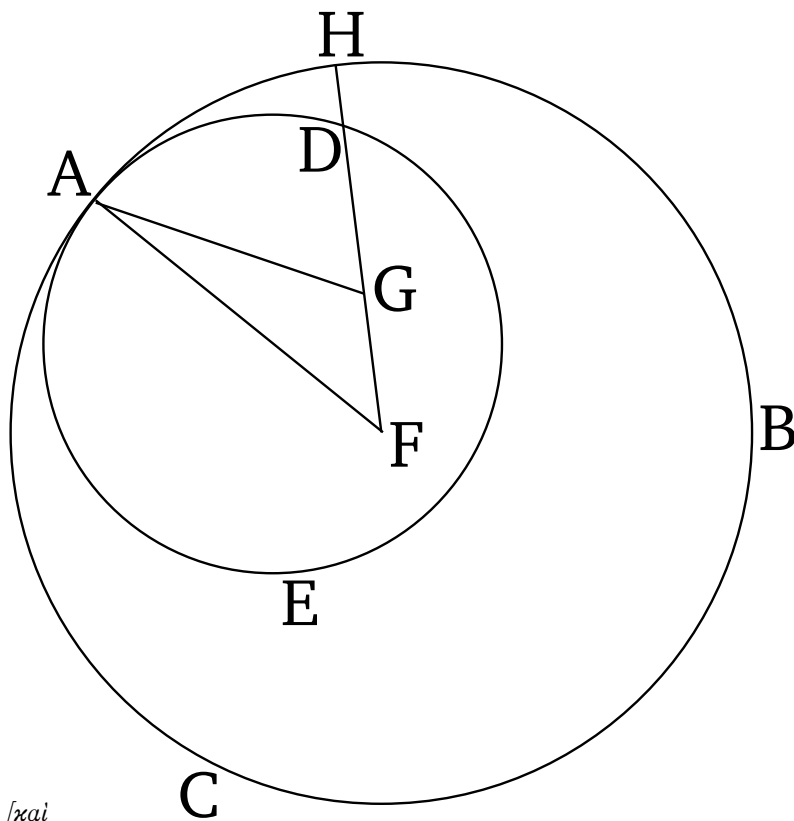
If two circles touch one another internally, and their centers are found, (then) the straight-line joining their centers, being produced, will fall upon the point of union of the circles.

For let two circles, ABC and ADE , touch one another internally at point A , and let the center F of circle ABC be found [Prop. 3.1], and (the center) G of (circle) ADE [Prop. 3.1]. I say that the straight-line joining G to F , being produced, will fall on A .

For (if) not (then), if possible, let it fall like FGH (in the figure), and let AF and AG be joined.

Therefore, since AG and GF is greater than FA , that is to say FH [Prop. 1.20], let FG be taken from both. Thus, the remainder AG is greater than the remainder GH . And AG (is) equal to GD . Thus, GD is also greater than GH , the lesser than the greater. The very thing is impossible. Thus, the straight-line joining F to G will not fall outside (one circle but inside the other). Thus, it will fall upon the point of union (of the

circles) at point A .



Ἐάν ἄρα δύο κύκλοι ἐφάπτονται ἀλλήλων ἐντός, [καὶ
ληφθῇ αὐτῶν τὰ κέντρα], ἡ ἐπὶ τὰ κέντρα αὐτῶν ἐπιζευγνυ-
μένη εὐθεῖα [καὶ ἐκβαλλομένη] ἐπὶ τὴν συναφὴν πεσεῖται τῶν
κύκλων· ὅπερ ἔδει δεῖχαι.

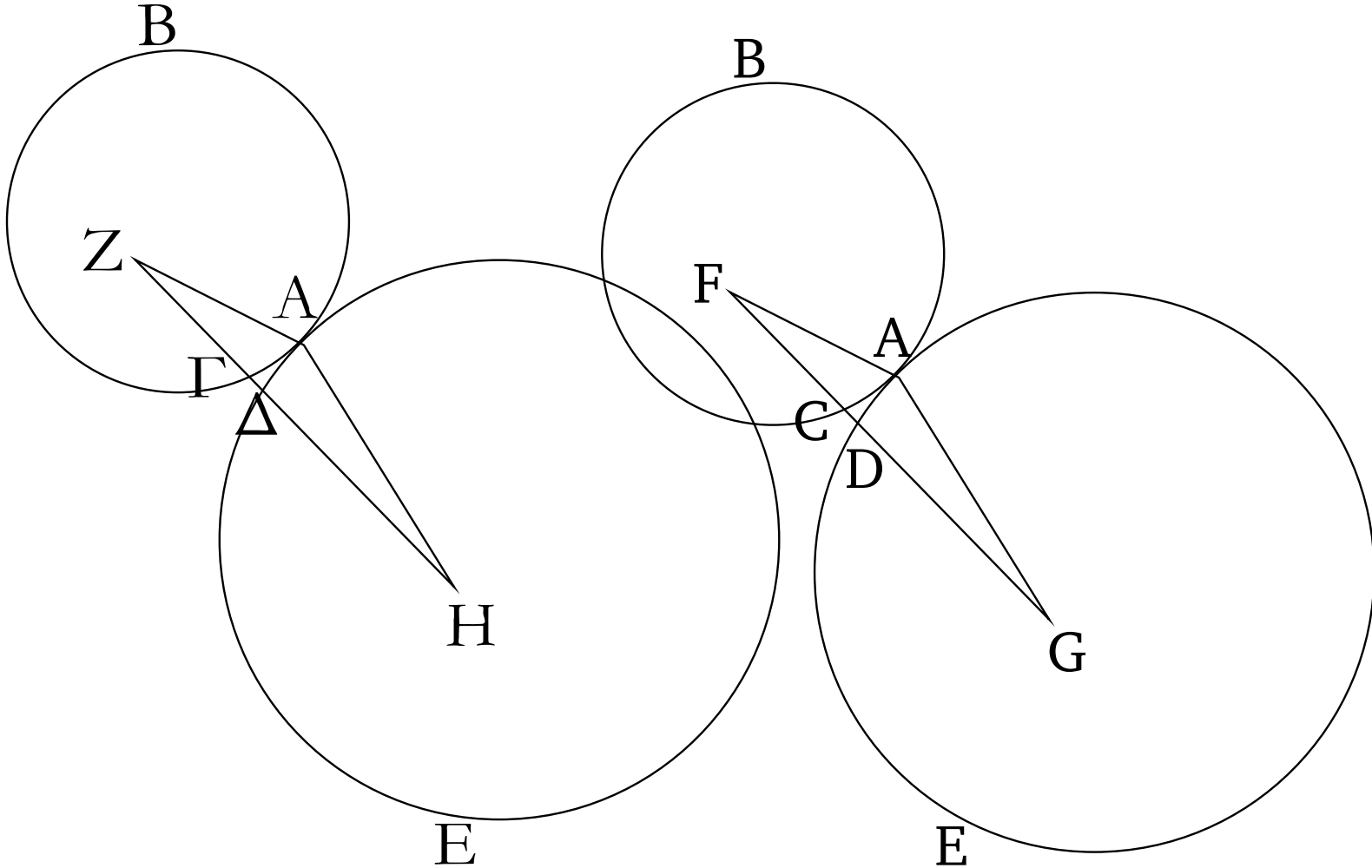
Thus, if two circles touch one another internally, [and their
centers are found], (then) the straight-line joining their cen-
ters, [being produced], will fall upon the point of union of the
circles. (Which is) the very thing it was required to show.

12.

Ἐάν δύο κύκλοι ἐφάπτονται ἀλλήλων ἐκτός, ἡ ἐπὶ τὰ
κέντρα αὐτῶν ἐπιζευγνυμένη διὰ τῆς ἐπαφῆς ἐλεύσεται.

Proposition 12

If two circles touch one another externally, (then) the
(straight-line) joining their centers will go through the point
of union.



Δύο γάρ κύκλοι οἱ $ABΓ$, $AΔΕ$ ἐφαπτέσθωσαν ἀλλήλων ἐκτός κατὰ τὸ A σημεῖον, καὶ εἰλήφθω τοῦ μὲν $ABΓ$ κέντρον τὸ Z , τοῦ δὲ $AΔΕ$ τὸ H . λέγω, ὅτι ἡ ἀπὸ τοῦ Z ἐπὶ τὸ H ἐπιξεννυμένη εὐθεῖα διὰ τῆς κατὰ τὸ A ἐπαφῆς ἐλεύσεται.

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἐρθέσθω ὡς ἡ $ZΓΔH$, καὶ ἐπεξεύθθωσαν αἱ AZ , AH .

Ἐπεὶ οὖν τὸ Z σημεῖον κέντρον ἐστὶ τοῦ $ABΓ$ κύκλου, ἴση ἐστὶν ἡ ZA τῇ $ZΓ$. πάλιν, ἐπεὶ τὸ H σημεῖον κέντρον ἐστὶ τοῦ $AΔΕ$ κύκλου, ἴση ἐστὶν ἡ HA τῇ $HΔ$. ἐδείκθη δὲ καὶ ἡ ZA τῇ $ZΓ$ ἴση· αἱ ἄρα ZA , AH ταῖς $ZΓ$, $HΔ$ ἴσαι εἰσὶν· ὥστε ὅλη ἡ ZH τῶν ZA , AH μείζων ἐστίν· ἀλλὰ καὶ ἐλάττων· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ Z ἐπὶ τὸ H ἐπιξεννυμένη εὐθεῖα διὰ τῆς κατὰ τὸ A ἐπαφῆς οὐκ ἐλεύσεται· δι' αὐτῆς ἄρα.

Ἐάν ἄρα δύο κύκλοι ἐφάπτονται ἀλλήλων ἐκτός, ἡ ἐπὶ τὰ κέντρα αὐτῶν ἐπιξεννυμένη [εὐθεῖα] διὰ τῆς ἐπαφῆς ἐλεύσεται· ὅπερ ἔδει δεῖξαι.

For let two circles, ABC and ADE , touch one another externally at point A , and let the center F of ABC be found [Prop. 3.1], and (the center) G of ADE [Prop. 3.1]. I say that the straight-line joining F to G will go through the point of union at A .

For (if) not (then), if possible, let it go like $FCDG$ (in the figure), and let AF and AG be joined.

Therefore, since point F is the center of circle ABC , FA is equal to FC . Again, since point G is the center of circle ADE , GA is equal to GD . And FA was also shown (to be) equal to FC . Thus, the (straight-lines) FA and AG are equal to the (straight-lines) FC and GD . So the whole of FG is greater than FA and AG . But, (it is) also less [Prop. 1.20]. The very thing is impossible. Thus, the straight-line joining F to G cannot not go through the point of union at A . Thus, (it will go) through it.

Thus, if two circles touch one another externally, (then) the [straight-line] joining their centers will go through the point of

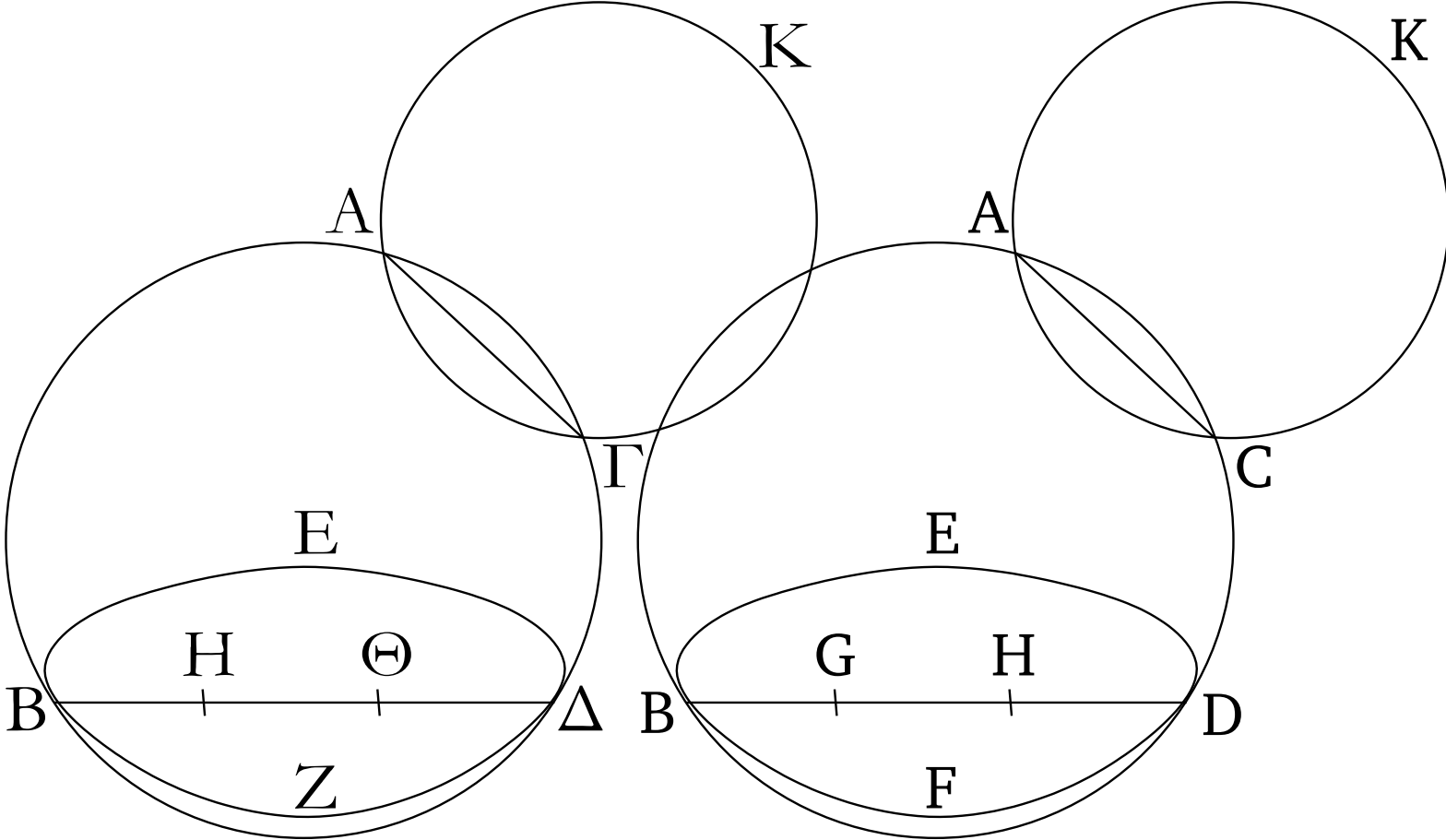
union. (Which is) the very thing it was required to show.

13.

Proposition 13

Κύκλος κύκλον οὐκ ἐφάπτεται κατὰ πλείονα σημεία ἢ καθ' ἓν, ἔαν τε ἐντὸς ἔαν τε ἐκτὸς ἐφάπτηται.

A circle does not touch a(nother) circle at more than one point, whether they touch internally or externally.



Εἰ γὰρ δυνατόν, κύκλος ὁ ABΓΔ κύκλον τοῦ EBZΔ ἐφάπτεσθω πρότερον ἐντὸς κατὰ πλείονα σημεία ἢ ἐν τὰ Δ, B.

Καὶ εἰλήφθω τοῦ μὲν ABΓΔ κύκλου κέντρον τὸ H, τοῦ δὲ EBZΔ τὸ Θ.

Ἡ ἄρα ἀπὸ τοῦ H ἐπὶ τὸ Θ ἐπιξεννυμένη ἐπὶ τὰ B, Δ πεσεῖται. πιπτέτω ὡς ἡ BHΘΔ. καὶ ἐπεὶ τὸ H σημεῖον κέντρον ἐστὶ τοῦ ABΓΔ κύκλου, ἴση ἐστὶν ἡ BH τῇ HD· μείζων ἄρα ἡ BH τῇς ΘΔ· πολλῶν ἄρα μείζων ἡ BΘ τῇς ΘΔ. πάλιν, ἐπεὶ τὸ Θ σημεῖον κέντρον ἐστὶ τοῦ EBZΔ κύκλου, ἴση ἐστὶν ἡ BΘ τῇ ΘΔ· ἐδείκθη δὲ αὐτῇς καὶ πολλῶν μείζων· ὅπερ ἀδύνατον· οὐκ ἄρα κύκλος κύκλον ἐφάπτεται ἐντὸς κατὰ πλείονα σημεία ἢ ἓν.

Λέγω δὴ, ὅτι οὐδὲ ἐκτὸς.

Εἰ γὰρ δυνατόν, κύκλος ὁ AΓΚ κύκλον τοῦ ABΓΔ ἐφάπτεσθω ἐκτὸς κατὰ πλείονα σημεία ἢ ἐν τὰ A, Γ, καὶ ἐπεξεύθθω ἡ AΓ.

For, if possible, let circle ABDC[†] touch circle EBFD—first of all, internally—at more than one point, D and B.

And let the center G of circle ABDC be found [Prop. 3.1], and (the center) H of EBFD [Prop. 3.1].

Thus, the (straight-line) joining G and H will fall on B and D [Prop. 3.11]. Let it fall like BGHD (in the figure). And since point G is the center of circle ABDC, BG is equal to GD. Thus, BG (is) greater than HD. Thus, BH (is) much greater than HD. Again, since point H is the center of circle EBFD, BH is equal to HD. But it was also shown (to be) much greater than it. The very thing (is) impossible. Thus, a circle does not touch a(nother) circle internally at more than one point.

So, I say that neither (does it touch) externally (at more than one point).

For, if possible, let circle ACK touch circle ABDC externally at more than one point, A and C. And let AC be joined.

Ἐπεὶ οὖν κύκλων τῶν $AB\Gamma\Delta$, $ΑΓΚ$ ἔλληπται ἐπὶ τῇς περιφερείας ἑκατέρον δύο τυθόντα σημεῖα τὰ A , Γ , ἢ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐντὸς ἑκατέρον πεσεῖται· ἀλλὰ τοῦ μὲν $AB\Gamma\Delta$ ἐντὸς ἔπεσεν, τοῦ δὲ $ΑΓΚ$ ἐκτός· ὅπερ ἄτοπον· οὐκ ἄρα κύκλος κύκλον ἐφάπτεται ἐκτός κατὰ πλείονα σημεῖα ἢ ἓν. ἐδείκθη δέ, ὅτι οὐδὲ ἐντός.

Κύκλος ἄρα κύκλον οὐκ ἐφάπτεται κατὰ πλείονα σημεῖα ἢ [καθ'] ἓν, ἔαν τε ἐντός ἔαν τε ἐκτός ἐφάπτηται· ὅπερ ἔδει δεῖχαι.

Therefore, since two points, A and C , be taken at random on the circumference of each of the circles $ABDC$ and ACK , the straight-line joining the points will fall inside each (circle) [Prop. 3.2]. But, it fell inside $ABDC$, and outside ACK [Def. 3.3]. The very thing (is) absurd. Thus, a circle does not touch a(nother) circle externally at more than one point. And it was shown that neither (does it) internally.

Thus, a circle does not touch a(nother) circle at more than one point, whether they touch internally or externally. (Which is) the very thing it was required to show.

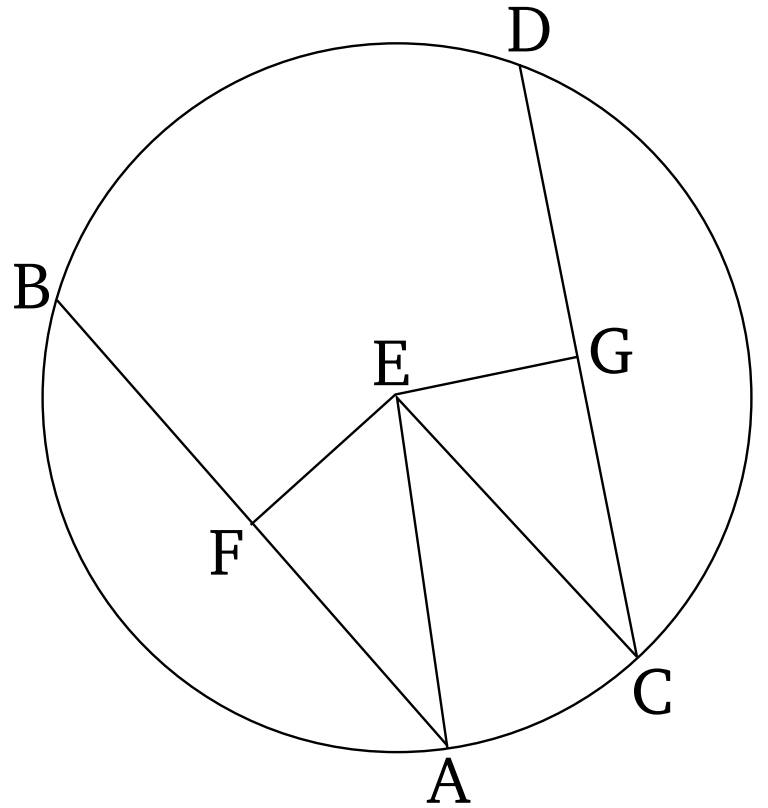
† The Greek text has “ $ABCD$ ”, which is obviously a mistake.

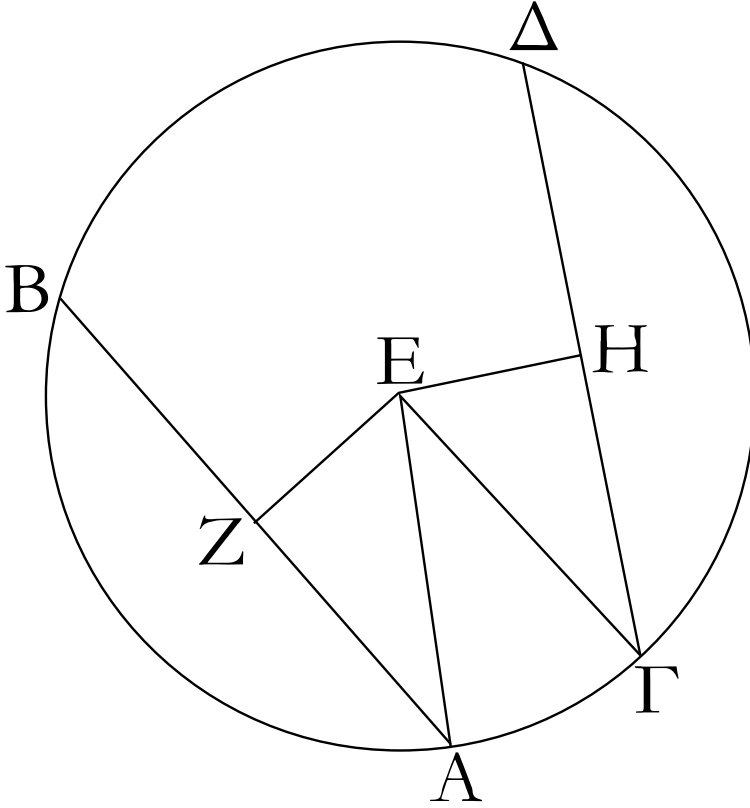
14.

Ἐν κύκλῳ αἱ ἴσαι εὐθεῖαι ἴσον ἀπέθουσιν ἀπὸ τοῦ κέντρου, καὶ αἱ ἴσον ἀπέθουσαι ἀπὸ τοῦ κέντρου ἴσαι ἀλλήλαις εἶσιν.

Proposition 14

In a circle, equal straight-lines are equally far from the center, and (straight-lines) which are equally far from the center are equal to one another.





Ἐστω κύκλος ὁ $AB\Gamma\Delta$, καὶ ἐν αὐτῷ ἴσαι εὐθεῖαι ἔστωσαν αἱ AB , $\Gamma\Delta$ · λέγω, ὅτι αἱ AB , $\Gamma\Delta$ ἴσον ἀπέθουσιν ἀπὸ τοῦ κέντρου.

Εἰλήφθω γάρ τὸ κέντρον τοῦ $AB\Gamma\Delta$ κύκλου καὶ ἔστω τὸ E , καὶ ἀπὸ τοῦ E ἐπὶ τὰς AB , $\Gamma\Delta$ κάθετοι ἡθθωσαν αἱ EZ , EH , καὶ ἐπεξεύθωσαν αἱ AE , EG .

Ἐπεὶ οὖν εὐθεῖα τις δια τοῦ κέντρου ἡ EZ εὐθεῖάν τινα μὴ διὰ τοῦ κέντρου τὴν AB πρὸς ὀρθὰς τέμνει, καὶ δίχα αὐτὴν τέμνει. ἴση ἄρα ἡ AZ τῇ ZB · διπλὴ ἄρα ἡ AB τῆς AZ . διὰ τὰ αὐτὰ δὴ καὶ ἡ $\Gamma\Delta$ τῆς GH ἐστι διπλὴ· καὶ ἐστὶν ἴση ἡ AB τῇ $\Gamma\Delta$ · ἴση ἄρα καὶ ἡ AZ τῇ GH . καὶ ἐπεὶ ἴση ἐστὶν ἡ AE τῇ EG , ἴσον καὶ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς EG . ἀλλὰ τῷ μὲν ἀπὸ τῆς AE ἴσα τὰ ἀπὸ τῶν AZ , EZ · ὀρθὴ γάρ ἡ πρὸς τῷ Z γωνία· τῷ δὲ ἀπὸ τῆς EG ἴσα τὰ ἀπὸ τῶν EH , HG · ὀρθὴ γάρ ἡ πρὸς τῷ H γωνία· τὰ ἄρα ἀπὸ τῶν AZ , ZE ἴσα ἐστὶ τοῖς ἀπὸ τῶν GH , HE , ὧν τὸ ἀπὸ τῆς AZ ἴσον ἐστὶ τῷ ἀπὸ τῆς GH · ἴση γάρ ἐστιν ἡ AZ τῇ GH · λοιπὸν ἄρα τὸ ἀπὸ τῆς ZE τῷ ἀπὸ τῆς EH ἴσον ἐστὶν· ἴση ἄρα ἡ EZ τῇ EH . ἐν δὲ κύκλῳ ἴσον ἀπέθειν ἀπὸ τοῦ κέντρου εὐθεῖαι λέγονται, ὅταν αἱ ἀπὸ τοῦ κέντρου ἐπ' αὐτὰς κάθετοι ἀγόμεναι ἴσαι ᾖσιν· αἱ ἄρα AB , $\Gamma\Delta$ ἴσον ἀπέθουσιν ἀπὸ τοῦ κέντρου.

Ἀλλὰ δὴ αἱ AB , $\Gamma\Delta$ εὐθεῖαι ἴσον ἀπεθέτωσαν ἀπὸ τοῦ κέντρου, τουτέστιν ἴση ἔστω ἡ EZ τῇ EH . λέγω, ὅτι ἴση ἐστὶ καὶ ἡ AB τῇ $\Gamma\Delta$.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχνομεν, ὅτι

Let $ABDC^{\dagger}$ be a circle, and let AB and CD be equal straight-lines within it. I say that AB and CD are equally far from the center.

For let the center of circle $ABDC$ be found [Prop. 3.1], and let it be (at) E . And let EF and EG be drawn from (point) E , perpendicular to AB and CD (respectively) [Prop. 1.12]. And let AE and EC be joined.

Therefore, since some straight-line, EF , through the center (of the circle), cuts some (other) straight-line, AB , not through the center, at right-angles, it also cuts it in half [Prop. 3.3]. Thus, AF (is) equal to FB . Thus, AB (is) double AF . So, for the same (reasons), CD is also double CG . And AB is equal to CD . Thus, AF (is) also equal to CG . And since AE is equal to EC , the (square) on AE (is) also equal to the (square) on EC . But, the (sum of the squares) on AF and EF (is) equal to the (square) on AE . For the angle at F (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on EG and GC (is) equal to the (square) on EC . For the angle at G (is) a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AF and FE is equal to the (sum of the squares) on CG and GE , of which the (square) on AF is equal to the (square) on CG . For AF is equal to CG . Thus, the remaining (square) on FE is equal to the (remaining square) on EG . Thus, EF (is) equal to EG . And straight-lines in a circle are said to be equally far from the center when perpendicular (straight-lines) which are drawn to them from the center are equal [Def. 3.4]. Thus, AB

διπλὴ ἐστὶν ἡ μὲν AB τῆς AZ , ἡ δὲ $\Gamma\Delta$ τῆς ΓH · καὶ ἐπεὶ ἴση ἐστὶν ἡ AE τῇ GE , ἴσον ἐστὶ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς GE · ἀλλὰ τῷ μὲν ἀπὸ τῆς AE ἴσα ἐστὶ τὰ ἀπὸ τῶν EZ , ZA , τῷ δὲ ἀπὸ τῆς GE ἴσα τὰ ἀπὸ τῶν EH , HG . τὰ ἄρα ἀπὸ τῶν EZ , ZA ἴσα ἐστὶ τοῖς ἀπὸ τῶν EH , HG . ὣν τὸ ἀπὸ τῆς EZ τῷ ἀπὸ τῆς EH ἐστὶν ἴσον· ἴση γάρ ἡ EZ τῇ EH . λοιπὸν ἄρα τὸ ἀπὸ τῆς AZ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΓH . ἴση ἄρα ἡ AZ τῇ ΓH · καὶ ἐστὶ τῆς μὲν AZ διπλὴ ἡ AB , τῆς δὲ ΓH διπλὴ ἡ $\Gamma\Delta$. ἴση ἄρα ἡ AB τῇ $\Gamma\Delta$.

Ἐν κύκλῳ ἄρα αἱ ἴσαι εὐθεῖαι ἴσον ἀπέθουσι ἀπὸ τοῦ κέντρου, καὶ αἱ ἴσον ἀπέθουσαι ἀπὸ τοῦ κέντρου ἴσαι ἀλλήλαις εἰσὶν· ὅπερ ἔδει δεῖξαι.

and CD are equally far from the center.

So, let the straight-lines AB and CD be equally far from the center. That is to say, let EF be equal to EG . I say that AB is also equal to CD .

For, with the same construction, we can, similarly, show that AB is double AF , and CD (double) CG . And since AE is equal to CE , the (square) on AE is equal to the (square) on CE . But, the (sum of the squares) on EF and FA is equal to the (square) on AE [Prop. 1.47]. And the (sum of the squares) on EG and GC (is) equal to the (square) on CE [Prop. 1.47]. Thus, the (sum of the squares) on EF and FA is equal to the (sum of the squares) on EG and GC , of which the (square) on EF is equal to the (square) on EG . For EF (is) equal to EG . Thus, the remaining (square) on AF is equal to the (remaining square) on CG . Thus, AF (is) equal to CG . And AB is double AF , and CD double CG . Thus, AB (is) equal to CD .

Thus, in a circle, equal straight-lines are equally far from the center, and (straight-lines) which are equally far from the center are equal to one another. (Which is) the very thing it was required to show.

† The Greek text has “ $ABCD$ ”, which is obviously a mistake.

15.

Ἐν κύκλῳ μεγίστη μὲν ἡ διάμετρος, τῶν δὲ ἄλλων αἰεὶ ἡ ἕγγιον τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν.

Ἐστω κύκλος ὁ $AB\Gamma\Delta$, διάμετρος δὲ αὐτοῦ ἔστω ἡ $A\Delta$, κέντρον δὲ τὸ E , καὶ ἕγγιον μὲν τῆς $A\Delta$ διαμέτρον ἔστω ἡ $B\Gamma$, ἀπώτερον δὲ ἡ ZH . λέγω, ὅτι μεγίστη μὲν ἐστὶν ἡ $A\Delta$, μείζων δὲ ἡ $B\Gamma$ τῆς ZH .

Ἦθθωσαν γάρ ἀπὸ τοῦ E κέντρου ἐπὶ τὰς $B\Gamma$, ZH κάθετοι αἱ $E\Theta$, $E\Kappa$. καὶ ἐπεὶ ἕγγιον μὲν τοῦ κέντρου ἐστὶν ἡ $B\Gamma$, ἀπώτερον δὲ ἡ ZH , μείζων ἄρα ἡ $E\Kappa$ τῆς $E\Theta$. κείσθω τῇ $E\Theta$ ἴση ἡ $ΕΛ$, καὶ διὰ τοῦ Λ τῇ $E\Kappa$ πρὸς ὀρθὰς ἀθθεῖσα ἡ ΛM διήθθω ἐπὶ τὸ N , καὶ ἐπεξέυθθωσαν αἱ ME , EN , ZE , EH .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ $E\Theta$ τῇ $ΕΛ$, ἴση ἐστὶ καὶ ἡ $B\Gamma$ τῇ MN . πάλιν, ἐπεὶ ἴση ἐστὶν ἡ μὲν AE τῇ EM , ἡ δὲ $ΕΔ$ τῇ EN , ἡ ἄρα $A\Delta$ ταῖς ME , EN ἴση ἐστίν. ἀλλ’ αἱ μὲν ME , EN τῆς MN μείζονές εἰσιν [καὶ ἡ $A\Delta$ τῆς MN μείζων ἐστίν], ἴση δὲ ἡ MN τῇ $B\Gamma$. ἡ $A\Delta$ ἄρα τῆς $B\Gamma$ μείζων ἐστίν. καὶ ἐπεὶ δύο αἱ ME , EN δύο ταῖς ZE , EH ἴσαι εἰσὶν, καὶ γωνία ἡ ὑπὸ MEN γωνία τῆς ὑπὸ ZEH μείζων [ἐστίν], βάσις ἄρα ἡ MN βάσεως τῆς ZH μείζων ἐστίν. ἀλλὰ ἡ MN τῇ $B\Gamma$ ἐδείχθη ἴση [καὶ ἡ $B\Gamma$ τῆς ZH μείζων ἐστίν]. μεγίστη μὲν ἄρα ἡ $A\Delta$ διάμετρος, μείζων δὲ ἡ $B\Gamma$ τῆς ZH .

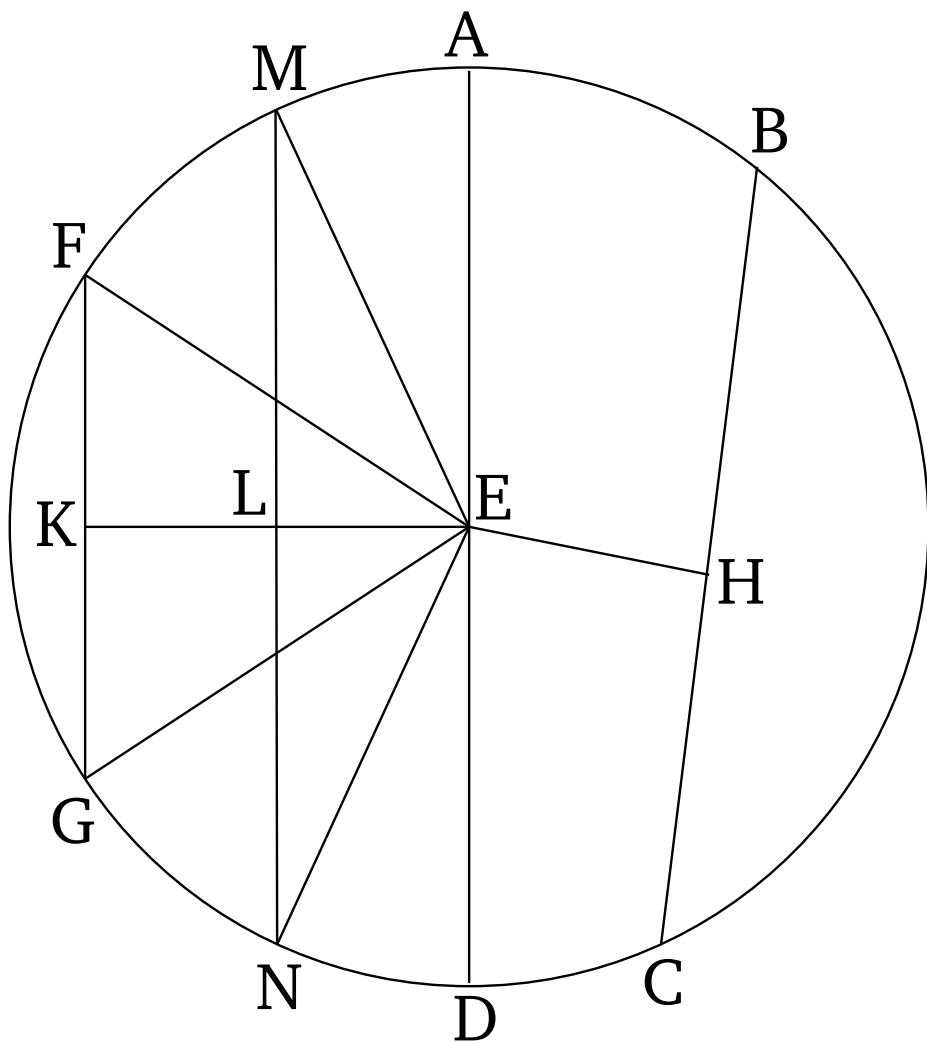
Proposition 15

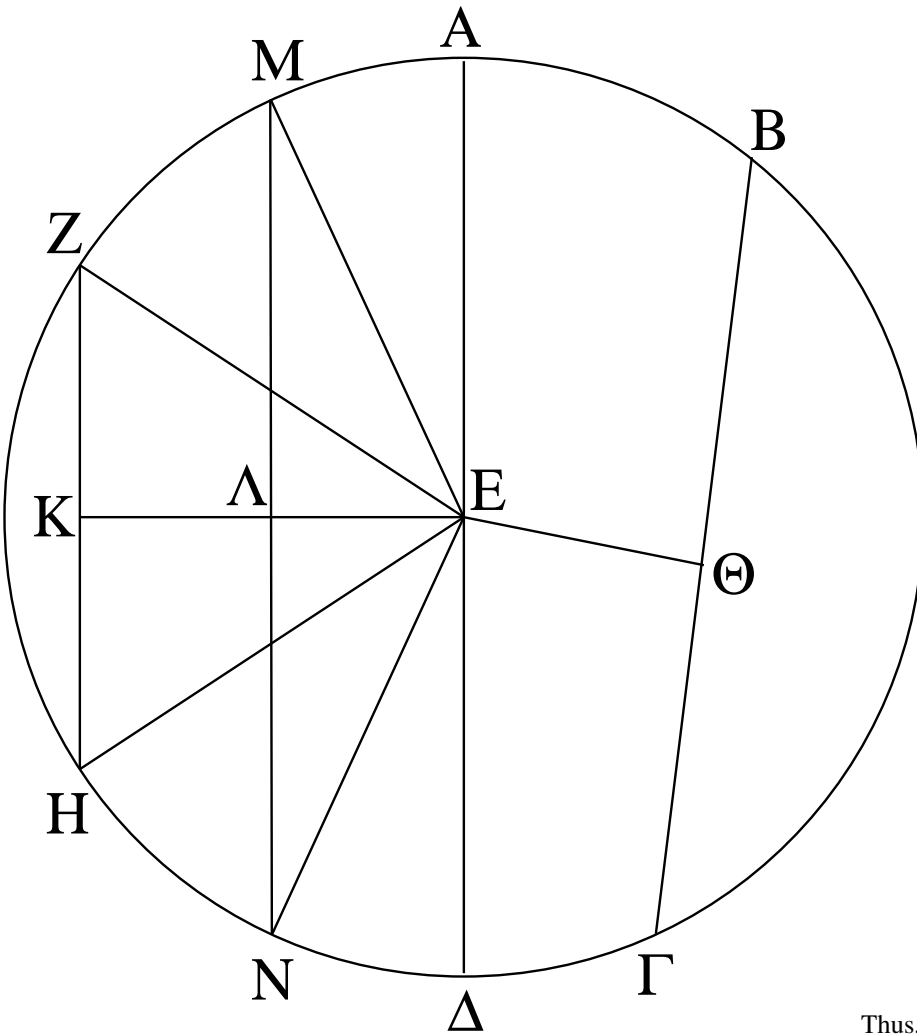
In a circle, a diameter (is) the greatest (straight-line), and for the others, a (straight-line) nearer to the center is always greater than one further away.

Let $ABCD$ be a circle, and let AD be its diameter, and E (its) center. And let BC be nearer to the diameter AD ,[†] and FG further away. I say that AD is the greatest (straight-line), and BC (is) greater than FG .

For let EH and $E\Kappa$ be drawn from the center E , at right-angles to BC and FG (respectively) [Prop. 1.12]. And since BC is nearer to the center, and FG further away, $E\Kappa$ (is) thus greater than EH [Def. 3.5]. Let EL be made equal to EH [Prop. 1.3]. And LM being drawn through L , at right-angles to $E\Kappa$ [Prop. 1.11], let it be drawn through to N . And let ME , EN , FE , and EG be joined.

And since EH is equal to EL , BC is also equal to MN [Prop. 3.14]. Again, since AE is equal to EM , and ED to EN , AD is thus equal to ME and EN . But, ME and EN is greater than MN [Prop. 1.20] [also AD is greater than MN], and MN (is) equal to BC . Thus, AD is greater than BC . And since the two (straight-lines) ME , EN are equal to the two (straight-lines) FE , EG (respectively), and angle MEN [is] greater than angle FEG ,[‡] the base MN is thus greater than the base FG [Prop. 1.24]. But, MN was shown (to be) equal to BC [(so) BC is also greater than FG]. Thus, the diameter AD (is) the greatest (straight-line), and BC (is) greater than FG .





Ἐν κύκλῳ ἄρα μέγιστη μὲν ἐστὶν ἡ διάμετρος, τῶν δὲ ἄλλων αἰεὶ ἡ ἔγγιον τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν· ὅπερ ἔδει δεῖξαι.

Thus, in a circle, a diameter (is) the greatest (straight-line), and for the others, a (straight-line) nearer to the center is always greater than one further away. (Which is) the very thing it was required to show.

[†] Euclid should have said “to the center”, rather than “to the diameter AD”, since BC, AD and FG are not necessarily parallel.

[‡] This is not proved, except by reference to the figure.

16.

Ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ’ ἀκρᾶς ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου, καὶ εἷς τὸν μεταχὺ τόπον τῆς τε εὐθείας καὶ τῆς περιφερείας ἑτέρα εὐθεῖα οὐ παρεμπεσεῖται, καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἀπάσης γωνίας ὀχείας εὐθυγράμμων μείζων ἐστίν, ἡ δὲ λοιπὴ ἐλάττων.

Ἐστω κύκλος ὁ ABΓ περὶ κέντρον τὸ Δ καὶ διάμετρον τὴν AB· λέγω, ὅτι ἡ ἀπὸ τοῦ A τῇ AB πρὸς ὀρθὰς ἀπ’ ἀκρᾶς ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου.

μὴ γάρ, ἀλλ’ εἰ δυνατόν, πιπτέτω ἐντὸς ὡς ἡ ΓΑ, καὶ ἐπεζεύθθω ἡ ΔΓ.

Proposition 16

A (straight-line) drawn at right-angles to the diameter of a circle, from its end, will fall outside the circle. And another straight-line cannot be inserted into the space between the (aforementioned) straight-line and the circumference. And the angle of the semi-circle is greater than any acute rectilinear angle whatsoever, and the remaining (angle is) less (than any acute rectilinear angle).

Let ABC be a circle around the center D and the diameter AB. I say that the (straight-line) drawn from A, at right-angles to AB [Prop 1.11], from its end, will fall outside the circle.

Ἐπεὶ ἴση ἐστὶν ἡ ΔA τῇ $\Delta \Gamma$, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $\Delta A \Gamma$ γωνία τῇ ὑπὸ $A \Gamma \Delta$. ὀρθὴ δὲ ἡ ὑπὸ $\Delta A \Gamma$. ὀρθὴ ἄρα καὶ ἡ ὑπὸ $A \Gamma \Delta$. τριγώνου δὲ τοῦ $A \Gamma \Delta$ αἱ δύο γωνίαι αἱ ὑπὸ $\Delta A \Gamma$, $A \Gamma \Delta$ δύο ὀρθαῖς ἴσαι εἰσὶν· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ A σημείου τῇ BA πρὸς ὀρθὰς ἀγομένη ἐντὸς πεσεῖται τοῦ κύκλου. ὁμοίως δὲ δεῖχμεν, ὅτι οὐδ' ἐπὶ τῇς περιφερείας· ἐκτὸς ἄρα.

Πιπτέτω ὡς ἡ AE . λέγω δὲ, ὅτι εἰς τὸν μεταχὺν τόπον τῇς τε AE εὐθείας καὶ τῇς $\Gamma\Theta A$ περιφερείας ἑτέρα εὐθεῖα οὐ παρεμπεσείται.

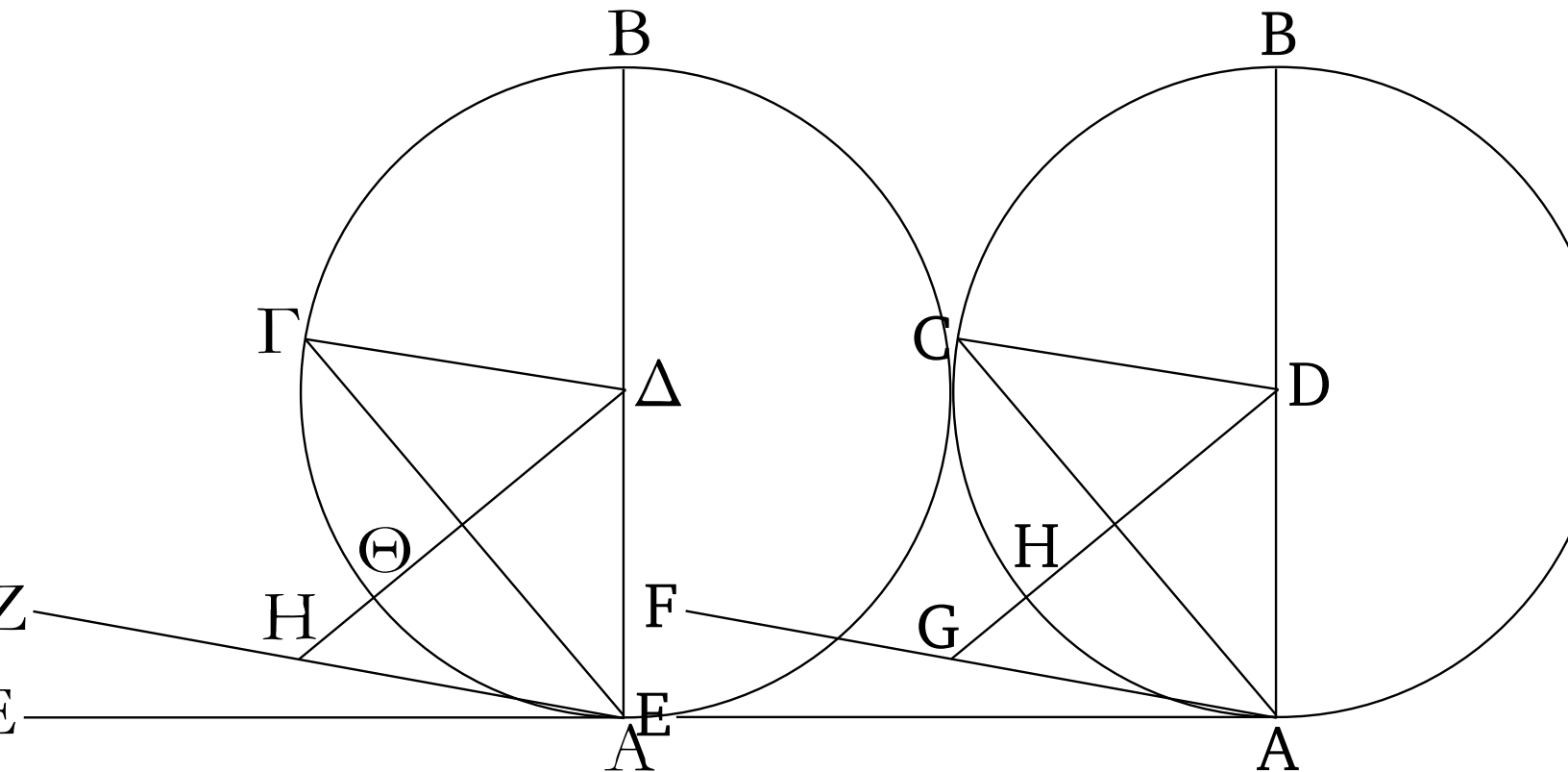
Εἰ γὰρ δυνατόν, παρεμπίπτέτω ὡς ἡ ZA , καὶ ἤθθω ἀπὸ τοῦ Δ σημείου ἐπὶ τὴν ZA κάθετος ἡ ΔH . καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ ὑπὸ $AH\Delta$, ἐλάττων δὲ ὀρθῇς ἡ ὑπὸ $\Delta A H$, μείζων ἄρα ἡ $A\Delta$ τῇς ΔH . ἴση δὲ ἡ ΔA τῇ $\Delta\Theta$. μείζων ἄρα ἡ $\Delta\Theta$ τῇς ΔH , ἢ ἐλάττων τῇς μείζονος· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα εἰς τὸν μεταχὺν τόπον τῇς τε εὐθείας καὶ τῇς περιφερείας ἑτέρα εὐθεῖα παρεμπεσείται.

For (if) not (then), if possible, let it fall inside, like CA (in the figure), and let DC be joined.

Since DA is equal to DC , angle DAC is also equal to angle ACD [Prop. 1.5]. And DAC (is) a right-angle. Thus, ACD (is) also a right-angle. So, in triangle ACD , the two angles DAC and ACD are equal to two right-angles. The very thing is impossible [Prop. 1.17]. Thus, the (straight-line) drawn from point A , at right-angles to BA , will not fall inside the circle. So, similarly, we can show that neither (will it fall) on the circumference. Thus, (it will fall) outside (the circle).

Let it fall like AE (in the figure). So, I say that another straight-line cannot be inserted into the space between the straight-line AE and the circumference CHA .

For, if possible, let it be inserted like FA (in the figure), and let DG be drawn from point D , perpendicular to FA [Prop. 1.12]. And since AGD is a right-angle, and DAG (is) less than a right-angle, AD (is) thus greater than DG [Prop. 1.19]. And DA (is) equal to DH . Thus, DH (is) greater than DG , the lesser than the greater. The very thing is impossible. Thus, another straight-line cannot be inserted into the space between the straight-line (AE) and the circumference.



Λέγω, ὅτι καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἡ περιεθρομένη ὑπὸ τε τῇς BA εὐθείας καὶ τῇς $\Gamma\Theta A$ περιφερείας ἀπόσης γωνίαξ ὀχλείας εὐθυγράμμου μείζων ἐστίν, ἢ δὲ λοιπὴ ἡ περιεθρομένη ὑπὸ τε τῇς $\Gamma\Theta A$ περιφερείας καὶ τῇς AE εὐθείας

And I also say that the semi-circular angle contained by the straight-line BA and the circumference CHA is greater than any acute rectilinear angle whatsoever, and the remaining (angle) contained by the circumference CHA and the straight-line

ἀπάσης γωνίας ὀξείας εὐθυγράμμων ἐλάττων ἐστίν.

Εἰ γὰρ ἐστί τις γωνία εὐθύγραμμος μείζων μὲν τῆς περιεθρομένης ὑπὸ τε τῆς BA εὐθείας καὶ τῆς $ΓΘΑ$ περιφερείας, ἐλάττων δὲ τῆς περιεθρομένης ὑπὸ τε τῆς $ΓΘΑ$ περιφερείας καὶ τῆς $ΑΕ$ εὐθείας, εἰς τὸν μεταχὺ τόπον τῆς τε $ΓΘΑ$ περιφερείας καὶ τῆς $ΑΕ$ εὐθείας εὐθεῖα παρεμπεσέται, ἥτις ποιήσει μείζονα μὲν τῆς περιεθρομένης ὑπὸ τε τῆς BA εὐθείας καὶ τῆς $ΓΘΑ$ περιφερείας ὑπὸ εὐθειῶν περιεθρομένην, ἐλάττωνα δὲ τῆς περιεθρομένης ὑπὸ τε τῆς $ΓΘΑ$ περιφερείας καὶ τῆς $ΑΕ$ εὐθείας. οὐ παρεμπίπτει δέ· οὐκ ἄρα τῆς περιεθρομένης γωνίας ὑπὸ τε τῆς BA εὐθείας καὶ τῆς $ΓΘΑ$ περιφερείας ἔσται μείζων ὀξεία ὑπὸ εὐθειῶν περιεθρομένη, οὐδὲ μὴν ἐλάττων τῆς περιεθρομένης ὑπὸ τε τῆς $ΓΘΑ$ περιφερείας καὶ τῆς $ΑΕ$ εὐθείας.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἁκρᾶς ἀγομένη ἐφάπτεται τοῦ κύκλου [καὶ ὅτι εὐθεῖα κύκλου καθ' ἓν μόνον ἐφάπτεται σημεῖον, ἐπειδήπερ καὶ ἡ κατὰ δύο αὐτῶν συμβάλλονσα ἐντὸς αὐτοῦ πίπτουσα ἐδείχθη]· ὅπερ ἔδει δεῖχαι.

17.

Ἀπὸ τοῦ δοθέντος σημείου τοῦ δοθέντος κύκλου ἐφαπτομένην εὐθεῖαν γραμμὴν ἀγαγεῖν.

Ἔστω τὸ μὲν δοθὲν σημεῖον τὸ A , ὃ δὲ δοθεὶς κύκλος ὁ $ΒΓΔ$ · δεῖ δὴ ἀπὸ τοῦ A σημείου τοῦ $ΒΓΔ$ κύκλου ἐφαπτομένην εὐθεῖαν γραμμὴν ἀγαγεῖν.

AE is less than any acute rectilinear angle whatsoever.

For if any rectilinear angle is greater than the (angle) contained by the straight-line BA and the circumference CHA , or less than the (angle) contained by the circumference CHA and the straight-line AE , (then) a straight-line can be inserted into the space between the circumference CHA and the straight-line AE —anything which will make (an angle) contained by straight-lines greater than the angle contained by the straight-line BA and the circumference CHA , or less than the (angle) contained by the circumference CHA and the straight-line AE . But (such a straight-line) cannot be inserted. Thus, an acute (angle) contained by straight-lines cannot be greater than the angle contained by the straight-line BA and the circumference CHA , neither (can it be) less than the (angle) contained by the circumference CHA and the straight-line AE .

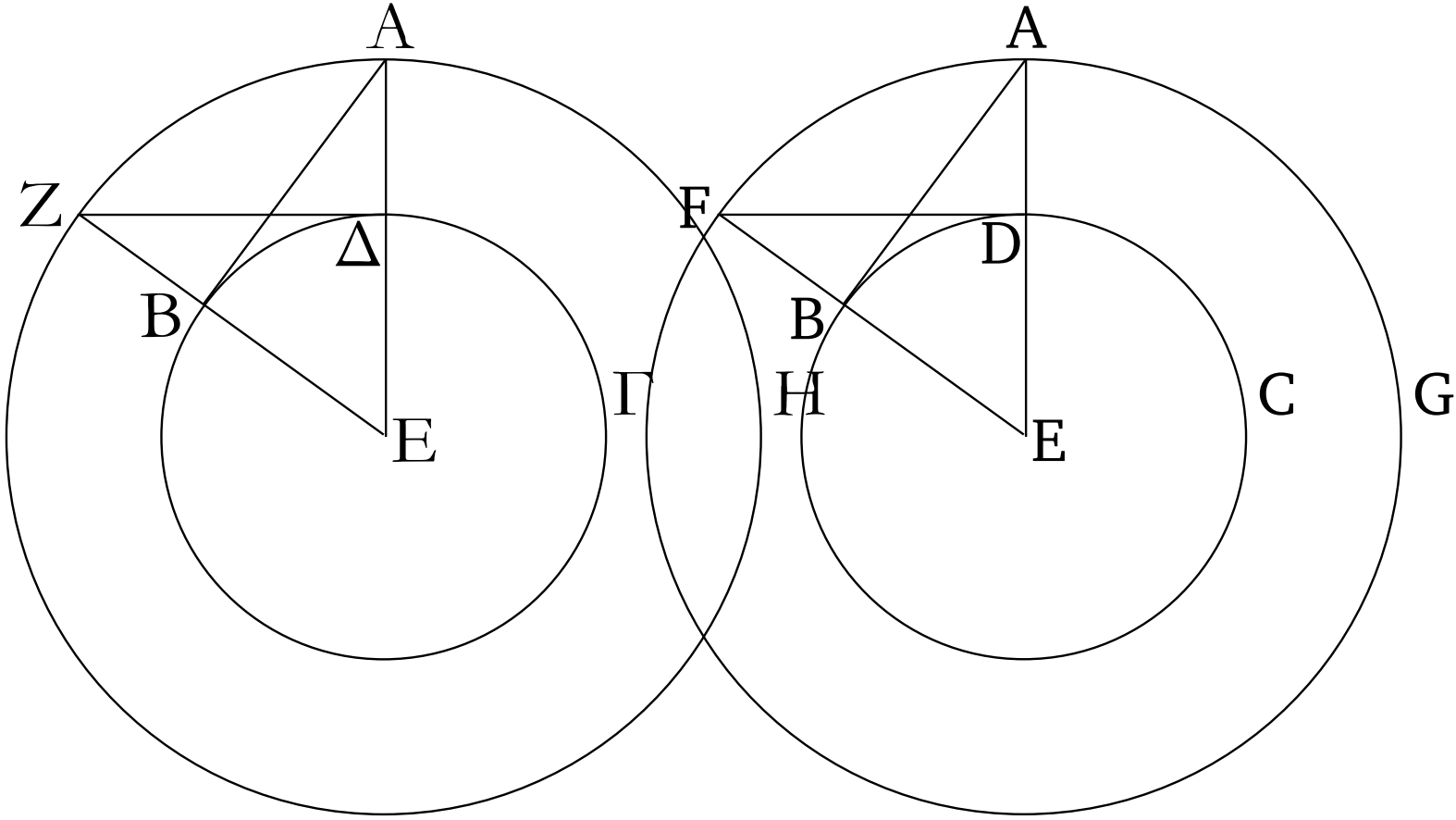
Corollary

So, from this, (it is) manifest that a (straight-line) drawn at right-angles to the diameter of a circle, from its extremity, touches the circle [and that the straight-line touches the circle at a single point, inasmuch as it was also shown that a (straight-line) meeting (the circle) at two (points) falls inside it [Prop. 3.2]]. (Which is) the very thing it was required to show.

Proposition 17

To draw a straight-line touching a given circle from a given point.

Let A be the given point, and BCD the given circle. So it is required to draw a straight-line touching circle BCD from point A .



Εἰλήφθω γάρ τὸ κέντρον τοῦ κύκλου τὸ E , καὶ ἐπεξεύθθω ἡ AE , καὶ κέντρῳ μὲν τῷ E διαστήματι δὲ τῷ EA κύκλῳ γεγράφθω ὁ AZH , καὶ ἀπὸ τοῦ Δ τῇ EA πρὸς ὀρθὰς ῥηθῇ ἡ ΔZ , καὶ ἐπεξεύθθωσαν αἱ EZ , AB . λέγω, ὅτι ἀπὸ τοῦ A σημείου τοῦ $B\Gamma\Delta$ κύκλου ἐφαπτομένη ἦκται ἡ AB .

Ἐπεὶ γάρ τὸ E κέντρον ἐστὶ τῶν $B\Gamma\Delta$, AZH κύκλων, ἴση ἄρα ἐστὶν ἡ μὲν EA τῇ EZ , ἡ δὲ $E\Delta$ τῇ EB . δύο δὲ αἱ AE , EB δύο ταῖς ZE , $E\Delta$ ἴσαι εἰσὶν· καὶ γωνίαν κοινὴν περιέθουσι τὴν πρὸς τῷ E . βάσεις ἄρα ἡ ΔZ βάσει τῇ AB ἴση ἐστίν, καὶ τὸ ΔEZ τρίγωνον τῷ EBA τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις· ἴση ἄρα ἡ ὑπὸ $E\Delta Z$ τῇ ὑπὸ EBA . ὀρθὴ δὲ ἡ ὑπὸ $E\Delta Z$. ὀρθὴ ἄρα καὶ ἡ ὑπὸ EBA . καὶ ἐστὶν ἡ EB ἐκ τοῦ κέντρον· ἡ δὲ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἀκραξ ἀγομένη ἐφάπτεται τοῦ κύκλου· ἡ AB ἄρα ἐφάπτεται τοῦ $B\Gamma\Delta$ κύκλου.

Ἀπὸ τοῦ ἄρα δοθέντος σημείου τοῦ A τοῦ δοθέντος κύκλου τοῦ $B\Gamma\Delta$ ἐφαπτομένη εὐθεῖα γραμμὴ ἦκται ἡ AB . ὅπερ ἔδει ποιῆσαι.

For let the center E of the circle be found [Prop. 3.1], and let AE be joined. And let (the circle) AFG be drawn with center E and radius EA . And let DF be drawn from (point) D , at right-angles to EA [Prop. 1.11]. And let EF and AB be joined. I say that the (straight-line) AB has been drawn from point A touching circle BCD .

For since E is the center of circles BCD and AFG , EA is thus equal to EF , and ED to EB . So the two (straight-lines) AE , EB are equal to the two (straight-lines) FE , ED (respectively). And they contain a common angle at E . Thus, the base DF is equal to the base AB , and triangle DEF is equal to triangle EBA , and the remaining angles (are equal) to the (corresponding) remaining angles [Prop. 1.4]. Thus, (angle) EDF (is) equal to EBA . And EDF (is) a right-angle. Thus, EBA (is) also a right-angle. And EB is a radius. And a (straight-line) drawn at right-angles to the diameter of a circle, from its extremity, touches the circle [Prop. 3.16 corr.]. Thus, AB touches circle BCD .

Thus, the straight-line AB has been drawn touching the given circle BCD from the given point A . (Which is) the very thing it was required to do.

Ἐάν κύκλον ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τοῦ κέντρου ἐπὶ τὴν ἀφὴν ἐπιξενθῇ τις εὐθεΐα, ἡ ἐπιξενθθεῖσα κάθετος ἔσται ἐπὶ τὴν ἐφαπτομένην.

Κύκλον γάρ τοῦ $ABΓ$ ἐφαπτέσθω τις εὐθεΐα ἡ $ΔΕ$ κατὰ τὸ $Γ$ σημεῖον, καὶ εἰλήφθω τὸ κέντρον τοῦ $ABΓ$ κύκλου τὸ Z , καὶ ἀπὸ τοῦ Z ἐπὶ τὸ $Γ$ ἐπεξενθῇ ἡ $ZΓ$. λέγω, ὅτι ἡ $ZΓ$ κάθετός ἐστιν ἐπὶ τὴν $ΔΕ$.

Εἰ γὰρ μή, ἦθθω ἀπὸ τοῦ Z ἐπὶ τὴν $ΔΕ$ κάθετος ἡ $ZΗ$.

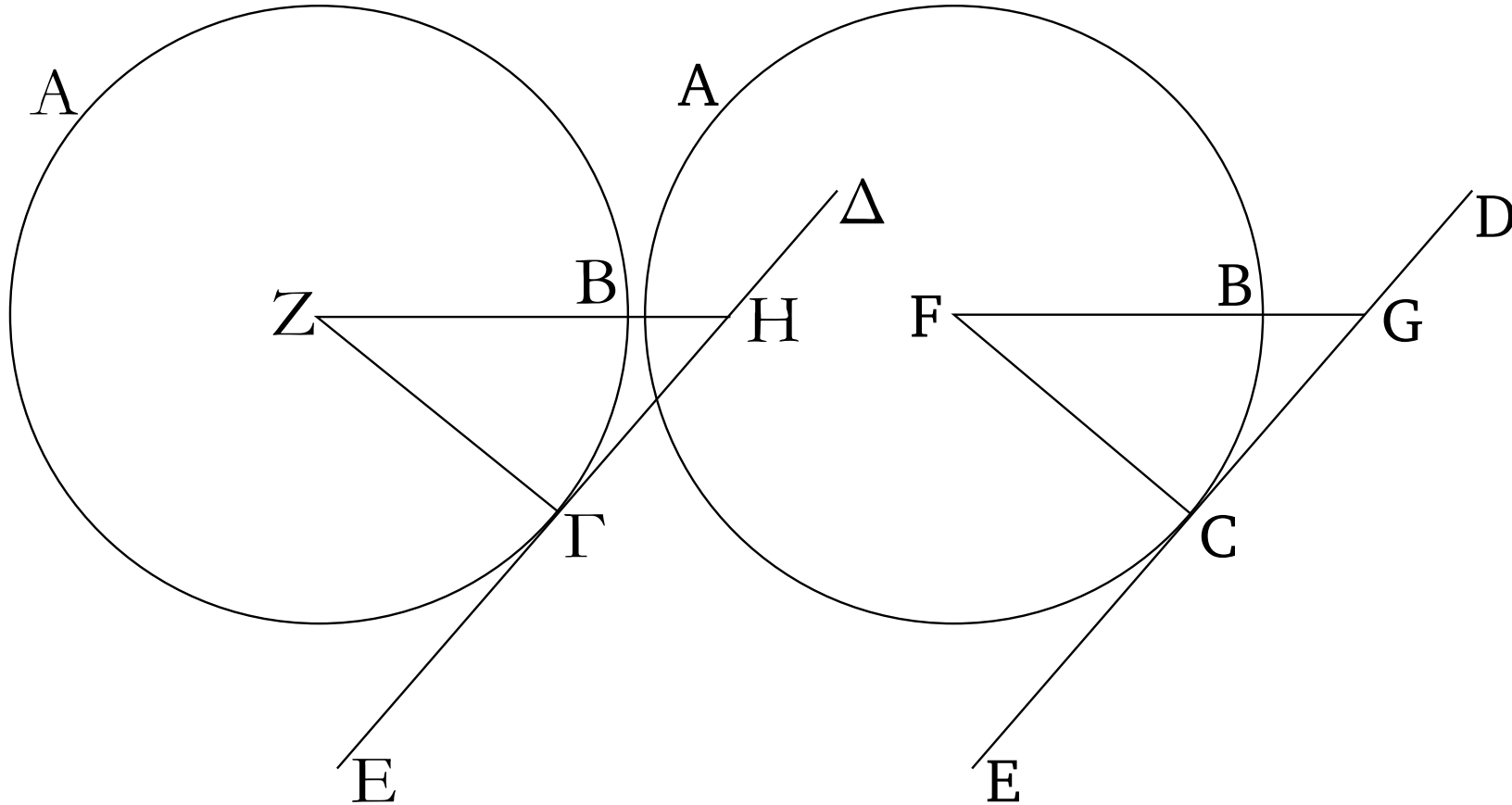
Ἐπεὶ οὖν ἡ ὑπὸ $ZΗΓ$ γωνία ὀρθή ἐστιν, ὁχεῖα ἄρα ἐστὶν ἡ ὑπὸ $ZΓΗ$. ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρά ὑποτείνει· μείζων ἄρα ἡ $ZΓ$ τῆς $ZΗ$. ἴση δὲ ἡ $ZΓ$ τῇ $ZΒ$. μείζων ἄρα καὶ ἡ $ZΒ$ τῆς $ZΗ$ ἢ ἐλάττων τῆς μείζονος· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ $ZΗ$ κάθετός ἐστιν ἐπὶ τὴν $ΔΕ$. ὁμοίως δὴ δεῖχμεν, ὅτι οὐδ' ἄλλη τις πλὴν τῆς $ZΓ$. ἡ $ZΓ$ ἄρα κάθετός ἐστιν ἐπὶ τὴν $ΔΕ$.

If some straight-line touches a circle, and some (other) straight-line is joined from the center (of the circle) to the point of contact, (then) the (straight-line) so joined will be perpendicular to the tangent.

For let some straight-line DE touch the circle ABC at point C , and let the center F of circle ABC be found [Prop. 3.1], and let FC be joined from F to C . I say that FC is perpendicular to DE .

For if not, let FG be drawn from F , perpendicular to DE [Prop. 1.12].

Therefore, since angle FGC is a right-angle, (angle) FCG is thus acute [Prop. 1.17]. And the greater angle is subtended by the greater side [Prop. 1.19]. Thus, FC (is) greater than FG . And FC (is) equal to FB . Thus, FB (is) also greater than FG , the lesser than the greater. The very thing is impossible. Thus, FG is not perpendicular to DE . So, similarly, we can show that neither (is) any other (straight-line) except FC . Thus, FC is perpendicular to DE .



Ἐάν ἄρα κύκλον ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τοῦ κέντρου ἐπὶ τὴν ἀφὴν ἐπιξενθῇ τις εὐθεΐα, ἡ ἐπιξενθθεῖσα κάθετος ἔσται ἐπὶ τὴν ἐφαπτομένην· ὅπερ ἔδει δεῖχαι.

Thus, if some straight-line touches a circle, and some (other) straight-line is joined from the center (of the circle) to the point of contact, (then) the (straight-line) so joined will be perpendicular to the tangent. (Which is) the very thing it

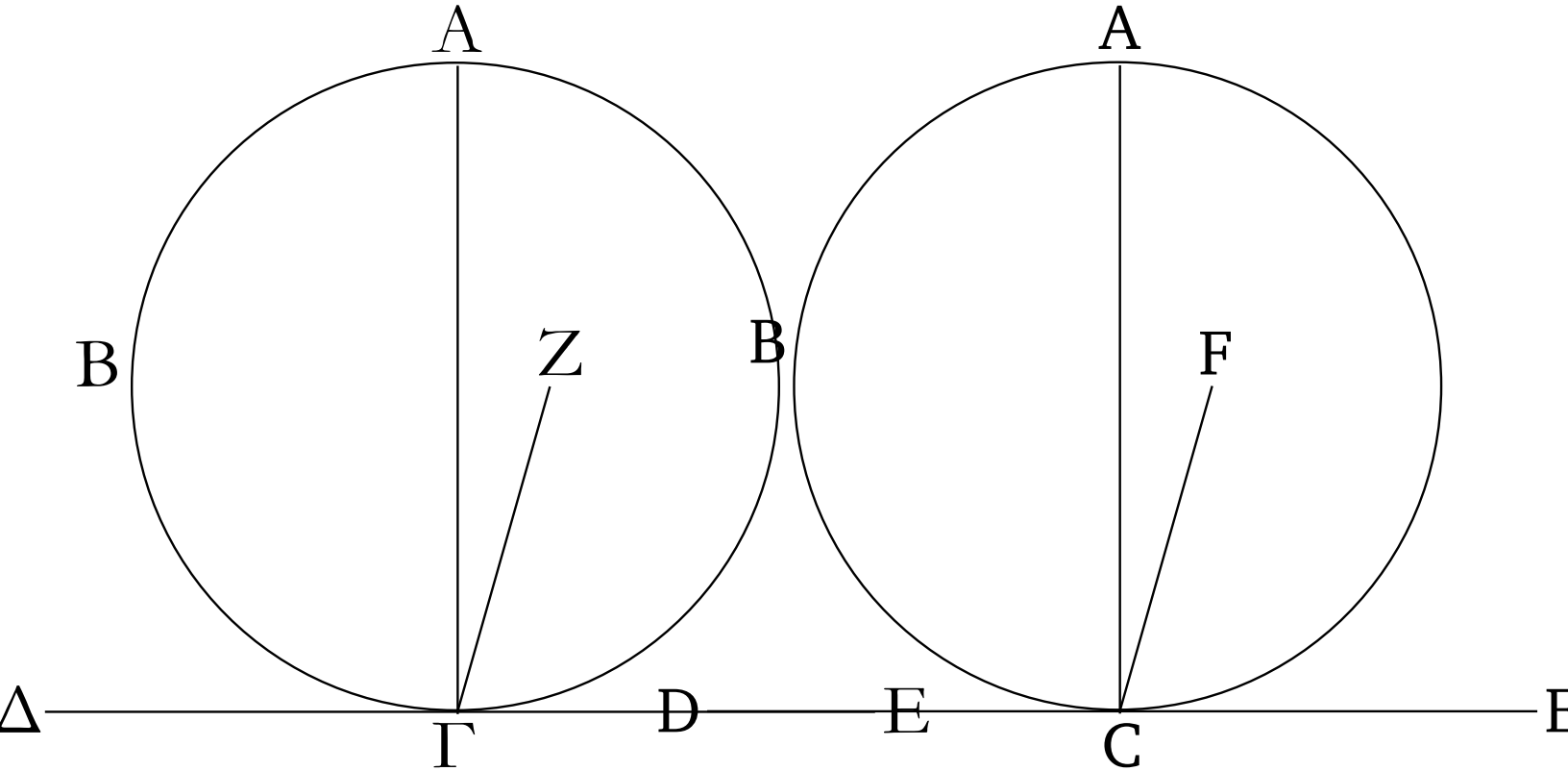
was required to show.

19.

Proposition 19

Ἐάν κύκλον ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τῆς ἀφ᾽ ἧς τῇ ἐφαπτομένην πρὸς ὀρθάξ [γωνίαξ] εὐθεΐα γραμμὴ ἀθθῇ, ἐπὶ τῆς ἀθθείσης ἔσται τὸ κέντρον τοῦ κύκλου.

If some straight-line touches a circle, and a straight-line is drawn from the point of contact, at right-[angles] to the tangent, (then) the center (of the circle) will be on the (straight-line) so drawn.



Κύκλον γάρ τοῦ ΑΒΓ ἐφαπτέσθω τις εὐθεΐα ἡ ΔΕ κατὰ τὸ Γ σημεῖον, καὶ ἀπὸ τοῦ Γ τῇ ΔΕ πρὸς ὀρθάξ ἡθθω ἡ ΓΑ· λέγω, ὅτι ἐπὶ τῆς ΑΓ ἔστι τὸ κέντρον τοῦ κύκλου.

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἔστω τὸ Ζ, καὶ ἐπέξενθω ἡ ΓΖ.

Ἐπεὶ [οὔν] κύκλον τοῦ ΑΒΓ ἐφάπτεται τις εὐθεΐα ἡ ΔΕ, ἀπὸ δὲ τοῦ κέντρον ἐπὶ τὴν ἀφῆν ἐπέξενεται ἡ ΖΓ, ἡ ΖΓ ἄρα κάθετόξ ἐστιν ἐπὶ τὴν ΔΕ· ὀρθή ἄρα ἐστὶν ἡ ὑπὸ ΖΓΕ. ἐστὶ δὲ καὶ ἡ ὑπὸ ΑΓΕ ὀρθή· ἴση ἄρα ἐστὶν ἡ ὑπὸ ΖΓΕ τῇ ὑπὸ ΑΓΕ ἡ ἐλάττων τῇ μείζονι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὸ Ζ κέντρον ἐστὶ τοῦ ΑΒΓ κύκλου. ὁμοίωξ δὲ δείχομεν, ὅτι οὐδ' ἄλλο τι πλὴν ἐπὶ τῆς ΑΓ.

Ἐάν ἄρα κύκλον ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τῆς ἀφ᾽ ἧς τῇ ἐφαπτομένην πρὸς ὀρθάξ εὐθεΐα γραμμὴ ἀθθῇ, ἐπὶ τῆς ἀθθείσης ἔσται τὸ κέντρον τοῦ κύκλου· ὅπερ ἔδει δείχαι.

For let some straight-line DE touch the circle ABC at point C . And let CA be drawn from C , at right-angles to DE [Prop. 1.11]. I say that the center of the circle is on AC .

For (if) not, if possible, let F be (the center of the circle), and let CF be joined.

[Therefore], since some straight-line DE touches the circle ABC , and FC has been joined from the center to the point of contact, FC is thus perpendicular to DE [Prop. 3.18]. Thus, FCE is a right-angle. And ACE is also a right-angle. Thus, FCE is equal to ACE , the lesser to the greater. The very thing is impossible. Thus, F is not the center of circle ABC . So, similarly, we can show that neither is any (point) other (than one) on AC .

Thus, if some straight-line touches a circle, and a straight-line is drawn from the point of contact, at right-angles to the tangent, (then) the center (of the circle) will be on the (straight-line) so drawn. (Which is) the very thing it was required to

show.

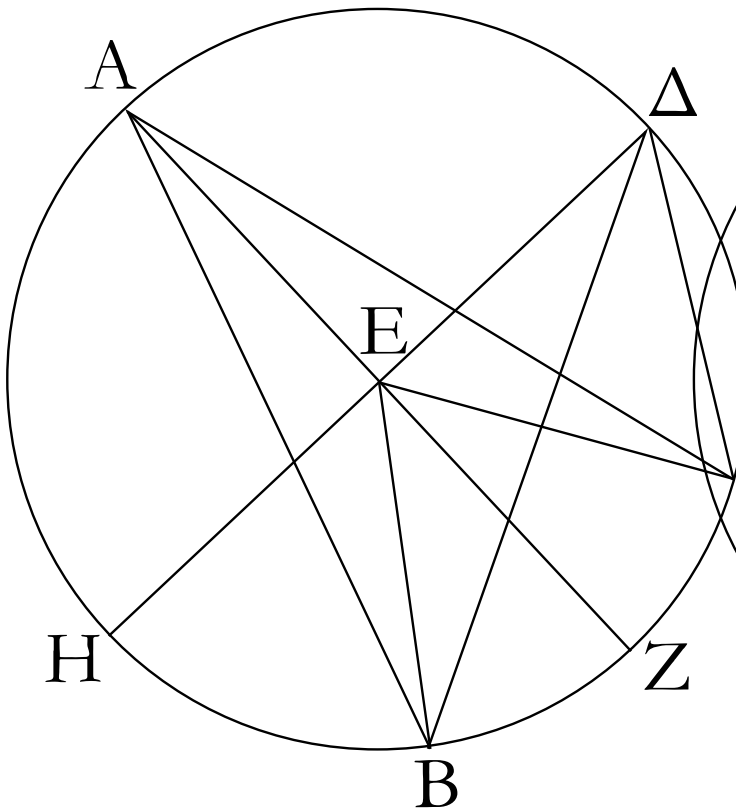
20.

Ἐν κύκλῳ ἡ πρὸς τῷ κέντρῳ γωνία διπλασίῳ ἐστὶ τῆς πρὸς τῇ περιφερείᾳ, ὅταν τὴν αὐτὴν περιφέρειαν βάσιν ἔθωσιν αἱ γωνίαι.

Ἐστω κύκλος ὁ $ABΓ$, καὶ πρὸς μὲν τῷ κέντρῳ αὐτοῦ γωνία ἔστω ἡ ὑπὸ BEG , πρὸς δὲ τῇ περιφερείᾳ ἡ ὑπὸ BAG , ἐθέτωσαν δὲ τὴν αὐτὴν περιφέρειαν βάσιν τὴν $BΓ$. λέγω, ὅτι διπλασίῳ ἐστὶν ἡ ὑπὸ BEG γωνία τῆς ὑπὸ BAG .

Ἐπιξευθθεῖσα γάρ ἡ AE διήθθη ἐπὶ τὸ Z .

Ἐπεὶ οὖν ἴση ἐστὶν ἡ EA τῇ EB , ἴση καὶ γωνία ἡ ὑπὸ EAB τῇ ὑπὸ EBA . αἱ ἄρα ὑπὸ EAB , EBA γωνίαι τῆς ὑπὸ EAB διπλασίονες εἰσιν. ἴση δὲ ἡ ὑπὸ BEZ ταῖς ὑπὸ EAB , EBA . καὶ ἡ ὑπὸ BEZ ἄρα τῆς ὑπὸ EAB ἐστὶ διπλῇ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ZEG τῆς ὑπὸ EAG ἐστὶ διπλῇ. ὅλη ἄρα ἡ ὑπὸ BEG ὅλης τῆς ὑπὸ BAG ἐστὶ διπλῇ.



Κεκλάσθω δὴ πάλιν, καὶ ἔστω ἑτέρα γωνία ἡ ὑπὸ $BΔΓ$, καὶ ἐπιξευθθεῖσα ἡ $ΔE$ ἐκβεβλήσθω ἐπὶ τὸ H . ὁμοίως δὲ δεῖξομεν, ὅτι διπλῇ ἐστὶν ἡ ὑπὸ $HEΓ$ γωνία τῆς ὑπὸ $EΔΓ$, ὣν ἡ ὑπὸ HEB διπλῇ ἐστὶ τῆς ὑπὸ $EΔB$. λοιπὴ ἄρα ἡ ὑπὸ BEG διπλῇ ἐστὶ τῆς ὑπὸ $BΔΓ$.

Ἐν κύκλῳ ἄρα ἡ πρὸς τῷ κέντρῳ γωνία διπλασίῳ ἐστὶ τῆς πρὸς τῇ περιφερείᾳ, ὅταν τὴν αὐτὴν περιφέρειαν βάσιν ἔθωσιν [αἱ γωνίαι]. ὅπερ ἔδει δεῖξαι.

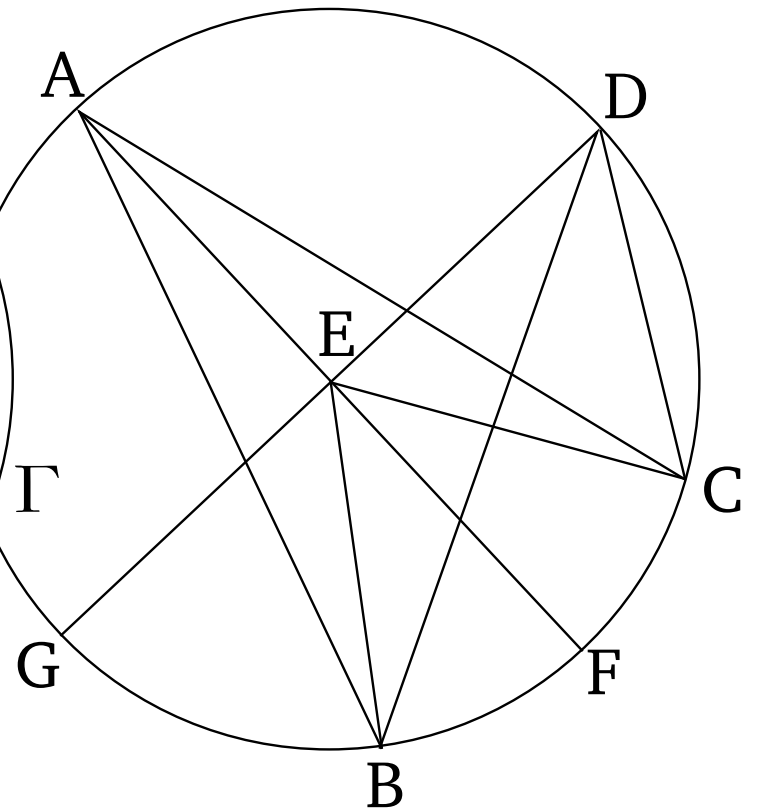
Proposition 20

In a circle, the angle at the center is double that at the circumference, when the angles have the same circumference base.

Let ABC be a circle, and let BEC be an angle at its center, and BAC (one) at (its) circumference. And let them have the same circumference base BC . I say that angle BEC is double (angle) BAC .

For being joined, let AE be drawn through to F .

Therefore, since EA is equal to EB , angle EAB (is) also equal to EBA [Prop. 1.5]. Thus, angle EAB and EBA is double (angle) EAB . And BEF (is) equal to EAB and EBA [Prop. 1.32]. Thus, BEF is also double EAB . So, for the same (reasons), FEC is also double EAC . Thus, the whole (angle) BEC is double the whole (angle) BAC .

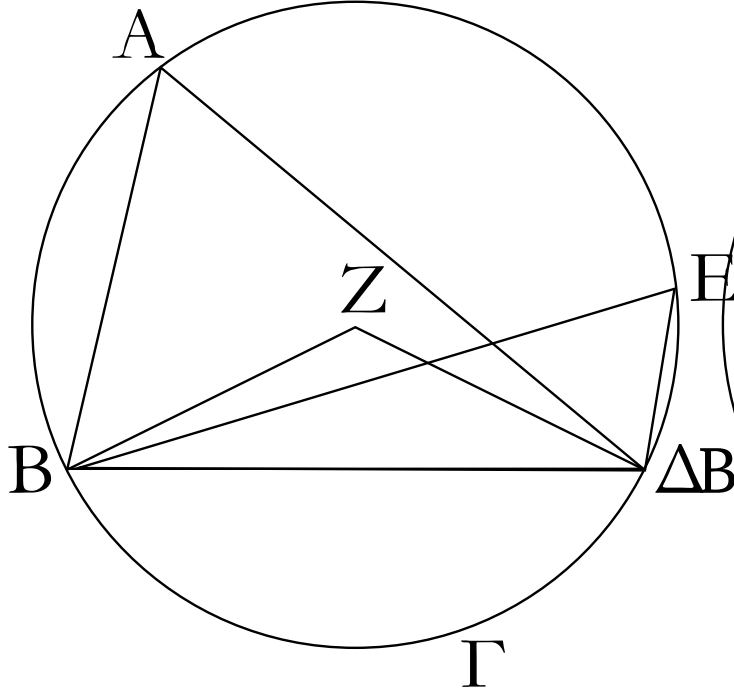


So let another (straight-line) be inflected, and let there be another angle, BDC . And DE being joined, let it be produced to G . So, similarly, we can show that angle GEC is double EDC , of which GEB is double EDB . Thus, the remaining (angle) BEC is double the (remaining angle) BDC .

Thus, in a circle, the angle at the center is double that at the circumference, when [the angles] have the same circumference base. (Which is) the very thing it was required to show.

21.

Ἐν κύκλῳ αἱ ἐν τῷ αὐτῷ τμήματι γωνίαι ἴσαι ἀλλήλαις εἰσίν.



Ἐστω κύκλος ὁ ΑΒΓΔ, καὶ ἐν τῷ αὐτῷ τμήματι τῷ ΒΑ-ΕΔ γωνίαι ἕστωσαν αἱ ὑπὸ ΒΑΔ, ΒΕΔ· λέγω, ὅτι αἱ ὑπὸ ΒΑΔ, ΒΕΔ γωνίαι ἴσαι ἀλλήλαις εἰσίν.

Εἰλήφθω γάρ τοῦ ΑΒΓΔ κύκλου τὸ κέντρον, καὶ ἕστω τὸ Ζ, καὶ ἐπεζεύθωσαν αἱ ΒΖ, ΖΔ.

Καὶ ἐπεὶ ἡ μὲν ὑπὸ ΒΖΔ γωνία πρὸς τῷ κέντρῳ ἐστίν, ἡ δὲ ὑπὸ ΒΑΔ πρὸς τῇ περιφερείᾳ, καὶ ἔθουσι τὴν αὐτὴν περιφέρειαν βάσιν τὴν ΒΓΔ, ἡ ἄρα ὑπὸ ΒΖΔ γωνία διπλασίον ἐστὶ τῆς ὑπὸ ΒΑΔ. διὰ τὰ αὐτὰ δὲ ἡ ὑπὸ ΒΖΔ καὶ τῆς ὑπὸ ΒΕΔ ἐστὶ διπλασίον· ἴση ἄρα ἡ ὑπὸ ΒΑΔ τῇ ὑπὸ ΒΕΔ.

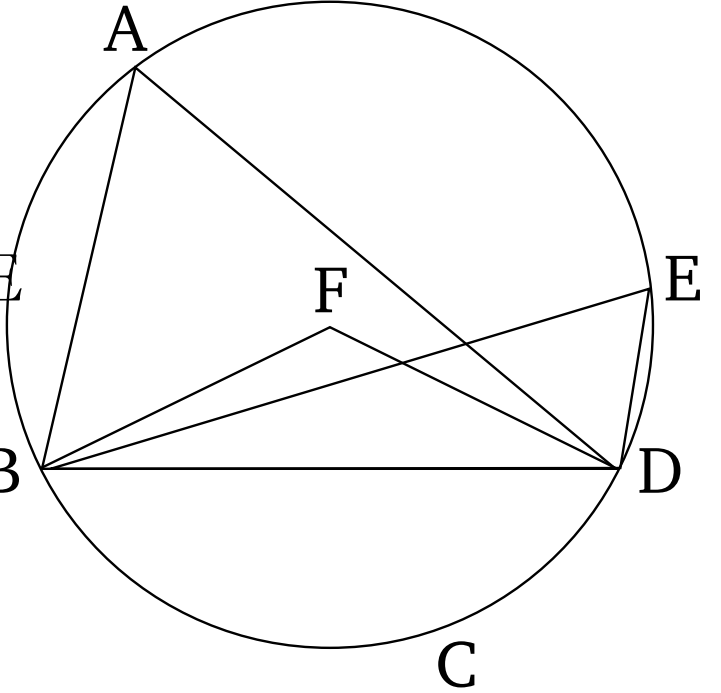
Ἐν κύκλῳ ἄρα αἱ ἐν τῷ αὐτῷ τμήματι γωνίαι ἴσαι ἀλλήλαις εἰσίν· ὅπερ ἔδει δεῖξαι.

22.

Τῶν ἐν τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυοὶν ὀρθαῖς ἴσαι εἰσίν.

Proposition 21

In a circle, angles in the same segment are equal to one another.



Let $ABCD$ be a circle, and let BAD and BED be angles in the same segment $BAED$. I say that angles BAD and BED are equal to one another.

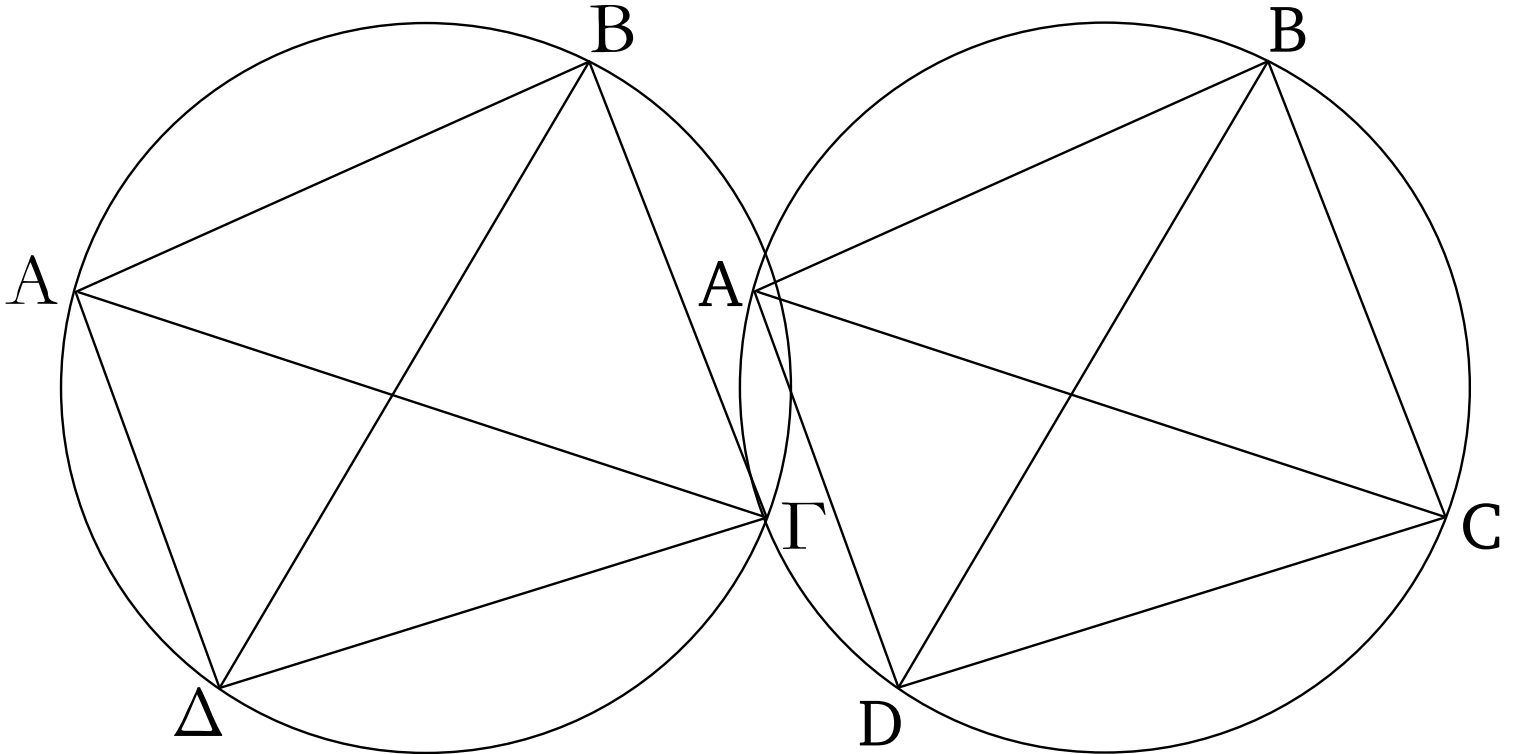
For let the center of circle $ABCD$ be found [Prop. 3.1], and let it be (at point) F . And let BF and FD be joined.

And since angle BFD is at the center, and BAD at the circumference, and they have the same circumference base BCD , angle BFD is thus double BAD [Prop. 3.20]. So, for the same (reasons), BFD is also double BED . Thus, BAD (is) equal to BED .

Thus, in a circle, angles in the same segment are equal to one another. (Which is) the very thing it was required to show.

Proposition 22

For quadrilaterals within circles, the (sum of the) opposite angles is equal to two right-angles.



Ἐστω κύκλος ὁ $AB\Gamma\Delta$, καὶ ἐν αὐτῷ τετραπλευρον ἔστω τὸ $AB\Gamma\Delta$. λέγω, ὅτι αἱ ἀπεναντίον γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσίν.

Ἐπεξεύϑθωσαν αἱ $A\Gamma$, $B\Delta$.

Ἐπεὶ οὖν παντὸς τριγώνου αἱ τρεῖς γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσίν, τοῦ $AB\Gamma$ ἄρα τριγώνου αἱ τρεῖς γωνίαι αἱ ὑπὸ ΓAB , $AB\Gamma$, $B\Gamma A$ δυοῖν ὀρθαῖς ἴσαι εἰσίν. ἴση δὲ ἡ μὲν ὑπὸ ΓAB τῇ ὑπὸ $B\Delta\Gamma$. ἐν γὰρ τῷ αὐτῷ τμήματι εἰσι τῷ $BA\Delta\Gamma$. ἡ δὲ ὑπὸ $A\Gamma B$ τῇ ὑπὸ $A\Delta B$. ἐν γὰρ τῷ αὐτῷ τμήματι εἰσι τῷ $A\Delta\Gamma B$. ὅλη ἄρα ἡ ὑπὸ $A\Delta\Gamma$ ταῖς ὑπὸ $BA\Gamma$, $A\Gamma B$ ἴση ἐστίν. κοινὴ προσκείσθω ἡ ὑπὸ $AB\Gamma$. αἱ ἄρα ὑπὸ $AB\Gamma$, $BA\Gamma$, $A\Gamma B$ ταῖς ὑπὸ $AB\Gamma$, $A\Delta\Gamma$ ἴσαι εἰσίν. ἀλλ' αἱ ὑπὸ $AB\Gamma$, $BA\Gamma$, $A\Gamma B$ δυοῖν ὀρθαῖς ἴσαι εἰσίν. καὶ αἱ ὑπὸ $AB\Gamma$, $A\Delta\Gamma$ ἄρα δυοῖν ὀρθαῖς ἴσαι εἰσίν. ὁμοίως δὲ δείχομεν, ὅτι καὶ αἱ ὑπὸ $BA\Delta$, $\Delta\Gamma B$ γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσίν.

Τῶν ἄρα ἐν τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσίν· ὅπερ ἔδει δεῖξαι.

23.

Ἐπὶ τῆς αὐτῆς εὐθείας δύο τμήματα κύκλων ὅμοια καὶ ἅμισα οὐ συνσταθήσεται ἐπὶ τὰ αὐτὰ μέρος.

Εἰ γὰρ δυνατόν, ἐπὶ τῆς αὐτῆς εὐθείας τῆς AB δύο τμήματα κύκλων ὅμοια καὶ ἅμισα συνεστιάτω ἐπὶ τὰ αὐτὰ μέρος τὰ $A\Gamma B$, $A\Delta B$, καὶ διήϑθω ἡ $A\Gamma\Delta$, καὶ ἐπεξεύϑθωσαν

Let $ABCD$ be a circle, and let $ABCD$ be a quadrilateral within it. I say that the (sum of the) opposite angles is equal to two right-angles.

Let AC and BD be joined.

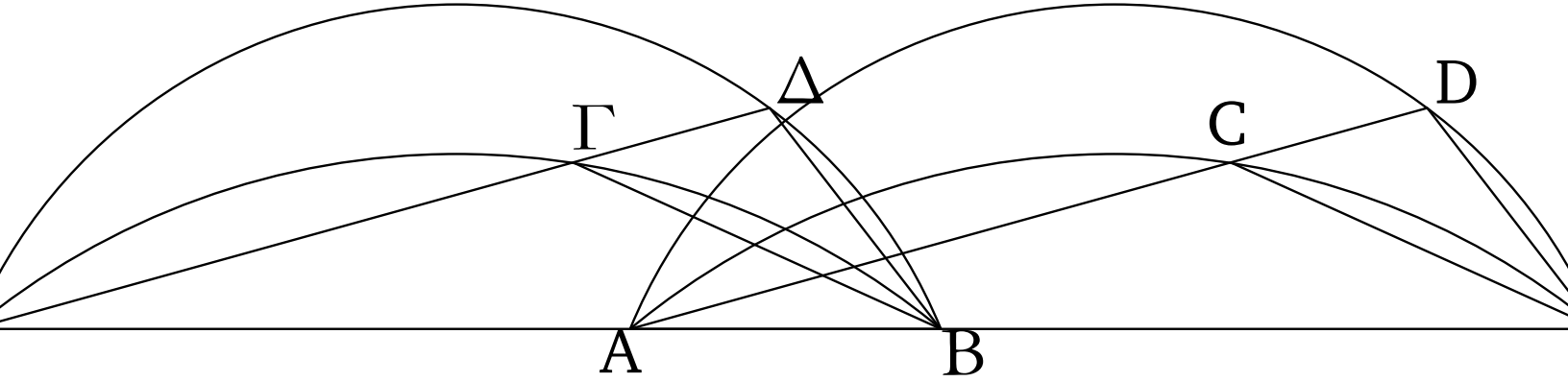
Therefore, since the three angles of any triangle are equal to two right-angles [Prop. 1.32], the three angles CAB , ABC , and BCA of triangle ABC are thus equal to two right-angles. And CAB (is) equal to BDC . For they are in the same segment $BADC$ [Prop. 3.21]. And ACB (is equal) to ADB . For they are in the same segment $ADCB$ [Prop. 3.21]. Thus, the whole of ADC is equal to BAC and ACB . Let ABC be added to both. Thus, ABC , BAC , and ACB are equal to ABC and ADC . But, ABC , BAC , and ACB are equal to two right-angles. Thus, ABC and ADC are also equal to two right-angles. Similarly, we can show that angles BAD and DCB are also equal to two right-angles.

Thus, for quadrilaterals within circles, the (sum of the) opposite angles is equal to two right-angles. (Which is) the very thing it was required to show.

Proposition 23

Two similar and unequal segments of circles cannot be constructed on the same side of the same straight-line.

For, if possible, let the two similar and unequal segments of circles, ACB and ADB , be constructed on the same side of the same straight-line AB . And let ACD be drawn through (the

αἱ GB , ΔB .segments), and let CB and DB be joined.

Ἐπεὶ οὖν ὁμοίον ἐστὶ τὸ AGB τμήμα τῷ $A\Delta B$ τμήματι, ὅμοια δὲ τμήματα κύκλων ἐστὶ τὰ δευόμενα γωνίαξ ἴσαξ, ἴση ἄρα ἐστὶν ἡ ὑπὸ AGB γωνία τῇ ὑπὸ $A\Delta B$ ἢ ἐκτός τῇ ἐντός· ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἐπὶ τῆς αὐτῆς εὐθείας δύο τμήματα κύκλων ὅμοια καὶ ἄνισα συσταθήσεται ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἔδει δεῖχαι.

Therefore, since segment ACB is similar to segment ADB , and similar segments of circles are those accepting equal angles [Def. 3.11], angle ACB is thus equal to ADB , the external to the internal. The very thing is impossible [Prop. 1.16].

Thus, two similar and unequal segments of circles cannot be constructed on the same side of the same straight-line.

24.

Proposition 24

Τὰ ἐπὶ ἴσων εὐθειῶν ὅμοια τμήματα κύκλων ἴσα ἀλλήλοις ἐστίν.

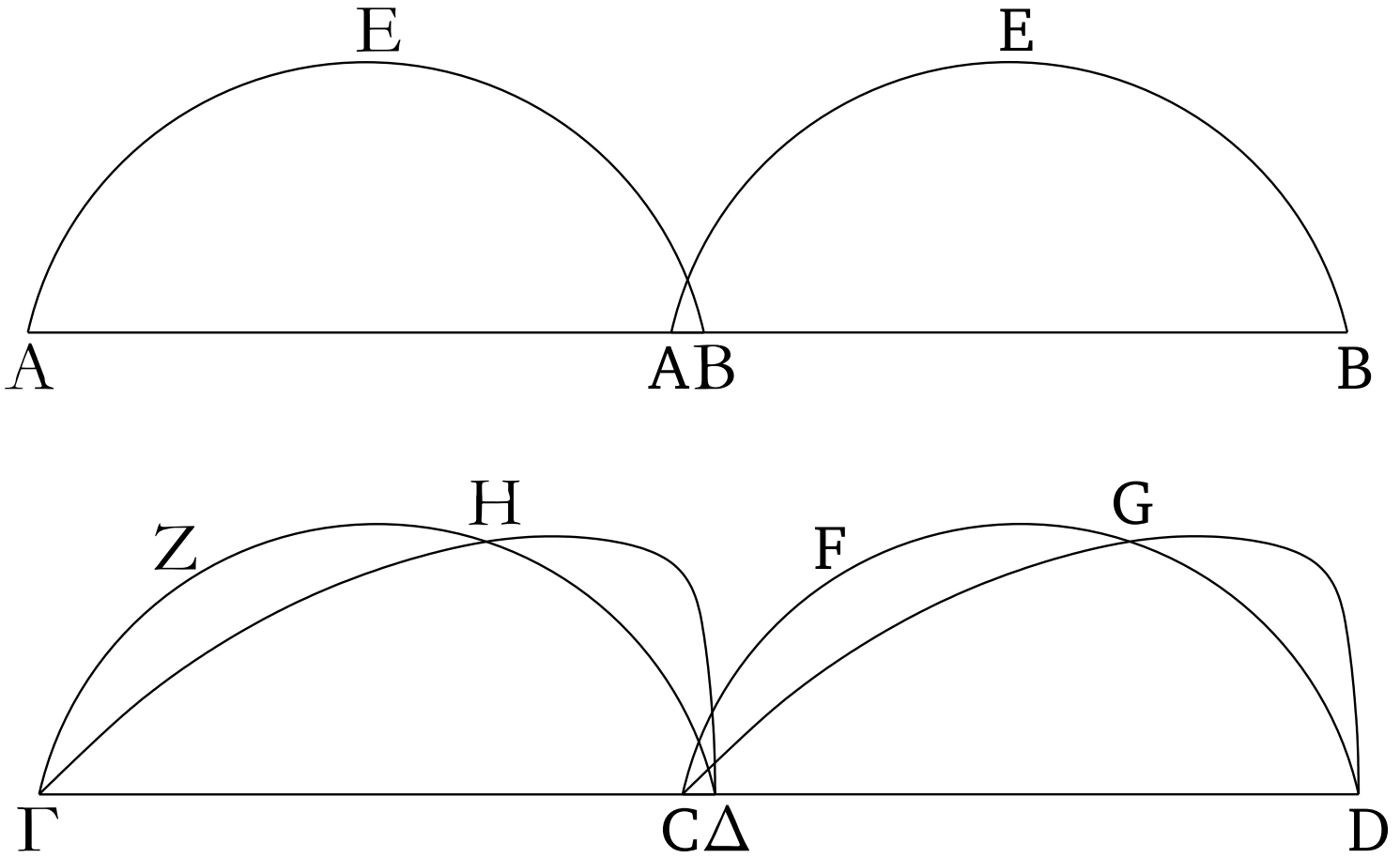
Ἐστωσαν γὰρ ἐπὶ ἴσων εὐθειῶν τῶν AB , $ΓΔ$ ὅμοια τμήματα κύκλων τὰ AEB , $ΓΖΔ$. λέγω, ὅτι ἴσον ἐστὶ τὸ AEB τμήμα τῷ $ΓΖΔ$ τμήματι.

Ἐφαρμοζόμενον γὰρ τοῦ AEB τμήματος ἐπὶ τὸ $ΓΖΔ$ καὶ τιθεμένον τοῦ μὲν A σημείου ἐπὶ τὸ $Γ$ τῆς δὲ AB εὐθείας ἐπὶ τὴν $ΓΔ$, ἐφαρμόσει καὶ τὸ B σημεῖον ἐπὶ τὸ $Δ$ σημεῖον διὰ τὸ ἴσων εἶναι τὴν AB τῇ $ΓΔ$. τῆς δὲ AB ἐπὶ τὴν $ΓΔ$ ἐφαρμοσάσης ἐφαρμόσει καὶ τὸ AEB τμήμα ἐπὶ τὸ $ΓΖΔ$. εἰ γὰρ ἡ AB εὐθεῖα ἐπὶ τὴν $ΓΔ$ ἐφαρμόσει, τὸ δὲ AEB τμήμα ἐπὶ τὸ $ΓΖΔ$ μὴ ἐφαρμόσει, ἥτοι ἐντός αὐτοῦ πεσεῖται ἢ ἐκτός ἢ παραλλάξει, ὥς τὸ $ΓΗΔ$, καὶ κύκλος κύκλον τέμνει κατὰ πλείονα σημεῖα ἢ δύο· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐφαρμολομένης τῆς AB εὐθείας ἐπὶ τὴν $ΓΔ$ οὐκ ἐφαρμόσει καὶ τὸ AEB τμήμα ἐπὶ τὸ $ΓΖΔ$. ἐφαρμόσει ἄρα, καὶ ἴσον αὐτῷ ἔσται.

Similar segments of circles on equal straight-lines are equal to one another.

For let AEB and CFD be similar segments of circles on the equal straight-lines AB and CD (respectively). I say that segment AEB is equal to segment CFD .

For if the segment AEB is applied to the segment CFD , and point A is placed on (point) C , and the straight-line AB on CD , (then) point B will also coincide with point D , on account of AB being equal to CD . And if AB coincides with CD , (then) the segment AEB will also coincide with CFD . For if the straight-line AB coincides with CD , and the segment AEB does not coincide with CFD , (then) it will surely either fall inside it, outside (it),[†] or it will miss like CGD (in the figure), and a circle (will) cut (another) circle at more than two points. The very thing is impossible [Prop. 3.10]. Thus, if the straight-line AB is applied to CD , the segment AEB cannot not also coincide with CFD . Thus, it will coincide, and will be equal to it [C.N. 4].



Τὰ ἄρα ἐπὶ ἴσων εὐθειῶν ὅμοια τμήματα κύκλων ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖχαι.

Thus, similar segments of circles on equal straight-lines are equal to one another. (Which is) the very thing it was required to show.

† Both this possibility, and the previous one, are precluded by Prop. 3.23.

25.

Κύκλον τμήματος δοθέντος προσαναγράψαι τὸν κύκλον, ὃν ὑπὲρ ἐστὶ τμήμα.

Ἐστω τὸ δοθέν τμήμα κύκλου τὸ $AB\Gamma$. δεῖ δὴ τοῦ $AB\Gamma$ τμήματος προσαναγράψαι τὸν κύκλον, ὃν ὑπὲρ ἐστὶ τμήμα.

Τετμήσθω γάρ ἡ AG δίῃα κατὰ τὸ Δ , καὶ ῥηθῇ ἀπὸ τοῦ Δ σημείου τῇ AG πρὸς ὀρθὰς ἡ ΔB , καὶ ἐπεξεύθῃ ἡ AB . ἡ ὑπὸ $AB\Delta$ γωνία ἄρα τῆς ὑπὸ $BA\Delta$ ἥτοι μείζων ἐστὶν ἢ ἴση ἢ ἐλάττω.

Ἐστω πρότερον μείζων, καὶ συνεστάτω πρὸς τῇ BA εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ ὑπὸ $AB\Delta$ γωνίᾳ ἴση ἢ ὑπὸ BAE , καὶ διήθῃ ἡ ΔB ἐπὶ τὸ E , καὶ ἐπεξεύθῃ ἡ EF . ἐπεὶ οὖν ἴση ἐστὶν ἡ ὑπὸ ABE γωνία τῇ ὑπὸ BAE , ἴση ἄρα ἐστὶ καὶ ἡ EB εὐθεῖα τῇ EA . καὶ ἐπεὶ ἴση ἐστὶν ἡ $A\Delta$ τῇ $\Delta\Gamma$, κοινὴ δὲ ἡ ΔE , δύο δὲ αἱ $A\Delta$, ΔE δύο ταῖς $\Gamma\Delta$, ΔE ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἢ ὑπὸ $A\Delta E$

Proposition 25

For a given segment of a circle, to complete the circle, the very one of which it is a segment.

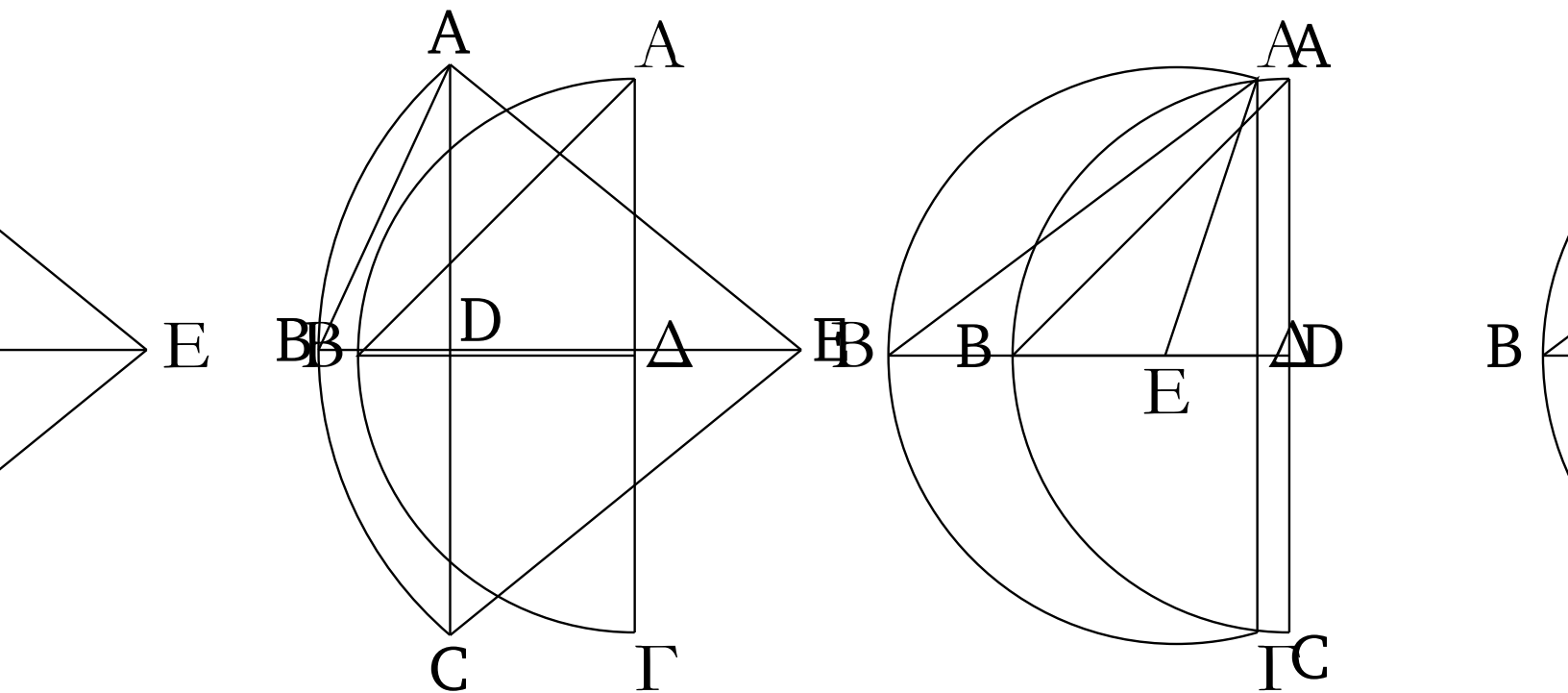
Let ABC be the given segment of a circle. So it is required to complete the circle for segment ABC , the very one of which it is a segment.

For let AC be cut in half at (point) D [Prop. 1.10], and let DB be drawn from point D , at right-angles to AC [Prop. 1.11]. And let AB be joined. Thus, angle ABD is surely either greater than, equal to, or less than (angle) BAD .

First of all, let it be greater. And let (angle) BAE , equal to angle ABD , be constructed on the straight-line BA , at the point A on it [Prop. 1.23]. And let DB be drawn through to E , and let EC be joined. Therefore, since angle ABE is equal to BAE , the straight-line EB is thus also equal to EA [Prop. 1.6]. And since AD is equal to DC , and DE (is) common, the two

γωνία τῇ ὑπὸ $\Gamma\Delta E$ ἔστιν ἴση· ὁρθή γάρ ἐκατέρα· βάσει ἄρα ἡ AE βάσει τῇ ΓE ἔστιν ἴση. ἀλλὰ ἡ AE τῇ BE ἐδείχθη ἴση· καὶ ἡ BE ἄρα τῇ ΓE ἔστιν ἴση· αἱ τρεῖς ἄρα αἱ AE , EB , ΓE ἴσαι ἀλλήλαις εἰσίν· ὁ ἄρα κέντρον τῷ E διαστήματι δὲ ἐνὶ τῶν AE , EB , ΓE κύκλος γραφόμενος ἔχει καὶ διὰ τῶν λοιπῶν σημείων καὶ ἔσται προσαναγεγραμμένος· κύκλον ἄρα τμήματος δοθέντος προσαναγράφεται ὁ κύκλος· καὶ δῆλον, ὥς τὸ $AB\Gamma$ τμήμα ἑλάττων ἔστιν ἡμικυκλίον διὰ τὸ τὸ E κέντρον ἐκτὸς αὐτοῦ τυγθάνειν.

(straight-lines) AD , DE are equal to the two (straight-lines) CD , DE , respectively. And angle ADE is equal to angle CDE . For each (is) a right-angle. Thus, the base AE is equal to the base CE [Prop. 1.4]. But, AE was shown (to be) equal to BE . Thus, BE is also equal to CE . Thus, the three (straight-lines) AE , EB , and EC are equal to one another. Thus, if a circle is drawn with center E , and radius one of AE , EB , or EC , it will also go through the remaining points (of the segment), and the (associated circle) will be completed [Prop. 3.9]. Thus, a circle has been completed from the given segment of a circle. And (it is) clear that the segment ABC is less than a semi-circle, because the center E happens to lie outside it.



Ὅμοίως [δὲ] κ' ἂν ἡ ὑπὸ $AB\Delta$ γωνία ἴση τῇ ὑπὸ $BA\Delta$, τῇς $A\Delta$ ἴσης γενομένης ἐκατέρω τῶν $B\Delta$, $\Delta\Gamma$ αἱ τρεῖς αἱ ΔA , ΔB , $\Delta\Gamma$ ἴσαι ἀλλήλαις ἔσονται, καὶ ἔσται τὸ Δ κέντρον τοῦ προσαναπεπληρωμένου κύκλου, καὶ δηλαδὴ ἔσται τὸ $AB\Gamma$ ἡμικύκλιον.

Ἐάν δὲ ἡ ὑπὸ $AB\Delta$ ἐλάττων ἢ τῇς ὑπὸ $BA\Delta$, καὶ συστησώμεθα πρὸς τῇ BA εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ ὑπὸ $AB\Delta$ γωνίᾳ ἴσην, ἐντὸς τοῦ $AB\Gamma$ τμήματος πεσεῖται τὸ κέντρον ἐπὶ τῇς ΔB , καὶ ἔσται δηλαδὴ τὸ $AB\Gamma$ τμήμα μείζον ἡμικυκλίον.

Κύκλον ἄρα τμήματος δοθέντος προσαναγράφεται ὁ κύκλος· ὅπερ ἔδει ποιῆσαι.

[And], similarly, even if angle ABD is equal to BAD , (since) AD becomes equal to each of BD [Prop. 1.6] and DC , the three (straight-lines) DA , DB , and DC will be equal to one another. And point D will be the center of the completed circle. And ABC will manifestly be a semi-circle.

And if ABD is less than BAD , and we construct (angle BAE), equal to angle ABD , on the straight-line BA , at the point A on it [Prop. 1.23], then the center will fall on DB , inside the segment ABC . And segment ABC will manifestly be greater than a semi-circle.

Thus, a circle has been completed from the given segment of a circle. (Which is) the very thing it was required to do.

ρειῶν βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὥσι βεβηκνῶται.

Ἐστωσαν ἴσοι κύκλοι οἱ $ABΓ$, $ΔΕΖ$ καὶ ἐν αὐτοῖς ἴσαι γωνίαι ἔστωσαν πρὸς μὲν τοῖς κέντροις αἱ ὑπὸ $BΗΓ$, $ΕΘΖ$, πρὸς δὲ ταῖς περιφερείαις αἱ ὑπὸ $ΒΑΓ$, $ΕΔΖ$. λέγω, ὅτι ἴση ἐστὶν ἡ $ΒΚΓ$ περιφέρεια τῇ $ΕΛΖ$ περιφερείᾳ.

Ἐπεξένθθωσαν γάρ αἱ $ΒΓ$, $ΕΖ$.

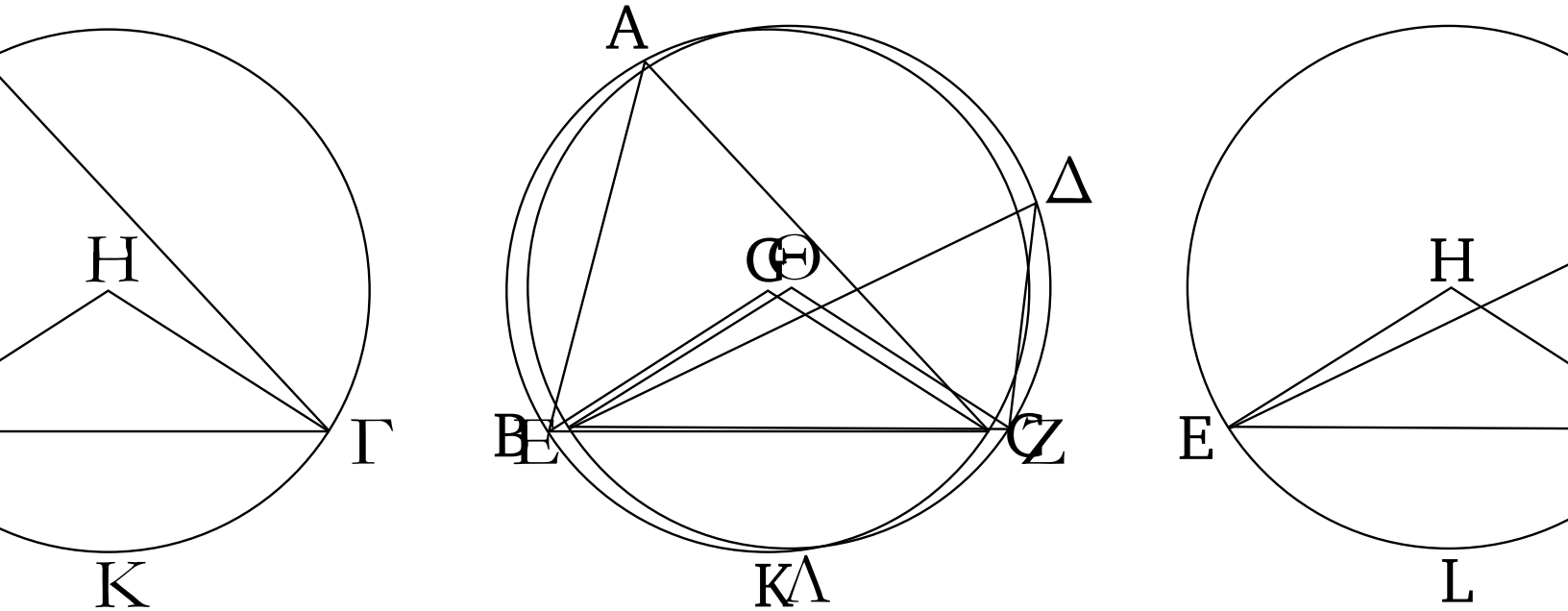
Καὶ ἐπεὶ ἴσοι εἰσὶν οἱ $ABΓ$, $ΔΕΖ$ κύκλοι, ἴσαι εἰσὶν αἱ ἐκ τῶν κέντρων δύο δὴ αἱ $BΗ$, $ΗΓ$ δύο ταῖς $ΕΘ$, $ΘΖ$ ἴσαι· καὶ γωνία ἡ πρὸς τῷ H γωνία τῇ πρὸς τῷ $Θ$ ἴση· βάσει ἄρα ἡ $ΒΓ$ βάσει τῇ $ΕΖ$ ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ πρὸς τῷ A γωνία τῇ πρὸς τῷ $Δ$, ὅμοιον ἄρα ἐστὶ τὸ $ΒΑΓ$ τμήμα τῷ $ΕΔΖ$ τμήματι· καὶ εἰσὶν ἐπὶ ἴσων εὐθειῶν [τῶν $ΒΓ$, $ΕΖ$]· τὰ δὲ ἐπὶ ἴσων εὐθειῶν ὅμοια τμήματα κύκλων ἴσα ἀλλήλοις ἐστὶν· ἴσον ἄρα τὸ $ΒΑΓ$ τμήμα τῷ $ΕΔΖ$. ἔστι δὲ καὶ ὅλος ὁ $ABΓ$ κύκλος ὅλῳ τῷ $ΔΕΖ$ κύκλῳ ἴσος· λοιπὴ ἄρα ἡ $ΒΚΓ$ περιφέρεια τῇ $ΕΛΖ$ περιφερείᾳ ἐστὶν ἴση.

ences whether they are standing at the center or at the circumference.

Let ABC and DEF be equal circles, and within them let BGC and EHF be equal angles at the center, and BAC and EDF (equal angles) at the circumference. I say that circumference BKC is equal to circumference ELF .

For let BC and EF be joined.

And since circles ABC and DEF are equal, their radii are equal. So the two (straight-lines) BG , GC (are) equal to the two (straight-lines) EH , HF (respectively). And the angle at G (is) equal to the angle at H . Thus, the base BC is equal to the base EF [Prop. 1.4]. And since the angle at A is equal to the (angle) at D , the segment BAC is thus similar to the segment EDF [Def. 3.11]. And they are on equal straight-lines [BC and EF]. And similar segments of circles on equal straight-lines are equal to one another [Prop. 3.24]. Thus, segment BAC is equal to (segment) EDF . And the whole circle ABC is also equal to the whole circle DEF . Thus, the remaining circumference BKC is equal to the (remaining) circumference ELF .



Ἐν ἄρα τοῖς ἴσοις κύκλοις αἱ ἴσαι γωνίαι ἐπὶ ἴσων περιφερειῶν βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὥσι βεβηκνῶται· ὅπερ ἔδει δεῖξαι.

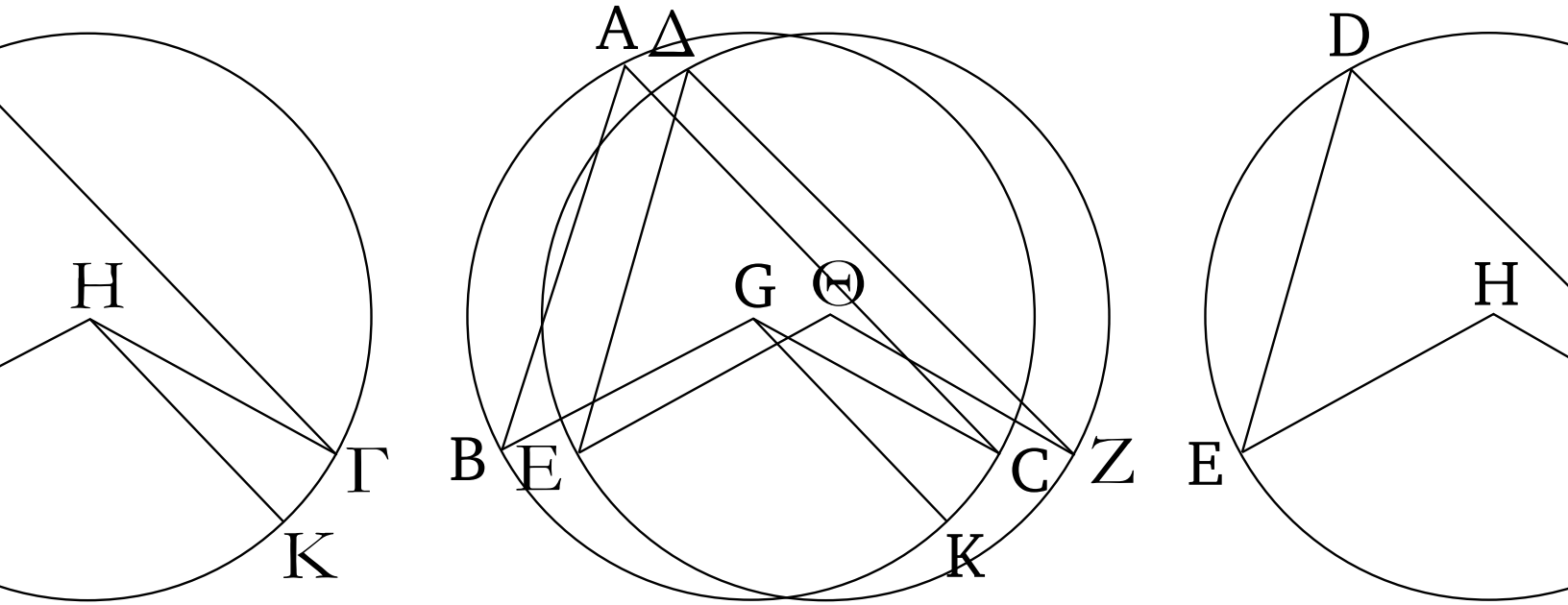
Thus, in equal circles, equal angles stand upon equal circumferences, whether they are standing at the center or at the circumference. (Which is) the very thing which it was required to show.

27.

Proposition 27

Ἐν τοῖς ἴσοις κύκλοις αἱ ἐπὶ ἴσων περιφερειῶν βεβηκνῶται γωνίαι ἴσαι ἀλλήλαις εἰσὶν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὥσι βεβηκνῶται.

In equal circles, angles standing upon equal circumferences are equal to one another, whether they are standing at the center or at the circumference.



Ἐν γάρ ἴσοις κύκλοις τοῖς $ABΓ$, $ΔΕΖ$ ἐπὶ ἴσων περιφερειῶν τῶν $ΒΓ$, $ΕΖ$ πρὸς μὲν τοῖς $Η$, $Θ$ κέντροις γωνίαι βεβηκέτωσαν αἱ ὑπὸ $ΒΗΓ$, $ΕΘΖ$, πρὸς δὲ ταῖς περιφερείαις αἱ ὑπὸ $ΒΑΓ$, $ΕΔΖ$ · λέγω, ὅτι ἡ μὲν ὑπὸ $ΒΗΓ$ γωνία τῇ ὑπὸ $ΕΘΖ$ ἐστὶν ἴση, ἡ δὲ ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΕΔΖ$ ἐστὶν ἴση.

Εἰ γὰρ ἄνισός ἐστιν ἡ ὑπὸ $ΒΗΓ$ τῇ ὑπὸ $ΕΘΖ$, μία αὐτῶν μείζων ἐστίν. ἔστω μείζων ἡ ὑπὸ $ΒΗΓ$, καὶ συνεστήτω πρὸς τῇ $ΒΗ$ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ $Η$ τῇ ὑπὸ $ΕΘΖ$ γωνίᾳ ἴση ἡ ὑπὸ $ΒΗΚ$ · αἱ δὲ ἴσαι γωνίαι ἐπὶ ἴσων περιφερειῶν βεβήκασι, ὅταν πρὸς τοῖς κέντροις ᾧσιν· ἴση ἄρα ἡ $ΒΚ$ περιφέρεια τῇ $ΕΖ$ περιφερείᾳ. ἀλλὰ ἡ $ΕΖ$ τῇ $ΒΓ$ ἐστὶν ἴση· καὶ ἡ $ΒΚ$ ἄρα τῇ $ΒΓ$ ἐστὶν ἴση ἢ ἐλάττω τῇ μείζον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἄνισός ἐστιν ἡ ὑπὸ $ΒΗΓ$ γωνία τῇ ὑπὸ $ΕΘΖ$ · ἴση ἄρα. καὶ ἐστὶ τῆς μὲν ὑπὸ $ΒΗΓ$ ἡμίσεια ἡ πρὸς τῷ $Α$, τῆς δὲ ὑπὸ $ΕΘΖ$ ἡμίσεια ἡ πρὸς τῷ $Δ$ · ἴση ἄρα καὶ ἡ πρὸς τῷ $Α$ γωνία τῇ πρὸς τῷ $Δ$.

Ἐν ἄρα τοῖς ἴσοις κύκλοις αἱ ἐπὶ ἴσων περιφερειῶν βεβηκνῦσαι γωνίαι ἴσαι ἀλλήλαις εἰσίν, ἥαν τε πρὸς τοῖς κέντροις ἥαν τε πρὸς ταῖς περιφερείαις ᾧσι βεβηκνῦναι· ὅπερ ἔδει δεῖξαι.

28.

Ἐν τοῖς ἴσοις κύκλοις αἱ ἴσαι εὐθεῖαι ἴσαξ περιφερείας ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττωνα τῇ ἐλάττω.

Ἐστῶσαν ἴσοι κύκλοι οἱ $ABΓ$, $ΔΕΖ$, καὶ ἐν τοῖς κύκλοις ἴσαι εὐθεῖαι ἔστωσαν αἱ AB , $ΔΕ$ τὰς μὲν $ΑΓΒ$, $ΑΖΕ$ περιφερείας μείζονας ἀφαιροῦσαι τὰς δὲ $ΑΗΒ$, $ΔΘΕ$ ἐλάττωνας·

For let the angles BGC and EHF at the centers G and H , and the (angles) BAC and EDF at the circumferences, stand upon the equal circumferences BC and EF , in the equal circles ABC and DEF (respectively). I say that angle BGC is equal to (angle) EHF , and BAC is equal to EDF .

For if BGC is unequal to EHF , one of them is greater. Let BGC be greater, and let the (angle) BGK , equal to angle EHF , be constructed on the straight-line BG , at the point G on it [Prop. 1.23]. But equal angles (in equal circles) stand upon equal circumferences, when they are at the centers [Prop. 3.26]. Thus, circumference BK (is) equal to circumference EF . But, EF is equal to BC . Thus, BK is also equal to BC , the lesser to the greater. The very thing is impossible. Thus, angle BGC is not unequal to EHF . Thus, (it is) equal. And the (angle) at A is half BGC , and the (angle) at D half EHF [Prop. 3.20]. Thus, the angle at A (is) also equal to the (angle) at D .

Thus, in equal circles, angles standing upon equal circumferences are equal to one another, whether they are standing at the center or at the circumference. (Which is) the very thing it was required to show.

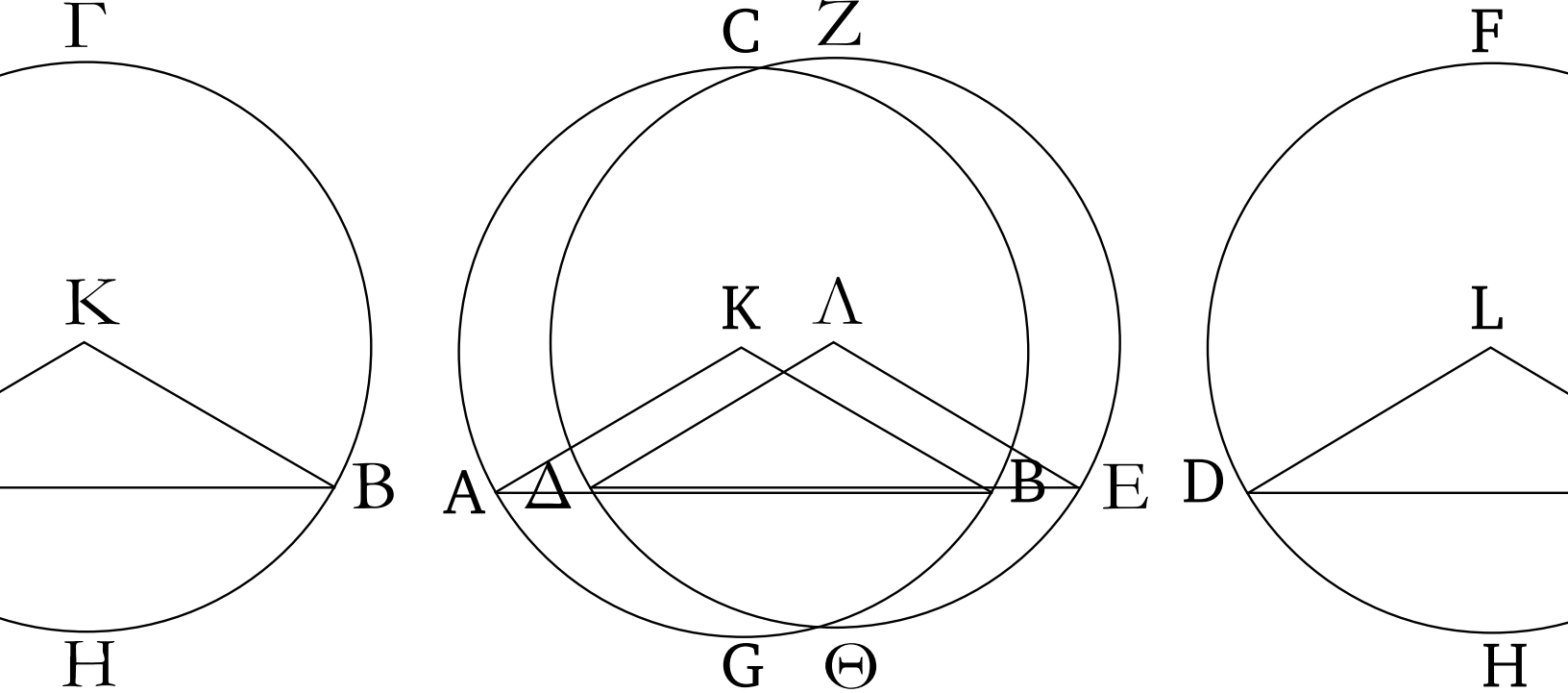
Proposition 28

In equal circles, equal straight-lines cut off equal circumferences, the greater (circumference being equal) to the greater, and the lesser to the lesser.

Let ABC and DEF be equal circles, and let AB and DE be equal straight-lines in these circles, cutting off the greater circumferences ACB and DFE , and the lesser (circumferences)

λέγω, ὅτι ἡ μὲν $ΑΓΒ$ μείζων περιφέρεια ἴση ἐστὶ τῇ $ΔΖΕ$ μείζονι περιφερείᾳ ἡ δὲ $ΑΗΒ$ ἐλάττων περιφέρεια τῇ $ΔΘΕ$.

$ΑΓΒ$ and $ΔΗΕ$ (respectively). I say that the greater circumference $ΑΒΓ$ is equal to the greater circumference $ΔΕΖ$, and the lesser circumference $ΑΒΗ$ to (the lesser) $ΔΗΕ$.



Εἰλήφθω γὰρ τὰ κέντρα τῶν κύκλων τὰ $K, Λ$, καὶ ἐπεξ-
ὑθῆσαν αἱ $AK, KB, ΔΛ, ΛΕ$.

Καὶ ἐπεὶ ἴσοι κύκλοι εἰσὶν, ἴσαι εἰσὶ καὶ αἱ ἐκ τῶν κέντρων
δύο δὴ αἱ AK, KB δυοὶ ταῖς $ΔΛ, ΛΕ$ ἴσαι εἰσὶν καὶ βάσει
ἡ AB βάσει τῇ $ΔΕ$ ἴση· γωνία ἄρα ἡ ὑπὸ AKB γωνία τῇ
ὑπὸ $ΔΛΕ$ ἴση ἐστίν. αἱ δὲ ἴσαι γωνίαι ἐπὶ ἴσων περιφερειῶν
βεβήκασιν, ὅταν πρὸς τοῖς κέντροις ᾗσιν ἴση ἄρα ἡ $ΑΗΒ$
περιφέρεια τῇ $ΔΘΕ$. ἐστὶ δὲ καὶ ὅλος ὁ $ΑΒΓ$ κύκλος ὅλος τῷ
 $ΔΕΖ$ κύκλῳ ἴσος· καὶ λοιπὴ ἄρα ἡ $ΑΓΒ$ περιφέρεια λοιπῇ
τῇ $ΔΖΕ$ περιφερείᾳ ἴση ἐστίν.

Ἐν ἄρα τοῖς ἴσοις κύκλοις αἱ ἴσαι εὐθεῖαι ἴσαξ περιφε-
ρείαξ ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττονα
τῇ ἐλάττονι· ὅπερ ἔδει δεῖχαι.

For let the centers of the circles, K and L , be found
[Prop. 3.1], and let AK, KB, DL , and LE be joined.

And since $(ABC$ and $DEF)$ are equal circles, their radii
are also equal [Def. 3.1]. So the two (straight-lines) AK, KB
are equal to the two (straight-lines) DL, LE (respectively).
And the base AB (is) equal to the base DE . Thus, angle
 AKB is equal to angle DLE [Prop. 1.8]. And equal angles
stand upon equal circumferences, when they are at the centers
[Prop. 3.26]. Thus, circumference $ΑΒΓ$ (is) equal to $ΔΗΕ$.
And the whole circle ABC is also equal to the whole circle
 DEF . Thus, the remaining circumference ACB is also equal
to the remaining circumference DFE .

Thus, in equal circles, equal straight-lines cut off equal cir-
cumferences, the greater (circumference being equal) to the
greater, and the lesser to the lesser. (Which is) the very thing
it was required to show.

29.

Proposition 29

Ἐν τοῖς ἴσοις κύκλοις τὰς ἴσαξ περιφερείας ἴσαι εὐθεῖαι
ὑποτείνουσιν.

Ἐστωσαν ἴσοι κύκλοι οἱ $ΑΒΓ, ΔΕΖ$, καὶ ἐν αὐτοῖς ἴσαι
περιφέρειαι ἀπειλήφθωσαν αἱ $ΒΗΓ, ΕΘΖ$, καὶ ἐπεξῆνθωσαν
αἱ $ΒΓ, ΕΖ$ εὐθεῖαι· λέγω, ὅτι ἴση ἐστὶν ἡ $ΒΓ$ τῇ $ΕΖ$.

Εἰλήφθω γὰρ τὰ κέντρα τῶν κύκλων, καὶ ἔστω τὰ $K, Λ$,

In equal circles, equal straight-lines subtend equal cir-
cumferences.

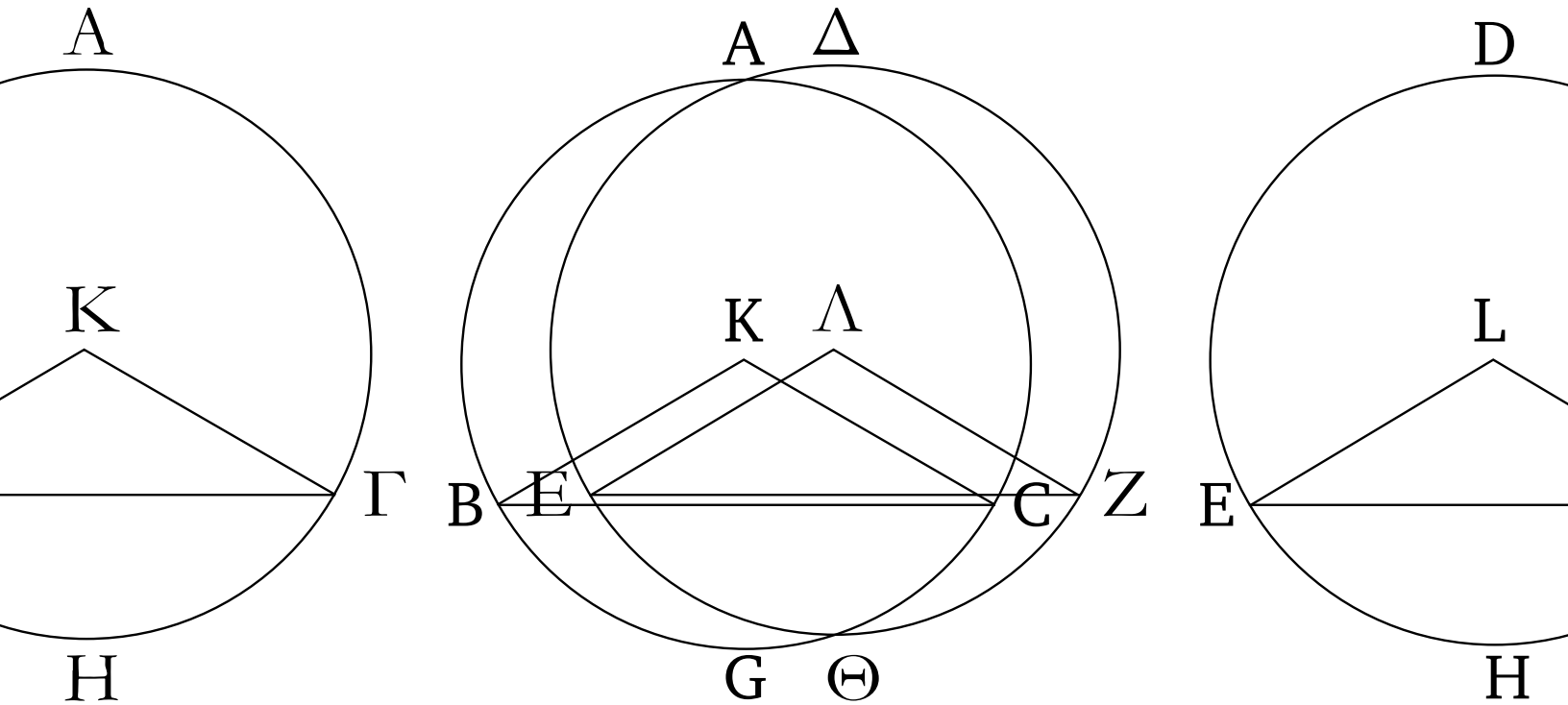
Let ABC and DEF be equal circles, and within them let
the equal circumferences BGC and EHF be cut off. And let
the straight-lines BC and EF be joined. I say that BC is equal
to EF .

καὶ ἐπεξεύχθησαν αἱ BK , $K\Gamma$, $ΕΛ$, $ΛΖ$.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ $BH\Gamma$ περιφέρεια τῇ $ΕΘΖ$ περιφερείᾳ, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $BK\Gamma$ τῇ ὑπὸ $ΕΛΖ$. καὶ ἐπεὶ ἴσοι εἰσὶν οἱ $AB\Gamma$, $ΔΕΖ$ κύκλοι, ἴσαι εἰσὶ καὶ αἱ ἐκ τῶν κέντρων δύο δὴ αἱ BK , $K\Gamma$ δυοὶ ταῖς $ΕΛ$, $ΛΖ$ ἴσαι εἰσὶν καὶ γωνίαξ ἴσαξ περιέθουσιν βάσιξ ἄρα ἡ $B\Gamma$ βάσει τῇ $ΕΖ$ ἴση ἐστίν.

For let the centers of the circles be found [Prop. 3.1], and let them be (at) K and L . And let BK , KC , EL , and LF be joined.

And since the circumference BGC is equal to the circumference EHF , the angle BKC is also equal to (angle) ELF [Prop. 3.27]. And since the circles ABC and DEF are equal, their radii are also equal [Def. 3.1]. So the two (straight-lines) BK , KC are equal to the two (straight-lines) EL , LF (respectively). And they contain equal angles. Thus, the base BC is equal to the base EF [Prop. 1.4].



Ἐν ἄρα τοῖς ἴσοις κύκλοις τὰξ ἴσαξ περιφερείας ἴσαι εὐθείαι ὑποτείνουσιν ὅπερ ἔδει δεῖχαι.

Thus, in equal circles, equal straight-lines subtend equal circumferences. (Which is) the very thing it was required to show.

30.

Proposition 30

Τὴν δοθεῖσαν περιφέρειαν διῖα τεμεῖν.

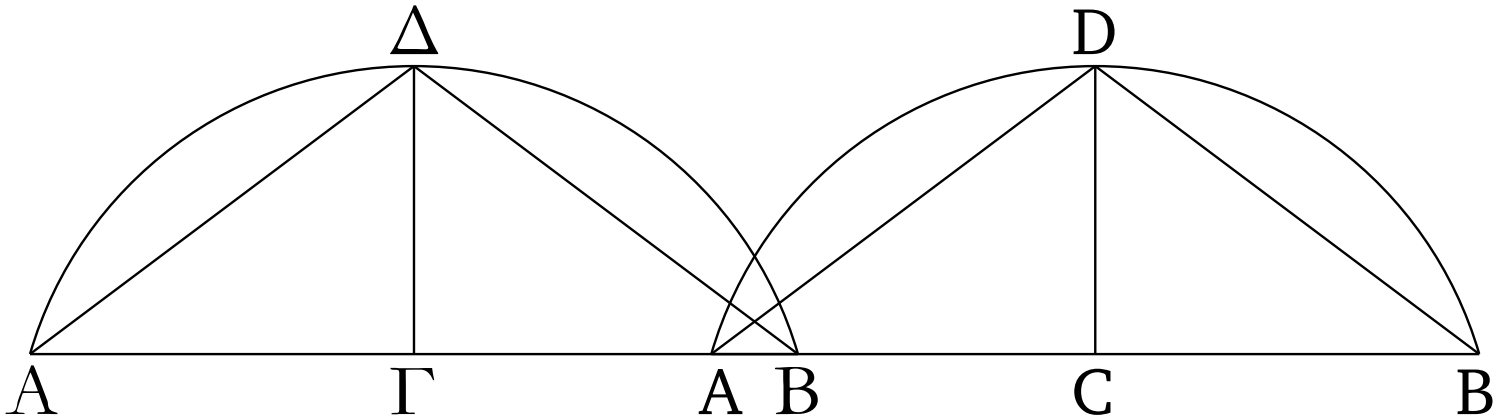
Ἐστω ἡ δοθεῖσα περιφέρεια ἡ $A\Delta B$. δεῖ δὴ τὴν $A\Delta B$ περιφέρειαν διῖα τεμεῖν.

Ἐπεξεύχθω ἡ AB , καὶ τετμήσθω διῖα κατὰ τὸ Γ , καὶ ἀπὸ τοῦ Γ σημείου τῇ AB εὐθείᾳ πρὸξ ὀρθὰξ ῥηθῶ ἡ $\Gamma\Delta$, καὶ ἐπεξεύχθησαν αἱ $A\Delta$, ΔB .

To cut a given circumference in half.

Let ADB be the given circumference. So it is required to cut circumference ADB in half.

Let AB be joined, and let it be cut in half at (point) C [Prop. 1.10]. And let CD be drawn from point C , at right-angles to AB [Prop. 1.11]. And let AD , and DB be joined.



Καὶ ἐπεὶ ἴση ἐστὶν ἡ AG τῇ GB , κοινὴ δὲ ἡ GD , δύο δὲ αἱ AG , GD δυοὶ ταῖς BG , GD ἴσαι εἰσὶν· καὶ γωνία ἡ ὑπὸ AGD γωνία τῇ ὑπὸ BGD ἴση· ὀρθή γάρ ἐκατέρα· βάσις ἄρα ἡ AD βάσει τῇ DB ἴση ἐστίν. αἱ δὲ ἴσαι εὐθεῖαι ἴσαξ περιφερεῖαξ ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττωνα τῇ ἐλάττω· καὶ ἐστὶν ἐκατέρα τῶν AD , DB περιφερειῶν ἐλάττων ἡμικυκλίον· ἴση ἄρα ἡ AD περιφέρεια τῇ DB περιφερείᾳ.

Ἡ ἄρα δοθεῖσα περιφέρεια διῖθα τέτμηται κατὰ τὸ D σημείον· ὅπερ ἔδει ποιῆσαι.

And since AC is equal to CB , and CD (is) common, the two (straight-lines) AC , CD are equal to the two (straight-lines) BC , CD (respectively). And angle ACD (is) equal to angle BCD . For (they are) each right-angles. Thus, the base AD is equal to the base DB [Prop. 1.4]. And equal straight-lines cut off equal circumferences, the greater (circumference being equal) to the greater, and the lesser to the lesser [Prop. 1.28]. And the circumferences AD and DB are each less than a semi-circle. Thus, circumference AD (is) equal to circumference DB .

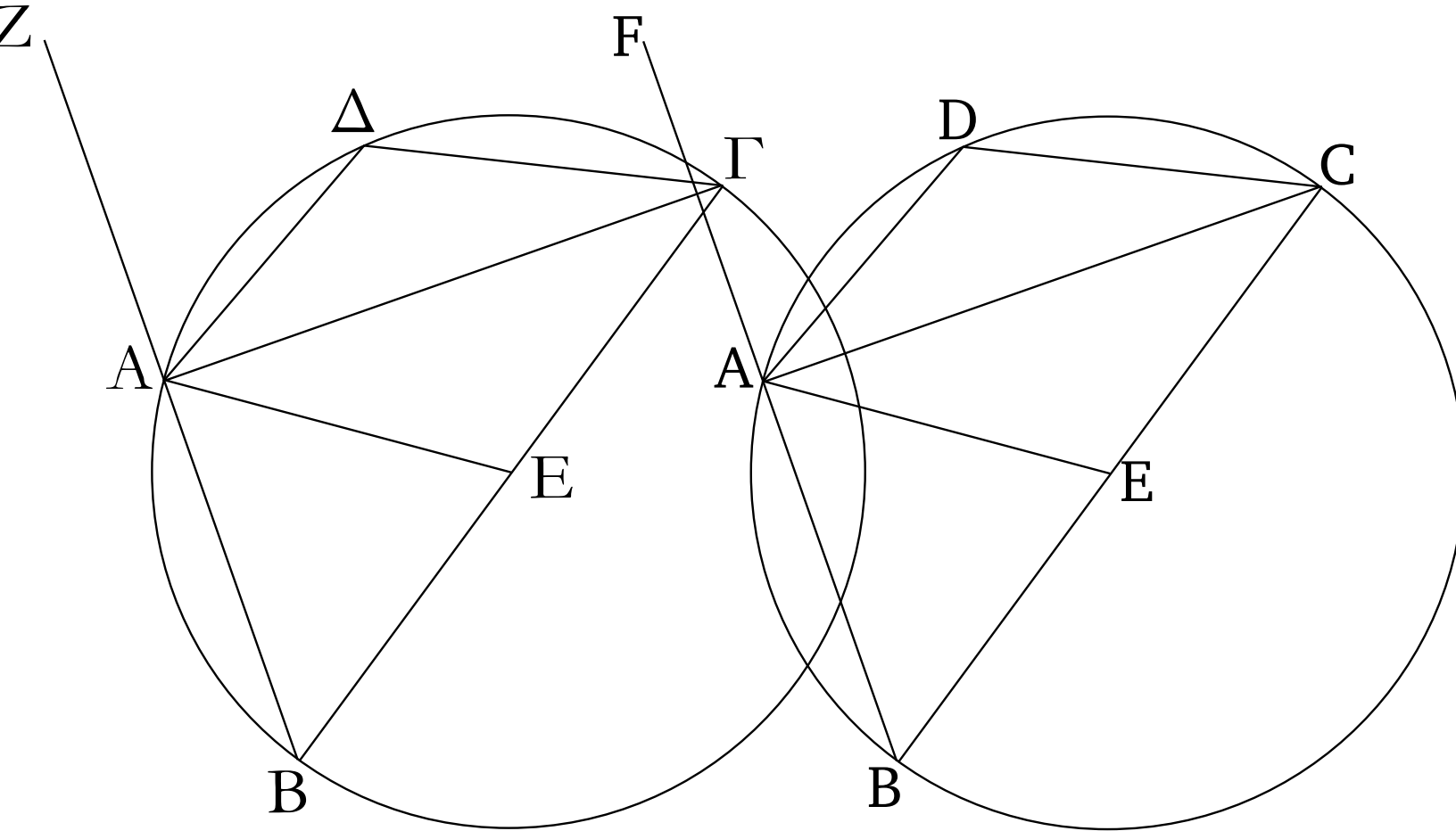
Thus, the given circumference has been cut in half at point D . (Which is) the very thing it was required to do.

31.

Ἐν κύκλῳ ἡ μὲν ἐν τῷ ἡμικυκλίῳ γωνία ὀρθή ἐστίν, ἡ δὲ ἐν τῷ μείζονι τμήματι ἐλάττων ὀρθῆς, ἡ δὲ ἐν τῷ ἐλάττωι τμήματι μείζων ὀρθῆς· καὶ ἔπι ἡ μὲν τοῦ μείζονος τμήματος γωνία μείζων ἐστὶν ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος γωνία ἐλάττων ὀρθῆς.

Proposition 31

In a circle, the angle in a semi-circle is a right-angle, and that in a greater segment (is) less than a right-angle, and that in a lesser segment (is) greater than a right-angle. And, further, the angle of a segment greater (than a semi-circle) is greater than a right-angle, and the angle of a segment less (than a semi-circle) is less than a right-angle.



Ἐστω κύκλος ὁ $ABΓΔ$, διάμετρος δὲ αὐτοῦ ἔστω ἡ $ΒΓ$, κέντρον δὲ τὸ E , καὶ ἐπεξεύθωσαν αἱ BA , $ΑΓ$, $ΑΔ$, $ΔΓ$. λέγω, ὅτι ἡ μὲν ἐν τῷ $ΒΑΓ$ ἡμικυκλίῳ γωνία ἡ ὑπὸ $ΒΑΓ$ ὀρθή ἐστιν, ἡ δὲ ἐν τῷ $ΑΒΓ$ μείζων τοῦ ἡμικυκλίου τμήματι γωνία ἡ ὑπὸ $ΑΒΓ$ ἐλάττων ἐστὶν ὀρθῆς, ἡ δὲ ἐν τῷ $ΑΔΓ$ ἐλάττων τοῦ ἡμικυκλίου τμήματι γωνία ἡ ὑπὸ $ΑΔΓ$ μείζων ἐστὶν ὀρθῆς.

Ἐπεξεύθω ἡ $ΑΕ$, καὶ διήθω ἡ $ΒΑ$ ἐπὶ τὸ Z .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΒΕ$ τῇ $ΕΑ$, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $ΑΒΕ$ τῇ ὑπὸ $ΒΑΕ$. πάλιν, ἐπεὶ ἴση ἐστὶν ἡ $ΓΕ$ τῇ $ΕΑ$, ἴση ἐστὶ καὶ ἡ ὑπὸ $ΑΓΕ$ τῇ ὑπὸ $ΓΑΕ$. ὅλη ἄρα ἡ ὑπὸ $ΒΑΓ$ δυοὶ ταῖς ὑπὸ $ΑΒΓ$, $ΑΓΒ$ ἴση ἐστίν. ἐστὶ δὲ καὶ ἡ ὑπὸ $ΖΑΓ$ ἐκτὸς τοῦ $ΑΒΓ$ τριγώνου δυοὶ ταῖς ὑπὸ $ΑΒΓ$, $ΑΓΒ$ γωνίαις ἴση· ἴση ἄρα καὶ ἡ ὑπὸ $ΒΑΓ$ γωνία τῇ ὑπὸ $ΖΑΓ$. ὀρθή ἄρα ἑκάτερα· ἡ ἄρα ἐν τῷ $ΒΑΓ$ ἡμικυκλίῳ γωνία ἡ ὑπὸ $ΒΑΓ$ ὀρθή ἐστίν.

Καὶ ἐπεὶ τοῦ $ΑΒΓ$ τριγώνου δύο γωνίαι αἱ ὑπὸ $ΑΒΓ$, $ΒΑΓ$ δύο ὀρθῶν ἐλάττονές εἰσιν, ὀρθή δὲ ἡ ὑπὸ $ΒΑΓ$, ἐλάττων ἄρα ὀρθῆς ἐστὶν ἡ ὑπὸ $ΑΒΓ$ γωνία· καὶ ἐστὶν ἐν τῷ $ΑΒΓ$ μείζων τοῦ ἡμικυκλίου τμήματι.

Καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρόν ἐστι τὸ $ΑΒΓΔ$, τῶν δὲ ἐν

Let $ABCD$ be a circle, and let BC be its diameter, and E its center. And let BA , AC , AD , and DC be joined. I say that the angle BAC in the semi-circle BAC is a right-angle, and the angle ABC in the segment ABC , (which is) greater than a semi-circle, is less than a right-angle, and the angle ADC in the segment ADC , (which is) less than a semi-circle, is greater than a right-angle.

Let AE be joined, and let BA be drawn through to F .

And since BE is equal to EA , angle ABE is also equal to BAE [Prop. 1.5]. Again, since CE is equal to EA , ACE is also equal to CAE [Prop. 1.5]. Thus, the whole (angle) BAC is equal to the two (angles) ABC and ACB . And FAC , (which is) external to triangle ABC , is also equal to the two angles ABC and ACB [Prop. 1.32]. Thus, angle BAC (is) also equal to FAC . Thus, (they are) each right-angles. [Def. 1.10]. Thus, the angle BAC in the semi-circle BAC is a right-angle.

And since the two angles ABC and BAC of triangle ABC are less than two right-angles [Prop. 1.17], and BAC is a right-angle, angle ABC is thus less than a right-angle. And it is in segment ABC , (which is) greater than a semi-circle.

And since $ABCD$ is a quadrilateral within a circle, and for

τοῖς κύκλοις τετραπλεύρων αἱ ἀπεναντίον γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν [αἱ ἄρα ὑπὸ $AB\Gamma$, $A\Delta\Gamma$ γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν], καὶ ἔστιν ἡ ὑπὸ $AB\Gamma$ ἐλάττων ὀρθῆς· λοιπὴ ἄρα ἡ ὑπὸ $A\Delta\Gamma$ γωνία μείζων ὀρθῆς ἔστιν· καὶ ἔστιν ἐν τῷ $A\Delta\Gamma$ ἐλάττονι τοῦ ἡμικυκλίου τμήματι.

Λέγω, ὅτι καὶ ἡ μὲν τοῦ μείζονος τμήματος γωνία ἡ περιεθρομένη ὑπὸ [τε] τῆς $AB\Gamma$ περιφερείας καὶ τῆς $A\Gamma$ εὐθείας μείζων ἔστιν ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος γωνία ἡ περιεθρομένη ὑπὸ [τε] τῆς $A\Delta\Gamma$ περιφερείας καὶ τῆς $A\Gamma$ εὐθείας ἐλάττων ἔστιν ὀρθῆς. καὶ ἔστιν αὐτόθεν φανερόν. ἐπεὶ γάρ ἡ ὑπὸ τῶν BA , $A\Gamma$ εὐθειῶν ὀρθή ἐστιν, ἡ ἄρα ὑπὸ τῆς $AB\Gamma$ περιφερείας καὶ τῆς $A\Gamma$ εὐθείας περιεθρομένη μείζων ἔστιν ὀρθῆς. πάλιν, ἐπεὶ ἡ ὑπὸ τῶν $A\Gamma$, AZ εὐθειῶν ὀρθή ἐστιν, ἡ ἄρα ὑπὸ τῆς ΓA εὐθείας καὶ τῆς $A\Delta\Gamma$ περιφερείας περιεθρομένη ἐλάττων ἔστιν ὀρθῆς.

Ἐν κύκλῳ ἄρα ἡ μὲν ἐν τῷ ἡμικυκλίῳ γωνία ὀρθή ἐστιν, ἡ δὲ ἐν τῷ μείζονι τμήματι ἐλάττων ὀρθῆς, ἡ δὲ ἐν τῷ ἐλάττονι [τμήματι] μείζων ὀρθῆς· καὶ ἔπι ἡ μὲν τοῦ μείζονος τμήματος [γωνία] μείζων [ἔστιν] ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος [γωνία] ἐλάττων ὀρθῆς· ὅπερ ἔδει δεῖχαι.

quadrilaterals within circles the (sum of the) opposite angles is equal to two right-angles [Prop. 3.22] [angles ABC and ADC are thus equal to two right-angles], and (angle) ABC is less than a right-angle. The remaining angle ADC is thus greater than a right-angle. And it is in segment ADC , (which is) less than a semi-circle.

I also say that the angle of the greater segment, (namely) that contained by the circumference ABC and the straight-line AC , is greater than a right-angle. And the angle of the lesser segment, (namely) that contained by the circumference $AD[C]$ and the straight-line AC , is less than a right-angle. And this is immediately apparent. For since the (angle contained by) the two straight-lines BA and AC is a right-angle, the (angle) contained by the circumference ABC and the straight-line AC is thus greater than a right-angle. Again, since the (angle contained by) the straight-lines AC and AF is a right-angle, the (angle) contained by the circumference $AD[C]$ and the straight-line CA is thus less than a right-angle.

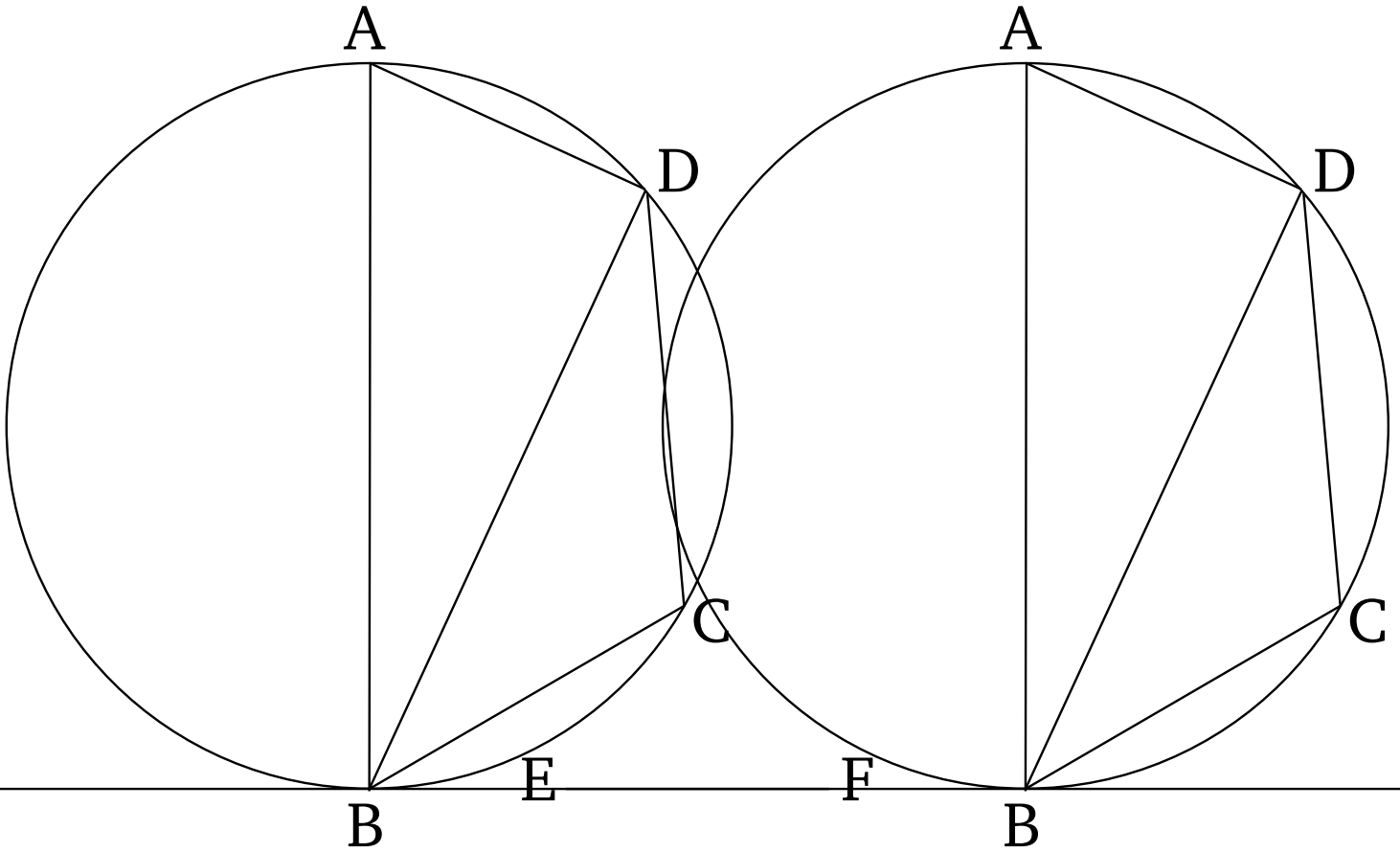
Thus, in a circle, the angle in a semi-circle is a right-angle, and that in a greater segment (is) less than a right-angle, and that in a lesser [segment] (is) greater than a right-angle. And, further, the [angle] of a segment greater (than a semi-circle) [is] greater than a right-angle, and the [angle] of a segment less (than a semi-circle) is less than a right-angle. (Which is) the very thing it was required to show.

32.

Proposition 32

Ἐὰν κύκλον ἐφάπτηται τις εὐθεῖα, ἀπὸ δὲ τῆς ἀφ᾽ ἧς εἰς τὸν κύκλον διαθῇ τις εὐθεῖα τέμνονσα τὸν κύκλον, ἃς ποιῇ γωνίας πρὸς τῇ ἐφαπτομένῃ, ἴσαι ἔσονται ταῖς ἐν τοῖς ἐναλλάχ τοῦ κύκλου τμήμασι γωνίαις.

If some straight-line touches a circle, and some (other) straight-line is drawn across, from the point of contact into the circle, cutting the circle (in two), (then) those angles the (straight-line) makes with the tangent will be equal to the angles in the alternate segments of the circle.



Κύκλον γάρ τοῦ $AB\Gamma\Delta$ ἐφαπτέσθω τις εὐθεΐα ἡ EZ κατὰ τὸ B σημεῖον, καὶ ἀπὸ τοῦ B σημείου διήθθω τις εὐθεΐα εἰς τὸν $AB\Gamma\Delta$ κύκλον τέμνουσα αὐτὸν ἡ $B\Delta$. λέγω, ὅτι ἅξ ποιεῖ γωνίαξ ἡ $B\Delta$ μετὰ τῆξ EZ ἐφαπτομένηξ, ἴσαξ ἔσσονται ταῖξ ἐν τοῖξ ἐναλλάξ τμήμασι τοῦ κύκλου γωνίαιξ, τουτέστιν, ὅτι ἡ μὲν ὑπὸ $ZB\Delta$ γωνία ἴση ἐστὶ τῇ ἐν τῷ $BA\Delta$ τμήματι συνισταμένη γωνία, ἡ δὲ ὑπὸ $EB\Delta$ γωνία ἴση ἐστὶ τῇ ἐν τῷ $\Delta\Gamma B$ τμήματι συνισταμένη γωνία.

Ἦθθω γάρ ἀπὸ τοῦ B τῇ EZ πρόξ ὀρθάξ ἡ BA , καὶ εἰλήφθω ἐπὶ τῆξ $B\Delta$ περιφερείαξ τυθὸν σημεῖον τὸ Γ , καὶ ἐπεξεύθθωσαν αἱ $A\Delta$, $\Delta\Gamma$, ΓB .

Καὶ ἐπεὶ κύκλον τοῦ $AB\Gamma\Delta$ ἐφάπτεται τις εὐθεΐα ἡ EZ κατὰ τὸ B , καὶ ἀπὸ τῆξ ἀφῆξ ἦκται τῇ ἐφαπτομένη πρόξ ὀρθάξ ἡ BA , ἐπὶ τῆξ BA ἄρα τὸ κέντρον ἐστὶ τοῦ $AB\Gamma\Delta$ κύκλου. ἡ BA ἄρα διάμετόξ ἐστὶ τοῦ $AB\Gamma\Delta$ κύκλου. ἡ ἄρα ὑπὸ $A\Delta B$ γωνία ἐν ἡμικυκλίῳ οὔσα ὀρθή ἐστιν. λοιπαὶ ἄρα αἱ ὑπὸ $BA\Delta$, $AB\Delta$ μᾶ ὀρθῇ ἴσαι εἰσίν. ἐστὶ δὲ καὶ ἡ ὑπὸ ABZ ὀρθή· ἡ ἄρα ὑπὸ ABZ ἴση ἐστὶ ταῖξ ὑπὸ $BA\Delta$, $AB\Delta$. κοινή ἀφηρήσθω ἡ ὑπὸ $AB\Delta$. λοιπή ἄρα ἡ ὑπὸ ΔBZ γωνία ἴση ἐστὶ τῇ ἐν τῷ ἐναλλάξ τμήματι τοῦ κύκλου γωνία τῇ ὑπὸ $BA\Delta$. καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρόν ἐστὶ τὸ $AB\Gamma\Delta$, αἱ ἀπεναντίον αὐτοῦ γωνίαι δυσὶν ὀρθαῖξ ἴσαι εἰσίν. εἰσὶ δὲ καὶ αἱ ὑπὸ ΔBZ , ΔBE δυσὶν ὀρθαῖξ ἴσαι· αἱ ἄρα ὑπὸ ΔBZ ,

For let some straight-line EF touch the circle $ABCD$ at the point B , and let some (other) straight-line BD be drawn from point B into the circle $ABCD$, cutting it (in two). I say that the angles BD makes with the tangent EF will be equal to the angles in the alternate segments of the circle. That is to say, that angle FBD is equal to the angle constructed in segment BAD , and angle EBD is equal to the angle constructed in segment DCB .

For let BA be drawn from B , at right-angles to EF [Prop. 1.11]. And let the point C be taken at random on the circumference BD . And let AD , DC , and CB be joined.

And since some straight-line EF touches the circle $ABCD$ at point B , and BA has been drawn from the point of contact, at right-angles to the tangent, the center of circle $ABCD$ is thus on BA [Prop. 3.19]. Thus, BA is a diameter of circle $ABCD$. Thus, angle ADB , being in a semi-circle, is a right-angle [Prop. 3.31]. Thus, the remaining angles (of triangle ADB) BAD and ABD are equal to one right-angle [Prop. 1.32]. And ABF is also a right-angle. Thus, ABF is equal to BAD and ABD . Let ABD be subtracted from both. Thus, the remaining angle DBF is equal to the angle BAD in the alternate segment of the circle. And since $ABCD$ is a quadrilateral in a circle, (the sum of) its opposite angles is equal to two right-angles

ΔBE ταῖς ὑπὸ $BA\Delta$, $B\Gamma\Delta$ ἴσαι εἰσίν, ἢ ὑπὸ $BA\Delta$ τῇ ὑπὸ ΔBZ ἐδιδυθῇ ἴση· λοιπὴ ἄρα ἡ ὑπὸ ΔBE τῇ ἐν τῷ ἐναλλάξ τοῦ κύκλου τμήματι τῷ $\Delta\Gamma B$ τῇ ὑπὸ $\Delta\Gamma B$ γωνίᾳ ἐστὶν ἴση.

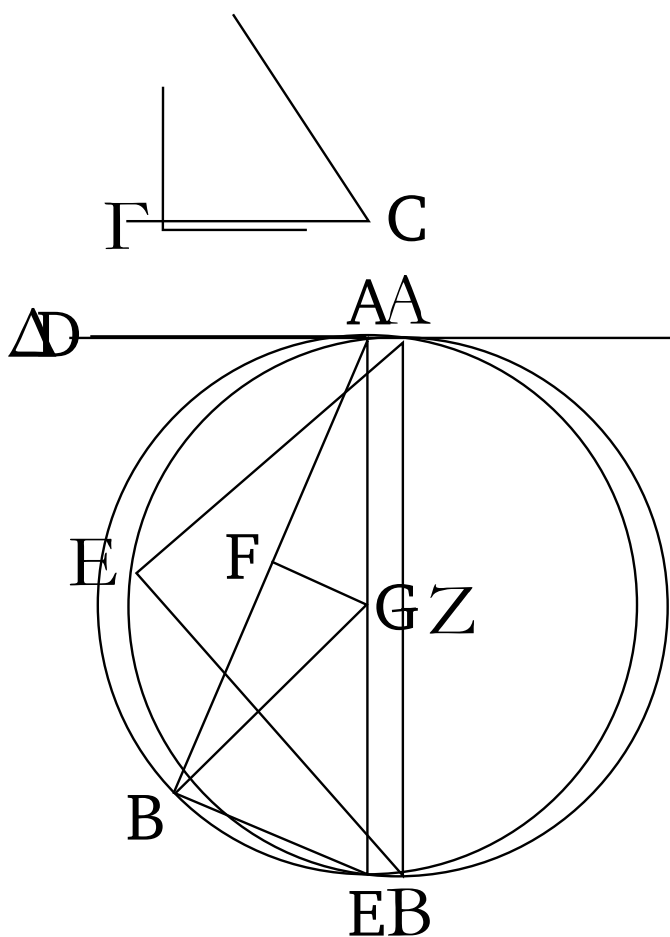
Ἐὰν ἄρα κύκλον ἐφάπτηται τις εὐθεΐα, ἀπὸ δὲ τῆς ἀφ᾽ ἑἰς τὸν κύκλον διαθεθῇ τις εὐθεΐα τέμνουσα τὸν κύκλον, ἅς ποιεῖ γωνίας πρὸς τῇ ἐφαπτομένῃ, ἴσαι ἔσονται ταῖς ἐν τοῖς ἐναλλάξ τοῦ κύκλου τμήμασι γωνίαις· ὅπερ ἔδει δεῖξαι.

[Prop. 3.22]. And DBF and DBE is also equal to two right-angles [Prop. 1.13]. Thus, DBF and DBE is equal to BAD and BCD , of which BAD was shown (to be) equal to DBF . Thus, the remaining (angle) DBE is equal to the angle DCB in the alternate segment DCB of the circle.

Thus, if some straight-line touches a circle, and some (other) straight-line is drawn across, from the point of contact into the circle, cutting the circle (in two), (then) those angles the (straight-line) makes with the tangent will be equal to the angles in the alternate segments of the circle. (Which is) the very thing it was required to show.

33.

Ἐπὶ τῆς δοθείσης εὐθείας γράψαι τμήμα κύκλον δευτέρου γωνίαν ἴσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμω.

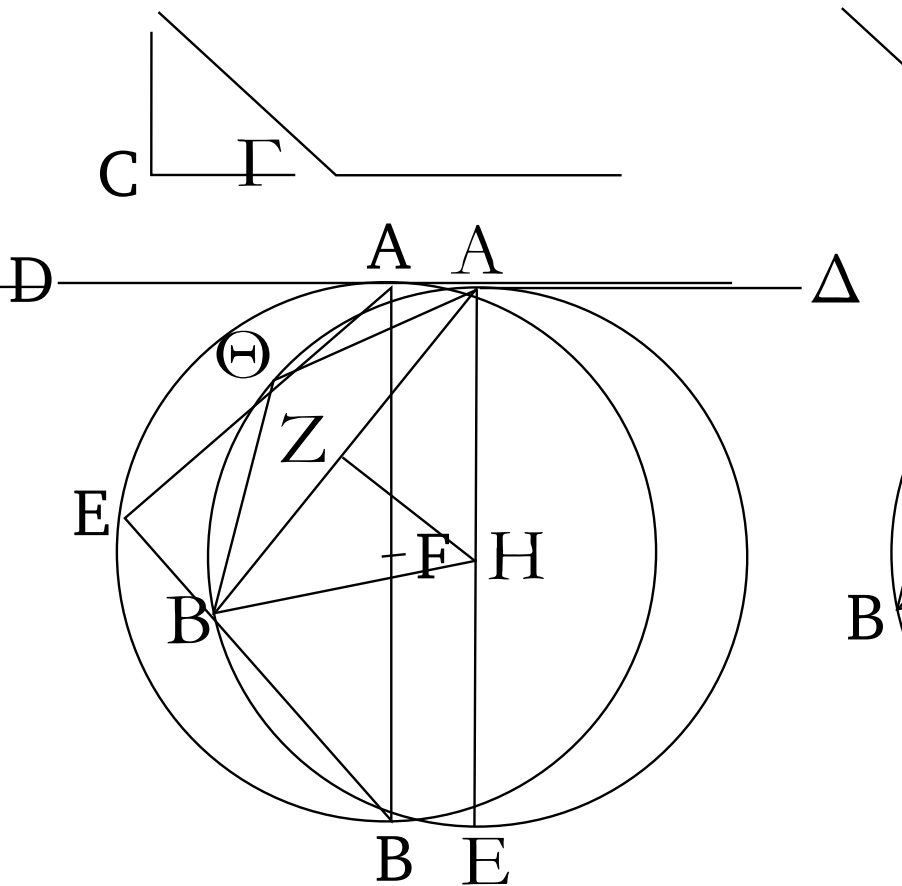


Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ AB , ἡ δὲ δοθεῖσα γωνία ἐκ-
 τὸν A ἡ Γ . δεῖ δὴ ἐπὶ τῆς δοθεῖσης εὐθείας
 τῆς AB γράψαι τμήμα κύκλου δεσθόμενον γωνίαν ἴσην τῇ
 ἐκ τὸν A ἡ Γ .

Ἡ δὴ πρὸς τῷ Γ [γωνία] ἥτοι ὀξεῖα ἐστὶν ἢ ὀρθή ἢ ἀμβλεῖα· ἔστω πρότερον ὀξεῖα, καὶ ὥς ἐπὶ τῇς πρώτης κατα-

Proposition 33

To draw a segment of a circle, accepting an angle equal to a given rectilinear angle, on a given straight-line.



Let AB be the given straight-line, and C the given rectilinear angle. So it is required to draw a segment of a circle, accepting an angle equal to C , on the given straight-line AB .

So the [angle] C is surely either acute, a right-angle, or obtuse. First of all, let it be acute. And, as in the first diagram (from the left), let (angle) BAD , equal to angle C ,

γραφήσιν συνεστάτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ A σημείῳ τῇ πρὸς τῷ Γ γωνία ἴση ἢ ὑπὸ $BA\Delta$. ὅχεια ἄρα ἐστὶ καὶ ἡ ὑπὸ $BA\Delta$. ἤθθω τῇ ΔA πρὸς ὀρθὰς ἡ AE , καὶ τετμήσθω ἡ AB δίθῃ κατὰ τὸ Z , καὶ ἤθθω ἀπὸ τοῦ Z σημείου τῇ AB πρὸς ὀρθὰς ἡ ZH , καὶ ἐπεξεύθθω ἡ HB .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ AZ τῇ ZB , κοινὴ δὲ ἡ ZH , δύο δὲ αἱ AZ , ZH δύο ταῖς BZ , ZH ἴσαι εἰσὶν καὶ γωνία ἡ ὑπὸ AZH [γωνία] τῇ ὑπὸ BZH ἴση· βάσει ἄρα ἡ AH βάσει τῇ BH ἴση ἐστίν. ὁ ἄρα κέντρον μὲν τῷ H διαστήματι δὲ τῷ HA κύκλος γραφόμενος ἔχει καὶ διὰ τοῦ B . γεγράφθω καὶ ἔστω ὁ ABE , καὶ ἐπεξεύθθω ἡ EB . ἐπεὶ οὖν ἀπ' ἀκρᾶς τῆς AE διαμέτρον ἀπὸ τοῦ A τῇ AE πρὸς ὀρθὰς ἐστὶν ἡ $A\Delta$, ἡ $A\Delta$ ἄρα ἐφάπτεται τοῦ ABE κύκλου· ἐπεὶ οὖν κύκλον τοῦ ABE ἐφάπτεται τις εὐθεῖα ἡ $A\Delta$, καὶ ἀπὸ τῆς κατὰ τὸ A ἀφῆς εἰς τὸν ABE κύκλον διηκταὶ τις εὐθεῖα ἡ AB , ἡ ἄρα ὑπὸ ΔAB γωνία ἴση ἐστὶ τῇ ἐν τῷ ἐναλλάχ τοῦ κύκλου τμήματι γωνία τῇ ὑπὸ AEB . ἀλλ' ἡ ὑπὸ ΔAB τῇ πρὸς τῷ Γ ἐστὶν ἴση· καὶ ἡ πρὸς τῷ Γ ἄρα γωνία ἴση ἐστὶ τῇ ὑπὸ AEB .

Ἐπὶ τῆς δοθείσης ἄρα εὐθείας τῆς AB τμήμα κύκλου γέγραπται τὸ ABE δεθόμενον γωνίαν τὴν ὑπὸ AEB ἴσην τῇ δοθείσῃ τῇ πρὸς τῷ Γ .

Ἀλλὰ δὴ ὀρθὴ ἔστω ἡ πρὸς τῷ Γ · καὶ δεόν πάλιν ἔστω ἐπὶ τῆς AB γράψαι τμήμα κύκλου δεθόμενον γωνίαν ἴσην τῇ πρὸς τῷ Γ ὀρθῇ [γωνία]. συνεστάτω [πάλιν] τῇ πρὸς τῷ Γ ὀρθῇ γωνία ἴση ἢ ὑπὸ $BA\Delta$, ὡς ἔθει ἐπὶ τῆς δευτέρας καταγραφῆς, καὶ τετμήσθω ἡ AB δίθῃ κατὰ τὸ Z , καὶ κέντρον τῷ Z , διαστήματι δὲ ὁποτέρῳ τῶν ZA , ZB , κύκλος γεγράφθω ὁ AEB .

Ἐφάπτεται ἄρα ἡ $A\Delta$ εὐθεῖα τοῦ ABE κύκλου διὰ τὸ ὀρθὴν εἶναι τὴν πρὸς τῷ A γωνίαν. καὶ ἴση ἐστὶν ἡ ὑπὸ $BA\Delta$ γωνία τῇ ἐν τῷ AEB τμήματι· ὀρθὴ γάρ καὶ αὐτὴ ἐν ἡμικυκλίῳ οὔσα. ἀλλὰ καὶ ἡ ὑπὸ $BA\Delta$ τῇ πρὸς τῷ Γ ἴση ἐστίν. καὶ ἡ ἐν τῷ AEB ἄρα ἴση ἐστὶ τῇ πρὸς τῷ Γ .

Γέγραπται ἄρα πάλιν ἐπὶ τῆς AB τμήμα κύκλου τὸ ABE δεθόμενον γωνίαν ἴσην τῇ πρὸς τῷ Γ .

Ἀλλὰ δὴ ἡ πρὸς τῷ Γ ἀμβλεία ἔστω· καὶ συνεστάτω αὐτῇ ἴση πρὸς τῇ AB εὐθείᾳ καὶ τῷ A σημείῳ ἡ ὑπὸ $BA\Delta$, ὡς ἔθει ἐπὶ τῆς τρίτης καταγραφῆς, καὶ τῇ $A\Delta$ πρὸς ὀρθὰς ἤθθω ἡ AE , καὶ τετμήσθω πάλιν ἡ AB δίθῃ κατὰ τὸ Z , καὶ τῇ AB πρὸς ὀρθὰς ἤθθω ἡ ZH , καὶ ἐπεξεύθθω ἡ HB .

Καὶ ἐπεὶ πάλιν ἴση ἐστὶν ἡ AZ τῇ ZB , καὶ κοινὴ ἡ ZH , δύο δὲ αἱ AZ , ZH δύο ταῖς BZ , ZH ἴσαι εἰσὶν καὶ γωνία ἡ ὑπὸ AZH γωνία τῇ ὑπὸ BZH ἴση· βάσει ἄρα ἡ AH βάσει τῇ BH ἴση ἐστίν. ὁ ἄρα κέντρον μὲν τῷ H διαστήματι δὲ τῷ HA κύκλος γραφόμενος ἔχει καὶ διὰ τοῦ B . ἐρθέσθω ὡς ὁ AEB . καὶ ἐπεὶ τῇ AE διαμέτρῳ ἀπ' ἀκρᾶς πρὸς ὀρθὰς ἐστὶν ἡ $A\Delta$, ἡ $A\Delta$ ἄρα ἐφάπτεται τοῦ ABE κύκλου. καὶ ἀπὸ τῆς κατὰ τὸ A ἐπαφῆς διηκταὶ ἡ AB · ἡ ἄρα ὑπὸ $BA\Delta$ γωνία ἴση ἐστὶ τῇ ἐν τῷ ἐναλλάχ τοῦ κύκλου τμήματι τῷ $A\Theta B$ συνισταμένη γωνία. ἀλλ' ἡ ὑπὸ $BA\Delta$ γωνία τῇ πρὸς τῷ Γ ἴση ἐστίν. καὶ ἡ ἐν τῷ $A\Theta B$ ἄρα τμήματι γωνία ἴση ἐστὶ τῇ πρὸς τῷ Γ .

be constructed on the straight-line AB , at the point A (on it) [Prop. 1.23]. Thus, BAD is also acute. Let AE be drawn, at right-angles to DA [Prop. 1.11]. And let AB be cut in half at F [Prop. 1.10]. And let FG be drawn from point F , at right-angles to AB [Prop. 1.11]. And let GB be joined.

And since AF is equal to FB , and FG (is) common, the two (straight-lines) AF , FG are equal to the two (straight-lines) BF , FG (respectively). And angle AFG (is) equal to [angle] BFG . Thus, the base AG is equal to the base BG [Prop. 1.4]. Thus, the circle drawn with center G , and radius GA , will also go through B (as well as A). Let it be drawn, and let it be (denoted) ABE . And let EB be joined. Therefore, since AD is at the extremity of diameter AE , (namely, point) A , at right-angles to AE , the (straight-line) AD thus touches the circle ABE [Prop. 3.16 corr.]. Therefore, since some straight-line AD touches the circle ABE , and some (other) straight-line AB has been drawn across from the point of contact A into circle ABE , angle DAB is thus equal to the angle AEB in the alternate segment of the circle [Prop. 3.32]. But, DAB is equal to C . Thus, angle C is also equal to AEB .

Thus, a segment AEB of a circle, accepting the angle AEB (which is) equal to the given (angle) C , has been drawn on the given straight-line AB .

And so let C be a right-angle. And let it again be necessary to draw a segment of a circle on AB , accepting an angle equal to the right-[angle] C . Let the (angle) BAD [again] be constructed, equal to the right-angle C [Prop. 1.23], as in the second diagram (from the left). And let AB be cut in half at F [Prop. 1.10]. And let the circle AEB be drawn with center F , and radius either FA or FB .

Thus, the straight-line AD touches the circle ABE , on account of the angle at A being a right-angle [Prop. 3.16 corr.]. And angle BAD is equal to the angle in segment AEB . For (the latter angle), being in a semi-circle, is also a right-angle [Prop. 3.31]. But, BAD is also equal to C . Thus, the (angle) in (segment) AEB is also equal to C .

Thus, a segment AEB of a circle, accepting an angle equal to C , has again been drawn on AB .

And so let (angle) C be obtuse. And let (angle) BAD , equal to (C), be constructed on the straight-line AB , at the point A (on it) [Prop. 1.23], as in the third diagram (from the left). And let AE be drawn, at right-angles to AD [Prop. 1.11]. And let AB again be cut in half at F [Prop. 1.10]. And let FG be drawn, at right-angles to AB [Prop. 1.10]. And let GB be joined.

And again, since AF is equal to FB , and FG (is) common, the two (straight-lines) AF , FG are equal to the two (straight-lines) BF , FG (respectively). And angle AFG (is) equal to angle BFG . Thus, the base AG is equal to the base BG [Prop. 1.4]. Thus, a circle of center G , and radius GA , being drawn, will also go through B (as well as A). Let it go like AEB (in the third diagram from the left). And since AD

Ἐπὶ τῆς ἄρα δοθείσης εὐθείας τῆς AB γέγραπται τμήμα κύκλον τὸ $A\Theta B$ δεθόμενον γωνίαν ἴσην τῇ πρὸς τῷ Γ . ὅπερ ἔδει ποιῆσαι.

is at right-angles to the diameter AE , at its extremity, AD thus touches circle AEB [Prop. 3.16 corr.]. And AB has been drawn across (the circle) from the point of contact A . Thus, angle BAD is equal to the angle constructed in the alternate segment AHB of the circle [Prop. 3.32]. But, angle BAD is equal to C . Thus, the angle in segment AHB is also equal to C .

Thus, a segment AHB of a circle, accepting an angle equal to C , has been drawn on the given straight-line AB . (Which is) the very thing it was required to do.

34.

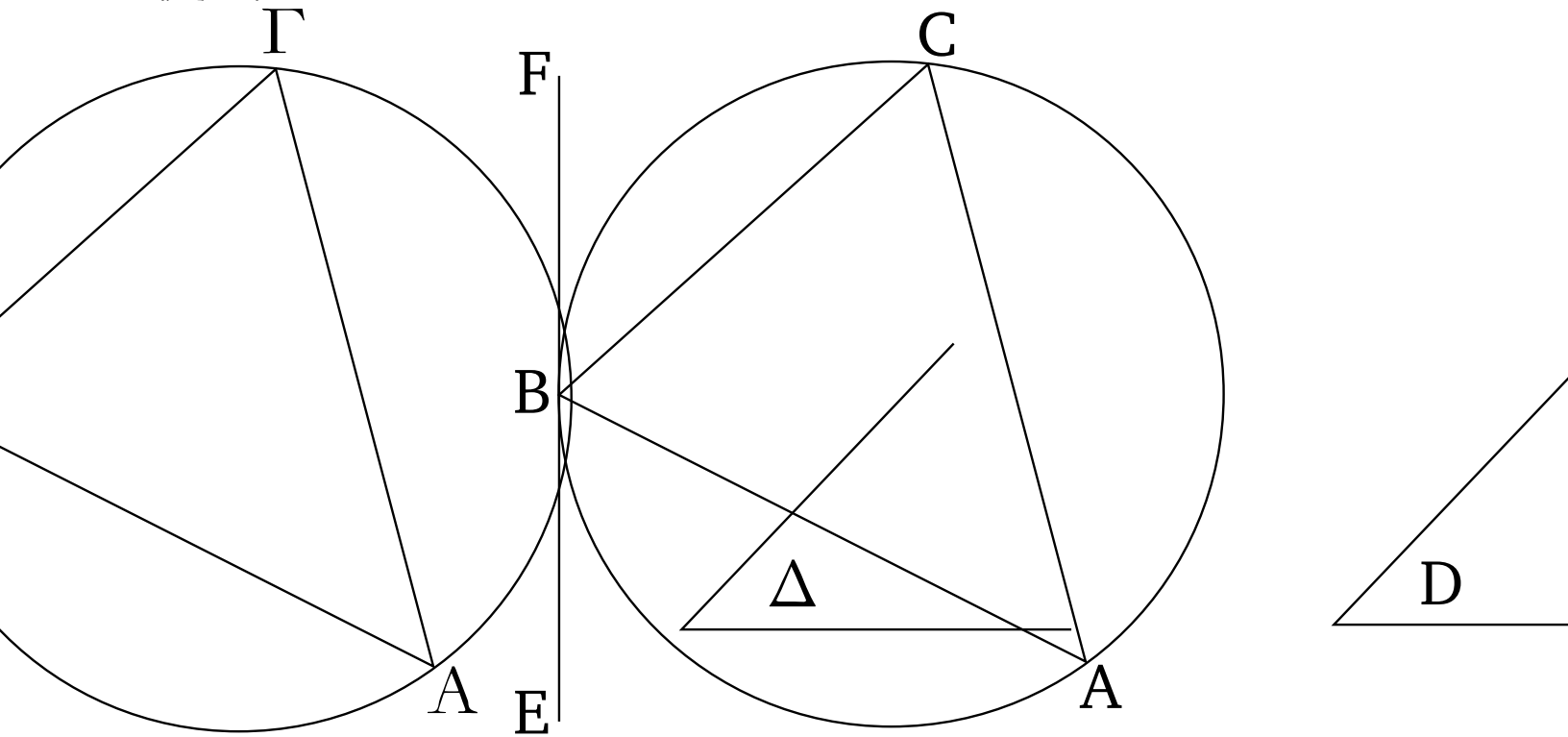
Ἀπὸ τοῦ δοθέντος κύκλου τμήμα ἀφελεῖν δεθόμενον γωνίαν ἴσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ.

Ἐστω ὁ δοθείς κύκλος ὁ $AB\Gamma$, ἡ δὲ δοθεῖσα γωνία εὐθυγράμμοις ἡ πρὸς τῷ Δ . δεῖ δὲ ἀπὸ τοῦ $AB\Gamma$ κύκλου τμήμα ἀφελεῖν δεθόμενον γωνίαν ἴσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ πρὸς τῷ Δ .

Proposition 34

To cut off a segment, accepting an angle equal to a given rectilinear angle, from a given circle.

Let ABC be the given circle, and D the given rectilinear angle. So it is required to cut off a segment, accepting an angle equal to the given rectilinear angle D , from the given circle ABC .



Ἦθθω τοῦ $AB\Gamma$ ἐφαπτομένη ἡ EZ κατὰ τὸ B σημεῖον, καὶ συνεστώτω πρὸς τῇ ZB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ B τῇ πρὸς τῷ Δ γωνίᾳ ἴση ἡ ὑπὸ $ZB\Gamma$.

Ἐπεὶ ὁ ὅν κύκλον τοῦ $AB\Gamma$ ἐφάπτεται τις εὐθεία ἡ EZ , καὶ ἀπὸ τῆς κατὰ τὸ B ἐπαφῆς διήκται ἡ $B\Gamma$, ἡ ὑπὸ $ZB\Gamma$ ἄρα γωνία ἴση ἐστὶ τῇ ἐν τῷ BAG ἐναλλάξ τμήματι συνισταμένη γωνίᾳ. ἀλλ' ἡ ὑπὸ $ZB\Gamma$ τῇ πρὸς τῷ Δ ἐστὶν ἴση· καὶ ἡ ἐν τῷ BAG ἄρα τμήματι ἴση ἐστὶ τῇ πρὸς τῷ Δ [γωνίᾳ].

Ἀπὸ τοῦ δοθέντος ἄρα κύκλου τοῦ $AB\Gamma$ τμήμα ἀφήρηται

Let EF be drawn touching ABC at point B .[†] And let (angle) FBC , equal to angle D , be constructed on the straight-line FB , at the point B on it [Prop. 1.23].

Therefore, since some straight-line EF touches the circle ABC , and BC has been drawn across (the circle) from the point of contact B , angle FBC is thus equal to the angle constructed in the alternate segment BAC [Prop. 1.32]. But, FBC is equal to D . Thus, the (angle) in the segment BAC is also equal to [angle] D .

τὸ BAG δεθόμενον γωνίαν ἴσην τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ τῇ πρὸς τῷ Δ . ὅπερ ἔδει ποιῆσαι.

Thus, the segment BAC , accepting an angle equal to the given rectilinear angle D , has been cut off from the given circle ABC . (Which is) the very thing it was required to do.

† Presumably, by finding the center of ABC [Prop. 3.1], drawing a straight-line between the center and point B , and then drawing EF through point B , at right-angles to the aforementioned straight-line [Prop. 1.11].

35.

Ἐάν ἐν κύκλῳ δύο εὐθεῖαι τέμνωσιν ἀλλήλας, τὸ ὑπὸ τῶν τῆς μιᾶς τμημάτων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν τῆς ἑτέρας τμημάτων περιεχομένῳ ὀρθογώνιῳ.

Ἐν γὰρ κύκλῳ τῷ $AB\Gamma\Delta$ δύο εὐθεῖαι αἱ AG , BD τέμνεται ἀλλήλας κατὰ τὸ E σημεῖον· λέγω, ὅτι τὸ ὑπὸ τῶν AE , EG περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν DE , EB περιεχομένῳ ὀρθογώνιῳ.

Εἰ μὲν οὖν αἱ AG , BD διὰ τοῦ κέντρου εἰσὶν ὥστε τὸ E κέντρον εἶναι τοῦ $AB\Gamma\Delta$ κύκλου, φανερόν, ὅτι ἴσων οὐσῶν τῶν AE , EG , DE , EB καὶ τὸ ὑπὸ τῶν AE , EG περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν DE , EB περιεχομένῳ ὀρθογώνιῳ.

μὴ ἔστωσαν δὴ αἱ AG , BD διὰ τοῦ κέντρου, καὶ εἰλήφθω τὸ κέντρον τοῦ $AB\Gamma\Delta$, καὶ ἔστω τὸ Z , καὶ ἀπὸ τοῦ Z ἐπὶ τὰς AG , BD εὐθείας κάθετοι ἡθῆσαν αἱ ZH , $Z\Theta$, καὶ ἐπεξέσθωσαν αἱ ZB , ZI , ZE .

Καὶ ἐπεὶ εὐθεῖα τις διὰ τοῦ κέντρου ἡ HZ εὐθειάν τινα μὴ διὰ τοῦ κέντρου τὴν AG πρὸς ὀρθὰς τέμνει, καὶ δίθα αὐτὴν τέμνει· ἴση ἄρα ἡ AH τῇ HG . ἐπεὶ οὖν εὐθεῖα ἡ AG τέμνεται εἰς μὲν ἴσα κατὰ τὸ H , εἰς δὲ ἄνισα κατὰ τὸ E , τὸ ἄρα ὑπὸ τῶν AE , EG περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς EH τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς HG · [κοινόν] προσκείσθω τὸ ἀπὸ τῆς HZ · τὸ ἄρα ὑπὸ τῶν AE , EG μετὰ τῶν ἀπὸ τῶν HE , HZ ἴσον ἐστὶ τοῖς ἀπὸ τῶν GH , HZ . ἀλλὰ τοῖς μὲν ἀπὸ τῶν EH , HZ ἴσον ἐστὶ τὸ ἀπὸ τῆς ZE , τοῖς δὲ ἀπὸ τῶν GH , HZ ἴσον ἐστὶ τὸ ἀπὸ τῆς ZI · τὸ ἄρα ὑπὸ τῶν AE , EG μετὰ τοῦ ἀπὸ τῆς ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς ZI . ἴση δὲ ἡ ZI τῇ ZB · τὸ ἄρα ὑπὸ τῶν AE , EG μετὰ τοῦ ἀπὸ τῆς ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς ZB . διὰ τὰ αὐτὰ δὴ καὶ τὸ ὑπὸ τῶν DE , EB μετὰ τοῦ ἀπὸ τῆς ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς ZB . ἐδείκθη δὲ καὶ τὸ ὑπὸ τῶν AE , EG μετὰ τοῦ ἀπὸ τῆς ZE ἴσον τῷ ἀπὸ τῆς ZB · τὸ ἄρα ὑπὸ τῶν AE , EG μετὰ τοῦ ἀπὸ τῆς ZE ἴσον ἐστὶ τῷ ὑπὸ τῶν DE , EB μετὰ τοῦ ἀπὸ τῆς ZE . κοινὸν ἀφῆρησθω τὸ ἀπὸ τῆς ZE · λοιπὸν ἄρα τὸ ὑπὸ τῶν AE , EG περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν DE , EB περιεχομένῳ ὀρθογώνιῳ.

Proposition 35

If two straight-lines in a circle cut one another, (then) the rectangle contained by the pieces of one is equal to the rectangle contained by the pieces of the other.

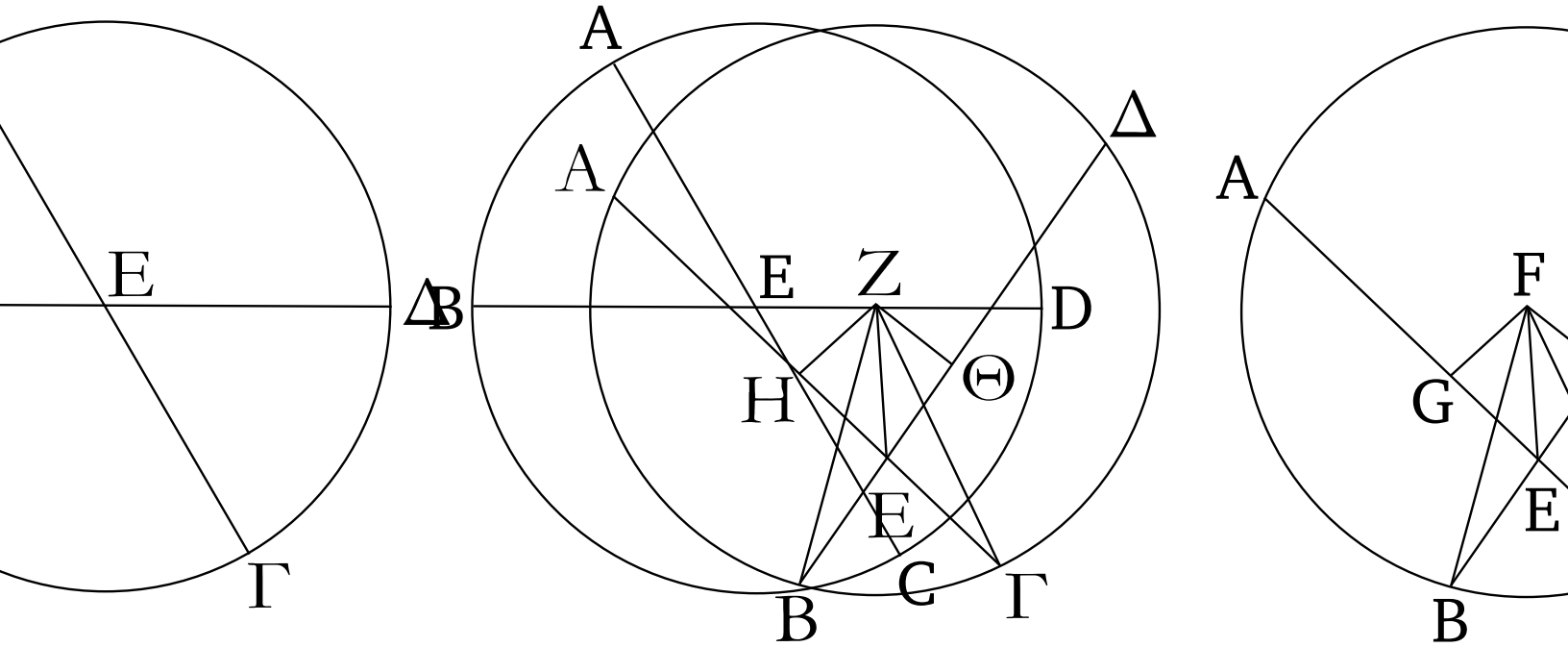
For let the two straight-lines AC and BD , in the circle $ABCD$, cut one another at point E . I say that the rectangle contained by AE and EC is equal to the rectangle contained by DE and EB .

In fact, if AC and BD are through the center (as in the first diagram from the left), so that E is the center of circle $ABCD$, (then it is) clear that, AE , EC , DE , and EB being equal, the rectangle contained by AE and EC is also equal to the rectangle contained by DE and EB .

So let AC and BD not be through the center (as in the second diagram from the left), and let the center of $ABCD$ be found [Prop. 3.1], and let it be (at) F . And let FG and FH be drawn from F , perpendicular to the straight-lines AC and BD (respectively) [Prop. 1.12]. And let FB , FC , and FE be joined.

And since some straight-line, GF , through the center, cuts at right-angles some (other) straight-line, AC , not through the center, (then) it also cuts it in half [Prop. 3.3]. Thus, AG (is) equal to GC . Therefore, since the straight-line AC is cut equally at G , and unequally at E , the rectangle contained by AE and EC plus the square on EG is thus equal to the (square) on GC [Prop. 2.5]. Let the (square) on GF be added [to both]. Thus, the (rectangle contained) by AE and EC plus the (sum of the squares) on GE and GF is equal to the (sum of the squares) on CG and GF . But, the (square) on FE is equal to the (sum of the squares) on EG and GF [Prop. 1.47], and the (square) on FC is equal to the (sum of the squares) on CG and GF [Prop. 1.47]. Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (square) on FC . And FC (is) equal to FB . Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (square) on FB . So, for the same (reasons), the (rectangle contained) by DE and EB plus the (square) on FE is equal to the (square) on FB . And the (rectangle contained) by AE and EC plus the (square) on FE was also shown (to be) equal to the (square) on FB . Thus, the (rectangle contained) by AE and EC plus the (square) on FE is equal to the (rectangle contained) by DE and EB plus the (square) on FE . Let the (square) on FE be taken from both. Thus, the remaining rectangle contained by

AE and EC is equal to the rectangle contained by DE and EB .



Ἐάν ἄρα ἐν κύκλῳ εὐθεῖαι δύο τέμνωσιν ἀλλήλας, τὸ ὑπὸ τῶν τῆς μᾶξ τιμημάτων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν τῆς ἑτέρας τιμημάτων περιεχομένῳ ὀρθογώνιῳ· ὅπερ ἔδει δεῖξαι.

Thus, if two straight-lines in a circle cut one another, (then) the rectangle contained by the pieces of one is equal to the rectangle contained by the pieces of the other. (Which is) the very thing it was required to show.

36.

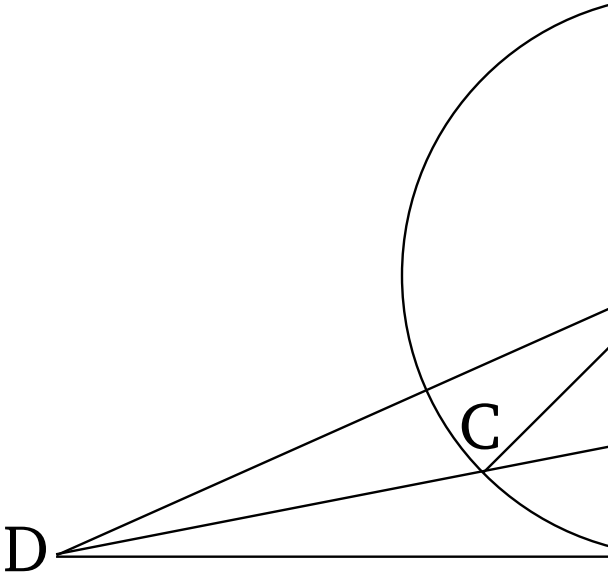
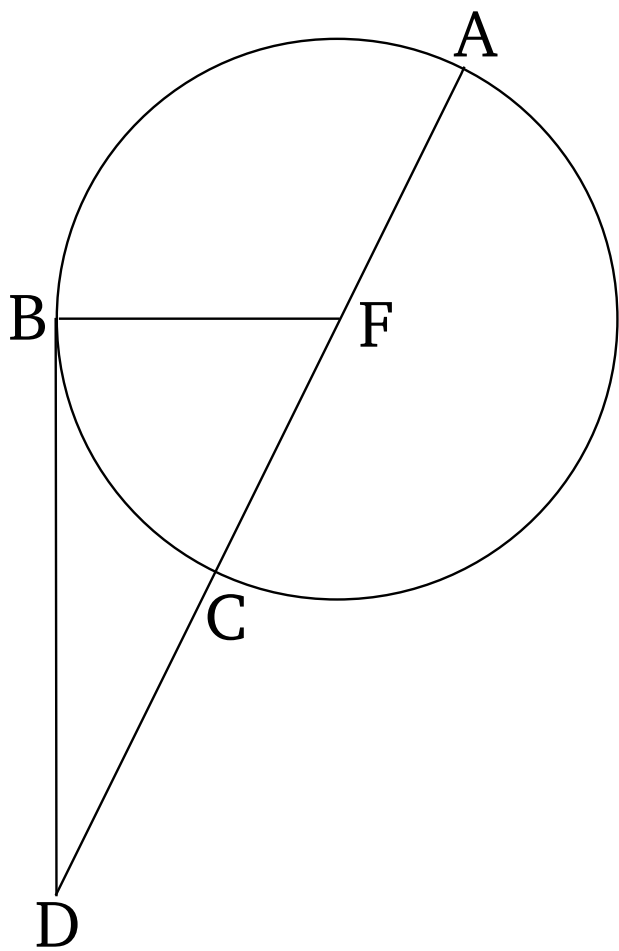
Proposition 36

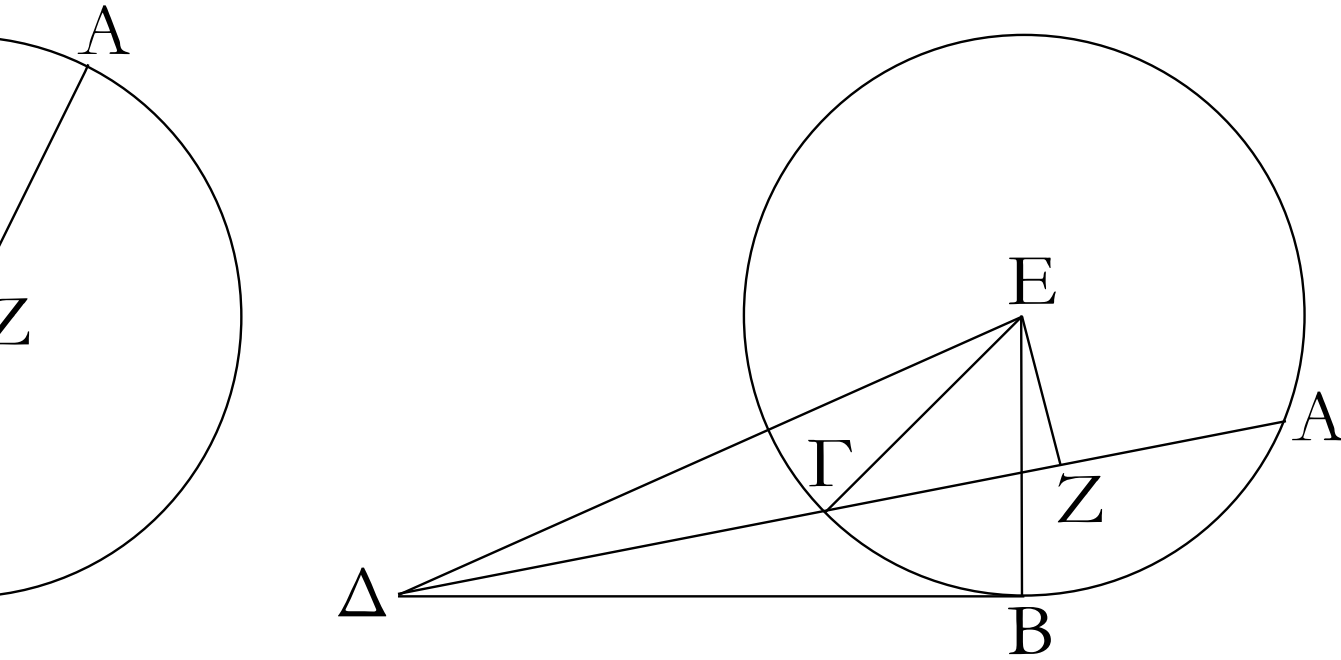
Ἐάν κύκλον ληφθῇ τι σημεῖον ἐκτός, καὶ ἀπ' αὐτοῦ πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ ἐφάπτεται, ἔσται τὸ ὑπὸ ὅλης τῆς τεμνοῦσης καὶ τῆς ἐκτός ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ἴσον τῷ ἀπὸ τῆς ἐφαπτομένης τετραγώνῳ.

Κύκλον γάρ τοῦ $AB\Gamma$ εἰλήφθω τι σημεῖον ἐκτός τὸ Δ , καὶ ἀπὸ τοῦ Δ πρὸς τὸν $AB\Gamma$ κύκλον προσπιπτέτωσαν δύο εὐθεῖαι αἱ $\Delta\Gamma[A]$, ΔB · καὶ ἡ μὲν $\Delta\Gamma A$ τεμνέτω τὸν $AB\Gamma$ κύκλον, ἡ δὲ $B\Delta$ ἐφαπτέσθω· λέγω, ὅτι τὸ ὑπὸ τῶν $A\Delta$, $\Delta\Gamma$ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔB τετραγώνῳ.

If some point is taken outside a circle, and two straight-lines radiate from it towards the circle, and (one) of them cuts the circle, and the (other) touches (it), (then) the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, will be equal to the square on the tangent (line).

For let some point D be taken outside circle ABC , and let two straight-lines, $DC[A]$ and DB , radiate from D towards circle ABC . And let DCA cut circle ABC , and let BD touch (it). I say that the rectangle contained by AD and DC is equal to the square on DB .





Ἡ ἄρα $[\Delta]ΓΑ$ ἤτοι διὰ τοῦ κέντρου ἐστὶν ἡ οὐ. ἔστω πρότερον διὰ τοῦ κέντρου, καὶ ἔστω τὸ Z κέντρον τοῦ $ABΓ$ κύκλου, καὶ ἐπεζεύθω ἡ ZB . ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ $ZBΔ$. καὶ ἐπεὶ εὐθεῖα ἡ $ΑΓ$ δίδια τέτμηται κατὰ τὸ Z , πρόσκειται δὲ αὐτῇ ἡ $ΓΔ$, τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τοῦ ἀπὸ τῆς $ZΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ZΔ$. ἴση δὲ ἡ $ZΓ$ τῇ ZB . τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τοῦ ἀπὸ τῆς ZB ἴσον ἐστὶ τῷ ἀπὸ τῆς $ZΔ$. τῷ δὲ ἀπὸ τῆς $ZΔ$ ἴσα ἐστὶ τὰ ἀπὸ τῶν ZB , $BΔ$. τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τοῦ ἀπὸ τῆς ZB ἴσον ἐστὶ τοῖς ἀπὸ τῶν ZB , $BΔ$. κοινὸν ἀφηγήσθω τὸ ἀπὸ τῆς ZB . λοιπὸν ἄρα τὸ ὑπὸ τῶν $ΑΔ$, $ΔΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΔB$ ἐφαπτομένης.

Ἀλλὰ δὴ ἡ $ΔΓΑ$ μὴ ἔστω διὰ τοῦ κέντρου τοῦ $ABΓ$ κύκλου, καὶ εἰλήφθω τὸ κέντρον τὸ E , καὶ ἀπὸ τοῦ E ἐπὶ τὴν $ΑΓ$ κάθετος ἡθῶ ἡ EZ , καὶ ἐπεζεύθωσαν αἱ EB , $ΕΓ$, $ΕΔ$. ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ $EBΔ$. καὶ ἐπεὶ εὐθεῖα τις διὰ τοῦ κέντρου ἡ EZ εὐθειάν τινα μὴ διὰ τοῦ κέντρου τὴν $ΑΓ$ πρὸς ὀρθὰς τέμνει, καὶ δίδια αὐτὴν τέμνει· ἡ AZ ἄρα τῇ $ZΓ$ ἐστὶν ἴση. καὶ ἐπεὶ εὐθεῖα ἡ $ΑΓ$ τέτμηται δίδια κατὰ τὸ Z σημείον, πρόσκειται δὲ αὐτῇ ἡ $ΓΔ$, τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τοῦ ἀπὸ τῆς $ZΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ZΔ$. κοινὸν προσκείσθω τὸ ἀπὸ τῆς ZE . τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ μετὰ τῶν ἀπὸ τῶν

$[D]CA$ is surely either through the center, or not. Let it first of all be through the center, and let F be the center of circle ABC , and let FB be joined. Thus, (angle) FBD is a right-angle [Prop. 3.18]. And since straight-line AC is cut in half at F , let CD be added to it. Thus, the (rectangle contained) by AD and DC plus the (square) on FC is equal to the (square) on FD [Prop. 2.6]. And FC (is) equal to FB . Thus, the (rectangle contained) by AD and DC plus the (square) on FB is equal to the (square) on FD . And the (square) on FD is equal to the (sum of the squares) on FB and BD [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on FB is equal to the (sum of the squares) on FB and BD . Let the (square) on FB be subtracted from both. Thus, the remaining (rectangle contained) by AD and DC is equal to the (square) on the tangent DB .

And so let DCA not be through the center of circle ABC , and let the center E be found, and let EF be drawn from E , perpendicular to AC [Prop. 1.12]. And let EB , EC , and ED be joined. (Angle) EBD (is) thus a right-angle [Prop. 3.18]. And since some straight-line, EF , through the center, cuts some (other) straight-line, AC , not through the center, at right-angles, it also cuts it in half [Prop. 3.3]. Thus, AF is equal to

FZ , ZE ἴσον ἐστὶ τοῖς ἀπὸ τῶν $Z\Delta$, ZE . τοῖς δὲ ἀπὸ τῶν FZ , ZE ἴσον ἐστὶ τὸ ἀπὸ τῆς EF . ὁρθὴ γὰρ [ἐστίν] ἡ ὑπὸ EZF [γωνία]. τοῖς δὲ ἀπὸ τῶν ΔZ , ZE ἴσον ἐστὶ τὸ ἀπὸ τῆς $E\Delta$. τὸ ἄρα ὑπὸ τῶν $A\Delta$, ΔF μετὰ τοῦ ἀπὸ τῆς EF ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Delta$. ἴση δὲ ἡ EF τῇ EB . τὸ ἄρα ὑπὸ τῶν $A\Delta$, ΔF μετὰ τοῦ ἀπὸ τῆς EB ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Delta$. τῷ δὲ ἀπὸ τῆς $E\Delta$ ἴσα ἐστὶ τὰ ἀπὸ τῶν EB , $B\Delta$. ὁρθὴ γὰρ ἡ ὑπὸ $EB\Delta$ γωνία. τὸ ἄρα ὑπὸ τῶν $A\Delta$, ΔF μετὰ τοῦ ἀπὸ τῆς EB ἴσον ἐστὶ τοῖς ἀπὸ τῶν EB , $B\Delta$. κοινὸν ἀφηρήσθω τὸ ἀπὸ τῆς EB . λοιπὸν ἄρα τὸ ὑπὸ τῶν $A\Delta$, ΔF ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔB .

Ἐὰν ἄρα κύκλον ληφθῇ τι σημεῖον ἐκτός, καὶ ἀπ' αὐτοῦ πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ ἐφάπτεται, ἔσται τὸ ὑπὸ ὅλης τῆς τεμνούσης καὶ τῆς ἐκτός ἀπολαμβάνομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ἴσον τῷ ἀπὸ τῆς ἐφαπτομένης τετραγώνῳ· ὅπερ ἔδει δεῖξαι.

FC . And since the straight-line AC is cut in half at point F , let CD be added to it. Thus, the (rectangle contained) by AD and DC plus the (square) on FC is equal to the (square) on FD [Prop. 2.6]. Let the (square) on FE be added to both. Thus, the (rectangle contained) by AD and DC plus the (sum of the squares) on CF and FE is equal to the (sum of the squares) on FD and FE . But the (square) on EC is equal to the (sum of the squares) on CF and FE . For [angle] EFC [is] a right-angle [Prop. 1.47]. And the (square) on ED is equal to the (sum of the squares) on DF and FE [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on EC is equal to the (square) on ED . And EC (is) equal to EB . Thus, the (rectangle contained) by AD and DC plus the (square) on EB is equal to the (square) on ED . And the (sum of the squares) on EB and BD is equal to the (square) on ED . For EBD (is) a right-angle [Prop. 1.47]. Thus, the (rectangle contained) by AD and DC plus the (square) on EB is equal to the (sum of the squares) on EB and BD . Let the (square) on EB be subtracted from both. Thus, the remaining (rectangle contained) by AD and DC is equal to the (square) on BD .

Thus, if some point is taken outside a circle, and two straight-lines radiate from it towards the circle, and (one) of them cuts the circle, and (the other) touches (it), (then) the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, will be equal to the square on the tangent (line). (Which is) the very thing it was required to show.

37.

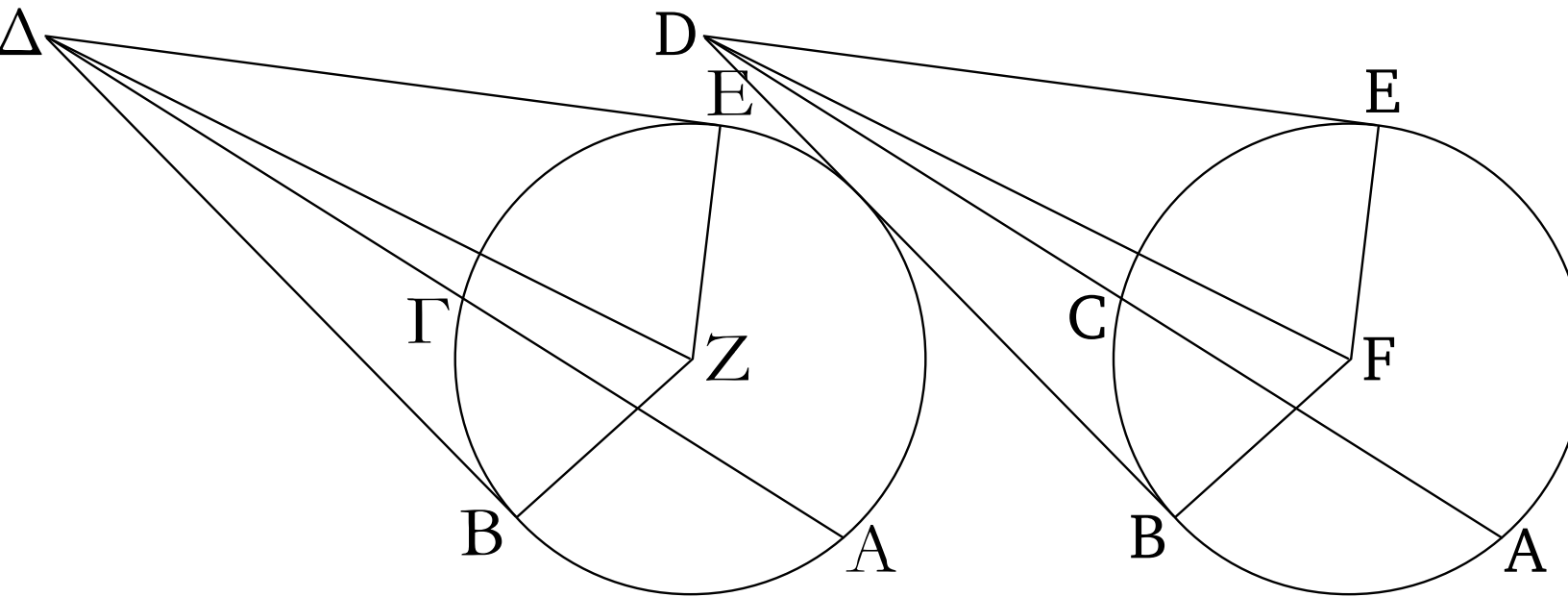
Proposition 37

Ἐὰν κύκλον ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσπίπτωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ προσπίπτῃ, ἢ δὲ τὸ ὑπὸ [τῆς] ὅλης τῆς τεμνούσης καὶ τῆς ἐκτός ἀπολαμβάνομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ἴσον τῷ ἀπὸ τῆς προσπίπτουσας, ἡ προσπίπτουσα ἐφάπτεται τοῦ κύκλου.

Κύκλον γὰρ τοῦ $AB\Gamma$ εἰλήφθω τι σημεῖον ἐκτός τὸ Δ , καὶ ἀπὸ τοῦ Δ πρὸς τὸν $AB\Gamma$ κύκλον προσπίπτέτωσαν δύο εὐθεῖαι αἱ $\Delta\Gamma A$, ΔB , καὶ ἡ μὲν $\Delta\Gamma A$ τεμνέτω τὸν κύκλον, ἡ δὲ ΔB προσπίπτέτω, ἔστω δὲ τὸ ὑπὸ τῶν $A\Delta$, $\Delta\Gamma$ ἴσον τῷ ἀπὸ τῆς ΔB . λέγω, ὅτι ἡ ΔB ἐφάπτεται τοῦ $AB\Gamma$ κύκλου.

If some point is taken outside a circle, and two straight-lines radiate from the point towards the circle, and one of them cuts the circle, and the (other) meets (it), and the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, is equal to the (square) on the (straight-line) meeting (the circle), (then) the (straight-line) meeting (the circle) will touch the circle.

For let some point D be taken outside circle ABC , and let two straight-lines, DCA and DB , radiate from D towards circle ABC , and let DCA cut the circle, and let DB meet (the circle). And let the (rectangle contained) by AD and DC be equal to the (square) on DB . I say that DB touches circle ABC .



Ἦθθω γάρ τοῦ $ABΓ$ ἐφαπτομένη ἡ $ΔΕ$, καὶ εἰλήφθω τὸ κέντρον τοῦ $ABΓ$ κύκλου, καὶ ἔστω τὸ Z , καὶ ἐπεζεύθωσαν αἱ ZE , ZB , $ZΔ$. ἡ ἄρα ὑπὸ $ZEΔ$ ὀρθή ἐστιν. καὶ ἐπεὶ ἡ $ΔΕ$ ἐφάπτεται τοῦ $ABΓ$ κύκλου, τέμνει δὲ ἡ $ΔΓΑ$, τὸ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΔΕ$. ἦν δὲ καὶ τὸ ὑπὸ τῶν $ΑΔ$, $ΔΓ$ ἴσον τῷ ἀπὸ τῆς $ΔΒ$. τὸ ἄρα ἀπὸ τῆς $ΔΕ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΔΒ$. ἴση ἄρα ἡ $ΔΕ$ τῇ $ΔΒ$. ἐστὶ δὲ καὶ ἡ ZE τῇ ZB ἴση· δύο δὲ αἱ $ΔΕ$, EZ δύο ταῖς $ΔΒ$, BZ ἴσαι εἰσὶν καὶ βάσεις αὐτῶν κοινὴ ἡ $ZΔ$. γωνία ἄρα ἡ ὑπὸ $ΔEZ$ γωνία τῇ ὑπὸ $ΔBZ$ ἐστὶν ἴση. ὀρθὴ δὲ ἡ ὑπὸ $ΔEZ$. ὀρθὴ ἄρα καὶ ἡ ὑπὸ $ΔBZ$. καὶ ἐστὶν ἡ ZB ἐκβαλλομένη διάμετρος· ἡ δὲ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένη ἐφάπτεται τοῦ κύκλου· ἡ $ΔΒ$ ἄρα ἐφάπτεται τοῦ $ABΓ$ κύκλου. ὁμοίως δὲ δειθθήσεται, κ' ἂν τὸ κέντρον ἐπὶ τῆς $ΑΓ$ τυγθάνῃ.

Ἐάν ἄρα κύκλον ληφθῇ τι σημεῖον ἐκτός, ἀπὸ δὲ τοῦ σημείου πρὸς τὸν κύκλον προσιπύωσι δύο εὐθεῖαι, καὶ ἡ μὲν αὐτῶν τέμνῃ τὸν κύκλον, ἡ δὲ προσιπύτῃ, ἥ δὲ τὸ ὑπὸ ὅλης τῆς τεμνούσης καὶ τῆς ἐκτός ἀπολαμβανομένης μεταχὺ τοῦ τε σημείου καὶ τῆς κυρτῆς περιφερείας ἴσον τῷ ἀπὸ τῆς προσιπυτούσης, ἡ προσιπύτουσα ἐφάπτεται τοῦ κύκλου· ὅπερ ἔδει δεῖξαι.

For let DE be drawn touching ABC [Prop. 3.17], and let the center of the circle ABC be found, and let it be (at) F . And let FE , FB , and FD be joined. (Angle) FED is thus a right-angle [Prop. 3.18]. And since DE touches circle ABC , and DCA cuts (it), the (rectangle contained) by AD and DC is thus equal to the (square) on DE [Prop. 3.36]. And the (rectangle contained) by AD and DC was also equal to the (square) on DB . Thus, the (square) on DE is equal to the (square) on DB . Thus, DE (is) equal to DB . And FE is also equal to FB . So the two (straight-lines) DE , EF are equal to the two (straight-lines) DB , BF (respectively). And their base, FD , is common. Thus, angle DEF is equal to angle DBF [Prop. 1.8]. And DEF (is) a right-angle. Thus, DBF (is) also a right-angle. And FB produced is a diameter, And a (straight-line) drawn at right-angles to a diameter of a circle, at its extremity, touches the circle [Prop. 3.16 corr.]. Thus, DB touches circle ABC . Similarly, (the same thing) can be shown, even if the center happens to be on AC .

Thus, if some point is taken outside a circle, and two straight-lines radiate from the point towards the circle, and one of them cuts the circle, and the (other) meets (it), and the (rectangle contained) by the whole (straight-line) cutting (the circle), and the (part of it) cut off outside (the circle), between the point and the convex circumference, is equal to the (square) on the (straight-line) meeting (the circle), (then) the (straight-line) meeting (the circle) will touch the circle. (Which is) the very thing it was required to show.

ELEMENTS BOOK 4

*Construction of Rectilinear Figures In and
Around Circles*

Ὅροι.

1. Σθῆμα εὐθύγραμμον εἷς σθῆμα εὐθύγραμμον ἐγγράφεισθαι λέγεται, ὅταν ἐκάστη τῶν τοῦ ἐγγραφομένου σθῆματοξ γωνιῶν ἐκάστης πλευρᾷ τοῦ, εἷς ὃ ἐγγράφεται, ἀπτηται.

2. Σθῆμα δὲ ὁμοίως περὶ σθῆμα περιγράφεισθαι λέγεται, ὅταν ἐκάστη πλευρὰ τοῦ περιγραφομένου ἐκάστης γωνίας τοῦ, περὶ ὃ περιγράφεται, ἀπτηται.

3. Σθῆμα εὐθύγραμμον εἷς κύκλον ἐγγράφεισθαι λέγεται, ὅταν ἐκάστη γωνία τοῦ ἐγγραφομένου ἀπτηται τῇς τοῦ κύκλου περιφερείᾳ.

4. Σθῆμα δὲ εὐθύγραμμον περὶ κύκλον περιγράφεισθαι λέγεται, ὅταν ἐκάστη πλευρὰ τοῦ περιγραφομένου ἐφάπτηται τῇς τοῦ κύκλου περιφερείᾳ.

5. Κύκλος δὲ εἷς σθῆμα ὁμοίως ἐγγράφεισθαι λέγεται, ὅταν ἢ τοῦ κύκλου περιφέρεια ἐκάστης πλευρᾷ τοῦ, εἷς ὃ ἐγγράφεται, ἀπτηται.

6. Κύκλος δὲ περὶ σθῆμα περιγράφεισθαι λέγεται, ὅταν ἢ τοῦ κύκλου περιφέρεια ἐκάστης γωνίας τοῦ, περὶ ὃ περιγράφεται, ἀπτηται.

7. Εὐθεῖα εἷς κύκλον ἐναρμόζεσθαι λέγεται, ὅταν τὰ πέρατα αὐτῆς ἐπὶ τῇς περιφερείᾳ ᾗ τοῦ κύκλου.

1.

Εἷς τὸν δοθέντα κύκλον τῇ δοθείσῃ εὐθείᾳ μὴ μείζονι οὕσῃ τῇς τοῦ κύκλου διαμέτρῳ ἴσην εὐθεῖαν ἐναρμόσαι.

Definitions

1. A rectilinear figure is said to be inscribed in a(nother) rectilinear figure when the respective angles of the inscribed figure touch the respective sides of the (figure) in which it is inscribed.

2. And, similarly, a (rectilinear) figure is said to be circumscribed about a(nother rectilinear) figure when the respective sides of the circumscribed (figure) touch the respective angles of the (figure) about which it is circumscribed.

3. A rectilinear figure is said to be inscribed in a circle when each angle of the inscribed (figure) touches the circumference of the circle.

4. And a rectilinear figure is said to be circumscribed about a circle when each side of the circumscribed (figure) touches the circumference of the circle.

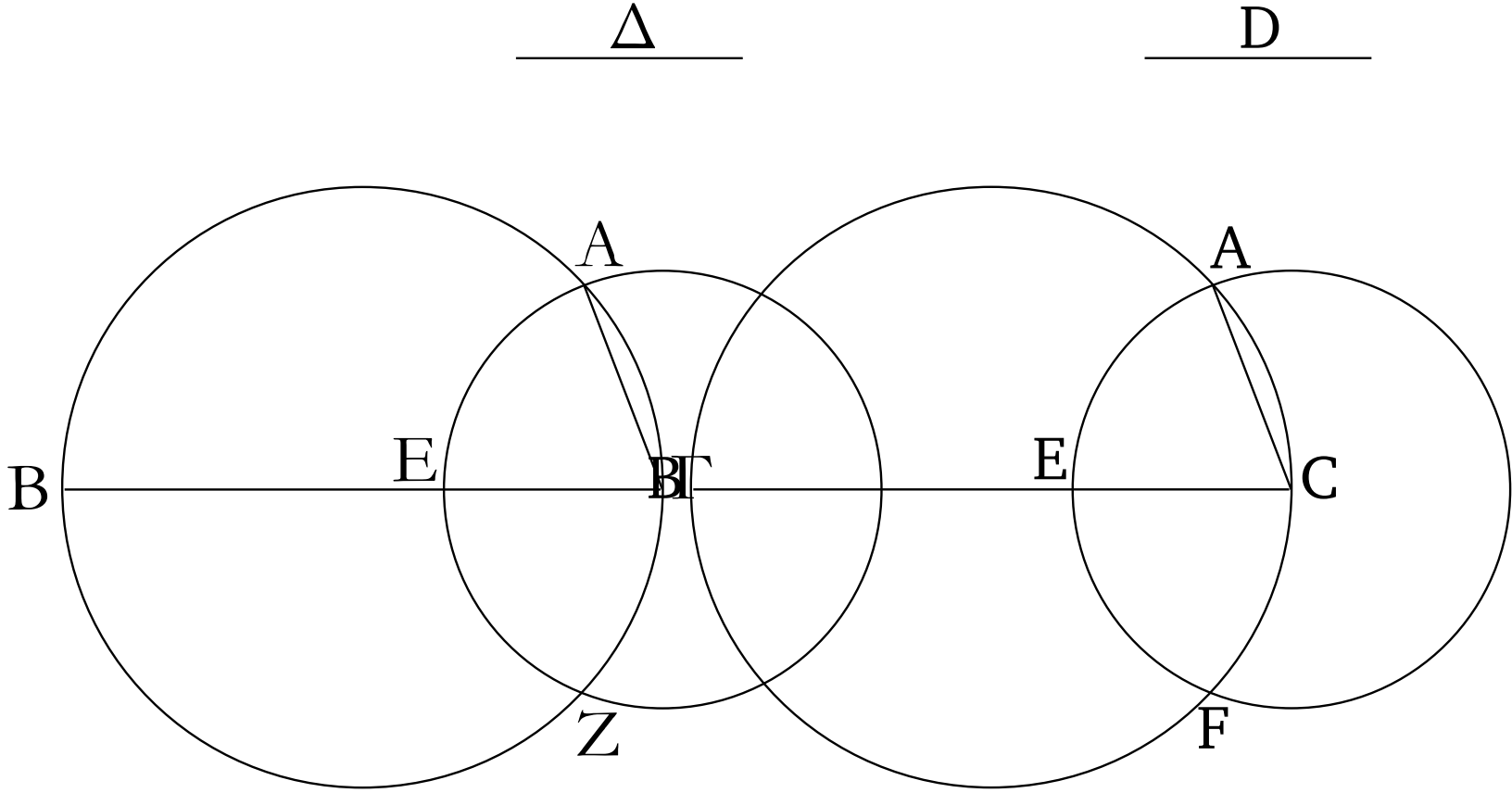
5. And, similarly, a circle is said to be inscribed in a (rectilinear) figure when the circumference of the circle touches each side of the (figure) in which it is inscribed.

6. And a circle is said to be circumscribed about a rectilinear (figure) when the circumference of the circle touches each angle of the (figure) about which it is circumscribed.

7. A straight-line is said to be inserted into a circle when its extremities are on the circumference of the circle.

Proposition 1

To insert a straight-line equal to a given straight-line into a circle, (the latter straight-line) not being greater than the diameter of the circle.



Ἐστω ὁ δοθείς κύκλος ὁ $ABΓ$, ἡ δὲ δοθεῖσα εὐθεῖα μὴ μείζων τῆς τοῦ κύκλου διαμέτρου ἢ $Δ$. δεῖ δὴ εἰς τὸν $ABΓ$ κύκλον τῇ $Δ$ εὐθείᾳ ἴσην εὐθεῖαν ἐναρμόσαι.

Ἦθθω τοῦ $ABΓ$ κύκλου διάμετρος ἡ $ΒΓ$. εἰ μὲν οὖν ἴση ἐστὶν ἡ $ΒΓ$ τῇ $Δ$, γεγονόξ ἂν εἴη τὸ ἐπιταθέν· ἐνήρμοσται γάρ εἰς τὸν $ABΓ$ κύκλον τῇ $Δ$ εὐθείᾳ ἴση ἡ $ΒΓ$. εἰ δὲ μείζων ἐστὶν ἡ $ΒΓ$ τῇ $Δ$, κείσθω τῇ $Δ$ ἴση ἡ $ΓΕ$, καὶ κέντρῳ τῷ $Γ$ διαστήματι δὲ τῷ $ΓΕ$ κύκλος γεγράφθω ὁ $ΕΑΖ$, καὶ ἐπεξεύθθω ἡ $ΓΑ$.

Ἐπεὶ οὖν το $Γ$ σημεῖον κέντρον ἐστὶ τοῦ $ΕΑΖ$ κύκλου, ἴση ἐστὶν ἡ $ΓΑ$ τῇ $ΓΕ$. ἀλλὰ τῇ $Δ$ ἡ $ΓΕ$ ἐστὶν ἴση· καὶ ἡ $Δ$ ἄρα τῇ $ΓΑ$ ἐστὶν ἴση.

Εἰς ἄρα τὸν δοθέντα κύκλον τὸν $ABΓ$ τῇ δοθείσῃ εὐθείᾳ τῇ $Δ$ ἴση ἐνήρμοσται ἡ $ΓΑ$ · ὅπερ ἔδει ποιῆσαι.

Let ABC be the given circle, and D the given straight-line (which is) not greater than the diameter of the circle. So it is required to insert a straight-line, equal to the straight-line D , into the circle ABC .

Let a diameter BC of circle ABC be drawn.[†] Therefore, if BC is equal to D , (then) that (which) was prescribed has taken place. For the (straight-line) BC , equal to the straight-line D , has been inserted into the circle ABC . And if BC is greater than D then let CE be made equal to D [Prop. 1.3], and let the circle EAF be drawn with center C and radius CE . And let CA be joined.

Therefore, since the point C is the center of circle EAF , CA is equal to CE . But, CE is equal to D . Thus, D is also equal to CA .

Thus, CA , equal to the given straight-line D , has been inserted into the given circle ABC . (Which is) the very thing it was required to do.

[†] Presumably, by finding the center of the circle [Prop. 3.1], and then drawing a line through it.

2.

Εἰς τὸν δοθέντα κύκλον τῷ δοθέντι τριγώνῳ ἰσογώνον τρίγωνον ἐγγράψαι.

Proposition 2

To inscribe a triangle, equiangular with a given triangle, in a given circle.

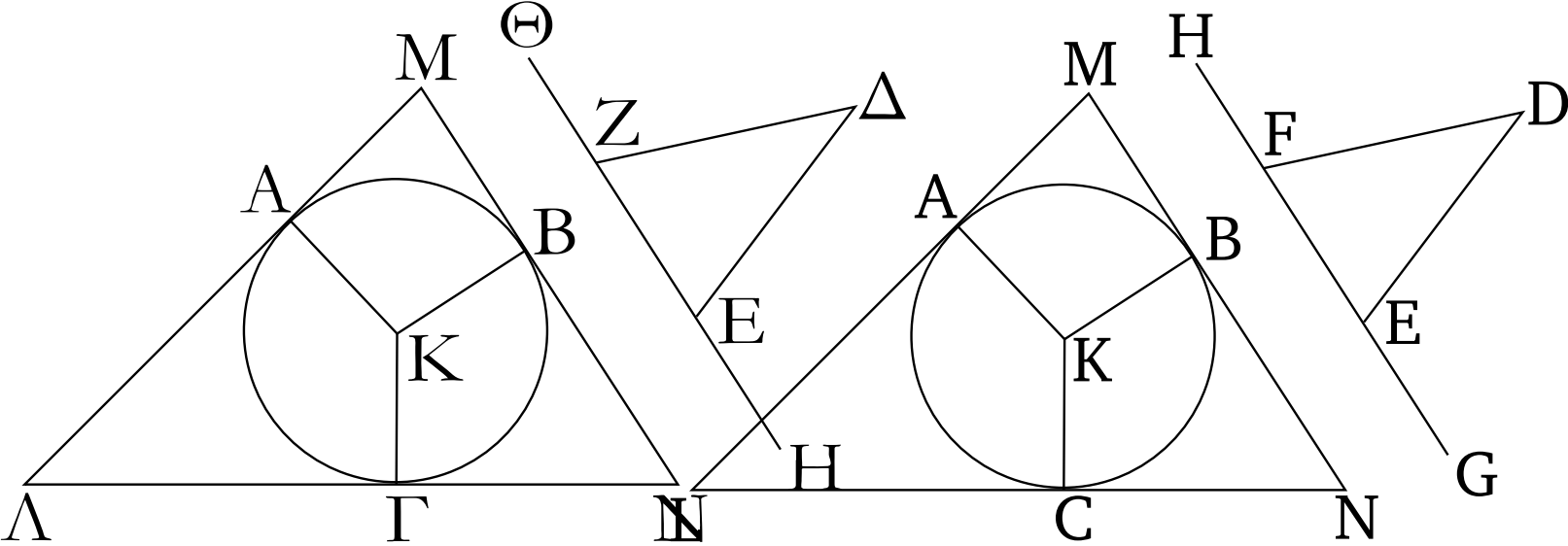
† See the footnote to Prop. 3.34.

3.

Proposition 3

Περὶ τὸν δοθέντα κύκλον τῷ δοθέντι τριγώνῳ ἰσογώνιον τρίγωνον περιγράφαι.

To circumscribe a triangle, equiangular with a given triangle, about a given circle.



Ἐστω ὁ δοθείς κύκλος ὁ $AB\Gamma$, τὸ δὲ δοθέν τρίγωνον τὸ ΔEZ . δεῖ δὴ περὶ τὸν $AB\Gamma$ κύκλον τῷ ΔEZ τριγώνῳ ἰσογώνιον τρίγωνον περιγράφαι.

Ἐκβεβλήσθω ἡ EZ ἐφ' ἐκάτερα τὰ μέρη κατὰ τὰ H , Θ σημεία, καὶ εἰλήφθω τοῦ $AB\Gamma$ κύκλου κέντρον τὸ K , καὶ διήθθω, ὥς ἔτυθεν, εὐθεῖα ἡ KB , καὶ συνεστάτω πρὸς τῇ KB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ K τῇ μὲν ὑπὸ ΔEH γωνία ἴση ἡ ὑπὸ BKA , τῇ δὲ ὑπὸ $\Delta Z\Theta$ ἴση ἡ ὑπὸ $BK\Gamma$, καὶ διὰ τῶν A , B , Γ σημείων ἡθθωσαν ἐφαπτόμεναι τοῦ $AB\Gamma$ κύκλου αἱ ΛAM , MBN , $N\Gamma\Lambda$.

Καὶ ἐπεὶ ἐφάπτονται τοῦ $AB\Gamma$ κύκλου αἱ ΛM , MN , $N\Lambda$ κατὰ τὰ A , B , Γ σημεία, ἀπὸ δὲ τοῦ K κέντρον ἐπὶ τὰ A , B , Γ σημεία ἐπεξενυγμέναι εἰσὶν αἱ KA , KB , $K\Gamma$, ὁρθαὶ ἄρα εἰσὶν αἱ πρὸς τοῖς A , B , Γ σημείοις γωνίαι. καὶ ἐπεὶ τοῦ $AMBK$ τετραπλεύρου αἱ τέσσαρες γωνίαι τέτρασιν ὁρθαῖς ἴσαι εἰσὶν, ἐπειδήπερ καὶ εἷς δύο τρίγωνα διαιρεῖται τὸ $AMBK$, καὶ εἰσὶν ὁρθαὶ αἱ ὑπὸ KAM , KBM γωνίαι, λοιπαὶ ἄρα αἱ ὑπὸ AKB , AMB δυοῖν ὁρθαῖς ἴσαι εἰσὶν. εἰσὶ δὲ καὶ αἱ ὑπὸ ΔEH , ΔEZ δυοῖν ὁρθαῖς ἴσαι· αἱ ἄρα ὑπὸ AKB , AMB ταῖς ὑπὸ ΔEH , ΔEZ ἴσαι εἰσὶν, ὧν ἡ ὑπὸ AKB τῇ ὑπὸ ΔEH ἔστιν ἴση· λοιπὴ ἄρα ἡ ὑπὸ AMB λοιπὴ τῇ ὑπὸ ΔEZ ἔστιν ἴση. ὁμοίως δὲ δευτῇσεται, ὅτι καὶ ἡ ὑπὸ ANB τῇ ὑπὸ ΔZE ἔστιν ἴση· καὶ λοιπὴ ἄρα ἡ ὑπὸ MAN [λοιπὴ] τῇ ὑπὸ $E\Delta Z$ ἔστιν ἴση. ἰσογώνιον ἄρα ἐστὶ τὸ ΛMN τρίγωνον τῷ ΔEZ τριγώνῳ· καὶ περιγράφεται περὶ τὸν $AB\Gamma$ κύκλον.

Περὶ τὸν δοθέντα ἄρα κύκλον τῷ δοθέντι τριγώνῳ ἰσογώνιον τρίγωνον περιγράφεται· ὅπερ ἔδει ποιῆσαι.

Let ABC be the given circle, and DEF the given triangle. So it is required to circumscribe a triangle, equiangular with triangle DEF , about circle ABC .

Let EF be produced in each direction to points G and H . And let the center K of circle ABC be found [Prop. 3.1]. And let the straight-line KB be drawn, at random, across (ABC). And let (angle) BKA , equal to angle DEG , be constructed on the straight-line KB at the point K on it, and (angle) BKC , equal to DFH [Prop. 1.23]. And let the (straight-lines) LAM , MBN , and NCL be drawn through the points A , B , and C (respectively), touching the circle ABC .†

And since LM , MN , and NL touch circle ABC at points A , B , and C (respectively), and KA , KB , and KC are joined from the center K to points A , B , and C (respectively), the angles at points A , B , and C are thus right-angles [Prop. 3.18]. And since the (sum of the) four angles of quadrilateral $AMBK$ is equal to four right-angles, inasmuch as $AMBK$ (can) also (be) divided into two triangles [Prop. 1.32], and angles KAM and KBM are (both) right-angles, the (sum of the) remaining (angles), AKB and AMB , is thus equal to two right-angles. And DEG and DEF is also equal to two right-angles [Prop. 1.13]. Thus, AKB and AMB is equal to DEG and DEF , of which AKB is equal to DEG . Thus, the remainder AMB is equal to the remainder DEF . So, similarly, it can be shown that ANB is also equal to DFE . Thus, the remaining (angle) MLN is also equal to the [remaining] (angle) EDF [Prop. 1.32]. Thus, triangle LMN is equiangular with triangle DEF . And it has

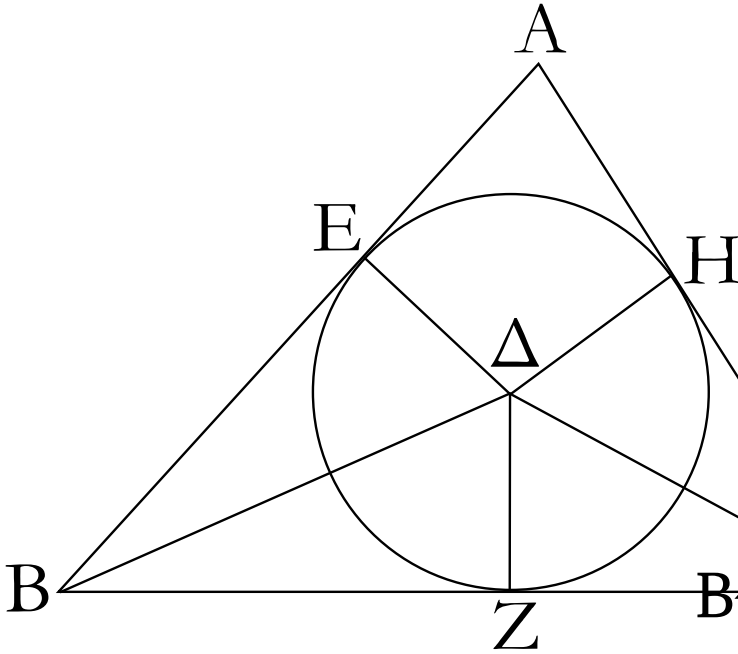
been drawn around circle ABC .

Thus, a triangle, equiangular with the given triangle, has been circumscribed about the given circle. (Which is) the very thing it was required to do.

[†] See the footnote to Prop. 3.34.

4.

Εἰς τὸ δοθὲν τρίγωνον κύκλον ἐγγράφαι.



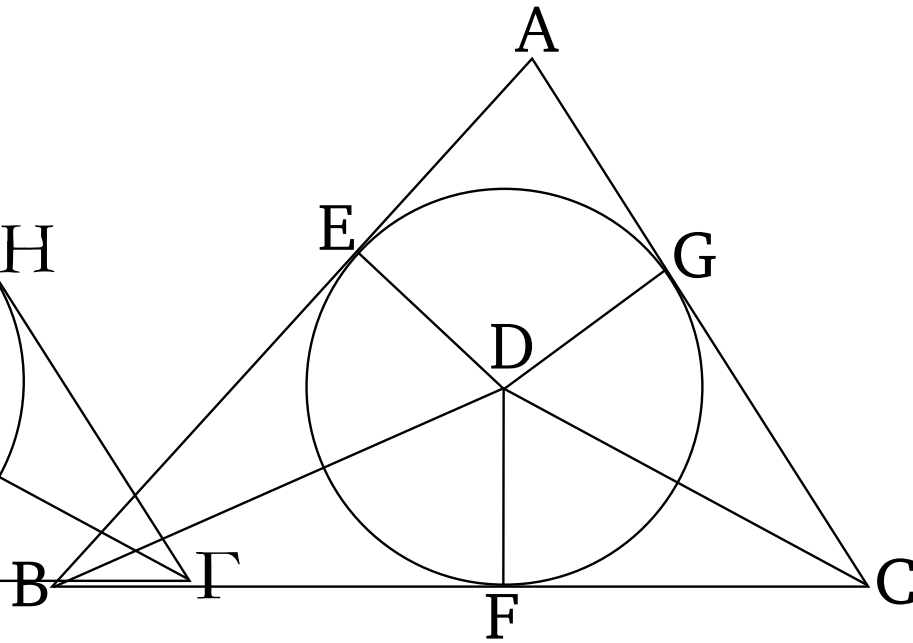
Ἐστω τὸ δοθὲν τρίγωνον τὸ $AB\Gamma$. δεῖ δὴ εἰς τὸ $AB\Gamma$ τρίγωνον κύκλον ἐγγράφαι.

Τετμήσθωσαν αἱ ὑπὸ $AB\Gamma$, $\Gamma B\Delta$ γωνίαι διῖθα ταῖς $B\Delta$, $\Gamma\Delta$ εὐθείαις, καὶ συμβαλλέτωσαν ἀλλήλαις κατὰ τὸ Δ σημειον, καὶ ῥηθῶσαν ἀπὸ τοῦ Δ ἐπὶ τὰς AB , $B\Gamma$, $\Gamma\Delta$ εὐθείας κάθετοι αἱ ΔE , ΔZ , ΔH .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ $AB\Delta$ γωνία τῇ ὑπὸ $\Gamma B\Delta$, ἐστὶ δὲ καὶ ὀρθὴ ἡ ὑπὸ $BE\Delta$ ὀρθὴ τῇ ὑπὸ $BZ\Delta$ ἴση, δύο δὲ γωνίαι ἐστὶ τὰ $EB\Delta$, $ZB\Delta$ τὰς δύο γωνίας ταῖς δυοῖ γωνίαις ἴσαις ἔχοντα καὶ μίαν πλευρὰν μὲν ΔE πλεονάζει ἴσην τὴν ὑποτείνουσαν ὑπὸ μίαν τῶν ἴσων γωνιῶν κοινὴν αὐτῶν τὴν $B\Delta$. καὶ τὰς λοιπὰς ἄρα πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαις ἔχουσιν. ἴση ἄρα ἡ ΔE τῇ ΔZ . διὰ τὰ αὐτὰ δὲ καὶ ἡ ΔH τῇ ΔZ ἐστὶν ἴση. αἱ τρεῖς ἄρα εὐθεῖαι αἱ ΔE , ΔZ , ΔH ἴσαι ἀλλήλαις εἰσὶν. ὁ ἄρα κέντρον τῷ Δ καὶ διαστήματι ἐνὶ τῶν E , Z , H κύκλος γραφόμενος ἔχει καὶ διὰ τῶν λοιπῶν σημείων καὶ ἐφάπεται τῶν AB , $B\Gamma$, $\Gamma\Delta$ εὐθειῶν διὰ τὸ ὀρθὰς εἶναι τὰς πρὸς τοῖς E , Z , H σημείοις γωνίας. εἰ γὰρ τεμεῖ αὐτάς, ἔσται ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἀκρᾶς ἀγομένη ἐντὸς πίπτουσα τοῦ κύκλου. ὅπερ ἄτοπον ἐδείχθη. οὐκ ἄρα ὁ κέντρον τῷ Δ διαστήματι δὲ ἐνὶ τῶν E , Z , H γραφόμενος

Proposition 4

To inscribe a circle in a given triangle.



Let ABC be the given triangle. So it is required to inscribe a circle in triangle ABC .

Let the angles ABC and ACB be cut in half by the straight-lines BD and CD (respectively) [Prop. 1.9], and let them meet one another at point D , and let DE , DF , and DG be drawn from point D , perpendicular to the straight-lines AB , BC , and CA (respectively) [Prop. 1.12].

And since angle ABD is equal to CBD , and the right-angle BED is also equal to the right-angle BFD , EBD and FBD are thus two triangles having two angles equal to two angles, and one side equal to one side—the (one) subtending one of the equal angles (which is) common to the (triangles)—(namely), BD . Thus, they will also have the remaining sides equal to the (corresponding) remaining sides [Prop. 1.26]. Thus, DE (is) equal to DF . So, for the same (reasons), DG is also equal to DF . Thus, the three straight-lines DE , DF , and DG are equal to one another. Thus, the circle drawn with center D , and radius one of E , F , or G ,[†] will also go through the remaining points, and will touch the straight-lines AB , BC , and CA , on account of the angles at E , F , and G being right-angles. For if it cuts (one of) them then it will be a (straight-line) drawn

κύκλος τεμεί τὰς AB , BF , $ΓA$ εὐθείας· ἐφάπεται ἄρα αὐτῶν, καὶ ἔσται ὁ κύκλος ἐγγεγραμμένος εἰς τὸ $ABΓ$ τρίγωνον. ἐγγεγράφθω ὡς ὁ ZHE .

Εἰς ἄρα τὸ δοθὲν τρίγωνον τὸ $ABΓ$ κύκλος ἐγγέγραπται ὁ EZH · ὅπερ ἔδει ποιῆσαι.

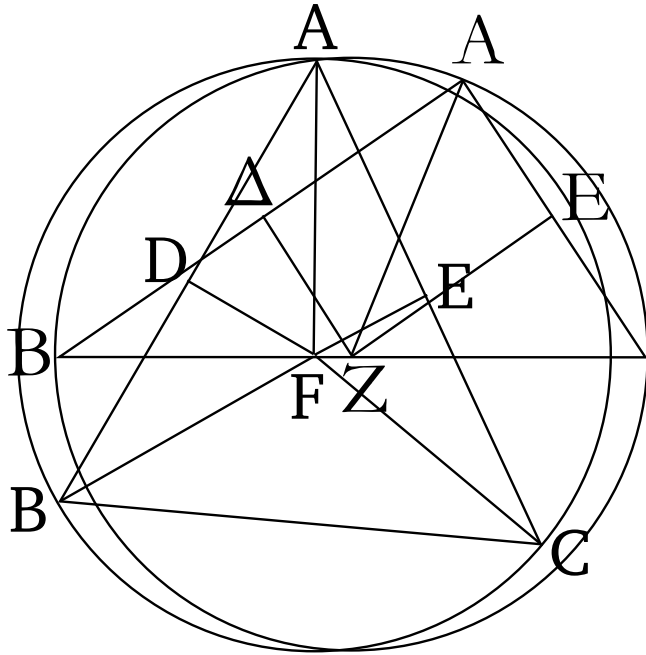
at right-angles to a diameter of the circle, from its extremity, falling inside the circle. The very thing was shown (to be) absurd [Prop. 3.16]. Thus, the circle drawn with center D , and radius one of E , F , or G , does not cut the straight-lines AB , BC , and CA . Thus, it will touch them, and will be the circle inscribed in triangle ABC . Let it be (so) inscribed, like FGE (in the figure).

Thus, the circle EFG has been inscribed in the given triangle ABC . (Which is) the very thing it was required to do.

† Here, and in the following propositions, it is understood that the radius is actually one of DE , DF , or DG .

5.

Περὶ τὸ δοθὲν τρίγωνον κύκλον περιγράφαι.



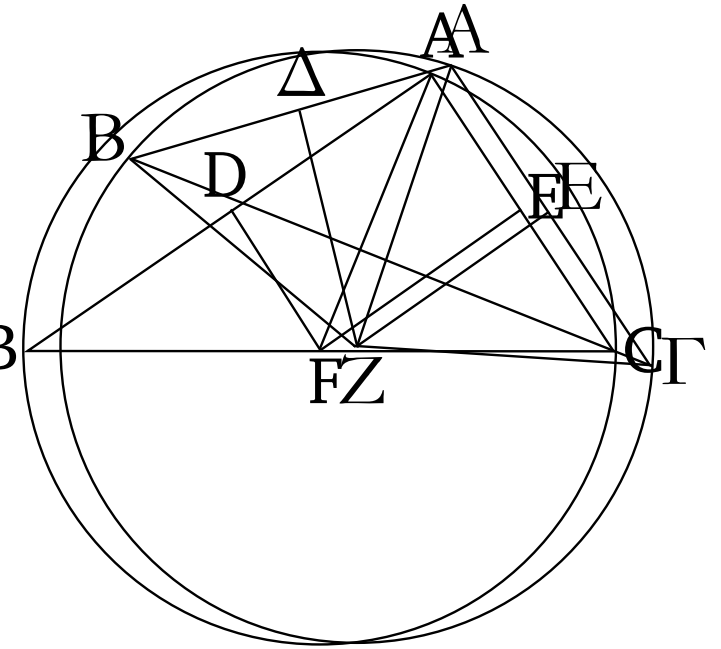
Ἐστω τὸ δοθὲν τρίγωνον τὸ $ABΓ$. δεῖ δὲ περὶ τὸ δοθὲν τρίγωνον τὸ $ABΓ$ κύκλον περιγράψαι.

Τετμήσθωσαν αἱ AB , AC εὐθεῖαι δίχα κατὰ τὰ D , E σημεία, καὶ ἀπὸ τῶν D , E σημείων ταῖς AB , AC πρὸς ὀρθὰς ἡθῇωσαν αἱ $ΔZ$, EZ · συμπεσοῦνται δὴ ἥτοι ἐντὸς τοῦ $ABΓ$ τριγώνου ἢ ἐπὶ τῆς BF εὐθείας ἢ ἐκτὸς τῆς BF .

Συμπιπτέτωσαν πρότερον ἐντὸς κατὰ τὸ Z , καὶ ἐπεξέσθωσαν αἱ ZB , $ZΓ$, ZA . καὶ ἐπεὶ ἴση ἐστὶν ἡ AD τῇ DB , κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ $ΔZ$, βάσεις ἄρα ἡ AZ βάσει τῇ ZB ἐστὶν ἴση. ὁμοίως δὴ δείχμεν, ὅτι καὶ ἡ $ΓZ$ τῇ AZ ἐστὶν ἴση· ὥστε καὶ ἡ ZB τῇ $ZΓ$ ἐστὶν ἴση· αἱ τρεῖς ἄρα αἱ ZA , ZB , $ZΓ$ ἴσαι ἀλλήλαις εἰσὶν. ὁ ἄρα κέντρον τῷ Z διαστήματι δὲ ἐνὶ τῶν A , B , $Γ$ κύκλος γραφόμενος ἔχει καὶ διὰ τῶν λοιπῶν σημείων, καὶ ἔσται περιγεγραμμένος ὁ κύκλος περὶ τὸ $ABΓ$ τρίγωνον. περιγεγράφθω ὡς ὁ $ABΓ$.

Proposition 5

To circumscribe a circle about a given triangle.



Let ABC be the given triangle. So it is required to circumscribe a circle about the given triangle ABC .

Let the straight-lines AB and AC be cut in half at points D and E (respectively) [Prop. 1.10]. And let DF and EF be drawn from points D and E , at right-angles to AB and AC (respectively) [Prop. 1.11]. So (DF and EF) will surely either meet inside triangle ABC , on the straight-line BC , or beyond BC .

Let them, first of all, meet inside (triangle ABC) at (point) F , and let FB , FC , and FA be joined. And since AD is equal to DB , and DF is common and at right-angles, the base AF is thus equal to the base FB [Prop. 1.4]. So, similarly, we can show that CF is also equal to AF . So that FB is also equal to FC . Thus, the three (straight-lines) FA , FB , and FC are equal to one another. Thus, the circle drawn with center F ,

Ἀλλὰ δὴ αἱ ΔZ , EZ συμπιπτεύουσιν ἐπὶ τῇ $B\Gamma$ εὐθείᾳ κατὰ τὸ Z , ὥς ἔδει ἐπὶ τῇ δευτέρᾳ καταγραφῇ, καὶ ἐπεξεύθθω ἡ AZ . ὁμοίως δὲ δείχομεν, ὅτι τὸ Z σημεῖον κέντρον ἐστὶ τοῦ περὶ τὸ $AB\Gamma$ τριγώνου περιγεγραμμένου κύκλου.

Ἀλλὰ δὴ αἱ ΔZ , EZ συμπιπτεύουσιν ἐκτὸς τοῦ $AB\Gamma$ τριγώνου κατὰ τὸ Z πάλιν, ὥς ἔδει ἐπὶ τῇ τρίτῃ καταγραφῇ, καὶ ἐπεξεύθθωσαν αἱ AZ , BZ , ΓZ . καὶ ἐπεὶ πάλιν ἴση ἐστὶν ἡ $A\Delta$ τῇ ΔB , κοινὴ δὲ καὶ πρόξ ὀρθὰς ἡ ΔZ , βάσις ἄρα ἡ AZ βάσει τῇ BZ ἐστὶν ἴση. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΓZ τῇ AZ ἐστὶν ἴση· ὥστε καὶ ἡ BZ τῇ ΓZ ἐστὶν ἴση· ὁ ἄρα [πάλιν] κέντρον τῷ Z διαστήματι δὲ ἐνὶ τῶν ZA , ZB , $Z\Gamma$ κύκλος γεγραμμένος ἥξει καὶ διὰ τῶν λοιπῶν σημείων, καὶ ἔσται περιγεγραμμένος περὶ τὸ $AB\Gamma$ τριγώνον.

Περὶ τὸ δοθέν ἄρα τριγώνον κύκλος περιγέγραπται· ὅπερ ἔδει ποιῆσαι.

6.

Εἰς τὸν δοθέντα κύκλον τετράγωνον ἐγγράψαι.

Ἐστω ἡ δοθεὶς κύκλος ὁ $AB\Gamma\Delta$. δεῖ δὲ εἰς τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον ἐγγράψαι.

Ἦθθωσαν τοῦ $AB\Gamma\Delta$ κύκλου δύο διαμέτροι πρόξ ὀρθὰς ἀλλήλαις αἱ $A\Gamma$, $B\Delta$, καὶ ἐπεξεύθθωσαν αἱ AB , $B\Gamma$, $\Gamma\Delta$, ΔA .

and radius one of A , B , or C , will also go through the remaining points. And the circle will be circumscribed about triangle ABC . Let it be (so) circumscribed, like ABC (in the first diagram from the left).

And so, let DF and EF meet on the straight-line BC at (point) F , like in the second diagram (from the left). And let AF be joined. So, similarly, we can show that point F is the center of the circle circumscribed about triangle ABC .

And so, let DF and EF meet outside triangle ABC , again at (point) F , like in the third diagram (from the left). And let AF , BF , and CF be joined. And, again, since AD is equal to DB , and DF is common and at right-angles, the base AF is thus equal to the base BF [Prop. 1.4]. So, similarly, we can show that CF is also equal to AF . So that BF is also equal to FC . Thus, [again] the circle drawn with center F , and radius one of FA , FB , and FC , will also go through the remaining points. And it will be circumscribed about triangle ABC .

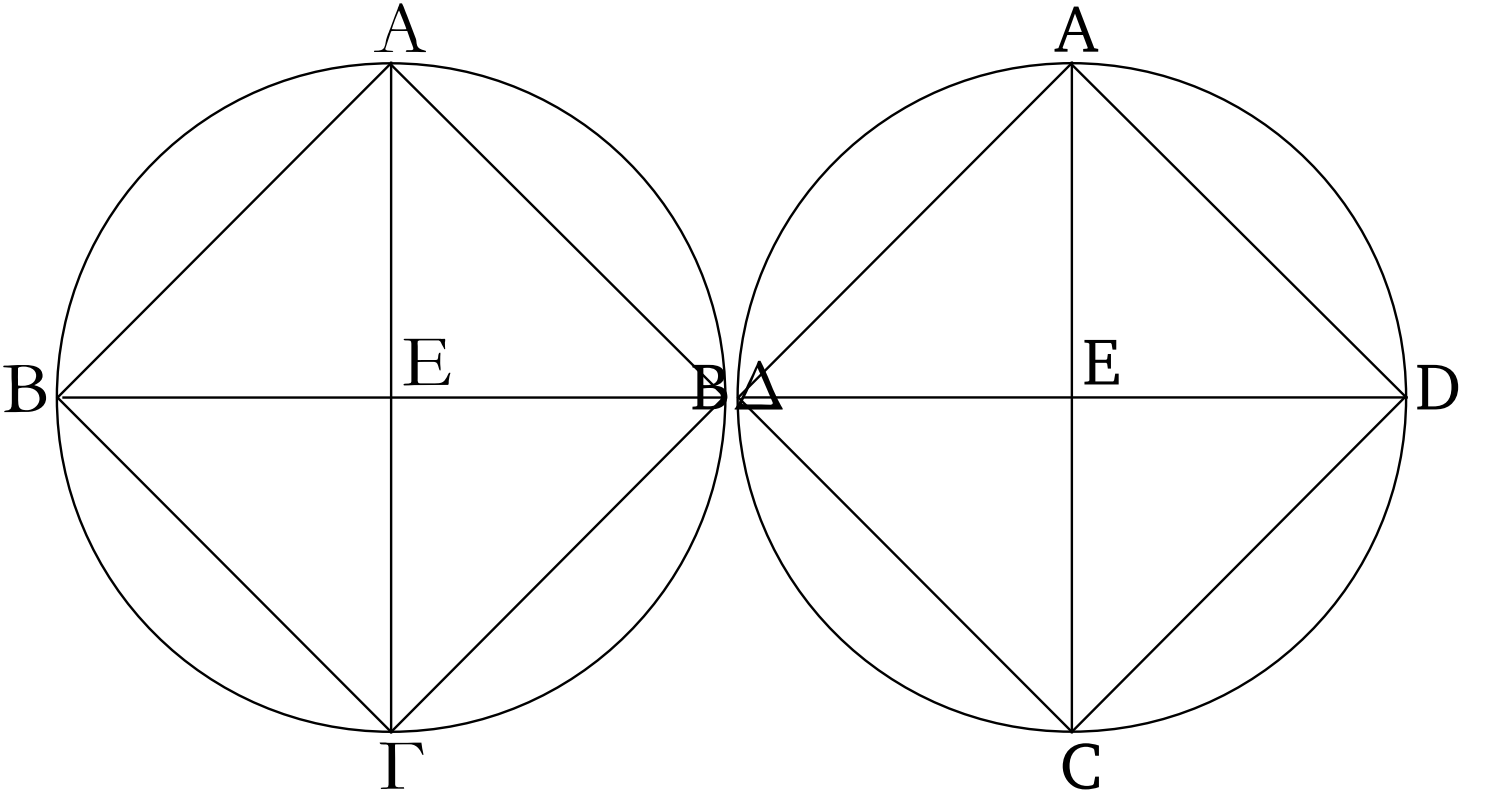
Thus, a circle has been circumscribed about the given triangle. (Which is) the very thing it was required to do.

Proposition 6

To inscribe a square in a given circle.

Let $ABCD$ be the given circle. So it is required to inscribe a square in circle $ABCD$.

Let two diameters of circle $ABCD$, AC and BD , be drawn at right-angles to one another.[†] And let AB , BC , CD , and DA be joined.



Καὶ ἐπεὶ ἴση ἐστὶν ἡ BE τῇ ED · κέντρον γάρ τὸ E · κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ EA , βάσεις ἄρα ἡ AB βάσει τῇ AD ἴση ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρω τῶν $BΓ$, $ΓΔ$ ἑκατέρω τῶν AB , AD ἴση ἐστίν· ἰσόπλευρον ἄρα ἐστὶ τὸ $ABΓΔ$ τετράπλευρον. λέγω δὴ, ὅτι καὶ ὀρθογώνιον. ἐπεὶ γάρ ἡ BD εὐθεῖα διάμετρος ἐστὶ τοῦ $ABΓΔ$ κύκλου, ἡμικύκλιον ἄρα ἐστὶ τὸ BAD · ὀρθὴ ἄρα ἡ ὑπὸ BAD γωνία. διὰ τὰ αὐτὰ δὴ καὶ ἑκάστη τῶν ὑπὸ $ABΓ$, $BΓΔ$, $ΓΔA$ ὀρθὴ ἐστίν· ὀρθογώνιον ἄρα ἐστὶ τὸ $ABΓΔ$ τετράπλευρον. ἐδείχθη δὲ καὶ ἰσόπλευρον τετράγωνον ἄρα ἐστίν. καὶ ἐγγέγραπται εἰς τὸν $ABΓΔ$ κύκλον.

Εἰς ἄρα τὸν δοθέντα κύκλον τετράγωνον ἐγγέγραπται τὸ $ABΓΔ$ · ὅπερ ἔδει ποιῆσαι.

And since BE is equal to ED , for E (is) the center (of the circle), and EA is common and at right-angles, the base AB is thus equal to the base AD [Prop. 1.4]. So, for the same (reasons), each of BC and CD is equal to each of AB and AD . Thus, the quadrilateral $ABCD$ is equilateral. So I say that (it is) also right-angled. For since the straight-line BD is a diameter of circle $ABCD$, BAD is thus a semi-circle. Thus, angle BAD (is) a right-angle [Prop. 3.31]. So, for the same (reasons), (angles) ABC , BCD , and CDA are also each right-angles. Thus, the quadrilateral $ABCD$ is right-angled. And it was also shown (to be) equilateral. Thus, it is a square [Def. 1.22]. And it has been inscribed in circle $ABCD$.

Thus, the square $ABCD$ has been inscribed in the given circle. (Which is) the very thing it was required to do.

† Presumably, by finding the center of the circle [Prop. 3.1], drawing a line through it, and then drawing a second line through it, at right-angles to the first [Prop. 1.11].

7.

Proposition 7

Περὶ τὸν δοθέντα κύκλον τετράγωνον περιγράψαι.

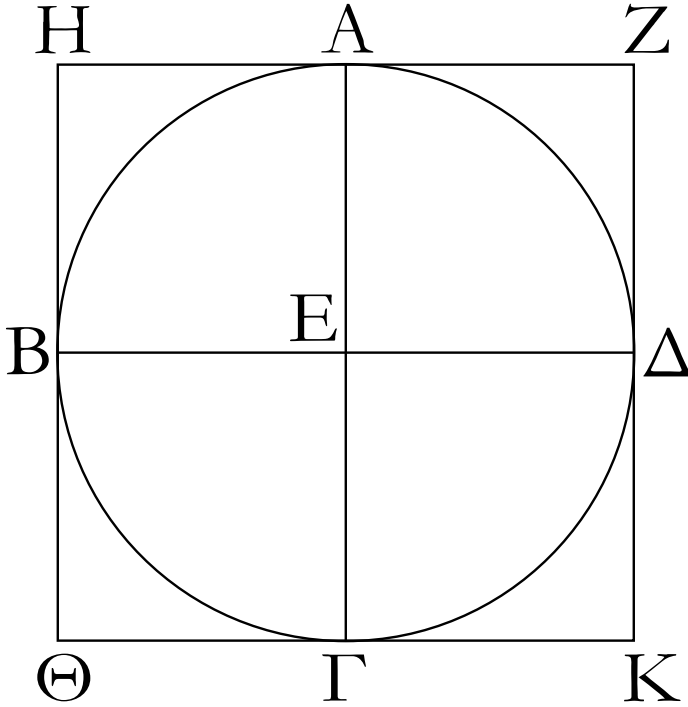
Ἐστω ὁ δοθεὶς κύκλος ὁ $ABΓΔ$ · δεῖ δὴ περὶ τὸν $ABΓΔ$ κύκλον τετράγωνον περιγράψαι.

Ἦθθωσαν τοῦ $ABΓΔ$ κύκλου δύο διαμέτροι πρὸς ὀρθὰς ἀλλήλαις αἱ AG , BD , καὶ διὰ τῶν A , B , $Γ$, $Δ$ σημείων ἡθθωσαν ἐφαπτόμεναι τοῦ $ABΓΔ$ κύκλου αἱ ZH , $HΘ$, $ΘK$, KZ .

To circumscribe a square about a given circle.

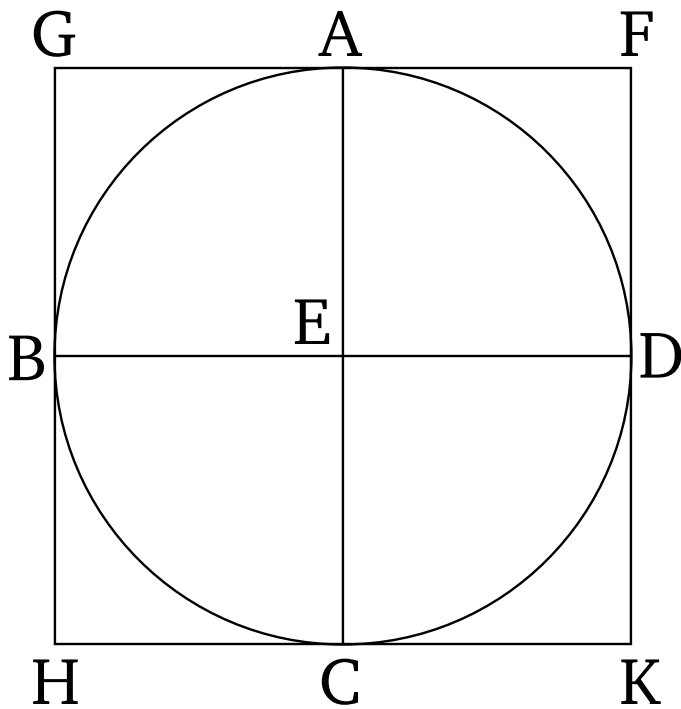
Let $ABCD$ be the given circle. So it is required to circumscribe a square about circle $ABCD$. Let two diameters of circle $ABCD$, AC and BD , be drawn at right-angles to one another.[†] And let FG , GH , HK , and KF be drawn through points A , B , C , and D (respectively), touching circle $ABCD$.[‡]

Ἐπεὶ οὖν ἐφάπτεται ἡ ZH τοῦ $AB\Gamma\Delta$ κύκλου, ἀπὸ δὲ τοῦ E κέντρου ἐπὶ τὴν κατὰ τὸ A ἐπαφὴν ἐπέξενεται ἡ EA , αἱ ἄρα πρὸς τῷ A γωνίαι ὀρθαί εἰσιν. διὰ τὰ αὐτὰ δὴ καὶ αἱ πρὸς τοῖς B , Γ , Δ σημείοις γωνίαι ὀρθαί εἰσιν. καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ ὑπὸ AEB γωνία, ἐστὶ δὲ ὀρθὴ καὶ ἡ ὑπὸ EBH , παράλληλος ἄρα ἐστὶν ἡ $H\Theta$ τῇ AG . διὰ τὰ αὐτὰ δὴ καὶ ἡ AG τῇ ZK ἐστὶ παράλληλος. ὥστε καὶ ἡ $H\Theta$ τῇ ZK ἐστὶ παράλληλος. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἑκάτερα τῶν HZ , ΘK τῇ $BE\Delta$ ἐστὶ παράλληλος. παραλληλόγραμμα ἄρα ἐστὶ τὰ HK , $H\Gamma$, AK , ZB , BK . ἴση ἄρα ἐστὶν ἡ μὲν HZ τῇ ΘK , ἡ δὲ $H\Theta$ τῇ ZK . καὶ ἐπεὶ ἴση ἐστὶν ἡ AG τῇ $B\Delta$, ἀλλὰ καὶ ἡ μὲν AG ἑκατέρα τῶν $H\Theta$, ZK , ἡ δὲ $B\Delta$ ἑκατέρα τῶν HZ , ΘK ἐστὶν ἴση [καὶ ἑκάτερα ἄρα τῶν $H\Theta$, ZK ἑκατέρα τῶν HZ , ΘK ἐστὶν ἴση], ἰσόπλευρον ἄρα ἐστὶ τὸ $ZH\Theta K$ τετράπλευρον. λέγω δὴ, ὅτι καὶ ὀρθογώνιον. ἐπεὶ γάρ παραλληλόγραμμόν ἐστὶ τὸ $HBEA$, καὶ ἐστὶν ὀρθὴ ἡ ὑπὸ AEB , ὀρθὴ ἄρα καὶ ἡ ὑπὸ AHB . ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ πρὸς τοῖς Θ , K , Z γωνίαι ὀρθαί εἰσιν. ὀρθογώνιον ἄρα ἐστὶ τὸ $ZH\Theta K$. ἐδείκθη δὲ καὶ ἰσόπλευρον· τετράγωνον ἄρα ἐστίν. καὶ περιγέγραπται περὶ τὸν $AB\Gamma\Delta$ κύκλον.



Therefore, since FG touches circle $ABCD$, and EA has been joined from the center E to the point of contact A , the angles at A are thus right-angles [Prop. 3.18]. So, for the same (reasons), the angles at points B , C , and D are also right-angles. And since angle AEB is a right-angle, and EBG is also a right-angle, GH is thus parallel to AC [Prop. 1.29]. So, for the same (reasons), AC is also parallel to FK . So that GH is also parallel to FK [Prop. 1.30]. So, similarly, we can show that GF and HK are each parallel to $BE\Delta$. Thus, GK , GC , AK , FB , and BK are (all) parallelograms. Thus, GF is equal to HK , and GH to FK [Prop. 1.34]. And since AC is equal to BD , but AC (is) also (equal) to each of GH and FK , and BD is equal to each of GF and HK [Prop. 1.34] [and each of GH and FK is thus equal to each of GF and HK], the quadrilateral $FGHK$ is thus equilateral. So I say that (it is) also right-angled. For since $GBEA$ is a parallelogram, and AEB is a right-angle, AGB is thus also a right-angle [Prop. 1.34]. So, similarly, we can show that the angles at H , K , and F are also right-angles. Thus, $FGHK$ is right-angled. And it was also shown (to be) equilateral. Thus, it is a square [Def. 1.22]. And

it has been circumscribed about circle $ABCD$.



Περὶ τὸν δοθέντα ἄρα κύκλον τετράγωνον περιγέγραπται·
ὅπερ ἔδει ποιῆσαι.

Thus, a square has been circumscribed about the given circle. (Which is) the very thing it was required to do.

† See the footnote to the previous proposition.

* See the footnote to Prop. 3.34.

8.

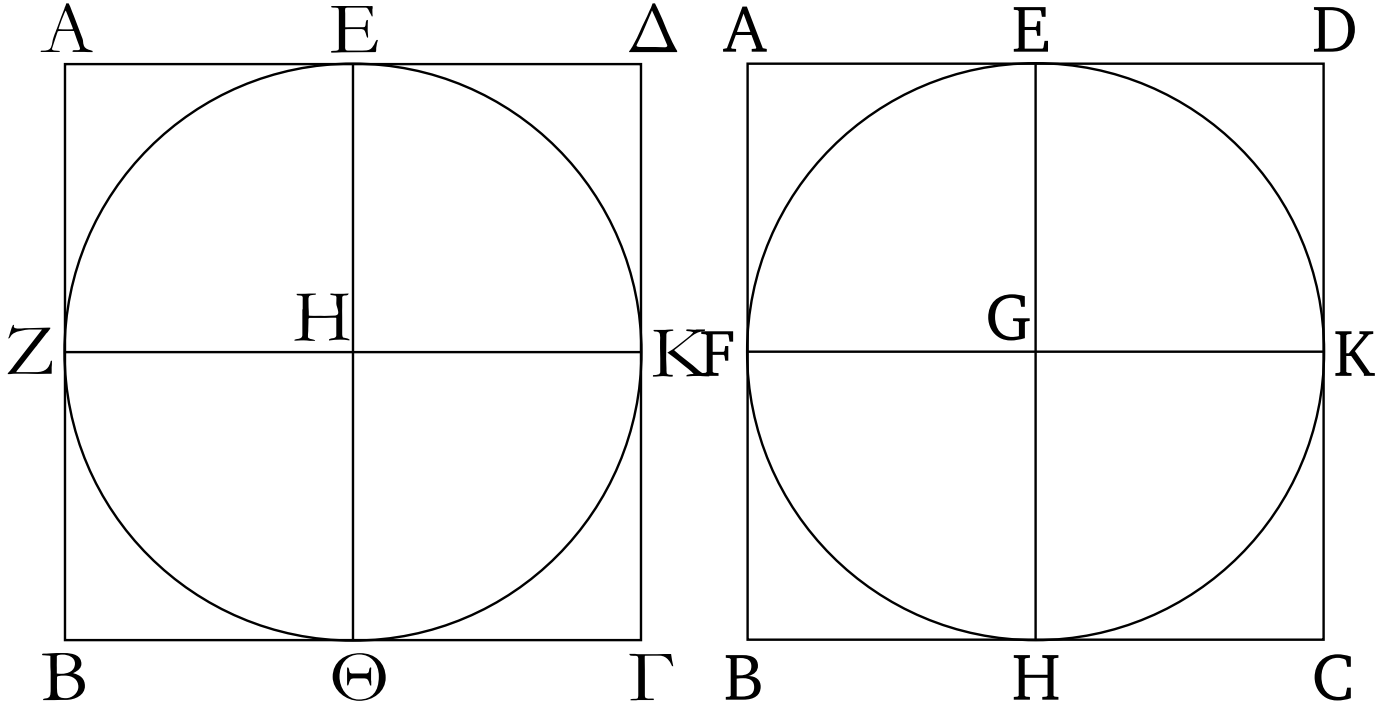
Proposition 8

Εἰς τὸ δοθέν τετράγωνον κύκλον ἐγγράψαι.

To inscribe a circle in a given square.

Ἐστω τὸ δοθέν τετράγωνον τὸ $AB\Gamma\Delta$. δεῖ δὴ εἰς τὸ $AB\Gamma\Delta$ τετράγωνον κύκλον ἐγγράψαι.

Let the given square be $ABCD$. So it is required to inscribe a circle in square $ABCD$.



Τετμήσθω ἑκατέρω τῶν AD , AB διῖθα κατὰ τὰ E , Z σημεία, καὶ διὰ μὲν τοῦ E ὁποτέρω τῶν AB , $ΓΔ$ παράλληλος ἦθθω ὁ $EΘ$, διὰ δὲ τοῦ Z ὁποτέρω τῶν AD , $BΓ$ παράλληλος ἦθθω ἡ ZK . παραλληλόγραμμοι ἄρα ἐστὶν ἕκαστος τῶν AK , KB , $AΘ$, $ΘΔ$, AH , $HΓ$, BH , $HΔ$, καὶ αἱ ἀπεναντίον αὐτῶν πλευραὶ δηλονότι ἴσαι [εἰσίν]. καὶ ἐπεὶ ἴση ἐστὶν ἡ AD τῇ AB , καὶ ἐστὶ τῆς μὲν AD ἡμίσεια ἡ AE , τῆς δὲ AB ἡμίσεια ἡ AZ , ἴση ἄρα καὶ ἡ AE τῇ AZ . ὥστε καὶ αἱ ἀπεναντίον ἴση ἄρα καὶ ἡ ZH τῇ HE . ὁμοίως δὲ δείχομεν, ὅτι καὶ ἑκατέρω τῶν $HΘ$, HK ἑκατέρω τῶν ZH , HE ἐστὶν ἴση. αἱ τέσσαρες ἄρα αἱ HE , HZ , $HΘ$, HK ἴσαι ἀλλήλαις [εἰσίν]. ὁ ἄρα κέντρον μὲν τῷ H διαστήματι δὲ ἐνὶ τῶν E , Z , $Θ$, K κύκλος γραφόμενος ἔχει καὶ διὰ τῶν λοιπῶν σημείων· καὶ ἐφάπεται τῶν AB , $BΓ$, $ΓΔ$, $ΔΑ$ εὐθειῶν διὰ τὸ ὀρθὰς εἶναι τὰς πρὸς τοῖς E , Z , $Θ$, K γωνίας· εἰ γάρ τεμεί ὁ κύκλος τὰς AB , $BΓ$, $ΓΔ$, $ΔΑ$, ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένη ἐντὸς πεσεῖται τοῦ κύκλου· ὅπερ ἄτοπον ἐδείχθη. οὐκ ἄρα ὁ κέντρον τῷ H διαστήματι δὲ ἐνὶ τῶν E , Z , $Θ$, K κύκλος γραφόμενος τεμεί τὰς AB , $BΓ$, $ΓΔ$, $ΔΑ$ εὐθείας. ἐφάπεται ἄρα αὐτῶν καὶ ἔσται ἐγγεγραμμένος εἰς τὸ $ABΓΔ$ τετράγωνον.

Εἰς ἄρα τὸ δοθὲν τετράγωνον κύκλος ἐγγέγραπται· ὅπερ ἔδει ποιῆσαι.

9.

Περὶ τὸ δοθὲν τετράγωνον κύκλον περιγράφαι.

Let AD and AB each be cut in half at points E and F (respectively) [Prop. 1.10]. And let EH be drawn through E , parallel to either of AB or CD , and let FK be drawn through F , parallel to either of AD or BC [Prop. 1.31]. Thus, AK , KB , AH , HD , AG , GC , BG , and GD are each parallelograms, and their opposite sides [are] manifestly equal [Prop. 1.34]. And since AD is equal to AB , and AE is half of AD , and AF half of AB , AE (is) thus also equal to AF . So that the opposite (sides are) also (equal). Thus, FG (is) also equal to GE . So, similarly, we can also show that each of GH and GK is equal to each of FG and GE . Thus, the four (straight-lines) GE , GF , GH , and GK [are] equal to one another. Thus, the circle drawn with center G , and radius one of E , F , H , or K , will also go through the remaining points. And it will touch the straight-lines AB , BC , CD , and DA , on account of the angles at E , F , H , and K being right-angles. For if the circle cuts AB , BC , CD , or DA , (then) a (straight-line) drawn at right-angles to a diameter of the circle, from its extremity, will fall inside the circle. The very thing was shown (to be) absurd [Prop. 3.16]. Thus, the circle drawn with center G , and radius one of E , F , H , or K , does not cut the straight-lines AB , BC , CD , or DA . Thus, it will touch them, and will be inscribed in the square $ABCD$.

Thus, a circle has been inscribed in the given square. (Which is) the very thing it was required to do.

Proposition 9

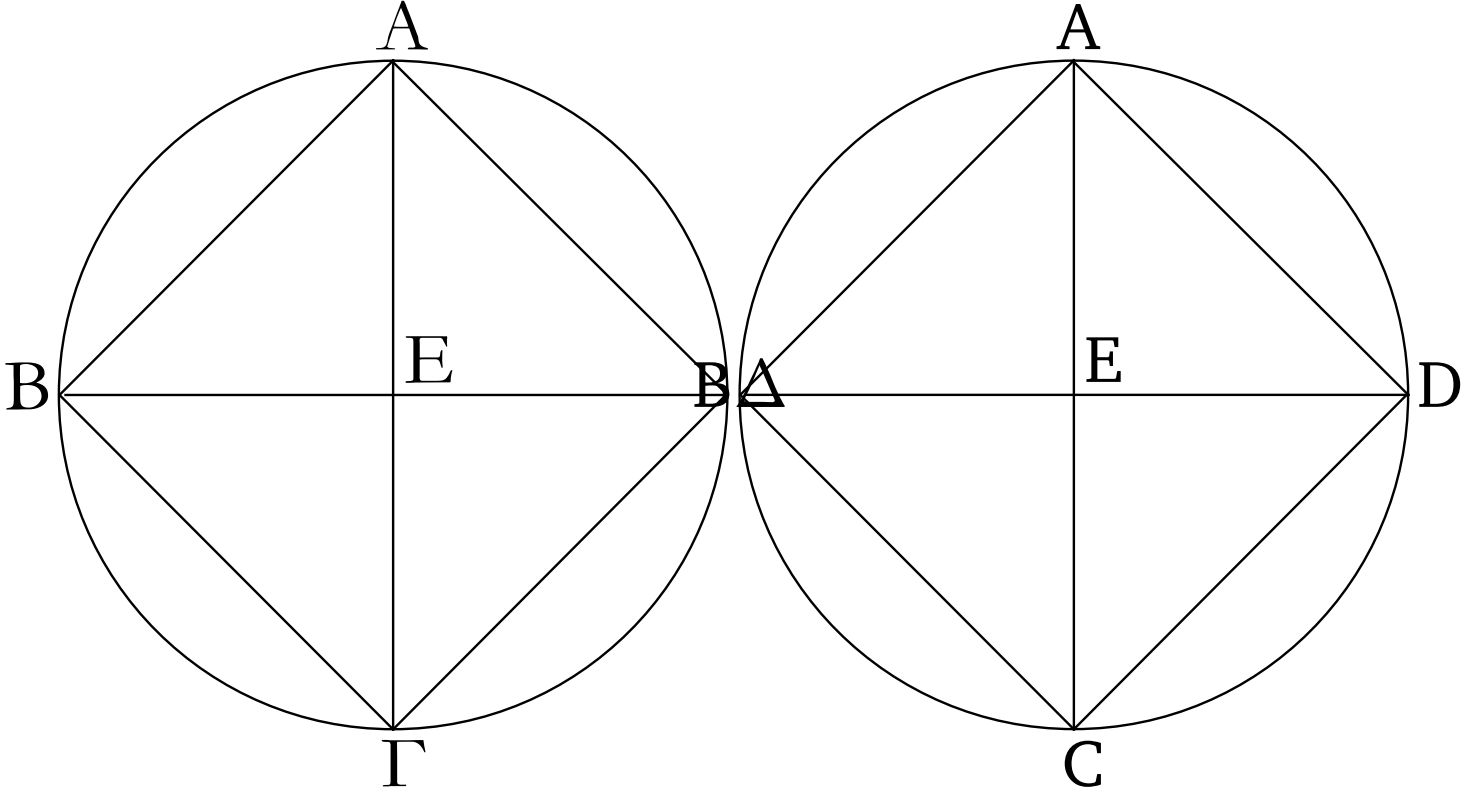
To circumscribe a circle about a given square.

Ἐστω τὸ δοθέν τετράγωνον τὸ $AB\Gamma\Delta$. δεῖ δὴ περὶ τὸ $AB\Gamma\Delta$ τετράγωνον κύκλον περιγράψαι.

Ἐπιξενθθεῖσαι γὰρ αἱ AG , $B\Delta$ τεμνέτωσαν ἀλλήλας κατὰ τὸ E .

Let $ABCD$ be the given square. So it is required to circumscribe a circle about square $ABCD$.

AC and BD being joined, let them cut one another at E .



Καὶ ἐπεὶ ἴση ἐστὶν ἡ ΔA τῇ AB , κοινὴ δὲ ἡ AG , δύο δὴ αἱ ΔA , AG δυοὶ ταῖς BA , AG ἴσαι εἰσὶν καὶ βάσις ἡ $\Delta\Gamma$ βάσει τῇ $B\Gamma$ ἴση· γωνία ἄρα ἡ ὑπὸ ΔAG γωνία τῇ ὑπὸ BAG ἴση ἐστὶν· ἡ ἄρα ὑπὸ ΔAB γωνία δίῃα τέτμηται ὑπὸ τῆς AG . ὁμοίως δὲ δείχμεν, ὅτι καὶ ἐκάστη τῶν ὑπὸ $AB\Gamma$, $B\Gamma\Delta$, $\Gamma\Delta A$ δίῃα τέτμηται ὑπὸ τῶν AG , ΔB εὐθειῶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ ΔAB γωνία τῇ ὑπὸ $AB\Gamma$, καὶ ἐστὶ τῆς μὲν ὑπὸ ΔAB ἡμίσεια ἡ ὑπὸ EAB , τῆς δὲ ὑπὸ $AB\Gamma$ ἡμίσεια ἡ ὑπὸ EBA , καὶ ἡ ὑπὸ EAB ἄρα τῇ ὑπὸ EBA ἐστὶν ἴση· ὥστε καὶ πλευρὰ ἡ EA τῇ EB ἐστὶν ἴση. ὁμοίως δὲ δείχμεν, ὅτι καὶ ἐκάτερα τῶν EA , EB [εὐθειῶν] ἐκάτερα τῶν EG , ED ἴση ἐστὶν. αἱ τέσσαρες ἄρα αἱ EA , EB , EG , ED ἴσαι ἀλλήλαις εἰσὶν. ὁ ἄρα κέντρον τῷ E καὶ διαστήματι ἐνὶ τῶν A , B , Γ , Δ κύκλος γραφόμενος ἥξει καὶ διὰ τῶν λοιπῶν σημείων καὶ ἔσται περιγεγραμμένος περὶ τὸ $AB\Gamma\Delta$ τετράγωνον. περιγεγράφθω ὡς ὁ $AB\Gamma\Delta$.

Περὶ τὸ δοθέν ἄρα τετράγωνον κύκλος περιγράφεται· ὅπερ ἔδει ποιῆσαι.

And since DA is equal to AB , and AC (is) common, the two (straight-lines) DA , AC are thus equal to the two (straight-lines) BA , AC . And the base DC (is) equal to the base BC . Thus, angle DAC is equal to angle BAC [Prop. 1.8]. Thus, the angle DAB has been cut in half by AC . So, similarly, we can show that ABC , BCD , and CDA have each been cut in half by the straight-lines AC and DB . And since angle DAB is equal to ABC , and EAB is half of DAB , and EBA half of ABC , EAB is thus also equal to EBA . So that side EA is also equal to EB [Prop. 1.6]. So, similarly, we can show that each of the [straight-lines] EA and EB are also equal to each of EC and ED . Thus, the four (straight-lines) EA , EB , EC , and ED are equal to one another. Thus, the circle drawn with center E , and radius one of A , B , C , or D , will also go through the remaining points, and will be circumscribed about the square $ABCD$. Let it be (so) circumscribed, like $ABCD$ (in the figure).

Thus, a circle has been circumscribed about the given square. (Which is) the very thing it was required to do.

10.

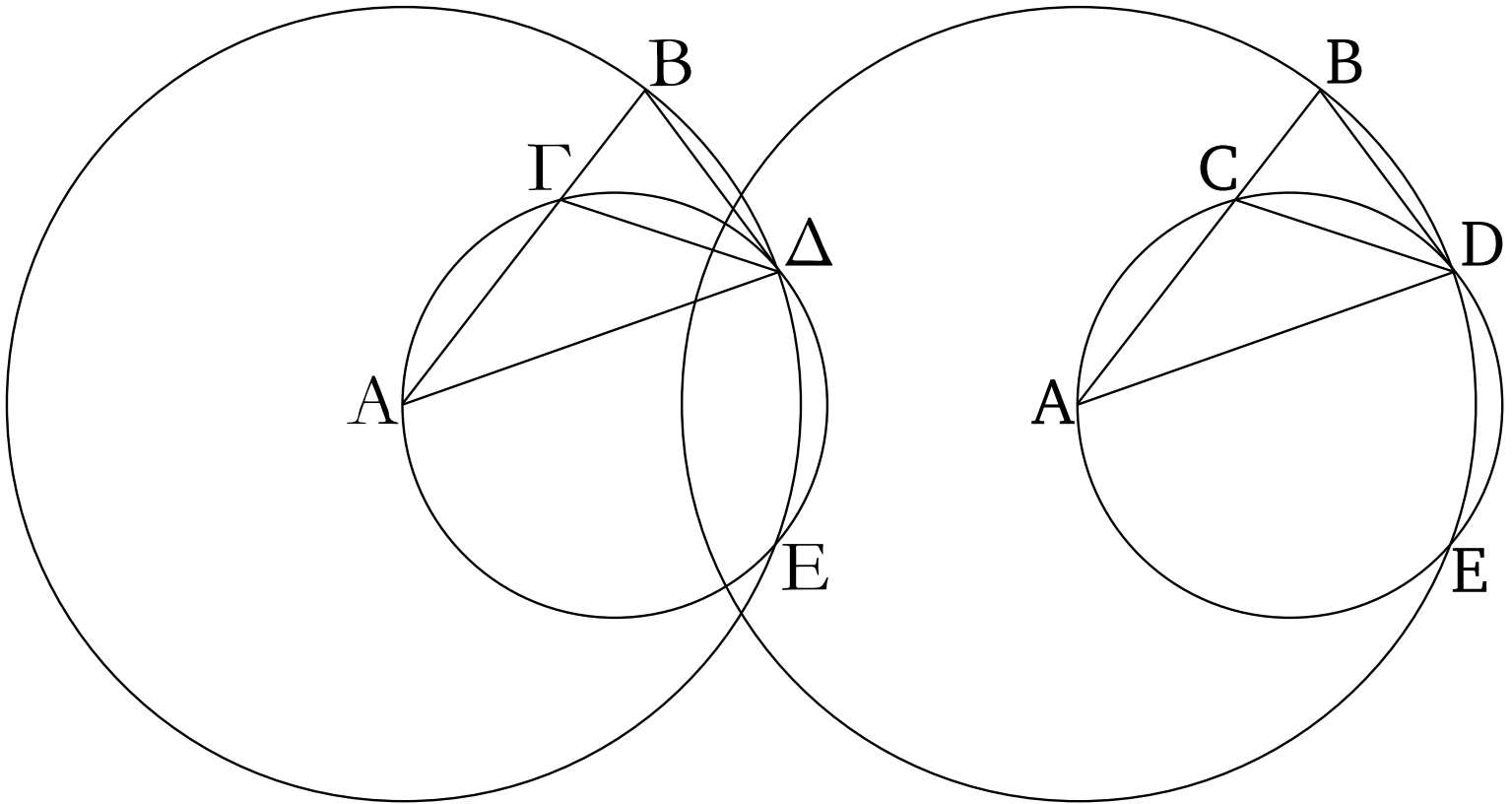
Proposition 10

Ἰσοσκελὲς τρίγωνον συστήσασθαι ἔθον ἐκατέραν τῶν

To construct an isosceles triangle having each of the angles

πρὸς τῇ βάσει γωνιῶν διπλασίονα τῇ λοιπῇ.

at the base double the remaining (angle).



Ἐκκείσθω τις εὐθεΐα ἡ AB , καὶ τετμήσθω κατὰ τὸ Γ σημεῖον, ὥστε τὸ ὑπὸ τῶν AB , $B\Gamma$ περιεχόμενον ὀρθογώνιον ἴσον εἶναι τῷ ἀπὸ τῆς ΓA τετραγώνῳ· καὶ κέντρῳ τῷ A καὶ διαστήματι τῷ AB κύκλος γεγράφθω ὁ $B\Delta E$, καὶ ἐνηρμόσθω εἰς τὸν $B\Delta E$ κύκλον τῇ AG εὐθείᾳ μὴ μείζονι ὅσῃ τῆς τοῦ $B\Delta E$ κύκλου διαμέτρου ἴση εὐθεΐα ἡ $B\Delta$ · καὶ ἐπεξεύθθωσαν αἱ AD , $\Delta\Gamma$, καὶ περιγεγράφθω περὶ τὸ $AG\Delta$ τρίγωνον κύκλος ὁ $AG\Delta$.

Καὶ ἐπεὶ τὸ ὑπὸ τῶν AB , $B\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AG , ἴση δὲ ἡ AG τῇ $B\Delta$, τὸ ἄρα ὑπὸ τῶν AB , $B\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $B\Delta$. καὶ ἐπεὶ κύκλον τοῦ $AG\Delta$ εἴληπται τι σημεῖον ἐκτὸς τὸ B , καὶ ἀπὸ τοῦ B πρὸς τὸν $AG\Delta$ κύκλον προσεπτώκασι δύο εὐθεΐαι αἱ BA , $B\Delta$, καὶ ἡ μὲν αὐτῶν τέμνει, ἡ δὲ προσπίπτει, καὶ ἐστὶ τὸ ὑπὸ τῶν AB , $B\Gamma$ ἴσον τῷ ἀπὸ τῆς $B\Delta$, ἡ $B\Delta$ ἄρα ἐφάπτεται τοῦ $AG\Delta$ κύκλου. ἐπεὶ οὖν ἐφάπτεται μὲν ἡ $B\Delta$, ἀπὸ δὲ τῆς κατὰ τὸ Δ ἐπαφῆς διῆκται ἡ $\Delta\Gamma$, ἡ ἄρα ὑπὸ $B\Delta\Gamma$ γωνία ἴση ἐστὶ τῇ ἐν τῷ ἐναλλάχ τοῦ κύκλου τμήματι γωνίᾳ τῇ ὑπὸ ΔAG . ἐπεὶ οὖν ἴση ἐστὶν ἡ ὑπὸ $B\Delta\Gamma$ τῇ ὑπὸ ΔAG , κοινὴ προσκείσθω ἡ ὑπὸ $\Gamma\Delta A$ · ὅλη ἄρα ἡ ὑπὸ $B\Delta A$ ἴση ἐστὶ δυοὶ ταῖς ὑπὸ $\Gamma\Delta A$, ΔAG . ἀλλὰ ταῖς ὑπὸ $\Gamma\Delta A$, ΔAG ἴση ἐστὶν ἡ ἐκτὸς ἡ ὑπὸ $B\Gamma\Delta$ · καὶ ἡ ὑπὸ $B\Delta A$ ἄρα ἴση ἐστὶ τῇ ὑπὸ $B\Gamma\Delta$. ἀλλὰ ἡ ὑπὸ $B\Delta A$ τῇ ὑπὸ $\Gamma B\Delta$ ἐστὶν ἴση, ἐπεὶ καὶ πλευρὰ ἡ AD τῇ AB ἐστὶν ἴση· ὥστε καὶ ἡ ὑπὸ $\Delta B A$ τῇ ὑπὸ $B\Gamma\Delta$ ἐστὶν ἴση. αἱ τρεῖς ἄρα

Let some straight-line AB be taken, and let it be cut at point C so that the rectangle contained by AB and BC is equal to the square on CA [Prop. 2.11]. And let the circle BDE be drawn with center A , and radius AB . And let the straight-line BD , equal to the straight-line AC , being not greater than the diameter of circle BDE , be inserted into circle BDE [Prop. 4.1]. And let AD and DC be joined. And let the circle ACD be circumscribed about triangle ACD [Prop. 4.5].

And since the (rectangle contained) by AB and BC is equal to the (square) on AC , and AC (is) equal to BD , the (rectangle contained) by AB and BC is thus equal to the (square) on BD . And since some point B has been taken outside of circle ACD , and two straight-lines BA and BD have radiated from B towards the circle ACD , and (one) of them cuts (the circle), and (the other) meets (the circle), and the (rectangle contained) by AB and BC is equal to the (square) on BD , BD thus touches circle ACD [Prop. 3.37]. Therefore, since BD touches (the circle), and DC has been drawn across (the circle) from the point of contact D , the angle BDC is thus equal to the angle DAC in the alternate segment of the circle [Prop. 3.32]. Therefore, since BDC is equal to DAC , let CDA be added to both. Thus, the whole of BDA is equal to the two (angles) CDA and DAC . But, the external (angle) BCD is equal to CDA and DAC [Prop. 1.32]. Thus, BDA is also equal to BCD . But,

αἱ ὑπὸ $B\Delta A$, $\Delta B A$, $B\Gamma A$ ἴσαι ἀλλήλαις εἰσὶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ $\Delta B\Gamma$ γωνία τῇ ὑπὸ $B\Gamma\Delta$, ἴση ἐστὶ καὶ πλευρὰ ἡ $B\Delta$ πλευρᾷ τῇ $\Delta\Gamma$. ἀλλὰ ἡ $B\Delta$ τῇ ΓA ὑπόκειται ἴση· καὶ ἡ ΓA ἄρα τῇ $\Gamma\Delta$ ἐστὶν ἴση· ὥστε καὶ γωνία ἡ ὑπὸ $\Gamma\Delta A$ γωνία τῇ ὑπὸ $\Delta A\Gamma$ ἐστὶν ἴση· αἱ ἄρα ὑπὸ $\Gamma\Delta A$, $\Delta A\Gamma$ τῆς ὑπὸ $\Delta A\Gamma$ εἰσι διπλασίονες. ἴση δὲ ἡ ὑπὸ $B\Gamma\Delta$ ταῖς ὑπὸ $\Gamma\Delta A$, $\Delta A\Gamma$ · καὶ ἡ ὑπὸ $B\Gamma\Delta$ ἄρα τῆς ὑπὸ $\Gamma A\Delta$ ἐστι διπλή. ἴση δὲ ἡ ὑπὸ $B\Gamma\Delta$ ἑκατέρω τῶν ὑπὸ $B\Delta A$, $\Delta B A$ · καὶ ἑκατέρα ἄρα τῶν ὑπὸ $B\Delta A$, $\Delta B A$ τῆς ὑπὸ $\Delta A B$ ἐστι διπλή.

Ἰσοσκελὲς ἄρα τρίγωνον συνέσταται τὸ $A B \Delta$ ἔθον ἑκατέρω τῶν πρὸς τῇ ΔB βάσει γωνιῶν διπλασίονα τῆς λοιπῆς· ὅπερ ἔδει ποιῆσαι.

BDA is equal to CBD , since the side AD is also equal to AB [Prop. 1.5]. So that DBA is also equal to BCD . Thus, the three (angles) BDA , DBA , and BCD are equal to one another. And since angle DBC is equal to BCD , side BD is also equal to side DC [Prop. 1.6]. But, BD was assumed (to be) equal to CA . Thus, CA is also equal to CD . So that angle CDA is also equal to angle DAC [Prop. 1.5]. Thus, CDA and DAC is double DAC . But BCD (is) equal to CDA and DAC . Thus, BCD is also double CAD . And BCD (is) equal to to each of BDA and DBA . Thus, BDA and DBA are each double DAB .

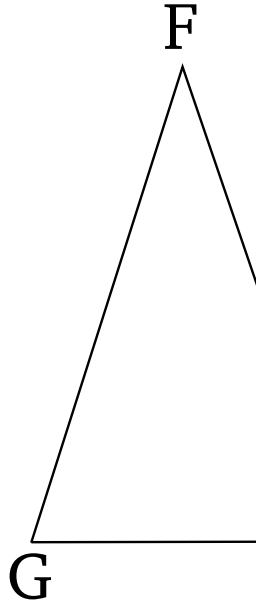
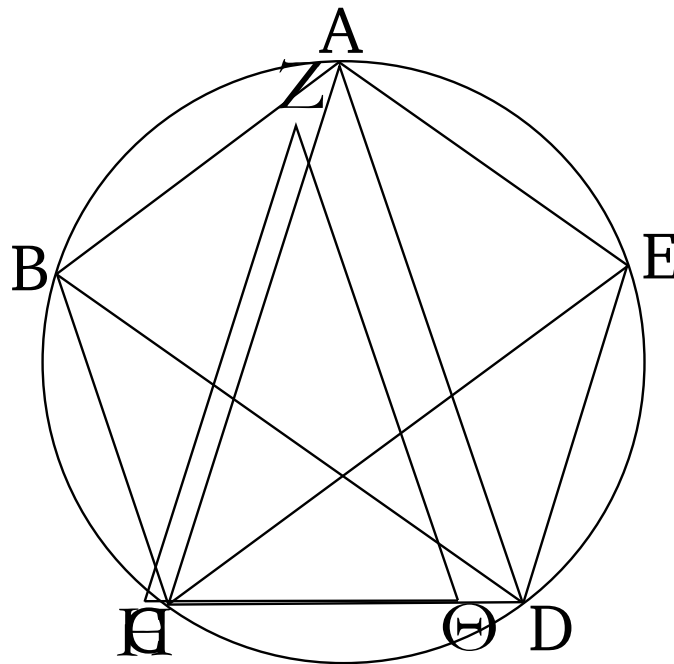
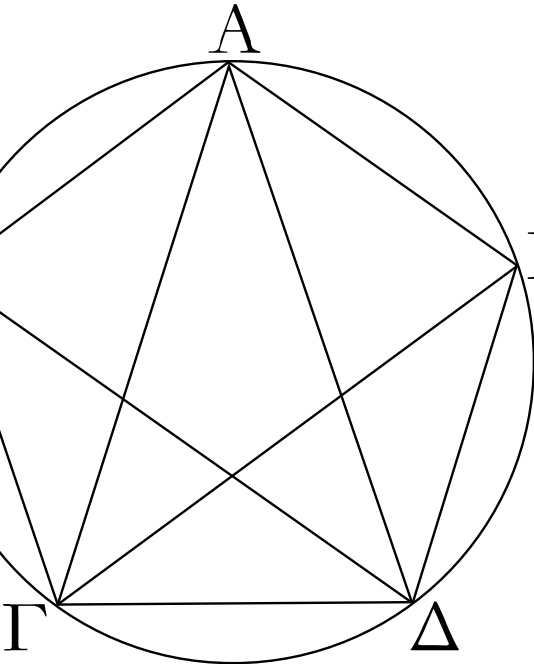
Thus, the isosceles triangle ABD has been constructed having each of the angles at the base BD double the remaining (angle). (Which is) the very thing it was required to do.

11.

Proposition 11

Εἰς τὸν δοθέντα κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι.

To inscribe an equilateral and equiangular pentagon in a given circle.



Ἐστω ὁ δοθείς κύκλος ὁ $AB\Gamma\Delta E$ · δεῖ δὲ εἰς τὸν $AB\Gamma\Delta E$ κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι.

Ἐκκείσθω τρίγωνον ἰσοσκελὲς τὸ $ZH\Theta$ διπλασίονα ἔθον ἑκατέρω τῶν πρὸς τοῖς H , Θ γωνιῶν τῆς πρὸς τῷ Z , καὶ ἐγγεγράφθω εἰς τὸν $AB\Gamma\Delta E$ κύκλον τῷ $ZH\Theta$ τριγώνῳ ἰσογώνιον τρίγωνον τὸ $A\Gamma\Delta$, ὥστε τῇ μὲν πρὸς τῷ Z γωνίᾳ ἴσην εἶναι τὴν ὑπὸ $\Gamma A\Delta$, ἑκατέρω δὲ τῶν πρὸς τοῖς H , Θ ἴσην ἑκατέρω τῶν ὑπὸ $A\Gamma\Delta$, $\Gamma\Delta A$ · καὶ ἑκατέρα ἄρα τῶν ὑπὸ $A\Gamma\Delta$, $\Gamma\Delta A$ τῆς ὑπὸ $\Gamma A\Delta$ ἐστι διπλή. τετμήσθω δὲ ἑκατέρα τῶν ὑπὸ $A\Gamma\Delta$, $\Gamma\Delta A$ δίῃα ὑπὸ ἑκατέρω τῶν ΓE , ΔB εὐθειῶν, καὶ ἐπεξέυθθωσαν αἱ AB , $B\Gamma$, ΔE , EA .

Ἐπεὶ οὖν ἑκατέρα τῶν ὑπὸ $A\Gamma\Delta$, $\Gamma\Delta A$ γωνιῶν διπλα-

Let $ABCDE$ be the given circle. So it is required to inscribed an equilateral and equiangular pentagon in circle $ABCDE$.

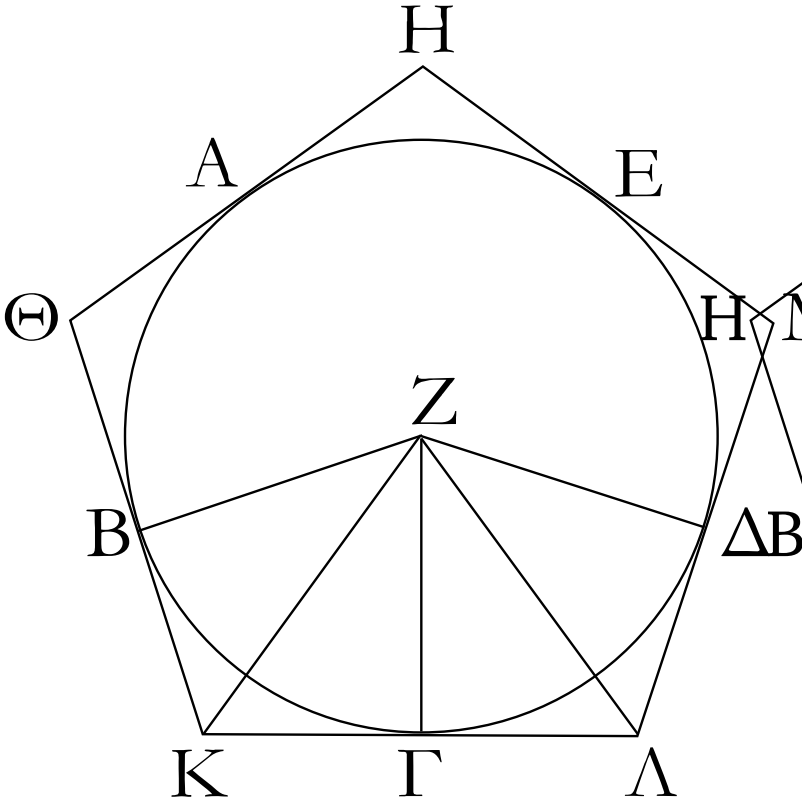
Let the the isosceles triangle FGH be set up having each of the angles at G and H double the (angle) at F [Prop. 4.10]. And let triangle ACD , equiangular to FGH , be inscribed in circle $ABCDE$, such that CAD is equal to the angle at F , and the (angles) at G and H (are) equal to ACD and CDA , respectively [Prop. 4.2]. Thus, ACD and CDA are each double CAD . So let ACD and CDA be cut in half by the straight-lines CE and DB , respectively [Prop. 1.9]. And let AB , BC , DE and EA be joined.

σίῳν ἐστὶ τῆς ὑπὸ $\Gamma\Lambda\Delta$, καὶ τετμημένοι εἰσὶ διὰ ὑπὸ τῶν $\Gamma\epsilon$, $\Delta\beta$ εὐθειῶν, αἱ πέντε ἄρα γωνίαι αἱ ὑπὸ $\Delta\Lambda\Gamma$, $\Lambda\Gamma\epsilon$, $\epsilon\Gamma\Delta$, $\Gamma\Delta\beta$, $\beta\Delta\Lambda$ ἴσαι ἀλλήλαις εἰσὶν. αἱ δὲ ἴσαι γωνίαι ἐπὶ ἴσων περιφερειῶν βεβήκασιν· αἱ πέντε ἄρα περιφέρειαι αἱ AB , $B\Gamma$, $\Gamma\Delta$, $\Delta\epsilon$, $\epsilon\Lambda$ ἴσαι ἀλλήλαις εἰσὶν. ὑπὸ δὲ τὰς ἴσας περιφερείας ἴσαι εὐθεῖαι ὑποτείνουσιν· αἱ πέντε ἄρα εὐθεῖαι αἱ AB , $B\Gamma$, $\Gamma\Delta$, $\Delta\epsilon$, $\epsilon\Lambda$ ἴσαι ἀλλήλαις εἰσὶν· ἰσόπλευρον ἄρα ἐστὶ τὸ $AB\Gamma\Delta\epsilon$ πεντάγωνον. λέγω δὴ, ὅτι καὶ ἰσογώνον. ἐπεὶ γὰρ ἡ AB περιφέρεια τῇ $\Delta\epsilon$ περιφερείᾳ ἐστὶν ἴση, κοινὴ προσκείσθω ἡ $B\Gamma\Delta$ · ὅλη ἄρα ἡ $AB\Gamma\Delta$ περιφέρεια ὅλη τῇ $\epsilon\Delta\Gamma\beta$ περιφερείᾳ ἐστὶν ἴση. καὶ βεβήκεν ἐπὶ μὲν τῇ $AB\Gamma\Delta$ περιφερείᾳ γωνία ἡ ὑπὸ $\Lambda\epsilon\Delta$, ἐπὶ δὲ τῇ $\epsilon\Delta\Gamma\beta$ περιφερείᾳ γωνία ἡ ὑπὸ $\beta\Lambda\epsilon$ · καὶ ἡ ὑπὸ $\beta\Lambda\epsilon$ ἄρα γωνία τῇ ὑπὸ $\Lambda\epsilon\Delta$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἑκάστη τῶν ὑπὸ $AB\Gamma$, $B\Gamma\Delta$, $\Gamma\Delta\epsilon$ γωνιῶν ἑκατέρω τῶν ὑπὸ $\beta\Lambda\epsilon$, $\Lambda\epsilon\Delta$ ἐστὶν ἴση· ἰσογώνον ἄρα ἐστὶ τὸ $AB\Gamma\Delta\epsilon$ πεντάγωνον. ἐδείχθη δὲ καὶ ἰσόπλευρον.

Εἰς ἄρα τὸν δοθέντα κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνον ἐγγράφεται· ὅπερ ἔδει ποιῆσαι.

12.

Περὶ τὸν δοθέντα κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνον περιγράφαι.



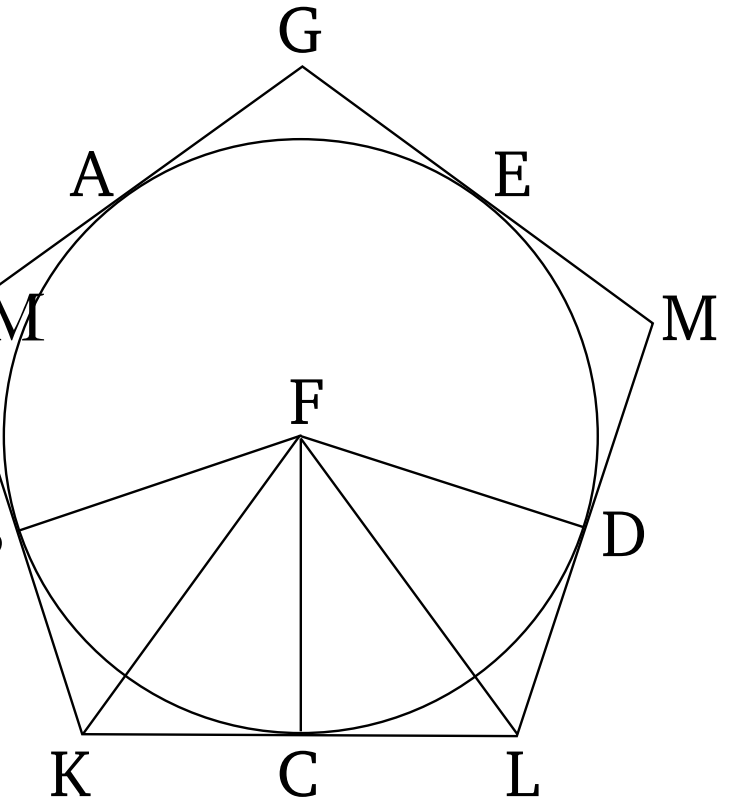
Ἐστω ὁ δοθείς κύκλος ὁ $AB\Gamma\Delta\epsilon$ · δεῖ δὲ περὶ τὸν $AB\Gamma\Delta\epsilon$ κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνον περι-

Therefore, since angles ACD and CDA are each double CAD , and are cut in half by the straight-lines CE and DB , the five angles DAC , ACE , ECD , CDB , and BDA are thus equal to one another. And equal angles stand upon equal circumferences [Prop. 3.26]. Thus, the five circumferences AB , BC , CD , DE , and EA are equal to one another [Prop. 3.29]. Thus, the pentagon $ABCDE$ is equilateral. So I say that (it is) also equiangular. For since the circumference AB is equal to the circumference DE , let BCD be added to both. Thus, the whole circumference $ABCD$ is equal to the whole circumference $EDCB$. And the angle AED stands upon circumference $ABCD$, and angle BAE upon circumference $EDCB$. Thus, angle BAE is also equal to AED [Prop. 3.27]. So, for the same (reasons), each of the angles ABC , BCD , and CDE is also equal to each of BAE and AED . Thus, pentagon $ABCDE$ is equiangular. And it was also shown (to be) equilateral.

Thus, an equilateral and equiangular pentagon has been inscribed in the given circle. (Which is) the very thing it was required to do.

Proposition 12

To circumscribe an equilateral and equiangular pentagon about a given circle.



Let $ABCDE$ be the given circle. So it is required to circumscribe an equilateral and equiangular pentagon about cir-

γράφαι.

Νενοήσθω τοῦ ἐγγεγραμμένου πενταγώνου τῶν γωνιῶν σημεία τὰ A, B, Γ, Δ, E , ὥστε ἴσαξ εἶναι τὰς $AB, B\Gamma, \Gamma\Delta, \Delta E, EA$ περιφερείας· καὶ διὰ τῶν A, B, Γ, Δ, E ἤθ-
θωσαν τοῦ κύκλου ἐφαπτόμεναι αἱ $H\Theta, \Theta K, K\Lambda, \Lambda M$, μὴ, καὶ εἰλήφθω τοῦ $AB\Gamma\Delta E$ κύκλου κέντρον τὸ Z , καὶ ἐπέξτε-
θωσαν αἱ $ZB, ZK, Z\Gamma, Z\Lambda, Z\Delta$.

Καὶ ἐπεὶ ἡ μὲν $K\Lambda$ εὐθεῖα ἐφάπτεται τοῦ $AB\Gamma\Delta E$ κατὰ τὸ Γ , ἀπὸ δὲ τοῦ Z κέντρον ἐπὶ τὴν κατὰ τὸ Γ ἐπαφὴν ἐπέξενεται ἡ $Z\Gamma$, ἡ $Z\Gamma$ ἄρα κάθετόξ ἐστὶν ἐπὶ τὴν $K\Lambda$ · ὁρθή ἄρα ἐστὶν ἑκατέρα τῶν πρὸς τῷ Γ γωνιών. διὰ τὰ αὐτὰ δὴ καὶ αἱ πρὸς τοῖς B, Δ σημείοις γωνίαι ὁρθαί εἰσιν. καὶ ἐπεὶ ὁρθή ἐστὶν ἡ ὑπὸ $Z\Gamma K$ γωνία, τὸ ἄρα ἀπὸ τῆς ZK ἴσον ἐστὶ τοῖς ἀπὸ τῶν $Z\Gamma, \Gamma K$. διὰ τὰ αὐτὰ δὴ καὶ τοῖς ἀπὸ τῶν ZB, BK ἴσον ἐστὶ τὸ ἀπὸ τῆς ZK · ὥστε τὰ ἀπὸ τῶν $Z\Gamma, \Gamma K$ τοῖς ἀπὸ τῶν ZB, BK ἐστὶν ἴσα, ὧν τὸ ἀπὸ τῆς $Z\Gamma$ τῷ ἀπὸ τῆς ZB ἐστὶν ἴσον· λοιπὸν ἄρα τὸ ἀπὸ τῆς ΓK τῷ ἀπὸ τῆς BK ἐστὶν ἴσον. ἴση ἄρα ἡ BK τῇ ΓK . καὶ ἐπεὶ ἴση ἐστὶν ἡ ZB τῇ $Z\Gamma$, καὶ κοινὴ ἡ ZK , δύο δὴ αἱ BZ, ZK δυοὶ ταῖς $\Gamma Z, ZK$ ἴσαι εἰσὶν· καὶ βάσεις ἡ BK βάσει τῇ ΓK [ἐστὶν] ἴση· γωνία ἄρα ἡ μὲν ὑπὸ BZK [γωνία] τῇ ὑπὸ $KZ\Gamma$ ἐστὶν ἴση· ἡ δὲ ὑπὸ BKZ τῇ ὑπὸ $ZK\Gamma$ διπλὴ ἄρα ἡ μὲν ὑπὸ $BZ\Gamma$ τῆς ὑπὸ $KZ\Gamma$, ἡ δὲ ὑπὸ $BK\Gamma$ τῆς ὑπὸ $ZK\Gamma$. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ $\Gamma Z\Delta$ τῆς ὑπὸ $\Gamma Z\Lambda$ ἐστὶ διπλὴ, ἡ δὲ ὑπὸ $\Delta\Lambda\Gamma$ τῆς ὑπὸ $Z\Lambda\Gamma$. καὶ ἐπεὶ ἴση ἐστὶν ἡ $B\Gamma$ περιφέρεια τῇ $\Gamma\Delta$, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ $BZ\Gamma$ τῇ ὑπὸ $\Gamma Z\Delta$. καὶ ἐστὶν ἡ μὲν ὑπὸ $BZ\Gamma$ τῆς ὑπὸ $KZ\Gamma$ διπλὴ, ἡ δὲ ὑπὸ $\Delta Z\Gamma$ τῆς ὑπὸ $\Lambda Z\Gamma$ · ἴση ἄρα καὶ ἡ ὑπὸ $KZ\Gamma$ τῇ ὑπὸ $\Lambda Z\Gamma$ · ἐστὶ δὲ καὶ ἡ ὑπὸ $Z\Gamma K$ γωνία τῇ ὑπὸ $Z\Gamma\Lambda$ ἴση. δύο δὴ τρίγωνά ἐστι τὰ $ZK\Gamma, Z\Lambda\Gamma$ τὰς δύο γωνίας ταῖς δυοὶ γωνίαις ἴσαξ ἔχοντα καὶ μίαν πλευράν μιᾷ πλευρᾷ ἴσην κοινὴν αὐτῶν τὴν $Z\Gamma$ · καὶ τὰς λοιπὰς ἄρα πλευράς ταῖς λοιπαῖς πλευραῖς ἴσαξ ἔχει καὶ τὴν λοιπὴν γωνίαν τῇ λοιπῇ γωνίᾳ· ἴση ἄρα ἡ μὲν $K\Gamma$ εὐθεῖα τῇ $\Gamma\Lambda$, ἡ δὲ ὑπὸ $ZK\Gamma$ γωνία τῇ ὑπὸ $Z\Lambda\Gamma$. καὶ ἐπεὶ ἴση ἐστὶν ἡ $K\Gamma$ τῇ $\Gamma\Lambda$, διπλὴ ἄρα ἡ $K\Lambda$ τῆς $K\Gamma$. διὰ τὰ αὐτὰ δὴ δευθθήσεται καὶ ἡ ΘK τῆς BK διπλὴ. καὶ ἐστὶν ἡ BK τῇ $K\Gamma$ ἴση· καὶ ἡ ΘK ἄρα τῇ $K\Lambda$ ἐστὶν ἴση. ὁμοίως δὴ δευθθήσεται καὶ ἑκάστη τῶν $\Theta H, H\Lambda, \Lambda M$ ἑκατέρω τῶν $\Theta K, K\Lambda$ ἴση· ἰσόπλευρον ἄρα ἐστὶ τὸ $H\Theta K\Lambda M$ πεντάγωνον. λέγω δὴ, ὅτι καὶ ἰσογώνιον. ἐπεὶ γάρ ἴση ἐστὶν ἡ ὑπὸ $ZK\Gamma$ γωνία τῇ ὑπὸ $Z\Lambda\Gamma$, καὶ ἐδείκθη τῆς μὲν ὑπὸ $ZK\Gamma$ διπλὴ ἡ ὑπὸ $\Theta K\Lambda$, τῆς δὲ ὑπὸ $Z\Lambda\Gamma$ διπλὴ ἡ ὑπὸ $K\Lambda M$, καὶ ἡ ὑπὸ $\Theta K\Lambda$ ἄρα τῇ ὑπὸ $K\Lambda M$ ἐστὶν ἴση. ὁμοίως δὴ δευθθήσεται καὶ ἑκάστη τῶν ὑπὸ $K\Theta H, \Theta H M, H\Lambda M$ ἑκατέρω τῶν ὑπὸ $\Theta K\Lambda, K\Lambda M$ ἴση· αἱ πέντε ἄρα γωνίαι αἱ ὑπὸ $H\Theta K, \Theta K\Lambda, K\Lambda M, \Lambda M\Theta$ ἴσαι ἀλλήλαις εἰσὶν. ἰσογώνιον ἄρα ἐστὶ τὸ $H\Theta K\Lambda M$ πεντάγωνον. ἐδείκθη δὲ καὶ ἰσόπλευρον, καὶ περιγράφεται περὶ τὸν $AB\Gamma\Delta E$ κύκλον.

[Περὶ τὸν δοθέντα ἄρα κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον περιγράφεται]· ὅπερ ἔδει ποιῆσαι.

cle $ABCDE$.

Let A, B, C, D , and E be conceived as the angular points of a pentagon having been inscribed (in circle $ABCDE$) [Prop. 3.11], such that the circumferences AB, BC, CD, DE , and EA are equal. And let GH, HK, KL, LM , and MG be drawn through (points) A, B, C, D , and E (respectively), touching the circle.[†] And let the center F of the circle $ABCDE$ be found [Prop. 3.1]. And let FB, FK, FC, FL , and FD be joined.

And since the straight-line KL touches (circle) $ABCDE$ at C , and FC has been joined from the center F to the point of contact C , FC is thus perpendicular to KL [Prop. 3.18]. Thus, each of the angles at C is a right-angle. So, for the same (reasons), the angles at B and D are also right-angles. And since angle FCK is a right-angle, the (square) on FK is thus equal to the (sum of the squares) on FC and CK [Prop. 1.47]. So, for the same (reasons), the (square) on FK is also equal to the (sum of the squares) on FB and BK . So that the (sum of the squares) on FC and CK is equal to the (sum of the squares) on FB and BK , of which the (square) on FC is equal to the (square) on FB . Thus, the remaining (square) on CK is equal to the remaining (square) on BK . Thus, BK (is) equal to CK . And since FB is equal to FC , and FK (is) common, the two (straight-lines) BF, FK are equal to the two (straight-lines) CF, FK . And the base BK [is] equal to the base CK . Thus, angle BFK is equal to [angle] KFC [Prop. 1.8]. And BKF (is equal) to FKC [Prop. 1.8]. Thus, BFC (is) double KFC , and BKC (is double) FKC . So, for the same (reasons), CFD is also double CFL , and DLC (is also double) FLC . And since circumference BC is equal to CD , angle BFC is also equal to CFD [Prop. 3.27]. And BFC is double KFC , and DFC (is double) LFC . Thus, KFC is also equal to LFC . And angle FCK is also equal to FCL . So, FKC and FLC are two triangles having two angles equal to two angles, and one side equal to one side, (namely) their common (side) FC . Thus, they will also have the remaining sides equal to the (corresponding) remaining sides, and the remaining angle to the remaining angle [Prop. 1.26]. Thus, the straight-line KC (is) equal to CL , and the angle FKC to FLC . And since KC is equal to CL , KL (is) thus double KC . So, for the same (reasons), it can be shown that HK (is) also double BK . And BK is equal to KC . Thus, HK is also equal to KL . So, similarly, each of HG, GM , and ML can also be shown (to be) equal to each of HK and KL . Thus, pentagon $GHKLM$ is equilateral. So I say that (it is) also equiangular. For since angle FKC is equal to FLC , and HKL was shown (to be) double FKC , and KLM double FLC , HKL is thus also equal to KLM . So, similarly, each of KHG, HGM , and GML can also be shown (to be) equal to each of HKL and KLM . Thus, the five angles GHK, HKL, KLM, LMG , and MGH are equal to one another. Thus, the pentagon $GHKLM$ is equiangular. And it was also shown (to be) equilateral, and

† See the footnote to Prop. 3.34.

13.

Εἰς τὸ δοθὲν πεντάγωνον, ὃ ἐστὶν ἰσόπλευρόν τε καὶ ἰσογώνιον, κύκλον ἐγγράψαι.

Ἐστω τὸ δοθὲν πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον τὸ $ABΓΔΕ$. δεῖ δὴ εἰς τὸ $ABΓΔΕ$ πεντάγωνον κύκλον ἐγγράψαι.

Τετμήσθω γὰρ ἑκατέρα τῶν ὑπὸ $ΒΓΔ$, $ΓΔΕ$ γωνιῶν διῖθα ὑπὸ ἑκατέρας τῶν $ΓΖ$, $ΔΖ$ εὐθειῶν· καὶ ἀπὸ τοῦ Z σημείου, καθ' ὃ συμβάλλουσιν ἀλλήλαις αἱ $ΓΖ$, $ΔΖ$ εὐθεῖαι, ἐπεξεύθωσαν αἱ ZB , ZA , ZE εὐθεῖαι. καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΒΓ$ τῇ $ΓΔ$, κοινὴ δὲ ἡ $ΓΖ$, δύο δὴ αἱ $ΒΓ$, $ΓΖ$ δυοὶ ταῖς $ΔΓ$, $ΓΖ$ ἴσαι εἰσὶν· καὶ γωνία ἡ ὑπὸ $ΒΓΖ$ γωνία τῇ ὑπὸ $ΔΓΖ$ [ἐστὶν] ἴση· βάσεις ἄρα ἡ BZ βάσει τῇ $ΔΖ$ ἐστὶν ἴση, καὶ τὸ $ΒΓΖ$ τρίγωνον τῷ $ΔΓΖ$ τριγώνῳ ἐστὶν ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἐξόνται, ὅφ' ἄξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἡ ὑπὸ $ΓΒΖ$ γωνία τῇ ὑπὸ $ΓΔΖ$. καὶ ἐπεὶ διπλὴ ἐστὶν ἡ ὑπὸ $ΓΔΕ$ τῇς ὑπὸ $ΓΔΖ$, ἴση δὲ ἡ μὲν ὑπὸ $ΓΔΕ$ τῇ ὑπὸ $ΑΒΓ$, ἡ δὲ ὑπὸ $ΓΔΖ$ τῇ ὑπὸ $ΓΒΖ$, καὶ ἡ ὑπὸ $ΓΒΑ$ ἄρα τῇς ὑπὸ $ΓΒΖ$ ἐστὶ διπλὴ· ἴση ἄρα ἡ ὑπὸ $ΑΒΖ$ γωνία τῇ ὑπὸ $ΖΒΓ$ · ἡ ἄρα ὑπὸ $ΑΒΓ$ γωνία διῖθα τέτμηται ὑπὸ τῇς BZ εὐθείας. ὁμοίως δὴ δειθθήσεται, ὅτι καὶ ἑκατέρα τῶν ὑπὸ $ΒΑΕ$, $ΑΕΔ$ διῖθα τέτμηται ὑπὸ ἑκατέρας τῶν $ΖΑ$, $ΖΕ$ εὐθειῶν. ἡθῶσαν δὴ ἀπὸ τοῦ Z σημείου ἐπὶ τὰς $ΑΒ$, $ΒΓ$, $ΓΔ$, $ΔΕ$, $ΕΑ$ εὐθείας κάθετοι αἱ ZH , $ZΘ$, ZK , $ΖΛ$, ZM . καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ $ΘΓΖ$ γωνία τῇ ὑπὸ $ΚΓΖ$, ἐστὶ δὲ καὶ ὀρθή ἡ ὑπὸ $ΖΘΓ$ [ὀρθή] τῇ ὑπὸ $ΖΚΓ$ ἴση, δύο δὴ τριγώνω ἐστὶ τὰ $ΖΘΓ$, $ΖΚΓ$ τὰς δύο γωνίας δυοὶ γωνίαις ἴσαξ ἔχοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην κοινήν αὐτῶν τὴν $ZΓ$ ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν· καὶ τὰς λοιπὰς ἄρα πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαξ ἔχει· ἴση ἄρα ἡ $ZΘ$ κάθετος τῇ ZK καθέτῳ. ὁμοίως δὴ δειθθήσεται, ὅτι καὶ ἑκάστη τῶν $ΖΛ$, ZM , ZH ἑκατέρω τῶν $ZΘ$, ZK ἴση ἐστὶν· αἱ πέντε ἄρα εὐθεῖαι αἱ ZH , $ZΘ$, ZK , $ΖΛ$, ZM ἴσαι ἀλλήλαις εἰσὶν. ὁ ἄρα κέντρον τῷ Z διαστήματι δὲ ἐνὶ τῶν H , $Θ$, K , $Λ$, M κύκλος γραφόμενος ἔχει καὶ διὰ τῶν λοιπῶν σημείων καὶ ἐφάπεται τῶν $ΑΒ$, $ΒΓ$, $ΓΔ$, $ΔΕ$, $ΕΑ$ εὐθειῶν διὰ τὸ ὀρθὰς εἶναι τὰς πρὸς τοῖς H , $Θ$, K , $Λ$, M σημείοις γωνίας. εἰ γὰρ οὐκ ἐφάπεται αὐτῶν, ἀλλὰ τεμεῖ αὐτάς, συμβήσεται τὴν τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένην ἐντὸς πίπτειν τοῦ κύκλου· ὅπερ ἄτοπον ἐδείχθη. οὐκ ἄρα ὁ κέντρον τῷ Z διαστήματι δὲ ἐνὶ τῶν H , $Θ$, K , $Λ$, M σημείων γραφόμενος κύκλος τεμεῖ τὰς $ΑΒ$, $ΒΓ$, $ΓΔ$, $ΔΕ$, $ΕΑ$ εὐθείας· ἐφάπεται ἄρα αὐτῶν. γεγράφθω ὡς ὁ $HΘΚΑΜ$.

has been circumscribed about circle $ABCDE$.

[Thus, an equilateral and equiangular pentagon has been circumscribed about the given circle]. (Which is) the very thing it was required to do.

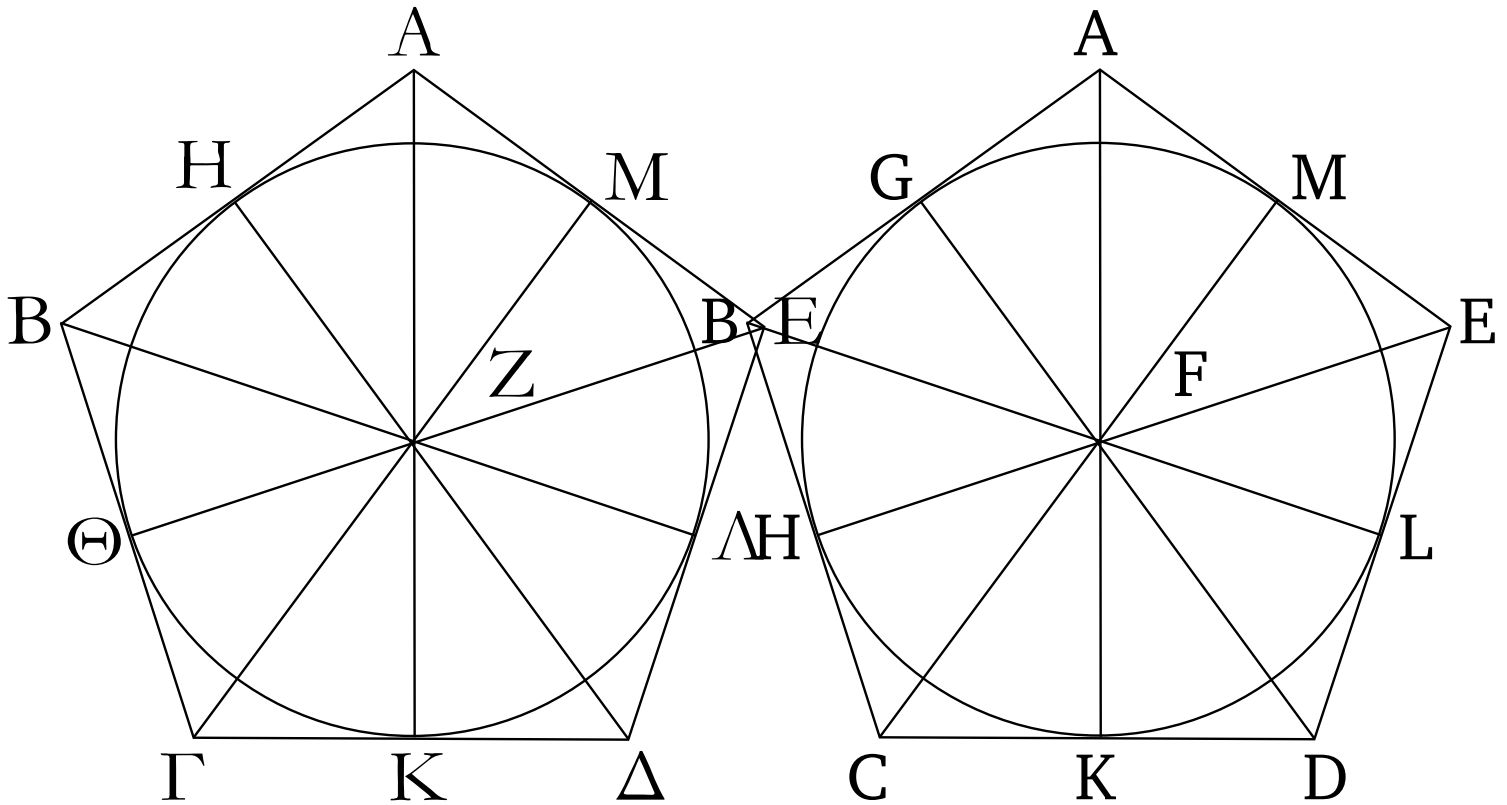
Proposition 13

To inscribe a circle in a given pentagon, which is equilateral and equiangular.

Let $ABCDE$ be the given equilateral and equiangular pentagon. So it is required to inscribe a circle in pentagon $ABCDE$.

For let angles BCD and CDE each be cut in half by each of the straight-lines CF and DF (respectively) [Prop. 1.9]. And from the point F , at which the straight-lines CF and DF meet one another, let the straight-lines FB , FA , and FE be joined. And since BC is equal to CD , and CF (is) common, the two (straight-lines) BC , CF are equal to the two (straight-lines) DC , CF . And angle BCF [is] equal to angle DCF . Thus, the base BF is equal to the base DF , and triangle BCF is equal to triangle DCF , and the remaining angles will be equal to the (corresponding) remaining angles which the equal sides subtend [Prop. 1.4]. Thus, angle CBF (is) equal to CDF . And since CDE is double CDF , and CDE (is) equal to ABC , and CDF to CBF , CBA is thus also double CBF . Thus, angle ABF is equal to FBC . Thus, angle ABC has been cut in half by the straight-line BF . So, similarly, it can be shown that BAE and AED have been cut in half by the straight-lines FA and FE , respectively. So let FG , FH , FK , FL , and FM be drawn from point F , perpendicular to the straight-lines AB , BC , CD , DE , and EA (respectively) [Prop. 1.12]. And since angle HCF is equal to KCF , and the right-angle FHC is also equal to the [right-angle] FKC , FHC and FKC are two triangles having two angles equal to two angles, and one side equal to one side, (namely) their common (side) FC , subtending one of the equal angles. Thus, they will also have the remaining sides equal to the (corresponding) remaining sides [Prop. 1.26]. Thus, the perpendicular FH (is) equal to the perpendicular FK . So, similarly, it can be shown that FL , FM , and FG are each equal to each of FH and FK . Thus, the five straight-lines FG , FH , FK , FL , and FM are equal to one another. Thus, the circle drawn with center F , and radius one of G , H , K , L , or M , will also go through the remaining points, and will touch the straight-lines AB , BC , CD , DE , and EA , on account of the angles at points G , H , K , L , and M being right-angles. For if it does not touch them, but cuts them, it follows that a (straight-line) drawn at right-angles to the diameter of the circle, from its extremity, falls inside the circle. The very thing was shown (to be) absurd [Prop. 3.16]. Thus, the circle drawn with center F , and

radius one of G, H, K, L , or M , does not cut the straight-lines AB, BC, CD, DE , or EA . Thus, it will touch them. Let it be drawn, like $GHKLM$ (in the figure).



Εἰς ἄρα τὸ δοθὲν πεντάγωνον, ὃ ἐστὶν ἰσόπλευρόν τε καὶ ἰσογώνιον, κύκλος ἐγγέγραπται· ὅπερ ἔδει ποιῆσαι.

Thus, a circle has been inscribed in the given pentagon which is equilateral and equiangular. (Which is) the very thing it was required to do.

14.

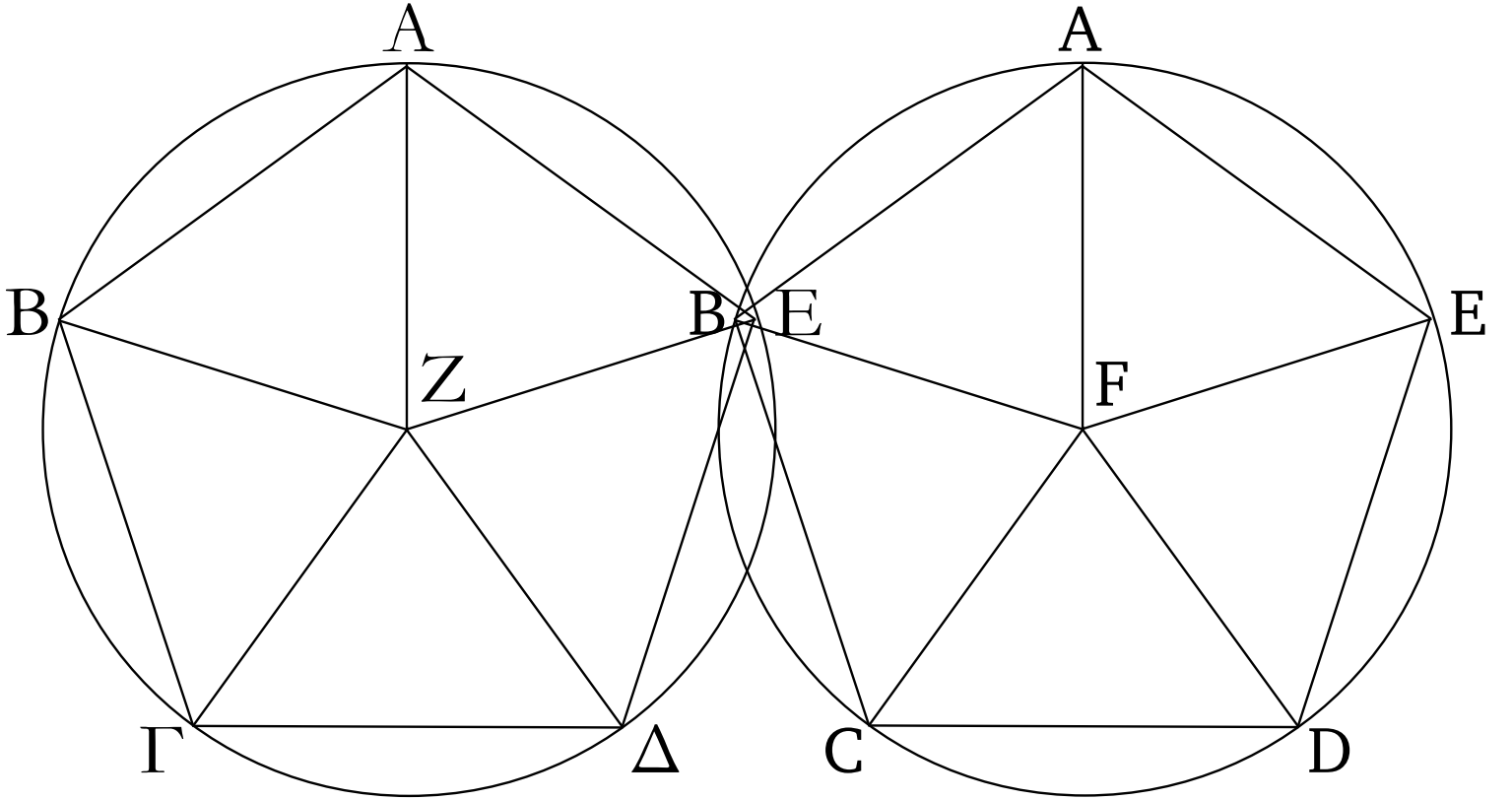
Proposition 14

Περὶ τὸ δοθὲν πεντάγωνον, ὃ ἐστὶν ἰσόπλευρόν τε καὶ ἰσογώνιον, κύκλον περιγράψαι.

To circumscribe a circle about a given pentagon which is equilateral and equiangular.

Ἐστω τὸ δοθὲν πεντάγωνον, ὃ ἐστὶν ἰσόπλευρόν τε καὶ ἰσογώνιον, τὸ $ABΓΔΕ$ · δεῖ δὴ περὶ τὸ $ABΓΔΕ$ πεντάγωνον κύκλον περιγράψαι.

Let $ABCDE$ be the given pentagon which is equilateral and equiangular. So it is required to circumscribe a circle about the pentagon $ABCDE$.



Τετμήσθω δὴ ἑκάτερα τῶν ὑπὸ $BΓΔ$, $ΓΔΕ$ γωνιῶν διὰ ὑπὸ ἑκατέρα τῶν $ΓΖ$, $ΔΖ$, καὶ ἀπὸ τοῦ Z σημείου, καθ' ὃ συμβάλλουσιν αἱ εὐθεῖαι, ἐπὶ τὰ B , A , E σημεία ἐπεξεύθωσαν εὐθεῖαι αἱ ZB , ZA , ZE . ὁμοίως δὴ τῷ πρὸ τούτου δειθῆσεται, ὅτι καὶ ἑκάστη τῶν ὑπὸ $ΓΒΑ$, $ΒΑΕ$, $ΑΕΔ$ γωνιῶν διὰ τέτμηται ὑπὸ ἑκάστης τῶν ZB , ZA , ZE εὐθειῶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ $BΓΔ$ γωνία τῇ ὑπὸ $ΓΔΕ$, καὶ ἐστὶ τῆς μὲν ὑπὸ $BΓΔ$ ἡμίσεια ἡ ὑπὸ $ΖΓΔ$, τῆς δὲ ὑπὸ $ΓΔΕ$ ἡμίσεια ἡ ὑπὸ $ΓΔΖ$, καὶ ἡ ὑπὸ $ΖΓΔ$ ἄρα τῇ ὑπὸ $ΖΔΓ$ ἐστὶν ἴση· ὥστε καὶ πλευρὰ ἡ $ΖΓ$ πλευρᾷ τῇ $ΖΔ$ ἐστὶν ἴση. ὁμοίως δὲ δειθῆσεται, ὅτι καὶ ἑκάστη τῶν ZB , ZA , ZE ἑκατέρῃ τῶν $ZΓ$, $ΖΔ$ ἐστὶν ἴση· αἱ πέντε ἄρα εὐθεῖαι αἱ ZA , ZB , $ZΓ$, $ΖΔ$, ZE ἴσαι ἀλλήλαις εἰσὶν. ὁ ἄρα κέντρῳ τῷ Z καὶ διαστήματι ἐνὶ τῶν ZA , ZB , $ZΓ$, $ΖΔ$, ZE κύκλος γραφόμενος ἔχει καὶ διὰ τῶν λοιπῶν σημείων καὶ ἔσται περιγεγραμμένος. περιγεγράφθω καὶ ἔστω ὁ $ΑΒΓΔΕ$.

Περὶ ἄρα τὸ δοθὲν πεντάγωνον, ὃ ἐστὶν ἰσόπλευρόν τε καὶ ἰσογώνιον, κύκλος περιέγρῃται· ὅπερ ἔδει ποιῆσαι.

15.

Εἰς τὸν δοθέντα κύκλον ἑχάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράψαι.

Ἐστω ὁ δοθείς κύκλος ὁ $ΑΒΓΔΕΖ$ · δεῖ δὴ εἰς τὸν $ΑΒΓΔΕΖ$ κύκλον ἑχάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγ-

So let angles BCD and CDE have been cut in half by the (straight-lines) CF and DF , respectively [Prop. 1.9]. And let the straight-lines FB , FA , and FE be joined from point F , at which the straight-lines meet, to the points B , A , and E (respectively). So, similarly, to the (proposition) before this (one), it can be shown that angles CBA , BAE , and AED have also been cut in half by the straight-lines FB , FA , and FE , respectively. And since angle BCD is equal to CDE , and FCD is half of BCD , and CDF half of CDE , FCD is thus also equal to FDC . So that side FC is also equal to side FD [Prop. 1.6]. So, similarly, it can be shown that FB , FA , and FE are also each equal to each of FC and FD . Thus, the five straight-lines FA , FB , FC , FD , and FE are equal to one another. Thus, the circle drawn with center F , and radius one of FA , FB , FC , FD , or FE , will also go through the remaining points, and will be circumscribed. Let it be (so) circumscribed, and let it be $ABCDE$.

Thus, a circle has been circumscribed about the given pentagon, which is equilateral and equiangular. (Which is) the very thing it was required to do.

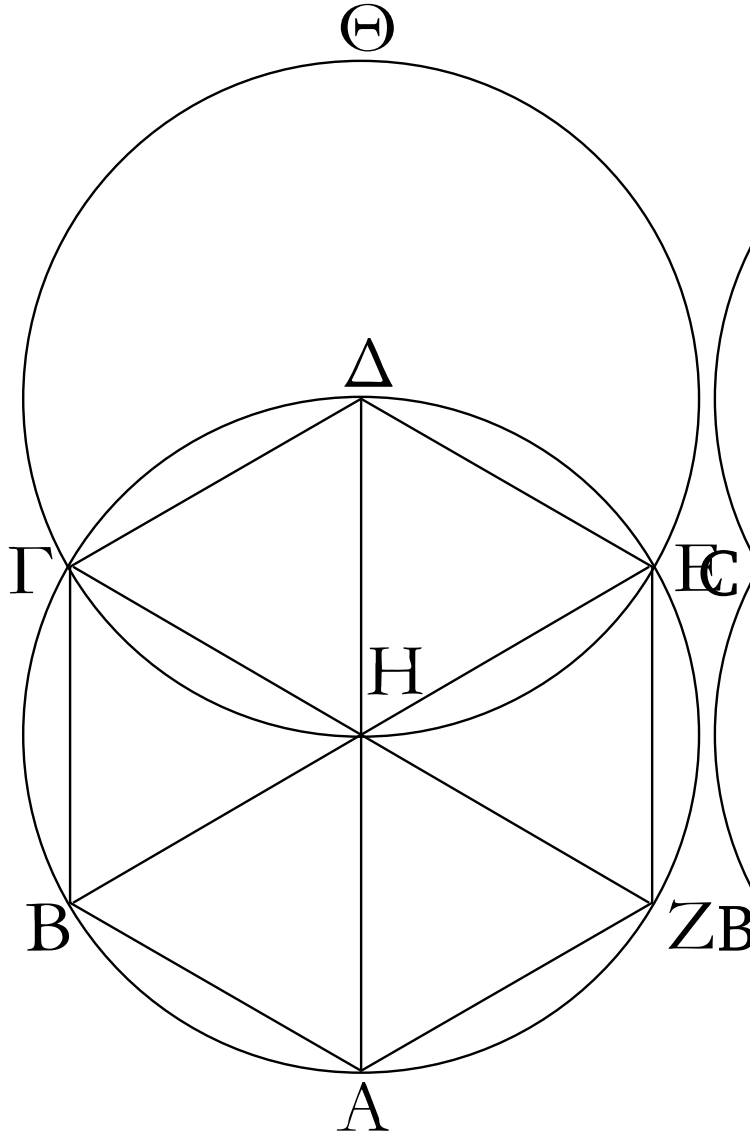
Proposition 15

To inscribe an equilateral and equiangular hexagon in a given circle.

Let $ABCDEF$ be the given circle. So it is required to inscribe an equilateral and equiangular hexagon in circle

γράφαι.

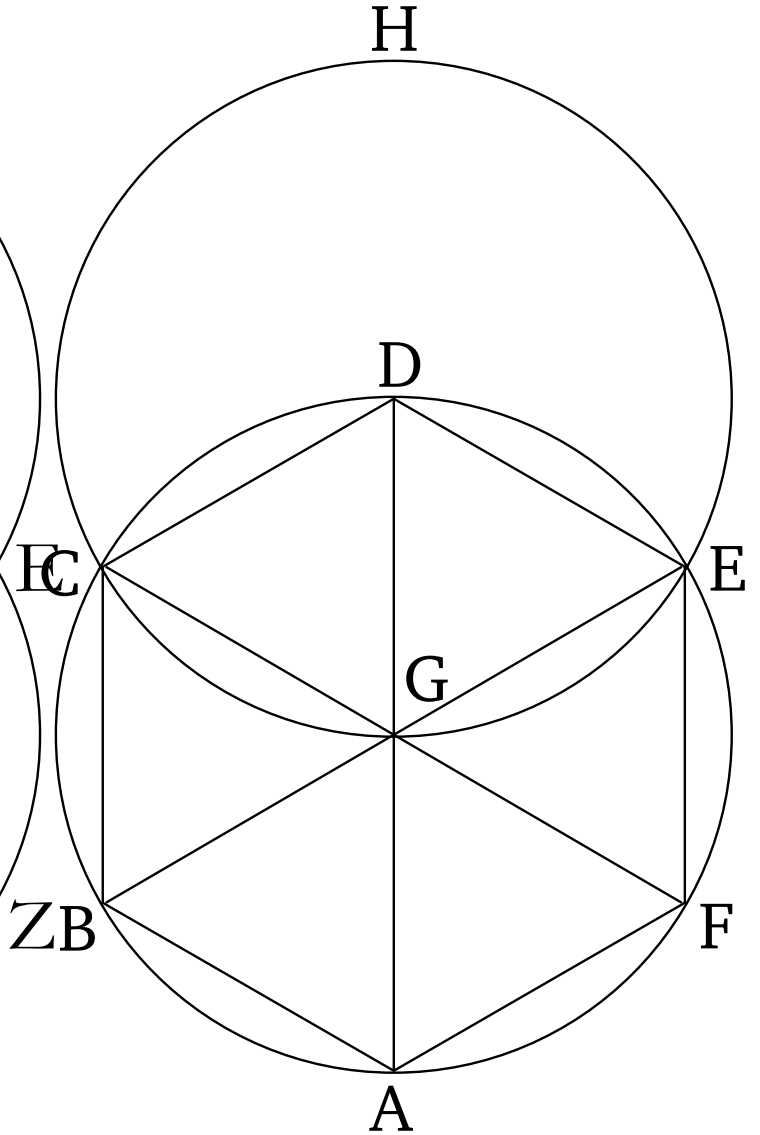
Ἦθθω τοῦ $ABΓΔΕΖ$ κύκλου διάμετρος ἡ $ΑΔ$, καὶ ἐ-
 ἰλήφθω τὸ κέντρον τοῦ κύκλου τὸ H , καὶ κέντρῳ μὲν τῷ $Δ$
 διαστήματι δὲ τῷ $ΔH$ κύκλος γεγράφθω ὁ $ΕΗΓΘ$, καὶ ἐπι-
 ζευθθεῖσαι αἱ $ΕΗ$, $ΓΗ$ διήθθωσαν ἐπὶ τὰ B , Z σημεία, καὶ
 ἐπεξεύθθωσαν αἱ $ΑΒ$, $ΒΓ$, $ΓΔ$, $ΔΕ$, $ΕΖ$, $ΖΑ$ · λέγω, ὅτι τὸ
 $ΑΒΓΔΕΖ$ ἑξάγωνον ἰσόπλευρόν τε ἐστὶ καὶ ἰσογώνιον.



Ἐπεὶ γὰρ τὸ H σημεῖον κέντρον ἐστὶ τοῦ $ΑΒΓΔΕΖ$
 κύκλου, ἴση ἐστὶν ἡ HE τῇ $HΔ$. πάλιν, ἐπεὶ τὸ $Δ$ σημεῖον
 κέντρον ἐστὶ τοῦ $ΗΓΘ$ κύκλου, ἴση ἐστὶν ἡ $ΔΕ$ τῇ $ΔΗ$. ἀλλ'
 ἡ HE τῇ $HΔ$ ἐδείχθη ἴση· καὶ ἡ HE ἄρα τῇ $ΕΔ$ ἴση ἐστὶν·
 ἰσόπλευρον ἄρα ἐστὶ τὸ $ΕΗΔ$ τρίγωνον· καὶ αἱ τρεῖς ἄρα
 αὐτοῦ γωνίαι αἱ ὑπὸ $ΕΗΔ$, $ΗΔΕ$, $ΔΕΗ$ ἴσαι ἀλλήλαις εἰσίν,
 ἐπειδὴ περ τῶν ἰσοσκελῶν τριγώνων αἱ πρὸς τῇ βάσει γωνίαι

$ΑΒCDEF$.

Let the diameter AD of circle $ABCDEF$ be drawn,[†] and
 let the center G of the circle be found [Prop. 3.1]. And let the
 circle $EGCH$ be drawn, with center D , and radius DG . And
 EG and CG being joined, let them be drawn across (the circle)
 to points B and F (respectively). And let AB , BC , CD , DE ,
 EF , and FA be joined. I say that the hexagon $ABCDEF$ is
 equilateral and equiangular.



For since point G is the center of circle $ABCDEF$, GE
 is equal to GD . Again, since point D is the center of circle
 GCH , DE is equal to DG . But, GE was shown (to be) equal
 to GD . Thus, GE is also equal to ED . Thus, triangle EGD
 is equilateral. Thus, its three angles EGD , GDE , and DEG
 are also equal to one another, inasmuch as the angles at the
 base of isosceles triangles are equal to one another [Prop. 1.5].

ἴσαι ἀλλήλαις εἰσὶν· καὶ εἰσὶν αἱ τρεῖς τοῦ τριγώνου γωνίαι δυνάμει ὀρθαῖς ἴσαι· ἡ ἄρα ὑπὸ EHD γωνία τρίτον ἐστὶ δύο ὀρθῶν. ὁμοίως δὲ δευτέρησεται καὶ ἡ ὑπὸ $\Delta H\Gamma$ τρίτον δύο ὀρθῶν. καὶ ἐπεὶ ἡ GH εὐθεῖα ἐπὶ τὴν EB σταθεῖσα τὰς ἐφεχῆς γωνίας τὰς ὑπὸ EHG , GHB δυνάμει ὀρθαῖς ἴσας ποιεῖ, καὶ λοιπὴ ἄρα ἡ ὑπὸ GHB τρίτον ἐστὶ δύο ὀρθῶν· αἱ ἄρα ὑπὸ EHD , $\Delta H\Gamma$, GHB γωνίαι ἴσαι ἀλλήλαις εἰσὶν· ὥστε καὶ αἱ κατὰ κορυφὴν αὐταῖς αἱ ὑπὸ BHA , AHZ , ZHE ἴσαι εἰσὶν [ταῖς ὑπὸ EHD , $\Delta H\Gamma$, GHB]. αἱ ἔχ' ἄρα γωνίαι αἱ ὑπὸ EHD , $\Delta H\Gamma$, GHB , BHA , AHZ , ZHE ἴσαι ἀλλήλαις εἰσὶν. αἱ δὲ ἴσαι γωνίαι ἐπὶ ἴσων περιφερειῶν βεβήκασιν· αἱ ἔχ' ἄρα περιφέρειαι αἱ AB , BF , FD , DE , EZ , ZA ἴσαι ἀλλήλαις εἰσὶν. ὑπὸ δὲ τὰς ἴσας περιφέρειας αἱ ἴσαι εὐθεῖαι ὑποτείνουσιν· αἱ ἔχ' ἄρα εὐθεῖαι ἴσαι ἀλλήλαις εἰσὶν· ἰσοπλευρὸν ἄρα ἐστὶ τὸ $AB\Gamma\Delta EZ$ ἑξάγωνον. λέγω δὴ, ὅτι καὶ ἰσογώνιον. ἐπεὶ γάρ ἴση ἐστὶν ἡ ZA περιφέρεια τῇ ED περιφερείᾳ, κοινὴ προσκείσθω ἡ $AB\Gamma\Delta$ περιφέρεια· ὅλη ἄρα ἡ $ZAB\Gamma\Delta$ ὅλη τῇ $E\Delta\Gamma BA$ ἐστὶν ἴση· καὶ βέβηκεν ἐπὶ μὲν τῇ $ZAB\Gamma\Delta$ περιφερείᾳ ἡ ὑπὸ $ZE\Delta$ γωνία, ἐπὶ δὲ τῇ $E\Delta\Gamma BA$ περιφερείᾳ ἡ ὑπὸ AZE γωνία· ἴση ἄρα ἡ ὑπὸ AZE γωνία τῇ ὑπὸ ΔEZ . ὁμοίως δὲ δευτέρησεται, ὅτι καὶ αἱ λοιπαὶ γωνίαι τοῦ $AB\Gamma\Delta EZ$ ἑξαγώνου κατὰ μίαν ἴσαι εἰσὶν ἑκατέρω τῶν ὑπὸ AZE , $ZE\Delta$ γωνιῶν ἰσογώνιον ἄρα ἐστὶ τὸ $AB\Gamma\Delta EZ$ ἑξάγωνον. ἐδείχθη δὲ καὶ ἰσοπλευρὸν· καὶ ἐγγράφεται εἰς τὸν $AB\Gamma\Delta EZ$ κύκλον.

Εἰς ἄρα τὸν δοθέντα κύκλον ἑξάγωνον ἰσοπλευρὸν τε καὶ ἰσογώνιον ἐγγράφεται· ὅπερ ἔδει ποιῆσαι.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἡ τοῦ ἑξαγώνου πλευρὰ ἴση ἐστὶ τῇ ἐκ τοῦ κέντρου τοῦ κύκλου.

Ὅμοίως δὲ τοῖς ἐπὶ τοῦ πενταγώνου ἔαν διὰ τῶν κατὰ τὸν κύκλον διαιρέσεων ἐφαπτομένας τοῦ κύκλου ἀγάγωμεν, περιγραφῆσεται περὶ τὸν κύκλον ἑξάγωνον ἰσοπλευρὸν τε καὶ ἰσογώνιον ἀκολούθως τοῖς ἐπὶ τοῦ πενταγώνου εἰρημένοις. καὶ ἔτι διὰ τῶν ὁμοίων τοῖς ἐπὶ τοῦ πενταγώνου εἰρημένοις εἰς τὸ δοθέν ἑξάγωνον κύκλον ἐγγράφομεν τε καὶ περιγράφομεν· ὅπερ ἔδει ποιῆσαι.

† See the footnote to Prop. 4.6.

16.

Εἰς τὸν δοθέντα κύκλον πεντεκαίδεκάγωνον ἰσοπλευρὸν τε καὶ ἰσογώνιον ἐγγράφαι.

And the three angles of the triangle are equal to two right-angles [Prop. 1.32]. Thus, angle EGD is one third of two right-angles. So, similarly, DGC can also be shown (to be) one third of two right-angles. And since the straight-line CG , standing on EB , makes adjacent angles EGC and CGB equal to two right-angles [Prop. 1.13], the remaining angle CGB is thus also one third of two right-angles. Thus, angles EGD , DGC , and CGB are equal to one another. And hence the (angles) opposite to them BGA , AGF , and FGE are also equal [to EGD , DGC , and CGB (respectively)] [Prop. 1.15]. Thus, the six angles EGD , DGC , CGB , BGA , AGF , and FGE are equal to one another. And equal angles stand on equal circumferences [Prop. 3.26]. Thus, the six circumferences AB , BC , CD , DE , EF , and FA are equal to one another. And equal circumferences are subtended by equal straight-lines [Prop. 3.29]. Thus, the six straight-lines (AB , BC , CD , DE , EF , and FA) are equal to one another. Thus, hexagon $ABCDEF$ is equilateral. So, I say that (it is) also equiangular. For since circumference FA is equal to circumference ED , let circumference $ABCD$ be added to both. Thus, the whole of $FABCD$ is equal to the whole of $EDCBA$. And angle FED stands on circumference $FABCD$, and angle AFE on circumference $EDCBA$. Thus, angle AFE is equal to DEF [Prop. 3.27]. Similarly, it can also be shown that the remaining angles of hexagon $ABCDEF$ are individually equal to each of the angles AFE and FED . Thus, hexagon $ABCDEF$ is equiangular. And it was also shown (to be) equilateral. And it has been inscribed in circle $ABCDE$.

Thus, an equilateral and equiangular hexagon has been inscribed in the given circle. (Which is) the very thing it was required to do.

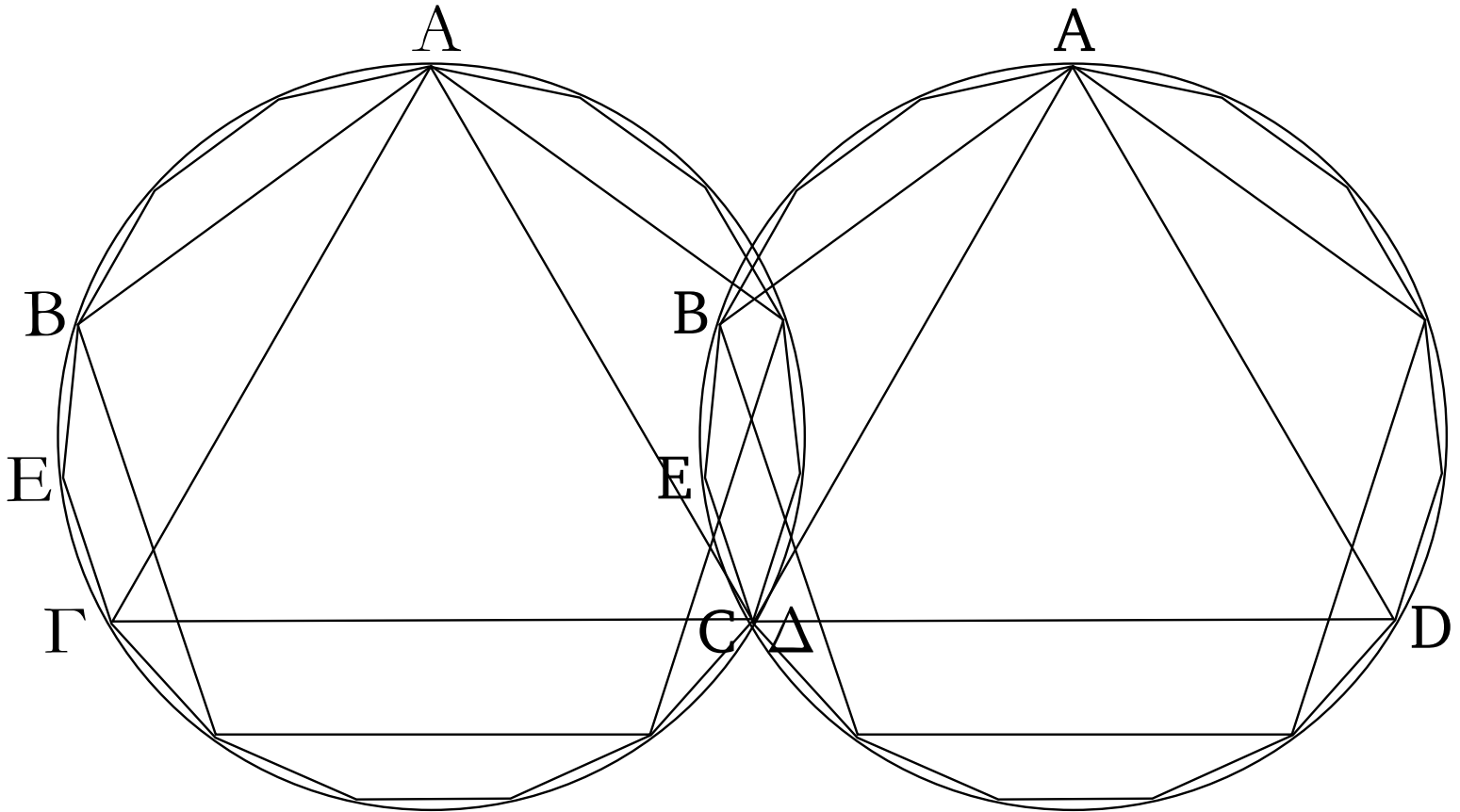
Corollary

So, from this, (it is) manifest that a side of the hexagon is equal to the radius of the circle.

And similarly to a pentagon, if we draw tangents to the circle through the (sixfold) divisions of the (circumference of the) circle, an equilateral and equiangular hexagon can be circumscribed about the circle, analogously to the aforementioned pentagon. And, further, by (means) similar to the aforementioned pentagon, we can inscribe and circumscribe a circle in (and about) a given hexagon. (Which is) the very thing it was required to do.

Proposition 16

To inscribe an equilateral and equiangular fifteen-sided figure in a given circle.



Ἐστω ὁ δοθείς κύκλος ὁ $ABΓΔ$. δεῖ δὴ εἰς τὸν $ABΓΔ$ κύκλον πεντεκαιδεκάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον ἐγγράφαι.

Ἐγγεγράφθω εἰς τὸν $ABΓΔ$ κύκλον τριγώνον μὲν ἰσοπλευρόν τοῦ εἰς αὐτὸν ἐγγραφομένου πλευρὰ ἢ $ΑΓ$, πενταγώνου δὲ ἰσοπλεύρου ἢ $ΑΒ$. οἷον ἄρα ἐστὶν ὁ $ABΓΔ$ κύκλος ἴσων τμημάτων δεκαπέντε, τοιούτων ἢ μὲν $ABΓ$ περιφέρεια τρίτον οὔσα τοῦ κύκλου ἔσται πέντε, ἢ δὲ AB περιφέρεια πέμpton οὔσα τοῦ κύκλου ἔσται τριῶν· λοιπὴ ἄρα ἢ $ΒΓ$ τῶν ἴσων δύο. τεμήσθω ἢ $ΒΓ$ δίδια κατὰ τὸ E . ἑκάτερα ἄρα τῶν BE , $ΕΓ$ περιφερειῶν πεντεκαιδεκάτον ἐστὶ τοῦ $ABΓΔ$ κύκλου.

Ἐὰν ἄρα ἐπιζεύχαντες τὰς BE , $ΕΓ$ ἴσαξ αὐταῖς κατὰ τὸ συνεθές εὐθείας ἐναρμόσωμεν εἰς τὸν $ABΓΔ[E]$ κύκλον, ἔσται εἰς αὐτὸν ἐγγεγραμμένον πεντεκαιδεκάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον· ὅπερ ἔδει ποιῆσαι.

Ὁμοίως δὲ τοῖς ἐπὶ τοῦ πενταγώνου ἔαν διὰ τῶν κατὰ τὸν κύκλον διαιρέσεων ἐφαπτομένας τοῦ κύκλου ἀγάγωμεν, περιγραφήσεται περὶ τὸν κύκλον πεντεκαιδεκάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον. ἔτι δὲ διὰ τῶν ὁμοίων τοῖς ἐπὶ τοῦ πενταγώνου δείξεω καὶ εἰς τὸ δοθὲν πεντεκαιδεκάγωνον κύκλον ἐγγράφομεν τε καὶ περιγράφομεν· ὅπερ ἔδει ποιῆσαι.

Let $ABCD$ be the given circle. So it is required to inscribe an equilateral and equiangular fifteen-sided figure in circle $ABCD$.

Let the side AC of an equilateral triangle inscribed in (the circle) [Prop. 4.2], and (the side) AB of an (inscribed) equilateral pentagon [Prop. 4.11], be inscribed in circle $ABCD$. Thus, just as the circle $ABCD$ is (made up) of fifteen equal pieces, the circumference ABC , being a third of the circle, will be (made up) of five such (pieces), and the circumference AB , being a fifth of the circle, will be (made up) of three. Thus, the remainder BC (will be made up) of two equal (pieces). Let (circumference) BC be cut in half at E [Prop. 3.30]. Thus, each of the circumferences BE and EC is one fifteenth of the circle $ABCDE$.

Thus, if, joining BE and EC , we continuously insert straight-lines equal to them into circle $ABCD[E]$ [Prop. 4.1], (then) an equilateral and equiangular fifteen-sided figure will be inserted into (the circle). (Which is) the very thing it was required to do.

And similarly to the pentagon, if we draw tangents to the circle through the (fifteenfold) divisions of the (circumference of the) circle, we can circumscribe an equilateral and equiangular fifteen-sided figure about the circle. And, further, through similar proofs to the pentagon, we can also inscribe

and circumscribe a circle in (and about) a given fifteen-sided figure. (Which is) the very thing it was required to do.

ELEMENTS BOOK 5

Proportion[†]

[†]The theory of proportion set out in this book is generally attributed to Eudoxus of Cnidus. The novel feature of this theory is its ability to deal with irrational magnitudes, which had hitherto been a major stumbling block for Greek mathematicians. Throughout the footnotes in this book, α, β, γ , etc., denote general (possibly irrational) magnitudes, whereas m, n, l , etc., denote positive integers.

Ὅροι.

Definitions

1. Μέρος ἐστὶ μέγεθος μεγέθους τὸ ἔλασσον τοῦ μείζονος, ὅταν καταμετρηῇ τὸ μείζον.

2. Πολλαπλάσιον δὲ τὸ μείζον τοῦ ἐλάττονος, ὅταν καταμετρηται ὑπὸ τοῦ ἐλάττονος.

3. Λόγος ἐστὶ δύο μεγεθῶν ὁμογενῶν ἢ κατὰ πληκίότητά ποια σθέσις.

4. Λόγον ἔθειν πρός ἄλληλα μεγέθη λέγεται, ἃ δύνανται πολλαπλασιαζόμενα ἀλλήλων ὑπερέθειν.

5. Ἐν τῷ αὐτῷ λόγῳ μεγέθη λέγεται εἶναι πρῶτον πρός δευτέρον καὶ τρίτον πρός τέταρτον, ὅταν τὰ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια τῶν τοῦ δευτέρου καὶ τετάρτου ἰσάκις πολλαπλασίων καθ' ὅποιον πολλαπλασιασμόν ἑκάτερον ἑκατέρου ἢ ἅμα ὑπερέθῃ ἢ ἅμα ἴσα ἢ ἢ ἅμα ἐλλείπῃ ληφθέντα κατάλληλα.

6. Τὰ δὲ τὸν αὐτὸν ἔθοντα λόγον μεγέθη ἀνάλογον καλεῖσθω.

7. Ὄταν δὲ τῶν ἰσάκις πολλαπλασίων τὸ μὲν τοῦ πρώτου πολλαπλάσιον ὑπερέθῃ τοῦ τοῦ δευτέρου πολλαπλασίου, τὸ δὲ τοῦ τρίτου πολλαπλάσιον μὴ ὑπερέθῃ τοῦ τοῦ τετάρτου πολλαπλασίου, τότε τὸ πρῶτον πρός τὸ δευτέρον μείζονα λόγον ἔθειν λέγεται, ἢ περὶ τὸ τρίτον πρός τὸ τέταρτον.

8. Ἀναλογία δὲ ἐν τρισὶν ὁροις ἐλαθίστη ἐστίν.

9. Ὄταν δὲ τρία μεγέθη ἀνάλογον ἢ, τὸ πρῶτον πρός τὸ τρίτον διπλασίονα λόγον ἔθειν λέγεται ἢ περὶ πρός τὸ δευτέρον.

10. Ὄταν δὲ τέσσαρα μεγέθη ἀνάλογον ἢ, τὸ πρῶτον πρός τὸ τέταρτον τριπλασίονα λόγον ἔθειν λέγεται ἢ περὶ πρός τὸ δευτέρον, καὶ αἱ ἐχῆς ὁμοίως, ὥς ἂν ἡ ἀναλογία ὑπάρθῃ.

11. Ὁμόλογα μεγέθη λέγεται τὰ μὲν ἡγούμενα τοῖς ἡγουμένοις τὰ δὲ ἐπόμενα τοῖς ἐπομένοις.

12. Ἐναλλάξ λόγος ἐστὶ λήψις τοῦ ἡγουμένου πρός τὸ ἡγούμενον καὶ τοῦ ἐπομένου πρός τὸ ἐπόμενον.

13. Ἀνάπαλιν λόγος ἐστὶ λήψις τοῦ ἐπομένου ὥς ἡγουμένου πρός τὸ ἡγούμενον ὥς ἐπόμενον.

14. Σύνθεσις λόγου ἐστὶ λήψις τοῦ ἡγουμένου μετὰ τοῦ ἐπομένου ὥς ἐνὸς πρός αὐτὸ τὸ ἐπόμενον.

15. Διαίρεσις λόγου ἐστὶ λήψις τῆς ὑπεροθῆς, ἢ ὑπερέθει τὸ ἡγούμενον τοῦ ἐπομένου, πρὸς αὐτὸ τὸ ἐπόμενον.

16. Ἀναστροφὴ λόγου ἐστὶ λήψις τοῦ ἡγουμένου πρός τὴν ὑπεροθὴν, ἢ ὑπερέθει τὸ ἡγούμενον τοῦ ἐπομένου.

17. Δι' ἴσον λόγος ἐστὶ πλειόνων ὄντων μεγεθῶν καὶ ἄλλων αὐτοῖς ἴσων τὸ πλῆθος σύνδυο λαμβανομένων καὶ ἐν τῷ αὐτῷ λόγῳ, ὅταν ἢ ὥς ἐν τοῖς πρώτοις μεγέθεσι τὸ πρῶτον πρός τὸ ἑσθαιον, ὁ ὅτις ἐν τοῖς δευτέροις μεγέθεσι τὸ πρῶτον πρός τὸ ἑσθαιον ἢ ἄλλως λήψις τῶν ἀκρῶν καθ' ὑπερχαίρεσιν τῶν μέσων.

18. Τεταραγμένη δὲ ἀναλογία ἐστίν, ὅταν τριῶν ὄντων μεγεθῶν καὶ ἄλλων αὐτοῖς ἴσων τὸ πλῆθος γίνηται ὥς μὲν ἐν

1. A magnitude is a part of a(nother) magnitude, the lesser of the greater, when it measures the greater.[†]

2. And the greater (magnitude is) a multiple of the lesser when it is measured by the lesser.

3. A ratio is a certain type of relation with respect to size of two magnitudes of the same kind.[‡]

4. (Those) magnitudes are said to have a ratio with respect to one another which, being multiplied, are capable of exceeding one another.[§]

5. Magnitudes are said to be in the same ratio, the first to the second, and the third to the fourth, when equal multiples of the first and the third either both exceed, are both equal to, or are both less than, equal multiples of the second and the fourth, respectively, being taken in corresponding order, according to any kind of multiplication whatever.[¶]

6. And let magnitudes having the same ratio be called proportional.*

7. And when for equal multiples (as in Def. 5), the multiple of the first (magnitude) exceeds the multiple of the second, and the multiple of the third (magnitude) does not exceed the multiple of the fourth, (then) the first (magnitude) is said to have a greater ratio to the second than the third (magnitude has) to the fourth.

8. And a proportion in three terms is the smallest (possible).[§]

9. And when three magnitudes are proportional, the first is said to have to the third the squared^b ratio of that (it has) to the second.^{††}

10. And when four magnitudes are (continuously) proportional, the first is said to have to the fourth the cubed^{‡‡} ratio of that (it has) to the second.^{§§} And so on, similarly, in successive order, whatever the (continuous) proportion might be.

11. These magnitudes are said to be corresponding (magnitudes): the leading to the leading (of two ratios), and the following to the following.

12. An alternate ratio is a taking of the (ratio of the) leading (magnitude) to the leading (of two equal ratios), and (setting it equal to) the (ratio of the) following (magnitude) to the following.^{¶¶}

13. An inverse ratio is a taking of the (ratio of the) following (magnitude) as the leading and the leading (magnitude) as the following.^{**}

14. A composition of a ratio is a taking of the (ratio of the) leading plus the following (magnitudes), as one, to the following (magnitude) by itself.^{§§}

15. A separation of a ratio is a taking of the (ratio of the) excess by which the leading (magnitude) exceeds the following to the following (magnitude) by itself.^{bb}

16. A conversion of a ratio is a taking of the (ratio of the)

τοῖς πρώτοις μεγέθεσιν ἡγούμενον πρόξ ἐπόμενον, οὔτως ἐν τοῖς δευτέροις μεγέθεσιν ἡγούμενον πρόξ ἐπόμενον, ὡς δὲ ἐν τοῖς πρώτοις μεγέθεσιν ἐπόμενον πρόξ ἄλλο τι, οὔτως ἐν τοῖς δευτέροις ἄλλο τι πρόξ ἡγούμενον.

leading (magnitude) to the excess by which the leading (magnitude) exceeds the following.^{†††}

17. There being several magnitudes, and other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, a ratio via equality (i.e., ex aequali) occurs when as the first is to the last in the first (set of) magnitudes, so the first (is) to the last in the second (set of) magnitudes. Or alternately, (it is) a taking of the (ratio of the) outer (magnitudes) by the removal of the inner (magnitudes).^{‡‡‡}

18. There being three magnitudes, and other (magnitudes) of equal number to them, a perturbed proportion occurs when as the leading is to the following in the first (set of) magnitudes, so the leading (is) to the following in the second (set of) magnitudes, and as the following (is) to some other (i.e., the remaining magnitude) in the first (set of) magnitudes, so some other (is) to the leading in the second (set of) magnitudes.^{§§§}

[†] In other words, α is said to be a part of β if $\beta = m\alpha$.

[‡] In modern notation, the ratio of two magnitudes, α and β , is denoted $\alpha : \beta$.

[§] In other words, α has a ratio with respect to β if $m\alpha > \beta$ and $n\beta > \alpha$, for some m and n .

[¶] In other words, $\alpha : \beta :: \gamma : \delta$ if and only if $m\alpha > n\beta$ whenever $m\gamma > n\delta$, and $m\alpha = n\beta$ whenever $m\gamma = n\delta$, and $m\alpha < n\beta$ whenever $m\gamma < n\delta$, for all m and n . This definition is the kernel of Eudoxus' theory of proportion, and is valid even if α, β , etc., are irrational.

^{*} Thus if α and β have the same ratio as γ and δ then they are proportional. In modern notation, $\alpha : \beta :: \gamma : \delta$.

[§] In modern notation, a proportion in three terms— α, β , and γ —is written: $\alpha : \beta :: \beta : \gamma$.

^b Literally, “double”.

^{††} In other words, if $\alpha : \beta :: \beta : \gamma$ then $\alpha : \gamma :: \alpha^2 : \beta^2$.

^{‡‡} Literally, “triple”.

^{§§} In other words, if $\alpha : \beta :: \beta : \gamma :: \gamma : \delta$ then $\alpha : \delta :: \alpha^3 : \beta^3$.

^{¶¶} In other words, if $\alpha : \beta :: \gamma : \delta$ then the alternate ratio corresponds to $\alpha : \gamma :: \beta : \delta$.

^{**} In other words, if $\alpha : \beta$ then the inverse ratio corresponds to $\beta : \alpha$.

^{§§} In other words, if $\alpha : \beta$ then the composed ratio corresponds to $\alpha + \beta : \beta$.

^{bb} In other words, if $\alpha : \beta$ then the separated ratio corresponds to $\alpha - \beta : \beta$.

^{†††} In other words, if $\alpha : \beta$ then the converted ratio corresponds to $\alpha : \alpha - \beta$.

^{‡‡‡} In other words, if α, β, γ are the first set of magnitudes, and δ, ϵ, ζ the second set, and $\alpha : \beta : \gamma :: \delta : \epsilon : \zeta$, then the ratio via equality (i.e., ex aequali) corresponds to $\alpha : \gamma :: \delta : \zeta$.

^{§§§} In other words, if α, β, γ are the first set of magnitudes, and δ, ϵ, ζ the second set, and $\alpha : \beta :: \delta : \epsilon$ as well as $\beta : \gamma :: \zeta : \delta$, then the proportion is said to be perturbed.

1.

Proposition 1[†]

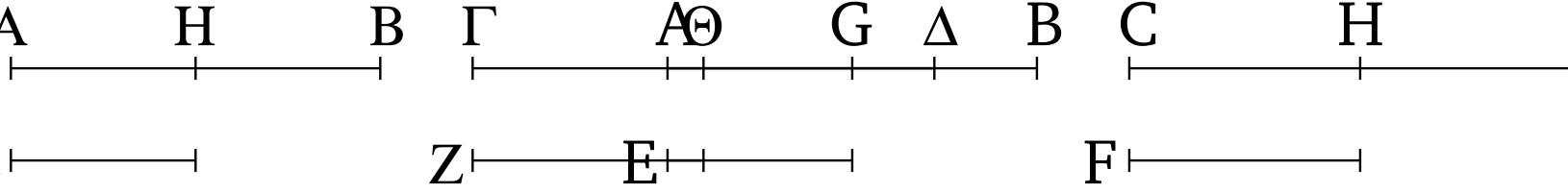
Ἐάν ᾗ ὁποσαοῦν μεγέθη ὁποσωνοῦν μεγεθῶν ἴσων τὸ πληθὸς ἑκάστον ἑκάστον ἰσάκις πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ἐν τῶν μεγεθῶν ἐνόξ, τοσαυταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων.

Ἐστω ὁποσαοῦν μεγέθη τὰ $AB, \Gamma\Delta$ ὁποσωνοῦν μεγεθῶν τῶν E, Z ἴσων τὸ πληθὸς ἑκάστον ἑκάστον ἰσάκις πολλαπλάσιον· λέγω, ὅτι ὁσαπλάσιόν ἐστι τὸ AB τοῦ E , τοσαυταπλάσια ἔσται καὶ τὰ $AB, \Gamma\Delta$ τῶν E, Z .

If there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second).

Let there be any number of magnitudes whatsoever, AB, CD , (which are) equal multiples, respectively, of some (other) magnitudes, E, F , of equal number (to them). I say that as

many times as AB is (divisible) by E , so many times will AB , CD also be (divisible) by E, F .



Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AB τοῦ E καὶ τὸ $\Gamma\Delta$ τοῦ Z , ὅσα ἄρα ἐστὶν ἐν τῷ AB μεγέθη ἴσα τῷ E , τοσαῦτα καὶ ἐν τῷ $\Gamma\Delta$ ἴσα τῷ Z . διηγήσθω τὸ μὲν AB εἰς τὰ τῷ E μεγέθη ἴσα τὰ AH, HB , τὸ δὲ $\Gamma\Delta$ εἰς τὰ τῷ Z ἴσα τὰ $\Gamma\Theta, \Theta\Delta$. ἔσται δὴ ἴσον τὸ πλῆθος τῶν AH, HB τῷ πλῆθει τῶν $\Gamma\Theta, \Theta\Delta$. καὶ ἐπεὶ ἴσον ἐστὶ τὸ μὲν AH τῷ E , τὸ δὲ $\Gamma\Theta$ τῷ Z , ἴσον ἄρα τὸ AH τῷ E , καὶ τὰ $AH, \Gamma\Theta$ τοῖς E, Z . διὰ τὰ αὐτὰ δὴ ἴσον ἐστὶ τὸ HB τῷ E , καὶ τὰ $HB, \Theta\Delta$ τοῖς E, Z . ὅσα ἄρα ἐστὶν ἐν τῷ AB ἴσα τῷ E , τοσαῦτα καὶ ἐν τοῖς $AB, \Gamma\Delta$ ἴσα τοῖς E, Z . ὁσαυτάσιον ἄρα ἐστὶ τὸ AB τοῦ E , τοσαυταπλάσια ἔσται καὶ τὰ $AB, \Gamma\Delta$ τῶν E, Z .

Ἐάν ἄρα ἡ ὁποσαοῦν μεγέθη ὁποσωνοῦν μεγεθῶν ἴσων τὸ πλῆθος ἕκαστον ἐκάστον ἰσάκις πολλαπλάσιον, ὁσαυτάσιον ἐστὶν ἐν τῶν μεγεθῶν ἐνόξ, τοσαυταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων ὅπερ ἔδει δεῖξαι.

For since AB, CD are equal multiples of E, F , thus as many magnitudes as (there) are in AB equal to E , so many (are there) also in CD equal to F . Let AB be divided into magnitudes AG, GB , equal to E , and CD into (magnitudes) CH, HD , equal to F . So, the number of (divisions) AG, GB will be equal to the number of (divisions) CH, HD . And since AG is equal to E , and CH to F , AG (is) thus equal to E , and AG, CH to E, F . So, for the same (reasons), GB is equal to E , and GB, HD to E, F . Thus, as many (magnitudes) as (there) are in AB equal to E , so many (are there) also in AB, CD equal to E, F . Thus, as many times as AB is (divisible) by E , so many times will AB, CD also be (divisible) by E, F .

Thus, if there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\alpha + m\beta + \dots = m(\alpha + \beta + \dots)$.

2.

Ἐάν πρῶτον δευτέρον ἰσάκις ἢ πολλαπλάσιον καὶ τρίτον τετάρτον, ἢ δὲ καὶ πέμπτον δευτέρον ἰσάκις πολλαπλάσιον καὶ ἕκτον τετάρτον, καὶ συντεθὲν πρῶτον καὶ πέμπτον δευτέρον ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τετάρτον.

Πρῶτον γὰρ τὸ AB δευτέρον τοῦ Γ ἰσάκις ἔστω πολλαπλάσιον καὶ τρίτον τὸ ΔE τετάρτον τοῦ Z , ἔστω δὲ καὶ πέμπτον τὸ BH δευτέρον τοῦ Γ ἰσάκις πολλαπλάσιον καὶ ἕκτον τὸ $E\Theta$ τετάρτον τοῦ Z . λέγω, ὅτι καὶ συντεθὲν πρῶτον καὶ πέμπτον τὸ AH δευτέρον τοῦ Γ ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τὸ $\Delta\Theta$ τετάρτον τοῦ Z .

Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AB τοῦ Γ καὶ τὸ ΔE τοῦ Z , ὅσα ἄρα ἐστὶν ἐν τῷ AB ἴσα τῷ Γ , τοσαῦτα καὶ ἐν τῷ ΔE ἴσα τῷ Z . διὰ τὰ αὐτὰ δὴ καὶ ὅσα ἐστὶν ἐν τῷ BH ἴσα τῷ Γ , τοσαῦτα καὶ ἐν τῷ $E\Theta$ ἴσα τῷ Z . ὅσα ἄρα ἐστὶν ἐν ὅλῳ τῷ AH ἴσα τῷ Γ , τοσαῦτα καὶ ἐν ὅλῳ τῷ $\Delta\Theta$ ἴσα τῷ Z . ὁσαυτάσιον ἄρα ἐστὶ τὸ AH τοῦ Γ , τοσαυταπλάσιον ἔσται καὶ τὸ $\Delta\Theta$ τοῦ Z . καὶ συντεθὲν ἄρα πρῶτον καὶ πέμπτον τὸ AH δευτέρον τοῦ Γ ἰσάκις ἔσται πολλαπλάσιον καὶ

Proposition 2†

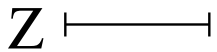
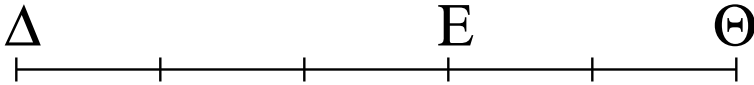
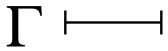
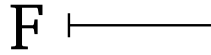
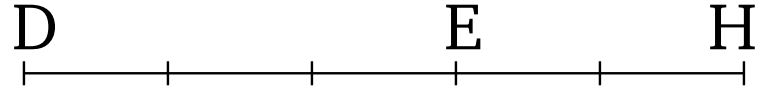
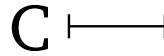
If a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and a fifth (magnitude) and a sixth (are) also equal multiples of the second and fourth (respectively), (then) the first (magnitude) and the fifth, being added together, and the third and the sixth, (being added together), will also be equal multiples of the second (magnitude) and the fourth (respectively).

For let a first (magnitude) AB and a third DE be equal multiples of a second C and a fourth F (respectively). And let a fifth (magnitude) BG and a sixth EH also be (other) equal multiples of the second C and the fourth F (respectively). I say that the first (magnitude) and the fifth, being added together, (to give) AG , and the third (magnitude) and the sixth, (being added together, to give) DH , will also be equal multiples of the second (magnitude) C and the fourth F (respectively).

For since AB and DE are equal multiples of C and F (respectively), thus as many (magnitudes) as (there) are in AB equal to C , so many (are there) also in DE equal to F . And so, for the same (reasons), as many (magnitudes) as (there) are

τρίτον καὶ ἕκτον τὸ ΔΘ τετάρτον τοῦ Ζ.

in BG equal to C , so many (are there) also in EH equal to F . Thus, as many (magnitudes) as (there) are in the whole of AG equal to C , so many (are there) also in the whole of DH equal to F . Thus, as many times as AG is (divisible) by C , so many times will DH also be divisible by F . Thus, the first (magnitude) and the fifth, being added together, (to give) AG , and the third (magnitude) and the sixth, (being added together, to give) DH , will also be equal multiples of the second (magnitude) C and the fourth F (respectively).



Ἐάν ἄρα πρῶτον δευτέρου ἰσάκις ἢ πολλαπλάσιον καὶ τρίτον τετάρτου, ἢ δὲ καὶ πέμπτον δευτέρου ἰσάκις πολλαπλάσιον καὶ ἕκτον τετάρτου, καὶ συντεθέν πρῶτον καὶ πέμπτον δευτέρου ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τετάρτου· ὅπερ ἔδει δεῖξαι.

Thus, if a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and a fifth (magnitude) and a sixth (are) also equal multiples of the second and fourth (respectively), (then) the first (magnitude) and the fifth, being added together, and the third and sixth, (being added together), will also be equal multiples of the second (magnitude) and the fourth (respectively). (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads $m\alpha + n\alpha = (m + n)\alpha$.

Ἐάν πρῶτον δευτέρου ἰσάκις ἢ πολλαπλάσιον καὶ τρίτον τετάρτου, ληφθῇ δὲ ἰσάκις πολλαπλάσια τοῦ τε πρώτου καὶ

If a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and equal multiples are

τρίτον, καὶ δι' ἴσον τῶν ληφθέντων ἑκάτερον ἑκατέρου ἰσάκις ἔσται πολλαπλάσιον τὸ μὲν τοῦ δευτέρου τὸ δὲ τοῦ τετάρτου.

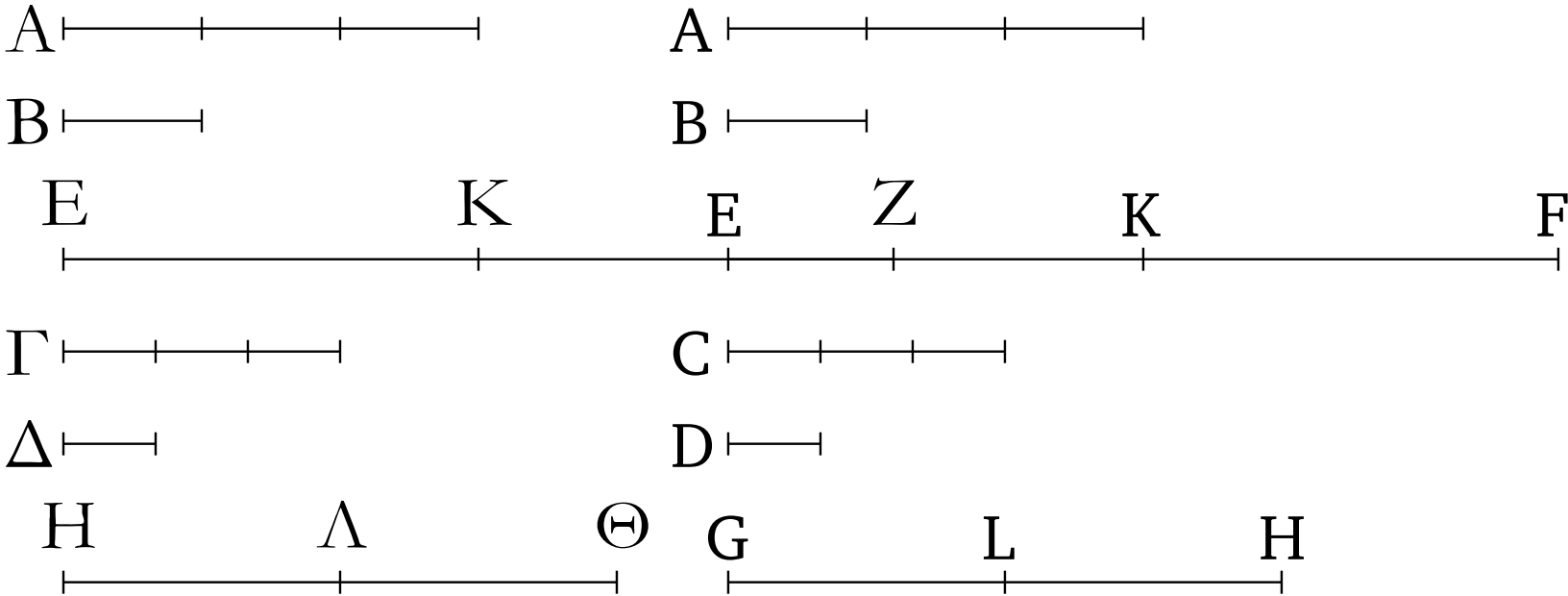
Πρῶτον γάρ τὸ A δευτέρου τοῦ B ἰσάκις ἔστω πολλαπλάσιον καὶ τρίτον τὸ Γ τετάρτου τοῦ Δ , καὶ εἰλήφθω τῶν A , Γ ἰσάκις πολλαπλάσια τὰ EZ , $H\Theta$. λέγω, ὅτι ἰσάκις ἔστι πολλαπλάσιον τὸ EZ τοῦ B καὶ τὸ $H\Theta$ τοῦ Δ .

Ἐπεὶ γὰρ ἰσάκις ἔστι πολλαπλάσιον τὸ EZ τοῦ A καὶ τὸ $H\Theta$ τοῦ Γ , ὅσα ἄρα ἔστιν ἐν τῷ EZ ἴσα τῷ A , τοσαῦτα καὶ ἐν τῷ $H\Theta$ ἴσα τῷ Γ . διηγήσθω τὸ μὲν EZ εἰς τὰ τῷ A μεγέθη ἴσα τὰ EK , KZ , τὸ δὲ $H\Theta$ εἰς τὰ τῷ Γ ἴσα τὰ $ΗΛ$, $\Lambda\Theta$. ἔσται δὴ ἴσον τὸ πλῆθος τῶν EK , KZ τῷ πλῆθει τῶν $ΗΛ$, $\Lambda\Theta$. καὶ ἐπεὶ ἰσάκις ἔστι πολλαπλάσιον τὸ A τοῦ B καὶ τὸ Γ τοῦ Δ , ἴσον δὲ τὸ μὲν EK τῷ A , τὸ δὲ $ΗΛ$ τῷ Γ , ἰσάκις ἄρα ἔστι πολλαπλάσιον τὸ EK τοῦ B καὶ τὸ $ΗΛ$ τοῦ Δ . διὰ τὰ αὐτὰ δὴ ἰσάκις ἔστι πολλαπλάσιον τὸ KZ τοῦ B καὶ τὸ $\Lambda\Theta$ τοῦ Δ . ἐπεὶ οὖν πρῶτον τὸ EK δευτέρου τοῦ B ἰσάκις ἔστι πολλαπλάσιον καὶ τρίτον τὸ $ΗΛ$ τετάρτου τοῦ Δ , ἔστι δὲ καὶ πέμπτον τὸ KZ δευτέρου τοῦ B ἰσάκις πολλαπλάσιον καὶ ἕκτον τὸ $\Lambda\Theta$ τετάρτου τοῦ Δ , καὶ συντεθέν ἄρα πρῶτον καὶ πέμπτον τὸ EZ δευτέρου τοῦ B ἰσάκις ἔστι πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τὸ $H\Theta$ τετάρτου τοῦ Δ .

taken of the first and the third, (then), via equality, the (magnitudes) taken will also be equal multiples of the second (magnitude) and the fourth, respectively.

For let a first (magnitude) A and a third C be equal multiples of a second B and a fourth D (respectively), and let the equal multiples EF and GH be taken of A and C (respectively). I say that EF and GH are equal multiples of B and D (respectively).

For since EF and GH are equal multiples of A and C (respectively), thus as many (magnitudes) as (there) are in EF equal to A , so many (are there) also in GH equal to C . Let EF be divided into magnitudes EK , KF equal to A , and GH into (magnitudes) GL , LH equal to C . So, the number of (magnitudes) EK , KF will be equal to the number of (magnitudes) GL , LH . And since A and C are equal multiples of B and D (respectively), and EK (is) equal to A , and GL to C , EK and GL are thus equal multiples of B and D (respectively). So, for the same (reasons), KF and LH are equal multiples of B and D (respectively). Therefore, since the first (magnitude) EK and the third GL are equal multiples of the second B and the fourth D (respectively), and the fifth (magnitude) KF and the sixth LH are also equal multiples of the second B and the fourth D (respectively), then the first (magnitude) and fifth, being added together, (to give) EF , and the third (magnitude) and sixth, (being added together, to give) GH , are thus also equal multiples of the second (magnitude) B and the fourth D (respectively) [Prop. 5.2].



Ἐὰν ἄρα πρῶτον δευτέρου ἰσάκις ἢ πολλαπλάσιον καὶ τρίτον τετάρτου, ληφθῇ δὲ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια, καὶ δι' ἴσον τῶν ληφθέντων ἑκάτερον ἑκατέρου ἰσάκις ἔσται πολλαπλάσιον τὸ μὲν τοῦ δευτέρου τὸ δὲ τοῦ

Thus, if a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and equal multiples are taken of the first and the third, (then), via equality, the (magnitudes) taken will also be equal multiples of the second

τετάρτον· ὅπερ ἔδει δεῖχαι.

(magnitude) and the fourth, respectively. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads $m(n\alpha) = (mn)\alpha$.

4.

Proposition 4[†]

Ἐάν πρῶτον πρόξ δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρόξ τέταρτον, καὶ τὰ ἰσάκις πολλαπλάσια τοῦ τε πρῶτου καὶ τρίτου πρόξ τὰ ἰσάκις πολλαπλάσια τοῦ δευτέρου καὶ τετάρτου καθ' ὁποιοῦν πολλαπλασιασμόν τὸν αὐτὸν ἔχει λόγον ληφθέντα κατάλληλα.

Πρῶτον γάρ τὸ A πρόξ δεύτερον τὸ B τὸν αὐτὸν ἐθέτω λόγον καὶ τρίτον τὸ Γ πρόξ τέταρτον τὸ Δ , καὶ εἰλήφθω τῶν μὲν A, Γ ἰσάκις πολλαπλάσια τὰ E, Z , τῶν δὲ B, Δ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ H, Θ . λέγω, ὅτι ἐστὶν ὥς τὸ E πρόξ τὸ H , οὕτως τὸ Z πρόξ τὸ Θ .

Εἰλήφθω γάρ τῶν μὲν E, Z ἰσάκις πολλαπλάσια τὰ K, Λ , τῶν δὲ H, Θ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ M, N .

[Καί] ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ μὲν E τοῦ A , τὸ δὲ Z τοῦ Γ , καὶ εἴληπται τῶν E, Z ἰσάκις πολλαπλάσια τὰ K, Λ , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ K τοῦ A καὶ τὸ Λ τοῦ Γ . διὰ τὰ αὐτὰ δὴ ἰσάκις ἐστὶ πολλαπλάσιον τὸ M τοῦ B καὶ τὸ N τοῦ Δ . καὶ ἐπεὶ ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , καὶ εἴληπται τῶν μὲν A, Γ ἰσάκις πολλαπλάσια τὰ K, Λ , τῶν δὲ B, Δ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ M, N , εἰ ἄρα ὑπερέχει τὸ K τοῦ M , ὑπερέχει καὶ τὸ Λ τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν K, Λ τῶν E, Z ἰσάκις πολλαπλάσια, τὰ δὲ M, N τῶν H, Θ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια· ἐστὶν ἄρα ὥς τὸ E πρὸς τὸ H , οὕτως τὸ Z πρὸς τὸ Θ .

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, (then) equal multiples of the first (magnitude) and the third will also have the same ratio to equal multiples of the second and the fourth, being taken in corresponding order, according to any kind of multiplication whatsoever.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D . And let equal multiples E and F be taken of A and C (respectively), and other random equal multiples G and H of B and D (respectively). I say that as E (is) to G , so F (is) to H .

For let equal multiples K and L be taken of E and F (respectively), and other random equal multiples M and N of G and H (respectively).

[And] since E and F are equal multiples of A and C (respectively), and the equal multiples K and L have been taken of E and F (respectively), K and L are thus equal multiples of A and C (respectively) [Prop. 5.3]. So, for the same (reasons), M and N are equal multiples of B and D (respectively). And since as A is to B , so C (is) to D , and the equal multiples K and L have been taken of A and C (respectively), and the other random equal multiples M and N of B and D (respectively), (then) if K exceeds M (then) L also exceeds N , and if (K is) equal (to M then L is also) equal (to N), and if (K is) less (than M then L is also) less (than N) [Def. 5.5]. And K and L are equal multiples of E and F (respectively), and M and N other random equal multiples of G and H (respectively). Thus, as E (is) to G , so F (is) to H [Def. 5.5].

A

B

E

H

K

M

Γ

Δ

Z

Θ

Λ

N

A

B

E

G

K

M

C

D

F

H

L

N

Ἐάν ᾗρα πρῶτον πρόξ δεύτερον τὸν αὐτὸν ἔθῃ λόγον
καὶ τρίτον πρόξ τέταρτον, καὶ τὰ ἰσάκις πολλαπλάσια τοῦ τε
πρώτου καὶ τρίτου πρόξ τὰ ἰσάκις πολλαπλάσια τοῦ δευτέρου
καὶ τετάρτου τὸν αὐτὸν ἔχει λόγον καθ' ὅποιον οὖν πολλαπλα-
σιασμὸν ληφθέντα κατάλληλα· ὅπερ ἔδει δεῖξαι.

Thus, if a first (magnitude) has the same ratio to a sec-
ond that a third (has) to a fourth, (then) equal multiples of the
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equal multiples of the second and the fourth, being taken in
corresponding order, according to any kind of multiplication
whatsoever. (Which is) the very thing it was required to show.

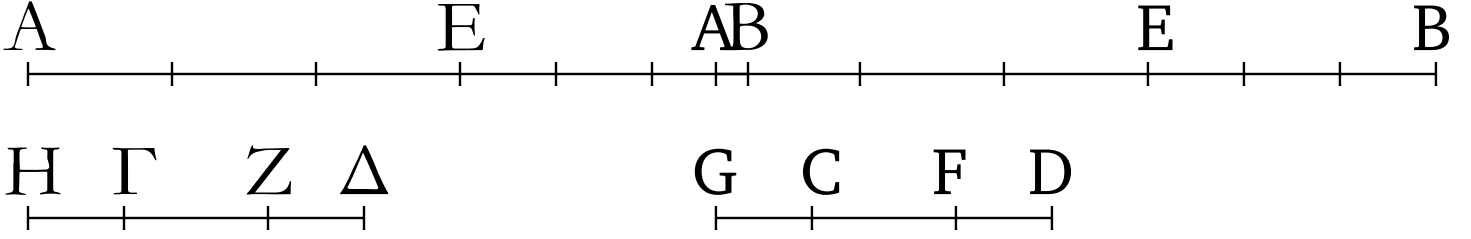
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $m\alpha : n\beta :: m\gamma : n\delta$, for all m and n .

5.

Proposition 5[†]

Ἐάν μέγεθος μέγεθος ἰσάκις ἢ πολλαπλάσιον, ὅπερ
ἀφαιρεθὲν ἀφαιρεθέντος, καὶ τὸ λοιπὸν τοῦ λοιποῦ ἰσάκις
ἔσται πολλαπλάσιον, ὅσαπλάσιόν ἐστι τὸ ὅλον τοῦ ὅλου.

If a magnitude is the same multiple of a magnitude that
a (part) taken away (is) of a (part) taken away (respectively),
(then) the remainder will also be the same multiple of the re-
mainder as that which the whole (is) of the whole (respec-
tively).



Μέγεθος γάρ τὸ AB μεγέθους τοῦ $\Gamma\Delta$ ἰσάκις ἔστω πολλαπλάσιον, ὅπερ ἀφαιρεθέν τὸ AE ἀφαιρεθέντος τοῦ ΓZ λέγω, ὅτι καὶ λοιπὸν τὸ EB λοιποῦ τοῦ $Z\Delta$ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ὅλον τὸ AB ὅλον τοῦ $\Gamma\Delta$.

Ὅσαπλάσιον γάρ ἐστι τὸ AE τοῦ ΓZ , τοσαυταπλάσιον γενοῦντο καὶ τὸ EB τοῦ ΓH .

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ EB τοῦ ΓH , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ AB τοῦ HZ . κεῖται δὲ ἰσάκις πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ AB τοῦ $\Gamma\Delta$. ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ AB ἑκατέρων τῶν HZ , $\Gamma\Delta$. ἴσον ἄρα τὸ HZ τῷ $\Gamma\Delta$. κοινὸν ἀφηρήσθω τὸ ΓZ . λοιπὸν ἄρα τὸ $H\Gamma$ λοιπῷ τῷ $Z\Delta$ ἴσον ἐστίν. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ EB τοῦ ΓH , ἴσον δὲ τὸ $H\Gamma$ τῷ $Z\Delta$, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ EB τοῦ $Z\Delta$. ἰσάκις δὲ ὑπόκειται πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ AB τοῦ $\Gamma\Delta$. ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ EB τοῦ $Z\Delta$ καὶ τὸ AB τοῦ $\Gamma\Delta$. καὶ λοιπὸν ἄρα τὸ EB λοιποῦ τοῦ $Z\Delta$ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ὅλον τὸ AB ὅλον τοῦ $\Gamma\Delta$.

Ἐὰν ἄρα μέγεθος μεγέθους ἰσάκις ἢ πολλαπλάσιον, ὅπερ ἀφαιρεθέν ἀφαιρεθέντος, καὶ τὸ λοιπὸν τοῦ λοιποῦ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστι καὶ τὸ ὅλον τοῦ ὅλου· ὅπερ ἔδει δεῖξαι.

For let the magnitude AB be the same multiple of the magnitude CD that the (part) taken away AE (is) of the (part) taken away CF (respectively). I say that the remainder EB will also be the same multiple of the remainder FD as that which the whole AB (is) of the whole CD (respectively).

For as many times as AE is (divisible) by CF , so many times let EB also be made (divisible) by CG .

And since AE and EB are equal multiples of CF and GC (respectively), AE and AB are thus equal multiples of CF and GF (respectively) [Prop. 5.1]. And AE and AB are assumed (to be) equal multiples of CF and CD (respectively). Thus, AB is an equal multiple of each of GF and CD . Thus, GF (is) equal to CD . Let CF be subtracted from both. Thus, the remainder GC is equal to the remainder FD . And since AE and EB are equal multiples of CF and GC (respectively), and GC (is) equal to DF , AE and EB are thus equal multiples of CF and FD (respectively). And AE and AB are assumed (to be) equal multiples of CF and CD (respectively). Thus, EB and AB are equal multiples of FD and CD (respectively). Thus, the remainder EB will also be the same multiple of the remainder FD as that which the whole AB (is) of the whole CD (respectively).

Thus, if a magnitude is the same multiple of a magnitude that a (part) taken away (is) of a (part) taken away (respectively), (then) the remainder will also be the same multiple of the remainder as that which the whole (is) of the whole (respectively). (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads $m\alpha - m\beta = m(\alpha - \beta)$.

6.

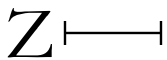
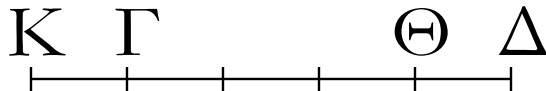
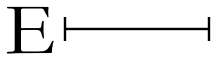
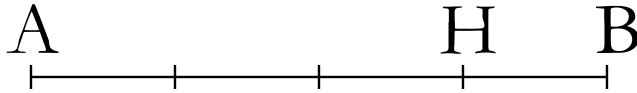
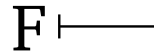
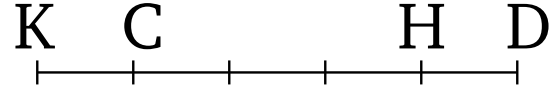
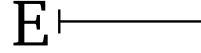
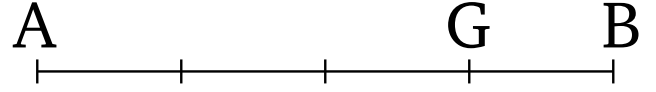
Ἐὰν δύο μεγέθη δύο μεγεθῶν ἰσάκις ἢ πολλαπλάσια, καὶ ἀφαιρεθέντα τινὰ τῶν αὐτῶν ἰσάκις ἢ πολλαπλάσια, καὶ τὰ λοιπὰ τοῖς αὐτοῖς ἴησι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια.

Δύο γάρ μεγέθη τὰ AB , $\Gamma\Delta$ δύο μεγεθῶν τῶν E , Z ἰσάκις ἔστω πολλαπλάσια, καὶ ἀφαιρεθέντα τὰ AH , $\Gamma\Theta$ τῶν αὐτῶν τῶν E , Z ἰσάκις ἔστω πολλαπλάσια· λέγω, ὅτι καὶ λοιπὰ τὰ HB , $\Theta\Delta$ τοῖς E , Z ἴησι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια.

Proposition 6[†]

If two magnitudes are equal multiples of two (other) magnitudes, and some (parts) taken away (from the former magnitudes) are equal multiples of the latter (magnitudes, respectively), (then) the remainders are also either equal to the latter (magnitudes), or (are) equal multiples of them (respectively).

For let two magnitudes AB and CD be equal multiples of two magnitudes E and F (respectively). And let the (parts) taken away (from the former) AG and CH be equal multiples of E and F (respectively). I say that the remainders GB and HD are also either equal to E and F (respectively), or (are) equal multiples of them.



Ἐστω γάρ πρότερον τὸ HB τῷ E ἴσον· λέγω, ὅτι καὶ τὸ $\Theta\Delta$ τῷ Z ἴσον ἐστίν.

Κείσθω γάρ τῷ Z ἴσον τὸ ΓK . ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AH τοῦ E καὶ τὸ $\Gamma\Theta$ τοῦ Z , ἴσον δὲ τὸ μὲν HB τῷ E , τὸ δὲ $K\Gamma$ τῷ Z , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ AB τοῦ E καὶ τὸ $K\Theta$ τοῦ Z . ἰσάκις δὲ ὑπόκειται πολλαπλάσιον τὸ AB τοῦ E καὶ τὸ $\Gamma\Delta$ τοῦ Z . ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ $K\Theta$ τοῦ Z καὶ τὸ $\Gamma\Delta$ τοῦ Z . ἐπεὶ οὖν ἐκάτερον τῶν $K\Theta$, $\Gamma\Delta$ τοῦ Z ἰσάκις ἐστὶ πολλαπλάσιον, ἴσον ἄρα ἐστὶ τὸ $K\Theta$ τῷ $\Gamma\Delta$. κοινὸν ἀφηρήσθω τὸ $\Gamma\Theta$. λοιπὸν ἄρα τὸ $K\Gamma$ λοιπὸν τῷ $\Theta\Delta$ ἴσον ἐστίν. ἀλλὰ τὸ Z τῷ $K\Gamma$ ἴσον· καὶ τὸ $\Theta\Delta$ ἄρα τῷ Z ἴσον ἐστίν. ὥστε εἰ τὸ HB τῷ E ἴσον ἐστίν, καὶ τὸ $\Theta\Delta$ ἴσον ἔσται τῷ Z .

Ὁμοίως δὴ δείξομεν, ὅτι, κ' ἂν πολλαπλάσιον ᾖ τὸ HB τοῦ E , τοσαυταπλάσιον ἔσται καὶ τὸ $\Theta\Delta$ τοῦ Z .

Ἐὰν ἄρα δύο μεγέθη δύο μεγεθῶν ἰσάκις ᾖ πολλαπλάσια, καὶ ἀφαιρεθέντα τινὰ τῶν αὐτῶν ἰσάκις ᾖ πολλαπλάσια, καὶ

For let GB be, first of all, equal to E . I say that HD is also equal to F .

For let CK be made equal to F . Since AG and CH are equal multiples of E and F (respectively), and GB (is) equal to E , and KC to F , AB and KH are thus equal multiples of E and F (respectively) [Prop. 5.2]. And AB and CD are assumed (to be) equal multiples of E and F (respectively). Thus, KH and CD are equal multiples of F and F (respectively). Therefore, KH and CD are each equal multiples of F . Thus, KH is equal to CD . Let CH be taken away from both. Thus, the remainder KC is equal to the remainder HD . But, F is equal to KC . Thus, HD is also equal to F . Hence, if GB is equal to E , (then) HD will also be equal to F .

So, similarly, we can show that even if GB is a multiple of E , (then) HD will also be the same multiple of F .

Thus, if two magnitudes are equal multiples of two (other) magnitudes, and some (parts) taken away (from the former magnitudes) are equal multiples of the latter (magnitudes, re-

τὰ λοιπὰ τοῖς αὐτοῖς ἤτοι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια· ὅπερ ἔδει δεῖξαι.

spectively), (then) the remainders are also either equal to the latter (magnitudes), or (are) equal multiples of them (respectively). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\alpha - n\alpha = (m - n)\alpha$.

7.

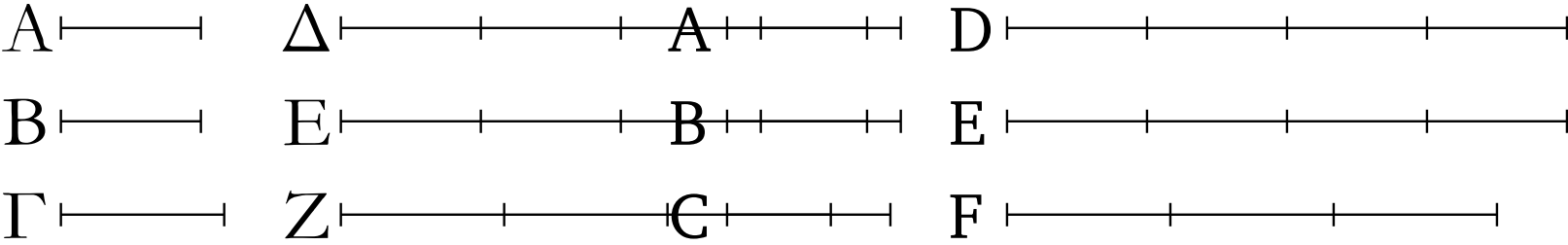
Proposition 7

Τὰ ἴσα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον καὶ τὸ αὐτὸ πρὸς τὰ ἴσα.

Ἐστω ἴσα μεγέθη τὰ A, B , ἄλλο δέ τι, ὃ ἔτυθεν, μέγεθος τὸ Γ · λέγω, ὅτι ἐκάτερον τῶν A, B πρὸς τὸ Γ τὸν αὐτὸν ἔθει λόγον, καὶ τὸ Γ πρὸς ἐκάτερον τῶν A, B .

Equal (magnitudes) have the same ratio to the same (magnitude), and the latter (magnitude has the same ratio) to the equal (magnitudes).

Let A and B be equal magnitudes, and C some other random magnitude. I say that A and B each have the same ratio to C , and (that) C (has the same ratio) to each of A and B .



Εἰλήφθω γὰρ τῶν μὲν A, B ἰσάκις πολλαπλάσια τὰ Δ, E , τοῦ δὲ Γ ἄλλο, ὃ ἔτυθεν, πολλαπλάσιον τὸ Z .

Ἐπεὶ οὖν ἰσάκις ἐστὶ πολλαπλάσιον τὸ Δ τοῦ A καὶ τὸ E τοῦ B , ἴσον δὲ τὸ A τῷ B , ἴσον ἄρα καὶ τὸ Δ τῷ E . ἄλλο δέ, ὃ ἔτυθεν, τὸ Z . εἰ ἄρα ὑπερέθει τὸ Δ τοῦ Z , ὑπερέθει καὶ τὸ E τοῦ Z , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν Δ, E τῶν A, B ἰσάκις πολλαπλάσια, τὸ δὲ Z τοῦ Γ ἄλλο, ὃ ἔτυθεν, πολλαπλάσιον· ἐστὶν ἄρα ὡς τὸ A πρὸς τὸ Γ , οὕτως τὸ B πρὸς τὸ Γ .

Λέγω [δή], ὅτι καὶ τὸ E πρὸς ἐκάτερον τῶν A, B τὸν αὐτὸν ἔθει λόγον.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἴσον ἐστὶ τὸ Δ τῷ E · ἄλλο δέ τι τὸ Z · εἰ ἄρα ὑπερέθει τὸ Z τοῦ Δ , ὑπερέθει καὶ τοῦ E , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὸ μὲν Z τοῦ Γ πολλαπλάσιον, τὰ δὲ Δ, E τῶν A, B ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια· ἐστὶν ἄρα ὡς τὸ Γ πρὸς τὸ A , οὕτως τὸ Γ πρὸς τὸ B .

Τὰ ἴσα ἄρα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον καὶ τὸ αὐτὸ πρὸς τὰ ἴσα.

For let the equal multiples D and E be taken of A and B (respectively), and the other random multiple F of C .

Therefore, since D and E are equal multiples of A and B (respectively), and A (is) equal to B , D (is) thus also equal to E . And F (is) different, at random. Thus, if D exceeds F (then) E also exceeds F , and if (D is) equal (to F then E is also) equal (to F), and if (D is) less (than F then E is also) less (than F). And D and E are equal multiples of A and B (respectively), and F another random multiple of C . Thus, as A (is) to C , so B (is) to C [Def. 5.5].

[So] I say that $C^\dagger$ also has the same ratio to each of A and B .

For, similarly, we can show, by the same construction, that D is equal to E . And F (has) some other (value). Thus, if F exceeds D then it also exceeds E , and if (F is) equal (to D then it is also) equal (to E), and if (F is) less (than D then it is also) less (than E). And F is a multiple of C , and D and E other random equal multiples of A and B . Thus, as C (is) to A , so C (is) to B [Def. 5.5].

Thus, equal (magnitudes) have the same ratio to the same (magnitude), and the latter (magnitude has the same ratio) to the equal (magnitudes).

Πόρισμα.

Corollary[‡]

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν μεγέθη τινὰ ἀνάλογον ᾗ, καὶ ἀνάπαλιν ἀνάλογον ᾖ. ὅπερ ἔδει δεῖξαι.

So (it is) clear, from this, that if some magnitudes are proportional, (then) they will also be proportional inversely. (Which is) the very thing it was required to show.

† The Greek text has “ E ”, which is obviously a mistake.

‡ In modern notation, this corollary reads that if $\alpha : \beta :: \gamma : \delta$ then $\beta : \alpha :: \delta : \gamma$.

8.

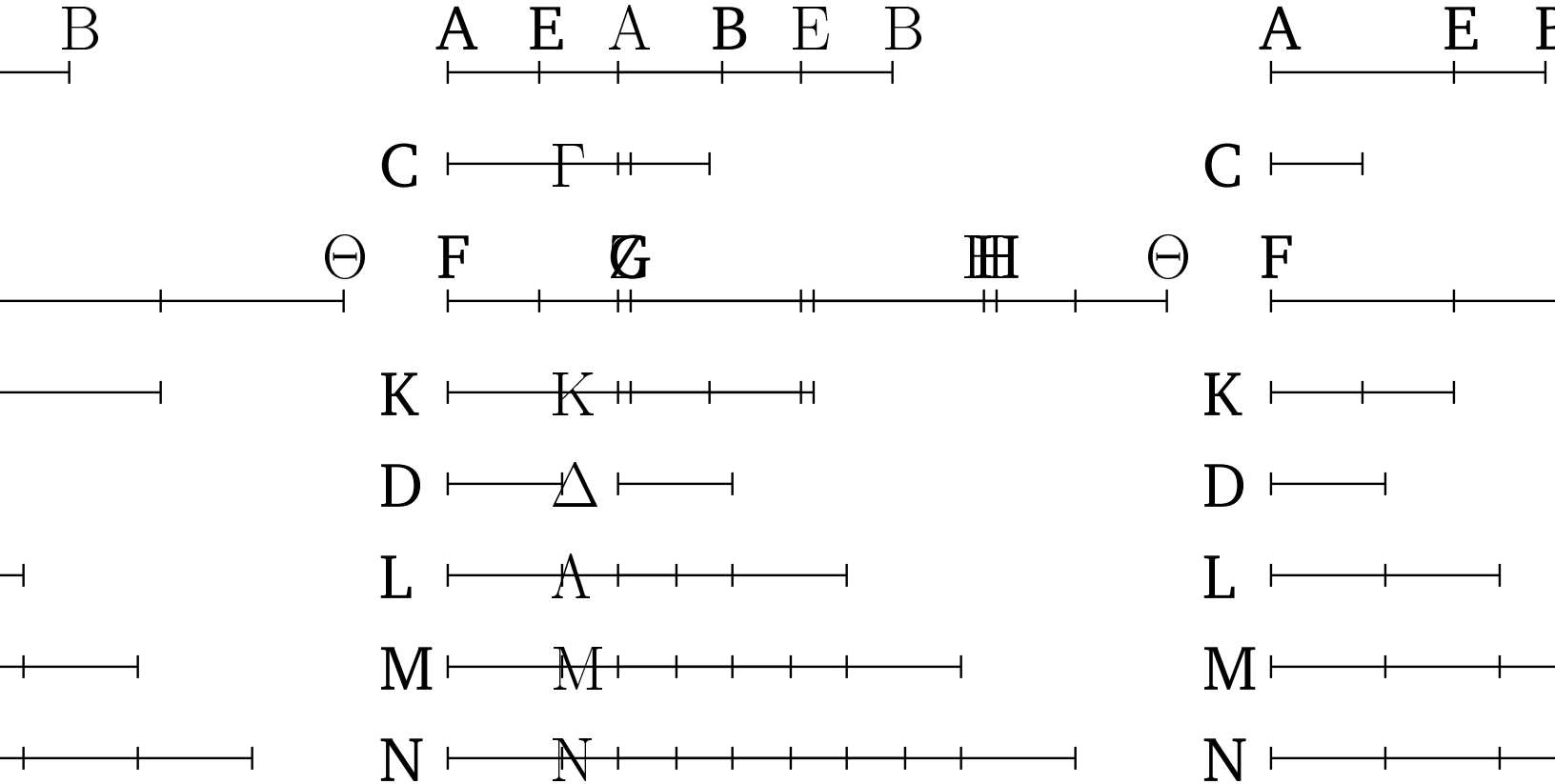
Proposition 8

Τῶν ἀνίσων μεγεθῶν τὸ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔθει ἥπερ τὸ ἔλαττον. καὶ τὸ αὐτὸ πρὸς τὸ ἔλαττον μείζονα λόγον ἔθει ἥπερ πρὸς τὸ μείζον.

Ἐστω ἄνισα μεγέθη τὰ AB , Γ , καὶ ἔστω μείζον τὸ AB , ἄλλο δέ, ὃ ἔτυθεν, τὸ Δ . λέγω, ὅτι τὸ AB πρὸς τὸ Δ μείζονα λόγον ἔθει ἥπερ τὸ Γ πρὸς τὸ Δ , καὶ τὸ Δ πρὸς τὸ Γ μείζονα λόγον ἔθει ἥπερ πρὸς τὸ AB .

For unequal magnitudes, the greater (magnitude) has a greater ratio than the lesser to the same (magnitude). And the latter (magnitude) has a greater ratio to the lesser (magnitude) than to the greater.

Let AB and C be unequal magnitudes, and let AB be the greater (of the two), and D another random magnitude. I say that AB has a greater ratio to D than C (has) to D , and (that) D has a greater ratio to C than (it has) to AB .



Ἐπεὶ γὰρ μείζον ἐστὶ τὸ AB τοῦ Γ , κείσθω τῷ Γ ἴσον τὸ BE . τὸ δὲ ἔλασσον τῶν AE , EB πολλαπλασιαζόμενον ἔσται ποτὲ τοῦ Δ μείζον. ἔστω πρότερον τὸ AE ἔλαττον τοῦ EB , καὶ πεπολλαπλασιάσθω τὸ AE , καὶ ἔστω αὐτοῦ πολλαπλάσιον τὸ ZH μείζον ὅν τοῦ Δ , καὶ ὁσαπλάσιόν ἐστι τὸ ZH τοῦ AE , τοσαυταπλάσιον γερονέτω καὶ τὸ μὲν $H\Theta$ τοῦ EB τὸ δὲ K τοῦ Γ . καὶ εἰλήφθω τοῦ Δ διπλάσιον μὲν τὸ Λ , τριπλάσιον δὲ τὸ M , καὶ ἕχῃς ἐνὶ πλείον, ἕως ἄν τὸ λαμβανόμενον πολλαπλάσιον μὲν γένηται τοῦ Δ , πρῶτως δὲ μείζον τοῦ K . εἰλήφθω, καὶ ἔστω τὸ N τετραπλάσιον μὲν τοῦ Δ , πρῶτως δὲ μείζον τοῦ K .

Ἐπεὶ οὖν τὸ K τοῦ N πρῶτως ἐστὶν ἔλαττον, τὸ K ἄρα τοῦ M οὐκ ἐστὶν ἔλαττον. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον

For since AB is greater than C , let BE be made equal to C . So, the lesser of AE and EB , being multiplied, will sometimes be greater than D [Def. 5.4]. First of all, let AE be less than EB , and let AE be multiplied, and let FG be a multiple of it which (is) greater than D . And as many times as FG is (divisible) by AE , so many times let GH also become (divisible) by EB , and K by C . And let the double multiple L of D be taken, and the triple multiple M , and several more, (each increasing) in order by one, until the (multiple) taken becomes the first multiple of D (which is) greater than K . Let it be taken, and let it also be the quadruple multiple N of D —the first (multiple) greater than K .

Therefore, since K is less than N first, K is thus not less

τὸ ZH τοῦ AE καὶ τὸ $H\Theta$ τοῦ EB , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ZH τοῦ AE καὶ τὸ $Z\Theta$ τοῦ AB . ἰσάκις δὲ ἐστὶ πολλαπλάσιον τὸ ZH τοῦ AE καὶ τὸ K τοῦ Γ . ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ $Z\Theta$ τοῦ AB καὶ τὸ K τοῦ Γ . τὰ $Z\Theta$, K ἄρα τῶν AB , Γ ἰσάκις ἐστὶ πολλαπλάσια. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ $H\Theta$ τοῦ EB καὶ τὸ K τοῦ Γ , ἴσον δὲ τὸ EB τῷ Γ , ἴσον ἄρα καὶ τὸ $H\Theta$ τῷ K . τὸ δὲ K τοῦ M οὐκ ἐστὶν ἑλάττω· οὐδ' ἄρα τὸ $H\Theta$ τοῦ M ἑλαττόν ἐστιν. μείζον δὲ τὸ ZH τοῦ Δ . ὅλον ἄρα τὸ $Z\Theta$ συναμφοτέρων τῶν Δ , M μείζον ἐστὶν. ἀλλὰ συναμφοτέρα τὰ Δ , M τῷ N ἐστὶν ἴσα, ἐπειδήπερ τὸ M τοῦ Δ τριπλάσιόν ἐστιν, συναμφοτέρα δὲ τὰ M , Δ τοῦ Δ ἐστὶ τετραπλάσια, ἐστὶ δὲ καὶ τὸ N τοῦ Δ τετραπλάσιον· συναμφοτέρα ἄρα τὰ M , Δ τῷ N ἴσα ἐστίν. ἀλλὰ τὸ $Z\Theta$ τῶν M , Δ μείζον ἐστὶν· τὸ $Z\Theta$ ἄρα τοῦ N ὑπερέθκει· τὸ δὲ K τοῦ N οὐθ' ὑπερέθκει. καὶ ἐστὶ τὰ μὲν $Z\Theta$, K τῶν AB , Γ ἰσάκις πολλαπλάσια, τὸ δὲ N τοῦ Δ ἄλλο, ὃ ἔτυθεν, πολλαπλάσιον· τὸ AB ἄρα πρὸς τὸ Δ μείζονα λόγον ἔθει ἥπερ τὸ Γ πρὸς τὸ Δ .

Λέγω δὴ, ὅτι καὶ τὸ Δ πρὸς τὸ Γ μείζονα λόγον ἔθει ἥπερ τὸ Δ πρὸς τὸ AB .

Τῶν γάρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι τὸ μὲν N τοῦ K ὑπερέθκει, τὸ δὲ N τοῦ $Z\Theta$ οὐθ' ὑπερέθκει. καὶ ἐστὶ τὸ μὲν N τοῦ Δ πολλαπλάσιον, τὰ δὲ $Z\Theta$, K τῶν AB , Γ ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια· τὸ Δ ἄρα πρὸς τὸ Γ μείζονα λόγον ἔθει ἥπερ τὸ Δ πρὸς τὸ AB .

Ἀλλὰ δὴ τὸ AE τοῦ EB μείζον ἐστὼ. τὸ δὴ ἑλάττω τὸ EB πολλαπλασιαζόμενον ἐστὶ ποτὲ τοῦ Δ μείζον. πεπολλαπλασιάσθω, καὶ ἐστὼ τὸ $H\Theta$ πολλαπλάσιον μὲν τοῦ EB , μείζον δὲ τοῦ Δ . καὶ ὁσαπλάσιόν ἐστὶ τὸ $H\Theta$ τοῦ EB , τοσαυταπλάσιον γερονέτω καὶ τὸ μὲν ZH τοῦ AE , τὸ δὲ K τοῦ Γ . ὁμοίως δὴ δείχομεν, ὅτι τὰ $Z\Theta$, K τῶν AB , Γ ἰσάκις ἐστὶ πολλαπλάσια· καὶ εἰλήφθω ὁμοίως τὸ N πολλαπλάσιον μὲν τοῦ Δ , πρῶτως δὲ μείζον τοῦ ZH . ὥστε πάλιν τὸ ZH τοῦ M οὐκ ἐστὶν ἑλασσον. μείζον δὲ τὸ $H\Theta$ τοῦ Δ . ὅλον ἄρα τὸ $Z\Theta$ τῶν Δ , M , τοντέστι τοῦ N , ὑπερέθκει. τὸ δὲ K τοῦ N οὐθ' ὑπερέθκει, ἐπειδήπερ καὶ τὸ ZH μείζον ὅν τοῦ $H\Theta$, τοντέστι τοῦ K , τοῦ N οὐθ' ὑπερέθκει. καὶ ὡσαύτως κατακολουθοῦντες τοῖς ἐπάνω περαινόμεν τὴν ἀπόδειξιν.

Τῶν ἄρα ἀνίσων μεγεθῶν τὸ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔθει ἥπερ τὸ ἑλάττω· καὶ τὸ αὐτὸ πρὸς τὸ ἑλάττω μείζονα λόγον ἔθει ἥπερ πρὸς τὸ μείζον· ὅπερ ἔδει δεῖξαι.

9.

Τὰ πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔχοντα λόγον ἴσα ἀλλήλοις ἐστίν· καὶ πρὸς ἃ τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον, ἐκεῖνα ἴσα ἐστίν.

than M . And since FG and GH are equal multiples of AE and EB (respectively), FG and FH are thus equal multiples of AE and AB (respectively) [Prop. 5.1]. And FG and K are equal multiples of AE and C (respectively). Thus, FH and K are equal multiples of AB and C (respectively). Thus, FH , K are equal multiples of AB , C . Again, since GH and K are equal multiples of EB and C , and EB (is) equal to C , GH (is) thus also equal to K . And K is not less than M . Thus, GH not less than M either. And FG (is) greater than D . Thus, the whole of FH is greater than D and M (added) together. But, D and M (added) together is equal to N , inasmuch as M is three times D , and M and D (added) together is four times D , and N is also four times D . Thus, M and D (added) together is equal to N . But, FH is greater than M and D . Thus, FH exceeds N . And K does not exceed N . And FH , K are equal multiples of AB , C , and N another random multiple of D . Thus, AB has a greater ratio to D than C (has) to D [Def. 5.7].

So, I say that D also has a greater ratio to C than D (has) to AB .

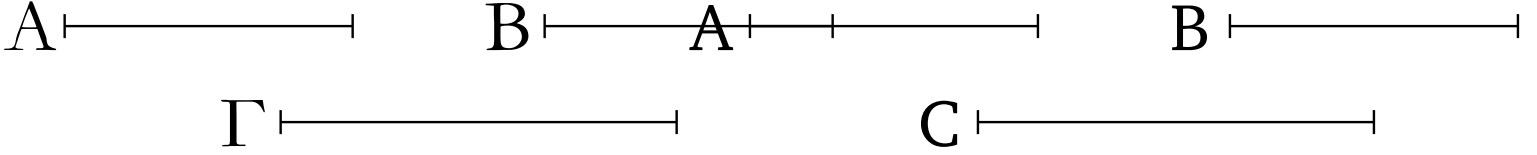
For, similarly, by the same construction, we can show that N exceeds K , and N does not exceed FH . And N is a multiple of D , and FH , K other random equal multiples of AB , C (respectively). Thus, D has a greater ratio to C than D (has) to AB [Def. 5.5].

And so let AE be greater than EB . So, the lesser, EB , being multiplied, will sometimes be greater than D . Let it be multiplied, and let GH be a multiple of EB (which is) greater than D . And as many times as GH is (divisible) by EB , so many times let FG also become (divisible) by AE , and K by C . So, similarly (to the above), we can show that FH and K are equal multiples of AB and C (respectively). And, similarly (to the above), let the multiple N of D , (which is) the first (multiple) greater than FG , be taken. So, FG is again not less than M . And GH (is) greater than D . Thus, the whole of FH exceeds D and M , that is to say N . And K does not exceed N , inasmuch as FG , which (is) greater than GH —that is to say, K —also does not exceed N . And, following the above (arguments), we (can) complete the proof in the same manner.

Thus, for unequal magnitudes, the greater (magnitude) has a greater ratio than the lesser to the same (magnitude). And the latter (magnitude) has a greater ratio to the lesser (magnitude) than to the greater. (Which is) the very thing it was required to show.

Proposition 9

(Magnitudes) having the same ratio to the same (magnitude) are equal to one another. And those (magnitudes) to which the same (magnitude) has the same ratio are equal.



Ἐθέτω γάρ ἑκάτερον τῶν A, B πρὸς τὸ Γ τὸν αὐτὸν λόγον· λέγω, ὅτι ἴσον ἐστὶ τὸ A τῷ B .

Εἰ γὰρ μή, οὐκ ἂν ἑκάτερον τῶν A, B πρὸς τὸ Γ τὸν αὐτὸν εἶθε λόγον· ἔθει δέ· ἴσον ἄρα ἐστὶ τὸ A τῷ B .

Ἐθέτω δὴ πάλιν τὸ Γ πρὸς ἑκάτερον τῶν A, B τὸν αὐτὸν λόγον· λέγω, ὅτι ἴσον ἐστὶ τὸ A τῷ B .

Εἰ γὰρ μή, οὐκ ἂν τὸ Γ πρὸς ἑκάτερον τῶν A, B τὸν αὐτὸν εἶθε λόγον· ἔθει δέ· ἴσον ἄρα ἐστὶ τὸ A τῷ B .

Τὰ ἄρα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθοντα λόγον ἴσα ἀλλήλοις ἐστίν· καὶ πρὸς ἃ τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον, ἐκεῖνα ἴσα ἐστίν· ὅπερ ἔδει δεῖχαι.

10.

Τῶν πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον ἔθον ἐκεῖνο μείζον ἐστίν· πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον ἔθει, ἐκεῖνο ἑλαττόν ἐστιν.

For let A and B each have the same ratio to C . I say that A is equal to B .

For if not, A and B would not each have the same ratio to C [Prop. 5.8]. But they do. Thus, A is equal to B .

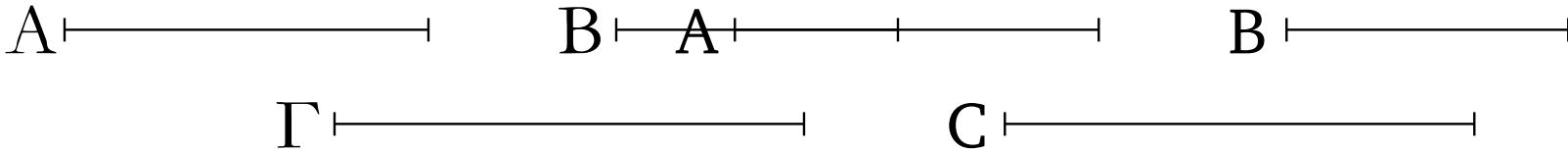
So, again, let C have the same ratio to each of A and B . I say that A is equal to B .

For if not, C would not have the same ratio to each of A and B [Prop. 5.8]. But it does. Thus, A is equal to B .

Thus, (magnitudes) having the same ratio to the same (magnitude) are equal to one another. And those (magnitudes) to which the same (magnitude) has the same ratio are equal. (Which is) the very thing it was required to show.

Proposition 10

For (magnitudes) having a ratio to the same (magnitude), that (magnitude which) has the greater ratio is (the) greater. And that (magnitude) to which the latter (magnitude) has a greater ratio is (the) lesser.



Ἐθέτω γάρ τὸ A πρὸς τὸ Γ μείζονα λόγον ἢπερ τὸ B πρὸς τὸ Γ · λέγω, ὅτι μείζον ἐστὶ τὸ A τοῦ B .

Εἰ γὰρ μή, ἦτοι ἴσον ἐστὶ τὸ A τῷ B ἢ ἔλασσον. ἴσον μὲν οὐκ ἔστι τὸ A τῷ B · ἑκάτερον γὰρ ἂν τῶν A, B πρὸς τὸ Γ τὸν αὐτὸν εἶθε λόγον. οὐκ ἔθει δέ· οὐκ ἄρα ἴσον ἐστὶ τὸ A τῷ B . οὐδὲ μὴν ἔλασσόν ἐστι τὸ A τοῦ B · τὸ A γὰρ ἂν πρὸς τὸ Γ ἐλάσσονα λόγον εἶθεν ἢπερ τὸ B πρὸς τὸ Γ . οὐκ ἔθει δέ· οὐκ ἄρα ἔλασσόν ἐστι τὸ A τοῦ B . ἐδείχθη δὲ οὐδὲ ἴσον μείζον ἄρα ἐστὶ τὸ A τοῦ B .

Ἐθέτω δὴ πάλιν τὸ Γ πρὸς τὸ B μείζονα λόγον ἢπερ τὸ Γ πρὸς τὸ A · λέγω, ὅτι ἔλασσόν ἐστι τὸ B τοῦ A .

Εἰ γὰρ μή, ἦτοι ἴσον ἐστὶν ἢ μείζον. ἴσον μὲν οὐκ ἔστι τὸ B τῷ A · τὸ Γ γὰρ ἂν πρὸς ἑκάτερον τῶν A, B τὸν αὐτὸν εἶθε λόγον. οὐκ ἔθει δέ· οὐκ ἄρα ἴσον ἐστὶ τὸ A τῷ B . οὐδὲ μὴν μείζον ἐστὶ τὸ B τοῦ A · τὸ Γ γὰρ ἂν πρὸς τὸ B ἐλάσσονα λόγον εἶθεν ἢπερ πρὸς τὸ A . οὐκ ἔθει δέ· οὐκ ἄρα μείζον ἐστὶ τὸ B τοῦ A . ἐδείχθη δέ, ὅτι οὐδὲ ἴσον· ἑλαττόν ἄρα ἐστὶ τὸ B τοῦ A .

Τῶν ἄρα πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον ἔθον μείζον ἐστίν· καὶ πρὸς ὃ τὸ αὐτὸ μείζονα λόγον ἔθει, ἐκεῖνο ἑλαττόν ἐστιν· ὅπερ ἔδει δεῖχαι.

For let A have a greater ratio to C than B (has) to C . I say that A is greater than B .

For if not, A is surely either equal to or less than B . In fact, A is not equal to B . For (then) A and B would each have the same ratio to C [Prop. 5.7]. But they do not. Thus, A is not equal to B . Neither, indeed, is A less than B . For (then) A would have a lesser ratio to C than B (has) to C [Prop. 5.8]. But it does not. Thus, A is not less than B . And it was shown not (to be) equal either. Thus, A is greater than B .

So, again, let C have a greater ratio to B than C (has) to A . I say that B is less than A .

For if not, (it is) surely either equal or greater. In fact, B is not equal to A . For (then) C would have the same ratio to each of A and B [Prop. 5.7]. But it does not. Thus, A is not equal to B . Neither, indeed, is B greater than A . For (then) C would have a lesser ratio to B than (it has) to A [Prop. 5.8]. But it does not. Thus, B is not greater than A . And it was shown that (it is) not equal (to A) either. Thus, B is less than A .

Thus, for (magnitudes) having a ratio to the same (magnitude), that (magnitude which) has the greater ratio is (the) greater. And that (magnitude) to which the latter (magnitude) has a greater ratio is (the) lesser. (Which is) the very thing it

was required to show.

11.

Proposition 11[†]

Οἱ τῶ αὐτῶ λόγῳ οἱ αὐτοὶ καὶ ἀλλήλοις εἰσὶν οἱ αὐτοί.

Ἐστωσαν γὰρ ὡς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , ὡς δὲ τὸ Γ πρὸς τὸ Δ , οὕτως τὸ E πρὸς τὸ Z . λέγω, ὅτι ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z .

Εἰλήφθω γὰρ τῶν A, Γ, E ἰσάκις πολλαπλάσια τὰ H, Θ, K , τῶν δὲ B, Δ, Z ἄλλα, ἃ ἐτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, M, N .

Γ ——— A ——— E ——— C ———

Δ ——— B ——— Z ——— D ———

Θ ——— G ——— K ——— H ———

M ——— L ——— N ——— M ———

E ———

F ———

K ———

N ———

Καὶ ἐπεὶ ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , καὶ ἐληπτὰ τῶν μὲν A, Γ ἰσάκις πολλαπλάσια τὰ H, Θ , τῶν δὲ B, Δ ἄλλα, ἃ ἐτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, M , εἰ ἄρα ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὸ Θ τοῦ M , καὶ εἰ ἴσον ἐστίν, ἴσον, καὶ εἰ ἐλλείπει, ἐλλείπει. πάλιν, ἐπεὶ ἐστὶν ὡς τὸ Γ πρὸς τὸ Δ , οὕτως τὸ E πρὸς τὸ Z , καὶ ἐληπτὰ τῶν Γ, E ἰσάκις πολλαπλάσια τὰ Θ, K , τῶν δὲ Δ, Z ἄλλα, ἃ ἐτυθεν, ἰσάκις πολλαπλάσια τὰ M, N , εἰ ἄρα ὑπερέθει τὸ Θ τοῦ M , ὑπερέθει καὶ τὸ K τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἐλάττον, ἐλάττον. ἀλλὰ εἰ ὑπερεῖθε τὸ Θ τοῦ M , ὑπερεῖθε καὶ τὸ H τοῦ Λ , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἐλάττον, ἐλάττον· ὥστε καὶ εἰ ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὸ K τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἐλάττον, ἐλάττον. καὶ ἐστὶ τὰ μὲν H, K τῶν A, E ἰσάκις πολλαπλάσια, τὰ δὲ Λ, N τῶν B, Z ἄλλα, ἃ ἐτυθεν, ἰσάκις πολλαπλάσια· ἐστὶν ἄρα ὡς τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z .

Οἱ ἄρα τῶ αὐτῶ λόγῳ οἱ αὐτοὶ καὶ ἀλλήλοις εἰσὶν οἱ αὐτοί· ὅπερ ἔδει δεῖχαι.

(Ratios which are) the same with the same ratio are also the same with one another.

For let it be that as A (is) to B , so C (is) to D , and as C (is) to D , so E (is) to F . I say that as A is to B , so E (is) to F .

For let the equal multiples G, H, K be taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively).

And since as A is to B , so C (is) to D , and the equal multiples G and H be taken of A and C (respectively), and the other random equal multiples L and M of B and D (respectively), thus if G exceeds L (then) H also exceeds M , and if (G is) equal (to L then H is also) equal (to M), and if (G is) less (than L then H is also) less (than M) [Def. 5.5]. Again, since as C is to D , so E (is) to F , and the equal multiples H and K have been taken of C and E (respectively), and the other random equal multiples M and N of D and F (respectively), thus if H exceeds M (then) K also exceeds N , and if (H is) equal (to M then K is also) equal (to N), and if (H is) less (than M then K is also) less (than N) [Def. 5.5]. But (we saw that) if H was exceeding M (then) G was also exceeding L , and if (H was) equal (to M then G was also) equal (to L), and if (H was) less (than M then G was also) less (than L). And, hence, if G exceeds L (then) K also exceeds N , and if (G is) equal (to L then K is also) equal (to N), and if (G is) less (than L then K is also) less (than N). And G and K are equal multiples of A and E (respectively), and L and N other random equal multiples of B and F (respectively). Thus, as A is to B , so E (is) to F [Def. 5.5].

Thus, (ratios which are) the same with the same ratio are also the same with one another. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\gamma : \delta :: \epsilon : \zeta$ then $\alpha : \beta :: \epsilon : \zeta$.

12.

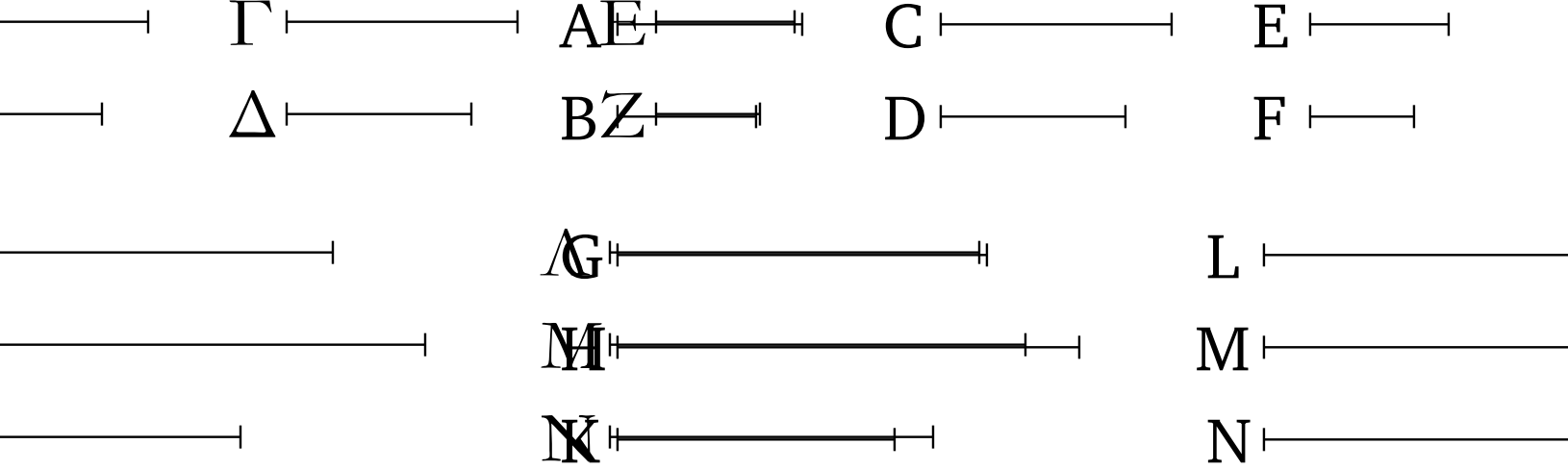
Proposition 12[†]

Ἐὰν ᾗ ὅποσαοῦν μεγέθη ἀνάλογον, ἔσται ὡς ἓν τῶν

If there are any number of magnitudes whatsoever (which

ἡγούμενων πρὸς ἓν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα.

are) proportional, (then) as one of the leading (magnitudes is) to one of the following, so will all of the leading (magnitudes) be to all of the following.



Ἐστωσαν ὅποσαοῦν μεγέθη ἀνάλογον τὰ Α, Β, Γ, Δ, Ε, Ζ, ὥς τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ, καὶ τὸ Ε πρὸς τὸ Ζ· λέγω, ὅτι ἐστὶν ὥς τὸ Α πρὸς τὸ Β, οὕτως τὰ Α, Γ, Ε πρὸς τὰ Β, Δ, Ζ.

Εἰλήφθω γὰρ τῶν μὲν Α, Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, Κ τῶν δὲ Β, Δ, Ζ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, Ν.

Καὶ ἐπεὶ ἐστὶν ὥς τὸ Α πρὸς τὸ Β, οὕτως τὸ Γ πρὸς τὸ Δ, καὶ τὸ Ε πρὸς τὸ Ζ, καὶ εἰληπται τῶν μὲν Α, Γ, Ε ἰσάκις πολλαπλάσια τὰ Η, Θ, Κ τῶν δὲ Β, Δ, Ζ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ Λ, Μ, Ν, εἰ ἄρα ὑπερέθει τὸ Η τοῦ Α, ὑπερέθει καὶ τὸ Θ τοῦ Μ, καὶ τὸ Κ τοῦ Ν, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. ὥστε καὶ εἰ ὑπερέθει τὸ Η τοῦ Α, ὑπερέθει καὶ τὰ Η, Θ, Κ τῶν Α, Μ, Ν, καὶ εἰ ἴσον, ἴσα, καὶ εἰ ἔλαττον, ἔλαττονα. καὶ ἐστὶ τὸ μὲν Η καὶ τὰ Η, Θ, Κ τοῦ Α καὶ τῶν Α, Γ, Ε ἰσάκις πολλαπλάσια, ἐπειδήπερ εἴαν ἡ ὅποσαοῦν μεγέθη ὅποσωνοῦν μεγεθῶν ἴσων τὸ πλῆθος ἑκάστον ἐκάστον ἰσάκις πολλαπλάσιον, ὅσαπλάσιόν ἐστιν ἐν τῶν μεγεθῶν ἐνόξ, τοσανταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων. διὰ τὰ αὐτὰ δὴ καὶ τὸ Α καὶ τὰ Α, Μ, Ν τοῦ Β καὶ τῶν Β, Δ, Ζ ἰσάκις ἐστὶ πολλαπλάσια· ἔστιν ἄρα ὥς τὸ Α πρὸς τὸ Β, οὕτως τὰ Α, Γ, Ε πρὸς τὰ Β, Δ, Ζ.

Ἐάν ἄρα ἡ ὅποσαοῦν μεγέθη ἀνάλογον, ἔσται ὥς ἐν τῶν ἡγούμενων πρὸς ἓν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα· ὅπερ ἔδει δεῖχαι.

Let there be any number of magnitudes whatsoever, Α, Β, C, D, Ε, F, (which are) proportional, (so that) as Α (is) to Β, so C (is) to D, and Ε to F. I say that as Α is to Β, so Α, C, Ε (are) to Β, D, F.

For let the equal multiples G, H, K be taken of Α, C, Ε (respectively), and the other random equal multiples L, M, N of Β, D, F (respectively).

And since as Α is to Β, so C (is) to D, and Ε to F, and the equal multiples G, H, K have been taken of Α, C, Ε (respectively), and the other random equal multiples L, M, N of Β, D, F (respectively), thus if G exceeds L (then) H also exceeds M, and K (exceeds) N, and if (G is) equal (to L then H is also) equal (to M, and K to N), and if (G is) less (than L then H is also) less (than M, and K than N) [Def. 5.5]. And, hence, if G exceeds L (then) G, H, K also exceed L, M, N, and if (G is) equal (to L then G, H, K are also) equal (to L, M, N) and if (G is) less (than L then G, H, K are also) less (than L, M, N). And G and G, H, K are equal multiples of Α and Α, C, Ε (respectively), inasmuch as if there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second) [Prop. 5.1]. So, for the same (reasons), L and L, M, N are also equal multiples of Β and Β, D, F (respectively). Thus, as Α is to Β, so Α, C, Ε (are) to Β, D, F (respectively).

Thus, if there are any number of magnitudes whatsoever (which are) proportional, (then) as one of the leading (magnitudes is) to one of the following, so will all of the leading (magnitudes) be to all of the following. (Which is) the very thing it was required to show.

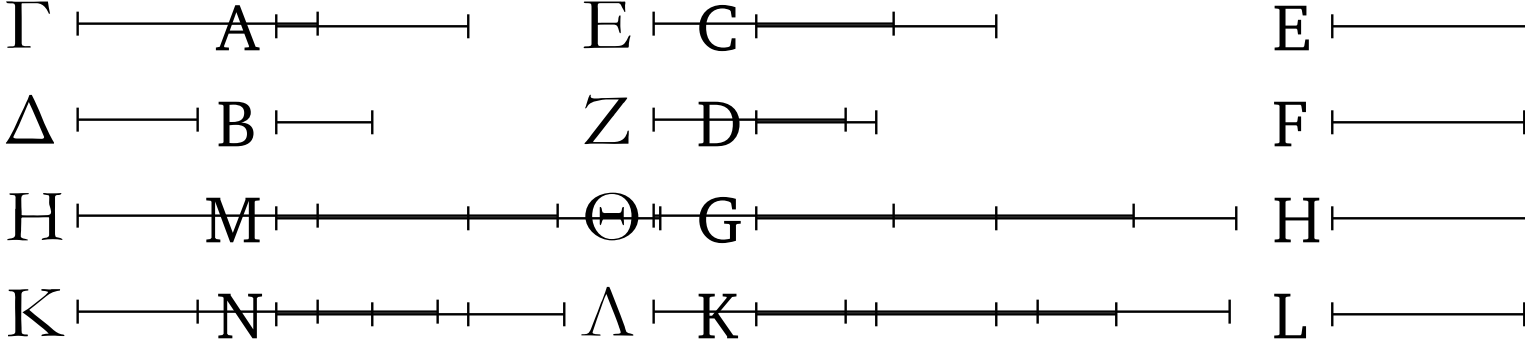
[†] In modern notation, this proposition reads that if $\alpha : \alpha' :: \beta : \beta' :: \gamma : \gamma'$ etc. then $\alpha : \alpha' :: (\alpha + \beta + \gamma + \dots) : (\alpha' + \beta' + \gamma' + \dots)$.

13.

Proposition 13[†]

Ἐάν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τρίτον δὲ πρὸς τέταρτον μείζονα λόγον ἔθῃ ἢ πέμπτον πρὸς ἕκτον, καὶ πρῶτον πρὸς δεύτερον μείζονα λόγον ἔχει ἢ πέμπτον πρὸς ἕκτον.

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the third (magnitude) has a greater ratio to the fourth than a fifth (has) to a sixth, (then) the first (magnitude) will also have a greater ratio to the second than the fifth (has) to the sixth.



Πρῶτον γάρ τὸ A πρὸς δεύτερον τὸ B τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ , τρίτον δὲ τὸ Γ πρὸς τέταρτον τὸ Δ μείζονα λόγον ἔθῃ ἢ πέμπτον τὸ E πρὸς ἕκτον τὸ Z . λέγω, ὅτι καὶ πρῶτον τὸ A πρὸς δεύτερον τὸ B μείζονα λόγον ἔχει ἢ περὶ πέμπτον τὸ E πρὸς ἕκτον τὸ Z .

Ἐπεὶ γάρ ἐστι τινὰ τῶν μὲν Γ , E ἰσάκις πολλαπλάσια, τῶν δὲ Δ , Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια, καὶ τὸ μὲν τοῦ Γ πολλαπλάσιον τοῦ τοῦ Δ πολλαπλασίον ὑπερέθει, τὸ δὲ τοῦ E πολλαπλάσιον τοῦ τοῦ Z πολλαπλασίον οὐθ' ὑπερέθει, εἰλήφθω, καὶ ἔστω τῶν μὲν Γ , E ἰσάκις πολλαπλάσια τὰ H , Θ , τῶν δὲ Δ , Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ K , Λ , ὥστε τὸ μὲν H τοῦ K ὑπερέθειν, τὸ δὲ Θ τοῦ Λ μὴ ὑπερέθειν καὶ ὁσαπλάσιον μὲν ἐστὶ τὸ H τοῦ Γ , τοσαυταπλάσιον ἔστω καὶ τὸ M τοῦ A , ὁσαπλάσιον δὲ τὸ K τοῦ Δ , τοσαυταπλάσιον ἔστω καὶ τὸ N τοῦ B .

Καὶ ἐπεὶ ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , καὶ εἰληπται τῶν μὲν A , Γ ἰσάκις πολλαπλάσια τὰ M , H , τῶν δὲ B , Δ ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ N , K , εἰ ἄρα ὑπερέθει τὸ M τοῦ N , ὑπερέθει καὶ τὸ H τοῦ K , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. ὑπερέθει δὲ τὸ H τοῦ K ὑπερέθει ἄρα καὶ τὸ M τοῦ N . τὸ δὲ Θ τοῦ Λ οὐθ' ὑπερέθει καὶ ἐστὶ τὰ μὲν M , Θ τῶν A , E ἰσάκις πολλαπλάσια, τὰ δὲ N , Λ τῶν B , Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια· τὸ ἄρα A πρὸς τὸ B μείζονα λόγον ἔθει ἢ περὶ τὸ E πρὸς τὸ Z .

Ἐάν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τρίτον δὲ πρὸς τέταρτον μείζονα λόγον ἔθῃ ἢ πέμπτον πρὸς ἕκτον, καὶ πρῶτον πρὸς δεύτερον μείζονα λόγον ἔχει ἢ πέμπτον πρὸς ἕκτον· ὅπερ εἶδει δεῖξαι.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D , and let the third (magnitude) C have a greater ratio to the fourth D than a fifth E (has) to a sixth F . I say that the first (magnitude) A will also have a greater ratio to the second B than the fifth E (has) to the sixth F .

For since there are some equal multiples of C and E , and other random equal multiples of D and F , (for which) the multiple of C exceeds the (multiple) of D , and the multiple of E does not exceed the multiple of F [Def. 5.7], let them be taken. And let G and H be equal multiples of C and E (respectively), and K and L other random equal multiples of D and F (respectively), such that G exceeds K , but H does not exceed L . And as many times as G is (divisible) by C , so many times let M be (divisible) by A . And as many times as K (is divisible) by D , so many times let N be (divisible) by B .

And since as A is to B , so C (is) to D , and the equal multiples M and G have been taken of A and C (respectively), and the other random equal multiples N and K of B and D (respectively), thus if M exceeds N (then) G exceeds K , and if (M is) equal (to N then G is also) equal (to K), and if (M is) less (than N then G is also) less (than K) [Def. 5.5]. And G exceeds K . Thus, M also exceeds N . And H does not exceed L . And M and H are equal multiples of A and E (respectively), and N and L other random equal multiples of B and F (respectively). Thus, A has a greater ratio to B than E (has) to F [Def. 5.7].

Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and a third (magnitude) has a greater ratio to a fourth than a fifth (has) to a sixth, (then) the first (magnitude) will also have a greater ratio to the second than the fifth (has) to the sixth. (Which is) the very thing it

was required to show.

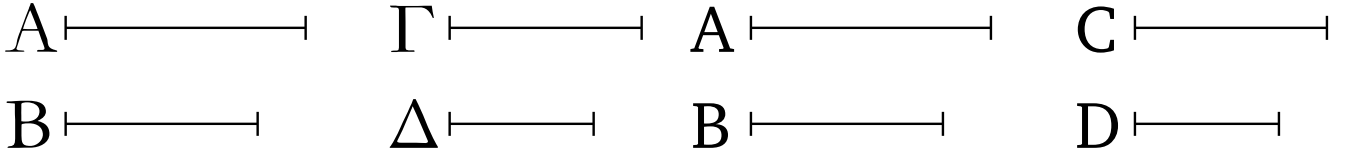
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\gamma : \delta > \epsilon : \zeta$ then $\alpha : \beta > \epsilon : \zeta$.

14.

Proposition 14[†]

Ἐάν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τὸ δὲ πρῶτον τοῦ τρίτου μείζον ἢ, καὶ τὸ δεύτερον τοῦ τετάρτου μείζον ἔσται, καὶ ἴσον, ἴσον, καὶ ἔλαττον, ἔλαττον.

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the first (magnitude) is greater than the third, (then) the second will also be greater than the fourth. And if (the first magnitude is) equal (to the third then the second will also be) equal (to the fourth). And if (the first magnitude is) less (than the third then the second will also be) less (than the fourth).



Πρῶτον γάρ τὸ A πρὸς δεύτερον τὸ B τὸν αὐτὸν ἔθεται λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ , μείζον δὲ ἔστω τὸ A τοῦ Γ . λέγω, ὅτι καὶ τὸ B τοῦ Δ μείζον ἔστιν.

Ἐπεὶ γὰρ τὸ A τοῦ Γ μείζον ἔστιν, ἄλλο δέ, ὃ ἔτυθεν, [μέγεθος] τὸ B , τὸ A ἄρα πρὸς τὸ B μείζονα λόγον ἔθει ἢ περ τὸ Γ πρὸς τὸ B . ὥς δὲ τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ . καὶ τὸ Γ ἄρα πρὸς τὸ Δ μείζονα λόγον ἔθει ἢ περ τὸ Γ πρὸς τὸ B . πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον ἔθει, ἐκεῖνο ἔλασσόν ἔστιν· ἔλασσον ἄρα τὸ Δ τοῦ B . ὥστε μείζον ἔστι τὸ B τοῦ Δ .

Ὀμοίως δὴ δεῖχμεν, ὅτι καὶ ἴσον ἢ τὸ A τῷ Γ , ἴσον ἔσται καὶ τὸ B τῷ Δ , καὶ ἔλασσον ἢ τὸ A τοῦ Γ , ἔλασσον ἔσται καὶ τὸ B τοῦ Δ .

Ἐάν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τὸ δὲ πρῶτον τοῦ τρίτου μείζον ἢ, καὶ τὸ δεύτερον τοῦ τετάρτου μείζον ἔσται, καὶ ἴσον, ἴσον, καὶ ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖχαι.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D . And let A be greater than C . I say that B is also greater than D .

For since A is greater than C , and B (is) another random [magnitude], A thus has a greater ratio to B than C (has) to B [Prop. 5.8]. And as A (is) to B , so C (is) to D . Thus, C also has a greater ratio to D than C (has) to B . And that (magnitude) to which the same (magnitude) has a greater ratio is the lesser [Prop. 5.10]. Thus, D (is) less than B . Hence, B is greater than D .

So, similarly, we can show that even if A is equal to C , (then) B will also be equal to D , and even if A is less than C , (then) B will also be less than D .

Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the first (magnitude) is greater than the third, (then) the second will also be greater than the fourth. And if (the first magnitude is) equal (to the third then the second will also be) equal (to the fourth). And if (the first magnitude is) less (than the third then the second will also be) less (than the fourth). (Which is) the very thing it was required to show.

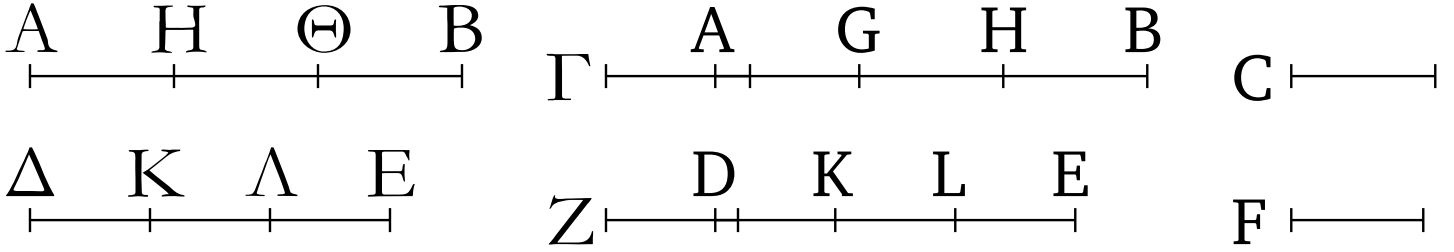
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha \gtrless \gamma$ as $\beta \gtrless \delta$.

15.

Proposition 15[†]

Τὰ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον ληφθέντα κατάλληλα.

Parts have the same ratio as similar multiples, taken in corresponding order.



Ἐστω γάρ ἰσάκις πολλαπλάσιον τὸ AB τοῦ Γ καὶ τὸ ΔE τοῦ Z · λέγω, ὅτι ἐστὶν ὡς τὸ Γ πρὸς τὸ Z , οὕτως τὸ AB πρὸς τὸ ΔE .

Ἐπεὶ γάρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AB τοῦ Γ καὶ τὸ ΔE τοῦ Z , ὅσα ἄρα ἐστὶν ἐν τῷ AB μεγέθει ἴσα τῷ Γ , τοσαῦτα καὶ ἐν τῷ ΔE ἴσα τῷ Z . διηγήσθω τὸ μὲν AB εἰς τὰ τῷ Γ ἴσα τὰ $AH, H\Theta, \Theta B$, τὸ δὲ ΔE εἰς τὰ τῷ Z ἴσα τὰ $\Delta K, KA, AE$ · ἔσται δὴ ἴσον τὸ πλῆθος τῶν $AH, H\Theta, \Theta B$ τῷ πλῆθει τῶν $\Delta K, KA, AE$. καὶ ἐπεὶ ἴσα ἐστὶ τὰ $AH, H\Theta, \Theta B$ ἀλλήλοις, ἔστι δὲ καὶ τὰ $\Delta K, KA, AE$ ἴσα ἀλλήλοις, ἔστιν ἄρα ὡς τὸ AH πρὸς τὸ ΔK , οὕτως τὸ $H\Theta$ πρὸς τὸ KA , καὶ τὸ ΘB πρὸς τὸ AE . ἔσται ἄρα καὶ ὡς ἓν τῶν ἡγουμένων πρὸς ἓν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγουμένα πρὸς ἅπαντα τὰ ἐπόμενα· ἔστιν ἄρα ὡς τὸ AH πρὸς τὸ ΔK , οὕτως τὸ AB πρὸς τὸ ΔE . ἴσον δὲ τὸ μὲν AH τῷ Γ , τὸ δὲ ΔK τῷ Z · ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ Z οὕτως τὸ AB πρὸς τὸ ΔE .

Τὰ ἄρα μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔδει λόγον ληφθέντα κατάλληλα· ὅπερ ἔδει δεῖξαι.

For let AB and DE be equal multiples of C and F (respectively). I say that as C is to F , so AB (is) to DE .

For since AB and DE are equal multiples of C and F (respectively), thus as many magnitudes as there are in AB equal to C , so many (are there) also in DE equal to F . Let AB be divided into (magnitudes) AG, GH, HB , equal to C , and DE into (magnitudes) DK, KL, LE , equal to F . So, the number of (magnitudes) AG, GH, HB will equal the number of (magnitudes) DK, KL, LE . And since AG, GH, HB are equal to one another, and DK, KL, LE are also equal to one another, thus as AG is to DK , so GH (is) to KL , and HB to LE [Prop. 5.7]. And, thus (for proportional magnitudes), as one of the leading (magnitudes) will be to one of the following, so all of the leading (magnitudes will be) to all of the following [Prop. 5.12]. Thus, as AG is to DK , so AB (is) to DE . And AG is equal to C , and DK to F . Thus, as C is to F , so AB (is) to DE .

Thus, parts have the same ratio as similar multiples, taken in corresponding order. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that $\alpha : \beta :: m\alpha : m\beta$.

16.

Ἐὰν τέσσαρα μεγέθη ἀνάλογον ᾗ, καὶ ἐναλλάξ ἀνάλογον ἔσται.

Ἐστω τέσσαρα μεγέθη ἀνάλογον τὰ A, B, Γ, Δ , ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ · λέγω, ὅτι καὶ ἐναλλάξ [ἀνάλογον] ἔσται, ὡς τὸ A πρὸς τὸ Γ , οὕτως τὸ B πρὸς τὸ Δ .

Εἰλήφθω γάρ τῶν μὲν A, B ἰσάκις πολλαπλάσια τὰ E, Z , τῶν δὲ Γ, Δ ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ H, Θ .

Proposition 16[†]

If four magnitudes are proportional, (then) they will also be proportional alternately.

Let A, B, C and D be four proportional magnitudes, (such that) as A (is) to B , so C (is) to D . I say that they will also be [proportional] alternately, (so that) as A (is) to C , so B (is) to D .

For let the equal multiples E and F be taken of A and B (respectively), and the other random equal multiples G and H of C and D (respectively).

A ———

Γ —A———

C ———

B ———

Δ —B———

D ———

E ———|———|———|———

H —E———||———|———|———

G ———|———|———|———

Z ———|———|———|———

Θ —F———|———|———|———|———

H ———|———|———|———

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ E τοῦ A καὶ τὸ Z τοῦ B , τὰ δὲ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὡς τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z . ὡς δὲ τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ . καὶ ὡς ἄρα τὸ Γ πρὸς τὸ Δ , οὕτως τὸ E πρὸς τὸ Z . πάλιν, ἐπεὶ τὰ H , Θ τῶν Γ , Δ ἰσάκις ἐστὶ πολλαπλάσια, ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ Δ , οὕτως τὸ H πρὸς τὸ Θ . ὡς δὲ τὸ Γ πρὸς τὸ Δ , [οὕτως] τὸ E πρὸς τὸ Z καὶ ὡς ἄρα τὸ E πρὸς τὸ Z , οὕτως τὸ H πρὸς τὸ Θ . ἔαν δὲ τέσσαρα μεγέθη ἀνάλογον ᾗ, τὸ δὲ πρῶτον τοῦ τρίτου μείζον ᾗ, καὶ τὸ δεύτερον τοῦ τετάρτου μείζον ἔσται, κ' ἂν ἴσον, ἴσον, κ' ἂν ἔλαττον, ἔλαττον. εἰ ἄρα ὑπερέθει τὸ E τοῦ H , ὑπερέθει καὶ τὸ Z τοῦ Θ , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν E , Z τῶν A , B ἰσάκις πολλαπλάσια, τὰ δὲ H , Θ τῶν Γ , Δ ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὡς τὸ A πρὸς τὸ Γ , οὕτως τὸ B πρὸς τὸ Δ .

Ἐὰν ἄρα τέσσαρα μεγέθη ἀνάλογον ᾗ, καὶ ἐναλλάχ ἀνάλογον ἔσται· ὅπερ ἔδει δεῖχαι.

And since E and F are equal multiples of A and B (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as A is to B , so E (is) to F . But as A (is) to B , so C (is) to D . And, thus, as C (is) to D , so E (is) to F [Prop. 5.11]. Again, since G and H are equal multiples of C and D (respectively), thus as C is to D , so G (is) to H [Prop. 5.15]. But as C (is) to D , [so] E (is) to F . And, thus, as E (is) to F , so G (is) to H [Prop. 5.11]. And if four magnitudes are proportional, and the first is greater than the third, (then) the second will also be greater than the fourth, and if (the first is) equal (to the third then the second will also be) equal (to the fourth), and if (the first is) less (than the third then the second will also be) less (than the fourth) [Prop. 5.14]. Thus, if E exceeds G (then) F also exceeds H , and if (E is) equal (to G then F is also) equal (to H), and if (E is) less (than G then F is also) less (than H). And E and F are equal multiples of A and B (respectively), and G and H other random equal multiples of C and D (respectively). Thus, as A is to C , so B (is) to D [Def. 5.5].

Thus, if four magnitudes are proportional, (then) they will also be proportional alternately. (Which is) the very thing it was required to show.

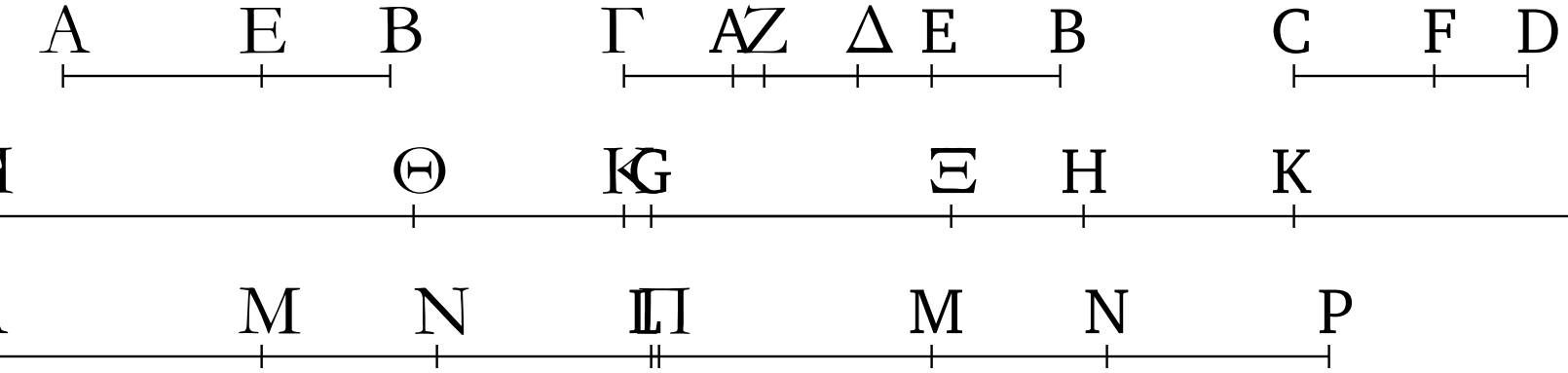
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \gamma :: \beta : \delta$.

17.

Ἐὰν συγκείμενα μεγέθη ἀνάλογον ᾗ, καὶ διαιρεθέντα ἀνάλογον ἔσται.

Proposition 17†

If composed magnitudes are proportional, (then) they will also be proportional (when) separated.



Ἐστω συγκείμενα μεγέθη ἀνάλογον τὰ AB , BE , $\Gamma\Delta$, ΔZ , ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ ΔZ · λέγω, ὅτι καὶ διαιρεθέντα ἀνάλογον ἔσται, ὥς τὸ AE πρὸς τὸ EB , οὕτως τὸ ΓZ πρὸς τὸ ΔZ .

Εἰλήφθω γὰρ τῶν μὲν AE , EB , ΓZ , ΔZ ἰσάκις πολλαπλάσια τὰ $H\Theta$, ΘK , ΛM , MN , τῶν δὲ EB , ΔZ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ KX , NI .

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ $H\Theta$ τοῦ AE καὶ τὸ ΘK τοῦ EB , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ $H\Theta$ τοῦ AE καὶ τὸ HK τοῦ AB . ἰσάκις δὲ ἐστὶ πολλαπλάσιον τὸ $H\Theta$ τοῦ AE καὶ τὸ ΛM τοῦ ΓZ · ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ HK τοῦ AB καὶ τὸ ΛM τοῦ ΓZ . πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΛM τοῦ ΓZ καὶ τὸ MN τοῦ ΔZ , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΛM τοῦ ΓZ καὶ τὸ ΛN τοῦ $\Gamma\Delta$. ἰσάκις δὲ ἦν πολλαπλάσιον τὸ ΛM τοῦ ΓZ καὶ τὸ HK τοῦ AB · ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ HK τοῦ AB καὶ τὸ ΛN τοῦ $\Gamma\Delta$. τὰ HK , ΛN ἄρα τῶν AB , $\Gamma\Delta$ ἰσάκις ἐστὶ πολλαπλάσια. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΘK τοῦ EB καὶ τὸ MN τοῦ ΔZ , ἔστι δὲ καὶ τὸ KX τοῦ EB ἰσάκις πολλαπλάσιον καὶ τὸ NI τοῦ ΔZ , καὶ συντεθέν τὸ ΘX τοῦ EB ἰσάκις ἐστὶ πολλαπλάσιον καὶ τὸ MI τοῦ ΔZ . καὶ ἐπεὶ ἐστὶν ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ ΔZ , καὶ εἰληπται τῶν μὲν AB , $\Gamma\Delta$ ἰσάκις πολλαπλάσια τὰ HK , ΛN , τῶν δὲ EB , ΔZ ἰσάκις πολλαπλάσια τὰ ΘX , MI , εἰ ἄρα ὑπερέθει τὸ HK τοῦ ΘX , ὑπερέθει καὶ τὸ ΛN τοῦ MI , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. ὑπερεθέτω δὴ τὸ HK τοῦ ΘX , καὶ κοινοῦ ἀφαιρεθέντος τοῦ ΘK ὑπερέθει ἄρα καὶ τὸ $H\Theta$ τοῦ KX . ἄλλα εἰ ὑπερεῖθε τὸ HK τοῦ ΘX ὑπερεῖθε καὶ τὸ ΛN τοῦ MI · ὑπερέθει ἄρα καὶ τὸ ΛN τοῦ MI , καὶ κοινοῦ ἀφαιρεθέντος τοῦ MN ὑπερέθει καὶ τὸ ΛM τοῦ NI · ὥστε εἰ ὑπερέθει τὸ $H\Theta$ τοῦ KX , ὑπερέθει καὶ τὸ ΛM τοῦ NI . ὁμοίως δὴ δεῖχουμεν, ὅτι κ' ἂν ἴσον ἢ τὸ $H\Theta$ τῶν KX , ἴσον ἔσται καὶ τὸ ΛM τῶν NI , κ' ἂν ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν $H\Theta$, ΛM τῶν AE , ΓZ ἰσάκις πολλαπλάσια, τὰ δὲ KX , NI τῶν EB , ΔZ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὥς τὸ AE πρὸς τὸ EB , οὕτως τὸ ΓZ πρὸς τὸ ΔZ .

Ἐὰν ἄρα συγκείμενα μεγέθη ἀνάλογον ἦ, καὶ διαιρεθέντα ἀνάλογον ἔσται· ὅπερ ἔδει δεῖχαι.

Let AB , BE , CD , and DF be composed magnitudes (which are) proportional, (so that) as AB (is) to BE , so CD (is) to DF . I say that they will also be proportional (when) separated, (so that) as AE (is) to EB , so CF (is) to DF .

For let the equal multiples GH , HK , LM , and MN be taken of AE , EB , CF , and FD (respectively), and the other random equal multiples KO and NP of EB and FD (respectively).

And since GH and HK are equal multiples of AE and EB (respectively), GH and GK are thus equal multiples of AE and AB (respectively) [Prop. 5.1]. But GH and LM are equal multiples of AE and CF (respectively). Thus, GK and LM are equal multiples of AB and CF (respectively). Again, since LM and MN are equal multiples of CF and FD (respectively), LM and LN are thus equal multiples of CF and CD (respectively) [Prop. 5.1]. And LM and GK were equal multiples of CF and AB (respectively). Thus, GK and LN are equal multiples of AB and CD (respectively). Thus, GK , LN are equal multiples of AB , CD . Again, since HK and MN are equal multiples of EB and FD (respectively), and KO and NP are also equal multiples of EB and FD (respectively), (then), added together, HO and MP are also equal multiples of EB and FD (respectively) [Prop. 5.2]. And since as AB (is) to BE , so CD (is) to DF , and the equal multiples GK , LN be taken of AB , CD , and the equal multiples HO , MP of EB , FD , thus if GK exceeds HO (then) LN also exceeds MP , and if (GK is) equal (to HO then LN is also) equal (to MP), and if (GK is) less (than HO then LN is also) less (than MP) [Def. 5.5]. So let GK exceed HO , and thus, HK being taken away from both, GH exceeds KO . But (we saw that) if GK was exceeding HO (then) LN was also exceeding MP . Thus, LN also exceeds MP , and, MN being taken away from both, LM also exceeds NP . Hence, if GH exceeds KO (then) LM also exceeds NP . So, similarly, we can show that even if GH is equal to KO (then) LM will also be equal to NP , and even if (GH is) less (than KO then LM will also be) less (than NP). And GH , LM are equal multiples of AE , CF , and KO , NP other random equal multiples of EB , FD . Thus, as AE is to EB , so CF (is) to FD [Def. 5.5].

Thus, if composed magnitudes are proportional, (then) they will also be proportional (when) separated. (Which is)

the very thing it was required to show.

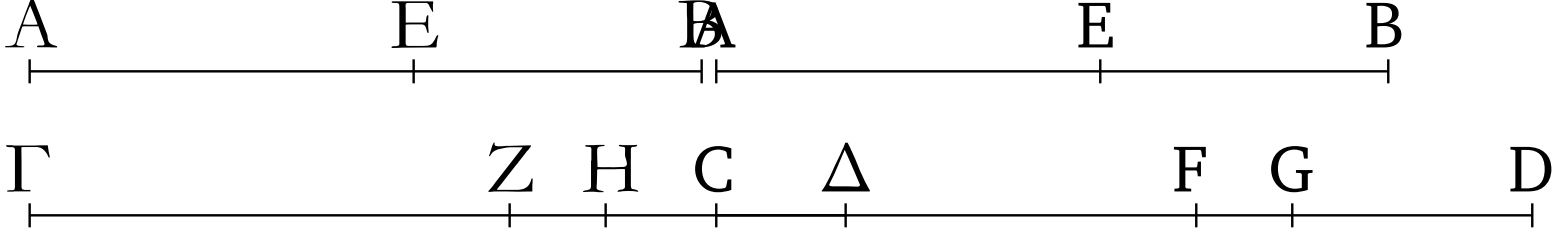
[†] In modern notation, this proposition reads that if $\alpha + \beta : \beta :: \gamma + \delta : \delta$ then $\alpha : \beta :: \gamma : \delta$.

18.

Proposition 18[†]

Ἐάν διηρημένα μεγέθη ἀνάλογον ᾗ, καὶ συντεθέντα ἀνάλογον ἔσται.

If separated magnitudes are proportional, (then) they will also be proportional (when) composed.



Ἐστω διηρημένα μεγέθη ἀνάλογον τὰ $AE, EB, \Gamma Z, Z\Delta$, ὥς τὸ AE πρὸς τὸ EB , οὕτως τὸ ΓZ πρὸς τὸ $Z\Delta$. λέγω, ὅτι καὶ συντεθέντα ἀνάλογον ἔσται, ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ $Z\Delta$.

Εἰ γὰρ μὴ ἔστιν ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ ΔZ , ἔσται ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ ᾗτοι πρὸς ἑλασσόν τι τοῦ ΔZ ἢ πρὸς μείζον.

Ἐστω πρότερον πρὸς ἑλασσον τὸ ΔH . καὶ ἐπεὶ ἔστιν ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ ΔH , συγκεείμενα μεγέθη ἀναλόγον ἔστιν· ὥστε καὶ διαιρεθέντα ἀνάλογον ἔσται. ἔστιν ἄρα ὥς τὸ AE πρὸς τὸ EB , οὕτως τὸ ΓH πρὸς τὸ $H\Delta$. ὑπόκειται δὲ καὶ ὥς τὸ AE πρὸς τὸ EB , οὕτως τὸ ΓZ πρὸς τὸ $Z\Delta$. καὶ ὥς ἄρα τὸ ΓH πρὸς τὸ $H\Delta$, οὕτως τὸ ΓZ πρὸς τὸ $Z\Delta$. μείζον δὲ τὸ πρῶτον τὸ ΓH τοῦ τρίτου τοῦ ΓZ . μείζον ἄρα καὶ τὸ δεύτερον τὸ $H\Delta$ τοῦ τετάρτου τοῦ $Z\Delta$. ἀλλὰ καὶ ἑλάττον· ὅπερ ἔστιν ἀδύνατον· οὐκ ἄρα ἔστιν ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς ἑλασσον τοῦ $Z\Delta$. ὁμοίως δὲ δείκνυμεν, ὅτι οὐδὲ πρὸς μείζον πρὸς αὐτὸ ἄρα.

Ἐάν ἄρα διηρημένα μεγέθη ἀνάλογον ᾗ, καὶ συντεθέντα ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.

Let AE, EB, CF , and FD be separated magnitudes (which are) proportional, (so that) as AE (is) to EB , so CF (is) to FD . I say that they will also be proportional (when) composed, (so that) as AB (is) to BE , so CD (is) to FD .

For if (it is) not (the case that) as AB is to BE , so CD (is) to FD , (then) it will surely be (the case that) as AB (is) to BE , so CD is either to some (magnitude) less than DF , or (some magnitude) greater (than DF).[‡]

Let it, first of all, be to (some magnitude) less (than DF), (namely) DG . And since composed magnitudes are proportional, (so that) as AB is to BE , so CD (is) to DG , they will thus also be proportional (when) separated [Prop. 5.17]. Thus, as AE is to EB , so CG (is) to GD . But it was also assumed that as AE (is) to EB , so CF (is) to FD . Thus, (it is) also (the case that) as CG (is) to GD , so CF (is) to FD [Prop. 5.11]. And the first (magnitude) CG (is) greater than the third CF . Thus, the second (magnitude) GD (is) also greater than the fourth FD [Prop. 5.14]. But (it is) also less. The very thing is impossible. Thus, (it is) not (the case that) as AB is to BE , so CD (is) to less than FD . Similarly, we can show that neither (is it the case) to greater (than FD). Thus, (it is the case) to the same (as FD).

Thus, if separated magnitudes are proportional, (then) they will also be proportional (when) composed. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha + \beta : \beta :: \gamma + \delta : \delta$.

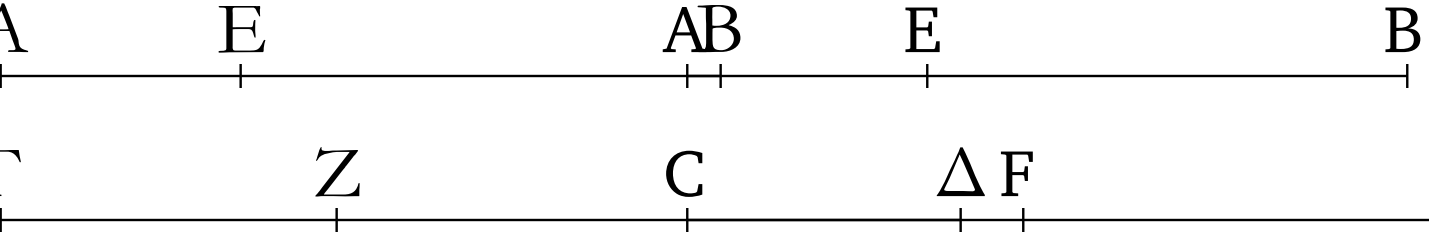
[‡] Here, Euclid assumes, without proof, that a fourth magnitude proportional to three given magnitudes can always be found.

19.

Proposition 19[†]

Ἐάν ᾗ ὅς ὅλον πρὸς ὅλον, οὕτως ἀφαιρεθέν πρὸς ἀφαιρεθέν, καὶ τὸ λοιπὸν πρὸς τὸ λοιπὸν ἔσται ὥς ὅλον πρὸς ὅλον.

If as the whole is to the whole so the (part) taken away is to the (part) taken away, (then) the remainder to the remainder will also be as the whole (is) to the whole.



Ἐστω γάρ ὡς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$, οὕτως ἀφαιρεθὲν τὸ AE πρὸς ἀφαιρεθὲν τὸ ΓZ · λέγω, ὅτι καὶ λοιπὸν τὸ EB πρὸς λοιπὸν τὸ $Z\Delta$ ἔσται ὡς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$.

Ἐπεὶ γάρ ἐστιν ὡς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ AE πρὸς τὸ ΓZ , καὶ ἐναλλάξ ὡς τὸ BA πρὸς τὸ AE , οὕτως τὸ $\Delta\Gamma$ πρὸς τὸ ΓZ . καὶ ἐπεὶ συγκείμενα μεγέθη ἀνάλογόν ἐστιν, καὶ διαιρεθέντα ἀνάλογον ἔσται, ὡς τὸ BE πρὸς τὸ EA , οὕτως τὸ ΔZ πρὸς τὸ ΓZ · καὶ ἐναλλάξ, ὡς τὸ BE πρὸς τὸ ΔZ , οὕτως τὸ EA πρὸς τὸ $Z\Gamma$. ὡς δὲ τὸ AE πρὸς τὸ ΓZ , οὕτως ὑπόκειται ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$. καὶ λοιπὸν ἄρα τὸ EB πρὸς λοιπὸν τὸ $Z\Delta$ ἔσται ὡς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$.

Ἐὰν ἄρα ἦ ὡς ὅλον πρὸς ὅλον, οὕτως ἀφαιρεθὲν πρὸς ἀφαιρεθὲν, καὶ τὸ λοιπὸν πρὸς τὸ λοιπὸν ἔσται ὡς ὅλον πρὸς ὅλον [ὅπερ ἔδει δεῖξαι].

[Καὶ ἐπεὶ ἐδείχθη ὡς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ EB πρὸς τὸ $Z\Delta$, καὶ ἐναλλάξ ὡς τὸ AB πρὸς τὸ BE οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ $Z\Delta$, συγκείμενα ἄρα μεγέθη ἀνάλογόν ἐστιν· ἐδείχθη δὲ ὡς τὸ BA πρὸς τὸ AE , οὕτως τὸ $\Delta\Gamma$ πρὸς τὸ ΓZ · καὶ ἐστὶν ἀναστρέφαντι].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν συγκείμενα μεγέθη ἀνάλογον ἦ, καὶ ἀναστρέφαντι ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \beta :: \alpha - \gamma : \beta - \delta$.

‡ In modern notation, this corollary reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \alpha - \beta :: \gamma : \gamma - \delta$.

20.

Ἐὰν ἦ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύν-
δυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι' ἴσον δὲ τὸ πρῶτον
τοῦ τρίτου μείζον ἢ, καὶ τὸ τέταρτον τοῦ ἑκτου μείζον ἔσται,
κ' ἂν ἴσον, ἴσον, κ' ἂν ἑλάττων, ἑλάττων.

For let the whole AB be to the whole CD as the (part) taken away AE (is) to the (part) taken away CF . I say that the remainder EB to the remainder FD will also be as the whole AB (is) to the whole CD .

For since as AB is to CD , so AE (is) to CF , (it is) also (the case), alternately, (that) as BA (is) to AE , so DC (is) to CF [Prop. 5.16]. And since composed magnitudes are proportional, (then) they will also be proportional (when) separated, (so that) as BE (is) to EA , so DF (is) to CF [Prop. 5.17]. Also, alternately, as BE (is) to DF , so EA (is) to FC [Prop. 5.16]. And it was assumed that as AE (is) to CF , so the whole AB (is) to the whole CD . And, thus, as the remainder EB (is) to the remainder FD , so the whole AB will be to the whole CD .

Thus, if as the whole is to the whole so the (part) taken away is to the (part) taken away, (then) the remainder to the remainder will also be as the whole (is) to the whole. [(Which is) the very thing it was required to show.]

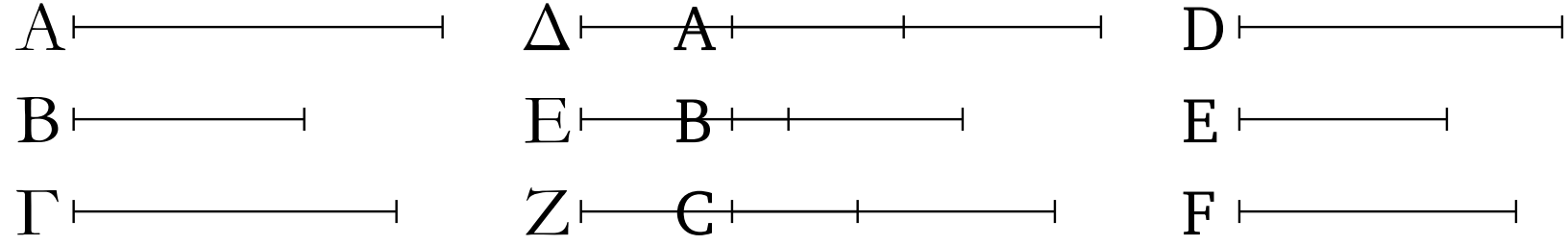
[And since it was shown (that) as AB (is) to CD , so EB (is) to FD , (it is) also (the case), alternately, (that) as AB (is) to BE , so CD (is) to FD . Thus, composed magnitudes are proportional. And it was shown (that) as BA (is) to AE , so DC (is) to CF . And (the latter) is converted (from the former).]

Corollary‡

So (it is) clear, from this, that if composed magnitudes are proportional, (then) they will also be proportional (when) converted. (Which is) the very thing it was required to show.

Proposition 20†

If there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth).



Ἐστω τρία μεγέθη τὰ A, B, Γ , καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, E, Z , σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ Δ πρὸς τὸ E , ὥς δὲ τὸ B πρὸς τὸ Γ , οὕτως τὸ E πρὸς τὸ Z , δι' ἴσον δὲ μείζον ἔστω τὸ A τοῦ Γ λέγω, ὅτι καὶ τὸ Δ τοῦ Z μείζον ἔσται, κ' ἂν ἴσον, ἴσον, κ' ἂν ἔλαττον, ἔλαττον.

Ἐπεὶ γὰρ μείζον ἐστὶ τὸ A τοῦ Γ , ἄλλο δέ τι τὸ B , τὸ δὲ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔθει ἥπερ τὸ ἔλαττον, τὸ A ἄρα πρὸς τὸ B μείζονα λόγον ἔθει ἥπερ τὸ Γ πρὸς τὸ B . ἀλλ' ὥς μὲν τὸ A πρὸς τὸ B [οὕτως] τὸ Δ πρὸς τὸ E , ὥς δὲ τὸ Γ πρὸς τὸ B , ἀνάπαλιν οὕτως τὸ Z πρὸς τὸ E · καὶ τὸ Δ ἄρα πρὸς τὸ E μείζονα λόγον ἔθει ἥπερ τὸ Z πρὸς τὸ E . τῶν δὲ πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον ἔθον μείζον ἐστίν. μείζον ἄρα τὸ Δ τοῦ Z . ὁμοίως δὲ δείχουμεν, ὅτι κ' ἂν ἴσον ᾖ τὸ A τῷ Γ , ἴσον ἔσται καὶ τὸ Δ τῷ Z , κ' ἂν ἔλαττον, ἔλαττον.

Ἐὰν ἄρα ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι' ἴσον δὲ τὸ πρῶτον τοῦ τρίτου μείζον ᾖ, καὶ τὸ τέταρτον τοῦ ἑκτου μείζον ἔσται, κ' ἂν ἴσον, ἴσον, κ' ἂν ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖχαι.

Let A, B , and C be three magnitudes, and D, E, F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two, (so that) as A (is) to B , so D (is) to E , and as B (is) to C , so E (is) to F . And let A be greater than C , via equality. I say that D will also be greater than F . And if (A is) equal (to C then D will also be) equal (to F). And if (A is) less (than C then D will also be) less (than F).

For since A is greater than C , and B some other (magnitude), and the greater (magnitude) has a greater ratio than the lesser to the same (magnitude) [Prop. 5.8], A thus has a greater ratio to B than C (has) to B . But as A (is) to B , [so] D (is) to E . And, inversely, as C (is) to B , so F (is) to E [Prop. 5.7 corr.]. Thus, D also has a greater ratio to E than F (has) to E [Prop. 5.13]. And for (magnitudes) having a ratio to the same (magnitude), that having the greater ratio is greater [Prop. 5.10]. Thus, D (is) greater than F . Similarly, we can show that even if A is equal to C (then) D will also be equal to F , and even if (A is) less (than C then D will also be) less (than F).

Thus, if there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if), via equality, the first is greater than the third, (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And (if the first is) less (than the third then the fourth will also be) less (than the sixth). (Which is) the very thing it was required to show.

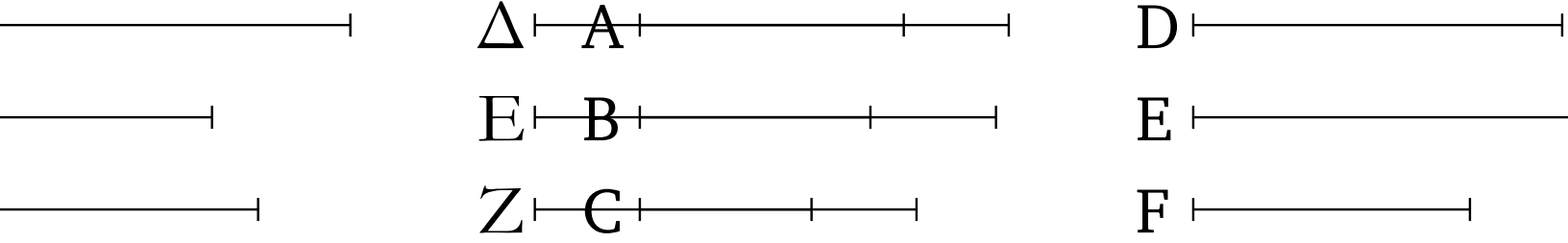
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \delta : \epsilon$ and $\beta : \gamma :: \epsilon : \zeta$ then $\alpha \gtrless \gamma$ as $\delta \gtrless \zeta$.

21.

Ἐὰν ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ᾗ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, δι' ἴσον δὲ τὸ πρῶτον τοῦ τρίτου μείζον ᾖ, καὶ τὸ τέταρτον τοῦ ἑκτου μείζον ἔσται, κ' ἂν ἴσον, ἴσον, κ' ἂν ἔλαττον, ἔλαττον.

Proposition 21[†]

If there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if) their proportion (is) perturbed, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth).



Ἐστω τρία μεγέθη τὰ A, B, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, E, Z , σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ἔστω δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z , ὥς δὲ τὸ B πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ E , δι' ἴσον δὲ τὸ A τοῦ Γ μείζον ἔστω· λέγω, ὅτι καὶ τὸ Δ τοῦ Z μείζον ἔσται, καὶ ἴσον, ἴσον, καὶ ἔλαττον, ἔλαττον.

Ἐπεὶ γὰρ μείζον ἐστὶ τὸ A τοῦ Γ , ἄλλο δέ τι τὸ B , τὸ A ἄρα πρὸς τὸ B μείζονα λόγον ἔθει ἢ περὶ τὸ Γ πρὸς τὸ B . ἀλλ' ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z , ὥς δὲ τὸ Γ πρὸς τὸ B , ἀνάπαλιν οὕτως τὸ E πρὸς τὸ Δ . καὶ τὸ E ἄρα πρὸς τὸ Z μείζονα λόγον ἔθει ἢ περὶ τὸ E πρὸς τὸ Δ . πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον ἔθει, ἐκείνο ἔλασσόν ἐστιν· ἔλασσον ἄρα ἐστὶ τὸ Z τοῦ Δ · μείζον ἄρα ἐστὶ τὸ Δ τοῦ Z . ὁμοίως δὲ δείκνυμεν, ὅτι καὶ ἴσον ἢ τὸ A τῷ Γ , ἴσον ἔσται καὶ τὸ Δ τῷ Z , καὶ ἔλαττον, ἔλαττον.

Ἐὰν ἄρα ἢ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ἢ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, δι' ἴσον δὲ τὸ πρῶτον τοῦ τρίτου μείζον ἢ, καὶ τὸ τέταρτον τοῦ ἑκτου μείζον ἔσται, καὶ ἴσον, ἴσον, καὶ ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖχαι.

Let A, B , and C be three magnitudes, and D, E, F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two. And let their proportion be perturbed, (so that) as A (is) to B , so E (is) to F , and as B (is) to C , so D (is) to E . And let A be greater than C , via equality. I say that D will also be greater than F . And if $(A$ is) equal (to C then D will also be) equal (to F). And if $(A$ is) less (than C then D will also be) less (than F).

For since A is greater than C , and B some other (magnitude), A thus has a greater ratio to B than C (has) to B [Prop. 5.8]. But as A (is) to B , so E (is) to F . And, inversely, as C (is) to B , so E (is) to D [Prop. 5.7 corr.]. Thus, E also has a greater ratio to F than E (has) to D [Prop. 5.13]. And that (magnitude) to which the same (magnitude) has a greater ratio is (the) lesser (magnitude) [Prop. 5.10]. Thus, F is less than D . Thus, D is greater than F . Similarly, we can show that even if A is equal to C (then) D will also be equal to F , and even if $(A$ is) less (than C then D will also be) less (than F).

Thus, if there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if) their proportion (is) perturbed, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth). (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \delta : \epsilon$ then $\alpha \gtrless \gamma$ as $\delta \gtrless \zeta$.

22.

Ἐὰν ἢ ὅποσαοῦν μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται.

Proposition 22[†]

If there are any number of magnitudes whatsoever, and (some) other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, (then) they will also be in the same ratio via equality.

B A

BΓ

C

E D

EZ

F

K G

KM

M

Λ H

LN

N

Ἐστω ὅποσαοῦν μεγέθη τὰ A, B, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, E, Z , σύνδνο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ Δ πρὸς τὸ E , ὥς δὲ τὸ B πρὸς τὸ Γ , οὕτως τὸ E πρὸς τὸ Z . λέγω, ὅτι καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται.

Εἰλήφθω γάρ τῶν μὲν A, Δ ἰσάκις πολλαπλάσια τὰ H, Θ , τῶν δὲ B, E ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ K, Λ , καὶ ἔτι τῶν Γ, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ M, N .

Καὶ ἐπεὶ ἔστιν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ Δ πρὸς τὸ E , καὶ εἰληπται τῶν μὲν A, Δ ἰσάκις πολλαπλάσια τὰ H, Θ , τῶν δὲ B, E ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ K, Λ , ἔστιν ἄρα ὥς τὸ H πρὸς τὸ K , οὕτως τὸ Θ πρὸς τὸ Λ . διὰ τὰ αὐτὰ δὴ καὶ ὥς τὸ K πρὸς τὸ M , οὕτως τὸ Λ πρὸς τὸ N . ἐπεὶ οὖν τρία μεγέθη ἐστί τὰ H, K, M , καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Θ, Λ, N , σύνδνο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι' ἴσον ἄρα, εἰ ὑπερέθει τὸ H τοῦ M , ὑπερέθει καὶ τὸ Θ τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν H, Θ τῶν A, Δ ἰσάκις πολλαπλάσια, τὰ δὲ M, N τῶν Γ, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια. ἔστιν ἄρα ὥς τὸ A πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ Z .

Ἐὰν ἄρα ἢ ὅποσαοῦν μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδνο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται. ὅπερ ἔδει δεῖχαι.

Let there be any number of magnitudes whatsoever, A, B, C , and (some) other (magnitudes), D, E, F , of equal number to them, (which are) in the same ratio taken two by two, (so that) as A (is) to B , so D (is) to E , and as B (is) to C , so E (is) to F . I say that they will also be in the same ratio via equality. (That is, as A is to C , so D is to F .)

For let the equal multiples G and H be taken of A and D (respectively), and the other random equal multiples K and L of B and E (respectively), and the yet other random equal multiples M and N of C and F (respectively).

And since as A is to B , so D (is) to E , and the equal multiples G and H have been taken of A and D (respectively), and the other random equal multiples K and L of B and E (respectively), thus as G is to K , so H (is) to L [Prop. 5.4]. And, so, for the same (reasons), as K (is) to M , so L (is) to N . Therefore, since G, K , and M are three magnitudes, and H, L , and N other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, thus, via equality, if G exceeds M (then) H also exceeds N , and if (G is) equal (to M then H is also) equal (to N), and if (G is) less (than M then H is also) less (than N) [Prop. 5.20]. And G and H are equal multiples of A and D (respectively), and M and N other random equal multiples of C and F (respectively). Thus, as A is to C , so D (is) to F [Def. 5.5].

Thus, if there are any number of magnitudes whatsoever, and (some) other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, (then) they will also be in the same ratio via equality. (Which is) the very thing it was required to show.

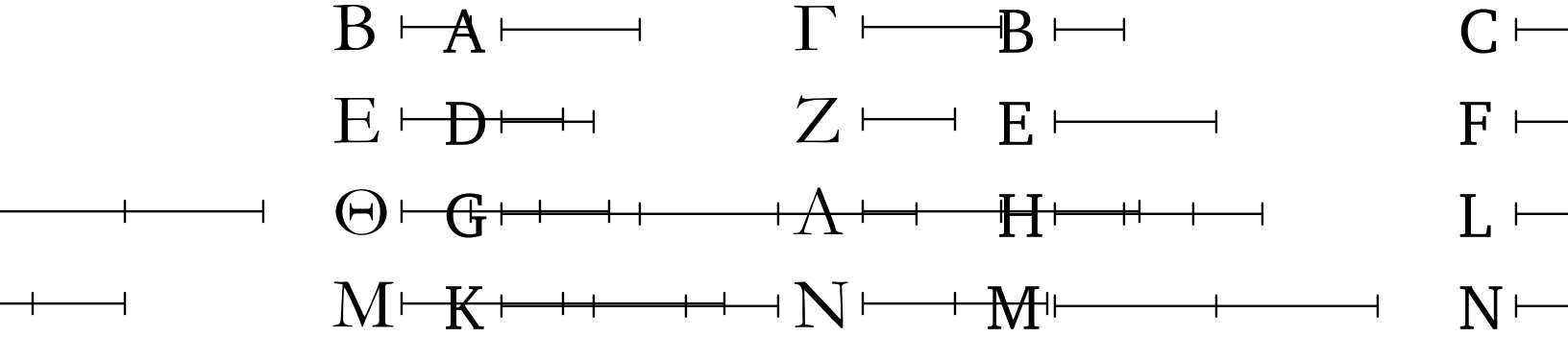
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \zeta : \eta$ and $\gamma : \delta :: \eta : \theta$ then $\alpha : \delta :: \epsilon : \theta$.

23.

Proposition 23[†]

Ἐὰν ἢ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδνο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ἢ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται.

If there are three magnitudes, and others of equal number to them, (being) in the same ratio taken two by two, and (if) their proportion is perturbed, (then) they will also be in the same ratio via equality.



Ἐστω τρία μεγέθη τὰ A, B, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ τὰ Δ, E, Z , ἔστω δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z , ὥς δὲ τὸ B πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ E . λέγω, ὅτι ἔστιν ὥς τὸ A πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ Z .

Εἰλήφθω τῶν μὲν A, B, Δ ἰσάκις πολλαπλάσια τὰ H, Θ, K , τῶν δὲ Γ, E, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, M, N .

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσια τὰ H, Θ τῶν A, B , τὰ δὲ μέρη τοῖς ὡσάντως πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὥς τὸ A πρὸς τὸ B , οὕτως τὸ H πρὸς τὸ Θ . διὰ τὰ αὐτὰ δὴ καὶ ὥς τὸ E πρὸς τὸ Z , οὕτως τὸ M πρὸς τὸ N . καὶ ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z . καὶ ὥς ἄρα τὸ H πρὸς τὸ Θ , οὕτως τὸ M πρὸς τὸ N . καὶ ἐπεὶ ἐστὶν ὥς τὸ B πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ E , καὶ ἐναλλάχ ὥς τὸ B πρὸς τὸ Δ , οὕτως τὸ Γ πρὸς τὸ E . καὶ ἐπεὶ τὰ Θ, K τῶν B, Δ ἰσάκις ἐστὶ πολλαπλάσια, τὰ δὲ μέρη τοῖς ἰσάκις πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὥς τὸ B πρὸς τὸ Δ , οὕτως τὸ Θ πρὸς τὸ K . ἀλλ' ὥς τὸ B πρὸς τὸ Δ , οὕτως τὸ Γ πρὸς τὸ E . καὶ ὥς ἄρα τὸ Θ πρὸς τὸ K , οὕτως τὸ Γ πρὸς τὸ E . πάλιν, ἐπεὶ τὰ Λ, M τῶν Γ, E ἰσάκις ἐστὶ πολλαπλάσια, ἔστιν ἄρα ὥς τὸ Γ πρὸς τὸ E , οὕτως τὸ Λ πρὸς τὸ M . ἀλλ' ὥς τὸ Γ πρὸς τὸ E , οὕτως τὸ Θ πρὸς τὸ K . καὶ ὥς ἄρα τὸ Θ πρὸς τὸ K , οὕτως τὸ Λ πρὸς τὸ M , καὶ ἐναλλάχ ὥς τὸ Θ πρὸς τὸ Λ , τὸ K πρὸς τὸ M . ἐδείχθη δὲ καὶ ὥς τὸ H πρὸς τὸ Θ , οὕτως τὸ M πρὸς τὸ N . ἐπεὶ οὖν τρία μεγέθη ἐστὶ τὰ H, Θ, Λ , καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ K, M, N σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, καὶ ἐστὶν αὐτῶν τεταραγμένη ἡ ἀναλογία, δι' ἴσον ἄρα, εἰ ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὸ K τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν H, K τῶν A, Δ ἰσάκις πολλαπλάσια, τὰ δὲ Λ, N τῶν Γ, Z . ἔστιν ἄρα ὥς τὸ A πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ Z .

Ἐὰν ἄρα ἢ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ἢ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται ὅπερ ἔδει δεῖξαι.

Let A, B , and C be three magnitudes, and D, E and F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two. And let their proportion be perturbed, (so that) as A (is) to B , so E (is) to F , and as B (is) to C , so D (is) to E . I say that as A is to C , so D (is) to F .

Let the equal multiples G, H , and K be taken of A, B , and D (respectively), and the other random equal multiples L, M , and N of C, E , and F (respectively).

And since G and H are equal multiples of A and B (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as A (is) to B , so G (is) to H . And, so, for the same (reasons), as E (is) to F , so M (is) to N . And as A is to B , so E (is) to F . And, thus, as G (is) to H , so M (is) to N [Prop. 5.11]. And since as B is to C , so D (is) to E , also, alternately, as B (is) to D , so C (is) to E [Prop. 5.16]. And since H and K are equal multiples of B and D (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as B is to D , so H (is) to K . But, as B (is) to D , so C (is) to E . And, thus, as H (is) to K , so C (is) to E [Prop. 5.11]. Again, since L and M are equal multiples of C and E (respectively), thus as C is to E , so L (is) to M [Prop. 5.15]. But, as C (is) to E , so H (is) to K . And, thus, as H (is) to K , so L (is) to M [Prop. 5.11]. Also, alternately, as H (is) to L , so K (is) to M [Prop. 5.16]. And it was also shown (that) as G (is) to H , so M (is) to N . Therefore, since G, H , and L are three magnitudes, and K, M , and N other (magnitudes) of equal number to them, (being) in the same ratio taken two by two, and their proportion is perturbed, thus, via equality, if G exceeds L (then) K also exceeds N , and if (G is) equal (to L then K is also) equal (to N), and if (G is) less (than L then K is also) less (than N) [Prop. 5.21]. And G and K are equal multiples of A and D (respectively), and L and N of C and F (respectively). Thus, as A (is) to C , so D (is) to F [Def. 5.5].

Thus, if there are three magnitudes, and others of equal number to them, (being) in the same ratio taken two by two, and (if) their proportion is perturbed, (then) they will also be in the same ratio via equality. (Which is) the very thing it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \delta : \epsilon$ then $\alpha : \gamma :: \delta : \zeta$.

24.

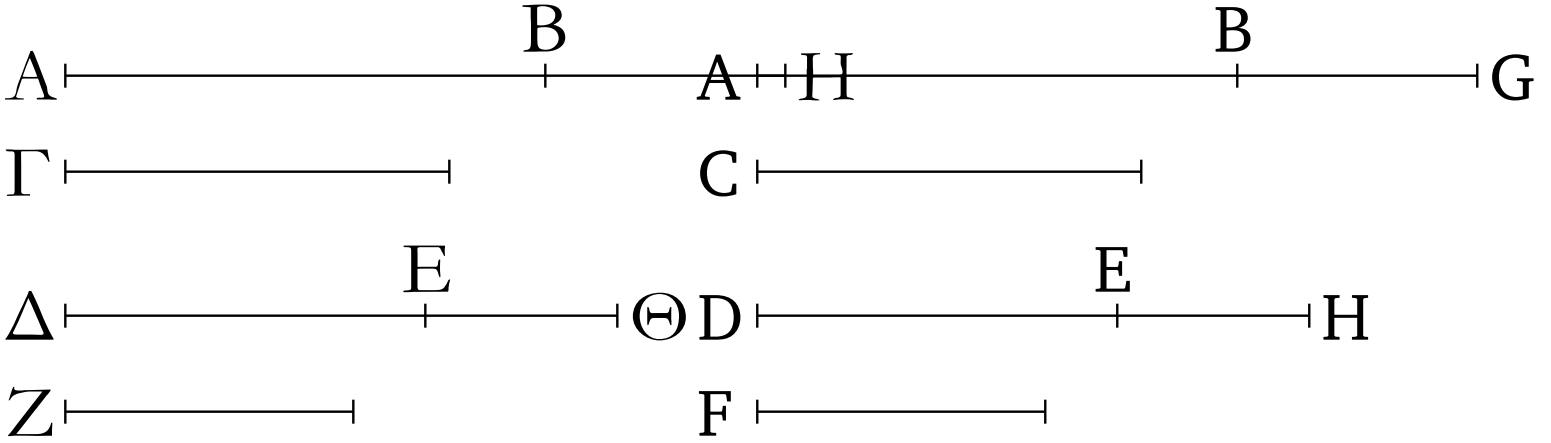
Proposition 24†

Ἐάν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, ἔθῃ δὲ καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν λόγον καὶ ἕκτον πρὸς τέταρτον, καὶ συντεθέν πρῶτον καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν ἔχει λόγον καὶ τρίτον καὶ ἕκτον πρὸς τέταρτον.

Πρῶτον γάρ τὸ AB πρὸς δεύτερον τὸ Γ τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον τὸ ΔE πρὸς τέταρτον τὸ Z , ἔθῃ δὲ καὶ πέμπτον τὸ BH πρὸς δεύτερον τὸ Γ τὸν αὐτὸν λόγον καὶ ἕκτον τὸ $E\Theta$ πρὸς τέταρτον τὸ Z . λέγω, ὅτι καὶ συντεθέν πρῶτον καὶ πέμπτον τὸ AH πρὸς δεύτερον τὸ Γ τὸν αὐτὸν ἔχει λόγον, καὶ τρίτον καὶ ἕκτον τὸ $\Delta\Theta$ πρὸς τέταρτον τὸ Z .

If a first (magnitude) has to a second the same ratio that third (has) to a fourth, and a fifth (magnitude) also has to the second the same ratio that a sixth (has) to the fourth, (then) the first (magnitude) and the fifth, added together, will also have the same ratio to the second that the third (magnitude) and sixth (added together, have) to the fourth.

For let a first (magnitude) AB have the same ratio to a second C that a third DE (has) to a fourth F . And let a fifth (magnitude) BG also have the same ratio to the second C that a sixth EH (has) to the fourth F . I say that the first (magnitude) and the fifth, added together, AG , will also have the same ratio to the second C that the third (magnitude) and the sixth, (added together), DH , (has) to the fourth F .



Ἐπεὶ γάρ ἐστιν ὡς τὸ BH πρὸς τὸ Γ , οὕτως τὸ $E\Theta$ πρὸς τὸ Z , ἀνάπαλιν ἄρα ὡς τὸ Γ πρὸς τὸ BH , οὕτως τὸ Z πρὸς τὸ $E\Theta$. Ἐπεὶ οὖν ἐστιν ὡς τὸ AB πρὸς τὸ Γ , οὕτως τὸ ΔE πρὸς τὸ Z , ὡς δὲ τὸ Γ πρὸς τὸ BH , οὕτως τὸ Z πρὸς τὸ $E\Theta$, δι' ἴσον ἄρα ἐστὶν ὡς τὸ AB πρὸς τὸ BH , οὕτως τὸ ΔE πρὸς τὸ $E\Theta$. καὶ ἐπεὶ διηρημένα μεγέθη ἀνάλογόν ἐστιν, καὶ συντεθέντα ἀνάλογον ἐστὶν. ἔστιν ἄρα ὡς τὸ AH πρὸς τὸ HB , οὕτως τὸ $\Delta\Theta$ πρὸς τὸ ΘE . ἔστι δὲ καὶ ὡς τὸ BH πρὸς τὸ Γ , οὕτως τὸ $E\Theta$ πρὸς τὸ Z . δι' ἴσον ἄρα ἐστὶν ὡς τὸ AH πρὸς τὸ Γ , οὕτως τὸ $\Delta\Theta$ πρὸς τὸ Z .

Ἐάν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, ἔθῃ δὲ καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν λόγον καὶ ἕκτον πρὸς τέταρτον, καὶ συντεθέν πρῶτον καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν ἔχει λόγον καὶ τρίτον καὶ ἕκτον πρὸς τέταρτον. ὅπερ ἔδει δεῖχαι.

For since as BG is to C , so EH (is) to F , thus, inversely, as C (is) to BG , so F (is) to EH [Prop. 5.7 corr.]. Therefore, since as AB is to C , so DE (is) to F , and as C (is) to BG , so F (is) to EH , thus, via equality, as AB is to BG , so DE (is) to EH [Prop. 5.22]. And since separated magnitudes are proportional, (then) they will also be proportional (when) composed [Prop. 5.18]. Thus, as AG is to GB , so DH (is) to HE . And, also, as BG is to C , so EH (is) to F . Thus, via equality, as AG is to C , so DH (is) to F [Prop. 5.22].

Thus, if a first (magnitude) has to a second the same ratio that a third (has) to a fourth, and a fifth (magnitude) also has to the second the same ratio that a sixth (has) to the fourth, (then) the first (magnitude) and the fifth, added together, will also have the same ratio to the second that the third (magnitude) and the sixth (added together, have) to the fourth. (Which is) the very thing it was required to show.

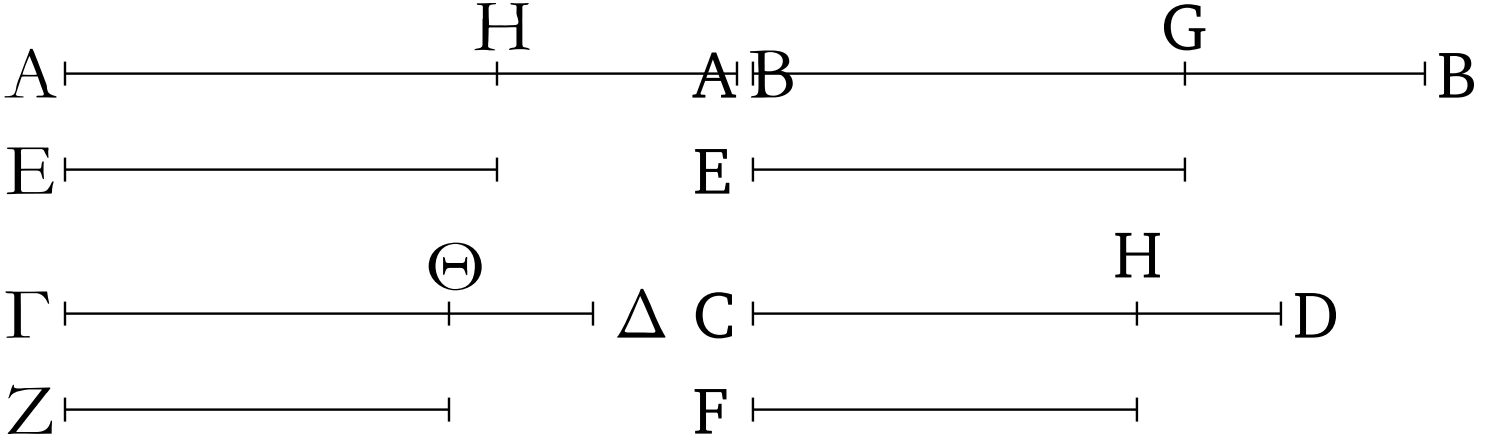
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\epsilon : \beta :: \zeta : \delta$ then $\alpha + \epsilon : \beta :: \gamma + \zeta : \delta$.

25.

Proposition 25†

Ἐάν τέσσαρα μεγέθη ἀνάλογον ᾗ, τὸ μέγιστον [αὐτῶν]
καὶ τὸ ἐλάχιστον δύο τῶν λοιπῶν μείζονά ἐστιν.

If four magnitudes are proportional, (then) the (sum of the)
largest and the smallest [of them] is greater than the (sum of
the) remaining two (magnitudes).



Ἐστω τέσσαρα μεγέθη ἀνάλογον τὰ AB , $\Gamma\Delta$, E , Z , ὥς τὸ
 AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ E πρὸς τὸ Z , ἔστω δὲ μέγιστον
μὲν αὐτῶν τὸ AB , ἐλάχιστον δὲ τὸ Z . λέγω, ὅτι τὰ AB , Z
τῶν $\Gamma\Delta$, E μείζονά ἐστιν.

Κείσθω γὰρ τῷ μὲν E ἴσον τὸ AH , τῷ δὲ Z ἴσον τὸ $\Gamma\Theta$.

Ἐπεὶ [οὕτως] ἐστὶν ὥς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ E πρὸς
τὸ Z , ἴσον δὲ τὸ μὲν E τῷ AH , τὸ δὲ Z τῷ $\Gamma\Theta$, ἔστιν ἄρα
ὥς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ AH πρὸς τὸ $\Gamma\Theta$. καὶ ἐπεὶ
ἐστὶν ὥς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$, οὕτως ἀφαιρεθέν
τὸ AH πρὸς ἀφαιρεθέν τὸ $\Gamma\Theta$, καὶ λοιπὸν ἄρα τὸ HB πρὸς
λοιπὸν τὸ $\Theta\Delta$ ἔσται ὥς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$. μείζον
δὲ τὸ AB τοῦ $\Gamma\Delta$ · μείζον ἄρα καὶ τὸ HB τοῦ $\Theta\Delta$. καὶ ἐπεὶ
ἴσον ἐστὶ τὸ μὲν AH τῷ E , τὸ δὲ $\Gamma\Theta$ τῷ Z , τὰ ἄρα AH , Z
ἴσα ἐστὶ τοῖς $\Gamma\Theta$, E . καὶ [ἐπεὶ] ἔαν [ἀνίστοιξ ἴσα προστεθῇ, τὰ
ὅλα ἴσος ἐστὶν, ἔαν ἄρα] τῶν HB , $\Theta\Delta$ ἀνίσων ὄντων καὶ
μείζονος τοῦ HB τῷ μὲν HB προστεθῇ τὰ AH , Z , τῷ δὲ $\Theta\Delta$
προστεθῇ τὰ $\Gamma\Theta$, E , συνάγεται τὰ AB , Z μείζονα τῶν $\Gamma\Delta$,
 E .

Ἐάν ἄρα τέσσαρα μεγέθη ἀνάλογον ᾗ, τὸ μέγιστον αὐτῶν
καὶ τὸ ἐλάχιστον δύο τῶν λοιπῶν μείζονά ἐστιν. ὅπερ ἔδει
δεῖξαι.

Let AB , CD , E , and F be four proportional magnitudes,
(such that) as AB (is) to CD , so E (is) to F . And let AB be the
greatest of them, and F the least. I say that AB and F is greater
than CD and E .

For let AG be made equal to E , and CH equal to F .

[In fact,] since as AB is to CD , so E (is) to F , and E (is)
equal to AG , and F to CH , thus as AB is to CD , so AG (is)
to CH . And since the whole AB is to the whole CD as the
(part) taken away AG (is) to the (part) taken away CH , thus
the remainder GB will also be to the remainder HD as the
whole AB (is) to the whole CD [Prop. 5.19]. And AB (is)
greater than CD . Thus, GB (is) also greater than HD . And
since AG is equal to E , and CH to F , thus AG and F is equal
to CH and E . And [since] if [equal (magnitudes) are added to
unequal (magnitudes then) the wholes are unequal, thus if] AG
and F are added to GB , and CH and E to HD — GB and HD
(is) greater—it is inferred that AB and F (is) greater than CD and E .

Thus, if four magnitudes are proportional, (then) the (sum
of the) largest and the smallest of them is greater than the (sum
of the) remaining two (magnitudes). (Which is) the very thing
it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$, and α is the greatest and δ the least, then $\alpha + \delta > \beta + \gamma$.

ELEMENTS BOOK 5

Proportion[†]

[†]The theory of proportion set out in this book is generally attributed to Eudoxus of Cnidus. The novel feature of this theory is its ability to deal with irrational magnitudes, which had hitherto been a major stumbling block for Greek mathematicians. Throughout the footnotes in this book, α, β, γ , etc., denote general (possibly irrational) magnitudes, whereas m, n, l , etc., denote positive integers.

Ὅροι.

Definitions

1. Μέρος ἐστὶ μέγεθος μεγέθους τὸ ἔλασσον τοῦ μείζονος, ὅταν καταμετρηῇ τὸ μείζον.

2. Πολλαπλάσιον δὲ τὸ μείζον τοῦ ἐλάττονος, ὅταν καταμετρηῇ τὸ ἐλάττονον.

3. Λόγος ἐστὶ δύο μεγεθῶν ὁμογενῶν ἢ κατὰ πληκίότητά ποια σθέσις.

4. Λόγον ἔθειν πρός ἄλληλα μεγέθη λέγεται, ἃ δύναται πολλαπλασιαζόμενα ἀλλήλων ὑπερέθειν.

5. Ἐν τῷ αὐτῷ λόγῳ μεγέθη λέγεται εἶναι πρῶτον πρός δευτέρον καὶ τρίτον πρός τέταρτον, ὅταν τὰ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια τῶν τοῦ δευτέρου καὶ τετάρτου ἰσάκις πολλαπλασίων καθ' ὅποιον πολλαπλασιασμόν ἑκάτερον ἑκατέρου ἢ ἅμα ὑπερέθῃ ἢ ἅμα ἴσα ἢ ἢ ἅμα ἐλλείπῃ ληφθέντα κατάλληλα.

6. Τὰ δὲ τὸν αὐτὸν ἔθοντα λόγον μεγέθη ἀνάλογον καλεῖσθω.

7. Ὄταν δὲ τῶν ἰσάκις πολλαπλασίων τὸ μὲν τοῦ πρώτου πολλαπλάσιον ὑπερέθῃ τοῦ τοῦ δευτέρου πολλαπλασίου, τὸ δὲ τοῦ τρίτου πολλαπλάσιον μὴ ὑπερέθῃ τοῦ τοῦ τετάρτου πολλαπλασίου, τότε τὸ πρῶτον πρός τὸ δευτέρον μείζονα λόγον ἔθειν λέγεται, ἢ περὶ τὸ τρίτον πρός τὸ τέταρτον.

8. Ἀναλογία δὲ ἐν τρισὶν ὁροις ἐλαθίστη ἐστίν.

9. Ὄταν δὲ τρία μεγέθη ἀνάλογον ἢ, τὸ πρῶτον πρός τὸ τρίτον διπλασίονα λόγον ἔθειν λέγεται ἢ περὶ τὸ δευτέρον.

10. Ὄταν δὲ τέσσαρα μεγέθη ἀνάλογον ἢ, τὸ πρῶτον πρός τὸ τέταρτον τριπλασίονα λόγον ἔθειν λέγεται ἢ περὶ τὸ δευτέρον, καὶ αἱ ἐχρήσθαι ὁμοίως, ὥς ἂν ἡ ἀναλογία ὑπάρθῃ.

11. Ὁμόλογα μεγέθη λέγεται τὰ μὲν ἡγούμενα τοῖς ἡγουμένοις τὰ δὲ ἐπόμενα τοῖς ἐπομένοις.

12. Ἐναλλάξ λόγος ἐστὶ λήψις τοῦ ἡγουμένου πρός τὸ ἡγούμενον καὶ τοῦ ἐπομένου πρός τὸ ἐπόμενον.

13. Ἀνάπαλιν λόγος ἐστὶ λήψις τοῦ ἐπομένου ὡς ἡγουμένου πρός τὸ ἡγούμενον ὡς ἐπόμενον.

14. Σύνθεσις λόγου ἐστὶ λήψις τοῦ ἡγουμένου μετὰ τοῦ ἐπομένου ὡς ἐνὸς πρός αὐτὸ τὸ ἐπόμενον.

15. Διαίρεσις λόγου ἐστὶ λήψις τῆς ὑπεροθῆς, ἢ ὑπερέθει τὸ ἡγούμενον τοῦ ἐπομένου, πρὸς αὐτὸ τὸ ἐπόμενον.

16. Ἀναστροφὴ λόγου ἐστὶ λήψις τοῦ ἡγουμένου πρός τὴν ὑπεροθὴν, ἢ ὑπερέθει τὸ ἡγούμενον τοῦ ἐπομένου.

17. Δι' ἴσον λόγος ἐστὶ πλειόνων ὄντων μεγεθῶν καὶ ἄλλων αὐτοῖς ἴσων τὸ πλῆθος σύνδυο λαμβανομένων καὶ ἐν τῷ αὐτῷ λόγῳ, ὅταν ἢ ὡς ἐν τοῖς πρώτοις μεγέθεσι τὸ πρῶτον πρός τὸ ἑσθαιον, ὁὕτως ἐν τοῖς δευτέροις μεγέθεσι τὸ πρῶτον πρός τὸ ἑσθαιον ἢ ἄλλως λήψις τῶν ἀκρῶν καθ' ὑπερχαίρεσιν τῶν μέσων.

18. Τεταραγμένη δὲ ἀναλογία ἐστίν, ὅταν τριῶν ὄντων μεγεθῶν καὶ ἄλλων αὐτοῖς ἴσων τὸ πλῆθος γίνηται ὡς μὲν ἐν

1. A magnitude is a part of a(nother) magnitude, the lesser of the greater, when it measures the greater.[†]

2. And the greater (magnitude is) a multiple of the lesser when it is measured by the lesser.

3. A ratio is a certain type of relation with respect to size of two magnitudes of the same kind.[‡]

4. (Those) magnitudes are said to have a ratio with respect to one another which, being multiplied, are capable of exceeding one another.[§]

5. Magnitudes are said to be in the same ratio, the first to the second, and the third to the fourth, when equal multiples of the first and the third either both exceed, are both equal to, or are both less than, equal multiples of the second and the fourth, respectively, being taken in corresponding order, according to any kind of multiplication whatever.[¶]

6. And let magnitudes having the same ratio be called proportional.*

7. And when for equal multiples (as in Def. 5), the multiple of the first (magnitude) exceeds the multiple of the second, and the multiple of the third (magnitude) does not exceed the multiple of the fourth, (then) the first (magnitude) is said to have a greater ratio to the second than the third (magnitude has) to the fourth.

8. And a proportion in three terms is the smallest (possible).[§]

9. And when three magnitudes are proportional, the first is said to have to the third the squared^b ratio of that (it has) to the second.^{††}

10. And when four magnitudes are (continuously) proportional, the first is said to have to the fourth the cubed^{‡‡} ratio of that (it has) to the second.^{§§} And so on, similarly, in successive order, whatever the (continuous) proportion might be.

11. These magnitudes are said to be corresponding (magnitudes): the leading to the leading (of two ratios), and the following to the following.

12. An alternate ratio is a taking of the (ratio of the) leading (magnitude) to the leading (of two equal ratios), and (setting it equal to) the (ratio of the) following (magnitude) to the following.^{¶¶}

13. An inverse ratio is a taking of the (ratio of the) following (magnitude) as the leading and the leading (magnitude) as the following.^{**}

14. A composition of a ratio is a taking of the (ratio of the) leading plus the following (magnitudes), as one, to the following (magnitude) by itself.^{§§}

15. A separation of a ratio is a taking of the (ratio of the) excess by which the leading (magnitude) exceeds the following to the following (magnitude) by itself.^{bb}

16. A conversion of a ratio is a taking of the (ratio of the)

τοῖς πρώτοις μεγέθεσιν ἡγούμενον πρόξ ἐπόμενον, οὔτως ἐν τοῖς δευτέροις μεγέθεσιν ἡγούμενον πρόξ ἐπόμενον, ὡς δὲ ἐν τοῖς πρώτοις μεγέθεσιν ἐπόμενον πρόξ ἄλλο τι, οὔτως ἐν τοῖς δευτέροις ἄλλο τι πρόξ ἡγούμενον.

leading (magnitude) to the excess by which the leading (magnitude) exceeds the following.^{†††}

17. There being several magnitudes, and other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, a ratio via equality (i.e., ex aequali) occurs when as the first is to the last in the first (set of) magnitudes, so the first (is) to the last in the second (set of) magnitudes. Or alternately, (it is) a taking of the (ratio of the) outer (magnitudes) by the removal of the inner (magnitudes).^{‡‡‡}

18. There being three magnitudes, and other (magnitudes) of equal number to them, a perturbed proportion occurs when as the leading is to the following in the first (set of) magnitudes, so the leading (is) to the following in the second (set of) magnitudes, and as the following (is) to some other (i.e., the remaining magnitude) in the first (set of) magnitudes, so some other (is) to the leading in the second (set of) magnitudes.^{§§§}

[†] In other words, α is said to be a part of β if $\beta = m\alpha$.

[‡] In modern notation, the ratio of two magnitudes, α and β , is denoted $\alpha : \beta$.

[§] In other words, α has a ratio with respect to β if $m\alpha > \beta$ and $n\beta > \alpha$, for some m and n .

[¶] In other words, $\alpha : \beta :: \gamma : \delta$ if and only if $m\alpha > n\beta$ whenever $m\gamma > n\delta$, and $m\alpha = n\beta$ whenever $m\gamma = n\delta$, and $m\alpha < n\beta$ whenever $m\gamma < n\delta$, for all m and n . This definition is the kernel of Eudoxus' theory of proportion, and is valid even if α, β , etc., are irrational.

^{*} Thus if α and β have the same ratio as γ and δ then they are proportional. In modern notation, $\alpha : \beta :: \gamma : \delta$.

[§] In modern notation, a proportion in three terms— α, β , and γ —is written: $\alpha : \beta :: \beta : \gamma$.

^b Literally, “double”.

^{††} In other words, if $\alpha : \beta :: \beta : \gamma$ then $\alpha : \gamma :: \alpha^2 : \beta^2$.

^{‡‡} Literally, “triple”.

^{§§} In other words, if $\alpha : \beta :: \beta : \gamma :: \gamma : \delta$ then $\alpha : \delta :: \alpha^3 : \beta^3$.

^{¶¶} In other words, if $\alpha : \beta :: \gamma : \delta$ then the alternate ratio corresponds to $\alpha : \gamma :: \beta : \delta$.

^{**} In other words, if $\alpha : \beta$ then the inverse ratio corresponds to $\beta : \alpha$.

^{§§} In other words, if $\alpha : \beta$ then the composed ratio corresponds to $\alpha + \beta : \beta$.

^{bb} In other words, if $\alpha : \beta$ then the separated ratio corresponds to $\alpha - \beta : \beta$.

^{†††} In other words, if $\alpha : \beta$ then the converted ratio corresponds to $\alpha : \alpha - \beta$.

^{‡‡‡} In other words, if α, β, γ are the first set of magnitudes, and δ, ϵ, ζ the second set, and $\alpha : \beta : \gamma :: \delta : \epsilon : \zeta$, then the ratio via equality (i.e., ex aequali) corresponds to $\alpha : \gamma :: \delta : \zeta$.

^{§§§} In other words, if α, β, γ are the first set of magnitudes, and δ, ϵ, ζ the second set, and $\alpha : \beta :: \delta : \epsilon$ as well as $\beta : \gamma :: \zeta : \delta$, then the proportion is said to be perturbed.

1.

Proposition 1[†]

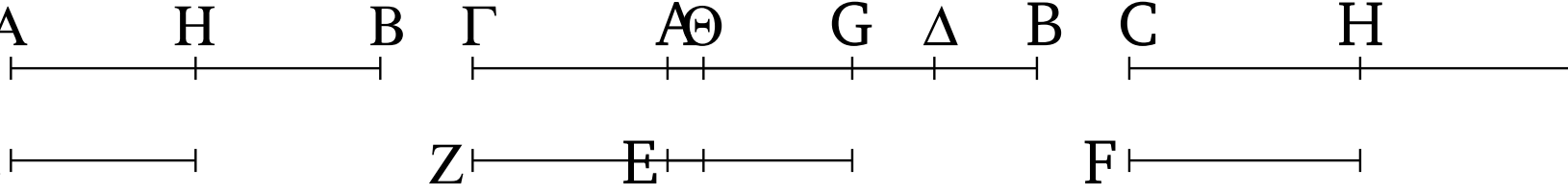
Ἐάν ᾗ ὁποσαοῦν μεγέθη ὁποσωνοῦν μεγεθῶν ἴσων τὸ πληθὸς ἕκαστον ἐκάστον ἰσάκις πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ἐν τῶν μεγεθῶν ἐνόξ, τοσαυταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων.

Ἐστω ὁποσαοῦν μεγέθη τὰ $AB, \Gamma\Delta$ ὁποσωνοῦν μεγεθῶν τῶν E, Z ἴσων τὸ πληθὸς ἕκαστον ἐκάστον ἰσάκις πολλαπλάσιον· λέγω, ὅτι ὁσαπλάσιόν ἐστι τὸ AB τοῦ E , τοσαυταπλάσια ἔσται καὶ τὰ $AB, \Gamma\Delta$ τῶν E, Z .

If there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second).

Let there be any number of magnitudes whatsoever, AB, CD , (which are) equal multiples, respectively, of some (other) magnitudes, E, F , of equal number (to them). I say that as

many times as AB is (divisible) by E , so many times will AB , CD also be (divisible) by E, F .



Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AB τοῦ E καὶ τὸ $\Gamma\Delta$ τοῦ Z , ὅσα ἄρα ἐστὶν ἐν τῷ AB μεγέθη ἴσα τῷ E , τοσαῦτα καὶ ἐν τῷ $\Gamma\Delta$ ἴσα τῷ Z . διηγήσθω τὸ μὲν AB εἰς τὰ τῷ E μεγέθη ἴσα τὰ AH, HB , τὸ δὲ $\Gamma\Delta$ εἰς τὰ τῷ Z ἴσα τὰ $\Gamma\Theta, \Theta\Delta$. ἔσται δὴ ἴσον τὸ πλῆθος τῶν AH, HB τῷ πλῆθει τῶν $\Gamma\Theta, \Theta\Delta$. καὶ ἐπεὶ ἴσον ἐστὶ τὸ μὲν AH τῷ E , τὸ δὲ $\Gamma\Theta$ τῷ Z , ἴσον ἄρα τὸ AH τῷ E , καὶ τὰ $AH, \Gamma\Theta$ τοῖς E, Z . διὰ τὰ αὐτὰ δὴ ἴσον ἐστὶ τὸ HB τῷ E , καὶ τὰ $HB, \Theta\Delta$ τοῖς E, Z . ὅσα ἄρα ἐστὶν ἐν τῷ AB ἴσα τῷ E , τοσαῦτα καὶ ἐν τοῖς $AB, \Gamma\Delta$ ἴσα τοῖς E, Z . ὁσαυτάσιον ἄρα ἐστὶ τὸ AB τοῦ E , τοσαυταπλάσια ἔσται καὶ τὰ $AB, \Gamma\Delta$ τῶν E, Z .

Ἐάν ἄρα ἡ ὁποσαοῦν μεγέθη ὁποσωνοῦν μεγεθῶν ἴσων τὸ πλῆθος ἕκαστον ἐκάστων ἰσάκις πολλαπλάσιον, ὁσαυτάσιον ἐστὶν ἐν τῶν μεγεθῶν ἐνόξ, τοσαυταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων. ὅπερ ἔδει δεῖξαι.

For since AB, CD are equal multiples of E, F , thus as many magnitudes as (there) are in AB equal to E , so many (are there) also in CD equal to F . Let AB be divided into magnitudes AG, GB , equal to E , and CD into (magnitudes) CH, HD , equal to F . So, the number of (divisions) AG, GB will be equal to the number of (divisions) CH, HD . And since AG is equal to E , and CH to F , AG (is) thus equal to E , and AG, CH to E, F . So, for the same (reasons), GB is equal to E , and GB, HD to E, F . Thus, as many (magnitudes) as (there) are in AB equal to E , so many (are there) also in AB, CD equal to E, F . Thus, as many times as AB is (divisible) by E , so many times will AB, CD also be (divisible) by E, F .

Thus, if there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\alpha + m\beta + \dots = m(\alpha + \beta + \dots)$.

2.

Ἐάν πρῶτον δευτέρον ἰσάκις ἢ πολλαπλάσιον καὶ τρίτον τετάρτον, ἢ δὲ καὶ πέμπτον δευτέρον ἰσάκις πολλαπλάσιον καὶ ἕκτον τετάρτον, καὶ συντεθὲν πρῶτον καὶ πέμπτον δευτέρον ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τετάρτον.

Πρῶτον γὰρ τὸ AB δευτέρον τοῦ Γ ἰσάκις ἔστω πολλαπλάσιον καὶ τρίτον τὸ ΔE τετάρτον τοῦ Z , ἔστω δὲ καὶ πέμπτον τὸ BH δευτέρον τοῦ Γ ἰσάκις πολλαπλάσιον καὶ ἕκτον τὸ $E\Theta$ τετάρτον τοῦ Z . λέγω, ὅτι καὶ συντεθὲν πρῶτον καὶ πέμπτον τὸ AH δευτέρον τοῦ Γ ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τὸ $\Delta\Theta$ τετάρτον τοῦ Z .

Ἐπεὶ γὰρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AB τοῦ Γ καὶ τὸ ΔE τοῦ Z , ὅσα ἄρα ἐστὶν ἐν τῷ AB ἴσα τῷ Γ , τοσαῦτα καὶ ἐν τῷ ΔE ἴσα τῷ Z . διὰ τὰ αὐτὰ δὴ καὶ ὅσα ἐστὶν ἐν τῷ BH ἴσα τῷ Γ , τοσαῦτα καὶ ἐν τῷ $E\Theta$ ἴσα τῷ Z . ὅσα ἄρα ἐστὶν ἐν ὅλῳ τῷ AH ἴσα τῷ Γ , τοσαῦτα καὶ ἐν ὅλῳ τῷ $\Delta\Theta$ ἴσα τῷ Z . ὁσαυτάσιον ἄρα ἐστὶ τὸ AH τοῦ Γ , τοσαυταπλάσιον ἔσται καὶ τὸ $\Delta\Theta$ τοῦ Z . καὶ συντεθὲν ἄρα πρῶτον καὶ πέμπτον τὸ AH δευτέρον τοῦ Γ ἰσάκις ἔσται πολλαπλάσιον καὶ

Proposition 2†

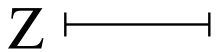
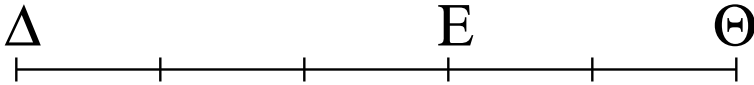
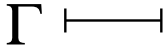
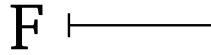
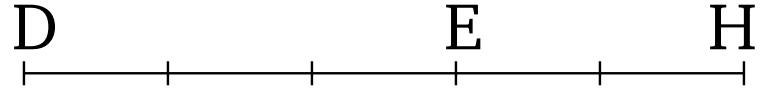
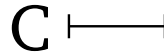
If a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and a fifth (magnitude) and a sixth (are) also equal multiples of the second and fourth (respectively), (then) the first (magnitude) and the fifth, being added together, and the third and the sixth, (being added together), will also be equal multiples of the second (magnitude) and the fourth (respectively).

For let a first (magnitude) AB and a third DE be equal multiples of a second C and a fourth F (respectively). And let a fifth (magnitude) BG and a sixth EH also be (other) equal multiples of the second C and the fourth F (respectively). I say that the first (magnitude) and the fifth, being added together, (to give) AG , and the third (magnitude) and the sixth, (being added together, to give) DH , will also be equal multiples of the second (magnitude) C and the fourth F (respectively).

For since AB and DE are equal multiples of C and F (respectively), thus as many (magnitudes) as (there) are in AB equal to C , so many (are there) also in DE equal to F . And so, for the same (reasons), as many (magnitudes) as (there) are

τρίτον καὶ ἕκτον τὸ ΔΘ τετάρτον τοῦ Ζ.

in BG equal to C , so many (are there) also in EH equal to F . Thus, as many (magnitudes) as (there) are in the whole of AG equal to C , so many (are there) also in the whole of DH equal to F . Thus, as many times as AG is (divisible) by C , so many times will DH also be divisible by F . Thus, the first (magnitude) and the fifth, being added together, (to give) AG , and the third (magnitude) and the sixth, (being added together, to give) DH , will also be equal multiples of the second (magnitude) C and the fourth F (respectively).



Ἐάν ἄρα πρῶτον δευτέρου ἰσάκις ἢ πολλαπλάσιον καὶ τρίτον τετάρτου, ἢ δὲ καὶ πέμπτον δευτέρου ἰσάκις πολλαπλάσιον καὶ ἕκτον τετάρτου, καὶ συντεθέν πρῶτον καὶ πέμπτον δευτέρου ἰσάκις ἔσται πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τετάρτου· ὅπερ ἔδει δεῖξαι.

Thus, if a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and a fifth (magnitude) and a sixth (are) also equal multiples of the second and fourth (respectively), (then) the first (magnitude) and the fifth, being added together, and the third and sixth, (being added together), will also be equal multiples of the second (magnitude) and the fourth (respectively). (Which is) the very thing it was required to show.

† In modern notation, this proposition reads $m\alpha + n\alpha = (m+n)\alpha$.

Ἐάν πρῶτον δευτέρου ἰσάκις ἢ πολλαπλάσιον καὶ τρίτον τετάρτου, ληφθῇ δὲ ἰσάκις πολλαπλάσια τοῦ τε πρώτου καὶ

If a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and equal multiples are

τρίτον, καὶ δι' ἴσον τῶν ληφθέντων ἑκάτερον ἑκατέρου ἰσάκις ἔσται πολλαπλάσιον τὸ μὲν τοῦ δευτέρου τὸ δὲ τοῦ τετάρτου.

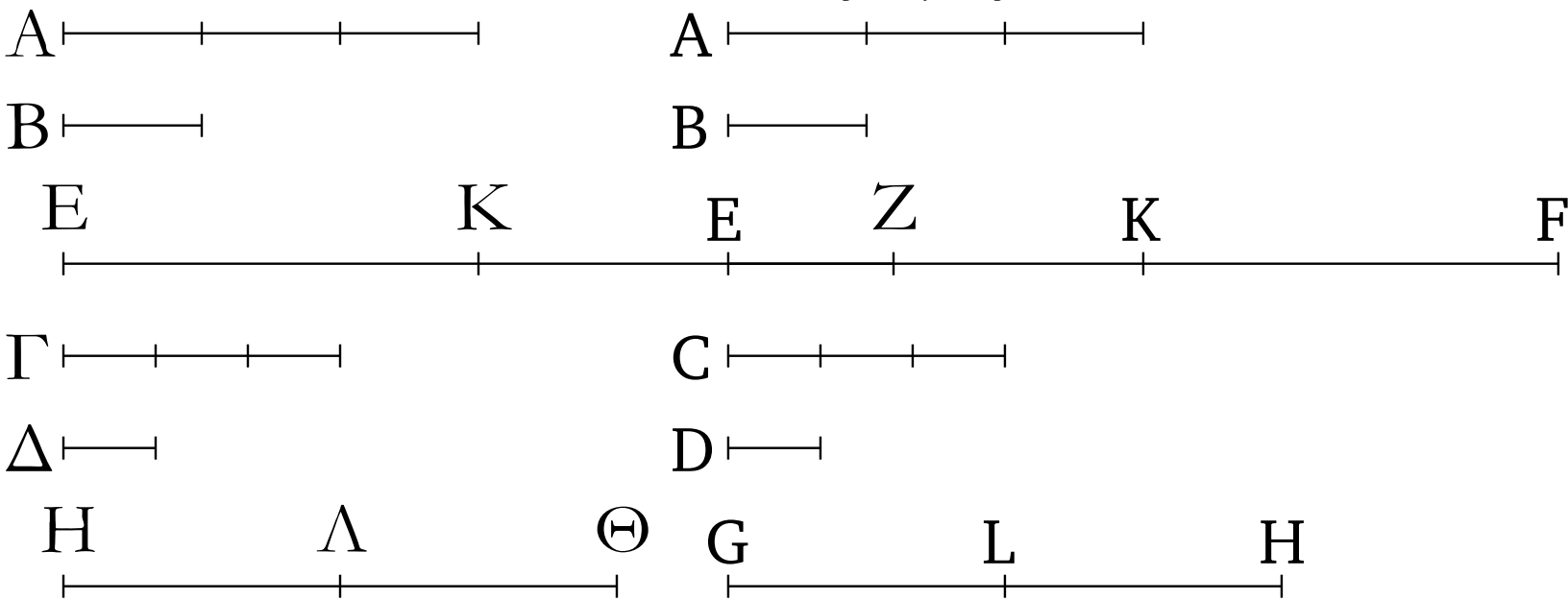
Πρῶτον γάρ τὸ A δευτέρου τοῦ B ἰσάκις ἔστω πολλαπλάσιον καὶ τρίτον τὸ Γ τετάρτου τοῦ Δ , καὶ εἰλήφθω τῶν A , Γ ἰσάκις πολλαπλάσια τὰ EZ , $H\Theta$. λέγω, ὅτι ἰσάκις ἔστι πολλαπλάσιον τὸ EZ τοῦ B καὶ τὸ $H\Theta$ τοῦ Δ .

Ἐπεὶ γὰρ ἰσάκις ἔστι πολλαπλάσιον τὸ EZ τοῦ A καὶ τὸ $H\Theta$ τοῦ Γ , ὅσα ἄρα ἔστιν ἐν τῷ EZ ἴσα τῷ A , τοσαῦτα καὶ ἐν τῷ $H\Theta$ ἴσα τῷ Γ . διηρῆσθω τὸ μὲν EZ εἰς τὰ τῷ A μεγέθη ἴσα τὰ EK , KZ , τὸ δὲ $H\Theta$ εἰς τὰ τῷ Γ ἴσα τὰ $ΗΛ$, $\Lambda\Theta$. ἔσται δὴ ἴσον τὸ πλῆθος τῶν EK , KZ τῷ πλῆθει τῶν $ΗΛ$, $\Lambda\Theta$. καὶ ἐπεὶ ἰσάκις ἔστι πολλαπλάσιον τὸ A τοῦ B καὶ τὸ Γ τοῦ Δ , ἴσον δὲ τὸ μὲν EK τῷ A , τὸ δὲ $ΗΛ$ τῷ Γ , ἰσάκις ἄρα ἔστι πολλαπλάσιον τὸ EK τοῦ B καὶ τὸ $ΗΛ$ τοῦ Δ . διὰ τὰ αὐτὰ δὴ ἰσάκις ἔστι πολλαπλάσιον τὸ KZ τοῦ B καὶ τὸ $\Lambda\Theta$ τοῦ Δ . ἐπεὶ οὖν πρῶτον τὸ EK δευτέρου τοῦ B ἰσάκις ἔστι πολλαπλάσιον καὶ τρίτον τὸ $ΗΛ$ τετάρτου τοῦ Δ , ἔστι δὲ καὶ πέμπτον τὸ KZ δευτέρου τοῦ B ἰσάκις πολλαπλάσιον καὶ ἕκτον τὸ $\Lambda\Theta$ τετάρτου τοῦ Δ , καὶ συντεθέν ἄρα πρῶτον καὶ πέμπτον τὸ EZ δευτέρου τοῦ B ἰσάκις ἔστι πολλαπλάσιον καὶ τρίτον καὶ ἕκτον τὸ $H\Theta$ τετάρτου τοῦ Δ .

taken of the first and the third, (then), via equality, the (magnitudes) taken will also be equal multiples of the second (magnitude) and the fourth, respectively.

For let a first (magnitude) A and a third C be equal multiples of a second B and a fourth D (respectively), and let the equal multiples EF and GH be taken of A and C (respectively). I say that EF and GH are equal multiples of B and D (respectively).

For since EF and GH are equal multiples of A and C (respectively), thus as many (magnitudes) as (there) are in EF equal to A , so many (are there) also in GH equal to C . Let EF be divided into magnitudes EK , KF equal to A , and GH into (magnitudes) GL , LH equal to C . So, the number of (magnitudes) EK , KF will be equal to the number of (magnitudes) GL , LH . And since A and C are equal multiples of B and D (respectively), and EK (is) equal to A , and GL to C , EK and GL are thus equal multiples of B and D (respectively). So, for the same (reasons), KF and LH are equal multiples of B and D (respectively). Therefore, since the first (magnitude) EK and the third GL are equal multiples of the second B and the fourth D (respectively), and the fifth (magnitude) KF and the sixth LH are also equal multiples of the second B and the fourth D (respectively), then the first (magnitude) and fifth, being added together, (to give) EF , and the third (magnitude) and sixth, (being added together, to give) GH , are thus also equal multiples of the second (magnitude) B and the fourth D (respectively) [Prop. 5.2].



Ἐὰν ἄρα πρῶτον δευτέρου ἰσάκις ᾗ πολλαπλάσιον καὶ τρίτον τετάρτου, ληφθῇ δὲ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια, καὶ δι' ἴσον τῶν ληφθέντων ἑκάτερον ἑκατέρου ἰσάκις ἔσται πολλαπλάσιον τὸ μὲν τοῦ δευτέρου τὸ δὲ τοῦ

Thus, if a first (magnitude) and a third are equal multiples of a second and a fourth (respectively), and equal multiples are taken of the first and the third, (then), via equality, the (magnitudes) taken will also be equal multiples of the second

τετάρτον· ὅπερ ἔδει δεῖχαι.

(magnitude) and the fourth, respectively. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads $m(n\alpha) = (mn)\alpha$.

4.

Proposition 4[†]

Ἐάν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, καὶ τὰ ἰσάκις πολλαπλάσια τοῦ τε πρῶτου καὶ τρίτου πρὸς τὰ ἰσάκις πολλαπλάσια τοῦ δευτέρου καὶ τετάρτου καθ' ὁποιοῦν πολλαπλασιασμόν τὸν αὐτὸν ἔχει λόγον ληφθέντα κατάλληλα.

Πρῶτον γάρ τὸ A πρὸς δεύτερον τὸ B τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ , καὶ εἰλήφθω τῶν μὲν A, Γ ἰσάκις πολλαπλάσια τὰ E, Z , τῶν δὲ B, Δ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ H, Θ . λέγω, ὅτι ἐστὶν ὥς τὸ E πρὸς τὸ H , οὕτως τὸ Z πρὸς τὸ Θ .

Εἰλήφθω γάρ τῶν μὲν E, Z ἰσάκις πολλαπλάσια τὰ K, Λ , τῶν δὲ H, Θ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ M, N .

[Καί] ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ μὲν E τοῦ A , τὸ δὲ Z τοῦ Γ , καὶ εἴληπται τῶν E, Z ἰσάκις πολλαπλάσια τὰ K, Λ , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ K τοῦ A καὶ τὸ Λ τοῦ Γ . διὰ τὰ αὐτὰ δὴ ἰσάκις ἐστὶ πολλαπλάσιον τὸ M τοῦ B καὶ τὸ N τοῦ Δ . καὶ ἐπεὶ ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , καὶ εἴληπται τῶν μὲν A, Γ ἰσάκις πολλαπλάσια τὰ K, Λ , τῶν δὲ B, Δ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ M, N , εἰ ἄρα ὑπερέχει τὸ K τοῦ M , ὑπερέχει καὶ τὸ Λ τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν K, Λ τῶν E, Z ἰσάκις πολλαπλάσια, τὰ δὲ M, N τῶν H, Θ ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια· ἐστὶν ἄρα ὥς τὸ E πρὸς τὸ H , οὕτως τὸ Z πρὸς τὸ Θ .

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, (then) equal multiples of the first (magnitude) and the third will also have the same ratio to equal multiples of the second and the fourth, being taken in corresponding order, according to any kind of multiplication whatsoever.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D . And let equal multiples E and F be taken of A and C (respectively), and other random equal multiples G and H of B and D (respectively). I say that as E (is) to G , so F (is) to H .

For let equal multiples K and L be taken of E and F (respectively), and other random equal multiples M and N of G and H (respectively).

[And] since E and F are equal multiples of A and C (respectively), and the equal multiples K and L have been taken of E and F (respectively), K and L are thus equal multiples of A and C (respectively) [Prop. 5.3]. So, for the same (reasons), M and N are equal multiples of B and D (respectively). And since as A is to B , so C (is) to D , and the equal multiples K and L have been taken of A and C (respectively), and the other random equal multiples M and N of B and D (respectively), (then) if K exceeds M (then) L also exceeds N , and if (K is) equal (to M then L is also) equal (to N), and if (K is) less (than M then L is also) less (than N) [Def. 5.5]. And K and L are equal multiples of E and F (respectively), and M and N other random equal multiples of G and H (respectively). Thus, as E (is) to G , so F (is) to H [Def. 5.5].

A

B

E

H

K

M

Γ

Δ

Z

Θ

Λ

N

A

B

E

G

K

M

C

D

F

H

L

N

Ἐάν ᾗρα πρῶτον πρόξ δεύτερον τὸν αὐτὸν ἔθῃ λόγον
καὶ τρίτον πρόξ τέταρτον, καὶ τὰ ἰσάκις πολλαπλάσια τοῦ τε
πρώτου καὶ τρίτου πρόξ τὰ ἰσάκις πολλαπλάσια τοῦ δευτέρου
καὶ τετάρτου τὸν αὐτὸν ἔχει λόγον καθ' ὅποιον οὖν πολλαπλα-
σιασμὸν ληφθέντα κατάλληλα· ὅπερ ἔδει δεῖξαι.

Thus, if a first (magnitude) has the same ratio to a sec-
ond that a third (has) to a fourth, (then) equal multiples of the
first (magnitude) and the third will also have the same ratio to
equal multiples of the second and the fourth, being taken in
corresponding order, according to any kind of multiplication
whatsoever. (Which is) the very thing it was required to show.

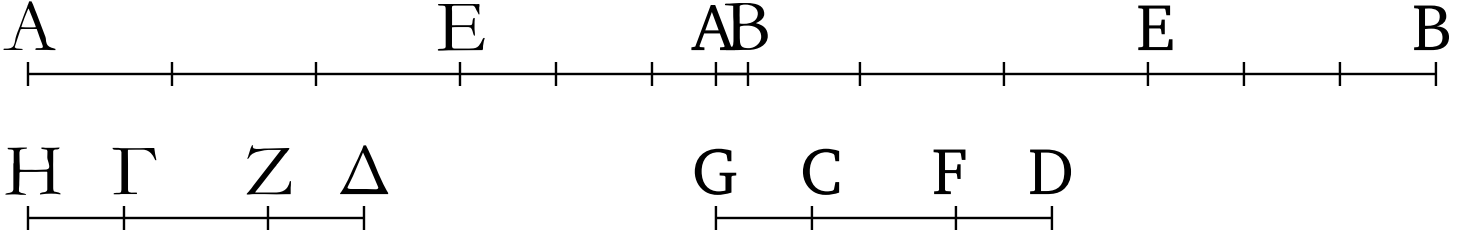
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $m\alpha : n\beta :: m\gamma : n\delta$, for all m and n .

5.

Proposition 5[†]

Ἐάν μέγεθος μέγεθος ἰσάκις ᾗ πολλαπλάσιον, ὅπερ
ἀφαιρεθὲν ἀφαιρεθέντος, καὶ τὸ λοιπὸν τοῦ λοιποῦ ἰσάκις
ἔσται πολλαπλάσιον, ὅσαπλάσιόν ἐστι τὸ ὅλον τοῦ ὅλου.

If a magnitude is the same multiple of a magnitude that
a (part) taken away (is) of a (part) taken away (respectively),
(then) the remainder will also be the same multiple of the re-
mainder as that which the whole (is) of the whole (respec-
tively).



Μέγεθος γάρ τὸ AB μεγέθους τοῦ $\Gamma\Delta$ ἰσάκις ἔστω πολλαπλάσιον, ὅπερ ἀφαιρεθέν τὸ AE ἀφαιρεθέντος τοῦ ΓZ λέγω, ὅτι καὶ λοιπὸν τὸ EB λοιποῦ τοῦ $Z\Delta$ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ὅλον τὸ AB ὅλον τοῦ $\Gamma\Delta$.

Ὅσαπλάσιον γάρ ἐστι τὸ AE τοῦ ΓZ , τοσαυταπλάσιον γενοῦντο καὶ τὸ EB τοῦ ΓH .

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ EB τοῦ ΓH , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ AB τοῦ HZ . κεῖται δὲ ἰσάκις πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ AB τοῦ $\Gamma\Delta$. ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ AB ἑκατέρων τῶν HZ , $\Gamma\Delta$. ἴσον ἄρα τὸ HZ τῷ $\Gamma\Delta$. κοινὸν ἀφηρήσθω τὸ ΓZ . λοιπὸν ἄρα τὸ $H\Gamma$ λοιπῷ τῷ $Z\Delta$ ἴσον ἐστίν. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ EB τοῦ ΓH , ἴσον δὲ τὸ $H\Gamma$ τῷ $Z\Delta$, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ EB τοῦ $Z\Delta$. ἰσάκις δὲ ὑπόκειται πολλαπλάσιον τὸ AE τοῦ ΓZ καὶ τὸ AB τοῦ $\Gamma\Delta$. ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ EB τοῦ $Z\Delta$ καὶ τὸ AB τοῦ $\Gamma\Delta$. καὶ λοιπὸν ἄρα τὸ EB λοιποῦ τοῦ $Z\Delta$ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ὅλον τὸ AB ὅλον τοῦ $\Gamma\Delta$.

Ἐὰν ἄρα μέγεθος μεγέθους ἰσάκις ἢ πολλαπλάσιον, ὅπερ ἀφαιρεθέν ἀφαιρεθέντος, καὶ τὸ λοιπὸν τοῦ λοιποῦ ἰσάκις ἔσται πολλαπλάσιον, ὁσαπλάσιόν ἐστι καὶ τὸ ὅλον τοῦ ὅλου· ὅπερ ἔδει δεῖξαι.

For let the magnitude AB be the same multiple of the magnitude CD that the (part) taken away AE (is) of the (part) taken away CF (respectively). I say that the remainder EB will also be the same multiple of the remainder FD as that which the whole AB (is) of the whole CD (respectively).

For as many times as AE is (divisible) by CF , so many times let EB also be made (divisible) by CG .

And since AE and EB are equal multiples of CF and GC (respectively), AE and AB are thus equal multiples of CF and GF (respectively) [Prop. 5.1]. And AE and AB are assumed (to be) equal multiples of CF and CD (respectively). Thus, AB is an equal multiple of each of GF and CD . Thus, GF (is) equal to CD . Let CF be subtracted from both. Thus, the remainder GC is equal to the remainder FD . And since AE and EB are equal multiples of CF and GC (respectively), and GC (is) equal to DF , AE and EB are thus equal multiples of CF and FD (respectively). And AE and AB are assumed (to be) equal multiples of CF and CD (respectively). Thus, EB and AB are equal multiples of FD and CD (respectively). Thus, the remainder EB will also be the same multiple of the remainder FD as that which the whole AB (is) of the whole CD (respectively).

Thus, if a magnitude is the same multiple of a magnitude that a (part) taken away (is) of a (part) taken away (respectively), (then) the remainder will also be the same multiple of the remainder as that which the whole (is) of the whole (respectively). (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads $m\alpha - m\beta = m(\alpha - \beta)$.

6.

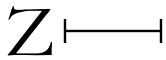
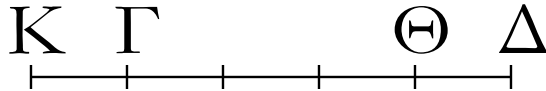
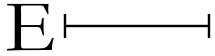
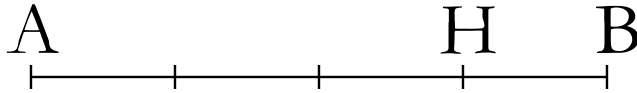
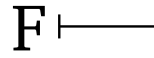
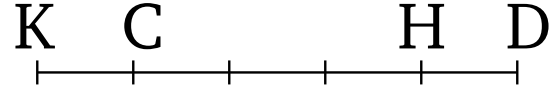
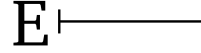
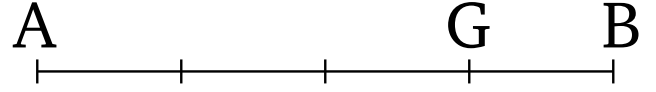
Ἐὰν δύο μεγέθη δύο μεγεθῶν ἰσάκις ἢ πολλαπλάσια, καὶ ἀφαιρεθέντα τινὰ τῶν αὐτῶν ἰσάκις ἢ πολλαπλάσια, καὶ τὰ λοιπὰ τοῖς αὐτοῖς ἴησι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια.

Δύο γάρ μεγέθη τὰ AB , $\Gamma\Delta$ δύο μεγεθῶν τῶν E , Z ἰσάκις ἔστω πολλαπλάσια, καὶ ἀφαιρεθέντα τὰ AH , $\Gamma\Theta$ τῶν αὐτῶν τῶν E , Z ἰσάκις ἔστω πολλαπλάσια· λέγω, ὅτι καὶ λοιπὰ τὰ HB , $\Theta\Delta$ τοῖς E , Z ἴησι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια.

Proposition 6[†]

If two magnitudes are equal multiples of two (other) magnitudes, and some (parts) taken away (from the former magnitudes) are equal multiples of the latter (magnitudes, respectively), (then) the remainders are also either equal to the latter (magnitudes), or (are) equal multiples of them (respectively).

For let two magnitudes AB and CD be equal multiples of two magnitudes E and F (respectively). And let the (parts) taken away (from the former) AG and CH be equal multiples of E and F (respectively). I say that the remainders GB and HD are also either equal to E and F (respectively), or (are) equal multiples of them.



Ἐστω γάρ πρότερον τὸ HB τῷ E ἴσον· λέγω, ὅτι καὶ τὸ $\Theta\Delta$ τῷ Z ἴσον ἐστίν.

Κείσθω γάρ τῷ Z ἴσον τὸ ΓK . ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AH τοῦ E καὶ τὸ $\Gamma\Theta$ τοῦ Z , ἴσον δὲ τὸ μὲν HB τῷ E , τὸ δὲ $K\Gamma$ τῷ Z , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ AB τοῦ E καὶ τὸ $K\Theta$ τοῦ Z . ἰσάκις δὲ ὑπόκειται πολλαπλάσιον τὸ AB τοῦ E καὶ τὸ $\Gamma\Delta$ τοῦ Z . ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ $K\Theta$ τοῦ Z καὶ τὸ $\Gamma\Delta$ τοῦ Z . ἐπεὶ οὖν ἐκάτερον τῶν $K\Theta$, $\Gamma\Delta$ τοῦ Z ἰσάκις ἐστὶ πολλαπλάσιον, ἴσον ἄρα ἐστὶ τὸ $K\Theta$ τῷ $\Gamma\Delta$. κοινὸν ἀφηρήσθω τὸ $\Gamma\Theta$. λοιπὸν ἄρα τὸ $K\Gamma$ λοιπὸν τῷ $\Theta\Delta$ ἴσον ἐστίν. ἀλλὰ τὸ Z τῷ $K\Gamma$ ἴσον· καὶ τὸ $\Theta\Delta$ ἄρα τῷ Z ἴσον ἐστίν. ὥστε εἰ τὸ HB τῷ E ἴσον ἐστίν, καὶ τὸ $\Theta\Delta$ ἴσον ἔσται τῷ Z .

Ὁμοίως δὴ δείξομεν, ὅτι, κ' ἂν πολλαπλάσιον ᾖ τὸ HB τοῦ E , τοσαυταπλάσιον ἔσται καὶ τὸ $\Theta\Delta$ τοῦ Z .

Ἐὰν ἄρα δύο μεγέθη δύο μεγεθῶν ἰσάκις ᾖ πολλαπλάσια, καὶ ἀφαιρεθέντα τινὰ τῶν αὐτῶν ἰσάκις ᾖ πολλαπλάσια, καὶ

For let GB be, first of all, equal to E . I say that HD is also equal to F .

For let CK be made equal to F . Since AG and CH are equal multiples of E and F (respectively), and GB (is) equal to E , and KC to F , AB and KH are thus equal multiples of E and F (respectively) [Prop. 5.2]. And AB and CD are assumed (to be) equal multiples of E and F (respectively). Thus, KH and CD are equal multiples of F and F (respectively). Therefore, KH and CD are each equal multiples of F . Thus, KH is equal to CD . Let CH be taken away from both. Thus, the remainder KC is equal to the remainder HD . But, F is equal to KC . Thus, HD is also equal to F . Hence, if GB is equal to E , (then) HD will also be equal to F .

So, similarly, we can show that even if GB is a multiple of E , (then) HD will also be the same multiple of F .

Thus, if two magnitudes are equal multiples of two (other) magnitudes, and some (parts) taken away (from the former magnitudes) are equal multiples of the latter (magnitudes, re-

τὰ λοιπὰ τοῖς αὐτοῖς ἦτοι ἴσα ἐστὶν ἢ ἰσάκις αὐτῶν πολλαπλάσια· ὅπερ ἔδει δεῖξαι.

spectively), (then) the remainders are also either equal to the latter (magnitudes), or (are) equal multiples of them (respectively). (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads $m\alpha - n\alpha = (m - n)\alpha$.

7.

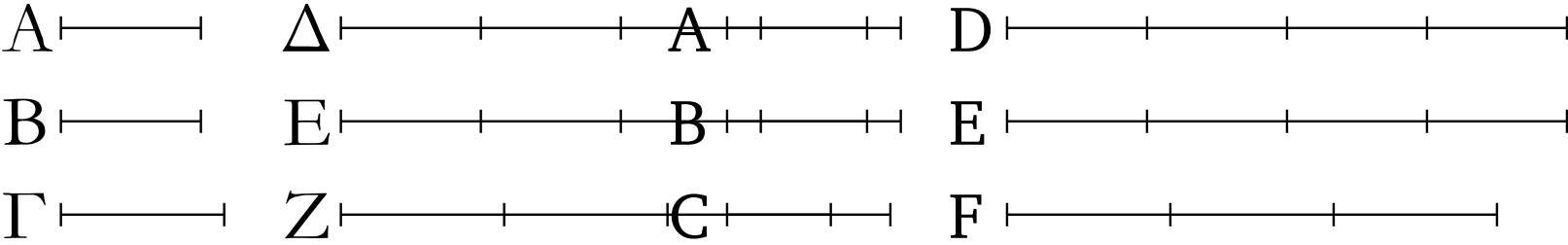
Proposition 7

Τὰ ἴσα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον καὶ τὸ αὐτὸ πρὸς τὰ ἴσα.

Ἐστω ἴσα μεγέθη τὰ A, B , ἄλλο δέ τι, ὃ ἔτυθεν, μέγεθος τὸ Γ · λέγω, ὅτι ἐκάτερον τῶν A, B πρὸς τὸ Γ τὸν αὐτὸν ἔθει λόγον, καὶ τὸ Γ πρὸς ἐκάτερον τῶν A, B .

Equal (magnitudes) have the same ratio to the same (magnitude), and the latter (magnitude has the same ratio) to the equal (magnitudes).

Let A and B be equal magnitudes, and C some other random magnitude. I say that A and B each have the same ratio to C , and (that) C (has the same ratio) to each of A and B .



Εἰλήφθω γὰρ τῶν μὲν A, B ἰσάκις πολλαπλάσια τὰ Δ, E , τοῦ δὲ Γ ἄλλο, ὃ ἔτυθεν, πολλαπλάσιον τὸ Z .

Ἐπεὶ οὖν ἰσάκις ἐστὶ πολλαπλάσιον τὸ Δ τοῦ A καὶ τὸ E τοῦ B , ἴσον δὲ τὸ A τῷ B , ἴσον ἄρα καὶ τὸ Δ τῷ E . ἄλλο δέ, ὃ ἔτυθεν, τὸ Z . εἰ ἄρα ὑπερέθει τὸ Δ τοῦ Z , ὑπερέθει καὶ τὸ E τοῦ Z , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν Δ, E τῶν A, B ἰσάκις πολλαπλάσια, τὸ δὲ Z τοῦ Γ ἄλλο, ὃ ἔτυθεν, πολλαπλάσιον· ἐστὶν ἄρα ὡς τὸ A πρὸς τὸ Γ , οὕτως τὸ B πρὸς τὸ Γ .

Λέγω [δή], ὅτι καὶ τὸ E πρὸς ἐκάτερον τῶν A, B τὸν αὐτὸν ἔθει λόγον.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἴσον ἐστὶ τὸ Δ τῷ E · ἄλλο δέ τι τὸ Z · εἰ ἄρα ὑπερέθει τὸ Z τοῦ Δ , ὑπερέθει καὶ τοῦ E , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὸ μὲν Z τοῦ Γ πολλαπλάσιον, τὰ δὲ Δ, E τῶν A, B ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια· ἐστὶν ἄρα ὡς τὸ Γ πρὸς τὸ A , οὕτως τὸ Γ πρὸς τὸ B .

Τὰ ἴσα ἄρα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον καὶ τὸ αὐτὸ πρὸς τὰ ἴσα.

For let the equal multiples D and E be taken of A and B (respectively), and the other random multiple F of C .

Therefore, since D and E are equal multiples of A and B (respectively), and A (is) equal to B , D (is) thus also equal to E . And F (is) different, at random. Thus, if D exceeds F (then) E also exceeds F , and if (D is) equal (to F then E is also) equal (to F), and if (D is) less (than F then E is also) less (than F). And D and E are equal multiples of A and B (respectively), and F another random multiple of C . Thus, as A (is) to C , so B (is) to C [Def. 5.5].

[So] I say that C [†] also has the same ratio to each of A and B .

For, similarly, we can show, by the same construction, that D is equal to E . And F (has) some other (value). Thus, if F exceeds D then it also exceeds E , and if (F is) equal (to D then it is also) equal (to E), and if (F is) less (than D then it is also) less (than E). And F is a multiple of C , and D and E other random equal multiples of A and B . Thus, as C (is) to A , so C (is) to B [Def. 5.5].

Thus, equal (magnitudes) have the same ratio to the same (magnitude), and the latter (magnitude has the same ratio) to the equal (magnitudes).

Πόρισμα.

Corollary[‡]

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν μεγέθη τινὰ ἀνάλογον ᾗ, καὶ ἀνάπαλιν ἀνάλογον ᾖ. ὅπερ ἔδει δεῖξαι.

So (it is) clear, from this, that if some magnitudes are proportional, (then) they will also be proportional inversely. (Which is) the very thing it was required to show.

[†] The Greek text has “ E ”, which is obviously a mistake.

‡ In modern notation, this corollary reads that if $\alpha : \beta :: \gamma : \delta$ then $\beta : \alpha :: \delta : \gamma$.

8.

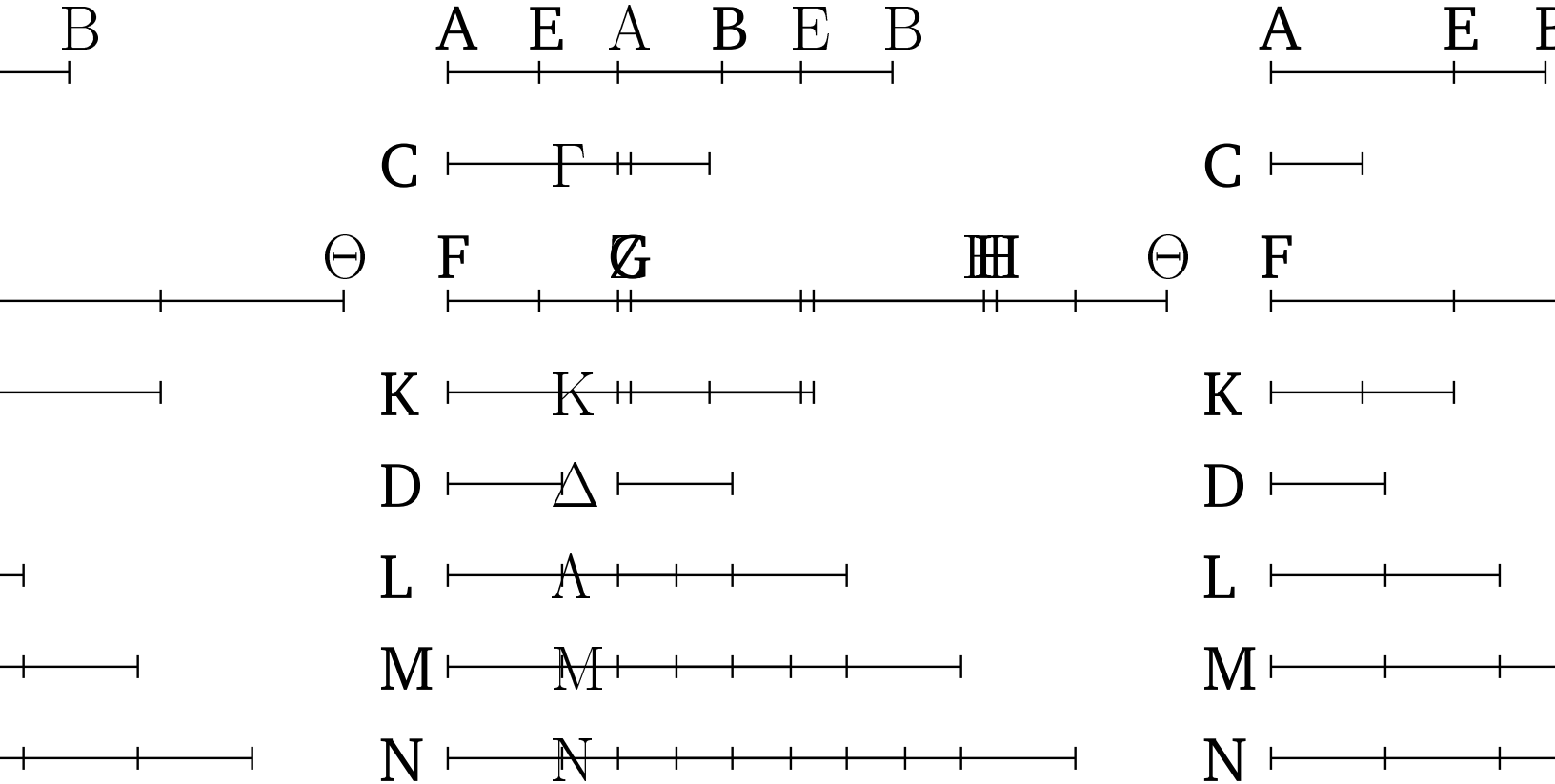
Proposition 8

Τῶν ἀνίσων μεγεθῶν τὸ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔθει ἥπερ τὸ ἔλαττον. καὶ τὸ αὐτὸ πρὸς τὸ ἔλαττον μείζονα λόγον ἔθει ἥπερ πρὸς τὸ μείζον.

Ἐστω ἄνισα μεγέθη τὰ AB , Γ , καὶ ἔστω μείζον τὸ AB , ἄλλο δέ, ὃ ἔτυθεν, τὸ Δ . λέγω, ὅτι τὸ AB πρὸς τὸ Δ μείζονα λόγον ἔθει ἥπερ τὸ Γ πρὸς τὸ Δ , καὶ τὸ Δ πρὸς τὸ Γ μείζονα λόγον ἔθει ἥπερ πρὸς τὸ AB .

For unequal magnitudes, the greater (magnitude) has a greater ratio than the lesser to the same (magnitude). And the latter (magnitude) has a greater ratio to the lesser (magnitude) than to the greater.

Let AB and C be unequal magnitudes, and let AB be the greater (of the two), and D another random magnitude. I say that AB has a greater ratio to D than C (has) to D , and (that) D has a greater ratio to C than (it has) to AB .



Ἐπεὶ γὰρ μείζον ἐστὶ τὸ AB τοῦ Γ , κείσθω τῷ Γ ἴσον τὸ BE . τὸ δὲ ἔλασσον τῶν AE , EB πολλαπλασιαζόμενον ἔσται ποτὲ τοῦ Δ μείζον. ἔστω πρότερον τὸ AE ἔλαττον τοῦ EB , καὶ πεπολλαπλασιάσθω τὸ AE , καὶ ἔστω αὐτοῦ πολλαπλάσιον τὸ ZH μείζον ὃν τοῦ Δ , καὶ ὁσαπλάσιόν ἐστι τὸ ZH τοῦ AE , τοσαυταπλάσιον γερονέτω καὶ τὸ μὲν $H\Theta$ τοῦ EB τὸ δὲ K τοῦ Γ . καὶ εἰλήφθω τοῦ Δ διπλάσιον μὲν τὸ Λ , τριπλάσιον δὲ τὸ M , καὶ ἐχῆς ἐνὶ πλεον, ἕως ἄν τὸ λαμβανόμενον πολλαπλάσιον μὲν γένηται τοῦ Δ , πρῶτως δὲ μείζον τοῦ K . εἰλήφθω, καὶ ἔστω τὸ N τετραπλάσιον μὲν τοῦ Δ , πρῶτως δὲ μείζον τοῦ K .

Ἐπεὶ οὖν τὸ K τοῦ N πρῶτως ἐστὶν ἔλαττον, τὸ K ἄρα τοῦ M οὐκ ἐστὶν ἔλαττον. καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον

For since AB is greater than C , let BE be made equal to C . So, the lesser of AE and EB , being multiplied, will sometimes be greater than D [Def. 5.4]. First of all, let AE be less than EB , and let AE be multiplied, and let FG be a multiple of it which (is) greater than D . And as many times as FG is (divisible) by AE , so many times let GH also become (divisible) by EB , and K by C . And let the double multiple L of D be taken, and the triple multiple M , and several more, (each increasing) in order by one, until the (multiple) taken becomes the first multiple of D (which is) greater than K . Let it be taken, and let it also be the quadruple multiple N of D —the first (multiple) greater than K .

Therefore, since K is less than N first, K is thus not less

τὸ ZH τοῦ AE καὶ τὸ $H\Theta$ τοῦ EB , ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ZH τοῦ AE καὶ τὸ $Z\Theta$ τοῦ AB . ἰσάκις δὲ ἐστὶ πολλαπλάσιον τὸ ZH τοῦ AE καὶ τὸ K τοῦ Γ . ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ $Z\Theta$ τοῦ AB καὶ τὸ K τοῦ Γ . τὰ $Z\Theta$, K ἄρα τῶν AB , Γ ἰσάκις ἐστὶ πολλαπλάσια. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ $H\Theta$ τοῦ EB καὶ τὸ K τοῦ Γ , ἴσον δὲ τὸ EB τῷ Γ , ἴσον ἄρα καὶ τὸ $H\Theta$ τῷ K . τὸ δὲ K τοῦ M οὐκ ἐστὶν ἑλάττω· οὐδ' ἄρα τὸ $H\Theta$ τοῦ M ἑλαττόν ἐστιν. μείζον δὲ τὸ ZH τοῦ Δ · ὅλον ἄρα τὸ $Z\Theta$ συναμφοτέρων τῶν Δ , M μείζον ἐστὶν. ἀλλὰ συναμφοτέρα τὰ Δ , M τῷ N ἐστὶν ἴσα, ἐπειδήπερ τὸ M τοῦ Δ τριπλάσιόν ἐστιν, συναμφοτέρα δὲ τὰ M , Δ τοῦ Δ ἐστὶ τετραπλάσια, ἐστὶ δὲ καὶ τὸ N τοῦ Δ τετραπλάσιον· συναμφοτέρα ἄρα τὰ M , Δ τῷ N ἴσα ἐστίν. ἀλλὰ τὸ $Z\Theta$ τῶν M , Δ μείζον ἐστὶν· τὸ $Z\Theta$ ἄρα τοῦ N ὑπερέθκει· τὸ δὲ K τοῦ N οὐθ' ὑπερέθκει. καὶ ἐστὶ τὰ μὲν $Z\Theta$, K τῶν AB , Γ ἰσάκις πολλαπλάσια, τὸ δὲ N τοῦ Δ ἄλλο, ὃ ἐτύθη, πολλαπλάσιον· τὸ AB ἄρα πρὸς τὸ Δ μείζονα λόγον ἔθει ἥπερ τὸ Γ πρὸς τὸ Δ .

Λέγω δὴ, ὅτι καὶ τὸ Δ πρὸς τὸ Γ μείζονα λόγον ἔθει ἥπερ τὸ Δ πρὸς τὸ AB .

Τῶν γάρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι τὸ μὲν N τοῦ K ὑπερέθκει, τὸ δὲ N τοῦ $Z\Theta$ οὐθ' ὑπερέθκει. καὶ ἐστὶ τὸ μὲν N τοῦ Δ πολλαπλάσιον, τὰ δὲ $Z\Theta$, K τῶν AB , Γ ἄλλα, ἃ ἐτύθη, ἰσάκις πολλαπλάσια· τὸ Δ ἄρα πρὸς τὸ Γ μείζονα λόγον ἔθει ἥπερ τὸ Δ πρὸς τὸ AB .

Ἀλλὰ δὴ τὸ AE τοῦ EB μείζον ἐστὼ. τὸ δὴ ἑλάττω τὸ EB πολλαπλασιαζόμενον ἐστὶ ποτὲ τοῦ Δ μείζον. πεπολλαπλασιάσθω, καὶ ἐστὼ τὸ $H\Theta$ πολλαπλάσιον μὲν τοῦ EB , μείζον δὲ τοῦ Δ · καὶ ὁσαπλάσιόν ἐστὶ τὸ $H\Theta$ τοῦ EB , τοσαυταπλάσιον γερονέτω καὶ τὸ μὲν ZH τοῦ AE , τὸ δὲ K τοῦ Γ . ὁμοίως δὴ δείχομεν, ὅτι τὰ $Z\Theta$, K τῶν AB , Γ ἰσάκις ἐστὶ πολλαπλάσια· καὶ εἰλήφθω ὁμοίως τὸ N πολλαπλάσιον μὲν τοῦ Δ , πρῶτως δὲ μείζον τοῦ ZH · ὥστε πάλιν τὸ ZH τοῦ M οὐκ ἐστὶν ἑλασσον. μείζον δὲ τὸ $H\Theta$ τοῦ Δ · ὅλον ἄρα τὸ $Z\Theta$ τῶν Δ , M , τοντέστι τοῦ N , ὑπερέθκει. τὸ δὲ K τοῦ N οὐθ' ὑπερέθκει, ἐπειδήπερ καὶ τὸ ZH μείζον ὅν τοῦ $H\Theta$, τοντέστι τοῦ K , τοῦ N οὐθ' ὑπερέθκει. καὶ ὡσαύτως κατακολουθοῦντες τοῖς ἐπάνω περαινόμεν τὴν ἀπόδειξιν.

Τῶν ἄρα ἀνίσων μεγεθῶν τὸ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔθει ἥπερ τὸ ἑλάττω· καὶ τὸ αὐτὸ πρὸς τὸ ἑλάττω μείζονα λόγον ἔθει ἥπερ πρὸς τὸ μείζον· ὅπερ ἔδει δεῖξαι.

9.

Τὰ πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔχοντα λόγον ἴσα ἀλλήλοις ἐστίν· καὶ πρὸς ἃ τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον, ἐκεῖνα ἴσα ἐστίν.

than M . And since FG and GH are equal multiples of AE and EB (respectively), FG and FH are thus equal multiples of AE and AB (respectively) [Prop. 5.1]. And FG and K are equal multiples of AE and C (respectively). Thus, FH and K are equal multiples of AB and C (respectively). Thus, FH , K are equal multiples of AB , C . Again, since GH and K are equal multiples of EB and C , and EB (is) equal to C , GH (is) thus also equal to K . And K is not less than M . Thus, GH not less than M either. And FG (is) greater than D . Thus, the whole of FH is greater than D and M (added) together. But, D and M (added) together is equal to N , inasmuch as M is three times D , and M and D (added) together is four times D , and N is also four times D . Thus, M and D (added) together is equal to N . But, FH is greater than M and D . Thus, FH exceeds N . And K does not exceed N . And FH , K are equal multiples of AB , C , and N another random multiple of D . Thus, AB has a greater ratio to D than C (has) to D [Def. 5.7].

So, I say that D also has a greater ratio to C than D (has) to AB .

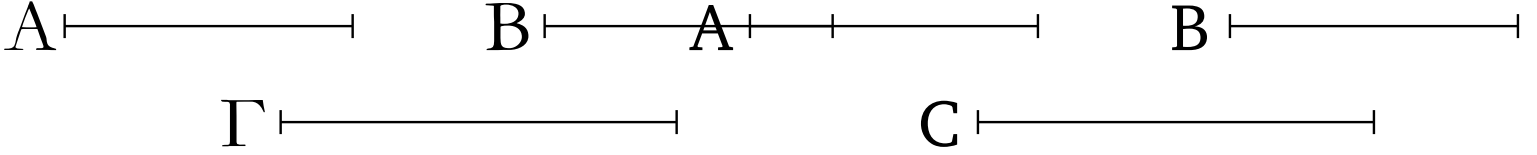
For, similarly, by the same construction, we can show that N exceeds K , and N does not exceed FH . And N is a multiple of D , and FH , K other random equal multiples of AB , C (respectively). Thus, D has a greater ratio to C than D (has) to AB [Def. 5.5].

And so let AE be greater than EB . So, the lesser, EB , being multiplied, will sometimes be greater than D . Let it be multiplied, and let GH be a multiple of EB (which is) greater than D . And as many times as GH is (divisible) by EB , so many times let FG also become (divisible) by AE , and K by C . So, similarly (to the above), we can show that FH and K are equal multiples of AB and C (respectively). And, similarly (to the above), let the multiple N of D , (which is) the first (multiple) greater than FG , be taken. So, FG is again not less than M . And GH (is) greater than D . Thus, the whole of FH exceeds D and M , that is to say N . And K does not exceed N , inasmuch as FG , which (is) greater than GH —that is to say, K —also does not exceed N . And, following the above (arguments), we (can) complete the proof in the same manner.

Thus, for unequal magnitudes, the greater (magnitude) has a greater ratio than the lesser to the same (magnitude). And the latter (magnitude) has a greater ratio to the lesser (magnitude) than to the greater. (Which is) the very thing it was required to show.

Proposition 9

(Magnitudes) having the same ratio to the same (magnitude) are equal to one another. And those (magnitudes) to which the same (magnitude) has the same ratio are equal.



Ἐθέτω γάρ ἐκάτερον τῶν A, B πρὸς τὸ Γ τὸν αὐτὸν λόγον· λέγω, ὅτι ἴσον ἐστὶ τὸ A τῷ B .

Εἰ γὰρ μή, οὐκ ἂν ἐκάτερον τῶν A, B πρὸς τὸ Γ τὸν αὐτὸν εἶθε λόγον· ἔθει δέ· ἴσον ἄρα ἐστὶ τὸ A τῷ B .

Ἐθέτω δὴ πάλιν τὸ Γ πρὸς ἐκάτερον τῶν A, B τὸν αὐτὸν λόγον· λέγω, ὅτι ἴσον ἐστὶ τὸ A τῷ B .

Εἰ γὰρ μή, οὐκ ἂν τὸ Γ πρὸς ἐκάτερον τῶν A, B τὸν αὐτὸν εἶθε λόγον· ἔθει δέ· ἴσον ἄρα ἐστὶ τὸ A τῷ B .

Τὰ ἄρα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθοντα λόγον ἴσα ἀλλήλοις ἐστίν· καὶ πρὸς ἃ τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον, ἐκεῖνα ἴσα ἐστίν· ὅπερ ἔδει δεῖχαι.

10.

Τῶν πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον ἔθον ἐκεῖνο μείζον ἐστίν· πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον ἔθει, ἐκεῖνο ἔλαττόν ἐστιν.

For let A and B each have the same ratio to C . I say that A is equal to B .

For if not, A and B would not each have the same ratio to C [Prop. 5.8]. But they do. Thus, A is equal to B .

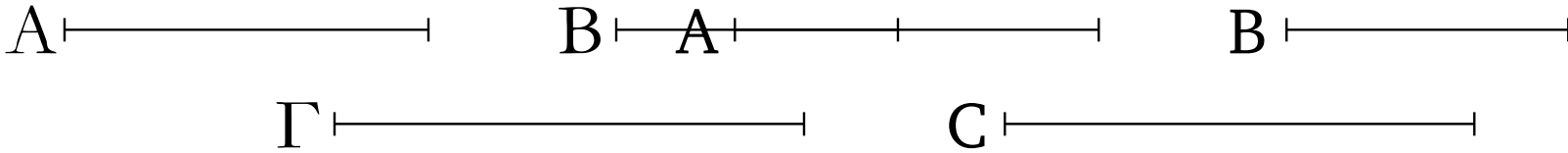
So, again, let C have the same ratio to each of A and B . I say that A is equal to B .

For if not, C would not have the same ratio to each of A and B [Prop. 5.8]. But it does. Thus, A is equal to B .

Thus, (magnitudes) having the same ratio to the same (magnitude) are equal to one another. And those (magnitudes) to which the same (magnitude) has the same ratio are equal. (Which is) the very thing it was required to show.

Proposition 10

For (magnitudes) having a ratio to the same (magnitude), that (magnitude which) has the greater ratio is (the) greater. And that (magnitude) to which the latter (magnitude) has a greater ratio is (the) lesser.



Ἐθέτω γάρ τὸ A πρὸς τὸ Γ μείζονα λόγον ἢπερ τὸ B πρὸς τὸ Γ · λέγω, ὅτι μείζον ἐστὶ τὸ A τοῦ B .

Εἰ γὰρ μή, ἦτοι ἴσον ἐστὶ τὸ A τῷ B ἢ ἔλασσον. ἴσον μὲν οὐκ ἔστι τὸ A τῷ B · ἐκάτερον γὰρ ἂν τῶν A, B πρὸς τὸ Γ τὸν αὐτὸν εἶθε λόγον. οὐκ ἔθει δέ· οὐκ ἄρα ἴσον ἐστὶ τὸ A τῷ B . οὐδὲ μὴν ἔλασσόν ἐστι τὸ A τοῦ B · τὸ A γὰρ ἂν πρὸς τὸ Γ ἐλάσσονα λόγον εἶθεν ἢπερ τὸ B πρὸς τὸ Γ . οὐκ ἔθει δέ· οὐκ ἄρα ἔλασσόν ἐστι τὸ A τοῦ B . ἐδείκθη δὲ οὐδὲ ἴσον μείζον ἄρα ἐστὶ τὸ A τοῦ B .

Ἐθέτω δὴ πάλιν τὸ Γ πρὸς τὸ B μείζονα λόγον ἢπερ τὸ A πρὸς τὸ A · λέγω, ὅτι ἔλασσόν ἐστι τὸ B τοῦ A .

Εἰ γὰρ μή, ἦτοι ἴσον ἐστὶν ἢ μείζον. ἴσον μὲν οὐκ ἔστι τὸ B τῷ A · τὸ Γ γὰρ ἂν πρὸς ἐκάτερον τῶν A, B τὸν αὐτὸν εἶθε λόγον. οὐκ ἔθει δέ· οὐκ ἄρα ἴσον ἐστὶ τὸ A τῷ B . οὐδὲ μὴν μείζον ἐστὶ τὸ B τοῦ A · τὸ Γ γὰρ ἂν πρὸς τὸ B ἐλάσσονα λόγον εἶθεν ἢπερ πρὸς τὸ A . οὐκ ἔθει δέ· οὐκ ἄρα μείζον ἐστὶ τὸ B τοῦ A . ἐδείκθη δέ, ὅτι οὐδὲ ἴσον· ἔλαττον ἄρα ἐστὶ τὸ B τοῦ A .

Τῶν ἄρα πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον ἔθον μείζον ἐστίν· καὶ πρὸς ὃ τὸ αὐτὸ μείζονα λόγον ἔθει, ἐκεῖνο ἔλαττόν ἐστιν· ὅπερ ἔδει δεῖχαι.

For let A have a greater ratio to C than B (has) to C . I say that A is greater than B .

For if not, A is surely either equal to or less than B . In fact, A is not equal to B . For (then) A and B would each have the same ratio to C [Prop. 5.7]. But they do not. Thus, A is not equal to B . Neither, indeed, is A less than B . For (then) A would have a lesser ratio to C than B (has) to C [Prop. 5.8]. But it does not. Thus, A is not less than B . And it was shown not (to be) equal either. Thus, A is greater than B .

So, again, let C have a greater ratio to B than C (has) to A . I say that B is less than A .

For if not, (it is) surely either equal or greater. In fact, B is not equal to A . For (then) C would have the same ratio to each of A and B [Prop. 5.7]. But it does not. Thus, A is not equal to B . Neither, indeed, is B greater than A . For (then) C would have a lesser ratio to B than (it has) to A [Prop. 5.8]. But it does not. Thus, B is not greater than A . And it was shown that (it is) not equal (to A) either. Thus, B is less than A .

Thus, for (magnitudes) having a ratio to the same (magnitude), that (magnitude which) has the greater ratio is (the) greater. And that (magnitude) to which the latter (magnitude) has a greater ratio is (the) lesser. (Which is) the very thing it

was required to show.

11.

Proposition 11[†]

Οἱ τῶ αὐτῶ λόγῳ οἱ αὐτοὶ καὶ ἀλλήλοις εἰσὶν οἱ αὐτοί.

Ἐστωσαν γὰρ ὡς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , ὡς δὲ τὸ Γ πρὸς τὸ Δ , οὕτως τὸ E πρὸς τὸ Z . λέγω, ὅτι ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z .

Εἰλήφθω γὰρ τῶν A, Γ, E ἰσάκις πολλαπλάσια τὰ H, Θ , K , τῶν δὲ B, Δ, Z ἄλλα, ἃ ἐτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, M, N .

Γ ——— A ——— E ——— C ———

Δ ——— B ——— Z ——— D ———

Θ ——— G ——— K ——— H ———

M ——— L ——— N ——— M ———

E ———

F ———

K ———

N ———

Καὶ ἐπεὶ ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , καὶ ἐληπτὰ τῶν μὲν A, Γ ἰσάκις πολλαπλάσια τὰ H, Θ , τῶν δὲ B, Δ ἄλλα, ἃ ἐτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, M , εἰ ἄρα ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὸ Θ τοῦ M , καὶ εἰ ἴσον ἐστίν, ἴσον, καὶ εἰ ἐλλείπει, ἐλλείπει. πάλιν, ἐπεὶ ἐστὶν ὡς τὸ Γ πρὸς τὸ Δ , οὕτως τὸ E πρὸς τὸ Z , καὶ ἐληπτὰ τῶν Γ, E ἰσάκις πολλαπλάσια τὰ Θ, K , τῶν δὲ Δ, Z ἄλλα, ἃ ἐτυθεν, ἰσάκις πολλαπλάσια τὰ M, N , εἰ ἄρα ὑπερέθει τὸ Θ τοῦ M , ὑπερέθει καὶ τὸ K τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἐλάττον, ἐλάττον. ἀλλὰ εἰ ὑπερεῖθε τὸ Θ τοῦ M , ὑπερεῖθε καὶ τὸ H τοῦ Λ , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἐλάττον, ἐλάττον· ὥστε καὶ εἰ ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὸ K τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἐλάττον, ἐλάττον. καὶ ἐστὶ τὰ μὲν H, K τῶν A, E ἰσάκις πολλαπλάσια, τὰ δὲ Λ, N τῶν B, Z ἄλλα, ἃ ἐτυθεν, ἰσάκις πολλαπλάσια· ἐστὶν ἄρα ὡς τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z .

Οἱ ἄρα τῶ αὐτῶ λόγῳ οἱ αὐτοὶ καὶ ἀλλήλοις εἰσὶν οἱ αὐτοί· ὅπερ ἔδει δεῖχαι.

(Ratios which are) the same with the same ratio are also the same with one another.

For let it be that as A (is) to B , so C (is) to D , and as C (is) to D , so E (is) to F . I say that as A is to B , so E (is) to F .

For let the equal multiples G, H, K be taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively).

And since as A is to B , so C (is) to D , and the equal multiples G and H be taken of A and C (respectively), and the other random equal multiples L and M of B and D (respectively), thus if G exceeds L (then) H also exceeds M , and if (G is) equal (to L then H is also) equal (to M), and if (G is) less (than L then H is also) less (than M) [Def. 5.5]. Again, since as C is to D , so E (is) to F , and the equal multiples H and K have been taken of C and E (respectively), and the other random equal multiples M and N of D and F (respectively), thus if H exceeds M (then) K also exceeds N , and if (H is) equal (to M then K is also) equal (to N), and if (H is) less (than M then K is also) less (than N) [Def. 5.5]. But (we saw that) if H was exceeding M (then) G was also exceeding L , and if (H was) equal (to M then G was also) equal (to L), and if (H was) less (than M then G was also) less (than L). And, hence, if G exceeds L (then) K also exceeds N , and if (G is) equal (to L then K is also) equal (to N), and if (G is) less (than L then K is also) less (than N). And G and K are equal multiples of A and E (respectively), and L and N other random equal multiples of B and F (respectively). Thus, as A is to B , so E (is) to F [Def. 5.5].

Thus, (ratios which are) the same with the same ratio are also the same with one another. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\gamma : \delta :: \epsilon : \zeta$ then $\alpha : \beta :: \epsilon : \zeta$.

12.

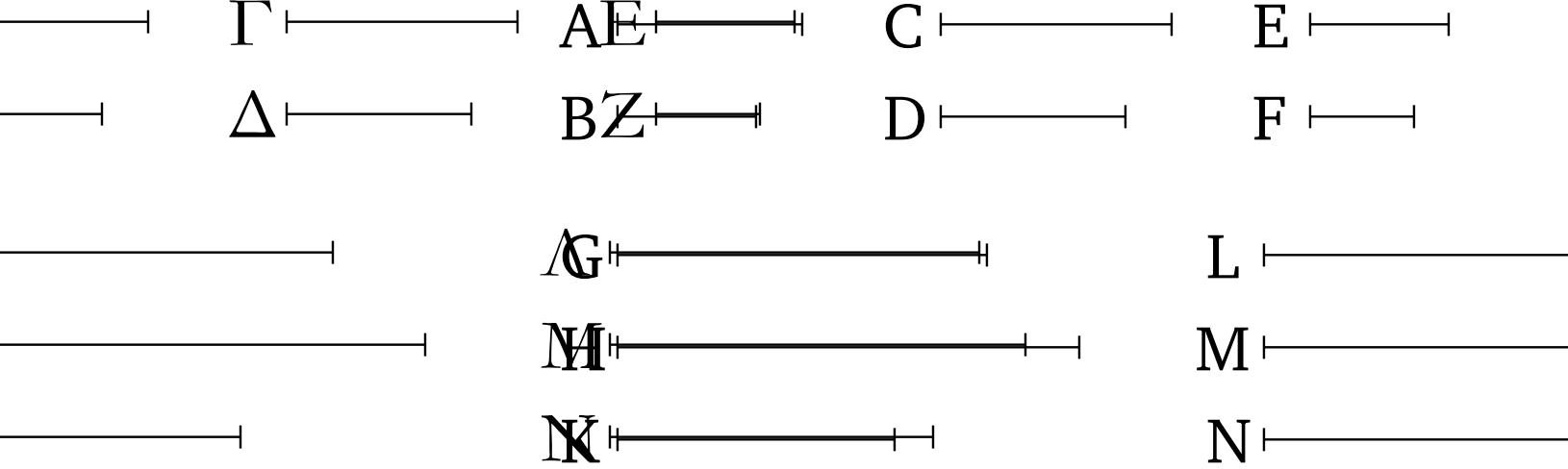
Proposition 12[†]

Ἐὰν ᾗ ὅποσαοῦν μεγέθη ἀνάλογον, ἔσται ὡς ἓν τῶν

If there are any number of magnitudes whatsoever (which

ἡγουμένων πρὸς ἓν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα.

are) proportional, (then) as one of the leading (magnitudes is) to one of the following, so will all of the leading (magnitudes) be to all of the following.



Ἐστωσαν ὅποσαοῦν μεγέθη ἀνάλογον τὰ $A, B, \Gamma, \Delta, E, Z$, ὥς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , καὶ τὸ E πρὸς τὸ Z . λέγω, ὅτι ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως τὰ A, Γ, E πρὸς τὰ B, Δ, Z .

Εἰλήφθω γὰρ τῶν μὲν A, Γ, E ἰσάκις πολλαπλάσια τὰ H, Θ, K , τῶν δὲ B, Δ, Z ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ Λ, M, N .

Καὶ ἐπεὶ ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , καὶ τὸ E πρὸς τὸ Z , καὶ εἰληπτὰ τῶν μὲν A, Γ, E ἰσάκις πολλαπλάσια τὰ H, Θ, K τῶν δὲ B, Δ, Z ἄλλα, ἃ ἔτυνθεν, ἰσάκις πολλαπλάσια τὰ Λ, M, N , εἴ ἄρα ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὸ Θ τοῦ M , καὶ τὸ K τοῦ N , καὶ εἴ ἴσον, ἴσον, καὶ εἴ ἔλαττον, ἔλαττον. ὥστε καὶ εἴ ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὰ H, Θ, K τῶν Λ, M, N , καὶ εἴ ἴσον, ἴσα, καὶ εἴ ἔλαττον, ἔλαττονα. καὶ ἐστὶ τὸ μὲν H καὶ τὰ H, Θ, K τοῦ A καὶ τῶν A, Γ, E ἰσάκις πολλαπλάσια, ἐπειδήπερ εἴαν ἡ ὅποσαοῦν μεγέθη ὅποσωνοῦν μεγεθῶν ἴσων τὸ πλῆθος ἑκάστον ἐκάστον ἰσάκις πολλαπλάσιον, ὁσαπλάσιόν ἐστιν ἐν τῶν μεγεθῶν ἐνόξ, τοσανταπλάσια ἔσται καὶ τὰ πάντα τῶν πάντων. διὰ τὰ αὐτὰ δὴ καὶ τὸ Λ καὶ τὰ Λ, M, N τοῦ B καὶ τῶν B, Δ, Z ἰσάκις ἐστὶ πολλαπλάσια· ἔστιν ἄρα ὥς τὸ A πρὸς τὸ B , οὕτως τὰ A, Γ, E πρὸς τὰ B, Δ, Z .

Ἐάν ἄρα ἡ ὅποσαοῦν μεγέθη ἀνάλογον, ἔσται ὥς ἐν τῶν ἡγουμένων πρὸς ἓν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα· ὅπερ ἔδει δεῖχαι.

Let there be any number of magnitudes whatsoever, A, B, C, D, E, F , (which are) proportional, (so that) as A (is) to B , so C (is) to D , and E to F . I say that as A is to B , so A, C, E (are) to B, D, F .

For let the equal multiples G, H, K be taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively).

And since as A is to B , so C (is) to D , and E to F , and the equal multiples G, H, K have been taken of A, C, E (respectively), and the other random equal multiples L, M, N of B, D, F (respectively), thus if G exceeds L (then) H also exceeds M , and K (exceeds) N , and if (G is) equal (to L then H is also) equal (to M , and K to N), and if (G is) less (than L then H is also) less (than M , and K than N) [Def. 5.5]. And, hence, if G exceeds L (then) G, H, K also exceed L, M, N , and if (G is) equal (to L then G, H, K are also) equal (to L, M, N) and if (G is) less (than L then G, H, K are also) less (than L, M, N). And G and G, H, K are equal multiples of A and A, C, E (respectively), inasmuch as if there are any number of magnitudes whatsoever (which are) equal multiples, respectively, of some (other) magnitudes, of equal number (to them), (then) as many times as one of the (first) magnitudes is (divisible) by one (of the second), so many times will all (of the first magnitudes) also (be divisible) by all (of the second) [Prop. 5.1]. So, for the same (reasons), L and L, M, N are also equal multiples of B and B, D, F (respectively). Thus, as A is to B , so A, C, E (are) to B, D, F (respectively).

Thus, if there are any number of magnitudes whatsoever (which are) proportional, (then) as one of the leading (magnitudes is) to one of the following, so will all of the leading (magnitudes) be to all of the following. (Which is) the very thing it was required to show.

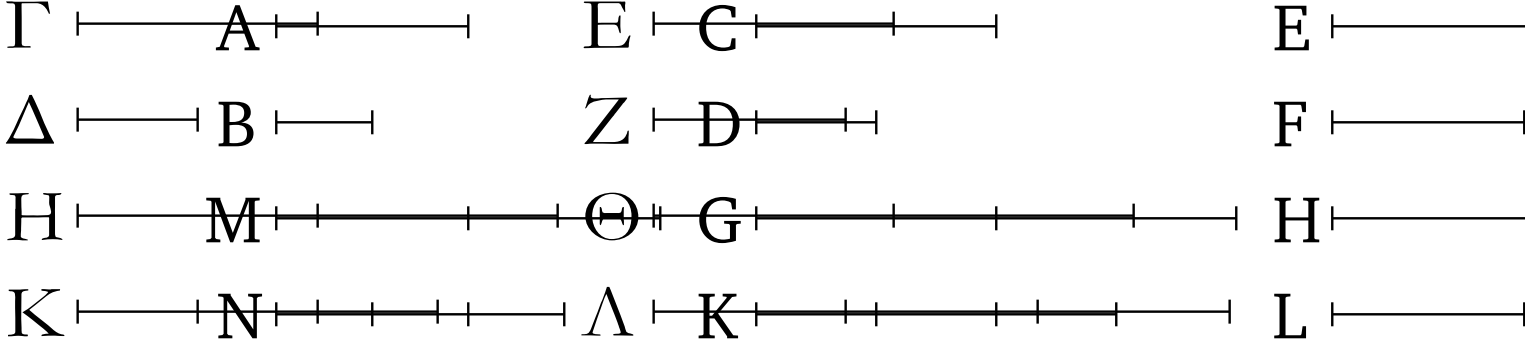
† In modern notation, this proposition reads that if $\alpha : \alpha' :: \beta : \beta' :: \gamma : \gamma'$ etc. then $\alpha : \alpha' :: (\alpha + \beta + \gamma + \dots) : (\alpha' + \beta' + \gamma' + \dots)$.

13.

Proposition 13†

Ἐάν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τρίτον δὲ πρὸς τέταρτον μείζονα λόγον ἔθῃ ἢ πέμπτον πρὸς ἕκτον, καὶ πρῶτον πρὸς δεύτερον μείζονα λόγον ἔχει ἢ πέμπτον πρὸς ἕκτον.

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the third (magnitude) has a greater ratio to the fourth than a fifth (has) to a sixth, (then) the first (magnitude) will also have a greater ratio to the second than the fifth (has) to the sixth.



Πρῶτον γάρ τὸ A πρὸς δεύτερον τὸ B τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ , τρίτον δὲ τὸ Γ πρὸς τέταρτον τὸ Δ μείζονα λόγον ἔθῃ ἢ πέμπτον τὸ E πρὸς ἕκτον τὸ Z . λέγω, ὅτι καὶ πρῶτον τὸ A πρὸς δεύτερον τὸ B μείζονα λόγον ἔχει ἢ περὶ πέμπτον τὸ E πρὸς ἕκτον τὸ Z .

Ἐπεὶ γὰρ ἔστι τινα τῶν μὲν Γ, E ἰσάκις πολλαπλάσια, τῶν δὲ Δ, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια, καὶ τὸ μὲν τοῦ Γ πολλαπλάσιον τοῦ τοῦ Δ πολλαπλασίον ὑπερέθει, τὸ δὲ τοῦ E πολλαπλάσιον τοῦ τοῦ Z πολλαπλασίον οὐθ' ὑπερέθει, εἰλήφθω, καὶ ἔστω τῶν μὲν Γ, E ἰσάκις πολλαπλάσια τὰ H, Θ , τῶν δὲ Δ, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ K, Λ , ὥστε τὸ μὲν H τοῦ K ὑπερέθειν, τὸ δὲ Θ τοῦ Λ μὴ ὑπερέθειν καὶ ὁσαπλάσιον μὲν ἔστι τὸ H τοῦ Γ , τοσαυταπλάσιον ἔστω καὶ τὸ M τοῦ A , ὁσαπλάσιον δὲ τὸ K τοῦ Δ , τοσαυταπλάσιον ἔστω καὶ τὸ N τοῦ B .

Καὶ ἐπεὶ ἔστιν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , καὶ εἰληπται τῶν μὲν A, Γ ἰσάκις πολλαπλάσια τὰ M, H , τῶν δὲ B, Δ ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ N, K , εἰ ἄρα ὑπερέθει τὸ M τοῦ N , ὑπερέθει καὶ τὸ H τοῦ K , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. ὑπερέθει δὲ τὸ H τοῦ K ὑπερέθει ἄρα καὶ τὸ M τοῦ N . τὸ δὲ Θ τοῦ Λ οὐθ' ὑπερέθει καὶ ἔστι τὰ μὲν M, Θ τῶν A, E ἰσάκις πολλαπλάσια, τὰ δὲ N, Λ τῶν B, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια· τὸ ἄρα A πρὸς τὸ B μείζονα λόγον ἔθει ἢ περὶ τὸ E πρὸς τὸ Z .

Ἐάν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τρίτον δὲ πρὸς τέταρτον μείζονα λόγον ἔθῃ ἢ πέμπτον πρὸς ἕκτον, καὶ πρῶτον πρὸς δεύτερον μείζονα λόγον ἔχει ἢ πέμπτον πρὸς ἕκτον· ὅπερ ἔδει δεῖξαι.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D , and let the third (magnitude) C have a greater ratio to the fourth D than a fifth E (has) to a sixth F . I say that the first (magnitude) A will also have a greater ratio to the second B than the fifth E (has) to the sixth F .

For since there are some equal multiples of C and E , and other random equal multiples of D and F , (for which) the multiple of C exceeds the (multiple) of D , and the multiple of E does not exceed the multiple of F [Def. 5.7], let them be taken. And let G and H be equal multiples of C and E (respectively), and K and L other random equal multiples of D and F (respectively), such that G exceeds K , but H does not exceed L . And as many times as G is (divisible) by C , so many times let M be (divisible) by A . And as many times as K (is divisible) by D , so many times let N be (divisible) by B .

And since as A is to B , so C (is) to D , and the equal multiples M and G have been taken of A and C (respectively), and the other random equal multiples N and K of B and D (respectively), thus if M exceeds N (then) G exceeds K , and if (M is) equal (to N then G is also) equal (to K), and if (M is) less (than N then G is also) less (than K) [Def. 5.5]. And G exceeds K . Thus, M also exceeds N . And H does not exceed L . And M and H are equal multiples of A and E (respectively), and N and L other random equal multiples of B and F (respectively). Thus, A has a greater ratio to B than E (has) to F [Def. 5.7].

Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and a third (magnitude) has a greater ratio to a fourth than a fifth (has) to a sixth, (then) the first (magnitude) will also have a greater ratio to the second than the fifth (has) to the sixth. (Which is) the very thing it

was required to show.

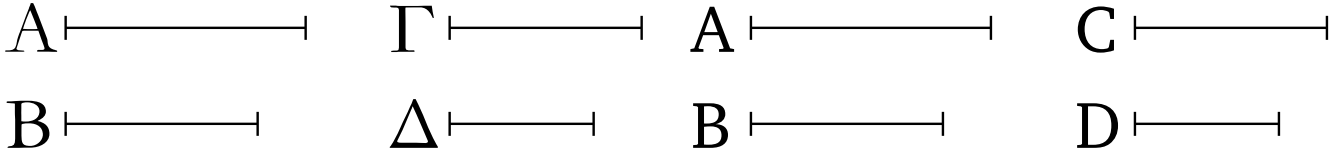
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\gamma : \delta > \epsilon : \zeta$ then $\alpha : \beta > \epsilon : \zeta$.

14.

Proposition 14†

Ἐάν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τὸ δὲ πρῶτον τοῦ τρίτου μείζον ἢ, καὶ τὸ δεύτερον τοῦ τετάρτου μείζον ἔσται, καὶ ἴσον, ἴσον, καὶ ἔλαττον, ἔλαττον.

If a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the first (magnitude) is greater than the third, (then) the second will also be greater than the fourth. And if (the first magnitude is) equal (to the third then the second will also be) equal (to the fourth). And if (the first magnitude is) less (than the third then the second will also be) less (than the fourth).



Πρῶτον γάρ τὸ A πρὸς δεύτερον τὸ B τὸν αὐτὸν ἔθεται λόγον καὶ τρίτον τὸ Γ πρὸς τέταρτον τὸ Δ , μείζον δὲ ἔστω τὸ A τοῦ Γ . λέγω, ὅτι καὶ τὸ B τοῦ Δ μείζον ἔστιν.

Ἐπεὶ γὰρ τὸ A τοῦ Γ μείζον ἔστιν, ἄλλο δέ, ὃ ἔτυθεν, [μέγεθος] τὸ B , τὸ A ἄρα πρὸς τὸ B μείζονα λόγον ἔθει ἢ περ τὸ Γ πρὸς τὸ B . ὥς δὲ τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ . καὶ τὸ Γ ἄρα πρὸς τὸ Δ μείζονα λόγον ἔθει ἢ περ τὸ Γ πρὸς τὸ B . πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον ἔθει, ἐκεῖνο ἔλασσόν ἔστιν· ἔλασσον ἄρα τὸ Δ τοῦ B . ὥστε μείζον ἔστι τὸ B τοῦ Δ .

Ὅμοιως δὴ δεῖχμεν, ὅτι καὶ ἴσον ἢ τὸ A τῷ Γ , ἴσον ἔσται καὶ τὸ B τῷ Δ , καὶ ἔλασσον ἢ τὸ A τοῦ Γ , ἔλασσον ἔσται καὶ τὸ B τοῦ Δ .

Ἐάν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, τὸ δὲ πρῶτον τοῦ τρίτου μείζον ἢ, καὶ τὸ δεύτερον τοῦ τετάρτου μείζον ἔσται, καὶ ἴσον, ἴσον, καὶ ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖχαι.

For let a first (magnitude) A have the same ratio to a second B that a third C (has) to a fourth D . And let A be greater than C . I say that B is also greater than D .

For since A is greater than C , and B (is) another random [magnitude], A thus has a greater ratio to B than C (has) to B [Prop. 5.8]. And as A (is) to B , so C (is) to D . Thus, C also has a greater ratio to D than C (has) to B . And that (magnitude) to which the same (magnitude) has a greater ratio is the lesser [Prop. 5.10]. Thus, D (is) less than B . Hence, B is greater than D .

So, similarly, we can show that even if A is equal to C , (then) B will also be equal to D , and even if A is less than C , (then) B will also be less than D .

Thus, if a first (magnitude) has the same ratio to a second that a third (has) to a fourth, and the first (magnitude) is greater than the third, (then) the second will also be greater than the fourth. And if (the first magnitude is) equal (to the third then the second will also be) equal (to the fourth). And if (the first magnitude is) less (than the third then the second will also be) less (than the fourth). (Which is) the very thing it was required to show.

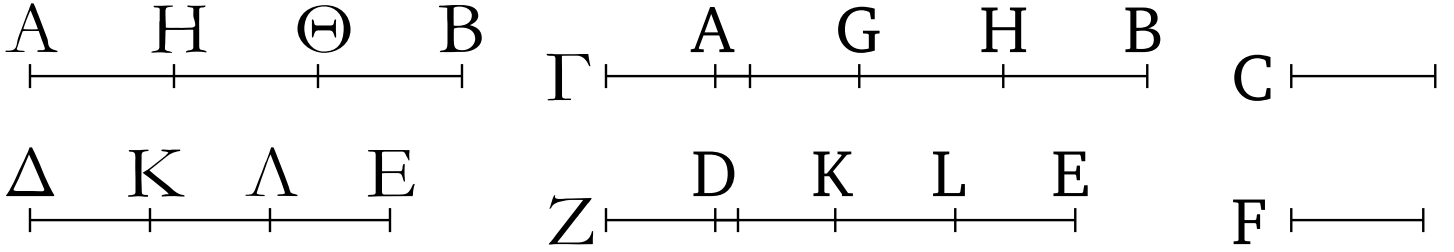
† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha \gtrless \gamma$ as $\beta \gtrless \delta$.

15.

Proposition 15†

Τὰ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον ληφθέντα κατάλληλα.

Parts have the same ratio as similar multiples, taken in corresponding order.



Ἐστω γάρ ἰσάκις πολλαπλάσιον τὸ AB τοῦ Γ καὶ τὸ ΔE τοῦ Z · λέγω, ὅτι ἐστὶν ὡς τὸ Γ πρὸς τὸ Z , οὕτως τὸ AB πρὸς τὸ ΔE .

Ἐπεὶ γάρ ἰσάκις ἐστὶ πολλαπλάσιον τὸ AB τοῦ Γ καὶ τὸ ΔE τοῦ Z , ὅσα ἄρα ἐστὶν ἐν τῷ AB μεγέθει ἴσα τῷ Γ , τοσαῦτα καὶ ἐν τῷ ΔE ἴσα τῷ Z . διηγήσθω τὸ μὲν AB εἰς τὰ τῷ Γ ἴσα τὰ $AH, H\Theta, \Theta B$, τὸ δὲ ΔE εἰς τὰ τῷ Z ἴσα τὰ $\Delta K, KA, AE$ · ἐστὶ δὴ ἴσον τὸ πλῆθος τῶν $AH, H\Theta, \Theta B$ τῷ πλῆθει τῶν $\Delta K, KA, AE$. καὶ ἐπεὶ ἴσα ἐστὶ τὰ $AH, H\Theta, \Theta B$ ἀλλήλοις, ἐστὶ δὲ καὶ τὰ $\Delta K, KA, AE$ ἴσα ἀλλήλοις, ἐστὶν ἄρα ὡς τὸ AH πρὸς τὸ ΔK , οὕτως τὸ $H\Theta$ πρὸς τὸ KA , καὶ τὸ ΘB πρὸς τὸ AE . ἐστὶ ἄρα καὶ ὡς ἓν τῶν ἡγουμένων πρὸς ἓν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγουμένα πρὸς ἅπαντα τὰ ἐπόμενα· ἐστὶν ἄρα ὡς τὸ AH πρὸς τὸ ΔK , οὕτως τὸ AB πρὸς τὸ ΔE . ἴσον δὲ τὸ μὲν AH τῷ Γ , τὸ δὲ ΔK τῷ Z · ἐστὶν ἄρα ὡς τὸ Γ πρὸς τὸ Z οὕτως τὸ AB πρὸς τὸ ΔE .

Τὰ ἄρα μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔδει λόγον ληφθέντα κατάλληλα· ὅπερ ἔδει δεῖξαι.

For let AB and DE be equal multiples of C and F (respectively). I say that as C is to F , so AB (is) to DE .

For since AB and DE are equal multiples of C and F (respectively), thus as many magnitudes as there are in AB equal to C , so many (are there) also in DE equal to F . Let AB be divided into (magnitudes) AG, GH, HB , equal to C , and DE into (magnitudes) DK, KL, LE , equal to F . So, the number of (magnitudes) AG, GH, HB will equal the number of (magnitudes) DK, KL, LE . And since AG, GH, HB are equal to one another, and DK, KL, LE are also equal to one another, thus as AG is to DK , so GH (is) to KL , and HB to LE [Prop. 5.7]. And, thus (for proportional magnitudes), as one of the leading (magnitudes) will be to one of the following, so all of the leading (magnitudes will be) to all of the following [Prop. 5.12]. Thus, as AG is to DK , so AB (is) to DE . And AG is equal to C , and DK to F . Thus, as C is to F , so AB (is) to DE .

Thus, parts have the same ratio as similar multiples, taken in corresponding order. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that $\alpha : \beta :: m\alpha : m\beta$.

16.

Ἐὰν τέσσαρα μεγέθη ἀνάλογον ᾗ, καὶ ἐναλλάξ ἀνάλογον ἔσται.

Ἐστω τέσσαρα μεγέθη ἀνάλογον τὰ A, B, Γ, Δ , ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ · λέγω, ὅτι καὶ ἐναλλάξ [ἀνάλογον] ἔσται, ὡς τὸ A πρὸς τὸ Γ , οὕτως τὸ B πρὸς τὸ Δ .

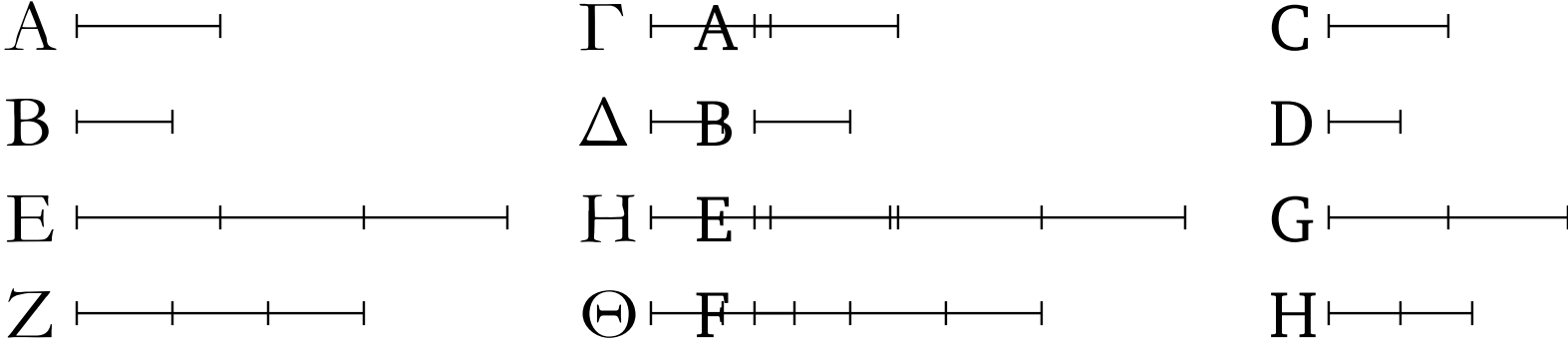
Εἰλήφθω γάρ τῶν μὲν A, B ἰσάκις πολλαπλάσια τὰ E, Z , τῶν δὲ Γ, Δ ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ H, Θ .

Proposition 16[†]

If four magnitudes are proportional, (then) they will also be proportional alternately.

Let A, B, C and D be four proportional magnitudes, (such that) as A (is) to B , so C (is) to D . I say that they will also be [proportional] alternately, (so that) as A (is) to C , so B (is) to D .

For let the equal multiples E and F be taken of A and B (respectively), and the other random equal multiples G and H of C and D (respectively).



Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ E τοῦ A καὶ τὸ Z τοῦ B , τὰ δὲ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὡς τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z . ὡς δὲ τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ . καὶ ὡς ἄρα τὸ Γ πρὸς τὸ Δ , οὕτως τὸ E πρὸς τὸ Z . πάλιν, ἐπεὶ τὰ H , Θ τῶν Γ , Δ ἰσάκις ἐστὶ πολλαπλάσια, ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ Δ , οὕτως τὸ H πρὸς τὸ Θ . ὡς δὲ τὸ Γ πρὸς τὸ Δ , [οὕτως] τὸ E πρὸς τὸ Z καὶ ὡς ἄρα τὸ E πρὸς τὸ Z , οὕτως τὸ H πρὸς τὸ Θ . ἔαν δὲ τέσσαρα μεγέθη ἀνάλογον ᾗ, τὸ δὲ πρῶτον τοῦ τρίτου μείζον ᾗ, καὶ τὸ δεύτερον τοῦ τετάρτου μείζον ἔσται, καὶ ἂν ἴσον, ἴσον, καὶ ἂν ἑλάττω, ἑλάττω. εἰ ἄρα ὑπερέθει τὸ E τοῦ H , ὑπερέθει καὶ τὸ Z τοῦ Θ , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἑλάττω, ἑλάττω. καὶ ἐστὶ τὰ μὲν E , Z τῶν A , B ἰσάκις πολλαπλάσια, τὰ δὲ H , Θ τῶν Γ , Δ ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὡς τὸ A πρὸς τὸ Γ , οὕτως τὸ B πρὸς τὸ Δ .

Ἐὰν ἄρα τέσσαρα μεγέθη ἀνάλογον ᾗ, καὶ ἐναλλάξ ἀνάλογον ἔσται· ὅπερ ἔδει δεῖχαι.

And since E and F are equal multiples of A and B (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as A is to B , so E (is) to F . But as A (is) to B , so C (is) to D . And, thus, as C (is) to D , so E (is) to F [Prop. 5.11]. Again, since G and H are equal multiples of C and D (respectively), thus as C is to D , so G (is) to H [Prop. 5.15]. But as C (is) to D , [so] E (is) to F . And, thus, as E (is) to F , so G (is) to H [Prop. 5.11]. And if four magnitudes are proportional, and the first is greater than the third, (then) the second will also be greater than the fourth, and if (the first is) equal (to the third then the second will also be) equal (to the fourth), and if (the first is) less (than the third then the second will also be) less (than the fourth) [Prop. 5.14]. Thus, if E exceeds G (then) F also exceeds H , and if (E is) equal (to G then F is also) equal (to H), and if (E is) less (than G then F is also) less (than H). And E and F are equal multiples of A and B (respectively), and G and H other random equal multiples of C and D (respectively). Thus, as A is to C , so B (is) to D [Def. 5.5].

Thus, if four magnitudes are proportional, (then) they will also be proportional alternately. (Which is) the very thing it was required to show.

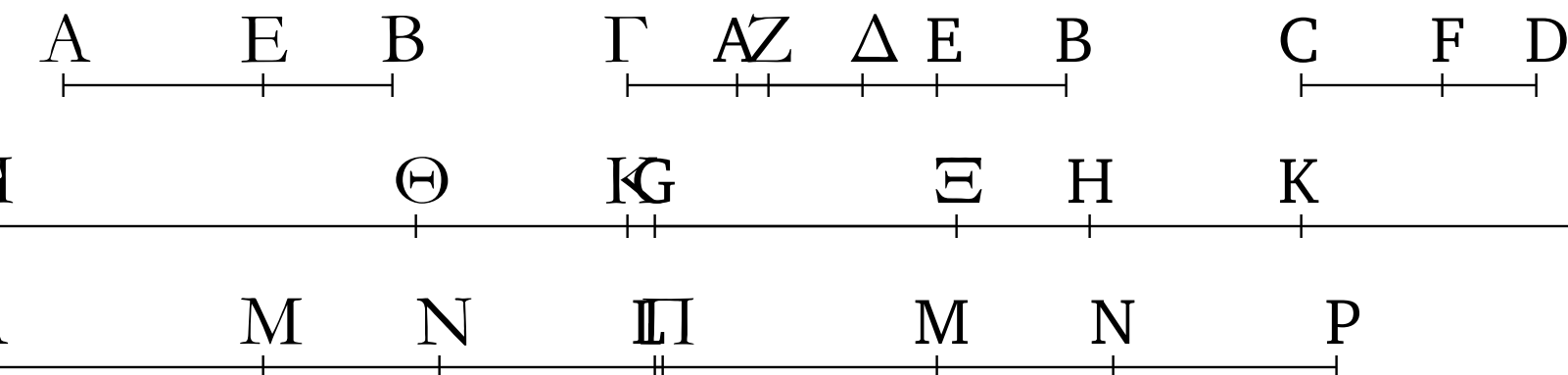
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \gamma :: \beta : \delta$.

17.

Ἐὰν συγκείμενα μεγέθη ἀνάλογον ᾗ, καὶ διαιρεθέντα ἀνάλογον ἔσται.

Proposition 17[†]

If composed magnitudes are proportional, (then) they will also be proportional (when) separated.



Ἐστω συγκείμενα μεγέθη ἀνάλογον τὰ $AB, BE, ΓΔ, ΔΖ$,
ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $ΓΔ$ πρὸς τὸ $ΔΖ$. λέγω,
ὅτι καὶ διαιρεθέντα ἀνάλογον ἔσται, ὥς τὸ AE πρὸς τὸ EB ,
οὕτως τὸ $ΓΖ$ πρὸς τὸ $ΔΖ$.

Εἰλήφθω γὰρ τῶν μὲν $AE, EB, \Gamma Z, Z\Delta$ ἰσάκις πολλαπλάσια τὰ $H\Theta, \Theta K, \Lambda M, MN$, τῶν δὲ $EB, Z\Delta$ ἄλλα, ἃ ἔττυθεν, ἰσάκις πολλαπλάσια τὰ $KX, N\Pi$.

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ ΑΕ καὶ τὸ ΘΚ τοῦ ΕΒ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ ΑΕ καὶ τὸ ΗΚ τοῦ ΑΒ. ἰσάκις δὲ ἐστὶ πολλαπλάσιον τὸ ΗΘ τοῦ ΑΕ καὶ τὸ ΑΜ τοῦ ΓΖ· ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΗΚ τοῦ ΑΒ καὶ τὸ ΑΜ τοῦ ΓΖ. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΑΜ τοῦ ΓΖ καὶ τὸ ΜΝ τοῦ ΖΔ, ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΑΜ τοῦ ΓΖ καὶ τὸ ΑΝ τοῦ ΓΔ. ἰσάκις δὲ ἦν πολλαπλάσιον τὸ ΑΜ τοῦ ΓΖ καὶ τὸ ΗΚ τοῦ ΑΒ· ἰσάκις ἄρα ἐστὶ πολλαπλάσιον τὸ ΗΚ τοῦ ΑΒ καὶ τὸ ΑΝ τοῦ ΓΔ. τὰ ΗΚ, ΑΝ ἄρα τῶν ΑΒ, ΓΔ ἰσάκις ἐστὶ πολλαπλάσια. πάλιν, ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσιον τὸ ΘΚ τοῦ ΕΒ καὶ τὸ ΜΝ τοῦ ΖΔ, ἔστι δὲ καὶ τὸ ΚΧ τοῦ ΕΒ ἰσάκις πολλαπλάσιον καὶ τὸ ΝΠ τοῦ ΖΔ, καὶ συντεθέν τὸ ΘΧ τοῦ ΕΒ ἰσάκις ἐστὶ πολλαπλάσιον καὶ τὸ ΜΠ τοῦ ΖΔ. καὶ ἐπεὶ ἐστὶν ὥς τὸ ΑΒ πρὸς τὸ ΒΕ, οὕτως τὸ ΓΔ πρὸς τὸ ΔΖ, καὶ εἴληπται τῶν μὲν ΑΒ, ΓΔ ἰσάκις πολλαπλάσια τὰ ΗΚ, ΑΝ, τῶν δὲ ΕΒ, ΖΔ ἰσάκις πολλαπλάσια τὰ ΘΧ, ΜΠ, εἰ ἄρα ὑπερέθει τὸ ΗΚ τοῦ ΘΧ, ὑπερέθει καὶ τὸ ΑΝ τοῦ ΜΠ, καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. ὑπερεθέτω δὴ τὸ ΗΚ τοῦ ΘΧ, καὶ κοινοῦ ἀφαιρεθέντος τοῦ ΘΚ ὑπερέθει ἄρα καὶ τὸ ΗΘ τοῦ ΚΧ. ἀλλὰ εἰ ὑπερεῖθε τὸ ΗΚ τοῦ ΘΧ ὑπερεῖθε καὶ τὸ ΑΝ τοῦ ΜΠ· ὑπερέθει ἄρα καὶ τὸ ΑΝ τοῦ ΜΠ, καὶ κοινοῦ ἀφαιρεθέντος τοῦ ΜΝ ὑπερέθει καὶ τὸ ΑΜ τοῦ ΝΠ· ὥστε εἰ ὑπερέθει τὸ ΗΘ τοῦ ΚΧ, ὑπερέθει καὶ τὸ ΑΜ τοῦ ΝΠ. ὁμοίως δὴ δεῖχμεν, ὅτι κ' ἂν ἴσον ἦ τὸ ΗΘ τῷ ΚΧ, ἴσον ἔσται καὶ τὸ ΑΜ τῷ ΝΠ, κ' ἂν ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν ΗΘ, ΑΜ τῶν ΑΕ, ΓΖ ἰσάκις πολλαπλάσια, τὰ δὲ ΚΧ, ΝΠ τῶν ΕΒ, ΖΔ ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια· ἔστιν ἄρα ὥς τὸ ΑΕ πρὸς τὸ ΕΒ, οὕτως τὸ ΓΖ πρὸς τὸ ΖΔ.

Ἐὰν ἄρα συγκείμενα μερέθη ἀνάλογον ἢ, καὶ διαιρεθέντα ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.

Let AB , BE , CD , and DF be composed magnitudes (which are) proportional, (so that) as AB (is) to BE , so CD (is) to DF . I say that they will also be proportional (when) separated, (so that) as AE (is) to EB , so CF (is) to DF .

For let the equal multiples GH, HK, LM , and MN be taken of AE, EB, CF , and FD (respectively), and the other random equal multiples KO and NP of EB and FD (respectively).

And since GH and HK are equal multiples of AE and EB (respectively), GH and GK are thus equal multiples of AE and AB (respectively) [Prop. 5.1]. But GH and LM are equal multiples of AE and CF (respectively). Thus, GK and LM are equal multiples of AB and CF (respectively). Again, since LM and MN are equal multiples of CF and FD (respectively), LM and LN are thus equal multiples of CF and CD (respectively) [Prop. 5.1]. And LM and GK were equal multiples of CF and AB (respectively). Thus, GK and LN are equal multiples of AB and CD (respectively). Thus, GK , LN are equal multiples of AB , CD . Again, since HK and MN are equal multiples of EB and FD (respectively), and KO and NP are also equal multiples of EB and FD (respectively), (then), added together, HO and MP are also equal multiples of EB and FD (respectively) [Prop. 5.2]. And since as AB (is) to BE , so CD (is) to DF , and the equal multiples GK , LN be taken of AB , CD , and the equal multiples HO , MP of EB , FD , thus if GK exceeds HO (then) LN also exceeds MP , and if (GK is) equal (to HO then LN is also) equal (to MP), and if (GK is) less (than HO then LN is also) less (than MP) [Def. 5.5]. So let GK exceed HO , and thus, HK being taken away from both, GH exceeds KO . But (we saw that) if GK was exceeding HO (then) LN was also exceeding MP . Thus, LN also exceeds MP , and, MN being taken away from both, LM also exceeds NP . Hence, if GH exceeds KO (then) LM also exceeds NP . So, similarly, we can show that even if GH is equal to KO (then) LM will also be equal to NP , and even if (GH is) less (than KO then LM will also be) less (than NP). And GH , LM are equal multiples of AE , CF , and KO , NP other random equal multiples of EB , FD . Thus, as AE is to EB , so CF (is) to FD [Def. 5.5].

Thus, if composed magnitudes are proportional, (then) they will also be proportional (when) separated. (Which is)

the very thing it was required to show.

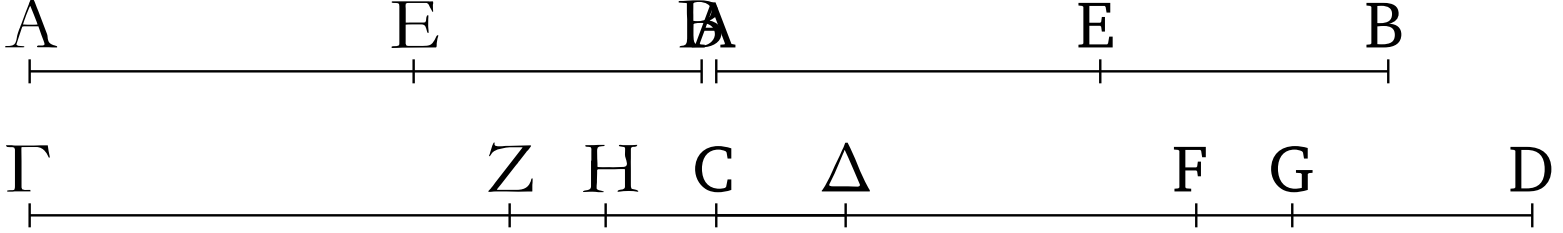
[†] In modern notation, this proposition reads that if $\alpha + \beta : \beta :: \gamma + \delta : \delta$ then $\alpha : \beta :: \gamma : \delta$.

18.

Proposition 18[†]

Ἐάν διηρημένα μεγέθη ἀνάλογον ᾗ, καὶ συντεθέντα ἀνάλογον ἔσται.

If separated magnitudes are proportional, (then) they will also be proportional (when) composed.



Ἐστω διηρημένα μεγέθη ἀνάλογον τὰ $AE, EB, \Gamma Z, Z\Delta$, ὥς τὸ AE πρὸς τὸ EB , οὕτως τὸ ΓZ πρὸς τὸ $Z\Delta$. λέγω, ὅτι καὶ συντεθέντα ἀνάλογον ἔσται, ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ $Z\Delta$.

Εἰ γὰρ μὴ ἔστιν ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ ΔZ , ἔσται ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ ᾗτοι πρὸς ἑλασσόν τι τοῦ ΔZ ἢ πρὸς μείζον.

Ἐστω πρότερον πρὸς ἑλασσον τὸ ΔH . καὶ ἐπεὶ ἔστιν ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ ΔH , συγκεείμενα μεγέθη ἀναλόγον ἔστιν· ὥστε καὶ διαιρεθέντα ἀνάλογον ἔσται. ἔστιν ἄρα ὥς τὸ AE πρὸς τὸ EB , οὕτως τὸ ΓH πρὸς τὸ $H\Delta$. ὑπόκειται δὲ καὶ ὥς τὸ AE πρὸς τὸ EB , οὕτως τὸ ΓZ πρὸς τὸ $Z\Delta$. καὶ ὥς ἄρα τὸ ΓH πρὸς τὸ $H\Delta$, οὕτως τὸ ΓZ πρὸς τὸ $Z\Delta$. μείζον δὲ τὸ πρῶτον τὸ ΓH τοῦ τρίτου τοῦ ΓZ . μείζον ἄρα καὶ τὸ δεύτερον τὸ $H\Delta$ τοῦ τετάρτου τοῦ $Z\Delta$. ἀλλὰ καὶ ἑλάττον· ὅπερ ἔστιν ἀδύνατον· οὐκ ἄρα ἔστιν ὥς τὸ AB πρὸς τὸ BE , οὕτως τὸ $\Gamma\Delta$ πρὸς ἑλασσον τοῦ $Z\Delta$. ὁμοίως δὲ δείκνυμεν, ὅτι οὐδὲ πρὸς μείζον πρὸς αὐτὸ ἄρα.

Ἐάν ἄρα διηρημένα μεγέθη ἀνάλογον ᾗ, καὶ συντεθέντα ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.

Let AE, EB, CF , and FD be separated magnitudes (which are) proportional, (so that) as AE (is) to EB , so CF (is) to FD . I say that they will also be proportional (when) composed, (so that) as AB (is) to BE , so CD (is) to FD .

For if (it is) not (the case that) as AB is to BE , so CD (is) to FD , (then) it will surely be (the case that) as AB (is) to BE , so CD is either to some (magnitude) less than DF , or (some magnitude) greater (than DF).[‡]

Let it, first of all, be to (some magnitude) less (than DF), (namely) DG . And since composed magnitudes are proportional, (so that) as AB is to BE , so CD (is) to DG , they will thus also be proportional (when) separated [Prop. 5.17]. Thus, as AE is to EB , so CG (is) to GD . But it was also assumed that as AE (is) to EB , so CF (is) to FD . Thus, (it is) also (the case that) as CG (is) to GD , so CF (is) to FD [Prop. 5.11]. And the first (magnitude) CG (is) greater than the third CF . Thus, the second (magnitude) GD (is) also greater than the fourth FD [Prop. 5.14]. But (it is) also less. The very thing is impossible. Thus, (it is) not (the case that) as AB is to BE , so CD (is) to less than FD . Similarly, we can show that neither (is it the case) to greater (than FD). Thus, (it is the case) to the same (as FD).

Thus, if separated magnitudes are proportional, (then) they will also be proportional (when) composed. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha + \beta : \beta :: \gamma + \delta : \delta$.

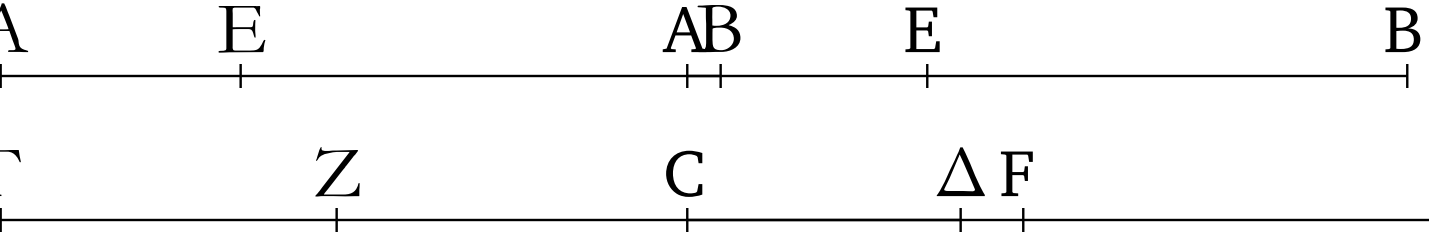
[‡] Here, Euclid assumes, without proof, that a fourth magnitude proportional to three given magnitudes can always be found.

19.

Proposition 19[†]

Ἐάν ᾗ ὅς ὅλον πρὸς ὅλον, οὕτως ἀφαιρεθέν πρὸς ἀφαιρεθέν, καὶ τὸ λοιπὸν πρὸς τὸ λοιπὸν ἔσται ὥς ὅλον πρὸς ὅλον.

If as the whole is to the whole so the (part) taken away is to the (part) taken away, (then) the remainder to the remainder will also be as the whole (is) to the whole.



Ἐστω γάρ ὡς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$, οὕτως ἀφαιρεθέν τὸ AE πρὸς ἀφαιρεθέν τὸ ΓZ · λέγω, ὅτι καὶ λοιπὸν τὸ EB πρὸς λοιπὸν τὸ $Z\Delta$ ἔσται ὡς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$.

Ἐπεὶ γάρ ἐστιν ὡς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ AE πρὸς τὸ ΓZ , καὶ ἐναλλάξ ὡς τὸ BA πρὸς τὸ AE , οὕτως τὸ $\Delta\Gamma$ πρὸς τὸ ΓZ . καὶ ἐπεὶ συγκείμενα μεγέθη ἀνάλογόν ἐστιν, καὶ διαιρεθέντα ἀνάλογον ἔσται, ὡς τὸ BE πρὸς τὸ EA , οὕτως τὸ ΔZ πρὸς τὸ ΓZ · καὶ ἐναλλάξ, ὡς τὸ BE πρὸς τὸ ΔZ , οὕτως τὸ EA πρὸς τὸ $Z\Gamma$. ὡς δὲ τὸ AE πρὸς τὸ ΓZ , οὕτως ὑπόκειται ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$. καὶ λοιπὸν ἄρα τὸ EB πρὸς λοιπὸν τὸ $Z\Delta$ ἔσται ὡς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$.

Ἐὰν ἄρα ᾗ ὡς ὅλον πρὸς ὅλον, οὕτως ἀφαιρεθέν πρὸς ἀφαιρεθέν, καὶ τὸ λοιπὸν πρὸς τὸ λοιπὸν ἔσται ὡς ὅλον πρὸς ὅλον [ὅπερ ἔδει δεῖξαι].

[Καὶ ἐπεὶ ἐδείχθη ὡς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ EB πρὸς τὸ $Z\Delta$, καὶ ἐναλλάξ ὡς τὸ AB πρὸς τὸ BE οὕτως τὸ $\Gamma\Delta$ πρὸς τὸ $Z\Delta$, συγκείμενα ἄρα μεγέθη ἀνάλογόν ἐστιν· ἐδείχθη δὲ ὡς τὸ BA πρὸς τὸ AE , οὕτως τὸ $\Delta\Gamma$ πρὸς τὸ ΓZ · καὶ ἐστὶν ἀναστρέφαντι].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν συγκείμενα μεγέθη ἀνάλογον ᾗ, καὶ ἀναστρέφαντι ἀνάλογον ἔσται· ὅπερ ἔδει δεῖξαι.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \beta :: \alpha - \gamma : \beta - \delta$.

‡ In modern notation, this corollary reads that if $\alpha : \beta :: \gamma : \delta$ then $\alpha : \alpha - \beta :: \gamma : \gamma - \delta$.

20.

Ἐὰν ᾗ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύν-
δυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι' ἴσον δὲ τὸ πρῶτον
τοῦ τρίτου μείζον ᾗ, καὶ τὸ τέταρτον τοῦ ἑκτου μείζον ἔσται,
κ' ἂν ἴσον, ἴσον, κ' ἂν ἕλαττον, ἕλαττον.

For let the whole AB be to the whole CD as the (part) taken away AE (is) to the (part) taken away CF . I say that the remainder EB to the remainder FD will also be as the whole AB (is) to the whole CD .

For since as AB is to CD , so AE (is) to CF , (it is) also (the case), alternately, (that) as BA (is) to AE , so DC (is) to CF [Prop. 5.16]. And since composed magnitudes are proportional, (then) they will also be proportional (when) separated, (so that) as BE (is) to EA , so DF (is) to CF [Prop. 5.17]. Also, alternately, as BE (is) to DF , so EA (is) to FC [Prop. 5.16]. And it was assumed that as AE (is) to CF , so the whole AB (is) to the whole CD . And, thus, as the remainder EB (is) to the remainder FD , so the whole AB will be to the whole CD .

Thus, if as the whole is to the whole so the (part) taken away is to the (part) taken away, (then) the remainder to the remainder will also be as the whole (is) to the whole. [(Which is) the very thing it was required to show.]

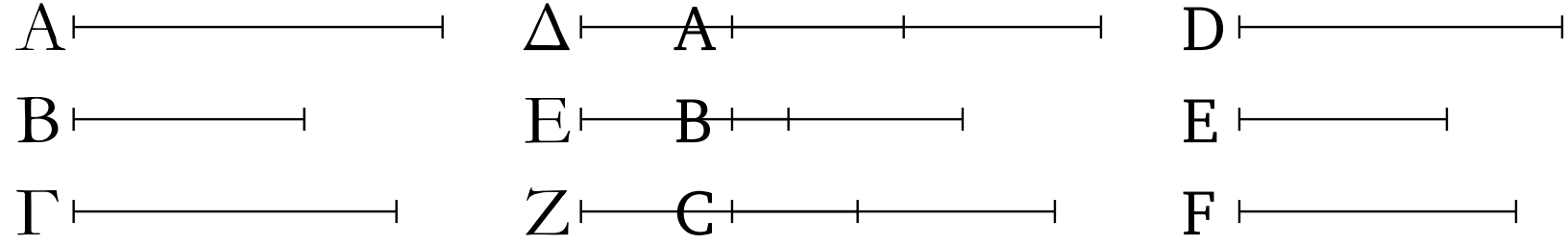
[And since it was shown (that) as AB (is) to CD , so EB (is) to FD , (it is) also (the case), alternately, (that) as AB (is) to BE , so CD (is) to FD . Thus, composed magnitudes are proportional. And it was shown (that) as BA (is) to AE , so DC (is) to CF . And (the latter) is converted (from the former).]

Corollary‡

So (it is) clear, from this, that if composed magnitudes are proportional, (then) they will also be proportional (when) converted. (Which is) the very thing it was required to show.

Proposition 20†

If there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth).



Ἐστω τρία μεγέθη τὰ A, B, Γ , καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, E, Z , σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ Δ πρὸς τὸ E , ὥς δὲ τὸ B πρὸς τὸ Γ , οὕτως τὸ E πρὸς τὸ Z , δι' ἴσον δὲ μείζον ἔστω τὸ A τοῦ Γ λέγω, ὅτι καὶ τὸ Δ τοῦ Z μείζον ἔσται, κ' ἂν ἴσον, ἴσον, κ' ἂν ἔλαττον, ἔλαττον.

Ἐπεὶ γὰρ μείζον ἐστὶ τὸ A τοῦ Γ , ἄλλο δέ τι τὸ B , τὸ δὲ μείζον πρὸς τὸ αὐτὸ μείζονα λόγον ἔθει ἥπερ τὸ ἔλαττον, τὸ A ἄρα πρὸς τὸ B μείζονα λόγον ἔθει ἥπερ τὸ Γ πρὸς τὸ B . ἀλλ' ὥς μὲν τὸ A πρὸς τὸ B [οὕτως] τὸ Δ πρὸς τὸ E , ὥς δὲ τὸ Γ πρὸς τὸ B , ἀνάπαλιν οὕτως τὸ Z πρὸς τὸ E · καὶ τὸ Δ ἄρα πρὸς τὸ E μείζονα λόγον ἔθει ἥπερ τὸ Z πρὸς τὸ E . τῶν δὲ πρὸς τὸ αὐτὸ λόγον ἐθόντων τὸ μείζονα λόγον ἔθον μείζον ἐστίν. μείζον ἄρα τὸ Δ τοῦ Z . ὁμοίως δὲ δείχουμεν, ὅτι κ' ἂν ἴσον ᾖ τὸ A τῷ Γ , ἴσον ἔσται καὶ τὸ Δ τῷ Z , κ' ἂν ἔλαττον, ἔλαττον.

Ἐὰν ἄρα ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι' ἴσον δὲ τὸ πρῶτον τοῦ τρίτου μείζον ᾖ, καὶ τὸ τέταρτον τοῦ ἑκτου μείζον ἔσται, κ' ἂν ἴσον, ἴσον, κ' ἂν ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖξαι.

Let A, B , and C be three magnitudes, and D, E, F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two, (so that) as A (is) to B , so D (is) to E , and as B (is) to C , so E (is) to F . And let A be greater than C , via equality. I say that D will also be greater than F . And if (A is) equal (to C then D will also be) equal (to F). And if (A is) less (than C then D will also be) less (than F).

For since A is greater than C , and B some other (magnitude), and the greater (magnitude) has a greater ratio than the lesser to the same (magnitude) [Prop. 5.8], A thus has a greater ratio to B than C (has) to B . But as A (is) to B , [so] D (is) to E . And, inversely, as C (is) to B , so F (is) to E [Prop. 5.7 corr.]. Thus, D also has a greater ratio to E than F (has) to E [Prop. 5.13]. And for (magnitudes) having a ratio to the same (magnitude), that having the greater ratio is greater [Prop. 5.10]. Thus, D (is) greater than F . Similarly, we can show that even if A is equal to C (then) D will also be equal to F , and even if (A is) less (than C then D will also be) less (than F).

Thus, if there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if), via equality, the first is greater than the third, (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And (if the first is) less (than the third then the fourth will also be) less (than the sixth). (Which is) the very thing it was required to show.

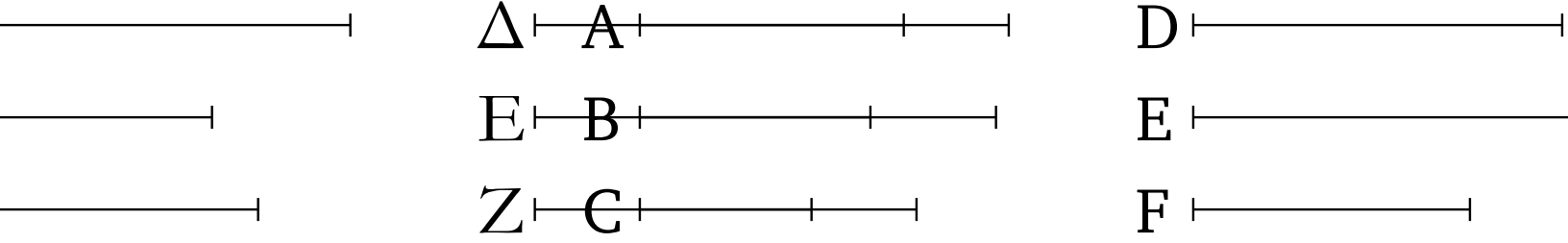
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \delta : \epsilon$ and $\beta : \gamma :: \epsilon : \zeta$ then $\alpha \gtrless \gamma$ as $\delta \gtrless \zeta$.

21.

Ἐὰν ᾖ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ᾗ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, δι' ἴσον δὲ τὸ πρῶτον τοῦ τρίτου μείζον ᾖ, καὶ τὸ τέταρτον τοῦ ἑκτου μείζον ἔσται, κ' ἂν ἴσον, ἴσον, κ' ἂν ἔλαττον, ἔλαττον.

Proposition 21[†]

If there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if) their proportion (is) perturbed, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth).



Ἐστω τρία μεγέθη τὰ A, B, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, E, Z , σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ἔστω δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z , ὥς δὲ τὸ B πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ E , δι' ἴσον δὲ τὸ A τοῦ Γ μείζον ἔστω· λέγω, ὅτι καὶ τὸ Δ τοῦ Z μείζον ἔσται, καὶ ἴσον, ἴσον, καὶ ἔλαττον, ἔλαττον.

Ἐπεὶ γὰρ μείζον ἐστὶ τὸ A τοῦ Γ , ἄλλο δέ τι τὸ B , τὸ A ἄρα πρὸς τὸ B μείζονα λόγον ἔθει ἢ περὶ τὸ Γ πρὸς τὸ B . ἀλλ' ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z , ὥς δὲ τὸ Γ πρὸς τὸ B , ἀνάπαλιν οὕτως τὸ E πρὸς τὸ Δ . καὶ τὸ E ἄρα πρὸς τὸ Z μείζονα λόγον ἔθει ἢ περὶ τὸ E πρὸς τὸ Δ . πρὸς ὃ δὲ τὸ αὐτὸ μείζονα λόγον ἔθει, ἐκείνο ἔλασσόν ἐστιν· ἔλασσον ἄρα ἐστὶ τὸ Z τοῦ Δ · μείζον ἄρα ἐστὶ τὸ Δ τοῦ Z . ὁμοίως δὲ δείκνυμεν, ὅτι καὶ ἴσον ἢ τὸ A τῷ Γ , ἴσον ἔσται καὶ τὸ Δ τῷ Z , καὶ ἔλαττον, ἔλαττον.

Ἐὰν ἄρα ἢ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, ἢ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, δι' ἴσον δὲ τὸ πρῶτον τοῦ τρίτου μείζον ἢ, καὶ τὸ τέταρτον τοῦ ἑκτου μείζον ἔσται, καὶ ἴσον, ἴσον, καὶ ἔλαττον, ἔλαττον· ὅπερ ἔδει δεῖχαι.

Let A, B , and C be three magnitudes, and D, E, F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two. And let their proportion be perturbed, (so that) as A (is) to B , so E (is) to F , and as B (is) to C , so D (is) to E . And let A be greater than C , via equality. I say that D will also be greater than F . And if $(A$ is) equal (to C then D will also be) equal (to F). And if $(A$ is) less (than C then D will also be) less (than F).

For since A is greater than C , and B some other (magnitude), A thus has a greater ratio to B than C (has) to B [Prop. 5.8]. But as A (is) to B , so E (is) to F . And, inversely, as C (is) to B , so E (is) to D [Prop. 5.7 corr.]. Thus, E also has a greater ratio to F than E (has) to D [Prop. 5.13]. And that (magnitude) to which the same (magnitude) has a greater ratio is (the) lesser (magnitude) [Prop. 5.10]. Thus, F is less than D . Thus, D is greater than F . Similarly, we can show that even if A is equal to C (then) D will also be equal to F , and even if $(A$ is) less (than C then D will also be) less (than F).

Thus, if there are three magnitudes, and others of equal number to them, (being) also in the same ratio taken two by two, and (if) their proportion (is) perturbed, and (if), via equality, the first is greater than the third (then) the fourth will also be greater than the sixth. And if (the first is) equal (to the third then the fourth will also be) equal (to the sixth). And if (the first is) less (than the third then the fourth will also be) less (than the sixth). (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \delta : \epsilon$ then $\alpha \gtrless \gamma$ as $\delta \gtrless \zeta$.

22.

Ἐὰν ἢ ὅποσαοῦν μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδυο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται.

Proposition 22[†]

If there are any number of magnitudes whatsoever, and (some) other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, (then) they will also be in the same ratio via equality.

B A

BΓ

C

E D

EZ

F

K G

KM

M

Λ H

LN

N

Ἐστω ὅποσαοῦν μεγέθη τὰ A, B, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Δ, E, Z , σύνδνο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ Δ πρὸς τὸ E , ὥς δὲ τὸ B πρὸς τὸ Γ , οὕτως τὸ E πρὸς τὸ Z . λέγω, ὅτι καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται.

Εἰλήφθω γάρ τῶν μὲν A, Δ ἰσάκις πολλαπλάσια τὰ H, Θ , τῶν δὲ B, E ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ K, Λ , καὶ ἔτι τῶν Γ, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ M, N .

Καὶ ἐπεὶ ἔστιν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ Δ πρὸς τὸ E , καὶ εἰληπται τῶν μὲν A, Δ ἰσάκις πολλαπλάσια τὰ H, Θ , τῶν δὲ B, E ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ K, Λ , ἔστιν ἄρα ὥς τὸ H πρὸς τὸ K , οὕτως τὸ Θ πρὸς τὸ Λ . διὰ τὰ αὐτὰ δὴ καὶ ὥς τὸ K πρὸς τὸ M , οὕτως τὸ Λ πρὸς τὸ N . ἐπεὶ οὖν τρία μεγέθη ἐστί τὰ H, K, M , καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ Θ, Λ, N , σύνδνο λαμβανόμενα καὶ ἐν τῷ αὐτῷ λόγῳ, δι' ἴσον ἄρα, εἰ ὑπερέθει τὸ H τοῦ M , ὑπερέθει καὶ τὸ Θ τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν H, Θ τῶν A, Δ ἰσάκις πολλαπλάσια, τὰ δὲ M, N τῶν Γ, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια. ἔστιν ἄρα ὥς τὸ A πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ Z .

Ἐὰν ἄρα ἢ ὅποσαοῦν μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος, σύνδνο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται. ὅπερ ἔδει δεῖχαι.

Let there be any number of magnitudes whatsoever, A, B, C , and (some) other (magnitudes), D, E, F , of equal number to them, (which are) in the same ratio taken two by two, (so that) as A (is) to B , so D (is) to E , and as B (is) to C , so E (is) to F . I say that they will also be in the same ratio via equality. (That is, as A is to C , so D is to F .)

For let the equal multiples G and H be taken of A and D (respectively), and the other random equal multiples K and L of B and E (respectively), and the yet other random equal multiples M and N of C and F (respectively).

And since as A is to B , so D (is) to E , and the equal multiples G and H have been taken of A and D (respectively), and the other random equal multiples K and L of B and E (respectively), thus as G is to K , so H (is) to L [Prop. 5.4]. And, so, for the same (reasons), as K (is) to M , so L (is) to N . Therefore, since G, K , and M are three magnitudes, and H, L , and N other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, thus, via equality, if G exceeds M (then) H also exceeds N , and if (G is) equal (to M then H is also) equal (to N), and if (G is) less (than M then H is also) less (than N) [Prop. 5.20]. And G and H are equal multiples of A and D (respectively), and M and N other random equal multiples of C and F (respectively). Thus, as A is to C , so D (is) to F [Def. 5.5].

Thus, if there are any number of magnitudes whatsoever, and (some) other (magnitudes) of equal number to them, (which are) also in the same ratio taken two by two, (then) they will also be in the same ratio via equality. (Which is) the very thing it was required to show.

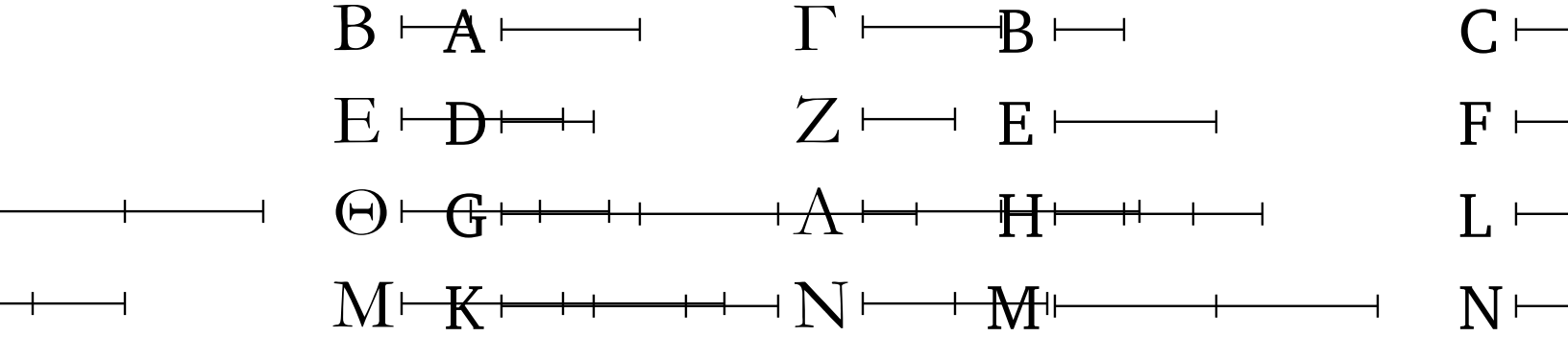
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \zeta : \eta$ and $\gamma : \delta :: \eta : \theta$ then $\alpha : \delta :: \epsilon : \theta$.

23.

Proposition 23[†]

Ἐὰν ἢ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδνο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ἢ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται.

If there are three magnitudes, and others of equal number to them, (being) in the same ratio taken two by two, and (if) their proportion is perturbed, (then) they will also be in the same ratio via equality.



Ἐστω τρία μεγέθη τὰ A, B, Γ καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ τὰ Δ, E, Z , ἔστω δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, ὥς μὲν τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z , ὥς δὲ τὸ B πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ E . λέγω, ὅτι ἔστιν ὥς τὸ A πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ Z .

Εἰλήφθω τῶν μὲν A, B, Δ ἰσάκις πολλαπλάσια τὰ H, Θ, K , τῶν δὲ Γ, E, Z ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια τὰ Λ, M, N .

Καὶ ἐπεὶ ἰσάκις ἐστὶ πολλαπλάσια τὰ H, Θ τῶν A, B , τὰ δὲ μέρη τοῖς ὡσάντως πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὥς τὸ A πρὸς τὸ B , οὕτως τὸ H πρὸς τὸ Θ . διὰ τὰ αὐτὰ δὴ καὶ ὥς τὸ E πρὸς τὸ Z , οὕτως τὸ M πρὸς τὸ N . καὶ ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως τὸ E πρὸς τὸ Z . καὶ ὥς ἄρα τὸ H πρὸς τὸ Θ , οὕτως τὸ M πρὸς τὸ N . καὶ ἐπεὶ ἐστὶν ὥς τὸ B πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ E , καὶ ἐναλλάχ ὥς τὸ B πρὸς τὸ Δ , οὕτως τὸ Γ πρὸς τὸ E . καὶ ἐπεὶ τὰ Θ, K τῶν B, Δ ἰσάκις ἐστὶ πολλαπλάσια, τὰ δὲ μέρη τοῖς ἰσάκις πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὥς τὸ B πρὸς τὸ Δ , οὕτως τὸ Θ πρὸς τὸ K . ἀλλ' ὥς τὸ B πρὸς τὸ Δ , οὕτως τὸ Γ πρὸς τὸ E . καὶ ὥς ἄρα τὸ Θ πρὸς τὸ K , οὕτως τὸ Γ πρὸς τὸ E . πάλιν, ἐπεὶ τὰ Λ, M τῶν Γ, E ἰσάκις ἐστὶ πολλαπλάσια, ἔστιν ἄρα ὥς τὸ Γ πρὸς τὸ E , οὕτως τὸ Λ πρὸς τὸ M . ἀλλ' ὥς τὸ Γ πρὸς τὸ E , οὕτως τὸ Θ πρὸς τὸ K . καὶ ὥς ἄρα τὸ Θ πρὸς τὸ K , οὕτως τὸ Λ πρὸς τὸ M , καὶ ἐναλλάχ ὥς τὸ Θ πρὸς τὸ Λ , τὸ K πρὸς τὸ M . ἐδείχθη δὲ καὶ ὥς τὸ H πρὸς τὸ Θ , οὕτως τὸ M πρὸς τὸ N . ἐπεὶ οὖν τρία μεγέθη ἐστὶ τὰ H, Θ, Λ , καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος τὰ K, M, N σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, καὶ ἐστὶν αὐτῶν τεταραγμένη ἡ ἀναλογία, δι' ἴσον ἄρα, εἰ ὑπερέθει τὸ H τοῦ Λ , ὑπερέθει καὶ τὸ K τοῦ N , καὶ εἰ ἴσον, ἴσον, καὶ εἰ ἔλαττον, ἔλαττον. καὶ ἐστὶ τὰ μὲν H, K τῶν A, Δ ἰσάκις πολλαπλάσια, τὰ δὲ Λ, N τῶν Γ, Z . ἔστιν ἄρα ὥς τὸ A πρὸς τὸ Γ , οὕτως τὸ Δ πρὸς τὸ Z .

Ἐὰν ἄρα ἢ τρία μεγέθη καὶ ἄλλα αὐτοῖς ἴσα τὸ πλῆθος σύνδυο λαμβανόμενα ἐν τῷ αὐτῷ λόγῳ, ἢ δὲ τεταραγμένη αὐτῶν ἡ ἀναλογία, καὶ δι' ἴσον ἐν τῷ αὐτῷ λόγῳ ἔσται ὅπερ ἔδει δεῖξαι.

Let A, B , and C be three magnitudes, and D, E and F other (magnitudes) of equal number to them, (being) in the same ratio taken two by two. And let their proportion be perturbed, (so that) as A (is) to B , so E (is) to F , and as B (is) to C , so D (is) to E . I say that as A is to C , so D (is) to F .

Let the equal multiples G, H , and K be taken of A, B , and D (respectively), and the other random equal multiples L, M , and N of C, E , and F (respectively).

And since G and H are equal multiples of A and B (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as A (is) to B , so G (is) to H . And, so, for the same (reasons), as E (is) to F , so M (is) to N . And as A is to B , so E (is) to F . And, thus, as G (is) to H , so M (is) to N [Prop. 5.11]. And since as B is to C , so D (is) to E , also, alternately, as B (is) to D , so C (is) to E [Prop. 5.16]. And since H and K are equal multiples of B and D (respectively), and parts have the same ratio as similar multiples [Prop. 5.15], thus as B is to D , so H (is) to K . But, as B (is) to D , so C (is) to E . And, thus, as H (is) to K , so C (is) to E [Prop. 5.11]. Again, since L and M are equal multiples of C and E (respectively), thus as C is to E , so L (is) to M [Prop. 5.15]. But, as C (is) to E , so H (is) to K . And, thus, as H (is) to K , so L (is) to M [Prop. 5.11]. Also, alternately, as H (is) to L , so K (is) to M [Prop. 5.16]. And it was also shown (that) as G (is) to H , so M (is) to N . Therefore, since G, H , and L are three magnitudes, and K, M , and N other (magnitudes) of equal number to them, (being) in the same ratio taken two by two, and their proportion is perturbed, thus, via equality, if G exceeds L (then) K also exceeds N , and if (G is) equal (to L then K is also) equal (to N), and if (G is) less (than L then K is also) less (than N) [Prop. 5.21]. And G and K are equal multiples of A and D (respectively), and L and N of C and F (respectively). Thus, as A (is) to C , so D (is) to F [Def. 5.5].

Thus, if there are three magnitudes, and others of equal number to them, (being) in the same ratio taken two by two, and (if) their proportion is perturbed, (then) they will also be in the same ratio via equality. (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $\alpha : \beta :: \epsilon : \zeta$ and $\beta : \gamma :: \delta : \epsilon$ then $\alpha : \gamma :: \delta : \zeta$.

24.

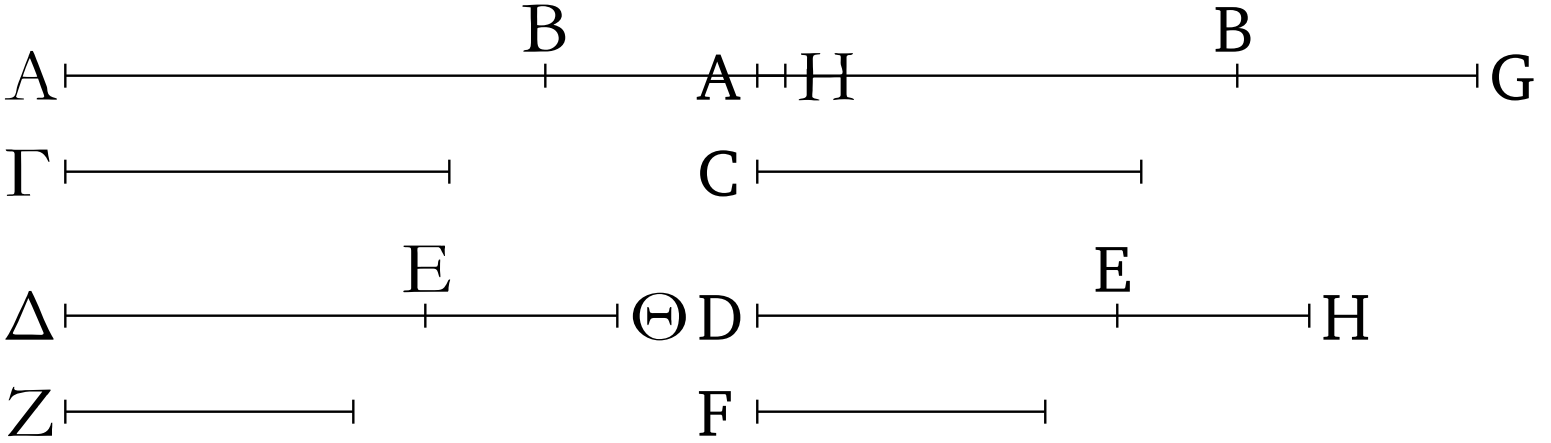
Proposition 24[†]

Ἐάν πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, ἔθῃ δὲ καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν λόγον καὶ ἕκτον πρὸς τέταρτον, καὶ συντεθέν πρῶτον καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν ἔχει λόγον καὶ τρίτον καὶ ἕκτον πρὸς τέταρτον.

Πρῶτον γάρ τὸ AB πρὸς δεύτερον τὸ Γ τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον τὸ ΔE πρὸς τέταρτον τὸ Z , ἔθῃ δὲ καὶ πέμπτον τὸ BH πρὸς δεύτερον τὸ Γ τὸν αὐτὸν λόγον καὶ ἕκτον τὸ $E\Theta$ πρὸς τέταρτον τὸ Z . λέγω, ὅτι καὶ συντεθέν πρῶτον καὶ πέμπτον τὸ AH πρὸς δεύτερον τὸ Γ τὸν αὐτὸν ἔχει λόγον, καὶ τρίτον καὶ ἕκτον τὸ $\Delta\Theta$ πρὸς τέταρτον τὸ Z .

If a first (magnitude) has to a second the same ratio that third (has) to a fourth, and a fifth (magnitude) also has to the second the same ratio that a sixth (has) to the fourth, (then) the first (magnitude) and the fifth, added together, will also have the same ratio to the second that the third (magnitude) and sixth (added together, have) to the fourth.

For let a first (magnitude) AB have the same ratio to a second C that a third DE (has) to a fourth F . And let a fifth (magnitude) BG also have the same ratio to the second C that a sixth EH (has) to the fourth F . I say that the first (magnitude) and the fifth, added together, AG , will also have the same ratio to the second C that the third (magnitude) and the sixth, (added together), DH , (has) to the fourth F .



Ἐπεὶ γάρ ἐστιν ὡς τὸ BH πρὸς τὸ Γ , οὕτως τὸ $E\Theta$ πρὸς τὸ Z , ἀνάπαλιν ἄρα ὡς τὸ Γ πρὸς τὸ BH , οὕτως τὸ Z πρὸς τὸ $E\Theta$. Ἐπεὶ οὖν ἐστιν ὡς τὸ AB πρὸς τὸ Γ , οὕτως τὸ ΔE πρὸς τὸ Z , ὡς δὲ τὸ Γ πρὸς τὸ BH , οὕτως τὸ Z πρὸς τὸ $E\Theta$, δι' ἴσον ἄρα ἐστὶν ὡς τὸ AB πρὸς τὸ BH , οὕτως τὸ ΔE πρὸς τὸ $E\Theta$. καὶ ἐπεὶ διηρημένα μεγέθη ἀνάλογόν ἐστιν, καὶ συντεθέντα ἀνάλογον ἔσται· ἔστιν ἄρα ὡς τὸ AH πρὸς τὸ HB , οὕτως τὸ $\Delta\Theta$ πρὸς τὸ ΘE . ἔστι δὲ καὶ ὡς τὸ BH πρὸς τὸ Γ , οὕτως τὸ $E\Theta$ πρὸς τὸ Z . δι' ἴσον ἄρα ἐστὶν ὡς τὸ AH πρὸς τὸ Γ , οὕτως τὸ $\Delta\Theta$ πρὸς τὸ Z .

Ἐάν ἄρα πρῶτον πρὸς δεύτερον τὸν αὐτὸν ἔθῃ λόγον καὶ τρίτον πρὸς τέταρτον, ἔθῃ δὲ καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν λόγον καὶ ἕκτον πρὸς τέταρτον, καὶ συντεθέν πρῶτον καὶ πέμπτον πρὸς δεύτερον τὸν αὐτὸν ἔχει λόγον καὶ τρίτον καὶ ἕκτον πρὸς τέταρτον· ὅπερ ἔδει δεῖχαι.

For since as BG is to C , so EH (is) to F , thus, inversely, as C (is) to BG , so F (is) to EH [Prop. 5.7 corr.]. Therefore, since as AB is to C , so DE (is) to F , and as C (is) to BG , so F (is) to EH , thus, via equality, as AB is to BG , so DE (is) to EH [Prop. 5.22]. And since separated magnitudes are proportional, (then) they will also be proportional (when) composed [Prop. 5.18]. Thus, as AG is to GB , so DH (is) to HE . And, also, as BG is to C , so EH (is) to F . Thus, via equality, as AG is to C , so DH (is) to F [Prop. 5.22].

Thus, if a first (magnitude) has to a second the same ratio that a third (has) to a fourth, and a fifth (magnitude) also has to the second the same ratio that a sixth (has) to the fourth, (then) the first (magnitude) and the fifth, added together, will also have the same ratio to the second that the third (magnitude) and the sixth (added together, have) to the fourth. (Which is) the very thing it was required to show.

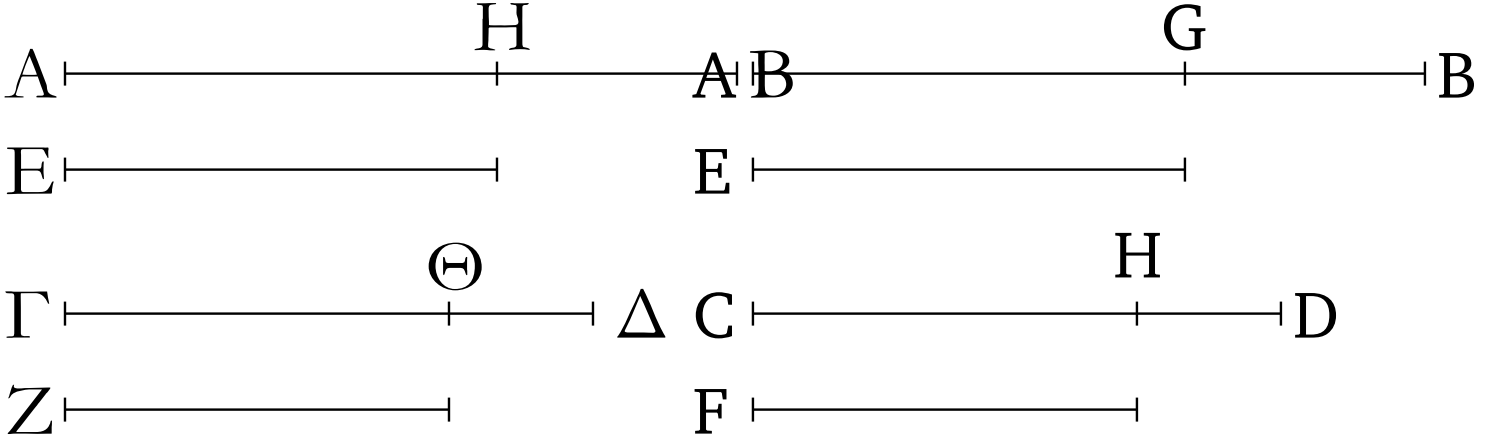
[†] In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$ and $\epsilon : \beta :: \zeta : \delta$ then $\alpha + \epsilon : \beta :: \gamma + \zeta : \delta$.

25.

Proposition 25[†]

Ἐάν τέσσαρα μεγέθη ἀνάλογον ᾗ, τὸ μέγιστον [αὐτῶν]
καὶ τὸ ἐλάχιστον δύο τῶν λοιπῶν μείζονά ἐστιν.

If four magnitudes are proportional, (then) the (sum of the)
largest and the smallest [of them] is greater than the (sum of
the) remaining two (magnitudes).



Ἐστω τέσσαρα μεγέθη ἀνάλογον τὰ AB , $\Gamma\Delta$, E , Z , ὥς τὸ
 AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ E πρὸς τὸ Z , ἔστω δὲ μέγιστον
μὲν αὐτῶν τὸ AB , ἐλάχιστον δὲ τὸ Z . λέγω, ὅτι τὰ AB , Z
τῶν $\Gamma\Delta$, E μείζονά ἐστιν.

Κείσθω γάρ τῳ μὲν E ἴσον τὸ AH , τῳ δὲ Z ἴσον τὸ $\Gamma\Theta$.

Ἐπεὶ [οὕτως] ἐστὶν ὥς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ E πρὸς
τὸ Z , ἴσον δὲ τὸ μὲν E τῳ AH , τὸ δὲ Z τῳ $\Gamma\Theta$, ἔστιν ἄρα
ὥς τὸ AB πρὸς τὸ $\Gamma\Delta$, οὕτως τὸ AH πρὸς τὸ $\Gamma\Theta$. καὶ ἐπεὶ
ἐστὶν ὥς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$, οὕτως ἀφαιρεθέν
τὸ AH πρὸς ἀφαιρεθέν τὸ $\Gamma\Theta$, καὶ λοιπὸν ἄρα τὸ HB πρὸς
λοιπὸν τὸ $\Theta\Delta$ ἔσται ὥς ὅλον τὸ AB πρὸς ὅλον τὸ $\Gamma\Delta$. μείζον
δὲ τὸ AB τοῦ $\Gamma\Delta$ · μείζον ἄρα καὶ τὸ HB τοῦ $\Theta\Delta$. καὶ ἐπεὶ
ἴσον ἐστὶ τὸ μὲν AH τῳ E , τὸ δὲ $\Gamma\Theta$ τῳ Z , τὰ ἄρα AH , Z
ἴσα ἐστὶ τοῖς $\Gamma\Theta$, E . καὶ [ἐπεὶ] ἔαν [ἀνίστοιξ ἴσα προστεθῇ, τὰ
ὅλα ἴσος ἐστὶν, ἔαν ἄρα] τῶν HB , $\Theta\Delta$ ἀνίσων ὄντων καὶ
μείζονος τοῦ HB τῳ μὲν HB προστεθῇ τὰ AH , Z , τῳ δὲ $\Theta\Delta$
προστεθῇ τὰ $\Gamma\Theta$, E , συνάγεται τὰ AB , Z μείζονα τῶν $\Gamma\Delta$,
 E .

Ἐάν ἄρα τέσσαρα μεγέθη ἀνάλογον ᾗ, τὸ μέγιστον αὐτῶν
καὶ τὸ ἐλάχιστον δύο τῶν λοιπῶν μείζονά ἐστιν. ὅπερ ἔδει
δεῖξαι.

Let AB , CD , E , and F be four proportional magnitudes,
(such that) as AB (is) to CD , so E (is) to F . And let AB be the
greatest of them, and F the least. I say that AB and F is greater
than CD and E .

For let AG be made equal to E , and CH equal to F .

[In fact,] since as AB is to CD , so E (is) to F , and E (is)
equal to AG , and F to CH , thus as AB is to CD , so AG (is)
to CH . And since the whole AB is to the whole CD as the
(part) taken away AG (is) to the (part) taken away CH , thus
the remainder GB will also be to the remainder HD as the
whole AB (is) to the whole CD [Prop. 5.19]. And AB (is)
greater than CD . Thus, GB (is) also greater than HD . And
since AG is equal to E , and CH to F , thus AG and F is equal
to CH and E . And [since] if [equal (magnitudes) are added to
unequal (magnitudes then) the wholes are unequal, thus if] AG
and F are added to GB , and CH and E to HD — GB and HD
(is) greater—it is inferred that AB and F (is) greater than CD and E .

Thus, if four magnitudes are proportional, (then) the (sum
of the) largest and the smallest of them is greater than the (sum
of the) remaining two (magnitudes). (Which is) the very thing
it was required to show.

† In modern notation, this proposition reads that if $\alpha : \beta :: \gamma : \delta$, and α is the greatest and δ the least, then $\alpha + \delta > \beta + \gamma$.

ELEMENTS BOOK 6

Similar Figures

Ὅροι.

1. Ὅμοια σθήματα εὐθύγραμμά ἐστιν, ὅσα τάς τε γωνίας ἴσαξ ἔδει κατά μίαν καὶ τὰς περὶ τὰς ἴσαξ γωνίας πλευράς ἀνάλογον.

2. Ἄκρον καὶ μέσον λόγον εὐθεῖα τετμησθαι λέγεται, ὅταν ἦ ὡς ἡ ὅλη πρὸς τὸ μείζον τμήμα, οὕτως τὸ μείζον πρὸς τὸ ἐλάττω.

3. Ὑψὸς ἐστὶ πάντοξ σθήματος ἡ ἀπὸ τῆς κορυφῆς ἐπὶ τὴν βάσιν κάθετος ἀγομένη.

Definitions

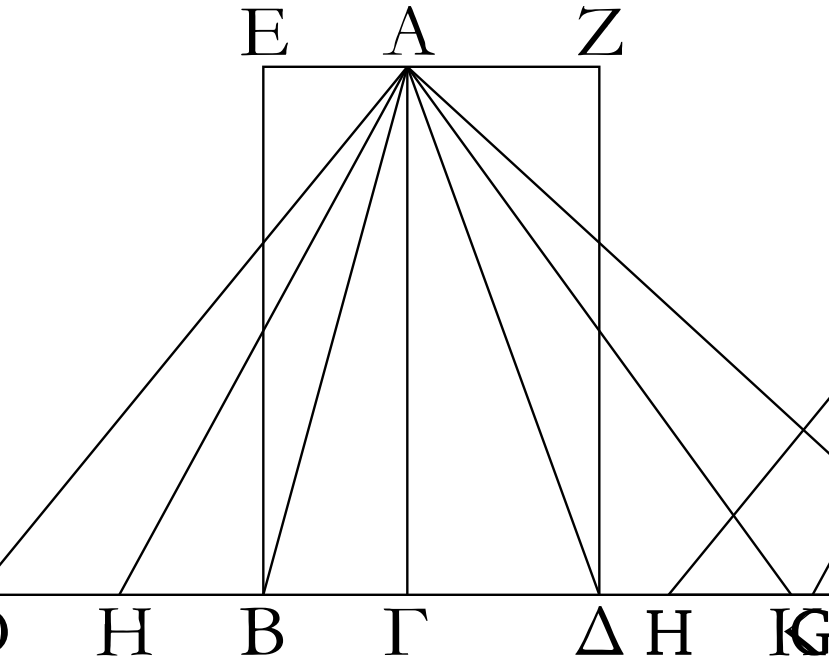
1. Similar rectilinear figures are those (which) have (their) angles separately equal, and the (corresponding) sides about the equal angles proportional.

2. A straight-line is said to be cut in extreme and mean ratio when as the whole is to the greater segment so the greater (segment is) to the lesser.

3. The height of any figure is the (straight-line) drawn from the vertex perpendicular to the base.

1.

Τὰ τρίγωνα καὶ τὰ παραλληλόγραμμα τὰ ὑπὸ τὸ αὐτὸ ὕψος ὄντα πρὸς ἄλληλά ἐστιν ὡς αἱ βάσεις.



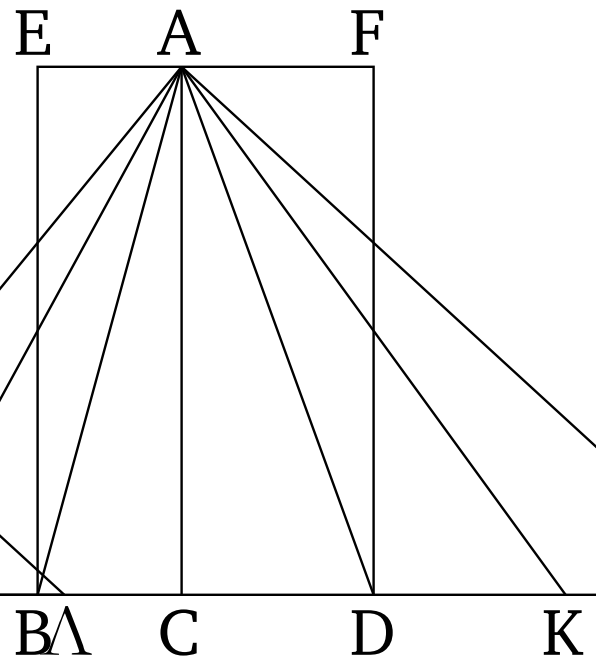
Ἐστω τρίγωνα μὲν τὰ $ABΓ$, $ACΔ$, παραλληλόγραμμα δὲ τὰ $EBΓ$, $ICΔ$ ὑπὸ τὸ αὐτὸ ὕψος τὸ AC · λέγω, ὅτι ἐστὶν ὡς ἡ BC βάσις πρὸς τὴν CD βάσιν, οὕτως τὸ $ABΓ$ τρίγωνον πρὸς τὸ $ACΔ$ τρίγωνον, καὶ τὸ $EBΓ$ παραλληλόγραμμον πρὸς τὸ $ICΔ$ παραλληλόγραμμον.

Ἐκβεβλήσθω γάρ ἡ BD ἐφ' ἑκάτερον τὰ μέρη ἐπὶ τὰ Θ , Λ σημεία, καὶ κείσθωσαν τῇ μὲν BC βάσει ἴσαι [ὁσαιδηποτοῦν] αἱ BH , $H\Theta$, τῇ δὲ CD βάσει ἴσαι ὁσαιδηποτοῦν αἱ ΔK , $K\Lambda$, καὶ ἐπεξεύθθωσαν αἱ AH , $A\Theta$, AK , $A\Lambda$.

Καὶ ἐπεὶ ἴσαι εἰσὶν αἱ GB , BH , $H\Theta$ ἀλλήλαις, ἴσα ἐστὶ καὶ τὰ $A\Theta H$, AHB , $ABΓ$ τρίγωνα ἀλλήλοις. ὁσαπλασίον ἄρα ἐστὶν ἡ $\Theta\Gamma$ βάσις τῆς BC βάσεως, τοσαυταπλάσιόν ἐστι καὶ τὸ $A\Theta\Gamma$ τρίγωνον τοῦ $ABΓ$ τριγώνου. διὰ τὰ αὐτὰ δὴ ὁσαπλασίον ἐστὶν ἡ AC βάσις τῆς CD βάσεως, τοσαυταπλάσιόν

Proposition 1[†]

Triangles and parallelograms which are of the same height are to one another as their bases.



Let ABC and ACD be triangles, and EC and CF parallelograms, of the same height AC . I say that as base BC is to base CD , so triangle ABC (is) to triangle ACD , and parallelogram EC to parallelogram CF .

For let the (straight-line) BD be produced in each direction to points H and L , and let [any number] (of straight-lines) BG and GH be made equal to base BC , and any number (of straight-lines) DK and KL equal to base CD . And let AG , AH , AK , and AL be joined.

And since CB , BG , and GH are equal to one another, triangles AHG , AGB , and ABC are also equal to one another [Prop. 1.38]. Thus, as many times as base HC is (divisible by) base BC , so many times is triangle AHC also (divisible by) triangle ABC . So, for the same (reasons), as many times as

ἔστι καὶ τὸ $ΑΛΓ$ τρίγωνον τοῦ $ΑΓΔ$ τριγώνου· καὶ εἰ ἴση ἔστιν ἡ $ΘΓ$ βάσις τῇ $ΓΛ$ βάσει, ἴσον ἐστὶ καὶ τὸ $ΑΘΓ$ τρίγωνον τῷ $ΑΓΛ$ τριγώνῳ, καὶ εἰ ὑπερέθει ἡ $ΘΓ$ βάσις τῇς $ΓΛ$ βάσεως, ὑπερέθει καὶ τὸ $ΑΘΓ$ τρίγωνον τοῦ $ΑΓΛ$ τριγώνου, καὶ εἰ ἐλάσσων, ἔλασσον. τεσσάρων δὲ ὄντων μεγεθῶν δύο μὲν βάσεων τῶν $ΒΓ$, $ΓΔ$, δύο δὲ τριγώνων τῶν $ΑΒΓ$, $ΑΓΔ$ εἴληπται ἰσάκις πολλαπλάσια τῇς μὲν $ΒΓ$ βάσεως καὶ τοῦ $ΑΒΓ$ τριγώνου ἢ τε $ΘΓ$ βάσις καὶ τὸ $ΑΘΓ$ τρίγωνον, τῇς δὲ $ΓΔ$ βάσεως καὶ τοῦ $ΑΔΓ$ τριγώνου ἄλλα, ἃ ἔτυθεν, ἰσάκις πολλαπλάσια ἢ τε $ΛΓ$ βάσις καὶ τὸ $ΑΛΓ$ τρίγωνον· καὶ δέδεικται, ὅτι, εἰ ὑπερέθει ἡ $ΘΓ$ βάσις τῇς $ΓΛ$ βάσεως, ὑπερέθει καὶ τὸ $ΑΘΓ$ τρίγωνον τοῦ $ΑΛΓ$ τριγώνου, καὶ εἰ ἴση, ἴσον, καὶ εἰ ἐλάσσων, ἔλασσον· ἔστιν ἄρα ὡς ἡ $ΒΓ$ βάσις πρὸς τὴν $ΓΔ$ βάσιν, οὕτως τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΑΓΔ$ τρίγωνον.

Καὶ ἐπεὶ τοῦ μὲν $ΑΒΓ$ τριγώνου διπλάσιόν ἐστι τὸ $ΕΓ$ παραλληλόγραμμον, τοῦ δὲ $ΑΓΔ$ τριγώνου διπλάσιόν ἐστι τὸ $ΖΓ$ παραλληλόγραμμον, τὰ δὲ μέρη τοῖς ὡσαύτως πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὡς τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΑΓΔ$ τρίγωνον, οὕτως τὸ $ΕΓ$ παραλληλόγραμμον πρὸς τὸ $ΖΓ$ παραλληλόγραμμον. ἐπεὶ οὖν ἐδείχθη, ὡς μὲν ἡ $ΒΓ$ βάσις πρὸς τὴν $ΓΔ$, οὕτως τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΑΓΔ$ τρίγωνον, ὡς δὲ τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΑΓΔ$ τρίγωνον, οὕτως τὸ $ΕΓ$ παραλληλόγραμμον πρὸς τὸ $ΓΖ$ παραλληλόγραμμον, καὶ ὡς ἄρα ἡ $ΒΓ$ βάσις πρὸς τὴν $ΓΔ$ βάσιν, οὕτως τὸ $ΕΓ$ παραλληλόγραμμον πρὸς τὸ $ΖΓ$ παραλληλόγραμμον.

Τὰ ἄρα τρίγωνα καὶ τὰ παραλληλόγραμμα τὰ ὑπὸ τὸ αὐτὸ ὕψος ὄντα πρὸς ἀλλήλα ἐστὶν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖξαι.

base LC is (divisible) by base CD , so many times is triangle ALC also (divisible) by triangle ACD . And if base HC is equal to base CL , (then) triangle AHC is also equal to triangle ACL [Prop. 1.38]. And if base HC exceeds base CL , (then) triangle AHC also exceeds triangle ACL .[‡] And if (HC is) less (than CL then AHC is also) less (than ACL). So, their being four magnitudes, two bases, BC and CD , and two triangles, ABC and ACD , equal multiples have been taken of base BC and triangle ABC —(namely), base HC and triangle AHC —and other random equal multiples of base CD and triangle ADC —(namely), base LC and triangle ALC . And it has been shown that if base HC exceeds base CL , (then) triangle AHC also exceeds triangle ALC , and if (HC is) equal (to CL then AHC is also) equal (to ALC), and if (HC is) less (than CL then AHC is also) less (than ALC). Thus, as base BC is to base CD , so triangle ABC (is) to triangle ACD [Def. 5.5]. And since parallelogram EC is double triangle ABC , and parallelogram FC is double triangle ACD [Prop. 1.34], and parts have the same ratio as similar multiples [Prop. 5.15], thus as triangle ABC is to triangle ACD , so parallelogram EC (is) to parallelogram FC . In fact, since it was shown that as base BC (is) to CD , so triangle ABC (is) to triangle ACD , and as triangle ABC (is) to triangle ACD , so parallelogram EC (is) to parallelogram CF , thus, also, as base BC (is) to base CD , so parallelogram EC (is) to parallelogram FC [Prop. 5.11].

Thus, triangles and parallelograms which are of the same height are to one another as their bases. (Which is) the very thing it was required to show.

[†] As is easily demonstrated, this proposition holds even when the triangles, or parallelograms, do not share a common side, and/or are not right-angled.

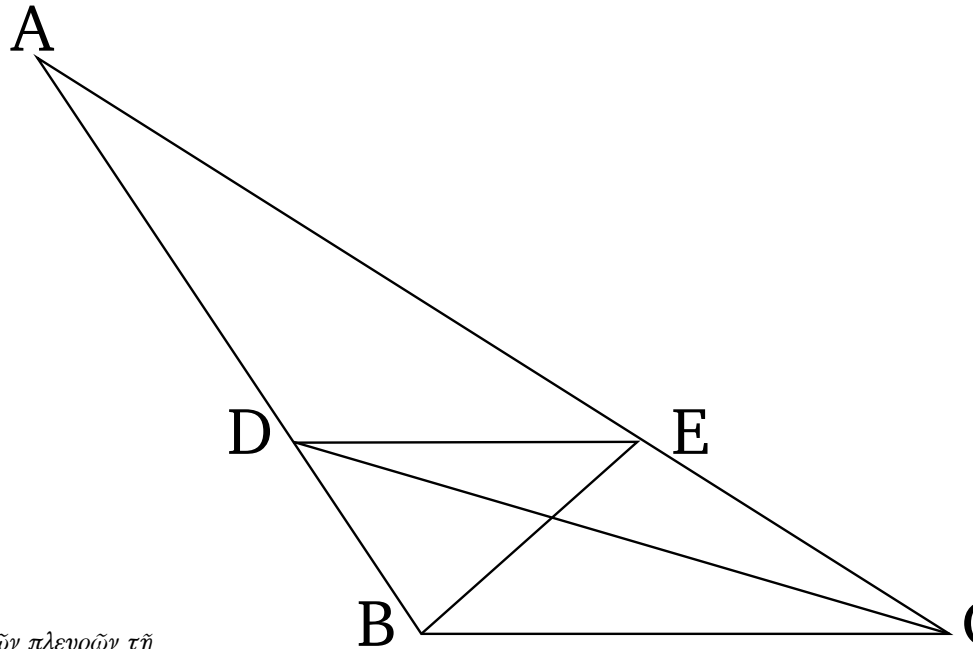
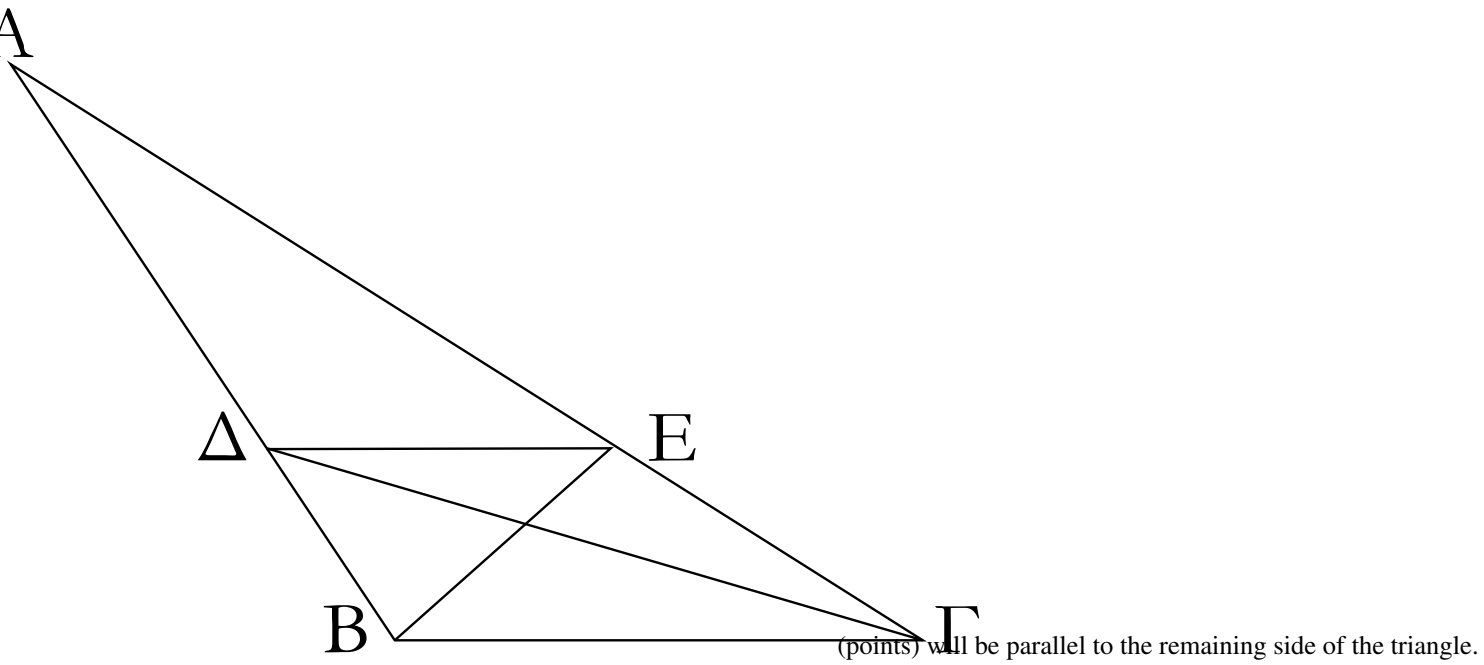
[‡] This is a straight-forward generalization of Prop. 1.38.

2.

Ἐάν τριγώνον παρὰ μίαν τῶν πλευρῶν ἀθθῇ τις εὐθεῖα, ἀνάλογον τεμεῖ τὰς τοῦ τριγώνου πλευράς· καὶ ἕαν αἱ τοῦ τριγώνου πλευραὶ ἀνάλογον τμηθῶσιν, ἢ ἐπὶ τὰς τομὰς ἐπιζευγνυμένη εὐθεῖα παρὰ τὴν λοιπὴν ἔσται τοῦ τριγώνου πλευράν.

Proposition 2

If some straight-line is drawn parallel to one of the sides of a triangle, (then) it will cut the (other) sides of the triangle proportionally. And if (two of) the sides of a triangle are cut proportionally, (then) the straight-line joining the cutting



Τριγώνον γάρ τοῦ $AB\Gamma$ παράλληλοξ μιᾶ τῶν πλευρῶν τῇ $B\Gamma$ ἤθθω ἡ ΔE · λέγω, ὅτι ἐστὶν ὡς ἡ $B\Delta$ πρὸς τὴν ΔA , οὔτως ἡ ΓE πρὸς τὴν EA .

Ἐπεξεύθθωσαν γὰρ αἱ BE , $\Gamma\Delta$.

Ἰσον ἄρα ἐστὶ τὸ $B\Delta E$ τριγώνον τῷ $\Gamma\Delta E$ τριγώνῳ· ἐπὶ γὰρ τῇ αὐτῇ βάσει ὡς ἐστὶ τῇ ΔE καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς ΔE , $B\Gamma$ · ἄλλο δέ τι τὸ $A\Delta E$ τριγώνον. τὰ δὲ ἴσα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον· ἐστὶν ἄρα ὡς τὸ $B\Delta E$ τριγώνον πρὸς τὸ $A\Delta E$ [τριγώνον], οὔτως τὸ $\Gamma\Delta E$ τριγώνον πρὸς τὸ $A\Delta E$ τριγώνον. ἀλλ' ὡς μὲν τὸ $B\Delta E$ τριγώνον πρὸς τὸ $A\Delta E$, οὔτως ἡ $B\Delta$ πρὸς τὴν ΔA · ὑπὸ γὰρ τὸ αὐτὸ ὕψος

For let DE be drawn parallel to one of the sides BC of triangle ABC . I say that as BD is to DA , so CE (is) to EA .

For let BE and CD be joined.

Thus, triangle BDE is equal to triangle CDE . For they are on the same base DE and between the same parallels DE and BC [Prop. 1.38]. And ADE is some other triangle. And equal (magnitudes) have the same ratio to the same (magnitude) [Prop. 5.7]. Thus, as triangle BDE is to [triangle] ADE , so triangle CDE (is) to triangle ADE . But, as triangle BDE (is) to triangle ADE , so (is) BD to DA . For, having the same

ὄντα τὴν ἀπὸ τοῦ E ἐπὶ τὴν AB κάθετον ἀγομένην πρὸς ἄλληλᾱ εἰσιν ὡς αἱ βάσεις. διὰ τὰ αὐτὰ δὴ ὡς τὸ $\Gamma\Delta E$ τρίγωνον πρὸς τὸ $A\Delta E$, οὕτως ἡ ΓE πρὸς τὴν EA . καὶ ὡς ἄρα ἡ $B\Delta$ πρὸς τὴν ΔA , οὕτως ἡ ΓE πρὸς τὴν EA .

Ἀλλὰ δὴ αἱ τοῦ $AB\Gamma$ τριγώνων πλευραὶ αἱ AB , AG ἀνάλογον τετμήσθωσαν, ὡς ἡ $B\Delta$ πρὸς τὴν ΔA , οὕτως ἡ ΓE πρὸς τὴν EA , καὶ ἐπεξεύθθω ἡ ΔE . λέγω, ὅτι παράλληλός ἐστιν ἡ ΔE τῇ $B\Gamma$.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἐστὶν ὡς ἡ $B\Delta$ πρὸς τὴν ΔA , οὕτως ἡ ΓE πρὸς τὴν EA , ἀλλ' ὡς μὲν ἡ $B\Delta$ πρὸς τὴν ΔA , οὕτως τὸ $B\Delta E$ τρίγωνον πρὸς τὸ $A\Delta E$ τρίγωνον, ὡς δὲ ἡ ΓE πρὸς τὴν EA , οὕτως τὸ $\Gamma\Delta E$ τρίγωνον πρὸς τὸ $A\Delta E$ τρίγωνον, καὶ ὡς ἄρα τὸ $B\Delta E$ τρίγωνον πρὸς τὸ $A\Delta E$ τρίγωνον, οὕτως τὸ $\Gamma\Delta E$ τρίγωνον πρὸς τὸ $A\Delta E$ τρίγωνον. ἐκάτερον ἄρα τῶν $B\Delta E$, $\Gamma\Delta E$ τριγώνων πρὸς τὸ $A\Delta E$ τὸν αὐτὸν ἔθει λόγον. ἴσον ἄρα ἐστὶ τὸ $B\Delta E$ τρίγωνον τῷ $\Gamma\Delta E$ τριγώνῳ· καὶ εἰσιν ἐπὶ τῆς αὐτῆς βάσεως τῆς ΔE . τὰ δὲ ἴσα τρίγωνα καὶ ἐπὶ τῆς αὐτῆς βάσεως ὄντα καὶ ἐν ταῖς αὐταῖς παραλλήλοις ἐστίν. παράλληλός ἄρα ἐστὶν ἡ ΔE τῇ $B\Gamma$.

Ἐὰν ἄρα τρίγωνον παρὰ μίαν τῶν πλευρῶν ἀθῆται εὐθεῖα, ἀνάλογον τεμεί τὰς τοῦ τριγώνου πλευράς· καὶ ἔαν αἱ τοῦ τριγώνου πλευραὶ ἀνάλογον τμηθῶσιν, ἡ ἐπὶ τὰς τομάς ἐπιξυρνημένη εὐθεῖα παρὰ τὴν λοιπὴν ἔσται τοῦ τριγώνου πλευρᾶν ὅπερ ἔδει δεῖξαι.

3.

Ἐὰν τρίγωνον ἡ γωνία διῇται τμηθῇ, ἡ δὲ τέμνουσα τὴν γωνίαν εὐθεῖα τέμνη καὶ τὴν βάσιν, τὰ τῆς βάσεως τμήματα τὸν αὐτὸν ἔχει λόγον ταῖς λοιπαῖς τοῦ τριγώνου πλευραῖς· καὶ ἔαν τὰ τῆς βάσεως τμήματα τὸν αὐτὸν ἔθῃ λόγον ταῖς λοιπαῖς τοῦ τριγώνου πλευραῖς, ἡ ἀπὸ τῆς κορυφῆς ἐπὶ τὴν τομὴν ἐπιξυρνημένη εὐθεῖα διῇται τεμεί τὴν τοῦ τριγώνου γωνίαν.

Ἔστω τρίγωνον τὸ $AB\Gamma$, καὶ τετμήσθω ἡ ὑπὸ $BA\Gamma$ γωνία διῇται ὑπὸ τῆς AD εὐθείας· λέγω, ὅτι ἐστὶν ὡς ἡ $B\Delta$ πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ BA πρὸς τὴν AG .

Ἦθθω γὰρ διὰ τοῦ Γ τῇ ΔA παράλληλός ἡ ΓE , καὶ διαθθεῖσα ἡ BA συμπίπτει αὐτῇ κατὰ τὸ E .

Καὶ ἐπεὶ εἷς παραλλήλους τὰς AD , EG εὐθεῖα ἐνέπεσεν ἡ AG , ἡ ἄρα ὑπὸ AGE γωνία ἴση ἐστὶ τῇ ὑπὸ ΓAD . ἀλλ' ἡ ὑπὸ ΓAD τῇ ὑπὸ $BA\Delta$ ὑπόκειται ἴση· καὶ ἡ ὑπὸ $BA\Delta$ ἄρα τῇ ὑπὸ AGE ἐστὶν ἴση. πάλιν, ἐπεὶ εἷς παραλλήλους τὰς AD , EG εὐθεῖα ἐνέπεσεν ἡ BAE , ἡ ἐκτὸς γωνία ἡ ὑπὸ $BA\Delta$ ἴση ἐστὶ τῇ ἐντὸς τῇ ὑπὸ AEG . ἐδείχθη δὲ καὶ ἡ ὑπὸ AGE τῇ ὑπὸ $BA\Delta$ ἴση· καὶ ἡ ὑπὸ AGE ἄρα γωνία τῇ ὑπὸ AEG ἐστὶν ἴση· ὥστε καὶ πλευρὰ ἡ AE πλευρᾷ τῇ AG ἐστὶν ἴση. καὶ ἐπεὶ τρίγωνον τοῦ $B\Gamma E$ παρὰ μίαν τῶν πλευρῶν τὴν EG ἥκται ἡ AD , ἀνάλογον ἄρα ἐστὶν ὡς ἡ $B\Delta$ πρὸς τὴν $\Delta\Gamma$, οὕτως ἡ BA πρὸς τὴν AE . ἴση δὲ ἡ AE τῇ AG · ὡς ἄρα ἡ $B\Delta$ πρὸς τὴν $\Delta\Gamma$, οὕτως ἡ BA πρὸς τὴν AG .

height—(namely), the (straight-line) drawn from E perpendicular to AB —they are to one another as their bases [Prop. 6.1]. So, for the same (reasons), as triangle CDE (is) to ADE , so CE (is) to EA . And, thus, as BD (is) to DA , so CE (is) to EA [Prop. 5.11].

And so, let the sides AB and AC of triangle ABC be cut proportionally (such that) as BD (is) to DA , so CE (is) to EA . And let DE be joined. I say that DE is parallel to BC .

For, by the same construction, since as BD is to DA , so CE (is) to EA , but as BD (is) to DA , so triangle BDE (is) to triangle ADE , and as CE (is) to EA , so triangle CDE (is) to triangle ADE [Prop. 6.1], thus, also, as triangle BDE (is) to triangle ADE , so triangle CDE (is) to triangle ADE [Prop. 5.11]. Thus, triangles BDE and CDE each have the same ratio to ADE . Thus, triangle BDE is equal to triangle CDE [Prop. 5.9]. And they are on the same base DE . And equal triangles, which are also on the same base, are also between the same parallels [Prop. 1.39]. Thus, DE is parallel to BC .

Thus, if some straight-line is drawn parallel to one of the sides of a triangle, (then) it will cut the (other) sides of the triangle proportionally. And if (two of) the sides of a triangle are cut proportionally, (then) the straight-line joining the cutting (points) will be parallel to the remaining side of the triangle. (Which is) the very thing it was required to show.

Proposition 3

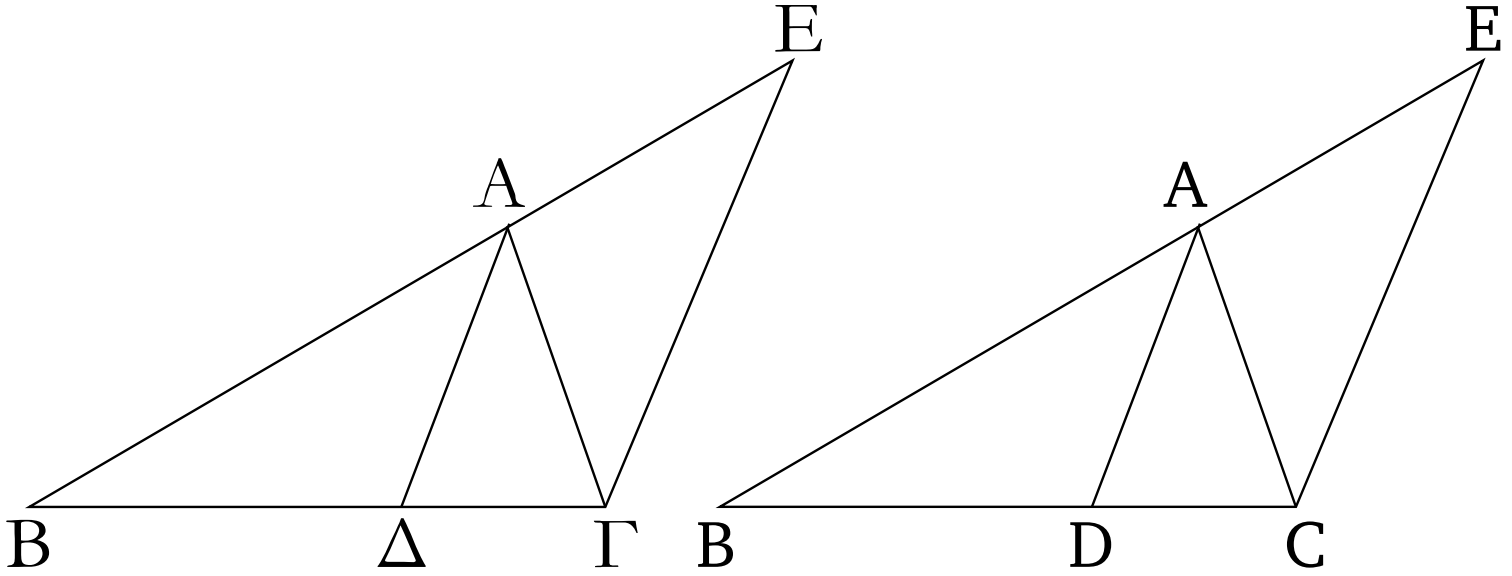
If an angle of a triangle is cut in half, and the straight-line cutting the angle also cuts the base, (then) the segments of the base will have the same ratio as the remaining sides of the triangle. And if the segments of the base have the same ratio as the remaining sides of the triangle, (then) the straight-line joining the vertex to the cutting (point) will cut the angle of the triangle in half.

Let ABC be a triangle. And let the angle BAC be cut in half by the straight-line AD . I say that as BD is to CD , so BA (is) to AC .

For let CE be drawn through (point) C parallel to DA . And, BA being drawn through, let it meet (CE) at (point) E .[†]

And since the straight-line AC falls across the parallel (straight-lines) AD and EC , angle ACE is thus equal to CAD [Prop. 1.29]. But, (angle) CAD is assumed (to be) equal to BAD . Thus, (angle) BAD is also equal to ACE . Again, since the straight-line BAE falls across the parallel (straight-lines) AD and EC , the external angle BAD is equal to the internal (angle) AEC [Prop. 1.29]. And (angle) ACE was also shown (to be) equal to BAD . Thus, angle ACE is also equal to AEC . And, hence, side AE is equal to side AC [Prop. 1.6]. And since AD has been drawn parallel to one of the sides EC of triangle BCE , thus, proportionally, as BD is to DC , so BA (is) to AE

[Prop. 6.2]. And AE (is) equal to AC . Thus, as BD (is) to DC , so BA (is) to AC .



Ἀλλὰ δὴ ἔστω ὥς ἡ BD πρὸς τὴν $\Delta\Gamma$, οὕτως ἡ BA πρὸς τὴν $ΑΓ$, καὶ ἐπεζεύθθω ἡ $ΑΔ$ · λέγω, ὅτι διῖτα τέτμηται ἡ ὑπὸ $BA\Gamma$ γωνία ὑπὸ τῆς $ΑΔ$ εὐθείας.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἐστὶν ὥς ἡ BD πρὸς τὴν $\Delta\Gamma$, οὕτως ἡ BA πρὸς τὴν $ΑΓ$, ἀλλὰ καὶ ὥς ἡ BD πρὸς τὴν $\Delta\Gamma$, οὕτως ἐστὶν ἡ BA πρὸς τὴν AE · τριγώνων γὰρ τοῦ $B\Gamma E$ παρὰ μίαν τὴν $E\Gamma$ ἥκται ἡ $ΑΔ$ · καὶ ὥς ἄρα ἡ BA πρὸς τὴν $ΑΓ$, οὕτως ἡ BA πρὸς τὴν AE . ἴση ἄρα ἡ $ΑΓ$ τῇ AE · ὥστε καὶ γωνία ἡ ὑπὸ $AE\Gamma$ τῇ ὑπὸ $ΑΓE$ ἐστὶν ἴση. ἀλλ' ἡ μὲν ὑπὸ $AE\Gamma$ τῇ ἐκτὸς τῇ ὑπὸ $BA\Delta$ [ἐστὶν] ἴση, ἡ δὲ ὑπὸ $ΑΓE$ τῇ ἐναλλάχ τῇ ὑπὸ $\Gamma A\Delta$ ἐστὶν ἴση· καὶ ἡ ὑπὸ $BA\Delta$ ἄρα τῇ ὑπὸ $\Gamma A\Delta$ ἐστὶν ἴση. ἡ ἄρα ὑπὸ $BA\Gamma$ γωνία διῖτα τέτμηται ὑπὸ τῆς $ΑΔ$ εὐθείας.

Ἐὰν ἄρα τριγώνων ἡ γωνία διῖτα τμηθῇ, ἡ δὲ τέμνουσα τὴν γωνίαν εὐθεία τέμνῃ καὶ τὴν βάσιν, τὰ τῆς βάσεως τμήματα τὸν αὐτὸν ἔχει λόγον ταῖς λοιπαῖς τοῦ τριγώνου πλευραῖς· καὶ εἰ τὰ τῆς βάσεως τμήματα τὸν αὐτὸν ἔθῃ λόγον ταῖς λοιπαῖς τοῦ τριγώνου πλευραῖς, ἡ ἀπὸ τῆς κορυφῆς ἐπὶ τὴν τομὴν ἐπιζευγνυμένη εὐθεία διῖτα τέμνει τὴν τοῦ τριγώνου γωνίαν· ὅπερ ἔδει δεῖχαι.

And so, let BD be to DC , as BA (is) to AC . And let AD be joined. I say that angle BAC has been cut in half by the straight-line AD .

For, by the same construction, since as BD is to DC , so BA (is) to AC , (then) also as BD (is) to DC , so BA is to AE . For AD has been drawn parallel to one (of the sides) EC of triangle BCE [Prop. 6.2]. Thus, also, as BA (is) to AC , so BA (is) to AE [Prop. 5.11]. Thus, AC (is) equal to AE [Prop. 5.9]. And, hence, angle AEC is equal to ACE [Prop. 1.5]. But, AEC [is] equal to the external (angle) BAD , and ACE is equal to the alternate (angle) CAD [Prop. 1.29]. Thus, (angle) BAD is also equal to CAD . Thus, angle BAC has been cut in half by the straight-line AD .

Thus, if an angle of a triangle is cut in half, and the straight-line cutting the angle also cuts the base, (then) the segments of the base will have the same ratio as the remaining sides of the triangle. And if the segments of the base have the same ratio as the remaining sides of the triangle, (then) the straight-line joining the vertex to the cutting (point) will cut the angle of the triangle in half. (Which is) the very thing it was required to show.

† The fact that the two straight-lines meet follows because the sum of ACE and CAE is less than two right-angles, as can easily be demonstrated. See Post. 5.

4.

Proposition 4

Τῶν ἰσογώνων τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας καὶ ὁμόλογοι αἱ ὑπὸ τὰς ἴσας γωνίας ὑποτείνουσιν.

Ἐστω ἰσογώνια τρίγωνα τὰ $AB\Gamma$, $\Delta\Gamma E$ ἴσην ἔχοντα τὴν μὲν ὑπὸ $AB\Gamma$ γωνίαν τῇ ὑπὸ $\Delta\Gamma E$, τὴν δὲ ὑπὸ $BA\Gamma$ τῇ ὑπὸ

In equiangular triangles the sides about the equal angles are proportional, and those (sides) subtending equal angles correspond.

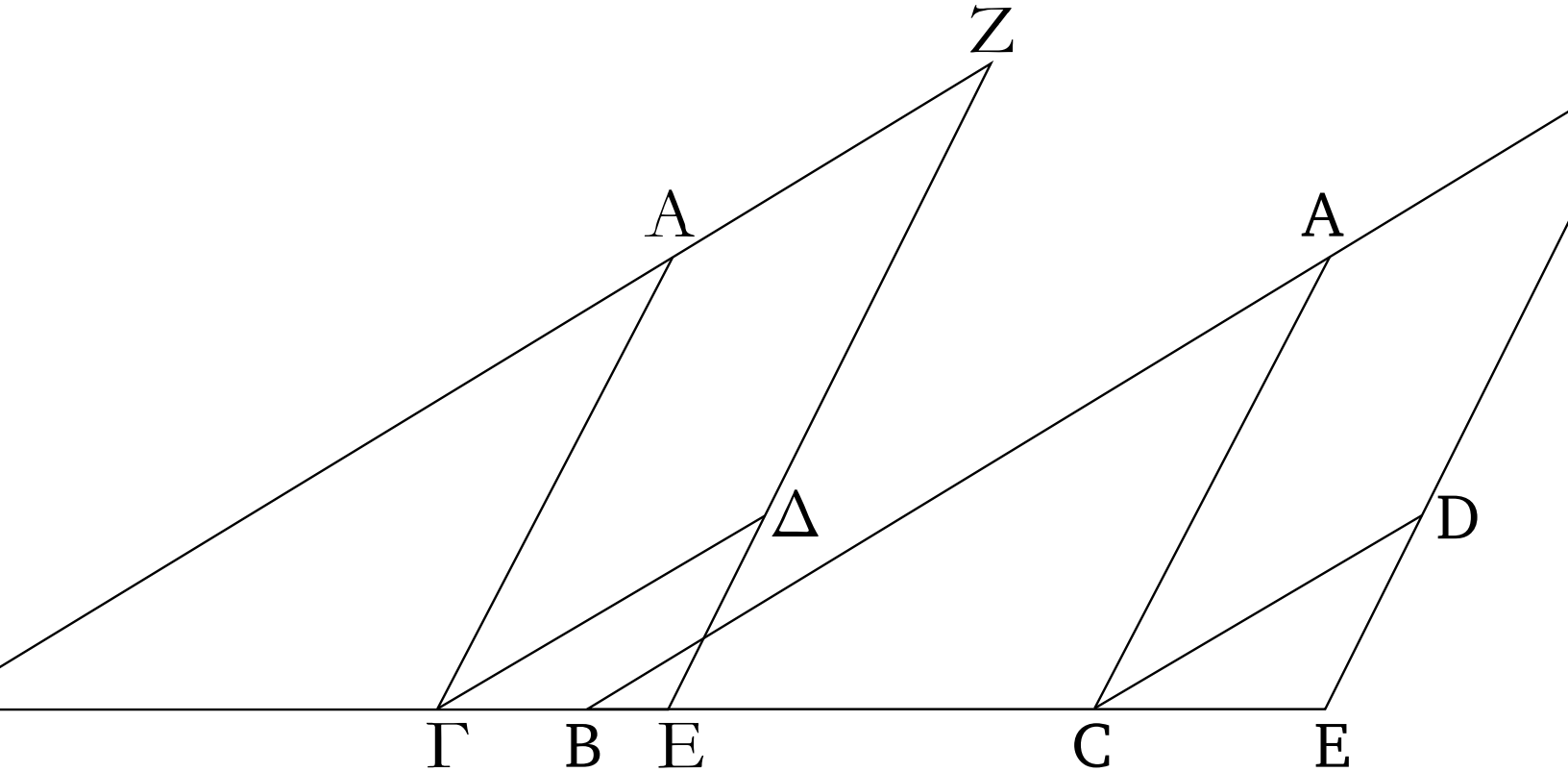
Let ABC and DCE be equiangular triangles, having angle ABC equal to DCE , and (angle) BAC to CDE , and, further,

ΓΔΕ καὶ ἔτι τὴν ὑπὸ ΑΓΒ τῇ ὑπὸ ΓΕΔ· λέγω, ὅτι τῶν ΑΒΓ, ΔΓΕ τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας καὶ ὁμόλογοι αἱ ὑπὸ τὰς ἴσας γωνίας ὑποτείνουσαι.

Κεῖσθω γάρ ἐπ' εὐθείας ἡ ΒΓ τῇ ΓΕ. καὶ ἐπεὶ αἱ ὑπὸ ΑΒΓ, ΑΓΒ γωνίαι δύο ὀρθῶν ἐλάττωνές εἰσιν, ἴση δὲ ἡ ὑπὸ ΑΓΒ τῇ ὑπὸ ΔΕΓ, αἱ ἄρα ὑπὸ ΑΒΓ, ΔΕΓ δύο ὀρθῶν ἐλάττωνές εἰσιν· αἱ ΒΑ, ΕΔ ἄρα ἐκβαλλόμεναι συμπεσοῦνται. ἐκβεβλήσθωσαν καὶ συμπιπτέωσαν κατὰ τὸ Ζ.

(angle) ACB to CED . I say that in triangles ABC and DCE the sides about the equal angles are proportional, and those (sides) subtending equal angles correspond.

Let BC be placed straight-on to CE . And since angles ABC and ACB are less than two right-angles [Prop 1.17], and ACB (is) equal to DEC , thus ABC and DEC are less than two right-angles. Thus, BA and ED , being produced, will meet [C.N. 5]. Let them be produced, and let them meet at (point) F .



Καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ ΔΓΕ γωνία τῇ ὑπὸ ΑΒΓ, παράλληλός ἐστιν ἡ ΒΖ τῇ ΓΔ. πάλιν, ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ ΑΓΒ τῇ ὑπὸ ΔΕΓ, παράλληλός ἐστιν ἡ ΑΓ τῇ ΖΕ. παραλληλόγραμμον ἄρα ἐστὶ τὸ ΖΑΓΔ· ἴση ἄρα ἡ μὲν ΖΑ τῇ ΔΓ, ἡ δὲ ΑΓ τῇ ΖΔ. καὶ ἐπεὶ τριγώνου τοῦ ΖΒΕ παρὰ μίαν τὴν ΖΕ ἥκται ἡ ΑΓ, ἐστὶν ἄρα ὡς ἡ ΒΑ πρὸς τὴν ΑΖ, οὕτως ἡ ΒΓ πρὸς τὴν ΓΕ. ἴση δὲ ἡ ΑΖ τῇ ΓΔ· ὡς ἄρα ἡ ΒΑ πρὸς τὴν ΓΔ, οὕτως ἡ ΒΓ πρὸς τὴν ΓΕ, καὶ ἐναλλάξ ὡς ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως ἡ ΔΓ πρὸς τὴν ΓΕ. πάλιν, ἐπεὶ παράλληλός ἐστιν ἡ ΓΔ τῇ ΒΖ, ἐστὶν ἄρα ὡς ἡ ΒΓ πρὸς τὴν ΓΕ, οὕτως ἡ ΖΔ πρὸς τὴν ΔΕ. ἴση δὲ ἡ ΖΔ τῇ ΑΓ· ὡς ἄρα ἡ ΒΓ πρὸς τὴν ΓΕ, οὕτως ἡ ΑΓ πρὸς τὴν ΔΕ, καὶ ἐναλλάξ ὡς ἡ ΒΓ πρὸς τὴν ΓΑ, οὕτως ἡ ΓΕ πρὸς τὴν ΕΔ. ἐπεὶ οὖν ἐδείχθη ὡς μὲν ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως ἡ ΔΓ πρὸς τὴν ΓΕ, ὡς δὲ ἡ ΒΓ πρὸς τὴν ΓΑ, οὕτως ἡ ΓΕ πρὸς τὴν ΕΔ, δι' ἴσου ἄρα ὡς ἡ ΒΑ πρὸς τὴν ΑΓ, οὕτως ἡ ΓΔ πρὸς τὴν ΔΕ.

Τῶν ἄρα ἰσογώνων τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ

And since angle DCE is equal to ABC , BF is parallel to CD [Prop. 1.28]. Again, since (angle) ACB is equal to DEC , AC is parallel to FE [Prop. 1.28]. Thus, $FACD$ is a parallelogram. Thus, FA is equal to DC , and AC to FD [Prop. 1.34]. And since AC has been drawn parallel to one (of the sides) FE of triangle FBE , thus as BA is to AF , so BC (is) to CE [Prop. 6.2]. And AF (is) equal to CD . Thus, as BA (is) to CD , so BC (is) to CE , and, alternately, as AB (is) to BC , so DC (is) to CE [Prop. 5.16]. Again, since CD is parallel to BF , thus as BC (is) to CE , so FD (is) to DE [Prop. 6.2]. And FD (is) equal to AC . Thus, as BC is to CE , so AC (is) to DE , and, alternately, as BC (is) to CA , so CE (is) to ED [Prop. 6.2]. Therefore, since it was shown that as AB (is) to BC , so DC (is) to CE , and as BC (is) to CA , so CE (is) to ED , thus, via equality, as BA (is) to AC , so CD (is) to DE [Prop. 5.22].

Thus, in equiangular triangles the sides about the equal angles are proportional, and those (sides) subtending equal an-

αἱ περὶ τὰς ἴσαξ γωνίας καὶ ὁμόλογοι αἱ ὑπὸ τὰς ἴσαξ γωνίας ὑποτείνουσιν· ὅπερ ἔδει δεῖχαι.

gles correspond. (Which is) the very thing it was required to show.

5.

Proposition 5

Ἐάν δύο τρίγωνα τὰς πλευράς ἀνάλογον ἔθῃ, ἰσογώνια ἔσται τὰ τρίγωνα καὶ ἴσαξ ἔχει τὰς γωνίας, ὅφ' ἂξ αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν.

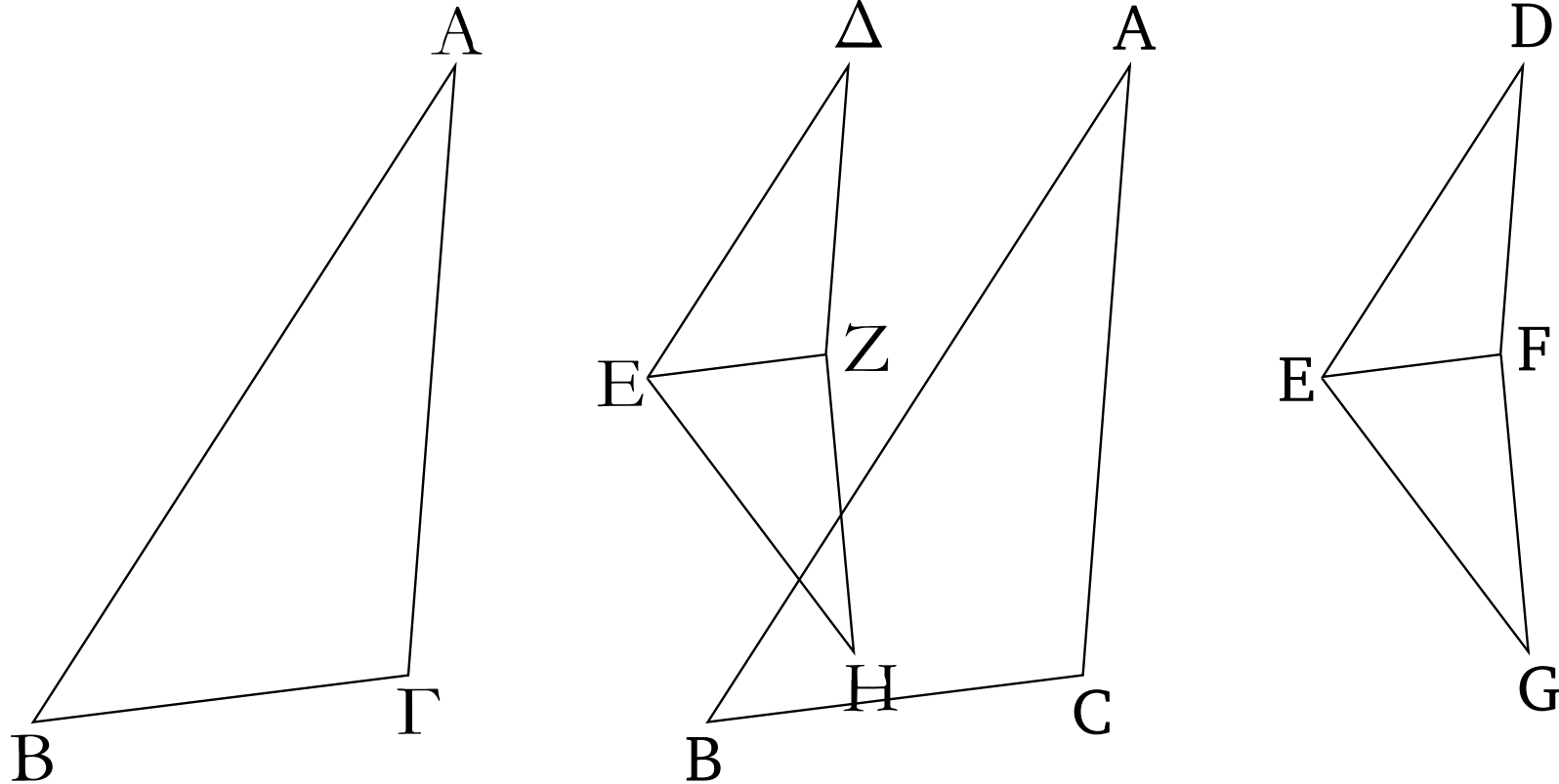
If two triangles have proportional sides, (then) the triangles will be equiangular, and will have the angles which corresponding sides subtend equal.

Ἐστω δύο τρίγωνα τὰ $AB\Gamma$, ΔEZ τὰς πλευράς ἀνάλογον ἔθοντα, ὥς μὲν τὴν AB πρὸς τὴν $B\Gamma$, οὕτως τὴν ΔE πρὸς τὴν EZ , ὥς δὲ τὴν $B\Gamma$ πρὸς τὴν ΓA , οὕτως τὴν EZ πρὸς τὴν $Z\Delta$, καὶ ἔτι ὥς τὴν BA πρὸς τὴν $A\Gamma$, οὕτως τὴν $E\Delta$ πρὸς τὴν ΔZ . λέγω, ὅτι ἰσογώνιον ἔστι τὸ $AB\Gamma$ τρίγωνον τῷ ΔEZ τριγώνῳ καὶ ἴσαξ ἔχουσι τὰς γωνίας, ὅφ' ἂξ αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν, τὴν μὲν ὑπὸ $AB\Gamma$ τῇ ὑπὸ ΔEZ , τὴν δὲ ὑπὸ $B\Gamma A$ τῇ ὑπὸ $EZ\Delta$ καὶ ἔτι τὴν ὑπὸ $B A \Gamma$ τῇ ὑπὸ $E \Delta Z$.

Let ABC and DEF be two triangles having proportional sides, (so that) as AB (is) to BC , so DE (is) to EF , and as BC (is) to CA , so EF (is) to FD , and, further, as BA (is) to AC , so ED (is) to DF . I say that triangle ABC is equiangular to triangle DEF , and (that the triangles) will have the angles which corresponding sides subtend equal. (That is), (angle) ABC (equal) to DEF , BCA to EFD , and, further, BAC to EDF .

Συνεστήτω γάρ πρὸς τῇ EZ εὐθείᾳ καὶ τοῖς πρὸς αὐτῇ σημείοις τοῖς E , Z τῇ μὲν ὑπὸ $AB\Gamma$ γωνίᾳ ἴση ἡ ὑπὸ ZEH , τῇ δὲ ὑπὸ $A\Gamma B$ ἴση ἡ ὑπὸ EZH . λοιπὴ ἄρα ἡ πρὸς τῷ A λοιπὴ τῇ πρὸς τῷ H ἔστιν ἴση.

For let (angle) FEG , equal to angle ABC , and (angle) EFG , equal to ACB , be constructed on the straight-line EF at the points E and F on it (respectively) [Prop. 1.23]. Thus, the remaining (angle) at A is equal to the remaining (angle) at G [Prop. 1.32].



Ἰσογώνιον ἄρα ἔστι τὸ $AB\Gamma$ τρίγωνον τῷ EZH [τριγώνῳ]. τῶν ἄρα $AB\Gamma$, EZH τριγώνων ἀναλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὰς ἴσαξ γωνίας καὶ ὁμόλογοι αἱ ὑπὸ τὰς ἴσαξ γωνίας ὑποτείνουσιν· ἔστιν ἄρα ὥς ἡ AB πρὸς τὴν $B\Gamma$, [οὕτως] ἡ HE πρὸς τὴν EZ . ἀλλ' ὥς ἡ AB πρὸς τὴν $B\Gamma$, οὕτως ὑπόκειται ἡ ΔE πρὸς τὴν EZ · ὥς ἄρα ἡ ΔE πρὸς τὴν EZ ,

Thus, triangle ABC is equiangular to [triangle] EGF . Thus, for triangles ABC and EGF , the sides about the equal angles are proportional, and (those) sides subtending equal angles correspond [Prop. 6.4]. Thus, as AB is to BC , [so] GE (is) to EF . But, as AB (is) to BC , so, it was assumed, (is) DE to EF . Thus, as DE (is) to EF , so GE (is) to EF [Prop. 5.11].

οὕτως ἡ HE πρὸς τὴν EZ . ἑκατέρα ἄρα τῶν DE , HE πρὸς τὴν EZ τὸν αὐτὸν ἔδει λόγον· ἴση ἄρα ἐστὶν ἡ DE τῇ HE . διὰ τὰ αὐτὰ δὴ καὶ ἡ ΔZ τῇ HZ ἐστὶν ἴση. ἐπεὶ οὖν ἴση ἐστὶν ἡ DE τῇ EH , κοινὴ δὲ ἡ EZ , δύο δὴ αἱ DE , EZ δυοὶ ταῖς HE , EZ ἴσαι εἰσὶν καὶ βάσει ἡ ΔZ βάσει τῇ ZH [ἐστὶν] ἴση· γωνία ἄρα ἡ ὑπὸ ΔEZ γωνία τῇ ὑπὸ HEZ ἐστὶν ἴση, καὶ τὸ ΔEZ τρίγωνον τῷ HEZ τριγώνῳ ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν. ἴση ἄρα ἐστὶ καὶ ἡ μὲν ὑπὸ ΔZE γωνία τῇ ὑπὸ HZE , ἡ δὲ ὑπὸ $E\Delta Z$ τῇ ὑπὸ EHZ . καὶ ἐπεὶ ἡ μὲν ὑπὸ $ZE\Delta$ τῇ ὑπὸ HEZ ἐστὶν ἴση, ἀλλ' ἡ ὑπὸ HEZ τῇ ὑπὸ $AB\Gamma$, καὶ ἡ ὑπὸ $AB\Gamma$ ἄρα γωνία τῇ ὑπὸ ΔEZ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $A\Gamma B$ τῇ ὑπὸ ΔZE ἐστὶν ἴση, καὶ ἔτι ἡ πρὸς τῷ A τῇ πρὸς τῷ Δ ἰσογώνιον ἄρα ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ ΔEZ τριγώνῳ.

Ἐὰν ἄρα δύο τρίγωνα τὰς πλευρὰς ἀνάλογον ἔθῃ, ἰσώνα ἐστὶ τὰ τρίγωνα καὶ ἴσαξ ἔχει τὰς γωνίας, ὅφ' ἂν αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν· ὅπερ ἔδει δεῖξαι.

6.

Ἐὰν δύο τρίγωνα μίαν γωνίαν μὲν γωνία ἴσην ἔθῃ, περὶ δὲ τὰς ἴσαξ γωνίας τὰς πλευρὰς ἀνάλογον, ἰσογώνα ἐστὶ τὰ τρίγωνα καὶ ἴσαξ ἔχει τὰς γωνίας, ὅφ' ἂν αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν.

Ἐστω δύο τρίγωνα τὰ $AB\Gamma$, ΔEZ μίαν γωνίαν τὴν ὑπὸ $BA\Gamma$ μὲν γωνία τῇ ὑπὸ $E\Delta Z$ ἴσην ἔθοντα, περὶ δὲ τὰς ἴσαξ γωνίας τὰς πλευρὰς ἀνάλογον, ὥς τὴν BA πρὸς τὴν $A\Gamma$, οὕτως τὴν $E\Delta$ πρὸς τὴν ΔZ . λέγω, ὅτι ἰσογώνιον ἐστὶ τὸ $AB\Gamma$ τρίγωνον τῷ ΔEZ τριγώνῳ καὶ ἴσην ἔχει τὴν ὑπὸ $AB\Gamma$ γωνίαν τῇ ὑπὸ ΔEZ , τὴν δὲ ὑπὸ $A\Gamma B$ τῇ ὑπὸ ΔZE .

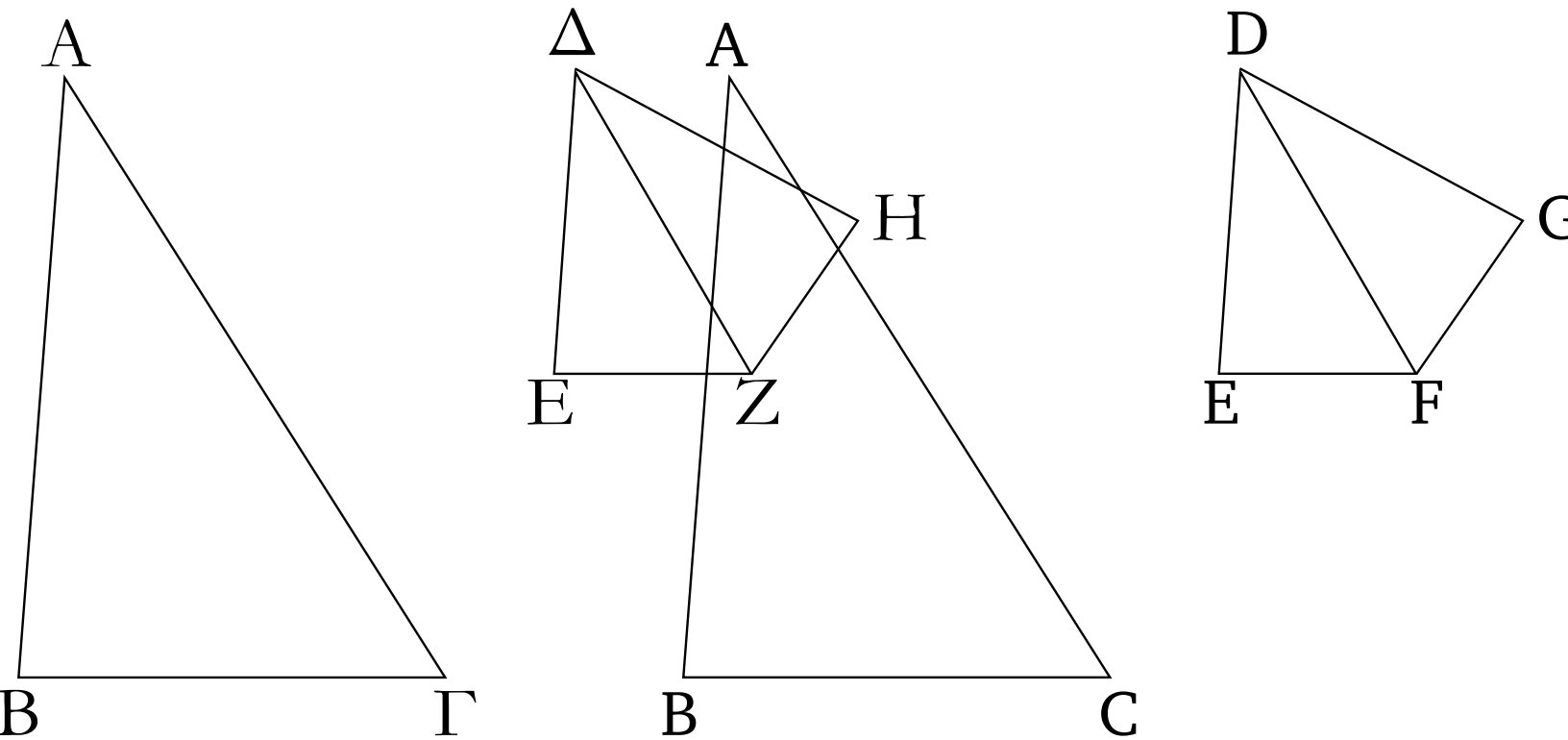
Thus, DE and GE each have the same ratio to EF . Thus, DE is equal to GE [Prop. 5.9]. So, for the same (reasons), DF is also equal to GF . Therefore, since DE is equal to EG , and EF (is) common, the two (sides) DE , EF are equal to the two (sides) GE , EF (respectively). And base DF [is] equal to base FG . Thus, angle DEF is equal to angle GEF [Prop. 1.8], and triangle DEF (is) equal to triangle GEF , and the remaining angles (are) equal to the remaining angles which the equal sides subtend [Prop. 1.4]. Thus, angle DFE is also equal to GFE , and (angle) EDF to EGF . And since (angle) FED is equal to GEF , and (angle) GEF to ABC , angle ABC is thus also equal to DEF . So, for the same (reasons), (angle) ACB is also equal to DFE , and, further, the (angle) at A to the (angle) at D . Thus, triangle ABC is equiangular to triangle DEF .

Thus, if two triangles have proportional sides, (then) the triangles will be equiangular, and will have the angles which corresponding sides subtend equal. (Which is) the very thing it was required to show.

Proposition 6

If two triangles have one angle equal to one angle, and the sides about the equal angles proportional, (then) the triangles will be equiangular, and will have the angles which corresponding sides subtend equal.

Let ABC and DEF be two triangles having one angle, BAC , equal to one angle, EDF (respectively), and the sides about the equal angles proportional, (so that) as BA (is) to AC , so ED (is) to DF . I say that triangle ABC is equiangular to triangle DEF , and will have angle ABC equal to DEF , and (angle) ACB to DFE .



Συνεστάτω γάρ πρὸς τῇ ΔΖ εὐθείᾳ καὶ τοῖς πρὸς αὐτῇ σημείοις τοῖς Δ, Ζ ὁποτέρῳ μὲν τῶν ὑπὸ ΒΑΓ, ΕΔΖ ἴση ἢ ὑπὸ ΖΔΗ, τῇ δὲ ὑπὸ ΑΓΒ ἴση ἢ ὑπὸ ΔΖΗ· λοιπὴ ἄρα ἢ πρὸς τῷ Β γωνία λοιπὴ τῇ πρὸς τῷ Η ἴση ἐστίν.

Ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΔΗΖ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ ΒΑ πρὸς τὴν ΑΓ, οὕτως ἡ ΗΔ πρὸς τὴν ΔΖ. ὑπόκειται δὲ καὶ ὡς ἡ ΒΑ πρὸς τὴν ΑΓ, οὕτως ἡ ΕΔ πρὸς τὴν ΔΖ· καὶ ὡς ἄρα ἡ ΕΔ πρὸς τὴν ΔΖ, οὕτως ἡ ΗΔ πρὸς τὴν ΔΖ. ἴση ἄρα ἡ ΕΔ τῇ ΔΗ· καὶ κοινὴ ἡ ΔΖ· δύο δὴ αἱ ΕΔ, ΔΖ δυοὶ ταῖς ΗΔ, ΔΖ ἴσας εἰσὶν· καὶ γωνία ἡ ὑπὸ ΕΔΖ γωνία τῇ ὑπὸ ΗΔΖ [ἐστίν] ἴση· βάσεις ἄρα ἡ ΕΖ βάσει τῇ ΗΖ ἐστὶν ἴση, καὶ τὸ ΔΕΖ τρίγωνον τῷ ΗΔΖ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωναὶ ταῖς λοιπαῖς γωνίαις ἴσας ἔσονται, ὅφ' ἂς ἴσας πλευραὶ ὑποτείνουσιν. ἴση ἄρα ἐστὶν ἡ μὲν ὑπὸ ΔΖΗ τῇ ὑπὸ ΔΖΕ, ἡ δὲ ὑπὸ ΔΗΖ τῇ ὑπὸ ΔΕΖ. ἀλλ' ἡ ὑπὸ ΔΖΗ τῇ ὑπὸ ΑΓΒ ἐστὶν ἴση· καὶ ἡ ὑπὸ ΑΓΒ ἄρα τῇ ὑπὸ ΔΖΕ ἐστὶν ἴση. ὑπόκειται δὲ καὶ ἡ ὑπὸ ΒΑΓ τῇ ὑπὸ ΕΔΖ ἴση· καὶ λοιπὴ ἄρα ἢ πρὸς τῷ Β λοιπὴ τῇ πρὸς τῷ Ε ἴση ἐστίν· ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΔΕΖ τριγώνῳ.

Ἐάν ἄρα δύο τρίγωνα μίαν γωνίαν μιᾷ γωνίᾳ ἴσην ἔθῃ, περὶ δὲ τὰς ἴσας γωνίας τὰς πλευρὰς ἀνάλογον, ἰσογώνια ἔσται τὰ τρίγωνα καὶ ἴσας ἔχει τὰς γωνίας, ὅφ' ἂς αἱ ὁμόλογοι πλευραὶ ὑποτείνουσιν· ὅπερ ἔδει δεῖχαι.

For let (angle) FDG , equal to each of BAC and EDF , and (angle) DFG , equal to ACB , be constructed on the straight-line AF at the points D and F on it (respectively) [Prop. 1.23]. Thus, the remaining angle at B is equal to the remaining angle at G [Prop. 1.32].

Thus, triangle ABC is equiangular to triangle DGF . Thus, proportionally, as BA (is) to AC , so GD (is) to DF [Prop. 6.4]. And it was also assumed that as BA (is) to AC , so ED (is) to DF . And, thus, as ED (is) to DF , so GD (is) to DF [Prop. 5.11]. Thus, ED (is) equal to DG [Prop. 5.9]. And DF (is) common. So, the two (sides) ED , DF are equal to the two (sides) GD , DF (respectively). And angle EDF [is] equal to angle GDF . Thus, base EF is equal to base GF , and triangle DEF is equal to triangle GDF , and the remaining angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4]. Thus, (angle) DFG is equal to DFE , and (angle) DGF to DEF . But, (angle) DFG is equal to ACB . Thus, (angle) ACB is also equal to DFE . And (angle) BAC was also assumed (to be) equal to EDF . Thus, the remaining (angle) at B is equal to the remaining (angle) at E [Prop. 1.32]. Thus, triangle ABC is equiangular to triangle DEF .

Thus, if two triangles have one angle equal to one angle, and the sides about the equal angles proportional, (then) the triangles will be equiangular, and will have the angles which corresponding sides subtend equal. (Which is) the very thing it was required to show.

7.

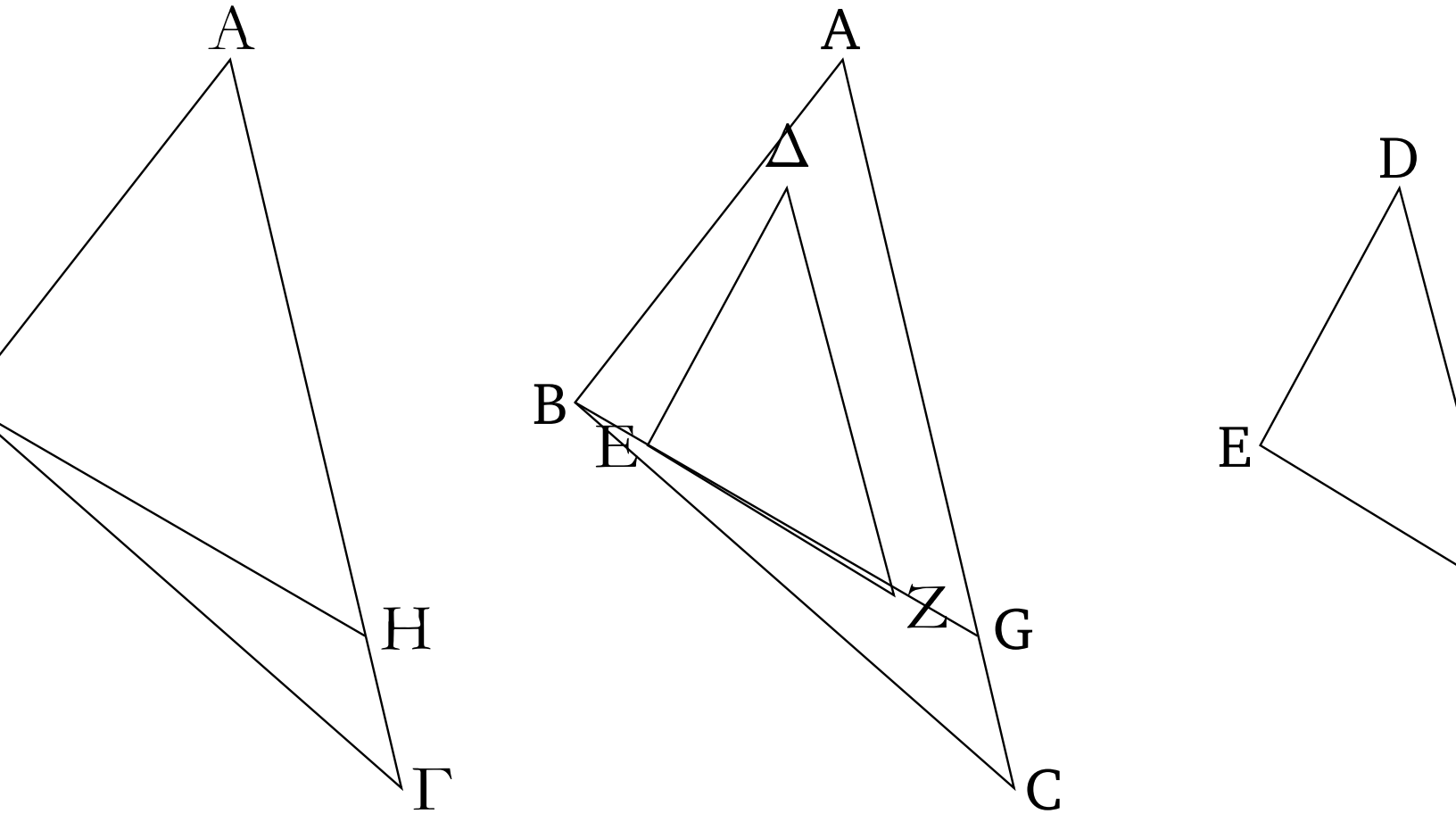
Proposition 7

Ἐάν δύο τρίγωνα μίαν γωνίαν μιᾷ γωνίᾳ ἴσην ἔθῃ, περί δέ ἄλλαξ γωνίαξ τὰς πλευράξ ἀνάλογον, τῶν δέ λοιπῶν ἑκατέραν ἅμα ἤτοι ἐλάσσονα ἢ μὴ ἐλάσσονα ὀρθῆξ, ἰσογώνια ἔσται τὰ τρίγωνα καὶ ἴσαξ ἔχει τὰς γωνίαξ, περί ἃξ ἀνάλογόν εἰσιν αἱ πλευραί.

Ἐστω δύο τρίγωνα τὰ $AB\Gamma$, ΔEZ μίαν γωνίαν μιᾷ γωνίᾳ ἴσην ἔθοντα τὴν ὑπὸ BAG τῇ ὑπὸ $E\Delta Z$, περί δέ ἄλλαξ γωνίαξ τὰς ὑπὸ $AB\Gamma$, ΔEZ τὰς πλευράξ ἀνάλογον, ὥς τὴν AB πρὸς τὴν $B\Gamma$, οὕτως τὴν ΔE πρὸς τὴν EZ , τῶν δέ λοιπῶν τῶν πρὸς τοῖς Γ , Z πρότερον ἑκατέραν ἅμα ἐλάσσονα ὀρθῆξ· λέγω, ὅτι ἰσογώνιον ἔστι τὸ $AB\Gamma$ τρίγωνον τῷ ΔEZ τριγώνῳ, καὶ ἴση ἔσται ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ ΔEZ , καὶ λοιπὴ δηλονότι ἡ πρὸς τῷ Γ λοιπῇ τῇ πρὸς τῷ Z ἴση.

If two triangles have one angle equal to one angle, and the sides about other angles proportional, and the remaining angles either both less than, or both not less than, right-angles, (then) the triangles will be equiangular, and will have the angles about which the sides are proportional equal.

Let ABC and DEF be two triangles having one angle, BAC , equal to one angle, EDF (respectively), and the sides about (some) other angles, ABC and DEF (respectively), proportional, (so that) as AB (is) to BC , so DE (is) to EF , and the remaining (angles) at C and F , first of all, both less than right-angles. I say that triangle ABC is equiangular to triangle DEF , and (that) angle ABC will be equal to DEF , and (that) the remaining (angle) at C (will be) manifestly equal to the remaining (angle) at F .



Εἰ γὰρ ἄνισός ἐστιν ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ ΔEZ , μία αὐτῶν μείζων ἔσται. Ἐστω μείζων ἡ ὑπὸ $AB\Gamma$. καὶ συνεστιάτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ B τῇ ὑπὸ ΔEZ γωνίᾳ ἴση ἡ ὑπὸ ABH .

Καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν A γωνία τῇ Δ , ἡ δὲ ὑπὸ ABH τῇ ὑπὸ ΔEZ , λοιπὴ ἄρα ἡ ὑπὸ AHB λοιπῇ τῇ ὑπὸ ΔZE

For if angle ABC is not equal to (angle) DEF , (then) one of them is greater. Let ABC be greater. And let (angle) ABG , equal to (angle) DEF , be constructed on the straight-line AB at the point B on it [Prop. 1.23].

And since angle A is equal to (angle) D , and (angle) ABG to DEF , the remaining (angle) AGB is thus equal to the re-

ἔστιν ἴση. ἰσογώνιον ἄρα ἐστὶ τὸ ABH τριγώνον τῷ ΔEZ τριγώνῳ. ἔστιν ἄρα ὥς ἡ AB πρὸς τὴν BH , οὕτως ἡ ΔE πρὸς τὴν EZ . ὥς δὲ ἡ ΔE πρὸς τὴν EZ , [οὕτως] ὑπόκειται ἡ AB πρὸς τὴν $B\Gamma$. ἡ AB ἄρα πρὸς ἑκατέραν τῶν $B\Gamma$, BH τὸν αὐτὸν ἔθει λόγον. ἴση ἄρα ἡ $B\Gamma$ τῇ BH . ὥστε καὶ γωνία ἡ πρὸς τῷ Γ γωνία τῇ ὑπὸ BHG ἐστὶν ἴση. ἐλάττων δὲ ὀρθῆς ὑπόκειται ἡ πρὸς τῷ Γ . ἐλάττων ἄρα ἐστὶν ὀρθῆς καὶ ὑπὸ BHG . ὥστε ἡ ἐφεχῆς αὐτῇ γωνία ἡ ὑπὸ AHB μείζων ἐστὶν ὀρθῆς. καὶ ἐδείκθη ἴση οὕσα τῇ πρὸς τῷ Z . καὶ ἡ πρὸς τῷ Z ἄρα μείζων ἐστὶν ὀρθῆς. ὑπόκειται δὲ ἐλάσσων ὀρθῆς. ὅπερ ἐστὶν ἄτοπον. οὐκ ἄρα ἄνισός ἐστιν ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ ΔEZ . ἴση ἄρα. ἔστι δὲ καὶ ἡ πρὸς τῷ A ἴση τῇ πρὸς τῷ Δ . καὶ λοιπὴ ἄρα ἡ πρὸς τῷ Γ λοιπῇ τῇ πρὸς τῷ Z ἴση ἐστίν. ἰσογώνιον ἄρα ἐστὶ τὸ $AB\Gamma$ τριγώνον τῷ ΔEZ τριγώνῳ.

Ἀλλὰ δὴ πάλιν ὑποκείσθω ἑκατέρα τῶν πρὸς τοῖς Γ , Z μὴ ἐλάσσων ὀρθῆς. λέγω πάλιν, ὅτι καὶ οὕτως ἐστὶν ἰσογώνιον τὸ $AB\Gamma$ τριγώνον τῷ ΔEZ τριγώνῳ.

Τῶν γάρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἴση ἐστὶν ἡ $B\Gamma$ τῇ BH . ὥστε καὶ γωνία ἡ πρὸς τῷ Γ τῇ ὑπὸ BHG ἴση ἐστίν. οὐκ ἐλάττων δὲ ὀρθῆς ἡ πρὸς τῷ Γ . οὐκ ἐλάττων ἄρα ὀρθῆς οὐδὲ ἡ ὑπὸ BHG . τριγώνον δὲ τοῦ BHG αἱ δύο γωνίαι δύο ὀρθῶν οὐκ εἰσιν ἐλάττονες. ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα πάλιν ἄνισός ἐστιν ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ ΔEZ . ἴση ἄρα. ἔστι δὲ καὶ ἡ πρὸς τῷ A τῇ πρὸς τῷ Δ ἴση. λοιπὴ ἄρα ἡ πρὸς τῷ Γ λοιπῇ τῇ πρὸς τῷ Z ἴση ἐστίν. ἰσογώνιον ἄρα ἐστὶ τὸ $AB\Gamma$ τριγώνον τῷ ΔEZ τριγώνῳ.

Ἐὰν ἄρα δύο τρίγωνα μίαν γωνίαν μιᾷ γωνίᾳ ἴσῃ ἔθῃ, περὶ δὲ ἄλλας γωνίας τὰς πλευρὰς ἀνάλογον, τῶν δὲ λοιπῶν ἑκατέραν ἅμα ἐλάττονα ἢ μὴ ἐλάττονα ὀρθῆς, ἰσογώνια ἔσται τὰ τρίγωνα καὶ ἴσα ἔχει τὰς γωνίας, περὶ ἃς ἀνάλογόν εἰσιν αἱ πλευραί. ὅπερ ἔδει δεῖχαι.

8.

Ἐὰν ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετοῦ ἀθῇ, τὰ πρὸς τῇ καθέτῳ τρίγωνα ὁμοιά ἐστι τῷ τε ὅλῳ καὶ ἀλλήλοις.

Ἔστω τριγώνον ὀρθογώνιον τὸ $AB\Gamma$ ὀρθὴν ἔθον τὴν ὑπὸ $BA\Gamma$ γωνίαν, καὶ ἦθῃ ἀπὸ τοῦ A ἐπὶ τὴν $B\Gamma$ κάθετοῦ ἡ AD . λέγω, ὅτι ὁμοιόν ἐστιν ἑκάτερον τῶν $AB\Delta$, $AD\Gamma$ τριγώνων ὅλῳ τῷ $AB\Gamma$ καὶ ἑτὶ ἀλλήλοις.

maining (angle) DFE [Prop. 1.32]. Thus, triangle ABG is equiangular to triangle DEF . Thus, as AB is to BG , so DE (is) to EF [Prop. 6.4]. And as DE (is) to EF , [so] it was assumed (is) AB to BC . Thus, AB has the same ratio to each of BC and BG [Prop. 5.11]. Thus, BC (is) equal to BG [Prop. 5.9]. And, hence, the angle at C is equal to angle BGC [Prop. 1.5]. And the angle at C was assumed (to be) less than a right-angle. Thus, (angle) BGC is also less than a right-angle. Hence, the adjacent angle to it, AGB , is greater than a right-angle [Prop. 1.13]. And (AGB) was shown to be equal to the (angle) at F . Thus, the (angle) at F is also greater than a right-angle. But it was assumed (to be) less than a right-angle. The very thing is absurd. Thus, angle ABC is not unequal to (angle) DEF . Thus, (it is) equal. And the (angle) at A is also equal to the (angle) at D . And thus the remaining (angle) at C is equal to the remaining (angle) at F [Prop. 1.32]. Thus, triangle ABC is equiangular to triangle DEF .

But, again, let each of the (angles) at C and F be assumed (to be) not less than a right-angle. I say, again, that triangle ABC is equiangular to triangle DEF in this case also.

For, with the same construction, we can similarly show that BC is equal to BG . Hence, also, the angle at C is equal to (angle) BGC . And the (angle) at C (is) not less than a right-angle. Thus, BGC (is) not less than a right-angle either. So, in triangle BGC the (sum of) two angles is not less than two right-angles. The very thing is impossible [Prop. 1.17]. Thus, again, angle ABC is not unequal to DEF . Thus, (it is) equal. And the (angle) at A is also equal to the (angle) at D . Thus, the remaining (angle) at C is equal to the remaining (angle) at F [Prop. 1.32]. Thus, triangle ABC is equiangular to triangle DEF .

Thus, if two triangles have one angle equal to one angle, and the sides about other angles proportional, and the remaining angles both less than, or both not less than, right-angles, (then) the triangles will be equiangular, and will have the angles about which the sides (are) proportional equal. (Which is) the very thing it was required to show.

Proposition 8

If, in a right-angled triangle, a (straight-line) is drawn from the right-angle perpendicular to the base, (then) the triangles around the perpendicular are similar to the whole (triangle), and to one another.

Let ABC be a right-angled triangle having the angle BAC a right-angle, and let AD be drawn from A , perpendicular to BC [Prop. 1.12]. I say that triangles ABD and ADC are each similar to the whole (triangle) ABC and, further, to one another.

(triangle), and to one another. [(Which is) the very thing it was required to show.]

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ἀθθῇ, ἡ ἀθθεῖσα τῶν τῆς βάσεως τμημάτων μέση ἀνάλογόν ἐστιν· ὅπερ ἔδει δεῖξαι.

Corollary

So (it is) clear, from this, that if, in a right-angled triangle, a (straight-line) is drawn from the right-angle perpendicular to the base, (then) the (straight-line so) drawn is in mean proportion to the pieces of the base.[†] (Which is) the very thing it was required to show.

[†] In other words, the perpendicular is the geometric mean of the pieces.

9.

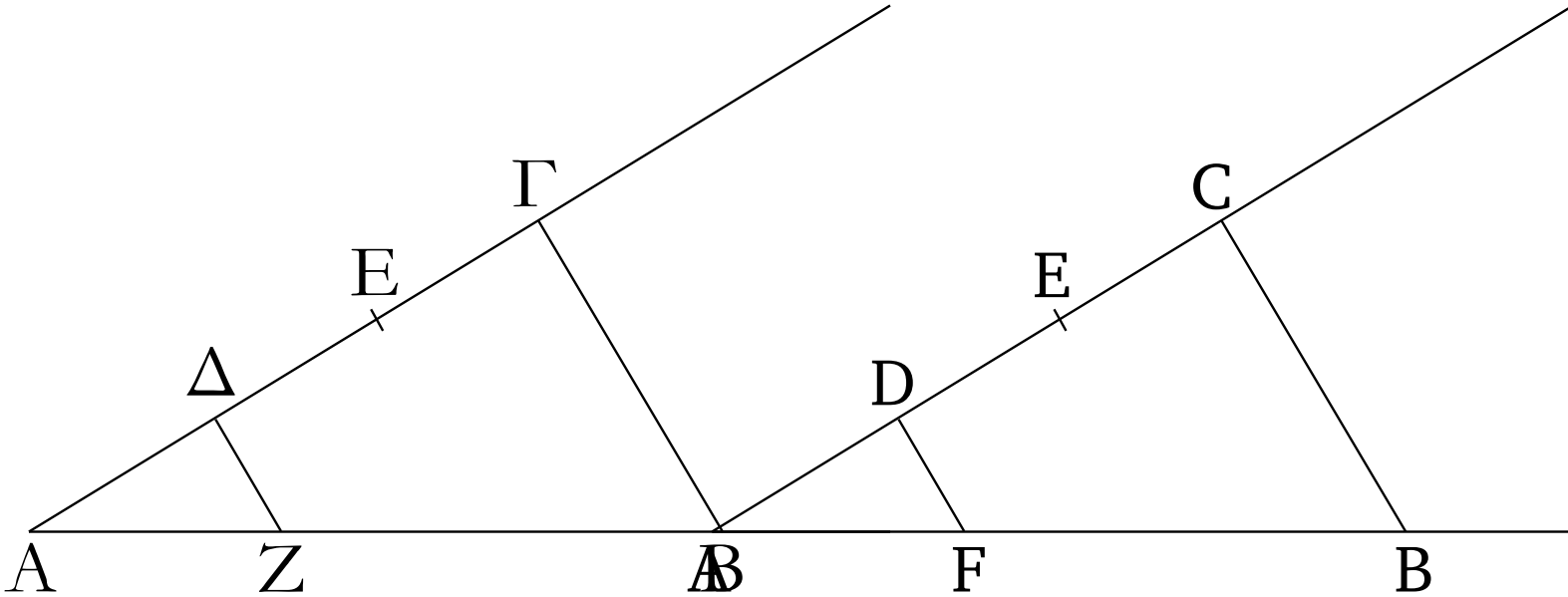
Τῆς δοθείσης εὐθείας τὸ προσταθὲν μέρος ἀφελεῖν.

Ἐστω ἡ δοθεῖσα εὐθεῖα ἡ AB . δεῖ δὴ τῆς AB τὸ προσταθὲν μέρος ἀφελεῖν.

Proposition 9

To cut off a prescribed part from a given straight-line.

Let AB be the given straight-line. So it is required to cut off a prescribed part from AB .



Ἐπιτετάθτω δὴ τὸ τρίτον. [καὶ] διήθτω τις ἀπὸ τοῦ A εὐθεῖα ἡ AC γωνίαν περιέχουσα μετὰ τῆς AB τυθοῦσαν· καὶ εἰλήφθω τυθὸν σημεῖον ἐπὶ τῆς AC τὸ Δ , καὶ κείσθωσαν τῇ AD ἴσαι αἱ DE , EC . καὶ ἐπεζεύθτω ἡ BC , καὶ διὰ τοῦ Δ παράλληλος αὐτῇ ἡθθω ἡ DZ .

Ἐπεὶ οὖν τριγώνον τοῦ ABC παρὰ μίαν τῶν πλευρῶν τὴν BC ἡκται ἡ DZ , ἀνάλογον ἄρα ἐστὶν ὡς ἡ GD πρὸς τὴν DA , οὕτως ἡ BZ πρὸς τὴν ZA . διπλὴ δὲ ἡ GD τῆς DA · διπλὴ ἄρα καὶ ἡ BZ τῆς ZA · τριπλὴ ἄρα ἡ BA τῆς AZ .

Τῆς ἄρα δοθείσης εὐθείας τῆς AB τὸ ἐπιταθὲν τρίτον μέρος ἀφῆρηται τὸ AZ · ὅπερ ἔδει ποιῆσαι.

So let a third (part) be prescribed. [And] let some straight-line AC be drawn from (point) A , encompassing a random angle with AB . And let a random point D be taken on AC . And let DE and EC be made equal to AD [Prop. 1.3]. And let BC be joined. And let DF be drawn through D parallel to it [Prop. 1.31].

Therefore, since FD has been drawn parallel to one of the sides, BC , of triangle ABC , then, proportionally, as CD is to DA , so BF (is) to FA [Prop. 6.2]. And CD (is) double DA . Thus, BF (is) also double FA . Thus, BA (is) triple AF .

Thus, the prescribed third part, AF , has been cut off from the given straight-line, AB . (Which is) the very thing it was required to do.

10.

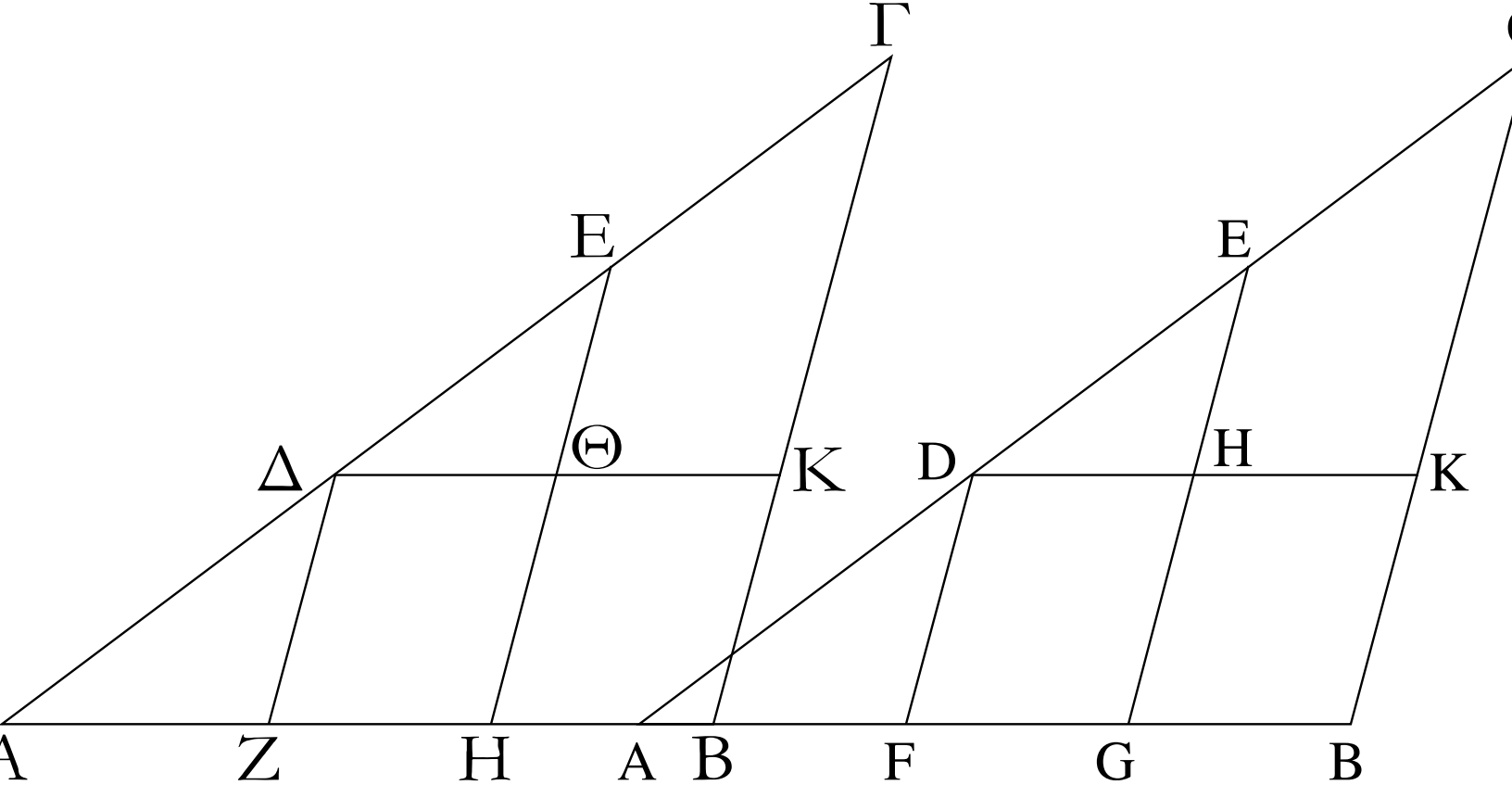
Proposition 10

Τὴν δοθεῖσαν εὐθεῖαν ἄτμητον τῇ δοθείσῃ τετμημένῃ ὁμοίως τεμεῖν.

Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἄτμητος ἡ AB , ἡ δὲ τετμημένη ἡ AG κατὰ τὰ Δ , E σημεία, καὶ κείσθωσαν ὥστε γωνίαν τυθοῦσαν περιέθιν, καὶ ἐπεξεύθθω ἡ GB , καὶ διὰ τῶν Δ , E τῇ BG παράλληλοι ἡθθωσαν αἱ ΔZ , EH , διὰ δὲ τοῦ Δ τῇ AB παράλληλος ἡθθω ἡ $\Delta\Theta K$.

To cut a given uncut straight-line similarly to a given cut (straight-line).

Let AB be the given uncut straight-line, and AC a (straight-line) cut at points D and E , and let (AC) be laid down so as to encompass a random angle (with AB). And let CB be joined. And let DF and EG be drawn through (points) D and E (respectively), parallel to BC , and let DHK be drawn through (point) D , parallel to AB [Prop. 1.31].



Παράλληλόγραμμον ἄρα ἐστὶν ἑκάτερον τῶν $Z\Theta$, ΘB · ἴση ἄρα ἡ μὲν $\Delta\Theta$ τῇ ZH , ἡ δὲ ΘK τῇ HB . καὶ ἐπεὶ τριγώνων τοῦ $\Delta K\Gamma$ παρὰ μίαν τῶν πλευρῶν τὴν $K\Gamma$ εὐθεῖα ἡκται ἡ ΘE , ἀνάλογον ἄρα ἐστὶν ὡς ἡ GE πρὸς τὴν $E\Delta$, οὕτως ἡ $K\Theta$ πρὸς τὴν $\Theta\Delta$. ἴση δὲ ἡ μὲν $K\Theta$ τῇ BH , ἡ δὲ $\Theta\Delta$ τῇ HZ . ἔστιν ἄρα ὡς ἡ GE πρὸς τὴν $E\Delta$, οὕτως ἡ BH πρὸς τὴν HZ . πάλιν, ἐπεὶ τριγώνων τοῦ AHE παρὰ μίαν τῶν πλευρῶν τὴν HE ἡκται ἡ $Z\Delta$, ἀνάλογον ἄρα ἐστὶν ὡς ἡ $E\Delta$ πρὸς τὴν ΔA , οὕτως ἡ HZ πρὸς τὴν ZA . ἐδείκθη δὲ καὶ ὡς ἡ GE πρὸς τὴν $E\Delta$, οὕτως ἡ BH πρὸς τὴν HZ · ἔστιν ἄρα ὡς μὲν ἡ GE πρὸς τὴν $E\Delta$, οὕτως ἡ BH πρὸς τὴν HZ , ὡς δὲ ἡ $E\Delta$ πρὸς τὴν ΔA , οὕτως ἡ HZ πρὸς τὴν ZA .

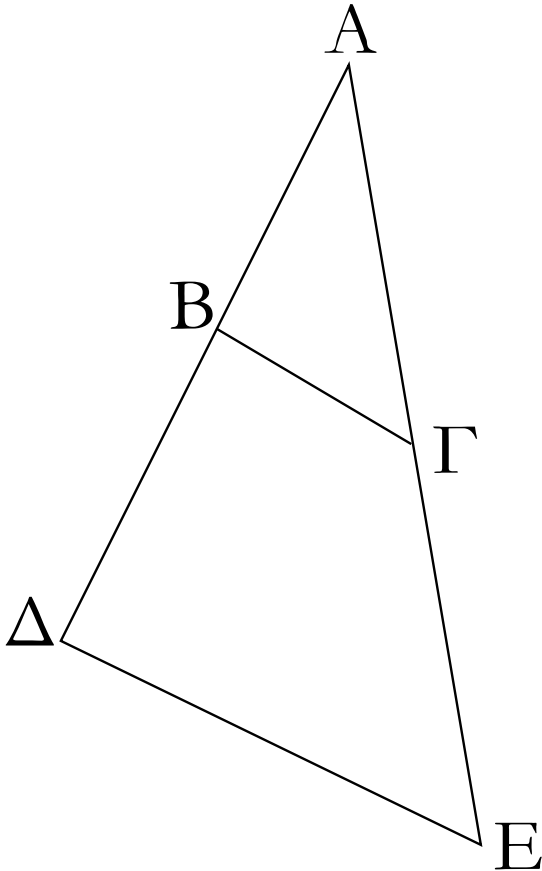
Ἡ ἄρα δοθεῖσα εὐθεῖα ἄτμητος ἡ AB τῇ δοθείσῃ εὐθείᾳ τετμημένῃ τῇ AG ὁμοίως τέτμηται· ὅπερ ἔδει ποιῆσαι.

Thus, FH and HB are each parallelograms. Thus, DH (is) equal to FG , and HK to GB [Prop. 1.34]. And since the straight-line HE has been drawn parallel to one of the sides, KC , of triangle DKC , thus, proportionally, as CE is to ED , so KH (is) to HD [Prop. 6.2]. And KH (is) equal to BG , and HD to GF . Thus, as CE is to ED , so BG (is) to GF . Again, since FD has been drawn parallel to one of the sides, GE , of triangle AGE , thus, proportionally, as ED is to DA , so GF (is) to FA [Prop. 6.2]. And it was also shown that as CE (is) to ED , so BG (is) to GF . Thus, as CE is to ED , so BG (is) to GF , and as ED (is) to DA , so GF (is) to FA .

Thus, the given uncut straight-line, AB , has been cut similarly to the given cut straight-line, AC . (Which is) the very thing it was required to do.

Δύο δοθεισών εὐθειῶν τρίτην ἀνάλογον προσευρεῖν.

Ἐστωσαν αἱ δοθεῖσαι [δύο εὐθεῖαι] αἱ BA , AG καὶ κείσθωσαν γωνίαν περιέθουσαι τυθοῦσαν. δεῖ δὴ τῶν BA , AG τρίτην ἀνάλογον προσευρεῖν. ἐκβεβλήσθωσαν γὰρ ἐπὶ τὰ Δ , E σημεία, καὶ κείσθω τῇ AG ἴση ἡ $B\Delta$, καὶ ἐπεζεύθθω ἡ $B\Gamma$, καὶ διὰ τοῦ Δ παράλληλοξ αὐτῇ ἡθθῶ ἡ ΔE .



Ἐπεὶ οὖν τρίγωνον τοῦ ADE παρὰ μίαν τῶν πλευρῶν τὴν DE ἦκται ἡ $B\Gamma$, ἀναλόγον ἐστιν ὥς ἡ AB πρὸς τὴν $B\Delta$, οὕτως ἡ AG πρὸς τὴν GE . ἴση δὲ ἡ $B\Delta$ τῇ AG . ἔστιν ἄρα ὥς ἡ AB πρὸς τὴν AG , οὕτως ἡ AG πρὸς τὴν GE .

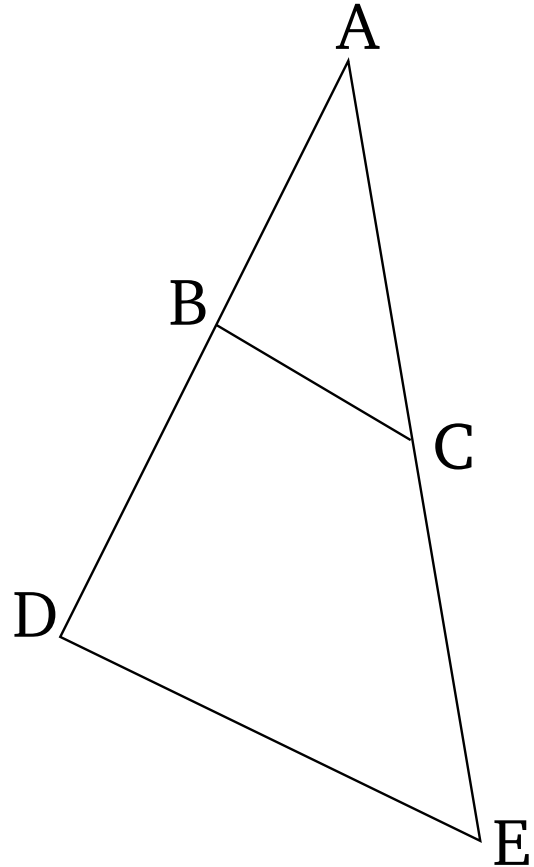
Δύο ἄρα δοθεισών εὐθειῶν τῶν AB , AG τρίτη ἀνάλογον αὐταῖς προσεύρηται ἡ GE . ὅπερ ἔδει ποιῆσαι.

12.

Τριῶν δοθεισών εὐθειῶν τετάρτην ἀνάλογον προσευρεῖν.

To find a third (straight-line) proportional to two given straight-lines.

Let BA and AC be the [two] given [straight-lines], and let them be laid down encompassing a random angle. So it is required to find a third (straight-line) proportional to BA and AC . For let (BA and AC) be produced to points D and E (respectively), and let BD be made equal to AC [Prop. 1.3]. And let BC be joined. And let DE be drawn through (point) D parallel to it [Prop. 1.31].

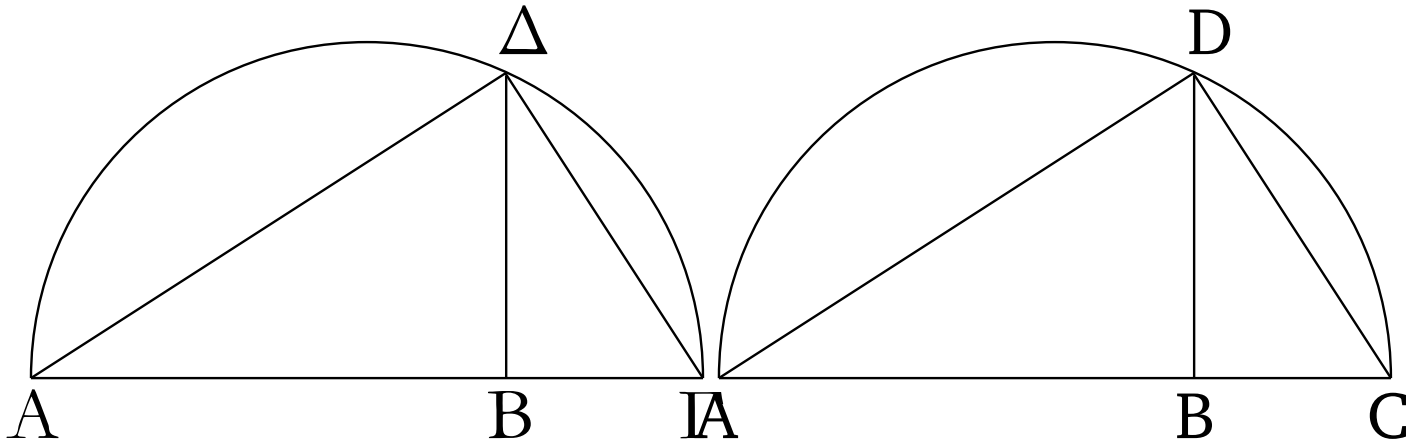


Therefore, since BC has been drawn parallel to one of the sides DE of triangle ADE , proportionally, as AB is to BD , so AC (is) to CE [Prop. 6.2]. And BD (is) equal to AC . Thus, as AB is to AC , so AC (is) to CE .

Thus, a third (straight-line), CE , has been found (which is) proportional to the two given straight-lines, AB and AC . (Which is) the very thing it was required to do.

Proposition 12

To find a fourth (straight-line) proportional to three given straight-lines.



Ἐστωσαν αἱ δοθεῖσαι δύο εὐθείαι αἱ AB , BF . δεῖ δὴ τῶν AB , BF μέσην ἀνάλογον προσευρεῖν.

Κείσθωσαν ἐπ' εὐθείας, καὶ γεγράφθω ἐπὶ τῆς AF ἡμικύκλιον τὸ $ADΓ$, καὶ ἦθθω ἀπὸ τοῦ B σημείου τῇ AF εὐθείᾳ πρὸς ὀρθὰς ἡ BD , καὶ ἐπεζεύθωσαν αἱ AD , $ΔΓ$.

Ἐπεὶ ἐν ἡμικυκλίῳ γωνία ἐστὶν ἡ ὑπὸ $ADΓ$, ὀρθή ἐστίν. καὶ ἐπεὶ ἐν ὀρθογωνίῳ τριγώνῳ τῷ $ADΓ$ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ἦκται ἡ DB , ἡ DB ἄρα τῶν τῆς βάσεως τμημάτων τῶν AB , BF μέση ἀνάλογόν ἐστιν.

Δύο ἄρα δοθεισῶν εὐθειῶν τῶν AB , BF μέση ἀνάλογον προσεύρηται ἡ DB . ὅπερ ἔδει ποιῆσαι.

Let AB and BC be the two given straight-lines. So it is required to find the (straight-line) in mean proportion to AB and BC .

Let (AB and BC) be laid down straight-on (with respect to one another), and let the semi-circle ADC be drawn on AC [Prop. 1.10]. And let BD be drawn from (point) B , at right-angles to AC [Prop. 1.11]. And let AD and DC be joined.

And since ADC is an angle in a semi-circle, it is a right-angle [Prop. 3.31]. And since, in the right-angled triangle ADC , the (straight-line) DB has been drawn from the right-angle perpendicular to the base, DB is thus the mean proportional to the pieces of the base, AB and BC [Prop. 6.8 corr.].

Thus, DB has been found (which is) in mean proportion to the two given straight-lines, AB and BC . (Which is) the very thing it was required to do.

† In other words, to find the geometric mean of two given straight-lines.

14.

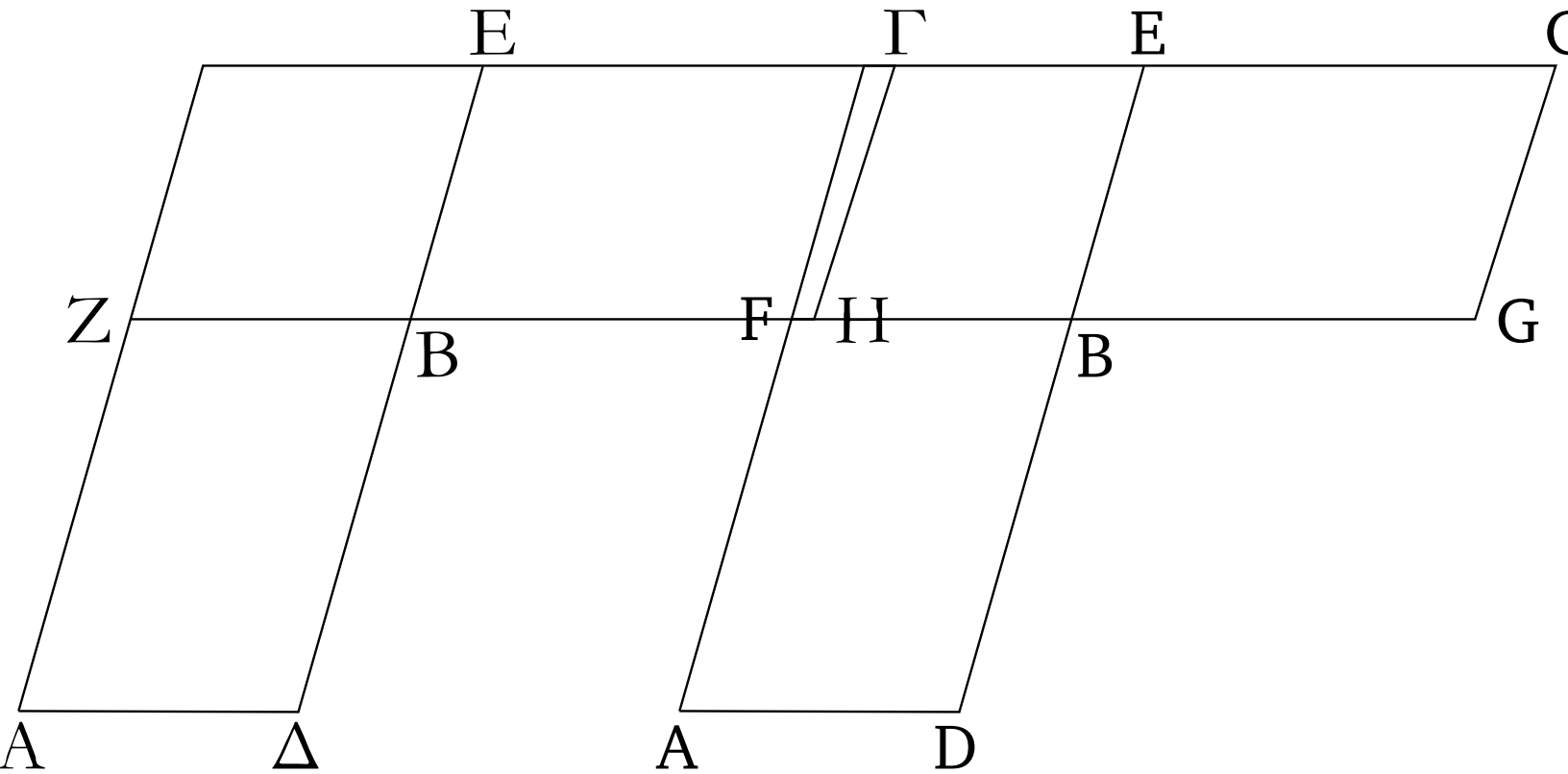
Τῶν ἴσων τε καὶ ἰσογωνίων παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· καὶ ὧν ἰσογωνίων παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἴσα ἐστὶν ἐκεῖνα.

Ἐστω ἴσα τε καὶ ἰσογώνια παραλληλόγραμμα τὰ AB , BF ἴσαξ ἔχοντα τὰς πρὸς τῷ B γωνίας, καὶ κείσθωσαν ἐπ' εὐθείας αἱ DB , BE · ἐπ' εὐθείας ἄρα εἰσὶ καὶ αἱ ZB , BH . λέγω, ὅτι τῶν AB , BF ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, τουτέστιν, ὅτι ἐστὶν ὡς ἡ DB πρὸς τὴν BE , ὅτι ὡς ἡ HB πρὸς τὴν BZ .

Proposition 14

In equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional. And those equiangular parallelograms in which the sides about the equal angles are reciprocally proportional are equal.

Let AB and BC be equal and equiangular parallelograms having the angles at B equal. And let DB and BE be laid down straight-on (with respect to one another). Thus, FB and BG are also straight-on (with respect to one another) [Prop. 1.14]. I say that the sides of AB and BC about the equal angles are reciprocally proportional, that is to say, that as DB is to BE , so GB (is) to BF .



Συμπεληρώσωθω γάρ τὸ ZE παραλληλόγραμμον. ἐπεὶ οὖν ἴσον ἐστὶ τὸ AB παραλληλόγραμμον τῷ BG παραλληλογράμμῳ, ἄλλο δέ τι τὸ ZE, ἔστιν ἄρα ὥς τὸ AB πρὸς τὸ ZE, οὕτως τὸ BG πρὸς τὸ ZE. ἀλλ' ὥς μὲν τὸ AB πρὸς τὸ ZE, οὕτως ἡ ΔB πρὸς τὴν BE, ὥς δὲ τὸ BG πρὸς τὸ ZE, οὕτως ἡ HB πρὸς τὴν BZ· καὶ ὥς ἄρα ἡ ΔB πρὸς τὴν BE, οὕτως ἡ HB πρὸς τὴν BZ. τῶν ἄρα AB, BG παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας.

Ἀλλὰ δὴ ἔστω ὥς ἡ ΔB πρὸς τὴν BE, οὕτως ἡ HB πρὸς τὴν BZ· λέγω, ὅτι ἴσον ἐστὶ τὸ AB παραλληλόγραμμον τῷ BG παραλληλογράμμῳ.

Ἐπεὶ γάρ ἐστιν ὥς ἡ ΔB πρὸς τὴν BE, οὕτως ἡ HB πρὸς τὴν BZ, ἀλλ' ὥς μὲν ἡ ΔB πρὸς τὴν BE, οὕτως τὸ AB παραλληλόγραμμον πρὸς τὸ ZE παραλληλόγραμμον, ὥς δὲ ἡ HB πρὸς τὴν BZ, οὕτως τὸ BG παραλληλόγραμμον πρὸς τὸ ZE παραλληλόγραμμον, καὶ ὥς ἄρα τὸ AB πρὸς τὸ ZE, οὕτως τὸ BG πρὸς τὸ ZE· ἴσον ἄρα ἐστὶ τὸ AB παραλληλόγραμμον τῷ BG παραλληλογράμμῳ.

Τῶν ἄρα ἴσων τε καὶ ἰσογωνίων παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· καὶ ὧν ἰσογωνίων παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἴσα ἐστὶν ἐκείνα· ὅπερ ἔδει δεῖχαι.

For let the parallelogram FE be completed. Therefore, since parallelogram AB is equal to parallelogram BC , and FE (is) some other (parallelogram), thus as (parallelogram) AB is to FE , so (parallelogram) BC (is) to FE [Prop. 5.7]. But, as (parallelogram) AB (is) to FE , so DB (is) to BE , and as (parallelogram) BC (is) to FE , so GB (is) to BF [Prop. 6.1]. Thus, also, as DB (is) to BE , so GB (is) to BF . Thus, in parallelograms AB and BC the sides about the equal angles are reciprocally proportional.

And so, let DB be to BE , as GB (is) to BF . I say that parallelogram AB is equal to parallelogram BC .

For since as DB is to BE , so GB (is) to BF , but as DB (is) to BE , so parallelogram AB (is) to parallelogram FE , and as GB (is) to BF , so parallelogram BC (is) to parallelogram FE [Prop. 6.1], thus, also, as (parallelogram) AB (is) to FE , so (parallelogram) BC (is) to FE [Prop. 5.11]. Thus, parallelogram AB is equal to parallelogram BC [Prop. 5.9].

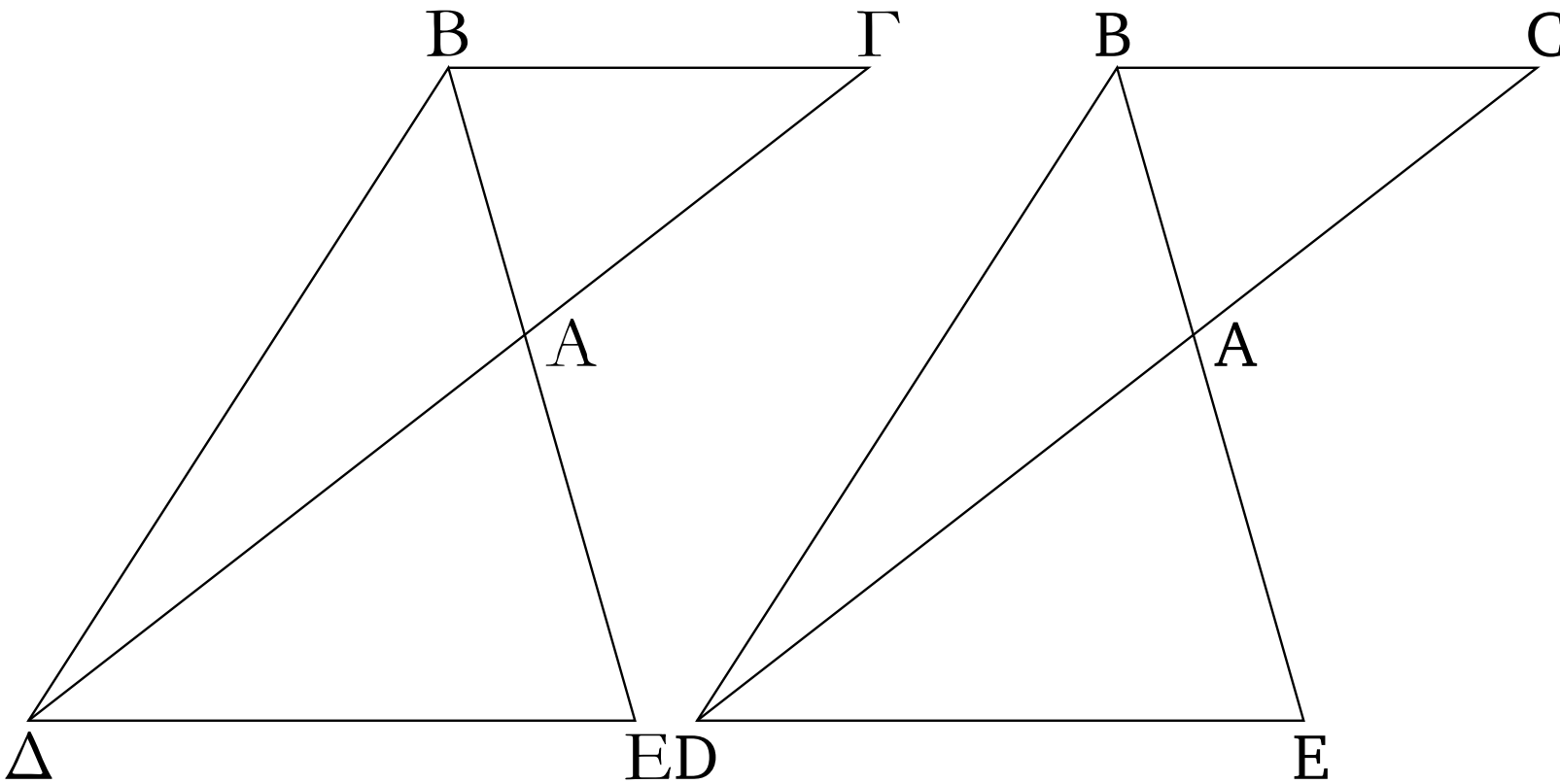
Thus, in equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional. And those equiangular parallelograms in which the sides about the equal angles are reciprocally proportional are equal. (Which is) the very thing it was required to show.

Τῶν ἴσων καὶ μίαν μὲ ἴσην ἐθόντων γωνίαν τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· καὶ ὧν μίαν μὲ ἴσην ἐθόντων γωνίαν τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἴσα ἐστὶν ἐκεῖνα.

Ἐστω ἴσα τρίγωνα τὰ $AB\Gamma$, $A\Delta E$ μίαν μὲ ἴσην ἐθόντα γωνίαν τὴν ὑπὸ BAG τῇ ὑπὸ ΔAE · λέγω, ὅτι τῶν $AB\Gamma$, $A\Delta E$ τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, τουτέστιν, ὅτι ἐστὶν ὡς ἡ ΓA πρὸς τὴν $A\Delta$, οὕτως ἡ EA πρὸς τὴν AB .

In equal triangles also having one angle equal to one (angle), the sides about the equal angles are reciprocally proportional. And those triangles having one angle equal to one angle for which the sides about the equal angles (are) reciprocally proportional are equal.

Let ABC and ADE be equal triangles having one angle equal to one (angle), (namely) BAC (equal) to DAE . I say that, in triangles ABC and ADE , the sides about the equal angles are reciprocally proportional, that is to say, that as CA is to AD , so EA (is) to AB .



Κείσθω γάρ ὥστε ἐπ' εὐθείᾳ εἶναι τὴν ΓA τῇ $A\Delta$ · ἐπ' εὐθείᾳ ἄρα ἐστὶ καὶ ἡ EA τῇ AB . καὶ ἐπεζεύθθω ἡ $B\Delta$.

Ἐπεὶ οὖν ἴσον ἐστὶ τὸ $AB\Gamma$ τριγώνον τῷ $A\Delta E$ τριγώνῳ, ἄλλο δέ τι τὸ $BA\Delta$, ἔστιν ἄρα ὡς τὸ ΓAB τριγώνον πρὸς τὸ $BA\Delta$ τριγώνον, οὕτως τὸ $EA\Delta$ τριγώνον πρὸς τὸ $BA\Delta$ τριγώνον. ἀλλ' ὡς μὲν τὸ ΓAB πρὸς τὸ $BA\Delta$, οὕτως ἡ ΓA πρὸς τὴν $A\Delta$, ὡς δέ τὸ $EA\Delta$ πρὸς τὸ $BA\Delta$, οὕτως ἡ EA πρὸς τὴν AB . καὶ ὡς ἄρα ἡ ΓA πρὸς τὴν $A\Delta$, οὕτως ἡ EA πρὸς τὴν AB . τῶν $AB\Gamma$, $A\Delta E$ ἄρα τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας.

Ἀλλὰ δὴ ἀντιπεπονητέωσαν αἱ πλευραὶ τῶν $AB\Gamma$, $A\Delta E$ τριγώνων, καὶ ἔστω ὡς ἡ ΓA πρὸς τὴν $A\Delta$, οὕτως ἡ EA πρὸς τὴν AB · λέγω, ὅτι ἴσον ἐστὶ τὸ $AB\Gamma$ τριγώνον τῷ $A\Delta E$ τριγώνῳ.

Ἐπιευνθείσῃ γὰρ πάλιν τῇς $B\Delta$, ἐπεὶ ἐστὶν ὡς ἡ ΓA

For let CA be laid down so as to be straight-on (with respect) to AD . Thus, EA is also straight-on (with respect) to AB [Prop. 1.14]. And let BD be joined.

Therefore, since triangle ABC is equal to triangle ADE , and BAD (is) some other (triangle), thus as triangle CAB is to triangle BAD , so triangle EAD (is) to triangle BAD [Prop. 5.7]. But, as (triangle) CAB (is) to BAD , so CA (is) to AD , and as (triangle) EAD (is) to BAD , so EA (is) to AB [Prop. 6.1]. And thus, as CA (is) to AD , so EA (is) to AB . Thus, in triangles ABC and ADE the sides about the equal angles (are) reciprocally proportional.

And so, let the sides of triangles ABC and ADE be reciprocally proportional, and (thus) let CA be to AD , as EA (is) to AB . I say that triangle ABC is equal to triangle ADE .

For, BD again being joined, since as CA is to AD , so EA

πρὸς τὴν AD , οὕτως ἡ EA πρὸς τὴν AB , ἀλλ' ὥς μὲν ἡ $ΓA$ πρὸς τὴν AD , οὕτως τὸ $ABΓ$ τρίγωνον πρὸς τὸ $BAΔ$ τρίγωνον, ὥς δὲ ἡ EA πρὸς τὴν AB , οὕτως τὸ $EAΔ$ τρίγωνον πρὸς τὸ $BAΔ$ τρίγωνον, ὥς ἄρα τὸ $ABΓ$ τρίγωνον πρὸς τὸ $BAΔ$ τρίγωνον, οὕτως τὸ $EAΔ$ τρίγωνον πρὸς τὸ $BAΔ$ τρίγωνον. ἐκάτερον ἄρα τῶν $ABΓ$, $EAΔ$ πρὸς τὸ $BAΔ$ τὸν αὐτὸν ἔθει λόγον. ἴσων ἄρα ἐστὶ τὸ $ABΓ$ [τρίγωνον] τῷ $EAΔ$ τριγώνῳ.

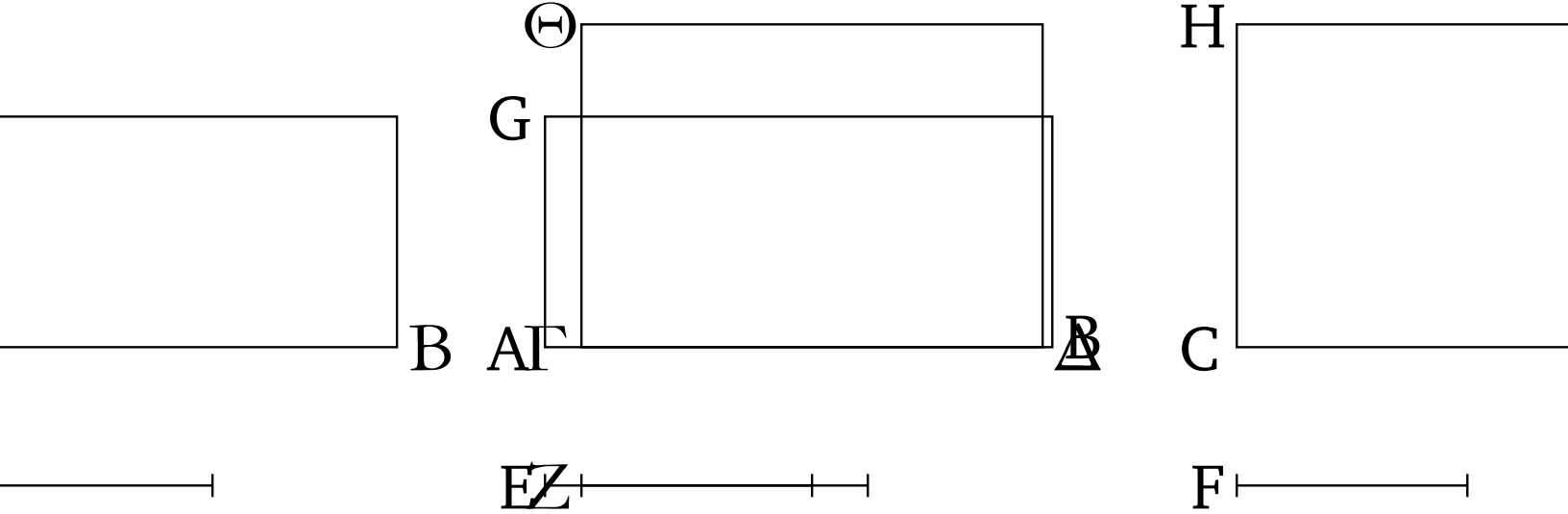
Τῶν ἄρα ἴσων καὶ μίαν μὲν ἴσην ἐθόντων γωνίαν τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· καὶ ὥς μίαν μὲν ἴσην ἐθόντων γωνίαν τριγώνων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἐκεῖνα ἴσα ἐστὶν· ὅπερ ἔδει δεῖχαι.

16.

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ᾖσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν μέσων περιεχόμενῳ ὀρθογώνῳ· κ' ἂν τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ᾖ τῷ ὑπὸ τῶν μέσων περιεχόμενῳ ὀρθογώνῳ, αἱ τέσσαρες εὐθεῖαι ἀνάλογον ἔσονται.

Proposition 16

If four straight-lines are proportional, (then) the rectangle contained by the (two) outermost is equal to the rectangle contained by the middle (two). And if the rectangle contained by the (two) outermost is equal to the rectangle contained by the middle (two), (then) the four straight-lines will be proportional.



Ἐστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ AB , $ΓΔ$, E , Z , ὥς ἡ AB πρὸς τὴν $ΓΔ$, οὕτως ἡ E πρὸς τὴν Z . λέγω, ὅτι τὸ ὑπὸ τῶν AB , Z περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν $ΓΔ$, E περιεχόμενῳ ὀρθογώνῳ.

Ἦθθωσαν [γάρ] ἀπὸ τῶν A , $Γ$ σημείων ταῖς AB , $ΓΔ$ εὐθείαις πρὸς ὀρθὰς αἱ AH , $ΓΘ$, καὶ κείσθω τῇ μὲν Z ἴση ἡ AH , τῇ δὲ E ἴση ἡ $ΓΘ$. καὶ συμπληρώσω τὰ BH , $ΔΘ$ παραλληλόγραμμα.

Καὶ ἐπεὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $ΓΔ$, οὕτως ἡ E πρὸς τὴν Z , ἴση δὲ ἡ μὲν E τῇ $ΓΘ$, ἡ δὲ Z τῇ AH , ἐστὶν ἄρα ὥς ἡ AB πρὸς τὴν $ΓΔ$, οὕτως ἡ $ΓΘ$ πρὸς τὴν AH . τῶν BH , $ΔΘ$ ἄρα παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ

Let AB , CD , E , and F be four proportional straight-lines, (such that) as AB (is) to CD , so E (is) to F . I say that the rectangle contained by AB and F is equal to the rectangle contained by CD and E .

[For] let AG and CH be drawn from points A and C at right-angles to the straight-lines AB and CD (respectively) [Prop. 1.11]. And let AG be made equal to F , and CH to E [Prop. 1.3]. And let the parallelograms BG and DH be completed.

And since as AB is to CD , so E (is) to F , and E (is) equal CH , and F to AG , thus as AB is to CD , so CH (is) to AG . Thus, in the parallelograms BG and DH , the sides about the

περὶ τὰς ἴσας γωνίας. ὧν δὲ ἰσογώνων παραλληλογράμμων ἀντιπεπνόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας, ἴσα ἐστὶν ἐκεῖνα· ἴσον ἄρα ἐστὶ τὸ BH παραλληλόγραμμον τῷ $\Delta\Theta$ παραλληλογράμμῳ. καὶ ἐστὶ τὸ μὲν BH τὸ ὑπὸ τῶν AB , Z · ἴση γάρ ἢ AH τῇ Z · τὸ δὲ $\Delta\Theta$ τὸ ὑπὸ τῶν $\Gamma\Delta$, E · ἴση γάρ ἢ E τῇ $\Gamma\Theta$ · τὸ ἄρα ὑπὸ τῶν AB , Z περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν $\Gamma\Delta$, E περιεχομένῳ ὀρθογώνιῳ.

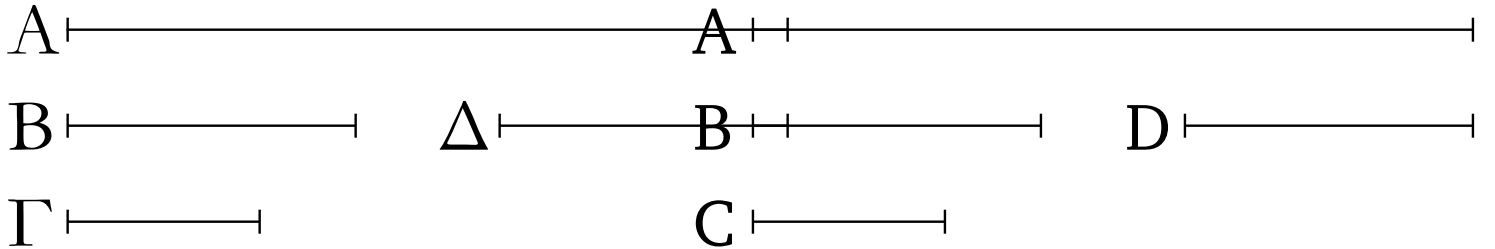
Ἀλλὰ δὴ τὸ ὑπὸ τῶν AB , Z περιεχόμενον ὀρθογώνιον ἴσον ἔστω τῷ ὑπὸ τῶν $\Gamma\Delta$, E περιεχομένῳ ὀρθογώνιῳ. λέγω, ὅτι αἱ τέσσαρες εὐθεῖαι ἀνάλογον ἔσονται, ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἢ E πρὸς τὴν Z .

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ τὸ ὑπὸ τῶν AB , Z ἴσον ἐστὶ τῷ ὑπὸ τῶν $\Gamma\Delta$, E , καὶ ἐστὶ τὸ μὲν ὑπὸ τῶν AB , Z τὸ BH · ἴση γάρ ἐστὶν ἢ AH τῇ Z · τὸ δὲ ὑπὸ τῶν $\Gamma\Delta$, E τὸ $\Delta\Theta$ · ἴση γάρ ἢ $\Gamma\Theta$ τῇ E · τὸ ἄρα BH ἴσον ἐστὶ τῷ $\Delta\Theta$. καὶ ἐστὶν ἰσογώνια. τῶν δὲ ἴσων καὶ ἰσογώνων παραλληλογράμμων ἀντιπεπνόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας. ἔστιν ἄρα ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἢ $\Gamma\Theta$ πρὸς τὴν AH . ἴση δὲ ἢ μὲν $\Gamma\Theta$ τῇ E , ἢ δὲ AH τῇ Z · ἔστιν ἄρα ὥς ἢ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἢ E πρὸς τὴν Z .

Ἐὰν ἄρα τέσσαρες εὐθεῖαι ἀνάλογον ᾖσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ὑπὸ τῶν μέσων περιεχομένῳ ὀρθογώνιῳ· καὶ τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ᾗ τῷ ὑπὸ τῶν μέσων περιεχομένῳ ὀρθογώνιῳ, αἱ τέσσαρες εὐθεῖαι ἀνάλογον ἔσονται· ὅπερ ἔδει δεῖχαι.

17.

Ἐὰν τρεῖς εὐθεῖαι ἀνάλογον ᾖσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς μέσης τετραγώνῳ· καὶ τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ᾗ τῷ ἀπὸ τῆς μέσης τετραγώνῳ, αἱ τρεῖς εὐθεῖαι ἀνάλογον ἔσονται.



Ἐστωσαν τρεῖς εὐθεῖαι ἀνάλογον αἱ A , B , Γ , ὥς ἢ A πρὸς τὴν B , οὕτως ἢ B πρὸς τὴν Γ . λέγω, ὅτι τὸ ὑπὸ τῶν A , Γ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς B τετραγώνῳ.

Κείσθω τῇ B ἴση ἢ Δ .

Καὶ ἐπεὶ ἐστὶν ὥς ἢ A πρὸς τὴν B , οὕτως ἢ B πρὸς τὴν Γ , ἴση δὲ ἢ B τῇ Δ , ἔστιν ἄρα ὥς ἢ A πρὸς τὴν B , ἢ Δ πρὸς τὴν Γ . ἔαν δὲ τέσσαρες εὐθεῖαι ἀνάλογον ᾖσιν, τὸ ὑπὸ

equal angles are reciprocally proportional. And those equiangular parallelograms in which the sides about the equal angles are reciprocally proportional are equal [Prop. 6.14]. Thus, parallelogram BG is equal to parallelogram DH . And BG is the (rectangle contained) by AB and F . For AG (is) equal to F . And DH (is) the (rectangle contained) by CD and E . For E (is) equal to CH . Thus, the rectangle contained by AB and F is equal to the rectangle contained by CD and E .

And so, let the rectangle contained by AB and F be equal to the rectangle contained by CD and E . I say that the four straight-lines will be proportional, (so that) as AB (is) to CD , so E (is) to F .

For, with the same construction, since the (rectangle contained) by AB and F is equal to the (rectangle contained) by CD and E . And BG is the (rectangle contained) by AB and F . For AG is equal to F . And DH (is) the (rectangle contained) by CD and E . For CH (is) equal to E . BG is thus equal to DH . And they are equiangular. And in equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional [Prop. 6.14]. Thus, as AB is to CD , so CH (is) to AG . And CH (is) equal to E , and AG to F . Thus, as AB is to CD , so E (is) to F .

Thus, if four straight-lines are proportional, (then) the rectangle contained by the (two) outermost is equal to the rectangle contained by the middle (two). And if the rectangle contained by the (two) outermost is equal to the rectangle contained by the middle (two) then the four straight-lines will be proportional. (Which is) the very thing it was required to show.

Proposition 17

If three straight-lines are proportional, (then) the rectangle contained by the (two) outermost is equal to the square on the middle (one). And if the rectangle contained by the (two) outermost is equal to the square on the middle (one), (then) the three straight-lines will be proportional.

Let A , B and C be three proportional straight-lines, (such that) as A (is) to B , so B (is) to C . I say that the rectangle contained by A and C is equal to the square on B .

Let D be made equal to B [Prop. 1.3].

And since as A is to B , so B (is) to C , and B (is) equal to D , thus as A is to B , (so) D (is) to C . And if four straight-lines are proportional, (then) the [rectangle] contained by the (two)

τῶν ἄκρων περιεχόμενον [ὀρθογώνιον] ἴσον ἐστὶ τῷ ὑπὸ τῶν μέσων περιεχομένῳ ὀρθογώνῳ. τὸ ἄρα ὑπὸ τῶν A, Γ ἴσον ἐστὶ τῷ ὑπὸ τῶν B, Δ . ἀλλὰ τὸ ὑπὸ τῶν B, Δ τὸ ἀπὸ τῆς B ἐστίν· ἴση γάρ ἡ B τῇ Δ . τὸ ἄρα ὑπὸ τῶν A, Γ περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς B τετραγώνῳ.

Ἀλλὰ δὴ τὸ ὑπὸ τῶν A, Γ ἴσον ἔστω τῷ ἀπὸ τῆς B . λέγω, ὅτι ἐστὶν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ .

Τῶν γάρ αὐτῶν κατασκευασθέντων, ἐπεὶ τὸ ὑπὸ τῶν A, Γ ἴσον ἐστὶ τῷ ἀπὸ τῆς B , ἀλλὰ τὸ ἀπὸ τῆς B τὸ ὑπὸ τῶν B, Δ ἐστίν· ἴση γάρ ἡ B τῇ Δ . τὸ ἄρα ὑπὸ τῶν A, Γ ἴσον ἐστὶ τῷ ὑπὸ τῶν B, Δ . ἔαν δὲ τὸ ὑπὸ τῶν ἄκρων ἴσον ᾗ τῷ ὑπὸ τῶν μέσων, αἱ τέσσαρες εὐθεῖαι ἀνάλογόν εἰσιν. ἔστιν ἄρα ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Δ πρὸς τὴν Γ . ἴση δὲ ἡ B τῇ Δ . ὥς ἄρα ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ .

Ἐάν ἄρα τρεῖς εὐθεῖαι ἀνάλογον ᾖσιν, τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ἐστὶ τῷ ἀπὸ τῆς μέσης τετραγώνῳ· κ' ἂν τὸ ὑπὸ τῶν ἄκρων περιεχόμενον ὀρθογώνιον ἴσον ᾗ τῷ ἀπὸ τῆς μέσης τετραγώνῳ, αἱ τρεῖς εὐθεῖαι ἀνάλογον ἔσονται· ὅπερ ἔδει δεῖχαι.

outermost is equal to the rectangle contained by the middle (two) [Prop. 6.16]. Thus, the (rectangle contained) by A and C is equal to the (rectangle contained) by B and D . But, the (rectangle contained) by B and D is the (square) on B . For B (is) equal to D . Thus, the rectangle contained by A and C is equal to the square on B .

And so, let the (rectangle contained) by A and C be equal to the (square) on B . I say that as A is to B , so B (is) to C .

For, with the same construction, since the (rectangle contained) by A and C is equal to the (square) on B . But, the (square) on B is the (rectangle contained) by B and D . For B (is) equal to D . The (rectangle contained) by A and C is thus equal to the (rectangle contained) by B and D . And if the (rectangle contained) by the (two) outermost is equal to the (rectangle contained) by the middle (two), (then) the four straight-lines are proportional [Prop. 6.16]. Thus, as A is to B , so D (is) to C . And B (is) equal to D . Thus, as A (is) to B , so B (is) to C .

Thus, if three straight-lines are proportional, (then) the rectangle contained by the (two) outermost is equal to the square on the middle (one). And if the rectangle contained by the (two) outermost is equal to the square on the middle (one), (then) the three straight-lines will be proportional. (Which is) the very thing it was required to show.

18.

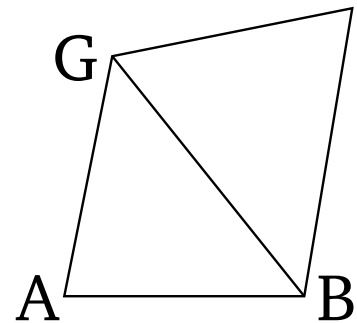
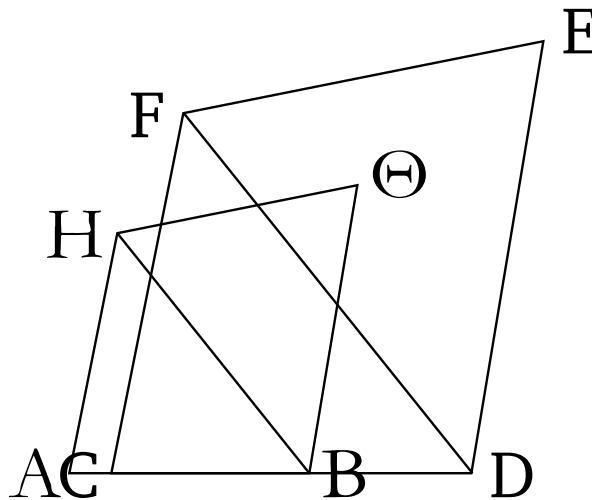
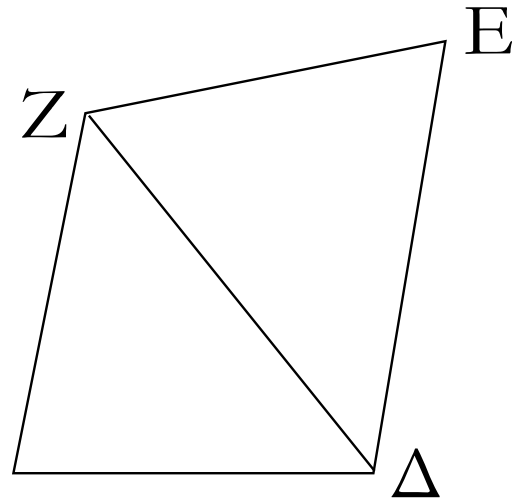
Ἀπὸ τῆς δοθείσης εὐθείας τῷ δοθέντι εὐθυγράμμῳ ὁμοίον τε καὶ ὁμοίως κείμενον εὐθύγραμμον ἀναγράψαι.

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθὲν εὐθύγραμμον τὸ ΓE . δεῖ δὴ ἀπὸ τῆς AB εὐθείας τῷ ΓE εὐθυγράμμῳ ὁμοίον τε καὶ ὁμοίως κείμενον εὐθύγραμμον ἀναγράψαι.

Proposition 18

To describe a rectilinear figure similar, and similarly laid down, to a given rectilinear figure on a given straight-line.

Let AB be the given straight-line, and CE the given rectilinear figure. So it is required to describe a rectilinear figure similar, and similarly laid down, to the rectilinear figure CE on the straight-line AB .



Ἐπεξεύθητο ἡ ΔZ , καὶ συνεστάτω πρὸς τῇ AB εὐθείᾳ καὶ τοῖς πρὸς αὐτῇ σημείοις τοῖς A, B τῇ μὲν πρὸς τῷ Γ γωνίᾳ

Let DF be joined, and let GAB , equal to the angle at C , and ABG , equal to (angle) CDF , be constructed on the

ἴση ἢ ὑπὸ HAB , τῇ δὲ ὑπὸ $\Gamma\Delta Z$ ἴση ἢ ὑπὸ ABH . λοιπὴ ἄρα ἢ ὑπὸ $\Gamma Z\Delta$ τῇ ὑπὸ AHB ἐστὶν ἴση· ἰσογώνιον ἄρα ἐστὶ τὸ $Z\Gamma\Delta$ τρίγωνον τῷ HAB τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ $Z\Delta$ πρὸς τὴν HB , οὕτως ἡ $Z\Gamma$ πρὸς τὴν HA , καὶ ἡ $\Gamma\Delta$ πρὸς τὴν AB . πάλιν συνεστάτω πρὸς τῇ BH εὐθείᾳ καὶ τοῖς πρὸς αὐτῇ σημείοις τοῖς B , H τῇ μὲν ὑπὸ ΔZE γωνίᾳ ἴση ἢ ὑπὸ $BH\Theta$, τῇ δὲ ὑπὸ $Z\Delta E$ ἴση ἢ ὑπὸ $HB\Theta$. λοιπὴ ἄρα ἢ πρὸς τῷ E λοιπῇ τῇ πρὸς τῷ Θ ἐστὶν ἴση· ἰσογώνιον ἄρα ἐστὶ τὸ $Z\Delta E$ τρίγωνον τῷ $H\Theta B$ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ $Z\Delta$ πρὸς τὴν HB , οὕτως ἡ ZE πρὸς τὴν $H\Theta$ καὶ ἡ $E\Delta$ πρὸς τὴν ΘB . ἐδείκθη δὲ καὶ ὡς ἡ $Z\Delta$ πρὸς τὴν HB , οὕτως ἡ $Z\Gamma$ πρὸς τὴν HA καὶ ἡ $\Gamma\Delta$ πρὸς τὴν AB · καὶ ὡς ἄρα ἡ $Z\Gamma$ πρὸς τὴν AH , οὕτως ἡ $\Gamma\Delta$ πρὸς τὴν AB καὶ ἡ ZE πρὸς τὴν $H\Theta$ καὶ ἔτι ἡ EA πρὸς τὴν ΘB . καὶ ἐπεὶ ἴση ἐστὶν ἢ μὲν ὑπὸ $\Gamma Z\Delta$ γωνία τῇ ὑπὸ AHB , ἢ δὲ ὑπὸ ΔZE τῇ ὑπὸ $BH\Theta$, ὅλη ἄρα ἢ ὑπὸ ΓZE ὅλη τῇ ὑπὸ $AH\Theta$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $\Gamma\Delta E$ τῇ ὑπὸ $AB\Theta$ ἐστὶν ἴση. ἔστι δὲ καὶ ἡ μὲν πρὸς τῷ Γ τῇ πρὸς τῷ A ἴση, ἢ δὲ πρὸς τῷ E τῇ πρὸς τῷ Θ . ἰσογώνιον ἄρα ἐστὶ τὸ $A\Theta$ τῷ ΓE · καὶ τὰς περὶ τὰς ἴσας γωνίας αὐτῶν πλευρὰς ἀνάλογον ἔθει· ὅμοιον ἄρα ἐστὶ τὸ $A\Theta$ εὐθύγραμμον τῷ ΓE εὐθύγραμμῳ.

Ἀπὸ τῆς δοθείσης ἄρα εὐθείας τῆς AB τῷ δοθέντι εὐθύγραμμῳ τῷ ΓE ὁμοίον τε καὶ ὁμοίως κείμενον εὐθύγραμμον ἀναγράσσεται τὸ $A\Theta$ · ὅπερ ἔδει ποιῆσαι.

19.

Τὰ ὅμοια τρίγωνα πρὸς ἄλληλα ἐν διπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν.

Ἐστω ὅμοια τρίγωνα τὰ $AB\Gamma$, ΔEZ ἴσην ἔχοντα τὴν πρὸς τῷ B γωνίαν τῇ πρὸς τῷ E , ὡς δὲ τὴν AB πρὸς τὴν $B\Gamma$, οὕτως τὴν ΔE πρὸς τὴν EZ , ὥστε ὁμολόγον εἶναι τὴν $B\Gamma$ τῇ EZ · λέγω, ὅτι τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ ΔEZ τρίγωνον διπλασίονα λόγον ἔθει ἢ περὶ ἡ $B\Gamma$ πρὸς τὴν EZ .

Εἰλήφθω γὰρ τῶν $B\Gamma$, EZ τρίτη ἀνάλογον ἡ BH , ὥστε εἶναι ὡς τὴν $B\Gamma$ πρὸς τὴν EZ , οὕτως τὴν EZ πρὸς τὴν BH · καὶ ἐπεξεύθθω ἡ AH .

straight-line AB at the points A and B on it (respectively) [Prop. 1.23]. Thus, the remaining (angle) CFD is equal to AGB [Prop. 1.32]. Thus, triangle FCD is equiangular to triangle GAB . Thus, proportionally, as FD is to GB , so FC (is) to GA , and CD to AB [Prop. 6.4]. Again, let BGH , equal to angle DFE , and GBH equal to (angle) FDE , be constructed on the straight-line BG at the points G and B on it (respectively) [Prop. 1.23]. Thus, the remaining (angle) at E is equal to the remaining (angle) at H [Prop. 1.32]. Thus, triangle FDE is equiangular to triangle GHB . Thus, proportionally, as FD is to GB , so FE (is) to GH , and ED to HB [Prop. 6.4]. And it was also shown (that) as FD (is) to GB , so FC (is) to GA , and CD to AB . Thus, also, as FC (is) to AG , so CD (is) to AB , and FE to GH , and, further, ED to HB . And since angle CFD is equal to AGB , and DFE to BGH , thus the whole (angle) CFE is equal to the whole (angle) AGH . So, for the same (reasons), (angle) CDE is also equal to ABH . And the (angle) at C is also equal to the (angle) at A , and the (angle) at E to the (angle) at H . Thus, (figure) AH is equiangular to CE . And (the two figures) have the sides about their equal angles proportional. Thus, the rectilinear figure AH is similar to the rectilinear figure CE [Def. 6.1].

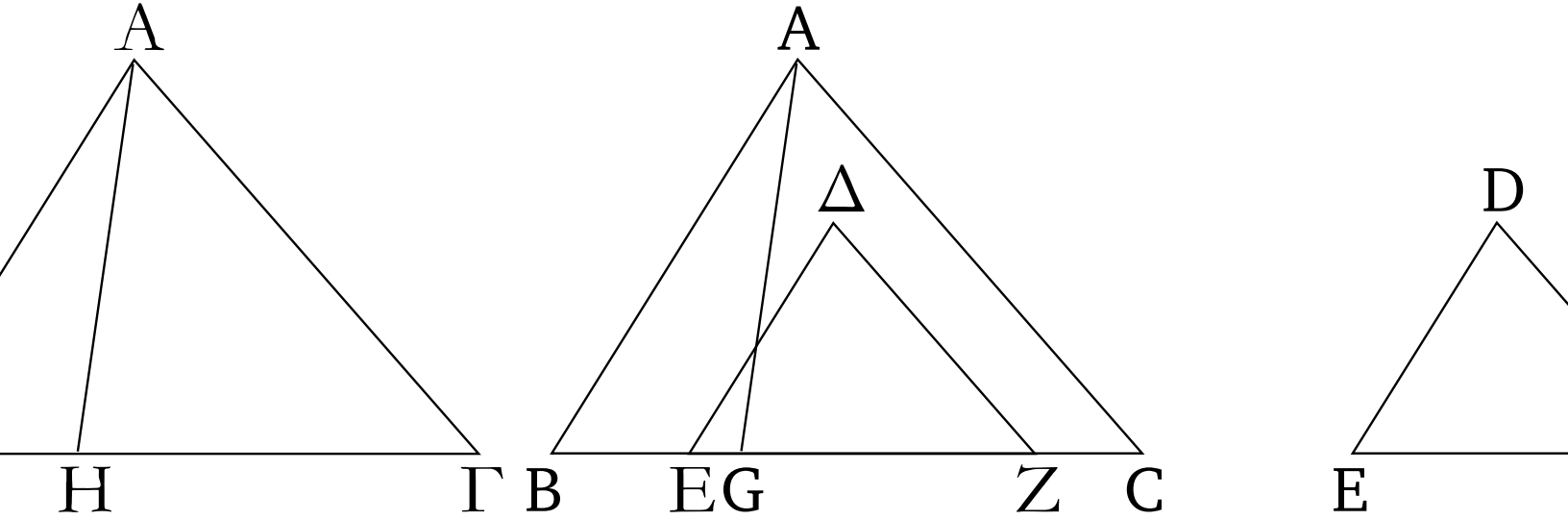
Thus, the rectilinear figure AH , similar, and similarly laid down, to the given rectilinear figure CE has been constructed on the given straight-line AB . (Which is) the very thing it was required to do.

Proposition 19

Similar triangles are to one another in the squared[†] ratio of (their) corresponding sides.

Let ABC and DEF be similar triangles having the angle at B equal to the (angle) at E , and AB to BC , as DE (is) to EF , such that BC corresponds to EF . I say that triangle ABC has a squared ratio to triangle DEF with respect to (that side) BC (has) to EF .

For let a third (straight-line), BG , be taken (which is) proportional to BC and EF , so that as BC (is) to EF , so EF (is) to BG [Prop. 6.11]. And let AG be joined.



Ἐπεὶ οὖν ἔστιν ὡς ἡ AB πρὸς τὴν $BΓ$, οὕτως ἡ $ΔΕ$ πρὸς τὴν $ΕΖ$, ἐναλλάξ ἄρα ἔστιν ὡς ἡ AB πρὸς τὴν $ΔΕ$, οὕτως ἡ $BΓ$ πρὸς τὴν $ΕΖ$. ἀλλ' ὡς ἡ $BΓ$ πρὸς $ΕΖ$, οὕτως ἔστιν ἡ $ΕΖ$ πρὸς $BΗ$. καὶ ὡς ἄρα ἡ AB πρὸς $ΔΕ$, οὕτως ἡ $ΕΖ$ πρὸς $BΗ$. τῶν ABH , $ΔΕΖ$ ἄρα τριγώνων ἀντιπεπόνθασιν αἱ πλευраὶ αἱ περὶ τὰς ἴσας γωνίας. ὦν δὲ μίαν μὲν ἴσην ἐθόντων γωνίαν τριγώνων ἀντιπεπόνθασιν αἱ πλευраὶ αἱ περὶ τὰς ἴσας γωνίας, ἴσα ἔστιν ἐκεῖνα. ἴσον ἄρα ἔστι τὸ ABH τρίγωνον τῷ $ΔΕΖ$ τριγώνῳ. καὶ ἔπει ἔστιν ὡς ἡ $BΓ$ πρὸς τὴν $ΕΖ$, οὕτως ἡ $ΕΖ$ πρὸς τὴν $BΗ$, ἔαν δὲ τρεῖς εὐθεῖαι ἀνάλογον ᾗσιν, ἡ πρώτη πρὸς τὴν τρίτην διπλασίονα λόγον ἔθει ἢ περ πρὸς τὴν δευτέραν, ἡ $BΓ$ ἄρα πρὸς τὴν $BΗ$ διπλασίονα λόγον ἔθει ἢ περ ἡ $ΓΒ$ πρὸς τὴν $ΕΖ$. ὡς δὲ ἡ $ΓΒ$ πρὸς τὴν $BΗ$, οὕτως τὸ $ABΓ$ τρίγωνον πρὸς τὸ ABH τρίγωνον· καὶ τὸ $ABΓ$ ἄρα τρίγωνον πρὸς τὸ ABH διπλασίονα λόγον ἔθει ἢ περ ἡ $BΓ$ πρὸς τὴν $ΕΖ$. ἴσον δὲ τὸ ABH τρίγωνον τῷ $ΔΕΖ$ τριγώνῳ· καὶ τὸ $ABΓ$ ἄρα τρίγωνον πρὸς τὸ $ΔΕΖ$ τρίγωνον διπλασίονα λόγον ἔθει ἢ περ ἡ $BΓ$ πρὸς τὴν $ΕΖ$.

Τὰ ἄρα ὅμοια τρίγωνα πρὸς ἀλλήλα ἐν διπλασίονι λόγῳ ἔστι τῶν ὁμολόγων πλευρῶν. [ὅπερ ἔδει δεῖξαι.]

Therefore, since as AB is to BC , so DE (is) to EF , thus, alternately, as AB is to DE , so BC (is) to EF [Prop. 5.16]. But, as BC (is) to EF , so EF is to BG . And, thus, as AB (is) to DE , so EF (is) to BG . Thus, for triangles ABG and DEF , the sides about the equal angles are reciprocally proportional. And those triangles having one (angle) equal to one (angle) for which the sides about the equal angles are reciprocally proportional are equal [Prop. 6.15]. Thus, triangle ABG is equal to triangle DEF . And since as BC (is) to EF , so EF (is) to BG , and if three straight-lines are proportional, (then) the first has a squared ratio to the third with respect to the second [Def. 5.9], BC thus has a squared ratio to BG with respect to (that) CB (has) to EF . And as CB (is) to BG , so triangle ABC (is) to triangle ABG [Prop. 6.1]. Thus, triangle ABC also has a squared ratio to (triangle) ABG with respect to (that side) BC (has) to EF . And triangle ABG (is) equal to triangle DEF . Thus, triangle ABC also has a squared ratio to triangle DEF with respect to (that side) BC (has) to EF .

Thus, similar triangles are to one another in the squared ratio of (their) corresponding sides. [(Which is) the very thing it was required to show].

Πόρισμα.

Ἐκ δὲ τούτου φανερόν, ὅτι, ἔαν τρεῖς εὐθεῖαι ἀνάλογον ᾗσιν, ἔστιν ὡς ἡ πρώτη πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς πρώτης εἰδος πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ ὅμοιον καὶ ὁμοίως ἀναγραφόμενον. ὅπερ ἔδει δεῖξαι.

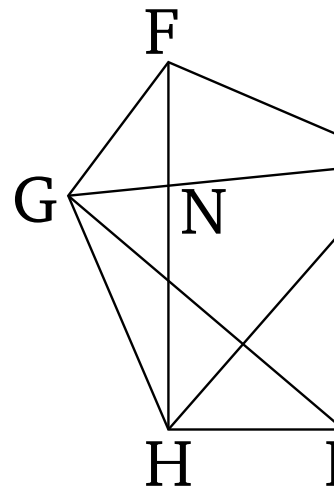
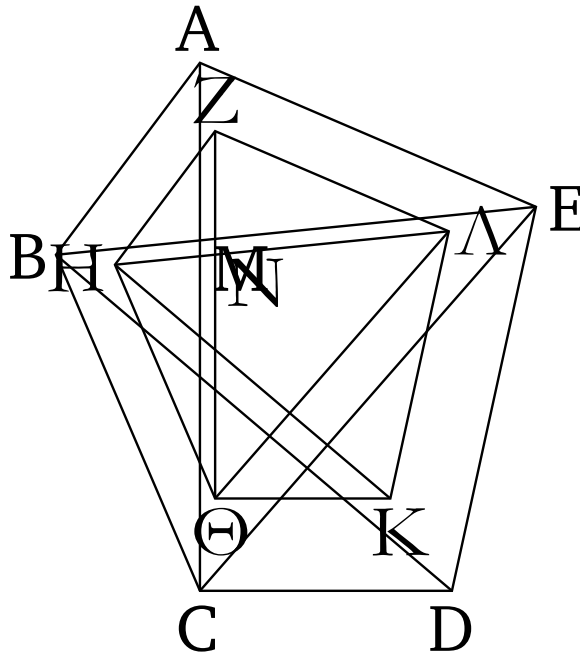
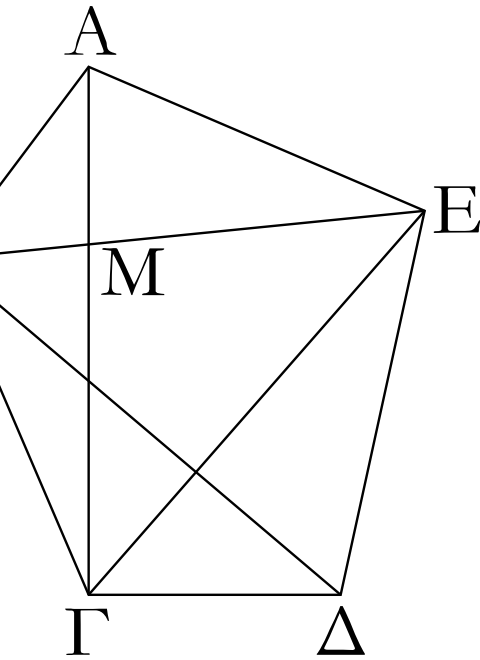
Corollary

So it is clear, from this, that if three straight-lines are proportional, (then) as the first is to the third, so the figure (described) on the first (is) to the similar, and similarly described, (figure) on the second. (Which is) the very thing it was required to show.

† Literally, "double".

εἰς ἴσα τὸ πλῆθος καὶ ὁμόλογα τοῖς ὅλοις, καὶ τὸ πολὺγωνον πρὸς τὸ πολὺγωνον διπλασίονα λόγον ἔθει ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν.

Ἐστω ὅμοια πολὺγωνα τὰ $ABΓΔΕ$, $ZHΘΚΛ$, ὁμόλογος δὲ ἔστω ἡ AB τῇ ZH . λέγω, ὅτι τὰ $ABΓΔΕ$, $ZHΘΚΛ$ πολὺγωνα εἰς τε ὅμοια τρίγωνα διαιρεῖται καὶ εἰς ἴσα τὸ πλῆθος καὶ ὁμόλογα τοῖς ὅλοις, καὶ τὸ $ABΓΔΕ$ πολὺγωνον πρὸς τὸ $ZHΘΚΛ$ πολὺγωνον διπλασίονα λόγον ἔθει ἥπερ ἡ AB πρὸς τὴν ZH .



Ἐπεξεύθυσαν αἱ BE , EG , HA , $ΛΘ$.

Καὶ ἐπεὶ ὁμοῖόν ἐστι τὸ $ABΓΔΕ$ πολὺγωνον τῷ $ZHΘΚΛ$ πολὺγῳ, ἴση ἐστὶν ἡ ὑπὸ BAE γωνία τῇ ὑπὸ HZA . καὶ ἐστὶν ὡς ἡ BA πρὸς AE , οὕτως ἡ HZ πρὸς ZA . ἐπεὶ οὖν δύο τρίγωνα ἐστὶ τὰ ABE , ZHA μίαν γωνίαν μὲν γωνία ἴσην ἔχοντα, περὶ δὲ τὰς ἴσας γωνίας τὰς πλευρὰς ἀνάλογον, ἰσογώνιον ἄρα ἐστὶ τὸ ABE τρίγωνον τῷ ZHA τριγῳ. ὥστε καὶ ὁμοῖον. ἴση ἄρα ἐστὶν ἡ ὑπὸ ABE γωνία τῇ ὑπὸ ZHA . ἐστὶ δὲ καὶ ὅλη ἡ ὑπὸ $ABΓ$ ὅλη τῇ ὑπὸ $ZHΘ$ ἴση διὰ τὴν ὁμοιότητα τῶν πολυγώνων. λοιπὴ ἄρα ἡ ὑπὸ $EBΓ$ γωνία τῇ ὑπὸ $ΛΗΘ$ ἐστὶν ἴση. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ABE , ZHA τριγώνων ἐστὶν ὡς ἡ EB πρὸς BA , οὕτως ἡ $ΛΗ$ πρὸς HZ , ἀλλὰ μὴν καὶ διὰ τὴν ὁμοιότητα τῶν πολυγώνων ἐστὶν ὡς ἡ AB πρὸς $ΒΓ$, οὕτως ἡ ZH πρὸς $ΗΘ$, δι' ἴσον ἄρα ἐστὶν ὡς ἡ EB πρὸς $ΒΓ$, οὕτως ἡ $ΛΗ$ πρὸς $ΗΘ$, καὶ περὶ τὰς ἴσας γωνίας τὰς ὑπὸ $EBΓ$, $ΛΗΘ$ αἱ πλευραὶ ἀνάλογόν εἰσιν ἰσογώνιον ἄρα ἐστὶ τὸ $EBΓ$ τρίγωνον τῷ $ΛΗΘ$ τριγῳ. ὥστε καὶ ὁμοῖόν ἐστι τὸ $EBΓ$ τρίγωνον τῷ $ΛΗΘ$ τριγῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $ΕΓΔ$ τρίγωνον ὁμοῖόν ἐστι τῷ $ΑΘΚ$ τριγῳ. τὰ ἄρα ὅμοια πολὺγωνα τὰ $ABΓΔΕ$, $ZHΘΚΛ$ εἰς τε ὅμοια

similar triangles corresponding (in proportion) to the wholes, and one polygon has to the (other) polygon a squared ratio with respect to (that) a corresponding side (has) to a corresponding side.

Let $ABCDE$ and $FGHKL$ be similar polygons, and let AB correspond to FG . I say that polygons $ABCDE$ and $FGHKL$ can be divided into equal numbers of similar triangles corresponding (in proportion) to the wholes, and (that) polygon $ABCDE$ has a squared ratio to polygon $FGHKL$ with respect to that AB (has) to FG .

Let BE , EC , GL , and LH be joined.

And since polygon $ABCDE$ is similar to polygon $FGHKL$, angle BAE is equal to angle GFL , and as BA is to AE , so GF (is) to FL [Def. 6.1]. Therefore, since ABE and FGL are two triangles having one angle equal to one angle and the sides about the equal angles proportional, triangle ABE is thus equiangular to triangle FGL [Prop. 6.6]. Hence, (they are) also similar [Prop. 6.4, Def. 6.1]. Thus, angle ABE is equal to (angle) FGL . And the whole (angle) ABC is equal to the whole (angle) FGH , on account of the similarity of the polygons. Thus, the remaining angle EBC is equal to LGH . And since, on account of the similarity of triangles ABE and FGL , as EB is to BA , so LG (is) to GF , but also, on account of the similarity of the polygons, as AB is to BC , so FG (is) to GH , thus, via equality, as EB is to BC , so LG (is) to GH [Prop. 5.22], and the sides about the equal angles, EBC and LGH , are proportional. Thus, triangle EBC is equiangular to triangle LGH [Prop. 6.6]. Hence, triangle EBC is also similar to triangle LGH [Prop. 6.4, Def. 6.1]. So, for the same (reasons), triangle ECD is also similar to triangle LHK . Thus, the similar

τρίγωνα διήρηται καὶ εἰς ἴσα τὸ πλῆθος.

Λέγω, ὅτι καὶ ὁμόλογα τοῖς ὅλοις, τουτέστιν ὥστε ἀνάλογον εἶναι τὰ τρίγωνα, καὶ ἡγούμενα μὲν εἶναι τὰ ABE , $EBΓ$, $ΕΓΔ$, ἐπόμενα δὲ αὐτῶν τὰ ZHA , $ΛΗΘ$, $ΛΘΚ$, καὶ ὅτι τὸ $ABΓΔΕ$ πολύγωνον πρὸς τὸ $ZHΘΚΛ$ πολύγωνον διπλασίονα λόγον ἔθει ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἡ AB πρὸς τὴν ZH .

Ἐπεξέυθθωσαν γάρ αἱ $ΑΓ$, $ΖΘ$. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν πολυγώνων ἴση ἐστὶν ἡ ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ZHΘ$, καὶ ἐστὶν ὡς ἡ AB πρὸς $ΒΓ$, οὕτως ἡ ZH πρὸς $ΗΘ$, ἰσογώνιον ἐστὶ τὸ $ABΓ$ τρίγωνον τῷ $ZHΘ$ τριγώνῳ· ἴση ἄρα ἐστὶν ἡ μὲν ὑπὸ $BAΓ$ γωνία τῇ ὑπὸ $HZΘ$, ἡ δὲ ὑπὸ $ΒΓΑ$ τῇ ὑπὸ $ΗΘΖ$. καὶ ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ BAM γωνία τῇ ὑπὸ HZN , ἐστὶ δὲ καὶ ἡ ὑπὸ ABM τῇ ὑπὸ ZHN ἴση, καὶ λοιπὴ ἄρα ἡ ὑπὸ AMB λοιπὴ τῇ ὑπὸ ZNH ἴση ἐστὶν· ἰσογώνιον ἄρα ἐστὶ τὸ ABM τρίγωνον τῷ ZHN τριγώνῳ. ὁμοίως δὲ δεῖχμεν, ὅτι καὶ τὸ $BMΓ$ τρίγωνον ἰσογώνιον ἐστὶ τῷ $HNΘ$ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν, ὡς μὲν ἡ AM πρὸς MB , οὕτως ἡ ZN πρὸς NH , ὡς δὲ ἡ BM πρὸς $ΜΓ$, οὕτως ἡ HN πρὸς $NΘ$. ὥστε καὶ δι' ἴσον, ὡς ἡ AM πρὸς $ΜΓ$, οὕτως ἡ ZN πρὸς $NΘ$. ἀλλ' ὡς ἡ AM πρὸς $ΜΓ$, οὕτως τὸ ABM [τρίγωνον] πρὸς τὸ $MBΓ$, καὶ τὸ AME πρὸς τὸ $EMΓ$ · πρὸς ἄλληλα γάρ εἰσιν ὡς αἱ βάσεις. καὶ ὡς ἄρα ἐν τῶν ἡγουμένων πρὸς ἐν τῶν ἐπόμενων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα· ὡς ἄρα τὸ AMB τρίγωνον πρὸς τὸ $BMΓ$, οὕτως τὸ ABE πρὸς τὸ $ΓBE$. ἀλλ' ὡς τὸ AMB πρὸς τὸ $BMΓ$, οὕτως ἡ AM πρὸς $ΜΓ$ · καὶ ὡς ἄρα ἡ AM πρὸς $ΜΓ$, οὕτως τὸ ABE τρίγωνον πρὸς τὸ $EBΓ$ τρίγωνον. διὰ τὰ αὐτὰ δὲ καὶ ὡς ἡ ZN πρὸς $NΘ$, οὕτως τὸ ZHA τρίγωνον πρὸς τὸ $ΗΛΘ$ τρίγωνον. καὶ ἐστὶν ὡς ἡ AM πρὸς $ΜΓ$, οὕτως ἡ ZN πρὸς $NΘ$ · καὶ ὡς ἄρα τὸ ABE τρίγωνον πρὸς τὸ $BEΓ$ τρίγωνον, οὕτως τὸ ZHA τρίγωνον πρὸς τὸ $ΗΛΘ$ τρίγωνον, καὶ ἐναλλάξ ὡς τὸ ABE τρίγωνον πρὸς τὸ ZHA τρίγωνον, οὕτως τὸ $BEΓ$ τρίγωνον πρὸς τὸ $ΗΛΘ$ τρίγωνον. ὁμοίως δὲ δεῖχμεν ἐπιξευθθεισῶν τῶν $ΒΔ$, $ΗΚ$, ὅτι καὶ ὡς τὸ $BEΓ$ τρίγωνον πρὸς τὸ $ΛΗΘ$ τρίγωνον, οὕτως τὸ $ΕΓΔ$ τρίγωνον πρὸς τὸ $ΛΘΚ$ τρίγωνον. καὶ ἐπεὶ ἐστὶν ὡς τὸ ABE τρίγωνον πρὸς τὸ ZHA τρίγωνον, οὕτως τὸ $EBΓ$ πρὸς τὸ $ΛΗΘ$, καὶ ἔτι τὸ $ΕΓΔ$ πρὸς τὸ $ΛΘΚ$, καὶ ὡς ἄρα ἐν τῶν ἡγουμένων πρὸς ἐν τῶν ἐπόμενων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα· ἐστὶν ἄρα ὡς τὸ ABE τρίγωνον πρὸς τὸ ZHA τρίγωνον, οὕτως τὸ $ABΓΔΕ$ πολύγωνον πρὸς τὸ $ZHΘΚΛ$ πολύγωνον. ἀλλὰ τὸ ABE τρίγωνον πρὸς τὸ ZHA τρίγωνον διπλασίονα λόγον ἔθει ἥπερ ἡ AB ὁμόλογος πλευρὰ πρὸς τὴν ZH ὁμόλογον πλευράν· τὰ γάρ ὅμοια τρίγωνα ἐν διπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν. καὶ τὸ $ABΓΔΕ$ ἄρα πολύγωνον πρὸς τὸ $ZHΘΚΛ$ πολύγωνον διπλασίονα λόγον ἔθει ἥπερ ἡ AB ὁμόλογος πλευρὰ πρὸς τὴν ZH ὁμόλογον πλευράν.

Τὰ ἄρα ὅμοια πολύγωνα εἰς τε ὅμοια τρίγωνα διαιρεῖται καὶ εἰς ἴσα τὸ πλῆθος καὶ ὁμόλογα τοῖς ὅλοις, καὶ τὸ πολύγωνον πρὸς τὸ πολύγωνον διπλασίονα λόγον ἔθει ἥπερ

polygons $ABCDE$ and $FGHKL$ have been divided into equal numbers of similar triangles.

I also say that (the triangles) correspond (in proportion) to the wholes. That is to say, the triangles are proportional: ABE , EBC , and ECD are the leading (magnitudes), and their (associated) following (magnitudes are) FGL , LGH , and LHK (respectively). (I) also (say) that polygon $ABCDE$ has a squared ratio to polygon $FGHKL$ with respect to (that) a corresponding side (has) to a corresponding side—that is to say, (side) AB to FG .

For let AC and FH be joined. And since angle ABC is equal to FGH , and as AB is to BC , so FG (is) to GH , on account of the similarity of the polygons, triangle ABC is equiangular to triangle FGH [Prop. 6.6]. Thus, angle BAC is equal to $G FH$, and (angle) BCA to $G HF$. And since angle BAM is equal to $G FN$, and (angle) ABM is also equal to FGN (see earlier), the remaining (angle) AMB is thus also equal to the remaining (angle) FNG [Prop. 1.32]. Thus, triangle ABM is equiangular to triangle FGN . So, similarly, we can show that triangle BMC is also equiangular to triangle GNH . Thus, proportionally, as AM is to MB , so FN (is) to NG , and as BM (is) to MC , so GN (is) to NH [Prop. 6.4]. Hence, also, via equality, as AM (is) to MC , so FN (is) to NH [Prop. 5.22]. But, as AM (is) to MC , so [triangle] ABM is to MBC , and AME to EMC . For they are to one another as their bases [Prop. 6.1]. And as one of the leading (magnitudes) is to one of the following (magnitudes), so (the sum of) all the leading (magnitudes) is to (the sum of) all the following (magnitudes) [Prop. 5.12]. Thus, as triangle AMB (is) to BMC , so (triangle) ABE (is) to CBE . But, as (triangle) AMB (is) to BMC , so AM (is) to MC . Thus, also, as AM (is) to MC , so triangle ABE (is) to triangle EBC . And so, for the same (reasons), as FN (is) to NH , so triangle FGL (is) to triangle GLH . And as AM is to MC , so FN (is) to NH . Thus, also, as triangle ABE (is) to triangle BEC , so triangle FGL (is) to triangle GLH , and, alternately, as triangle ABE (is) to triangle FGL , so triangle BEC (is) to triangle GLH [Prop. 5.16]. So, similarly, we can also show, by joining BD and GK , (that) as triangle BEC (is) to triangle LGH , so triangle ECD (is) to triangle LHK . And since as triangle ABE is to triangle FGL , so (triangle) EBC (is) to LGH , and, further, (triangle) ECD to LHK , and also as one of the leading (magnitudes is) to one of the following, so (the sum of) all the leading (magnitudes is) to (the sum of) all the following [Prop. 5.12], thus as triangle ABE is to triangle FGL , so polygon $ABCDE$ (is) to polygon $FGHKL$. But, triangle ABE has a squared ratio to triangle FGL with respect to (that) the corresponding side AB (has) to the corresponding side FG . For, similar triangles are in the squared ratio of corresponding sides [Prop. 6.14]. Thus, polygon $ABCDE$ also has a squared ratio to polygon $FGHKL$ with respect to (that) the corresponding side AB (has) to the corresponding side FG .

ἡ ὁμόλογος πλευρά πρόξ τὴν ὁμόλογον πλευράν [ὅπερ ἔδει δείχαι].

Thus, similar polygons can be divided into equal numbers of similar triangles corresponding (in proportion) to the wholes, and one polygon has to the (other) polygon a squared ratio with respect to (that) a corresponding side (has) to a corresponding side. [(Which is) the very thing it was required to show].

Πόρισμα.

Ἦσάντως δὲ καὶ ἐπὶ τῶν [ὁμοίων] τετραπλεύρων δειθθήσεται, ὅτι ἐν διπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. ἐδείχθη δὲ καὶ ἐπὶ τῶν τριγώνων· ὥστε καὶ καθόλου τὰ ὅμοια εὐθύγραμμα σήματα πρόξ ἄλληλα ἐν διπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. ὅπερ ἔδει δείχαι.

Corollary

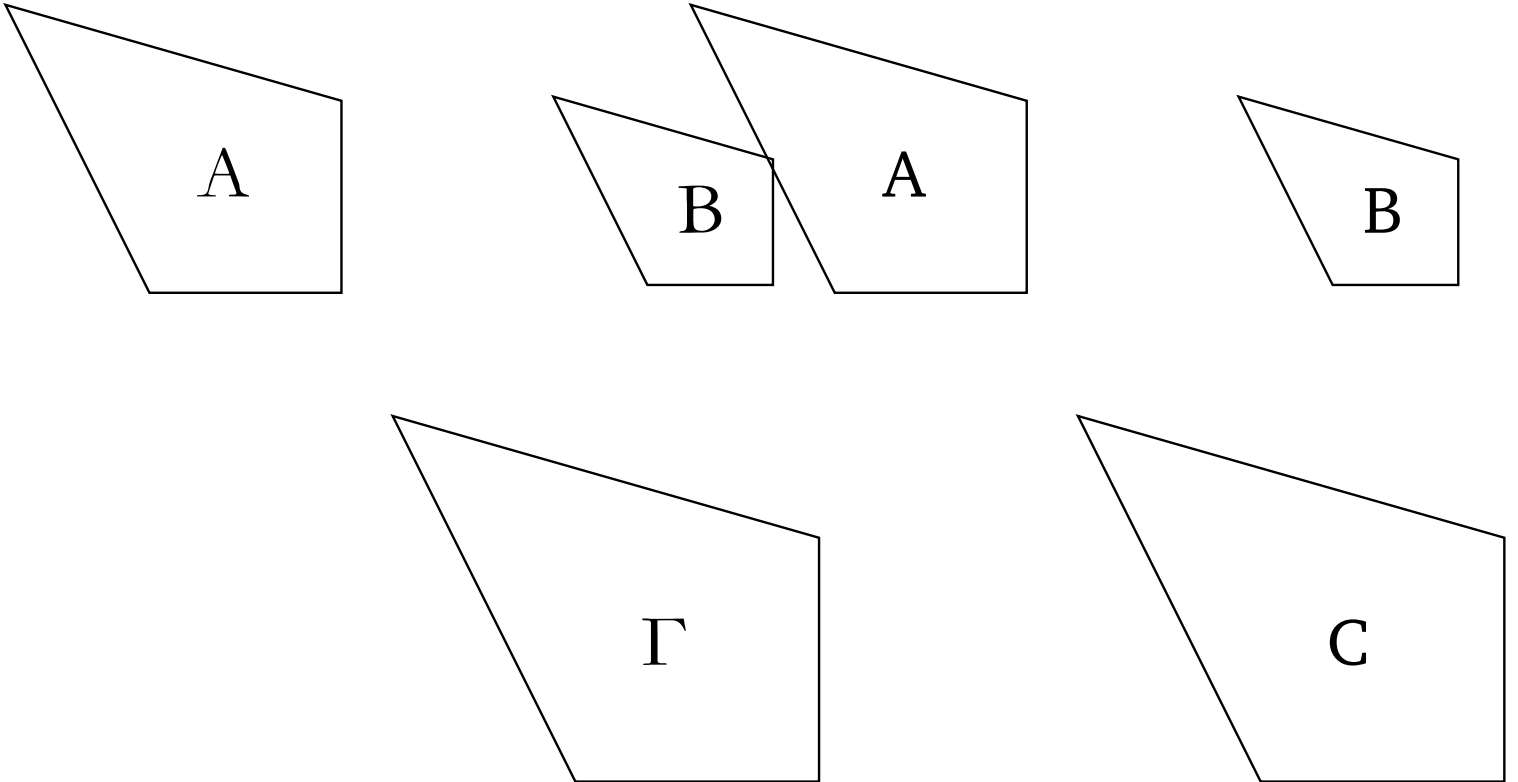
And, in the same manner, it can also be shown for [similar] quadrilaterals, (that) they are in the squared ratio of (their) corresponding sides. And it was also shown for triangles. Hence, in general, similar rectilinear figures are also to one another in the squared ratio of (their) corresponding sides. (Which is) the very thing it was required to show.

21.

Τὰ τῶ αὐτῷ εὐθύγραμμῳ ὅμοια καὶ ἀλλήλοις ἐστὶν ὅμοια.

Proposition 21

(Rectilinear figures) similar to the same rectilinear figure are also similar to one another.



Ἐστω γάρ ἐκάτερον τῶν A, B εὐθύγραμμων τῷ Γ ὅμοιον λέγω, ὅτι καὶ τὸ A τῷ B ἐστὶν ὅμοιον.

Ἐπεὶ γὰρ ὅμοιον ἐστὶ τὸ A τῷ Γ , ἰσογώνιον τέ ἐστὶν αὐτῷ καὶ τὰς περὶ τὰς ἴσας γωνίας πλευράς ἀνάλογον ἔθει. πάλιν, ἐπεὶ ὅμοιον ἐστὶ τὸ B τῷ Γ , ἰσογώνιον τέ ἐστὶν αὐτῷ καὶ τὰς περὶ τὰς ἴσας γωνίας πλευράς ἀνάλογον ἔθει. ἐκάτερον ἄρα

Let each of the rectilinear figures A and B be similar to (the rectilinear figure) C . I say that A is also similar to B .

For since A is similar to C , (A) is equiangular to (C), and has the sides about the equal angles proportional [Def. 6.1]. Again, since B is similar to C , (B) is equiangular to (C), and has the sides about the equal angles proportional [Def. 6.1].

τῶν A, B τῶ Γ ἰσογώνιον τέ ἐστι καὶ τὰς περὶ τὰς ἴσας γωνίας πλευράς ἀνάλογον ἔθει [ὥστε καὶ τὸ A τῶ B ἰσογώνιον τέ ἐστι καὶ τὰς περὶ τὰς ἴσας γωνίας πλευράς ἀνάλογον ἔθει]. ὅμοιον ἄρα ἐστὶ τὸ A τῶ B · ὅπερ ἔδει δεῖχαι.

Thus, A and B are each equiangular to C , and have the sides about the equal angles proportional [hence, A is also equiangular to B , and has the sides about the equal angles proportional]. Thus, A is similar to B [Def. 6.1]. (Which is) the very thing it was required to show.

22.

Proposition 22

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ᾧσιν, καὶ τὰ ἀπ' αὐτῶν εὐθύγραμμα ὅμοιά τε καὶ ὁμοίως ἀναγεγραμμένα ἀνάλογον ἔσται· κ' ἂν τὰ ἀπ' αὐτῶν εὐθύγραμμα ὅμοιά τε καὶ ὁμοίως ἀναγεγραμμένα ἀνάλογον ᾧ, καὶ αὐταὶ αἱ εὐθεῖαι ἀνάλογον ἔσονται.

If four straight-lines are proportional, (then) similar, and similarly described, rectilinear figures (drawn) on them will also be proportional. And if similar, and similarly described, rectilinear figures (drawn) on them are proportional, (then) the straight-lines themselves will also be proportional.

Ἔστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ $AB, \Gamma\Delta, EZ, H\Theta$, ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$, καὶ ἀναγεγράφθωσαν ἀπὸ μὲν τῶν $AB, \Gamma\Delta$ ὅμοιά τε καὶ ὁμοίως κείμενα εὐθύγραμμα τὰ $KAB, \Lambda\Gamma\Delta$, ἀπὸ δὲ τῶν $EZ, H\Theta$ ὅμοιά τε καὶ ὁμοίως κείμενα εὐθύγραμμα τὰ $MZ, N\Theta$ · λέγω, ὅτι ἐστὶν ὥς τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, οὕτως τὸ MZ πρὸς τὸ $N\Theta$.

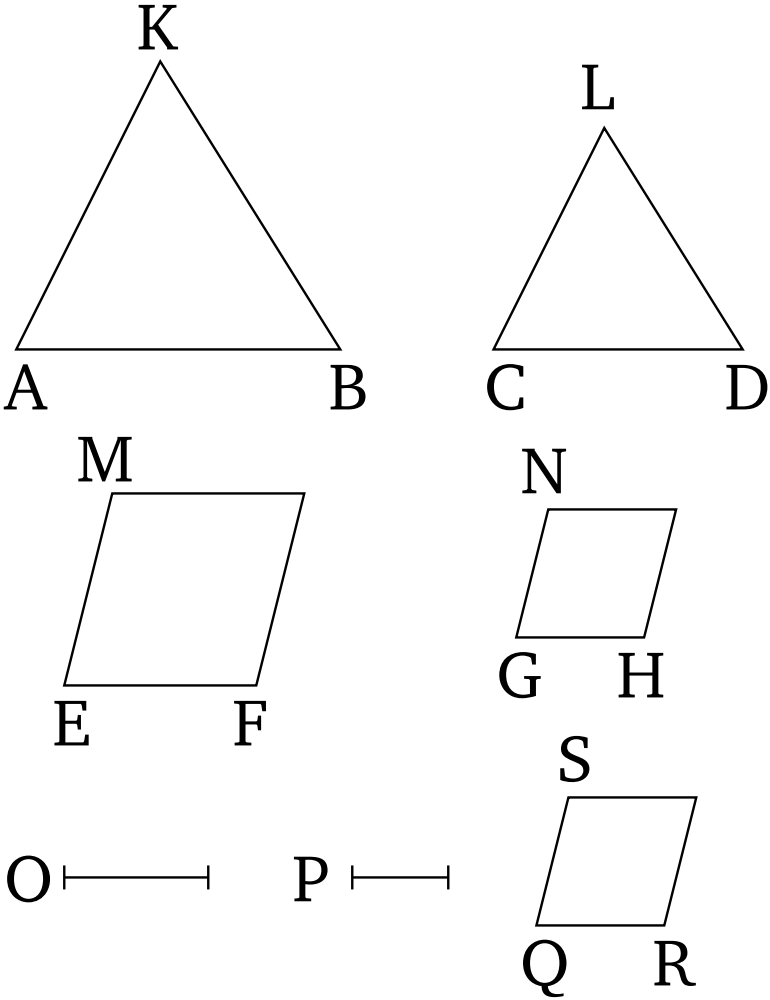
Let AB, CD, EF , and GH be four proportional straight-lines, (such that) as AB (is) to CD , so EF (is) to GH . And let the similar, and similarly laid out, rectilinear figures KAB and LCD be described on AB and CD (respectively), and the similar, and similarly laid out, rectilinear figures MF and NH on EF and GH (respectively). I say that as KAB is to LCD , so MF (is) to NH .

Εἰλήφθω γὰρ τῶν μὲν $AB, \Gamma\Delta$ τρίτη ἀνάλογον ἡ X , τῶν δὲ $EZ, H\Theta$ τρίτη ἀνάλογον ἡ O . καὶ ἐπεὶ ἐστὶν ὥς μὲν ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$, ὥς δὲ ἡ $\Gamma\Delta$ πρὸς τὴν X , οὕτως ἡ $H\Theta$ πρὸς τὴν O , δι' ἴσον ἄρα ἐστὶν ὥς ἡ AB πρὸς τὴν X , οὕτως ἡ EZ πρὸς τὴν O . ἀλλ' ὥς μὲν ἡ AB πρὸς τὴν X , οὕτως [καὶ] τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, ὥς δὲ ἡ EZ πρὸς τὴν O , οὕτως τὸ MZ πρὸς τὸ $N\Theta$ · καὶ ὥς ἄρα τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, οὕτως τὸ MZ πρὸς τὸ $N\Theta$.

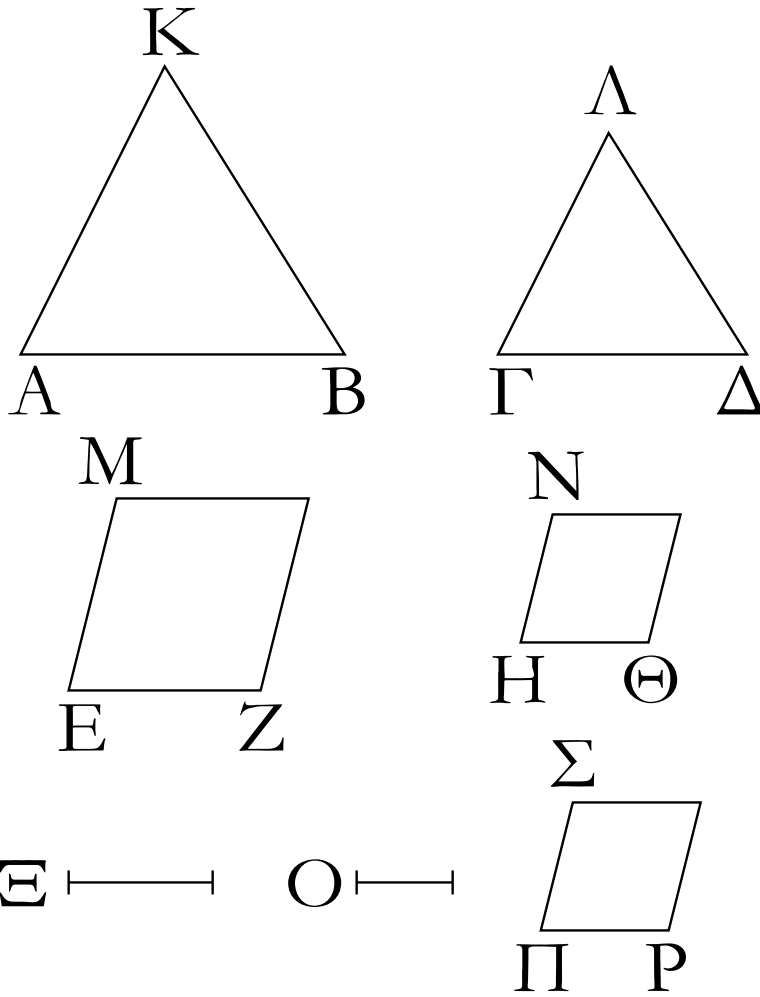
For let a third (straight-line) O be taken (which is) proportional to AB and CD , and a third (straight-line) P proportional to EF and GH [Prop. 6.11]. And since as AB is to CD , so EF (is) to GH , and as CD (is) to O , so GH (is) to P , thus, via equality, as AB is to O , so EF (is) to P [Prop. 5.22]. But, as AB (is) to O , so [also] KAB (is) to LCD , and as EF (is) to P , so MF (is) to NH [Prop. 5.19 corr.]. And, thus, as KAB (is) to LCD , so MF (is) to NH .

Ἀλλὰ δὴ ἔστω ὥς τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, οὕτως τὸ MZ πρὸς τὸ $N\Theta$ · λέγω, ὅτι ἐστὶ καὶ ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$. εἰ γὰρ μὴ ἐστὶν, ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$, ἔστω ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν ΠP , καὶ ἀναγεγράφθω ἀπὸ τῆς ΠP ὁποτέρῳ τῶν $MZ, N\Theta$ ὁμοίον τε καὶ ὁμοίως κείμενον

And so let KAB be to LCD , as MF (is) to NH . I say also that as AB is to CD , so EF (is) to GH . For if as AB is to CD , so EF (is) not to GH , let AB be to CD , as EF (is) to QR [Prop. 6.12]. And let the rectilinear figure SR , similar, and similarly laid down, to either of MF or NH , be described on QR [Props. 6.18, 6.21].



εὐθύγραμμον τὸ ΣΡ.



Ἐπεὶ οὖν ἔστιν ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν ΠP , καὶ ἀναγράφεται ἀπὸ μὲν τῶν AB , $\Gamma\Delta$ ὁμοίως τε καὶ ὁμοίως κείμενα τὰ KAB , $\Lambda\Gamma\Delta$, ἀπὸ δὲ τῶν EZ , ΠP ὁμοίως τε καὶ ὁμοίως κείμενα τὰ MZ , ΣP , ἔστιν ἄρα ὥς τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, οὕτως τὸ MZ πρὸς τὸ ΣP . ὑπόκειται δὲ καὶ ὥς τὸ KAB πρὸς τὸ $\Lambda\Gamma\Delta$, οὕτως τὸ MZ πρὸς τὸ $\text{N}\Theta$ · καὶ ὥς ἄρα τὸ MZ πρὸς τὸ ΣP , οὕτως τὸ MZ πρὸς τὸ $\text{N}\Theta$. τὸ MZ ἄρα πρὸς ἑκάτερον τῶν $\text{N}\Theta$, ΣP τὸν αὐτὸν ἔθει λόγον· ἴσον ἄρα ἐστὶ τὸ $\text{N}\Theta$ τῷ ΣP . ἔστι δὲ αὐτῷ καὶ ὅμοιον καὶ ὁμοίως κείμενον· ἴση ἄρα ἡ $\text{H}\Theta$ τῇ ΠP . καὶ ἐπεὶ ἔστιν ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν ΠP , ἴση δὲ ἡ ΠP τῇ $\text{H}\Theta$, ἔστιν ἄρα ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $\text{H}\Theta$.

Ἐάν ἄρα τέσσαρες εὐθεῖαι ἀνάλογον ᾧσιν, καὶ τὰ ἀπ' αὐτῶν εὐθύγραμμα ὁμοία τε καὶ ὁμοίως ἀναγεγραμμένα ἀνάλογον ἔσται· κ' ἂν τὰ ἀπ' αὐτῶν εὐθύγραμμα ὁμοία τε καὶ ὁμοίως ἀναγεγραμμένα ἀνάλογον ᾧ, καὶ αὐταὶ αἱ εὐθεῖαι ἀνάλογον ἔσονται· ὅπερ ἔδει δεῖξαι.

Therefore, since as AB is to CD , so EF (is) to QR , and the similar, and similarly laid out, (rectilinear figures) KAB and LCD have been described on AB and CD (respectively), and the similar, and similarly laid out, (rectilinear figures) MF and SR on EF and QR (respectively), thus as KAB is to LCD , so MF (is) to SR (see above). And it was also assumed that as KAB (is) to LCD , so MF (is) to NH . Thus, also, as MF (is) to SR , so MF (is) to NH [Prop. 5.11]. Thus, MF has the same ratio to each of NH and SR . Thus, NH is equal to SR [Prop. 5.9]. And it is also similar, and similarly laid out, to it. Thus, GH (is) equal to QR .[†] And since AB is to CD , as EF (is) to QR , and QR (is) equal to GH , thus as AB is to CD , so EF (is) to GH .

Thus, if four straight-lines are proportional, (then) similar, and similarly described, rectilinear figures (drawn) on them will also be proportional. And if similar, and similarly described, rectilinear figures (drawn) on them are proportional, (then) the straight-lines themselves will also be proportional. (Which is) the very thing it was required to show.

† Here, Euclid assumes, without proof, that if two similar figures are equal then any pair of corresponding sides is also equal.

23.

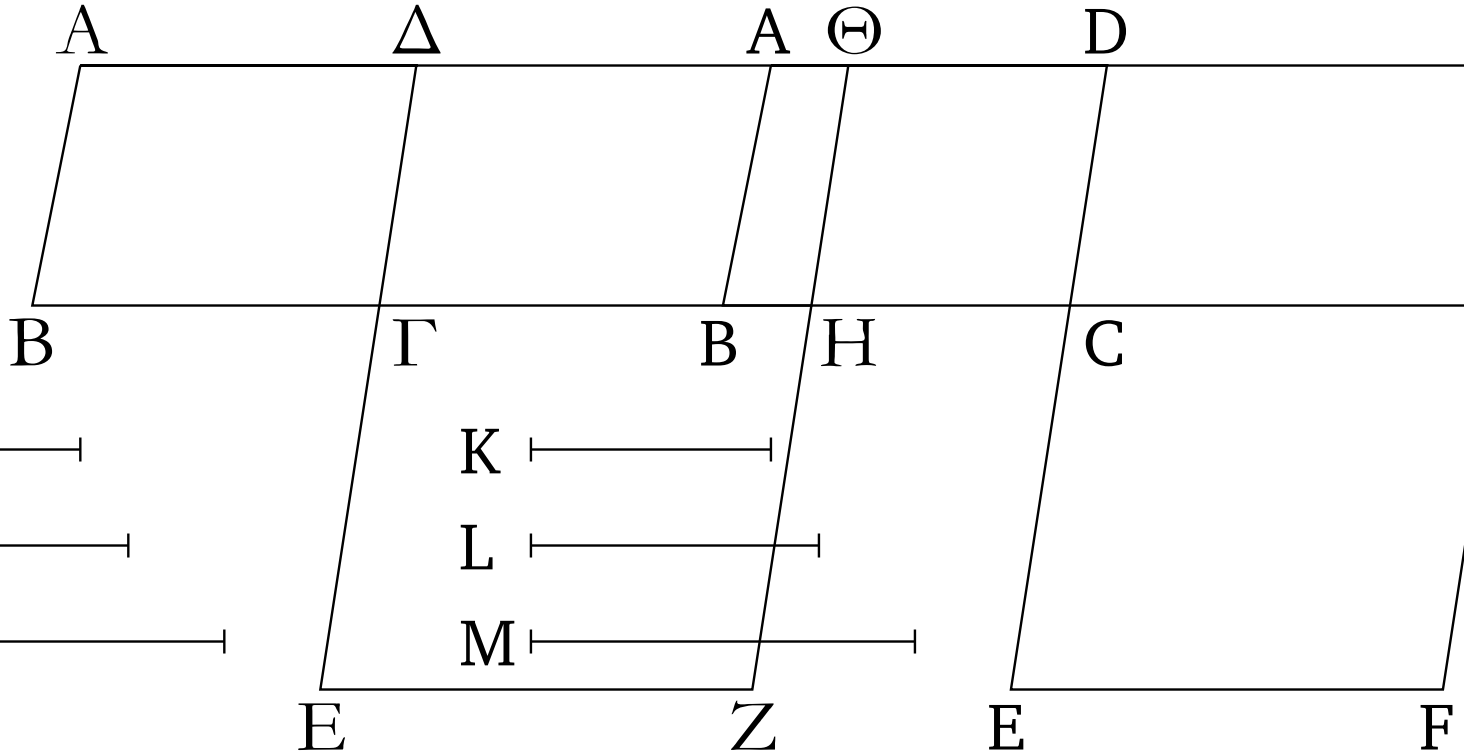
Proposition 23

Τὰ ἰσογώνια παραλληλόγραμμα πρὸς ἄλληλα λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν.

Ἐστω ἰσογώνια παραλληλόγραμμα τὰ $ΑΓ$, $ΓΖ$ ἴσην ἔθοντα τὴν ὑπὸ $ΒΓΔ$ γωνίαν τῇ ὑπὸ $ΕΓΗ$ · λέγω, ὅτι τὸ $ΑΓ$ παραλληλόγραμμον πρὸς τὸ $ΓΖ$ παραλληλόγραμμον λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν.

Equiangular parallelograms have to one another the ratio compounded† out of (the ratios of) their sides.

Let AC and CF be equiangular parallelograms having angle BCD equal to ECG . I say that parallelogram AC has to parallelogram CF the ratio compounded out of (the ratios of) their sides.



Κεῖσθω γὰρ ὥστε ἐπ' εὐθείᾳ εἶναι τὴν $ΒΓ$ τῇ $ΓΗ$ · ἐπ' εὐθείᾳ ἄρα ἐστὶ καὶ ἡ $ΔΓ$ τῇ $ΓΕ$. καὶ συμπληρώσθω τὸ $ΔΗ$ παραλληλόγραμμον, καὶ ἐκκεῖσθω τις εὐθεῖα ἡ K , καὶ γερονέτω ὥς μὲν ἡ $ΒΓ$ πρὸς τὴν $ΓΗ$, οὕτως ἡ K πρὸς τὴν $Λ$, ὥς δὲ ἡ $ΔΓ$ πρὸς τὴν $ΓΕ$, οὕτως ἡ $Λ$ πρὸς τὴν $Μ$.

Οἱ ἄρα λόγοι τῆς τε K πρὸς τὴν $Λ$ καὶ τῆς $Λ$ πρὸς τὴν $Μ$ οἱ αὐτοὶ εἰσι τοῖς λόγοις τῶν πλευρῶν, τῆς τε $ΒΓ$ πρὸς τὴν $ΓΗ$ καὶ τῆς $ΔΓ$ πρὸς τὴν $ΓΕ$. ἀλλ' ὁ τῆς K πρὸς $Μ$ λόγος σύγκειται ἐκ τε τοῦ τῆς K πρὸς $Λ$ λόγον καὶ τοῦ τῆς $Λ$ πρὸς $Μ$ · ὥστε καὶ ἡ K πρὸς τὴν $Μ$ λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν. καὶ ἐπεὶ ἐστὶν ὥς ἡ $ΒΓ$ πρὸς τὴν $ΓΗ$, οὕτως τὸ $ΑΓ$ παραλληλόγραμμον πρὸς τὸ $ΓΘ$, ἀλλ' ὥς ἡ $ΒΓ$ πρὸς τὴν $ΓΗ$, οὕτως ἡ K πρὸς τὴν $Λ$, καὶ ὥς ἄρα ἡ K πρὸς τὴν $Λ$, οὕτως τὸ $ΑΓ$ πρὸς τὸ $ΓΘ$. πάλιν, ἐπεὶ ἐστὶν ὥς ἡ $ΔΓ$ πρὸς τὴν $ΓΕ$, οὕτως τὸ $ΓΘ$ παραλληλόγραμμον πρὸς τὸ $ΓΖ$, ἀλλ' ὥς ἡ $ΔΓ$ πρὸς τὴν $ΓΕ$, οὕτως ἡ $Λ$ πρὸς τὴν $Μ$, καὶ ὥς ἄρα ἡ $Λ$ πρὸς τὴν $Μ$, οὕτως τὸ $ΓΘ$ παραλληλόγραμμον

For let BC be laid down so as to be straight-on to CG . Thus, DC is also straight-on to CE [Prop. 1.14]. And let the parallelogram DG be completed. And let some straight-line K be laid down. And let it be contrived that as BC (is) to CG , so K (is) to L , and as DC (is) to CE , so L (is) to M [Prop. 6.12].

Thus, the ratios of K to L and of L to M are the same as the ratios of the sides, (namely), BC to CG and DC to CE (respectively). But, the ratio of K to M is compounded out of the ratio of K to L and (the ratio) of L to M . Hence, K also has to M the ratio compounded out of (the ratios of) the sides (of the parallelograms). And since as BC is to CG , so parallelogram AC (is) to CH [Prop. 6.1], but as BC (is) to CG , so K (is) to L , thus, also, as K (is) to L , so (parallelogram) AC (is) to CH . Again, since as DC (is) to CE , so parallelogram CH (is) to CF [Prop. 6.1], but as DC (is) to CE , so L (is) to M , thus, also, as L (is) to M , so parallelogram CH (is) to parallelogram CF . Therefore, since it was shown that as K (is) to

πρὸς τὸ ΓΖ παραλληλόγραμμον. ἐπεὶ οὖν ἐδείκθη, ὥς μὲν ἡ Κ πρὸς τὴν Α, οὕτως τὸ ΑΓ παραλληλόγραμμον πρὸς τὸ ΓΘ παραλληλόγραμμον, ὥς δὲ ἡ Α πρὸς τὴν Μ, οὕτως τὸ ΓΘ παραλληλόγραμμον πρὸς τὸ ΓΖ παραλληλόγραμμον, δι' ἴσον ἄρα ἐστὶν ὥς ἡ Κ πρὸς τὴν Μ, οὕτως τὸ ΑΓ πρὸς τὸ ΓΖ παραλληλόγραμμον. ἡ δὲ Κ πρὸς τὴν Μ λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν· καὶ τὸ ΑΓ ἄρα πρὸς τὸ ΓΖ λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν.

Τὰ ἄρα ἰσογώνια παραλληλόγραμμα πρὸς ἄλληλα λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν· ὅπερ ἔδει δεῖξαι.

L , so parallelogram AC (is) to parallelogram CH , and as L (is) to M , so parallelogram CH (is) to parallelogram CF , thus, via equality, as K is to M , so (parallelogram) AC (is) to parallelogram CF [Prop. 5.22]. And K has to M the ratio compounded out of (the ratios of) the sides (of the parallelograms). Thus, (parallelogram) AC also has to (parallelogram) CF the ratio compounded out of (the ratio of) their sides.

Thus, equiangular parallelograms have to one another the ratio compounded out of (the ratio of) their sides. (Which is) the very thing it was required to show.

† In modern terminology, if two ratios are “compounded” then they are multiplied together.

24.

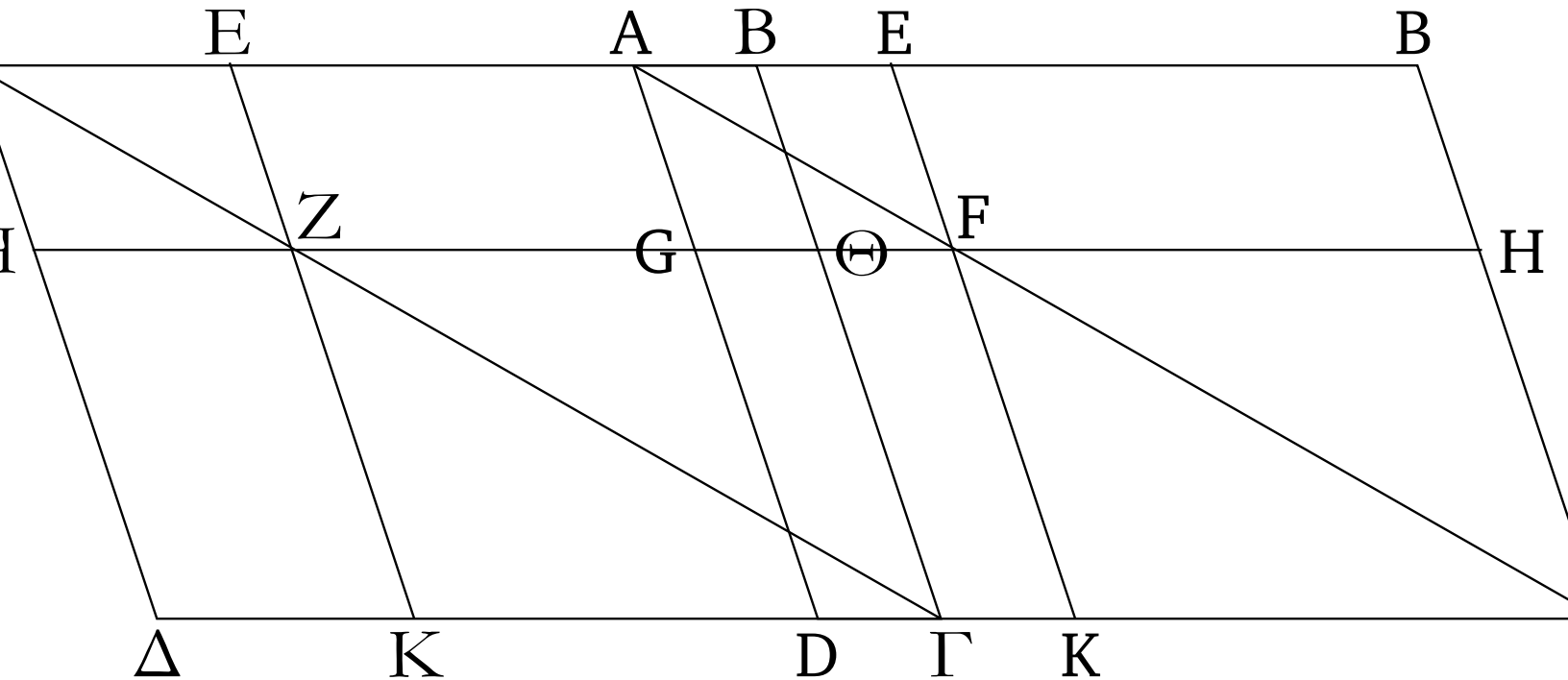
Proposition 24

Παντὸς παραλληλογράμμου τὰ περὶ τὴν διάμετρον παραλληλόγραμμα ὁμοία ἐστὶ τῷ τε ὅλῳ καὶ ἀλλήλοις.

Ἐστω παραλληλόγραμμον τὸ ΑΒΓΔ, διάμετρος δὲ αὐτοῦ ἡ ΑΓ, περὶ δὲ τὴν ΑΓ παραλληλόγραμμα ἔστω τὰ ΕΗ, ΘΚ· λέγω, ὅτι ἐκάτερον τῶν ΕΗ, ΘΚ παραλληλογράμμων ὁμοίον ἐστὶ ὅλῳ τῷ ΑΒΓΔ καὶ ἀλλήλοις.

In any parallelogram, the parallelograms about the diagonal are similar to the whole, and to one another.

Let $ABCD$ be a parallelogram, and AC its diagonal. And let EG and HK be parallelograms about AC . I say that the parallelograms EG and HK are each similar to the whole (parallelogram) $ABCD$, and to one another.



Ἐπεὶ γὰρ τριγώνου τοῦ ΑΒΓ παρὰ μίαν τῶν πλευρῶν τὴν ΒΓ ἤκται ἡ ΕΖ, ἀνάλογόν ἐστιν ὥς ἡ ΒΕ πρὸς τὴν ΕΑ, οὕτως ἡ ΓΖ πρὸς τὴν ΖΑ. πάλιν, ἐπεὶ τριγώνου τοῦ ΑΓΔ παρὰ μίαν τὴν ΓΔ ἤκται ἡ ΖΗ, ἀνάλογόν ἐστιν ὥς ἡ ΓΖ πρὸς τὴν ΖΑ, οὕτως ἡ ΔΗ πρὸς τὴν ΗΑ. ἀλλ' ὥς ἡ ΓΖ πρὸς τὴν ΖΑ, οὕτως ἐδείκθη καὶ ἡ ΒΕ πρὸς τὴν ΕΑ· καὶ ὥς ἄρα ἡ ΒΕ πρὸς τὴν ΕΑ, οὕτως ἡ ΔΗ πρὸς τὴν ΗΑ, καὶ συνθέντι ἄρα ὥς ἡ ΒΑ πρὸς ΑΕ, οὕτως ἡ ΔΑ πρὸς ΑΗ, καὶ ἐναλλάξ

For since EF has been drawn parallel to one of the sides BC of triangle ABC , proportionally, as BE is to EA , so CF (is) to FA [Prop. 6.2]. Again, since FG has been drawn parallel to one (of the sides) CD of triangle ACD , proportionally, as CF is to FA , so DG (is) to GA [Prop. 6.2]. But, as CF (is) to FA , so it was also shown (is) BE to EA . And thus as BE (is) to EA , so DG (is) to GA . And, thus, compounding, as BA (is) to AE , so DA (is) to AG [Prop. 5.18]. And, alternately, as

ὥς ἡ BA πρὸς τὴν AD , οὕτως ἡ EA πρὸς τὴν AH . τῶν ἄρα $ABΓΔ$, EH παραλληλογράμων ἀνάλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὴν κοινὴν γωνίαν τὴν ὑπὸ $BAΔ$. καὶ ἐπεὶ παράλληλός ἐστιν ἡ HZ τῇ $ΔΓ$, ἴση ἐστὶν ἡ μὲν ὑπὸ AZH γωνία τῇ ὑπὸ $ΔΓΑ$ · καὶ κοινὴ τῶν δύο τριγώνων τῶν $ΑΔΓ$, AHZ ἡ ὑπὸ $ΔΑΓ$ γωνία· ἰσογώνιον ἄρα ἐστὶ τὸ $ΑΔΓ$ τρίγωνον τῷ AHZ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $ΑΓΒ$ τρίγωνον ἰσογώνιον ἐστὶ τῷ AZE τριγώνῳ, καὶ ὅλον τὸ $ABΓΔ$ παραλληλόγραμμον τῷ EH παραλληλογράμῳ ἰσογώνιον ἐστίν. ἀνάλογον ἄρα ἐστὶν ὥς ἡ $ΑΔ$ πρὸς τὴν $ΔΓ$, οὕτως ἡ AH πρὸς τὴν HZ , ὥς δὲ ἡ $ΔΓ$ πρὸς τὴν $ΓΑ$, οὕτως ἡ HZ πρὸς τὴν $ΖΑ$, ὥς δὲ ἡ $ΑΓ$ πρὸς τὴν $ΓΒ$, οὕτως ἡ AZ πρὸς τὴν ZE , καὶ ἔτι ὥς ἡ $ΓΒ$ πρὸς τὴν BA , οὕτως ἡ ZE πρὸς τὴν EA . καὶ ἐπεὶ ἐδείχθη ὥς μὲν ἡ $ΔΓ$ πρὸς τὴν $ΓΑ$, οὕτως ἡ HZ πρὸς τὴν $ΖΑ$, ὥς δὲ ἡ $ΑΓ$ πρὸς τὴν $ΓΒ$, οὕτως ἡ AZ πρὸς τὴν ZE , δι' ἴσον ἄρα ἐστὶν ὥς ἡ $ΔΓ$ πρὸς τὴν $ΓΒ$, οὕτως ἡ HZ πρὸς τὴν ZE . τῶν ἄρα $ABΓΔ$, EH παραλληλογράμων ἀνάλογόν εἰσιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· ὅμοιον ἄρα ἐστὶ τὸ $ABΓΔ$ παραλληλόγραμμον τῷ EH παραλληλογράμῳ. διὰ τὰ αὐτὰ δὴ τὸ $ABΓΔ$ παραλληλόγραμμον καὶ τῷ $KΘ$ παραλληλογράμῳ ὅμοιον ἐστίν· ἐκάτερον ἄρα τῶν EH , $ΘΚ$ παραλληλογράμων τῷ $ABΓΔ$ [παραλληλογράμῳ] ὅμοιον ἐστίν. τὰ δὲ τῶν αὐτῶν εὐθυγράμμω ὅμοια καὶ ἀλλήλοις ἐστὶν ὅμοια· καὶ τὸ EH ἄρα παραλληλόγραμμον τῷ $ΘΚ$ παραλληλογράμῳ ὅμοιον ἐστίν.

Παντὸς ἄρα παραλληλογράμῳ τὰ περὶ τὴν διάμετρον παραλληλόγραμμοι ὅμοιοι ἐστὶ τῷ τε ὅλῳ καὶ ἀλλήλοις· ὅπερ ἔδει δεῖξαι.

25.

Τῷ δοθέντι εὐθυγράμῳ ὅμοιον καὶ ἄλλω τῷ δοθέντι ἴσον τὸ αὐτὸ συστήσασθαι.

BA (is) to AD , so EA (is) to AG [Prop. 5.16]. Thus, in parallelograms $ABCD$ and EG , the sides about the common angle BAD are proportional. And since GF is parallel to DC , angle AFG is equal to DCA [Prop. 1.29]. And angle DAC (is) common to the two triangles ADC and AGF . Thus, triangle ADC is equiangular to triangle AGF [Prop. 1.32]. So, for the same (reasons), triangle ACB is equiangular to triangle AFE , and the whole parallelogram $ABCD$ is equiangular to parallelogram EG . Thus, proportionally, as AD (is) to DC , so AG (is) to GF , and as DC (is) to CA , so GF (is) to FA , and as AC (is) to CB , so AF (is) to FE , and, further, as CB (is) to BA , so FE (is) to EA [Prop. 6.4]. And since it was shown that as DC is to CA , so GF (is) to FA , and as AC (is) to CB , so AF (is) to FE , thus, via equality, as DC is to CB , so GF (is) to FE [Prop. 5.22]. Thus, in parallelograms $ABCD$ and EG , the sides about the equal angles are proportional. Thus, parallelogram $ABCD$ is similar to parallelogram EG [Def. 6.1]. So, for the same (reasons), parallelogram $ABCD$ is also similar to parallelogram KH . Thus, parallelograms EG and HK are each similar to [parallelogram] $ABCD$. And (rectilinear figures) similar to the same rectilinear figure are also similar to one another [Prop. 6.21]. Thus, parallelogram EG is also similar to parallelogram HK .

Thus, in any parallelogram, the parallelograms about the diagonal are similar to the whole, and to one another. (Which is) the very thing it was required to show.

Proposition 25

To construct a single (rectilinear figure) similar to a given rectilinear figure, and equal to a different given rectilinear figure.

the very thing it was required to do.

26.

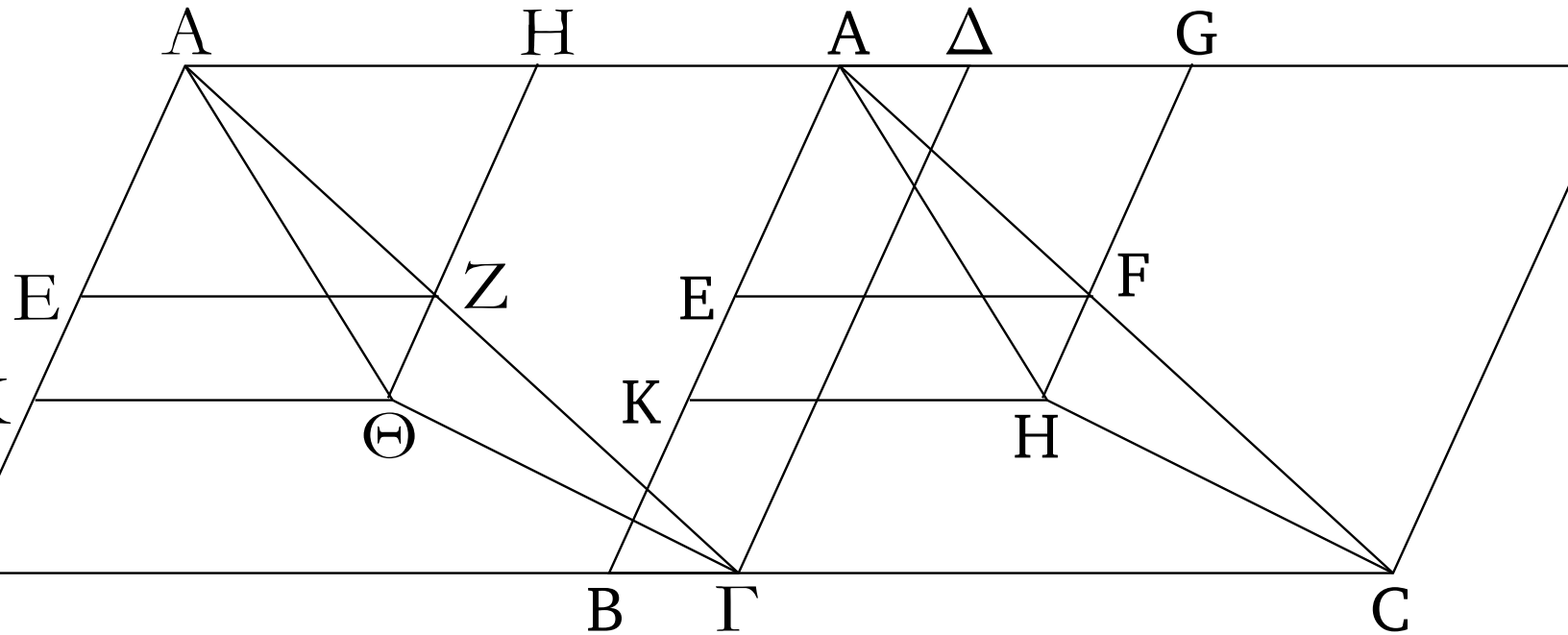
Proposition 26

Ἐάν ἀπὸ παραλληλογράμμου παραλληλόγραμμον ἀφαιρεθῇ ὁμοίον τε τῷ ὅλῳ καὶ ὁμοίως κείμενον κοινὴν γωνίαν ἔθον αὐτῷ, περὶ τὴν αὐτὴν διάμετρόν ἐστι τῷ ὅλῳ.

Ἀπὸ γὰρ παραλληλογράμμου τοῦ $ABΓΔ$ παραλληλόγραμμον ἀφαιρεθῇ τὸ AZ ὁμοίον τῷ $ABΓΔ$ καὶ ὁμοίως κείμενον κοινὴν γωνίαν ἔθον αὐτῷ τὴν ὑπὸ $ΔAB$ · λέγω, ὅτι περὶ τὴν αὐτὴν διάμετρόν ἐστι τὸ $ABΓΔ$ τῷ AZ .

If from a parallelogram a(nother) parallelogram is subtracted (which is) similar, and similarly laid out, to the whole, having a common angle with it, (then the subtracted parallelogram) is about the same diagonal as the whole.

For, from parallelogram $ABCD$, let (parallelogram) AF be subtracted (which is) similar, and similarly laid out, to $ABCD$, having the common angle DAB with it. I say that $ABCD$ is about the same diagonal as AF .



μή γάρ, ἀλλ' εἰ δυνατόν, ἔστω [αὐτῶν] διάμετρος ἡ $AΘΓ$, καὶ ἐκβληθεῖσα ἡ HZ διήθθω ἐπὶ τὸ $Θ$, καὶ ἦθθω διὰ τοῦ $Θ$ ὁπορέρα τῶν $AΔ$, $BΓ$ παράλληλος ἡ $ΘΚ$.

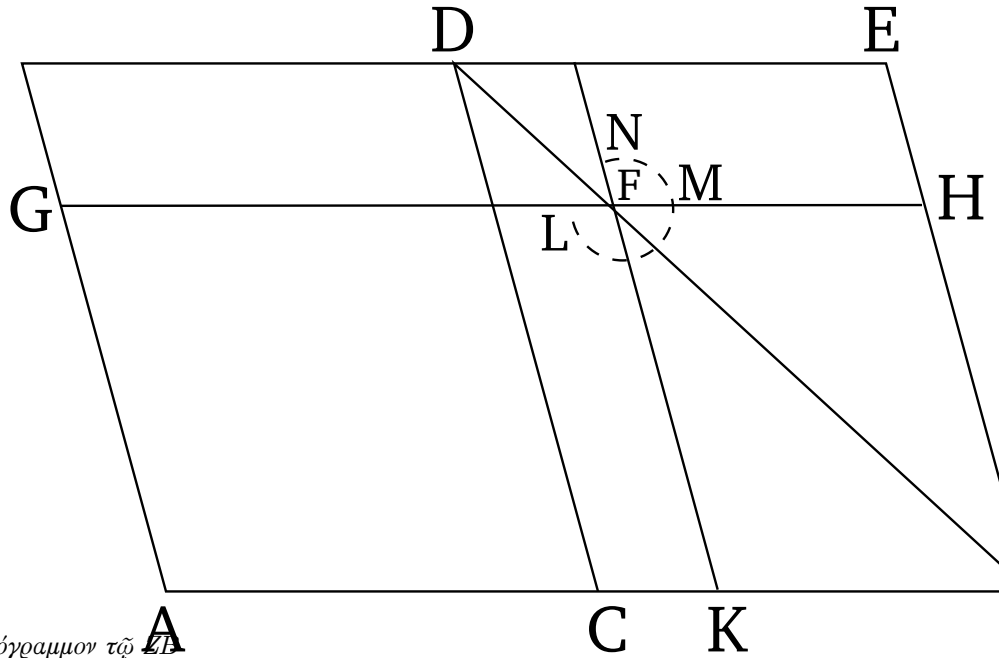
Ἐπεὶ οὖν περὶ τὴν αὐτὴν διάμετρόν ἐστι τὸ $ABΓΔ$ τῷ KH , ἔστιν ἄρα ὡς ἡ $ΔA$ πρὸς τὴν AB , οὕτως ἡ HA πρὸς τὴν AK . ἔστι δὲ καὶ διὰ τὴν ὁμοιότητα τῶν $ABΓΔ$, EH καὶ ὡς ἡ $ΔA$ πρὸς τὴν AB , οὕτως ἡ HA πρὸς τὴν AE · καὶ ὡς ἄρα ἡ HA πρὸς τὴν AK , οὕτως ἡ HA πρὸς τὴν AE . ἡ HA ἄρα πρὸς ἑκατέραν τῶν AK , AE τὸν αὐτὸν ἔθει λόγον. ἴση ἄρα ἐστὶν ἡ AE τῇ AK ἡ ἐλάττων τῇ μείζονι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οὐκ ἐστὶ περὶ τὴν αὐτὴν διάμετρον τὸ $ABΓΔ$ τῷ AZ · περὶ τὴν αὐτὴν ἄρα ἐστὶ διάμετρον τὸ $ABΓΔ$ παραλληλόγραμμον τῷ AZ παραλληλογράμμῳ.

Ἐάν ἄρα ἀπὸ παραλληλογράμμου παραλληλόγραμμον ἀφαιρεθῇ ὁμοίον τε τῷ ὅλῳ καὶ ὁμοίως κείμενον κοινὴν γωνίαν ἔθον αὐτῷ, περὶ τὴν αὐτὴν διάμετρόν ἐστι τῷ ὅλῳ· ὅπερ ἔδει δεῖξαι.

For (if) not, (then), if possible, let AHC be [$ABCD$'s] diagonal. And producing GF , let it be drawn through to (point) H . And let HK be drawn through (point) H , parallel to either of AD or BC [Prop. 1.31].

Therefore, since $ABCD$ is about the same diagonal as KG , thus as DA is to AB , so GA (is) to AK [Prop. 6.24]. And, on account of the similarity of $ABCD$ and EG , also, as DA (is) to AB , so GA (is) to AE . Thus, also, as GA (is) to AK , so GA (is) to AE . Thus, GA has the same ratio to each of AK and AE . Thus, AE is equal to AK [Prop. 5.9], the lesser to the greater. The very thing is impossible. Thus, $ABCD$ is not not about the same diagonal as AF . Thus, parallelogram $ABCD$ is about the same diagonal as parallelogram AF .

Thus, if from a parallelogram a(nother) parallelogram is subtracted (which is) similar, and similarly laid out, to the whole, having a common angle with it, (then the subtracted parallelogram) is about the same diagonal as the whole.



Ἐπεὶ γὰρ ὅμοιόν ἐστι τὸ ΔB παραλληλόγραμμον τῷ ZE παραλληλογράμμῳ, περὶ τὴν αὐτὴν εἶσι διάμετρον. ἦθθω αὐτῶν διάμετρος ἡ ΔB , καὶ καταγεγράφθω τὸ στήμα.

Ἐπεὶ οὖν ἴσον ἐστὶ τὸ ΓZ τῷ ZE , κοινὸν δὲ τὸ ZB , ὅλον ἄρα τὸ $\Gamma\Theta$ ὅλῳ τῷ KE ἔστιν ἴσον. ἀλλὰ τὸ $\Gamma\Theta$ τῷ ΓH ἔστιν ἴσον, ἐπεὶ καὶ ἡ AF τῇ GB . καὶ τὸ $H\Gamma$ ἄρα τῷ EK ἔστιν ἴσον. κοινὸν προσκείσθω τὸ ΓZ . ὅλον ἄρα τὸ AZ τῷ AMN γνῶμόνι ἔστιν ἴσον· ὥστε τὸ ΔB παραλληλόγραμμον, τουτέστι τὸ ΔD , τοῦ AZ παραλληλογράμμου μείζον ἐστίν.

Πάντων ἄρα τῶν παρὰ τὴν αὐτὴν εὐθεῖαν παραβαλλομένων παραλληλογράμμων καὶ ἐλλειπόντων εἴδει παραλληλογράμμοις ὁμοίοις τε καὶ ὁμοίως κεμένοις τῷ ἀπὸ τῆς ἡμισείας ἀναγραφόμενῳ μέγιστόν ἐστι τὸ ἀπὸ τῆς ἡμισείας παραβληθέν· ὅπερ ἔδει δεῖξαι.

For since parallelogram DB is similar to parallelogram FB , they are about the same diagonal [Prop. 6.26]. Let their (common) diagonal DB be drawn, and let the (rest of the) figure be described.

Therefore, since (complement) CF is equal to (complement) FE [Prop. 1.43], and (parallelogram) FB is common, the whole (parallelogram) CH is thus equal to the whole (parallelogram) KE . But, (parallelogram) CH is equal to CG , since AC (is) also (equal) to CB [Prop. 6.1]. Thus, (parallelogram) GC is also equal to EK . Let (parallelogram) CF be added to both. Thus, the whole (parallelogram) AF is equal to the gnomon LMN . Hence, parallelogram DB —that is to say, AD —is greater than parallelogram AF .

Thus, for all parallelograms applied to the same straight-line, and falling short by a parallelogrammic figure similar, and similarly laid out, to the (parallelogram) described on half (the straight-line), the greatest is the [parallelogram] applied to half (the straight-line). (Which is) the very thing it was required to show.

28.

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι εὐθύγραμμῳ ἴσον παραλληλόγραμμον παραβαλεῖν ἐλλείπον εἴδει παραλληλογράμμῳ ὁμοίῳ τῷ δοθέντι· δεῖ δὲ τὸ διδόμενον εὐθύγραμμον [ᾧ δεῖ ἴσον παραβαλεῖν] μὴ μείζον εἶναι τοῦ ἀπὸ τῆς ἡμισείας ἀναγραφόμενου ὁμοίου τῷ ἐλλείμματι [τοῦ τε ἀπὸ τῆς ἡμισείας καὶ ᾧ δεῖ ὅμοιον ἐλλείπειν].

Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθέν εὐθύγραμμον, ᾧ δεῖ ἴσον παρὰ τὴν AB παραβαλεῖν, τὸ Γ μὴ μείζον [ὄν] τοῦ ἀπὸ τῆς ἡμισείας τῆς AB ἀναγραφόμενου ὁμοίου τῷ ἐλ-

Proposition 28[†]

To apply a parallelogram, equal to a given rectilinear figure, to a given straight-line, (the applied parallelogram) falling short by a parallelogrammic figure similar to a given (parallelogram). It is necessary for the given rectilinear figure [to which it is required to apply an equal (parallelogram)] not to be greater than the (parallelogram) described on half (of the straight-line) and similar to the deficit.

Let AB be the given straight-line, and C the given rectilinear figure to which the (parallelogram) applied to AB is re-

ἴσον ἄρα καὶ ὅμοιον ἐστὶ [τὸ $HΠ$] τῷ KM [ἀλλὰ τὸ KM τῷ HB ὅμοιον ἐστίν]. καὶ τὸ $HΠ$ ἄρα τῷ HB ὅμοιον ἐστίν· περὶ τὴν αὐτὴν ἄρα διάμετρον ἐστὶ τὸ $HΠ$ τῷ HB . ἔστω αὐτῶν διάμετρος ἡ $ΗΠΒ$, καὶ καταγεγράφθω τὸ σθῆμα.

Ἐπεὶ οὖν ἴσον ἐστὶ τὸ BH τοῖς $Γ$, KM , ὣν τὸ $HΠ$ τῷ KM ἐστὶν ἴσον, λοιπὸς ἄρα ὁ $ΥΘΦ$ γνόμεον λοιπῷ τῷ $Γ$ ἴσος ἐστίν. καὶ ἐπεὶ ἴσον ἐστὶ τὸ OP τῷ $XΣ$, κοινὸν προσκείσθω τὸ $ΠΒ$ · ὅλον ἄρα τὸ OB ὅλῳ τῷ XB ἴσον ἐστίν. ἀλλὰ τὸ XB τῷ TE ἐστὶν ἴσον, ἐπεὶ καὶ πλευρὰ ἡ AE πλευρᾷ τῇ EB ἐστὶν ἴση· καὶ τὸ TE ἄρα τῷ OB ἐστὶν ἴσον. κοινὸν προσκείσθω τὸ $XΣ$ · ὅλον ἄρα τὸ $TΣ$ ὅλῳ τῷ $ΦΘΥ$ γνόμεον ἐστὶν ἴσον. ἀλλ' ὁ $ΦΘΥ$ γνόμεον τῷ $Γ$ ἐδείκθη ἴσος· καὶ τὸ $TΣ$ ἄρα τῷ $Γ$ ἐστὶν ἴσον.

Παρὰ τὴν δοθεῖσαν ἄρα εὐθεῖαν τὴν AB τῷ δοθέντι εὐθυγράμμῳ τῷ $Γ$ ἴσον παραλληλόγραμμον παραβέβληται τὸ $ΣΤ$ ἐλλείπον εἶδει παραλληλογράμμῳ τῷ $ΠΒ$ ὁμοίῳ ὄντι τῷ $Δ$ [ἐπειδήπερ τὸ $ΠΒ$ τῷ $HΠ$ ὅμοιον ἐστίν]· ὅπερ ἔδει ποιῆσαι.

[Prop. 1.3]. And let the parallelogram $OGPQ$ be completed. Thus, $[GQ]$ is equal and similar to KM [but, KM is similar to GB]. Thus, GQ is also similar to GB [Prop. 6.21]. Thus, GQ and GB are about the same diagonal [Prop. 6.26]. Let GQB be their (common) diagonal, and let the (remainder of the) figure be described.

Therefore, since BG is equal to C and KM , of which GQ is equal to KM , the remaining gnomon UWV is thus equal to the remainder C . And since (the complement) PR is equal to (the complement) OS [Prop. 1.43], let (parallelogram) QB be added to both. Thus, the whole (parallelogram) PB is equal to the whole (parallelogram) OB . But, OB is equal to TE , since side AE is equal to side EB [Prop. 6.1]. Thus, TE is also equal to PB . Let (parallelogram) OS be added to both. Thus, the whole (parallelogram) TS is equal to the gnomon VWU . But, gnomon VWU was shown (to be) equal to C . Therefore, (parallelogram) TS is also equal to (figure) C .

Thus, the parallelogram ST , equal to the given rectilinear figure C , has been applied to the given straight-line AB , falling short by the parallelogramic figure QB , which is similar to D [inasmuch as QB is similar to GQ [Prop. 6.24]]. (Which is) the very thing it was required to do.

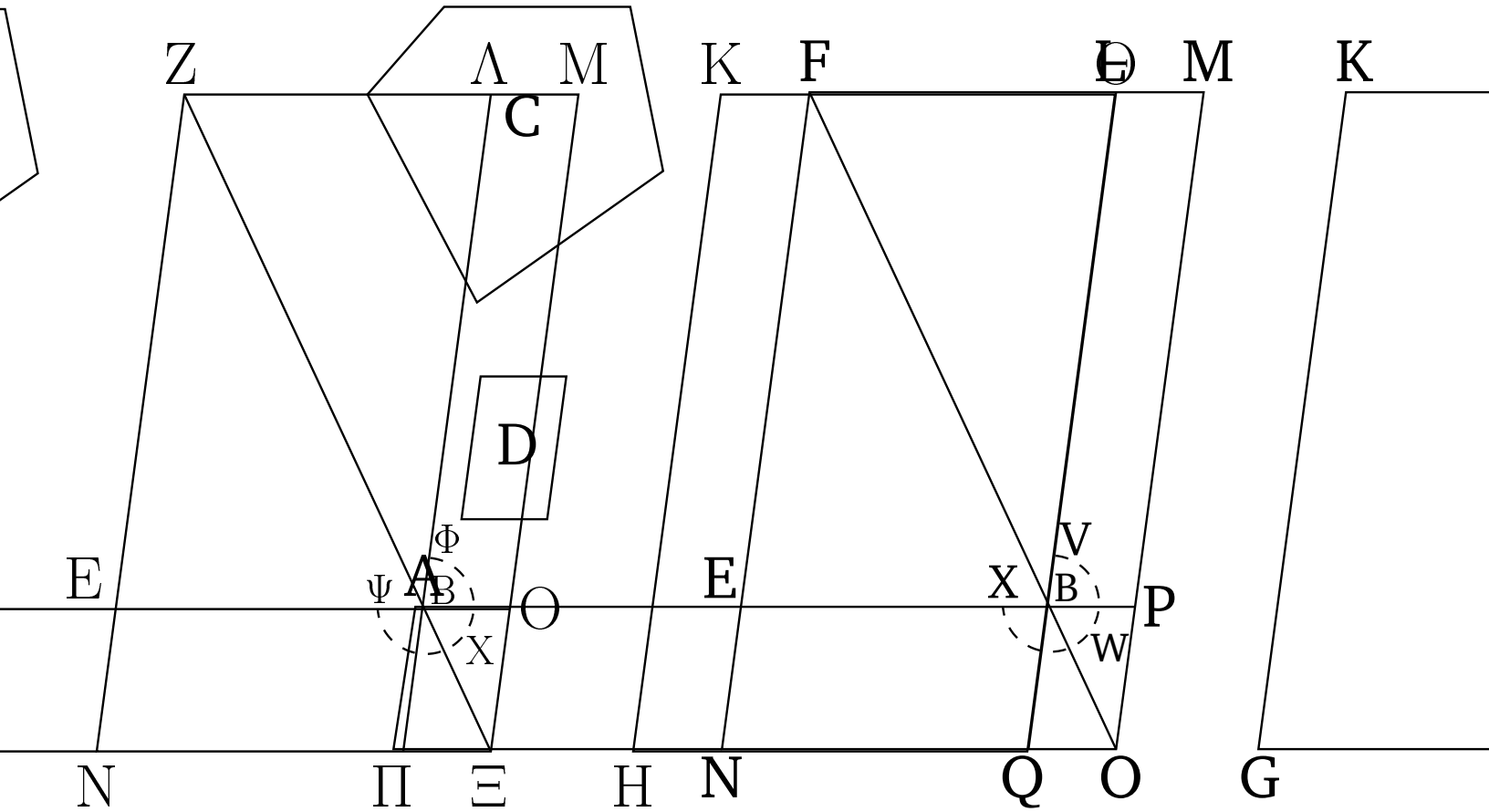
[†] This proposition is a geometric solution of the quadratic equation $x^2 - \alpha x + \beta = 0$. Here, x is the ratio of a side of the deficit to the corresponding side of figure D , α is the ratio of the length of AB to the length of that side of figure D which corresponds to the side of the deficit running along AB , and β is the ratio of the areas of figures C and D . The constraint corresponds to the condition $\beta < \alpha^2/4$ for the equation to have real roots. Only the smaller root of the equation is found. The larger root can be found by a similar method.

29.

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι εὐθυγράμμῳ ἴσον παραλληλόγραμμον παραβαλεῖν ὑπερβάλλον εἶδει παραλληλογράμμῳ ὁμοίῳ τῷ δοθέντι.

Proposition 29[†]

To apply a parallelogram, equal to a given rectilinear figure, to a given straight-line, (the applied parallelogram) overshooting by a parallelogramic figure similar to a given (parallelogram).



Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθὲν εὐθύγραμμον, ὃν δεῖ ἴσον παρὰ τὴν AB παραβαλεῖν, τὸ Γ , ὃν δὲ δεῖ ὅμοιον ὑπερβάλλειν, τὸ Δ . δεῖ δὴ παρὰ τὴν AB εὐθεῖαν τῷ Γ εὐθύγραμμῳ ἴσον παραλληλόγραμμον παραβαλεῖν ὑπερβάλλον εἶδει παραλληλογράμμῳ ὁμοίῳ τῷ Δ .

Τετμήσθω ἡ AB διῖθα κατὰ τὸ E , καὶ ἀναγεγράθω ἀπὸ τῆς EB τῷ Δ ὅμοιον καὶ ὁμοίωξ κείμενον παραλληλόγραμμον τὸ BZ , καὶ συναμφοτέροις μὲν τοῖς BZ , Γ ἴσον, τῷ δὲ Δ ὅμοιον καὶ ὁμοίωξ κείμενον τὸ αὐτὸ συνεστάτω τὸ $H\Theta$. ὁμολογοῦ δὲ ἔστω ἡ μὲν $K\Theta$ τῇ ZA , ἡ δὲ KH τῇ ZE . καὶ ἐπεὶ μείζον ἔστι τὸ $H\Theta$ τοῦ ZB , μείζων ἄρα ἔστι καὶ ἡ μὲν $K\Theta$ τῇ ZA , ἡ δὲ KH τῇ ZE . ἐκβεβλήσθωσαν αἱ ZA , ZE , καὶ τῇ μὲν $K\Theta$ ἴση ἔστω ἡ ZAM , τῇ δὲ KH ἴση ἡ ZEN , καὶ συμπληρώσθω τὸ MN . τὸ MN ἄρα τῷ $H\Theta$ ἴσον τέ ἐστι καὶ ὅμοιον. ἀλλὰ τὸ $H\Theta$ τῷ EA ἔστιν ὅμοιον· καὶ τὸ MN ἄρα τῷ EA ὅμοιον ἔστιν· περὶ τὴν αὐτὴν ἄρα διάμετρον ἔστι τὸ EA τῷ MN . ἦρθω αὐτῶν διάμετρος ἡ ZX , καὶ καταγεγράψθω τὸ σθῆμα.

Ἐπεὶ ἴσον ἐστὶ τὸ $H\Theta$ τοῖς EA , Γ , ἀλλὰ τὸ $H\Theta$ τῷ MN ἴσον ἐστίν, καὶ τὸ MN ἄρα τοῖς EA , Γ ἴσον ἐστίν. κοινὸν ἀφηρήσθω τὸ EA . λοιπὸς ἄρα ὁ $\Psi\Theta\Phi$ γνῶμων τῷ Γ ἔστιν ἴσος. καὶ ἐπεὶ ἴση ἐστὶν ἡ AE τῇ EB , ἴσον ἐστὶ καὶ τὸ AN τῷ NB , τοντέστι τῷ AO . κοινὸν προσκείσθω τὸ EX . ὅλον ἄρα τὸ AX ἴσον ἐστὶ τῷ $\Phi\Theta\Psi$ γνῶμονι. ἀλλὰ ὁ $\Phi\Theta\Psi$ γνῶμων τῷ

Let AB be the given straight-line, and C the given rectilinear figure to which the (parallelogram) applied to AB is required (to be) equal, and D the (parallelogram) to which the excess is required (to be) similar. So it is required to apply a parallelogram, equal to the given rectilinear figure C , to the given straight-line AB , overshooting by a parallelogrammic figure similar to D .

Let AB be cut in half at (point) E [Prop. 1.10], and let the parallelogram BF , (which is) similar, and similarly laid out, to D , be described on EB [Prop. 6.18]. And let (parallelogram) GH be constructed (so as to be) both similar, and similarly laid out, to D , and equal to the sum of BF and C [Prop. 6.25]. And let KH correspond to FL , and KG to FE . And since (parallelogram) GH is greater than (parallelogram) FB , KH is thus also greater than FL , and KG than FE . Let FL and FE be produced, and let FLM be (made) equal to KH , and FEN to KG [Prop. 1.3]. And let (parallelogram) MN be completed. Thus, MN is equal and similar to GH . But, GH is similar to EL . Thus, MN is also similar to EL [Prop. 6.21]. EL is thus about the same diagonal as MN [Prop. 6.26]. Let their (common) diagonal FO be drawn, and let the (remainder of the) figure be described.

And since (parallelogram) GH is equal to (parallelogram)

Γ ἴσοξ ἐστίν· καὶ τὸ AX ἄρα τῷ Γ ἴσον ἐστίν.

Παρά τὴν δοθεῖσαν ἄρα εὐθεῖαν τὴν AB τῷ δοθέντι εὐθυγράμμῳ τῷ Γ ἴσον παραλληλόγραμμον παραβέβληται τὸ AX ὑπερβάλλον εἶδει παραλληλογράμμῳ τῷ ΠO ὁμοίῳ ὄντι τῷ Δ , ἐπεὶ καὶ τῷ EA ἐστὶν ὁμοίον τὸ $O\Pi$ · ὅπερ ἔδει ποιῆσαι.

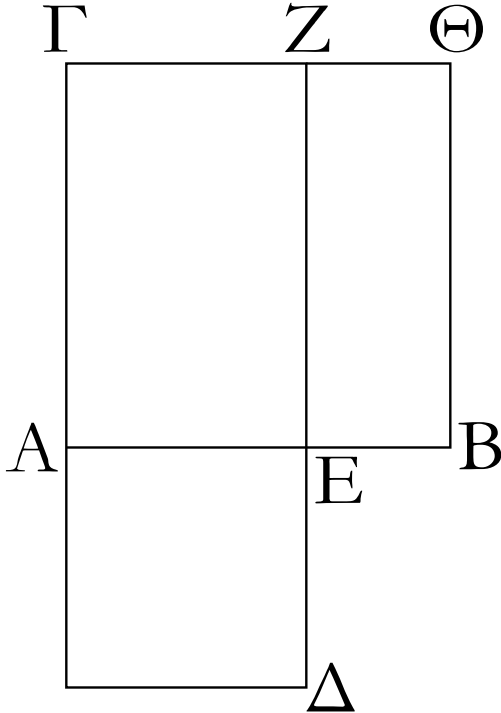
EL and (figure) C , but GH is equal to (parallelogram) MN , MN is thus also equal to EL and C . Let EL be subtracted from both. Thus, the remaining gnomon XWV is equal to (figure) C . And since AE is equal to EB , (parallelogram) AN is also equal to (parallelogram) NB [Prop. 6.1], that is to say, (parallelogram) LP [Prop. 1.43]. Let (parallelogram) EO be added to both. Thus, the whole (parallelogram) AO is equal to the gnomon VWX . But, the gnomon VWX is equal to (figure) C . Thus, (parallelogram) AO is also equal to (figure) C .

Thus, the parallelogram AO , equal to the given rectilinear figure C , has been applied to the given straight-line AB , overshooting by the parallelogramic figure QP which is similar to D , since PQ is also similar to EL [Prop. 6.24]. (Which is) the very thing it was required to do.

† This proposition is a geometric solution of the quadratic equation $x^2 + \alpha x - \beta = 0$. Here, x is the ratio of a side of the excess to the corresponding side of figure D , α is the ratio of the length of AB to the length of that side of figure D which corresponds to the side of the excess running along AB , and β is the ratio of the areas of figures C and D . Only the positive root of the equation is found.

30.

Τὴν δοθεῖσαν εὐθεῖαν πεπερασμένην ἄκρον καὶ μέσον λόγον τεμεῖν.



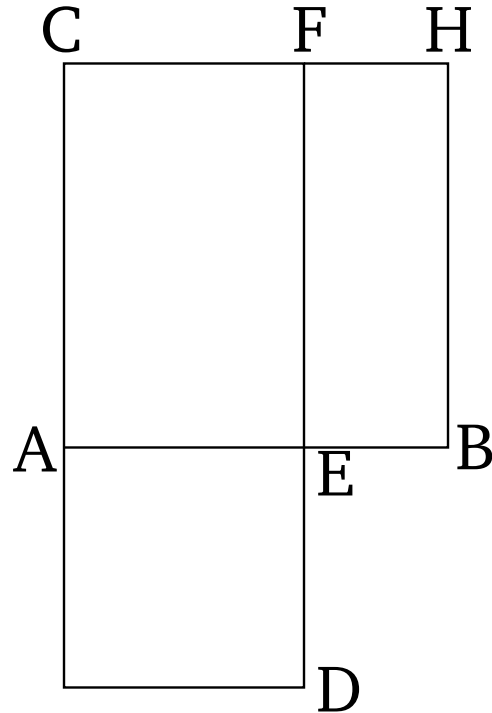
Ἐστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ AB · δεῖ δὴ τὴν AB εὐθεῖαν ἄκρον καὶ μέσον λόγον τεμεῖν.

Ἀναγεγράφθω ἀπὸ τῆς AB τετράγωνον τὸ $BΓ$, καὶ παραβέβλησθω παρὰ τὴν $ΑΓ$ τῷ $BΓ$ ἴσον παραλληλόγραμμον τὸ $ΓΔ$ ὑπερβάλλον εἶδει τῷ $ΑΔ$ ὁμοίῳ τῷ $BΓ$.

Τετράγωνον δέ ἐστι τὸ $BΓ$ · τετράγωνον ἄρα ἐστὶ καὶ τὸ

Proposition 30†

To cut a given finite straight-line in extreme and mean ratio.



Let AB be the given finite straight-line. So it is required to cut the straight-line AB in extreme and mean ratio.

Let the square BC be described on AB [Prop. 1.46], and let the parallelogram CD , equal to BC , be applied to AC , overshooting by the figure AD (which is) similar to BC [Prop. 6.29].

ΑΔ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΒΓ τῷ ΓΔ, κοινὸν ἀφηρήσθω τὸ ΓΕ· λοιπὸν ἄρα τὸ ΒΖ λοιπὸν τῷ ΑΔ ἐστὶν ἴσον. ἔστι δὲ αὐτῷ καὶ ἰσογώνιον· τῶν ΒΖ, ΑΔ ἄρα ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· ἔστιν ἄρα ὡς ἡ ΖΕ πρὸς τὴν ΕΔ, οὕτως ἡ ΑΕ πρὸς τὴν ΕΒ. ἴση δὲ ἡ μὲν ΖΕ τῇ ΑΒ, ἡ δὲ ΕΔ τῇ ΑΕ. ἔστιν ἄρα ὡς ἡ ΒΑ πρὸς τὴν ΑΕ, οὕτως ἡ ΑΕ πρὸς τὴν ΕΒ. μείζων δὲ ἡ ΑΒ τῇ ΑΕ· μείζων ἄρα καὶ ἡ ΑΕ τῇ ΕΒ.

Ἡ ἄρα ΑΒ εὐθεῖα ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Ε, καὶ τὸ μείζον αὐτῆς τμήμα ἐστὶ τὸ ΑΕ· ὅπερ ἔδει ποιῆσαι.

And BC is a square. Thus, AD is also a square. And since BC is equal to CD , let (rectangle) CE be subtracted from both. Thus, the remaining (rectangle) BF is equal to the remaining (square) AD . And it is also equiangular to it. Thus, the sides of BF and AD about the equal angles are reciprocally proportional [Prop. 6.14]. Thus, as FE is to ED , so AE (is) to EB . And FE (is) equal to AB , and ED to AE . Thus, as BA is to AE , so AE (is) to EB . And AB (is) greater than AE . Thus, AE (is) also greater than EB [Prop. 5.14].

Thus, the straight-line AB has been cut in extreme and mean ratio at E , and AE is its greater piece. (Which is) the very thing it was required to do.

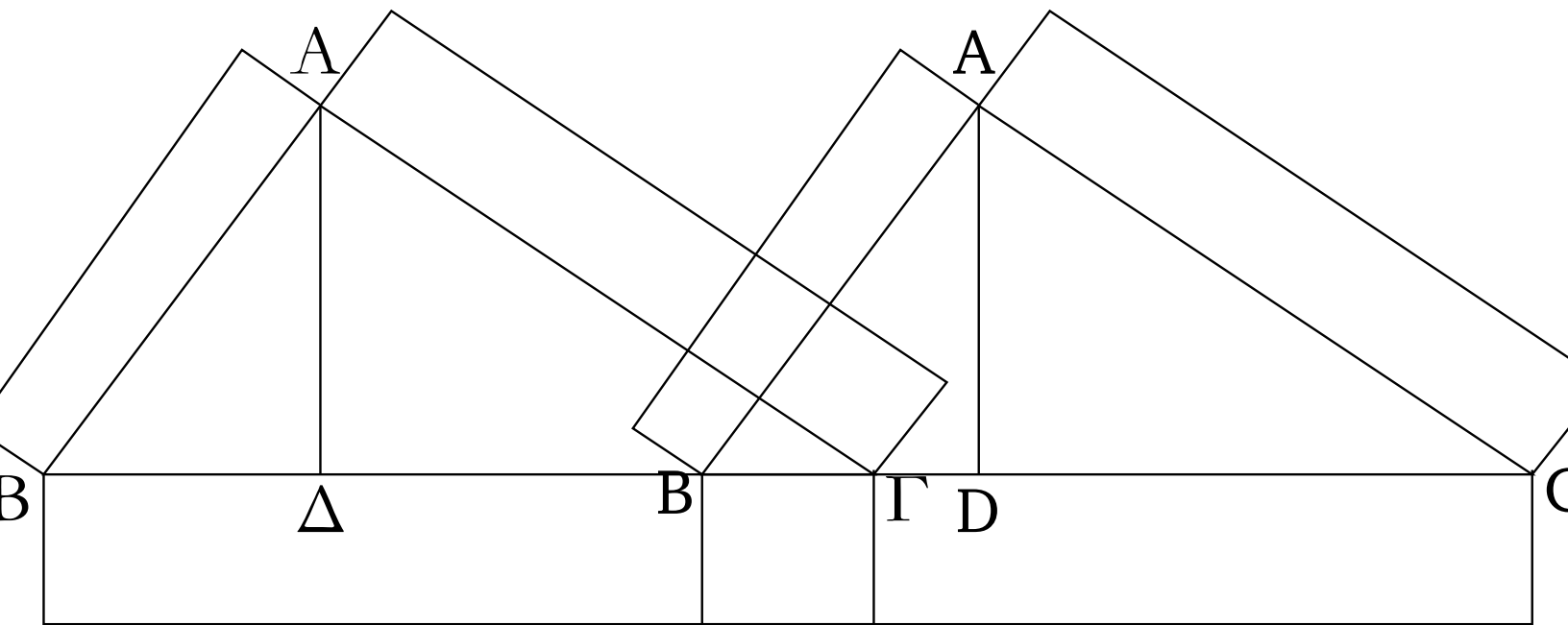
† This method of cutting a straight-line is sometimes called the “Golden Section”—see Prop. 2.11.

31.

Ἐν τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τήν ὀρθὴν γωνίαν ὑποτείνουσας πλευρᾶς εἶδος ἴσον ἐστὶ τοῖς ἀπὸ τῶν τήν ὀρθὴν γωνίαν περιεθουσῶν πλευρῶν εἶδει τοῖς ὁμοίοις τε καὶ ὁμοίως ἀναγραφόμενοις.

Proposition 31

In right-angled triangles, the figure (drawn) on the side subtending the right-angle is equal to the (sum of the) similar, and similarly described, figures on the sides surrounding the right-angle.



Ἐστω τρίγωνον ὀρθογώνιον τὸ ΑΒΓ ὀρθὴν ἔχον τὴν ὑπὸ ΒΑΓ γωνίαν λέγω, ὅτι τὸ ἀπὸ τῆς ΒΓ εἶδος ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΒΑ, ΑΓ εἶδει τοῖς ὁμοίοις τε καὶ ὁμοίως ἀναγραφόμενοις.

Ἡθθω κάθετος ἡ ΑΔ.

Ἐπεὶ οὖν ἐν ὀρθογωνίῳ τριγώνῳ τῷ ΑΒΓ ἀπὸ τῆς πρὸς τῷ Α ὀρθῆς γωνίας ἐπὶ τὴν ΒΓ βάσιν κάθετος ἦκται ἡ ΑΔ, τὰ ΑΒΔ, ΑΔΓ πρὸς τῇ καθετῷ τρίγωνα ὁμοία ἐστὶ τῷ τε ὅλῳ τῷ ΑΒΓ καὶ ἀλλήλοις. καὶ ἐπεὶ ὁμοίων ἐστὶ τὸ ΑΒΓ τῷ ΑΒΔ, ἔστιν ἄρα ὡς ἡ ΓΒ πρὸς τὴν ΒΑ, οὕτως ἡ ΑΒ πρὸς

Let ABC be a right-angled triangle having the angle BAC a right-angle. I say that the figure (drawn) on BC is equal to the (sum of the) similar, and similarly described, figures on BA and AC .

Let the perpendicular AD be drawn [Prop. 1.12].

Therefore, since, in the right-angled triangle ABC , the (straight-line) AD has been drawn from the right-angle at A perpendicular to the base BC , the triangles ABD and ADC about the perpendicular are similar to the whole (triangle) ABC , and to one another [Prop. 6.8]. And since ABC is similar

τὴν $B\Delta$. καὶ ἐπεὶ τρεῖς εὐθεῖαι ἀνάλογόν εἰσιν, ἔστιν ὥς ἡ πρώτη πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς πρώτης εἶδος πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ ὅμοιον καὶ ὁμοίως ἀναγραφόμενον. ὥς ἄρα ἡ ΓB πρὸς τὴν $B\Delta$, οὕτως τὸ ἀπὸ τῆς ΓB εἶδος πρὸς τὸ ἀπὸ τῆς BA τὸ ὅμοιον καὶ ὁμοίως ἀναγραφόμενον. διὰ τὰ αὐτὰ δὴ καὶ ὥς ἡ $B\Gamma$ πρὸς τὴν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς $B\Gamma$ εἶδος πρὸς τὸ ἀπὸ τῆς ΓA . ὥστε καὶ ὥς ἡ $B\Gamma$ πρὸς τὰς $B\Delta$, $\Delta\Gamma$, οὕτως τὸ ἀπὸ τῆς $B\Gamma$ εἶδος πρὸς τὰ ἀπὸ τῶν BA , $A\Gamma$ τὰ ὅμοια καὶ ὁμοίως ἀναγραφόμενα. ἴση δὲ ἡ $B\Gamma$ ταῖς $B\Delta$, $\Delta\Gamma$. ἴσον ἄρα καὶ τὸ ἀπὸ τῆς $B\Gamma$ εἶδος τοῖς ἀπὸ τῶν BA , $A\Gamma$ εἶδει τοῖς ὁμοίοις τε καὶ ὁμοίως ἀναγραφόμενοις.

Ἐν ἄρα τοῖς ὀρθογωνίοις τριγώνοις τὸ ἀπὸ τῆς τὴν ὀρθὴν γωνίαν ὑποτείνουσας πλευρᾶς εἶδος ἴσον ἐστὶ τοῖς ἀπὸ τῶν τὴν ὀρθὴν γωνίαν περιεθουσῶν πλευρῶν εἶδει τοῖς ὁμοίοις τε καὶ ὁμοίως ἀναγραφόμενοις· ὅπερ ἔδει δεῖχαι.

to ABD , thus as CB is to BA , so AB (is) to BD [Def. 6.1]. And since three straight-lines are proportional, as the first is to the third, so the figure (drawn) on the first is to the similar, and similarly described, (figure) on the second [Prop. 6.19 corr.]. Thus, as CB (is) to BD , so the figure (drawn) on CB (is) to the similar, and similarly described, (figure) on BA . And so, for the same (reasons), as BC (is) to CD , so the figure (drawn) on BC (is) to the (figure) on CA . Hence, also, as BC (is) to BD and DC , so the figure (drawn) on BC (is) to the (sum of the) similar, and similarly described, (figures) on BA and AC [Prop. 5.24]. And BC is equal to BD and DC . Thus, the figure (drawn) on BC (is) also equal to the (sum of the) similar, and similarly described, figures on BA and AC [Prop. 5.9].

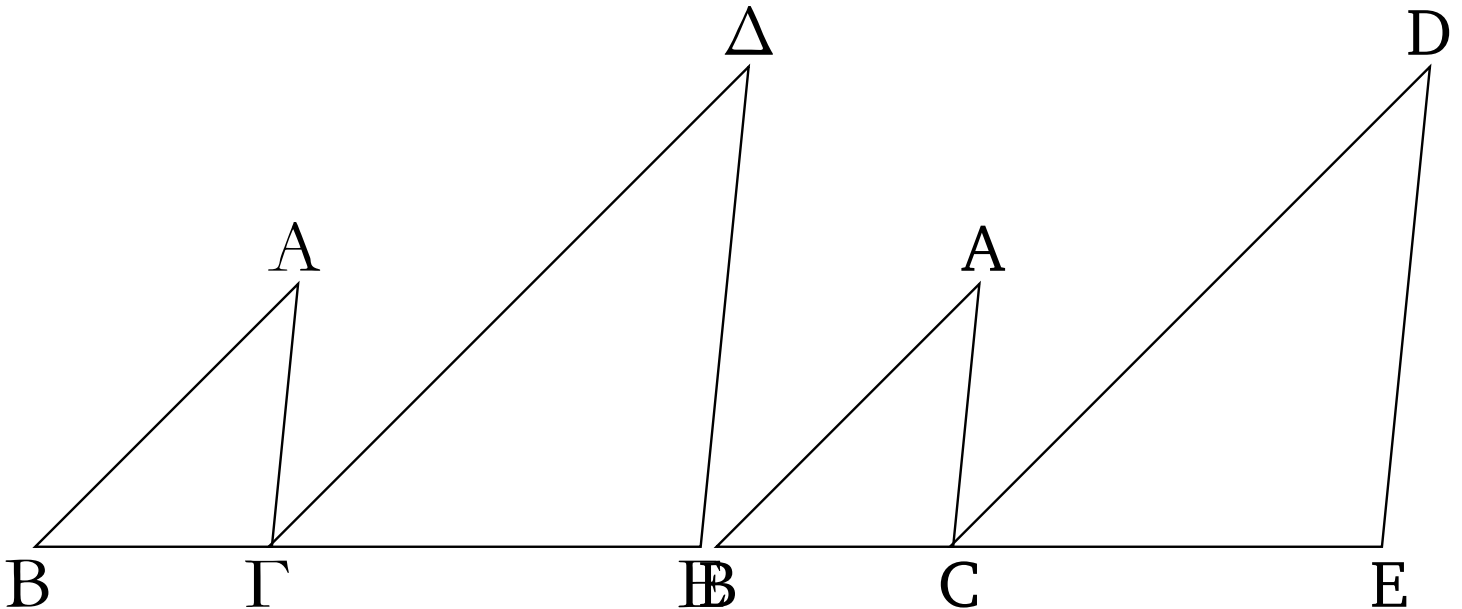
Thus, in right-angled triangles, the figure (drawn) on the side subtending the right-angle is equal to the (sum of the) similar, and similarly described, figures on the sides surrounding the right-angle. (Which is) the very thing it was required to show.

32.

Ἐὰν δύο τρίγωνα συντεθῇ κατὰ μίαν γωνίαν τὰς δύο πλευρὰς ταῖς δυοῖ πλευραῖς ἀνάλογον ἔθοντα ὥστε τὰς ὁμολόγους αὐτῶν πλευρὰς καὶ παραλλήλους εἶναι, αἱ λοιπαὶ τῶν τριγώνων πλευραὶ ἐπ' εὐθείας ἔσονται.

Proposition 32

If two triangles, having two sides proportional to two sides, are placed together at a single angle such that the corresponding sides are also parallel, (then) the remaining sides of the triangles will be straight-on (with respect to one another).



Ἐστω δύο τρίγωνα τὰ $AB\Gamma$, $\Delta\Gamma E$ τὰς δύο πλευρὰς τὰς BA , $A\Gamma$ ταῖς δυοῖ πλευραῖς ταῖς $\Delta\Gamma$, ΔE ἀνάλογον ἔθοντα, ὥς μὲν τὴν AB πρὸς τὴν $A\Gamma$, οὕτως τὴν $\Delta\Gamma$ πρὸς τὴν ΔE , παράλληλον δὲ τὴν μὲν AB τῇ $\Delta\Gamma$, τὴν δὲ $A\Gamma$ τῇ ΔE . λέγω, ὅτι ἐπ' εὐθείας ἐστὶν ἡ $B\Gamma$ τῇ ΓE .

Let ABC and DCE be two triangles having the two sides BA and AC proportional to the two sides DC and DE —so that as AB (is) to AC , so DC (is) to DE —and (having side) AB parallel to DC , and AC to DE . I say that (side) BC is straight-on to CE .

Ἐπεὶ γὰρ παράλληλός ἐστιν ἡ AB τῇ $\Delta\Gamma$, καὶ εἰς αὐτὰς ἐμπίπτωκεν εὐθεῖα ἡ $ΑΓ$, αἱ ἐναλλάχ γωνίαι αἱ ὑπὸ $ΒΑΓ$, $ΑΓΔ$ ἴσαι ἀλλήλαις εἰσίν. διὰ τὰ αὐτὰ δὲ καὶ ἡ ὑπὸ $ΓΔΕ$ τῇ ὑπὸ $ΑΓΔ$ ἴση ἐστίν. ὥστε καὶ ἡ ὑπὸ $ΒΑΓ$ τῇ ὑπὸ $ΓΔΕ$ ἐστιν ἴση. καὶ ἐπεὶ δύο τρίγωνά ἐστι τὰ $ΑΒΓ$, $ΔΓΕ$ μίαν γωνίαν τὴν πρὸς τῷ $Α$ μὲν γωνία τῇ πρὸς τῷ $Δ$ ἴσην ἔθοντα, περὶ δὲ τὰς ἴσας γωνίας τὰς πλευρὰς ἀνάλογον, ὥς τὴν $ΒΑ$ πρὸς τὴν $ΑΓ$, οὕτως τὴν $ΓΔ$ πρὸς τὴν $ΔΕ$, ἰσογώνιον ἄρα ἐστὶ τὸ $ΑΒΓ$ τρίγωνον τῷ $ΔΓΕ$ τριγώνῳ. ἴση ἄρα ἡ ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΔΓΕ$. ἐδείκθη δὲ καὶ ἡ ὑπὸ $ΑΓΔ$ τῇ ὑπὸ $ΒΑΓ$ ἴση. ὅλη ἄρα ἡ ὑπὸ $ΑΓΕ$ δυοὶ ταῖς ὑπὸ $ΑΒΓ$, $ΒΑΓ$ ἴση ἐστίν. κοινὴ προσκείσθω ἡ ὑπὸ $ΑΓΒ$. αἱ ἄρα ὑπὸ $ΑΓΕ$, $ΑΓΒ$ ταῖς ὑπὸ $ΒΑΓ$, $ΑΓΒ$, $ΓΒΑ$ ἴσαι εἰσίν. ἀλλ' αἱ ὑπὸ $ΒΑΓ$, $ΑΒΓ$, $ΑΓΒ$ δυοὶν ὀρθαῖς ἴσαι εἰσίν. καὶ αἱ ὑπὸ $ΑΓΕ$, $ΑΓΒ$ ἄρα δυοὶν ὀρθαῖς ἴσαι εἰσίν. πρὸς δὲ τινι εὐθείᾳ τῇ $ΑΓ$ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ $Γ$ δύο εὐθεῖαι αἱ $ΒΓ$, $ΓΕ$ μὴ ἐπὶ τὰ αὐτὰ μέρη κείμεναι τὰς ἐφεχῆς γωνίας τὰς ὑπὸ $ΑΓΕ$, $ΑΓΒ$ δυοὶν ὀρθαῖς ἴσας ποιοῦσιν. ἐπ' εὐθείας ἄρα ἐστίν ἡ $ΒΓ$ τῇ $ΓΕ$.

Ἐὰν ἄρα δύο τρίγωνα συντεθῇ κατὰ μίαν γωνίαν τὰς δύο πλευρὰς ταῖς δυοὶ πλευραῖς ἀνάλογον ἔθοντα ὥστε τὰς ὁμολόγων ἀντῶν πλευρὰς καὶ παραλλήλους εἶναι, αἱ λοιπαὶ τῶν τριγώνων πλευραὶ ἐπ' εὐθείας ἔσσονται. ὅπερ ἔδει δεῖξαι.

33.

Ἐν τοῖς ἴσοις κύκλοις αἱ γωνίαι τὸν αὐτὸν ἔθουσι λόγον ταῖς περιφερείαις, ἐφ' ᾧ βεβήκασιν, ἔαν τε πρὸς τοῖς κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὥσι βεβηκνῶσι.

Ἐστωσαν ἴσοι κύκλοι οἱ $ΑΒΓ$, $ΔΕΖ$, καὶ πρὸς μὲν τοῖς κέντροις αὐτῶν τοῖς $Η$, $Θ$ γωνίαι ἔστωσαν αἱ ὑπὸ $ΒΗΓ$, $ΕΘΖ$, πρὸς δὲ ταῖς περιφερείαις αἱ ὑπὸ $ΒΑΓ$, $ΕΔΖ$. λέγω, ὅτι ἐστὶν ὥς ἡ $ΒΓ$ περιφέρεια πρὸς τὴν $ΕΖ$ περιφέρειαν, οὕτως ἡ τε ὑπὸ $ΒΗΓ$ γωνία πρὸς τὴν ὑπὸ $ΕΘΖ$ καὶ ἡ ὑπὸ $ΒΑΓ$ πρὸς τὴν ὑπὸ $ΕΔΖ$.

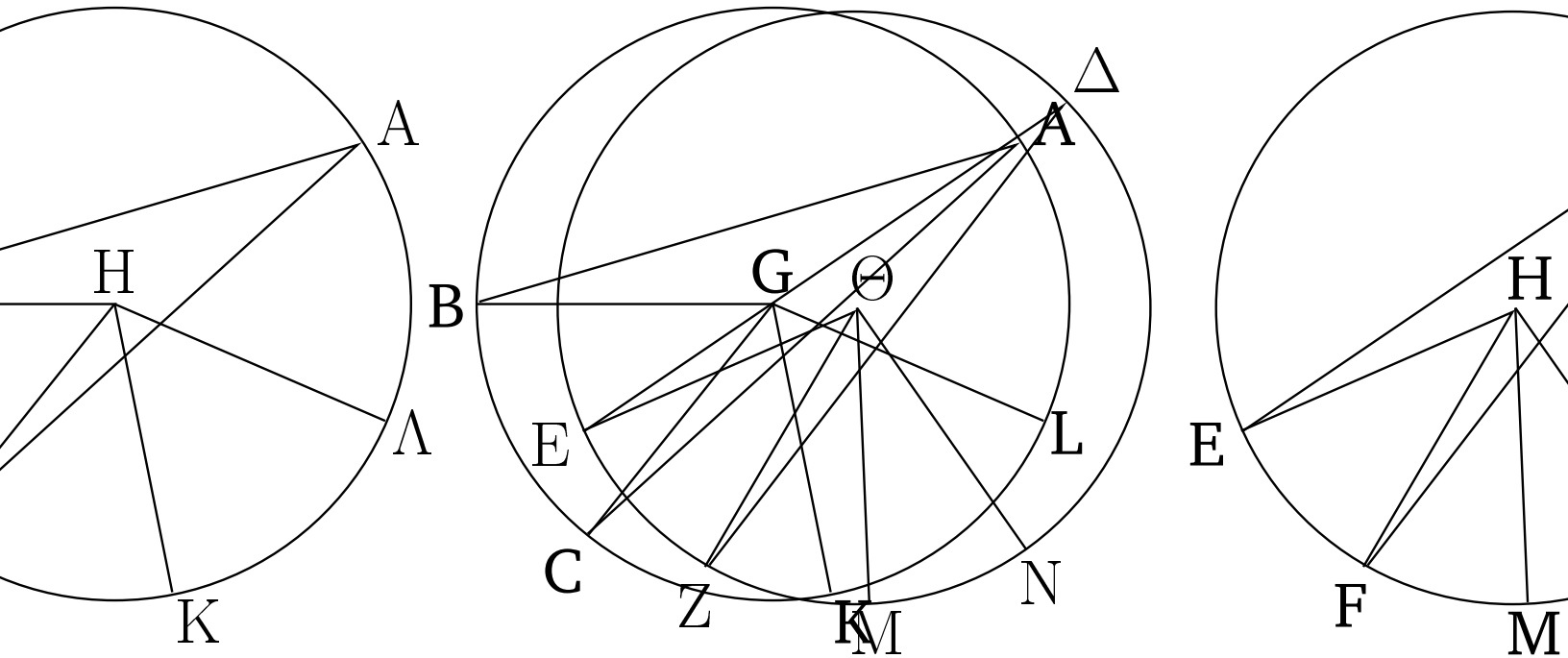
For since AB is parallel to DC , and the straight-line AC has fallen across them, the alternate angles BAC and ACD are equal to one another [Prop. 1.29]. So, for the same (reasons), CDE is also equal to ACD . And, hence, BAC is equal to CDE . And since ABC and DCE are two triangles having the one angle at A equal to the one angle at D , and the sides about the equal angles proportional, (so that) as BA (is) to AC , so CD (is) to DE , triangle ABC is thus equiangular to triangle DCE [Prop. 6.6]. Thus, angle ABC is equal to DCE . And (angle) ACD was also shown (to be) equal to BAC . Thus, the whole (angle) ACE is equal to the two (angles) ABC and BAC . Let ACB be added to both. Thus, ACE and ACB are equal to BAC , ACB , and CBA . But, BAC , ABC , and ACB are equal to two right-angles [Prop. 1.32]. Thus, ACE and ACB are also equal to two right-angles. Thus, the two straight-lines BC and CE , not lying on the same side, make adjacent angles ACE and ACB (whose sum is) equal to two right-angles with some straight-line AC , at the point C on it. Thus, BC is straight-on to CE [Prop. 1.14].

Thus, if two triangles, having two sides proportional to two sides, are placed together at a single angle such that the corresponding sides are also parallel, (then) the remaining sides of the triangles will be straight-on (with respect to one another). (Which is) the very thing it was required to show.

Proposition 33

In equal circles, angles have the same ratio as the (ratio of the) circumferences on which they stand, whether they are standing at the centers (of the circles) or at the circumferences.

Let ABC and DEF be equal circles, and let BGC and EHF be angles at their centers, G and H (respectively), and BAC and EDF (angles) at their circumferences. I say that as circumference BC is to circumference EF , so angle BGC (is) to EHF , and (angle) BAC to EDF .



Κείσθωσαν γάρ τῃ μὲν $B\Gamma$ περιφέρεια ἴσαι κατὰ τὸ ἐχθῆς ὁσαιδηποτοῦν αἱ ΓK , $K\Lambda$, τῇ δὲ EZ περιφέρεια ἴσαι ὁσαιδηποτοῦν αἱ ZM , MN , καὶ ἐπεξεύθωσαν αἱ HK , HL , ΘM , ΘN .

Ἐπεὶ οὖν ἴσαι εἰσὶν αἱ $B\Gamma$, ΓK , $K\Lambda$ περιφέρειαι ἀλλήλαις, ἴσαι εἰσὶ καὶ αἱ ὑπὸ BHG , ΓHK , KHL γωνίαι ἀλλήλαις· ὁσαπλασίων ἄρα ἐστὶν ἡ BA περιφέρεια τῆς $B\Gamma$, τοσανταπλασίων ἐστὶ καὶ ἡ ὑπὸ BHL γωνία τῆς ὑπὸ BHG . διὰ τὰ αὐτὰ δὴ καὶ ὁσαπλασίων ἐστὶν ἡ NE περιφέρεια τῆς EZ , τοσανταπλασίων ἐστὶ καὶ ἡ ὑπὸ $N\Theta E$ γωνία τῆς ὑπὸ $E\Theta Z$. εἰ ἄρα ἴση ἐστὶν ἡ BA περιφέρεια τῇ EN περιφέρειᾳ, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ BHL τῇ ὑπὸ $E\Theta N$, καὶ εἰ μείζων ἐστὶν ἡ BA περιφέρεια τῆς EN περιφερείας, μείζων ἐστὶ καὶ ἡ ὑπὸ BHL γωνία τῆς ὑπὸ $E\Theta N$, καὶ εἰ ἐλάσσων, ἐλάσσων. τεσσάρων δὴ ὄντων μεγεθῶν, δύο μὲν περιφερειῶν τῶν $B\Gamma$, EZ , δύο δὲ γωνιῶν τῶν ὑπὸ BHG , $E\Theta Z$, ἐληπτὰ τῆς μὲν $B\Gamma$ περιφερείας καὶ τῆς ὑπὸ BHG γωνίας ἰσάκις πολλαπλασίων ἢ τε BA περιφέρεια καὶ ἡ ὑπὸ BHL γωνία, τῆς δὲ EZ περιφερείας καὶ τῆς ὑπὸ $E\Theta Z$ γωνίας ἢ τε EN περιφέρεια καὶ ἡ ὑπὸ $E\Theta N$ γωνία. καὶ δέδεικται, ὅτι εἰ ὑπερέθῃ ἡ BA περιφέρεια τῆς EN περιφερείας, ὑπερέθῃ καὶ ἡ ὑπὸ BHL γωνία τῆς ὑπὸ $E\Theta N$ γωνίας, καὶ εἰ ἴση, ἴση, καὶ εἰ ἐλάσσων, ἐλάσσων. ἐστὶν ἄρα, ὥς ἡ $B\Gamma$ περιφέρεια πρὸς τὴν EZ , οὕτως ἡ ὑπὸ BHG γωνία πρὸς τὴν ὑπὸ $E\Theta Z$. ἀλλ' ὥς ἡ ὑπὸ BHG γωνία πρὸς τὴν ὑπὸ $E\Theta Z$, οὕτως ἡ ὑπὸ BAG πρὸς τὴν ὑπὸ $E\Delta Z$. διπλασία γὰρ ἑκάτερα ἑκάτερας. καὶ ὥς ἄρα ἡ $B\Gamma$ περιφέρεια πρὸς τὴν EZ περιφέρειαν, οὕτως ἢ τε ὑπὸ BHG γωνία πρὸς τὴν ὑπὸ $E\Theta Z$ καὶ ἡ ὑπὸ BAG πρὸς τὴν ὑπὸ $E\Delta Z$.

Ἐν ἄρα τοῖς ἴσοις κύκλοις αἱ γωνίαι τὸν αὐτὸν ἔθουσι

For let any number whatsoever of consecutive (circumferences), CK and KL , be made equal to circumference BC , and any number whatsoever, FM and MN , to circumference EF . And let GK , GL , HM , and HN be joined.

Therefore, since circumferences BC , CK , and KL are equal to one another, angles BGC , CGK , and KGL are also equal to one another [Prop. 3.27]. Thus, as many times as circumference BL is (divisible) by BC , so many times is angle BGL also (divisible) by BGC . And so, for the same (reasons), as many times as circumference NE is (divisible) by EF , so many times is angle NHE also (divisible) by EHF . Thus, if circumference BL is equal to circumference EN (then) angle BGL is also equal to EHN [Prop. 3.27], and if circumference BL is greater than circumference EN (then) angle BGL is also greater than EHN ,[†] and if (BL is) less (than EN then BGL is also) less (than EHN). So there are four magnitudes, two circumferences BC and EF , and two angles BGC and EHF . And equal multiples have been taken of circumference BC and angle BGC , (namely) circumference BL and angle BGL , and of circumference EF and angle EHF , (namely) circumference EN and angle EHN . And it has been shown that if circumference BL exceeds circumference EN (then) angle BGL also exceeds angle EHN , and if (BL is) equal (to EN then BGL is also) equal (to EHN), and if (BL is) less (than EN then BGL is also) less (than EHN). Thus, as circumference BC (is) to EF , so angle BGC (is) to EHF [Def. 5.5]. But as angle BGC (is) to EHF , so (angle) BAC (is) to EDF [Prop. 5.15]. For the former (are) double the latter (respectively) [Prop. 3.20]. Thus, also, as circumference BC (is) to circumference EF , so

λόγον ταῖς περιφερείαις, ἐφ' ὧν βεβήκασιν, ἔαν τε πρὸς τοῖς
κέντροις ἔαν τε πρὸς ταῖς περιφερείαις ὥσι βεβηκνῶσι· ὅπερ
ἔδει δεῖξαι.

angle BGC (is) to EHF , and BAC to EDF .

Thus, in equal circles, angles have the same ratio as the (ratio of the) circumferences on which they stand, whether they are standing at the centers (of the circles) or at the circumferences. (Which is) the very thing it was required to show.

[†] This is a straight-forward generalization of Prop. 3.27

ELEMENTS BOOK 7

Elementary Number Theory[†]

[†]The propositions contained in Books 7–9 are generally attributed to the school of Pythagoras.

Ὅροι.

1. Μονάς ἐστίν, καθ' ἣν ἕκαστον τῶν ὄντων ἐν λέγεται.
2. Ἀριθμός δὲ τὸ ἐκ μονάδων συγκείμενον πλῆθος.
3. Μέρος ἐστὶν ἀριθμὸς ἀριθμοῦ ὁ ἐλάσσων τοῦ μείζονος, ὅταν καταμετρηῇ τὸν μείζονα.
4. Μέρη δέ, ὅταν μὴ καταμετρηῇ.
5. Πολλαπλάσιος δὲ ὁ μείζων τοῦ ἐλάσσονος, ὅταν καταμετρηῇ ὑπὸ τοῦ ἐλάσσονος.
6. Ἄρτιος ἀριθμὸς ἐστὶν ὁ δίῃ διαιρούμενος.
7. Περισσὸς δὲ ὁ μὴ διαιρούμενος δίῃ ἢ [ὁ] μονάδι διαφέρων ἀρτίον ἀριθμοῦ.
8. Ἀρτιάκις ἄρτιος ἀριθμὸς ἐστὶν ὁ ὑπὸ ἀρτίον ἀριθμοῦ μετρούμενος κατὰ ἄρτιον ἀριθμόν.
9. Ἀρτιάκις δὲ περισσὸς ἐστὶν ὁ ὑπὸ ἀρτίον ἀριθμοῦ μετρούμενος κατὰ περισσὸν ἀριθμόν.
10. Περισσάκις δὲ περισσὸς ἀριθμὸς ἐστὶν ὁ ὑπὸ περισσοῦ ἀριθμοῦ μετρούμενος κατὰ περισσὸν ἀριθμόν.
11. Πρῶτος ἀριθμὸς ἐστὶν ὁ μονάδι μόνῃ μετρούμενος.
12. Πρῶτοι πρὸς ἀλλήλους ἀριθμοὶ εἰσιν οἱ μονάδι μόνῃ μετρούμενοι κοινῷ μέτρῳ.
13. Σύνθετος ἀριθμὸς ἐστὶν ὁ ἀριθμῷ τινι μετρούμενος.
14. Σύνθετοι δὲ πρὸς ἀλλήλους ἀριθμοὶ εἰσιν οἱ ἀριθμῷ τινι μετρούμενοι κοινῷ μέτρῳ.
15. Ἀριθμὸς ἀριθμόν πολλαπλασιάζειν λέγεται, ὅταν, ὅσαι εἰσὶν ἐν αὐτῷ μονάδες, τοσαντάκις συντεθῇ ὁ πολλαπλασιαζόμενος, καὶ γένηται τις.
16. Ὅταν δὲ δύο ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλους ποιῶσι τινα, ὁ γενόμενος ἐπίπεδος καλεῖται, πλευραὶ δὲ αὐτοῦ οἱ πολλαπλασιάσαντες ἀλλήλους ἀριθμοί.
17. Ὅταν δὲ τρεῖς ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλους ποιῶσι τινα, ὁ γενόμενος στερεὸς ἐστίν, πλευραὶ δὲ αὐτοῦ οἱ πολλαπλασιάσαντες ἀλλήλους ἀριθμοί.
18. Τετράγωνος ἀριθμὸς ἐστὶν ὁ ἰσάκις ἴσος ἢ [ὁ] ὑπὸ δύο ἴσων ἀριθμῶν περιεθόμενος.
19. Κύβος δὲ ὁ ἰσάκις ἴσος ἰσάκις ἢ [ὁ] ὑπὸ τριῶν ἴσων ἀριθμῶν περιεθόμενος.
20. Ἀριθμοὶ ἀνάλογόν εἰσιν, ὅταν ὁ πρῶτος τοῦ δευτέρου καὶ ὁ τρίτος τοῦ τετάρτου ἰσάκις ἢ πολλαπλάσιος ἢ τὸ αὐτὸ μέρος ἢ τὰ αὐτὰ μέρη ᾧσιν.
21. Ὅμοιοι ἐπίπεδοι καὶ στερεοὶ ἀριθμοὶ εἰσιν οἱ ἀνάλογον ἔχοντες τὰς πλευράς.
22. Τέλειος ἀριθμὸς ἐστὶν ὁ τοῖς ἑαυτοῦ μέρεσιν ἴσος ᾧν.

Definitions

1. A unit is (that) according to which each existing (thing) is said (to be) one.
2. And a number (is) a multitude composed of units.[†]
3. A number is part of a(nother) number, the lesser of the greater, when it measures the greater.[‡]
4. But (the lesser is) parts (of the greater) when it does not measure it.[§]
5. And the greater (number is) a multiple of the lesser when it is measured by the lesser.
6. An even number is one (which can be) divided in half.
7. And an odd number is one (which can)not (be) divided in half, or which differs from an even number by a unit.
8. An even-times-even number is one (which is) measured by an even number according to an even number.[¶]
9. And an even-times-odd number is one (which is) measured by an even number according to an odd number.*
10. And an odd-times-odd number is one (which is) measured by an odd number according to an odd number.[§]
11. A prime^b number is one (which is) measured by a unit alone.
12. Numbers prime to one another are those (which are) measured by a unit alone as a common measure.
13. A composite number is one (which is) measured by some number.
14. And numbers composite to one another are those (which are) measured by some number as a common measure.
15. A number is said to multiply a(nother) number when the (number being) multiplied is added (to itself) as many times as there are units in the former (number), and (thereby) some (other number) is produced.
16. And when two numbers multiplying one another make some (other number then) the (number so) created is called plane, and its sides (are) the numbers which multiply one another.
17. And when three numbers multiplying one another make some (other number then) the (number so) created is (called) solid, and its sides (are) the numbers which multiply one another.
18. A square number is an equal times an equal, or (a plane number) contained by two equal numbers.
19. And a cube (number) is an equal times an equal times an equal, or (a solid number) contained by three equal numbers.
20. Numbers are proportional when the first is the same multiple, or the same part, or the same parts, of the second that the third (is) of the fourth.
21. Similar plane and solid numbers are those having proportional sides.

22. A perfect number is that which is equal to its own parts.^{††}

[†] In other words, a “number” is a positive integer greater than unity.

[‡] In other words, a number a is part of another number b if there exists some number n such that $na = b$.

[§] In other words, a number a is parts of another number b (where $a < b$) if there exist distinct numbers, m and n , such that $na = mb$.

[¶] In other words, an even-times-even number is the product of two even numbers.

^{*} In other words, an even-times-odd number is the product of an even and an odd number.

[§] In other words, an odd-times-odd number is the product of two odd numbers.

^ᵇ Literally, “first”.

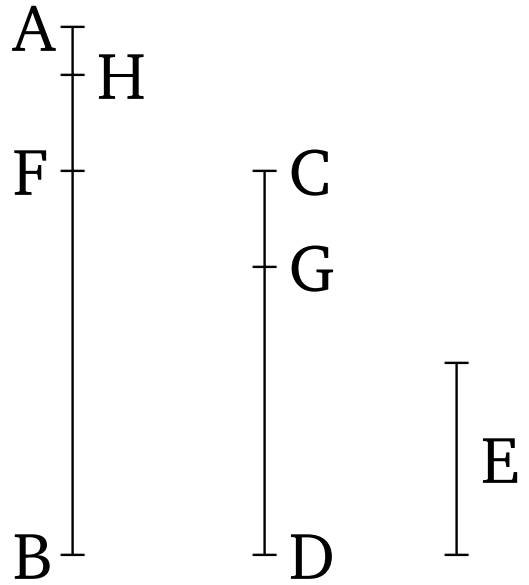
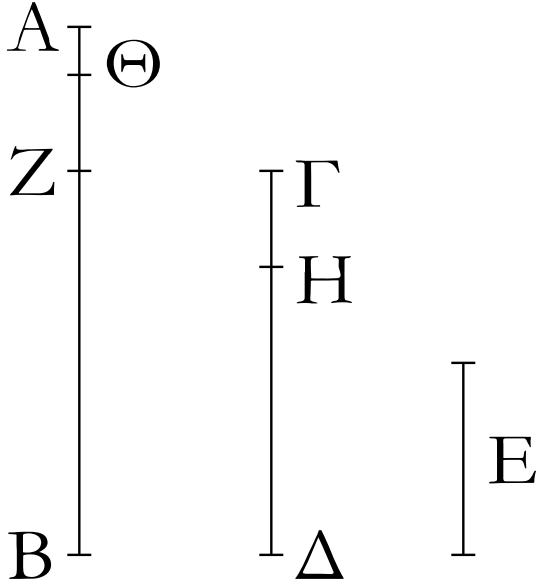
^{††} In other words, a perfect number is equal to the sum of its own factors.

1.

Proposition 1

Δύο ἀριθμῶν ἀνίσων ἐκκειμένων, ἀνθυφαιρουμένον δὲ αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος, ἔαν ὁ λειπόμενος μηδέποτε καταμετρῇ τὸν πρὸ ἑαυτοῦ, ἕως ὅτ' ὑ λειφθῇ μονάξ, οἱ ἐξ ἀρ-
θῆς ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους ἔσονται.

Two unequal numbers (being) laid down, and the lesser being continually subtracted, in turn, from the greater, if the remainder never measures the (number) preceding it, until a unit remains, (then) the original numbers will be prime to one another.



Δύο γὰρ [ἀνίσων] ἀριθμῶν τῶν AB , $\Gamma\Delta$ ἀνθυφαιρουμένον αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος ὁ λειπόμενος μηδέποτε καταμετρεῖται τὸν πρὸ ἑαυτοῦ, ἕως ὅτ' ὑ λειφθῇ μονάξ· λέγω, ὅτι οἱ AB , $\Gamma\Delta$ πρῶτοι πρὸς ἀλλήλους εἰσίν, τουτέστιν ὅτι τούξ AB , $\Gamma\Delta$ μονάξ μόνη μετρεῖ.

Εἰ γὰρ μή εἰσιν οἱ AB , $\Gamma\Delta$ πρῶτοι πρὸς ἀλλήλους, με-
τρήσει τις αὐτοῦξ ἀριθμός. μετρεῖται, καὶ ἔστω ὁ E · καὶ ὁ
μὲν $\Gamma\Delta$ τὸν BZ μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν ZA , ὁ
δὲ AZ τὸν ΔH μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν $H\Gamma$, ὁ
δὲ $H\Gamma$ τὸν $Z\Theta$ μετρῶν λειπέτω μονάδα τὴν ΘA .

Ἐπεὶ οὖν ὁ E τὸν $\Gamma\Delta$ μετρεῖ, ὁ δὲ $\Gamma\Delta$ τὸν BZ μετρεῖ,
καὶ ὁ E ἄρα τὸν BZ μετρεῖ· μετρεῖ δὲ καὶ ὅλον τὸν BA · καὶ
λοιπὸν ἄρα τὸν AZ μετρήσει. ὁ δὲ AZ τὸν ΔH μετρεῖ· καὶ ὁ

For two [unequal] numbers, AB and CD , the lesser being continually subtracted, in turn, from the greater, let the remainder never measure the (number) preceding it, until a unit remains. I say that AB and CD are prime to one another—that is to say, that a unit alone measures (both) AB and CD .

For if AB and CD are not prime to one another, (then) some number will measure them. Let (some number) measure them, and let it be E . And let CD measuring BF leave FA less than itself, and let AF measuring DG leave GC less than itself, and let GC measuring FH leave a unit, HA .

In fact, since E measures CD , and CD measures BF , E thus also measures BF .[†] And (E) also measures the whole of BA . Thus, (E) will also measure the remainder AF .[‡] And AF

E ἄρα τὸν ΔH μετρεῖ· μετρεῖ δὲ καὶ ὅλον τὸν $\Delta \Gamma$ · καὶ λοιπὸν ἄρα τὸν ΓH μετρήσει. ὁ δὲ ΓH τὸν $Z\Theta$ μετρεῖ· καὶ ὁ E ἄρα τὸν $Z\Theta$ μετρεῖ· μετρεῖ δὲ καὶ ὅλον τὸν ZA · καὶ λοιπὴν ἄρα τὴν $A\Theta$ μονάδα μετρήσει ἀριθμὸς ὧν ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τούξ AB , $\Gamma\Delta$ ἀριθμοὺς μετρήσει τις ἀριθμὸς· οἱ AB , $\Gamma\Delta$ ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν· ὅπερ ἔδει δεῖξαι.

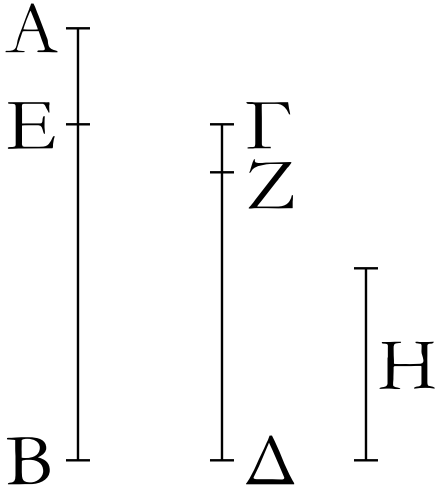
measures DG . Thus, E also measures DG . And (E) also measures the whole of DC . Thus, (E) will also measure the remainder CG . And CG measures FH . Thus, E also measures FH . And (E) also measures the whole of FA . Thus, (E) will also measure the remaining unit AH , (despite) being a number. The very thing is impossible. Thus, some number does not measure (both) the numbers AB and CD . Thus, AB and CD are prime to one another. (Which is) the very thing it was required to show.

† Here, use is made of the unstated common notion that if a measures b , and b measures c , then a also measures c , where all symbols denote numbers.

‡ Here, use is made of the unstated common notion that if a measures b , and a measures part of b , then a also measures the remainder of b , where all symbols denote numbers.

2.

Δύο ἀριθμῶν δοθέντων μὴ πρῶτων πρὸς ἀλλήλους τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.



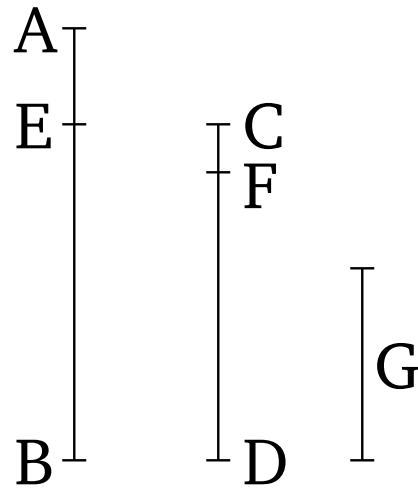
Ἐστωσαν οἱ δοθέντες δύο ἀριθμοὶ μὴ πρῶτοι πρὸς ἀλλήλους οἱ AB , $\Gamma\Delta$. δεῖ δὴ τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Εἰ μὲν οὖν ὁ $\Gamma\Delta$ τὸν AB μετρεῖ, μετρεῖ δὲ καὶ ἑαυτόν, ὁ $\Gamma\Delta$ ἄρα τῶν $\Gamma\Delta$, AB κοινὸν μέτρον ἐστίν. καὶ φανερόν, ὅτι καὶ μέγιστον· οὐδεὶς γάρ μείζων τοῦ $\Gamma\Delta$ τὸν $\Gamma\Delta$ μετρήσει.

Εἰ δὲ οὐ μετρεῖ ὁ $\Gamma\Delta$ τὸν AB , τῶν AB , $\Gamma\Delta$ ἀνθυφαίρουμένων ἀεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος λειψθήσεται τις ἀριθμὸς, ὃς μετρήσει τὸν πρὸ ἑαυτοῦ. μονὰς μὲν γὰρ οὐ λειψθήσεται· εἰ δὲ μή, ἔσονται οἱ AB , $\Gamma\Delta$ πρῶτοι πρὸς ἀλλήλους· ὅπερ οὐθὺν ὑπόκειται. λειψθήσεται τις ἄρα ἀριθμὸς, ὃς μετρήσει τὸν πρὸ ἑαυτοῦ. καὶ ὁ μὲν $\Gamma\Delta$ τὸν BE μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν EA , ὁ δὲ EA τὸν ΔZ μετρῶν λειπέτω ἑαυτοῦ ἐλάσσονα τὸν $Z\Gamma$, ὁ δὲ ΓZ τὸν AE μετρήτω. ἐπεὶ οὖν ὁ ΓZ τὸν AE μετρεῖ, ὁ δὲ AE τὸν ΔZ μετρεῖ, καὶ ὁ ΓZ ἄρα τὸν ΔZ μετρήσει. μετρεῖ δὲ καὶ ἑαυτόν· καὶ ὅλον ἄρα τὸν $\Gamma\Delta$ μετρήσει. ὁ δὲ $\Gamma\Delta$ τὸν BE μετρεῖ· καὶ ὁ ΓZ ἄρα τὸν BE μετρεῖ· μετρεῖ δὲ καὶ τὸν EA · καὶ ὅλον ἄρα τὸν

Proposition 2

To find the greatest common measure of two given numbers (which are) not prime to one another.



Let AB and CD be the two given numbers (which are) not prime to one another. So it is required to find the greatest common measure of AB and CD .

In fact, if CD measures AB , CD is thus a common measure of CD and AB , (since CD) also measures itself. And (it is) manifest that (it is) also the greatest (common measure). For nothing greater than CD can measure CD .

But if CD does not measure AB , (then) some number will remain from AB and CD , the lesser being continually subtracted, in turn, from the greater, which will measure the (number) preceding it. For a unit will not be left. But if not, AB and CD will be prime to one another [Prop. 7.1]. The very opposite thing was assumed. Thus, some number will remain which will measure the (number) preceding it. And let CD measuring BE leave EA less than itself, and let EA measuring DF leave FC less than itself, and let CF measure AE . Therefore, since CF measures AE , and AE measures DF , CF will thus also measure DF . And it also measures itself. Thus, it

BA μετρήσει· μετρεῖ δὲ καὶ τὸν $\Gamma\Delta$ · ὁ ΓZ ἄρα τοῖς AB , $\Gamma\Delta$ μετρεῖ. ὁ ΓZ ἄρα τῶν AB , $\Gamma\Delta$ κοινὸν μέτρον ἐστίν. λέγω δὴ, ὅτι καὶ μέγιστον. εἰ γὰρ μὴ ἐστὶν ὁ ΓZ τῶν AB , $\Gamma\Delta$ μέγιστον κοινὸν μέτρον, μετρήσει τις τοῖς AB , $\Gamma\Delta$ ἀριθμὸς ἀριθμὸς μείζων ὢν τοῦ ΓZ . μετρεῖτω, καὶ ἔστω ὁ H . καὶ ἐπεὶ ὁ H τὸν $\Gamma\Delta$ μετρεῖ, ὁ δὲ $\Gamma\Delta$ τὸν BE μετρεῖ, καὶ ὁ H ἄρα τὸν BE μετρεῖ· μετρεῖ δὲ καὶ ὅλον τὸν BA · καὶ λοιπὸν ἄρα τὸν AE μετρήσει. ὁ δὲ AE τὸν ΔZ μετρεῖ· καὶ ὁ H ἄρα τὸν ΔZ μετρήσει· μετρεῖ δὲ καὶ ὅλον τὸν $\Delta\Gamma$ · καὶ λοιπὸν ἄρα τὸν ΓZ μετρήσει ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα τοῖς AB , $\Gamma\Delta$ ἀριθμὸς ἀριθμὸς τις μετρήσει μείζων ὢν τοῦ ΓZ · ὁ ΓZ ἄρα τῶν AB , $\Gamma\Delta$ μέγιστόν ἐστι κοινὸν μέτρον [ὅπερ ἔδει δεῖχαι].

will also measure the whole of CD . And CD measures BE . Thus, CF also measures BE . And it also measures EA . Thus, it will also measure the whole of BA . And it also measures CD . Thus, CF measures (both) AB and CD . Thus, CF is a common measure of AB and CD . So I say that (it is) also the greatest (common measure). For if CF is not the greatest common measure of AB and CD , (then) some number which is greater than CF will measure the numbers AB and CD . Let it (so) measure (AB and CD), and let it be G . And since G measures CD , and CD measures BE , G thus also measures BE . And it also measures the whole of BA . Thus, it will also measure the remainder AE . And AE measures DF . Thus, G will also measure DF . And it also measures the whole of DC . Thus, it will also measure the remainder CF , the greater (measuring) the lesser. The very thing is impossible. Thus, some number which is greater than CF cannot measure the numbers AB and CD . Thus, CF is the greatest common measure of AB and CD . [(Which is) the very thing it was required to show].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν ἀριθμὸς δύο ἀριθμὸς μετρήῃ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσει· ὅπερ ἔδει δεῖχαι.

Corollary

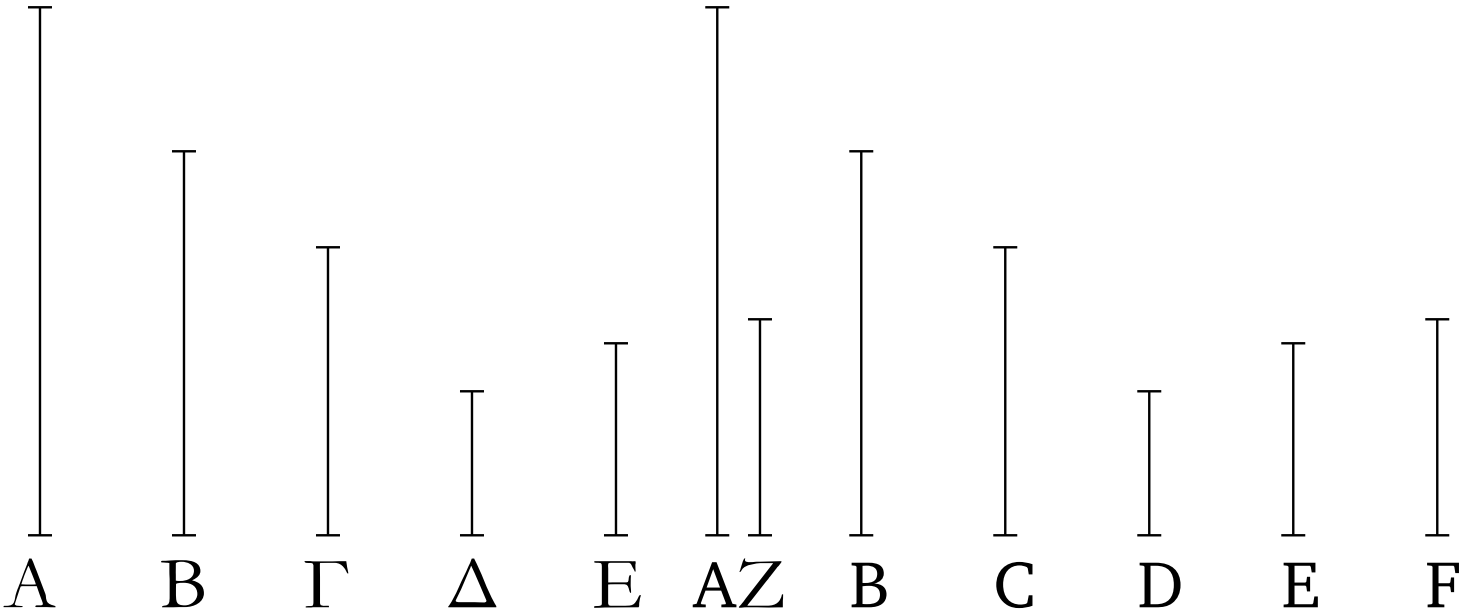
So it is manifest, from this, that if a number measures two numbers, (then) it will also measure their greatest common measure. (Which is) the very thing it was required to show.

3.

Τριῶν ἀριθμῶν δοθέντων μὴ πρώτων πρὸς ἀλλήλους τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

Proposition 3

To find the greatest common measure of three given numbers (which are) not prime to one another.



Ἐστωσαν οἱ δοθέντες τρεῖς ἀριθμοὶ μὴ πρῶτοι πρὸς ἀλλήλους οἱ A , B , Γ . δεῖ δὴ τῶν A , B , Γ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Let A , B , and C be the three given numbers (which are) not prime to one another. So it is required to find the greatest common measure of A , B , and C .

Εἰλήφθω γάρ δύο τῶν A, B τὸ μέγιστον κοινὸν μέτρον ὁ Δ . ὁ δὲ Δ τὸν Γ ἤτοι μετρεῖ ἢ οὐ μετρεῖ. μετρεῖτω πρότερον· μετρεῖ δὲ καὶ τοὺς A, B . ὁ Δ ἄρα τοὺς A, B, Γ μετρεῖ. ὁ Δ ἄρα τῶν A, B, Γ κοινὸν μέτρον ἐστίν. λέγω δὴ, ὅτι καὶ μέγιστον. εἰ γὰρ μὴ ἐστὶν ὁ Δ τῶν A, B, Γ μέγιστον κοινὸν μέτρον, μετρήσει τις τοὺς A, B, Γ ἀριθμὸν ἀριθμὸς μείζων ὢν τοῦ Δ . μετρεῖτω, καὶ ἔστω ὁ E . ἐπεὶ οὖν ὁ E τοὺς A, B, Γ μετρεῖ, καὶ τοὺς A, B ἄρα μετρήσει· καὶ τὸ τῶν A, B ἄρα μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν A, B μέγιστον κοινὸν μέτρον ἐστὶν ὁ Δ . ὁ E ἄρα τὸν Δ μετρεῖ ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τοὺς A, B, Γ ἀριθμὸν ἀριθμὸς τις μετρήσει μείζων ὢν τοῦ Δ . ὁ Δ ἄρα τῶν A, B, Γ μέγιστόν ἐστι κοινὸν μέτρον.

μὴ μετρεῖτω δὴ ὁ Δ τὸν Γ . λέγω πρῶτον, ὅτι οἱ Γ, Δ οὐκ εἰσι πρῶτοι πρὸς ἀλλήλους. ἐπεὶ γὰρ οἱ A, B, Γ οὐκ εἰσι πρῶτοι πρὸς ἀλλήλους, μετρήσει τις αὐτοὺς ἀριθμὸς. ὁ δὲ τοὺς A, B, Γ μετρῶν καὶ τοὺς A, B μετρήσει, καὶ τὸ τῶν A, B μέγιστον κοινὸν μέτρον τὸν Δ μετρήσει· μετρεῖ δὲ καὶ τὸν Γ . τοὺς Δ, Γ ἄρα ἀριθμὸν ἀριθμὸς τις μετρήσει. οἱ Δ, Γ ἄρα οὐκ εἰσι πρῶτοι πρὸς ἀλλήλους. εἰλήφθω οὖν αὐτῶν τὸ μέγιστον κοινὸν μέτρον ὁ E . καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ, ὁ δὲ Δ τοὺς A, B μετρεῖ, καὶ ὁ E ἄρα τοὺς A, B μετρεῖ· μετρεῖ δὲ καὶ τὸν Γ . ὁ E ἄρα τοὺς A, B, Γ μετρεῖ. ὁ E ἄρα τῶν A, B, Γ κοινὸν ἐστὶ μέτρον. λέγω δὴ, ὅτι καὶ μέγιστον. εἰ γὰρ μὴ ἐστὶν ὁ E τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον, μετρήσει τις τοὺς A, B, Γ ἀριθμὸν ἀριθμὸς μείζων ὢν τοῦ E . μετρεῖτω, καὶ ἔστω ὁ Z . καὶ ἐπεὶ ὁ Z τοὺς A, B, Γ μετρεῖ, καὶ τοὺς A, B μετρεῖ· καὶ τὸ τῶν A, B ἄρα μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν A, B μέγιστον κοινὸν μέτρον ἐστὶν ὁ Δ . ὁ Z ἄρα τὸν Δ μετρεῖ· μετρεῖ δὲ καὶ τὸν Γ . ὁ Z ἄρα τοὺς Δ, Γ μετρεῖ· καὶ τὸ τῶν Δ, Γ ἄρα μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν Δ, Γ μέγιστον κοινὸν μέτρον ἐστὶν ὁ E . ὁ Z ἄρα τὸν E μετρεῖ ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τοὺς A, B, Γ ἀριθμὸν ἀριθμὸς τις μετρήσει μείζων ὢν τοῦ E . ὁ E ἄρα τῶν A, B, Γ μέγιστόν ἐστι κοινὸν μέτρον· ὅπερ ἔδει δεῖξαι.

4.

Ἄπαξ ἀριθμὸς παντὸς ἀριθμοῦ ὁ ἐλάσσων τοῦ μείζονος ἤτοι μέρος ἐστὶν ἢ μέρος.

Ἐστωσαν δύο ἀριθμοὶ οἱ $A, B\Gamma$, καὶ ἔστω ἐλάσσων ὁ $B\Gamma$. λέγω, ὅτι ὁ $B\Gamma$ τοῦ A ἤτοι μέρος ἐστὶν ἢ μέρος.

Οἱ $A, B\Gamma$ γὰρ ἤτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν ἢ οὐ.

For let the greatest common measure, D , of the two (numbers) A and B be taken [Prop. 7.2]. So D either measures, or does not measure, C . First of all, let it measure (C). And it also measures A and B . Thus, D measures A, B , and C . Thus, D is a common measure of A, B , and C . So I say that (it is) also the greatest (common measure). For if D is not the greatest common measure of A, B , and C , (then) some number greater than D will measure the numbers A, B , and C . Let it (so) measure (A, B , and C), and let it be E . Therefore, since E measures A, B , and C , it will thus also measure A and B . Thus, it will also measure the greatest common measure of A and B [Prop. 7.2 corr.]. And D is the greatest common measure of A and B . Thus, E measures D , the greater (measuring) the lesser. The very thing is impossible. Thus, some number which is greater than D cannot measure the numbers A, B , and C . Thus, D is the greatest common measure of A, B , and C .

So let D not measure C . I say, first of all, that C and D are not prime to one another. For since A, B, C are not prime to one another, some number will measure them. So the (number) measuring A, B , and C will also measure A and B , and it will also measure the greatest common measure, D , of A and B [Prop. 7.2 corr.]. And it also measures C . Thus, some number will measure the numbers D and C . Thus, D and C are not prime to one another. Therefore, let their greatest common measure, E , be taken [Prop. 7.2]. And since E measures D , and D measures A and B , E thus also measures A and B . And it also measures C . Thus, E measures A, B , and C . Thus, E is a common measure of A, B , and C . So I say that (it is) also the greatest (common measure). For if E is not the greatest common measure of A, B , and C , (then) some number greater than E will measure the numbers A, B , and C . Let it (so) measure (A, B , and C), and let it be F . And since F measures A, B , and C , it also measures A and B . Thus, it will also measure the greatest common measure of A and B [Prop. 7.2 corr.]. And D is the greatest common measure of A and B . Thus, F measures D . And it also measures C . Thus, F measures D and C . Thus, it will also measure the greatest common measure of D and C [Prop. 7.2 corr.]. And E is the greatest common measure of D and C . Thus, F measures E , the greater (measuring) the lesser. The very thing is impossible. Thus, some number which is greater than E does not measure the numbers A, B , and C . Thus, E is the greatest common measure of A, B , and C . (Which is) the very thing it was required to show.

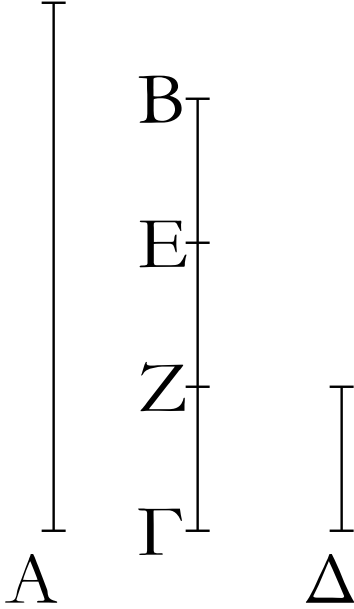
Proposition 4

Any number is either part or parts of any (other) number, the lesser of the greater.

Let A and BC be two numbers, and let BC be the lesser. I say that BC is either part or parts of A .

For A and BC are either prime to one another, or not. Let

ἔστωσαν πρότερον οἱ A , $B\Gamma$ πρώτοι πρὸς ἀλλήλους. διαιρεθέντος δὲ τοῦ $B\Gamma$ εἰς τὰς ἐν αὐτῷ μονάδας ἔσται ἑκάστη μονὰς τῶν ἐν τῷ $B\Gamma$ μέρος τι τοῦ A . ὥστε μέρη ἐστὶν ὁ $B\Gamma$ τοῦ A .



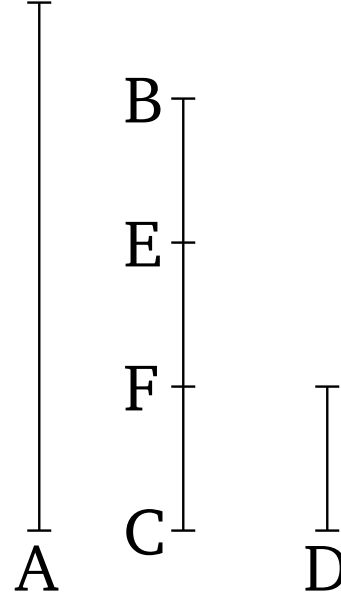
μὴ ἔστωσαν δὴ οἱ A , $B\Gamma$ πρώτοι πρὸς ἀλλήλους· ὁ δὲ $B\Gamma$ τὸν A ἢτοι μετρεῖ ἢ οὐ μετρεῖ. εἰ μὲν οὖν ὁ $B\Gamma$ τὸν A μετρεῖ, μέρος ἐστὶν ὁ $B\Gamma$ τοῦ A . εἰ δὲ οὐ, εἰλήφθω τῶν A , $B\Gamma$ μέγιστον κοινὸν μέτρον ὁ Δ , καὶ διηγήσθω ὁ $B\Gamma$ εἰς τοὺς τῷ Δ ἴσους τοὺς BE , EZ , $Z\Gamma$. καὶ ἐπεὶ ὁ Δ τὸν A μετρεῖ, μέρος ἐστὶν ὁ Δ τοῦ A . ἴσος δὲ ὁ Δ ἑκάστῳ τῶν BE , EZ , $Z\Gamma$. καὶ ἑκάστος ἄρα τῶν BE , EZ , $Z\Gamma$ τοῦ A μέρος ἐστὶν· ὥστε μέρη ἐστὶν ὁ $B\Gamma$ τοῦ A .

Ἄρα ἂν ἀριθμὸς παντὸς ἀριθμοῦ ὁ ἐλάσσων τοῦ μείζονος ἢτοι μέρος ἐστὶν ἢ μέρη· ὅπερ ἔδει δεῖξαι.

5.

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρος ᾗ, καὶ ἕτερος ἕτερον τὸ αὐτὸ μέρος ᾗ, καὶ συναμφοτέρως συναμφοτέρων τὸ αὐτὸ μέρος ἔσται, ὅπερ ὁ εἶξ τοῦ ἐνόξ.

A and BC , first of all, be prime to one another. So separating BC into its constituent units, each of the units in BC will be some part of A . Hence, BC is parts of A .

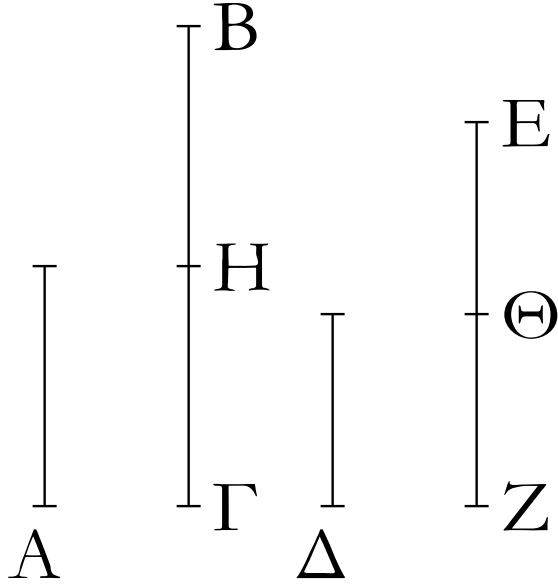


So let A and BC be not prime to one another. So BC either measures, or does not measure, A . Therefore, if BC measures A , (then) BC is part of A . And if not, let the greatest common measure, D , of A and BC be taken [Prop. 7.2], and let BC be divided into BE , EF , and FC , equal to D . And since D measures A , D is a part of A . And D is equal to each of BE , EF , and FC . Thus, BE , EF , and FC are also each part of A . Hence, BC is parts of A .

Thus, any number is either part or parts of any (other) number, the lesser of the greater. (Which is) the very thing it was required to show.

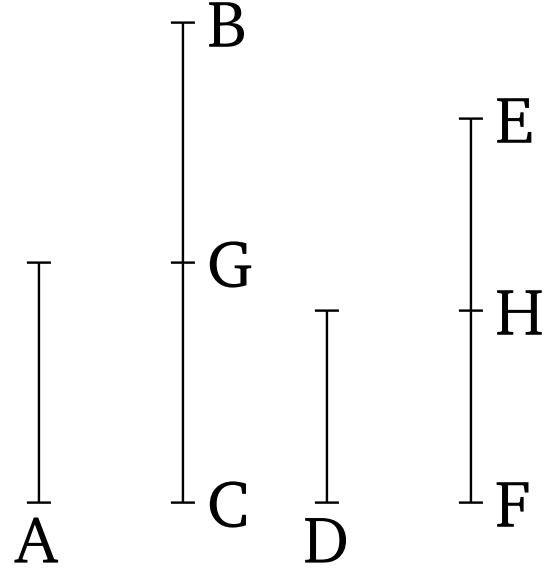
Proposition 5[†]

If a number is part of a number, and another (number) is the same part of another, (then) the sum (of the leading numbers) will also be the same part of the sum (of the following numbers) that one (number) is of another.



Ἀριθμὸς γάρ ὁ A [ἀριθμοῦ] τοῦ $BΓ$ μέρους ἔστω, καὶ ἕτερος ὁ $Δ$ ἑτέρου τοῦ $EΖ$ τὸ αὐτὸ μέρος, ὅπερ ὁ A τοῦ $BΓ$ λέγω, ὅτι καὶ συναμφοτέρως ὁ A , $Δ$ συναμφοτέρου τοῦ $BΓ$, $EΖ$ τὸ αὐτὸ μέρος ἐστίν, ὅπερ ὁ A τοῦ $BΓ$.

Ἐπεὶ γάρ, ὁ μέρος ἐστὶν ὁ A τοῦ $BΓ$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ $Δ$ τοῦ $EΖ$, ὅσοι ἄρα εἰσὶν ἐν τῷ $BΓ$ ἀριθμοὶ ἴσοι τῷ A , τοσοῦτοὶ εἰσι καὶ ἐν τῷ $EΖ$ ἀριθμοὶ ἴσοι τῷ $Δ$. διηγήσθω ὁ μὲν $BΓ$ εἰς τοὺς τῷ A ἴσους τοὺς $BΗ$, $ΗΓ$, ὁ δὲ $EΖ$ εἰς τοὺς τῷ $Δ$ ἴσους τοὺς $EΘ$, $ΘΖ$. ἔσται δὴ ἴσον τὸ πλῆθος τῶν $BΗ$, $ΗΓ$ τῷ πλῆθει τῶν $EΘ$, $ΘΖ$. καὶ ἐπεὶ ἴσοξ ἐστὶν ὁ μὲν $BΗ$ τῷ A , ὁ δὲ $EΘ$ τῷ $Δ$, καὶ οἱ $BΗ$, $EΘ$ ἄρα τοῖς A , $Δ$ ἴσοι. διὰ τὰ αὐτὰ δὴ καὶ οἱ $ΗΓ$, $ΘΖ$ τοῖς A , $Δ$. ὅσοι ἄρα [εἰσὶν] ἐν τῷ $BΓ$ ἀριθμοὶ ἴσοι τῷ A , τοσοῦτοὶ εἰσι καὶ ἐν τοῖς $BΓ$, $EΖ$ ἴσοι τοῖς A , $Δ$. ὁσαπλασίων ἄρα ἐστὶν ὁ $BΓ$ τοῦ A , τοσανταπλασίων ἐστὶ καὶ συναμφοτέρως ὁ $BΓ$, $EΖ$ συναμφοτέρου τοῦ A , $Δ$. ὁ ἄρα μέρος ἐστὶν ὁ A τοῦ $BΓ$, τὸ αὐτὸ μέρος ἐστὶ καὶ συναμφοτέρως ὁ A , $Δ$ συναμφοτέρου τοῦ $BΓ$, $EΖ$. ὅπερ ἔδει δεῖξαι.



For let a number A be part of a [number] BC , and another (number) D (be) the same part of another (number) EF that A (is) of BC . I say that the sum A , D is also the same part of the sum BC , EF that A (is) of BC .

For since which(ever) part A is of BC , D is the same part of EF , thus as many numbers as are in BC equal to A , so many numbers are also in EF equal to D . Let BC be divided into BG and GC , equal to A , and EF into EH and HF , equal to D . So the multitude of (divisions) BG , GC will be equal to the multitude of (divisions) EH , HF . And since BG is equal to A , and EH to D , thus BG , EH (is) also equal to A , D . So, for the same (reasons), GC , HF (is) also (equal) to A , D . Thus, as many numbers as [are] in BC equal to A , so many are also in BC , EF equal to A , D . Thus, as many times as BC is (divisible) by A , so many times is the sum BC , EF also (divisible) by the sum A , D . Thus, which(ever) part A is of BC , the sum A , D is also the same part of the sum BC , EF . (Which is) the very thing it was required to show.

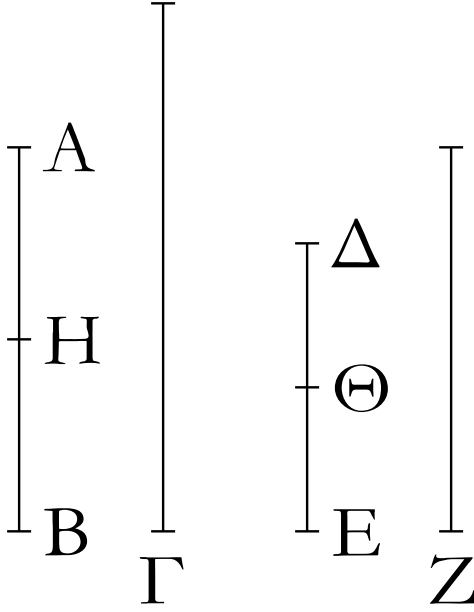
[†] In modern notation, this proposition states that if $a = (1/n)b$ and $c = (1/n)d$ then $(a + c) = (1/n)(b + d)$, where all symbols denote numbers.

6.

Proposition 6[†]

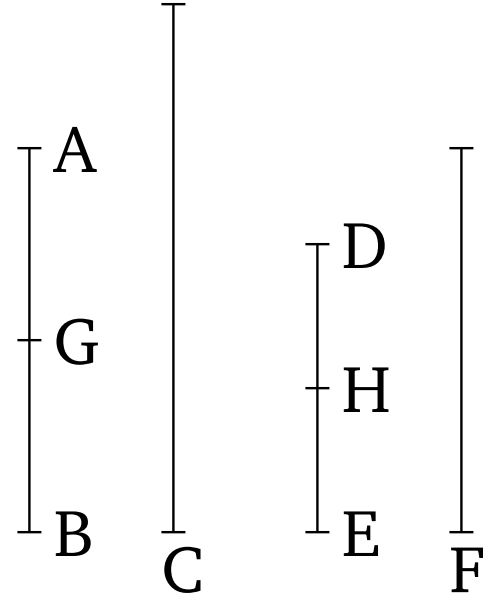
Ἐὰν ἀριθμὸς ἀριθμοῦ μέρη ᾗ, καὶ ἕτερος ἑτέρου τὰ αὐτὰ μέρη ᾗ, καὶ συναμφοτέρως συναμφοτέρου τὰ αὐτὰ μέρη ἔσται, ὅπερ ὁ εἶξ τοῦ ἐνόξ.

If a number is parts of a number, and another (number) is the same parts of another, (then) the sum (of the leading numbers) will also be the same parts of the sum (of the following numbers) that one (number) is of another.



Ἀριθμὸς γάρ ὁ AB ἀριθμοῦ τοῦ Γ μέρη ἔστω, καὶ ἕτεροξ ὁ ΔE ἑτέρου τοῦ Z τὰ αὐτὰ μέρη, ἅπερ ὁ AB τοῦ Γ · λέγω, ὅτι καὶ συναμφοτέροξ ὁ AB , ΔE συναμφοτέρου τοῦ Γ , Z τὰ αὐτὰ μέρη ἐστίν, ἅπερ ὁ AB τοῦ Γ .

Ἐπεὶ γάρ, ἃ μέρη ἐστὶν ὁ AB τοῦ Γ , τὰ αὐτὰ μέρη καὶ ὁ ΔE τοῦ Z , ὅσα ἄρα ἐστὶν ἐν τῷ AB μέρη τοῦ Γ , τοσαῦτά ἐστι καὶ ἐν τῷ ΔE μέρη τοῦ Z . διηγήσθω ὁ μὲν AB εἰς τὰ τοῦ Γ μέρη τὰ AH , HB , ὁ δὲ ΔE εἰς τὰ τοῦ Z μέρη τὰ $\Delta\Theta$, ΘE . ἔσται δὴ ἴσον τὸ πλῆθος τῶν AH , HB τῷ πλῆθει τῶν $\Delta\Theta$, ΘE . καὶ ἐπεὶ, ὃ μέρος ἐστὶν ὁ AH τοῦ Γ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ $\Delta\Theta$ τοῦ Z , ὃ ἄρα μέρος ἐστὶν ὁ AH τοῦ Γ , τὸ αὐτὸ μέρος ἐστὶ καὶ συναμφοτέροξ ὁ AH , $\Delta\Theta$ συναμφοτέρου τοῦ Γ , Z . διὰ τὰ αὐτὰ δὴ καὶ ὃ μέρος ἐστὶν ὁ HB τοῦ Γ , τὸ αὐτὸ μέρος ἐστὶ καὶ συναμφοτέροξ ὁ HB , ΘE συναμφοτέρου τοῦ Γ , Z . ἃ ἄρα μέρη ἐστὶν ὁ AB τοῦ Γ , τὰ αὐτὰ μέρη ἐστὶ καὶ συναμφοτέροξ ὁ AB , ΔE συναμφοτέρου τοῦ Γ , Z . ὅπερ ἔδει δεῖξαι.



For let a number AB be parts of a number C , and another (number) DE (be) the same parts of another (number) F that AB (is) of C . I say that the sum AB , DE is also the same parts of the sum C , F that AB (is) of C .

For since which(ever) parts AB is of C , DE (is) also the same parts of F , thus as many parts of C as are in AB , so many parts of F are also in DE . Let AB be divided into the parts of C , AG and GB , and DE into the parts of F , DH and HE . So the multitude of (divisions) AG , GB will be equal to the multitude of (divisions) DH , HE . And since which(ever) part AG is of C , DH is also the same part of F , thus which(ever) part AG is of C , the sum AG , DH is also the same part of the sum C , F [Prop. 7.5]. And so, for the same (reasons), which(ever) part GB is of C , the sum GB , HE is also the same part of the sum C , F . Thus, which(ever) parts AB is of C , the sum AB , DE is also the same parts of the sum C , F . (Which is) the very thing it was required to show.

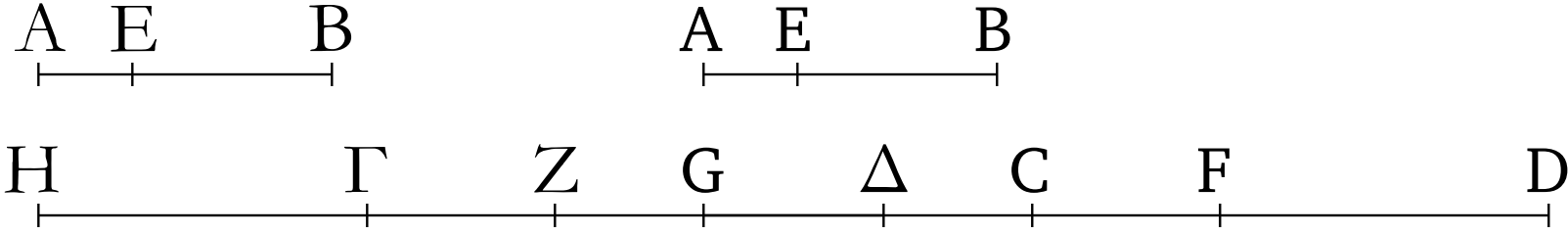
[†] In modern notation, this proposition states that if $a = (m/n)b$ and $c = (m/n)d$ then $(a + c) = (m/n)(b + d)$, where all symbols denote numbers.

7.

Proposition 7[†]

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρος ᾗ, ὅπερ ἀφαιρεθεῖς ἀφαιρεθέντος, καὶ ὁ λοιπὸς τοῦ λοιποῦ τὸ αὐτὸ μέρος ἔσται, ὅπερ ὁ ὅλος τοῦ ὅλου.

If a number is that part of a number that a (part) taken away (is) of a (part) taken away, (then) the remainder will also be the same part of the remainder that the whole (is) of the whole.



Ἀριθμὸς γὰρ ὁ AB ἀριθμοῦ τοῦ $\Gamma\Delta$ μέρος ἔστω, ὅπερ ἀφαιρεθεὶς ὁ AE ἀφαιρεθέντος τοῦ ΓZ λέγω, ὅτι καὶ λοιπὸς ὁ EB λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστίν, ὅπερ ὅλος ὁ AB ὅλον τοῦ $\Gamma\Delta$.

Ὁ γὰρ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἔστω καὶ ὁ EB τοῦ ΓH . καὶ ἐπεὶ, ὁ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ EB τοῦ ΓH , ὁ ἄρα μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ AB τοῦ HZ . ὁ δὲ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ὑπόκειται καὶ ὁ AB τοῦ $\Gamma\Delta$. ὁ ἄρα μέρος ἐστὶ καὶ ὁ AB τοῦ HZ , τὸ αὐτὸ μέρος ἐστὶ καὶ τοῦ $\Gamma\Delta$. ἴσος ἄρα ἐστὶν ὁ HZ τῷ $\Gamma\Delta$. κοινὸς ἀφηρήσθω ὁ ΓZ . λοιπὸς ἄρα ὁ $H\Gamma$ λοιπῷ τῷ $Z\Delta$ ἐστὶν ἴσος. καὶ ἐπεὶ, ὁ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος [ἐστὶ] καὶ ὁ EB τοῦ $H\Gamma$, ἴσος δὲ ὁ $H\Gamma$ τῷ $Z\Delta$, ὁ ἄρα μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ EB τοῦ $Z\Delta$. ἀλλὰ ὁ μέρος ἐστὶν ὁ AE τοῦ ΓZ , τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ AB τοῦ $\Gamma\Delta$. καὶ λοιπὸς ἄρα ὁ EB λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστίν, ὅπερ ὅλος ὁ AB ὅλον τοῦ $\Gamma\Delta$. ὅπερ ἔδει δεῖξαι.

For let a number AB be that part of a number CD that a (part) taken away AE (is) of a part taken away CF . I say that the remainder EB is also the same part of the remainder FD that the whole AB (is) of the whole CD .

For which(ever) part AE is of CF , let EB also be the same part of CG . And since which(ever) part AE is of CF , EB is also the same part of CG , thus which(ever) part AE is of CF , AB is also the same part of GF [Prop. 7.5]. And which(ever) part AE is of CF , AB is also assumed (to be) the same part of CD . Thus, also, which(ever) part AB is of GF , (AB) is also the same part of CD . Thus, GF is equal to CD . Let CF be subtracted from both. Thus, the remainder GC is equal to the remainder FD . And since which(ever) part AE is of CF , EB [is] also the same part of GC , and GC (is) equal to FD , thus which(ever) part AE is of CF , EB is also the same part of FD . But, which(ever) part AE is of CF , AB is also the same part of CD . Thus, the remainder EB is also the same part of the remainder FD that the whole AB (is) of the whole CD . (Which is) the very thing it was required to show.

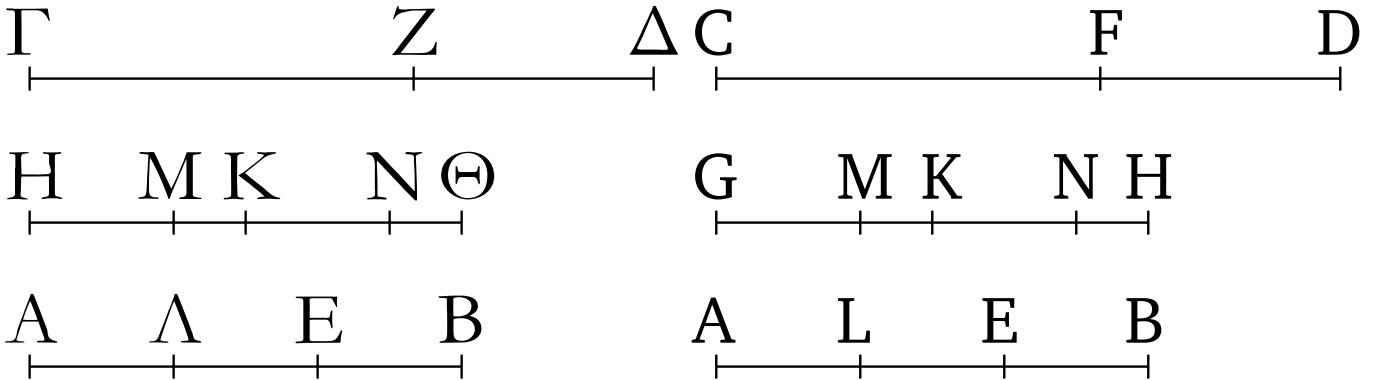
[†] In modern notation, this proposition states that if $a = (1/n)b$ and $c = (1/n)d$ then $(a - c) = (1/n)(b - d)$, where all symbols denote numbers.

8.

Proposition 8[†]

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρη ᾗ, ἅπερ ἀφαιρεθεὶς ἀφαιρεθέντος, καὶ ὁ λοιπὸς τοῦ λοιποῦ τὰ αὐτὰ μέρη ἔσται, ἅπερ ὁ ὅλος τοῦ ὅλου.

If a number is those parts of a number that a (part) taken away (is) of a (part) taken away, (then) the remainder will also be the same parts of the remainder that the whole (is) of the whole.



Ἀριθμὸς γὰρ ὁ AB ἀριθμοῦ τοῦ $\Gamma\Delta$ μέρη ἔστω, ἅπερ ἀφαιρεθεὶς ὁ AE ἀφαιρεθέντος τοῦ ΓZ λέγω, ὅτι καὶ λοιπὸς ὁ EB λοιποῦ τοῦ $Z\Delta$ τὰ αὐτὰ μέρη ἐστίν, ἅπερ ὅλος ὁ AB

For let a number AB be those parts of a number CD that a (part) taken away AE (is) of a (part) taken away CF . I say that the remainder EB is also the same parts of the remainder FD

ὅλον τοῦ $\Gamma\Delta$.

Κείσθω γάρ τῳ AB ἴσος ὁ $H\Theta$, ἃ ἄρα μέρη ἐστὶν ὁ $H\Theta$ τοῦ $\Gamma\Delta$, τὰ αὐτὰ μέρη ἐστὶ καὶ ὁ AE τοῦ ΓZ . διηγήσθω ὁ μὲν $H\Theta$ εἰς τὰ τοῦ $\Gamma\Delta$ μέρη τὰ HK , $K\Theta$, ὁ δὲ AE εἰς τὰ τοῦ ΓZ μέρη τὰ AL , LE . ἔσται δὴ ἴσον τὸ πλῆθος τῶν HK , $K\Theta$ τῳ πλῆθει τῶν AL , LE . καὶ ἐπεὶ, ὁ μέρος ἐστὶν ὁ HK τοῦ $\Gamma\Delta$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ AL τοῦ ΓZ , μείζων δὲ ὁ $\Gamma\Delta$ τοῦ ΓZ , μείζων ἄρα καὶ ὁ HK τοῦ AL . κείσθω τῳ AL ἴσος ὁ HM . ὁ ἄρα μέρος ἐστὶν ὁ HK τοῦ $\Gamma\Delta$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ HM τοῦ ΓZ . καὶ λοιπὸς ἄρα ὁ MK λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστὶν, ὅπερ ὅλος ὁ HK ὅλον τοῦ $\Gamma\Delta$. πάλιν ἐπεὶ, ὁ μέρος ἐστὶν ὁ $K\Theta$ τοῦ $\Gamma\Delta$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ EL τοῦ ΓZ , μείζων δὲ ὁ $\Gamma\Delta$ τοῦ ΓZ , μείζων ἄρα καὶ ὁ ΘK τοῦ EL . κείσθω τῳ EL ἴσος ὁ KN . ὁ ἄρα μέρος ἐστὶν ὁ $K\Theta$ τοῦ $\Gamma\Delta$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ KN τοῦ ΓZ . καὶ λοιπὸς ἄρα ὁ $N\Theta$ λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστὶν, ὅπερ ὅλος ὁ $K\Theta$ ὅλον τοῦ $\Gamma\Delta$. ἐδείχθη δὲ καὶ λοιπὸς ὁ MK λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ὡν, ὅπερ ὅλος ὁ HK ὅλον τοῦ $\Gamma\Delta$. καὶ συναμφοτέροξ ἄρα ὁ MK , $N\Theta$ τοῦ ΔZ τὰ αὐτὰ μέρη ἐστὶν, ἅπερ ὅλος ὁ ΘH ὅλον τοῦ $\Gamma\Delta$. ἴσος δὲ συναμφοτέροξ μὲν ὁ MK , $N\Theta$ τῳ EB , ὁ δὲ ΘH τῳ BA . καὶ λοιπὸς ἄρα ὁ EB λοιποῦ τοῦ $Z\Delta$ τὰ αὐτὰ μέρη ἐστὶν, ἅπερ ὅλος ὁ AB ὅλον τοῦ $\Gamma\Delta$. ὅπερ ἔδει δεῖχαι.

that the whole AB (is) of the whole CD .

For let GH be laid down equal to AB . Thus, which(ever) parts GH is of CD , AE is also the same parts of CF . Let GH be divided into the parts of CD , GK and KH , and AE into the parts of CF , AL and LE . So the multitude of (divisions) GK , KH will be equal to the multitude of (divisions) AL , LE . And since which(ever) part GK is of CD , AL is also the same part of CF , and CD (is) greater than CF , GK (is) thus also greater than AL . Let GM be made equal to AL . Thus, which(ever) part GK is of CD , GM is also the same part of CF . Thus, the remainder MK is also the same part of the remainder FD that the whole GK (is) of the whole CD [Prop. 7.5]. Again, since which(ever) part KH is of CD , EL is also the same part of CF , and CD (is) greater than CF , KH (is) thus also greater than EL . Let KN be made equal to EL . Thus, which(ever) part KH (is) of CD , KN is also the same part of CF . Thus, the remainder NH is also the same part of the remainder FD that the whole KH (is) of the whole CD [Prop. 7.5]. And the remainder MK was also shown to be the same part of the remainder FD that the whole GK (is) of the whole CD . Thus, the sum MK , NH is the same parts of DF that the whole HG (is) of the whole CD . And the sum MK , NH (is) equal to EB , and HG to BA . Thus, the remainder EB is also the same parts of the remainder FD that the whole AB (is) of the whole CD . (Which is) the very thing it was required to show.

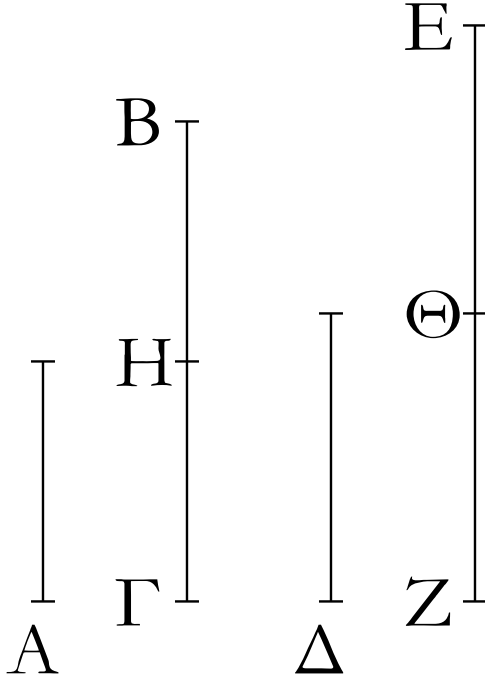
[†] In modern notation, this proposition states that if $a = (m/n)b$ and $c = (m/n)d$ then $(a - c) = (m/n)(b - d)$, where all symbols denote numbers.

9.

Ἐάν ἀριθμὸς ἀριθμοῦ μέρος ᾗ, καὶ ἕτεροξ ἑτέρου τὸ αὐτὸ μέρος ᾗ, καὶ ἐναλλάξ, ὁ μέρος ἐστὶν ᾗ μέρος ὁ πρωτοξ τοῦ τρίτου, τὸ αὐτὸ μέρος ἔσται ᾗ τὰ αὐτὰ μέρη καὶ ὁ δευτεροξ τοῦ τετάρτου.

Proposition 9[†]

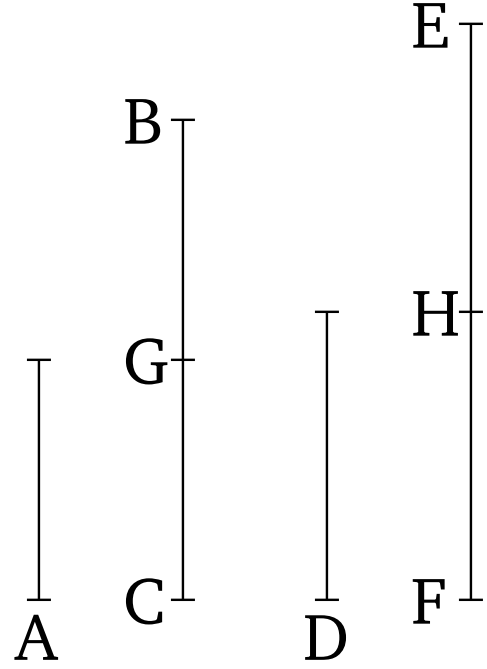
If a number is part of a number, and another (number) is the same part of another, also, alternately, which(ever) part, or parts, the first (number) is of the third, the second (number) will also be the same part, or the same parts, of the fourth.



Ἀριθμὸς γάρ ὁ A ἀριθμοῦ τοῦ $BΓ$ μέρος ἔστω, καὶ ἕτερος ὁ $Δ$ ἕτερον τοῦ EZ τὸ αὐτὸ μέρος, ὅπερ ὁ A τοῦ $BΓ$ λέγω, ὅτι καὶ ἐναλλάχ, ὁ μέρος ἐστὶν ὁ A τοῦ $Δ$ ἢ μέρη, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ $BΓ$ τοῦ EZ ἢ μέρη.

Ἐπεὶ γὰρ ὁ μέρος ἐστὶν ὁ A τοῦ $BΓ$, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ $Δ$ τοῦ EZ , ὅσοι ἄρα εἰσὶν ἐν τῷ $BΓ$ ἀριθμοὶ ἴσοι τῷ A , τοσοῦτοί εἰσι καὶ ἐν τῷ EZ ἴσοι τῷ $Δ$. διηγήσθω ὁ μὲν $BΓ$ εἰς τοὺς τῷ A ἴσους τοὺς BH , $HΓ$, ὁ δὲ EZ εἰς τοὺς τῷ $Δ$ ἴσους τοὺς $EΘ$, $ΘΖ$: ἔσται δὴ ἴσον τὸ πλῆθος τῶν BH , $HΓ$ τῷ πλῆθει τῶν $EΘ$, $ΘΖ$.

Καὶ ἐπεὶ ἴσοι εἰσὶν οἱ BH , $HΓ$ ἀριθμοὶ ἀλλήλοισι, εἰσὶ δὲ καὶ οἱ $EΘ$, $ΘΖ$ ἀριθμοὶ ἴσοι ἀλλήλοισι, καὶ ἐστὶν ἴσον τὸ πλῆθος τῶν BH , $HΓ$ τῷ πλῆθει τῶν $EΘ$, $ΘΖ$, ὁ ἄρα μέρος ἐστὶν ὁ BH τοῦ $EΘ$ ἢ μέρη, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ $HΓ$ τοῦ $ΘΖ$ ἢ τὰ αὐτὰ μέρη· ὥστε καὶ ὁ μέρος ἐστὶν ὁ BH τοῦ $EΘ$ ἢ μέρη, τὸ αὐτὸ μέρος ἐστὶ καὶ συναμφοτέροξ ὁ $BΓ$ συναμφοτέρον τοῦ EZ ἢ τὰ αὐτὰ μέρη. ἴσοξ δὲ ὁ μὲν BH τῷ A , ὁ δὲ $EΘ$ τῷ $Δ$: ὁ ἄρα μέρος ἐστὶν ὁ A τοῦ $Δ$ ἢ μέρη, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ $BΓ$ τοῦ EZ ἢ τὰ αὐτὰ μέρη· ὅπερ ἔδει δεῖχαι.



For let a number A be part of a number BC , and another (number) D (be) the same part of another EF that A (is) of BC . I say that, also, alternately, which(ever) part, or parts, A is of D , BC is also the same part, or parts, of EF .

For since which(ever) part A is of BC , D is also the same part of EF , thus as many numbers as are in BC equal to A , so many are also in EF equal to D . Let BC be divided into BG and GC , equal to A , and EF into EH and HF , equal to D . So the multitude of (divisions) BG , GC will be equal to the multitude of (divisions) EH , HF .

And since the numbers BG and GC are equal to one another, and the numbers EH and HF are also equal to one another, and the multitude of (divisions) BG , GC is equal to the multitude of (divisions) EH , HF , thus which(ever) part, or parts, BG is of EH , GC is also the same part, or the same parts, of HF . And hence, which(ever) part, or parts, BG is of EH , the sum BC is also the same part, or the same parts, of the sum EF [Props. 7.5, 7.6]. And BG (is) equal to A , and EH to D . Thus, which(ever) part, or parts, A is of D , BC is also the same part, or the same parts, of EF . (Which is) the very thing it was required to show.

[†] In modern notation, this proposition states that if $a = (1/n)b$ and $c = (1/n)d$ then if $a = (k/l)c$ then $b = (k/l)d$, where all symbols denote numbers.

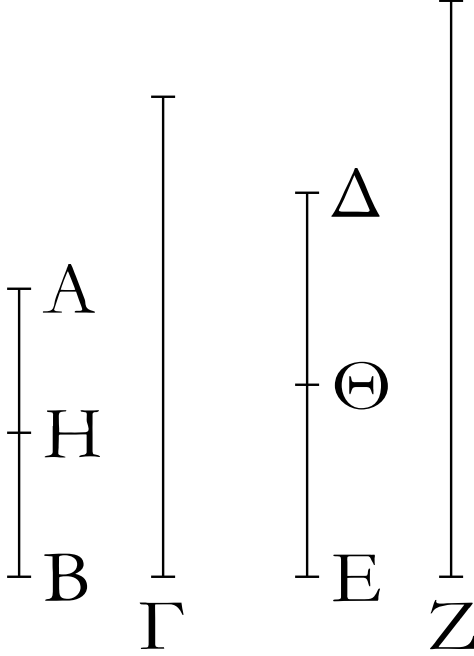
10.

Proposition 10[†]

Ἐὰν ἀριθμὸς ἀριθμοῦ μέρη ᾗ, καὶ ἕτερος ἕτερον τὰ αὐτὰ μέρη ᾗ, καὶ ἐναλλάχ, ἃ μέρη ἐστὶν ὁ πρώτοξ τοῦ τρίτου ἢ μέροςξ, τὰ αὐτὰ μέρη ἔσται καὶ ὁ δεῦτεροξ τοῦ τετάρτου ἢ τὸ αὐτὸ μέροςξ.

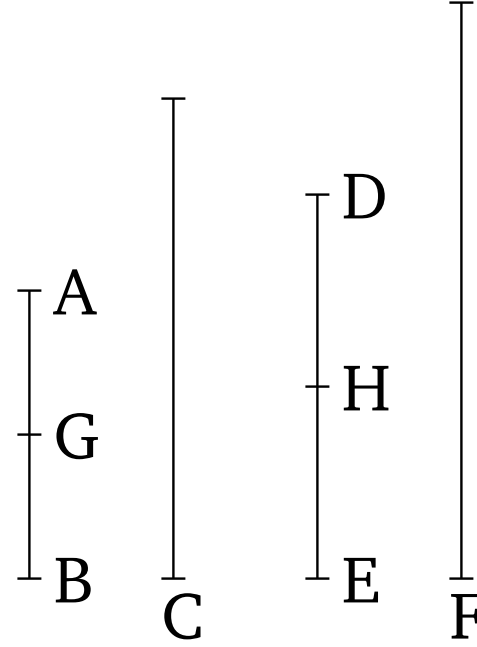
If a number is parts of a number, and another (number) is the same parts of another, also, alternately, which(ever) parts, or part, the first (number) is of the third, the second will also be the same parts, or the same part, of the fourth.

Ἀριθμὸς γάρ ὁ AB ἀριθμοῦ τοῦ Γ μέρη ἔστω, καὶ ἕτεροξ ὁ ΔE ἕτερον τοῦ Z τὰ αὐτὰ μέρη λέγω, ὅτι καὶ ἐναλλάχ, ἃ μέρη ἔστιν ὁ AB τοῦ ΔE ἢ μέρος, τὰ αὐτὰ μέρη ἔστι καὶ ὁ Γ τοῦ Z ἢ τὸ αὐτὸ μέρος.



Ἐπεὶ γάρ, ἃ μέρη ἔστιν ὁ AB τοῦ Γ , τὰ αὐτὰ μέρη ἔστι καὶ ὁ ΔE τοῦ Z , ὅσα ἄρα ἔστιν ἐν τῷ AB μέρη τοῦ Γ , τοσαῦτα καὶ ἐν τῷ ΔE μέρη τοῦ Z . διηγήσθω ὁ μὲν AB εἰς τὰ τοῦ Γ μέρη τὰ AH , HB , ὁ δὲ ΔE εἰς τὰ τοῦ Z μέρη τὰ $\Delta\Theta$, ΘE . ἔσται δὴ ἴσον τὸ πλῆθος τῶν AH , HB τῷ πλῆθει τῶν $\Delta\Theta$, ΘE . καὶ ἐπεὶ, ὁ μέρος ἔστιν ὁ AH τοῦ Γ , τὸ αὐτὸ μέρος ἔστι καὶ ὁ $\Delta\Theta$ τοῦ Z , καὶ ἐναλλάχ, ὁ μέρος ἔστιν ὁ AH τοῦ $\Delta\Theta$ ἢ μέρη, τὸ αὐτὸ μέρος ἔστι καὶ ὁ Γ τοῦ Z ἢ τὰ αὐτὰ μέρη. διὰ τὰ αὐτὰ δὴ καὶ, ὁ μέρος ἔστιν ὁ HB τοῦ ΘE ἢ μέρη, τὸ αὐτὸ μέρος ἔστι καὶ ὁ Γ τοῦ Z ἢ τὰ αὐτὰ μέρη· ὥστε καὶ [ὁ μέρος ἔστιν ὁ AH τοῦ $\Delta\Theta$ ἢ μέρη, τὸ αὐτὸ μέρος ἔστι καὶ ὁ HB τοῦ ΘE ἢ τὰ αὐτὰ μέρη· καὶ ὁ ἄρα μέρος ἔστιν ὁ AH τοῦ $\Delta\Theta$ ἢ μέρη, τὸ αὐτὸ μέρος ἔστι καὶ ὁ AB τοῦ ΔE ἢ τὰ αὐτὰ μέρη· ἀλλ' ὁ μέρος ἔστιν ὁ AH τοῦ $\Delta\Theta$ ἢ μέρη, τὸ αὐτὸ μέρος ἔδειχθη καὶ ὁ Γ τοῦ Z ἢ τὰ αὐτὰ μέρη, καὶ] ἃ [ἄρα] μέρη ἔστιν ὁ AB τοῦ ΔE ἢ μέρος, τὰ αὐτὰ μέρη ἔστι καὶ ὁ Γ τοῦ Z ἢ τὸ αὐτὸ μέρος· ὅπερ ἔδει δεῖχαι.

For let a number AB be parts of a number C , and another (number) DE (be) the same parts of another F . I say that, also, alternately, which(ever) parts, or part, AB is of DE , C is also the same parts, or the same part, of F .



For since which(ever) parts AB is of C , DE is also the same parts of F , thus as many parts of C as are in AB , so many parts of F (are) also in DE . Let AB be divided into the parts of C , AG and GB , and DE into the parts of F , DH and HE . So the multitude of (divisions) AG , GB will be equal to the multitude of (divisions) DH , HE . And since which(ever) part AG is of C , DH is also the same part of F , also, alternately, which(ever) part, or parts, AG is of DH , C is also the same parts, of F [Prop. 7.9]. And so, for the same (reasons), which(ever) part, or parts, GB is of HE , C is also the same part, or the same parts, of F [Prop. 7.9]. And so [which(ever) part, or parts, AG is of DH , GB is also the same part, or the same parts, of HE . And thus, which(ever) part, or parts, AG is of DH , AB is also the same part, or the same parts, of DE [Props. 7.5, 7.6]. But, which(ever) part, or parts, AG is of DH , C was also shown (to be) the same part, or the same parts, of F . And, thus] which(ever) parts, or part, AB is of DE , C is also the same parts, or the same part, of F . (Which is) the very thing it was required to show.

[†] In modern notation, this proposition states that if $a = (m/n)b$ and $c = (m/n)d$ then if $a = (k/l)c$ then $b = (k/l)d$, where all symbols denote numbers.

11.

Proposition 11

Ἐαν ἡ ὅλος πρὸς ὅλον, ὅντωξ ἀφαιρεθεῖς πρὸς ἀφαιρεθέντα, καὶ ὁ λοιπόξ πρὸς τὸν λοιπὸν ἔσται, ὡς ὅλος πρὸς

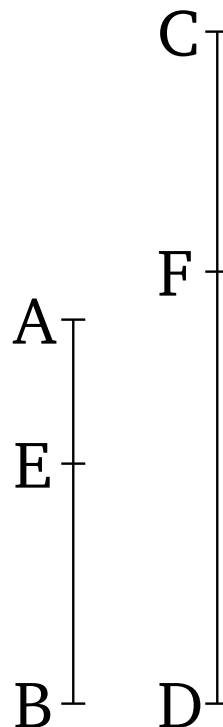
If as the whole (of a number) is to the whole (of another), so a (part) taken away (is) to a (part) taken away, (then) the

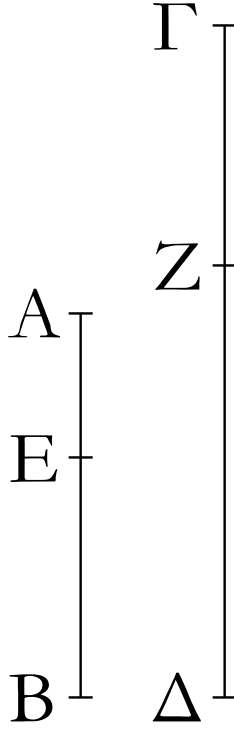
ὅλον.

Ἐστω ὡς ὅλος ὁ AB πρὸς ὅλον τὸν $\Gamma\Delta$, οὕτως ἀφαιρεθεὶς ὁ AE πρὸς ἀφαιρεθέντα τὸν ΓZ · λέγω, ὅτι καὶ λοιπὸς ὁ EB πρὸς λοιπὸν τὸν $Z\Delta$ ἐστίν, ὡς ὅλος ὁ AB πρὸς ὅλον τὸν $\Gamma\Delta$.

remainder will also be to the remainder as the whole (is) to the whole.

Let the whole AB be to the whole CD as the (part) taken away AE (is) to the (part) taken away CF . I say that the remainder EB is to the remainder FD as the whole AB (is) to the whole CD .





Ἐπεὶ ἐστὶν ὡς ὁ AB πρὸς τὸν $\Gamma\Delta$, οὕτως ὁ AE πρὸς τὸν ΓZ , ὃ ἄρα μέρος ἐστὶν ὁ AB τοῦ $\Gamma\Delta$ ἢ μέρος, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ AE τοῦ ΓZ ἢ τὰ αὐτὰ μέρος. καὶ λοιπὸν ἄρα ὁ EB λοιποῦ τοῦ $Z\Delta$ τὸ αὐτὸ μέρος ἐστὶν ἢ μέρος, ἄπερ ὁ AB τοῦ $\Gamma\Delta$. ἐστὶν ἄρα ὡς ὁ EB πρὸς τὸν $Z\Delta$, οὕτως ὁ AB πρὸς τὸν $\Gamma\Delta$. ὅπερ εἶδει δεῖχαι.

(For) since as AB is to CD , so AE (is) to CF , thus which(ever) part, or parts, AB is of CD , AE is also the same part, or the same parts, of CF [Def. 7.20]. Thus, the remainder EB is also the same part, or parts, of the remainder FD that AB (is) of CD [Props. 7.7, 7.8]. Thus, as EB is to FD , so AB (is) to CD [Def. 7.20]. (Which is) the very thing it was required to show.

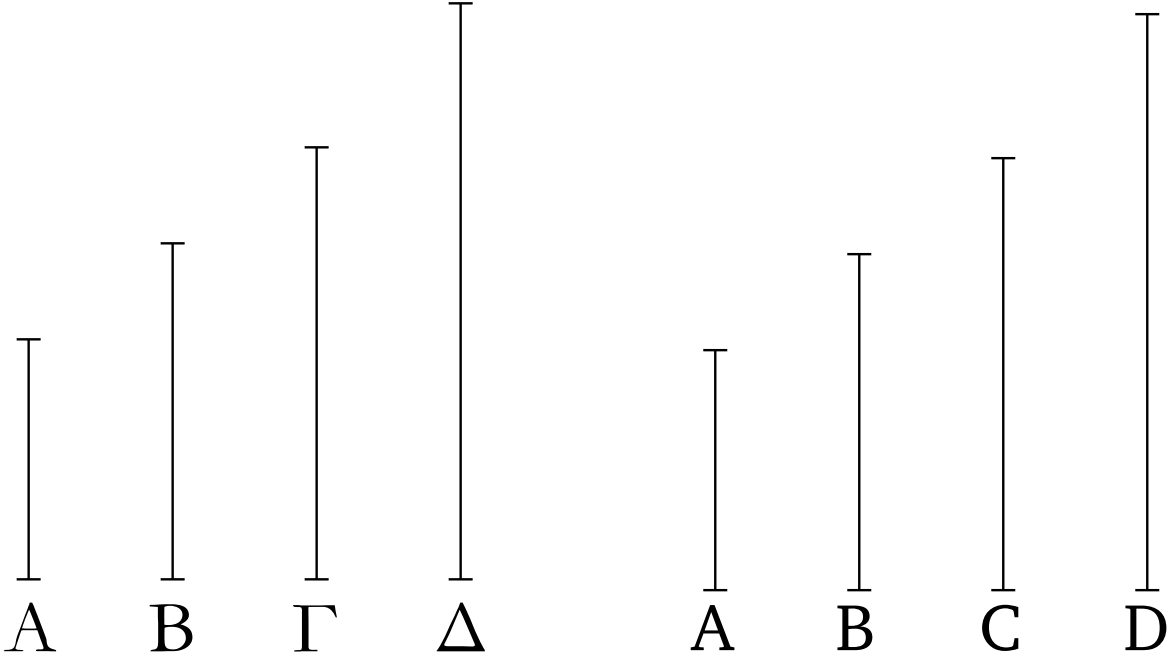
[†] In modern notation, this proposition states that if $a : b :: c : d$ then $a : b :: a - c : b - d$, where all symbols denote numbers.

12.

Ἐὰν ὥσιν ὁποσοιοῦν ἀριθμοὶ ἀνάλογον, ἐστὶ ὡς εἷς τῶν ἡγουμένων πρὸς ἓνα τῶν ἐπομένων, οὕτως ἅπαντες οἱ ἡγούμενοι πρὸς ἅπαντας τοὺς ἐπομένους.

Proposition 12[†]

If any multitude whatsoever of numbers are proportional, (then) as one of the leading (numbers is) to one of the following so (the sum of) all of the leading (numbers) will be to (the sum of) all of the following.



Ἐστωσαν ὅποιοι ἄριθμοι ἀνάλογον οἱ A, B, Γ, Δ , ὥς δA πρὸς τὸν B , οὕτως $\delta \Gamma$ πρὸς τὸν Δ . λέγω, ὅτι ἐστὶν ὥς δA πρὸς τὸν B , οὕτως οἱ A, Γ πρὸς τοὺς B, Δ .

Ἐπεὶ γάρ ἐστιν ὥς δA πρὸς τὸν B , οὕτως $\delta \Gamma$ πρὸς τὸν Δ , ὃ ἄρα μέρος ἐστὶν δA τοῦ B ἢ μέρος, τὸ αὐτὸ μέρος ἐστὶ καὶ $\delta \Gamma$ τοῦ Δ ἢ μέρος. καὶ συναμφότερος ἄρα $\delta A, \Gamma$ συναμφοτέρων τοῦ B, Δ τὸ αὐτὸ μέρος ἐστὶν ἢ τὰ αὐτὰ μέρος, ἅπερ δA τοῦ B . ἔστιν ἄρα ὥς δA πρὸς τὸν B , οὕτως οἱ A, Γ πρὸς τοὺς B, Δ . ὅπερ ἔδει δεῖξαι.

Let any multitude whatsoever of numbers, A, B, C, D , be proportional, (such that) as A (is) to B , so C (is) to D . I say that as A is to B , so A, C (is) to B, D .

For since as A is to B , so C (is) to D , thus which(ever) part, or parts, A is of B , C is also the same part, or parts, of D [Def. 7.20]. Thus, the sum A, C is also the same part, or the same parts, of the sum B, D that A (is) of B [Props. 7.5, 7.6]. Thus, as A is to B , so A, C (is) to B, D [Def. 7.20]. (Which is) the very thing it was required to show.

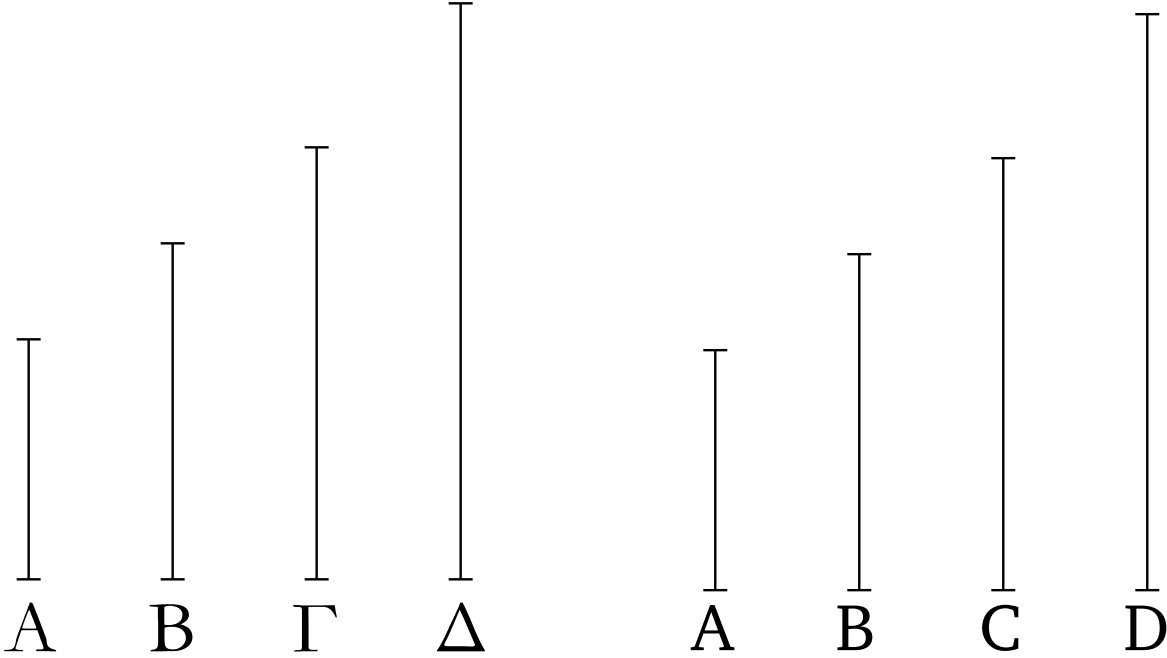
[†] In modern notation, this proposition states that if $a : b :: c : d$ then $a : b :: a + c : b + d$, where all symbols denote numbers.

13.

Ἐὰν τέσσαρες ἄριθμοι ἀνάλογον ᾗσιν, καὶ ἐναλλάχ ἀνάλογον ἔσονται.

Proposition 13[†]

If four numbers are proportional, (then) they will also be proportional alternately.



Ἐστωσαν τέσσαρες ἀριθμοὶ ἀνάλογον οἱ A, B, Γ, Δ , ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Γ πρὸς τὸν Δ . λέγω, ὅτι καὶ ἐναλλάξ ἀνάλογον ἔσονται, ὥς ὁ A πρὸς τὸν Γ , οὕτως ὁ B πρὸς τὸν Δ .

Ἐπεὶ γάρ ἐστιν ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Γ πρὸς τὸν Δ , ὃ ἄρα μέρος ἐστὶν ὁ A τοῦ B ἢ μέρος, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ Γ τοῦ Δ ἢ τὰ αὐτὰ μέρη. ἐναλλάξ ἄρα, ὃ μέρος ἐστὶν ὁ A τοῦ Γ ἢ μέρος, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ B τοῦ Δ ἢ τὰ αὐτὰ μέρη. ἔστιν ἄρα ὥς ὁ A πρὸς τὸν Γ , οὕτως ὁ B πρὸς τὸν Δ . ὅπερ εἶδει δεῖχαι.

Let the four numbers A, B, C , and D be proportional, (such that) as A (is) to B , so C (is) to D . I say that they will also be proportional alternately, (such that) as A (is) to C , so B (is) to D .

For since as A is to B , so C (is) to D , thus which(ever) part, or parts, A is of B , C is also the same part, or the same parts, of D [Def. 7.20]. Thus, alternately, which(ever) part, or parts, A is of C , B is also the same part, or the same parts, of D [Props. 7.9, 7.10]. Thus, as A is to C , so B (is) to D [Def. 7.20]. (Which is) the very thing it was required to show.

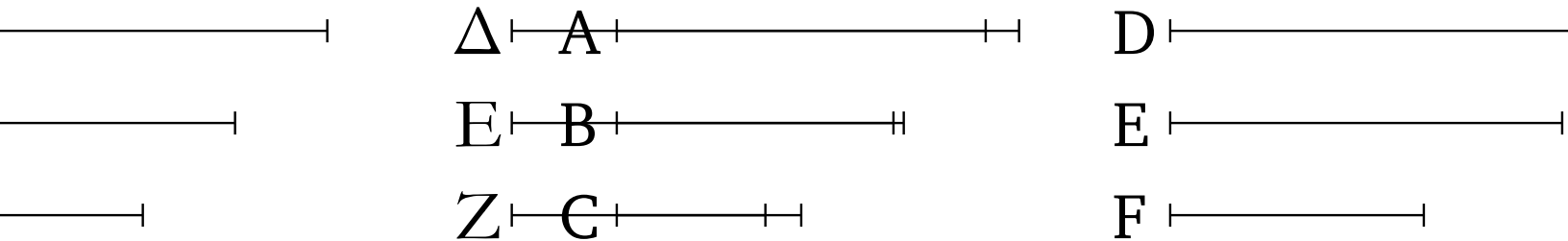
[†] In modern notation, this proposition states that if $a : b :: c : d$ then $a : c :: b : d$, where all symbols denote numbers.

14.

Proposition 14[†]

Ἐάν ὧσιν ὅποσοιῶν ἀριθμοὶ καὶ ἄλλοι αὐτοῖς ἴσοι τὸ πλήθος σύνδνο λαμβανόμενοι καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ δι' ἴσου ἐν τῷ αὐτῷ λόγῳ ἔσονται.

If there are any multitude of numbers whatsoever, and (some) other (numbers) of equal multitude to them, (which are) also in the same ratio taken two by two, (then) they will also be in the same ratio via equality.



Ἐστωσαν ὅποσοιῶν ἀριθμοὶ οἱ A, B, Γ καὶ ἄλλοι αὐτοῖς ἴσοι τὸ πλήθος σύνδνο λαμβανόμενοι ἐν τῷ αὐτῷ λόγῳ οἱ Δ, E, Z , ὥς μὲν ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E , ὥς

Let there be any multitude of numbers whatsoever, A, B, C , and (some) other (numbers), D, E, F , of equal multitude to them, (which are) in the same ratio taken two by two, (such

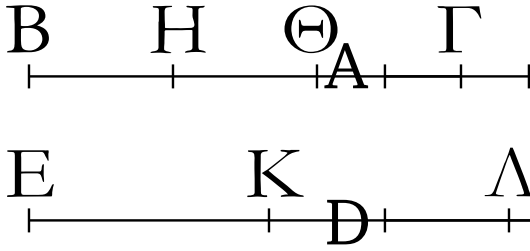
δὲ ὁ B πρὸς τὸν Γ , οὕτως ὁ E πρὸς τὸν Z · λέγω, ὅτι καὶ δι' ἴσον ἔστιν ὡς ὁ A πρὸς τὸν Γ , οὕτως ὁ Δ πρὸς τὸν Z .

Ἐπεὶ γάρ ἐστιν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E , ἐναλλάχ' ἄρα ἔστιν ὡς ὁ A πρὸς τὸν Δ , οὕτως ὁ B πρὸς τὸν E . πάλιν, ἐπεὶ ἔστιν ὡς ὁ B πρὸς τὸν Γ , οὕτως ὁ E πρὸς τὸν Z , ἐναλλάχ' ἄρα ἔστιν ὡς ὁ B πρὸς τὸν E , οὕτως ὁ Γ πρὸς τὸν Z . ὡς δὲ ὁ B πρὸς τὸν E , οὕτως ὁ A πρὸς τὸν Δ · καὶ ὡς ἄρα ὁ A πρὸς τὸν Δ , οὕτως ὁ Γ πρὸς τὸν Z · ἐναλλάχ' ἄρα ἔστιν ὡς ὁ A πρὸς τὸν Γ , οὕτως ὁ Δ πρὸς τὸν Z · ὅπερ ἔδει δεῖχαι.

† In modern notation, this proposition states that if $a : b :: d : e$ and $b : c :: e : f$ then $a : c :: d : f$, where all symbols denote numbers.

15.

Ἐὰν μονὰς ἀριθμὸν τινα μετρήῃ, ἰσακὶς δὲ ἕτεροξ ἀριθμὸς ἄλλον τινὰ ἀριθμὸν μετρήῃ, καὶ ἐναλλάχ' ἰσακὶς ἡ μονὰς τὸν τρίτον ἀριθμὸν μετρήσῃ καὶ ὁ δεύτεροξ τὸν τέταρτον.

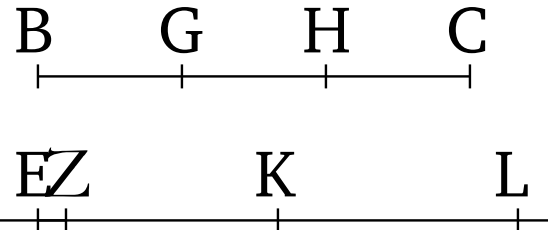


Μονὰς γάρ ἡ A ἀριθμὸν τινα τὸν $B\Gamma$ μετρεῖτω, ἰσακὶς δὲ ἕτεροξ ἀριθμὸς ὁ Δ ἄλλον τινὰ ἀριθμὸν τὸν $E\Lambda$ μετρεῖτω· λέγω, ὅτι καὶ ἐναλλάχ' ἰσακὶς ἡ A μονὰς τὸν Δ ἀριθμὸν μετρεῖ καὶ ὁ $B\Gamma$ τὸν $E\Lambda$.

Ἐπεὶ γάρ ἰσακὶς ἡ A μονὰς τὸν $B\Gamma$ ἀριθμὸν μετρεῖ καὶ ὁ Δ τὸν $E\Lambda$, ὅσαι ἄρα εἰσὶν ἐν τῷ $B\Gamma$ μονάδες, τοσοῦτοι εἰσὶ καὶ ἐν τῷ $E\Lambda$ ἀριθμοὶ ἴσοι τῷ Δ . διηγήσθω ὁ μὲν $B\Gamma$ εἰς τὰς ἐν ἑαυτῷ μονάδας τὰς BH , $H\Theta$, $\Theta\Gamma$, ὁ δὲ $E\Lambda$ εἰς τὰς τῷ Δ ἴσους τὰς EK , $K\Lambda$, $\Lambda\Lambda$. ἔσται δὲ ἴσον τὸ πλῆθος τῶν BH , $H\Theta$, $\Theta\Gamma$ τῷ πλῆθει τῶν EK , $K\Lambda$, $\Lambda\Lambda$. καὶ ἐπεὶ ἴσαι εἰσὶν αἱ BH , $H\Theta$, $\Theta\Gamma$ μονάδες ἀλλήλαις, εἰσὶ δὲ καὶ οἱ EK , $K\Lambda$, $\Lambda\Lambda$ ἀριθμοὶ ἴσοι ἀλλήλοις, καὶ ἔστιν ἴσον τὸ πλῆθος τῶν BH , $H\Theta$, $\Theta\Gamma$ μονάδων τῷ πλῆθει τῶν EK , $K\Lambda$, $\Lambda\Lambda$ ἀριθμῶν, ἔσται ἄρα ὡς ἡ BH μονὰς πρὸς τὸν EK ἀριθμὸν, οὕτως ἡ $H\Theta$ μονὰς πρὸς τὸν $K\Lambda$ ἀριθμὸν καὶ ἡ $\Theta\Gamma$ μονὰς πρὸς τὸν $\Lambda\Lambda$ ἀριθμὸν. ἔσται ἄρα καὶ ὡς εἷς τῶν ἡγουμένων πρὸς ἕνα τῶν ἐπομένων, οὕτως ἅπαντες οἱ ἡγούμενοι πρὸς ἅπαντας τὸν ἐπομένους· ἔστιν ἄρα ὡς ἡ BH μονὰς πρὸς τὸν EK ἀριθμὸν, οὕτως ὁ $B\Gamma$ πρὸς τὸν $E\Lambda$. ἴση δὲ ἡ BH μονὰς τῇ A μονάδι, ὁ δὲ EK ἀριθμὸς τῷ Δ ἀριθμῷ. ἔστιν ἄρα ὡς ἡ A μονὰς πρὸς τὸν Δ ἀριθμὸν, οὕτως ὁ $B\Gamma$ πρὸς τὸν $E\Lambda$. ἰσακὶς ἄρα ἡ A μονὰς τὸν Δ ἀριθμὸν μετρεῖ καὶ ὁ $B\Gamma$ τὸν $E\Lambda$ · ὅπερ ἔδει δεῖχαι.

Proposition 15

If a unit measures some number, and another number measures some other number as many times, (then), also, alternately, the unit will measure the third number as many times as the second (number measures) the fourth.



For let a unit A measure some number BC , and let another number D measure some other number EF as many times. I say that, also, alternately, the unit A also measures the number D as many times as BC (measures) EF .

For since the unit A measures the number BC as many times as D (measures) EF , thus as many units as are in BC , so many numbers are also in EF equal to D . Let BC be divided into its constituent units, BG , GH , and HC , and EF into the (divisions) EK , KL , and LF , equal to D . So the multitude of (units) BG , GH , HC will be equal to the multitude of (divisions) EK , KL , LF . And since the units BG , GH , and HC are equal to one another, and the numbers EK , KL , and LF are also equal to one another, and the multitude of the (units) BG , GH , HC is equal to the multitude of the numbers EK , KL , LF , thus as the unit BG (is) to the number EK , so the unit GH will be to the number KL , and the unit HC to the number LF . And thus, as one of the leading (numbers is) to one of the following, so (the sum of) all of the leading will be to (the sum of) all of the following [Prop. 7.12]. Thus, as the unit BG (is) to the number EK , so BC (is) to EF . And the unit BG (is) equal to the unit A , and the number EK to the number D . Thus, as the unit A is to the number D , so BC (is) to EF . Thus, the unit A measures the number D as many times as BC (measures) EF [Def. 7.20]. (Which is) the very thing it was required to show.

† This proposition is a special case of Prop. 7.9.

16.

Εάν δύο ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλων ποιῶσί τιναξ, οἱ γενόμενοι ἐκ αὐτῶν ἴσοι ἀλλήλοις ἔσονται.

A

B

Γ

Δ

E

Ἐστωσαν δύο ἀριθμοὶ οἱ A, B , καὶ ὁ μὲν A τὸν B πολλαπλασιάσας τὸν Γ ποιείτω, ὁ δὲ B τὸν A πολλαπλασιάσας τὸν Δ ποιείτω· λέγω, ὅτι ἴσος ἐστὶν ὁ Γ τῷ Δ .

Ἐπεὶ γὰρ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ B ἄρα τὸν Γ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας· μετρεῖ δὲ καὶ ἡ E μονὰξ τὸν A ἀριθμὸν κατὰ τὰς ἐν αὐτῷ μονάδας· ἰσάκις ἄρα ἡ E μονὰξ τὸν A ἀριθμὸν μετρεῖ καὶ ὁ B τὸν Γ . ἐναλλάξ ἄρα ἰσάκις ἡ E μονὰξ τὸν B ἀριθμὸν μετρεῖ καὶ ὁ A τὸν Γ . πάλιν, ἐπεὶ ὁ B τὸν A πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ A ἄρα τὸν Δ μετρεῖ κατὰ τὰς ἐν τῷ B μονάδας· μετρεῖ δὲ καὶ ἡ E μονὰξ τὸν B κατὰ τὰς ἐν αὐτῷ μονάδας· ἰσάκις ἄρα ἡ E μονὰξ τὸν B ἀριθμὸν μετρεῖ καὶ ὁ A τὸν Δ . ἰσάκις δὲ ἡ E μονὰξ τὸν B ἀριθμὸν ἐμέτρει καὶ ὁ A τὸν Γ · ἰσάκις ἄρα ὁ A ἐκάτερον τῶν Γ, Δ μετρεῖ. ἴσος ἄρα ἐστὶν ὁ Γ τῷ Δ · ὅπερ ἔδει δεῖχαι.

Proposition 16†

If two numbers multiplying one another make some (numbers, then) the (numbers) generated from them will be equal to one another.

A

B

C

D

E

Let A and B be two numbers. And let A make C (by) multiplying B , and let B make D (by) multiplying A . I say that C is equal to D .

For since A has made C (by) multiplying B , B thus measures C according to the units in A [Def. 7.15]. And the unit E also measures the number A according to the units in it. Thus, the unit E measures the number A as many times as B (measures) C . Thus, alternately, the unit E measures the number B as many times as A (measures) C [Prop. 7.15]. Again, since B has made D (by) multiplying A , A thus measures D according to the units in B [Def. 7.15]. And the unit E also measures B according to the units in it. Thus, the unit E measures the number B as many times as A (measures) D . And the unit E was measuring the number B as many times as A (measures) C . Thus, A measures each of C and D an equal number of times. Thus, C is equal to D . (Which is) the very thing it was required to show.

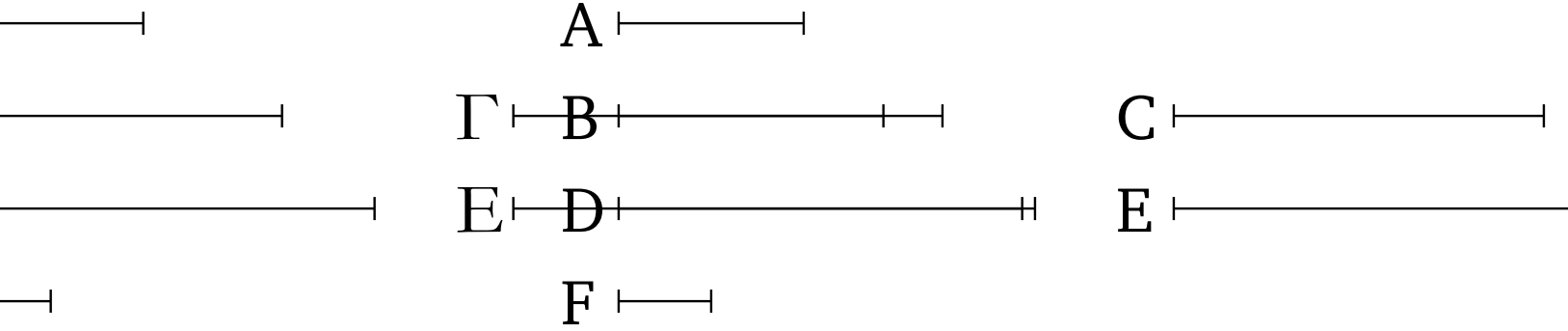
† In modern notation, this proposition states that $ab = ba$, where all symbols denote numbers.

17.

Εάν ἀριθμὸς δύο ἀριθμοὺς πολλαπλασιάσας ποιῇ τιναξ, οἱ γενόμενοι ἐκ αὐτῶν τὸν αὐτὸν ἔχουσι λόγον τοῖς πολλαπλασιασθεῖσιν.

Proposition 17†

If a number multiplying two numbers makes some (numbers, then) the (numbers) generated from them will have the same ratio as the multiplied (numbers).



Ἀριθμὸς γάρ ὁ A δύο ἀριθμοὺς τούς B, Γ πολλαπλασιάσας τούς Δ, E ποιεῖτω· λέγω, ὅτι ἐστὶν ὥς ὁ B πρὸς τὸν Γ , οὕτως ὁ Δ πρὸς τὸν E .

Ἐπεὶ γάρ ὁ A τὸν B πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ B ἄρα τὸν Δ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας· μετρεῖ δὲ καὶ ἡ Z μονὰς τὸν A ἀριθμὸν κατὰ τὰς ἐν αὐτῇ μονάδας· ἰσάκις ἄρα ἡ Z μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ B τὸν Δ . ἔστιν ἄρα ὥς ἡ Z μονὰς πρὸς τὸν A ἀριθμὸν, οὕτως ὁ B πρὸς τὸν Δ . διὰ τὰ αὐτὰ δὴ καὶ ὥς ἡ Z μονὰς πρὸς τὸν A ἀριθμὸν, οὕτως ὁ Γ πρὸς τὸν E · καὶ ὥς ἄρα ὁ B πρὸς τὸν Δ , οὕτως ὁ Γ πρὸς τὸν E . ἐναλλάξ ἄρα ἐστὶν ὥς ὁ B πρὸς τὸν Γ , οὕτως ὁ Δ πρὸς τὸν E · ὅπερ ἔδει δεῖχαι.

For let the number A make (the numbers) D and E (by) multiplying the two numbers B and C (respectively). I say that as B is to C , so D (is) to E .

For since A has made D (by) multiplying B , B thus measures D according to the units in A [Def. 7.15]. And the unit F also measures the number A according to the units in it. Thus, the unit F measures the number A as many times as B (measures) D . Thus, as the unit F is to the number A , so B (is) to D [Def. 7.20]. And so, for the same (reasons), as the unit F (is) to the number A , so C (is) to E . And thus, as B (is) to D , so C (is) to E . Thus, alternately, as B is to C , so D (is) to E [Prop. 7.13]. (Which is) the very thing it was required to show.

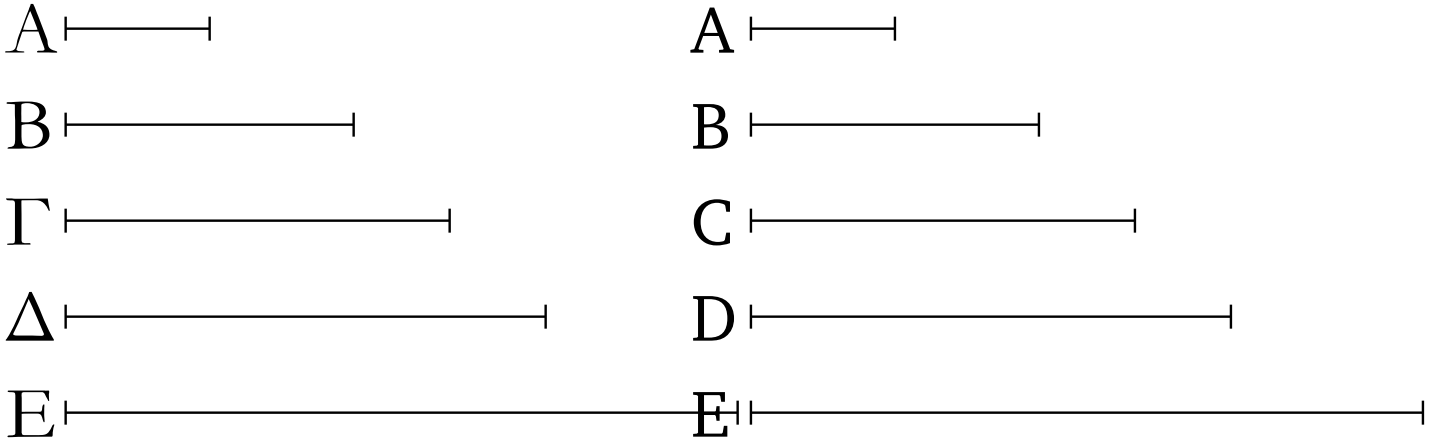
† In modern notation, this proposition states that if $d = ab$ and $e = ac$ then $d : e :: b : c$, where all symbols denote numbers.

18.

Proposition 18[†]

Ἐάν δύο ἀριθμοὶ ἀριθμὸν τινα πολλαπλασιάσαντες ποιῶσιν τιναξ, οἱ γενόμενοι ἐξ αὐτῶν τὸν αὐτὸν ἔχουσι λόγον τοῖς πολλαπλασιάσανσιν.

If two numbers multiplying some number make some (other numbers, then) the (numbers) generated from them will have the same ratio as the multiplying (numbers).



Δύο γάρ ἀριθμοὶ οἱ A, B ἀριθμὸν τινα τὸν Γ πολλαπλασιάσαντες τούς Δ, E ποιεῖτωσαν· λέγω, ὅτι ἐστὶν ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E .

Ἐπεὶ γάρ ὁ A τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν, καὶ ὁ Γ ἄρα τὸν A πολλαπλασιάσας τὸν Δ πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Γ τὸν B πολλαπλασιάσας τὸν E πεποίηκεν.

For let the two numbers A and B make (the numbers) D and E (respectively, by) multiplying some number C . I say that as A is to B , so D (is) to E .

For since A has made D (by) multiplying C , C has thus also made D (by) multiplying A [Prop. 7.16]. So, for the same (reasons), C has also made E (by) multiplying B . So the number C

ἀριθμὸς δὴ ὁ Γ δύο ἀριθμὸν τούξ A, B πολλαπλασιάσας τούξ Δ, E πεποίηκεν. ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E . ὅπερ ἔδει δεῖχαι.

has made D and E (by) multiplying the two numbers A and B (respectively). Thus, as A is to B , so D (is) to E [Prop. 7.17]. (Which is) the very thing it was required to show.

[†] In modern notation, this propositions states that if $ac = d$ and $bc = e$ then $a : b :: d : e$, where all symbols denote numbers.

19.

Proposition 19[†]

Ἐὰν τέσσαρες ἀριθμοὶ ἀνάλογον ᾶσιν, ὁ ἐκ πρώτον καὶ τετάρτον γενομένου ἀριθμὸς ἴσος ἔσται τῷ ἐκ δευτέρου καὶ τρίτου γενομένου ἀριθμῷ· καὶ ἔαν ὁ ἐκ πρώτον καὶ τετάρτον γενομένου ἀριθμὸς ἴσος ᾖ τῷ ἐκ δευτέρου καὶ τρίτου, οἱ τέσσαρες ἀριθμοὶ ἀνάλογον ἔσονται.

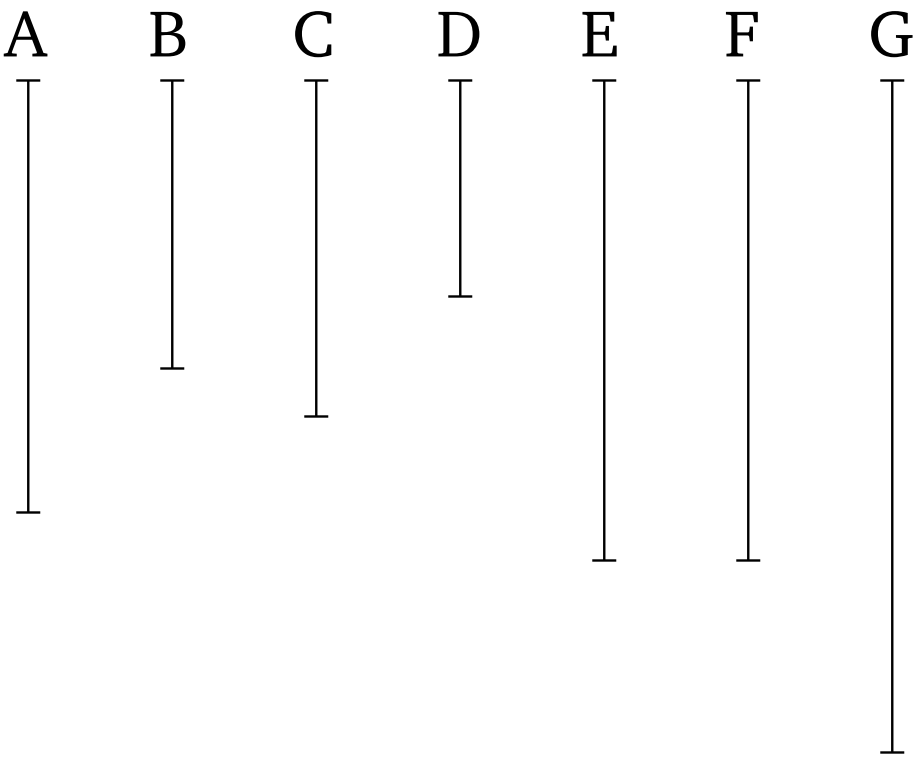
Ἐστωσαν τέσσαρες ἀριθμοὶ ἀνάλογον οἱ A, B, Γ, Δ , ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Γ πρὸς τὸν Δ , καὶ ὁ μὲν A τὸν Δ πολλαπλασιάσας τὸν E ποιέτω, ὁ δὲ B τὸν Γ πολλαπλασιάσας τὸν Z ποιέτω· λέγω, ὅτι ἴσος ἔστιν ὁ E τῷ Z .

Ὁ γὰρ A τὸν Γ πολλαπλασιάσας τὸν H ποιέτω. ἐπεὶ οὖν ὁ A τὸν Γ πολλαπλασιάσας τὸν H πεποίηκεν, τὸν δὲ Δ πολλαπλασιάσας τὸν E πεποίηκεν, ἀριθμὸς δὴ ὁ A δύο ἀριθμὸν τούξ Γ, Δ πολλαπλασιάσας τούξ H, E πεποίηκεν. ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ H πρὸς τὸν E . ἀλλ' ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν B . καὶ ὡς ἄρα ὁ A πρὸς τὸν B , οὕτως ὁ H πρὸς τὸν E . πάλιν, ἐπεὶ ὁ A τὸν Γ πολλαπλασιάσας τὸν H πεποίηκεν, ἀλλὰ μὴν καὶ ὁ B τὸν Γ πολλαπλασιάσας τὸν Z πεποίηκεν, δύο δὴ ἀριθμοὶ οἱ A, B ἀριθμὸν τινὰ τὸν Γ πολλαπλασιάσαντες τούξ H, Z πεποίηκασιν. ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ H πρὸς τὸν Z . ἀλλὰ μὴν καὶ ὡς ὁ A πρὸς τὸν B , οὕτως ὁ H πρὸς τὸν E . καὶ ὡς ἄρα ὁ H πρὸς τὸν E , οὕτως ὁ H πρὸς τὸν Z . ὁ H ἄρα πρὸς ἑκάτερον τῶν E, Z τὸν αὐτὸν ἔθει λόγον.

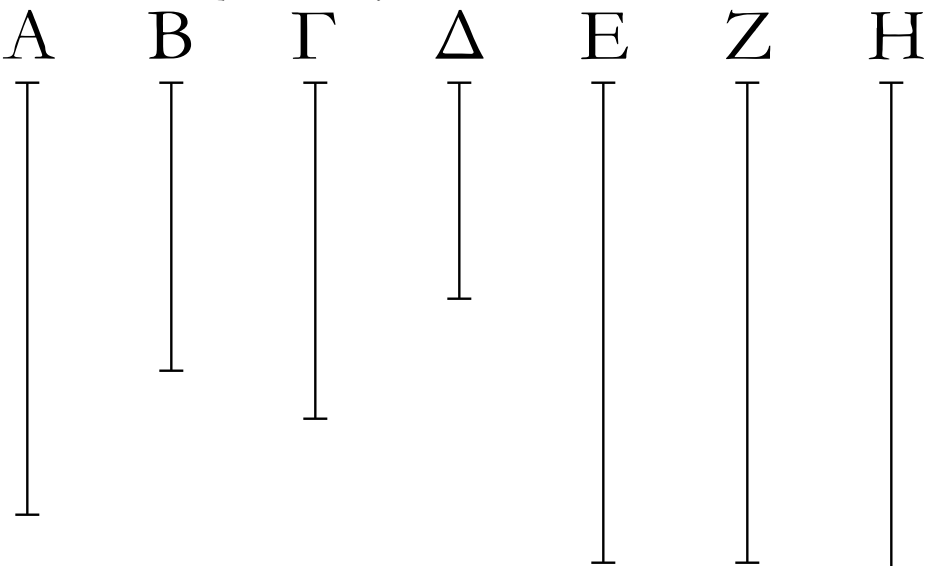
If four number are proportional, (then) the number created from (multiplying) the first and fourth will be equal to the number created from (multiplying) the second and third. And if the number created from (multiplying) the first and fourth is equal to the (number created) from (multiplying) the second and third, (then) the four numbers will be proportional.

Let A, B, C , and D be four proportional numbers, (such that) as A (is) to B , so C (is) to D . And let A make E (by) multiplying D , and let B make F (by) multiplying C . I say that E is equal to F .

For let A make G (by) multiplying C . Therefore, since A has made G (by) multiplying C , and has made E (by) multiplying D , the number A has made G and E by multiplying the two numbers C and D (respectively). Thus, as C is to D , so G (is) to E [Prop. 7.17]. But, as C (is) to D , so A (is) to B . Thus, also, as A (is) to B , so G (is) to E . Again, since A has made G (by) multiplying C , but, in fact, B has also made F (by) multiplying C , the two numbers A and B have made G and F (respectively, by) multiplying some number C . Thus, as A is to B , so G (is) to F [Prop. 7.18]. But, also, as A (is) to B , so G (is) to E . And thus, as G (is) to E , so G (is) to F . Thus, G has the same ratio to each of E and F . Thus, E is equal to F [Prop. 5.9].



ἴσοξ ἄρα ἐστὶν ὁ E τῷ Z.



Ἐστω δὴ πάλιν ἴσοξ ὁ E τῷ Z· λέγω, ὅτι ἐστὶν ὡς ὁ A πρὸς τὸν B, οὕτως ὁ Γ πρὸς τὸν Δ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἴσοξ ἐστὶν ὁ E

So, again, let E be equal to F . I say that as A is to B , so C (is) to D .

For, with the same construction, since E is equal to F , thus as G is to E , so G (is) to F [Prop. 5.7]. But, as G (is) to E ,

τῷ Z , ἔστιν ἄρα ὥς ὁ H πρὸς τὸν E , οὕτως ὁ H πρὸς τὸν Z . ἀλλ' ὥς μὲν ὁ H πρὸς τὸν E , οὕτως ὁ Γ πρὸς τὸν Δ , ὥς δὲ ὁ H πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν B . καὶ ὥς ἄρα ὁ A πρὸς τὸν B , οὕτως ὁ Γ πρὸς τὸν Δ . ὅπερ ἔδει δεῖχαι.

so C (is) to D [Prop. 7.17]. And as G (is) to F , so A (is) to B [Prop. 7.18]. And, thus, as A (is) to B , so C (is) to D . (Which is) the very thing it was required to show.

[†] In modern notation, this proposition reads that if $a : b :: c : d$ then $ad = bc$, and vice versa, where all symbols denote numbers.

20.

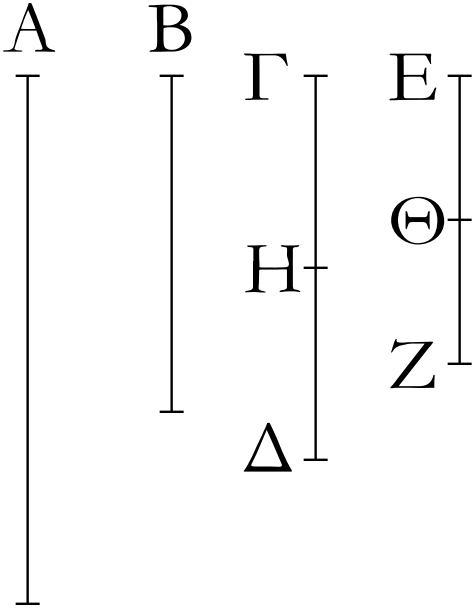
Proposition 20

Οἱ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς μετροῦσι τοὺς τὸν αὐτὸν λόγον ἐθονταξ ἰσάκεις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα.

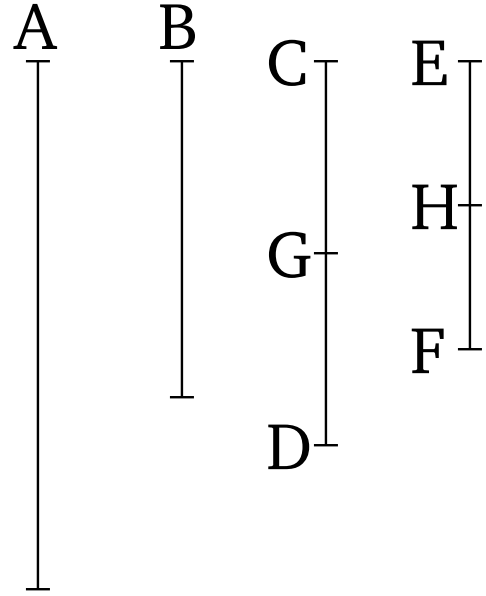
Ἐστωσαν γὰρ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B οἱ $\Gamma\Delta, EZ$ · λέγω, ὅτι ἰσάκεις ὁ $\Gamma\Delta$ τὸν A μετρεῖ καὶ ὁ EZ τὸν B .

The least numbers of those (numbers) having the same ratio measure those (numbers) having the same ratio as them an equal number of times, the greater (measuring) the greater, and the lesser the lesser.

For let CD and EF be the least numbers having the same ratio as A and B (respectively). I say that CD measures A the same number of times as EF (measures) B .



Ὁ $\Gamma\Delta$ γὰρ τοῦ A οὐκ ἔστι μέρος. εἰ γὰρ δυνατόν, ἔστω· καὶ ὁ EZ ἄρα τοῦ B τὰ αὐτὰ μέρος ἔστιν, ἅπερ ὁ $\Gamma\Delta$ τοῦ A . ὅσα ἄρα ἔστιν ἐν τῷ $\Gamma\Delta$ μέρη τοῦ A , τοσαῦτά ἐστι καὶ ἐν τῷ EZ μέρη τοῦ B . διηγήσθω ὁ μὲν $\Gamma\Delta$ εἰς τὰ τοῦ A μέρη τὰ $\Gamma\Theta, \Theta\Delta$, ὁ δὲ EZ εἰς τὰ τοῦ B μέρη τὰ $E\Theta, \Theta Z$. ἔσται δὴ ἴσον τὸ πλῆθος τῶν $\Gamma\Theta, \Theta\Delta$ τῷ πλῆθει τῶν $E\Theta, \Theta Z$. καὶ ἐπεὶ ἴσοι εἰσὶν οἱ $\Gamma\Theta, \Theta\Delta$ ἀριθμοὶ ἀλλήλοισι, εἰσὶ δὲ καὶ οἱ $E\Theta, \Theta Z$ ἀριθμοὶ ἴσοι ἀλλήλοισι, καὶ ἔστιν ἴσον τὸ πλῆθος τῶν $\Gamma\Theta, \Theta\Delta$ τῷ πλῆθει τῶν $E\Theta, \Theta Z$, ἔστιν ἄρα ὥς ὁ $\Gamma\Theta$ πρὸς τὸν $E\Theta$, οὕτως ὁ $\Theta\Delta$ πρὸς τὸν ΘZ . ἔσται ἄρα καὶ ὥς εἷς τῶν ἡγουμένων πρὸς ἓνα τῶν ἐπομένων, οὕτως ἅπαντες οἱ ἡγούμενοι πρὸς ἅπαντας τοὺς ἐπομένους. ἔστιν ἄρα ὥς ὁ $\Gamma\Theta$ πρὸς τὸν $E\Theta$, οὕτως ὁ $\Gamma\Delta$ πρὸς τὸν EZ . οἱ $\Gamma\Theta, E\Theta$ ἄρα τοῖς $\Gamma\Delta, EZ$ ἐν τῷ αὐτῷ λόγῳ εἰσὶν ἐλάσσονες ὄντες αὐτῶν ὅπερ ἔστιν ἀδύνατον· ὑπόκειται γὰρ οἱ $\Gamma\Delta, EZ$ ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς. οὐκ ἄρα μέρος ἔστιν ὁ



For CD is not parts of A . For, if possible, let it be (parts of A). Thus, EF is also the same parts of B that CD (is) of A [Def. 7.20, Prop. 7.13]. Thus, as many parts of A as are in CD , so many parts of B are also in EF . Let CD be divided into the parts of A , CG and GD , and EF into the parts of B , EH and HF . So the multitude of (divisions) CG, GD will be equal to the multitude of (divisions) EH, HF . And since the numbers CG and GD are equal to one another, and the numbers EH and HF are also equal to one another, and the multitude of (divisions) CG, GD is equal to the multitude of (divisions) EH, HF , thus as CG is to EH , so GD (is) to HF . Thus, as one of the leading (numbers is) to one of the following, so will (the sum of) all of the leading (numbers) be to (the sum of) all of the following [Prop. 7.12]. Thus, as CG is to EH , so CD (is) to EF . Thus, CG and EH are in the same ratio as CD and EF , being less than them. The very thing is impossible. For

ΓΔ τοῦ A μέρος ἄρα. καὶ ὁ ΕΖ τοῦ B τὸ αὐτὸ μέρος ἐστίν, ὅπερ ὁ ΓΔ τοῦ A ἰσάκεις ἄρα ὁ ΓΔ τὸν A μετρεῖ καὶ ὁ ΕΖ τὸν B · ὅπερ ἔδει δεῖχαι.

CD and EF were assumed (to be) the least of those (numbers) having the same ratio as them. Thus, CD is not parts of A . Thus, (it is) a part (of A) [Prop. 7.4]. And EF is the same part of B that CD (is) of A [Def. 7.20, Prop 7.13]. Thus, CD measures A the same number of times that EF (measures) B . (Which is) the very thing it was required to show.

21.

Οἱ πρῶτοι πρὸς ἀλλήλων ἀριθμοὶ ἐλάτισταί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖξ.

Ἐστωσαν πρῶτοι πρὸς ἀλλήλων ἀριθμοὶ οἱ A , B · λέγω, ὅτι οἱ A , B ἐλάτισταί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖξ.

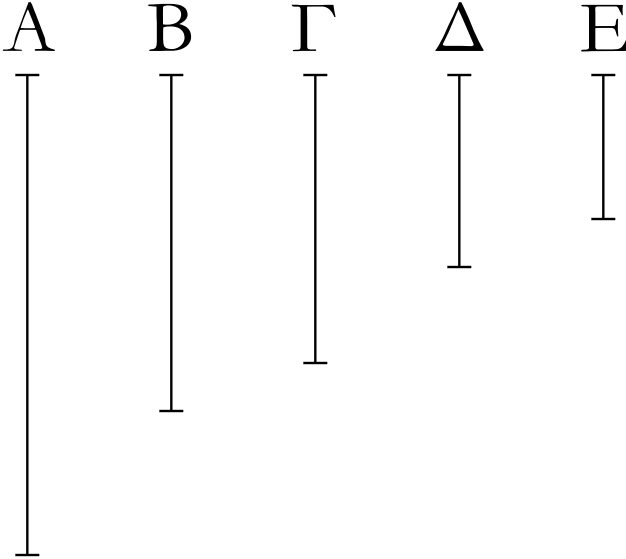
Εἰ γὰρ μή, ἔσονται τινεὶ τῶν A , B ἐλάσσονες ἀριθμοὶ ἐν τῷ αὐτῷ λόγῳ ὄντες τοῖς A , B . ἔστωσαν οἱ Γ , Δ .

Proposition 21

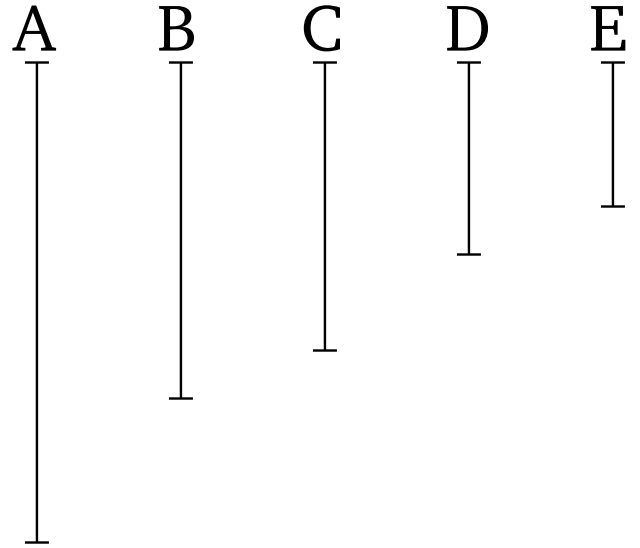
Numbers prime to one another are the least of those (numbers) having the same ratio as them.

Let A and B be numbers prime to one another. I say that A and B are the least of those (numbers) having the same ratio as them.

For if not then there will be some numbers less than A and B which are in the same ratio as A and B . Let them be C and D .



Ἐπεὶ οὖν οἱ ἐλάτισταί ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔθονταξ ἰσάκεις ὅ τε μεῖζων τὸν μεῖζονα καὶ ὁ ἐλάττων τὸν ἐλάττονα, τοντέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον, ἰσάκεις ἄρα ὁ Γ τὸν A μετρεῖ καὶ ὁ Δ τὸν B . ὁσάκεις δὴ ὁ Γ τὸν A μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ E . καὶ ὁ Δ ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν τῷ E μονάδας. καὶ ἐπεὶ ὁ Γ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ E μονάδας, καὶ ὁ E ἄρα τὸν A μετρεῖ κατὰ τὰς ἐν τῷ Γ μονάδας. διὰ τὰ αὐτὰ δὴ ὁ E καὶ τὸν B μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας. ὁ E ἄρα τοὺς A , B μετρεῖ πρώτους ὄντας πρὸς ἀλλήλους· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἔσονται τινεὶ τῶν A , B ἐλάσσονες ἀριθμοὶ ἐν τῷ αὐτῷ λόγῳ ὄντες τοῖς A , B . οἱ A , B ἄρα ἐλάτισταί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖξ· ὅπερ ἔδει δεῖχαι.

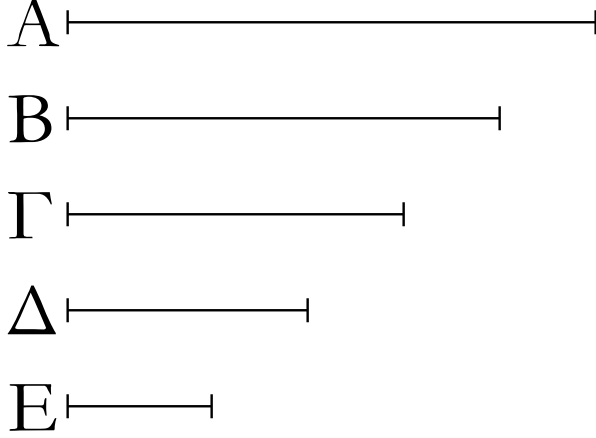


Therefore, since the least numbers of those (numbers) having the same ratio measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following— C thus measures A the same number of times that D (measures) B [Prop. 7.20]. So as many times as C measures A , so many units let there be in E . Thus, D also measures B according to the units in E . And since C measures A according to the units in E , E thus also measures A according to the units in C [Prop. 7.16]. So, for the same (reasons), E also measures B according to the units in D [Prop. 7.16]. Thus, E measures A and B , which are prime to one another. The very thing is impossible. Thus, there cannot be any numbers less than A and B which are in the same ratio as A and B . Thus, A and B are the least of those (numbers) having the same ratio as them.

(Which is) the very thing it was required to show.

22.

Οἱ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς
πρῶτοι πρὸς ἀλλήλους εἰσίν.



Ἐστωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων
αὐτοῖς οἱ A, B . λέγω, ὅτι οἱ A, B πρῶτοι πρὸς ἀλλήλους
εἰσίν.

Εἰ γὰρ μή εἰσι πρῶτοι πρὸς ἀλλήλους, μετρήσει τις αὐτοὺς
ἀριθμὸς. μετρεῖτω, καὶ ἔστω ὁ Γ . καὶ ὁσάκις μὲν ὁ Γ τὸν A
μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Δ , ὁσάκις δὲ ὁ Γ
τὸν B μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ E .

Ἐπεὶ ὁ Γ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας, ὁ Γ
ἄρα τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν. διὰ τὰ αὐτὰ
δὴ καὶ ὁ Γ τὸν E πολλαπλασιάσας τὸν B πεποίηκεν. ἀριθμὸς
δὴ ὁ Γ δύο ἀριθμὸν τὸν Δ , E πολλαπλασιάσας τὸν A ,
 B πεποίηκεν ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν E , οὕτως ὁ A
πρὸς τὸν B . οἱ Δ, E ἄρα τοῖς A, B ἐν τῷ αὐτῷ λόγῳ εἰσίν
ἐλάσσονες ὄντες αὐτῶν ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὸν A ,
 B ἀριθμὸν ἀριθμὸς τις μετρήσει. οἱ A, B ἄρα πρῶτοι
πρὸς ἀλλήλους εἰσίν ὅπερ ἔδει δεῖχαι.

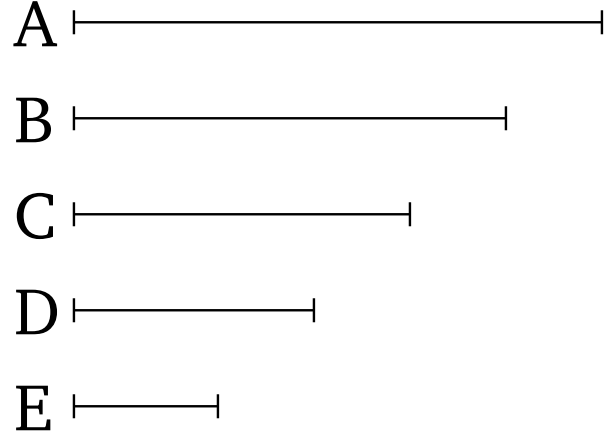
23.

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους ᾖσιν, ὁ τὸν ἓνα
αὐτῶν μετρῶν ἀριθμὸς πρὸς τὸν λοιπὸν πρῶτος ἔσται.

Ἐστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ A, B , τὸν
δὲ A μετρεῖτω τις ἀριθμὸς ὁ Γ . λέγω, ὅτι καὶ οἱ Γ, B πρῶτοι
πρὸς ἀλλήλους εἰσίν.

Proposition 22

The least numbers of those (numbers) having the same ra-
tio as them are prime to one another.



Let A and B be the least numbers of those (numbers) hav-
ing the same ratio as them. I say that A and B are prime to one
another.

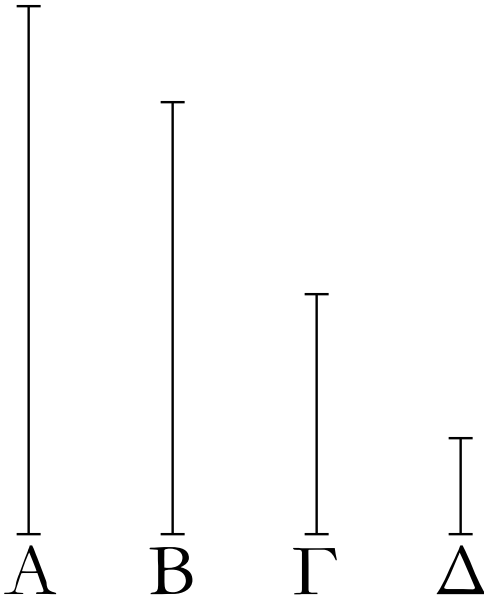
For if they are not prime to one another, (then) some num-
ber will measure them. Let it (so measure them), and let it be
 C . And as many times as C measures A , so many units let there
be in D . And as many times as C measures B , so many units
let there be in E .

Since C measures A according to the units in D , C has thus
made A (by) multiplying D [Def. 7.15]. So, for the same (rea-
sons), C has also made B (by) multiplying E . So the number C
has made A and B (by) multiplying the two numbers D and E
(respectively). Thus, as D is to E , so A (is) to B [Prop. 7.17].
Thus, D and E are in the same ratio as A and B , being less than
them. The very thing is impossible. Thus, some number does
not measure the numbers A and B . Thus, A and B are prime to
one another. (Which is) the very thing it was required to show.

Proposition 23

If two numbers are prime to one another, (then) a number
measuring one of them will be prime to the remaining (one).

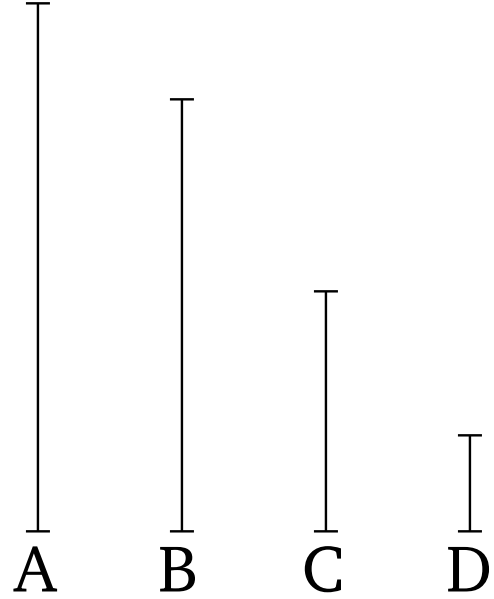
Let A and B be two numbers (which are) prime to one an-
other, and let some number C measure A . I say that C and B
are also prime to one another.



Εἰ γὰρ μὴ εἰσιν οἱ Γ , B πρῶτοι πρὸς ἀλλήλους, μετρήσει [τις] τοὺς Γ , B ἀριθμὸς. μετρεῖτω, καὶ ἔστω ὁ Δ . ἐπεὶ ὁ Δ τὸν Γ μετρεῖ, ὁ δὲ Γ τὸν A μετρεῖ, καὶ ὁ Δ ἄρα τὸν A μετρεῖ. μετρεῖ δὲ καὶ τὸν B . ὁ Δ ἄρα τοὺς A , B μετρεῖ πρῶτονξ ὄνταξ πρὸς ἀλλήλους· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τοὺς Γ , B ἀριθμὸνξ ἀριθμὸς τις μετρήσει. οἱ Γ , B ἄρα πρῶτοι πρὸς ἀλλήλους εἰσὶν· ὅπερ ἔδει δεῖχαι.

24.

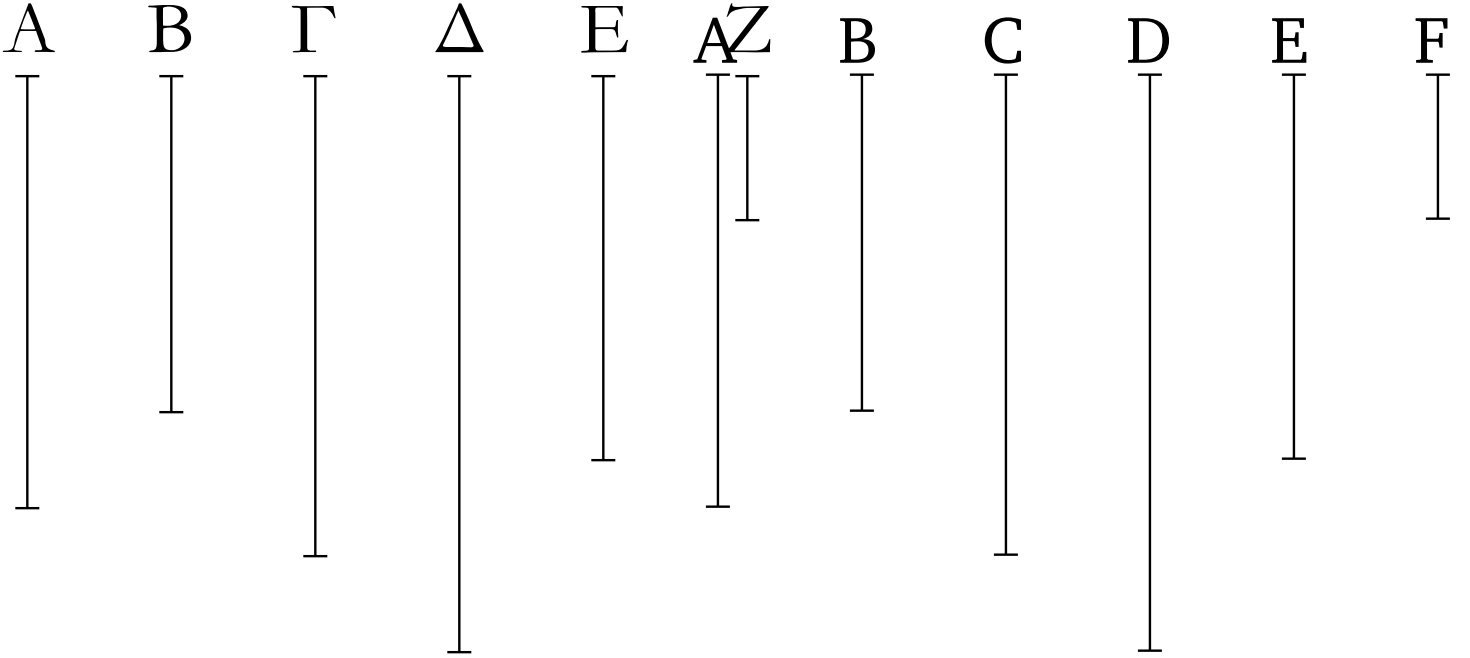
Ἐάν δύο ἀριθμοὶ πρὸς τινα ἀριθμὸν πρῶτοι ᾤων, καὶ ὁ ἐκ αὐτῶν γενόμενος πρὸς τὸν αὐτὸν πρῶτος ἔσται.



For if C and B are not prime to one another, (then) [some] number will measure C and B . Let it (so) measure (them), and let it be D . Since D measures C , and C measures A , D thus also measures A . And (D) also measures B . Thus, D measures A and B , which are prime to one another. The very thing is impossible. Thus, some number does not measure the numbers C and B . Thus, C and B are prime to one another. (Which is) the very thing it was required to show.

Proposition 24

If two numbers are prime to some number, (then) the number created from (multiplying) the former (two numbers) will also be prime to the latter (number).



Δύο γὰρ ἀριθμοὶ οἱ A, B πρὸς τινὰ ἀριθμὸν τὸν Γ πρῶτοι ἔστωσαν, καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Δ ποιείτω· λέγω, ὅτι οἱ Γ, Δ πρῶτοι πρὸς ἀλλήλους εἰσίν.

Εἰ γὰρ μὴ εἰσιν οἱ Γ, Δ πρῶτοι πρὸς ἀλλήλους, μετρήσει [τις] τοὺς Γ, Δ ἀριθμοὺς· μετρεῖτω, καὶ ἔστω ὁ E . καὶ ἐπεὶ οἱ Γ, Δ πρῶτοι πρὸς ἀλλήλους εἰσίν, τὸν δὲ Γ μετρεῖ τις ἀριθμὸς ὁ E , οἱ A, E ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν· ὁσάκις δὴ ὁ E τὸν Δ μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Z · καὶ ὁ Z ἄρα τὸν Δ μετρεῖ κατὰ τάξιν ἐν τῷ E μονάδαξ. ὁ E ἄρα τὸν B πολλαπλασιάσας τὸν Δ πεποίηκεν· ἀλλὰ μὴν καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Δ πεποίηκεν· ἴσοξ ἄρα ἐστὶν ὁ ἐκ τῶν E, Z τῷ ἐκ τῶν A, B . ἔαν δὲ ὁ ὑπὸ τῶν ἁκρῶν ἴσοξ ᾗ τῷ ὑπὸ τῶν μέσων, οἱ τέσσαρες ἀριθμοὶ ἀνάλογόν εἰσιν· ἔστιν ἄρα ὡς ὁ E πρὸς τὸν A , οὕτως ὁ B πρὸς τὸν Z . οἱ δὲ A, E πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔθονταξ ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὁ E ἄρα τὸν B μετρεῖ· μετρεῖ δὲ καὶ τὸν Γ · ὁ E ἄρα τοὺς B, Γ μετρεῖ πρώτουξ ὄνταξ πρὸς ἀλλήλους· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τοὺς Γ, Δ ἀριθμοὺς ἀριθμὸς τις μετρήσει· οἱ Γ, Δ ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν· ὅπερ ἔδει δεῖξαι.

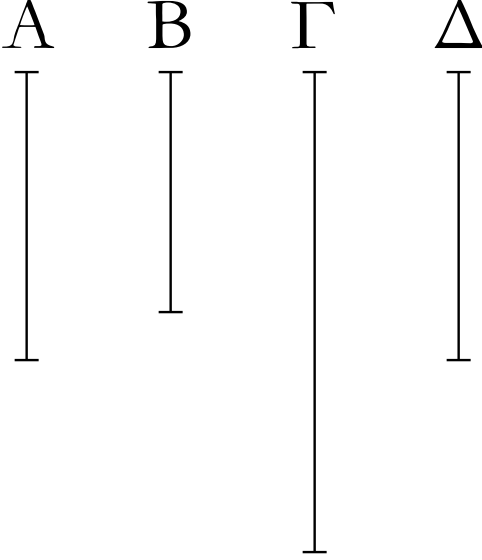
For let A and B be two numbers (which are both) prime to some number C . And let A make D (by) multiplying B . I say that C and D are prime to one another.

For if C and D are not prime to one another, (then) [some] number will measure C and D . Let it (so) measure them, and let it be E . And since C and A are prime to one another, and some number E measures C , A and E are thus prime to one another [Prop. 7.23]. So as many times as E measures D , so many units let there be in F . Thus, F also measures D according to the units in E [Prop. 7.16]. Thus, E has made D (by) multiplying F [Def. 7.15]. But, in fact, A has also made D (by) multiplying B . Thus, the (number created) from (multiplying) E and F is equal to the (number created) from (multiplying) A and B . And if the (rectangle contained) by the (two) outermost is equal to the (rectangle contained) by the middle (two) then the four numbers are proportional [Prop. 6.15]. Thus, as E is to A , so B (is) to F . And A and E (are) prime (to one another). And (numbers) prime (to one another) are also the least (of those numbers having the same ratio) [Prop. 7.21]. And the least numbers of those (numbers) having the same ratio measure those (numbers) having the same ratio as them an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures B . And it also measures C . Thus, E measures B and C , which are prime to one another. The very thing is impossible. Thus, some number cannot measure the numbers C and D . Thus, C and D are prime to one another. (Which is) the very thing it was required to show.

25.

Ἐάν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους ᾧσιν, ὁ ἐκ τοῦ ἐνόξ αὐτῶν γενόμενος πρὸς τὸν λοιπὸν πρῶτος ἔσται.

Ἔστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ A , B , καὶ ὁ A ἐαντὸν πολλαπλασιάσας τὸν Γ ποιείτω· λέγω, ὅτι οἱ B , Γ πρῶτοι πρὸς ἀλλήλους εἰσίν.



Κείσθω γάρ τῳ A ἴσος ὁ Δ . ἐπεὶ οἱ A , B πρῶτοι πρὸς ἀλλήλους εἰσίν, ἴσος δὲ ὁ A τῳ Δ , καὶ οἱ Δ , B ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν. ἐκάτερος ἄρα τῶν Δ , A πρὸς τὸν B πρῶτός ἐστιν· καὶ ὁ ἐκ τῶν Δ , A ἄρα γενόμενος πρὸς τὸν B πρῶτος ἔσται. ὁ δὲ ἐκ τῶν Δ , A γενόμενος ἀριθμὸς ἐστὶν ὁ Γ . οἱ Γ , B ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν· ὅπερ ἔδει δεῖξαι.

26.

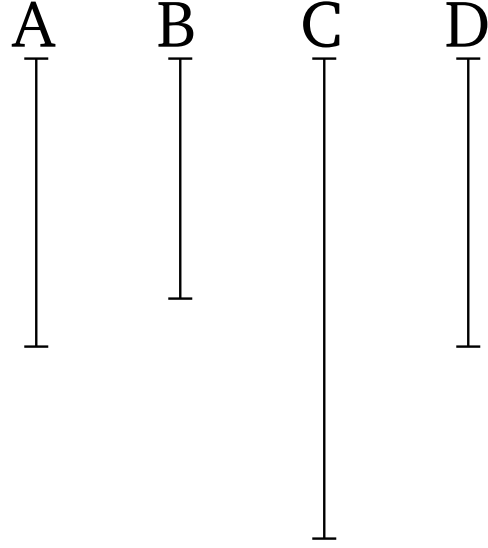
Ἐάν δύο ἀριθμοὶ πρὸς δύο ἀριθμὸν ἀμφοτέρωι πρὸς ἐκάτερον πρῶτοι ᾧσιν, καὶ οἱ ἐκ αὐτῶν γενόμενοι πρῶτοι πρὸς ἀλλήλους ἔσονται.

Δύο γάρ ἀριθμοὶ οἱ A , B πρὸς δύο ἀριθμὸν τὸν Γ , Δ ἀμφοτέρωι πρὸς ἐκάτερον πρῶτοι ἔστωσαν, καὶ ὁ μὲν A τὸν B πολλαπλασιάσας τὸν E ποιείτω, ὁ δὲ Γ τὸν Δ πολλαπλασιάσας τὸν Z ποιείτω· λέγω, ὅτι οἱ E , Z πρῶτοι πρὸς ἀλλήλους εἰσίν.

Proposition 25

If two numbers are prime to one another, (then) the number created from (squaring) one of them will be prime to the remaining (number).

Let A and B be two numbers (which are) prime to one another. And let A make C (by) multiplying itself. I say that B and C are prime to one another.

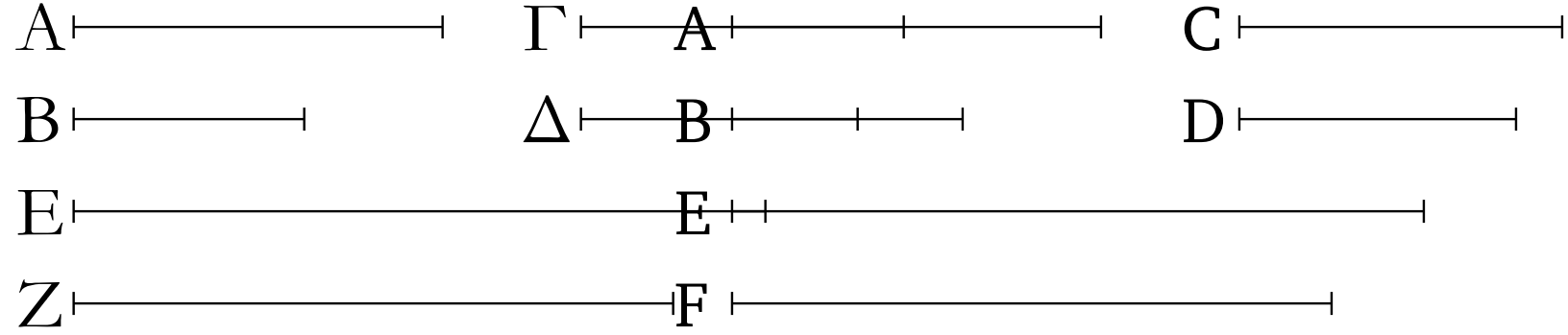


For let D be made equal to A . Since A and B are prime to one another, and A (is) equal to D , D and B are thus also prime to one another. Thus, D and A are each prime to B . Thus, the (number) created from (multiplying) D and A will also be prime to B [Prop. 7.24]. And C is the number created from (multiplying) D and A . Thus, C and B are prime to one another. (Which is) the very thing it was required to show.

Proposition 26

If two numbers are both prime to each of two numbers, (then) the (numbers) created from (multiplying) them will also be prime to one another.

For let two numbers, A and B , both be prime to each of two numbers, C and D . And let A make E (by) multiplying B , and let C make F (by) multiplying D . I say that E and F are prime to one another.



Ἐπεὶ γὰρ ἑκάτερος τῶν A, B πρὸς τὸν Γ πρῶτός ἐστιν, καὶ ὁ ἐκ τῶν A, B ἄρα γενόμενος πρὸς τὸν Γ πρῶτος ἔσται. ὁ δὲ ἐκ τῶν A, B γενόμενός ἐστιν ὁ E . οἱ E, Γ ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν. διὰ τὰ αὐτὰ δὴ καὶ οἱ E, Δ πρῶτοι πρὸς ἀλλήλους εἰσίν. ἑκάτερος ἄρα τῶν Γ, Δ πρὸς τὸν E πρῶτός ἐστιν. καὶ ὁ ἐκ τῶν Γ, Δ ἄρα γενόμενος πρὸς τὸν E πρῶτος ἔσται. ὁ δὲ ἐκ τῶν Γ, Δ γενόμενός ἐστιν ὁ Z . οἱ E, Z ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν· ὅπερ ἔδει δεῖχαι.

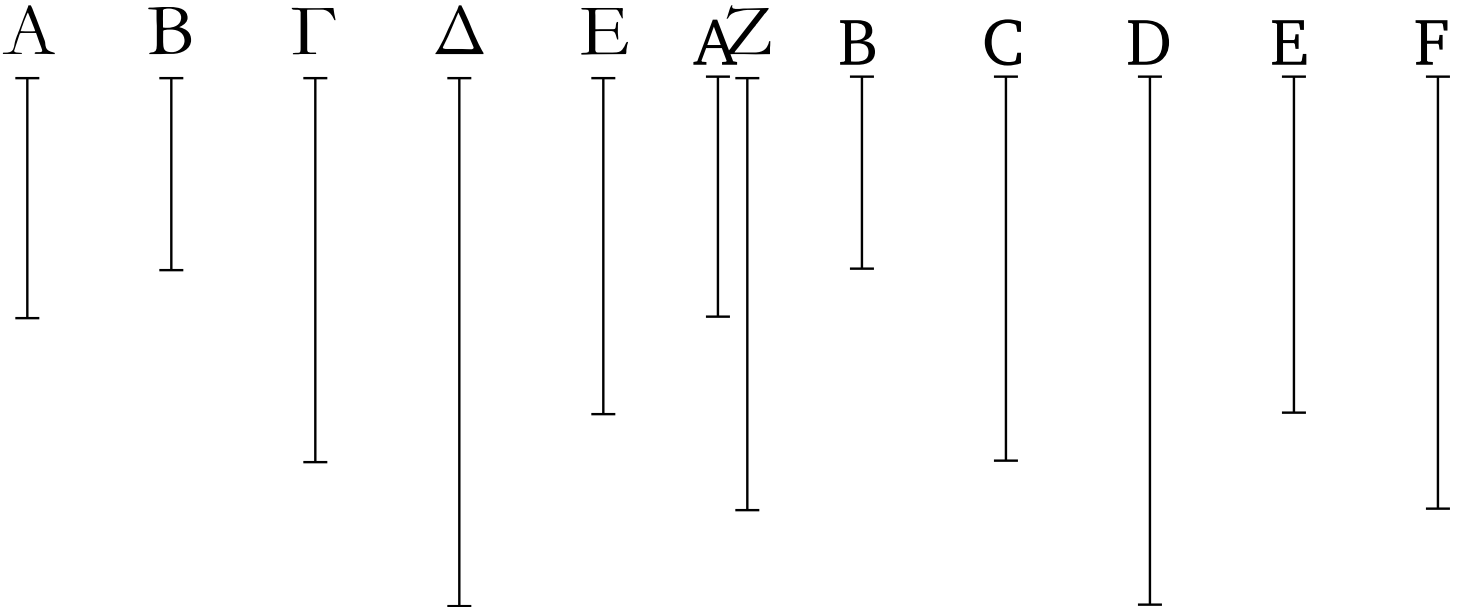
For since A and B are each prime to C , the (number) created from (multiplying) A and B will thus also be prime to C [Prop. 7.24]. And E is the (number) created from (multiplying) A and B . Thus, E and C are prime to one another. So, for the same (reasons), E and D are also prime to one another. Thus, C and D are each prime to E . Thus, the (number) created from (multiplying) C and D will also be prime to E [Prop. 7.24]. And F is the (number) created from (multiplying) C and D . Thus, E and F are prime to one another. (Which is) the very thing it was required to show.

27.

Proposition 27[†]

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους ᾖσιν, καὶ πολλαπλασιάσας ἑκάτερος ἑαυτὸν ποιῇ τινα, οἱ γενόμενοι ἐκ αὐτῶν πρῶτοι πρὸς ἀλλήλους ἔσονται, κ' ἂν οἱ ἐκ ἀρθῆς τούτῳ γενομένον πολλαπλασιάσαντες ποιῶσι τινας, καὶ κείνοι πρῶτοι πρὸς ἀλλήλους ἔσονται [καὶ αἰεὶ περὶ τούτῳ ἄκρον τοῦτο συμβαίνει].

If two numbers are prime to one another and each makes some (number by) multiplying itself, (then) the numbers created from them will be prime to one another, and if the original (numbers) make some (more numbers by) multiplying the created (numbers, then) these will also be prime to one another [and this always happens with the extremes].



Ἐστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ A, B , καὶ ὁ A ἑαυτὸν μὲν πολλαπλασιάσας τὸν Γ ποιείτω, τὸν δὲ Γ

Let A and B be two numbers prime to one another, and let A make C (by) multiplying itself, and let it make D (by)

πολλαπλασιάσας τὸν Δ ποιεῖτω, ὁ δὲ B ἑαυτὸν μὲν πολλαπλασιάσας τὸν E ποιεῖτω, τὸν δὲ E πολλαπλασιάσας τὸν Z ποιεῖτω· λέγω, ὅτι οἱ Γ , E καὶ οἱ Δ , Z πρῶτοι πρὸς ἀλλήλους εἰσίν.

Ἐπεὶ γὰρ οἱ A , B πρῶτοι πρὸς ἀλλήλους εἰσίν, καὶ ὁ A ἑαυτὸν πολλαπλασιάσας τὸν Γ πεποίηκεν, οἱ Γ , B ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν. ἐπεὶ οὖν οἱ Γ , B πρῶτοι πρὸς ἀλλήλους εἰσίν, καὶ ὁ B ἑαυτὸν πολλαπλασιάσας τὸν E πεποίηκεν, οἱ Γ , E ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν. πάλιν, ἐπεὶ οἱ A , B πρῶτοι πρὸς ἀλλήλους εἰσίν, καὶ ὁ B ἑαυτὸν πολλαπλασιάσας τὸν E πεποίηκεν, οἱ A , E ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν. ἐπεὶ οὖν δύο ἀριθμοὶ οἱ A , Γ πρὸς δύο ἀριθμοὺς τούτους B , E ἀμφοτέρω πρὸς ἑκάτερον πρῶτοι εἰσίν, καὶ ὁ ἐκ τῶν A , Γ ἄρα γενομένος πρὸς τὸν ἐκ τῶν B , E πρῶτός ἐστιν. καὶ ἐστὶν ὁ μὲν ἐκ τῶν A , Γ ὁ Δ , ὁ δὲ ἐκ τῶν B , E ὁ Z . οἱ Δ , Z ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν· ὅπερ ἔδει δεῖχαι.

multiplying C . And let B make E (by) multiplying itself, and let it make F by multiplying E . I say that C and E , and D and F , are prime to one another.

For since A and B are prime to one another, and A has made C (by) multiplying itself, C and B are thus prime to one another [Prop. 7.25]. Therefore, since C and B are prime to one another, and B has made E (by) multiplying itself, C and E are thus prime to one another [Prop. 7.25]. Again, since A and B are prime to one another, and B has made E (by) multiplying itself, A and E are thus prime to one another [Prop. 7.25]. Therefore, since the two numbers A and C are both prime to each of the two numbers B and E , the (number) created from (multiplying) A and C is thus prime to the (number created) from (multiplying) B and E [Prop. 7.26]. And D is the (number created) from (multiplying) A and C , and F the (number created) from (multiplying) B and E . Thus, D and F are prime to one another. (Which is) the very thing it was required to show.

† In modern notation, this proposition states that if a is prime to b , then a^2 is also prime to b^2 , as well as a^3 to b^3 , etc., where all symbols denote numbers.

28.

Proposition 28

Ἐὰν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους ᾖσιν, καὶ συναμφοτέροισι πρὸς ἑκάτερον αὐτῶν πρῶτος ᾖ· καὶ ἔαν συναμφοτέροισι πρὸς ἑνα τινὰ αὐτῶν πρῶτος ᾖ, καὶ οἱ ἐκ ἀριθμῶν ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους ᾖσονται.

If two numbers are prime to one another, (then) their sum will also be prime to each of them. And if the sum (of two numbers) is prime to any one of them, (then) the original numbers will also be prime to one another.



Δ

D

Συγκείσθωσαν γὰρ δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ AB , $B\Gamma$. λέγω, ὅτι καὶ συναμφοτέροισι ὁ $A\Gamma$ πρὸς ἑκάτερον τῶν AB , $B\Gamma$ πρῶτός ἐστιν.

Εἰ γὰρ μὴ εἰσιν οἱ ΓA , AB πρῶτοι πρὸς ἀλλήλους, μετρήσει τις τούτους ΓA , AB ἀριθμός. μετρεῖτω, καὶ ἔστω ὁ Δ . ἐπεὶ οὖν ὁ Δ τούτους ΓA , AB μετρεῖ, καὶ λοιπὸν ἄρα τὸν $B\Gamma$ μετρήσει. μετρεῖ δὲ καὶ τὸν BA . ὁ Δ ἄρα τούτους AB , $B\Gamma$ μετρεῖ πρῶτον ἐξ ὧν ταῦτα πρὸς ἀλλήλους· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τούτους ΓA , AB ἀριθμὸς ἀριθμός τις μετρήσει· οἱ ΓA , AB ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν. διὰ τὰ αὐτὰ δὴ καὶ οἱ $A\Gamma$, ΓB πρῶτοι πρὸς ἀλλήλους εἰσίν. ὁ ΓA ἄρα πρὸς ἑκάτερον τῶν AB , $B\Gamma$ πρῶτός ἐστιν.

Ἔστωσαν δὴ πάλιν οἱ ΓA , AB πρῶτοι πρὸς ἀλλήλους· λέγω, ὅτι καὶ οἱ AB , $B\Gamma$ πρῶτοι πρὸς ἀλλήλους εἰσίν.

Εἰ γὰρ μὴ εἰσιν οἱ AB , $B\Gamma$ πρῶτοι πρὸς ἀλλήλους, μετρήσει τις τούτους AB , $B\Gamma$ ἀριθμός. μετρεῖτω, καὶ ἔστω ὁ Δ . καὶ ἐπεὶ ὁ Δ ἑκάτερον τῶν AB , $B\Gamma$ μετρεῖ, καὶ ὅλον ἄρα τὸν ΓA μετρήσει. μετρεῖ δὲ καὶ τὸν AB . ὁ Δ ἄρα τούτους ΓA , AB

For let the two numbers, AB and BC , (which are) prime to one another, be laid down together. I say that their sum AC is also prime to each of AB and BC .

For if CA and AB are not prime to one another, (then) some number will measure CA and AB . Let it (so) measure (them), and let it be D . Therefore, since D measures CA and AB , it will thus also measure the remainder BC . And it also measures BA . Thus, D measures AB and BC , which are prime to one another. The very thing is impossible. Thus, some number cannot measure (both) the numbers CA and AB . Thus, CA and AB are prime to one another. So, for the same (reasons), AC and CB are also prime to one another. Thus, CA is prime to each of AB and BC .

So, again, let CA and AB be prime to one another. I say that AB and BC are also prime to one another.

For if AB and BC are not prime to one another, (then) some number will measure AB and BC . Let it (so) measure (them), and let it be D . And since D measures each of AB and BC , it

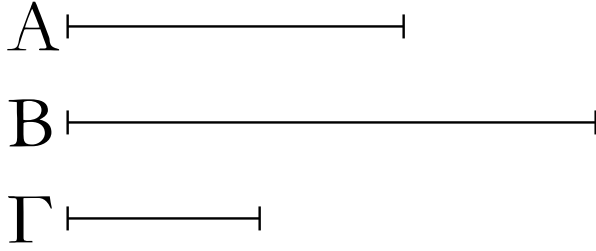
μετρεῖ πρώτων ὄνταξ πρὸς ἀλλήλων· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τοὺς AB , $BΓ$ ἀριθμὸς ἀριθμὸς τις μετρήσει. οἱ AB , $BΓ$ ἄρα πρώτοι πρὸς ἀλλήλων εἰσὶν· ὅπερ ἔδει δεῖχαι.

will thus also measure the whole of CA . And it also measures AB . Thus, D measures CA and AB , which are prime to one another. The very thing is impossible. Thus, some number cannot measure (both) the numbers AB and BC . Thus, AB and BC are prime to one another. (Which is) the very thing it was required to show.

29.

Ἄπαξ πρώτος ἀριθμὸς πρὸς ἅπαντα ἀριθμόν, ὃν μὴ μετρεῖ, πρώτός ἐστιν.

Ἐστω πρώτος ἀριθμὸς ὁ A καὶ τὸν B μὴ μετρεῖτω· λέγω, ὅτι οἱ B , A πρώτοι πρὸς ἀλλήλων εἰσὶν.

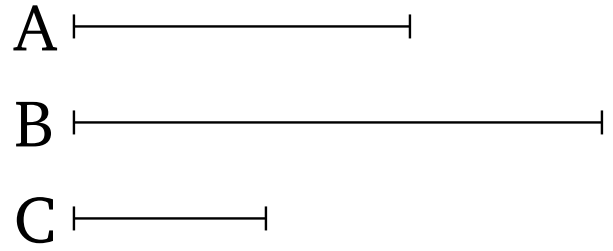


Εἰ γὰρ μὴ εἰσὶν οἱ B , A πρώτοι πρὸς ἀλλήλων, μετρήσει τις αὐτοὺς ἀριθμὸς. μετρεῖτω ὁ $Γ$. ἐπεὶ ὁ $Γ$ τὸν B μετρεῖ, ὁ δὲ A τὸν B οὐ μετρεῖ, ὁ $Γ$ ἄρα τῷ A ὅν ἐστιν ὁ αὐτός. καὶ ἐπεὶ ὁ $Γ$ τοὺς B , A μετρεῖ, καὶ τὸν A ἄρα μετρεῖ πρώτον ὄντα μὴ ὡν αὐτῷ ὁ αὐτός· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τοὺς B , A μετρήσει τις ἀριθμὸς. οἱ A , B ἄρα πρώτοι πρὸς ἀλλήλων εἰσὶν· ὅπερ ἔδει δεῖχαι.

Proposition 29

Every prime number is prime to every number which it does not measure.

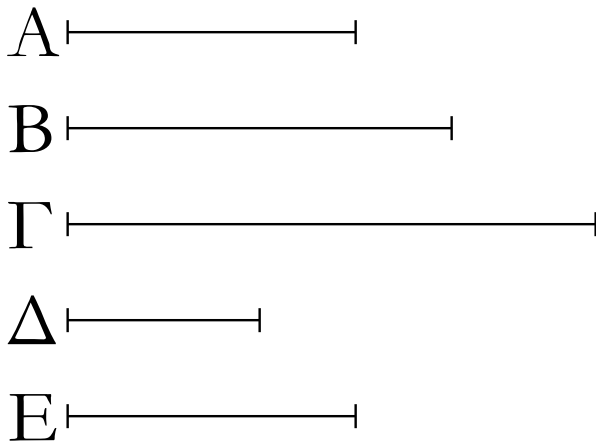
Let A be a prime number, and let it not measure B . I say that B and A are prime to one another.



For if B and A are not prime to one another, (then) some number will measure them. Let C measure (them). Since C measures B , and A does not measure B , C is thus not the same as A . And since C measures B and A , it thus also measures A , which is prime, (despite) not being the same as it. The very thing is impossible. Thus, some number cannot measure (both) B and A . Thus, A and B are prime to one another. (Which is) the very thing it was required to show.

30.

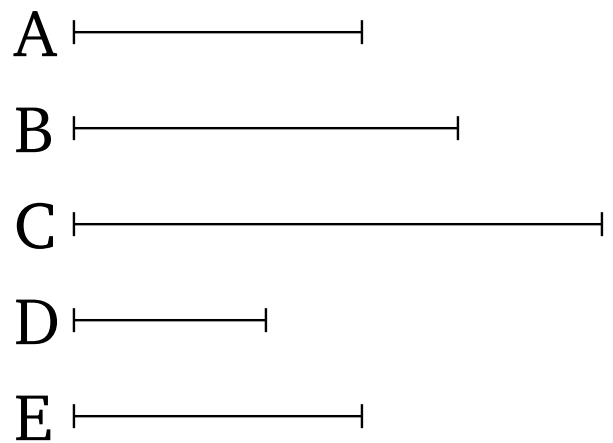
Ἐὰν δύο ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλων ποιῶσί τινα, τὸν δὲ γενόμενον ἔχ αὐτῶν μετρητὴ τις πρώτος ἀριθμὸς, καὶ ἓνα τῶν ἔχ ἀριθμὸς μετρήσει.



Δύο γὰρ ἀριθμοὶ οἱ A , B πολλαπλασιάσαντες ἀλλήλων

Proposition 30

If two numbers make some (number by) multiplying one another, and some prime number measures the number (so) created from them, (then) it will also measure one of the original (numbers).



For let two numbers A and B make C (by) multiplying one

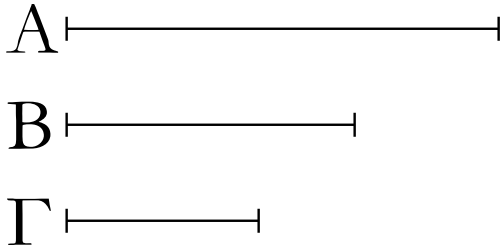
τὸν Γ ποιείτωσαν, τὸν δὲ Γ μετρεῖτω τις πρῶτος ἀριθμὸς ὁ Δ . λέγω, ὅτι ὁ Δ ἓνα τῶν A, B μετρεῖ.

Τὸν γὰρ A μὴ μετρεῖτω· καὶ ἐστὶ πρῶτος ὁ Δ . οἱ A, Δ ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ὁσάκις ὁ Δ τὸν Γ μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ E . ἐπεὶ οὖν ὁ Δ τὸν Γ μετρεῖ κατὰ τὰς ἐν τῷ E μονάδας, ὁ Δ ἄρα τὸν E πολλαπλασιάσας τὸν Γ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν· ἴσος ἄρα ἐστὶν ὁ ἐκ τῶν Δ, E τῷ ἐκ τῶν A, B . ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν A , οὕτως ὁ B πρὸς τὸν E . οἱ δὲ Δ, A πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον· ὁ Δ ἄρα τὸν B μετρεῖ. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἕαν τὸν B μὴ μετρήῃ, τὸν A μετρήσει. ὁ Δ ἄρα ἓνα τῶν A, B μετρεῖ· ὅπερ ἔδει δεῖχαι.

31.

Ἄπαξ σύνθεντος ἀριθμοῦ ὑπὸ πρῶτον τινὸς ἀριθμοῦ μετρεῖται.

Ἐστω σύνθεντος ἀριθμὸς ὁ A . λέγω, ὅτι ὁ A ὑπὸ πρῶτον τινὸς ἀριθμοῦ μετρεῖται.



Ἐπεὶ γὰρ σύνθετός ἐστιν ὁ A , μετρήσει τις αὐτὸν ἀριθμὸς. μετρεῖτω, καὶ ἔστω ὁ B . καὶ εἰ μὲν πρῶτός ἐστιν ὁ B , γερονὸς ἂν εἴη τὸ ἐπιταθέν. εἰ δὲ σύνθετος, μετρήσει τις αὐτὸν ἀριθμὸς. μετρεῖτω, καὶ ἔστω ὁ Γ . καὶ ἐπεὶ ὁ Γ τὸν B μετρεῖ, ὁ δὲ B τὸν A μετρεῖ, καὶ ὁ Γ ἄρα τὸν A μετρεῖ. καὶ εἰ μὲν πρῶτός ἐστιν ὁ Γ , γερονὸς ἂν εἴη τὸ ἐπιταθέν. εἰ δὲ σύνθετος, μετρήσει τις αὐτὸν ἀριθμὸς. τοιαύτης δὲ γινομένης ἐπισκέψεως ληφθήσεται τις πρῶτος ἀριθμὸς, ὃς μετρήσει. εἰ γὰρ οὐ ληφθήσεται, μετρήσουσι τὸν A ἀριθμὸν ἄπειροι ἀριθμοί, ὧν ἕτερος ἑτέρου ἐλάσσων ἐστίν· ὅπερ ἐστὶν ἀδύνατον ἐν ἀριθμοῖς. ληφθήσεται τις ἄρα πρῶτος ἀριθμὸς, ὃς μετρήσει τὸν πρὸ ἐαυτοῦ, ὃς καὶ τὸν A μετρήσει.

Ἄπαξ ἄρα σύνθεντος ἀριθμοῦ ὑπὸ πρῶτον τινὸς ἀριθμοῦ μετρεῖται· ὅπερ ἔδει δεῖχαι.

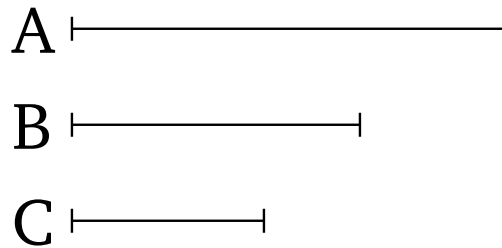
another, and let some prime number D measure C . I say that D measures one of A and B .

For let it not measure A . And since D is prime, A and D are thus prime to one another [Prop. 7.29]. And as many times as D measures C , so many units let there be in E . Therefore, since D measures C according to the units E , D has thus made C (by multiplying E [Def. 7.15]. But, in fact, A has also made C (by multiplying B . Thus, the (number created) from (multiplying) D and E is equal to the (number created) from (multiplying) A and B . Thus, as D is to A , so B (is) to E [Prop. 7.19]. And D and A (are) prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, D measures B . So, similarly, we can also show that if (D) does not measure B , (then) it will measure A . Thus, D measures one of A and B . (Which is) the very thing it was required to show.

Proposition 31

Every composite number is measured by some prime number.

Let A be a composite number. I say that A is measured by some prime number.

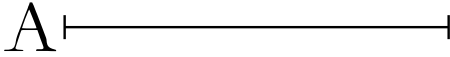


For since A is composite, some number will measure it. Let it (so) measure (A), and let it be B . And if B is prime then that which was prescribed has happened. And if (B is) composite, (then) some number will measure it. Let it (so) measure (B), and let it be C . And since C measures B , and B measures A , C thus also measures A . And if C is prime then that which was prescribed has happened. And if (C is) composite, (then) some number will measure it. So, in this manner of continued investigation, some prime number will be found which will measure (the number preceding it, which will also measure A). And if (such a number) cannot be found, (then) an infinite (series of) numbers, each of which is less than the preceding, will measure the number A . The very thing is impossible for numbers. Thus, some prime number will (eventually) be found which will measure the (number) preceding it, which will also measure A .

Thus, every composite number is measured by some prime number. (Which is) the very thing it was required to show.

32.

Ἄπαξ ἀριθμὸς ἥτοι πρῶτός ἐστιν ἢ ὑπὸ πρῶτον τινὸς ἀριθμοῦ μετρεῖται.

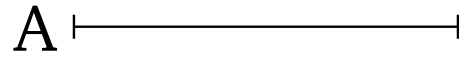


Ἐστω ἀριθμὸς ὁ A · λέγω, ὅτι ὁ A ἥτοι πρῶτός ἐστιν ἢ ὑπὸ πρῶτον τινὸς ἀριθμοῦ μετρεῖται.

Εἰ μὲν οὖν πρῶτός ἐστιν ὁ A , γεγονός ἂν εἴη τό ἐπιταθέν. εἰ δὲ σύνθετος, μετρήσει τις αὐτὸν πρῶτος ἀριθμός.

Ἄπαξ ἄρα ἀριθμὸς ἥτοι πρῶτός ἐστιν ἢ ὑπὸ πρῶτον τινὸς ἀριθμοῦ μετρεῖται· ὅπερ ἔδει δεῖξαι.

Every number is either prime, or is measured by some prime number.



Let A be a number. I say that A is either prime, or is measured by some prime number.

In fact, if A is prime, (then) that which was prescribed has happened. And if (it is) composite, (then) some prime number will measure it [Prop. 7.31].

Thus, every number is either prime, or is measured by some prime number. (Which is) the very thing it was required to show.

Proposition 33

33.

Ἀριθμῶν δοθέντων ὁποσωνοῦν εὑρεῖν τοὺς ἐλαθίστους τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

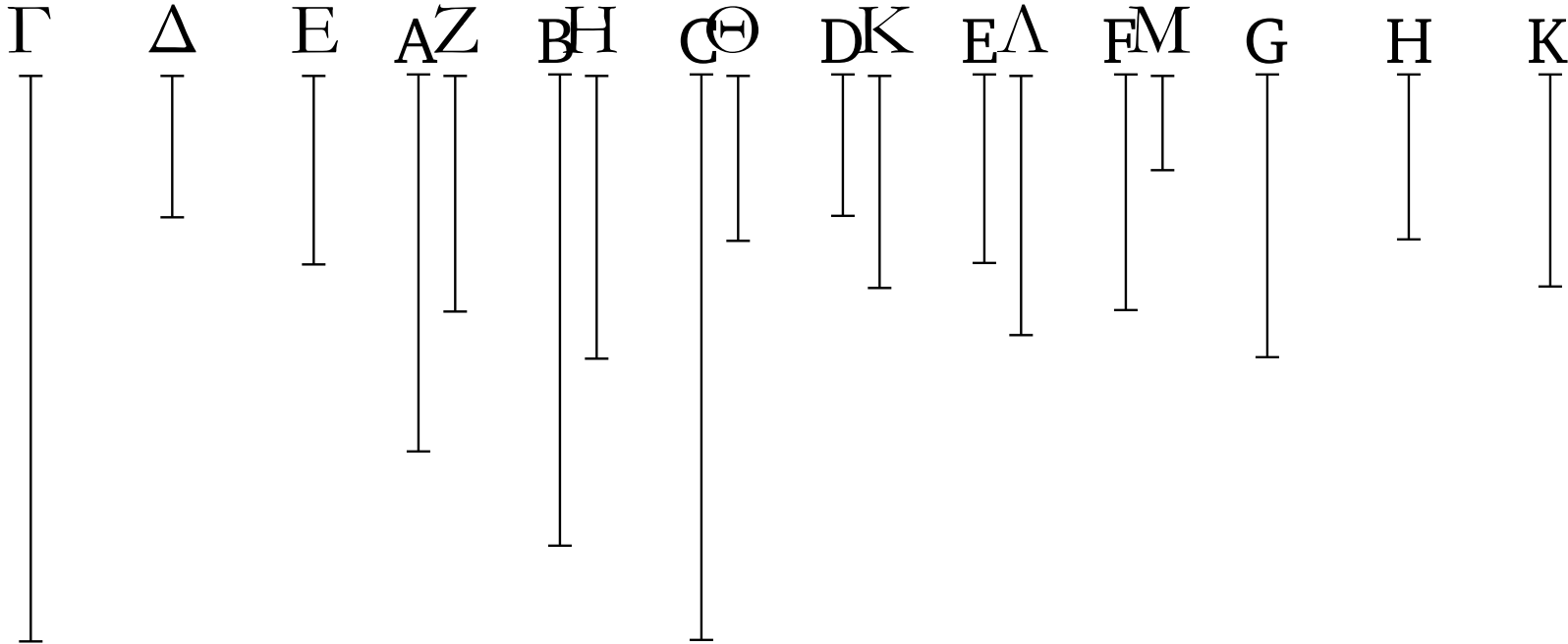
Ἐστωσαν οἱ δοθέντες ὁποσοιοῦν ἀριθμοὶ οἱ A, B, Γ · δεῖ δὴ εὑρεῖν τοὺς ἐλαθίστους τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ .

Οἱ A, B, Γ γὰρ ἥτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν ἢ οὐ. εἰ μὲν οὖν οἱ A, B, Γ πρῶτοι πρὸς ἀλλήλους εἰσὶν, ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

To find the least of those (numbers) having the same ratio as any given multitude of numbers.

Let A, B , and C be any given multitude of numbers. So it is required to find the least of those (numbers) having the same ratio as A, B , and C .

For A, B , and C are either prime to one another, or not. In fact, if A, B , and C are prime to one another, (then) they are the least of those (numbers) having the same ratio as them [Prop. 7.22].



Εἰ δὲ οὐ, εἰλήφθω τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον ὁ Δ , καὶ ὅσάκις ὁ Δ ἑκάστον τῶν A, B, Γ μετρεῖ, τοσαῦται

And if not, let the greatest common measure, D , of A, B , and C have been taken [Prop. 7.3]. And as many times as D

μονάδες ἔστωσαν ἐν ἑκάστῳ τῶν E, Z, H . καὶ ἑκάστοξ ἄρα τῶν E, Z, H ἑκάστον τῶν A, B, Γ μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδαξ. οἱ E, Z, H ἄρα τοῖξ A, B, Γ ἰσάκιξ μετροῦσιν· οἱ E, Z, H ἄρα τοῖξ A, B, Γ ἐν τῷ αὐτῷ λόγῳ εἰσίν. λέγω δὴ, ὅτι καὶ ἐλάχιστοι. εἰ γὰρ μὴ εἰσιν οἱ E, Z, H ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖξ A, B, Γ , ἔσονται [τινεξ] τῶν E, Z, H ἐλάσσονεξ ἀριθμοὶ ἐν τῷ αὐτῷ λόγῳ ὄντεξ τοῖξ A, B, Γ . ἔστωσαν οἱ Θ, K, Λ ἰσάκιξ ἄρα ὁ Θ τὸν A μετρεῖ καὶ ἑκάτεροξ τῶν K, Λ ἑκάτερον τῶν B, Γ . ὁσάκιξ δὲ ὁ Θ τὸν A μετρεῖ, τοσαῦται μονάδεξ ἔστωσαν ἐν τῷ M · καὶ ἑκάτεροξ ἄρα τῶν K, Λ ἑκάτερον τῶν B, Γ μετρεῖ κατὰ τὰς ἐν τῷ M μονάδαξ. καὶ ἐπεὶ ὁ Θ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ M μονάδαξ, καὶ ὁ M ἄρα τὸν A μετρεῖ κατὰ τὰς ἐν τῷ Θ μονάδαξ. διὰ τὰ αὐτὰ δὴ ὁ M καὶ ἑκάτερον τῶν B, Γ μετρεῖ κατὰ τὰς ἐν ἑκατέρῳ τῶν K, Λ μονάδαξ· ὁ M ἄρα τοῖξ A, B, Γ μετρεῖ. καὶ ἐπεὶ ὁ Θ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ M μονάδαξ, ὁ Θ ἄρα τὸν M πολλαπλασιάσαξ τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ E τὸν Δ πολλαπλασιάσαξ τὸν A πεποίηκεν. ἴσοξ ἄρα ἐστὶν ὁ ἐκ τῶν E, Δ τῷ ἐκ τῶν Θ, M . ἔστιν ἄρα ὡξ ὁ E πρὸξ τὸν Θ , οὔτως ὁ M πρὸξ τὸν Δ . μείζων δὲ ὁ E τοῦ Θ · μείζων ἄρα καὶ ὁ M τοῦ Δ . καὶ μετρεῖ τοῖξ A, B, Γ · ὅπερ ἐστὶν ἀδύνατον· ὑπόκειται γὰρ ὁ Δ τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον. οὐκ ἄρα ἔσονται τινεξ τῶν E, Z, H ἐλάσσονεξ ἀριθμοὶ ἐν τῷ αὐτῷ λόγῳ ὄντεξ τοῖξ A, B, Γ . οἱ E, Z, H ἄρα ἐλάχιστοὶ εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖξ A, B, Γ · ὅπερ ἔδει δεῖχαι.

measures A, B, C , so many units let there be in E, F, G , respectively. And thus E, F, G measure A, B, C , respectively, according to the units in D [Prop. 7.15]. Thus, E, F, G measure A, B, C (respectively) an equal number of times. Thus, E, F, G are in the same ratio as A, B, C (respectively) [Def. 7.20]. So I say that (they are) also the least (of those numbers having the same ratio as A, B, C). For if E, F, G are not the least of those (numbers) having the same ratio as A, B, C (respectively, then) there will be [some] numbers less than E, F, G which are in the same ratio as A, B, C (respectively). Let them be H, K, L . Thus, H measures A the same number of times that K, L also measure B, C , respectively. And as many times as H measures A , so many units let there be in M . Thus, K, L measure B, C , respectively, according to the units in M . And since H measures A according to the units in M , M thus also measures A according to the units in H [Prop. 7.15]. So, for the same (reasons), M also measures B, C according to the units in K, L , respectively. Thus, M measures A, B , and C . And since H measures A according to the units in M , H has thus made A (by) multiplying M . So, for the same (reasons), E has also made A (by) multiplying D . Thus, the (number created) from (multiplying) E and D is equal to the (number created) from (multiplying) H and M . Thus, as E (is) to H , so M (is) to D [Prop. 7.19]. And E (is) greater than H . Thus, M (is) also greater than D [Prop. 5.13]. And (M) measures A, B , and C . The very thing is impossible. For D was assumed (to be) the greatest common measure of A, B , and C . Thus, there cannot be any numbers less than E, F, G which are in the same ratio as A, B, C (respectively). Thus, E, F, G are the least of (those numbers) having the same ratio as A, B, C (respectively). (Which is) the very thing it was required to show.

34.

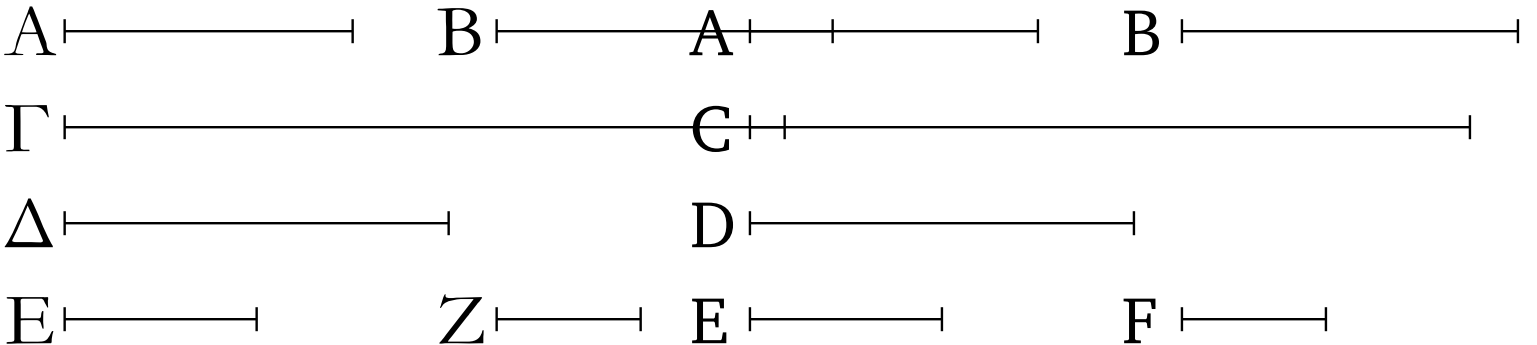
Proposition 34

Δύο ἀριθμῶν δοθέντων εὐρεῖν, ὃν ἐλάχιστον μετροῦσιν ἀριθμόν.

Ἔστωσαν οἱ δοθέντεξ δύο ἀριθμοὶ οἱ A, B · δεῖ δὴ εὐρεῖν, ὃν ἐλάχιστον μετροῦσιν ἀριθμόν.

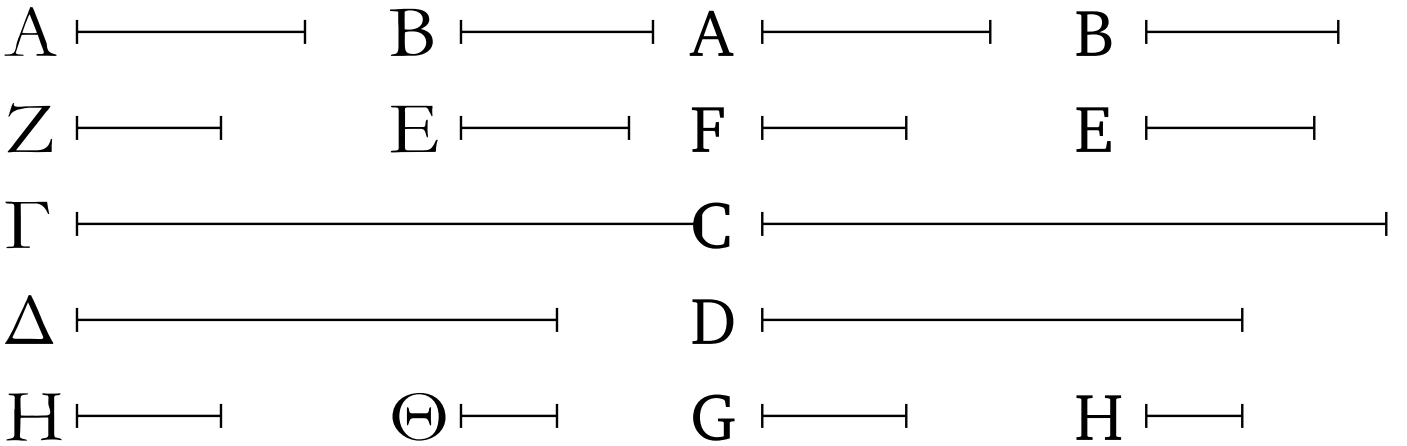
To find the least number which two given numbers (both) measure.

Let A and B be the two given numbers. So it is required to find the least number which they (both) measure.



Οἱ A, B γάρ ἤτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν ἢ οὐ. ἔστωσαν πρότερον οἱ A, B πρῶτοι πρὸς ἀλλήλους, καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ ποιείτω· καὶ ὁ B ἄρα τὸν A πολλαπλασιάσας τὸν Γ πεποιήκεν. οἱ A, B ἄρα τὸν Γ μετροῦσιν. λέγω δὴ, ὅτι καὶ ἐλάχιστον. εἰ γὰρ μή, μετρήσουσιν τινα ἀριθμὸν οἱ A, B ἐλάσσονα ὄντα τοῦ Γ . μετρεῖτωσαν τὸν Δ . καὶ ὁσάκις ὁ A τὸν Δ μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ E , ὁσάκις δὲ ὁ B τὸν Δ μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Z . ὁ μὲν A ἄρα τὸν E πολλαπλασιάσας τὸν Δ πεποιήκεν, ὁ δὲ B τὸν Z πολλαπλασιάσας τὸν Δ πεποιήκεν· ἴσοις ἄρα ἐστὶν ὁ ἐκ τῶν A, E τῷ ἐκ τῶν B, Z . ἐστὶν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Z πρὸς τὸν E . οἱ δὲ A, B πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσιν τοὺς τὸν αὐτὸν λόγον ἔχοντα ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα· ὁ B ἄρα τὸν E μετρεῖ, ὡς ἐπόμενος ἐπόμενον. καὶ ἐπεὶ ὁ A τοὺς B, E πολλαπλασιάσας τοὺς Γ, Δ πεποιήκεν, ἐστὶν ἄρα ὡς ὁ B πρὸς τὸν E , οὕτως ὁ Γ πρὸς τὸν Δ . μετρεῖ δὲ ὁ B τὸν E · μετρεῖ ἄρα καὶ ὁ Γ τὸν Δ ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οἱ A, B μετροῦσιν τινα ἀριθμὸν ἐλάσσονα ὄντα τοῦ Γ . ὁ Γ ἄρα ἐλάχιστος ὡς ὑπὸ τῶν A, B μετρεῖται.

For A and B are either prime to one another, or not. Let them, first of all, be prime to one another. And let A make C (by) multiplying B . Thus, B has also made C (by) multiplying A [Prop. 7.16]. Thus, A and B (both) measure C . So I say that (C) is also the least (number which they both measure). For if not, A and B will (both) measure some (other) number which is less than C . Let them (both) measure D (which is less than C). And as many times as A measures D , so many units let there be in E . And as many times as B measures D , so many units let there be in F . Thus, A has made D (by) multiplying E , and B has made D (by) multiplying F . Thus, the (number created) from (multiplying) A and E is equal to the (number created) from (multiplying) B and F . Thus, as A (is) to B , so F (is) to E [Prop. 7.19]. And A and B are prime (to one another), and prime (numbers) are the least (of those numbers having the same ratio) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser [Prop. 7.20]. Thus, B measures E , as the following (number measuring) the following. And since A has made C and D (by) multiplying B and E (respectively), thus as B is to E , so C (is) to D [Prop. 7.17]. And B measures E . Thus, C also measures D , the greater (measuring) the lesser. The very thing is impossible. Thus, A and B do not (both) measure some number which is less than C . Thus, C is the least (number) which is measured by (both) A and B .



μή ἔστωσαν δὴ οἱ A, B πρῶτοι πρὸς ἀλλήλους, καὶ ἐ-
 ιλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἔχόντων
 τοῖς A, B οἱ Z, E · ἴσοις ἄρα ἐστὶν ὁ ἐκ τῶν A, E τῷ ἐκ
 τῶν B, Z . καὶ ὁ A τὸν E πολλαπλασιάσας τὸν Γ ποιείτω·
 καὶ ὁ B ἄρα τὸν Z πολλαπλασιάσας τὸν Γ πεποιήκεν· οἱ A, B
 ἄρα τὸν Γ μετροῦσιν. λέγω δὴ, ὅτι καὶ ἐλάχιστον. ε-
 ἰ γὰρ μή, μετρήσουσιν τινα ἀριθμὸν οἱ A, B ἐλάσσονα ὄντα
 τοῦ Γ . μετρεῖτωσαν τὸν Δ . καὶ ὁσάκις μὲν ὁ A τὸν Δ μετρεῖ,
 τοσαῦται μονάδες ἔστωσαν ἐν τῷ H , ὁσάκις δὲ ὁ B τὸν Δ
 μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Θ . ὁ μὲν A ἄρα τὸν
 H πολλαπλασιάσας τὸν Δ πεποιήκεν, ὁ δὲ B τὸν Θ πολλα-

So let A and B be not prime to one another. And let the
 least numbers, F and E , be taken having the same ratio as A
 and B (respectively) [Prop. 7.33]. Thus, the (number created)
 from (multiplying) A and E is equal to the (number created)
 from (multiplying) B and F [Prop. 7.19]. And let A make C
 (by) multiplying E . Thus, B has also made C (by) multiply-
 ing F . Thus, A and B (both) measure C . So I say that (C) is
 also the least (number which they both measure). For if not,
 A and B will (both) measure some number which is less than
 C . Let them (both) measure D (which is less than C). And
 as many times as A measures D , so many units let there be in

πλασιάσας τὸν Δ πεποίηκεν. ἴσοξ ἄρα ἐστὶν ὁ ἐκ τῶν A, H τῷ ἐκ τῶν B, Θ . ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Θ πρὸς τὸν H . ὡς δὲ ὁ A πρὸς τὸν B , οὕτως ὁ Z πρὸς τὸν E . καὶ ὡς ἄρα ὁ Z πρὸς τὸν E , οὕτως ὁ Θ πρὸς τὸν H . οἱ δὲ Z, E ἐλάτισται, οἱ δὲ ἐλάτισται μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχονταξ ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα. ὁ E ἄρα τὸν H μετρεῖ. καὶ ἐπεὶ ὁ A τοὺς E, H πολλαπλασιάσας τοὺς Γ, Δ πεποίηκεν, ἔστιν ἄρα ὡς ὁ E πρὸς τὸν H , οὕτως ὁ Γ πρὸς τὸν Δ . ὁ δὲ E τὸν H μετρεῖ· καὶ ὁ Γ ἄρα τὸν Δ μετρεῖ ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οἱ A, B μετρήσουσιν τινα ἀριθμὸν ἐλάσσονα ὄντα τοῦ Γ . ὁ Γ ἄρα ἐλάτισταξ ὡν ὑπὸ τῶν A, B μετρεῖται· ὅπερ ἔπει δεῖχαι.

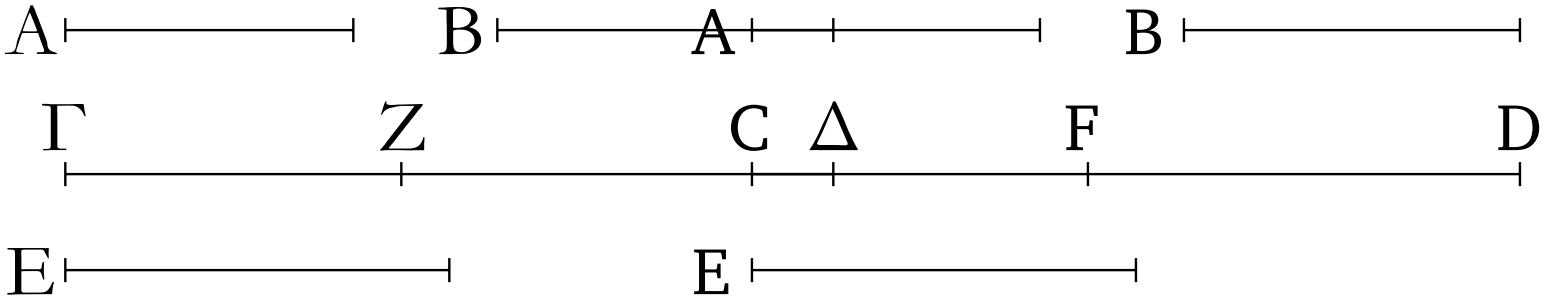
G . And as many times as B measures D , so many units let there be in H . Thus, A has made D (by) multiplying G , and B has made D (by) multiplying H . Thus, the (number created) from (multiplying) A and G is equal to the (number created) from (multiplying) B and H . Thus, as A is to B , so H (is) to G [Prop. 7.19]. And as A (is) to B , so F (is) to E . Thus, also, as F (is) to E , so H (is) to G . And F and E are the least (numbers having the same ratio as A and B), and the least (numbers) measure those (numbers) having the same ratio an equal number of times, the greater (measuring) the greater, and the lesser the lesser [Prop. 7.20]. Thus, E measures G . And since A has made C and D (by) multiplying E and G (respectively), thus as E is to G , so C (is) to D [Prop. 7.17]. And E measures G . Thus, C also measures D , the greater (measuring) the lesser. The very thing is impossible. Thus, A and B do not (both) measure some (number) which is less than C . Thus, C (is) the least (number) which is measured by (both) A and B . (Which is) the very thing it was required to show.

35.

Proposition 35

Ἐάν δύο ἀριθμοὶ ἀριθμὸν τινα μετρῶσιν, καὶ ὁ ἐλάτισταξ ὑπ' αὐτῶν μετρούμενοξ τὸν αὐτὸν μετρήσει.

If two numbers (both) measure some number, (then) the least (number) measured by them will also measure the same (number).



Δύο γὰρ ἀριθμοὶ οἱ A, B ἀριθμὸν τινα τὸν $\Gamma\Delta$ μετρεῖτωσαν, ἐλάτισταξ δὲ τὸν E . λέγω, ὅτι καὶ ὁ E τὸν $\Gamma\Delta$ μετρεῖ.

Εἰ γὰρ οὐ μετρεῖ ὁ E τὸν $\Gamma\Delta$, ὁ E τὸν ΔZ μετρῶν λειπέτω εἰς τοῦ ἐλάσσονα τὸν ΓZ . καὶ ἐπεὶ οἱ A, B τὸν E μετροῦσιν, ὁ δὲ E τὸν ΔZ μετρεῖ, καὶ οἱ A, B ἄρα τὸν ΔZ μετρήσουσιν. μετροῦσι δὲ καὶ ὅλον τὸν $\Gamma\Delta$. καὶ λοιπὸν ἄρα τὸν ΓZ μετρήσουσιν ἐλάσσονα ὄντα τοῦ E . ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οὐ μετρεῖ ὁ E τὸν $\Gamma\Delta$. μετρεῖ ἄρα· ὅπερ ἔδει δεῖχαι.

For let two numbers, A and B , (both) measure some number CD , and (let) E (be the) least (number measured by both A and B). I say that E also measures CD .

For if E does not measure CD , (then) let E leave CF less than itself (in) measuring DF . And since A and B (both) measure E , and E measures DF , A and B will thus also measure DF . And (A and B) also measure the whole of CD . Thus, they will also measure the remainder CF , which is less than E . The very thing is impossible. Thus, E cannot not measure CD . Thus, (E) measures (CD). (Which is) the very thing it was required to show.

36.

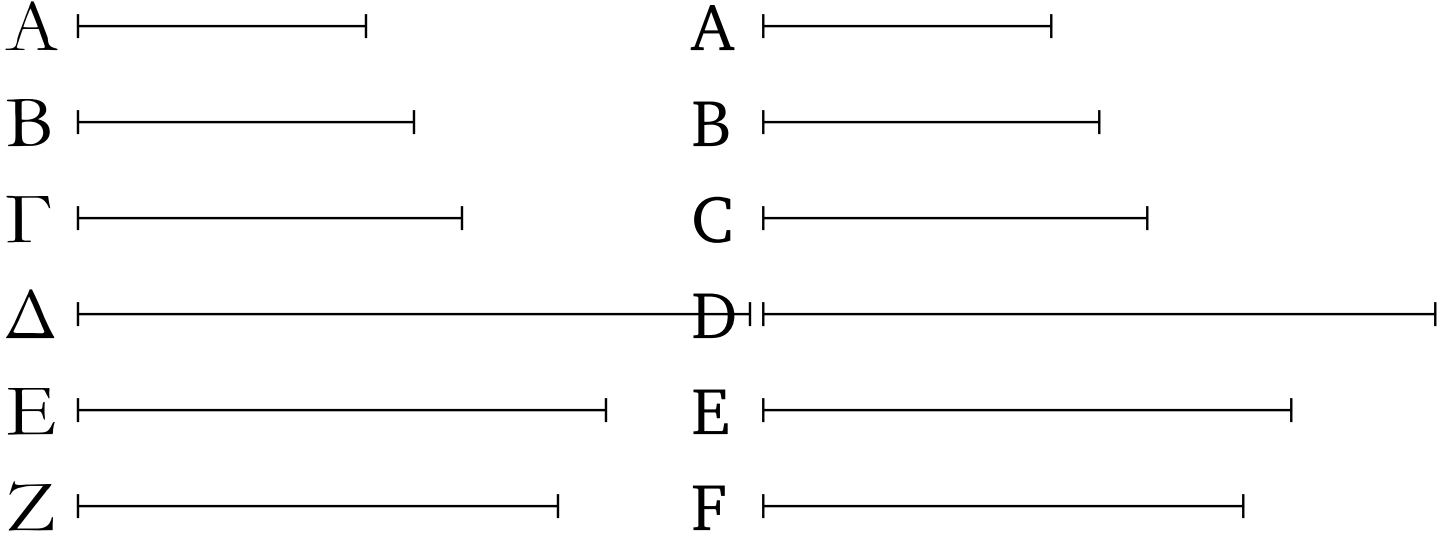
Proposition 36

Τριῶν ἀριθμῶν δοθέντων εὑρεῖν, ὃν ἐλάτισταξ μετροῦσιν ἀριθμὸν.

Ἔστωσαν οἱ δοθέντεξ τρεῖξ ἀριθμοὶ οἱ A, B, Γ . δεῖ δὴ εὑρεῖν, ὃν ἐλάτισταξ μετροῦσιν ἀριθμὸν.

To find the least number which three given numbers (all) measure.

Let A, B , and C be the three given numbers. So it is required to find the least number which they (all) measure.



Εἰλήφθω γὰρ ὑπὸ δύο τῶν A, B ἐλάχιστος μετρούμενος ὁ Δ . ὁ δὲ Γ τὸν Δ ἤτοι μετρεῖ ἢ οὐ μετρεῖ. μετρεῖτω πρότερον. μετροῦσι δὲ καὶ οἱ A, B τὸν Δ . οἱ A, B, Γ ἄρα τὸν Δ μετροῦσιν. λέγω δὴ, ὅτι καὶ ἐλάχιστον. εἰ γὰρ μή, μετρήσουσιν [τινα] ἀριθμὸν οἱ A, B, Γ ἐλάσσονα ὄντα τοῦ Δ . μετρεῖτωσαν τὸν E . ἐπεὶ οἱ A, B, Γ τὸν E μετροῦσιν, καὶ οἱ A, B ἄρα τὸν E μετροῦσιν. καὶ ὁ ἐλάχιστος ἄρα ὑπὸ τῶν A, B μετρούμενος [τὸν E] μετρήσει. ἐλάχιστος δὲ ὑπὸ τῶν A, B μετρούμενός ἐστιν ὁ Δ . ὁ Δ ἄρα τὸν E μετρήσει ὁ μείζων τὸν ἐλάσσονα. ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οἱ A, B, Γ μετρήσουσιν τινα ἀριθμὸν ἐλάσσονα ὄντα τοῦ Δ . οἱ A, B, Γ ἄρα ἐλάχιστον τὸν Δ μετροῦσιν.

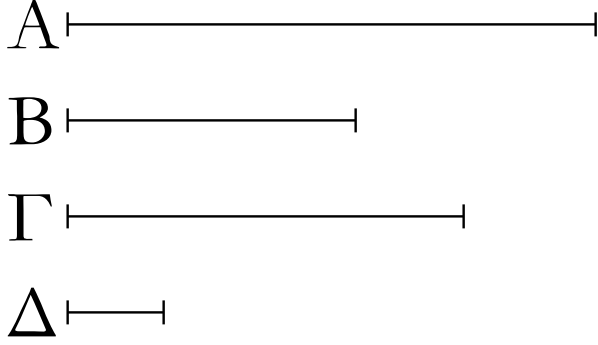
μή μετρεῖτω δὴ πάλιν ὁ Γ τὸν Δ , καὶ εἰλήφθω ὑπὸ τῶν Γ, Δ ἐλάχιστος μετρούμενος ἀριθμὸς ὁ E . ἐπεὶ οἱ A, B τὸν Δ μετροῦσιν, ὁ δὲ Δ τὸν E μετρεῖ, καὶ οἱ A, B ἄρα τὸν E μετροῦσιν. μετρεῖ δὲ καὶ ὁ Γ [τὸν E καὶ] οἱ A, B, Γ ἄρα τὸν E μετροῦσιν. λέγω δὴ, ὅτι καὶ ἐλάχιστον. εἰ γὰρ μή, μετρήσουσιν τινα οἱ A, B, Γ ἐλάσσονα ὄντα τοῦ E . μετρεῖτωσαν τὸν Z . ἐπεὶ οἱ A, B, Γ τὸν Z μετροῦσιν, καὶ οἱ A, B ἄρα τὸν Z μετροῦσιν· καὶ ὁ ἐλάχιστος ἄρα ὑπὸ τῶν A, B μετρούμενος τὸν Z μετρήσει. ἐλάχιστος δὲ ὑπὸ τῶν A, B μετρούμενός ἐστιν ὁ Δ . ὁ Δ ἄρα τὸν Z μετρεῖ. μετρεῖ δὲ καὶ ὁ Γ τὸν Z . οἱ Δ, Γ ἄρα τὸν Z μετροῦσιν· ὥστε καὶ ὁ ἐλάχιστος ὑπὸ τῶν Δ, Γ μετρούμενος τὸν Z μετρήσει. ὁ δὲ ἐλάχιστος ὑπὸ τῶν Γ, Δ μετρούμενός ἐστιν ὁ E . ὁ E ἄρα τὸν Z μετρεῖ ὁ μείζων τὸν ἐλάσσονα. ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οἱ A, B, Γ μετρήσουσιν τινα ἀριθμὸν ἐλάσσονα ὄντα τοῦ E . ὁ E ἄρα ἐλάχιστος ὡν ὑπὸ τῶν A, B, Γ μετρεῖται· ὅπερ ἔδει δεῖξαι.

For let the least (number), D , measured by the two (numbers) A and B be taken [Prop. 7.34]. So C either measures, or does not measure, D . Let it, first of all, measure (D). And A and B also measure D . Thus, A, B , and C (all) measure D . So I say that (D is) also the least (number measured by A, B , and C). For if not, A, B , and C will (all) measure [some] number which is less than D . Let them measure E (which is less than D). Since A, B , and C (all) measure E , (then) A and B thus also measure E . Thus, the least (number) measured by A and B will also measure [E] [Prop. 7.35]. And D is the least (number) measured by A and B . Thus, D will measure E , the greater (measuring) the lesser. The very thing is impossible. Thus, A, B , and C cannot (all) measure some number which is less than D . Thus, A, B , and C (all) measure the least (number) D .

So, again, let C not measure D . And let the least number, E , measured by C and D be taken [Prop. 7.34]. Since A and B measure D , and D measures E , A and B thus also measure E . And C also measures [E]. Thus, A, B , and C [also] measure E . So I say that (E is) also the least (number measured by A, B , and C). For if not, A, B , and C will (all) measure some (number) which is less than E . Let them measure F (which is less than E). Since A, B , and C (all) measure F , A and B thus also measure F . Thus, the least (number) measured by A and B will also measure F [Prop. 7.35]. And D is the least (number) measured by A and B . Thus, D measures F . And C also measures F . Thus, D and C (both) measure F . Hence, the least (number) measured by D and C will also measure F [Prop. 7.35]. And E is the least (number) measured by C and D . Thus, E measures F , the greater (measuring) the lesser. The very thing is impossible. Thus, A, B , and C cannot measure some number which is less than E . Thus, E (is) the least (number) which is measured by A, B , and C . (Which is) the very thing it was required to show.

37.

Ἐάν ἀριθμὸς ὑπὸ τινος ἀριθμοῦ μετρηῇται, ὁ μετρούμενος ὁμώνυμον μέρος ἔχει τῷ μετρούντι.

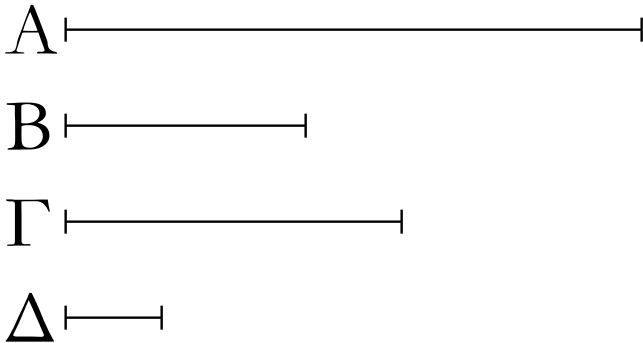


Ἀριθμὸς γάρ ὁ A ὑπὸ τινος ἀριθμοῦ τοῦ B μετρεῖσθω· λέγω, ὅτι ὁ A ὁμώνυμον μέρος ἔχει τῷ B .

Ὅσάκις γάρ ὁ B τὸν A μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Γ . Ἐπεὶ ὁ B τὸν A μετρεῖ κατὰ τὰς ἐν τῷ Γ μονάδας, μετρεῖ δὲ καὶ ἡ Δ μονὰς τὸν Γ ἀριθμὸν κατὰ τὰς ἐν αὐτῷ μονάδας, ἰσάκις ἄρα ἡ Δ μονὰς τὸν Γ ἀριθμὸν μετρεῖ καὶ ὁ B τὸν A . ἐναλλάξ ἄρα ἰσάκις ἡ Δ μονὰς τὸν B ἀριθμὸν μετρεῖ καὶ ὁ Γ τὸν A . ὁ ἄρα μέρος ἐστὶν ἡ Δ μονὰς τοῦ B ἀριθμοῦ, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ Γ τοῦ A . ἡ δὲ Δ μονὰς τοῦ B ἀριθμοῦ μέρος ἐστὶν ὁμώνυμον αὐτῷ· καὶ ὁ Γ ἄρα τοῦ A μέρος ἐστὶν ὁμώνυμον τῷ B . ὥστε ὁ A μέρος ἔχει τὸν Γ ὁμώνυμον ὄντα τῷ B . ὅπερ ἔδει δεῖχαι.

38.

Ἐάν ἀριθμὸς μέρος ἔθῃ οἷον, ὑπὸ ὁμωνύμου ἀριθμοῦ μετρηθήσεται τῷ μέρει.

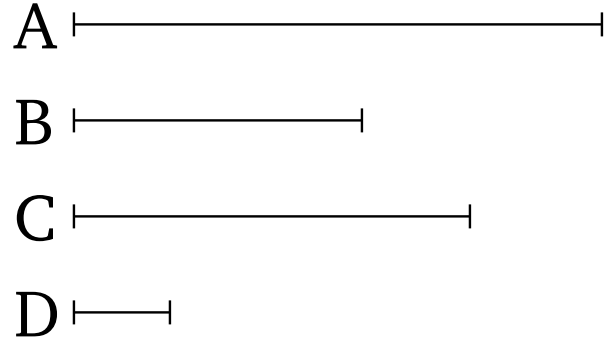


Ἀριθμὸς γάρ ὁ A μέρος ἐθέτω οἷον τὸν B , καὶ τῷ B μέρει ὁμώνυμος ἔστω [ἀριθμὸς] ὁ Γ . λέγω, ὅτι ὁ Γ τὸν A μετρεῖ.

Ἐπεὶ γάρ ὁ B τοῦ A μέρος ἐστὶν ὁμώνυμον τῷ Γ , ἔστι δὲ καὶ ἡ Δ μονὰς τοῦ Γ μέρος ὁμώνυμον αὐτῷ, ὁ ἄρα μέρος

Proposition 37

If a number is measured by some number, (then) the (number) measured will have a part called the same as the measuring (number).

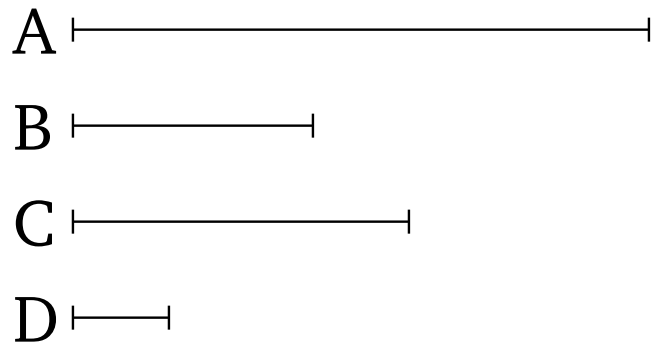


For let the number A be measured by some number B . I say that A has a part called the same as B .

For as many times as B measures A , so many units let there be in C . Since B measures A according to the units in C , and the unit D also measures C according to the units in it, the unit D thus measures the number C as many times as B (measures) A . Thus, alternately, the unit D measures the number B as many times as C (measures) A [Prop. 7.15]. Thus, which(ever) part the unit D is of the number B , C is also the same part of A . And the unit D is a part of the number B called the same as it (i.e., a B th part). Thus, C is also a part of A called the same as B (i.e., C is the B th part of A). Hence, A has a part C which is called the same as B (i.e., A has a B th part). (Which is) the very thing it was required to show.

Proposition 38

If a number has any part whatever, (then) it will be measured by a number called the same as the part.



For let the number A have any part whatever, B . And let the [number] C be called the same as the part B (i.e., B is the C th part of A). I say that C measures A .

For since B is a part of A called the same as C , and the unit D is also a part of C called the same as it (i.e., D is the C th part

ἐστὶν ἡ Δ μονὰξ τοῦ Γ ἀριθμοῦ, τὸ αὐτὸ μέρος ἐστὶ καὶ ὁ B τοῦ A : ἰσάκις ἄρα ἡ Δ μονὰξ τὸν Γ ἀριθμὸν μετρεῖ καὶ ὁ B τὸν A . ἐναλλάξ ἄρα ἰσάκις ἡ Δ μονὰξ τὸν B ἀριθμὸν μετρεῖ καὶ ὁ Γ τὸν A . ὁ Γ ἄρα τὸν A μετρεῖ· ὅπερ ἔδει δεῖχαι.

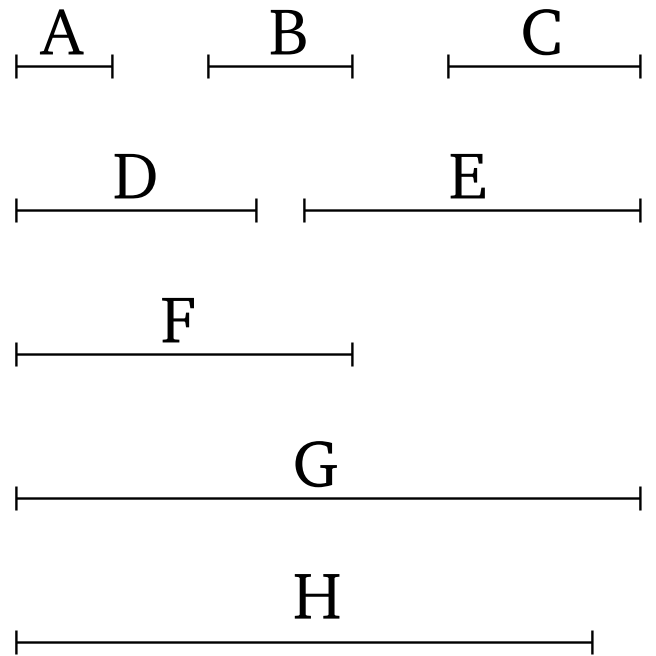
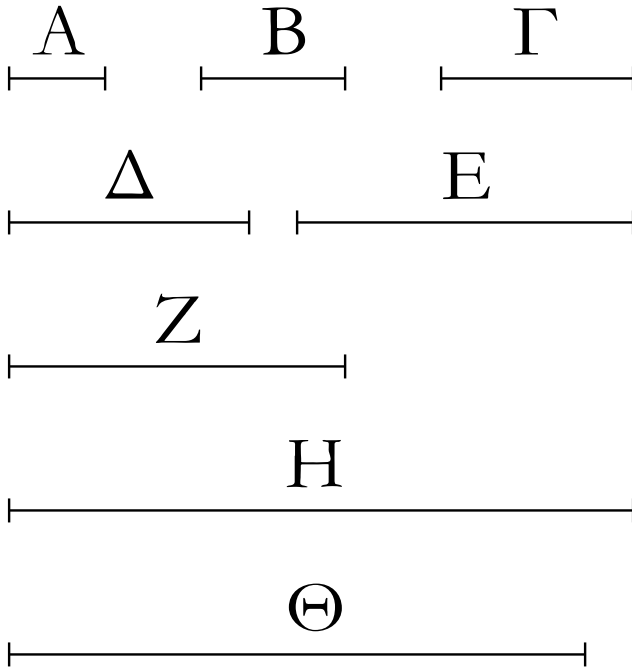
of C), thus which(ever) part the unit D is of the number C , B is also the same part of A . Thus, the unit D measures the number C as many times as B (measures) A . Thus, alternately, the unit D measures the number B as many times as C (measures) A [Prop. 7.15]. Thus, C measures A . (Which is) the very thing it was required to show.

39.

Ἀριθμὸν εὐρεῖν, ὃς ἐλάχιστος ὢν ἔχει τὰ δοθέντα μέρη.

Proposition 39

To find the least number that will have given parts.



Ἐστω τὰ δοθέντα μέρη τὰ A, B, Γ · δεῖ δὴ ἀριθμὸν εὐρεῖν, ὃς ἐλάχιστος ὢν ἔχει τὰ A, B, Γ μέρη.

Ἐστωσαν γὰρ τοῖς A, B, Γ μέρεσιν ὁμώνυμοι ἀριθμοὶ οἱ Δ, E, Z , καὶ εἰλήφθω ὑπὸ τῶν Δ, E, Z ἐλάχιστος μετρούμενος ἀριθμὸς ὁ H .

Ὁ H ἄρα ὁμώνυμα μέρη ἔθει τοῖς Δ, E, Z . τοῖς δὲ Δ, E, Z ὁμώνυμα μέρη ἐστὶ τὰ A, B, Γ · ὁ H ἄρα ἔθει τὰ A, B, Γ μέρη. λέγω δὴ, ὅτι καὶ ἐλάχιστος ὢν, εἰ γὰρ μή, ἔσται τις τοῦ H ἐλάσσων ἀριθμὸς, ὃς ἔχει τὰ A, B, Γ μέρη. ἔστω ὁ Θ . ἐπεὶ ὁ Θ ἔθει τὰ A, B, Γ μέρη, ὁ Θ ἄρα ὑπὸ ὁμωνύμων ἀριθμῶν μετρηθήσεται τοῖς A, B, Γ μέρεσιν. τοῖς δὲ A, B, Γ μέρεσιν ὁμώνυμοι ἀριθμοὶ εἰσιν οἱ Δ, E, Z · ὁ Θ ἄρα ὑπὸ τῶν Δ, E, Z μετρεῖται. καὶ ἐστὶν ἐλάσσων τοῦ H · ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἔσται τις τοῦ H ἐλάσσων ἀριθμὸς, ὃς ἔχει τὰ A, B, Γ μέρη· ὅπερ ἔδει δεῖχαι.

Let A, B , and C be the given parts. So it is required to find the least number which will have the parts A, B , and C (i.e., an A th part, a B th part, and a C th part).

For let D, E , and F be numbers having the same names as the parts A, B , and C (respectively). And let the least number, G , measured by D, E , and F , be taken [Prop. 7.36].

Thus, G has parts called the same as D, E , and F [Prop. 7.37]. And A, B , and C are parts called the same as D, E , and F (respectively). Thus, G has the parts A, B , and C . So I say that (G) is also the least (number having the parts A, B , and C). For if not, there will be some number less than G which will have the parts A, B , and C . Let it be H . Since H has the parts A, B , and C , H will thus be measured by numbers called the same as the parts A, B , and C [Prop. 7.38]. And D, E , and F are numbers called the same as the parts A, B , and C (respectively). Thus, H is measured by D, E , and F . And (H) is less than G . The very thing is impossible. Thus, there cannot be some number less than G which will have the parts

A, *B*, and *C*. (Which is) the very thing it was required to show.

ELEMENTS BOOK 8

Continued Proportion[†]

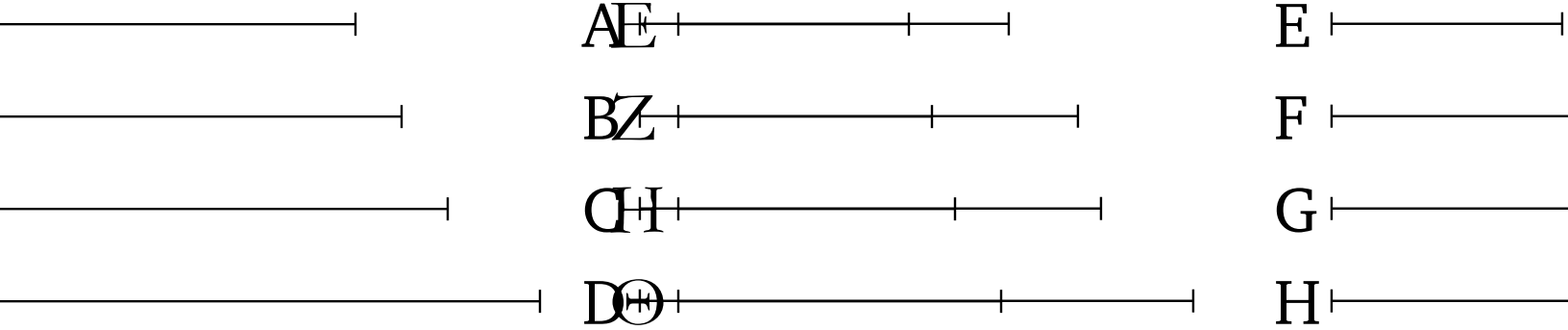
[†]The propositions contained in Books 7–9 are generally attributed to the school of Pythagoras.

1.

Proposition 1

Ἐάν ὧσιν ὁσοιδηποτοῦν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ δὲ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους ὧσιν, ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

If there are any multitude whatsoever of continuously proportional numbers, and the outermost of them are prime to one another, then the (numbers) are the least of those (numbers) having the same ratio as them.



Ἔστωσαν ὅποιοι ἄριθμοι ἐχῇς ἀνάλογον οἱ A, B, Γ, Δ , οἱ δὲ ἄκροι αὐτῶν οἱ A, Δ , πρῶτοι πρὸς ἀλλήλους ἔστωσαν λέγω, ὅτι οἱ A, B, Γ, Δ ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

Let A, B, C, D be any multitude whatsoever of continuously proportional numbers. And let the outermost of them, A and D , be prime to one another. I say that A, B, C, D are the least of those (numbers) having the same ratio as them.

Εἰ γὰρ μή, ἔστωσαν ἐλάττωρες τῶν A, B, Γ, Δ οἱ E, Z, H, Θ ἐν τῷ αὐτῷ λόγῳ ὄντες αὐτοῖς. καὶ ἐπεὶ οἱ A, B, Γ, Δ ἐν τῷ αὐτῷ λόγῳ εἰσὶ τοῖς E, Z, H, Θ , καὶ ἐστὶν ἴσον τὸ πλῆθος [τῶν A, B, Γ, Δ] τῷ πλῆθει [τῶν E, Z, H, Θ], δι' ἴσον ἄρα ἐστὶν ὡς ὁ A πρὸς τὸν Δ , ὁ E πρὸς τὸν Θ . οἱ δὲ A, Δ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἐθονταῖς ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. μετρεῖ ἄρα ὁ A τὸν E ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οἱ E, Z, H, Θ ἐλάσσονες ὄντες τῶν A, B, Γ, Δ ἐν τῷ αὐτῷ λόγῳ εἰσὶν αὐτοῖς. οἱ A, B, Γ, Δ ἄρα ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς· ὅπερ ἔδει δεῖξαι.

For if not, let E, F, G, H be less than A, B, C, D (respectively), being in the same ratio as them. And since A, B, C, D are in the same ratio as E, F, G, H , and the multitude [of A, B, C, D] is equal to the multitude [of E, F, G, H], thus, via equality, as A is to D , (so) E (is) to H [Prop. 7.14]. And A and D (are) prime (to one another). And prime (numbers are) also the least of those (numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures E , the greater (measuring) the lesser. The very thing is impossible. Thus, E, F, G, H , being less than A, B, C, D , are not in the same ratio as them. Thus, A, B, C, D are the least of those (numbers) having the same ratio as them. (Which is) the very thing it was required to show.

2.

Proposition 2

Αριθμοὺς εὐρεῖν ἐχῇς ἀνάλογον ἐλαθίστους, ὅσους ἂν ἐπιτάχῃ τις, ἐν τῷ δοθέντι λόγῳ.

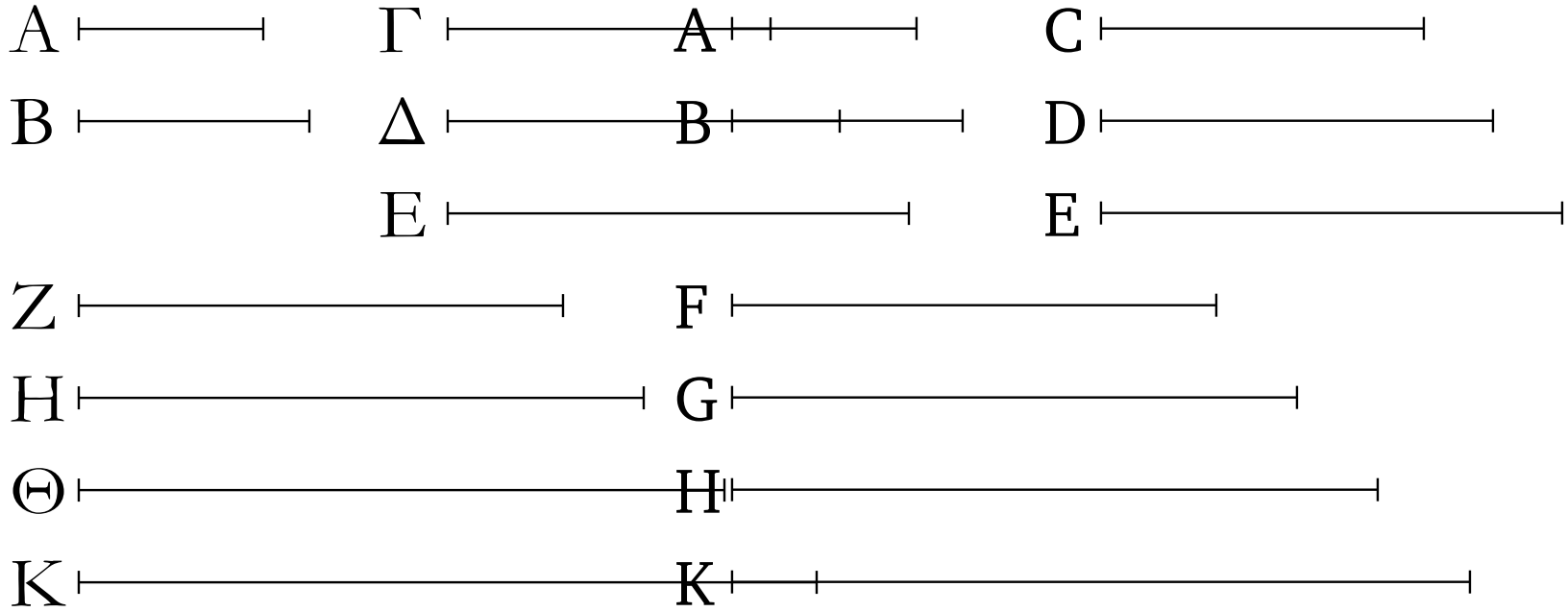
To find the least numbers, as many as may be prescribed, (which are) continuously proportional in a given ratio.

Ἔστω ὁ δοθείς λόγος ἐν ἐλαθίστοις ἀριθμοῖς ὁ τοῦ A πρὸς τὸν B . δεῖ δὴ ἀριθμοὺς εὐρεῖν ἐχῇς ἀνάλογον ἐλαθίστους, ὅσους ἂν τις ἐπιτάχῃ, ἐν τῷ τοῦ A πρὸς τὸν B λόγῳ.

Let the given ratio, (expressed) in the least numbers, be that of A to B . So it is required to find the least numbers, as many as may be prescribed, (which are) in the ratio of A to B .

Ἐπιτετάθωσαν δὴ τέσσαρες, καὶ ὁ A ἐαντὸν πολλαπλασιάσας τὸν Γ ποιέτω, τὸν δὲ B πολλαπλασιάσας τὸν Δ ποιέτω, καὶ ἔτι ὁ B ἐαντὸν πολλαπλασιάσας τὸν E ποιέτω, καὶ ἔτι ὁ A τοὺς Γ, Δ, E πολλαπλασιάσας τοὺς Z, H, Θ ποιέτω, ὁ δὲ B τὸν E πολλαπλασιάσας τὸν K ποιέτω.

Let four (numbers) be prescribed. And let A make C (by) multiplying itself, and let it make D (by) multiplying B . And, further, let B make E (by) multiplying itself. And, further, let A make F, G, H (by) multiplying C, D, E . And let B make K (by) multiplying E .



Καὶ ἐπεὶ ὁ A ἐαυτὸν μὲν πολλαπλασιάσας τὸν Γ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Δ πεποίηκεν, ἔστιν ἄρα ὥς ὁ A πρὸς τὸν B , [οὕτως] ὁ Γ πρὸς τὸν Δ . πάλιν, ἐπεὶ ὁ μὲν A τὸν B πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ δὲ B ἐαυτὸν πολλαπλασιάσας τὸν E πεποίηκεν, ἐκάτερος ἄρα τῶν A , B τὸν B πολλαπλασιάσας ἐκάτερον τῶν Δ , E πεποίηκεν. ἔστιν ἄρα ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E . ἀλλ' ὥς ὁ A πρὸς τὸν B , ὁ Γ πρὸς τὸν Δ · καὶ ὥς ἄρα ὁ Γ πρὸς τὸν Δ , ὁ Δ πρὸς τὸν E . καὶ ἐπεὶ ὁ A τοὺς Γ , Δ πολλαπλασιάσας τοὺς Z , H πεποίηκεν, ἔστιν ἄρα ὥς ὁ Γ πρὸς τὸν Δ , [οὕτως] ὁ Z πρὸς τὸν H . ὥς δὲ ὁ Γ πρὸς τὸν Δ , οὕτως ἦν ὁ A πρὸς τὸν B · καὶ ὥς ἄρα ὁ A πρὸς τὸν B , ὁ Z πρὸς τὸν H . πάλιν, ἐπεὶ ὁ A τοὺς Δ , E πολλαπλασιάσας τοὺς H , Θ πεποίηκεν, ἔστιν ἄρα ὥς ὁ Δ πρὸς τὸν E , ὁ H πρὸς τὸν Θ . ἀλλ' ὥς ὁ Δ πρὸς τὸν E , ὁ A πρὸς τὸν B . καὶ ὥς ἄρα ὁ A πρὸς τὸν B , οὕτως ὁ H πρὸς τὸν Θ . καὶ ἐπεὶ οἱ A , B τὸν E πολλαπλασιάσαντες τοὺς Θ , K πεποιήκασιν, ἔστιν ἄρα ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Θ πρὸς τὸν K . ἀλλ' ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Z πρὸς τὸν H καὶ ὁ H πρὸς τὸν Θ . καὶ ὥς ἄρα ὁ Z πρὸς τὸν H , οὕτως ὁ Θ πρὸς τὸν K . οἱ Γ , Δ , E ἄρα καὶ οἱ Z , H , Θ , K ἀνάλογόν εἰσιν ἐν τῷ τοῦ A πρὸς τὸν B λόγῳ. λέγω δὴ, ὅτι καὶ ἐλάχιστοι. ἐπεὶ γὰρ οἱ A , B ἐλάχιστοι εἰσὶ τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ δὲ ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων πρῶτοι πρὸς ἀλλήλους εἰσὶν, οἱ A , B ἄρα πρῶτοι πρὸς ἀλλήλους εἰσὶν. καὶ ἐκάτερος μὲν τῶν A , B ἐαυτὸν πολλαπλασιάσας ἐκάτερον τῶν Γ , E πεποίηκεν, ἐκάτερον δὲ τῶν Γ , E πολλαπλασιάσας ἐκάτερον τῶν Z , K πεποίηκεν· οἱ Γ , E ἄρα καὶ οἱ Z , K πρῶτοι πρὸς ἀλλήλους εἰσὶν. ἔαν δὲ ὥσιν ὁποιοῖσιν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ δὲ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους ὦσιν, ἐλάχιστοι εἰσὶ τῶν τὸν αὐτὸν λόγον ἐθόν-

And since A has made C (by) multiplying itself, and has made D (by) multiplying B , thus as A is to B , [so] C (is) to D [Prop. 7.17]. Again, since A has made D (by) multiplying B , and B has made E (by) multiplying itself, A , B have thus made D , E , respectively, (by) multiplying B . Thus, as A is to B , so D (is) to E [Prop. 7.18]. But, as A (is) to B , (so) C (is) to D . And thus as C (is) to D , (so) D (is) to E . And since A has made F , G (by) multiplying C , D , thus as C is to D , [so] F (is) to G [Prop. 7.17]. And as C (is) to D , so A was to B . And thus as A (is) to B , (so) F (is) to G . Again, since A has made G , H (by) multiplying D , E , thus as D is to E , (so) G (is) to H [Prop. 7.17]. But, as D (is) to E , (so) A (is) to B . And thus as A (is) to B , so G (is) to H . And since A , B have made H , K (by) multiplying E , thus as A is to B , so H (is) to K . But, as A (is) to B , so F (is) to G , and G to H . And thus as F (is) to G , so G (is) to H , and H to K . Thus, C , D , E and F , G , H , K are (both continuously) proportional in the ratio of A to B . So I say that (they are) also the least (sets of numbers continuously proportional in that ratio). For since A and B are the least of those (numbers) having the same ratio as them, and the least of those (numbers) having the same ratio are prime to one another [Prop. 7.22], A and B are thus prime to one another. And A , B have made C , E , respectively, (by) multiplying themselves, and have made F , K by multiplying C , E , respectively. Thus, C , E and F , K are prime to one another [Prop. 7.27]. And if there are any multitude whatsoever of continuously proportional numbers, and the outermost of them are prime to one another, then the (numbers) are the least of those (numbers) having the same ratio as them [Prop. 8.1]. Thus, C , D , E and F , G , H , K are the least of those (continuously proportional sets of numbers) having the same ratio as

των αὐτοῖς. οἱ Γ , Δ , E ἄρα καὶ οἱ Z , H , Θ , K ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A , B · ὅπερ ἔδει δεῖχαι.

A and B . (Which is) the very thing it was required to show.

Πόρισμα.

Corollary

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ἐλάχιστοι ᾧσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ ἄκρον αὐτῶν τετράγωνοί εἰσιν, ἔαν δὲ τέσσαρες, κύβοι.

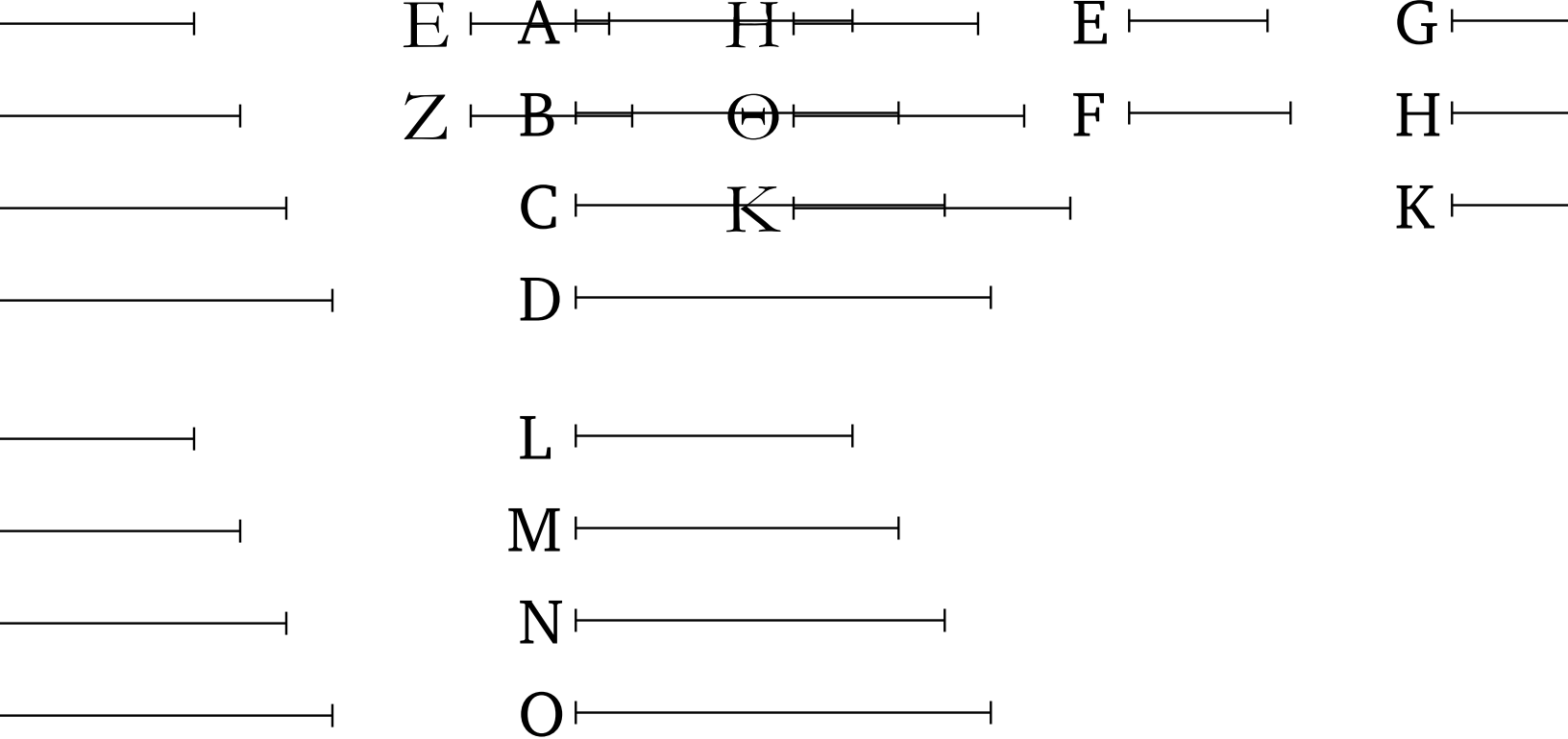
So it is clear, from this, that if three continuously proportional numbers are the least of those (numbers) having the same ratio as them then the outermost of them are square, and, if four (numbers), cube.

3.

Proposition 3

Ἐάν ᾧσιν ὅποιοι ἄριθμοι ἐχῇς ἀνάλογον ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν.

If there are any multitude whatsoever of continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them then the outermost of them are prime to one another.



Ἔστωσαν ὅποιοι ἄριθμοι ἐχῇς ἀνάλογον ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς οἱ A , B , Γ , Δ · λέγω, ὅτι οἱ ἄκροι αὐτῶν οἱ A , Δ πρῶτοι πρὸς ἀλλήλους εἰσίν.

Εἰλήφθωσαν γάρ δύο μὲν ἀριθμοὶ ἐλάχιστοι ἐν τῷ τῶν A , B , Γ , Δ λόγῳ οἱ E , Z , τρεῖς δὲ οἱ H , Θ , K , καὶ ἐχῇς ἐνὶ πλείονι, ἕως τὸ λαμβανόμενον πλήθος ἴσον γένηται τῷ πλήθει τῶν A , B , Γ , Δ . εἰλήφθωσαν καὶ ἔστωσαν οἱ L , M , N , X .

Καὶ ἐπεὶ οἱ E , Z ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ ἑκάτερος τῶν E , Z ἑαυτὸν μὲν πολλαπλασιάσας ἑκάτερον τῶν H , K πεποίηκεν, ἑκάτερον δὲ τῶν H , K πολλαπλασιάσας ἑκάτερον τῶν

Let A , B , C , D be any multitude whatsoever of continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them. I say that the outermost of them, A and D , are prime to one another.

For let the two least (numbers) E , F (which are) in the same ratio as A , B , C , D be taken [Prop. 7.33]. And the three (least numbers) G , H , K [Prop. 8.2]. And (so on), successively increasing by one, until the multitude of (numbers) taken is made equal to the multitude of A , B , C , D . Let them be taken, and let them be L , M , N , O .

And since E and F are the least of those (numbers) hav-

A, X πεποιήκεν, καὶ οἱ H, K ἄρα καὶ οἱ A, X πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ οἱ A, B, Γ, Δ ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, εἰσὶ δὲ καὶ οἱ A, M, N, X ἐλάχιστοι ἐν τῷ αὐτῷ λόγῳ ὄντες τοῖς A, B, Γ, Δ , καὶ ἐστὶν ἴσον τὸ πλῆθος τῶν A, B, Γ, Δ τῷ πλῆθει τῶν A, M, N, X , ἕκαστος ἄρα τῶν A, B, Γ, Δ ἐκάστω τῶν A, M, N, X ἴσος ἐστίν· ἴσος ἄρα ἐστὶν ὁ μὲν A τῷ A , ὁ δὲ Δ τῷ X . καὶ εἰσιν οἱ A, X πρῶτοι πρὸς ἀλλήλους. καὶ οἱ A, Δ ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν· ὅπερ εἶδει δεῖχαι.

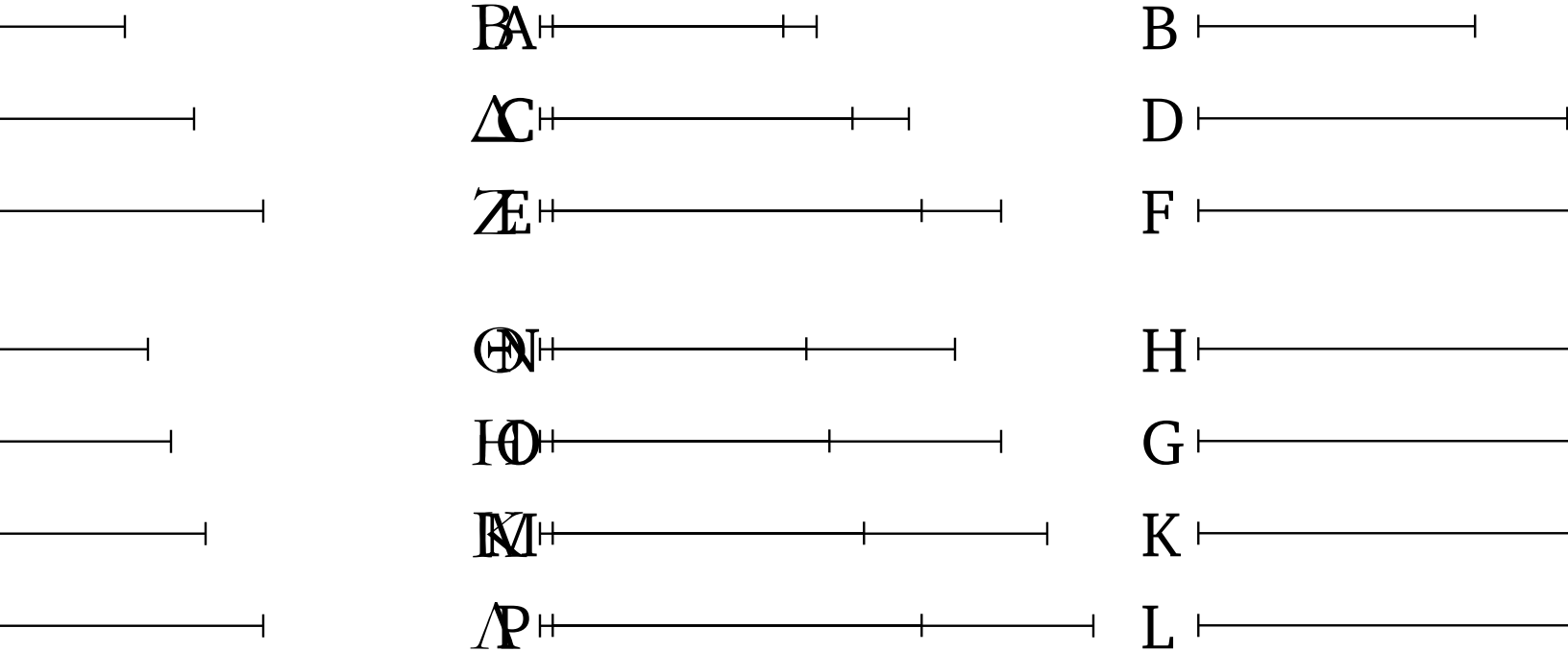
ing the same ratio as them they are prime to one another [Prop. 7.22]. And since E, F have made G, K , respectively, (by) multiplying themselves [Prop. 8.2 corr.], and have made L, O (by) multiplying G, K , respectively, G, K and L, O are thus also prime to one another [Prop. 7.27]. And since A, B, C, D are the least of those (numbers) having the same ratio as them, and L, M, N, O are also the least (of those numbers having the same ratio as them), being in the same ratio as A, B, C, D , and the multitude of A, B, C, D is equal to the multitude of L, M, N, O , thus A, B, C, D are equal to L, M, N, O , respectively. Thus, A is equal to L , and D to O . And L and O are prime to one another. Thus, A and D are also prime to one another. (Which is) the very thing it was required to show.

4.

Proposition 4

Λόγων δοθέντων ὅποσωνοῦν ἐν ἐλάχιστοις ἀριθμοῖς ἀριθμοὺς εὑρεῖν ἐχῆς ἀνάλογον ἐλάχιστους ἐν τοῖς δοθεῖσι λόγοις.

For any multitude whatsoever of given ratios, (expressed) in the least numbers, to find the least numbers continuously proportional in these given ratios.



Ἐστωσαν οἱ δοθέντες λόγοι ἐν ἐλάχιστοις ἀριθμοῖς ὅ τε τοῦ A πρὸς τὸν B καὶ ὁ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι ὁ τοῦ E πρὸς τὸν Z . δεῖ δὲ ἀριθμοὺς εὑρεῖν ἐχῆς ἀνάλογον ἐλάχιστους ἐν τε τῷ τοῦ A πρὸς τὸν B λόγῳ καὶ ἐν τῷ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι τῷ τοῦ E πρὸς τὸν Z .

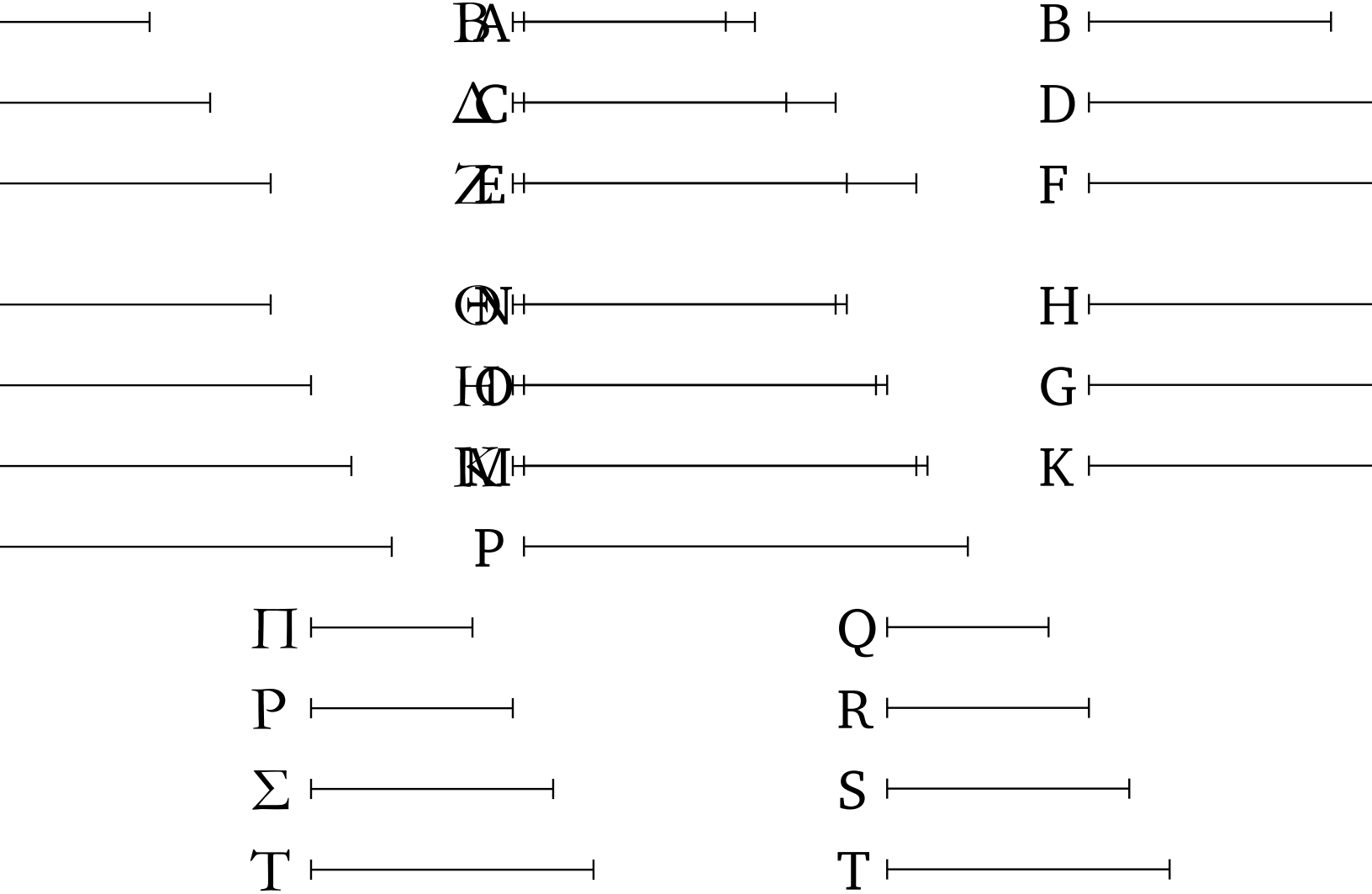
Εἰλήφθω γὰρ ὁ ὑπὸ τῶν B, Γ ἐλάχιστος μετρούμενος ἀριθμὸς ὁ H . καὶ ὁσάκις μὲν ὁ B τὸν H μετρεῖ, τοσαντάκις καὶ ὁ A τὸν Θ μετρεῖτω, ὁσάκις δὲ ὁ Γ τὸν H μετρεῖ, τοσαντάκις καὶ ὁ Δ τὸν K μετρεῖτω. ὁ δὲ E τὸν K ἤτοι μετρεῖ ἢ οὐ μετρεῖ. μετρεῖτω πρότερον. καὶ ὁσάκις ὁ E τὸν K μετρεῖ, τοσαντάκις καὶ ὁ Z τὸν Λ μετρεῖτω. καὶ ἐπεὶ ἰσάκις ὁ A τὸν Θ μετρεῖ καὶ

Let the given ratios, (expressed) in the least numbers, be the (ratios) of A to B , and of C to D , and, further, of E to F . So it is required to find the least numbers continuously proportional in the ratio of A to B , and of C to B , and, further, of E to F .

For let the least number, G , measured by (both) B and C have been taken [Prop. 7.34]. And as many times as B measures G , so many times let A also measure H . And as many times as C measures G , so many times let D also measure K . And E either measures, or does not measure, K . Let it, first of all, measure (K). And as many times as E measures K , so

ὁ B τὸν H , ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Θ πρὸς τὸν H . διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ H πρὸς τὸν K , καὶ ἔτι ὡς ὁ E πρὸς τὸν Z , οὕτως ὁ K πρὸς τὸν Λ . οἱ Θ , H , K , Λ ἄρα ἐχῆς ἀνάλογόν εἰσιν ἐν τε τῷ τοῦ A πρὸς τὸν B καὶ ἐν τῷ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι ἐν τῷ τοῦ E πρὸς τὸν Z λόγῳ. λέγω δὴ, ὅτι καὶ ἐλάχιστοι. εἰ γὰρ μὴ εἰσιν οἱ Θ , H , K , Λ ἐχῆς ἀνάλογον ἐλάχιστοι ἐν τε τοῖς τοῦ A πρὸς τὸν B καὶ τοῦ Γ πρὸς τὸν Δ καὶ ἐν τῷ τοῦ E πρὸς τὸν Z λόγοις, ἔστωσαν οἱ N , X , M , O . καὶ ἐπεὶ ἐστὶν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ N πρὸς τὸν X , οἱ δὲ A , B ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔθονταξ ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τοντέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον, ὁ B ἄρα τὸν X μετρεῖ. διὰ τὰ αὐτὰ δὴ καὶ ὁ Γ τὸν X μετρεῖ· οἱ B , Γ ἄρα τὸν X μετροῦσιν· καὶ ὁ ἐλάχιστος ἄρα ὑπὸ τῶν B , Γ μετρούμενος τὸν X μετρήσει. ἐλάχιστος δὲ ὑπὸ τῶν B , Γ μετρεῖται ὁ H . ὁ H ἄρα τὸν X μετρεῖ ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἔσονται τινες τῶν Θ , H , K , Λ ἐλάσσονες ἀριθμοὶ ἐχῆς ἐν τε τῷ τοῦ A πρὸς τὸν B καὶ τῷ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι τῷ τοῦ E πρὸς τὸν Z λόγῳ.

many times let F also measure L . And since A measures H the same number of times that B also (measures) G , thus as A is to B , so H (is) to G [Def. 7.20, Prop. 7.13]. And so, for the same (reasons), as C (is) to D , so G (is) to K , and, further, as E (is) to F , so K (is) to L . Thus, H , G , K , L are continuously proportional in the ratio of A to B , and of C to D , and, further, of E to F . So I say that (they are) also the least (numbers continuously proportional in these ratios). For if H , G , K , L are not the least numbers continuously proportional in the ratios of A to B , and of C to D , and of E to F , let N , O , M , P be (the least such numbers). And since as A is to B , so N (is) to O , and A and B are the least (numbers which have the same ratio as them), and the least (numbers) measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20], B thus measures O . So, for the same (reasons), C also measures O . Thus, B and C (both) measure O . Thus, the least number measured by (both) B and C will also measure O [Prop. 7.35]. And G (is) the least number measured by (both) B and C . Thus, G measures O , the greater (measuring) the lesser. The very thing is impossible. Thus, there cannot be any numbers less than H , G , K , L (which are) continuously (proportional) in the ratio of A to B , and of C to D , and, further, of E to F .



μή μετρεῖτω δὴ ὁ E τὸν K , καὶ εἰλήφθω ὑπὸ τῶν E , K ἐλάχιστος μετρούμενος ἀριθμὸς ὁ M . καὶ ὁσάκις μὲν ὁ K τὸν M μετρεῖ, τοσαντάκις καὶ ἑκάτερος τῶν Θ , H ἑκάτερον τῶν N , X μετρεῖτω, ὁσάκις δὲ ὁ E τὸν M μετρεῖ, τοσαντάκις καὶ ὁ Z τὸν O μετρεῖτω. ἐπεὶ ἰσάκις ὁ Θ τὸν N μετρεῖ καὶ ὁ H τὸν X , ἔστιν ἄρα ὡς ὁ Θ πρὸς τὸν H , οὕτως ὁ N πρὸς τὸν X . ὡς δὲ ὁ Θ πρὸς τὸν H , οὕτως ὁ A πρὸς τὸν B . καὶ ὡς ἄρα ὁ A πρὸς τὸν B , οὕτως ὁ N πρὸς τὸν X . διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ X πρὸς τὸν M . πάλιν, ἐπεὶ ἰσάκις ὁ E τὸν M μετρεῖ καὶ ὁ Z τὸν O , ἔστιν ἄρα ὡς ὁ E πρὸς τὸν Z , οὕτως ὁ M πρὸς τὸν O . οἱ N , X , M , O ἄρα ἐχῆς ἀνάλογόν εἰσιν ἐν τοῖς τοῦ τε A πρὸς τὸν B καὶ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι τοῦ E πρὸς τὸν Z λόγοις. λέγω δὴ, ὅτι καὶ ἐλάχιστοι ἐν τοῖς $A B$, $\Gamma \Delta$, $E Z$ λόγοις. εἰ γὰρ μή, ἔσσονται τιναξ τῶν N , X , M , O ἐλάσσονες ἀριθμοὶ ἐχῆς ἀνάλογον ἐν τοῖς $A B$, $\Gamma \Delta$, $E Z$ λόγοις. ἔστωσαν οἱ Π , P , Σ , T . καὶ ἐπεὶ ἔστιν ὡς ὁ Π πρὸς τὸν P , οὕτως ὁ A πρὸς τὸν B , οἱ δὲ A ,

So let E not measure K . And let the least number, M , measured by (both) E and K be taken [Prop. 7.34]. And as many times as K measures M , so many times let H , G also measure N , O , respectively. And as many times as E measures M , so many times let F also measure P . Since H measures N the same number of times as G (measures) O , thus as H is to G , so N (is) to O [Def. 7.20, Prop. 7.13]. And as H (is) to G , so A (is) to B . And thus as A (is) to B , so N (is) to O . And so, for the same (reasons), as C (is) to D , so O (is) to M . Again, since E measures M the same number of times as F (measures) P , thus as E is to F , so M (is) to P [Def. 7.20, Prop. 7.13]. Thus, N , O , M , P are continuously proportional in the ratios of A to B , and of C to D , and, further, of E to F . So I say that (they are) also the least (numbers) in the ratios of $A B$, $C D$, $E F$. For if not, then there will be some numbers less than N , O , M , P (which are) continuously proportional in the ratios of $A B$, $C D$, $E F$. Let them be Q , R , S , T . And since as Q is to

B ἐλάτισται, οἱ δὲ ἐλάτισται μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχονταξ αὐτοῖς ἰσάκις ὅ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον, ὁ B ἄρα τὸν P μετρῇ. διὰ τὰ αὐτὰ δὴ καὶ ὁ Γ τὸν P μετρῇ· οἱ B , Γ ἄρα τὸν P μετροῦσιν. καὶ ὁ ἐλάτισταξ ἄρα ὑπὸ τῶν B , Γ μετρούμενοξ τὸν P μετρήσει. ἐλάτισταξ δὲ ὑπὸ τῶν B , Γ μετρούμενοξ ἐστὶν ὁ H · ὁ H ἄρα τὸν P μετρῇ. καὶ ἐστὶν ὡς ὁ H πρὸξ τὸν P , οὕτως ὁ K πρὸξ τὸν Σ · καὶ ὁ K ἄρα τὸν Σ μετρῇ. μετρῇ δὲ καὶ ὁ E τὸν Σ · οἱ E , K ἄρα τὸν Σ μετροῦσιν. καὶ ὁ ἐλάτισταξ ἄρα ὑπὸ τῶν E , K μετρούμενοξ τὸν Σ μετρήσει. ἐλάτισταξ δὲ ὑπὸ τῶν E , K μετρούμενοξ ἐστὶν ὁ M · ὁ M ἄρα τὸν Σ μετρῇ ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἔσονται τινεξ τῶν N , X , M , O ἐλάσσονεξ ἀριθμοὶ ἐχῆξ ἀνάλογον ἔν τε τοῖξ τοῦ A πρὸξ τὸν B καὶ τοῦ Γ πρὸξ τὸν Δ καὶ ἔτι τοῦ E πρὸξ τὸν Z λόγοιξ· οἱ N , X , M , O ἄρα ἐχῆξ ἀνάλογον ἐλάτισταί εἰσιν ἐν τοῖξ A , B , Γ , Δ , E , Z λόγοιξ· ὅπερ ἔδει δεῖχαι.

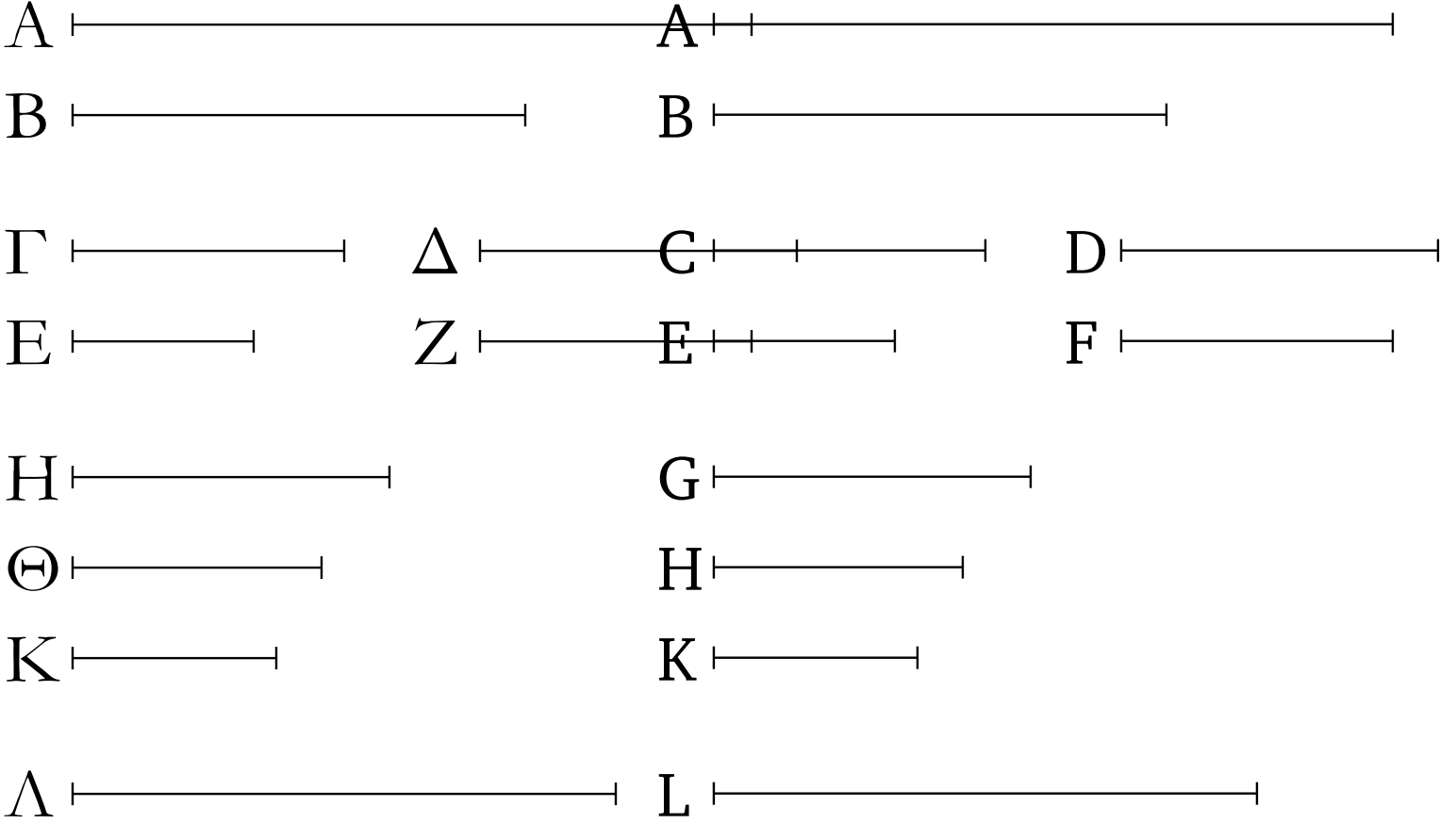
5.

Οἱ ἐπίπεδοι ἀριθμοὶ πρὸξ ἀλλήλων λόγον ἔχουσι τὸν συγκείμενον ἐκ τῶν πλευρῶν.

R , so A (is) to B , and A and B (are) the least (numbers having the same ratio as them), and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20], B thus measures R . So, for the same (reasons), C also measures R . Thus, B and C (both) measure R . Thus, the least (number) measured by (both) B and C will also measure R [Prop. 7.35]. And G is the least number measured by (both) B and C . Thus, G measures R . And as G is to R , so K (is) to S . Thus, K also measures S [Def. 7.20]. And E also measures S [Prop. 7.20]. Thus, E and K (both) measure S . Thus, the least (number) measured by (both) E and K will also measure S [Prop. 7.35]. And M is the least (number) measured by (both) E and K . Thus, M measures S , the greater (measuring) the lesser. The very thing is impossible. Thus there cannot be any numbers less than N , O , M , P (which are) continuously proportional in the ratios of A to B , and of C to D , and, further, of E to F . Thus, N , O , M , P are the least (numbers) continuously proportional in the ratios of A , B , C , D , E , F . (Which is) the very thing it was required to show.

Proposition 5

Plane numbers have to one another the ratio compounded[†] out of (the ratios of) their sides.



Ἐστωσαν ἐπίπεδοι ἀριθμοὶ οἱ A, B , καὶ τοῦ μὲν A πλευραὶ ἔστωσαν οἱ Γ, Δ ἀριθμοί, τοῦ δὲ B οἱ E, Z λέγω, ὅτι ὁ A πρὸς τὸν B λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν.

Λόγων γὰρ δοθέντων τοῦ τε ὄν ἔθει ὁ Γ πρὸς τὸν E καὶ ὁ Δ πρὸς τὸν Z εἰλήφθωσαν ἀριθμοὶ ἐχῆς ἐλάχιστοι ἐν τοῖς $\Gamma E, \Delta Z$ λόγοις, οἱ H, Θ, K , ὥστε εἶναι ὡς μὲν τὸν Γ πρὸς τὸν E , οὕτως τὸν H πρὸς τὸν Θ , ὡς δὲ τὸν Δ πρὸς τὸν Z , οὕτως τὸν Θ πρὸς τὸν K . καὶ ὁ Δ τὸν E πολλαπλασιάσας τὸν Λ ποιείτω.

Καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ E πολλαπλασιάσας τὸν Λ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν E , οὕτως ὁ A πρὸς τὸν Λ . ὡς δὲ ὁ Γ πρὸς τὸν E , οὕτως ὁ H πρὸς τὸν Θ . καὶ ὡς ἄρα ὁ H πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν Λ . πάλιν, ἐπεὶ ὁ E τὸν Δ πολλαπλασιάσας τὸν Λ πεποίηκεν, ἀλλὰ μὴν καὶ τὸν Z πολλαπλασιάσας τὸν B πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ Λ πρὸς τὸν B . ἀλλ' ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ Θ πρὸς τὸν K . καὶ ὡς ἄρα ὁ Θ πρὸς τὸν K , οὕτως ὁ Λ πρὸς τὸν B . ἐδείκθη δὲ καὶ ὡς ὁ H πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν Λ . δι' ἴσον ἄρα ἐστὶν ὡς ὁ H πρὸς τὸν K , [οὕτως] ὁ A πρὸς τὸν B . ὁ δὲ H πρὸς τὸν K λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν καὶ ὁ A ἄρα πρὸς τὸν B λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν ὅπερ ἔδει δεῖξαι.

Let A and B be plane numbers, and let the numbers C, D be the sides of A , and (the numbers) E, F (the sides) of B . I say that A has to B the ratio compounded out of (the ratios of) their sides.

For given the ratios which C has to E , and D (has) to F , let the least numbers, G, H, K , continuously proportional in the ratios $C E, D F$ be taken [Prop. 8.4], so that as C is to E , so G (is) to H , and as D (is) to F , so H (is) to K . And let D make L (by) multiplying E .

And since D has made A (by) multiplying C , and has made L (by) multiplying E , thus as C is to E , so A (is) to L [Prop. 7.17]. And as C (is) to E , so G (is) to H . And thus as G (is) to H , so A (is) to L . Again, since E has made L (by) multiplying D [Prop. 7.16], but, in fact, has also made B (by) multiplying F , thus as D is to F , so L (is) to B [Prop. 7.17]. But, as D (is) to F , so H (is) to K . And thus as H (is) to K , so L (is) to B . And it was also shown that as G (is) to H , so A (is) to L . Thus, via equality, as G is to K , [so] A (is) to B [Prop. 7.14]. And G has to K the ratio compounded out of (the ratios of) the sides (of A and B). Thus, A also has to B the ratio compounded out of (the ratios of) the sides (of A and B). (Which is) the very thing it was required to show.

[†] i.e., multiplied.

6.

Ἐάν ὥσιν ὁποσοῖον ἀριθμοὶ ἐχῇς ἀνάλογον, ὁ δὲ πρώτοξ τὸν δεύτερον μὴ μετρήῃ, οὐδὲ ἄλλοξ οὐδείξ οὐδένα μετρήσει.

Ἐστῶσαν ὁποσοῖον ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A, B, Γ, Δ, E , ὁ δὲ A τὸν B μὴ μετρεῖτω· λέγω, ὅτι οὐδὲ ἄλλοξ οὐδείξ οὐδένα μετρήσει.

A —————

B —————

Γ —————

Δ —————

E —————

Z —————

H —————

Θ —————

A —————

B —————

C —————

D —————

E —————

F —————

G —————

H —————

Ὅτι μὲν οὖν οἱ A, B, Γ, Δ, E ἐχῇς ἀλλήλων οὐ μετροῦσιν, φανερόν· οὐδὲ γάρ ὁ A τὸν B μετρεῖ. λέγω δὴ, ὅτι οὐδὲ ἄλλοξ οὐδείξ οὐδένα μετρήσει. εἰ γὰρ δυνατόν, μετρεῖτω ὁ A τὸν Γ . καὶ ὅσοι εἰσὶν οἱ A, B, Γ , τοσοῦτοι εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ οἱ Z, H, Θ . καὶ ἐπεὶ οἱ Z, H, Θ ἐν τῷ αὐτῷ λόγῳ εἰσὶ τοῖς A, B, Γ , καὶ ἐστὶν ἴσον τὸ πλῆθος τῶν A, B, Γ τῷ πλήθει τῶν Z, H, Θ , δι' ἴσον ἄρα ἐστὶν ὡς ὁ A πρὸς τὸν Γ , οὕτως ὁ Z πρὸς τὸν Θ . καὶ ἐπεὶ ἐστὶν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Z πρὸς τὸν H , οὐ μετρεῖ δὲ ὁ A τὸν B , οὐ μετρεῖ ἄρα οὐδὲ ὁ Z τὸν H . οὐκ ἄρα μονὰξ ἐστὶν ὁ Z · ἢ γὰρ μονὰξ πάντα ἀριθμὸν μετρεῖ. καὶ εἰσὶν οἱ Z, Θ πρώτοι πρὸς ἀλλήλους [οὐδὲ ὁ Z ἄρα τὸν Θ μετρεῖ]. καὶ ἐστὶν ὡς ὁ Z πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν Γ · οὐδὲ ὁ A ἄρα τὸν Γ μετρεῖ. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ἄλλοξ οὐδείξ οὐδένα μετρήσει· ὅπερ ἔδει δεῖξαι.

Proposition 6

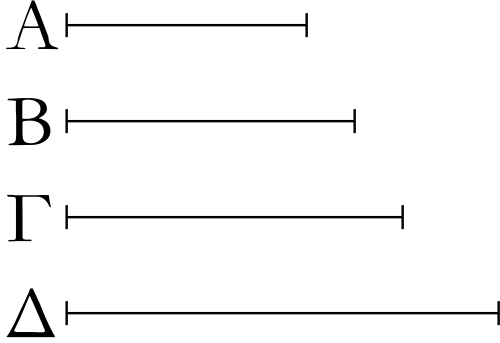
If there are any multitude whatsoever of continuously proportional numbers, and the first does not measure the second, then no other (number) will measure any other (number) either.

Let A, B, C, D, E be any multitude whatsoever of continuously proportional numbers, and let A not measure B . I say that no other (number) will measure any other (number) either.

Now, (it is) clear that A, B, C, D, E do not successively measure one another. For A does not even measure B . So I say that no other (number) will measure any other (number) either. For, if possible, let A measure C . And as many (numbers) as are A, B, C , let so many of the least numbers, F, G, H , be taken of those (numbers) having the same ratio as A, B, C [Prop. 7.33]. And since F, G, H are in the same ratio as A, B, C , and the multitude of A, B, C is equal to the multitude of F, G, H , thus, via equality, as A is to C , so F (is) to H [Prop. 7.14]. And since as A is to B , so F (is) to G , and A does not measure B , F does not measure G either [Def. 7.20]. Thus, F is not a unit. For a unit measures all numbers. And F and H are prime to one another [Prop. 8.3] [and thus F does not measure H]. And as F is to H , so A (is) to C . And thus A does not measure C either [Def. 7.20]. So, similarly, we can show that no other (number) can measure any other (number) either.

7.

Ἐάν ὥσιν ὅποσοιὺν ἀριθμοὶ [ἐχῇ] ἀνάλογον, ὁ δὲ πρῶτος τὸν ἑσθαιον μετρή, καὶ τὸν δεύτερον μετρήσει.



Ἐστωσαν ὅποσοιὺν ἀριθμοὶ ἐχῇ ἀνάλογον οἱ A, B, Γ, Δ , ὁ δὲ A τὸν Δ μετρεῖτω· λέγω, ὅτι καὶ ὁ A τὸν B μετρεῖ.

Εἰ γὰρ οὐ μετρεῖ ὁ A τὸν B , οὐδὲ ἄλλος οὐδεὶς οὐδένα μετρήσει· μετρεῖ δὲ ὁ A τὸν Δ . μετρεῖ ἄρα καὶ ὁ A τὸν B · ὅπερ ἔδει δεῖχαι.

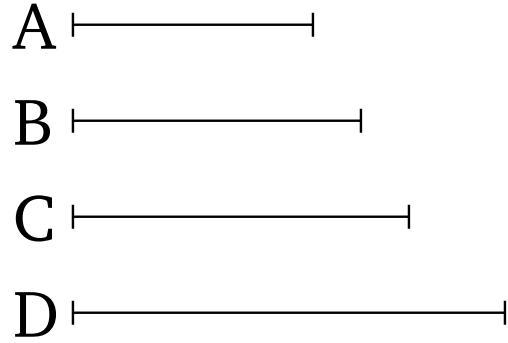
8.

Ἐάν δύο ἀριθμῶν μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅσοι εἰς αὐτοὺς μεταχὺ κατὰ τὸ συνεθὲς ἀνόλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοὺς τὸν αὐτὸν λόγον ἔχονταξ [αὐτοῖς] μεταχὺ κατὰ τὸ συνέθεξ ἀνάλογον ἐμπεσοῦνται

(Which is) the very thing it was required to show.

Proposition 7

If there are any multitude whatsoever of [continuously] proportional numbers, and the first measures the last, then (the first) will also measure the second.

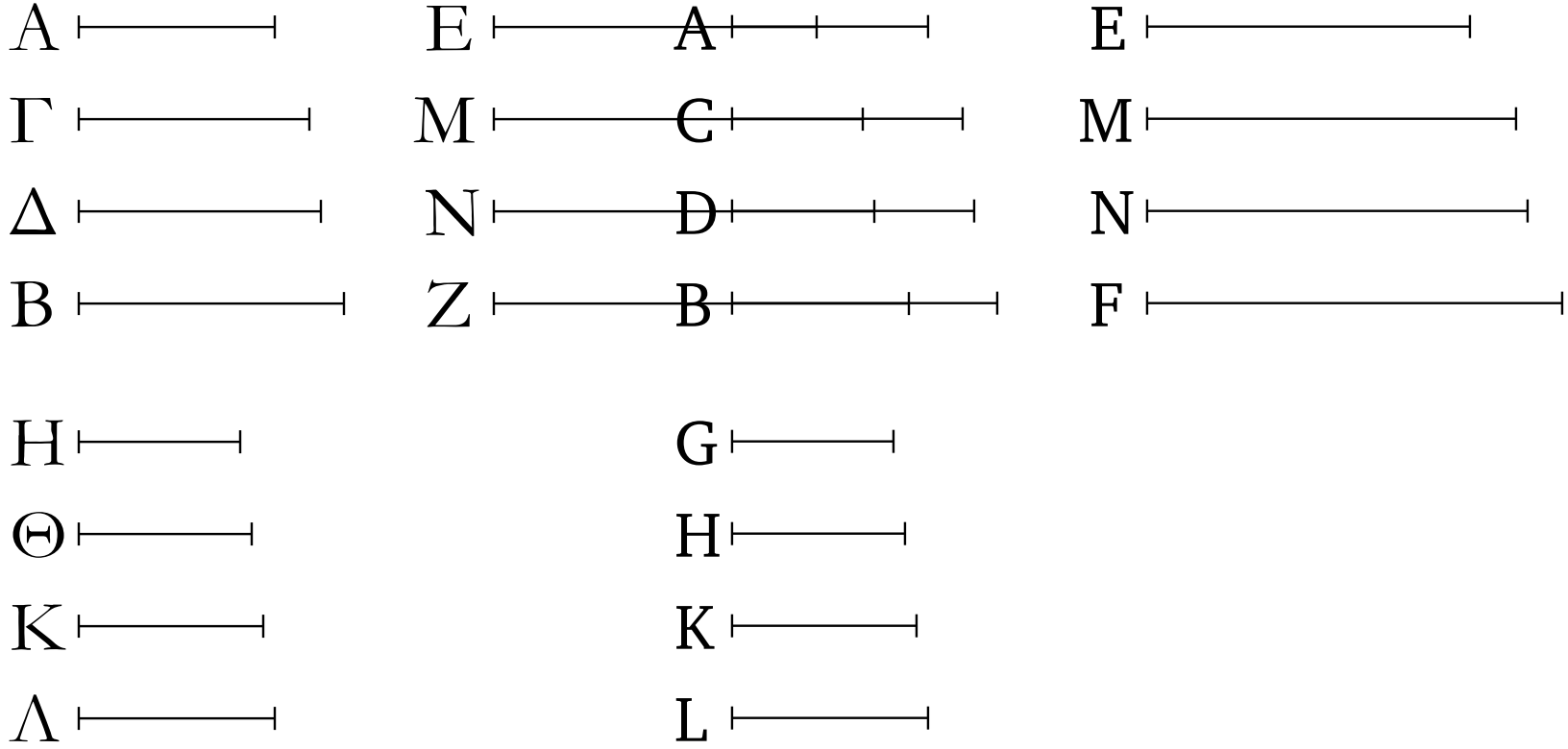


Let A, B, C, D be any number whatsoever of continuously proportional numbers. And let A measure D . I say that A also measures B .

For if A does not measure B then no other (number) will measure any other (number) either [Prop. 8.6]. But A measures D . Thus, A also measures B . (Which is) the very thing it was required to show.

Proposition 8

If between two numbers there fall (some) numbers in continued proportion then, as many numbers as fall in between them in continued proportion, so many (numbers) will also fall in between (any two numbers) having the same ratio [as them] in continued proportion.



Δύο γάρ ἀριθμῶν τῶν A, B μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπιπτέωσαν ἀριθμοὶ οἱ Γ, Δ , καὶ πεποιήσθω ὥς ὁ A πρὸς τὸν B , οὕτως ὁ E πρὸς τὸν Z . λέγω, ὅτι ὅσοι εἰς τὸν A, B μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τὸν E, Z μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

Ὅσοι γάρ εἰσι τῶν πλήθει οἱ A, B, Γ, Δ , τοσοῦτοι εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἔχόντων τοῖς A, Γ, Δ, B οἱ H, Θ, K, Λ . οἱ ἄρα ἄκροι αὐτῶν οἱ H, Λ πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ οἱ A, Γ, Δ, B τοῖς H, Θ, K, Λ ἐν τῶ αὐτῷ λόγῳ εἰσίν, καὶ ἔστιν ἴσον τὸ πλῆθος τῶν A, Γ, Δ, B τῶν πλήθει τῶν H, Θ, K, Λ , δι' ἴσον ἄρα ἔστιν ὥς ὁ A πρὸς τὸν B , οὕτως ὁ H πρὸς τὸν Λ . ὥς δὲ ὁ A πρὸς τὸν B , οὕτως ὁ E πρὸς τὸν Z . οἱ δὲ H, Λ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντα ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. ἰσάκις ἄρα ὁ H τὸν E μετρεῖ καὶ ὁ Λ τὸν Z . ὁσάκις δὴ ὁ H τὸν E μετρεῖ, τοσαυτάκις καὶ ἑκάτερος τῶν Θ, K ἑκάτερον τῶν M, N μετρεῖτω. οἱ H, Θ, K, Λ ἄρα τοὺς E, M, N, Z ἰσάκις μετροῦσιν. οἱ H, Θ, K, Λ ἄρα τοῖς E, M, N, Z ἐν τῶ αὐτῷ λόγῳ εἰσίν. ἀλλὰ οἱ H, Θ, K, Λ τοῖς A, Γ, Δ, B ἐν τῶ αὐτῷ λόγῳ εἰσίν· καὶ οἱ A, Γ, Δ, B ἄρα τοῖς E, M, N, Z ἐν τῶ αὐτῷ λόγῳ εἰσίν. οἱ δὲ A, Γ, Δ, B ἐχῆς ἀνάλογόν εἰσιν καὶ οἱ E, M, N, Z ἄρα ἐχῆς ἀνάλογόν

For let the numbers, C and D , fall between two numbers, A and B , in continued proportion, and let it be contrived (that) as A (is) to B , so E (is) to F . I say that, as many numbers as have fallen in between A and B in continued proportion, so many (numbers) will also fall in between E and F in continued proportion.

For as many as A, B, C, D are in multitude, let so many of the least numbers, G, H, K, L , having the same ratio as A, B, C, D , be taken [Prop. 7.33]. Thus, the outermost of them, G and L , are prime to one another [Prop. 8.3]. And since A, B, C, D are in the same ratio as G, H, K, L , and the multitude of A, B, C, D is equal to the multitude of G, H, K, L , thus, via equality, as A is to B , so G (is) to L [Prop. 7.14]. And as A (is) to B , so E (is) to F . And thus as G (is) to L , so E (is) to F . And G and L (are) prime (to one another). And (numbers) prime (to one another are) also the least (numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, G measures E the same number of times as L (measures) F . So as many times as G measures E , so many times let H, K also measure M, N , respectively. Thus, G, H, K, L measure E, M, N, F (respectively) an equal number of times. Thus, G, H, K, L are in the same ratio as E, M, N, F [Def. 7.20]. But, $G, H,$

εἶσιν. ὅσοι ἄρα εἰς τοὺς A, B μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοὺς E, Z μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί· ὅπερ ἔδει δεῖχαι.

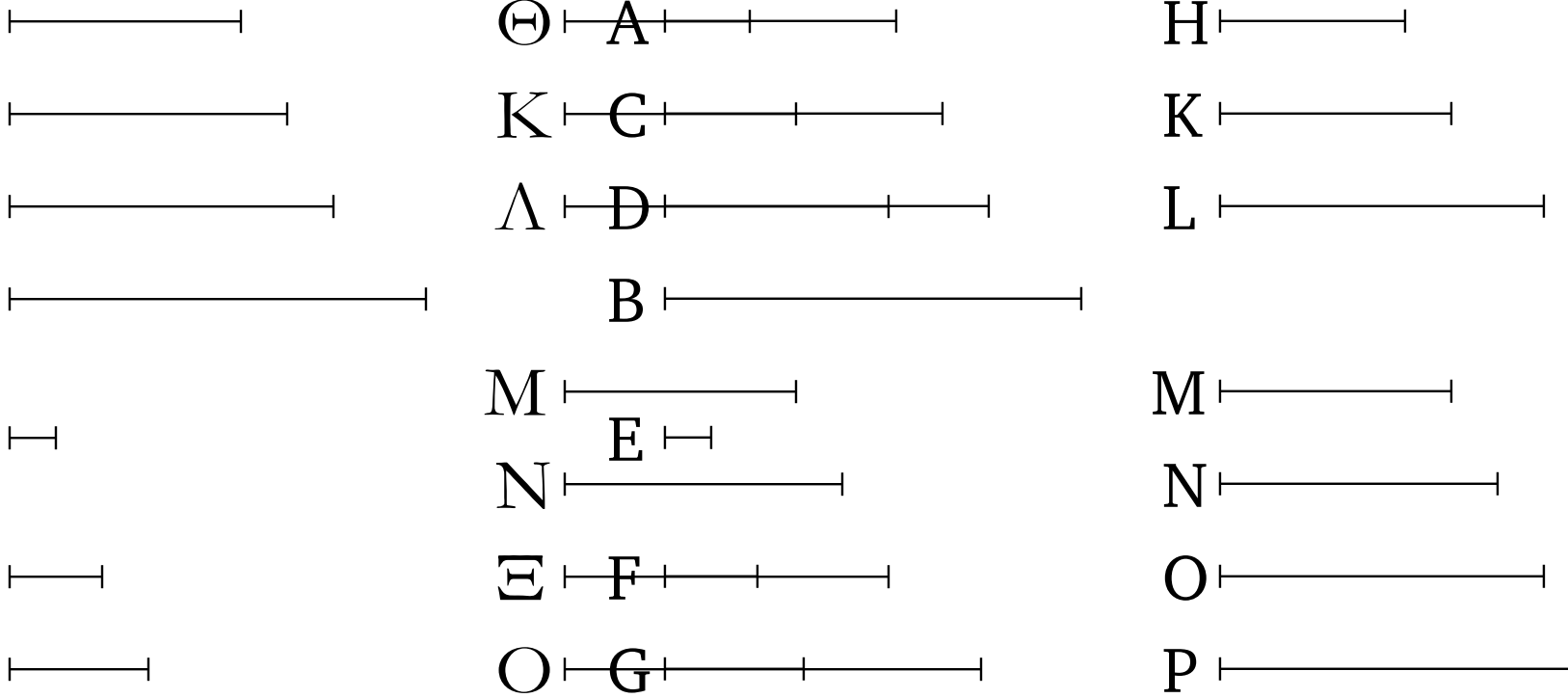
K, L are in the same ratio as A, C, D, B . Thus, A, C, D, B are also in the same ratio as E, M, N, F . And A, C, D, B are continuously proportional. Thus, E, M, N, F are also continuously proportional. Thus, as many numbers as have fallen in between A and B in continued proportion, so many numbers have also fallen in between E and F in continued proportion. (Which is) the very thing it was required to show.

9.

Proposition 9

Ἐάν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους ᾧσιν, καὶ εἰς αὐτοὺς μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅσοι εἰς αὐτοὺς μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρωσιν αὐτῶν καὶ μονάδοξ μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

If two numbers are prime to one another and there fall in between them (some) numbers in continued proportion then, as many numbers as fall in between them in continued proportion, so many (numbers) will also fall between each of them and a unit in continued proportion.



Ἐστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ A, B , καὶ εἰς αὐτοὺς μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτέωσιν οἱ Γ, Δ , καὶ ἐκκείσθω ἡ E μονάξ· λέγω, ὅτι ὅσοι εἰς τοὺς A, B μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρωσιν τῶν A, B καὶ τῆς μονάδοξ μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

Εἰλήφθωσαν γάρ δύο μὲν ἀριθμοὶ ἐλάδιστοι ἐν τῷ τῶν A, Γ, Δ, B λόγῳ ὄντες οἱ Z, H , τρεῖς δὲ οἱ Θ, K, Λ , καὶ αἰεὶ ἐχῆς ἐνὶ πλείονξ, ἕως ἂν ἴσον γένηται τὸ πλῆθος αὐτῶν τῷ πλήθει τῶν A, Γ, Δ, B . εἰλήφθωσαν, καὶ ἔστωσαν οἱ M, N, X, O . φανερόν δὲ, ὅτι ὁ μὲν Z ἐαντὸν πολλαπλασιάσας τὸν Θ πεποίηκεν, τὸν δὲ Θ πολλαπλασιάσας τὸν M πεποίηκεν, καὶ ὁ H ἐαντὸν μὲν πολλαπλασιάσας τὸν Λ πεποίηκεν, τὸν δὲ Λ

Let A and B be two numbers (which are) prime to one another, and let the (numbers) C and D fall in between them in continued proportion. And let the unit E be set out. I say that, as many numbers as have fallen in between A and B in continued proportion, so many (numbers) will also fall between each of A and B and the unit in continued proportion.

For let the least two numbers, F and G , which are in the ratio of A, C, D, B , be taken [Prop. 7.33]. And the (least) three (numbers), H, K, L . And so on, successively increasing by one, until the multitude of the (least numbers taken) is made equal to the multitude of A, C, D, B [Prop. 8.2]. Let them be taken, and let them be M, N, O, P . So (it is) clear that F has made H (by) multiplying itself, and has made M (by)

πολλαπλασιάσας τὸν O πεποίηκεν. καὶ ἐπεὶ οἱ M, N, X, O ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Z, H , εἰσὶ δὲ καὶ οἱ A, Γ, Δ, B ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Z, H , καὶ ἔστιν ἴσον τὸ πλῆθος τῶν M, N, X, O τῷ πλῆθει τῶν A, Γ, Δ, B , ἕκαστος ἄρα τῶν M, N, X, O ἐκάστω τῶν A, Γ, Δ, B ἴσος ἐστίν· ἴσος ἄρα ἐστὶν ὁ μὲν M τῷ A , ὁ δὲ O τῷ B . καὶ ἐπεὶ ὁ Z ἐαυτὸν πολλαπλασιάσας τὸν Θ πεποίηκεν, ὁ Z ἄρα τὸν Θ μετρεῖ κατὰ τὰς ἐν τῷ Z μονάδας. μετρεῖ δὲ καὶ ἡ E μονάξ τὸν Z κατὰ τὰς ἐν αὐτῷ μονάδας· ἰσάκις ἄρα ἡ E μονάξ τὸν Z ἀριθμὸν μετρεῖ καὶ ὁ Z τὸν Θ . ἔστιν ἄρα ὡς ἡ E μονάξ πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Z πρὸς τὸν Θ . πάλιν, ἐπεὶ ὁ Z τὸν Θ πολλαπλασιάσας τὸν M πεποίηκεν, ὁ Θ ἄρα τὸν M μετρεῖ κατὰ τὰς ἐν τῷ Z μονάδας. μετρεῖ δὲ καὶ ἡ E μονάξ τὸν Z ἀριθμὸν κατὰ τὰς ἐν αὐτῷ μονάδας· ἰσάκις ἄρα ἡ E μονάξ τὸν Z ἀριθμὸν μετρεῖ καὶ ὁ Θ τὸν M . ἔστιν ἄρα ὡς ἡ E μονάξ πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Θ πρὸς τὸν M . ἐδείκθη δὲ καὶ ὡς ἡ E μονάξ πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Z πρὸς τὸν Θ · καὶ ὡς ἄρα ἡ E μονάξ πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Z πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν M . ἴσος δὲ ὁ M τῷ A · ἔστιν ἄρα ὡς ἡ E μονάξ πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Z πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν A . διὰ τὰ αὐτὰ δὴ καὶ ὡς ἡ E μονάξ πρὸς τὸν H ἀριθμὸν, οὕτως ὁ H πρὸς τὸν Λ καὶ ὁ Λ πρὸς τὸν B . ὅσοι ἄρα εἰς τούς A, B μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρων τῶν A, B καὶ μονάδος τῆς E μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί· ὅπερ ἔδει δεῖχαι.

10.

Ἐάν δύο ἀριθμῶν ἑκατέρων καὶ μονάδος μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅσοι ἑκατέρων αὐτῶν καὶ μονάδος μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς αὐτοὺς μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

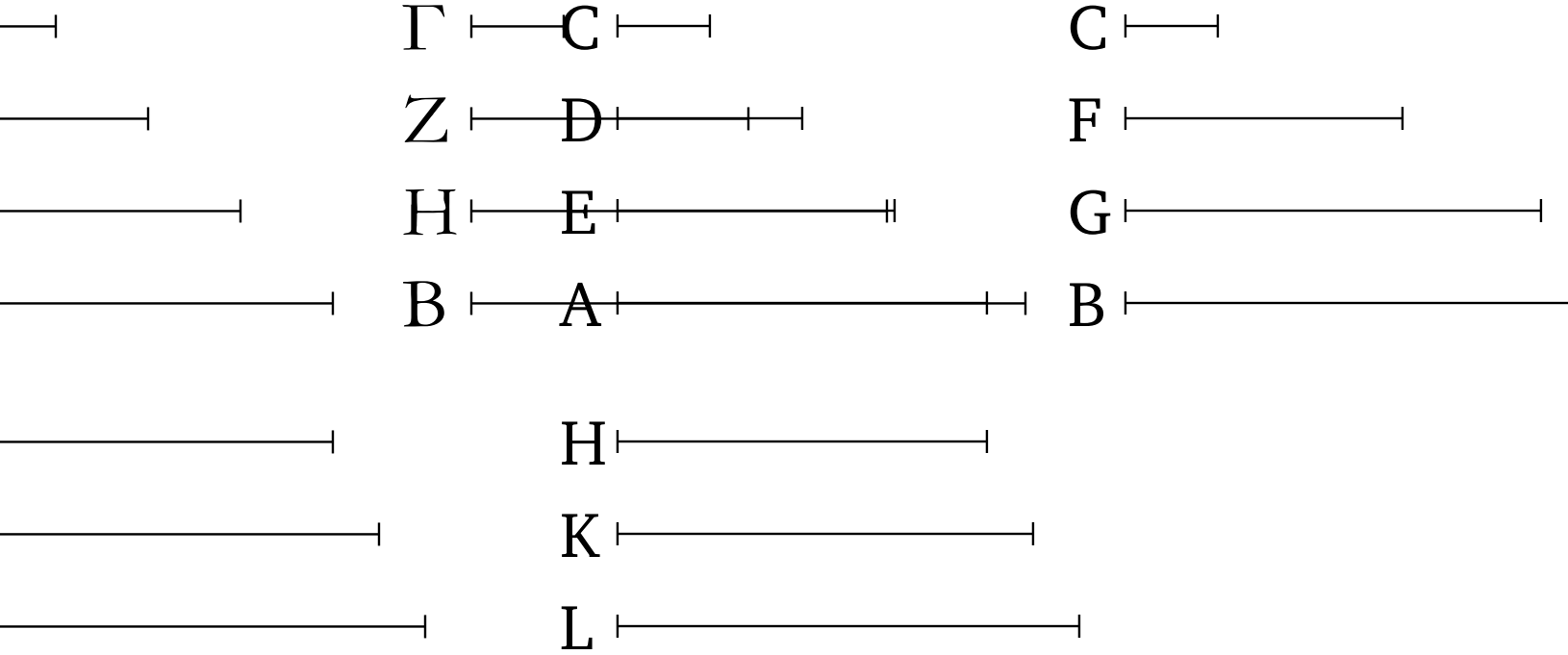
Δύο γάρ ἀριθμῶν τῶν A, B καὶ μονάδος τῆς Γ μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτέτωσιν ἀριθμοί οἱ τε Δ, E καὶ οἱ Z, H · λέγω, ὅτι ὅσοι ἑκατέρων τῶν A, B καὶ μονάδος τῆς Γ μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τούς A, B μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

multiplying H . And G has made L (by) multiplying itself, and has made P (by) multiplying L [Prop. 8.2 corr.]. And since M, N, O, P are the least of those (numbers) having the same ratio as F, G , and A, C, D, B are also the least of those (numbers) having the same ratio as F, G [Prop. 8.2], and the multitude of M, N, O, P is equal to the multitude of A, C, D, B , thus M, N, O, P are equal to A, C, D, B , respectively. Thus, M is equal to A , and P to B . And since F has made H (by) multiplying itself, F thus measures H according to the units in F [Def. 7.15]. And the unit E also measures F according to the units in it. Thus, the unit E measures the number F as many times as F (measures) H . Thus, as the unit E is to the number F , so F (is) to H [Def. 7.20]. Again, since F has made M (by) multiplying H , H thus measures M according to the units in F [Def. 7.15]. And the unit E also measures the number F according to the units in it. Thus, the unit E measures the number F as many times as H (measures) M . Thus, as the unit E is to the number F , so H (is) to M [Prop. 7.20]. And it was shown that as the unit E (is) to the number F , so F (is) to H . And thus as the unit E (is) to the number F , so F (is) to H , and H (is) to M . And M (is) equal to A . Thus, as the unit E is to the number F , so F (is) to H , and H to A . And so, for the same (reasons), as the unit E (is) to the number G , so G (is) to L , and L to B . Thus, as many (numbers) as have fallen in between A and B in continued proportion, so many numbers have also fallen between each of A and B and the unit E in continued proportion. (Which is) the very thing it was required to show.

Proposition 10

If (some) numbers fall between each of two numbers and a unit in continued proportion then, as many (numbers) as fall between each of the (two numbers) and the unit in continued proportion, so many (numbers) will also fall in between the (two numbers) themselves in continued proportion.

For let the numbers D, E and F, G fall between the numbers A and B (respectively) and the unit C in continued proportion. I say that, as many numbers as have fallen between each of A and B and the unit C in continued proportion, so many will also fall in between A and B in continued proportion.



Ὁ Δ γὰρ τὸν Ζ πολλαπλασιάσας τὸν Θ ποιεῖτω, ἑκάτε-
ροξ δὲ τῶν Δ, Ζ τὸν Θ πολλαπλασιάσας ἑκάτερον τῶν Κ, Α
ποιεῖτω.

Καὶ ἐπεὶ ἔστιν ὡς ἡ Γ μονὰξ πρὸς τὸν Δ ἀριθμὸν, οὕτως
ὁ Δ πρὸς τὸν Ε, ἰσάκις ἄρα ἡ Γ μονὰξ τὸν Δ ἀριθμὸν μετρεῖ
καὶ ὁ Δ τὸν Ε. ἡ δὲ Γ μονὰξ τὸν Δ ἀριθμὸν μετρεῖ κατὰ
τάξιν ἐν τῷ Δ μονάδαξ· καὶ ὁ Δ ἄρα ἀριθμὸς τὸν Ε μετρεῖ
κατὰ τάξιν ἐν τῷ Δ μονάδαξ· ὁ Δ ἄρα ἐαυτὸν πολλαπλασιάσας
τὸν Ε πεποίηκεν. πάλιν, ἐπεὶ ἔστιν ὡς ἡ Γ [μονὰξ] πρὸς τὸν
Δ ἀριθμὸν, οὕτως ὁ Ε πρὸς τὸν Α, ἰσάκις ἄρα ἡ Γ μονὰξ
τὸν Δ ἀριθμὸν μετρεῖ καὶ ὁ Ε τὸν Α. ἡ δὲ Γ μονὰξ τὸν Δ
ἀριθμὸν μετρεῖ κατὰ τάξιν ἐν τῷ Δ μονάδαξ· καὶ ὁ Ε ἄρα
τὸν Α μετρεῖ κατὰ τάξιν ἐν τῷ Δ μονάδαξ· ὁ Δ ἄρα τὸν Ε
πολλαπλασιάσας τὸν Α πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ
μὲν Ζ ἐαυτὸν πολλαπλασιάσας τὸν Η πεποίηκεν, τὸν δὲ Η
πολλαπλασιάσας τὸν Β πεποίηκεν. καὶ ἐπεὶ ὁ Δ ἐαυτὸν μὲν
πολλαπλασιάσας τὸν Ε πεποίηκεν, τὸν δὲ Ζ πολλαπλασιάσας
τὸν Θ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Ε
πρὸς τὸν Θ. διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ
Θ πρὸς τὸν Η. καὶ ὡς ἄρα ὁ Ε πρὸς τὸν Θ, οὕτως ὁ Θ πρὸς
τὸν Η. πάλιν, ἐπεὶ ὁ Δ ἑκάτερον τῶν Ε, Θ πολλαπλασιάσας
ἑκάτερον τῶν Α, Κ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Ε πρὸς τὸν
Θ, οὕτως ὁ Α πρὸς τὸν Κ. ἀλλ' ὡς ὁ Ε πρὸς τὸν Θ, οὕτως ὁ
Δ πρὸς τὸν Ζ· καὶ ὡς ἄρα ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Α πρὸς
τὸν Κ. πάλιν, ἐπεὶ ἑκάτεροξ τῶν Δ, Ζ τὸν Θ πολλαπλασιάσας
ἑκάτερον τῶν Κ, Α πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν
Ζ, οὕτως ὁ Κ πρὸς τὸν Α. ἀλλ' ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως
ὁ Α πρὸς τὸν Κ· καὶ ὡς ἄρα ὁ Α πρὸς τὸν Κ, οὕτως ὁ Κ
πρὸς τὸν Α. ἔτι ἐπεὶ ὁ Ζ ἑκάτερον τῶν Θ, Η πολλαπλασιάσας
ἑκάτερον τῶν Α, Β πεποίηκεν, ἔστιν ἄρα ὡς ὁ Θ πρὸς τὸν

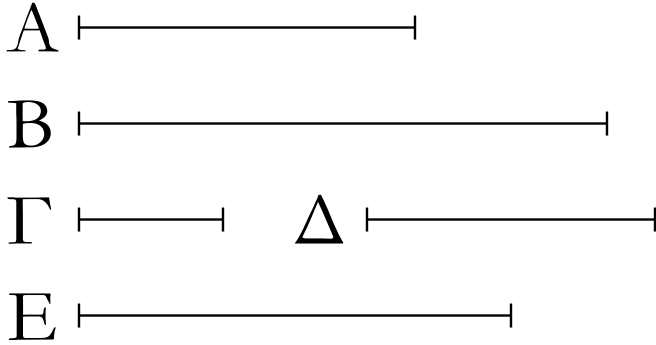
For let D make H (by) multiplying F . And let D, F make
 K, L , respectively, by multiplying H .

As since as the unit C is to the number D , so D (is) to E ,
the unit C thus measures the number D as many times as D
(measures) E [Def. 7.20]. And the unit C measures the num-
ber D according to the units in D . Thus, the number D also
measures E according to the units in D . Thus, D has made E
(by) multiplying itself. Again, since as the [unit] C is to the
number D , so E (is) to A , the unit C thus measures the number
 D as many times as E (measures) A [Def. 7.20]. And the unit
 C measures the number D according to the units in D . Thus,
 E also measures A according to the units in D . Thus, D has
made A (by) multiplying E . And so, for the same (reasons), F
has made G (by) multiplying itself, and has made B (by) mul-
tiplying G . And since D has made E (by) multiplying itself,
and has made H (by) multiplying F , thus as D is to F , so E (is)
to H [Prop 7.17]. And so, for the same reasons, as D (is) to F ,
so H (is) to G [Prop. 7.18]. And thus as E (is) to H , so H (is)
to G . Again, since D has made A, K (by) multiplying E, H ,
respectively, thus as E is to H , so A (is) to K [Prop 7.17]. But,
as E (is) to H , so D (is) to F . And thus as D (is) to F , so A
(is) to K . Again, since D, F have made K, L , respectively, (by)
multiplying H , thus as D is to F , so K (is) to L [Prop. 7.18].
But, as D (is) to F , so A (is) to K . And thus as A (is) to K , so
 K (is) to L . Further, since F has made L, B (by) multiplying
 H, G , respectively, thus as H is to G , so L (is) to B [Prop 7.17].
And as H (is) to G , so D (is) to F . And thus as D (is) to F , so L
(is) to B . And it was also shown that as D (is) to F , so A (is) to
 K , and K to L . And thus as A (is) to K , so K (is) to L , and L to
 B . Thus, A, K, L, B are successively in continued proportion.

H , οὕτως ὁ Λ πρὸς τὸν B . ὥς δὲ ὁ Θ πρὸς τὸν H , οὕτως ὁ Δ πρὸς τὸν Z : καὶ ὥς ἄρα ὁ Δ πρὸς τὸν Z , οὕτως ὁ Λ πρὸς τὸν B . ἐδείκθη δὲ καὶ ὥς ὁ Δ πρὸς τὸν Z , οὕτως ὁ Λ πρὸς τὸν K καὶ ὁ K πρὸς τὸν Λ : καὶ ὥς ἄρα ὁ A πρὸς τὸν K , οὕτως ὁ K πρὸς τὸν Λ καὶ ὁ Λ πρὸς τὸν B . οἱ A , K , Λ , B ἄρα κατὰ τὸ συνεθὲς ἐχῆς εἶναι ἀνάλογον. ὅσοι ἄρα ἑκατέρων τῶν A , B καὶ τῆς Γ μονάδος μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς τὸν A , B μεταχὺ κατὰ τὸ συνεθὲς ἐμπεσοῦνται· ὅπερ ἔδει δεῖξαι.

11.

Δύο τετραγώνων ἀριθμῶν εἷς μέσοξ ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ τετράγωνος πρὸς τὸν τετράγωνον διπλασίονα λόγον ἔθει ἢ περὶ ἢ πλευρὰ πρὸς τὴν πλευράν.



Ἐστωσαν τετράγωνοι ἀριθμοὶ οἱ A , B , καὶ τοῦ μὲν A πλευρὰ ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ : λέγω, ὅτι τῶν A , B εἷς μέσοξ ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἢ περὶ ὁ Γ πρὸς τὸν Δ .

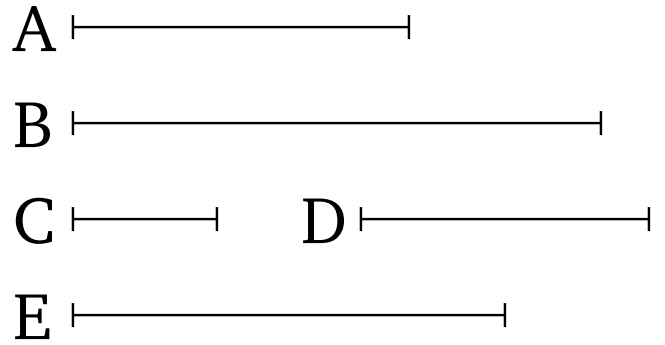
Ὁ Γ γάρ τὸν Δ πολλαπλασιάσας τὸν E ποιεῖτω. καὶ ἐπεὶ τετράγωνός ἐστιν ὁ A , πλευρὰ δὲ αὐτοῦ ἐστιν ὁ Γ , ὁ Γ ἄρα ἑαυτὸν πολλαπλασιάσας τὸν A πεποιήκεν. διὰ τὰ αὐτὰ δὲ καὶ ὁ Δ ἑαυτὸν πολλαπλασιάσας τὸν B πεποιήκεν. ἐπεὶ οὖν ὁ Γ ἑκάτερον τῶν Γ , Δ πολλαπλασιάσας ἑκάτερον τῶν A , E πεποιήκεν, ἔστιν ἄρα ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν E . διὰ τὰ αὐτὰ δὲ καὶ ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν B . καὶ ὥς ἄρα ὁ A πρὸς τὸν E , οὕτως ὁ E πρὸς τὸν B . τῶν A , B ἄρα εἷς μέσοξ ἀνάλογόν ἐστιν ἀριθμός.

Λέγω δὲ, ὅτι καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἢ περὶ ὁ Γ πρὸς τὸν Δ . ἐπεὶ γὰρ τρεῖς ἀριθμοὶ ἀνάλογόν εἰσιν οἱ A , E , B , ὁ A ἄρα πρὸς τὸν B διπλασίονα λόγον ἔθει ἢ περὶ ὁ A πρὸς τὸν E . ὥς δὲ ὁ A πρὸς τὸν E , οὕτως ὁ Γ πρὸς τὸν Δ . ὁ A ἄρα πρὸς τὸν B διπλασίονα λόγον ἔθει ἢ περὶ ὁ Γ πρὸς τὸν Δ : ὅπερ ἔδει δεῖξαι.

Thus, as many numbers as fall between each of A and B and the unit C in continued proportion, so many will also fall between A and B in continued proportion. (Which is) the very thing it was required to show.

Proposition 11

There exists one number in mean proportion to two (given) square numbers.[†] And (one) square (number) has to the (other) square (number) a squared[‡] ratio with respect to (that) the side (of the former has) to the side (of the latter).



Let A and B be square numbers, and let C be the side of A , and D (the side) of B . I say that there exists one number in mean proportion to A and B , and that A has to B a squared ratio with respect to (that) C (has) to D .

For let C make E (by) multiplying D . And since A is square, and C is its side, C has thus made A (by) multiplying itself. And so, for the same (reasons), D has made B (by) multiplying itself. Therefore, since C has made A , E (by) multiplying C , D , respectively, thus as C is to D , so A (is) to E [Prop. 7.17]. And so, for the same (reasons), as C (is) to D , so E (is) to B [Prop. 7.18]. And thus as A (is) to E , so E (is) to B . Thus, one number (namely, E) is in mean proportion to A and B .

So I say that A also has to B a squared ratio with respect to (that) C (has) to D . For since A , E , B are three (continuously) proportional numbers, A thus has to B a squared ratio with respect to (that) A (has) to E [Def. 5.9]. And as A (is) to E , so C (is) to D . Thus, A has to B a squared ratio with respect to (that) side C (has) to (side) D . (Which is) the very thing it was required to show.

[†] In other words, between two given square numbers there exists a number in continued proportion.

[‡] Literally, “double”.

12.

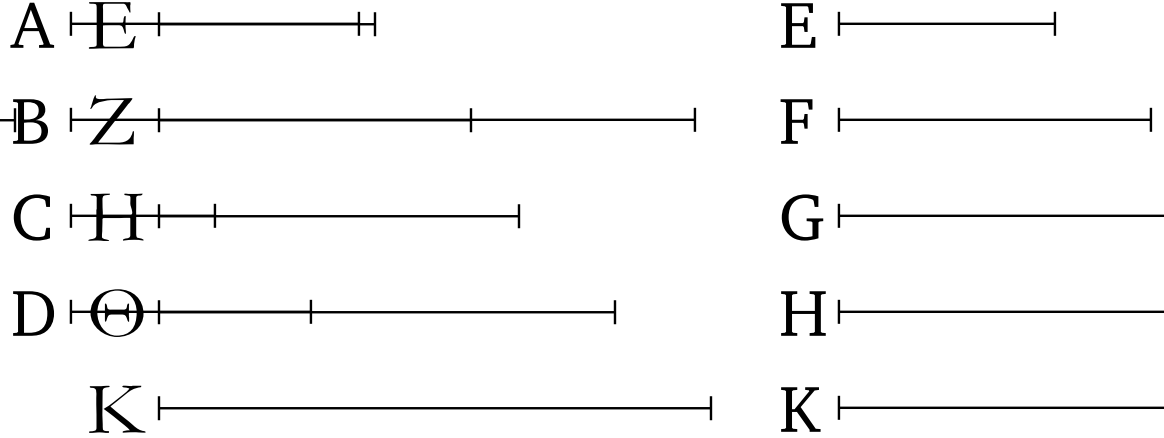
Proposition 12

Δύο κύβων ἀριθμῶν δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοί, καὶ ὁ κύβος πρὸς τὸν κύβον τριπλασίονα λόγον ἔθει ἥπερ ἡ πλευρὰ πρὸς τὴν πλευράν.

Ἐστωσαν κύβοι ἀριθμοὶ οἱ A, B καὶ τοῦ μὲν A πλευρὰ ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ . λέγω, ὅτι τῶν A, B δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοί, καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν Δ .

There exist two numbers in mean proportion to two (given) cube numbers.[†] And (one) cube (number) has to the (other) cube (number) a cubed[‡] ratio with respect to (that) the side (of the former has) to the side (of the latter).

Let A and B be cube numbers, and let C be the side of A , and D (the side) of B . I say that there exist two numbers in mean proportion to A and B , and that A has to B a cubed ratio with respect to (that) C (has) to D .



Ὁ γὰρ Γ ἑαυτὸν μὲν πολλαπλασιάσας τὸν E ποιεῖτω, τὸν δὲ Δ πολλαπλασιάσας τὸν Z ποιεῖτω, ὁ δὲ Δ ἑαυτὸν πολλαπλασιάσας τὸν H ποιεῖτω, ἐκάτεροξ δὲ τῶν Γ, Δ τὸν Z πολλαπλασιάσας ἐκάτερον τῶν Θ, K ποιεῖτω.

Καὶ ἐπεὶ κύβος ἐστὶν ὁ A , πλευρὰ δὲ αὐτοῦ ὁ Γ , καὶ ὁ Γ ἑαυτὸν μὲν πολλαπλασιάσας τὸν E πεποίηκεν, ὁ Γ ἄρα ἑαυτὸν μὲν πολλαπλασιάσας τὸν E πεποίηκεν, τὸν δὲ E πολλαπλασιάσας τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Δ ἑαυτὸν μὲν πολλαπλασιάσας τὸν H πεποίηκεν, τὸν δὲ H πολλαπλασιάσας τὸν B πεποίηκεν. καὶ ἐπεὶ ὁ Γ ἐκάτερον τῶν Γ, Δ πολλαπλασιάσας ἐκάτερον τῶν E, Z πεποίηκεν, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν Z . διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ Z πρὸς τὸν H . πάλιν, ἐπεὶ ὁ Γ ἐκάτερον τῶν E, Z πολλαπλασιάσας ἐκάτερον τῶν Θ, K πεποίηκεν, ἔστιν ἄρα ὡς ὁ E πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν Θ . ὡς δὲ ὁ E πρὸς τὸν Z , οὕτως ὁ Γ πρὸς τὸν Δ . καὶ ὡς ἄρα ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν Θ . πάλιν, ἐπεὶ ἐκάτεροξ τῶν Γ, Δ τὸν Z πολλαπλασιάσας ἐκάτερον τῶν Θ, K πεποίηκεν, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ Θ πρὸς τὸν K . πάλιν, ἐπεὶ ὁ Δ ἐκάτερον τῶν Z, H πολλαπλασιάσας ἐκάτερον τῶν K, B πεποίηκεν, ἔστιν ἄρα ὡς ὁ Z πρὸς τὸν H , οὕτως ὁ K πρὸς τὸν B . ὡς δὲ ὁ Z πρὸς τὸν H , οὕτως ὁ Γ πρὸς τὸν Δ . καὶ ὡς ἄρα ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν K καὶ ὁ K πρὸς τὸν B . τῶν A, B ἄρα δύο μέσοι ἀνάλογόν εἰσιν οἱ Θ, K .

Λέγω δὴ, ὅτι καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν Δ . ἐπεὶ γὰρ τέσσαρες ἀριθμοὶ ἀνάλογόν

For let C make E (by) multiplying itself, and let it make F (by) multiplying D . And let D make G (by) multiplying itself, and let C, D make H, K , respectively, (by) multiplying F .

And since A is cube, and C (is) its side, and C has made E (by) multiplying itself, C has thus made E (by) multiplying itself, and has made A (by) multiplying E . And so, for the same (reasons), D has made G (by) multiplying itself, and has made B (by) multiplying G . And since C has made E, F (by) multiplying C, D , respectively, thus as C is to D , so E (is) to F [Prop. 7.17]. And so, for the same (reasons), as C (is) to D , so F (is) to G [Prop. 7.18]. Again, since C has made A, H (by) multiplying E, F , respectively, thus as E is to F , so A (is) to H [Prop. 7.17]. And as E (is) to F , so C (is) to D . And thus as C (is) to D , so A (is) to H . Again, since C, D have made H, K , respectively, (by) multiplying F , thus as C is to D , so H (is) to K [Prop. 7.18]. Again, since D has made K, B (by) multiplying F, G , respectively, thus as F is to G , so K (is) to B [Prop. 7.17]. And as F (is) to G , so C (is) to D . And thus as C (is) to D , so A (is) to H , and H to K , and K to B . Thus, H and K are two (numbers) in mean proportion to A and B .

So I say that A also has to B a cubed ratio with respect to (that) C (has) to D . For since A, H, K, B are four (continuously) proportional numbers, A thus has to B a cubed ratio with respect to (that) A (has) to H [Def. 5.10]. And as A (is) to H , so C (is) to D . And [thus] A has to B a cubed ratio with respect to (that) C (has) to D . (Which is) the very thing it was required to show.

εἰσιν οἱ A, Θ, K, B , ὁ A ἄρα πρὸς τὸν B τριπλασίονα λόγον ἔθει ἤπερ ὁ A πρὸς τὸν Θ . ὥς δὲ ὁ A πρὸς τὸν Θ , οὕτως ὁ Γ πρὸς τὸν Δ · καὶ ὁ A [ἄρα] πρὸς τὸν B τριπλασίονα λόγον ἔθει ἤπερ ὁ Γ πρὸς τὸν Δ · ὅπερ ἔδει δεῖχαι.

† In other words, between two given cube numbers there exist two numbers in continued proportion.

‡ Literally, “triple”.

13.

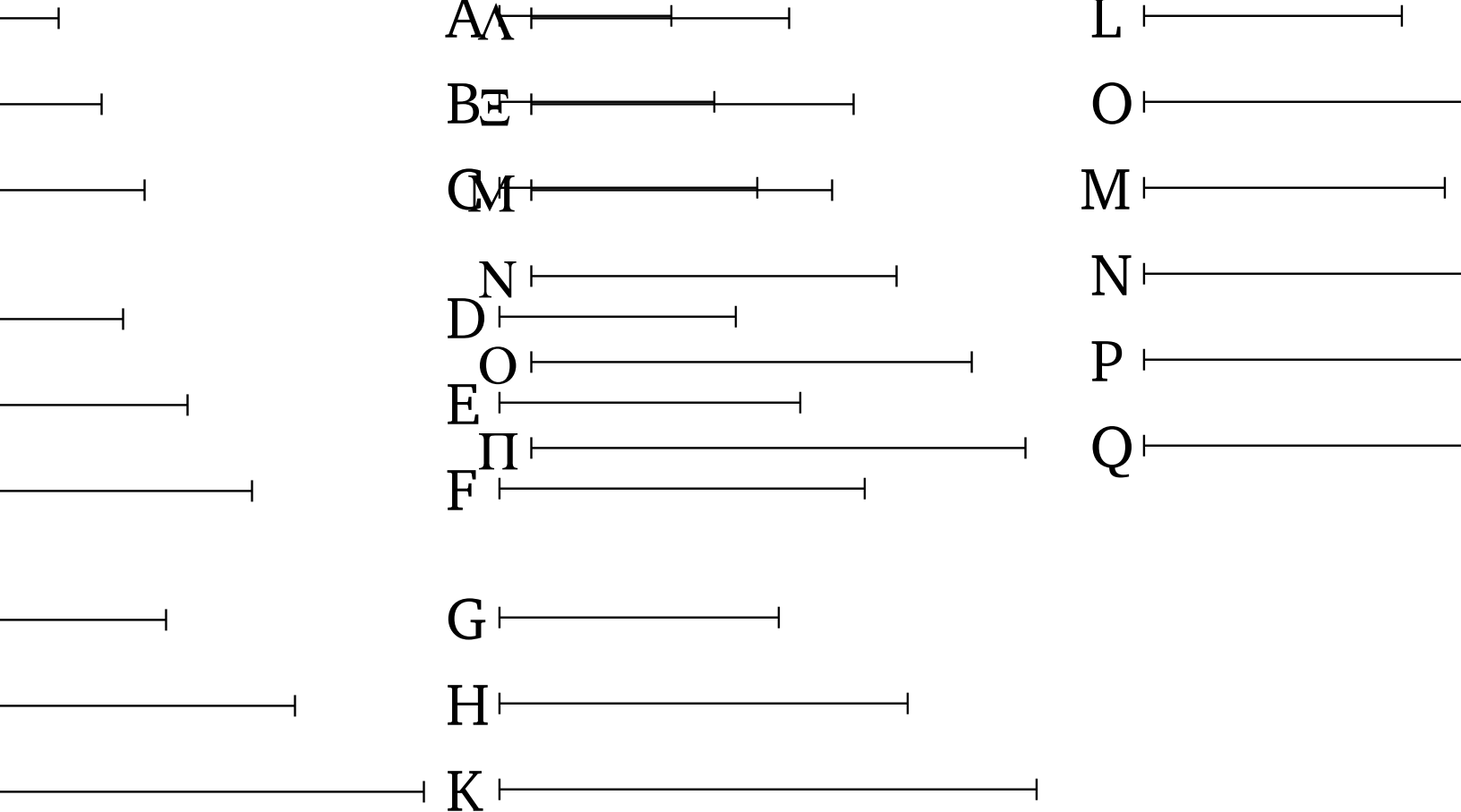
Proposition 13

Ἐὰν ὥσιν ὅσοιδηποῦν ἀριθμοὶ ἐχῇς ἀνάλογον, καὶ πολλαπλασιάσας ἕκαστος ἐαυτὸν ποιῇ τινα, οἱ γεγόμενοι ἐκ αὐτῶν ἀνάλογον ἔσονται· καὶ ἔαν οἱ ἐκ ἀρθῇς τούς γενομένους πολλαπλασιάσαντες ποιῶσί τιναξ, καὶ αὐτοὶ ἀνάλογον ἔσονται [καὶ αἰεὶ περὶ τούς ἄκρονξ τοῦτο συμβαίνει].

Ἐστωσαν ὅποιοιῦν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ A, B, Γ , ὥς ὁ A πρὸς τὸν B , οὕτως ὁ B πρὸς τὸν Γ , καὶ οἱ A, B, Γ ἐαντούξ μὲν πολλαπλασιάσαντες τούξ Δ, E, Z ποιείτωσαν, τούξ δὲ Δ, E, Z πολλαπλασιάσαντες τούξ H, Θ, K ποιείτωσαν· λέγω, ὅτι οἱ τε Δ, E, Z καὶ οἱ H, Θ, K ἐχῇς ἀνάλογον εἰσιν.

If there are any multitude whatsoever of continuously proportional numbers, and each makes some (number by) multiplying itself, then the (numbers) created from them will (also) be (continuously) proportional. And if the original (numbers) make some (more numbers by) multiplying the created (numbers) then these will also be (continuously) proportional [and this always happens with the extremes].

Let A, B, C be any multitude whatsoever of continuously proportional numbers, (such that) as A (is) to B , so B (is) to C . And let A, B, C make D, E, F (by) multiplying themselves, and let them make G, H, K (by) multiplying D, E, F . I say that D, E, F and G, H, K are continuously proportional.



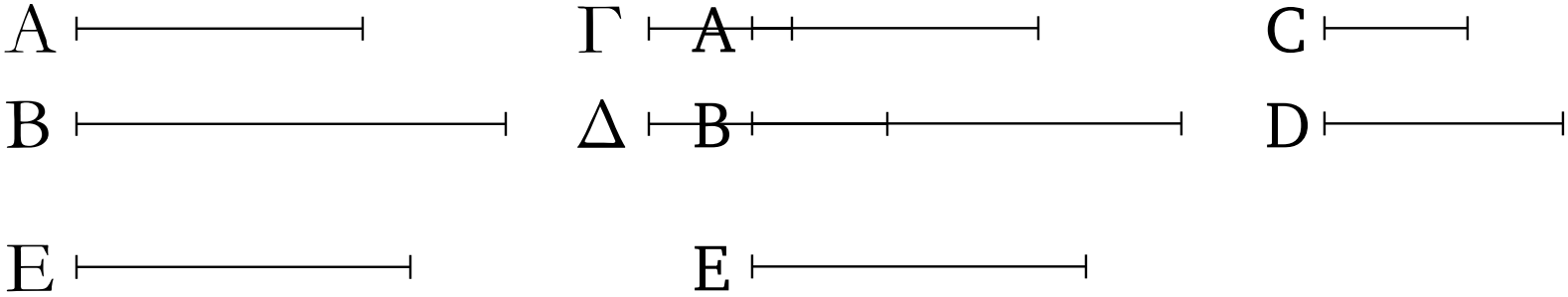
Ὅ μὲν γάρ A τὸν B πολλαπλασιάσας τὸν Λ ποιεῖτω, ἑκάτερος δὲ τῶν A, B τὸν Λ πολλαπλασιάσας ἑκάτερον τῶν M, N ποιεῖτω. καὶ πάλιν ὁ μὲν B τὸν Γ πολλαπλασιάσας τὸν X ποιεῖτω, ἑκάτερος δὲ τῶν B, Γ τὸν X πολλαπλασιάσας ἑκάτερον τῶν O, Π ποιεῖτω.

Ὅμοιως δὴ τοῖς ἐπάνω δεῖχμεν, ὅτι οἱ Δ, Λ, E καὶ οἱ H, M, N, Θ ἐχῆς εἶναι ἀνάλογον ἐν τῷ τοῦ A πρὸς τὸν B λόγῳ, καὶ ἔτι οἱ E, X, Z καὶ οἱ Θ, O, Π, K ἐχῆς εἶναι ἀνάλογον ἐν τῷ τοῦ B πρὸς τὸν Γ λόγῳ. καὶ ἐστὶν ὥς ὁ A πρὸς τὸν B , οὕτως ὁ B πρὸς τὸν Γ · καὶ οἱ Δ, Λ, E ἄρα τοῖς E, X, Z ἐν τῷ αὐτῷ λόγῳ εἰσὶ καὶ ἔτι οἱ H, M, N, Θ τοῖς Θ, O, Π, K . καὶ ἐστὶν ἴσον τὸ μὲν τῶν Δ, Λ, E πλῆθος τῷ τῶν E, X, Z πλήθει, τὸ δὲ τῶν H, M, N, Θ τῷ τῶν Θ, O, Π, K δι' ἴσον ἄρα ἐστὶν ὥς μὲν ὁ Δ πρὸς τὸν E , οὕτως ὁ E πρὸς τὸν Z , ὥς δὲ ὁ H πρὸς τὸν Θ , οὕτως ὁ Θ πρὸς τὸν K · ὅπερ ἔδει δεῖχαι.

14.

Ἐὰν τετράγωνος τετράγωνον μετρή, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσει· καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρή, καὶ ὁ τετράγωνος τὸν τετράγωνον μετρήσει.

Ἔστωσαν τετράγωνοι ἀριθμοὶ οἱ A, B , πλευραὶ δὲ αὐτῶν ἔστωσαν οἱ Γ, Δ , ὁ δὲ A τὸν B μετρεῖτω· λέγω, ὅτι καὶ ὁ Γ τὸν Δ μετρεῖ.



Ὅ Γ γάρ τὸν Δ πολλαπλασιάσας τὸν E ποιεῖτω· οἱ A, E, B ἄρα ἐχῆς ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ οἱ A, E, B ἐχῆς ἀνάλογόν εἰσιν, καὶ μετρεῖ ὁ A τὸν B , μετρεῖ ἄρα καὶ ὁ A τὸν E . καὶ ἐστὶν ὥς ὁ A πρὸς τὸν E , οὕτως ὁ Γ πρὸς τὸν Δ · μετρεῖ ἄρα καὶ ὁ Γ τὸν Δ .

Πάλιν δὴ ὁ Γ τὸν Δ μετρεῖτω· λέγω, ὅτι καὶ ὁ A τὸν B μετρεῖ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δεῖχμεν, ὅτι οἱ A, E, B ἐχῆς ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ ἐστὶν ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν E , μετρεῖ δὲ ὁ Γ τὸν Δ , μετρεῖ ἄρα καὶ ὁ A τὸν E . καὶ εἰσιν οἱ A, E, B ἐχῆς ἀνάλογον μετρεῖ ἄρα καὶ ὁ A τὸν B .

Ἐὰν ἄρα τετράγωνος τετράγωνον μετρή, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσει· καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρή, καὶ ὁ τετράγωνος τὸν τετράγωνον μετρήσει· ὅπερ ἔδει δεῖχαι.

For let A make L (by) multiplying B . And let A, B make M, N , respectively, (by) multiplying L . And, again, let B make O (by) multiplying C . And let B, C make P, Q , respectively, (by) multiplying O .

So, similarly to the above, we can show that D, L, E and G, M, N, H are continuously proportional in the ratio of A to B , and, further, (that) E, O, F and H, P, Q, K are continuously proportional in the ratio of B to C . And as A is to B , so B (is) to C . And thus D, L, E are in the same ratio as E, O, F , and, further, G, M, N, H (are in the same ratio) as H, P, Q, K . And the multitude of D, L, E is equal to the multitude of E, O, F , and that of G, M, N, H to that of H, P, Q, K . Thus, via equality, as D is to E , so E (is) to F , and as G (is) to H , so H (is) to K [Prop. 7.14]. (Which is) the very thing it was required to show.

Proposition 14

If a square (number) measures a(nother) square (number) then the side (of the former) will also measure the side (of the latter). And if the side (of a square number) measures the side (of another square number) then the (former) square (number) will also measure the (latter) square (number).

Let A and B be square numbers, and let C and D be their sides (respectively). And let A measure B . I say that C also measures D .

For let C make E (by) multiplying D . Thus, A, E, B are continuously proportional in the ratio of C to D [Prop. 8.11]. And since A, E, B are continuously proportional, and A measures B , A thus also measures E [Prop. 8.7]. And as A is to E , so C (is) to D . Thus, C also measures D [Def. 7.20].

So, again, let C measure D . I say that A also measures B .

For similarly, with the same construction, we can show that A, E, B are continuously proportional in the ratio of C to D . And since as C is to D , so A (is) to E , and C measures D , A thus also measures E [Def. 7.20]. And A, E, B are continuously proportional. Thus, A also measures B .

Thus, if a square (number) measures a(nother) square (number) then the side (of the former) will also measure the side (of the latter). And if the side (of a square number) measures the side (of another square number) then the (former)

square (number) will also measure the (latter) square (number). (Which is) the very thing it was required to show.

15.

Ἐὰν κύβος ἀριθμὸς κύβον ἀριθμὸν μετρήῃ, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσῃ· καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρήῃ, καὶ ὁ κύβος τὸν κύβον μετρήσῃ.

Κύβος γὰρ ἀριθμὸς ὁ A κύβον τὸν B μετρεῖτω, καὶ τοῦ μὲν A πλευρὰ ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ · λέγω, ὅτι ὁ Γ τὸν Δ μετρεῖ.

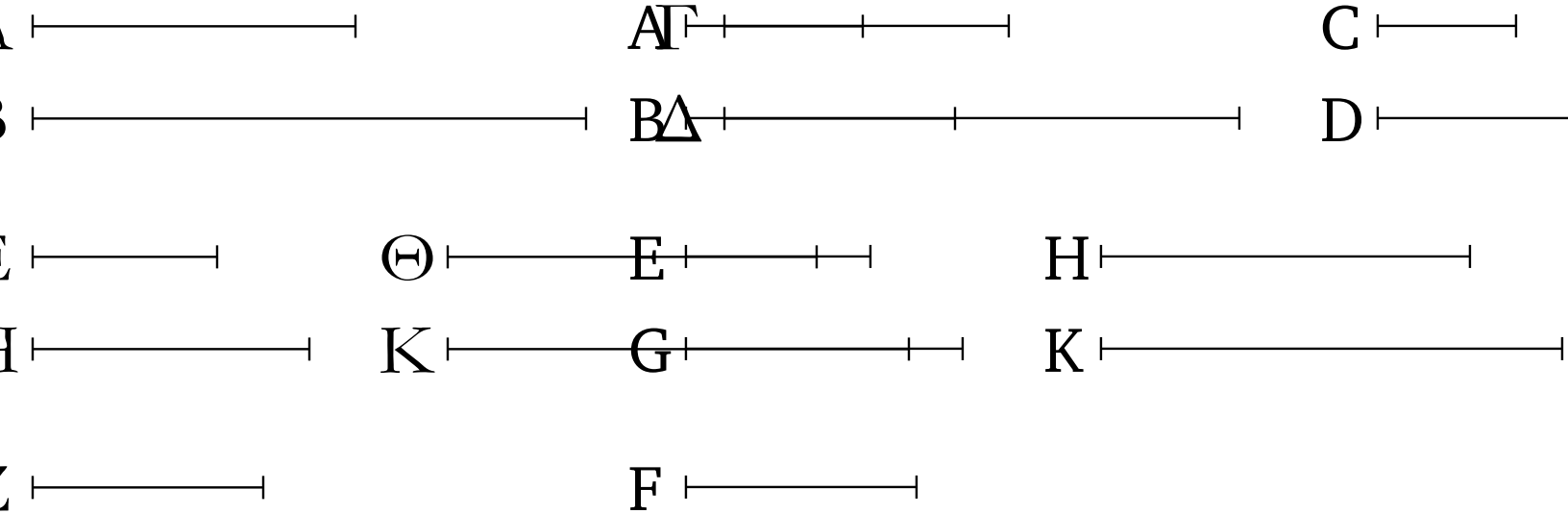
Ὁ Γ γὰρ ἑαυτὸν πολλαπλασιάσας τὸν E ποιεῖτω, ὁ δὲ Δ ἑαυτὸν πολλαπλασιάσας τὸν H ποιεῖτω, καὶ ἔτι ὁ Γ τὸν Δ πολλαπλασιάσας τὸν Z [ποιεῖτω], ἐκάτερος δὲ τῶν Γ , Δ τὸν Z πολλαπλασιάσας ἐκάτερον τῶν Θ , K ποιεῖτω. φανερόν δὴ, ὅτι οἱ E , Z , H καὶ οἱ A , Θ , K , B ἐχῆς ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ οἱ A , Θ , K , B ἐχῆς ἀνάλογόν εἰσιν, καὶ μετρεῖ ὁ A τὸν B , μετρεῖ ἄρα καὶ τὸν Θ . καὶ ἔστιν ὥς ὁ A πρὸς τὸν Θ , οὕτως ὁ Γ πρὸς τὸν Δ · μετρεῖ ἄρα καὶ ὁ Γ τὸν Δ .

Proposition 15

If a cube number measures a(nother) cube number then the side (of the former) will also measure the side (of the latter). And if the side (of a cube number) measures the side (of another cube number) then the (former) cube (number) will also measure the (latter) cube (number).

For let the cube number A measure the cube (number) B , and let C be the side of A , and D (the side) of B . I say that C measures D .

For let C make E (by) multiplying itself. And let D make G (by) multiplying itself. And, further, [let] C [make] F (by) multiplying D , and let C , D make H , K , respectively, (by) multiplying F . So it is clear that E , F , G and A , H , K , B are continuously proportional in the ratio of C to D [Prop. 8.12]. And since A , H , K , B are continuously proportional, and A measures B , (A) thus also measures H [Prop. 8.7]. And as A is to H , so C (is) to D . Thus, C also measures D [Def. 7.20].



Ἀλλὰ δὴ μετρεῖτω ὁ Γ τὸν Δ · λέγω, ὅτι καὶ ὁ A τὸν B μετρήσῃ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δὴ δείκμεν, ὅτι οἱ A , Θ , K , B ἐχῆς ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ ὁ Γ τὸν Δ μετρεῖ, καὶ ἔστιν ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν Θ , καὶ ὁ A ἄρα τὸν Θ μετρεῖ· ὥστε καὶ τὸν B μετρεῖ ὁ A · ὅπερ ἔδει δεῖξαι.

And so let C measure D . I say that A will also measure B . For similarly, with the same construction, we can show that A , H , K , B are continuously proportional in the ratio of C to D . And since C measures D , and as C is to D , so A (is) to H , A thus also measures H [Def. 7.20]. Hence, A also measures B . (Which is) the very thing it was required to show.

16.

Ἐὰν τετράγωνος ἀριθμὸς τετράγωνον ἀριθμὸν μὴ μετρήῃ, οὐδὲ ἡ πλευρὰ τὴν πλευρὰν μετρήσῃ· κ' ἂν ἡ πλευρὰ τὴν πλευ-

Proposition 16

If a square number does not measure a(nother) square number then the side (of the former) will not measure the side

ἐὰν μὴ μετρήῃ, οὐδὲ ὁ τετράγωνος τὸν τετράγωνον μετρήσει.

(of the latter) either. And if the side (of a square number) does not measure the side (of another square number) then the (former) square (number) will not measure the (latter) square

A —————

Γ —————

B —————

Δ ————— (number) either.

A —————

C —————

B —————

D —————

Ἐστωσαν τετράγωνοι ἀριθμοὶ οἱ A, B , πλευραὶ δὲ αὐτῶν ἔστωσαν οἱ Γ, Δ , καὶ μὴ μετρεῖται ὁ A τὸν B · λέγω, ὅτι οὐδὲ ὁ Γ τὸν Δ μετρεῖ.

Let A and B be square numbers, and let C and D be their sides (respectively). And let A not measure B . I say that C does not measure D either.

Εἰ γὰρ μετρεῖ ὁ Γ τὸν Δ , μετρήσει καὶ ὁ A τὸν B . οὐ μετρεῖ δὲ ὁ A τὸν B · οὐδὲ ἄρα ὁ Γ τὸν Δ μετρήσει.

For if C measures D then A will also measure B [Prop. 8.14].

μὴ μετρεῖται [δὴ] πάλιν ὁ Γ τὸν Δ · λέγω, ὅτι οὐδὲ ὁ A τὸν B μετρήσει.

And A does not measure B . Thus, C will not measure D either.

Εἰ γὰρ μετρεῖ ὁ A τὸν B , μετρήσει καὶ ὁ Γ τὸν Δ . οὐ μετρεῖ δὲ ὁ Γ τὸν Δ · οὐδὲ ἄρα ὁ A τὸν B μετρήσει· ὅπερ ἔδει δεῖξαι.

[So], again, let C not measure D . I say that A will not measure B either.

For if A measures B then C will also measure D [Prop. 8.14]. And C does not measure D . Thus, A will not measure B either. (Which is) the very thing it was required to show.

17.

Proposition 17

Ἐὰν κύβος ἀριθμὸς κύβον ἀριθμὸν μὴ μετρήῃ, οὐδὲ ἡ πλευρὰ τὴν πλευρὰν μετρήσει· καὶ ἡ πλευρὰ τὴν πλευρὰν μὴ μετρήῃ, οὐδὲ ὁ κύβος τὸν κύβον μετρήσει.

If a cube number does not measure a(nother) cube number then the side (of the former) will not measure the side (of the latter) either. And if the side (of a cube number) does not measure the side (of another cube number) then the (former) cube (number) will not measure the (latter) cube (number) either.

A —————

Γ ————— A —————

C —————

B —————

Δ ————— B —————

D —————

Κύβος γὰρ ἀριθμὸς ὁ A κύβον ἀριθμὸν τὸν B μὴ μετρεῖται, καὶ τοῦ μὲν A πλευρὰ ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ · λέγω, ὅτι ὁ Γ τὸν Δ οὐ μετρήσει.

For let the cube number A not measure the cube number B . And let C be the side of A , and D (the side) of B . I say that C will not measure D .

Εἰ γὰρ μετρεῖ ὁ Γ τὸν Δ , καὶ ὁ A τὸν B μετρήσει. οὐ μετρεῖ δὲ ὁ A τὸν B · οὐδὲ ἄρα ὁ Γ τὸν Δ μετρεῖ.

For if C measures D then A will also measure B [Prop. 8.15].

Ἀλλὰ δὴ μὴ μετρεῖται ὁ Γ τὸν Δ · λέγω, ὅτι οὐδὲ ὁ A τὸν B μετρήσει.

And A does not measure B . Thus, C does not measure D either.

Εἰ γὰρ ὁ A τὸν B μετρεῖ, καὶ ὁ Γ τὸν Δ μετρήσει. οὐ μετρεῖ δὲ ὁ Γ τὸν Δ · οὐδὲ ἄρα ὁ A τὸν B μετρήσει· ὅπερ ἔδει δεῖξαι.

And so let C not measure D . I say that A will not measure B either.

For if A measures B then C will also measure D [Prop. 8.15]. And C does not measure D . Thus, A will not measure B either. (Which is) the very thing it was required to show.

18.

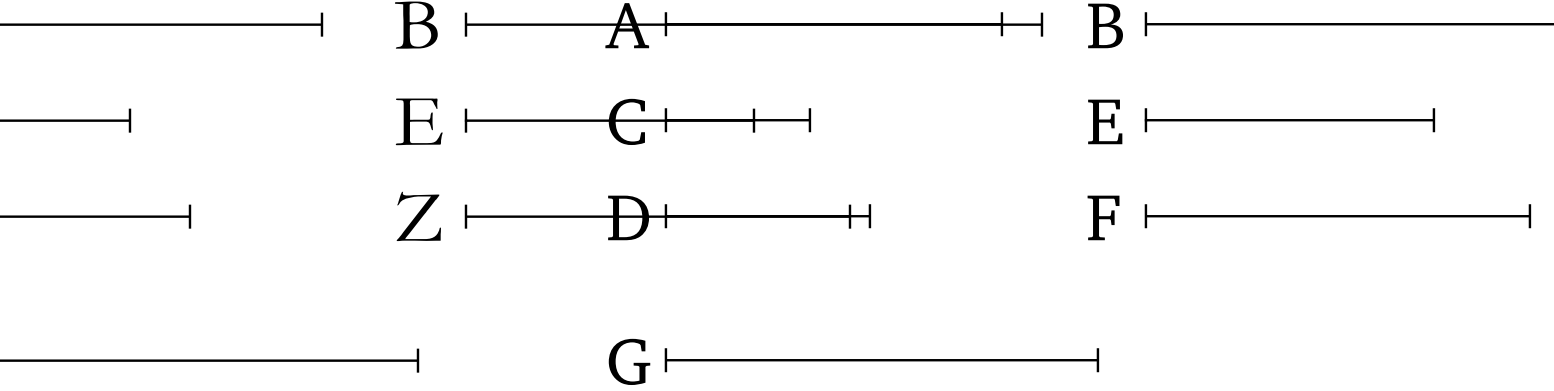
Proposition 18

Δύο ὁμοίων ἐπιπέδων ἀριθμῶν εἷξ μέσοξ ἀνάλογόν ἐστιν ἀριθμὸς· καὶ ὁ ἐπίπεδος πρόξ τὸν ἐπίπεδον διπλασίονα λόγον

There exists one number in mean proportion to two similar plane numbers. And (one) plane (number) has to the (other)

ἔθει ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν.

plane (number) a squared[†] ratio with respect to (that) a corresponding side (of the former has) to a corresponding side (of the latter).



Ἐστωσαν δύο ὅμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B , καὶ τοῦ μὲν A πλευραὶ ἔστωσαν οἱ Γ, Δ ἀριθμοί, τοῦ δὲ B οἱ E, Z . καὶ ἐπεὶ ὅμοιοι ἐπίπεδοι εἰσιν οἱ ἀνάλογον ἔθοντες τὰς πλευράς, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν Z . λέγω οὖν, ὅτι τῶν A, B εἷς μέσος ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν E ἢ ὁ Δ πρὸς τὸν Z , τουτέστιν ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον [πλευράν].

Καὶ ἐπεὶ ἐστὶν ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν Z , ἐναλλάξ ἄρα ἐστὶν ὡς ὁ Γ πρὸς τὸν E , ὁ Δ πρὸς τὸν Z . καὶ ἐπεὶ ἐπίπεδός ἐστιν ὁ A , πλευραὶ δὲ αὐτοῦ οἱ Γ, Δ , ὁ Δ ἄρα τὸν Γ πολλαπλασιάσας τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ E τὸν Z πολλαπλασιάσας τὸν B πεποίηκεν. ὁ Δ δὴ τὸν E πολλαπλασιάσας τὸν H ποιεῖτω. καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ E πολλαπλασιάσας τὸν H πεποίηκεν, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν E , οὕτως ὁ A πρὸς τὸν H . ἀλλ' ὡς ὁ Γ πρὸς τὸν E , [οὕτως] ὁ Δ πρὸς τὸν Z . καὶ ὡς ἄρα ὁ Δ πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν H . πάλιν, ἐπεὶ ὁ E τὸν μὲν Δ πολλαπλασιάσας τὸν H πεποίηκεν, τὸν δὲ Z πολλαπλασιάσας τὸν B πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ H πρὸς τὸν B . ἐδείκθη δὲ καὶ ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν H . καὶ ὡς ἄρα ὁ A πρὸς τὸν H , οὕτως ὁ H πρὸς τὸν B . οἱ A, H, B ἄρα ἐχῆς ἀνάλογόν εἰσιν. τῶν A, B ἄρα εἷς μέσος ἀνάλογόν ἐστιν ἀριθμός.

Λέγω δὴ, ὅτι καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἥπερ ὁ Γ πρὸς τὸν E ἢ ὁ Δ πρὸς τὸν Z . ἐπεὶ γάρ οἱ A, H, B ἐχῆς ἀνάλογόν εἰσιν, ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἥπερ πρὸς τὸν H . καὶ ἐστὶν ὡς ὁ A πρὸς τὸν H , οὕτως ὁ Γ πρὸς τὸν E καὶ ὁ Δ πρὸς τὸν Z . καὶ ὁ A ἄρα πρὸς τὸν B διπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν E ἢ ὁ Δ πρὸς τὸν Z . ὅπερ ἔδει δεῖξαι.

Let A and B be two similar plane numbers. And let the numbers C, D be the sides of A , and E, F (the sides) of B . And since similar numbers are those having proportional sides [Def. 7.21], thus as C is to D , so E (is) to F . Therefore, I say that there exists one number in mean proportion to A and B , and that A has to B a squared ratio with respect to that C (has) to E , or D to F —that is to say, with respect to (that) a corresponding side (has) to a corresponding [side].

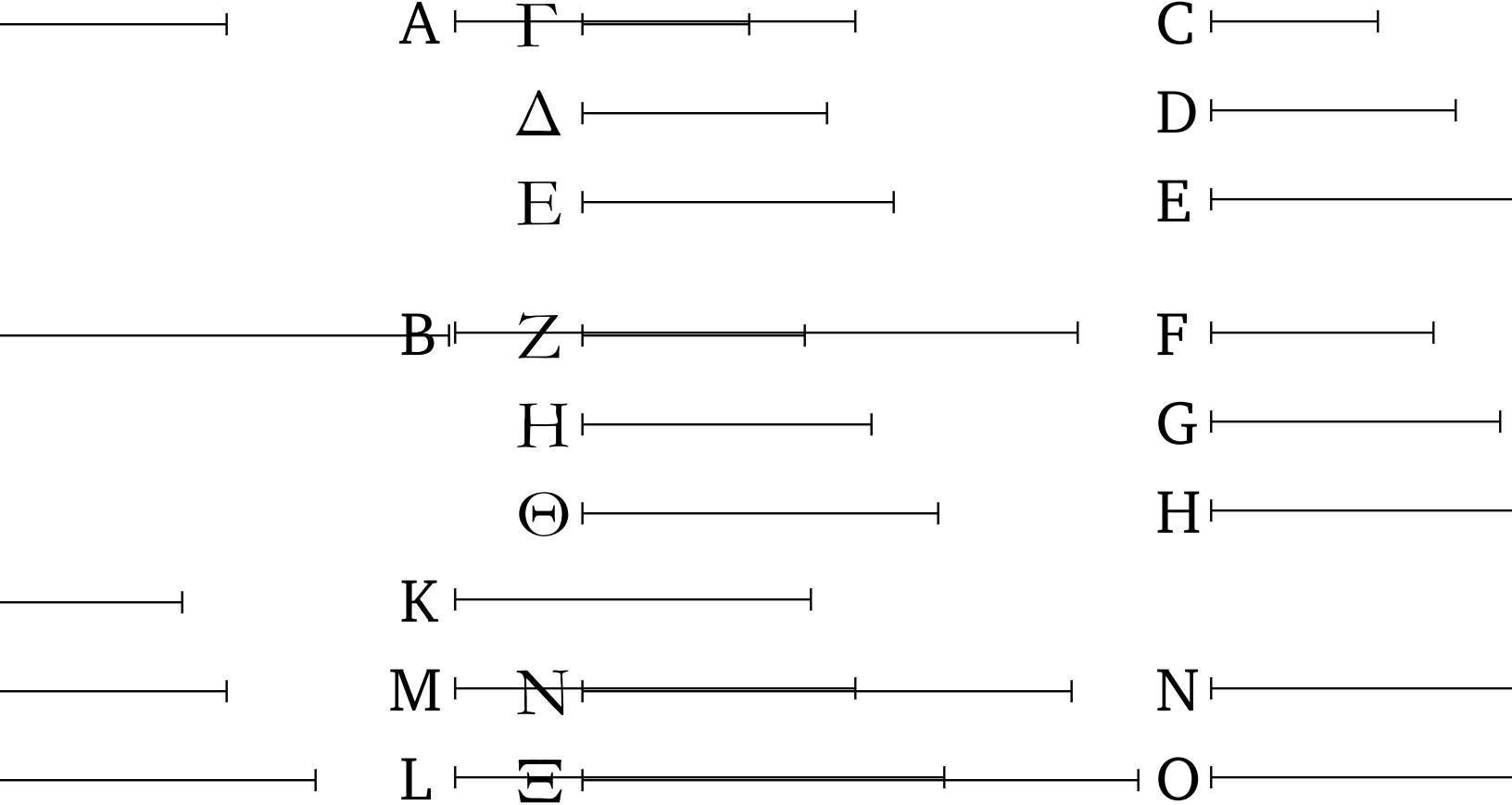
For since as C is to D , so E (is) to F , thus, alternately, as C is to E , so D (is) to F [Prop. 7.13]. And since A is plane, and C, D its sides, D has thus made A (by) multiplying C . And so, for the same (reasons), E has made B (by) multiplying F . So let D make G (by) multiplying E . And since D has made A (by) multiplying C , and has made G (by) multiplying E , thus as C is to E , so A (is) to G [Prop. 7.17]. But as C (is) to E , [so] D (is) to F . And thus as D (is) to F , so A (is) to G . Again, since E has made G (by) multiplying D , and has made B (by) multiplying F , thus as D is to F , so G (is) to B [Prop. 7.17]. And it was also shown that as D (is) to F , so A (is) to G . And thus as A (is) to G , so G (is) to B . Thus, A, G, B are continuously proportional. Thus, there exists one number (namely, G) in mean proportion to A and B .

So I say that A also has to B a squared ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) C (has) to E , or D to F . For since A, G, B are continuously proportional, A has to B a squared ratio with respect to (that A has) to G [Prop. 5.9]. And as A is to G , so C (is) to E , and D to F . And thus A has to B a squared ratio with respect to (that) C (has) to E , or D to F . (Which is) the very thing it was required to show.

[†] Literally, “double”.

Δύο ὁμοίων στερεῶν ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί· καὶ ὁ στερεὸς πρὸς τὸν ὅμοιον στερεὸν τριπλασίονα λόγον ἔθει ἢπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν.

Two numbers fall (between) two similar solid numbers in mean proportion. And a solid (number) has to a similar solid (number) a cubed[†] ratio with respect to (that) a corresponding side (has) to a corresponding side.



Ἐστωσαν δύο ὅμοιοι στερεοὶ οἱ A, B , καὶ τοῦ μὲν A πλευραὶ ἔστωσαν οἱ Γ, Δ, E , τοῦ δὲ B οἱ Z, H, Θ . καὶ ἐπεὶ ὅμοιοι στερεοὶ εἰσιν οἱ ἀνάλογον ἔθοντες τὰς πλευράς, ἔστιν ἄρα ὥς μὲν ὁ Γ πρὸς τὸν Δ , οὕτως ὁ Z πρὸς τὸν H , ὥς δὲ ὁ Δ πρὸς τὸν E , οὕτως ὁ H πρὸς τὸν Θ . λέγω, ὅτι τῶν A, B δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί, καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον ἔθει ἢπερ ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ .

Ὁ Γ γὰρ τὸν Δ πολλαπλασιάσας τὸν K ποιεῖτω, ὁ δὲ Z τὸν H πολλαπλασιάσας τὸν Λ ποιεῖτω. καὶ ἐπεὶ οἱ Γ, Δ τοίς Z, H ἐν τῷ αὐτῷ λόγῳ εἰσίν, καὶ ἐκ μὲν τῶν Γ, Δ ἔστιν ὁ K , ἐκ δὲ τῶν Z, H ὁ Λ , οἱ K, Λ [ἄρα] ὅμοιοι ἐπίπεδοι εἰσιν ἀριθμοί· τῶν K, Λ ἄρα εἴς μέσοις ἀνάλογον ἔστιν ἀριθμός. ἔστω ὁ M . ὁ M ἄρα ἔστιν ὁ ἐκ τῶν Δ, Z , ὥς ἐν τῷ πρὸ τούτου θεωρήματι ἐδείχθη. καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσας τὸν K πεποίηκεν, τὸν δὲ Z πολλαπλασιάσας τὸν M πεποίηκεν, ἔστιν ἄρα ὥς ὁ Γ πρὸς τὸν Z , οὕτως ὁ K πρὸς τὸν M . ἀλλ' ὥς ὁ K πρὸς τὸν M , ὁ M πρὸς τὸν Λ . οἱ K, M, Λ ἄρα ἐχθῆς εἰσιν ἀνάλογον ἐν τῷ τοῦ Γ πρὸς τὸν Z λόγῳ. καὶ ἐπεὶ ἔστιν

Let A and B be two similar solid numbers, and let C, D, E be the sides of A , and F, G, H (the sides) of B . And since similar solid (numbers) are those having proportional sides [Def. 7.21], thus as C is to D , so F (is) to G , and as D (is) to E , so G (is) to H . I say that two numbers fall (between) A and B in mean proportion, and (that) A has to B a cubed ratio with respect to (that) C (has) to F , and D to G , and, further, E to H .

For let C make K (by) multiplying D , and let F make L (by) multiplying G . And since C, D are in the same ratio as F, G , and K is the (number created) from (multiplying) C, D , and L the (number created) from (multiplying) F, G , [thus] K and L are similar plane numbers [Def. 7.21]. Thus, there exists one number in mean proportion to K and L [Prop. 8.18]. Let it be M . Thus, M is the (number created) from (multiplying) D, F , as shown in the theorem before this (one). And since D has made K (by) multiplying C , and has made M (by) multiplying F , thus as C is to F , so K (is) to M [Prop. 7.17]. But, as K (is) to M , (so) M (is) to L . Thus, K, M, L are continuously

ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ Z πρὸς τὸν H , ἐναλλάχ' ἄρα ἐστὶν ὥς ὁ Γ πρὸς τὸν Z , οὕτως ὁ Δ πρὸς τὸν H . διὰ τὰ αὐτὰ δὴ καὶ ὥς ὁ Δ πρὸς τὸν H , οὕτως ὁ E πρὸς τὸν Θ . οἱ K , M , Λ ἄρα ἐχῆς εἶσιν ἀνάλογον ἐν τε τῷ τοῦ Γ πρὸς τὸν Z λόγῳ καὶ τῷ τοῦ Δ πρὸς τὸν H καὶ ἔτι τῷ τοῦ E πρὸς τὸν Θ . ἑκάτερος δὴ τῶν E , Θ τὸν M πολλαπλασιάσας ἑκάτερον τῶν N , X ποιείτω. καὶ ἐπεὶ στερεὸς ἐστὶν ὁ A , πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Γ , Δ , E , ὁ E ἄρα τὸν ἐκ τῶν Γ , Δ πολλαπλασιάσας τὸν A πεποιήκεν. ὁ δὲ ἐκ τῶν Γ , Δ ἐστὶν ὁ K . ὁ E ἄρα τὸν K πολλαπλασιάσας τὸν A πεποιήκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Θ τὸν Λ πολλαπλασιάσας τὸν B πεποιήκεν. καὶ ἐπεὶ ὁ E τὸν K πολλαπλασιάσας τὸν A πεποιήκεν, ἀλλὰ μὴν καὶ τὸν M πολλαπλασιάσας τὸν N πεποιήκεν, ἐστὶν ἄρα ὥς ὁ K πρὸς τὸν M , οὕτως ὁ A πρὸς τὸν N . ὥς δὲ ὁ K πρὸς τὸν M , οὕτως ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ . καὶ ὥς ἄρα ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ὁ E πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν N . πάλιν, ἐπεὶ ἑκάτερος τῶν E , Θ τὸν M πολλαπλασιάσας ἑκάτερον τῶν N , X πεποιήκεν, ἐστὶν ἄρα ὥς ὁ E πρὸς τὸν Θ , οὕτως ὁ N πρὸς τὸν X . ἀλλ' ὥς ὁ E πρὸς τὸν Θ , οὕτως ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H . καὶ ὥς ἄρα ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ὁ E πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν N καὶ ὁ N πρὸς τὸν X . πάλιν, ἐπεὶ ὁ Θ τὸν M πολλαπλασιάσας τὸν B πεποιήκεν, ἀλλὰ μὴν καὶ τὸν Λ πολλαπλασιάσας τὸν B πεποιήκεν, ἐστὶν ἄρα ὥς ὁ M πρὸς τὸν Λ , οὕτως ὁ X πρὸς τὸν B . ἀλλ' ὥς ὁ M πρὸς τὸν Λ , οὕτως ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ὁ E πρὸς τὸν Θ . καὶ ὥς ἄρα ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ὁ E πρὸς τὸν Θ , οὕτως οὐ μόνον ὁ X πρὸς τὸν B , ἀλλὰ καὶ ὁ A πρὸς τὸν N καὶ ὁ N πρὸς τὸν X . οἱ A , N , X , B ἄρα ἐχῆς εἰσιν ἀνάλογον ἐν τοῖς εἰρημένοις τῶν πλευρῶν λόγοις.

Λέγω, ὅτι καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον ἔθει ἢ περ ἢ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἢ περ ὁ Γ ἀριθμὸς πρὸς τὸν Z ἢ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ . ἐπεὶ γὰρ τέσσαρες ἀριθμοὶ ἐχῆς ἀνάλογον εἰσιν οἱ A , N , X , B , ὁ A ἄρα πρὸς τὸν B τριπλασίονα λόγον ἔθει ἢ περ ὁ A πρὸς τὸν N . ἀλλ' ὥς ὁ A πρὸς τὸν N , οὕτως ἐδείχθη ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ . καὶ ὁ A ἄρα πρὸς τὸν B τριπλασίονα λόγον ἔθει ἢ περ ἢ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἢ περ ὁ Γ ἀριθμὸς πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ . ὁ περ ἔδει δεῖχαι.

† Literally, "triple".

20.

Ἐάν δύο ἀριθμῶν εἷς μέσος ἀνάλογον ἐμπίπτῃ ἀριθμὸς, ὅμοιοι ἐπίπεδοι ἔσονται οἱ ἀριθμοί.

Δύο γὰρ ἀριθμῶν τῶν A , B εἷς μέσος ἀνάλογον ἐμπέτω ἀριθμὸς ὁ Γ . λέγω, ὅτι οἱ A , B ὅμοιοι ἐπίπεδοι εἰσιν

proportional in the ratio of C to F . And since as C is to D , so F (is) to G , thus, alternately, as C is to F , so D (is) to G [Prop. 7.13]. And so, for the same (reasons), as D (is) to G , so E (is) to H . Thus, K , M , L are continuously proportional in the ratio of C to F , and of D to G , and, further, of E to H . So let E , H make N , O , respectively, (by) multiplying M . And since A is solid, and C , D , E are its sides, E has thus made A (by) multiplying the (number created) from (multiplying) C , D . And K is the (number created) from (multiplying) C , D . Thus, E has made A (by) multiplying K . And so, for the same (reasons), H has made B (by) multiplying L . And since E has made A (by) multiplying K , but has, in fact, also made N (by) multiplying M , thus as K is to M , so A (is) to N [Prop. 7.17]. And as K (is) to M , so C (is) to F , and D to G , and, further, E to H . And thus as C (is) to F , and D to G , and E to H , so A (is) to N . Again, since E , H have made N , O , respectively, (by) multiplying M , thus as E is to H , so N (is) to O [Prop. 7.18]. But, as E (is) to H , so C (is) to F , and D to G . And thus as C (is) to F , and D to G , and E to H , so (is) A to N , and N to O . Again, since H has made O (by) multiplying M , but has, in fact, also made B (by) multiplying L , thus as M (is) to L , so O (is) to B [Prop. 7.17]. But, as M (is) to L , so C (is) to F , and D to G , and E to H . And thus as C (is) to F , and D to G , and E to H , so not only (is) O to B , but also A to N , and N to O . Thus, A , N , O , B are continuously proportional in the aforementioned ratios of the sides.

So I say that A also has to B a cubed ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) the number C (has) to F , or D to G , and, further, E to H . For since A , N , O , B are four continuously proportional numbers, A thus has to B a cubed ratio with respect to (that) A (has) to N [Def. 5.10]. But, as A (is) to N , so it was shown (is) C to F , and D to G , and, further, E to H . And thus A has to B a cubed ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) the number C (has) to F , and D to G , and, further, E to H . (Which is) the very thing it was required to show.

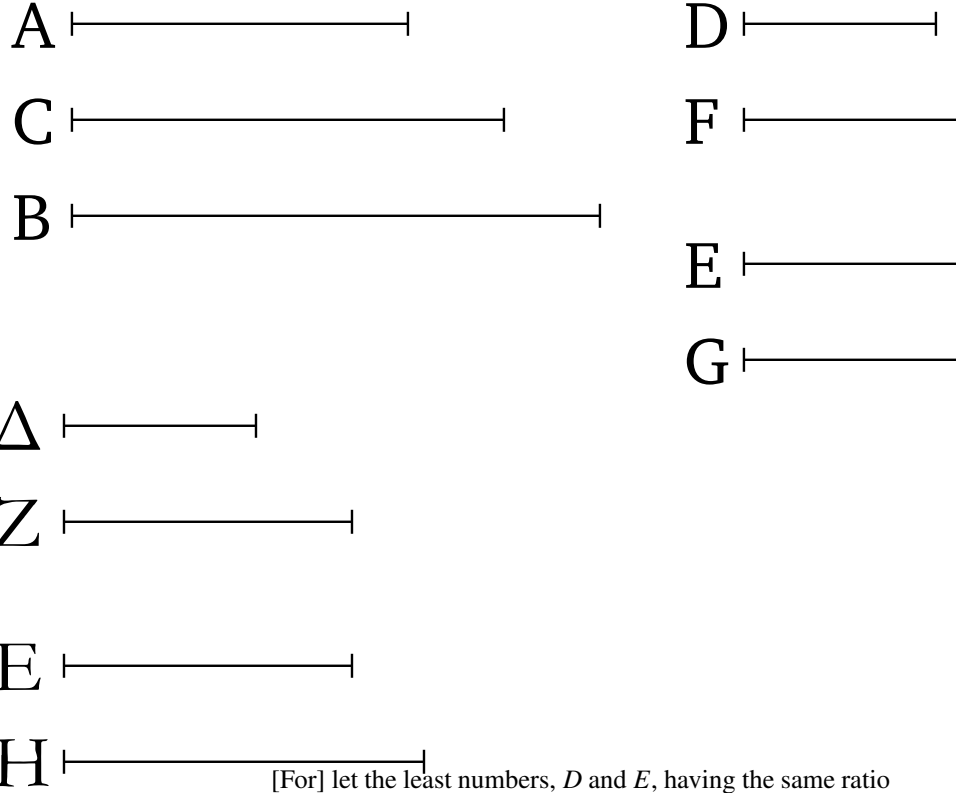
Proposition 20

If one number falls between two numbers in mean proportion then the numbers will be similar plane (numbers).

For let one number C fall between the two numbers A and B in mean proportion. I say that A and B are similar plane

ἀριθμοί.

numbers.



Εἰλήφθωσαν [γάρ] ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ οἱ Δ, E · ἰσάκις ἄρα ὁ Δ τὸν A μετρεῖ καὶ ὁ E τὸν Γ . ὁσάκις δὴ ὁ Δ τὸν A μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Z · ὁ Z ἄρα τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν. ὥστε ὁ A ἐπίπεδός ἐστιν, πλευραὶ δὲ αὐτοῦ οἱ Δ, Z . πάλιν, ἐπεὶ οἱ Δ, E ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Γ, B , ἰσάκις ἄρα ὁ Δ τὸν Γ μετρεῖ καὶ ὁ E τὸν B . ὁσάκις δὴ ὁ E τὸν B μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ H . ὁ E ἄρα τὸν B μετρεῖ κατὰ τάξιν ἐν τῷ H μονάδας· ὁ H ἄρα τὸν E πολλαπλασιάσας τὸν B πεποίηκεν. ὁ B ἄρα ἐπίπεδος ἐστι, πλευραὶ δὲ αὐτοῦ εἰσι οἱ E, H . οἱ A, B ἄρα ἐπίπεδοι εἰσι ἀριθμοί. λέγω δὴ, ὅτι καὶ ὅμοιοι. ἐπεὶ γάρ ὁ Z τὸν μὲν Δ πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ E πολλαπλασιάσας τὸν Γ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν E , οὕτως ὁ A πρὸς τὸν Γ , τοντέστιν ὁ Γ πρὸς τὸν B . πάλιν, ἐπεὶ ὁ E ἐκάτερον τῶν Z, H πολλαπλασιάσας τοῖς Γ, B πεποίηκεν, ἔστιν ἄρα ὡς ὁ Z πρὸς τὸν H , οὕτως ὁ Γ πρὸς τὸν B . ὡς δὲ ὁ Γ πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E · καὶ ὡς ἄρα ὁ Δ πρὸς τὸν E , οὕτως ὁ Z πρὸς τὸν H · καὶ ἐναλλάξ ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ E πρὸς τὸν H . οἱ A, B ἄρα ὅμοιοι ἐπίπεδοι ἀριθμοὶ εἰσιν· αἱ γὰρ πλευραὶ αὐτῶν ἀνάλογόν εἰσιν· ὅπερ ἔδει δεῖξαι.

[For] let the least numbers, D and E , having the same ratio as A and C be taken [Prop. 7.33]. Thus, D measures A as many times as E (measures) C [Prop. 7.20]. So as many times as D measures A , so many units let there be in F . Thus, F has made A (by) multiplying D [Def. 7.15]. Hence, A is plane, and D, F (are) its sides. Again, since D and E are the least of those (numbers) having the same ratio as C and B , D thus measures C as many times as E (measures) B [Prop. 7.20]. So as many times as E measures B , so many units let there be in G . Thus, E measures B according to the units in G . Thus, G has made B (by) multiplying E [Def. 7.15]. Thus, B is plane, and E, G are its sides. Thus, A and B are (both) plane numbers. So I say that (they are) also similar. For since F has made A (by) multiplying D , and has made C (by) multiplying E , thus as D is to E , so A (is) to C —that is to say, C to B [Prop. 7.17].[†] Again, since E has made C, B (by) multiplying F, G , respectively, thus as F is to G , so C (is) to B [Prop. 7.17]. And as C (is) to B , so D (is) to E . And thus as D (is) to E , so F (is) to G . And, alternately, as D (is) to F , so E (is) to G [Prop. 7.13]. Thus, A and B are similar plane numbers. For their sides are proportional [Def. 7.21]. (Which is) the very thing it was required to show.

[†] This part of the proof is defective, since it is not demonstrated that $F \times E = C$. Furthermore, it is not necessary to show that $D : E :: A : C$, because this is

true by hypothesis.

21.

Ἐάν δύο ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅμοιοι στερεοί εἰσιν οἱ ἀριθμοί.

Δύο γάρ ἀριθμῶν τῶν A, B δύο μέσοι ἀνάλογον ἐμπιπτέωσαν ἀριθμοί οἱ Γ, Δ . λέγω, ὅτι οἱ A, B ὅμοιοι στερεοί εἰσιν.

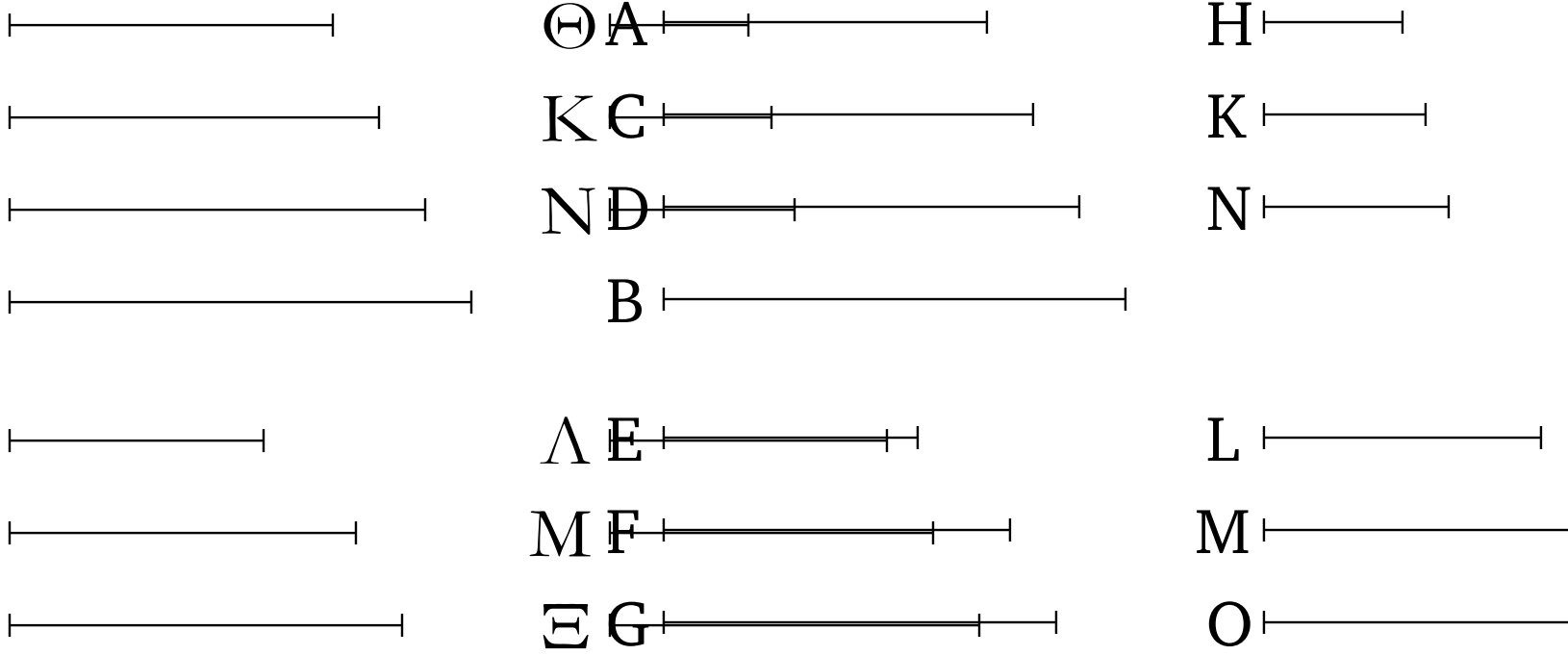
Εἰλήφθωσαν γάρ ἐλάθιστοι ἀριθμοί τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ, Δ τρεῖς οἱ E, Z, H . οἱ ἄρα ἄκροι αὐτῶν οἱ E, H πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ τῶν E, H εἴς μέσοξ ἀνάλογον ἐμπέπτωκεν ἀριθμὸξ ὁ Z , οἱ E, H ἄρα ἀριθμοί ὅμοιοι ἐπίπεδοι εἰσιν. ἔστωσαν οὖν τοῦ μὲν E πλευραὶ οἱ Θ, K , τοῦ δὲ H οἱ Λ, M . φανερόν ἄρα ἐστὶν ἐκ τοῦ πρὸ τούτου, ὅτι οἱ E, Z, H ἐχῆξ εἰσιν ἀνάλογον ἐν τε τῷ τοῦ Θ πρὸξ τὸν Λ λόγῳ καὶ τῷ τοῦ K πρὸξ τὸν M . καὶ ἐπεὶ οἱ E, Z, H ἐλάθιστοι εἰσὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ, Δ , καὶ ἐστὶν ἴσον τὸ πλῆθος τῶν E, Z, H τῷ πλῆθει τῶν A, Γ, Δ , δι' ἴσον ἄρα ἐστὶν ὥς ὁ E πρὸξ τὸν H , οὕτως ὁ A πρὸξ τὸν Δ . οἱ δὲ E, H πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάθιστοι, οἱ δὲ ἐλάθιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχονταξ αὐτοῖς ἰσάκιξ ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὅ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον· ἰσάκιξ ἄρα ὁ E τὸν A μετρεῖ καὶ ὁ H τὸν Δ . ὁσάκιξ δὴ ὁ E τὸν A μετρεῖ, τοσαῦται μονάδεξ ἔστωσαν ἐν τῷ N . ὁ N ἄρα τὸν E πολλαπλασιάσαξ τὸν A πεποίηκεν. ὁ δὲ E ἐστὶν ὁ ἐκ τῶν Θ, K . ὁ N ἄρα τὸν ἐκ τῶν Θ, K πολλαπλασιάσαξ τὸν A πεποίηκεν. στερεὸξ ἄρα ἐστὶν ὁ A , πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Θ, K, N . πάλιν, ἐπεὶ οἱ E, Z, H ἐλάθιστοί εἰσὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Γ, Δ, B , ἰσάκιξ ἄρα ὁ E τὸν Γ μετρεῖ καὶ ὁ H τὸν B . ὁσάκιξ δὴ ὁ E τὸν Γ μετρεῖ, τοσαῦται μονάδεξ ἔστωσαν ἐν τῷ X . ὁ X ἄρα τὸν Γ πολλαπλασιάσαξ τὸν B πεποίηκεν. ὁ δὲ H ἐστὶν ὁ ἐκ τῶν Λ, M . ὁ X ἄρα τὸν ἐκ τῶν Λ, M πολλαπλασιάσαξ τὸν B πεποίηκεν. στερεὸξ ἄρα ἐστὶν ὁ B , πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Λ, M, X . οἱ A, B ἄρα στερεοί εἰσιν.

Proposition 21

If two numbers fall between two numbers in mean proportion then the (latter) are similar solid (numbers).

For let the two numbers C and D fall between the two numbers A and B in mean proportion. I say that A and B are similar solid (numbers).

For let the three least numbers E, F, G having the same ratio as A, C, D be taken [Prop. 8.2]. Thus, the outermost of them, E and G , are prime to one another [Prop. 8.3]. And since one number, F , has fallen (between) E and G in mean proportion, E and G are thus similar plane numbers [Prop. 8.20]. Therefore, let H, K be the sides of E , and L, M (the sides) of G . Thus, it is clear from the (proposition) before this (one) that E, F, G are continuously proportional in the ratio of H to L , and of K to M . And since E, F, G are the least (numbers) having the same ratio as A, C, D , and the multitude of E, F, G is equal to the multitude of A, C, D , thus, via equality, as E is to G , so A (is) to D [Prop. 7.14]. And E and G (are) prime (to one another), and prime (numbers) are also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures A the same number of times as G (measures) D . So as many times as E measures A , so many units let there be in N . Thus, N has made A (by) multiplying E [Def. 7.15]. And E is the (number created) from (multiplying) H and K . Thus, N has made A (by) multiplying the (number created) from (multiplying) H and K . Thus, A is solid, and its sides are H, K, N . Again, since E, F, G are the least (numbers) having the same ratio as C, D, B , thus E measures C the same number of times as G (measures) B [Prop. 7.20]. So as many times as E measures C , so many units let there be in O . Thus, G measures B according to the units in O . Thus, O has made B (by) multiplying G . And G is the (number created) from (multiplying) L and M . Thus, O has made B (by) multiplying the (number created) from (multiplying) L and M . Thus, B is solid, and its sides are L, M, O . Thus, A and B are (both) solid.



Λέγω [δὴ], ὅτι καὶ ὅμοιοι. ἐπεὶ γὰρ οἱ N , X τὸν E πολλαπλασιάσαντες τοὺς A , Γ πεποιήκασιν, ἔστιν ἄρα ὡς ὁ N πρὸς τὸν X , ὁ A πρὸς τὸν Γ , τουτέστιν ὁ E πρὸς τὸν Z . ἀλλ' ὡς ὁ E πρὸς τὸν Z , ὁ Θ πρὸς τὸν Λ καὶ ὁ K πρὸς τὸν M καὶ ὡς ἄρα ὁ Θ πρὸς τὸν Λ , οὕτως ὁ K πρὸς τὸν M καὶ ὁ N πρὸς τὸν X . καὶ εἰσιν οἱ μὲν Θ , K , N πλευραὶ τοῦ A , οἱ δὲ X , Λ , M πλευραὶ τοῦ B . οἱ A , B ἄρα ἀριθμοὶ ὅμοιοι στερεοὶ εἰσιν· ὅπερ ἔδει δεῖξαι.

[So] I say that (they are) also similar. For since N , O have made A , C (by) multiplying E , thus as N is to O , so A (is) to C —that is to say, E to F [Prop. 7.18]. But, as E (is) to F , so H (is) to L , and K to M . And thus as H (is) to L , so K (is) to M , and N to O . And H , K , N are the sides of A , and L , M , O the sides of B . Thus, A and B are similar solid numbers [Def. 7.21]. (Which is) the very thing it was required to show.

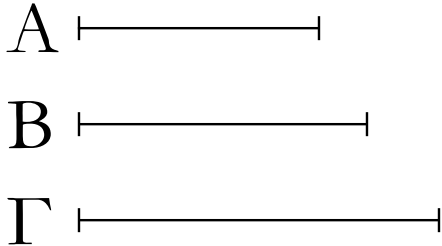
† The Greek text has “ O , L , M ”, which is obviously a mistake.

22.

Proposition 22

Ἐὰν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ὦσιν, ὁ δὲ πρῶτος τετράγωνός ᾗ, καὶ ὁ τρίτος τετράγωνός ἔσται.

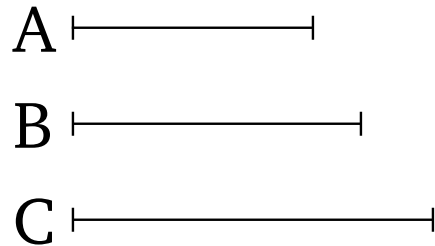
Ἐστωσαν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A , B , Γ , ὁ δὲ πρῶτος ὁ A τετράγωνός ἔστω· λέγω, ὅτι καὶ ὁ τρίτος ὁ Γ τετράγωνός ἔστιν.



Ἐπεὶ γὰρ τῶν A , Γ εἴς μέσοξ ἀνάλογόν ἐστιν ἀριθμὸς ὁ B , οἱ A , Γ ἄρα ὅμοιοι ἐπίπεδοι εἰσιν. τετράγωνός δὲ ὁ A · τετράγωνός ἄρα καὶ ὁ Γ · ὅπερ ἔδει δεῖξαι.

If three numbers are continuously proportional, and the first is square, then the third will also be square.

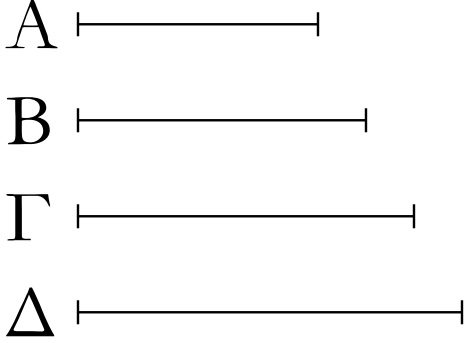
Let A , B , C be three continuously proportional numbers, and let the first A be square. I say that the third C is also square.



For since one number, B , is in mean proportion to A and C , A and C are thus similar plane (numbers) [Prop. 8.20]. And A is square. Thus, C is also square [Def. 7.21]. (Which is) the

23.

Ἐάν τέσσαρες ἀριθμοὶ ἐχῇς ἀνάλογον ὦσιν, ὁ δὲ πρῶτος κύβοξ ᾗ, καὶ ὁ τέταρτος κύβοξ ἔσται.



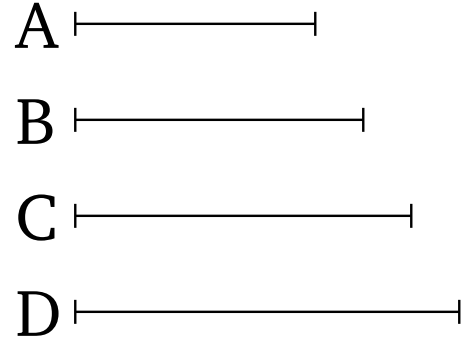
Ἐστωσαν τέσσαρες ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A, B, Γ, Δ , ὁ δὲ A κύβοξ ἔστω· λέγω, ὅτι καὶ ὁ Δ κύβοξ ἔστί.

Ἐπεὶ γὰρ τῶν A, Δ δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοὶ οἱ B, Γ , οἱ A, Δ ἄρα ὅμοιοί εἰσι στερεοὶ ἀριθμοί. κύβοξ δὲ ὁ A κύβοξ ἄρα καὶ ὁ Δ · ὅπερ ἔδει δεῖξαι.

very thing it was required to show.

Proposition 23

If four numbers are continuously proportional, and the first is cube, then the fourth will also be cube.

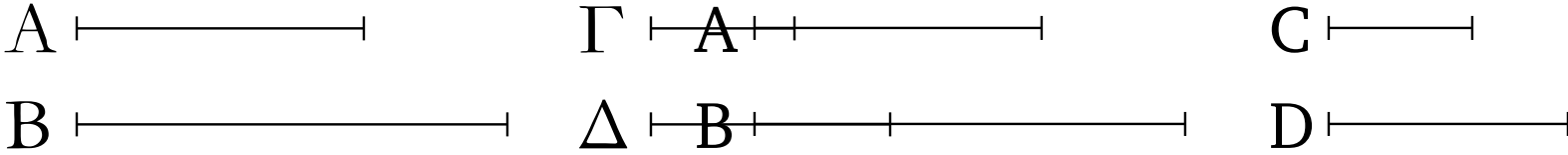


Let A, B, C, D be four continuously proportional numbers, and let A be cube. I say that D is also cube.

For since two numbers, B and C , are in mean proportion to A and D , A and D are thus similar solid numbers [Prop. 8.21]. And A (is) cube. Thus, D (is) also cube [Def. 7.21]. (Which is) the very thing it was required to show.

24.

Ἐάν δύο ἀριθμοὶ πρὸς ἀλλήλους λόγον ἔθωσιν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, ὁ δὲ πρῶτος τετράγωνος ᾗ, καὶ ὁ δεύτερος τετράγωνος ἔσται.



Δύο γὰρ ἀριθμοὶ οἱ A, B πρὸς ἀλλήλους λόγον ἔθέτωσαν, ὃν τετράγωνος ἀριθμὸς ὁ Γ πρὸς τετράγωνον ἀριθμόν τὸν Δ , ὁ δὲ A τετράγωνος ἔστω· λέγω, ὅτι καὶ ὁ B τετράγωνός ἐστι.

Ἐπεὶ γὰρ οἱ Γ, Δ τετράγωνοί εἰσιν, οἱ Γ, Δ ἄρα ὅμοιοι ἐπίπεδοι εἰσιν. τῶν Γ, Δ ἄρα εἷς μέσος ἀνάλογον ἐμπίπτει ἀριθμὸς. καὶ ἐστὶν ὥς ὁ Γ πρὸς τὸν Δ , ὁ A πρὸς τὸν B . καὶ τῶν A, B ἄρα εἷς μέσος ἀνάλογον ἐμπίπτει ἀριθμὸς. καὶ ἐστὶν ὁ A τετράγωνος· καὶ ὁ B ἄρα τετράγωνός ἐστιν· ὅπερ ἔδει δεῖξαι.

If two numbers have to one another the ratio which a square number (has) to a(nother) square number, and the first is square, then the second will also be square.

Proposition 24

For let two numbers, A and B , have to one another the ratio which the square number C (has) to the square number D . And let A be square. I say that B is also square.

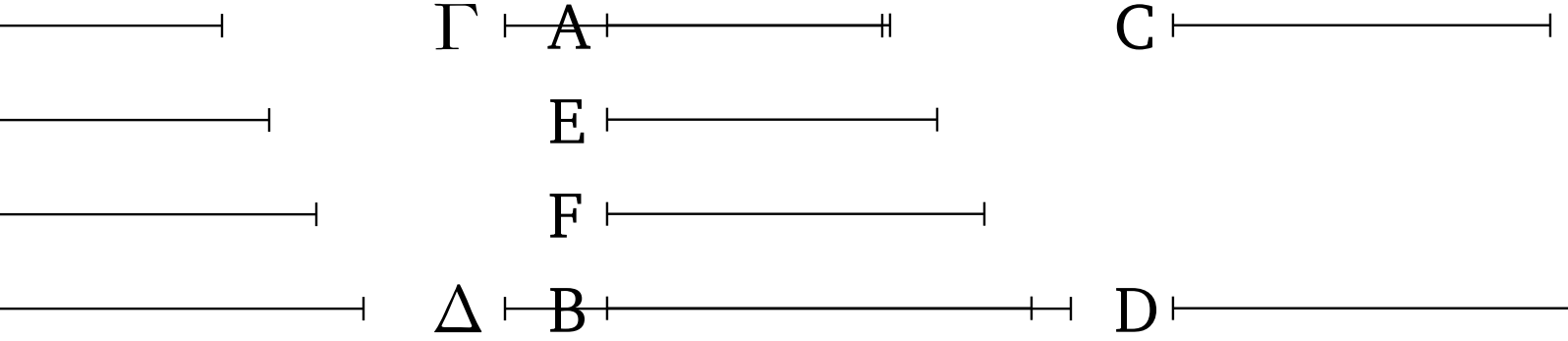
For since C and D are square, C and D are thus similar plane (numbers). Thus, one number falls (between) C and D in mean proportion [Prop. 8.18]. And as C is to D , (so) A (is) to B . Thus, one number also falls (between) A and B in mean proportion [Prop. 8.8]. And A is square. Thus, B is also square [Prop. 8.22]. (Which is) the very thing it was required to show.

25.

Ἐάν δύο ἀριθμοὶ πρὸς ἀλλήλους λόγον ἔθωσιν, ὃν κύβοξ ἀριθμὸς πρὸς κύβον ἀριθμόν, ὁ δὲ πρῶτος κύβοξ ᾗ, καὶ ὁ δεύτερος κύβοξ ἔσται.

If two numbers have to one another the ratio which a cube number (has) to a(nother) cube number, and the first is cube, then the second will also be cube.

Proposition 25



Δύο γὰρ ἀριθμοὶ οἱ A, B πρὸς ἀλλήλων λόγον ἐθέτωσαν, ὃν κύβοξ ἀριθμὸς ὁ Γ πρὸς κύβον ἀριθμὸν τὸν Δ , κύβοξ δὲ ἔστω ὁ A · λέγω [δὴ], ὅτι καὶ ὁ B κύβοξ ἐστίν.

Ἐπεὶ γὰρ οἱ Γ, Δ κύβοι εἰσίν, οἱ Γ, Δ ὅμοιοι στερεοὶ εἰσιν τῶν Γ, Δ ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. ὅσοι δὲ εἰς τοὺς Γ, Δ μεταχὺ κατὰ τὸ συννεθὲς ἀνάλογον ἐμπίπτουσιν, τοσοῦτοι καὶ εἰς τοὺς τὸν αὐτὸν λόγον ἔθονταξ αὐτοῖς· ὥστε καὶ τῶν A, B δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. ἐμπίπτέτωσαν οἱ E, Z . ἐπεὶ οὖν τέσσαρες ἀριθμοὶ οἱ A, E, Z, B ἐχῆξ ἀνάλογόν εἰσιν, καὶ ἐστι κύβοξ ὁ A , κύβοξ ἄρα καὶ ὁ B · ὅπερ ἔδει δεῖχαι.

For let two numbers, A and B , have to one another the ratio which the cube number C (has) to the cube number D . And let A be cube. [So] I say that B is also cube.

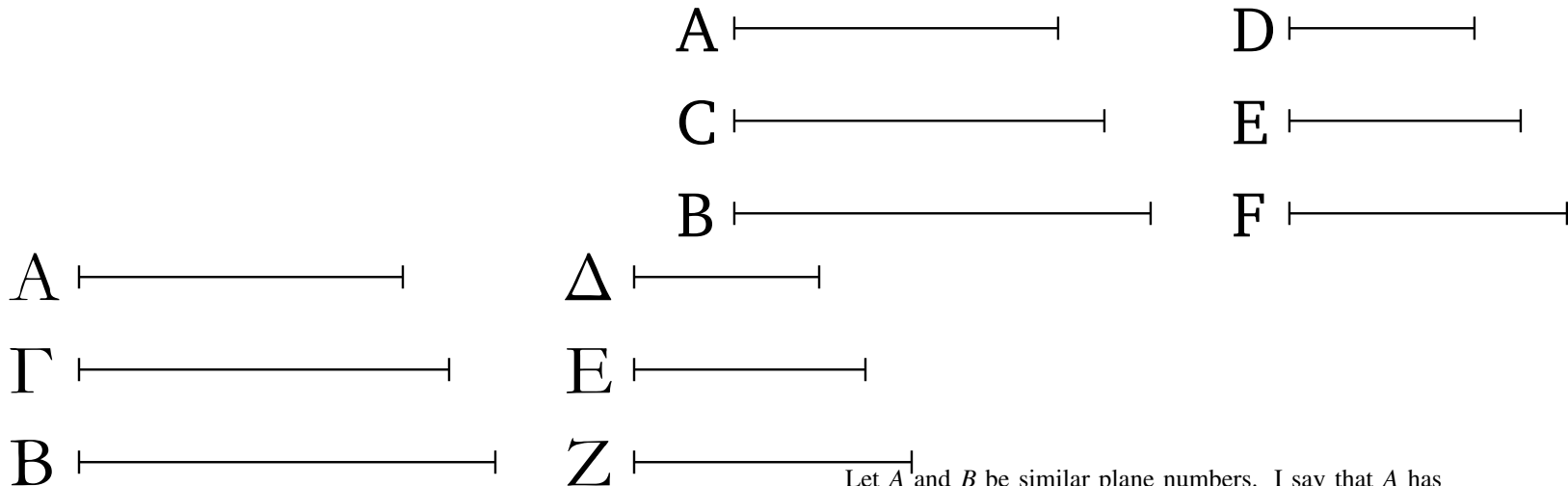
For since C and D are cube (numbers), C and D are (thus) similar solid (numbers). Thus, two numbers fall (between) C and D in mean proportion [Prop. 8.19]. And as many (numbers) as fall in between C and D in continued proportion, so many also (fall) in (between) those (numbers) having the same ratio as them (in continued proportion) [Prop. 8.8]. And hence two numbers fall (between) A and B in mean proportion. Let E and F (so) fall. Therefore, since the four numbers A, E, F, B are continuously proportional, and A is cube, B (is) thus also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

26.

Proposition 26

Οἱ ὅμοιοι ἐπίπεδοι ἀριθμοὶ πρὸς ἀλλήλων λόγον ἔθουσιν, ὃν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν.

Similar plane numbers have to one another the ratio which (some) square number (has) to a(nother) square number.



Ἐστωσαν ὅμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B · λέγω, ὅτι ὁ A πρὸς τὸν B λόγον ἔθει, ὃν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν.

Ἐπεὶ γὰρ οἱ A, B ὅμοιοι ἐπίπεδοι εἰσιν, τῶν A, B ἄρα εἷξ μέσοξ ἀνάλογον ἐμπίπτει ἀριθμὸξ. ἐμπίπτέτω καὶ ἔστω

Let A and B be similar plane numbers. I say that A has to B the ratio which (some) square number (has) to a(nother) square number.

For since A and B are similar plane numbers, one number thus falls (between) A and B in mean proportion [Prop. 8.18]. Let it (so) fall, and let it be C . And let the least numbers, D, E ,

ὁ Γ , καὶ εἰλήφθωσαν ἐλάττωστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ, B οἱ Δ, E, Z : οἱ ἄρα ἄκροι αὐτῶν οἱ Δ, Z τετράγωνοι εἰσιν. καὶ ἐπεὶ ἐστὶν ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν B , καὶ εἰσιν οἱ Δ, Z τετράγωνοι, ὁ A ἄρα πρὸς τὸν B λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ὅπερ ἔδει δεῖχαι.

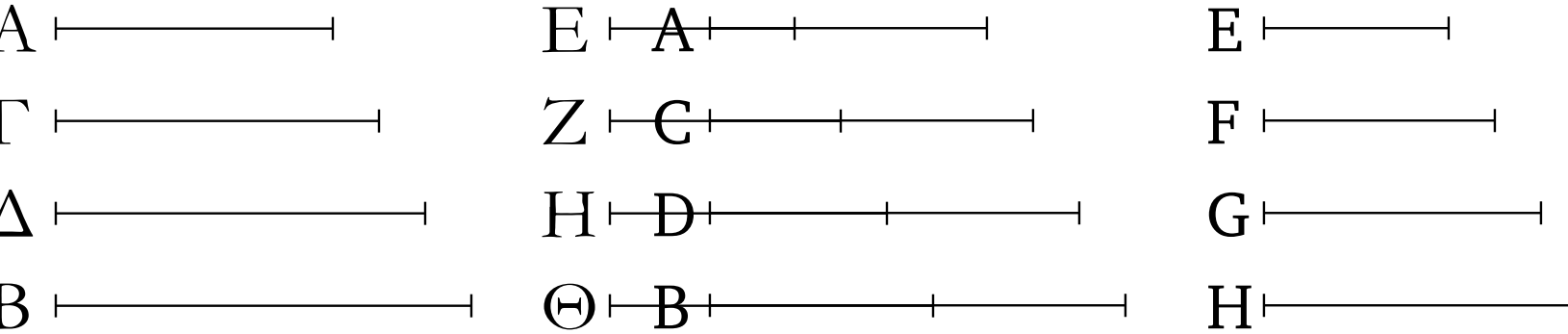
F , having the same ratio as A, C, B be taken [Prop. 8.2]. The outermost of them, D and F , are thus square [Prop. 8.2 corr.]. And since as D is to F , so A (is) to B , and D and F are square, A thus has to B the ratio which (some) square number (has) to a(nother) square number. (Which is) the very thing it was required to show.

27.

Proposition 27

Οἱ ὅμοιοι στερεοὶ ἀριθμοὶ πρὸς ἀλλήλων λόγον ἔθουσιν, ὃν κύβοξ ἀριθμὸς πρὸς κύβον ἀριθμὸν.

Similar solid numbers have to one another the ratio which (some) cube number (has) to a(nother) cube number.



Ἐστωσαν ὅμοιοι στερεοὶ ἀριθμοὶ οἱ A, B . λέγω, ὅτι ὁ A πρὸς τὸν B λόγον ἔθει, ὃν κύβοξ ἀριθμὸς πρὸς κύβον ἀριθμὸν.

Ἐπεὶ γὰρ οἱ A, B ὅμοιοι στερεοὶ εἰσιν, τῶν A, B ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. ἐμπίπτετωσαν οἱ Γ, Δ , καὶ εἰλήφθωσαν ἐλάττωστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ, Δ, B ἴσοι αὐτοῖς τὸ πλεονάζον οἱ E, Z, H, Θ : οἱ ἄρα ἄκροι αὐτῶν οἱ E, Θ κύβοι εἰσίν. καὶ ἐστὶν ὡς ὁ E πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν B : καὶ ὁ A ἄρα πρὸς τὸν B λόγον ἔθει, ὃν κύβοξ ἀριθμὸς πρὸς κύβον ἀριθμὸν ὅπερ ἔδει δεῖχαι.

Let A and B be similar solid numbers. I say that A has to B the ratio which (some) cube number (has) to a(nother) cube number.

For since A and B are similar solid (numbers), two numbers thus fall (between) A and B in mean proportion [Prop. 8.19]. Let C and D have (so) fallen. And let the least numbers, E, F, G, H , having the same ratio as A, C, D, B , (and) equal in multitude to them, be taken [Prop. 8.2]. Thus, the outermost of them, E and H , are cube [Prop. 8.2 corr.]. And as E is to H , so A (is) to B . And thus A has to B the ratio which (some) cube number (has) to a(nother) cube number. (Which is) the very thing it was required to show.

ELEMENTS BOOK 8

Continued Proportion[†]

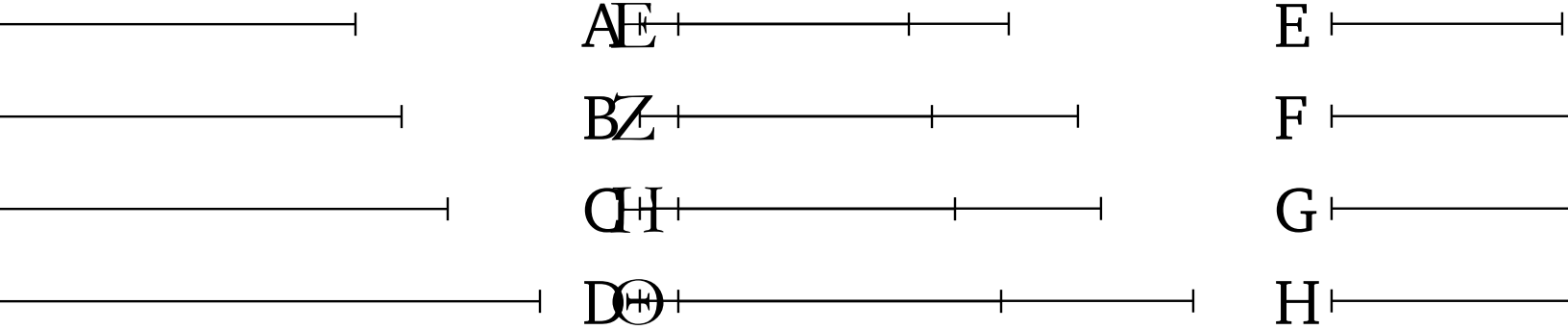
[†]The propositions contained in Books 7–9 are generally attributed to the school of Pythagoras.

1.

Proposition 1

Ἐάν ὧσιν ὁσοιδηποτοῦν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ δὲ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους ὧσιν, ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

If there are any multitude whatsoever of continuously proportional numbers, and the outermost of them are prime to one another, then the (numbers) are the least of those (numbers) having the same ratio as them.



Ἐστωσαν ὅποιοι οὖν ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A, B, Γ, Δ , οἱ δὲ ἄκροι αὐτῶν οἱ A, Δ , πρῶτοι πρὸς ἀλλήλους ἔστωσαν λέγω, ὅτι οἱ A, B, Γ, Δ ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς.

Let A, B, C, D be any multitude whatsoever of continuously proportional numbers. And let the outermost of them, A and D , be prime to one another. I say that A, B, C, D are the least of those (numbers) having the same ratio as them.

Εἰ γὰρ μή, ἔστωσαν ἐλάττωρες τῶν A, B, Γ, Δ οἱ E, Z, H, Θ ἐν τῷ αὐτῷ λόγῳ ὄντες αὐτοῖς. καὶ ἐπεὶ οἱ A, B, Γ, Δ ἐν τῷ αὐτῷ λόγῳ εἰσὶ τοῖς E, Z, H, Θ , καὶ ἐστὶν ἴσον τὸ πλῆθος [τῶν A, B, Γ, Δ] τῷ πλῆθει [τῶν E, Z, H, Θ], δι' ἴσον ἄρα ἐστὶν ὡς ὁ A πρὸς τὸν Δ , ὁ E πρὸς τὸν Θ . οἱ δὲ A, Δ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἐθόντας ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. μετρεῖ ἄρα ὁ A τὸν E ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οἱ E, Z, H, Θ ἐλάσσονες ὄντες τῶν A, B, Γ, Δ ἐν τῷ αὐτῷ λόγῳ εἰσὶν αὐτοῖς. οἱ A, B, Γ, Δ ἄρα ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς· ὅπερ ἔδει δεῖξαι.

For if not, let E, F, G, H be less than A, B, C, D (respectively), being in the same ratio as them. And since A, B, C, D are in the same ratio as E, F, G, H , and the multitude [of A, B, C, D] is equal to the multitude [of E, F, G, H], thus, via equality, as A is to D , (so) E (is) to H [Prop. 7.14]. And A and D (are) prime (to one another). And prime (numbers are) also the least of those (numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures E , the greater (measuring) the lesser. The very thing is impossible. Thus, E, F, G, H , being less than A, B, C, D , are not in the same ratio as them. Thus, A, B, C, D are the least of those (numbers) having the same ratio as them. (Which is) the very thing it was required to show.

2.

Proposition 2

Αριθμοὺς εὐρεῖν ἐχῇς ἀνάλογον ἐλαθίστους, ὅσους ἂν ἐπιτάχῃ τις, ἐν τῷ δοθέντι λόγῳ.

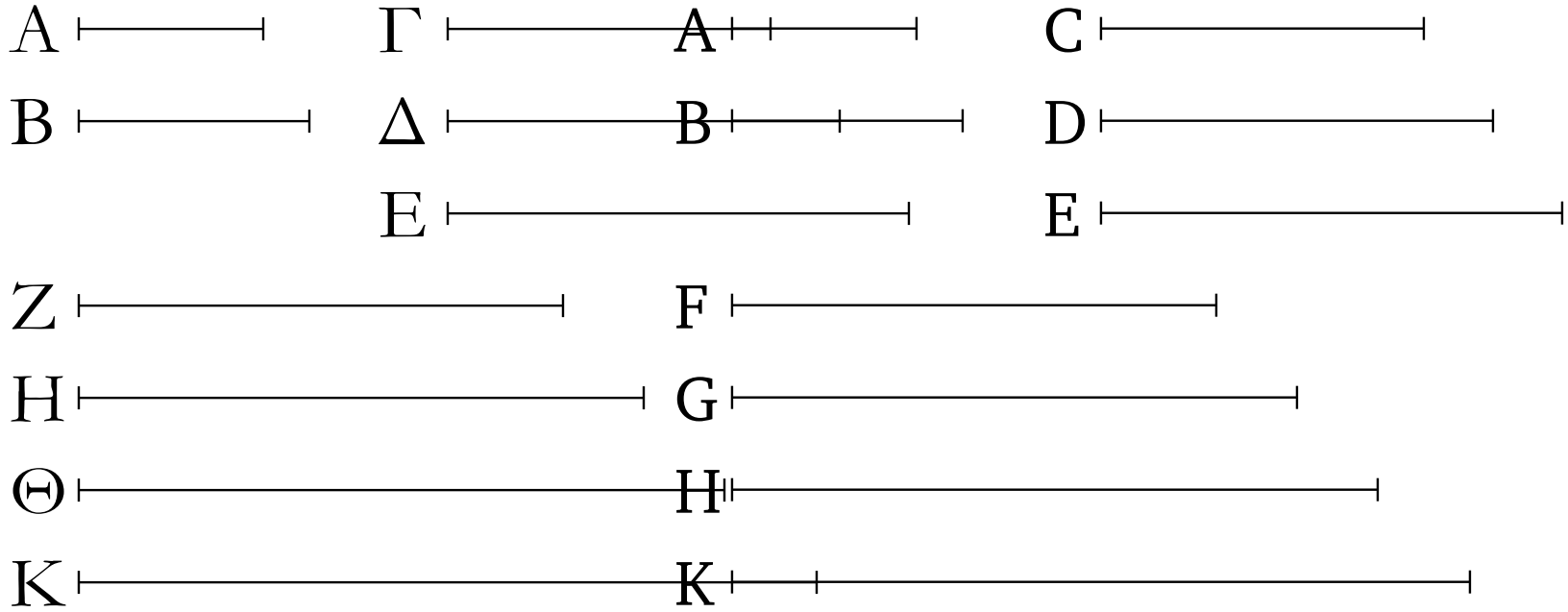
To find the least numbers, as many as may be prescribed, (which are) continuously proportional in a given ratio.

Ἐστω ὁ δοθείς λόγος ἐν ἐλαθίστοις ἀριθμοῖς ὁ τοῦ A πρὸς τὸν B . δεῖ δὴ ἀριθμοὺς εὐρεῖν ἐχῇς ἀνάλογον ἐλαθίστους, ὅσους ἂν τις ἐπιτάχῃ, ἐν τῷ τοῦ A πρὸς τὸν B λόγῳ.

Let the given ratio, (expressed) in the least numbers, be that of A to B . So it is required to find the least numbers, as many as may be prescribed, (which are) in the ratio of A to B .

Ἐπιτετάθωσαν δὴ τέσσαρες, καὶ ὁ A ἐαντὸν πολλαπλασιάσας τὸν Γ ποιέτω, τὸν δὲ B πολλαπλασιάσας τὸν Δ ποιέτω, καὶ ἔτι ὁ B ἐαντὸν πολλαπλασιάσας τὸν E ποιέτω, καὶ ἔτι ὁ A τοὺς Γ, Δ, E πολλαπλασιάσας τοὺς Z, H, Θ ποιέτω, ὁ δὲ B τὸν E πολλαπλασιάσας τὸν K ποιέτω.

Let four (numbers) be prescribed. And let A make C (by) multiplying itself, and let it make D (by) multiplying B . And, further, let B make E (by) multiplying itself. And, further, let A make F, G, H (by) multiplying C, D, E . And let B make K (by) multiplying E .



Καὶ ἐπεὶ ὁ A ἐαυτὸν μὲν πολλαπλασιάσας τὸν Γ πεποιήκεν, τὸν δὲ B πολλαπλασιάσας τὸν Δ πεποιήκεν, ἔστιν ἄρα ὥς ὁ A πρὸς τὸν B , [οὕτως] ὁ Γ πρὸς τὸν Δ . πάλιν, ἐπεὶ ὁ μὲν A τὸν B πολλαπλασιάσας τὸν Δ πεποιήκεν, ὁ δὲ B ἐαυτὸν πολλαπλασιάσας τὸν E πεποιήκεν, ἐκάτερος ἄρα τῶν A , B τὸν B πολλαπλασιάσας ἐκάτερον τῶν Δ , E πεποιήκεν. ἔστιν ἄρα ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E . ἀλλ' ὥς ὁ A πρὸς τὸν B , ὁ Γ πρὸς τὸν Δ · καὶ ὥς ἄρα ὁ Γ πρὸς τὸν Δ , ὁ Δ πρὸς τὸν E . καὶ ἐπεὶ ὁ A τοὺς Γ , Δ πολλαπλασιάσας τοὺς Z , H πεποιήκεν, ἔστιν ἄρα ὥς ὁ Γ πρὸς τὸν Δ , [οὕτως] ὁ Z πρὸς τὸν H . ὥς δὲ ὁ Γ πρὸς τὸν Δ , οὕτως ἦν ὁ A πρὸς τὸν B · καὶ ὥς ἄρα ὁ A πρὸς τὸν B , ὁ Z πρὸς τὸν H . πάλιν, ἐπεὶ ὁ A τοὺς Δ , E πολλαπλασιάσας τοὺς H , Θ πεποιήκεν, ἔστιν ἄρα ὥς ὁ Δ πρὸς τὸν E , ὁ H πρὸς τὸν Θ . ἀλλ' ὥς ὁ Δ πρὸς τὸν E , ὁ A πρὸς τὸν B . καὶ ὥς ἄρα ὁ A πρὸς τὸν B , οὕτως ὁ H πρὸς τὸν Θ . καὶ ἐπεὶ οἱ A , B τὸν E πολλαπλασιάσαντες τοὺς Θ , K πεποιήκασιν, ἔστιν ἄρα ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Θ πρὸς τὸν K . ἀλλ' ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Z πρὸς τὸν H καὶ ὁ H πρὸς τὸν Θ . καὶ ὥς ἄρα ὁ Z πρὸς τὸν H , οὕτως ὁ Θ πρὸς τὸν K . οἱ Γ , Δ , E ἄρα καὶ οἱ Z , H , Θ , K ἀνάλογόν εἰσιν ἐν τῷ τοῦ A πρὸς τὸν B λόγῳ. λέγω δὴ, ὅτι καὶ ἐλάχιστοι. ἐπεὶ γὰρ οἱ A , B ἐλάχιστοι εἰσὶ τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ δὲ ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων πρῶτοι πρὸς ἀλλήλους εἰσὶν, οἱ A , B ἄρα πρῶτοι πρὸς ἀλλήλους εἰσὶν. καὶ ἐκάτερος μὲν τῶν A , B ἐαυτὸν πολλαπλασιάσας ἐκάτερον τῶν Γ , E πεποιήκεν, ἐκάτερον δὲ τῶν Γ , E πολλαπλασιάσας ἐκάτερον τῶν Z , K πεποιήκεν· οἱ Γ , E ἄρα καὶ οἱ Z , K πρῶτοι πρὸς ἀλλήλους εἰσὶν. ἔαν δὲ ὥσιν ὁποιοῖσιν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ δὲ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους ὦσιν, ἐλάχιστοι εἰσὶ τῶν τὸν αὐτὸν λόγον ἐθόν-

And since A has made C (by) multiplying itself, and has made D (by) multiplying B , thus as A is to B , [so] C (is) to D [Prop. 7.17]. Again, since A has made D (by) multiplying B , and B has made E (by) multiplying itself, A , B have thus made D , E , respectively, (by) multiplying B . Thus, as A is to B , so D (is) to E [Prop. 7.18]. But, as A (is) to B , (so) C (is) to D . And thus as C (is) to D , (so) D (is) to E . And since A has made F , G (by) multiplying C , D , thus as C is to D , [so] F (is) to G [Prop. 7.17]. And as C (is) to D , so A was to B . And thus as A (is) to B , (so) F (is) to G . Again, since A has made G , H (by) multiplying D , E , thus as D is to E , (so) G (is) to H [Prop. 7.17]. But, as D (is) to E , (so) A (is) to B . And thus as A (is) to B , so G (is) to H . And since A , B have made H , K (by) multiplying E , thus as A is to B , so H (is) to K . But, as A (is) to B , so F (is) to G , and G to H . And thus as F (is) to G , so G (is) to H , and H to K . Thus, C , D , E and F , G , H , K are (both continuously) proportional in the ratio of A to B . So I say that (they are) also the least (sets of numbers continuously proportional in that ratio). For since A and B are the least of those (numbers) having the same ratio as them, and the least of those (numbers) having the same ratio are prime to one another [Prop. 7.22], A and B are thus prime to one another. And A , B have made C , E , respectively, (by) multiplying themselves, and have made F , K by multiplying C , E , respectively. Thus, C , E and F , K are prime to one another [Prop. 7.27]. And if there are any multitude whatsoever of continuously proportional numbers, and the outermost of them are prime to one another, then the (numbers) are the least of those (numbers) having the same ratio as them [Prop. 8.1]. Thus, C , D , E and F , G , H , K are the least of those (continuously proportional sets of numbers) having the same ratio as

των αὐτοῖς. οἱ Γ , Δ , E ἄρα καὶ οἱ Z , H , Θ , K ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A , B · ὅπερ ἔδει δεῖχαι.

A and B . (Which is) the very thing it was required to show.

Πόρισμα.

Corollary

Ἐκ δὴ τούτου φανερόν, ὅτι ἔαν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ἐλάχιστοι ᾧσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ ἄκρον αὐτῶν τετράγωνοί εἰσιν, ἔαν δὲ τέσσαρες, κύβοι.

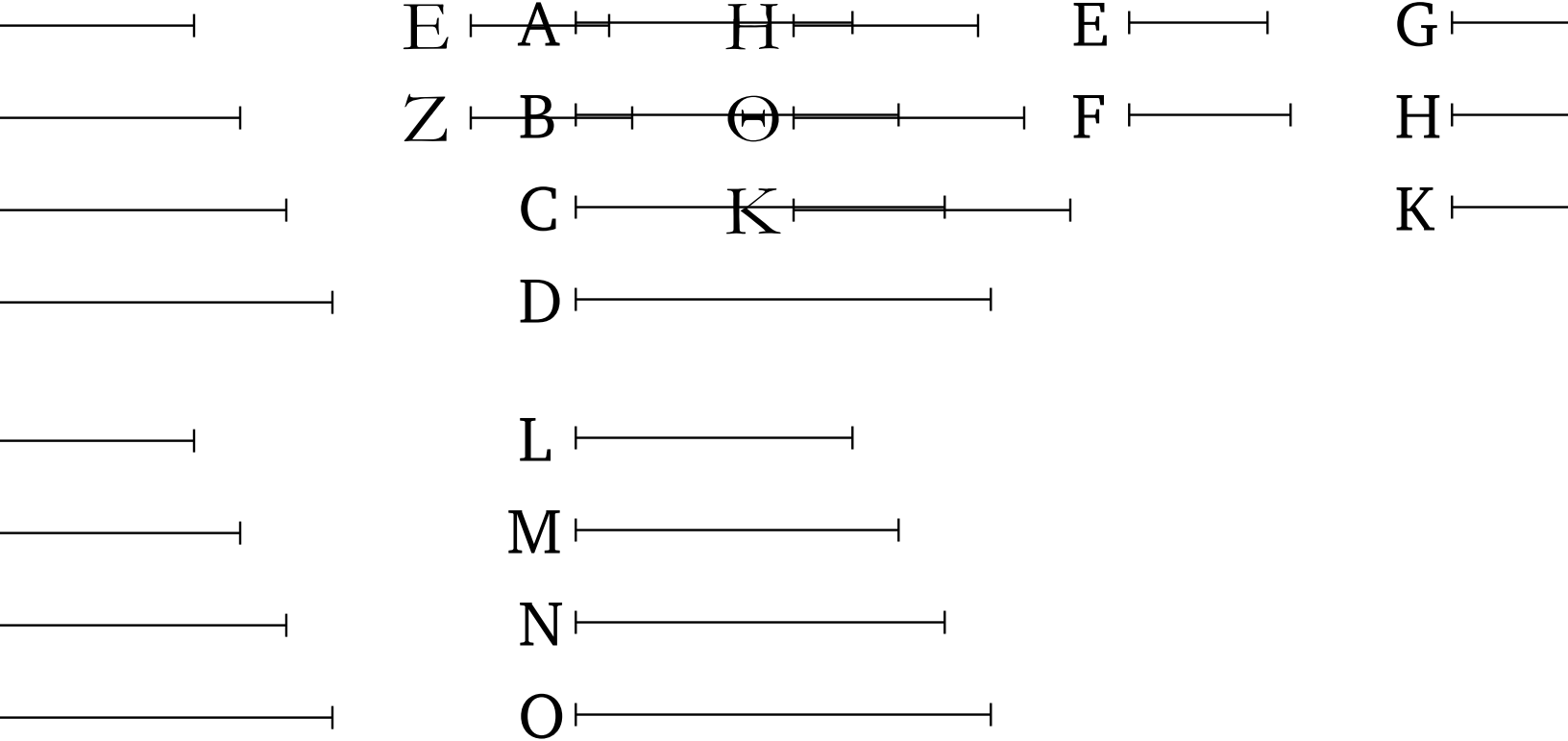
So it is clear, from this, that if three continuously proportional numbers are the least of those (numbers) having the same ratio as them then the outermost of them are square, and, if four (numbers), cube.

3.

Proposition 3

Ἐάν ᾧσιν ὅποιοι ἄριθμοι ἐχῇς ἀνάλογον ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, οἱ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν.

If there are any multitude whatsoever of continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them then the outermost of them are prime to one another.



Ἐστωσαν ὅποιοι ἄριθμοι ἐχῇς ἀνάλογον ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς οἱ A , B , Γ , Δ · λέγω, ὅτι οἱ ἄκροι αὐτῶν οἱ A , Δ πρῶτοι πρὸς ἀλλήλους εἰσίν.

Εἰλήφθωσαν γάρ δύο μὲν ἀριθμοὶ ἐλάχιστοι ἐν τῷ τῶν A , B , Γ , Δ λόγῳ οἱ E , Z , τρεῖς δὲ οἱ H , Θ , K , καὶ ἐχῇς ἐνὶ πλείονι, ἕως τὸ λαμβανόμενον πλήθος ἴσον γένηται τῷ πλήθει τῶν A , B , Γ , Δ . εἰλήφθωσαν καὶ ἕστωσαν οἱ L , M , N , X .

Καὶ ἐπεὶ οἱ E , Z ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ ἑκάτερος τῶν E , Z ἑαυτὸν μὲν πολλαπλασιάσας ἑκάτερον τῶν H , K πεποίηκεν, ἑκάτερον δὲ τῶν H , K πολλαπλασιάσας ἑκάτερον τῶν

Let A , B , C , D be any multitude whatsoever of continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them. I say that the outermost of them, A and D , are prime to one another.

For let the two least (numbers) E , F (which are) in the same ratio as A , B , C , D be taken [Prop. 7.33]. And the three (least numbers) G , H , K [Prop. 8.2]. And (so on), successively increasing by one, until the multitude of (numbers) taken is made equal to the multitude of A , B , C , D . Let them be taken, and let them be L , M , N , O .

And since E and F are the least of those (numbers) hav-

A, X πεποιήκεν, καὶ οἱ H, K ἄρα καὶ οἱ A, X πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ οἱ A, B, Γ, Δ ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, εἰσὶ δὲ καὶ οἱ A, M, N, X ἐλάχιστοι ἐν τῷ αὐτῷ λόγῳ ὄντες τοῖς A, B, Γ, Δ , καὶ ἐστὶν ἴσον τὸ πλῆθος τῶν A, B, Γ, Δ τῷ πλῆθει τῶν A, M, N, X , ἕκαστος ἄρα τῶν A, B, Γ, Δ ἐκάστῳ τῶν A, M, N, X ἴσος ἐστίν· ἴσος ἄρα ἐστὶν ὁ μὲν A τῷ A , ὁ δὲ Δ τῷ X . καὶ εἰσὶν οἱ A, X πρῶτοι πρὸς ἀλλήλους. καὶ οἱ A, Δ ἄρα πρῶτοι πρὸς ἀλλήλους εἰσίν· ὅπερ εἶδει δεῖχαι.

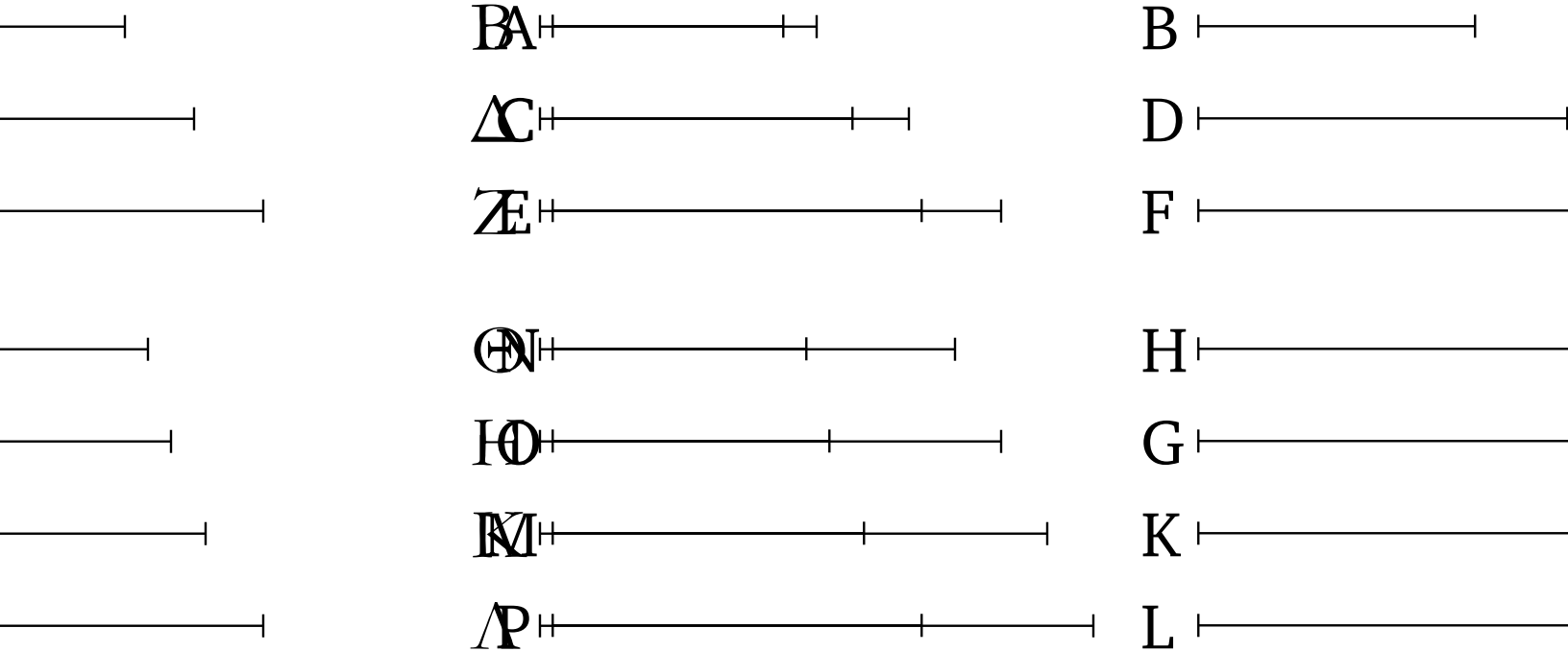
ing the same ratio as them they are prime to one another [Prop. 7.22]. And since E, F have made G, K , respectively, (by) multiplying themselves [Prop. 8.2 corr.], and have made L, O (by) multiplying G, K , respectively, G, K and L, O are thus also prime to one another [Prop. 7.27]. And since A, B, C, D are the least of those (numbers) having the same ratio as them, and L, M, N, O are also the least (of those numbers having the same ratio as them), being in the same ratio as A, B, C, D , and the multitude of A, B, C, D is equal to the multitude of L, M, N, O , thus A, B, C, D are equal to L, M, N, O , respectively. Thus, A is equal to L , and D to O . And L and O are prime to one another. Thus, A and D are also prime to one another. (Which is) the very thing it was required to show.

4.

Proposition 4

Λόγων δοθέντων ὅποσωνοῦν ἐν ἐλάχιστοις ἀριθμοῖς ἀριθμοὺς εὑρεῖν ἐχῆς ἀνάλογον ἐλάχιστους ἐν τοῖς δοθεῖσι λόγοις.

For any multitude whatsoever of given ratios, (expressed) in the least numbers, to find the least numbers continuously proportional in these given ratios.



Ἐστωσαν οἱ δοθέντες λόγοι ἐν ἐλάχιστοις ἀριθμοῖς ὁ τε τοῦ A πρὸς τὸν B καὶ ὁ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι ὁ τοῦ E πρὸς τὸν Z . δεῖ δὲ ἀριθμοὺς εὑρεῖν ἐχῆς ἀνάλογον ἐλάχιστους ἐν τε τῷ τοῦ A πρὸς τὸν B λόγῳ καὶ ἐν τῷ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι τῷ τοῦ E πρὸς τὸν Z .

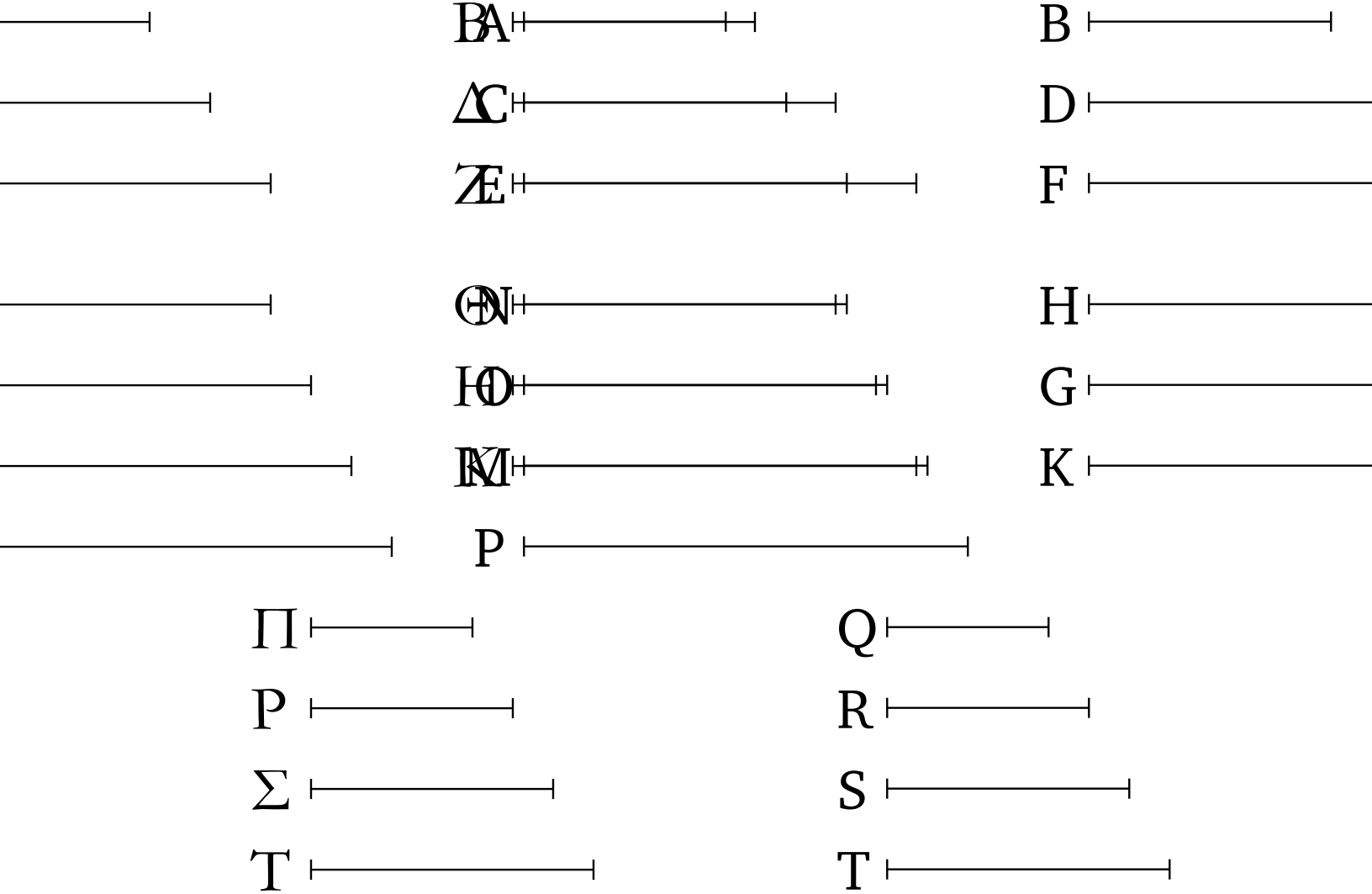
Εἰλήφθω γὰρ ὁ ὑπὸ τῶν B, Γ ἐλάχιστος μετρούμενος ἀριθμὸς ὁ H . καὶ ὁσάκις μὲν ὁ B τὸν H μετρεῖ, τοσαντάκις καὶ ὁ A τὸν Θ μετρεῖτω, ὁσάκις δὲ ὁ Γ τὸν H μετρεῖ, τοσαντάκις καὶ ὁ Δ τὸν K μετρεῖτω. ὁ δὲ E τὸν K ἤτοι μετρεῖ ἢ οὐ μετρεῖ. μετρεῖτω πρότερον. καὶ ὁσάκις ὁ E τὸν K μετρεῖ, τοσαντάκις καὶ ὁ Z τὸν Λ μετρεῖτω. καὶ ἐπεὶ ἰσάκις ὁ A τὸν Θ μετρεῖ καὶ

Let the given ratios, (expressed) in the least numbers, be the (ratios) of A to B , and of C to D , and, further, of E to F . So it is required to find the least numbers continuously proportional in the ratio of A to B , and of C to B , and, further, of E to F .

For let the least number, G , measured by (both) B and C have been taken [Prop. 7.34]. And as many times as B measures G , so many times let A also measure H . And as many times as C measures G , so many times let D also measure K . And E either measures, or does not measure, K . Let it, first of all, measure (K). And as many times as E measures K , so

ὁ B τὸν H , ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Θ πρὸς τὸν H . διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ H πρὸς τὸν K , καὶ ἔτι ὡς ὁ E πρὸς τὸν Z , οὕτως ὁ K πρὸς τὸν Λ . οἱ Θ , H , K , Λ ἄρα ἐχῆς ἀνάλογόν εἰσιν ἐν τε τῷ τοῦ A πρὸς τὸν B καὶ ἐν τῷ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι ἐν τῷ τοῦ E πρὸς τὸν Z λόγῳ. λέγω δὴ, ὅτι καὶ ἐλάχιστοι. εἰ γὰρ μὴ εἰσιν οἱ Θ , H , K , Λ ἐχῆς ἀνάλογον ἐλάχιστοι ἐν τε τοῖς τοῦ A πρὸς τὸν B καὶ τοῦ Γ πρὸς τὸν Δ καὶ ἐν τῷ τοῦ E πρὸς τὸν Z λόγοις, ἔστωσαν οἱ N , X , M , O . καὶ ἐπεὶ ἐστὶν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ N πρὸς τὸν X , οἱ δὲ A , B ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔθονταξ ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τοντέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον, ὁ B ἄρα τὸν X μετρεῖ. διὰ τὰ αὐτὰ δὴ καὶ ὁ Γ τὸν X μετρεῖ· οἱ B , Γ ἄρα τὸν X μετροῦσιν· καὶ ὁ ἐλάχιστος ἄρα ὑπὸ τῶν B , Γ μετρούμενος τὸν X μετρήσει. ἐλάχιστος δὲ ὑπὸ τῶν B , Γ μετρεῖται ὁ H . ὁ H ἄρα τὸν X μετρεῖ ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἔσονται τινες τῶν Θ , H , K , Λ ἐλάσσονες ἀριθμοὶ ἐχῆς ἐν τε τῷ τοῦ A πρὸς τὸν B καὶ τῷ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι τῷ τοῦ E πρὸς τὸν Z λόγῳ.

many times let F also measure L . And since A measures H the same number of times that B also (measures) G , thus as A is to B , so H (is) to G [Def. 7.20, Prop. 7.13]. And so, for the same (reasons), as C (is) to D , so G (is) to K , and, further, as E (is) to F , so K (is) to L . Thus, H , G , K , L are continuously proportional in the ratio of A to B , and of C to D , and, further, of E to F . So I say that (they are) also the least (numbers continuously proportional in these ratios). For if H , G , K , L are not the least numbers continuously proportional in the ratios of A to B , and of C to D , and of E to F , let N , O , M , P be (the least such numbers). And since as A is to B , so N (is) to O , and A and B are the least (numbers which have the same ratio as them), and the least (numbers) measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20], B thus measures O . So, for the same (reasons), C also measures O . Thus, B and C (both) measure O . Thus, the least number measured by (both) B and C will also measure O [Prop. 7.35]. And G (is) the least number measured by (both) B and C . Thus, G measures O , the greater (measuring) the lesser. The very thing is impossible. Thus, there cannot be any numbers less than H , G , K , L (which are) continuously (proportional) in the ratio of A to B , and of C to D , and, further, of E to F .



μή μετρεῖται δὴ ὁ E τὸν K , καὶ εἰλήφθω ὑπὸ τῶν E , K ἐλάχιστος μετρούμενος ἀριθμὸς ὁ M . καὶ ὁσάκις μὲν ὁ K τὸν M μετρεῖ, τοσαντάκις καὶ ἑκάτερος τῶν Θ , H ἑκάτερον τῶν N , X μετρεῖται, ὁσάκις δὲ ὁ E τὸν M μετρεῖ, τοσαντάκις καὶ ὁ Z τὸν O μετρεῖται. ἐπεὶ ἰσάκις ὁ Θ τὸν N μετρεῖ καὶ ὁ H τὸν X , ἔστιν ἄρα ὡς ὁ Θ πρὸς τὸν H , οὕτως ὁ N πρὸς τὸν X . ὡς δὲ ὁ Θ πρὸς τὸν H , οὕτως ὁ A πρὸς τὸν B . καὶ ὡς ἄρα ὁ A πρὸς τὸν B , οὕτως ὁ N πρὸς τὸν X . διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ X πρὸς τὸν M . πάλιν, ἐπεὶ ἰσάκις ὁ E τὸν M μετρεῖ καὶ ὁ Z τὸν O , ἔστιν ἄρα ὡς ὁ E πρὸς τὸν Z , οὕτως ὁ M πρὸς τὸν O . οἱ N , X , M , O ἄρα ἐχῆς ἀνάλογόν εἰσιν ἐν τοῖς τοῦ τε A πρὸς τὸν B καὶ τοῦ Γ πρὸς τὸν Δ καὶ ἔτι τοῦ E πρὸς τὸν Z λόγοις. λέγω δὴ, ὅτι καὶ ἐλάχιστοι ἐν τοῖς $A B$, $\Gamma \Delta$, $E Z$ λόγοις. εἰ γὰρ μή, ἔσονται τιναξ τῶν N , X , M , O ἐλάσσονες ἀριθμοὶ ἐχῆς ἀνάλογον ἐν τοῖς $A B$, $\Gamma \Delta$, $E Z$ λόγοις. ἔστωσαν οἱ Π , P , Σ , T . καὶ ἐπεὶ ἔστιν ὡς ὁ Π πρὸς τὸν P , οὕτως ὁ A πρὸς τὸν B , οἱ δὲ A ,

So let E not measure K . And let the least number, M , measured by (both) E and K be taken [Prop. 7.34]. And as many times as K measures M , so many times let H , G also measure N , O , respectively. And as many times as E measures M , so many times let F also measure P . Since H measures N the same number of times as G (measures) O , thus as H is to G , so N (is) to O [Def. 7.20, Prop. 7.13]. And as H (is) to G , so A (is) to B . And thus as A (is) to B , so N (is) to O . And so, for the same (reasons), as C (is) to D , so O (is) to M . Again, since E measures M the same number of times as F (measures) P , thus as E is to F , so M (is) to P [Def. 7.20, Prop. 7.13]. Thus, N , O , M , P are continuously proportional in the ratios of A to B , and of C to D , and, further, of E to F . So I say that (they are) also the least (numbers) in the ratios of $A B$, $C D$, $E F$. For if not, then there will be some numbers less than N , O , M , P (which are) continuously proportional in the ratios of $A B$, $C D$, $E F$. Let them be Q , R , S , T . And since as Q is to

B ἐλάθιστοι, οἱ δὲ ἐλάθιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχονταξ αὐτοῖς ἰσάκις ὅ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον, ὁ B ἄρα τὸν P μετρῇ. διὰ τὰ αὐτὰ δὴ καὶ ὁ Γ τὸν P μετρῇ· οἱ B, Γ ἄρα τὸν P μετροῦσιν. καὶ ὁ ἐλάθιστοξ ἄρα ὑπὸ τῶν B, Γ μετρούμενοξ τὸν P μετρήσει. ἐλάθιστοξ δὲ ὑπὸ τῶν B, Γ μετρούμενοξ ἐστὶν ὁ H · ὁ H ἄρα τὸν P μετρῇ. καὶ ἐστὶν ὥξ ὁ H πρὸξ τὸν P , ὅντωξ ὁ K πρὸξ τὸν Σ · καὶ ὁ K ἄρα τὸν Σ μετρῇ. μετρῇ δὲ καὶ ὁ E τὸν Σ · οἱ E, K ἄρα τὸν Σ μετροῦσιν. καὶ ὁ ἐλάθιστοξ ἄρα ὑπὸ τῶν E, K μετρούμενοξ τὸν Σ μετρήσει. ἐλάθιστοξ δὲ ὑπὸ τῶν E, K μετρούμενοξ ἐστὶν ὁ M · ὁ M ἄρα τὸν Σ μετρῇ ὁ μείζων τὸν ἐλάσσονα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἔσονται τινεξ τῶν N, X, M, O ἐλάσσονεξ ἀριθμοὶ ἐχῆξ ἀνάλογον ἔν τε τοῖξ τοῦ A πρὸξ τὸν B καὶ τοῦ Γ πρὸξ τὸν Δ καὶ ἔτι τοῦ E πρὸξ τὸν Z λόγοιξ· οἱ N, X, M, O ἄρα ἐχῆξ ἀνάλογον ἐλάθιστοί εἰσιν ἐν τοῖξ $A B, \Gamma \Delta, E Z$ λόγοιξ· ὅπερ ἔδει δεῖχαι.

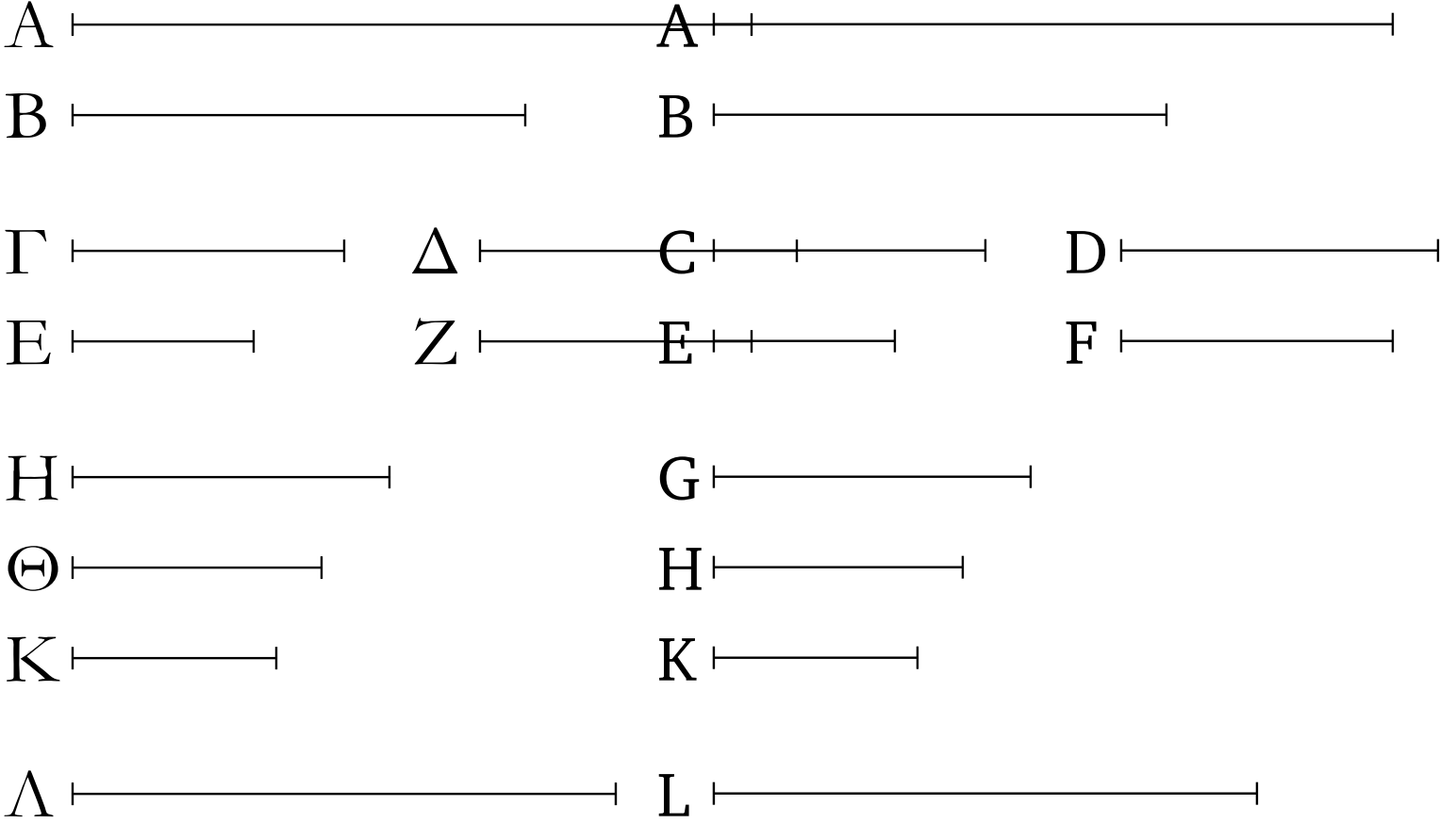
5.

Οἱ ἐπίπεδοι ἀριθμοὶ πρὸξ ἀλλήλωνξ λόγον ἔχουσι τὸν συγκείμενον ἐκ τῶν πλευρῶν.

R , so A (is) to B , and A and B (are) the least (numbers having the same ratio as them), and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20], B thus measures R . So, for the same (reasons), C also measures R . Thus, B and C (both) measure R . Thus, the least (number) measured by (both) B and C will also measure R [Prop. 7.35]. And G is the least number measured by (both) B and C . Thus, G measures R . And as G is to R , so K (is) to S . Thus, K also measures S [Def. 7.20]. And E also measures S [Prop. 7.20]. Thus, E and K (both) measure S . Thus, the least (number) measured by (both) E and K will also measure S [Prop. 7.35]. And M is the least (number) measured by (both) E and K . Thus, M measures S , the greater (measuring) the lesser. The very thing is impossible. Thus there cannot be any numbers less than N, O, M, P (which are) continuously proportional in the ratios of A to B , and of C to D , and, further, of E to F . Thus, N, O, M, P are the least (numbers) continuously proportional in the ratios of $A B, C D, E F$. (Which is) the very thing it was required to show.

Proposition 5

Plane numbers have to one another the ratio compounded[†] out of (the ratios of) their sides.



Ἐστωσαν ἐπίπεδοι ἀριθμοὶ οἱ A, B , καὶ τοῦ μὲν A πλευραὶ ἔστωσαν οἱ Γ, Δ ἀριθμοί, τοῦ δὲ B οἱ E, Z λέγω, ὅτι ὁ A πρὸς τὸν B λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν.

Λόγων γὰρ δοθέντων τοῦ τε ὄν ἔθει ὁ Γ πρὸς τὸν E καὶ ὁ Δ πρὸς τὸν Z εἰλήφθωσαν ἀριθμοὶ ἐχῆς ἐλάχιστοι ἐν τοῖς $\Gamma E, \Delta Z$ λόγοις, οἱ H, Θ, K , ὥστε εἶναι ὡς μὲν τὸν Γ πρὸς τὸν E , οὕτως τὸν H πρὸς τὸν Θ , ὡς δὲ τὸν Δ πρὸς τὸν Z , οὕτως τὸν Θ πρὸς τὸν K . καὶ ὁ Δ τὸν E πολλαπλασιάσας τὸν Λ ποιείτω.

Καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ E πολλαπλασιάσας τὸν Λ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν E , οὕτως ὁ A πρὸς τὸν Λ . ὡς δὲ ὁ Γ πρὸς τὸν E , οὕτως ὁ H πρὸς τὸν Θ . καὶ ὡς ἄρα ὁ H πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν Λ . πάλιν, ἐπεὶ ὁ E τὸν Δ πολλαπλασιάσας τὸν Λ πεποίηκεν, ἀλλὰ μὴν καὶ τὸν Z πολλαπλασιάσας τὸν B πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ Λ πρὸς τὸν B . ἀλλ' ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ Θ πρὸς τὸν K . καὶ ὡς ἄρα ὁ Θ πρὸς τὸν K , οὕτως ὁ Λ πρὸς τὸν B . ἐδείκθη δὲ καὶ ὡς ὁ H πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν Λ . δι' ἴσον ἄρα ἐστὶν ὡς ὁ H πρὸς τὸν K , [οὕτως] ὁ A πρὸς τὸν B . ὁ δὲ H πρὸς τὸν K λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν καὶ ὁ A ἄρα πρὸς τὸν B λόγον ἔθει τὸν συγκείμενον ἐκ τῶν πλευρῶν ὅπερ ἔδει δεῖξαι.

Let A and B be plane numbers, and let the numbers C, D be the sides of A , and (the numbers) E, F (the sides) of B . I say that A has to B the ratio compounded out of (the ratios of) their sides.

For given the ratios which C has to E , and D (has) to F , let the least numbers, G, H, K , continuously proportional in the ratios $C E, D F$ be taken [Prop. 8.4], so that as C is to E , so G (is) to H , and as D (is) to F , so H (is) to K . And let D make L (by) multiplying E .

And since D has made A (by) multiplying C , and has made L (by) multiplying E , thus as C is to E , so A (is) to L [Prop. 7.17]. And as C (is) to E , so G (is) to H . And thus as G (is) to H , so A (is) to L . Again, since E has made L (by) multiplying D [Prop. 7.16], but, in fact, has also made B (by) multiplying F , thus as D is to F , so L (is) to B [Prop. 7.17]. But, as D (is) to F , so H (is) to K . And thus as H (is) to K , so L (is) to B . And it was also shown that as G (is) to H , so A (is) to L . Thus, via equality, as G is to K , [so] A (is) to B [Prop. 7.14]. And G has to K the ratio compounded out of (the ratios of) the sides (of A and B). Thus, A also has to B the ratio compounded out of (the ratios of) the sides (of A and B). (Which is) the very thing it was required to show.

[†] i.e., multiplied.

6.

Ἐάν ὥσιν ὁποσοῖον ἀριθμοὶ ἐχῇς ἀνάλογον, ὁ δὲ πρώτοξ τὸν δεύτερον μὴ μετρήῃ, οὐδὲ ἄλλοξ οὐδεὶς οὐδένα μετρήσει.

Ἐστῶσαν ὁποσοῖον ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A, B, Γ, Δ, E , ὁ δὲ A τὸν B μὴ μετρεῖτω· λέγω, ὅτι οὐδὲ ἄλλοξ οὐδεὶς οὐδένα μετρήσει.

A —————

B —————

Γ —————

Δ —————

E —————

Z —————

H —————

Θ —————

A —————

B —————

C —————

D —————

E —————

F —————

G —————

H —————

Ὅτι μὲν οὖν οἱ A, B, Γ, Δ, E ἐχῇς ἀλλήλων οὐ μετροῦσιν, φανερόν· οὐδὲ γάρ ὁ A τὸν B μετρεῖ. λέγω δὴ, ὅτι οὐδὲ ἄλλοξ οὐδεὶς οὐδένα μετρήσει. εἰ γὰρ δυνατόν, μετρεῖτω ὁ A τὸν Γ . καὶ ὅσοι εἰσὶν οἱ A, B, Γ , τοσοῦτοι εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ οἱ Z, H, Θ . καὶ ἐπεὶ οἱ Z, H, Θ ἐν τῷ αὐτῷ λόγῳ εἰσὶ τοῖς A, B, Γ , καὶ ἐστὶν ἴσον τὸ πλῆθος τῶν A, B, Γ τῷ πλήθει τῶν Z, H, Θ , δι' ἴσον ἄρα ἐστὶν ὥς ὁ A πρὸς τὸν Γ , οὕτως ὁ Z πρὸς τὸν Θ . καὶ ἐπεὶ ἐστὶν ὥς ὁ A πρὸς τὸν B , οὕτως ὁ Z πρὸς τὸν H , οὐ μετρεῖ δὲ ὁ A τὸν B , οὐ μετρεῖ ἄρα οὐδὲ ὁ Z τὸν H . οὐκ ἄρα μονὰξ ἐστὶν ὁ Z · ἢ γὰρ μονὰξ πάντα ἀριθμὸν μετρεῖ. καὶ εἰσὶν οἱ Z, Θ πρῶτοι πρὸς ἀλλήλους [οὐδὲ ὁ Z ἄρα τὸν Θ μετρεῖ]. καὶ ἐστὶν ὥς ὁ Z πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν Γ · οὐδὲ ὁ A ἄρα τὸν Γ μετρεῖ. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ἄλλοξ οὐδεὶς οὐδένα μετρήσει· ὅπερ ἔδει δεῖξαι.

Proposition 6

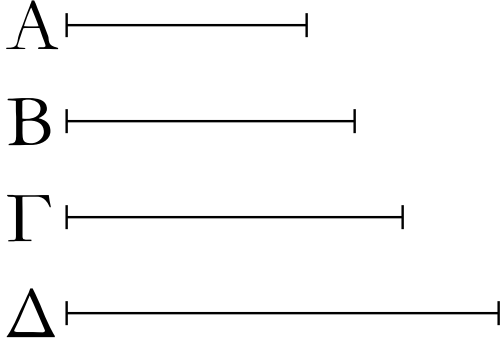
If there are any multitude whatsoever of continuously proportional numbers, and the first does not measure the second, then no other (number) will measure any other (number) either.

Let A, B, C, D, E be any multitude whatsoever of continuously proportional numbers, and let A not measure B . I say that no other (number) will measure any other (number) either.

Now, (it is) clear that A, B, C, D, E do not successively measure one another. For A does not even measure B . So I say that no other (number) will measure any other (number) either. For, if possible, let A measure C . And as many (numbers) as are A, B, C , let so many of the least numbers, F, G, H , be taken of those (numbers) having the same ratio as A, B, C [Prop. 7.33]. And since F, G, H are in the same ratio as A, B, C , and the multitude of A, B, C is equal to the multitude of F, G, H , thus, via equality, as A is to C , so F (is) to H [Prop. 7.14]. And since as A is to B , so F (is) to G , and A does not measure B , F does not measure G either [Def. 7.20]. Thus, F is not a unit. For a unit measures all numbers. And F and H are prime to one another [Prop. 8.3] [and thus F does not measure H]. And as F is to H , so A (is) to C . And thus A does not measure C either [Def. 7.20]. So, similarly, we can show that no other (number) can measure any other (number) either.

7.

Ἐάν ὥσιν ὅποσσιον ἀριθμοὶ [ἐχῇ] ἀνάλογον, ὁ δὲ πρῶτος τὸν ἑσθαιον μετρή, καὶ τὸν δεύτερον μετρήσει.



Ἐστωσαν ὅποσσιον ἀριθμοὶ ἐχῇ ἀνάλογον οἱ A, B, Γ, Δ , ὁ δὲ A τὸν Δ μετρεῖτω· λέγω, ὅτι καὶ ὁ A τὸν B μετρεῖ.

Εἰ γὰρ οὐ μετρεῖ ὁ A τὸν B , οὐδὲ ἄλλος οὐδεὶς οὐδένα μετρήσει· μετρεῖ δὲ ὁ A τὸν Δ . μετρεῖ ἄρα καὶ ὁ A τὸν B · ὅπερ ἔδει δεῖχαι.

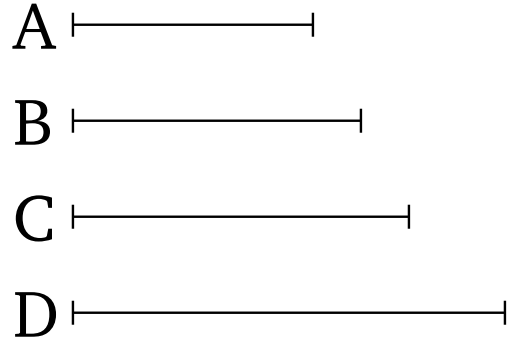
8.

Ἐάν δύο ἀριθμῶν μεταχὺ κατὰ τὸ συνεθεῖ ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅσοι εἰς αὐτοὺς μεταχὺ κατὰ τὸ συνεθεῖ ἀνόλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοὺς τὸν αὐτὸν λόγον ἔχονταξ [αὐτοῖς] μεταχὺ κατὰ τὸ συνεθεῖ ἀνάλογον ἐμπεσοῦνται

(Which is) the very thing it was required to show.

Proposition 7

If there are any multitude whatsoever of [continuously] proportional numbers, and the first measures the last, then (the first) will also measure the second.

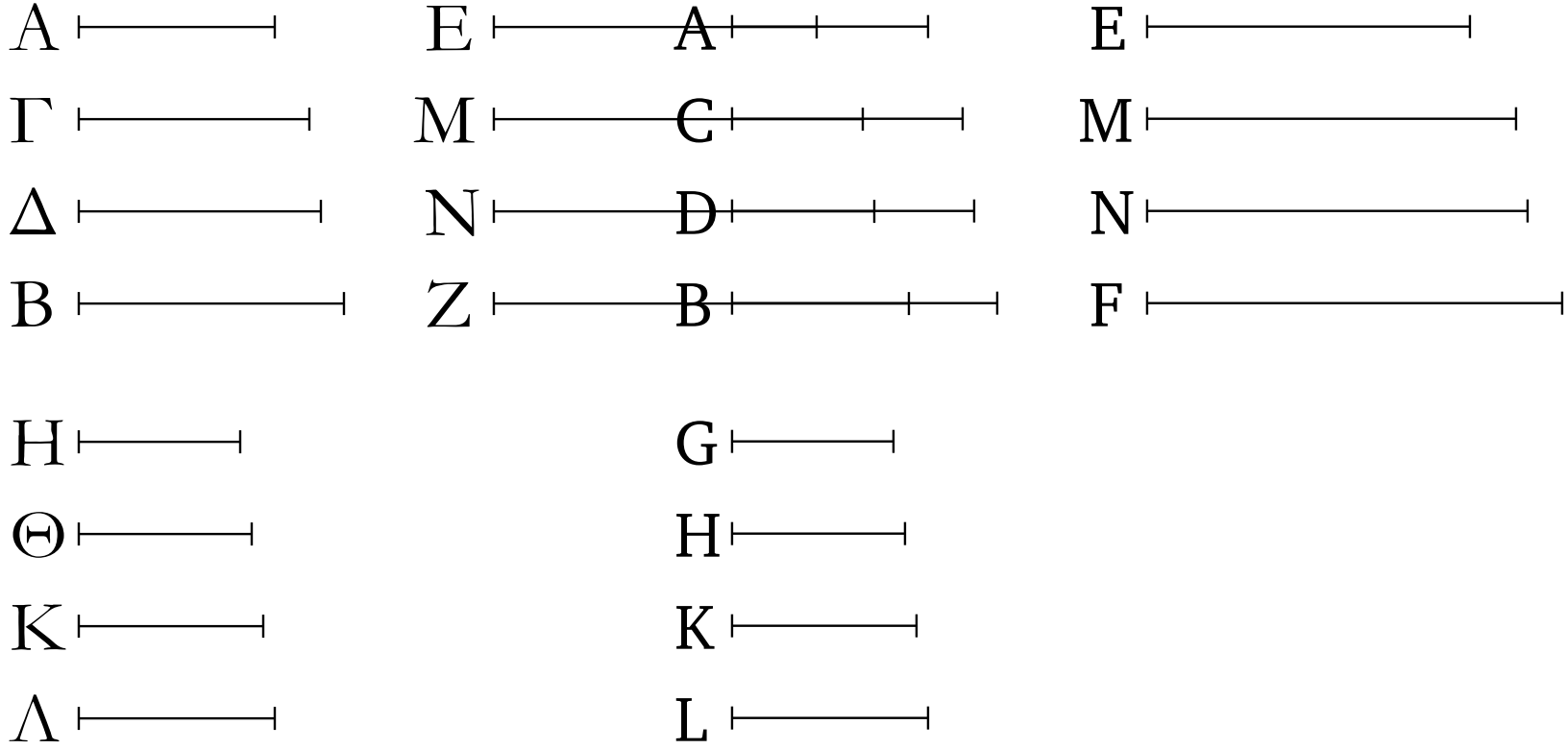


Let A, B, C, D be any number whatsoever of continuously proportional numbers. And let A measure D . I say that A also measures B .

For if A does not measure B then no other (number) will measure any other (number) either [Prop. 8.6]. But A measures D . Thus, A also measures B . (Which is) the very thing it was required to show.

Proposition 8

If between two numbers there fall (some) numbers in continued proportion then, as many numbers as fall in between them in continued proportion, so many (numbers) will also fall in between (any two numbers) having the same ratio [as them] in continued proportion.



Δύο γάρ ἀριθμῶν τῶν A, B μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπιπτέωσαν ἀριθμοὶ οἱ Γ, Δ , καὶ πεποιήσθω ὥς ὁ A πρὸς τὸν B , οὕτως ὁ E πρὸς τὸν Z . λέγω, ὅτι ὅσοι εἰς τούξ A, B μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τούξ E, Z μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

Ὅσοι γάρ εἰσι τῷ πλήθει οἱ A, B, Γ, Δ , τοσοῦτοι εἰλήφθωσαν ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἔθόντων τοῖς A, Γ, Δ, B οἱ H, Θ, K, Λ . οἱ ἄρα ἄκροι αὐτῶν οἱ H, Λ πρῶτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ οἱ A, Γ, Δ, B τοῖς H, Θ, K, Λ ἐν τῷ αὐτῷ λόγῳ εἰσίν, καὶ ἔστιν ἴσον τὸ πλῆθος τῶν A, Γ, Δ, B τῷ πλήθει τῶν H, Θ, K, Λ , δι' ἴσον ἄρα ἔστιν ὥς ὁ A πρὸς τὸν B , οὕτως ὁ H πρὸς τὸν Λ . ὥς δὲ ὁ A πρὸς τὸν B , οὕτως ὁ E πρὸς τὸν Z . οἱ δὲ H, Λ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τούξ τὸν αὐτὸν λόγον ἔθονταξ ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. ἰσάκις ἄρα ὁ H τὸν E μετρεῖ καὶ ὁ Λ τὸν Z . ὁσάκις δὴ ὁ H τὸν E μετρεῖ, τοσαντάκις καὶ ἑκάτεροξ τῶν Θ, K ἑκάτερον τῶν M, N μετρεῖτω· οἱ H, Θ, K, Λ ἄρα τούξ E, M, N, Z ἰσάκις μετροῦσιν. οἱ H, Θ, K, Λ ἄρα τοῖς E, M, N, Z ἐν τῷ αὐτῷ λόγῳ εἰσίν. ἀλλὰ οἱ H, Θ, K, Λ τοῖς A, Γ, Δ, B ἐν τῷ αὐτῷ λόγῳ εἰσίν· καὶ οἱ A, Γ, Δ, B ἄρα τοῖς E, M, N, Z ἐν τῷ αὐτῷ λόγῳ εἰσίν. οἱ δὲ A, Γ, Δ, B ἐχῆξ ἀνάλογόν εἰσιν· καὶ οἱ E, M, N, Z ἄρα ἐχῆξ ἀνάλογόν

For let the numbers, C and D , fall between two numbers, A and B , in continued proportion, and let it be contrived (that) as A (is) to B , so E (is) to F . I say that, as many numbers as have fallen in between A and B in continued proportion, so many (numbers) will also fall in between E and F in continued proportion.

For as many as A, B, C, D are in multitude, let so many of the least numbers, G, H, K, L , having the same ratio as A, B, C, D , be taken [Prop. 7.33]. Thus, the outermost of them, G and L , are prime to one another [Prop. 8.3]. And since A, B, C, D are in the same ratio as G, H, K, L , and the multitude of A, B, C, D is equal to the multitude of G, H, K, L , thus, via equality, as A is to B , so G (is) to L [Prop. 7.14]. And as A (is) to B , so E (is) to F . And thus as G (is) to L , so E (is) to F . And G and L (are) prime (to one another). And (numbers) prime (to one another are) also the least (numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, G measures E the same number of times as L (measures) F . So as many times as G measures E , so many times let H, K also measure M, N , respectively. Thus, G, H, K, L measure E, M, N, F (respectively) an equal number of times. Thus, G, H, K, L are in the same ratio as E, M, N, F [Def. 7.20]. But, $G, H,$

είσιν. ὅσοι ἄρα εἰς τοὺς A, B μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τοὺς E, Z μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί· ὅπερ ἔδει δεῖχαι.

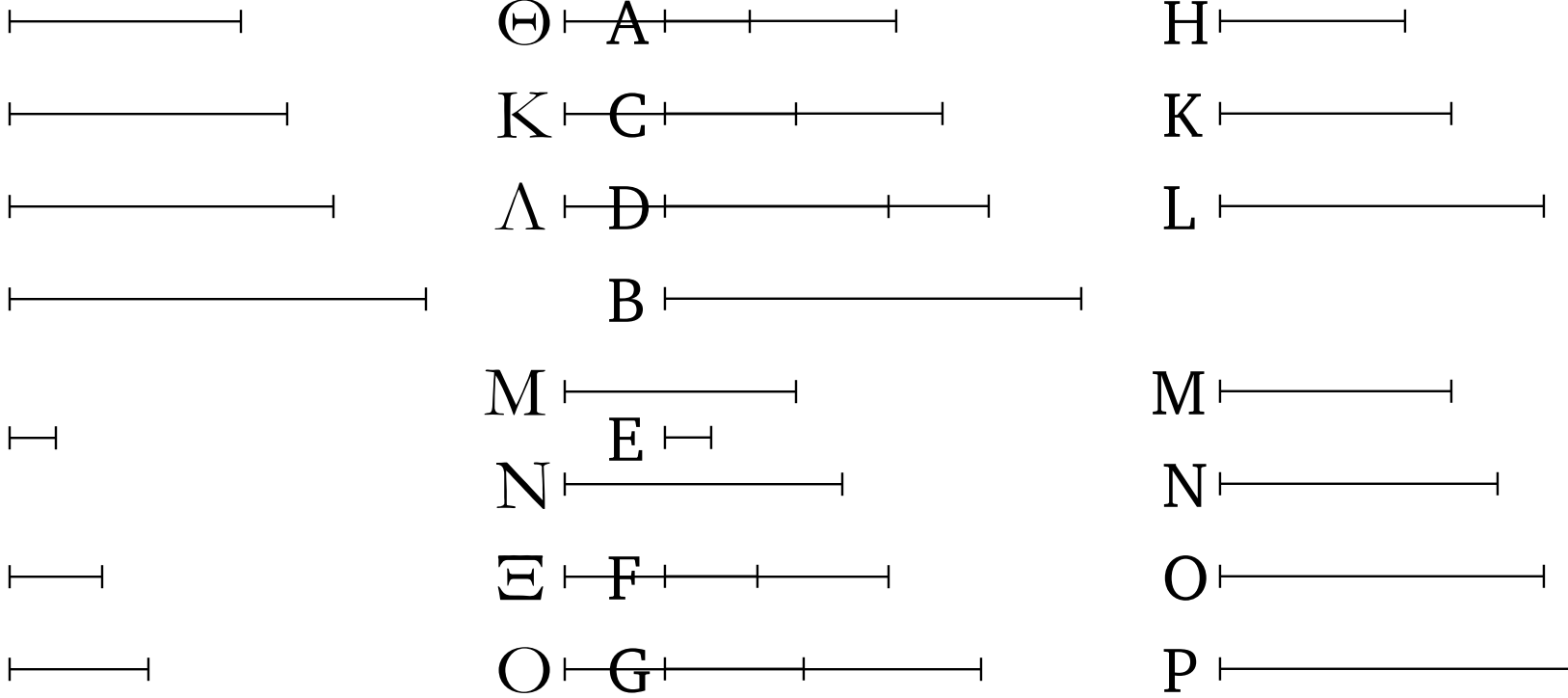
K, L are in the same ratio as A, C, D, B . Thus, A, C, D, B are also in the same ratio as E, M, N, F . And A, C, D, B are continuously proportional. Thus, E, M, N, F are also continuously proportional. Thus, as many numbers as have fallen in between A and B in continued proportion, so many numbers have also fallen in between E and F in continued proportion. (Which is) the very thing it was required to show.

9.

Proposition 9

Ἐάν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους ᾧσιν, καὶ εἰς αὐτοὺς μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅσοι εἰς αὐτοὺς μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρωσιν αὐτῶν καὶ μονάδοξ μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

If two numbers are prime to one another and there fall in between them (some) numbers in continued proportion then, as many numbers as fall in between them in continued proportion, so many (numbers) will also fall between each of them and a unit in continued proportion.



Ἐστωσαν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλους οἱ A, B , καὶ εἰς αὐτοὺς μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτέωσιν οἱ Γ, Δ , καὶ ἐκκείσθω ἡ E μονάξ· λέγω, ὅτι ὅσοι εἰς τοὺς A, B μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρωσιν τῶν A, B καὶ τῆς μονάδοξ μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

Εἰλήφθωσαν γάρ δύο μὲν ἀριθμοὶ ἐλάδιστοι ἐν τῷ τῶν A, Γ, Δ, B λόγῳ ὄντες οἱ Z, H , τρεῖς δὲ οἱ Θ, K, Λ , καὶ αἰεὶ ἐχῆς ἐνὶ πλείονξ, ἕως ἂν ἴσον γένηται τὸ πλῆθος αὐτῶν τῷ πλήθει τῶν A, Γ, Δ, B . εἰλήφθωσαν, καὶ ἔστωσαν οἱ M, N, X, O . φανερόν δὲ, ὅτι ὁ μὲν Z ἐαντὸν πολλαπλασιάσας τὸν Θ πεποίηκεν, τὸν δὲ Θ πολλαπλασιάσας τὸν M πεποίηκεν, καὶ ὁ H ἐαντὸν μὲν πολλαπλασιάσας τὸν Λ πεποίηκεν, τὸν δὲ Λ

Let A and B be two numbers (which are) prime to one another, and let the (numbers) C and D fall in between them in continued proportion. And let the unit E be set out. I say that, as many numbers as have fallen in between A and B in continued proportion, so many (numbers) will also fall between each of A and B and the unit in continued proportion.

For let the least two numbers, F and G , which are in the ratio of A, C, D, B , be taken [Prop. 7.33]. And the (least) three (numbers), H, K, L . And so on, successively increasing by one, until the multitude of the (least numbers taken) is made equal to the multitude of A, C, D, B [Prop. 8.2]. Let them be taken, and let them be M, N, O, P . So (it is) clear that F has made H (by) multiplying itself, and has made M (by)

πολλαπλασιάσας τὸν O πεποίηκεν. καὶ ἐπεὶ οἱ M, N, X, O ἐλάχιστοι εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Z, H , εἰσὶ δὲ καὶ οἱ A, Γ, Δ, B ἐλάχιστοι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Z, H , καὶ ἔστιν ἴσον τὸ πλῆθος τῶν M, N, X, O τῷ πλῆθει τῶν A, Γ, Δ, B , ἕκαστος ἄρα τῶν M, N, X, O ἐκάστω τῶν A, Γ, Δ, B ἴσος ἐστίν· ἴσος ἄρα ἐστὶν ὁ μὲν M τῷ A , ὁ δὲ O τῷ B . καὶ ἐπεὶ ὁ Z ἐαυτὸν πολλαπλασιάσας τὸν Θ πεποίηκεν, ὁ Z ἄρα τὸν Θ μετρεῖ κατὰ τὰς ἐν τῷ Z μονάδας. μετρεῖ δὲ καὶ ἡ E μονὰς τὸν Z κατὰ τὰς ἐν αὐτῷ μονάδας· ἰσάκις ἄρα ἡ E μονὰς τὸν Z ἀριθμὸν μετρεῖ καὶ ὁ Z τὸν Θ . ἔστιν ἄρα ὡς ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Z πρὸς τὸν Θ . πάλιν, ἐπεὶ ὁ Z τὸν Θ πολλαπλασιάσας τὸν M πεποίηκεν, ὁ Θ ἄρα τὸν M μετρεῖ κατὰ τὰς ἐν τῷ Z μονάδας. μετρεῖ δὲ καὶ ἡ E μονὰς τὸν Z ἀριθμὸν κατὰ τὰς ἐν αὐτῷ μονάδας· ἰσάκις ἄρα ἡ E μονὰς τὸν Z ἀριθμὸν μετρεῖ καὶ ὁ Θ τὸν M . ἔστιν ἄρα ὡς ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Θ πρὸς τὸν M . ἐδείκθη δὲ καὶ ὡς ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Z πρὸς τὸν Θ · καὶ ὡς ἄρα ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Z πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν M . ἴσος δὲ ὁ M τῷ A · ἔστιν ἄρα ὡς ἡ E μονὰς πρὸς τὸν Z ἀριθμὸν, οὕτως ὁ Z πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν A . διὰ τὰ αὐτὰ δὴ καὶ ὡς ἡ E μονὰς πρὸς τὸν H ἀριθμὸν, οὕτως ὁ H πρὸς τὸν Λ καὶ ὁ Λ πρὸς τὸν B . ὅσοι ἄρα εἰς τούξ A, B μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ ἑκατέρων τῶν A, B καὶ μονάδος τῆς E μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί· ὅπερ ἔδει δεῖχαι.

10.

Ἐάν δύο ἀριθμῶν ἑκατέρων καὶ μονάδος μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅσοι ἑκατέρων αὐτῶν καὶ μονάδος μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς αὐτοῦξ μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

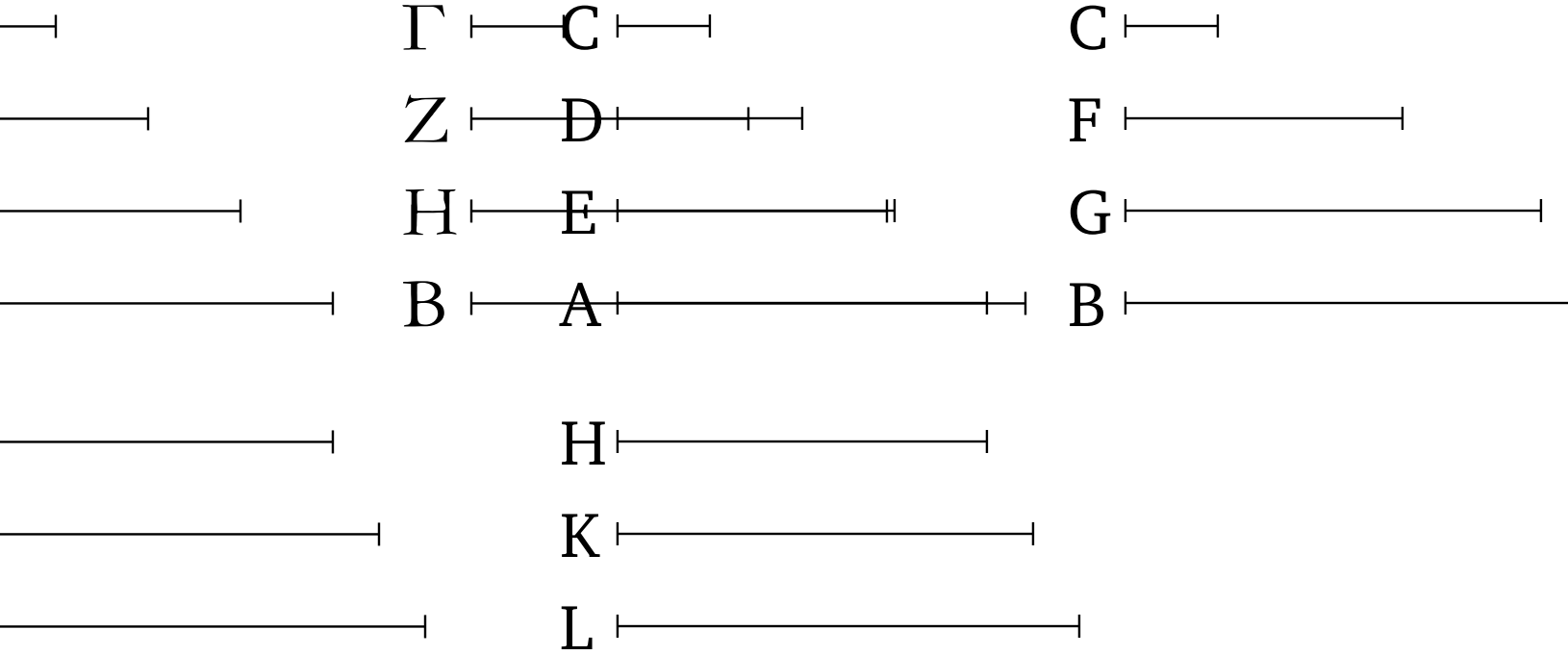
Δύο γάρ ἀριθμῶν τῶν A, B καὶ μονάδος τῆς Γ μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτέτωσαν ἀριθμοί οἱ τε Δ, E καὶ οἱ Z, H · λέγω, ὅτι ὅσοι ἑκατέρων τῶν A, B καὶ μονάδος τῆς Γ μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεπτώκασιν ἀριθμοί, τοσοῦτοι καὶ εἰς τούξ A, B μεταχὺν κατὰ τὸ συνεθὲς ἀνάλογον ἐμπεσοῦνται.

multiplying H . And G has made L (by) multiplying itself, and has made P (by) multiplying L [Prop. 8.2 corr.]. And since M, N, O, P are the least of those (numbers) having the same ratio as F, G , and A, C, D, B are also the least of those (numbers) having the same ratio as F, G [Prop. 8.2], and the multitude of M, N, O, P is equal to the multitude of A, C, D, B , thus M, N, O, P are equal to A, C, D, B , respectively. Thus, M is equal to A , and P to B . And since F has made H (by) multiplying itself, F thus measures H according to the units in F [Def. 7.15]. And the unit E also measures F according to the units in it. Thus, the unit E measures the number F as many times as F (measures) H . Thus, as the unit E is to the number F , so F (is) to H [Def. 7.20]. Again, since F has made M (by) multiplying H , H thus measures M according to the units in F [Def. 7.15]. And the unit E also measures the number F according to the units in it. Thus, the unit E measures the number F as many times as H (measures) M . Thus, as the unit E is to the number F , so H (is) to M [Prop. 7.20]. And it was shown that as the unit E (is) to the number F , so F (is) to H . And thus as the unit E (is) to the number F , so F (is) to H , and H (is) to M . And M (is) equal to A . Thus, as the unit E is to the number F , so F (is) to H , and H to A . And so, for the same (reasons), as the unit E (is) to the number G , so G (is) to L , and L to B . Thus, as many (numbers) as have fallen in between A and B in continued proportion, so many numbers have also fallen between each of A and B and the unit E in continued proportion. (Which is) the very thing it was required to show.

Proposition 10

If (some) numbers fall between each of two numbers and a unit in continued proportion then, as many (numbers) as fall between each of the (two numbers) and the unit in continued proportion, so many (numbers) will also fall in between the (two numbers) themselves in continued proportion.

For let the numbers D, E and F, G fall between the numbers A and B (respectively) and the unit C in continued proportion. I say that, as many numbers as have fallen between each of A and B and the unit C in continued proportion, so many will also fall in between A and B in continued proportion.



Ὁ Δ γὰρ τὸν Ζ πολλαπλασιάσας τὸν Θ ποιεῖτω, ἑκάτε-
ροξ δὲ τῶν Δ, Ζ τὸν Θ πολλαπλασιάσας ἑκάτερον τῶν Κ, Α
ποιεῖτω.

Καὶ ἐπεὶ ἔστιν ὡς ἡ Γ μονὰξ πρὸς τὸν Δ ἀριθμόν, οὕτως
ὁ Δ πρὸς τὸν Ε, ἰσάκις ἄρα ἡ Γ μονὰξ τὸν Δ ἀριθμόν μετρεῖ
καὶ ὁ Δ τὸν Ε. ἡ δὲ Γ μονὰξ τὸν Δ ἀριθμόν μετρεῖ κατὰ
τάξιν ἐν τῷ Δ μονάδαξ· καὶ ὁ Δ ἄρα ἀριθμὸς τὸν Ε μετρεῖ
κατὰ τάξιν ἐν τῷ Δ μονάδαξ· ὁ Δ ἄρα ἐαυτὸν πολλαπλασιάσας
τὸν Ε πεποίηκεν. πάλιν, ἐπεὶ ἔστιν ὡς ἡ Γ [μονὰξ] πρὸς τὸν
Δ ἀριθμόν, οὕτως ὁ Ε πρὸς τὸν Α, ἰσάκις ἄρα ἡ Γ μονὰξ
τὸν Δ ἀριθμόν μετρεῖ καὶ ὁ Ε τὸν Α. ἡ δὲ Γ μονὰξ τὸν Δ
ἀριθμόν μετρεῖ κατὰ τάξιν ἐν τῷ Δ μονάδαξ· καὶ ὁ Ε ἄρα
τὸν Α μετρεῖ κατὰ τάξιν ἐν τῷ Δ μονάδαξ· ὁ Δ ἄρα τὸν Ε
πολλαπλασιάσας τὸν Α πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ
μὲν Ζ ἐαυτὸν πολλαπλασιάσας τὸν Η πεποίηκεν, τὸν δὲ Η
πολλαπλασιάσας τὸν Β πεποίηκεν. καὶ ἐπεὶ ὁ Δ ἐαυτὸν μὲν
πολλαπλασιάσας τὸν Ε πεποίηκεν, τὸν δὲ Ζ πολλαπλασιάσας
τὸν Θ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Ε
πρὸς τὸν Θ. διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ
Θ πρὸς τὸν Η. καὶ ὡς ἄρα ὁ Ε πρὸς τὸν Θ, οὕτως ὁ Θ πρὸς
τὸν Η. πάλιν, ἐπεὶ ὁ Δ ἑκάτερον τῶν Ε, Θ πολλαπλασιάσας
ἑκάτερον τῶν Α, Κ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Ε πρὸς τὸν
Θ, οὕτως ὁ Α πρὸς τὸν Κ. ἀλλ' ὡς ὁ Ε πρὸς τὸν Θ, οὕτως ὁ
Δ πρὸς τὸν Ζ· καὶ ὡς ἄρα ὁ Δ πρὸς τὸν Ζ, οὕτως ὁ Α πρὸς
τὸν Κ. πάλιν, ἐπεὶ ἑκάτεροξ τῶν Δ, Ζ τὸν Θ πολλαπλασιάσας
ἑκάτερον τῶν Κ, Α πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν
Ζ, οὕτως ὁ Κ πρὸς τὸν Α. ἀλλ' ὡς ὁ Δ πρὸς τὸν Ζ, οὕτως
ὁ Α πρὸς τὸν Κ· καὶ ὡς ἄρα ὁ Α πρὸς τὸν Κ, οὕτως ὁ Κ
πρὸς τὸν Α. ἔτι ἐπεὶ ὁ Ζ ἑκάτερον τῶν Θ, Η πολλαπλασιάσας
ἑκάτερον τῶν Α, Β πεποίηκεν, ἔστιν ἄρα ὡς ὁ Θ πρὸς τὸν

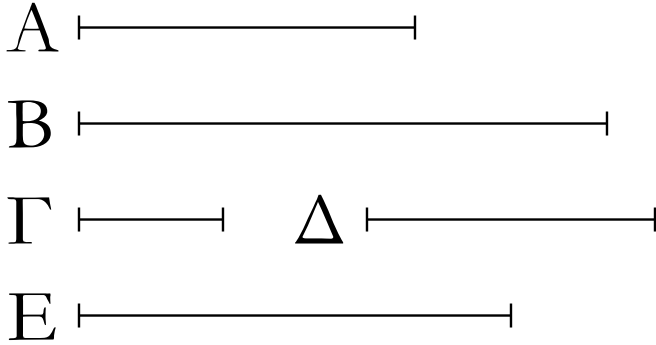
For let D make H (by) multiplying F . And let D, F make
 K, L , respectively, by multiplying H .

As since as the unit C is to the number D , so D (is) to E ,
the unit C thus measures the number D as many times as D
(measures) E [Def. 7.20]. And the unit C measures the num-
ber D according to the units in D . Thus, the number D also
measures E according to the units in D . Thus, D has made E
(by) multiplying itself. Again, since as the [unit] C is to the
number D , so E (is) to A , the unit C thus measures the number
 D as many times as E (measures) A [Def. 7.20]. And the unit
 C measures the number D according to the units in D . Thus,
 E also measures A according to the units in D . Thus, D has
made A (by) multiplying E . And so, for the same (reasons), F
has made G (by) multiplying itself, and has made B (by) mul-
tiplying G . And since D has made E (by) multiplying itself,
and has made H (by) multiplying F , thus as D is to F , so E (is)
to H [Prop 7.17]. And so, for the same reasons, as D (is) to F ,
so H (is) to G [Prop. 7.18]. And thus as E (is) to H , so H (is)
to G . Again, since D has made A, K (by) multiplying E, H ,
respectively, thus as E is to H , so A (is) to K [Prop 7.17]. But,
as E (is) to H , so D (is) to F . And thus as D (is) to F , so A
(is) to K . Again, since D, F have made K, L , respectively, (by)
multiplying H , thus as D is to F , so K (is) to L [Prop. 7.18].
But, as D (is) to F , so A (is) to K . And thus as A (is) to K , so
 K (is) to L . Further, since F has made L, B (by) multiplying
 H, G , respectively, thus as H is to G , so L (is) to B [Prop 7.17].
And as H (is) to G , so D (is) to F . And thus as D (is) to F , so L
(is) to B . And it was also shown that as D (is) to F , so A (is) to
 K , and K to L . And thus as A (is) to K , so K (is) to L , and L to
 B . Thus, A, K, L, B are successively in continued proportion.

H , οὕτως ὁ Λ πρὸς τὸν B . ὥς δὲ ὁ Θ πρὸς τὸν H , οὕτως ὁ Δ πρὸς τὸν Z : καὶ ὥς ἄρα ὁ Δ πρὸς τὸν Z , οὕτως ὁ Λ πρὸς τὸν B . ἐδείκθη δὲ καὶ ὥς ὁ Δ πρὸς τὸν Z , οὕτως ὁ Λ πρὸς τὸν K καὶ ὁ K πρὸς τὸν Λ : καὶ ὥς ἄρα ὁ A πρὸς τὸν K , οὕτως ὁ K πρὸς τὸν Λ καὶ ὁ Λ πρὸς τὸν B . οἱ A , K , Λ , B ἄρα κατὰ τὸ συνεθὲς ἐχῆς εἶναι ἀνάλογον. ὅσοι ἄρα ἑκατέρων τῶν A , B καὶ τῆς Γ μονάδος μεταχὺ κατὰ τὸ συνεθὲς ἀνάλογον ἐμπίπτουσιν ἀριθμοί, τοσοῦτοι καὶ εἰς τὸν A , B μεταχὺ κατὰ τὸ συνεθὲς ἐμπεσοῦνται· ὅπερ ἔδει δεῖξαι.

11.

Δύο τετραγώνων ἀριθμῶν εἷς μέσοξ ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ τετράγωνος πρὸς τὸν τετράγωνον διπλασίονα λόγον ἔθει ἢ περὶ ἢ πλευρὰ πρὸς τὴν πλευράν.



Ἐστωσαν τετράγωνοι ἀριθμοὶ οἱ A , B , καὶ τοῦ μὲν A πλευρὰ ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ : λέγω, ὅτι τῶν A , B εἷς μέσοξ ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἢ περὶ ὁ Γ πρὸς τὸν Δ .

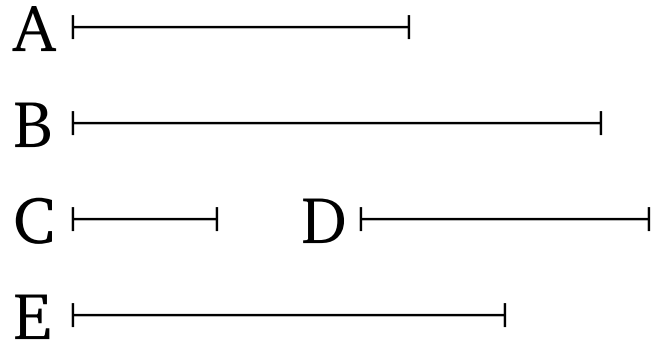
Ὁ Γ γάρ τὸν Δ πολλαπλασιάσας τὸν E ποιεῖτω. καὶ ἐπεὶ τετράγωνός ἐστιν ὁ A , πλευρὰ δὲ αὐτοῦ ἐστιν ὁ Γ , ὁ Γ ἄρα ἑαυτὸν πολλαπλασιάσας τὸν A πεποιήκεν. διὰ τὰ αὐτὰ δὲ καὶ ὁ Δ ἑαυτὸν πολλαπλασιάσας τὸν B πεποιήκεν. ἐπεὶ οὖν ὁ Γ ἑκάτερον τῶν Γ , Δ πολλαπλασιάσας ἑκάτερον τῶν A , E πεποιήκεν, ἔστιν ἄρα ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν E . διὰ τὰ αὐτὰ δὲ καὶ ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν B . καὶ ὥς ἄρα ὁ A πρὸς τὸν E , οὕτως ὁ E πρὸς τὸν B . τῶν A , B ἄρα εἷς μέσοξ ἀνάλογόν ἐστιν ἀριθμός.

Λέγω δὲ, ὅτι καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἢ περὶ ὁ Γ πρὸς τὸν Δ . ἐπεὶ γὰρ τρεῖς ἀριθμοὶ ἀνάλογόν εἰσιν οἱ A , E , B , ὁ A ἄρα πρὸς τὸν B διπλασίονα λόγον ἔθει ἢ περὶ ὁ A πρὸς τὸν E . ὥς δὲ ὁ A πρὸς τὸν E , οὕτως ὁ Γ πρὸς τὸν Δ . ὁ A ἄρα πρὸς τὸν B διπλασίονα λόγον ἔθει ἢ περὶ ὁ Γ πρὸς τὸν Δ : ὅπερ ἔδει δεῖξαι.

Thus, as many numbers as fall between each of A and B and the unit C in continued proportion, so many will also fall between A and B in continued proportion. (Which is) the very thing it was required to show.

Proposition 11

There exists one number in mean proportion to two (given) square numbers.[†] And (one) square (number) has to the (other) square (number) a squared[‡] ratio with respect to (that) the side (of the former has) to the side (of the latter).



Let A and B be square numbers, and let C be the side of A , and D (the side) of B . I say that there exists one number in mean proportion to A and B , and that A has to B a squared ratio with respect to (that) C (has) to D .

For let C make E (by) multiplying D . And since A is square, and C is its side, C has thus made A (by) multiplying itself. And so, for the same (reasons), D has made B (by) multiplying itself. Therefore, since C has made A , E (by) multiplying C , D , respectively, thus as C is to D , so A (is) to E [Prop. 7.17]. And so, for the same (reasons), as C (is) to D , so E (is) to B [Prop. 7.18]. And thus as A (is) to E , so E (is) to B . Thus, one number (namely, E) is in mean proportion to A and B .

So I say that A also has to B a squared ratio with respect to (that) C (has) to D . For since A , E , B are three (continuously) proportional numbers, A thus has to B a squared ratio with respect to (that) A (has) to E [Def. 5.9]. And as A (is) to E , so C (is) to D . Thus, A has to B a squared ratio with respect to (that) side C (has) to (side) D . (Which is) the very thing it was required to show.

[†] In other words, between two given square numbers there exists a number in continued proportion.

[‡] Literally, “double”.

12.

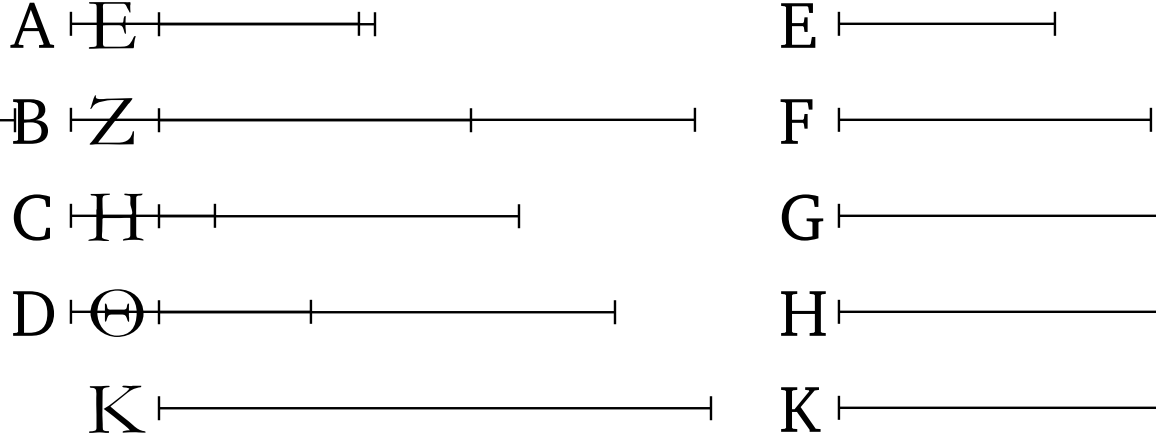
Proposition 12

Δύο κύβων ἀριθμῶν δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοί, καὶ ὁ κύβοξ πρὸς τὸν κύβον τριπλασίονα λόγον ἔθει ἥπερ ἡ πλευρὰ πρὸς τὴν πλευράν.

Ἐστωσαν κύβοι ἀριθμοὶ οἱ A, B καὶ τοῦ μὲν A πλευρὰ ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ . λέγω, ὅτι τῶν A, B δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοί, καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν Δ .

There exist two numbers in mean proportion to two (given) cube numbers.[†] And (one) cube (number) has to the (other) cube (number) a cubed[‡] ratio with respect to (that) the side (of the former has) to the side (of the latter).

Let A and B be cube numbers, and let C be the side of A , and D (the side) of B . I say that there exist two numbers in mean proportion to A and B , and that A has to B a cubed ratio with respect to (that) C (has) to D .



Ὁ γὰρ Γ ἑαυτὸν μὲν πολλαπλασιάσας τὸν E ποιεῖτω, τὸν δὲ Δ πολλαπλασιάσας τὸν Z ποιεῖτω, ὁ δὲ Δ ἑαυτὸν πολλαπλασιάσας τὸν H ποιεῖτω, ἐκάτεροξ δὲ τῶν Γ, Δ τὸν Z πολλαπλασιάσας ἐκάτερον τῶν Θ, K ποιεῖτω.

Καὶ ἐπεὶ κύβοξ ἐστὶν ὁ A , πλευρὰ δὲ αὐτοῦ ὁ Γ , καὶ ὁ Γ ἑαυτὸν μὲν πολλαπλασιάσας τὸν E πεποίηκεν, ὁ Γ ἄρα ἑαυτὸν μὲν πολλαπλασιάσας τὸν E πεποίηκεν, τὸν δὲ E πολλαπλασιάσας τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Δ ἑαυτὸν μὲν πολλαπλασιάσας τὸν H πεποίηκεν, τὸν δὲ H πολλαπλασιάσας τὸν B πεποίηκεν. καὶ ἐπεὶ ὁ Γ ἐκάτερον τῶν Γ, Δ πολλαπλασιάσας ἐκάτερον τῶν E, Z πεποίηκεν, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν Z . διὰ τὰ αὐτὰ δὴ καὶ ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ Z πρὸς τὸν H . πάλιν, ἐπεὶ ὁ Γ ἐκάτερον τῶν E, Z πολλαπλασιάσας ἐκάτερον τῶν Θ, K πεποίηκεν, ἔστιν ἄρα ὡς ὁ E πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν Θ . ὡς δὲ ὁ E πρὸς τὸν Z , οὕτως ὁ Γ πρὸς τὸν Δ . καὶ ὡς ἄρα ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν Θ . πάλιν, ἐπεὶ ἐκάτεροξ τῶν Γ, Δ τὸν Z πολλαπλασιάσας ἐκάτερον τῶν Θ, K πεποίηκεν, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ Θ πρὸς τὸν K . πάλιν, ἐπεὶ ὁ Δ ἐκάτερον τῶν Z, H πολλαπλασιάσας ἐκάτερον τῶν K, B πεποίηκεν, ἔστιν ἄρα ὡς ὁ Z πρὸς τὸν H , οὕτως ὁ K πρὸς τὸν B . ὡς δὲ ὁ Z πρὸς τὸν H , οὕτως ὁ Γ πρὸς τὸν Δ . καὶ ὡς ἄρα ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν Θ καὶ ὁ Θ πρὸς τὸν K καὶ ὁ K πρὸς τὸν B . τῶν A, B ἄρα δύο μέσοι ἀνάλογόν εἰσιν οἱ Θ, K .

Λέγω δὴ, ὅτι καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν Δ . ἐπεὶ γὰρ τέσσαρες ἀριθμοὶ ἀνάλογόν

For let C make E (by) multiplying itself, and let it make F (by) multiplying D . And let D make G (by) multiplying itself, and let C, D make H, K , respectively, (by) multiplying F .

And since A is cube, and C (is) its side, and C has made E (by) multiplying itself, C has thus made E (by) multiplying itself, and has made A (by) multiplying E . And so, for the same (reasons), D has made G (by) multiplying itself, and has made B (by) multiplying G . And since C has made E, F (by) multiplying C, D , respectively, thus as C is to D , so E (is) to F [Prop. 7.17]. And so, for the same (reasons), as C (is) to D , so F (is) to G [Prop. 7.18]. Again, since C has made A, H (by) multiplying E, F , respectively, thus as E is to F , so A (is) to H [Prop. 7.17]. And as E (is) to F , so C (is) to D . And thus as C (is) to D , so A (is) to H . Again, since C, D have made H, K , respectively, (by) multiplying F , thus as C is to D , so H (is) to K [Prop. 7.18]. Again, since D has made K, B (by) multiplying F, G , respectively, thus as F is to G , so K (is) to B [Prop. 7.17]. And as F (is) to G , so C (is) to D . And thus as C (is) to D , so A (is) to H , and H to K , and K to B . Thus, H and K are two (numbers) in mean proportion to A and B .

So I say that A also has to B a cubed ratio with respect to (that) C (has) to D . For since A, H, K, B are four (continuously) proportional numbers, A thus has to B a cubed ratio with respect to (that) A (has) to H [Def. 5.10]. And as A (is) to H , so C (is) to D . And [thus] A has to B a cubed ratio with respect to (that) C (has) to D . (Which is) the very thing it was required to show.

εἰσιν οἱ A, Θ, K, B , ὁ A ἄρα πρὸς τὸν B τριπλασίονα λόγον ἔθει ἥπερ ὁ A πρὸς τὸν Θ . ὥς δὲ ὁ A πρὸς τὸν Θ , οὕτως ὁ Γ πρὸς τὸν Δ · καὶ ὁ A [ἄρα] πρὸς τὸν B τριπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν Δ · ὅπερ ἔδει δεῖχαι.

† In other words, between two given cube numbers there exist two numbers in continued proportion.

‡ Literally, “triple”.

13.

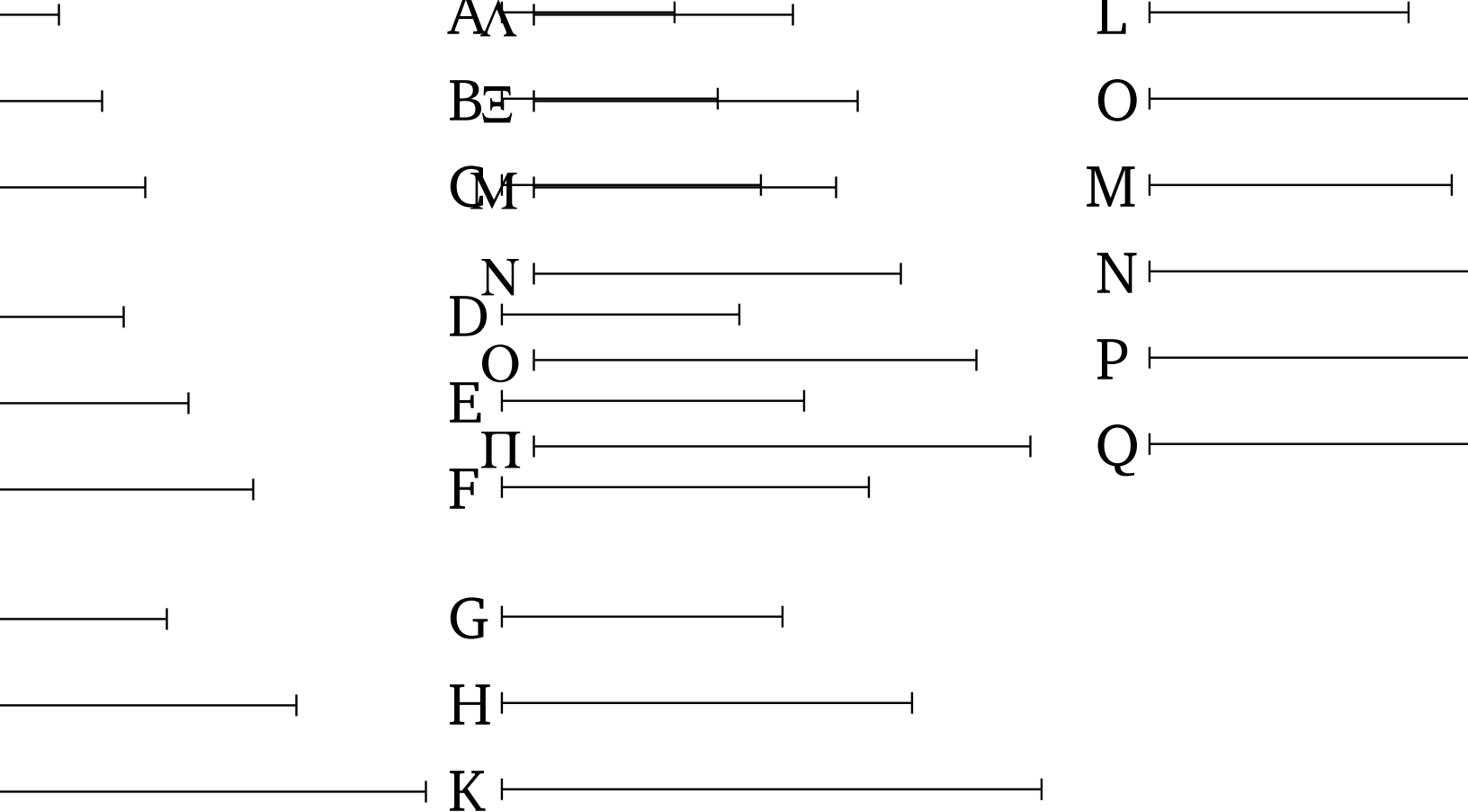
Proposition 13

Ἐάν ὥσιν ὅσοιδηποῦν ἀριθμοὶ ἐχῇς ἀνάλογον, καὶ πολλαπλασιάσας ἕκαστοξ ἐαντὸν ποιῇ τινα, οἱ γεγόμενοι ἐξ αὐτῶν ἀνάλογον ἔσονται· καὶ ἔαν οἱ ἐξ ἀρθῇς τούξ γεγομένονξ πολλαπλασιάσαντεξ ποιῶσί τιναξ, καὶ αὐτοὶ ἀνάλογον ἔσονται [καὶ αἰεὶ περὶ τούξ ἄκρονξ τοῦτο συμβαίνει].

Ἐστωσαν ὅποιοιῦν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ A, B, Γ , ὥς ὁ A πρὸς τὸν B , οὕτως ὁ B πρὸς τὸν Γ , καὶ οἱ A, B, Γ ἐαντούξ μὲν πολλαπλασιάσαντεξ τούξ Δ, E, Z ποιείτωσαν, τούξ δὲ Δ, E, Z πολλαπλασιάσαντεξ τούξ H, Θ, K ποιείτωσαν· λέγω, ὅτι οἱ τε Δ, E, Z καὶ οἱ H, Θ, K ἐχῇς ἀνάλογον εἰσιν.

If there are any multitude whatsoever of continuously proportional numbers, and each makes some (number by) multiplying itself, then the (numbers) created from them will (also) be (continuously) proportional. And if the original (numbers) make some (more numbers by) multiplying the created (numbers) then these will also be (continuously) proportional [and this always happens with the extremes].

Let A, B, C be any multitude whatsoever of continuously proportional numbers, (such that) as A (is) to B , so B (is) to C . And let A, B, C make D, E, F (by) multiplying themselves, and let them make G, H, K (by) multiplying D, E, F . I say that D, E, F and G, H, K are continuously proportional.



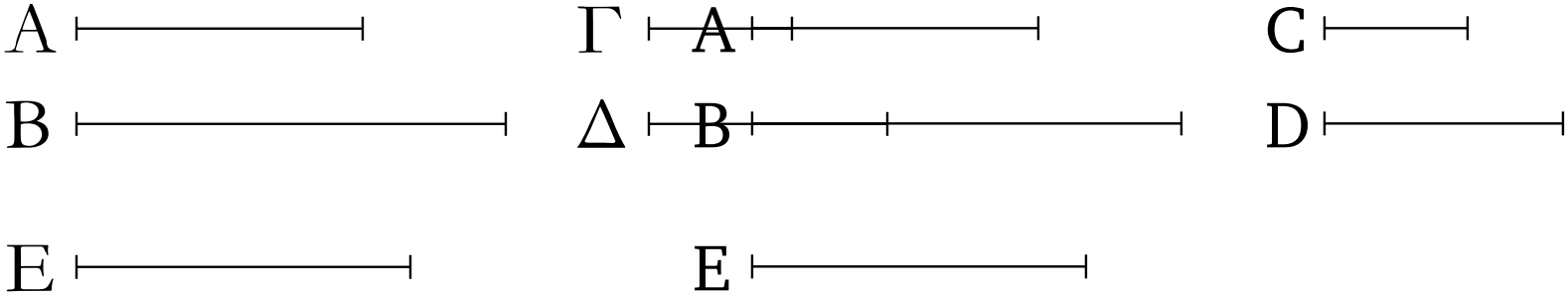
Ὅ μὲν γὰρ A τὸν B πολλαπλασιάσας τὸν Λ ποιεῖτω, ἑκάτερος δὲ τῶν A, B τὸν Λ πολλαπλασιάσας ἑκάτερον τῶν M, N ποιεῖτω. καὶ πάλιν ὁ μὲν B τὸν Γ πολλαπλασιάσας τὸν X ποιεῖτω, ἑκάτερος δὲ τῶν B, Γ τὸν X πολλαπλασιάσας ἑκάτερον τῶν O, Π ποιεῖτω.

Ὅμοιως δὴ τοῖς ἐπάνω δεῖχμεν, ὅτι οἱ Δ, Λ, E καὶ οἱ H, M, N, Θ ἐχῆς εἶναι ἀνάλογον ἐν τῷ τοῦ A πρὸς τὸν B λόγῳ, καὶ ἔτι οἱ E, X, Z καὶ οἱ Θ, O, Π, K ἐχῆς εἶναι ἀνάλογον ἐν τῷ τοῦ B πρὸς τὸν Γ λόγῳ. καὶ ἐστὶν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ B πρὸς τὸν Γ · καὶ οἱ Δ, Λ, E ἄρα τοῖς E, X, Z ἐν τῷ αὐτῷ λόγῳ εἰσὶ καὶ ἔτι οἱ H, M, N, Θ τοῖς Θ, O, Π, K . καὶ ἐστὶν ἴσον τὸ μὲν τῶν Δ, Λ, E πλῆθος τῷ τῶν E, X, Z πλήθει, τὸ δὲ τῶν H, M, N, Θ τῷ τῶν Θ, O, Π, K δι' ἴσον ἄρα ἐστὶν ὡς μὲν ὁ Δ πρὸς τὸν E , οὕτως ὁ E πρὸς τὸν Z , ὡς δὲ ὁ H πρὸς τὸν Θ , οὕτως ὁ Θ πρὸς τὸν K · ὅπερ ἔδει δεῖχαι.

14.

Ἐὰν τετράγωνος τετράγωνον μετρή, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσει· καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρή, καὶ ὁ τετράγωνος τὸν τετράγωνον μετρήσει.

Ἔστωσαν τετράγωνοι ἀριθμοὶ οἱ A, B , πλευραὶ δὲ αὐτῶν ἔστωσαν οἱ Γ, Δ , ὁ δὲ A τὸν B μετρεῖτω· λέγω, ὅτι καὶ ὁ Γ τὸν Δ μετρεῖ.



Ὅ Γ γὰρ τὸν Δ πολλαπλασιάσας τὸν E ποιεῖτω· οἱ A, E, B ἄρα ἐχῆς ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ οἱ A, E, B ἐχῆς ἀνάλογόν εἰσιν, καὶ μετρεῖ ὁ A τὸν B , μετρεῖ ἄρα καὶ ὁ A τὸν E . καὶ ἐστὶν ὡς ὁ A πρὸς τὸν E , οὕτως ὁ Γ πρὸς τὸν Δ · μετρεῖ ἄρα καὶ ὁ Γ τὸν Δ .

Πάλιν δὴ ὁ Γ τὸν Δ μετρεῖτω· λέγω, ὅτι καὶ ὁ A τὸν B μετρεῖ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δεῖχμεν, ὅτι οἱ A, E, B ἐχῆς ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ ἐστὶν ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν E , μετρεῖ δὲ ὁ Γ τὸν Δ , μετρεῖ ἄρα καὶ ὁ A τὸν E . καὶ εἰσιν οἱ A, E, B ἐχῆς ἀνάλογον μετρεῖ ἄρα καὶ ὁ A τὸν B .

Ἐὰν ἄρα τετράγωνος τετράγωνον μετρή, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσει· καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρή, καὶ ὁ τετράγωνος τὸν τετράγωνον μετρήσει· ὅπερ ἔδει δεῖχαι.

For let A make L (by) multiplying B . And let A, B make M, N , respectively, (by) multiplying L . And, again, let B make O (by) multiplying C . And let B, C make P, Q , respectively, (by) multiplying O .

So, similarly to the above, we can show that D, L, E and G, M, N, H are continuously proportional in the ratio of A to B , and, further, (that) E, O, F and H, P, Q, K are continuously proportional in the ratio of B to C . And as A is to B , so B (is) to C . And thus D, L, E are in the same ratio as E, O, F , and, further, G, M, N, H (are in the same ratio) as H, P, Q, K . And the multitude of D, L, E is equal to the multitude of E, O, F , and that of G, M, N, H to that of H, P, Q, K . Thus, via equality, as D is to E , so E (is) to F , and as G (is) to H , so H (is) to K [Prop. 7.14]. (Which is) the very thing it was required to show.

Proposition 14

If a square (number) measures a(nother) square (number) then the side (of the former) will also measure the side (of the latter). And if the side (of a square number) measures the side (of another square number) then the (former) square (number) will also measure the (latter) square (number).

Let A and B be square numbers, and let C and D be their sides (respectively). And let A measure B . I say that C also measures D .

For let C make E (by) multiplying D . Thus, A, E, B are continuously proportional in the ratio of C to D [Prop. 8.11]. And since A, E, B are continuously proportional, and A measures B , A thus also measures E [Prop. 8.7]. And as A is to E , so C (is) to D . Thus, C also measures D [Def. 7.20].

So, again, let C measure D . I say that A also measures B .

For similarly, with the same construction, we can show that A, E, B are continuously proportional in the ratio of C to D . And since as C is to D , so A (is) to E , and C measures D , A thus also measures E [Def. 7.20]. And A, E, B are continuously proportional. Thus, A also measures B .

Thus, if a square (number) measures a(nother) square (number) then the side (of the former) will also measure the side (of the latter). And if the side (of a square number) measures the side (of another square number) then the (former)

square (number) will also measure the (latter) square (number). (Which is) the very thing it was required to show.

15.

Ἐὰν κύβος ἀριθμὸς κύβον ἀριθμὸν μετρήῃ, καὶ ἡ πλευρὰ τὴν πλευρὰν μετρήσῃ· καὶ ἔαν ἡ πλευρὰ τὴν πλευρὰν μετρήῃ, καὶ ὁ κύβος τὸν κύβον μετρήσῃ.

Κύβος γὰρ ἀριθμὸς ὁ A κύβον τὸν B μετρεῖτω, καὶ τοῦ μὲν A πλευρὰ ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ · λέγω, ὅτι ὁ Γ τὸν Δ μετρεῖ.

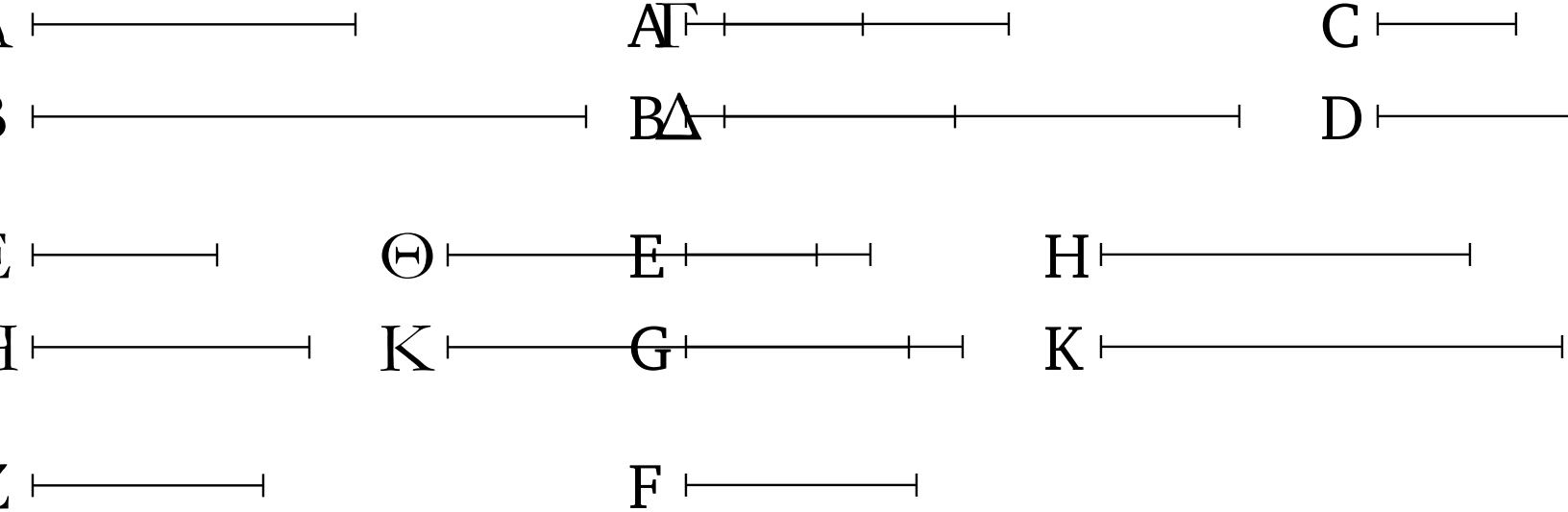
Ὁ Γ γὰρ ἑαυτὸν πολλαπλασιάσας τὸν E ποιεῖτω, ὁ δὲ Δ ἑαυτὸν πολλαπλασιάσας τὸν H ποιεῖτω, καὶ ἔτι ὁ Γ τὸν Δ πολλαπλασιάσας τὸν Z [ποιεῖτω], ἐκάτερος δὲ τῶν Γ , Δ τὸν Z πολλαπλασιάσας ἐκάτερον τῶν Θ , K ποιεῖτω. φανερόν δὴ, ὅτι οἱ E , Z , H καὶ οἱ A , Θ , K , B ἐχῆς ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ οἱ A , Θ , K , B ἐχῆς ἀνάλογόν εἰσιν, καὶ μετρεῖ ὁ A τὸν B , μετρεῖ ἄρα καὶ τὸν Θ . καὶ ἔστιν ὥς ὁ A πρὸς τὸν Θ , οὕτως ὁ Γ πρὸς τὸν Δ · μετρεῖ ἄρα καὶ ὁ Γ τὸν Δ .

Proposition 15

If a cube number measures a(nother) cube number then the side (of the former) will also measure the side (of the latter). And if the side (of a cube number) measures the side (of another cube number) then the (former) cube (number) will also measure the (latter) cube (number).

For let the cube number A measure the cube (number) B , and let C be the side of A , and D (the side) of B . I say that C measures D .

For let C make E (by) multiplying itself. And let D make G (by) multiplying itself. And, further, [let] C [make] F (by) multiplying D , and let C , D make H , K , respectively, (by) multiplying F . So it is clear that E , F , G and A , H , K , B are continuously proportional in the ratio of C to D [Prop. 8.12]. And since A , H , K , B are continuously proportional, and A measures B , (A) thus also measures H [Prop. 8.7]. And as A is to H , so C (is) to D . Thus, C also measures D [Def. 7.20].



Ἀλλὰ δὴ μετρεῖτω ὁ Γ τὸν Δ · λέγω, ὅτι καὶ ὁ A τὸν B μετρήσῃ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δὴ δείκμεν, ὅτι οἱ A , Θ , K , B ἐχῆς ἀνάλογόν εἰσιν ἐν τῷ τοῦ Γ πρὸς τὸν Δ λόγῳ. καὶ ἐπεὶ ὁ Γ τὸν Δ μετρεῖ, καὶ ἔστιν ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ A πρὸς τὸν Θ , καὶ ὁ A ἄρα τὸν Θ μετρεῖ· ὥστε καὶ τὸν B μετρεῖ ὁ A · ὅπερ ἔδει δεῖξαι.

And so let C measure D . I say that A will also measure B . For similarly, with the same construction, we can show that A , H , K , B are continuously proportional in the ratio of C to D . And since C measures D , and as C is to D , so A (is) to H , A thus also measures H [Def. 7.20]. Hence, A also measures B . (Which is) the very thing it was required to show.

16.

Ἐὰν τετράγωνος ἀριθμὸς τετράγωνον ἀριθμὸν μὴ μετρήῃ, οὐδὲ ἡ πλευρὰ τὴν πλευρὰν μετρήσῃ· κ' ἂν ἡ πλευρὰ τὴν πλευ-

Proposition 16

If a square number does not measure a(nother) square number then the side (of the former) will not measure the side

ἐὰν μὴ μετρήῃ, οὐδὲ ὁ τετράγωνος τὸν τετράγωνον μετρήσει.

(of the latter) either. And if the side (of a square number) does not measure the side (of another square number) then the (former) square (number) will not measure the (latter) square

A —————

Γ —————

B —————

Δ ————— (number) either.

A —————

C —————

B —————

D —————

Ἐστωσαν τετράγωνοι ἀριθμοὶ οἱ A, B , πλευραὶ δὲ αὐτῶν ἔστωσαν οἱ Γ, Δ , καὶ μὴ μετρεῖται ὁ A τὸν B · λέγω, ὅτι οὐδὲ ὁ Γ τὸν Δ μετρεῖ.

Let A and B be square numbers, and let C and D be their sides (respectively). And let A not measure B . I say that C does not measure D either.

Εἰ γὰρ μετρεῖ ὁ Γ τὸν Δ , μετρήσει καὶ ὁ A τὸν B . οὐ μετρεῖ δὲ ὁ A τὸν B · οὐδὲ ἄρα ὁ Γ τὸν Δ μετρήσει.

For if C measures D then A will also measure B [Prop. 8.14].

μὴ μετρεῖται [δὴ] πάλιν ὁ Γ τὸν Δ · λέγω, ὅτι οὐδὲ ὁ A τὸν B μετρήσει.

And A does not measure B . Thus, C will not measure D either.

Εἰ γὰρ μετρεῖ ὁ A τὸν B , μετρήσει καὶ ὁ Γ τὸν Δ . οὐ μετρεῖ δὲ ὁ Γ τὸν Δ · οὐδὲ ἄρα ὁ A τὸν B μετρήσει· ὅπερ ἔδει δεῖξαι.

[So], again, let C not measure D . I say that A will not measure B either.

For if A measures B then C will also measure D [Prop. 8.14]. And C does not measure D . Thus, A will not measure B either. (Which is) the very thing it was required to show.

17.

Proposition 17

Ἐὰν κύβος ἀριθμὸς κύβον ἀριθμὸν μὴ μετρήῃ, οὐδὲ ἡ πλευρὰ τὴν πλευρὰν μετρήσει· καὶ ἡ πλευρὰ τὴν πλευρὰν μὴ μετρήῃ, οὐδὲ ὁ κύβος τὸν κύβον μετρήσει.

If a cube number does not measure a(nother) cube number then the side (of the former) will not measure the side (of the latter) either. And if the side (of a cube number) does not measure the side (of another cube number) then the (former) cube (number) will not measure the (latter) cube (number) either.

A —————

Γ ————— A —————

C —————

B —————

Δ ————— B —————

D —————

Κύβος γὰρ ἀριθμὸς ὁ A κύβον ἀριθμὸν τὸν B μὴ μετρεῖται, καὶ τοῦ μὲν A πλευρὰ ἔστω ὁ Γ , τοῦ δὲ B ὁ Δ · λέγω, ὅτι ὁ Γ τὸν Δ οὐ μετρήσει.

For let the cube number A not measure the cube number B . And let C be the side of A , and D (the side) of B . I say that C will not measure D .

Εἰ γὰρ μετρεῖ ὁ Γ τὸν Δ , καὶ ὁ A τὸν B μετρήσει. οὐ μετρεῖ δὲ ὁ A τὸν B · οὐδὲ ἄρα ὁ Γ τὸν Δ μετρεῖ.

For if C measures D then A will also measure B [Prop. 8.15].

Ἀλλὰ δὴ μὴ μετρεῖται ὁ Γ τὸν Δ · λέγω, ὅτι οὐδὲ ὁ A τὸν B μετρήσει.

And A does not measure B . Thus, C does not measure D either.

Εἰ γὰρ ὁ A τὸν B μετρεῖ, καὶ ὁ Γ τὸν Δ μετρήσει. οὐ μετρεῖ δὲ ὁ Γ τὸν Δ · οὐδὲ ἄρα ὁ A τὸν B μετρήσει· ὅπερ ἔδει δεῖξαι.

And so let C not measure D . I say that A will not measure B either.

For if A measures B then C will also measure D [Prop. 8.15]. And C does not measure D . Thus, A will not measure B either. (Which is) the very thing it was required to show.

18.

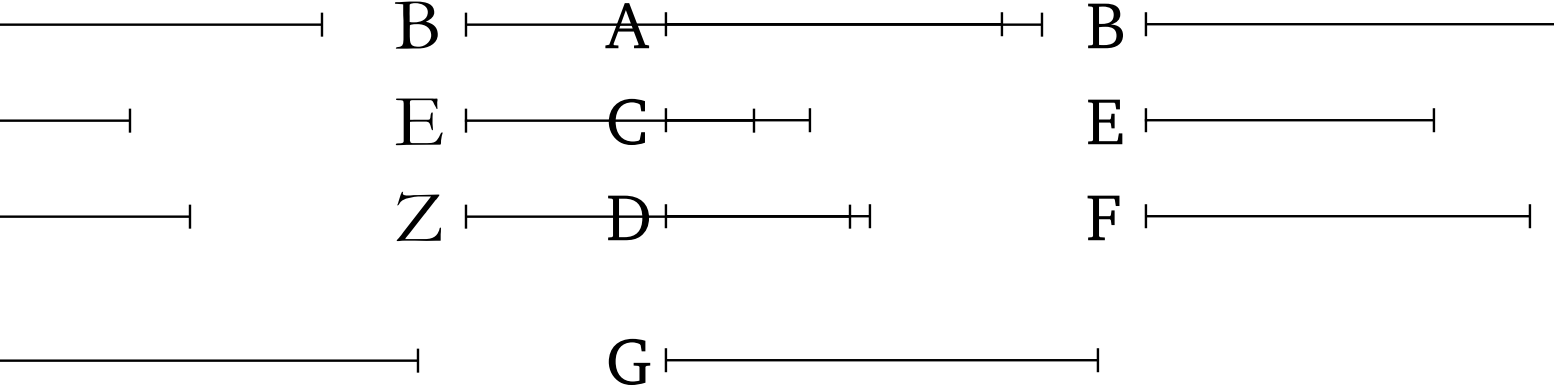
Proposition 18

Δύο ὁμοίων ἐπιπέδων ἀριθμῶν εἷξ μέσοξ ἀνάλογόν ἐστιν ἀριθμὸς· καὶ ὁ ἐπίπεδος πρὸξ τὸν ἐπίπεδον διπλασίονα λόγον

There exists one number in mean proportion to two similar plane numbers. And (one) plane (number) has to the (other)

ἔθει ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν.

plane (number) a squared[†] ratio with respect to (that) a corresponding side (of the former has) to a corresponding side (of the latter).



Ἐστωσαν δύο ὅμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B , καὶ τοῦ μὲν A πλευραὶ ἔστωσαν οἱ Γ, Δ ἀριθμοί, τοῦ δὲ B οἱ E, Z . καὶ ἐπεὶ ὅμοιοι ἐπίπεδοι εἰσιν οἱ ἀνάλογον ἔθοντες τὰς πλευράς, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν Z . λέγω οὖν, ὅτι τῶν A, B εἷς μέσος ἀνάλογόν ἐστιν ἀριθμός, καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν E ἢ ὁ Δ πρὸς τὸν Z , τουτέστιν ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον [πλευράν].

Καὶ ἐπεὶ ἐστὶν ὡς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν Z , ἐναλλάξ ἄρα ἐστὶν ὡς ὁ Γ πρὸς τὸν E , ὁ Δ πρὸς τὸν Z . καὶ ἐπεὶ ἐπίπεδός ἐστιν ὁ A , πλευραὶ δὲ αὐτοῦ οἱ Γ, Δ , ὁ Δ ἄρα τὸν Γ πολλαπλασιάσας τὸν A πεποίηκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ E τὸν Z πολλαπλασιάσας τὸν B πεποίηκεν. ὁ Δ δὴ τὸν E πολλαπλασιάσας τὸν H ποιεῖτω. καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ E πολλαπλασιάσας τὸν H πεποίηκεν, ἔστιν ἄρα ὡς ὁ Γ πρὸς τὸν E , οὕτως ὁ A πρὸς τὸν H . ἀλλ' ὡς ὁ Γ πρὸς τὸν E , [οὕτως] ὁ Δ πρὸς τὸν Z . καὶ ὡς ἄρα ὁ Δ πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν H . πάλιν, ἐπεὶ ὁ E τὸν μὲν Δ πολλαπλασιάσας τὸν H πεποίηκεν, τὸν δὲ Z πολλαπλασιάσας τὸν B πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ H πρὸς τὸν B . ἐδείκθη δὲ καὶ ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν H . καὶ ὡς ἄρα ὁ A πρὸς τὸν H , οὕτως ὁ H πρὸς τὸν B . οἱ A, H, B ἄρα ἐχῆς ἀνάλογόν εἰσιν. τῶν A, B ἄρα εἷς μέσος ἀνάλογόν ἐστιν ἀριθμός.

Λέγω δὴ, ὅτι καὶ ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἥπερ ὁ Γ πρὸς τὸν E ἢ ὁ Δ πρὸς τὸν Z . ἐπεὶ γάρ οἱ A, H, B ἐχῆς ἀνάλογόν εἰσιν, ὁ A πρὸς τὸν B διπλασίονα λόγον ἔθει ἥπερ πρὸς τὸν H . καὶ ἐστὶν ὡς ὁ A πρὸς τὸν H , οὕτως ὁ Γ πρὸς τὸν E καὶ ὁ Δ πρὸς τὸν Z . καὶ ὁ A ἄρα πρὸς τὸν B διπλασίονα λόγον ἔθει ἥπερ ὁ Γ πρὸς τὸν E ἢ ὁ Δ πρὸς τὸν Z . ὅπερ ἔδει δεῖξαι.

Let A and B be two similar plane numbers. And let the numbers C, D be the sides of A , and E, F (the sides) of B . And since similar numbers are those having proportional sides [Def. 7.21], thus as C is to D , so E (is) to F . Therefore, I say that there exists one number in mean proportion to A and B , and that A has to B a squared ratio with respect to that C (has) to E , or D to F —that is to say, with respect to (that) a corresponding side (has) to a corresponding [side].

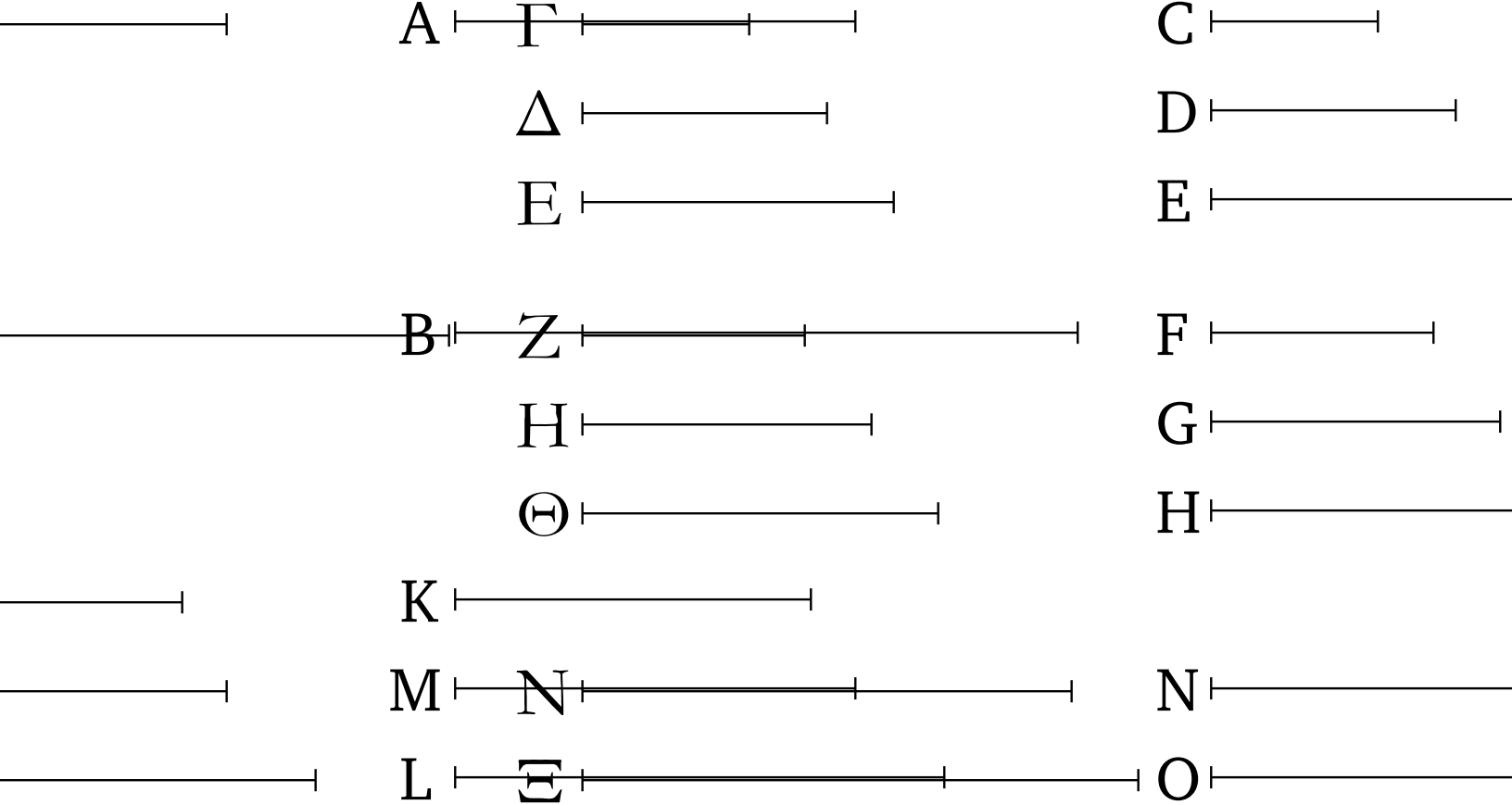
For since as C is to D , so E (is) to F , thus, alternately, as C is to E , so D (is) to F [Prop. 7.13]. And since A is plane, and C, D its sides, D has thus made A (by) multiplying C . And so, for the same (reasons), E has made B (by) multiplying F . So let D make G (by) multiplying E . And since D has made A (by) multiplying C , and has made G (by) multiplying E , thus as C is to E , so A (is) to G [Prop. 7.17]. But as C (is) to E , [so] D (is) to F . And thus as D (is) to F , so A (is) to G . Again, since E has made G (by) multiplying D , and has made B (by) multiplying F , thus as D is to F , so G (is) to B [Prop. 7.17]. And it was also shown that as D (is) to F , so A (is) to G . And thus as A (is) to G , so G (is) to B . Thus, A, G, B are continuously proportional. Thus, there exists one number (namely, G) in mean proportion to A and B .

So I say that A also has to B a squared ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) C (has) to E , or D to F . For since A, G, B are continuously proportional, A has to B a squared ratio with respect to (that A has) to G [Prop. 5.9]. And as A is to G , so C (is) to E , and D to F . And thus A has to B a squared ratio with respect to (that) C (has) to E , or D to F . (Which is) the very thing it was required to show.

[†] Literally, “double”.

Δύο ὁμοίων στερεῶν ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί· καὶ ὁ στερεὸς πρὸς τὸν ὅμοιον στερεὸν τριπλασίονα λόγον ἔθει ἢπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν.

Two numbers fall (between) two similar solid numbers in mean proportion. And a solid (number) has to a similar solid (number) a cubed[†] ratio with respect to (that) a corresponding side (has) to a corresponding side.



Ἐστωσαν δύο ὅμοιοι στερεοὶ οἱ A, B , καὶ τοῦ μὲν A πλευραὶ ἕστωσαν οἱ Γ, Δ, E , τοῦ δὲ B οἱ Z, H, Θ . καὶ ἐπεὶ ὅμοιοι στερεοὶ εἰσιν οἱ ἀνάλογον ἔχοντες τὰς πλευράς, ἔστιν ἄρα ὥς μὲν ὁ Γ πρὸς τὸν Δ , οὕτως ὁ Z πρὸς τὸν H , ὥς δὲ ὁ Δ πρὸς τὸν E , οὕτως ὁ H πρὸς τὸν Θ . λέγω, ὅτι τῶν A, B δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί, καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον ἔθει ἢπερ ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ .

Ὁ Γ γὰρ τὸν Δ πολλαπλασιάσας τὸν K ποιεῖτω, ὁ δὲ Z τὸν H πολλαπλασιάσας τὸν Λ ποιεῖτω. καὶ ἐπεὶ οἱ Γ, Δ τοῖς Z, H ἐν τῷ αὐτῷ λόγῳ εἰσίν, καὶ ἐκ μὲν τῶν Γ, Δ ἔστιν ὁ K , ἐκ δὲ τῶν Z, H ὁ Λ , οἱ K, Λ [ἄρα] ὅμοιοι ἐπίπεδοι εἰσιν ἀριθμοί· τῶν K, Λ ἄρα εἴς μέσος ἀνάλογον ἔστιν ἀριθμὸς. ἔστω ὁ M . ὁ M ἄρα ἔστιν ὁ ἐκ τῶν Δ, Z , ὥς ἐν τῷ πρὸ τούτου θεωρήματι ἐδείχθη. καὶ ἐπεὶ ὁ Δ τὸν μὲν Γ πολλαπλασιάσας τὸν K πεποίηκεν, τὸν δὲ Z πολλαπλασιάσας τὸν M πεποίηκεν, ἔστιν ἄρα ὥς ὁ Γ πρὸς τὸν Z , οὕτως ὁ K πρὸς τὸν M . ἀλλ' ὥς ὁ K πρὸς τὸν M , ὁ M πρὸς τὸν Λ . οἱ K, M, Λ ἄρα ἐχθῆς εἰσιν ἀνάλογον ἐν τῷ τοῦ Γ πρὸς τὸν Z λόγῳ. καὶ ἐπεὶ ἔστιν

Let A and B be two similar solid numbers, and let C, D, E be the sides of A , and F, G, H (the sides) of B . And since similar solid (numbers) are those having proportional sides [Def. 7.21], thus as C is to D , so F (is) to G , and as D (is) to E , so G (is) to H . I say that two numbers fall (between) A and B in mean proportion, and (that) A has to B a cubed ratio with respect to (that) C (has) to F , and D to G , and, further, E to H .

For let C make K (by) multiplying D , and let F make L (by) multiplying G . And since C, D are in the same ratio as F, G , and K is the (number created) from (multiplying) C, D , and L the (number created) from (multiplying) F, G , [thus] K and L are similar plane numbers [Def. 7.21]. Thus, there exists one number in mean proportion to K and L [Prop. 8.18]. Let it be M . Thus, M is the (number created) from (multiplying) D, F , as shown in the theorem before this (one). And since D has made K (by) multiplying C , and has made M (by) multiplying F , thus as C is to F , so K (is) to M [Prop. 7.17]. But, as K (is) to M , (so) M (is) to L . Thus, K, M, L are continuously

ὥς ὁ Γ πρὸς τὸν Δ , οὕτως ὁ Z πρὸς τὸν H , ἐναλλάχ' ἄρα ἐστὶν ὥς ὁ Γ πρὸς τὸν Z , οὕτως ὁ Δ πρὸς τὸν H . διὰ τὰ αὐτὰ δὴ καὶ ὥς ὁ Δ πρὸς τὸν H , οὕτως ὁ E πρὸς τὸν Θ . οἱ K , M , Λ ἄρα ἐχῆς εἶσιν ἀνάλογον ἐν τε τῷ τοῦ Γ πρὸς τὸν Z λόγῳ καὶ τῷ τοῦ Δ πρὸς τὸν H καὶ ἔτι τῷ τοῦ E πρὸς τὸν Θ . ἑκάτερος δὴ τῶν E , Θ τὸν M πολλαπλασιάσας ἑκάτερον τῶν N , X ποιείτω. καὶ ἐπεὶ στερεὸς ἐστὶν ὁ A , πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Γ , Δ , E , ὁ E ἄρα τὸν ἐκ τῶν Γ , Δ πολλαπλασιάσας τὸν A πεποιήκεν. ὁ δὲ ἐκ τῶν Γ , Δ ἐστὶν ὁ K . ὁ E ἄρα τὸν K πολλαπλασιάσας τὸν A πεποιήκεν. διὰ τὰ αὐτὰ δὴ καὶ ὁ Θ τὸν Λ πολλαπλασιάσας τὸν B πεποιήκεν. καὶ ἐπεὶ ὁ E τὸν K πολλαπλασιάσας τὸν A πεποιήκεν, ἀλλὰ μὴν καὶ τὸν M πολλαπλασιάσας τὸν N πεποιήκεν, ἐστὶν ἄρα ὥς ὁ K πρὸς τὸν M , οὕτως ὁ A πρὸς τὸν N . ὥς δὲ ὁ K πρὸς τὸν M , οὕτως ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ . καὶ ὥς ἄρα ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ὁ E πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν N . πάλιν, ἐπεὶ ἑκάτερος τῶν E , Θ τὸν M πολλαπλασιάσας ἑκάτερον τῶν N , X πεποιήκεν, ἐστὶν ἄρα ὥς ὁ E πρὸς τὸν Θ , οὕτως ὁ N πρὸς τὸν X . ἀλλ' ὥς ὁ E πρὸς τὸν Θ , οὕτως ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H . καὶ ὥς ἄρα ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ὁ E πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν N καὶ ὁ N πρὸς τὸν X . πάλιν, ἐπεὶ ὁ Θ τὸν M πολλαπλασιάσας τὸν B πεποιήκεν, ἀλλὰ μὴν καὶ τὸν Λ πολλαπλασιάσας τὸν B πεποιήκεν, ἐστὶν ἄρα ὥς ὁ M πρὸς τὸν Λ , οὕτως ὁ X πρὸς τὸν B . ἀλλ' ὥς ὁ M πρὸς τὸν Λ , οὕτως ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ὁ E πρὸς τὸν Θ . καὶ ὥς ἄρα ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ὁ E πρὸς τὸν Θ , οὕτως οὐ μόνον ὁ X πρὸς τὸν B , ἀλλὰ καὶ ὁ A πρὸς τὸν N καὶ ὁ N πρὸς τὸν X . οἱ A , N , X , B ἄρα ἐχῆς εἰσιν ἀνάλογον ἐν τοῖς εἰρημένοις τῶν πλευρῶν λόγοις.

Λέγω, ὅτι καὶ ὁ A πρὸς τὸν B τριπλασίονα λόγον ἔθει ἢ περ ἢ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἢ περ ὁ Γ ἀριθμὸς πρὸς τὸν Z ἢ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ . ἐπεὶ γὰρ τέσσαρες ἀριθμοὶ ἐχῆς ἀνάλογον εἰσιν οἱ A , N , X , B , ὁ A ἄρα πρὸς τὸν B τριπλασίονα λόγον ἔθει ἢ περ ὁ A πρὸς τὸν N . ἀλλ' ὥς ὁ A πρὸς τὸν N , οὕτως ἐδείχθη ὁ Γ πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ . καὶ ὁ A ἄρα πρὸς τὸν B τριπλασίονα λόγον ἔθει ἢ περ ἢ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἢ περ ὁ Γ ἀριθμὸς πρὸς τὸν Z καὶ ὁ Δ πρὸς τὸν H καὶ ἔτι ὁ E πρὸς τὸν Θ . ὁ περ ἔδει δεῖχαι.

† Literally, "triple".

20.

Ἐάν δύο ἀριθμῶν εἷς μέσος ἀνάλογον ἐμπίπτῃ ἀριθμὸς, ὅμοιοι ἐπίπεδοι ἔσονται οἱ ἀριθμοί.

Δύο γὰρ ἀριθμῶν τῶν A , B εἷς μέσος ἀνάλογον ἐμπέτω ἀριθμὸς ὁ Γ . λέγω, ὅτι οἱ A , B ὅμοιοι ἐπίπεδοι εἰσιν

proportional in the ratio of C to F . And since as C is to D , so F (is) to G , thus, alternately, as C is to F , so D (is) to G [Prop. 7.13]. And so, for the same (reasons), as D (is) to G , so E (is) to H . Thus, K , M , L are continuously proportional in the ratio of C to F , and of D to G , and, further, of E to H . So let E , H make N , O , respectively, (by) multiplying M . And since A is solid, and C , D , E are its sides, E has thus made A (by) multiplying the (number created) from (multiplying) C , D . And K is the (number created) from (multiplying) C , D . Thus, E has made A (by) multiplying K . And so, for the same (reasons), H has made B (by) multiplying L . And since E has made A (by) multiplying K , but has, in fact, also made N (by) multiplying M , thus as K is to M , so A (is) to N [Prop. 7.17]. And as K (is) to M , so C (is) to F , and D to G , and, further, E to H . And thus as C (is) to F , and D to G , and E to H , so A (is) to N . Again, since E , H have made N , O , respectively, (by) multiplying M , thus as E is to H , so N (is) to O [Prop. 7.18]. But, as E (is) to H , so C (is) to F , and D to G . And thus as C (is) to F , and D to G , and E to H , so (is) A to N , and N to O . Again, since H has made O (by) multiplying M , but has, in fact, also made B (by) multiplying L , thus as M (is) to L , so O (is) to B [Prop. 7.17]. But, as M (is) to L , so C (is) to F , and D to G , and E to H . And thus as C (is) to F , and D to G , and E to H , so not only (is) O to B , but also A to N , and N to O . Thus, A , N , O , B are continuously proportional in the aforementioned ratios of the sides.

So I say that A also has to B a cubed ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) the number C (has) to F , or D to G , and, further, E to H . For since A , N , O , B are four continuously proportional numbers, A thus has to B a cubed ratio with respect to (that) A (has) to N [Def. 5.10]. But, as A (is) to N , so it was shown (is) C to F , and D to G , and, further, E to H . And thus A has to B a cubed ratio with respect to (that) a corresponding side (has) to a corresponding side—that is to say, with respect to (that) the number C (has) to F , and D to G , and, further, E to H . (Which is) the very thing it was required to show.

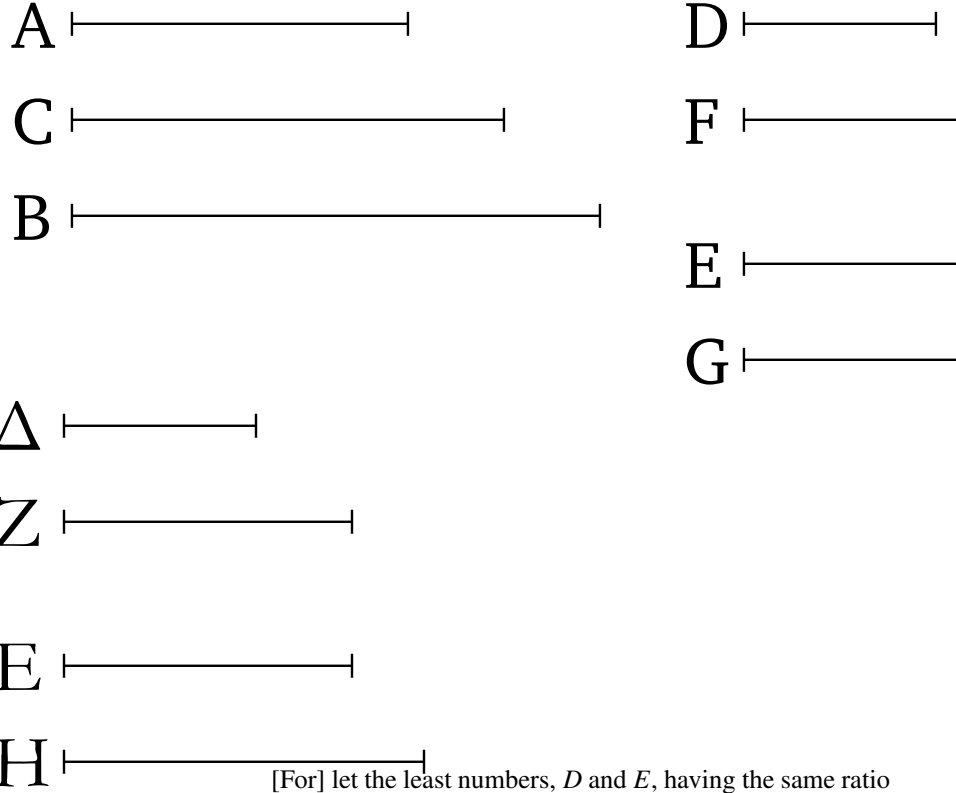
Proposition 20

If one number falls between two numbers in mean proportion then the numbers will be similar plane (numbers).

For let one number C fall between the two numbers A and B in mean proportion. I say that A and B are similar plane

ἀριθμοί.

numbers.



Εἰλήφθωσαν [γάρ] ἐλάχιστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ οἱ Δ, E · ἰσάκις ἄρα ὁ Δ τὸν A μετρεῖ καὶ ὁ E τὸν Γ . ὁσάκις δὴ ὁ Δ τὸν A μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Z · ὁ Z ἄρα τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν. ὥστε ὁ A ἐπίπεδός ἐστιν, πλευραὶ δὲ αὐτοῦ οἱ Δ, Z . πάλιν, ἐπεὶ οἱ Δ, E ἐλάχιστοί εἰσι τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Γ, B , ἰσάκις ἄρα ὁ Δ τὸν Γ μετρεῖ καὶ ὁ E τὸν B . ὁσάκις δὴ ὁ E τὸν B μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ H . ὁ E ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν τῷ H μονάδας· ὁ H ἄρα τὸν E πολλαπλασιάσας τὸν B πεποίηκεν. ὁ B ἄρα ἐπίπεδος ἐστι, πλευραὶ δὲ αὐτοῦ εἰσι οἱ E, H . οἱ A, B ἄρα ἐπίπεδοι εἰσι ἀριθμοί. λέγω δὴ, ὅτι καὶ ὅμοιοι. ἐπεὶ γάρ ὁ Z τὸν μὲν Δ πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ E πολλαπλασιάσας τὸν Γ πεποίηκεν, ἔστιν ἄρα ὡς ὁ Δ πρὸς τὸν E , οὕτως ὁ A πρὸς τὸν Γ , τοντέστιν ὁ Γ πρὸς τὸν B . πάλιν, ἐπεὶ ὁ E ἐκάτερον τῶν Z, H πολλαπλασιάσας τοῖς Γ, B πεποίηκεν, ἔστιν ἄρα ὡς ὁ Z πρὸς τὸν H , οὕτως ὁ Γ πρὸς τὸν B . ὡς δὲ ὁ Γ πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E · καὶ ὡς ἄρα ὁ Δ πρὸς τὸν E , οὕτως ὁ Z πρὸς τὸν H · καὶ ἐναλλάξ ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ E πρὸς τὸν H . οἱ A, B ἄρα ὅμοιοι ἐπίπεδοι ἀριθμοὶ εἰσιν· αἱ γὰρ πλευραὶ αὐτῶν ἀνάλογόν εἰσιν· ὅπερ ἔδει δεῖχαι.

[For] let the least numbers, D and E , having the same ratio as A and C be taken [Prop. 7.33]. Thus, D measures A as many times as E (measures) C [Prop. 7.20]. So as many times as D measures A , so many units let there be in F . Thus, F has made A (by) multiplying D [Def. 7.15]. Hence, A is plane, and D, F (are) its sides. Again, since D and E are the least of those (numbers) having the same ratio as C and B , D thus measures C as many times as E (measures) B [Prop. 7.20]. So as many times as E measures B , so many units let there be in G . Thus, E measures B according to the units in G . Thus, G has made B (by) multiplying E [Def. 7.15]. Thus, B is plane, and E, G are its sides. Thus, A and B are (both) plane numbers. So I say that (they are) also similar. For since F has made A (by) multiplying D , and has made C (by) multiplying E , thus as D is to E , so A (is) to C —that is to say, C to B [Prop. 7.17].[†] Again, since E has made C, B (by) multiplying F, G , respectively, thus as F is to G , so C (is) to B [Prop. 7.17]. And as C (is) to B , so D (is) to E . And thus as D (is) to E , so F (is) to G . And, alternately, as D (is) to F , so E (is) to G [Prop. 7.13]. Thus, A and B are similar plane numbers. For their sides are proportional [Def. 7.21]. (Which is) the very thing it was required to show.

[†] This part of the proof is defective, since it is not demonstrated that $F \times E = C$. Furthermore, it is not necessary to show that $D : E :: A : C$, because this is

true by hypothesis.

21.

Ἐάν δύο ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅμοιοι στερεοί εἰσιν οἱ ἀριθμοί.

Δύο γάρ ἀριθμῶν τῶν A, B δύο μέσοι ἀνάλογον ἐμπιπτέωσαν ἀριθμοί οἱ Γ, Δ . λέγω, ὅτι οἱ A, B ὅμοιοι στερεοί εἰσιν.

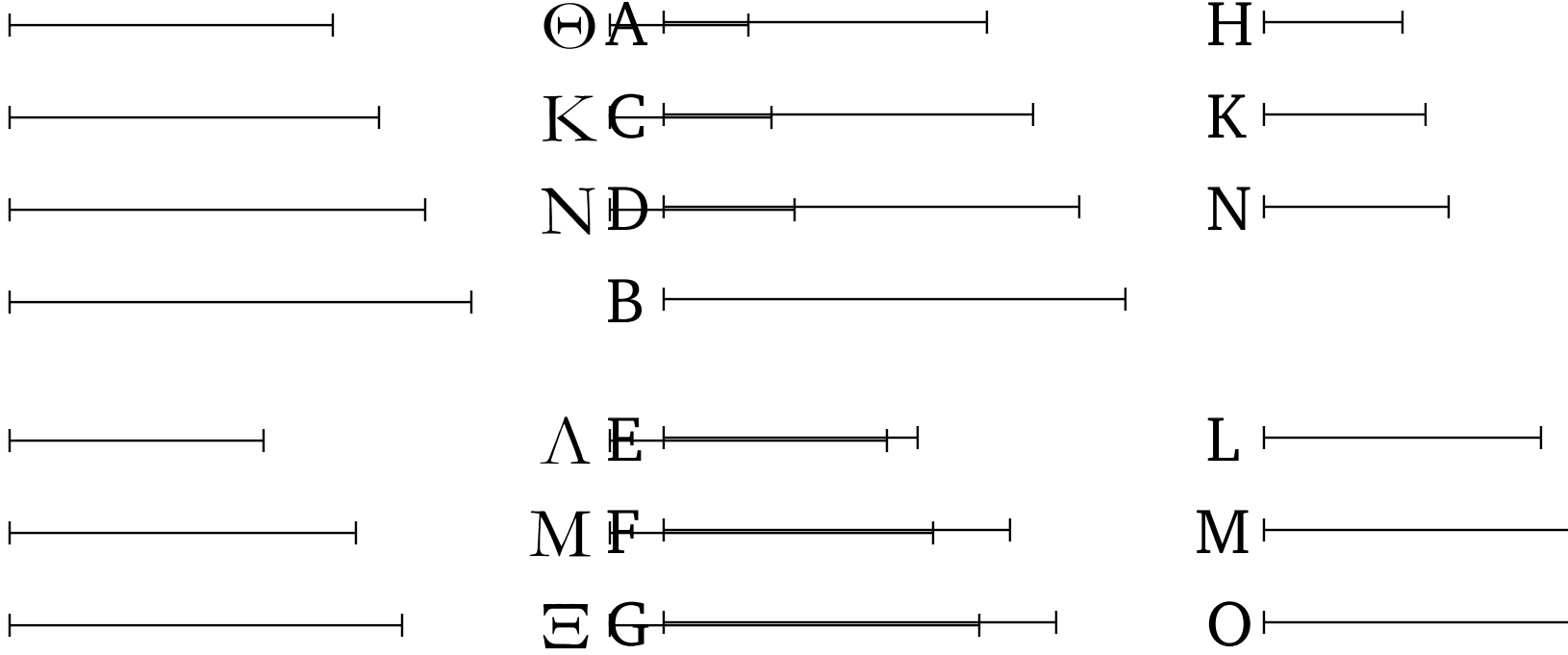
Εἰλήφθωσαν γάρ ἐλάθιστοι ἀριθμοί τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ, Δ τρεῖς οἱ E, Z, H . οἱ ἄρα ἄκροι αὐτῶν οἱ E, H πρῶτοι πρὸς ἀλλήλους εἰσιν. καὶ ἐπεὶ τῶν E, H εἴς μέσοξ ἀνάλογον ἐμπέπτωκεν ἀριθμὸς ὁ Z , οἱ E, H ἄρα ἀριθμοί ὅμοιοι ἐπίπεδοι εἰσιν. ἔστωσαν οὖν τοῦ μὲν E πλευραὶ οἱ Θ, K , τοῦ δὲ H οἱ Λ, M . φανερόν ἄρα ἐστὶν ἐκ τοῦ πρὸ τούτου, ὅτι οἱ E, Z, H ἐχῆξ εἰσιν ἀνάλογον ἐν τε τῷ τοῦ Θ πρὸς τὸν Λ λόγῳ καὶ τῷ τοῦ K πρὸς τὸν M . καὶ ἐπεὶ οἱ E, Z, H ἐλάθιστοι εἰσὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ, Δ , καὶ ἐστὶν ἴσον τὸ πλῆθος τῶν E, Z, H τῷ πλῆθει τῶν A, Γ, Δ , δι' ἴσον ἄρα ἐστὶν ὥς ὁ E πρὸς τὸν H , οὕτως ὁ A πρὸς τὸν Δ . οἱ δὲ E, H πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάθιστοι, οἱ δὲ ἐλάθιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχονταξ αὐτοῖς ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὅ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον· ἰσάκις ἄρα ὁ E τὸν A μετρεῖ καὶ ὁ H τὸν Δ . ὁσάκις δὴ ὁ E τὸν A μετρεῖ, τοσαῦται μονάδεξ ἔστωσαν ἐν τῷ N . ὁ N ἄρα τὸν E πολλαπλασιάσαξ τὸν A πεποίηκεν. ὁ δὲ E ἐστὶν ὁ ἐκ τῶν Θ, K . ὁ N ἄρα τὸν ἐκ τῶν Θ, K πολλαπλασιάσαξ τὸν A πεποίηκεν. στερεὸξ ἄρα ἐστὶν ὁ A , πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Θ, K, N . πάλιν, ἐπεὶ οἱ E, Z, H ἐλάθιστοί εἰσὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς Γ, Δ, B , ἰσάκις ἄρα ὁ E τὸν Γ μετρεῖ καὶ ὁ H τὸν B . ὁσάκις δὴ ὁ E τὸν Γ μετρεῖ, τοσαῦται μονάδεξ ἔστωσαν ἐν τῷ X . ὁ X ἄρα τὸν Γ πολλαπλασιάσαξ τὸν B πεποίηκεν. ὁ δὲ H ἐστὶν ὁ ἐκ τῶν Λ, M . ὁ X ἄρα τὸν ἐκ τῶν Λ, M πολλαπλασιάσαξ τὸν B πεποίηκεν. στερεὸξ ἄρα ἐστὶν ὁ B , πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Λ, M, X . οἱ A, B ἄρα στερεοί εἰσιν.

Proposition 21

If two numbers fall between two numbers in mean proportion then the (latter) are similar solid (numbers).

For let the two numbers C and D fall between the two numbers A and B in mean proportion. I say that A and B are similar solid (numbers).

For let the three least numbers E, F, G having the same ratio as A, C, D be taken [Prop. 8.2]. Thus, the outermost of them, E and G , are prime to one another [Prop. 8.3]. And since one number, F , has fallen (between) E and G in mean proportion, E and G are thus similar plane numbers [Prop. 8.20]. Therefore, let H, K be the sides of E , and L, M (the sides) of G . Thus, it is clear from the (proposition) before this (one) that E, F, G are continuously proportional in the ratio of H to L , and of K to M . And since E, F, G are the least (numbers) having the same ratio as A, C, D , and the multitude of E, F, G is equal to the multitude of A, C, D , thus, via equality, as E is to G , so A (is) to D [Prop. 7.14]. And E and G (are) prime (to one another), and prime (numbers) are also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the greater (measuring) the greater, and the lesser the lesser—that is to say, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures A the same number of times as G (measures) D . So as many times as E measures A , so many units let there be in N . Thus, N has made A (by) multiplying E [Def. 7.15]. And E is the (number created) from (multiplying) H and K . Thus, N has made A (by) multiplying the (number created) from (multiplying) H and K . Thus, A is solid, and its sides are H, K, N . Again, since E, F, G are the least (numbers) having the same ratio as C, D, B , thus E measures C the same number of times as G (measures) B [Prop. 7.20]. So as many times as E measures C , so many units let there be in O . Thus, G measures B according to the units in O . Thus, O has made B (by) multiplying G . And G is the (number created) from (multiplying) L and M . Thus, O has made B (by) multiplying the (number created) from (multiplying) L and M . Thus, B is solid, and its sides are L, M, O . Thus, A and B are (both) solid.



Λέγω [δή], ὅτι καὶ ὅμοιοι. ἐπεὶ γάρ οἱ N , X τὸν E πολλαπλασιάσαντες τοὺς A , Γ πεποιήκασιν, ἔστιν ἄρα ὡς ὁ N πρὸς τὸν X , ὁ A πρὸς τὸν Γ , τουτέστιν ὁ E πρὸς τὸν Z . ἀλλ' ὡς ὁ E πρὸς τὸν Z , ὁ Θ πρὸς τὸν Λ καὶ ὁ K πρὸς τὸν M καὶ ὡς ἄρα ὁ Θ πρὸς τὸν Λ , οὕτως ὁ K πρὸς τὸν M καὶ ὁ N πρὸς τὸν X . καὶ εἰσιν οἱ μὲν Θ , K , N πλευραὶ τοῦ A , οἱ δὲ X , Λ , M πλευραὶ τοῦ B . οἱ A , B ἄρα ἀριθμοὶ ὅμοιοι στερεοὶ εἰσιν· ὅπερ ἔδει δεῖξαι.

[So] I say that (they are) also similar. For since N , O have made A , C (by) multiplying E , thus as N is to O , so A (is) to C —that is to say, E to F [Prop. 7.18]. But, as E (is) to F , so H (is) to L , and K to M . And thus as H (is) to L , so K (is) to M , and N to O . And H , K , N are the sides of A , and L , M , O the sides of B . Thus, A and B are similar solid numbers [Def. 7.21]. (Which is) the very thing it was required to show.

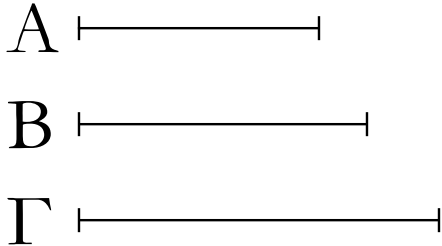
† The Greek text has “ O , L , M ”, which is obviously a mistake.

22.

Proposition 22

Ἐὰν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ὦσιν, ὁ δὲ πρῶτος τετράγωνός ᾗ, καὶ ὁ τρίτος τετράγωνός ἔσται.

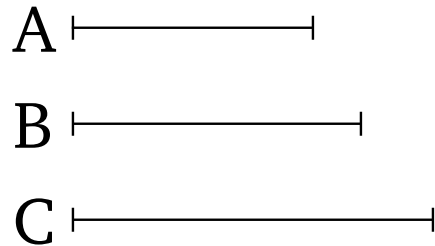
Ἐστώσαν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A , B , Γ , ὁ δὲ πρῶτος ὁ A τετράγωνός ἔστω· λέγω, ὅτι καὶ ὁ τρίτος ὁ Γ τετράγωνός ἔστιν.



Ἐπεὶ γάρ τῶν A , Γ εἷς μέσος ἀνάλογόν ἐστιν ἀριθμὸς ὁ B , οἱ A , Γ ἄρα ὅμοιοι ἐπίπεδοι εἰσιν. τετράγωνός δὲ ὁ A · τετράγωνός ἄρα καὶ ὁ Γ · ὅπερ ἔδει δεῖξαι.

If three numbers are continuously proportional, and the first is square, then the third will also be square.

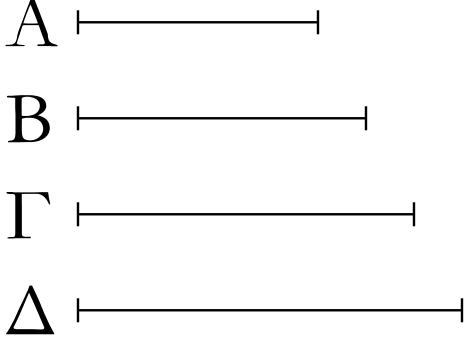
Let A , B , C be three continuously proportional numbers, and let the first A be square. I say that the third C is also square.



For since one number, B , is in mean proportion to A and C , A and C are thus similar plane (numbers) [Prop. 8.20]. And A is square. Thus, C is also square [Def. 7.21]. (Which is) the

23.

Ἐάν τέσσαρες ἀριθμοὶ ἐχῇς ἀνάλογον ὦσιν, ὁ δὲ πρῶτος κύβοξ ᾗ, καὶ ὁ τέταρτος κύβοξ ἔσται.



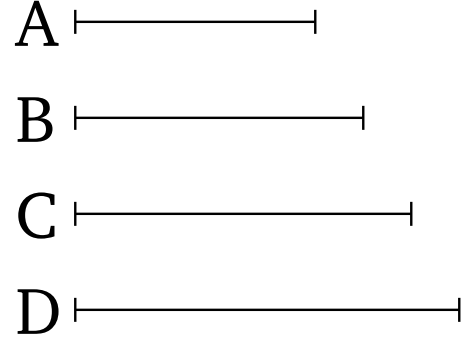
Ἐστωσαν τέσσαρες ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A, B, Γ, Δ , ὁ δὲ A κύβοξ ἔστω· λέγω, ὅτι καὶ ὁ Δ κύβοξ ἔστί.

Ἐπεὶ γὰρ τῶν A, Δ δύο μέσοι ἀνάλογόν εἰσιν ἀριθμοὶ οἱ B, Γ , οἱ A, Δ ἄρα ὅμοιοί εἰσι στερεοὶ ἀριθμοί. κύβοξ δὲ ὁ A κύβοξ ἄρα καὶ ὁ Δ · ὅπερ ἔδει δεῖξαι.

very thing it was required to show.

Proposition 23

If four numbers are continuously proportional, and the first is cube, then the fourth will also be cube.

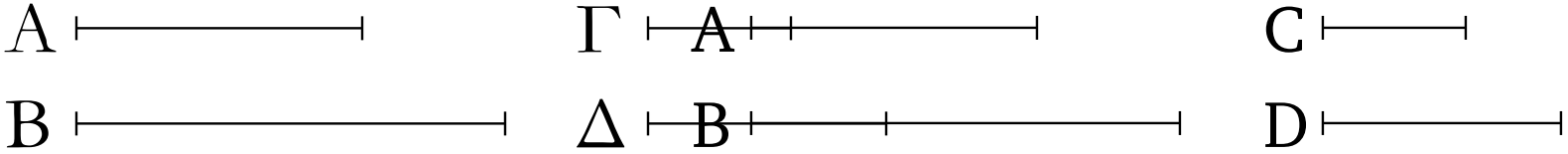


Let A, B, C, D be four continuously proportional numbers, and let A be cube. I say that D is also cube.

For since two numbers, B and C , are in mean proportion to A and D , A and D are thus similar solid numbers [Prop. 8.21]. And A (is) cube. Thus, D (is) also cube [Def. 7.21]. (Which is) the very thing it was required to show.

24.

Ἐάν δύο ἀριθμοὶ πρὸς ἀλλήλους λόγον ἔθωσιν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, ὁ δὲ πρῶτος τετράγωνος ᾗ, καὶ ὁ δεύτερος τετράγωνος ἔσται.



Δύο γὰρ ἀριθμοὶ οἱ A, B πρὸς ἀλλήλους λόγον ἔθέτωσαν, ὃν τετράγωνος ἀριθμὸς ὁ Γ πρὸς τετράγωνον ἀριθμόν τὸν Δ , ὁ δὲ A τετράγωνος ἔστω· λέγω, ὅτι καὶ ὁ B τετράγωνός ἐστι.

Ἐπεὶ γὰρ οἱ Γ, Δ τετράγωνοί εἰσιν, οἱ Γ, Δ ἄρα ὅμοιοι ἐπίπεδοι εἰσιν. τῶν Γ, Δ ἄρα εἷς μέσος ἀνάλογον ἐμπίπτει ἀριθμὸς. καὶ ἐστὶν ὡς ὁ Γ πρὸς τὸν Δ , ὁ A πρὸς τὸν B . καὶ τῶν A, B ἄρα εἷς μέσος ἀνάλογον ἐμπίπτει ἀριθμὸς. καὶ ἐστὶν ὁ A τετράγωνος· καὶ ὁ B ἄρα τετράγωνός ἐστιν· ὅπερ ἔδει δεῖξαι.

If two numbers have to one another the ratio which a square number (has) to a(nother) square number, and the first is square, then the second will also be square.

Proposition 24

For let two numbers, A and B , have to one another the ratio which the square number C (has) to the square number D . And let A be square. I say that B is also square.

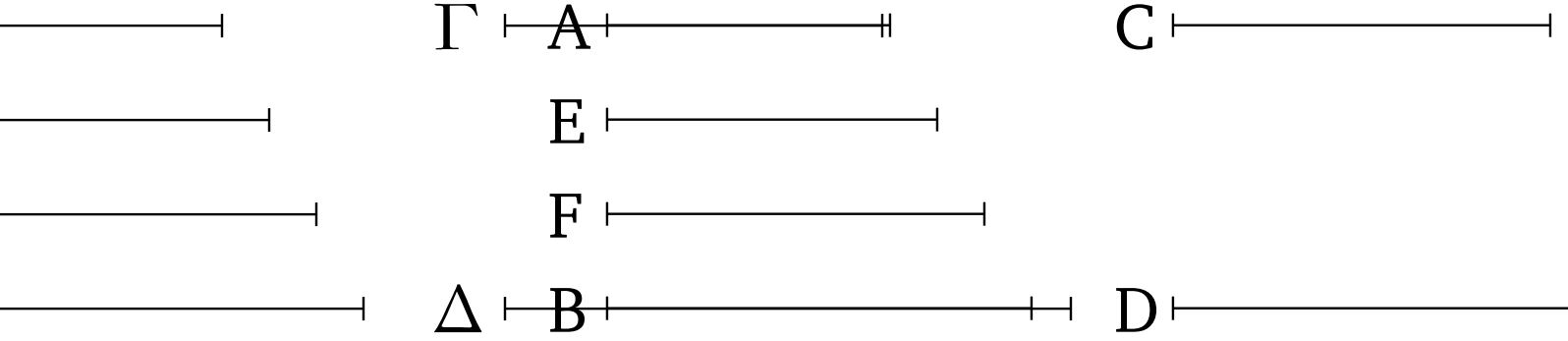
For since C and D are square, C and D are thus similar plane (numbers). Thus, one number falls (between) C and D in mean proportion [Prop. 8.18]. And as C is to D , (so) A (is) to B . Thus, one number also falls (between) A and B in mean proportion [Prop. 8.8]. And A is square. Thus, B is also square [Prop. 8.22]. (Which is) the very thing it was required to show.

25.

Ἐάν δύο ἀριθμοὶ πρὸς ἀλλήλους λόγον ἔθωσιν, ὃν κύβοξ ἀριθμὸς πρὸς κύβον ἀριθμόν, ὁ δὲ πρῶτος κύβοξ ᾗ, καὶ ὁ δεύτερος κύβοξ ἔσται.

If two numbers have to one another the ratio which a cube number (has) to a(nother) cube number, and the first is cube, then the second will also be cube.

Proposition 25



Δύο γὰρ ἀριθμοὶ οἱ A, B πρὸς ἀλλήλων λόγον ἐθέτωσαν, ὃν κύβοξ ἀριθμὸς ὁ Γ πρὸς κύβον ἀριθμὸν τὸν Δ , κύβοξ δὲ ἔστω ὁ A · λέγω [δη], ὅτι καὶ ὁ B κύβοξ ἐστίν.

Ἐπεὶ γὰρ οἱ Γ, Δ κύβοι εἰσίν, οἱ Γ, Δ ὅμοιοι στερεοὶ εἰσιν τῶν Γ, Δ ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. ὅσοι δὲ εἰς τοὺς Γ, Δ μεταχὺ κατὰ τὸ συνενθὲς ἀνάλογον ἐμπίπτουσιν, τοσοῦτοι καὶ εἰς τοὺς τὸν αὐτὸν λόγον ἔθονταξ αὐτοῖς· ὥστε καὶ τῶν A, B δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. ἐμπίπτέτωσαν οἱ E, Z . ἐπεὶ οὖν τέσσαρες ἀριθμοὶ οἱ A, E, Z, B ἐχῆξ ἀνάλογόν εἰσιν, καὶ ἐστι κύβοξ ὁ A , κύβοξ ἄρα καὶ ὁ B · ὅπερ ἔδει δεῖχαι.

For let two numbers, A and B , have to one another the ratio which the cube number C (has) to the cube number D . And let A be cube. [So] I say that B is also cube.

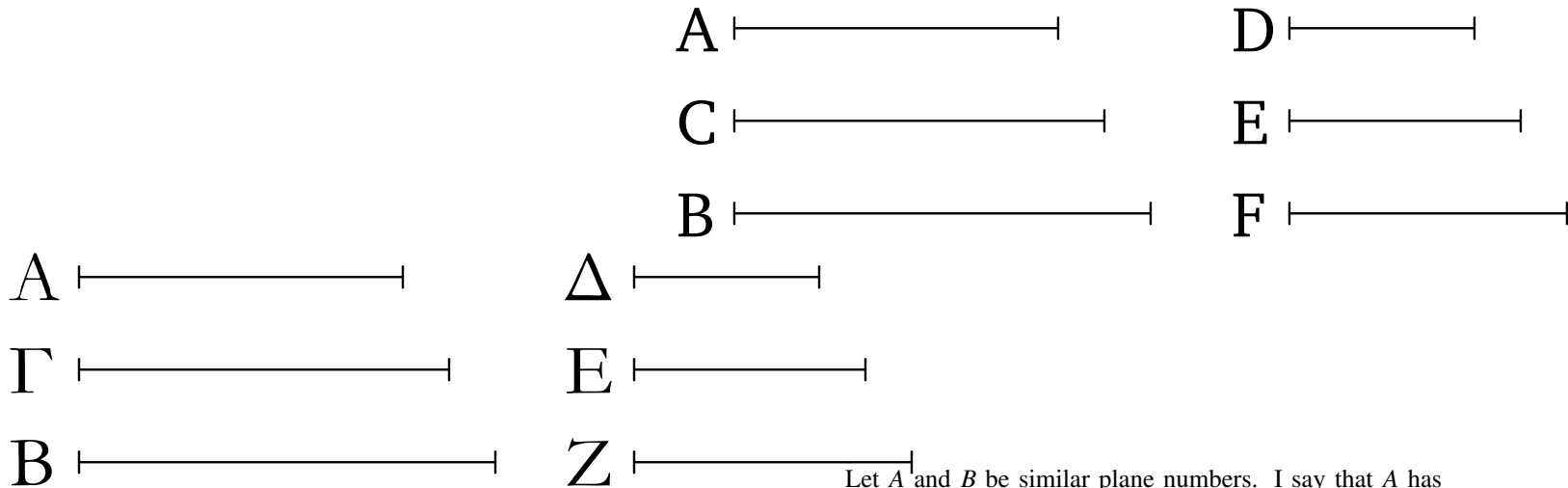
For since C and D are cube (numbers), C and D are (thus) similar solid (numbers). Thus, two numbers fall (between) C and D in mean proportion [Prop. 8.19]. And as many (numbers) as fall in between C and D in continued proportion, so many also (fall) in (between) those (numbers) having the same ratio as them (in continued proportion) [Prop. 8.8]. And hence two numbers fall (between) A and B in mean proportion. Let E and F (so) fall. Therefore, since the four numbers A, E, F, B are continuously proportional, and A is cube, B (is) thus also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

26.

Proposition 26

Οἱ ὅμοιοι ἐπίπεδοι ἀριθμοὶ πρὸς ἀλλήλων λόγον ἔθουσιν, ὃν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν.

Similar plane numbers have to one another the ratio which (some) square number (has) to a(nother) square number.



Ἐστωσαν ὅμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B · λέγω, ὅτι ὁ A πρὸς τὸν B λόγον ἔθει, ὃν τετράγωνοξ ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν.

Ἐπεὶ γὰρ οἱ A, B ὅμοιοι ἐπίπεδοι εἰσιν, τῶν A, B ἄρα εἷξ μέσοξ ἀνάλογον ἐμπίπτει ἀριθμὸξ. ἐμπίπτέτω καὶ ἔστω

Let A and B be similar plane numbers. I say that A has to B the ratio which (some) square number (has) to a(nother) square number.

For since A and B are similar plane numbers, one number thus falls (between) A and B in mean proportion [Prop. 8.18]. Let it (so) fall, and let it be C . And let the least numbers, D, E ,

ὁ Γ , καὶ εἰλήφθωσαν ἐλάττωστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ, B οἱ Δ, E, Z : οἱ ἄρα ἄκροι αὐτῶν οἱ Δ, Z τετράγωνοι εἰσιν. καὶ ἐπεὶ ἐστὶν ὡς ὁ Δ πρὸς τὸν Z , οὕτως ὁ A πρὸς τὸν B , καὶ εἰσιν οἱ Δ, Z τετράγωνοι, ὁ A ἄρα πρὸς τὸν B λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ὅπερ ἔδει δεῖχαι.

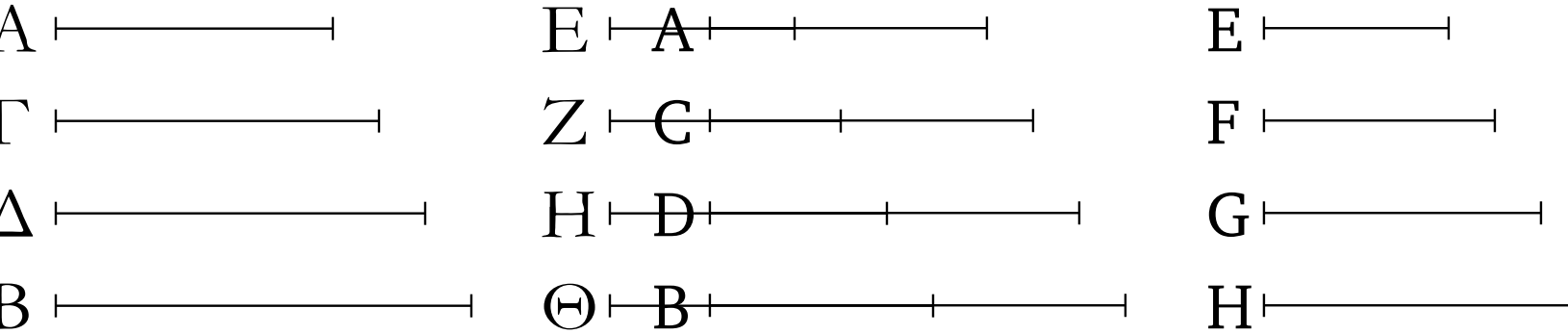
F , having the same ratio as A, C, B be taken [Prop. 8.2]. The outermost of them, D and F , are thus square [Prop. 8.2 corr.]. And since as D is to F , so A (is) to B , and D and F are square, A thus has to B the ratio which (some) square number (has) to a(nother) square number. (Which is) the very thing it was required to show.

27.

Proposition 27

Οἱ ὅμοιοι στερεοὶ ἀριθμοὶ πρὸς ἀλλήλων λόγον ἔθουσιν, ὃν κύβοξ ἀριθμὸς πρὸς κύβον ἀριθμόν.

Similar solid numbers have to one another the ratio which (some) cube number (has) to a(nother) cube number.



Ἐστωσαν ὅμοιοι στερεοὶ ἀριθμοὶ οἱ A, B . λέγω, ὅτι ὁ A πρὸς τὸν B λόγον ἔθει, ὃν κύβοξ ἀριθμὸς πρὸς κύβον ἀριθμόν.

Ἐπεὶ γὰρ οἱ A, B ὅμοιοι στερεοὶ εἰσιν, τῶν A, B ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. ἐμπίπτεωσαν οἱ Γ, Δ , καὶ εἰλήφθωσαν ἐλάττωστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, Γ, Δ, B ἴσοι αὐτοῖς τὸ πλῆθος οἱ E, Z, H, Θ : οἱ ἄρα ἄκροι αὐτῶν οἱ E, Θ κύβοι εἰσίν. καὶ ἐστὶν ὡς ὁ E πρὸς τὸν Θ , οὕτως ὁ A πρὸς τὸν B : καὶ ὁ A ἄρα πρὸς τὸν B λόγον ἔθει, ὃν κύβοξ ἀριθμὸς πρὸς κύβον ἀριθμόν· ὅπερ ἔδει δεῖχαι.

Let A and B be similar solid numbers. I say that A has to B the ratio which (some) cube number (has) to a(nother) cube number.

For since A and B are similar solid (numbers), two numbers thus fall (between) A and B in mean proportion [Prop. 8.19]. Let C and D have (so) fallen. And let the least numbers, E, F, G, H , having the same ratio as A, C, D, B , (and) equal in multitude to them, be taken [Prop. 8.2]. Thus, the outermost of them, E and H , are cube [Prop. 8.2 corr.]. And as E is to H , so A (is) to B . And thus A has to B the ratio which (some) cube number (has) to a(nother) cube number. (Which is) the very thing it was required to show.

ELEMENTS BOOK 9

Applications of Number Theory[†]

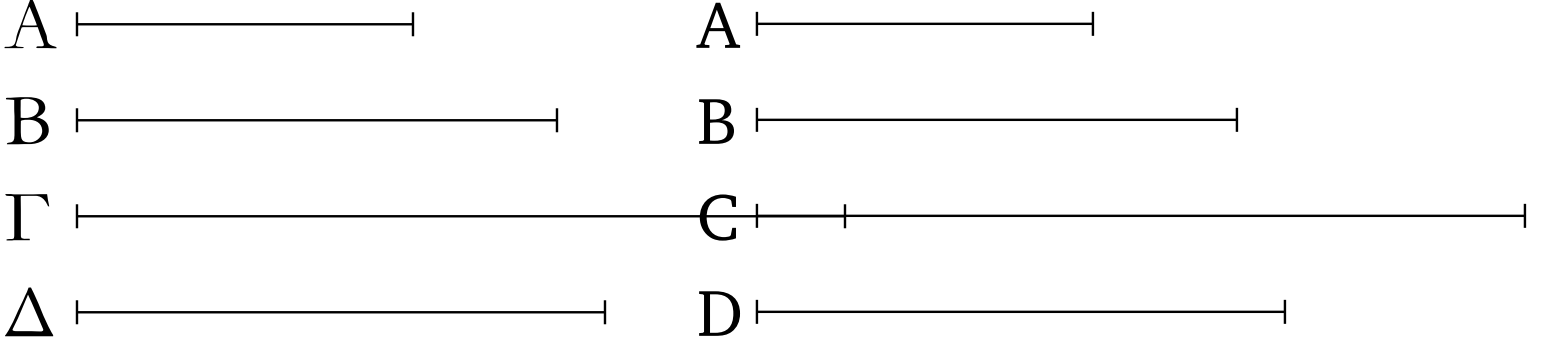
[†]The propositions contained in Books 7–9 are generally attributed to the school of Pythagoras.

1.

Proposition 1

Ἐάν δύο ὅμοιοι ἐπίπεδοι ἀριθμοὶ πολλαπλασιάσαντεξ ἀλλήλων ποιῶσι τινα, ὁ γενόμενος τετράγωνος ἔσται.

If two similar plane numbers make some (number by) multiplying one another then the created (number) will be square.



Ἐστωσαν δύο ὅμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B , καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ ποιείτω· λέγω, ὅτι ὁ Γ τετράγωνός ἐστιν.

Let A and B be two similar plane numbers, and let A make C (by) multiplying B . I say that C is square.

Ὁ γὰρ A ἐαντὸν πολλαπλασιάσας τὸν Δ ποιείτω. ὁ Δ ἄρα τετράγωνός ἐστιν. ἐπεὶ οὖν ὁ A ἐαντὸν μὲν πολλαπλασιάσας τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν Γ . καὶ ἐπεὶ οἱ A, B ὅμοιοι ἐπίπεδοι εἰσιν ἀριθμοί, τῶν A, B ἄρα εἷξ μέσοξ ἀνάλογον ἐμπίπτει ἀριθμός. ἔαν δὲ δύο ἀριθμῶν μεταχὺν κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅσοι εἷξ αὐτοῦξ ἐμπίπτουσι, τοσοῦτοι καὶ εἷξ τοῦξ τὸν αὐτὸν λόγον ἔθονταξ· ὥστε καὶ τῶν Δ, Γ εἷξ μέσοξ ἀνάλογον ἐμπίπτει ἀριθμός. καὶ ἐστι τετράγωνος ὁ Δ · τετράγωνος ἄρα καὶ ὁ Γ · ὅπερ ἔδει δεῖχαι.

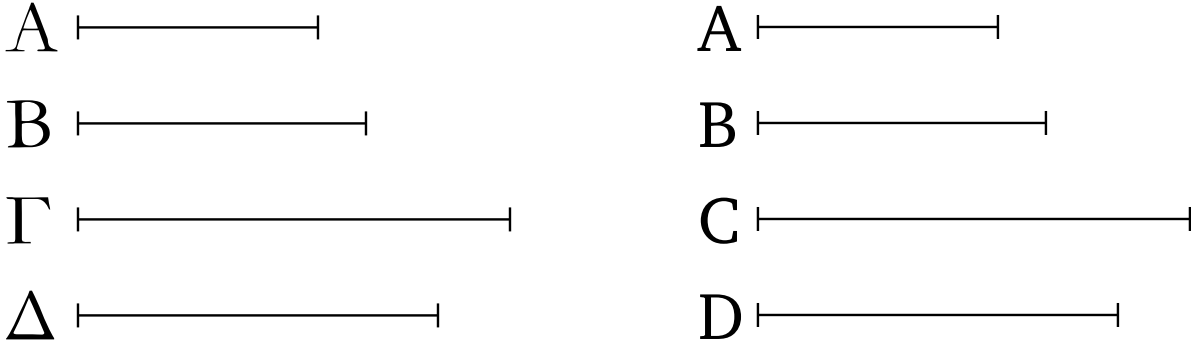
For let A make D (by) multiplying itself. D is thus square. Therefore, since A has made D (by) multiplying itself, and has made C (by) multiplying B , thus as A is to B , so D (is) to C [Prop. 7.17]. And since A and B are similar plane numbers, one number thus falls (between) A and B in mean proportion [Prop. 8.18]. And if (some) numbers fall between two numbers in continued proportion then, as many (numbers) as fall in (between) them (in continued proportion), so many also (fall) in (between numbers) having the same ratio (as them in continued proportion) [Prop. 8.8]. And hence one number falls (between) D and C in mean proportion. And D is square. Thus, C (is) also square [Prop. 8.22]. (Which is) the very thing it was required to show.

2.

Proposition 2

Ἐάν δύο ἀριθμοὶ πολλαπλασιάσαντεξ ἀλλήλων ποιῶσι τετράγωνον, ὅμοιοι ἐπίπεδοι εἰσιν ἀριθμοί.

If two numbers make a square (number by) multiplying one another then they are similar plane numbers.



Ἐστωσαν δύο ἀριθμοὶ οἱ A, B , καὶ ὁ A τὸν B πολλαπλασιάσας τετράγωνον τὸν Γ ποιείτω· λέγω, ὅτι οἱ A, B ὅμοιοι ἐπίπεδοι εἰσιν ἀριθμοί.

Let A and B be two numbers, and let A make the square (number) C (by) multiplying B . I say that A and B are similar plane numbers.

Ὁ γὰρ A ἐαντὸν πολλαπλασιάσας τὸν Δ ποιεῖτω· ὁ Δ ἄρα τετράγωνός ἐστιν. καὶ ἐπεὶ ὁ A ἐαντὸν μὲν πολλαπλασιάσας τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , ὁ Δ πρὸς τὸν Γ . καὶ ἐπεὶ ὁ Δ τετράγωνός ἐστιν, ἀλλὰ καὶ ὁ Γ , οἱ Δ , Γ ἄρα ὅμοιοι ἐπίπεδοι εἰσιν. τῶν Δ , Γ ἄρα εἷς μέσος ἀνάλογον ἐμπίπτει. καὶ ἐστιν ὡς ὁ Δ πρὸς τὸν Γ , οὕτως ὁ A πρὸς τὸν B · καὶ τῶν A , B ἄρα εἷς μέσος ἀνάλογον ἐμπίπτει. ἔαν δὲ δύο ἀριθμῶν εἷς μέσος ἀνάλογον ἐμπίπτῃ, ὅμοιοι ἐπίπεδοι εἰσιν [οἱ] ἀριθμοί· οἱ ἄρα A , B ὅμοιοι εἰσιν ἐπίπεδοι· ὅπερ ἔδει δεῖχαι.

3.

Ἐάν κύβος ἀριθμὸς ἐαντὸν πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενος κύβος ἔσται.

A —————

B —————

Γ —————

Δ —————

Κύβος γὰρ ἀριθμὸς ὁ A ἐαντὸν πολλαπλασιάσας τὸν B ποιεῖτω· λέγω, ὅτι ὁ B κύβος ἐστίν.

Εἰλήφθω γὰρ τοῦ A πλεονὰς ὁ Γ , καὶ ὁ Γ ἐαντὸν πολλαπλασιάσας τὸν Δ ποιεῖτω. φανερόν δὲ ἔστιν, ὅτι ὁ Γ τὸν Δ πολλαπλασιάσας τὸν Δ πεποίηκεν, καὶ ἐπεὶ ὁ Γ ἐαντὸν πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ Γ ἄρα τὸν Δ μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας. ἀλλὰ μὴν καὶ ἡ μονὰς τὸν Γ μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· ἔστιν ἄρα ὡς ἡ μονὰς πρὸς τὸν Γ , ὁ Γ πρὸς τὸν Δ . πάλιν, ἐπεὶ ὁ Γ τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν, ὁ Δ ἄρα τὸν A μετρεῖ κατὰ τὰς ἐν τῷ Γ μονάδας. μετρεῖ δὲ καὶ ἡ μονὰς τὸν Γ κατὰ τὰς ἐν αὐτῷ μονάδας· ἔστιν ἄρα ὡς ἡ μονὰς πρὸς τὸν Γ , ὁ Δ πρὸς τὸν A . ἀλλ' ὡς ἡ μονὰς πρὸς τὸν Γ , ὁ Γ πρὸς τὸν Δ · καὶ ὡς ἄρα ἡ μονὰς πρὸς τὸν Γ , οὕτως ὁ Γ πρὸς τὸν Δ καὶ ὁ Δ πρὸς τὸν A . τῆς ἄρα μονάδος καὶ τοῦ A ἀριθμοῦ δύο μέσοι ἀνάλογον κατὰ τὸ συνεθὲς ἐμπεπτώκασιν ἀριθμοί οἱ Γ , Δ . πάλιν, ἐπεὶ ὁ A ἐαντὸν πολλαπλασιάσας τὸν B πεποίηκεν, ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· μετρεῖ δὲ καὶ ἡ μονὰς τὸν A κατὰ τὰς ἐν αὐτῷ μονάδας· ἔστιν ἄρα ὡς ἡ μονὰς πρὸς τὸν A , ὁ A πρὸς τὸν B . τῆς δὲ μονάδος καὶ τοῦ A δύο μέσοι ἀνάλογον ἐμπεπτώκασιν ἀριθμοί· καὶ τῶν A , B ἄρα δύο μέσοι ἀνάλογον ἐμπεσύνται ἀριθμοί. ἔαν δὲ δύο ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτωσιν, ὁ δὲ πρῶτος κύβος ᾗ, καὶ ὁ δεύτερος

For let A make D (by) multiplying itself. Thus, D is square. And since A has made D (by) multiplying itself, and has made C (by) multiplying B , thus as A is to B , so D (is) to C [Prop. 7.17]. And since D is square, and C (is) also, D and C are thus similar plane numbers. Thus, one (number) falls (between) D and C in mean proportion [Prop. 8.18]. And as D is to C , so A (is) to B . Thus, one (number) also falls (between) A and B in mean proportion [Prop. 8.8]. And if one (number) falls (between) two numbers in mean proportion then [the] numbers are similar plane (numbers) [Prop. 8.20]. Thus, A and B are similar plane (numbers). (Which is) the very thing it was required to show.

Proposition 3

If a cube number makes some (number by) multiplying itself then the created (number) will be cube.

A —————

B —————

C —————

D —————

For let the cube number A make B (by) multiplying itself. I say that B is cube.

For let the side C of A be taken. And let C make D by multiplying itself. So it is clear that C has made A (by) multiplying D . And since C has made D (by) multiplying itself, C thus measures D according to the units in it [Def. 7.15]. But, in fact, a unit also measures C according to the units in it [Def. 7.20]. Thus, as a unit is to C , so C (is) to D . Again, since C has made A (by) multiplying D , D thus measures A according to the units in C . And a unit also measures C according to the units in it. Thus, as a unit is to C , so D (is) to A . But, as a unit (is) to C , so C (is) to D . And thus as a unit (is) to C , so C (is) to D , and D to A . Thus, two numbers, C and D , have fallen (between) a unit and the number A in continued mean proportion. Again, since A has made B (by) multiplying itself, A thus measures B according to the units in it. And a unit also measures A according to the units in it. Thus, as a unit is to A , so A (is) to B . And two numbers have fallen (between) a unit and A in mean proportion. Thus two numbers will also fall (between) A and B in mean proportion [Prop. 8.8]. And if two (numbers) fall (between) two numbers in mean proportion, and the first (number) is cube, then the second will also be cube [Prop. 8.23]. And A is cube. Thus, B is also cube.

κύβοξ ἔσται. καὶ ἐστὶν ὁ A κύβοξ· καὶ ὁ B ἄρα κύβοξ ἐστίν· (Which is) the very thing it was required to show.
ὅπερ ἔδει δεῖχαι.

4.

Ἐὰν κύβοξ ἀριθμὸς κύβον ἀριθμὸν πολλαπλασιάσας ποιῇ
τινα, ὁ γενόμενος κύβοξ ἔσται.

Κύβοξ γὰρ ἀριθμὸς ὁ A κύβον ἀριθμὸν τὸν B πολλαπλα-
σιάσας τὸν Γ ποιεῖτω· λέγω, ὅτι ὁ Γ κύβοξ ἐστίν.

A

B

Γ

Δ

Ὅ γὰρ A ἑαυτὸν πολλαπλασιάσας τὸν Δ ποιεῖτω· ὁ Δ
ἄρα κύβοξ ἐστίν. καὶ ἐπεὶ ὁ A ἑαυτὸν μὲν πολλαπλασιάσας
τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν,
ἐστὶν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν Γ . καὶ
ἐπεὶ οἱ A , B κύβοι εἰσὶν, ὅμοιοι στερεοὶ εἰσιν οἱ A , B . τῶν
 A , B ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί· ὥστε καὶ
τῶν Δ , Γ δύο μέσοι ἀνάλογον ἐμπεσοῦνται ἀριθμοί. καὶ ἐστὶ
κύβοξ ὁ Δ · κύβοξ ἄρα καὶ ὁ Γ · ὅπερ ἔδει δεῖχαι.

5.

Ἐὰν κύβοξ ἀριθμὸς ἀριθμὸν τινα πολλαπλασιάσας κύβον
ποιῇ, καὶ ὁ πολλαπλασιασθεὶς κύβοξ ἔσται.

A

B

Γ

Δ

Κύβοξ γὰρ ἀριθμὸς ὁ A ἀριθμὸν τινα τὸν B πολλαπλασι-
άσας κύβον τὸν Γ ποιεῖτω· λέγω, ὅτι ὁ B κύβοξ ἐστίν.

Ὅ γὰρ A ἑαυτὸν πολλαπλασιάσας τὸν Δ ποιεῖτω· κύβοξ
ἄρα ἐστὶν ὁ Δ . καὶ ἐπεὶ ὁ A ἑαυτὸν μὲν πολλαπλασιάσας
τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν,

Proposition 4

If a cube number makes some (number by) multiplying
a(nother) cube number then the created (number) will be cube.

For let the cube number A make C (by) multiplying the
cube number B . I say that C is cube.

A

B

C

D

For let A make D (by) multiplying itself. Thus, D is cube
[Prop. 9.3]. And since A has made D (by) multiplying itself,
and has made C (by) multiplying B , thus as A is to B , so D
(is) to C [Prop. 7.17]. And since A and B are cube, A and
 B are similar solid (numbers). Thus, two numbers fall (be-
tween) A and B in mean proportion [Prop. 8.19]. Hence, two
numbers will also fall (between) D and C in mean proportion
[Prop. 8.8]. And D is cube. Thus, C (is) also cube [Prop. 8.23].
(Which is) the very thing it was required to show.

Proposition 5

If a cube number makes a(nother) cube number (by) mul-
tiplying some (number) then the (number) multiplied will also
be cube.

A

B

C

D

For let the cube number A make the cube (number) C (by)
multiplying some number B . I say that B is cube.

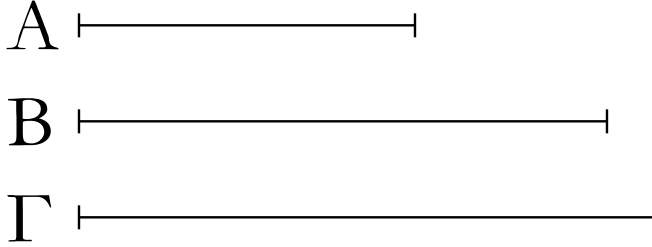
For let A make D (by) multiplying itself. D is thus cube
[Prop. 9.3]. And since A has made D (by) multiplying itself,
and has made C (by) multiplying B , thus as A is to B , so D (is)

ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , ὁ Δ πρὸς τὸν Γ . καὶ ἐπεὶ οἱ Δ , Γ κύβοι εἰσὶν, ὅμοιοι στερεοὶ εἰσιν. τῶν Δ , Γ ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. καὶ ἔστιν ὡς ὁ Δ πρὸς τὸν Γ , οὕτως ὁ A πρὸς τὸν B . καὶ τῶν A , B ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. καὶ ἐστὶ κύβοξ ὁ A . κύβοξ ἄρα ἐστὶ καὶ ὁ B . ὅπερ ἔδει δεῖχαι.

6.

Ἐάν ἀριθμὸς ἑαυτὸν πολλαπλασιάσας κύβον ποιῇ, καὶ αὐτὸς κύβοξ ἔσται.

Ἀριθμὸς γάρ ὁ A ἑαυτὸν πολλαπλασιάσας κύβον τὸν B ποιείτω· λέγω, ὅτι καὶ ὁ A κύβοξ ἐστίν.



Ὅ γάρ A τὸν B πολλαπλασιάσας τὸν Γ ποιείτω. ἐπεὶ οὖν ὁ A ἑαυτὸν μὲν πολλαπλασιάσας τὸν B πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ Γ ἄρα κύβοξ ἐστίν. καὶ ἐπεὶ ὁ A ἑαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν, ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας. μετρεῖ δὲ καὶ ἡ μονὰξ τὸν A κατὰ τὰς ἐν αὐτῷ μονάδας. ἔστιν ἄρα ὡς ἡ μονὰξ πρὸς τὸν A , οὕτως ὁ A πρὸς τὸν B . καὶ ἐπεὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ B ἄρα τὸν Γ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας. μετρεῖ δὲ καὶ ἡ μονὰξ τὸν A κατὰ τὰς ἐν αὐτῷ μονάδας. ἔστιν ἄρα ὡς ἡ μονὰξ πρὸς τὸν A , οὕτως ὁ B πρὸς τὸν Γ . ἀλλ' ὡς ἡ μονὰξ πρὸς τὸν A , οὕτως ὁ A πρὸς τὸν B . καὶ ὡς ἄρα ὁ A πρὸς τὸν B , ὁ B πρὸς τὸν Γ . καὶ ἐπεὶ οἱ B , Γ κύβοι εἰσὶν, ὅμοιοι στερεοὶ εἰσιν. τῶν B , Γ ἄρα δύο μέσοι ἀνάλογον εἰσιν ἀριθμοί. καὶ ἔστιν ὡς ὁ B πρὸς τὸν Γ , ὁ A πρὸς τὸν B . καὶ τῶν A , B ἄρα δύο μέσοι ἀνάλογον εἰσιν ἀριθμοί. καὶ ἐστὶν κύβοξ ὁ B . κύβοξ ἄρα ἐστὶ καὶ ὁ A . ὅπερ ἔδει δεῖχαι.

7.

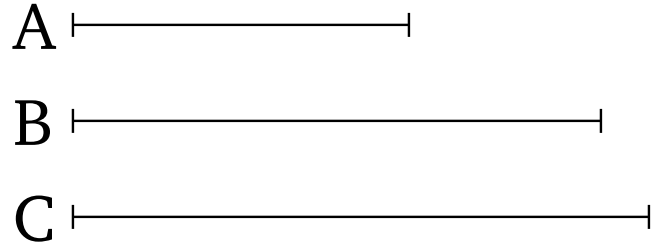
Ἐάν σύνθετος ἀριθμὸς ἀριθμὸν τινα πολλαπλασιάσας ποιῇ τινα, ὁ γεγόμενος στερεὸς ἔσται.

to C [Prop. 7.17]. And since D and C are (both) cube, they are similar solid (numbers). Thus, two numbers fall (between) D and C in mean proportion [Prop. 8.19]. And as D is to C , so A (is) to B . Thus, two numbers also fall (between) A and B in mean proportion [Prop. 8.8]. And A is cube. Thus, B is also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

Proposition 6

If a number makes a cube (number by) multiplying itself then it itself will also be cube.

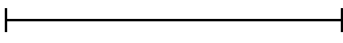
For let the number A make the cube (number) B (by) multiplying itself. I say that A is also cube.

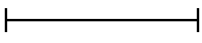


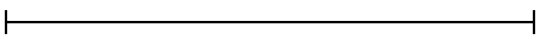
For let A make C (by) multiplying B . Therefore, since A has made B (by) multiplying itself, and has made C (by) multiplying B , C is thus cube. And since A has made B (by) multiplying itself, A thus measures B according to the units in (A). And a unit also measures A according to the units in it. Thus, as a unit is to A , so A (is) to B . And since A has made C (by) multiplying B , B thus measures C according to the units in A . And a unit also measures A according to the units in it. Thus, as a unit is to A , so B (is) to C . But, as a unit (is) to A , so A (is) to B . And thus as A (is) to B , (so) B (is) to C . And since B and C are cube, they are similar solid (numbers). Thus, there exist two numbers in mean proportion (between) B and C [Prop. 8.19]. And as B is to C , (so) A (is) to B . Thus, there also exist two numbers in mean proportion (between) A and B [Prop. 8.8]. And B is cube. Thus, A is also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

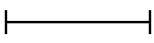
Proposition 7

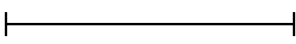
If a composite number makes some (number by) multiplying some (other) number then the created (number) will be solid.

A 

B 

Γ 

Δ 

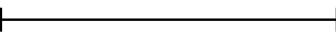
E 

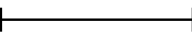
Σύνθετος γὰρ ἀριθμὸς ὁ A ἀριθμὸν τινα τὸν B πολλαπλασιάσας τὸν Γ ποιεῖτω· λέγω, ὅτι ὁ Γ στερεὸς ἐστίν.

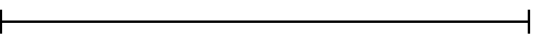
Ἐπεὶ γὰρ ὁ A σύνθετός ἐστιν, ὑπὸ ἀριθμοῦ τινος μετρηθήσεται. μετρείσθω ὑπὸ τοῦ Δ , καὶ ὁσάκις ὁ Δ τὸν A μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ E . ἔπει οὖν ὁ Δ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ E μονάδας, ὁ E ἄρα τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν. καὶ ἐπεὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ δὲ A ἐστίν ὁ ἐκ τῶν Δ , E , ὁ ἄρα ἐκ τῶν Δ , E τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν. ὁ Γ ἄρα στερεὸς ἐστίν, πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Δ , E , B . ὅπερ ἔδει δεῖχαι.

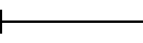
8.

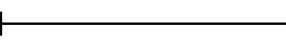
Ἐάν ἀπὸ μονάδος ὅποσοιὺν ἀριθμοὶ ἐχῇς ἀνάλογον ᾗσιν, ὁ μὲν τρίτος ἀπὸ τῆς μονάδος τετράγωνος ἔσται καὶ οἱ ἓνα διαλείποντες, ὁ δὲ τέταρτος κύβος καὶ οἱ δύο διαλείποντες πάντες, ὁ δὲ ἑβδομὸς κύβος ἅμα καὶ τετράγωνος καὶ οἱ πέντε διαλείποντες.

A 

B 

C 

D 

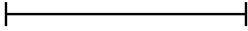
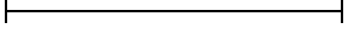
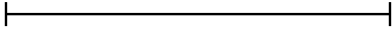
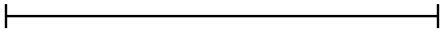
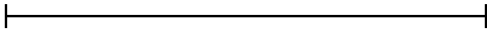
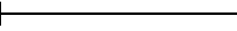
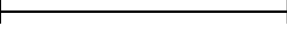
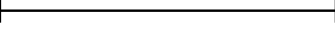
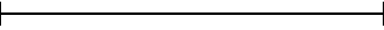
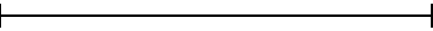
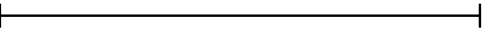
E 

For let the composite number A make C (by) multiplying some number B . I say that C is solid.

For since A is a composite (number), it will be measured by some number. Let it be measured by D . And, as many times as D measures A , so many units let there be in E . Therefore, since D measures A according to the units in E , E has thus made A (by) multiplying D [Def. 7.15]. And since A has made C (by) multiplying B , and A is the (number created) from (multiplying) D , E , the (number created) from (multiplying) D , E has thus made C (by) multiplying B . Thus, C is solid, and its sides are D , E , B . (Which is) the very thing it was required to show.

Proposition 8

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then the third from the unit will be square, and (all) those (numbers after that) which leave an interval of one (number), and the fourth (will be) cube, and all those (numbers after that) which leave an interval of two (numbers), and the seventh (will be) both cube and square, and (all) those (numbers after that) which leave an interval of five (numbers).

A B Γ Δ E Z A B C D E F 

Ἐστωσαν ἀπὸ μονάδος ὅποσοιὸν ἀριθμοὶ ἐχῆς ἀνάλογον οἱ $A, B, \Gamma, \Delta, E, Z$ · λέγω, ὅτι ὁ μὲν τρίτος ἀπὸ τῆς μονάδος ὁ B τετράγωνός ἐστι καὶ οἱ ἑνα διαλείποντες πάντες, ὁ δὲ τέταρτος ὁ Γ κύβος καὶ οἱ δύο διαλείποντες πάντες, ὁ δὲ ἑβδομος ὁ Z κύβος ἅμα καὶ τετράγωνος καὶ οἱ πέντε διαλείποντες πάντες.

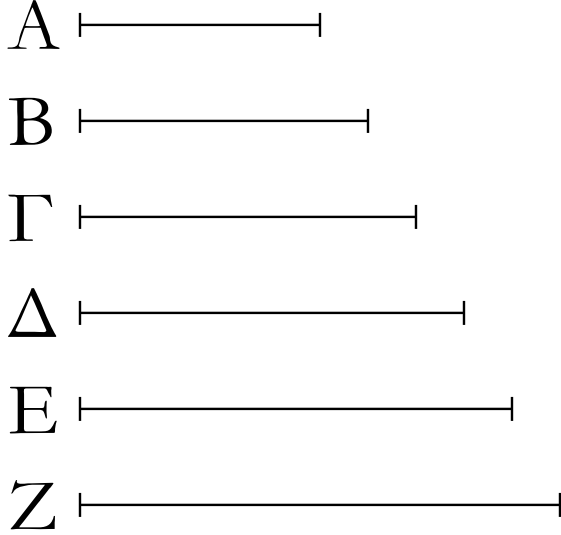
Ἐπεὶ γάρ ἐστιν ὡς ἡ μονὰς πρὸς τὸν A , οὕτως ὁ A πρὸς τὸν B , ἰσάκις ἄρα ἡ μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ A τὸν B . ἡ δὲ μονὰς τὸν A ἀριθμὸν μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· καὶ ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας. ὁ A ἄρα ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν· τετράγωνος ἄρα ἐστὶν ὁ B . καὶ ἐπεὶ οἱ B, Γ, Δ ἐχῆς ἀνάλογόν εἰσιν, ὁ δὲ B τετράγωνός ἐστιν, καὶ ὁ Δ ἄρα τετράγωνός ἐστιν ἡμιά τὰ αὐτὰ δὴ καὶ ὁ Z τετράγωνός ἐστιν. ὁμοίως δὴ δείχομεν, ὅτι καὶ οἱ ἑνα διαλείποντες πάντες τετράγωνοί εἰσιν. λέγω δὴ, ὅτι καὶ ὁ τέταρτος ἀπὸ τῆς μονάδος ὁ Γ κύβος ἐστὶ καὶ οἱ δύο διαλείποντες πάντες. ἐπεὶ γάρ ἐστιν ὡς ἡ μονὰς πρὸς τὸν A , οὕτως ὁ B πρὸς τὸν Γ , ἰσάκις ἄρα ἡ μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ B τὸν Γ . ἡ δὲ μονὰς τὸν A ἀριθμὸν μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας· καὶ ὁ B ἄρα τὸν Γ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας· ὁ A ἄρα τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν. ἐπεὶ οὖν ὁ A ἐαυτὸν μὲν πολλαπλασιάσας τὸν B πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, κύβος ἄρα ἐστὶν ὁ Γ . καὶ ἐπεὶ οἱ Γ, Δ, E, Z ἐχῆς ἀνάλογόν εἰσιν, ὁ δὲ Γ κύβος ἐστίν, καὶ ὁ Z ἄρα κύβος ἐστίν. ἐδείκθη δὲ καὶ τετράγωνος· ὁ ἄρα ἑβδομος ἀπὸ τῆς μονάδος κύβος τέ ἐστι καὶ τετράγωνος. ὁμοίως δὴ δείχομεν, ὅτι καὶ οἱ πέντε διαλείποντες πάντες κύβοι τέ εἰσι καὶ τετράγωνοι· ὅπερ ἔδει δείχαι.

Let any multitude whatsoever of numbers, A, B, C, D, E, F , be continuously proportional, (starting) from a unit. I say that the third from the unit, B , is square, and all those (numbers after that) which leave an interval of one (number). And the fourth (from the unit), C , (is) cube, and all those (numbers after that) which leave an interval of two (numbers). And the seventh (from the unit), F , (is) both cube and square, and all those (numbers after that) which leave an interval of five (numbers).

For since as the unit is to A , so A (is) to B , the unit thus measures the number A the same number of times as A (measures) B [Def. 7.20]. And the unit measures the number A according to the units in it. Thus, A also measures B according to the units in A . A has thus made B (by) multiplying itself [Def. 7.15]. Thus, B is square. And since B, C, D are continuously proportional, and B is square, D is thus also square [Prop. 8.22]. So, for the same (reasons), F is also square. So, similarly, we can also show that all those (numbers after that) which leave an interval of one (number) are square. So I also say that the fourth (number) from the unit, C , is cube, and all those (numbers after that) which leave an interval of two (numbers). For since as the unit is to A , so B (is) to C , the unit thus measures the number A the same number of times that B (measures) C . And the unit measures the number A according to the units in A . And thus B measures C according to the units in A . A has thus made C (by) multiplying B . Therefore, since A has made B (by) multiplying itself, and has made C (by) multiplying B , C is thus cube. And since C, D, E, F are continuously proportional, and C is cube, F is thus also cube [Prop. 8.23]. And it was also shown (to be) square. Thus, the seventh (number) from the unit is (both) cube and square. So, similarly, we can show that all those (numbers after that) which leave an interval of five (numbers) are (both) cube and

9.

Ἐάν ἀπὸ μονάδος ὁποσοῖον ἐχῇς κατὰ τὸ συνεθές ἀριθμοὶ ἀνάλογον ᾧσιν, ὁ δὲ μετὰ τὴν μονάδα τετράγωνος ᾗ, καὶ οἱ λοιποὶ πάντες τετράγωνοι ἔσονται. καὶ ἔαν ὁ μετὰ τὴν μονάδα κύβος ᾗ, καὶ οἱ λοιποὶ πάντες κύβοι ἔσονται.



Ἔστωσαν ἀπὸ μονάδος ἐχῇς ἀνάλογον ὅσοιδηποῦν ἀριθμοὶ οἱ $A, B, \Gamma, \Delta, E, Z$, ὁ δὲ μετὰ τὴν μονάδα ὁ A τετράγωνος ἔστω· λέγω, ὅτι καὶ οἱ λοιποὶ πάντες τετράγωνοι ἔσονται.

Ὅτι μὲν οὖν ὁ τρίτος ἀπὸ τῆς μονάδος ὁ B τετράγωνός ἐστι καὶ οἱ ἑνα διαπλείοντες πάντες, δέδεικται· λέγω [δή], ὅτι καὶ οἱ λοιποὶ πάντες τετράγωνοι εἰσιν. ἐπεὶ γάρ οἱ A, B, Γ ἐχῇς ἀνάλογόν εἰσιν, καὶ ἔστιν ὁ A τετράγωνος, καὶ ὁ Γ [ἄρα] τετράγωνος ἐστίν. πάλιν, ἐπεὶ [καὶ] οἱ B, Γ, Δ ἐχῇς ἀνάλογόν εἰσιν, καὶ ἔστιν ὁ B τετράγωνος, καὶ ὁ Δ [ἄρα] τετράγωνός ἐστιν. ὁμοίως δὲ δείχμεν, ὅτι καὶ οἱ λοιποὶ πάντες τετράγωνοι εἰσιν.

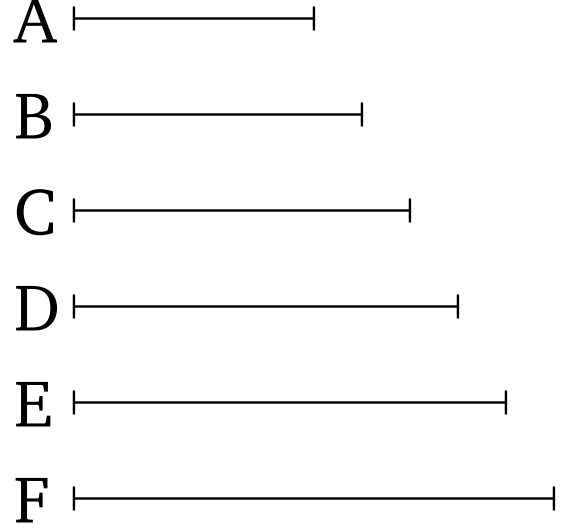
Ἀλλὰ δὴ ἔστω ὁ A κύβος· λέγω, ὅτι καὶ οἱ λοιποὶ πάντες κύβοι εἰσιν.

Ὅτι μὲν οὖν ὁ τέταρτος ἀπὸ τῆς μονάδος ὁ Γ κύβος ἐστὶ καὶ οἱ δύο διαλείποντες πάντες, δέδεικται· λέγω [δή], ὅτι καὶ οἱ λοιποὶ πάντες κύβοι εἰσιν. ἐπεὶ γάρ ἐστιν ὡς ἡ μονὰς πρὸς τὸν A , οὕτως ὁ A πρὸς τὸν B , ἰσάκις ἄρα ἡ μονὰς τὸν A μετρεῖ καὶ ὁ A τὸν B . ἡ δὲ μονὰς τὸν A μετρεῖ κατὰ τὰς ἐν αὐτῇ μονάδας· καὶ ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν αὐτῇ μονάδας· ὁ A ἄρα ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν. καὶ ἔστιν ὁ A κύβος. ἔαν δὲ κύβος ἀριθμὸς ἐαυτὸν πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενος κύβος ἐστίν· καὶ ὁ B ἄρα κύβος ἐστίν. καὶ ἐπεὶ τέσσαρες ἀριθμοὶ οἱ A, B, Γ, Δ ἐχῇς ἀνάλογόν εἰσιν, καὶ ἔστιν ὁ A κύβος, καὶ ὁ Δ ἄρα κύβος ἐστίν. διὰ

square. (Which is) the very thing it was required to show.

Proposition 9

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, and the (number) after the unit is square, then all the remaining (numbers) will also be square. And if the (number) after the unit is cube, then all the remaining (numbers) will also be cube.



Let any multitude whatsoever of numbers, A, B, C, D, E, F , be continuously proportional, (starting) from a unit. And let the (number) after the unit, A , be square. I say that all the remaining (numbers) will also be square.

In fact, it has (already) been shown that the third (number) from the unit, B , is square, and all those (numbers after that) which leave an interval of one (number) [Prop. 9.8]. [So] I say that all the remaining (numbers) are also square. For since A, B, C are continuously proportional, and A (is) square, C is [thus] also square [Prop. 8.22]. Again, since B, C, D are [also] continuously proportional, and B is square, D is [thus] also square [Prop. 8.22]. So, similarly, we can show that all the remaining (numbers) are also square.

And so let A be cube. I say that all the remaining (numbers) are also cube.

In fact, it has (already) been shown that the fourth (number) from the unit, C , is cube, and all those (numbers after that) which leave an interval of two (numbers) [Prop. 9.8]. [So] I say that all the remaining (numbers) are also cube. For since as the unit is to A , so A (is) to B , the unit thus measures A the same number of times as A (measures) B . And the unit measures A according to the units in it. Thus, A also measures B according to the units in (A). A has thus made B (by) multiplying itself. And A is cube. And if a cube number makes some (number by) multiplying itself then the created (num-

τὰ αὐτὰ δὴ καὶ ὁ E κύβος ἐστίν, καὶ ὁμοίως οἱ λοιποὶ πάντες κύβοι εἰσὶν· ὅπερ ἔδει δεῖχαι.

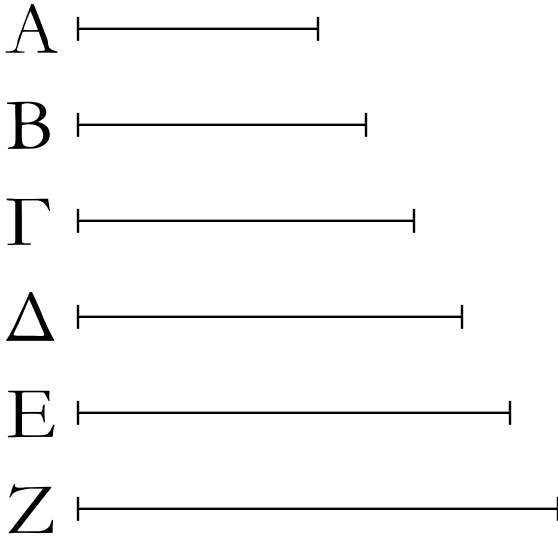
ber) is cube [Prop. 9.3]. Thus, B is also cube. And since the four numbers A, B, C, D are continuously proportional, and A is cube, D is thus also cube [Prop. 8.23]. So, for the same (reasons), E is also cube, and, similarly, all the remaining (numbers) are cube. (Which is) the very thing it was required to show.

10.

Proposition 10

Ἐάν ἀπὸ μονάδος ὁποσοιοῦν ἀριθμοὶ [ἐχῇ] ἀνάλογον ᾧσιν, ὁ δὲ μετὰ τὴν μονάδα μὴ ἢ τετράγωνος, οὐδ' ἄλλος οὐδεὶς τετράγωνος ἔσται θωρίξ τοῦ τρίτου ἀπὸ τῆς μονάδος καὶ τῶν ἑνα διαλειπόντων πάντων. καὶ ἔαν ὁ μετὰ τὴν μονάδα κύβος μὴ ἢ, οὐδὲ ἄλλος οὐδεὶς κύβος ἔσται θωρίξ τοῦ τετάρτου ἀπὸ τῆς μονάδος καὶ τῶν δύο διαλειπόντων πάντων.

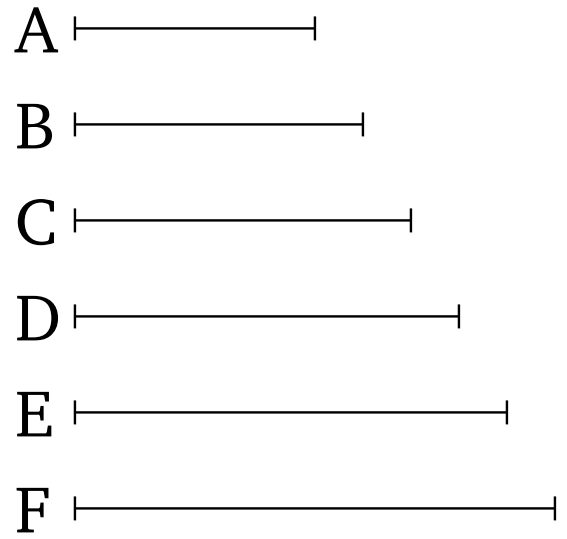
If any multitude whatsoever of numbers is [continuously] proportional, (starting) from a unit, and the (number) after the unit is not square, then no other (number) will be square either, apart from the third from the unit, and all those (numbers after that) which leave an interval of one (number). And if the (number) after the unit is not cube, then no other (number) will be cube either, apart from the fourth from the unit, and all those (numbers after that) which leave an interval of two (numbers).



Ἐστωσαν ἀπὸ μονάδος ἐχῇ ἀνάλογον ὅσοιδηποτοῦν ἀριθμοὶ οἱ $A, B, \Gamma, \Delta, E, Z$, ὁ μετὰ τὴν μονάδα ὁ A μὴ ἔστω τετράγωνος· λέγω, ὅτι οὐδὲ ἄλλος οὐδεὶς τετράγωνος ἔσται θωρίξ τοῦ τρίτου ἀπὸ τῆς μονάδος [καὶ τῶν ἑνα διαλειπόντων].

Εἰ γὰρ δυνατόν, ἔστω ὁ Γ τετράγωνος. ἔστι δὲ καὶ ὁ B τετράγωνος· οἱ B, Γ ἄρα πρὸς ἀλλήλους λόγον ἔθουσιν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ ἐστὶν ὡς ὁ B πρὸς τὸν Γ , ὁ A πρὸς τὸν B · οἱ A, B ἄρα πρὸς ἀλλήλους λόγον ἔθουσιν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ὥστε οἱ A, B ὅμοιοι ἐπίπεδοι εἰσιν. καὶ ἐστὶ τετράγωνος ὁ B · τετράγωνος ἄρα ἐστὶ καὶ ὁ A · ὅπερ οὐθ' ὑπέκειτο. οὐκ ἄρα ὁ Γ τετράγωνος ἐστίν. ὁμοίως δὲ δείχομεν, ὅτι οὐδ' ἄλλος οὐδεὶς τετράγωνος ἐστὶ θωρίξ τοῦ τρίτου ἀπὸ τῆς μονάδος καὶ τῶν ἑνα διαλειπόντων.

Ἀλλὰ δὴ μὴ ἔστω ὁ A κύβος. λέγω, ὅτι οὐδ' ἄλλος οὐδεὶς κύβος ἔσται θωρίξ τοῦ τετάρτου ἀπὸ τῆς μονάδος καὶ



Let any multitude whatsoever of numbers, A, B, C, D, E, F , be continuously proportional, (starting) from a unit. And let the (number) after the unit, A , not be square. I say that no other (number) will be square either, apart from the third from the unit [and (all) those (numbers after that) which leave an interval of one (number)].

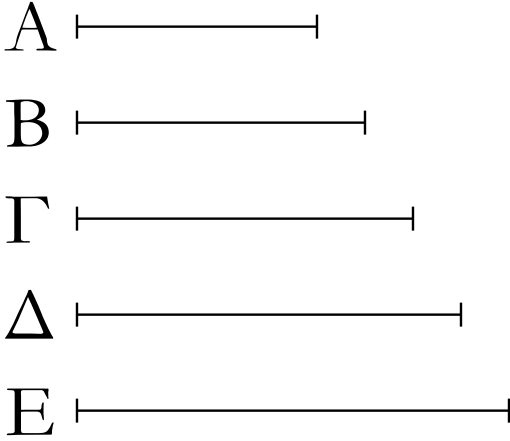
For, if possible, let C be square. And B is also square [Prop. 9.8]. Thus, B and C have to one another (the) ratio which (some) square number (has) to (some other) square number. And as B is to C , (so) A (is) to B . Thus, A and B have to one another (the) ratio which (some) square number has to (some other) square number. Hence, A and B are similar plane (numbers) [Prop. 8.26]. And B is square. Thus, A is also square. The very opposite thing was assumed. C is thus not square. So, similarly, we can show that no other (number is) square either, apart from the third from the unit, and

τῶν δύο διαλειπόντων.

Εἰ γὰρ δυνατόν, ἔστω ὁ Δ κύβος. ἔστι δὲ καὶ ὁ Γ κύβος· τέταρτος γάρ ἐστιν ἀπὸ τῆς μονάδος. καὶ ἐστὶν ὥς ὁ Γ πρὸς τὸν Δ , ὁ B πρὸς τὸν Γ · καὶ ὁ B ἄρα πρὸς τὸν Γ λόγον ἔθει, ὃν κύβος πρὸς κύβον. καὶ ἐστὶν ὁ Γ κύβος· καὶ ὁ B ἄρα κύβος ἐστίν. καὶ ἐπεὶ ἐστὶν ὥς ἡ μονὰς πρὸς τὸν A , ὁ A πρὸς τὸν B , ἡ δὲ μονὰς τὸν A μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας, καὶ ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· ὁ A ἄρα ἐαντὸν πολλαπλασιάσας κύβον τὸν B πεποίηκεν. ἔαν δὲ ἀριθμὸς ἐαντὸν πολλαπλασιάσας κύβον ποιῇ, καὶ αὐτὸς κύβος ἔσται. κύβος ἄρα καὶ ὁ A · ὅπερ οὐθ' ὑπόκειται. οὐκ ἄρα ὁ Δ κύβος ἐστίν. ὁμοίως δὲ δείκνυμεν, ὅτι οὐδ' ἄλλος οὐδείς κύβος ἐστὶ θωρίξ τοῦ τετάρτου ἀπὸ τῆς μονάδος καὶ τῶν δύο διαλειπόντων· ὅπερ ἔδει δεῖξαι.

11.

Ἐὰν ἀπὸ μονάδος ὁποσοῖον ἀριθμοὶ ἐχῇς ἀνάλογον ᾧσιν, ὁ ἐλάττων τὸν μείζονα μετρεῖ κατὰ τινὰ τῶν ὑπαρθόντων ἐν τοῖς ἀνάλογον ἀριθμοῖς.



Ἐστωσαν ἀπὸ μονάδος τῆς A ὁποσοῖον ἀριθμοὶ ἐχῇς ἀνάλογον οἱ B, Γ, Δ, E . λέγω, ὅτι τῶν B, Γ, Δ, E ὁ ἐλάχιστος ὁ B τὸν E μετρεῖ κατὰ τινὰ τῶν Γ, Δ .

Ἐπεὶ γάρ ἐστιν ὥς ἡ μονὰς πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E , ἰσάκις ἄρα ἡ A μονὰς τὸν B ἀριθμὸν μετρεῖ καὶ ὁ Δ τὸν E . ἐναλλάξ ἄρα ἰσάκις ἡ A μονὰς τὸν Δ μετρεῖ καὶ ὁ B τὸν E . ἡ δὲ A μονὰς τὸν Δ μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· καὶ ὁ B ἄρα τὸν E μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας· ὥστε ὁ ἐλάττωσιν ὁ B τὸν μείζονα τὸν E μετρεῖ κατὰ τινὰ ἀριθμὸν

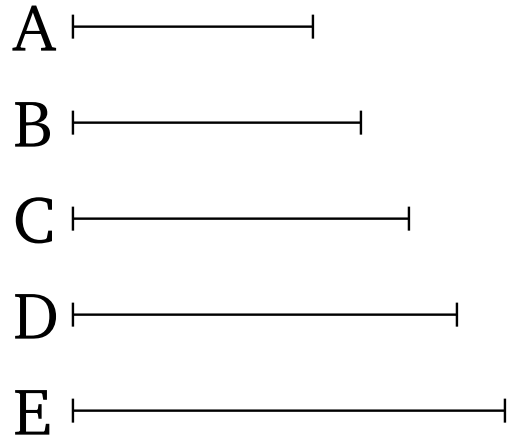
(all) those (numbers after that) which leave an interval of one (number).

And so let A not be cube. I say that no other (number) will be cube either, apart from the fourth from the unit, and (all) those (numbers after that) which leave an interval of two (numbers).

For, if possible, let D be cube. And C is also cube [Prop. 9.8]. For it is the fourth (number) from the unit. And as C is to D , (so) B (is) to C . And B thus has to C the ratio which (some) cube (number has) to (some other) cube (number). And C is cube. Thus, B is also cube [Props. 7.13, 8.25]. And since as the unit is to A , (so) A (is) to B , and the unit measures A according to the units in it, A thus also measures B according to the units in (A). Thus, A has made the cube (number) B (by) multiplying itself. And if a number makes a cube (number by) multiplying itself then it itself will be cube [Prop. 9.6]. Thus, A (is) also cube. The very opposite thing was assumed. Thus, D is not cube. So, similarly, we can show that no other (number) is cube either, apart from the fourth from the unit, and (all) those (numbers after that) which leave an interval of two (numbers). (Which is) the very thing it was required to show.

Proposition 11

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then a lesser (number) measures a greater according to some existing (number) among the proportional numbers.



Let any multitude whatsoever of numbers, B, C, D, E , be continuously proportional, (starting) from the unit A . I say that, for B, C, D, E , the least (number), B , measures E according to some (one) of C, D .

For since as the unit A is to B , so D (is) to E , the unit A thus measures the number B the same number of times as D (measures) E . Thus, alternately, the unit A measures D the same number of times as B (measures) E [Prop. 7.15]. And the unit A measures D according to the units in it. Thus, B

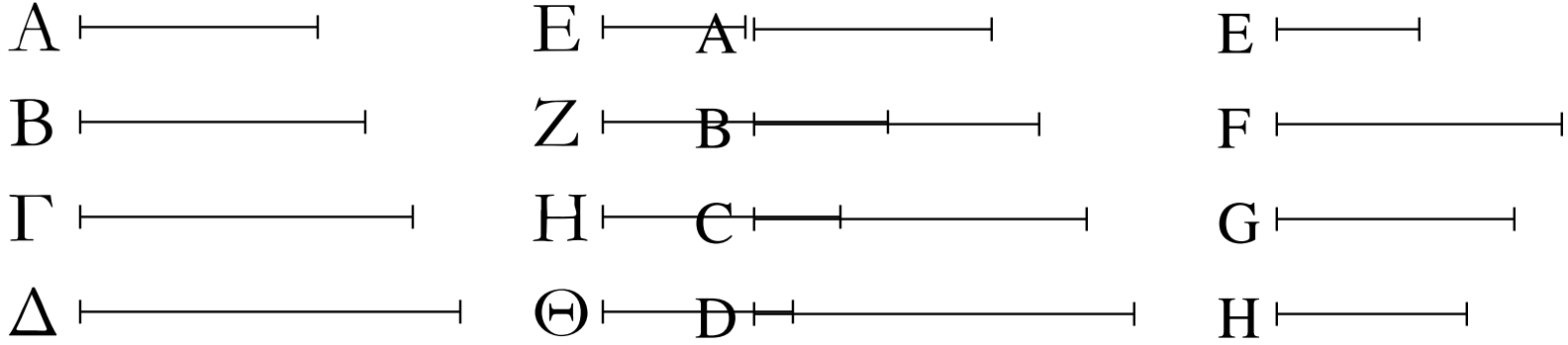
τῶν ὑπαρθόντων ἐν τοῖς ἀνάλογον ἀριθμοῖς.

Πόρισμα.

Καὶ φανερόν, ὅτι ἤν ἔθει τάχιν ὁ μετρῶν ἀπὸ μονάδος, τὴν αὐτὴν ἔθει καὶ ὁ καθ' ὃν μετρεῖ ἀπὸ τοῦ μετρούμενου ἐπὶ τὸ πρὸ αὐτοῦ. ὅπερ ἔδει δεῖξαι.

12.

Ἐάν ἀπὸ μονάδος ὁποσοῖον ἀριθμοὶ ἐχῇ ἀνάλογον ὦσιν, ὅφ' ὅσων ἂν ὁ ἕσθατος πρώτων ἀριθμῶν μετρηται, ὑπὸ τῶν αὐτῶν καὶ ὁ παρὰ τὴν μονάδα μετρηθήσεται.



Ἔστωσαν ἀπὸ μονάδος ὁποσοιδηποῦν ἀριθμοὶ ἀνάλογον οἱ A, B, Γ, Δ λέγω, ὅτι ὅφ' ὅσων ἂν ὁ Δ πρώτων ἀριθμῶν μετρηται, ὑπὸ τῶν αὐτῶν καὶ ὁ A μετρηθήσεται.

Μετρεῖσθω γάρ ὁ Δ ὑπὸ τινος πρώτου ἀριθμοῦ τοῦ E . λέγω, ὅτι ὁ E τὸν A μετρεῖ. μὴ γάρ· καὶ ἐστὶν ὁ E πρώτος, ἅπασι δὲ πρώτος ἀριθμὸς πρὸς ἅπαντα, ὃν μὴ μετρεῖ, πρῶτός ἐστιν· οἱ E, A ἄρα πρώτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Z . ὁ E ἄρα τὸν Z πολλαπλασιάσας τὸν Δ πεποίηκεν. πάλιν, ἐπεὶ ὁ E τὸν Δ μετρεῖ κατὰ τὰς ἐν τῷ Γ μονάδας, ὁ A ἄρα τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ E τὸν Z πολλαπλασιάσας τὸν Δ πεποίηκεν· ὁ ἄρα ἐκ τῶν A, Γ ἴσος ἐστὶ τῷ ἐκ τῶν E, Z . ἐστὶν ἄρα ὡς ὁ A πρὸς τὸν E , ὁ Z πρὸς τὸν Γ . οἱ δὲ A, E πρώτοι, οἱ δὲ πρώτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ἰσάκις ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον· μετρεῖ ἄρα ὁ E τὸν Γ . μετρεῖτω αὐτὸν κατὰ τὸν H . ὁ E ἄρα τὸν H πολλαπλασιάσας τὸν Γ πεποίηκεν. ἀλλὰ μὴν διὰ τὸ πρὸ τούτου καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν. ὁ ἄρα ἐκ τῶν A, B ἴσος ἐστὶ τῷ ἐκ τῶν E, H . ἐστὶν ἄρα ὡς ὁ A πρὸς τὸν E , ὁ H πρὸς τὸν B . οἱ δὲ A, E πρώτοι, οἱ δὲ πρώτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας αὐτοῖς ἰσάκις ὅ τε ἡγούμενος τὸν ἡγούμενον

also measures E according to the units in D . Hence, the lesser (number) B measures the greater E according to some existing number among the proportional numbers (namely, D).

Corollary

And (it is) clear that what(ever relative) place the measuring (number) has from the unit, the (number) according to which it measures has the same (relative) place from the measured (number), in (the direction of the number) before it. (Which is) the very thing it was required to show.

Proposition 12

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then however many prime numbers the last (number) is measured by, the (number) next to the unit will also be measured by the same (prime numbers).

Let any multitude whatsoever of numbers, A, B, C, D , be (continuously) proportional, (starting) from a unit. I say that however many prime numbers D is measured by, A will also be measured by the same (prime numbers).

For let D be measured by some prime number E . I say that E measures A . For (suppose it does) not. E is prime, and every prime number is prime to every number which it does not measure [Prop. 7.29]. Thus, E and A are prime to one another. And since E measures D , let it measure it according to F . Thus, E has made D (by) multiplying F . Again, since A measures D according to the units in C [Prop. 9.11 corr.], A has thus made D (by) multiplying C . But, in fact, E has also made D (by) multiplying F . Thus, the (number created) from (multiplying) E, C is equal to the (number created) from (multiplying) E, F . Thus, as A is to E , (so) F (is) to C [Prop. 7.19]. And A and E (are) prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures C . Let it measure it according to G . Thus, E has made C (by) multiplying G . But, in fact, via the (proposition) before this, A

καὶ ὁ ἐπόμενος τὸν ἐπόμενον μετρεῖ ἄρα ὁ E τὸν B . μετρεῖται αὐτὸν κατὰ τὸν Θ . ὁ E ἄρα τὸν Θ πολλαπλασιάσας τὸν B πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν. ὁ ἄρα ἐκ τῶν E , Θ ἴσος ἐστὶ τῷ ἀπὸ τοῦ A . ἔστιν ἄρα ὡς ὁ E πρὸς τὸν A , ὁ A πρὸς τὸν Θ . οἱ δὲ A , E πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάτισται, οἱ δὲ ἐλάτισται μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ἰσάκις ὁ ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον μετρεῖ ἄρα ὁ E τὸν A ὡς ἡγούμενος ἡγούμενον. ἀλλὰ μὴν καὶ οὐ μετρεῖ. ὅπερ ἀδύνατον. οὐκ ἄρα οἱ E , A πρῶτοι πρὸς ἀλλήλους εἰσίν. σύνθετοι ἄρα. οἱ δὲ σύνθετοι ὑπὸ [πρώτου] ἀριθμοῦ τινος μετροῦνται. καὶ ἐπεὶ ὁ E πρῶτος ὑπόκειται, ὁ δὲ πρῶτος ὑπὸ ἐτέρου ἀριθμοῦ οὐ μετρεῖται ἢ ὑφ' ἐαυτοῦ, ὁ E ἄρα τοὺς A , E μετρεῖ. ὥστε ὁ E τὸν A μετρεῖ. μετρεῖ δὲ καὶ τὸν Δ . ὁ E ἄρα τοὺς A , Δ μετρεῖ. ὁμοίως δὲ δείχομεν, ὅτι ὑφ' ὅσων ἂν ὁ Δ πρώτων ἀριθμῶν μετρηται, ὑπὸ τῶν αὐτῶν καὶ ὁ A μετρηθήσεται. ὅπερ ἔδει δεῖχαι.

has also made C (by) multiplying B [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) A , B is equal to the (number created) from (multiplying) E , G . Thus, as A is to E , (so) G (is) to B [Prop. 7.19]. And A and E (are) prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures B . Let it measure it according to H . Thus, E has made B (by) multiplying H . But, in fact, A has also made B (by) multiplying itself [Prop. 9.8]. Thus, the (number created) from (multiplying) E , H is equal to the (square) on A . Thus, as E is to A , (so) A (is) to H [Prop. 7.19]. And A and E are prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures A , as the leading (measuring the) leading. But, in fact, (E) also does not measure (A). The very thing (is) impossible. Thus, E and A are not prime to one another. Thus, (they are) composite (to one another). And (numbers) composite (to one another) are (both) measured by some [prime] number [Def. 7.14]. And since E is assumed (to be) prime, and a prime (number) is not measured by another number (other) than itself [Def. 7.11], E thus measures (both) A and E . Hence, E measures A . And it also measures D . Thus, E measures (both) A and D . So, similarly, we can show that however many prime numbers D is measured by, A will also be measured by the same (prime numbers). (Which is) the very thing it was required to show.

13.

Ἐάν ἀπὸ μονάδος ὅποσοιὼν ἀριθμοὶ ἐχῇς ἀνάλογον ᾧσιν, ὁ δὲ μετὰ τὴν μονάδα πρῶτος ᾗ, ὁ μέγιστος ὑπ' οὐδενὸς [ἀλλου] μετρηθήσεται παρέχ τῶν ὑπαρθόντων ἐν τοῖς ἀνάλογον ἀριθμοῖς.

Ἐστῶσαν ἀπὸ μονάδος ὅποσοιὼν ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A , B , Γ , Δ , ὁ δὲ μετὰ τὴν μονάδα ὁ A πρῶτος ἔστω. λέγω, ὅτι ὁ μέγιστος αὐτῶν ὁ Δ ὑπ' οὐδενὸς ἄλλον μετρηθήσεται παρέχ τῶν A , B , Γ .

Proposition 13

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, and the (number) after the unit is prime, then the greatest (number) will be measured by no [other] (numbers) except (numbers) existing among the proportional numbers.

Let any multitude whatsoever of numbers, A , B , C , D , be continuously proportional, (starting) from a unit. And let the (number) after the unit, A , be prime. I say that the greatest of them, D , will be measured by no other (numbers) except A , B , C .

A ———|

E ———|A|———|

E ———|

B ———|

Z ———|B|———|

F ———|

Γ ———|

H ———|Ε|———|

G ———|

Δ ———|

Θ ———|D|———|

H ———|

Εἴ γάρ δυνατόν, μετρεῖσθω ὑπὸ τοῦ E , καὶ ὁ E μηδενὶ τῶν A, B, Γ ἔστω ὁ αὐτόξ. φανερόν δὴ, ὅτι ὁ E πρωτοξ οὐκ ἔστιν. εἰ γάρ ὁ E πρωτόξ ἐστὶ καὶ μετρεῖ τὸν Δ , καὶ τὸν A μετρήσει πρωτὸν ὄντα μὴ ὡν αὐτῷ ὁ αὐτόξ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ E πρωτόξ ἐστὶν. σύνθετοξ ἄρα. πᾶξ δὲ σύνθετοξ ἀριθμὸξ ὑπὸ πρώτου τινὸξ ἀριθμοῦ μετρεῖται· ὁ E ἄρα ὑπὸ πρώτου τινὸξ ἀριθμοῦ μετρεῖται. λέγω δὴ, ὅτι ὑπ' οὐδενὸξ ἄλλου πρώτου μετρηθήσεται πλὴν τοῦ A . εἰ γάρ ὑφ' ἑτέρου μετρεῖται ὁ E , ὁ δὲ E τὸν Δ μετρεῖ, καὶ κεῖνοξ ἄρα τὸν Δ μετρήσει· ὥστε καὶ τὸν A μετρήσει πρωτὸν ὄντα μὴ ὡν αὐτῷ ὁ αὐτόξ· ὅπερ ἐστὶν ἀδύνατον. ὁ A ἄρα τὸν E μετρεῖ. καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Z . λέγω, ὅτι ὁ Z οὐδενὶ τῶν A, B, Γ ἐστὶν ὁ αὐτόξ. εἰ γάρ ὁ Z ἐνὶ τῶν A, B, Γ ἐστὶν ὁ αὐτόξ καὶ μετρεῖ τὸν Δ κατὰ τὸν E , καὶ εἷξ ἄρα τῶν A, B, Γ τὸν Δ μετρεῖ κατὰ τὸν E . ἀλλὰ εἷξ τῶν A, B, Γ τὸν Δ μετρεῖ κατὰ τινὰ τῶν A, B, Γ · καὶ ὁ E ἄρα ἐνὶ τῶν A, B, Γ ἐστὶν ὁ αὐτόξ· ὅπερ οὐθ' ὑπόκειται. οὐκ ἄρα ὁ Z ἐνὶ τῶν A, B, Γ ἐστὶν ὁ αὐτόξ. ὁμοίωξ δὴ δείχομεν, ὅτι μετρεῖται ὁ Z ὑπὸ τοῦ A , δεικνύντεξ πάλιν, ὅτι ὁ Z οὐκ ἐστὶ πρωτόξ. εἰ γάρ, καὶ μετρεῖ τὸν Δ , καὶ τὸν A μετρήσει πρωτὸν ὄντα μὴ ὡν αὐτῷ ὁ αὐτόξ· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα πρωτόξ ἐστὶν ὁ Z · σύνθετοξ ἄρα. ἅπαξ δὲ σύνθετοξ ἀριθμὸξ ὑπὸ πρώτου τινὸξ ἀριθμοῦ μετρεῖται· ὁ Z ἄρα ὑπὸ πρώτου τινὸξ ἀριθμοῦ μετρεῖται. λέγω δὴ, ὅτι ὑφ' ἑτέρου πρώτου οὐ μετρηθήσεται πλὴν τοῦ A . εἰ γάρ ἑτεροξ τιξ πρωτοξ τὸν Z μετρεῖ, ὁ δὲ Z τὸν Δ μετρεῖ, καὶ κεῖνοξ ἄρα τὸν Δ μετρήσει· ὥστε καὶ τὸν A μετρήσει πρωτὸν ὄντα μὴ ὡν αὐτῷ ὁ αὐτόξ· ὅπερ ἐστὶν ἀδύνατον. ὁ A ἄρα τὸν Z μετρεῖ. καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ κατὰ τὸν Z , ὁ E ἄρα τὸν Z πολλαπλασιάσας τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν· ὁ ἄρα ἐκ τῶν A, Γ ἴσοξ ἐστὶ τῷ ἐκ τῶν E, Z . ἀνάλογον ἄρα ἐστὶν ὡς ὁ A πρὸξ τὸν E , οὕτωξ ὁ Z πρὸξ τὸν Γ . ὁ δὲ A τὸν E μετρεῖ· καὶ ὁ Z ἄρα τὸν Γ μετρεῖ. μετρεῖτω αὐτὸν κατὰ τὸν H . ὁμοίωξ δὴ δείχομεν, ὅτι ὁ H οὐδενὶ τῶν A, B ἐστὶν ὁ αὐτόξ, καὶ ὅτι μετρεῖται ὑπὸ τοῦ A . καὶ ἐπεὶ ὁ Z τὸν Γ μετρεῖ κατὰ τὸν H , ὁ Z ἄρα τὸν H πολλαπλασιάσας τὸν Γ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν· ὁ ἄρα ἐκ τῶν A, B ἴσοξ ἐστὶ τῷ ἐκ τῶν Z, H . ἀνάλογον ἄρα ὡς ὁ A πρὸξ τὸν Z , ὁ H πρὸξ τὸν B . μετρεῖ δὲ ὁ A τὸν Z · μετρεῖ ἄρα καὶ ὁ H τὸν B . μετρεῖτω αὐτὸν

For, if possible, let it be measured by E , and let E not be the same as one of A, B, C . So it is clear that E is not prime. For if E is prime, and measures D , then it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, E is not prime. Thus, (it is) composite. And every composite number is measured by some prime number [Prop. 7.31]. Thus, E is measured by some prime number. So I say that it will be measured by no other prime number than A . For if E is measured by another (prime number), and E measures D , then this (prime number) will thus also measure D . Hence, it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, A measures E . And since E measures D , let it measure it according to F . I say that F is not the same as one of A, B, C . For if F is the same as one of A, B, C , and measures D according to E , then one of A, B, C thus also measures D according to E . But one of A, B, C (only) measures D according to some (one) of A, B, C [Prop. 9.11]. And thus E is the same as one of A, B, C . The very opposite thing was assumed. Thus, F is not the same as one of A, B, C . Similarly, we can show that F is measured by A , (by) again showing that F is not prime. For if (F is prime), and measures D , then it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, F is not prime. Thus, (it is) composite. And every composite number is measured by some prime number [Prop. 7.31]. Thus, F is measured by some prime number. So I say that it will be measured by no other prime number than A . For if some other prime (number) measures F , and F measures D , then this (prime number) will thus also measure D . Hence, it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, A measures F . And since E measures D according to F , E has thus made D (by) multiplying F . But, in fact, A has also made D (by) multiplying C [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) A, C is equal to the (number created) from (multiplying) E, F . Thus, proportionally, as A is to E , so F (is) to C [Prop. 7.19]. And A measures E . Thus, F also measures C . Let it measure it according to G . So, similarly,

κατὰ τὸν Θ . ὁμοίως δὴ δείχομεν, ὅτι ὁ Θ τῷ A οὐκ ἔστιν ὁ αὐτός. καὶ ἐπεὶ ὁ H τὸν B μετρεῖ κατὰ τὸν Θ , ὁ H ἄρα τὸν Θ πολλαπλασιάσας τὸν B πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν· ὁ ἄρα ὑπὸ Θ , H ἴσος ἐστὶ τῷ ἀπὸ τοῦ A τετραγώνῳ· ἔστιν ἄρα ὡς ὁ Θ πρὸς τὸν A , ὁ A πρὸς τὸν H . μετρεῖ δὲ ὁ A τὸν H · μετρεῖ ἄρα καὶ ὁ Θ τὸν A πρῶτον ὄντα μὴ ὡν αὐτῷ ὁ αὐτός· ὅπερ ἄτοπον. οὐκ ἄρα ὁ μέγιστος ὁ Δ ὑπὸ ἐτέρου ἀριθμοῦ μετρηθήσεται παρέχ τῶν A , B , Γ . ὅπερ ἔδει δεῖχαι.

we can show that G is not the same as one of A , B , and that it is measured by A . And since F measures C according to G , F has thus made C (by) multiplying G . But, in fact, A has also made C (by) multiplying B [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) A , B is equal to the (number created) from (multiplying) F , G . Thus, proportionally, as A (is) to F , so G (is) to B [Prop. 7.19]. And A measures F . Thus, G also measures B . Let it measure it according to H . So, similarly, we can show that H is not the same as A . And since G measures B according to H , G has thus made B (by) multiplying H . But, in fact, A has also made B (by) multiplying itself [Prop. 9.8]. Thus, the (number created) from (multiplying) H , G is equal to the square on A . Thus, as H is to A , (so) A (is) to G [Prop. 7.19]. And A measures G . Thus, H also measures A , (despite A) being prime (and) not being the same as it. The very thing (is) absurd. Thus, the greatest (number) D cannot be measured by another (number) except (one of) A , B , C . (Which is) the very thing it was required to show.

14.

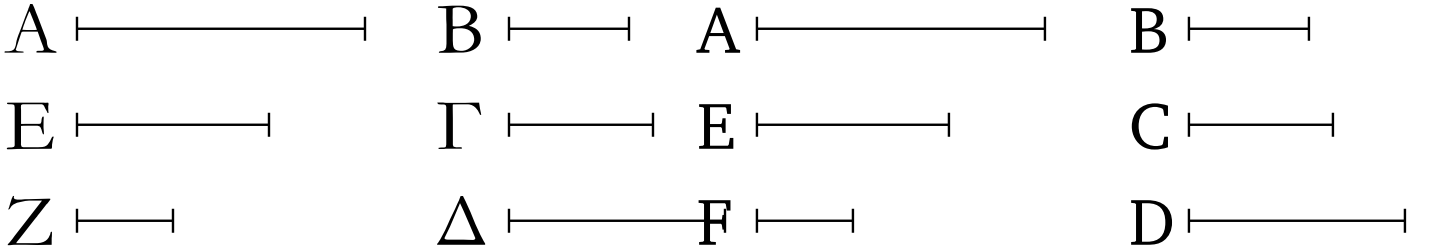
Proposition 14

Ἐὰν ἐλάχιστος ἀριθμὸς ὑπὸ πρώτων ἀριθμῶν μετρήται, ὑπ' οὐδενὸς ἄλλον πρώτου ἀριθμοῦ μετρηθήσεται παρέχ τῶν ἐκ ἀρθῆς μετρούντων.

If a least number is measured by (some) prime numbers then it will not be measured by any other prime number except (one of) the original measuring (numbers).

Ἐλάχιστος γάρ ἀριθμὸς ὁ A ὑπὸ πρώτων ἀριθμῶν τῶν B , Γ , Δ μετρεῖσθω· λέγω, ὅτι ὁ A ὑπ' οὐδενὸς ἄλλον πρώτου ἀριθμοῦ μετρηθήσεται παρέχ τῶν B , Γ , Δ .

For let A be the least number measured by the prime numbers B , C , D . I say that A will not be measured by any other prime number except (one of) B , C , D .



Εἰ γὰρ δυνατόν, μετρεῖσθω ὑπὸ πρώτου τοῦ E , καὶ ὁ E μηδενὶ τῶν B , Γ , Δ ἔστω ὁ αὐτός. καὶ ἐπεὶ ὁ E τὸν A μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Z · ὁ E ἄρα τὸν Z πολλαπλασιάσας τὸν A πεποίηκεν. καὶ μετρεῖται ὁ A ὑπὸ πρώτων ἀριθμῶν τῶν B , Γ , Δ . ἐὰν δὲ δύο ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλους ποιῶσί τινα, τὸν δὲ γεγόμενον ἐκ αὐτῶν μετρή τις πρῶτος ἀριθμὸς, καὶ ἓνα τῶν ἐκ ἀρθῆς μετρήσει· οἱ B , Γ , Δ ἄρα ἓνα τῶν E , Z μετρήσουσιν. τὸν μὲν οὖν E οὐ μετρήσουσιν· ὁ γὰρ E πρῶτός ἐστι καὶ οὐδενὶ τῶν B , Γ , Δ ὁ αὐτός. τὸν Z ἄρα μετροῦσιν ἐλάσσονα ὄντα τοῦ A · ὅπερ ἀδύνατον. ὁ γὰρ A ὑπόκειται ἐλάχιστος ὑπὸ τῶν B , Γ , Δ μετρούμενος. οὐκ ἄρα τὸν A μετρήσει πρῶτος ἀριθμὸς παρέχ τῶν B , Γ , Δ · ὅπερ ἔδει δεῖχαι.

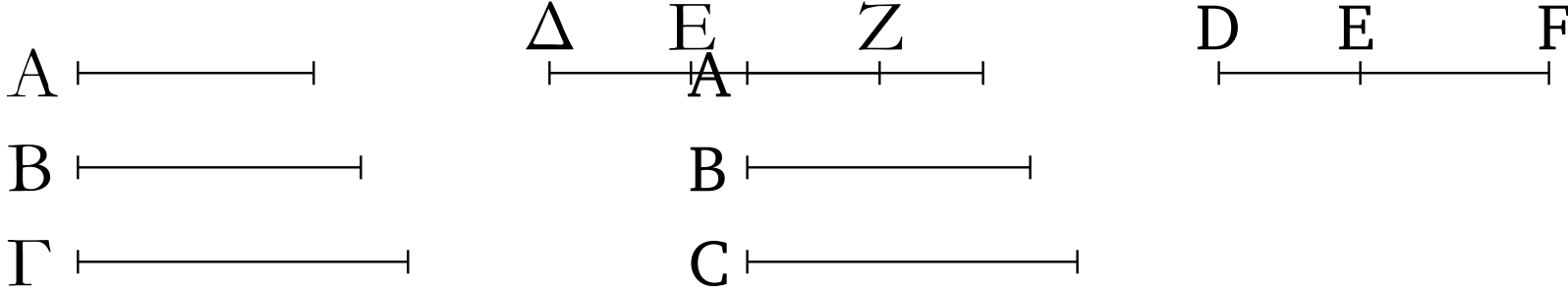
For, if possible, let it be measured by the prime (number) E . And let E not be the same as one of B , C , D . And since E measures A , let it measure it according to F . Thus, E has made A (by) multiplying F . And A is measured by the prime numbers B , C , D . And if two numbers make some (number by) multiplying one another, and some prime number measures the number created from them, then (the prime number) will also measure one of the original (numbers) [Prop. 7.30]. Thus, B , C , D will measure one of E , F . In fact, they do not measure E . For E is prime, and not the same as one of B , C , D . Thus, they (all) measure F , which is less than A . The very thing (is) impossible. For A was assumed (to be) the least (number) measured by B , C , D . Thus, no prime number can measure A except (one of) B , C , D . (Which is) the very thing it was required to show.

15.

Proposition 15

Ἐάν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ὥσιν ἐλάτισται τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, δύο ὁποιοῦν συντεθέντες πρὸς τὸν λοιπὸν πρῶτοί εἰσιν.

If three continuously proportional numbers are the least of those (numbers) having the same ratio as them then two (of them) added together in any way are prime to the remaining (one).



Ἐστωσαν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ἐλάτισται τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς οἱ A, B, Γ . λέγω, ὅτι τῶν A, B, Γ δύο ὁποιοῦν συντεθέντες πρὸς τὸν λοιπὸν πρῶτοί εἰσιν, οἱ μὲν A, B πρὸς τὸν Γ , οἱ δὲ B, Γ πρὸς τὸν A καὶ ἔτι οἱ A, Γ πρὸς τὸν B .

Let A, B, C be three continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them. I say that two of A, B, C added together in any way are prime to the remaining (one), (that is) A and B (prime) to C , B and C to A , and, further, A and C to B .

Εἰλήφθωσαν γὰρ ἐλάτισται ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ δύο οἱ $\Delta E, EZ$. φανερόν δὲ, ὅτι ὁ μὲν ΔE ἐαντὸν πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ EZ πολλαπλασιάσας τὸν B πεποίηκεν, καὶ ἔτι ὁ EZ ἐαντὸν πολλαπλασιάσας τὸν Γ πεποίηκεν. καὶ ἐπεὶ οἱ $\Delta E, EZ$ ἐλάτισταί εἰσιν, πρῶτοί πρὸς ἀλλήλους εἰσίν. ἔαν δὲ δύο ἀριθμοὶ πρῶτοί πρὸς ἀλλήλους ὦσιν, καὶ συναμφότεροξ πρὸς ἑκάτερον πρῶτός ἐστιν· καὶ ὁ ΔZ ἄρα πρὸς ἑκάτερον τῶν $\Delta E, EZ$ πρῶτός ἐστιν. ἀλλὰ μὴν καὶ ὁ ΔE πρὸς τὸν EZ πρῶτός ἐστιν· οἱ $\Delta Z, \Delta E$ ἄρα πρὸς τὸν EZ πρῶτοί εἰσιν. ἔαν δὲ δύο ἀριθμοὶ πρὸς τινὰ ἀριθμὸν πρῶτοί ὦσιν, καὶ ὁ ἐκ αὐτῶν γενόμενος πρὸς τὸν λοιπὸν πρῶτός ἐστιν· ὥστε ὁ ἐκ τῶν $Z\Delta, \Delta E$ πρὸς τὸν EZ πρῶτός ἐστιν· ὥστε καὶ ὁ ἐκ τῶν $Z\Delta, \Delta E$ πρὸς τὸν ἀπὸ τοῦ EZ πρῶτός ἐστιν. [ἔαν γὰρ δύο ἀριθμοὶ πρῶτοί πρὸς ἀλλήλους ὦσιν, ὁ ἐκ τοῦ ἐνόξ αὐτῶν γενόμενος πρὸς τὸν λοιπὸν πρῶτός ἐστιν]. ἀλλ' ὁ ἐκ τῶν $Z\Delta, \Delta E$ ὁ ἀπὸ τοῦ ΔE ἐστὶ μετὰ τοῦ ἐκ τῶν $\Delta E, EZ$ · ὁ ἄρα ἀπὸ τοῦ ΔE μετὰ τοῦ ἐκ τῶν $\Delta E, EZ$ πρὸς τὸν ἀπὸ τοῦ EZ πρῶτός ἐστιν. καὶ ἐστὶν ὁ μὲν ἀπὸ τοῦ ΔE ὁ A , ὁ δὲ ἐκ τῶν $\Delta E, EZ$ ὁ B , ὁ δὲ ἀπὸ τοῦ EZ ὁ Γ . οἱ A, B ἄρα συντεθέντες πρὸς τὸν Γ πρῶτοί εἰσιν. ὁμοίως δὲ δείχουμεν, ὅτι καὶ οἱ B, Γ πρὸς τὸν A πρῶτοί εἰσιν. λέγω δὲ, ὅτι καὶ οἱ A, Γ πρὸς τὸν B πρῶτοί εἰσιν. ἐπεὶ γὰρ ὁ ΔZ πρὸς ἑκάτερον τῶν $\Delta E, EZ$ πρῶτός ἐστιν, καὶ ὁ ἀπὸ τοῦ ΔZ πρὸς τὸν ἐκ τῶν $\Delta E, EZ$ πρῶτός ἐστιν. ἀλλὰ τῷ ἀπὸ τοῦ ΔZ ἴσοι εἰσίν οἱ ἀπὸ τῶν $\Delta E, EZ$ μετὰ τοῦ διξ ἐκ τῶν $\Delta E, EZ$ · καὶ οἱ ἀπὸ τῶν $\Delta E, EZ$ ἄρα μετὰ τοῦ διξ ὑπὸ τῶν $\Delta E, EZ$ πρὸς τὸν ὑπὸ τῶν $\Delta E, EZ$ πρῶτοί [εἰσι]. διελόντι οἱ ἀπὸ τῶν $\Delta E, EZ$ μετὰ τοῦ ἀπαχ ὑπὸ $\Delta E, EZ$ πρὸς τὸν ὑπὸ $\Delta E, EZ$ πρῶτοί εἰσιν. ἔτι διελόντι οἱ ἀπὸ τῶν $\Delta E, EZ$ ἄρα πρὸς τὸν ὑπὸ $\Delta E, EZ$ πρῶτοί εἰσιν. καὶ ἐστὶν ὁ μὲν ἀπὸ τοῦ ΔE ὁ A , ὁ δὲ ὑπὸ τῶν $\Delta E, EZ$ ὁ B ,

Let the two least numbers, DE and EF , having the same ratio as A, B, C , be taken [Prop. 8.2]. So it is clear that DE has made A (by) multiplying itself, and has made B (by) multiplying EF , and, further, EF has made C (by) multiplying itself [Prop. 8.2]. And since DE, EF are the least (of those numbers having the same ratio as them), they are prime to one another [Prop. 7.22]. And if two numbers are prime to one another then the sum (of them) is also prime to each [Prop. 7.28]. Thus, DF is also prime to each of DE, EF . But, in fact, DE is also prime to EF . Thus, DF, DE are (both) prime to EF . And if two numbers are (both) prime to some number then the (number) created from (multiplying) them is also prime to the remaining (number) [Prop. 7.24]. Hence, the (number created) from (multiplying) FD, DE is prime to EF . Hence, the (number created) from (multiplying) FD, DE is also prime to the (square) on EF [Prop. 7.25]. [For if two numbers are prime to one another then the (number) created from (squaring) one of them is prime to the remaining (number).] But the (number created) from (multiplying) FD, DE is the (square) on DE plus the (number created) from (multiplying) DE, EF [Prop. 2.3]. Thus, the (square) on DE plus the (number created) from (multiplying) DE, EF is prime to the (square) on EF . And the (square) on DE is A , and the (number created) from (multiplying) DE, EF (is) B , and the (square) on EF (is) C . Thus, A, B summed is prime to C . So, similarly, we can show that B, C (summed) is also prime to A . So I say that A, C (summed) is also prime to B . For since DF is prime to each of DE, EF then the (square) on DF is also prime to the (number created) from (multiplying) DE, EF [Prop. 7.25]. But, the (sum of the squares) on DE, EF plus twice the (number created) from (multiplying) DE, EF is equal to the (square) on

ὁ δὲ ἀπὸ τοῦ EZ ὁ Γ . οἱ A , Γ ἄρα συντεθέντες πρὸς τὸν B πρῶτοι εἰσιν· ὅπερ ἔδει δεῖχαι.

DF [Prop. 2.4]. And thus the (sum of the squares) on DE , EF plus twice the (rectangle contained) by DE , EF [is] prime to the (rectangle contained) by DE , EF . By separation, the (sum of the squares) on DE , EF plus once the (rectangle contained) by DE , EF is prime to the (rectangle contained) by DE , EF .[†] Again, by separation, the (sum of the squares) on DE , EF is prime to the (rectangle contained) by DE , EF . And the (square) on DE is A , and the (rectangle contained) by DE , EF (is) B , and the (square) on EF (is) C . Thus, A , C summed is prime to B . (Which is) the very thing it was required to show.

[†] Since if $\alpha\beta$ measures $\alpha^2 + \beta^2 + 2\alpha\beta$ then it also measures $\alpha^2 + \beta^2 + \alpha\beta$, and vice versa.

16.

Ἐάν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλων ᾧσιν, οὐκ ἔσται ὡς ὁ πρῶτος πρὸς τὸν δεύτερον, οὕτως ὁ δεύτερος πρὸς ἄλλον τινά.



Δύο γὰρ ἀριθμοὶ οἱ A , B πρῶτοι πρὸς ἀλλήλων ἔστωσαν· λέγω, ὅτι οὐκ ἔστιν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ B πρὸς ἄλλον τινά.

Εἰ γὰρ δυνατόν, ἔστω ὡς ὁ A πρὸς τὸν B , ὁ B πρὸς τὸν Γ . οἱ δὲ A , B πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ἰσάκεις ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον· μετρεῖ ἄρα ὁ A τὸν B ὡς ἡγούμενος ἡγούμενον. μετρεῖ δὲ καὶ ἐαντόν· ὁ A ἄρα τοὺς A , B μετρεῖ πρῶτον· ὅντα πρὸς ἀλλήλους· ὅπερ ἄτοπον. οὐκ ἄρα ἔσται ὡς ὁ A πρὸς τὸν B , οὕτως ὁ B πρὸς τὸν Γ . ὅπερ ἔδει δεῖχαι.

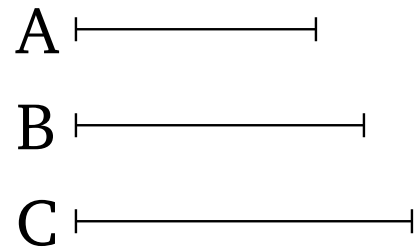
17.

Ἐάν ᾧσιν ὁσοιδηποτοῖν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ δὲ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους ᾧσιν, οὐκ ἔσται ὡς ὁ πρῶτος πρὸς τὸν δεύτερον, οὕτως ὁ ἔσθατος πρὸς ἄλλον τινά.

Ἐστωσαν ὁσοιδηποτοῖν ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A , B , Γ , Δ , οἱ δὲ ἄκροι αὐτῶν οἱ A , Δ πρῶτοι πρὸς ἀλλήλους ἔστωσαν· λέγω, ὅτι οὐκ ἔστιν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς ἄλλον τινά.

Proposition 16

If two numbers are prime to one another then as the first is to the second, so the second (will) not (be) to some other (number).



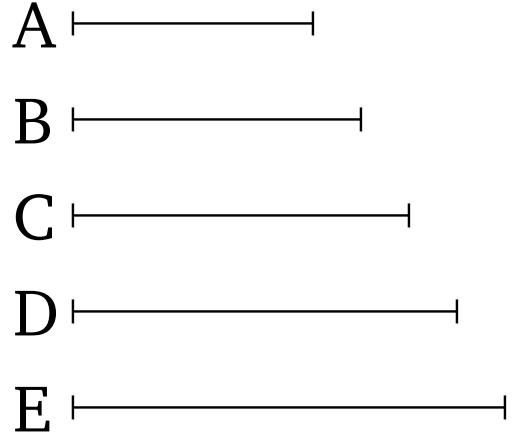
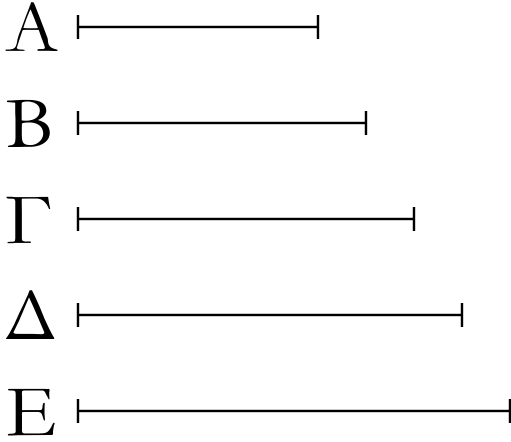
For let the two numbers A and B be prime to one another. I say that as A is to B , so B is not to some other (number).

For, if possible, let it be that as A (is) to B , (so) B (is) to C . And A and B (are) prime (to one another). And (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures B , as the leading (measuring) the leading. And (A) also measures itself. Thus, A measures A and B , which are prime to one another. The very thing (is) absurd. Thus, as A (is) to B , so B cannot be to C . (Which is) the very thing it was required to show.

Proposition 17

If any multitude whatsoever of numbers is continuously proportional, and the outermost of them are prime to one another, then as the first (is) to the second, so the last will not be to some other (number).

Let A , B , C , D be any multitude whatsoever of continuously proportional numbers. And let the outermost of them, A and D , be prime to one another. I say that as A is to B , so D (is) not to some other (number).



Εἰ γὰρ δυνατόν, ἔστω ὥς ὁ A πρόξ τὸν B , οὕτως ὁ Δ πρόξ τὸν E . ἐναλλάξ ἄρα ἐστὶν ὥς ὁ A πρόξ τὸν Δ , ὁ B πρόξ τὸν E . οἱ δὲ A , Δ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔθονταξ ἰσάκις ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. μετρεῖ ἄρα ὁ A τὸν B . καὶ ἐστὶν ὥς ὁ A πρόξ τὸν B , ὁ B πρόξ τὸν Γ . καὶ ὁ B ἄρα τὸν Γ μετρεῖ· ὥστε καὶ ὁ A τὸν Γ μετρεῖ. καὶ ἐπεὶ ἐστὶν ὥς ὁ B πρόξ τὸν Γ , ὁ Γ πρόξ τὸν Δ , μετρεῖ δὲ ὁ B τὸν Γ , μετρεῖ ἄρα καὶ ὁ Γ τὸν Δ . ἀλλ' ὁ A τὸν Γ ἐμέτρει· ὥστε ὁ A καὶ τὸν Δ μετρεῖ. μετρεῖ δὲ καὶ ἑαυτόν. ὁ A ἄρα τοὺς A , Δ μετρεῖ πρῶτονξ ὄνταξ πρόξ ἀλλήλων· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἔσται ὥς ὁ A πρόξ τὸν B , οὕτως ὁ Δ πρόξ ἄλλον τινά· ὅπερ ἔδει δεῖχαι.

For, if possible, let it be that as A (is) to B , so D (is) to E . Thus, alternately, as A is to D , (so) B (is) to E [Prop. 7.13]. And A and D are prime (to one another). And (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures B . And as A is to B , (so) B (is) to C . Thus, B also measures C . And hence A measures C [Def. 7.20]. And since as B is to C , (so) C (is) to D , and B measures C , C thus also measures D [Def. 7.20]. But, A was (found to be) measuring C . And hence A also measures D . And (A) also measures itself. Thus, A measures A and D , which are prime to one another. The very thing is impossible. Thus, as A (is) to B , so D cannot be to some other (number). (Which is) the very thing it was required to show.

18.

Δύο ἀριθμῶν δοθέντων ἐπισκέψασθαι, εἰ δυνατόν ἐστὶν αὐτοῖξ τρίτον ἀνάλογον προσευρεῖν.

Proposition 18

For two given numbers, to investigate whether it is possible to find a third (number) proportional to them.

Ἐστωσαν οἱ δοθέντες δύο ἀριθμοὶ οἱ A, B , καὶ δέον ἔστω ἐπισκέψασθαι, εἰ δυνατόν ἐστιν αὐτοῖς τρίτον ἀνάλογον προσευρεῖν.

Οἱ δὴ A, B ἢτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν ἢ οὐ. καὶ εἰ πρῶτοι πρὸς ἀλλήλους εἰσὶν, δέδεικται, ὅτι ἀδύνατόν ἐστιν αὐτοῖς τρίτον ἀνάλογον προσευρεῖν.

Ἀλλὰ δὴ μὴ ἔστωσαν οἱ A, B πρῶτοι πρὸς ἀλλήλους, καὶ ὁ B ἐαυτὸν πολλαπλασιάσας τὸν Γ ποιεῖτω. ὁ A δὴ τὸν Γ ἢτοι μετρεῖ ἢ οὐ μετρεῖ. μετρεῖτω πρότερον κατὰ τὸν Δ . ὁ A ἄρα τὸν Δ πολλαπλασιάσας τὸν Γ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ B ἐαυτὸν πολλαπλασιάσας τὸν Γ πεποίηκεν. ὁ ἄρα ἐκ τῶν A, Δ ἴσος ἐστὶ τῷ ἀπὸ τοῦ B . ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , ὁ B πρὸς τὸν Δ . τοῖς A, B ἄρα τρίτος ἀριθμὸς ἀνάλογον προσηγήσεται ὁ Δ .

Ἀλλὰ δὴ μὴ μετρεῖτω ὁ A τὸν Γ . λέγω, ὅτι τοῖς A, B ἀδύνατόν ἐστι τρίτον ἀνάλογον προσευρεῖν ἀριθμόν. εἰ γάρ δυνατόν, προσηγήσθω ὁ Δ . ὁ ἄρα ἐκ τῶν A, Δ ἴσος ἐστὶ τῷ ἀπὸ τοῦ B . ὁ δὲ ἀπὸ τοῦ B ἐστὶν ὁ Γ . ὁ ἄρα ἐκ τῶν A, Δ ἴσος ἐστὶ τῷ Γ . ὥστε ὁ A τὸν Δ πολλαπλασιάσας τὸν Γ πεποίηκεν. ὁ A ἄρα τὸν Γ μετρεῖ κατὰ τὸν Δ . ἀλλὰ μὴν ὑπόκειται καὶ μὴ μετρῶν. ὅπερ ἄτοπον. οὐκ ἄρα δυνατόν ἐστι τοῖς A, B τρίτον ἀνάλογον προσευρεῖν ἀριθμόν, ὅταν ὁ A τὸν Γ μὴ μετρῇ. ὅπερ ἔδει δεῖχαι.

Let A and B be the two given numbers. And let it be required to investigate whether it is possible to find a third (number) proportional to them.

So A and B are either prime to one another, or not. And if they are prime to one another then it has (already) been shown that it is impossible to find a third (number) proportional to them [Prop. 9.16].

And so let A and B not be prime to one another. And let B make C (by) multiplying itself. So A either measures, or does not measure, C . Let it first of all measure (C) according to D . Thus, A has made C (by) multiplying D . But, in fact, B has also made C (by) multiplying itself. Thus, the (number created) from (multiplying) A, D is equal to the (square) on B . Thus, as A is to B , (so) B (is) to D [Prop. 7.19]. Thus, a third number has been found proportional to A, B , (namely) D .

And so let A not measure C . I say that it is impossible to find a third number proportional to A, B . For, if possible, let it be found, (and let it be) D . Thus, the (number created) from (multiplying) A, D is equal to the (square) on B [Prop. 7.19]. And the (square) on B is C . Thus, the (number created) from (multiplying) A, D is equal to C . Hence, A has made C (by) multiplying D . Thus, A measures C according to D . But (A) was, in fact, also assumed (to be) not measuring (C). The very thing (is) absurd. Thus, it is not possible to find a third number proportional to A, B when A does not measure C . (Which is) the very thing it was required to show.

19.

Τριῶν ἀριθμῶν δοθέντων ἐπισκέψασθαι, πότε δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσευρεῖν.

Ἐστωσαν οἱ δοθέντες τρεῖς ἀριθμοὶ οἱ A, B, Γ , κα-

Proposition 19[†]

For three given numbers, to investigate when it is possible to find a fourth (number) proportional to them.

Let A, B, C be the three given numbers. And let it be re-

ἰ δέον ἔστω επισκέψασθαι, πότε δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσσευρεῖν.

Ἦτοι οὖν οὐκ εἰσιν ἐχῆς ἀνάλογον, καὶ οἱ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν, ἢ ἐχῆς εἰσιν ἀνάλογον, καὶ οἱ ἄκροι αὐτῶν οὐκ εἰσιν πρῶτοι πρὸς ἀλλήλους, ἢ οὔτε ἐχῆς εἰσιν ἀνάλογον, οὔτε οἱ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν, ἢ καὶ ἐχῆς εἰσιν ἀνάλογον, καὶ οἱ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν.

Εἰ μὲν οὖν οἱ A, B, Γ ἐχῆς εἰσιν ἀνάλογον, καὶ οἱ ἄκροι αὐτῶν οἱ A, Γ πρῶτοι πρὸς ἀλλήλους εἰσίν, δέδεικται, ὅτι ἀδύνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσσευρεῖν ἀριθμόν. μὴ ἔστωσαν δὴ οἱ A, B, Γ ἐχῆς ἀνάλογον τῶν ἀκρῶν πάλιν ὄντων πρῶτων πρὸς ἀλλήλους. λέγω, ὅτι καὶ οὔτως ἀδύνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσσευρεῖν. εἰ γὰρ δυνατόν, προσευρήσθω ὁ Δ , ὥστε εἶναι ὡς τὸν A πρὸς τὸν B , τὸν Γ πρὸς τὸν Δ , καὶ γεγονέτω ὡς ὁ B πρὸς τὸν Γ , ὁ Δ πρὸς τὸν E . καὶ ἐπεὶ ἐστὶν ὡς μὲν ὁ A πρὸς τὸν B , ὁ Γ πρὸς τὸν Δ , ὡς δὲ ὁ B πρὸς τὸν Γ , ὁ Δ πρὸς τὸν E , δι' ἴσον ἄρα ὡς ὁ A πρὸς τὸν Γ , ὁ Γ πρὸς τὸν E . οἱ δὲ A, Γ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάθιστοι, οἱ δὲ ἐλάθιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. μετρεῖ ἄρα ὁ A τὸν Γ ὡς ἡγούμενος ἡγούμενον. μετρεῖ δὲ καὶ ἑαυτὸν ὁ A ἄρα τοὺς A, Γ μετρεῖ πρῶτον ὄντας πρὸς ἀλλήλους· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τοῖς A, B, Γ δυνατόν ἐστι τέταρτον ἀνάλογον προσσευρεῖν.

Ἀλλὰ δὴ πάλιν ἔστωσαν οἱ A, B, Γ ἐχῆς ἀνάλογον, οἱ δὲ A, Γ μὴ ἔστωσαν πρῶτοι πρὸς ἀλλήλους. λέγω, ὅτι δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσσευρεῖν. ὁ γὰρ B τὸν Γ πολλαπλασιάσας τὸν Δ ποιεῖτω· ὁ A ἄρα τὸν Δ ἤτοι μετρεῖ ἢ οὐ μετρεῖ. μετρεῖτω αὐτὸν πρότερον κατὰ τὸν E · ὁ A ἄρα τὸν E πολλαπλασιάσας τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ B τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν· ὁ ἄρα ἐκ τῶν A, E ἴσος ἐστὶ τῷ ἐκ τῶν B, Γ . ἀνάλογον ἄρα [ἐστίν] ὡς ὁ A πρὸς τὸν B , ὁ Γ πρὸς τὸν E · τοῖς A, B, Γ ἄρα τέταρτος ἀνάλογον προσηγύρεται ὁ E .

Ἀλλὰ δὴ μὴ μετρεῖτω ὁ A τὸν Δ · λέγω, ὅτι ἀδύνατόν ἐστι τοῖς A, B, Γ τέταρτον ἀνάλογον προσσευρεῖν ἀριθμόν. εἰ γὰρ δυνατόν, προσευρήσθω ὁ E · ὁ ἄρα ἐκ τῶν A, E ἴσος ἐστὶ τῷ ἐκ τῶν B, Γ . ἀλλὰ ὁ ἐκ τῶν B, Γ ἐστὶν ὁ Δ · καὶ ὁ ἐκ τῶν A, E ἄρα ἴσος ἐστὶ τῷ Δ . ὁ A ἄρα τὸν E πολλαπλασιάσας τὸν Δ πεποίηκεν· ὁ A ἄρα τὸν Δ μετρεῖ κατὰ τὸν E · ὥστε μετρεῖ ὁ A τὸν Δ . ἀλλὰ καὶ οὐ μετρεῖ· ὅπερ ἄτοπον. οὐκ ἄρα δυνατόν ἐστι τοῖς A, B, Γ τέταρτον ἀνάλογον προσσευρεῖν ἀριθμόν, ὅταν ὁ A τὸν Δ μὴ μετρή. ἀλλὰ δὴ οἱ A, B, Γ μήτε ἐχῆς ἔστωσαν ἀνάλογον μήτε οἱ ἄκροι πρῶτοι πρὸς ἀλλήλους. καὶ ὁ B τὸν Γ πολλαπλασιάσας τὸν Δ ποιεῖτω. ὁμοίως δὴ δευθθήσεται, ὅτι εἰ μὲν μετρεῖ ὁ A τὸν Δ , δυνατόν ἐστιν αὐτοῖς ἀνάλογον προσσευρεῖν, εἰ δὲ οὐ μετρεῖ, ἀδύνατον· ὅπερ ἔδει δεῖχαι.

quired to investigate when it is possible to find a fourth (number) proportional to them.

In fact, (A, B, C) are either not continuously proportional and the outermost of them are prime to one another, or are continuously proportional and the outermost of them are not prime to one another, or are neither continuously proportional nor are the outermost of them prime to one another, or are continuously proportional and the outermost of them are prime to one another.

In fact, if A, B, C are continuously proportional, and the outermost of them, A and C , are prime to one another, (then) it has (already) been shown that it is impossible to find a fourth number proportional to them [Prop. 9.17]. So let A, B, C not be continuously proportional, (with) the outermost of them again being prime to one another. I say that, in this case, it is also impossible to find a fourth (number) proportional to them. For, if possible, let it be found, (and let it be) D . Hence, it will be that as A (is) to B , (so) C (is) to D . And let it be contrived that as B (is) to C , (so) D (is) to E . And since as A is to B , (so) C (is) to D , and as B (is) to C , (so) D (is) to E , thus, via equality, as A (is) to C , (so) C (is) to E [Prop. 7.14]. And A and C (are) prime (to one another). And (numbers) prime (to one another are) also the least (numbers having the same ratio as them) [Prop. 7.21]. And the least (numbers) measure those numbers having the same ratio as them (the same number of times), the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures C , (as) the leading (measuring) the leading. And it also measures itself. Thus, A measures A and C , which are prime to one another. The very thing is impossible. Thus, it is not possible to find a fourth (number) proportional to A, B, C .

And so let A, B, C again be continuously proportional, and let A and C not be prime to one another. I say that it is possible to find a fourth (number) proportional to them. For let B make D (by) multiplying C . Thus, A either measures or does not measure D . Let it, first of all, measure (D) according to E . Thus, A has made D (by) multiplying E . But, in fact, B has also made D (by) multiplying C . Thus, the (number created) from (multiplying) A, E is equal to the (number created) from (multiplying) B, C . Thus, proportionally, as A [is] to B , (so) C (is) to E [Prop. 7.19]. Thus, a fourth (number) proportional to A, B, C has been found, (namely) E .

And so let A not measure D . I say that it is impossible to find a fourth number proportional to A, B, C . For, if possible, let it be found, (and let it be) E . Thus, the (number created) from (multiplying) A, E is equal to the (number created) from (multiplying) B, C . But, the (number created) from (multiplying) B, C is D . And thus the (number created) from (multiplying) A, E is equal to D . Thus, A has made D (by) multiplying E . Thus, A measures D according to E . Hence, A measures D . But, it also does not measure (D). The very thing (is) absurd.

Thus, it is not possible to find a fourth number proportional to A, B, C when A does not measure D . And so (let) A, B, C (be) neither continuously proportional, nor (let) the outermost of them (be) prime to one another. And let B make D (by) multiplying C . So, similarly, it can be show that if A measures D then it is possible to find a fourth (number) proportional to (A, B, C) , and impossible if (A) does not measure (D) . (Which is) the very thing it was required to show.

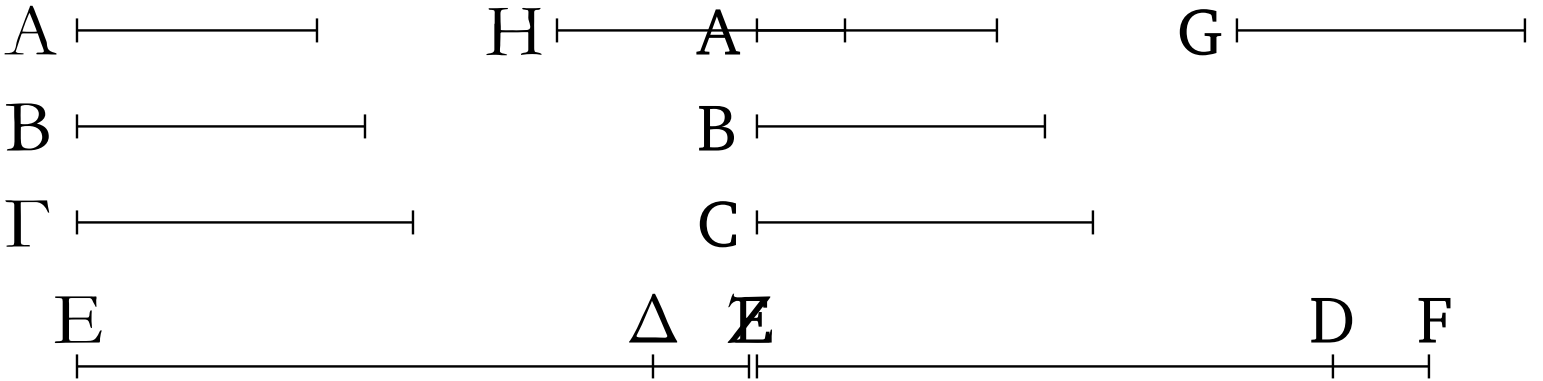
† The proof of this proposition is incorrect. There are, in fact, only two cases. Either A, B, C are continuously proportional, with A and C prime to one another, or not. In the first case, it is impossible to find a fourth proportional number. In the second case, it is possible to find a fourth proportional number provided that A measures B times C . Of the four cases considered by Euclid, the proof given in the second case is incorrect, since it only demonstrates that if $A : B :: C : D$ then a number E cannot be found such that $B : C :: D : E$. The proofs given in the other three cases are correct.

20.

Proposition 20

Οἱ πρώτοι ἀριθμοὶ πλείονες εἰσὶ παντὸς τοῦ προτεθέντος
πλήθους πρώτων ἀριθμῶν.

The (set of all) prime numbers is more numerous than any
assigned multitude of prime numbers.



Ἐστωσαν οἱ προτεθέντες πρώτοι ἀριθμοὶ οἱ A, B, Γ λέγω,
ὅτι τῶν A, B, Γ πλείονες εἰσὶ πρώτοι ἀριθμοί.

Εἰλήφθω γάρ ὁ ὑπὸ τῶν A, B, Γ ἐλάχιστος μετρούμενος
καὶ ἔστω ΔE , καὶ προσκείσθω τῷ ΔE μονὰς ἡ ΔZ . ὁ δὲ EZ
ἢτοι πρώτος ἔστιν ἢ οὐ. ἔστω πρότερον πρώτος· εὐρημένοι
ἄρα εἰσὶ πρώτοι ἀριθμοὶ οἱ A, B, Γ, EZ πλείονες τῶν A, B, Γ .

Ἀλλὰ δὴ μὴ ἔστω ὁ EZ πρώτος· ὑπὸ πρώτον ἄρα τινὸς
ἀριθμοῦ μετρεῖται. μετρεῖσθω ὑπὸ πρώτον τοῦ H λέγω, ὅτι ὁ
 H οὐδενὶ τῶν A, B, Γ ἔστιν ὁ αὐτός. εἰ γὰρ δυνατόν, ἔστω. οἱ
δὲ A, B, Γ τὸν ΔE μετροῦσιν καὶ ὁ H ἄρα τὸν ΔE μετρήσει.
μετρεῖ δὲ καὶ τὸν EZ · καὶ λοιπὴν τὴν ΔZ μονάδα μετρήσει ὁ
 H ἀριθμὸς ὧν ὅπερ ἄτοπον. οὐκ ἄρα ὁ H ἐνὶ τῶν A, B, Γ
ἔστιν ὁ αὐτός. καὶ ὑπόκειται πρώτος. εὐρημένοι ἄρα εἰσὶ
πρώτοι ἀριθμοὶ πλείονες τοῦ προτεθέντος πλήθους τῶν A, B, Γ
οἱ A, B, Γ, H · ὅπερ ἔδει δεῖχαι.

Let A, B, C be the assigned prime numbers. I say that the
(set of all) primes numbers is more numerous than A, B, C .

For let the least number measured by A, B, C be taken, and
let it be DE [Prop. 7.36]. And let the unit DF be added to
 DE . So EF is either prime, or not. Let it, first of all, be prime.
Thus, the (set of) prime numbers A, B, C, EF , (which is) more
numerous than A, B, C , has been found.

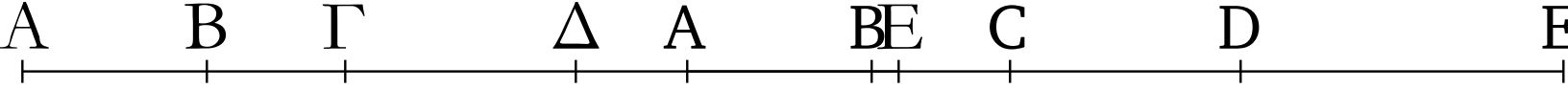
And so let EF not be prime. Thus, it is measured by some
prime number [Prop. 7.31]. Let it be measured by the prime
(number) G . I say that G is not the same as any of A, B, C . For,
if possible, let it be (the same). And A, B, C (all) measure DE .
Thus, G will also measure DE . And it also measures EF . (So)
 G will also measure the remainder, unit DF , (despite) being a
number [Prop. 7.28]. The very thing (is) absurd. Thus, G is not
the same as one of A, B, C . And it was assumed (to be) prime.
Thus, the (set of) prime numbers A, B, C, G , (which is) more
numerous than the assigned multitude (of prime numbers), A, B, C ,
has been found. (Which is) the very thing it was required
to show.

21.

Proposition 21

Ἐάν ἄρτιοι ἀριθμοὶ ὅποσοιὸν συντεθῶσιν, ὁ ὅλος ἄρτιός ἐστιν.

If any multitude whatsoever of even numbers is added together then the whole is even.



Συγκείσθωσαν γὰρ ἄρτιοι ἀριθμοὶ ὅποσοιὸν οἱ AB , $BΓ$, $ΓΔ$, $ΔΕ$ · λέγω, ὅτι ὅλος ὁ $ΑΕ$ ἄρτιός ἐστιν.

For let any multitude whatsoever of even numbers, AB , BC , CD , DE , lie together. I say that the whole, AE , is even.

Ἐπεὶ γὰρ ἕκαστος τῶν AB , $BΓ$, $ΓΔ$, $ΔΕ$ ἄρτιός ἐστιν, ἔθει μέρος ἥμισυν· ὥστε καὶ ὅλος ὁ $ΑΕ$ ἔθει μέρος ἥμισυν. ἄρτιος δὲ ἀριθμός ἐστιν ὁ δίπλα διαιρούμενος· ἄρτιος ἄρα ἐστὶν ὁ $ΑΕ$ · ὅπερ ἔδει δεῖξαι.

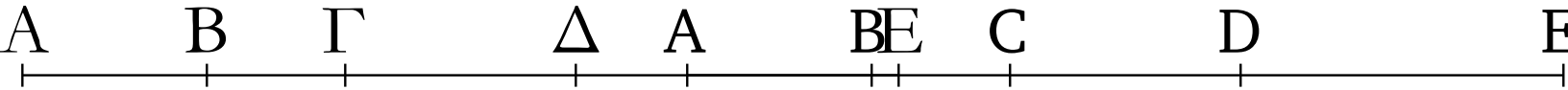
For since everyone of AB , BC , CD , DE is even, it has a half part [Def. 7.6]. And hence the whole AE has a half part. And an even number is one (which can be) divided in half [Def. 7.6]. Thus, AE is even. (Which is) the very thing it was required to show.

22.

Proposition 22

Ἐάν περισσοὶ ἀριθμοὶ ὅποσοιὸν συντεθῶσιν, τὸ δὲ πλῆθος αὐτῶν ἄρτιον ἢ, ὁ ὅλος ἄρτιος ἐσται.

If any multitude whatsoever of odd numbers is added together, and the multitude of them is even, then the whole will be even.



Συγκείσθωσαν γὰρ περισσοὶ ἀριθμοὶ ὅσοιδηποτοῦν ἄρτιοι τὸ πλῆθος οἱ AB , $BΓ$, $ΓΔ$, $ΔΕ$ · λέγω, ὅτι ὅλος ὁ $ΑΕ$ ἄρτιός ἐστιν.

For let any even multitude whatsoever of odd numbers, AB , BC , CD , DE , lie together. I say that the whole, AE , is even.

Ἐπεὶ γὰρ ἕκαστος τῶν AB , $BΓ$, $ΓΔ$, $ΔΕ$ περιττός ἐστιν, ἀφαιρεθείσῃ μονάδῳ ἀφ' ἑκάστου ἕκαστος τῶν λοιπῶν ἄρτιος ἐσται· ὥστε καὶ ὁ συγκεείμενος ἐξ αὐτῶν ἄρτιος ἐσται. ἔστι δὲ καὶ τὸ πλῆθος τῶν μονάδων ἄρτιον. καὶ ὅλος ἄρα ὁ $ΑΕ$ ἄρτιός ἐστιν· ὅπερ ἔδει δεῖξαι.

For since everyone of AB , BC , CD , DE is odd then, a unit being subtracted from each, everyone of the remainders will be (made) even [Def. 7.7]. And hence the sum of them will be even [Prop. 9.21]. And the multitude of the units is even. Thus, the whole AE is also even [Prop. 9.21]. (Which is) the very thing it was required to show.

23.

Proposition 23

Ἐάν περισσοὶ ἀριθμοὶ ὅποσοιὸν συντεθῶσιν, τὸ δὲ πλῆθος αὐτῶν περισόν ἢ, καὶ ὁ ὅλος περισός ἐσται.

If any multitude whatsoever of odd numbers is added together, and the multitude of them is odd, then the whole will also be odd.



Συγκείσθωσαν γὰρ ὅποσοιὸν περισσοὶ ἀριθμοί, ὧν τὸ πλῆθος περισόν ἐστω, οἱ AB , $BΓ$, $ΓΔ$ · λέγω, ὅτι καὶ ὅλος ὁ $ΑΔ$ περισός ἐστιν.

For let any multitude whatsoever of odd numbers, AB , BC , CD , lie together, and let the multitude of them be odd. I say that the whole, AD , is also odd.

Ἀφηρήσθω ἀπὸ τοῦ $ΓΔ$ μονὰς ἡ $ΔΕ$ · λοιπὸς ἄρα ὁ $ΓΕ$ ἄρτιός ἐστιν. ἔστι δὲ καὶ ὁ $ΓΑ$ ἄρτιος· καὶ ὅλος ἄρα ὁ $ΑΕ$ ἄρτιός ἐστιν. καὶ ἐστὶ μονὰς ἡ $ΔΕ$. περισός ἄρα ἐστὶν ὁ $ΑΔ$ · ὅπερ ἔδει δεῖξαι.

For let the unit DE be subtracted from CD . The remainder, CE , is thus even [Def. 7.7]. And CA is also even [Prop. 9.22]. Thus, the whole, AE , is also even [Prop. 9.21]. And DE is a unit. Thus, AD is odd [Def. 7.7]. (Which is) the very thing it was required to show.

24.

Proposition 24

Ἐάν ἀπὸ ἁρτίου ἀριθμοῦ ἄρτιος ἀφαιρεθῇ, ὁ λοιπὸς ἄρτιος ἐσται.

If an even (number) is subtracted from an (other) even number then the remainder will be even.



Ἀπὸ γὰρ ἄρτιον τοῦ AB ἄρτιος ἀφηρήσθω ὁ $BΓ$. λέγω, ὅτι ὁ λοιπὸς ὁ $ΓΑ$ ἄρτιός ἐστιν.

Ἐπεὶ γὰρ ὁ AB ἄρτιός ἐστιν, ἔθει μέρος ἥμισυν. διὰ τὰ αὐτὰ δὴ καὶ ὁ $BΓ$ ἔθει μέρος ἥμισυν· ὥστε καὶ λοιπὸς [ὁ $ΓΑ$ ἔθει μέρος ἥμισυν] ἄρτιος [ἄρα] ἐστὶν ὁ $ΑΓ$. ὅπερ ἔδει δεῖχαι.

For let the even (number) BC be subtracted from the even number AB . I say that the remainder, CA , is even.

For since AB is even, it has a half part [Def. 7.6]. So, for the same (reasons), BC also has a half part. And hence the remainder [CA has a half part]. [Thus,] AC is even. (Which is) the very thing it was required to show.

25.

Proposition 25

Ἐὰν ἀπὸ ἄρτιον ἀριθμοῦ περισσόξ ἀφαιρεθῇ, ὁ λοιπὸς περισσόξ ἔσται.

If an odd (number) is subtracted from an even number then the remainder will be odd.



Ἀπὸ γὰρ ἄρτιον τοῦ AB περισσόξ ἀφηρήσθω ὁ $BΓ$. λέγω, ὅτι ὁ λοιπὸς ὁ $ΓΑ$ περισσόξ ἐστίν.

Ἀφηρήσθω γὰρ ἀπὸ τοῦ $BΓ$ μονάξ ἡ $ΓΔ$. ὁ $ΔB$ ἄρα ἄρτιός ἐστιν. ἔστι δὲ καὶ ὁ AB ἄρτιος· καὶ λοιπὸς ἄρα ὁ $ΑΔ$ ἄρτιός ἐστιν. καὶ ἐστὶ μονάξ ἡ $ΓΔ$. ὁ $ΓΑ$ ἄρα περισσόξ ἐστίν· ὅπερ ἔδει δεῖχαι.

For let the odd (number) BC be subtracted from the even number AB . I say that the remainder CA is odd.

For let the unit CD be subtracted from BC . DB is thus even [Def. 7.7]. And AB is also even. And thus the remainder AD is even [Prop. 9.24]. And CD is a unit. Thus, CA is odd [Def. 7.7]. (Which is) the very thing it was required to show.

26.

Proposition 26

Ἐὰν ἀπὸ περισσοῦ ἀριθμοῦ περισσόξ ἀφαιρεθῇ, ὁ λοιπὸς ἄρτιος ἔσται.

If an odd (number) is subtracted from an odd number then the remainder will be even.



Ἀπὸ γὰρ περισσοῦ τοῦ AB περισσόξ ἀφηρήσθω ὁ $BΓ$. λέγω, ὅτι ὁ λοιπὸς ὁ $ΓΑ$ ἄρτιός ἐστιν.

Ἐπεὶ γὰρ ὁ AB περισσόξ ἐστίν, ἀφηρήσθω μονάξ ἡ $ΒΔ$. λοιπὸς ἄρα ὁ $ΑΔ$ ἄρτιός ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ὁ $ΓΔ$ ἄρτιός ἐστιν· ὥστε καὶ λοιπὸς ὁ $ΓΑ$ ἄρτιός ἐστιν· ὅπερ ἔδει δεῖχαι.

For let the odd (number) BC be subtracted from the odd (number) AB . I say that the remainder, CA , is even.

For since AB is odd, let the unit BD be subtracted (from it). Thus, the remainder AD is even [Def. 7.7]. So, for the same (reasons), CD is also even. And hence the remainder CA is even [Prop. 9.24]. (Which is) the very thing it was required to show.

27.

Proposition 27

Ἐὰν ἀπὸ περισσοῦ ἀριθμοῦ ἄρτιος ἀφαιρεθῇ, ὁ λοιπὸς περισσόξ ἔσται.

If an even (number) is subtracted from an odd number then the remainder will be odd.



Ἀπὸ γὰρ περισσοῦ τοῦ AB ἄρτιος ἀφηρήσθω ὁ $BΓ$. λέγω, ὅτι ὁ λοιπὸς ὁ $ΓΑ$ περισσόξ ἐστίν.

Ἀφηρήσθω [γὰρ] μονάξ ἡ $ΑΔ$. ὁ $ΔB$ ἄρα ἄρτιός ἐστιν. ἔστι δὲ καὶ ὁ $BΓ$ ἄρτιος· καὶ λοιπὸς ἄρα ὁ $ΓΔ$ ἄρτιός ἐστιν. περισσόξ ἄρα ὁ $ΓΑ$. ὅπερ ἔδει δεῖχαι.

For let the even (number) BC be subtracted from the odd (number) AB . I say that the remainder, CA , is odd.

[For] let the unit AD be subtracted (from AB). DB is thus even [Def. 7.7]. And BC is also even. Thus, the remainder CD is also even [Prop. 9.24]. CA (is) thus odd [Def. 7.7]. (Which is) the very thing it was required to show.

is) the very thing it was required to show.

28.

Proposition 28

Ἐάν περισσόξ ἀριθμόξ ἄρτιον πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενος ἄρτιος ἔσται.

If an odd number makes some (number by) multiplying an even (number) then the created (number) will be even.

A |—————|

A |—————|

B |—————|

B |—————|

Γ |—————| C |—————|

Περισσόξ γάρ ἀριθμός ὁ A ἄρτιον τὸν B πολλαπλασιάσας τὸν Γ ποιείτω· λέγω, ὅτι ὁ Γ ἄρτιός ἐστιν.

For let the odd number A make C (by) multiplying the even (number) B . I say that C is even.

Ἐπεὶ γάρ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ Γ ἄρα σύγκειται ἐκ τοσούτων ἴσων τῷ B , ὅσαι εἰσὶν ἐν τῷ A μονάδες. καὶ ἐστὶν ὁ B ἄρτιος· ὁ Γ ἄρα σύγκειται ἐξ ἄρτίων. ἔαν δὲ ἄρτιοι ἀριθμοὶ ὁποσοιοῦν συντεθῶσιν, ὁ ὅλος ἄρτιός ἐστιν. ἄρτιος ἄρα ἐστὶν ὁ Γ · ὅπερ ἔδει δεῖχαι.

For since A has made C (by) multiplying B , C is thus composed out of so many (magnitudes) equal to B , as many as (there) are units in A [Def. 7.15]. And B is even. Thus, C is composed out of even (numbers). And if any multitude whatsoever of even numbers is added together then the whole is even [Prop. 9.21]. Thus, C is even. (Which is) the very thing it was required to show.

29.

Proposition 29

Ἐάν περισσόξ ἀριθμός περισσὸν ἀριθμὸν πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενος περισσόξ ἔσται.

If an odd number makes some (number by) multiplying an odd (number) then the created (number) will be odd.

A |—————|

A |—————|

B |—————|

B |—————|

Γ |—————| C |—————|

Περισσόξ γάρ ἀριθμός ὁ A περισσὸν τὸν B πολλαπλασιάσας τὸν Γ ποιείτω· λέγω, ὅτι ὁ Γ περισσόξ ἐστιν.

For let the odd number A make C (by) multiplying the odd (number) B . I say that C is odd.

Ἐπεὶ γάρ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ Γ ἄρα σύγκειται ἐκ τοσούτων ἴσων τῷ B , ὅσαι εἰσὶν ἐν τῷ A μονάδες. καὶ ἐστὶν ἐκάτερος τῶν A , B περισσόξ· ὁ Γ ἄρα σύγκειται ἐκ περισσῶν ἀριθμῶν, ὧν τὸ πλῆθος περισσόν ἐστιν. ὥστε ὁ Γ περισσόξ ἐστὶν· ὅπερ ἔδει δεῖχαι.

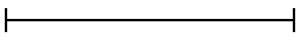
For since A has made C (by) multiplying B , C is thus composed out of so many (magnitudes) equal to B , as many as (there) are units in A [Def. 7.15]. And each of A , B is odd. Thus, C is composed out of odd (numbers), (and) the multitude of them is odd. Hence C is odd [Prop. 9.23]. (Which is) the very thing it was required to show.

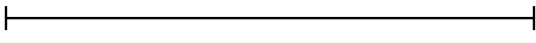
30.

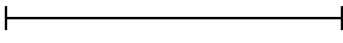
Proposition 30

Ἐάν περισσόξ ἀριθμός ἄρτιον ἀριθμὸν μετρήῃ, καὶ τὸν ἡμισὴν αὐτοῦ μετρήσῃ.

If an odd number measures an even number then it will also measure (one) half of it.

A 

B 

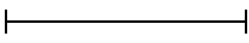
Γ 

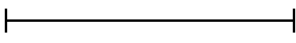
Περισσός γάρ ἀριθμός ὁ *A* ἄρτιον τὸν *B* μετρεῖτω· λέγω, ὅτι καὶ τὸν ἥμισυν αὐτοῦ μετρήσει.

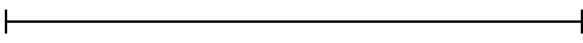
Ἐπεὶ γὰρ ὁ *A* τὸν *B* μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν *Γ*· λέγω, ὅτι ὁ *Γ* οὐκ ἔστι περισσός. εἰ γὰρ δυνατόν, ἔστω. καὶ ἐπεὶ ὁ *A* τὸν *B* μετρεῖ κατὰ τὸν *Γ*, ὁ *A* ἄρα τὸν *Γ* πολλαπλασιάσας τὸν *B* πεποίηκεν. ὁ *B* ἄρα σύγκειται ἐκ περισσῶν ἀριθμῶν, ὧν τὸ πλῆθος περισσόν ἐστιν. ὁ *B* ἄρα περισσός ἐστιν· ὅπερ ἄτοπον· ὑπόκειται γὰρ ἄρτιος. οὐκ ἄρα ὁ *Γ* περισσός ἐστιν· ἄρτιος ἄρα ἐστὶν ὁ *Γ*. ὥστε ὁ *A* τὸν *B* μετρεῖ ἀρτιάκις. διὰ δὴ τοῦτο καὶ τὸν ἥμισυν αὐτοῦ μετρήσει· ὅπερ ἔδει δεῖχαι.

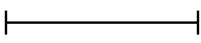
31.

Ἐὰν περισσός ἀριθμός πρός τινα ἀριθμὸν πρώτος ᾖ, καὶ πρός τὸν διπλασίονα αὐτοῦ πρώτος ἔσται.

A 

B 

Γ 

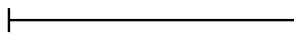
Δ 

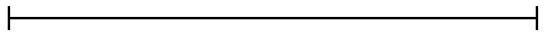
Περισσός γάρ ἀριθμός ὁ *A* πρός τινα ἀριθμὸν τὸν *B* πρώτος ἔστω, τοῦ δὲ *B* διπλασίων ἔστω ὁ *Γ*. λέγω, ὅτι ὁ *A* [καὶ] πρός τὸν *Γ* πρώτος ἐστίν.

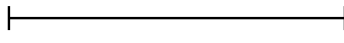
Εἰ γὰρ μὴ εἰσιν [οἱ *A*, *Γ*] πρώτοι, μετρήσει τις αὐτοὺς ἀριθμός. μετρεῖτω, καὶ ἔστω ὁ *Δ*. καὶ ἐστὶν ὁ *A* περισσός· περισσός ἄρα καὶ ὁ *Δ*. καὶ ἐπεὶ ὁ *Δ* περισσός ὧν τὸν *Γ* μετρεῖ, καὶ ἐστὶν ὁ *Γ* ἄρτιος, καὶ τὸν ἥμισυν ἄρα τοῦ *Γ* μετρήσει [ὁ *Δ*]. τοῦ δὲ *Γ* ἥμισυν ἐστὶν ὁ *B*. ὁ *Δ* ἄρα τὸν *B* μετρεῖ. μετρεῖ δὲ καὶ τὸν *A*. ὁ *Δ* ἄρα τοὺς *A*, *B* μετρεῖ πρώτους ὄντας πρός ἀλλήλους· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ *A* πρός τὸν *Γ* πρώτος οὐκ ἐστίν. οἱ *A*, *Γ* ἄρα πρώτοι πρός ἀλλήλους εἰσὶν· ὅπερ ἔδει δεῖχαι.

32.

Τῶν ἀπὸ δυνάδος διπλασιαζομένων ἀριθμῶν ἕκαστος ἀρ-

A 

B 

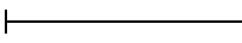
C 

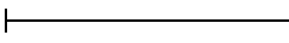
For let the odd number *A* measure the even (number) *B*. I say that (*A*) will also measure (one) half of (*B*).

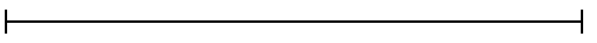
For since *A* measures *B*, let it measure it according to *C*. I say that *C* is not odd. For, if possible, let it be (odd). And since *A* measures *B* according to *C*, *A* has thus made *B* (by) multiplying *C*. Thus, *B* is composed out of odd numbers, (and) the multitude of them is odd. *B* is thus odd [Prop. 9.23]. The very thing (is) absurd. For (*B*) was assumed (to be) even. Thus, *C* is not odd. Thus, *C* is even. Hence, *A* measures *B* an even number of times. So, on account of this, (*A*) will also measure (one) half of (*B*). (Which is) the very thing it was required to show.

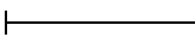
Proposition 31

If an odd number is prime to some number then it will also be prime to its double.

A 

B 

C 

D 

For let the odd number *A* be prime to some number *B*. And let *C* be double *B*. I say that *A* is [also] prime to *C*.

For if [*A* and *C*] are not prime (to one another) then some number will measure them. Let it measure (them), and let it be *D*. And *A* is odd. Thus, *D* (is) also odd. And since *D*, which is odd, measures *C*, and *C* is even, [*D*] will thus also measure half of *C* [Prop. 9.30]. And *B* is half of *C*. Thus, *D* measures *B*. And it also measures *A*. Thus, *D* measures (both) *A* and *B*, (despite) them being prime to one another. The very thing is impossible. Thus, *A* is not unprime to *C*. Thus, *A* and *C* are prime to one another. (Which is) the very thing it was required to show.

Proposition 32

Each of the numbers (which is continually) doubled, (start-

τιάκις ἄρτιός ἐστι μόνον.

A ————

B ————

Γ ————

Δ ————

Ἀπὸ γὰρ δυνάδοξ τῆς A δεδιπλασιάσθωσαν ὅσοιδηποτοῦν ἀριθμοὶ οἱ B, Γ, Δ· λέγω, ὅτι οἱ B, Γ, Δ ἀρτιάκις ἄρτιοὶ εἰσι μόνον.

Ὅτι μὲν οὖν ἕκαστος [τῶν B, Γ, Δ] ἀρτιάκις ἄρτιός ἐστιν, φανερόν· ἀπὸ γὰρ δυνάδοξ ἐστὶ διπλασιασθείς. λέγω, ὅτι καὶ μόνον. ἐκκείσθω γὰρ μονάξ. ἐπεὶ οὖν ἀπὸ μονάδοξ ὅποσοιὺν ἀριθμοὶ ἐχῆξ ἀνάλογόν εἰσιν, ὁ δὲ μετὰ τὴν μονάδα ὁ A πρῶτός ἐστιν, ὁ μέγιστος τῶν A, B, Γ, Δ ὁ Δ ὑπ' οὐδενός ἄλλον μετρηθήσεται παρέχ τῶν A, B, Γ. καὶ ἐστὶν ἕκαστος τῶν A, B, Γ ἄρτιος· ὁ Δ ἄρα ἀρτιάκις ἄρτιός ἐστι μόνον. ὁμοίως δὲ δείκνυμεν, ὅτι [καὶ] ἑκάτερος τῶν B, Γ ἀρτιάκις ἄρτιός ἐστι μόνον· ὅπερ ἔδει δεῖξαι.

33.

Ἐὰν ἀριθμὸς τὸν ἥμισυν ἔθῃ περισσόν, ἀρτιάκις περισσός ἐστι μόνον.

A ————

Ἀριθμὸς γὰρ ὁ A τὸν ἥμισυν ἐθέτω περισσόν· λέγω, ὅτι ὁ A ἀρτιάκις περισσός ἐστι μόνον.

Ὅτι μὲν οὖν ἀρτιάκις περισσός ἐστιν, φανερόν· ὁ γὰρ ἥμισυν αὐτοῦ περισσός ὡς μετρεῖ αὐτὸν ἀρτιάκις, λέγω δὲ, ὅτι καὶ μόνον. εἰ γὰρ ἔσται ὁ A καὶ ἀρτιάκις ἄρτιος, μετρηθήσεται ὑπὸ ἀρτίου κατὰ ἄρτιον ἀριθμόν· ὥστε καὶ ὁ ἥμισυν αὐτοῦ μετρηθήσεται ὑπὸ ἀρτίου ἀριθμοῦ περισσός ὡς ὅπερ ἐστὶν ἄτοπον. ὁ A ἄρα ἀρτιάκις περισσός ἐστι μόνον· ὅπερ ἔδει δεῖξαι.

34.

Ἐὰν ἀριθμὸς μῆτε τῶν ἀπὸ δυνάδοξ διπλασιαζομένων ἢ, μῆτε τὸν ἥμισυν ἔθῃ περισσόν, ἀρτιάκις τε ἄρτιός ἐστι καὶ ἀρτιάκις περισσός.

ing) from a dyad, is an even-times-even (number) only.

A ————

B ————

C ————

D ————

For let any multitude of numbers whatsoever, B, C, D, be (continually) doubled, (starting) from the dyad A. I say that B, C, D are even-times-even (numbers) only.

In fact, (it is) clear that each [of B, C, D] is an even-times-even (number). For it is doubled from a dyad [Def. 7.8]. I also say that (they are even-times-even numbers) only. For let a unit be laid down. Therefore, since any multitude of numbers whatsoever are continuously proportional, starting from a unit, and the (number) A after the unit is prime, the greatest of A, B, C, D, (namely) D, will not be measured by any other (numbers) except A, B, C [Prop. 9.13]. And each of A, B, C is even. Thus, D is an even-time-even (number) only [Def. 7.8]. So, similarly, we can show that each of B, C is [also] an even-time-even (number) only. (Which is) the very thing it was required to show.

Proposition 33

If a number has an odd half then it is an even-time-odd (number) only.

A ————

For let the number A have an odd half. I say that A is an even-times-odd (number) only.

In fact, (it is) clear that (A) is an even-times-odd (number). For its half, being odd, measures it an even number of times [Def. 7.9]. So I also say that (it is an even-times-odd number) only. For if A is also an even-times-even (number) then it will be measured by an even (number) according to an even number [Def. 7.8]. Hence, its half will also be measured by an even number, (despite) being odd. The very thing is absurd. Thus, A is an even-times-odd (number) only. (Which is) the very thing it was required to show.

Proposition 34

If a number is neither (one) of the (numbers) doubled from a dyad, nor has an odd half, then it is (both) an even-times-even and an even-times-odd (number).

A —————

Ἀριθμὸς γὰρ ὁ A μήτε τῶν ἀπὸ δυνάδοξ διπλασιαζομένων ἔστω μήτε τὸν ἥμισυν ἐθέτω περισσόν· λέγω, ὅτι ὁ A ἀρτιάκις τέ ἐστιν ἄρτιος καὶ ἀρτιάκις περισσός.

Ὅτι μὲν οὖν ὁ A ἀρτιάκις ἐστὶν ἄρτιος, φανερόν· τὸν γὰρ ἥμισυν οὐκ ἔθει περισσόν. λέγω δὲ, ὅτι καὶ ἀρτιάκις περισσός ἐστιν. ἔαν γὰρ τὸν A τέμνωμεν δίθα καὶ τὸν ἥμισυν αὐτοῦ δίθα καὶ τοῦτο αἰεὶ ποιῶμεν, καταστήσομεν εἷς τινα ἀριθμὸν περισσόν, ὃς μετρήσει τὸν A κατὰ ἄρτιον ἀριθμόν. εἰ γὰρ οὐ, καταστήσομεν εἷς δυνάδα, καὶ ἔσται ὁ A τῶν ἀπὸ δυνάδοξ διπλασιαζομένων· ὅπερ οὐθ' ὑπόκειται. ὥστε ὁ A ἀρτιάκις περισσόν ἐστιν. ἐδείχθη δὲ καὶ ἀρτιάκις ἄρτιος. ὁ A ἄρα ἀρτιάκις τε ἄρτιός ἐστι καὶ ἀρτιάκις περισσός· ὅπερ ἔδει δεῖχαι.

A —————

For let the number A neither be (one) of the (numbers) doubled from a dyad, nor let it have an odd half. I say that A is (both) an even-times-even and an even-times-odd (number).

In fact, (it is) clear that A is an even-times-even (number) [Def. 7.8]. For it does not have an odd half. So I say that it is also an even-times-odd (number). For if we cut A in half, and (then cut) its half in half, and we do this continually, then we will arrive at some odd number which will measure A according to an even number. For if not, we will arrive at a dyad, and A will be (one) of the (numbers) doubled from a dyad. The very opposite thing (was) assumed. Hence, A is an even-times-odd (number) [Def. 7.9]. And it was also shown (to be) an even-times-even (number). Thus, A is (both) an even-times-even and an even-times-odd (number). (Which is) the very thing it was required to show.

35.

Proposition 35[†]

Ἐάν ᾧσιν ὅσοιδηποτοῦν ἀριθμοὶ ἐχῇς ἀνάλογον, ἀφαιρεθῶσι δὲ ἀπὸ τε τοῦ δευτέρου καὶ τοῦ ἐσθάτου ἴσοι τῷ πρώτῳ, ἔσται ὥς ἡ τοῦ δευτέρου ὑπεροθὴ πρὸς τὸν πρῶτον, οὕτως ἡ τοῦ ἐσθάτου ὑπεροθὴ πρὸς τὸν πρὸ ἐαυτοῦ πάντα.

If there is any multitude whatsoever of continually proportional numbers, and (numbers) equal to the first are subtracted from (both) the second and the last, then as the excess of the second (number is) to the first, so the excess of the last will be to (the sum of) all those (numbers) before it.

A —————

B ————— H Γ

A —————

B ————— G C

Δ —————

D —————

E Λ K Θ

Ξ L K H F

Ἔστωσαν ὅποσοιδηποτοῦν ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A , $BΓ$, $Δ$, $EΖ$ ἀφθόμενοι ἀπὸ ἐλαθίστου τοῦ A , καὶ ἀφηρήσθω ἀπὸ τοῦ $BΓ$ καὶ τοῦ $EΖ$ τῷ A ἴσοις ἐκάτερος τῶν $BΗ$, $ΖΘ$. λέγω, ὅτι ἐστὶν ὥς ὁ $HΓ$ πρὸς τὸν A , οὕτως ὁ $EΘ$ πρὸς τὸν $Δ$, $BΓ$, $Δ$.

Κείσθω γὰρ τῷ μὲν $BΓ$ ἴσος ὁ ZK , τῷ δὲ $Δ$ ἴσος ὁ $ΖΛ$. καὶ ἐπεὶ ὁ ZK τῷ $BΓ$ ἴσος ἐστίν, ὦν ὁ $ΖΘ$ τῷ $BΗ$ ἴσος ἐστίν, λοιπὸς ἄρα ὁ $ΘK$ λοιπῷ τῷ $HΓ$ ἐστὶν ἴσος. καὶ ἐπεὶ ἐστὶν ὥς ὁ $EΖ$ πρὸς τὸν $Δ$, οὕτως ὁ $Δ$ πρὸς τὸν $BΓ$ καὶ ὁ $BΓ$ πρὸς τὸν A , ἴσος δὲ ὁ μὲν $Δ$ τῷ $ΖΛ$, ὁ δὲ $BΓ$ τῷ ZK , ὁ δὲ A τῷ $ΖΘ$, ἐστὶν ἄρα ὥς ὁ $EΖ$ πρὸς τὸν $ΖΛ$, οὕτως ὁ $ΛΖ$ πρὸς τὸν

Let A , BC , D , EF be any multitude whatsoever of continually proportional numbers, beginning from the least A . And let BG and FH , each equal to A , be subtracted from BC and EF (respectively). I say that as GC is to A , so EH is to A , BC , D .

For let FK be made equal to BC , and FL to D . And since FK is equal to BC , of which FH is equal to BG , the remainder, HK , is thus equal to the remainder, GC . And since as EF is to D , so D (is) to BC , and BC to A [Prop. 7.13], and D (is) equal to FL , and BC to FK , and A to FH , thus as EF is to FL , so LF (is) to FK , and FK to FH . By separation, as EL (is) to

ZK καὶ ὁ ZK πρὸς τὸν $Z\Theta$. διελόντι, ὡς ὁ EA πρὸς τὸν AZ , οὕτως ὁ AK πρὸς τὸν ZK καὶ ὁ $K\Theta$ πρὸς τὸν $Z\Theta$. ἔστιν ἄρα καὶ ὡς εἶξ τῶν ἡγούμενων πρὸς ἓνα τῶν ἐπομένων, οὕτως ἅπαντες οἱ ἡγούμενοι πρὸς ἅπαντας τοὺς ἐπομένους· ἔστιν ἄρα ὡς ὁ $K\Theta$ πρὸς τὸν $Z\Theta$, οὕτως οἱ EA , AK , $K\Theta$ πρὸς τοὺς AZ , ZK , ΘZ . ἴσος δὲ ὁ μὲν $K\Theta$ τῷ $ΓH$, ὁ δὲ $Z\Theta$ τῷ A , οἱ δὲ AZ , ZK , ΘZ τοῖς Δ , $BΓ$, A . ἔστιν ἄρα ὡς ὁ $ΓH$ πρὸς τὸν A , οὕτως ὁ $E\Theta$ πρὸς τοὺς Δ , $BΓ$, A . ἔστιν ἄρα ὡς ἡ τοῦ δευτέρου ὑπεροθὴ πρὸς τὸν πρῶτον, οὕτως ἡ τοῦ ἐσθαιτον ὑπεροθὴ πρὸς τοὺς πρὸ ἐαυτοῦ πάντα· ὅπερ εἶδει δεῖχαι.

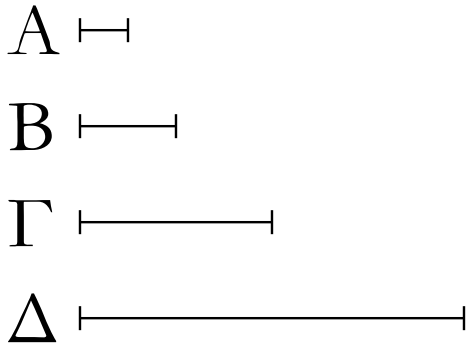
LF , so LK (is) to FK , and KH to FH [Props. 7.11, 7.13]. And thus as one of the leading (numbers) is to one of the following, so (the sum of) all of the leading (numbers is) to (the sum of) all of the following [Prop. 7.12]. Thus, as KH is to FH , so EL , LK , KH (are) to LF , FK , HF . And KH (is) equal to CG , and FH to A , and LF , FK , HF to D , BC , A . Thus, as CG is to A , so EH (is) to D , BC , A . Thus, as the excess of the second (number) is to the first, so the excess of the last (is) to (the sum of) all those (numbers) before it. (Which is) the very thing it was required to show.

[†] This proposition allows us to sum a geometric series of the form $a, ar, ar^2, ar^3, \dots, ar^{n-1}$. According to Euclid, the sum, S_n , satisfies $(ar - a)/a = (ar^n - a)/S_n$. Hence, $S_n = a(r^n - 1)/(r - 1)$.

36.

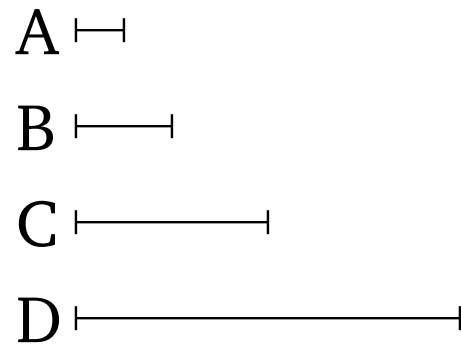
Ἐὰν ἀπὸ μονάδος ὅποιοι ἄριθμοι ἐχῇς ἐκτεθῶσιν ἐν τῇ διπλασίονι ἀναλογίᾳ, ἕως οὔ ὁ σύμπαξ συντεθείς πρῶτος γένηται, καὶ ὁ σύμπαξ ἐπὶ τὸν ἐσθαιτον πολλαπλασιασθεὶς ποιῇ τινα, ὁ γεγόμενος τέλειος ἔσται.

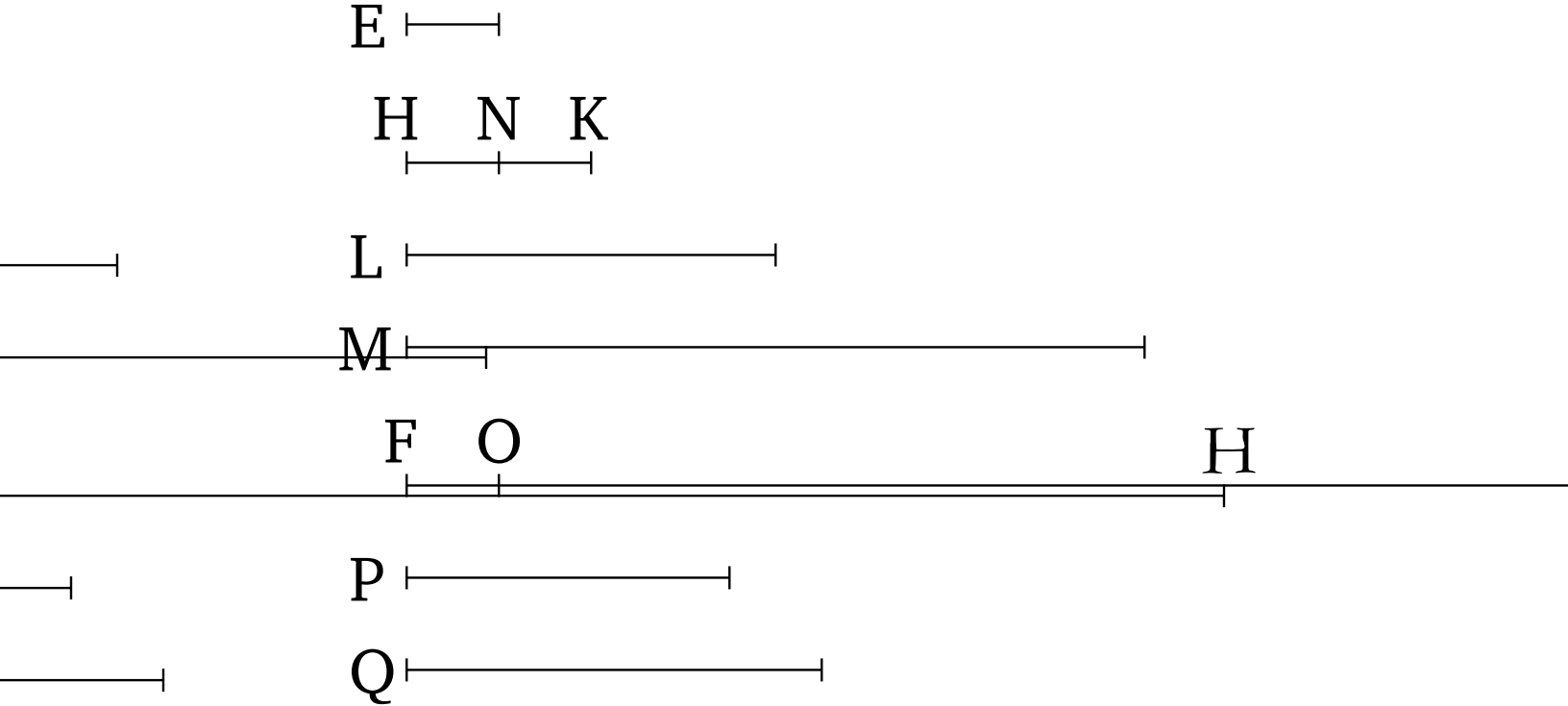
Ἀπὸ γὰρ μονάδος ἐκκείσθωσαν ὁσοιδηποτοῦν ἄριθμοι ἐν τῇ διπλασίονι ἀναλογίᾳ, ἕως οὔ ὁ σύμπαξ συντεθείς πρῶτος γένηται, οἱ A , B , Γ , Δ , καὶ τῷ σύμπαντι ἴσος ἔστω ὁ E , καὶ ὁ E τὸν Δ πολλαπλασιάσας τὸν ZH ποιείτω. λέγω, ὅτι ὁ ZH τέλειός ἐστιν.

Proposition 36[†]

If any multitude whatsoever of numbers is set out continuously in a double proportion, (starting) from a unit, until the whole sum added together becomes prime, and the sum multiplied into the last (number) makes some (number), then the (number so) created will be perfect.

For let any multitude of numbers, A , B , C , D , be set out (continuously) in a double proportion, until the whole sum added together is made prime. And let E be equal to the sum. And let E make FG (by) multiplying D . I say that FG is a perfect (number).





Ὅσοι γάρ εἰσιν οἱ A, B, Γ, Δ τῶ πλήθει, τοσοῦτοι ἀπὸ τοῦ E εἰλήφθωσαν ἐν τῇ διπλασίονι ἀναλογίᾳ οἱ $E, \Theta K, \Lambda, M$. δι' ἴσον ἄρα ἐστὶν ὡς ὁ A πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν M . ὁ ἄρα ἐκ τῶν E, Δ ἴσοξ ἐστὶ τῶ ἐκ τῶν A, M . καὶ ἐστὶν ὁ ἐκ τῶν E, Δ ὁ ZH · καὶ ὁ ἐκ τῶν A, M ἄρα ἐστὶν ὁ ZH . ὁ A ἄρα τὸν M πολλαπλασιάσας τὸν ZH πεποίηκεν· ὁ M ἄρα τὸν ZH μετρεῖ κατὰ τὰς ἐν τῶ A μονάδας. καὶ ἐστὶ δυνάξ ὁ A διπλασίωξ ἄρα ἐστὶν ὁ ZH τοῦ M . εἰσὶ δὲ καὶ οἱ $M, \Lambda, \Theta K, E$ ἐχῆς διπλασίοι ἀλλήλων· οἱ $E, \Theta K, \Lambda, M, ZH$ ἄρα ἐχῆς ἀνάλογόν εἰσιν ἐν τῇ διπλασίονι ἀναλογίᾳ. ἀφηγήσθω δὲ ἀπὸ τοῦ δευτέρου τοῦ ΘK καὶ τοῦ ἐσθάτου τοῦ ZH τῶ πρώτῳ τῶ E ἴσοξ ἐκάτεροξ τῶν $\Theta N, ZX$ · ἐστὶν ἄρα ὡς ἡ τοῦ δευτέρου ἀριθμοῦ ὑπεροθὴ πρὸς τὸν πρώτον, οὕτως ἡ τοῦ ἐσθάτου ὑπεροθὴ πρὸς τοῦξ πρὸ ἐαυτοῦ πάνταξ. ἐστὶν ἄρα ὡς ὁ NK πρὸς τὸν E , οὕτως ὁ XH πρὸς τοῦξ $M, \Lambda, K\Theta, E$. καὶ ἐστὶν ὁ NK ἴσοξ τῶ E · καὶ ὁ XH ἄρα ἴσοξ ἐστὶ τοῖξ $M, \Lambda, \Theta K, E$. ἐστὶ δὲ καὶ ὁ ZX τῶ E ἴσοξ, ὁ δὲ E τοῖξ A, B, Γ, Δ καὶ τῇ μονάδι. ὅλοξ ἄρα ὁ ZH ἴσοξ ἐστὶ τοῖξ τε $E, \Theta K, \Lambda, M$ καὶ τοῖξ A, B, Γ, Δ καὶ τῇ μονάδι· καὶ μετρεῖται ὑπ' αὐτῶν. λέγω, ὅτι καὶ ὁ ZH ὑπ' οὐδενὸς ἄλλον μετρηθήσεται παρέχ τῶν $A, B, \Gamma, \Delta, E, \Theta K, \Lambda, M$ καὶ τῆξ μονάδοξ. εἰ γάρ δυνατόν, μετρεῖτω τιξ τὸν ZH ὁ O , καὶ ὁ O μηδενὶ τῶν $A, B, \Gamma, \Delta, E, \Theta K, \Lambda, M$ ἔστω ὁ αὐτόξ. καὶ ὁσάκιξ ὁ O τὸν ZH μετρεῖ, τοσαῦται μονάδεξ ἔστωσαν ἐν τῶ Π · ὁ Π ἄρα τὸν O πολλαπλασιάσας τὸν ZH πεποίηκεν. ἀλλὰ μὴν καὶ ὁ E τὸν Δ πολλαπλασιάσας τὸν ZH πεποίηκεν· ἐστὶν ἄρα ὡς ὁ E πρὸς τὸν Π , ὁ O πρὸς τὸν Δ . καὶ ἐπεὶ ἀπὸ μονάδοξ ἐχῆς ἀνάλογόν εἰσιν οἱ A, B, Γ, Δ , ὁ Δ ἄρα ὑπ' οὐδενὸς ἄλλον

For as many as is the multitude of A, B, C, D , let so many (numbers), E, HK, L, M , be taken in a double proportion, (starting) from E . Thus, via equality, as A is to D , so E (is) to M [Prop. 7.14]. Thus, the (number created) from (multiplying) E, D is equal to the (number created) from (multiplying) A, M . And FG is the (number created) from (multiplying) E, D . Thus, FG is also the (number created) from (multiplying) A, M [Prop. 7.19]. Thus, A has made FG (by) multiplying M . Thus, M measures FG according to the units in A . And A is a dyad. Thus, FG is double M . And M, L, HK, E are also continuously double one another. Thus, E, HK, L, M, FG are continuously proportional in a double proportion. So let HN and FO , each equal to the first (number) E , be subtracted from the second (number) HK and the last FG (respectively). Thus, as the excess of the second number is to the first, so the excess of the last (is) to (the sum of) all those (numbers) before it [Prop. 9.35]. Thus, as NK is to E , so OG (is) to M, L, KH, E . And NK is equal to E . And thus OG is equal to M, L, HK, E . And FO is also equal to E , and E to A, B, C, D , and a unit. Thus, the whole of FG is equal to E, HK, L, M , and A, B, C, D , and a unit. And it is measured by them. I also say that FG will be measured by no other (numbers) except A, B, C, D, E, HK, L, M , and a unit. For, if possible, let some (number) P measure FG , and let P not be the same as any of A, B, C, D, E, HK, L, M . And as many times as P measures FG , so many units let there be in Q . Thus, Q has made FG (by) multiplying P . But, in fact, E has also made FG (by) multiplying D . Thus, as E is to Q , so P (is) to D [Prop. 7.19]. And since A ,

ἀριθμοῦ μετρηθήσεται παρέχ τῶν A, B, Γ . καὶ ὑπόκειται ὁ O οὐδενὶ τῶν A, B, Γ ὁ αὐτόξ· οὐκ ἄρα μετρήσει ὁ O τὸν Δ . ἀλλ' ὥς ὁ O πρὸς τὸν Δ , ὁ E πρὸς τὸν Π · οὐδὲ ὁ E ἄρα τὸν Π μετρεῖ. καὶ ἐστὶν ὁ E πρωτοξ· πᾶς δὲ πρωτοξ ἀριθμὸς πρὸς ἅπαντα, ὃν μὴ μετρεῖ, πρωτόξ [ἐστίν]. οἱ E, Π ἄρα πρωτοὶ πρὸς ἀλλήλων εἰσίν. οἱ δὲ πρωτοὶ καὶ ἐλάθιστοι, οἱ δὲ ἐλάθιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔθονταξ ἰσάκις ὅ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον· καὶ ἐστὶν ὥς ὁ E πρὸς τὸν Π , ὁ O πρὸς τὸν Δ · ἰσάκιξ ἄρα ὁ E τὸν O μετρεῖ καὶ ὁ Π τὸν Δ . ὁ δὲ Δ ὑπ' οὐδενὸς ἄλλου μετρεῖται παρέχ τῶν A, B, Γ · ὁ Π ἄρα ἐν τῶν A, B, Γ ἐστὶν ὁ αὐτόξ. ἔστω τῷ B ὁ αὐτόξ. καὶ ὅσοι εἰσὶν οἱ B, Γ, Δ τῷ πλήθει τοσοῦτοι εἰλήφθωσαν ἀπὸ τοῦ E οἱ $E, \Theta K, \Lambda$. καὶ εἰσὶν οἱ $E, \Theta K, \Lambda$ τοῖξ B, Γ, Δ ἐν τῷ αὐτῷ λόγῳ· δι' ἴσον ἄρα ἐστὶν ὥς ὁ B πρὸς τὸν Δ , ὁ E πρὸς τὸν Λ . ὁ ἄρα ἐκ τῶν B, Λ ἴσοξ ἐστὶ τῷ ἐκ τῶν Δ, E · ἀλλ' ὁ ἐκ τῶν Δ, E ἴσοξ ἐστὶ τῷ ἐκ τῶν Π, O · καὶ ὁ ἐκ τῶν Π, O ἄρα ἴσοξ ἐστὶ τῷ ἐκ τῶν B, Λ . ἐστὶν ἄρα ὥς ὁ Π πρὸς τὸν B , ὁ Λ πρὸς τὸν O . καὶ ἐστὶν ὁ Π τῷ B ὁ αὐτόξ· καὶ ὁ Λ ἄρα τῷ O ἐστὶν ὁ αὐτόξ· ὅπερ ἀδύνατον· ὁ γὰρ O ὑπόκειται μηδενὶ τῶν ἐκκειμένων ὁ αὐτόξ· οὐκ ἄρα τὸν ZH μετρήσει τιξ ἀριθμὸς παρέχ τῶν $A, B, \Gamma, \Delta, E, \Theta K, \Lambda, M$ καὶ τῆξ μονάδοξ. καὶ ἐδείθη ὁ ZH τοῖξ $A, B, \Gamma, \Delta, E, \Theta K, \Lambda, M$ καὶ τῇ μονάδι ἴσοξ. τέλειοξ δὲ ἀριθμὸξ ἐστὶν ὁ τοῖξ ἐαντοῦ μέρεσιν ἴσοξ ὢν· τέλειοξ ἄρα ἐστὶν ὁ ZH · ὅπερ ἔδει δεῖχαι.

B, C, D are continually proportional, (starting) from a unit, D will thus not be measured by any other numbers except A, B, C [Prop. 9.13]. And P was assumed not (to be) the same as any of A, B, C . Thus, P does not measure D . But, as P (is) to D , so E (is) to Q . Thus, E does not measure Q either [Def. 7.20]. And E is a prime (number). And every prime number [is] prime to every (number) which it does not measure [Prop. 7.29]. Thus, E and Q are prime to one another. And (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. And as E is to Q , (so) P (is) to D . Thus, E measures P the same number of times as Q (measures) D . And D is not measured by any other (numbers) except A, B, C . Thus, Q is the same as one of A, B, C . Let it be the same as B . And as many as is the multitude of B, C, D , let so many (of the set out numbers) be taken, (starting) from E , (namely) E, HK, L . And E, HK, L are in the same ratio as B, C, D . Thus, via equality, as B (is) to D , (so) E (is) to L [Prop. 7.14]. Thus, the (number created) from (multiplying) B, L is equal to the (number created) from multiplying D, E [Prop. 7.19]. But, the (number created) from (multiplying) D, E is equal to the (number created) from (multiplying) Q, P . Thus, the (number created) from (multiplying) Q, P is equal to the (number created) from (multiplying) B, L . Thus, as Q is to B , (so) L (is) to P [Prop. 7.19]. And Q is the same as B . Thus, L is also the same as P . The very thing (is) impossible. For P was assumed not (to be) the same as any of the (numbers) set out. Thus, FG cannot be measured by any number except A, B, C, D, E, HK, L, M , and a unit. And FG was shown (to be) equal to (the sum of) A, B, C, D, E, HK, L, M , and a unit. And a perfect number is one which is equal to (the sum of) its own parts [Def. 7.22]. Thus, FG is a perfect (number). (Which is) the very thing it was required to show.

[†] This proposition demonstrates that perfect numbers take the form $2^{n-1}(2^n - 1)$ provided that $2^n - 1$ is a prime number. The ancient Greeks knew of four perfect numbers: 6, 28, 496, and 8128, which correspond to $n = 2, 3, 5$, and 7, respectively.

ELEMENTS BOOK 9

Applications of Number Theory[†]

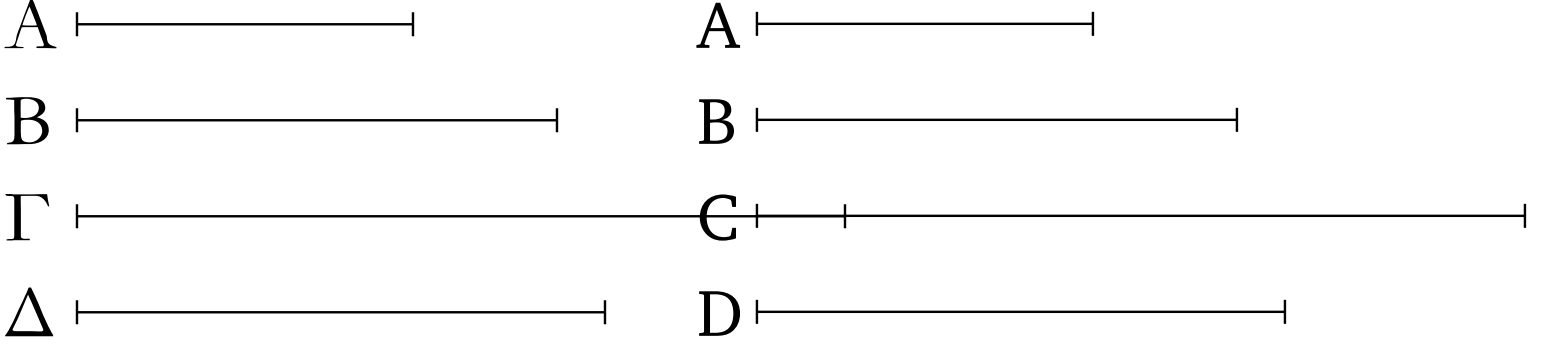
[†]The propositions contained in Books 7–9 are generally attributed to the school of Pythagoras.

1.

Proposition 1

Ἐάν δύο ὅμοιοι ἐπίπεδοι ἀριθμοὶ πολλαπλασιάσαντεξ ἀλλήλων ποιῶσι τινα, ὁ γενόμενος τετράγωνος ἔσται.

If two similar plane numbers make some (number by) multiplying one another then the created (number) will be square.



Ἐστωσαν δύο ὅμοιοι ἐπίπεδοι ἀριθμοὶ οἱ A, B , καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ ποιείτω· λέγω, ὅτι ὁ Γ τετράγωνός ἐστιν.

Let A and B be two similar plane numbers, and let A make C (by) multiplying B . I say that C is square.

Ὁ γὰρ A ἐαυτὸν πολλαπλασιάσας τὸν Δ ποιείτω. ὁ Δ ἄρα τετράγωνός ἐστιν. ἐπεὶ οὖν ὁ A ἐαυτὸν μὲν πολλαπλασιάσας τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν Γ . καὶ ἐπεὶ οἱ A, B ὅμοιοι ἐπίπεδοι εἰσιν ἀριθμοί, τῶν A, B ἄρα εἷξ μέσοξ ἀνάλογον ἐμπίπτει ἀριθμός. ἔαν δὲ δύο ἀριθμῶν μεταχὺν κατὰ τὸ συνεθὲξ ἀνάλογον ἐμπίπτωσιν ἀριθμοί, ὅσοι εἷξ αὐτοῦξ ἐμπίπτουσι, τοσοῦτοι καὶ εἷξ τοῦξ τὸν αὐτὸν λόγον ἔθονταξ· ὥστε καὶ τῶν Δ, Γ εἷξ μέσοξ ἀνάλογον ἐμπίπτει ἀριθμός. καὶ ἐστι τετράγωνος ὁ Δ · τετράγωνος ἄρα καὶ ὁ Γ · ὅπερ ἔδει δεῖχαι.

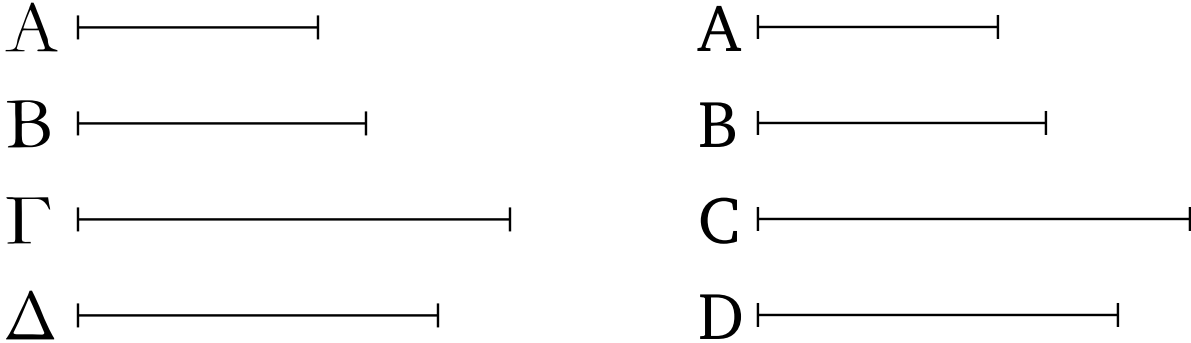
For let A make D (by) multiplying itself. D is thus square. Therefore, since A has made D (by) multiplying itself, and has made C (by) multiplying B , thus as A is to B , so D (is) to C [Prop. 7.17]. And since A and B are similar plane numbers, one number thus falls (between) A and B in mean proportion [Prop. 8.18]. And if (some) numbers fall between two numbers in continued proportion then, as many (numbers) as fall in (between) them (in continued proportion), so many also (fall) in (between numbers) having the same ratio (as them in continued proportion) [Prop. 8.8]. And hence one number falls (between) D and C in mean proportion. And D is square. Thus, C (is) also square [Prop. 8.22]. (Which is) the very thing it was required to show.

2.

Proposition 2

Ἐάν δύο ἀριθμοὶ πολλαπλασιάσαντεξ ἀλλήλων ποιῶσι τετράγωνον, ὅμοιοι ἐπίπεδοι εἰσιν ἀριθμοί.

If two numbers make a square (number by) multiplying one another then they are similar plane numbers.



Ἐστωσαν δύο ἀριθμοὶ οἱ A, B , καὶ ὁ A τὸν B πολλαπλασιάσας τετράγωνον τὸν Γ ποιείτω· λέγω, ὅτι οἱ A, B ὅμοιοι ἐπίπεδοι εἰσιν ἀριθμοί.

Let A and B be two numbers, and let A make the square (number) C (by) multiplying B . I say that A and B are similar plane numbers.

Ὁ γὰρ A ἐαντὸν πολλαπλασιάσας τὸν Δ ποιεῖτω· ὁ Δ ἄρα τετράγωνός ἐστιν. καὶ ἐπεὶ ὁ A ἐαντὸν μὲν πολλαπλασιάσας τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , ὁ Δ πρὸς τὸν Γ . καὶ ἐπεὶ ὁ Δ τετράγωνός ἐστιν, ἀλλὰ καὶ ὁ Γ , οἱ Δ , Γ ἄρα ὅμοιοι ἐπίπεδοι εἰσιν. τῶν Δ , Γ ἄρα εἷς μέσος ἀνάλογον ἐμπίπτει. καὶ ἐστιν ὡς ὁ Δ πρὸς τὸν Γ , οὕτως ὁ A πρὸς τὸν B . καὶ τῶν A , B ἄρα εἷς μέσος ἀνάλογον ἐμπίπτει. ἔαν δὲ δύο ἀριθμῶν εἷς μέσος ἀνάλογον ἐμπίπτῃ, ὅμοιοι ἐπίπεδοι εἰσιν [οἱ] ἀριθμοί· οἱ ἄρα A , B ὅμοιοι εἰσιν ἐπίπεδοι· ὅπερ ἔδει δεῖχαι.

3.

Ἐάν κύβος ἀριθμὸς ἐαντὸν πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενος κύβος ἔσται.

A —————

B —————

Γ —————

Δ —————

Κύβος γὰρ ἀριθμὸς ὁ A ἐαντὸν πολλαπλασιάσας τὸν B ποιεῖτω· λέγω, ὅτι ὁ B κύβος ἐστίν.

Εἰλήφθω γὰρ τοῦ A πλεονὰς ὁ Γ , καὶ ὁ Γ ἐαντὸν πολλαπλασιάσας τὸν Δ ποιεῖτω. φανερόν δὲ ἔστιν, ὅτι ὁ Γ τὸν Δ πολλαπλασιάσας τὸν Δ πεποίηκεν, καὶ ἐπεὶ ὁ Γ ἐαντὸν πολλαπλασιάσας τὸν Δ πεποίηκεν, ὁ Γ ἄρα τὸν Δ μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας. ἀλλὰ μὴν καὶ ἡ μονὰς τὸν Γ μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· ἔστιν ἄρα ὡς ἡ μονὰς πρὸς τὸν Γ , ὁ Γ πρὸς τὸν Δ . πάλιν, ἐπεὶ ὁ Γ τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν, ὁ Δ ἄρα τὸν A μετρεῖ κατὰ τὰς ἐν τῷ Γ μονάδας. μετρεῖ δὲ καὶ ἡ μονὰς τὸν Γ κατὰ τὰς ἐν αὐτῷ μονάδας· ἔστιν ἄρα ὡς ἡ μονὰς πρὸς τὸν Γ , ὁ Δ πρὸς τὸν A . ἀλλ' ὡς ἡ μονὰς πρὸς τὸν Γ , ὁ Γ πρὸς τὸν Δ · καὶ ὡς ἄρα ἡ μονὰς πρὸς τὸν Γ , οὕτως ὁ Γ πρὸς τὸν Δ καὶ ὁ Δ πρὸς τὸν A . τῆς ἄρα μονάδος καὶ τοῦ A ἀριθμοῦ δύο μέσοι ἀνάλογον κατὰ τὸ συνεθὲς ἐμπεπτώκασιν ἀριθμοί οἱ Γ , Δ . πάλιν, ἐπεὶ ὁ A ἐαντὸν πολλαπλασιάσας τὸν B πεποίηκεν, ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· μετρεῖ δὲ καὶ ἡ μονὰς τὸν A κατὰ τὰς ἐν αὐτῷ μονάδας· ἔστιν ἄρα ὡς ἡ μονὰς πρὸς τὸν A , ὁ A πρὸς τὸν B . τῆς δὲ μονάδος καὶ τοῦ A δύο μέσοι ἀνάλογον ἐμπεπτώκασιν ἀριθμοί· καὶ τῶν A , B ἄρα δύο μέσοι ἀνάλογον ἐμπεσύνται ἀριθμοί. ἔαν δὲ δύο ἀριθμῶν δύο μέσοι ἀνάλογον ἐμπίπτωσιν, ὁ δὲ πρῶτος κύβος ᾗ, καὶ ὁ δεύτερος

For let A make D (by) multiplying itself. Thus, D is square. And since A has made D (by) multiplying itself, and has made C (by) multiplying B , thus as A is to B , so D (is) to C [Prop. 7.17]. And since D is square, and C (is) also, D and C are thus similar plane numbers. Thus, one (number) falls (between) D and C in mean proportion [Prop. 8.18]. And as D is to C , so A (is) to B . Thus, one (number) also falls (between) A and B in mean proportion [Prop. 8.8]. And if one (number) falls (between) two numbers in mean proportion then [the] numbers are similar plane (numbers) [Prop. 8.20]. Thus, A and B are similar plane (numbers). (Which is) the very thing it was required to show.

Proposition 3

If a cube number makes some (number by) multiplying itself then the created (number) will be cube.

A —————

B —————

C —————

D —————

For let the cube number A make B (by) multiplying itself. I say that B is cube.

For let the side C of A be taken. And let C make D by multiplying itself. So it is clear that C has made A (by) multiplying D . And since C has made D (by) multiplying itself, C thus measures D according to the units in it [Def. 7.15]. But, in fact, a unit also measures C according to the units in it [Def. 7.20]. Thus, as a unit is to C , so C (is) to D . Again, since C has made A (by) multiplying D , D thus measures A according to the units in C . And a unit also measures C according to the units in it. Thus, as a unit is to C , so D (is) to A . But, as a unit (is) to C , so C (is) to D . And thus as a unit (is) to C , so C (is) to D , and D to A . Thus, two numbers, C and D , have fallen (between) a unit and the number A in continued mean proportion. Again, since A has made B (by) multiplying itself, A thus measures B according to the units in it. And a unit also measures A according to the units in it. Thus, as a unit is to A , so A (is) to B . And two numbers have fallen (between) a unit and A in mean proportion. Thus two numbers will also fall (between) A and B in mean proportion [Prop. 8.8]. And if two (numbers) fall (between) two numbers in mean proportion, and the first (number) is cube, then the second will also be cube [Prop. 8.23]. And A is cube. Thus, B is also cube.

κύβοξ ἔσται. καὶ ἐστὶν ὁ A κύβοξ· καὶ ὁ B ἄρα κύβοξ ἐστίν· (Which is) the very thing it was required to show.
ὅπερ ἔδει δεῖχαι.

4.

Ἐὰν κύβοξ ἀριθμὸς κύβον ἀριθμὸν πολλαπλασιάσας ποιῇ
τινα, ὁ γενόμενος κύβοξ ἔσται.

Κύβοξ γὰρ ἀριθμὸς ὁ A κύβον ἀριθμὸν τὸν B πολλαπλα-
σιάσας τὸν Γ ποιεῖτω· λέγω, ὅτι ὁ Γ κύβοξ ἐστίν.

A

B

Γ

Δ

Ὁ γὰρ A ἐαυτὸν πολλαπλασιάσας τὸν Δ ποιεῖτω· ὁ Δ
ἄρα κύβοξ ἐστίν. καὶ ἐπεὶ ὁ A ἐαυτὸν μὲν πολλαπλασιάσας
τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν,
ἐστὶν ἄρα ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν Γ . καὶ
ἐπεὶ οἱ A , B κύβοι εἰσὶν, ὅμοιοι στερεοὶ εἰσιν οἱ A , B . τῶν
 A , B ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί· ὥστε καὶ
τῶν Δ , Γ δύο μέσοι ἀνάλογον ἐμπεσοῦνται ἀριθμοί. καὶ ἐστὶ
κύβοξ ὁ Δ · κύβοξ ἄρα καὶ ὁ Γ · ὅπερ ἔδει δεῖχαι.

5.

Ἐὰν κύβοξ ἀριθμὸς ἀριθμὸν τινα πολλαπλασιάσας κύβον
ποιῇ, καὶ ὁ πολλαπλασιασθεὶς κύβοξ ἔσται.

A

B

Γ

Δ

Κύβοξ γὰρ ἀριθμὸς ὁ A ἀριθμὸν τινα τὸν B πολλαπλασι-
άσας κύβον τὸν Γ ποιεῖτω· λέγω, ὅτι ὁ B κύβοξ ἐστίν.

Ὁ γὰρ A ἐαυτὸν πολλαπλασιάσας τὸν Δ ποιεῖτω· κύβοξ
ἄρα ἐστὶν ὁ Δ . καὶ ἐπεὶ ὁ A ἐαυτὸν μὲν πολλαπλασιάσας
τὸν Δ πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν,

Proposition 4

If a cube number makes some (number by) multiplying
a(nother) cube number then the created (number) will be cube.

For let the cube number A make C (by) multiplying the
cube number B . I say that C is cube.

A

B

C

D

For let A make D (by) multiplying itself. Thus, D is cube
[Prop. 9.3]. And since A has made D (by) multiplying itself,
and has made C (by) multiplying B , thus as A is to B , so D
(is) to C [Prop. 7.17]. And since A and B are cube, A and
 B are similar solid (numbers). Thus, two numbers fall (be-
tween) A and B in mean proportion [Prop. 8.19]. Hence, two
numbers will also fall (between) D and C in mean proportion
[Prop. 8.8]. And D is cube. Thus, C (is) also cube [Prop. 8.23].
(Which is) the very thing it was required to show.

Proposition 5

If a cube number makes a(nother) cube number (by) mul-
tiplying some (number) then the (number) multiplied will also
be cube.

A

B

C

D

For let the cube number A make the cube (number) C (by)
multiplying some number B . I say that B is cube.

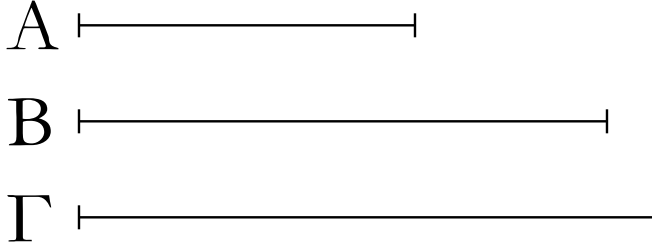
For let A make D (by) multiplying itself. D is thus cube
[Prop. 9.3]. And since A has made D (by) multiplying itself,
and has made C (by) multiplying B , thus as A is to B , so D (is)

ἔστιν ἄρα ὥς ὁ A πρός τὸν B , ὁ Δ πρός τὸν Γ . καὶ ἐπεὶ οἱ Δ , Γ κύβοι εἰσὶν, ὅμοιοι στερεοὶ εἰσιν. τῶν Δ , Γ ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. καὶ ἔστιν ὥς ὁ Δ πρός τὸν Γ , οὕτως ὁ A πρός τὸν B . καὶ τῶν A , B ἄρα δύο μέσοι ἀνάλογον ἐμπίπτουσιν ἀριθμοί. καὶ ἐστὶ κύβος ὁ A . κύβος ἄρα ἐστὶ καὶ ὁ B . ὅπερ ἔδει δεῖχαι.

6.

Ἐάν ἀριθμὸς ἑαυτὸν πολλαπλασιάσας κύβον ποιῇ, καὶ αὐτὸς κύβος ἔσται.

Ἀριθμὸς γάρ ὁ A ἑαυτὸν πολλαπλασιάσας κύβον τὸν B ποιείτω· λέγω, ὅτι καὶ ὁ A κύβος ἐστίν.



Ὅ γάρ A τὸν B πολλαπλασιάσας τὸν Γ ποιείτω. ἐπεὶ οὖν ὁ A ἑαυτὸν μὲν πολλαπλασιάσας τὸν B πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ Γ ἄρα κύβος ἐστίν. καὶ ἐπεὶ ὁ A ἑαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν, ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας. μετρεῖ δὲ καὶ ἡ μονὰς τὸν A κατὰ τὰς ἐν αὐτῷ μονάδας. ἔστιν ἄρα ὥς ἡ μονὰς πρός τὸν A , οὕτως ὁ A πρός τὸν B . καὶ ἐπεὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ B ἄρα τὸν Γ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας. μετρεῖ δὲ καὶ ἡ μονὰς τὸν A κατὰ τὰς ἐν αὐτῷ μονάδας. ἔστιν ἄρα ὥς ἡ μονὰς πρός τὸν A , οὕτως ὁ B πρός τὸν Γ . ἀλλ' ὥς ἡ μονὰς πρός τὸν A , οὕτως ὁ A πρός τὸν B . καὶ ὥς ἄρα ὁ A πρός τὸν B , ὁ B πρός τὸν Γ . καὶ ἐπεὶ οἱ B , Γ κύβοι εἰσὶν, ὅμοιοι στερεοὶ εἰσιν. τῶν B , Γ ἄρα δύο μέσοι ἀνάλογον εἰσιν ἀριθμοί. καὶ ἔστιν ὥς ὁ B πρός τὸν Γ , ὁ A πρός τὸν B . καὶ τῶν A , B ἄρα δύο μέσοι ἀνάλογον εἰσιν ἀριθμοί. καὶ ἐστὶν κύβος ὁ B . κύβος ἄρα ἐστὶ καὶ ὁ A . ὅπερ ἔδει δεῖχαι.

7.

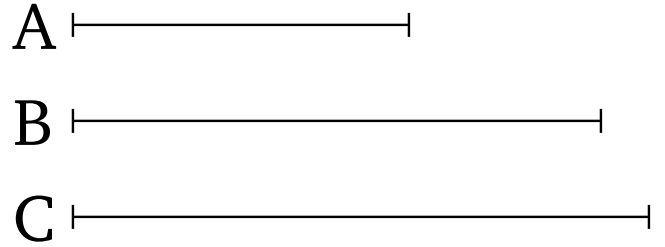
Ἐάν σύνθετος ἀριθμὸς ἀριθμὸν τινα πολλαπλασιάσας ποιῇ τινα, ὁ γεγόμενος στερεὸς ἔσται.

to C [Prop. 7.17]. And since D and C are (both) cube, they are similar solid (numbers). Thus, two numbers fall (between) D and C in mean proportion [Prop. 8.19]. And as D is to C , so A (is) to B . Thus, two numbers also fall (between) A and B in mean proportion [Prop. 8.8]. And A is cube. Thus, B is also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

Proposition 6

If a number makes a cube (number by) multiplying itself then it itself will also be cube.

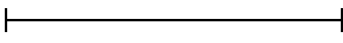
For let the number A make the cube (number) B (by) multiplying itself. I say that A is also cube.

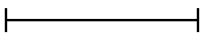


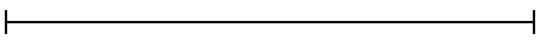
For let A make C (by) multiplying B . Therefore, since A has made B (by) multiplying itself, and has made C (by) multiplying B , C is thus cube. And since A has made B (by) multiplying itself, A thus measures B according to the units in (A). And a unit also measures A according to the units in it. Thus, as a unit is to A , so A (is) to B . And since A has made C (by) multiplying B , B thus measures C according to the units in A . And a unit also measures A according to the units in it. Thus, as a unit is to A , so B (is) to C . But, as a unit (is) to A , so A (is) to B . And thus as A (is) to B , (so) B (is) to C . And since B and C are cube, they are similar solid (numbers). Thus, there exist two numbers in mean proportion (between) B and C [Prop. 8.19]. And as B is to C , (so) A (is) to B . Thus, there also exist two numbers in mean proportion (between) A and B [Prop. 8.8]. And B is cube. Thus, A is also cube [Prop. 8.23]. (Which is) the very thing it was required to show.

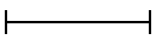
Proposition 7

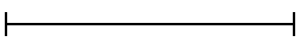
If a composite number makes some (number by) multiplying some (other) number then the created (number) will be solid.

A 

B 

Γ 

Δ 

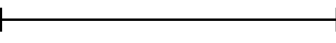
E 

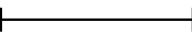
Σύνθετος γὰρ ἀριθμὸς ὁ A ἀριθμὸν τινα τὸν B πολλαπλασιάσας τὸν Γ ποιεῖτω· λέγω, ὅτι ὁ Γ στερεὸς ἐστίν.

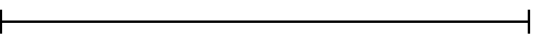
Ἐπεὶ γὰρ ὁ A σύνθετός ἐστιν, ὑπὸ ἀριθμοῦ τινος μετρηθήσεται. μετρείσθω ὑπὸ τοῦ Δ , καὶ ὁσάκις ὁ Δ τὸν A μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ E . ἔπει οὖν ὁ Δ τὸν A μετρεῖ κατὰ τὰς ἐν τῷ E μονάδας, ὁ E ἄρα τὸν Δ πολλαπλασιάσας τὸν A πεποίηκεν. καὶ ἐπεὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ δὲ A ἐστίν ὁ ἐκ τῶν Δ , E , ὁ ἄρα ἐκ τῶν Δ , E τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν. ὁ Γ ἄρα στερεὸς ἐστίν, πλευραὶ δὲ αὐτοῦ εἰσιν οἱ Δ , E , B . ὅπερ ἔδει δεῖχαι.

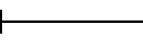
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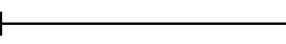
Ἐάν ἀπὸ μονάδος ὅποσοιὺν ἀριθμοὶ ἐχῇς ἀνάλογον ᾗσιν, ὁ μὲν τρίτος ἀπὸ τῆς μονάδος τετράγωνος ἔσται καὶ οἱ ἕνα διαλείποντες, ὁ δὲ τέταρτος κύβος καὶ οἱ δύο διαλείποντες πάντες, ὁ δὲ ἑβδομὸς κύβος ἅμα καὶ τετράγωνος καὶ οἱ πέντε διαλείποντες.

A 

B 

C 

D 

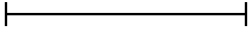
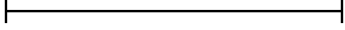
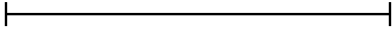
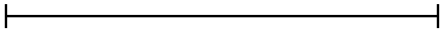
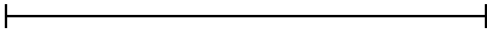
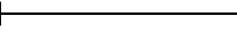
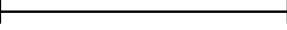
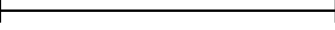
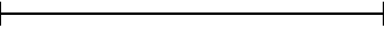
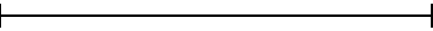
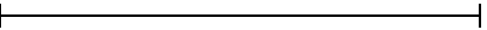
E 

For let the composite number A make C (by) multiplying some number B . I say that C is solid.

For since A is a composite (number), it will be measured by some number. Let it be measured by D . And, as many times as D measures A , so many units let there be in E . Therefore, since D measures A according to the units in E , E has thus made A (by) multiplying D [Def. 7.15]. And since A has made C (by) multiplying B , and A is the (number created) from (multiplying) D , E , the (number created) from (multiplying) D , E has thus made C (by) multiplying B . Thus, C is solid, and its sides are D , E , B . (Which is) the very thing it was required to show.

Proposition 8

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then the third from the unit will be square, and (all) those (numbers after that) which leave an interval of one (number), and the fourth (will be) cube, and all those (numbers after that) which leave an interval of two (numbers), and the seventh (will be) both cube and square, and (all) those (numbers after that) which leave an interval of five (numbers).

A B Γ Δ E Z A B C D E F 

Ἐστωσαν ἀπὸ μονάδος ὅποσοιὸν ἀριθμοὶ ἐχῆς ἀνάλογον οἱ $A, B, \Gamma, \Delta, E, Z$ · λέγω, ὅτι ὁ μὲν τρίτος ἀπὸ τῆς μονάδος ὁ B τετράγωνός ἐστι καὶ οἱ ἑνα διαλείποντες πάντες, ὁ δὲ τέταρτος ὁ Γ κύβος καὶ οἱ δύο διαλείποντες πάντες, ὁ δὲ ἑβδομος ὁ Z κύβος ἅμα καὶ τετράγωνος καὶ οἱ πέντε διαλείποντες πάντες.

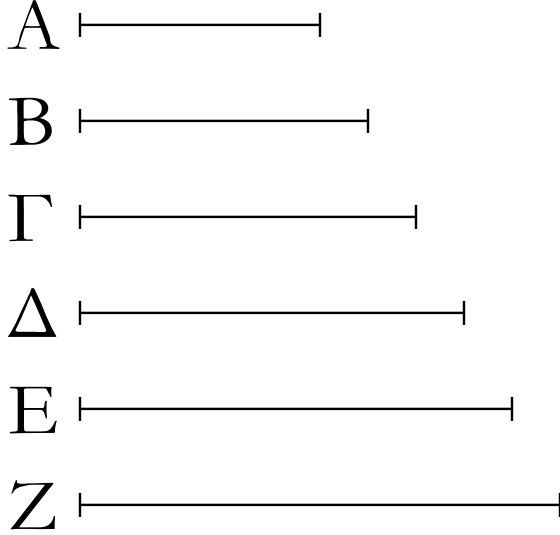
Ἐπεὶ γάρ ἐστιν ὥς ἡ μονὰς πρὸς τὸν A , οὕτως ὁ A πρὸς τὸν B , ἰσάκις ἄρα ἡ μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ A τὸν B . ἡ δὲ μονὰς τὸν A ἀριθμὸν μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· καὶ ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας. ὁ A ἄρα ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν· τετράγωνος ἄρα ἐστὶν ὁ B . καὶ ἐπεὶ οἱ B, Γ, Δ ἐχῆς ἀνάλογόν εἰσιν, ὁ δὲ B τετράγωνός ἐστιν, καὶ ὁ Δ ἄρα τετράγωνός ἐστιν ἡμιά τὰ αὐτὰ δὴ καὶ ὁ Z τετράγωνός ἐστιν. ὁμοίως δὴ δείχομεν, ὅτι καὶ οἱ ἑνα διαλείποντες πάντες τετράγωνοί εἰσιν. λέγω δὴ, ὅτι καὶ ὁ τέταρτος ἀπὸ τῆς μονάδος ὁ Γ κύβος ἐστὶ καὶ οἱ δύο διαλείποντες πάντες. ἐπεὶ γάρ ἐστιν ὥς ἡ μονὰς πρὸς τὸν A , οὕτως ὁ B πρὸς τὸν Γ , ἰσάκις ἄρα ἡ μονὰς τὸν A ἀριθμὸν μετρεῖ καὶ ὁ B τὸν Γ . ἡ δὲ μονὰς τὸν A ἀριθμὸν μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας· καὶ ὁ B ἄρα τὸν Γ μετρεῖ κατὰ τὰς ἐν τῷ A μονάδας· ὁ A ἄρα τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν. ἐπεὶ οὖν ὁ A ἐαυτὸν μὲν πολλαπλασιάσας τὸν B πεποίηκεν, τὸν δὲ B πολλαπλασιάσας τὸν Γ πεποίηκεν, κύβος ἄρα ἐστὶν ὁ Γ . καὶ ἐπεὶ οἱ Γ, Δ, E, Z ἐχῆς ἀνάλογόν εἰσιν, ὁ δὲ Γ κύβος ἐστίν, καὶ ὁ Z ἄρα κύβος ἐστίν. ἐδείκθη δὲ καὶ τετράγωνος· ὁ ἄρα ἑβδομος ἀπὸ τῆς μονάδος κύβος τέ ἐστι καὶ τετράγωνος. ὁμοίως δὴ δείχομεν, ὅτι καὶ οἱ πέντε διαλείποντες πάντες κύβοι τέ εἰσι καὶ τετράγωνοι· ὅπερ ἔδει δεῖχαι.

Let any multitude whatsoever of numbers, A, B, C, D, E, F , be continuously proportional, (starting) from a unit. I say that the third from the unit, B , is square, and all those (numbers after that) which leave an interval of one (number). And the fourth (from the unit), C , (is) cube, and all those (numbers after that) which leave an interval of two (numbers). And the seventh (from the unit), F , (is) both cube and square, and all those (numbers after that) which leave an interval of five (numbers).

For since as the unit is to A , so A (is) to B , the unit thus measures the number A the same number of times as A (measures) B [Def. 7.20]. And the unit measures the number A according to the units in it. Thus, A also measures B according to the units in A . A has thus made B (by) multiplying itself [Def. 7.15]. Thus, B is square. And since B, C, D are continuously proportional, and B is square, D is thus also square [Prop. 8.22]. So, for the same (reasons), F is also square. So, similarly, we can also show that all those (numbers after that) which leave an interval of one (number) are square. So I also say that the fourth (number) from the unit, C , is cube, and all those (numbers after that) which leave an interval of two (numbers). For since as the unit is to A , so B (is) to C , the unit thus measures the number A the same number of times that B (measures) C . And the unit measures the number A according to the units in A . And thus B measures C according to the units in A . A has thus made C (by) multiplying B . Therefore, since A has made B (by) multiplying itself, and has made C (by) multiplying B , C is thus cube. And since C, D, E, F are continuously proportional, and C is cube, F is thus also cube [Prop. 8.23]. And it was also shown (to be) square. Thus, the seventh (number) from the unit is (both) cube and square. So, similarly, we can show that all those (numbers after that) which leave an interval of five (numbers) are (both) cube and

9.

Ἐάν ἀπὸ μονάδος ὁποσοῖον ἔχη κατὰ τὸ συνεθές ἀριθμοὶ ἀνάλογον ᾧσιν, ὁ δὲ μετὰ τὴν μονάδα τετράγωνος ᾗ, καὶ οἱ λοιποὶ πάντες τετράγωνοι ἔσονται. καὶ ἔαν ὁ μετὰ τὴν μονάδα κύβος ᾗ, καὶ οἱ λοιποὶ πάντες κύβοι ἔσονται.



Ἔστωσαν ἀπὸ μονάδος ἔχῃς ἀνάλογον ὅσοιδηποῦν ἀριθμοὶ οἱ $A, B, \Gamma, \Delta, E, Z$, ὁ δὲ μετὰ τὴν μονάδα ὁ A τετράγωνος ἔστω· λέγω, ὅτι καὶ οἱ λοιποὶ πάντες τετράγωνοι ἔσονται.

Ὅτι μὲν οὖν ὁ τρίτος ἀπὸ τῆς μονάδος ὁ B τετράγωνός ἐστι καὶ οἱ ἑνα διαπλείοντες πάντες, δέδεικται· λέγω [δή], ὅτι καὶ οἱ λοιποὶ πάντες τετράγωνοι εἰσιν. ἐπεὶ γάρ οἱ A, B, Γ ἔχῃς ἀνάλογόν εἰσιν, καὶ ἔστιν ὁ A τετράγωνος, καὶ ὁ Γ [ἄρα] τετράγωνος ἐστίν. πάλιν, ἐπεὶ [καὶ] οἱ B, Γ, Δ ἔχῃς ἀνάλογόν εἰσιν, καὶ ἔστιν ὁ B τετράγωνος, καὶ ὁ Δ [ἄρα] τετράγωνός ἐστιν. ὁμοίως δὲ δείχμεν, ὅτι καὶ οἱ λοιποὶ πάντες τετράγωνοι εἰσιν.

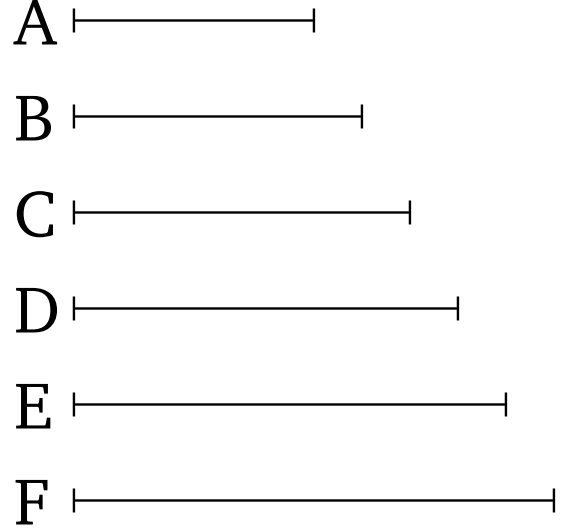
Ἀλλὰ δὴ ἔστω ὁ A κύβος· λέγω, ὅτι καὶ οἱ λοιποὶ πάντες κύβοι εἰσιν.

Ὅτι μὲν οὖν ὁ τέταρτος ἀπὸ τῆς μονάδος ὁ Γ κύβος ἐστὶ καὶ οἱ δύο διαλείποντες πάντες, δέδεικται· λέγω [δή], ὅτι καὶ οἱ λοιποὶ πάντες κύβοι εἰσιν. ἐπεὶ γάρ ἐστιν ὡς ἡ μονὰς πρὸς τὸν A , οὕτως ὁ A πρὸς τὸν B , ἰσάκις ἄρα ἡ μονὰς τὸν A μετρεῖ καὶ ὁ A τὸν B . ἡ δὲ μονὰς τὸν A μετρεῖ κατὰ τὰς ἐν αὐτῇ μονάδας· καὶ ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν αὐτῇ μονάδας· ὁ A ἄρα ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν. καὶ ἔστιν ὁ A κύβος. ἔαν δὲ κύβος ἀριθμὸς ἐαυτὸν πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενος κύβος ἐστίν· καὶ ὁ B ἄρα κύβος ἐστίν. καὶ ἐπεὶ τέσσαρες ἀριθμοὶ οἱ A, B, Γ, Δ ἔχῃς ἀνάλογόν εἰσιν, καὶ ἔστιν ὁ A κύβος, καὶ ὁ Δ ἄρα κύβος ἐστίν. διὰ

square. (Which is) the very thing it was required to show.

Proposition 9

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, and the (number) after the unit is square, then all the remaining (numbers) will also be square. And if the (number) after the unit is cube, then all the remaining (numbers) will also be cube.



Let any multitude whatsoever of numbers, A, B, C, D, E, F , be continuously proportional, (starting) from a unit. And let the (number) after the unit, A , be square. I say that all the remaining (numbers) will also be square.

In fact, it has (already) been shown that the third (number) from the unit, B , is square, and all those (numbers after that) which leave an interval of one (number) [Prop. 9.8]. [So] I say that all the remaining (numbers) are also square. For since A, B, C are continuously proportional, and A (is) square, C is [thus] also square [Prop. 8.22]. Again, since B, C, D are [also] continuously proportional, and B is square, D is [thus] also square [Prop. 8.22]. So, similarly, we can show that all the remaining (numbers) are also square.

And so let A be cube. I say that all the remaining (numbers) are also cube.

In fact, it has (already) been shown that the fourth (number) from the unit, C , is cube, and all those (numbers after that) which leave an interval of two (numbers) [Prop. 9.8]. [So] I say that all the remaining (numbers) are also cube. For since as the unit is to A , so A (is) to B , the unit thus measures A the same number of times as A (measures) B . And the unit measures A according to the units in it. Thus, A also measures B according to the units in (A). A has thus made B (by) multiplying itself. And A is cube. And if a cube number makes some (number by) multiplying itself then the created (num-

τὰ αὐτὰ δὴ καὶ ὁ E κύβος ἐστίν, καὶ ὁμοίως οἱ λοιποὶ πάντες κύβοι εἰσὶν· ὅπερ ἔδει δεῖχαι.

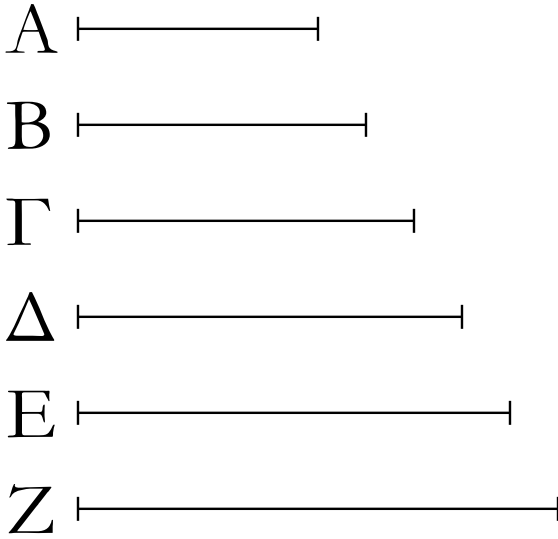
ber) is cube [Prop. 9.3]. Thus, B is also cube. And since the four numbers A, B, C, D are continuously proportional, and A is cube, D is thus also cube [Prop. 8.23]. So, for the same (reasons), E is also cube, and, similarly, all the remaining (numbers) are cube. (Which is) the very thing it was required to show.

10.

Proposition 10

Ἐάν ἀπὸ μονάδος ὁποσοιοῦν ἀριθμοὶ [ἐχῇ] ἀνάλογον ᾧσιν, ὁ δὲ μετὰ τὴν μονάδα μὴ ἢ τετράγωνος, οὐδ' ἄλλος οὐδεὶς τετράγωνος ἔσται θωρίξ τοῦ τρίτου ἀπὸ τῆς μονάδος καὶ τῶν ἑνα διαλειπόντων πάντων. καὶ ἔαν ὁ μετὰ τὴν μονάδα κύβος μὴ ἢ, οὐδὲ ἄλλος οὐδεὶς κύβος ἔσται θωρίξ τοῦ τετάρτου ἀπὸ τῆς μονάδος καὶ τῶν δύο διαλειπόντων πάντων.

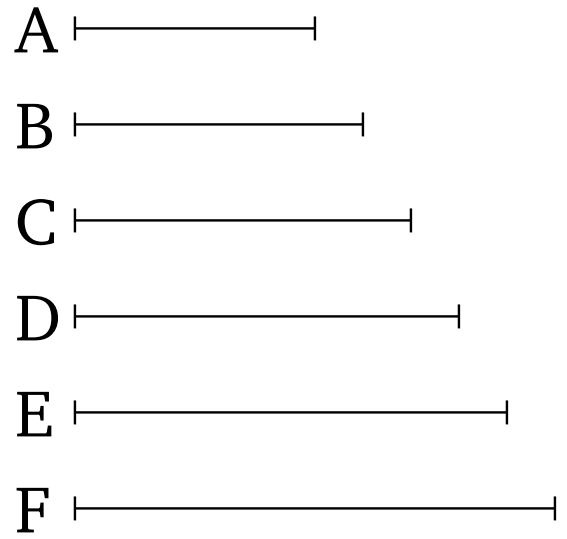
If any multitude whatsoever of numbers is [continuously] proportional, (starting) from a unit, and the (number) after the unit is not square, then no other (number) will be square either, apart from the third from the unit, and all those (numbers after that) which leave an interval of one (number). And if the (number) after the unit is not cube, then no other (number) will be cube either, apart from the fourth from the unit, and all those (numbers after that) which leave an interval of two (numbers).



Ἐστωσαν ἀπὸ μονάδος ἐχῇ ἀνάλογον ὅσοιδηποτοῦν ἀριθμοὶ οἱ $A, B, \Gamma, \Delta, E, Z$, ὁ μετὰ τὴν μονάδα ὁ A μὴ ἔστω τετράγωνος· λέγω, ὅτι οὐδὲ ἄλλος οὐδεὶς τετράγωνος ἔσται θωρίξ τοῦ τρίτου ἀπὸ τῆς μονάδος [καὶ τῶν ἑνα διαλειπόντων].

Εἰ γὰρ δυνατόν, ἔστω ὁ Γ τετράγωνος. ἔστι δὲ καὶ ὁ B τετράγωνος· οἱ B, Γ ἄρα πρὸς ἀλλήλους λόγον ἔθουσιν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ ἐστὶν ὡς ὁ B πρὸς τὸν Γ , ὁ A πρὸς τὸν B · οἱ A, B ἄρα πρὸς ἀλλήλους λόγον ἔθουσιν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ὥστε οἱ A, B ὅμοιοι ἐπίπεδοι εἰσιν. καὶ ἐστὶ τετράγωνος ὁ B · τετράγωνος ἄρα ἐστὶ καὶ ὁ A · ὅπερ οὐθ' ὑπέκειτο. οὐκ ἄρα ὁ Γ τετράγωνος ἐστίν. ὁμοίως δὲ δείχομεν, ὅτι οὐδ' ἄλλος οὐδεὶς τετράγωνος ἐστὶ θωρίξ τοῦ τρίτου ἀπὸ τῆς μονάδος καὶ τῶν ἑνα διαλειπόντων.

Ἀλλὰ δὴ μὴ ἔστω ὁ A κύβος. λέγω, ὅτι οὐδ' ἄλλος οὐδεὶς κύβος ἔσται θωρίξ τοῦ τετάρτου ἀπὸ τῆς μονάδος καὶ



Let any multitude whatsoever of numbers, A, B, C, D, E, F , be continuously proportional, (starting) from a unit. And let the (number) after the unit, A , not be square. I say that no other (number) will be square either, apart from the third from the unit [and (all) those (numbers after that) which leave an interval of one (number)].

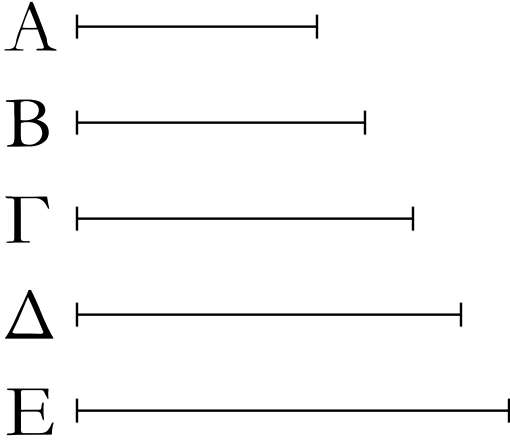
For, if possible, let C be square. And B is also square [Prop. 9.8]. Thus, B and C have to one another (the) ratio which (some) square number (has) to (some other) square number. And as B is to C , (so) A (is) to B . Thus, A and B have to one another (the) ratio which (some) square number has to (some other) square number. Hence, A and B are similar plane (numbers) [Prop. 8.26]. And B is square. Thus, A is also square. The very opposite thing was assumed. C is thus not square. So, similarly, we can show that no other (number is) square either, apart from the third from the unit, and

τῶν δύο διαλειπόντων.

Εἰ γὰρ δυνατόν, ἔστω ὁ Δ κύβος. ἔστι δὲ καὶ ὁ Γ κύβος· τέταρτος γάρ ἐστιν ἀπὸ τῆς μονάδος. καὶ ἐστὶν ὥς ὁ Γ πρὸς τὸν Δ , ὁ B πρὸς τὸν Γ · καὶ ὁ B ἄρα πρὸς τὸν Γ λόγον ἔθει, ὃν κύβος πρὸς κύβον. καὶ ἐστὶν ὁ Γ κύβος· καὶ ὁ B ἄρα κύβος ἐστίν. καὶ ἐπεὶ ἐστὶν ὥς ἡ μονὰς πρὸς τὸν A , ὁ A πρὸς τὸν B , ἡ δὲ μονὰς τὸν A μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας, καὶ ὁ A ἄρα τὸν B μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· ὁ A ἄρα ἐαντὸν πολλαπλασιάσας κύβον τὸν B πεποίηκεν. ἔαν δὲ ἀριθμὸς ἐαντὸν πολλαπλασιάσας κύβον ποιῇ, καὶ αὐτὸς κύβος ἔσται. κύβος ἄρα καὶ ὁ A · ὅπερ οὐθ' ὑπόκειται. οὐκ ἄρα ὁ Δ κύβος ἐστίν. ὁμοίως δὲ δείκνυμεν, ὅτι οὐδ' ἄλλος οὐδείς κύβος ἐστὶ θωρίξ τοῦ τετάρτου ἀπὸ τῆς μονάδος καὶ τῶν δύο διαλειπόντων· ὅπερ ἔδει δεῖξαι.

11.

Ἐὰν ἀπὸ μονάδος ὁποσοῖον ἀριθμοὶ ἐχῇς ἀνάλογον ᾧσιν, ὁ ἐλάττων τὸν μείζονα μετρεῖ κατὰ τινὰ τῶν ὑπαρθόντων ἐν τοῖς ἀνάλογον ἀριθμοῖς.



Ἐστωσαν ἀπὸ μονάδος τῆς A ὁποσοῖον ἀριθμοὶ ἐχῇς ἀνάλογον οἱ B, Γ, Δ, E . λέγω, ὅτι τῶν B, Γ, Δ, E ὁ ἐλάττωτος ὁ B τὸν E μετρεῖ κατὰ τινὰ τῶν Γ, Δ .

Ἐπεὶ γάρ ἐστιν ὥς ἡ μονὰς πρὸς τὸν B , οὕτως ὁ Δ πρὸς τὸν E , ἰσάκις ἄρα ἡ A μονὰς τὸν B ἀριθμὸν μετρεῖ καὶ ὁ Δ τὸν E . ἐναλλάξ ἄρα ἰσάκις ἡ A μονὰς τὸν Δ μετρεῖ καὶ ὁ B τὸν E . ἡ δὲ A μονὰς τὸν Δ μετρεῖ κατὰ τὰς ἐν αὐτῷ μονάδας· καὶ ὁ B ἄρα τὸν E μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας· ὥστε ὁ ἐλάττωτος ὁ B τὸν μείζονα τὸν E μετρεῖ κατὰ τινὰ ἀριθμὸν

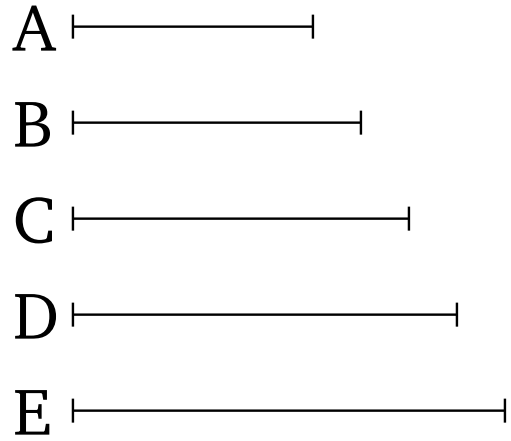
(all) those (numbers after that) which leave an interval of one (number).

And so let A not be cube. I say that no other (number) will be cube either, apart from the fourth from the unit, and (all) those (numbers after that) which leave an interval of two (numbers).

For, if possible, let D be cube. And C is also cube [Prop. 9.8]. For it is the fourth (number) from the unit. And as C is to D , (so) B (is) to C . And B thus has to C the ratio which (some) cube (number has) to (some other) cube (number). And C is cube. Thus, B is also cube [Props. 7.13, 8.25]. And since as the unit is to A , (so) A (is) to B , and the unit measures A according to the units in it, A thus also measures B according to the units in (A). Thus, A has made the cube (number) B (by) multiplying itself. And if a number makes a cube (number by) multiplying itself then it itself will be cube [Prop. 9.6]. Thus, A (is) also cube. The very opposite thing was assumed. Thus, D is not cube. So, similarly, we can show that no other (number) is cube either, apart from the fourth from the unit, and (all) those (numbers after that) which leave an interval of two (numbers). (Which is) the very thing it was required to show.

Proposition 11

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then a lesser (number) measures a greater according to some existing (number) among the proportional numbers.



Let any multitude whatsoever of numbers, B, C, D, E , be continuously proportional, (starting) from the unit A . I say that, for B, C, D, E , the least (number), B , measures E according to some (one) of C, D .

For since as the unit A is to B , so D (is) to E , the unit A thus measures the number B the same number of times as D (measures) E . Thus, alternately, the unit A measures D the same number of times as B (measures) E [Prop. 7.15]. And the unit A measures D according to the units in it. Thus, B

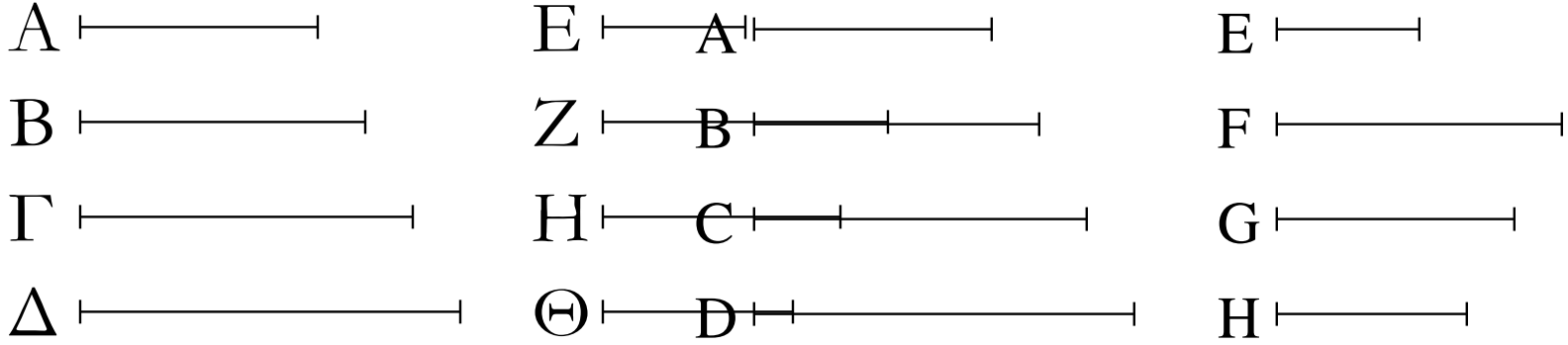
τῶν ὑπαρθόντων ἐν τοῖς ἀνάλογον ἀριθμοῖς.

Πόρισμα.

Καὶ φανερόν, ὅτι ἤν ἔθει τάχιν ὁ μετρῶν ἀπὸ μονάδος, τὴν αὐτὴν ἔθει καὶ ὁ καθ' ὃν μετρεῖ ἀπὸ τοῦ μετρούμενου ἐπὶ τὸ πρὸ αὐτοῦ. ὅπερ ἔδει δεῖξαι.

12.

Ἐάν ἀπὸ μονάδος ὁποσοῖον ἀριθμοὶ ἐχῇ ἀνάλογον ὦσιν, ὅφ' ὅσων ἂν ὁ ἕσθατος πρώτων ἀριθμῶν μετρηται, ὑπὸ τῶν αὐτῶν καὶ ὁ παρὰ τὴν μονάδα μετρηθήσεται.



Ἐστωσαν ἀπὸ μονάδος ὁποσοιδηποτὸν ἀριθμοὶ ἀνάλογον οἱ A, B, Γ, Δ λέγω, ὅτι ὅφ' ὅσων ἂν ὁ Δ πρώτων ἀριθμῶν μετρηται, ὑπὸ τῶν αὐτῶν καὶ ὁ A μετρηθήσεται.

Μετρεῖσθω γάρ ὁ Δ ὑπὸ τινος πρώτου ἀριθμοῦ τοῦ E . λέγω, ὅτι ὁ E τὸν A μετρεῖ. μὴ γάρ· καὶ ἐστὶν ὁ E πρώτος, ἅπασι δὲ πρώτος ἀριθμὸς πρὸς ἅπαντα, ὃν μὴ μετρεῖ, πρῶτός ἐστιν· οἱ E, A ἄρα πρώτοι πρὸς ἀλλήλους εἰσίν. καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Z . ὁ E ἄρα τὸν Z πολλαπλασιάσας τὸν Δ πεποίηκεν. πάλιν, ἐπεὶ ὁ E τὸν Δ μετρεῖ κατὰ τὰς ἐν τῷ Γ μονάδας, ὁ A ἄρα τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ E τὸν Z πολλαπλασιάσας τὸν Δ πεποίηκεν· ὁ ἄρα ἐκ τῶν A, Γ ἴσος ἐστὶ τῷ ἐκ τῶν E, Z . ἐστὶν ἄρα ὡς ὁ A πρὸς τὸν E , ὁ Z πρὸς τὸν Γ . οἱ δὲ A, E πρώτοι, οἱ δὲ πρώτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ἰσάκις ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον· μετρεῖ ἄρα ὁ E τὸν Γ . μετρεῖτω αὐτὸν κατὰ τὸν H . ὁ E ἄρα τὸν H πολλαπλασιάσας τὸν Γ πεποίηκεν. ἀλλὰ μὴν διὰ τὸ πρὸ τούτου καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν. ὁ ἄρα ἐκ τῶν A, B ἴσος ἐστὶ τῷ ἐκ τῶν E, H . ἐστὶν ἄρα ὡς ὁ A πρὸς τὸν E , ὁ H πρὸς τὸν B . οἱ δὲ A, E πρώτοι, οἱ δὲ πρώτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας αὐτοῖς ἰσάκις ὅ τε ἡγούμενος τὸν ἡγούμενον

also measures E according to the units in D . Hence, the lesser (number) B measures the greater E according to some existing number among the proportional numbers (namely, D).

Corollary

And (it is) clear that what(ever relative) place the measuring (number) has from the unit, the (number) according to which it measures has the same (relative) place from the measured (number), in (the direction of the number) before it. (Which is) the very thing it was required to show.

Proposition 12

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, then however many prime numbers the last (number) is measured by, the (number) next to the unit will also be measured by the same (prime numbers).

Let any multitude whatsoever of numbers, A, B, C, D , be (continuously) proportional, (starting) from a unit. I say that however many prime numbers D is measured by, A will also be measured by the same (prime numbers).

For let D be measured by some prime number E . I say that E measures A . For (suppose it does) not. E is prime, and every prime number is prime to every number which it does not measure [Prop. 7.29]. Thus, E and A are prime to one another. And since E measures D , let it measure it according to F . Thus, E has made D (by) multiplying F . Again, since A measures D according to the units in C [Prop. 9.11 corr.], A has thus made D (by) multiplying C . But, in fact, E has also made D (by) multiplying F . Thus, the (number created) from (multiplying) E, C is equal to the (number created) from (multiplying) E, F . Thus, as A is to E , (so) F (is) to C [Prop. 7.19]. And A and E (are) prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures C . Let it measure it according to G . Thus, E has made C (by) multiplying G . But, in fact, via the (proposition) before this, A

καὶ ὁ ἐπόμενος τὸν ἐπόμενον μετρεῖ ἄρα ὁ E τὸν B . μετρεῖται αὐτὸν κατὰ τὸν Θ . ὁ E ἄρα τὸν Θ πολλαπλασιάσας τὸν B πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν. ὁ ἄρα ἐκ τῶν E , Θ ἴσος ἐστὶ τῷ ἀπὸ τοῦ A . ἔστιν ἄρα ὡς ὁ E πρὸς τὸν A , ὁ A πρὸς τὸν Θ . οἱ δὲ A , E πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάτισται, οἱ δὲ ἐλάτισται μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ἰσάκις ὁ ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον μετρεῖ ἄρα ὁ E τὸν A ὡς ἡγούμενος ἡγούμενον. ἀλλὰ μὴν καὶ οὐ μετρεῖ. ὅπερ ἀδύνατον. οὐκ ἄρα οἱ E , A πρῶτοι πρὸς ἀλλήλους εἰσίν. σύνθετοι ἄρα. οἱ δὲ σύνθετοι ὑπὸ [πρώτου] ἀριθμοῦ τινος μετροῦνται. καὶ ἐπεὶ ὁ E πρῶτος ὑπόκειται, ὁ δὲ πρῶτος ὑπὸ ἐτέρου ἀριθμοῦ οὐ μετρεῖται ἢ ὑφ' ἐαυτοῦ, ὁ E ἄρα τοὺς A , E μετρεῖ. ὥστε ὁ E τὸν A μετρεῖ. μετρεῖ δὲ καὶ τὸν Δ . ὁ E ἄρα τοὺς A , Δ μετρεῖ. ὁμοίως δὲ δείχομεν, ὅτι ὑφ' ὅσων ἂν ὁ Δ πρώτων ἀριθμῶν μετρηται, ὑπὸ τῶν αὐτῶν καὶ ὁ A μετρηθήσεται. ὅπερ ἔδει δεῖχαι.

has also made C (by) multiplying B [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) A , B is equal to the (number created) from (multiplying) E , G . Thus, as A is to E , (so) G (is) to B [Prop. 7.19]. And A and E (are) prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures B . Let it measure it according to H . Thus, E has made B (by) multiplying H . But, in fact, A has also made B (by) multiplying itself [Prop. 9.8]. Thus, the (number created) from (multiplying) E , H is equal to the (square) on A . Thus, as E is to A , (so) A (is) to H [Prop. 7.19]. And A and E are prime (to one another), and (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, E measures A , as the leading (measuring the) leading. But, in fact, (E) also does not measure (A). The very thing (is) impossible. Thus, E and A are not prime to one another. Thus, (they are) composite (to one another). And (numbers) composite (to one another) are (both) measured by some [prime] number [Def. 7.14]. And since E is assumed (to be) prime, and a prime (number) is not measured by another number (other) than itself [Def. 7.11], E thus measures (both) A and E . Hence, E measures A . And it also measures D . Thus, E measures (both) A and D . So, similarly, we can show that however many prime numbers D is measured by, A will also be measured by the same (prime numbers). (Which is) the very thing it was required to show.

13.

Proposition 13

Ἐὰν ἀπὸ μονάδος ὅποσοιὼν ἀριθμοὶ ἐχῇς ἀνάλογον ᾗσιν, ὁ δὲ μετὰ τὴν μονάδα πρῶτος ᾗ, ὁ μέγιστος ὑπ' οὐδενὸς [ἀλλου] μετρηθήσεται παρέχ τῶν ὑπαρθόντων ἐν τοῖς ἀνάλογον ἀριθμοῖς.

Ἐστῶσαν ἀπὸ μονάδος ὅποσοιὼν ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A , B , Γ , Δ , ὁ δὲ μετὰ τὴν μονάδα ὁ A πρῶτος ἔστω. λέγω, ὅτι ὁ μέγιστος αὐτῶν ὁ Δ ὑπ' οὐδενὸς ἄλλον μετρηθήσεται παρέχ τῶν A , B , Γ .

If any multitude whatsoever of numbers is continuously proportional, (starting) from a unit, and the (number) after the unit is prime, then the greatest (number) will be measured by no [other] (numbers) except (numbers) existing among the proportional numbers.

Let any multitude whatsoever of numbers, A , B , C , D , be continuously proportional, (starting) from a unit. And let the (number) after the unit, A , be prime. I say that the greatest of them, D , will be measured by no other (numbers) except A , B , C .

A ———|

E ———|A|———|

E ———|

B ———|

Z ———|B|———|

F ———|

Γ ———|

H ———|Ε|———|

G ———|

Δ ———|

Θ ———|D|———|

H ———|

Εἴ γάρ δυνατόν, μετρεῖσθω ὑπὸ τοῦ E , καὶ ὁ E μηδενὶ τῶν A, B, Γ ἔστω ὁ αὐτόξ. φανερόν δὴ, ὅτι ὁ E πρωτοξ οὐκ ἔστιν. εἰ γάρ ὁ E πρωτόξ ἐστὶ καὶ μετρεῖ τὸν Δ , καὶ τὸν A μετρήσει πρωτον ὄντα μὴ ὡν αὐτῷ ὁ αὐτόξ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ E πρωτόξ ἐστὶν. σύνθετοξ ἄρα. πᾶξ δὲ σύνθετοξ ἀριθμὸξ ὑπὸ πρώτου τινὸξ ἀριθμοῦ μετρεῖται· ὁ E ἄρα ὑπὸ πρώτου τινὸξ ἀριθμοῦ μετρεῖται. λέγω δὴ, ὅτι ὑπ' οὐδενὸξ ἄλλου πρώτου μετρηθήσεται πλὴν τοῦ A . εἰ γάρ ὑφ' ἑτέρου μετρεῖται ὁ E , ὁ δὲ E τὸν Δ μετρεῖ, καὶ κεῖνοξ ἄρα τὸν Δ μετρήσει· ὥστε καὶ τὸν A μετρήσει πρωτον ὄντα μὴ ὡν αὐτῷ ὁ αὐτόξ· ὅπερ ἐστὶν ἀδύνατον. ὁ A ἄρα τὸν E μετρεῖ. καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Z . λέγω, ὅτι ὁ Z οὐδενὶ τῶν A, B, Γ ἐστὶν ὁ αὐτόξ. εἰ γάρ ὁ Z ἐνὶ τῶν A, B, Γ ἐστὶν ὁ αὐτόξ καὶ μετρεῖ τὸν Δ κατὰ τὸν E , καὶ εἰξ ἄρα τῶν A, B, Γ τὸν Δ μετρεῖ κατὰ τὸν E . ἀλλὰ εἰξ τῶν A, B, Γ τὸν Δ μετρεῖ κατὰ τινὰ τῶν A, B, Γ · καὶ ὁ E ἄρα ἐνὶ τῶν A, B, Γ ἐστὶν ὁ αὐτόξ· ὅπερ οὐθ' ὑπόκειται. οὐκ ἄρα ὁ Z ἐνὶ τῶν A, B, Γ ἐστὶν ὁ αὐτόξ. ὁμοίωξ δὴ δείχομεν, ὅτι μετρεῖται ὁ Z ὑπὸ τοῦ A , δεικνύντεξ πάλιν, ὅτι ὁ Z οὐκ ἐστὶ πρωτόξ. εἰ γάρ, καὶ μετρεῖ τὸν Δ , καὶ τὸν A μετρήσει πρωτον ὄντα μὴ ὡν αὐτῷ ὁ αὐτόξ· ὅπερ ἐστὶν ἀδύνατον· οὐκ ἄρα πρωτόξ ἐστὶν ὁ Z · σύνθετοξ ἄρα. ἅπαξ δὲ σύνθετοξ ἀριθμὸξ ὑπὸ πρώτου τινὸξ ἀριθμοῦ μετρεῖται· ὁ Z ἄρα ὑπὸ πρώτου τινὸξ ἀριθμοῦ μετρεῖται. λέγω δὴ, ὅτι ὑφ' ἑτέρου πρώτου οὐ μετρηθήσεται πλὴν τοῦ A . εἰ γάρ ἑτεροξ τιξ πρωτοξ τὸν Z μετρεῖ, ὁ δὲ Z τὸν Δ μετρεῖ, καὶ κεῖνοξ ἄρα τὸν Δ μετρήσει· ὥστε καὶ τὸν A μετρήσει πρωτον ὄντα μὴ ὡν αὐτῷ ὁ αὐτόξ· ὅπερ ἐστὶν ἀδύνατον. ὁ A ἄρα τὸν Z μετρεῖ. καὶ ἐπεὶ ὁ E τὸν Δ μετρεῖ κατὰ τὸν Z , ὁ E ἄρα τὸν Z πολλαπλασιάσας τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν· ὁ ἄρα ἐκ τῶν A, Γ ἴσοξ ἐστὶ τῷ ἐκ τῶν E, Z . ἀνάλογον ἄρα ἐστὶν ὡξ ὁ A πρὸξ τὸν E , οὕτωξ ὁ Z πρὸξ τὸν Γ . ὁ δὲ A τὸν E μετρεῖ· καὶ ὁ Z ἄρα τὸν Γ μετρεῖ. μετρεῖτω αὐτὸν κατὰ τὸν H . ὁμοίωξ δὴ δείχομεν, ὅτι ὁ H οὐδενὶ τῶν A, B ἐστὶν ὁ αὐτόξ, καὶ ὅτι μετρεῖται ὑπὸ τοῦ A . καὶ ἐπεὶ ὁ Z τὸν Γ μετρεῖ κατὰ τὸν H , ὁ Z ἄρα τὸν H πολλαπλασιάσας τὸν Γ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν· ὁ ἄρα ἐκ τῶν A, B ἴσοξ ἐστὶ τῷ ἐκ τῶν Z, H . ἀνάλογον ἄρα ὡξ ὁ A πρὸξ τὸν Z , ὁ H πρὸξ τὸν B . μετρεῖ δὲ ὁ A τὸν Z · μετρεῖ ἄρα καὶ ὁ H τὸν B . μετρεῖτω αὐτὸν

For, if possible, let it be measured by E , and let E not be the same as one of A, B, C . So it is clear that E is not prime. For if E is prime, and measures D , then it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, E is not prime. Thus, (it is) composite. And every composite number is measured by some prime number [Prop. 7.31]. Thus, E is measured by some prime number. So I say that it will be measured by no other prime number than A . For if E is measured by another (prime number), and E measures D , then this (prime number) will thus also measure D . Hence, it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, A measures E . And since E measures D , let it measure it according to F . I say that F is not the same as one of A, B, C . For if F is the same as one of A, B, C , and measures D according to E , then one of A, B, C thus also measures D according to E . But one of A, B, C (only) measures D according to some (one) of A, B, C [Prop. 9.11]. And thus E is the same as one of A, B, C . The very opposite thing was assumed. Thus, F is not the same as one of A, B, C . Similarly, we can show that F is measured by A , (by) again showing that F is not prime. For if (F is prime), and measures D , then it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, F is not prime. Thus, (it is) composite. And every composite number is measured by some prime number [Prop. 7.31]. Thus, F is measured by some prime number. So I say that it will be measured by no other prime number than A . For if some other prime (number) measures F , and F measures D , then this (prime number) will thus also measure D . Hence, it will also measure A , (despite A) being prime (and) not being the same as it [Prop. 9.12]. The very thing is impossible. Thus, A measures F . And since E measures D according to F , E has thus made D (by) multiplying F . But, in fact, A has also made D (by) multiplying C [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) A, C is equal to the (number created) from (multiplying) E, F . Thus, proportionally, as A is to E , so F (is) to C [Prop. 7.19]. And A measures E . Thus, F also measures C . Let it measure it according to G . So, similarly,

κατὰ τὸν Θ . ὁμοίως δὴ δείχομεν, ὅτι ὁ Θ τῷ A οὐκ ἔστιν ὁ αὐτός. καὶ ἐπεὶ ὁ H τὸν B μετρεῖ κατὰ τὸν Θ , ὁ H ἄρα τὸν Θ πολλαπλασιάσας τὸν B πεποίηκεν. ἀλλὰ μὴν καὶ ὁ A ἐαυτὸν πολλαπλασιάσας τὸν B πεποίηκεν· ὁ ἄρα ὑπὸ Θ , H ἴσος ἐστὶ τῷ ἀπὸ τοῦ A τετραγώνῳ· ἔστιν ἄρα ὡς ὁ Θ πρὸς τὸν A , ὁ A πρὸς τὸν H . μετρεῖ δὲ ὁ A τὸν H · μετρεῖ ἄρα καὶ ὁ Θ τὸν A πρῶτον ὄντα μὴ ὡν αὐτῷ ὁ αὐτός· ὅπερ ἄτοπον. οὐκ ἄρα ὁ μέγιστος ὁ Δ ὑπὸ ἐτέρου ἀριθμοῦ μετρηθήσεται παρέχ τῶν A , B , Γ . ὅπερ ἔδει δεῖχαι.

we can show that G is not the same as one of A , B , and that it is measured by A . And since F measures C according to G , F has thus made C (by) multiplying G . But, in fact, A has also made C (by) multiplying B [Prop. 9.11 corr.]. Thus, the (number created) from (multiplying) A , B is equal to the (number created) from (multiplying) F , G . Thus, proportionally, as A (is) to F , so G (is) to B [Prop. 7.19]. And A measures F . Thus, G also measures B . Let it measure it according to H . So, similarly, we can show that H is not the same as A . And since G measures B according to H , G has thus made B (by) multiplying H . But, in fact, A has also made B (by) multiplying itself [Prop. 9.8]. Thus, the (number created) from (multiplying) H , G is equal to the square on A . Thus, as H is to A , (so) A (is) to G [Prop. 7.19]. And A measures G . Thus, H also measures A , (despite A) being prime (and) not being the same as it. The very thing (is) absurd. Thus, the greatest (number) D cannot be measured by another (number) except (one of) A , B , C . (Which is) the very thing it was required to show.

14.

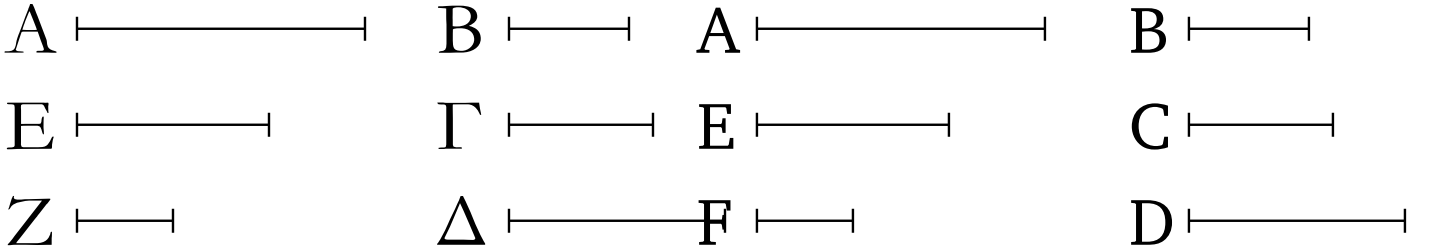
Proposition 14

Ἐὰν ἐλάχιστος ἀριθμὸς ὑπὸ πρώτων ἀριθμῶν μετρήται, ὑπ' οὐδενὸς ἄλλον πρώτου ἀριθμοῦ μετρηθήσεται παρέχ τῶν ἐκ ἀρθῆς μετρούντων.

If a least number is measured by (some) prime numbers then it will not be measured by any other prime number except (one of) the original measuring (numbers).

Ἐλάχιστος γάρ ἀριθμὸς ὁ A ὑπὸ πρώτων ἀριθμῶν τῶν B , Γ , Δ μετρεῖσθω· λέγω, ὅτι ὁ A ὑπ' οὐδενὸς ἄλλον πρώτου ἀριθμοῦ μετρηθήσεται παρέχ τῶν B , Γ , Δ .

For let A be the least number measured by the prime numbers B , C , D . I say that A will not be measured by any other prime number except (one of) B , C , D .



Εἰ γὰρ δυνατόν, μετρεῖσθω ὑπὸ πρώτου τοῦ E , καὶ ὁ E μηδενὶ τῶν B , Γ , Δ ἔστω ὁ αὐτός. καὶ ἐπεὶ ὁ E τὸν A μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν Z · ὁ E ἄρα τὸν Z πολλαπλασιάσας τὸν A πεποίηκεν. καὶ μετρεῖται ὁ A ὑπὸ πρώτων ἀριθμῶν τῶν B , Γ , Δ . ἔαν δὲ δύο ἀριθμοὶ πολλαπλασιάσαντες ἀλλήλους ποιῶσί τινα, τὸν δὲ γεγόμενον ἐκ αὐτῶν μετρή τις πρῶτος ἀριθμὸς, καὶ ἓνα τῶν ἐκ ἀρθῆς μετρήσει· οἱ B , Γ , Δ ἄρα ἓνα τῶν E , Z μετρήσουσιν. τὸν μὲν οὖν E οὐ μετρήσουσιν· ὁ γὰρ E πρῶτός ἐστι καὶ οὐδενὶ τῶν B , Γ , Δ ὁ αὐτός. τὸν Z ἄρα μετροῦσιν ἐλάσσονα ὄντα τοῦ A · ὅπερ ἀδύνατον. ὁ γὰρ A ὑπόκειται ἐλάχιστος ὑπὸ τῶν B , Γ , Δ μετρούμενος. οὐκ ἄρα τὸν A μετρήσει πρῶτος ἀριθμὸς παρέχ τῶν B , Γ , Δ · ὅπερ ἔδει δεῖχαι.

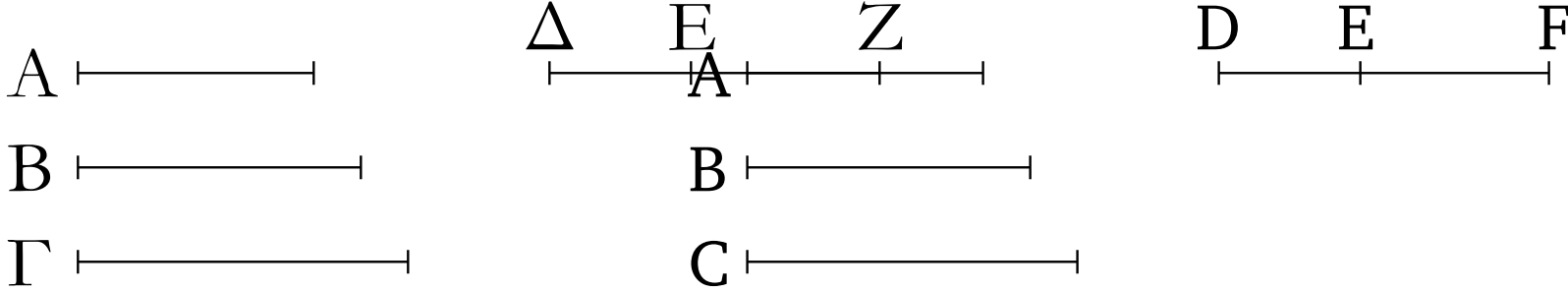
For, if possible, let it be measured by the prime (number) E . And let E not be the same as one of B , C , D . And since E measures A , let it measure it according to F . Thus, E has made A (by) multiplying F . And A is measured by the prime numbers B , C , D . And if two numbers make some (number by) multiplying one another, and some prime number measures the number created from them, then (the prime number) will also measure one of the original (numbers) [Prop. 7.30]. Thus, B , C , D will measure one of E , F . In fact, they do not measure E . For E is prime, and not the same as one of B , C , D . Thus, they (all) measure F , which is less than A . The very thing (is) impossible. For A was assumed (to be) the least (number) measured by B , C , D . Thus, no prime number can measure A except (one of) B , C , D . (Which is) the very thing it was required to show.

15.

Proposition 15

Ἐάν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ὥσιν ἐλάττωστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς, δύο ὁποιοῦν συντεθέντες πρὸς τὸν λοιπὸν πρῶτοί εἰσιν.

If three continuously proportional numbers are the least of those (numbers) having the same ratio as them then two (of them) added together in any way are prime to the remaining (one).



Ἐστωσαν τρεῖς ἀριθμοὶ ἐχῇς ἀνάλογον ἐλάττωστοι τῶν τὸν αὐτὸν λόγον ἐθόντων αὐτοῖς οἱ A, B, Γ . λέγω, ὅτι τῶν A, B, Γ δύο ὁποιοῦν συντεθέντες πρὸς τὸν λοιπὸν πρῶτοί εἰσιν, οἱ μὲν A, B πρὸς τὸν Γ , οἱ δὲ B, Γ πρὸς τὸν A καὶ ἔτι οἱ A, Γ πρὸς τὸν B .

Let A, B, C be three continuously proportional numbers (which are) the least of those (numbers) having the same ratio as them. I say that two of A, B, C added together in any way are prime to the remaining (one), (that is) A and B (prime) to C , B and C to A , and, further, A and C to B .

Εἰλήφθωσαν γὰρ ἐλάττωστοι ἀριθμοὶ τῶν τὸν αὐτὸν λόγον ἐθόντων τοῖς A, B, Γ δύο οἱ $\Delta E, EZ$. φανερόν δὲ, ὅτι ὁ μὲν ΔE ἐαντὸν πολλαπλασιάσας τὸν A πεποίηκεν, τὸν δὲ EZ πολλαπλασιάσας τὸν B πεποίηκεν, καὶ ἔτι ὁ EZ ἐαντὸν πολλαπλασιάσας τὸν Γ πεποίηκεν. καὶ ἐπεὶ οἱ $\Delta E, EZ$ ἐλάττωστοί εἰσιν, πρῶτοί πρὸς ἀλλήλους εἰσίν. ἔαν δὲ δύο ἀριθμοὶ πρῶτοί πρὸς ἀλλήλους ὦσιν, καὶ συναμφότεροξ πρὸς ἑκάτερον πρῶτός ἐστιν· καὶ ὁ ΔZ ἄρα πρὸς ἑκάτερον τῶν $\Delta E, EZ$ πρῶτός ἐστιν. ἀλλὰ μὴν καὶ ὁ ΔE πρὸς τὸν EZ πρῶτός ἐστιν· οἱ $\Delta Z, \Delta E$ ἄρα πρὸς τὸν EZ πρῶτοί εἰσιν. ἔαν δὲ δύο ἀριθμοὶ πρὸς τινὰ ἀριθμὸν πρῶτοί ὦσιν, καὶ ὁ ἐκ αὐτῶν γενόμενος πρὸς τὸν λοιπὸν πρῶτός ἐστιν· ὥστε ὁ ἐκ τῶν $Z\Delta, \Delta E$ πρὸς τὸν EZ πρῶτός ἐστιν· ὥστε καὶ ὁ ἐκ τῶν $Z\Delta, \Delta E$ πρὸς τὸν ἀπὸ τοῦ EZ πρῶτός ἐστιν. [ἔαν γὰρ δύο ἀριθμοὶ πρῶτοί πρὸς ἀλλήλους ὦσιν, ὁ ἐκ τοῦ ἐνόξ αὐτῶν γενόμενος πρὸς τὸν λοιπὸν πρῶτός ἐστιν]. ἀλλ' ὁ ἐκ τῶν $Z\Delta, \Delta E$ ὁ ἀπὸ τοῦ ΔE ἐστὶ μετὰ τοῦ ἐκ τῶν $\Delta E, EZ$ · ὁ ἄρα ἀπὸ τοῦ ΔE μετὰ τοῦ ἐκ τῶν $\Delta E, EZ$ πρὸς τὸν ἀπὸ τοῦ EZ πρῶτός ἐστιν. καὶ ἐστὶν ὁ μὲν ἀπὸ τοῦ ΔE ὁ A , ὁ δὲ ἐκ τῶν $\Delta E, EZ$ ὁ B , ὁ δὲ ἀπὸ τοῦ EZ ὁ Γ . οἱ A, B ἄρα συντεθέντες πρὸς τὸν Γ πρῶτοί εἰσιν. ὁμοίως δὲ δείχομεν, ὅτι καὶ οἱ B, Γ πρὸς τὸν A πρῶτοί εἰσιν. λέγω δὲ, ὅτι καὶ οἱ A, Γ πρὸς τὸν B πρῶτοί εἰσιν. ἐπεὶ γὰρ ὁ ΔZ πρὸς ἑκάτερον τῶν $\Delta E, EZ$ πρῶτός ἐστιν, καὶ ὁ ἀπὸ τοῦ ΔZ πρὸς τὸν ἐκ τῶν $\Delta E, EZ$ πρῶτός ἐστιν. ἀλλὰ τῷ ἀπὸ τοῦ ΔZ ἴσοι εἰσίν οἱ ἀπὸ τῶν $\Delta E, EZ$ μετὰ τοῦ διξ ἐκ τῶν $\Delta E, EZ$ · καὶ οἱ ἀπὸ τῶν $\Delta E, EZ$ ἄρα μετὰ τοῦ διξ ὑπὸ τῶν $\Delta E, EZ$ πρὸς τὸν ὑπὸ τῶν $\Delta E, EZ$ πρῶτοί [εἰσι]. διελόντι οἱ ἀπὸ τῶν $\Delta E, EZ$ μετὰ τοῦ ἀπαχ ὑπὸ $\Delta E, EZ$ πρὸς τὸν ὑπὸ $\Delta E, EZ$ πρῶτοί εἰσιν. ἔτι διελόντι οἱ ἀπὸ τῶν $\Delta E, EZ$ ἄρα πρὸς τὸν ὑπὸ $\Delta E, EZ$ πρῶτοί εἰσιν. καὶ ἐστὶν ὁ μὲν ἀπὸ τοῦ ΔE ὁ A , ὁ δὲ ὑπὸ τῶν $\Delta E, EZ$ ὁ B ,

Let the two least numbers, DE and EF , having the same ratio as A, B, C , be taken [Prop. 8.2]. So it is clear that DE has made A (by) multiplying itself, and has made B (by) multiplying EF , and, further, EF has made C (by) multiplying itself [Prop. 8.2]. And since DE, EF are the least (of those numbers having the same ratio as them), they are prime to one another [Prop. 7.22]. And if two numbers are prime to one another then the sum (of them) is also prime to each [Prop. 7.28]. Thus, DF is also prime to each of DE, EF . But, in fact, DE is also prime to EF . Thus, DF, DE are (both) prime to EF . And if two numbers are (both) prime to some number then the (number) created from (multiplying) them is also prime to the remaining (number) [Prop. 7.24]. Hence, the (number created) from (multiplying) FD, DE is prime to EF . Hence, the (number created) from (multiplying) FD, DE is also prime to the (square) on EF [Prop. 7.25]. [For if two numbers are prime to one another then the (number) created from (squaring) one of them is prime to the remaining (number).] But the (number created) from (multiplying) FD, DE is the (square) on DE plus the (number created) from (multiplying) DE, EF [Prop. 2.3]. Thus, the (square) on DE plus the (number created) from (multiplying) DE, EF is prime to the (square) on EF . And the (square) on DE is A , and the (number created) from (multiplying) DE, EF (is) B , and the (square) on EF (is) C . Thus, A, B summed is prime to C . So, similarly, we can show that B, C (summed) is also prime to A . So I say that A, C (summed) is also prime to B . For since DF is prime to each of DE, EF then the (square) on DF is also prime to the (number created) from (multiplying) DE, EF [Prop. 7.25]. But, the (sum of the squares) on DE, EF plus twice the (number created) from (multiplying) DE, EF is equal to the (square) on

ὁ δὲ ἀπὸ τοῦ EZ ὁ Γ . οἱ A , Γ ἄρα συντεθέντες πρὸς τὸν B πρῶτοι εἰσιν· ὅπερ ἔδει δεῖχαι.

DF [Prop. 2.4]. And thus the (sum of the squares) on DE , EF plus twice the (rectangle contained) by DE , EF [is] prime to the (rectangle contained) by DE , EF . By separation, the (sum of the squares) on DE , EF plus once the (rectangle contained) by DE , EF is prime to the (rectangle contained) by DE , EF .[†] Again, by separation, the (sum of the squares) on DE , EF is prime to the (rectangle contained) by DE , EF . And the (square) on DE is A , and the (rectangle contained) by DE , EF (is) B , and the (square) on EF (is) C . Thus, A , C summed is prime to B . (Which is) the very thing it was required to show.

[†] Since if $\alpha\beta$ measures $\alpha^2 + \beta^2 + 2\alpha\beta$ then it also measures $\alpha^2 + \beta^2 + \alpha\beta$, and vice versa.

16.

Ἐάν δύο ἀριθμοὶ πρῶτοι πρὸς ἀλλήλων ᾧσιν, οὐκ ἔσται ὡς ὁ πρῶτος πρὸς τὸν δεύτερον, οὕτως ὁ δεύτερος πρὸς ἄλλον τινά.



Δύο γὰρ ἀριθμοὶ οἱ A , B πρῶτοι πρὸς ἀλλήλων ἔστωσαν· λέγω, ὅτι οὐκ ἔστιν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ B πρὸς ἄλλον τινά.

Εἰ γὰρ δυνατόν, ἔστω ὡς ὁ A πρὸς τὸν B , ὁ B πρὸς τὸν Γ . οἱ δὲ A , B πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ἰσάκεις ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον· μετρεῖ ἄρα ὁ A τὸν B ὡς ἡγούμενος ἡγούμενον. μετρεῖ δὲ καὶ ἐαντόν· ὁ A ἄρα τοὺς A , B μετρεῖ πρῶτον· ὅντα πρὸς ἀλλήλους· ὅπερ ἄτοπον. οὐκ ἄρα ἔσται ὡς ὁ A πρὸς τὸν B , οὕτως ὁ B πρὸς τὸν Γ . ὅπερ ἔδει δεῖχαι.

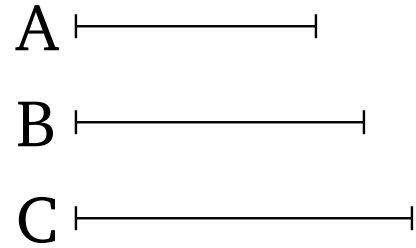
17.

Ἐάν ᾧσιν ὁσοιδηποτοῖν ἀριθμοὶ ἐχῇς ἀνάλογον, οἱ δὲ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους ᾧσιν, οὐκ ἔσται ὡς ὁ πρῶτος πρὸς τὸν δεύτερον, οὕτως ὁ ἔσθατος πρὸς ἄλλον τινά.

Ἐστωσαν ὁσοιδηποτοῖν ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A , B , Γ , Δ , οἱ δὲ ἄκροι αὐτῶν οἱ A , Δ πρῶτοι πρὸς ἀλλήλους ἔστωσαν· λέγω, ὅτι οὐκ ἔστιν ὡς ὁ A πρὸς τὸν B , οὕτως ὁ Δ πρὸς ἄλλον τινά.

Proposition 16

If two numbers are prime to one another then as the first is to the second, so the second (will) not (be) to some other (number).



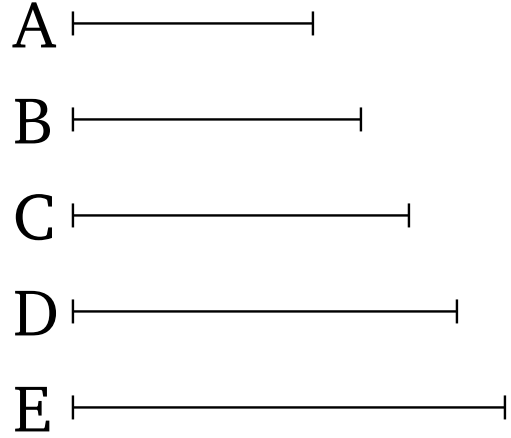
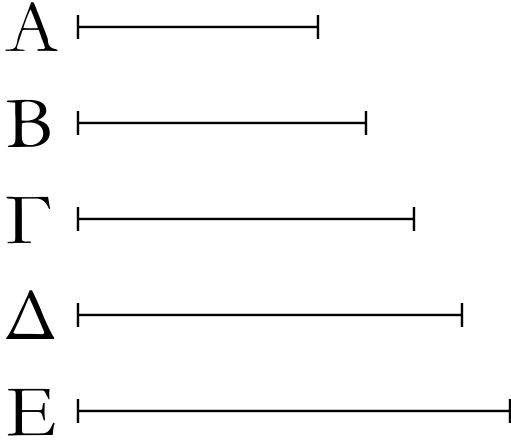
For let the two numbers A and B be prime to one another. I say that as A is to B , so B is not to some other (number).

For, if possible, let it be that as A (is) to B , (so) B (is) to C . And A and B (are) prime (to one another). And (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures B , as the leading (measuring) the leading. And (A) also measures itself. Thus, A measures A and B , which are prime to one another. The very thing (is) absurd. Thus, as A (is) to B , so B cannot be to C . (Which is) the very thing it was required to show.

Proposition 17

If any multitude whatsoever of numbers is continuously proportional, and the outermost of them are prime to one another, then as the first (is) to the second, so the last will not be to some other (number).

Let A , B , C , D be any multitude whatsoever of continuously proportional numbers. And let the outermost of them, A and D , be prime to one another. I say that as A is to B , so D (is) not to some other (number).



Εἰ γὰρ δυνατόν, ἔστω ὡς ὁ A πρόξ τὸν B , οὕτως ὁ Δ πρόξ τὸν E : ἐναλλάξ ἄρα ἐστὶν ὡς ὁ A πρόξ τὸν Δ , ὁ B πρόξ τὸν E . οἱ δὲ A , Δ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι ἀριθμοὶ μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔθονταξ ἰσάκις ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. μετρεῖ ἄρα ὁ A τὸν B . καὶ ἐστὶν ὡς ὁ A πρόξ τὸν B , ὁ B πρόξ τὸν Γ . καὶ ὁ B ἄρα τὸν Γ μετρεῖ· ὥστε καὶ ὁ A τὸν Γ μετρεῖ. καὶ ἐπεὶ ἐστὶν ὡς ὁ B πρόξ τὸν Γ , ὁ Γ πρόξ τὸν Δ , μετρεῖ δὲ ὁ B τὸν Γ , μετρεῖ ἄρα καὶ ὁ Γ τὸν Δ . ἀλλ' ὁ A τὸν Γ ἐμέτρει· ὥστε ὁ A καὶ τὸν Δ μετρεῖ. μετρεῖ δὲ καὶ ἑαυτόν. ὁ A ἄρα τοὺς A , Δ μετρεῖ πρῶτονξ ὄνταξ πρόξ ἀλλήλων· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἔσται ὡς ὁ A πρόξ τὸν B , οὕτως ὁ Δ πρόξ ἄλλον τινά· ὅπερ ἔδει δεῖχαι.

For, if possible, let it be that as A (is) to B , so D (is) to E . Thus, alternately, as A is to D , (so) B (is) to E [Prop. 7.13]. And A and D are prime (to one another). And (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21]. And the least numbers measure those (numbers) having the same ratio (as them) an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures B . And as A is to B , (so) B (is) to C . Thus, B also measures C . And hence A measures C [Def. 7.20]. And since as B is to C , (so) C (is) to D , and B measures C , C thus also measures D [Def. 7.20]. But, A was (found to be) measuring C . And hence A also measures D . And (A) also measures itself. Thus, A measures A and D , which are prime to one another. The very thing is impossible. Thus, as A (is) to B , so D cannot be to some other (number). (Which is) the very thing it was required to show.

18.

Δύο ἀριθμῶν δοθέντων ἐπισκέψασθαι, εἰ δυνατόν ἐστὶν αὐτοῖξ τρίτον ἀνάλογον προσευρεῖν.

Proposition 18

For two given numbers, to investigate whether it is possible to find a third (number) proportional to them.

Ἐστωσαν οἱ δοθέντες δύο ἀριθμοὶ οἱ A, B , καὶ δέον ἔστω ἐπισκέψασθαι, εἰ δυνατόν ἐστιν αὐτοῖς τρίτον ἀνάλογον προσευρεῖν.

Οἱ δὴ A, B ἢτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν ἢ οὐ. καὶ εἰ πρῶτοι πρὸς ἀλλήλους εἰσὶν, δέδεικται, ὅτι ἀδύνατόν ἐστιν αὐτοῖς τρίτον ἀνάλογον προσευρεῖν.

Ἀλλὰ δὴ μὴ ἔστωσαν οἱ A, B πρῶτοι πρὸς ἀλλήλους, καὶ ὁ B ἐαυτὸν πολλαπλασιάσας τὸν Γ ποιεῖτω. ὁ A δὴ τὸν Γ ἢτοι μετρεῖ ἢ οὐ μετρεῖ. μετρεῖτω πρότερον κατὰ τὸν Δ . ὁ A ἄρα τὸν Δ πολλαπλασιάσας τὸν Γ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ B ἐαυτὸν πολλαπλασιάσας τὸν Γ πεποίηκεν. ὁ ἄρα ἐκ τῶν A, Δ ἴσος ἐστὶ τῷ ἀπὸ τοῦ B . ἔστιν ἄρα ὡς ὁ A πρὸς τὸν B , ὁ B πρὸς τὸν Δ . τοῖς A, B ἄρα τρίτος ἀριθμὸς ἀνάλογον προσηγήσεται ὁ Δ .

Ἀλλὰ δὴ μὴ μετρεῖτω ὁ A τὸν Γ . λέγω, ὅτι τοῖς A, B ἀδύνατόν ἐστι τρίτον ἀνάλογον προσευρεῖν ἀριθμόν. εἰ γάρ δυνατόν, προσηγήσθω ὁ Δ . ὁ ἄρα ἐκ τῶν A, Δ ἴσος ἐστὶ τῷ ἀπὸ τοῦ B . ὁ δὲ ἀπὸ τοῦ B ἐστὶν ὁ Γ . ὁ ἄρα ἐκ τῶν A, Δ ἴσος ἐστὶ τῷ Γ . ὥστε ὁ A τὸν Δ πολλαπλασιάσας τὸν Γ πεποίηκεν. ὁ A ἄρα τὸν Γ μετρεῖ κατὰ τὸν Δ . ἀλλὰ μὴν ὑπόκειται καὶ μὴ μετρῶν. ὅπερ ἄτοπον. οὐκ ἄρα δυνατόν ἐστι τοῖς A, B τρίτον ἀνάλογον προσευρεῖν ἀριθμόν, ὅταν ὁ A τὸν Γ μὴ μετρῇ. ὅπερ ἔδει δεῖχαι.

Let A and B be the two given numbers. And let it be required to investigate whether it is possible to find a third (number) proportional to them.

So A and B are either prime to one another, or not. And if they are prime to one another then it has (already) been shown that it is impossible to find a third (number) proportional to them [Prop. 9.16].

And so let A and B not be prime to one another. And let B make C (by) multiplying itself. So A either measures, or does not measure, C . Let it first of all measure (C) according to D . Thus, A has made C (by) multiplying D . But, in fact, B has also made C (by) multiplying itself. Thus, the (number created) from (multiplying) A, D is equal to the (square) on B . Thus, as A is to B , (so) B (is) to D [Prop. 7.19]. Thus, a third number has been found proportional to A, B , (namely) D .

And so let A not measure C . I say that it is impossible to find a third number proportional to A, B . For, if possible, let it be found, (and let it be) D . Thus, the (number created) from (multiplying) A, D is equal to the (square) on B [Prop. 7.19]. And the (square) on B is C . Thus, the (number created) from (multiplying) A, D is equal to C . Hence, A has made C (by) multiplying D . Thus, A measures C according to D . But (A) was, in fact, also assumed (to be) not measuring (C). The very thing (is) absurd. Thus, it is not possible to find a third number proportional to A, B when A does not measure C . (Which is) the very thing it was required to show.

19.

Τριῶν ἀριθμῶν δοθέντων ἐπισκέψασθαι, πότε δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσευρεῖν.

Ἐστωσαν οἱ δοθέντες τρεῖς ἀριθμοὶ οἱ A, B, Γ , κα-

Proposition 19[†]

For three given numbers, to investigate when it is possible to find a fourth (number) proportional to them.

Let A, B, C be the three given numbers. And let it be re-

ἰ δέον ἔστω επισκέψασθαι, πότε δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσευρεῖν.

Ἦτοι οὖν οὐκ εἰσιν ἐχῆς ἀνάλογον, καὶ οἱ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν, ἢ ἐχῆς εἰσιν ἀνάλογον, καὶ οἱ ἄκροι αὐτῶν οὐκ εἰσιν πρῶτοι πρὸς ἀλλήλους, ἢ οὔτε ἐχῆς εἰσιν ἀνάλογον, οὔτε οἱ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν, ἢ καὶ ἐχῆς εἰσιν ἀνάλογον, καὶ οἱ ἄκροι αὐτῶν πρῶτοι πρὸς ἀλλήλους εἰσίν.

Εἰ μὲν οὖν οἱ A, B, Γ ἐχῆς εἰσιν ἀνάλογον, καὶ οἱ ἄκροι αὐτῶν οἱ A, Γ πρῶτοι πρὸς ἀλλήλους εἰσίν, δέδεικται, ὅτι ἀδύνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσευρεῖν ἀριθμόν. μὴ ἔστωσαν δὴ οἱ A, B, Γ ἐχῆς ἀνάλογον τῶν ἀκρῶν πάλιν ὄντων πρῶτων πρὸς ἀλλήλους. λέγω, ὅτι καὶ οὔτως ἀδύνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσευρεῖν. εἰ γὰρ δυνατόν, προσευρήσθω ὁ Δ , ὥστε εἶναι ὡς τὸν A πρὸς τὸν B , τὸν Γ πρὸς τὸν Δ , καὶ γερονέτω ὡς ὁ B πρὸς τὸν Γ , ὁ Δ πρὸς τὸν E . καὶ ἐπεὶ ἐστὶν ὡς μὲν ὁ A πρὸς τὸν B , ὁ Γ πρὸς τὸν Δ , ὡς δὲ ὁ B πρὸς τὸν Γ , ὁ Δ πρὸς τὸν E , δι' ἴσον ἄρα ὡς ὁ A πρὸς τὸν Γ , ὁ Γ πρὸς τὸν E . οἱ δὲ A, Γ πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάθιστοι, οἱ δὲ ἐλάθιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἐπόμενος τὸν ἐπόμενον. μετρεῖ ἄρα ὁ A τὸν Γ ὡς ἡγούμενος ἡγούμενον. μετρεῖ δὲ καὶ ἑαυτὸν ὁ A ἄρα τοὺς A, Γ μετρεῖ πρῶτον ὄντας πρὸς ἀλλήλους· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τοῖς A, B, Γ δυνατόν ἐστι τέταρτον ἀνάλογον προσευρεῖν.

Ἀλλὰ δὴ πάλιν ἔστωσαν οἱ A, B, Γ ἐχῆς ἀνάλογον, οἱ δὲ A, Γ μὴ ἔστωσαν πρῶτοι πρὸς ἀλλήλους. λέγω, ὅτι δυνατόν ἐστιν αὐτοῖς τέταρτον ἀνάλογον προσευρεῖν. ὁ γὰρ B τὸν Γ πολλαπλασιάσας τὸν Δ ποιεῖτω· ὁ A ἄρα τὸν Δ ἤτοι μετρεῖ ἢ οὐ μετρεῖ. μετρεῖτω αὐτὸν πρότερον κατὰ τὸν E · ὁ A ἄρα τὸν E πολλαπλασιάσας τὸν Δ πεποίηκεν. ἀλλὰ μὴν καὶ ὁ B τὸν Γ πολλαπλασιάσας τὸν Δ πεποίηκεν· ὁ ἄρα ἐκ τῶν A, E ἴσος ἐστὶ τῷ ἐκ τῶν B, Γ . ἀνάλογον ἄρα [ἐστίν] ὡς ὁ A πρὸς τὸν B , ὁ Γ πρὸς τὸν E · τοῖς A, B, Γ ἄρα τέταρτος ἀνάλογον προσηύρηται ὁ E .

Ἀλλὰ δὴ μὴ μετρεῖτω ὁ A τὸν Δ · λέγω, ὅτι ἀδύνατόν ἐστι τοῖς A, B, Γ τέταρτον ἀνάλογον προσευρεῖν ἀριθμόν. εἰ γὰρ δυνατόν, προσευρήσθω ὁ E · ὁ ἄρα ἐκ τῶν A, E ἴσος ἐστὶ τῷ ἐκ τῶν B, Γ . ἀλλὰ ὁ ἐκ τῶν B, Γ ἐστὶν ὁ Δ · καὶ ὁ ἐκ τῶν A, E ἄρα ἴσος ἐστὶ τῷ Δ . ὁ A ἄρα τὸν E πολλαπλασιάσας τὸν Δ πεποίηκεν· ὁ A ἄρα τὸν Δ μετρεῖ κατὰ τὸν E · ὥστε μετρεῖ ὁ A τὸν Δ . ἀλλὰ καὶ οὐ μετρεῖ· ὅπερ ἄτοπον. οὐκ ἄρα δυνατόν ἐστι τοῖς A, B, Γ τέταρτον ἀνάλογον προσευρεῖν ἀριθμόν, ὅταν ὁ A τὸν Δ μὴ μετρή. ἀλλὰ δὴ οἱ A, B, Γ μήτε ἐχῆς ἔστωσαν ἀνάλογον μήτε οἱ ἄκροι πρῶτοι πρὸς ἀλλήλους. καὶ ὁ B τὸν Γ πολλαπλασιάσας τὸν Δ ποιεῖτω. ὁμοίως δὴ δευθθήσεται, ὅτι εἰ μὲν μετρεῖ ὁ A τὸν Δ , δυνατόν ἐστιν αὐτοῖς ἀνάλογον προσευρεῖν, εἰ δὲ οὐ μετρεῖ, ἀδύνατον· ὅπερ ἔδει δεῖχαι.

quired to investigate when it is possible to find a fourth (number) proportional to them.

In fact, (A, B, C) are either not continuously proportional and the outermost of them are prime to one another, or are continuously proportional and the outermost of them are not prime to one another, or are neither continuously proportional nor are the outermost of them prime to one another, or are continuously proportional and the outermost of them are prime to one another.

In fact, if A, B, C are continuously proportional, and the outermost of them, A and C , are prime to one another, (then) it has (already) been shown that it is impossible to find a fourth number proportional to them [Prop. 9.17]. So let A, B, C not be continuously proportional, (with) the outermost of them again being prime to one another. I say that, in this case, it is also impossible to find a fourth (number) proportional to them. For, if possible, let it be found, (and let it be) D . Hence, it will be that as A (is) to B , (so) C (is) to D . And let it be contrived that as B (is) to C , (so) D (is) to E . And since as A is to B , (so) C (is) to D , and as B (is) to C , (so) D (is) to E , thus, via equality, as A (is) to C , (so) C (is) to E [Prop. 7.14]. And A and C (are) prime (to one another). And (numbers) prime (to one another are) also the least (numbers having the same ratio as them) [Prop. 7.21]. And the least (numbers) measure those numbers having the same ratio as them (the same number of times), the leading (measuring) the leading, and the following the following [Prop. 7.20]. Thus, A measures C , (as) the leading (measuring) the leading. And it also measures itself. Thus, A measures A and C , which are prime to one another. The very thing is impossible. Thus, it is not possible to find a fourth (number) proportional to A, B, C .

And so let A, B, C again be continuously proportional, and let A and C not be prime to one another. I say that it is possible to find a fourth (number) proportional to them. For let B make D (by) multiplying C . Thus, A either measures or does not measure D . Let it, first of all, measure (D) according to E . Thus, A has made D (by) multiplying E . But, in fact, B has also made D (by) multiplying C . Thus, the (number created) from (multiplying) A, E is equal to the (number created) from (multiplying) B, C . Thus, proportionally, as A [is] to B , (so) C (is) to E [Prop. 7.19]. Thus, a fourth (number) proportional to A, B, C has been found, (namely) E .

And so let A not measure D . I say that it is impossible to find a fourth number proportional to A, B, C . For, if possible, let it be found, (and let it be) E . Thus, the (number created) from (multiplying) A, E is equal to the (number created) from (multiplying) B, C . But, the (number created) from (multiplying) B, C is D . And thus the (number created) from (multiplying) A, E is equal to D . Thus, A has made D (by) multiplying E . Thus, A measures D according to E . Hence, A measures D . But, it also does not measure (D). The very thing (is) absurd.

Thus, it is not possible to find a fourth number proportional to A, B, C when A does not measure D . And so (let) A, B, C (be) neither continuously proportional, nor (let) the outermost of them (be) prime to one another. And let B make D (by) multiplying C . So, similarly, it can be show that if A measures D then it is possible to find a fourth (number) proportional to (A, B, C) , and impossible if (A) does not measure (D) . (Which is) the very thing it was required to show.

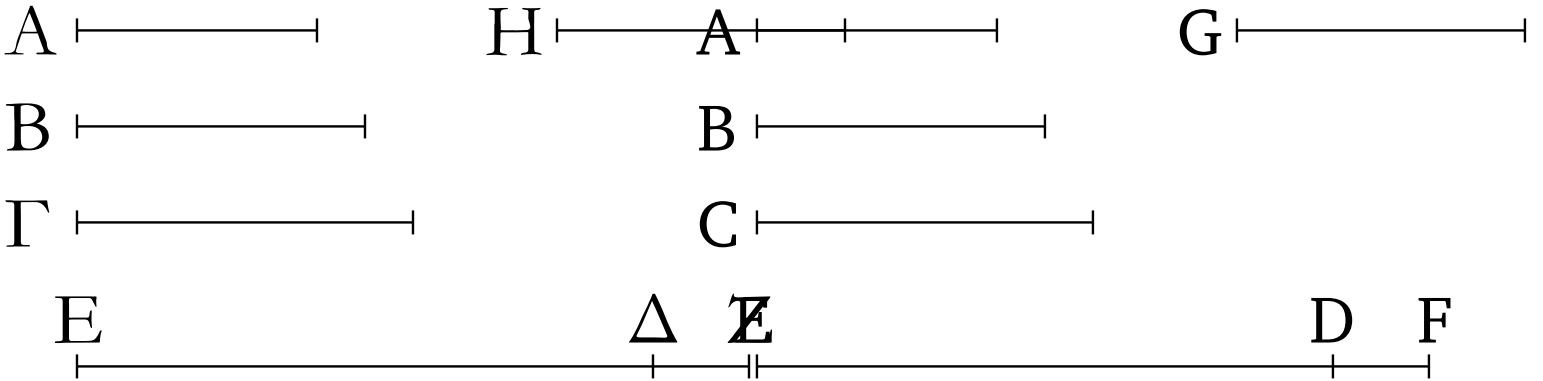
† The proof of this proposition is incorrect. There are, in fact, only two cases. Either A, B, C are continuously proportional, with A and C prime to one another, or not. In the first case, it is impossible to find a fourth proportional number. In the second case, it is possible to find a fourth proportional number provided that A measures B times C . Of the four cases considered by Euclid, the proof given in the second case is incorrect, since it only demonstrates that if $A : B :: C : D$ then a number E cannot be found such that $B : C :: D : E$. The proofs given in the other three cases are correct.

20.

Proposition 20

Οἱ πρώτοι ἀριθμοὶ πλείονες εἰσὶ παντὸς τοῦ προτεθέντος
πλήθους πρώτων ἀριθμῶν.

The (set of all) prime numbers is more numerous than any
assigned multitude of prime numbers.



Ἐστωσαν οἱ προτεθέντες πρώτοι ἀριθμοὶ οἱ A, B, Γ λέγω,
ὅτι τῶν A, B, Γ πλείονες εἰσὶ πρώτοι ἀριθμοί.

Εἰλήφθω γάρ ὁ ὑπὸ τῶν A, B, Γ ἐλάχιστος μετρούμενος
καὶ ἔστω ΔE , καὶ προσκείσθω τῷ ΔE μονὰς ἡ ΔZ . ὁ δὲ EZ
ἢτοι πρώτος ἔστιν ἢ οὐ. ἔστω πρότερον πρώτος· εὐρημένοι
ἄρα εἰσὶ πρώτοι ἀριθμοὶ οἱ A, B, Γ, EZ πλείονες τῶν A, B, Γ .

Ἀλλὰ δὴ μὴ ἔστω ὁ EZ πρώτος· ὑπὸ πρώτον ἄρα τινὸς
ἀριθμοῦ μετρεῖται. μετρεῖσθω ὑπὸ πρώτον τοῦ H λέγω, ὅτι ὁ
 H οὐδενὶ τῶν A, B, Γ ἔστιν ὁ αὐτός. εἰ γὰρ δυνατόν, ἔστω. οἱ
δὲ A, B, Γ τὸν ΔE μετροῦσιν καὶ ὁ H ἄρα τὸν ΔE μετρήσει.
μετρεῖ δὲ καὶ τὸν EZ · καὶ λοιπὴν τὴν ΔZ μονάδα μετρήσει ὁ
 H ἀριθμὸς ὧν ὅπερ ἄτοπον. οὐκ ἄρα ὁ H ἐνὶ τῶν A, B, Γ
ἔστιν ὁ αὐτός. καὶ ὑπόκειται πρώτος. εὐρημένοι ἄρα εἰσὶ
πρώτοι ἀριθμοὶ πλείονες τοῦ προτεθέντος πλήθους τῶν A, B, Γ
οἱ A, B, Γ, H · ὅπερ ἔδει δεῖχαι.

Let A, B, C be the assigned prime numbers. I say that the
(set of all) primes numbers is more numerous than A, B, C .

For let the least number measured by A, B, C be taken, and
let it be DE [Prop. 7.36]. And let the unit DF be added to
 DE . So EF is either prime, or not. Let it, first of all, be prime.
Thus, the (set of) prime numbers A, B, C, EF , (which is) more
numerous than A, B, C , has been found.

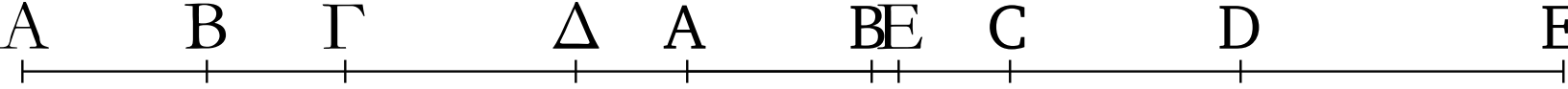
And so let EF not be prime. Thus, it is measured by some
prime number [Prop. 7.31]. Let it be measured by the prime
(number) G . I say that G is not the same as any of A, B, C . For,
if possible, let it be (the same). And A, B, C (all) measure DE .
Thus, G will also measure DE . And it also measures EF . (So)
 G will also measure the remainder, unit DF , (despite) being a
number [Prop. 7.28]. The very thing (is) absurd. Thus, G is not
the same as one of A, B, C . And it was assumed (to be) prime.
Thus, the (set of) prime numbers A, B, C, G , (which is) more
numerous than the assigned multitude (of prime numbers), A, B, C ,
has been found. (Which is) the very thing it was required
to show.

21.

Proposition 21

Ἐάν ἄρτιοι ἀριθμοὶ ὅποσοιὸν συντεθῶσιν, ὁ ὅλος ἄρτιός ἐστιν.

If any multitude whatsoever of even numbers is added together then the whole is even.



Συγκείσθωσαν γὰρ ἄρτιοι ἀριθμοὶ ὅποσοιὸν οἱ AB, BΓ, ΓΔ, ΔΕ· λέγω, ὅτι ὅλος ὁ ΑΕ ἄρτιός ἐστιν.

For let any multitude whatsoever of even numbers, AB, BC, CD, DE, lie together. I say that the whole, AE, is even.

Ἐπεὶ γὰρ ἕκαστος τῶν AB, BΓ, ΓΔ, ΔΕ ἄρτιός ἐστιν, ἔθει μέρος ἥμισυν· ὥστε καὶ ὅλος ὁ ΑΕ ἔθει μέρος ἥμισυν. ἄρτιος δὲ ἀριθμός ἐστιν ὁ δίπλα διαιρούμενος· ἄρτιος ἄρα ἐστὶν ὁ ΑΕ· ὅπερ ἔδει δεῖλαι.

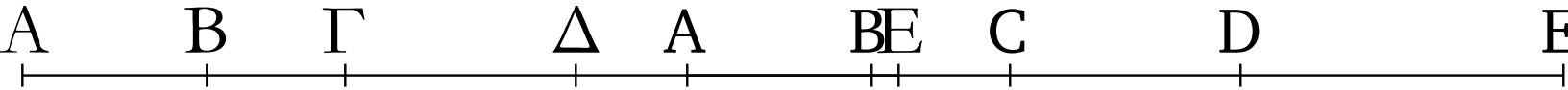
For since everyone of AB, BC, CD, DE is even, it has a half part [Def. 7.6]. And hence the whole AE has a half part [Def. 7.6]. Thus, AE is even. (Which is) the very thing it was required to show.

22.

Proposition 22

Ἐάν περισσοὶ ἀριθμοὶ ὅποσοιὸν συντεθῶσιν, τὸ δὲ πλῆθος αὐτῶν ἄρτιον ἢ, ὁ ὅλος ἄρτιος ἐσται.

If any multitude whatsoever of odd numbers is added together, and the multitude of them is even, then the whole will be even.



Συγκείσθωσαν γὰρ περισσοὶ ἀριθμοὶ ὅσοιδηποτοῦν ἄρτιον τὸ πλῆθος οἱ AB, BΓ, ΓΔ, ΔΕ· λέγω, ὅτι ὅλος ὁ ΑΕ ἄρτιός ἐστιν.

For let any even multitude whatsoever of odd numbers, AB, BC, CD, DE, lie together. I say that the whole, AE, is even.

Ἐπεὶ γὰρ ἕκαστος τῶν AB, BΓ, ΓΔ, ΔΕ περιττός ἐστιν, ἀφαιρεθείσῃ μονάδῳ ἀφ' ἑκάστου ἕκαστος τῶν λοιπῶν ἄρτιος ἐσται· ὥστε καὶ ὁ συγκεείμενος ἐξ αὐτῶν ἄρτιος ἐσται. ἔστι δὲ καὶ τὸ πλῆθος τῶν μονάδων ἄρτιον. καὶ ὅλος ἄρα ὁ ΑΕ ἄρτιός ἐστιν· ὅπερ ἔδει δεῖλαι.

For since everyone of AB, BC, CD, DE is odd then, a unit being subtracted from each, everyone of the remainders will be (made) even [Def. 7.7]. And hence the sum of them will be even [Prop. 9.21]. And the multitude of the units is even. Thus, the whole AE is also even [Prop. 9.21]. (Which is) the very thing it was required to show.

23.

Proposition 23

Ἐάν περισσοὶ ἀριθμοὶ ὅποσοιὸν συντεθῶσιν, τὸ δὲ πλῆθος αὐτῶν περισόν ἢ, καὶ ὁ ὅλος περισός ἐσται.

If any multitude whatsoever of odd numbers is added together, and the multitude of them is odd, then the whole will also be odd.



Συγκείσθωσαν γὰρ ὅποσοιὸν περισσοὶ ἀριθμοί, ὧν τὸ πλῆθος περισόν ἐστω, οἱ AB, BΓ, ΓΔ· λέγω, ὅτι καὶ ὅλος ὁ ΑΔ περισός ἐστιν.

For let any multitude whatsoever of odd numbers, AB, BC, CD, lie together, and let the multitude of them be odd. I say that the whole, AD, is also odd.

Ἀφηρήσθω ἀπὸ τοῦ ΓΔ μονάξ ἡ ΔΕ· λοιπὸς ἄρα ὁ ΓΕ ἄρτιός ἐστιν. ἔστι δὲ καὶ ὁ ΓΑ ἄρτιος· καὶ ὅλος ἄρα ὁ ΑΕ ἄρτιός ἐστιν. καὶ ἐστὶ μονάξ ἡ ΔΕ. περισός ἄρα ἐστὶν ὁ ΑΔ· ὅπερ ἔδει δεῖλαι.

For let the unit DE be subtracted from CD. The remainder, CE, is thus even [Def. 7.7]. And CA is also even [Prop. 9.22]. Thus, the whole, AE, is also even [Prop. 9.21]. And DE is a unit. Thus, AD is odd [Def. 7.7]. (Which is) the very thing it was required to show.

24.

Proposition 24

Ἐάν ἀπὸ ἁρτίου ἀριθμοῦ ἄρτιος ἀφαιρεθῇ, ὁ λοιπὸς ἄρτιος ἐσται.

If an even (number) is subtracted from an (other) even number then the remainder will be even.



Ἀπὸ γὰρ ἄρτιον τοῦ AB ἄρτιος ἀφηρήσθω ὁ $BΓ$. λέγω, ὅτι ὁ λοιπὸς ὁ $ΓΑ$ ἄρτιός ἐστιν.

Ἐπεὶ γὰρ ὁ AB ἄρτιός ἐστιν, ἔθει μέρος ἡμισυν. διὰ τὰ αὐτὰ δὴ καὶ ὁ $BΓ$ ἔθει μέρος ἡμισυν· ὥστε καὶ λοιπὸς [ὁ $ΓΑ$ ἔθει μέρος ἡμισυν] ἄρτιος [ἄρα] ἐστὶν ὁ $ΑΓ$. ὅπερ ἔδει δεῖχαι.

For let the even (number) BC be subtracted from the even number AB . I say that the remainder, CA , is even.

For since AB is even, it has a half part [Def. 7.6]. So, for the same (reasons), BC also has a half part. And hence the remainder [CA has a half part]. [Thus,] AC is even. (Which is) the very thing it was required to show.

25.

Proposition 25

Ἐὰν ἀπὸ ἄρτιον ἀριθμοῦ περισσόξ ἀφαιρεθῇ, ὁ λοιπὸς περισσόξ ἔσται.

If an odd (number) is subtracted from an even number then the remainder will be odd.



Ἀπὸ γὰρ ἄρτιον τοῦ AB περισσόξ ἀφηρήσθω ὁ $BΓ$. λέγω, ὅτι ὁ λοιπὸς ὁ $ΓΑ$ περισσόξ ἐστὶν.

Ἀφηρήσθω γὰρ ἀπὸ τοῦ $BΓ$ μονάξ ἡ $ΓΔ$. ὁ $ΔB$ ἄρα ἄρτιός ἐστιν. ἔστι δὲ καὶ ὁ AB ἄρτιος· καὶ λοιπὸς ἄρα ὁ $ΑΔ$ ἄρτιός ἐστιν. καὶ ἐστὶ μονάξ ἡ $ΓΔ$. ὁ $ΓΑ$ ἄρα περισσόξ ἐστὶν· ὅπερ ἔδει δεῖχαι.

For let the odd (number) BC be subtracted from the even number AB . I say that the remainder CA is odd.

For let the unit CD be subtracted from BC . DB is thus even [Def. 7.7]. And AB is also even. And thus the remainder AD is even [Prop. 9.24]. And CD is a unit. Thus, CA is odd [Def. 7.7]. (Which is) the very thing it was required to show.

26.

Proposition 26

Ἐὰν ἀπὸ περισσοῦ ἀριθμοῦ περισσόξ ἀφαιρεθῇ, ὁ λοιπὸς ἄρτιος ἔσται.

If an odd (number) is subtracted from an odd number then the remainder will be even.



Ἀπὸ γὰρ περισσοῦ τοῦ AB περισσόξ ἀφηρήσθω ὁ $BΓ$. λέγω, ὅτι ὁ λοιπὸς ὁ $ΓΑ$ ἄρτιός ἐστιν.

Ἐπεὶ γὰρ ὁ AB περισσόξ ἐστὶν, ἀφηρήσθω μονάξ ἡ $ΒΔ$. λοιπὸς ἄρα ὁ $ΑΔ$ ἄρτιός ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ὁ $ΓΔ$ ἄρτιός ἐστιν· ὥστε καὶ λοιπὸς ὁ $ΓΑ$ ἄρτιός ἐστιν· ὅπερ ἔδει δεῖχαι.

For let the odd (number) BC be subtracted from the odd (number) AB . I say that the remainder, CA , is even.

For since AB is odd, let the unit BD be subtracted (from it). Thus, the remainder AD is even [Def. 7.7]. So, for the same (reasons), CD is also even. And hence the remainder CA is even [Prop. 9.24]. (Which is) the very thing it was required to show.

27.

Proposition 27

Ἐὰν ἀπὸ περισσοῦ ἀριθμοῦ ἄρτιος ἀφαιρεθῇ, ὁ λοιπὸς περισσόξ ἔσται.

If an even (number) is subtracted from an odd number then the remainder will be odd.



Ἀπὸ γὰρ περισσοῦ τοῦ AB ἄρτιος ἀφηρήσθω ὁ $BΓ$. λέγω, ὅτι ὁ λοιπὸς ὁ $ΓΑ$ περισσόξ ἐστὶν.

Ἀφηρήσθω [γὰρ] μονάξ ἡ $ΑΔ$. ὁ $ΔB$ ἄρα ἄρτιός ἐστιν. ἔστι δὲ καὶ ὁ $BΓ$ ἄρτιος· καὶ λοιπὸς ἄρα ὁ $ΓΔ$ ἄρτιός ἐστιν. περισσόξ ἄρα ὁ $ΓΑ$. ὅπερ ἔδει δεῖχαι.

For let the even (number) BC be subtracted from the odd (number) AB . I say that the remainder, CA , is odd.

[For] let the unit AD be subtracted (from AB). DB is thus even [Def. 7.7]. And BC is also even. Thus, the remainder CD is also even [Prop. 9.24]. CA (is) thus odd [Def. 7.7]. (Which is) the very thing it was required to show.

is) the very thing it was required to show.

28.

Proposition 28

Ἐάν περισσόξ ἀριθμόξ ἄρτιον πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενος ἄρτιος ἔσται.

If an odd number makes some (number by) multiplying an even (number) then the created (number) will be even.

A |—————|

A |—————|

B |—————|

B |—————|

Γ |—————| C |—————|

Περισσόξ γάρ ἀριθμός ὁ A ἄρτιον τὸν B πολλαπλασιάσας τὸν Γ ποιείτω· λέγω, ὅτι ὁ Γ ἄρτιός ἐστιν.

For let the odd number A make C (by) multiplying the even (number) B . I say that C is even.

Ἐπεὶ γάρ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ Γ ἄρα σύγκειται ἐκ τοσούτων ἴσων τῷ B , ὅσαι εἰσὶν ἐν τῷ A μονάδες. καὶ ἐστὶν ὁ B ἄρτιος· ὁ Γ ἄρα σύγκειται ἐξ ἄρτίων. ἔαν δὲ ἄρτιοι ἀριθμοὶ ὁποσοῖσιν συντεθῶσιν, ὁ ὅλος ἄρτιός ἐστιν. ἄρτιος ἄρα ἐστὶν ὁ Γ · ὅπερ ἔδει δεῖχαι.

For since A has made C (by) multiplying B , C is thus composed out of so many (magnitudes) equal to B , as many as (there) are units in A [Def. 7.15]. And B is even. Thus, C is composed out of even (numbers). And if any multitude whatsoever of even numbers is added together then the whole is even [Prop. 9.21]. Thus, C is even. (Which is) the very thing it was required to show.

29.

Proposition 29

Ἐάν περισσόξ ἀριθμός περισσὸν ἀριθμὸν πολλαπλασιάσας ποιῇ τινα, ὁ γενόμενος περισσόξ ἔσται.

If an odd number makes some (number by) multiplying an odd (number) then the created (number) will be odd.

A |—————|

A |—————|

B |—————|

B |—————|

Γ |—————| C |—————|

Περισσόξ γάρ ἀριθμός ὁ A περισσὸν τὸν B πολλαπλασιάσας τὸν Γ ποιείτω· λέγω, ὅτι ὁ Γ περισσόξ ἐστιν.

For let the odd number A make C (by) multiplying the odd (number) B . I say that C is odd.

Ἐπεὶ γάρ ὁ A τὸν B πολλαπλασιάσας τὸν Γ πεποίηκεν, ὁ Γ ἄρα σύγκειται ἐκ τοσούτων ἴσων τῷ B , ὅσαι εἰσὶν ἐν τῷ A μονάδες. καὶ ἐστὶν ἐκάτερος τῶν A , B περισσόξ· ὁ Γ ἄρα σύγκειται ἐκ περισσῶν ἀριθμῶν, ὧν τὸ πλῆθος περισσόν ἐστιν. ὥστε ὁ Γ περισσόξ ἐστὶν· ὅπερ ἔδει δεῖχαι.

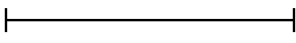
For since A has made C (by) multiplying B , C is thus composed out of so many (magnitudes) equal to B , as many as (there) are units in A [Def. 7.15]. And each of A , B is odd. Thus, C is composed out of odd (numbers), (and) the multitude of them is odd. Hence C is odd [Prop. 9.23]. (Which is) the very thing it was required to show.

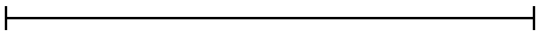
30.

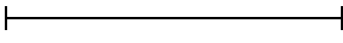
Proposition 30

Ἐάν περισσόξ ἀριθμός ἄρτιον ἀριθμὸν μετρήῃ, καὶ τὸν ἡμισὴν αὐτοῦ μετρήσῃ.

If an odd number measures an even number then it will also measure (one) half of it.

A 

B 

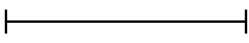
Γ 

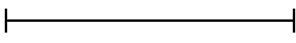
Περισσός γάρ ἀριθμός ὁ *A* ἄρτιον τὸν *B* μετρεῖτω· λέγω, ὅτι καὶ τὸν ἥμισυν αὐτοῦ μετρήσει.

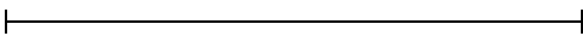
Ἐπεὶ γὰρ ὁ *A* τὸν *B* μετρεῖ, μετρεῖτω αὐτὸν κατὰ τὸν *Γ*· λέγω, ὅτι ὁ *Γ* οὐκ ἔστι περισσός. εἰ γὰρ δυνατόν, ἔστω. καὶ ἐπεὶ ὁ *A* τὸν *B* μετρεῖ κατὰ τὸν *Γ*, ὁ *A* ἄρα τὸν *Γ* πολλαπλασιάσας τὸν *B* πεποίηκεν. ὁ *B* ἄρα σύγκειται ἐκ περισσῶν ἀριθμῶν, ὧν τὸ πλῆθος περισσόν ἐστιν. ὁ *B* ἄρα περισσός ἐστιν· ὅπερ ἄτοπον· ὑπόκειται γὰρ ἄρτιος. οὐκ ἄρα ὁ *Γ* περισσός ἐστιν· ἄρτιος ἄρα ἐστὶν ὁ *Γ*. ὥστε ὁ *A* τὸν *B* μετρεῖ ἀρτιάκις. διὰ δὴ τοῦτο καὶ τὸν ἥμισυν αὐτοῦ μετρήσει· ὅπερ ἔδει δεῖχαι.

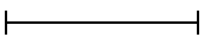
31.

Ἐὰν περισσός ἀριθμός πρὸς τινα ἀριθμὸν πρῶτος ᾖ, καὶ πρὸς τὸν διπλασίονα αὐτοῦ πρῶτος ἔσται.

A 

B 

Γ 

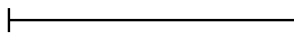
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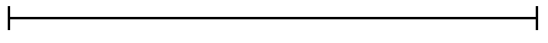
Περισσός γάρ ἀριθμός ὁ *A* πρὸς τινα ἀριθμὸν τὸν *B* πρῶτος ἔστω, τοῦ δὲ *B* διπλασίων ἔστω ὁ *Γ*. λέγω, ὅτι ὁ *A* [καὶ] πρὸς τὸν *Γ* πρῶτος ἐστίν.

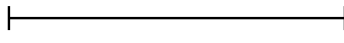
Εἰ γὰρ μὴ εἰσιν [οἱ *A*, *Γ*] πρῶτοι, μετρήσει τις αὐτοὺς ἀριθμός. μετρεῖτω, καὶ ἔστω ὁ *Δ*. καὶ ἐστὶν ὁ *A* περισσός· περισσός ἄρα καὶ ὁ *Δ*. καὶ ἐπεὶ ὁ *Δ* περισσός ὧν τὸν *Γ* μετρεῖ, καὶ ἐστὶν ὁ *Γ* ἄρτιος, καὶ τὸν ἥμισυν ἄρα τοῦ *Γ* μετρήσει [ὁ *Δ*]. τοῦ δὲ *Γ* ἥμισυν ἐστὶν ὁ *B*. ὁ *Δ* ἄρα τὸν *B* μετρεῖ. μετρεῖ δὲ καὶ τὸν *A*. ὁ *Δ* ἄρα τοὺς *A*, *B* μετρεῖ πρῶτον· ὅντα γὰρ πρὸς ἀλλήλους· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ *A* πρὸς τὸν *Γ* πρῶτος οὐκ ἐστίν. οἱ *A*, *Γ* ἄρα πρῶτοι πρὸς ἀλλήλους εἰσὶν· ὅπερ ἔδει δεῖχαι.

32.

Τῶν ἀπὸ δυνάδος διπλασιαζομένων ἀριθμῶν ἕκαστος ἀρ-

A 

B 

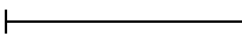
C 

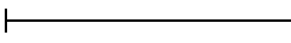
For let the odd number *A* measure the even (number) *B*. I say that (*A*) will also measure (one) half of (*B*).

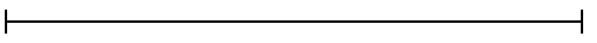
For since *A* measures *B*, let it measure it according to *C*. I say that *C* is not odd. For, if possible, let it be (odd). And since *A* measures *B* according to *C*, *A* has thus made *B* (by) multiplying *C*. Thus, *B* is composed out of odd numbers, (and) the multitude of them is odd. *B* is thus odd [Prop. 9.23]. The very thing (is) absurd. For (*B*) was assumed (to be) even. Thus, *C* is not odd. Thus, *C* is even. Hence, *A* measures *B* an even number of times. So, on account of this, (*A*) will also measure (one) half of (*B*). (Which is) the very thing it was required to show.

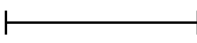
Proposition 31

If an odd number is prime to some number then it will also be prime to its double.

A 

B 

C 

D 

For let the odd number *A* be prime to some number *B*. And let *C* be double *B*. I say that *A* is [also] prime to *C*.

For if [*A* and *C*] are not prime (to one another) then some number will measure them. Let it measure (them), and let it be *D*. And *A* is odd. Thus, *D* (is) also odd. And since *D*, which is odd, measures *C*, and *C* is even, [*D*] will thus also measure half of *C* [Prop. 9.30]. And *B* is half of *C*. Thus, *D* measures *B*. And it also measures *A*. Thus, *D* measures (both) *A* and *B*, (despite) them being prime to one another. The very thing is impossible. Thus, *A* is not unprime to *C*. Thus, *A* and *C* are prime to one another. (Which is) the very thing it was required to show.

Proposition 32

Each of the numbers (which is continually) doubled, (start-

τιάκις ἄρτιός ἐστι μόνον.

A ————

B ————

Γ ————

Δ ————

Ἀπὸ γὰρ δυνάδοξ τῆς A δεδιπλασιάσθωσαν ὅσοιδηποτοῦν ἀριθμοὶ οἱ B, Γ, Δ· λέγω, ὅτι οἱ B, Γ, Δ ἀρτιάκις ἄρτιοὶ εἰσι μόνον.

Ὅτι μὲν οὖν ἕκαστος [τῶν B, Γ, Δ] ἀρτιάκις ἄρτιός ἐστιν, φανερόν· ἀπὸ γὰρ δυνάδοξ ἐστὶ διπλασιασθείς. λέγω, ὅτι καὶ μόνον. ἐκκείσθω γὰρ μονάξ. ἐπεὶ οὖν ἀπὸ μονάδοξ ὅποσοιὺν ἀριθμοὶ ἐχῆς ἀνάλογόν εἰσιν, ὁ δὲ μετὰ τὴν μονάδα ὁ A πρῶτός ἐστιν, ὁ μέγιστος τῶν A, B, Γ, Δ ὁ Δ ὑπ' οὐδενός ἄλλον μετρηθήσεται παρέχ τῶν A, B, Γ. καὶ ἐστὶν ἕκαστος τῶν A, B, Γ ἄρτιος· ὁ Δ ἄρα ἀρτιάκις ἄρτιός ἐστι μόνον. ὁμοίως δὲ δείκμεν, ὅτι [καὶ] ἑκάτερος τῶν B, Γ ἀρτιάκις ἄρτιός ἐστι μόνον· ὅπερ ἔδει δεῖχαι.

33.

Ἐὰν ἀριθμὸς τὸν ἥμισυν ἔθῃ περισσόν, ἀρτιάκις περισσός ἐστι μόνον.

A ————

Ἀριθμὸς γὰρ ὁ A τὸν ἥμισυν ἐθέτω περισσόν· λέγω, ὅτι ὁ A ἀρτιάκις περισσός ἐστι μόνον.

Ὅτι μὲν οὖν ἀρτιάκις περισσός ἐστιν, φανερόν· ὁ γὰρ ἥμισυν αὐτοῦ περισσός ὡν μετρεῖ αὐτὸν ἀρτιάκις, λέγω δὴ, ὅτι καὶ μόνον. εἰ γὰρ ἔσται ὁ A καὶ ἀρτιάκις ἄρτιος, μετρηθήσεται ὑπὸ ἀρτίου κατὰ ἄρτιον ἀριθμόν· ὥστε καὶ ὁ ἥμισυν αὐτοῦ μετρηθήσεται ὑπὸ ἀρτίου ἀριθμοῦ περισσός ὡν· ὅπερ ἐστὶν ἄτοπον. ὁ A ἄρα ἀρτιάκις περισσός ἐστι μόνον· ὅπερ ἔδει δεῖχαι.

34.

Ἐὰν ἀριθμὸς μῆτε τῶν ἀπὸ δυνάδοξ διπλασιαζομένων ἢ, μῆτε τὸν ἥμισυν ἔθῃ περισσόν, ἀρτιάκις τε ἄρτιός ἐστι καὶ ἀρτιάκις περισσός.

ing) from a dyad, is an even-times-even (number) only.

A ————

B ————

C ————

D ————

For let any multitude of numbers whatsoever, B, C, D, be (continually) doubled, (starting) from the dyad A. I say that B, C, D are even-times-even (numbers) only.

In fact, (it is) clear that each [of B, C, D] is an even-times-even (number). For it is doubled from a dyad [Def. 7.8]. I also say that (they are even-times-even numbers) only. For let a unit be laid down. Therefore, since any multitude of numbers whatsoever are continuously proportional, starting from a unit, and the (number) A after the unit is prime, the greatest of A, B, C, D, (namely) D, will not be measured by any other (numbers) except A, B, C [Prop. 9.13]. And each of A, B, C is even. Thus, D is an even-time-even (number) only [Def. 7.8]. So, similarly, we can show that each of B, C is [also] an even-time-even (number) only. (Which is) the very thing it was required to show.

Proposition 33

If a number has an odd half then it is an even-time-odd (number) only.

A ————

For let the number A have an odd half. I say that A is an even-times-odd (number) only.

In fact, (it is) clear that (A) is an even-times-odd (number). For its half, being odd, measures it an even number of times [Def. 7.9]. So I also say that (it is an even-times-odd number) only. For if A is also an even-times-even (number) then it will be measured by an even (number) according to an even number [Def. 7.8]. Hence, its half will also be measured by an even number, (despite) being odd. The very thing is absurd. Thus, A is an even-times-odd (number) only. (Which is) the very thing it was required to show.

Proposition 34

If a number is neither (one) of the (numbers) doubled from a dyad, nor has an odd half, then it is (both) an even-times-even and an even-times-odd (number).

A —————

Ἀριθμὸς γὰρ ὁ A μήτε τῶν ἀπὸ δυνάδοξ διπλασιαζομένων ἔστω μήτε τὸν ἥμισυν ἐθέτω περισσόν· λέγω, ὅτι ὁ A ἀρτιάκις τέ ἐστιν ἄρτιος καὶ ἀρτιάκις περισσός.

Ὅτι μὲν οὖν ὁ A ἀρτιάκις ἐστὶν ἄρτιος, φανερόν· τὸν γὰρ ἥμισυν οὐκ ἔθει περισσόν. λέγω δὲ, ὅτι καὶ ἀρτιάκις περισσός ἐστιν. ἔαν γὰρ τὸν A τέμνωμεν δίθα καὶ τὸν ἥμισυν αὐτοῦ δίθα καὶ τοῦτο αἰεὶ ποιῶμεν, καταστήσομεν εἷς τινα ἀριθμὸν περισσόν, ὃς μετρήσει τὸν A κατὰ ἄρτιον ἀριθμόν. εἰ γὰρ οὐ, καταστήσομεν εἷς δυνάδα, καὶ ἔσται ὁ A τῶν ἀπὸ δυνάδοξ διπλασιαζομένων· ὅπερ οὐθ' ὑπόκειται. ὥστε ὁ A ἀρτιάκις περισσόν ἐστιν. ἐδείχθη δὲ καὶ ἀρτιάκις ἄρτιος. ὁ A ἄρα ἀρτιάκις τε ἄρτιός ἐστι καὶ ἀρτιάκις περισσός· ὅπερ ἔδει δεῖχαι.

A —————

For let the number A neither be (one) of the (numbers) doubled from a dyad, nor let it have an odd half. I say that A is (both) an even-times-even and an even-times-odd (number).

In fact, (it is) clear that A is an even-times-even (number) [Def. 7.8]. For it does not have an odd half. So I say that it is also an even-times-odd (number). For if we cut A in half, and (then cut) its half in half, and we do this continually, then we will arrive at some odd number which will measure A according to an even number. For if not, we will arrive at a dyad, and A will be (one) of the (numbers) doubled from a dyad. The very opposite thing (was) assumed. Hence, A is an even-times-odd (number) [Def. 7.9]. And it was also shown (to be) an even-times-even (number). Thus, A is (both) an even-times-even and an even-times-odd (number). (Which is) the very thing it was required to show.

35.

Proposition 35[†]

Ἐάν ᾧσιν ὅσοιδηποτοῦν ἀριθμοὶ ἐχῇς ἀνάλογον, ἀφαιρεθῶσι δὲ ἀπὸ τε τοῦ δευτέρου καὶ τοῦ ἐσθάτου ἴσοι τῷ πρώτῳ, ἔσται ὥς ἡ τοῦ δευτέρου ὑπεροθὴ πρὸς τὸν πρῶτον, οὕτως ἡ τοῦ ἐσθάτου ὑπεροθὴ πρὸς τὸν πρὸ ἐαυτοῦ πάντα.

If there is any multitude whatsoever of continually proportional numbers, and (numbers) equal to the first are subtracted from (both) the second and the last, then as the excess of the second (number is) to the first, so the excess of the last will be to (the sum of) all those (numbers) before it.

A —————

B ————— H Γ

A —————

B ————— G C

Δ —————

D —————

E Λ K Θ

Ξ L K H F

Ἔστωσαν ὅποσοιδηποτοῦν ἀριθμοὶ ἐχῇς ἀνάλογον οἱ A , $BΓ$, $Δ$, $EΖ$ ἀφθόμενοι ἀπὸ ἐλαθίστου τοῦ A , καὶ ἀφηρήσθω ἀπὸ τοῦ $BΓ$ καὶ τοῦ $EΖ$ τῷ A ἴσοις ἐκάτερος τῶν $BΗ$, $ΖΘ$. λέγω, ὅτι ἐστὶν ὥς ὁ $HΓ$ πρὸς τὸν A , οὕτως ὁ $EΘ$ πρὸς τὸν A , $BΓ$, $Δ$.

Κείσθω γὰρ τῷ μὲν $BΓ$ ἴσος ὁ ZK , τῷ δὲ $Δ$ ἴσος ὁ $ΖΛ$. καὶ ἐπεὶ ὁ ZK τῷ $BΓ$ ἴσος ἐστίν, ὦν ὁ $ΖΘ$ τῷ $BΗ$ ἴσος ἐστίν, λοιπὸς ἄρα ὁ $ΘK$ λοιπῷ τῷ $HΓ$ ἐστὶν ἴσος. καὶ ἐπεὶ ἐστὶν ὥς ὁ $EΖ$ πρὸς τὸν $Δ$, οὕτως ὁ $Δ$ πρὸς τὸν $BΓ$ καὶ ὁ $BΓ$ πρὸς τὸν A , ἴσος δὲ ὁ μὲν $Δ$ τῷ $ΖΛ$, ὁ δὲ $BΓ$ τῷ ZK , ὁ δὲ A τῷ $ΖΘ$, ἐστὶν ἄρα ὥς ὁ $EΖ$ πρὸς τὸν $ΖΛ$, οὕτως ὁ $ΛΖ$ πρὸς τὸν

Let A , BC , D , EF be any multitude whatsoever of continually proportional numbers, beginning from the least A . And let $BΓ$ and $FΗ$, each equal to A , be subtracted from BC and EF (respectively). I say that as GC is to A , so EH is to A , BC , D .

For let FK be made equal to BC , and FL to D . And since FK is equal to BC , of which $FΗ$ is equal to $BΓ$, the remainder, HK , is thus equal to the remainder, GC . And since as EF is to D , so D (is) to BC , and BC to A [Prop. 7.13], and D (is) equal to FL , and BC to FK , and A to $FΗ$, thus as EF is to FL , so LF (is) to FK , and FK to $FΗ$. By separation, as EL (is) to

ZK καὶ ὁ ZK πρὸς τὸν $Z\Theta$. διελόντι, ὥς ὁ EA πρὸς τὸν AZ , οὕτως ὁ AK πρὸς τὸν ZK καὶ ὁ $K\Theta$ πρὸς τὸν $Z\Theta$. ἔστιν ἄρα καὶ ὥς εἶξ τῶν ἡγούμενων πρὸς ἓνα τῶν ἐπομένων, οὕτως ἅπαντες οἱ ἡγούμενοι πρὸς ἅπαντας τοὺς ἐπομένους· ἔστιν ἄρα ὥς ὁ $K\Theta$ πρὸς τὸν $Z\Theta$, οὕτως οἱ EA , AK , $K\Theta$ πρὸς τοὺς AZ , ZK , ΘZ . ἴσος δὲ ὁ μὲν $K\Theta$ τῷ $ΓH$, ὁ δὲ $Z\Theta$ τῷ A , οἱ δὲ AZ , ZK , ΘZ τοῖς Δ , $BΓ$, A . ἔστιν ἄρα ὥς ὁ $ΓH$ πρὸς τὸν A , οὕτως ὁ $E\Theta$ πρὸς τοὺς Δ , $BΓ$, A . ἔστιν ἄρα ὥς ἡ τοῦ δευτέρου ὑπεροθὴ πρὸς τὸν πρῶτον, οὕτως ἡ τοῦ ἐσθαιτον ὑπεροθὴ πρὸς τοὺς πρὸ ἐαυτοῦ πάντα· ὅπερ εἶδει δεῖχαι.

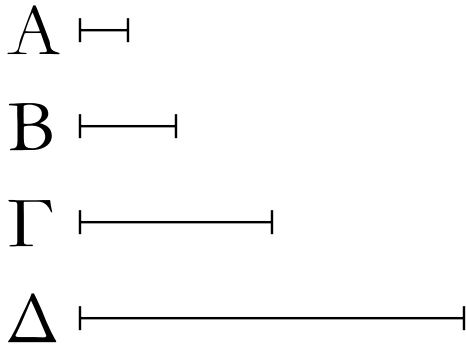
LF , so LK (is) to FK , and KH to FH [Props. 7.11, 7.13]. And thus as one of the leading (numbers) is to one of the following, so (the sum of) all of the leading (numbers is) to (the sum of) all of the following [Prop. 7.12]. Thus, as KH is to FH , so EL , LK , KH (are) to LF , FK , HF . And KH (is) equal to CG , and FH to A , and LF , FK , HF to D , BC , A . Thus, as CG is to A , so EH (is) to D , BC , A . Thus, as the excess of the second (number) is to the first, so the excess of the last (is) to (the sum of) all those (numbers) before it. (Which is) the very thing it was required to show.

[†] This proposition allows us to sum a geometric series of the form $a, ar, ar^2, ar^3, \dots, ar^{n-1}$. According to Euclid, the sum, S_n , satisfies $(ar - a)/a = (ar^n - a)/S_n$. Hence, $S_n = a(r^n - 1)/(r - 1)$.

36.

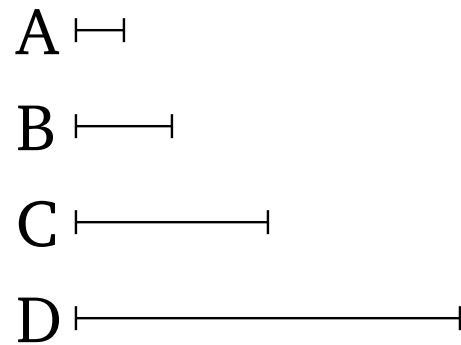
Ἐὰν ἀπὸ μονάδος ὁποσοιῶν ἀριθμοὶ ἐχῇς ἐκτεθῶσιν ἐν τῇ διπλασίῳ ἀναλογίᾳ, ἕως οὔ ὁ σύμπαξ συντεθείς πρῶτος γένηται, καὶ ὁ σύμπαξ ἐπὶ τὸν ἐσθαιτον πολλαπλασιασθείς ποιῇ τινα, ὁ γεγόμενος τέλειος ἔσται.

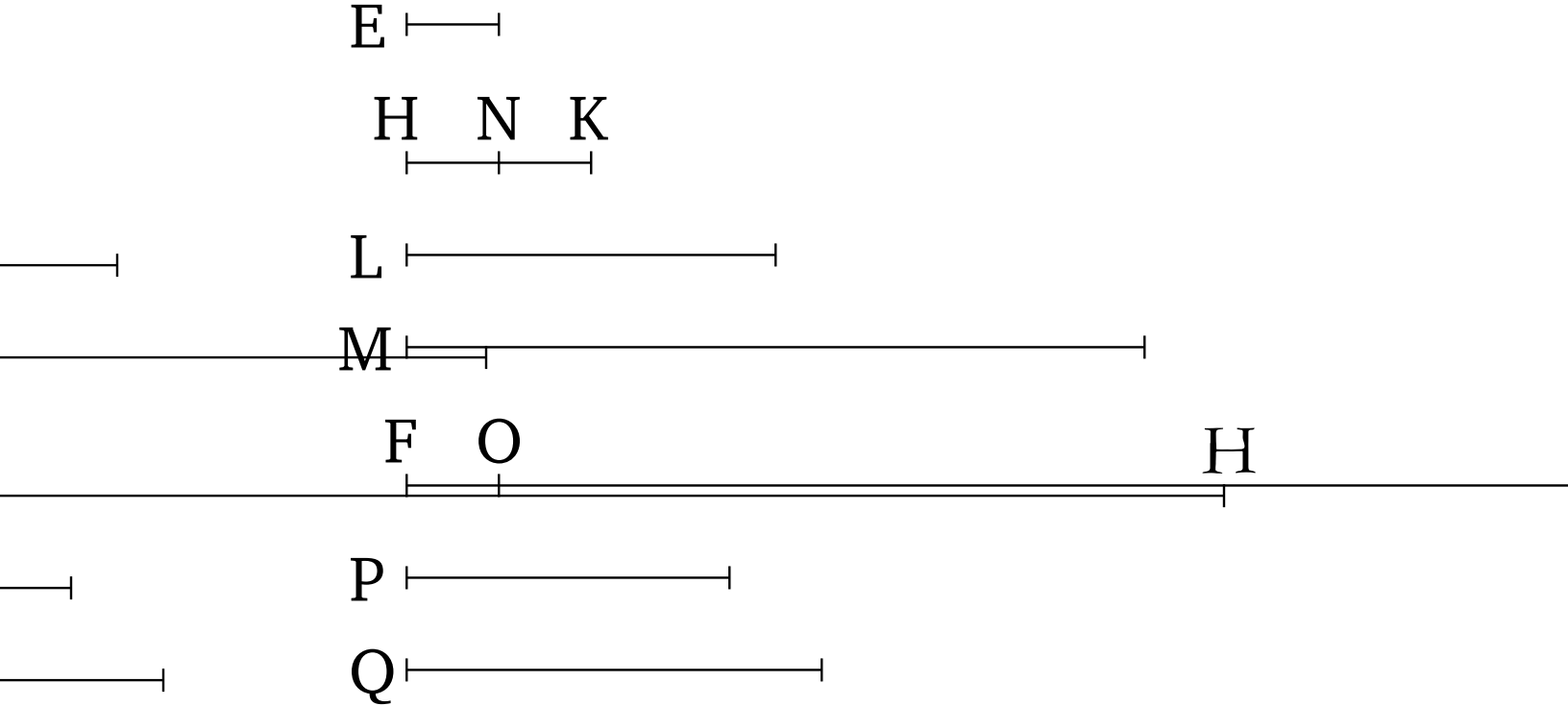
Ἀπὸ γὰρ μονάδος ἐκκείσθωσαν ὁσοιδηποτοῦν ἀριθμοὶ ἐν τῇ διπλασίῳ ἀναλογίᾳ, ἕως οὔ ὁ σύμπαξ συντεθείς πρῶτος γένηται, οἱ A , B , Γ , Δ , καὶ τῷ σύμπαντι ἴσος ἔστω ὁ E , καὶ ὁ E τὸν Δ πολλαπλασιάσας τὸν ZH ποιείτω. λέγω, ὅτι ὁ ZH τέλειός ἐστιν.

Proposition 36[†]

If any multitude whatsoever of numbers is set out continuously in a double proportion, (starting) from a unit, until the whole sum added together becomes prime, and the sum multiplied into the last (number) makes some (number), then the (number so) created will be perfect.

For let any multitude of numbers, A , B , C , D , be set out (continuously) in a double proportion, until the whole sum added together is made prime. And let E be equal to the sum. And let E make FG (by) multiplying D . I say that FG is a perfect (number).





Ὅσοι γάρ εἰσιν οἱ A, B, Γ, Δ τῶ πλήθει, τοσοῦτοι ἀπὸ τοῦ E εἰλήφθωσαν ἐν τῇ διπλασίονι ἀναλογίᾳ οἱ $E, \Theta K, \Lambda, M$. δι' ἴσον ἄρα ἐστὶν ὡς ὁ A πρὸς τὸν Δ , οὕτως ὁ E πρὸς τὸν M . ὁ ἄρα ἐκ τῶν E, Δ ἴσος ἐστὶ τῶ ἐκ τῶν A, M . καὶ ἐστὶν ὁ ἐκ τῶν E, Δ ὁ ZH · καὶ ὁ ἐκ τῶν A, M ἄρα ἐστὶν ὁ ZH . ὁ A ἄρα τὸν M πολλαπλασιάσας τὸν ZH πεποίηκεν· ὁ M ἄρα τὸν ZH μετρεῖ κατὰ τὰς ἐν τῶ A μονάδας. καὶ ἐστὶ δυνάξ ὁ A διπλασίως ἄρα ἐστὶν ὁ ZH τοῦ M . εἰσὶ δὲ καὶ οἱ $M, \Lambda, \Theta K, E$ ἐχῆς διπλασίοι ἀλλήλων· οἱ $E, \Theta K, \Lambda, M, ZH$ ἄρα ἐχῆς ἀνάλογόν εἰσιν ἐν τῇ διπλασίονι ἀναλογίᾳ. ἀφηγήσθω δὲ ἀπὸ τοῦ δευτέρου τοῦ ΘK καὶ τοῦ ἐσθάτου τοῦ ZH τῶ πρώτῳ τῶ E ἴσος ἐκάτερος τῶν $\Theta N, ZX$ · ἐστὶν ἄρα ὡς ἡ τοῦ δευτέρου ἀριθμοῦ ὑπεροθὴ πρὸς τὸν πρώτον, οὕτως ἡ τοῦ ἐσθάτου ὑπεροθὴ πρὸς τοὺς πρὸ ἐαυτοῦ πάντα. ἐστὶν ἄρα ὡς ὁ NK πρὸς τὸν E , οὕτως ὁ XH πρὸς τοὺς $M, \Lambda, K\Theta, E$. καὶ ἐστὶν ὁ NK ἴσος τῶ E · καὶ ὁ XH ἄρα ἴσος ἐστὶ τοῖς $M, \Lambda, \Theta K, E$. ἐστὶ δὲ καὶ ὁ ZX τῶ E ἴσος, ὁ δὲ E τοῖς A, B, Γ, Δ καὶ τῇ μονάδι. ὅλος ἄρα ὁ ZH ἴσος ἐστὶ τοῖς τε $E, \Theta K, \Lambda, M$ καὶ τοῖς A, B, Γ, Δ καὶ τῇ μονάδι· καὶ μετρεῖται ὑπ' αὐτῶν. λέγω, ὅτι καὶ ὁ ZH ὑπ' οὐδενὸς ἄλλον μετρηθήσεται παρέχ τῶν $A, B, \Gamma, \Delta, E, \Theta K, \Lambda, M$ καὶ τῆς μονάδος. εἰ γάρ δυνατόν, μετρεῖτω τις τὸν ZH ὁ O , καὶ ὁ O μηδενὶ τῶν $A, B, \Gamma, \Delta, E, \Theta K, \Lambda, M$ ἔστω ὁ αὐτός. καὶ ὁσάκις ὁ O τὸν ZH μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῶ Π · ὁ Π ἄρα τὸν O πολλαπλασιάσας τὸν ZH πεποίηκεν. ἀλλὰ μὴν καὶ ὁ E τὸν Δ πολλαπλασιάσας τὸν ZH πεποίηκεν· ἐστὶν ἄρα ὡς ὁ E πρὸς τὸν Π , ὁ O πρὸς τὸν Δ . καὶ ἐπεὶ ἀπὸ μονάδος ἐχῆς ἀνάλογόν εἰσιν οἱ A, B, Γ, Δ , ὁ Δ ἄρα ὑπ' οὐδενὸς ἄλλον

For as many as is the multitude of A, B, C, D , let so many (numbers), E, HK, L, M , be taken in a double proportion, (starting) from E . Thus, via equality, as A is to D , so E (is) to M [Prop. 7.14]. Thus, the (number created) from (multiplying) E, D is equal to the (number created) from (multiplying) A, M . And FG is the (number created) from (multiplying) E, D . Thus, FG is also the (number created) from (multiplying) A, M [Prop. 7.19]. Thus, A has made FG (by) multiplying M . Thus, M measures FG according to the units in A . And A is a dyad. Thus, FG is double M . And M, L, HK, E are also continuously double one another. Thus, E, HK, L, M, FG are continuously proportional in a double proportion. So let HN and FO , each equal to the first (number) E , be subtracted from the second (number) HK and the last FG (respectively). Thus, as the excess of the second number is to the first, so the excess of the last (is) to (the sum of) all those (numbers) before it [Prop. 9.35]. Thus, as NK is to E , so OG (is) to M, L, KH, E . And NK is equal to E . And thus OG is equal to M, L, HK, E . And FO is also equal to E , and E to A, B, C, D , and a unit. Thus, the whole of FG is equal to E, HK, L, M , and A, B, C, D , and a unit. And it is measured by them. I also say that FG will be measured by no other (numbers) except A, B, C, D, E, HK, L, M , and a unit. For, if possible, let some (number) P measure FG , and let P not be the same as any of A, B, C, D, E, HK, L, M . And as many times as P measures FG , so many units let there be in Q . Thus, Q has made FG (by) multiplying P . But, in fact, E has also made FG (by) multiplying D . Thus, as E is to Q , so P (is) to D [Prop. 7.19]. And since A ,

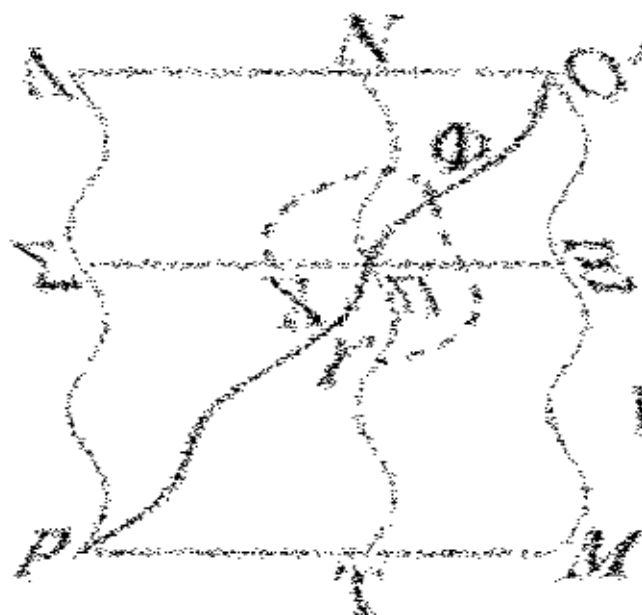
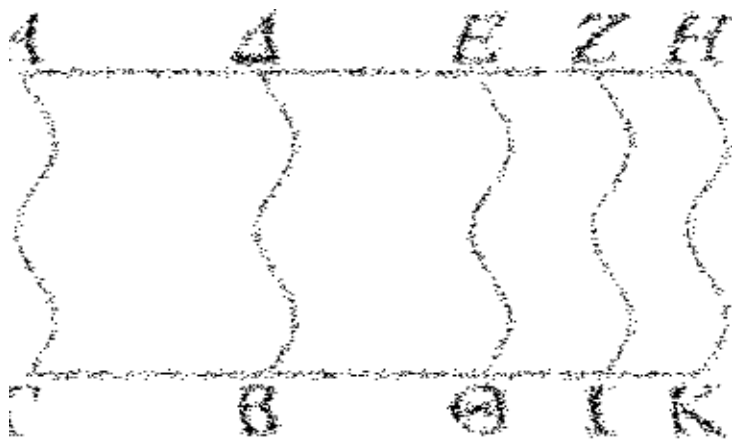
ἀριθμοῦ μετρηθήσεται παρέχ τῶν A, B, Γ . καὶ ὑπόκειται ὁ O οὐδενὶ τῶν A, B, Γ ὁ αὐτόξ· οὐκ ἄρα μετρήσει ὁ O τὸν Δ . ἀλλ' ὥς ὁ O πρὸς τὸν Δ , ὁ E πρὸς τὸν Π · οὐδὲ ὁ E ἄρα τὸν Π μετρεῖ. καὶ ἐστὶν ὁ E πρωτοξ· πᾶς δὲ πρωτοξ ἀριθμὸς πρὸς ἅπαντα, ὃν μὴ μετρεῖ, πρωτόξ [ἐστίν]. οἱ E, Π ἄρα πρωτοὶ πρὸς ἀλλήλων εἰσίν. οἱ δὲ πρωτοὶ καὶ ἐλάθιστοι, οἱ δὲ ἐλάθιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔθονταξ ἰσάκις ὅ τε ἡγούμενοξ τὸν ἡγούμενον καὶ ὁ ἐπόμενοξ τὸν ἐπόμενον· καὶ ἐστὶν ὥς ὁ E πρὸς τὸν Π , ὁ O πρὸς τὸν Δ · ἰσάκιξ ἄρα ὁ E τὸν O μετρεῖ καὶ ὁ Π τὸν Δ . ὁ δὲ Δ ὑπ' οὐδενὸς ἄλλου μετρεῖται παρέχ τῶν A, B, Γ · ὁ Π ἄρα ἐν τῶν A, B, Γ ἐστὶν ὁ αὐτόξ. ἔστω τῷ B ὁ αὐτόξ. καὶ ὅσοι εἰσὶν οἱ B, Γ, Δ τῷ πλήθει τοσοῦτοι εἰλήφθωσαν ἀπὸ τοῦ E οἱ $E, \Theta K, \Lambda$. καὶ εἰσὶν οἱ $E, \Theta K, \Lambda$ τοῖξ B, Γ, Δ ἐν τῷ αὐτῷ λόγῳ· δι' ἴσον ἄρα ἐστὶν ὥς ὁ B πρὸς τὸν Δ , ὁ E πρὸς τὸν Λ . ὁ ἄρα ἐκ τῶν B, Λ ἴσοξ ἐστὶ τῷ ἐκ τῶν Δ, E · ἀλλ' ὁ ἐκ τῶν Δ, E ἴσοξ ἐστὶ τῷ ἐκ τῶν Π, O · καὶ ὁ ἐκ τῶν Π, O ἄρα ἴσοξ ἐστὶ τῷ ἐκ τῶν B, Λ . ἐστὶν ἄρα ὥς ὁ Π πρὸς τὸν B , ὁ Λ πρὸς τὸν O . καὶ ἐστὶν ὁ Π τῷ B ὁ αὐτόξ· καὶ ὁ Λ ἄρα τῷ O ἐστὶν ὁ αὐτόξ· ὅπερ ἀδύνατον· ὁ γὰρ O ὑπόκειται μηδενὶ τῶν ἐκκειμένων ὁ αὐτόξ· οὐκ ἄρα τὸν ZH μετρήσει τιξ ἀριθμὸς παρέχ τῶν $A, B, \Gamma, \Delta, E, \Theta K, \Lambda, M$ καὶ τῆξ μονάδοξ. καὶ ἐδείθη ὁ ZH τοῖξ $A, B, \Gamma, \Delta, E, \Theta K, \Lambda, M$ καὶ τῇ μονάδι ἴσοξ. τέλειοξ δὲ ἀριθμὸξ ἐστὶν ὁ τοῖξ ἐαντοῦ μέρεσιν ἴσοξ ὢν· τέλειοξ ἄρα ἐστὶν ὁ ZH · ὅπερ ἔδει δεῖχαι.

B, C, D are continually proportional, (starting) from a unit, D will thus not be measured by any other numbers except A, B, C [Prop. 9.13]. And P was assumed not (to be) the same as any of A, B, C . Thus, P does not measure D . But, as P (is) to D , so E (is) to Q . Thus, E does not measure Q either [Def. 7.20]. And E is a prime (number). And every prime number [is] prime to every (number) which it does not measure [Prop. 7.29]. Thus, E and Q are prime to one another. And (numbers) prime (to one another are) also the least (of those numbers having the same ratio as them) [Prop. 7.21], and the least (numbers) measure those (numbers) having the same ratio as them an equal number of times, the leading (measuring) the leading, and the following the following [Prop. 7.20]. And as E is to Q , (so) P (is) to D . Thus, E measures P the same number of times as Q (measures) D . And D is not measured by any other (numbers) except A, B, C . Thus, Q is the same as one of A, B, C . Let it be the same as B . And as many as is the multitude of B, C, D , let so many (of the set out numbers) be taken, (starting) from E , (namely) E, HK, L . And E, HK, L are in the same ratio as B, C, D . Thus, via equality, as B (is) to D , (so) E (is) to L [Prop. 7.14]. Thus, the (number created) from (multiplying) B, L is equal to the (number created) from multiplying D, E [Prop. 7.19]. But, the (number created) from (multiplying) D, E is equal to the (number created) from (multiplying) Q, P . Thus, the (number created) from (multiplying) Q, P is equal to the (number created) from (multiplying) B, L . Thus, as Q is to B , (so) L (is) to P [Prop. 7.19]. And Q is the same as B . Thus, L is also the same as P . The very thing (is) impossible. For P was assumed not (to be) the same as any of the (numbers) set out. Thus, FG cannot be measured by any number except A, B, C, D, E, HK, L, M , and a unit. And FG was shown (to be) equal to (the sum of) A, B, C, D, E, HK, L, M , and a unit. And a perfect number is one which is equal to (the sum of) its own parts [Def. 7.22]. Thus, FG is a perfect (number). (Which is) the very thing it was required to show.

[†] This proposition demonstrates that perfect numbers take the form $2^{n-1}(2^n - 1)$ provided that $2^n - 1$ is a prime number. The ancient Greeks knew of four perfect numbers: 6, 28, 496, and 8128, which correspond to $n = 2, 3, 5$, and 7, respectively.

ELEMENTS BOOK 10

Incommensurable Magnitudes[†]



[†]The theory of incommensurable magnitudes set out in this book is generally attributed to Theaetetus of Athens. In the footnotes throughout this book, k , k' , etc. stand for distinct ratios of positive integers.

Ὅροι.

Definitions

1. Σύμμετρα μεγέθη λέγεται τὰ τῷ αὐτῷ μετρῶ μετρούμενα, ἀσύμμετρα δέ, ὧν μηδὲν ἐνδέχεται κοινὸν μέτρον γενέσθαι.

2. Εὐθείαι δυνάμει σύμμετροί εἰσιν, ὅταν τὰ ἀπ' αὐτῶν τετράγωνα τῷ αὐτῷ θωρίῳ μετρηῇται, ἀσύμμετροι δέ, ὅταν τοῖς ἀπ' αὐτῶν τετραγώνοις μηδὲν ἐνδέχεται θωρίον κοινὸν μέτρον γενέσθαι.

3. Τούτων ὑποκειμένων δείκνυται, ὅτι τῇ προτεθείσῃ εὐθείᾳ ὑπάρχουσιν εὐθείαι πλήθει ἅπειροι σύμμετροί τε καὶ ἀσύμμετροι αἱ μὲν μήκει μόνον, αἱ δὲ καὶ δυνάμει. καλείσθω ὅν ἡ μὲν προτεθείσα εὐθεῖα ῥητή, καὶ αἱ ταύτης σύμμετροι εἴτε μήκει καὶ δυνάμει εἴτε δυνάμει μόνον ῥηταί, αἱ δὲ ταύτης ἀσύμμετροι ἄλογοι καλείσθωσαν.

4. Καὶ τὸ μὲν ἀπὸ τῆς προτεθείσης εὐθείας τετράγωνον ῥητόν, καὶ τὰ τούτῳ σύμμετρα ῥητά, τὰ δὲ τούτῳ ἀσύμμετρα ἄλογα καλείσθω, καὶ αἱ δυνάμεναι αὐτὰ ἄλογοι, εἰ μὲν τετράγωνα εἴη, αὐταὶ αἱ πλευραί, εἰ δὲ ἕτερά τινα εὐθύγραμμα, αἱ ἴσα αὐτοῖς τετράγωνα ἀναγράφουσαι.

1. Those magnitudes measured by the same measure are said (to be) commensurable, but (those) of which no (magnitude) admits to be a common measure (are said to be) incommensurable.[†]

2. (Two) straight-lines are commensurable in square[‡] when the squares on them are measured by the same area, but (are) incommensurable (in square) when no area admits to be a common measure of the squares on them.[§]

3. These things being assumed, it is proved that there exist an infinite multitude of straight-lines commensurable and incommensurable with an assigned straight-line—those (incommensurable) in length only, and those also (commensurable or incommensurable) in square.[¶] Therefore, let the assigned straight-line be called rational. And (let) the (straight-lines) commensurable with it, either in length and square, or in square only, (also be called) rational. But let the (straight-lines) incommensurable with it be called irrational.*

4. And let the square on the assigned straight-line be called rational. And (let areas) commensurable with it (also be called) rational. But (let areas) incommensurable with it (be called) irrational, and (let) their square-roots[§] (also be called) irrational—the sides themselves, if the (areas) are squares, and the (straight-lines) describing squares equal to them, if the (areas) are some other rectilinear (figure).^b

[†] In other words, two magnitudes α and β are commensurable if $\alpha : \beta :: 1 : k$, and incommensurable otherwise.

[‡] Literally, “in power”.

[§] In other words, two straight-lines of length α and β are commensurable in square if $\alpha : \beta :: 1 : k^{1/2}$, and incommensurable in square otherwise. Likewise, the straight-lines are commensurable in length if $\alpha : \beta :: 1 : k$, and incommensurable in length otherwise.

[¶] To be more exact, straight-lines can either be commensurable in square only, incommensurable in length only, or commensurable/incommensurable in both length and square, with an assigned straight-line.

* Let the length of the assigned straight-line be unity. Then rational straight-lines have lengths expressible as k or $k^{1/2}$, depending on whether the lengths are commensurable in length, or in square only, respectively, with unity. All other straight-lines are irrational.

[§] The square-root of an area is the length of the side of an equal area square.

^b The area of the square on the assigned straight-line is unity. Rational areas are expressible as k . All other areas are irrational. Thus, squares whose sides are of rational length have rational areas, and vice versa.

1.

Proposition 1[†]

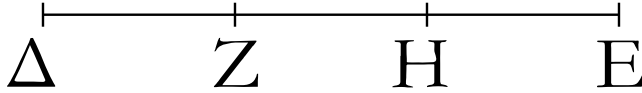
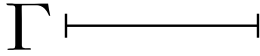
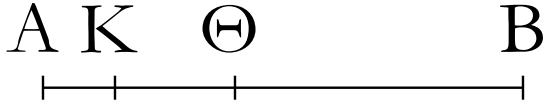
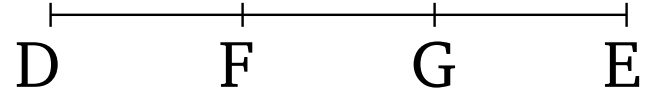
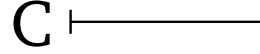
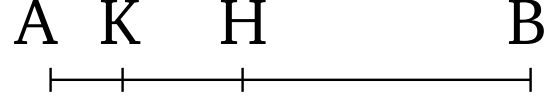
Δύο μεγεθῶν ἀνόσων ἐκκεκμένων, ἔαν ἀπὸ τοῦ μείζονος ἀφαιρεθῇ μείζον ἢ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μείζον ἢ τὸ ἥμισυ, καὶ τοῦτο αἰ γίνηται, λειφθήσεται τι μέγεθος, ὃ ἔσται ἔλασσον τοῦ ἐκκεκμένου ἐλάσσονος μεγέθους.

Ἐστω δύο μεγέθη ἅπσα τὰ AB , Γ , ὧν μείζον τὸ AB . λέγω, ὅτι, ἔαν ἀπὸ τοῦ AB ἀφαιρεθῇ μείζον ἢ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μείζον ἢ τὸ ἥμισυ, καὶ τοῦτο αἰ γίνηται, λειφθήσεται τι μέγεθος, ὃ ἔσται ἔλασσον τοῦ Γ μεγέθους.

If, from the greater of two unequal magnitudes (which are) laid out, (a part) greater than half is subtracted, and (if from) the remainder (a part) greater than half (is subtracted), and (if) this happens continually, then some magnitude will (eventually) be left which will be less than the lesser laid out magnitude.

Let AB and C be two unequal magnitudes, of which (let) AB (be) the greater. I say that if (a part) greater than half is subtracted from AB , and (if a part) greater than half (is sub-

tracted) from the remainder, and (if) this happens continually, then some magnitude will (eventually) be left which will be less than the magnitude C .



Τὸ Γ γὰρ πολλαπλασιαζόμενον ἔσται ποτὲ τοῦ AB μείζον. πεπολλαπλασιάσθω, καὶ ἔστω τὸ ΔE τοῦ μὲν Γ πολλαπλασίον, τοῦ δὲ AB μείζον, καὶ διηρῆσθω τὸ ΔE εἰς τὰ τῷ Γ ἴσα τὰ ΔZ , ZH , HE , καὶ ἀφηρήσθω ἀπὸ μὲν τοῦ AB μείζον ἢ τὸ ἥμισυν τὸ $B\Theta$, ἀπὸ δὲ τοῦ $A\Theta$ μείζον ἢ τὸ ἥμισυν τὸ ΘK , καὶ τοῦτο αἰ γινέσθω, ἕως ἄν αἱ ἐν τῷ AB διαιρέσεις ἰσοπληθεῖς γένωνται ταῖς ἐν τῷ ΔE διαιρέσεσιν.

Ἐστωσαν οὖν αἱ AK , $K\Theta$, ΘB διαιρέσεις ἰσοπληθεῖς οὗσαι ταῖς ΔZ , ZH , HE · καὶ ἐπεὶ μείζον ἐστὶ τὸ ΔE τοῦ AB , καὶ ἀφήρηται ἀπὸ μὲν τοῦ ΔE ἔλασσον τοῦ ἡμίσεως τὸ EH , ἀπὸ δὲ τοῦ AB μείζον ἢ τὸ ἥμισυν τὸ $B\Theta$, λοιπὸν ἄρα τὸ $H\Delta$ λοιποῦ τοῦ ΘA μείζον ἐστίν. καὶ ἐπεὶ μείζον ἐστὶ τὸ $H\Delta$ τοῦ ΘA , καὶ ἀφήρηται τοῦ μὲν $H\Delta$ ἥμισυν τὸ HZ , τοῦ δὲ ΘA μείζον ἢ τὸ ἥμισυν τὸ ΘK , λοιπὸν ἄρα τὸ ΔZ λοιποῦ τοῦ AK μείζον ἐστίν. ἴσον δὲ τὸ ΔZ τῷ Γ · καὶ τὸ Γ ἄρα τοῦ AK μείζον ἐστίν. ἔλασσον ἄρα τὸ AK τοῦ Γ .

Καταλείπεται ἄρα ἀπὸ τοῦ AB μεγέθονος τὸ AK μέγεθος ἔλασσον ὃν τοῦ ἐκκειμένου ἐλάσσονος μεγέθονος τοῦ Γ · ὅπερ ἔδει δεῖξαι.—ὁμοίως δὲ δειθθήσεται, κ' ἂν ἡμίση ἢ τὰ ἀφαιρούμενα.

For C , when multiplied (by some number), will sometimes be greater than AB [Def. 5.4]. Let it be (so) multiplied. And let DE be (both) a multiple of C , and greater than AB . And let DE be divided into the (divisions) DF , FG , GE , equal to C . And let BH , (which is) greater than half, be subtracted from AB . And (let) HK , (which is) greater than half, (be subtracted) from AH . And let this happen continually, until the divisions in AB become equal in number to the divisions in DE .

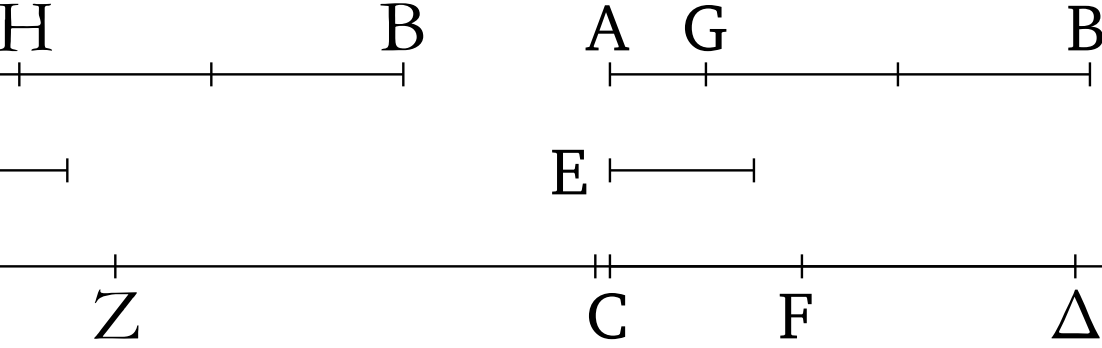
Therefore, let the divisions (in AB) be AK , KH , HB , being equal in number to DF , FG , GE . And since DE is greater than AB , and EG , (which is) less than half, has been subtracted from DE , and BH , (which is) greater than half, from AB , the remainder GD is thus greater than the remainder HA . And since GD is greater than HA , and the half GF has been subtracted from GD , and HK , (which is) greater than half, from HA , the remainder DF is thus greater than the remainder AK . And DF (is) equal to C . C is thus also greater than AK . Thus, AK (is) less than C .

Thus, the magnitude AK , which is less than the lesser laid out magnitude C , is left over from the magnitude AB . (Which is) the very thing it was required to show. — (The theorem) can similarly be proved even if the (parts) subtracted are halves.

† This theorem is the basis of the so-called method of exhaustion, and is generally attributed to Eudoxus of Cnidus.

Ἐάν δύο μεγεθῶν [ἐκκειμένων] ἀνίσων ἀνθυφαιρουμένων αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος τὸ καταλειπόμενον μηδέποτε καταμετρῇ τὸ πρὸ ἑαυτοῦ, ἀσύμμετρα ἔσται τὰ μεγέθη.

If the remainder of two unequal magnitudes (which are) [laid out] never measures the (magnitude) before it, (when) the lesser (magnitude is) continually subtracted in turn from the greater, then the (original) magnitudes will be incommensurable.



Δύο γὰρ μεγεθῶν ὄντων ἀνίσων τῶν AB , $\Gamma\Delta$ καὶ ἐλάσσονος τοῦ AB ἀνθυφαιρουμένων αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος τὸ περιλειπόμενον μηδέποτε καταμετρεῖται τὸ πρὸ ἑαυτοῦ· λέγω, ὅτι ἀσύμμετρά ἐσσι τὰ AB , $\Gamma\Delta$ μεγέθη.

Εἰ γὰρ ἔστι σύμμετρα, μετρήσει τι αὐτὰ μέγεθος· μετρεῖται, εἰ δυνατόν, καὶ ἔστω τὸ E · καὶ τὸ μὲν AB τὸ ΔZ καταμετροῦν λειπέτω ἑαυτοῦ ἕλασσον τὸ ΓZ , τὸ δὲ ΓZ τὸ BH καταμετροῦν λειπέτω ἑαυτοῦ ἕλασσον τὸ AH , καὶ τοῦτο αἰ γνώσθω, ὥς οὗ λειφθῇ τι μέγεθος, ὃ ἐστιν ἕλασσον τοῦ E . γεγονέντω, καὶ λελείφθω τὸ AH ἕλασσον τοῦ E . ἐπεὶ οὖν τὸ E τὸ AB μετρεῖ, ἀλλὰ τὸ AB τὸ ΔZ μετρεῖ, καὶ τὸ E ἄρα τὸ ΔZ μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ $\Gamma\Delta$ · καὶ λοιπὸν ἄρα τὸ ΓZ μετρήσει. ἀλλὰ τὸ ΓZ τὸ BH μετρεῖ· καὶ τὸ E ἄρα τὸ BH μετρεῖ. μετρεῖ δὲ καὶ ὅλον τὸ AB · καὶ λοιπὸν ἄρα τὸ AH μετρήσει, τὸ μείζον τὸ ἕλασσον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ AB , $\Gamma\Delta$ μεγέθη μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐσσι τὰ AB , $\Gamma\Delta$ μεγέθη.

Ἐάν ἄρα δύο μεγεθῶν ἀνίσων, καὶ τὰ ἐχῇς.

For, AB and CD being two unequal magnitudes, and AB (being) the lesser, let the remainder never measure the (magnitude) before it, (when) the lesser (magnitude is) continually subtracted in turn from the greater. I say that the magnitudes AB and CD are incommensurable.

For if they are commensurable then some magnitude will measure them (both). If possible, let it (so) measure (them), and let it be E . And let AB leave CF less than itself (in) measuring FD , and let CF leave AG less than itself (in) measuring BG , and let this happen continually, until some magnitude which is less than E is left. Let (this) have occurred,[†] and let AG , (which is) less than E , be left. Therefore, since E measures AB , but AB measures DF , E will thus also measure FD . And it also measures the whole (of) CD . Thus, it will also measure the remainder CF . But, CF measures BG . Thus, E also measures BG . And it also measures the whole (of) AB . Thus, it will also measure the remainder AG , the greater (measuring) the lesser. The very thing is impossible. Thus, some magnitude cannot measure (both) the magnitudes AB and CD . Thus, the magnitudes AB and CD are incommensurable [Def. 10.1].

Thus, if ... of two unequal magnitudes, and so on ...

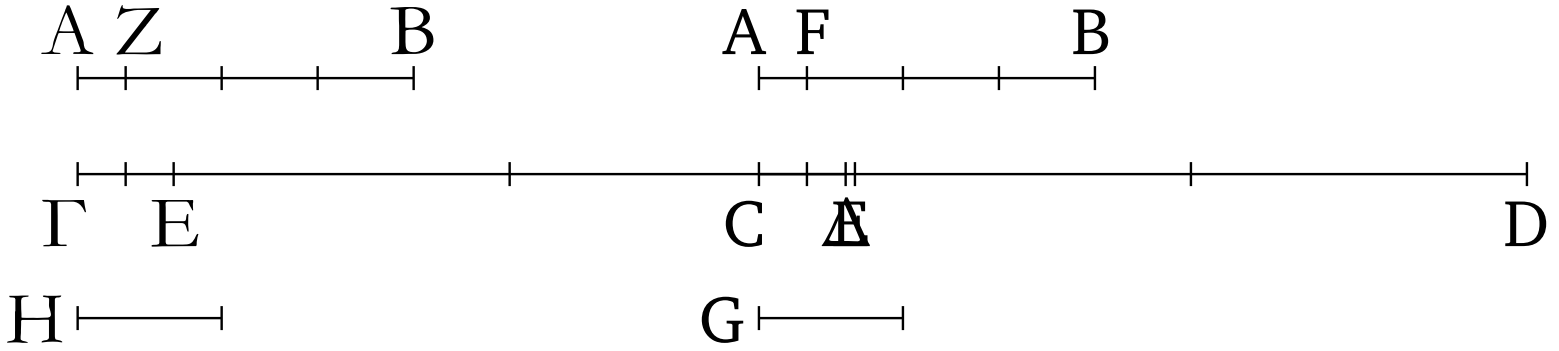
[†] The fact that this will eventually occur is guaranteed by Prop. 10.1.

3.

Δύο μεγεθῶν συμμέτρων δοθέντων τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

Proposition 3

To find the greatest common measure of two given commensurable magnitudes.



Ἐστω τὰ δοθέντα δύο μεγέθη σύμμετρα τὰ AB , $\Gamma\Delta$, ὧν ἔλασσον τὸ AB . δεῖ δὴ τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Τὸ AB γὰρ μέγεθος ἢτοι μετρεῖ τὸ $\Gamma\Delta$ ἢ οὐ. εἰ μὲν οὖν μετρεῖ, μετρεῖ δὲ καὶ ἑαυτό, τὸ AB ἄρα τῶν AB , $\Gamma\Delta$ κοινὸν μέτρον ἐστίν· καὶ φανερόν, ὅτι καὶ μέγιστον. μείζον γὰρ τοῦ AB μεγέθους τὸ AB οὐ μετρήσει.

μὴ μετρεῖτω δὴ τὸ AB τὸ $\Gamma\Delta$. καὶ ἀνθυφαιρουμένον αἰεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος, τὸ περιλειπόμενον μετρήσει ποτὲ τὸ πρὸ ἑαυτοῦ διὰ τὸ μὴ εἶναι ἀσύμμετρα τὰ AB , $\Gamma\Delta$. καὶ τὸ μὲν AB τὸ $E\Delta$ καταμετροῦν λειπέτω ἑαυτοῦ ἔλασσον τὸ EF , τὸ δὲ EF τὸ ZB καταμετροῦν λειπέτω ἑαυτοῦ ἔλασσον τὸ AZ , τὸ δὲ AZ τὸ ΓE μετρεῖτω.

Ἐπεὶ οὖν τὸ AZ τὸ ΓE μετρεῖ, ἀλλὰ τὸ ΓE τὸ ZB μετρεῖ, καὶ τὸ AZ ἄρα τὸ ZB μετρήσει. μετρεῖ δὲ καὶ ἑαυτό· καὶ ὅλον ἄρα τὸ AB μετρήσει τὸ AZ . ἀλλὰ τὸ AB τὸ ΔE μετρεῖ· καὶ τὸ AZ ἄρα τὸ $E\Delta$ μετρήσει. μετρεῖ δὲ καὶ τὸ ΓE · καὶ ὅλον ἄρα τὸ $\Gamma\Delta$ μετρεῖ· τὸ AZ ἄρα τῶν AB , $\Gamma\Delta$ κοινὸν μέτρον ἐστίν. λέγω δὴ, ὅτι καὶ μέγιστον. εἰ γὰρ μή, ἔσται τι μέγεθος μείζον τοῦ AZ , ὃ μετρήσει τὰ AB , $\Gamma\Delta$. ἔστω τὸ H . ἐπεὶ οὖν τὸ H τὸ AB μετρεῖ, ἀλλὰ τὸ AB τὸ $E\Delta$ μετρεῖ, καὶ τὸ H ἄρα τὸ $E\Delta$ μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ $\Gamma\Delta$. καὶ λοιπὸν ἄρα τὸ ΓE μετρήσει τὸ H . ἀλλὰ τὸ ΓE τὸ ZB μετρεῖ· καὶ τὸ H ἄρα τὸ ZB μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ AB , καὶ λοιπὸν τὸ AZ μετρήσει, τὸ μείζον τὸ ἔλασσον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα μείζον τι μέγεθος τοῦ AZ τὰ AB , $\Gamma\Delta$ μετρήσει· τὸ AZ ἄρα τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον ἐστίν.

Δύο ἄρα μεγεθῶν συμμέτρων δοθέντων τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον ἡύρηται· ὅπερ ἔδει δεῖχαι.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἕαν μέγεθος δύο μεγέθη μετρήῃ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσει.

Let AB and CD be the two given magnitudes, of which (let) AB (be) the lesser. So, it is required to find the greatest common measure of AB and CD .

For the magnitude AB either measures, or (does) not (measure), CD . Therefore, if it measures (CD), and (since) it also measures itself, AB is thus a common measure of AB and CD . And (it is) clear that (it is) also (the) greatest. For a (magnitude) greater than magnitude AB cannot measure AB .

So let AB not measure CD . And continually subtracting in turn the lesser (magnitude) from the greater, the remaining (magnitude) will (at) some time measure the (magnitude) before it, on account of AB and CD not being incommensurable [Prop. 10.2]. And let AB leave EC less than itself (in) measuring ED , and let EC leave AF less than itself (in) measuring FB , and let AF measure CE .

Therefore, since AF measures CE , but CE measures FB , AF will thus also measure FB . And it also measures itself. Thus, AF will also measure the whole (of) AB . But, AB measures DE . Thus, AF will also measure ED . And it also measures CE . Thus, it also measures the whole of CD . Thus, AF is a common measure of AB and CD . So I say that (it is) also (the) greatest (common measure). For, if not, there will be some magnitude, greater than AF , which will measure (both) AB and CD . Let it be G . Therefore, since G measures AB , but AB measures ED , G will thus also measure ED . And it also measures the whole of CD . Thus, G will also measure the remainder CE . But CE measures FB . Thus, G will also measure FB . And it also measures the whole (of) AB . And (so) it will measure the remainder AF , the greater (measuring) the lesser. The very thing is impossible. Thus, some magnitude greater than AF cannot measure (both) AB and CD . Thus, AF is the greatest common measure of AB and CD .

Thus, the greatest common measure of two given commensurable magnitudes, AB and CD , has been found. (Which is) the very thing it was required to show.

Corollary

So (it is) clear, from this, that if a magnitude measures two magnitudes then it will also measure their greatest common

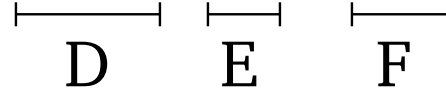
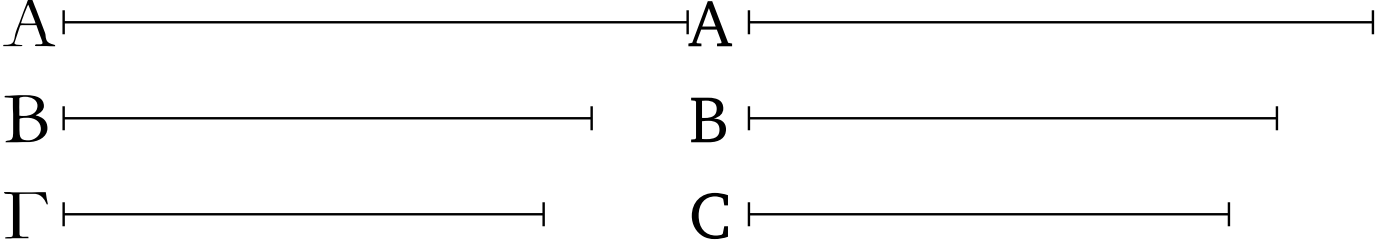
measure.

4.

Proposition 4

Τριῶν μεγεθῶν συμμετρων δοθέντων τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

To find the greatest common measure of three given commensurable magnitudes.



Ἐστω τὰ δοθέντα τρία μεγέθη σύμμετρα τὰ A, B, Γ · δεῖ δὴ τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Let A, B, C be the three given commensurable magnitudes. So it is required to find the greatest common measure of A, B, C .

Εἰλήφθω γάρ δύο τῶν A, B τὸ μέγιστον κοινὸν μέτρον, καὶ ἔστω τὸ Δ · τὸ δὴ Δ τὸ Γ ἤτοι μετρεῖ ἢ οὐ [μετρεῖ]. μετρεῖτω πρότερον. ἐπεὶ οὖν τὸ Δ τὸ Γ μετρεῖ, μετρεῖ δὲ καὶ τὰ A, B , τὸ Δ ἄρα τὰ A, B, Γ μετρεῖ· τὸ Δ ἄρα τῶν A, B, Γ κοινὸν μέτρον ἐστίν. καὶ φανερόν, ὅτι καὶ μέγιστον· μείζον γὰρ τοῦ Δ μεγέθους τὰ A, B οὐ μετρεῖ.

For let the greatest common measure of the two (magnitudes) A and B be taken [Prop. 10.3], and let it be D . So D either measures, or [does] not [measure], C . Let it, first of all, measure (C). Therefore, since D measures C , and it also measures A and B , D thus measures A, B, C . Thus, D is a common measure of A, B, C . And (it is) clear that (it is) also (the) greatest (common measure). For no magnitude larger than D measures (both) A and B .

μὴ μετρεῖτω δὴ τὸ Δ τὸ Γ . λέγω πρότερον, ὅτι σύμμετρά ἐστι τὰ Γ, Δ . ἐπεὶ γὰρ σύμμετρά ἐστι τὰ A, B, Γ , μετρήσει τι αὐτὰ μέγεθος, ὃ δηλαδὴ καὶ τὰ A, B μετρήσει· ὥστε καὶ τὸ τῶν A, B μέγιστον κοινὸν μέτρον τὸ Δ μετρήσει. μετρεῖ δὲ καὶ τὸ Γ · ὥστε τὸ εἰρημένον μέγεθος μετρήσει τὰ Γ, Δ · σύμμετρα ἄρα ἐστὶ τὰ Γ, Δ . εἰλήφθω οὖν αὐτῶν τὸ μέγιστον κοινὸν μέτρον, καὶ ἔστω τὸ E . ἐπεὶ οὖν τὸ E τὸ Δ μετρεῖ, ἀλλὰ τὸ Δ τὰ A, B μετρεῖ, καὶ τὸ E ἄρα τὰ A, B μετρήσει. μετρεῖ δὲ καὶ τὸ Γ . τὸ E ἄρα τὰ A, B, Γ μετρεῖ· τὸ E ἄρα τῶν A, B, Γ κοινὸν ἐστὶ μέτρον. λέγω δὴ, ὅτι καὶ μέγιστον. εἰ γὰρ δυνατόν, ἔστω τι τοῦ E μείζον μέγεθος τὸ Z , καὶ μετρεῖτω τὰ A, B, Γ . καὶ ἐπεὶ τὸ Z τὰ A, B, Γ μετρεῖ, καὶ τὰ A, B ἄρα μετρήσει καὶ τὸ τῶν A, B μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν A, B μέγιστον κοινὸν μέτρον ἐστὶ τὸ Δ · τὸ Z ἄρα τὸ Δ μετρεῖ. μετρεῖ δὲ καὶ τὸ Γ · τὸ Z ἄρα τὰ Γ, Δ μετρεῖ· καὶ τὸ τῶν Γ, Δ ἄρα μέγιστον κοινὸν μέτρον μετρήσει τὸ Z . ἐστὶ δὲ τὸ E · τὸ Z ἄρα τὸ E μετρήσει, τὸ μείζον τὸ ἔλασσον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα μείζον τι τοῦ E μεγέθους [μέγεθος] τὰ A, B, Γ μετρεῖ· τὸ E ἄρα τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον ἐστίν, ἕαν μὴ μετρηῇ τὸ Δ τὸ Γ , ἕαν δὲ μετρηῇ, αὐτὸ τὸ Δ .

So let D not measure C . I say, first, that C and D are commensurable. For if A, B, C are commensurable then some magnitude will measure them which will clearly also measure A and B . Hence, it will also measure D , the greatest common measure of A and B [Prop. 10.3 corr.]. And it also measures C . Hence, the aforementioned magnitude will measure (both) C and D . Thus, C and D are commensurable [Def. 10.1]. Therefore, let their greatest common measure be taken [Prop. 10.3], and let it be E . Therefore, since E measures D , but D measures (both) A and B , E will thus also measure A and B . And it also measures C . Thus, E measures A, B, C . Thus, E is a common measure of A, B, C . So I say that (it is) also (the) greatest (common measure). For, if possible, let F be some magnitude greater than E , and let it measure A, B, C . And since F measures A, B, C , it will thus also measure A and B , and will (thus) measure the greatest common measure of A and B [Prop. 10.3 corr.]. And D is the greatest common measure of A and B . Thus, F measures D . And it also measures C . Thus, F measures (both) C and D . Thus, F will also measure the greatest common measure of C and D [Prop. 10.3 corr.]. And it is E . Thus, F will measure E , the greater (measuring) the lesser. The very thing is impossible. Thus, some [magnitude]

Τριῶν ἄρα μεγεθῶν συμμετρων δοθέντων τὸ μέγιστον κοινὸν μέτρον ἠΰρηται [ὅπερ ἔδει δεῖχαι].

greater than the magnitude E cannot measure A, B, C . Thus, if D does not measure C then E is the greatest common measure of A, B, C . And if it does measure (C) then D itself (is the greatest common measure).

Thus, the greatest common measure of three given commensurable magnitudes has been found. [(Which is) the very thing it was required to show.]

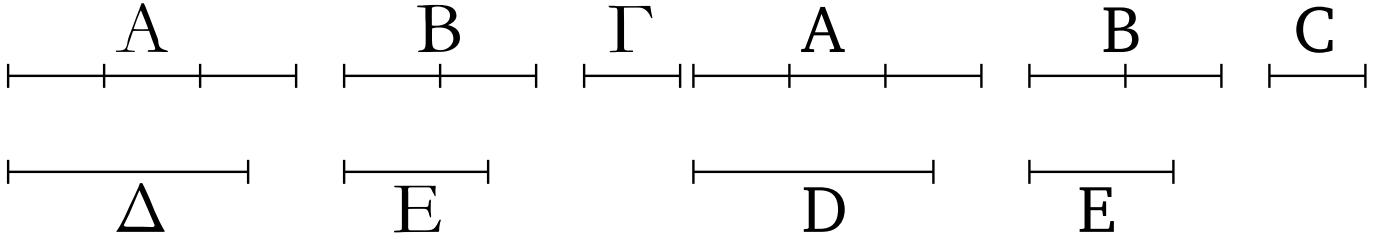
Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἔαν μέγεθος τρία μεγέθη μετρήῃ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσῃ.

Ὅμοίως δὴ καὶ ἐπὶ πλείονων τὸ μέγιστον κοινὸν μέτρον ληφθήσεται, καὶ τὸ πόρισμα προθωρήσῃ. ὅπερ ἔδει δεῖχαι.

5.

Τὰ σύμμετρα μεγέθη πρὸς ἄλληλα λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.



Ἐστω σύμμετρα μεγέθη τὰ A, B · λέγω, ὅτι τὸ A πρὸς τὸ B λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.

Ἐπεὶ γὰρ σύμμετρά ἐστι τὰ A, B , μετρήσῃ τι αὐτὰ μέγεθος. μετρεῖτω, καὶ ἔστω τὸ Γ . καὶ ὁσάκις τὸ Γ τὸ A μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Δ , ὁσάκις δὲ τὸ Γ τὸ B μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ E .

Ἐπεὶ οὖν τὸ Γ τὸ A μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας, μετρεῖ δὲ καὶ ἡ μονὰς τὸν Δ κατὰ τὰς ἐν αὐτῷ μονάδας, ἰσάκις ἄρα ἡ μονὰς τὸν Δ μετρεῖ ἀριθμόν καὶ τὸ Γ μέγεθος τὸ A · ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ A , οὕτως ἡ μονὰς πρὸς τὸν Δ · ἀνάπαλιν ἄρα, ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ πρὸς τὴν μονάδα. πάλιν ἐπεὶ τὸ Γ τὸ B μετρεῖ κατὰ τὰς ἐν τῷ E μονάδας, μετρεῖ δὲ καὶ ἡ μονὰς τὸν E κατὰ τὰς ἐν αὐτῷ μονάδας, ἰσάκις ἄρα ἡ μονὰς τὸν E μετρεῖ καὶ τὸ Γ τὸ B · ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ B , οὕτως ἡ μονὰς πρὸς τὸν E . ἐδείκθη δὲ καὶ ὡς τὸ A πρὸς τὸ Γ , ὁ Δ πρὸς τὴν μονάδα· δι' ἴσον ἄρα ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως ὁ Δ ἀριθμὸς πρὸς τὸν E .

Τὰ ἄρα σύμμετρα μεγέθη τὰ A, B πρὸς ἄλληλα λόγον ἔθει, ὃν ἀριθμὸς ὁ Δ πρὸς ἀριθμόν τὸν E · ὅπερ ἔδει δεῖχαι.

Corollary

So (it is) clear, from this, that if a magnitude measures three magnitudes then it will also measure their greatest common measure.

So, similarly, the greatest common measure of more (magnitudes) can also be taken, and the (above) corollary will go forward. (Which is) the very thing it was required to show.

Proposition 5

Commensurable magnitudes have to one another the ratio which (some) number (has) to (some) number.

Let A and B be commensurable magnitudes. I say that A has to B the ratio which (some) number (has) to (some) number.

For if A and B are commensurable (magnitudes) then some magnitude will measure them. Let it (so) measure (them), and let it be C . And as many times as C measures A , so many units let there be in D . And as many times as C measures B , so many units let there be in E .

Therefore, since C measures A according to the units in D , and a unit also measures D according to the units in it, a unit thus measures the number D as many times as the magnitude C (measures) A . Thus, as C is to A , so a unit (is) to D [Def. 7.20].[†] Thus, inversely, as A (is) to C , so D (is) to a unit [Prop. 5.7 corr.]. Again, since C measures B according to the units in E , and a unit also measures E according to the units in it, a unit thus measures E the same number of times that C (measures) B . Thus, as C is to B , so a unit (is) to E [Def. 7.20]. And it was also shown that as A (is) to C , so D (is) to a unit. Thus, via equality, as A is to B , so the number D (is) to the (number) E [Prop. 5.22].

Thus, the commensurable magnitudes A and B have to one another the ratio which the number D (has) to the number E . (Which is) the very thing it was required to show.

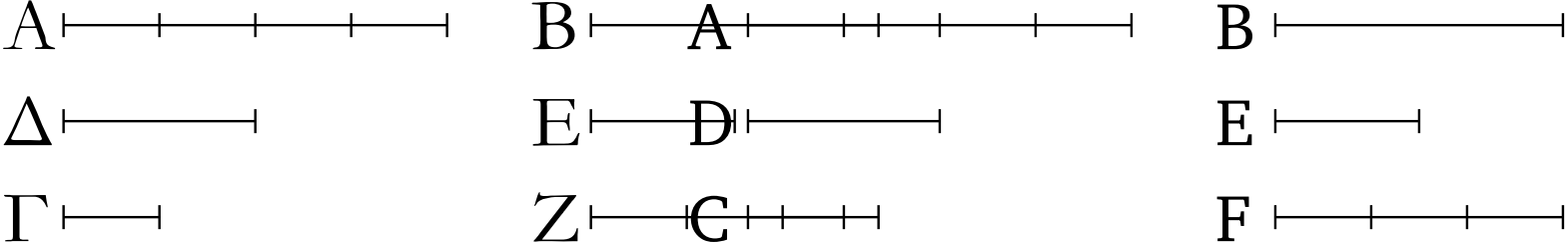
† There is a slight logical gap here, since Def. 7.20 applies to four numbers, rather than two number and two magnitudes.

6.

Proposition 6

Ἐάν δύο μεγέθη πρὸς ἄλληλα λόγον ἔθῃ, ὃν ἀριθμὸς πρὸς ἀριθμὸν, σύμμετρα ἔσται τὰ μεγέθη.

If two magnitudes have to one another the ratio which (some) number (has) to (some) number then the magnitudes will be commensurable.



Δύο γάρ μεγέθη τὰ A, B πρὸς ἄλληλα λόγον ἔθῃτω, ὃν ἀριθμὸς ὁ Δ πρὸς ἀριθμὸν τὸν E · λέγω, ὅτι σύμμετρά ἐστι τὰ A, B μεγέθη.

Ὅσαι γάρ εἰσιν ἐν τῷ Δ μονάδες, εἰς τοσαῦτα ἴσα διηγήσθω τὸ A , καὶ ἐν αὐτῶν ἴσον ἔστω τὸ Γ · ὅσαι δὲ εἰσιν ἐν τῷ E μονάδες, ἐκ τοσούτων μεγεθῶν ἴσων τῷ Γ συγκείσθω τὸ Z .

Ἐπεὶ οὖν, ὅσαι εἰσιν ἐν τῷ Δ μονάδες, τοσαῦτά εἰσι καὶ ἐν τῷ A μεγέθει ἴσα τῷ Γ , ὃ ἄρα μέρος ἐστὶν ἡ μονὰς τοῦ Δ , τὸ αὐτὸ μέρος ἐστὶ καὶ τὸ Γ τοῦ A · ἐστὶν ἄρα ὡς τὸ Γ πρὸς τὸ A , οὕτως ἡ μονὰς πρὸς τὸν Δ . μετρεῖ δὲ ἡ μονὰς τὸν Δ ἀριθμὸν μετρεῖ ἄρα καὶ τὸ Γ τὸ A . καὶ ἐπεὶ ἐστὶν ὡς τὸ Γ πρὸς τὸ A , οὕτως ἡ μονὰς πρὸς τὸν Δ [ἀριθμὸν], ἀνάπαλιν ἄρα ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ ἀριθμὸς πρὸς τὴν μονάδα. πάλιν ἐπεὶ, ὅσαι εἰσιν ἐν τῷ E μονάδες, τοσαῦτά εἰσι καὶ ἐν τῷ Z ἴσα τῷ Γ , ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ Z , οὕτως ἡ μονὰς πρὸς τὸν E [ἀριθμὸν]. ἐδείκθη δὲ καὶ ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ πρὸς τὴν μονάδα· δι' ἴσον ἄρα ἐστὶν ὡς τὸ A πρὸς τὸ Z , οὕτως ὁ Δ πρὸς τὸν E . ἀλλ' ὡς ὁ Δ πρὸς τὸν E , οὕτως ἐστὶ τὸ A πρὸς τὸ B · καὶ ὡς ἄρα τὸ A πρὸς τὸ B , οὕτως καὶ πρὸς τὸ Z . τὸ A ἄρα πρὸς ἐκάτερον τῶν B, Z τὸν αὐτὸν ἔχει λόγον· ἴσον ἄρα ἐστὶ τὸ B τῷ Z . μετρεῖ δὲ τὸ Γ τὸ Z · μετρεῖ ἄρα καὶ τὸ B . ἀλλὰ μὴν καὶ τὸ A · τὸ Γ ἄρα τὰ A, B μετρεῖ. σύμμετρον ἄρα ἐστὶ τὸ A τῷ B .

Ἐάν ἄρα δύο μεγέθη πρὸς ἄλληλα, καὶ τὰ ἐχῇς.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἔαν ᾧσι δύο ἀριθμοί, ὡς οἱ Δ, E , καὶ εὐθεῖα, ὡς ἡ A , δυνατόν ἐστι ποιῆσαι ὡς ὁ Δ ἀριθμὸς πρὸς τὸν E ἀριθμὸν, οὕτως τὴν εὐθεῖαν πρὸς εὐθεῖαν. ἔαν δὲ καὶ τῶν A, Z μέση ἀνάλογον ληφθῇ, ὡς ἡ B , ἔσται ὡς ἡ A πρὸς τὴν Z , οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς B , τουτέστιν ὡς ἡ πρώτη πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ ὅμοιον καὶ ὁμοίως

For let the two magnitudes A and B have to one another the ratio which the number D (has) to the number E . I say that the magnitudes A and B are commensurable.

For, as many units as there are in D , let A be divided into so many equal (divisions). And let C be equal to one of them. And as many units as there are in E , let F be the sum of so many magnitudes equal to C .

Therefore, since as many units as there are in D , so many magnitudes equal to C are also in A , therefore whichever part a unit is of D , C is also the same part of A . Thus, as C is to A , so a unit (is) to D [Def. 7.20]. And a unit measures the number D . Thus, C also measures A . And since as C is to A , so a unit (is) to the [number] D , thus, inversely, as A (is) to C , so the number D (is) to a unit [Prop. 5.7 corr.]. Again, since as many units as there are in E , so many (magnitudes) equal to C are also in F , there as C is to F , so a unit (is) to the [number] E [Def. 7.20]. And it was also shown that as A (is) to C , so D (is) to a unit. Thus, via equality, as A is to F , so D (is) to E [Prop. 5.22]. But, as D (is) to E , so A is to B . And thus as A (is) to B , so (it) also is to F [Prop. 5.11]. Thus, A has the same ratio to each of B and F . Thus, B is equal to F [Prop. 5.9]. And C measures F . Thus, it also measures B . But, in fact, (it) also (measures) A . Thus, C measures (both) A and B . Thus, A is commensurable with B [Def. 10.1].

Thus, if two magnitudes ... to one another, and so on

Corollary

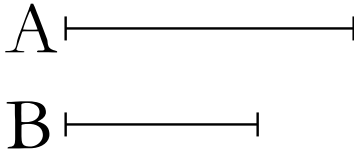
So it is clear, from this, that if there are two numbers, like D and E , and a straight-line, like A , then it is possible to contrive that as the number D (is) to the number E , so the straight-line (is) to (another) straight-line (i.e., F). And if the mean proportion, (say) B , is taken of A and F , then as A is to F , so the (square) on A (will be) to the (square) on B . That is to say, as the first (is) to the third, so the (figure) on the first (is) to the similar, and similarly described, (figure) on the second

ἀναγραφόμενον. ἀλλ' ὥς ἡ A πρὸς τὴν Z , οὕτως ἐστὶν ὁ Δ ἀριθμὸς πρὸς τὸν E ἀριθμόν· γέγονεν ἄρα καὶ ὥς ὁ Δ ἀριθμὸς πρὸς τὸν E ἀριθμόν, οὕτως τὸ ἀπὸ τῆς A εὐθείας πρὸς τὸ ἀπὸ τῆς B εὐθείας· ὅπερ ἔδει δεῖξαι.

7.

Τὰ ἀσύμμετρα μεγέθη πρὸς ἄλληλα λόγον οὐκ ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.

Ἐστω ἀσύμμετρα μεγέθη τὰ A , B · λέγω, ὅτι τὸ A πρὸς τὸ B λόγον οὐκ ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.

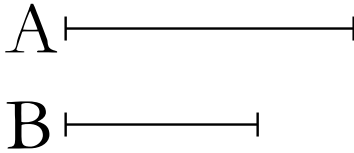


Εἰ γὰρ ἔθει τὸ A πρὸς τὸ B λόγον, ὃν ἀριθμὸς πρὸς ἀριθμόν, σύμμετρον ἔσται τὸ A τῷ B . οὐκ ἔστι δέ· οὐκ ἄρα τὸ A πρὸς τὸ B λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.

Τὰ ἄρα ἀσύμμετρα μεγέθη πρὸς ἄλληλα λόγον οὐκ ἔθει, καὶ τὰ ἐχῆξ.

8.

Ἐὰν δύο μεγέθη πρὸς ἄλληλα λόγον μὴ ἔθῃ, ὃν ἀριθμὸς πρὸς ἀριθμόν, ἀσύμμετρα ἔσται τὰ μεγέθη.



Δύο γὰρ μεγέθη τὰ A , B πρὸς ἄλληλα λόγον μὴ ἔθῃτω, ὃν ἀριθμὸς πρὸς ἀριθμόν· λέγω, ὅτι ἀσύμμετρά ἐσσι τὰ A , B μεγέθη.

Εἰ γὰρ ἔσται σύμμετρα, τὸ A πρὸς τὸ B λόγον ἔχει, ὃν ἀριθμὸς πρὸς ἀριθμόν. οὐκ ἔθει δέ· ἀσύμμετρα ἄρα ἐσσι τὰ A , B μεγέθη.

Ἐὰν ἄρα δύο μεγέθη πρὸς ἄλληλα, καὶ τὰ ἐχῆξ.

9.

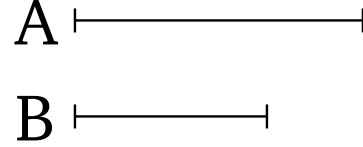
Τὰ ἀπὸ τῶν μήκει συμμέτρων εὐθειῶν τετράγωνα πρὸς ἄλληλα λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὰ τετράγωνα τὰ πρὸς ἄλληλα λόγον ἔθοντα, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ τὰς

[Prop. 6.19 corr.]. But, as A (is) to F , so the number D is to the number E . Thus, it has also been contrived that as the number D (is) to the number E , so the (figure) on the straight-line A (is) to the (similar figure) on the straight-line B . (Which is) the very thing it was required to show.

Proposition 7

Incommensurable magnitudes do not have to one another the ratio which (some) number (has) to (some) number.

Let A and B be incommensurable magnitudes. I say that A does not have to B the ratio which (some) number (has) to (some) number.

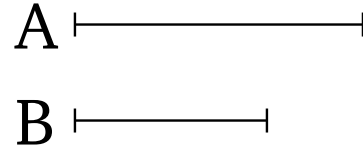


For if A has to B the ratio which (some) number (has) to (some) number then A will be commensurable with B [Prop. 10.6]. But it is not. Thus, A does not have to B the ratio which (some) number (has) to (some) number.

Thus, incommensurable numbers do not have to one another, and so on . . .

Proposition 8

If two magnitudes do not have to one another the ratio which (some) number (has) to (some) number then the magnitudes will be incommensurable.



For let the two magnitudes A and B not have to one another the ratio which (some) number (has) to (some) number. I say that the magnitudes A and B are incommensurable.

For if they are commensurable, A will have to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. But it does not have (such a ratio). Thus, the magnitudes A and B are incommensurable.

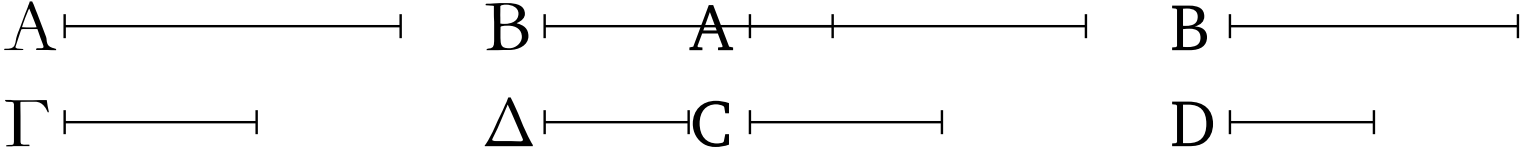
Thus, if two magnitudes . . . to one another, and so on . . .

Proposition 9

Squares on straight-lines (which are) commensurable in length have to one another the ratio which (some) square number (has) to (some) square number. And squares having to one another the ratio which (some) square number (has) to (some)

πλευράς ἔχει μήκει συμμετρουνς. τὰ δὲ ἀπὸ τῶν μήκει ἀσυμμετρων εὐθειῶν τετράγωνα πρὸς ἄλληλα λόγον οὐκ ἔθει, ὅνπερ τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὰ τετράγωνα τὰ πρὸς ἄλληλα λόγον μὴ ἔθοντα, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὰς πλευράς ἔχει μήκει συμμετρουνς.

square number will also have sides (which are) commensurable in length. But squares on straight-lines (which are) incommensurable in length do not have to one another the ratio which (some) square number (has) to (some) square number. And squares not having to one another the ratio which (some) square number (has) to (some) square number will not have sides (which are) commensurable in length either.



Ἐστωσαν γάρ αἱ A, B μήκει σύμμετροι· λέγω, ὅτι τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B τετράγωνον λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

For let A and B be (straight-lines which are) commensurable in length. I say that the square on A has to the square on B the ratio which (some) square number (has) to (some) square number.

Ἐπεὶ γὰρ σύμμετρος ἐστὶν ἡ A τῇ B μήκει, ἡ A ἄρα πρὸς τὴν B λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, ὃν ὁ Γ πρὸς τὸν Δ . ἐπεὶ οὖν ἐστὶν ὡς ἡ A πρὸς τὴν B , οὕτως ὁ Γ πρὸς τὸν Δ , ἀλλὰ τοῦ μὲν τῆς A πρὸς τὴν B λόγον διπλασίον ἐστὶν ὁ τοῦ ἀπὸ τῆς A τετραγώνου πρὸς τὸ ἀπὸ τῆς B τετράγωνον· τὰ γὰρ ὅμοια στήματα ἐν διπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν· τοῦ δὲ τοῦ Γ [ἀριθμοῦ] πρὸς τὸν Δ [ἀριθμόν] λόγον διπλασίον ἐστὶν ὁ τοῦ ἀπὸ τοῦ Γ τετραγώνου πρὸς τὸν ἀπὸ τοῦ Δ τετράγωνον· δύο γὰρ τετραγώνων ἀριθμῶν εἷς μέσος ἀνάλογόν ἐστιν ἀριθμὸς, καὶ ὁ τετράγωνος πρὸς τὸν τετράγωνον [ἀριθμόν] διπλασίονα λόγον ἔθει, ἥπερ ἡ πλευρὰ πρὸς τὴν πλευράν· ἐστὶν ἄρα καὶ ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B τετράγωνον, οὕτως ὁ ἀπὸ τοῦ Γ τετράγωνος [ἀριθμοῦ] πρὸς τὸν ἀπὸ τοῦ Δ [ἀριθμοῦ] τετράγωνον [ἀριθμόν].

For since A is commensurable in length with B , A thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (that) which C (has) to D . Therefore, since as A is to B , so C (is) to D . But the (ratio) of the square on A to the square on B is the square of the ratio of A to B . For similar figures are in the squared ratio of (their) corresponding sides [Prop. 6.20 corr.]. And the (ratio) of the square on C to the square on D is the square of the ratio of the [number] C to the [number] D . For there exists one number in mean proportion to two square numbers, and (one) square (number) has to the (other) square [number] a squared ratio with respect to (that) the side (of the former has) to the side (of the latter) [Prop. 8.11]. And, thus, as the square on A is to the square on B , so the square [number] on the (number) C (is) to the square [number] on the [number] D .[†]

Ἀλλὰ δὴ ἔστω ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B , οὕτως ὁ ἀπὸ τοῦ Γ τετράγωνος πρὸς τὸν ἀπὸ τοῦ Δ [τετράγωνον]· λέγω, ὅτι σύμμετρος ἐστὶν ἡ A τῇ B μήκει.

And so let the square on A be to the (square) on B as the square (number) on C (is) to the [square] (number) on D . I say that A is commensurable in length with B .

Ἐπεὶ γὰρ ἐστὶν ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον], οὕτως ὁ ἀπὸ τοῦ Γ τετράγωνος πρὸς τὸν ἀπὸ τοῦ Δ [τετράγωνον], ἀλλ' ὁ μὲν τοῦ ἀπὸ τῆς A τετραγώνου πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγος διπλασίον ἐστὶ τοῦ τῆς A πρὸς τὴν B λόγον, ὁ δὲ τοῦ ἀπὸ τοῦ Γ [ἀριθμοῦ] τετραγώνου [ἀριθμοῦ] πρὸς τὸν ἀπὸ τοῦ Δ [ἀριθμοῦ] τετράγωνον [ἀριθμόν] λόγος διπλασίον ἐστὶ τοῦ τοῦ Γ [ἀριθμοῦ] πρὸς τὸν Δ [ἀριθμόν] λόγον, ἐστὶν ἄρα καὶ ὡς ἡ A πρὸς τὴν B , οὕτως ὁ Γ [ἀριθμοῦ] πρὸς τὸν Δ [ἀριθμόν]. ἡ A ἄρα πρὸς τὴν B λόγον ἔθει, ὃν ἀριθμὸς ὁ Γ πρὸς ἀριθμόν τὸν Δ · σύμμετρος ἄρα ἐστὶν ἡ A τῇ B μήκει.

For since as the square on A is to the [square] on B , so the square (number) on C (is) to the [square] (number) on D . But, the ratio of the square on A to the (square) on B is the square of the (ratio) of A to B [Prop. 6.20 corr.]. And the (ratio) of the square [number] on the [number] C to the square [number] on the [number] D is the square of the ratio of the [number] C to the [number] D [Prop. 8.11]. Thus, as A is to B , so the [number] C also (is) to the [number] D . A , thus, has to B the ratio which the number C has to the number D . Thus, A is commensurable in length with B [Prop. 10.6].[‡]

Ἀλλὰ δὴ ἀσύμμετρος ἔστω ἡ A τῇ B μήκει· λέγω, ὅτι τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

And so let A be incommensurable in length with B . I say that the square on A does not have to the [square] on B the ratio which (some) square number (has) to (some) square number.

Εἰ γὰρ ἔθει τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, σύμμετρος ἔσται ἡ A τῇ B . οὐκ ἔστι δέ· οὐκ ἄρα τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον

For if the square on A has to the [square] on B the ratio which (some) square number (has) to (some) square number then A will be commensurable (in length) with B . But it is not. Thus, the square on A does not have to the [square] on the B

ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

Πάλιν δὴ τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον μὴ ἐθέτω, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· λέγω, ὅτι ἀσύμμετρος ἔστιν ἡ A τῇ B μήκει.

Εἰ γὰρ ἔστι σύμμετρος ἡ A τῇ B , ἔχει τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς B λόγον, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. οὐκ ἔθει δέ· οὐκ ἄρα σύμμετρος ἔστιν ἡ A τῇ B μήκει.

Τὰ ἄρα ἀπὸ τῶν μήκει συμμετρων, καὶ τὰ ἐχῆς.

Πόρισμα.

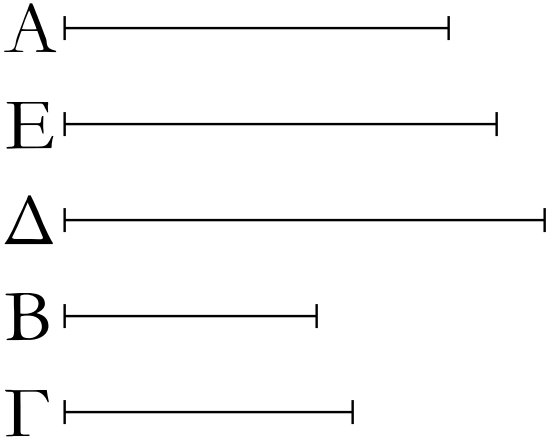
Καὶ φανερόν ἐκ τῶν δεδειγμένων ἔσται, ὅτι αἱ μήκει σύμμετροι πάντως καὶ δυνάμει, αἱ δὲ δυνάμει οὐ πάντως καὶ μήκει.

† There is an unstated assumption here that if $\alpha : \beta :: \gamma : \delta$ then $\alpha^2 : \beta^2 :: \gamma^2 : \delta^2$.

* There is an unstated assumption here that if $\alpha^2 : \beta^2 :: \gamma^2 : \delta^2$ then $\alpha : \beta :: \gamma : \delta$.

10.

Τῇ προτεθείσῃ εὐθείᾳ προσενεῖν δύο εὐθείας ἀσυμμέτρους, τὴν μὲν μήκει μόνον, τὴν δὲ καὶ δυνάμει.



Ἐστω ἡ προτεθείσα εὐθεῖα ἡ A · δεῖ δὴ τῇ A προσενεῖν δύο εὐθείας ἀσυμμέτρους, τὴν μὲν μήκει μόνον, τὴν δὲ καὶ δυνάμει.

Ἐκκείσθωσαν γὰρ δύο ἀριθμοὶ οἱ B, Γ πρὸς ἀλλήλους λόγον μὴ ἔχοντες, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, τουτέστι μὴ ὅμοιοι ἐπίπεδοι, καὶ γερονέτω ὡς ὁ B πρὸς τὸν Γ , οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς Δ τετράγωνον· ἐμάθομεν γάρ· σύμμετρον ἄρα τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς Δ . καὶ ἐπεὶ ὁ B πρὸς τὸν Γ λόγον οὐκ ἔθει,

the ratio which (some) square number (has) to (some) square number.

So, again, let the square on A not have to the [square] on B the ratio which (some) square number (has) to (some) square number. I say that A is incommensurable in length with B .

For if A is commensurable (in length) with B then the (square) on A will have to the (square) on B the ratio which (some) square number (has) to (some) square number. But it does not have (such a ratio). Thus, A is not commensurable in length with B .

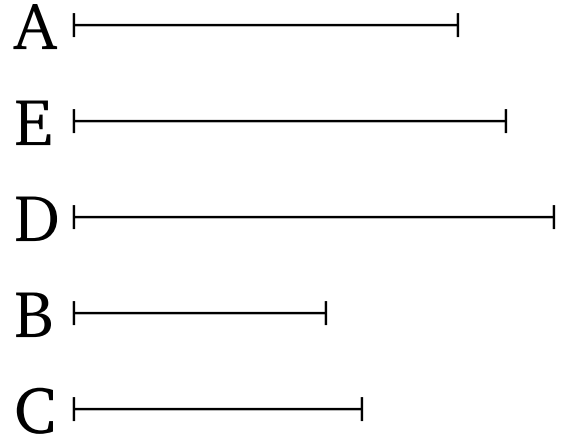
Thus, (squares) on (straight-lines which are) commensurable in length, and so on . . .

Corollary

And it will be clear, from (what) has been demonstrated, that (straight-lines) commensurable in length (are) always also (commensurable) in square, but (straight-lines commensurable) in square (are) not always also (commensurable) in length.

Proposition 10†

To find two straight-lines incommensurable with a given straight-line, the one (incommensurable) in length only, the other also (incommensurable) in square.



Let A be the given straight-line. So it is required to find two straight-lines incommensurable with A , the one (incommensurable) in length only, the other also (incommensurable) in square.

For let two numbers, B and C , not having to one another the ratio which (some) square number (has) to (some) square number—that is to say, not (being) similar plane (numbers)—be taken. And let it be contrived that as B (is) to C , so the square on A (is) to the square on D . For we learned (how to

ὅν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς Δ λόγον ἔθει, ὅν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ Δ μήκει. εἰλήφθω τῶν A , Δ μέση ἀνάλογον ἡ E · ἐστὶν ἄρα ὡς ἡ A πρὸς τὴν Δ , οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς E . ἀσύμμετρος δέ ἐστιν ἡ A τῇ Δ μήκει· ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς A τετράγωνον τῷ ἀπὸ τῆς E τετραγώνῳ· ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ E δυνάμει.

Τῇ ἄρα προτεθείσῃ εὐθείᾳ τῇ A προσεύρηται δύο εὐθεῖαι ἀσύμμετροι αἱ Δ , E , μήκει μὲν μόνον ἡ Δ , δυνάμει δέ καὶ μήκει δηλαδὴ ἡ E [ὅπερ ἔδει δεῖξαι].

do this [Prop. 10.6 corr.]. Thus, the (square) on A (is) commensurable with the (square) on D [Prop. 10.6]. And since B does not have to C the ratio which (some) square number (has) to (some) square number, the (square) on A thus does not have to the (square) on D the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with D [Prop. 10.9]. Let the (straight-line) E (which is) in mean proportion to A and D be taken [Prop. 6.13]. Thus, as A is to D , so the square on A (is) to the (square) on E [Def. 5.9]. And A is incommensurable in length with D . Thus, the square on A is also incommensurable with the square on E [Prop. 10.11]. Thus, A is incommensurable in square with E .

Thus, two straight-lines, D and E , (which are) incommensurable with the given straight-line A , be found, the one, D , (incommensurable) in length only, the other, E , (incommensurable) in square, and, clearly, also in length. [(Which is) the very thing it was required to show.]

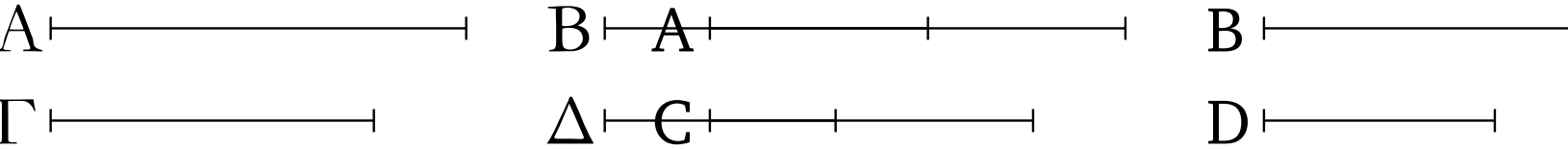
† This whole proposition is regarded by Heiberg as an interpolation into the original text.

11.

Proposition 11

Ἐὰν τέσσαρα μεγέθη ἀνάλογον ᾖ, τὸ δὲ πρῶτον τῷ δευτέρῳ σύμμετρον ᾖ, καὶ τὸ τρίτον τῷ τετάρτῳ σύμμετρον ἔσται· κ' ἂν τὸ πρῶτον τῷ δευτέρῳ ἀσύμμετρον ᾖ, καὶ τὸ τρίτον τῷ τετάρτῳ ἀσύμμετρον ἔσται.

If four magnitudes are proportional, and the first is commensurable with the second, then the third will also be commensurable with the fourth. And if the first is incommensurable with the second, then the third will also be incommensurable with the fourth.



Ἐστωσαν τέσσαρα μεγέθη ἀνάλογον τὰ A , B , Γ , Δ , ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , τὸ A δὲ τῷ B σύμμετρον ἔστω· λέγω, ὅτι καὶ τὸ Γ τῷ Δ σύμμετρον ἔσται.

Ἐπεὶ γὰρ σύμμετρον ἐστὶ τὸ A τῷ B , τὸ A ἄρα πρὸς τὸ B λόγον ἔθει, ὅν ἀριθμὸς πρὸς ἀριθμόν. καὶ ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ · καὶ τὸ Γ ἄρα πρὸς τὸ Δ λόγον ἔθει, ὅν ἀριθμὸς πρὸς ἀριθμόν· σύμμετρον ἄρα ἐστὶ τὸ Γ τῷ Δ .

Ἀλλὰ δὴ τὸ A τῷ B ἀσύμμετρον ἔστω· λέγω, ὅτι καὶ τὸ Γ τῷ Δ ἀσύμμετρον ἔσται. ἐπεὶ γὰρ ἀσύμμετρον ἐστὶ τὸ A τῷ B , τὸ A ἄρα πρὸς τὸ B λόγον οὐκ ἔθει, ὅν ἀριθμὸς πρὸς ἀριθμόν. καὶ ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ · οὐδὲ τὸ Γ ἄρα πρὸς τὸ Δ λόγον ἔθει, ὅν ἀριθμὸς πρὸς ἀριθμόν· ἀσύμμετρον ἄρα ἐστὶ τὸ Γ τῷ Δ .

Ἐὰν ἄρα τέσσαρα μεγέθη, καὶ τὰ ἐχῇς.

Let A , B , C , D be four proportional magnitudes, (such that) as A (is) to B , so C (is) to D . And let A be commensurable with B . I say that C will also be commensurable with D .

For since A is commensurable with B , A thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. And as A is to B , so C (is) to D . Thus, C also has to D the ratio which (some) number (has) to (some) number. Thus, C is commensurable with D [Prop. 10.6].

And so let A be incommensurable with B . I say that C will also be incommensurable with D . For since A is incommensurable with B , A thus does not have to B the ratio which (some) number (has) to (some) number [Prop. 10.7]. And as A is to B , so C (is) to D . Thus, C does not have to D the ratio which (some) number (has) to (some) number either. Thus, C is incommensurable with D [Prop. 10.8].

Thus, if four magnitudes, and so on . . .

12.

Proposition 12

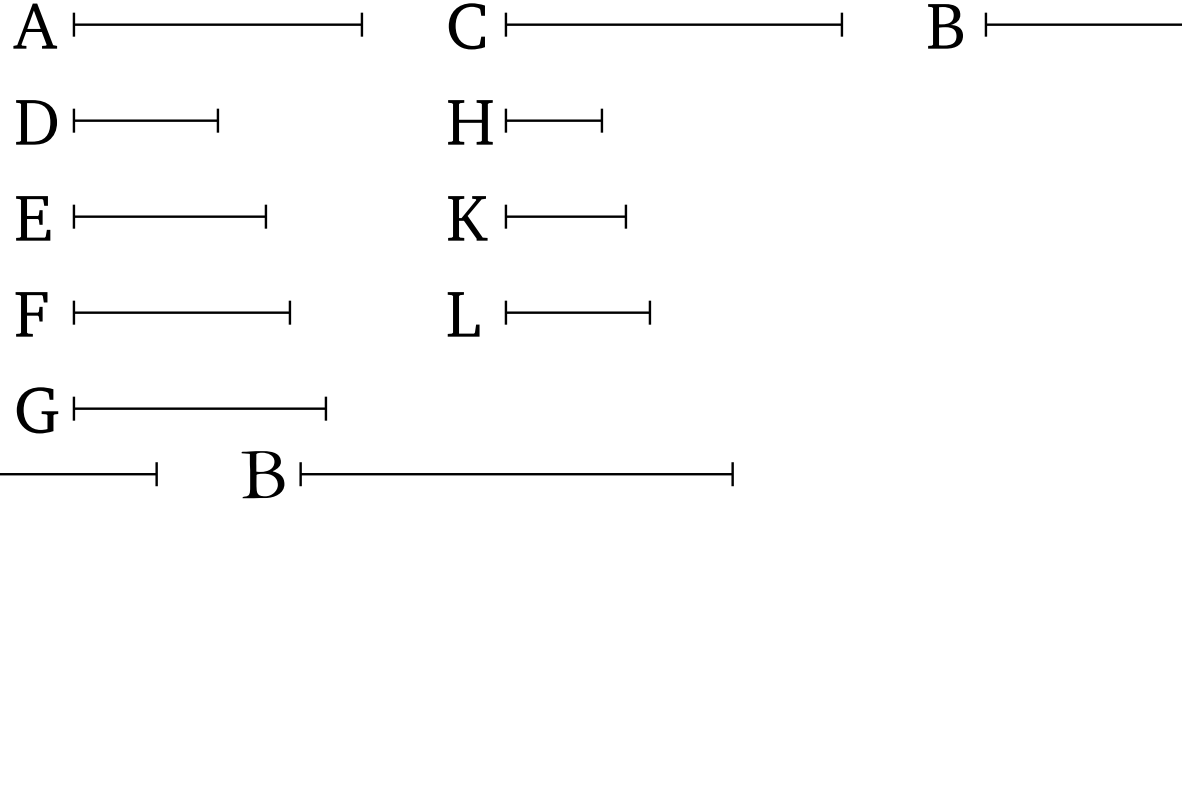
Τὰ τῷ αὐτῷ μεγέθει σύμμετρα καὶ ἀλλήλοισι ἐστὶ σύμμε-

(Magnitudes) commensurable with the same magnitude

τρα.

Ἐκάτερον γὰρ τῶν A, B τῷ Γ ἔστω σύμμετρον. λέγω, ὅτι καὶ τὸ A τῷ B ἐστὶ σύμμετρον.

Ἐπεὶ γὰρ σύμμετρον ἐστὶ τὸ A τῷ Γ , τὸ A ἄρα πρὸς τὸ Γ λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, ὃν ὁ Δ πρὸς τὸν E . πάλιν, ἐπεὶ σύμμετρον ἐστὶ τὸ Γ τῷ B , τὸ Γ ἄρα πρὸς τὸ B λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, ὃν ὁ Z πρὸς τὸν H . καὶ λόγων δοθέντων ὁποσωνοῦν τοῦ τε, ὃν ἔθει ὁ Δ πρὸς τὸν E , καὶ ὁ Z πρὸς τὸν H εἰλήφθωσαν ἀριθμοὶ ἐχῆς ἐν τοῖς δοθείσι λόγοις οἱ Θ, K, Λ ὥστε εἶναι ὡς μὲν τὸν Δ πρὸς τὸν E , οὕτως τὸν Θ πρὸς τὸν K , ὡς δὲ τὸν Z πρὸς τὸν H , οὕτως τὸν K πρὸς τὸν Λ .



Ἐπεὶ οὖν ἐστὶν ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ πρὸς τὸν E , ἀλλ' ὡς ὁ Δ πρὸς τὸν E , οὕτως ὁ Θ πρὸς τὸν K , ἔστιν ἄρα καὶ ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Θ πρὸς τὸν K . πάλιν, ἐπεὶ ἐστὶν ὡς τὸ Γ πρὸς τὸ B , οὕτως ὁ Z πρὸς τὸν H , ἀλλ' ὡς ὁ Z πρὸς τὸν H , [οὕτως] ὁ K πρὸς τὸν Λ , καὶ ὡς ἄρα τὸ Γ πρὸς τὸ B , οὕτως ὁ K πρὸς τὸν Λ . ἔστι δὲ καὶ ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Θ πρὸς τὸν K δι' ἴσον ἄρα ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως ὁ Θ πρὸς τὸν Λ . τὸ A ἄρα πρὸς τὸ B λόγον ἔθει, ὃν ἀριθμὸς ὁ Θ πρὸς ἀριθμόν τὸν Λ . σύμμετρον

are also commensurable with one another.

For let A and B each be commensurable with C . I say that A is also commensurable with B .

For since A is commensurable with C , A thus has to C the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (the ratio) which D (has) to E . Again, since C is commensurable with B , C thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (the ratio) which F (has) to G . And for any multitude whatsoever of given ratios—(namely,) those which D has to E , and F to G —let the numbers H, K, L (which are) continuously (proportional) in the(se) given ratios be taken [Prop. 8.4]. Hence, as D is to E , so H (is) to K , and as F (is) to G , so K (is) to L .

Therefore, since as A is to C , so D (is) to E , but as D (is) to E , so H (is) to K , thus also as A is to C , so H (is) to K [Prop. 5.11]. Again, since as C is to B , so F (is) to G , but as F (is) to G , [so] K (is) to L , thus also as C (is) to B , so K (is) to L [Prop. 5.11]. And also as A is to C , so H (is) to K . Thus, via equality, as A is to B , so H (is) to L [Prop. 5.22]. Thus, A has to B the ratio which the number H (has) to the number L . Thus, A is commensurable with B [Prop. 10.6].

Thus, (magnitudes) commensurable with the same magnitude are also commensurable with one another. (Which is) the

ἄρα ἐστὶ τὸ A τῷ B .

Τὰ ἄρα τῷ αὐτῷ μεγέθει σύμμετρα καὶ ἀλλήλοις ἐστὶ σύμμετρα· ὅπερ ἔδει δεῖχαι.

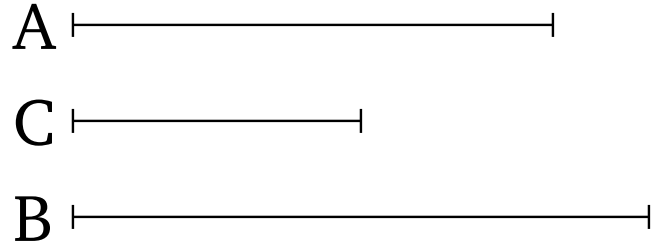
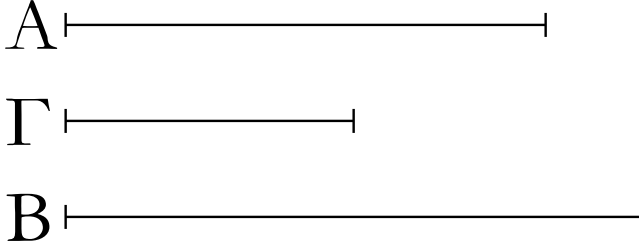
very thing it was required to show.

13.

Proposition 13

Ἐάν ἡ δύο μεγέθη σύμμετρα, τὸ δὲ ἕτερον αὐτῶν μεγέθει τινὶ ἀσύμμετρον ᾗ, καὶ τὸ λοιπὸν τῷ αὐτῷ ἀσύμμετρον ἔσται.

If two magnitudes are commensurable, and one of them is incommensurable with some magnitude, then the remaining (magnitude) will also be incommensurable with it.



Ἐστω δύο μεγέθη σύμμετρα τὰ A, B , τὸ δὲ ἕτερον αὐτῶν τὸ A ἄλλῳ τινὶ τῷ $Γ$ ἀσύμμετρον ἔστω· λέγω, ὅτι καὶ τὸ λοιπὸν τὸ B τῷ $Γ$ ἀσύμμετρον ἔστιν.

Let A and B be two commensurable magnitudes, and let one of them, A , be incommensurable with some other (magnitude), C . I say that the remaining (magnitude), B , is also incommensurable with C .

Εἰ γάρ ἐστι σύμμετρον τὸ B τῷ $Γ$, ἀλλὰ καὶ τὸ A τῷ B σύμμετρον ἔστιν, καὶ τὸ A ἄρα τῷ $Γ$ σύμμετρον ἔστιν. ἀλλὰ καὶ ἀσύμμετρον· ὅπερ ἀδύνατον. οὐκ ἄρα σύμμετρον ἔστι τὸ B τῷ $Γ$. ἀσύμμετρον ἄρα.

For if B is commensurable with C , but A is also commensurable with B , A is thus also commensurable with C [Prop. 10.12]. But, (it is) also incommensurable (with C). The very thing (is) impossible. Thus, B is not commensurable with C . Thus, (it is) incommensurable.

Ἐάν ἄρα ἡ δύο μεγέθη σύμμετρα, καὶ τὰ ἐχῇ.

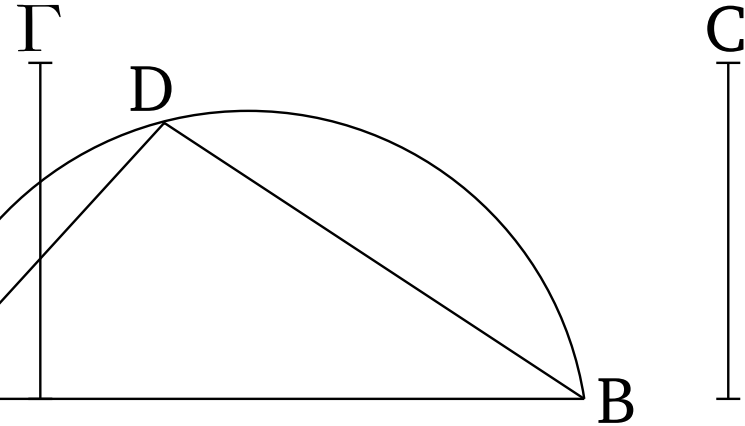
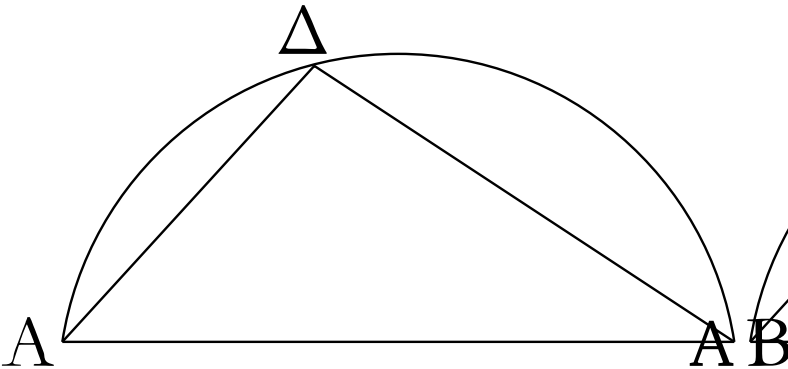
Thus, if two magnitudes are commensurable, and so on
....

Λήμμα.

Lemma

Δύο δοθεισῶν εὐθειῶν ἀνίσων εὐρεῖν, τίνι μείζον δύναται ἢ μείζων τῇς ἐλάσσονος.

For two given unequal straight-lines, to find by (the square on) which (straight-line) the square on the greater (straight-line is) larger than (the square on) the lesser.[†]



Ἐστωσαν αἱ δοθεῖσαι δύο ἄνισοι εὐθεῖαι αἱ $AB, Γ$, ὧν μείζων ἔστω ἡ AB · δεῖ δὴ εὐρεῖν, τίνι μείζον δύναται ἡ AB τῇς $Γ$.

Let AB and C be the two given unequal straight-lines, and let AB be the greater of them. So it is required to find by (the square on) which (straight-line) the square on AB (is) greater than (the square on) C .

Γεγραφθῶ ἐπὶ τῇς AB ἡμικύκλιον τὸ $AΔB$, καὶ εἰξ αὐτό

ἐνηρμόσθω τῇ Γ ἴση ἡ $A\Delta$, καὶ ἐπεξεύθθω ἡ ΔB . φανερόν δὲ, ὅτι ὀρθή ἐστιν ἡ ὑπὸ $A\Delta B$ γωνία, καὶ ὅτι ἡ AB τῆς $A\Delta$, τουτέστι τῆς Γ , μείζον δύνатаι τῇ ΔB .

Ὅμοίωξ δὲ καὶ δύο δοθεῖσων εὐθειῶν ἡ δυναμένη αὐτὰς εὐρίσκεται οὕτως.

Ἐστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι αἱ $A\Delta$, ΔB , καὶ δέον ἔστω εὐρεῖν τὴν δυναμένην αὐτάξ. κείσθωσαν γάρ, ὥστε ὀρθὴν γωνίαν περιέθιν τὴν ὑπὸ $A\Delta$, ΔB , καὶ ἐπεξεύθθω ἡ AB . φανερόν πάλιν, ὅτι ἡ τὰς $A\Delta$, ΔB δυναμένη ἐστὶν ἡ AB . ὅπερ ἔδει δεῖχαι.

Let the semi-circle ADB be described on AB . And let AD , equal to C , be inserted into it [Prop. 4.1]. And let DB be joined. So (it is) clear that the angle ADB is a right-angle [Prop. 3.31], and that the square on AB (is) greater than (the square on) AD —that is to say, (the square on) C —by (the square on) DB [Prop. 1.47].

And, similarly, the square-root of (the sum of the squares on) two given straight-lines is also found likewise.

Let AD and DB be the two given straight-lines. And let it be necessary to find the square-root of (the sum of the squares on) them. For let them be laid down such as to encompass a right-angle—(namely), that (angle encompassed) by AD and DB . And let AB be joined. (It is) again clear that AB is the square-root of (the sum of the squares on) AD and DB [Prop. 1.47]. (Which is) the very thing it was required to show.

† That is, if α and β are the lengths of two given straight-lines, with α being greater than β , to find a straight-line of length γ such that $\alpha^2 = \beta^2 + \gamma^2$. Similarly, we can also find γ such that $\gamma^2 = \alpha^2 + \beta^2$.

14.

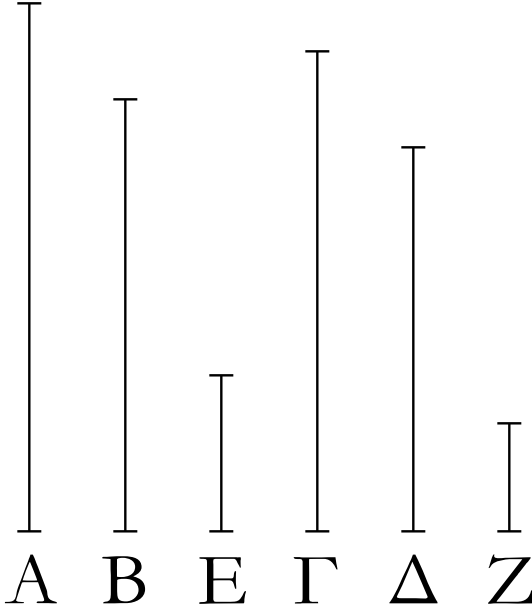
Proposition 14

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ᾗσιν, δύνηται δὲ ἡ πρώτη τῆς δευτέρας μείζον τῷ ἀπὸ συμμετρων ἑαυτῇ [μήκει], καὶ ἡ τρίτη τῆς τετάρτης μείζον δυνήσεται τῷ ἀπὸ συμμετρων ἑαυτῇ [μήκει]. καὶ ἔαν ἡ πρώτη τῆς δευτέρας μείζον δύνηται τῷ ἀπὸ ἀσυμμέτρων ἑαυτῇ [μήκει], καὶ ἡ τρίτη τῆς τετάρτης μείζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρων ἑαυτῇ [μήκει].

Ἐστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ A , B , Γ , Δ , ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ , καὶ ἡ A μὲν τῆς B μείζον δυνάσθω τῷ ἀπὸ τῆς E , ἡ δὲ Γ τῆς Δ μείζον δυνάσθω τῷ ἀπὸ τῆς Z . λέγω, ὅτι, εἴτε σύμμετρος ἐστὶν ἡ A τῇ E , σύμμετρος ἐστὶ καὶ ἡ Γ τῇ Z , εἴτε ἀσύμμετρος ἐστὶν ἡ A τῇ E , ἀσύμμετρος ἐστὶ καὶ ὁ Γ τῇ Z .

If four straight-lines are proportional, and the square on the first is greater than (the square on) the second by the (square) on (some straight-line) commensurable [in length] with the first, then the square on the third will also be greater than (the square on) the fourth by the (square) on (some straight-line) commensurable [in length] with the third. And if the square on the first is greater than (the square on) the second by the (square) on (some straight-line) incommensurable [in length] with the first, then the square on the third will also be greater than (the square on) the fourth by the (square) on (some straight-line) incommensurable [in length] with the third.

Let A , B , C , D be four proportional straight-lines, (such that) as A (is) to B , so C (is) to D . And let the square on A be greater than (the square on) B by the (square) on E , and let the square on C be greater than (the square on) D by the (square) on F . I say that A is either commensurable (in length) with E , and C is also commensurable with F , or A is incommensurable (in length) with E , and C is also incommensurable with F .



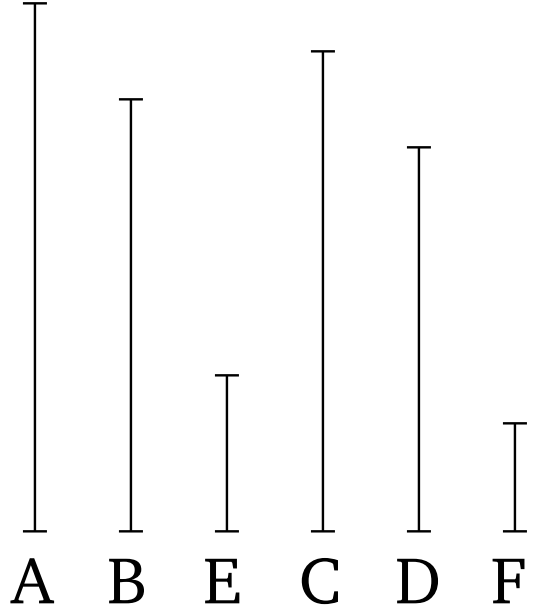
Ἐπεὶ γάρ ἐστιν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ , ἐστὶν ἄρα καὶ ὥς τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς B , οὕτως τὸ ἀπὸ τῆς Γ πρὸς τὸ ἀπὸ τῆς Δ . ἀλλὰ τῶ μὲν ἀπὸ τῆς A ἴσα ἐστὶ τὰ ἀπὸ τῶν E, B , τῶ δὲ ἀπὸ τῆς Γ ἴσα ἐστὶ τὰ ἀπὸ τῶν Δ, Z . ἐστὶν ἄρα ὥς τὰ ἀπὸ τῶν E, B πρὸς τὸ ἀπὸ τῆς B , οὕτως τὰ ἀπὸ τῶν Δ, Z πρὸς τὸ ἀπὸ τῆς Δ · διελόντι ἄρα ἐστὶν ὥς τὸ ἀπὸ τῆς E πρὸς τὸ ἀπὸ τῆς B , οὕτως τὸ ἀπὸ τῆς Z πρὸς τὸ ἀπὸ τῆς Δ · ἐστὶν ἄρα καὶ ὥς ἡ E πρὸς τὴν B , οὕτως ἡ Z πρὸς τὴν Δ · ἀνάπαλιν ἄρα ἐστὶν ὥς ἡ B πρὸς τὴν E , οὕτως ἡ Δ πρὸς τὴν Z . ἐστὶ δὲ καὶ ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ · δι' ἴσον ἄρα ἐστὶν ὥς ἡ A πρὸς τὴν E , οὕτως ἡ Γ πρὸς τὴν Z . εἴτε οὖν σύμμετρος ἐστὶν ἡ A τῇ E , σύμμετρος ἐστὶ καὶ ἡ Γ τῇ Z , εἴτε ἀσύμμετρος ἐστὶν ἡ A τῇ E , ἀσύμμετρος ἐστὶ καὶ ἡ Γ τῇ Z .

Ἐὰν ἄρα, καὶ τὰ ἐχῇς.

15.

Ἐὰν δύο μεγέθη σύμμετρα συντεθῇ, καὶ τὸ ὅλον ἐκατέρω αὐτῶν σύμμετρον ἔσται· καὶ τὸ ὅλον ἐν αὐτῶν σύμμετρον ἔσται, καὶ τὰ ἐκ ἀρθῆς μεγέθη σύμμετρα ἔσται.

Συγκείσθω γάρ δύο μεγέθη σύμμετρα τὰ AB, BG · λέγω, ὅτι καὶ ὅλον τὸ AG ἐκατέρω τῶν AB, BG ἐστὶ σύμμετρον.



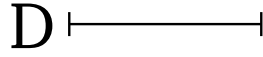
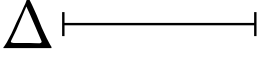
For since as A is to B , so C (is) to D , thus as the (square) on A is to the (square) on B , so the (square) on C (is) to the (square) on D [Prop. 6.22]. But the (sum of the squares) on E and B is equal to the (square) on A , and the (sum of the squares) on D and F is equal to the (square) on C . Thus, as the (sum of the squares) on E and B is to the (square) on B , so the (sum of the squares) on D and F (is) to the (square) on D . Thus, via separation, as the (square) on E is to the (square) on B , so the (square) on F (is) to the (square) on D [Prop. 5.17]. Thus, also, as E is to B , so F (is) to D [Prop. 6.22]. Thus, inversely, as B is to E , so D (is) to F [Prop. 5.7 corr.]. But, as A is to B , so C also (is) to D . Thus, via equality, as A is to E , so C (is) to F [Prop. 5.22]. Therefore, A is either commensurable (in length) with E , and C is also commensurable with F , or A is incommensurable (in length) with E , and C is also incommensurable with F [Prop. 10.11].

Thus, if, and so on . . .

Proposition 15

If two commensurable magnitudes are added together then the whole will also be commensurable with each of them. And if the whole is commensurable with one of them then the original magnitudes will also be commensurable (with one another).

For let the two commensurable magnitudes AB and BC be laid down together. I say that the whole AC is also commensurable with each of AB and BC .



Ἐπεὶ γὰρ σύμμετρά ἐστι τὰ AB , $B\Gamma$, μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ ἔστω τὸ Δ . ἐπεὶ οὖν τὸ Δ τὰ AB , $B\Gamma$ μετρεῖ, καὶ ὅλον τὸ $A\Gamma$ μετρήσει. μετρεῖ δὲ καὶ τὰ AB , $B\Gamma$. τὸ Δ ἄρα τὰ AB , $B\Gamma$, $A\Gamma$ μετρεῖ· σύμμετρον ἄρα ἐστὶ τὸ $A\Gamma$ ἐκατέρω τῶν AB , $B\Gamma$.

Ἀλλὰ δὴ τὸ $A\Gamma$ ἔστω σύμμετρον τῷ AB . λέγω δὴ, ὅτι καὶ τὰ AB , $B\Gamma$ σύμμετρά ἐστιν.

Ἐπεὶ γὰρ σύμμετρά ἐστι τὰ $A\Gamma$, AB , μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ ἔστω τὸ Δ . ἐπεὶ οὖν τὸ Δ τὰ $A\Gamma$, AB μετρεῖ, καὶ λοιπὸν ἄρα τὸ $B\Gamma$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB . τὸ Δ ἄρα τὰ AB , $B\Gamma$ μετρήσει· σύμμετρα ἄρα ἐστὶ τὰ AB , $B\Gamma$.

Ἐὰν ἄρα δύο μεγέθη, καὶ τὰ ἐχῇ.

For since AB and BC are commensurable, some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) AB and BC , it will also measure the whole AC . And it also measures AB and BC . Thus, D measures AB , BC , and AC . Thus, AC is commensurable with each of AB and BC [Def. 10.1].

And so let AC be commensurable with AB . I say that AB and BC are also commensurable.

For since AC and AB are commensurable, some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) CA and AB , it will thus also measure the remainder BC . And it also measures AB . Thus, D will measure (both) AB and BC . Thus, AB and BC are commensurable [Def. 10.1].

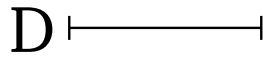
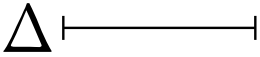
Thus, if two magnitudes, and so on

16.

Proposition 16

Ἐὰν δύο μεγέθη ἀσύμμετρα συντεθῇ, καὶ τὸ ὅλον ἐκατέρω αὐτῶν ἀσύμμετρον ἔσται· καὶ τὸ ὅλον ἐνὶ αὐτῶν ἀσύμμετρον ἦ, καὶ τὰ ἐκ ἀρθῆς μεγέθη ἀσύμμετρα ἔσται.

If two incommensurable magnitudes are added together then the whole will also be incommensurable with each of them. And if the whole is incommensurable with one of them then the original magnitudes will also be incommensurable (with one another).



Συγκείσθω γὰρ δύο μεγέθη ἀσύμμετρα τὰ AB , $B\Gamma$. λέγω, ὅτι καὶ ὅλον τὸ $A\Gamma$ ἐκατέρω τῶν AB , $B\Gamma$ ἀσύμμετρον ἐστίν.

Εἰ γὰρ μὴ ἐστὶν ἀσύμμετρα τὰ $A\Gamma$, AB , μετρήσει τι [αὐτὰ] μέγεθος. μετρεῖτω, εἰ δυνατόν, καὶ ἔστω τὸ Δ . ἐπεὶ οὖν τὸ Δ τὰ $A\Gamma$, AB μετρεῖ, καὶ λοιπὸν ἄρα τὸ $B\Gamma$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB . τὸ Δ ἄρα τὰ AB , $B\Gamma$ μετρεῖ. σύμμετρα ἄρα ἐστὶ τὰ AB , $B\Gamma$. ὑπέκειντο δὲ καὶ ἀσύμμετρα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ $A\Gamma$, AB μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐστὶ τὰ $A\Gamma$, AB . ὁμοίως δὲ δείχμεν, ὅτι καὶ τὰ $A\Gamma$, $B\Gamma$ ἀσύμμετρά ἐστιν. τὸ $A\Gamma$ ἄρα ἐκατέρω τῶν AB , $B\Gamma$ ἀσύμμετρον ἐστίν.

Ἀλλὰ δὴ τὸ $A\Gamma$ ἐνὶ τῶν AB , $B\Gamma$ ἀσύμμετρον ἔστω. ἔστω δὴ πρότερον τῷ AB . λέγω, ὅτι καὶ τὰ AB , $B\Gamma$ ἀσύμμετρά ἐστιν. εἰ γὰρ ἔσται σύμμετρα, μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ ἔστω τὸ Δ . ἐπεὶ οὖν τὸ Δ τὰ AB , $B\Gamma$ μετρεῖ,

For let the two incommensurable magnitudes AB and BC be laid down together. I say that that the whole AC is also incommensurable with each of AB and BC .

For if CA and AB are not incommensurable then some magnitude will measure [them]. If possible, let it (so) measure (them), and let it be D . Therefore, since D measures (both) CA and AB , it will thus also measure the remainder BC . And it also measures AB . Thus, D measures (both) AB and BC . Thus, AB and BC are commensurable [Def. 10.1]. But they were also assumed (to be) incommensurable. The very thing is impossible. Thus, some magnitude cannot measure (both) CA and AB . Thus, CA and AB are incommensurable [Def. 10.1]. So, similarly, we can show that AC and CB are also incommensurable. Thus, AC is incommensurable with each of AB and BC .

καὶ ὅλον ἄρα τὸ AG μετρήσει. μετρεῖ δὲ καὶ τὸ AB · τὸ Δ ἄρα τὰ GA , AB μετρεῖ. σύμμετρα ἄρα ἐστὶ τὰ GA , AB · ὑπέκειτο δὲ καὶ ἀσύμμετρα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ AB , BG μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐστὶ τὰ AB , BG .

Ἐάν ἄρα δύο μεγέθη, καὶ τὰ ἐχῆς.

And so let AC be incommensurable with one of AB and BC . So let it, first of all, be incommensurable with AB . I say that AB and BC are also incommensurable. For if they are commensurable then some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) AB and BC , it will thus also measure the whole AC . And it also measures AB . Thus, D measures (both) CA and AB . Thus, CA and AB are commensurable [Def. 10.1]. But they were also assumed (to be) incommensurable. The very thing is impossible. Thus, some magnitude cannot measure (both) AB and BC . Thus, AB and BC are incommensurable [Def. 10.1].

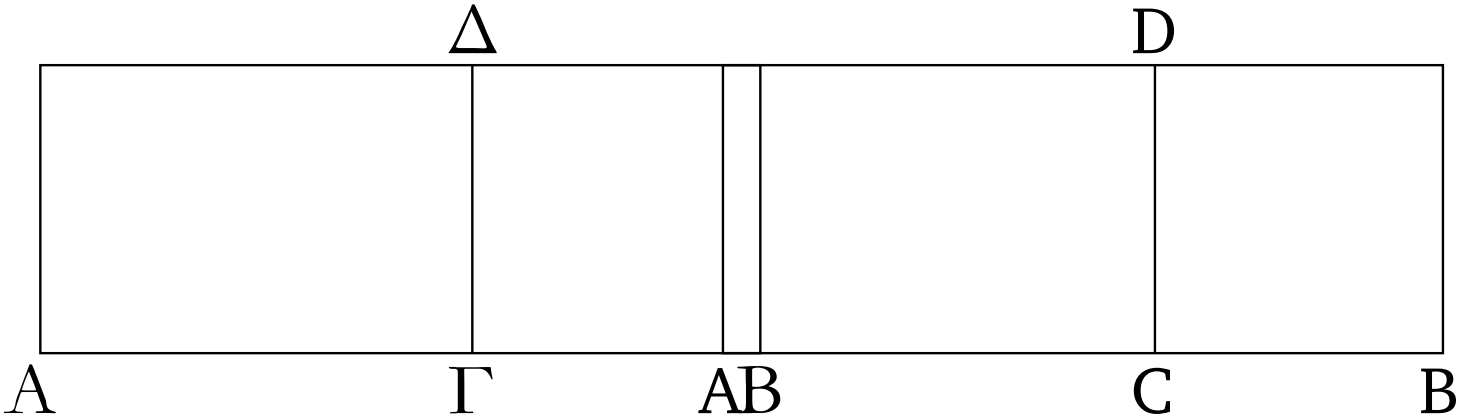
Thus, if two... magnitudes, and so on....

Λήμμα.

Ἐάν παρὰ τινα εὐθεῖαν παραβληθῇ παραλληλόγραμμον ἐλλείπον εἶδει τετραγώνῳ, τὸ παραβληθέν ἴσον ἐστὶ τῷ ὑπὸ τῶν ἐκ τῆς παραβολῆς γενομένων τμημάτων τῆς εὐθείας.

Lemma

If a parallelogram,[†] falling short by a square figure, is applied to some straight-line then the applied (parallelogram) is equal (in area) to the (rectangle contained) by the pieces of the straight-line created via the application (of the parallelogram).



Παρά γάρ εὐθεῖαν τὴν AB παραβεβλήσθω παραλληλόγραμμον τὸ AD ἐλλείπον εἶδει τετραγώνῳ τῷ DB · λέγω, ὅτι ἴσον ἐστὶ τὸ AD τῷ ὑπὸ τῶν AG , GB .

Καὶ ἐστὶν αὐτόθεν φανερόν· ἐπεὶ γάρ τετράγωνόν ἐστι τὸ DB , ἴση ἐστὶν ἡ ΔG τῇ GB , καὶ ἐστὶ τὸ AD τὸ ὑπὸ τῶν AG , GD , τουτέστι τὸ ὑπὸ τῶν AG , GB .

Ἐάν ἄρα παρὰ τινα εὐθεῖαν, καὶ τὰ ἐχῆς.

For let the parallelogram AD , falling short by the square figure DB , be applied to the straight-line AB . I say that AD is equal to the (rectangle contained) by AC and CB .

And it is immediately obvious. For since DB is a square, DC is equal to CB . And AD is the (rectangle contained) by AC and CD —that is to say, by AC and CB .

Thus, if ... to some straight-line, and so on....

[†] Note that this lemma only applies to rectangular parallelograms.

17.

Ἐάν ὥσι δύο εὐθεῖαι ἄνισοι, τῷ δὲ τετράτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ καὶ εἷς σύμμετρα αὐτὴν διαιρῇ μήκει, ἢ μείζων τῆς ἐλάσσονος μείζων δυνήσεται τῷ ἀπὸ συμμετρου ἐαυτῇ [μήκει]. καὶ ἔαν ἡ μείζων τῆς ἐλάσσονος μείζων δύνῃται τῷ ἀπὸ συμμετρου ἐαυτῇ [μήκει], τῷ δὲ τετράτῳ τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον εἶδει

Proposition 17[†]

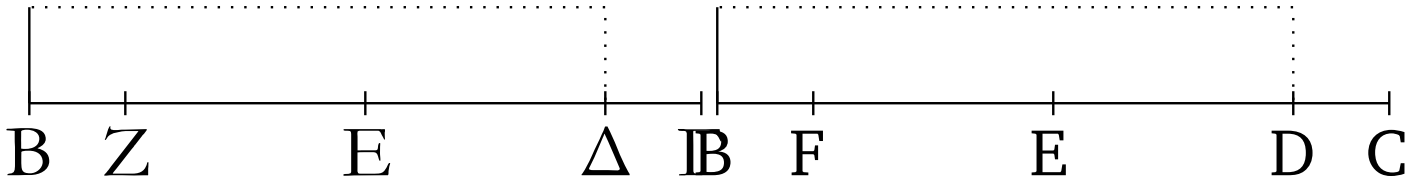
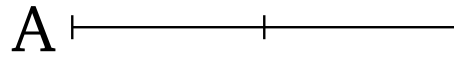
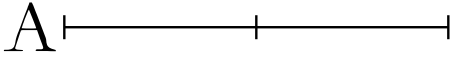
If there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) commensurable in length, then the square on the greater will be larger than (the square on) the lesser by the (square) on (some straight-line) commensurable [in length] with the greater. And if the square on the greater is

τετραγώνω, εἷς σύμμετρα αὐτὴν διαιρεῖ μήκει.

Ἐστωσαν δύο εὐθεῖαι ἄνισοι αἱ A , $B\Gamma$, ὧν μείζων ἡ $B\Gamma$, τῷ δὲ τετράρτῳ μέρει τοῦ ἀπὸ ἐλάσσονος τῆς A , τοντέστι τῷ ἀπὸ τῆς ἡμισείας τῆς A , ἴσον παρὰ τὴν $B\Gamma$ παραβεβλήσθω ἐλλείπον εἶδει τετραγώνω, καὶ ἔστω τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$, σύμμετρος δὲ ἔστω ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει· λέγω, ὅτι ἡ $B\Gamma$ τῆς A μείζων δύναται τῷ ἀπὸ συμμέτρων ἑαυτῇ.

larger than (the square on) the lesser by the (square) on (some straight-line) commensurable [in length] with the greater, and a (rectangle) equal to the fourth (part) of the (square) on the lesser, falling short by a square figure, is applied to the greater, then it divides it into (parts which are) commensurable in length.

Let A and BC be two unequal straight-lines, of which (let) BC (be) the greater. And let a (rectangle) equal to the fourth part of the (square) on the lesser, A —that is, (equal) to the (square) on half of A —falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC [see previous lemma]. And let BD be commensurable in length with DC . I say that that the square on BC is greater than the (square on) A by (the square on some straight-line) commensurable (in length) with (BC).



Τετμήσθω γάρ ἡ $B\Gamma$ δίδυα κατὰ τὸ E σημεῖον, καὶ κείσθω τῇ ΔE ἴση ἡ EZ . λοιπὴ ἄρα ἡ $\Delta\Gamma$ ἴση ἐστὶ τῇ BZ . καὶ ἐπεὶ εὐθεῖα ἡ $B\Gamma$ τέτμηται εἷς μὲν ἴσα κατὰ τὸ E , εἷς δὲ ἄνισα κατὰ τὸ Δ , τὸ ἄρα ὑπὸ $B\Delta$, $\Delta\Gamma$ περιεθόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς $E\Delta$ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Gamma$ τετραγώνῳ· καὶ τὰ τετραπλάσια· τὸ ἄρα τετράκις ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ μετὰ τοῦ τετραπλασίου τοῦ ἀπὸ τῆς ΔE ἴσον ἐστὶ τῷ τετράκις ἀπὸ τῆς $E\Gamma$ τετραγώνῳ. ἀλλὰ τῷ μὲν τετραπλασίῳ τοῦ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ ἴσον ἐστὶ τὸ ἀπὸ τῆς A τετράγωνον, τῷ δὲ τετραπλασίῳ τοῦ ἀπὸ τῆς ΔE ἴσον ἐστὶ τὸ ἀπὸ τῆς ΔZ τετράγωνον· διπλασίων γάρ ἐστιν ἡ ΔZ τῆς ΔE . τῷ δὲ τετραπλασίῳ τοῦ ἀπὸ τῆς $E\Gamma$ ἴσον ἐστὶ τὸ ἀπὸ τῆς $B\Gamma$ τετράγωνον· διπλασίων γάρ ἐστι πάλιν ἡ $B\Gamma$ τῆς ΓE . τὰ ἄρα ἀπὸ τῶν A , ΔZ τετράγωνα ἴσα ἐστὶ τῷ ἀπὸ τῆς $B\Gamma$ τετράγωνῳ· ὥστε τὸ ἀπὸ τῆς $B\Gamma$ τοῦ ἀπὸ τῆς A μείζον ἐστὶ τῷ ἀπὸ τῆς ΔZ · ἡ $B\Gamma$ ἄρα τῆς A μείζων δύναται τῇ ΔZ . δεικτέον, ὅτι καὶ σύμμετρος ἐστὶν ἡ $B\Gamma$ τῇ ΔZ . ἐπεὶ γὰρ σύμμετρος ἐστὶν ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει, σύμμετρος ἄρα ἐστὶ καὶ ἡ $B\Gamma$ τῇ $\Gamma\Delta$ μήκει. ἀλλὰ ἡ $\Gamma\Delta$ ταῖς $\Gamma\Delta$, BZ ἐστὶ σύμμετρος μήκει· ἴση γάρ ἐστιν ἡ $\Gamma\Delta$ τῇ BZ . καὶ ἡ $B\Gamma$ ἄρα σύμμετρος ἐστὶ ταῖς BZ , $\Gamma\Delta$ μήκει· ὥστε καὶ λοιπὴ τῇ $Z\Delta$ σύμμετρος ἐστὶν ἡ $B\Gamma$ μήκει· ἡ $B\Gamma$ ἄρα τῆς A μείζων δύναται τῷ ἀπὸ συμμέτρων ἑαυτῇ.

Ἀλλὰ δὴ ἡ $B\Gamma$ τῆς A μείζων δυνάσθω τῷ ἀπὸ συμμέτρων ἑαυτῇ, τῷ δὲ τετράρτῳ τοῦ ἀπὸ τῆς A ἴσον παρὰ τὴν $B\Gamma$

For let BC be cut in half at the point E [Prop. 1.10]. And let EF be made equal to DE [Prop. 1.3]. Thus, the remainder DC is equal to BF . And since the straight-line BC has been cut into equal (pieces) at E , and into unequal (pieces) at D , the rectangle contained by BD and DC , plus the square on ED , is thus equal to the square on EC [Prop. 2.5]. (The same) also (for) the quadruples. Thus, four times the (rectangle contained) by BD and DC , plus the quadruple of the (square) on DE , is equal to four times the square on EC . But, the square on A is equal to the quadruple of the (rectangle contained) by BD and DC , and the square on DF is equal to the quadruple of the (square) on DE . For DF is double DE . And the square on BC is equal to the quadruple of the (square) on EC . For, again, BC is double CE . Thus, the (sum of the) squares on A and DF is equal to the square on BC . Hence, the (square) on BC is greater than the (square) on A by the (square) on DF . Thus, BC is greater in square than A by DF . It must also be shown that BC is commensurable (in length) with DF . For since BD is commensurable in length with DC , BC is thus also commensurable in length with CD [Prop. 10.15]. But, CD is commensurable in length with CD plus BF . For CD is equal to BF [Prop. 10.6]. Thus, BC is also commensurable in length with the remainder FD [Prop. 10.15]. Thus, the square on BC is greater than (the square on) A by the (square)

παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$. δεικτέον, ὅτι σύμμετρος ἔστιν ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἡ $B\Gamma$ τῆς A μείζον δύνатаι τῷ ἀπὸ τῆς $Z\Delta$. δύνатаι δὲ ἡ $B\Gamma$ τῆς A μείζον τῷ ἀπὸ συμμέτρον ἑαντῇ. σύμμετρος ἄρα ἔστιν ἡ $B\Gamma$ τῇ $Z\Delta$ μήκει· ὥστε καὶ λοιπὴ συναμφοτέρῳ τῇ BZ , $\Delta\Gamma$ σύμμετρος ἔστιν ἡ $B\Gamma$ μήκει. ἀλλὰ συναμφοτέρως ἡ BZ , $\Delta\Gamma$ σύμμετρος ἔστι τῇ $\Delta\Gamma$ [μήκει]. ὥστε καὶ ἡ $B\Gamma$ τῇ $\Gamma\Delta$ σύμμετρος ἔστι μήκει· καὶ διελόντι ἄρα ἡ $B\Delta$ τῇ $\Delta\Gamma$ ἔστι σύμμετρος μήκει.

Ἐάν ἄρα ὦσι δύο εὐθεῖαι ἄνισοι, καὶ τὰ ἐχθῆς.

on (some straight-line) commensurable (in length) with (BC) .

And so let the square on BC be greater than the (square on) A by the (square) on (some straight-line) commensurable (in length) with (BC) . And let a (rectangle) equal to the fourth (part) of the (square) on A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC . It must be shown that BD is commensurable in length with DC .

For, similarly, by the same construction, we can show that the square on BC is greater than the (square on) A by the (square) on FD . And the square on BC is greater than the (square on) A by the (square) on (some straight-line) commensurable (in length) with (BC) . Thus, BC is commensurable in length with FD . Hence, BC is also commensurable in length with the remaining sum of BF and DC [Prop. 10.15]. But, the sum of BF and DC is commensurable [in length] with DC [Prop. 10.6]. Hence, BC is also commensurable in length with CD [Prop. 10.12]. Thus, via separation, BD is also commensurable in length with DC [Prop. 10.15].

Thus, if there are two unequal straight-lines, and so on . . .

[†] This proposition states that if $\alpha x - x^2 = \beta^2/4$ (where $\alpha = BC$, $x = DC$, and $\beta = A$) then α and $\sqrt{\alpha^2 - \beta^2}$ are commensurable when $\alpha - x$ are x are commensurable, and vice versa.

18.

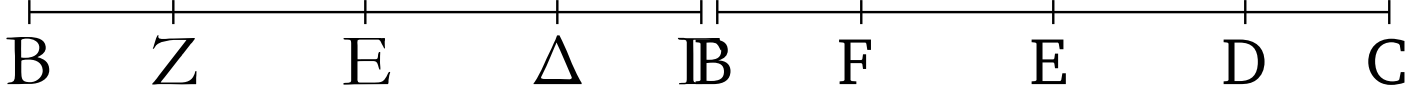
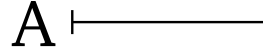
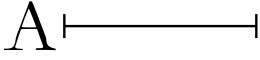
Proposition 18[†]

Ἐάν ὦσι δύο εὐθεῖαι ἄνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, καὶ εἰς ἀσύμμετρα αὐτὴν διαιρῇ [μήκει], ἡ μείζων τῆς ἐλάσσονος μείζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρον ἑαντῇ. καὶ ἔαν ἡ μείζων τῆς ἐλάσσονος μείζον δύνηται τῷ ἀπὸ ἀσυμμέτρον ἑαντῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, εἰς ἀσύμμετρα αὐτὴν διαιρεῖ [μήκει].

Ἐστῶσαν δύο εὐθεῖαι ἄνισοι αἱ A , $B\Gamma$, ὧν μείζων ἡ $B\Gamma$, τῷ δὲ τετάρτῳ [μέρει] τοῦ ἀπὸ τῆς ἐλάσσονος τῆς A ἴσον παρὰ τὴν $B\Gamma$ παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν $B\Delta\Gamma$, ἀσύμμετρος δὲ ἔστω ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει· λέγω, ὅτι ἡ $B\Gamma$ τῆς A μείζον δύνатаι τῷ ἀπὸ ἀσυμμέτρον ἑαντῇ.

If there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) incommensurable [in length], then the square on the greater will be larger than the (square on the) lesser by the (square) on (some straight-line) incommensurable (in length) with the greater. And if the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) incommensurable (in length) with the greater, and a (rectangle) equal to the fourth (part) of the (square) on the lesser, falling short by a square figure, is applied to the greater, then it divides it into (parts which are) incommensurable [in length].

Let A and BC be two unequal straight-lines, of which (let) BC (be) the greater. And let a (rectangle) equal to the fourth [part] of the (square) on the lesser, A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BDC . And let BD be incommensurable in length with DC . I say that that the square on BC is greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC) .



Τῶν γὰρ αὐτῶν κατασκευασθέντων τῷ πρότερον ὁμοίως δείχομεν, ὅτι ἡ $BΓ$ τῆς A μείζον δύναται τῷ ἀπὸ τῆς $ZΔ$. δεικτέον [οὔν], ὅτι ἀσύμμετρος ἐστὶν ἡ $BΓ$ τῇ $ΔZ$ μήκει. ἐπεὶ γὰρ ἀσύμμετρος ἐστὶν ἡ $BΔ$ τῇ $ΔΓ$ μήκει, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ $BΓ$ τῇ $ΓΔ$ μήκει. ἀλλὰ ἡ $ΔΓ$ σύμμετρος ἐστὶ συναμφοτέραις ταῖς BZ , $ΔΓ$. καὶ ἡ $BΓ$ ἄρα ἀσύμμετρος ἐστὶ συναμφοτέραις ταῖς BZ , $ΔΓ$. ὥστε καὶ λοιπὴ τῇ $ZΔ$ ἀσύμμετρος ἐστὶν ἡ $BΓ$ μήκει. καὶ ἡ $BΓ$ τῆς A μείζον δύναται τῷ ἀπὸ τῆς $ZΔ$. ἡ $BΓ$ ἄρα τῆς A μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαντῇ.

Δυνάσθω δὴ πάλιν ἡ $BΓ$ τῆς A μείζον τῷ ἀπὸ ἀσυμμέτρου ἑαντῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς A ἴσον παρὰ τὴν $BΓ$ παραβεβλήσθω ἐλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν $BΔ$, $ΔΓ$. δεικτέον, ὅτι ἀσύμμετρος ἐστὶν ἡ $BΔ$ τῇ $ΔΓ$ μήκει.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἡ $BΓ$ τῆς A μείζον δύναται τῷ ἀπὸ τῆς $ZΔ$. ἀλλὰ ἡ $BΓ$ τῆς A μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαντῇ. ἀσύμμετρος ἄρα ἐστὶν ἡ $BΓ$ τῇ $ZΔ$ μήκει. ὥστε καὶ λοιπὴ συναμφοτέρῳ τῇ BZ , $ΔΓ$ ἀσύμμετρος ἐστὶν ἡ $BΓ$. ἀλλὰ συναμφοτέροξ ἡ BZ , $ΔΓ$ τῇ $ΔΓ$ σύμμετρος ἐστὶ μήκει. καὶ ἡ $BΓ$ ἄρα τῇ $ΔΓ$ ἀσύμμετρος ἐστὶ μήκει. ὥστε καὶ διελόντι ἡ $BΔ$ τῇ $ΔΓ$ ἀσύμμετρος ἐστὶ μήκει.

Ἐὰν ἄρα ὡσι δύο εὐθεῖαι, καὶ τὰ ἐχθῆς.

For, similarly, by the same construction as before, we can show that the square on BC is greater than the (square on) A by the (square) on FD . [Therefore] it must be shown that BC is incommensurable in length with DF . For since BD is incommensurable in length with DC , BC is thus also incommensurable in length with CD [Prop. 10.16]. But, DC is commensurable (in length) with the sum of BF and DC [Prop. 10.6]. And, thus, BC is incommensurable (in length) with the sum of BF and DC [Prop. 10.13]. Hence, BC is also incommensurable in length with the remainder FD [Prop. 10.16]. And the square on BC is greater than the (square on) A by the (square) on FD . Thus, the square on BC is greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC) .

So, again, let the square on BC be greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC) . And let a (rectangle) equal to the fourth [part] of the (square) on A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC . It must be shown that BD is incommensurable in length with DC .

For, similarly, by the same construction, we can show that the square on BC is greater than the (square) on A by the (square) on FD . But, the square on BC is greater than the (square) on A by the (square) on (some straight-line) incommensurable (in length) with (BC) . Thus, BC is incommensurable in length with FD . Hence, BC is also incommensurable (in length) with the remaining sum of BF and DC [Prop. 10.16]. But, the sum of BF and DC is commensurable in length with DC [Prop. 10.6]. Thus, BC is also incommensurable in length with DC [Prop. 10.13]. Hence, via separation, BD is also incommensurable in length with DC [Prop. 10.16].

Thus, if there are two ... straight-lines, and so on ...

† This proposition states that if $\alpha x - x^2 = \beta^2/4$ (where $\alpha = BC$, $x = DC$, and $\beta = A$) then α and $\sqrt{\alpha^2 - \beta^2}$ are incommensurable when $\alpha - x$ are x are incommensurable, and vice versa.

19.

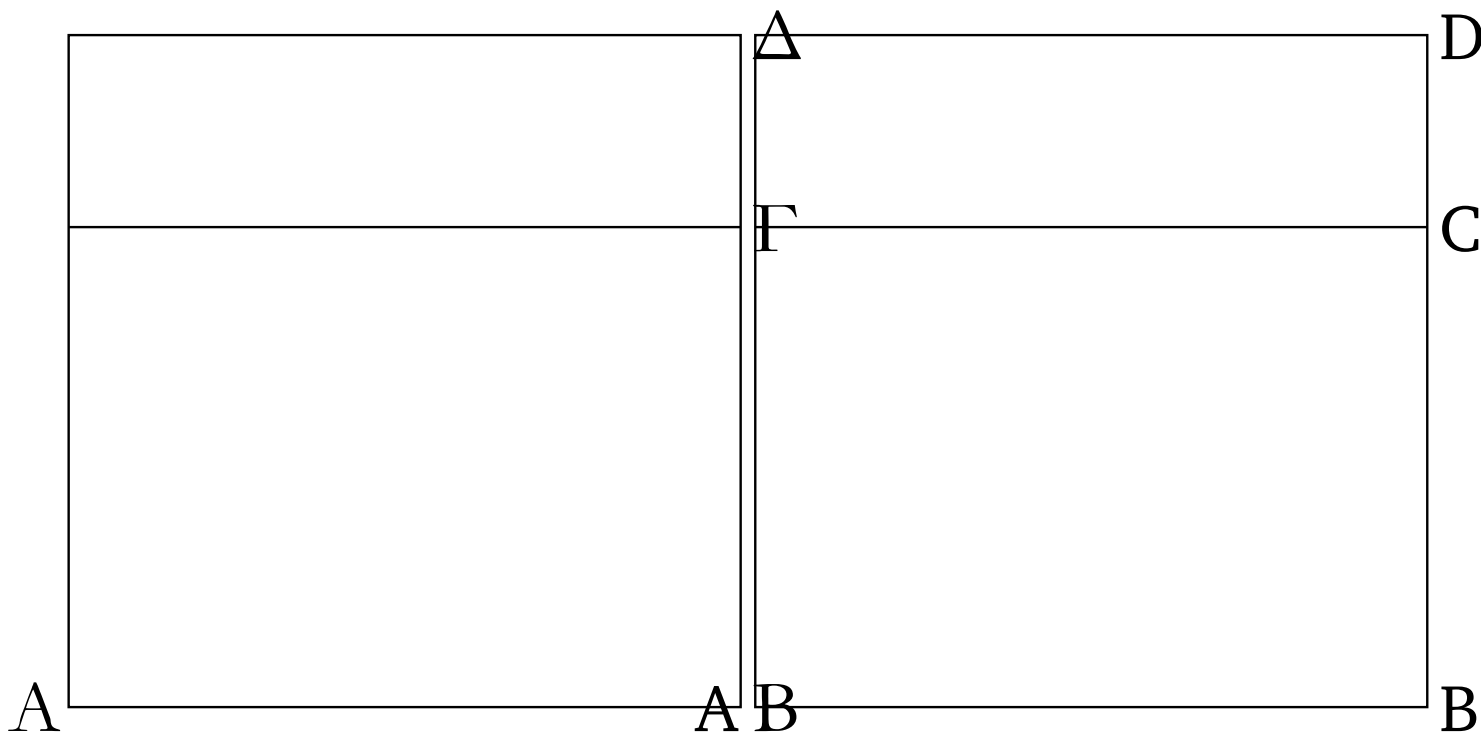
Proposition 19

Τὸ ὑπὸ ῥητῶν μήκει συμμετρῶν εὐθειῶν περιεχόμενον ὀρθογώνιον ῥητόν ἐστιν.

Ὑπὸ γὰρ ῥητῶν μήκει συμμετρῶν εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιεθέσθω τὸ $ΑΓ$. λέγω, ὅτι ῥητόν ἐστὶ τὸ $ΑΓ$.

The rectangle contained by rational straight-lines (which are) commensurable in length is rational.

For let the rectangle AC be enclosed by the rational straight-lines AB and BC (which are) commensurable in length. I say that AC is rational.



Ἀναγεγράφθω γάρ ἀπὸ τῆς AB τετράγωνον τὸ $A\Delta$. ῥητὸν ἄρα ἐστὶ τὸ $A\Delta$. καὶ ἐπεὶ σύμμετρόξ ἐστιν ἡ AB τῇ $B\Gamma$ μήκει, ἴση δέ ἐστιν ἡ AB τῇ $B\Delta$, σύμμετροξ ἄρα ἐστὶν ἡ $B\Delta$ τῇ $B\Gamma$ μήκει. καὶ ἐστὶν ὥς ἡ $B\Delta$ πρὸς τὴν $B\Gamma$, οὕτως τὸ ΔA πρὸς τὸ $A\Gamma$. σύμμετρον ἄρα ἐστὶ τὸ ΔA τῷ $A\Gamma$. ῥητὸν δὲ τὸ ΔA . ῥητὸν ἄρα ἐστὶ καὶ τὸ $A\Gamma$.

Τὸ ἄρα ὑπὸ ῥητῶν μήκει συμμέτρων, καὶ τὰ ἐχῆξ.

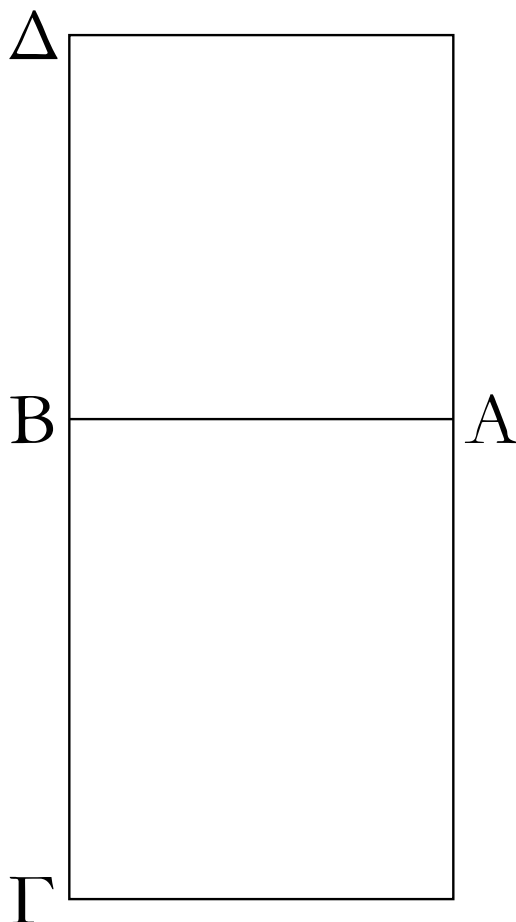
For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And since AB is commensurable in length with BC , and AB is equal to BD , BD is thus commensurable in length with BC . And as BD is to BC , so DA (is) to AC [Prop. 6.1]. Thus, DA is commensurable with AC [Prop. 10.11]. And DA (is) rational. Thus, AC is also rational [Def. 10.4]. Thus, the ...by rational straight-lines ... commensurable, and so on

20.

Ἐάν ῥητὸν παρὰ ῥητὴν παραβληθῇ, πλάτος ποιῇ ῥητὴν καὶ σύμμετρον τῇ, παρ' ἣν παράκειται, μήκει.

Proposition 20

If a rational (area) is applied to a rational (straight-line) then it produces as breadth a (straight-line which is) rational, and commensurable in length with the (straight-line) to which it is applied.



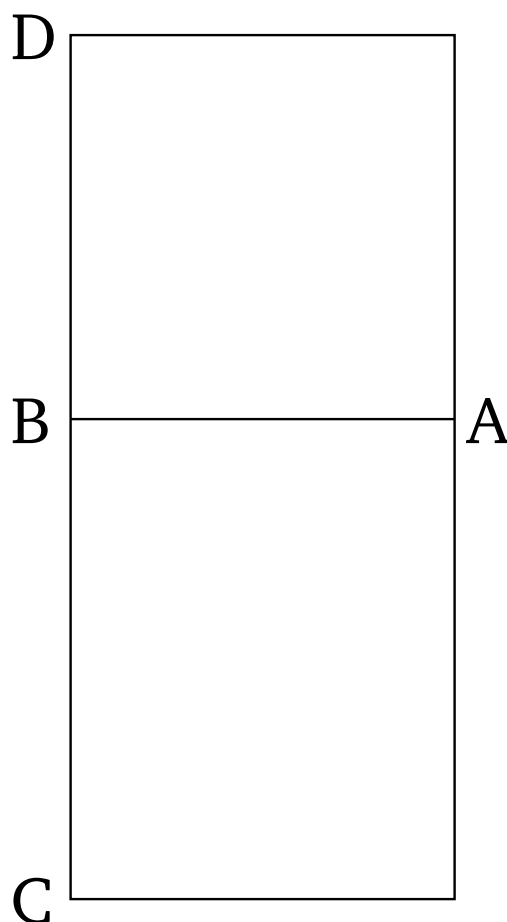
Ῥητόν γάρ τὸ $ΑΓ$ παρὰ ῥητὴν τὴν $ΑΒ$ παραβεβλήσθω πλάτος ποιῶν τὴν $ΒΓ$. λέγω, ὅτι ῥητὴ ἐστὶν ἡ $ΒΓ$ καὶ σύμμετρος τῇ $ΒΑ$ μήκει.

Ἀναγεγράφθω γάρ ἀπὸ τῆς $ΑΒ$ τετράγωνον τὸ $ΑΔ$. ῥητόν ἄρα ἐστὶ τὸ $ΑΔ$. ῥητόν δὲ καὶ τὸ $ΑΓ$. σύμμετρον ἄρα ἐστὶ τὸ $ΔΑ$ τῷ $ΑΓ$. καὶ ἐστὶν ὥς τὸ $ΔΑ$ πρὸς τὸ $ΑΓ$, οὕτως ἡ $ΔΒ$ πρὸς τὴν $ΒΓ$. σύμμετρος ἄρα ἐστὶ καὶ ἡ $ΔΒ$ τῇ $ΒΓ$. ἴση δὲ ἡ $ΔΒ$ τῇ $ΒΑ$. σύμμετρος ἄρα καὶ ἡ $ΑΒ$ τῇ $ΒΓ$. ῥητὴ δὲ ἐστὶν ἡ $ΑΒ$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΒΓ$ καὶ σύμμετρος τῇ $ΑΒ$ μήκει.

Ἐάν ἄρα ῥητόν παρὰ ῥητὴν παραβληθῇ, καὶ τὰ ἐχθῇ.

21.

Τὸ ὑπὸ ῥητῶν δυνάμει μόνον συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν, καὶ ἡ δυναμένη αὐτοῦ ἄλογός ἐστιν, καλεῖσθω δὲ μέση.



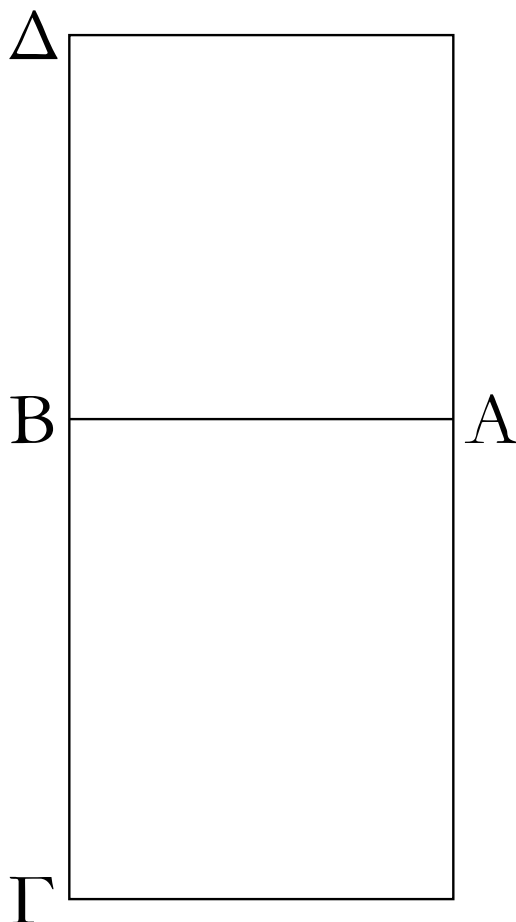
For let the rational (area) AC be applied to the rational (straight-line) AB , producing the (straight-line) BC as breadth. I say that BC is rational, and commensurable in length with BA .

For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And AC (is) also rational. DA is thus commensurable with AC . And as DA is to AC , so DB (is) to BC [Prop. 6.1]. Thus, DB is also commensurable (in length) with BC [Prop. 10.11]. And DB (is) equal to BA . Thus, AB (is) also commensurable (in length) with BC . And AB is rational. Thus, BC is also rational, and commensurable in length with AB [Def. 10.3].

Thus, if a rational (area) is applied to a rational (straight-line), and so on . . .

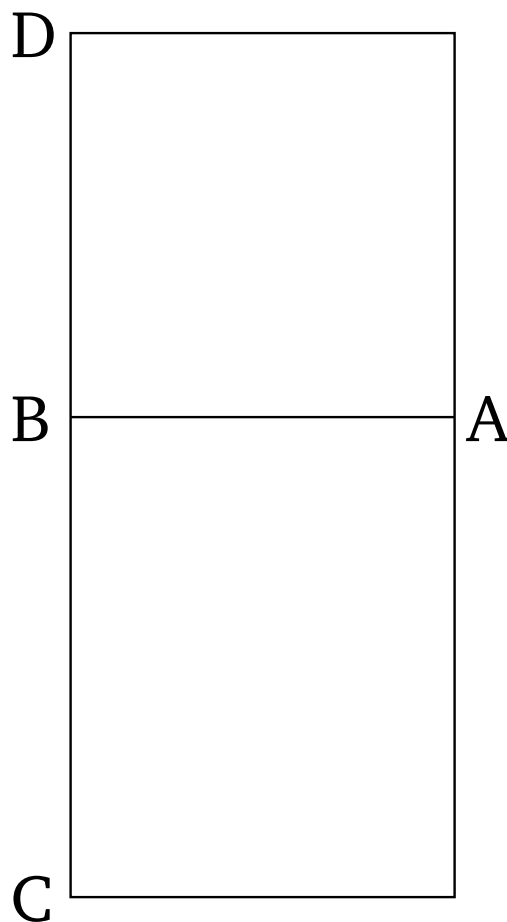
Proposition 21

The rectangle contained by rational straight-lines (which are) commensurable in square only is irrational, and its square-root is irrational—let it be called medial.[†]



Ὑπὸ γὰρ ῥητῶν δυνάμει μόνον συμμετρῶν εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιεθέσθω τὸ $ΑΓ$. λέγω, ὅτι ἄλογόν ἐστι τὸ $ΑΓ$, καὶ ἡ δυναμένη αὐτὸ ἄλογός ἐστιν, καλεῖσθω δὲ μέση.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $ΑΔ$. ῥητὸν ἄρα ἐστὶ τὸ $ΑΔ$. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει· δυνάμει γὰρ μόνον ὑπόκεινται σύμμετροι· ἴση δὲ ἡ AB τῇ $BΔ$, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ $ΔB$ τῇ $BΓ$ μήκει. καὶ ἐστὶν ὥς ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως τὸ $ΑΔ$ πρὸς τὸ $ΑΓ$. ἀσύμμετρον ἄρα [ἐστὶ] τὸ $ΔΑ$ τῷ $ΑΓ$. ῥητὸν δὲ τὸ $ΔΑ$. ἄλογον ἄρα ἐστὶ τὸ $ΑΓ$. ὥστε καὶ ἡ δυναμένη τὸ $ΑΓ$ [τουτέστιν ἡ ἴσον αὐτῷ τετράγωνον δυναμένη] ἄλογός ἐστιν, καλεῖσθω δε μέση· ὅπερ ἔδει δεῖχαι.



For let the rectangle AC be contained by the rational straight-lines AB and BC (which are) commensurable in square only. I say that AC is irrational, and its square-root is irrational—let it be called medial.

For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And since AB is incommensurable in length with BC . For they were assumed to be commensurable in square only. And AB (is) equal to BD . DB is thus also incommensurable in length with BC . And as DB is to BC , so AD (is) to AC [Prop. 6.1]. Thus, DA [is] incommensurable with AC [Prop. 10.11]. And DA (is) rational. Thus, AC is irrational [Def. 10.4]. Hence, its square-root [that is to say, the square-root of the square equal to it] is also irrational [Def. 10.4]—let it be called medial. (Which is) the very thing it was required to show.

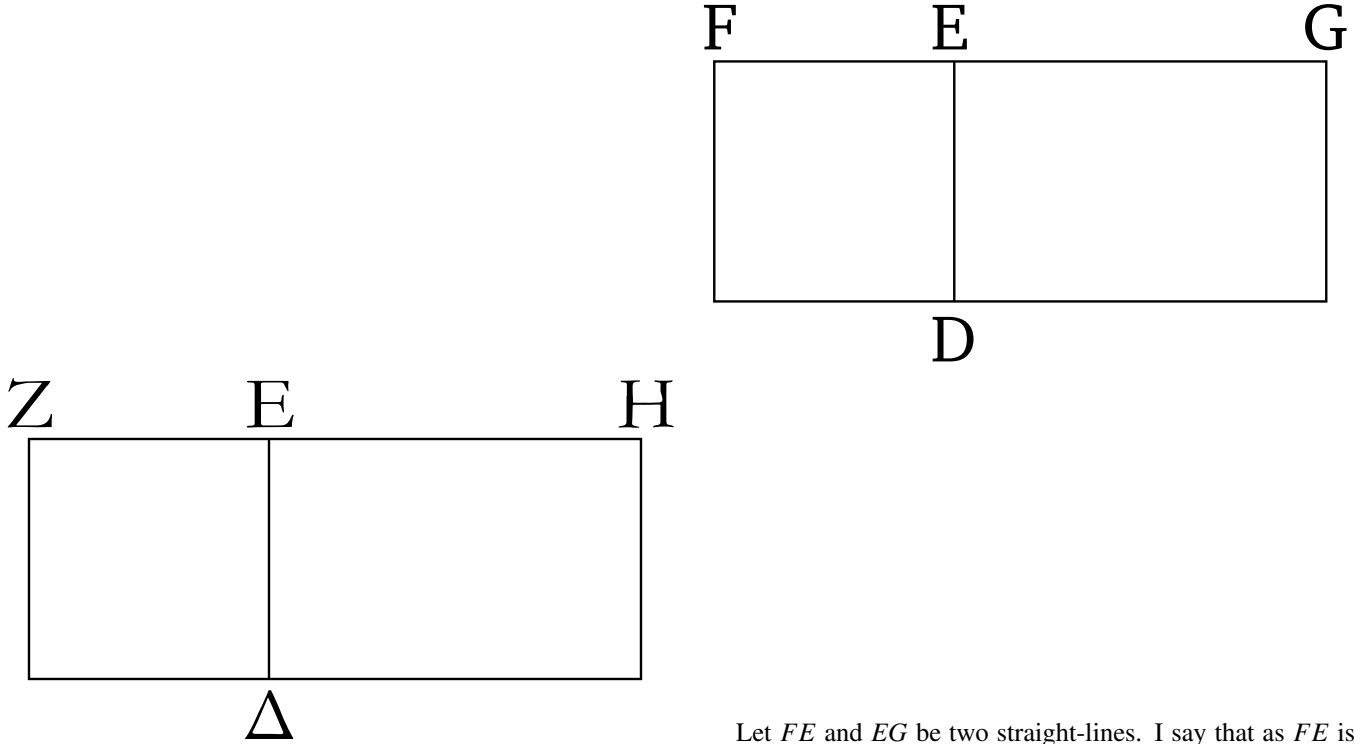
[†] Thus, a medial straight-line has a length expressible as $k^{1/4}$.

Λήμμα.

Ἐάν ᾧσι δύο εὐθεῖαι, ἔστιν ὥς ἡ πρώτη πρὸς τὴν δευτέραν, οὕτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ὑπὸ τῶν δύο εὐθειῶν.

Lemma

If there are two straight-lines then as the first is to the second, so the (square) on the first (is) to the (rectangle contained) by the two straight-lines.



Ἐστωσαν δύο εὐθεῖαι αἱ ZE , EH . λέγω, ὅτι ἐστὶν ὥς ἡ ZE πρὸς τὴν EH , οὕτως τὸ ἀπὸ τῆς ZE πρὸς τὸ ὑπὸ τῶν ZE , EH .

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ZE τετράγωνον τὸ ΔZ , καὶ συμπληρώσθω τὸ $H\Delta$. ἐπεὶ οὖν ἐστὶν ὥς ἡ ZE πρὸς τὴν EH , οὕτως τὸ $Z\Delta$ πρὸς τὸ ΔH , καὶ ἐστὶ τὸ μὲν $Z\Delta$ τὸ ἀπὸ τῆς ZE , τὸ δὲ ΔH τὸ ὑπὸ τῶν ΔE , EH , τουτέστι τὸ ὑπὸ τῶν ZE , EH , ἐστὶν ἄρα ὥς ἡ ZE πρὸς τὴν EH , οὕτως τὸ ἀπὸ τῆς ZE πρὸς τὸ ὑπὸ τῶν ZE , EH . ὁμοίως δὲ καὶ ὥς τὸ ὑπὸ τῶν HE , EZ πρὸς τὸ ἀπὸ τῆς EZ , τουτέστιν ὥς τὸ $H\Delta$ πρὸς τὸ $Z\Delta$, οὕτως ἡ HE πρὸς τὴν EZ . ὅπερ ἔδει δεῖξαι.

22.

Τὸ ἀπὸ μέσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ῥητὴν καὶ ἀσύμμετρον τῇ, παρ' ἣν παράκειται, μήκει.

Ἐστω μέση μὲν ἡ A , ῥητὴ δὲ ἡ GB , καὶ τῷ ἀπὸ τῆς A ἴσον παρὰ τὴν $B\Gamma$ παραβεβλήσθω θωρίον ὀρθογώνιον τὸ $B\Delta$ πλάτος ποιῶν τὴν $\Gamma\Delta$. λέγω, ὅτι ῥητὴ ἐστὶν ἡ $\Gamma\Delta$ καὶ ἀσύμμετρος τῇ GB μήκει.

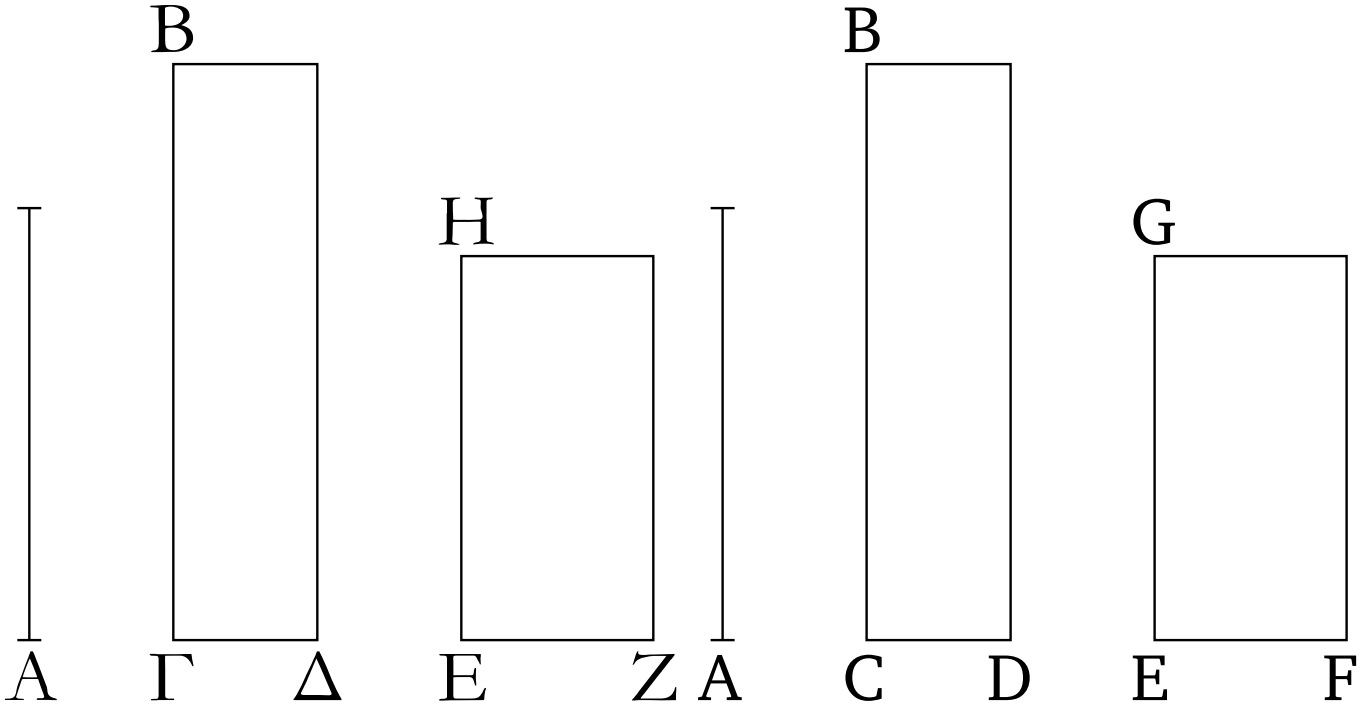
Let FE and EG be two straight-lines. I say that as FE is to EG , so the (square) on FE (is) to the (rectangle contained) by FE and EG .

For let the square DF be described on FE . And let GD be completed. Therefore, since as FE is to EG , so FD (is) to DG [Prop. 6.1], and FD is the (square) on FE , and DG the (rectangle contained) by DE and EG —that is to say, the (rectangle contained) by FE and EG —thus as FE is to EG , so the (square) on FE (is) to the (rectangle contained) by FE and EG . And also, similarly, as the (rectangle contained) by GE and EF is to the (square on) EF —that is to say, as GD (is) to FD —so GE (is) to EF . (Which is) the very thing it was required to show.

Proposition 22

The square on a medial (straight-line), being applied to a rational (straight-line), produces as breadth a (straight-line which is) rational, and incommensurable in length with the (straight-line) to which it is applied.

Let A be a medial (straight-line), and CB a rational (straight-line), and let the rectangular area BD , equal to the (square) on A , be applied to BC , producing CD as breadth. I say that CD is rational, and incommensurable in length with CB .



Ἐπεὶ γὰρ μέση ἐστὶν ἡ A , δύνανται θεωρεῖν περιεχόμενον ὑπὸ ῥητῶν δυνάμει μόνον συμμετρῶν. δυνάσθω τὸ HZ . δύνανται δὲ καὶ τὸ BD . ἴσον ἄρα ἐστὶ τὸ BD τῷ HZ . ἐστὶ δὲ αὐτῷ καὶ ἰσογώνιον· τῶν δὲ ἴσων τε καὶ ἰσογώνων παραλληλογράμμων ἀντιπεπρόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· ἀνάλογον ἄρα ἐστὶν ὡς ἡ $BΓ$ πρὸς τὴν EH , οὕτως ἡ EZ πρὸς τὴν $ΓΔ$. ἐστὶν ἄρα καὶ ὡς τὸ ἀπὸ τῆς $BΓ$ πρὸς τὸ ἀπὸ τῆς EH , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς $ΓΔ$. σύμμετρον δὲ ἐστὶ τὸ ἀπὸ τῆς $ΓΒ$ τῷ ἀπὸ τῆς EH . ῥητὴ γάρ ἐστιν ἑκατέρα αὐτῶν· σύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς EZ τῷ ἀπὸ τῆς $ΓΔ$. ῥητὸν δὲ ἐστὶ τὸ ἀπὸ τῆς EZ . ῥητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $ΓΔ$. ῥητὴ ἄρα ἐστὶν ἡ $ΓΔ$. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ EZ τῇ EH μήκει· δυνάμει γὰρ μόνον εἰσὶ σύμμετροι· ὡς δὲ ἡ EZ πρὸς τὴν EH , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ὑπὸ τῶν ZE, EH , ἀσύμμετρον ἄρα [ἐστὶ] τὸ ἀπὸ τῆς EZ τῷ ὑπὸ τῶν ZE, EH . ἀλλὰ τῷ μὲν ἀπὸ τῆς EZ σύμμετρον ἐστὶ τὸ ἀπὸ τῆς $ΓΔ$. ῥηταὶ γὰρ εἰσι δυνάμει· τῷ δὲ ὑπὸ τῶν ZE, EH σύμμετρον ἐστὶ τὸ ὑπὸ τῶν $ΔΓ, ΓΒ$. ἴσα γὰρ ἐστὶ τῷ ἀπὸ τῆς A . ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $ΓΔ$ τῷ ὑπὸ τῶν $ΔΓ, ΓΒ$. ὡς δὲ τὸ ἀπὸ τῆς $ΓΔ$ πρὸς τὸ ὑπὸ τῶν $ΔΓ, ΓΒ$, οὕτως ἐστὶν ἡ $ΔΓ$ πρὸς τὴν $ΓΒ$. ἀσύμμετρος ἄρα ἐστὶν ἡ $ΔΓ$ τῇ $ΓΒ$ μήκει. ῥητὴ ἄρα ἐστὶν ἡ $ΓΔ$ καὶ ἀσύμμετρος τῇ $ΓΒ$ μήκει· ὅπερ εἶδει δεῖξαι.

For since A is medial, the square on it is equal to a (rectangular) area contained by rational (straight-lines which are) commensurable in square only [Prop. 10.21]. Let the square on (A) be equal to GF . And the square on (A) is also equal to BD . Thus, BD is equal to GF . And (BD) is also equiangular with (GF) . And for equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional [Prop. 6.14]. Thus, proportionally, as BC is to EG , so EF (is) to CD . And, also, as the (square) on BC is to the (square) on EG , so the (square) on EF (is) to the (square) on CD [Prop. 6.22]. And the (square) on CB is commensurable with the (square) on EG . For they are each rational. Thus, the (square) on EF is also commensurable with the (square) on CD [Prop. 10.11]. And the (square) on EF is rational. Thus, the (square) on CD is also rational [Def. 10.4]. Thus, CD is rational. And since EF is incommensurable in length with EG . For they are commensurable in square only. And as EF (is) to EG , so the (square) on EF (is) to the (rectangle contained) by FE and EG [see previous lemma]. The (square) on EF [is] thus incommensurable with the (rectangle contained) by FE and EG [Prop. 10.11]. But, the (square) on CD is commensurable with the (square) on EF . For they are rational in square. And the (rectangle contained) by DC and CB is commensurable with the (rectangle contained) by FE and EG . For they are (both) equal to the (square) on A . Thus, the (square) on CD is also incommensurable with the (rectangle contained) by DC and CB [Prop. 10.13]. And as the (square) on CD (is) to the (rectangle contained) by DC and CB , so DC is to CB [see previous lemma]. Thus, DC is incommensurable in length with

CB [Prop. 10.11]. Thus, CD is rational, and incommensurable in length with CB . (Which is) the very thing it was required to show.

[†] Literally, “rational”.

23.

Ἡ τῇ μέσῃ σύμμετρος μέση ἐστίν.

Ἐστω μέση ἡ A , καὶ τῇ A σύμμετρος ἔστω ἡ B . λέγω, ὅτι καὶ ἡ B μέση ἐστίν.

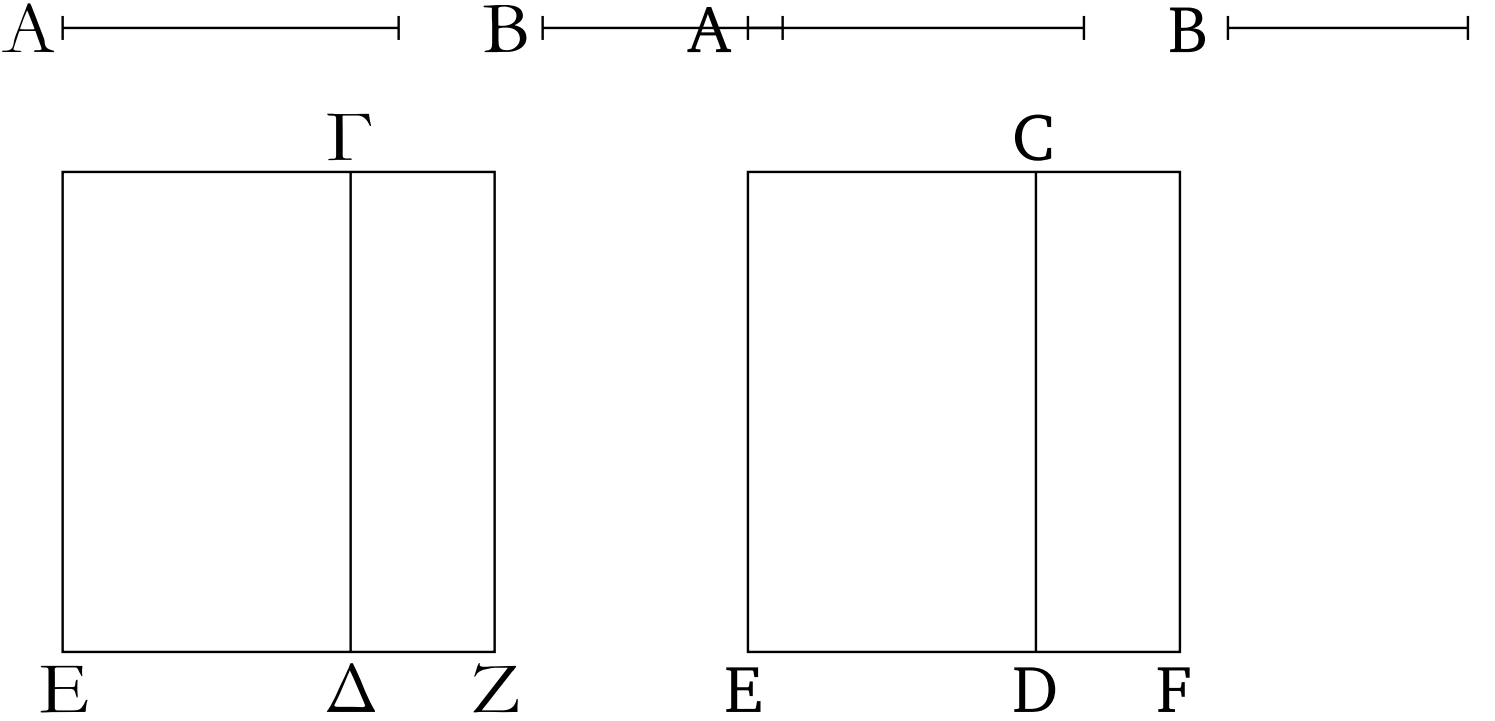
Ἐκκείσθω γὰρ ῥητὴ ἡ $\Gamma\Delta$, καὶ τῷ μὲν ἀπὸ τῆς A ἴσον παρὰ τὴν $\Gamma\Delta$ παραβελήσθω θωρίον ὀρθογώνιον τὸ ΓE πλάτος ποιοῦν τὴν $E\Delta$. ῥητὴ ἄρα ἐστὶν ἡ $E\Delta$ καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. τῷ δὲ ἀπὸ τῆς B ἴσον παρὰ τὴν $\Gamma\Delta$ παραβελήσθω θωρίον ὀρθογώνιον τὸ ΓZ πλάτος ποιοῦν τὴν ΔZ . ἐπεὶ οὖν σύμμετρος ἐστὶν ἡ A τῇ B , σύμμετρον ἐστὶ καὶ τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς B . ἀλλὰ τῷ μὲν ἀπὸ τῆς A ἴσον ἐστὶ τὸ $E\Gamma$, τῷ δὲ ἀπὸ τῆς B ἴσον ἐστὶ τὸ ΓZ . σύμμετρον ἄρα ἐστὶ τὸ $E\Gamma$ τῷ ΓZ . καὶ ἐστὶν ὥς τὸ $E\Gamma$ πρὸς τὸ ΓZ , οὕτως ἡ $E\Delta$ πρὸς τὴν ΔZ . σύμμετρος ἄρα ἐστὶν ἡ $E\Delta$ τῇ ΔZ μήκει. ῥητὴ δὲ ἐστὶν ἡ $E\Delta$ καὶ ἀσύμμετρος τῇ ΔZ μήκει. ῥητὴ ἄρα ἐστὶ καὶ ἡ ΔZ καὶ ἀσύμμετρος τῇ $\Delta\Gamma$ μήκει· αἱ $\Gamma\Delta$, ΔZ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἡ δὲ τὸ ὑπὸ ῥητῶν δυνάμει μόνον συμμέτρων δυνάμενη μέση ἐστίν. ἡ ἄρα τὸ ὑπὸ τῶν $\Gamma\Delta$, ΔZ δυνάμενη μέση ἐστίν· καὶ δύναται τὸ ὑπὸ τῶν $\Gamma\Delta$, ΔZ ἡ B μέση ἄρα ἐστὶν ἡ B .

Proposition 23

A (straight-line) commensurable with a medial (straight-line) is medial.

Let A be a medial (straight-line), and let B be commensurable with A . I say that B is also a medial (straight-line).

Let the rational (straight-line) CD be set out, and let the rectangular area CE , equal to the (square) on A , be applied to CD , producing ED as width. ED is thus rational, and incommensurable in length with CD [Prop. 10.22]. And let the rectangular area CF , equal to the (square) on B , be applied to CD , producing DF as width. Therefore, since A is commensurable with B , the (square) on A is also commensurable with the (square) on B . But, EC is equal to the (square) on A , and CF is equal to the (square) on B . Thus, EC is commensurable with CF . And as EC is to CF , so ED (is) to DF [Prop. 6.1]. Thus, ED is commensurable in length with DF [Prop. 10.11]. And ED is rational, and incommensurable in length with CD . DF is thus also rational [Def. 10.3], and incommensurable in length with DC [Prop. 10.13]. Thus, CD and DF are rational, and commensurable in square only. And the square-root of a (rectangle contained) by rational (straight-lines which are) commensurable in square only is medial [Prop. 10.21]. Thus, the square-root of the (rectangle contained) by CD and DF is medial. And the square on B is equal to the (rectangle contained) by CD and DF . Thus, B is a medial (straight-line).



Πόρισμα.

Corollary

Ἐκ δὴ τούτων φανερόν, ὅτι τὸ τῶ μέσῳ θεωρῶ σύμμετρον μέσον ἐστίν.

And (it is) clear, from this, that an (area) commensurable with a medial area[†] is medial.

[†] A medial area is equal to the square on some medial straight-line. Hence, a medial area is expressible as $k^{1/2}$.

24.

Proposition 24

Τὸ ὑπὸ μέσων μήκει συμμετρων ἐνθειῶν περιεχόμενον ὀρθογώνιον μέσον ἐστίν.

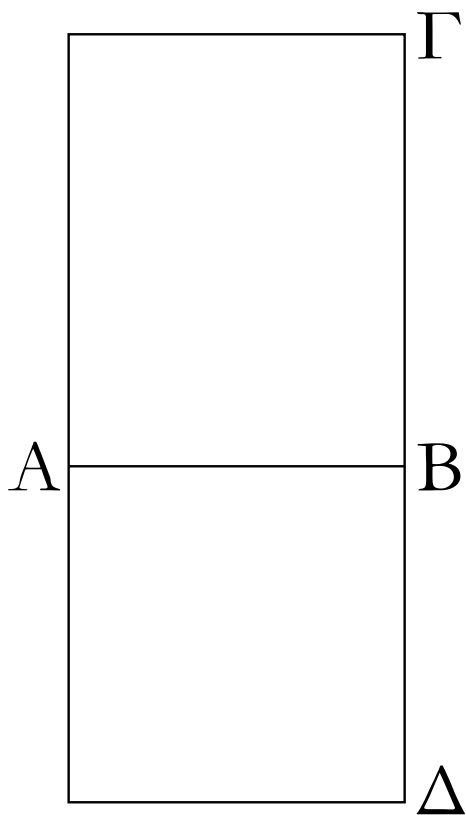
A rectangle contained by medial straight-lines (which are) commensurable in length is medial.

Ὑπὸ γὰρ μέσων μήκει συμμετρων ἐνθειῶν τῶν AB , $BΓ$ περιεθέσθω ὀρθογώνιον τὸ $ΑΓ$. λέγω, ὅτι τὸ $ΑΓ$ μέσον ἐστίν.

For let the rectangle AC be contained by the medial straight-lines AB and BC (which are) commensurable in length. I say that AC is medial.

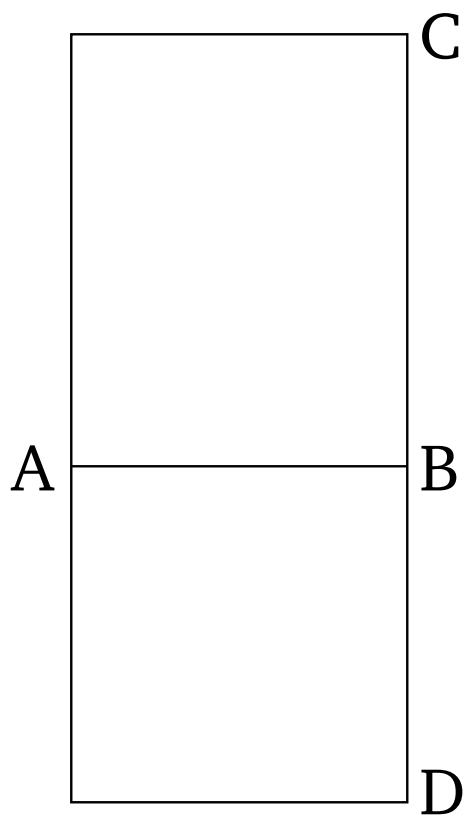
Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $ΑΔ$. μέσον ἄρα ἐστὶ τὸ $ΑΔ$. καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει, ἴση δὲ ἡ AB τῇ $BΔ$, σύμμετρος ἄρα ἐστὶ καὶ ἡ $ΔB$ τῇ $BΓ$ μήκει· ὥστε καὶ τὸ $ΔA$ τῶ $ΑΓ$ σύμμετρον ἐστίν. μέσον δὲ τὸ $ΔA$. μέσον ἄρα καὶ τὸ $ΑΓ$. ὅπερ ἔδει δεῖξαι.

For let the square AD be described on AB . AD is thus medial [see previous footnote]. And since AB is commensurable in length with BC , and AB (is) equal to BD , DB is thus also commensurable in length with BC . Hence, DA is also commensurable with AC [Props. 6.1, 10.11]. And DA (is) medial. Thus, AC (is) also medial [Prop. 10.23 corr.]. (Which is) the very thing it was required to show.



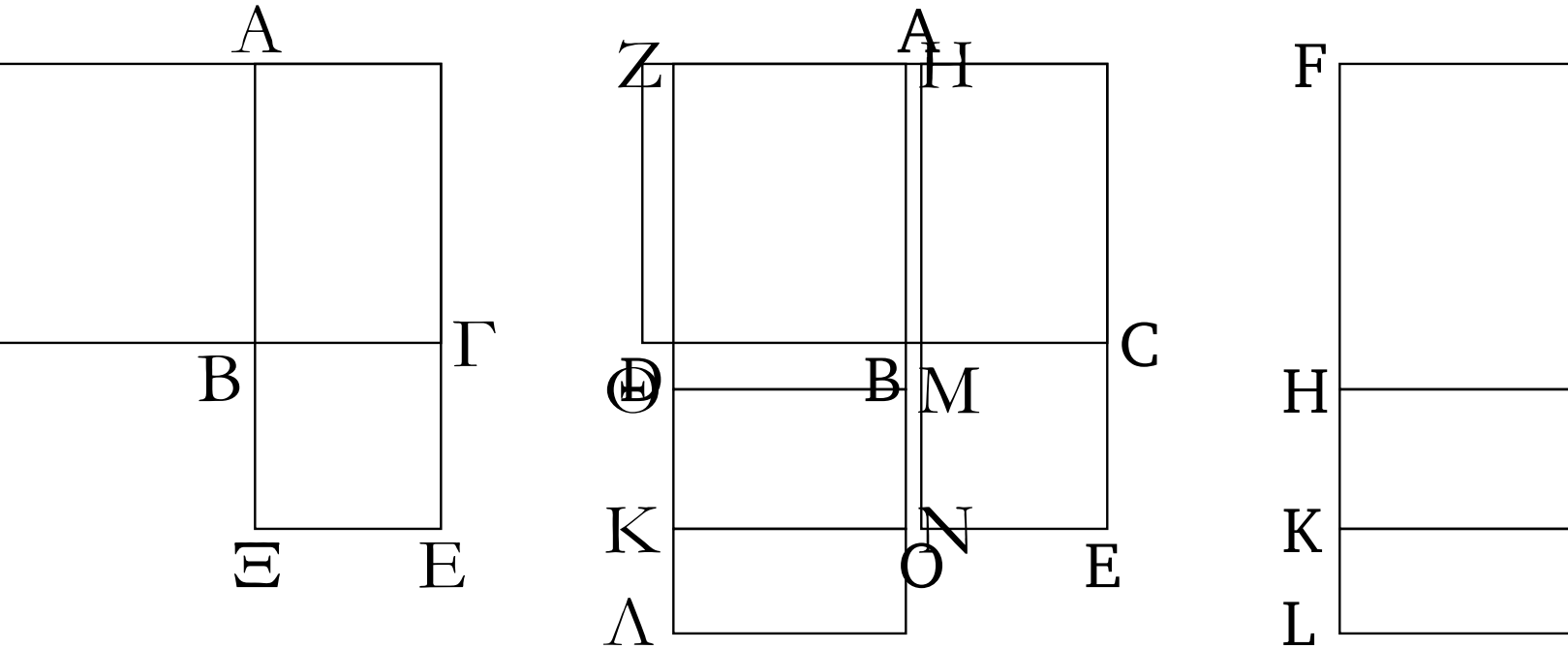
25.

*Τὸ ὑπὸ μέσων δυνάμει μόνον συμμέτρων εὐθειῶν περιε-
χόμενον ὀρθογώνιον ἤτοι ῥητὸν ἢ μέσον ἐστίν.*



Proposition 25

The rectangle contained by medial straight-lines (which are) commensurable in square only is either rational or medial.



Ὑπὸ γὰρ μέσων δυνάμει μόνον συμμετρων εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιεθέσθω τὸ $ΑΓ$. λέγω, ὅτι τὸ $ΑΓ$ ἥτοι ῥητὸν ἢ μέσον ἐστίν.

Ἀναγεγράφθω γὰρ ἀπὸ τῶν AB , $BΓ$ τετράγωνα τὰ $ΑΔ$, BE . μέσον ἄρα ἐστὶν ἑκάτερον τῶν $ΑΔ$, BE . καὶ ἐκκείσθω ῥητὴ ἡ ZH , καὶ τῷ μὲν $ΑΔ$ ἴσον παρὰ τὴν ZH παραβεβλήσθω ὀρθογώνιον παραλληλόγραμμον τὸ $HΘ$ πλάτος ποιῶν τὴν $ZΘ$, τῷ δὲ $ΑΓ$ ἴσον παρὰ τὴν $ΘM$ παραβεβλήσθω ὀρθογώνιον παραλληλόγραμμον τὸ MK πλάτος ποιῶν τὴν $ΘK$, καὶ ἔτι τῷ BE ἴσον ὁμοίως παρὰ τὴν KN παραβεβλήσθω τὸ $NΛ$ πλάτος ποιῶν τὴν KL . ἐπ' εὐθείᾳς ἄρα εἰσὶν αἱ $ZΘ$, $ΘK$, KL . ἐπεὶ οὖν μέσον ἐστὶν ἑκάτερον τῶν $ΑΔ$, BE , καὶ ἐστὶν ἴσον τὸ μὲν $ΑΔ$ τῷ $HΘ$, τὸ δὲ BE τῷ $NΛ$, μέσον ἄρα καὶ ἑκάτερον τῶν $HΘ$, $NΛ$. καὶ παρὰ ῥητὴν τὴν ZH παράκειται ῥητὴ ἄρα ἐστὶν ἑκάτερα τῶν $ZΘ$, KL καὶ ἀσύμμετρος τῇ ZH μήκει. καὶ ἐπεὶ σύμμετρόν ἐστι τὸ $ΑΔ$ τῷ BE , σύμμετρον ἄρα ἐστὶ καὶ τὸ $HΘ$ τῷ $NΛ$. καὶ ἐστὶν ὥς τὸ $HΘ$ πρὸς τὸ $NΛ$, οὕτως ἡ $ZΘ$ πρὸς τὴν KL . σύμμετρος ἄρα ἐστὶν ἡ $ZΘ$ τῇ KL μήκει. αἱ $ZΘ$, KL ἄρα ῥηταὶ εἰσι μήκει σύμμετροι. ῥητὸν ἄρα ἐστὶ τὸ ὑπὸ τῶν $ZΘ$, KL . καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν $ΔB$ τῇ BA , ἡ δὲ XB τῇ $BΓ$, ἔστιν ἄρα ὥς ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως ἡ AB πρὸς τὴν BX . ἀλλ' ὥς μὲν ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως τὸ $ΔA$ πρὸς τὸ $ΑΓ$. ὥς δὲ ἡ AB πρὸς τὴν BX , οὕτως τὸ $ΑΓ$ πρὸς τὸ $ΓX$. ἔστιν ἄρα ὥς τὸ $ΔA$ πρὸς τὸ $ΑΓ$, οὕτως τὸ $ΑΓ$ πρὸς τὸ $ΓX$. ἴσον δὲ ἐστὶ τὸ μὲν $ΑΔ$ τῷ $HΘ$, τὸ δὲ $ΑΓ$ τῷ MK , τὸ δὲ $ΓX$ τῷ $NΛ$. ἔστιν ἄρα ὥς τὸ $HΘ$ πρὸς τὸ MK , οὕτως τὸ MK πρὸς τὸ $NΛ$. ἔστιν ἄρα καὶ ὥς ἡ $ZΘ$ πρὸς τὴν $ΘK$, οὕτως ἡ $ΘK$ πρὸς τὴν KL . τὸ ἄρα ὑπὸ τῶν $ZΘ$, KL ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΘK$. ῥητὸν δὲ τὸ ὑπὸ τῶν $ZΘ$, KL . ῥητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $ΘK$. ῥητὴ ἄρα

For let the rectangle AC be contained by the medial straight-lines AB and BC (which are) commensurable in square only. I say that AC is either rational or medial.

For let the squares AD and BE be described on (the straight-lines) AB and BC (respectively). AD and BE are thus each medial. And let the rational (straight-line) FG be laid out. And let the rectangular parallelogram GH , equal to AD , be applied to FG , producing FH as breadth. And let the rectangular parallelogram MK , equal to AC , be applied to HM , producing HK as breadth. And, finally, let NL , equal to BE , have similarly been applied to KN , producing KL as breadth. Thus, FH , HK , and KL are in a straight-line. Therefore, since AD and BE are each medial, and AD is equal to GH , and BE to NL , GH and NL (are) thus each also medial. And they are applied to the rational (straight-line) FG . FH and KL are thus each rational, and incommensurable in length with FG [Prop. 10.22]. And since AD is commensurable with BE , GH is thus also commensurable with NL . And as GH is to NL , so FH (is) to KL [Prop. 6.1]. Thus, FH is commensurable in length with KL [Prop. 10.11]. Thus, FH and KL are rational (straight-lines which are) commensurable in length. Thus, the (rectangle contained) by FH and KL is rational [Prop. 10.19]. And since DB is equal to BA , and OB to BC , thus as DB is to BC , so AB (is) to BO . But, as DB (is) to BC , so DA (is) to AC [Props. 6.1]. And as AB (is) to BO , so AC (is) to CO [Prop. 6.1]. Thus, as DA is to AC , so AC (is) to CO . And AD is equal to GH , and AC to MK , and CO to NL . Thus, as GH is to MK , so MK (is) to NL . Thus, also, as FH is to HK , so HK (is) to KL [Props. 6.1, 5.11]. Thus, the (rectangle contained) by FH and KL is equal to the (square) on HK

ἐστὶν ἡ ΘK . καὶ εἰ μὲν σύμμετρος ἐστί τῇ ZH μήκει, ῥητόν ἐστι τὸ ΘN . εἰ δὲ ἀσύμμετρος ἐστί τῇ ZH μήκει, αἱ $K\Theta$, ΘM ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα τὸ ΘN . τὸ ΘN ἄρα ἥτοι ῥητόν ἢ μέσον ἐστίν. ἴσον δὲ τὸ ΘN τῷ AG . τὸ AG ἄρα ἥτοι ῥητόν ἢ μέσον ἐστίν.

Τὸ ἄρα ὑπὸ μέσων δυνάμει μόνον συμμέτρων, καὶ τὰ ἐχῆξ.

[Prop. 6.17]. And the (rectangle contained) by FH and KL (is) rational. Thus, the (square) on HK is also rational. Thus, HK is rational. And if it is commensurable in length with FG then HN is rational [Prop. 10.19]. And if it is incommensurable in length with FG then KH and HM are rational (straight-lines which are) commensurable in square only: thus, HN is medial [Prop. 10.21]. Thus, HN is either rational or medial. And HN (is) equal to AC . Thus, AC is either rational or medial.

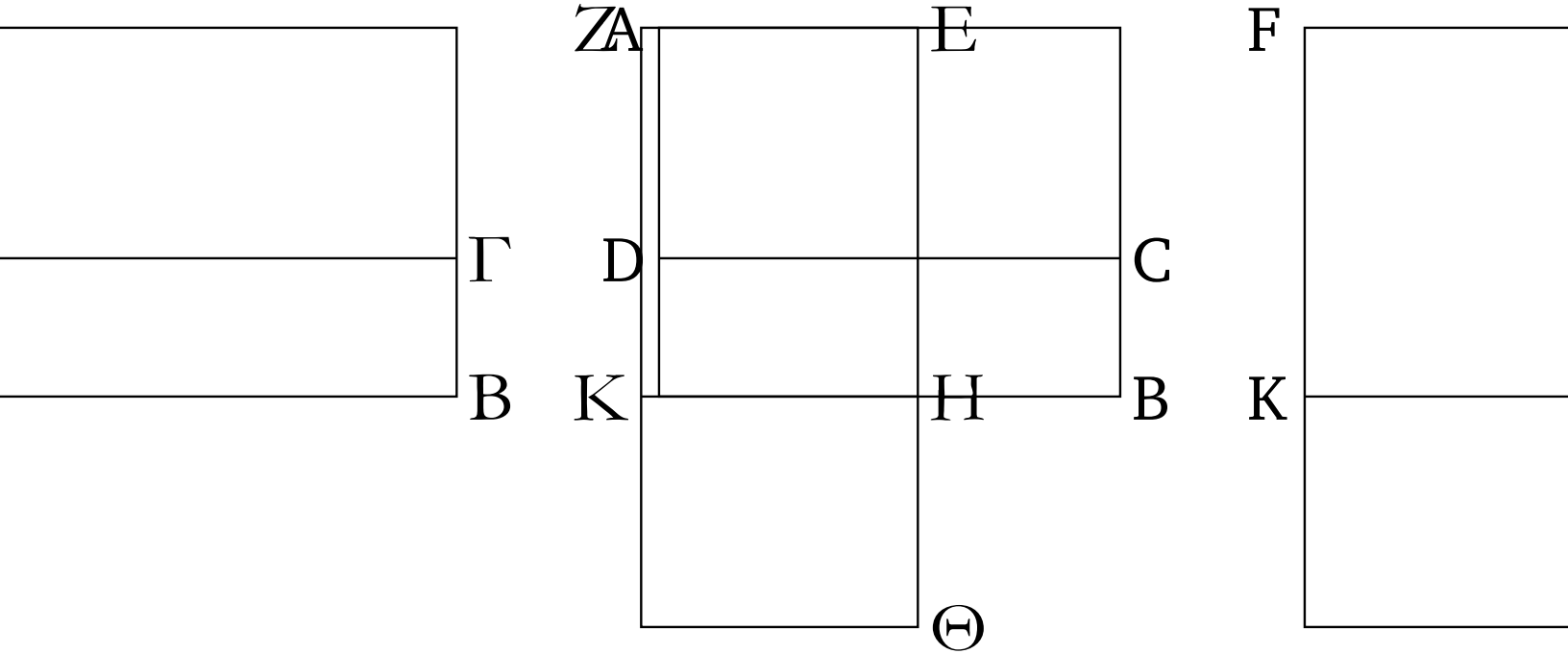
Thus, the ... by medial straight-lines (which are) commensurable in square only, and so on

26.

Μέσον μέσον οὐδ' ὑπερέχει ῥητῶ.

Proposition 26

A medial (area) does not exceed a medial (area) by a rational (area).[†]



Εἰ γὰρ δυνατόν, μέσον τὸ AB μέσον τοῦ AG ὑπερεθέτω ῥητῶ τῷ ΔB , καὶ ἐκκείσθω ῥητὴ ἡ EZ , καὶ τῷ AB ἴσον παρὰ τὴν EZ παραβελήσθω παραλληλόγραμμον ὀρθογώνιον τὸ $Z\Theta$ πλάτος ποιῶν τὴν $E\Theta$, τῷ δὲ AG ἴσον ἀφηγήσθω τὸ ZH . λοιπὸν ἄρα τὸ $B\Delta$ λοιπῶ τῷ $K\Theta$ ἐστὶν ἴσον. ῥητόν δὲ ἐστί τὸ ΔB . ῥητόν ἄρα ἐστὶ καὶ τὸ $K\Theta$. ἐπεὶ οὖν μέσον ἐστὶν ἐκάτερον τῶν AB , AG , καὶ ἐστί τὸ μὲν AB τῷ $Z\Theta$ ἴσον, τὸ δὲ AG τῷ ZH , μέσον ἄρα καὶ ἐκάτερον τῶν $Z\Theta$, ZH . καὶ παρὰ ῥητὴν τὴν EZ παράκειται ῥητὴ ἄρα ἐστὶν ἐκάτερα τῶν ΘE , EH καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ῥητόν ἐστί τὸ ΔB καὶ ἐστὶν ἴσον τῷ $K\Theta$, ῥητόν ἄρα ἐστὶ καὶ τὸ $K\Theta$. καὶ παρὰ ῥητὴν τὴν EZ παράκειται ῥητὴ ἄρα ἐστὶν ἡ $H\Theta$ καὶ σύμμετρος τῇ EZ μήκει. ἀλλὰ καὶ ἡ EH ῥητὴ ἐστί καὶ ἀσύμμετρος τῇ EZ μήκει· ἀσύμμετρος ἄρα ἐστὶν ἡ EH τῇ $H\Theta$ μήκει. καὶ ἐστὶν ὥς ἡ EH πρὸς τὴν $H\Theta$, οὕτως τὸ ἀπὸ

For, if possible, let the medial (area) AB exceed the medial (area) AC by the rational (area) DB . And let the rational (straight-line) EF be laid down. And let the rectangular parallelogram FH , equal to AB , be applied to EF , producing EH as breadth. And let FG , equal to AC , be cut off (from FH). Thus, the remainder BD is equal to the remainder KH . And DB is rational. Thus, KH is also rational. Therefore, since AB and AC are each medial, and AB is equal to FH , and AC to FG , FH and FG are thus each also medial. And they are applied to the rational (straight-line) EF . Thus, HE and EG are each rational, and incommensurable in length with EF [Prop. 10.22]. And since DB is rational, and is equal to KH , KH is thus also rational. And (KH) is applied to the rational (straight-line) EF . GH is thus rational, and commensurable in length with EF [Prop. 10.20]. But, EG is also rational, and

τῆς EH πρὸς τὸ ὑπὸ τῶν $EH, H\Theta$ ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς EH τῷ ὑπὸ τῶν $EH, H\Theta$. ἀλλὰ τῷ μὲν ἀπὸ τῆς EH σύμμετρόν ἐστι τὰ ἀπὸ τῶν $EH, H\Theta$ τετράγωνα· ῥητὰ γὰρ ἀμφοτέρω· τῷ δὲ ὑπὸ τῶν $EH, H\Theta$ σύμμετρόν ἐστι τὸ διξ ὑπὸ τῶν $EH, H\Theta$ διπλάσιον γὰρ ἐστὶν αὐτοῦ· ἀσύμμετρον ἄρα ἐστὶ τὰ ἀπὸ τῶν $EH, H\Theta$ τῷ διξ ὑπὸ τῶν $EH, H\Theta$ καὶ συναμφοτέρω ἄρα τὰ τε ἀπὸ τῶν $EH, H\Theta$ καὶ τὸ διξ ὑπὸ τῶν $EH, H\Theta$, ὅπερ ἐστὶ τὸ ἀπὸ τῆς $E\Theta$, ἀσύμμετρόν ἐστι τοῖς ἀπὸ τῶν $EH, H\Theta$. ῥητὰ δὲ τὰ ἀπὸ τῶν $EH, H\Theta$ ἄλογον ἄρα τὸ ἀπὸ τῆς $E\Theta$. ἄλογοξ ἄρα ἐστὶν ἡ $E\Theta$. ἀλλὰ καὶ ῥηρή· ὅπερ ἐστὶν ἀδύνατον.

Μέσον ἄρα μέσον οὐθ' ὑπερέθει ῥητῷ· ὅπερ εἶδει δεῖχαι.

incommensurable in length with EF . Thus, EG is incommensurable in length with GH [Prop. 10.13]. And as EG is to GH , so the (square) on EG (is) to the (rectangle contained) by EG and GH [Prop. 10.13 lem.]. Thus, the (square) on EG is incommensurable with the (rectangle contained) by EG and GH [Prop. 10.11]. But, the (sum of the) squares on EG and GH is commensurable with the (square) on EG . For (EG and GH are) both rational. And twice the (rectangle contained) by EG and GH is commensurable with the (rectangle contained) by EG and GH [Prop. 10.6]. For (the former) is double the latter. Thus, the (sum of the squares) on EG and GH is incommensurable with twice the (rectangle contained) by EG and GH [Prop. 10.13]. And thus the sum of the (squares) on EG and GH plus twice the (rectangle contained) by EG and GH , that is the (square) on EH [Prop. 2.4], is incommensurable with the (sum of the squares) on EG and GH [Prop. 10.16]. And the (sum of the squares) on EG and GH (is) rational. Thus, the (square) on EH is irrational [Def. 10.4]. Thus, EH is irrational [Def. 10.4]. But, (it is) also rational. The very thing is impossible.

Thus, a medial (area) does not exceed a medial (area) by a rational (area). (Which is) the very thing it was required to show.

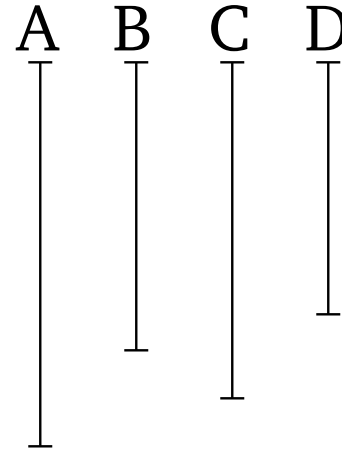
[†] In other words, $\sqrt{k} - \sqrt{k'} \neq k''$.

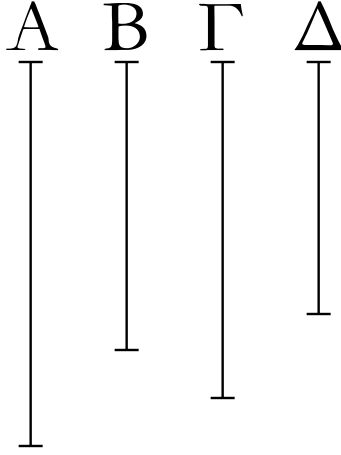
27.

Μέσαξ εὐρεῖν δυνάμει μόνον συμμέτρους ῥητὸν περιεθούσαξ.

Proposition 27

To find (two) medial (straight-lines), containing a rational (area), (which are) commensurable in square only.





Ἐκκεισθῶσαν δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ A , B , καὶ εἰλήφθω τῶν A , B μέση ἀνάλογον ἡ Γ , καὶ γερονέτω ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ .

Καὶ ἐπεὶ αἱ A , B ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, τὸ ἄρα ὑπὸ τῶν A , B , τουτέστι τὸ ἀπὸ τῆς Γ , μέσον ἐστίν. μέση ἄρα ἡ Γ . καὶ ἐπεὶ ἐστὶν ὥς ἡ A πρὸς τὴν B , [οὕτως] ἡ Γ πρὸς τὴν Δ , αἱ δὲ A , B δυνάμει μόνον [εἰσὶ] σύμμετροι, καὶ αἱ Γ , Δ ἄρα δυνάμει μόνον εἰσι σύμμετροι. καὶ ἐστὶ μέση ἡ Γ μέση ἄρα καὶ ἡ Δ . αἱ Γ , Δ ἄρα μέσαι εἰσι δυνάμει μόνον σύμμετροι. λέγω, ὅτι καὶ ῥητὸν περιέθουσιν. ἐπεὶ γάρ ἐστὶν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ , ἐναλλάξ ἄρα ἐστὶν ὥς ἡ A πρὸς τὴν Γ , ἡ B πρὸς τὴν Δ . ἀλλ' ὥς ἡ A πρὸς τὴν Γ , ἡ Γ πρὸς τὴν B . καὶ ὥς ἄρα ἡ Γ πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Δ . τὸ ἄρα ὑπὸ τῶν Γ , Δ ἴσον ἐστὶ τῷ ἀπὸ τῆς B . ῥητὸν δὲ τὸ ἀπὸ τῆς B . ῥητὸν ἄρα [ἐστὶ] καὶ τὸ ὑπὸ τῶν Γ , Δ .

Εὐρηνται ἄρα μέσαι δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι· ὅπερ ἔδει δεῖχαι.

Let the two rational (straight-lines) A and B , (which are) commensurable in square only, be laid down. And let C —the mean proportional (straight-line) to A and B —be taken [Prop. 6.13]. And let it be contrived that as A (is) to B , so C (is) to D [Prop. 6.12].

And since the rational (straight-lines) A and B are commensurable in square only, the (rectangle contained) by A and B —that is to say, the (square) on C [Prop. 6.17]—is thus medial [Prop. 10.21]. Thus, C is medial [Prop. 10.21]. And since as A is to B , [so] C (is) to D , and A and B [are] commensurable in square only, C and D are thus also commensurable in square only [Prop. 10.11]. And C is medial. Thus, D is also medial [Prop. 10.23]. Thus, C and D are medial (straight-lines which are) commensurable in square only. I say that they also contain a rational (area). For since as A is to B , so C (is) to D , thus, alternately, as A is to C , so B (is) to D [Prop. 5.16]. But, as A (is) to C , (so) C (is) to B . And thus as C (is) to B , so B (is) to D [Prop. 5.11]. Thus, the (rectangle contained) by C and D is equal to the (square) on B [Prop. 6.17]. And the (square) on B (is) rational. Thus, the (rectangle contained) by C and D [is] also rational.

Thus, (two) medial (straight-lines, C and D), containing a rational (area), (which are) commensurable in square only, be found.[†] (Which is) the very thing it was required to show.

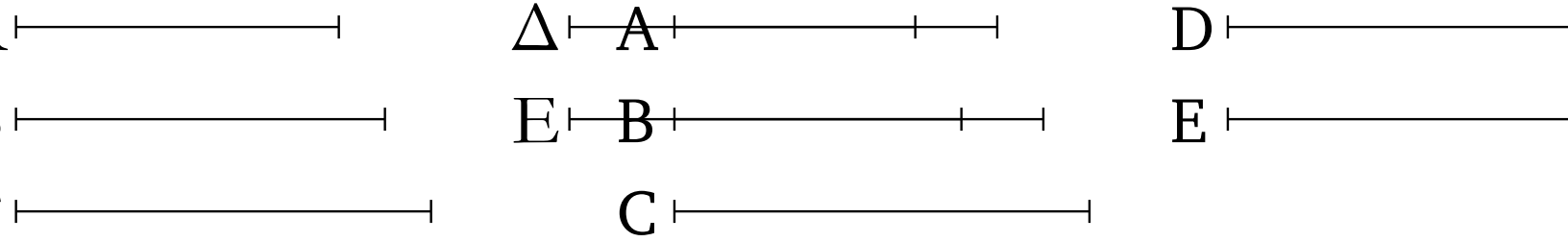
[†] C and D have lengths $k^{1/4}$ and $k^{3/4}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A .

28.

Proposition 28

Μέσαξ εὐρεῖν δυνάμει μόνον συμμέτρον μέσον περιεθούσαξ.

To find (two) medial (straight-lines), containing a medial (area), (which are) commensurable in square only.



Ἐκκεισθώσαν [τρεῖς] ῥηταὶ δυνάμει μόνον σύμμετροι αἱ A , B , Γ , καὶ εἰλήφθω τῶν A , B μέση ἀνάλογον ἡ Δ , καὶ γερονέτω ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E .

Ἐπεὶ αἱ A , B ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, τὸ ἄρα ὑπὸ τῶν A , B , τουτέστι τὸ ἀπὸ τῆς Δ , μέσον ἐστίν. μέση ἄρα ἡ Δ . καὶ ἐπεὶ αἱ B , Γ δυνάμει μόνον εἰσι σύμμετροι, καὶ ἐστὶν ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E , καὶ αἱ Δ , E ἄρα δυνάμει μόνον εἰσι σύμμετροι. μέση δὲ ἡ Δ · μέση ἄρα καὶ ἡ E . αἱ Δ , E ἄρα μέσαι εἰσι δυνάμει μόνον σύμμετροι. λέγω δὴ, ὅτι καὶ μέσον περιέθουσιν. ἐπεὶ γάρ ἐστὶν ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E , ἐναλλάξ ἄρα ὥς ἡ B πρὸς τὴν Δ , ἡ Γ πρὸς τὴν E . ὥς δὲ ἡ B πρὸς τὴν Δ , ἡ Δ πρὸς τὴν A · καὶ ὥς ἄρα ἡ Δ πρὸς τὴν A , ἡ Γ πρὸς τὴν E . τὸ ἄρα ὑπὸ τῶν A , Γ ἴσον ἐστὶ τῷ ὑπὸ τῶν Δ , E . μέσον δὲ τὸ ὑπὸ τῶν A , Γ · μέσον ἄρα καὶ τὸ ὑπὸ τῶν Δ , E .

Ἐύρηγνται ἄρα μέσαι δυνάμει μόνον σύμμετροι μέσον περιέθουσαι· ὅπερ ἔδει δεῖχαι.

Let the [three] rational (straight-lines) A , B , and C , (which are) commensurable in square only, be laid down. And let, D , the mean proportional (straight-line) to A and B , be taken [Prop. 6.13]. And let it be contrived that as B (is) to C , (so) D (is) to E [Prop. 6.12].

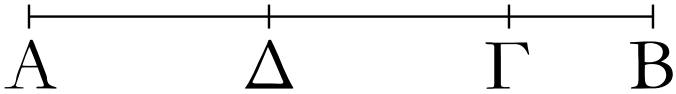
Since the rational (straight-lines) A and B are commensurable in square only, the (rectangle contained) by A and B —that is to say, the (square) on D [Prop. 6.17]—is medial [Prop. 10.21]. Thus, D (is) medial [Prop. 10.21]. And since B and C are commensurable in square only, and as B is to C , (so) D (is) to E , D and E are thus commensurable in square only [Prop. 10.11]. And D (is) medial. E (is) thus also medial [Prop. 10.23]. Thus, D and E are medial (straight-lines which are) commensurable in square only. So, I say that they also enclose a medial (area). For since as B is to C , (so) D (is) to E , thus, alternately, as B (is) to D , (so) C (is) to E [Prop. 5.16]. And as B (is) to D , (so) D (is) to A . And thus as D (is) to A , (so) C (is) to E . Thus, the (rectangle contained) by A and C is equal to the (rectangle contained) by D and E [Prop. 6.16]. And the (rectangle contained) by A and C is medial [Prop. 10.21]. Thus, the (rectangle contained) by D and E (is) also medial.

Thus, (two) medial (straight-lines, D and E), containing a medial (area), (which are) commensurable in square only, be found. (Which is) the very thing it was required to show.

[†] D and E have lengths $k^{1/4}$ and $k'^{1/2}/k^{1/4}$ times that of A , respectively, where the lengths of B and C are $k^{1/2}$ and $k'^{1/2}$ times that of A , respectively.

Λήμμα 1.

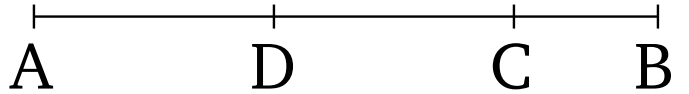
Εύρειν δύο τετραγώνους ἀριθμούς, ὥστε καὶ τὸν συγκεκριμένον ἔχ αὐτῶν εἶναι τετράγωνον.



Ἐκκεισθώσαν δύο ἀριθμοὶ οἱ AB , $B\Gamma$, ἔστωσαν δὲ ἦτοι ἄρτιοι ἢ περιττοί. καὶ ἐπεὶ, ἔαν τε ἀπὸ ἀρτίου ἄρτιος ἀφαιρεθῇ, ἔαν τε ἀπὸ περισσοῦ περισσόξ, ὁ λοιπὸξ ἄρτιός ἐστιν, ὁ λοιπὸξ ἄρα ὁ $A\Gamma$ ἄρτιός ἐστιν. τεμῆσθω ὁ $A\Gamma$ διὰ κατὰ τὸ Δ . ἔστωσαν δὲ καὶ οἱ AB , $B\Gamma$ ἦτοι ὅμοιοι ἐπίπεδοι ἢ τετράγωνοι, οἳ καὶ αὐτοὶ ὅμοιοι εἰσιν ἐπίπεδοι· ὁ ἄρα ἐκ τῶν AB , $B\Gamma$ μετὰ τοῦ ἀπὸ [τοῦ] $\Gamma\Delta$ τετραγώνου ἴσοξ ἐστὶ τῷ ἀπὸ τοῦ $B\Delta$ τετραγώνῳ. καὶ ἐστὶ τετράγωνος ὁ ἐκ τῶν AB , $B\Gamma$, ἐπειδήπερ ἐδείχθη, ὅτι, ἔαν δύο ὅμοιοι ἐπίπεδοι πολλαπλασι-

Lemma I

To find two square numbers such that the sum of them is also square.



Let the two numbers AB and BC be laid down. And let them be either (both) even or (both) odd. And since, if an even (number) is subtracted from an even (number), or if an odd (number is subtracted) from an odd (number), then the remainder is even [Props. 9.24, 9.26], the remainder AC is thus even. Let AC be cut in half at D . And let AB and BC also be either similar plane (numbers), or square (numbers)—which are themselves also similar plane (numbers). Thus, the (number created) from (multiplying) AB and BC , plus the square on

άσαντες ἀλλήλους ποιῶσι τινα, ὁ γεγόμενος τετράγωνός ἐστιν. ἐύρηται ἄρα δύο τετράγωνοι ἀριθμοὶ ὅ τε ἐκ τῶν AB , BF καὶ ὁ ἀπὸ τοῦ $\Gamma\Delta$, οἱ συντεθέντες ποιοῦσι τὸν ἀπὸ τοῦ $B\Delta$ τετράγωνον.

Καὶ φανερόν, ὅτι ἐύρηται πάλιν δύο τετράγωνοι ὅ τε ἀπὸ τοῦ $B\Delta$ καὶ ὁ ἀπὸ τοῦ $\Gamma\Delta$, ὥστε τὴν ὑπεροθὴν αὐτῶν τὸν ὑπὸ AB , BF εἶναι τετράγωνον, ὅταν οἱ AB , BF ὅμοιοι ᾖσιν ἐπίπεδοι. ὅταν δὲ μὴ ᾖσιν ὅμοιοι ἐπίπεδοι, ἐύρηται δύο τετράγωνοι ὅ τε ἀπὸ τοῦ $B\Delta$ καὶ ὁ ἀπὸ τοῦ $\Delta\Gamma$, ὧν ἡ ὑπεροθὴ ὁ ὑπὸ τῶν AB , BF οὐκ ἔστι τετράγωνος· ὅπερ ἔδει δεῖξαι.

CD , is equal to the square on BD [Prop. 2.6]. And the (number created) from (multiplying) AB and BC is square—inasmuch as it was shown that if two similar plane (numbers) make some (number) by multiplying one another then the (number so) created is square [Prop. 9.1]. Thus, two square numbers be found—(namely,) the (number created) from (multiplying) AB and BC , and the (square) on CD —which, (when) added (together), make the square on BD .

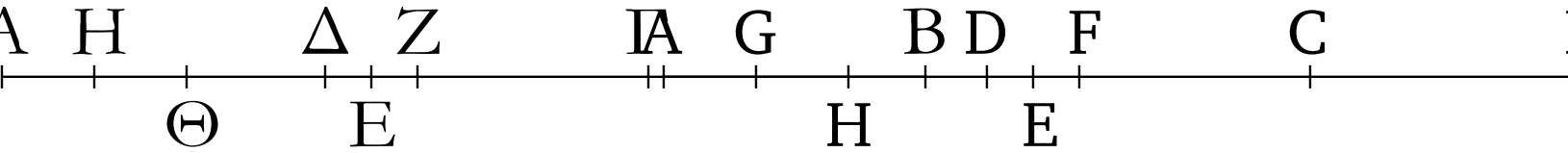
And (it is) clear that two square (numbers) have again been found—(namely,) the (square) on BD , and the (square) on CD —such that their difference—(namely,) the (rectangle) contained by AB and BC —is square whenever AB and BC are similar plane (numbers). But, when they are not similar plane numbers, two square (numbers) be found—(namely,) the (square) on BD , and the (square) on DC —between which the difference—(namely,) the (rectangle) contained by AB and BC —is not square. (Which is) the very thing it was required to show.

Λήμμα 2.

Εὐρεῖν δύο τετραγώνους ἀριθμούς, ὥστε τὸν ἐκ αὐτῶν συγκείμενον μὴ εἶναι τετράγωνον.

Lemma II

To find two square numbers such that the sum of them is not square.



Ἐστω γάρ ὁ ἐκ τῶν AB , BF , ὡς ἔφαμεν, τετράγωνος, καὶ ἄρτιος ὁ ΓA , καὶ τεμήσθω ὁ ΓA διὰ τῷ Δ . φανερόν δὲ, ὅτι ὁ ἐκ τῶν AB , BF τετράγωνος μετὰ τοῦ ἀπὸ [τοῦ] $\Gamma\Delta$ τετραγώνου ἴσος ἐστὶ τῷ ἀπὸ [τοῦ] $B\Delta$ τετραγώνῳ. ἀφηρήσθω μονὰξ ἡ ΔE . ὁ ἄρα ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ [τοῦ] ΓE ἐλάσσων ἐστὶ τοῦ ἀπὸ [τοῦ] $B\Delta$ τετραγώνου. λέγω οὖν, ὅτι ὁ ἐκ τῶν AB , BF τετράγωνος μετὰ τοῦ ἀπὸ [τοῦ] ΓE οὐκ ἔσται τετράγωνος.

Εἰ γὰρ ἔσται τετράγωνος, ἦτοι ἴσος ἐστὶ τῷ ἀπὸ [τοῦ] BE ἢ ἐλάσσων τοῦ ἀπὸ [τοῦ] BE , οὐκέτι δὲ καὶ μείζων, ἵνα μὴ τηθη ἡ μονὰξ. ἔστω, εἰ δυνατόν, πρότερον ὁ ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ ΓE ἴσος τῷ ἀπὸ BE , καὶ ἔστω τῆς ΔE μονάδος διπλασίον ὁ HA . ἐπεὶ οὖν ὁλόξ ὁ AG ὅλον τοῦ $\Gamma\Delta$ ἐστὶ διπλασίον, ὧν ὁ AH τοῦ ΔE ἐστὶ διπλασίον, καὶ λοιπὸς ἄρα ὁ HG λοιποῦ τοῦ EG ἐστὶ διπλασίον διὰ ἄρα τέμνεται ὁ HG τῷ E . ὁ ἄρα ἐκ τῶν HB , BF μετὰ τοῦ ἀπὸ ΓE ἴσος ἐστὶ τῷ ἀπὸ BE τετραγώνῳ. ἀλλὰ καὶ ὁ ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ ΓE ἴσος ὑπόκειται τῷ ἀπὸ [τοῦ] BE τετραγώνῳ· ὁ ἄρα ἐκ τῶν HB , BF μετὰ τοῦ ἀπὸ ΓE ἴσος ἐστὶ τῷ ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ ΓE . καὶ κοινοῦ ἀφαιρεθέντος τοῦ ἀπὸ ΓE συνάγεται ὁ AB ἴσος τῷ HB . ὅπερ ἄτοπον. οὐκ ἄρα ὁ ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ [τοῦ] ΓE ἴσος ἐστὶ τῷ ἀπὸ BE . λέγω δὲ, ὅτι οὐδὲ ἐλάσσων τοῦ ἀπὸ BE . εἰ γὰρ δυνατόν,

For let the (number created) from (multiplying) AB and BC , as we said, be square. And (let) CA (be) even. And let CA be cut in half at D . So it is clear that the square (number created) from (multiplying) AB and BC , plus the square on CD , is equal to the square on BD [see previous lemma]. Let the unit DE be subtracted (from BD). Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is less than the square on BD . I say, therefore, that the square (number created) from (multiplying) AB and BC , plus the (square) on CE , is not square.

For if it is square, it is either equal to the (square) on BE , or less than the (square) on BE , but cannot any more be greater (than the square on BE), lest the unit be divided. First of all, if possible, let the (number created) from (multiplying) AB and BC , plus the (square) on CE , be equal to the (square) on BE . And let GA be double the unit DE . Therefore, since the whole of AC is double the whole of CD , of which AG is double DE , the remainder GC is thus double the remainder EC . Thus, GC has been cut in half at E . Thus, the (number created) from (multiplying) GB and BC , plus the (square) on CE , is equal to the square on BE [Prop. 2.6]. But, the (number created) from (multiplying) AB and BC , plus the (square) on CE , was also assumed (to be) equal to the square on BE . Thus, the (number

ἔστω τῷ ἀπὸ BZ ἴσος, καὶ τοῦ ΔZ διπλασίων ὁ ΘA . καὶ συναθροίσεται πάλιν διπλασίων ὁ $\Theta \Gamma$ τοῦ ΓZ . ὥστε καὶ τὸν $\Gamma \Theta$ διδα τεμῆσθαι κατὰ τὸ Z , καὶ διὰ τοῦτο τὸν ἐκ τῶν ΘB , $B \Gamma$ μετὰ τοῦ ἀπὸ $Z \Gamma$ ἴσον γίνεσθαι τῷ ἀπὸ BZ . ὑπόκειται δὲ καὶ ὁ ἐκ τῶν AB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓE ἴσος τῷ ἀπὸ BZ . ὥστε καὶ ὁ ἐκ τῶν ΘB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓZ ἴσος ἔσται τῷ ἐκ τῶν AB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓE . ὅπερ ἄτοπον. οὐκ ἄρα ὁ ἐκ τῶν AB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓE ἴσος ἐστὶ [τῷ] ἐλάσσονι τοῦ ἀπὸ BE . ἐδείκθη δέ, ὅτι οὐδὲ [αὐτῷ] τῷ ἀπὸ BE . οὐκ ἄρα ὁ ἐκ τῶν AB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓE τετραγώνος ἐστιν. ὅπερ ἔδει δεῖχαι.

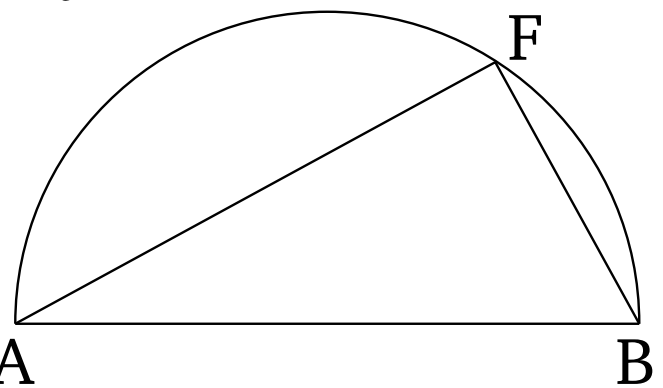
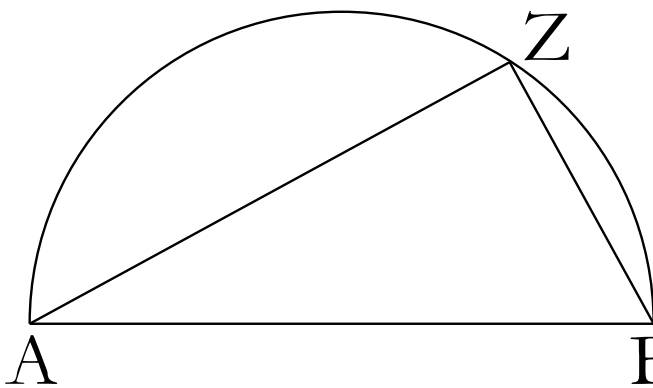
created) from (multiplying) GB and BC , plus the (square) on CE , is equal to the (number created) from (multiplying) AB and BC , plus the (square) on CE . And subtracting the (square) on CE from both, AB is inferred (to be) equal to GB . The very thing is absurd. Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is not equal to the (square) on BE . So I say that (it is) not less than the (square) on BE either. For, if possible, let it be equal to the (square) on BF . And (let) HA (be) double DF . And it can again be inferred that HC (is) double CF . Hence, CH has also been cut in half at F . And, on account of this, the (number created) from (multiplying) HB and BC , plus the (square) on FC , becomes equal to the (square) on BF [Prop. 2.6]. And the (number created) from (multiplying) AB and BC , plus the (square) on CE , was also assumed (to be) equal to the (square) on BF . Hence, the (number created) from (multiplying) HB and BC , plus the (square) on CF , will also be equal to the (number created) from (multiplying) AB and BC , plus the (square) on CE . The very thing is absurd. Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is not equal to less than the (square) on BE . And it was shown that (is it) not equal to the (square) on BE either. Thus, the (number created) from (multiplying) AB and BC , plus the square on CE , is not square. (Which is) the very thing it was required to show.

29.

Proposition 29

Εὕρεῖν δύο ῥητάς δυνάμει μόνον συμμετρωνς, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετρων εαυτῇ μήκει.

To find two rational (straight-lines which are) commensurable in square only, such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line which is) commensurable in length with the greater.



Ἐκκείσθω γάρ τις ῥητὴ ἡ AB καὶ δύο τετράγωνοι ἀριθμοὶ οἱ $\Gamma \Delta$, ΔE , ὥστε τὴν ὑπεροθὴν αὐτῶν τὸν ΓE μὴ εἶναι

For let some rational (straight-line) AB be laid down, and two square numbers, CD and DE , such that the difference be-

τετράγωνον, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AZB , καὶ πεποιήσθω ὥς ὁ $\Delta\Gamma$ πρὸς τὸν $ΓΕ$, οὕτως τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς AZ τετράγωνον, καὶ ἐπεξέσθω ἡ ZB .

Ἐπεὶ [οὕτως] ἔστιν ὥς τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , οὕτως ὁ $\Delta\Gamma$ πρὸς τὸν $ΓΕ$, τὸ ἀπὸ τῆς BA ἄρα πρὸς τὸ ἀπὸ τῆς AZ λόγον ἔθει, ὃν ἀριθμὸς ὁ $\Delta\Gamma$ πρὸς ἀριθμὸν τὸν $ΓΕ$ · σύμμετρον ἄρα ἔστι τὸ ἀπὸ τῆς BA τῷ ἀπὸ τῆς AZ . ῥητὸν δὲ τὸ ἀπὸ τῆς AB · ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς AZ · ῥητὴ ἄρα καὶ ἡ AZ . καὶ ἐπεὶ ὁ $\Delta\Gamma$ πρὸς τὸν $ΓΕ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς BA ἄρα πρὸς τὸ ἀπὸ τῆς AZ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἔστιν ἡ AB τῇ AZ μήκει· αἱ BA , AZ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ [ἔστιν] ὥς ὁ $\Delta\Gamma$ πρὸς τὸν $ΓΕ$, οὕτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , ἀναστρέψαντι ἄρα ὥς ὁ $\Gamma\Delta$ πρὸς τὸν $\DeltaΕ$, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ . ὁ δὲ $\Gamma\Delta$ πρὸς τὸν $\DeltaΕ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· καὶ τὸ ἀπὸ τῆς AB ἄρα πρὸς τὸ ἀπὸ τῆς BZ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· σύμμετρος ἄρα ἔστιν ἡ AB τῇ BZ μήκει. καὶ ἔστι τὸ ἀπὸ τῆς AB ἴσον τοῖς ἀπὸ τῶν AZ , ZB · ἡ AB ἄρα τῆς AZ μείζον δύναται τῇ BZ συμμέτρῳ ἑαυτῇ.

Ἐύρεται ἄρα δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ BA , AZ , ὥστε τὴν μείζονα τὴν AB τῆς ἐλάσσονος τῆς AZ μείζον δύνασθαι τῷ ἀπὸ τῆς BZ συμμέτρῳ ἑαυτῇ μήκει· ὅπερ ἔδει δεῖχαι.

tween them, CE , is not square [Prop. 10.28 lem. I]. And let the semi-circle AFB be drawn on AB . And let it be contrived that as DC (is) to CE , so the square on BA (is) to the square on AF [Prop. 10.6 corr.]. And let FB be joined.

[Therefore,] since as the (square) on BA is to the (square) on AF , so DC (is) to CE , the (square) on BA thus has to the (square) on AF the ratio which the number DC (has) to the number CE . Thus, the (square) on BA is commensurable with the (square) on AF [Prop. 10.6]. And the (square) on AB (is) rational [Def. 10.4]. Thus, the (square) on AF (is) also rational. Thus, AF (is) also rational. And since DC does not have to CE the ratio which (some) square number (has) to (some) square number, the (square) on BA thus does not have to the (square) on AF the ratio which (some) square number has to (some) square number either. Thus, AB is incommensurable in length with AF [Prop. 10.9]. Thus, the rational (straight-lines) BA and AF are commensurable in square only. And since as DC [is] to CE , so the (square) on BA (is) to the (square) on AF , thus, via conversion, as CD (is) to DE , so the (square) on AB (is) to the (square) on BF [Props. 5.19 corr., 3.31, 1.47]. And CD has to DE the ratio which (some) square number (has) to (some) square number. Thus, the (square) on AB also has to the (square) on BF the ratio which (some) square number has to (some) square number. AB is thus commensurable in length with BF [Prop. 10.9]. And the (square) on AB is equal to the (sum of the squares) on AF and FB [Prop. 1.47]. Thus, the square on AB is greater than (the square on) AF by (the square on) BF , (which is) commensurable (in length) with (AB).

Thus, two rational (straight-lines), BA and AF , commensurable in square only, be found such that the square on the greater, AB , is larger than (the square on) the lesser, AF , by the (square) on BF , (which is) commensurable in length with (AB).[†] (Which is) the very thing it was required to show.

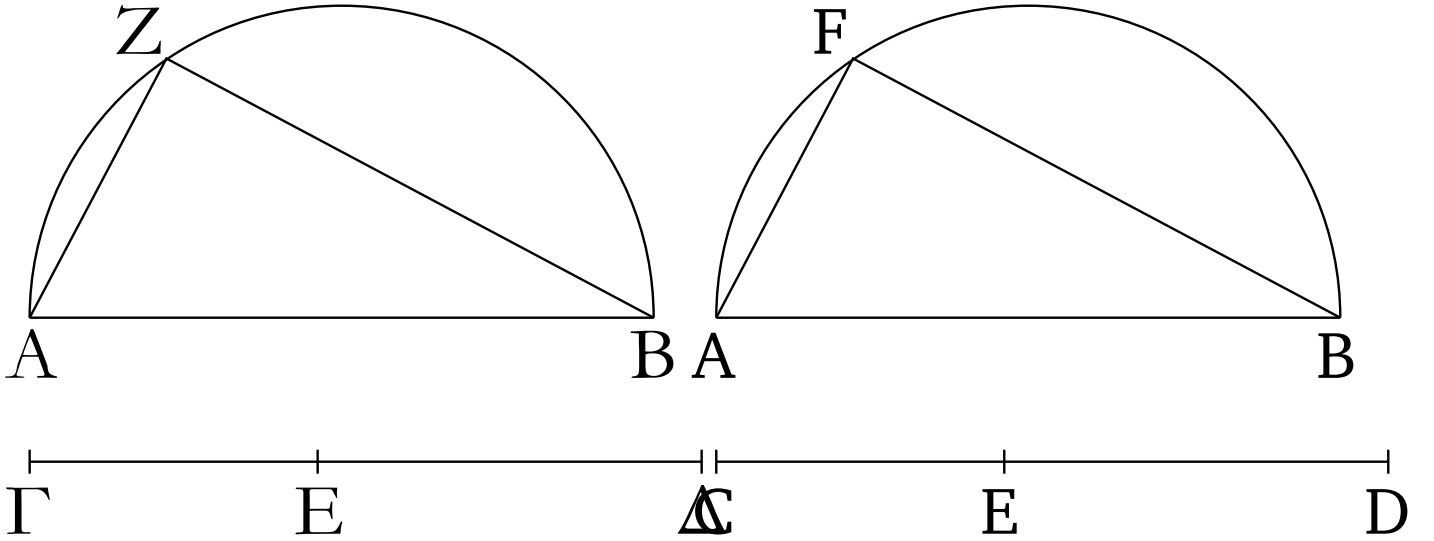
[†] BA and AF have lengths 1 and $\sqrt{1-k^2}$ times that of AB , respectively, where $k = \sqrt{DE/CD}$.

30.

Εύρεῖν δύο ῥητάς δυνάμει μόνον συμμέτρους, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ ἀσυνμέτρῳ ἑαυτῇ μήκει.

Proposition 30

To find two rational (straight-lines which are) commensurable in square only, such that the square on the greater is larger than the (the square on) lesser by the (square) on (some straight-line which is) incommensurable in length with the greater.



Ἐκκείσθω ῥητὴ ἡ AB καὶ δύο τετράγωνοι ἀριθμοὶ οἱ $ΓΕ$, $ΕΔ$, ὥστε τὸν συγκείμενον ἔχ' αὐτῶν τὸν $ΓΔ$ μὴ εἶναι τετράγωνον, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AZB , καὶ πεποιήσθω ὥς ὁ $ΔΓ$ πρὸς τὸν $ΓΕ$, οὔτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , καὶ ἐπεζεύθθω ἡ ZB .

Ὅμοίως δὲ δείχομεν τῷ πρὸ τούτου, ὅτι αἱ BA , AZ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ ἐστὶν ὥς ὁ $ΔΓ$ πρὸς τὸν $ΓΕ$, οὔτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , ἀναστρέφαντι ἄρα ὥς ὁ $ΓΔ$ πρὸς τὸν $ΔΕ$, οὔτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ . ὁ δὲ $ΓΔ$ πρὸς τὸν $ΔΕ$ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν οὐδ' ἄρα τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ἀσύμμετρος ἄρα ἐστὶν ἡ AB τῇ BZ μήκει. καὶ δύναται ἡ AB τῇ AZ μείζον τῷ ἀπὸ τῆς ZB ἀσυνμέτρον ἑαυτῇ.

Αἱ AB , AZ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AB τῇ AZ μείζον δύναται τῷ ἀπὸ τῆς ZB ἀσυνμέτρον ἑαυτῇ μήκει· ὅπερ ἔδει δεῖξαι.

Let the rational (straight-line) AB be laid out, and the two square numbers, CE and ED , such that the sum of them, CD , is not square [Prop. 10.28 lem. II]. And let the semi-circle AFB be drawn on AB . And let it be contrived that as DC (is) to CE , so the (square) on BA (is) to the (square) on AF [Prop. 10.6 corr]. And let FB be joined.

So, similarly to the (proposition) before this, we can show that BA and AF are rational (straight-lines which are) commensurable in square only. And since as DC is to CE , so the (square) on BA (is) to the (square) on AF , thus, via conversion, as CD (is) to DE , so the (square) on AB (is) to the (square) on BF [Props. 5.19 corr., 3.31, 1.47]. And CD does not have to DE the ratio which (some) square number (has) to (some) square number. Thus, the (square) on AB does not have to the (square) on BF the ratio which (some) square number has to (some) square number either. Thus, AB is incommensurable in length with BF [Prop. 10.9]. And the square on AB is greater than the (square) on AF by the (square) on FB [Prop. 1.47], (which is) incommensurable (in length) with (AB) .

Thus, AB and AF are rational (straight-lines which are) commensurable in square only, and the square on AB is greater than (the square on) AF by the (square) on FB , (which is) incommensurable (in length) with (AB) .[†] (Which is) the very thing it was required to show.

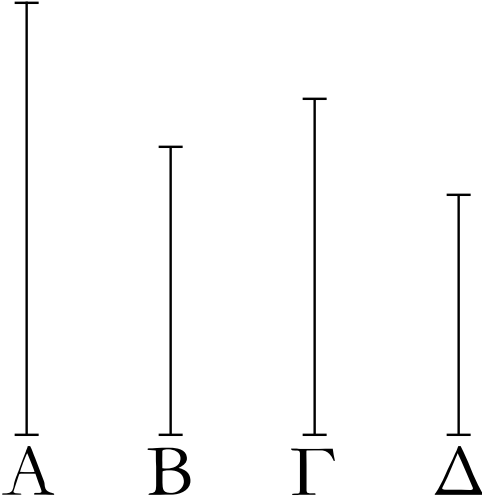
[†] AB and AF have lengths 1 and $1/\sqrt{1+k^2}$ times that of AB , respectively, where $k = \sqrt{DE/CE}$.

31.

Εὑρεῖν δύο μέσας δυνάμει μόνον συμμέτρον ῥητὸν περιεούσας, ὥστε τὴν μείζονα τῇ ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμέτρον ἑαυτῇ μήκει.

Proposition 31

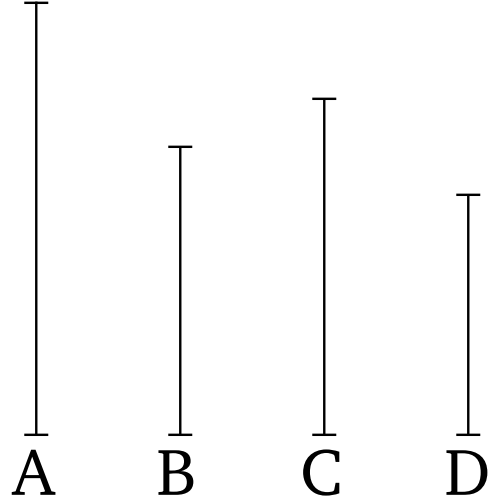
To find two medial (straight-lines), commensurable in square only, (and) containing a rational (area), such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable in length with the greater.



Ἐκκεισθώσαν δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ A , B , ὥστε τὴν A μείζονα οὕσαν τῆς ἐλάσσονος τῆς B μείζον δύνασθαι τῷ ἀπὸ συμμέτρου ἐναντὶ μήκει. καὶ τῷ ὑπὸ τῶν A , B ἴσον ἔστω τὸ ἀπὸ τῆς Γ . μέσον δὲ τὸ ὑπὸ τῶν A , B μέσον ἄρα καὶ τὸ ἀπὸ τῆς Γ μέση ἄρα καὶ ἡ Γ . τῷ δὲ ἀπὸ τῆς B ἴσον ἔστω τὸ ὑπὸ τῶν Γ , Δ . ῥητὸν δὲ τὸ ἀπὸ τῆς B ῥητὸν ἄρα καὶ τὸ ὑπὸ τῶν Γ , Δ . καὶ ἐπεὶ ἔστιν ὥς ἡ A πρὸς τὴν B , οὕτως τὸ ὑπὸ τῶν A , B πρὸς τὸ ἀπὸ τῆς B , ἀλλὰ τῷ μὲν ὑπὸ τῶν A , B ἴσον ἐστὶ τὸ ἀπὸ τῆς Γ , τῷ δὲ ἀπὸ τῆς B ἴσον τὸ ὑπὸ τῶν Γ , Δ , ὥς ἄρα ἡ A πρὸς τὴν B , οὕτως τὸ ἀπὸ τῆς Γ πρὸς τὸ ὑπὸ τῶν Γ , Δ . ὥς δὲ τὸ ἀπὸ τῆς Γ πρὸς τὸ ὑπὸ τῶν Γ , Δ , οὕτως ἡ Γ πρὸς τὴν Δ . καὶ ὥς ἄρα ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ . σύμμετρος δὲ ἡ A τῇ B δυνάμει μόνον· σύμμετρος ἄρα καὶ ἡ Γ τῇ Δ δυνάμει μόνον. καὶ ἐστὶ μέση ἡ Γ · μέση ἄρα καὶ ἡ Δ . καὶ ἐπεὶ ἔστιν ὥς ἡ A πρὸς τὴν B , ἡ Γ πρὸς τὴν Δ , ἡ δὲ A τῆς B μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντὶ, καὶ ἡ Γ ἄρα τῆς Δ μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντὶ.

Εὐρηγται ἄρα δύο μέσαι δυνάμει μόνον σύμμετροι αἱ Γ , Δ ῥητὸν περιέθουσαι, καὶ ἡ Γ τῆς Δ μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντὶ μήκει.

Ὅμοίως δὴ δειθθήσεται καὶ τῷ ἀπὸ ἀσυνμέτρου, ὅταν ἡ A τῆς B μείζον δύνηται τῷ ἀπὸ ἀσυνμέτρου ἐναντὶ.



Let two rational (straight-lines), A and B , commensurable in square only, be laid out, such that the square on the greater A is larger than the (square on the) lesser B by the (square) on (some straight-line) commensurable in length with (A) [Prop. 10.29]. And let the (square) on C be equal to the (rectangle contained) by A and B . And the (rectangle contained) by A and B (is) medial [Prop. 10.21]. Thus, the (square) on C (is) also medial. Thus, C (is) also medial [Prop. 10.21]. And let the (rectangle contained) by C and D be equal to the (square) on B . And the (square) on B (is) rational. Thus, the (rectangle contained) by C and D (is) also rational. And since as A is to B , so the (rectangle contained) by A and B (is) to the (square) on B [Prop. 10.21 lem.], but the (square) on C is equal to the (rectangle contained) by A and B , and the (rectangle contained) by C and D to the (square) on B , thus as A (is) to B , so the (square) on C (is) to the (rectangle contained) by C and D . And as the (square) on C (is) to the (rectangle contained) by C and D , so C (is) to D [Prop. 10.21 lem.]. And thus as A (is) to B , so C (is) to D . And A is commensurable in square only with B . Thus, C (is) also commensurable in square only with D [Prop. 10.11]. And C is medial. Thus, D (is) also medial [Prop. 10.23]. And since as A is to B , (so) C (is) to D , and the square on A is greater than (the square on) B by the (square) on (some straight-line) commensurable (in length) with (A), the square on C is thus also greater than (the square on) D by the (square) on (some straight-line) commensurable (in length) with (C) [Prop. 10.14].

Thus, two medial (straight-lines), C and D , commensurable in square only, (and) containing a rational (area), be found. And the square on C is greater than (the square on) D by the (square) on (some straight-line) commensurable in length with (C).[†]

So, similarly, (the proposition) can also be demonstrated for (some straight-line) incommensurable (in length with C), provided that the square on A is greater than (the square on B)

by the (square) on (some straight-line) incommensurable (in length) with (A) [Prop. 10.30].[‡]

[†] C and D have lengths $(1 - k^2)^{1/4}$ and $(1 - k^2)^{3/4}$ times that of A, respectively, where k is defined in the footnote to Prop. 10.29.

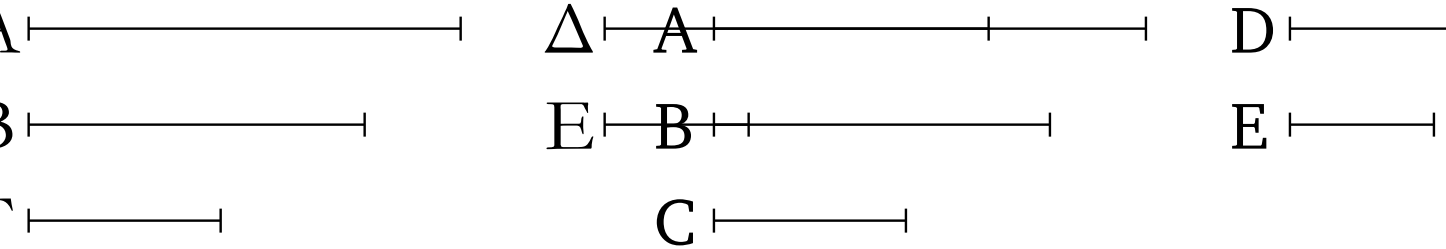
[‡] C and D would have lengths $1/(1 + k^2)^{1/4}$ and $1/(1 + k^2)^{3/4}$ times that of A, respectively, where k is defined in the footnote to Prop. 10.30.

32.

Proposition 32

Εὕρεῖν δύο μέσας δυνάμει μόνον συμμετρους μέσον περιεθούσας, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετρων ἑαντῇ.

To find two medial (straight-lines), commensurable in square only, (and) containing a medial (area), such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable (in length) with the greater.



Ἐκκείσθωσαν τρεῖς ῥηταὶ δυνάμει μόνον σύμμετροι αἱ A, B, Γ, ὥστε τὴν A τῆς Γ μείζον δύνασθαι τῷ ἀπὸ συμμετρων ἑαντῇ, καὶ τῷ μὲν ὑπὸ τῶν A, B ἴσον ἔστω τὸ ἀπὸ τῆς Δ. μέσον ἄρα τὸ ἀπὸ τῆς Δ· καὶ ἡ Δ ἄρα μέση ἐστίν. τῷ δὲ ὑπὸ τῶν B, Γ ἴσον ἔστω τὸ ὑπὸ τῶν Δ, Ε. καὶ ἐπεὶ ἐστὶν ὡς τὸ ὑπὸ τῶν A, B πρὸς τὸ ὑπὸ τῶν B, Γ, οὕτως ἡ A πρὸς τὴν Γ, ἀλλὰ τῷ μὲν ὑπὸ τῶν A, B ἴσον ἐστὶ τὸ ἀπὸ τῆς Δ, τῷ δὲ ὑπὸ τῶν B, Γ ἴσον τὸ ὑπὸ τῶν Δ, Ε, ἔστιν ἄρα ὡς ἡ A πρὸς τὴν Γ, οὕτως τὸ ἀπὸ τῆς Δ πρὸς τὸ ὑπὸ τῶν Δ, Ε. ὡς δὲ τὸ ἀπὸ τῆς Δ πρὸς τὸ ὑπὸ τῶν Δ, Ε, οὕτως ἡ Δ πρὸς τὴν Ε· καὶ ὡς ἄρα ἡ A πρὸς τὴν Γ, οὕτως ἡ Δ πρὸς τὴν Ε. σύμμετρος δὲ ἡ A τῇ Γ δυνάμει [μόνον]. σύμμετρος ἄρα καὶ ἡ Δ τῇ Ε δυνάμει μόνον. μέση δὲ ἡ Δ· μέση ἄρα καὶ ἡ Ε. καὶ ἐπεὶ ἐστὶν ὡς ἡ A πρὸς τὴν Γ, ἡ Δ πρὸς τὴν Ε, ἡ δὲ A τῆς Γ μείζον δύναται τῷ ἀπὸ συμμετρων ἑαντῇ, καὶ ἡ Δ ἄρα τῆς Ε μείζον δυνήσεται τῷ ἀπὸ συμμετρων ἑαντῇ. λέγω δὴ, ὅτι καὶ μέσον ἐστὶ τὸ ὑπὸ τῶν Δ, Ε. ἐπεὶ γὰρ ἴσον ἐστὶ τὸ ὑπὸ τῶν B, Γ τῷ ὑπὸ τῶν Δ, Ε, μέσον δὲ τὸ ὑπὸ τῶν B, Γ [αἱ γὰρ B, Γ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι], μέσον ἄρα καὶ τὸ ὑπὸ τῶν Δ, Ε.

Εὕρεται ἄρα δύο μέσαι δυνάμει μόνον σύμμετροι αἱ Δ, Ε μέσον περιέθουσας, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετρων ἑαντῇ.

Ὁμοίως δὴ πάλιν διεθθήσεται καὶ τῷ ἀπὸ ἀσύμμετρων, ὅταν ἡ A τῆς Γ μείζον δύνῃται τῷ ἀπὸ ἀσύμμετρων ἑαντῇ.

Let three rational (straight-lines), A, B and C, commensurable in square only, be laid out such that the square on A is greater than (the square on C) by the (square) on (some straight-line) commensurable (in length) with (A) [Prop. 10.29]. And let the (square) on D be equal to the (rectangle contained) by A and B. Thus, the (square) on D (is) medial. Thus, D is also medial [Prop. 10.21]. And let the (rectangle contained) by D and E be equal to the (rectangle contained) by B and C. And since as the (rectangle contained) by A and B is to the (rectangle contained) by B and C, so A (is) to C [Prop. 10.21 lem.], but the (square) on D is equal to the (rectangle contained) by A and B, and the (rectangle contained) by D and E to the (rectangle contained) by B and C, thus as A is to C, so the (square) on D (is) to the (rectangle contained) by D and E. And as the (square) on D (is) to the (rectangle contained) by D and E, so D (is) to E [Prop. 10.21 lem.]. And thus as A (is) to C, so D (is) to E. And A (is) commensurable in square [only] with C. Thus, D (is) also commensurable in square only with E [Prop. 10.11]. And D (is) medial. Thus, E (is) also medial [Prop. 10.23]. And since as A is to C, (so) D (is) to E, and the square on A is greater than (the square on) C by the (square) on (some straight-line) commensurable (in length) with (A), the square on D will thus also be greater than (the square on) E by the (square) on (some straight-line) commensurable (in length) with (D) [Prop. 10.14]. So, I also say that the (rectangle contained) by D and E is medial. For since the (rectangle contained) by B and C is equal to the (rectangle contained) by D and E, and the (rectangle contained) by B and C (is) medial [for B and C are rational (straight-lines which are) commensurable in square only] [Prop. 10.21], the

(rectangle contained) by D and E (is) thus also medial.

Thus, two medial (straight-lines), D and E , commensurable in square only, (and) containing a medial (area), be found such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable (in length) with the greater.[†]

So, similarly, (the proposition) can again also be demonstrated for (some straight-line) incommensurable (in length with the greater), provided that the square on A is greater than (the square on) C by the (square) on (some straight-line) incommensurable (in length) with (A) [Prop. 10.30].[‡]

[†] D and E have lengths $k^{1/4}$ and $k^{1/4} \sqrt{1 - k^2}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A , and k is defined in the footnote to Prop. 10.29.

[‡] D and E would have lengths $k^{1/4}$ and $k^{1/4} / \sqrt{1 + k^2}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A , and k is defined in the footnote to Prop. 10.30.

Λήμμα.

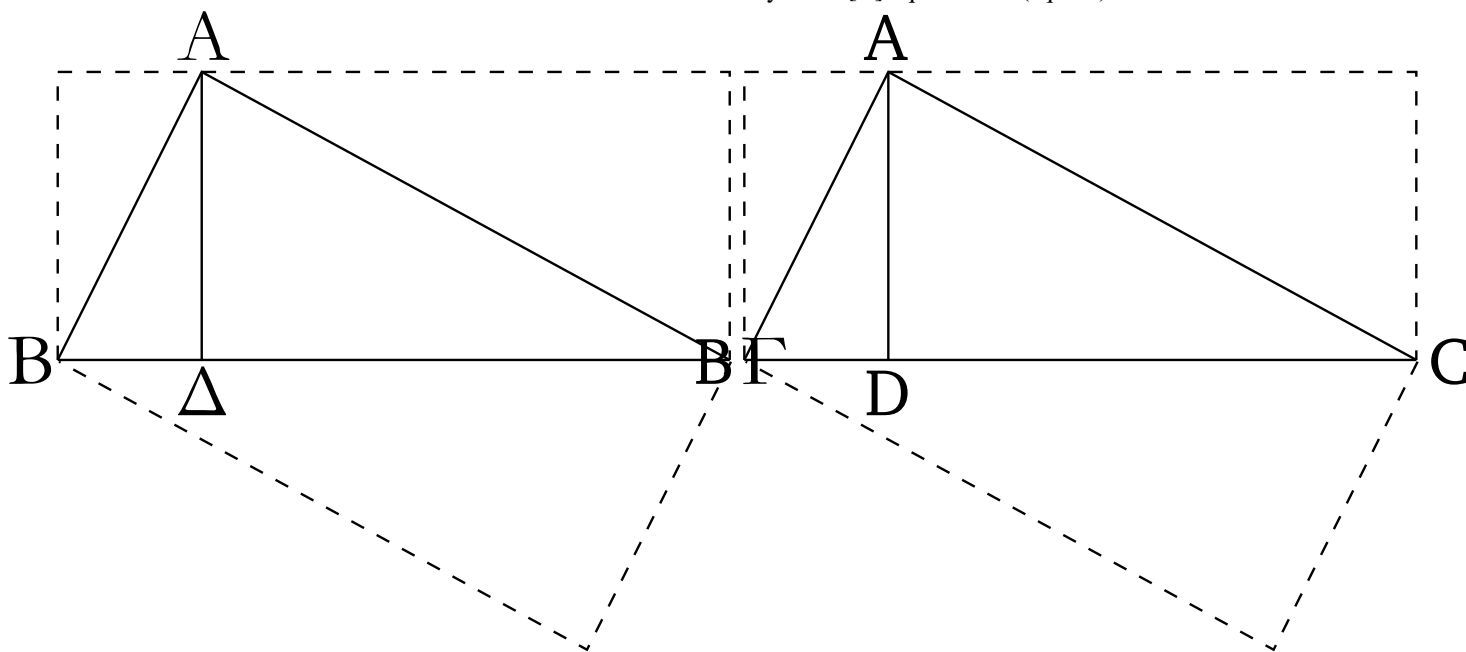
Ἐστω τρίγωνον ὀρθογώνιον τὸ $AB\Gamma$ ὀρθὴν ἔχον τὴν A , καὶ ἤθθω κάθετος ἡ $A\Delta$. λέγω, ὅτι τὸ μὲν ὑπὸ τῶν $\Gamma B\Delta$ ἴσον ἐστὶ τῷ ἀπὸ τῆς BA , τὸ δὲ ὑπὸ τῶν $B\Gamma A$ ἴσον τῷ ἀπὸ τῆς ΓA , καὶ τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ ἴσον τῷ ἀπὸ τῆς $A\Delta$, καὶ ἔτι τὸ ὑπὸ τῶν $B\Gamma$, $A\Delta$ ἴσον [ἐστὶ] τῷ ὑπὸ τῶν BA , $A\Gamma$.

Καὶ πρῶτον, ὅτι τὸ ὑπὸ τῶν $\Gamma B\Delta$ ἴσον [ἐστὶ] τῷ ἀπὸ τῆς BA .

Lemma

Let ABC be a right-angled triangle having the (angle) A a right-angle. And let the perpendicular AD be drawn. I say that the (rectangle contained) by CBD is equal to the (square) on BA , and the (rectangle contained) by BCD (is) equal to the (square) on CA , and the (rectangle contained) by BD and DC (is) equal to the (square) on AD , and, further, the (rectangle contained) by BC and AD [is] equal to the (rectangle contained) by BA and AC .

And, first of all, (let us prove) that the (rectangle contained) by CBD [is] equal to the (square) on BA .



Ἐπεὶ γὰρ ἐν ὀρθογώνιῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ἦκται ἡ $A\Delta$, τὰ $AB\Delta$, $A\Delta\Gamma$ ἄρα

For since AD has been drawn from the right-angle in a right-angled triangle, perpendicular to the base, ABD and

τρίγωνα ὁμοιά ἐστι τῷ τε ὅλῳ τῷ $ABΓ$ καὶ ἀλλήλοις. καὶ ἐπεὶ ὁμοιόν ἐστι τὸ $ABΓ$ τριγώνων τῷ $ABΔ$ τριγώνῳ, ἔστιν ἄρα ὡς ἡ $ΓΒ$ πρὸς τὴν $ΒΑ$, οὕτως ἡ $ΒΑ$ πρὸς τὴν $ΒΔ$. τὸ ἄρα ὑπὸ τῶν $ΓΒΔ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB .

Διὰ τὰ αὐτὰ δὴ καὶ τὸ ὑπὸ τῶν $ΒΓΔ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΑΓ$.

Καὶ ἐπεὶ, ἔαν ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ἀθῇ, ἡ ἀθθεῖσα τῶν τῆς βάσεως τμημάτων μέση ἀνάλογόν ἐστιν, ἔστιν ἄρα ὡς ἡ $ΒΑ$ πρὸς τὴν $ΔΑ$, οὕτως ἡ $ΑΔ$ πρὸς τὴν $ΔΓ$. τὸ ἄρα ὑπὸ τῶν $ΒΔ, ΔΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΔΑ$.

Λέγω, ὅτι καὶ τὸ ὑπὸ τῶν $ΒΓ, ΑΔ$ ἴσον ἐστὶ τῷ ὑπὸ τῶν $ΒΑ, ΑΓ$. ἐπεὶ γάρ, ὡς ἔφαμεν, ὁμοιόν ἐστι τὸ $ABΓ$ τῷ $ABΔ$, ἔστιν ἄρα ὡς ἡ $ΒΓ$ πρὸς τὴν $ΓΑ$, οὕτως ἡ $ΒΑ$ πρὸς τὴν $ΑΔ$. τὸ ἄρα ὑπὸ τῶν $ΒΓ, ΑΔ$ ἴσον ἐστὶ τῷ ὑπὸ τῶν $ΒΑ, ΑΓ$. ὁπερ ἔδει δεῖξαι.

ADC are thus triangles (which are) similar to the whole, ABC , and to one another [Prop. 6.8]. And since triangle ABC is similar to triangle ABD , thus as CB is to BA , so BA (is) to BD [Prop. 6.4]. Thus, the (rectangle contained) by CBD is equal to the (square) on AB [Prop. 6.17].

So, for the same (reasons), the (rectangle contained) by BCD is also equal to the (square) on AC .

And since if a (straight-line) is drawn from the right-angle in a right-angled triangle, perpendicular to the base, the (straight-line so) drawn is the mean proportional to the pieces of the base [Prop. 6.8 corr.], thus as BD is to DA , so AD (is) to DC . Thus, the (rectangle contained) by BD and DC is equal to the (square) on DA [Prop. 6.17].

I also say that the (rectangle contained) by BC and AD is equal to the (rectangle contained) by BA and AC . For since, as we said, ABC is similar to ABD , thus as BC is to CA , so BA (is) to AD [Prop. 6.4]. Thus, the (rectangle contained) by BC and AD is equal to the (rectangle contained) by BA and AC [Prop. 6.16]. (Which is) the very thing it was required to show.

33.

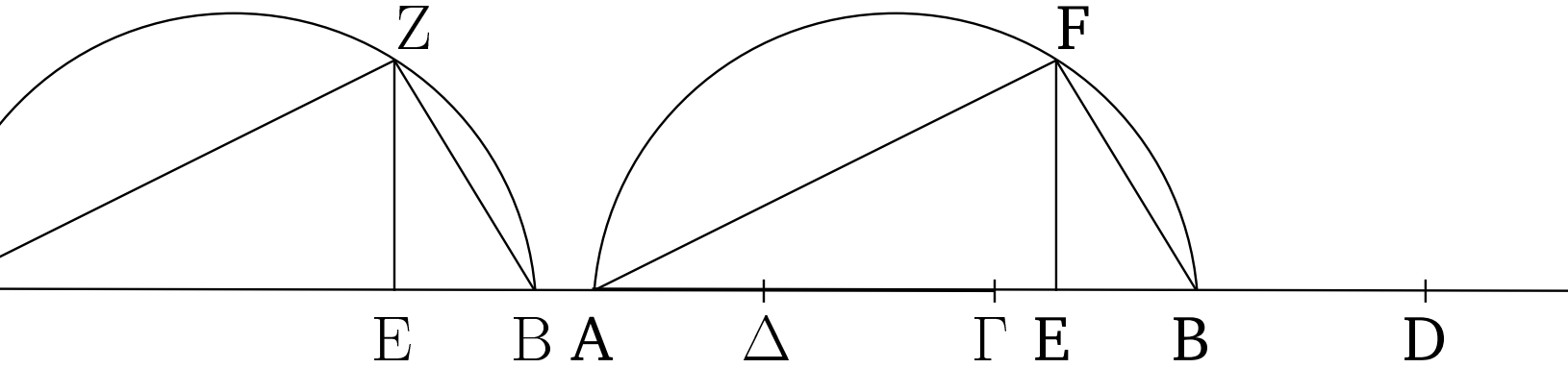
Proposition 33

Εὐρεῖν δύο εὐθείας δυνάμει ἀσυμμέτρους ποιούσας τὸ μὲν συγκεκριμένον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον.

Ἐκκεῖσθωσαν δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ $AB, ΒΓ$, ὥστε τὴν μείζονα τὴν AB τῆς ἐλάσσονος τῆς $ΒΓ$ μείζον δύνασθαι τῷ ἀπὸ ἀσυμμέτρου ἑαντῇ, καὶ τεμήσθω ἡ $ΒΓ$ διῖθα κατὰ τὸ $Δ$, καὶ τῷ ἀφ' ὁποτέρᾳ τῶν $ΒΔ, ΔΓ$ ἴσον παρὰ τὴν AB παραβεβλήσθω παραλληλόγραμμον ἐλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν $ΑΕΒ$, καὶ γεγράφθω ἐπὶ τῆς AB ημικύκλιον τὸ AZB , καὶ ῥηθῇ τῇ AB πρὸς ὀρθὰς ἡ EZ , καὶ ἐπεζεύθωσαν αἱ AZ, ZB .

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial.

Let the two rational (straight-lines) AB and BC , (which are) commensurable in square only, be laid out such that the square on the greater, AB , is larger than (the square on) the lesser, BC , by the (square) on (some straight-line which is) incommensurable (in length) with (AB) [Prop. 10.30]. And let BC be cut in half at D . And let a parallelogram equal to the (square) on either of BD or DC , (and) falling short by a square figure, be applied to AB [Prop. 6.28], and let it be the (rectangle contained) by AEB . And let the semi-circle AFB be drawn on AB . And let EF be drawn at right-angles to AB . And let AF and FB be joined.



Καὶ ἐπεὶ [δύο] εὐθεῖαι ἄνισοί εἰσιν αἱ $AB, ΒΓ$, καὶ ἡ AB τῆς $ΒΓ$ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαντῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς $ΒΓ$, τουτέστι τῷ ἀπὸ τῆς ἡμισείας αὐτῆς,

And since AB and BC are [two] unequal straight-lines, and the square on AB is greater than (the square on) BC by the (square) on (some straight-line which is) incommensu-

ἴσον παρὰ τὴν AB παραβέβληται παραλληλόγραμμον ἑλλείπον εἶδει τετραγώνῳ καὶ ποιεῖ τὸ ὑπὸ τῶν AEB , ἀσύμμετρος ἄρα ἐστὶν ἡ AE τῇ EB . καὶ ἐστὶν ὥς ἡ AE πρὸς EB , οὕτως τὸ ὑπὸ τῶν BA , AE πρὸς τὸ ὑπὸ τῶν AB , BE , ἴσον δὲ τὸ μὲν ὑπὸ τῶν BA , AE τῷ ἀπὸ τῆς AZ , τὸ δὲ ὑπὸ τῶν AB , BE τῷ ἀπὸ τῆς BZ . ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AZ τῷ ἀπὸ τῆς BZ . αἱ AZ , BZ ἄρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ ἡ AB ῥητὴ ἐστὶν, ῥητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς AB . ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AZ , BZ ῥητὸν ἐστὶν. καὶ ἐπεὶ πάλιν τὸ ὑπὸ τῶν AE , EB ἴσον ἐστὶ τῷ ἀπὸ τῆς EZ , ὑπόκειται δὲ τὸ ὑπὸ τῶν AE , EB καὶ τῷ ἀπὸ τῆς BD ἴσον, ἴση ἄρα ἐστὶν ἡ ZE τῇ BD . διπλὴ ἄρα ἡ BF τῆς ZE . ὥστε καὶ τὸ ὑπὸ τῶν AB , BF σύμμετρον ἐστὶ τῷ ὑπὸ τῶν AB , EZ . μέσον δὲ τὸ ὑπὸ τῶν AB , BF . μέσον ἄρα καὶ τὸ ὑπὸ τῶν AB , EZ . ἴσον δὲ τὸ ὑπὸ τῶν AB , EZ τῷ ὑπὸ τῶν AZ , BZ . μέσον ἄρα καὶ τὸ ὑπὸ τῶν AZ , BZ . ἐδείχθη δὲ καὶ ῥητὸν τὸ συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων.

Εὐρένται ἄρα δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AZ , BZ ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ ὑπ' αὐτῶν μέσον. ὅπερ εἶδει δεῖχαι.

able (in length) with (AB) . And a parallelogram, equal to one quarter of the (square) on BC —that is to say, (equal) to the (square) on half of it—(and) falling short by a square figure, has been applied to AB , and makes the (rectangle contained) by AEB . AE is thus incommensurable (in length) with EB [Prop. 10.18]. And as AE is to EB , so the (rectangle contained) by BA and AE (is) to the (rectangle contained) by AB and BE . And the (rectangle contained) by BA and AE (is) equal to the (square) on AF , and the (rectangle contained) by AB and BE to the (square) on BF [Prop. 10.32 lem.]. The (square) on AF is thus incommensurable with the (square) on BF [Prop. 10.11]. Thus, AF and BF are incommensurable in square. And since AB is rational, the (square) on AB is also rational. Hence, the sum of the (squares) on AF and BF is also rational [Prop. 1.47]. And, again, since the (rectangle contained) by AE and EB is equal to the (square) on EF , and the (rectangle contained) by AE and EB was assumed (to be) equal to the (square) on BD , FE is thus equal to BD . Thus, BC is double FE . And hence the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and EF [Prop. 10.6]. And the (rectangle contained) by AB and BC (is) medial [Prop. 10.21]. Thus, the (rectangle contained) by AB and EF (is) also medial [Prop. 10.23 corr.]. And the (rectangle contained) by AB and EF (is) equal to the (rectangle contained) by AF and BF [Prop. 10.32 lem.]. Thus, the (rectangle contained) by AF and BF (is) also medial. And the sum of the squares on them was also shown (to be) rational.

Thus, the two straight-lines, AF and BF , (which are) incommensurable in square, be found, making the sum of the squares on them rational, and the (rectangle contained) by them medial. (Which is) the very thing it was required to show.

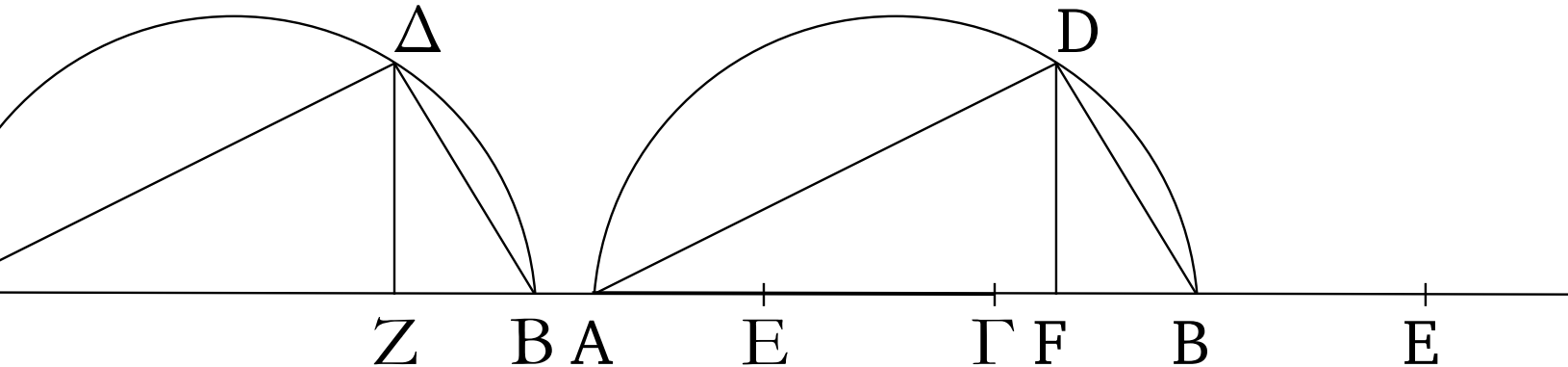
† AF and BF have lengths $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2}$ and $\sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ times that of AB , respectively, where k is defined in the footnote to Prop. 10.30.

34.

Proposition 34

Εὐρεῖν δύο εὐθείας δυνάμει ἀσύμμετρον ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητὸν.

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational.



Ἐκκείσθωσαν δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ ῥητὸν περιέθουσαι τὸ ὑπ' αὐτῶν, ὥστε τὴν AB τῆς $BΓ$ μείζον δύνασθαι τῷ ἀπὸ ἀσυνμέτρου εἰς αὐτῆς, καὶ γεγραφθῶ ἐπὶ τῆς AB τὸ $AΔB$ ἡμικύκλιον, καὶ τετμήσθω ἡ $BΓ$ διὰ κατὰ τὸ E , καὶ παραβέβλησθω παρὰ τὴν AB τῷ ἀπὸ τῆς BE ἴσον παραλληλόγραμμον ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν AZB · ἀσύμμετρος ἄρα [ἐστίν] ἡ AZ τῇ ZB μήκει. καὶ ἡθῆσθω ἀπὸ τοῦ Z τῇ AB πρὸς ὀρθὰς ἡ $ZΔ$, καὶ ἐπεξεύθωσαν αἱ $AΔ$, $ΔB$.

Ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AZ τῇ ZB , ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν BA , AZ τῷ ὑπὸ τῶν AB , BZ . ἴσον δὲ τὸ μὲν ὑπὸ τῶν BA , AZ τῷ ἀπὸ τῆς $AΔ$, τὸ δὲ ὑπὸ τῶν AB , BZ τῷ ἀπὸ τῆς $ΔB$ · ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $AΔ$ τῷ ἀπὸ τῆς $ΔB$. καὶ ἐπεὶ μέσον ἐστὶ τὸ ἀπὸ τῆς AB , μέσον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΔ$, $ΔB$. καὶ ἐπεὶ διπλὴ ἐστὶν ἡ $BΓ$ τῆς $ΔZ$, διπλάσιον ἄρα καὶ τὸ ὑπὸ τῶν AB , $BΓ$ τοῦ ὑπὸ τῶν AB , $ZΔ$. ῥητὸν δὲ τὸ ὑπὸ τῶν AB , $BΓ$ ῥητὸν ἄρα καὶ τὸ ὑπὸ τῶν AB , $ZΔ$. τὸ δὲ ὑπὸ τῶν AB , $ZΔ$ ἴσον τῷ ὑπὸ τῶν $AΔ$, $ΔB$ · ὥστε καὶ τὸ ὑπὸ τῶν $AΔ$, $ΔB$ ῥητὸν ἐστίν.

Ἐρρηγνύεται ἄρα δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ $AΔ$, $ΔB$ ποιοῦσαι τὸ [μὲν] συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητὸν· ὅπερ ἔδει δεῖχαι.

Let the two medial (straight-lines) AB and BC , (which are) commensurable in square only, be laid out having the (rectangle contained) by them rational, (and) such that the square on AB is greater than (the square on) BC by the (square) on (some straight-line) incommensurable (in length) with (AB) [Prop. 10.31]. And let the semi-circle ADB be drawn on AB . And let BC be cut in half at E . And let a (rectangular) parallelogram equal to the (square) on BE , (and) falling short by a square figure, be applied to AB , (and let it be) the (rectangle contained by) AFB [Prop. 6.28]. Thus, AF [is] incommensurable in length with FB [Prop. 10.18]. And let FD be drawn from F at right-angles to AB . And let AD and DB be joined.

Since AF is incommensurable (in length) with FB , the (rectangle contained) by BA and AF is thus also incommensurable with the (rectangle contained) by AB and BF [Prop. 10.11]. And the (rectangle contained) by BA and AF (is) equal to the (square) on AD , and the (rectangle contained) by AB and BF to the (square) on DB [Prop. 10.32 lem.]. Thus, the (square) on AD is also incommensurable with the (square) on DB . And since the (square) on AB is medial, the sum of the (squares) on AD and DB (is) thus also medial [Props. 3.31, 1.47]. And since BC is double DF [see previous proposition], the (rectangle contained) by AB and BC (is) thus also double the (rectangle contained) by AB and FD . And the (rectangle contained) by AB and BC (is) rational. Thus, the (rectangle contained) by AB and FD (is) also rational [Prop. 10.6, Def. 10.4]. And the (rectangle contained) by AB and FD (is) equal to the (rectangle contained) by AD and DB [Prop. 10.32 lem.]. And hence the (rectangle contained) by AD and DB is rational.

Thus, two straight-lines, AD and DB , (which are) incommensurable in square, be found, making the sum of the squares on them medial, and the (rectangle contained) by them rational.[†] (Which is) the very thing it was required to show.

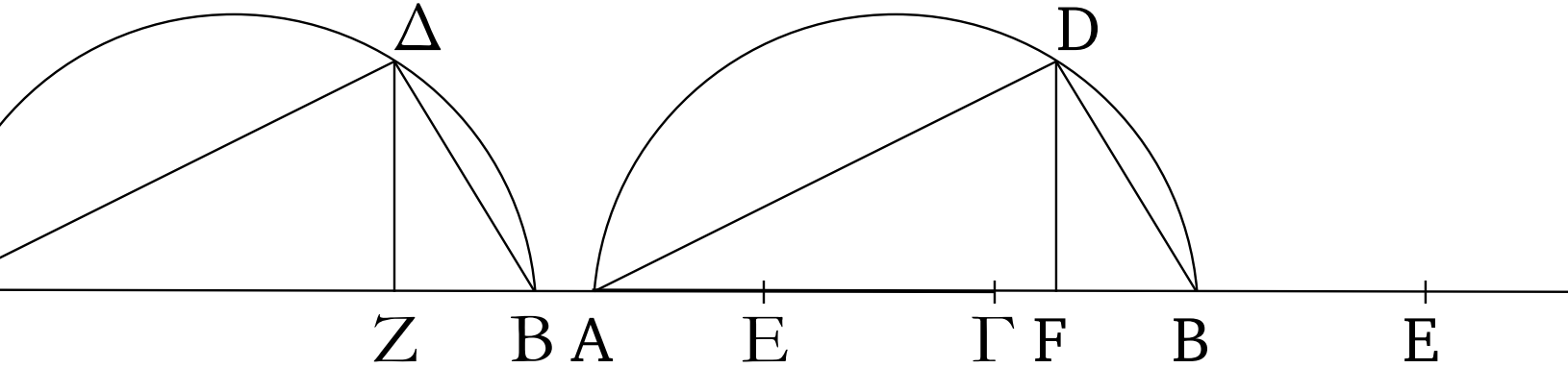
[†] AD and DB have lengths $\sqrt{[(1+k^2)^{1/2} + k]/[2(1+k^2)]}$ and $\sqrt{[(1+k^2)^{1/2} - k]/[2(1+k^2)]}$ times that of AB , respectively, where k is defined in the footnote to Prop. 10.29.

35.

Proposition 35

Ἐύρεῖν δύο εὐθείας δυνάμει ἀσυνμέτρον ποιούσας τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων.

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them.



Ἐκκείσθωσαν δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ μέσον περιέθουσαι, ὥστε τὴν AB τῆς $BΓ$ μείζον δύνασθαι τῷ ἀπὸ ἀσύμμετρον ἑαυτῇ, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $AΔB$, καὶ τὰ λοιπὰ γεγονέτω τοῖς ἐπάνω ὁμοίως.

Καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AZ τῇ ZB μήκει, ἀσύμμετρος ἐστὶ καὶ ἡ $AΔ$ τῇ $ΔB$ δυνάμει. καὶ ἐπεὶ μέσον ἐστὶ τὸ ἀπὸ τῆς AB , μέσον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΔ$, $ΔB$. καὶ ἐπεὶ τὸ ὑπὸ τῶν AZ , ZB ἴσον ἐστὶ τῷ ἀφ' ἑκατέρᾳ τῶν BE , $ΔZ$, ἴση ἄρα ἐστὶν ἡ BE τῇ $ΔZ$ διπλῇ ἄρα ἡ $BΓ$ τῆς $ΔZ$ · ὥστε καὶ τὸ ὑπὸ τῶν AB , $BΓ$ διπλάσιόν ἐστι τοῦ ὑπὸ τῶν AB , $ΔZ$. μέσον δὲ τὸ ὑπὸ τῶν AB , $BΓ$ μέσον ἄρα καὶ τὸ ὑπὸ τῶν AB , $ΔZ$. καὶ ἐστὶν ἴσον τῷ ὑπὸ τῶν $AΔ$, $ΔB$ · μέσον ἄρα καὶ τὸ ὑπὸ τῶν $AΔ$, $ΔB$. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει, σύμμετρος δὲ ἡ $ΓB$ τῇ BE , ἀσύμμετρος ἄρα καὶ ἡ AB τῇ BE μήκει· ὥστε καὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB , BE ἀσύμμετρόν ἐστιν. ἀλλὰ τῷ μὲν ἀπὸ τῆς AB ἴσα ἐστὶ τὰ ἀπὸ τῶν $AΔ$, $ΔB$, τῷ δὲ ὑπὸ τῶν AB , BE ἴσον ἐστὶ τὸ ὑπὸ τῶν AB , $ΔZ$, τουτέστι τὸ ὑπὸ τῶν $AΔ$, $ΔB$ · ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΔ$, $ΔB$ τῷ ὑπὸ τῶν $AΔ$, $ΔB$.

Εὐρήνεται ἄρα δύο εὐθεῖαι αἱ $AΔ$, $ΔB$ δυνάμει ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων· ὅπερ ἔδει δεῖξαι.

Let the two medial (straight-lines) AB and BC , (which are) commensurable in square only, be laid out containing a medial (area), such that the square on AB is greater than (the square on) BC by the (square) on (some straight-line) incommensurable (in length) with (AB) [Prop. 10.32]. And let the semicircle ADB be drawn on AB . And let the remainder (of the figure) be generated similarly to the above (proposition).

And since AF is incommensurable in length with FB [Prop. 10.18], AD is also incommensurable in square with DB [Prop. 10.11]. And since the (square) on AB is medial, the sum of the (squares) on AD and DB (is) thus also medial [Props. 3.31, 1.47]. And since the (rectangle contained) by AF and FB is equal to the (square) on each of BE and DF , BE is thus equal to DF . Thus, BC (is) double FD . And hence the (rectangle contained) by AB and BC is double the (rectangle) contained by AB and FD . And the (rectangle contained) by AB and BC (is) medial. Thus, the (rectangle contained) by AB and FD (is) also medial. And it is equal to the (rectangle contained) by AD and DB [Prop. 10.32 lem.]. Thus, the (rectangle contained) by AD and DB (is) also medial. And since AB is incommensurable in length with BC , and CB (is) commensurable (in length) with BE , AB (is) thus also incommensurable in length with BE [Prop. 10.13]. And hence the (square) on AB is also incommensurable with the (rectangle contained) by AB and BE [Prop. 10.11]. But the (sum of the squares) on AD and DB is equal to the (square) on AB [Prop. 1.47]. And the (rectangle contained) by AB and FD —that is to say, the (rectangle contained) by AD and DB —is equal to the (rectangle contained) by AB and BE . Thus, the sum of the (squares) on AD and DB is incommensurable with the (rectangle contained) by AD and DB .

Thus, two straight-lines, AD and DB , (which are) incommensurable in square, be found, making the sum of the (squares) on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them.[†] (Which is) the very thing it was required to show.

[†] AD and DB have lengths $k^{1/4} \sqrt{[1 + k/(1 + k^2)^{1/2}]/2}$ and $k'^{1/4} \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ times that of AB , respectively, where k and k' are defined in the footnote

to Prop. 10.32.

36.

Proposition 36

Ἐάν δύο ῥηταὶ δυνάμει μόνον σύμμετροι συντεθῶσιν, ἡ ὅλη ἄλογός ἐστιν, καλείσθω δὲ ἐκ δύο ὀνομάτων.

If two rational (straight-lines which are) commensurable in square only are added together then the whole (straight-line) is irrational—let it be called a binomial (straight-line).[†]



Συγκείσθωσαν γάρ δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$. λέγω, ὅτι ὅλη ἡ $ΑΓ$ ἄλογός ἐστιν.

Ἐπεὶ γάρ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει· δυνάμει γάρ μόνον εἰσὶ σύμμετροι· ὥς δὲ ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ὑπὸ τῶν $ΑΒΓ$ πρὸς τὸ ἀπὸ τῆς $BΓ$, ἀσύμμετρον ἄρα ἐστὶ τὸ ὑπὸ τῶν AB , $BΓ$ τῷ ἀπὸ τῆς $BΓ$. ἀλλὰ τῷ μὲν ὑπὸ τῶν AB , $BΓ$ σύμμετρον ἐστὶ τὸ διζ ὑπὸ τῶν AB , $BΓ$, τῷ δὲ ἀπὸ τῆς $BΓ$ σύμμετρον ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$. αἱ γὰρ AB , $BΓ$ ῥηταὶ εἰσὶ δυνάμει μόνον σύμμετροι· ἀσύμμετρον ἄρα ἐστὶ τὸ διζ ὑπὸ τῶν AB , $BΓ$ τοῖς ἀπὸ τῶν AB , $BΓ$. καὶ συνθέντι τὸ διζ ὑπὸ τῶν AB , $BΓ$ μετὰ τῶν ἀπὸ τῶν AB , $BΓ$, τοντέστι τὸ ἀπὸ τῆς $ΑΓ$, ἀσύμμετρον ἐστὶ τῷ συγκεκλιμένῳ ἐκ τῶν ἀπὸ τῶν AB , $BΓ$. ῥητὸν δὲ τὸ συγκεκλιμένον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$. ἄλογον ἄρα [ἐστὶ] τὸ ἀπὸ τῆς $ΑΓ$. ὥστε καὶ ἡ $ΑΓ$ ἄλογός ἐστιν, καλείσθω δὲ ἐκ δύο ὀνομάτων· ὅπερ ἔδει δεῖχαι.

For let the two rational (straight-lines), AB and BC , (which are) commensurable in square only, be laid down together. I say that the whole (straight-line), AC , is irrational. For since AB is incommensurable in length with BC —for they are commensurable in square only—and as AB (is) to BC , so the (rectangle contained) by ABC (is) to the (square) on BC , the (rectangle contained) by AB and BC is thus incommensurable with the (square) on BC [Prop. 10.11]. But, twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. And (the sum of) the (squares) on AB and BC is commensurable with the (square) on BC —for the rational (straight-lines) AB and BC are commensurable in square only [Prop. 10.15]. Thus, twice the (rectangle contained) by AB and BC is incommensurable with (the sum of) the (squares) on AB and BC [Prop. 10.13]. And, via composition, twice the (rectangle contained) by AB and BC , plus (the sum of) the (squares) on AB and BC —that is to say, the (square) on AC [Prop. 2.4]—is incommensurable with the sum of the (squares) on AB and BC [Prop. 10.16]. And the sum of the (squares) on AB and BC (is) rational. Thus, the (square) on AC [is] irrational [Def. 10.4]. Hence, AC is also irrational [Def. 10.4]—let it be called a binomial (straight-line).[‡] (Which is) the very thing it was required to show.

[†] Literally, “from two names”.

[‡] Thus, a binomial straight-line has a length expressible as $1 + k^{1/2}$ [or, more generally, $\rho(1 + k^{1/2})$, where ρ is rational—the same proviso applies to the definitions in the following propositions]. The binomial and the corresponding apotome, whose length is expressible as $1 - k^{1/2}$ (see Prop. 10.73), are the positive roots of the quartic $x^4 - 2(1 + k)x^2 + (1 - k)^2 = 0$.

37.

Proposition 37

Ἐάν δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι ῥητὸν περιέθουσαι, ἡ ὅλη ἄλογός ἐστιν, καλείσθω δὲ ἐκ δύο μέσων πρώτης.

If two medial (straight-lines), commensurable in square only, which contain a rational (area), are added together then the whole (straight-line) is irrational—let it be called a first bimedral (straight-line).[†]



Συγκείσθωσαν γάρ δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ ῥητὸν περιέθουσαι· λέγω, ὅτι ὅλη ἡ $ΑΓ$ ἄλογός ἐστιν.

For let the two medial (straight-lines), AB and BC , commensurable in square only, (and) containing a rational (area), be laid down together. I say that the whole (straight-line), AC ,

Ἐπεὶ γὰρ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει, καὶ τὰ ἀπὸ τῶν AB , $BΓ$ ἄρα ἀσύμμετρα ἐστὶ τῷ διξ ὑπὸ τῶν AB , $BΓ$ · καὶ συνθέντι τὰ ἀπὸ τῶν AB , $BΓ$ μετὰ τοῦ διξ ὑπὸ τῶν AB , $BΓ$, ὅπερ ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$, ἀσύμμετρον ἐστὶ τῷ ὑπὸ τῶν AB , $BΓ$. ῥητόν δὲ τὸ ὑπὸ τῶν AB , $BΓ$ · ὑπόκειται γὰρ αἱ AB , $BΓ$ ῥητόν περιέθουσαι· ἄλογον ἄρα τὸ ἀπὸ τῆς $ΑΓ$ · ἄλογός ἄρα ἡ $ΑΓ$, καλείσθω δὲ ἐκ δύο μέσων πρῶτη· ὅπερ ἔδει δεῖχαι.

is irrational.

For since AB is incommensurable in length with BC , (the sum of) the (squares) on AB and BC is thus also incommensurable with twice the (rectangle contained) by AB and BC [see previous proposition]. And, via composition, (the sum of) the (squares) on AB and BC , plus twice the (rectangle contained) by AB and BC —that is, the (square) on AC [Prop. 2.4]—is incommensurable with the (rectangle contained) by AB and BC [Prop. 10.16]. And the (rectangle contained) by AB and BC (is) rational—for AB and BC were assumed to enclose a rational (area). Thus, the (square) on AC (is) irrational. Thus, AC (is) irrational [Def. 10.4]—let it be called a first bimedral (straight-line).[‡] (Which is) the very thing it was required to show.

[†] Literally, “first from two medials”.

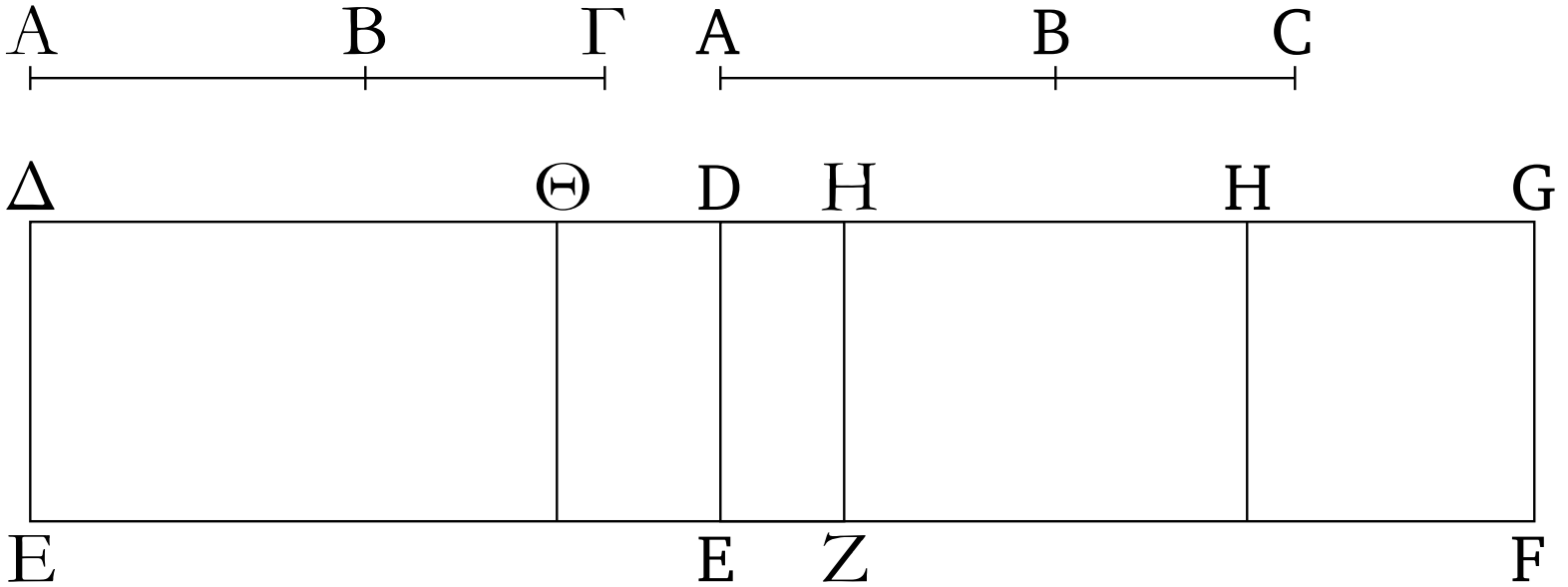
[‡] Thus, a first bimedral straight-line has a length expressible as $k^{1/4} + k^{3/4}$. The first bimedral and the corresponding first apotome of a medial, whose length is expressible as $k^{1/4} - k^{3/4}$ (see Prop. 10.74), are the positive roots of the quartic $x^4 - 2\sqrt{k}(1+k)x^2 + k(1-k)^2 = 0$.

38.

Proposition 38

Ἐὰν δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι μέσον περιέθουσαι, ἡ ὅλη ἄλογός ἐστιν, καλείσθω δὲ ἐκ δύο μέσων δευτέρα.

If two medial (straight-lines), commensurable in square only, which contain a medial (area), are added together then the whole (straight-line) is irrational—let it be called a second bimedral (straight-line).



Συγκείσθωσαν γὰρ δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ μέσον περιέθουσαι· λέγω, ὅτι ἄλογός ἐστιν ἡ $ΑΓ$.

Ἐκκείσθω γὰρ ῥητὴ ἡ $ΔΕ$, καὶ τῷ ἀπὸ τῆς $ΑΓ$ ἴσον παρὰ τὴν $ΔΕ$ παραβεβλήσθω τὸ $ΔΖ$ πλάτος ποιῶν τὴν $ΔΗ$. καὶ ἐπεὶ τὸ ἀπὸ τῆς $ΑΓ$ ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν AB , $BΓ$ καὶ τῷ διξ ὑπὸ τῶν AB , $BΓ$, παραβεβλήσθω δὴ τοῖς ἀπὸ τῶν AB , $BΓ$ παρὰ τὴν $ΔΕ$ ἴσον τὸ $ΕΘ$ · λοιπὸν ἄρα τὸ $ΘΖ$ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AB , $BΓ$. καὶ ἐπεὶ μέση ἐστὶν ἑκατέρα τῶν

For let the two medial (straight-lines), AB and BC , commensurable in square only, (and) containing a medial (area), be laid down together [Prop. 10.28]. I say that AC is irrational.

For let the rational (straight-line) DE be laid down, and let (the rectangle) DF , equal to the (square) on AC , be applied to DE , making DG as breadth [Prop. 1.44]. And since the (square) on AC is equal to (the sum of) the (squares) on AB and BC , plus twice the (rectangle contained) by AB and BC

AB , BF , μέσα ἄρα ἐστὶ καὶ τὰ ἀπὸ τῶν AB , BF . μέσον δὲ ὑπόκειται καὶ τὸ διζ ὑπὸ τῶν AB , BF . καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν AB , BF ἴσον τὸ $E\Theta$, τῷ δὲ διζ ὑπὸ τῶν AB , BF ἴσον τὸ $Z\Theta$. μέσον ἄρα ἐκάτερον τῶν $E\Theta$, ΘZ . καὶ παρὰ ῥητὴν τὴν ΔE παράκειται ῥητὴ ἄρα ἐστὶν ἐκάτερα τῶν $\Delta\Theta$, ΘH καὶ ἀσύμμετρος τῇ ΔE μήκει. ἐπεὶ οὖν ἀσύμμετρος ἐστὶν ἡ AB τῇ BF μήκει, καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν BF , οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ὑπὸ τῶν AB , BF , ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB , BF . ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BF τετραγώνων, τῷ δὲ ὑπὸ τῶν AB , BF σύμμετρον ἐστὶ τὸ διζ ὑπὸ τῶν AB , BF . ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BF τῷ διζ ὑπὸ τῶν AB , BF . ἀλλὰ τοῖς μὲν ἀπὸ τῶν AB , BF ἴσον ἐστὶ τὸ $E\Theta$, τῷ δὲ διζ ὑπὸ τῶν AB , BF ἴσον ἐστὶ τὸ ΘZ . ἀσύμμετρον ἄρα ἐστὶ τὸ $E\Theta$ τῷ ΘZ . ὥστε καὶ ἡ $\Delta\Theta$ τῇ ΘH ἐστὶν ἀσύμμετρος μήκει. αἱ $\Delta\Theta$, ΘH ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι. ὥστε ἡ ΔH ἄλογός ἐστιν. ῥητὴ δὲ ἡ ΔE . τὸ δὲ ὑπὸ ἀλόγον καὶ ῥητῆς περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν. ἄλογον ἄρα ἐστὶ τὸ ΔZ θωρίον, καὶ ἡ δυνάμεν [αὐτὸ] ἄλογός ἐστιν. δύναται δὲ τὸ ΔZ ἢ AG . ἄλογός ἄρα ἐστὶν ἡ AG , καλεῖσθω δὲ ἐκ δύο μέσων δευτέρα. ὅπερ ἔδει δεῖξαι.

[Prop. 2.4], so let (the rectangle) EH , equal to (the sum of) the squares on AB and BC , be applied to DE . The remainder HF is thus equal to twice the (rectangle contained) by AB and BC . And since AB and BC are each medial, (the sum of) the squares on AB and BC is thus also medial.[‡] And twice the (rectangle contained) by AB and BC was also assumed (to be) medial. And EH is equal to (the sum of) the squares on AB and BC , and HF (is) equal to twice the (rectangle contained) by AB and BC . Thus, EH and HF (are) each medial. And they were applied to the rational (straight-line) DE . Thus, DH and HG are each rational, and incommensurable in length with DE [Prop. 10.22]. Therefore, since AB is incommensurable in length with BC , and as AB is to BC , so the (square) on AB (is) to the (rectangle contained) by AB and BC [Prop. 10.21 lem.], the (square) on AB is thus incommensurable with the (rectangle contained) by AB and BC [Prop. 10.11]. But, the sum of the squares on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. Thus, the sum of the (squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC [Prop. 10.13]. But, EH is equal to (the sum of) the squares on AB and BC , and HF is equal to twice the (rectangle) contained by AB and BC . Thus, EH is incommensurable with HF . Hence, DH is also incommensurable in length with HG [Props. 6.1, 10.11]. Thus, DH and HG are rational (straight-lines which are) commensurable in square only. Hence, DG is irrational [Prop. 10.36]. And DE (is) rational. And the rectangle contained by irrational and rational (straight-lines) is irrational [Prop. 10.20]. The area DF is thus irrational, and (so) the square-root [of it] is irrational [Def. 10.4]. And AC is the square-root of DF . AC is thus irrational—let it be called a second bimedial (straight-line).[§] (Which is) the very thing it was required to show.

[†] Literally, “second from two medials”.

[‡] Since, by hypothesis, the squares on AB and BC are commensurable—see Props. 10.15, 10.23.

[§] Thus, a second bimedial straight-line has a length expressible as $k^{1/4} + k^{1/2}/k^{1/4}$. The second bimedial and the corresponding second apotome of a medial, whose length is expressible as $k^{1/4} - k^{1/2}/k^{1/4}$ (see Prop. 10.75), are the positive roots of the quartic $x^4 - 2[(k + k')/\sqrt{k}]x^2 + [(k - k')^2/k] = 0$.

39.

Proposition 39

Ἐάν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον, ἡ ὅλη εὐθεῖα ἄλογός ἐστιν, καλεῖσθω δὲ μείζων.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial, are added together then the whole straight-line is irrational—let it be called a major (straight-line).



Συνγείσθωσαν γὰρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι α-

For let the two straight-lines, AB and BC , incommensu-

ἡ AB , $B\Gamma$ ποιῶσαι τὰ προκείμενα· λέγω, ὅτι ἄλογός ἐστιν ἡ $ΑΓ$.

Ἐπεὶ γάρ τὸ ὑπὸ τῶν AB , $B\Gamma$ μέσον ἐστίν, καὶ τὸ διξ [ἄρα] ὑπὸ τῶν AB , $B\Gamma$ μέσον ἐστίν. τὸ δὲ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ ῥητόν· ἀσύμμετρον ἄρα ἐστὶ τὸ διξ ὑπὸ τῶν AB , $B\Gamma$ τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ · ὥστε καὶ τὰ ἀπὸ τῶν AB , $B\Gamma$ μετὰ τοῦ διξ ὑπὸ τῶν AB , $B\Gamma$, ὅπερ ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$, ἀσύμμετρόν ἐστι τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ [ῥητόν δὲ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$]· ἄλογον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$. ὥστε καὶ ἡ $ΑΓ$ ἄλογός ἐστιν, καλείσθω δὲ μείζων. ὅπερ ἔδει δεῖξαι.

able in square, and fulfilling the prescribed (conditions), be laid down together [Prop. 10.33]. I say that AC is irrational.

For since the (rectangle contained) by AB and BC is medial, twice the (rectangle contained) by AB and BC is [thus] also medial [Props. 10.6, 10.23 corr.]. And the sum of the (squares) on AB and BC (is) rational. Thus, twice the (rectangle contained) by AB and BC is incommensurable with the sum of the (squares) on AB and BC [Def. 10.4]. Hence, (the sum of) the squares on AB and BC , plus twice the (rectangle contained) by AB and BC —that is, the (square) on AC [Prop. 2.4]—is also incommensurable with the sum of the (squares) on AB and BC [Prop. 10.16] [and the sum of the (squares) on AB and BC (is) rational]. Thus, the (square) on AC is irrational. Hence, AC is also irrational [Def. 10.4]—let it be called a major (straight-line).[†] (Which is) the very thing it was required to show.

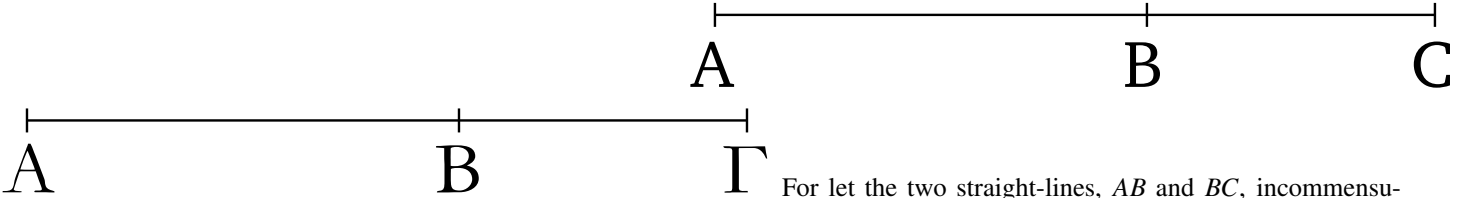
[†] Thus, a major straight-line has a length expressible as $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$. The major and the corresponding minor, whose length is expressible as $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} - \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ (see Prop. 10.76), are the positive roots of the quartic $x^4 - 2x^2 + k^2/(1 + k^2) = 0$.

40.

Proposition 40

Ἐάν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν, ἡ ὅλη εὐθεῖα ἄλογός ἐστιν, καλείσθω δὲ ῥητόν καὶ μέσον δυναμένη.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational, are added together then the whole straight-line is irrational—let it be called the square-root of a rational plus a medial (area).



Συγκείσθωσαν γάρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι α-
ἡ AB , $B\Gamma$ ποιῶσαι τὰ προκείμενα· λέγω, ὅτι ἄλογός ἐστιν ἡ $ΑΓ$.

Ἐπεὶ γάρ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ μέσον ἐστίν, τὸ δὲ διξ ὑπὸ τῶν AB , $B\Gamma$ ῥητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ τῷ διξ ὑπὸ τῶν AB , $B\Gamma$ · ὥστε καὶ τὸ ἀπὸ τῆς $ΑΓ$ ἀσύμμετρόν ἐστι τῷ διξ ὑπὸ τῶν AB , $B\Gamma$. ῥητόν δὲ τὸ διξ ὑπὸ τῶν AB , $B\Gamma$ · ἄλογον ἄρα τὸ ἀπὸ τῆς $ΑΓ$. ἄλογοξ ἄρα ἡ $ΑΓ$, καλείσθω δὲ ῥητόν καὶ μέσον δυναμένη. ὅπερ ἔδει δεῖξαι.

For let the two straight-lines, AB and BC , incommensurable in square, (and) fulfilling the prescribed (conditions), be laid down together [Prop. 10.34]. I say that AC is irrational.

For since the sum of the (squares) on AB and BC is medial, and twice the (rectangle contained) by AB and BC (is) rational, the sum of the (squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . Hence, the (square) on AC is also incommensurable with twice the (rectangle contained) by AB and BC [Prop. 10.16]. And twice the (rectangle contained) by AB and BC (is) rational. The (square) on AC (is) thus irrational. Thus, AC (is) irrational [Def. 10.4]—let it be called the square-root of a rational plus a medial (area).[†] (Which is) the very thing it was required to show.

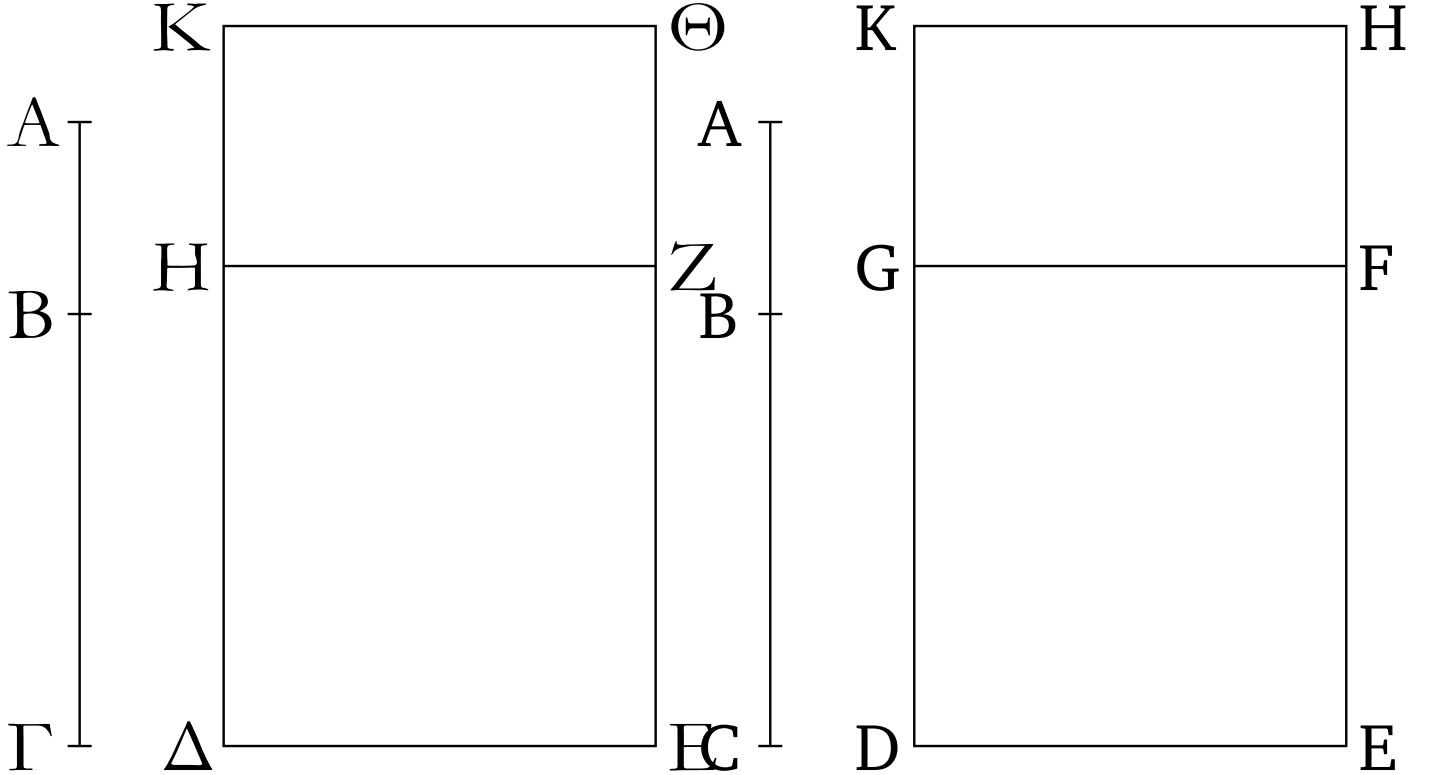
[†] Thus, the square-root of a rational plus a medial (area) has a length expressible as $\sqrt{[(1 + k^2)^{1/2} + k]/[2(1 + k^2)]} + \sqrt{[(1 + k^2)^{1/2} - k]/[2(1 + k^2)]}$. This and the corresponding irrational with a minus sign, whose length is expressible as $\sqrt{[(1 + k^2)^{1/2} + k]/[2(1 + k^2)]} - \sqrt{[(1 + k^2)^{1/2} - k]/[2(1 + k^2)]}$ (see Prop. 10.77), are the positive roots of the quartic $x^4 - (2/\sqrt{1 + k^2})x^2 + k^2/(1 + k^2)^2 = 0$.

41.

Proposition 41

Ἐάν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καί τὸ ὑπ' αὐτῶν μέσον καί ἔτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων, ἡ ὅλη εὐθεῖα ἄλογός ἐστιν, καλείσθω δὲ δύο μέσα δυναμένη.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them, are added together then the whole straight-line is irrational—let it be called the square-root of (the sum of) two medial (areas).



Συγκείσθωσαν γάρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AB , BF ποιοῦσαι τὰ προκείμενα· λέγω, ὅτι ἡ AG ἄλογός ἐστιν.

For let the two straight-lines, AB and BC , incommensurable in square, (and) fulfilling the prescribed (conditions), be laid down together [Prop. 10.35]. I say that AC is irrational.

Ἐκκείσθω ῥητὴ ἡ DE , καὶ παραβελθήσθω παρὰ τὴν DE τοῖς μὲν ἀπὸ τῶν AB , BF ἴσον τὸ ΔZ , τῷ δὲ διξ ὑπὸ τῶν AB , BF ἴσον τὸ $H\Theta$. ὅλον ἄρα τὸ $\Delta\Theta$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AG τετραγώνῳ. καὶ ἐπεὶ μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BF , καὶ ἐστιν ἴσον τῷ ΔZ , μέσον ἄρα ἐστὶ καὶ τὸ ΔZ . καὶ παρὰ ῥητὴν τὴν DE παράκειται ῥητὴ ἄρα ἐστὶν ἡ ΔH καὶ ἀσύμμετρος τῇ DE μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ HK ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ HZ , τουτέστι τῇ DE , μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB , BF τῷ διξ ὑπὸ τῶν AB , BF , ἀσύμμετρόν ἐστι τὸ ΔZ τῷ $H\Theta$. ὥστε καὶ ἡ ΔH τῇ HK ἀσύμμετρος ἐστὶν. καὶ εἰσι ῥηταί· αἱ ΔH , HK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἄλογος ἄρα ἐστὶν ἡ ΔK ἡ καλουμένη ἐκ δύο ὀνομάτων. ῥητὴ δὲ ἡ DE · ἄλογον ἄρα ἐστὶ τὸ $\Delta\Theta$ καὶ ἡ δυναμένη αὐτὸ ἄλογός ἐστιν. δύναται δὲ τὸ $\Theta\Delta$ ἡ AG . ἄλογος ἄρα ἐστὶν ἡ AG , καλείσθω δὲ δύο μέσα δυναμένη. ὅπερ ἔδει δεῖξαι.

Let the rational (straight-line) DE be laid out, and let (the rectangle) DF , equal to (the sum of) the (squares) on AB and BC , and (the rectangle) GH , equal to twice the (rectangle contained) by AB and BC , be applied to DE . Thus, the whole of DH is equal to the square on AC [Prop. 2.4]. And since the sum of the (squares) on AB and BC is medial, and is equal to DF , DF is thus also medial. And it is applied to the rational (straight-line) DE . Thus, DG is rational, and incommensurable in length with DE [Prop. 10.22]. So, for the same (reasons), GK is also rational, and incommensurable in length with GF —that is to say, DE . And since (the sum of) the (squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC , DF is incommensurable with GH . Hence, DG is also incommensurable (in length) with GK [Props. 6.1, 10.11]. And they are rational. Thus, DG and GK are rational (straight-lines which are) commensurable in

square only. Thus, DK is irrational, and that (straight-line which is) called binomial [Prop. 10.36]. And DE (is) rational. Thus, DH is irrational, and its square-root is irrational [Def. 10.4]. And AC (is) the square-root of HD . Thus, AC is irrational—let it be called the square-root of (the sum of) two medial (areas).[†] (Which is) the very thing it was required to show.

[†] Thus, the square-root of (the sum of) two medial (areas) has a length expressible as $k^{1/4} \left(\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} \right)$. This and the corresponding irrational with a minus sign, whose length is expressible as $k^{1/4} \left(\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} - \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} \right)$ (see Prop. 10.78), are the positive roots of the quartic $x^4 - 2k^{1/2}x^2 + k'k^2/(1 + k^2) = 0$.

Λήμμα.

Lemma

Ὅτι δὲ αἱ εἰρημέναι ἄλογοι μοναθῶς διαιροῦνται εἰς τὰς εὐθείας, ἐχ' ὧν σύγκεινται ποιουσῶν τὰ προκείμενα εἶδη, δεῖχόμεν ἥδη προεκθέμενοι λημμάτων τοιοῦτον·

We will now demonstrate that the aforementioned irrational (straight-lines) are uniquely divided into the straight-lines of which they are the sum, and which produce the prescribed types, (after) setting forth the following lemma.



Ἐκκείσθω εὐθεῖα ἡ AB καὶ τετμήσθω ἡ ὅλη εἰς ἅμισα καθ' ἑκάτερον τῶν Γ , Δ , ὑποκείσθω δὲ μείζων ἡ AG τῆς ΔB . λέγω, ὅτι τὰ ἀπὸ τῶν AG , GB μείζονά ἐστι τῶν ἀπὸ τῶν AD , ΔB .

Let the straight-line AB be laid out, and let the whole (straight-line) be cut into unequal parts at each of the (points) C and D . And let AC be assumed (to be) greater than DB . I say that (the sum of) the (squares) on AC and CB is greater than (the sum of) the (squares) on AD and DB .

Τετμήσθω γάρ ἡ AB διὰ κατὰ τὸ E . καὶ ἐπεὶ μείζων ἐστὶν ἡ AG τῆς ΔB , κοινὴ ἀφηρήσθω ἡ $\Delta\Gamma$. λοιπὴ ἄρα ἡ $A\Delta$ λοιπῆς τῆς GB μείζων ἐστίν. ἴση δὲ ἡ AE τῇ EB . ἐλάττων ἄρα ἡ ΔE τῆς $E\Gamma$. τὰ Γ , Δ ἄρα σημεία οὐκ ἴσον ἀπέθουσι τῆς διχοτομίας. καὶ ἐπεὶ τὸ ὑπὸ τῶν AG , GB μετὰ τοῦ ἀπὸ τῆς $E\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς EB , ἀλλὰ μὴν καὶ τὸ ὑπὸ τῶν $A\Delta$, ΔB μετὰ τοῦ ἀπὸ ΔE ἴσον ἐστὶ τῷ ἀπὸ τῆς EB , τὸ ἄρα ὑπὸ τῶν AG , GB μετὰ τοῦ ἀπὸ τῆς $E\Gamma$ ἴσον ἐστὶ τῷ ὑπὸ τῶν $A\Delta$, ΔB μετὰ τοῦ ἀπὸ τῆς ΔE . ὧν τὸ ἀπὸ τῆς ΔE ἔλασσόν ἐστι τοῦ ἀπὸ τῆς $E\Gamma$. καὶ λοιπὸν ἄρα τὸ ὑπὸ τῶν AG , GB ἔλασσόν ἐστι τοῦ ὑπὸ τῶν $A\Delta$, ΔB . ὥστε καὶ τὸ διζὺ ὑπὸ τῶν AG , GB ἔλασσόν ἐστι τοῦ διζὺ ὑπὸ τῶν $A\Delta$, ΔB . καὶ λοιπὸν ἄρα τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μείζον ἐστὶ τοῦ συγκειμένου ἐκ τῶν ἀπὸ τῶν $A\Delta$, ΔB . ὅπερ ἔδει δεῖχαι.

For let AB be cut in half at E . And since AC is greater than DB , let DC be subtracted from both. Thus, the remainder AD is greater than the remainder CB . And AE (is) equal to EB . Thus, DE (is) less than EC . Thus, points C and D are not equally far from the point of bisection. And since the (rectangle contained) by AC and CB , plus the (square) on EC , is equal to the (square) on EB [Prop. 2.5], but, moreover, the (rectangle contained) by AD and DB , plus the (square) on DE , is also equal to the (square) on EB [Prop. 2.5], the (rectangle contained) by AC and CB , plus the (square) on EC , is thus equal to the (rectangle contained) by AD and DB , plus the (square) on DE . And, of these, the (square) on DE is less than the (square) on EC . And, thus, the remaining (rectangle contained) by AC and CB is less than the (rectangle contained) by AD and DB . And, hence, twice the (rectangle contained) by AC and CB is less than twice the (rectangle contained) by AD and DB . And thus the remaining sum of the (squares) on AC and CB is greater than the sum of the (squares) on AD and DB .[†] (Which is) the very thing it was required to show.

[†] Since, $AC^2 + CB^2 + 2AC \cdot CB = AD^2 + DB^2 + 2AD \cdot DB = AB^2$.

42.

Proposition 42

Ἡ ἐκ δύο ὀνομάτων κατὰ ἓν μόνον σημεῖον διαιρεῖται εἰς τὰ ὀνόματα.

A binomial (straight-line) can be divided into its (component) terms at one point only.[†]



Ἐστω ἐκ δύο ὀνομάτων ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ . αἱ AG , GB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται εἰς δύο ῥητάς δυνάμει μόνον συμμέτρους.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ , ὥστε καὶ τὰς $A\Delta$, ΔB ῥητάς εἶναι δυνάμει μόνον συμμέτρους. φανερόν δὲ, ὅτι ἡ AG τῇ ΔB οὐκ ἔστιν ἡ αὐτή. εἰ γὰρ δυνατόν, ἔστω. ἔσται δὲ καὶ ἡ $A\Delta$ τῇ GB ἡ αὐτή· καὶ ἔσται ὥς ἡ AG πρὸς τὴν GB , οὕτως ἡ $B\Delta$ πρὸς τὴν ΔA , καὶ ἔσται ἡ AB κατὰ τὸ αὐτὸ τῇ κατὰ τὸ Γ διαιρέσει διαιρεθεῖσα καὶ κατὰ τὸ Δ . ὅπερ οὐθ' ὑπόκειται. οὐκ ἄρα ἡ AG τῇ ΔB ἔστιν ἡ αὐτή. διὰ δὲ τοῦτο καὶ τὰ Γ , Δ σημεῖα οὐκ ἴσον ἀπέθουσι τῆς διχοτομίας. ὥ' ἄρα διαφέρει τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν $A\Delta$, ΔB , τούτω διαφέρει καὶ τὸ διξ ὑπὸ τῶν $A\Delta$, ΔB τοῦ διξ ὑπὸ τῶν AG , GB διὰ τὸ καὶ τὰ ἀπὸ τῶν AG , GB μετὰ τοῦ διξ ὑπὸ τῶν AG , GB καὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB μετὰ τοῦ διξ ὑπὸ τῶν $A\Delta$, ΔB ἴσα εἶναι τῶν ἀπὸ τῆς AB . ἀλλὰ τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν $A\Delta$, ΔB διαφέρει ῥητῶ· ῥητὰ γὰρ ἀμφοτέρω· καὶ τὸ διξ ἄρα ὑπὸ τῶν $A\Delta$, ΔB τοῦ διξ ὑπὸ τῶν AG , GB διαφέρει ῥητῶ μέσα ὄντα· ὅπερ ἄτοπον· μέσον γὰρ μέσον οὐθ' ὑπερέχει ῥητῶ.

Οὐθ' ἄρα ἡ ἐκ δύο ὀνομάτων κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται καθ' ἐν ἄρα μόνον· ὅπερ ἔδει δεῖχαι.

Let AB be a binomial (straight-line) which has been divided into its (component) terms at C . AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. I say that AB cannot be divided at another point into two rational (straight-lines which are) commensurable in square only.

For, if possible, let it also be divided at D , such that AD and DB are also rational (straight-lines which are) commensurable in square only. So, (it is) clear that AC is not the same as DB . For, if possible, let it be (the same). So, AD will also be the same as CB . And as AC will be to CB , so BD (will be) to DA . And AB will (thus) also be divided at D in the same (manner) as the division at C . The very opposite was assumed. Thus, AC is not the same as DB . So, on account of this, points C and D are not equally far from the point of bisection. Thus, by whatever (amount the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB , twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by this (same amount)—on account of both (the sum of) the (squares) on AC and CB , plus twice the (rectangle contained) by AC and CB , and (the sum of) the (squares) on AD and DB , plus twice the (rectangle contained) by AD and DB , being equal to the (square) on AB [Prop. 2.4]. But, (the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB by a rational (area). For (they are) both rational (areas). Thus, twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by a rational (area, despite both being medial (areas) [Prop. 10.21]. The very thing is absurd. For a medial (area) cannot exceed a medial (area) by a rational (area) [Prop. 10.26].

Thus, a binomial (straight-line) cannot be divided (into its component terms) at different points. Thus, (it can be so divided) at one point only. (Which is) the very thing it was required to show.

[†] In other words, $k + k^{1/2} = k'' + k'''^{1/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$. Likewise, $k^{1/2} + k'^{1/2} = k''^{1/2} + k'''^{1/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$ (or, equivalently, $k'' = k'$ and $k''' = k$).

43.

Proposition 43

Ἡ ἐκ δύο μέσων πρώτη καθ' ἐν μόνον σημεῖον διαιρεῖται.

A first bimedral (straight-line) can be divided (into its component terms) at one point only.[†]



Ἐστω ἐκ δύο μέσων πρώτη ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε τὰς AG , GB μέσας εἶναι δυνάμει μόνον συμμέτρους

Let AB be a first bimedral (straight-line) which has been divided at C , such that AC and CB are medial (straight-lines),

ῥητὸν περιεθούσαξ· λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν διηγήσθω καὶ κατὰ τὸ Δ , ὥστε καὶ τὰς AD , DB μέσαξ εἶναι δυνάμει μόνον συμμετρωνξ ῥητὸν περιεθούσαξ. ἐπεὶ οὖν, ὧ διαφέρει τὸ διξ ὑπὸ τῶν AD , DB τοῦ διξ ὑπὸ τῶν AG , GB , τούτω διαφέρει τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν AD , DB , ῥητῷ δὲ διαφέρει τὸ διξ ὑπὸ τῶν AD , DB τοῦ διξ ὑπὸ τῶν AG , GB . ῥητὰ γὰρ ἀμφοτέρω· ῥητῷ ἄρα διαφέρει καὶ τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν AD , DB μέσα ὄντα· ὅπερ ἄτοπον.

Οὐκ ἄρα ἡ ἐκ δύο μέσων πρώτη κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται εἰς τὰ ὀνόματα· καθ' ἐν ἄρα μόνον· ὅπερ ἔδει δεῖξαι.

commensurable in square only, (and) containing a rational (area) [Prop. 10.37]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , such that AD and DB are also medial (straight-lines), commensurable in square only, (and) containing a rational (area). Since, therefore, by whatever (amount) twice the (rectangle contained) by AD and DB differs from twice the (rectangle contained) by AC and CB , (the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB by this (same amount) [Prop. 10.41 lem.]. And twice the (rectangle contained) by AD and DB differs from twice the (rectangle contained) by AC and CB by a rational (area). For (they are) both rational (areas). (The sum of) the (squares) on AC and CB thus differs from (the sum of) the (squares) on AD and DB by a rational (area, despite both) being medial (areas). The very thing is absurd [Prop. 10.26].

Thus, a first bimedral (straight-line) cannot be divided into its (component) terms at different points. Thus, (it can be so divided) at one point only. (Which is) the very thing it was required to show.

[†] In other words, $k^{1/4} + k^{3/4} = k'^{1/4} + k'^{3/4}$ has only one solution: i.e., $k' = k$.

44.

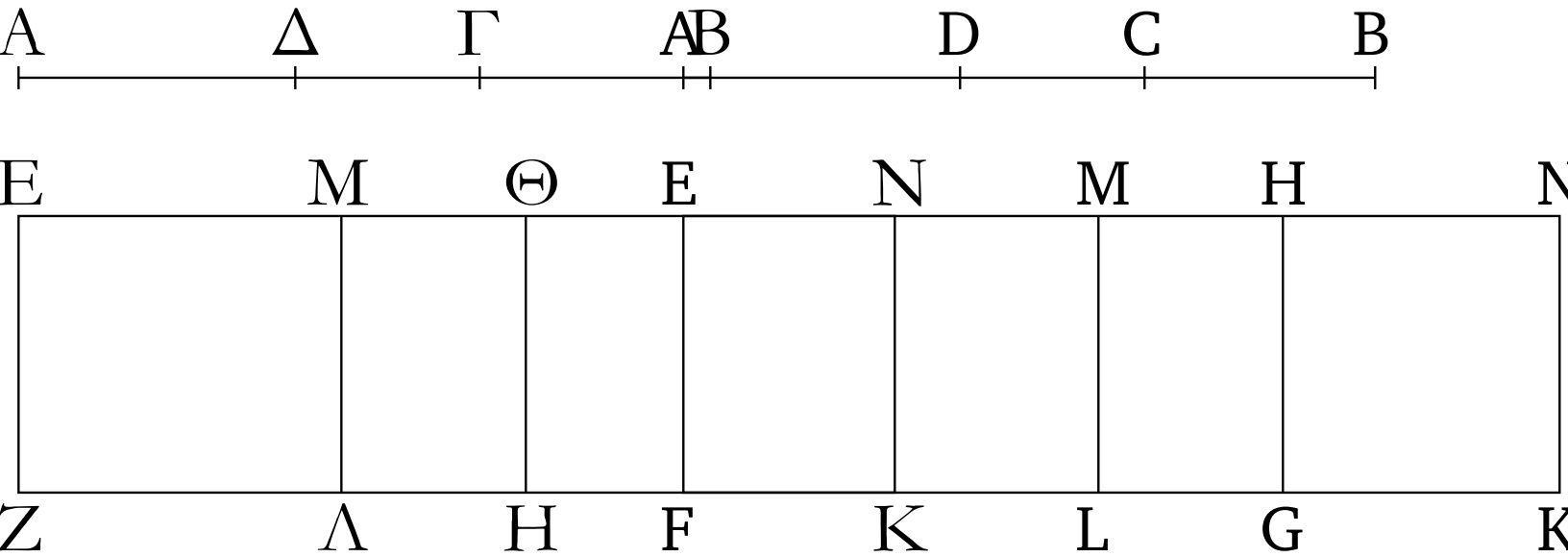
Ἡ ἐκ δύο μέσων δευτέρα καθ' ἐν μόνον σημεῖον διαιρεῖται.

Ἐστω ἐκ δύο μέσων δευτέρα ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε τὰς AG , GB μέσαξ εἶναι δυνάμει μόνον συμμετρωνξ μέσον περιεθούσαξ· φανερόν δὲ, ὅτι τὸ Γ οὐκ ἔστι κατὰ τῆς διχοτομίας, ὅτι οὐκ εἰσὶ μήκει σύμμετροι. λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται.

Proposition 44

A second bimedral (straight-line) can be divided (into its component terms) at one point only.[†]

Let AB be a second bimedral (straight-line) which has been divided at C , so that AC and BC are medial (straight-lines), commensurable in square only, (and) containing a medial (area) [Prop. 10.38]. So, (it is) clear that C is not (located) at the point of bisection, since (AC and BC) are not commensurable in length. I say that AB cannot be (so) divided at another point.



Εἰ γὰρ δυνατόν, διηρησθῶ καὶ κατὰ τὸ Δ, ὥστε τὴν ΑΓ τῇ ΔΒ μὴ εἶναι τὴν αὐτὴν, ἀλλὰ μείζονα καθ' ὑπόθεσιν τὴν ΑΓ· δῆλον δὲ, ὅτι καὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ, ὡς ἐπάνω ἐδείξαμεν, ἐλάσσονα τῶν ἀπὸ τῶν ΑΓ, ΓΒ· καὶ τὰς ΑΔ, ΔΒ μέσα εἶναι δυνάμει μόνον συμμετρους μέσον περιεχούσας. καὶ ἐκκεῖσθω ῥητὴ ἡ ΕΖ, καὶ τῷ μὲν ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΕΖ παραλληλόγραμμον ὀρθογώνιον παραβεβλήσθω τὸ ΕΚ, τοῖς δὲ ἀπὸ τῶν ΑΓ, ΓΒ ἴσον ἀφηρησθῶ τὸ ΕΗ· λοιπὸν ἄρα τὸ ΘΚ ἴσον ἐστὶ τῷ διζῷ ὑπὸ τῶν ΑΓ, ΓΒ. πάλιν δὲ τοῖς ἀπὸ τῶν ΑΔ, ΔΒ, ἄπερ ἐλάσσονα ἐδείχθη τῶν ἀπὸ τῶν ΑΓ, ΓΒ, ἴσον ἀφηρησθῶ τὸ ΕΛ· καὶ λοιπὸν ἄρα τὸ ΜΚ ἴσον τῷ διζῷ ὑπὸ τῶν ΑΔ, ΔΒ. καὶ ἐπεὶ μέσα ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ, μέσον ἄρα [καὶ] τὸ ΕΗ. καὶ παρὰ ῥητὴν τὴν ΕΖ παράκειται· ῥητὴ ἄρα ἐστὶν ἡ ΕΘ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. διὰ τὰ αὐτὰ δὲ καὶ ἡ ΘΝ ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. καὶ ἐπεὶ αἱ ΑΓ, ΓΒ μέσα εἰσὶ δυνάμει μόνον σύμμετροι, ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΓ τῇ ΓΒ μήκει. ὡς δὲ ἡ ΑΓ πρὸς τὴν ΓΒ, οὕτως τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ὑπὸ τῶν ΑΓ, ΓΒ· ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΓ τῷ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΓ σύμμετρά ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ· δυνάμει γὰρ εἰσὶ σύμμετροι αἱ ΑΓ, ΓΒ. τῷ δὲ ὑπὸ τῶν ΑΓ, ΓΒ σύμμετρόν ἐστὶ τὸ διζῷ ὑπὸ τῶν ΑΓ, ΓΒ. καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ ἄρα ἀσύμμετρά ἐστὶ τῷ διζῷ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ ἴσον ἐστὶ τὸ ΕΗ, τῷ δὲ διζῷ ὑπὸ τῶν ΑΓ, ΓΒ ἴσον τὸ ΘΚ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΕΗ τῷ ΘΚ· ὥστε καὶ ἡ ΕΘ τῇ ΘΝ ἀσύμμετρος ἐστὶ μήκει. καὶ εἰσὶ ῥηταί· αἱ ΕΘ, ΘΝ ἄρα ῥηταὶ εἰσὶ δυνάμει μόνον σύμμετροι. ἔαν δὲ δύο ῥηταὶ δυνάμει μόνον σύμμετροι συντεθῶσιν, ἡ ὅλη ἄλογός ἐστὶν ἡ καλουμένη ἐκ δύο ὀνομάτων· ἡ ΕΝ ἄρα ἐκ δύο ὀνομάτων ἐστὶ διηρημένη κατὰ τὸ Θ. κατὰ τὰ αὐτὰ δὲ δευθῆσονται καὶ αἱ ΕΜ, ΜΝ ῥηταὶ δυνάμει μόνον σύμμετροι· καὶ ἔσται ἡ ΕΝ ἐκ δύο ὀνομάτων κατ' ἄλλο καὶ ἄλλο διηρημένη τὸ τε Θ καὶ τὸ Μ, καὶ οὐκ ἔστιν ἡ ΕΘ τῇ ΜΝ ἢ αὐτῇ, ὅτι τὰ ἀπὸ τῶν

For, if possible, let it also have been (so) divided at D , so that AC is not the same as DB , but AC (is), by hypothesis, greater. So, (it is) clear that (the sum of) the (squares) on AD and DB is also less than (the sum of) the (squares) on AC and CB , as we showed above [Prop. 10.41 lem.]. And AD and DB are medial (straight-lines), commensurable in square only, (and) containing a medial (area). And let the rational (straight-line) EF be laid down. And let the rectangular parallelogram EK , equal to the (square) on AB , be applied to EF . And let EG , equal to (the sum of) the (squares) on AC and CB , be cut off (from EK). Thus, the remainder, HK , is equal to twice the (rectangle contained) by AC and CB [Prop. 2.4]. So, again, let EL , equal to (the sum of) the (squares) on AD and DB —which was shown (to be) less than (the sum of) the (squares) on AC and CB —be cut off (from EK). And, thus, the remainder, MK , (is) equal to twice the (rectangle contained) by AD and DB . And since (the sum of) the (squares) on AC and CB is medial, EG (is) thus [also] medial. And it is applied to the rational (straight-line) EF . Thus, EH is rational, and incommensurable in length with EF [Prop. 10.22]. So, for the same (reasons), HN is also rational, and incommensurable in length with EF . And since AC and CB are medial (straight-lines which are) commensurable in square only, AC is thus incommensurable in length with CB . And as AC (is) to CB , so the (square) on AC (is) to the (rectangle contained) by AC and CB [Prop. 10.21 lem.]. Thus, the (square) on AC is incommensurable with the (rectangle contained) by AC and CB [Prop. 10.11]. But, (the sum of) the (squares) on AC and CB is commensurable with the (square) on AC . For, AC and CB are commensurable in square [Prop. 10.15]. And twice the (rectangle contained) by AC and CB is commensurable with the (rectangle contained) by AC and CB [Prop. 10.6]. And thus (the sum of) the squares on AC and CB is incommen-

$ΑΓ, ΓΒ$ μείζονά ἐστι τῶν ἀπὸ τῶν $ΑΔ, ΔΒ$. ἀλλὰ τὰ ἀπὸ τῶν $ΑΔ, ΔΒ$ μείζονά ἐστι τοῦ διξ ὑπὸ $ΑΔ, ΔΒ$. πολλῶν ἄρα καὶ τὰ ἀπὸ τῶν $ΑΓ, ΓΒ$, τουτέστι τὸ $ΕΗ$, μείζον ἐστι τοῦ διξ ὑπὸ τῶν $ΑΔ, ΔΒ$, τουτέστι τοῦ $ΜΚ$. ὥστε καὶ ἡ $ΕΘ$ τῆς $ΜΝ$ μείζων ἐστίν. ἡ ἄρα $ΕΘ$ τῇ $ΜΝ$ οὐκ ἔστιν ἡ αὐτή· ὅπερ ἔδει δεῖχαι.

surable with twice the (rectangle contained) by AC and CB [Prop. 10.13]. But, EG is equal to (the sum of) the (squares) on AC and CB , and HK equal to twice the (rectangle contained) by AC and CB . Thus, EG is incommensurable with HK . Hence, EH is also incommensurable in length with HN [Props. 6.1, 10.11]. And (they are) rational (straight-lines). Thus, EH and HN are rational (straight-lines which are) commensurable in square only. And if two rational (straight-lines which are) commensurable in square only are added together then the whole (straight-line) is that irrational called binomial [Prop. 10.36]. Thus, EN is a binomial (straight-line) which has been divided (into its component terms) at H . So, according to the same (reasoning), EM and MN can be shown (to be) rational (straight-lines which are) commensurable in square only. And EN will (thus) be a binomial (straight-line) which has been divided (into its component terms) at the different (points) H and M (which is absurd [Prop. 10.42]). And EH is not the same as MN , since (the sum of) the (squares) on AC and CB is greater than (the sum of) the (squares) on AD and DB . But, (the sum of) the (squares) on AD and DB is greater than twice the (rectangle contained) by AD and DB [Prop. 10.59 lem.]. Thus, (the sum of) the (squares) on AC and CB —that is to say, EG —is also much greater than twice the (rectangle contained) by AD and DB —that is to say, MK . Hence, EH is also greater than MN [Prop. 6.1]. Thus, EH is not the same as MN . (Which is) the very thing it was required to show.

† In other words, $k^{1/4} + k^{1/2}/k^{1/4} = k'^{1/4} + k'^{3/4}/k'^{1/4}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$.

45.

Proposition 45

Ἡ μείζων κατὰ τὸ αὐτὸ μόνον σημεῖον διαιρεῖται.

A major (straight-line) can only be divided (into its component terms) at the same point.†



Ἐστω μείζων ἡ $ΑΒ$ διηρημένη κατὰ τὸ $Γ$, ὥστε τὰς $ΑΓ, ΓΒ$ δυνάμει ἀσυμμέτρον εἶναι ποιούσας τὸ μὲν συγκεκείμενον ἐκ τῶν ἀπὸ τῶν $ΑΓ, ΓΒ$ τετραγώνων ῥητόν, τὸ δ' ὑπὸ τῶν $ΑΓ, ΓΒ$ μέσον· λέγω, ὅτι ἡ $ΑΒ$ κατ' ἄλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ $Δ$, ὥστε καὶ τὰς $ΑΔ, ΔΒ$ δυνάμει ἀσυμμέτρον εἶναι ποιούσας τὸ μὲν συγκεκείμενον ἐκ τῶν ἀπὸ τῶν $ΑΔ, ΔΒ$ ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον. καὶ ἐπεὶ, ὥς διαφέρει τὰ ἀπὸ τῶν $ΑΓ, ΓΒ$ τῶν ἀπὸ τῶν $ΑΔ, ΔΒ$, τούτῳ διαφέρει καὶ τὸ διξ ὑπὸ τῶν $ΑΔ, ΔΒ$ τοῦ διξ ὑπὸ τῶν $ΑΓ, ΓΒ$, ἀλλὰ τὰ ἀπὸ τῶν $ΑΓ, ΓΒ$ τῶν ἀπὸ τῶν $ΑΔ, ΔΒ$ ὑπερέθει ῥητῶ· ῥητὰ γὰρ ἀμφοτέρω· καὶ τὸ διξ ὑπὸ τῶν $ΑΔ, ΔΒ$ ἄρα τοῦ διξ ὑπὸ τῶν $ΑΓ, ΓΒ$ ὑπερέθει ῥητῶ μέσα ὄντα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ μείζων κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται· κατὰ τὸ αὐτὸ ἄρα μόνον διαιρεῖται.

Let AB be a major (straight-line) which has been divided at C , so that AC and CB are incommensurable in square, making the sum of the squares on AC and CB rational, and the (rectangle contained) by AC and CD medial [Prop. 10.39]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , such that AD and DB are also incommensurable in square, making the sum of the (squares) on AD and DB rational, and the (rectangle contained) by them medial. And since, by whatever (amount the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB , twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by this (same amount). But, (the sum of) the (squares) on AC and CB exceeds (the sum of) the

ὅπερ ἔδει δεῖχαι.

(squares) on AD and DB by a rational (area). For (they are) both rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area), (despite both) being medial (areas). The very thing is impossible [Prop. 10.26]. Thus, a major (straight-line) cannot be divided (into its component terms) at different points. Thus, it can only be (so) divided at the same (point). (Which is) the very thing it was required to show.

[†] In other words, $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} = \sqrt{[1 + k'/(1 + k'^2)^{1/2}]/2} + \sqrt{[1 - k'/(1 + k'^2)^{1/2}]/2}$ has only one solution: i.e., $k' = k$.

46.

Proposition 46

Ἡ ῥητὸν καὶ μέσον δυναμένη καθ' ἓν μόνον σημεῖον διαιρεῖται.

The square-root of a rational plus a medial (area) can be divided (into its component terms) at one point only.[†]



Ἐστω ῥητὸν καὶ μέσον δυναμένη ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε τὰς AG , GB δυνάμει ἀσυνμέτρους εἶναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσον, τὸ δὲ διξ ὑπὸ τῶν AG , GB ῥητὸν· λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ , ὥστε καὶ τὰς $A\Delta$, ΔB δυνάμει ἀσυνμέτρους εἶναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $A\Delta$, ΔB μέσον, τὸ δὲ διξ ὑπὸ τῶν $A\Delta$, ΔB ῥητὸν. ἐπεὶ οὖν, ὅσον διαφέρει τὸ διξ ὑπὸ τῶν AG , GB τοῦ διξ ὑπὸ τῶν $A\Delta$, ΔB , τούτω διαφέρει καὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB τῶν ἀπὸ τῶν AG , GB , τὸ δὲ διξ ὑπὸ τῶν AG , GB τοῦ διξ ὑπὸ τῶν $A\Delta$, ΔB ὑπερέθει ῥητῶ, καὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB ἄρα τῶν ἀπὸ τῶν AG , GB ὑπερέθει ῥητῶ μέσα ὄντα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ῥητὸν καὶ μέσον δυναμένη κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται. κατὰ ἓν ἄρα σημεῖον διαιρεῖται· ὅπερ ἔδει δεῖχαι.

Let AB be the square-root of a rational plus a medial (area) which has been divided at C , so that AC and CB are incommensurable in square, making the sum of the (squares) on AC and CB medial, and twice the (rectangle contained) by AC and CB rational [Prop. 10.40]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , so that AD and DB are also incommensurable in square, making the sum of the (squares) on AD and DB medial, and twice the (rectangle contained) by AD and DB rational. Therefore, since by whatever (amount) twice the (rectangle contained) by AC and CB differs from twice the (rectangle contained) by AD and DB , (the sum of) the (squares) on AD and DB also differs from (the sum of) the (squares) on AC and CB by this (same amount). And twice the (rectangle contained) by AC and CB exceeds twice the (rectangle contained) by AD and DB by a rational (area). (The sum of) the (squares) on AD and DB thus also exceeds (the sum of) the (squares) on AC and CB by a rational (area), (despite both) being medial (areas). The very thing is impossible [Prop. 10.26]. Thus, the square-root of a rational plus a medial (area) cannot be divided (into its component terms) at different points. Thus, it can be (so) divided at one point (only). (Which is) the very thing it was required to show.

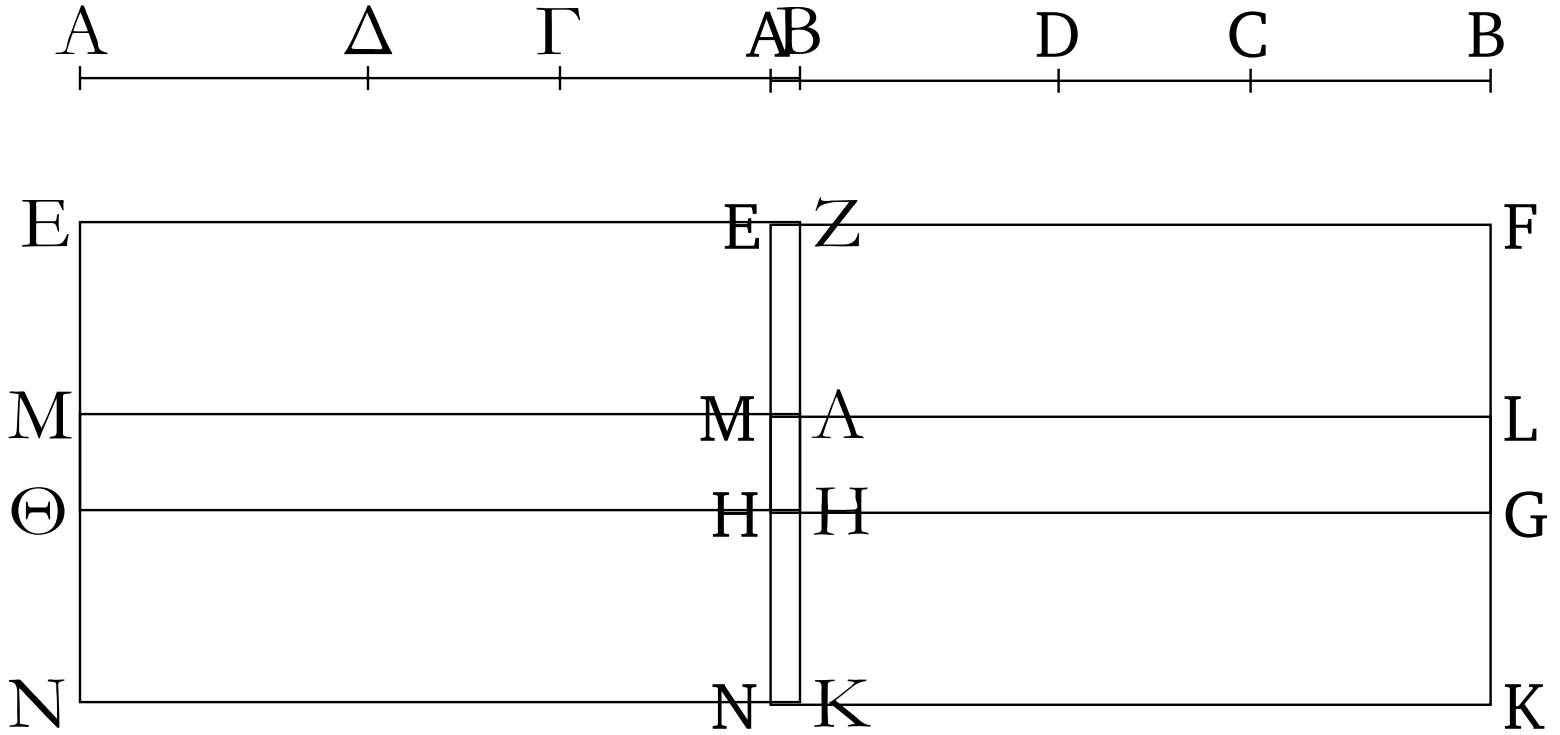
[†] In other words, $\sqrt{[(1 + k^2)^{1/2} + k]/[2(1 + k^2)]} + \sqrt{[(1 + k^2)^{1/2} - k]/[2(1 + k^2)]} = \sqrt{[(1 + k'^2)^{1/2} + k']/[2(1 + k'^2)]} + \sqrt{[(1 + k'^2)^{1/2} - k']/[2(1 + k'^2)]}$ has only one solution: i.e., $k' = k$.

47.

Proposition 47

Ἡ δύο μέσα δυναμένη καθ' ἓν μόνον σημεῖον διαιρεῖται.

The square-root of (the sum of) two medial (areas) can be divided (into its component terms) at one point only.[†]



Ἐστω [δύο μέσα δυναμένη] ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε τὰς AG , GB δυνάμει ἀσυνμέτρους εἶναι ποιοῦσαξ τὸ τε συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσων καὶ τὸ ὑπὸ τῶν AG , GB μέσων καὶ ἔτι ἀσύμμετρον τῷ συγκεμένῳ ἐκ τῶν ἀπ' αὐτῶν. λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται ποιοῦσα τὰ προκείμενα.

Εἰ γὰρ δυνατόν, διηρήσθω κατὰ τὸ Δ , ὥστε πάλιν δηλονότι τὴν AG τῇ ΔB μὴ εἶναι τὴν αὐτήν, ἀλλὰ μείζονα καθ' ὑπόθεσιν τὴν AG , καὶ ἐκλείσθω ῥητὴ ἡ EZ , καὶ παραβεβλήσθω παρὰ τὴν EZ τοῖς μὲν ἀπὸ τῶν AG , GB ἴσον τὸ EH , τῷ δὲ διξ ὑπὸ τῶν AG , GB ἴσον τὸ ΘK . ὅλον ἄρα τὸ EK ἴσον ἐστὶ τῷ ἀπὸ τῆς AB τετραγώνῳ. πάλιν δὲ παραβεβλήσθω παρὰ τὴν EZ τοῖς ἀπὸ τῶν $A\Delta$, ΔB ἴσον τὸ EL . λοιπὸν ἄρα τὸ διξ ὑπὸ τῶν $A\Delta$, ΔB λοιπῷ τῷ MK ἴσον ἐστίν. καὶ ἐπεὶ μέσον ὑπόκειται τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσων ἄρα ἐστὶ καὶ τὸ EH . καὶ παρὰ ῥητὴν τὴν EZ παράκειται ῥητὴ ἄρα ἐστὶν ἡ ΘE καὶ ἀσύμμετρος τῇ EZ μήκει. διὰ τὰ αὐτὰ δὲ καὶ ἡ ΘN ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB τῷ διξ ὑπὸ τῶν AG , GB , καὶ τὸ EH ἄρα τῷ HN ἀσύμμετρόν ἐστιν. ὥστε καὶ ἡ $E\Theta$ τῇ ΘN ἀσύμμετρός ἐστιν. καὶ εἰσι ῥηταί· αἱ $E\Theta$, ΘN ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ EN ἄρα ἐκ δύο ὀνομάτων ἐστὶ διηρημένη κατὰ τὸ Θ . ὁμοίως δὲ δείχουμεν, ὅτι καὶ κατὰ τὸ M διήρηται. καὶ οὐκ ἐστὶν ἡ $E\Theta$ τῇ MN ἡ αὐτή· ἡ ἄρα ἐκ δύο ὀνομάτων κατ' ἄλλο καὶ ἄλλο σημεῖον διήρηται. ὅπερ ἐστὶν ἄτοπον. οὐκ ἄρα ἡ δύο μέσα δυναμένη κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται· καθ' ἓν ἄρα μόνον [σημεῖον] διαιρεῖται.

Let AB be [the square-root of (the sum of) two medial (areas)] which has been divided at C , such that AC and CB are incommensurable in square, making the sum of the (squares) on AC and CB medial, and the (rectangle contained) by AC and CB medial, and, moreover, incommensurable with the sum of the (squares) on (AC and CB) [Prop. 10.41]. I say that AB cannot be divided at another point fulfilling the prescribed (conditions).

For, if possible, let it be divided at D , such that AC is again manifestly not the same as DB , but AC (is), by hypothesis, greater. And let the rational (straight-line) EF be laid down. And let EG , equal to (the sum of) the (squares) on AC and CB , and HK , equal to twice the (rectangle contained) by AC and CB , be applied to EF . Thus, the whole of EK is equal to the square on AB [Prop. 2.4]. So, again, let EL , equal to (the sum of) the (squares) on AD and DB , be applied to EF . Thus, the remainder—twice the (rectangle contained) by AD and DB —is equal to the remainder, MK . And since the sum of the (squares) on AC and CB was assumed (to be) medial, EG is also medial. And it is applied to the rational (straight-line) EF . HE is thus rational, and incommensurable in length with EF [Prop. 10.22]. So, for the same (reasons), HN is also rational, and incommensurable in length with EF . And since the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , EG is thus also incommensurable with GN . Hence, EH is also incommensurable with HN [Props. 6.1, 10.11]. And they are (both) rational (straight-lines). Thus, EH and HN are rational (straight-lines).

which are) commensurable in square only. Thus, EN is a binomial (straight-line) which has been divided (into its component terms) at H [Prop. 10.36]. So, similarly, we can show that it has also been (so) divided at M . And EH is not the same as MN . Thus, a binomial (straight-line) has been divided (into its component terms) at different points. The very thing is absurd [Prop. 10.42]. Thus, the square-root of (the sum of) two medial (areas) cannot be divided (into its component terms) at different points. Thus, it can be (so) divided at one [point] only.

[†] In other words, $k'^{1/4} \sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + k'^{1/4} \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} = k''^{1/4} \sqrt{[1 + k''/(1 + k''^2)^{1/2}]/2} + k''^{1/4} \sqrt{[1 - k''/(1 + k''^2)^{1/2}]/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$.

Ὅροι δεῦτεροι.

5. Ὑποκειμένης ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων διηρημένης εἰς τὰ ὀνόματα, ἥς τὸ μείζον ὄνομα τοῦ ἐλάσσονος μείζον δύναται τῷ ἀπὸ συμμετρον ἑαυτῇ μήκει, ἔαν μὲν τὸ μείζον ὄνομα σύμμετρον ᾗ μήκει τῇ ἐκκειμένη ῥητῇ, καλείσθω [ἡ ὅλη] ἐκ δύο ὀνομάτων πρώτη.

6. Ἐάν δὲ τὸ ἐλάσσον ὄνομα σύμμετρον ᾗ μήκει τῇ ἐκκειμένη ῥητῇ, καλείσθω ἐκ δύο ὀνομάτων δευτέρα.

7. Ἐάν δὲ μηδέτερον τῶν ὀνομάτων σύμμετρον ᾗ μήκει τῇ ἐκκειμένη ῥητῇ, καλείσθω ἐκ δύο ὀνομάτων τρίτη.

8. Πάλιν δὲ ἔαν τὸ μείζον ὄνομα [τοῦ ἐλάσσονος] μείζον δύνῃται τῷ ἀπὸ ἀσυμμετρον ἑαυτῇ μήκει, ἔαν μὲν τὸ μείζον ὄνομα σύμμετρον ᾗ μήκει τῇ ἐκκειμένη ῥητῇ, καλείσθω ἐκ δύο ὀνομάτων τετάρτη.

9. Ἐάν δὲ τὸ ἐλάσσον, πέμπτη.

10. Ἐάν δὲ μηδέτερον, ἕκτη.

Definitions II

5. Given a rational (straight-line), and a binomial (straight-line) which has been divided into its (component) terms, of which the square on the greater term is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable in length with (the greater) then, if the greater term is commensurable in length with the rational (straight-line previously) laid out, let [the whole] (straight-line) be called a first binomial (straight-line).

6. And if the lesser term is commensurable in length with the rational (straight-line previously) laid out then let (the whole straight-line) be called a second binomial (straight-line).

7. And if neither of the terms is commensurable in length with the rational (straight-line previously) laid out then let (the whole straight-line) be called a third binomial (straight-line).

8. So, again, if the square on the greater term is larger than (the square on) [the lesser] by the (square) on (some straight-line) incommensurable in length with (the greater) then, if the greater term is commensurable in length with the rational (straight-line previously) laid out, let (the whole straight-line) be called a fourth binomial (straight-line).

9. And if the lesser (term is commensurable), a fifth (binomial straight-line).

10. And if neither (term is commensurable), a sixth (binomial straight-line).

48.

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων πρώτην.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν συγκεῖμενον ἔχ αὐτῶν τὸν $ΑΒ$ πρὸς μὲν τὸν $ΒΓ$ λόγον ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν $ΓΑ$ λόγον μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ ἐκκείσθω τις ῥητὴ ἡ $Δ$, καὶ τῇ $Δ$ σύμμετρος ἔστω μήκει ἡ $ΕΖ$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΕΖ$. καὶ γερονέτω ὥς ὁ $ΒΑ$ ἀριθμὸς πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΕΖ$ πρὸς

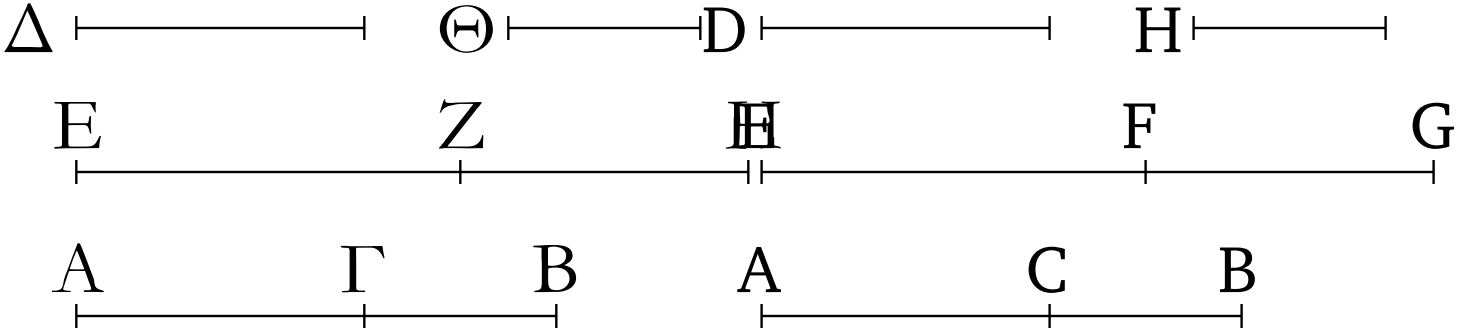
Proposition 48

To find a first binomial (straight-line).

Let two numbers AC and CB be laid down such that their sum AB has to BC the ratio which (some) square number (has) to (some) square number, and does not have to CA the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let some rational (straight-line) D be laid down. And let EF be commensurable in length with D . EF is thus also rational [Def. 10.3]. And let it be contrived

τὸ ἀπὸ τῆς ZH . ὁ δὲ AB πρὸς τὸν AG λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμὸν· καὶ τὸ ἀπὸ τῆς EZ ἄρα πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμὸν· ὥστε σύμμετρον ἐστὶ τὸ ἀπὸ τῆς EZ τῷ ἀπὸ τῆς ZH . καὶ ἐστὶ ῥητὴ ἡ EZ · ῥητὴ ἄρα καὶ ἡ ZH . καὶ ἐπεὶ ὁ BA πρὸς τὸν AG λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς EZ ἄρα πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἐστὶν ἡ EZ τῇ ZH μήκει. αἱ EZ , ZH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EH . λέγω, ὅτι καὶ πρώτη.

that as the number BA (is) to AC , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. And AB has to AC the ratio which (some) number (has) to (some) number. Thus, the (square) on EF also has to the (square) on FG the ratio which (some) number (has) to (some) number. Hence, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. And EF is rational. Thus, FG (is) also rational. And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, thus the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop 10.9]. EF and FG are thus rational (straight-lines which are) commensurable in square only. Thus, EG is a binomial (straight-line) [Prop. 10.36]. I say that (it is) also a first (binomial straight-line).



Ἐπεὶ γάρ ἐστιν ὡς ὁ BA ἀριθμὸς πρὸς τὸν AG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH , μείζων δὲ ὁ BA τοῦ AG , μείζων ἄρα καὶ τὸ ἀπὸ τῆς EZ τοῦ ἀπὸ τῆς ZH . ἔστω οὖν τῷ ἀπὸ τῆς EZ ἴσα τὰ ἀπὸ τῶν ZH , Θ . καὶ ἐπεὶ ἐστὶν ὡς ὁ BA πρὸς τὸν AG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH , ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ AB πρὸς τὸν BG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς Θ . ὁ δὲ AB πρὸς τὸν BG λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. καὶ τὸ ἀπὸ τῆς EZ ἄρα πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. σύμμετρος ἄρα ἐστὶν ἡ EZ τῇ Θ μήκει· ἡ EZ ἄρα τῆς ZH μείζων δύναται τῷ ἀπὸ συμμέτρον ἑαυτῇ. καὶ εἰσι ῥηταὶ αἱ EZ , ZH , καὶ σύμμετρος ἡ EZ τῇ Δ μήκει.

Ἡ EH ἄρα ἐκ δύο ὀνομάτων ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

For since as the number BA is to AC , so the (square) on EF (is) to the (square) on FG , and BA (is) greater than AC , the (square) on EF (is) thus also greater than the (square) on FG [Prop. 5.14]. Therefore, let (the sum of) the (squares) on FG and H be equal to the (square) on EF . And since as BA is to AC , so the (square) on EF (is) to the (square) on FG , thus, via conversion, as AB is to BC , so the (square) on EF (is) to the (square) on H [Prop. 5.19 corr.]. And AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, EF is commensurable in length with H [Prop. 10.9]. Thus, the square on EF is greater than (the square on) FG by the (square) on (some straight-line) commensurable (in length) with (EF). And EF and FG are rational (straight-lines). And EF (is) commensurable in length with D .

Thus, EG is a first binomial (straight-line) [Def. 10.5].[†] (Which is) the very thing it was required to show.

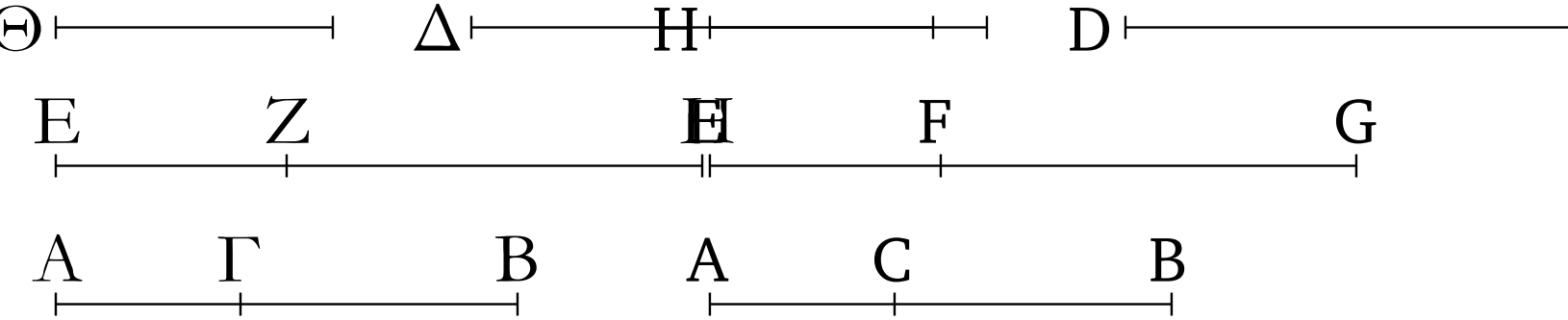
[†]If the rational straight-line has unit length then the length of a first binomial straight-line is $k + k\sqrt{1 - k'^2}$. This, and the first apotome, whose length is $k - k\sqrt{1 - k'^2}$ [Prop. 10.85], are the roots of $x^2 - 2kx + k^2k'^2 = 0$.

Εὑρεῖν τὴν ἐκ δύο ὀνομάτων δευτέραν.

To find a second binomial (straight-line).

Ἐκκεισθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν συγκε-
 ῖμενον ἔχ αὐτῶν τὸν $ΑΒ$ πρὸς μὲν τὸν $ΒΓ$ λόγον ἔθειν, ὃν
 τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν
 $ΑΓ$ λόγον μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον
 ἀριθμόν, καὶ ἐκκεισθω ῥητὴ ἡ $Δ$, καὶ τῇ $Δ$ σύμμετρος ἔστω
 ἡ $ΕΖ$ μήκει· ῥητὴ ἄρα ἐστὶν ἡ $ΕΖ$. γεγονέτω δὴ καὶ ὡς ὁ $ΓΑ$
 ἀριθμὸς πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $ΕΖ$ πρὸς τὸ ἀπὸ
 τῆς $ΖΗ$ · σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΕΖ$ τῷ ἀπὸ τῆς $ΖΗ$.
 ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΖΗ$. καὶ ἐπεὶ ὁ $ΓΑ$ ἀριθμὸς πρὸς τὸν
 $ΑΒ$ λόγον οὐκ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον
 ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς $ΕΖ$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$ λόγον ἔθειν,
 ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμε-
 τρος ἄρα ἐστὶν ἡ $ΕΖ$ τῇ $ΖΗ$ μήκει· αἱ $ΕΖ$, $ΖΗ$ ἄρα ῥηταὶ
 εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ
 $ΕΗ$. δεικτέον δὴ, ὅτι καὶ δευτέρα.

Let the two numbers AC and CB be laid down such that
 their sum AB has to BC the ratio which (some) square number
 (has) to (some) square number, and does not have to AC the ra-
 tio which (some) square number (has) to (some) square num-
 ber [Prop. 10.28 lem. I]. And let the rational (straight-line) D
 be laid down. And let EF be commensurable in length with D .
 EF is thus a rational (straight-line). So, let it also be contrived
 that as the number CA (is) to AB , so the (square) on EF (is)
 to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on
 EF is commensurable with the (square) on FG [Prop. 10.6].
 Thus, FG is also a rational (straight-line). And since the num-
 ber CA does not have to AB the ratio which (some) square
 number (has) to (some) square number, the (square) on EF
 does not have to the (square) on FG the ratio which (some)
 square number (has) to (some) square number either. Thus,
 EF is incommensurable in length with FG [Prop. 10.9]. EF
 and FG are thus rational (straight-lines which are) commen-
 surable in square only. Thus, EG is a binomial (straight-line)
 [Prop. 10.36]. So, we must show that (it is) also a second (bi-
 nomial straight-line).



Ἐπεὶ γὰρ ἀνάπαλιν ἐστὶν ὡς ὁ $ΒΑ$ ἀριθμὸς πρὸς τὸν $ΑΓ$,
 οὕτως τὸ ἀπὸ τῆς $ΗΖ$ πρὸς τὸ ἀπὸ τῆς $ΖΕ$, μείζων δὲ ὁ $ΒΑ$
 τοῦ $ΑΓ$, μείζων ἄρα [καὶ] τὸ ἀπὸ τῆς $ΗΖ$ τοῦ ἀπὸ τῆς $ΖΕ$.
 ἔστω τῷ ἀπὸ τῆς $ΗΖ$ ἴσα τὰ ἀπὸ τῶν $ΕΖ$, Θ· ἀναστρέψαντι
 ἄρα ἐστὶν ὡς ὁ $ΑΒ$ πρὸς τὸν $ΒΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς
 τὸ ἀπὸ τῆς Θ. ἀλλ' ὁ $ΑΒ$ πρὸς τὸν $ΒΓ$ λόγον ἔθειν, ὃν τε-
 τράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὸ ἀπὸ τῆς
 $ΖΗ$ ἄρα πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔθειν, ὃν τετράγωνος ἀρι-
 θμὸς πρὸς τετράγωνον ἀριθμόν. σύμμετρος ἄρα ἐστὶν ἡ $ΖΗ$
 τῇ Θ μήκει· ὥστε ἡ $ΖΗ$ τῆς $ΖΕ$ μείζων δύνатаι τῷ ἀπὸ συμ-
 μέτρον ἑαυτῇ. καὶ εἰσι ῥηταὶ αἱ $ΖΗ$, $ΖΕ$ δυνάμει μόνον σύμμε-
 τροι, καὶ τὸ $ΕΖ$ ἔλασσον ὄνομα τῇ ἐκκειμένῃ ῥητῇ σύμμετρόν
 ἐστὶ τῇ $Δ$ μήκει.

Ἡ $ΕΗ$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ δευτέρα· ὅπερ ἔδει
 δεῖξαι.

For since, inversely, as the number BA is to AC , so the
 (square) on GF (is) to the (square) on FE [Prop. 5.7 corr.],
 and BA (is) greater than AC , the (square) on GF (is) thus [also]
 greater than the (square) on FE [Prop. 5.14]. Let (the sum of)
 the (squares) on EF and H be equal to the (square) on GF .
 Thus, via conversion, as AB is to BC , so the (square) on FG
 (is) to the (square) on H [Prop. 5.19 corr.]. But, AB has to BC
 the ratio which (some) square number (has) to (some) square
 number. Thus, the (square) on FG also has to the (square)
 on H the ratio which (some) square number (has) to (some)
 square number. Thus, FG is commensurable in length with
 H [Prop. 10.9]. Hence, the square on FG is greater than (the
 square on) FE by the (square) on (some straight-line) com-
 mensurable in length with (FG). And FG and FE are rational
 (straight-lines which are) commensurable in square only. And
 the lesser term EF is commensurable in length with the ratio-
 nal (straight-line) D (previously) laid down.

Thus, EG is a second binomial (straight-line) [Def. 10.6].[†]
 (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then the length of a second binomial straight-line is $k/\sqrt{1-k'^2} + k$. This, and the second apotome, whose length is $k/\sqrt{1-k'^2} - k$ [Prop. 10.86], are the roots of $x^2 - (2k/\sqrt{1-k'^2})x + k^2/[k'^2/(1-k'^2)] = 0$.

50.

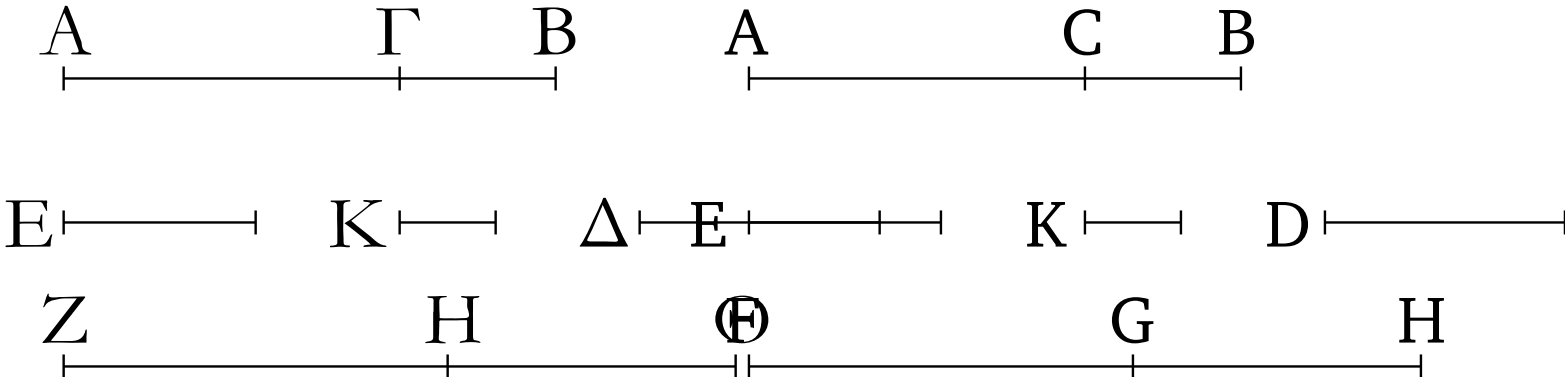
Proposition 50

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων τρίτην.

Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν συγκε-
 ῖμενον ἔχ αὐτῶν τὸν $ΑΒ$ πρὸς μὲν τὸν $ΒΓ$ λόγον ἔθειν, ὃν
 τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν
 $ΑΓ$ λόγον μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον
 ἀριθμόν. ἐκκεῖσθω δὲ τις καὶ ἄλλος μὴ τετράγωνος ἀριθμὸς
 ὁ $Δ$, καὶ πρὸς ἐκάτερον τῶν $ΒΑ$, $ΑΓ$ λόγον μὴ ἐθέτω, ὃν
 τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ ἐκκεῖσθω
 τις ῥητὴ εὐθεῖα ἡ $Ε$, καὶ γερονέτω ὡς ὁ $Δ$ πρὸς τὸν $ΑΒ$,
 οὕτως τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$ · σύμμετρον ἄρα
 ἐστὶ τὸ ἀπὸ τῆς $Ε$ τῷ ἀπὸ τῆς $ΖΗ$. καὶ ἐστὶ ῥητὴ ἡ $Ε$ · ῥητὴ
 ἄρα ἐστὶ καὶ ἡ $ΖΗ$. καὶ ἐπεὶ ὁ $Δ$ πρὸς τὸν $ΑΒ$ λόγον οὐκ
 ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ
 τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$ λόγον ἔθει, ὃν τετράγωνος
 ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ
 $Ε$ τῇ $ΖΗ$ μήκει. γερονέτω δὴ πάλιν ὡς ἡ $ΒΑ$ ἀριθμὸς πρὸς
 τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$ · σύμ-
 μετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΖΗ$ τῷ ἀπὸ τῆς $ΗΘ$. ῥητὴ δὲ ἡ
 $ΖΗ$ · ῥητὴ ἄρα καὶ ἡ $ΗΘ$. καὶ ἐπεὶ ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$ λόγον
 οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν,
 οὐδὲ τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΘΗ$ λόγον ἔθει, ὃν
 τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος
 ἄρα ἐστὶν ἡ $ΖΗ$ τῇ $ΗΘ$ μήκει. αἱ $ΖΗ$, $ΗΘ$ ἄρα ῥηταὶ εἰσι
 δυνάμει μόνον σύμμετροι· ἡ $ΖΘ$ ἄρα ἐκ δύο ὀνομάτων ἐστίν.
 λέγω δὴ, ὅτι καὶ τρίτη.

To find a third binomial (straight-line).

Let the two numbers AC and CB be laid down such that
 their sum AB has to BC the ratio which (some) square num-
 ber (has) to (some) square number, and does not have to AC
 the ratio which (some) square number (has) to (some) square
 number. And let some other non-square number D also be
 laid down, and let it not have to each of BA and AC the ratio
 which (some) square number (has) to (some) square number.
 And let some rational straight-line E be laid down, and let it
 be contrived that as D (is) to AB , so the (square) on E (is)
 to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square)
 on E is commensurable with the (square) on FG [Prop. 10.6].
 And E is a rational (straight-line). Thus, FG is also a ratio-
 nal (straight-line). And since D does not have to AB the ra-
 tio which (some) square number has to (some) square num-
 ber, the (square) on E does not have to the (square) on FG
 the ratio which (some) square number (has) to (some) square
 number either. E is thus incommensurable in length with FG
 [Prop. 10.9]. So, again, let it be contrived that as the number
 BA (is) to AC , so the (square) on FG (is) to the (square) on GH
 [Prop. 10.6 corr.]. Thus, the (square) on FG is commensurable
 with the (square) on GH [Prop. 10.6]. And FG (is) a rational
 (straight-line). Thus, GH (is) also a rational (straight-line).
 And since BA does not have to AC the ratio which (some)
 square number (has) to (some) square number, the (square)
 on FG does not have to the (square) on HG the ratio which
 (some) square number (has) to (some) square number either.
 Thus, FG is incommensurable in length with GH [Prop. 10.9].
 FG and GH are thus rational (straight-lines which are) com-
 mensurable in square only. Thus, FH is a binomial (straight-
 line) [Prop. 10.36]. So, I say that (it is) also a third (binomial
 straight-line).



Ἐπεὶ γάρ ἐστιν ὡς ὁ $Δ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς
 $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$, ὡς δὲ ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως

For since as D is to AB , so the (square) on E (is) to the
 (square) on FG , and as BA (is) to AC , so the (square) on FG

τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, δι' ἴσον ἄρα ἐστὶν ὡς ὁ Δ πρὸς τὸν $ΑΓ$, οὔτως τὸ ἀπὸ τῆς E πρὸς τὸ ἀπὸ τῆς $H\Theta$. ὁ δὲ Δ πρὸς τὸν $ΑΓ$ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὲ τὸ ἀπὸ τῆς E ἄρα πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ E τῇ $H\Theta$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ὁ BA πρὸς τὸν $ΑΓ$, οὔτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, μείζον ἄρα τὸ ἀπὸ τῆς ZH τοῦ ἀπὸ τῆς $H\Theta$. ἔστω οὖν τῶ ἀπὸ τῆς ZH ἴσα τὰ ἀπὸ τῶν $H\Theta$, K · ἀναστρέψαντι ἄρα [ἐστὶν] ὡς ὁ AB πρὸς τὸν $BΓ$, οὔτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ AB πρὸς τὸν $BΓ$ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὸ ἀπὸ τῆς ZH ἄρα πρὸς τὸ ἀπὸ τῆς K λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα [ἐστὶν] ἡ ZH τῇ K μήκει. ἡ ZH ἄρα τῆς $H\Theta$ μείζον δύναται τῶ ἀπὸ συμμέτρου ἐαντιῇ. καὶ εἰσιν αἱ ZH , $H\Theta$ ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα αὐτῶν σύμμετρος ἐστὶ τῇ E μήκει.

Ἡ $Z\Theta$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη· ὅπερ ἔδει δεῖξαι.

(is) to the (square) on GH , thus, via equality, as D (is) to AC , so the (square) on E (is) to the (square) on GH [Prop. 5.22]. And D does not have to AC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on E does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, E is incommensurable in length with GH [Prop. 10.9]. And since as BA is to AC , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus greater than the (square) on GH [Prop. 5.14]. Therefore, let (the sum of) the (squares) on GH and K be equal to the (square) on FG . Thus, via conversion, as AB [is] to BC , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on K the ratio which (some) square number (has) to (some) square number. Thus, FG [is] commensurable in length with K [Prop. 10.9]. Thus, the square on FG is greater than (the square on) GH by the (square) on (some straight-line) commensurable (in length) with (FG) . And FG and GH are rational (straight-lines which are) commensurable in square only, and neither of them is commensurable in length with E .

Thus, FH is a third binomial (straight-line) [Def. 10.7].[†] (Which is) the very thing it was required to show.

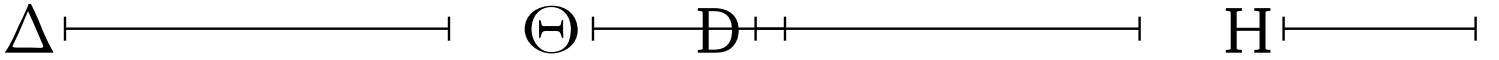
[†] If the rational straight-line has unit length then the length of a third binomial straight-line is $k^{1/2}(1 + \sqrt{1 - k'^2})$. This, and the third apotome, whose length is $k^{1/2}(1 - \sqrt{1 - k'^2})$ [Prop. 10.87], are the roots of $x^2 - 2k^{1/2}x + k'^2 = 0$.

51.

Proposition 51

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων τετάρτην.

To find a fourth binomial (straight-line).



Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν AB πρὸς τὸν $BΓ$ λόγον μὴ ἔχειν μήτε μὴν πρὸς τὸν $ΑΓ$, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ ἐκκεῖσθω ῥητὴ ἡ Δ , καὶ τῇ Δ σύμμετρος ἔστω μήκει ἡ EZ · ῥητὴ ἄρα ἐστὶ καὶ ἡ EZ . καὶ γεγονέντω ὡς ὁ BA ἀριθμὸς πρὸς τὸν $ΑΓ$, οὔτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH · σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ

Let the two numbers AC and CB be laid down such that AB does not have to BC , or to AC either, the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let the rational (straight-line) D be laid down. And let EF be commensurable in length with D . Thus, EF is also a rational (straight-line). And let it be

τῆς EZ τῷ ἀπὸ τῆς ZH ῥητῇ ἄρα ἐστὶ καὶ ἡ ZH . καὶ ἐπεὶ ὁ BA πρὸς τὸν AG λόγον οὐκ ἔθλει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔθλει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ EZ τῇ ZH μήκει. αἱ EZ , ZH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ὥστε ἡ EH ἐκ δύο ὀνομάτων ἐστίν. λέγω δὴ, ὅτι καὶ τετάρτη.

Ἐπεὶ γάρ ἐστιν ὡς ὁ BA πρὸς τὸν AG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH [μείζων δὲ ὁ BA τοῦ AG], μείζον ἄρα τὸ ἀπὸ τῆς EZ τοῦ ἀπὸ τῆς ZH . ἔστω οὖν τῷ ἀπὸ τῆς EZ ἴσα τὰ ἀπὸ τῶν ZH , Θ · ἀναστρέψαντι ἄρα ὡς ὁ AB ἀριθμὸς πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς Θ . ὁ δὲ AB πρὸς τὸν $B\Gamma$ λόγον οὐκ ἔθλει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδ' ἄρα τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔθλει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ EZ τῇ Θ μήκει· ἡ EZ ἄρα τῆς HZ μείζον δύνανται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ. καὶ εἰσιν αἱ EZ , ZH ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ ἡ EZ τῇ Δ σύμμετρος ἐστὶ μήκει.

Ἡ EH ἄρα ἐκ δύο ὀνομάτων ἐστὶ τετάρτη· ὅπερ ἔδει δεῖν.

contrived that as the number BA (is) to AC , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. Thus, FG is also a rational (straight-line). And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop. 10.9]. Thus, EF and FG are rational (straight-lines which are) commensurable in square only. Hence, EG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fourth (binomial straight-line).

For since as BA is to AC , so the (square) on EF (is) to the (square) on FG [and BA (is) greater than AC], the (square) on EF (is) thus greater than the (square) on FG [Prop. 5.14]. Therefore, let (the sum of) the squares on FG and H be equal to the (square) on EF . Thus, via conversion, as the number AB (is) to BC , so the (square) on EF (is) to the (square) on H [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with H [Prop. 10.9]. Thus, the square on EF is greater than (the square on) GF by the (square) on (some straight-line) incommensurable (in length) with (EF). And EF and FG are rational (straight-lines which are) commensurable in square only. And EF is commensurable in length with D .

Thus, EG is a fourth binomial (straight-line) [Def. 10.8].[†] (Which is) the very thing it was required to show.

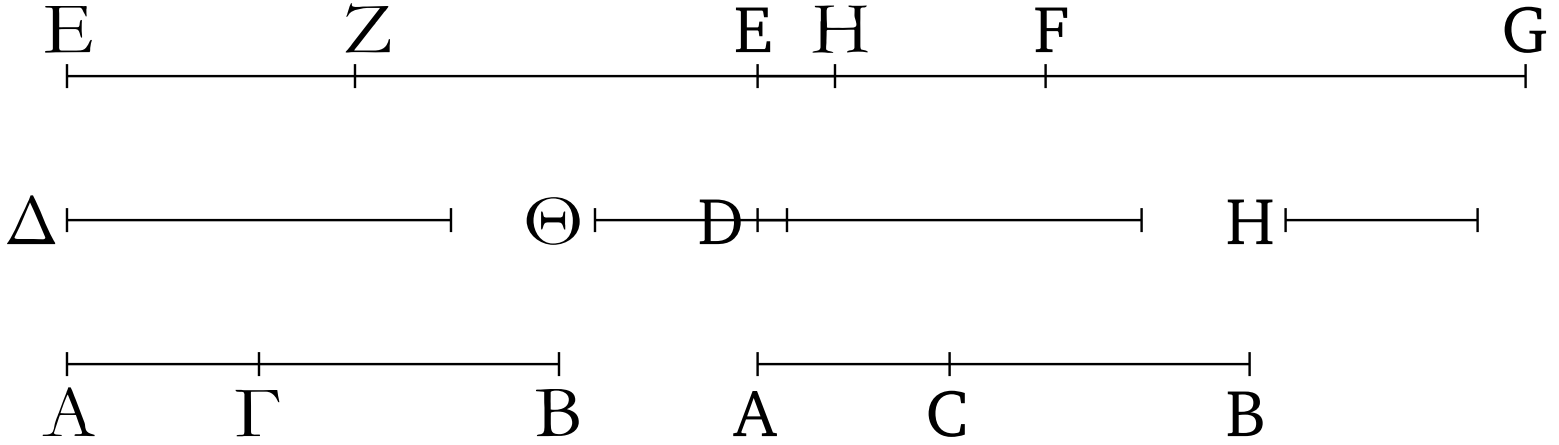
[†] If the rational straight-line has unit length then the length of a fourth binomial straight-line is $k(1 + 1/\sqrt{1+k'})$. This, and the fourth apotome, whose length is $k(1 - 1/\sqrt{1+k'})$ [Prop. 10.88], are the roots of $x^2 - 2kx + k^2k'/(1+k') = 0$.

52.

Εὕρεῖν τὴν ἐκ δύο ὀνομάτων πέμπτην.

Proposition 52

To find a fifth binomial straight-line.



Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν $ΑΒ$ πρὸς ἑκάτερον αὐτῶν λόγον μὴ ἔδειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ ἐκκείσθω ῥητὴ τις εὐθεῖα ἡ $Δ$, καὶ τῇ $Δ$ σύμμετρος ἔστω [μήκει] ἡ $ΕΖ$. ῥητὴ ἄρα ἡ $ΕΖ$. καὶ γεγονέτω ὥς ὁ $ΓΑ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $ΕΖ$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$. ὁ δὲ $ΓΑ$ πρὸς τὸν $ΑΒ$ λόγον οὐκ ἔδει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὲ τὸ ἀπὸ τῆς $ΕΖ$ ἄρα πρὸς τὸ ἀπὸ τῆς $ΖΗ$ λόγον ἔδει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. αἱ $ΕΖ$, $ΖΗ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $ΕΗ$. λέγω δὴ, ὅτι καὶ πέμπτη.

Ἐπεὶ γάρ ἐστιν ὥς ὁ $ΓΑ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $ΕΖ$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$, ἀνάπαλιν ὥς ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΖΕ$. μείζον ἄρα τὸ ἀπὸ τῆς $ΗΖ$ τοῦ ἀπὸ τῆς $ΖΕ$. ἔστω οὖν τῶ ἀπὸ τῆς $ΗΖ$ ἴσα τὰ ἀπὸ τῶν $ΕΖ$, $Θ$. ἀναστρέψαντι ἄρα ἐστὶν ὥς ὁ $ΑΒ$ ἀριθμὸς πρὸς τὸν $ΒΓ$, οὕτως τὸ ἀπὸ τῆς $ΗΖ$ πρὸς τὸ ἀπὸ τῆς $Θ$. ὁ δὲ $ΑΒ$ πρὸς τὸν $ΒΓ$ λόγον οὐκ ἔδει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδ' ἄρα τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔδει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ $ΖΗ$ τῇ $Θ$ μήκει· ὥστε ἡ $ΖΗ$ τῆς $ΖΕ$ μείζον δύναται τῶ ἀπὸ ἀσυνμέτρων ἑαυτῇ. καὶ εἰσιν αἱ $ΗΖ$, $ΖΕ$ ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ τὸ $ΕΖ$ ἕλαττον ὄνομα σύμμετρόν ἐστι τῇ ἐκκειμένῃ ῥητῇ τῇ $Δ$ μήκει.

Ἡ $ΕΗ$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ πέμπτη· ὅπερ ἔδει δεῖχαι.

Let the two numbers AC and CB be laid down such that AB does not have to either of them the ratio which (some) square number (has) to (some) square number [Prop. 10.38 lem.]. And let some rational straight-line D be laid down. And let EF be commensurable [in length] with D . Thus, EF (is) a rational (straight-line). And let it be contrived that as CA (is) to AB , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. And CA does not have to AB the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF and FG are rational (straight-lines which are) commensurable in square only [Prop. 10.9]. Thus, EG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fifth (binomial straight-line).

For since as CA is to AB , so the (square) on EF (is) to the (square) on FG , inversely, as BA (is) to AC , so the (square) on FG (is) to the (square) on FE [Prop. 5.7 corr.]. Thus, the (square) on GF (is) greater than the (square) on FE [Prop. 5.14]. Therefore, let (the sum of) the (squares) on EF and H be equal to the (square) on GF . Thus, via conversion, as the number AB is to BC , so the (square) on GF (is) to the (square) on H [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with H [Prop. 10.9]. Hence, the square on FG is greater than (the square on) FE by the (square) on (some straight-line) incommensurable (in length) with (FG). And GF and FE are rational (straight-lines which are) commensurable in square only. And the lesser term EF is commensurable in length with the rational (straight-line previously) laid down, D .

Thus, EG is a fifth binomial (straight-line).[†] (Which is) the very thing it was required to show.

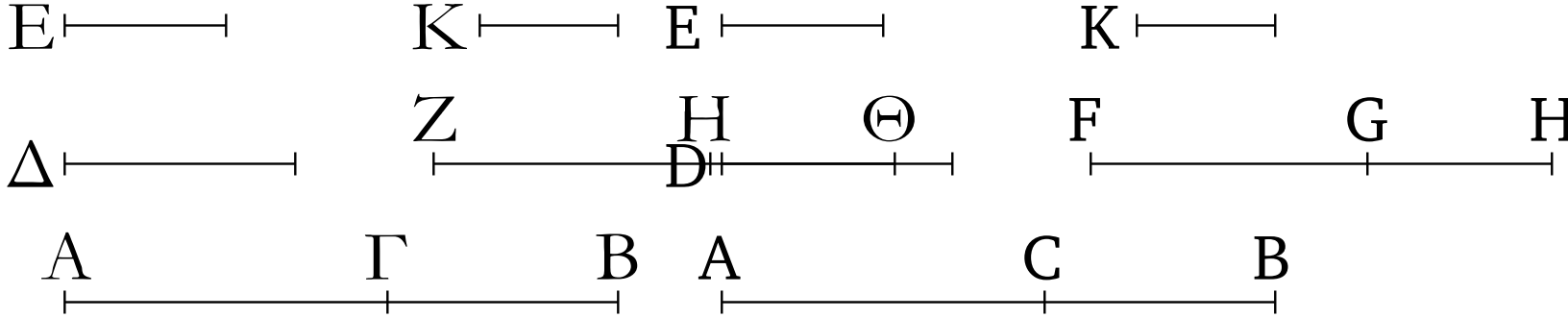
[†] If the rational straight-line has unit length then the length of a fifth binomial straight-line is $k(\sqrt{1+k'} + 1)$. This, and the fifth apotome, whose length is $k(\sqrt{1+k'} - 1)$ [Prop. 10.89], are the roots of $x^2 - 2k\sqrt{1+k'}x + k^2k' = 0$.

53.

Proposition 53

Εὕρεῖν τὴν ἐκ δύο ὀνομάτων ἑκτὴν.

To find a sixth binomial (straight-line).



Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν $ΑΒ$ πρὸς ἑκάτερον αὐτῶν λόγον μὴ ἔδειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἔστω δὲ καὶ ἕτερος ἀριθμὸς ὁ $Δ$ μὴ τετράγωνος ὡς μηδὲ πρὸς ἑκάτερον τῶν $ΒΑ$, $ΑΓ$ λόγον ἔθων, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· καὶ ἐκκεῖσθω τις ῥητὴ εὐθεῖα ἡ $Ε$, καὶ γερονέτω ὡς ὁ $Δ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$ · σύμμετρον ἄρα τὸ ἀπὸ τῆς $Ε$ τῷ ἀπὸ τῆς $ΖΗ$. καὶ ἐστὶ ῥητὴ ἡ $Ε$ · ῥητὴ ἄρα καὶ ἡ $ΖΗ$. καὶ ἐπεὶ οὐκ ἔθει ὁ $Δ$ πρὸς τὸν $ΑΒ$ λόγον, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς $Ε$ ἄρα πρὸς τὸ ἀπὸ τῆς $ΖΗ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἡ $Ε$ τῇ $ΖΗ$ μήκει. γερονέτω δὲ πάλιν ὡς ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$. σύμμετρον ἄρα τὸ ἀπὸ τῆς $ΖΗ$ τῷ ἀπὸ τῆς $ΗΘ$. ῥητὸν ἄρα τὸ ἀπὸ τῆς $ΗΘ$ · ῥητὴ ἄρα ἡ $ΗΘ$. καὶ ἐπεὶ ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἐστὶν ἡ $ΖΗ$ τῇ $ΗΘ$ μήκει. αἱ $ΖΗ$, $ΗΘ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $ΖΘ$. δεικτέον δὲ, ὅτι καὶ ἑκτὴ.

Ἐπεὶ γάρ ἐστιν ὡς ὁ $Δ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$, ἐστὶ δὲ καὶ ὡς ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$, δι' ἴσον ἄρα ἐστὶν ὡς ὁ $Δ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$. ὁ δὲ $Δ$ πρὸς τὸν $ΑΓ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· οὐδὲ τὸ ἀπὸ τῆς $Ε$ ἄρα πρὸς τὸ ἀπὸ τῆς $ΗΘ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἐστὶν ἡ $Ε$ τῇ $ΗΘ$ μήκει. ἐδείχθη δὲ καὶ τῇ $ΖΗ$ ἀσύμμετρος· ἑκατέρα ἄρα τῶν $ΖΗ$, $ΗΘ$ ἀσύμμετρος ἐστὶ τῇ $Ε$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$, μείζον ἄρα τὸ ἀπὸ τῆς $ΖΗ$ τοῦ ἀπὸ τῆς $ΗΘ$. ἔστω οὖν

Let the two numbers AC and CB be laid down such that AB does not have to each of them the ratio which (some) square number (has) to (some) square number. And let D also be another number, which is not square, and does not have to each of BA and AC the ratio which (some) square number (has) to (some) square number either [Prop. 10.28 lem. I]. And let some rational straight-line E be laid down. And let it be contrived that as D (is) to AB , so the (square) on E (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on E (is) commensurable with the (square) on FG [Prop. 10.6]. And E is rational. Thus, FG (is) also rational. And since D does not have to AB the ratio which (some) square number (has) to (some) square number, the (square) on E thus does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, E (is) incommensurable in length with FG [Prop. 10.9]. So, again, let it have be contrived that as BA (is) to AC , so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.]. The (square) on FG (is) thus commensurable with the (square) on HG [Prop. 10.6]. The (square) on HG (is) thus rational. Thus, HG (is) rational. And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on FG does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. Thus, FG and GH are rational (straight-lines which are) commensurable in square only. Thus, FH is a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a sixth (binomial straight-line).

For since as D is to AB , so the (square) on E (is) to the (square) on FG , and also as BA is to AC , so the (square) on FG (is) to the (square) on GH , thus, via equality, as D is to AC , so the (square) on E (is) to the (square) on GH [Prop. 5.22]. And D does not have to AC the ratio which (some) square

τῷ ἀπὸ [τῆς] ZH ἴσα τὰ ἀπὸ τῶν $H\Theta$, K ἀναστρέψαντι ἄρα ὥς ὁ AB πρὸς $B\Gamma$, οὕτως τὸ ἀπὸ ZH πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ AB πρὸς τὸν $B\Gamma$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ὥστε οὐδὲ τὸ ἀπὸ ZH πρὸς τὸ ἀπὸ τῆς K λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἐστὶν ἡ ZH τῇ K μήκει· ἡ ZH ἄρα τῆς $H\Theta$ μείζον δύναιται τῷ ἀπὸ ἀσυμμέτρον ἐαντῇ. καὶ εἰσιν αἱ ZH , $H\Theta$ ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ οὐδετέρᾳ αὐτῶν σύμμετρος ἐστὶ μήκει τῇ ἐκκειμένη ῥῆτι τῇ E .

Ἡ $Z\Theta$ ἄρα ἐκ δύο ὀνομάτων ἐστὶν ἑκτῇ· ὅπερ ἔδει δεῖξαι.

number (has) to (some) square number. Thus, the (square) on E does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. E is thus incommensurable in length with GH [Prop. 10.9]. And (E) was also shown (to be) incommensurable (in length) with FG . Thus, FG and GH are each incommensurable in length with E . And since as BA is to AC , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus greater than the (square) on GH [Prop. 5.14]. Therefore, let (the sum of) the (squares) on GH and K be equal to the (square) on FG . Thus, via conversion, as AB (is) to BC , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Hence, the (square) on FG does not have to the (square) on K the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with K [Prop. 10.9]. The square on FG is thus greater than (the square on) GH by the (square) on (some straight-line which is) incommensurable (in length) with (FG). And FG and GH are rational (straight-lines which are) commensurable in square only, and neither of them is commensurable in length with the rational (straight-line) E (previously) laid down.

Thus, FH is a sixth binomial (straight-line) [Def. 10.10].[†] (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then the length of a sixth binomial straight-line is $\sqrt{k} + \sqrt{k'}$. This, and the sixth apotome, whose length is $\sqrt{k} - \sqrt{k'}$ [Prop. 10.90], are the roots of $x^2 - 2\sqrt{k}x + (k - k') = 0$.

Λήμμα.

Ἐστω δύο τετράγωνα τὰ AB , $B\Gamma$ καὶ κείσθωσαν ὥστε ἐπ' ἐνθείᾳ εἶναι τὴν ΔB τῇ BE · ἐπ' ἐνθείᾳ ἄρα ἐστὶ καὶ ἡ ZB τῇ BH . καὶ συμπεπληρώσθω τὸ AG παραλληλόγραμμον· λέγω, ὅτι τετράγωνόν ἐστι τὸ AG , καὶ ὅτι τῶν AB , $B\Gamma$ μέσον ἀνάλογόν ἐστι τὸ ΔH , καὶ ἔτι τῶν AG , ΓB μέσον ἀνάλογόν ἐστι τὸ $\Delta\Gamma$.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ μὲν ΔB τῇ BZ , ἡ δὲ BE τῇ BH , ὅλη ἄρα ἡ ΔE ὅλη τῇ ZH ἐστὶν ἴση. ἀλλ' ἡ μὲν ΔE ἐκατέρᾳ τῶν $A\Theta$, $K\Gamma$ ἐστὶν ἴση, ἡ δὲ ZH ἐκατέρᾳ τῶν AK , $\Theta\Gamma$ ἐστὶν ἴση· καὶ ἐκατέρᾳ ἄρα τῶν $A\Theta$, $K\Gamma$ ἐκατέρᾳ τῶν AK , $\Theta\Gamma$ ἐστὶν ἴση. ἰσόπλευρον ἄρα ἐστὶ τὸ AG παραλληλόγραμμον· ἐστὶ δὲ καὶ ὀρθογώνιον· τετράγωνον ἄρα ἐστὶ τὸ AG .

Καὶ ἐπεὶ ἐστὶν ὥς ἡ ZB πρὸς τὴν BH , οὕτως ἡ ΔB πρὸς τὴν BE , ἀλλ' ὥς μὲν ἡ ZB πρὸς τὴν BH , οὕτως τὸ AB πρὸς τὸ ΔH , ὥς δὲ ἡ ΔB πρὸς τὴν BE , οὕτως τὸ ΔH πρὸς τὸ $B\Gamma$, καὶ ὥς ἄρα τὸ AB πρὸς τὸ ΔH , οὕτως τὸ ΔH πρὸς τὸ $B\Gamma$. τῶν AB , $B\Gamma$ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΔH .

Λέγω δὴ, ὅτι καὶ τῶν AG , ΓB μέσον ἀνάλογόν [ἐστὶ] τὸ $\Delta\Gamma$.

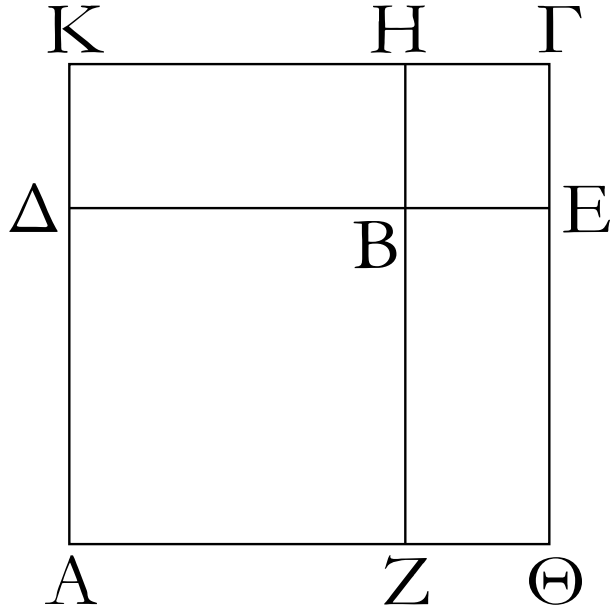
Lemma

Let AB and BC be two squares, and let them be laid down such that DB is straight-on to BE . FB is, thus, also straight-on to BG . And let the parallelogram AC be completed. I say that AC is a square, and that DG is the mean proportional to AB and BC , and, moreover, DC is the mean proportional to AC and CB .

For since DB is equal to BF , and BE to BG , the whole of DE is thus equal to the whole of FG . But DE is equal to each of AH and KC , and FG is equal to each of AK and HC [Prop. 1.34]. Thus, AH and KC are also equal to AK and HC , respectively. Thus, the parallelogram AC is equilateral. And (it is) also right-angled. Thus, AC is a square.

And since as FB is to BG , so DB (is) to BE , but as FB (is) to BG , so AB (is) to DG , and as DB (is) to BE , so DG (is) to BC [Prop. 6.1], thus also as AB (is) to DG , so DG (is) to BC [Prop. 5.11]. Thus, DG is the mean proportional to AB and BC .

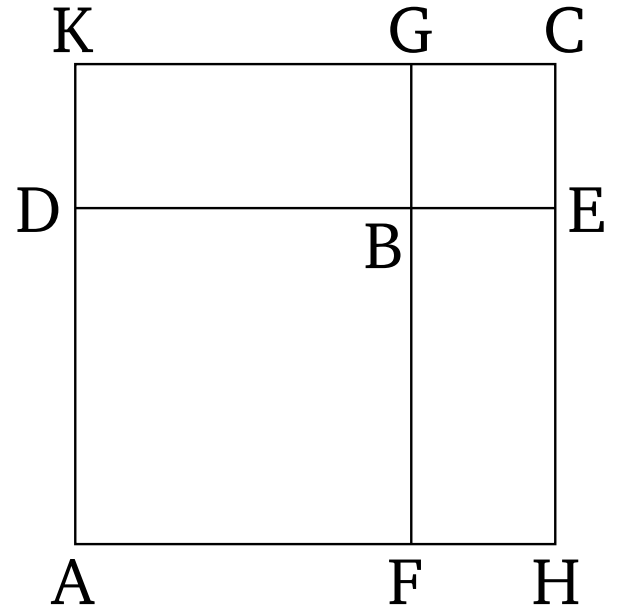
So I also say that DC [is] the mean proportional to AC and CB .



Ἐπεὶ γάρ ἐστιν ὡς ἡ AD πρὸς τὴν $ΔK$, οὕτως ἡ KH πρὸς τὴν $HΓ$. ἴση γάρ [ἐστιν] ἑκατέρω ἑκατέρω· καὶ συνθέντι ὡς ἡ AK πρὸς $KΔ$, οὕτως ἡ $KΓ$ πρὸς $ΓH$, ἀλλ' ὡς μὲν ἡ AK πρὸς $KΔ$, οὕτως τὸ $ΑΓ$ πρὸς τὸ $ΓΔ$, ὡς δὲ ἡ $KΓ$ πρὸς $ΓH$, οὕτως τὸ $ΔΓ$ πρὸς $ΓB$, καὶ ὡς ἄρα τὸ $ΑΓ$ πρὸς $ΔΓ$, οὕτως τὸ $ΔΓ$ πρὸς τὸ $BΓ$. τῶν $ΑΓ$, $ΓB$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $ΔΓ$ · ἃ προέκειτο δεῖξαι.

54.

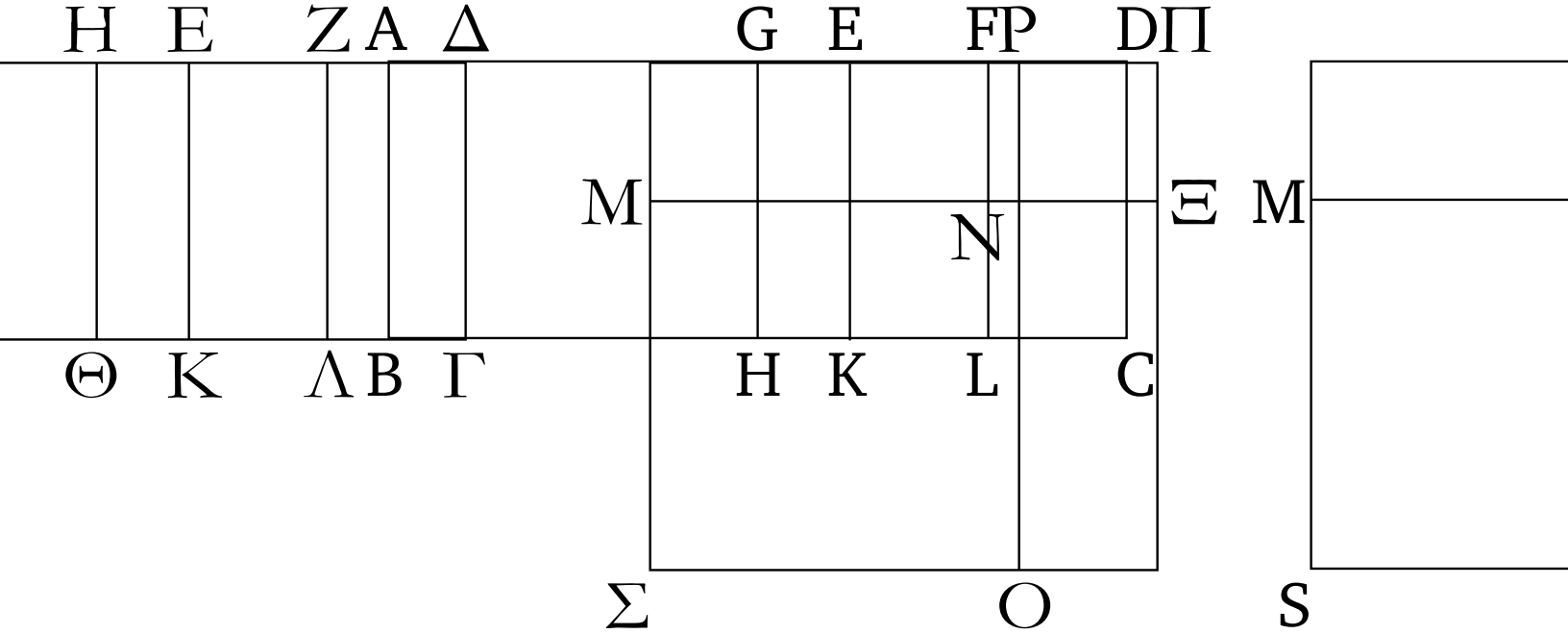
Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πρώτης, ἡ τὸ θωρίον δυναμένη ἀλογόξ ἐστιν ἡ καλονμένη ἐκ δύο ὀνομάτων.



For since as AD is to DK , so KG (is) to GC . For [they are] respectively equal. And, via composition, as AK (is) to KD , so KC (is) to CG [Prop. 5.18]. But as AK (is) to KD , so AC (is) to CD , and as KC (is) to CG , so DC (is) to CB [Prop. 6.1]. Thus, also, as AC (is) to DC , so DC (is) to BC [Prop. 5.11]. Thus, DC is the mean proportional to AC and CB . Which (is the very thing) it was prescribed to show.

Proposition 54

If an area is contained by a rational (straight-line) and a first binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called binomial.[†]



Θωρίον γάρ τὸ ΑΓ περιεθέσθω ὑπὸ ῥητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων πρώτης τῆς ΑΔ· λέγω, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλουμένη ἐκ δύο ὀνομάτων.

Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων ἐστὶ πρώτη ἡ ΑΔ, διηγήσθω εἰς τὰ ὀνόματα κατὰ τὸ Ε, καὶ ἔστω τὸ μείζον ὄνομα τὸ ΑΕ. φανερόν δὴ, ὅτι αἱ ΑΕ, ΕΔ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ, καὶ ἡ ΑΕ σύμμετρος ἐστὶ τῇ ἐκκεκμένη ῥητὴ τῇ ΑΒ μήκει. τετμήσθω δὴ ἡ ΕΔ δίδια κατὰ τὸ Ζ σημείον. καὶ ἐπεὶ ἡ ΑΕ τῆς ΕΔ μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος, τοντέστι τῷ ἀπὸ τῆς ΕΖ, ἴσον παρὰ τὴν μείζονα τὴν ΑΕ παραβληθῇ ἐλλείπων εἶδει τετραγώνῳ, εἰς σύμμετρα αὐτὴν διαιρεῖ. παραβελήσθω οὖν παρὰ τὴν ΑΕ τῷ ἀπὸ τῆς ΕΖ ἴσον τὸ ὑπὸ ΑΗ, ΗΕ· σύμμετρος ἄρα ἐστὶν ἡ ΑΗ τῇ ΕΗ μήκει. καὶ ἤθθωσαν ἀπὸ τῶν Η, Ε, Ζ ὁποτέρᾳ τῶν ΑΒ, ΓΔ παράλληλοι αἱ ΗΘ, ΕΚ, ΖΛ· καὶ τῷ μὲν ΑΘ παραλληλογράμῳ ἴσον τετράγωνον συνεστάτω τὸ ΣΝ, τῷ δὲ ΗΚ ἴσον τὸ ΝΠ, καὶ κείσθω ὥστε ἐπ' εὐθείᾳ εἶναι τὴν ΜΝ τῇ ΝΧ· ἐπ' εὐθείᾳ ἄρα ἐστὶ καὶ ἡ ΡΝ τῇ ΝΟ. καὶ συμπεπληρώσθω τὸ ΣΠ παραλληλόγραμμον· τετράγωνον ἄρα ἐστὶ τὸ ΣΠ. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΑΗ, ΗΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΖ, ἔστιν ἄρα ὡς ἡ ΑΗ πρὸς ΕΖ, οὕτως ἡ ΖΕ πρὸς ΕΗ· καὶ ὡς ἄρα τὸ ΑΘ πρὸς ΕΛ, τὸ ΕΛ πρὸς ΚΗ· τῶν ΑΘ, ΗΚ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΕΛ. ἀλλὰ τὸ μὲν ΑΘ ἴσον ἐστὶ τῷ ΣΝ, τὸ δὲ ΗΚ ἴσον τῷ ΝΠ· τῶν ΣΝ, ΝΠ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΕΛ. ἔστι δὲ τῶν αὐτῶν τῶν ΣΝ, ΝΠ μέσον ἀνάλογον καὶ τὸ ΜΡ· ἴσον ἄρα ἐστὶ τὸ ΕΛ τῷ ΜΡ· ὥστε καὶ τῷ ΟΧ ἴσον ἐστίν. ἔστι δὲ καὶ τὰ ΑΘ, ΗΚ τοῖς ΣΝ, ΝΠ ἴσα· ὅλον ἄρα τὸ ΑΓ ἴσον ἐστὶν ὅλῳ τῷ ΣΠ, τοντέστι τῷ ἀπὸ τῆς ΜΧ τετραγώνῳ· τὸ ΑΓ

For let the area AC be contained by the rational (straight-line) AB and by the first binomial (straight-line) AD. I say that square-root of area AC is the irrational (straight-line which is) called binomial.

For since AD is a first binomial (straight-line), let it be divided into its (component) terms at E, and let AE be the greater term. So, (it is) clear that AE and ED are rational (straight-lines which are) commensurable in square only, and that the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE), and that AE is commensurable (in length) with the rational (straight-line) AB (first) laid out [Def. 10.5]. So, let ED be cut in half at point F. And since the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE), thus if a (rectangle) equal to the fourth part of the (square) on the lesser (term)—that is to say, the (square) on EF—falling short by a square figure, is applied to the greater (term) AE, then it divides it into (terms which are) commensurable (in length) [Prop 10.17]. Therefore, let the (rectangle contained) by AG and GE, equal to the (square) on EF, be applied to AE. AG is thus commensurable in length with EG. And let GH, EK, and FL be drawn from (points) G, E, and F (respectively), parallel to either of AB or CD. And let the square SN, equal to the parallelogram AH, be constructed, and (the square) NQ, equal to (the parallelogram) GK [Prop. 2.14]. And let MN be laid down so as to be straight-on to NO. RN is thus also straight-on to NP. And let the parallelogram SQ be completed. SQ is thus a square [Prop. 10.53 lem.]. And since the (rectangle contained) by AG and GE is equal to the (square) on EF, thus

ἄρα δύναται ἡ MX . λέγω, ὅτι ἡ MX ἐκ δύο ὀνομάτων ἐστίν.

Ἐπεὶ γὰρ σύμμετρος ἐστὶν ἡ AH τῇ HE , σύμμετρος ἐστὶ καὶ ἡ AE ἑκατέρᾳ τῶν AH , HE . ὑπόκειται δὲ καὶ ἡ AE τῇ AB σύμμετρος· καὶ αἱ AH , HE ἄρα τῇ AB σύμμετροί εἰσιν. καὶ ἐστὶ ῥητὴ ἡ AB · ῥητὴ ἄρα ἐστὶ καὶ ἑκάτερα τῶν AH , HE · ῥητὸν ἄρα ἐστὶν ἑκάτερον τῶν $A\Theta$, HK , καὶ ἐστὶ σύμμετρον τὸ $A\Theta$ τῷ HK . ἀλλὰ τὸ μὲν $A\Theta$ τῷ ΣN ἴσον ἐστίν, τὸ δὲ HK τῷ $N\Pi$ · καὶ τὰ ΣN , $N\Pi$ ἄρα, τοντέστι τὰ ἀπὸ τῶν MN , NX , ῥητά ἐστὶ καὶ σύμμετρα. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AE τῇ $E\Delta$ μήκει, ἀλλ' ἡ μὲν AE τῇ AH ἐστὶ σύμμετρος, ἡ δὲ ΔE τῇ EZ σύμμετρος, ἀσύμμετρος ἄρα καὶ ἡ AH τῇ EZ · ὥστε καὶ τὸ $A\Theta$ τῷ $E\Lambda$ ἀσύμμετρον ἐστίν. ἀλλὰ τὸ μὲν $A\Theta$ τῷ ΣN ἐστὶν ἴσον, τὸ δὲ $E\Lambda$ τῷ MP · καὶ τὸ ΣN ἄρα τῷ MP ἀσύμμετρον ἐστίν. ἀλλ' ὥς τὸ ΣN προξ MP , ἡ ON προξ τῇν NP · ἀσύμμετρος ἄρα ἐστὶν ἡ ON τῇ NP . ἴση δὲ ἡ μὲν ON τῇ MN , ἡ δὲ NP τῇ NX · ἀσύμμετρος ἄρα ἐστὶν ἡ MN τῇ NX . καὶ ἐστὶ τὸ ἀπὸ τῆς MN σύμμετρον τῷ ἀπὸ τῆς NX , καὶ ῥητὸν ἑκάτερον· αἱ MN , NX ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι.

Ἡ MX ἄρα ἐκ δύο ὀνομάτων ἐστὶ καὶ δύναται τὸ AG · ὅπερ ἔδει δεῖξαι.

as AG is to EF , so FE (is) to EG [Prop. 6.17]. And thus as AH (is) to EL , (so) EL (is) to KG [Prop. 6.1]. Thus, EL is the mean proportional to AH and GK . But, AH is equal to SN , and GK (is) equal to NQ . EL is thus the mean proportional to SN and NQ . And MR is also the mean proportional to the same—(namely), SN and NQ [Prop. 10.53 lem.]. EL is thus equal to MR . Hence, it is also equal to PO [Prop. 1.43]. And AH plus GK is equal to SN plus NQ . Thus, the whole of AC is equal to the whole of SQ —that is to say, to the square on MO . Thus, MO (is) the square-root of (area) AC . I say that MO is a binomial (straight-line).

For since AG is commensurable (in length) with GE , AE is also commensurable (in length) with each of AG and GE [Prop. 10.15]. And AE was also assumed (to be) commensurable (in length) with AB . Thus, AG and GE are also commensurable (in length) with AB [Prop. 10.12]. And AB is rational. AG and GE are thus each also rational. Thus, AH and GK are each rational (areas), and AH is commensurable with GK [Prop. 10.19]. But, AH is equal to SN , and GK to NQ . SN and NQ —that is to say, the (squares) on MN and NO (respectively)—are thus also rational and commensurable. And since AE is incommensurable in length with ED , but AE is commensurable (in length) with AG , and DE (is) commensurable (in length) with EF , AG (is) thus also incommensurable (in length) with EF [Prop. 10.13]. Hence, AH is also incommensurable with EL [Props. 6.1, 10.11]. But, AH is equal to SN , and EL to MR . Thus, SN is also incommensurable with MR . But, as SN (is) to MR , (so) PN (is) to NR [Prop. 6.1]. PN is thus incommensurable (in length) with NR [Prop. 10.11]. And PN (is) equal to MN , and NR to NO . Thus, MN is incommensurable (in length) with NO . And the (square) on MN is commensurable with the (square) on NO , and each (is) rational. MN and NO are thus rational (straight-lines which are) commensurable in square only.

Thus, MO is (both) a binomial (straight-line) [Prop. 10.36], and the square-root of AC . (Which is) the very thing it was required to show.

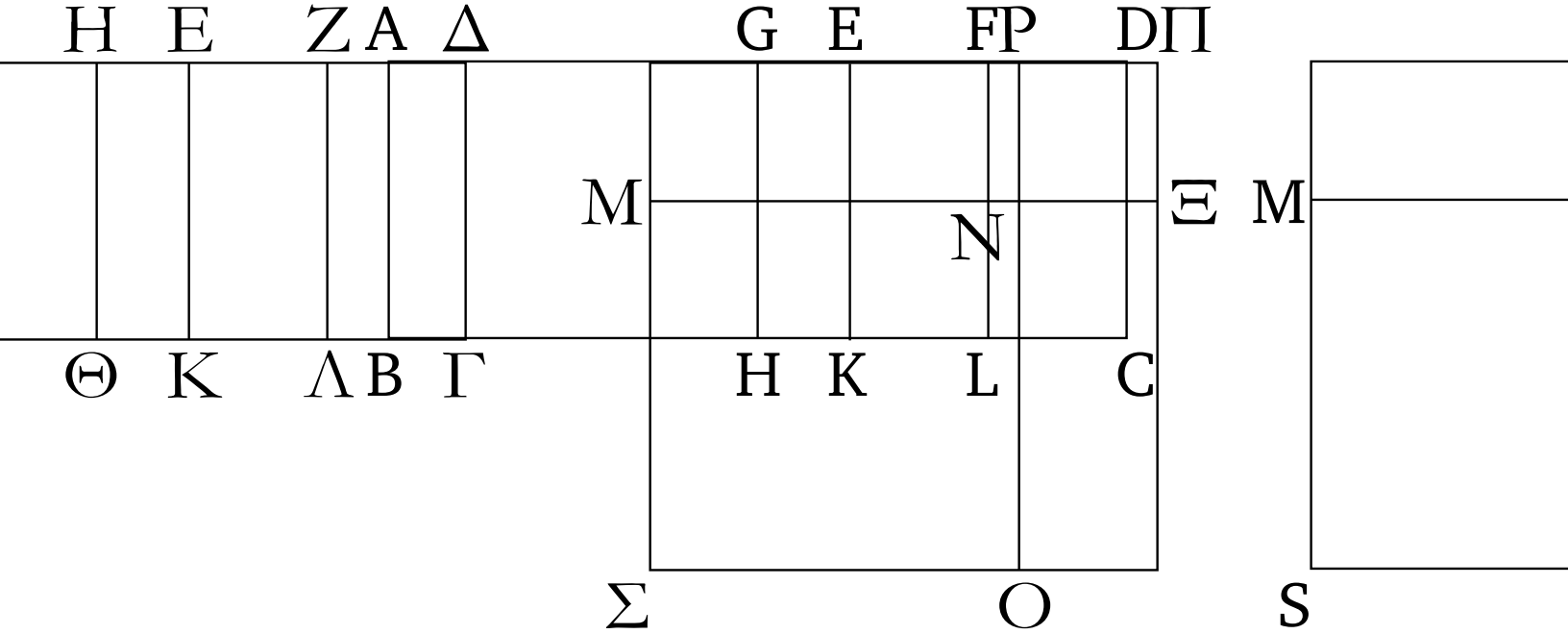
[†] If the rational straight-line has unit length then this proposition states that the square-root of a first binomial straight-line is a binomial straight-line: i.e., a first binomial straight-line has a length $k + k\sqrt{1 - k'^2}$ whose square-root can be written $\rho(1 + \sqrt{k''})$, where $\rho = \sqrt{k(1 + k')}/2$ and $k'' = (1 - k')/(1 + k')$. This is the length of a binomial straight-line (see Prop. 10.36), since ρ is rational.

55.

Proposition 55

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας, ἡ τὸ θωρίον δυνάμενη ἄλογός ἐστιν ἡ καλουμένη ἐκ δύο μέσων πρώτη.

If an area is contained by a rational (straight-line) and a second binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called first bimedial.[†]



Περιεθέσθω γάρ θωρίον τὸ $AB\Gamma\Delta$ ὑπὸ ῥητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας τῆς $A\Delta$ · λέγω, ὅτι ἡ τὸ AF θωρίον δυναμένη ἐκ δύο μέσων πρώτη ἐστίν.

Ἐπεὶ γάρ ἐκ δύο ὀνομάτων δευτέρα ἐστὶν ἡ $A\Delta$, διηγήσθω εἰς τὰ ὀνόματα κατὰ τὸ E , ὥστε τὸ μείζον ὄνομα εἶναι τὸ AE · αἱ AE , $E\Delta$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AE τῆς $E\Delta$ μείζον δύναται τῷ ἀπὸ συμμετρῶν ἑαυτῇ, καὶ τὸ ἔλαττον ὄνομα ἡ $E\Delta$ σύμμετρόν ἐστι τῇ AB μήκει. τεμήσθω ἡ $E\Delta$ δίχα κατὰ τὸ Z , καὶ τῷ ἀπὸ τῆς EZ ἴσον παρὰ τὴν AE παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν AHE · σύμμετρος ἄρα ἡ AH τῇ HE μήκει. καὶ διὰ τῶν H , E , Z παράλληλοι ἤθθωσαν ταῖς AB , $\Gamma\Delta$ αἱ $H\Theta$, EK , $Z\Lambda$, καὶ τῷ μὲν $A\Theta$ παραλληλογράμμῳ ἴσον τετράγωνον συνεστιάτω τὸ SN , τῷ δὲ HK ἴσον τετράγωνον τὸ $N\Pi$, καὶ κείσθω ὥστε ἐπ' εὐθείας εἶναι τὴν MN τῇ NX · ἐπ' εὐθείας ἄρα [ἐστὶ] καὶ ἡ PN τῇ NO . καὶ συμπληρώσθω τὸ $\Sigma\Pi$ τετράγωνον· φανερόν δὲ ἐκ τοῦ προδεδειγμένου, ὅτι τὸ MP μέσον ἀνάλογόν ἐστι τῶν SN , $N\Pi$, καὶ ἴσον τῷ EL , καὶ ὅτι τὸ AF θωρίον δύναται ἡ MX . δεικτέον δὲ, ὅτι ἡ MX ἐκ δύο μέσων ἐστὶ πρώτη.

Ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AE τῇ $E\Delta$ μήκει, σύμμετρος δὲ ἡ $E\Delta$ τῇ AB , ἀσύμμετρος ἄρα ἡ AE τῇ AB . καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ AH τῇ EH , σύμμετρος ἐστὶ καὶ ἡ AE ἑκάτερά τῶν AH , HE . ἀλλὰ ἡ AE ἀσύμμετρος τῇ AB μήκει· καὶ αἱ AH , HE ἄρα ἀσύμμετροί εἰσι τῇ AB . αἱ BA , AH , HE ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ὥστε μέσον ἐστὶν ἑκάτερον τῶν $A\Theta$, HK . ὥστε καὶ ἑκάτερον τῶν SN , $N\Pi$ μέσον ἐστίν. καὶ αἱ MN , NX ἄρα μέσαι εἰσίν. καὶ ἐπεὶ σύμμετρος ἡ AH τῇ HE μήκει, σύμμετρόν ἐστι καὶ τὸ $A\Theta$ τῷ HK , τοντέστι τὸ SN τῷ $N\Pi$, τοντέστι τὸ ἀπὸ τῆς MN τῷ

For let the area $ABCD$ be contained by the rational (straight-line) AB and by the second binomial (straight-line) AD . I say that the square-root of area AC is a first binomial (straight-line).

For since AD is a second binomial (straight-line), let it be divided into its (component) terms at E , such that AE is the greater term. Thus, AE and ED are rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE), and the lesser term ED is commensurable in length with AB [Def. 10.6]. Let ED be cut in half at F . And let the (rectangle contained) by AGE , equal to the (square) on EF , be applied to AE , falling short by a square figure. AG (is) thus commensurable in length with GE [Prop. 10.17]. And let GH , EK , and FL be drawn through (points) G , E , and F (respectively), parallel to AB and CD . And let the square SN , equal to the parallelogram AH , be constructed, and the square NQ , equal to GK . And let MN be laid down so as to be straight-on to NO . Thus, RN [is] also straight-on to NP . And let the square SQ be completed. So, (it is) clear from what has been previously demonstrated [Prop. 10.53 lem.] that MR is the mean proportional to SN and NQ , and (is) equal to EL , and that MO is the square-root of the area AC . So, we must show that MO is a first binomial (straight-line).

Since AE is incommensurable in length with ED , and ED (is) commensurable (in length) with AB , AE (is) thus incommensurable (in length) with AB [Prop. 10.13]. And since AG is commensurable (in length) with EG , AE is also commensurable (in length) with each of AG and GE [Prop. 10.15].

ἀπὸ τῆς NX [ὥστε δυνάμει εἰσὶ σύμμετροι αἱ MN , NX]. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AE τῇ $EΔ$ μήκει, ἀλλ' ἡ μὲν AE σύμμετρος ἐστὶ τῇ AH , ἡ δὲ $EΔ$ τῇ EZ σύμμετρος, ἀσύμμετρος ἄρα ἡ AH τῇ EZ . ὥστε καὶ τὸ $AΘ$ τῷ $ΕΛ$ ἀσύμμετρόν ἐστιν, τοντέστι τὸ $ΣΝ$ τῷ MP , τοντέστιν ὁ ON τῇ NP , τοντέστιν ἡ MN τῇ NX ἀσύμμετρος ἐστὶ μήκει. ἐδείκθησαν δὲ αἱ MN , NX καὶ μέσαι οὕσαι καὶ δυνάμει σύμμετροι· αἱ MN , NX ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δὴ, ὅτι καὶ ῥητὸν περιέθουσιν. ἐπεὶ γάρ ἡ $ΔΕ$ ὑπόκειται ἐκατέρῃ τῶν AB , EZ σύμμετρος, σύμμετρος ἄρα καὶ ἡ EZ τῇ $EΚ$. καὶ ῥητὴ ἐκατέρα αὐτῶν ῥητὸν ἄρα τὸ $ΕΛ$, τοντέστι τὸ MP . τὸ δὲ MP ἐστὶ τὸ ὑπὸ τῶν MNX . ἔαν δὲ δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι ῥητὸν περιέθουσαι, ἡ ὅλη ἄλογός ἐστιν, καλεῖται δὲ ἐκ δύο μέσων πρώτη.

Ἡ ἄρα MX ἐκ δύο μέσων ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

But, AE is incommensurable in length with AB . Thus, AG and GE are also (both) incommensurable (in length) with AB [Prop. 10.13]. Thus, BA , AG , and (BA , and) GE are (pairs of) rational (straight-lines which are) commensurable in square only. And, hence, each of AH and GK is a medial (area) [Prop. 10.21]. Hence, each of SN and NQ is also a medial (area). Thus, MN and NO are medial (straight-lines). And since AG (is) commensurable in length with GE , AH is also commensurable with GK —that is to say, SN with NQ —that is to say, the (square) on MN with the (square) on NO [hence, MN and NO are commensurable in square] [Props. 6.1, 10.11]. And since AE is incommensurable in length with ED , but AE is commensurable (in length) with AG , and ED commensurable (in length) with EF , AG (is) thus incommensurable (in length) with EF [Prop. 10.13]. Hence, AH is also incommensurable with EL —that is to say, SN with MR —that is to say, PN with NR —that is to say, MN is incommensurable in length with NO [Props. 6.1, 10.11]. But MN and NO have also been shown to be medial (straight-lines) which are commensurable in square. Thus, MN and NO are medial (straight-lines which are) commensurable in square only. So, I say that they also contain a rational (area). For since DE was assumed (to be) commensurable (in length) with each of AB and EF , EF (is) thus also commensurable with EK [Prop. 10.12]. And they (are) each rational. Thus, EL —that is to say, MR —(is) rational [Prop. 10.19]. And MR is the (rectangle contained) by MNO . And if two medial (straight-lines), commensurable in square only, which contain a rational (area), are added together, then the whole is (that) irrational (straight-line which is) called first bimedial [Prop. 10.37].

Thus, MO is a first bimedial (straight-line). (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a second binomial straight-line is a first bimedial straight-line: i.e., a second binomial straight-line has a length $k/\sqrt{1-k'^2} + k$ whose square-root can be written $\rho(k'^{1/4} + k'^{3/4})$, where $\rho = \sqrt{(k/2)(1+k')/(1-k')}$ and $k'' = (1-k')/(1+k')$. This is the length of a first bimedial straight-line (see Prop. 10.37), since ρ is rational.

56.

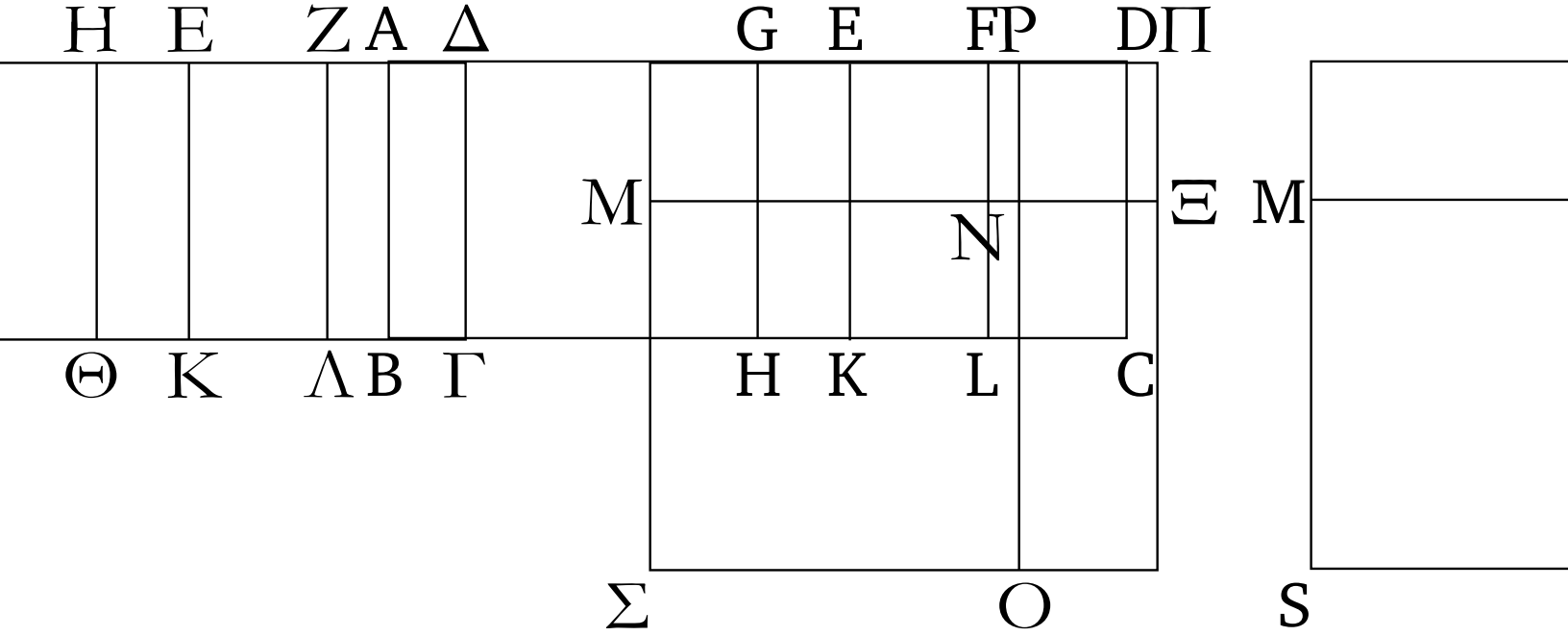
Proposition 56

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων τρίτης, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλουμένη ἐκ δύο μέσων δευτέρα.

Θωρίον γάρ τὸ $ABΓΔ$ περιέθῃται ὑπὸ ῥητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων τρίτης τῆς $ΑΔ$ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ E , ὧν μείζον ἐστὶ τὸ AE . λέγω, ὅτι ἡ τὸ $ΑΓ$ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλουμένη ἐκ δύο μέσων δευτέρα.

If an area is contained by a rational (straight-line) and a third binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called second bimedial.†

For let the area $ABCD$ be contained by the rational (straight-line) AB and by the third binomial (straight-line) AD , which has been divided into its (component) terms at E , of which AE is the greater. I say that the square-root of area AC is the irrational (straight-line which is) called second bimedial.



Κατεσκευάσθω γάρ τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶ τρίτη ἡ $ΑΔ$, αἱ $ΑΕ$, $ΕΔ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ $ΑΕ$ τῇ $ΕΔ$ μείζον δύνανται τῷ ἀπὸ συμέτρου ἐαντῇ, καὶ οὐδετέρω τῶν $ΑΕ$, $ΕΔ$ σύμμετρόξ [ἐστὶ] τῇ $ΑΒ$ μήκει. ὁμοίως δὲ τοῖς προοδεδειγμένοις δείχομεν, ὅτι ἡ $ΜΧ$ ἐστὶν ἡ τὸ $ΑΓ$ θωρίον δυναμένη, καὶ αἱ $ΜΝ$, $ΝΧ$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· ὥστε ἡ $ΜΧ$ ἐκ δύο μέσων ἐστίν. δεικτέον δὲ, ὅτι καὶ δευτέρα.

[Καί] ἐπεὶ ἀσύμμετρός ἐστὶν ἡ $ΔΕ$ τῇ $ΑΒ$ μήκει, τοντέστι τῇ $ΕΚ$, σύμμετρος δὲ ἡ $ΔΕ$ τῇ $ΕΖ$, ἀσύμμετρος ἄρα ἐστὶν ἡ $ΕΖ$ τῇ $ΕΚ$ μήκει. καὶ εἰσι ῥηταὶ· αἱ $ΖΕ$, $ΕΚ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. μέσον ἄρα [ἐστὶ] τὸ $ΕΛ$, τοντέστι τὸ $ΜΡ$ · καὶ περιέχεται ὑπὸ τῶν $ΜΝΞ$ · μέσον ἄρα ἐστὶ τὸ ὑπὸ τῶν $ΜΝΞ$.

Ἡ $ΜΧ$ ἄρα ἐκ δύο μέσων ἐστὶ δευτέρα· ὅπερ ἔδει δεῖξαι.

For let the same construction be made as previously. And since AD is a third binomial (straight-line), AE and ED are thus rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE) , and neither of AE and ED [is] commensurable in length with AB [Def. 10.7]. So, similarly to that which has been previously demonstrated, we can show that MO is the square-root of area AC , and MN and NO are medial (straight-lines which are) commensurable in square only. Hence, MO is bimedral. So, we must show that (it is) also second (bimedial).

[And] since DE is incommensurable in length with AB —that is to say, with EK —and DE (is) commensurable (in length) with EF , EF is thus incommensurable in length with EK [Prop. 10.13]. And they are (both) rational (straight-lines). Thus, FE and EK are rational (straight-lines which are) commensurable in square only. EL —that is to say, MR —[is] thus medial [Prop. 10.21]. And it is contained by MNO . Thus, the (rectangle contained) by MNO is medial.

Thus, MO is a second bimedral (straight-line) [Prop. 10.38]. (Which is) the very thing it was required to show.

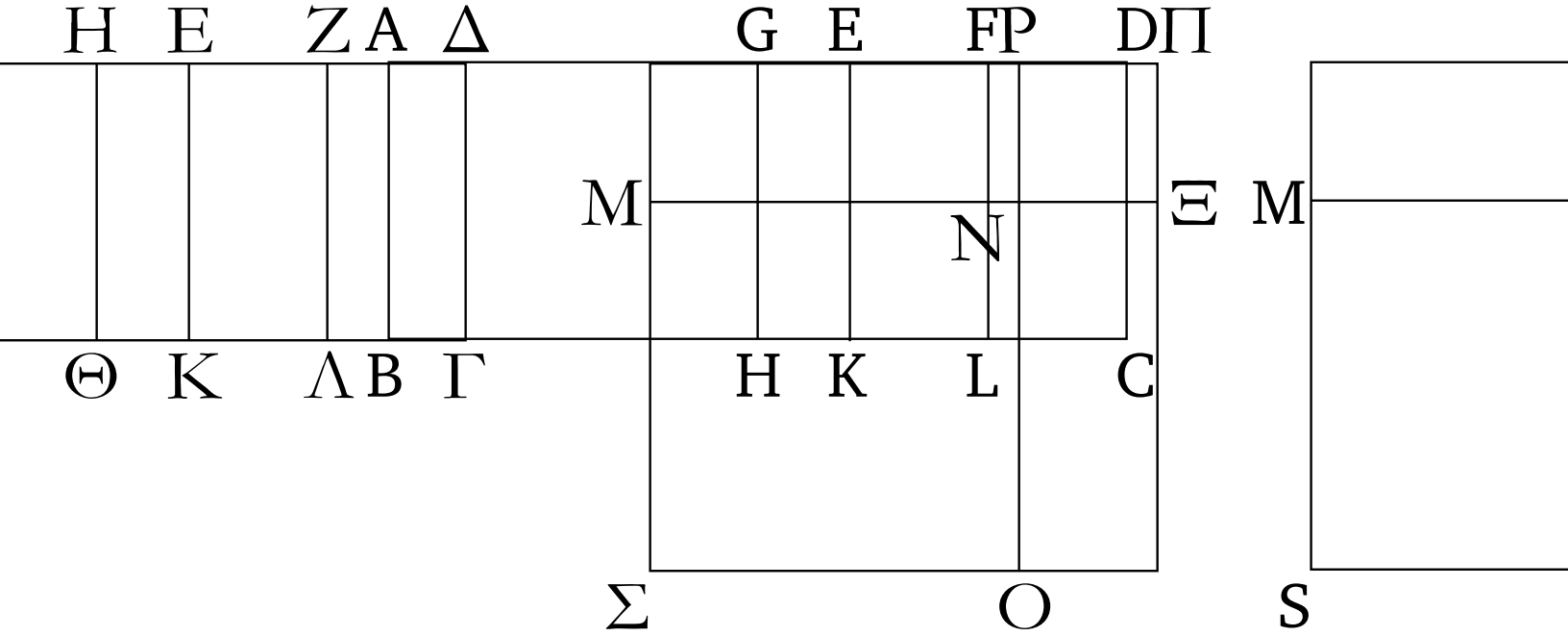
† If the rational straight-line has unit length then this proposition states that the square-root of a third binomial straight-line is a second bimedral straight-line: i.e., a third binomial straight-line has a length $k^{1/2}(1 + \sqrt{1 - k'^2})$ whose square-root can be written $\rho(k^{1/4} + k'^{1/2}/k^{1/4})$, where $\rho = \sqrt{(1 + k')/2}$ and $k' = k(1 - k')/(1 + k')$. This is the length of a second bimedral straight-line (see Prop. 10.38), since ρ is rational.

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῇς ἐκ δύο ὀνομάτων τετάρτης, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἡ κα-

If an area is contained by a rational (straight-line) and a fourth binomial (straight-line) then the square-root of the area

λογυμένη μείζων.

is the irrational (straight-line which is) called major.[†]



Θωρίον γάρ τὸ ΑΓ περιεθέσθω ὑπὸ ῥητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης τῆς ΑΔ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ Ε, ὧν μείζον ἔστω τὸ ΑΕ· λέγω, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλουμένη μείζων.

Ἐπεὶ γὰρ ἡ ΑΔ ἐκ δύο ὀνομάτων ἐστὶ τετάρτη, αἱ ΑΕ, ΕΔ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μείζον δύναται τῷ ἀπὸ ἀσύμμετρον ἐναντῇ, καὶ ἡ ΑΕ τῇ ΑΒ σύμμετρος [ἐστὶ] μήκει. τετμήσθω ἡ ΔΕ δίχα κατὰ τὸ Ζ, καὶ τῷ ἀπὸ τῆς ΕΖ ἴσον παρὰ τὴν ΑΕ παραβελήσθω παραλληλόγραμμον τὸ ὑπὸ ΑΗ, ΗΕ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΗ τῇ ΗΕ μήκει. ἤθθωσαν παράλληλοι τῇ ΑΒ αἱ ΗΘ, ΕΚ, ΖΑ, καὶ τὰ λοιπὰ τὰ αὐτὰ τοῖς πρὸ τούτου γερονέτω· φανερόν δὴ, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἐστὶν ἡ ΜΧ. δεικτέον δὴ, ὅτι ἡ ΜΧ ἄλογός ἐστιν ἡ καλουμένη μείζων.

Ἐπεὶ ἀσύμμετρος ἐστὶν ἡ ΑΗ τῇ ΕΗ μήκει, ἀσύμμετρόν ἐστι καὶ τὸ ΑΘ τῷ ΗΚ, τοντέστι τὸ ΣΝ τῷ ΝΠ· αἱ ΜΝ, ΝΧ ἄρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ ΑΕ τῇ ΑΒ μήκει, ῥητόν ἐστι τὸ ΑΚ· καὶ ἐστὶν ἴσον τοῖς ἀπὸ τῶν ΜΝ, ΝΧ· ῥητόν ἄρα [ἐστὶ] καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ. καὶ ἐπεὶ ἀσύμμετρος [ἐστὶν] ἡ ΔΕ τῇ ΑΒ μήκει, τοντέστι τῇ ΕΚ, ἀλλὰ ἡ ΔΕ σύμμετρος ἐστὶ τῇ ΕΖ, ἀσύμμετρος ἄρα ἡ ΕΖ τῇ ΕΚ μήκει. αἱ ΕΚ, ΕΖ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα τὸ ΑΕ, τοντέστι τὸ ΜΡ. καὶ περιέχεται ὑπὸ τῶν ΜΝ, ΝΧ· μέσον ἄρα ἐστὶ τὸ ὑπὸ τῶν ΜΝ, ΝΧ. καὶ ῥητόν τὸ [συνκείμενον] ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ, καὶ εἰσὶν ἀσύμμετροι αἱ ΜΝ, ΝΧ δυνάμει. ἔαν δὲ δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον, ἡ ὅλη ἄλογός ἐστιν, καλεῖται δὲ μείζων.

For let the area AC be contained by the rational (straight-line) AB and the fourth binomial (straight-line) AD, which has been divided into its (component) terms at E, of which let AE be the greater. I say that the square-root of AC is the irrational (straight-line which is) called major.

For since AD is a fourth binomial (straight-line), AE and ED are thus rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) incommensurable (in length) with (AE), and AE [is] commensurable in length with AB [Def. 10.8]. Let DE be cut in half at F, and let the parallelogram (contained by) AG and GE, equal to the (square) on EF, (and falling short by a square figure) be applied to AE. AG is thus incommensurable in length with GE [Prop. 10.18]. Let GH, EK, and FL be drawn parallel to AB, and let the rest (of the construction) be made the same as the (proposition) before this. So, it is clear that MO is the square-root of area AC. So, we must show that MO is the irrational (straight-line which is) called major.

Since AG is incommensurable in length with EG, AH is also incommensurable with GK—that is to say, SN with NQ [Props. 6.1, 10.11]. Thus, MN and NO are incommensurable in square. And since AE is commensurable in length with AB, AK is rational [Prop. 10.19]. And it is equal to the (sum of the squares) on MN and NO. Thus, the sum of the (squares) on MN and NO [is] also rational. And since DE [is] incommensurable in length with AB [Prop. 10.13]—that is to say, with EK—but DE is commensurable (in length) with EF, EF (is) thus incommensurable in length with EK [Prop. 10.13].

Ἡ MX ἄρα ἄλογός ἐστιν ἡ καλονμένη μείζων, καὶ δύναται τὸ AG θωρίον· ὅπερ ἔδει δεῖχαι.

Thus, EK and EF are rational (straight-lines which are) commensurable in square only. LE —that is to say, MR —(is) thus medial [Prop. 10.21]. And it is contained by MN and NO . The (rectangle contained) by MN and NO is thus medial. And the [sum] of the (squares) on MN and NO (is) rational, and MN and NO are incommensurable in square. And if two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial, are added together, then the whole is the irrational (straight-line which is) called major [Prop. 10.39].

Thus, MO is the irrational (straight-line which is) called major. And (it is) the square-root of area AC . (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then this proposition states that the square-root of a fourth binomial straight-line is a major straight-line: i.e., a fourth binomial straight-line has a length $k(1 + 1/\sqrt{1+k'})$ whose square-root can be written $\rho\sqrt{[1+k''/(1+k''^2)^{1/2}]/2} + \rho\sqrt{[1-k''/(1+k''^2)^{1/2}]/2}$, where $\rho = \sqrt{k}$ and $k''^2 = k'$. This is the length of a major straight-line (see Prop. 10.39), since ρ is rational.

58.

Proposition 58

Ἐάν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πέμπτης, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλονμένη ῥητὸν καὶ μέσον δυναμένη.

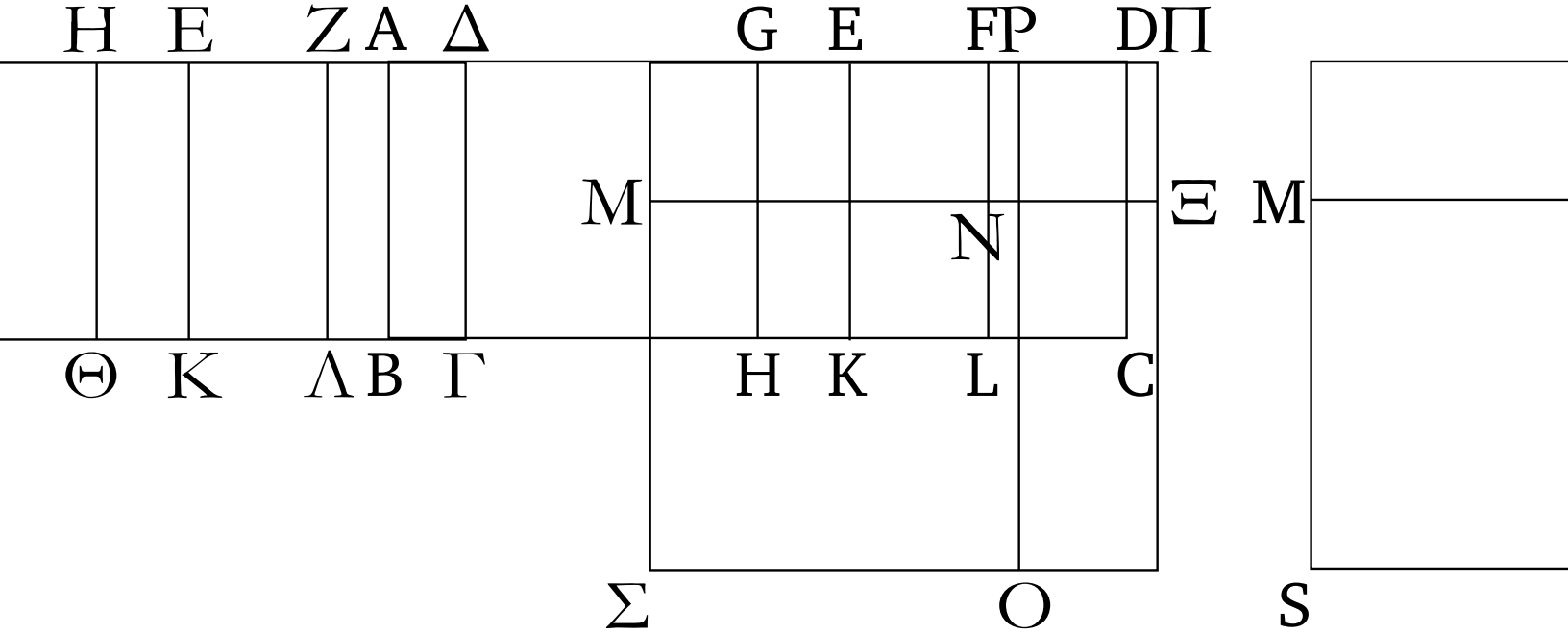
Θωρίον γάρ τὸ AG περιεθέσθω ὑπὸ ῥητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων πέμπτης τῆς AD διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ E , ὥστε τὸ μείζον ὄνομα εἶναι τὸ AE · λέγω [δὴ], ὅτι ἡ τὸ AG θωρίον δυναμένη ἄλογός ἐστιν ἡ καλονμένη ῥητὸν καὶ μέσον δυναμένη.

Κατεσκευάσθω γάρ τὰ αὐτὰ τοῖς πρότερον δεδειγμένοις· φανερόν δὴ, ὅτι ἡ τὸ AG θωρίον δυναμένη ἐστὶν ἡ MX . Δείκτεον δὴ, ὅτι ἡ MX ἐστὶν ἡ ῥητὸν καὶ μέσον δυναμένη.

If an area is contained by a rational (straight-line) and a fifth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called the square-root of a rational plus a medial (area).[†]

For let the area AC be contained by the rational (straight-line) AB and the fifth binomial (straight-line) AD , which has been divided into its (component) terms at E , such that AE is the greater term. [So] I say that the square-root of area AC is the irrational (straight-line which is) called the square-root of a rational plus a medial (area).

For let the same construction be made as that shown previously. So, (it is) clear that MO is the square-root of area AC . So, we must show that MO is the square-root of a rational plus a medial (area).



Ἐπεὶ γὰρ ἀσύμμετρος ἐστὶν ἡ AH τῇ HE , ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ $A\Theta$ τῷ ΘE , τουτέστι τὸ ἀπὸ τῆς MN τῷ ἀπὸ τῆς NX . αἱ MN , NX ἄρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ ἡ $A\Delta$ ἐκ δύο ὀνομάτων ἐστὶ πέμπτη, καὶ [ἐστὶν] ἔλασσον αὐτῆς τμήμα τὸ $E\Delta$, σύμμετρος ἄρα ἡ $E\Delta$ τῇ AB μήκει. ἀλλὰ ἡ AE τῇ $E\Delta$ ἐστὶν ἀσύμμετρος· καὶ ἡ AB ἄρα τῇ AE ἐστὶν ἀσύμμετρος μήκει [αἱ BA , AE ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι]· μέσον ἄρα ἐστὶ τὸ AK , τουτέστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν MN , NX . καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ ΔE τῇ AB μήκει, τουτέστι τῇ EK , ἀλλὰ ἡ ΔE τῇ EZ σύμμετρος ἐστὶν, καὶ ἡ EZ ἄρα τῇ EK σύμμετρος ἐστὶν. καὶ ῥητὴ ἡ EK · ῥητὸν ἄρα καὶ τὸ $E\Lambda$, τουτέστι τὸ MP , τουτέστι τὸ ὑπὸ MNX . αἱ MN , NX ἄρα δυνάμει ἀσύμμετροί εἰσι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν.

Ἡ MX ἄρα ῥητόν καὶ μέσον δυναμένη ἐστὶ καὶ δύναται τὸ $A\Gamma$ θωρίον· ὅπερ ἔδει δεῖχαι.

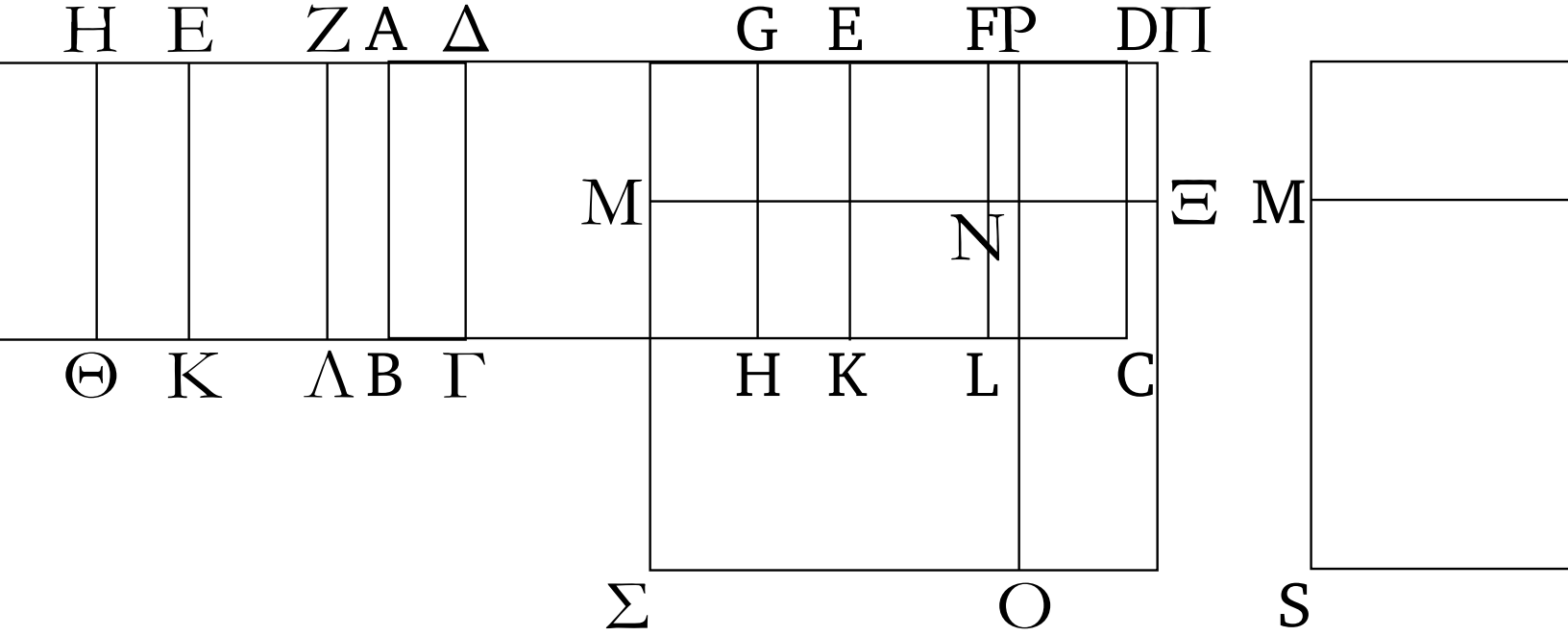
For since AG is incommensurable (in length) with GE [Prop. 10.18], AH is thus also incommensurable with HE —that is to say, the (square) on MN with the (square) on NO [Props. 6.1, 10.11]. Thus, MN and NO are incommensurable in square. And since AD is a fifth binomial (straight-line), and ED [is] its lesser segment, ED (is) thus commensurable in length with AB [Def. 10.9]. But, AE is incommensurable (in length) with ED . Thus, AB is also incommensurable in length with AE [BA and AE are rational (straight-lines which are) commensurable in square only] [Prop. 10.13]. Thus, AK —that is to say, the sum of the (squares) on MN and NO —is medial [Prop. 10.21]. And since DE is commensurable in length with AB —that is to say, with EK —but, DE is commensurable (in length) with EF , EF is thus also commensurable (in length) with EK [Prop. 10.12]. And EK (is) rational. Thus, EL —that is to say, MR —that is to say, the (rectangle contained) by MNO —(is) also rational [Prop. 10.19]. MN and NO are thus (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational.

Thus, MO is the square-root of a rational plus a medial (area) [Prop. 10.40]. And (it is) the square-root of area AC . (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then this proposition states that the square-root of a fifth binomial straight-line is the square root of a rational plus a medial area: i.e., a fifth binomial straight-line has a length $k(\sqrt{1+k'}+1)$ whose square-root can be written $\rho\sqrt{[(1+k''^2)^{1/2}+k'']/[2(1+k''^2)]}+\rho\sqrt{[(1+k''^2)^{1/2}-k'']/[2(1+k''^2)]}$, where $\rho=\sqrt{k(1+k''^2)}$ and $k''^2=k'$. This is the length of the square root of a rational plus a medial area (see Prop. 10.40), since ρ is rational.

Ἐάν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων ἑκτῆς, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλονμένη δύο μέσα δυναμένη.

If an area is contained by a rational (straight-line) and a sixth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called the square-root of (the sum of) two medial (areas).[†]



Θωρίον γάρ τὸ ΑΒΓΔ περιεθέσθω ὑπὸ ῥητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων ἑκτῆς τῆς ΑΔ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ Ε, ὥστε τὸ μείζον ὄνομα εἶναι τὸ ΑΕ· λέγω, ὅτι ἡ τὸ ΑΓ δυναμένη ἡ δύο μέσα δυναμένη ἐστίν.

Κατεσκευάσθω [γάρ] τὰ αὐτὰ τοῖς προοδευμένοις. φανερόν δὲ, ὅτι [ἡ] τὸ ΑΓ δυναμένη ἐστίν ἡ ΜΧ, καὶ ὅτι ἀσύμμετρος ἐστὶν ἡ ΜΝ τῇ ΝΧ δυνάμει. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ ΕΑ τῇ ΑΒ μήκει, αἱ ΕΑ, ΑΒ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα ἐστὶ τὸ ΑΚ, τοντέστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ. πάλιν, ἐπεὶ ἀσύμμετρος ἐστὶν ἡ ΕΔ τῇ ΑΒ μήκει, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ΖΕ τῇ ΕΚ· αἱ ΖΕ, ΕΚ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα ἐστὶ τὸ ΕΛ, τοντέστι τὸ ΜΡ, τοντέστι τὸ ὑπὸ τῶν ΜΝΧ. καὶ ἐπεὶ ἀσύμμετρος ἡ ΑΕ τῇ ΕΖ, καὶ τὸ ΑΚ τῷ ΕΛ ἀσύμμετρόν ἐστιν. ἀλλὰ τὸ μὲν ΑΚ ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ, τὸ δὲ ΕΛ ἐστὶ τὸ ὑπὸ τῶν ΜΝΧ· ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝΧ τῷ ὑπὸ τῶν ΜΝΧ. καὶ ἐστὶ μέσον ἐκάτερον αὐτῶν, καὶ αἱ ΜΝ, ΝΧ δυνάμει εἰσὶν ἀσύμμετροι.

Ἡ ΜΧ ἄρα δύο μέσα δυναμένη ἐστὶ καὶ δύναται τὸ ΑΓ· ὅπερ ἔδει δεῖξαι.

For let the area $ABCD$ be contained by the rational (straight-line) AB and the sixth binomial (straight-line) AD , which has been divided into its (component) terms at E , such that AE is the greater term. So, I say that the square-root of AC is the square-root of (the sum of) two medial (areas).

[For] let the same construction be made as that shown previously. So, (it is) clear that MO is the square-root of AC , and that MN is incommensurable in square with NO . And since EA is incommensurable in length with AB [Def. 10.10], EA and AB are thus rational (straight-lines which are) commensurable in square only. Thus, AK —that is to say, the sum of the (squares) on MN and NO —is medial [Prop. 10.21]. Again, since ED is incommensurable in length with AB [Def. 10.10], FE is thus also incommensurable (in length) with EK [Prop. 10.13]. Thus, FE and EK are rational (straight-lines which are) commensurable in square only. Thus, EL —that is to say, MR —that is to say, the (rectangle contained) by MNO —is medial [Prop. 10.21]. And since AE is incommensurable (in length) with EF , AK is also incommensurable with EL [Props. 6.1, 10.11]. But, AK is the sum of the (squares) on MN and NO , and EL is the (rectangle contained) by MNO . Thus, the sum of the (squares) on MNO is incommensurable with the (rectangle contained) by MNO . And each of them is medial. And MN and NO are incommensurable in square.

Thus, MO is the square-root of (the sum of) two me-

dial (areas) [Prop. 10.41]. And (it is) the square-root of AC . (Which is) the very thing it was required to show.

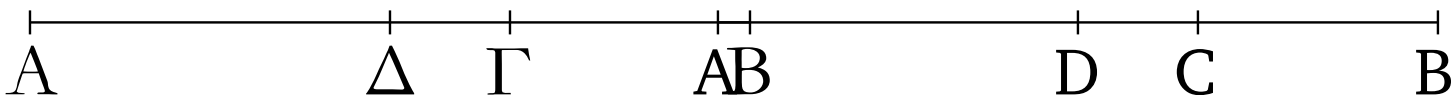
† If the rational straight-line has unit length then this proposition states that the square-root of a sixth binomial straight-line is the square root of the sum of two medial areas: i.e., a sixth binomial straight-line has a length $\sqrt{k} + \sqrt{k'}$ whose square-root can be written $k^{1/4} \left(\sqrt{[1 + k''/(1 + k''^2)^{1/2}]/2} + \sqrt{[1 - k''/(1 + k''^2)^{1/2}]/2} \right)$, where $k''^2 = (k - k')/k'$. This is the length of the square-root of the sum of two medial areas (see Prop. 10.41).

Λήμμα.

Lemma

Ἐάν εὐθεῖα γραμμὴ τμηθῇ εἰς ἄνισα, τὰ ἀπὸ τῶν ἀνίσων τετράγωνα μείζονά ἐστι τοῦ διζ ὑπὸ τῶν ἀνίσων περιεθρομένου ὀρθογωνίου.

If a straight-line is cut unequally then (the sum of) the squares on the unequal (parts) is greater than twice the rectangle contained by the unequal (parts).



Ἐστω εὐθεῖα ἡ AB καὶ τετμήσθω εἰς ἄνισα κατὰ τὸ Γ , καὶ ἔστω μείζων ἡ AG . λέγω, ὅτι τὰ ἀπὸ τῶν AG , GB μείζονά ἐστι τοῦ διζ ὑπὸ τῶν AG , GB .

Let AB be a straight-line, and let it be cut unequally at C , and let AC be greater (than CB). I say that (the sum of) the (squares) on AC and CB is greater than twice the (rectangle contained) by AC and CB .

Τετμήσθω γάρ ἡ AB δίχα κατὰ τὸ Δ . ἐπεὶ οὖν εὐθεῖα γραμμὴ τέτμηται εἰς μὲν ἴσα κατὰ τὸ Δ , εἰς δὲ ἄνισα κατὰ τὸ Γ , τὸ ἄρα ὑπὸ τῶν AG , GB μετὰ τοῦ ἀπὸ $\Gamma\Delta$ ἴσον ἐστὶ τῷ ἀπὸ $A\Delta$. ὥστε τὸ ὑπὸ τῶν AG , GB ἔλαττον ἐστι τοῦ ἀπὸ $A\Delta$. τὸ ἄρα διζ ὑπὸ τῶν AG , GB ἔλαττον ἢ διπλάσιόν ἐστι τοῦ ἀπὸ $A\Delta$. ἀλλὰ τὰ ἀπὸ τῶν AG , GB διπλάσιά [ἐστι] τῶν ἀπὸ τῶν $A\Delta$, $\Delta\Gamma$. τὰ ἄρα ἀπὸ τῶν AG , GB μείζονά ἐστι τοῦ διζ ὑπὸ τῶν AG , GB . ὅπερ ἔδει δεῖξαι.

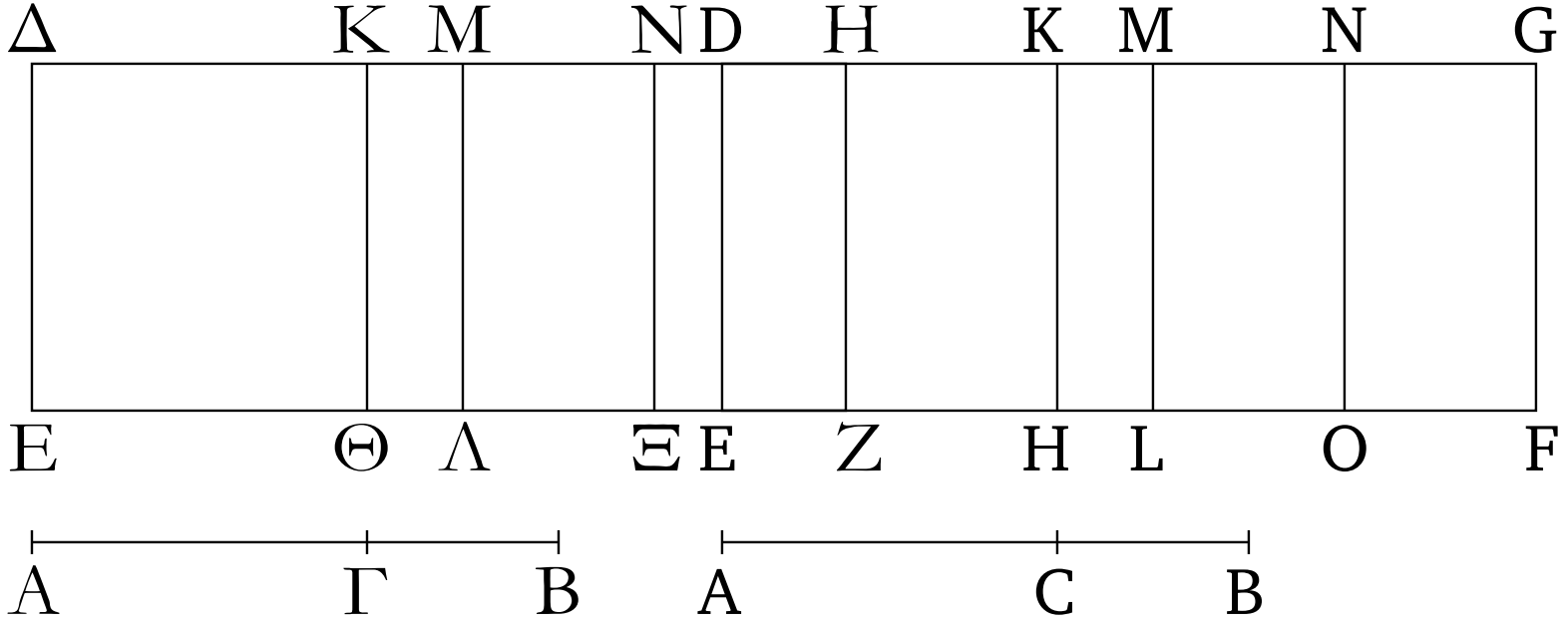
For let AB be cut in half at D . Therefore, since a straight-line has been cut into equal (parts) at D , and into unequal (parts) at C , the (rectangle contained) by AC and CB , plus the (square) on CD , is thus equal to the (square) on AD [Prop. 2.5]. Hence, the (rectangle contained) by AC and CB is less than the (square) on AD . Thus, twice the (rectangle contained) by AC and CB is less than double the (square) on AD . But, (the sum of) the (squares) on AC and CB [is] double (the sum of) the (squares) on AD and DC [Prop. 2.9]. Thus, (the sum of) the (squares) on AC and CB is greater than twice the (rectangle contained) by AC and CB . (Which is) the very thing it was required to show.

60.

Proposition 60

Τὸ ἀπὸ τῆς ἐκ δύο ὀνομάτων παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πρῶτην.

The square on a binomial (straight-line) applied to a rational (straight-line) produces as breadth a first binomial (straight-line).†



Ἐστω ἐκ δύο ὀνομάτων ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ , ὥστε τὸ μείζον ὄνομα εἶναι τὸ AG , καὶ ἐκκείσθω ῥητὴ ἡ DE , καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν DE παραβεβλήσθω τὸ $DEZH$ πλάτος ποιῶν τὴν DH . λέγω, ὅτι ἡ DH ἐκ δύο ὀνομάτων ἐστὶ πρώτη.

Παραβεβλήσθω γάρ παρὰ τὴν DE τῷ μὲν ἀπὸ τῆς AG ἴσον τὸ $\Delta\Theta$, τῷ δὲ ἀπὸ τῆς $B\Gamma$ ἴσον τὸ $ΚΛ$. λοιπὸν ἄρα τὸ διζ ὑπὸ τῶν AG , GB ἴσον ἐστὶ τῷ MZ . τετμήσθω ἡ MH διὰ κατὰ τὸ N , καὶ παράλληλος ἡθῆθω ἡ NX [ἐκατέρω τῶν ML , HZ]. ἐκάτερον ἄρα τῶν MX , NZ ἴσον ἐστὶ τῷ ἅπασι ὑπὸ τῶν AGB . καὶ ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶν ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ , αἱ AG , GB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· τὰ ἄρα ἀπὸ τῶν AG , GB ῥητὰ ἐστὶ καὶ σύμμετρα ἀλλήλοισι· ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB . καὶ ἐστὶν ἴσον τῷ $\Delta\Lambda$. ῥητὸν ἄρα ἐστὶ τὸ $\Delta\Lambda$. καὶ παρὰ ῥητὴν τὴν DE παράκειται ῥητὴ ἄρα ἐστὶν ἡ ΔM καὶ σύμμετρος τῇ DE μήκει. πάλιν, ἐπεὶ αἱ AG , GB ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, μέσον ἄρα ἐστὶ τὸ διζ ὑπὸ τῶν AG , GB , τουτέστι τὸ MZ . καὶ παρὰ ῥητὴν τὴν ML παράκειται ῥητὴ ἄρα καὶ ἡ MH καὶ ἀσύμμετρος τῇ ML , τουτέστι τῇ DE , μήκει. ἐστὶ δὲ καὶ ἡ $M\Delta$ ῥητὴ καὶ τῇ DE μήκει σύμμετρος· ἀσύμμετρος ἄρα ἐστὶν ἡ ΔM τῇ MH μήκει. καὶ εἰσι ῥηταὶ αἱ ΔM , MH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ DH . δεικτέον δὴ, ὅτι καὶ πρώτη.

Ἐπεὶ τῶν ἀπὸ τῶν AG , GB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AGB , καὶ τῶν $\Delta\Theta$, $ΚΛ$ ἄρα μέσον ἀνάλογόν ἐστι τὸ MX . ἐστὶν ἄρα ὥς τὸ $\Delta\Theta$ πρὸς τὸ MX , οὕτως τὸ MX πρὸς τὸ $ΚΛ$, τουτέστιν ὥς ἡ ΔK πρὸς τὴν MN , ἡ MN πρὸς τὴν MK . τὸ ἄρα ὑπὸ τῶν ΔK , KM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN . καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AG τῷ ἀπὸ τῆς GB , σύμμετρόν ἐστι καὶ τὸ $\Delta\Theta$ τῷ $ΚΛ$. ὥστε καὶ ἡ ΔK τῇ KM

Let AB be a binomial (straight-line), having been divided into its (component) terms at C , such that AC is the greater term. And let the rational (straight-line) DE be laid down. And let the (rectangle) $DEFG$, equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a first binomial (straight-line).

For let DH , equal to the (square) on AC , and KL , equal to the (square) on BC , be applied to DE . Thus, the remaining twice the (rectangle contained) by AC and CB is equal to MF [Prop. 2.4]. Let MG be cut in half at N , and let NO be drawn parallel [to each of ML and GF]. MO and NF are thus each equal to once the (rectangle contained) by ACB . And since AB is a binomial (straight-line), having been divided into its (component) terms at C , AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. Thus, the (squares) on AC and CB are rational, and commensurable with one another. And hence the sum of the (squares) on AC and CB (is rational) [Prop. 10.15], and is equal to DL . Thus, DL is rational. And it is applied to the rational (straight-line) DE . DM is thus rational, and commensurable in length with DE [Prop. 10.20]. Again, since AC and CB are rational (straight-lines which are) commensurable in square only, twice the (rectangle contained) by AC and CB —that is to say, MF —is thus medial [Prop. 10.21]. And it is applied to the rational (straight-line) ML . MG is thus also rational, and incommensurable in length with ML —that is to say, with DE [Prop. 10.22]. And MD is also rational, and commensurable in length with DE . Thus, DM is incommensurable in length with MG [Prop. 10.13]. And they are rational. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line)

σύμμετρος ἔστιν. καὶ ἐπεὶ μείζονά ἐστι τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$ τοῦ διζ ὑπὸ τῶν $ΑΓ$, $ΓΒ$, μείζων ἄρα καὶ τὸ $ΔΑ$ τοῦ $ΜΖ$. ὥστε καὶ ἡ $ΔΜ$ τῆς $ΜΗ$ μείζων ἐστίν. καὶ ἔστιν ἴσον τὸ ὑπὸ τῶν $ΔΚ$, $ΚΜ$ τῷ ἀπὸ τῆς $ΜΝ$, τουτέστι τῷ τετάρτῳ τοῦ ἀπὸ τῆς $ΜΗ$ —.7πτ η, καὶ σύμμετρος ἡ $ΔΚ$ τῇ $ΚΜ$. ἔαν δὲ ὥσι δύο εὐθεῖαι ἄνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ καὶ εἷς σύμμετρα αὐτὴν διαιρῇ, ἡ μείζων τῆς ἐλάσσονος μείζων δύναται τῷ ἀπὸ συμμέτρου ἐκείνης· ἡ $ΔΜ$ ἄρα τῆς $ΜΗ$ μείζων δύναται τῷ ἀπὸ συμμέτρου ἐκείνης. καὶ εἰσι ῥηταὶ αἱ $ΔΜ$, $ΜΗ$, καὶ ἡ $ΔΜ$ μείζων ὄνομα ὅσα σύμμετρος ἔστι τῇ ἐκκειμένῃ ῥητῇ τῇ $ΔΕ$ μήκει.

Ἡ $ΔΗ$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

[Prop. 10.36]. So, we must show that (it is) also a first (binomial straight-line).

Since the (rectangle contained) by ACB is the mean proportional to the squares on AC and CB [Prop. 10.53 lem.], MO is thus also the mean proportional to DH and KL . Thus, as DH is to MO , so MO (is) to KL —that is to say, as DK (is) to MN , (so) MN (is) to MK [Prop. 6.1]. Thus, the (rectangle contained) by DK and KM is equal to the (square) on MN [Prop. 6.17]. And since the (square) on AC is commensurable with the (square) on CB , DH is also commensurable with KL . Hence, DK is also commensurable with KM [Props. 6.1, 10.11]. And since (the sum of) the squares on AC and CB is greater than twice the (rectangle contained) by AC and CB [Prop. 10.59 lem.], DL (is) thus also greater than MF . Hence, DM is also greater than MG [Props. 6.1, 5.14]. And the (rectangle contained) by DK and KM is equal to the (square) on MN —that is to say, to one quarter the (square) on MG . And DK (is) commensurable (in length) with KM . And if there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) commensurable (in length), then the square on the greater is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable (in length) with the greater [Prop. 10.17]. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM). And DM and MG are rational. And DM , which is the greater term, is commensurable in length with the (previously) laid down rational (straight-line) DE .

Thus, DG is a first binomial (straight-line) [Def. 10.5]. (Which is) the very thing it was required to show.

† In other words, the square of a binomial is a first binomial. See Prop. 10.54.

61.

Τὸ ἀπὸ τῆς ἐκ δύο μέσων πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων δευτέραν.

Ἐστω ἐκ δύο μέσων πρώτη ἡ $ΑΒ$ διηρημένη εἰς τὰς μέσας κατὰ τὸ $Γ$, ὧν μείζων ἡ $ΑΓ$, καὶ ἐκκείσθω ῥητὴ ἡ $ΔΕ$, καὶ παραβεβλήσθω παρὰ τὴν $ΔΕ$ τῷ ἀπὸ τῆς $ΑΒ$ ἴσον παραλληλόγραμμον τὸ $ΔΖ$ πλάτος ποιῶν τὴν $ΔΗ$. λέγω, ὅτι ἡ $ΔΗ$ ἐκ δύο ὀνομάτων ἐστὶ δευτέρα.

Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρὸ τούτου. καὶ ἐπεὶ ἡ $ΑΒ$ ἐκ δύο μέσων ἐστὶ πρώτη διηρημένη κατὰ τὸ $Γ$, αἱ $ΑΓ$, $ΓΒ$ ἄρα μέσαι εἰσι δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι· ὥστε καὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$ μέσα ἐστίν. μέσον ἄρα ἐστὶ τὸ $ΔΑ$. καὶ παρὰ ῥητὴν τὴν $ΔΕ$ παραβεβλήται· ῥητὴ ἄρα ἐστίν ἡ $ΜΔ$ καὶ ἀσύμμετρος τῇ $ΔΕ$ μήκει. πάλιν, ἐπεὶ ῥητὸν ἐστὶ τὸ διζ ὑπὸ τῶν $ΑΓ$, $ΓΒ$, ῥητὸν ἐστὶ καὶ τὸ $ΜΖ$.

Proposition 61

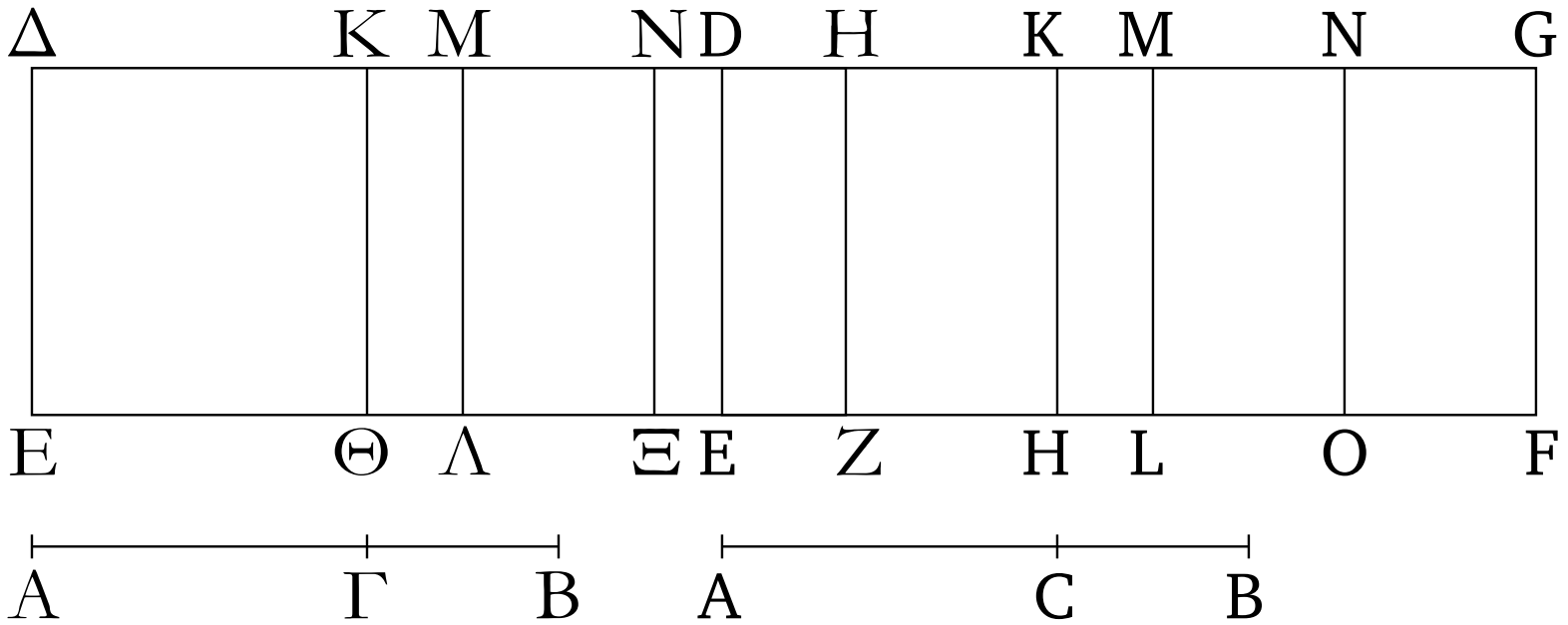
The square on a first binomial (straight-line) applied to a rational (straight-line) produces as breadth a second binomial (straight-line).[†]

Let AB be a first binomial (straight-line) having been divided into its (component) medial (straight-lines) at C , of which AC (is) the greater. And let the rational (straight-line) DE be laid down. And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a second binomial (straight-line).

For let the same construction be made as in the (proposition) before this. And since AB is a first binomial (straight-line), having been divided at C , AC and CB are thus medial (straight-lines) commensurable in square only, and containing a rational (area) [Prop. 10.37]. Hence, the (squares) on

καὶ παρὰ ῥητὴν τὴν $ΜΛ$ παράκειται ῥητὴ ἄρα [ἐστὶ] καὶ ἡ $ΜΗ$ καὶ μήκει σύμμετρος τῇ $ΜΛ$, τούτεστι τῇ $ΔΕ$. ἀσύμμετρος ἄρα ἐστὶν ἡ $ΔΜ$ τῇ $ΜΗ$ μήκει. καὶ εἰσι ῥηταί· αἱ $ΔΜ$, $ΜΗ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $ΔΗ$. δεικτέον δὴ, ὅτι καὶ δευτέρα.

AC and CB are also medial [Prop. 10.21]. Thus, DL is medial [Props. 10.15, 10.23 corr.]. And it has been applied to the rational (straight-line) DE . MD is thus rational, and incommensurable in length with DE [Prop. 10.22]. Again, since twice the (rectangle contained) by AC and CB is rational, MF is also rational. And it is applied to the rational (straight-line) ML . Thus, MG [is] also rational, and commensurable in length with ML —that is to say, with DE [Prop. 10.20]. DM is thus incommensurable in length with MG [Prop. 10.13]. And they are rational. DM and MG are thus rational, and commensurable in square only. DG is thus a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a second (binomial straight-line).



Ἐπεὶ γὰρ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$ μείζονά ἐστι τοῦ δι᾽ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ μείζον ἄρα καὶ τὸ $ΔΑ$ τοῦ $ΜΖ$. ὥστε καὶ ἡ $ΔΜ$ τῇς $ΜΗ$. καὶ ἐπεὶ σύμμετρον ἐστὶ τὸ ἀπὸ τῇς $ΑΓ$ τῷ ἀπὸ τῇς $ΓΒ$, σύμμετρον ἐστὶ καὶ τὸ $ΔΘ$ τῷ $ΚΛ$. ὥστε καὶ ἡ $ΔΚ$ τῇ $ΚΜ$ σύμμετρος ἐστίν. καὶ ἐστὶ τὸ ὑπὸ τῶν $ΔΚΜ$ ἴσον τῷ ἀπὸ τῇς $ΜΝ$. ἡ $ΔΜ$ ἄρα τῇς $ΜΗ$ μείζον δύναται τῷ ἀπὸ συμμέτρον ἐαντῇ. καὶ ἐστὶν ἡ $ΜΗ$ σύμμετρος τῇ $ΔΕ$ μήκει.

Ἡ $ΔΗ$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ δευτέρα.

For since (the sum of) the squares on AC and CB is greater than twice the (rectangle contained) by AC and CB [Prop. 10.59], DL (is) thus also greater than MF . Hence, DM (is) also (greater) than MG [Prop. 6.1]. And since the (square) on AC is commensurable with the (square) on CB , DH is also commensurable with KL . Hence, DK is also commensurable (in length) with KM [Props. 6.1, 10.11]. And the (rectangle contained) by DKM is equal to the (square) on MN . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM) [Prop. 10.17]. And MG is commensurable in length with DE .

Thus, DG is a second binomial (straight-line) [Def. 10.6].

† In other words, the square of a first binomial is a second binomial. See Prop. 10.55.

Τὸ ἀπὸ τῇς ἐκ δύο μέσων δευτέρας παρὰ ῥητὴν παραβαλ-

The square on a second binomial (straight-line) applied to

λόμενον πλάτος ποιῇ τὴν ἐκ δύο ὀνομάτων τρίτην.

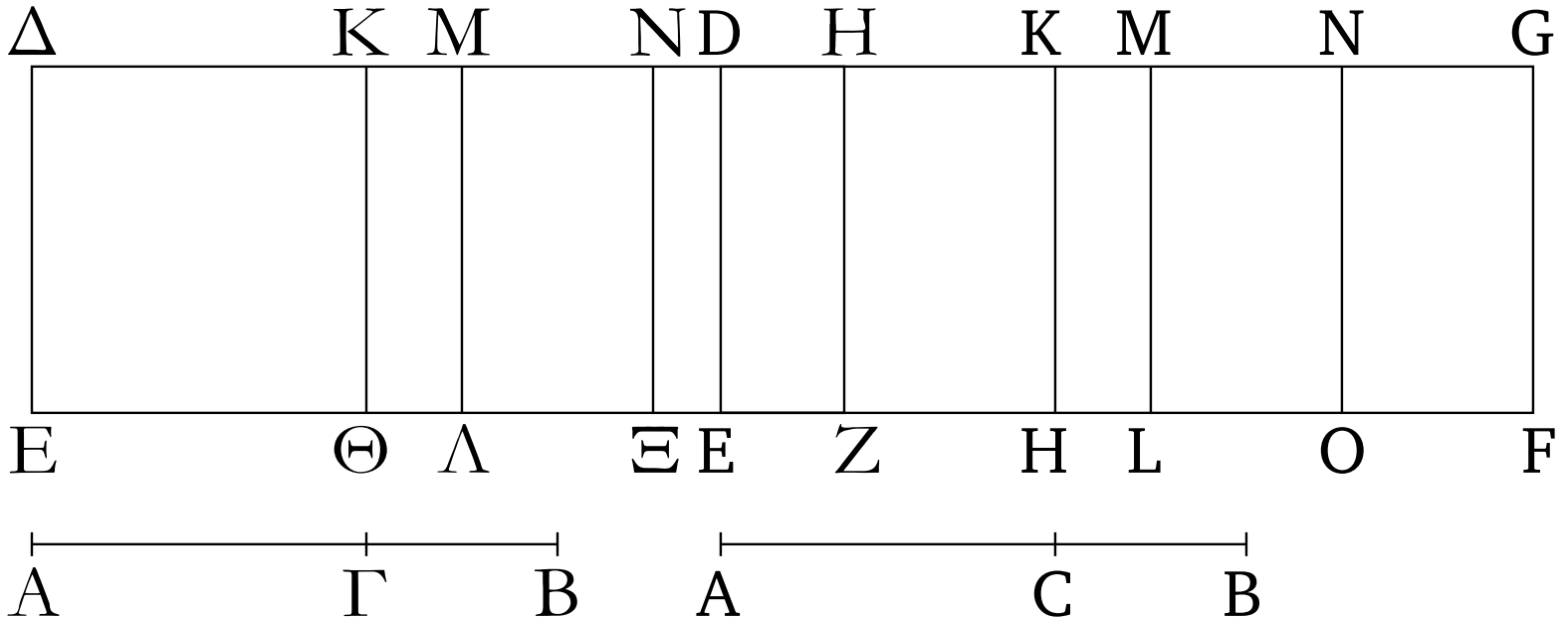
Ἐστω ἐκ δύο μέσων δευτέρα ἡ AB διηρημένη εἰς τὰς μέσας κατὰ τὸ Γ , ὥστε τὸ μείζον τμήμα εἶναι τὸ AG , ῥητὴ δὲ τις ἔστω ἡ ΔE , καὶ παρὰ τὴν ΔE τῷ ἀπὸ τῆς AB ἴσον παραλληλόγραμμον παραβεβλήσθω τὸ ΔZ πλάτος ποιῶν τὴν ΔH . λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶ τρίτη.

Κατεσκευάσθω τὰ αὐτὰ τοῖς προοδευμένοις. καὶ ἐπεὶ ἐκ δύο μέσων δευτέρα ἐστὶν ἡ AB διηρημένη κατὰ τὸ Γ , αἱ AG , GB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι. ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσον ἐστίν. καὶ ἐστὶν ἴσον τῷ $\Delta\Lambda$. μέσον ἄρα καὶ τὸ $\Delta\Lambda$. καὶ παράκειται παρὰ ῥητὴν τὴν ΔE . ῥητὴ ἄρα ἐστὶ καὶ ἡ $M\Delta$ καὶ ἀσύμμετρος τῇ ΔE μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ MH ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ $M\Lambda$, τοντέστι τῇ ΔE , μήκει. ῥητὴ ἄρα ἐστὶν ἑκατέρα τῶν ΔM , MH καὶ ἀσύμμετρος τῇ ΔE μήκει. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AG τῇ GB μήκει, ὡς δὲ ἡ AG πρὸς τὴν GB , οὕτως τὸ ἀπὸ τῆς AG πρὸς τὸ ὑπὸ τῶν AGB , ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς AG τῷ ὑπὸ τῶν AGB . ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB τῷ δι᾽ ὑπὸ τῶν AGB ἀσύμμετρόν ἐστιν, τοντέστι τὸ $\Delta\Lambda$ τῷ MZ . ὥστε καὶ ἡ ΔM τῷ MH ἀσύμμετρος ἐστίν. καὶ εἰσι ῥηταί. ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH . δεικτέον [δὴ], ὅτι καὶ τρίτη.

a rational (straight-line) produces as breadth a third binomial (straight-line).[†]

Let AB be a second binomial (straight-line) having been divided into its (component) medial (straight-lines) at C , such that AC is the greater segment. And let DE be some rational (straight-line). And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a third binomial (straight-line).

Let the same construction be made as that shown previously. And since AB is a second binomial (straight-line), having been divided at C , AC and CB are thus medial (straight-lines) commensurable in square only, and containing a medial (area) [Prop. 10.38]. Hence, the sum of the (squares) on AC and CB is also medial [Props. 10.15, 10.23 corr.]. And it is equal to DL . Thus, DL (is) also medial. And it is applied to the rational (straight-line) DE . MD is thus also rational, and incommensurable in length with DE [Prop. 10.22]. So, for the same (reasons), MG is also rational, and incommensurable in length with ML —that is to say, with DE . Thus, DM and MG are each rational, and incommensurable in length with DE . And since AC is incommensurable in length with CB , and as AC (is) to CB , so the (square) on AC (is) to the (rectangle contained) by ACB [Prop. 10.21 lem.], the (square) on AC (is) also incommensurable with the (rectangle contained) by ACB [Prop. 10.11]. And hence the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by ACB —that is to say, DL with MF [Props. 10.12, 10.13]. Hence, DM is also incommensurable (in length) with MG [Props. 6.1, 10.11]. And they are rational. DG is thus a binomial (straight-line) [Prop. 10.36]. [So] we must show that (it is) also a third (binomial straight-line).



Ὁμοίως δὴ τοῖς προτέροις ἐπιλογισμένοι, ὅτι μείζων ἐστὶν ἡ ΔM τῆς MH , καὶ σύμμετρος ἡ ΔK τῇ KM . καὶ ἐστὶ τὸ ὑπὸ τῶν ΔKM ἴσον τῷ ἀπὸ τῆς MN . ἡ ΔM ἄρα τῆς MH μείζων δύνатаι τῷ ἀπὸ συμμέτρου ἐαυτῆς. καὶ οὐδετέρα τῶν ΔM , MH σύμμετρος ἐστὶ τῇ ΔE μήκει.

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη· ὅπερ ἔδει δεῖχαι.

So, similarly to the previous (propositions), we can conclude that DM is greater than MG , and DK (is) commensurable (in length) with KM . And the (rectangle contained) by DKM is equal to the (square) on MN . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM) [Prop. 10.17]. And neither of DM and MG is commensurable in length with DE .

Thus, DG is a third binomial (straight-line) [Def. 10.7]. (Which is) the very thing it was required to show.

† In other words, the square of a second binomial is a third binomial. See Prop. 10.56.

63.

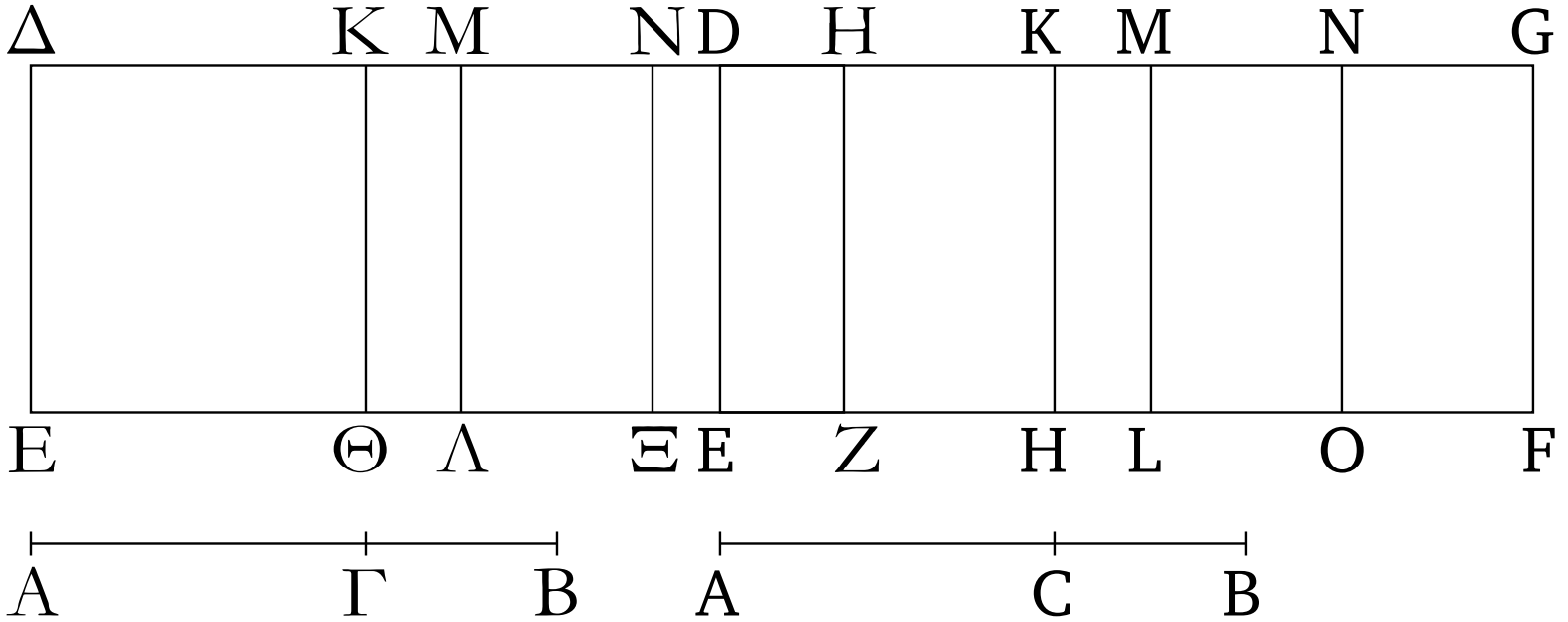
Proposition 63

Τὸ ἀπὸ τῆς μείζονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τετάρτην.

Ἐστω μείζων ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε μείζονα εἶναι τὴν AG τῆς GB , ῥητὴ δὲ ἡ ΔE , καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν ΔE παραβεβλήσθω τὸ ΔZ παραλληλόγραμμον πλάτος ποιοῦν τὴν ΔH . λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶ τετάρτη.

The square on a major (straight-line) applied to a rational (straight-line) produces as breadth a fourth binomial (straight-line).†

Let AB be a major (straight-line) having been divided at C , such that AC is greater than CB , and (let) DE (be) a rational (straight-line). And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a fourth binomial (straight-line).



Κατεσκευάσθω τὰ αὐτὰ τοῖς προοδευσμένοις. καὶ ἐπεὶ μείζων ἐστὶν ἡ AB διηρημένη κατὰ τὸ Γ , αἱ AG , GB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δὲ ὑπ' αὐτῶν μέσον. ἐπεὶ οὖν ῥητόν ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB , ῥητόν ἄρα ἐστὶ τὸ ΔA . ῥητὴ ἄρα καὶ ἡ ΔM καὶ σύμμετρος τῇ ΔE μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ διξ ὑπὸ τῶν AG , GB , τουτέστι τὸ MZ , καὶ παρὰ ῥητὴν ἐστὶ τὴν MA , ῥητὴ ἄρα ἐστὶ καὶ ἡ MH καὶ ἀσύμμετρος τῇ ΔE μήκει· ἀσύμμετρος

Let the same construction be made as that shown previously. And since AB is a major (straight-line), having been divided at C , AC and CB are incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial [Prop. 10.39]. Therefore, since the sum of the (squares) on AC and CB is rational, DL is thus rational. Thus, DM (is) also rational, and commensurable in length with DE [Prop. 10.20]. Again, since twice the (rectangle contained) by AC and CB —that is to say,

ἄρα ἐστὶ καὶ ἡ ΔM τῇ MH μήκει. αἱ ΔM , MH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH . δεικτέον [δὴ], ὅτι καὶ τετάρτη.

Ὁμοίως δὴ δείχομεν τοῖς πρότερον, ὅτι μείζων ἐστὶν ἡ ΔM τῇ MH , καὶ ὅτι τὸ ὑπὸ ΔKM ἴσον ἐστὶ τῷ ἀπὸ τῇ MN . ἐπεὶ οὖν ἀσύμμετρόν ἐστι τὸ ἀπὸ τῇ AF τῷ ἀπὸ τῇ GB , ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ $\Delta \Theta$ τῷ KL . ὥστε ἀσύμμετρος καὶ ἡ ΔK τῇ KM ἐστίν. ἔαν δὲ ὥσι δύο εὐθεῖαι ἄνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῇ ἐλάσσονος ἴσον παραλληλόγραμμον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ καὶ εἰς ἀσύμμετρα αὐτὴν διαιρῇ, ἡ μείζων τῇ ἐλάσσονος μείζον δυνήσεται τῷ ἀπὸ ἀσύμμετρον ἐαυτῇ μήκει· ἡ ΔM ἄρα τῇ MH μείζον δύναται τῷ ἀπὸ ἀσύμμετρον ἐαυτῇ. καὶ εἰσιν αἱ ΔM , MH ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ ἡ ΔM σύμμετρος ἐστὶ τῇ ἐκκειμένῃ ῥητῇ τῇ ΔE .

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶ τετάρτη· ὅπερ ἔδει δεῖχαι.

MF —is medial, and is (applied to) the rational (straight-line) ML , MG is thus also rational, and incommensurable in length with DE [Prop. 10.22]. DM is thus also incommensurable in length with MG [Prop. 10.13]. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. [So] we must show that (it is) also a fourth (binomial straight-line).

So, similarly to the previous (propositions), we can show that DM is greater than MG , and that the (rectangle contained) by DKM is equal to the (square) on MN . Therefore, since the (square) on AC is incommensurable with the (square) on CB , DH is also incommensurable with KL . Hence, DK is also incommensurable with KM [Props. 6.1, 10.11]. And if there are two unequal straight-lines, and a parallelogram equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) incommensurable (in length), then the square on the greater will be larger than (the square on) the lesser by the (square) on (some straight-line) incommensurable in length with the greater [Prop. 10.18]. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) incommensurable (in length) with (DM). And DM and MG are rational (straight-lines which are) commensurable in square only. And DM is commensurable (in length) with the (previously) laid down rational (straight-line) DE .

Thus, DG is a fourth binomial (straight-line) [Def. 10.8]. (Which is) the very thing it was required to show.

† In other words, the square of a major is a fourth binomial. See Prop. 10.57.

64.

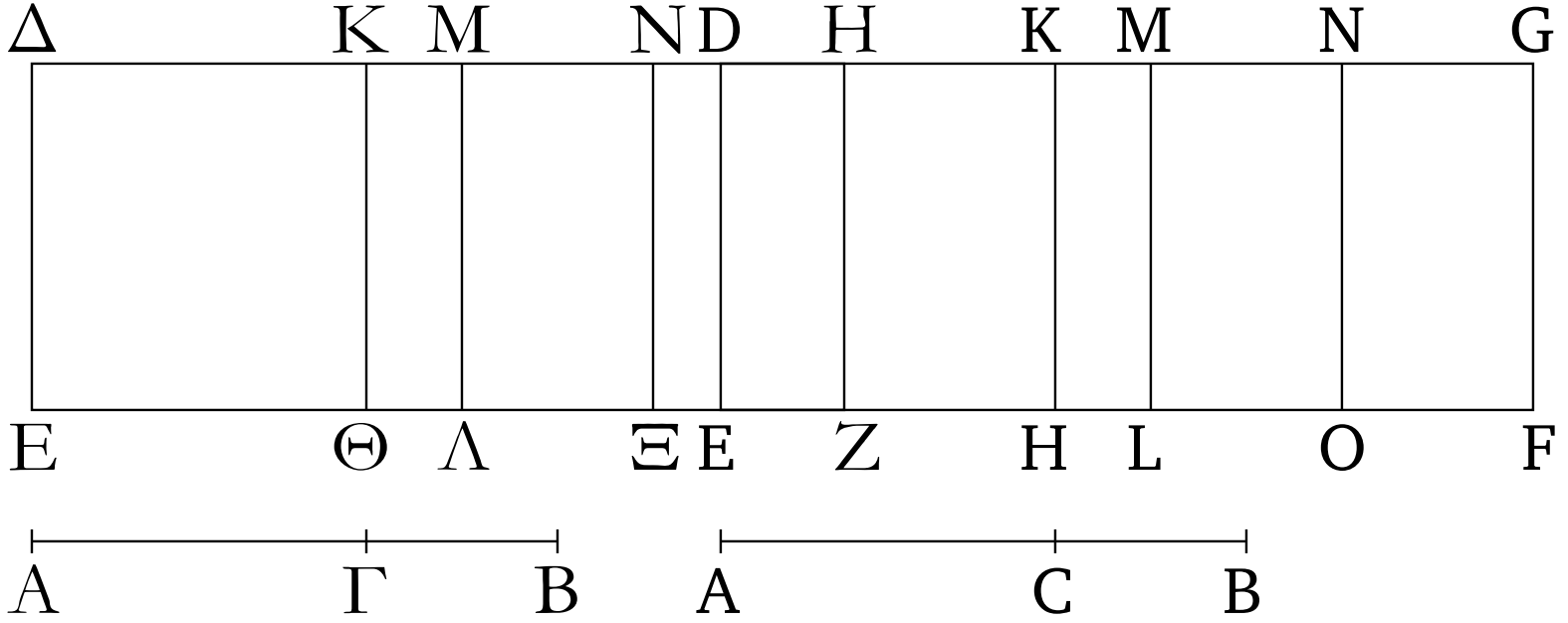
Τὸ ἀπὸ τῇ ῥητὸν καὶ μέσον δυναμένης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πέμπτην.

Ἐστω ῥητὸν καὶ μέσον δυναμένη ἡ AB διηρημένη εἰς τὰς εὐθείας κατὰ τὸ Γ , ὥστε μείζονα εἶναι τὴν AG , καὶ ἐκκείσθω ῥητὴ ἡ ΔE , καὶ τῷ ἀπὸ τῇ AB ἴσον παρὰ τὴν ΔE παραβλήσθω τὸ ΔZ πλάτος ποιοῦν τὴν ΔH . λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶ πέμπτη.

Proposition 64

The square on the square-root of a rational plus a medial (area) applied to a rational (straight-line) produces as breadth a fifth binomial (straight-line).[†]

Let AB be the square-root of a rational plus a medial (area) having been divided into its (component) straight-lines at C , such that AC is greater. And let the rational (straight-line) DE be laid down. And let the (parallelogram) DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a fifth binomial straight-line.



Κατεσκευάσθω τὰ αὐτὰ τοῖς πρὸ τούτων. ἐπεὶ οὖν ῥητόν καὶ μέσον δυναμένη ἐστὶν ἡ AB διηρημένη κατὰ τὸ Γ , αἱ AG , GB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν. ἐπεὶ οὖν μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB , μέσον ἄρα ἐστὶ τὸ ΔA . ὥστε ῥητὴ ἐστὶν ἡ ΔM καὶ μήκει ἀσύμμετρος τῇ ΔE . πάλιν, ἐπεὶ ῥητόν ἐστὶ τὸ διξ ὑπὸ τῶν AGB , τοντέστι τὸ MZ , ῥητὴ ἄρα ἡ MH καὶ σύμμετρος τῇ ΔE . ἀσύμμετρος ἄρα ἡ ΔM τῇ MH . αἱ ΔM , MH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH . λέγω δὴ, ὅτι καὶ πέμπτη.

Ὅμοιως γὰρ διεθθήσεται, ὅτι τὸ ὑπὸ τῶν ΔKM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN , καὶ ἀσύμμετρος ἡ ΔK τῇ KM μήκει· ἡ ΔM ἄρα τῆς MH μείζον δύνανται τῷ ἀπὸ ἀσυμμέτρον ἑαυτῇ. καὶ εἰσιν αἱ ΔM , MH [ῥηταὶ] δυνάμει μόνον σύμμετροι, καὶ ἡ ἐλάσσων ἡ MH σύμμετρος τῇ ΔE μήκει.

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶ πέμπτη· ὅπερ ἔδει δεῖξαι.

Let the same construction be made as in the (propositions) before this. Therefore, since AB is the square-root of a rational plus a medial (area), having been divided at C , AC and CB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational [Prop. 10.40]. Therefore, since the sum of the (squares) on AC and CB is medial, DL is thus medial. Hence, DM is rational and incommensurable in length with DE [Prop. 10.22]. Again, since twice the (rectangle contained) by ACB —that is to say, MF —is rational, MG (is) thus rational and commensurable (in length) with DE [Prop. 10.20]. DM (is) thus incommensurable (in length) with MG [Prop. 10.13]. Thus, DM and MG are rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fifth (binomial straight-line).

For, similarly (to the previous propositions), it can be shown that the (rectangle contained) by DKM is equal to the (square) on MN , and DK (is) incommensurable in length with KM . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) incommensurable (in length) with (DM) [Prop. 10.18]. And DM and MG are [rational] (straight-lines which are) commensurable in square only, and the lesser MG is commensurable in length with DE .

Thus, DG is a fifth binomial (straight-line) [Def. 10.9]. (Which is) the very thing it was required to show.

† In other words, the square of the square-root of a rational plus medial is a fifth binomial. See Prop. 10.58.

Τὸ ἀπὸ τῆς δύο μέσα δυναμένης παρὰ ῥητὴν παραβαλ-

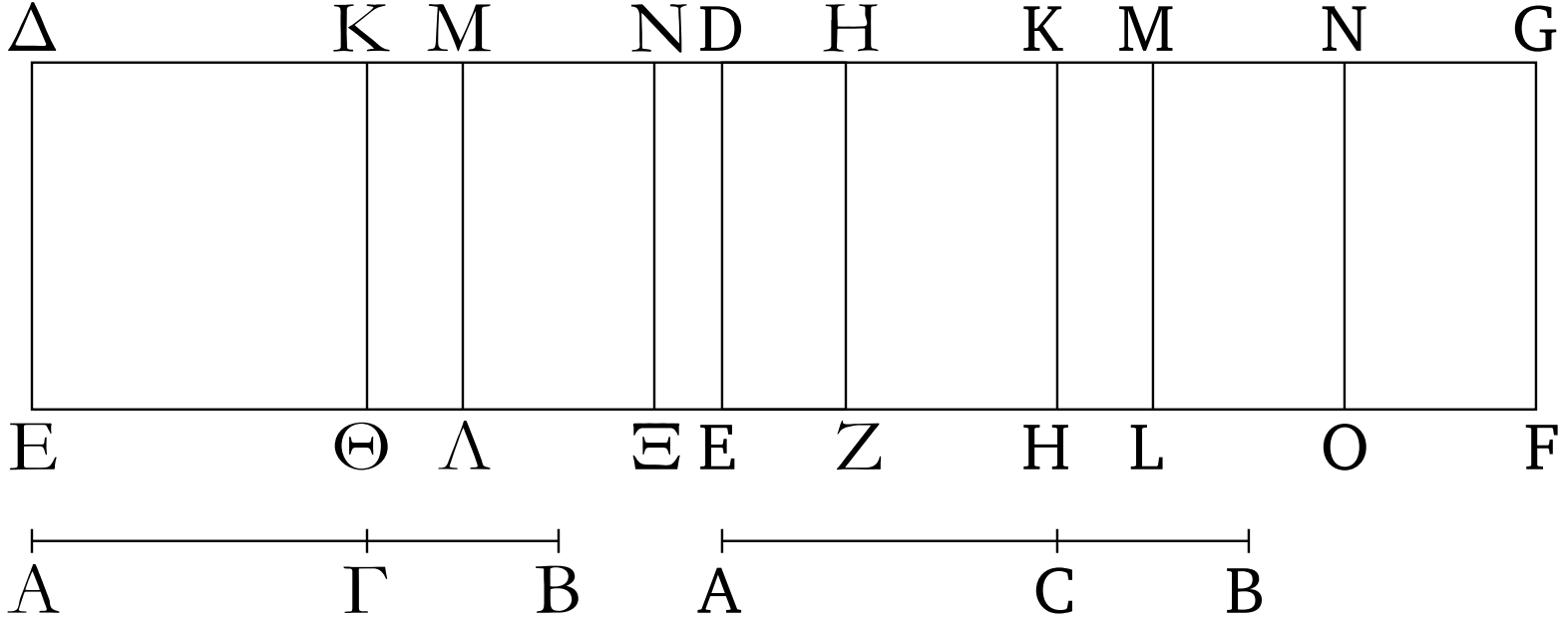
The square on the square-root of (the sum of) two medial

λόμενον πλάτος ποιῇ τὴν ἐκ δύο ὀνομάτων ἑκτὴν.

Ἐστω δύο μέσα δυναμένη ἡ AB διηρημένη κατὰ τὸ Γ , ῥητὴ δὲ ἔστω ἡ ΔE , καὶ παρὰ τὴν ΔE τῷ ἀπὸ τῆς AB ἴσον παραβεβλήσθω τὸ ΔZ πλάτος ποιοῦν τὴν ΔH . λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶν ἑκτὴ.

(areas) applied to a rational (straight-line) produces as breadth a sixth binomial (straight-line).[†]

Let AB be the square-root of (the sum of) two medial (areas), having been divided at C . And let DE be a rational (straight-line). And let the (parallelogram) DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a sixth binomial (straight-line).



Κατεσκευάσθω γάρ τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἡ AB δύο μέσα δυναμένη ἐστὶ διηρημένη κατὰ τὸ Γ , αἱ AG, GB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τὸ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων συγκείμενον τῷ ὑπ' αὐτῶν ὥστε κατὰ τὰ προδεδειγμένα μέσον ἐστὶν ἑκάτερον τῶν $\Delta\Lambda, MZ$. καὶ παρὰ ῥητὴν τὴν ΔE παράκειται ῥητὴ ἄρα ἐστὶν ἑκάτερον τῶν $\Delta M, MH$ καὶ ἀσύμμετρος τῇ ΔE μήκει. καὶ ἐπεὶ ἀσύμμετρον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG, GB τῷ διξ ὑπὸ τῶν AG, GB , ἀσύμμετρον ἄρα ἐστὶ τὸ $\Delta\Lambda$ τῷ MZ . ἀσύμμετρος ἄρα καὶ ἡ ΔM τῇ MH . αἱ $\Delta M, MH$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH . λέγω δὴ, ὅτι καὶ ἑκτὴ.

Ὁμοίως δὴ πάλιν δεῖχμεν, ὅτι τὸ ὑπὸ τῶν ΔKM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN , καὶ ὅτι ἡ ΔK τῇ KM μήκει ἐστὶν ἀσύμμετρος· καὶ διὰ τὰ αὐτὰ δὴ ἡ ΔM τῇ MH μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἐναντὶ μήκει. καὶ οὐδετέρα τῶν $\Delta M, MH$ σύμμετρος ἐστὶ τῇ ἐκκεκλιμένη ῥητὴ τῇ ΔE μήκει.

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶν ἑκτὴ. ὅπερ ἔδει δεῖξαι.

For let the same construction be made as in the previous (propositions). And since AB is the square-root of (the sum of) two medial (areas), having been divided at C , AC and CB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, the sum of the squares on them incommensurable with the (rectangle contained) by them [Prop. 10.41]. Hence, according to what has been previously demonstrated, DL and MF are each medial. And they are applied to the rational (straight-line) DE . Thus, DM and MG are each rational, and incommensurable in length with DE [Prop. 10.22]. And since the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , DL is thus incommensurable with MF . Thus, DM (is) also incommensurable (in length) with MG [Props. 6.1, 10.11]. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a sixth (binomial straight-line).

So, similarly (to the previous propositions), we can again show that the (rectangle contained) by DKM is equal to the (square) on MN , and that DK is incommensurable in length with KM . And so, for the same (reasons), the square on DM is greater than (the square on) MG by the (square) on

(some straight-line) incommensurable in length with (DM) [Prop. 10.18]. And neither of DM and MG is commensurable in length with the (previously) laid down rational (straight-line) DE .

Thus, DG is a sixth binomial (straight-line) [Def. 10.10]. (Which is) the very thing it was required to show.

† In other words, the square of the square-root of two medials is a sixth binomial. See Prop. 10.59.

66.

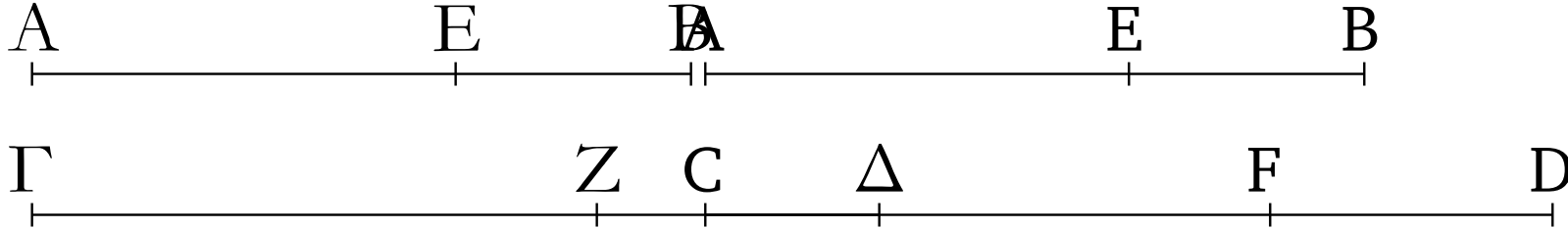
Proposition 66

Ἡ τῇ ἐκ δύο ὀνομάτων μήκει σύμμετρος καὶ αὐτῇ ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ τάχει ἡ αὐτή.

Ἐστω ἐκ δύο ὀνομάτων ἡ AB , καὶ τῇ AB μήκει σύμμετρος ἔστω ἡ $\Gamma\Delta$. λέγω, ὅτι ἡ $\Gamma\Delta$ ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ τάχει ἡ αὐτῇ τῇ AB .

A (straight-line) commensurable in length with a binomial (straight-line) is itself also binomial, and the same in order.

Let AB be a binomial (straight-line), and let CD be commensurable in length with AB . I say that CD is a binomial (straight-line), and (is) the same in order as AB .



Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων ἐστὶν ἡ AB , διηγήσθω εἷξ τὰ ὀνόματα κατὰ τὸ E , καὶ ἔστω μείζων ὄνομα τὸ AE . αἱ AE , EB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. γερονέτω ὡς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ AE πρὸς τὴν ΓZ καὶ λοιπὴ ἄρα ἡ EB πρὸς λοιπὴν τὴν $Z\Delta$ ἐστίν, ὡς ἡ AB πρὸς τὴν $\Gamma\Delta$. σύμμετρος δὲ ἡ AB τῇ $\Gamma\Delta$ μήκει· σύμμετρος ἄρα ἐστὶ καὶ ἡ μὲν AE τῇ ΓZ , ἡ δὲ EB τῇ $Z\Delta$. καὶ εἰσι ῥηταὶ αἱ AE , EB . ῥηταὶ ἄρα εἰσι καὶ αἱ ΓZ , $Z\Delta$. καὶ ἐστὶν ὡς ἡ AE πρὸς ΓZ , ἡ EB πρὸς $Z\Delta$. ἐναλλάξ ἄρα ἐστὶν ὡς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$. αἱ δὲ AE , EB δυνάμει μόνον [εἰσὶ] σύμμετροι· καὶ αἱ ΓZ , $Z\Delta$ ἄρα δυνάμει μόνον εἰσὶ σύμμετροι. καὶ εἰσι ῥηταὶ· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $\Gamma\Delta$. λέγω δὴ, ὅτι τῇ τάχει ἐστὶν ἡ αὐτῇ τῇ AB .

Ἡ γὰρ AE τῇ EB μείζων δύναται ἥτοι τῷ ἀπὸ συμμέτρου ἐαντῇ ἢ τῷ ἀπὸ ἀσυνμέτρου. εἰ μὲν οὖν ἡ AE τῇ EB μείζων δύναται τῷ ἀπὸ συμμέτρου ἐαντῇ, καὶ ἡ ΓZ τῇ $Z\Delta$ μείζων δυνήσεται τῷ ἀπὸ συμμέτρου ἐαντῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ AE τῇ ἐκκεκμένη ῥητῇ, καὶ ἡ ΓZ σύμμετρος αὐτῇ ἔσται, καὶ διὰ τοῦτο ἑκάτερα τῶν AB , $\Gamma\Delta$ ἐκ δύο ὀνομάτων ἐστὶ πρώτη, τουτέστι τῇ τάχει ἡ αὐτή. εἰ δὲ ἡ EB σύμμετρος ἐστὶ τῇ ἐκκεκμένη ῥητῇ, καὶ ἡ $Z\Delta$ σύμμετρος αὐτῇ, καὶ διὰ τοῦτο πάλιν τῇ τάχει ἡ αὐτῇ ἔσται τῇ AB . ἑκάτερα γὰρ αὐτῶν ἔσται ἐκ δύο ὀνομάτων δευτέρα. εἰ δὲ οὐδετέρα τῶν AE , EB σύμμετρος ἐστὶ τῇ ἐκκεκμένη ῥητῇ, οὐδετέρα τῶν ΓZ , $Z\Delta$ σύμμετρος αὐτῇ ἔσται, καὶ ἐστὶν ἑκάτερα τρίτη. εἰ δὲ ἡ AE τῇ EB μείζων δύναται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ, καὶ ἡ ΓZ τῇ $Z\Delta$ μείζων δύναται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ.

For since AB is a binomial (straight-line), let it be divided into its (component) terms at E , and let AE be the greater term. AE and EB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. Let it be contrived that as AB (is) to CD , so AE (is) to CF [Prop. 6.12]. Thus, the remainder EB is also to the remainder FD , as AB (is) to CD [Props. 6.16, 5.19 corr.]. And AB (is) commensurable in length with CD . Thus, AE is also commensurable (in length) with CF , and EB with FD [Prop. 10.11]. And AE and EB are rational. Thus, CF and FD are also rational. And as AE is to CF , (so) EB (is) to FD [Prop. 5.11]. Thus, alternately, as AE is to EB , (so) CF (is) to FD [Prop. 5.16]. And AE and EB [are] commensurable in square only. Thus, CF and FD are also commensurable in square only [Prop. 10.11]. And they are rational. CD is thus a binomial (straight-line) [Prop. 10.36]. So, I say that it is the same in order as AB .

For the square on AE is greater than (the square on) EB by the (square) on (some straight-line) either commensurable or incommensurable (in length) with (AE). Therefore, if the square on AE is greater than (the square on) EB by the (square) on (some straight-line) commensurable (in length) with (AE) then the square on CF will also be greater than (the square on) FD by the (square) on (some straight-line) commensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable (in length) with (some previously) laid down rational (straight-line) then CF will also be commensurable (in length) with it [Prop. 10.12]. And, on account of this, AB and CD are each

καὶ εἰ μὲν ἡ AE σύμμετρος ἔστω τῇ ἐκκεκλιμένη ῥήτῃ, καὶ ἡ ΓZ σύμμετρος ἔστω αὐτῇ, καὶ ἔστω ἑκατέρα τετάρτη. εἰ δὲ ἡ EB , καὶ ἡ $Z\Delta$, καὶ ἔστω ἑκατέρα πέμπτη. εἰ δὲ οὐδετέρα τῶν AE , EB , καὶ τῶν ΓZ , $Z\Delta$ οὐδετέρα σύμμετρος ἔστω τῇ ἐκκεκλιμένη ῥήτῃ, καὶ ἔστω ἑκατέρα ἕκτη.

Ὡστε ἡ τῇ ἐκ δύο ὀνομάτων μήκει σύμμετρος ἐκ δύο ὀνομάτων ἔστι καὶ τῇ τάχει ἡ αὐτή· ὅπερ ἔδει δεῖξαι.

first binomial (straight-lines) [Def. 10.5]—that is to say, the same in order. And if EB is commensurable (in length) with the (previously) laid down rational (straight-line) then FD is also commensurable (in length) with it [Prop. 10.12], and, again, on account of this, (CD) will be the same in order as AB . For each of them will be second binomial (straight-lines) [Def. 10.6]. And if neither of AE and EB is commensurable (in length) with the (previously) laid down rational (straight-line) then neither of CF and FD will be commensurable (in length) with it [Prop. 10.13], and each (of AB and CD) is a third (binomial straight-line) [Def. 10.7]. And if the square on AE is greater than (the square on) EB by the (square) on (some straight-line) incommensurable (in length) with (AE) then the square on CF is also greater than (the square on) FD by the (square) on (some straight-line) incommensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable (in length) with the (previously) laid down rational (straight-line) then CF is also commensurable (in length) with it [Prop. 10.12], and each (of AB and CD) is a fourth (binomial straight-line) [Def. 10.8]. And if EB (is commensurable in length with the previously laid down rational straight-line) then FD (is) also (commensurable in length with it), and each (of AB and CD) will be a fifth (binomial straight-line) [Def. 10.9]. And if neither of AE and EB (is commensurable in length with the previously laid down rational straight-line) then also neither of CF and FD is commensurable (in length) with the laid down rational (straight-line), and each (of AB and CD) will be a sixth (binomial straight-line) [Def. 10.10].

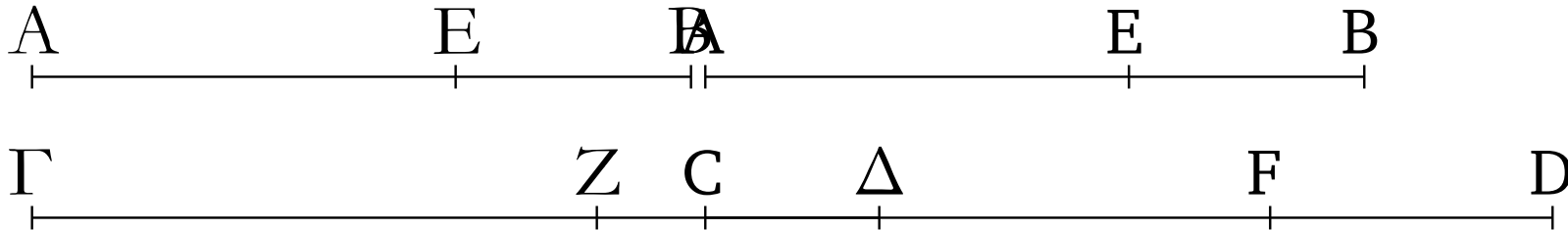
Hence, a (straight-line) commensurable in length with a binomial (straight-line) is a binomial (straight-line), and the same in order. (Which is) the very thing it was required to show.

67.

Proposition 67

Ἡ τῇ ἐκ δύο μέσων μήκει σύμμετρος καὶ αὐτῇ ἐκ δύο μέσων ἔστι καὶ τῇ τάχει ἡ αὐτή.

A (straight-line) commensurable in length with a bimedral (straight-line) is itself also bimedral, and the same in order.



Ἐστω ἐκ δύο μέσων ἡ AB , καὶ τῇ AB σύμμετρος ἔστω μήκει ἡ $\Gamma\Delta$. λέγω, ὅτι ἡ $\Gamma\Delta$ ἐκ δύο μέσων ἔστι καὶ τῇ τάχει ἡ αὐτῇ τῇ AB .

Ἐπεὶ γὰρ ἐκ δύο μέσων ἐστὶν ἡ AB , διηγήσθω εἰς τὰς μέσας κατὰ τὸ E αἱ AE , EB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ γεγονένω ὥς ἡ AB πρός $\Gamma\Delta$, ἡ AE πρός

Let AB be a bimedral (straight-line), and let CD be commensurable in length with AB . I say that CD is bimedral, and the same in order as AB .

For since AB is a bimedral (straight-line), let it be divided into its (component) medial (straight-lines) at E . Thus, AE and EB are medial (straight-lines which are) commensurable

ΓZ · καὶ λοιπὴ ἄρα ἡ EB πρὸς λοιπὴν τὴν $Z\Delta$ ἐστίν, ὥς ἡ AB πρὸς $\Gamma\Delta$. σύμμετρος δὲ ἡ AB τῇ $\Gamma\Delta$ μήκει· σύμμετρος ἄρα καὶ ἑκατέρα τῶν AE , EB ἑκατέρῃ τῶν ΓZ , $Z\Delta$. μέσαι δὲ αἱ AE , EB · μέσαι ἄρα καὶ αἱ ΓZ , $Z\Delta$. καὶ ἐπεὶ ἐστὶν ὥς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$, αἱ δὲ AE , EB δυνάμει μόνον σύμμετροί εἰσιν, καὶ αἱ ΓZ , $Z\Delta$ [ἄρα] δυνάμει μόνον σύμμετροί εἰσιν, ἐδείκθησαν δὲ καὶ μέσαι· ἡ $\Gamma\Delta$ ἄρα ἐκ δύο μέσων ἐστίν. λέγω δὴ, ὅτι καὶ τῇ τάχει ἡ αὐτὴ ἐστὶ τῇ AB .

Ἐπεὶ γάρ ἐστιν ὥς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$, καὶ ὥς ἄρα τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AEB , οὕτως τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν $\Gamma Z\Delta$ · ἐναλλάξ ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς ΓZ , οὕτως τὸ ὑπὸ τῶν AEB πρὸς τὸ ὑπὸ τῶν $\Gamma Z\Delta$. σύμμετρον δὲ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς ΓZ · σύμμετρον ἄρα καὶ τὸ ὑπὸ τῶν AEB τῷ ὑπὸ τῶν $\Gamma Z\Delta$. εἴτε οὖν ῥητὸν ἐστὶ τὸ ὑπὸ τῶν AEB , καὶ τὸ ὑπὸ τῶν $\Gamma Z\Delta$ ῥητὸν ἐστὶν [καὶ διὰ τοῦτο ἐστὶν ἐκ δύο μέσων πρῶτη]. εἴτε μέσον, καὶ ἐστὶν ἑκατέρα δευτέρα.

Καὶ διὰ τοῦτο ἔσται ἡ $\Gamma\Delta$ τῇ AB τῇ τάχει ἡ αὐτὴ· ὅπερ ἔδει δεῖχαι.

in square only [Props. 10.37, 10.38]. And let it be contrived that as AB (is) to CD , (so) AE (is) to CF [Prop. 6.12]. And thus as the remainder EB is to the remainder FD , so AB (is) to CD [Props. 5.19 corr., 6.16]. And AB (is) commensurable in length with CD . Thus, AE and EB are also commensurable (in length) with CF and FD , respectively [Prop. 10.11]. And AE and EB (are) medial. Thus, CF and FD (are) also medial [Prop. 10.23]. And since as AE is to EB , (so) CF (is) to FD , and AE and EB are commensurable in square only, CF and FD are [thus] also commensurable in square only [Prop. 10.11]. And they were also shown (to be) medial. Thus, CD is a bimedral (straight-line). So, I say that it is also the same in order as AB .

For since as AE is to EB , (so) CF (is) to FD , thus also as the (square) on AE (is) to the (rectangle contained) by AEB , so the (square) on CF (is) to the (rectangle contained) by CFD [Prop. 10.21 lem.]. Alternately, as the (square) on AE (is) to the (square) on CF , so the (rectangle contained) by AEB (is) to the (rectangle contained) by CFD [Prop. 5.16]. And the (square) on AE (is) commensurable with the (square) on CF . Thus, the (rectangle contained) by AEB (is) also commensurable with the (rectangle contained) by CFD [Prop. 10.11]. Therefore, either the (rectangle contained) by AEB is rational, and the (rectangle contained) by CFD is rational [and, on account of this, (AE and CD) are first bimedral (straight-lines)], or (the rectangle contained by AEB is) medial, and (the rectangle contained by CFD is) medial, and (AB and CD) are each second (bimedral straight-lines) [Props. 10.23, 10.37, 10.38].

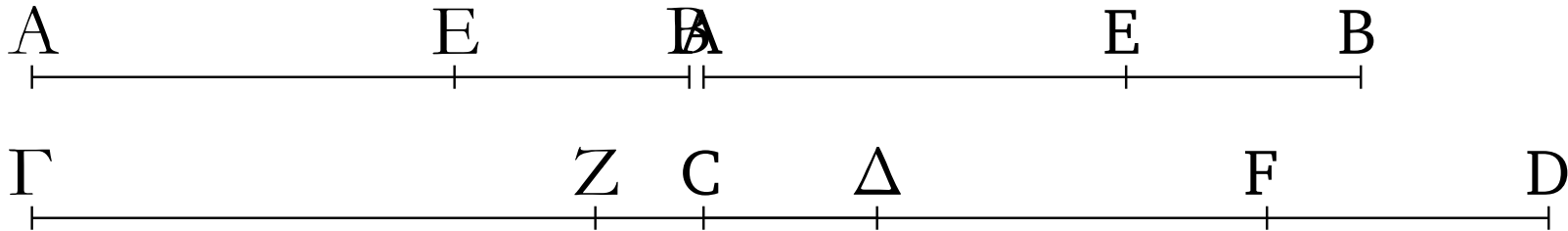
And, on account of this, CD will be the same in order as AB . (Which is) the very thing it was required to show.

68.

Ἡ τῇ μείζονι σύμμετρος καὶ αὐτὴ μείζων ἐστίν.

Proposition 68

A (straight-line) commensurable (in length) with a major (straight-line) is itself also major.



Ἐστω μείζων ἡ AB , καὶ τῇ AB σύμμετρος ἔστω ἡ $\Gamma\Delta$. λέγω, ὅτι ἡ $\Gamma\Delta$ μείζων ἐστίν.

Διηγήσθω ἡ AB κατὰ τὸ E · αἱ AE , EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητὸν, τὸ δ' ὑπ' αὐτῶν μέσον· καὶ γερονέτω τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἢ τε AE πρὸς τὴν ΓZ καὶ ἡ EB πρὸς τὴν $Z\Delta$, καὶ ὥς ἄρα ἡ AE πρὸς τὴν ΓZ , οὕτως ἡ EB πρὸς τὴν $Z\Delta$.

Let AB be a major (straight-line), and let CD be commensurable (in length) with AB . I say that CD is a major (straight-line).

Let AB be divided (into its component terms) at E . AE and EB are thus incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial [Prop. 10.39]. And let (the) same (things) be contrived as in the previous (propositions). And since as

σύμμετρος δὲ ἡ AB τῇ $\Gamma\Delta$. σύμμετρος ἄρα καὶ ἑκάτερα τῶν AE , EB ἑκάτερα τῶν ΓZ , $Z\Delta$. καὶ ἐπεὶ ἐστὶν ὡς ἡ AE πρὸς τὴν ΓZ , οὕτως ἡ EB πρὸς τὴν $Z\Delta$, καὶ ἐναλλάξ ὡς ἡ AE πρὸς EB , οὕτως ἡ ΓZ πρὸς $Z\Delta$, καὶ συνθέντι ἄρα ἐστὶν ὡς ἡ AB πρὸς τὴν BE , οὕτως ἡ $\Gamma\Delta$ πρὸς τὴν ΔZ . καὶ ὡς ἄρα τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BE , οὕτως τὸ ἀπὸ τῆς $\Gamma\Delta$ πρὸς τὸ ἀπὸ τῆς ΔZ . ὁμοίως δὲ δείχομεν, ὅτι καὶ ὡς τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς AE , οὕτως τὸ ἀπὸ τῆς $\Gamma\Delta$ πρὸς τὸ ἀπὸ τῆς ΓZ . καὶ ὡς ἄρα τὸ ἀπὸ τῆς AB πρὸς τὰ ἀπὸ τῶν AE , EB , οὕτως τὸ ἀπὸ τῆς $\Gamma\Delta$ πρὸς τὰ ἀπὸ τῶν ΓZ , $Z\Delta$. καὶ ἐναλλάξ ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς $\Gamma\Delta$, οὕτως τὰ ἀπὸ τῶν AE , EB πρὸς τὰ ἀπὸ τῶν ΓZ , $Z\Delta$. σύμμετρον δὲ τὸ ἀπὸ τῆς AB τῷ ἀπὸ τῆς $\Gamma\Delta$. σύμμετρα ἄρα καὶ τὰ ἀπὸ τῶν AE , EB τοῖς ἀπὸ τῶν ΓZ , $Z\Delta$. καὶ ἐστὶ τὰ ἀπὸ τῶν AE , EB ἅμα ῥητόν, καὶ τὰ ἀπὸ τῶν ΓZ , $Z\Delta$ ἅμα ῥητόν ἐστὶν. ὁμοίως δὲ καὶ τὸ διξ ὑπὸ τῶν AE , EB σύμμετρόν ἐστὶ τῷ διξ ὑπὸ τῶν ΓZ , $Z\Delta$. καὶ ἐστὶ μέσον τὸ διξ ὑπὸ τῶν AE , EB . μέσον ἄρα καὶ τὸ διξ ὑπὸ τῶν ΓZ , $Z\Delta$. αἱ ΓZ , $Z\Delta$ ἄρα δυνάμει ἀσύμμετροί εἰσι ποιῶσαι τὸ μὲν συγκαείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ἅμα ῥητόν, τὸ δὲ διξ ὑπ' αὐτῶν μέσον· ὅλη ἄρα ἡ $\Gamma\Delta$ ἀλογόξ ἐστὶν ἡ καλουμένη μείζων.

Ἡ ἄρα τῇ μείζονι σύμμετρος μείζων ἐστίν· ὅπερ ἔδει δεῖξαι.

AB is to CD , so AE (is) to CF and EB to FD , thus also as AE (is) to CF , so EB (is) to FD [Prop. 5.11]. And AB (is) commensurable (in length) with CD . Thus, AE and EB (are) also commensurable (in length) with CF and FD , respectively [Prop. 10.11]. And since as AE is to CF , so EB (is) to FD , also, alternately, as AE (is) to EB , so CF (is) to FD [Prop. 5.16], and thus, via composition, as AB is to BE , so CD (is) to DF [Prop. 5.18]. And thus as the (square) on AB (is) to the (square) on BE , so the (square) on CD (is) to the (square) on DF [Prop. 6.20]. So, similarly, we can also show that as the (square) on AB (is) to the (square) on AE , so the (square) on CD (is) to the (square) on CF . And thus as the (square) on AB (is) to (the sum of) the (squares) on AE and EB , so the (square) on CD (is) to (the sum of) the (squares) on CF and FD . And thus, alternately, as the (square) on AB is to the (square) on CD , so (the sum of) the (squares) on AE and EB (is) to (the sum of) the (squares) on CF and FD [Prop. 5.16]. And the (square) on AB (is) commensurable with the (square) on CD . Thus, (the sum of) the (squares) on AE and EB (is) also commensurable with (the sum of) the (squares) on CF and FD [Prop. 10.11]. And the (squares) on AE and EB (added) together are rational. The (squares) on CF and FD (added) together (are) thus also rational. So, similarly, twice the (rectangle contained) by AE and EB is also commensurable with twice the (rectangle contained) by CF and FD . And twice the (rectangle contained) by AE and EB is medial. Therefore, twice the (rectangle contained) by CF and FD (is) also medial [Prop. 10.23 corr.]. CF and FD are thus (straight-lines which are) incommensurable in square [Prop 10.13], simultaneously making the sum of the squares on them rational, and twice the (rectangle contained) by them medial. The whole, CD , is thus that irrational (straight-line) called major [Prop. 10.39].

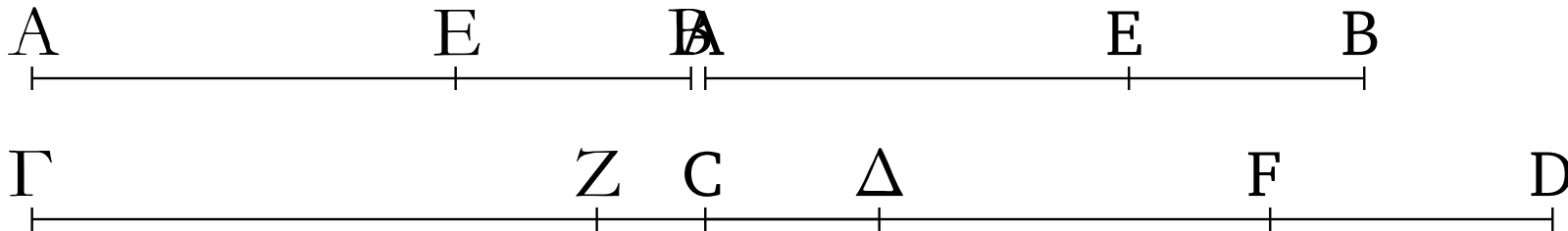
Thus, a (straight-line) commensurable (in length) with a major (straight-line) is major. (Which is) the very thing it was required to show.

69.

Proposition 69

Ἡ τῇ ῥητόν καὶ μέσον δυναμένη σύμμετρος [καὶ αὐτὴ] ῥητόν καὶ μέσον δυναμένη ἐστίν.

A (straight-line) commensurable (in length) with the square-root of a rational plus a medial (area) is [itself also] the square-root of a rational plus a medial (area).



Ἐστω ῥητόν καὶ μέσον δυναμένη ἡ AB , καὶ τῇ AB σύμμετρος ἔστω ἡ $\Gamma\Delta$. δεικτέον, ὅτι καὶ ἡ $\Gamma\Delta$ ῥητόν καὶ μέσον

Let AB be the square-root of a rational plus a medial (area), and let CD be commensurable (in length) with AB . We must

δυναμένη ἐστίν.

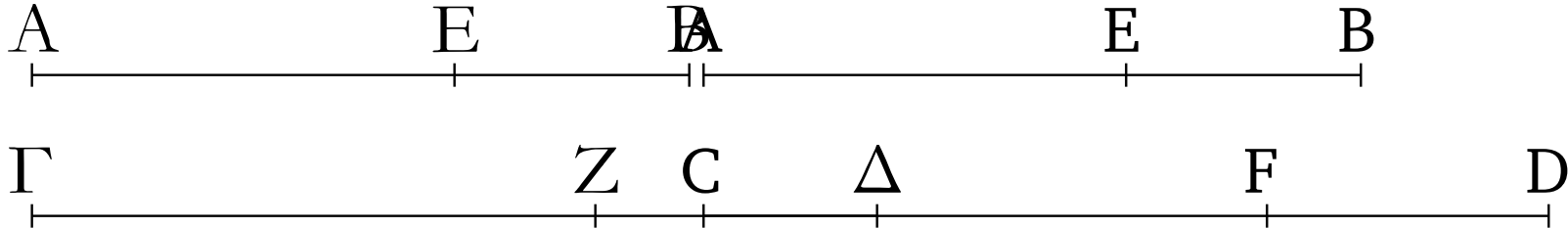
Διηγήσθω ἡ AB εἰς τὰς εὐθείας κατὰ τὸ E · αἱ AE , EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν· καὶ τὰ αὐτὰ κατεσκευάσθω τοῖς πρότερον. ὁμοίως δὲ δείχομεν, ὅτι καὶ αἱ ΓZ , $Z\Delta$ δυνάμει εἰσὶν ἀσύμμετροι, καὶ σύμμετρον τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τῶν συγκειμένων ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$, τὸ δὲ ὑπὸ AE , EB τῶν ὑπὸ ΓZ , $Z\Delta$ · ὥστε καὶ τὸ [μὲν] συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων ἐστὶ μέσον, τὸ δ' ὑπὸ τῶν ΓZ , $Z\Delta$ ῥητόν.

Ῥητόν ἄρα καὶ μέσον δυναμένη ἐστίν ἡ $\Gamma\Delta$ · ὅπερ ἔδει δεῖχαι.

70.

Ἡ τῇ δύο μέσα δυναμένη σύμμετρος δύο μέσα δυναμένη ἐστίν.

Ἐστω δύο μέσα δυναμένη ἡ AB , καὶ τῇ AB σύμμετρος ἡ $\Gamma\Delta$ · δεικτέον, ὅτι καὶ ἡ $\Gamma\Delta$ δύο μέσα δυναμένη ἐστίν.



Ἐπεὶ γὰρ δύο μέσα δυναμένη ἐστίν ἡ AB , διηγήσθω εἰς τὰς εὐθείας κατὰ τὸ E · αἱ AE , EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν [τετραγώνων] μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῶν ὑπὸ τῶν AE , EB · καὶ κατεσκευάσθω τὰ αὐτὰ τοῖς πρότερον. ὁμοίως δὲ δείχομεν, ὅτι καὶ αἱ ΓZ , $Z\Delta$ δυνάμει εἰσὶν ἀσύμμετροι καὶ σύμμετρον τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τῶν συγκειμένων ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$, τὸ δὲ ὑπὸ τῶν AE , EB τῶν ὑπὸ τῶν ΓZ , $Z\Delta$ · ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων μέσον ἐστὶ καὶ τὸ ὑπὸ τῶν ΓZ , $Z\Delta$ μέσον καὶ ἔτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων τῶν ὑπὸ τῶν ΓZ , $Z\Delta$.

Ἡ ἄρα $\Gamma\Delta$ δύο μέσα δυναμένη ἐστίν· ὅπερ ἔδει δεῖχαι.

show that CD is also the square-root of a rational plus a medial (area).

Let AB be divided into its (component) straight-lines at E . AE and EB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational [Prop. 10.40]. And let the same construction be made as in the previous (propositions). So, similarly, we can show that CF and FD are also incommensurable in square, and that the sum of the (squares) on AE and EB (is) commensurable with the sum of the (squares) on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . And hence the sum of the squares on CF and FD is medial, and the (rectangle contained) by CF and FD (is) rational.

Thus, CD is the square-root of a rational plus a medial (area) [Prop. 10.40]. (Which is) the very thing it was required to show.

Proposition 70

A (straight-line) commensurable (in length) with the square-root of (the sum of) two medial (areas) is (itself also) the square-root of (the sum of) two medial (areas).

Let AB be the square-root of (the sum of) two medial (areas), and (let) CD (be) commensurable (in length) with AB . We must show that CD is also the square-root of (the sum of) two medial (areas).

For since AB is the square-root of (the sum of) two medial (areas), let it be divided into its (component) straight-lines at E . Thus, AE and EB are incommensurable in square, making the sum of the [squares] on them medial, and the (rectangle contained) by them medial, and, moreover, the sum of the (squares) on AE and EB incommensurable with the (rectangle) contained by AE and EB [Prop. 10.41]. And let the same construction be made as in the previous (propositions). So, similarly, we can show that CF and FD are also incommensurable in square, and (that) the sum of the (squares) on AE and EB (is) commensurable with the sum of the (squares) on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Hence, the sum of the squares on CF and FD is also medial, and the (rectangle contained) by CF and FD (is) medial, and, moreover, the sum of the squares on CF and FD (is) incommensurable with the

71.

Ῥητοῦ καὶ μέσον συντιθεμένου τέσσαρες ἄλογοι γίνονται ἥτοι ἐκ δύο ὀνομάτων ἢ ἐκ δύο μέσων πρώτη ἢ μείζων ἢ ῥητόν καὶ μέσον δυναμένη.

Ἐστω ῥητόν μὲν τὸ AB , μέσον δὲ τὸ $\Gamma\Delta$. λέγω, ὅτι ἡ τὸ $A\Delta$ θωρίον δυναμένη ἥτοι ἐκ δύο ὀνομάτων ἐστὶν ἢ ἐκ δύο μέσων πρώτη ἢ μείζων ἢ ῥητόν καὶ μέσον δυναμένη.

Τὸ γὰρ AB τοῦ $\Gamma\Delta$ ἥτοι μείζον ἐστὶν ἢ ἔλασσον. ἔστω πρότερον μείζον· καὶ ἐκκείσθω ῥητὴ ἡ EZ , καὶ παραβελθήσθω παρὰ τὴν EZ τῷ AB ἴσον τὸ EH πλάτος ποιοῦν τὴν $E\Theta$. τῷ δὲ $\Delta\Gamma$ ἴσον παρὰ τὴν EZ παραβελθήσθω τὸ ΘI πλάτος ποιοῦν τὴν ΘK . καὶ ἐπεὶ ῥητόν ἐστι τὸ AB καὶ ἐστιν ἴσον τῷ EH , ῥητόν ἄρα καὶ τὸ EH . καὶ παρὰ [ῥητὴν] τὴν EZ παραβελθήσθω πλάτος ποιοῦν τὴν $E\Theta$. ἡ $E\Theta$ ἄρα ῥητὴ ἐστὶ καὶ σύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ $\Gamma\Delta$ καὶ ἐστὶν ἴσον τῷ ΘI , μέσον ἄρα ἐστὶ καὶ τὸ ΘI . καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν ΘK . ῥητὴ ἄρα ἐστὶν ἡ ΘK καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ μέσον ἐστὶ τὸ $\Gamma\Delta$, ῥητόν δὲ τὸ AB , ἀσύμμετρον ἄρα ἐστὶ τὸ AB τῷ $\Gamma\Delta$. ὥστε καὶ τὸ EH ἀσύμμετρον ἐστὶ τῷ ΘI . ὥς δὲ τὸ EH πρὸς τὸ ΘI , οὕτως ἐστὶν ἡ $E\Theta$ πρὸς τὴν ΘK . ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ $E\Theta$ τῇ ΘK μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ $E\Theta$, ΘK ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EK διηρημένη κατὰ τὸ Θ . καὶ ἐπεὶ μείζον ἐστὶ τὸ AB τοῦ $\Gamma\Delta$, ἴσον δὲ τὸ μὲν AB τῷ EH , τὸ δὲ $\Gamma\Delta$ τῷ ΘI , μείζον ἄρα καὶ τὸ EH τοῦ ΘI . καὶ ἡ $E\Theta$ ἄρα μείζων ἐστὶ τῆς ΘK . ἥτοι οὗν ἡ $E\Theta$ τῆς ΘK μείζον δύναται τῷ ἀπὸ συμέτρου ἐαντὶ μήκει ἢ τῷ ἀπὸ ἀσυμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμέτρου ἐαντὶ· καὶ ἐστὶν ἡ μείζων ἡ ΘE σύμμετρος τῇ ἐκκευμένη ῥητὴ τῇ EZ . ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶ πρώτη. ῥητὴ δὲ ἡ EZ . ἔαν δὲ θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πρώτης, ἡ τὸ θωρίον δυναμένη ἐκ δύο ὀνομάτων ἐστὶν. ἡ ἄρα τὸ EI δυναμένη ἐκ δύο ὀνομάτων ἐστὶν. ὥστε καὶ ἡ τὸ $A\Delta$ δυναμένη ἐκ δύο ὀνομάτων ἐστὶν. ἀλλὰ δὴ δυνάσθω ἡ $E\Theta$ τῆς ΘK μείζον τῷ ἀπὸ ἀσυμέτρου ἐαντὶ· καὶ ἐστὶν ἡ μείζων ἡ $E\Theta$ σύμμετρος τῇ ἐκκευμένη ῥητὴ τῇ EZ μήκει. ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶ τετάρτη. ῥητὴ δὲ ἡ EZ . ἔαν δὲ θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἢ καλουμένη μείζων. ἡ ἄρα τὸ EI θωρίον δυναμένη μείζων ἐστὶν. ὥστε καὶ ἡ τὸ $A\Delta$ δυναμένη μείζων ἐστὶν.

Ἀλλὰ δὴ ἔστω ἔλασσον τὸ AB τοῦ $\Gamma\Delta$. καὶ τὸ EH ἄρα ἔλασσόν ἐστι τοῦ ΘI . ὥστε καὶ ἡ $E\Theta$ ἐλάσσων ἐστὶ τῆς ΘK . ἥτοι δὲ ἡ ΘK τῆς $E\Theta$ μείζον δύναται τῷ ἀπὸ συμέτρου ἐαντὶ ἢ τῷ ἀπὸ ἀσυμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμέτρου

(rectangle contained) by CF and FD .

Thus, CD is the square-root of (the sum of) two medial (areas) [Prop. 10.41]. (Which is) the very thing it was required to show.

Proposition 71

When a rational and a medial (area) are added together, four irrational (straight-lines) arise (as the square-roots of the total area)—either a binomial, or a first bimedial, or a major, or the square-root of a rational plus a medial (area).

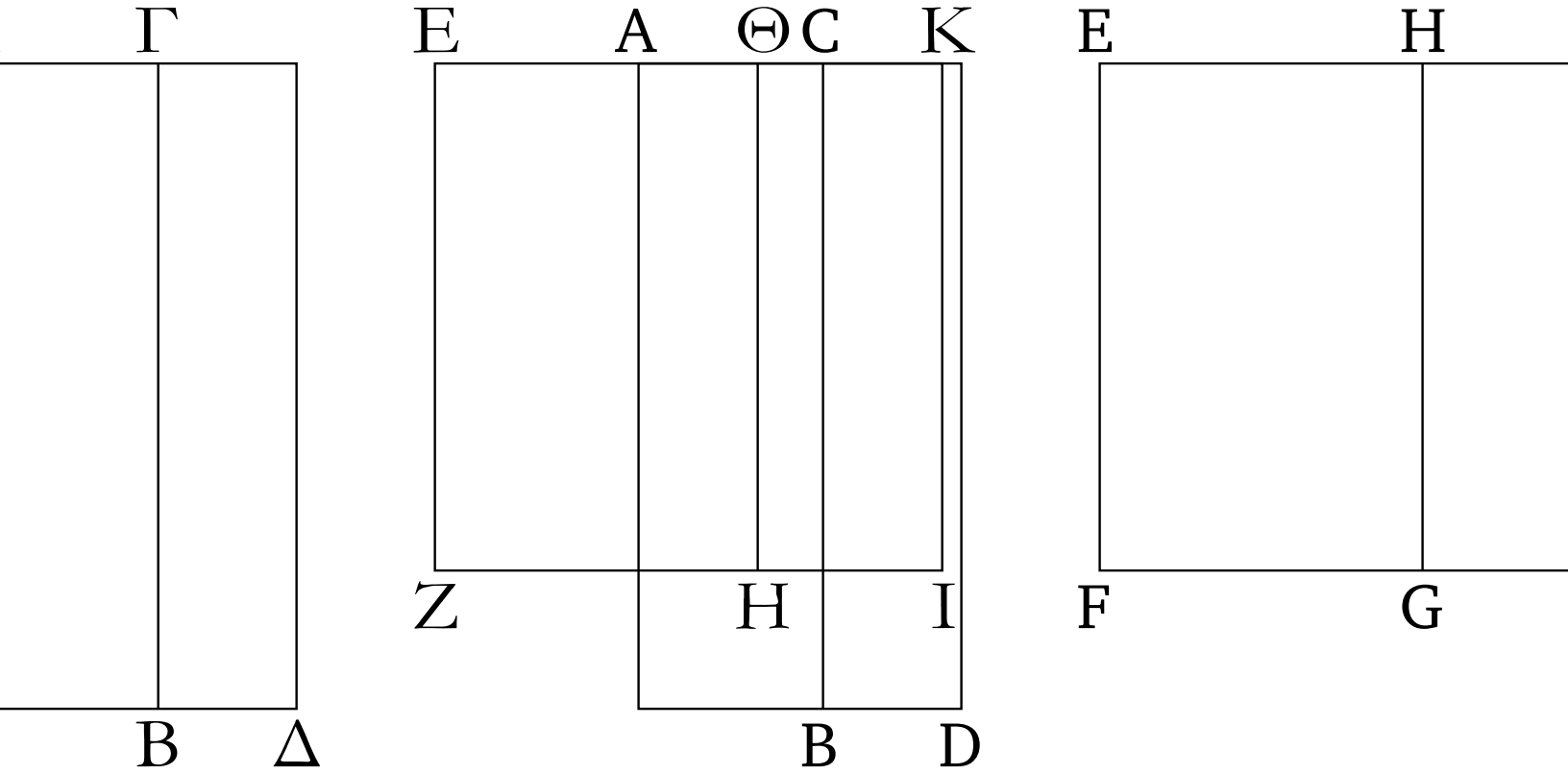
Let AB be a rational (area), and CD a medial (area). I say that the square-root of area AD is either binomial, or first bimedial, or major, or the square-root of a rational plus a medial (area).

For AB is either greater or less than CD . Let it, first of all, be greater. And let the rational (straight-line) EF be laid down. And let (the rectangle) EG , equal to AB , be applied to EF , producing EH as breadth. And let (the rectangle) HI , equal to DC , be applied to EF , producing HK as breadth. And since AB is rational, and is equal to EG , EG is thus also rational. And it has been applied to the [rational] (straight-line) EF , producing EH as breadth. EH is thus rational, and commensurable in length with EF [Prop. 10.20]. Again, since CD is medial, and is equal to HI , HI is thus also medial. And it is applied to the rational (straight-line) EF , producing HK as breadth. HK is thus rational, and incommensurable in length with EF [Prop. 10.22]. And since CD is medial, and AB rational, AB is thus incommensurable with CD . Hence, EG is also incommensurable with HI . And as EG (is) to HI , so EH is to HK [Prop. 6.1]. Thus, EH is also incommensurable in length with HK [Prop. 10.11]. And they are both rational. Thus, EH and HK are rational (straight-lines which are) commensurable in square only. EK is thus a binomial (straight-line), having been divided (into its component terms) at H [Prop. 10.36]. And since AB is greater than CD , and AB (is) equal to EG , and CD to HI , EG (is) thus also greater than HI . Thus, EH is also greater than HK [Prop. 5.14]. Therefore, the square on EH is greater than (the square on) HK either by the (square) on (some straight-line) commensurable in length with (EH) , or by the (square) on (some straight-line) incommensurable (in length with EH). Let it, first of all, be greater by the (square) on (some straight-line) commensurable (in length with EH). And the greater (of the two components of EK) HE is commensurable (in length) with the (previously) laid down (straight-line) EF . EK is thus a first binomial (straight-line) [Def. 10.5]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a first bimedial (straight-line) then the square-root of the area is a binomial (straight-line) [Prop. 10.54]. Thus, the square-root of EI is a binomial (straight-line). Hence the square-root of

ἐαντὴ μήκει· καὶ ἐστὶν ἡ ἐλάσσων ἢ $E\Theta$ σύμμετρος τῇ ἐκκειμένη ῥητῇ τῇ EZ μήκει· ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶ δευτέρα. ῥητὴ δὲ ἡ EZ · ἔαν δὲ θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας, ἡ τὸ θωρίον δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη· ὥστε καὶ ἡ τὸ $A\Delta$ δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη. ἀλλὰ δὴ ἡ ΘK τῆς ΘE μείζον δυνάσθω τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ. καὶ ἐστὶν ἡ ἐλάσσων ἢ $E\Theta$ σύμμετρος τῇ ἐκκειμένη ῥητῇ τῇ EZ · ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶ πέμπτη. ῥητὴ δὲ ἡ EZ · ἔαν δὲ θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πέμπτης, ἡ τὸ θωρίον δυναμένη ῥητόν καὶ μέσον δυναμένη ἐστίν· ὥστε καὶ ἡ τὸ $A\Delta$ θωρίον δυναμένη ῥητόν καὶ μέσον δυναμένη ἐστίν.

AD is also a binomial (straight-line). And, so, let the square on EH be greater than (the square on) HK by the (square) on (some straight-line) incommensurable (in length) with (EH). And the greater (of the two components of EK) EH is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a fourth binomial (straight-line) [Def. 10.8]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a fourth binomial (straight-line) then the square-root of the area is the irrational (straight-line) called major [Prop. 10.57]. Thus, the square-root of area EI is a major (straight-line). Hence, the square-root of AD is also major.

And so, let AB be less than CD . Thus, EG is also less than HI . Hence, EH is also less than HK [Props. 6.1, 5.14]. And the square on HK is greater than (the square on) EH either by the (square) on (some straight-line) commensurable (in length) with (HK), or by the (square) on (some straight-line) incommensurable (in length) with (HK). Let it, first of all, be greater by the square on (some straight-line) commensurable in length with (HK). And the lesser (of the two components of EK) EH is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a second binomial (straight-line) [Def. 10.6]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a second binomial (straight-line) then the square-root of the area is a first bimedral (straight-line) [Prop. 10.55]. Thus, the square-root of area EI is a first bimedral (straight-line). Hence, the square-root of AD is also a first bimedral (straight-line). And so, let the square on HK be greater than (the square on) HE by the (square) on (some straight-line) incommensurable (in length) with (HK). And the lesser (of the two components of EK) EH is commensurable (in length) with the (previously) laid down rational (straight-line) EF . Thus, EK is a fifth binomial (straight-line) [Def. 10.9]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a fifth binomial (straight-line) then the square-root of the area is the square-root of a rational plus a medial (area) [Prop. 10.58]. Thus, the square-root of area EI is the square-root of a rational plus a medial (area). Hence, the square-root of area AD is also the square-root of a rational plus a medial (area).



Πρὸ τοῦ ἄρα καὶ μέσον συντιθεμένων τέσσαρες ἄλογοι γίνονται ἥτοι ἐκ δύο ὀνομάτων ἢ ἐκ δύο μέσων πρώτη ἢ μείζων ἢ ῥητὸν καὶ μέσον δυναμένη· ὅπερ ἔδει δεῖξαι.

Thus, when a rational and a medial area are added together, four irrational (straight-lines) arise (as the square-roots of the total area)—either a binomial, or a first bimedral, or a major, or the square-root of a rational plus a medial (area). (Which is) the very thing it was required to show.

62.

Proposition 72

Δύο μέσων ἀσυμμέτρων ἀλλήλοις συντιθεμένων αἱ λοιπαὶ δύο ἄλογοι γίνονται ἥτοι ἐκ δύο μέσων δευτέρα ἢ [ἡ] δύο μέσα δυναμένη.

Συγκείσθω γάρ δύο μέσα ἀσύμμετρα ἀλλήλοις τὰ AB, ΓΔ· λέγω, ὅτι ἡ τὸ AD θωρίον δυναμένη ἥτοι ἐκ δύο μέσων ἐστὶ δευτέρα ἢ δύο μέσα δυναμένη.

Τὸ γὰρ AB τοῦ ΓΔ ἥτοι μείζον ἐστὶν ἢ ἕλασσον. ἔστω, εἰ τύχον, πρότερον μείζον τὸ AB τοῦ ΓΔ· καὶ ἐκκείσθω ῥητὴ ἢ EZ, καὶ τῷ μὲν AB ἴσον παρὰ τὴν EZ παραβελθήσθω τὸ EH πλάτοξ ποιῶν τὴν EΘ, τῷ δὲ ΓΔ ἴσον τὸ ΘΙ πλάτοξ ποιῶν τὴν ΘΚ. καὶ ἐπεὶ μέσον ἐστὶν ἑκάτερον τῶν AB, ΓΔ, μέσον ἄρα καὶ ἑκάτερον τῶν EH, ΘΙ. καὶ παρὰ ῥητὴν τὴν ZE παράκειται πλάτοξ ποιῶν τὰς EΘ, ΘΚ· ἑκάτερα ἄρα τῶν EΘ, ΘΚ ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρον ἐστὶ τὸ AB τῷ ΓΔ, καὶ ἐστὶν ἴσον τὸ μὲν AB τῷ EH, τὸ δὲ ΓΔ τῷ ΘΙ, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ EH τῷ ΘΙ. ὥς δὲ τὸ EH πρὸς τὸ ΘΙ, οὕτως ἐστὶν ἡ EΘ πρὸς ΘΚ· ἀσύμμετρος ἄρα ἐστὶν ἡ EΘ τῇ ΘΚ μήκει. αἱ EΘ, ΘΚ ἄρα

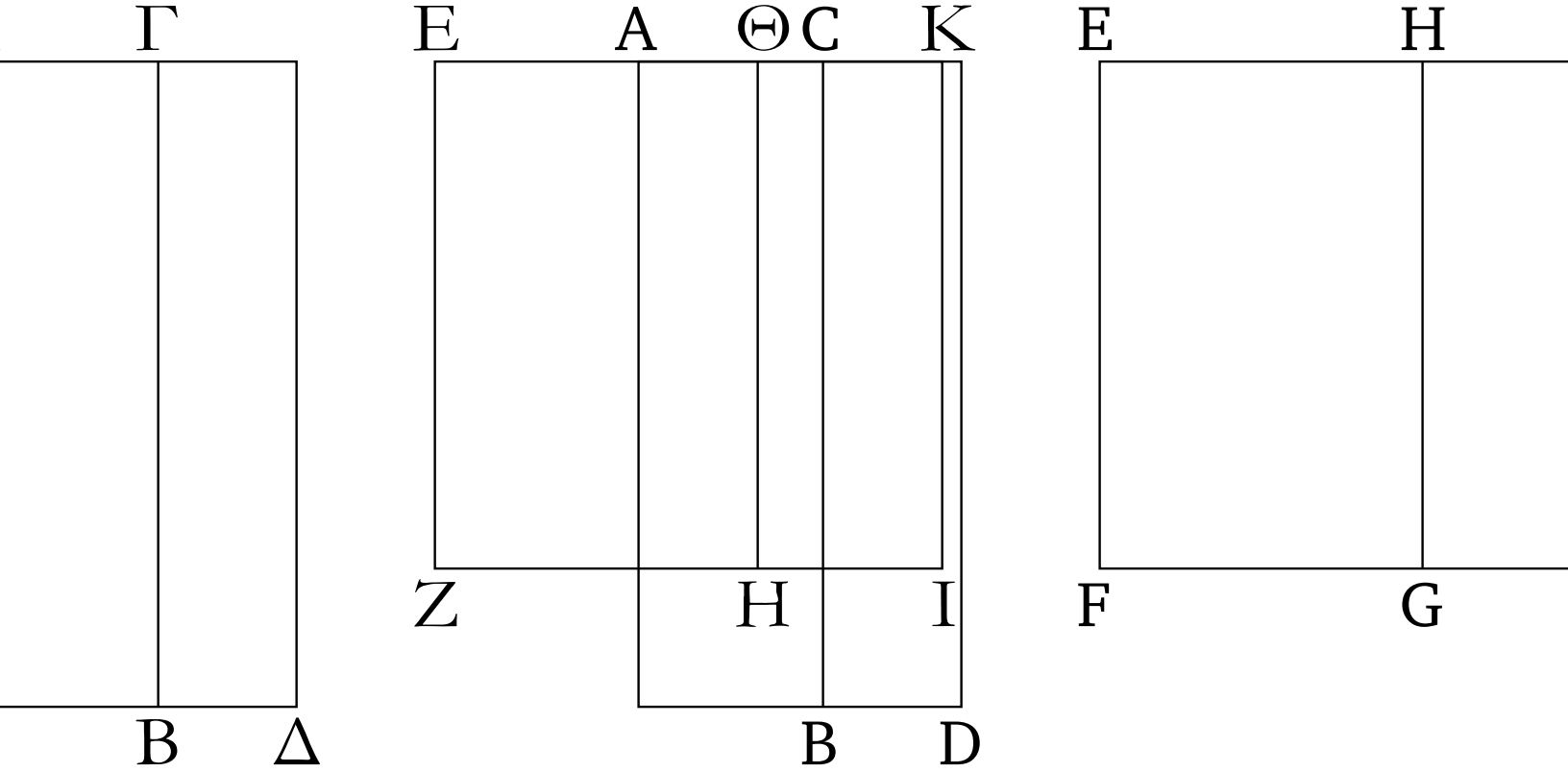
When two medial (areas which are) incommensurable with one another are added together, the remaining two irrational (straight-lines) arise (as the square-roots of the total area)—either a second bimedral, or the square-root of (the sum of) two medial (areas).

For let the two medial (areas) AB and CD, (which are) incommensurable with one another, be added together. I say that the square-root of area AD is either a second bimedral, or the square-root of (the sum of) two medial (areas).

For AB is either greater than or less than CD. By chance, let AB, first of all, be greater than CD. And let the rational (straight-line) EF be laid down. And let EG, equal to AB, be applied to EF, producing EH as breadth, and HI, equal to CD, producing HK as breadth. And since AB and CD are each medial, EG and HI (are) thus also each medial. And they are applied to the rational straight-line FE, producing EH and HK (respectively) as breadth. Thus, EH and HK are each rational (straight-lines which are) incommensurable in length with EF

ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EK . ἤτοι δὲ ἡ $E\Theta$ τῆς ΘK μείζον δύναιται τῷ ἀπὸ συμμέτρου ἐναντὶ ἢ τῷ ἀπὸ ἀσυνμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμμέτρου ἐναντὶ μήκει· καὶ οὐδετέρα τῶν $E\Theta$, ΘK σύμμετρόξ ἐστι τῇ ἐκκεκμένη ῥητῇ τῇ EZ μήκει· ἡ EK ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη. ῥητὴ δὲ ἡ EZ · ἔαν δὲ θεωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων τρίτης, ἡ τὸ θεωρίον δυναμένη ἐκ δύο μέσων ἐστὶ δευτέρα· ἡ ἄρα τὸ EI , τουτέστι τὸ $A\Delta$, δυναμένη ἐκ δύο μέσων ἐστὶ δευτέρα. ἀλλὰ δὴ ἡ $E\Theta$ τῆς ΘK μείζον δυνάσθω τῷ ἀπὸ ἀσυνμέτρου ἐναντὶ μήκει· καὶ ἀσύμμετρόξ ἐστὶν ἑκάτερα τῶν $E\Theta$, ΘK τῇ EZ μήκει· ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶν ἕκτη. ἔαν δὲ θεωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων ἕκτης, ἡ τὸ θεωρίον δυναμένη ἢ δύο μέσα δυναμένη ἐστὶν· ὥστε καὶ ἡ τὸ $A\Delta$ θεωρίον δυναμένη ἢ δύο μέσα δυναμένη ἐστὶν.

[Prop. 10.22]. And since AB is incommensurable with CD , and AB is equal to EG , and CD to HI , EG is thus also incommensurable with HI . And as EG (is) to HI , so EH is to HK [Prop. 6.1]. EH is thus incommensurable in length with HK [Prop. 10.11]. Thus, EH and HK are rational (straight-lines which are) commensurable in square only. EK is thus a binomial (straight-line) [Prop. 10.36]. And the square on EH is greater than (the square on) HK either by the (square) on (some straight-line) commensurable (in length) with (EH), or by the (square) on (some straight-line) incommensurable (in length with EH). Let it, first of all, be greater by the square on (some straight-line) commensurable in length with (EH). And neither of EH or HK is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a third binomial (straight-line) [Def. 10.7]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a third binomial (straight-line) then the square-root of the area is a second bimedral (straight-line) [Prop. 10.56]. Thus, the square-root of EI —that is to say, of AD —is a second bimedral. And so, let the square on EH be greater than (the square) on HK by the (square) on (some straight-line) incommensurable in length with (EH). And EH and HK are each incommensurable in length with EF . Thus, EK is a sixth binomial (straight-line) [Def. 10.10]. And if an area is contained by a rational (straight-line) and a sixth binomial (straight-line) then the square-root of the area is the square-root of (the sum of) two medial (areas) [Prop. 10.59]. Hence, the square-root of area AD is also the square-root of (the sum of) two medial (areas).



[Ὁμοίως δὴ δείξομεν, ὅτι κἂν ἔλαττον ᾖ τὸ AB τοῦ $\Gamma\Delta$, ἢ τὸ $A\Delta$ χωρίον δυναμένη ᾖ ἐκ δύο μέσων δευτέρα ἐστὶν ἥτοι δύο μέσα δυναμένη].

Δύο ἄρα μέσων ἀσυμμέτρων ἀλλήλοις συντιθεμένων αἰ λοιπαὶ δύο ἄλογοι γίνονται ἥτοι ἐκ δύο μέσων δευτέρα ἢ δύο μέσα δυναμένη.

Ἡ ἐκ δύο ὀνομάτων καὶ αἱ μετ' αὐτὴν ἄλογοι οὔτε τῇ μέσῃ οὔτε ἀλλήλαις εἰσὶν αἱ αὐταί. τὸ μὲν γὰρ ἀπὸ μέσῃ παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ῥητὴν καὶ ἀσύμμετρον τῇ παρ' ἣν παράκειται μήκει. τὸ δὲ ἀπὸ τῆς ἐκ δύο ὀνομάτων παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πρώτην. τὸ δὲ ἀπὸ τῆς ἐκ δύο μέσων πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων δευτέρα. τὸ δὲ ἀπὸ τῆς ἐκ δύο μέσων δευτέρας παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τρίτην. τὸ δὲ ἀπὸ τῆς μείζονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τετάρτην. τὸ δὲ ἀπὸ τῆς ῥητὸν καὶ μέσον δυναμένης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πέμπτην. τὸ δὲ ἀπὸ τῆς δύο μέσα δυναμένης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων ἕκτην. τὰ δ' εἰρημένα πλάτη διαφέρει τοῦ τε πρώτου καὶ ἀλλήλων, τοῦ μὲν πρώτου, ὅτι ῥητὴ ἐστίν, ἀλλήλων δέ, ὅτι τῇ τάχει οὐκ εἰσὶν αἱ αὐταί· ὥστε καὶ αὐταὶ αἱ ἄλογοι διαφέρουσιν ἀλλήλων.

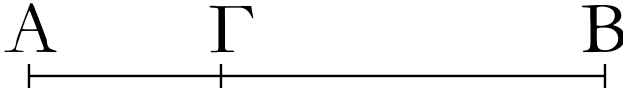
[So, similarly, we can show that, even if AB is less than CD , the square-root of area AD is either a second binomial or the square-root of (the sum of) two medial (areas).]

Thus, when two medial (areas which are) incommensurable with one another are added together, the remaining two irrational (straight-lines) arise (as the square-roots of the total area)—either a second binomial, or the square-root of (the sum of) two medial (areas).

A binomial (straight-line), and the (other) irrational (straight-lines) after it, are neither the same as a medial (straight-line) nor (the same) as one another. For the (square) on a medial (straight-line), applied to a rational (straight-line), produces as breadth a rational (straight-line which is) also incommensurable in length with (the straight-line) to which it is applied [Prop. 10.22]. And the (square) on a binomial (straight-line), applied to a rational (straight-line), produces as breadth a first binomial [Prop. 10.60]. And the (square) on a first binomial (straight-line), applied to a rational (straight-line), produces as breadth a second binomial [Prop. 10.61]. And the (square) on a second binomial (straight-line), applied to a rational (straight-line), produces as breadth a third binomial [Prop. 10.62]. And the (square) on a major (straight-line), applied to a rational (straight-line), produces as breadth a fourth binomial [Prop. 10.63]. And the (square) on the square-root of a rational plus a medial (area), applied to a rational (straight-line),

73.

Ἐάν ἀπὸ ῥητῆς ῥητῆ ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ, ἡ λοιπὴ ἄλογός ἐστιν· καλεῖσθω δὲ ἀποτομή.

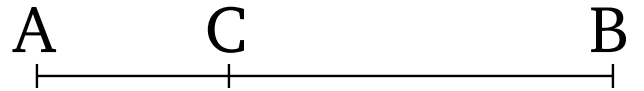


Ἀπὸ γὰρ ῥητῆς τῆς AB ῥητῆ ἀφαιρεθῶ ἡ $BΓ$ δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ· λέγω, ὅτι ἡ λοιπὴ ἡ $AΓ$ ἄλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐπεὶ γὰρ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει, καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ὑπὸ τῶν AB , $BΓ$, ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB , $BΓ$. ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρά ἐστι τὰ ἀπὸ τῶν AB , $BΓ$ τετράγωνα, τῷ δὲ ὑπὸ τῶν AB , $BΓ$ σύμμετρόν ἐστι τὸ δις ὑπὸ τῶν AB , $BΓ$. καὶ ἐπειδήπερ τὰ ἀπὸ τῶν AB , $BΓ$ ἴσα ἐστὶ τῷ δις ὑπὸ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓA$, καὶ λοιπῷ ἄρα τῷ ἀπὸ τῆς $AΓ$ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB , $BΓ$. ῥητὰ δὲ τὰ ἀπὸ τῶν AB , $BΓ$ ἄλογος ἄρα ἐστὶν ἡ $AΓ$ · καλεῖσθω δὲ ἀποτομή. ὅπερ ἔδει δεῖχαι.

Proposition 73

If a rational (straight-line), which is commensurable in square only with the whole, is subtracted from a(nother) rational (straight-line) then the remainder is an irrational (straight-line). Let it be called an apotome.



For let the rational (straight-line) BC , which commensurable in square only with the whole, be subtracted from the rational (straight-line) AB . I say that the remainder AC is that irrational (straight-line) called an apotome.

For since AB is incommensurable in length with BC , and as AB is to BC , so the (square) on AB (is) to the (rectangle contained) by AB and BC [Prop. 10.21 lem.], the (square) on AB is thus incommensurable with the (rectangle contained) by AB and BC [Prop. 10.11]. But, the (sum of the) squares on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. And, inasmuch as the (sum of the) squares on AB and BC is equal to twice the (rectangle contained) by AB and BC plus the (square) on CA [Prop. 2.7], the (sum of the) squares on AB and BC is thus also incommensurable with the remaining (square) on AC [Props. 10.13, 10.16]. And the (sum of the) squares on AB and BC is rational. AC is thus an irrational (straight-line) [Def. 10.4]. And let it be called an apotome.[†] (Which is) the very thing it was required to show.

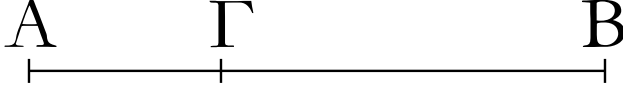
[†] See footnote to Prop. 10.36.

74.

Ἐάν ἀπὸ μέσης μέσης ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ῥητὸν περιέθουσα, ἡ λοιπὴ ἄλογός ἐστιν· καλεῖσθω δὲ μέσης ἀποτομή πρώτη.

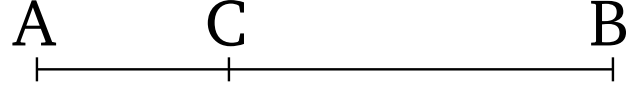
Proposition 74

If a medial (straight-line), which is commensurable in square only with the whole, and which contains a rational (area) with the whole, is subtracted from a(nother) medial (straight-line) then the remainder is an irrational (straight-line). Let it be called a first apotome of a medial (straight-line).



Ἀπὸ γὰρ μέσης τῆς AB μέση ἀφηγήσθω ἡ $BΓ$ δυνάμει μόνον σύμμετρος ὅσα τῇ AB , μετὰ δὲ τῆς AB ῥητόν ποιοῦσα τὸ ὑπὸ τῶν AB , $BΓ$ λέγω, ὅτι ἡ λοιπὴ ἡ $ΑΓ$ ἄλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ πρώτη.

Ἐπεὶ γὰρ αἱ AB , $BΓ$ μέσαι εἰσὶν, μέσα ἐστὶ καὶ τὰ ἀπὸ τῶν AB , $BΓ$. ῥητόν δὲ τὸ διξ ὑπὸ τῶν AB , $BΓ$ · ἀσύμμετρα ἄρα τὰ ἀπὸ τῶν AB , $BΓ$ τῷ διξ ὑπὸ τῶν AB , $BΓ$ · καὶ λοιπῶν ἄρα τῷ ἀπὸ τῆς $ΑΓ$ ἀσύμμετρον ἐστὶ τὸ διξ ὑπὸ τῶν AB , $BΓ$, ἐπεὶ κ' ἂν τὸ ὅλον ἐν αὐτῶν ἀσύμμετρον ᾖ, καὶ τὰ ἐχ ἀρθῆς μεγέθη ἀσύμμετρα ἐσταί. ῥητόν δὲ τὸ διξ ὑπὸ τῶν AB , $BΓ$ · ἄλογον ἄρα τὸ ἀπὸ τῆς $ΑΓ$ · ἄλογος ἄρα ἐστὶν ἡ $ΑΓ$ · καλείσθω δὲ μέσης ἀποτομὴ πρώτη.



For let the medial (straight-line) BC , which is commensurable in square only with AB , and which makes with AB the rational (rectangle contained) by AB and BC , be subtracted from the medial (straight-line) AB [Prop. 10.27]. I say that the remainder AC is an irrational (straight-line). Let it be called the first apotome of a medial (straight-line).

For since AB and BC are medial (straight-lines), the (sum of the squares) on AB and BC is also medial. And twice the (rectangle contained) by AB and BC (is) rational. The (sum of the squares) on AB and BC (is) thus incommensurable with twice the (rectangle contained) by AB and BC . Thus, twice the (rectangle contained) by AB and BC is also incommensurable with the remaining (square) on AC [Prop. 2.7], since if the whole is incommensurable with one of the (constituent magnitudes) then the original magnitudes will also be incommensurable (with one another) [Prop. 10.16]. And twice the (rectangle contained) by AB and BC (is) rational. Thus, the (square) on AC is irrational. Thus, AC is an irrational (straight-line) [Def. 10.4]. Let it be called a first apotome of a medial (straight-line).[†]

[†] See footnote to Prop. 10.37.

75.

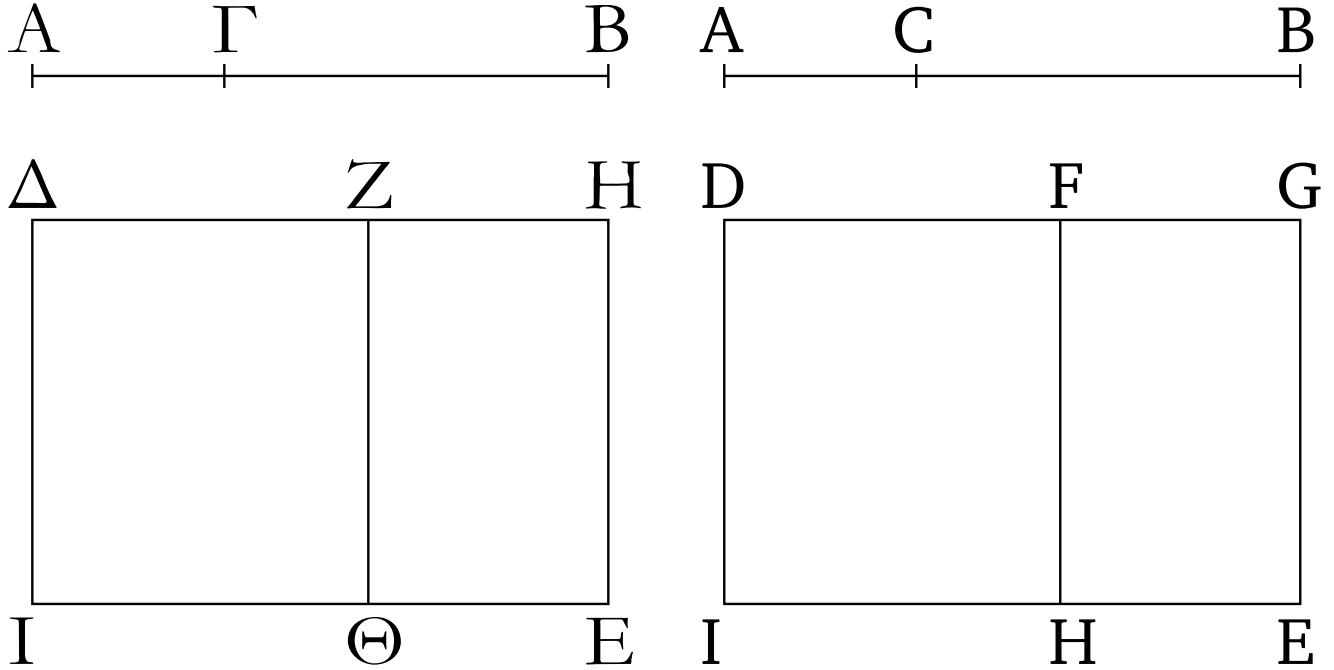
Ἐὰν ἀπὸ μέσης μέση ἀφαιρεθῇ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης μέσον περιέθουσα, ἡ λοιπὴ ἄλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ δευτέρα.

Ἀπὸ γὰρ μέσης τῆς AB μέση ἀφηγήσθω ἡ $ΓΒ$ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ τῇ AB , μετὰ δὲ τῆς ὅλης τῆς AB μέσον περιέθουσα τὸ ὑπὸ τῶν AB , $BΓ$ λέγω, ὅτι ἡ λοιπὴ ἡ $ΑΓ$ ἄλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ δευτέρα.

Proposition 75

If a medial (straight-line), which is commensurable in square only with the whole, and which contains a medial (area) with the whole, is subtracted from a(nother) medial (straight-line) then the remainder is an irrational (straight-line). Let it be called a second apotome of a medial (straight-line).

For let the medial (straight-line) CB , which is commensurable in square only with the whole, AB , and which contains with the whole, AB , the medial (rectangle contained) by AB and BC , be subtracted from the medial (straight-line) AB [Prop. 10.28]. I say that the remainder AC is an irrational (straight-line). Let it be called a second apotome of a medial (straight-line).



Ἐκκείσθω γάρ ῥητὴ ἡ ΔI , καὶ τοῖς μὲν ἀπὸ τῶν AB , BG ἴσον παρὰ τὴν ΔI παραβεβλήσθω τὸ ΔE πλάτος ποιοῦν τὴν ΔH , τῷ δὲ διζῷ ὑπὸ τῶν AB , BG ἴσον παρὰ τὴν ΔI παραβεβλήσθω τὸ $\Delta \Theta$ πλάτος ποιοῦν τὴν ΔZ . λοιπὸν ἄρα τὸ ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς AG . καὶ ἐπεὶ μέσα καὶ σύμμετρά ἐστι τὰ ἀπὸ τῶν AB , BG , μέσον ἄρα καὶ τὸ ΔE . καὶ παρὰ ῥητὴν τὴν ΔI παρὰκείται πλάτος ποιοῦν τὴν ΔH . ῥητὴ ἄρα ἐστὶν ἡ ΔH καὶ ἀσύμμετρος τῇ ΔI μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ὑπὸ τῶν AB , BG , καὶ τὸ διζῷ ἄρα ὑπὸ τῶν AB , BG μέσον ἐστίν. καὶ ἐστὶν ἴσον τῷ $\Delta \Theta$. καὶ τὸ $\Delta \Theta$ ἄρα μέσον ἐστίν. καὶ παρὰ ῥητὴν τὴν ΔI παραβεβλήται πλάτος ποιοῦν τὴν ΔZ . ῥητὴ ἄρα ἐστὶν ἡ ΔZ καὶ ἀσύμμετρος τῇ ΔI μήκει. καὶ ἐπεὶ αἱ AB , BG δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶν ἡ AB τῇ BG μήκει. ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς AB τετράγωνον τῷ ὑπὸ τῶν AB , BG . ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρά ἐστι τὰ ἀπὸ τῶν AB , BG , τῷ δὲ ὑπὸ τῶν AB , BG σύμμετρον ἐστὶ τὸ διζῷ ὑπὸ τῶν AB , BG . ἀσύμμετρον ἄρα ἐστὶ τὸ διζῷ ὑπὸ τῶν AB , BG τοῖς ἀπὸ τῶν AB , BG . ἴσον δὲ τοῖς μὲν ἀπὸ τῶν AB , BG τὸ ΔE , τῷ δὲ διζῷ ὑπὸ τῶν AB , BG τὸ $\Delta \Theta$. ἀσύμμετρον ἄρα [ἐστὶ] τὸ ΔE τῷ $\Delta \Theta$. ὥς δὲ τὸ ΔE πρὸς τὸ $\Delta \Theta$, οὕτως ἡ $H\Delta$ πρὸς τὴν ΔZ . ἀσύμμετρος ἄρα ἐστὶν ἡ $H\Delta$ τῇ ΔZ . καὶ εἰσιν ἀμφοτέραι ῥηταί. αἱ ἄρα $H\Delta$, ΔZ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἡ ZH ἄρα ἀποτομή ἐστίν. ῥητὴ δὲ ἡ ΔI . τὸ δὲ ὑπὸ ῥητῆς καὶ ἀλόγου περιεχόμενον ἄλογόν ἐστιν, καὶ ἡ δυνάμεν αὐτὸ ἄλογός ἐστιν. καὶ δύναται τὸ ZE ἢ AG . ἡ AG ἄρα ἄλογός ἐστιν. καλεῖσθω δὲ μέσης ἀποτομὴ δευτέρα. ὅπερ ἔδει δεῖξαι.

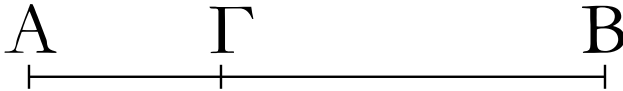
For let the rational (straight-line) DI be laid down. And let DE , equal to the (sum of the squares) on AB and BC , be applied to DI , producing DG as breadth. And let DH , equal to twice the (rectangle contained) by AB and BC , be applied to DI , producing DF as breadth. The remainder FE is thus equal to the (square) on AC [Prop. 2.7]. And since the (squares) on AB and BC are medial and commensurable (with one another), DE (is) thus also medial [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) DI , producing DG as breadth. Thus, DG is rational, and incommensurable in length with DI [Prop. 10.22]. Again, since the (rectangle contained) by AB and BC is medial, twice the (rectangle contained) by AB and BC is thus also medial [Prop. 10.23 corr.]. And it is equal to DH . Thus, DH is also medial. And it has been applied to the rational (straight-line) DI , producing DF as breadth. DF is thus rational, and incommensurable in length with DI [Prop. 10.22]. And since AB and BC are commensurable in square only, AB is thus incommensurable in length with BC . Thus, the square on AB (is) also incommensurable with the (rectangle contained) by AB and BC [Props. 10.21 lem., 10.11]. But, the (sum of the squares) on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. Thus, twice the (rectangle contained) by AB and BC is incommensurable with the (sum of the squares) on AB and BC [Prop. 10.13]. And DE is equal to the (sum of the squares) on AB and BC , and DH to twice the (rectangle contained) by AB and BC . Thus, DE [is] incommensurable with DH . And as DE (is) to DH , so GD (is) to DF [Prop. 6.1]. Thus, GD is in-

commensurable with DF [Prop. 10.11]. And they are both rational (straight-lines). Thus, GD and DF are rational (straight-lines which are) commensurable in square only. Thus, FG is an apotome [Prop. 10.73]. And DI (is) rational. And the (area) contained by a rational and an irrational (straight-line) is irrational [Prop. 10.20], and its square-root is irrational. And AC is the square-root of FE . Thus, AC is an irrational (straight-line) [Def. 10.4]. And let it be called the second apotome of a medial (straight-line).[†] (Which is) the very thing it was required to show.

[†] See footnote to Prop. 10.38.

76.

Ἐάν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιούσα τὰ μὲν ἀπ' αὐτῶν ἅμα ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον, ἡ λοιπὴ ἄλογός ἐστιν· καλείσθω δὲ ἐλάσσων.

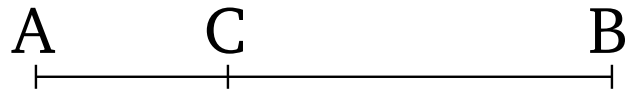


Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφαιρήσθω ἡ $BΓ$ δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ ποιούσα τὰ προκείμενα. λέγω, ὅτι ἡ λοιπὴ ἡ $ΑΓ$ ἄλογός ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ τετραγώνων ῥητόν ἐστιν, τὸ δὲ διξ ὑπὸ τῶν AB , $BΓ$ μέσον, ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$ τῷ διξ ὑπὸ τῶν AB , $BΓ$ · καὶ ἀναστρέψαντι λοιπῷ τῷ ἀπὸ τῆς $ΑΓ$ ἀσύμμετρά ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$. ῥητὰ δὲ τὰ ἀπὸ τῶν AB , $BΓ$ ἄλογον ἄρα τὸ ἀπὸ τῆς $ΑΓ$ ἄλογος ἄρα ἡ $ΑΓ$ · καλείσθω δὲ ἐλάσσων. ὅπερ ἔδει δεῖχαι.

Proposition 76

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the (squares) on them (added) together rational, and the (rectangle contained) by them medial, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called a minor (straight-line).



For let the straight-line BC , which is incommensurable in square with the whole, and fulfils the (other) prescribed (conditions), be subtracted from the straight-line AB [Prop. 10.33]. I say that the remainder AC is that irrational (straight-line) called minor.

For since the sum of the squares on AB and BC is rational, and twice the (rectangle contained) by AB and BC (is) medial, the (sum of the squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . And, via conversion, the (sum of the squares) on AB and BC is incommensurable with the remaining (square) on AC [Props. 2.7, 10.16]. And the (sum of the squares) on AB and BC (is) rational. The (square) on AC (is) thus irrational. Thus, AC (is) an irrational (straight-line) [Def. 10.4]. Let it be called a minor (straight-line).[†] (Which is) the very thing it was required to show.

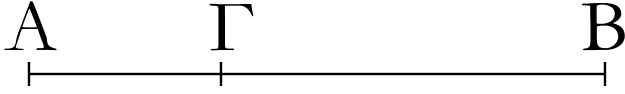
[†] See footnote to Prop. 10.39.

77.

Ἐάν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιούσα τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διξ ὑπ' αὐτῶν ῥητόν, ἡ λοιπὴ ἄλογός ἐστιν· καλείσθω δὲ ἡ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα.

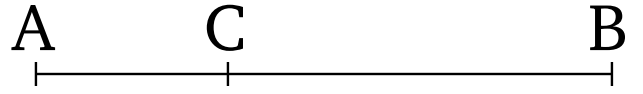
Proposition 77

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them rational, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called that which makes with a rational (area) a medial whole.



Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφηρήσθω ἡ $BΓ$ δυνάμει ἀσύμμετος ὅσα τῇ AB ποιοῦσα τὰ προκείμενα· λέγω, ὅτι ἡ λοιπὴ ἡ $ΑΓ$ ἄλογός ἐστιν ἡ προειρημένη.

Ἐπεὶ γὰρ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ τετραγώνων μέσον ἐστίν, τὸ δὲ διξ ὑπὸ τῶν AB , $BΓ$ ῥητόν, ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$ τῷ διξ ὑπὸ τῶν AB , $BΓ$ · καὶ λοιπὸν ἄρα τὸ ἀπὸ τῆς $ΑΓ$ ἀσύμμετρόν ἐστι τῷ διξ ὑπὸ τῶν AB , $BΓ$. καὶ ἐστὶ τὸ διξ ὑπὸ τῶν AB , $BΓ$ ῥητόν· τὸ ἄρα ἀπὸ τῆς $ΑΓ$ ἄλογόν ἐστιν· ἄλογος ἄρα ἐστὶν ἡ $ΑΓ$ · καλεῖσθω δὲ ἡ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσα. ὅπερ ἔδει δεῖξαι.



For let the straight-line BC , which is incommensurable in square with AB , and fulfils the (other) prescribed (conditions), be subtracted from the straight-line AB [Prop. 10.34]. I say that the remainder AC is the aforementioned irrational (straight-line).

For since the sum of the squares on AB and BC is medial, and twice the (rectangle contained) by AB and BC rational, the (sum of the squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . Thus, the remaining (square) on AC is also incommensurable with twice the (rectangle contained) by AB and BC [Props. 2.7, 10.16]. And twice the (rectangle contained) by AB and BC is rational. Thus, the (square) on AC is irrational. Thus, AC is an irrational (straight-line) [Def. 10.4]. And let it be called that which makes with a rational (area) a medial whole.[†] (Which is) the very thing it was required to show.

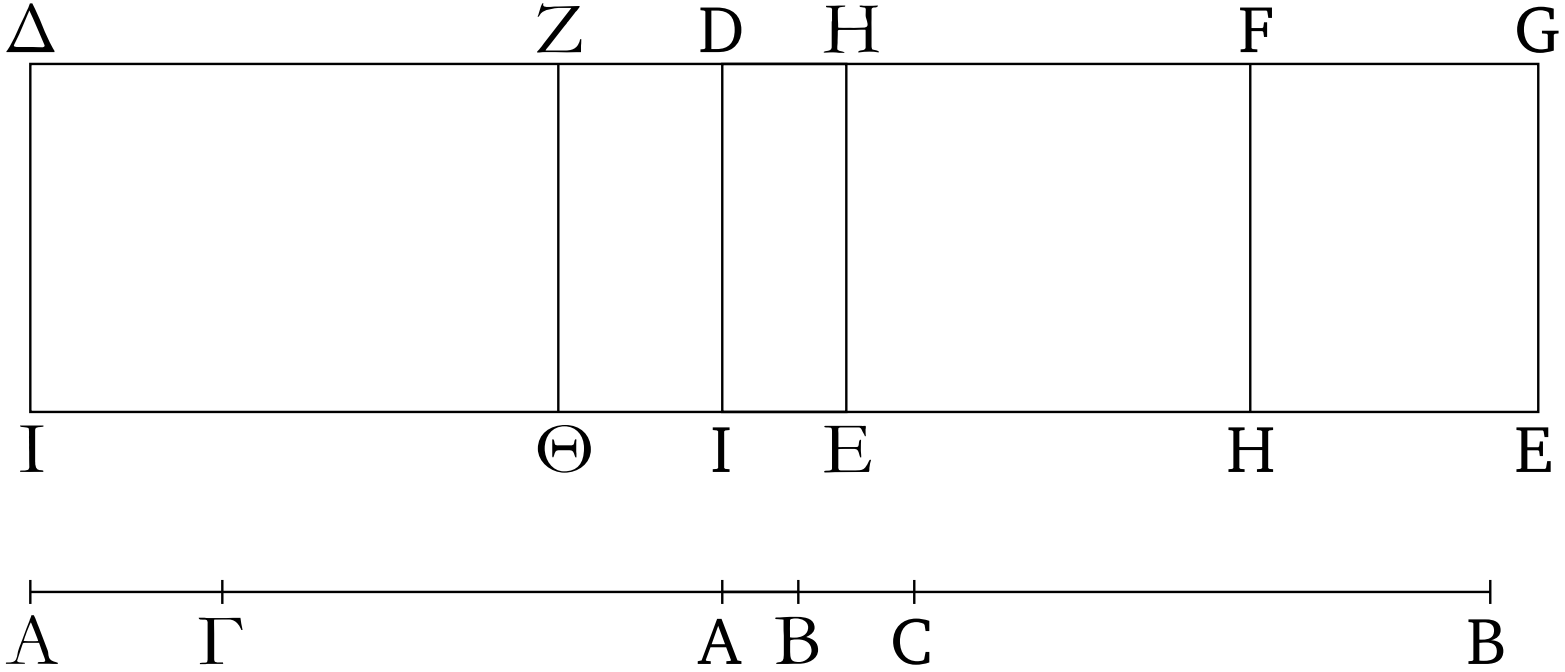
[†] See footnote to Prop. 10.40.

78.

Proposition 78

Ἐὰν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος ὅσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον τὸ τε διξ ὑπ' αὐτῶν μέσον καὶ ἔτι τὰ ἀπ' αὐτῶν τετράγωνα ἀσύμμετρα τῷ διξ ὑπ' αὐτῶν, ἡ λοιπὴ ἄλογός ἐστιν· καλεῖσθω δὲ ἡ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα.

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them medial, and, moreover, the (sum of the) squares on them incommensurable with twice the (rectangle contained) by them, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called that which makes with a medial (area) a medial whole.



Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφηρήσθω ἡ BI δυνάμει ἀσύμμετρος ὅσα τῇ AB ποιοῦσα τὰ προκείμενα· λέγω, ὅτι ἡ λοιπὴ ἡ AI ἄλογός ἐστιν ἡ καλουμένη ἡ μετὰ μέσον μέσον τὸ ὅλον ποιοῦσα.

Ἐκκείσθω γὰρ ῥητὴ ἡ DI , καὶ τοῖς μὲν ἀπὸ τῶν AB , BI ἴσον παρὰ τὴν DI παραβεβλήσθω τὸ DE πλάτος ποιοῦν τὴν DH , τῷ δὲ διζῷ ὑπὸ τῶν AB , BI ἴσον ἀφηρήσθω τὸ $ΔΘ$ [πλάτος ποιοῦν τὴν $ΔΖ$]. λοιπὸν ἄρα τὸ ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς AI · ὥστε ἡ AI δύναται τὸ ZE . καὶ ἐπεὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BI τετραγώνων μέσον ἐστὶ καὶ ἐστὶν ἴσον τῷ DE , μέσον ἄρα [ἐστὶ] τὸ DE . καὶ παρὰ ῥητὴν τὴν DI παράκειται πλάτος ποιοῦν τὴν DH · ῥητὴ ἄρα ἐστὶν ἡ DH καὶ ἀσύμμετρος τῇ DI μήκει. πάλιν, ἐπεὶ τὸ διζῷ ὑπὸ τῶν AB , BI μέσον ἐστὶ καὶ ἐστὶν ἴσον τῷ $ΔΘ$, τὸ ἄρα $ΔΘ$ μέσον ἐστὶν. καὶ παρὰ ῥητὴν τὴν DI παράκειται πλάτος ποιοῦν τὴν $ΔΖ$ · ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΔΖ$ καὶ ἀσύμμετρος τῇ DI μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB , BI τῷ διζῷ ὑπὸ τῶν AB , BI , ἀσύμμετρον ἄρα καὶ τὸ DE τῷ $ΔΘ$. ὥς δὲ τὸ DE πρὸς τὸ $ΔΘ$, οὕτως ἐστὶ καὶ ἡ DH πρὸς τὴν $ΔΖ$ · ἀσύμμετρος ἄρα ἡ DH τῇ $ΔΖ$. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ HD , $ΔΖ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἀποτομὴ ἄρα ἐστὶν ἡ ZH · ῥητὴ δὲ ἡ $ZΘ$. τὸ δὲ ὑπὸ ῥητῆς καὶ ἀποτομῆς περιεχόμενον [ὀρθογώνιον] ἄλογόν ἐστιν, καὶ ἡ δυνάμεν αὐτὸ ἄλογός ἐστιν· καὶ δύναται τὸ ZE ἡ AI · ἡ AI ἄρα ἄλογός ἐστιν· καλεῖσθω δὲ ἡ μετὰ μέσον μέσον τὸ ὅλον ποιοῦσα. ὅπερ ἔδει δεῖχαι.

For let the straight-line BC , which is incommensurable in square AB , and fulfils the (other) prescribed (conditions), be subtracted from the (straight-line) AB [Prop. 10.35]. I say that the remainder AC is the irrational (straight-line) called that which makes with a medial (area) a medial whole.

For let the rational (straight-line) DI be laid down. And let DE , equal to the (sum of the squares) on AB and BC , be applied to DI , producing DG as breadth. And let DH , equal to twice the (rectangle contained) by AB and BC , be subtracted (from DE) [producing DF as breadth]. Thus, the remainder FE is equal to the (square) on AC [Prop. 2.7]. Hence, AC is the square-root of FE . And since the sum of the squares on AB and BC is medial, and is equal to DE , DE [is] thus medial. And it is applied to the rational (straight-line) DI , producing DG as breadth. Thus, DG is rational, and incommensurable in length with DI [Prop 10.22]. Again, since twice the (rectangle contained) by AB and BC is medial, and is equal to DH , DH is thus medial. And it is applied to the rational (straight-line) DI , producing DF as breadth. Thus, DF is also rational, and incommensurable in length with DI [Prop. 10.22]. And since the (sum of the squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC , DE (is) also incommensurable with DH . And as DE (is) to DH , so DG also is to DF [Prop. 6.1]. Thus, DG (is) incommensurable (in length) with DF [Prop. 10.11]. And they are both rational. Thus, GD and DF are rational (straight-lines which are) commensurable in square only. Thus, FG is an apotome [Prop. 10.73]. And FH (is) rational. And the [rectangle] contained by a rational (straight-line) and an apotome is irrational [Prop. 10.20], and its square-root is irrational. And AC is the

square-root of FE . Thus, AC is irrational. Let it be called that which makes with a medial (area) a medial whole.[†] (Which is) the very thing it was required to show.

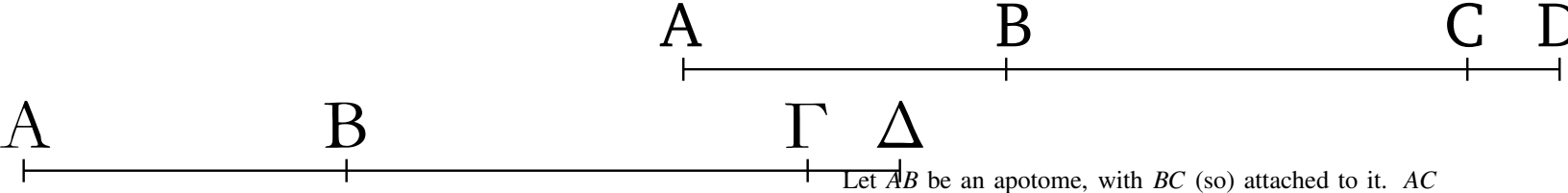
[†] See footnote to Prop. 10.41.

79.

Proposition 79

Τῇ ἀποτομῇ μία [μόνον] προσαρμόζει εὐθεῖα ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ.

[Only] one rational straight-line, which is commensurable in square only with the whole, can be attached to an apotome.[†]



Ἐστω ἀποτομή ἡ AB , προσαρμόζουσα δὲ αὐτῇ ἡ $BΓ$. αἱ $ΑΓ$, $ΓΒ$ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμόζει ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ.

Εἰ γὰρ δυνατόν, προσαρμόξέτω ἡ $ΒΔ$ · καὶ αἱ $ΑΔ$, $ΔΒ$ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ, ὥς ὑπερέθει τὰ ἀπὸ τῶν $ΑΔ$, $ΔΒ$ τοῦ διξ ὑπὸ τῶν $ΑΔ$, $ΔΒ$, τούτῳ ὑπερέθει καὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$ τοῦ διξ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ · τῷ γὰρ αὐτῷ τῷ ἀπὸ τῆς AB ἀμφοτέρα ὑπερέθει· ἐναλλάξ ἄρα, ὥς ὑπερέθει τὰ ἀπὸ τῶν $ΑΔ$, $ΔΒ$ τῶν ἀπὸ τῶν $ΑΓ$, $ΓΒ$, τούτῳ ὑπερέθει [καὶ] τὸ διξ ὑπὸ τῶν $ΑΔ$, $ΔΒ$ τοῦ διξ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. τὰ δὲ ἀπὸ τῶν $ΑΔ$, $ΔΒ$ τῶν ἀπὸ τῶν $ΑΓ$, $ΓΒ$ ὑπερέθει ῥητῶν· ῥητὰ γὰρ ἀμφοτέρα. καὶ τὸ διξ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΒ$ τοῦ διξ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ ὑπερέθει ῥητῶν· ὅπερ ἐστὶν ἀδύνατον· μέσα γὰρ ἀμφοτέρα, μέσον δὲ μέσον οὐδ' ὑπερέθει ῥητῶν. τῇ ἄρα AB ἑτέρα οὐ προσαρμόζει ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ.

Μία ἄρα μόνη τῇ ἀποτομῇ προσαρμόζει ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ· ὅπερ ἔδει δεῖξαι.

Let AB be an apotome, with BC (so) attached to it. AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.73]. I say that another rational (straight-line), which is commensurable in square only with the whole, cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds twice the (rectangle contained) by AD and DB , the (sum of the squares) on AC and CB also exceeds twice the (rectangle contained) by AC and CB by this (same area). For both exceed by the same (area)—(namely), the (square) on AB [Prop. 2.7]. Thus, alternately, by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB [also] exceeds twice the (rectangle contained) by AC and CB by this (same area). And the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB by a rational (area). For both (are) rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.21], and a medial (area) cannot exceed a(nother) medial (area) by a rational (area) [Prop. 10.26]. Thus, another rational (straight-line), which is commensurable in square only with the whole, cannot be attached to AB .

Thus, only one rational (straight-line), which is commensurable in square only with the whole, can be attached to an apotome. (Which is) the very thing it was required to show.

[†] This proposition is equivalent to Prop. 10.42, with minus signs instead of plus signs.

80.

Proposition 80

Τῇ μέσῃ ἀποτομῇ πρώτη μία μόνον προσαρμόζει εὐθεῖα

Only one medial straight-line, which is commensurable

μέση δυνάμει μόνον σύμμετροξ ὅσα τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃς ῥητὸν περιέθουσα.

in square only with the whole, and contains a rational (area) with the whole, can be attached to a first apotome of a medial (straight-line).[†]



Ἐστω γάρ μέσηξ ἀποτομή πρώτη ἡ AB , καὶ τῇ AB προσαρμύζετω ἡ BF . αἱ AF , FB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι τὸ ὑπὸ τῶν AF , FB . λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμύζει μέση δυνάμει μόνον σύμμετροξ ὅσα τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃς ῥητὸν περιέθουσα.

Εἰ γὰρ δυνατόν, προσαρμύζετω καὶ ἡ $ΔB$. αἱ ἄρα AD , $ΔB$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι τὸ ὑπὸ τῶν AD , $ΔB$. καὶ ἐπεὶ, ὥς ὑπερέθει τὰ ἀπὸ τῶν AD , $ΔB$ τοῦ διξ ὑπὸ τῶν AD , $ΔB$, τούτῳ ὑπερέθει καὶ τὰ ἀπὸ τῶν AF , FB τοῦ διξ ὑπὸ τῶν AF , FB . τῷ γὰρ αὐτῷ [πάλιν] ὑπερέθουσι τῷ ἀπὸ τῇξ AB . ἐναλλάχ ἄρα, ὥς ὑπερέθει τὰ ἀπὸ τῶν AD , $ΔB$ τῶν ἀπὸ τῶν AF , FB , τούτῳ ὑπερέθει καὶ τὸ διξ ὑπὸ τῶν AD , $ΔB$ τοῦ διξ ὑπὸ τῶν AF , FB . τὸ δὲ διξ ὑπὸ τῶν AD , $ΔB$ τοῦ διξ ὑπὸ τῶν AF , FB ὑπερέθει ῥητῷ ῥητὰ γὰρ ἀμφοτέρω. καὶ τὰ ἀπὸ τῶν AD , $ΔB$ ἄρα τῶν ἀπὸ τῶν AF , FB [τετραγώνων] ὑπερέθει ῥητῷ. ὅπερ ἐστὶν ἀδύνατον· μέσα γὰρ ἐστὶν ἀμφοτέρω, μέσον δὲ μέσον οὐδ' ὑπερέθει ῥητῷ.

Τῇ ἄρα μέσηξ ἀποτομῇ πρώτῃ μία μόνον προσαρμύζει ἐνθεῖα μέση δυνάμει μόνον σύμμετροξ ὅσα τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃς ῥητὸν περιέθουσα. ὅπερ ἔδει δεῖξαι.

For let AB be a first apotome of a medial (straight-line), and let BC be (so) attached to AB . Thus, AC and CB are medial (straight-lines which are) commensurable in square only, containing a rational (area)—(namely, that contained) by AC and CB [Prop. 10.74]. I say that a(nother) medial (straight-line), which is commensurable in square only with the whole, and contains a rational (area) with the whole, cannot be attached to AB .

For, if possible, let DB also be (so) attached to AB . Thus, AD and DB are medial (straight-lines which are) commensurable in square only, containing a rational (area)—(namely, that) contained by AD and DB [Prop. 10.74]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds twice the (rectangle contained) by AD and DB , the (sum of the squares) on AC and CB also exceeds twice the (rectangle contained) by AC and CB by this (same area). For [again] both exceed by the same (area)—(namely), the (square) on AB [Prop. 2.7]. Thus, alternately, by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area). And twice the (rectangle contained) by AD and DB exceeds twice the (rectangle contained) by AC and CB by a rational (area). For both (are) rational (areas). Thus, the (sum of the squares) on AD and DB also exceeds the (sum of the) [squares] on AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Props. 10.15, 10.23 corr.], and a medial (area) cannot exceed a(nother) medial (area) by a rational (area) [Prop. 10.26].

Thus, only one medial (straight-line), which is commensurable in square only with the whole, and contains a rational (area) with the whole, can be attached to a first apotome of a medial (straight-line). (Which is) the very thing it was required to show.

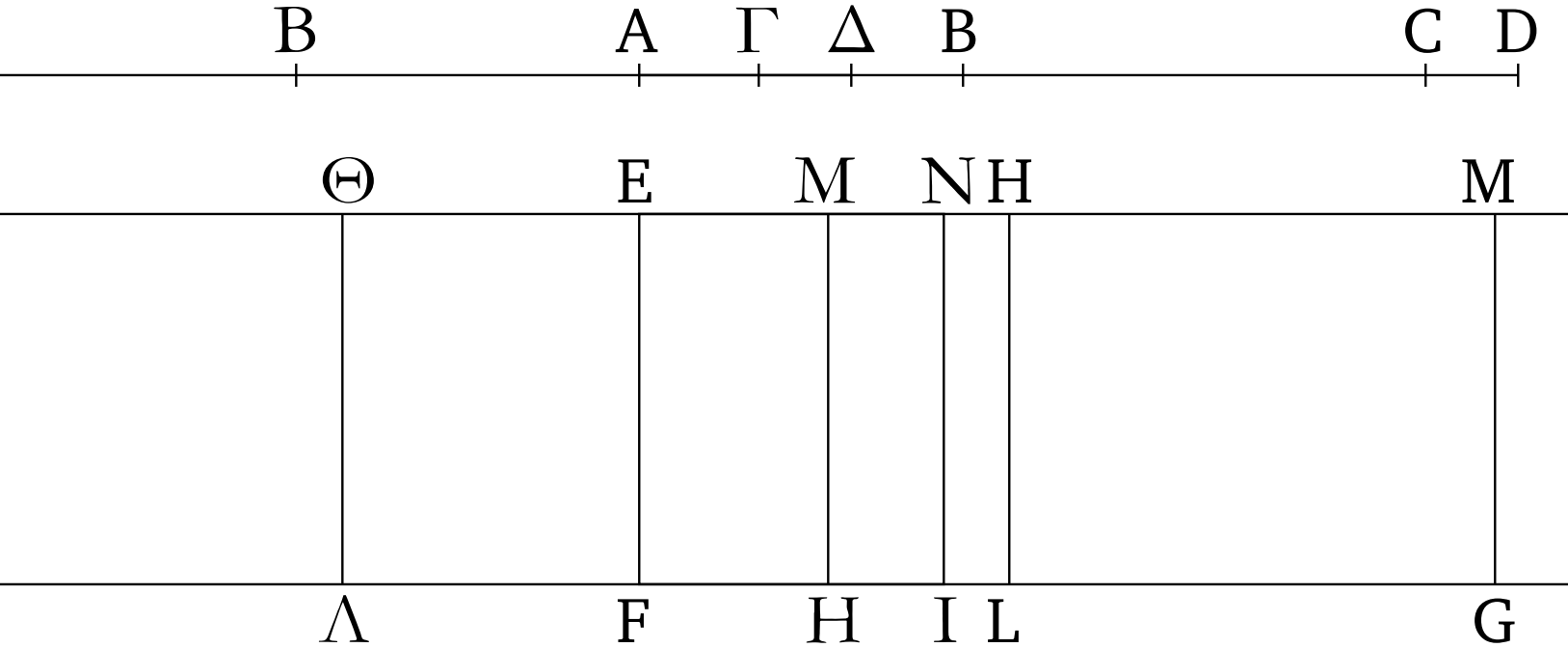
[†] This proposition is equivalent to Prop. 10.43, with minus signs instead of plus signs.

81.

Proposition 81

Τῇ μέσηξ ἀποτομῇ δευτέρῃ μία μόνον προσαρμύζει ἐνθεῖα μέση δυνάμει μόνον σύμμετροξ τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃς μέσον περιέθουσα.

Only one medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, can be attached to a second apotome of a medial (straight-line).[†]



Ἐστω μέσης ἀποτομή δευτέρα ἡ AB καὶ τῇ AB προσαρμόζονσα ἡ $BΓ$. αἱ ἄρα $ΑΓ$, $ΓΒ$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι τὸ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμόσει εὐθεῖα μέση δυνάμει μόνον σύμμετρος οὕσα τῇ ὅλῃ, μετὰ δὲ τῇς ὅλης μέσον περιέθουσα.

Εἰ γὰρ δυνατόν, προσαρμόξέτω ἡ $BΔ$. καὶ αἱ $ΑΔ$, $ΔΒ$ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι τὸ ὑπὸ τῶν $ΑΔ$, $ΔΒ$. καὶ ἐκκείσθω ῥητὴ ἡ EZ , καὶ τοῖς μὲν ἀπὸ τῶν $ΑΓ$, $ΓΒ$ ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ $ΕΗ$ πλάτος ποιοῦν τὴν $ΕΜ$. τῷ δὲ διζῷ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ ἴσον ἀφηρήσθω τὸ $ΘΗ$ πλάτος ποιοῦν τὴν $ΘΜ$. λοιπὸν ἄρα τὸ $ΕΛ$ ἴσον ἐστὶ τῷ ἀπὸ τῇς AB . ὥστε ἡ AB δύναται τὸ $ΕΛ$. πάλιν δὲ τοῖς ἀπὸ τῶν $ΑΔ$, $ΔΒ$ ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ $ΕΙ$ πλάτος ποιοῦν τὴν $ΕΝ$. ἐστὶ δὲ καὶ τὸ $ΕΛ$ ἴσον τῷ ἀπὸ τῇς AB τετραγώνῳ. λοιπὸν ἄρα τὸ $ΘΙ$ ἴσον ἐστὶ τῷ διζῷ ὑπὸ τῶν $ΑΔ$, $ΔΒ$. καὶ ἐπεὶ μέσαι εἰσὶν αἱ $ΑΓ$, $ΓΒ$, μέσα ἄρα ἐστὶ καὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$. καὶ ἐστὶν ἴσα τῷ $ΕΗ$. μέσον ἄρα καὶ τὸ $ΕΗ$. καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν $ΕΜ$. ῥητὴ ἄρα ἐστὶν ἡ $ΕΜ$ καὶ ἀσύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ὑπὸ τῶν $ΑΓ$, $ΓΒ$, καὶ τὸ διζῷ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ μέσον ἐστίν. καὶ ἐστὶν ἴσον τῷ $ΘΗ$. καὶ τὸ $ΘΗ$ ἄρα μέσον ἐστίν. καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν $ΘΜ$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΘΜ$ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ αἱ $ΑΓ$, $ΓΒ$ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶν ἡ $ΑΓ$ τῇ $ΓΒ$ μήκει. ὥς δὲ ἡ $ΑΓ$ πρὸς τὴν $ΓΒ$, οὕτως ἐστὶ τὸ ἀπὸ τῇς $ΑΓ$ πρὸς τὸ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῇς $ΑΓ$ τῷ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. ἀλλὰ τῷ μὲν ἀπὸ τῇς $ΑΓ$ σύμμετρον ἐστὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$, τῷ δὲ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ σύμμετρον ἐστὶ τὸ διζῷ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν $ΑΓ$,

Let AB be a second apotome of a medial (straight-line), with BC (so) attached to AB . Thus, AC and CB are medial (straight-lines which are) commensurable in square only, containing a medial (area)—(namely, that contained) by AC and CB [Prop. 10.75]. I say that a(nother) medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, cannot be attached to AB .

For, if possible, let BD be (so) attached. Thus, AD and DB are also medial (straight-lines which are) commensurable in square only, containing a medial (area)—(namely, that contained) by AD and DB [Prop. 10.75]. And let the rational (straight-line) EF be laid down. And let EG , equal to the (sum of the squares) on AC and CB , be applied to EF , producing EM as breadth. And let HG , equal to twice the (rectangle contained) by AC and CB , be subtracted (from EG), producing HM as breadth. The remainder EL is thus equal to the (square) on AB [Prop. 2.7]. Hence, AB is the square-root of EL . So, again, let EI , equal to the (sum of the squares) on AD and DB be applied to EF , producing EN as breadth. And EL is also equal to the square on AB . Thus, the remainder HI is equal to twice the (rectangle contained) by AD and DB [Prop. 2.7]. And since AC and CB are (both) medial (straight-lines), the (sum of the squares) on AC and CB is also medial. And it is equal to EG . Thus, EG is also medial [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) EF , producing EM as breadth. Thus, EM is rational, and incommensurable in length with EF [Prop. 10.22]. Again, since the (rectangle contained) by AC and CB is medial, twice the (rectangle contained) by AC and CB is also me-

GB τῷ διζ ὑπὸ τῶν AG , GB . καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν AG , GB ἴσον τὸ EH , τῷ δὲ διζ ὑπὸ τῶν AG , GB ἴσον τὸ $H\Theta$. ἀσύμμετρον ἄρα ἐστὶ τὸ EH τῷ ΘH . ὥς δὲ τὸ EH πρὸς τὸ ΘH , οὕτως ἐστὶν ἡ EM πρὸς τὴν ΘM . ἀσύμμετρος ἄρα ἐστὶν ἡ EM τῇ $M\Theta$ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ EM , $M\Theta$ ἄρα ῥηταὶ εἰςὶ δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ $E\Theta$, προσαρμόζουσα δὲ αὐτῇ ἡ ΘM . ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΘN αὐτῇ προσαρμόζει· τῇ ἄρα ἀποτομῇ ἄλλη καὶ ἄλλη προσαρμόζει εὐθεῖα δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ· ὅπερ ἐστὶν ἀδύνατον.

Τῇ ἄρα μέσῃ ἀποτομῇ δευτέρᾳ μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῇς ὅλης μέσον περιέθουσα· ὅπερ ἔδει δεῖχαι.

dial [Prop. 10.23 corr.]. And it is equal to HG . Thus, HG is also medial. And it is applied to the rational (straight-line) EF , producing HM as breadth. Thus, HM is also rational, and incommensurable in length with EF [Prop. 10.22]. And since AC and CB are commensurable in square only, AC is thus incommensurable in length with CB . And as AC (is) to CB , so the (square) on AC is to the (rectangle contained) by AC and CB [Prop. 10.21 corr.]. Thus, the (square) on AC is incommensurable with the (rectangle contained) by AC and CB [Prop. 10.11]. But, the (sum of the squares) on AC and CB is commensurable with the (square) on AC , and twice the (rectangle contained) by AC and CB is commensurable with the (rectangle contained) by AC and CB [Prop. 10.6]. Thus, the (sum of the squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB [Prop. 10.13]. And EG is equal to the (sum of the squares) on AC and CB . And GH is equal to twice the (rectangle contained) by AC and CB . Thus, EG is incommensurable with HG . And as EG (is) to HG , so EM is to HM [Prop. 6.1]. Thus, EM is incommensurable in length with MH [Prop. 10.11]. And they are both rational (straight-lines). Thus, EM and MH are rational (straight-lines which are) commensurable in square only. Thus, EH is an apotome [Prop. 10.73], and HM (is) attached to it. So, similarly, we can show that HN (is) also (commensurable in square only with EN and is) attached to (EH). Thus, different straight-lines, which are commensurable in square only with the whole, are attached to an apotome. The very thing is impossible [Prop. 10.79].

Thus, only one medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, can be attached to a second apotome of a medial (straight-line). (Which is) the very thing it was required to show.

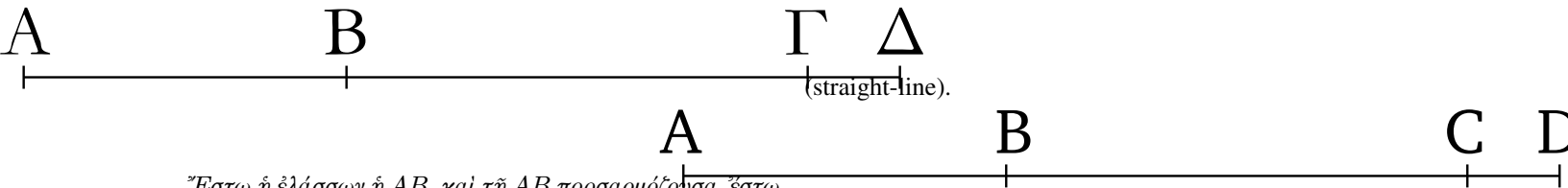
† This proposition is equivalent to Prop. 10.44, with minus signs instead of plus signs.

82.

Proposition 82

Τῇ ἐλάσσονι μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ ποιοῦσα μετὰ τῇς ὅλης τὸ μὲν ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δὲ διζ ὑπ' αὐτῶν μέσον.

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the (sum of the) squares on them rational, and twice the (rectangle contained) by them medial, can be attached to a minor



Ἐστω ἡ ἐλάσσων ἡ AB , καὶ τῇ AB προσαρμόζουσα ἔστω ἡ $B\Gamma$. αἱ ἄρα AG , GB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκεκλιμένον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δὲ διζ ὑπ' αὐτῶν μέσον· λέγω, ὅτι τῇ AB ἑτέρα εὐθεῖα οὐ

Let AB be a minor (straight-line), and let BC be (so) attached to AB . Thus, AC and CB are (straight-lines which are) incommensurable in square, making the sum of the squares

προσαρμόσει τὰ αὐτὰ ποιῶσα.

Εἰ γὰρ δυνατόν, προσαρμोजέτω ἡ BD · καὶ αἱ AD , DB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὰ προειρημένα. καὶ ἐπεὶ, ὥς ὑπερέθει τὰ ἀπὸ τῶν AD , DB τῶν ἀπὸ τῶν AG , GB , τούτῳ ὑπερέθει καὶ τὸ διξ ὑπὸ τῶν AD , DB τοῦ διξ ὑπὸ τῶν AG , GB , τὰ δὲ ἀπὸ τῶν AD , DB τετράγωνα τῶν ἀπὸ τῶν AG , GB τετραγώνων ὑπερέθει ῥητῶ· ῥητὰ γάρ ἐστιν ἀμφότερα· καὶ τὸ διξ ὑπὸ τῶν AD , DB ἄρα τοῦ διξ ὑπὸ τῶν AG , GB ὑπερέθει ῥητῶ· ὅπερ ἐστὶν ἀδύνατον· μέσα γάρ ἐστιν ἀμφότερα.

Τῇ ἄρα ἐλάσσονι μία μόνον προσαρμोजεῖ εὐθεῖα δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ καὶ ποιῶσα τὰ μὲν ἀπ' αὐτῶν τετράγωνα ἅμα ῥητόν, τὸ δὲ διξ ὑπ' αὐτῶν μέσον· ὅπερ ἔδει δεῖχαι.

on them rational, and twice the (rectangle contained) by them medial [Prop. 10.76]. I say that another another straight-line fulfilling the same (conditions) cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also (straight-lines which are) incommensurable in square, fulfilling the (other) aforementioned (conditions) [Prop. 10.76]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area) [Prop. 2.7]. And the (sum of the) squares on AD and DB exceeds the (sum of the) squares on AC and CB by a rational (area). For both are rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.26].

Thus, only one straight-line, which is incommensurable in square with the whole, and (with the whole) makes the squares on them (added) together rational, and twice the (rectangle contained) by them medial, can be attached to a minor (straight-line). (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.45, with minus signs instead of plus signs.

83.

Proposition 83

Τῇ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσῃ μία μόνον προσαρμोजεῖ εὐθεῖα δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῇ ὅλῃ ποιῶσα τὸ μὲν συγκεείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διξ ὑπ' αὐτῶν ῥητόν.

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them rational, can be attached to that (straight-line) which with a rational (area) makes a medial whole.†



Ἐστω ἡ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα ἡ AB , καὶ τῇ AB προσαρμोजέτω ἡ $BΓ$ · αἱ ἄρα AG , GB δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὰ προκείμενα· λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμोजεῖ τὰ αὐτὰ ποιῶσα.

Εἰ γὰρ δυνατόν, προσαρμोजέτω ἡ BD · καὶ αἱ AD , DB ἄρα εὐθεῖαι δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὰ προκείμενα. ἐπεὶ οὖν, ὥς ὑπερέθει τὰ ἀπὸ τῶν AD , DB τῶν ἀπὸ τῶν AG , GB , τούτῳ ὑπερέθει καὶ τὸ διξ ὑπὸ τῶν AD , DB τοῦ διξ ὑπὸ τῶν AG , GB ἀκολούθως τοῖς πρὸ αὐτοῦ, τὸ δὲ διξ ὑπὸ τῶν AD , DB τοῦ διξ ὑπὸ τῶν AG , GB ὑπερέθει ῥητῶ· ῥητὰ γάρ ἐστιν ἀμφότερα· καὶ τὰ ἀπὸ τῶν AD , DB ἄρα τῶν ἀπὸ τῶν AG , GB ὑπερέθει ῥητῶ· ὅπερ ἐστὶν ἀδύνατον· μέσα γάρ ἐστιν ἀμφότερα.

Οὐκ ἄρα τῇ AB ἑτέρα προσαρμोजεῖ εὐθεῖα δυνάμει ἀσύμ-

Let AB be a (straight-line) which with a rational (area) makes a medial whole, and let BC be (so) attached to AB . Thus, AC and CB are (straight-lines which are) incommensurable in square, fulfilling the (other) proscribed (conditions) [Prop. 10.77]. I say that another (straight-line) fulfilling the same (conditions) cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also straight-lines (which are) incommensurable in square, fulfilling the (other) proscribed (conditions) [Prop. 10.77]. Therefore, analogously to the (propositions) before this, since by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this

μετροξ οὔσα τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃξ ποιούσα τὰ προειρη-
μένα· μία ἄρα μόνον προσαρμόσει· ὅπερ ἔδει δεῖχαι.

(same area). And twice the (rectangle contained) by AD and DB exceeds twice the (rectangle contained) by AC and CB by a rational (area). For they are (both) rational (areas). Thus, the (sum of the squares) on AD and DB also exceeds the (sum of the squares) on AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.26].

Thus, another straight-line cannot be attached to AB , which is incommensurable in square with the whole, and fulfills the (other) aforementioned (conditions) with the whole. Thus, only one (such straight-line) can be (so) attached. (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.46, with minus signs instead of plus signs.

84.

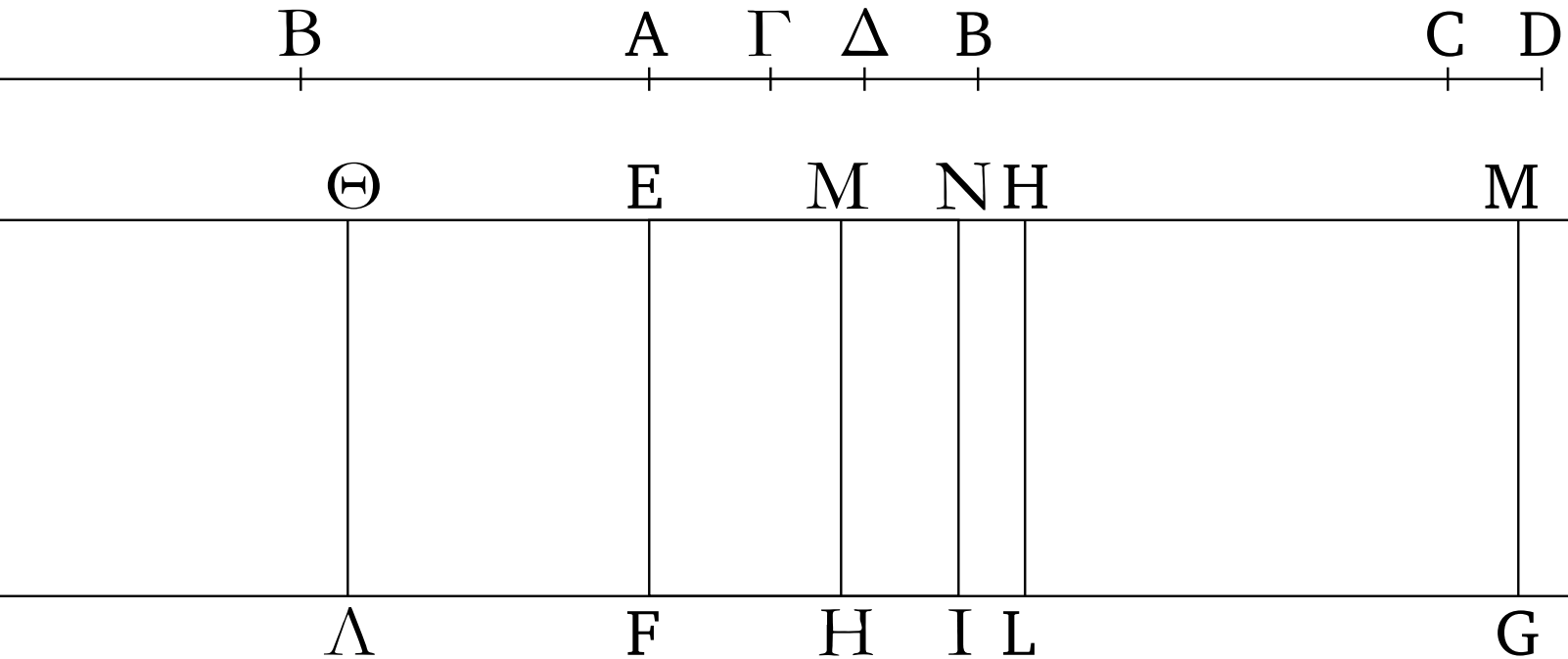
Proposition 84

Τῇ μετὰ μέσον μέσον τὸ ὅλον ποιούσῃ μία μόνῃ προσ-
αρμόζει εὐθεῖα δυνάμει ἀσύμμετροξ οὔσα τῇ ὅλῃ, μετὰ δὲ
τῇξ ὅλῃξ ποιούσα τό τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετρα-
γώνων μέσον τό τε διξ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον
τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν.

Ἐστω ἡ μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἡ AB , προσ-
αρμόζονσα δὲ αὐτῇ ἡ $BΓ$. αἱ ἄρα $ΑΓ$, $ΓΒ$ δυνάμει εἰσὶν ἀσύμ-
μετροι ποιούσαι τὰ προειρημένα. λέγω, ὅτι τῇ AB ἑτέρα οὐ
προσαρμόσει ποιούσα προειρημένα.

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the (squares) on them, can be attached to that (straight-line) which with a medial (area) makes a medial whole.†

Let AB be a (straight-line) which with a medial (area) makes a medial whole, BC being (so) attached to it. Thus, AC and CB are incommensurable in square, fulfilling the (other) aforementioned (conditions) [Prop. 10.78]. I say that a(nother) (straight-line) fulfilling the aforementioned (conditions) cannot be attached to AB .



Εἰ γὰρ δυνατόν, προσαρμोजέτω ἡ $BΔ$, ὥστε καὶ τὰξ $ΑΔ$,

For, if possible, let BD be (so) attached. Hence, AD and

ΔB δυνάμει ἀσύμμετρον εἶναι ποιούσαξ τά τε ἀπὸ τῶν ΔA , ΔB τετράγωνα ἅμα μέσον καὶ τὸ διζ ὑπὸ τῶν ΔA , ΔB μέσον καὶ ἔτι τὰ ἀπὸ τῶν ΔA , ΔB ἀσύμμετρα τῷ διζ ὑπὸ τῶν ΔA , ΔB καὶ ἐκκείσθω ῥητὴ ἡ EZ , καὶ τοῖς μὲν ἀπὸ τῶν AG , GB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EH πλάτος ποιῶν τὴν EM , τῷ δὲ διζ ὑπὸ τῶν AG , GB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ ΘH πλάτος ποιῶν τὴν ΘM . λοιπὸν ἄρα τὸ ἀπὸ τῆς AB ἴσον ἐστὶ τῷ EL ἢ ἄρα AB δύνανται τὸ EL . πάλιν τοῖς ἀπὸ τῶν ΔA , ΔB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EI πλάτος ποιῶν τὴν EN . ἐστὶ δὲ καὶ τὸ ἀπὸ τῆς AB ἴσον τῷ EL λοιπὸν ἄρα τὸ διζ ὑπὸ τῶν ΔA , ΔB ἴσον [ἐστὶ] τῷ ΘI . καὶ ἐπεὶ μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB καὶ ἐστὶν ἴσον τῷ EH , μέσον ἄρα ἐστὶ καὶ τὸ EH . καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιῶν τὴν EM ῥητὴ ἄρα ἐστὶν ἡ EM καὶ ἀσύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ διζ ὑπὸ τῶν AG , GB καὶ ἐστὶν ἴσον τῷ ΘH , μέσον ἄρα καὶ τὸ ΘH . καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιῶν τὴν ΘM ῥητὴ ἄρα ἐστὶν ἡ ΘM καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AG , GB τῷ διζ ὑπὸ τῶν AG , GB , ἀσύμμετρόν ἐστι καὶ τὸ EH τῷ ΘH . ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ EM τῇ $M\Theta$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ἄρα EM , $M\Theta$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ $E\Theta$, προσαρμόζονσα δὲ αὐτῇ ἡ ΘM . ὁμοίως δὲ δείχμεν, ὅτι ἡ $E\Theta$ πάλιν ἀποτομὴ ἐστὶν, προσαρμόζονσα δὲ αὐτῇ ἡ ΘN . τῇ ἄρα ἀποτομῇ ἄλλη καὶ ἄλλη προσαρμόζει ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ· ὅπερ εἰδείσθαι ἀδύνατον. οὐκ ἄρα τῇ AB ἑτέρα προσαρμόσει εὐθεῖα.

Τῇ ἄρα AB μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος ὅσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιούσα τά τε ἀπ' αὐτῶν τετράγωνα ἅμα μέσον καὶ τὸ διζ ὑπ' αὐτῶν μέσον καὶ ἔτι τὰ ἀπ' αὐτῶν τετράγωνα ἀσύμμετρα τῷ διζ ὑπ' αὐτῶν· ὅπερ εἶδει δεῖχαι.

DB are also (straight-lines which are) incommensurable in square, making the squares on AD and DB (added) together medial, and twice the (rectangle contained) by AD and DB medial, and, moreover, the (sum of the squares) on AD and DB incommensurable with twice the (rectangle contained) by AD and DB [Prop. 10.78]. And let the rational (straight-line) EF be laid down. And let EG , equal to the (sum of the squares) on AC and CB , be applied to EF , producing EM as breadth. And let HG , equal to twice the (rectangle contained) by AC and CB , be applied to EF , producing HM as breadth. Thus, the remaining (square) on AB is equal to EL [Prop. 2.7]. Thus, AB is the square-root of EL . Again, let EI , equal to the (sum of the squares) on AD and DB , be applied to EF , producing EN as breadth. And the (square) on AB is also equal to EL . Thus, the remaining twice the (rectangle contained) by AD and DB [is] equal to HI [Prop. 2.7]. And since the sum of the (squares) on AC and CB is medial, and is equal to EG , EG is thus also medial. And it is applied to the rational (straight-line) EF , producing EM as breadth. EM is thus rational, and incommensurable in length with EF [Prop. 10.22]. Again, since twice the (rectangle contained) by AC and CB is medial, and is equal to HG , HG is thus also medial. And it is applied to the rational (straight-line) EF , producing HM as breadth. HM is thus rational, and incommensurable in length with EF [Prop. 10.22]. And since the (sum of the squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , EG is also incommensurable with HG . Thus, EM is also incommensurable in length with MH [Props. 6.1, 10.11]. And they are both rational (straight-lines). Thus, EM and MH are rational (straight-lines which are) commensurable in square only. Thus, EH is an apotome [Prop. 10.73], with HM attached to it. So, similarly, we can show that EH is again an apotome, with HN attached to it. Thus, different rational (straight-lines), which are commensurable in square only with the whole, are attached to an apotome. The very thing was shown (to be) impossible [Prop. 10.79]. Thus, another straight-line cannot be (so) attached to AB .

Thus, only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the squares on them (added) together medial, and twice the (rectangle contained) by them medial, and, moreover, the (sum of the) squares on them incommensurable with the (rectangle contained) by them, can be attached to AB . (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.47, with minus signs instead of plus signs.

Ὅροι τρίτοι.

Definitions III

11. Ὑποκειμένης ῥητῆς καὶ ἀποτομῆς, εἰ μὲν ἡ ὅλη τῆς προσαρμόζουσῃς μείζον δύνηται τῷ ἀπὸ συμμέτρον ἐαυτῇ

11. Given a rational (straight-line) and an apotome, if the square on the whole is greater than the (square on a straight-line) attached (to the apotome) by the (square) on (some

μήκει, καὶ ἡ ὅλη σύμμετρος ἢ τῇ ἐκκειμένῃ ῥητῇ μήκει, καλείσθω ἀποτομή πρώτη.

12. Ἐὰν δὲ ἡ προσαρμόζουσα σύμμετρος ἢ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ ὅλη τῆς προσαρμοζούσης μείζον δύνηται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καλείσθω ἀποτομή δευτέρα.

13. Ἐὰν δὲ μηδετέρα σύμμετρος ἢ τῇ ἐκκειμένῃ ῥητῇ μήκει, ἡ δὲ ὅλη τῆς προσαρμοζούσης μείζον δύνηται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καλείσθω ἀποτομή τρίτη.

14. Πάλιν, ἔαν ἡ ὅλη τῆς προσαρμοζούσης μείζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ [μήκει], ἔαν μὲν ἡ ὅλη σύμμετρος ἢ τῇ ἐκκειμένῃ ῥητῇ μήκει, καλείσθω ἀποτομή τετάρτη.

15. Ἐὰν δὲ ἡ προσαρμόζουσα, πέμπτη.

16. Ἐὰν δὲ μηδετέρα, ἕκτη.

straight-line) commensurable in length with (the whole), and the whole is commensurable in length with the (previously) laid down rational (straight-line), then let the (apotome) be called a first apotome.

12. And if the attached (straight-line) is commensurable in length with the (previously) laid down rational (straight-line), and the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) commensurable (in length) with (the whole), then let the (apotome) be called a second apotome.

13. And if neither of (the whole or the attached straight-line) is commensurable in length with the (previously) laid down rational (straight-line), and the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) commensurable (in length) with (the whole), then let the (apotome) be called a third apotome.

14. Again, if the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) incommensurable [in length] with (the whole), and the whole is commensurable in length with the (previously) laid down rational (straight-line), then let the (apotome) be called a fourth apotome.

15. And if the attached (straight-line is commensurable), a fifth (apotome).

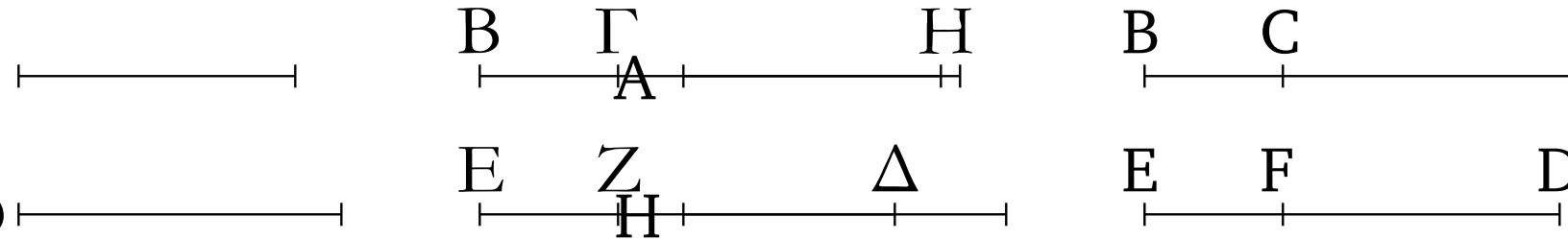
16. And if neither (the whole nor the attached straight-line is commensurable), a sixth (apotome).

85.

Proposition 85

Εὕρεῖν τὴν πρώτην ἀποτομήν.

To find a first apotome.



Ἐκκείσθω ῥητὴ ἡ A , καὶ τῇ A μήκει σύμμετρος ἔστω ἡ BH . ῥητὴ ἄρα ἐστὶ καὶ ἡ BH . καὶ ἐκκείσθωσαν δύο τετράγωνοι ἀριθμοὶ οἱ ΔE , EZ , ὧν ἡ ὑπεροχὴ ὁ $Z\Delta$ μὴ ἔστω τετράγωνος· οὐδ' ἄρα ὁ $E\Delta$ πρὸς τὸν ΔZ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ πεποιήσθω ὥς ὁ $E\Delta$ πρὸς τὸν ΔZ , οὕτως τὸ ἀπὸ τῆς BH τετράγωνον πρὸς τὸ ἀπὸ τῆς $H\Gamma$ τετράγωνον· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς BH τῷ ἀπὸ τῆς $H\Gamma$. ῥητὸν δὲ τὸ ἀπὸ τῆς BH . ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $H\Gamma$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $H\Gamma$. καὶ ἐπεὶ ὁ $E\Delta$ πρὸς τὸν ΔZ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' ἄρα τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $H\Gamma$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

Let the rational (straight-line) A be laid down. And let BG be commensurable in length with A . BG is thus also a rational (straight-line). And let two square numbers DE and EF be laid down, and let their difference FD be not square [Prop. 10.28 lem. I]. Thus, ED does not have to DF the ratio which (some) square number (has) to (some) square number. And let it be contrived that as ED (is) to DF , so the square on BG (is) to the square on GC [Prop. 10.6. corr.]. Thus, the (square) on BG is commensurable with the (square) on GC [Prop. 10.6]. And the (square) on BG (is) rational. Thus, the (square) on GC (is) also rational. Thus, GC is also rational. And since ED does not have to DF the ratio which (some)

νον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ BH τῇ $HΓ$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ BH , $HΓ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ ἄρα $ΒΓ$ ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ πρώτη.

Ἦ γὰρ μείζων ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $HΓ$, ἔστω τὸ ἀπὸ τῆς $Θ$. καὶ ἐπεὶ ἐστὶν ὡς ὁ $ΕΔ$ πρὸς τὸν $ΖΔ$, οὕτως τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$, καὶ ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ $ΔΕ$ πρὸς τὸν $ΕΖ$, οὕτως τὸ ἀπὸ τῆς HB πρὸς τὸ ἀπὸ τῆς $Θ$. ὁ δὲ $ΔΕ$ πρὸς τὸν $ΕΖ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἐκάτερος γὰρ τετράγωνός ἐστιν· καὶ τὸ ἀπὸ τῆς HB ἄρα πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα ἐστὶν ἡ BH τῇ $Θ$ μήκει. καὶ δύναιται ἡ BH τῆς $HΓ$ μείζον τῷ ἀπὸ τῆς $Θ$ · ἡ BH ἄρα τῆς $HΓ$ μείζον δύναιται τῷ ἀπὸ συμμετρὸν ἐαυτῇ μήκει. καὶ ἐστὶν ἡ ὅλη ἡ BH σύμμετρος τῇ ἐκκειμένῃ ῥητῇ μήκει τῇ A . ἡ $ΒΓ$ ἄρα ἀποτομή ἐστὶ πρώτη.

Εὐρεται ἄρα ἡ πρώτη ἀποτομή ἡ $ΒΓ$ · ὅπερ ἔδει εὐρεῖν.

square number (has) to (some) square number, the (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). Thus, BG and GC are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that (it is) also a first (apotome).

Let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.]. And since as ED is to FD , so the (square) on BG (is) to the (square) on GC , thus, via conversion, as DE is to EF , so the (square) on GB (is) to the (square) on H [Prop. 5.19 corr.]. And DE has to EF the ratio which (some) square-number (has) to (some) square-number. For each is a square (number). Thus, the (square) on GB also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, BG is commensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on BG is greater than (the square on) GC by the (square) on (some straight-line) commensurable in length with (BG). And the whole, BG , is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is a first apotome [Def. 10.11].

Thus, the first apotome BC has been found. (Which is) the very thing it was required to find.

† See footnote to Prop. 10.48.

86.

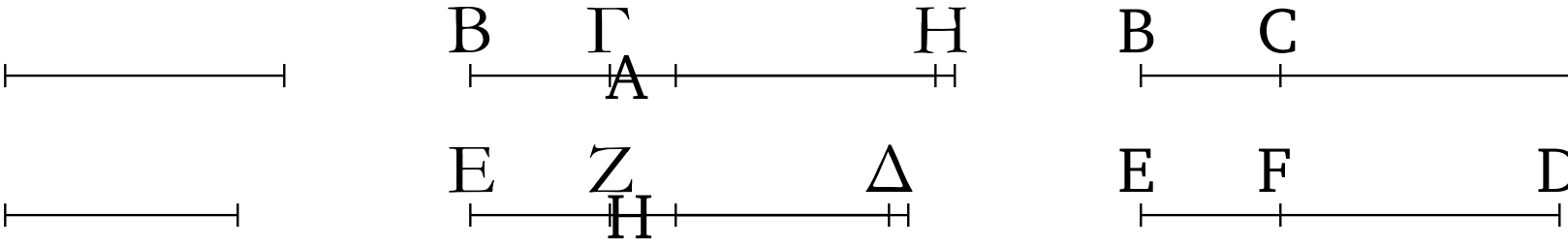
Εὐρεῖν τὴν δευτέραν ἀποτομήν.

Ἐκκείσθω ῥητὴ ἡ A καὶ τῇ A σύμμετρος μήκει ἡ $HΓ$. ῥητὴ ἄρα ἐστὶν ἡ $HΓ$. καὶ ἐκκείσθωσαν δύο τετράγωνοι ἀριθμοὶ οἱ $ΔΕ$, $ΕΖ$, ὧν ἡ ὑπεροβὴ ὁ $ΔΖ$ μὴ ἔστω τετράγωνος. καὶ πεποιήσθω ὡς ὁ $ΖΔ$ πρὸς τὸν $ΔΕ$, οὕτως τὸ ἀπὸ τῆς $ΓΗ$ τετράγωνον πρὸς τὸ ἀπὸ τῆς HB τετράγωνον. σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΓΗ$ τετράγωνον τῷ ἀπὸ τῆς HB τετραγώνῳ. ῥητὸν δὲ τὸ ἀπὸ τῆς $ΓΗ$. ῥητὸν ἄρα [ἐστὶ] καὶ τὸ ἀπὸ τῆς HB · ῥητὴ ἄρα ἐστὶν ἡ BH . καὶ ἐπεὶ τὸ ἀπὸ τῆς $HΓ$ τετράγωνον πρὸς τὸ ἀπὸ τῆς HB λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, ἀσύμμετρος ἐστὶν ἡ $ΓΗ$ τῇ HB μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ $ΓΗ$, HB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ $ΒΓ$ ἄρα ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ δευτέρα.

Proposition 86

To find a second apotome.

Let the rational (straight-line) A , and GC (which is) commensurable in length with A , be laid down. Thus, GC is a rational (straight-line). And let the two square numbers DE and EF be laid down, and let their difference DF be not square [Prop. 10.28 lem. I]. And let it be contrived that as FD (is) to DE , so the square on CG (is) to the square on GB [Prop. 10.6 corr.]. Thus, the square on CG is commensurable with the square on GB [Prop. 10.6]. And the (square) on CG (is) rational. Thus, the (square) on GB [is] also rational. Thus, BG is a rational (straight-line). And since the square on GC does not have to the (square) on GB the ratio which (some) square number (has) to (some) square number, CG is incommensurable in length with GB [Prop. 10.9]. And they are both rational (straight-lines). Thus, CG and GB are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that it is also a second (apotome).



ἵ γὰρ μείζον ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $HΓ$, ἔστω τὸ ἀπὸ τῆς $Θ$. ἐπεὶ οὖν ἐστὶν ὡς τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$, οὕτως ὁ $EΔ$ ἀριθμὸς πρὸς τὸν $ΔΖ$ ἀριθμόν, ἀναστρέψαντι ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $Θ$, οὕτως ὁ $ΔΕ$ πρὸς τὸν $EΖ$. καὶ ἐστὶν ἐκάτερος τῶν $ΔΕ$, $EΖ$ τετράγωνος· τὸ ἄρα ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα ἐστὶν ἡ BH τῇ $Θ$ μήκει. καὶ δύνатаι ἡ BH τῆς $HΓ$ μείζον τῷ ἀπὸ τῆς $Θ$ · ἡ BH ἄρα τῆς $HΓ$ μείζον δύνатаι τῷ ἀπὸ συμέτρου ἐναυτῇ μήκει. καὶ ἐστὶν ἡ προσαρμοζούσα ἡ $ΓΗ$ τῇ ἐκκειμένῃ ῥητῇ σύμμετρος τῇ A . ἡ $BΓ$ ἄρα ἀποτομή ἐστὶ δευτέρα.

Εὐρίηται ἄρα δευτέρα ἀποτομή ἡ $BΓ$. ὅπερ ἔδει δεῖξαι.

For let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.1 lem.]. Therefore, since as the (square) on BG is to the (square) on GC , so the number ED (is) to the number DF , thus, also, via conversion, as the (square) on BG is to the (square) on H , so DE (is) to EF [Prop. 5.19 corr.]. And DE and EF are each square (numbers). Thus, the (square) on BG has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, BG is commensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on BG is greater than (the square on) GC by the (square) on (some straight-line) commensurable in length with (BG). And the attachment CG is commensurable (in length) with the (previously) laid down rational (straight-line) A . Thus, BC is a second apotome [Def. 10.12].[†]

Thus, the second apotome BC has been found. (Which is) the very thing it was required to show.

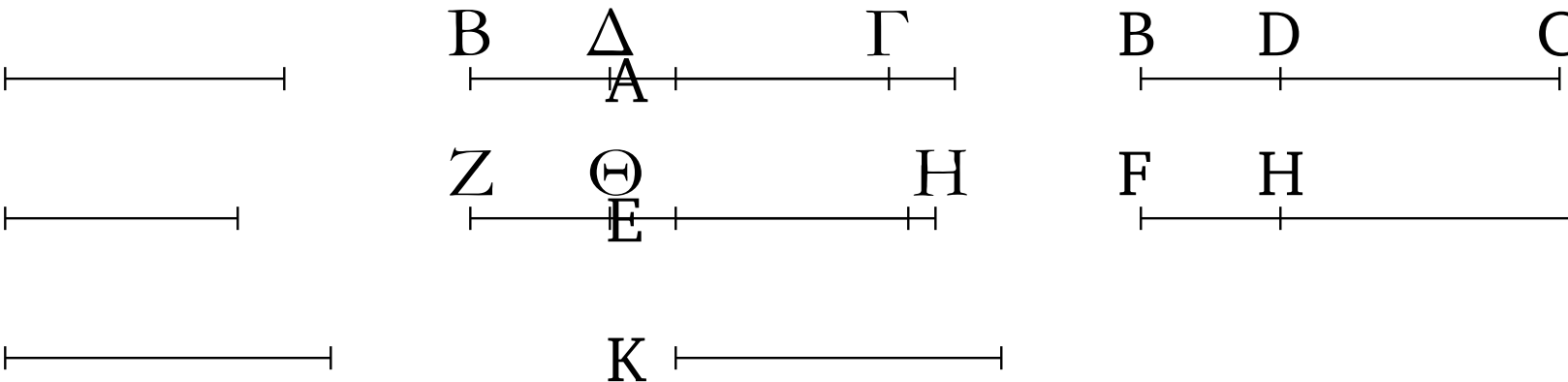
[†] See footnote to Prop. 10.49.

87.

Proposition 87

Εὐρεῖν τὴν τρίτην ἀποτομήν.

To find a third apotome.



Ἐκκείσθω ῥητὴ ἡ A , καὶ ἐκκείσθωσαν τρεῖς ἀριθμοὶ οἱ E , $BΓ$, $ΓΔ$ λόγον μὴ ἔχοντες πρὸς ἀλλήλους, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, ὃ δὲ $ΓΒ$ πρὸς τὸν $ΒΔ$ λόγον ἔθεται, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ πεποιήσθω ὡς μὲν ὁ E πρὸς τὸν $BΓ$, οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς ZH τετράγωνον, ὡς δὲ ὁ $BΓ$ πρὸς τὸν $ΓΔ$, οὕτως τὸ ἀπὸ τῆς ZH τετράγωνον πρὸς

Let the rational (straight-line) A be laid down. And let the three numbers, E , BC , and CD , not having to one another the ratio which (some) square number (has) to (some) square number, be laid down. And let CB have to BD the ratio which (some) square number (has) to (some) square number. And let it be contrived that as E (is) to BC , so the square on A (is) to the square on FG , and as BC (is) to CD , so the square on

τὸ ἀπὸ τῆς $H\Theta$. ἐπεὶ οὖν ἔστιν ὡς ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς ZH τετράγωνον, σύμμετρον ἄρα ἔστι τὸ ἀπὸ τῆς A τετράγωνον τῷ ἀπὸ τῆς ZH τετραγώνῳ. ῥητὸν δὲ τὸ ἀπὸ τῆς A τετράγωνον. ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς ZH . ῥητὴ ἄρα ἔστιν ἡ ZH . καὶ ἐπεὶ ὁ E πρὸς τὸν $B\Gamma$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ' ἄρα τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς ZH [τετράγωνον] λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἔστιν ἡ A τῇ ZH μήκει. πάλιν, ἐπεὶ ἔστιν ὡς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH τετράγωνον πρὸς τὸ ἀπὸ τῆς $H\Theta$, σύμμετρον ἄρα ἔστι τὸ ἀπὸ τῆς ZH τῷ ἀπὸ τῆς $H\Theta$. ῥητὸν δὲ τὸ ἀπὸ τῆς ZH . ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $H\Theta$. ῥητὴ ἄρα ἔστιν ἡ $H\Theta$. καὶ ἐπεὶ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ' ἄρα τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἔστιν ἡ ZH τῇ $H\Theta$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ZH , $H\Theta$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἔστιν ἡ $Z\Theta$. λέγω δὴ, ὅτι καὶ τρίτη.

Ἐπεὶ γάρ ἔστιν ὡς μὲν ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς ZH , ὡς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς ΘH , δι' ἴσου ἄρα ἔστιν ὡς ὁ E πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ΘH . ὁ δὲ E πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. οὐδ' ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἡ A τῇ $H\Theta$ μήκει. οὐδετέρα ἄρα τῶν ZH , $H\Theta$ σύμμετρος ἔστι τῇ ἐκκειμένῃ ῥητῇ τῇ A μήκει. ὥ οὖν μείζον ἔστι τὸ ἀπὸ τῆς ZH τοῦ ἀπὸ τῆς $H\Theta$, ἔστω τὸ ἀπὸ τῆς K . ἐπεὶ οὖν ἔστιν ὡς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, ἀναστρέψαντι ἄρα ἔστιν ὡς ὁ $B\Gamma$ πρὸς τὸν $B\Delta$, οὕτως τὸ ἀπὸ τῆς ZH τετράγωνον πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ $B\Gamma$ πρὸς τὸν $B\Delta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. καὶ τὸ ἀπὸ τῆς ZH ἄρα πρὸς τὸ ἀπὸ τῆς K λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. σύμμετρος ἄρα ἔστιν ἡ ZH τῇ K μήκει, καὶ δύναται ἡ ZH τῆς $H\Theta$ μείζον τῷ ἀπὸ συμμέτρων ἑαυτῇ. καὶ οὐδετέρα τῶν ZH , $H\Theta$ σύμμετρος ἔστι τῇ ἐκκειμένῃ ῥητῇ τῇ A μήκει· ἡ $Z\Theta$ ἄρα ἀποτομή ἔστι τρίτη.

Ἐύρηται ἄρα ἡ τρίτη ἀποτομή ἡ $Z\Theta$. ὅπερ ἔδει δεῖχαι.

FG (is) to the (square) on GH [Prop. 10.6 corr.]. Therefore, since as E is to BC , so the square on A (is) to the square on FG , the square on A is thus commensurable with the square on FG [Prop. 10.6]. And the square on A (is) rational. Thus, the (square) on FG (is) also rational. Thus, FG is a rational (straight-line). And since E does not have to BC the ratio which (some) square number (has) to (some) square number, the square on A thus does not have to the [square] on FG the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with FG [Prop. 10.9]. Again, since as BC is to CD , so the square on FG is to the (square) on GH , the square on FG is thus commensurable with the (square) on GH [Prop. 10.6]. And the (square) on FG (is) rational. Thus, the (square) on GH (is) also rational. Thus, GH is a rational (straight-line). And since BC does not have to CD the ratio which (some) square number (has) to (some) square number, the (square) on FG thus does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. And both are rational (straight-lines). FG and GH are thus rational (straight-lines which are) commensurable in square only. Thus, FH is an apotome [Prop. 10.73]. So, I say that (it is) also a third (apotome).

For since as E is to BC , so the square on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on HG , thus, via equality, as E is to CD , so the (square) on A (is) to the (square) on HG [Prop. 5.22]. And E does not have to CD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on A does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. A (is) thus incommensurable in length with GH [Prop. 10.9]. Thus, neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Therefore, let the (square) on K be that (area) by which the (square) on FG is greater than the (square) on GH [Prop. 10.13 lem.]. Therefore, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , thus, via conversion, as BC is to BD , so the square on FG (is) to the square on K [Prop. 5.19 corr.]. And BC has to BD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on K the ratio which (some) square number (has) to (some) square number. FG is thus commensurable in length with K [Prop. 10.9]. And the square on FG is (thus) greater than (the square on) GH by the (square) on (some straight-line) commensurable (in length) with (FG). And neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, FH is a third apotome [Def. 10.13].

Thus, the third apotome FH has been found. (Which is)

very thing it was required to show.

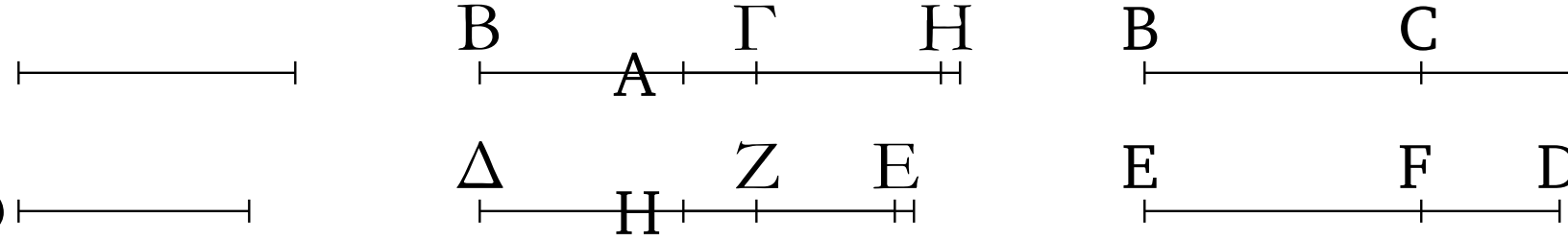
† See footnote to Prop. 10.50.

88.

Proposition 88

Εὐρεῖν τὴν τετάρτην ἀποτομήν.

To find a fourth apotome.



Ἐκκείσθω ῥητὴ ἡ A καὶ τῇ A μήκει σύμμετρος ἡ BH . ῥητὴ ἄρα ἐστὶ καὶ ἡ BH . καὶ ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΔZ , ZE , ὥστε τὸν ΔE ὅλον πρὸς ἑκάτερον τῶν ΔZ , EZ λόγον μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ πεποιήσθω ὥς ὁ ΔE πρὸς τὸν EZ , οὕτως τὸ ἀπὸ τῆς BH τετράγωνον πρὸς τὸ ἀπὸ τῆς $HΓ$. σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς BH τῷ ἀπὸ τῆς $HΓ$. ῥητὸν δὲ τὸ ἀπὸ τῆς BH . ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $HΓ$. ῥητὴ ἄρα ἐστὶν ἡ $HΓ$. καὶ ἐπεὶ ὁ ΔE πρὸς τὸν EZ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' ἄρα τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ BH τῇ $HΓ$ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ BH , $HΓ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἀποτομὴ ἄρα ἐστὶν ἡ $BΓ$. [λέγω δὴ, ὅτι καὶ τετάρτη.]

Ἴ οὖν μείζον ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $HΓ$, ἔστω τὸ ἀπὸ τῆς Θ . ἐπεὶ οὖν ἐστὶν ὥς ὁ ΔE πρὸς τὸν EZ , οὕτως τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$, καὶ ἀναστρέψαντι ἄρα ἐστὶν ὥς ὁ $E\Delta$ πρὸς τὸν ΔZ , οὕτως τὸ ἀπὸ τῆς HB πρὸς τὸ ἀπὸ τῆς Θ . ὁ δὲ $E\Delta$ πρὸς τὸν ΔZ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. οὐδ' ἄρα τὸ ἀπὸ τῆς HB πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ BH τῇ Θ μήκει. καὶ δύναται ἡ BH τῇ $HΓ$ μείζον τῷ ἀπὸ τῆς Θ . ἡ ἄρα BH τῇ $HΓ$ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἐαντῇ. καὶ ἐστὶν ὅλη ἡ BH σύμμετρος τῇ ἐκκεκμένη ῥητῇ μήκει τῇ A . ἡ ἄρα $BΓ$ ἀποτομὴ ἐστὶ τετάρτη.

Εὐρηται ἄρα ἡ τετάρτη ἀποτομή· ὅπερ ἔδει δεῖξαι.

Let the rational (straight-line) A , and BG (which is) commensurable in length with A , be laid down. Thus, BG is also a rational (straight-line). And let the two numbers DF and FE be laid down such that the whole, DE , does not have to each of DF and EF the ratio which (some) square number (has) to (some) square number. And let it be contrived that as DE (is) to EF , so the square on BG (is) to the (square) on GC [Prop. 10.6 corr.]. The (square) on BG is thus commensurable with the (square) on GC [Prop. 10.6]. And the (square) on BG (is) rational. Thus, the (square) on GC (is) also rational. Thus, GC (is) a rational (straight-line). And since DE does not have to EF the ratio which (some) square number (has) to (some) square number, the (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). Thus, BG and GC are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. [So, I say that (it is) also a fourth (apotome).]

Now, let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.]. Therefore, since as DE is to EF , so the (square) on BG (is) to the (square) on GC , thus, also, via conversion, as ED is to DF , so the (square) on GB (is) to the (square) on H [Prop. 5.19 corr.]. And ED does not have to DF the ratio which (some) square number (has) to (some) square number. Thus, the (square) on GB does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on BG is greater than (the square) on GC by the (square) on (some straight-line) incommensurable (in length) with (BG). And the whole, BG , is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is

a fourth apotome [Def. 10.14].[†]

Thus, a fourth apotome has been found. (Which is) the very thing it was required to show.

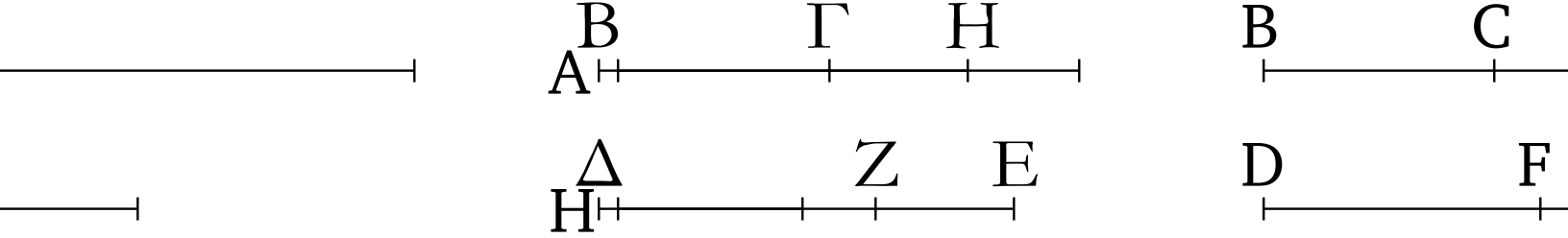
[†] See footnote to Prop. 10.51.

89.

Proposition 89

Εὕρεῖν τὴν πέμπτην ἀποτομήν.

To find a fifth apotome.



Ἐκκείσθω ῥητὴ ἡ A , καὶ τῇ A μήκει σύμμετρος ἔστω ἡ $ΓΗ$. ῥητὴ ἄρα [ἐστίν] ἡ $ΓΗ$. καὶ ἐκκείσθωσαν δύο ἀριθμοὶ οἱ $ΔΖ$, $ΖΕ$, ὥστε τὸν $ΔΕ$ πρὸς ἑκάτερον τῶν $ΔΖ$, $ΖΕ$ λόγον πάλιν μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ πεποιήσθω ὥς ὁ $ΖΕ$ πρὸς τὸν $ΕΔ$, οὕτως τὸ ἀπὸ τῆς $ΓΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΒ$. ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $ΗΒ$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΒΗ$. καὶ ἐπεὶ ἐστὶν ὥς ὁ $ΔΕ$ πρὸς τὸν $ΕΖ$, οὕτως τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΓ$, ὁ δὲ $ΔΕ$ πρὸς τὸν $ΕΖ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' ἄρα τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΓ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ $ΒΗ$ τῇ $ΗΓ$ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ $ΒΗ$, $ΗΓ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ $ΒΓ$ ἄρα ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ πέμπτη.

Ἦ γὰρ μείζον ἐστὶ τὸ ἀπὸ τῆς $ΒΗ$ τοῦ ἀπὸ τῆς $ΗΓ$, ἔστω τὸ ἀπὸ τῆς $Θ$. ἐπεὶ οὖν ἐστὶν ὥς τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΓ$, οὕτως ὁ $ΔΕ$ πρὸς τὸν $ΕΖ$, ἀναστρέψαντι ἄρα ἐστὶν ὥς ὁ $ΕΔ$ πρὸς τὸν $ΔΖ$, οὕτως τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $Θ$, ὁ δὲ $ΕΔ$ πρὸς τὸν $ΔΖ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδ' ἄρα τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ $ΒΗ$ τῇ $Θ$ μήκει. καὶ δύναται ἡ $ΒΗ$ τῆς $ΗΓ$ μείζον τῷ ἀπὸ τῆς $Θ$ · ἡ $ΗΒ$ ἄρα τῆς $ΗΓ$ μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἐαντὶ μήκει. καὶ ἐστὶν ἡ προσαρμόζουσα ἡ $ΓΗ$ σύμμετρος τῇ ἐκκεκλιμένη ῥητὶ τῇ A μήκει· ἡ ἄρα $ΒΓ$ ἀποτομή ἐστὶ πέμπτη.

Εὐρίηται ἄρα ἡ πέμπτη ἀποτομή ἡ $ΒΓ$. ὅπερ ἔδει δεῖχαι.

Let the rational (straight-line) A be laid down, and let CG be commensurable in length with A . Thus, CG [is] a rational (straight-line). And let the two numbers DF and FE be laid down such that DE again does not have to each of DF and FE the ratio which (some) square number (has) to (some) square number. And let it be contrived that as FE (is) to ED , so the (square) on CG (is) to the (square) on GB . Thus, the (square) on GB (is) also rational [Prop. 10.6]. Thus, BG is also rational. And since as DE is to EF , so the (square) on BG (is) to the (square) on GC . And DE does not have to EF the ratio which (some) square number (has) to (some) square number. The (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). BG and GC are thus rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that (it is) also a fifth (apotome).

For, let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.]. Therefore, since as the (square) on BG (is) to the (square) on GC , so DE (is) to EF , thus, via conversion, as ED is to DF , so the (square) on BG (is) to the (square) on H [Prop. 5.19 corr.]. And ED does not have to DF the ratio which (some) square number (has) to (some) square number. Thus, the (square) on BG does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on GB is greater than (the square on) GC by the (square) on (some straight-line) incommensurable in length with (GB). And the attachment CG is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is a fifth

apotome [Def. 10.15].[†]

Thus, the fifth apotome BC has been found. (Which is) the very thing it was required to show.

[†] See footnote to Prop. 10.52.

90.

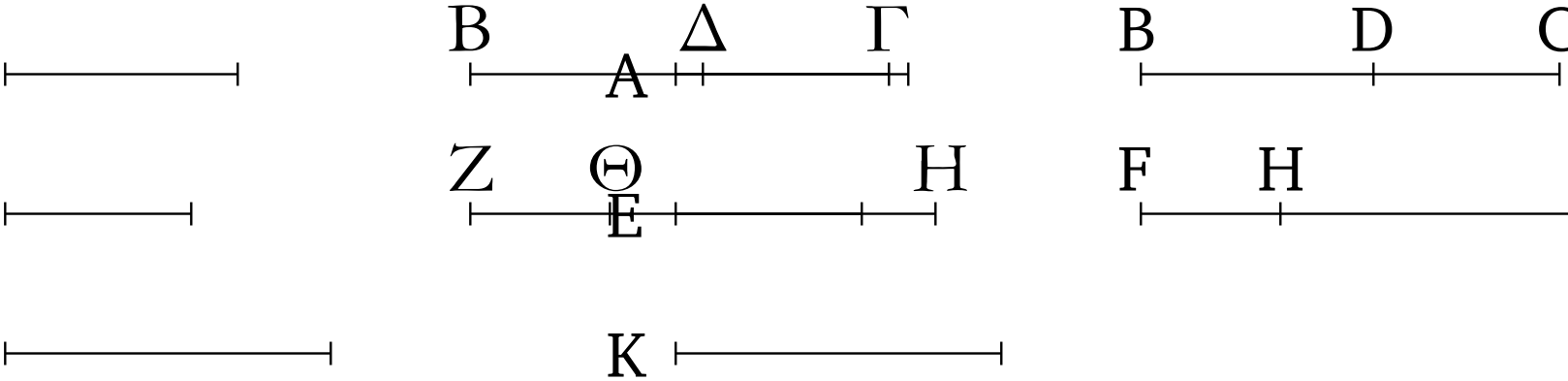
Εὐρεῖν τὴν ἕκτην ἀποτομήν.

Ἐκκείσθω ῥητὴ ἡ A καὶ τρεῖς ἀριθμοὶ οἱ E , $B\Gamma$, $\Gamma\Delta$ λόγον μὴ ἔθοντες πρὸς ἀλλήλους, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ἔτι δὲ καὶ ὁ ΓB πρὸς τὸν $B\Delta$ λόγον μὴ ἔθετώ, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν καὶ πεποιήσθω ὥς μὲν ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH , ὥς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$.

Proposition 90

To find a sixth apotome.

Let the rational (straight-line) A , and the three numbers E , BC , and CD , not having to one another the ratio which (some) square number (has) to (some) square number, be laid down. Furthermore, let CB also not have to BD the ratio which (some) square number (has) to (some) square number. And let it be contrived that as E (is) to BC , so the (square) on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.].



Ἐπεὶ οὖν ἔστιν ὥς ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH , σύμμετρον ἄρα τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς ZH . ῥητὸν δὲ τὸ ἀπὸ τῆς A ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς ZH . ῥητὴ ἄρα ἐστὶ καὶ ἡ ZH . καὶ ἐπεὶ ὁ E πρὸς τὸν $B\Gamma$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ' ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ ZH μήκει. πάλιν, ἐπεὶ ἔστιν ὥς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, σύμμετρον ἄρα τὸ ἀπὸ τῆς ZH τῷ ἀπὸ τῆς $H\Theta$. ῥητὸν δὲ τὸ ἀπὸ τῆς ZH . ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $H\Theta$. ῥητὴ ἄρα καὶ ἡ $H\Theta$. καὶ ἐπεὶ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ' ἄρα τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἐστὶν ἡ ZH τῇ $H\Theta$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ZH , $H\Theta$ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ἄρα $Z\Theta$ ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ ἕκτη.

Ἐπεὶ γάρ ἐστιν ὥς μὲν ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH , ὥς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, δι' ἴσον ἄρα ἐστὶν ὥς ὁ E πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $H\Theta$. ὁ δὲ E πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος

Therefore, since as E is to BC , so the (square) on A (is) to the (square) on FG , the (square) on A (is) thus commensurable with the (square) on FG [Prop. 10.6]. And the (square) on A (is) rational. Thus, the (square) on FG (is) also rational. Thus, FG is also a rational (straight-line). And since E does not have to BC the ratio which (some) square number (has) to (some) square number, the (square) on A thus does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with FG [Prop. 10.9]. Again, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus commensurable with the (square) on GH [Prop. 10.6]. And the (square) on FG (is) rational. Thus, the (square) on GH (is) also rational. Thus, GH (is) also rational. And since BC does not have to CD the ratio which (some) square number (has) to (some) square number, the (square) on FG thus does not have to the (square) on GH the ratio which (some) square (number) has to (some) square (number) either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. And both are rational (straight-lines). Thus, FG and GH are rational (straight-lines which are) commensurable in square only. Thus, FH is an apotome [Prop. 10.73]. So, I say that (it is) also a sixth (apotome).

ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν οὐδ' ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ $H\Theta$ μήκει· οὐδετέρα ἄρα τῶν ZH , $H\Theta$ σύμμετρος ἐστὶ τῇ A ῥητῇ μήκει. ὥ οὖν μείζον ἐστὶ τὸ ἀπὸ τῆς ZH τοῦ ἀπὸ τῆς $H\Theta$, ἔστω τὸ ἀπὸ τῆς K . ἐπεὶ οὖν ἐστὶν ὡς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ ΓB πρὸς τὸν $B\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ ΓB πρὸς τὸν $B\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν οὐδ' ἄρα τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς K λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ἀσύμμετρος ἄρα ἐστὶν ἡ ZH τῇ K μήκει. καὶ δύναιται ἡ ZH τῆς $H\Theta$ μείζον τῷ ἀπὸ τῆς K · ἡ ZH ἄρα τῆς $H\Theta$ μείζον δύναιται τῷ ἀπὸ ἀσυνμέτρου ἐαυτῇ μήκει. καὶ οὐδετέρα τῶν ZH , $H\Theta$ σύμμετρος ἐστὶ τῇ $\epsilon\kappa\kappa\epsilon\mu\epsilon\acute{\nu}\eta\eta$ ῥητῇ μήκει τῇ A . ἡ ἄρα $Z\Theta$ ἀποτομή ἐστὶν ἑκτῇ.

E ῥηται ἄρα ἡ ἑκτῇ ἀποτομή ἡ $Z\Theta$ · ὅπερ ἔδει δεῖξαι.

For since as E is to BC , so the (square) on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on GH , thus, via equality, as E is to CD , so the (square) on A (is) to the (square) on GH [Prop. 5.22]. And E does not have to CD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on A does not have to the (square) GH the ratio which (some) square number (has) to (some) square number either. A is thus incommensurable in length with GH [Prop. 10.9]. Thus, neither of FG and GH is commensurable in length with the rational (straight-line) A . Therefore, let the (square) on K be that (area) by which the (square) on FG is greater than the (square) on GH [Prop. 10.13 lem.]. Therefore, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , thus, via conversion, as CB is to BD , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And CB does not have to BD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG does not have to the (square) on K the ratio which (some) square number (has) to (some) square number either. FG is thus incommensurable in length with K [Prop. 10.9]. And the square on FG is greater than (the square on) GH by the (square) on K . Thus, the square on FG is greater than (the square on) GH by the (square) on (some straight-line) incommensurable in length with (FG). And neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, FH is a sixth apotome [Def. 10.16].

Thus, the sixth apotome FH has been found. (Which is) the very thing it was required to show.

† See footnote to Prop. 10.53.

91.

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς πρώτης, ἡ τὸ θωρίον δυναμένη ἀπορομή ἐστὶν.

Περιεθέσθω γὰρ θωρίον τὸ AB ὑπὸ ῥητῆς τῆς AG καὶ ἀποτομῆς πρώτης τῆς AD · λέγω, ὅτι ἡ τὸ AB θωρίον δυναμένη ἀποτομή ἐστὶν.

Ἐπεὶ γὰρ ἀποτομή ἐστὶ πρώτη ἡ AD , ἔστω αὐτῇ προσαρμόζουσα ἡ ΔH · αἱ AH , $H\Delta$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ ὅλη ἡ AH σύμμετρος ἐστὶ τῇ ἑκκεμμένῃ ῥητῇ τῇ AG , καὶ ἡ AH τῆς $H\Delta$ μείζον δύναιται τῷ ἀπὸ συμέτρου ἐαυτῇ μήκει· ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔH ἴσον παρὰ τὴν AH παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, εἷς σύμμετρα αὐτῇ διαιρεῖ. τετμήσθω ἡ ΔH δίῃα κατὰ τὸ E , καὶ τῷ ἀπὸ τῆς EH ἴσον παρὰ τὴν AH παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν AZ , ZH · σύμμετρος ἄρα ἐστὶν ἡ AZ τῇ ZH . καὶ διὰ τῶν E , Z , H σημείων τῇ AG παράλληλοι ἡθῇσαν αἱ $E\Theta$, ZI , HK .

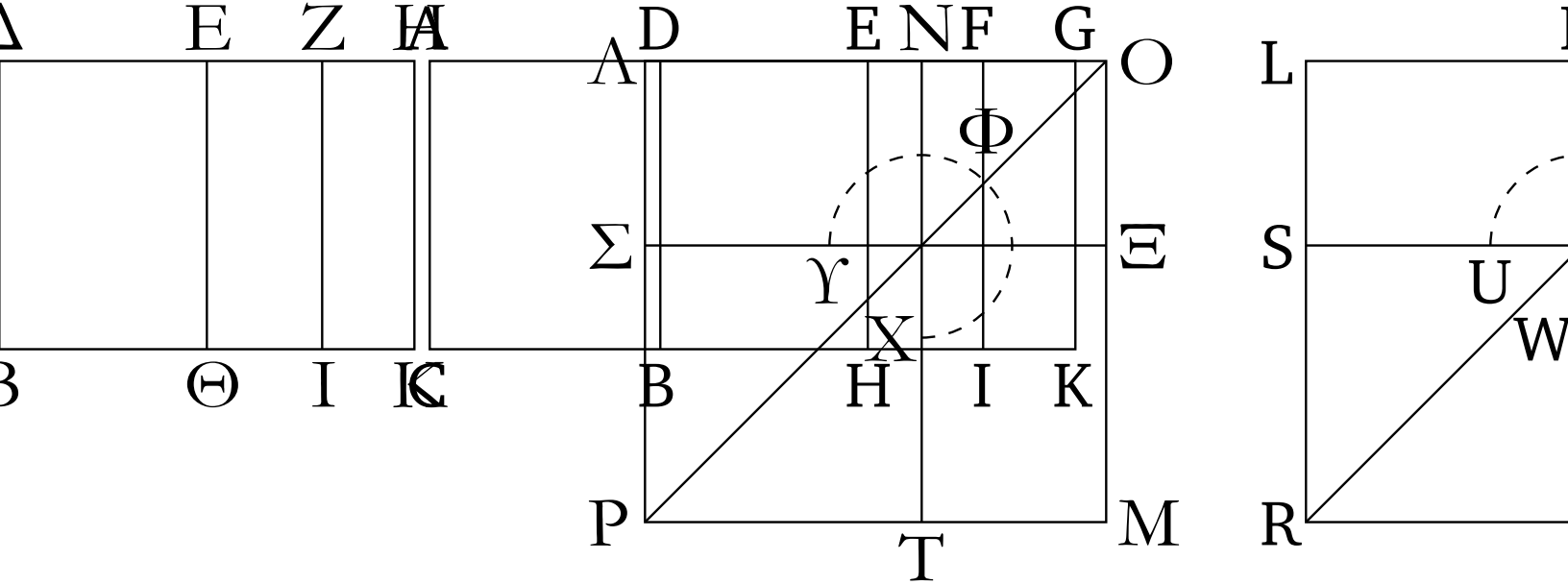
Proposition 91

If an area is contained by a rational (straight-line) and a first apotome then the square-root of the area is an apotome.

For let the area AB be contained by the rational (straight-line) AC and the first apotome AD . I say that the square-root of area AB is an apotome.

For since AD is a first apotome, let DG be its attachment. Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And the whole, AG , is commensurable (in length) with the (previously) laid down rational (straight-line) AC , and the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable in length with (AG) [Def. 10.11]. Thus, if (an area) equal to the fourth part of the (square) on DG is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short

by a square figure. And let it be the (rectangle contained) by AF and FG . AF is thus commensurable (in length) with FG . And let EH , FI , and GK be drawn through points E , F , and G (respectively), parallel to AC .



Καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ AZ τῇ ZH μήκει, καὶ ἡ AH ἄρα ἐκατέρω τῶν AZ , ZH σύμμετρος ἐστὶ μήκει. ἀλλὰ ἡ AH σύμμετρος ἐστὶ τῇ AG · καὶ ἐκατέρα ἄρα τῶν AZ , ZH σύμμετρος ἐστὶ τῇ AG μήκει. καὶ ἐστὶ ῥητὴ ἡ AG · ῥητὴ ἄρα καὶ ἐκατέρα τῶν AZ , ZH · ὥστε καὶ ἐκάτερον τῶν AI , ZK ῥητόν ἐστιν. καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ DE τῇ EH μήκει, καὶ ἡ DH ἄρα ἐκατέρω τῶν DE , EH σύμμετρος ἐστὶ μήκει. ῥητὴ δὲ ἡ DH καὶ ἀσύμμετρος τῇ AG μήκει· ῥητὴ ἄρα καὶ ἐκατέρα τῶν DE , EH καὶ ἀσύμμετρος τῇ AG μήκει· ἐκάτερον ἄρα τῶν $\Delta\Theta$, EK μέσον ἐστίν.

Κεῖσθω δὴ τῷ μὲν AI ἴσον τετράγωνον τὸ ΛM , τῷ δὲ ZK ἴσον τετράγωνον ἀφηρήσθω κοινὴν γωνίαν ἔθον αὐτῶ τὴν ὑπὸ ΛOM τὸ NX · περὶ τὴν αὐτὴν ἄρα διάμετρον ἐστὶ τὰ ΛM , NX τετράγωνα. ἔστω αὐτῶν διάμετρος ἡ OP , καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ οὖν ἴσον ἐστὶ τὸ ὑπὸ τῶν AZ , ZH περιεχόμενον ὀρθογώνιον τῷ ἀπὸ τῆς EH τετραγώνῳ, ἔστιν ἄρα ὥς ἡ AZ πρὸς τὴν EH , οὕτως ἡ EH πρὸς τὴν ZH . ἀλλ' ὥς μὲν ἡ AZ πρὸς τὴν EH , οὕτως τὸ AI πρὸς τὸ EK , ὥς δὲ ἡ EH πρὸς τὴν ZH , οὕτως ἐστὶ τὸ EK πρὸς τὸ KZ · τῶν ἄρα AI , KZ μέσον ἀνάλογόν ἐστι τὸ EK . ἔστι δὲ καὶ τῶν ΛM , NX μέσον ἀνάλογον τὸ MN , ὥς ἐν τοῖς ἔμπροσθεν ἐδείχθη, καὶ ἐστὶ τὸ [μὲν] AI τῷ ΛM τετραγώνῳ ἴσον, τὸ δὲ KZ τῷ NX · καὶ τὸ MN ἄρα τῷ EK ἴσον ἐστίν. ἀλλὰ τὸ μὲν EK τῷ $\Delta\Theta$ ἐστὶν ἴσον, τὸ δὲ MN τῷ AX · τὸ ἄρα ΔK ἴσον ἐστὶ τῷ $Y\Phi\Theta$ γνώμονι καὶ τῷ NX . ἔστι δὲ καὶ τὸ AK ἴσον τοῖς ΛM , NX τετραγώνοις· λοιπὸν ἄρα τὸ AB ἴσον ἐστὶ τῷ ΣT . τὸ δὲ ΣT τὸ ἀπὸ τῆς ΛN ἐστὶ τετράγωνον· τὸ ἄρα ἀπὸ

And since AF is commensurable in length with FG , AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. But AG is commensurable (in length) with AC . Thus, each of AF and FG is also commensurable in length with AC [Prop. 10.12]. And AC is a rational (straight-line). Thus, AF and FG (are) each also rational (straight-lines). Hence, AI and FK are also each rational (areas) [Prop. 10.19]. And since DE is commensurable in length with EG , DG is thus also commensurable in length with each of DE and EG [Prop. 10.15]. And DG (is) rational, and incommensurable in length with AC . DE and EG (are) thus each rational, and incommensurable in length with AC [Prop. 10.13]. Thus, DH and EK are each medial (areas) [Prop. 10.21].

So let the square LM , equal to AI , be laid down. And let the square NO , equal to FK , be subtracted (from LM), having with it the common angle LPM . Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the rectangle contained by AF and FG is equal to the square EG , thus as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG , so AI (is) to EK , and as EG (is) to FG , so EK is to KF [Prop. 6.1]. Thus, EK is the mean proportional to AI and KF [Prop. 5.11]. And MN is also the mean proportional to LM and NO , as shown before [Prop. 10.53 lem.]. And AI is equal to the square LM , and KF to NO . Thus, MN is also equal to EK . But, EK is equal to DH , and MN to LO [Prop. 1.43]. Thus, DK is equal to the

τῆς AN τετραγώνων ἴσον ἐστὶ τῷ AB . ἡ AN ἄρα δύναται τὸ AB . λέγω δὴ, ὅτι ἡ AN ἀποτομή ἐστίν.

Ἐπεὶ γὰρ ῥητόν ἐστιν ἑκάτερον τῶν AI , ZK , καὶ ἐστὶν ἴσον τοῖς AM , NX , καὶ ἑκάτερον ἄρα τῶν AM , NX ῥητόν ἐστιν, τουτέστι τὸ ἀπὸ ἑκατέρωξ τῶν AO , ON . καὶ ἑκατέρα ἄρα τῶν AO , ON ῥητὴ ἐστίν. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ $\Delta\Theta$ καὶ ἐστὶν ἴσον τῷ AX , μέσον ἄρα ἐστὶ καὶ τὸ AX . ἐπεὶ οὖν τὸ μὲν AX μέσον ἐστίν, τὸ δὲ NX ῥητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ AX τῷ NX . ὥς δὲ τὸ AX πρὸς τὸ NX , οὕτως ἐστὶν ἡ AO πρὸς τὴν ON . ἀσύμμετρος ἄρα ἐστὶν ἡ AO τῇ ON μήκει. καὶ εἰσὶν ἀμφοτέρω ῥηταί. αἱ AO , ON ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι. ἀποτομή ἄρα ἐστὶν ἡ AN . καὶ δύναται τὸ AB θωρίον ἡ ἄρα τὸ AB θωρίον δυνάμενη ἀποτομή ἐστίν.

Ἐὰν ἄρα θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τὰ ἐχθῆς.

gnomon UVW and NO . And AK is also equal to (the sum of) the squares LM and NO . Thus, the remainder AB is equal to ST . And ST is the square on LN . Thus, the square on LN is equal to AB . Thus, LN is the square-root of AB . So, I say that LN is an apotome.

For since AI and FK are each rational (areas), and are equal to LM and NO (respectively), thus LM and NO —that is to say, the (squares) on each of LP and PN (respectively)—are also each rational (areas). Thus, LP and PN are also each rational (straight-lines). Again, since DH is a medial (area), and is equal to LO , LO is thus also a medial (area). Therefore, since LO is medial, and NO rational, LO is thus incommensurable with NO . And as LO (is) to NO , so LP is to PN [Prop. 6.1]. LP is thus incommensurable in length with PN [Prop. 10.11]. And they are both rational (straight-lines). Thus, LP and PN are rational (straight-lines which are) commensurable in square only. Thus, LN is an apotome [Prop. 10.73]. And it is the square-root of area AB . Thus, the square-root of area AB is an apotome.

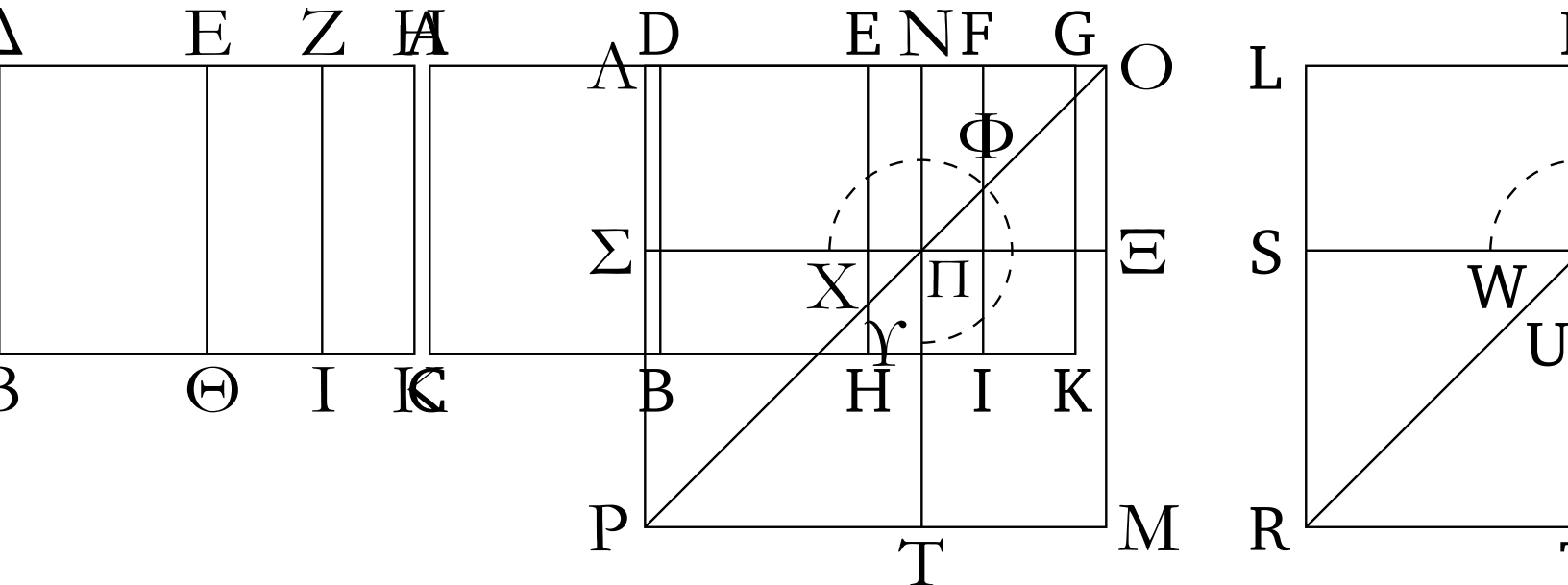
Thus, if an area is contained by a rational (straight-line), and so on . . .

92.

Proposition 92

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς δευτέρας, ἡ τὸ θωρίον δυνάμενη μέση ἀποτομή ἐστὶ πρώτη.

If an area is contained by a rational (straight-line) and a second apotome then the square-root of the area is a first apotome of a medial (straight-line).



Θωρίον γὰρ τὸ AB περιέθεσθαι ὑπὸ ῥητῆς τῆς AG καὶ ἀποτομῆς δευτέρας τῆς AD . λέγω, ὅτι ἡ τὸ AB θωρίον δυνάμενη μέση ἀποτομή ἐστὶ πρώτη.

Ἐστω γὰρ τῇ AD προσαρμόζουσα ἡ ΔH . αἱ ἄρα AH , $H\Delta$ ῥηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ προσαρμόζουσα

For let the area AB be contained by the rational (straight-line) AC and the second apotome AD . I say that the square-root of area AB is the first apotome of a medial (straight-line).

For let DG be an attachment to AD . Thus, AG and GD are rational (straight-lines which are) commensurable in

ἡ ΔH σύμμετρος ἐστὶ τῇ ἐκκεκμένη ῥητῇ τῇ AG , ἡ δὲ ὅλη ἡ AH τῇς προσαρμοζούσης τῇς $H\Delta$ μείζον δύναιται τῷ ἀπὸ συμμετρὸν ἑαυτῇ μήκει. ἐπεὶ οὖν ἡ AH τῇς $H\Delta$ μείζον δύναιται τῷ ἀπὸ συμμετρὸν ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῇς $H\Delta$ ἴσον παρὰ τὴν AH παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, εἷς σύμμετρος αὐτὴν διαιρεῖ. τετμήσθω οὖν ἡ ΔH διὰ κατὰ τὸ E . καὶ τῷ ἀπὸ τῇς EH ἴσον παρὰ τὴν AH παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν AZ , ZH . σύμμετρος ἄρα ἐστὶν ἡ AZ τῇ ZH μήκει. καὶ ἡ AH ἄρα ἑκατέρᾳ τῶν AZ , ZH σύμμετρος ἐστὶ μήκει. ῥητὴ δὲ ἡ AH καὶ ἀσύμμετρος τῇ AG μήκει· καὶ ἑκατέρᾳ ἄρα τῶν AZ , ZH ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ AG μήκει· ἑκάτερον ἄρα τῶν AI , ZK μέσον ἐστὶν. πάλιν, ἐπεὶ σύμμετρος ἐστὶν ἡ ΔE τῇ EH , καὶ ἡ ΔH ἄρα ἑκατέρᾳ τῶν ΔE , EH σύμμετρος ἐστὶν. ἀλλ' ἡ ΔH σύμμετρος ἐστὶ τῇ AG μήκει [ῥητὴ ἄρα καὶ ἑκατέρᾳ τῶν ΔE , EH καὶ σύμμετρος τῇ AG μήκει]. ἑκάτερον ἄρα τῶν $\Delta\Theta$, EK ῥητὸν ἐστὶν.

Συνεστάτω οὖν τῷ μὲν AI ἴσον τετραγώνον τὸ AM , τῷ δὲ ZK ἴσον ἀφῆρησθω τὸ NX περὶ τὴν αὐτὴν γωνίαν ὅν τῷ AM τὴν ὑπὸ τῶν $ΛΟΜ$. περὶ τὴν αὐτὴν ἄρα ἐστὶ διάμετρον τὰ AM , NX τετραγώνων. ἔστω αὐτῶν διάμετρος ἡ OP , καὶ καταγεγραμθῇ τὸ σθῆμα. ἐπεὶ οὖν τὰ AI , ZK μέσα ἐστὶ καὶ ἐστὶν ἴσα τοῖς ἀπὸ τῶν $ΛΟ$, $ΟΝ$, καὶ τὰ ἀπὸ τῶν $ΛΟ$, $ΟΝ$ [ἄρα] μέσα ἐστὶν· καὶ αἱ $ΛΟ$, $ΟΝ$ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ τὸ ὑπὸ τῶν AZ , ZH ἴσον ἐστὶ τῷ ἀπὸ τῇς EH , ἔστιν ἄρα ὡς ἡ AZ πρὸς τὴν EH , οὕτως ἡ EH πρὸς τὴν ZH . ἀλλ' ὡς μὲν ἡ AZ πρὸς τὴν EH , οὕτως [ἐστὶ] τὸ AI πρὸς τὸ EK . ὡς δὲ ἡ EH πρὸς τὴν ZH , οὕτως [ἐστὶ] τὸ EK πρὸς τὸ ZK . τῶν ἄρα AI , ZK μέσον ἀνάλογόν ἐστι τὸ EK . ἔστι δὲ καὶ τῶν AM , NX τετραγώνων μέσον ἀνάλογον τὸ MN . καὶ ἐστὶν ἴσον τὸ μὲν AI τῷ AM , τὸ δὲ ZK τῷ NX . καὶ τὸ MN ἄρα ἴσον ἐστὶ τῷ EK . ἀλλὰ τῷ μὲν EK ἴσον [ἐστὶ] τὸ $\Delta\Theta$, τῷ δὲ MN ἴσον τὸ AX . ὅλον ἄρα τὸ ΔK ἴσον ἐστὶ τῷ $Y\Phi\Theta$ γνώμονι καὶ τῷ NX . ἐπεὶ οὖν ὅλον τὸ AK ἴσον ἐστὶ τοῖς AM , NX , ὣν τὸ ΔK ἴσον ἐστὶ τῷ $Y\Phi\Theta$ γνώμονι καὶ τῷ NX , λοιπὸν ἄρα τὸ AB ἴσον ἐστὶ τῷ $T\Sigma$. τὸ δὲ $T\Sigma$ ἐστὶ τὸ ἀπὸ τῇς AN . τὸ ἀπὸ τῇς AN ἄρα ἴσον ἐστὶ τῷ AB θωρίῳ· ἡ AN ἄρα δύναιται τὸ AB θωρίον. λέγω [δή], ὅτι ἡ AN μέσης ἀποτομῆς ἐστὶ πρώτη.

Ἐπεὶ γὰρ ῥητὸν ἐστὶ τὸ EK καὶ ἐστὶν ἴσον τῷ AX , ῥητὸν ἄρα ἐστὶ τὸ AX , τουτέστι τὸ ὑπὸ τῶν $ΛΟ$, $ΟΝ$. μέσον δὲ εἰδείθῃ τὸ NX . ἀσύμμετρον ἄρα ἐστὶ τὸ AX τῷ NX . ὡς δὲ τὸ AX πρὸς τὸ NX , οὕτως ἐστὶν ἡ $ΛΟ$ πρὸς $ΟΝ$. αἱ $ΛΟ$, $ΟΝ$ ἄρα ἀσύμμετροί εἰσι μήκει. αἱ ἄρα $ΛΟ$, $ΟΝ$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι· ἡ AN ἄρα μέσης ἀποτομῆς ἐστὶ πρώτη καὶ δύναιται τὸ AB θωρίον.

Ἡ ἄρα τὸ AB θωρίον δυναμένη μέσης ἀποτομῆς ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

square only [Prop. 10.73], and the attachment DG is commensurable (in length) with the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, GD , by the (square) on (some straight-line) commensurable in length with (AG) [Def. 10.12]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable (in length) with (AG), thus if (an area) equal to the fourth part of the (square) on GD is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Therefore, let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short by a square figure. And let it be the (rectangle contained) by AF and FG . Thus, AF is commensurable in length with FG . AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. And AG (is) a rational (straight-line), and incommensurable in length with AC . AF and FG are thus also each rational (straight-lines), and incommensurable in length with AC [Prop. 10.13]. Thus, AI and FK are each medial (areas) [Prop. 10.21]. Again, since DE is commensurable (in length) with EG , thus DG is also commensurable (in length) with each of DE and EG [Prop. 10.15]. But, DG is commensurable in length with AC [thus, DE and EG are also each rational, and commensurable in length with AC]. Thus, DH and EK are each rational (areas) [Prop. 10.19].

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , which is about the same angle LPM as LM , be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since AI and FK are medial (areas), and are equal to the (squares) on LP and PN (respectively), [thus] the (squares) on LP and PN are also medial. Thus, LP and PN are also medial (straight-lines which are) commensurable in square only.[†] And since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus as AF is to EG , so EG (is) to FG [Prop. 10.17]. But, as AF (is) to EG , so AI (is) to EK . And as EG (is) to FG , so EK [is] to FK [Prop. 6.1]. Thus, EK is the mean proportional to AI and FK [Prop. 5.11]. And MN is also the mean proportional to the squares LM and NO [Prop. 10.53 lem.]. And AI is equal to LM , and FK to NO . Thus, MN is also equal to EK . But, DH [is] equal to EK , and LO equal to MN [Prop. 1.43]. Thus, the whole (of) DK is equal to the gnomon UVW and NO . Therefore, since the whole (of) AK is equal to LM and NO , of which DK is equal to the gnomon UVW and NO , the remainder AB is thus equal to TS . And TS is the (square) on LN . Thus, the (square) on LN is equal to the area AB . LN is thus the square-root of area AB . [So], I say that LN is the first apotome of a medial (straight-line).

For since EK is a rational (area), and is equal to LO , LO —that is to say, the (rectangle contained) by LP and PN —is thus a rational (area). And NO was shown (to be) a medial (area). Thus, LO is incommensurable with NO . And as LO (is) to NO , so LP is to PN [Prop. 6.1]. Thus, LP and PN are incommensurable in length [Prop. 10.11]. LP and PN are thus medial (straight-lines which are) commensurable in square only, and which contain a rational (area). Thus, LN is the first apotome of a medial (straight-line) [Prop. 10.74]. And it is the square-root of area AB .

Thus, the square root of area AB is the first apotome of a medial (straight-line). (Which is) the very thing it was required to show.

† There is an error in the argument here. It should just say that LP and PN are commensurable in square, rather than in square only, since LP and PN are only shown to be incommensurable in length later on.

93.

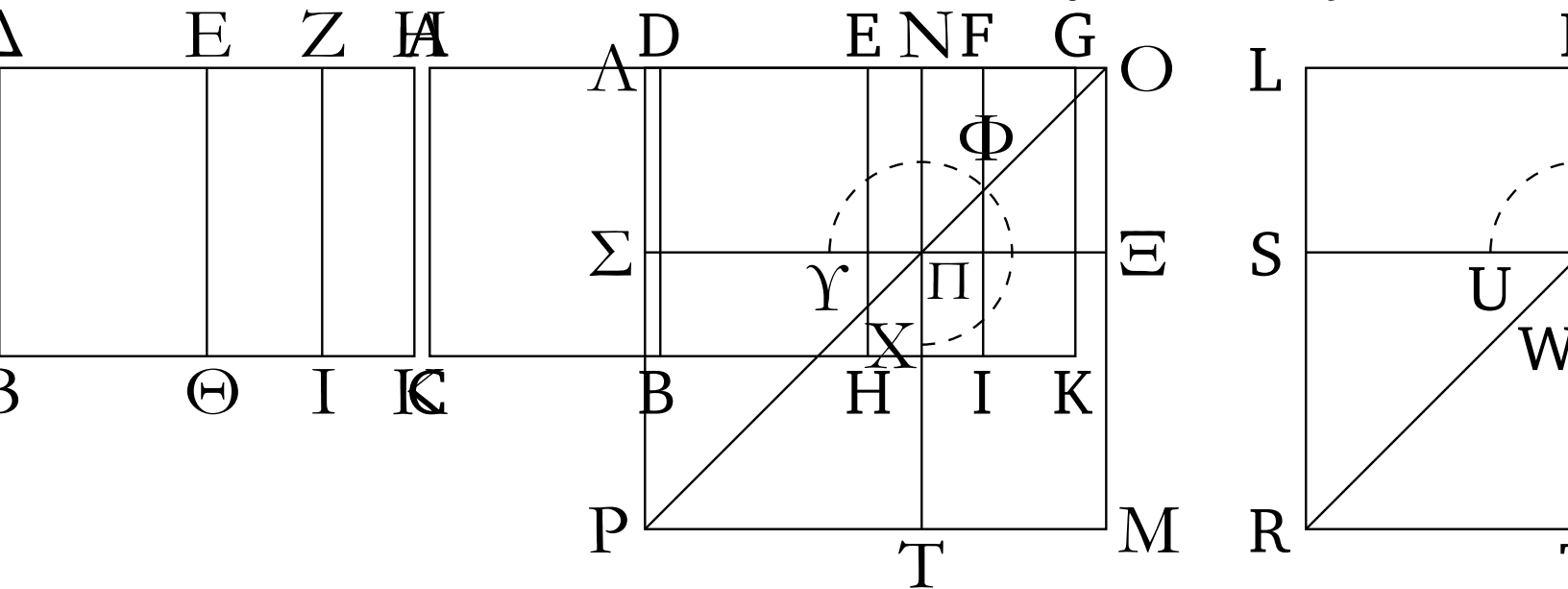
Ἐὰν θεωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς τρίτης,
ἢ τὸ θεωρίον δυναμένη μέσση ἀποτομή ἐστὶ δευτέρα.

Θωρίον γὰρ τὸ AB περιθεόσθω ὑπὸ ῥητῆς τῆς $ΑΓ$ καὶ ἀποτομῆς τρίτης τῆς $ΑΔ$: λέγω, ὅτι ἡ τὸ AB θωρίον δυναμένη μέσση ἀποτομή ἐστι δευτέρα.

Proposition 93

If an area is contained by a rational (straight-line) and a third apotome then the square-root of the area is a second apotome of a medial (straight-line).

For let the area AB be contained by the rational (straight-line) AC and the third apotome AD . I say that the square-root of area AB is the second apotome of a medial (straight-line).



Ἐστω γὰρ τῇ $\Delta\Delta$ προσαρμόζονσα ἡ ΔH . αἱ AH , $\text{H}\Delta$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα τῶν AH , $\text{H}\Delta$ σύμμετρος ἐστὶ μήκει τῇ ἐκκεκλεμένη ῥητῇ τῇ $\text{A}\Gamma$, ἥ δὲ ὅλη ἡ AH τῆς προσαρμοζούσης τῆς ΔH μείζον δύνатаι τῷ ἀπὸ συμμέτρου ἑαντῇ. ἐπεὶ οὖν ἡ AH τῆς $\text{H}\Delta$ μείζον δύνатаι τῷ ἀπὸ συμμέτρου ἑαντῇ, ἕαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔH ἴσον παρὰ τὴν AH παραβληθῇ ἐλλείπον

For let DG be an attachment to AD . Thus, AG and GD are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and neither of AG and GD is commensurable in length with the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, DG , by the (square) on (some straight-line) commensurable (in length) with (AG)

εἶδει τετραγώνω, εἰς σύμμετρα αὐτὴν διελεί. τετμήσθω οὖν ἡ ΔΗ διὰ κατὰ τὸ Ε, καὶ τῷ ἀπὸ τῆς ΕΗ ἴσον παρὰ τὴν ΑΗ παραβεβλήσθω ἐλλείπον εἶδει τετραγώνω, καὶ ἔστω τὸ ὑπὸ τῶν ΑΖ, ΖΗ. καὶ ἡθῶσαν διὰ τῶν Ε, Ζ, Η σημείων τῇ ΑΓ παράλληλοι αἱ ΕΘ, ΖΙ, ΗΚ· σύμμετροι ἄρα εἰσὶν αἱ ΑΖ, ΖΗ· σύμμετρον ἄρα καὶ τὸ ΑΙ τῷ ΖΚ. καὶ ἐπεὶ αἱ ΑΖ, ΖΗ σύμμετροί εἰσι μήκει, καὶ ἡ ΑΗ ἄρα ἐκατέρω τῶν ΑΖ, ΖΗ σύμμετρος ἔστι μήκει. ῥητὴ δὲ ἡ ΑΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ὥστε καὶ αἱ ΑΖ, ΖΗ· ἐκότερον ἄρα τῶν ΑΙ, ΖΚ μέσον ἐστίν. πάλιν, ἐπεὶ σύμμετρος ἔστιν ἡ ΔΕ τῇ ΕΗ μήκει, καὶ ἡ ΔΗ ἄρα ἐκατέρω τῶν ΔΕ, ΕΗ σύμμετρος ἔστι μήκει. ῥητὴ δὲ ἡ ΗΔ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ῥητὴ ἄρα καὶ ἐκατέρω τῶν ΔΕ, ΕΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ἐκότερον ἄρα τῶν ΔΘ, ΕΚ μέσον ἐστίν. καὶ ἐπεὶ αἱ ΑΗ, ΗΔ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἔστι μήκει ἡ ΑΗ τῇ ΗΔ. ἀλλ' ἡ μὲν ΑΗ τῇ ΑΖ σύμμετρος ἔστι μήκει ἡ δὲ ΔΗ τῇ ΕΗ· ἀσύμμετρος ἄρα ἔστιν ἡ ΑΖ τῇ ΕΗ μήκει. ὥς δὲ ἡ ΑΖ πρὸς τὴν ΕΗ, οὕτως ἔστι τὸ ΑΙ πρὸς τὸ ΕΚ· ἀσύμμετρον ἄρα ἔστι τὸ ΑΙ τῷ ΕΚ.

Συνεστάτω οὖν τῷ μὲν ΑΙ ἴσον τετραγώνον τὸ ΑΜ, τῷ δὲ ΖΚ ἴσον ἀφῆρῆσθω τὸ ΝΧ περὶ τὴν αὐτὴν γωνίαν ὅν τῷ ΑΜ· περὶ τὴν αὐτὴν ἄρα διάμετρον ἔστι τὰ ΑΜ, ΝΧ. ἔστω αὐτῶν διάμετρος ἡ ΟΡ, καὶ καταγεγραμθῶ τὸ σθῆμα. ἐπεὶ οὖν τὸ ὑπὸ τῶν ΑΖ, ΖΗ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΗ, ἔστιν ἄρα ὡς ἡ ΑΖ πρὸς τὴν ΕΗ, οὕτως ἡ ΕΗ πρὸς τὴν ΖΗ. ἀλλ' ὡς μὲν ἡ ΑΖ πρὸς τὴν ΕΗ, οὕτως ἔστι τὸ ΑΙ πρὸς τὸ ΕΚ· ὥς δὲ ἡ ΕΗ πρὸς τὴν ΖΗ, οὕτως ἔστι τὸ ΕΚ πρὸς τὸ ΖΚ· καὶ ὥς ἄρα τὸ ΑΙ πρὸς τὸ ΕΚ, οὕτως τὸ ΕΚ πρὸς τὸ ΖΚ· τῶν ἄρα ΑΙ, ΖΚ μέσον ἀνάλογον ἔστι τὸ ΕΚ. ἔστι δὲ καὶ τῶν ΑΜ, ΝΧ τετραγώνων μέσον ἀνάλογον τὸ ΜΝ· καὶ ἔστιν ἴσον τὸ μὲν ΑΙ τῷ ΑΜ, τὸ δὲ ΖΚ τῷ ΝΧ· καὶ τὸ ΕΚ ἄρα ἴσον ἐστὶ τῷ ΜΝ. ἀλλὰ τὸ μὲν ΜΝ ἴσον ἐστὶ τῷ ΑΧ, τὸ δὲ ΕΚ ἴσον [ἐστὶ] τῷ ΔΘ· καὶ ὅλον ἄρα τὸ ΔΚ ἴσον ἐστὶ τῷ ΥΦΘ γνώμῃ καὶ τῷ ΝΧ. ἔστι δὲ καὶ τὸ ΑΚ ἴσον τοῖς ΑΜ, ΝΧ· λοιπὸν ἄρα τὸ ΑΒ ἴσον ἐστὶ τῷ ΣΤ, τοντέστι τῷ ἀπὸ τῆς ΑΝ τετραγώνω· ἡ ΑΝ ἄρα δύναιται τὸ ΑΒ θωρίον. λέγω, ὅτι ἡ ΑΝ μέσης ἀποτομή ἐστὶ δευτέρα.

Ἐπεὶ γὰρ μέσα ἐδείχθη τὰ ΑΙ, ΖΚ καὶ ἔστιν ἴσα τοῖς ἀπὸ τῶν ΑΟ, ΟΝ, μέσον ἄρα καὶ ἐκότερον τῶν ἀπὸ τῶν ΑΟ, ΟΝ· μέση ἄρα ἐκατέρω τῶν ΑΟ, ΟΝ. καὶ ἐπεὶ σύμμετρον ἔστι τὸ ΑΙ τῷ ΖΚ, σύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς ΑΟ τῷ ἀπὸ τῆς ΟΝ. πάλιν, ἐπεὶ ἀσύμμετρον ἐδείχθη τὸ ΑΙ τῷ ΕΚ, ἀσύμμετρον ἄρα ἔστι καὶ τὸ ΑΜ τῷ ΜΝ, τοντέστι τὸ ἀπὸ τῆς ΑΟ τῷ ὑπὸ τῶν ΑΟ, ΟΝ· ὥστε καὶ ἡ ΑΟ ἀσύμμετρος ἔστι μήκει τῇ ΟΝ· αἱ ΑΟ, ΟΝ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δὴ, ὅτι καὶ μέσον περιέθουσιν.

Ἐπεὶ γὰρ μέσον ἐδείχθη τὸ ΕΚ καὶ ἔστιν ἴσον τῷ ὑπὸ τῶν ΑΟ, ΟΝ, μέσον ἄρα ἔστι καὶ τὸ ὑπὸ τῶν ΑΟ, ΟΝ· ὥστε αἱ ΑΟ, ΟΝ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσιν. ἡ ΑΝ ἄρα μέσης ἀποτομή ἐστὶ δευτέρα· καὶ δύναιται τὸ ΑΒ θωρίον.

[Def. 10.13]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable (in length) with (AG) , thus if (an area) equal to the fourth part of the square on DG is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Therefore, let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short by a square figure. And let it be the (rectangle contained) by AF and FG . And let EH , FI , and GK be drawn through points E , F , and G (respectively), parallel to AC . Thus, AF and FG are commensurable (in length). AI (is) thus also commensurable with FK [Props. 6.1, 10.11]. And since AF and FG are commensurable in length, AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. And AG (is) rational, and incommensurable in length with AC . Hence, AF and FG (are) also (rational, and incommensurable in length with AC) [Prop. 10.13]. Thus, AI and FK are each medial (areas) [Prop. 10.21]. Again, since DE is commensurable in length with EG , DG is also commensurable in length with each of DE and EG [Prop. 10.15]. And GD (is) rational, and incommensurable in length with AC . Thus, DE and EG (are) each also rational, and incommensurable in length with AC [Prop. 10.13]. DH and EK are thus each medial (areas) [Prop. 10.21]. And since AG and GD are commensurable in square only, AG is thus incommensurable in length with GD . But, AG is commensurable in length with AF , and DG with EG . Thus, AF is incommensurable in length with EG [Prop. 10.13]. And as AF (is) to EG , so AI is to EK [Prop. 6.1]. Thus, AI is incommensurable with EK [Prop. 10.11].

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , which is about the same angle as LM , be subtracted (from LM). Thus, LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG , so AI is to EK [Prop. 6.1]. And as EG (is) to FG , so EK is to FK [Prop. 6.1]. And thus as AI (is) to EK , so EK (is) to FK [Prop. 5.11]. Thus, EK is the mean proportional to AI and FK . And MN is also the mean proportional to the squares LM and NO [Prop. 10.53 lem.]. And AI is equal to LM , and FK to NO . Thus, EK is also equal to MN . But, MN is equal to LO , and EK [is] equal to DH [Prop. 1.43]. And thus the whole of DK is equal to the gnomon UVW and NO . And AK (is) also equal to LM and NO . Thus, the remainder AB is equal to ST —that is to say, to the square on LN . Thus, LN is the square-root of area AB . I say that LN is the second apotome of a medial (straight-line).

For since AI and FK were shown (to be) medial (areas), and are equal to the (squares) on LP and PN (respectively),

Ἡ ἄρα τὸ AB θωρίον δυναμένη μέση ἀποτομή ἐστὶ δυν-
τέρα· ὅπερ ἔδει δεῖχαι.

the (squares) on each of LP and PN (are) thus also medial. Thus, LP and PN (are) each medial (straight-lines). And since AI is commensurable with FK [Props. 6.1, 10.11], the (square) on LP (is) thus also commensurable with the (square) on PN . Again, since AI was shown (to be) incommensurable with EK , LM is thus also incommensurable with MN —that is to say, the (square) on LP with the (rectangle contained) by LP and PN . Hence, LP is also incommensurable in length with PN [Props. 6.1, 10.11]. Thus, LP and PN are medial (straight-lines which are) commensurable in square only. So, I say that they also contain a medial (area).

For since EK was shown (to be) a medial (area), and is equal to the (rectangle contained) by LP and PN , the (rectangle contained) by LP and PN is thus also medial. Hence, LP and PN are medial (straight-lines which are) commensurable in square only, and which contain a medial (area). Thus, LN is the second apotome of a medial (straight-line) [Prop. 10.75]. And it is the square-root of area AB .

Thus, the square-root of area AB is the second apotome of a medial (straight-line). (Which is) the very thing it was required to show.

94.

Proposition 94

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς τετάρ-
της, ἡ τὸ θωρίον δυναμένη ἐλάσσων ἐστίν.

Θωρίον γάρ τὸ AB περιέθῃται ὑπὸ ῥητῆς τῆς AG κα-
ὶ ἀποτομῆς τετάρτης τῆς AD . λέγω, ὅτι ἡ τὸ AB θωρίον
δυναμένη ἐλάσσων ἐστίν.

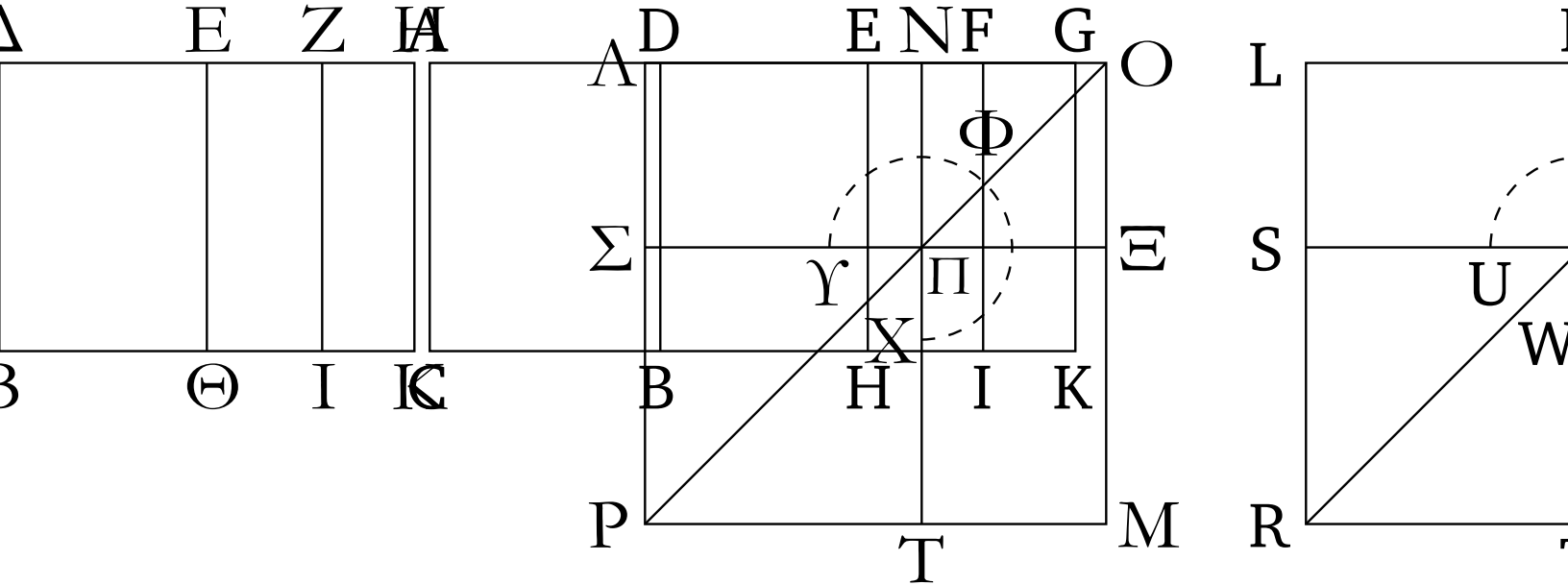
Ἐστω γάρ τῇ AD προσαρμόζουσα ἡ ΔH . αἱ ἄρα AH ,
 $H\Delta$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AH σύμμετρος
ἐστὶ τῇ ἐκκεκλιμένη ῥητῇ τῇ AG μήκει, ἡ δὲ ὅλη ἡ AH τῇ
προσαρμοζούσῃ τῇ ΔH μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου
ἐκκλιμένη μήκει. ἐπεὶ οὖν ἡ AH τῇ $H\Delta$ μείζον δύναται τῷ ἀπὸ
ἀσυμμέτρου ἐκκλιμένη μήκει, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ
τῆς ΔH ἴσον παρὰ τὴν AH παραβληθῇ ἐλλείπον εἶδει τε-
τραγώνῳ, εἷς ἀσύμμετρος αὐτὴν διελεί. τεμήσθω οὖν ἡ ΔH
δίδια κατὰ τὸ E , καὶ τῷ ἀπὸ τῆς EH ἴσον παρὰ τὴν AH πα-
ραβελήσθω ἐλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν
 AZ , ZH . ἀσύμμετρος ἄρα ἐστὶ μήκει ἡ AZ τῇ ZH . ἡθροισαν
οὖν διὰ τῶν E , Z , H παράλληλοι ταῖς AG , BD αἱ $E\Theta$, ZI ,
 HK . ἐπεὶ οὖν ῥητὴ ἐστὶν ἡ AH καὶ σύμμετρος τῇ AG μήκει,
ῥητὸν ἄρα ἐστὶν ὅλον τὸ AK . πάλιν, ἐπεὶ ἀσύμμετρος ἐστὶν
ἡ ΔH τῇ AG μήκει, καὶ εἰσιν ἀμφοτέραι ῥηταὶ, μέσον ἄρα
ἐστὶ τὸ ΔK . πάλιν, ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AZ τῇ ZH μήκει,
ἀσύμμετρον ἄρα καὶ τὸ AI τῷ ZK .

If an area is contained by a rational (straight-line) and a
fourth apotome then the square-root of the area is a minor
(straight-line).

For let the area AB be contained by the rational (straight-
line) AC and the fourth apotome AD . I say that the square-root
of area AB is a minor (straight-line).

For let DG be an attachment to AD . Thus, AG and DG are
rational (straight-lines which are) commensurable in square
only [Prop. 10.73], and AG is commensurable in length with
the (previously) laid down rational (straight-line) AC , and the
square on the whole, AG , is greater than (the square on) the
attachment, DG , by the square on (some straight-line) incom-
mensurable in length with (AG) [Def. 10.14]. Therefore, since
the square on AG is greater than (the square on) GD by the
(square) on (some straight-line) incommensurable in length
with (AG) , thus if (some area), equal to the fourth part of the
(square) on DG , is applied to AG , falling short by a square
figure, then it divides (AG) into (parts which are) incommen-
surable (in length) [Prop. 10.18]. Therefore, let DG be cut in
half at E , and let (some area), equal to the (square) on EG ,
be applied to AG , falling short by a square figure, and let it
be the (rectangle contained) by AF and FG . Thus, AF is in-
commensurable in length with FG . Therefore, let EH , FI , and
 GK be drawn through E , F , and G (respectively), parallel to
 AC and BD . Therefore, since AG is rational, and commensu-
rable in length with AC , the whole (area) AK is thus rational
[Prop. 10.19]. Again, since DG is incommensurable in length

with AC , and both are rational (straight-lines), DK is thus a medial (area) [Prop. 10.21]. Again, since AF is incommensurable in length with FG , AI (is) thus also incommensurable with FK [Props. 6.1, 10.11].



Συνεστάτω οὖν τῷ μὲν AI ἴσον τετράγωνον τὸ AM , τῷ δὲ ZK ἴσον ἀφηρήσθω περὶ τὴν αὐτὴν γωνίαν τὴν ὑπὸ τῶν $ΛΟΜ$ τὸ NX . περὶ τὴν αὐτὴν ἄρα διάμετόν ἐστι τὰ AM , NX τετράγωνα. ἔστω αὐτῶν διάμετοξ ἡ OP , καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ οὖν τὸ ὑπὸ τῶν AZ , ZH ἴσον ἐστὶ τῷ ἀπὸ τῆς EH , ἀνάλογον ἄρα ἐστὶν ὡς ἡ AZ πρὸς τὴν EH , οὕτως ἡ EH πρὸς τὴν ZH . ἀλλ' ὡς μὲν ἡ AZ πρὸς τὴν EH , οὕτως ἐστὶ τὸ AI πρὸς τὸ EK , ὡς δὲ ἡ EH πρὸς τὴν ZH , οὕτως ἐστὶ τὸ EK πρὸς τὸ ZK . τῶν ἄρα AI , ZK μέσον ἀνάλογόν ἐστι τὸ EK . ἔστι δὲ καὶ τῶν AM , NX τετραγώνων μέσον ἀνάλογον τὸ MN , καὶ ἐστὶν ἴσον τὸ μὲν AI τῷ AM , τὸ δὲ ZK τῷ NX . καὶ τὸ EK ἄρα ἴσον ἐστὶ τῷ MN . ἀλλὰ τῷ μὲν EK ἴσον ἐστὶ τὸ $ΔΘ$, τῷ δὲ MN ἴσον ἐστὶ τὸ $ΛΧ$. ὅλον ἄρα τὸ $ΔΚ$ ἴσον ἐστὶ τῷ $ΥΦΘ$ γνώμονι καὶ τῷ NX . ἐπεὶ οὖν ὅλον τὸ AK ἴσον ἐστὶ τοῖς AM , NX τετραγώνοις, ὧν τὸ $ΔΚ$ ἴσον ἐστὶ τῷ $ΥΦΘ$ γνώμονι καὶ τῷ NX τετραγώνῳ, λοιπὸν ἄρα τὸ AB ἴσον ἐστὶ τῷ $ΣΤ$, τουτέστι τῷ ἀπὸ τῆς AN τετραγώνῳ. ἡ AN ἄρα δύναται τὸ AB θωρίον. λέγω, ὅτι ἡ AN ἄλογός ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ ῥητόν ἐστι τὸ AK καὶ ἐστὶν ἴσον τοῖς ἀπὸ τῶν AO , ON τετραγώνοις, τὸ ἄρα συγκείμενον ἐκ τῶν ἀπὸ τῶν AO , ON ῥητόν ἐστιν. πάλιν, ἐπεὶ τὸ $ΔΚ$ μέσον ἐστίν, καὶ ἐστὶν ἴσον τὸ $ΔΚ$ τῷ διξ ὑπὸ τῶν AO , ON , τὸ ἄρα διξ ὑπὸ τῶν AO , ON μέσον ἐστίν. καὶ ἐπεὶ ἀσύμμετρον ἐδείχθη τὸ AI τῷ ZK , ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς AO τετράγωνον τῷ ἀπὸ τῆς ON τετραγώνῳ. αἱ AO , ON ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , (and) about the same angle, LPM , be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus, proportionally, as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG , so AI is to EK , and as EG (is) to FG , so EK is to FK [Prop. 6.1]. Thus, EK is the mean proportional to AI and FK [Prop. 5.11]. And MN is also the mean proportional to the squares LM and NO [Prop. 10.13 lem.], and AI is equal to LM , and FK to NO . EK is thus also equal to MN . But, DH is equal to EK , and LO is equal to MN [Prop. 1.43]. Thus, the whole of DK is equal to the gnomon UVW and NO . Therefore, since the whole of AK is equal to the (sum of the) squares LM and NO , of which DK is equal to the gnomon UVW and the square NO , the remainder AB is thus equal to ST —that is to say, to the square on LN . Thus, LN is the square-root of area AB . I say that LN is the irrational (straight-line which is) called minor.

For since AK is rational, and is equal to the (sum of the) squares LP and PN , the sum of the (squares) on LP and PN is thus rational. Again, since DK is medial, and DK is equal to twice the (rectangle contained) by LP and PN , thus twice the (rectangle contained) by LP and PN is medial. And since AI was shown (to be) incommensurable with FK , the square on LP (is) thus also incommensurable with the square on PN .

ῥητόν, τὸ δὲ διξ ὑπ' αὐτῶν μέσον. ἡ ΛN ἄρα ἄλογός ἐστιν ἡ καλουμένη ἐλάσσων καὶ δύναται τὸ AB θωρίον.

Ἡ ἄρα τὸ AB θαυρίον δυναμένη ἐλάσσων ἐστίν· ὅπερ ἔδει
δειῆναι.

Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and twice the (rectangle contained) by them medial. LN is thus the irrational (straight-line) called minor [Prop. 10.76]. And it is the square-root of area AB .

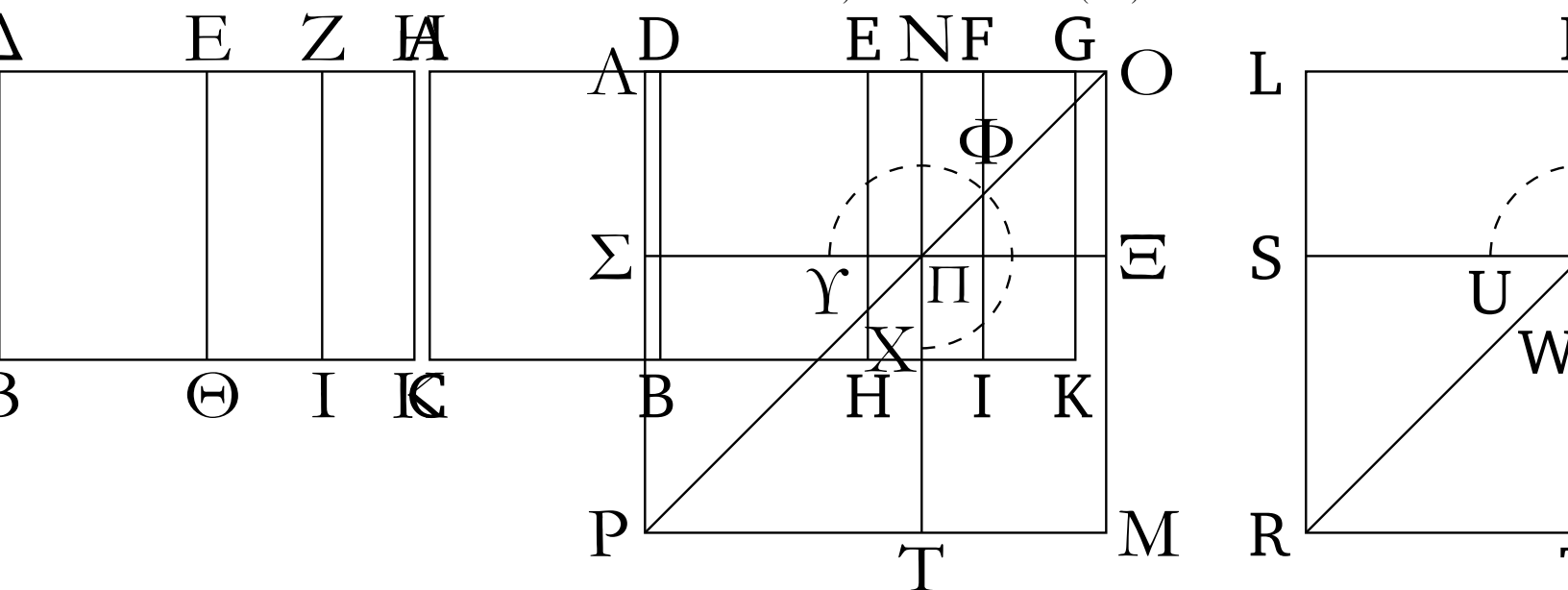
Thus, the square-root of area AB is a minor (straight-line). (Which is) the very thing it was required to show.

95.

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς πέμπτης,
ἢ τὸ θωρίον δυναμένη [ἢ] μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα
ἔστιν.

Proposition 95

If an area is contained by a rational (straight-line) and a fifth apotome then the square-root of the area is that (straight-line) which with a rational (area) makes a medial whole.



Θωρίον γὰρ τὸ AB περιθεόσθω ὑπὸ ῥητῆς τῆς AG καὶ ἀποτομῆς πέμπτης τῆς AD . λέγω, ὅτι ἡ τὸ AB θωρίον δυνάμενη [ή] μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσα ἔστιν.

Ἐστω γάρ τῇ $\Delta\Delta$ προσαρμόζουσα ἡ $\Delta\mathcal{H}$ · αἱ ἄρα $A\mathcal{H}$, $H\Delta$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ προσαρμόζουσα ἡ $H\Delta$ σύμμετροξ ἔστι μήκει τῇ ἐκκεκμένη ῥητῇ τῇ $A\Gamma$, ἡ δὲ ὅλη ἡ $A\mathcal{H}$ τῆς προσαρμόζουσῃς τῆς $\Delta\mathcal{H}$ μείζον δύναται τῷ ἀπὸ ἀσύμμετρον ἑαυτῇ. ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $\Delta\mathcal{H}$ ἴσον παρὰ τὴν $A\mathcal{H}$ παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, εἷς ἀσύμμετρα αὐτὴν διελεῖ. τετμήσθω οὖν ἡ $\Delta\mathcal{H}$ δίθῃ κατὰ τὸ E σημεῖον, καὶ τῷ ἀπὸ τῆς $E\mathcal{H}$ ἴσον παρὰ τὴν $A\mathcal{H}$ παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ καὶ ἔστω τὸ ὑπὸ τῶν AZ , $Z\mathcal{H}$ · ἀσύμμετροξ ἄρα ἐστὶν ἡ AZ τῇ $Z\mathcal{H}$ μήκει. καὶ ἐπεὶ ἀσύμμετροξ ἐστὶν ἡ $A\mathcal{H}$ τῇ ΓA μήκει, καὶ εἰσὶν ἀμφοτέραι ῥηταί, μέσον ἄρα ἐστὶ τὸ AK . πάλιν, ἐπεὶ ῥητὴ ἐστὶν ἡ $\Delta\mathcal{H}$ καὶ σύμμετροξ τῇ $A\Gamma$ μήκει, ῥητόν ἐστι τὸ ΔK .

Συνεστάτω οὖν τῷ μὲν AI ἴσον τετράγωνον τὸ AM , τῷ δὲ ZK ἴσον τετράγωνον ἀφηρήσθω τὸ NX περὶ τὴν αὐτὴν

For let the area AB be contained by the rational (straight-line) AC and the fifth apotome AD . I say that the square-root of area AB is that (straight-line) which with a rational (area) makes a medial whole.

For let DG be an attachment to AD . Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and the attachment GD is commensurable in length the the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, DG , by the (square) on (some straight-line) incommensurable (in length) with (AG) [Def. 10.15]. Thus, if (some area), equal to the fourth part of the (square) on DG , is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let DG be divided in half at point E , and let (some area), equal to the (square) on EG , be applied to AG , falling short by a square figure, and let it be the (rectangle contained) by AF and FG . Thus, AF is incommensurable in length with FG . And since AG is incom-

γωνίαν τὴν ὑπὸ ΛOM · περὶ τὴν αὐτὴν ἄρα διάμετρον ἔστι τὰ ΛM , NX τετράγωνα. ἔστω αὐτῶν διάμετρος ἡ OP , καὶ καταγεγράφθω τὸ σθῆμα. ὁμοίως δὲ δείχομεν, ὅτι ἡ ΛN δύναται τὸ AB θωρίον. λέγω, ὅτι ἡ ΛN ἢ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα ἔστιν.

Ἐπεὶ γὰρ μέσον ἐδείχθη τὸ AK καὶ ἔστιν ἴσον τοῖς ἀπὸ τῶν ΛO , ON , τὸ ἄρα συγκείμενον ἐκ τῶν ἀπὸ τῶν ΛO , ON μέσον ἔστιν. πάλιν, ἐπεὶ ῥητόν ἐστι τὸ ΔK καὶ ἔστιν ἴσον τῷ διζ ὑπὸ τῶν ΛO , ON , καὶ αὐτὸ ῥητόν ἐστιν. καὶ ἐπεὶ ἀσύμμετρον ἔστι τὸ AI τῷ ZK , ἀσύμμετρον ἄρα ἔστι καὶ τὸ ἀπὸ τῆς ΛO τῷ ἀπὸ τῆς ON · αἱ ΛO , ON ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διζ ὑπ' αὐτῶν ῥητόν. ἡ λοιπὴ ἄρα ἡ ΛN ἄλογός ἐστιν ἢ καλουμένη μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα· καὶ δύναται τὸ AB θωρίον.

Ἡ τὸ AB ἄρα θωρίον δυναμένη μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα ἔστιν· ὅπερ ἔδει δεῖχαι.

mensurable in length with CA , and both are rational (straight-lines), AK is thus a medial (area) [Prop. 10.21]. Again, since DG is rational, and commensurable in length with AC , DK is a rational (area) [Prop. 10.19].

Therefore, let the square LM , equal to AI , be constructed. And let the square NO , equal to FK , (and) about the same angle, LPM , be subtracted (from NO). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let (the rest of) the figure be drawn. So, similarly (to the previous propositions), we can show that LN is the square-root of area AB . I say that LN is that (straight-line) which with a rational (area) makes a medial whole.

For since AK was shown (to be) a medial (area), and is equal to (the sum of) the squares on LP and PN , the sum of the (squares) on LP and PN is thus medial. Again, since DK is rational, and is equal to twice the (rectangle contained) by LP and PN , (the latter) is also rational. And since AI is incommensurable with FK , the (square) on LP is thus also incommensurable with the (square) on PN . Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by them rational. Thus, the remainder LN is the irrational (straight-line) called that which with a rational (area) makes a medial whole [Prop. 10.77]. And it is the square-root of area AB .

Thus, the square-root of area AB is that (straight-line) which with a rational (area) makes a medial whole. (Which is) the very thing it was required to show.

96.

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς ἑκτῆς, ἡ τὸ θωρίον δυναμένη μετὰ μέσον μέσον τὸ ὅλον ποιῶσα ἔστιν.

Proposition 96

If an area is contained by a rational (straight-line) and a sixth apotome then the square-root of the area is that (straight-line) which with a medial (area) makes a medial whole.

ἐστίν. καὶ ἐπεὶ ἀσύμμετρον ἐδείχθη τὸ AK τῷ $ΔK$, ἀσύμμετρα [ἄρα] ἐστὶ καὶ τὰ ἀπὸ τῶν $ΛO$, ON τετράγωνα τῷ διζυπὸ τῶν $ΛO$, ON . καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ AI τῷ ZK , ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς $ΛO$ τῷ ἀπὸ τῆς ON αἱ $ΛO$, ON ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ διζυπὸ αὐτῶν μέσον ἔτι τε τὰ ἀπ' αὐτῶν τετράγωνα ἀσύμμετρα τῷ διζυπὸ αὐτῶν. ἡ ἄρα $ΛN$ ἀλογόξ ἐστίν ἢ καλουμένη μετὰ μέσον μέσον τὸ ὅλον ποιοῦσα· καὶ δύναται τὸ AB θεωρίον.

Ἡ ἄρα τὸ θεωρίον δυναμένη μετὰ μέσον μέσον τὸ ὅλον ποιοῦσά ἐστιν· ὅπερ ἔδει δεῖχαι.

And let NO , equal to FK , (and) about the same angle, be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let (the rest of) the figure be drawn. So, similarly to the above, we can show that LN is the square-root of area AB . I say that LN is that (straight-line) which with a medial (area) makes a medial whole.

For since AK was shown (to be) a medial (area), and is equal to the (sum of the) squares on LP and PN , the sum of the (squares) on LP and PN is medial. Again, since DK was shown (to be) a medial (area), and is equal to twice the (rectangle contained) by LP and PN , twice the (rectangle contained) by LP and PN is also medial. And since AK was shown (to be) incommensurable with DK , [thus] the (sum of the) squares on LP and PN is also incommensurable with twice the (rectangle contained) by LP and PN . And since AI is incommensurable with FK , the (square) on LP (is) thus also incommensurable with the (square) on PN . Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by medial, and, furthermore, the (sum of the) squares on them incommensurable with twice the (rectangle contained) by them. Thus, LN is the irrational (straight-line) called that which with a medial (area) makes a medial whole [Prop. 10.78]. And it is the square-root of area AB .

Thus, the square-root of area (AB) is that (straight-line) which with a medial (area) makes a medial whole. (Which is) the very thing it was required to show.

97.

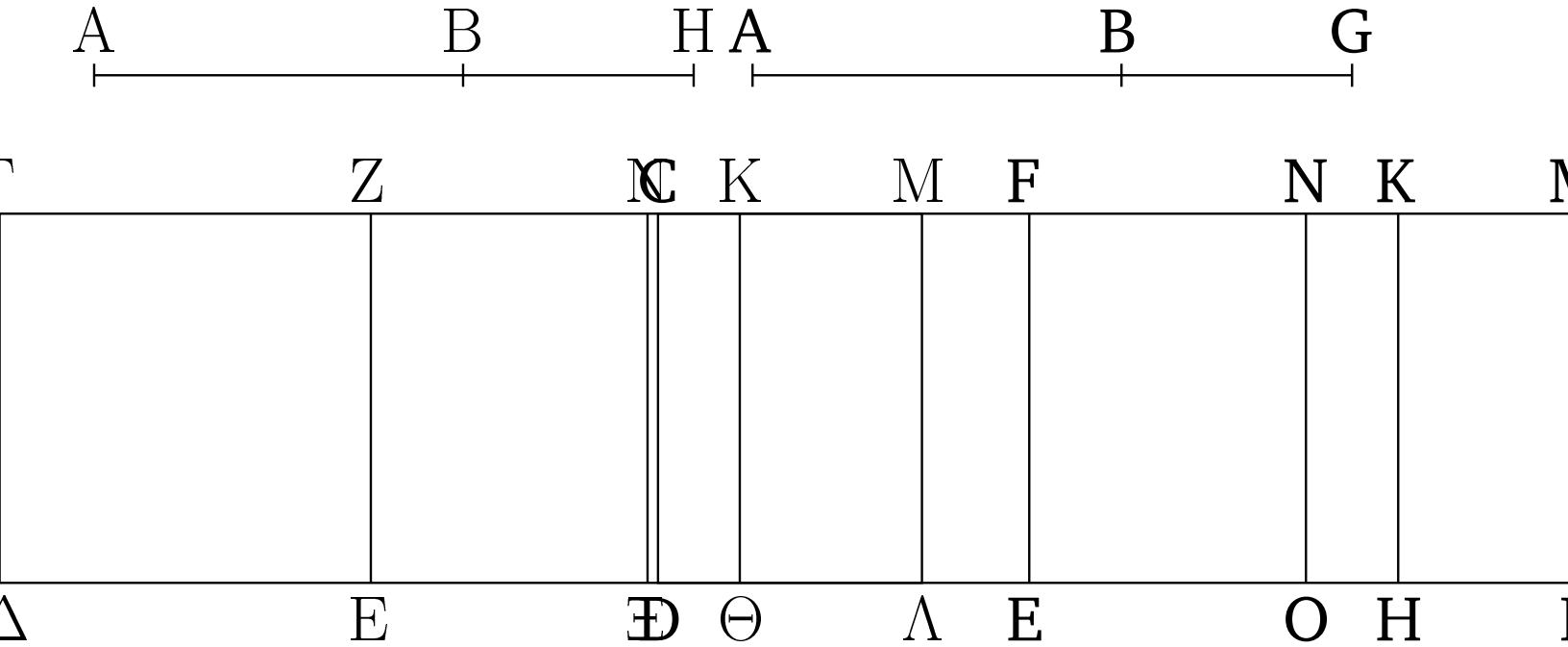
Proposition 97

Τὸ ἀπὸ ἀποτομῆς παρὰ ῥητὴν παραβαλλόμενον πλάτοξ ποιεῖ ἀποτομήν πρώτην.

Ἐστω ἀποτομή ἡ AB , ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $ΓΔ$ παραβελήσθω τὸ $ΓΕ$ πλάτοξ ποιῶν τὴν $ΓΖ$ · λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστὶ πρώτη.

The (square) on an apotome, applied to a rational (straight-line), produces a first apotome as breadth.

Let AB be an apotome, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a first apotome.



Ἐστω γάρ τῇ AB προσαρμόζονσα ἡ BH . αἱ ἄρα AH , HB ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΘ$, τῷ δὲ ἀπὸ τῆς BH τὸ $ΚΛ$. ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB . ὧν τὸ $ΓΕ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB . λοιπὸν ἄρα τὸ $ΖΛ$ ἴσον ἐστὶ τῷ διζ ὑπὸ τῶν AH , HB . τετμήσθω ἡ ZM διῖθα κατὰ τὸ N σημεῖον, καὶ ῥηθῶ διὰ τοῦ N τῇ $ΓΔ$ παράλληλος ἡ NX . ἐκάτερον ἄρα τῶν ZX , AN ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τὰ ἀπὸ τῶν AH , HB ῥητά ἐστιν, καὶ ἐστὶ τοῖς ἀπὸ τῶν AH , HB ἴσον τὸ $ΔΜ$, ῥητὸν ἄρα ἐστὶ τὸ $ΔΜ$. καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παραβέβληται πλάτος ποιοῦν τὴν $ΓΜ$. ῥητὴ ἄρα ἐστὶν ἡ $ΓΜ$ καὶ σύμμετρος τῇ $ΓΔ$ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ διζ ὑπὸ τῶν AH , HB , καὶ τῷ διζ ὑπὸ τῶν AH , HB ἴσον τὸ $ΖΛ$, μέσον ἄρα τὸ $ΖΛ$. καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παράκειται πλάτος ποιοῦν τὴν ZM . ῥητὴ ἄρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ τὰ μὲν ἀπὸ τῶν AH , HB ῥητά ἐστιν, τὸ δὲ διζ ὑπὸ τῶν AH , HB μέσον, ἀσύμμετρον ἄρα ἐστὶ τὰ ἀπὸ τῶν AH , HB τῷ διζ ὑπὸ τῶν AH , HB . καὶ τοῖς μὲν ἀπὸ τῶν AH , HB ἴσον ἐστὶ τὸ $ΓΛ$, τῷ δὲ διζ ὑπὸ τῶν AH , HB τὸ $ΖΛ$. ἀσύμμετρον ἄρα ἐστὶ τὸ $ΔΜ$ τῷ $ΖΛ$. ὥς δὲ τὸ $ΔΜ$ πρόξ τὸ $ΖΛ$, οὕτως ἐστὶν ἡ $ΓΜ$ πρόξ τὴν ZM . ἀσύμμετρος ἄρα ἐστὶν ἡ $ΓΜ$ τῇ ZM μήκει. καὶ εἰσιν ἀμφότεραι ῥηταί· αἱ ἄρα $ΓΜ$, MZ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ $ΓΖ$ ἄρα ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ πρώτη.

Ἐπεὶ γάρ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $ΓΘ$, τῷ δὲ ἀπὸ τῆς BH ἴσον τὸ $ΚΛ$, τῷ δὲ ὑπὸ τῶν AH , HB τὸ $ΝΛ$, καὶ τῶν $ΓΘ$, $ΚΛ$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $ΝΛ$. ἐστὶν ἄρα ὡς τὸ $ΓΘ$ πρόξ τὸ $ΝΛ$, οὕτως τὸ $ΝΛ$ πρόξ τὸ $ΚΛ$. ἀλλ’

For let BG be an attachment to AB . Thus, AG and GB are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let CH , equal to the (square) on AG , and KL , (equal) to the (square) on BG , be applied to CD . Thus, the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB . The remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Let FM be cut in half at point N . And let NO be drawn through N , parallel to CD . Thus, FO and LN are each equal to the (rectangle contained) by AG and GB . And since the (sum of the squares) on AG and GB is rational, and DM is equal to the (sum of the squares) on AG and GB , DM is thus rational. And it has been applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and commensurable in length with CD [Prop. 10.20]. Again, since twice the (rectangle contained) by AG and GB is medial, and FL (is) equal to twice the (rectangle contained) by AG and GB , FL (is) thus a medial (area). And it is applied to the rational (straight-line) CD , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since the (sum of the squares) on AG and GB is rational, and twice the (rectangle contained) by AG and GB medial, the (sum of the squares) on AG and GB is thus incommensurable with twice the (rectangle contained) by AG and GB . And CL is equal to the (sum of the squares) on AG and GB , and FL to twice the (rectangle contained) by AG and GB . DM is thus incommensurable with FL . And as DM (is) to FL , so CM is to FM [Prop. 6.1]. CM is thus incommensurable in length with FM [Prop. 10.11]. And both are rational (straight-lines). Thus, CM and MF are rational (straight-lines).

ὥς μὲν τὸ ΓΘ πρὸς τὸ ΝΛ, οὕτως ἐστὶν ἡ ΓΚ πρὸς τὴν ΝΜ· ὥς δὲ τὸ ΝΛ πρὸς τὸ ΚΛ, οὕτως ἐστὶν ἡ ΝΜ πρὸς τὴν ΚΜ· τὸ ἄρα ὑπὸ τῶν ΓΚ, ΚΜ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΝΜ, τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ. καὶ ἐπεὶ σύμμετρον ἐστὶ τὸ ἀπὸ τῆς ΑΗ τῷ ἀπὸ τῆς ΗΒ, σύμμετρον [ἐστὶ] καὶ τὸ ΓΘ τῷ ΚΛ. ὥς δὲ τὸ ΓΘ πρὸς τὸ ΚΛ, οὕτως ἡ ΓΚ πρὸς τὴν ΚΜ· σύμμετρος ἄρα ἐστὶν ἡ ΓΚ τῇ ΚΜ. ἐπεὶ οὖν δύο εὐθεῖαι ἄκισοι εἰσιν αἱ ΓΜ, ΜΖ, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ ἴσον παρὰ τὴν ΓΜ παραβέβληται ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν ΓΚ, ΚΜ, καὶ ἐστὶ σύμμετρος ἡ ΓΚ τῇ ΚΜ, ἡ ἄρα ΓΜ τῆς ΜΖ μείζον δύνανται τῷ ἀπὸ συμέτρου ἐαυτῇ μήκει. καὶ ἐστὶν ἡ ΓΜ σύμμετρος τῇ ἐκκειμένῃ ῥητῇ τῇ ΓΔ μήκει· ἡ ἄρα ΓΖ ἀποτομή ἐστὶ πρώτη.

Τὸ ἄρα ἀπὸ ἀποτομῆς παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην· ὅπερ ἔδει δεῖξαι.

98.

Τὸ ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν.

Ἐστω μέσης ἀποτομῆς πρώτη ἡ ΑΒ, ῥητὴ δὲ ἡ ΓΔ, καὶ τῷ ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΓΔ παραβέβλησθω τὸ ΓΕ πλάτος ποιοῦν τὴν ΓΖ· λέγω, ὅτι ἡ ΓΖ ἀποτομή ἐστὶ δευτέρα.

Ἐστω γάρ τῇ ΑΒ προσαρμόζουσα ἡ ΒΗ· αἱ ἄρα ΑΗ, ΗΒ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ῥητὸν περιέθονσαι. καὶ τῷ μὲν ἀπὸ τῆς ΑΗ ἴσον παρὰ τὴν ΓΔ παραβέβλησθω τὸ ΓΘ πλάτος ποιοῦν τὴν ΓΚ, τῷ δὲ ἀπὸ τῆς ΗΒ ἴσον τὸ ΚΛ πλάτος ποιοῦν τὴν ΚΜ· ὅλον ἄρα τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΑΗ, ΗΒ· μέσον ἄρα καὶ τὸ ΓΛ. καὶ παρὰ ῥητὴν τὴν ΓΔ παράκειται πλάτος ποιοῦν τὴν ΓΜ· ῥητὴ ἄρα ἐστὶν ἡ ΓΜ καὶ ἀσύμμετρος τῇ ΓΔ μήκει. καὶ ἐπεὶ τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΑΗ, ΗΒ, ὧν τὸ ἀπὸ τῆς ΑΒ ἴσον ἐστὶ τῷ ΓΕ, λοιπὸν ἄρα τὸ διζ ὑπὸ τῶν ΑΗ, ΗΒ ἴσον ἐστὶ τῷ ΖΛ. ῥητὸν δὲ [ἐστὶ] τὸ διζ ὑπὸ τῶν ΑΗ, ΗΒ· ῥητὸν ἄρα τὸ ΖΛ. καὶ παρὰ ῥητὴν τὴν ΖΕ παράκειται πλάτος ποιοῦν τὴν ΖΜ· ῥητὴ ἄρα ἐστὶ καὶ ἡ ΖΜ καὶ σύμμετρος τῇ ΓΔ μήκει. ἐπεὶ οὖν τὰ μὲν ἀπὸ τῶν ΑΗ, ΗΒ, τουτέστι τὸ ΓΛ, μέσον ἐστὶν, τὸ δὲ διζ ὑπὸ τῶν ΑΗ, ΗΒ, τουτέστι τὸ ΖΛ, ῥητὸν ἀσύμμετρον

which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a first (apotome).

For since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG , and KL equal to the (square) on GB , and NL to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to KM [Prop. 6.1]. Thus, the (rectangle contained) by CK and KM is equal to the (square) on NM —that is to say, to the fourth part of the (square) on FM [Prop. 6.17]. And since the (square) on AG is commensurable with the (square) on GB , CH [is] also commensurable with KL . And as CH (is) to KL , so CK (is) to KM [Prop. 6.1]. CK is thus commensurable (in length) with KM [Prop. 10.11]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on FM , has been applied to CM , falling short by a square figure, and CK is commensurable (in length) with KM , the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable in length with (CM) [Prop. 10.17]. And CM is commensurable in length with the (previously) laid down rational (straight-line) CD . Thus, CF is a first apotome [Def. 10.15].

Thus, the (square) on an apotome, applied to a rational (straight-line), produces a first apotome as breadth. (Which is) the very thing it was required to show.

Proposition 98

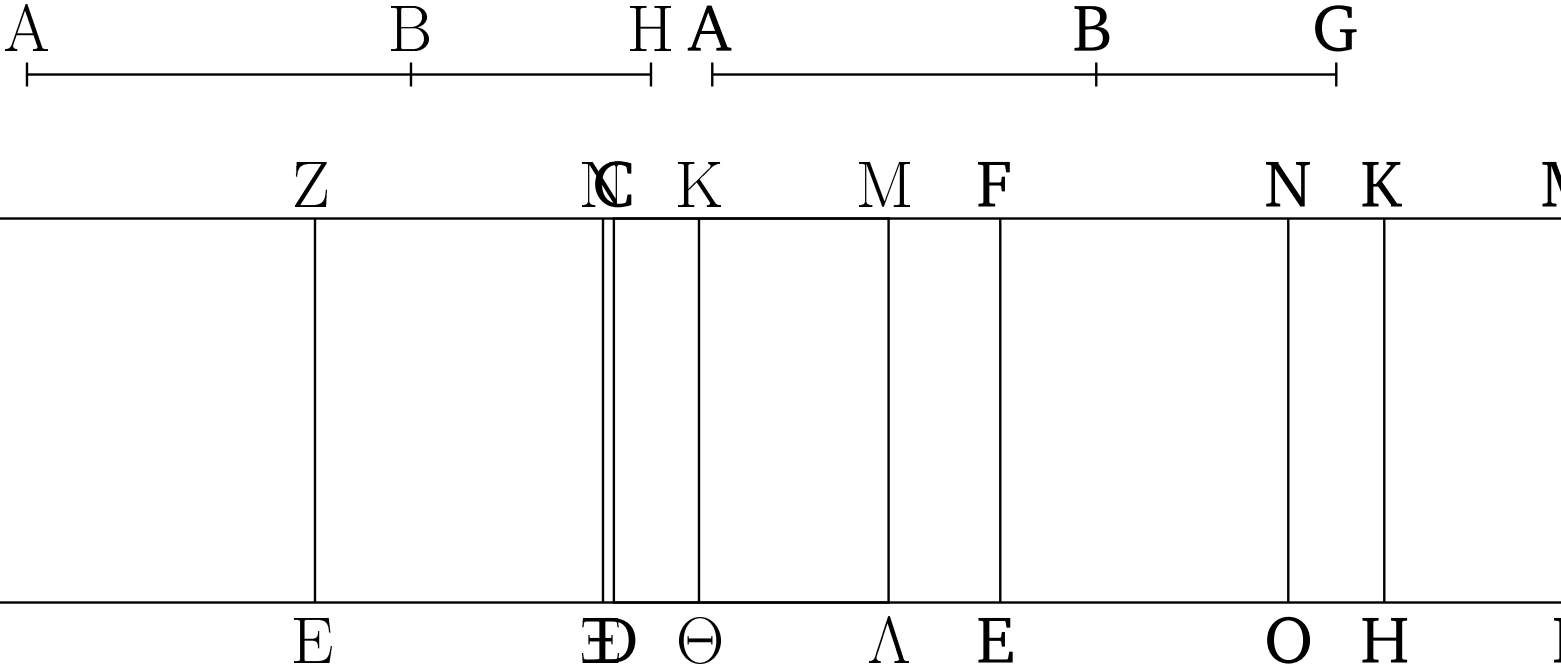
The (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces a second apotome as breadth.

Let AB be a first apotome of a medial (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a second apotome.

For let BG be an attachment to AB . Thus, AG and GB are medial (straight-lines which are) commensurable in square only, containing a rational (area) [Prop. 10.74]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on GB , producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . Thus, CL (is) also a medial (area) [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) CD , producing CM as breadth. CM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since CL is equal to the (sum of the squares) on AG and GB , of which the (square) on AB is equal to CE , the remainder, twice the (rectangle contained) by AG and GB , is thus equal to

ἄρα ἐστὶ τὸ $\Gamma\Lambda$ τῷ $Ζ\Lambda$. ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Ζ\Lambda$, οὕτως ἐστὶν ἡ $\Gamma\mathcal{M}$ πρὸς τὴν $Ζ\mathcal{M}$. ἀσύμμετρος ἄρα ἡ $\Gamma\mathcal{M}$ τῇ $Ζ\mathcal{M}$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ἄρα $\Gamma\mathcal{M}$, $\mathcal{M}Ζ$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ $\GammaΖ$ ἄρα ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ δευτέρα.

FL [Prop. 2.7]. And twice the (rectangle contained) by AG and GB [is] rational. Thus, FL (is) rational. And it is applied to the rational (straight-line) FE , producing FM as breadth. FM is thus also rational, and commensurable in length with CD [Prop. 10.20]. Therefore, since the (sum of the squares) on AG and GB —that is to say, CL —is medial, and twice the (rectangle contained) by AG and GB —that is to say, FL —(is) rational, CL is thus incommensurable with FL . And as CL (is) to FL , so CM is to FM [Prop. 6.1]. Thus, CM (is) incommensurable in length with FM [Prop. 10.11]. And they are both rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a second (apotome).



Τετμήσθω γὰρ ἡ $Ζ\mathcal{M}$ διῖθα κατὰ τὸ N , καὶ ἤθθω διὰ τοῦ N τῇ $\Gamma\Delta$ παράλληλοξ ἡ NX . ἐκάτερον ἄρα τῶν $ΖX$, $N\Lambda$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB τετραγώνων μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶν ἴσον τὸ μὲν ἀπὸ τῆς AH τῷ $\Gamma\Theta$, τὸ δὲ ὑπὸ τῶν AH , HB τῷ $N\Lambda$, τὸ δὲ ἀπὸ τῆς BH τῷ $K\Lambda$, καὶ τῶν $\Gamma\Theta$, $K\Lambda$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $N\Lambda$. ἔστιν ἄρα ὥς τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$. ἀλλ' ὥς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως ἐστὶν ἡ ΓK πρὸς τὴν $N\mathcal{M}$, ὥς δὲ τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$, οὕτως ἐστὶν ἡ $N\mathcal{M}$ πρὸς τὴν MK . ὥς ἄρα ἡ ΓK πρὸς τὴν $N\mathcal{M}$, οὕτως ἐστὶν ἡ $N\mathcal{M}$ πρὸς τὴν $K\mathcal{M}$. τὸ ἄρα ὑπὸ τῶν ΓK , $K\mathcal{M}$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $N\mathcal{M}$, τοντέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $Ζ\mathcal{M}$ [καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς BH , σύμμετρόν ἐστι καὶ τὸ $\Gamma\Theta$ τῷ $K\Lambda$, τοντέστιν ἡ ΓK τῇ $K\mathcal{M}$]. ἐπεὶ οὖν δύο εὐθεῖαι ἄνισοι εἰσιν αἱ $\Gamma\mathcal{M}$, $\mathcal{M}Ζ$, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $\mathcal{M}Ζ$ ἴσον παρὰ τὴν μείζονα

For let FM be cut in half at N . And let NO be drawn through (point) N , parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since the (rectangle contained) by AG and GB is the mean proportional to the squares on AG and GB [Prop. 10.21 lem.], and the (square) on AG is equal to CH , and the (rectangle contained) by AG and GB to NL , and the (square) on BG to KL , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL [Prop. 5.11]. But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to MK [Prop. 6.1]. Thus, as CK (is) to NM , so NM is to KM [Prop. 5.11]. The (rectangle contained) by CK and $K\mathcal{M}$ is thus equal to the (square) on $N\mathcal{M}$ [Prop. 6.17]—that is to say, to the fourth part of the (square) on FM [and since the (square) on AG is commensurable with the (square) on BG , CH is also commensurable with KL —that is to say, CK with

τὴν $ΓΜ$ παραβέβληται ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν $ΓΚ$, $ΚΜ$ καὶ εἷς σύμμετρα αὐτὴν διαιρεῖ, ἡ ἄρα $ΓΜ$ τῆς $ΜΖ$ μείζον δύνатаι τῷ ἀπὸ σύμμετρον ἐαντὴ μήκει. καὶ ἐστὶν ἡ προσαρμόζονσα ἡ $ΖΜ$ σύμμετρος μήκει τῇ ἐκκειμένῃ ῥητῇ τῇ $ΓΔ$. ἡ ἄρα $ΓΖ$ ἀποτομή ἐστὶ δευτέρα.

Τὸ ἄρα ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν· ὅπερ ἔδει δεῖχαι.

KM]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on MF , has been applied to the greater CM , falling short by a square figure, and divides it into commensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable in length with (CM) [Prop. 10.17]. The attachment FM is also commensurable in length with the (previously) laid down rational (straight-line) CD . CF is thus a second apotome [Def. 10.16].

Thus, the (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces a second apotome as breadth. (Which is) the very thing it was required to show.

99.

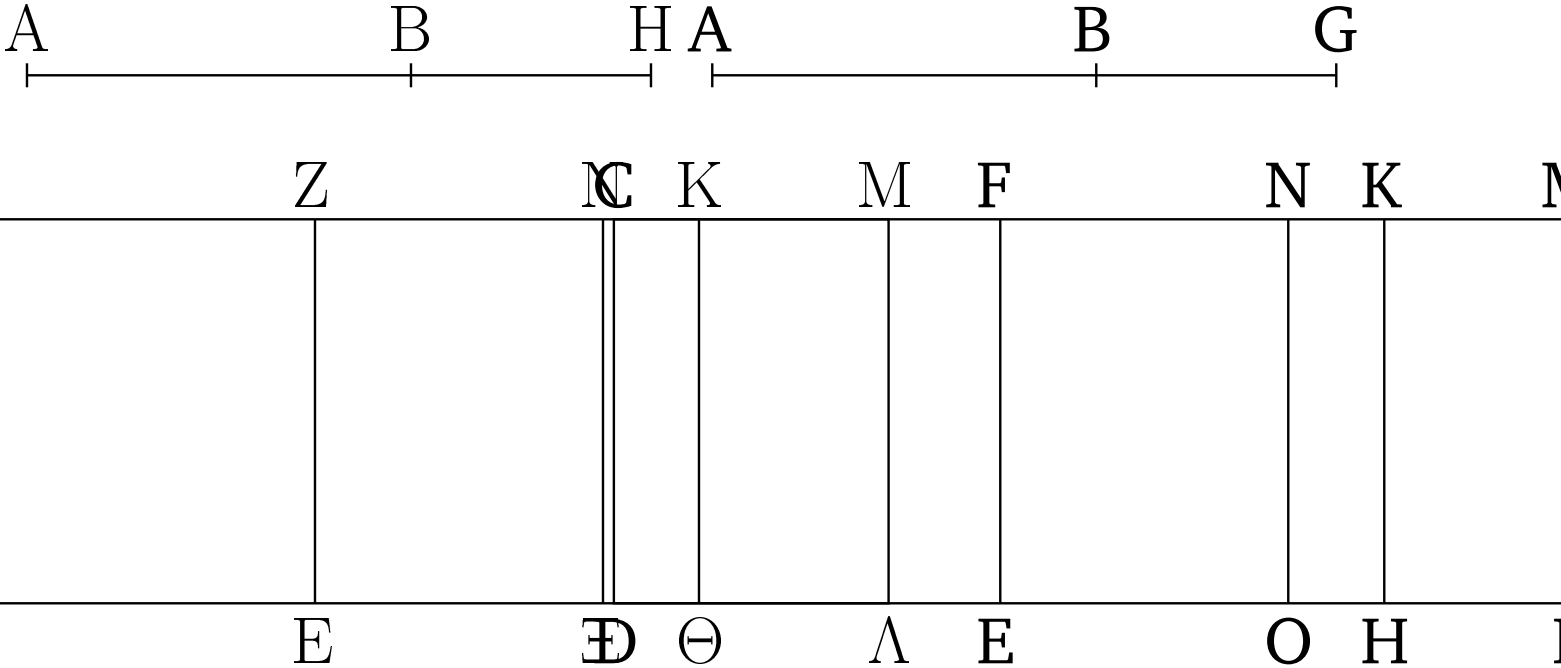
Proposition 99

Τὸ ἀπὸ μέσης ἀποτομῆς δευτέρας παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην.

Ἐστω μέσης ἀποτομῆς δευτέρας ἡ $ΑΒ$, ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς $ΑΒ$ ἴσον παρὰ τὴν $ΓΔ$ παραβελήσθω τὸ $ΓΕ$ πλάτος ποιοῦν τὴν $ΓΖ$. λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστὶ τρίτη.

The (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces a third apotome as breadth.

Let AB be the second apotome of a medial (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a third apotome.



Ἐστω γάρ τῇ $ΑΒ$ προσαρμόζονσα ἡ $ΒΗ$. αἱ ἄρα $ΑΗ$, $ΗΒ$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέχονσαι. καὶ τῷ μὲν ἀπὸ τῆς $ΑΗ$ ἴσον παρὰ τὴν $ΓΔ$ παραβελήσθω τὸ $ΓΘ$ πλάτος ποιοῦν τὴν $ΓΚ$, τῷ δὲ ἀπὸ τῆς $ΒΗ$ ἴσον παρὰ τὴν $ΚΘ$ παραβελήσθω τὸ $ΚΛ$ πλάτος ποιοῦν τὴν $ΚΜ$. ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΑΗ$, $ΗΒ$ [καὶ ἐστὶ μέσα τὰ ἀπὸ

For let BG be an attachment to AB . Thus, AG and GB are medial (straight-lines which are) commensurable in square only, containing a medial (area) [Prop. 10.75]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth. And let KL , equal to the (square) on BG , be applied to KH , producing KM as breadth. Thus, the whole of

τῶν AH, HB]· μέσον ἄρα καὶ τὸ $\Gamma\Lambda$. καὶ παρὰ ῥητὴν τὴν $\Gamma\Delta$ παραβέβληται πλάτος ποιοῦν τὴν ΓM · ῥητὴ ἄρα ἐστὶν ἡ ΓM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ ὅλον τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB , ὥν τὸ ΓE ἴσον ἐστὶ τῷ ἀπὸ τῆς AB , λοιπὸν ἄρα τὸ ΛZ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AH, HB . τετμήσθω οὖν ἡ ZM δίδια κατὰ τὸ N σημεῖον, καὶ τῇ $\Gamma\Delta$ παράλληλος ἡθῆτω ἡ NX · ἐκάτερον ἄρα τῶν $ZX, \Lambda\Lambda$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH, HB . μέσον δὲ τὸ ὑπὸ τῶν AH, HB · μέσον ἄρα ἐστὶ καὶ τὸ $Z\Lambda$. καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν ZM · ῥητὴ ἄρα καὶ ἡ ZM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ αἱ AH, HB δυνάμει μόνον εἰσὶ σύμμετροι, ἀσύμμετρος ἄρα [ἐστὶ] μήκει ἡ AH τῇ HB · ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς AH τῷ ὑπὸ τῶν AH, HB . ἀλλὰ τῷ μὲν ἀπὸ τῆς AH σύμμετρά ἐστὶ τὰ ἀπὸ τῶν AH, HB , τῷ δὲ ὑπὸ τῶν AH, HB τὸ διξ ὑπὸ τῶν AH, HB · ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AH, HB τῷ διξ ὑπὸ τῶν AH, HB . ἀλλὰ τοῖς μὲν ἀπὸ τῶν AH, HB ἴσον ἐστὶ τὸ $\Gamma\Lambda$, τῷ δὲ διξ ὑπὸ τῶν AH, HB ἴσον ἐστὶ τὸ $Z\Lambda$ · ἀσύμμετρον ἄρα ἐστὶ τὸ $\Gamma\Lambda$ τῷ $Z\Lambda$. ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Z\Lambda$, οὕτως ἐστὶν ἡ ΓM πρὸς τὴν ZM · ἀσύμμετρος ἄρα ἐστὶν ἡ ΓM τῇ ZM μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ ἄρα $\Gamma M, MZ$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ ΓZ . λέγω δὴ, ὅτι καὶ τρίτη.

Ἐπεὶ γὰρ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB , σύμμετρον ἄρα καὶ τὸ $\Gamma\Theta$ τῷ $ΚΛ$ · ὥστε καὶ ἡ ΓK τῇ KM . καὶ ἐπεὶ τῶν ἀπὸ τῶν AH, HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH, HB , καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $ΚΛ$, τῷ δὲ ὑπὸ τῶν AH, HB ἴσον τὸ $\Lambda\Lambda$, καὶ τῶν $\Gamma\Theta, ΚΛ$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $\Lambda\Lambda$ · ἐστὶν ἄρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $\Lambda\Lambda$, οὕτως τὸ $\Lambda\Lambda$ πρὸς τὸ $ΚΛ$. ἀλλ' ὡς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $\Lambda\Lambda$, οὕτως ἐστὶν ἡ ΓK πρὸς τὴν NM , ὡς δὲ τὸ $\Lambda\Lambda$ πρὸς τὸ $ΚΛ$, οὕτως ἐστὶν ἡ NM πρὸς τὴν KM · ὡς ἄρα ἡ ΓK πρὸς τὴν MN , οὕτως ἐστὶν ἡ MN πρὸς τὴν KM · τὸ ἄρα ὑπὸ τῶν $\Gamma K, KM$ ἴσον ἐστὶ τῷ [ἀπὸ τῆς MN , τουτέστι τῷ] τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . ἐπεὶ οὖν δύο εὐθεῖαι ἄνισοι εἰσιν αἱ $\Gamma M, MZ$, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM ἴσον παρὰ τὴν ΓM παραβέβληται ἐλλείπον εἶδει τετραγώνῳ καὶ εἰς σύμμετρα αὐτὴν διαιρεῖ, ἡ ΓM ἄρα τῆς MZ μείζον δύναται τῷ ἀπὸ συμέτρου ἐαντῇ. καὶ οὐδετέρα τῶν $\Gamma M, MZ$ σύμμετρος ἐστὶ μήκει τῇ ἐκκειμένη ῥητῇ τῇ $\Gamma\Delta$ · ἡ ἄρα ΓZ ἀποτομή ἐστὶ τρίτη.

Τὸ ἄρα ἀπὸ μέσης ἀποτομῆς δευτέραξ παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην· ὅπερ ἔδει δεῖξαι.

CL is equal to the (sum of the squares) on AG and GB [and the (sum of the squares) on AG and GB is medial]. CL (is) thus also medial [Props. 10.15, 10.23 corr.]. And it has been applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and incommensurable in length with CD [Prop. 10.22]. And since the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB , the remainder LF is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at point N . And let NO be drawn parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And the (rectangle contained) by AG and GB (is) medial. Thus, FL is also medial. And it is applied to the rational (straight-line) EF , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since AG and GB are commensurable in square only, AG [is] thus incommensurable in length with GB . Thus, the (square) on AG is also incommensurable with the (rectangle contained) by AG and GB [Props. 6.1, 10.11]. But, the (sum of the squares) on AG and GB is commensurable with the (square) on AG , and twice the (rectangle contained) by AG and GB with the (rectangle contained) by AG and GB . The (sum of the squares) on AG and GB is thus incommensurable with twice the (rectangle contained) by AG and GB [Prop. 10.13]. But, CL is equal to the (sum of the squares) on AG and GB , and FL is equal to twice the (rectangle contained) by AG and GB . Thus, CL is incommensurable with FL . And as CL (is) to FL , so CM is to FM [Prop. 6.1]. CM is thus incommensurable in length with FM [Prop. 10.11]. And they are both rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a third (apotome).

For since the (square) on AG is commensurable with the (square) on GB , CH (is) thus also commensurable with KL . Hence, CK (is) also (commensurable in length) with KM [Props. 6.1, 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG , and KL equal to the (square) on GB , and NL equal to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM (is) to KM [Prop. 6.1]. Thus, as CK (is) to MN , so MN is to KM [Prop. 5.11]. Thus, the (rectangle contained) by CK and KM is equal to the [(square) on MN —that is to say, to the] fourth part of the (square) on FM [Prop. 6.17]. Therefore, since CM and MF are two unequal straight-lines, and (some area), equal to the fourth part of the (square) on FM , has been applied to CM , falling short by a square figure, and divides it into commensurable (parts),

the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable (in length) with (CM) [Prop. 10.17]. And neither of CM and MF is commensurable in length with the (previously) laid down rational (straight-line) CD . CF is thus a third apotome [Def. 10.13].

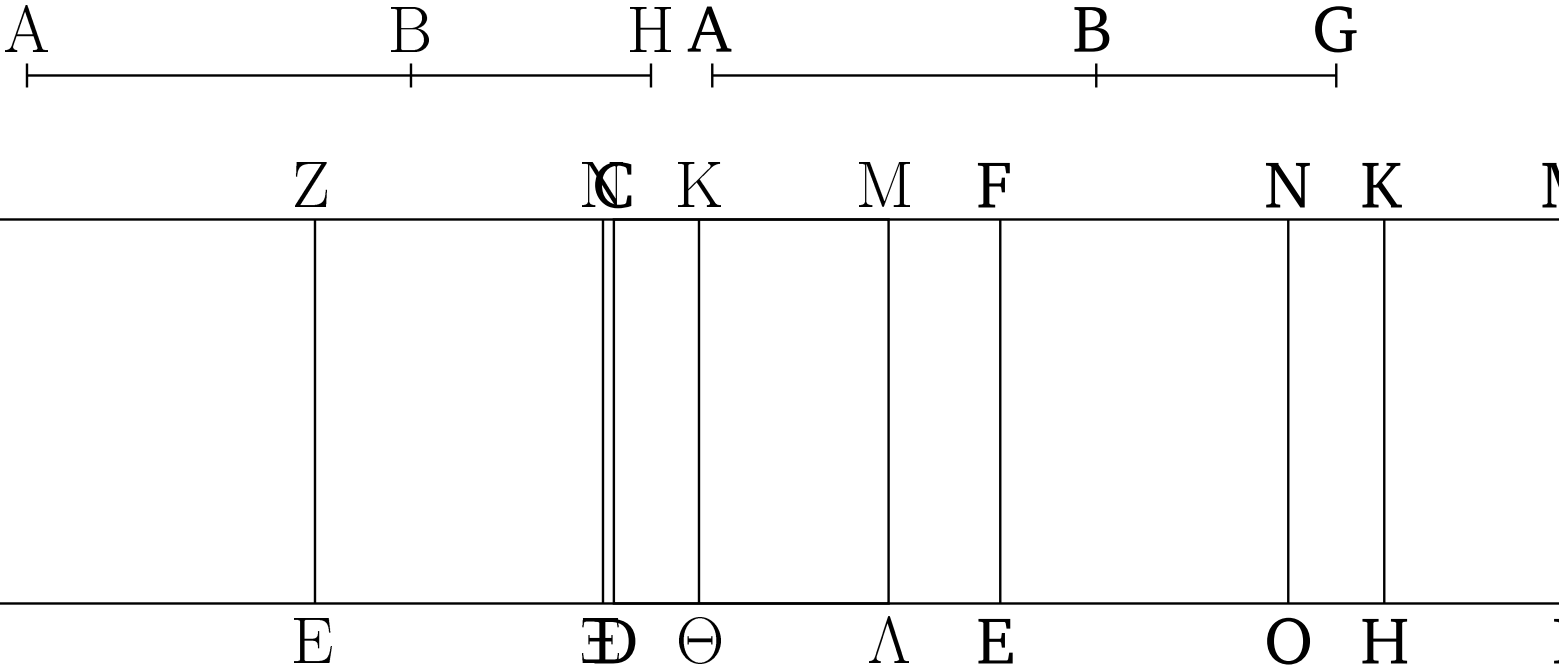
Thus, the (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces a third apotome as breadth. (Which is) the very thing it was required to show.

100.

Proposition 100

Τὸ ἀπὸ ἐλάσσονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν τετάρτην.

The (square) on a minor (straight-line), applied to a rational (straight-line), produces a fourth apotome as breadth.



Ἐστω ἐλάσσων ἡ AB , ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ ῥητὴν τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΕ$ πλάτος ποιοῦν τὴν $ΓΖ$ · λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστὶ τετάρτη.

Ἐστω γάρ τῇ AB προσαρμόζονσα ἡ BH · αἱ ἄρα AH , HB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB τετραγώνων ῥητόν, τὸ δὲ διξ ὑπὸ τῶν AH , HB μέσον. καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΘ$ πλάτος ποιοῦν τὴν $ΓΚ$, τῷ δὲ ἀπὸ τῆς BH ἴσον τὸ $ΚΛ$ πλάτος ποιοῦν τὴν $ΚΜ$ · ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB . καὶ ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB ῥητόν· ῥητόν ἄρα ἐστὶ καὶ τὸ $ΓΛ$. καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παράκειται πλάτος ποιοῦν τὴν $ΓΜ$ · ῥητὴ ἄρα καὶ ἡ $ΓΜ$ καὶ σύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ ὅλον τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , ὧν τὸ $ΓΕ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB , λοιπὸν ἄρα τὸ $ΖΛ$ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AH , HB . τετραγώνω οὖν ἡ $ΖΜ$ δίθω κατὰ τὸ $Ν$ σημείον, καὶ

Let AB be a minor (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to the rational (straight-line) CD , producing CF as breadth. I say that CF is a fourth apotome.

For let BG be an attachment to AB . Thus, AG and GB are incommensurable in square, making the sum of the squares on AG and GB rational, and twice the (rectangle contained) by AG and GB medial [Prop. 10.76]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on BG , producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . And the sum of the (squares) on AG and GB is rational. CL is thus also rational. And it is applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM (is) also rational, and commensurable in length with CD [Prop. 10.20]. And since the whole of CL is

ἤθθω δια τοῦ N ὁποτέρᾳ τῶν $\Gamma\Delta$, $M\Lambda$ παράλληλος ἢ NX · ἑκάτερον ἄρα τῶν ZX , $N\Lambda$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τὸ διζ ὑπὸ τῶν AH , HB μέσον ἐστὶ καὶ ἐστὶν ἴσον τῷ $Z\Lambda$, καὶ τὸ $Z\Lambda$ ἄρα μέσον ἐστίν. καὶ παρὰ ῥητὴν τὴν ZE παράκειται πλάτος ποιῶν τὴν ZM · ῥητὴ ἄρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB ῥητόν ἐστιν, τὸ δὲ διζ ὑπὸ τῶν AH , HB μέσον, ἀσύμμετρον [ἄρα] ἐστὶ τὰ ἀπὸ τῶν AH , HB τῷ διζ ὑπὸ τῶν AH , HB . ἴσον δὲ [ἐστὶ] τὸ $\Gamma\Lambda$ τοῖς ἀπὸ τῶν AH , HB , τῷ δὲ διζ ὑπὸ τῶν AH , HB ἴσον τὸ $Z\Lambda$ · ἀσύμμετρον ἄρα [ἐστὶ] τὸ $\Gamma\Lambda$ τῷ $Z\Lambda$. ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Z\Lambda$, οὕτως ἐστὶν ἡ ΓM πρὸς τὴν MZ · ἀσύμμετρος ἄρα ἐστὶν ἡ ΓM τῇ MZ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ ἄρα ΓM , MZ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ ΓZ . λέγω [δὴ], ὅτι καὶ τετάρτη.

Ἐπεὶ γὰρ αἱ AH , HB δυνάμει εἰσὶν ἀσύμμετροι, ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB . καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $K\Lambda$ · ἀσύμμετρον ἄρα ἐστὶ τὸ $\Gamma\Theta$ τῷ $K\Lambda$. ὥς δὲ τὸ $\Gamma\Theta$ πρὸς τὸ $K\Lambda$, οὕτως ἐστὶν ἡ ΓK πρὸς τὴν KM · ἀσύμμετρος ἄρα ἐστὶν ἡ ΓK τῇ KM μήκει. καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶν ἴσον τὸ μὲν ἀπὸ τῆς AH τῷ $\Gamma\Theta$, τὸ δὲ ἀπὸ τῆς HB τῷ $K\Lambda$, τὸ δὲ ὑπὸ τῶν AH , HB τῷ $N\Lambda$, τῶν ἄρα $\Gamma\Theta$, $K\Lambda$ μέσον ἀνάλογόν ἐστι τὸ $N\Lambda$ · ἴσον ἄρα ὥς τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$. ἀλλ' ὥς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως ἐστὶν ἡ ΓK πρὸς τὴν NM , ὥς δὲ τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$, οὕτως ἐστὶν ἡ NM πρὸς τὴν KM · ὥς ἄρα ἡ ΓK πρὸς τὴν MN , οὕτως ἐστὶν ἡ MN πρὸς τὴν KM · τὸ ἄρα ὑπὸ τῶν ΓK , KM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN , τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . ἐπεὶ οὖν δύο εὐθεῖαι ἄνισοι εἰσὶν αἱ ΓM , MZ , καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς MZ ἴσον παρὰ τὴν ΓM παραβέβληται ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν ΓK , KM καὶ εἷς ἀσύμμετρον αὐτὴν διαιρεῖ, ἡ ἄρα ΓM τῆς MZ μείζον δύναιται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ ἐστὶν ὅλη ἡ ΓM σύμμετρος μήκει τῇ ἐκκειμένη ῥητῇ τῇ $\Gamma\Delta$ · ἡ ἄρα ΓZ ἀποτομή ἐστὶ τετάρτη.

Τὸ ἄρα ἀπὸ ἐλάσσονος καὶ τὰ ἐχθξ.

equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB , the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at point N . And let NO be drawn through N , parallel to either of CD or ML . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since twice the (rectangle contained) by AG and GB is medial, and is equal to FL , FL is thus also medial. And it is applied to the rational (straight-line) FE , producing FM as breadth. Thus, FM is rational, and incommensurable in length with CD [Prop. 10.22]. And since the sum of the (squares) on AG and GB is rational, and twice the (rectangle contained) by AG and GB medial, the (sum of the squares) on AG and GB is [thus] incommensurable with twice the (rectangle contained) by AG and GB . And CL (is) equal to the (sum of the squares) on AG and GB , and FL equal to twice the (rectangle contained) by AG and GB . CL [is] thus incommensurable with FL . And as CL (is) to FL , so CM is to MF [Prop. 6.1]. CM is thus incommensurable in length with MF [Prop. 10.11]. And both are rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. [So], I say that (it is) also a fourth (apotome).

For since AG and GB are incommensurable in square, the (square) on AG (is) thus also incommensurable with the (square) on GB . And CH is equal to the (square) on AG , and KL equal to the (square) on GB . Thus, CH is incommensurable with KL . And as CH (is) to KL , so CK is to KM [Prop. 6.1]. CK is thus incommensurable in length with KM [Prop. 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and the (square) on AG is equal to CH , and the (square) on GB to KL , and the (rectangle contained) by AG and GB to NL , NL is thus the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to KM [Prop. 6.1]. Thus, as CK (is) to NM , so NM is to KM [Prop. 5.11]. The (rectangle contained) by CK and KM is thus equal to the (square) on MN —that is to say, to the fourth part of the (square) on FM [Prop. 6.17]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on MF , has been applied to CM , falling short by a square figure, and divides it into incommensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And the whole of CM is commensurable in length with the (previously) laid down rational (straight-line) CD . Thus, CF is a fourth apotome [Def. 10.14].

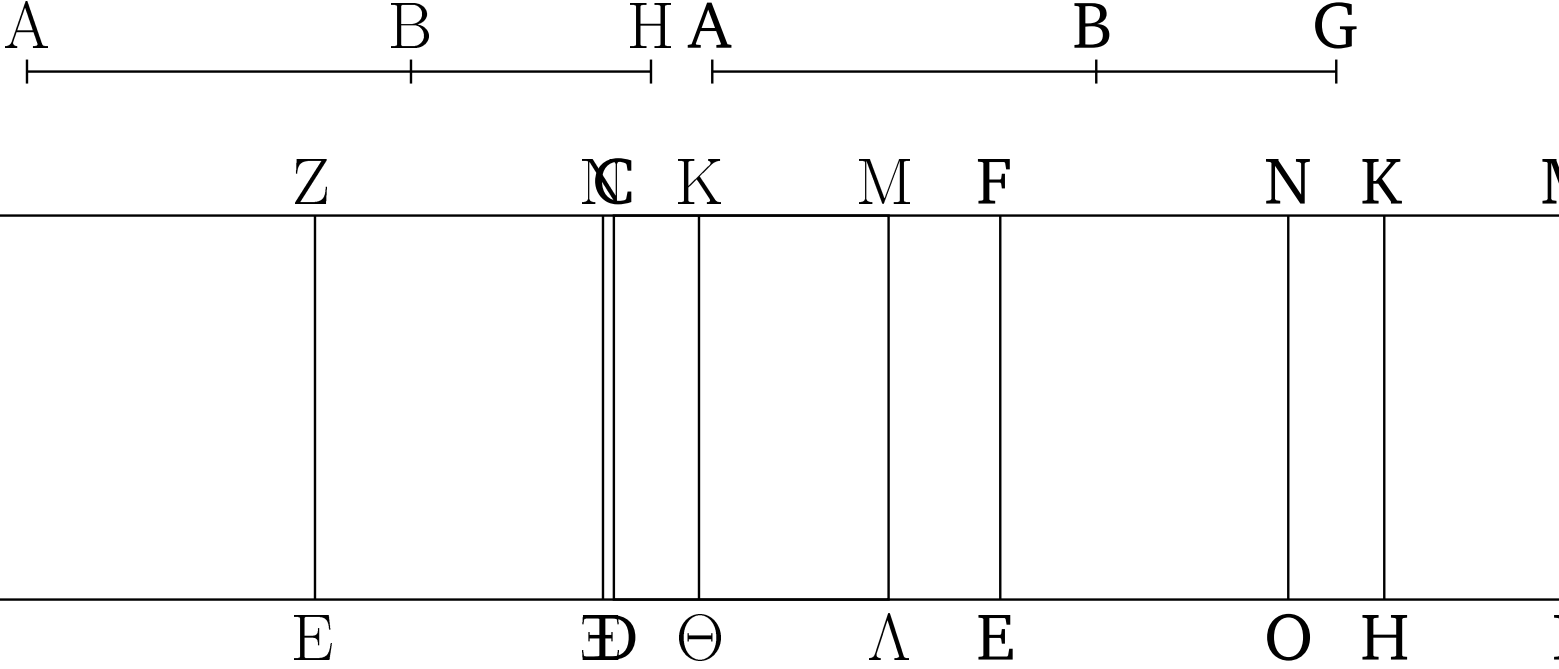
Thus, the (square) on a minor, and so on ...

101.

Proposition 101

Τὸ ἀπὸ τῆς μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσῃ παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν πέμπτην.

The (square) on that (straight-line) which with a rational (area) makes a medial whole, applied to a rational (straight-line), produces a fifth apotome as breadth.



Ἐστω ἡ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἡ AB , ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΕ$ πλάτος ποιοῦν τὴν $ΓΖ$ · λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστι πέμπτη.

Ἐστω γάρ τῃ AB προσαρμόζονσα ἡ BH · αἱ ἄρα AH , HB εὐθεῖαι δυνάμει εἰσὶν ἀσύμμετροι ποιούσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διξὺ ὑπ' αὐτῶν ῥητόν, καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΘ$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $ΚΛ$ · ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB . τὸ δὲ συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB ἅμα μέσον ἐστὶν μέσον ἄρα ἐστὶ τὸ $ΓΛ$. καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παράκειται πλάτος ποιοῦν τὴν $ΓΜ$ · ῥητὴ ἄρα ἐστὶν ἡ $ΓΜ$ καὶ ἀσύμμετρος τῇ $ΓΔ$. καὶ ἐπεὶ ὅλον τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , ὧν τὸ $ΓΕ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB , λοιπὸν ἄρα τὸ $ΖΛ$ ἴσον ἐστὶ τῷ διξὺ ὑπὸ τῶν AH , HB . τετμήσθω οὖν ἡ ZM διὰ κατὰ τὸ N , καὶ ἤθθω διὰ τοῦ N ὁποτέρῃ τῶν $ΓΔ$, $ΜΛ$ παράλληλος ἡ NX · ἐκάτερον ἄρα τῶν ZX , $NΛ$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB , καὶ ἐπεὶ τὸ διξὺ ὑπὸ τῶν AH , HB ῥητόν ἐστι καὶ [ἐστίν] ἴσον τῷ $ΖΛ$, ῥητόν ἄρα ἐστὶ τὸ $ΖΛ$. καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν ZM · ῥητὴ ἄρα ἐστὶν ἡ ZM καὶ σύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ τὸ μὲν $ΓΛ$ μέσον ἐστὶν, τὸ δὲ $ΖΛ$ ῥητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ $ΓΛ$ τῷ $ΖΛ$. ὥς δὲ τὸ $ΓΛ$ πρόξ τὸ $ΖΛ$, οὕτως ἡ $ΓΜ$ πρόξ τὴν MZ · ἀσύμμετρος ἄρα ἐστὶν ἡ $ΓΜ$ τῇ MZ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ ἄρα $ΓΜ$,

Let AB be that (straight-line) which with a rational (area) makes a medial whole, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a fifth apotome.

Let BG be an attachment to AB . Thus, the straight-lines AG and GB are incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by them rational [Prop. 10.77]. And let CH , equal to the (square) on AG , be applied to CD , and KL , equal to the (square) on GB . The whole of CL is thus equal to the (sum of the squares) on AG and GB . And the sum of the (squares) on AG and GB together is medial. Thus, CL is medial. And it has been applied to the rational (straight-line) CD , producing CM as breadth. CM is thus rational, and incommensurable (in length) with CD [Prop. 10.22]. And since the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB , the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at N . And let NO be drawn through N , parallel to either of CD or ML . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since twice the (rectangle contained) by AG and GB is rational, and [is] equal to FL , FL is thus rational. And it is applied to the rational (straight-line) EF , producing FM as breadth. Thus, FM is rational, and commensurable in length with CD .

MZ ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἢ FZ . λέγω δὴ, ὅτι καὶ πέμπτη.

Ὅμοίως γὰρ δείχομεν, ὅτι τὸ ὑπὸ τῶν $ΓΚΜ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς NM , τοντέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB , ἴσον δὲ τὸ μὲν ἀπὸ τῆς AH τῷ $ΓΘ$, τὸ δὲ ἀπὸ τῆς HB τῷ $ΚΛ$, ἀσύμμετρον ἄρα τὸ $ΓΘ$ τῷ $ΚΛ$. ὥς δὲ τὸ $ΓΘ$ πρὸς τὸ $ΚΛ$, οὕτως ἢ $ΓΚ$ πρὸς τὴν KM · ἀσύμμετρος ἄρα ἢ $ΓΚ$ τῇ KM μήκει. ἐπεὶ οὖν δύο εὐθεῖαι ἄνισοί εἰσιν αἱ $ΓΜ$, MZ , καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM ἴσον παρὰ τὴν $ΓΜ$ παραβέβληται ἐλλείπον εἶδει τετραγώνῳ καὶ εἰς ἀσύμμετρα αὐτὴν διαιρεῖ, ἢ ἄρα $ΓΜ$ τῆς MZ μείζον δύναται τῷ ἀπὸ ἀσύμμετρον ἐαντῇ. καὶ ἐστὶν ἢ προσαρμόζουσα ἢ ZM σύμμετρος τῇ ἐκκειμένῃ ῥητῇ τῇ $ΓΔ$ · ἢ ἄρα $ΓΖ$ ἀποτομή ἐστὶ πέμπτη· ὅπερ ἔδει δεῖχαι.

[Prop. 10.20]. And since CL is medial, and FL rational, CL is thus incommensurable with FL . And as CL (is) to FL , so CM (is) to MF [Prop. 6.1]. CM is thus incommensurable in length with MF [Prop. 10.11]. And both are rational. Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a fifth (apotome).

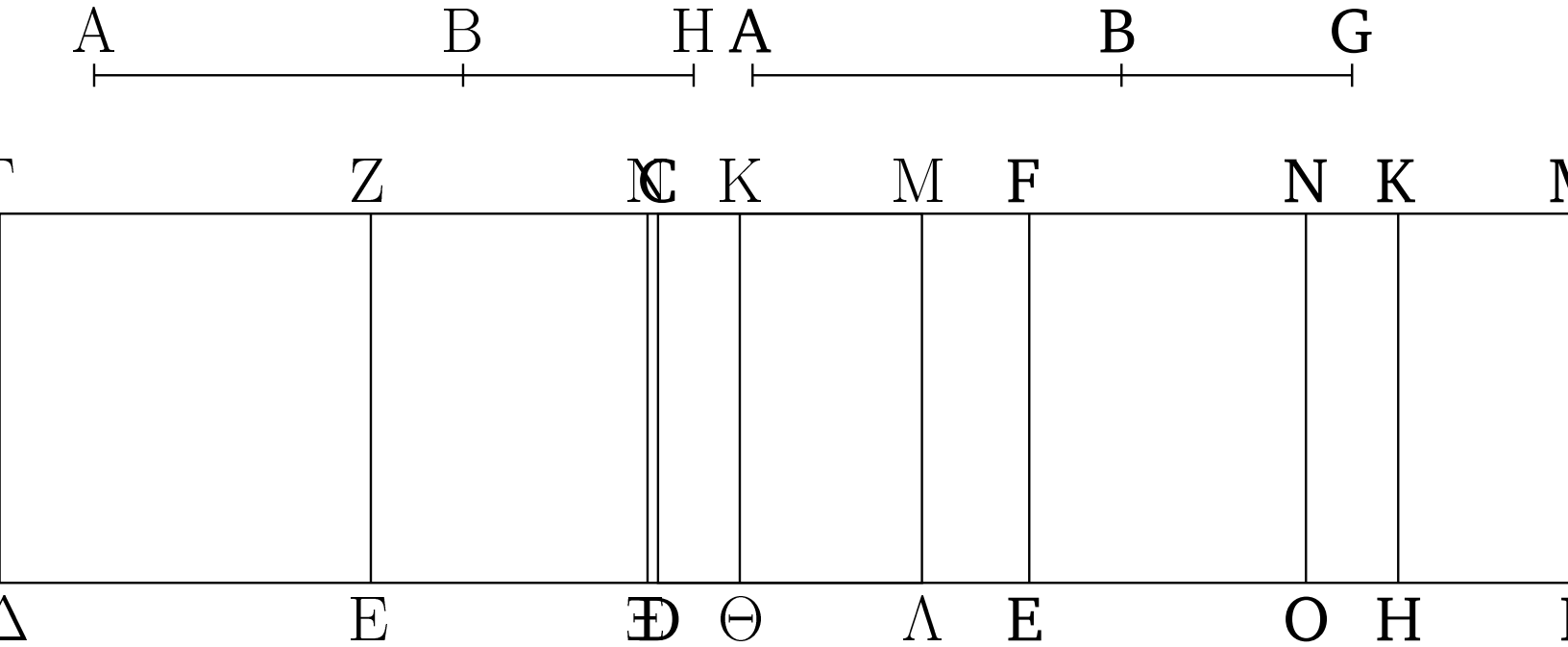
For, similarly (to the previous propositions), we can show that the (rectangle contained) by CKM is equal to the (square) on NM —that is to say, to the fourth part of the (square) on FM . And since the (square) on AG is incommensurable with the (square) on GB , and the (square) on AG (is) equal to CH , and the (square) on GB to KL , CH (is) thus incommensurable with KL . And as CH (is) to KL , so CK (is) to KM [Prop. 6.1]. Thus, CK (is) incommensurable in length with KM [Prop. 10.11]. Therefore, since CM and MF are two unequal straight-lines, and (some area), equal to the fourth part of the (square) on FM , has been applied to CM , falling short by a square figure, and divides it into incommensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And the attachment FM is commensurable with the (previously) laid down rational (straight-line) CD . Thus, CF is a fifth apotome [Def. 10.15]. (Which is) the very thing it was required to show.

102.

Τὸ ἀπὸ τῆς μετὰ μέσον μέσον τὸ ὅλον ποιούσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν ἑκτὴν.

Proposition 102

The (square) on that (straight-line) which with a medial (area) makes a medial whole, applied to a rational (straight-line), produces a sixth apotome as breadth.



Ἐστω ἡ μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἡ AB , ῥητὴ δὲ ἡ $\Gamma\Delta$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω τὸ ΓE πλάτος ποιῶν τὴν ΓZ · λέγω, ὅτι ἡ ΓZ ἀποτομή ἐστὶν ἕκτη.

Ἐστω γάρ τῃ AB προσαρμόζονσα ἡ BH · αἱ ἄρα AH , HB δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ διζὺν ὑπὸ τῶν AH , HB μέσον καὶ ἀσύμμετρον τὰ ἀπὸ τῶν AH , HB τῷ διζὺν ὑπὸ τῶν AH , HB · παραβεβλήσθω οὖν παρὰ τὴν $\Gamma\Delta$ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$ πλάτος ποιῶν τὴν ΓK , τῷ δὲ ἀπὸ τῆς BH τὸ $ΚΛ$ · ὅλον ἄρα τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB · μέσον ἄρα [ἐστὶ] καὶ τὸ $\Gamma\Lambda$ · καὶ παρὰ ῥητὴν τὴν $\Gamma\Delta$ παράκειται πλάτος ποιῶν τὴν ΓM · ῥητὴ ἄρα ἐστὶν ἡ ΓM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει· ἐπεὶ οὖν τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , ὣν τὸ ΓE ἴσον τῷ ἀπὸ τῆς AB , λοιπὸν ἄρα τὸ $Z\Lambda$ ἴσον ἐστὶ τῷ διζὺν ὑπὸ τῶν AH , HB · καὶ ἐστὶ τὸ διζὺν ὑπὸ τῶν AH , HB μέσον καὶ τὸ $Z\Lambda$ ἄρα μέσον ἐστίν· καὶ παρὰ ῥητὴν τὴν ZE παράκειται πλάτος ποιῶν τὴν ZM · ῥητὴ ἄρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει· καὶ ἐπεὶ τὰ ἀπὸ τῶν AH , HB ἀσύμμετρά ἐστι τῷ διζὺν ὑπὸ τῶν AH , HB , καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν AH , HB ἴσον τὸ $\Gamma\Lambda$, τῷ δὲ διζὺν ὑπὸ τῶν AH , HB ἴσον τὸ $Z\Lambda$, ἀσύμμετρος ἄρα [ἐστὶ] τὸ $\Gamma\Lambda$ τῷ $Z\Lambda$ · ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Z\Lambda$, οὕτως ἐστὶν ἡ ΓM πρὸς τὴν MZ · ἀσύμμετρος ἄρα ἐστὶν ἡ ΓM τῇ MZ μήκει· καὶ εἰσὶν ἀμφότεραι ῥηταί· αἱ ΓM , MZ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ ΓZ · λέγω δὴ, ὅτι καὶ ἕκτη.

Ἐπεὶ γάρ τὸ $Z\Lambda$ ἴσον ἐστὶ τῷ διζὺν ὑπὸ τῶν AH , HB , τετμήσθω δίθια ἡ ZM κατὰ τὸ N , καὶ ἡθθω διὰ τοῦ N τῇ $\Gamma\Delta$ παράλληλος ἡ NX · ἐκάτερον ἄρα τῶν ZX , NA ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB · καὶ ἐπεὶ αἱ AH , HB δυνάμει εἰσὶν ἀσύμμε-

Let AB be that (straight-line) which with a medial (area) makes a medial whole, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a sixth apotome.

For let BG be an attachment to AB . Thus, AG and GB are incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by AG and GB medial, and the (sum of the squares) on AG and GB incommensurable with twice the (rectangle contained) by AG and GB [Prop. 10.78]. Therefore, let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on BG . Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . CL [is] thus also medial. And it is applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and incommensurable in length with CD [Prop. 10.22]. Therefore, since CL is equal to the (sum of the squares) on AG and GB , of which CE (is) equal to the (square) on AB , the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. And twice the (rectangle contained) by AG and GB (is) medial. Thus, FL is also medial. And it is applied to the rational (straight-line) FE , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since the (sum of the squares) on AG and GB is incommensurable with twice the (rectangle contained) by AG and GB , and CL equal to the (sum of the squares) on AG and GB , and FL equal to twice the (rectangle contained) by AG and GB , CL [is] thus incommensurable with FL . And as CL (is) to FL , so CM is to MF [Prop. 6.1]. Thus, CM is incommensurable in length with MF [Prop. 10.11]. And they

τροι, ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB . ἀλλὰ τῷ μὲν ἀπὸ τῆς AH ἴσον ἐστὶ τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον ἐστὶ τὸ $ΚΛ$. ἀσύμμετρον ἄρα ἐστὶ τὸ $\Gamma\Theta$ τῷ $ΚΛ$. ὥς δὲ τὸ $\Gamma\Theta$ πρὸς τὸ $ΚΛ$, οὕτως ἐστὶν ἡ $\GammaΚ$ πρὸς τὴν $ΚΜ$. ἀσύμμετρος ἄρα ἐστὶν ἡ $\GammaΚ$ τῇ $ΚΜ$. καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $ΚΛ$, τῷ δὲ ὑπὸ τῶν AH , HB ἴσον τὸ $ΝΛ$, καὶ τῶν ἄρα $\Gamma\Theta$, $ΚΛ$ μέσον ἀνάλογόν ἐστι τὸ $ΝΛ$. ἐστὶν ἄρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $ΝΛ$, οὕτως τὸ $ΝΛ$ πρὸς τὸ $ΚΛ$. καὶ διὰ τὰ αὐτὰ ἡ $\GammaΜ$ τῆς $ΜΖ$ μείζον δύνανται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ. καὶ οὐδετέρω αὐτῶν σύμμετρος ἐστὶ τῇ ἐκκειμένη ῥητῇ τῇ $\GammaΔ$. ἡ $\GammaΖ$ ἄρα ἀποτομή ἐστὶν ἕκτη· ὅπερ εἶδει δεῖχαι.

are both rational. Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a sixth (apotome).

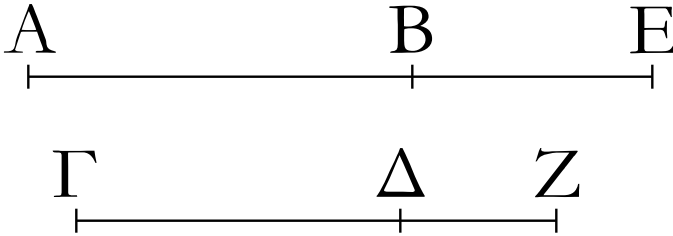
For since FL is equal to twice the (rectangle contained) by AG and GB , let FM be cut in half at N , and let NO be drawn through N , parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since AG and GB are incommensurable in square, the (square) on AG is thus incommensurable with the (square) on GB . But, CH is equal to the (square) on AG , and KL is equal to the (square) on GB . Thus, CH is incommensurable with KL . And as CH (is) to KL , so CK is to KM [Prop. 6.1]. Thus, CK is incommensurable (in length) with KM [Prop. 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG , and KL equal to the (square) on GB , and NL equal to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . And for the same (reasons as the preceding propositions), the square on CM is greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And neither of them is commensurable with the (previously) laid down rational (straight-line) CD . Thus, CF is a sixth apotome [Def. 10.16]. (Which is) the very thing it was required to show.

103.

Proposition 103

Ἡ τῇ ἀποτομῇ μήκει σύμμετρος ἀποτομή ἐστὶ καὶ τῇ τάξει ἡ αὐτή.

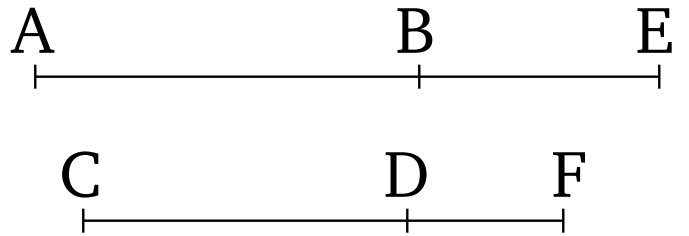
Ἐστω ἀποτομή ἡ AB , καὶ τῇ AB μήκει σύμμετρος ἔστω ἡ $\GammaΔ$. λέγω, ὅτι καὶ ἡ $\GammaΔ$ ἀποτομή ἐστὶ καὶ τῇ τάξει ἡ αὐτή τῇ AB .



Ἐπεὶ γὰρ ἀποτομή ἐστὶν ἡ AB , ἔστω αὐτῇ προσαρμόζουσα ἡ BE . αἱ AE , EB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ τῆς AB πρὸς τὴν $\GammaΔ$ λόγῳ ὁ αὐτὸς γερονέτω ὁ τῆς BE πρὸς τὴν $\Delta Ζ$. καὶ ὥς ἐν ἄρα πρὸς ἓν, πάντα [ἐστὶ] πρὸς πάντα. ἐστὶν ἄρα καὶ ὥς ὅλη ἡ AE πρὸς ὅλην τὴν $\GammaΖ$, οὕτως ἡ AB πρὸς τὴν $\GammaΔ$. σύμμετρος δὲ ἡ AB τῇ $\GammaΔ$ μήκει. σύμμετρος ἄρα καὶ ἡ AE μὲν τῇ $\GammaΖ$, ἡ δὲ BE τῇ $\Delta Ζ$. καὶ αἱ AE , EB ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ αἱ $\GammaΖ$, $\Delta Ζ$

A (straight-line) commensurable in length with an apotome is an apotome, and (is) the same in order.

Let AB be an apotome, and let CD be commensurable in length with AB . I say that CD is also an apotome, and (is) the same in order as AB .



For since AB is an apotome, let BE be an attachment to it. Thus, AE and EB are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let it be contrived that the (ratio) of BE to DF is the same as the ratio of AB to CD [Prop. 6.12]. Thus, also, as one is to one, (so) all [are] to all [Prop. 5.12]. And thus as the whole AE is to the whole CF , so AB (is) to CD . And AB (is) commensurable in length with CD . AE (is) thus also commensurable (in

ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι [ἀποτομή ἄρα ἐστὶν ἡ ΓΔ. λέγω δὴ, ὅτι καὶ τῇ τάχει ἡ αὐτὴ τῇ ΑΒ].

Ἐπεὶ οὖν ἐστὶν ὡς ἡ ΑΕ πρὸς τὴν ΓΖ, οὕτως ἡ ΒΕ πρὸς τὴν ΔΖ, ἐναλλάξ ἄρα ἐστὶν ὡς ἡ ΑΕ πρὸς τὴν ΕΒ, οὕτως ἡ ΓΖ πρὸς τὴν ΖΔ. ἦτοι δὴ ἡ ΑΕ τῇ ΕΒ μείζον δύνανται τῷ ἀπὸ συμμέτρου ἐαντῇ ἢ τῷ ἀπὸ ἀσυνμέτρου. εἰ μὲν οὖν ἡ ΑΕ τῇ ΕΒ μείζον δύνανται τῷ ἀπὸ συμμέτρου ἐαντῇ, καὶ ἡ ΓΖ τῇ ΖΔ μείζον δυνήσεται τῷ ἀπὸ συμμέτρου ἐαντῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ ΑΕ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ ΓΖ, εἰ δὲ ἡ ΒΕ, καὶ ἡ ΔΖ, εἰ δὲ οὐδετέρα τῶν ΑΕ, ΕΒ, καὶ οὐδετέρα τῶν ΓΖ, ΖΔ. εἰ δὲ ἡ ΑΕ [τῇ ΕΒ] μείζον δύνανται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ, καὶ ἡ ΓΖ τῇ ΖΔ μείζον δυνήσεται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ ΑΕ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ ΓΖ, εἰ δὲ ἡ ΒΕ, καὶ ἡ ΔΖ, εἰ δὲ οὐδετέρα τῶν ΑΕ, ΕΒ, οὐδετέρα τῶν ΓΖ, ΖΔ.

Ἀποτομή ἄρα ἐστὶν ἡ ΓΔ καὶ τῇ τάχει ἡ αὐτὴ τῇ ΑΒ· ὅπερ ἔδει δεῖξαι.

length) with CF , and BE with DF [Prop. 10.11]. And AE and BE are rational (straight-lines which are) commensurable in square only. Thus, CF and FD are also rational (straight-lines which are) commensurable in square only [Prop. 10.13]. [CD is thus an apotome. So, I say that (it is) also the same in order as AB .]

Therefore, since as AE is to CF , so BE (is) to DF , thus, alternately, as AE is to EB , so CF (is) to FD [Prop. 5.16]. So, the square on AE is greater than (the square on) EB either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (AE). Therefore, if the (square) on AE is greater than (the square on) EB by the (square) on (some straight-line) commensurable (in length) with (AE) then the square on CF will also be greater than (the square on) FD by the (square) on (some straight-line) commensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable in length with a (previously) laid down rational (straight-line) then so (is) CF [Prop. 10.12], and if BE (is commensurable), so (is) DF , and if neither of AE or EB (are commensurable), neither (are) either of CF or FD [Prop. 10.13]. And if the (square) on AE is greater [than (the square on) EB] by the (square) on (some straight-line) incommensurable (in length) with (AE) then the (square) on CF will also be greater than (the square on) FD by the (square) on (some straight-line) incommensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable in length with a (previously) laid down rational (straight-line), so (is) CF [Prop. 10.12], and if BE (is commensurable), so (is) DF , and if neither of AE or EB (are commensurable), neither (are) either of CF or FD [Prop. 10.13].

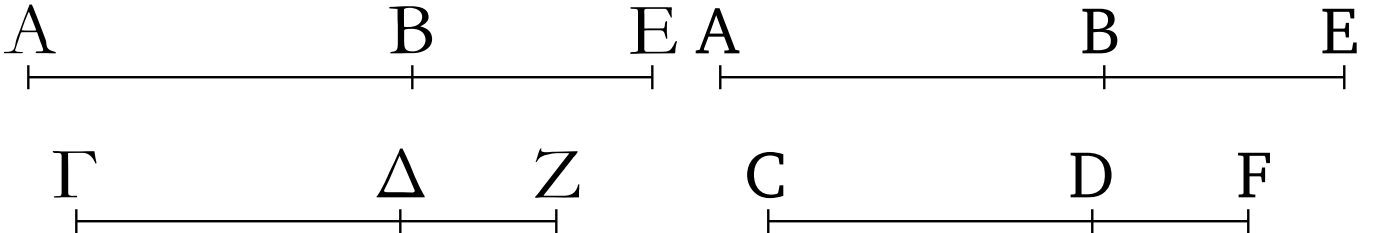
Thus, CD is an apotome, and (is) the same in order as AB [Defs. 10.11—10.16]. (Which is) the very thing it was required to show.

104.

Proposition 104

Ἡ τῇ μέσῃ ἀποτομῇ σύμμετρος μέσῃ ἀποτομή ἐστὶ καὶ τῇ τάχει ἡ αὐτή.

A (straight-line) commensurable (in length) with an apotome of a medial (straight-line) is an apotome of a medial (straight-line), and (is) the same in order.



Ἐστω μέσῃ ἀποτομῇ ἡ ΑΒ, καὶ τῇ ΑΒ μήκει σύμμετρος ἔστω ἡ ΓΔ· λέγω, ὅτι καὶ ἡ ΓΔ μέσῃ ἀποτομῇ ἐστὶ καὶ τῇ τάχει ἡ αὐτὴ τῇ ΑΒ.

Ἐπεὶ γὰρ μέσῃ ἀποτομῇ ἐστὶν ἡ ΑΒ, ἔστω αὐτῇ προσ-

Let AB be an apotome of a medial (straight-line), and let CD be commensurable in length with AB . I say that CD is also an apotome of a medial (straight-line), and (is) the same in order as AB .

αρμόζονσα ἡ EB . αἱ AE , EB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ γερονέτω ὡς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ BE πρὸς τὴν ΔZ . σύμμετρος ἄρα [ἐστὶ] καὶ ἡ AE τῇ ΓZ , ἡ δὲ BE τῇ ΔZ . αἱ δὲ AE , EB μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· καὶ αἱ ΓZ , $Z\Delta$ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· μέσης ἄρα ἀποτομή ἐστὶν ἡ $\Gamma\Delta$. λέγω δὴ, ὅτι καὶ τῇ τάχει ἐστὶν ἡ αὐτὴ τῇ AB .

Ἐπεὶ [γάρ] ἐστὶν ὡς ἡ AE πρὸς τὴν EB , οὕτως ἡ ΓZ πρὸς τὴν $Z\Delta$ [ἀλλ' ὡς μὲν ἡ AE πρὸς τὴν EB , οὕτως τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE , EB , ὡς δὲ ἡ ΓZ πρὸς τὴν $Z\Delta$, οὕτως τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν ΓZ , $Z\Delta$], ἐστὶν ἄρα καὶ ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE , EB , οὕτως τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν ΓZ , $Z\Delta$ [καὶ ἐναλλάχῃ ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς ΓZ , οὕτως τὸ ὑπὸ τῶν AE , EB πρὸς τὸ ὑπὸ τῶν ΓZ , $Z\Delta$]. σύμμετρον δὲ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς ΓZ · σύμμετρον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν AE , EB τῷ ὑπὸ τῶν ΓZ , $Z\Delta$. εἴτε οὖν ῥητόν ἐστι τὸ ὑπὸ τῶν AE , EB , ῥητόν ἐστὶ καὶ τὸ ὑπὸ τῶν ΓZ , $Z\Delta$, εἴτε μέσον [ἐστὶ] τὸ ὑπὸ τῶν AE , EB , μέσον [ἐστὶ] καὶ τὸ ὑπὸ τῶν ΓZ , $Z\Delta$.

Μέσης ἄρα ἀποτομή ἐστὶν ἡ $\Gamma\Delta$ καὶ τῇ τάχει ἡ αὐτὴ τῇ AB . ὅπερ ἔδει δεῖχαι.

For since AB is an apotome of a medial (straight-line), let EB be an attachment to it. Thus, AE and EB are medial (straight-lines which are) commensurable in square only [Props. 10.74, 10.75]. And let it be contrived that as AB is to CD , so BE (is) to DF [Prop. 6.12]. Thus, AE [is] also commensurable (in length) with CF , and BE with DF [Props. 5.12, 10.11]. And AE and EB are medial (straight-lines which are) commensurable in square only. CF and FD are thus also medial (straight-lines which are) commensurable in square only [Props. 10.23, 10.13]. Thus, CD is an apotome of a medial (straight-line) [Props. 10.74, 10.75]. So, I say that it is also the same in order as AB .

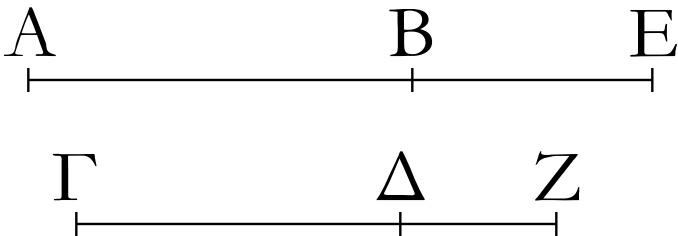
[For] since as AE is to EB , so CF (is) to FD [Props. 5.12, 5.16] [but as AE (is) to EB , so the (square) on AE (is) to the (rectangle contained) by AE and EB , and as CF (is) to FD , so the (square) on CF (is) to the (rectangle contained) by CF and FD], thus as the (square) on AE is to the (rectangle contained) by AE and EB , so the (square) on CF also (is) to the (rectangle contained) by CF and FD [Prop. 10.21 lem.] [and, alternately, as the (square) on AE (is) to the (square) on CF , so the (rectangle contained) by AE and EB (is) to the (rectangle contained) by CF and FD]. And the (square) on AE (is) commensurable with the (square) on CF . Thus, the (rectangle contained) by AE and EB is also commensurable with the (rectangle contained) by CF and FD [Props. 5.16, 10.11]. Therefore, either the (rectangle contained) by AE and EB is rational, and the (rectangle contained) by CF and FD will also be rational [Def. 10.4], or the (rectangle contained) by AE and EB [is] medial, and the (rectangle contained) by CF and FD [is] also medial [Prop. 10.23 corr.].

Therefore, CD is the apotome of a medial (straight-line), and is the same in order as AB [Props. 10.74, 10.75]. (Which is) the very thing it was required to show.

105.

Ἡ τῇ ἐλάσσονι σύμμετρος ἐλάσσων ἐστίν.

Ἐστω γὰρ ἐλάσσων ἡ AB καὶ τῇ AB σύμμετρος ἡ $\Gamma\Delta$. λέγω, ὅτι καὶ ἡ $\Gamma\Delta$ ἐλάσσων ἐστίν.

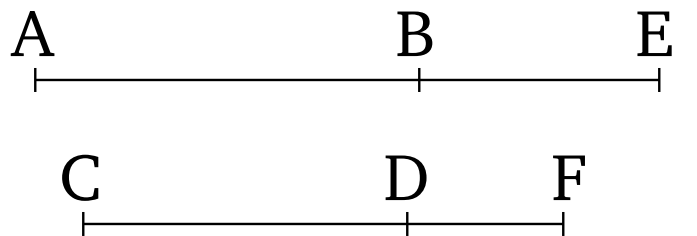


Γερονέτω γὰρ τὰ αὐτά· καὶ ἐπεὶ αἱ AE , EB δυνάμει εἰσὶν ἀσύμμετροι, καὶ αἱ ΓZ , $Z\Delta$ ἄρα δυνάμει εἰσὶν ἀσύμμετροι.

Proposition 105

A (straight-line) commensurable (in length) with a minor (straight-line) is a minor (straight-line).

For let AB be a minor (straight-line), and (let) CD (be) commensurable (in length) with AB . I say that CD is also a minor (straight-line).



For let the same things be contrived (as in the former proposition). And since AE and EB are (straight-lines which

ἐπεὶ οὖν ἔστιν ὥς ἡ AE πρὸς τὴν EB , οὕτως ἡ GZ πρὸς τὴν ZD , ἔστιν ἄρα καὶ ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς EB , οὕτως τὸ ἀπὸ τῆς GZ πρὸς τὸ ἀπὸ τῆς ZD . συνθέντι ἄρα ἔστιν ὥς τὰ ἀπὸ τῶν AE , EB πρὸς τὸ ἀπὸ τῆς EB , οὕτως τὰ ἀπὸ τῶν GZ , ZD πρὸς τὸ ἀπὸ τῆς ZD [καὶ ἐναλλάχ]. σύμμετρον δὲ ἔστι τὸ ἀπὸ τῆς BE τῷ ἀπὸ τῆς ΔZ · σύμμετρον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν GZ , ZD τετραγώνων. ῥητόν δὲ ἔστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων· ῥητόν ἄρα ἔστι καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν GZ , ZD τετραγώνων. πάλιν, ἐπεὶ ἔστιν ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE , EB , οὕτως τὸ ἀπὸ τῆς GZ πρὸς τὸ ὑπὸ τῶν GZ , ZD , σύμμετρον δὲ τὸ ἀπὸ τῆς AE τετράγωνον τῷ ἀπὸ τῆς GZ τετραγώνῳ, σύμμετρον ἄρα ἔστι καὶ τὸ ὑπὸ τῶν AE , EB τῷ ὑπὸ τῶν GZ , ZD . μέσον δὲ τὸ ὑπὸ τῶν AE , EB · μέσον ἄρα καὶ τὸ ὑπὸ τῶν GZ , ZD · αἱ GZ , ZD ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον.

Ἐλάσσων ἄρα ἔστιν ἡ $\Gamma\Delta$ · ὅπερ ἔδει δεῖλαι.

106.

Ἡ τῇ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσῃ σύμμετρος μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσά ἐστιν.

Ἐστω μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἡ AB καὶ τῇ AB σύμμετρος ἡ $\Gamma\Delta$ · λέγω, ὅτι καὶ ἡ $\Gamma\Delta$ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσά ἐστιν.

Ἐστω γὰρ τῇ AB προσαρμόζουσα ἡ BE · αἱ AE , EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν. καὶ τὰ αὐτὰ κατεσκευάσθω. ὁμοίως δὲ δείχομεν τοῖς πρότερον, ὅτι αἱ GZ , ZD ἐν τῷ αὐτῷ λόγῳ εἰσὶ ταῖς AE , EB , καὶ σύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν GZ , ZD τετραγώνων, τὸ δὲ ὑπὸ τῶν AE , EB τῷ ὑπὸ τῶν GZ , ZD · ὥστε καὶ αἱ GZ , ZD δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν GZ , ZD τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν.

are) incommensurable in square [Prop. 10.76], CF and FD are thus also (straight-lines which are) incommensurable in square [Prop. 10.13]. Therefore, since as AE is to EB , so CF (is) to FD [Props. 5.12, 5.16], thus also as the (square) on AE is to the (square) on EB , so the (square) on CF (is) to the (square) on FD [Prop. 6.22]. Thus, via composition, as the (sum of the squares) on AE and EB is to the (square) on EB , so the (sum of the squares) on CF and FD (is) to the (square) on FD [Prop. 5.18], [also alternately]. And the (square) on BE is commensurable with the (square) on DF [Prop. 10.104]. The sum of the squares on AE and EB (is) thus also commensurable with the sum of the squares on CF and FD [Prop. 5.16, 10.11]. And the sum of the (squares) on AE and EB is rational [Prop. 10.76]. Thus, the sum of the (squares) on CF and FD is also rational [Def. 10.4]. Again, since as the (square) on AE is to the (rectangle contained) by AE and EB , so the (square) on CF (is) to the (rectangle contained) by CF and FD [Prop. 10.21 lem.], and the square on AE (is) commensurable with the square on CF , the (rectangle contained) by AE and EB is thus also commensurable with the (rectangle contained) by CF and FD . And the (rectangle contained) by AE and EB (is) medial [Prop. 10.76]. Thus, the (rectangle contained) by CF and FD (is) also medial [Prop. 10.23 corr.]. CF and FD are thus (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial.

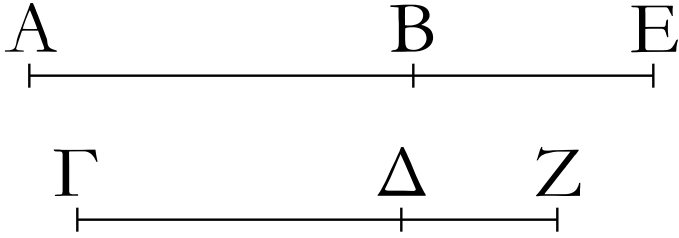
Thus, CD is a minor (straight-line) [Prop. 10.76]. (Which is) the very thing it was required to show.

Proposition 106

A (straight-line) commensurable (in length) with a (straight-line) which with a rational (area) makes a medial whole is a (straight-line) which with a rational (area) makes a medial whole.

Let AB be a (straight-line) which with a rational (area) makes a medial whole, and (let) CD (be) commensurable (in length) with AB . I say that CD is also a (straight-line) which with a rational (area) makes a medial (whole).

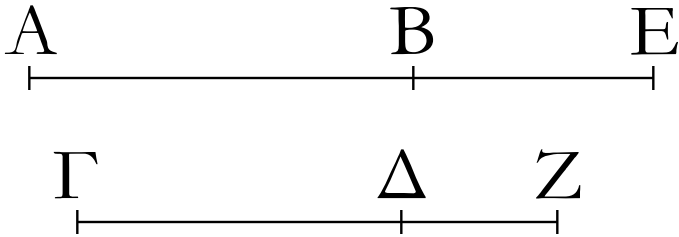
For let BE be an attachment to AB . Thus, AE and EB are (straight-lines which are) incommensurable in square, making the sum of the squares on AE and EB medial, and the (rectangle contained) by them rational [Prop. 10.77]. And let the same construction be made (as in the previous propositions). So, similarly to the previous (propositions), we can show that CF and FD are in the same ratio as AE and EB , and the sum of the squares on AE and EB is commensurable with the sum of the squares on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Hence, CF and FD are also (straight-lines which are) incommensurable in square, making the sum of the squares on CF



Ἡ ΓΔ ἄρα μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἐστίν· ὅπερ ἔδει δεῖχαι.

107.

Ἡ τῇ μετὰ μέσον μέσον τὸ ὅλον ποιούση σύμμετρος καὶ αὐτὴ μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἐστίν.

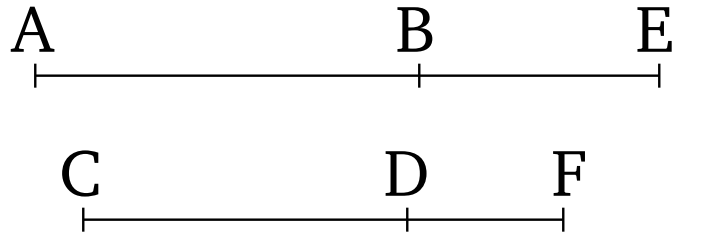


Ἐστω μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἡ ΑΒ, καὶ τῇ ΑΒ ἔστω σύμμετρος ἡ ΓΔ· λέγω, ὅτι καὶ ἡ ΓΔ μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἐστίν.

Ἐστω γάρ τῇ ΑΒ προσαρμόζονσα ἡ ΒΕ, καὶ τὰ αὐτὰ κατεσκευάσθω· αἱ ΑΕ, ΕΒ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιούσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων τῷ ὑπ' αὐτῶν. καὶ εἰσιν, ὥς ἐδείκθη, αἱ ΑΕ, ΕΒ σύμμετροι ταῖς ΓΖ, ΖΔ, καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΕ, ΕΒ τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν ΓΖ, ΖΔ, τὸ δὲ ὑπὸ τῶν ΑΕ, ΕΒ τῷ ὑπὸ τῶν ΓΖ, ΖΔ· καὶ αἱ ΓΖ, ΖΔ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιούσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπ' αὐτῶν [τετραγώνων] τῷ ὑπ' αὐτῶν.

Ἡ ΓΔ ἄρα μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἐστίν· ὅπερ ἔδει δεῖχαι.

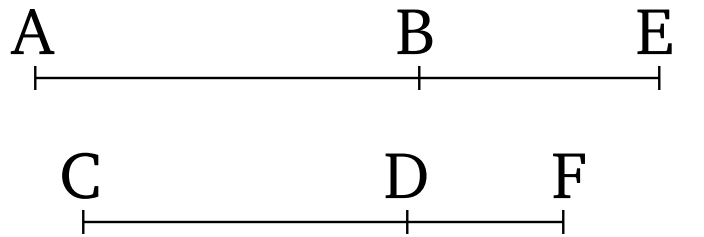
and FD medial, and the (rectangle contained) by them rational.



CD is thus a (straight-line) which with a rational (area) makes a medial whole [Prop. 10.77]. (Which is) the very thing it was required to show.

Proposition 107

A (straight-line) commensurable (in length) with a (straight-line) which with a medial (area) makes a medial whole is itself also a (straight-line) which with a medial (area) makes a medial whole.



Let AB be a (straight-line) which with a medial (area) makes a medial whole, and let CD be commensurable (in length) with AB . I say that CD is also a (straight-line) which with a medial (area) makes a medial whole.

For let BE be an attachment to AB . And let the same construction be made (as in the previous propositions). Thus, AE and EB are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, further, the sum of the squares on them incommensurable with the (rectangle contained) by them [Prop. 10.78]. And, as was shown (previously), AE and EB are commensurable (in length) with CF and FD (respectively), and the sum of the squares on AE and EB with the sum of the squares on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Thus, CF and FD are also (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, further, the sum of the [squares] on them incommensurable with the (rectangle contained) by them.

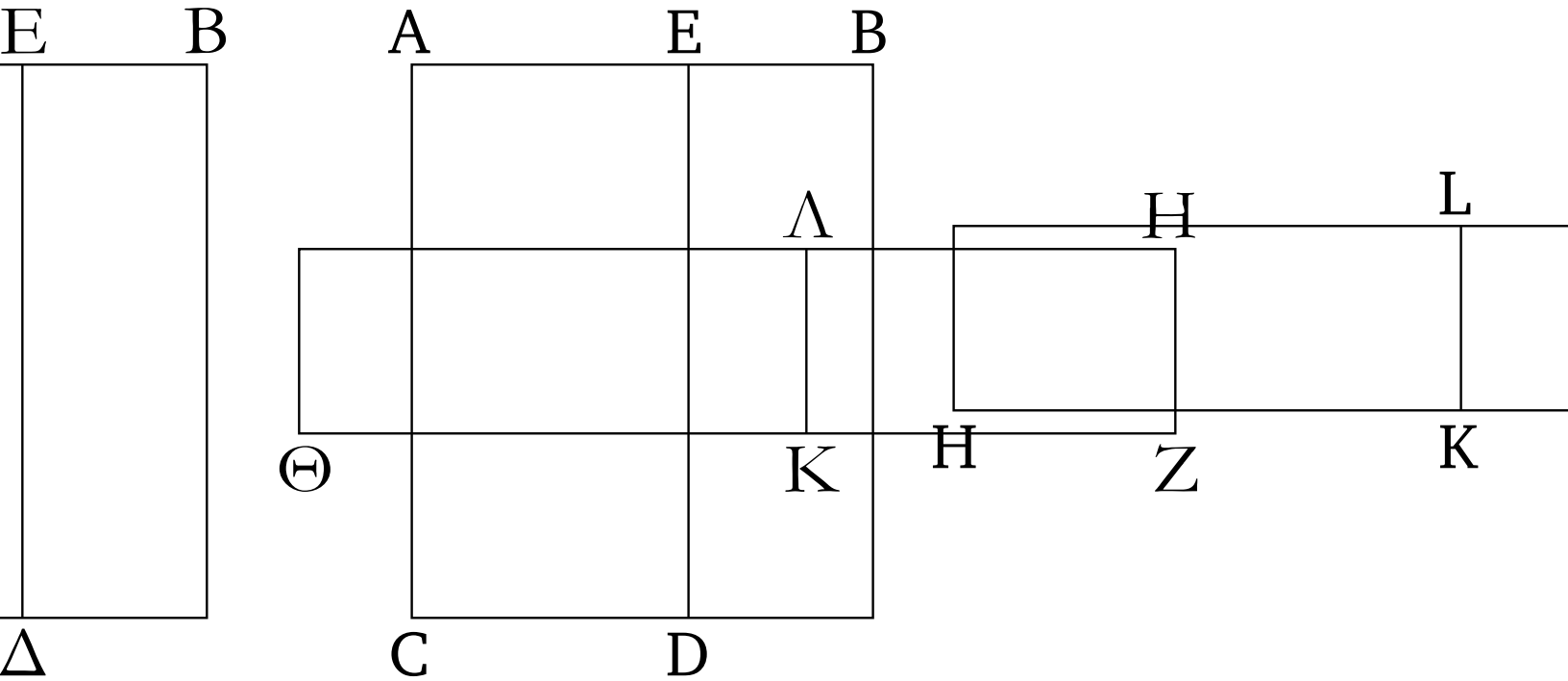
Thus, CD is a (straight-line) which with a medial (area) makes a medial whole [Prop. 10.78]. (Which is) the very thing it was required to show.

108.

Proposition 108

Ἀπὸ ῥητοῦ μέσον ἀφαιρουμένον ἢ τὸ λοιπὸν θεωρίον δυναμένη μία δύο ἀλόγων γίνεται ἥτοι ἀποτομή ἢ ἐλάσσων.

A medial (area) being subtracted from a rational (area), one of two irrational (straight-lines) arise (as) the square-root of the remaining area—either an apotome, or a minor (straight-line).



Ἀπὸ γάρ ῥητοῦ τοῦ BF μέσον ἀφηρήσθω τὸ $BΔ$ · λέγω, ὅτι ἢ τὸ λοιπὸν δυναμένη τὸ EF μία δύο ἀλόγων γίνεται ἥτοι ἀποτομή ἢ ἐλάσσων.

Ἐκκείσθω γάρ ῥητὴ ἡ ZH , καὶ τῷ μὲν BF ἴσον παρὰ τὴν ZH παραβελήσθω ὀρθογώνιον παραλληλόγραμμον τὸ $HΘ$, τῷ δὲ $ΔB$ ἴσον ἀφηρήσθω τὸ HK · λοιπὸν ἄρα τὸ EF ἴσον ἐστὶ τῷ $ΛΘ$. ἐπεὶ οὖν ῥητὸν μὲν ἐστὶ τὸ BF , μέσον δὲ τὸ $BΔ$, ἴσον δὲ τὸ μὲν BF τῷ $HΘ$, τὸ δὲ $BΔ$ τῷ HK , ῥητὸν μὲν ἄρα ἐστὶ τὸ $HΘ$, μέσον δὲ τὸ HK . καὶ παρὰ ῥητὴν τὴν ZH παράκειται· ῥητὴ μὲν ἄρα ἡ $ZΘ$ καὶ σύμμετρος τῇ ZH μήκει, ῥητὴ δὲ ἡ ZK καὶ ἀσύμμετρος τῇ ZH μήκει· ἀσύμμετρος ἄρα ἐστὶν ἡ $ZΘ$ τῇ ZK μήκει. αἱ $ZΘ$, ZK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ $KΘ$, προσαρμύζουσα δὲ αὐτῇ ἡ KZ . ἥτοι δὴ ἡ $ΘZ$ τῇ ZK μείζον δύναται τῷ ἀπὸ συμέτρον ἢ οὐ.

Δυνάσθω πρότερον τῷ ἀπὸ συμέτρον. καὶ ἐστὶν ὅλη ἡ $ΘZ$ σύμμετρος τῇ ἐκκειμένη ῥητῇ μήκει τῇ ZH · ἀποτομή ἄρα πρώτη ἐστὶν ἡ $KΘ$. τὸ δ' ὑπὸ ῥητῆς καὶ ἀποτομῆς πρώτης περιεχόμενον ἢ δυναμένη ἀποτομή ἐστὶν. ἢ ἄρα τὸ $ΛΘ$, τοντέστι τὸ EF , δυναμένη ἀποτομή ἐστὶν.

Εἰ δὲ ἡ $ΘZ$ τῇ ZK μείζον δύναται τῷ ἀπὸ ἀσυμέτρον ἐαντῇ, καὶ ἐστὶν ὅλη ἡ $ZΘ$ σύμμετρος τῇ ἐκκειμένη ῥητῇ μήκει τῇ ZH , ἀποτομή τετάρτη ἐστὶν ἡ $KΘ$. τὸ δ' ὑπὸ ῥητῆς καὶ

For let the medial (area) BD be subtracted from the rational (area) BC . I say that one of two irrational (straight-lines) arise (as) the square-root of the remaining (area), EC —either an apotome, or a minor (straight-line).

For let the rational (straight-line) FG be laid out, and let the right-angled parallelogram GH , equal to BC , be applied to FG , and let GK , equal to DB , be subtracted (from GH). Thus, the remainder EC is equal to LH . Therefore, since BC is a rational (area), and BD a medial (area), and BC (is) equal to GH , and BD to GK , GH is thus a rational (area), and GK a medial (area). And they are applied to the rational (straight-line) FG . Thus, FH (is) rational, and commensurable in length with FG [Prop. 10.20], and FK (is) also rational, and incommensurable in length with FG [Prop. 10.22]. Thus, FH is incommensurable in length with FK [Prop. 10.13]. FH and FK are thus rational (straight-lines which are) commensurable in square only. Thus, KH is an apotome [Prop. 10.73], and KF an attachment to it. So, the square on HF is greater than (the square on) FK by the (square) on (some straight-line which is) either commensurable, or not (commensurable), (in length with HF).

First, let the square (on it) be (greater) by the (square) on (some straight-line which is) commensurable (in length with

ἀποτομῇς τετάρτης περιεχόμενον ἢ δυναμένη ἐλάσσων ἐστίν· ὅπερ ἔδει δεῖχαι.

HF). And the whole of HF is commensurable in length with the (previously) laid down rational (straight-line) FG . Thus, KH is a first apotome [Def. 10.1]. And the square-root of an (area) contained by a rational (straight-line) and a first apotome is an apotome [Prop. 10.91]. Thus, the square-root of LH —that is to say, (of) EC —is an apotome.

And if the square on HF is greater than (the square on) FK by the (square) on (some straight-line which is) incommensurable (in length) with (HF) , and (since) the whole of FH is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a fourth apotome [Prop. 10.14]. And the square-root of an (area) contained by a rational (straight-line) and a fourth apotome is a minor (straight-line) [Prop. 10.94]. (Which is) the very thing it was required to show.

109.

Proposition 109

Ἀπὸ μέσον ῥητοῦ ἀφαιρουμένον ἄλλαι δύο ἄλογοι γίνονται ἥτοι μέση ἀποτομή πρώτη ἢ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσα.

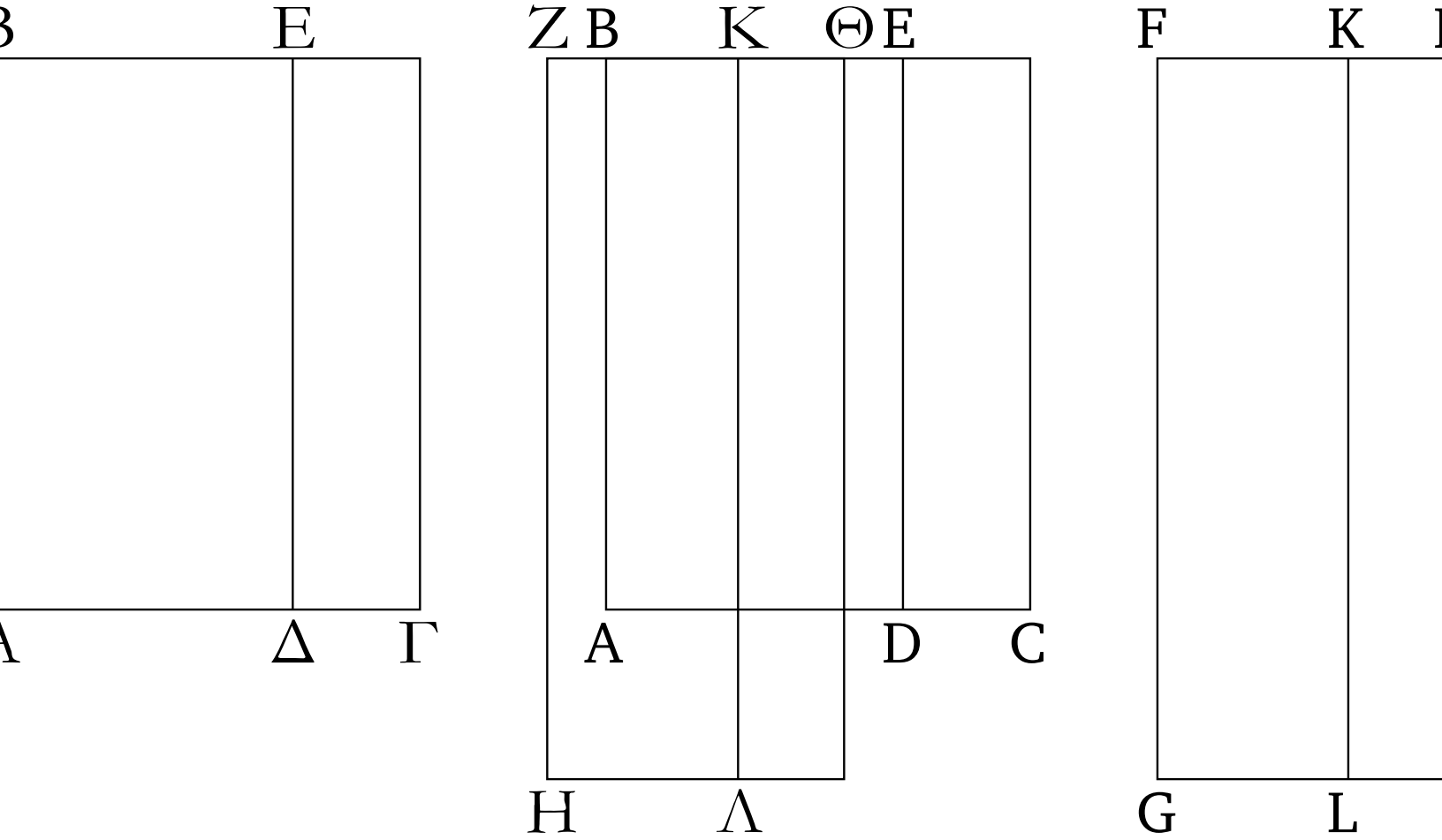
Ἀπὸ γὰρ μέσον τοῦ $BΓ$ ῥητὸν ἀφηρήσθω τὸ $ΒΔ$. λέγω, ὅτι ἡ τὸ λοιπὸν τὸ $ΕΓ$ δυναμένη μία δύο ἀλόγων γίνεται ἥτοι μέση ἀποτομή πρώτη ἢ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσα.

Ἐκκείσθω γὰρ ῥητὴ ἡ ZH , καὶ παραβελλήσθω ὁμοίως τὰ $θωρία$. ἔστι δὴ ἀκολουθῶς ῥητὴ μὲν ἡ $ZΘ$ καὶ ἀσύμμετρος τῇ ZH μήκει, ῥητὴ δὲ ἡ KZ καὶ σύμμετρος τῇ ZH μήκει· αἱ $ZΘ$, ZK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ $KΘ$, προσαρμόζουσα δὲ ταύτῃ ἡ ZK . ἥτοι δὴ ἡ $ΘΖ$ τῇ ZK μείζον δύναιται τῷ ἀπὸ συμμέτρον ἑαυτῇ ἢ τῷ ἀπὸ ἀσύμμετρον.

A rational (area) being subtracted from a medial (area), two other irrational (straight-lines) arise (as the square-root of the remaining area)—either a first apotome of a medial (straight-line), or that (straight-line) which with a rational (area) makes a medial whole.

For let the rational (area) BD be subtracted from the medial (area) BC . I say that one of two irrational (straight-lines) arise (as) the square-root of the remaining (area), EC —either a first apotome of a medial (straight-line), or that (straight-line) which with a rational (area) makes a medial whole.

For let the rational (straight-line) FG be laid down, and let similar areas (to the preceding proposition) have been applied (to it). So, accordingly, FH is rational, and incommensurable in length with FG , and KF (is) also rational, and commensurable in length with FG . Thus, FH and FK are rational (straight-lines which are) commensurable in square only [Prop. 10.13]. KH is thus an apotome [Prop. 10.73], and FK an attachment to it. So, the square on HF is greater than (the square on) FK either by the (square) on (some straight-line) commensurable (in length) with (HF) , or by the (square) on (some straight-line) incommensurable (in length with HF).



Εἰ μὲν οὖν ἡ ΘΖ τῆς ΖΚ μείζον δύναται τῷ ἀπὸ συμμετρων ἑαυτῇ, καὶ ἐστὶν ἡ προσαρμόζουσα ἡ ΖΚ σύμμετρος τῇ ἐκκειμένῃ ῥητῇ μῆκει τῇ ΖΗ, ἀποτομή δευτέρα ἐστὶν ἡ ΚΘ. ῥητὴ δὲ ἡ ΖΗ· ὥστε ἡ τὸ ΛΘ, τουτέστι τὸ ΕΓ, δυναμένη μέσης ἀποτομῇ πρώτη ἐστίν.

Εἰ δὲ ἡ ΘΖ τῆς ΖΚ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρων, καὶ ἐστὶν ἡ προσαρμόζουσα ἡ ΖΚ σύμμετρος τῇ ἐκκειμένῃ ῥητῇ μῆκει τῇ ΖΗ, ἀποτομή πέμπτη ἐστὶν ἡ ΚΘ· ὥστε ἡ τὸ ΕΓ δυναμένη μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἐστίν· ὅπερ ἔδει δεῖξαι.

Therefore, if the square on HF is greater than (the square on) FK by the (square) on (some straight-line) commensurable (in length) with (HF) , and (since) the attachment FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a second apotome [Def. 10.12]. And FG (is) rational. Hence, the square-root of LH —that is to say, (of) EC —is a first apotome of a medial (straight-line) [Prop. 10.92].

And if the square on HF is greater than (the square on) FK by the (square) on (some straight-line) incommensurable (in length with HF), and (since) the attachment FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a fifth apotome [Def. 10.15]. Hence, the square-root of EC is that (straight-line) which with a rational (area) makes a medial whole [Prop. 10.95]. (Which is) the very thing it was required to show.

110.

Proposition 110

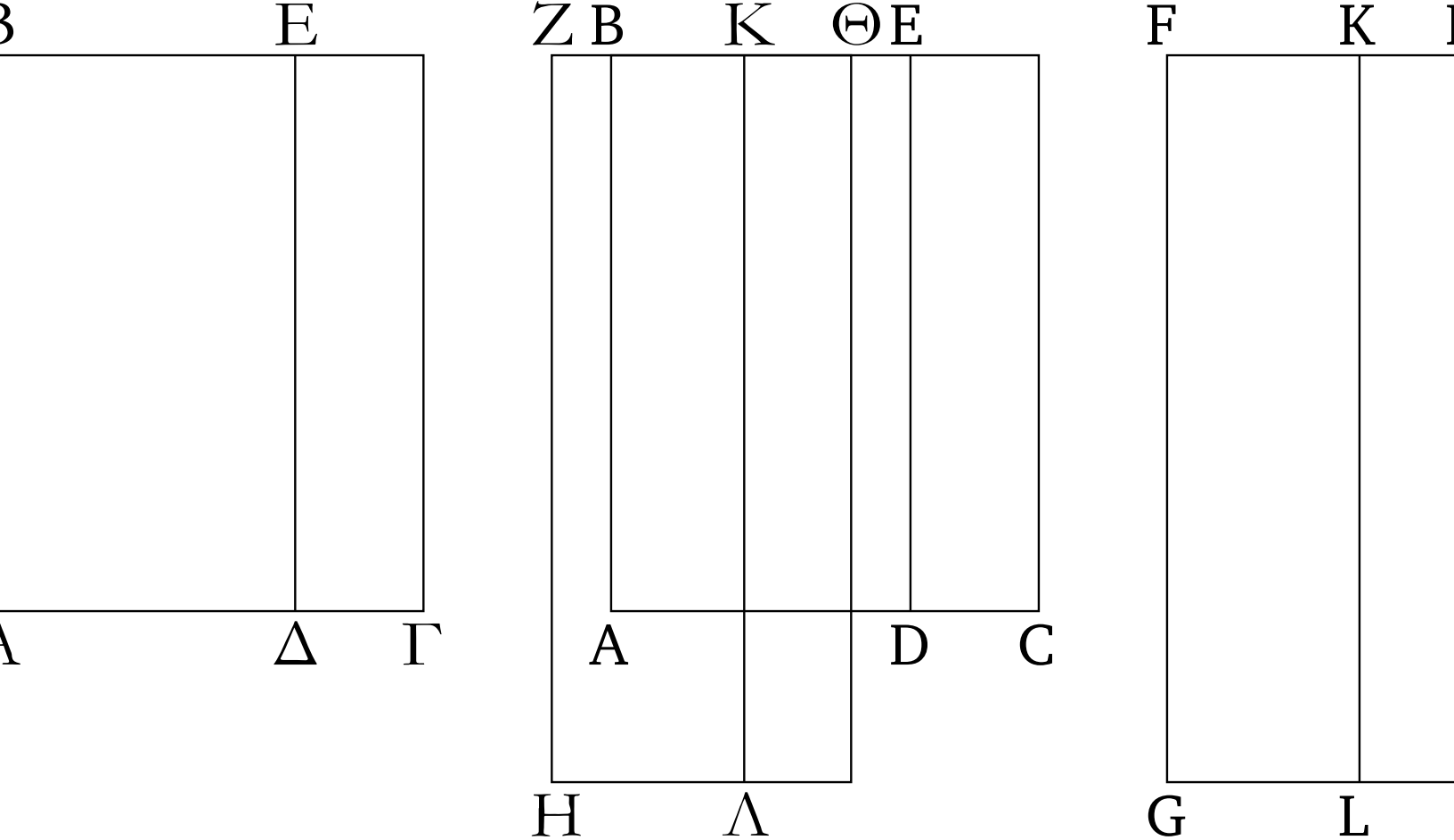
Ἀπὸ μέσον μέσον ἀφαιρουμένον ἀσυμμέτρων τῷ ὅλῳ αἰλοιπαὶ δύο ἄλογοι γίνονται ἥτοι μέσης ἀποτομῇ δευτέρα ἢ μετὰ μέσον μέσον τὸ ὅλον ποιούσα.

A medial (area), incommensurable with the whole, being subtracted from a medial (area), the two remaining irrational (straight-lines) arise (as) the (square-root of the area)—either

Ἀφρηθήσθω γάρ ὡς ἐπὶ τῶν προκειμένων καταγραφῶν ἀπὸ μέσον τοῦ $B\Gamma$ μέσον τὸ $B\Delta$ ἀσύμμετρον τῷ ὅλῳ· λέγω, ὅτι ἡ τὸ $E\Gamma$ δυναμένη μία ἐστὶ δύο ἀλόγων ἥτοι μέσῃς ἀποτομῇ δευτέρᾳ ἢ μετὰ μέσον μέσον τὸ ὅλον ποιούσα.

a second apotome of a medial (straight-line), or that (straight-line) which with a medial (area) makes a medial whole.

For, as in the previous figures, let the medial (area) BD , incommensurable with the whole, be subtracted from the medial (area) BC . I say that the square-root of EC is one of two irrational (straight-lines)—either a second apotome of a medial (straight-line), or that (straight-line) which with a medial (area) makes a medial whole.



Ἐπεὶ γὰρ μέσον ἐστὶν ἑκάτερον τῶν $B\Gamma$, $B\Delta$, καὶ ἀσύμμετρον τὸ $B\Gamma$ τῷ $B\Delta$, ἔσται ἀκολούθως ῥητὴ ἑκάτερα τῶν $Z\Theta$, ZK καὶ ἀσύμμετροξ τῇ ZH μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ $B\Gamma$ τῷ $B\Delta$, τοντέστι τὸ $H\Theta$ τῷ HK , ἀσύμμετροξ καὶ ἡ ΘZ τῇ ZK . αἱ $Z\Theta$, ZK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομῇ ἄρα ἐστὶν ἡ $K\Theta$ [προσαρμόζουσα δὲ ἡ ZK . ἥτοι δὴ ἡ $Z\Theta$ τῇ ZK μείζον δύνανται τῷ ἀπὸ συμμέτρον ἢ τῷ ἀπὸ ἀσυμμέτρον ἑαυτῇ].

Εἰ μὲν δὴ ἡ $Z\Theta$ τῇ ZK μείζον δύνανται τῷ ἀπὸ συμμέτρον ἑαυτῇ, καὶ οὐθετέρα τῶν $Z\Theta$, ZK σύμμετροξ ἐστὶ τῇ ἑκκειμένη ῥητῇ μήκει τῇ ZH , ἀποτομῇ τρίτῃ ἐστὶν ἡ $K\Theta$. ῥητὴ δὲ ἡ $K\Lambda$, τὸ δ' ὑπὸ ῥητῇ καὶ ἀποτομῇ τρίτῃ περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ ἄλογόξ ἐστιν,

For since BC and BD are each medial (areas), and BC (is) incommensurable with BD , accordingly, FH and FK will each be rational (straight-lines), and incommensurable in length with FG [Prop. 10.22]. And since BC is incommensurable with BD —that is to say, GH with GK — HF (is) also incommensurable (in length) with FK [Props. 6.1, 10.11]. Thus, FH and FK are rational (straight-lines which are) commensurable in square only. KH is thus as apotome [Prop. 10.73], [and FK an attachment (to it). So, the square on FH is greater than (the square on) FK either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (FH) .]

So, if the square on FH is greater than (the square on)

καλεῖται δὲ μέση ἀποτομή δευτέρα· ὥστε ἡ τὸ $\Lambda\Theta$, τοντέστι τὸ $ΕΓ$, δυναμένη μέση ἀποτομή ἐστὶ δευτέρα.

Εἰ δὲ ἡ $Z\Theta$ τῆς ZK μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ [μήκει], καὶ οὐδετέρα τῶν ΘZ , ZK σύμμετρος ἐστὶ τῇ ZH μήκει, ἀποτομή ἕκτη ἐστὶν ἡ $K\Theta$. τὸ δ' ὑπὸ ῥητῆς καὶ ἀποτομῆς ἕκτης ἡ δυναμένη ἐστὶ μετὰ μέσον μέσον τὸ ὅλον ποιῶσα. ἡ τὸ $\Lambda\Theta$ ἄρα, τοντέστι τὸ $ΕΓ$, δυναμένη μετὰ μέσον μέσον τὸ ὅλον ποιῶσα ἐστὶν· ὅπερ ἔδει δεῖχαι.

FK by the (square) on (some straight-line) commensurable (in length) with (FH) , and (since) neither of FH and FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a third apotome [Def. 10.3]. And KL (is) rational. And the rectangle contained by a rational (straight-line) and a third apotome is irrational, and the square-root of it is that irrational (straight-line) called a second apotome of a medial (straight-line) [Prop. 10.93]. Hence, the square-root of LH —that is to say, (of) EC —is a second apotome of a medial (straight-line).

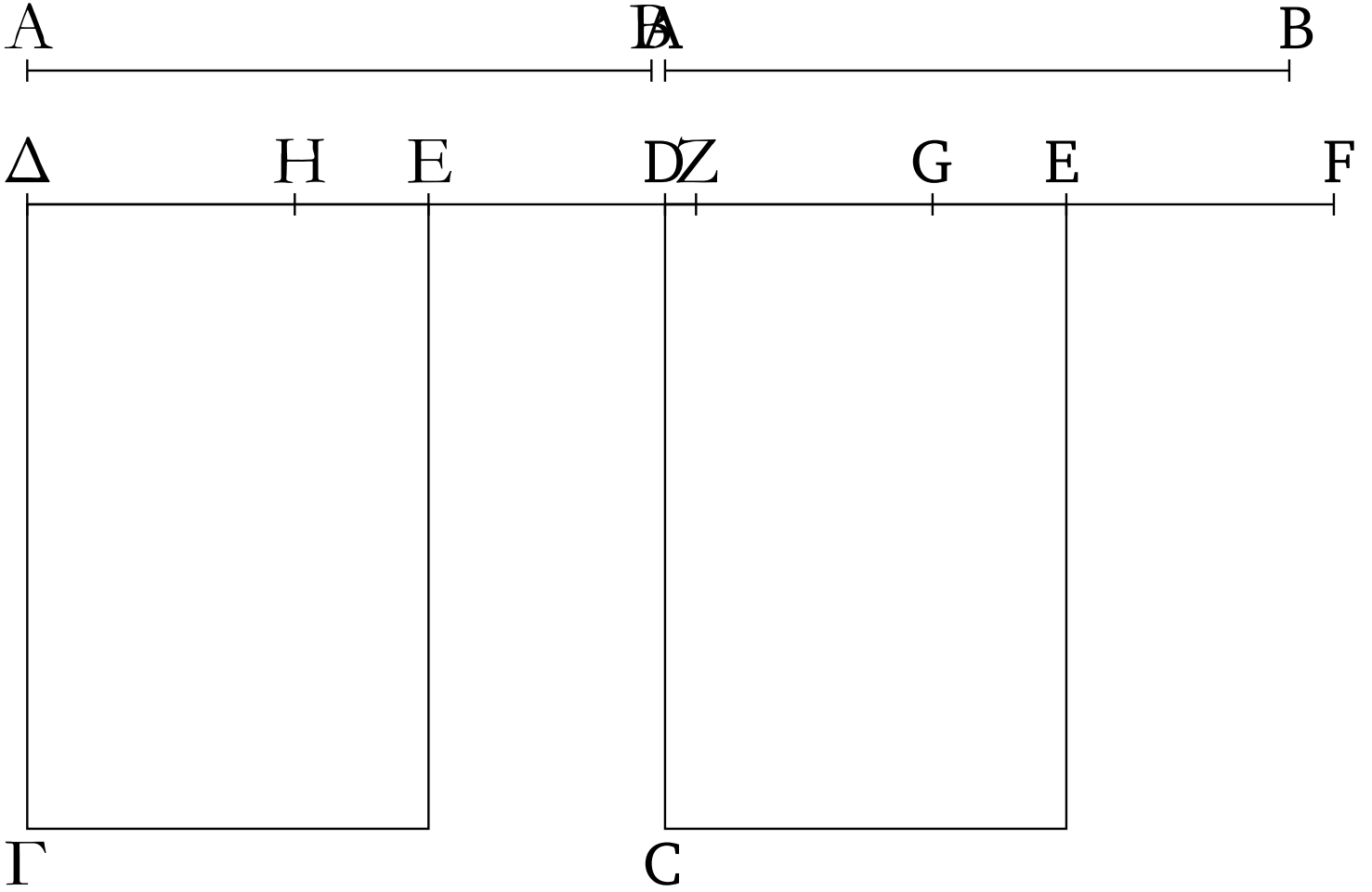
And if the square on FH is greater than (the square on) FK by the (square) on (some straight-line) incommensurable [in length] with (FH) , and (since) neither of HF and FK is commensurable in length with FG , KH is a sixth apotome [Def. 10.16]. And the square-root of the (rectangle contained) by a rational (straight-line) and a sixth apotome is that (straight-line) which with a medial (area) makes a medial whole [Prop. 10.96]. Thus, the square-root of LH —that is to say, (of) EC —is that (straight-line) which with a medial (area) makes a medial whole. (Which is) the very thing it was required to show.

111.

Ἡ ἀποτομή οὐκ ἔστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων.

Proposition 111

An apotome is not the same as a binomial.



Ἐστω ἀποτομή ἡ AB . λέγω, ὅτι ἡ AB οὐκ ἔστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων.

Εἰ γὰρ δυνατόν, ἔστω· καὶ ἐκκείσθω ῥητὴ ἡ $\Delta\Gamma$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω ὀρθογώνιον τὸ ΓE πλάτος ποιῶν τὴν ΔE . ἔπει οὖν ἀποτομή ἐστὶν ἡ AB , ἀποτομή πρώτη ἐστὶν ἡ ΔE . ἔστω αὐτῇ προσαρμύζουσα ἡ EZ . αἱ ΔZ , ZE ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΔZ τῆς ZE μείζον δύναται τῷ ἀπὸ συμέτρου ἐναντῆς, καὶ ἡ ΔZ σύμμετρος ἐστὶ τῇ ἐκκειμένῃ ῥητῇ μήκει τῇ $\Delta\Gamma$. πάλιν, ἔπει ἐκ δύο ὀνομάτων ἐστὶν ἡ AB , ἐκ δύο ἄρα ὀνομάτων πρώτη ἐστὶν ἡ ΔE . διηρήσθω εἰς τὰ ὀνόματα κατὰ τὸ H , καὶ ἔστω μείζον ὄνομα τὸ ΔH . αἱ ΔH , HE ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΔH τῆς HE μείζον δύναται τῷ ἀπὸ συμέτρου ἐναντῆς, καὶ τὸ μείζον ἡ ΔH σύμμετρος ἐστὶ τῇ ἐκκειμένῃ ῥητῇ μήκει τῇ $\Delta\Gamma$. καὶ ἡ ΔZ ἄρα τῇ ΔH σύμμετρος ἐστὶ μήκει· καὶ λοιπὴ ἄρα ἡ HZ σύμμετρος ἐστὶ τῇ ΔZ μήκει. [ἔπει οὖν σύμμετρος ἐστὶν ἡ ΔZ τῇ HZ , ῥητὴ δὲ ἐστὶν ἡ ΔZ , ῥητὴ ἄρα ἐστὶ καὶ ἡ HZ . ἔπει οὖν σύμμετρος ἐστὶν ἡ ΔZ τῇ HZ μήκει] ἀσύμμετρος δὲ ἡ ΔZ τῇ EZ μήκει. ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ZH τῇ EZ μήκει. αἱ HZ , ZE ἄρα ῥηταὶ [εἰσι] δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν

Let AB be an apotome. I say that AB is not the same as a binomial.

For, if possible, let it be (the same). And let a rational (straight-line) DC be laid down. And let the rectangle CE , equal to the (square) on AB , be applied to CD , producing DE as breadth. Therefore, since AB is an apotome, DE is a first apotome [Prop. 10.97]. Let EF be an attachment to it. Thus, DF and FE are rational (straight-lines which are) commensurable in square only, and the square on DF is greater than (the square on) FE by the (square) on (some straight-line) commensurable (in length) with (DF), and DF is commensurable in length with the (previously) laid down rational (straight-line) DC [Def. 10.10]. Again, since AB is a binomial, DE is thus a first binomial [Prop. 10.60]. Let (DE) be divided into its (component) terms at G , and let DG be the greater term. Thus, DG and GE are rational (straight-lines which are) commensurable in square only, and the square on DG is greater than (the square on) GE by the (square) on (some straight-line) commensurable (in length) with (DG), and the greater (term) DG is commensurable in length with the (previously) laid down rational (straight-line) DC [Def. 10.5]. Thus, DF is also com-

ἡ EH . ἀλλὰ καὶ ῥητὴ ὅπερ ἐστὶν ἀδύνατον.

Ἡ ἄρα ἀποτομή οὐκ ἔστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων ὅπερ ἔδει δεῖχαι.

measurable in length with DG [Prop. 10.12]. The remainder GF is thus commensurable in length with DF [Prop. 10.15]. [Therefore, since DF is commensurable with GF , and DF is rational, GF is thus also rational. Therefore, since DF is commensurable in length with GF ,] DF (is) incommensurable in length with EF . Thus, FG is also incommensurable in length with EF [Prop. 10.13]. GF and FE [are] thus rational (straight-lines which are) commensurable in square only. Thus, EG is an apotome [Prop. 10.73]. But, (it is) also rational. The very thing is impossible.

Thus, an apotome is not the same as a binomial. (Which is) the very thing it was required to show.

[Πόρισμα.]

Ἡ ἀποτομή καὶ αἱ μετ' αὐτὴν ἄλογοι οὔτε τῇ μέσῃ οὔτε ἀλλήλαις εἰσὶν αἱ αὐταί.

Τὸ μὲν γὰρ ἀπὸ μέσῃς παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ῥητὴν καὶ ἀσύμμετρον τῇ, παρ' ἣν παράκειται, μήκει, τὸ δὲ ἀπὸ ἀποτομῆς παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην, τὸ δὲ ἀπὸ μέσῃς ἀποτομῆς πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν, τὸ δὲ ἀπὸ μέσῃς ἀποτομῆς δευτέρας παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην, τὸ δὲ ἀπὸ ἐλάσσονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τετάρτην, τὸ δὲ ἀπὸ τῆς μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πέμπτην, τὸ δὲ ἀπὸ τῆς μετὰ μέσον μέσον τὸ ὅλον ποιούσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν ἕκτην. ἐπεὶ οὖν τὰ εἰρημένα πλάτη διαφέρει τοῦ τε πρώτου καὶ ἀλλήλων, τοῦ μὲν πρώτου, ὅτι ῥητὴ ἐστίν, ἀλλήλων δὲ, ἐπεὶ τῇ τάχει οὐκ εἰσὶν αἱ αὐταί, δῆλον, ὥς καὶ αὐταί αἱ ἄλογοι διαφέρουσιν ἀλλήλων. καὶ ἐπεὶ δέδεικται ἡ ἀποτομή οὐκ οὔσα ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων, ποιοῦσι δὲ πλάτη παρὰ ῥητὴν παραβαλλόμενα αἱ μετὰ τὴν ἀποτομὴν ἀποτομάς ἀκολουθῶς ἐκάστη τῇ τάχει τῇ καθ' αὐτήν, αἱ δὲ μετὰ τὴν ἐκ δύο ὀνομάτων τάξ ἐκ δύο ὀνομάτων καὶ αὐταί τῇ τάχει ἀκολουθῶς, ἔτεροι ἄρα εἰσὶν αἱ μετὰ τὴν ἀποτομὴν καὶ ἔτεροι αἱ μετὰ τὴν ἐκ δύο ὀνομάτων, ὥς εἶναι τῇ τάχει πάσας ἀλόγους γ,

[Corollary]

The apotome and the irrational (straight-lines) after it are neither the same as a medial (straight-line) nor (the same) as one another.

For the (square) on a medial (straight-line), applied to a rational (straight-line), produces as breadth a rational (straight-line) which is) incommensurable in length with the (straight-line) to which (the area) is applied [Prop. 10.22]. And the (square) on an apotome, applied to a rational (straight-line), produces as breadth a first apotome [Prop. 10.97]. And the (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces as breadth a second apotome [Prop. 10.98]. And the (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces as breadth a third apotome [Prop. 10.99]. And (square) on a minor (straight-line), applied to a rational (straight-line), produces as breadth a fourth apotome [Prop. 10.100]. And (square) on that (straight-line) which with a rational (area) produces a medial whole, applied to a rational (straight-line), produces as breadth a fifth apotome [Prop. 10.101]. And (square) on that (straight-line) which with a medial (area) produces a medial whole, applied to a rational (straight-line), produces as breadth a sixth apotome [Prop. 10.102]. Therefore, since the aforementioned breadths differ from the first (breadth), and from one another—from the first, because it is rational, and from one another since they are not the same in order—clearly, the irrational (straight-lines) themselves also differ from one another. And since it has been shown that an apotome is not the same as a binomial [Prop. 10.111], and (that) the (irrational straight-lines) after the apotome, being applied to a rational (straight-line), produce as breadth, each according to its own (order), apotomes, and (that) the (irrational straight-lines) after the binomial themselves also (produce as breadth), according (to their) order, binomials, the (irrational straight-lines) after the apotome are thus different, and the (irrational straight-lines) after the binomial (are also) different, so that there are, in order, 13 irrational (straight-lines) in all:

Μέσσην,
 Ἐκ δύο ὀνομάτων,
 Ἐκ δύο μέσων πρώτην,
 Ἐκ δύο μέσων δευτέραν,
 Μείζονα,
 μὴ ῥητὸν καὶ μέσον δυναμένην,
 Δύο μέσα δυναμένην,
 Ἀποτομήν,
 Μέσῃς ἀποτομὴν πρώτην,
 Μέσῃς ἀποτομὴν δευτέραν,
 Ἐλάσσονα,
 Μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσαν,
 Μετὰ μέσου μέσον τὸ ὅλον ποιοῦσαν.

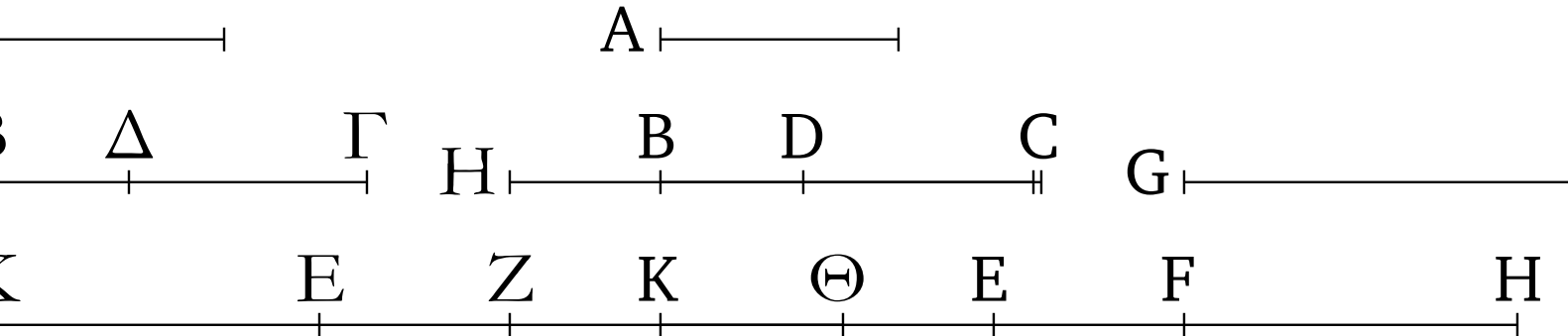
Medial,
 Binomial,
 First bimedral,
 Second bimedral,
 Major,
 Square-root of a rational plus a medial (area),
 Square-root of (the sum of) two medial (areas),
 Apotome,
 First apotome of a medial,
 Second apotome of a medial,
 Minor,
 That which with a rational (area) produces a medial whole,
 That which with a medial (area) produces a medial whole.

112.

Proposition 112[†]

Τὸ ἀπὸ ῥητῆς παρὰ τὴν ἐκ δύο ὀνομάτων παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν, ἥς τὰ ὀνόματα σύμμετρα ἔστι τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι καὶ ἔτι ἐν τῷ αὐτῷ λόγῳ, καὶ ἔτι ἡ γινόμενη ἀποτομή τὴν αὐτὴν ἔχει τάχιν τῇ ἐκ δύο ὀνομάτων.

The (square) on a rational (straight-line), applied to a binomial (straight-line), produces as breadth an apotome whose terms are commensurable (in length) with the terms of the binomial, and, furthermore, in the same ratio. Moreover, the created apotome will have the same order as the binomial.



Ἐστω ῥητὴ μὲν ἡ A , ἐκ δύο ὀνομάτων δὲ ἡ BF , ἥς μείζον ὄνομα ἔστω ἡ $\Delta\Gamma$, καὶ τῷ ἀπὸ τῆς A ἴσον ἔστω τὸ ὑπὸ τῶν $B\Gamma$, EZ · λέγω, ὅτι ἡ EZ ἀποτομή ἐστίν, ἥς τὰ ὀνόματα σύμμετρα ἔστι τοῖς $\Gamma\Delta$, ΔB , καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἔτι ἡ EZ τὴν αὐτὴν ἔχει τάχιν τῇ $B\Gamma$.

Ἐστω γάρ πάλιν τῷ ἀπὸ τῆς A ἴσον τὸ ὑπὸ τῶν $B\Delta$, H . ἐπεὶ οὖν τὸ ὑπὸ τῶν $B\Gamma$, EZ ἴσον ἐστὶ τῷ ὑπὸ τῶν $B\Delta$, H , ἔστιν ἄρα ὥς ἡ ΓB πρὸς τὴν $B\Delta$, οὕτως ἡ H πρὸς τὴν EZ . μείζων δὲ ἡ ΓB τῆς $B\Delta$ · μείζων ἄρα ἐστὶ καὶ ἡ H τῆς EZ . ἔστω τῇ H ἴση ἡ $E\Theta$ · ἔστιν ἄρα ὥς ἡ ΓB πρὸς τὴν $B\Delta$, οὕτως ἡ ΘE πρὸς τὴν EZ · διελόντι ἄρα ἐστὶν ὥς ἡ $\Gamma\Delta$ πρὸς τὴν $B\Delta$, οὕτως ἡ ΘZ πρὸς τὴν ZE . γεγονέτω ὥς ἡ ΘZ πρὸς τὴν ZE , οὕτως ἡ ZE πρὸς τὴν KE · καὶ ὅλη ἄρα ἡ ΘK πρὸς ὅλην τὴν KZ ἐστίν, ὥς ἡ ZK πρὸς KE · ὥς γάρ ἐν τῶν ἡγουμένων πρὸς ἐν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα. ὥς δὲ ἡ ZK πρὸς KE ,

Let A be a rational (straight-line), and BC a binomial (straight-line), of which let DC be the greater term. And let the (rectangle contained) by BC and EF be equal to the (square) on A . I say that EF is an apotome whose terms are commensurable (in length) with CD and DB , and in the same ratio, and, moreover, that EF will have the same order as BC .

For, again, let the (rectangle contained) by BD and G be equal to the (square) on A . Therefore, since the (rectangle contained) by BC and EF is equal to the (rectangle contained) by BD and G , thus as CB is to BD , so G (is) to EF [Prop. 6.16]. And CB (is) greater than BD . Thus, G is also greater than EF [Props. 5.16, 5.14]. Let EH be equal to G . Thus, as CB is to BD , so HE (is) to EF . Thus, via separation, as CD is to BD , so HF (is) to FE [Prop. 5.17]. Let it be contrived that as HF (is) to FE , so FK (is) to KE . And, thus, the whole HK is to the whole KF , as FK (is) to KE . For as one of the leading (pro-

οὕτως ἐστὶν ἡ $\Gamma\Delta$ πρὸς τὴν ΔB · καὶ ὥς ἄρα ἡ ΘK πρὸς KZ , οὕτως ἡ $\Gamma\Delta$ πρὸς τὴν ΔB . σύμμετρον δὲ τὸ ἀπὸ τῆς $\Gamma\Delta$ τῷ ἀπὸ τῆς ΔB · σύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΘK τῷ ἀπὸ τῆς KZ . καὶ ἐστὶν ὥς τὸ ἀπὸ τῆς ΘK πρὸς τὸ ἀπὸ τῆς KZ , οὕτως ἡ ΘK πρὸς τὴν KE , ἐπεὶ αἱ $\tau\rho\epsilon\acute{\iota}\varsigma$ αἱ ΘK , KZ , KE ἀνάλογόν εἰσιν. σύμμετρος ἄρα ἡ ΘK τῇ KE μήκει. ὥστε καὶ ἡ ΘE τῇ EK σύμμετρος ἐστὶ μήκει. καὶ ἐπεὶ τὸ ἀπὸ τῆς A ἴσον ἐστὶ τῷ ὑπὸ τῶν $E\Theta$, $B\Delta$, ῥητὸν δὲ ἐστὶ τὸ ἀπὸ τῆς A , ῥητὸν ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν $E\Theta$, $B\Delta$. καὶ παρὰ ῥητὴν τὴν $B\Delta$ παράκειται· ῥητὴ ἄρα ἐστὶν ἡ $E\Theta$ καὶ σύμμετρος τῇ $B\Delta$ μήκει· ὥστε καὶ ἡ σύμμετρος αὐτῇ ἡ EK ῥητὴ ἐστὶ καὶ σύμμετρος τῇ $B\Delta$ μήκει. ἐπεὶ οὖν ἐστὶν ὥς ἡ $\Gamma\Delta$ πρὸς ΔB , οὕτως ἡ ZK πρὸς KE , αἱ δὲ $\Gamma\Delta$, ΔB δυνάμει μόνον εἰσὶ σύμμετροι, καὶ αἱ ZK , KE δυνάμει μόνον εἰσὶ σύμμετροι. ῥητὴ δὲ ἐστὶν ἡ KE · ῥητὴ ἄρα ἐστὶ καὶ ἡ ZK . αἱ ZK , KE ἄρα ῥηταὶ δυνάμει μόνον εἰσὶ σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ EZ .

Ἦτοι δὲ ἡ $\Gamma\Delta$ τῆς ΔB μείζον δύνатаι τῷ ἀπὸ συμμέτρον ἑαντῇ ἢ τῷ ἀπὸ ἀσυνμέτρον.

Εἰ μὲν οὖν ἡ $\Gamma\Delta$ τῆς ΔB μείζον δύνатаι τῷ ἀπὸ συμμέτρον [ἑαντῇ], καὶ ἡ ZK τῆς KE μείζον δυνήσεται τῷ ἀπὸ συμμέτρον ἑαντῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ $\Gamma\Delta$ τῇ ἐκκεκλεμένη ῥητῇ μήκει, καὶ ἡ ZK · εἰ δὲ ἡ $B\Delta$, καὶ ἡ KE · εἰ δὲ οὐδετέρα τῶν $\Gamma\Delta$, ΔB , καὶ οὐδετέρα τῶν ZK , KE .

Εἰ δὲ ἡ $\Gamma\Delta$ τῆς ΔB μείζον δύνатаι τῷ ἀπὸ ἀσυνμέτρον ἑαντῇ, καὶ ἡ ZK τῆς KE μείζον δυνήσεται τῷ ἀπὸ ἀσυνμέτρον ἑαντῇ. καὶ εἰ μὲν ἡ $\Gamma\Delta$ σύμμετρος ἐστὶ τῇ ἐκκεκλεμένη ῥητῇ μήκει, καὶ ἡ ZK · εἰ δὲ ἡ $B\Delta$, καὶ ἡ KE · εἰ δὲ οὐδετέρα τῶν $\Gamma\Delta$, ΔB , καὶ οὐδετέρα τῶν ZK , KE · ὥστε ἀποτομή ἐστὶν ἡ ZE , ἥς τὰ ὀνόματα τὰ ZK , KE σύμμετρα ἐστὶ τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι τοῖς $\Gamma\Delta$, ΔB καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ τὴν αὐτὴν τάχιν ἔθει $B\Gamma$ · ὅπερ ἔδει δεῖξαι.

portional magnitudes is) to one of the following, so all of the leading (magnitudes) are to all of the following [Prop. 5.12]. And as FK (is) to KE , so CD is to DB [Prop. 5.11]. And, thus, as HK (is) to KF , so CD is to DB [Prop. 5.11]. And the (square) on CD (is) commensurable with the (square) on DB [Prop. 10.36]. The (square) on HK is thus also commensurable with the (square) on KF [Props. 6.22, 10.11]. And as the (square) on HK is to the (square) on KF , so HK (is) to KE , since the three (straight-lines) HK , KF , and KE are proportional [Def. 5.9]. HK is thus commensurable in length with KE [Prop. 10.11]. Hence, HE is also commensurable in length with EK [Prop. 10.15]. And since the (square) on A is equal to the (rectangle contained) by EH and BD , and the (square) on A is rational, the (rectangle contained) by EH and BD is thus also rational. And it is applied to the rational (straight-line) BD . Thus, EH is rational, and commensurable in length with BD [Prop. 10.20]. And, hence, the (straight-line) commensurable (in length) with it, EK , is also rational [Def. 10.3], and commensurable in length with BD [Prop. 10.12]. Therefore, since as CD is to DB , so FK (is) to KE , and CD and DB are (straight-lines which are) commensurable in square only, FK and KE are also commensurable in square only [Prop. 10.11]. And KE is rational. Thus, FK is also rational. FK and KE are thus rational (straight-lines which are) commensurable in square only. Thus, EF is an apotome [Prop. 10.73].

And the square on CD is greater than (the square on) DB either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (the square on) CD .

Therefore, if the square on CD is greater than (the square on) DB by the (square) on (some straight-line) commensurable (in length) with $[CD]$ then the square on FK will also be greater than (the square on) KE by the (square) on (some straight-line) commensurable (in length) with (FK) [Prop. 10.14]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FK [Props. 10.11, 10.12]. And if BD (is) commensurable, (so) also (is) KE [Prop. 10.12]. And if neither of CD or DB (is) commensurable, neither also (are) either of FK or KE .

And if the square on CD is greater than (the square on) DB by the (square) on (some straight-line) incommensurable (in length) with (CD) then the square on FK will also be greater than (the square on) KE by the (square) on (some straight-line) incommensurable (in length) with (FK) [Prop. 10.14]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FK [Props. 10.11, 10.12]. And if BD (is) commensurable, (so) also (is) KE [Prop. 10.12]. And if neither of CD or DB (is) commensurable, neither also (are) either of FK or KE . Hence, FE is an apotome whose terms, FK and KE , are commensurable (in length) with the terms, CD and DB , of the binomial, and in the same ratio. And

(FE) has the same order as BC [Defs. 10.5—10.10]. (Which is) the very thing it was required to show.

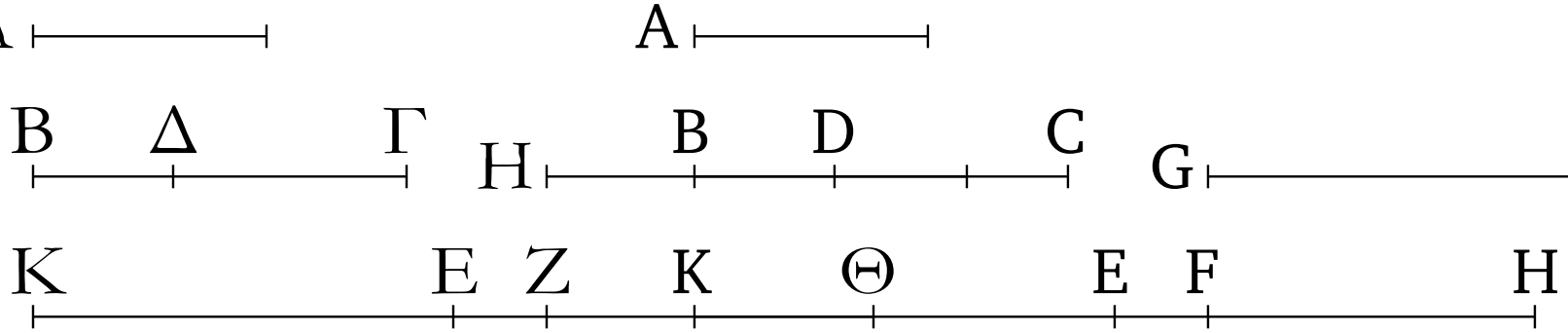
[†] Heiberg considers this proposition, and the succeeding ones, to be relatively early interpolations into the original text.

113.

Proposition 113

Τὸ ἀπὸ ῥητῆς παρὰ ἀποτομὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων, ἥξ τὰ ὀνόματα σύμμετρα ἔστι τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἔτι δὲ ἡ γινομένη ἐκ δύο ὀνομάτων τὴν αὐτὴν τάξιν ἔθει τῇ ἀποτομῇ.

The (square) on a rational (straight-line), applied to an apotome, produces as breadth a binomial whose terms are commensurable with the terms of the apotome, and in the same ratio. Moreover, the created binomial has the same order as the apotome.



Ἐστω ῥητὴ μὲν ἡ A , ἀποτομή δὲ ἡ BD , καὶ τῷ ἀπὸ τῆς A ἴσον ἔστω τὸ ὑπὸ τῶν BD , $KΘ$, ὥστε τὸ ἀπὸ τῆς A ῥητῆς παρὰ τὴν BD ἀποτομὴν παραβαλλόμενον πλάτος ποιεῖ τὴν $KΘ$. λέγω, ὅτι ἐκ δύο ὀνομάτων ἐστὶν ἡ $KΘ$, ἥξ τὰ ὀνόματα σύμμετρα ἔστι τοῖς τῆς BD ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἔτι ἡ $KΘ$ τὴν αὐτὴν ἔθει τάξιν τῇ BD .

Ἐστω γάρ τῇ BD προσαρμόζουσα ἡ $ΔΓ$. αἱ $BΓ$, $ΓΔ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ ἀπὸ τῆς A ἴσον ἔστω καὶ τὸ ὑπὸ τῶν $BΓ$, H . ῥητὸν δὲ τὸ ἀπὸ τῆς A ῥητὸν ἄρα καὶ τὸ ὑπὸ τῶν $BΓ$, H . καὶ παρὰ ῥητὴν τὴν $BΓ$ παραβέβληται ῥητὴ ἄρα ἐστὶν ἡ H καὶ σύμμετρος τῇ $BΓ$ μήκει. ἐπεὶ οὖν τὸ ὑπὸ τῶν $BΓ$, H ἴσον ἐστὶ τῷ ὑπὸ τῶν BD , $KΘ$, ἀνάλογον ἄρα ἐστὶν ὡς ἡ $ΓB$ πρὸς BD , οὕτως ἡ $KΘ$ πρὸς H . μείζων δὲ ἡ $BΓ$ τῆς BD . μείζων ἄρα καὶ ἡ $KΘ$ τῆς H . κείσθω τῇ H ἴση ἡ KE . σύμμετρος ἄρα ἐστὶν ἡ KE τῇ $BΓ$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ἡ $ΓB$ πρὸς BD , οὕτως ἡ $ΘK$ πρὸς KE , ἀναστρέψαντι ἄρα ἐστὶν ὡς ἡ $BΓ$ πρὸς τὴν $ΓΔ$, οὕτως ἡ $KΘ$ πρὸς $ΘE$. γεγονέτω ὡς ἡ $KΘ$ πρὸς $ΘE$, οὕτως ἡ $ΘZ$ πρὸς ZE . καὶ λοιπὴ ἄρα ἡ KZ πρὸς $ZΘ$ ἐστὶν, ὡς ἡ $KΘ$ πρὸς $ΘE$, τοντέστιν [ὡς] ἡ $BΓ$ πρὸς $ΓΔ$. αἱ δὲ $BΓ$, $ΓΔ$ δυνάμει μόνον [εἰσὶ] σύμμετροι. καὶ αἱ KZ , $ZΘ$ ἄρα δυνάμει μόνον εἰσὶ σύμμετροι. καὶ ἐπεὶ ἐστὶν ὡς ἡ $KΘ$ πρὸς $ΘE$, ἡ KZ πρὸς $ZΘ$, ἀλλ' ὡς ἡ $KΘ$ πρὸς $ΘE$, ἡ $ΘZ$ πρὸς ZE , καὶ ὡς ἄρα ἡ KZ πρὸς $ZΘ$, ἡ $ΘZ$ πρὸς ZE . ὥστε καὶ ὡς ἡ πρώτη πρὸς τὴν τρίτην, τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας. καὶ ὡς ἄρα ἡ KZ πρὸς ZE , οὕτως τὸ ἀπὸ τῆς KZ πρὸς τὸ ἀπὸ τῆς $ZΘ$. σύμμετρον δὲ ἐστὶ τὸ ἀπὸ τῆς KZ τῷ ἀπὸ τῆς $ZΘ$. αἱ γὰρ KZ , $ZΘ$ δυνάμει εἰσὶ σύμμετροι.

Let A be a rational (straight-line), and BD an apotome. And let the (rectangle contained) by BD and KH be equal to the (square) on A , such that the square on the rational (straight-line) A , applied to the apotome BD , produces KH as breadth. I say that KH is a binomial whose terms are commensurable with the terms of BD , and in the same ratio, and, moreover, that KH has the same order as BD .

For let DC be an attachment to BD . Thus, BC and CD are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let the (rectangle contained) by BC and G also be equal to the (square) on A . And the (square) on A (is) rational. The (rectangle contained) by BC and G (is) thus also rational. And it has been applied to the rational (straight-line) BC . Thus, G is rational, and commensurable in length with BC [Prop. 10.20]. Therefore, since the (rectangle contained) by BC and G is equal to the (rectangle contained) by BD and KH , thus, proportionally, as CB is to BD , so KH (is) to G [Prop. 6.16]. And BC (is) greater than BD . Thus, KH (is) also greater than G [Prop. 5.16, 5.14]. Let KE be made equal to G . KE is thus commensurable in length with BC . And since as CB is to BD , so HK (is) to KE , thus, via conversion, as BC (is) to CD , so KH (is) to HE [Prop. 5.19 corr.]. Let it be contrived that as KH (is) to HE , so HF (is) to FE . And thus the remainder KF is to FH , as KH (is) to HE —that is to say, [as] BC (is) to CD [Prop. 5.19]. And BC and CD [are] commensurable in square only. KF and FH are thus also commensurable in square only [Prop. 10.11]. And since as KH is to HE , (so) KF (is) to FH , but as KH (is) to HE ,

σύμμετρος ἄρα ἐστὶ καὶ ἡ KZ τῇ ZE μήκει· ὥστε ἡ KZ καὶ τῇ KE σύμμετρος [ἐστὶ] μήκει. ῥητὴ δὲ ἐστὶν ἡ KE καὶ σύμμετρος τῇ $BΓ$ μήκει. ῥητὴ ἄρα καὶ ἡ KZ καὶ σύμμετρος τῇ $BΓ$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ἡ $BΓ$ πρὸς $ΓΔ$, οὕτως ἡ KZ πρὸς $ZΘ$, ἐναλλάξ ὡς ἡ $BΓ$ πρὸς KZ , οὕτως ἡ $ΔΓ$ πρὸς $ZΘ$. σύμμετρος δὲ ἡ $BΓ$ τῇ KZ · σύμμετρος ἄρα καὶ ἡ $ZΘ$ τῇ $ΓΔ$ μήκει. αἱ $BΓ$, $ΓΔ$ δὲ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· καὶ αἱ KZ , $ZΘ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ὀνομάτων ἐστὶν ἄρα ἡ $KΘ$.

Εἰ μὲν οὖν ἡ $BΓ$ τῇ $ΓΔ$ μείζον δύνανται τῷ ἀπὸ συμμέτρον ἑαυτῇ, καὶ ἡ KZ τῇ $ZΘ$ μείζον δυνήσεται τῷ ἀπὸ συμμέτρον ἑαυτῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ $BΓ$ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ KZ , εἰ δὲ ἡ $ΓΔ$ σύμμετρος ἐστὶ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ $ZΘ$, εἰ δὲ οὐδετέρα τῶν $BΓ$, $ΓΔ$, οὐδετέρα τῶν KZ , $ZΘ$.

Εἰ δὲ ἡ $BΓ$ τῇ $ΓΔ$ μείζον δύνανται τῷ ἀπὸ ἀσυνμέτρον ἑαυτῇ, καὶ ἡ KZ τῇ $ZΘ$ μείζον δυνήσεται τῷ ἀπὸ ἀσυνμέτρον ἑαυτῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ $BΓ$ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ KZ , εἰ δὲ ἡ $ΓΔ$, καὶ ἡ $ZΘ$, εἰ δὲ οὐδετέρα τῶν $BΓ$, $ΓΔ$, οὐδετέρα τῶν KZ , $ZΘ$.

Ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $KΘ$, ἥξ τὰ ὀνόματα τὰ KZ , $ZΘ$ σύμμετρα [ἐστὶ] τοῖς τῇς ἀποτομῆς ὀνόμασι τοῖς $BΓ$, $ΓΔ$ καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἔτι ἡ $KΘ$ τῇ $BΓ$ τὴν αὐτὴν ἔχει τάχιν· ὅπερ ἔδει δεῖχαι.

(so) HF (is) to FE , thus, also as KF (is) to FH , (so) HF (is) to FE [Prop. 5.11]. And hence as the first (is) to the third, so the (square) on the first (is) to the (square) on the second [Def. 5.9]. And thus as KF (is) to FE , so the (square) on KF (is) to the (square) on FH . And the (square) on KF is commensurable with the (square) on FH . For KF and FH are commensurable in square. Thus, KF is also commensurable in length with FE [Prop. 10.11]. Hence, KF [is] also commensurable in length with KE [Prop. 10.15]. And KE is rational, and commensurable in length with BC . Thus, KF (is) also rational, and commensurable in length with BC [Prop. 10.12]. And since as BC is to CD , (so) KF (is) to FH , alternately, as BC (is) to KF , so DC (is) to FH [Prop. 5.16]. And BC (is) commensurable (in length) with KF . Thus, FH (is) also commensurable in length with CD [Prop. 10.11]. And BC and CD are rational (straight-lines which are) commensurable in square only. KF and FH are thus also rational (straight-lines which are) commensurable in square only [Def. 10.3, Prop. 10.13]. Thus, KH is a binomial [Prop. 10.36].

Therefore, if the square on BC is greater than (the square on) CD by the (square) on (some straight-line) commensurable (in length) with (BC), then the square on KF will also be greater than (the square on) FH by the (square) on (some straight-line) commensurable (in length) with (KF) [Prop. 10.14]. And if BC is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) KF [Prop. 10.12]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FH [Prop. 10.12]. And if neither of BC or CD (are) commensurable, neither also (are) either of KF or FH [Prop. 10.13].

And if the square on BC is greater than (the square on) CD by the (square) on (some straight-line) incommensurable (in length) with (BC) then the square on KF will also be greater than (the square on) FH by the (square) on (some straight-line) incommensurable (in length) with (KF) [Prop. 10.14]. And if BC is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) KF [Prop. 10.12]. And if CD is commensurable, (so) also (is) FH [Prop. 10.12]. And if neither of BC or CD (are) commensurable, neither also (are) either of KF or FH [Prop. 10.13].

KH is thus a binomial whose terms, KF and FH , [are] commensurable (in length) with the terms, BC and CD , of the apotome, and in the same ratio. Moreover, KH will have the same order as BC [Defs. 10.5—10.10]. (Which is) the very thing it was required to show.

114.

Proposition 114

Ἐάν θωρίον περιέθῃται ὑπὸ ἀποτομῆς καὶ τῇς ἐκ δύο ὀνομάτων, ἥξ τὰ ὀνόματα σύμμετρα τέ ἐστι τοῖς τῇς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἡ τὸ θωρίον δυναμένη ῥητὴ

If an area is contained by an apotome, and a binomial whose terms are commensurable with, and in the same ratio as, the terms of the apotome then the square-root of the area is

έστιν.

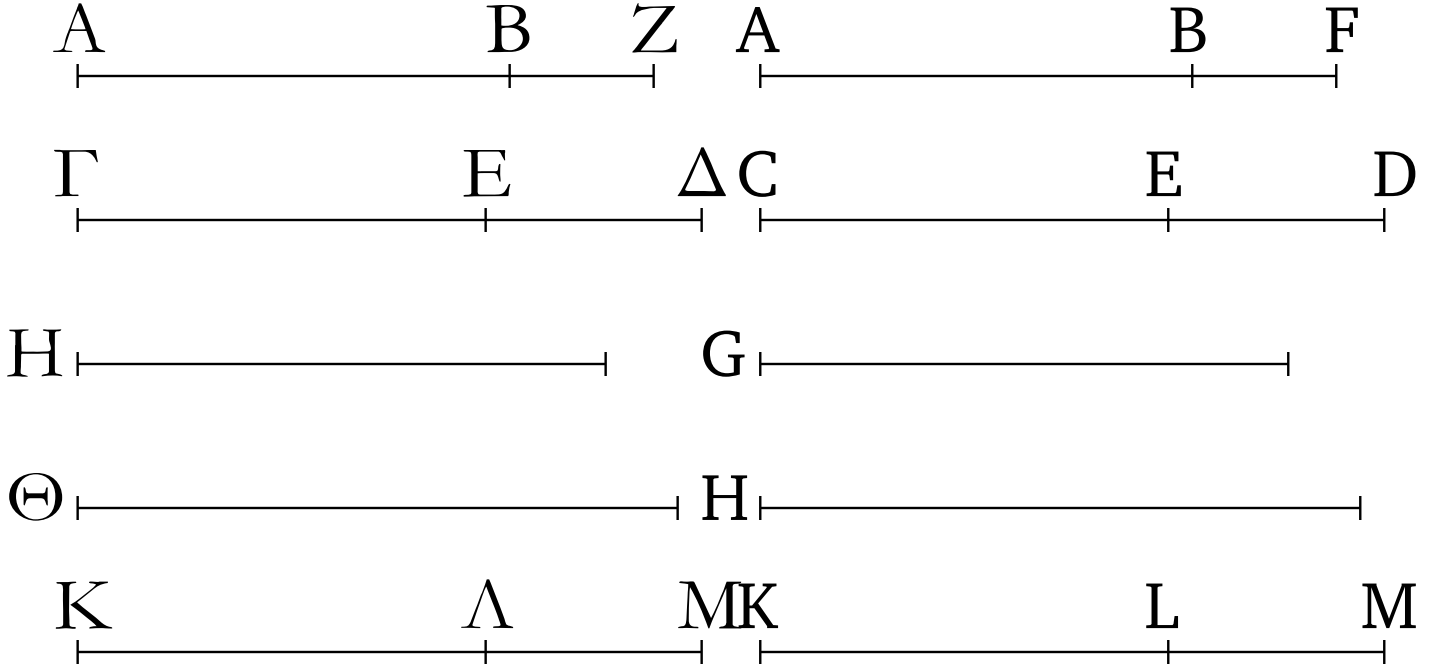
Περιεθέσθω γάρ θωρίον τὸ ὑπὸ τῶν AB , $\Gamma\Delta$ ὑπὸ ἀποτομῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων τῆς $\Gamma\Delta$, ἥς μείζον ὄνομα ἔστω τὸ ΓE , καὶ ἔστω τὰ ὀνόματα τῆς ἐκ δύο ὀνομάτων τὰ ΓE , $E\Delta$ σύμμετρά τε τοῖς τῆς ἀποτομῆς ὀνόμασι τοῖς AZ , ZB καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἔστω ἡ τὸ ὑπὸ τῶν AB , $\Gamma\Delta$ δυναμένη ἡ H · λέγω, ὅτι ῥητὴ ἔστιν ἡ H .

Ἐκκείσθω γάρ ῥητὴ ἡ Θ , καὶ τῷ ἀπὸ τῆς Θ ἴσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω πλάτος ποιῶν τὴν $ΚΛ$ · ἀποτομὴ ἄρα ἔστιν ἡ $ΚΛ$, ἥς τὰ ὀνόματα ἔστω τὰ KM , ML σύμμετρα τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι τοῖς ΓE , $E\Delta$ καὶ ἐν τῷ αὐτῷ λόγῳ. ἀλλὰ καὶ αἱ ΓE , $E\Delta$ σύμμετροί τε εἰσι ταῖς AZ , ZB καὶ ἐν τῷ αὐτῷ λόγῳ· ἔστιν ἄρα ὡς ἡ AZ πρὸς τὴν ZB , οὕτως ἡ KM πρὸς ML . ἐναλλάχ ἄρα ἔστιν ὡς ἡ AZ πρὸς τὴν KM , οὕτως ἡ BZ πρὸς τὴν $ΛΜ$ · καὶ λοιπὴ ἄρα ἡ AB πρὸς λοιπὴν τὴν $ΚΛ$ ἔστιν ὡς ἡ AZ πρὸς KM . σύμμετρος δὲ ἡ AZ τῇ KM · σύμμετρος ἄρα ἔστι καὶ ἡ AB τῇ $ΚΛ$. καὶ ἔστιν ὡς ἡ AB πρὸς $ΚΛ$, οὕτως τὸ ὑπὸ τῶν $\Gamma\Delta$, AB πρὸς τὸ ὑπὸ τῶν $\Gamma\Delta$, $ΚΛ$ · σύμμετρον ἄρα ἔστι καὶ τὸ ὑπὸ τῶν $\Gamma\Delta$, $ΚΛ$ · ἴσον δὲ τὸ ὑπὸ τῶν $\Gamma\Delta$, $ΚΛ$ τῷ ἀπὸ τῆς Θ · σύμμετρον ἄρα ἔστι τὸ ὑπὸ τῶν $\Gamma\Delta$, AB τῷ ἀπὸ τῆς Θ . τῷ δὲ ὑπὸ τῶν $\Gamma\Delta$, AB ἴσον ἔστι τὸ ἀπὸ τῆς H · σύμμετρον ἄρα ἔστι τὸ ἀπὸ τῆς H τῷ ἀπὸ τῆς Θ . ῥητὸν δὲ τὸ ἀπὸ τῆς Θ · ῥητὸν ἄρα ἔστι καὶ τὸ ἀπὸ τῆς H · ῥητὴ ἄρα ἔστιν ἡ H . καὶ δύναται τὸ ὑπὸ τῶν $\Gamma\Delta$, AB .

a rational (straight-line).

For let an area, the (rectangle contained) by AB and CD , be contained by the apotome AB , and the binomial CD , of which let the greater term be CE . And let the terms of the binomial, CE and ED , be commensurable with the terms of the apotome, AF and FB (respectively), and in the same ratio. And let the square-root of the (rectangle contained) by AB and CD be G . I say that G is a rational (straight-line).

For let the rational (straight-line) H be laid down. And let (some rectangle), equal to the (square) on H , be applied to CD , producing KL as breadth. Thus, KL is an apotome, of which let the terms, KM and ML , be commensurable with the terms of the binomial, CE and ED (respectively), and in the same ratio [Prop. 10.112]. But, CE and ED are also commensurable with AF and FB (respectively), and in the same ratio. Thus, as AF is to FB , so KM (is) to ML . Thus, alternately, as AF is to KM , so BF (is) to LM [Prop. 5.16]. Thus, the remainder AB is also to the remainder KL as AF (is) to KM [Prop. 5.19]. And AF (is) commensurable with KM [Prop. 10.12]. AB is thus also commensurable with KL [Prop. 10.11]. And as AB is to KL , so the (rectangle contained) by CD and AB (is) to the (rectangle contained) by CD and KL [Prop. 6.1]. Thus, the (rectangle contained) by CD and AB is also commensurable with the (rectangle contained) by CD and KL [Prop. 10.11]. And the (rectangle contained) by CD and KL (is) equal to the (square) on H . Thus, the (rectangle contained) by CD and AB is commensurable with the (square) on H . And the (square) on G is equal to the (rectangle contained) by CD and AB . The (square) on G is thus commensurable with the (square) on H . And the (square) on H (is) rational. Thus, the (square) on G is also rational. G is thus rational. And it is the square-root of the (rectangle contained) by CD and AB .



Ἐάν ἄρα θωρίον περιέθῃται ὑπὸ ἀποτομῆς καὶ τῆς ἐκ δύο ὀνομάτων, ἥς τὰ ὀνόματα σύμμετρα ἐστὶ τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἡ τὸ θωρίον δυναμένη ῥητὴ ἐστίν.

Thus, if an area is contained by an apotome, and a binomial whose terms are commensurable with, and in the same ratio as, the terms of the apotome, then the square-root of the area is a rational (straight-line).

Πόρισμα.

Καὶ γέγονεν ἡμῖν καὶ διὰ τούτου φανερόν, ὅτι δυνατόν ἐστι ῥητὸν θωρίον ὑπὸ ἀλόγων εὐθειῶν περιέθεσθαι. ὅπερ ἔδει δεῖξαι.

Corollary

And it has also been made clear to us, through this, that it is possible for a rational area to be contained by irrational straight-lines. (Which is) the very thing it was required to show.

115.

Ἀπὸ μέσης ἄπειροι ἄλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾶ τῶν πρότερον ἢ αὐτῇ.

Ἐστω μέση ἡ A· λέγω, ὅτι ἀπὸ τῆς A ἄπειροι ἄλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾶ τῶν πρότερον ἢ αὐτῇ.

Proposition 115

An infinite (series) of irrational (straight-lines) can be created from a medial (straight-line), and none of them is the same as any of the preceding (straight-lines).

Let A be a medial (straight-line). I say that an infinite (series) of irrational (straight-lines) can be created from A, and that none of them is the same as any of the preceding (straight-lines).

A ————

B —————

Γ —————

Δ —————

A ————

B —————

C —————

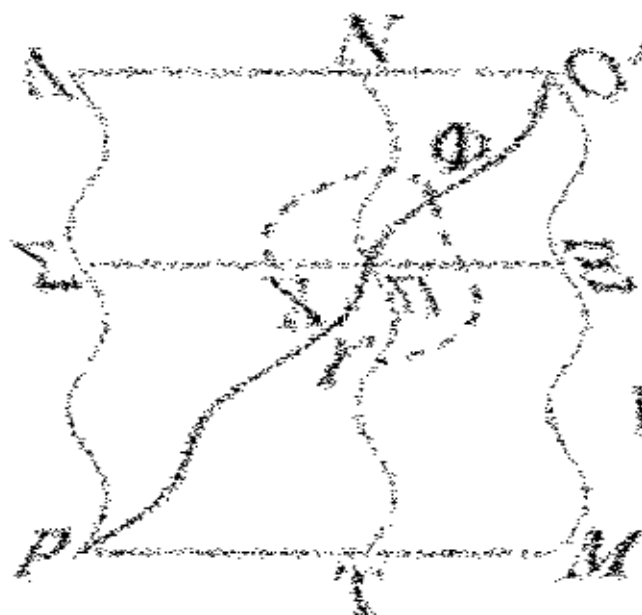
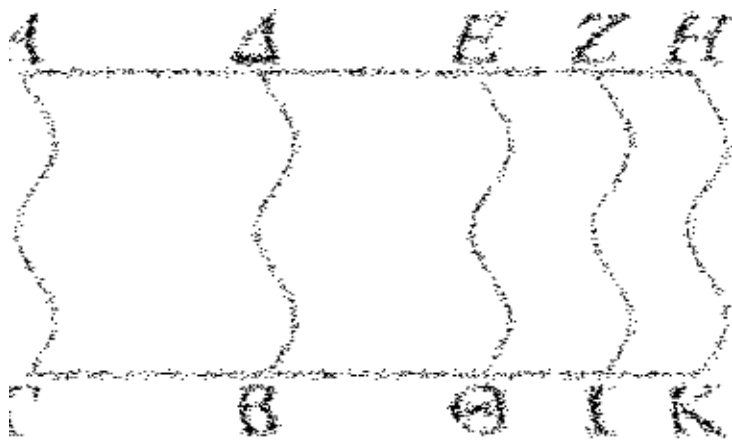
D —————

Ἐκκείσθω ῥητὴ ἡ B , καὶ τῷ ὑπὸ τῶν B , A ἴσον ἔστω τὸ ἀπὸ τῆς Γ · ἄλογος ἄρα ἐστὶν ἡ Γ · τὸ γὰρ ὑπὸ ἀλόγων καὶ ῥητῆς ἄλογόν ἐστιν. καὶ οὐδεμιᾶ τῶν πρότερον ἡ αὐτὴ· τὸ γὰρ ἀπ' οὐδεμιᾶς τῶν πρότερον παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ μέσην. πάλιν δὴ τῷ ὑπὸ τῶν B , Γ ἴσον ἔστω τὸ ἀπὸ τῆς Δ · ἄλογον ἄρα ἐστὶ τὸ ἀπὸ τῆς Δ · ἄλογος ἄρα ἐστὶν ἡ Δ · καὶ οὐδεμιᾶ τῶν πρότερον ἡ αὐτὴ· τὸ γὰρ ἀπ' οὐδεμιᾶς τῶν πρότερον παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν Γ . ὁμοίως δὴ τῆς τοιαύτης τάξεως ἐπ' ἄπειρον προβαينούσης φανερόν, ὅτι ἀπὸ τῆς μέσης ἄπειροι ἄλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾶ τῶν πρότερον ἡ αὐτὴ· ὅπερ ἔδει δεῖχαι.

Let the rational (straight-line) B be laid down. And let the (square) on C be equal to the (rectangle contained) by B and A . Thus, C is irrational [Def. 10.4]. For an (area contained) by an irrational and a rational (straight-line) is irrational [Prop. 10.20]. And (C is) not the same as any of the preceding (straight-lines). For the (square) on none of the preceding (straight-lines), applied to a rational (straight-line), produces a medial (straight-line) as breadth. So, again, let the (square) on D be equal to the (rectangle contained) by B and C . Thus, the (square) on D is irrational [Prop. 10.20]. D is thus irrational [Def. 10.4]. And (D is) not the same as any of the preceding (straight-lines). For the (square) on none of the preceding (straight-lines), applied to a rational (straight-line), produces C as breadth. So, similarly, this arrangement being advanced to infinity, it is clear that an infinite (series) of irrational (straight-lines) can be created from a medial (straight-line), and that none of them is the same as any of the preceding (straight-lines). (Which is) the very thing it was required to show.

ELEMENTS BOOK 10

Incommensurable Magnitudes[†]



[†]The theory of incommensurable magnitudes set out in this book is generally attributed to Theaetetus of Athens. In the footnotes throughout this book, k , k' , etc. stand for distinct ratios of positive integers.

Ὅροι.

Definitions

1. Σύμμετρα μεγέθη λέγεται τὰ τῷ αὐτῷ μετρῶ μετρούμενα, ἀσύμμετρα δέ, ὧν μηδὲν ἐνδέχεται κοινὸν μέτρον γενέσθαι.

2. Εὐθείαι δυνάμει σύμμετροί εἰσιν, ὅταν τὰ ἀπ' αὐτῶν τετράγωνα τῷ αὐτῷ θωρίῳ μετρήται, ἀσύμμετροι δέ, ὅταν τοῖς ἀπ' αὐτῶν τετραγώνοις μηδὲν ἐνδέχεται θωρίον κοινὸν μέτρον γενέσθαι.

3. Τούτων ὑποκειμένων δείκνυται, ὅτι τῇ προτεθείσῃ εὐθείᾳ ὑπάρχουσιν εὐθείαι πλήθει ἅπειροι σύμμετροί τε καὶ ἀσύμμετροι αἱ μὲν μήκει μόνον, αἱ δὲ καὶ δυνάμει. καλείσθω ὅν ἡ μὲν προτεθείσα εὐθεῖα ῥητή, καὶ αἱ ταύτης σύμμετροι εἴτε μήκει καὶ δυνάμει εἴτε δυνάμει μόνον ῥηταί, αἱ δὲ ταύτης ἀσύμμετροι ἄλογοι καλείσθωσαν.

4. Καὶ τὸ μὲν ἀπὸ τῆς προτεθείσης εὐθείας τετράγωνον ῥητόν, καὶ τὰ τούτῳ σύμμετρα ῥητά, τὰ δὲ τούτῳ ἀσύμμετρα ἄλογα καλείσθω, καὶ αἱ δυνάμεναι αὐτὰ ἄλογοι, εἰ μὲν τετράγωνα εἴη, αὐταὶ αἱ πλευραί, εἰ δὲ ἕτερά τινα εὐθύγραμμα, αἱ ἴσα αὐτοῖς τετράγωνα ἀναγράφουσαι.

1. Those magnitudes measured by the same measure are said (to be) commensurable, but (those) of which no (magnitude) admits to be a common measure (are said to be) incommensurable.[†]

2. (Two) straight-lines are commensurable in square[‡] when the squares on them are measured by the same area, but (are) incommensurable (in square) when no area admits to be a common measure of the squares on them.[§]

3. These things being assumed, it is proved that there exist an infinite multitude of straight-lines commensurable and incommensurable with an assigned straight-line—those (incommensurable) in length only, and those also (commensurable or incommensurable) in square.[¶] Therefore, let the assigned straight-line be called rational. And (let) the (straight-lines) commensurable with it, either in length and square, or in square only, (also be called) rational. But let the (straight-lines) incommensurable with it be called irrational.*

4. And let the square on the assigned straight-line be called rational. And (let areas) commensurable with it (also be called) rational. But (let areas) incommensurable with it (be called) irrational, and (let) their square-roots[§] (also be called) irrational—the sides themselves, if the (areas) are squares, and the (straight-lines) describing squares equal to them, if the (areas) are some other rectilinear (figure).^b

[†] In other words, two magnitudes α and β are commensurable if $\alpha : \beta :: 1 : k$, and incommensurable otherwise.

[‡] Literally, “in power”.

[§] In other words, two straight-lines of length α and β are commensurable in square if $\alpha : \beta :: 1 : k^{1/2}$, and incommensurable in square otherwise. Likewise, the straight-lines are commensurable in length if $\alpha : \beta :: 1 : k$, and incommensurable in length otherwise.

[¶] To be more exact, straight-lines can either be commensurable in square only, incommensurable in length only, or commensurable/incommensurable in both length and square, with an assigned straight-line.

* Let the length of the assigned straight-line be unity. Then rational straight-lines have lengths expressible as k or $k^{1/2}$, depending on whether the lengths are commensurable in length, or in square only, respectively, with unity. All other straight-lines are irrational.

[§] The square-root of an area is the length of the side of an equal area square.

^b The area of the square on the assigned straight-line is unity. Rational areas are expressible as k . All other areas are irrational. Thus, squares whose sides are of rational length have rational areas, and vice versa.

1.

Proposition 1[†]

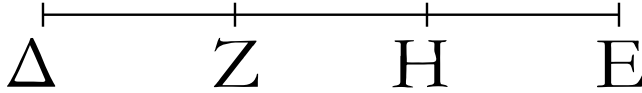
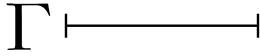
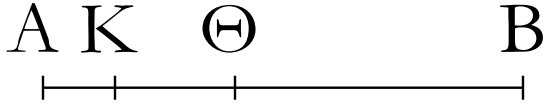
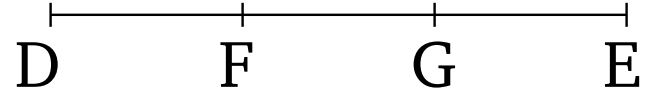
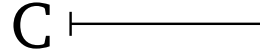
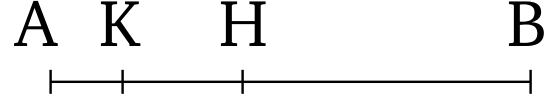
Δύο μεγεθῶν ἀνόσων ἐκκεκμένων, ἔαν ἀπὸ τοῦ μείζονος ἀφαιρεθῇ μείζον ἢ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μείζον ἢ τὸ ἥμισυ, καὶ τοῦτο αἰεὶ γίγνεται, λειφθήσεται τι μέγεθος, ὃ ἔσται ἔλασσον τοῦ ἐκκεκμένου ἐλάσσονος μεγέθους.

Ἐστω δύο μεγέθη ἅπασα τὰ AB , Γ , ὧν μείζον τὸ AB . λέγω, ὅτι, ἔαν ἀπὸ τοῦ AB ἀφαιρεθῇ μείζον ἢ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μείζον ἢ τὸ ἥμισυ, καὶ τοῦτο αἰεὶ γίγνεται, λειφθήσεται τι μέγεθος, ὃ ἔσται ἔλασσον τοῦ Γ μεγέθους.

If, from the greater of two unequal magnitudes (which are) laid out, (a part) greater than half is subtracted, and (if from) the remainder (a part) greater than half (is subtracted), and (if) this happens continually, then some magnitude will (eventually) be left which will be less than the lesser laid out magnitude.

Let AB and C be two unequal magnitudes, of which (let) AB (be) the greater. I say that if (a part) greater than half is subtracted from AB , and (if a part) greater than half (is sub-

tracted) from the remainder, and (if) this happens continually, then some magnitude will (eventually) be left which will be less than the magnitude C .



Τὸ Γ γὰρ πολλαπλασιαζόμενον ἔσται ποτὲ τοῦ AB μείζον. πεπολλαπλασιάσθω, καὶ ἔστω τὸ ΔE τοῦ μὲν Γ πολλαπλασίον, τοῦ δὲ AB μείζον, καὶ διηρῆσθω τὸ ΔE εἰς τὰ τῷ Γ ἴσα τὰ ΔZ , ZH , HE , καὶ ἀφηρήσθω ἀπὸ μὲν τοῦ AB μείζον ἢ τὸ ἥμισυν τὸ $B\Theta$, ἀπὸ δὲ τοῦ $A\Theta$ μείζον ἢ τὸ ἥμισυν τὸ ΘK , καὶ τοῦτο αἰ γινέσθω, ἕως ἄν αἱ ἐν τῷ AB διαιρέσεις ἰσοπληθεῖς γένωνται ταῖς ἐν τῷ ΔE διαιρέσεσιν.

Ἐστωσαν οὖν αἱ AK , $K\Theta$, ΘB διαιρέσεις ἰσοπληθεῖς οὔσαι ταῖς ΔZ , ZH , HE · καὶ ἐπεὶ μείζον ἔστι τὸ ΔE τοῦ AB , καὶ ἀφήρηται ἀπὸ μὲν τοῦ ΔE ἔλασσον τοῦ ἡμίσεως τὸ EH , ἀπὸ δὲ τοῦ AB μείζον ἢ τὸ ἥμισυν τὸ $B\Theta$, λοιπὸν ἄρα τὸ $H\Delta$ λοιποῦ τοῦ ΘA μείζον ἔστιν. καὶ ἐπεὶ μείζον ἔστι τὸ $H\Delta$ τοῦ ΘA , καὶ ἀφήρηται τοῦ μὲν $H\Delta$ ἥμισυν τὸ HZ , τοῦ δὲ ΘA μείζον ἢ τὸ ἥμισυν τὸ ΘK , λοιπὸν ἄρα τὸ ΔZ λοιποῦ τοῦ AK μείζον ἔστιν. ἴσον δὲ τὸ ΔZ τῷ Γ · καὶ τὸ Γ ἄρα τοῦ AK μείζον ἔστιν. ἔλασσον ἄρα τὸ AK τοῦ Γ .

Καταλείπεται ἄρα ἀπὸ τοῦ AB μεγέθονος τὸ AK μέγεθος ἔλασσον ὅν τοῦ ἐκκειμένου ἐλάσσονος μεγέθονος τοῦ Γ · ὅπερ ἔδει δεῖξαι.—ὁμοίως δὲ δειθθήσεται, κ' ἂν ἡμίση ἢ τὰ ἀφαιρούμενα.

For C , when multiplied (by some number), will sometimes be greater than AB [Def. 5.4]. Let it be (so) multiplied. And let DE be (both) a multiple of C , and greater than AB . And let DE be divided into the (divisions) DF , FG , GE , equal to C . And let BH , (which is) greater than half, be subtracted from AB . And (let) HK , (which is) greater than half, (be subtracted) from AH . And let this happen continually, until the divisions in AB become equal in number to the divisions in DE .

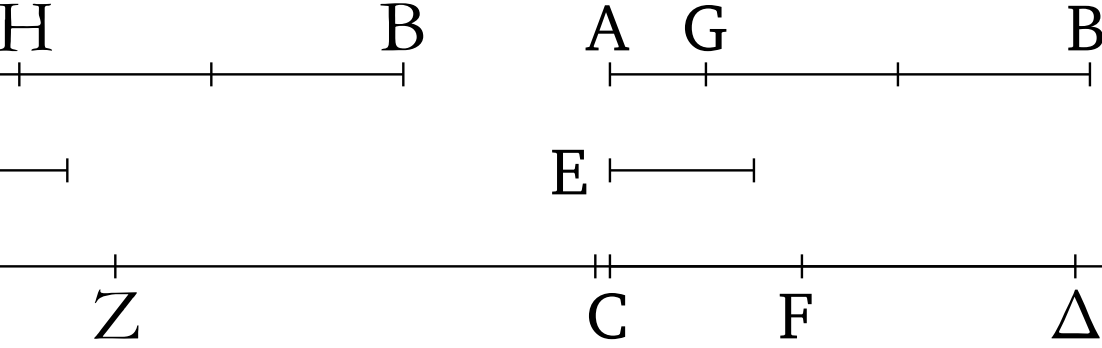
Therefore, let the divisions (in AB) be AK , KH , HB , being equal in number to DF , FG , GE . And since DE is greater than AB , and EG , (which is) less than half, has been subtracted from DE , and BH , (which is) greater than half, from AB , the remainder GD is thus greater than the remainder HA . And since GD is greater than HA , and the half GF has been subtracted from GD , and HK , (which is) greater than half, from HA , the remainder DF is thus greater than the remainder AK . And DF (is) equal to C . C is thus also greater than AK . Thus, AK (is) less than C .

Thus, the magnitude AK , which is less than the lesser laid out magnitude C , is left over from the magnitude AB . (Which is) the very thing it was required to show. — (The theorem) can similarly be proved even if the (parts) subtracted are halves.

† This theorem is the basis of the so-called method of exhaustion, and is generally attributed to Eudoxus of Cnidus.

Ἐάν δύο μεγεθῶν [ἐκκειμένων] ἀνίσων ἀνθυφαιρουμένων αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος τὸ καταλειπόμενον μηδέποτε καταμετρῇ τὸ πρὸ ἑαυτοῦ, ἀσύμμετρα ἔσται τὰ μεγέθη.

If the remainder of two unequal magnitudes (which are) [laid out] never measures the (magnitude) before it, (when) the lesser (magnitude is) continually subtracted in turn from the greater, then the (original) magnitudes will be incommensurable.



Δύο γὰρ μεγεθῶν ὄντων ἀνίσων τῶν AB , $\Gamma\Delta$ καὶ ἐλάσσονος τοῦ AB ἀνθυφαιρουμένου αἰ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος τὸ περιλειπόμενον μηδέποτε καταμετρεῖται τὸ πρὸ ἑαυτοῦ· λέγω, ὅτι ἀσύμμετρά ἐσσι τὰ AB , $\Gamma\Delta$ μεγέθη.

Εἰ γὰρ ἔστι σύμμετρα, μετρήσει τι αὐτὰ μέγεθος· μετρεῖται, εἰ δυνατόν, καὶ ἔστω τὸ E · καὶ τὸ μὲν AB τὸ ΔZ καταμετροῦν λειπέτω ἑαυτοῦ ἕλασσον τὸ ΓZ , τὸ δὲ ΓZ τὸ BH καταμετροῦν λειπέτω ἑαυτοῦ ἕλασσον τὸ AH , καὶ τοῦτο αἰ γνώσθω, ὅτι οὐ λειπθῇ τι μέγεθος, ὅ ἐστιν ἕλασσον τοῦ E . γεγονέντω, καὶ λελείφθω τὸ AH ἕλασσον τοῦ E . ἐπεὶ οὖν τὸ E τὸ AB μετρεῖ, ἀλλὰ τὸ AB τὸ ΔZ μετρεῖ, καὶ τὸ E ἄρα τὸ ΔZ μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ $\Gamma\Delta$ · καὶ λοιπὸν ἄρα τὸ ΓZ μετρήσει. ἀλλὰ τὸ ΓZ τὸ BH μετρεῖ· καὶ τὸ E ἄρα τὸ BH μετρεῖ. μετρεῖ δὲ καὶ ὅλον τὸ AB · καὶ λοιπὸν ἄρα τὸ AH μετρήσει, τὸ μείζον τὸ ἕλασσον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ AB , $\Gamma\Delta$ μεγέθη μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐσσι τὰ AB , $\Gamma\Delta$ μεγέθη.

Ἐάν ἄρα δύο μεγεθῶν ἀνίσων, καὶ τὰ ἐχῇς.

For, AB and CD being two unequal magnitudes, and AB (being) the lesser, let the remainder never measure the (magnitude) before it, (when) the lesser (magnitude is) continually subtracted in turn from the greater. I say that the magnitudes AB and CD are incommensurable.

For if they are commensurable then some magnitude will measure them (both). If possible, let it (so) measure (them), and let it be E . And let AB leave CF less than itself (in) measuring FD , and let CF leave AG less than itself (in) measuring BG , and let this happen continually, until some magnitude which is less than E is left. Let (this) have occurred,[†] and let AG , (which is) less than E , be left. Therefore, since E measures AB , but AB measures DF , E will thus also measure FD . And it also measures the whole (of) CD . Thus, it will also measure the remainder CF . But, CF measures BG . Thus, E also measures BG . And it also measures the whole (of) AB . Thus, it will also measure the remainder AG , the greater (measuring) the lesser. The very thing is impossible. Thus, some magnitude cannot measure (both) the magnitudes AB and CD . Thus, the magnitudes AB and CD are incommensurable [Def. 10.1].

Thus, if ... of two unequal magnitudes, and so on ...

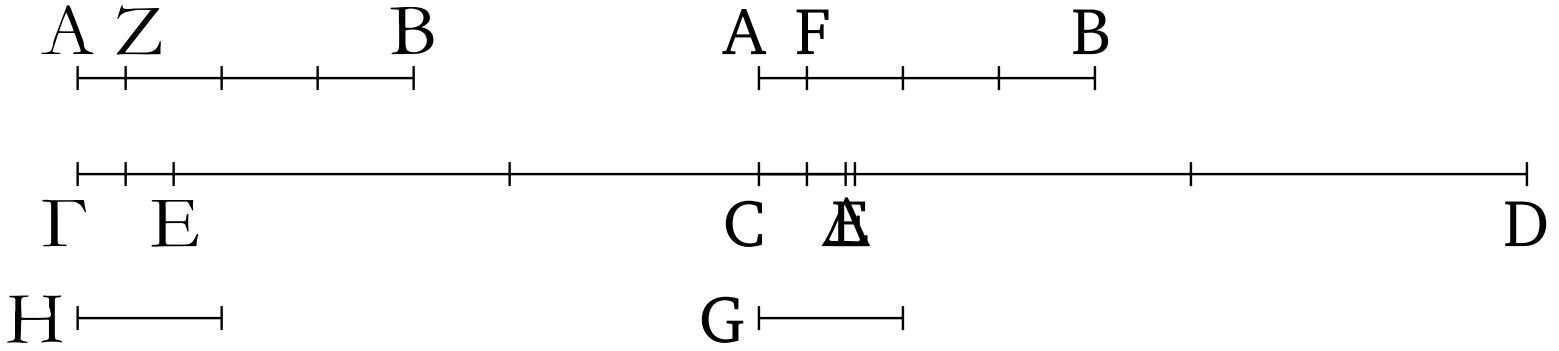
[†] The fact that this will eventually occur is guaranteed by Prop. 10.1.

3.

Δύο μεγεθῶν συμμέτρων δοθέντων τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

Proposition 3

To find the greatest common measure of two given commensurable magnitudes.



Ἐστω τὰ δοθέντα δύο μεγέθη σύμμετρα τὰ AB , $\Gamma\Delta$, ὧν ἔλασσον τὸ AB . δεῖ δὴ τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Τὸ AB γὰρ μέγεθος ἢτοι μετρεῖ τὸ $\Gamma\Delta$ ἢ οὐ. εἰ μὲν οὖν μετρεῖ, μετρεῖ δὲ καὶ ἑαυτό, τὸ AB ἄρα τῶν AB , $\Gamma\Delta$ κοινὸν μέτρον ἐστίν· καὶ φανερόν, ὅτι καὶ μέγιστον. μείζον γὰρ τοῦ AB μεγέθους τὸ AB οὐ μετρήσει.

μὴ μετρεῖτω δὴ τὸ AB τὸ $\Gamma\Delta$. καὶ ἀνθυφαιρουμένον αἰεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος, τὸ περιλειπόμενον μετρήσει ποτὲ τὸ πρὸ ἑαυτοῦ διὰ τὸ μὴ εἶναι ἀσύμμετρα τὰ AB , $\Gamma\Delta$. καὶ τὸ μὲν AB τὸ $E\Delta$ καταμετροῦν λειπέτω ἑαυτοῦ ἔλασσον τὸ EF , τὸ δὲ EF τὸ ZB καταμετροῦν λειπέτω ἑαυτοῦ ἔλασσον τὸ AZ , τὸ δὲ AZ τὸ ΓE μετρεῖτω.

Ἐπεὶ οὖν τὸ AZ τὸ ΓE μετρεῖ, ἀλλὰ τὸ ΓE τὸ ZB μετρεῖ, καὶ τὸ AZ ἄρα τὸ ZB μετρήσει. μετρεῖ δὲ καὶ ἑαυτό· καὶ ὅλον ἄρα τὸ AB μετρήσει τὸ AZ . ἀλλὰ τὸ AB τὸ ΔE μετρεῖ· καὶ τὸ AZ ἄρα τὸ $E\Delta$ μετρήσει. μετρεῖ δὲ καὶ τὸ ΓE · καὶ ὅλον ἄρα τὸ $\Gamma\Delta$ μετρεῖ· τὸ AZ ἄρα τῶν AB , $\Gamma\Delta$ κοινὸν μέτρον ἐστίν. λέγω δὴ, ὅτι καὶ μέγιστον. εἰ γὰρ μή, ἔσται τι μέγεθος μείζον τοῦ AZ , ὃ μετρήσει τὰ AB , $\Gamma\Delta$. ἔστω τὸ H . ἐπεὶ οὖν τὸ H τὸ AB μετρεῖ, ἀλλὰ τὸ AB τὸ $E\Delta$ μετρεῖ, καὶ τὸ H ἄρα τὸ $E\Delta$ μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ $\Gamma\Delta$. καὶ λοιπὸν ἄρα τὸ ΓE μετρήσει τὸ H . ἀλλὰ τὸ ΓE τὸ ZB μετρεῖ· καὶ τὸ H ἄρα τὸ ZB μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ AB , καὶ λοιπὸν τὸ AZ μετρήσει, τὸ μείζον τὸ ἔλασσον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα μείζον τι μέγεθος τοῦ AZ τὰ AB , $\Gamma\Delta$ μετρήσει· τὸ AZ ἄρα τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον ἐστίν.

Δύο ἄρα μεγεθῶν συμμέτρων δοθέντων τῶν AB , $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον ἡύρηται· ὅπερ ἔδει δεῖχαι.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἕαν μέγεθος δύο μεγέθη μετρήῃ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσει.

Let AB and CD be the two given magnitudes, of which (let) AB (be) the lesser. So, it is required to find the greatest common measure of AB and CD .

For the magnitude AB either measures, or (does) not (measure), CD . Therefore, if it measures (CD), and (since) it also measures itself, AB is thus a common measure of AB and CD . And (it is) clear that (it is) also (the) greatest. For a (magnitude) greater than magnitude AB cannot measure AB .

So let AB not measure CD . And continually subtracting in turn the lesser (magnitude) from the greater, the remaining (magnitude) will (at) some time measure the (magnitude) before it, on account of AB and CD not being incommensurable [Prop. 10.2]. And let AB leave EC less than itself (in) measuring ED , and let EC leave AF less than itself (in) measuring FB , and let AF measure CE .

Therefore, since AF measures CE , but CE measures FB , AF will thus also measure FB . And it also measures itself. Thus, AF will also measure the whole (of) AB . But, AB measures DE . Thus, AF will also measure ED . And it also measures CE . Thus, it also measures the whole of CD . Thus, AF is a common measure of AB and CD . So I say that (it is) also (the) greatest (common measure). For, if not, there will be some magnitude, greater than AF , which will measure (both) AB and CD . Let it be G . Therefore, since G measures AB , but AB measures ED , G will thus also measure ED . And it also measures the whole of CD . Thus, G will also measure the remainder CE . But CE measures FB . Thus, G will also measure FB . And it also measures the whole (of) AB . And (so) it will measure the remainder AF , the greater (measuring) the lesser. The very thing is impossible. Thus, some magnitude greater than AF cannot measure (both) AB and CD . Thus, AF is the greatest common measure of AB and CD .

Thus, the greatest common measure of two given commensurable magnitudes, AB and CD , has been found. (Which is) the very thing it was required to show.

Corollary

So (it is) clear, from this, that if a magnitude measures two magnitudes then it will also measure their greatest common

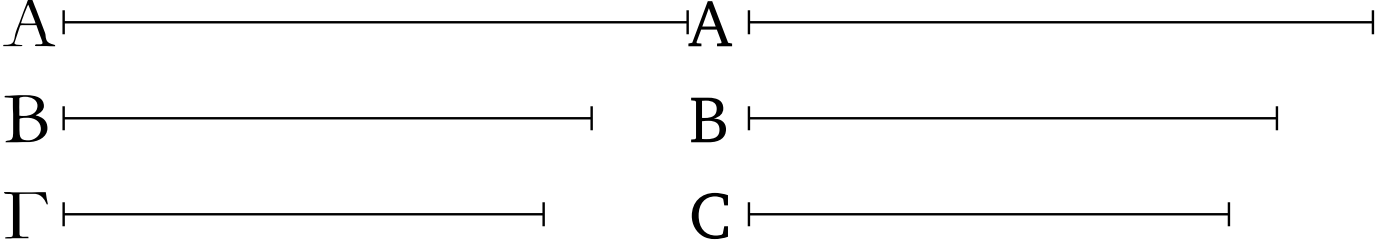
measure.

4.

Proposition 4

Τριῶν μεγεθῶν συμμετρων δοθέντων τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

To find the greatest common measure of three given commensurable magnitudes.



Ἐστω τὰ δοθέντα τρία μεγέθη σύμμετρα τὰ A, B, Γ · δεῖ δὴ τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Εἰλήφθω γάρ δύο τῶν A, B τὸ μέγιστον κοινὸν μέτρον, καὶ ἔστω τὸ Δ · τὸ δὴ Δ τὸ Γ ἤτοι μετρεῖ ἢ οὐ [μετρεῖ]. μετρεῖτω πρότερον. ἐπεὶ οὖν τὸ Δ τὸ Γ μετρεῖ, μετρεῖ δὲ καὶ τὰ A, B , τὸ Δ ἄρα τὰ A, B, Γ μετρεῖ· τὸ Δ ἄρα τῶν A, B, Γ κοινὸν μέτρον ἐστίν. καὶ φανερόν, ὅτι καὶ μέγιστον· μείζον γάρ τοῦ Δ μεγέθους τὰ A, B οὐ μετρεῖ.

μὴ μετρεῖτω δὴ τὸ Δ τὸ Γ . λέγω πρότερον, ὅτι σύμμετρά ἐστι τὰ Γ, Δ . ἐπεὶ γάρ σύμμετρά ἐστι τὰ A, B, Γ , μετρήσει τι αὐτὰ μέγεθος, ὃ δηλαδὴ καὶ τὰ A, B μετρήσει· ὥστε καὶ τὸ τῶν A, B μέγιστον κοινὸν μέτρον τὸ Δ μετρήσει. μετρεῖ δὲ καὶ τὸ Γ · ὥστε τὸ εἰρημένον μέγεθος μετρήσει τὰ Γ, Δ · σύμμετρα ἄρα ἐστὶ τὰ Γ, Δ . εἰλήφθω οὖν αὐτῶν τὸ μέγιστον κοινὸν μέτρον, καὶ ἔστω τὸ E . ἐπεὶ οὖν τὸ E τὸ Δ μετρεῖ, ἀλλὰ τὸ Δ τὰ A, B μετρεῖ, καὶ τὸ E ἄρα τὰ A, B μετρήσει. μετρεῖ δὲ καὶ τὸ Γ . τὸ E ἄρα τὰ A, B, Γ μετρεῖ· τὸ E ἄρα τῶν A, B, Γ κοινὸν ἐστὶ μέτρον. λέγω δὴ, ὅτι καὶ μέγιστον. εἰ γάρ δυνατόν, ἔστω τι τοῦ E μείζον μέγεθος τὸ Z , καὶ μετρεῖτω τὰ A, B, Γ . καὶ ἐπεὶ τὸ Z τὰ A, B, Γ μετρεῖ, καὶ τὰ A, B ἄρα μετρήσει καὶ τὸ τῶν A, B μέγιστον κοινὸν μέτρον μετρήσει. τὸ δὲ τῶν A, B μέγιστον κοινὸν μέτρον ἐστὶ τὸ Δ · τὸ Z ἄρα τὸ Δ μετρεῖ. μετρεῖ δὲ καὶ τὸ Γ · τὸ Z ἄρα τὰ Γ, Δ μετρεῖ· καὶ τὸ τῶν Γ, Δ ἄρα μέγιστον κοινὸν μέτρον μετρήσει τὸ Z . ἐστὶ δὲ τὸ E · τὸ Z ἄρα τὸ E μετρήσει, τὸ μείζον τὸ ἔλασσον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα μείζον τι τοῦ E μεγέθους [μέγεθος] τὰ A, B, Γ μετρεῖ· τὸ E ἄρα τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον ἐστίν, ἕαν μὴ μετρηῇ τὸ Δ τὸ Γ , ἕαν δὲ μετρηῇ, αὐτὸ τὸ Δ .

Τριῶν ἄρα μεγεθῶν συμμετρων δοθέντων τὸ μέγιστον κοινὸν μέτρον ἠΰρηται [ὅπερ ἔδει δεῖχαι].

Let A, B, C be the three given commensurable magnitudes. So it is required to find the greatest common measure of A, B, C .

For let the greatest common measure of the two (magnitudes) A and B be taken [Prop. 10.3], and let it be D . So D either measures, or [does] not [measure], C . Let it, first of all, measure (C). Therefore, since D measures C , and it also measures A and B , D thus measures A, B, C . Thus, D is a common measure of A, B, C . And (it is) clear that (it is) also (the) greatest (common measure). For no magnitude larger than D measures (both) A and B .

So let D not measure C . I say, first, that C and D are commensurable. For if A, B, C are commensurable then some magnitude will measure them which will clearly also measure A and B . Hence, it will also measure D , the greatest common measure of A and B [Prop. 10.3 corr.]. And it also measures C . Hence, the aforementioned magnitude will measure (both) C and D . Thus, C and D are commensurable [Def. 10.1]. Therefore, let their greatest common measure be taken [Prop. 10.3], and let it be E . Therefore, since E measures D , but D measures (both) A and B , E will thus also measure A and B . And it also measures C . Thus, E measures A, B, C . Thus, E is a common measure of A, B, C . So I say that (it is) also (the) greatest (common measure). For, if possible, let F be some magnitude greater than E , and let it measure A, B, C . And since F measures A, B, C , it will thus also measure A and B , and will (thus) measure the greatest common measure of A and B [Prop. 10.3 corr.]. And D is the greatest common measure of A and B . Thus, F measures D . And it also measures C . Thus, F measures (both) C and D . Thus, F will also measure the greatest common measure of C and D [Prop. 10.3 corr.]. And it is E . Thus, F will measure E , the greater (measuring) the lesser. The very thing is impossible. Thus, some [magnitude]

greater than the magnitude E cannot measure A, B, C . Thus, if D does not measure C then E is the greatest common measure of A, B, C . And if it does measure (C) then D itself (is the greatest common measure).

Thus, the greatest common measure of three given commensurable magnitudes has been found. [(Which is) the very thing it was required to show.]

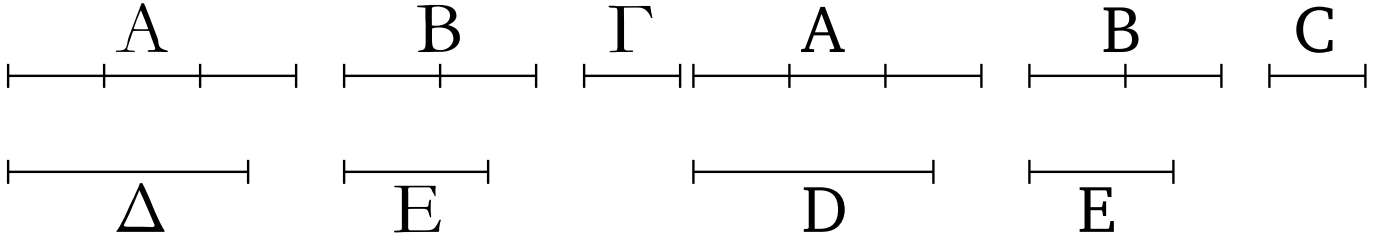
Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἔαν μέγεθος τρία μεγέθη μετρήῃ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσῃ.

Ὅμοίως δὴ καὶ ἐπὶ πλείονων τὸ μέγιστον κοινὸν μέτρον ληφθήσεται, καὶ τὸ πόρισμα προθωρήσῃ. ὅπερ ἔδει δεῖχαι.

5.

Τὰ σύμμετρα μεγέθη πρὸς ἄλληλα λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.



Ἐστω σύμμετρα μεγέθη τὰ A, B · λέγω, ὅτι τὸ A πρὸς τὸ B λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.

Ἐπεὶ γὰρ σύμμετρά ἐστι τὰ A, B , μετρήσῃ τι αὐτὰ μέγεθος. μετρεῖτω, καὶ ἔστω τὸ Γ . καὶ ὁσάκις τὸ Γ τὸ A μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ Δ , ὁσάκις δὲ τὸ Γ τὸ B μετρεῖ, τοσαῦται μονάδες ἔστωσαν ἐν τῷ E .

Ἐπεὶ οὖν τὸ Γ τὸ A μετρεῖ κατὰ τὰς ἐν τῷ Δ μονάδας, μετρεῖ δὲ καὶ ἡ μονὰς τὸν Δ κατὰ τὰς ἐν αὐτῷ μονάδας, ἰσάκις ἄρα ἡ μονὰς τὸν Δ μετρεῖ ἀριθμόν καὶ τὸ Γ μέγεθος τὸ A · ἔστιν ἄρα ὥς τὸ Γ πρὸς τὸ A , οὕτως ἡ μονὰς πρὸς τὸν Δ · ἀνάπαλιν ἄρα, ὥς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ πρὸς τὴν μονάδα. πάλιν ἐπεὶ τὸ Γ τὸ B μετρεῖ κατὰ τὰς ἐν τῷ E μονάδας, μετρεῖ δὲ καὶ ἡ μονὰς τὸν E κατὰ τὰς ἐν αὐτῷ μονάδας, ἰσάκις ἄρα ἡ μονὰς τὸν E μετρεῖ καὶ τὸ Γ τὸ B · ἔστιν ἄρα ὥς τὸ Γ πρὸς τὸ B , οὕτως ἡ μονὰς πρὸς τὸν E . ἐδείκθη δὲ καὶ ὥς τὸ A πρὸς τὸ Γ , ὁ Δ πρὸς τὴν μονάδα· δι' ἴσον ἄρα ἐστὶν ὥς τὸ A πρὸς τὸ B , οὕτως ὁ Δ ἀριθμὸς πρὸς τὸν E .

Τὰ ἄρα σύμμετρα μεγέθη τὰ A, B πρὸς ἄλληλα λόγον ἔθει, ὃν ἀριθμὸς ὁ Δ πρὸς ἀριθμόν τὸν E · ὅπερ ἔδει δεῖχαι.

Corollary

So (it is) clear, from this, that if a magnitude measures three magnitudes then it will also measure their greatest common measure.

So, similarly, the greatest common measure of more (magnitudes) can also be taken, and the (above) corollary will go forward. (Which is) the very thing it was required to show.

Proposition 5

Commensurable magnitudes have to one another the ratio which (some) number (has) to (some) number.

Let A and B be commensurable magnitudes. I say that A has to B the ratio which (some) number (has) to (some) number.

For if A and B are commensurable (magnitudes) then some magnitude will measure them. Let it (so) measure (them), and let it be C . And as many times as C measures A , so many units let there be in D . And as many times as C measures B , so many units let there be in E .

Therefore, since C measures A according to the units in D , and a unit also measures D according to the units in it, a unit thus measures the number D as many times as the magnitude C (measures) A . Thus, as C is to A , so a unit (is) to D [Def. 7.20].[†] Thus, inversely, as A (is) to C , so D (is) to a unit [Prop. 5.7 corr.]. Again, since C measures B according to the units in E , and a unit also measures E according to the units in it, a unit thus measures E the same number of times that C (measures) B . Thus, as C is to B , so a unit (is) to E [Def. 7.20]. And it was also shown that as A (is) to C , so D (is) to a unit. Thus, via equality, as A is to B , so the number D (is) to the (number) E [Prop. 5.22].

Thus, the commensurable magnitudes A and B have to one another the ratio which the number D (has) to the number E . (Which is) the very thing it was required to show.

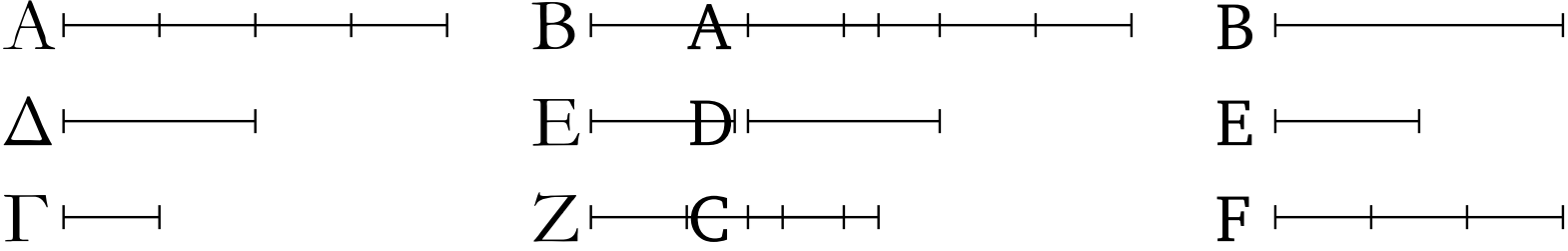
† There is a slight logical gap here, since Def. 7.20 applies to four numbers, rather than two number and two magnitudes.

6.

Proposition 6

Ἐάν δύο μεγέθη πρὸς ἄλληλα λόγον ἔθῃ, ὃν ἀριθμὸς πρὸς ἀριθμὸν, σύμμετρα ἔσται τὰ μεγέθη.

If two magnitudes have to one another the ratio which (some) number (has) to (some) number then the magnitudes will be commensurable.



Δύο γὰρ μεγέθη τὰ A, B πρὸς ἄλληλα λόγον ἔθῃτω, ὃν ἀριθμὸς ὁ Δ πρὸς ἀριθμὸν τὸν E · λέγω, ὅτι σύμμετρά ἐστι τὰ A, B μεγέθη.

Ὅσαι γάρ εἰσιν ἐν τῷ Δ μονάδες, εἰς τοσαῦτα ἴσα διηγήσθω τὸ A , καὶ ἐν αὐτῶν ἴσον ἔστω τὸ Γ · ὅσαι δὲ εἰσιν ἐν τῷ E μονάδες, ἐκ τοσούτων μεγεθῶν ἴσων τῷ Γ συγκείσθω τὸ Z .

Ἐπεὶ οὖν, ὅσαι εἰσιν ἐν τῷ Δ μονάδες, τοσαῦτά εἰσι καὶ ἐν τῷ A μεγέθει ἴσα τῷ Γ , ὃ ἄρα μέρος ἐστὶν ἡ μονὰς τοῦ Δ , τὸ αὐτὸ μέρος ἐστὶ καὶ τὸ Γ τοῦ A · ἐστὶν ἄρα ὡς τὸ Γ πρὸς τὸ A , οὕτως ἡ μονὰς πρὸς τὸν Δ . μετρεῖ δὲ ἡ μονὰς τὸν Δ ἀριθμὸν μετρεῖ ἄρα καὶ τὸ Γ τὸ A . καὶ ἐπεὶ ἐστὶν ὡς τὸ Γ πρὸς τὸ A , οὕτως ἡ μονὰς πρὸς τὸν Δ [ἀριθμὸν], ἀνάπαλιν ἄρα ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ ἀριθμὸς πρὸς τὴν μονάδα. πάλιν ἐπεὶ, ὅσαι εἰσιν ἐν τῷ E μονάδες, τοσαῦτά εἰσι καὶ ἐν τῷ Z ἴσα τῷ Γ , ἔστιν ἄρα ὡς τὸ Γ πρὸς τὸ Z , οὕτως ἡ μονὰς πρὸς τὸν E [ἀριθμὸν]. ἐδείχθη δὲ καὶ ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ πρὸς τὴν μονάδα· δι' ἴσον ἄρα ἐστὶν ὡς τὸ A πρὸς τὸ Z , οὕτως ὁ Δ πρὸς τὸν E . ἀλλ' ὡς ὁ Δ πρὸς τὸν E , οὕτως ἐστὶ τὸ A πρὸς τὸ B · καὶ ὡς ἄρα τὸ A πρὸς τὸ B , οὕτως καὶ πρὸς τὸ Z . τὸ A ἄρα πρὸς ἐκάτερον τῶν B, Z τὸν αὐτὸν ἔχει λόγον ἴσον ἄρα ἐστὶ τὸ B τῷ Z . μετρεῖ δὲ τὸ Γ τὸ Z · μετρεῖ ἄρα καὶ τὸ B . ἀλλὰ μὴν καὶ τὸ A · τὸ Γ ἄρα τὰ A, B μετρεῖ. σύμμετρον ἄρα ἐστὶ τὸ A τῷ B .

Ἐάν ἄρα δύο μεγέθη πρὸς ἄλληλα, καὶ τὰ ἐχῇ.

For let the two magnitudes A and B have to one another the ratio which the number D (has) to the number E . I say that the magnitudes A and B are commensurable.

For, as many units as there are in D , let A be divided into so many equal (divisions). And let C be equal to one of them. And as many units as there are in E , let F be the sum of so many magnitudes equal to C .

Therefore, since as many units as there are in D , so many magnitudes equal to C are also in A , therefore whichever part a unit is of D , C is also the same part of A . Thus, as C is to A , so a unit (is) to D [Def. 7.20]. And a unit measures the number D . Thus, C also measures A . And since as C is to A , so a unit (is) to the [number] D , thus, inversely, as A (is) to C , so the number D (is) to a unit [Prop. 5.7 corr.]. Again, since as many units as there are in E , so many (magnitudes) equal to C are also in F , there as C is to F , so a unit (is) to the [number] E [Def. 7.20]. And it was also shown that as A (is) to C , so D (is) to a unit. Thus, via equality, as A is to F , so D (is) to E [Prop. 5.22]. But, as D (is) to E , so A is to B . And thus as A (is) to B , so (it) also is to F [Prop. 5.11]. Thus, A has the same ratio to each of B and F . Thus, B is equal to F [Prop. 5.9]. And C measures F . Thus, it also measures B . But, in fact, (it) also (measures) A . Thus, C measures (both) A and B . Thus, A is commensurable with B [Def. 10.1].

Thus, if two magnitudes ... to one another, and so on

Corollary

So it is clear, from this, that if there are two numbers, like

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἔαν ᾧσι δύο ἀριθμοί, ὡς οἱ Δ, E , καὶ εὐθεῖα, ὡς ἡ A , δυνατόν ἐστι ποιῆσαι ὡς ὁ Δ ἀριθμὸς πρὸς τὸν E ἀριθμὸν, οὕτως τὴν εὐθεῖαν πρὸς εὐθεῖαν. ἔαν δὲ καὶ τῶν A, Z μέση ἀνάλογον ληφθῇ, ὡς ἡ B , ἔσται ὡς ἡ A πρὸς τὴν Z , οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς B , τουτέστιν ὡς ἡ πρώτη πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ ὅμοιον καὶ ὁμοίως

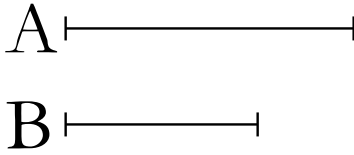
D and E , and a straight-line, like A , then it is possible to contrive that as the number D (is) to the number E , so the straight-line (is) to (another) straight-line (i.e., F). And if the mean proportion, (say) B , is taken of A and F , then as A is to F , so the (square) on A (will be) to the (square) on B . That is to say, as the first (is) to the third, so the (figure) on the first (is) to the similar, and similarly described, (figure) on the second

ἀναγραφόμενον. ἀλλ' ὥς ἡ A πρὸς τὴν Z , οὕτως ἐστὶν ὁ Δ ἀριθμὸς πρὸς τὸν E ἀριθμόν· γέγονεν ἄρα καὶ ὥς ὁ Δ ἀριθμὸς πρὸς τὸν E ἀριθμόν, οὕτως τὸ ἀπὸ τῆς A εὐθείας πρὸς τὸ ἀπὸ τῆς B εὐθείας· ὅπερ ἔδει δεῖξαι.

7.

Τὰ ἀσύμμετρα μεγέθη πρὸς ἄλληλα λόγον οὐκ ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.

Ἐστω ἀσύμμετρα μεγέθη τὰ A , B · λέγω, ὅτι τὸ A πρὸς τὸ B λόγον οὐκ ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.

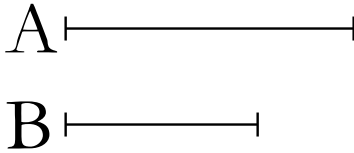


Εἰ γὰρ ἔθει τὸ A πρὸς τὸ B λόγον, ὃν ἀριθμὸς πρὸς ἀριθμόν, σύμμετρον ἔσται τὸ A τῷ B . οὐκ ἔστι δέ· οὐκ ἄρα τὸ A πρὸς τὸ B λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν.

Τὰ ἄρα ἀσύμμετρα μεγέθη πρὸς ἄλληλα λόγον οὐκ ἔθει, καὶ τὰ ἐχῆξ.

8.

Ἐὰν δύο μεγέθη πρὸς ἄλληλα λόγον μὴ ἔθῃ, ὃν ἀριθμὸς πρὸς ἀριθμόν, ἀσύμμετρα ἔσται τὰ μεγέθη.



Δύο γὰρ μεγέθη τὰ A , B πρὸς ἄλληλα λόγον μὴ ἔθῃτω, ὃν ἀριθμὸς πρὸς ἀριθμόν· λέγω, ὅτι ἀσύμμετρά ἐσσι τὰ A , B μεγέθη.

Εἰ γὰρ ἔσται σύμμετρα, τὸ A πρὸς τὸ B λόγον ἔχει, ὃν ἀριθμὸς πρὸς ἀριθμόν. οὐκ ἔθει δέ· ἀσύμμετρα ἄρα ἐσσι τὰ A , B μεγέθη.

Ἐὰν ἄρα δύο μεγέθη πρὸς ἄλληλα, καὶ τὰ ἐχῆξ.

9.

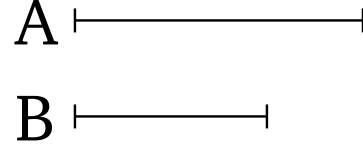
Τὰ ἀπὸ τῶν μήκει συμμέτρων εὐθειῶν τετράγωνα πρὸς ἄλληλα λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὰ τετράγωνα τὰ πρὸς ἄλληλα λόγον ἔθοντα, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ τὰς

[Prop. 6.19 corr.]. But, as A (is) to F , so the number D is to the number E . Thus, it has also been contrived that as the number D (is) to the number E , so the (figure) on the straight-line A (is) to the (similar figure) on the straight-line B . (Which is) the very thing it was required to show.

Proposition 7

Incommensurable magnitudes do not have to one another the ratio which (some) number (has) to (some) number.

Let A and B be incommensurable magnitudes. I say that A does not have to B the ratio which (some) number (has) to (some) number.

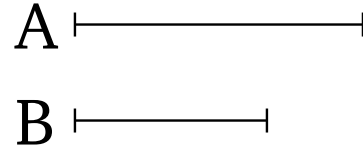


For if A has to B the ratio which (some) number (has) to (some) number then A will be commensurable with B [Prop. 10.6]. But it is not. Thus, A does not have to B the ratio which (some) number (has) to (some) number.

Thus, incommensurable numbers do not have to one another, and so on . . .

Proposition 8

If two magnitudes do not have to one another the ratio which (some) number (has) to (some) number then the magnitudes will be incommensurable.



For let the two magnitudes A and B not have to one another the ratio which (some) number (has) to (some) number. I say that the magnitudes A and B are incommensurable.

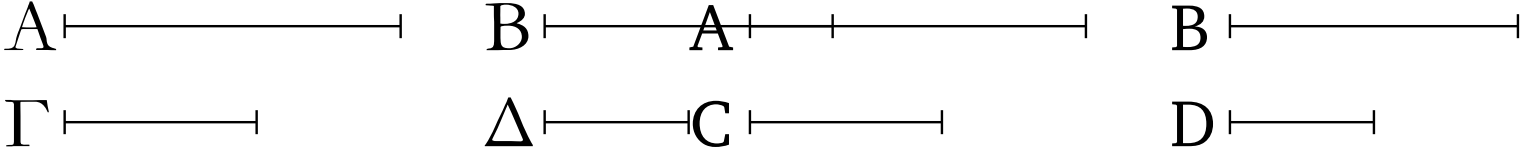
For if they are commensurable, A will have to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. But it does not have (such a ratio). Thus, the magnitudes A and B are incommensurable.

Thus, if two magnitudes . . . to one another, and so on . . .

Proposition 9

Squares on straight-lines (which are) commensurable in length have to one another the ratio which (some) square number (has) to (some) square number. And squares having to one another the ratio which (some) square number (has) to (some)

πλευράς ἔχει μήκει συμμετρουνς. τὰ δὲ ἀπὸ τῶν μήκει ἀσυμμετρων εὐθειῶν τετράγωνα πρὸς ἄλληλα λόγον οὐκ ἔθει, ὅνπερ τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὰ τετράγωνα τὰ πρὸς ἄλληλα λόγον μὴ ἔθοντα, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὰς πλευράς ἔχει μήκει συμμετρουνς.



Ἐστωσαν γάρ αἱ A, B μήκει σύμμετροι· λέγω, ὅτι τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B τετράγωνον λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

Ἐπεὶ γὰρ σύμμετρος ἐστὶν ἡ A τῇ B μήκει, ἡ A ἄρα πρὸς τὴν B λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, ὃν ὁ Γ πρὸς τὸν Δ . ἐπεὶ οὖν ἐστὶν ὡς ἡ A πρὸς τὴν B , οὕτως ὁ Γ πρὸς τὸν Δ , ἀλλὰ τοῦ μὲν τῆς A πρὸς τὴν B λόγον διπλασίον ἐστὶν ὁ τοῦ ἀπὸ τῆς A τετραγώνου πρὸς τὸ ἀπὸ τῆς B τετράγωνον· τὰ γὰρ ὅμοια στήματα ἐν διπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν· τοῦ δὲ τοῦ Γ [ἀριθμοῦ] πρὸς τὸν Δ [ἀριθμόν] λόγον διπλασίον ἐστὶν ὁ τοῦ ἀπὸ τοῦ Γ τετραγώνου πρὸς τὸν ἀπὸ τοῦ Δ τετράγωνον· δύο γὰρ τετραγώνων ἀριθμῶν εἷς μέσος ἀνάλογόν ἐστιν ἀριθμὸς, καὶ ὁ τετράγωνος πρὸς τὸν τετράγωνον [ἀριθμόν] διπλασίονα λόγον ἔθει, ἥπερ ἡ πλευρὰ πρὸς τὴν πλευράν· ἐστὶν ἄρα καὶ ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B τετράγωνον, οὕτως ὁ ἀπὸ τοῦ Γ τετράγωνος [ἀριθμὸς] πρὸς τὸν ἀπὸ τοῦ Δ [ἀριθμοῦ] τετράγωνον [ἀριθμόν].

Ἀλλὰ δὴ ἔστω ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B , οὕτως ὁ ἀπὸ τοῦ Γ τετράγωνος πρὸς τὸν ἀπὸ τοῦ Δ [τετράγωνον]· λέγω, ὅτι σύμμετρος ἐστὶν ἡ A τῇ B μήκει.

Ἐπεὶ γὰρ ἐστὶν ὡς τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον], οὕτως ὁ ἀπὸ τοῦ Γ τετράγωνος πρὸς τὸν ἀπὸ τοῦ Δ [τετράγωνον], ἀλλ' ὁ μὲν τοῦ ἀπὸ τῆς A τετραγώνου πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγος διπλασίον ἐστὶ τοῦ τῆς A πρὸς τὴν B λόγον, ὁ δὲ τοῦ ἀπὸ τοῦ Γ [ἀριθμοῦ] τετραγώνου [ἀριθμοῦ] πρὸς τὸν ἀπὸ τοῦ Δ [ἀριθμοῦ] τετράγωνον [ἀριθμόν] λόγος διπλασίον ἐστὶ τοῦ τοῦ Γ [ἀριθμοῦ] πρὸς τὸν Δ [ἀριθμόν] λόγον, ἐστὶν ἄρα καὶ ὡς ἡ A πρὸς τὴν B , οὕτως ὁ Γ [ἀριθμὸς] πρὸς τὸν Δ [ἀριθμόν]. ἡ A ἄρα πρὸς τὴν B λόγον ἔθει, ὃν ἀριθμὸς ὁ Γ πρὸς ἀριθμόν τὸν Δ · σύμμετρος ἄρα ἐστὶν ἡ A τῇ B μήκει.

Ἀλλὰ δὴ ἀσύμμετρος ἔστω ἡ A τῇ B μήκει· λέγω, ὅτι τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

Εἰ γὰρ ἔθει τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, σύμμετρος ἔσται ἡ A τῇ B . οὐκ ἔστι δέ· οὐκ ἄρα τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον

square number will also have sides (which are) commensurable in length. But squares on straight-lines (which are) incommensurable in length do not have to one another the ratio which (some) square number (has) to (some) square number. And squares not having to one another the ratio which (some) square number (has) to (some) square number will not have sides (which are) commensurable in length either.

For let A and B be (straight-lines which are) commensurable in length. I say that the square on A has to the square on B the ratio which (some) square number (has) to (some) square number.

For since A is commensurable in length with B , A thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (that) which C (has) to D . Therefore, since as A is to B , so C (is) to D . But the (ratio) of the square on A to the square on B is the square of the ratio of A to B . For similar figures are in the squared ratio of (their) corresponding sides [Prop. 6.20 corr.]. And the (ratio) of the square on C to the square on D is the square of the ratio of the [number] C to the [number] D . For there exists one number in mean proportion to two square numbers, and (one) square (number) has to the (other) square [number] a squared ratio with respect to (that) the side (of the former has) to the side (of the latter) [Prop. 8.11]. And, thus, as the square on A is to the square on B , so the square [number] on the (number) C (is) to the square [number] on the [number] D .[†]

And so let the square on A be to the (square) on B as the square (number) on C (is) to the [square] (number) on D . I say that A is commensurable in length with B .

For since as the square on A is to the [square] on B , so the square (number) on C (is) to the [square] (number) on D . But, the ratio of the square on A to the (square) on B is the square of the (ratio) of A to B [Prop. 6.20 corr.]. And the (ratio) of the square [number] on the [number] C to the square [number] on the [number] D is the square of the ratio of the [number] C to the [number] D [Prop. 8.11]. Thus, as A is to B , so the [number] C also (is) to the [number] D . A , thus, has to B the ratio which the number C has to the number D . Thus, A is commensurable in length with B [Prop. 10.6].[‡]

And so let A be incommensurable in length with B . I say that the square on A does not have to the [square] on B the ratio which (some) square number (has) to (some) square number.

For if the square on A has to the [square] on B the ratio which (some) square number (has) to (some) square number then A will be commensurable (in length) with B . But it is not. Thus, the square on A does not have to the [square] on the B

ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

Πάλιν δὴ τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς B [τετράγωνον] λόγον μὴ ἐθέτω, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· λέγω, ὅτι ἀσύμμετρος ἔστιν ἡ A τῇ B μήκει.

Εἰ γὰρ ἔστι σύμμετρος ἡ A τῇ B , ἔχει τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς B λόγον, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. οὐκ ἔθει δέ· οὐκ ἄρα σύμμετρος ἔστιν ἡ A τῇ B μήκει.

Τὰ ἄρα ἀπὸ τῶν μήκει συμμετρων, καὶ τὰ ἐχῆς.

Πόρισμα.

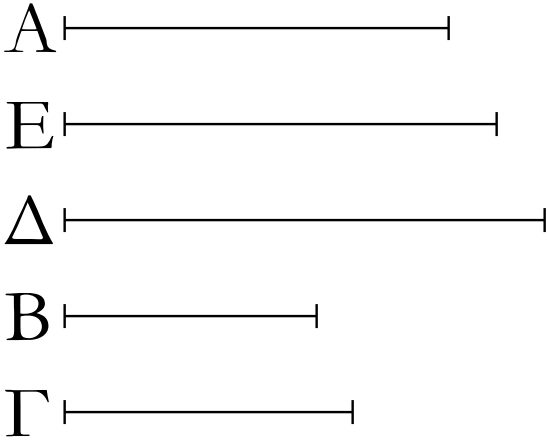
Καὶ φανερόν ἐκ τῶν δεδειγμένων ἔσται, ὅτι αἱ μήκει σύμμετροι πάντως καὶ δυνάμει, αἱ δὲ δυνάμει οὐ πάντως καὶ μήκει.

† There is an unstated assumption here that if $\alpha : \beta :: \gamma : \delta$ then $\alpha^2 : \beta^2 :: \gamma^2 : \delta^2$.

* There is an unstated assumption here that if $\alpha^2 : \beta^2 :: \gamma^2 : \delta^2$ then $\alpha : \beta :: \gamma : \delta$.

10.

Τῇ προτεθείσῃ εὐθείᾳ προσενεῖν δύο εὐθείας ἀσυμμέτρους, τὴν μὲν μήκει μόνον, τὴν δὲ καὶ δυνάμει.



Ἐστω ἡ προτεθείσα εὐθεῖα ἡ A · δεῖ δὴ τῇ A προσενεῖν δύο εὐθείας ἀσυμμέτρους, τὴν μὲν μήκει μόνον, τὴν δὲ καὶ δυνάμει.

Ἐκκείσθωσαν γὰρ δύο ἀριθμοὶ οἱ B, Γ πρὸς ἀλλήλους λόγον μὴ ἔχοντες, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, τουτέστι μὴ ὅμοιοι ἐπίπεδοι, καὶ γερονέτω ὡς ὁ B πρὸς τὸν Γ , οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς Δ τετράγωνον· ἐμάθομεν γάρ· σύμμετρον ἄρα τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς Δ . καὶ ἐπεὶ ὁ B πρὸς τὸν Γ λόγον οὐκ ἔθει,

the ratio which (some) square number (has) to (some) square number.

So, again, let the square on A not have to the [square] on B the ratio which (some) square number (has) to (some) square number. I say that A is incommensurable in length with B .

For if A is commensurable (in length) with B then the (square) on A will have to the (square) on B the ratio which (some) square number (has) to (some) square number. But it does not have (such a ratio). Thus, A is not commensurable in length with B .

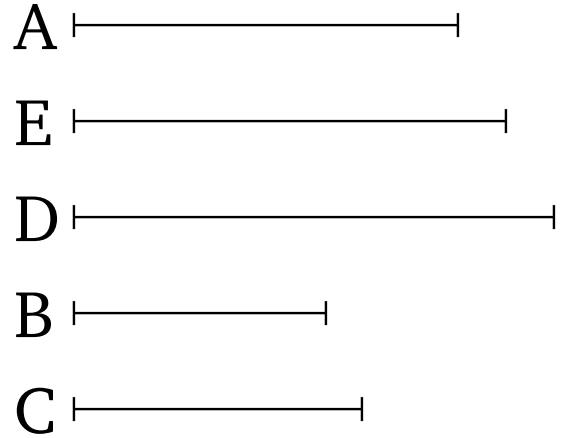
Thus, (squares) on (straight-lines which are) commensurable in length, and so on . . .

Corollary

And it will be clear, from (what) has been demonstrated, that (straight-lines) commensurable in length (are) always also (commensurable) in square, but (straight-lines commensurable) in square (are) not always also (commensurable) in length.

Proposition 10†

To find two straight-lines incommensurable with a given straight-line, the one (incommensurable) in length only, the other also (incommensurable) in square.



Let A be the given straight-line. So it is required to find two straight-lines incommensurable with A , the one (incommensurable) in length only, the other also (incommensurable) in square.

For let two numbers, B and C , not having to one another the ratio which (some) square number (has) to (some) square number—that is to say, not (being) similar plane (numbers)—be taken. And let it be contrived that as B (is) to C , so the square on A (is) to the square on D . For we learned (how to

ὅν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς Δ λόγον ἔθει, ὅν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ Δ μήκει. εἰλήφθω τῶν A , Δ μέση ἀνάλογον ἡ E · ἐστὶν ἄρα ὡς ἡ A πρὸς τὴν Δ , οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς E . ἀσύμμετρος δέ ἐστιν ἡ A τῇ Δ μήκει· ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς A τετράγωνον τῷ ἀπὸ τῆς E τετραγώνῳ· ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ E δυνάμει.

Τῇ ἄρα προτεθείσῃ εὐθείᾳ τῇ A προσεύρηται δύο εὐθεῖαι ἀσύμμετροι αἱ Δ , E , μήκει μὲν μόνον ἡ Δ , δυνάμει δέ καὶ μήκει δηλαδὴ ἡ E [ὅπερ ἔδει δεῖχαι].

do this [Prop. 10.6 corr.]. Thus, the (square) on A (is) commensurable with the (square) on D [Prop. 10.6]. And since B does not have to C the ratio which (some) square number (has) to (some) square number, the (square) on A thus does not have to the (square) on D the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with D [Prop. 10.9]. Let the (straight-line) E (which is) in mean proportion to A and D be taken [Prop. 6.13]. Thus, as A is to D , so the square on A (is) to the (square) on E [Def. 5.9]. And A is incommensurable in length with D . Thus, the square on A is also incommensurable with the square on E [Prop. 10.11]. Thus, A is incommensurable in square with E .

Thus, two straight-lines, D and E , (which are) incommensurable with the given straight-line A , be found, the one, D , (incommensurable) in length only, the other, E , (incommensurable) in square, and, clearly, also in length. [(Which is) the very thing it was required to show.]

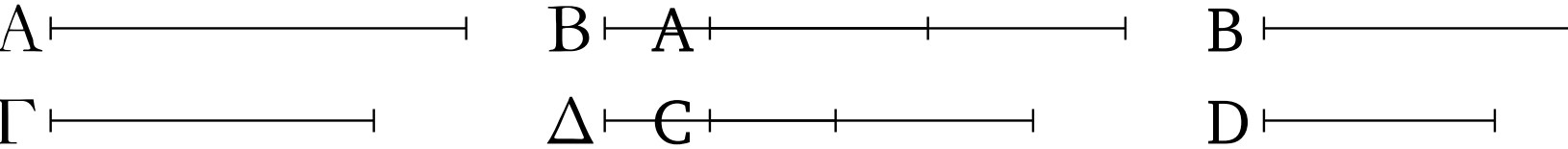
† This whole proposition is regarded by Heiberg as an interpolation into the original text.

11.

Proposition 11

Ἐὰν τέσσαρα μεγέθη ἀνάλογον ᾖ, τὸ δὲ πρῶτον τῷ δευτέρῳ σύμμετρον ᾖ, καὶ τὸ τρίτον τῷ τετάρτῳ σύμμετρον ᾖ· ἐστὶ· κ' ἂν τὸ πρῶτον τῷ δευτέρῳ ἀσύμμετρον ᾖ, καὶ τὸ τρίτον τῷ τετάρτῳ ἀσύμμετρον ᾖ.

If four magnitudes are proportional, and the first is commensurable with the second, then the third will also be commensurable with the fourth. And if the first is incommensurable with the second, then the third will also be incommensurable with the fourth.



Ἐστωσαν τέσσαρα μεγέθη ἀνάλογον τὰ A , B , Γ , Δ , ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ , τὸ A δὲ τῷ B σύμμετρον ἔστω· λέγω, ὅτι καὶ τὸ Γ τῷ Δ σύμμετρον ἔσται.

Ἐπεὶ γὰρ σύμμετρόν ἐστι τὸ A τῷ B , τὸ A ἄρα πρὸς τὸ B λόγον ἔθει, ὅν ἀριθμὸς πρὸς ἀριθμόν. καὶ ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ · καὶ τὸ Γ ἄρα πρὸς τὸ Δ λόγον ἔθει, ὅν ἀριθμὸς πρὸς ἀριθμόν· σύμμετρον ἄρα ἐστὶ τὸ Γ τῷ Δ .

Ἀλλὰ δὴ τὸ A τῷ B ἀσύμμετρον ἔστω· λέγω, ὅτι καὶ τὸ Γ τῷ Δ ἀσύμμετρον ἔσται. ἐπεὶ γὰρ ἀσύμμετρόν ἐστι τὸ A τῷ B , τὸ A ἄρα πρὸς τὸ B λόγον οὐκ ἔθει, ὅν ἀριθμὸς πρὸς ἀριθμόν. καὶ ἐστὶν ὡς τὸ A πρὸς τὸ B , οὕτως τὸ Γ πρὸς τὸ Δ · οὐδὲ τὸ Γ ἄρα πρὸς τὸ Δ λόγον ἔθει, ὅν ἀριθμὸς πρὸς ἀριθμόν· ἀσύμμετρον ἄρα ἐστὶ τὸ Γ τῷ Δ .

Ἐὰν ἄρα τέσσαρα μεγέθη, καὶ τὰ ἐχῇ.

Let A , B , C , D be four proportional magnitudes, (such that) as A (is) to B , so C (is) to D . And let A be commensurable with B . I say that C will also be commensurable with D .

For since A is commensurable with B , A thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. And as A is to B , so C (is) to D . Thus, C also has to D the ratio which (some) number (has) to (some) number. Thus, C is commensurable with D [Prop. 10.6].

And so let A be incommensurable with B . I say that C will also be incommensurable with D . For since A is incommensurable with B , A thus does not have to B the ratio which (some) number (has) to (some) number [Prop. 10.7]. And as A is to B , so C (is) to D . Thus, C does not have to D the ratio which (some) number (has) to (some) number either. Thus, C is incommensurable with D [Prop. 10.8].

Thus, if four magnitudes, and so on . . .

12.

Proposition 12

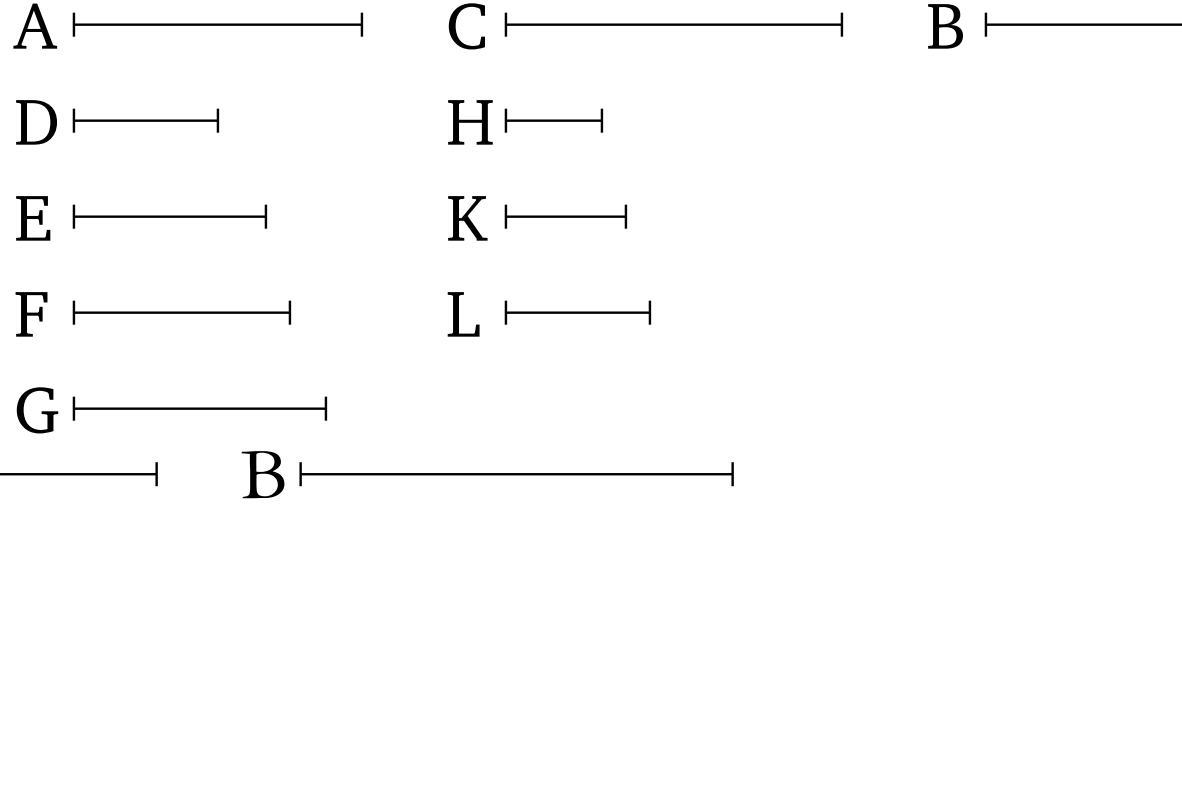
Τὰ τῷ αὐτῷ μεγέθει σύμμετρα καὶ ἀλλήλοις ἐστὶ σύμμε-

(Magnitudes) commensurable with the same magnitude

τρα.

Ἐκάτερον γὰρ τῶν A, B τῷ Γ ἔστω σύμμετρον. λέγω, ὅτι καὶ τὸ A τῷ B ἔστι σύμμετρον.

Ἐπεὶ γὰρ σύμμετρόν ἐστι τὸ A τῷ Γ , τὸ A ἄρα πρὸς τὸ Γ λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, ὃν ὁ Δ πρὸς τὸν E . πάλιν, ἐπεὶ σύμμετρόν ἐστι τὸ Γ τῷ B , τὸ Γ ἄρα πρὸς τὸ B λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν. ἐθέτω, ὃν ὁ Z πρὸς τὸν H . καὶ λόγων δοθέντων ὁποσωνοῦν τοῦ τε, ὃν ἔθει ὁ Δ πρὸς τὸν E , καὶ ὁ Z πρὸς τὸν H εἰλήφθωσαν ἀριθμοὶ ἐχῆς ἐν τοῖς δοθείσι λόγοις οἱ Θ, K, Λ ὥστε εἶναι ὡς μὲν τὸν Δ πρὸς τὸν E , οὕτως τὸν Θ πρὸς τὸν K , ὡς δὲ τὸν Z πρὸς τὸν H , οὕτως τὸν K πρὸς τὸν Λ .



Ἐπεὶ οὖν ἔστιν ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Δ πρὸς τὸν E , ἀλλ' ὡς ὁ Δ πρὸς τὸν E , οὕτως ὁ Θ πρὸς τὸν K , ἔστιν ἄρα καὶ ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Θ πρὸς τὸν K . πάλιν, ἐπεὶ ἔστιν ὡς τὸ Γ πρὸς τὸ B , οὕτως ὁ Z πρὸς τὸν H , ἀλλ' ὡς ὁ Z πρὸς τὸν H , [οὕτως] ὁ K πρὸς τὸν Λ , καὶ ὡς ἄρα τὸ Γ πρὸς τὸ B , οὕτως ὁ K πρὸς τὸν Λ . ἔστι δὲ καὶ ὡς τὸ A πρὸς τὸ Γ , οὕτως ὁ Θ πρὸς τὸν K δι' ἴσον ἄρα ἔστιν ὡς τὸ A πρὸς τὸ B , οὕτως ὁ Θ πρὸς τὸν Λ . τὸ A ἄρα πρὸς τὸ B λόγον ἔθει, ὃν ἀριθμὸς ὁ Θ πρὸς ἀριθμόν τὸν Λ . σύμμετρον

are also commensurable with one another.

For let A and B each be commensurable with C . I say that A is also commensurable with B .

For since A is commensurable with C , A thus has to C the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (the ratio) which D (has) to E . Again, since C is commensurable with B , C thus has to B the ratio which (some) number (has) to (some) number [Prop. 10.5]. Let it have (the ratio) which F (has) to G . And for any multitude whatsoever of given ratios—(namely,) those which D has to E , and F to G —let the numbers H, K, L (which are) continuously (proportional) in the(se) given ratios be taken [Prop. 8.4]. Hence, as D is to E , so H (is) to K , and as F (is) to G , so K (is) to L .

Therefore, since as A is to C , so D (is) to E , but as D (is) to E , so H (is) to K , thus also as A is to C , so H (is) to K [Prop. 5.11]. Again, since as C is to B , so F (is) to G , but as F (is) to G , [so] K (is) to L , thus also as C (is) to B , so K (is) to L [Prop. 5.11]. And also as A is to C , so H (is) to K . Thus, via equality, as A is to B , so H (is) to L [Prop. 5.22]. Thus, A has to B the ratio which the number H (has) to the number L . Thus, A is commensurable with B [Prop. 10.6].

Thus, (magnitudes) commensurable with the same magnitude are also commensurable with one another. (Which is) the

ἄρα ἐστὶ τὸ A τῷ B .

Τὰ ἄρα τῷ αὐτῷ μεγέθει σύμμετρα καὶ ἀλλήλοις ἐστὶ σύμμετρα· ὅπερ ἔδει δεῖχαι.

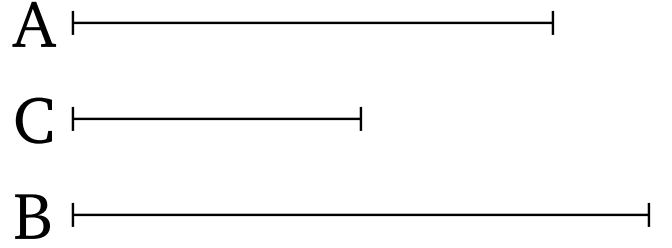
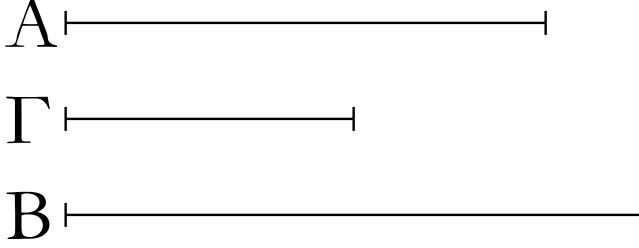
very thing it was required to show.

13.

Proposition 13

Ἐάν ἡ δύο μεγέθη σύμμετρα, τὸ δὲ ἕτερον αὐτῶν μεγέθει τινὶ ἀσύμμετρον ᾗ, καὶ τὸ λοιπὸν τῷ αὐτῷ ἀσύμμετρον ἔσται.

If two magnitudes are commensurable, and one of them is incommensurable with some magnitude, then the remaining (magnitude) will also be incommensurable with it.



Ἐστω δύο μεγέθη σύμμετρα τὰ A , B , τὸ δὲ ἕτερον αὐτῶν τὸ A ἄλλῳ τινὶ τῷ $Γ$ ἀσύμμετρον ἔστω· λέγω, ὅτι καὶ τὸ λοιπὸν τὸ B τῷ $Γ$ ἀσύμμετρον ἔστιν.

Let A and B be two commensurable magnitudes, and let one of them, A , be incommensurable with some other (magnitude), C . I say that the remaining (magnitude), B , is also incommensurable with C .

Εἰ γάρ ἐστι σύμμετρον τὸ B τῷ $Γ$, ἀλλὰ καὶ τὸ A τῷ B σύμμετρον ἔστιν, καὶ τὸ A ἄρα τῷ $Γ$ σύμμετρον ἔστιν. ἀλλὰ καὶ ἀσύμμετρον· ὅπερ ἀδύνατον. οὐκ ἄρα σύμμετρον ἔστι τὸ B τῷ $Γ$ · ἀσύμμετρον ἄρα.

For if B is commensurable with C , but A is also commensurable with B , A is thus also commensurable with C [Prop. 10.12]. But, (it is) also incommensurable (with C). The very thing (is) impossible. Thus, B is not commensurable with C . Thus, (it is) incommensurable.

Ἐάν ἄρα ἡ δύο μεγέθη σύμμετρα, καὶ τὰ ἐχῇ.

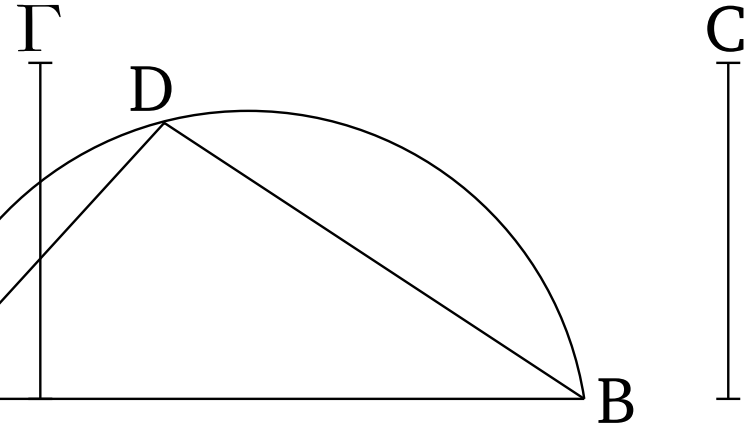
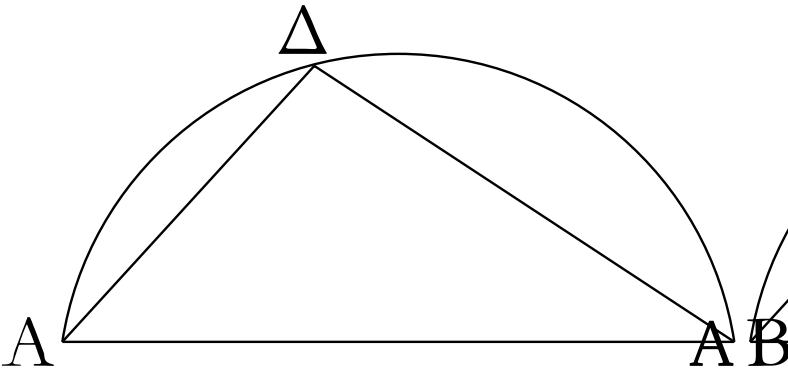
Thus, if two magnitudes are commensurable, and so on
....

Λήμμα.

Lemma

Δύο δοθεῖσων εὐθειῶν ἀνίσων εὐρεῖν, τίνι μείζον δύναται ἢ μείζων τῇς ἐλάσσονος.

For two given unequal straight-lines, to find by (the square on) which (straight-line) the square on the greater (straight-line is) larger than (the square on) the lesser.[†]



Ἐστωσαν αἱ δοθεῖσαι δύο ἄισοι εὐθεῖαι αἱ AB , $Γ$, ὧν μείζων ἔστω ἢ AB · δεῖ δὴ εὐρεῖν, τίνι μείζον δύναται ἢ AB τῇς $Γ$.

Let AB and C be the two given unequal straight-lines, and let AB be the greater of them. So it is required to find by (the square on) which (straight-line) the square on AB (is) greater than (the square on) C .

Γεγραφθῶ ἐπὶ τῇς AB ἡμικύκλιον τὸ $AΔB$, καὶ εἰξ αὐτὸ

ἐνηρμόσθω τῇ Γ ἴση ἡ $A\Delta$, καὶ ἐπεξεύθθω ἡ ΔB . φανερόν δὲ, ὅτι ὀρθή ἐστὶν ἡ ὑπὸ $A\Delta B$ γωνία, καὶ ὅτι ἡ AB τῆς $A\Delta$, τουτέστι τῆς Γ , μείζον δύνатаι τῇ ΔB .

Ὅμοίωξ δὲ καὶ δύο δοθεῖσων εὐθειῶν ἡ δυναμένη αὐτὰς εὐρίσκεται οὕτως.

Ἐστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι αἱ $A\Delta$, ΔB , καὶ δέον ἔστω εὐρεῖν τὴν δυναμένην αὐτάς. κείσθωσαν γάρ, ὥστε ὀρθὴν γωνίαν περιέθην τὴν ὑπὸ $A\Delta$, ΔB , καὶ ἐπεξεύθθω ἡ AB . φανερόν πάλιν, ὅτι ἡ τὰς $A\Delta$, ΔB δυναμένη ἐστὶν ἡ AB . ὅπερ ἔδει δεῖχαι.

Let the semi-circle ADB be described on AB . And let AD , equal to C , be inserted into it [Prop. 4.1]. And let DB be joined. So (it is) clear that the angle ADB is a right-angle [Prop. 3.31], and that the square on AB (is) greater than (the square on) AD —that is to say, (the square on) C —by (the square on) DB [Prop. 1.47].

And, similarly, the square-root of (the sum of the squares on) two given straight-lines is also found likewise.

Let AD and DB be the two given straight-lines. And let it be necessary to find the square-root of (the sum of the squares on) them. For let them be laid down such as to encompass a right-angle—(namely), that (angle encompassed) by AD and DB . And let AB be joined. (It is) again clear that AB is the square-root of (the sum of the squares on) AD and DB [Prop. 1.47]. (Which is) the very thing it was required to show.

† That is, if α and β are the lengths of two given straight-lines, with α being greater than β , to find a straight-line of length γ such that $\alpha^2 = \beta^2 + \gamma^2$. Similarly, we can also find γ such that $\gamma^2 = \alpha^2 + \beta^2$.

14.

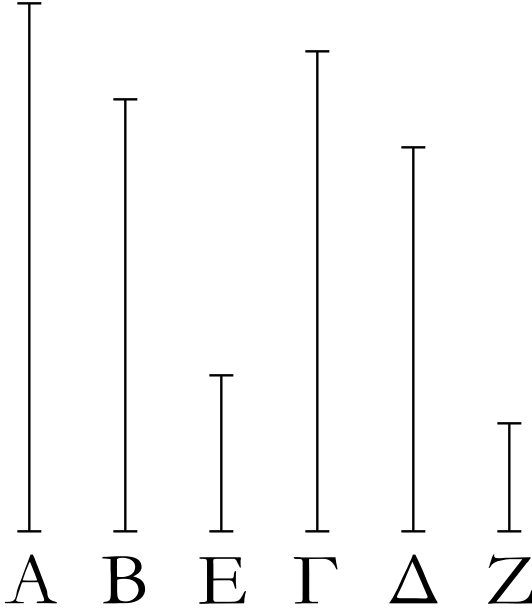
Proposition 14

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ᾗσιν, δύνῃται δὲ ἡ πρώτη τῆς δευτέρας μείζον τῷ ἀπὸ συμμετρων ἑαυτῇ [μήκει], καὶ ἡ τρίτη τῆς τετάρτης μείζον δυνήσεται τῷ ἀπὸ συμμετρων ἑαυτῇ [μήκει]. καὶ ἔαν ἡ πρώτη τῆς δευτέρας μείζον δύνῃται τῷ ἀπὸ ἀσυμμέτρων ἑαυτῇ [μήκει], καὶ ἡ τρίτη τῆς τετάρτης μείζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρων ἑαυτῇ [μήκει].

Ἐστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ A , B , Γ , Δ , ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ , καὶ ἡ A μὲν τῆς B μείζον δυνάσθω τῷ ἀπὸ τῆς E , ἡ δὲ Γ τῆς Δ μείζον δυνάσθω τῷ ἀπὸ τῆς Z . λέγω, ὅτι, εἴτε σύμμετρος ἐστὶν ἡ A τῇ E , σύμμετρος ἐστὶ καὶ ἡ Γ τῇ Z , εἴτε ἀσύμμετρος ἐστὶν ἡ A τῇ E , ἀσύμμετρος ἐστὶ καὶ ὁ Γ τῇ Z .

If four straight-lines are proportional, and the square on the first is greater than (the square on) the second by the (square) on (some straight-line) commensurable [in length] with the first, then the square on the third will also be greater than (the square on) the fourth by the (square) on (some straight-line) commensurable [in length] with the third. And if the square on the first is greater than (the square on) the second by the (square) on (some straight-line) incommensurable [in length] with the first, then the square on the third will also be greater than (the square on) the fourth by the (square) on (some straight-line) incommensurable [in length] with the third.

Let A , B , C , D be four proportional straight-lines, (such that) as A (is) to B , so C (is) to D . And let the square on A be greater than (the square on) B by the (square) on E , and let the square on C be greater than (the square on) D by the (square) on F . I say that A is either commensurable (in length) with E , and C is also commensurable with F , or A is incommensurable (in length) with E , and C is also incommensurable with F .



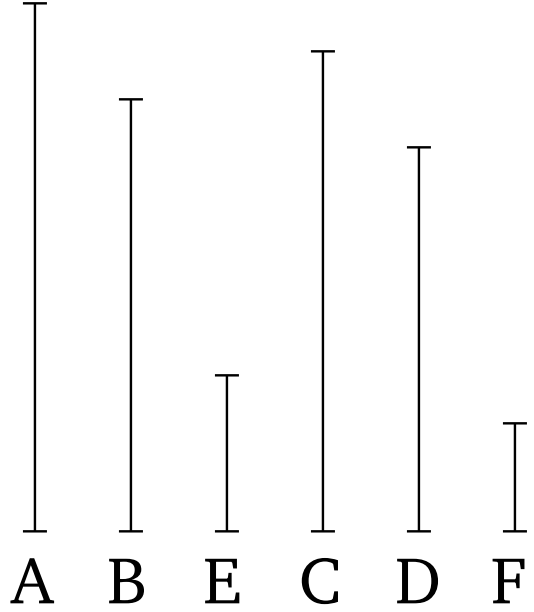
Ἐπεὶ γὰρ ἔστιν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ , ἔστιν ἄρα καὶ ὥς τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς B , οὕτως τὸ ἀπὸ τῆς Γ πρὸς τὸ ἀπὸ τῆς Δ . ἀλλὰ τῶ μὲν ἀπὸ τῆς A ἴσα ἐστὶ τὰ ἀπὸ τῶν E, B , τῶ δὲ ἀπὸ τῆς Γ ἴσα ἐστὶ τὰ ἀπὸ τῶν Δ, Z . ἔστιν ἄρα ὥς τὰ ἀπὸ τῶν E, B πρὸς τὸ ἀπὸ τῆς B , οὕτως τὰ ἀπὸ τῶν Δ, Z πρὸς τὸ ἀπὸ τῆς Δ : διελόντι ἄρα ἐστὶν ὥς τὸ ἀπὸ τῆς E πρὸς τὸ ἀπὸ τῆς B , οὕτως τὸ ἀπὸ τῆς Z πρὸς τὸ ἀπὸ τῆς Δ : ἔστιν ἄρα καὶ ὥς ἡ E πρὸς τὴν B , οὕτως ἡ Z πρὸς τὴν Δ : ἀνάπαλιν ἄρα ἐστὶν ὥς ἡ B πρὸς τὴν E , οὕτως ἡ Δ πρὸς τὴν Z . ἔστι δὲ καὶ ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ : δι' ἴσον ἄρα ἐστὶν ὥς ἡ A πρὸς τὴν E , οὕτως ἡ Γ πρὸς τὴν Z . εἴτε οὖν σύμμετρος ἔστιν ἡ A τῇ E , σύμμετρος ἔστι καὶ ἡ Γ τῇ Z , εἴτε ἀσύμμετρος ἔστιν ἡ A τῇ E , ἀσύμμετρος ἔστι καὶ ἡ Γ τῇ Z .

Ἐὰν ἄρα, καὶ τὰ ἐχῇς.

15.

Ἐὰν δύο μεγέθη σύμμετρα συντεθῇ, καὶ τὸ ὅλον ἐκατέρω αὐτῶν σύμμετρον ἔσται· καὶ τὸ ὅλον ἐν αὐτῶν σύμμετρον ἔσται, καὶ τὰ ἐκ ἀρθῆς μεγέθη σύμμετρα ἔσται.

Συγκείσθω γὰρ δύο μεγέθη σύμμετρα τὰ AB, BG · λέγω, ὅτι καὶ ὅλον τὸ AG ἐκατέρω τῶν AB, BG ἔστι σύμμετρον.



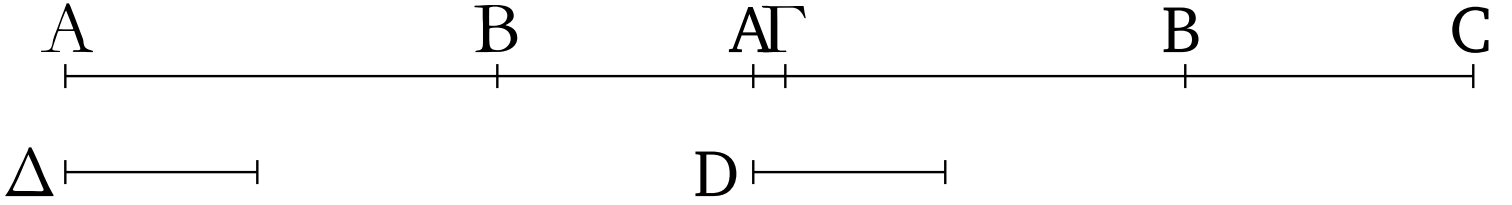
For since as A is to B , so C (is) to D , thus as the (square) on A is to the (square) on B , so the (square) on C (is) to the (square) on D [Prop. 6.22]. But the (sum of the squares) on E and B is equal to the (square) on A , and the (sum of the squares) on D and F is equal to the (square) on C . Thus, as the (sum of the squares) on E and B is to the (square) on B , so the (sum of the squares) on D and F (is) to the (square) on D . Thus, via separation, as the (square) on E is to the (square) on B , so the (square) on F (is) to the (square) on D [Prop. 5.17]. Thus, also, as E is to B , so F (is) to D [Prop. 6.22]. Thus, inversely, as B is to E , so D (is) to F [Prop. 5.7 corr.]. But, as A is to B , so C also (is) to D . Thus, via equality, as A is to E , so C (is) to F [Prop. 5.22]. Therefore, A is either commensurable (in length) with E , and C is also commensurable with F , or A is incommensurable (in length) with E , and C is also incommensurable with F [Prop. 10.11].

Thus, if, and so on . . .

Proposition 15

If two commensurable magnitudes are added together then the whole will also be commensurable with each of them. And if the whole is commensurable with one of them then the original magnitudes will also be commensurable (with one another).

For let the two commensurable magnitudes AB and BC be laid down together. I say that the whole AC is also commensurable with each of AB and BC .



Ἐπεὶ γὰρ σύμμετρά ἐστι τὰ AB , $BΓ$, μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ ἔστω τὸ Δ . ἐπεὶ οὖν τὸ Δ τὰ AB , $BΓ$ μετρεῖ, καὶ ὅλον τὸ $AΓ$ μετρήσει. μετρεῖ δὲ καὶ τὰ AB , $BΓ$. τὸ Δ ἄρα τὰ AB , $BΓ$, $AΓ$ μετρεῖ· σύμμετρον ἄρα ἐστὶ τὸ $AΓ$ ἐκατέρω τῶν AB , $BΓ$.

Ἀλλὰ δὴ τὸ $AΓ$ ἔστω σύμμετρον τῷ AB · λέγω δὴ, ὅτι καὶ τὰ AB , $BΓ$ σύμμετρά ἐστιν.

Ἐπεὶ γὰρ σύμμετρά ἐστι τὰ $AΓ$, AB , μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ ἔστω τὸ Δ . ἐπεὶ οὖν τὸ Δ τὰ $ΓA$, AB μετρεῖ, καὶ λοιπὸν ἄρα τὸ $BΓ$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB · τὸ Δ ἄρα τὰ AB , $BΓ$ μετρήσει· σύμμετρα ἄρα ἐστὶ τὰ AB , $BΓ$.

Ἐὰν ἄρα δύο μεγέθη, καὶ τὰ ἐχῇ.

For since AB and BC are commensurable, some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) AB and BC , it will also measure the whole AC . And it also measures AB and BC . Thus, D measures AB , BC , and AC . Thus, AC is commensurable with each of AB and BC [Def. 10.1].

And so let AC be commensurable with AB . I say that AB and BC are also commensurable.

For since AC and AB are commensurable, some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) CA and AB , it will thus also measure the remainder BC . And it also measures AB . Thus, D will measure (both) AB and BC . Thus, AB and BC are commensurable [Def. 10.1].

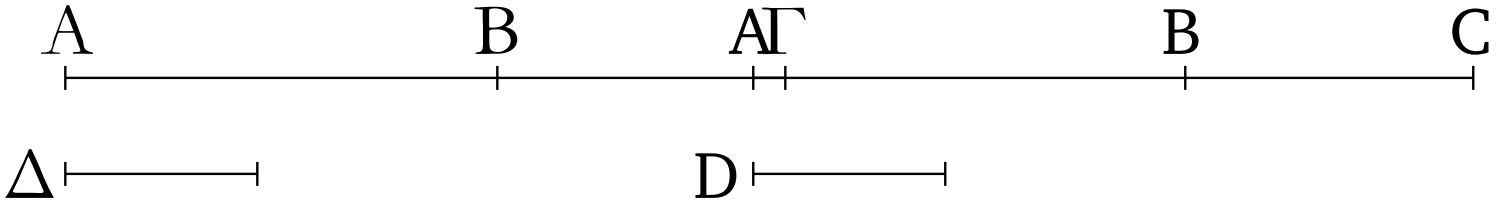
Thus, if two magnitudes, and so on . . .

16.

Proposition 16

Ἐὰν δύο μεγέθη ἀσύμμετρα συντεθῇ, καὶ τὸ ὅλον ἐκατέρω αὐτῶν ἀσύμμετρον ἔσται· καὶ τὸ ὅλον ἐνὶ αὐτῶν ἀσύμμετρον ἦ, καὶ τὰ ἐκ ἀρθῆς μεγέθη ἀσύμμετρα ἔσται.

If two incommensurable magnitudes are added together then the whole will also be incommensurable with each of them. And if the whole is incommensurable with one of them then the original magnitudes will also be incommensurable (with one another).



Συγκείσθω γὰρ δύο μεγέθη ἀσύμμετρα τὰ AB , $BΓ$ · λέγω, ὅτι καὶ ὅλον τὸ $AΓ$ ἐκατέρω τῶν AB , $BΓ$ ἀσύμμετρον ἐστίν.

Εἰ γὰρ μὴ ἐστὶν ἀσύμμετρα τὰ $ΓA$, AB , μετρήσει τι [αὐτὰ] μέγεθος. μετρεῖτω, εἰ δυνατόν, καὶ ἔστω τὸ Δ . ἐπεὶ οὖν τὸ Δ τὰ $ΓA$, AB μετρεῖ, καὶ λοιπὸν ἄρα τὸ $BΓ$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB · τὸ Δ ἄρα τὰ AB , $BΓ$ μετρεῖ. σύμμετρα ἄρα ἐστὶ τὰ AB , $BΓ$ · ὑπέκειντο δὲ καὶ ἀσύμμετρα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ $ΓA$, AB μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐστὶ τὰ $ΓA$, AB . ὁμοίως δὲ δείχμεν, ὅτι καὶ τὰ $AΓ$, $ΓB$ ἀσύμμετρά ἐστιν. τὸ $AΓ$ ἄρα ἐκατέρω τῶν AB , $BΓ$ ἀσύμμετρον ἐστίν.

Ἀλλὰ δὴ τὸ $AΓ$ ἐνὶ τῶν AB , $BΓ$ ἀσύμμετρον ἔστω. ἔστω δὴ πρότερον τῷ AB · λέγω, ὅτι καὶ τὰ AB , $BΓ$ ἀσύμμετρά ἐστιν. εἰ γὰρ ἔσται σύμμετρα, μετρήσει τι αὐτὰ μέγεθος. μετρεῖτω, καὶ ἔστω τὸ Δ . ἐπεὶ οὖν τὸ Δ τὰ AB , $BΓ$ μετρεῖ,

For let the two incommensurable magnitudes AB and BC be laid down together. I say that that the whole AC is also incommensurable with each of AB and BC .

For if CA and AB are not incommensurable then some magnitude will measure [them]. If possible, let it (so) measure (them), and let it be D . Therefore, since D measures (both) CA and AB , it will thus also measure the remainder BC . And it also measures AB . Thus, D measures (both) AB and BC . Thus, AB and BC are commensurable [Def. 10.1]. But they were also assumed (to be) incommensurable. The very thing is impossible. Thus, some magnitude cannot measure (both) CA and AB . Thus, CA and AB are incommensurable [Def. 10.1]. So, similarly, we can show that AC and CB are also incommensurable. Thus, AC is incommensurable with each of AB and BC .

καὶ ὅλον ἄρα τὸ AG μετρήσει. μετρεῖ δὲ καὶ τὸ AB · τὸ Δ ἄρα τὰ GA , AB μετρεῖ. σύμμετρα ἄρα ἐστὶ τὰ GA , AB · ὑπέκειτο δὲ καὶ ἀσύμμετρα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ AB , BG μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐστὶ τὰ AB , BG .

Ἐάν ἄρα δύο μεγέθη, καὶ τὰ ἐχῇς.

And so let AC be incommensurable with one of AB and BC . So let it, first of all, be incommensurable with AB . I say that AB and BC are also incommensurable. For if they are commensurable then some magnitude will measure them. Let it (so) measure (them), and let it be D . Therefore, since D measures (both) AB and BC , it will thus also measure the whole AC . And it also measures AB . Thus, D measures (both) CA and AB . Thus, CA and AB are commensurable [Def. 10.1]. But they were also assumed (to be) incommensurable. The very thing is impossible. Thus, some magnitude cannot measure (both) AB and BC . Thus, AB and BC are incommensurable [Def. 10.1].

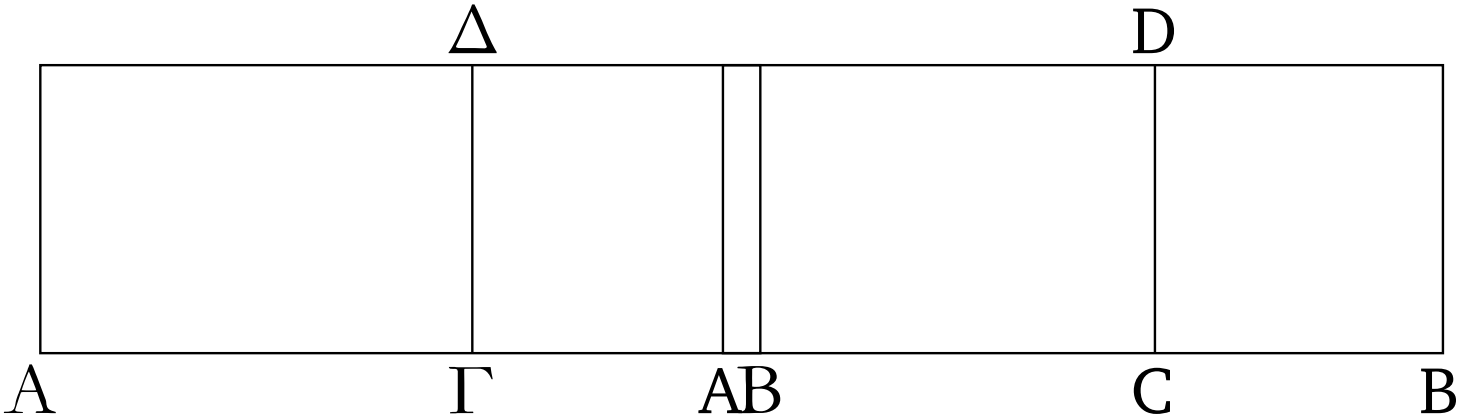
Thus, if two... magnitudes, and so on....

Λήμμα.

Ἐάν παρὰ τινα εὐθεῖαν παραβληθῇ παραλληλόγραμμον ἐλλείπον εἶδει τετραγώνῳ, τὸ παραβληθέν ἴσον ἐστὶ τῷ ὑπὸ τῶν ἐκ τῆς παραβολῆς γενομένων τμημάτων τῆς εὐθείας.

Lemma

If a parallelogram,[†] falling short by a square figure, is applied to some straight-line then the applied (parallelogram) is equal (in area) to the (rectangle contained) by the pieces of the straight-line created via the application (of the parallelogram).



Παρὰ γὰρ εὐθεῖαν τὴν AB παραβεβλήσθω παραλληλόγραμμον τὸ $A\Delta$ ἐλλείπον εἶδει τετραγώνῳ τῷ ΔB · λέγω, ὅτι ἴσον ἐστὶ τὸ $A\Delta$ τῷ ὑπὸ τῶν AG , GB .

Καὶ ἐστὶν αὐτόθεν φανερόν· ἐπεὶ γὰρ τετραγώνον ἐστὶ τὸ ΔB , ἴση ἐστὶν ἡ $\Delta\Gamma$ τῇ GB , καὶ ἐστὶ τὸ $A\Delta$ τὸ ὑπὸ τῶν AG , $\Gamma\Delta$, τουτέστι τὸ ὑπὸ τῶν AG , GB .

Ἐάν ἄρα παρὰ τινα εὐθεῖαν, καὶ τὰ ἐχῇς.

For let the parallelogram AD , falling short by the square figure DB , be applied to the straight-line AB . I say that AD is equal to the (rectangle contained) by AC and CB .

And it is immediately obvious. For since DB is a square, DC is equal to CB . And AD is the (rectangle contained) by AC and CD —that is to say, by AC and CB .

Thus, if ... to some straight-line, and so on....

[†] Note that this lemma only applies to rectangular parallelograms.

17.

Ἐάν ὥσι δύο εὐθεῖαι ἄνισοι, τῷ δὲ τετράτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ καὶ εἷς σύμμετρα αὐτὴν διαιρῇ μήκει, ἢ μείζων τῆς ἐλάσσονος μείζον δυνήσεται τῷ ἀπὸ συμμέτρου ἐαυτῇ [μήκει]. καὶ ἔαν ἡ μείζων τῆς ἐλάσσονος μείζον δύνῃται τῷ ἀπὸ συμμέτρου ἐαυτῇ [μήκει], τῷ δὲ τετράτῳ τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον εἶδει

Proposition 17[†]

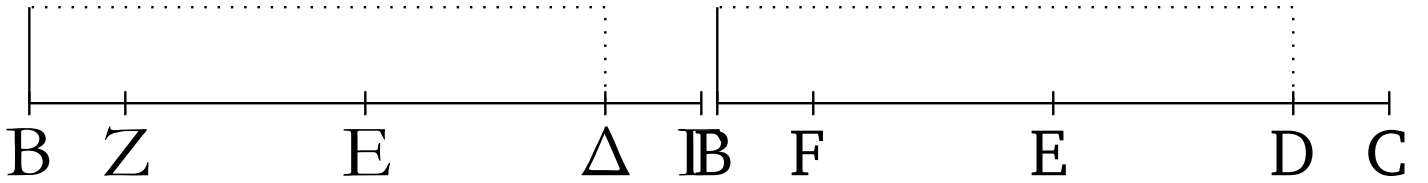
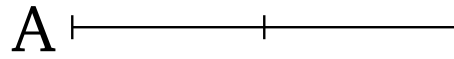
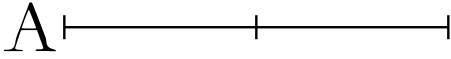
If there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) commensurable in length, then the square on the greater will be larger than (the square on) the lesser by the (square) on (some straight-line) commensurable [in length] with the greater. And if the square on the greater is

τετραγώνω, εἷς σύμμετρα αὐτὴν διαιρεῖ μήκει.

Ἐστωσαν δύο εὐθεῖαι ἄνισοι αἱ A , $B\Gamma$, ὧν μείζων ἡ $B\Gamma$, τῷ δὲ τετράρτῳ μέρει τοῦ ἀπὸ ἐλάσσονος τῆς A , τοντέστι τῷ ἀπὸ τῆς ἡμισείας τῆς A , ἴσον παρὰ τὴν $B\Gamma$ παραβεβλήσθω ἐλλείπον εἶδει τετραγώνω, καὶ ἔστω τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$, σύμμετρος δὲ ἔστω ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει· λέγω, ὅτι ἡ $B\Gamma$ τῆς A μείζων δύναται τῷ ἀπὸ συμμέτρων ἑαυτῇ.

larger than (the square on) the lesser by the (square) on (some straight-line) commensurable [in length] with the greater, and a (rectangle) equal to the fourth (part) of the (square) on the lesser, falling short by a square figure, is applied to the greater, then it divides it into (parts which are) commensurable in length.

Let A and BC be two unequal straight-lines, of which (let) BC (be) the greater. And let a (rectangle) equal to the fourth part of the (square) on the lesser, A —that is, (equal) to the (square) on half of A —falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC [see previous lemma]. And let BD be commensurable in length with DC . I say that that the square on BC is greater than the (square on) A by (the square on some straight-line) commensurable (in length) with (BC).



Τετμήσθω γάρ ἡ $B\Gamma$ δίῃα κατὰ τὸ E σημεῖον, καὶ κείσθω τῇ ΔE ἴση ἡ EZ . λοιπὴ ἄρα ἡ $\Delta\Gamma$ ἴση ἐστὶ τῇ BZ . καὶ ἐπεὶ εὐθεῖα ἡ $B\Gamma$ τέτμηται εἷς μὲν ἴσα κατὰ τὸ E , εἷς δὲ ἄνισα κατὰ τὸ Δ , τὸ ἄρα ὑπὸ $B\Delta$, $\Delta\Gamma$ περιεθόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς $E\Delta$ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Gamma$ τετραγώνῳ· καὶ τὰ τετραπλάσια τὸ ἄρα τετράκις ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ μετὰ τοῦ τετραπλασίου τοῦ ἀπὸ τῆς ΔE ἴσον ἐστὶ τῷ τετράκις ἀπὸ τῆς $E\Gamma$ τετραγώνῳ. ἀλλὰ τῷ μὲν τετραπλασίῳ τοῦ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ ἴσον ἐστὶ τὸ ἀπὸ τῆς A τετράγωνον, τῷ δὲ τετραπλασίῳ τοῦ ἀπὸ τῆς ΔE ἴσον ἐστὶ τὸ ἀπὸ τῆς ΔZ τετράγωνον· διπλασίων γάρ ἐστιν ἡ ΔZ τῆς ΔE . τῷ δὲ τετραπλασίῳ τοῦ ἀπὸ τῆς $E\Gamma$ ἴσον ἐστὶ τὸ ἀπὸ τῆς $B\Gamma$ τετράγωνον· διπλασίων γάρ ἐστι πάλιν ἡ $B\Gamma$ τῆς ΓE . τὰ ἄρα ἀπὸ τῶν A , ΔZ τετράγωνα ἴσα ἐστὶ τῷ ἀπὸ τῆς $B\Gamma$ τετράγωνῳ· ὥστε τὸ ἀπὸ τῆς $B\Gamma$ τοῦ ἀπὸ τῆς A μείζον ἐστὶ τῷ ἀπὸ τῆς ΔZ · ἡ $B\Gamma$ ἄρα τῆς A μείζων δύναται τῇ ΔZ . δεικτέον, ὅτι καὶ σύμμετρος ἐστὶν ἡ $B\Gamma$ τῇ ΔZ . ἐπεὶ γάρ σύμμετρος ἐστὶν ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει, σύμμετρος ἄρα ἐστὶ καὶ ἡ $B\Gamma$ τῇ $\Gamma\Delta$ μήκει. ἀλλὰ ἡ $\Gamma\Delta$ ταῖς $\Gamma\Delta$, BZ ἐστὶ σύμμετρος μήκει· ἴση γάρ ἐστιν ἡ $\Gamma\Delta$ τῇ BZ . καὶ ἡ $B\Gamma$ ἄρα σύμμετρος ἐστὶ ταῖς BZ , $\Gamma\Delta$ μήκει· ὥστε καὶ λοιπὴ τῇ $Z\Delta$ σύμμετρος ἐστὶν ἡ $B\Gamma$ μήκει· ἡ $B\Gamma$ ἄρα τῆς A μείζων δύναται τῷ ἀπὸ συμμέτρων ἑαυτῇ.

Ἀλλὰ δὴ ἡ $B\Gamma$ τῆς A μείζων δυνάσθω τῷ ἀπὸ συμμέτρων ἑαυτῇ, τῷ δὲ τετράρτῳ τοῦ ἀπὸ τῆς A ἴσον παρὰ τὴν $B\Gamma$

For let BC be cut in half at the point E [Prop. 1.10]. And let EF be made equal to DE [Prop. 1.3]. Thus, the remainder DC is equal to BF . And since the straight-line BC has been cut into equal (pieces) at E , and into unequal (pieces) at D , the rectangle contained by BD and DC , plus the square on ED , is thus equal to the square on EC [Prop. 2.5]. (The same) also (for) the quadruples. Thus, four times the (rectangle contained) by BD and DC , plus the quadruple of the (square) on DE , is equal to four times the square on EC . But, the square on A is equal to the quadruple of the (rectangle contained) by BD and DC , and the square on DF is equal to the quadruple of the (square) on DE . For DF is double DE . And the square on BC is equal to the quadruple of the (square) on EC . For, again, BC is double CE . Thus, the (sum of the) squares on A and DF is equal to the square on BC . Hence, the (square) on BC is greater than the (square) on A by the (square) on DF . Thus, BC is greater in square than A by DF . It must also be shown that BC is commensurable (in length) with DF . For since BD is commensurable in length with DC , BC is thus also commensurable in length with CD [Prop. 10.15]. But, CD is commensurable in length with CD plus BF . For CD is equal to BF [Prop. 10.6]. Thus, BC is also commensurable in length with the remainder FD [Prop. 10.15]. Thus, the square on BC is greater than (the square on) A by the (square)

παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$. δεικτέον, ὅτι σύμμετρος ἔστιν ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἡ $B\Gamma$ τῆς A μείζον δύνатаι τῷ ἀπὸ τῆς $Z\Delta$. δύνатаι δὲ ἡ $B\Gamma$ τῆς A μείζον τῷ ἀπὸ συμμέτρον ἑαντῇ. σύμμετρος ἄρα ἔστιν ἡ $B\Gamma$ τῇ $Z\Delta$ μήκει· ὥστε καὶ λοιπὴ συναμφοτέρῳ τῇ BZ , $\Delta\Gamma$ σύμμετρος ἔστιν ἡ $B\Gamma$ μήκει. ἀλλὰ συναμφοτέρως ἡ BZ , $\Delta\Gamma$ σύμμετρος ἔστι τῇ $\Delta\Gamma$ [μήκει]. ὥστε καὶ ἡ $B\Gamma$ τῇ $\Gamma\Delta$ σύμμετρος ἔστι μήκει· καὶ διελόντι ἄρα ἡ $B\Delta$ τῇ $\Delta\Gamma$ ἔστι σύμμετρος μήκει.

Ἐάν ἄρα ὥσι δύο εὐθεῖαι ἄνισοι, καὶ τὰ ἐχθῆς.

on (some straight-line) commensurable (in length) with (BC) .

And so let the square on BC be greater than the (square on) A by the (square) on (some straight-line) commensurable (in length) with (BC) . And let a (rectangle) equal to the fourth (part) of the (square) on A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC . It must be shown that BD is commensurable in length with DC .

For, similarly, by the same construction, we can show that the square on BC is greater than the (square on) A by the (square) on FD . And the square on BC is greater than the (square on) A by the (square) on (some straight-line) commensurable (in length) with (BC) . Thus, BC is commensurable in length with FD . Hence, BC is also commensurable in length with the remaining sum of BF and DC [Prop. 10.15]. But, the sum of BF and DC is commensurable [in length] with DC [Prop. 10.6]. Hence, BC is also commensurable in length with CD [Prop. 10.12]. Thus, via separation, BD is also commensurable in length with DC [Prop. 10.15].

Thus, if there are two unequal straight-lines, and so on . . .

[†] This proposition states that if $\alpha x - x^2 = \beta^2/4$ (where $\alpha = BC$, $x = DC$, and $\beta = A$) then α and $\sqrt{\alpha^2 - \beta^2}$ are commensurable when $\alpha - x$ are x are commensurable, and vice versa.

18.

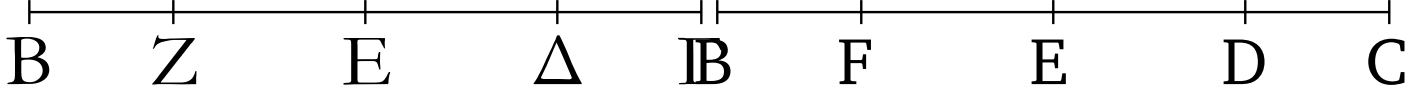
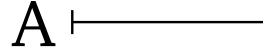
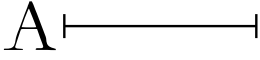
Proposition 18[†]

Ἐάν ὥσι δύο εὐθεῖαι ἄνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, καὶ εἰς ἀσύμμετρα αὐτὴν διαιρῇ [μήκει], ἡ μείζων τῆς ἐλάσσονος μείζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρον ἑαντῇ. καὶ ἔαν ἡ μείζων τῆς ἐλάσσονος μείζον δύνηται τῷ ἀπὸ ἀσυμμέτρον ἑαντῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, εἰς ἀσύμμετρα αὐτὴν διαιρεῖ [μήκει].

Ἐστωσαν δύο εὐθεῖαι ἄνισοι αἱ A , $B\Gamma$, ὧν μείζων ἡ $B\Gamma$, τῷ δὲ τετάρτῳ [μέρει] τοῦ ἀπὸ τῆς ἐλάσσονος τῆς A ἴσον παρὰ τὴν $B\Gamma$ παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν $B\Delta\Gamma$, ἀσύμμετρος δὲ ἔστω ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει· λέγω, ὅτι ἡ $B\Gamma$ τῆς A μείζον δύνатаι τῷ ἀπὸ ἀσυμμέτρον ἑαντῇ.

If there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) incommensurable [in length], then the square on the greater will be larger than the (square on the) lesser by the (square) on (some straight-line) incommensurable (in length) with the greater. And if the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) incommensurable (in length) with the greater, and a (rectangle) equal to the fourth (part) of the (square) on the lesser, falling short by a square figure, is applied to the greater, then it divides it into (parts which are) incommensurable [in length].

Let A and BC be two unequal straight-lines, of which (let) BC (be) the greater. And let a (rectangle) equal to the fourth [part] of the (square) on the lesser, A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BDC . And let BD be incommensurable in length with DC . I say that that the square on BC is greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC) .



Τῶν γὰρ αὐτῶν κατασκευασθέντων τῷ πρότερον ὁμοίως δείχομεν, ὅτι ἡ $BΓ$ τῆς A μείζον δύναται τῷ ἀπὸ τῆς $ZΔ$. δεικτέον [οὔν], ὅτι ἀσύμμετρος ἐστὶν ἡ $BΓ$ τῇ $ΔZ$ μήκει. ἐπεὶ γὰρ ἀσύμμετρος ἐστὶν ἡ $BΔ$ τῇ $ΔΓ$ μήκει, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ $BΓ$ τῇ $ΓΔ$ μήκει. ἀλλὰ ἡ $ΔΓ$ σύμμετρος ἐστὶ συναμφοτέραις ταῖς BZ , $ΔΓ$. καὶ ἡ $BΓ$ ἄρα ἀσύμμετρος ἐστὶ συναμφοτέραις ταῖς BZ , $ΔΓ$. ὥστε καὶ λοιπὴ τῇ $ZΔ$ ἀσύμμετρος ἐστὶν ἡ $BΓ$ μήκει. καὶ ἡ $BΓ$ τῆς A μείζον δύναται τῷ ἀπὸ τῆς $ZΔ$. ἡ $BΓ$ ἄρα τῆς A μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ.

Δυνάσθω δὴ πάλιν ἡ $BΓ$ τῆς A μείζον τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς A ἴσον παρὰ τὴν $BΓ$ παραβεβλήσθω ἐλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν $BΔ$, $ΔΓ$. δεικτέον, ὅτι ἀσύμμετρος ἐστὶν ἡ $BΔ$ τῇ $ΔΓ$ μήκει.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἡ $BΓ$ τῆς A μείζον δύναται τῷ ἀπὸ τῆς $ZΔ$. ἀλλὰ ἡ $BΓ$ τῆς A μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. ἀσύμμετρος ἄρα ἐστὶν ἡ $BΓ$ τῇ $ZΔ$ μήκει. ὥστε καὶ λοιπὴ συναμφοτέρῳ τῇ BZ , $ΔΓ$ ἀσύμμετρος ἐστὶν ἡ $BΓ$. ἀλλὰ συναμφοτέροξ ἡ BZ , $ΔΓ$ τῇ $ΔΓ$ σύμμετρος ἐστὶ μήκει. καὶ ἡ $BΓ$ ἄρα τῇ $ΔΓ$ ἀσύμμετρος ἐστὶ μήκει. ὥστε καὶ διελόντι ἡ $BΔ$ τῇ $ΔΓ$ ἀσύμμετρος ἐστὶ μήκει.

Ἐὰν ἄρα ὡσι δύο εὐθεῖαι, καὶ τὰ ἐχθῆς.

For, similarly, by the same construction as before, we can show that the square on BC is greater than the (square on) A by the (square) on FD . [Therefore] it must be shown that BC is incommensurable in length with DF . For since BD is incommensurable in length with DC , BC is thus also incommensurable in length with CD [Prop. 10.16]. But, DC is commensurable (in length) with the sum of BF and DC [Prop. 10.6]. And, thus, BC is incommensurable (in length) with the sum of BF and DC [Prop. 10.13]. Hence, BC is also incommensurable in length with the remainder FD [Prop. 10.16]. And the square on BC is greater than the (square on) A by the (square) on FD . Thus, the square on BC is greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC) .

So, again, let the square on BC be greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC) . And let a (rectangle) equal to the fourth [part] of the (square) on A , falling short by a square figure, be applied to BC . And let it be the (rectangle contained) by BD and DC . It must be shown that BD is incommensurable in length with DC .

For, similarly, by the same construction, we can show that the square on BC is greater than the (square) on A by the (square) on FD . But, the square on BC is greater than the (square) on A by the (square) on (some straight-line) incommensurable (in length) with (BC) . Thus, BC is incommensurable in length with FD . Hence, BC is also incommensurable (in length) with the remaining sum of BF and DC [Prop. 10.16]. But, the sum of BF and DC is commensurable in length with DC [Prop. 10.6]. Thus, BC is also incommensurable in length with DC [Prop. 10.13]. Hence, via separation, BD is also incommensurable in length with DC [Prop. 10.16].

Thus, if there are two ... straight-lines, and so on ...

† This proposition states that if $\alpha x - x^2 = \beta^2/4$ (where $\alpha = BC$, $x = DC$, and $\beta = A$) then α and $\sqrt{\alpha^2 - \beta^2}$ are incommensurable when $\alpha - x$ are x are incommensurable, and vice versa.

19.

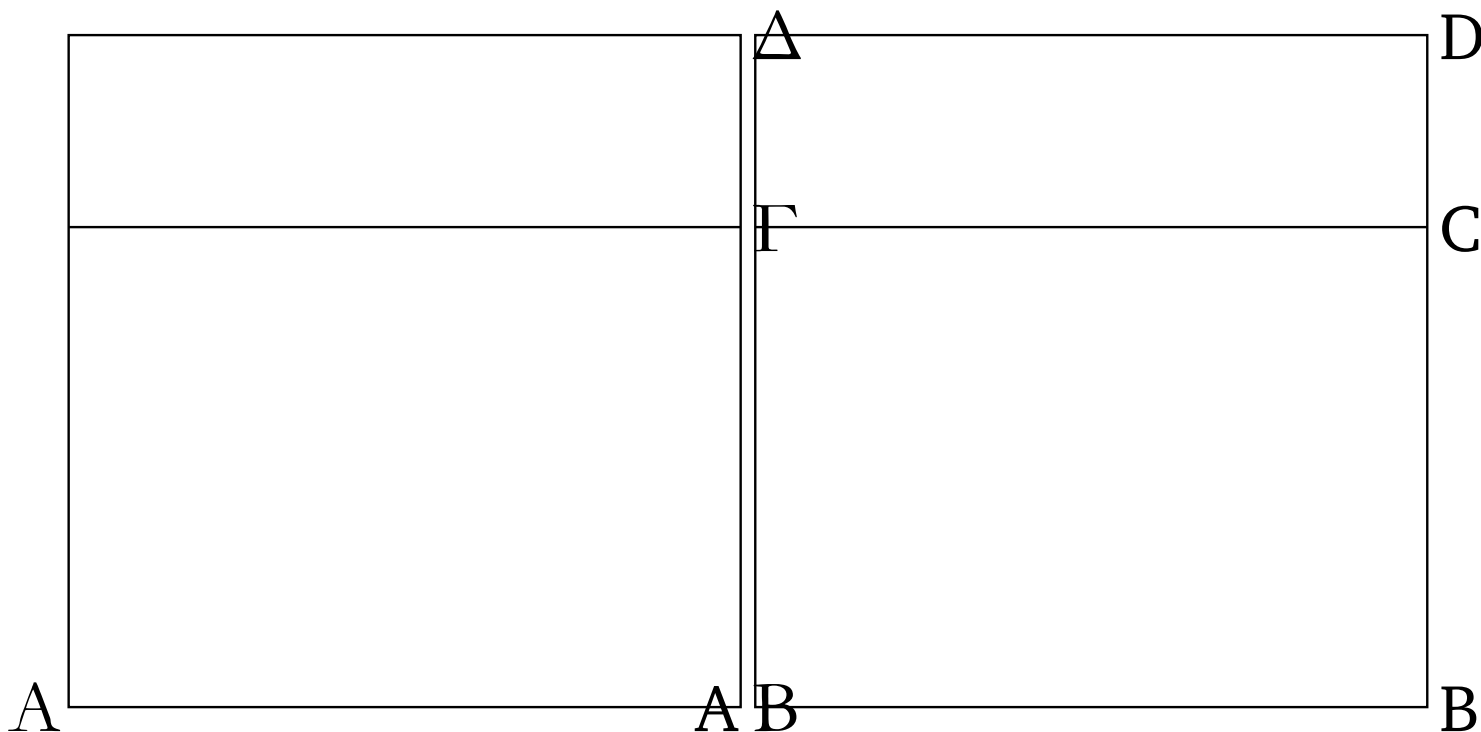
Proposition 19

Τὸ ὑπὸ ῥητῶν μήκει συμμετρῶν εὐθειῶν περιεχόμενον ὀρθογώνιον ῥητόν ἐστιν.

Ὑπὸ γὰρ ῥητῶν μήκει συμμετρῶν εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιεθέσθω τὸ $ΑΓ$. λέγω, ὅτι ῥητόν ἐστὶ τὸ $ΑΓ$.

The rectangle contained by rational straight-lines (which are) commensurable in length is rational.

For let the rectangle AC be enclosed by the rational straight-lines AB and BC (which are) commensurable in length. I say that AC is rational.



Ἀναγεγράφθω γάρ ἀπὸ τῆς AB τετράγωνον τὸ $A\Delta$. ῥητὸν ἄρα ἐστὶ τὸ $A\Delta$. καὶ ἐπεὶ σύμμετρόξ ἐστιν ἡ AB τῇ $B\Gamma$ μήκει, ἴση δέ ἐστιν ἡ AB τῇ $B\Delta$, σύμμετροξ ἄρα ἐστὶν ἡ $B\Delta$ τῇ $B\Gamma$ μήκει. καὶ ἐστὶν ὥς ἡ $B\Delta$ πρὸς τὴν $B\Gamma$, οὕτως τὸ ΔA πρὸς τὸ $A\Gamma$. σύμμετρον ἄρα ἐστὶ τὸ ΔA τῷ $A\Gamma$. ῥητὸν δὲ τὸ ΔA . ῥητὸν ἄρα ἐστὶ καὶ τὸ $A\Gamma$.

Τὸ ἄρα ὑπὸ ῥητῶν μήκει συμμέτρων, καὶ τὰ ἐχῆξ.

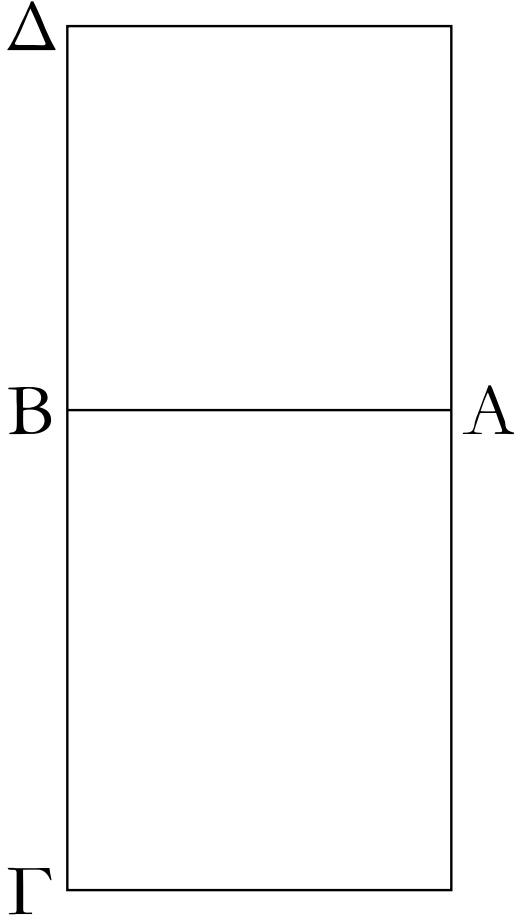
For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And since AB is commensurable in length with BC , and AB is equal to BD , BD is thus commensurable in length with BC . And as BD is to BC , so DA (is) to AC [Prop. 6.1]. Thus, DA is commensurable with AC [Prop. 10.11]. And DA (is) rational. Thus, AC is also rational [Def. 10.4]. Thus, the ...by rational straight-lines ... commensurable, and so on

20.

Ἐὰν ῥητὸν παρὰ ῥητὴν παραβληθῇ, πλάτος ποιῇ ῥητὴν καὶ σύμμετρον τῇ, παρ' ἣν παράκειται, μήκει.

Proposition 20

If a rational (area) is applied to a rational (straight-line) then it produces as breadth a (straight-line which is) rational, and commensurable in length with the (straight-line) to which it is applied.



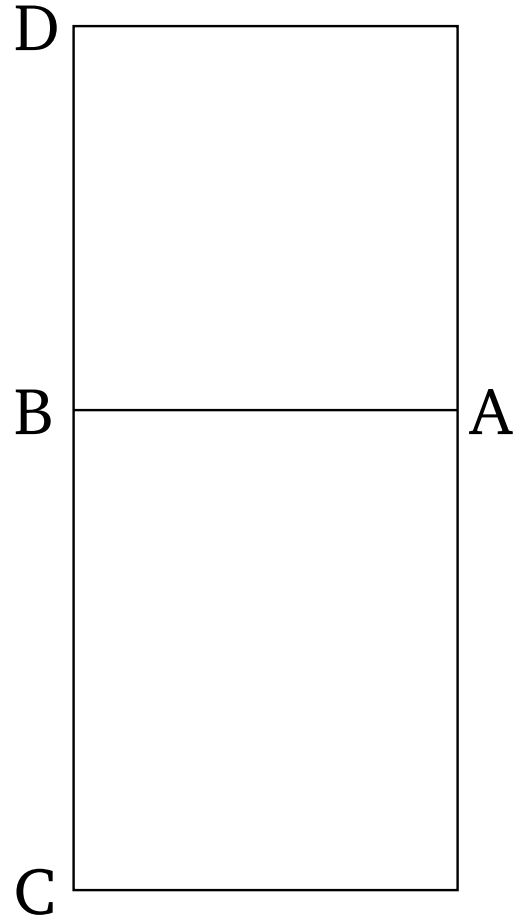
Ῥητόν γάρ τὸ $ΑΓ$ παρὰ ῥητὴν τὴν $ΑΒ$ παραβεβλήσθω πλάτος ποιῶν τὴν $ΒΓ$. λέγω, ὅτι ῥητὴ ἐστὶν ἡ $ΒΓ$ καὶ σύμμετρος τῇ $ΒΑ$ μήκει.

Ἀναγεγράφθω γάρ ἀπὸ τῆς $ΑΒ$ τετράγωνον τὸ $ΑΔ$. ῥητόν ἄρα ἐστὶ τὸ $ΑΔ$. ῥητόν δὲ καὶ τὸ $ΑΓ$. σύμμετρον ἄρα ἐστὶ τὸ $ΔΑ$ τῷ $ΑΓ$. καὶ ἐστὶν ὥς τὸ $ΔΑ$ πρὸς τὸ $ΑΓ$, οὕτως ἡ $ΔΒ$ πρὸς τὴν $ΒΓ$. σύμμετρος ἄρα ἐστὶ καὶ ἡ $ΔΒ$ τῇ $ΒΓ$. ἴση δὲ ἡ $ΔΒ$ τῇ $ΒΑ$. σύμμετρος ἄρα καὶ ἡ $ΑΒ$ τῇ $ΒΓ$. ῥητὴ δὲ ἐστὶν ἡ $ΑΒ$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΒΓ$ καὶ σύμμετρος τῇ $ΑΒ$ μήκει.

Ἐάν ἄρα ῥητόν παρὰ ῥητὴν παραβληθῇ, καὶ τὰ ἐχθῇ.

21.

Τὸ ὑπὸ ῥητῶν δυνάμει μόνον συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν, καὶ ἡ δυναμένη αὐτοῦ ἄλογός ἐστιν, καλεῖσθω δὲ μέση.



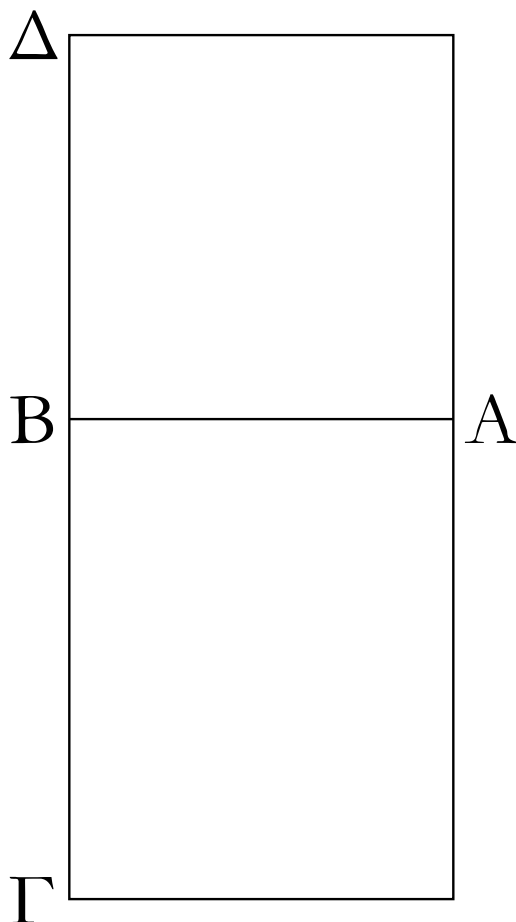
For let the rational (area) AC be applied to the rational (straight-line) AB , producing the (straight-line) BC as breadth. I say that BC is rational, and commensurable in length with BA .

For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And AC (is) also rational. DA is thus commensurable with AC . And as DA is to AC , so DB (is) to BC [Prop. 6.1]. Thus, DB is also commensurable (in length) with BC [Prop. 10.11]. And DB (is) equal to BA . Thus, AB (is) also commensurable (in length) with BC . And AB is rational. Thus, BC is also rational, and commensurable in length with AB [Def. 10.3].

Thus, if a rational (area) is applied to a rational (straight-line), and so on . . .

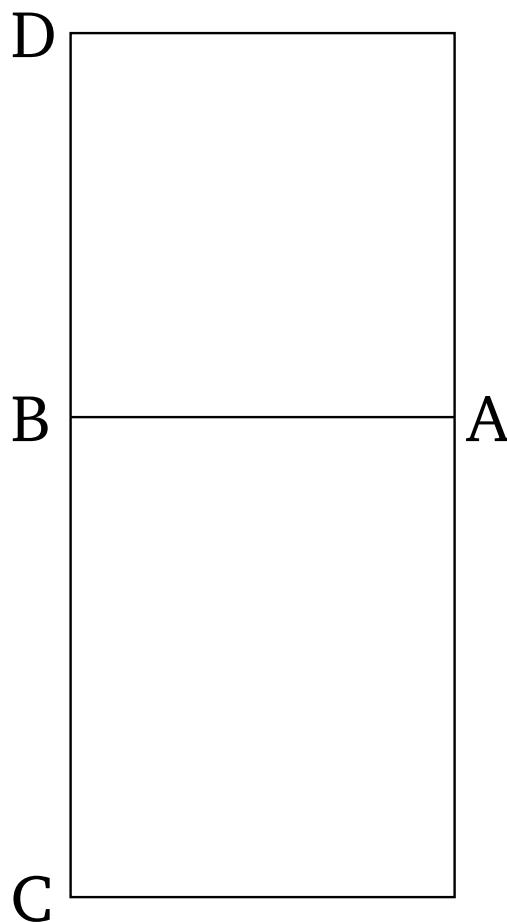
Proposition 21

The rectangle contained by rational straight-lines (which are) commensurable in square only is irrational, and its square-root is irrational—let it be called medial.[†]



Ὑπὸ γὰρ ῥητῶν δυνάμει μόνον συμμετρῶν εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιεθέσθω τὸ $ΑΓ$. λέγω, ὅτι ἄλογόν ἐστὶ τὸ $ΑΓ$, καὶ ἡ δυναμένη αὐτὸ ἄλογός ἐστιν, καλεῖσθω δὲ μέση.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $ΑΔ$. ῥητὸν ἄρα ἐστὶ τὸ $ΑΔ$. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει· δυνάμει γὰρ μόνον ὑπόκεινται σύμμετροι· ἴση δὲ ἡ AB τῇ $BΔ$, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ $ΔB$ τῇ $BΓ$ μήκει. καὶ ἐστὶν ὥς ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως τὸ $ΑΔ$ πρὸς τὸ $ΑΓ$. ἀσύμμετρον ἄρα [ἐστὶ] τὸ $ΔΑ$ τῷ $ΑΓ$. ῥητὸν δὲ τὸ $ΔΑ$. ἄλογον ἄρα ἐστὶ τὸ $ΑΓ$. ὥστε καὶ ἡ δυναμένη τὸ $ΑΓ$ [τουτέστιν ἡ ἴσον αὐτῷ τετράγωνον δυναμένη] ἄλογός ἐστιν, καλεῖσθω δε μέση· ὅπερ ἔδει δεῖχαι.



For let the rectangle AC be contained by the rational straight-lines AB and BC (which are) commensurable in square only. I say that AC is irrational, and its square-root is irrational—let it be called medial.

For let the square AD be described on AB . AD is thus rational [Def. 10.4]. And since AB is incommensurable in length with BC . For they were assumed to be commensurable in square only. And AB (is) equal to BD . DB is thus also incommensurable in length with BC . And as DB is to BC , so AD (is) to AC [Prop. 6.1]. Thus, DA [is] incommensurable with AC [Prop. 10.11]. And DA (is) rational. Thus, AC is irrational [Def. 10.4]. Hence, its square-root [that is to say, the square-root of the square equal to it] is also irrational [Def. 10.4]—let it be called medial. (Which is) the very thing it was required to show.

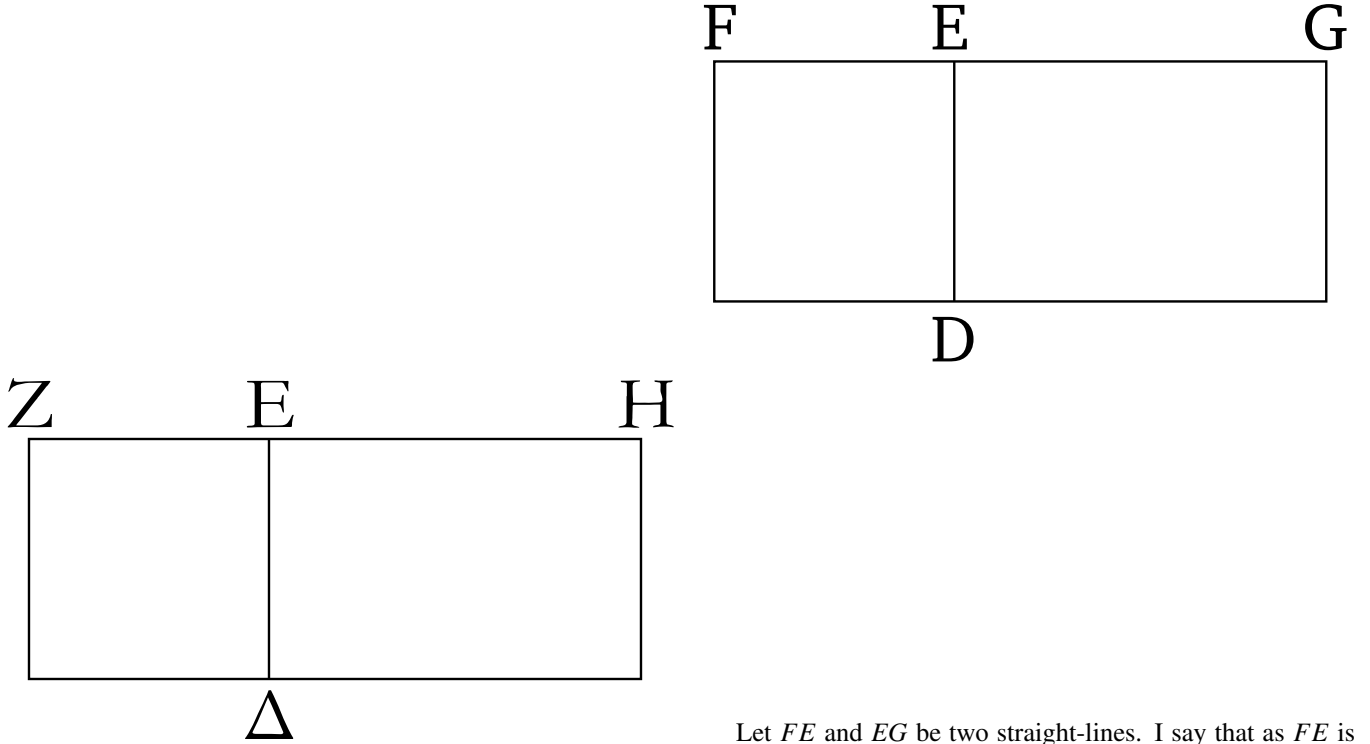
[†] Thus, a medial straight-line has a length expressible as $k^{1/4}$.

Λήμμα.

Ἐάν ᾧσι δύο εὐθεῖαι, ἔστιν ὥς ἡ πρώτη πρὸς τὴν δευτέραν, οὕτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ὑπὸ τῶν δύο εὐθειῶν.

Lemma

If there are two straight-lines then as the first is to the second, so the (square) on the first (is) to the (rectangle contained) by the two straight-lines.



Ἐστωσαν δύο εὐθεῖαι αἱ ZE , EH . λέγω, ὅτι ἐστὶν ὥς ἡ ZE πρὸς τὴν EH , οὕτως τὸ ἀπὸ τῆς ZE πρὸς τὸ ὑπὸ τῶν ZE , EH .

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ZE τετράγωνον τὸ ΔZ , καὶ συμπληρώσθω τὸ $H\Delta$. ἐπεὶ οὖν ἐστὶν ὥς ἡ ZE πρὸς τὴν EH , οὕτως τὸ $Z\Delta$ πρὸς τὸ ΔH , καὶ ἐστὶ τὸ μὲν $Z\Delta$ τὸ ἀπὸ τῆς ZE , τὸ δὲ ΔH τὸ ὑπὸ τῶν ΔE , EH , τουτέστι τὸ ὑπὸ τῶν ZE , EH , ἐστὶν ἄρα ὥς ἡ ZE πρὸς τὴν EH , οὕτως τὸ ἀπὸ τῆς ZE πρὸς τὸ ὑπὸ τῶν ZE , EH . ὁμοίως δὲ καὶ ὥς τὸ ὑπὸ τῶν HE , EZ πρὸς τὸ ἀπὸ τῆς EZ , τουτέστιν ὥς τὸ $H\Delta$ πρὸς τὸ $Z\Delta$, οὕτως ἡ HE πρὸς τὴν EZ . ὅπερ ἔδει δεῖξαι.

22.

Τὸ ἀπὸ μέσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ῥητὴν καὶ ἀσύμμετρον τῇ, παρ' ἣν παράκειται, μήκει.

Ἐστω μέση μὲν ἡ A , ῥητὴ δὲ ἡ GB , καὶ τῷ ἀπὸ τῆς A ἴσον παρὰ τὴν $B\Gamma$ παραβεβλήσθω θωρίον ὀρθογώνιον τὸ $B\Delta$ πλάτος ποιῶν τὴν $\Gamma\Delta$. λέγω, ὅτι ῥητὴ ἐστὶν ἡ $\Gamma\Delta$ καὶ ἀσύμμετρος τῇ GB μήκει.

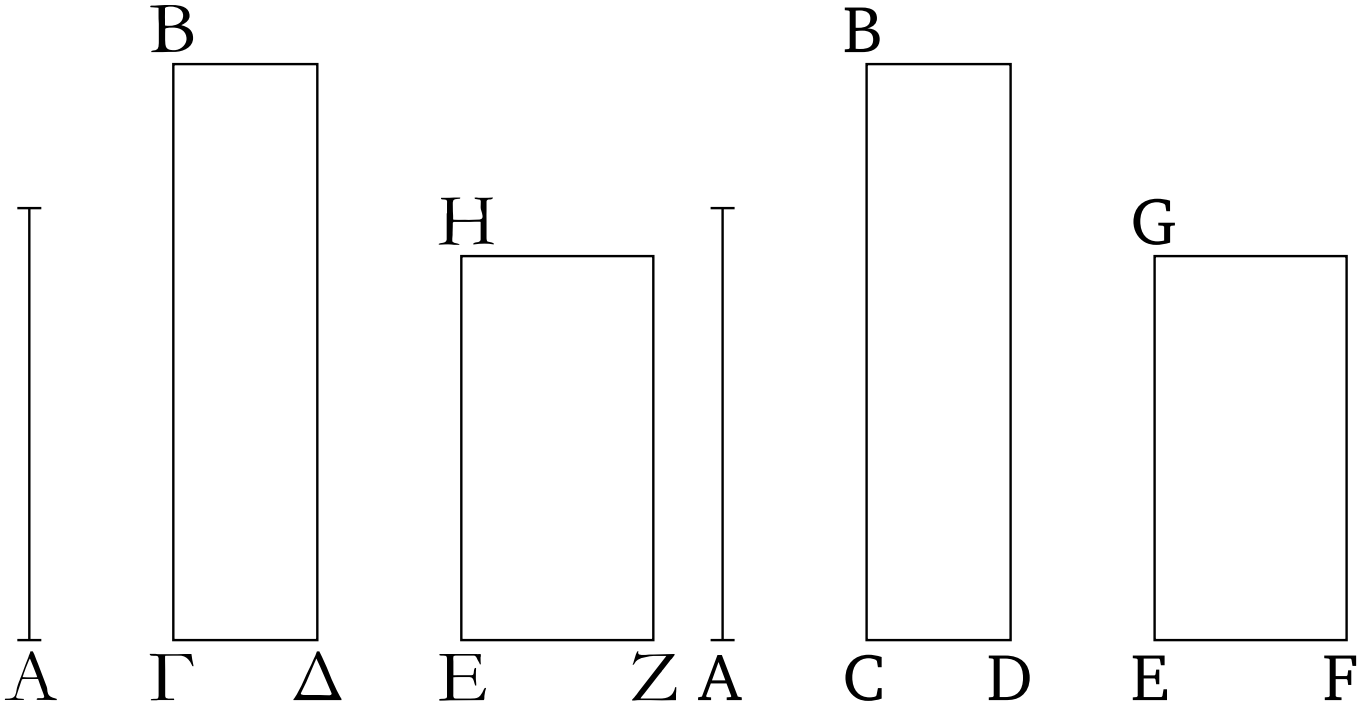
Let FE and EG be two straight-lines. I say that as FE is to EG , so the (square) on FE (is) to the (rectangle contained) by FE and EG .

For let the square DF be described on FE . And let GD be completed. Therefore, since as FE is to EG , so FD (is) to DG [Prop. 6.1], and FD is the (square) on FE , and DG the (rectangle contained) by DE and EG —that is to say, the (rectangle contained) by FE and EG —thus as FE is to EG , so the (square) on FE (is) to the (rectangle contained) by FE and EG . And also, similarly, as the (rectangle contained) by GE and EF is to the (square on) EF —that is to say, as GD (is) to FD —so GE (is) to EF . (Which is) the very thing it was required to show.

Proposition 22

The square on a medial (straight-line), being applied to a rational (straight-line), produces as breadth a (straight-line which is) rational, and incommensurable in length with the (straight-line) to which it is applied.

Let A be a medial (straight-line), and CB a rational (straight-line), and let the rectangular area BD , equal to the (square) on A , be applied to BC , producing CD as breadth. I say that CD is rational, and incommensurable in length with CB .



Ἐπεὶ γὰρ μέση ἐστὶν ἡ A , δύνатаι θεωρίον περιεχόμενον ὑπὸ ῥητῶν δυνάμει μόνον συμμετρῶν. δυνάσθω τὸ HZ . δύνатаι δὲ καὶ τὸ $B\Delta$. ἴσον ἄρα ἐστὶ τὸ $B\Delta$ τῷ HZ . ἐστὶ δὲ αὐτῷ καὶ ἰσογώνιον· τῶν δὲ ἴσων τε καὶ ἰσογώνων παραλληλογράμμων ἀντιπεπρόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· ἀνάλογον ἄρα ἐστὶν ὡς ἡ $B\Gamma$ πρὸς τὴν EH , οὕτως ἡ EZ πρὸς τὴν $\Gamma\Delta$. ἐστὶν ἄρα καὶ ὡς τὸ ἀπὸ τῆς $B\Gamma$ πρὸς τὸ ἀπὸ τῆς EH , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς $\Gamma\Delta$. σύμμετρον δὲ ἐστὶ τὸ ἀπὸ τῆς ΓB τῷ ἀπὸ τῆς EH . ῥητὴ γάρ ἐστιν ἑκατέρα αὐτῶν· σύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς EZ τῷ ἀπὸ τῆς $\Gamma\Delta$. ῥητὸν δὲ ἐστὶ τὸ ἀπὸ τῆς EZ . ῥητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $\Gamma\Delta$. ῥητὴ ἄρα ἐστὶν ἡ $\Gamma\Delta$. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ EZ τῇ EH μήκει· δυνάμει γὰρ μόνον εἰσὶ σύμμετροι· ὡς δὲ ἡ EZ πρὸς τὴν EH , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ὑπὸ τῶν ZE, EH , ἀσύμμετρον ἄρα [ἐστὶ] τὸ ἀπὸ τῆς EZ τῷ ὑπὸ τῶν ZE, EH . ἀλλὰ τῷ μὲν ἀπὸ τῆς EZ σύμμετρον ἐστὶ τὸ ἀπὸ τῆς $\Gamma\Delta$. ῥηταὶ γὰρ εἰσι δυνάμει· τῷ δὲ ὑπὸ τῶν ZE, EH σύμμετρον ἐστὶ τὸ ὑπὸ τῶν $\Delta\Gamma, \Gamma B$. ἴσα γὰρ ἐστὶ τῷ ἀπὸ τῆς A . ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $\Gamma\Delta$ τῷ ὑπὸ τῶν $\Delta\Gamma, \Gamma B$. ὡς δὲ τὸ ἀπὸ τῆς $\Gamma\Delta$ πρὸς τὸ ὑπὸ τῶν $\Delta\Gamma, \Gamma B$, οὕτως ἐστὶν ἡ $\Delta\Gamma$ πρὸς τὴν ΓB . ἀσύμμετρος ἄρα ἐστὶν ἡ $\Delta\Gamma$ τῇ ΓB μήκει. ῥητὴ ἄρα ἐστὶν ἡ $\Gamma\Delta$ καὶ ἀσύμμετρος τῇ ΓB μήκει· ὅπερ εἶδει δεῖχαι.

For since A is medial, the square on it is equal to a (rectangular) area contained by rational (straight-lines which are) commensurable in square only [Prop. 10.21]. Let the square on (A) be equal to GF . And the square on (A) is also equal to BD . Thus, BD is equal to GF . And (BD) is also equiangular with (GF) . And for equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional [Prop. 6.14]. Thus, proportionally, as BC is to EG , so EF (is) to CD . And, also, as the (square) on BC is to the (square) on EG , so the (square) on EF (is) to the (square) on CD [Prop. 6.22]. And the (square) on CB is commensurable with the (square) on EG . For they are each rational. Thus, the (square) on EF is also commensurable with the (square) on CD [Prop. 10.11]. And the (square) on EF is rational. Thus, the (square) on CD is also rational [Def. 10.4]. Thus, CD is rational. And since EF is incommensurable in length with EG . For they are commensurable in square only. And as EF (is) to EG , so the (square) on EF (is) to the (rectangle contained) by FE and EG [see previous lemma]. The (square) on EF [is] thus incommensurable with the (rectangle contained) by FE and EG [Prop. 10.11]. But, the (square) on CD is commensurable with the (square) on EF . For they are rational in square. And the (rectangle contained) by DC and CB is commensurable with the (rectangle contained) by FE and EG . For they are (both) equal to the (square) on A . Thus, the (square) on CD is also incommensurable with the (rectangle contained) by DC and CB [Prop. 10.13]. And as the (square) on CD (is) to the (rectangle contained) by DC and CB , so DC is to CB [see previous lemma]. Thus, DC is incommensurable in length with

CB [Prop. 10.11]. Thus, CD is rational, and incommensurable in length with CB . (Which is) the very thing it was required to show.

[†] Literally, “rational”.

23.

Ἡ τῇ μέσῃ σύμμετρος μέση ἐστίν.

Ἐστω μέση ἡ A , καὶ τῇ A σύμμετρος ἔστω ἡ B . λέγω, ὅτι καὶ ἡ B μέση ἐστίν.

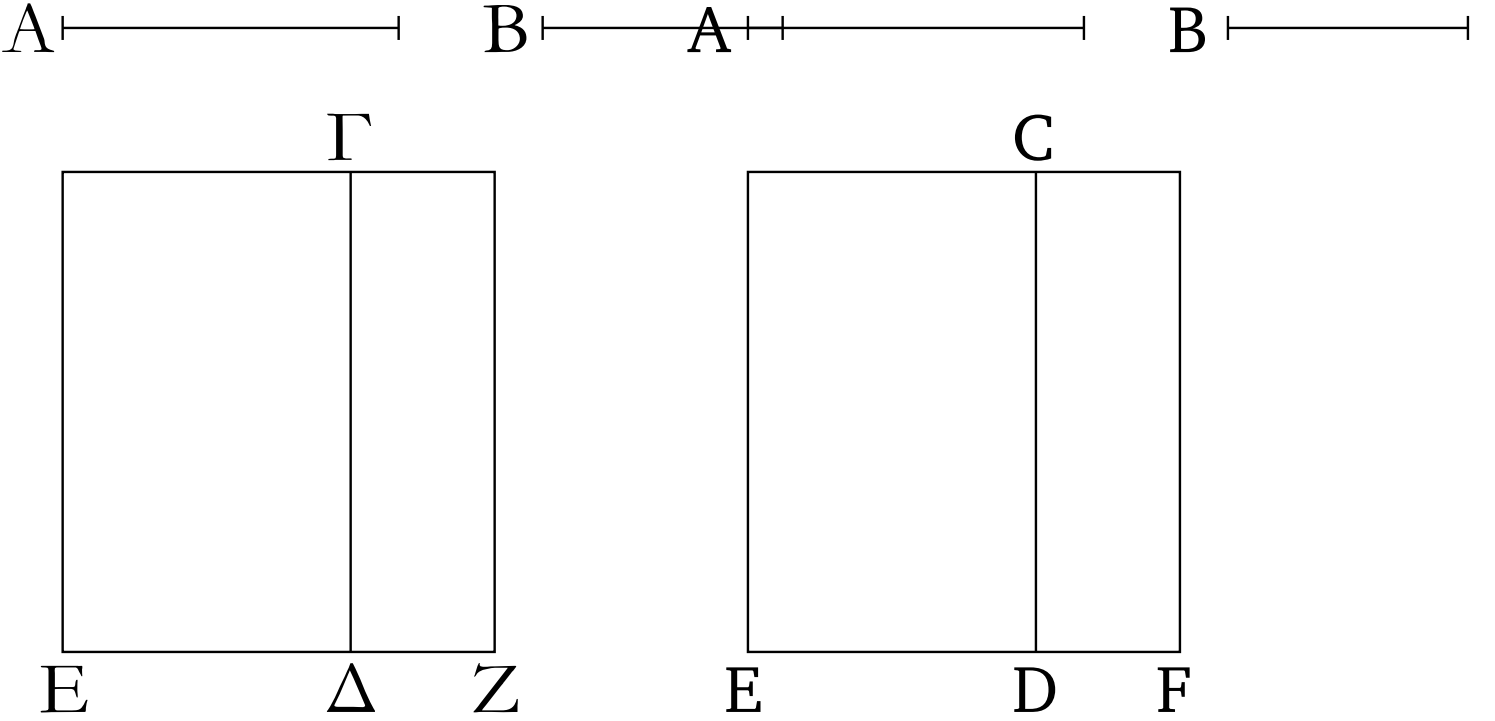
Ἐκκείσθω γὰρ ῥητὴ ἡ $\Gamma\Delta$, καὶ τῷ μὲν ἀπὸ τῆς A ἴσον παρὰ τὴν $\Gamma\Delta$ παραβελθήσθω θωρίον ὀρθογώνιον τὸ ΓE πλάτος ποιοῦν τὴν $E\Delta$. ῥητὴ ἄρα ἐστὶν ἡ $E\Delta$ καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. τῷ δὲ ἀπὸ τῆς B ἴσον παρὰ τὴν $\Gamma\Delta$ παραβελθήσθω θωρίον ὀρθογώνιον τὸ ΓZ πλάτος ποιοῦν τὴν ΔZ . ἐπεὶ οὖν σύμμετρος ἐστὶν ἡ A τῇ B , σύμμετρον ἐστὶ καὶ τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς B . ἀλλὰ τῷ μὲν ἀπὸ τῆς A ἴσον ἐστὶ τὸ $E\Gamma$, τῷ δὲ ἀπὸ τῆς B ἴσον ἐστὶ τὸ ΓZ . σύμμετρον ἄρα ἐστὶ τὸ $E\Gamma$ τῷ ΓZ . καὶ ἐστὶν ὥς τὸ $E\Gamma$ πρὸς τὸ ΓZ , οὕτως ἡ $E\Delta$ πρὸς τὴν ΔZ . σύμμετρος ἄρα ἐστὶν ἡ $E\Delta$ τῇ ΔZ μήκει. ῥητὴ δὲ ἐστὶν ἡ $E\Delta$ καὶ ἀσύμμετρος τῇ ΔZ μήκει. ῥητὴ ἄρα ἐστὶ καὶ ἡ ΔZ καὶ ἀσύμμετρος τῇ $\Delta\Gamma$ μήκει· αἱ $\Gamma\Delta$, ΔZ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἡ δὲ τὸ ὑπὸ ῥητῶν δυνάμει μόνον συμμέτρων δυνάμενη μέση ἐστίν. ἡ ἄρα τὸ ὑπὸ τῶν $\Gamma\Delta$, ΔZ δυνάμενη μέση ἐστίν· καὶ δύναται τὸ ὑπὸ τῶν $\Gamma\Delta$, ΔZ ἡ B μέση ἄρα ἐστὶν ἡ B .

Proposition 23

A (straight-line) commensurable with a medial (straight-line) is medial.

Let A be a medial (straight-line), and let B be commensurable with A . I say that B is also a medial (straight-line).

Let the rational (straight-line) CD be set out, and let the rectangular area CE , equal to the (square) on A , be applied to CD , producing ED as width. ED is thus rational, and incommensurable in length with CD [Prop. 10.22]. And let the rectangular area CF , equal to the (square) on B , be applied to CD , producing DF as width. Therefore, since A is commensurable with B , the (square) on A is also commensurable with the (square) on B . But, EC is equal to the (square) on A , and CF is equal to the (square) on B . Thus, EC is commensurable with CF . And as EC is to CF , so ED (is) to DF [Prop. 6.1]. Thus, ED is commensurable in length with DF [Prop. 10.11]. And ED is rational, and incommensurable in length with CD . DF is thus also rational [Def. 10.3], and incommensurable in length with DC [Prop. 10.13]. Thus, CD and DF are rational, and commensurable in square only. And the square-root of a (rectangle contained) by rational (straight-lines which are) commensurable in square only is medial [Prop. 10.21]. Thus, the square-root of the (rectangle contained) by CD and DF is medial. And the square on B is equal to the (rectangle contained) by CD and DF . Thus, B is a medial (straight-line).



Πόρισμα.

Corollary

Ἐκ δὴ τούτων φανερόν, ὅτι τὸ τῶ μέσῳ θεωρῶ σύμμετρον μέσον ἐστίν.

And (it is) clear, from this, that an (area) commensurable with a medial area[†] is medial.

[†] A medial area is equal to the square on some medial straight-line. Hence, a medial area is expressible as $k^{1/2}$.

24.

Proposition 24

Τὸ ὑπὸ μέσων μήκει συμμετρων ἐνθειῶν περιεχόμενον ὀρθογώνιον μέσον ἐστίν.

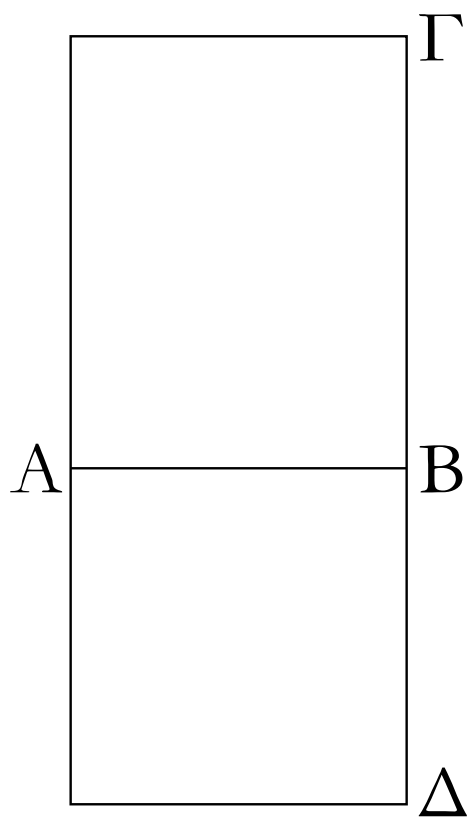
A rectangle contained by medial straight-lines (which are) commensurable in length is medial.

Ὑπὸ γὰρ μέσων μήκει συμμετρων ἐνθειῶν τῶν AB , BF περιεθέσθω ὀρθογώνιον τὸ AF . λέγω, ὅτι τὸ AF μέσον ἐστίν.

For let the rectangle AC be contained by the medial straight-lines AB and BC (which are) commensurable in length. I say that AC is medial.

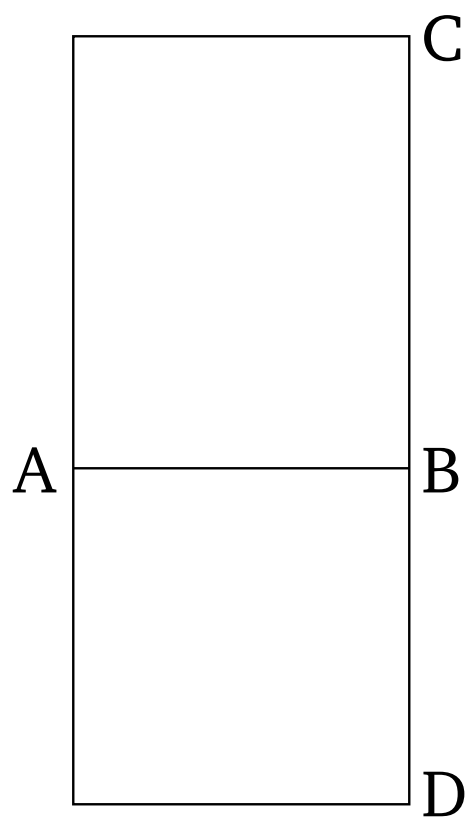
Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ AD . μέσον ἄρα ἐστὶ τὸ AD . καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ AB τῇ BF μήκει, ἴση δὲ ἡ AB τῇ BD , σύμμετρος ἄρα ἐστὶ καὶ ἡ DB τῇ BF μήκει· ὥστε καὶ τὸ DA τῶ AF σύμμετρον ἐστίν. μέσον δὲ τὸ DA . μέσον ἄρα καὶ τὸ AF . ὅπερ ἔδει δεῖξαι.

For let the square AD be described on AB . AD is thus medial [see previous footnote]. And since AB is commensurable in length with BC , and AB (is) equal to BD , DB is thus also commensurable in length with BC . Hence, DA is also commensurable with AC [Props. 6.1, 10.11]. And DA (is) medial. Thus, AC (is) also medial [Prop. 10.23 corr.]. (Which is) the very thing it was required to show.



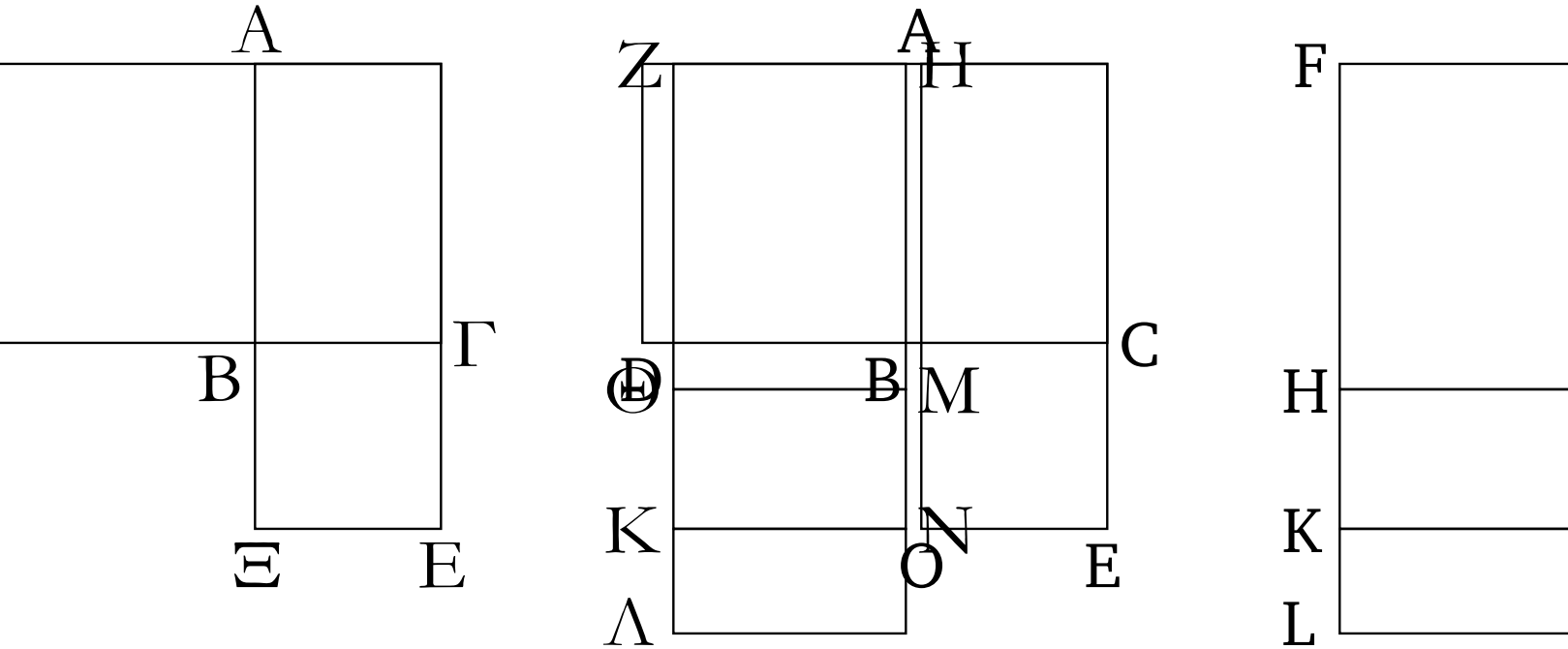
25.

*Τὸ ὑπὸ μέσων δυνάμει μόνον συμμέτρων εὐθειῶν περιε-
χόμενον ὀρθογώνιον ἤτοι ῥητὸν ἢ μέσον ἐστίν.*



Proposition 25

The rectangle contained by medial straight-lines (which are) commensurable in square only is either rational or medial.



Ὑπὸ γὰρ μέσων δυνάμει μόνον συμμετρων εὐθειῶν τῶν AB , $BΓ$ ὀρθογώνιον περιεθέσθω τὸ $AΓ$. λέγω, ὅτι τὸ $AΓ$ ἥτοι ῥητὸν ἢ μέσον ἐστίν.

Ἀναγεγράφθω γὰρ ἀπὸ τῶν AB , $BΓ$ τετράγωνα τὰ AD , BE . μέσον ἄρα ἐστὶν ἑκάτερον τῶν AD , BE . καὶ ἐκκείσθω ῥητὴ ἡ ZH , καὶ τῷ μὲν AD ἴσον παρὰ τὴν ZH παραβελθήσθω ὀρθογώνιον παραλληλόγραμμον τὸ $HΘ$ πλάτος ποιῶν τὴν $ZΘ$, τῷ δὲ $AΓ$ ἴσον παρὰ τὴν $ΘM$ παραβελθήσθω ὀρθογώνιον παραλληλόγραμμον τὸ MK πλάτος ποιῶν τὴν $ΘK$, καὶ ἔτι τῷ BE ἴσον ὁμοίως παρὰ τὴν KN παραβελθήσθω τὸ $NΛ$ πλάτος ποιῶν τὴν KL . ἐπ' εὐθείᾳς ἄρα εἰσὶν αἱ $ZΘ$, $ΘK$, KL . ἐπεὶ οὖν μέσον ἐστὶν ἑκάτερον τῶν AD , BE , καὶ ἐστὶν ἴσον τὸ μὲν AD τῷ $HΘ$, τὸ δὲ BE τῷ $NΛ$, μέσον ἄρα καὶ ἑκάτερον τῶν $HΘ$, $NΛ$. καὶ παρὰ ῥητὴν τὴν ZH παράκειται ῥητὴ ἄρα ἐστὶν ἑκάτερα τῶν $ZΘ$, KL καὶ ἀσύμμετρος τῇ ZH μήκει. καὶ ἐπεὶ σύμμετρόν ἐστι τὸ AD τῷ BE , σύμμετρον ἄρα ἐστὶ καὶ τὸ $HΘ$ τῷ $NΛ$. καὶ ἐστὶν ὥς τὸ $HΘ$ πρὸς τὸ $NΛ$, οὕτως ἡ $ZΘ$ πρὸς τὴν KL . σύμμετρος ἄρα ἐστὶν ἡ $ZΘ$ τῇ KL μήκει. αἱ $ZΘ$, KL ἄρα ῥηταὶ εἰσι μήκει σύμμετροι. ῥητὸν ἄρα ἐστὶ τὸ ὑπὸ τῶν $ZΘ$, KL . καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν $ΔB$ τῇ BA , ἡ δὲ XB τῇ $BΓ$, ἔστιν ἄρα ὥς ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως ἡ AB πρὸς τὴν BX . ἀλλ' ὥς μὲν ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως τὸ $ΔA$ πρὸς τὸ $AΓ$. ὥς δὲ ἡ AB πρὸς τὴν BX , οὕτως τὸ $AΓ$ πρὸς τὸ $ΓX$. ἔστιν ἄρα ὥς τὸ $ΔA$ πρὸς τὸ $AΓ$, οὕτως τὸ $AΓ$ πρὸς τὸ $ΓX$. ἴσον δὲ ἐστὶ τὸ μὲν AD τῷ $HΘ$, τὸ δὲ $AΓ$ τῷ MK , τὸ δὲ $ΓX$ τῷ $NΛ$. ἔστιν ἄρα ὥς τὸ $HΘ$ πρὸς τὸ MK , οὕτως τὸ MK πρὸς τὸ $NΛ$. ἔστιν ἄρα καὶ ὥς ἡ $ZΘ$ πρὸς τὴν $ΘK$, οὕτως ἡ $ΘK$ πρὸς τὴν KL . τὸ ἄρα ὑπὸ τῶν $ZΘ$, KL ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΘK$. ῥητὸν δὲ τὸ ὑπὸ τῶν $ZΘ$, KL . ῥητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $ΘK$. ῥητὴ ἄρα

For let the rectangle AC be contained by the medial straight-lines AB and BC (which are) commensurable in square only. I say that AC is either rational or medial.

For let the squares AD and BE be described on (the straight-lines) AB and BC (respectively). AD and BE are thus each medial. And let the rational (straight-line) FG be laid out. And let the rectangular parallelogram GH , equal to AD , be applied to FG , producing FH as breadth. And let the rectangular parallelogram MK , equal to AC , be applied to HM , producing HK as breadth. And, finally, let NL , equal to BE , have similarly been applied to KN , producing KL as breadth. Thus, FH , HK , and KL are in a straight-line. Therefore, since AD and BE are each medial, and AD is equal to GH , and BE to NL , GH and NL (are) thus each also medial. And they are applied to the rational (straight-line) FG . FH and KL are thus each rational, and incommensurable in length with FG [Prop. 10.22]. And since AD is commensurable with BE , GH is thus also commensurable with NL . And as GH is to NL , so FH (is) to KL [Prop. 6.1]. Thus, FH is commensurable in length with KL [Prop. 10.11]. Thus, FH and KL are rational (straight-lines which are) commensurable in length. Thus, the (rectangle contained) by FH and KL is rational [Prop. 10.19]. And since DB is equal to BA , and OB to BC , thus as DB is to BC , so AB (is) to BO . But, as DB (is) to BC , so DA (is) to AC [Props. 6.1]. And as AB (is) to BO , so AC (is) to CO [Prop. 6.1]. Thus, as DA is to AC , so AC (is) to CO . And AD is equal to GH , and AC to MK , and CO to NL . Thus, as GH is to MK , so MK (is) to NL . Thus, also, as FH is to HK , so HK (is) to KL [Props. 6.1, 5.11]. Thus, the (rectangle contained) by FH and KL is equal to the (square) on HK

ἐστὶν ἡ ΘK . καὶ εἰ μὲν σύμμετρος ἐστί τῇ ZH μήκει, ῥητόν ἐστι τὸ ΘN . εἰ δὲ ἀσύμμετρος ἐστί τῇ ZH μήκει, αἱ $K\Theta$, ΘM ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα τὸ ΘN . τὸ ΘN ἄρα ἥτοι ῥητόν ἢ μέσον ἐστίν. ἴσον δὲ τὸ ΘN τῷ AG . τὸ AG ἄρα ἥτοι ῥητόν ἢ μέσον ἐστίν.

Τὸ ἄρα ὑπὸ μέσων δυνάμει μόνον συμμέτρων, καὶ τὰ ἐχῆξ.

[Prop. 6.17]. And the (rectangle contained) by FH and KL (is) rational. Thus, the (square) on HK is also rational. Thus, HK is rational. And if it is commensurable in length with FG then HN is rational [Prop. 10.19]. And if it is incommensurable in length with FG then KH and HM are rational (straight-lines which are) commensurable in square only: thus, HN is medial [Prop. 10.21]. Thus, HN is either rational or medial. And HN (is) equal to AC . Thus, AC is either rational or medial.

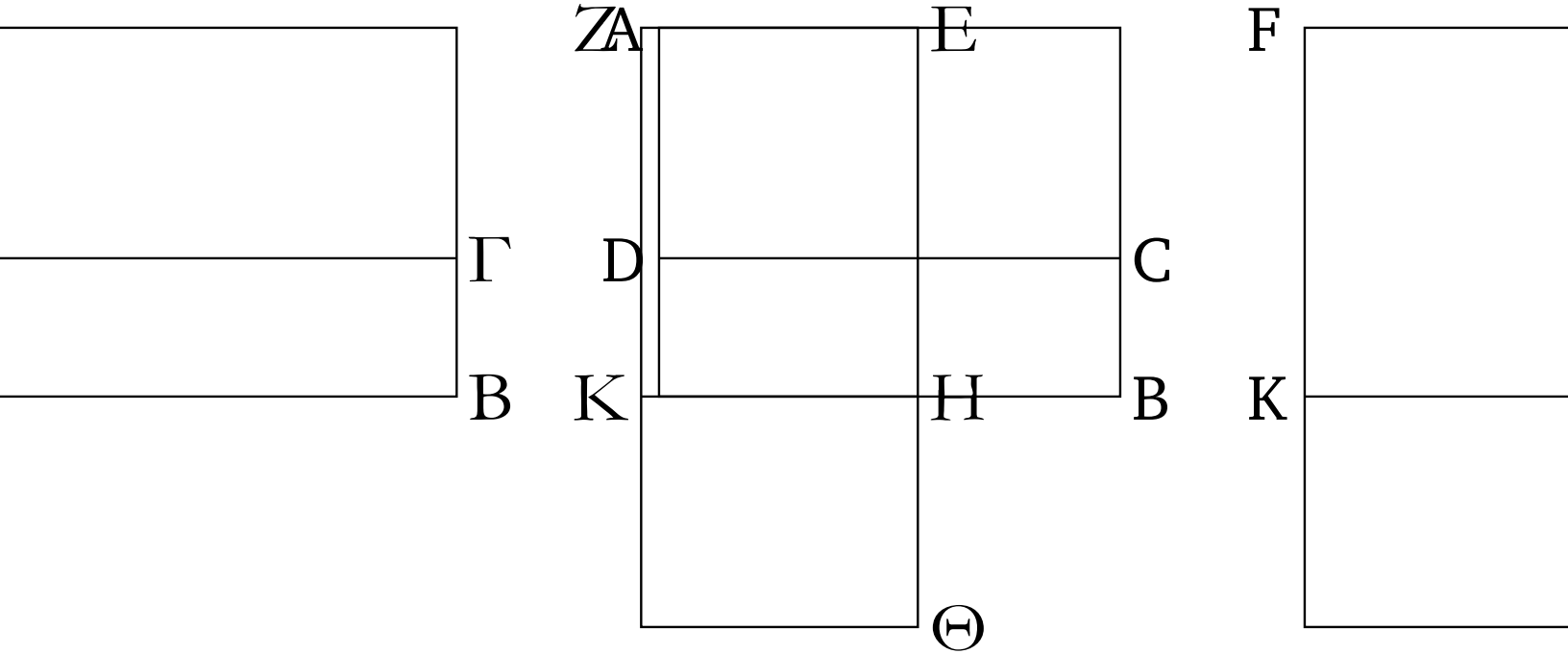
Thus, the ... by medial straight-lines (which are) commensurable in square only, and so on

26.

Μέσον μέσον οὐδ' ὑπερέχει ῥητῶ.

Proposition 26

A medial (area) does not exceed a medial (area) by a rational (area).[†]



Εἰ γὰρ δυνατόν, μέσον τὸ AB μέσον τοῦ AG ὑπερεθέτω ῥητῶ τῷ ΔB , καὶ ἐκκείσθω ῥητὴ ἡ EZ , καὶ τῷ AB ἴσον παρὰ τὴν EZ παραβελήσθω παραλληλόγραμμον ὀρθογώνιον τὸ $Z\Theta$ πλάτος ποιῶν τὴν $E\Theta$, τῷ δὲ AG ἴσον ἀφηγήσθω τὸ ZH . λοιπὸν ἄρα τὸ $B\Delta$ λοιπῶ τῷ $K\Theta$ ἐστὶν ἴσον. ῥητόν δὲ ἐστὶ τὸ ΔB . ῥητόν ἄρα ἐστὶ καὶ τὸ $K\Theta$. ἐπεὶ οὖν μέσον ἐστὶν ἐκάτερον τῶν AB , AG , καὶ ἐστὶ τὸ μὲν AB τῷ $Z\Theta$ ἴσον, τὸ δὲ AG τῷ ZH , μέσον ἄρα καὶ ἐκάτερον τῶν $Z\Theta$, ZH . καὶ παρὰ ῥητὴν τὴν EZ παράκειται· ῥητὴ ἄρα ἐστὶν ἐκάτερα τῶν ΘE , EH καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ῥητόν ἐστὶ τὸ ΔB καὶ ἐστὶν ἴσον τῷ $K\Theta$, ῥητόν ἄρα ἐστὶ καὶ τὸ $K\Theta$. καὶ παρὰ ῥητὴν τὴν EZ παράκειται· ῥητὴ ἄρα ἐστὶν ἡ $H\Theta$ καὶ σύμμετρος τῇ EZ μήκει. ἀλλὰ καὶ ἡ EH ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ EZ μήκει· ἀσύμμετρος ἄρα ἐστὶν ἡ EH τῇ $H\Theta$ μήκει. καὶ ἐστὶν ὥς ἡ EH πρὸς τὴν $H\Theta$, οὕτως τὸ ἀπὸ

For, if possible, let the medial (area) AB exceed the medial (area) AC by the rational (area) DB . And let the rational (straight-line) EF be laid down. And let the rectangular parallelogram FH , equal to AB , be applied to EF , producing EH as breadth. And let FG , equal to AC , be cut off (from FH). Thus, the remainder BD is equal to the remainder KH . And DB is rational. Thus, KH is also rational. Therefore, since AB and AC are each medial, and AB is equal to FH , and AC to FG , FH and FG are thus each also medial. And they are applied to the rational (straight-line) EF . Thus, HE and EG are each rational, and incommensurable in length with EF [Prop. 10.22]. And since DB is rational, and is equal to KH , KH is thus also rational. And (KH) is applied to the rational (straight-line) EF . GH is thus rational, and commensurable in length with EF [Prop. 10.20]. But, EG is also rational, and

τῆς EH πρὸς τὸ ὑπὸ τῶν $EH, H\Theta$ ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς EH τῷ ὑπὸ τῶν $EH, H\Theta$. ἀλλὰ τῷ μὲν ἀπὸ τῆς EH σύμμετρόν ἐστι τὰ ἀπὸ τῶν $EH, H\Theta$ τετράγωνα· ῥητὰ γὰρ ἀμφοτέρω· τῷ δὲ ὑπὸ τῶν $EH, H\Theta$ σύμμετρόν ἐστι τὸ διξ ὑπὸ τῶν $EH, H\Theta$ διπλάσιον γὰρ ἐστὶν αὐτοῦ· ἀσύμμετρον ἄρα ἐστὶ τὰ ἀπὸ τῶν $EH, H\Theta$ τῷ διξ ὑπὸ τῶν $EH, H\Theta$ καὶ συναμφοτέρω ἄρα τὰ τε ἀπὸ τῶν $EH, H\Theta$ καὶ τὸ διξ ὑπὸ τῶν $EH, H\Theta$, ὅπερ ἐστὶ τὸ ἀπὸ τῆς $E\Theta$, ἀσύμμετρόν ἐστι τοῖς ἀπὸ τῶν $EH, H\Theta$. ῥητὰ δὲ τὰ ἀπὸ τῶν $EH, H\Theta$ ἄλογον ἄρα τὸ ἀπὸ τῆς $E\Theta$. ἄλλοις ἄρα ἐστὶν ἡ $E\Theta$. ἀλλὰ καὶ ῥηρή· ὅπερ ἐστὶν ἀδύνατον.

Μέσον ἄρα μέσον οὐδ' ὑπερέθει ῥητῷ· ὅπερ εἶδει δεῖχαι.

incommensurable in length with EF . Thus, EG is incommensurable in length with GH [Prop. 10.13]. And as EG is to GH , so the (square) on EG (is) to the (rectangle contained) by EG and GH [Prop. 10.13 lem.]. Thus, the (square) on EG is incommensurable with the (rectangle contained) by EG and GH [Prop. 10.11]. But, the (sum of the) squares on EG and GH is commensurable with the (square) on EG . For (EG and GH are) both rational. And twice the (rectangle contained) by EG and GH is commensurable with the (rectangle contained) by EG and GH [Prop. 10.6]. For (the former) is double the latter. Thus, the (sum of the squares) on EG and GH is incommensurable with twice the (rectangle contained) by EG and GH [Prop. 10.13]. And thus the sum of the (squares) on EG and GH plus twice the (rectangle contained) by EG and GH , that is the (square) on EH [Prop. 2.4], is incommensurable with the (sum of the squares) on EG and GH [Prop. 10.16]. And the (sum of the squares) on EG and GH (is) rational. Thus, the (square) on EH is irrational [Def. 10.4]. Thus, EH is irrational [Def. 10.4]. But, (it is) also rational. The very thing is impossible.

Thus, a medial (area) does not exceed a medial (area) by a rational (area). (Which is) the very thing it was required to show.

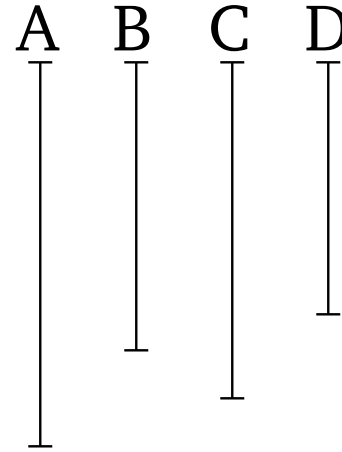
[†] In other words, $\sqrt{k} - \sqrt{k'} \neq k''$.

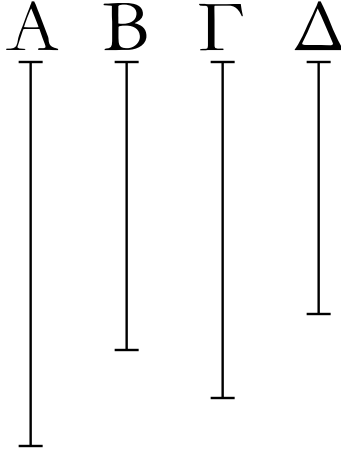
27.

Μέσαξ εὐρεῖν δυνάμει μόνον συμμέτρους ῥητὸν περιεθούσαξ.

Proposition 27

To find (two) medial (straight-lines), containing a rational (area), (which are) commensurable in square only.





Ἐκκεισθωσαν δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ A , B , καὶ εἰλήφθω τῶν A , B μέση ἀνάλογον ἡ Γ , καὶ γερονέτω ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ .

Καὶ ἐπεὶ αἱ A , B ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, τὸ ἄρα ὑπὸ τῶν A , B , τουτέστι τὸ ἀπὸ τῆς Γ , μέσον ἐστίν. μέση ἄρα ἡ Γ . καὶ ἐπεὶ ἐστὶν ὥς ἡ A πρὸς τὴν B , [οὕτως] ἡ Γ πρὸς τὴν Δ , αἱ δὲ A , B δυνάμει μόνον [εἰσὶ] σύμμετροι, καὶ αἱ Γ , Δ ἄρα δυνάμει μόνον εἰσι σύμμετροι. καὶ ἐστὶ μέση ἡ Γ μέση ἄρα καὶ ἡ Δ . αἱ Γ , Δ ἄρα μέσαι εἰσι δυνάμει μόνον σύμμετροι. λέγω, ὅτι καὶ ῥητὸν περιέθουσιν. ἐπεὶ γάρ ἐστὶν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ , ἐναλλάξ ἄρα ἐστὶν ὥς ἡ A πρὸς τὴν Γ , ἡ B πρὸς τὴν Δ . ἀλλ' ὥς ἡ A πρὸς τὴν Γ , ἡ Γ πρὸς τὴν B · καὶ ὥς ἄρα ἡ Γ πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Δ · τὸ ἄρα ὑπὸ τῶν Γ , Δ ἴσον ἐστὶ τῷ ἀπὸ τῆς B . ῥητὸν δὲ τὸ ἀπὸ τῆς B · ῥητὸν ἄρα [ἐστὶ] καὶ τὸ ὑπὸ τῶν Γ , Δ .

Εὐρηγεται ἄρα μέσαι δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι· ὅπερ ἔδει δεῖχαι.

Let the two rational (straight-lines) A and B , (which are) commensurable in square only, be laid down. And let C —the mean proportional (straight-line) to A and B —be taken [Prop. 6.13]. And let it be contrived that as A (is) to B , so C (is) to D [Prop. 6.12].

And since the rational (straight-lines) A and B are commensurable in square only, the (rectangle contained) by A and B —that is to say, the (square) on C [Prop. 6.17]—is thus medial [Prop. 10.21]. Thus, C is medial [Prop. 10.21]. And since as A is to B , [so] C (is) to D , and A and B [are] commensurable in square only, C and D are thus also commensurable in square only [Prop. 10.11]. And C is medial. Thus, D is also medial [Prop. 10.23]. Thus, C and D are medial (straight-lines which are) commensurable in square only. I say that they also contain a rational (area). For since as A is to B , so C (is) to D , thus, alternately, as A is to C , so B (is) to D [Prop. 5.16]. But, as A (is) to C , (so) C (is) to B . And thus as C (is) to B , so B (is) to D [Prop. 5.11]. Thus, the (rectangle contained) by C and D is equal to the (square) on B [Prop. 6.17]. And the (square) on B (is) rational. Thus, the (rectangle contained) by C and D [is] also rational.

Thus, (two) medial (straight-lines, C and D), containing a rational (area), (which are) commensurable in square only, be found.[†] (Which is) the very thing it was required to show.

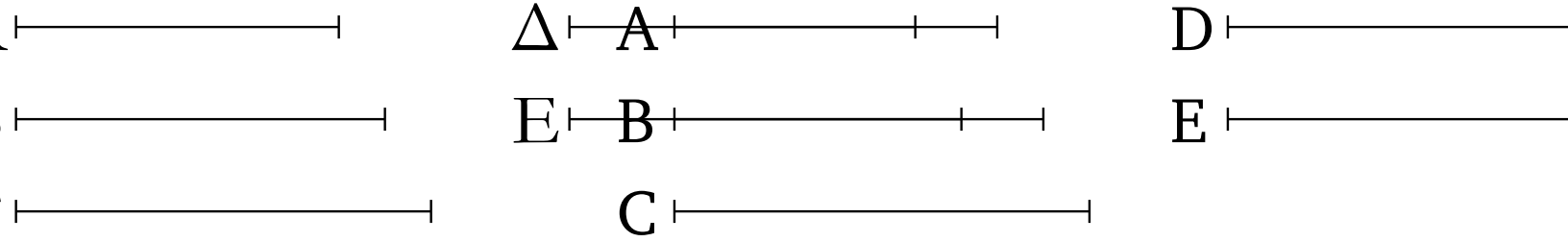
[†] C and D have lengths $k^{1/4}$ and $k^{3/4}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A .

28.

Proposition 28

Μέσαξ εὐρεῖν δυνάμει μόνον συμμέτρον μέσον περιεθούσαξ.

To find (two) medial (straight-lines), containing a medial (area), (which are) commensurable in square only.



Ἐκκεισθώσαν [τρεῖς] ῥηταὶ δυνάμει μόνον σύμμετροι αἱ A , B , Γ , καὶ εἰλήφθω τῶν A , B μέση ἀνάλογον ἡ Δ , καὶ γερονέτω ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E .

Ἐπεὶ αἱ A , B ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, τὸ ἄρα ὑπὸ τῶν A , B , τουτέστι τὸ ἀπὸ τῆς Δ , μέσον ἐστίν. μέση ἄρα ἡ Δ . καὶ ἐπεὶ αἱ B , Γ δυνάμει μόνον εἰσι σύμμετροι, καὶ ἐστὶν ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E , καὶ αἱ Δ , E ἄρα δυνάμει μόνον εἰσι σύμμετροι. μέση δὲ ἡ Δ · μέση ἄρα καὶ ἡ E . αἱ Δ , E ἄρα μέσαι εἰσι δυνάμει μόνον σύμμετροι. λέγω δὴ, ὅτι καὶ μέσον περιέθουσιν. ἐπεὶ γάρ ἐστιν ὥς ἡ B πρὸς τὴν Γ , ἡ Δ πρὸς τὴν E , ἐναλλάξ ἄρα ὥς ἡ B πρὸς τὴν Δ , ἡ Γ πρὸς τὴν E . ὥς δὲ ἡ B πρὸς τὴν Δ , ἡ Δ πρὸς τὴν A · καὶ ὥς ἄρα ἡ Δ πρὸς τὴν A , ἡ Γ πρὸς τὴν E . τὸ ἄρα ὑπὸ τῶν A , Γ ἴσον ἐστὶ τῷ ὑπὸ τῶν Δ , E . μέσον δὲ τὸ ὑπὸ τῶν A , Γ · μέσον ἄρα καὶ τὸ ὑπὸ τῶν Δ , E .

Ἐύρηγνται ἄρα μέσαι δυνάμει μόνον σύμμετροι μέσον περιέθουσαι· ὅπερ ἔδει δεῖξαι.

Let the [three] rational (straight-lines) A , B , and C , (which are) commensurable in square only, be laid down. And let, D , the mean proportional (straight-line) to A and B , be taken [Prop. 6.13]. And let it be contrived that as B (is) to C , (so) D (is) to E [Prop. 6.12].

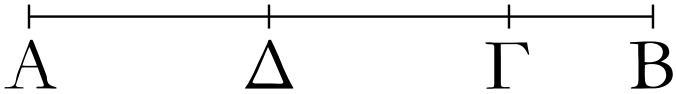
Since the rational (straight-lines) A and B are commensurable in square only, the (rectangle contained) by A and B —that is to say, the (square) on D [Prop. 6.17]—is medial [Prop. 10.21]. Thus, D (is) medial [Prop. 10.21]. And since B and C are commensurable in square only, and as B is to C , (so) D (is) to E , D and E are thus commensurable in square only [Prop. 10.11]. And D (is) medial. E (is) thus also medial [Prop. 10.23]. Thus, D and E are medial (straight-lines which are) commensurable in square only. So, I say that they also enclose a medial (area). For since as B is to C , (so) D (is) to E , thus, alternately, as B (is) to D , (so) C (is) to E [Prop. 5.16]. And as B (is) to D , (so) D (is) to A . And thus as D (is) to A , (so) C (is) to E . Thus, the (rectangle contained) by A and C is equal to the (rectangle contained) by D and E [Prop. 6.16]. And the (rectangle contained) by A and C is medial [Prop. 10.21]. Thus, the (rectangle contained) by D and E (is) also medial.

Thus, (two) medial (straight-lines, D and E), containing a medial (area), (which are) commensurable in square only, be found. (Which is) the very thing it was required to show.

[†] D and E have lengths $k^{1/4}$ and $k'^{1/2}/k^{1/4}$ times that of A , respectively, where the lengths of B and C are $k^{1/2}$ and $k'^{1/2}$ times that of A , respectively.

Λήμμα 1.

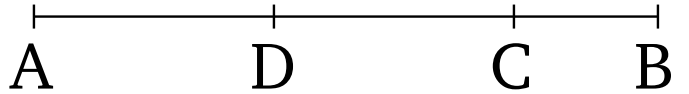
Εύρειν δύο τετραγώνους ἀριθμούς, ὥστε καὶ τὸν συγκεκριμένον ἔχει αὐτῶν εἶναι τετράγωνον.



Ἐκκεισθώσαν δύο ἀριθμοὶ οἱ AB , $B\Gamma$, ἔστωσαν δὲ ἦτοι ἄρτιοι ἢ περιττοί. καὶ ἐπεὶ, ἔαν τε ἀπὸ ἀρτίου ἄρτιος ἀφαιρεθῇ, ἔαν τε ἀπὸ περισσοῦ περισσόξ, ὁ λοιπὸς ἄρτιός ἐστιν, ὁ λοιπὸς ἄρα ὁ $A\Gamma$ ἄρτιός ἐστιν. τεμήσθω ὁ $A\Gamma$ διὰ κατὰ τὸ Δ . ἔστωσαν δὲ καὶ οἱ AB , $B\Gamma$ ἦτοι ὅμοιοι ἐπίπεδοι ἢ τετράγωνοι, οἱ καὶ αὐτοὶ ὅμοιοι εἰσιν ἐπίπεδοι· ὁ ἄρα ἐκ τῶν AB , $B\Gamma$ μετὰ τοῦ ἀπὸ [τοῦ] $\Gamma\Delta$ τετραγώνου ἴσοξ ἐστὶ τῷ ἀπὸ τοῦ $B\Delta$ τετραγώνῳ. καὶ ἐστὶ τετράγωνος ὁ ἐκ τῶν AB , $B\Gamma$, ἐπειδήπερ ἐδείχθη, ὅτι, ἔαν δύο ὅμοιοι ἐπίπεδοι πολλαπλασι-

Lemma I

To find two square numbers such that the sum of them is also square.



Let the two numbers AB and BC be laid down. And let them be either (both) even or (both) odd. And since, if an even (number) is subtracted from an even (number), or if an odd (number is subtracted) from an odd (number), then the remainder is even [Props. 9.24, 9.26], the remainder AC is thus even. Let AC be cut in half at D . And let AB and BC also be either similar plane (numbers), or square (numbers)—which are themselves also similar plane (numbers). Thus, the (number created) from (multiplying) AB and BC , plus the square on

άσαντες ἀλλήλους ποιῶσι τινα, ὁ γεγόμενος τετράγωνός ἐστιν. ἐύρηται ἄρα δύο τετράγωνοι ἀριθμοὶ ὅ τε ἐκ τῶν AB , BF καὶ ὁ ἀπὸ τοῦ $\Gamma\Delta$, οἱ συντεθέντες ποιοῦσι τὸν ἀπὸ τοῦ $B\Delta$ τετράγωνον.

Καὶ φανερόν, ὅτι ἐύρηται πάλιν δύο τετράγωνοι ὅ τε ἀπὸ τοῦ $B\Delta$ καὶ ὁ ἀπὸ τοῦ $\Gamma\Delta$, ὥστε τὴν ὑπεροθὴν αὐτῶν τὸν ὑπὸ AB , BF εἶναι τετράγωνον, ὅταν οἱ AB , BF ὅμοιοι ᾖσιν ἐπίπεδοι. ὅταν δὲ μὴ ᾖσιν ὅμοιοι ἐπίπεδοι, ἐύρηται δύο τετράγωνοι ὅ τε ἀπὸ τοῦ $B\Delta$ καὶ ὁ ἀπὸ τοῦ $\Delta\Gamma$, ὧν ἡ ὑπεροθὴ ὁ ὑπὸ τῶν AB , BF οὐκ ἔστι τετράγωνος· ὅπερ ἔδει δεῖξαι.

CD , is equal to the square on BD [Prop. 2.6]. And the (number created) from (multiplying) AB and BC is square—inasmuch as it was shown that if two similar plane (numbers) make some (number) by multiplying one another then the (number so) created is square [Prop. 9.1]. Thus, two square numbers be found—(namely,) the (number created) from (multiplying) AB and BC , and the (square) on CD —which, (when) added (together), make the square on BD .

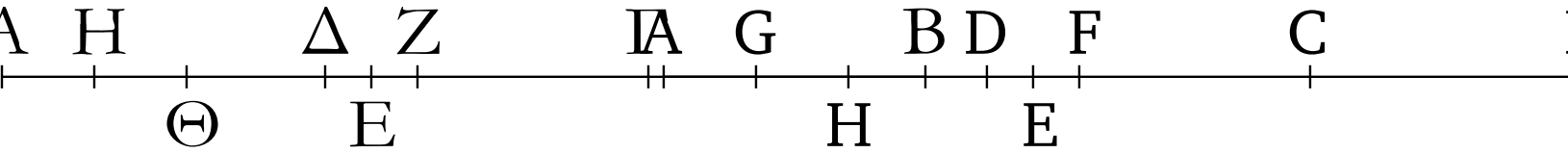
And (it is) clear that two square (numbers) have again been found—(namely,) the (square) on BD , and the (square) on CD —such that their difference—(namely,) the (rectangle) contained by AB and BC —is square whenever AB and BC are similar plane (numbers). But, when they are not similar plane numbers, two square (numbers) be found—(namely,) the (square) on BD , and the (square) on DC —between which the difference—(namely,) the (rectangle) contained by AB and BC —is not square. (Which is) the very thing it was required to show.

Λήμμα 2.

Εὐρεῖν δύο τετραγώνους ἀριθμούς, ὥστε τὸν ἐκ αὐτῶν συγκείμενον μὴ εἶναι τετράγωνον.

Lemma II

To find two square numbers such that the sum of them is not square.



Ἐστω γάρ ὁ ἐκ τῶν AB , BF , ὡς ἔφαμεν, τετράγωνος, καὶ ἄρτιος ὁ ΓA , καὶ τεμήσθω ὁ ΓA διῖθα τῷ Δ . φανερόν δὲ, ὅτι ὁ ἐκ τῶν AB , BF τετράγωνος μετὰ τοῦ ἀπὸ [τοῦ] $\Gamma\Delta$ τετραγώνου ἴσος ἐστὶ τῷ ἀπὸ [τοῦ] $B\Delta$ τετραγώνῳ. ἀφηρήσθω μονὰξ ἡ ΔE . ὁ ἄρα ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ [τοῦ] ΓE ἐλάσσων ἐστὶ τοῦ ἀπὸ [τοῦ] $B\Delta$ τετραγώνου. λέγω οὖν, ὅτι ὁ ἐκ τῶν AB , BF τετράγωνος μετὰ τοῦ ἀπὸ [τοῦ] ΓE οὐκ ἔσται τετράγωνος.

Εἰ γὰρ ἔσται τετράγωνος, ἦτοι ἴσος ἐστὶ τῷ ἀπὸ [τοῦ] BE ἢ ἐλάσσων τοῦ ἀπὸ [τοῦ] BE , οὐκέτι δὲ καὶ μείζων, ἵνα μὴ τηθηῇ ἡ μονὰξ. ἔστω, εἰ δυνατόν, πρότερον ὁ ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ ΓE ἴσος τῷ ἀπὸ BE , καὶ ἔστω τῆς ΔE μονάδος διπλασίον ὁ HA . ἐπεὶ οὖν ὁλόξ ὁ AG ὅλον τοῦ $\Gamma\Delta$ ἐστὶ διπλασίον, ὧν ὁ AH τοῦ ΔE ἐστὶ διπλασίον, καὶ λοιπὸς ἄρα ὁ $H\Gamma$ λοιποῦ τοῦ $E\Gamma$ ἐστὶ διπλασίον διῖθα ἄρα τέτμηται ὁ $H\Gamma$ τῷ E . ὁ ἄρα ἐκ τῶν HB , BF μετὰ τοῦ ἀπὸ ΓE ἴσος ἐστὶ τῷ ἀπὸ BE τετραγώνῳ. ἀλλὰ καὶ ὁ ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ ΓE ἴσος ὑπόκειται τῷ ἀπὸ [τοῦ] BE τετραγώνῳ· ὁ ἄρα ἐκ τῶν HB , BF μετὰ τοῦ ἀπὸ ΓE ἴσος ἐστὶ τῷ ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ ΓE . καὶ κοινοῦ ἀφαιρεθέντος τοῦ ἀπὸ ΓE συνάγεται ὁ AB ἴσος τῷ HB . ὅπερ ἄτοπον. οὐκ ἄρα ὁ ἐκ τῶν AB , BF μετὰ τοῦ ἀπὸ [τοῦ] ΓE ἴσος ἐστὶ τῷ ἀπὸ BE . λέγω δὲ, ὅτι οὐδὲ ἐλάσσων τοῦ ἀπὸ BE . εἰ γὰρ δυνατόν,

For let the (number created) from (multiplying) AB and BC , as we said, be square. And (let) CA (be) even. And let CA be cut in half at D . So it is clear that the square (number created) from (multiplying) AB and BC , plus the square on CD , is equal to the square on BD [see previous lemma]. Let the unit DE be subtracted (from BD). Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is less than the square on BD . I say, therefore, that the square (number created) from (multiplying) AB and BC , plus the (square) on CE , is not square.

For if it is square, it is either equal to the (square) on BE , or less than the (square) on BE , but cannot any more be greater (than the square on BE), lest the unit be divided. First of all, if possible, let the (number created) from (multiplying) AB and BC , plus the (square) on CE , be equal to the (square) on BE . And let GA be double the unit DE . Therefore, since the whole of AC is double the whole of CD , of which AG is double DE , the remainder GC is thus double the remainder EC . Thus, GC has been cut in half at E . Thus, the (number created) from (multiplying) GB and BC , plus the (square) on CE , is equal to the square on BE [Prop. 2.6]. But, the (number created) from (multiplying) AB and BC , plus the (square) on CE , was also assumed (to be) equal to the square on BE . Thus, the (number

ἔστω τῷ ἀπὸ BZ ἴσος, καὶ τοῦ ΔZ διπλασίων ὁ ΘA . καὶ συναθροίσεται πάλιν διπλασίων ὁ $\Theta \Gamma$ τοῦ ΓZ . ὥστε καὶ τὸν $\Gamma \Theta$ διδα τεμῆσθαι κατὰ τὸ Z , καὶ διὰ τοῦτο τὸν ἐκ τῶν ΘB , $B \Gamma$ μετὰ τοῦ ἀπὸ $Z \Gamma$ ἴσον γίνεσθαι τῷ ἀπὸ BZ . ὑπόκειται δὲ καὶ ὁ ἐκ τῶν AB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓE ἴσος τῷ ἀπὸ BZ . ὥστε καὶ ὁ ἐκ τῶν ΘB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓZ ἴσος ἔσται τῷ ἐκ τῶν AB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓE . ὅπερ ἄτοπον. οὐκ ἄρα ὁ ἐκ τῶν AB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓE ἴσος ἐστὶ [τῷ] ἐλάσσονι τοῦ ἀπὸ BE . ἐδείκθη δέ, ὅτι οὐδὲ [αὐτῷ] τῷ ἀπὸ BE . οὐκ ἄρα ὁ ἐκ τῶν AB , $B \Gamma$ μετὰ τοῦ ἀπὸ ΓE τετραγώνος ἐστιν. ὅπερ ἔδει δεῖχαι.

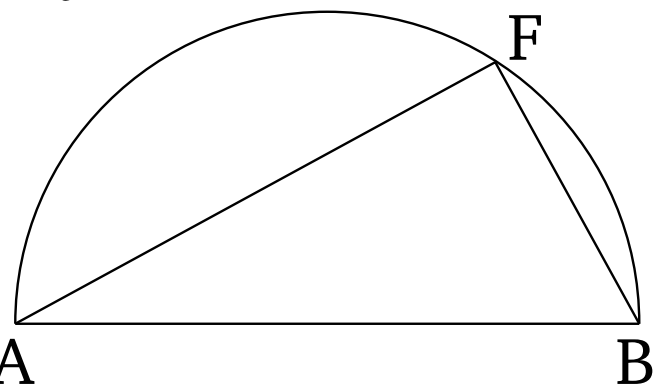
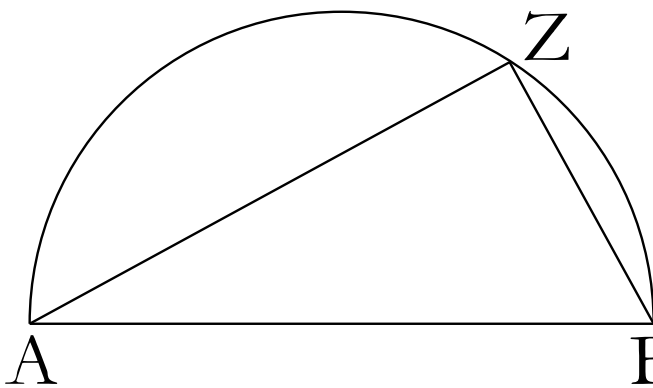
created) from (multiplying) GB and BC , plus the (square) on CE , is equal to the (number created) from (multiplying) AB and BC , plus the (square) on CE . And subtracting the (square) on CE from both, AB is inferred (to be) equal to GB . The very thing is absurd. Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is not equal to the (square) on BE . So I say that (it is) not less than the (square) on BE either. For, if possible, let it be equal to the (square) on BF . And (let) HA (be) double DF . And it can again be inferred that HC (is) double CF . Hence, CH has also been cut in half at F . And, on account of this, the (number created) from (multiplying) HB and BC , plus the (square) on FC , becomes equal to the (square) on BF [Prop. 2.6]. And the (number created) from (multiplying) AB and BC , plus the (square) on CE , was also assumed (to be) equal to the (square) on BF . Hence, the (number created) from (multiplying) HB and BC , plus the (square) on CF , will also be equal to the (number created) from (multiplying) AB and BC , plus the (square) on CE . The very thing is absurd. Thus, the (number created) from (multiplying) AB and BC , plus the (square) on CE , is not equal to less than the (square) on BE . And it was shown that (is it) not equal to the (square) on BE either. Thus, the (number created) from (multiplying) AB and BC , plus the square on CE , is not square. (Which is) the very thing it was required to show.

29.

Proposition 29

Εὕρεῖν δύο ῥητάς δυνάμει μόνον συμμετρωνς, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετρων εαυτῇ μήκει.

To find two rational (straight-lines which are) commensurable in square only, such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line which is) commensurable in length with the greater.



Ἐκκεῖσθω γάρ τις ῥητὴ ἡ AB καὶ δύο τετραγώνοι ἀριθμοὶ οἱ $\Gamma \Delta$, ΔE , ὥστε τὴν ὑπεροθὴν αὐτῶν τὸν ΓE μὴ εἶναι

For let some rational (straight-line) AB be laid down, and two square numbers, CD and DE , such that the difference be-

τετράγωνον, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AZB , καὶ πεποιήσθω ὡς ὁ $\Delta\Gamma$ πρὸς τὸν $ΓΕ$, οὕτως τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς AZ τετράγωνον, καὶ ἐπεξέσθω ἡ ZB .

Ἐπεὶ [οὕτως] ἔστιν ὡς τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , οὕτως ὁ $\Delta\Gamma$ πρὸς τὸν $ΓΕ$, τὸ ἀπὸ τῆς BA ἄρα πρὸς τὸ ἀπὸ τῆς AZ λόγον ἔθει, ὃν ἀριθμὸς ὁ $\Delta\Gamma$ πρὸς ἀριθμὸν τὸν $ΓΕ$ · σύμμετρον ἄρα ἔστι τὸ ἀπὸ τῆς BA τῷ ἀπὸ τῆς AZ . ῥητὸν δὲ τὸ ἀπὸ τῆς AB · ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς AZ · ῥητὴ ἄρα καὶ ἡ AZ . καὶ ἐπεὶ ὁ $\Delta\Gamma$ πρὸς τὸν $ΓΕ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς BA ἄρα πρὸς τὸ ἀπὸ τῆς AZ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἔστιν ἡ AB τῇ AZ μήκει· αἱ BA , AZ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ [ἔστιν] ὡς ὁ $\Delta\Gamma$ πρὸς τὸν $ΓΕ$, οὕτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , ἀναστρέψαντι ἄρα ὡς ὁ $\Gamma\Delta$ πρὸς τὸν $\DeltaΕ$, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ . ὁ δὲ $\Gamma\Delta$ πρὸς τὸν $\DeltaΕ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· καὶ τὸ ἀπὸ τῆς AB ἄρα πρὸς τὸ ἀπὸ τῆς BZ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· σύμμετρος ἄρα ἔστιν ἡ AB τῇ BZ μήκει. καὶ ἔστι τὸ ἀπὸ τῆς AB ἴσον τοῖς ἀπὸ τῶν AZ , ZB · ἡ AB ἄρα τῆς AZ μείζον δύναται τῇ BZ συμμέτρῳ ἑαυτῇ.

Ἐύρεται ἄρα δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ BA , AZ , ὥστε τὴν μείζονα τὴν AB τῆς ἐλάσσονος τῆς AZ μείζον δύνασθαι τῷ ἀπὸ τῆς BZ συμμέτρῳ ἑαυτῇ μήκει· ὅπερ ἔδει δεῖχαι.

tween them, CE , is not square [Prop. 10.28 lem. I]. And let the semi-circle AFB be drawn on AB . And let it be contrived that as DC (is) to CE , so the square on BA (is) to the square on AF [Prop. 10.6 corr.]. And let FB be joined.

[Therefore,] since as the (square) on BA is to the (square) on AF , so DC (is) to CE , the (square) on BA thus has to the (square) on AF the ratio which the number DC (has) to the number CE . Thus, the (square) on BA is commensurable with the (square) on AF [Prop. 10.6]. And the (square) on AB (is) rational [Def. 10.4]. Thus, the (square) on AF (is) also rational. Thus, AF (is) also rational. And since DC does not have to CE the ratio which (some) square number (has) to (some) square number, the (square) on BA thus does not have to the (square) on AF the ratio which (some) square number has to (some) square number either. Thus, AB is incommensurable in length with AF [Prop. 10.9]. Thus, the rational (straight-lines) BA and AF are commensurable in square only. And since as DC [is] to CE , so the (square) on BA (is) to the (square) on AF , thus, via conversion, as CD (is) to DE , so the (square) on AB (is) to the (square) on BF [Props. 5.19 corr., 3.31, 1.47]. And CD has to DE the ratio which (some) square number (has) to (some) square number. Thus, the (square) on AB also has to the (square) on BF the ratio which (some) square number has to (some) square number. AB is thus commensurable in length with BF [Prop. 10.9]. And the (square) on AB is equal to the (sum of the squares) on AF and FB [Prop. 1.47]. Thus, the square on AB is greater than (the square on) AF by (the square on) BF , (which is) commensurable (in length) with (AB).

Thus, two rational (straight-lines), BA and AF , commensurable in square only, be found such that the square on the greater, AB , is larger than (the square on) the lesser, AF , by the (square) on BF , (which is) commensurable in length with (AB).[†] (Which is) the very thing it was required to show.

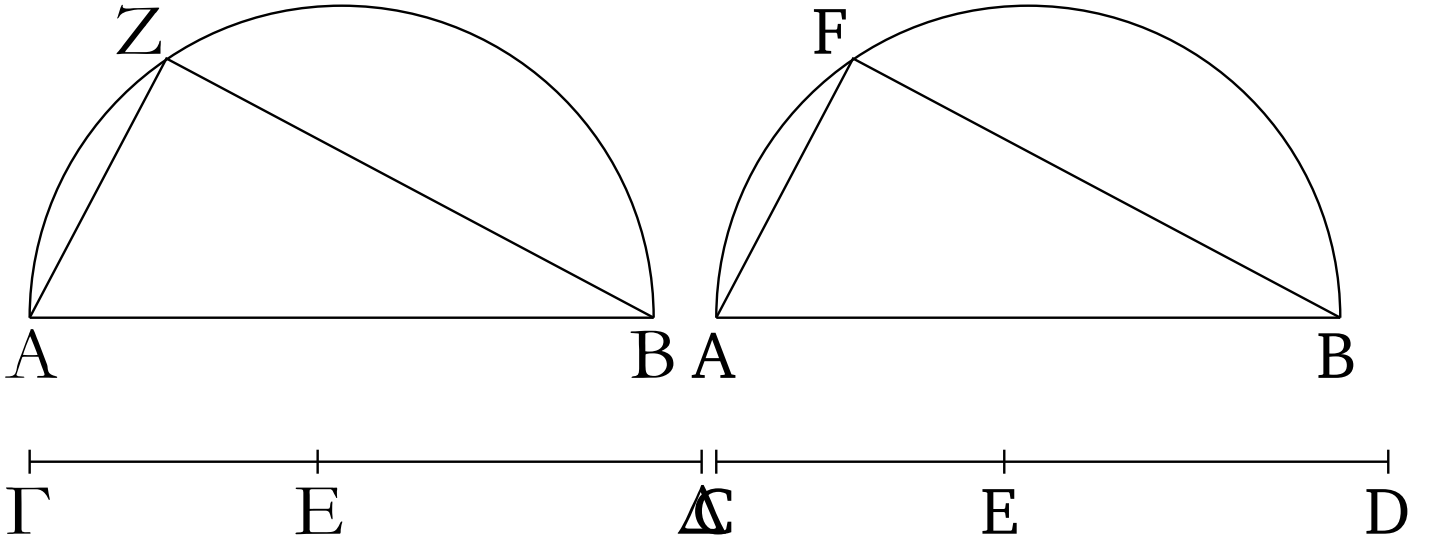
[†] BA and AF have lengths 1 and $\sqrt{1-k^2}$ times that of AB , respectively, where $k = \sqrt{DE/CD}$.

30.

Εύρεῖν δύο ῥητάς δυνάμει μόνον συμμέτρους, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ ἀσυνμέτρῳ ἑαυτῇ μήκει.

Proposition 30

To find two rational (straight-lines which are) commensurable in square only, such that the square on the greater is larger than the (the square on) lesser by the (square) on (some straight-line which is) incommensurable in length with the greater.



Ἐκκείσθω ῥητὴ ἡ AB καὶ δύο τετράγωνοι ἀριθμοὶ οἱ $ΓΕ$, $ΕΔ$, ὥστε τὸν συγκείμενον ἔχ' αὐτῶν τὸν $ΓΔ$ μὴ εἶναι τετράγωνον, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AZB , καὶ πεποιήσθω ὥς ὁ $ΔΓ$ πρὸς τὸν $ΓΕ$, οὔτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , καὶ ἐπεζεύθθω ἡ ZB .

Ὅμοίως δὲ δείχομεν τῷ πρὸ τούτου, ὅτι αἱ BA , AZ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ ἐστὶν ὥς ὁ $ΔΓ$ πρὸς τὸν $ΓΕ$, οὔτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ , ἀναστρέφαντι ἄρα ὥς ὁ $ΓΔ$ πρὸς τὸν $ΔΕ$, οὔτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ . ὁ δὲ $ΓΔ$ πρὸς τὸν $ΔΕ$ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν οὐδ' ἄρα τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ἀσύμμετρος ἄρα ἐστὶν ἡ AB τῇ BZ μήκει. καὶ δύναται ἡ AB τῇ AZ μείζον τῷ ἀπὸ τῆς ZB ἀσυνμέτρον ἑαυτῇ.

Αἱ AB , AZ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AB τῇ AZ μείζον δύναται τῷ ἀπὸ τῆς ZB ἀσυνμέτρον ἑαυτῇ μήκει· ὅπερ ἔδει δεῖξαι.

Let the rational (straight-line) AB be laid out, and the two square numbers, CE and ED , such that the sum of them, CD , is not square [Prop. 10.28 lem. II]. And let the semi-circle AFB be drawn on AB . And let it be contrived that as DC (is) to CE , so the (square) on BA (is) to the (square) on AF [Prop. 10.6 corr]. And let FB be joined.

So, similarly to the (proposition) before this, we can show that BA and AF are rational (straight-lines which are) commensurable in square only. And since as DC is to CE , so the (square) on BA (is) to the (square) on AF , thus, via conversion, as CD (is) to DE , so the (square) on AB (is) to the (square) on BF [Props. 5.19 corr., 3.31, 1.47]. And CD does not have to DE the ratio which (some) square number (has) to (some) square number. Thus, the (square) on AB does not have to the (square) on BF the ratio which (some) square number has to (some) square number either. Thus, AB is incommensurable in length with BF [Prop. 10.9]. And the square on AB is greater than the (square) on AF by the (square) on FB [Prop. 1.47], (which is) incommensurable (in length) with (AB) .

Thus, AB and AF are rational (straight-lines which are) commensurable in square only, and the square on AB is greater than (the square on) AF by the (square) on FB , (which is) incommensurable (in length) with (AB) .[†] (Which is) the very thing it was required to show.

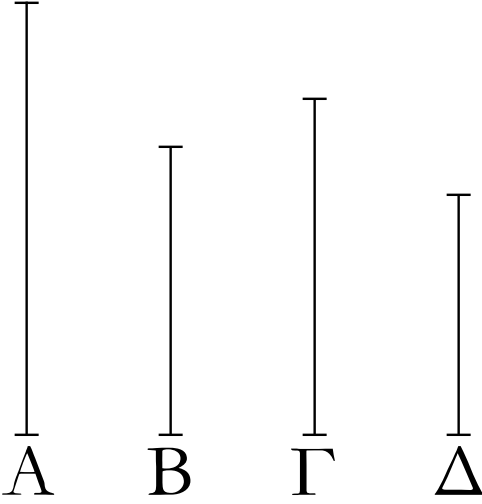
[†] AB and AF have lengths 1 and $1/\sqrt{1+k^2}$ times that of AB , respectively, where $k = \sqrt{DE/CE}$.

31.

Εὑρεῖν δύο μέσας δυνάμει μόνον συμμέτρον ῥητὸν περιεούσας, ὥστε τὴν μείζονα τῇ ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμέτρον ἑαυτῇ μήκει.

Proposition 31

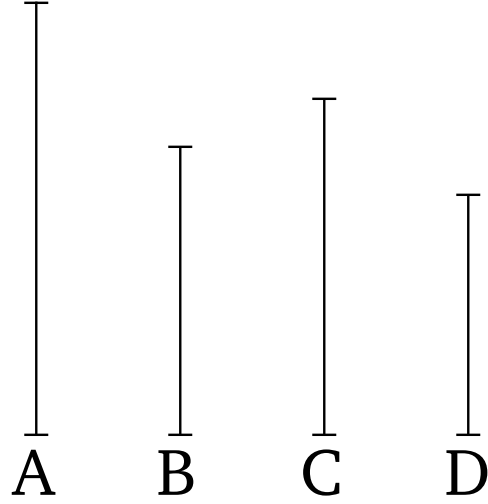
To find two medial (straight-lines), commensurable in square only, (and) containing a rational (area), such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable in length with the greater.



Ἐκκεισθώσαν δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ A , B , ὥστε τὴν A μείζονα οὕσαν τῆς ἐλάσσονος τῆς B μείζον δύνασθαι τῷ ἀπὸ συμμέτρου ἐναντὶ μήκει. καὶ τῷ ὑπὸ τῶν A , B ἴσον ἔστω τὸ ἀπὸ τῆς Γ . μέσον δὲ τὸ ὑπὸ τῶν A , B μέσον ἄρα καὶ τὸ ἀπὸ τῆς Γ · μέση ἄρα καὶ ἡ Γ . τῷ δὲ ἀπὸ τῆς B ἴσον ἔστω τὸ ὑπὸ τῶν Γ , Δ · ῥητὸν δὲ τὸ ἀπὸ τῆς B · ῥητὸν ἄρα καὶ τὸ ὑπὸ τῶν Γ , Δ . καὶ ἐπεὶ ἔστιν ὥς ἡ A πρὸς τὴν B , οὕτως τὸ ὑπὸ τῶν A , B πρὸς τὸ ἀπὸ τῆς B , ἀλλὰ τῷ μὲν ὑπὸ τῶν A , B ἴσον ἐστὶ τὸ ἀπὸ τῆς Γ , τῷ δὲ ἀπὸ τῆς B ἴσον τὸ ὑπὸ τῶν Γ , Δ , ὥς ἄρα ἡ A πρὸς τὴν B , οὕτως τὸ ἀπὸ τῆς Γ πρὸς τὸ ὑπὸ τῶν Γ , Δ . ὥς δὲ τὸ ἀπὸ τῆς Γ πρὸς τὸ ὑπὸ τῶν Γ , Δ , οὕτως ἡ Γ πρὸς τὴν Δ · καὶ ὥς ἄρα ἡ A πρὸς τὴν B , οὕτως ἡ Γ πρὸς τὴν Δ . σύμμετρος δὲ ἡ A τῇ B δυνάμει μόνον· σύμμετρος ἄρα καὶ ἡ Γ τῇ Δ δυνάμει μόνον. καὶ ἐστὶ μέση ἡ Γ · μέση ἄρα καὶ ἡ Δ . καὶ ἐπεὶ ἔστιν ὥς ἡ A πρὸς τὴν B , ἡ Γ πρὸς τὴν Δ , ἡ δὲ A τῆς B μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντὶ, καὶ ἡ Γ ἄρα τῆς Δ μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντὶ.

Εὐρηνται ἄρα δύο μέσαι δυνάμει μόνον σύμμετροι αἱ Γ , Δ ῥητὸν περιέθουσαι, καὶ ἡ Γ τῆς Δ μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντὶ μήκει.

Ὅμοίως δὴ δειθθήσεται καὶ τῷ ἀπὸ ἀσυνμέτρου, ὅταν ἡ A τῆς B μείζον δύνηται τῷ ἀπὸ ἀσυνμέτρου ἐναντὶ.



Let two rational (straight-lines), A and B , commensurable in square only, be laid out, such that the square on the greater A is larger than the (square on the) lesser B by the (square) on (some straight-line) commensurable in length with (A) [Prop. 10.29]. And let the (square) on C be equal to the (rectangle contained) by A and B . And the (rectangle contained) by A and B (is) medial [Prop. 10.21]. Thus, the (square) on C (is) also medial. Thus, C (is) also medial [Prop. 10.21]. And let the (rectangle contained) by C and D be equal to the (square) on B . And the (square) on B (is) rational. Thus, the (rectangle contained) by C and D (is) also rational. And since as A is to B , so the (rectangle contained) by A and B (is) to the (square) on B [Prop. 10.21 lem.], but the (square) on C is equal to the (rectangle contained) by A and B , and the (rectangle contained) by C and D to the (square) on B , thus as A (is) to B , so the (square) on C (is) to the (rectangle contained) by C and D . And as the (square) on C (is) to the (rectangle contained) by C and D , so C (is) to D [Prop. 10.21 lem.]. And thus as A (is) to B , so C (is) to D . And A is commensurable in square only with B . Thus, C (is) also commensurable in square only with D [Prop. 10.11]. And C is medial. Thus, D (is) also medial [Prop. 10.23]. And since as A is to B , (so) C (is) to D , and the square on A is greater than (the square on) B by the (square) on (some straight-line) commensurable (in length) with (A), the square on C is thus also greater than (the square on) D by the (square) on (some straight-line) commensurable (in length) with (C) [Prop. 10.14].

Thus, two medial (straight-lines), C and D , commensurable in square only, (and) containing a rational (area), be found. And the square on C is greater than (the square on) D by the (square) on (some straight-line) commensurable in length with (C).[†]

So, similarly, (the proposition) can also be demonstrated for (some straight-line) incommensurable (in length with C), provided that the square on A is greater than (the square on B)

by the (square) on (some straight-line) incommensurable (in length) with (A) [Prop. 10.30].[‡]

[†] C and D have lengths $(1 - k^2)^{1/4}$ and $(1 - k^2)^{3/4}$ times that of A, respectively, where k is defined in the footnote to Prop. 10.29.

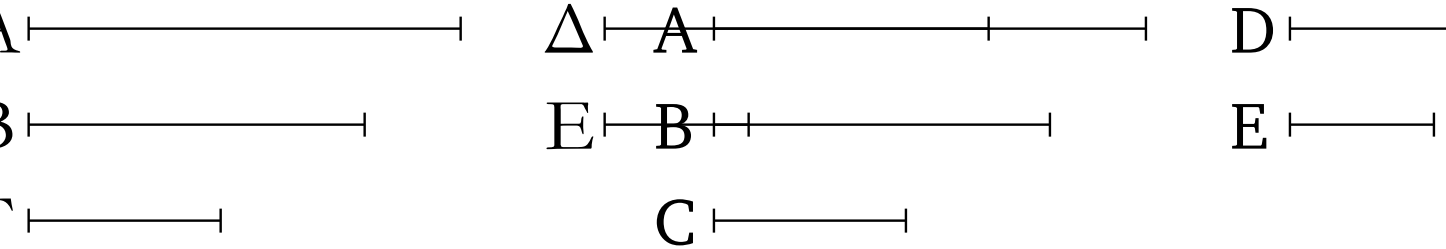
[‡] C and D would have lengths $1/(1 + k^2)^{1/4}$ and $1/(1 + k^2)^{3/4}$ times that of A, respectively, where k is defined in the footnote to Prop. 10.30.

32.

Proposition 32

Εὕρεῖν δύο μέσας δυνάμει μόνον συμμετρους μέσον περιεθούσας, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετρων ἑαντῇ.

To find two medial (straight-lines), commensurable in square only, (and) containing a medial (area), such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable (in length) with the greater.



Ἐκκείσθωσαν τρεῖς ῥηταὶ δυνάμει μόνον σύμμετροι αἱ A, B, Γ, ὥστε τὴν A τῆς Γ μείζον δύνασθαι τῷ ἀπὸ συμμετρων ἑαντῇ, καὶ τῷ μὲν ὑπὸ τῶν A, B ἴσον ἔστω τὸ ἀπὸ τῆς Δ. μέσον ἄρα τὸ ἀπὸ τῆς Δ· καὶ ἡ Δ ἄρα μέση ἐστίν. τῷ δὲ ὑπὸ τῶν B, Γ ἴσον ἔστω τὸ ὑπὸ τῶν Δ, E. καὶ ἐπεὶ ἐστὶν ὡς τὸ ὑπὸ τῶν A, B πρὸς τὸ ὑπὸ τῶν B, Γ, οὕτως ἡ A πρὸς τὴν Γ, ἀλλὰ τῷ μὲν ὑπὸ τῶν A, B ἴσον ἐστὶ τὸ ἀπὸ τῆς Δ, τῷ δὲ ὑπὸ τῶν B, Γ ἴσον τὸ ὑπὸ τῶν Δ, E, ἔστιν ἄρα ὡς ἡ A πρὸς τὴν Γ, οὕτως τὸ ἀπὸ τῆς Δ πρὸς τὸ ὑπὸ τῶν Δ, E. ὡς δὲ τὸ ἀπὸ τῆς Δ πρὸς τὸ ὑπὸ τῶν Δ, E, οὕτως ἡ Δ πρὸς τὴν E· καὶ ὡς ἄρα ἡ A πρὸς τὴν Γ, οὕτως ἡ Δ πρὸς τὴν E. σύμμετρος δὲ ἡ A τῇ Γ δυνάμει [μόνον]. σύμμετρος ἄρα καὶ ἡ Δ τῇ E δυνάμει μόνον. μέση δὲ ἡ Δ· μέση ἄρα καὶ ἡ E. καὶ ἐπεὶ ἐστὶν ὡς ἡ A πρὸς τὴν Γ, ἡ Δ πρὸς τὴν E, ἡ δὲ A τῆς Γ μείζον δύναται τῷ ἀπὸ συμμετρων ἑαντῇ, καὶ ἡ Δ ἄρα τῆς E μείζον δυνήσεται τῷ ἀπὸ συμμετρων ἑαντῇ. λέγω δὴ, ὅτι καὶ μέσον ἐστὶ τὸ ὑπὸ τῶν Δ, E. ἐπεὶ γὰρ ἴσον ἐστὶ τὸ ὑπὸ τῶν B, Γ τῷ ὑπὸ τῶν Δ, E, μέσον δὲ τὸ ὑπὸ τῶν B, Γ [αἱ γὰρ B, Γ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι], μέσον ἄρα καὶ τὸ ὑπὸ τῶν Δ, E.

Εὕρεται ἄρα δύο μέσαι δυνάμει μόνον σύμμετροι αἱ Δ, E μέσον περιέθουσας, ὥστε τὴν μείζονα τῆς ἐλάσσονος μείζον δύνασθαι τῷ ἀπὸ συμμετρων ἑαντῇ.

Ὁμοίως δὴ πάλιν διεθθήσεται καὶ τῷ ἀπὸ ἀσύμμετρων, ὅταν ἡ A τῆς Γ μείζον δύνῃται τῷ ἀπὸ ἀσύμμετρων ἑαντῇ.

Let three rational (straight-lines), A, B and C, commensurable in square only, be laid out such that the square on A is greater than (the square on C) by the (square) on (some straight-line) commensurable (in length) with (A) [Prop. 10.29]. And let the (square) on D be equal to the (rectangle contained) by A and B. Thus, the (square) on D (is) medial. Thus, D is also medial [Prop. 10.21]. And let the (rectangle contained) by D and E be equal to the (rectangle contained) by B and C. And since as the (rectangle contained) by A and B is to the (rectangle contained) by B and C, so A (is) to C [Prop. 10.21 lem.], but the (square) on D is equal to the (rectangle contained) by A and B, and the (rectangle contained) by D and E to the (rectangle contained) by B and C, thus as A is to C, so the (square) on D (is) to the (rectangle contained) by D and E. And as the (square) on D (is) to the (rectangle contained) by D and E, so D (is) to E [Prop. 10.21 lem.]. And thus as A (is) to C, so D (is) to E. And A (is) commensurable in square [only] with C. Thus, D (is) also commensurable in square only with E [Prop. 10.11]. And D (is) medial. Thus, E (is) also medial [Prop. 10.23]. And since as A is to C, (so) D (is) to E, and the square on A is greater than (the square on) C by the (square) on (some straight-line) commensurable (in length) with (A), the square on D will thus also be greater than (the square on) E by the (square) on (some straight-line) commensurable (in length) with (D) [Prop. 10.14]. So, I also say that the (rectangle contained) by D and E is medial. For since the (rectangle contained) by B and C is equal to the (rectangle contained) by D and E, and the (rectangle contained) by B and C (is) medial [for B and C are rational (straight-lines which are) commensurable in square only] [Prop. 10.21], the

(rectangle contained) by D and E (is) thus also medial.

Thus, two medial (straight-lines), D and E , commensurable in square only, (and) containing a medial (area), be found such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable (in length) with the greater.[†]

So, similarly, (the proposition) can again also be demonstrated for (some straight-line) incommensurable (in length with the greater), provided that the square on A is greater than (the square on) C by the (square) on (some straight-line) incommensurable (in length) with (A) [Prop. 10.30].[‡]

[†] D and E have lengths $k^{1/4}$ and $k^{1/4} \sqrt{1 - k^2}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A , and k is defined in the footnote to Prop. 10.29.

[‡] D and E would have lengths $k^{1/4}$ and $k^{1/4} / \sqrt{1 + k^2}$ times that of A , respectively, where the length of B is $k^{1/2}$ times that of A , and k is defined in the footnote to Prop. 10.30.

Λήμμα.

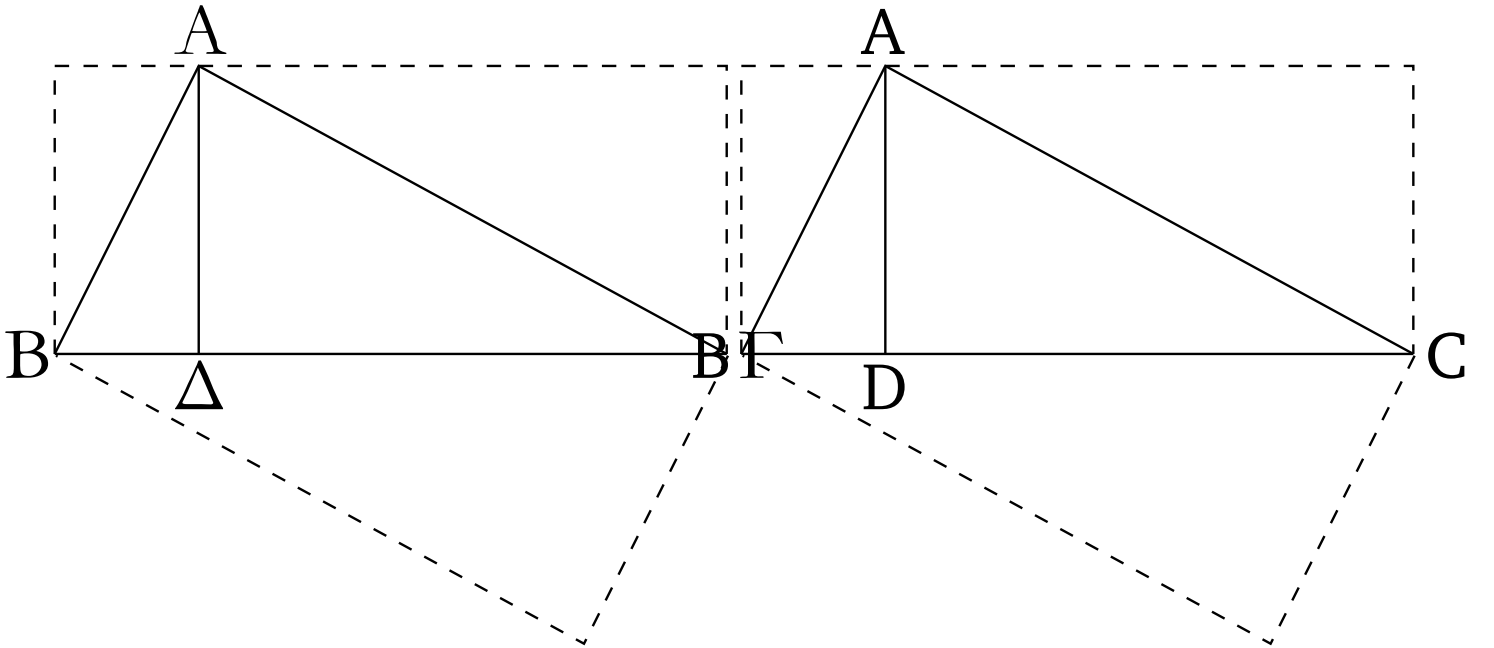
Ἐστω τρίγωνον ὀρθογώνιον τὸ $AB\Gamma$ ὀρθὴν ἔχον τὴν A , καὶ ἤθθω κάθετος ἡ $A\Delta$. λέγω, ὅτι τὸ μὲν ὑπὸ τῶν $\Gamma B\Delta$ ἴσον ἐστὶ τῷ ἀπὸ τῆς BA , τὸ δὲ ὑπὸ τῶν $B\Gamma A$ ἴσον τῷ ἀπὸ τῆς ΓA , καὶ τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ ἴσον τῷ ἀπὸ τῆς $A\Delta$, καὶ ἔτι τὸ ὑπὸ τῶν $B\Gamma$, $A\Delta$ ἴσον [ἐστὶ] τῷ ὑπὸ τῶν BA , $A\Gamma$.

Καὶ πρῶτον, ὅτι τὸ ὑπὸ τῶν $\Gamma B\Delta$ ἴσον [ἐστὶ] τῷ ἀπὸ τῆς BA .

Lemma

Let ABC be a right-angled triangle having the (angle) A a right-angle. And let the perpendicular AD be drawn. I say that the (rectangle contained) by CBD is equal to the (square) on BA , and the (rectangle contained) by BCD (is) equal to the (square) on CA , and the (rectangle contained) by BD and DC (is) equal to the (square) on AD , and, further, the (rectangle contained) by BC and AD [is] equal to the (rectangle contained) by BA and AC .

And, first of all, (let us prove) that the (rectangle contained) by CBD [is] equal to the (square) on BA .



Ἐπεὶ γὰρ ἐν ὀρθογώνιῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ἦκται ἡ $A\Delta$, τὰ $AB\Delta$, $A\Delta\Gamma$ ἄρα

For since AD has been drawn from the right-angle in a right-angled triangle, perpendicular to the base, ABD and

τρίγωνα ὁμοιά ἐστι τῷ τε ὅλῳ τῷ $AB\Gamma$ καὶ ἀλλήλοις. καὶ ἐπεὶ ὁμοιόν ἐστι τὸ $AB\Gamma$ τριγώνων τῷ $AB\Delta$ τριγώνῳ, ἔστιν ἄρα ὡς ἡ ΓB πρὸς τὴν BA , οὕτως ἡ BA πρὸς τὴν $B\Delta$. τὸ ἄρα ὑπὸ τῶν $\Gamma B\Delta$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB .

Διὰ τὰ αὐτὰ δὴ καὶ τὸ ὑπὸ τῶν $B\Gamma\Delta$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AG .

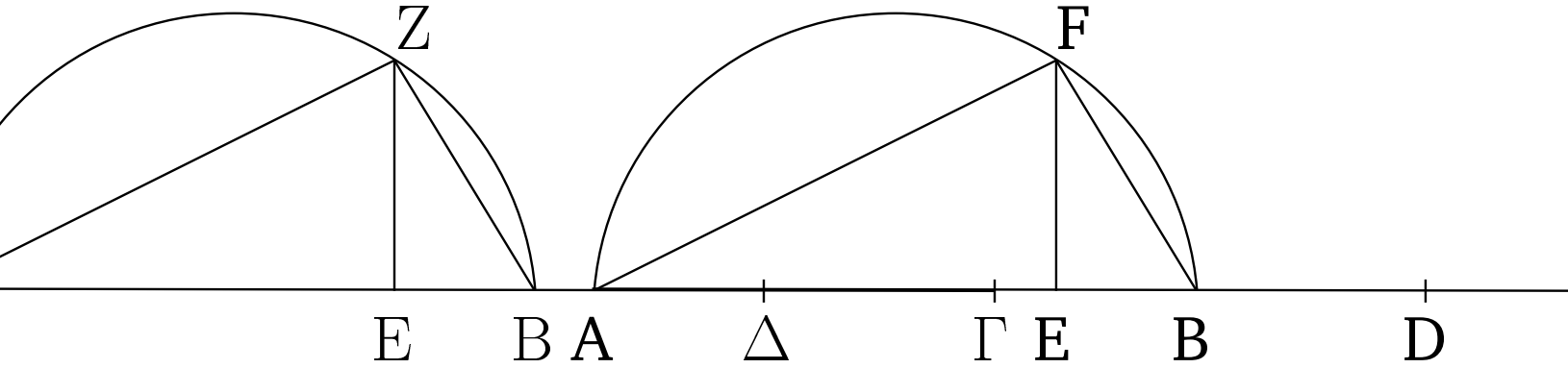
Καὶ ἐπεὶ, ἔαν ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ἀθῇ, ἡ ἀθθεῖσα τῶν τῆς βάσεως τμημάτων μέση ἀνάλογόν ἐστιν, ἔστιν ἄρα ὡς ἡ BA πρὸς τὴν ΔA , οὕτως ἡ ΔA πρὸς τὴν $\Delta\Gamma$. τὸ ἄρα ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔA .

Λέγω, ὅτι καὶ τὸ ὑπὸ τῶν $B\Gamma$, ΔA ἴσον ἐστὶ τῷ ὑπὸ τῶν BA , AG . ἐπεὶ γάρ, ὡς ἔφαμεν, ὁμοιόν ἐστι τὸ $AB\Gamma$ τῷ $AB\Delta$, ἔστιν ἄρα ὡς ἡ $B\Gamma$ πρὸς τὴν ΓA , οὕτως ἡ BA πρὸς τὴν AG . τὸ ἄρα ὑπὸ τῶν $B\Gamma$, ΔA ἴσον ἐστὶ τῷ ὑπὸ τῶν BA , AG . ὁπερ ἔδει δεῖξαι.

33.

Εὐρεῖν δύο εὐθείας δυνάμει ἀσυμμέτρους ποιούσας τὸ μὲν συγκεκριμένον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον.

Ἐκκείσθωσαν δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ AB , $B\Gamma$, ὥστε τὴν μείζονα τὴν AB τῆς ἐλάσσονος τῆς $B\Gamma$ μείζον δύνασθαι τῷ ἀπὸ ἀσυμμέτρου ἑαντῇ, καὶ τεμήσθω ἡ $B\Gamma$ διῖθα κατὰ τὸ Δ , καὶ τῷ ἀφ' ὁποτέρᾳ τῶν $B\Delta$, $\Delta\Gamma$ ἴσον παρὰ τὴν AB παραβεβλήσθω παραλληλόγραμμον ἐλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν AEB , καὶ γεγράφθω ἐπὶ τῆς AB ημικύκλιον τὸ AZB , καὶ ῥηθῇ τῇ AB πρὸς ὀρθὰς ἡ EZ , καὶ ἐπεζεύθωσαν αἱ AZ , ZB .



Καὶ ἐπεὶ [δύο] εὐθεῖαι ἄνισοί εἰσιν αἱ AB , $B\Gamma$, καὶ ἡ AB τῆς $B\Gamma$ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαντῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς $B\Gamma$, τουτέστι τῷ ἀπὸ τῆς ἡμισείας αὐτῆς,

ADC are thus triangles (which are) similar to the whole, ABC , and to one another [Prop. 6.8]. And since triangle ABC is similar to triangle ABD , thus as CB is to BA , so BA (is) to BD [Prop. 6.4]. Thus, the (rectangle contained) by CBD is equal to the (square) on AB [Prop. 6.17].

So, for the same (reasons), the (rectangle contained) by BCD is also equal to the (square) on AC .

And since if a (straight-line) is drawn from the right-angle in a right-angled triangle, perpendicular to the base, the (straight-line so) drawn is the mean proportional to the pieces of the base [Prop. 6.8 corr.], thus as BD is to DA , so AD (is) to DC . Thus, the (rectangle contained) by BD and DC is equal to the (square) on DA [Prop. 6.17].

I also say that the (rectangle contained) by BC and AD is equal to the (rectangle contained) by BA and AC . For since, as we said, ABC is similar to ABD , thus as BC is to CA , so BA (is) to AD [Prop. 6.4]. Thus, the (rectangle contained) by BC and AD is equal to the (rectangle contained) by BA and AC [Prop. 6.16]. (Which is) the very thing it was required to show.

Proposition 33

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial.

Let the two rational (straight-lines) AB and BC , (which are) commensurable in square only, be laid out such that the square on the greater, AB , is larger than (the square on) the lesser, BC , by the (square) on (some straight-line which is) incommensurable (in length) with (AB) [Prop. 10.30]. And let BC be cut in half at D . And let a parallelogram equal to the (square) on either of BD or DC , (and) falling short by a square figure, be applied to AB [Prop. 6.28], and let it be the (rectangle contained) by AEB . And let the semi-circle AFB be drawn on AB . And let EF be drawn at right-angles to AB . And let AF and FB be joined.

And since AB and BC are [two] unequal straight-lines, and the square on AB is greater than (the square on) BC by the (square) on (some straight-line which is) incommensu-

ἴσον παρὰ τὴν AB παραβέβληται παραλληλόγραμμον ἑλλείπον εἶδει τετραγώνῳ καὶ ποιεῖ τὸ ὑπὸ τῶν AEB , ἀσύμμετρος ἄρα ἐστὶν ἡ AE τῇ EB . καὶ ἐστὶν ὥς ἡ AE πρὸς EB , οὕτως τὸ ὑπὸ τῶν BA , AE πρὸς τὸ ὑπὸ τῶν AB , BE , ἴσον δὲ τὸ μὲν ὑπὸ τῶν BA , AE τῷ ἀπὸ τῆς AZ , τὸ δὲ ὑπὸ τῶν AB , BE τῷ ἀπὸ τῆς BZ . ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AZ τῷ ἀπὸ τῆς BZ . αἱ AZ , BZ ἄρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ ἡ AB ῥητὴ ἐστίν, ῥητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς AB . ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AZ , BZ ῥητὸν ἐστίν. καὶ ἐπεὶ πάλιν τὸ ὑπὸ τῶν AE , EB ἴσον ἐστὶ τῷ ἀπὸ τῆς EZ , ὑπόκειται δὲ τὸ ὑπὸ τῶν AE , EB καὶ τῷ ἀπὸ τῆς BD ἴσον, ἴση ἄρα ἐστὶν ἡ ZE τῇ BD . διπλὴ ἄρα ἡ BF τῆς ZE . ὥστε καὶ τὸ ὑπὸ τῶν AB , BF σύμμετρόν ἐστι τῷ ὑπὸ τῶν AB , EZ . μέσον δὲ τὸ ὑπὸ τῶν AB , BF . μέσον ἄρα καὶ τὸ ὑπὸ τῶν AB , EZ . ἴσον δὲ τὸ ὑπὸ τῶν AB , EZ τῷ ὑπὸ τῶν AZ , BZ . μέσον ἄρα καὶ τὸ ὑπὸ τῶν AZ , BZ . ἐδείχθη δὲ καὶ ῥητὸν τὸ συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων.

Εὐρένται ἄρα δύο εὐθείαι δυνάμει ἀσύμμετροι αἱ AZ , BZ ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ ὑπ' αὐτῶν μέσον. ὅπερ εἶδει δεῖχαι.

able (in length) with (AB) . And a parallelogram, equal to one quarter of the (square) on BC —that is to say, (equal) to the (square) on half of it—(and) falling short by a square figure, has been applied to AB , and makes the (rectangle contained) by AEB . AE is thus incommensurable (in length) with EB [Prop. 10.18]. And as AE is to EB , so the (rectangle contained) by BA and AE (is) to the (rectangle contained) by AB and BE . And the (rectangle contained) by BA and AE (is) equal to the (square) on AF , and the (rectangle contained) by AB and BE to the (square) on BF [Prop. 10.32 lem.]. The (square) on AF is thus incommensurable with the (square) on BF [Prop. 10.11]. Thus, AF and BF are incommensurable in square. And since AB is rational, the (square) on AB is also rational. Hence, the sum of the (squares) on AF and BF is also rational [Prop. 1.47]. And, again, since the (rectangle contained) by AE and EB is equal to the (square) on EF , and the (rectangle contained) by AE and EB was assumed (to be) equal to the (square) on BD , FE is thus equal to BD . Thus, BC is double FE . And hence the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and EF [Prop. 10.6]. And the (rectangle contained) by AB and BC (is) medial [Prop. 10.21]. Thus, the (rectangle contained) by AB and EF (is) also medial [Prop. 10.23 corr.]. And the (rectangle contained) by AB and EF (is) equal to the (rectangle contained) by AF and BF [Prop. 10.32 lem.]. Thus, the (rectangle contained) by AF and BF (is) also medial. And the sum of the squares on them was also shown (to be) rational.

Thus, the two straight-lines, AF and BF , (which are) incommensurable in square, be found, making the sum of the squares on them rational, and the (rectangle contained) by them medial. (Which is) the very thing it was required to show.

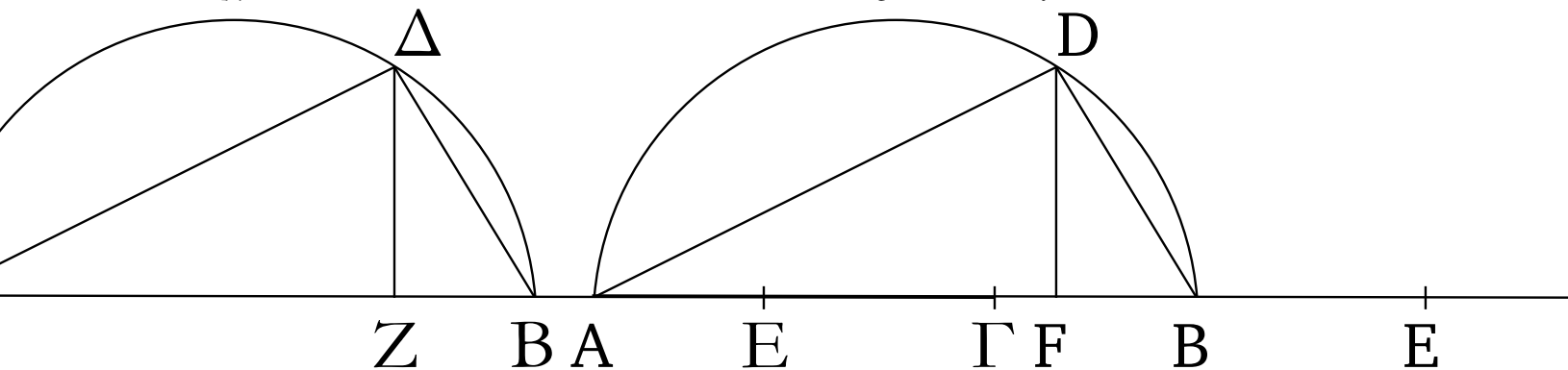
† AF and BF have lengths $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2}$ and $\sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ times that of AB , respectively, where k is defined in the footnote to Prop. 10.30.

34.

Proposition 34

Εὐρεῖν δύο εὐθείας δυνάμει ἀσύμμετρον ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν.

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational.



Ἐκκείσθωσαν δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ ῥητὸν περιέθουσαι τὸ ὑπ' αὐτῶν, ὥστε τὴν AB τῆς $BΓ$ μείζον δύνασθαι τῷ ἀπὸ ἀσυνμέτρου εἰς αὐτῆς, καὶ γεγραφθῶ ἐπὶ τῆς AB τὸ $AΔB$ ἡμικύκλιον, καὶ τετμήσθω ἡ $BΓ$ διὰ κατὰ τὸ E , καὶ παραβέβλησθω παρὰ τὴν AB τῷ ἀπὸ τῆς BE ἴσον παραλληλόγραμμον ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν AZB · ἀσύμμετρος ἄρα [ἐστίν] ἡ AZ τῇ ZB μήκει. καὶ ἤθθω ἀπὸ τοῦ Z τῇ AB πρὸς ὀρθὰς ἡ $ZΔ$, καὶ ἐπεξεύθωσαν αἱ $AΔ$, $ΔB$.

Ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AZ τῇ ZB , ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν BA , AZ τῷ ὑπὸ τῶν AB , BZ . ἴσον δὲ τὸ μὲν ὑπὸ τῶν BA , AZ τῷ ἀπὸ τῆς $AΔ$, τὸ δὲ ὑπὸ τῶν AB , BZ τῷ ἀπὸ τῆς $ΔB$ · ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $AΔ$ τῷ ἀπὸ τῆς $ΔB$. καὶ ἐπεὶ μέσον ἐστὶ τὸ ἀπὸ τῆς AB , μέσον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΔ$, $ΔB$. καὶ ἐπεὶ διπλὴ ἐστὶν ἡ $BΓ$ τῆς $ΔZ$, διπλάσιον ἄρα καὶ τὸ ὑπὸ τῶν AB , $BΓ$ τοῦ ὑπὸ τῶν AB , $ZΔ$. ῥητὸν δὲ τὸ ὑπὸ τῶν AB , $BΓ$ ῥητὸν ἄρα καὶ τὸ ὑπὸ τῶν AB , $ZΔ$. τὸ δὲ ὑπὸ τῶν AB , $ZΔ$ ἴσον τῷ ὑπὸ τῶν $AΔ$, $ΔB$ · ὥστε καὶ τὸ ὑπὸ τῶν $AΔ$, $ΔB$ ῥητὸν ἐστίν.

Ἐΐρηται ἄρα δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ $AΔ$, $ΔB$ ποιοῦσαι τὸ [μὲν] συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητὸν· ὅπερ ἔδει δεῖχαι.

Let the two medial (straight-lines) AB and BC , (which are) commensurable in square only, be laid out having the (rectangle contained) by them rational, (and) such that the square on AB is greater than (the square on) BC by the (square) on (some straight-line) incommensurable (in length) with (AB) [Prop. 10.31]. And let the semi-circle ADB be drawn on AB . And let BC be cut in half at E . And let a (rectangular) parallelogram equal to the (square) on BE , (and) falling short by a square figure, be applied to AB , (and let it be) the (rectangle contained by) AFB [Prop. 6.28]. Thus, AF [is] incommensurable in length with FB [Prop. 10.18]. And let FD be drawn from F at right-angles to AB . And let AD and DB be joined.

Since AF is incommensurable (in length) with FB , the (rectangle contained) by BA and AF is thus also incommensurable with the (rectangle contained) by AB and BF [Prop. 10.11]. And the (rectangle contained) by BA and AF (is) equal to the (square) on AD , and the (rectangle contained) by AB and BF to the (square) on DB [Prop. 10.32 lem.]. Thus, the (square) on AD is also incommensurable with the (square) on DB . And since the (square) on AB is medial, the sum of the (squares) on AD and DB (is) thus also medial [Props. 3.31, 1.47]. And since BC is double DF [see previous proposition], the (rectangle contained) by AB and BC (is) thus also double the (rectangle contained) by AB and FD . And the (rectangle contained) by AB and BC (is) rational. Thus, the (rectangle contained) by AB and FD (is) also rational [Prop. 10.6, Def. 10.4]. And the (rectangle contained) by AB and FD (is) equal to the (rectangle contained) by AD and DB [Prop. 10.32 lem.]. And hence the (rectangle contained) by AD and DB is rational.

Thus, two straight-lines, AD and DB , (which are) incommensurable in square, be found, making the sum of the squares on them medial, and the (rectangle contained) by them rational.[†] (Which is) the very thing it was required to show.

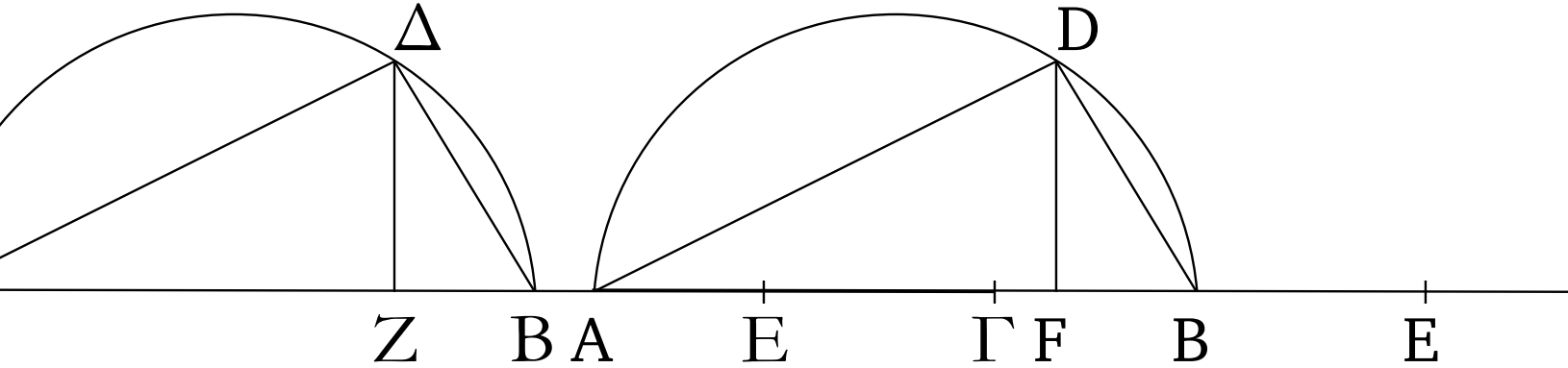
[†] AD and DB have lengths $\sqrt{[(1+k^2)^{1/2} + k]/[2(1+k^2)]}$ and $\sqrt{[(1+k^2)^{1/2} - k]/[2(1+k^2)]}$ times that of AB , respectively, where k is defined in the footnote to Prop. 10.29.

35.

Proposition 35

Εὕρεῖν δύο εὐθείας δυνάμει ἀσυνμέτρον ποιούσας τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων.

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them.



Ἐκκείσθωσαν δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ μέσον περιέθουσαι, ὥστε τὴν AB τῆς $BΓ$ μείζον δύνασθαι τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $AΔB$, καὶ τὰ λοιπὰ γεγονέτω τοῖς ἐπάνω ὁμοίως.

Καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AZ τῇ ZB μήκει, ἀσύμμετρος ἐστὶ καὶ ἡ $AΔ$ τῇ $ΔB$ δυνάμει. καὶ ἐπεὶ μέσον ἐστὶ τὸ ἀπὸ τῆς AB , μέσον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΔ$, $ΔB$. καὶ ἐπεὶ τὸ ὑπὸ τῶν AZ , ZB ἴσον ἐστὶ τῷ ἀφ' ἐκατέρᾳ τῶν BE , $ΔZ$, ἴση ἄρα ἐστὶν ἡ BE τῇ $ΔZ$ διπλῇ ἄρα ἡ $BΓ$ τῆς $ΔZ$ · ὥστε καὶ τὸ ὑπὸ τῶν AB , $BΓ$ διπλάσιόν ἐστι τοῦ ὑπὸ τῶν AB , $ΔZ$. μέσον δὲ τὸ ὑπὸ τῶν AB , $BΓ$ μέσον ἄρα καὶ τὸ ὑπὸ τῶν AB , $ΔZ$. καὶ ἐστὶν ἴσον τῷ ὑπὸ τῶν $AΔ$, $ΔB$ · μέσον ἄρα καὶ τὸ ὑπὸ τῶν $AΔ$, $ΔB$. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει, σύμμετρος δὲ ἡ $ΓB$ τῇ BE , ἀσύμμετρος ἄρα καὶ ἡ AB τῇ BE μήκει· ὥστε καὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB , BE ἀσύμμετρόν ἐστιν. ἀλλὰ τῷ μὲν ἀπὸ τῆς AB ἴσα ἐστὶ τὰ ἀπὸ τῶν $AΔ$, $ΔB$, τῷ δὲ ὑπὸ τῶν AB , BE ἴσον ἐστὶ τὸ ὑπὸ τῶν AB , $ΔZ$, τουτέστι τὸ ὑπὸ τῶν $AΔ$, $ΔB$ · ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΔ$, $ΔB$ τῷ ὑπὸ τῶν $AΔ$, $ΔB$.

Ἐνθρηνηται ἄρα δύο εὐθεῖαι αἱ $AΔ$, $ΔB$ δυνάμει ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων· ὅπερ ἔδει δεῖξαι.

Let the two medial (straight-lines) AB and BC , (which are) commensurable in square only, be laid out containing a medial (area), such that the square on AB is greater than (the square on) BC by the (square) on (some straight-line) incommensurable (in length) with (AB) [Prop. 10.32]. And let the semicircle ADB be drawn on AB . And let the remainder (of the figure) be generated similarly to the above (proposition).

And since AF is incommensurable in length with FB [Prop. 10.18], AD is also incommensurable in square with DB [Prop. 10.11]. And since the (square) on AB is medial, the sum of the (squares) on AD and DB (is) thus also medial [Props. 3.31, 1.47]. And since the (rectangle contained) by AF and FB is equal to the (square) on each of BE and DF , BE is thus equal to DF . Thus, BC (is) double FD . And hence the (rectangle contained) by AB and BC is double the (rectangle) contained by AB and FD . And the (rectangle contained) by AB and BC (is) medial. Thus, the (rectangle contained) by AB and FD (is) also medial. And it is equal to the (rectangle contained) by AD and DB [Prop. 10.32 lem.]. Thus, the (rectangle contained) by AD and DB (is) also medial. And since AB is incommensurable in length with BC , and CB (is) commensurable (in length) with BE , AB (is) thus also incommensurable in length with BE [Prop. 10.13]. And hence the (square) on AB is also incommensurable with the (rectangle contained) by AB and BE [Prop. 10.11]. But the (sum of the squares) on AD and DB is equal to the (square) on AB [Prop. 1.47]. And the (rectangle contained) by AB and FD —that is to say, the (rectangle contained) by AD and DB —is equal to the (rectangle contained) by AB and BE . Thus, the sum of the (squares) on AD and DB is incommensurable with the (rectangle contained) by AD and DB .

Thus, two straight-lines, AD and DB , (which are) incommensurable in square, be found, making the sum of the (squares) on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them.[†] (Which is) the very thing it was required to show.

[†] AD and DB have lengths $k^{1/4} \sqrt{[1 + k/(1 + k^2)^{1/2}]/2}$ and $k'^{1/4} \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ times that of AB , respectively, where k and k' are defined in the footnote

to Prop. 10.32.

36.

Proposition 36

Ἐάν δύο ῥηταὶ δυνάμει μόνον σύμμετροι συντεθῶσιν, ἡ ὅλη ἄλογός ἐστιν, καλείσθω δὲ ἐκ δύο ὀνομάτων.

If two rational (straight-lines which are) commensurable in square only are added together then the whole (straight-line) is irrational—let it be called a binomial (straight-line).[†]



Συγκείσθωσαν γάρ δύο ῥηταὶ δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$. λέγω, ὅτι ὅλη ἡ $ΑΓ$ ἄλογός ἐστιν.

Ἐπεὶ γάρ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει· δυνάμει γάρ μόνον εἰσὶ σύμμετροι· ὥς δὲ ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ὑπὸ τῶν $ΑΒΓ$ πρὸς τὸ ἀπὸ τῆς $BΓ$, ἀσύμμετρον ἄρα ἐστὶ τὸ ὑπὸ τῶν AB , $BΓ$ τῷ ἀπὸ τῆς $BΓ$. ἀλλὰ τῷ μὲν ὑπὸ τῶν AB , $BΓ$ σύμμετρον ἐστὶ τὸ διζ ὑπὸ τῶν AB , $BΓ$, τῷ δὲ ἀπὸ τῆς $BΓ$ σύμμετρόν ἐστι τὰ ἀπὸ τῶν AB , $BΓ$. αἱ γὰρ AB , $BΓ$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀσύμμετρον ἄρα ἐστὶ τὸ διζ ὑπὸ τῶν AB , $BΓ$ τοῖς ἀπὸ τῶν AB , $BΓ$. καὶ συνθέντι τὸ διζ ὑπὸ τῶν AB , $BΓ$ μετὰ τῶν ἀπὸ τῶν AB , $BΓ$, τοντέστι τὸ ἀπὸ τῆς $ΑΓ$, ἀσύμμετρον ἐστὶ τῷ συγκεκλιμένῳ ἐκ τῶν ἀπὸ τῶν AB , $BΓ$. ῥητὸν δὲ τὸ συγκεκλιμένον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ ἄλογον ἄρα [ἐστὶ] τὸ ἀπὸ τῆς $ΑΓ$. ὥστε καὶ ἡ $ΑΓ$ ἄλογός ἐστιν, καλείσθω δὲ ἐκ δύο ὀνομάτων· ὅπερ ἔδει δεῖχαι.

For let the two rational (straight-lines), AB and BC , (which are) commensurable in square only, be laid down together. I say that the whole (straight-line), AC , is irrational. For since AB is incommensurable in length with BC —for they are commensurable in square only—and as AB (is) to BC , so the (rectangle contained) by ABC (is) to the (square) on BC , the (rectangle contained) by AB and BC is thus incommensurable with the (square) on BC [Prop. 10.11]. But, twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. And (the sum of) the (squares) on AB and BC is commensurable with the (square) on BC —for the rational (straight-lines) AB and BC are commensurable in square only [Prop. 10.15]. Thus, twice the (rectangle contained) by AB and BC is incommensurable with (the sum of) the (squares) on AB and BC [Prop. 10.13]. And, via composition, twice the (rectangle contained) by AB and BC , plus (the sum of) the (squares) on AB and BC —that is to say, the (square) on AC [Prop. 2.4]—is incommensurable with the sum of the (squares) on AB and BC [Prop. 10.16]. And the sum of the (squares) on AB and BC (is) rational. Thus, the (square) on AC [is] irrational [Def. 10.4]. Hence, AC is also irrational [Def. 10.4]—let it be called a binomial (straight-line).[‡] (Which is) the very thing it was required to show.

[†] Literally, “from two names”.

[‡] Thus, a binomial straight-line has a length expressible as $1 + k^{1/2}$ [or, more generally, $\rho(1 + k^{1/2})$, where ρ is rational—the same proviso applies to the definitions in the following propositions]. The binomial and the corresponding apotome, whose length is expressible as $1 - k^{1/2}$ (see Prop. 10.73), are the positive roots of the quartic $x^4 - 2(1 + k)x^2 + (1 - k)^2 = 0$.

37.

Proposition 37

Ἐάν δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι ῥητὸν περιέθουσαι, ἡ ὅλη ἄλογός ἐστιν, καλείσθω δὲ ἐκ δύο μέσων πρώτης.

If two medial (straight-lines), commensurable in square only, which contain a rational (area), are added together then the whole (straight-line) is irrational—let it be called a first bimedral (straight-line).[†]



Συγκείσθωσαν γάρ δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ ῥητὸν περιέθουσαι· λέγω, ὅτι ὅλη ἡ $ΑΓ$ ἄλογός ἐστιν.

For let the two medial (straight-lines), AB and BC , commensurable in square only, (and) containing a rational (area), be laid down together. I say that the whole (straight-line), AC ,

Ἐπεὶ γὰρ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει, καὶ τὰ ἀπὸ τῶν AB , $BΓ$ ἄρα ἀσύμμετρα ἐστὶ τῷ διξ ὑπὸ τῶν AB , $BΓ$ · καὶ συνθέντι τὰ ἀπὸ τῶν AB , $BΓ$ μετὰ τοῦ διξ ὑπὸ τῶν AB , $BΓ$, ὅπερ ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$, ἀσύμμετρον ἐστὶ τῷ ὑπὸ τῶν AB , $BΓ$. ῥητόν δὲ τὸ ὑπὸ τῶν AB , $BΓ$ · ὑπόκειται γὰρ αἱ AB , $BΓ$ ῥητόν περιέθουσαι· ἄλογον ἄρα τὸ ἀπὸ τῆς $ΑΓ$ · ἄλογός ἄρα ἡ $ΑΓ$, καλείσθω δὲ ἐκ δύο μέσων πρῶτη· ὅπερ ἔδει δεῖχαι.

is irrational.

For since AB is incommensurable in length with BC , (the sum of) the (squares) on AB and BC is thus also incommensurable with twice the (rectangle contained) by AB and BC [see previous proposition]. And, via composition, (the sum of) the (squares) on AB and BC , plus twice the (rectangle contained) by AB and BC —that is, the (square) on AC [Prop. 2.4]—is incommensurable with the (rectangle contained) by AB and BC [Prop. 10.16]. And the (rectangle contained) by AB and BC (is) rational—for AB and BC were assumed to enclose a rational (area). Thus, the (square) on AC (is) irrational. Thus, AC (is) irrational [Def. 10.4]—let it be called a first bimedral (straight-line).[‡] (Which is) the very thing it was required to show.

[†] Literally, “first from two medials”.

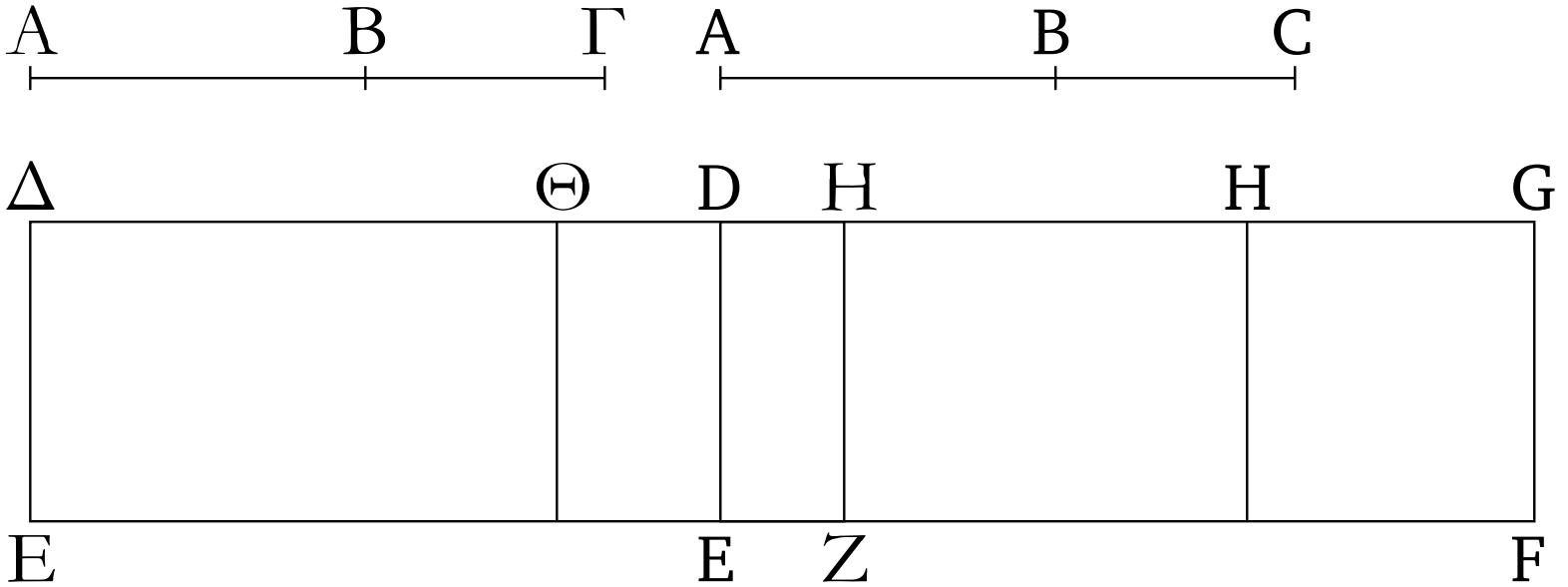
[‡] Thus, a first bimedral straight-line has a length expressible as $k^{1/4} + k^{3/4}$. The first bimedral and the corresponding first apotome of a medial, whose length is expressible as $k^{1/4} - k^{3/4}$ (see Prop. 10.74), are the positive roots of the quartic $x^4 - 2\sqrt{k}(1+k)x^2 + k(1-k)^2 = 0$.

38.

Proposition 38

Ἐὰν δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι μέσον περιέθουσαι, ἡ ὅλη ἄλογός ἐστιν, καλείσθω δὲ ἐκ δύο μέσων δευτέρα.

If two medial (straight-lines), commensurable in square only, which contain a medial (area), are added together then the whole (straight-line) is irrational—let it be called a second bimedral (straight-line).



Συγκείσθωσαν γὰρ δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB , $BΓ$ μέσον περιέθουσαι· λέγω, ὅτι ἄλογός ἐστιν ἡ $ΑΓ$.

Ἐκκείσθω γὰρ ῥητὴ ἡ $ΔΕ$, καὶ τῷ ἀπὸ τῆς $ΑΓ$ ἴσον παρὰ τὴν $ΔΕ$ παραβελθήσθω τὸ $ΔΖ$ πλάτος ποιῶν τὴν $ΔΗ$. καὶ ἐπεὶ τὸ ἀπὸ τῆς $ΑΓ$ ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν AB , $BΓ$ καὶ τῷ διξ ὑπὸ τῶν AB , $BΓ$, παραβελθήσθω δὴ τοῖς ἀπὸ τῶν AB , $BΓ$ παρὰ τὴν $ΔΕ$ ἴσον τὸ $ΕΘ$ · λοιπὸν ἄρα τὸ $ΘΖ$ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AB , $BΓ$. καὶ ἐπεὶ μέση ἐστὶν ἑκατέρα τῶν

For let the two medial (straight-lines), AB and BC , commensurable in square only, (and) containing a medial (area), be laid down together [Prop. 10.28]. I say that AC is irrational.

For let the rational (straight-line) DE be laid down, and let (the rectangle) DF , equal to the (square) on AC , be applied to DE , making DG as breadth [Prop. 1.44]. And since the (square) on AC is equal to (the sum of) the (squares) on AB and BC , plus twice the (rectangle contained) by AB and BC

AB , BF , μέσα ἄρα ἐστὶ καὶ τὰ ἀπὸ τῶν AB , BF . μέσον δὲ ὑπόκειται καὶ τὸ διζ ὑπὸ τῶν AB , BF . καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν AB , BF ἴσον τὸ $E\Theta$, τῷ δὲ διζ ὑπὸ τῶν AB , BF ἴσον τὸ $Z\Theta$. μέσον ἄρα ἐκάτερον τῶν $E\Theta$, ΘZ . καὶ παρὰ ῥητὴν τὴν ΔE παράκειται ῥητὴ ἄρα ἐστὶν ἐκάτερα τῶν $\Delta\Theta$, ΘH καὶ ἀσύμμετρος τῇ ΔE μήκει. ἐπεὶ οὖν ἀσύμμετρος ἐστὶν ἡ AB τῇ BF μήκει, καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν BF , οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ὑπὸ τῶν AB , BF , ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB , BF . ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BF τετραγώνων, τῷ δὲ ὑπὸ τῶν AB , BF σύμμετρον ἐστὶ τὸ διζ ὑπὸ τῶν AB , BF . ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BF τῷ διζ ὑπὸ τῶν AB , BF . ἀλλὰ τοῖς μὲν ἀπὸ τῶν AB , BF ἴσον ἐστὶ τὸ $E\Theta$, τῷ δὲ διζ ὑπὸ τῶν AB , BF ἴσον ἐστὶ τὸ ΘZ . ἀσύμμετρον ἄρα ἐστὶ τὸ $E\Theta$ τῷ ΘZ . ὥστε καὶ ἡ $\Delta\Theta$ τῇ ΘH ἐστὶν ἀσύμμετρος μήκει. αἱ $\Delta\Theta$, ΘH ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι. ὥστε ἡ ΔH ἄλογός ἐστιν. ῥητὴ δὲ ἡ ΔE . τὸ δὲ ὑπὸ ἀλόγον καὶ ῥητῆς περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν. ἄλογον ἄρα ἐστὶ τὸ ΔZ θωρίον, καὶ ἡ δυνάμεν [αὐτὸ] ἄλογός ἐστιν. δύναται δὲ τὸ ΔZ ἢ AG . ἄλογός ἄρα ἐστὶν ἡ AG , καλεῖσθω δὲ ἐκ δύο μέσων δευτέρα. ὅπερ ἔδει δεῖξαι.

[Prop. 2.4], so let (the rectangle) EH , equal to (the sum of) the squares on AB and BC , be applied to DE . The remainder HF is thus equal to twice the (rectangle contained) by AB and BC . And since AB and BC are each medial, (the sum of) the squares on AB and BC is thus also medial.[‡] And twice the (rectangle contained) by AB and BC was also assumed (to be) medial. And EH is equal to (the sum of) the squares on AB and BC , and HF (is) equal to twice the (rectangle contained) by AB and BC . Thus, EH and HF (are) each medial. And they were applied to the rational (straight-line) DE . Thus, DH and HG are each rational, and incommensurable in length with DE [Prop. 10.22]. Therefore, since AB is incommensurable in length with BC , and as AB is to BC , so the (square) on AB (is) to the (rectangle contained) by AB and BC [Prop. 10.21 lem.], the (square) on AB is thus incommensurable with the (rectangle contained) by AB and BC [Prop. 10.11]. But, the sum of the squares on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. Thus, the sum of the (squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC [Prop. 10.13]. But, EH is equal to (the sum of) the squares on AB and BC , and HF is equal to twice the (rectangle) contained by AB and BC . Thus, EH is incommensurable with HF . Hence, DH is also incommensurable in length with HG [Props. 6.1, 10.11]. Thus, DH and HG are rational (straight-lines which are) commensurable in square only. Hence, DG is irrational [Prop. 10.36]. And DE (is) rational. And the rectangle contained by irrational and rational (straight-lines) is irrational [Prop. 10.20]. The area DF is thus irrational, and (so) the square-root [of it] is irrational [Def. 10.4]. And AC is the square-root of DF . AC is thus irrational—let it be called a second bimedial (straight-line).[§] (Which is) the very thing it was required to show.

[†] Literally, “second from two medials”.

[‡] Since, by hypothesis, the squares on AB and BC are commensurable—see Props. 10.15, 10.23.

[§] Thus, a second bimedial straight-line has a length expressible as $k^{1/4} + k^{1/2}/k^{1/4}$. The second bimedial and the corresponding second apotome of a medial, whose length is expressible as $k^{1/4} - k^{1/2}/k^{1/4}$ (see Prop. 10.75), are the positive roots of the quartic $x^4 - 2[(k + k')/\sqrt{k}]x^2 + [(k - k')^2/k] = 0$.

39.

Proposition 39

Ἐάν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον, ἡ ὅλη εὐθεῖα ἄλογός ἐστιν, καλεῖσθω δὲ μείζων.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial, are added together then the whole straight-line is irrational—let it be called a major (straight-line).



Συνγείσθωσαν γὰρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι α-

For let the two straight-lines, AB and BC , incommensu-

ἡ AB , $B\Gamma$ ποιῶσαι τὰ προκείμενα· λέγω, ὅτι ἄλογός ἐστιν ἡ $ΑΓ$.

Ἐπεὶ γάρ τὸ ὑπὸ τῶν AB , $B\Gamma$ μέσον ἐστίν, καὶ τὸ διξ [ἄρα] ὑπὸ τῶν AB , $B\Gamma$ μέσον ἐστίν. τὸ δὲ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ ῥητόν· ἀσύμμετρον ἄρα ἐστὶ τὸ διξ ὑπὸ τῶν AB , $B\Gamma$ τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ · ὥστε καὶ τὰ ἀπὸ τῶν AB , $B\Gamma$ μετὰ τοῦ διξ ὑπὸ τῶν AB , $B\Gamma$, ὅπερ ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$, ἀσύμμετρόν ἐστι τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ [ῥητόν δὲ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$]. ἄλογον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΑΓ$. ὥστε καὶ ἡ $ΑΓ$ ἄλογός ἐστιν, καλείσθω δὲ μείζων. ὅπερ ἔδει δεῖξαι.

able in square, and fulfilling the prescribed (conditions), be laid down together [Prop. 10.33]. I say that AC is irrational.

For since the (rectangle contained) by AB and BC is medial, twice the (rectangle contained) by AB and BC is [thus] also medial [Props. 10.6, 10.23 corr.]. And the sum of the (squares) on AB and BC (is) rational. Thus, twice the (rectangle contained) by AB and BC is incommensurable with the sum of the (squares) on AB and BC [Def. 10.4]. Hence, (the sum of) the squares on AB and BC , plus twice the (rectangle contained) by AB and BC —that is, the (square) on AC [Prop. 2.4]—is also incommensurable with the sum of the (squares) on AB and BC [Prop. 10.16] [and the sum of the (squares) on AB and BC (is) rational]. Thus, the (square) on AC is irrational. Hence, AC is also irrational [Def. 10.4]—let it be called a major (straight-line).[†] (Which is) the very thing it was required to show.

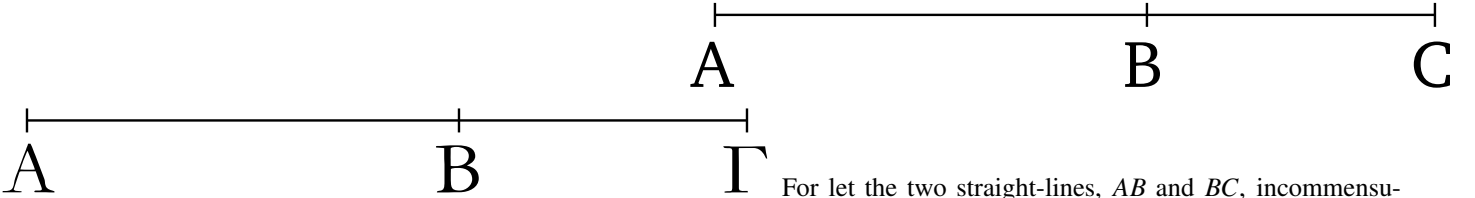
[†] Thus, a major straight-line has a length expressible as $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$. The major and the corresponding minor, whose length is expressible as $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} - \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ (see Prop. 10.76), are the positive roots of the quartic $x^4 - 2x^2 + k^2/(1 + k^2) = 0$.

40.

Proposition 40

Ἐάν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν, ἡ ὅλη εὐθεῖα ἄλογός ἐστιν, καλείσθω δὲ ῥητόν καὶ μέσον δυναμένη.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational, are added together then the whole straight-line is irrational—let it be called the square-root of a rational plus a medial (area).



Συγκείσθωσαν γάρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι α-
ἡ AB , $B\Gamma$ ποιῶσαι τὰ προκείμενα· λέγω, ὅτι ἄλογός ἐστιν ἡ $ΑΓ$.

Ἐπεὶ γάρ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ μέσον ἐστίν, τὸ δὲ διξ ὑπὸ τῶν AB , $B\Gamma$ ῥητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $B\Gamma$ τῷ διξ ὑπὸ τῶν AB , $B\Gamma$ · ὥστε καὶ τὸ ἀπὸ τῆς $ΑΓ$ ἀσύμμετρόν ἐστι τῷ διξ ὑπὸ τῶν AB , $B\Gamma$. ῥητόν δὲ τὸ διξ ὑπὸ τῶν AB , $B\Gamma$ · ἄλογον ἄρα τὸ ἀπὸ τῆς $ΑΓ$. ἄλογοξ ἄρα ἡ $ΑΓ$, καλείσθω δὲ ῥητόν καὶ μέσον δυναμένη. ὅπερ ἔδει δεῖξαι.

For let the two straight-lines, AB and BC , incommensurable in square, (and) fulfilling the prescribed (conditions), be laid down together [Prop. 10.34]. I say that AC is irrational.

For since the sum of the (squares) on AB and BC is medial, and twice the (rectangle contained) by AB and BC (is) rational, the sum of the (squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . Hence, the (square) on AC is also incommensurable with twice the (rectangle contained) by AB and BC [Prop. 10.16]. And twice the (rectangle contained) by AB and BC (is) rational. The (square) on AC (is) thus irrational. Thus, AC (is) irrational [Def. 10.4]—let it be called the square-root of a rational plus a medial (area).[†] (Which is) the very thing it was required to show.

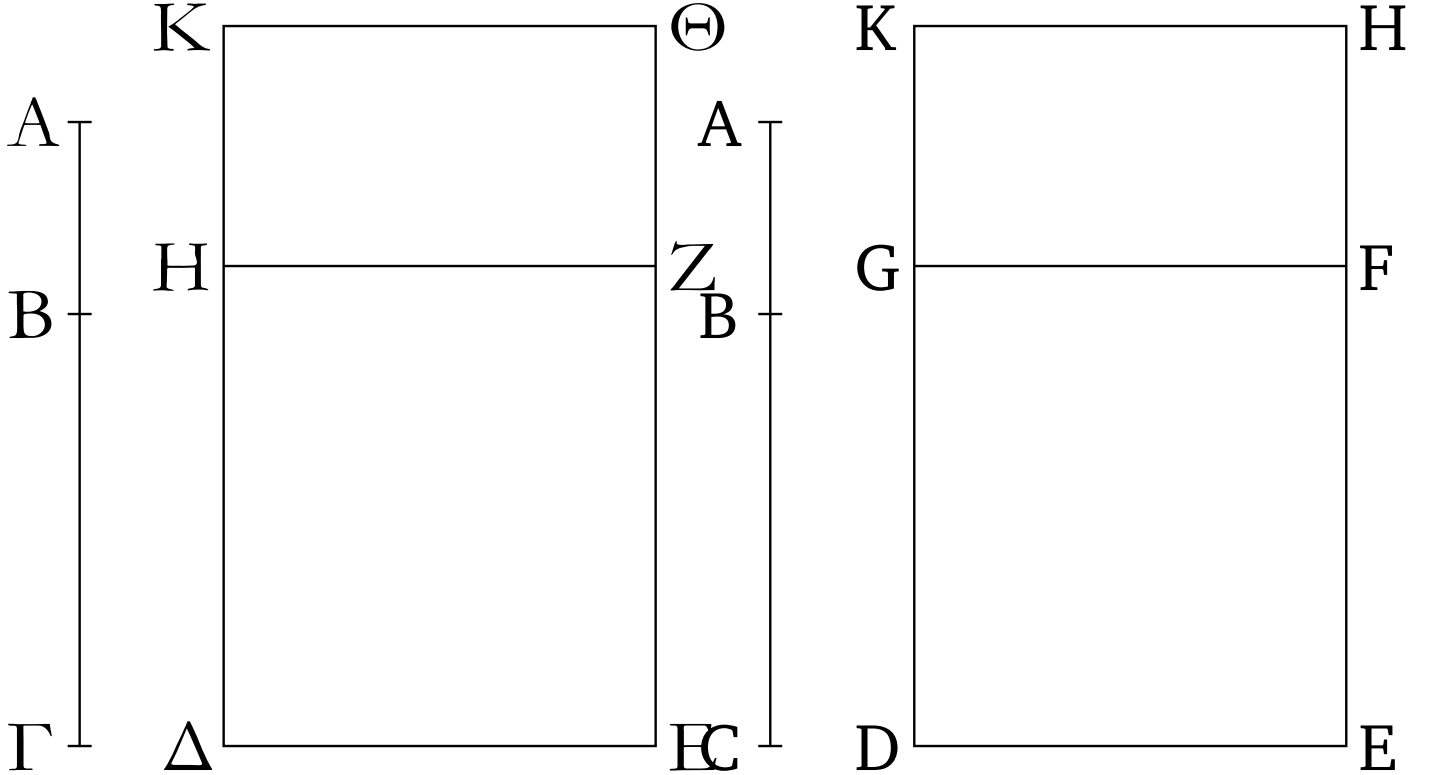
[†] Thus, the square-root of a rational plus a medial (area) has a length expressible as $\sqrt{[(1 + k^2)^{1/2} + k]/[2(1 + k^2)]} + \sqrt{[(1 + k^2)^{1/2} - k]/[2(1 + k^2)]}$. This and the corresponding irrational with a minus sign, whose length is expressible as $\sqrt{[(1 + k^2)^{1/2} + k]/[2(1 + k^2)]} - \sqrt{[(1 + k^2)^{1/2} - k]/[2(1 + k^2)]}$ (see Prop. 10.77), are the positive roots of the quartic $x^4 - (2/\sqrt{1 + k^2})x^2 + k^2/(1 + k^2)^2 = 0$.

41.

Proposition 41

Ἐάν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καί τὸ ὑπ' αὐτῶν μέσον καί ἔτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων, ἡ ὅλη εὐθεῖα ἄλογός ἐστιν, καλεῖσθω δὲ δύο μέσα δυναμένη.

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them, are added together then the whole straight-line is irrational—let it be called the square-root of (the sum of) two medial (areas).



Συγκείσθωσαν γάρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AB , BF ποιοῦσαι τὰ προκείμενα· λέγω, ὅτι ἡ AF ἄλογός ἐστιν.

Ἐκκείσθω ῥητὴ ἡ DE , καὶ παραβελθήσθω παρὰ τὴν DE τοῖς μὲν ἀπὸ τῶν AB , BF ἴσον τὸ ΔZ , τῷ δὲ διξ ὑπὸ τῶν AB , BF ἴσον τὸ $H\Theta$. ὅλον ἄρα τὸ $\Delta\Theta$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AF τετραγώνῳ. καὶ ἐπεὶ μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BF , καὶ ἐστιν ἴσον τῷ ΔZ , μέσον ἄρα ἐστὶ καὶ τὸ ΔZ . καὶ παρὰ ῥητὴν τὴν DE παράκειται ῥητὴ ἄρα ἐστὶν ἡ ΔH καὶ ἀσύμμετρος τῇ DE μήκει. διὰ τὰ αὐτὰ δὲ καὶ ἡ HK ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ HZ , τουτέστι τῇ DE , μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB , BF τῷ διξ ὑπὸ τῶν AB , BF , ἀσύμμετρον ἐστὶ τὸ ΔZ τῷ $H\Theta$. ὥστε καὶ ἡ ΔH τῇ HK ἀσύμμετρος ἐστὶν. καὶ εἰσι ῥηταί· αἱ ΔH , HK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἄλογος ἄρα ἐστὶν ἡ ΔK ἡ καλουμένη ἐκ δύο ὀνομάτων. ῥητὴ δὲ ἡ DE · ἄλογον ἄρα ἐστὶ τὸ $\Delta\Theta$ καὶ ἡ δυναμένη αὐτὸ ἄλογός ἐστιν. δύναται δὲ τὸ $\Theta\Delta$ ἡ AF · ἄλογος ἄρα ἐστὶν ἡ AF , καλεῖσθω δὲ δύο μέσα δυναμένη. ὅπερ ἔδει δεῖξαι.

For let the two straight-lines, AB and BC , incommensurable in square, (and) fulfilling the prescribed (conditions), be laid down together [Prop. 10.35]. I say that AC is irrational.

Let the rational (straight-line) DE be laid out, and let (the rectangle) DF , equal to (the sum of) the (squares) on AB and BC , and (the rectangle) GH , equal to twice the (rectangle contained) by AB and BC , be applied to DE . Thus, the whole of DH is equal to the square on AC [Prop. 2.4]. And since the sum of the (squares) on AB and BC is medial, and is equal to DF , DF is thus also medial. And it is applied to the rational (straight-line) DE . Thus, DG is rational, and incommensurable in length with DE [Prop. 10.22]. So, for the same (reasons), GK is also rational, and incommensurable in length with GF —that is to say, DE . And since (the sum of) the (squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC , DF is incommensurable with GH . Hence, DG is also incommensurable (in length) with GK [Props. 6.1, 10.11]. And they are rational. Thus, DG and GK are rational (straight-lines which are) commensurable in

square only. Thus, DK is irrational, and that (straight-line which is) called binomial [Prop. 10.36]. And DE (is) rational. Thus, DH is irrational, and its square-root is irrational [Def. 10.4]. And AC (is) the square-root of HD . Thus, AC is irrational—let it be called the square-root of (the sum of) two medial (areas).[†] (Which is) the very thing it was required to show.

[†] Thus, the square-root of (the sum of) two medial (areas) has a length expressible as $k^{1/4} \left(\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} \right)$. This and the corresponding irrational with a minus sign, whose length is expressible as $k^{1/4} \left(\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} - \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} \right)$ (see Prop. 10.78), are the positive roots of the quartic $x^4 - 2k^{1/2}x^2 + k'k^2/(1 + k^2) = 0$.

Λήμμα.

Lemma

Ὅτι δὲ αἱ εἰρημέναι ἄλογοι μοναθῶς διαιροῦνται εἰς τὰς εὐθείας, ἐχ' ὧν σύγκεινται ποιουσῶν τὰ προκείμενα εἶδη, δεῖχόμεν ἥδη προεκθέμενοι λημμάτων τοιοῦτον·

We will now demonstrate that the aforementioned irrational (straight-lines) are uniquely divided into the straight-lines of which they are the sum, and which produce the prescribed types, (after) setting forth the following lemma.



Ἐκκείσθω εὐθεῖα ἡ AB καὶ τετμήσθω ἡ ὅλη εἰς ἅμισα καθ' ἑκάτερον τῶν Γ , Δ , ὑποκείσθω δὲ μείζων ἡ AG τῆς ΔB . λέγω, ὅτι τὰ ἀπὸ τῶν AG , GB μείζονά ἐστι τῶν ἀπὸ τῶν AD , ΔB .

Let the straight-line AB be laid out, and let the whole (straight-line) be cut into unequal parts at each of the (points) C and D . And let AC be assumed (to be) greater than DB . I say that (the sum of) the (squares) on AC and CB is greater than (the sum of) the (squares) on AD and DB .

Τετμήσθω γάρ ἡ AB διὰ κατὰ τὸ E . καὶ ἐπεὶ μείζων ἐστὶν ἡ AG τῆς ΔB , κοινὴ ἀφρηρήσθω ἡ $\Delta\Gamma$. λοιπὴ ἄρα ἡ $A\Delta$ λοιπῆς τῆς GB μείζων ἐστίν. ἴση δὲ ἡ AE τῇ EB . ἐλάττων ἄρα ἡ ΔE τῆς $E\Gamma$. τὰ Γ , Δ ἄρα σημεία οὐκ ἴσον ἀπέθουσι τῆς διχοτομίας. καὶ ἐπεὶ τὸ ὑπὸ τῶν AG , GB μετὰ τοῦ ἀπὸ τῆς $E\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς EB , ἀλλὰ μὴν καὶ τὸ ὑπὸ τῶν $A\Delta$, ΔB μετὰ τοῦ ἀπὸ ΔE ἴσον ἐστὶ τῷ ἀπὸ τῆς EB , τὸ ἄρα ὑπὸ τῶν AG , GB μετὰ τοῦ ἀπὸ τῆς $E\Gamma$ ἴσον ἐστὶ τῷ ὑπὸ τῶν $A\Delta$, ΔB μετὰ τοῦ ἀπὸ τῆς ΔE . ὧν τὸ ἀπὸ τῆς ΔE ἐλασσόν ἐστι τοῦ ἀπὸ τῆς $E\Gamma$. καὶ λοιπὸν ἄρα τὸ ὑπὸ τῶν AG , GB ἐλασσόν ἐστι τοῦ ὑπὸ τῶν $A\Delta$, ΔB . ὥστε καὶ τὸ διζὺ ὑπὸ τῶν AG , GB ἐλασσόν ἐστι τοῦ διζὺ ὑπὸ τῶν $A\Delta$, ΔB . καὶ λοιπὸν ἄρα τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μείζον ἐστι τοῦ συγκειμένου ἐκ τῶν ἀπὸ τῶν $A\Delta$, ΔB . ὅπερ ἔδει δεῖχαι.

For let AB be cut in half at E . And since AC is greater than DB , let DC be subtracted from both. Thus, the remainder AD is greater than the remainder CB . And AE (is) equal to EB . Thus, DE (is) less than EC . Thus, points C and D are not equally far from the point of bisection. And since the (rectangle contained) by AC and CB , plus the (square) on EC , is equal to the (square) on EB [Prop. 2.5], but, moreover, the (rectangle contained) by AD and DB , plus the (square) on DE , is also equal to the (square) on EB [Prop. 2.5], the (rectangle contained) by AC and CB , plus the (square) on EC , is thus equal to the (rectangle contained) by AD and DB , plus the (square) on DE . And, of these, the (square) on DE is less than the (square) on EC . And, thus, the remaining (rectangle contained) by AC and CB is less than the (rectangle contained) by AD and DB . And, hence, twice the (rectangle contained) by AC and CB is less than twice the (rectangle contained) by AD and DB . And thus the remaining sum of the (squares) on AC and CB is greater than the sum of the (squares) on AD and DB .[†] (Which is) the very thing it was required to show.

[†] Since, $AC^2 + CB^2 + 2AC \cdot CB = AD^2 + DB^2 + 2AD \cdot DB = AB^2$.

42.

Proposition 42

Ἡ ἐκ δύο ὀνομάτων κατὰ ἓν μόνον σημεῖον διαιρεῖται εἰς τὰ ὀνόματα.

A binomial (straight-line) can be divided into its (component) terms at one point only.[†]



Ἐστω ἐκ δύο ὀνομάτων ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ . αἱ AG , GB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται εἰς δύο ῥητάς δυνάμει μόνον συμμέτρους.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ , ὥστε καὶ τὰς $A\Delta$, ΔB ῥητάς εἶναι δυνάμει μόνον συμμέτρους. φανερόν δὲ, ὅτι ἡ AG τῇ ΔB οὐκ ἔστιν ἡ αὐτή. εἰ γὰρ δυνατόν, ἔστω. ἔσται δὲ καὶ ἡ $A\Delta$ τῇ GB ἡ αὐτή· καὶ ἔσται ὥς ἡ AG πρὸς τὴν GB , οὕτως ἡ $B\Delta$ πρὸς τὴν ΔA , καὶ ἔσται ἡ AB κατὰ τὸ αὐτὸ τῇ κατὰ τὸ Γ διαιρέσει διαιρεθεῖσα καὶ κατὰ τὸ Δ . ὅπερ οὐθ' ὑπόκειται. οὐκ ἄρα ἡ AG τῇ ΔB ἔστιν ἡ αὐτή. διὰ δὲ τοῦτο καὶ τὰ Γ , Δ σημεῖα οὐκ ἴσον ἀπέθουσι τῆς διχοτομίας. ὥ' ἄρα διαφέρει τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν $A\Delta$, ΔB , τούτω διαφέρει καὶ τὸ διξ ὑπὸ τῶν $A\Delta$, ΔB τοῦ διξ ὑπὸ τῶν AG , GB διὰ τὸ καὶ τὰ ἀπὸ τῶν AG , GB μετὰ τοῦ διξ ὑπὸ τῶν AG , GB καὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB μετὰ τοῦ διξ ὑπὸ τῶν $A\Delta$, ΔB ἴσα εἶναι τῶν ἀπὸ τῆς AB . ἀλλὰ τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν $A\Delta$, ΔB διαφέρει ῥητῶ· ῥητὰ γὰρ ἀμφοτέρω· καὶ τὸ διξ ἄρα ὑπὸ τῶν $A\Delta$, ΔB τοῦ διξ ὑπὸ τῶν AG , GB διαφέρει ῥητῶ μέσα ὄντα· ὅπερ ἄτοπον· μέσον γὰρ μέσον οὐθ' ὑπερέχει ῥητῶ.

Οὐθ' ἄρα ἡ ἐκ δύο ὀνομάτων κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται καθ' ἐν ἄρα μόνον· ὅπερ ἔδει δεῖχαι.

Let AB be a binomial (straight-line) which has been divided into its (component) terms at C . AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. I say that AB cannot be divided at another point into two rational (straight-lines which are) commensurable in square only.

For, if possible, let it also be divided at D , such that AD and DB are also rational (straight-lines which are) commensurable in square only. So, (it is) clear that AC is not the same as DB . For, if possible, let it be (the same). So, AD will also be the same as CB . And as AC will be to CB , so BD (will be) to DA . And AB will (thus) also be divided at D in the same (manner) as the division at C . The very opposite was assumed. Thus, AC is not the same as DB . So, on account of this, points C and D are not equally far from the point of bisection. Thus, by whatever (amount the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB , twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by this (same amount)—on account of both (the sum of) the (squares) on AC and CB , plus twice the (rectangle contained) by AC and CB , and (the sum of) the (squares) on AD and DB , plus twice the (rectangle contained) by AD and DB , being equal to the (square) on AB [Prop. 2.4]. But, (the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB by a rational (area). For (they are) both rational (areas). Thus, twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by a rational (area, despite both being medial (areas) [Prop. 10.21]. The very thing is absurd. For a medial (area) cannot exceed a medial (area) by a rational (area) [Prop. 10.26].

Thus, a binomial (straight-line) cannot be divided (into its component terms) at different points. Thus, (it can be so divided) at one point only. (Which is) the very thing it was required to show.

[†] In other words, $k + k^{1/2} = k'' + k'''^{1/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$. Likewise, $k^{1/2} + k'^{1/2} = k''^{1/2} + k'''^{1/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$ (or, equivalently, $k'' = k'$ and $k''' = k$).

43.

Proposition 43

Ἡ ἐκ δύο μέσων πρώτη καθ' ἐν μόνον σημεῖον διαιρεῖται.

A first bimedral (straight-line) can be divided (into its component terms) at one point only.[†]



Ἐστω ἐκ δύο μέσων πρώτη ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε τὰς AG , GB μέσας εἶναι δυνάμει μόνον συμμέτρους

Let AB be a first bimedral (straight-line) which has been divided at C , such that AC and CB are medial (straight-lines),

ῥητὸν περιεθούσαξ· λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν διηγήσθω καὶ κατὰ τὸ Δ , ὥστε καὶ τὰς $A\Delta$, ΔB μέσαξ εἶναι δυνάμει μόνον συμμετρωνξ ῥητὸν περιεθούσαξ. ἐπεὶ οὖν, ὧ διαφέρει τὸ διξ ὑπὸ τῶν $A\Delta$, ΔB τοῦ διξ ὑπὸ τῶν AG , GB , τούτω διαφέρει τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν $A\Delta$, ΔB , ῥητῷ δὲ διαφέρει τὸ διξ ὑπὸ τῶν $A\Delta$, ΔB τοῦ διξ ὑπὸ τῶν AG , GB . ῥητὰ γὰρ ἀμφοτέρω· ῥητῷ ἄρα διαφέρει καὶ τὰ ἀπὸ τῶν AG , GB τῶν ἀπὸ τῶν $A\Delta$, ΔB μέσα ὄντα· ὅπερ ἄτοπον.

Οὐκ ἄρα ἡ ἐκ δύο μέσων πρώτη κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται εἰς τὰ ὀνόματα· καθ' ἐν ἄρα μόνον· ὅπερ ἔδει δεῖξαι.

commensurable in square only, (and) containing a rational (area) [Prop. 10.37]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , such that AD and DB are also medial (straight-lines), commensurable in square only, (and) containing a rational (area). Since, therefore, by whatever (amount) twice the (rectangle contained) by AD and DB differs from twice the (rectangle contained) by AC and CB , (the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB by this (same amount) [Prop. 10.41 lem.]. And twice the (rectangle contained) by AD and DB differs from twice the (rectangle contained) by AC and CB by a rational (area). For (they are) both rational (areas). (The sum of) the (squares) on AC and CB thus differs from (the sum of) the (squares) on AD and DB by a rational (area, despite both) being medial (areas). The very thing is absurd [Prop. 10.26].

Thus, a first bimedral (straight-line) cannot be divided into its (component) terms at different points. Thus, (it can be so divided) at one point only. (Which is) the very thing it was required to show.

[†] In other words, $k^{1/4} + k^{3/4} = k'^{1/4} + k'^{3/4}$ has only one solution: i.e., $k' = k$.

44.

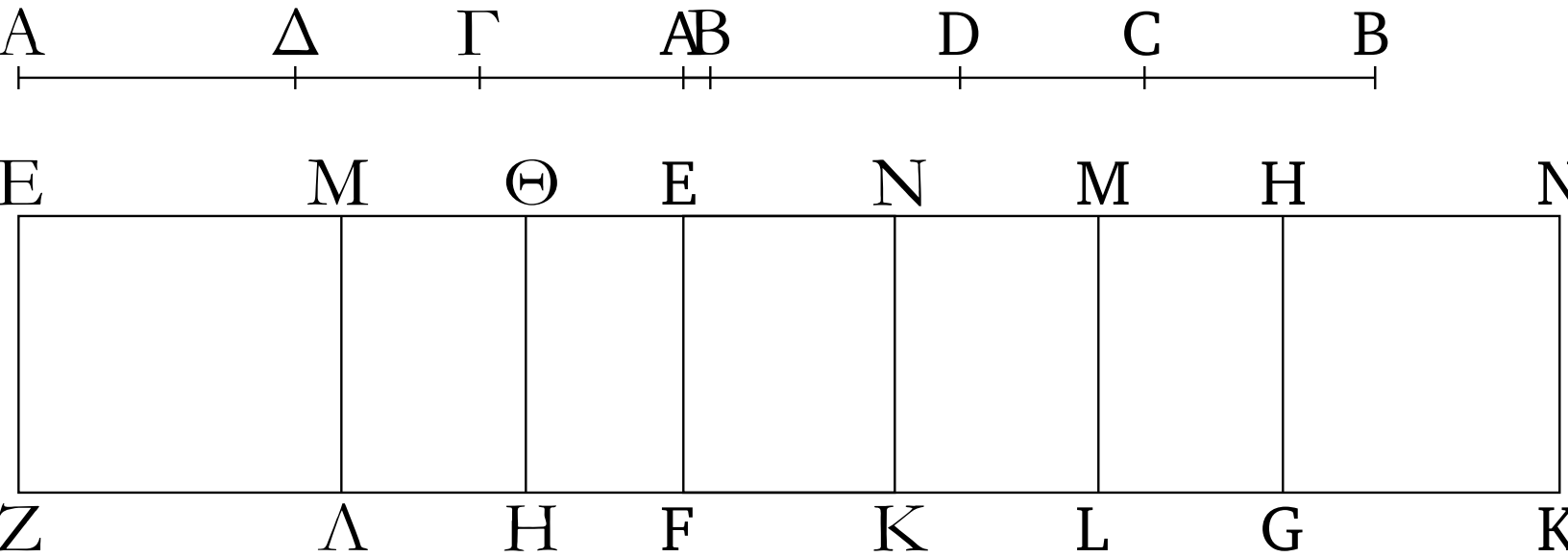
Ἡ ἐκ δύο μέσων δευτέρα καθ' ἐν μόνον σημεῖον διαιρεῖται.

Ἐστω ἐκ δύο μέσων δευτέρα ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε τὰς AG , GB μέσαξ εἶναι δυνάμει μόνον συμμετρωνξ μέσον περιεθούσαξ· φανερόν δὲ, ὅτι τὸ Γ οὐκ ἔστι κατὰ τῆς διχοτομίας, ὅτι οὐκ εἰσὶ μήκει σύμμετροι. λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται.

Proposition 44

A second bimedral (straight-line) can be divided (into its component terms) at one point only.[†]

Let AB be a second bimedral (straight-line) which has been divided at C , so that AC and BC are medial (straight-lines), commensurable in square only, (and) containing a medial (area) [Prop. 10.38]. So, (it is) clear that C is not (located) at the point of bisection, since (AC and BC) are not commensurable in length. I say that AB cannot be (so) divided at another point.



Εἰ γὰρ δυνατόν, διηρησθῶ καὶ κατὰ τὸ Δ, ὥστε τὴν ΑΓ τῇ ΔΒ μὴ εἶναι τὴν αὐτήν, ἀλλὰ μείζονα καθ' ὑπόθεσιν τὴν ΑΓ· δῆλον δὲ, ὅτι καὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ, ὡς ἐπάνω ἐδείξαμεν, ἐλάσσονα τῶν ἀπὸ τῶν ΑΓ, ΓΒ· καὶ τὰς ΑΔ, ΔΒ μέσα εἶναι δυνάμει μόνον συμμετρῶν μέσων περιεχούσας. καὶ ἐκκεῖσθω ῥητὴ ἡ ΕΖ, καὶ τῷ μὲν ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΕΖ παραλληλόγραμμον ὀρθογώνιον παραβεβλήσθω τὸ ΕΚ, τοῖς δὲ ἀπὸ τῶν ΑΓ, ΓΒ ἴσον ἀφηρησθῶ τὸ ΕΗ· λοιπὸν ἄρα τὸ ΘΚ ἴσον ἐστὶ τῷ διζῷ ὑπὸ τῶν ΑΓ, ΓΒ. πάλιν δὲ τοῖς ἀπὸ τῶν ΑΔ, ΔΒ, ἄπερ ἐλάσσονα ἐδείχθη τῶν ἀπὸ τῶν ΑΓ, ΓΒ, ἴσον ἀφηρησθῶ τὸ ΕΛ· καὶ λοιπὸν ἄρα τὸ ΜΚ ἴσον τῷ διζῷ ὑπὸ τῶν ΑΔ, ΔΒ. καὶ ἐπεὶ μέσα ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ, μέσων ἄρα [καὶ] τὸ ΕΗ. καὶ παρὰ ῥητὴν τὴν ΕΖ παράκειται· ῥητὴ ἄρα ἐστὶν ἡ ΕΘ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. διὰ τὰ αὐτὰ δὲ καὶ ἡ ΘΝ ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. καὶ ἐπεὶ αἱ ΑΓ, ΓΒ μέσα εἰσὶ δυνάμει μόνον σύμμετροι, ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΓ τῇ ΓΒ μήκει. ὥς δὲ ἡ ΑΓ πρὸς τὴν ΓΒ, οὕτως τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ὑπὸ τῶν ΑΓ, ΓΒ· ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΓ τῷ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΓ σύμμετρά ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ· δυνάμει γὰρ εἰσὶ σύμμετροι αἱ ΑΓ, ΓΒ. τῷ δὲ ὑπὸ τῶν ΑΓ, ΓΒ σύμμετρόν ἐστὶ τὸ διζῷ ὑπὸ τῶν ΑΓ, ΓΒ. καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ ἄρα ἀσύμμετρά ἐστὶ τῷ διζῷ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ ἴσον ἐστὶ τὸ ΕΗ, τῷ δὲ διζῷ ὑπὸ τῶν ΑΓ, ΓΒ ἴσον τὸ ΘΚ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΕΗ τῷ ΘΚ· ὥστε καὶ ἡ ΕΘ τῇ ΘΝ ἀσύμμετρος ἐστὶ μήκει. καὶ εἰσὶ ῥηταί· αἱ ΕΘ, ΘΝ ἄρα ῥηταὶ εἰσὶ δυνάμει μόνον σύμμετροι. ἔαν δὲ δύο ῥηταὶ δυνάμει μόνον σύμμετροι συντεθῶσιν, ἡ ὅλη ἄλογός ἐστιν ἡ καλουμένη ἐκ δύο ὀνομάτων· ἡ ΕΝ ἄρα ἐκ δύο ὀνομάτων ἐστὶ διηρημένη κατὰ τὸ Θ. κατὰ τὰ αὐτὰ δὲ δευθῆσονται καὶ αἱ ΕΜ, ΜΝ ῥηταὶ δυνάμει μόνον σύμμετροι· καὶ ἔσται ἡ ΕΝ ἐκ δύο ὀνομάτων κατ' ἄλλο καὶ ἄλλο διηρημένη τὸ τε Θ καὶ τὸ Μ, καὶ οὐκ ἔστιν ἡ ΕΘ τῇ ΜΝ ἢ αὐτῇ, ὅτι τὰ ἀπὸ τῶν

For, if possible, let it also have been (so) divided at D , so that AC is not the same as DB , but AC (is), by hypothesis, greater. So, (it is) clear that (the sum of) the (squares) on AD and DB is also less than (the sum of) the (squares) on AC and CB , as we showed above [Prop. 10.41 lem.]. And AD and DB are medial (straight-lines), commensurable in square only, (and) containing a medial (area). And let the rational (straight-line) EF be laid down. And let the rectangular parallelogram EK , equal to the (square) on AB , be applied to EF . And let EG , equal to (the sum of) the (squares) on AC and CB , be cut off (from EK). Thus, the remainder, HK , is equal to twice the (rectangle contained) by AC and CB [Prop. 2.4]. So, again, let EL , equal to (the sum of) the (squares) on AD and DB —which was shown (to be) less than (the sum of) the (squares) on AC and CB —be cut off (from EK). And, thus, the remainder, MK , (is) equal to twice the (rectangle contained) by AD and DB . And since (the sum of) the (squares) on AC and CB is medial, EG (is) thus [also] medial. And it is applied to the rational (straight-line) EF . Thus, EH is rational, and incommensurable in length with EF [Prop. 10.22]. So, for the same (reasons), HN is also rational, and incommensurable in length with EF . And since AC and CB are medial (straight-lines which are) commensurable in square only, AC is thus incommensurable in length with CB . And as AC (is) to CB , so the (square) on AC (is) to the (rectangle contained) by AC and CB [Prop. 10.21 lem.]. Thus, the (square) on AC is incommensurable with the (rectangle contained) by AC and CB [Prop. 10.11]. But, (the sum of) the (squares) on AC and CB is commensurable with the (square) on AC . For, AC and CB are commensurable in square [Prop. 10.15]. And twice the (rectangle contained) by AC and CB is commensurable with the (rectangle contained) by AC and CB [Prop. 10.6]. And thus (the sum of) the squares on AC and CB is incommen-

$ΑΓ, ΓΒ$ μείζονά ἐστι τῶν ἀπὸ τῶν $ΑΔ, ΔΒ$. ἀλλὰ τὰ ἀπὸ τῶν $ΑΔ, ΔΒ$ μείζονά ἐστι τοῦ διξ ὑπὸ $ΑΔ, ΔΒ$. πολλῶν ἄρα καὶ τὰ ἀπὸ τῶν $ΑΓ, ΓΒ$, τουτέστι τὸ $ΕΗ$, μείζον ἐστι τοῦ διξ ὑπὸ τῶν $ΑΔ, ΔΒ$, τουτέστι τοῦ $ΜΚ$. ὥστε καὶ ἡ $ΕΘ$ τῆς $ΜΝ$ μείζων ἐστίν. ἡ ἄρα $ΕΘ$ τῇ $ΜΝ$ οὐκ ἔστιν ἡ αὐτή· ὅπερ ἔδει δεῖχαι.

surable with twice the (rectangle contained) by AC and CB [Prop. 10.13]. But, EG is equal to (the sum of) the (squares) on AC and CB , and HK equal to twice the (rectangle contained) by AC and CB . Thus, EG is incommensurable with HK . Hence, EH is also incommensurable in length with HN [Props. 6.1, 10.11]. And (they are) rational (straight-lines). Thus, EH and HN are rational (straight-lines which are) commensurable in square only. And if two rational (straight-lines which are) commensurable in square only are added together then the whole (straight-line) is that irrational called binomial [Prop. 10.36]. Thus, EN is a binomial (straight-line) which has been divided (into its component terms) at H . So, according to the same (reasoning), EM and MN can be shown (to be) rational (straight-lines which are) commensurable in square only. And EN will (thus) be a binomial (straight-line) which has been divided (into its component terms) at the different (points) H and M (which is absurd [Prop. 10.42]). And EH is not the same as MN , since (the sum of) the (squares) on AC and CB is greater than (the sum of) the (squares) on AD and DB . But, (the sum of) the (squares) on AD and DB is greater than twice the (rectangle contained) by AD and DB [Prop. 10.59 lem.]. Thus, (the sum of) the (squares) on AC and CB —that is to say, EG —is also much greater than twice the (rectangle contained) by AD and DB —that is to say, MK . Hence, EH is also greater than MN [Prop. 6.1]. Thus, EH is not the same as MN . (Which is) the very thing it was required to show.

† In other words, $k^{1/4} + k^{1/2}/k^{1/4} = k'^{1/4} + k'^{3/4}/k'^{1/4}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$.

45.

Proposition 45

Ἡ μείζων κατὰ τὸ αὐτὸ μόνον σημεῖον διαιρεῖται.

A major (straight-line) can only be divided (into its component terms) at the same point.†



Ἐστω μείζων ἡ $ΑΒ$ διηρημένη κατὰ τὸ $Γ$, ὥστε τὰς $ΑΓ, ΓΒ$ δυνάμει ἀσυμμέτρον εἶναι ποιούσας τὸ μὲν συγκεκείμενον ἐκ τῶν ἀπὸ τῶν $ΑΓ, ΓΒ$ τετραγώνων ῥητόν, τὸ δ' ὑπὸ τῶν $ΑΓ, ΓΒ$ μέσον· λέγω, ὅτι ἡ $ΑΒ$ κατ' ἄλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ $Δ$, ὥστε καὶ τὰς $ΑΔ, ΔΒ$ δυνάμει ἀσυμμέτρον εἶναι ποιούσας τὸ μὲν συγκεκείμενον ἐκ τῶν ἀπὸ τῶν $ΑΔ, ΔΒ$ ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον. καὶ ἐπεὶ, ὥ διαφέρει τὰ ἀπὸ τῶν $ΑΓ, ΓΒ$ τῶν ἀπὸ τῶν $ΑΔ, ΔΒ$, τούτῳ διαφέρει καὶ τὸ διξ ὑπὸ τῶν $ΑΔ, ΔΒ$ τοῦ διξ ὑπὸ τῶν $ΑΓ, ΓΒ$, ἀλλὰ τὰ ἀπὸ τῶν $ΑΓ, ΓΒ$ τῶν ἀπὸ τῶν $ΑΔ, ΔΒ$ ὑπερέθει ῥητῶ· ῥητὰ γὰρ ἀμφοτέρω· καὶ τὸ διξ ὑπὸ τῶν $ΑΔ, ΔΒ$ ἄρα τοῦ διξ ὑπὸ τῶν $ΑΓ, ΓΒ$ ὑπερέθει ῥητῶ μέσα ὄντα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ μείζων κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται· κατὰ τὸ αὐτὸ ἄρα μόνον διαιρεῖται.

Let AB be a major (straight-line) which has been divided at C , so that AC and CB are incommensurable in square, making the sum of the squares on AC and CB rational, and the (rectangle contained) by AC and CD medial [Prop. 10.39]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , such that AD and DB are also incommensurable in square, making the sum of the (squares) on AD and DB rational, and the (rectangle contained) by them medial. And since, by whatever (amount the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB , twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by this (same amount). But, (the sum of) the (squares) on AC and CB exceeds (the sum of) the

ὅπερ ἔδει δεῖχαι.

(squares) on AD and DB by a rational (area). For (they are) both rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area), (despite both) being medial (areas). The very thing is impossible [Prop. 10.26]. Thus, a major (straight-line) cannot be divided (into its component terms) at different points. Thus, it can only be (so) divided at the same (point). (Which is) the very thing it was required to show.

[†] In other words, $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} = \sqrt{[1 + k'/(1 + k'^2)^{1/2}]/2} + \sqrt{[1 - k'/(1 + k'^2)^{1/2}]/2}$ has only one solution: i.e., $k' = k$.

46.

Proposition 46

Ἡ ῥητὸν καὶ μέσον δυναμένη καθ' ἓν μόνον σημεῖον διαιρεῖται.

The square-root of a rational plus a medial (area) can be divided (into its component terms) at one point only.[†]



Ἐστω ῥητὸν καὶ μέσον δυναμένη ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε τὰς AG , GB δυνάμει ἀσυνμέτρους εἶναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσον, τὸ δὲ διξ ὑπὸ τῶν AG , GB ῥητὸν· λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ , ὥστε καὶ τὰς $A\Delta$, ΔB δυνάμει ἀσυνμέτρους εἶναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $A\Delta$, ΔB μέσον, τὸ δὲ διξ ὑπὸ τῶν $A\Delta$, ΔB ῥητὸν. ἐπεὶ οὖν, ὧ διαφέρει τὸ διξ ὑπὸ τῶν AG , GB τοῦ διξ ὑπὸ τῶν $A\Delta$, ΔB , τούτῳ διαφέρει καὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB τῶν ἀπὸ τῶν AG , GB , τὸ δὲ διξ ὑπὸ τῶν AG , GB τοῦ διξ ὑπὸ τῶν $A\Delta$, ΔB ὑπερέθει ῥητῶ, καὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB ἄρα τῶν ἀπὸ τῶν AG , GB ὑπερέθει ῥητῶ μέσα ὄντα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ῥητὸν καὶ μέσον δυναμένη κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται. κατὰ ἓν ἄρα σημεῖον διαιρεῖται· ὅπερ ἔδει δεῖχαι.

Let AB be the square-root of a rational plus a medial (area) which has been divided at C , so that AC and CB are incommensurable in square, making the sum of the (squares) on AC and CB medial, and twice the (rectangle contained) by AC and CB rational [Prop. 10.40]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D , so that AD and DB are also incommensurable in square, making the sum of the (squares) on AD and DB medial, and twice the (rectangle contained) by AD and DB rational. Therefore, since by whatever (amount) twice the (rectangle contained) by AC and CB differs from twice the (rectangle contained) by AD and DB , (the sum of) the (squares) on AD and DB also differs from (the sum of) the (squares) on AC and CB by this (same amount). And twice the (rectangle contained) by AC and CB exceeds twice the (rectangle contained) by AD and DB by a rational (area). (The sum of) the (squares) on AD and DB thus also exceeds (the sum of) the (squares) on AC and CB by a rational (area), (despite both) being medial (areas). The very thing is impossible [Prop. 10.26]. Thus, the square-root of a rational plus a medial (area) cannot be divided (into its component terms) at different points. Thus, it can be (so) divided at one point (only). (Which is) the very thing it was required to show.

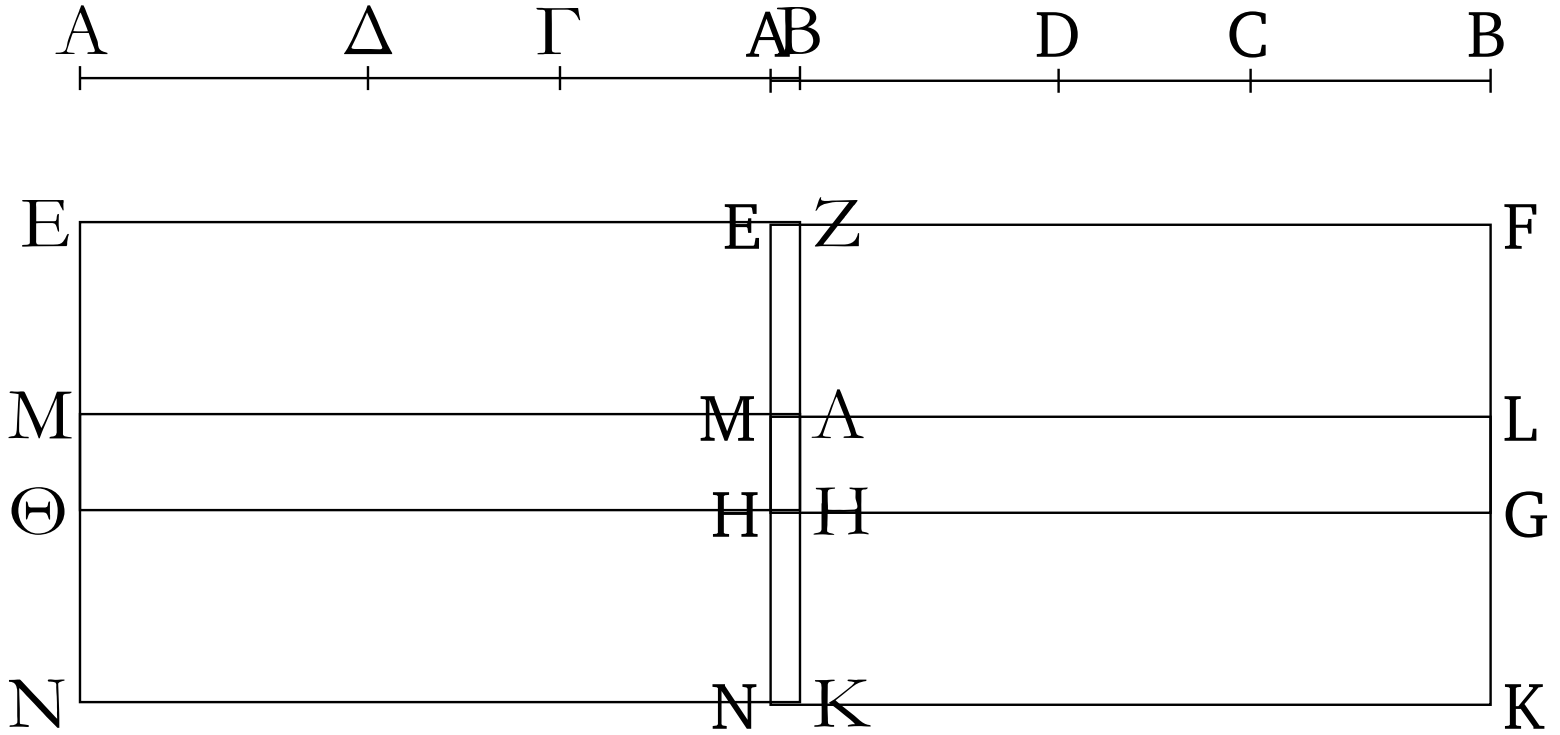
[†] In other words, $\sqrt{[(1 + k^2)^{1/2} + k]/[2(1 + k^2)]} + \sqrt{[(1 + k^2)^{1/2} - k]/[2(1 + k^2)]} = \sqrt{[(1 + k'^2)^{1/2} + k']/[2(1 + k'^2)]} + \sqrt{[(1 + k'^2)^{1/2} - k']/[2(1 + k'^2)]}$ has only one solution: i.e., $k' = k$.

47.

Proposition 47

Ἡ δύο μέσα δυναμένη καθ' ἓν μόνον σημεῖον διαιρεῖται.

The square-root of (the sum of) two medial (areas) can be divided (into its component terms) at one point only.[†]



Ἐστω [δύο μέσα δυναμένη] ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε τὰς AG , GB δυνάμει ἀσυνμέτρους εἶναι ποιοῦσας τὸ τε συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσων καὶ τὸ ὑπὸ τῶν AG , GB μέσων καὶ ἔτι ἀσύμμετρον τῷ συγκεμένῳ ἐκ τῶν ἀπ' αὐτῶν. λέγω, ὅτι ἡ AB κατ' ἄλλο σημεῖον οὐ διαιρεῖται ποιοῦσα τὰ προκείμενα.

Εἰ γὰρ δυνατόν, διηρήσθω κατὰ τὸ Δ , ὥστε πάλιν δηλονότι τὴν AG τῇ ΔB μὴ εἶναι τὴν αὐτήν, ἀλλὰ μείζονα καθ' ὑπόθεσιν τὴν AG , καὶ ἐκλείσθω ῥητὴ ἡ EZ , καὶ παραβεβλήσθω παρὰ τὴν EZ τοῖς μὲν ἀπὸ τῶν AG , GB ἴσον τὸ EH , τῷ δὲ διξ ὑπὸ τῶν AG , GB ἴσον τὸ ΘK . ὅλον ἄρα τὸ EK ἴσον ἐστὶ τῷ ἀπὸ τῆς AB τετραγώνῳ. πάλιν δὲ παραβεβλήσθω παρὰ τὴν EZ τοῖς ἀπὸ τῶν $A\Delta$, ΔB ἴσον τὸ EL . λοιπὸν ἄρα τὸ διξ ὑπὸ τῶν $A\Delta$, ΔB λοιπῷ τῷ MK ἴσον ἐστίν. καὶ ἐπεὶ μέσον ὑπόκειται τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσων ἄρα ἐστὶ καὶ τὸ EH . καὶ παρὰ ῥητὴν τὴν EZ παράκειται ῥητὴ ἄρα ἐστὶν ἡ ΘE καὶ ἀσύμμετρος τῇ EZ μήκει. διὰ τὰ αὐτὰ δὲ καὶ ἡ ΘN ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB τῷ διξ ὑπὸ τῶν AG , GB , καὶ τὸ EH ἄρα τῷ HN ἀσύμμετρόν ἐστιν. ὥστε καὶ ἡ $E\Theta$ τῇ ΘN ἀσύμμετρος ἐστίν. καὶ εἰσι ῥηταί· αἱ $E\Theta$, ΘN ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ EN ἄρα ἐκ δύο ὀνομάτων ἐστὶ διηρημένη κατὰ τὸ Θ . ὁμοίως δὲ δείχομεν, ὅτι καὶ κατὰ τὸ M διήρηται. καὶ οὐκ ἐστὶν ἡ $E\Theta$ τῇ MN ἡ αὐτή· ἡ ἄρα ἐκ δύο ὀνομάτων κατ' ἄλλο καὶ ἄλλο σημεῖον διήρηται. ὅπερ ἐστὶν ἄτοπον. οὐκ ἄρα ἡ δύο μέσα δυναμένη κατ' ἄλλο καὶ ἄλλο σημεῖον διαιρεῖται· καθ' ἓν ἄρα μόνον [σημεῖον] διαιρεῖται.

Let AB be [the square-root of (the sum of) two medial (areas)] which has been divided at C , such that AC and CB are incommensurable in square, making the sum of the (squares) on AC and CB medial, and the (rectangle contained) by AC and CB medial, and, moreover, incommensurable with the sum of the (squares) on (AC and CB) [Prop. 10.41]. I say that AB cannot be divided at another point fulfilling the prescribed (conditions).

For, if possible, let it be divided at D , such that AC is again manifestly not the same as DB , but AC (is), by hypothesis, greater. And let the rational (straight-line) EF be laid down. And let EG , equal to (the sum of) the (squares) on AC and CB , and HK , equal to twice the (rectangle contained) by AC and CB , be applied to EF . Thus, the whole of EK is equal to the square on AB [Prop. 2.4]. So, again, let EL , equal to (the sum of) the (squares) on AD and DB , be applied to EF . Thus, the remainder—twice the (rectangle contained) by AD and DB —is equal to the remainder, MK . And since the sum of the (squares) on AC and CB was assumed (to be) medial, EG is also medial. And it is applied to the rational (straight-line) EF . HE is thus rational, and incommensurable in length with EF [Prop. 10.22]. So, for the same (reasons), HN is also rational, and incommensurable in length with EF . And since the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , EG is thus also incommensurable with GN . Hence, EH is also incommensurable with HN [Props. 6.1, 10.11]. And they are (both) rational (straight-lines). Thus, EH and HN are rational (straight-lines).

which are) commensurable in square only. Thus, EN is a binomial (straight-line) which has been divided (into its component terms) at H [Prop. 10.36]. So, similarly, we can show that it has also been (so) divided at M . And EH is not the same as MN . Thus, a binomial (straight-line) has been divided (into its component terms) at different points. The very thing is absurd [Prop. 10.42]. Thus, the square-root of (the sum of) two medial (areas) cannot be divided (into its component terms) at different points. Thus, it can be (so) divided at one [point] only.

[†] In other words, $k'^{1/4} \sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + k'^{1/4} \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} = k''^{1/4} \sqrt{[1 + k''/(1 + k''^2)^{1/2}]/2} + k''^{1/4} \sqrt{[1 - k''/(1 + k''^2)^{1/2}]/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$.

Ὅροι δεῦτεροι.

5. Ὑποκειμένης ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων διηρημένης εἰς τὰ ὀνόματα, ἥς τὸ μείζον ὄνομα τοῦ ἐλάσσονος μείζον δύναται τῷ ἀπὸ συμμετρον ἑαυτῇ μήκει, ἔαν μὲν τὸ μείζον ὄνομα σύμμετρον ᾗ μήκει τῇ ἐκκειμένη ῥητῇ, καλείσθω [ἡ ὅλη] ἐκ δύο ὀνομάτων πρώτη.

6. Ἐάν δὲ τὸ ἐλάσσον ὄνομα σύμμετρον ᾗ μήκει τῇ ἐκκειμένη ῥητῇ, καλείσθω ἐκ δύο ὀνομάτων δευτέρα.

7. Ἐάν δὲ μηδέτερον τῶν ὀνομάτων σύμμετρον ᾗ μήκει τῇ ἐκκειμένη ῥητῇ, καλείσθω ἐκ δύο ὀνομάτων τρίτη.

8. Πάλιν δὲ ἔαν τὸ μείζον ὄνομα [τοῦ ἐλάσσονος] μείζον δύνηται τῷ ἀπὸ ἀσυμμετρον ἑαυτῇ μήκει, ἔαν μὲν τὸ μείζον ὄνομα σύμμετρον ᾗ μήκει τῇ ἐκκειμένη ῥητῇ, καλείσθω ἐκ δύο ὀνομάτων τετάρτη.

9. Ἐάν δὲ τὸ ἐλάσσον, πέμπτη.

10. Ἐάν δὲ μηδέτερον, ἕκτη.

Definitions II

5. Given a rational (straight-line), and a binomial (straight-line) which has been divided into its (component) terms, of which the square on the greater term is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable in length with (the greater) then, if the greater term is commensurable in length with the rational (straight-line previously) laid out, let [the whole] (straight-line) be called a first binomial (straight-line).

6. And if the lesser term is commensurable in length with the rational (straight-line previously) laid out then let (the whole straight-line) be called a second binomial (straight-line).

7. And if neither of the terms is commensurable in length with the rational (straight-line previously) laid out then let (the whole straight-line) be called a third binomial (straight-line).

8. So, again, if the square on the greater term is larger than (the square on) [the lesser] by the (square) on (some straight-line) incommensurable in length with (the greater) then, if the greater term is commensurable in length with the rational (straight-line previously) laid out, let (the whole straight-line) be called a fourth binomial (straight-line).

9. And if the lesser (term is commensurable), a fifth (binomial straight-line).

10. And if neither (term is commensurable), a sixth (binomial straight-line).

48.

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων πρώτην.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν συγκεῖμενον ἔχ αὐτῶν τὸν $ΑΒ$ πρὸς μὲν τὸν $ΒΓ$ λόγον ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν $ΓΑ$ λόγον μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ ἐκκείσθω τις ῥητὴ ἡ $Δ$, καὶ τῇ $Δ$ σύμμετρος ἔστω μήκει ἡ $ΕΖ$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΕΖ$. καὶ γερονέτω ὥς ὁ $ΒΑ$ ἀριθμὸς πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΕΖ$ πρὸς

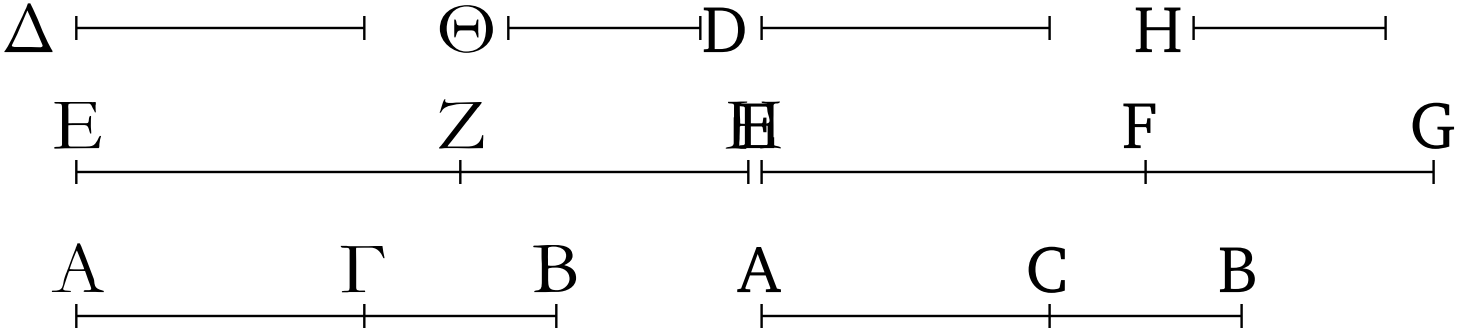
Proposition 48

To find a first binomial (straight-line).

Let two numbers AC and CB be laid down such that their sum AB has to BC the ratio which (some) square number (has) to (some) square number, and does not have to CA the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let some rational (straight-line) D be laid down. And let EF be commensurable in length with D . EF is thus also rational [Def. 10.3]. And let it be contrived

τὸ ἀπὸ τῆς ZH . ὁ δὲ AB πρὸς τὸν AG λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν· καὶ τὸ ἀπὸ τῆς EZ ἄρα πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν· ὥστε σύμμετρον ἐστὶ τὸ ἀπὸ τῆς EZ τῷ ἀπὸ τῆς ZH . καὶ ἐστὶ ῥητὴ ἡ EZ . ῥητὴ ἄρα καὶ ἡ ZH . καὶ ἐπεὶ ὁ BA πρὸς τὸν AG λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς EZ ἄρα πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ EZ τῇ ZH μήκει. αἱ EZ , ZH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EH . λέγω, ὅτι καὶ πρώτη.

that as the number BA (is) to AC , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. And AB has to AC the ratio which (some) number (has) to (some) number. Thus, the (square) on EF also has to the (square) on FG the ratio which (some) number (has) to (some) number. Hence, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. And EF is rational. Thus, FG (is) also rational. And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, thus the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop 10.9]. EF and FG are thus rational (straight-lines which are) commensurable in square only. Thus, EG is a binomial (straight-line) [Prop. 10.36]. I say that (it is) also a first (binomial straight-line).



Ἐπεὶ γάρ ἐστιν ὡς ὁ BA ἀριθμὸς πρὸς τὸν AG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH , μείζων δὲ ὁ BA τοῦ AG , μείζων ἄρα καὶ τὸ ἀπὸ τῆς EZ τοῦ ἀπὸ τῆς ZH . ἔστω οὖν τῷ ἀπὸ τῆς EZ ἴσα τὰ ἀπὸ τῶν ZH , Θ . καὶ ἐπεὶ ἐστὶν ὡς ὁ BA πρὸς τὸν AG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH , ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ AB πρὸς τὸν BG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς Θ . ὁ δὲ AB πρὸς τὸν BG λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ τὸ ἀπὸ τῆς EZ ἄρα πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. σύμμετρος ἄρα ἐστὶν ἡ EZ τῇ Θ μήκει· ἡ EZ ἄρα τῆς ZH μείζων δύναται τῷ ἀπὸ συμμέτρου ἐκαστῆς. καὶ εἰσι ῥηταὶ αἱ EZ , ZH , καὶ σύμμετρος ἡ EZ τῇ Δ μήκει.

Ἡ EH ἄρα ἐκ δύο ὀνομάτων ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

For since as the number BA is to AC , so the (square) on EF (is) to the (square) on FG , and BA (is) greater than AC , the (square) on EF (is) thus also greater than the (square) on FG [Prop. 5.14]. Therefore, let (the sum of) the (squares) on FG and H be equal to the (square) on EF . And since as BA is to AC , so the (square) on EF (is) to the (square) on FG , thus, via conversion, as AB is to BC , so the (square) on EF (is) to the (square) on H [Prop. 5.19 corr.]. And AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, EF is commensurable in length with H [Prop. 10.9]. Thus, the square on EF is greater than (the square on) FG by the (square) on (some straight-line) commensurable (in length) with (EF). And EF and FG are rational (straight-lines). And EF (is) commensurable in length with D .

Thus, EG is a first binomial (straight-line) [Def. 10.5].[†] (Which is) the very thing it was required to show.

[†]If the rational straight-line has unit length then the length of a first binomial straight-line is $k + k\sqrt{1 - k'^2}$. This, and the first apotome, whose length is $k - k\sqrt{1 - k'^2}$ [Prop. 10.85], are the roots of $x^2 - 2kx + k^2k'^2 = 0$.

[†] If the rational straight-line has unit length then the length of a second binomial straight-line is $k/\sqrt{1-k'^2} + k$. This, and the second apotome, whose length is $k/\sqrt{1-k'^2} - k$ [Prop. 10.86], are the roots of $x^2 - (2k/\sqrt{1-k'^2})x + k^2/[k'^2/(1-k'^2)] = 0$.

50.

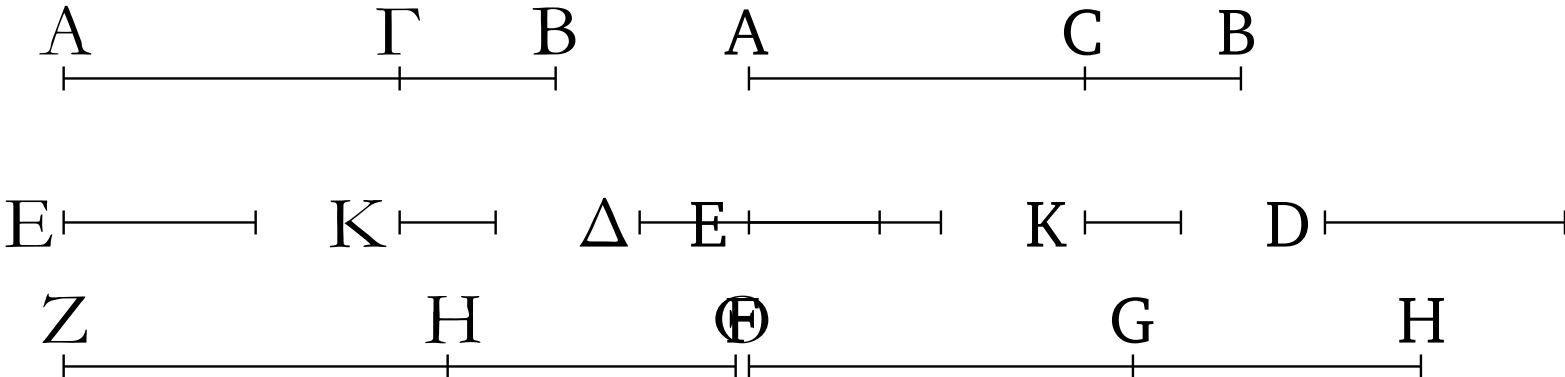
Proposition 50

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων τρίτην.

Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν συγκε-
 ῖμενον ἔχ αὐτῶν τὸν $ΑΒ$ πρὸς μὲν τὸν $ΒΓ$ λόγον ἔθειν, ὃν
 τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν
 $ΑΓ$ λόγον μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον
 ἀριθμόν. ἐκκεῖσθω δὲ τις καὶ ἄλλος μὴ τετράγωνος ἀριθμὸς
 ὁ $Δ$, καὶ πρὸς ἐκάτερον τῶν $ΒΑ$, $ΑΓ$ λόγον μὴ ἐθέτω, ὃν
 τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ ἐκκεῖσθω
 τις ῥητὴ εὐθεῖα ἡ $Ε$, καὶ γερονέτω ὡς ὁ $Δ$ πρὸς τὸν $ΑΒ$,
 οὕτως τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$ · σύμμετρον ἄρα
 ἐστὶ τὸ ἀπὸ τῆς $Ε$ τῷ ἀπὸ τῆς $ΖΗ$. καὶ ἐστὶ ῥητὴ ἡ $Ε$ · ῥητὴ
 ἄρα ἐστὶ καὶ ἡ $ΖΗ$. καὶ ἐπεὶ ὁ $Δ$ πρὸς τὸν $ΑΒ$ λόγον οὐκ
 ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ
 τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$ λόγον ἔθει, ὃν τετράγωνος
 ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ
 $Ε$ τῇ $ΖΗ$ μήκει. γερονέτω δὴ πάλιν ὡς ἡ $ΒΑ$ ἀριθμὸς πρὸς
 τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$ · σύμ-
 μετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΖΗ$ τῷ ἀπὸ τῆς $ΗΘ$. ῥητὴ δὲ ἡ
 $ΖΗ$ · ῥητὴ ἄρα καὶ ἡ $ΗΘ$. καὶ ἐπεὶ ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$ λόγον
 οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν,
 οὐδὲ τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΘΗ$ λόγον ἔθει, ὃν
 τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος
 ἄρα ἐστὶν ἡ $ΖΗ$ τῇ $ΗΘ$ μήκει. αἱ $ΖΗ$, $ΗΘ$ ἄρα ῥηταὶ εἰσι
 δυνάμει μόνον σύμμετροι· ἡ $ΖΘ$ ἄρα ἐκ δύο ὀνομάτων ἐστίν.
 λέγω δὴ, ὅτι καὶ τρίτην.

To find a third binomial (straight-line).

Let the two numbers AC and CB be laid down such that
 their sum AB has to BC the ratio which (some) square num-
 ber (has) to (some) square number, and does not have to AC
 the ratio which (some) square number (has) to (some) square
 number. And let some other non-square number D also be
 laid down, and let it not have to each of BA and AC the ratio
 which (some) square number (has) to (some) square number.
 And let some rational straight-line E be laid down, and let it
 be contrived that as D (is) to AB , so the (square) on E (is)
 to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square)
 on E is commensurable with the (square) on FG [Prop. 10.6].
 And E is a rational (straight-line). Thus, FG is also a ratio-
 nal (straight-line). And since D does not have to AB the ra-
 tio which (some) square number has to (some) square num-
 ber, the (square) on E does not have to the (square) on FG
 the ratio which (some) square number (has) to (some) square
 number either. E is thus incommensurable in length with FG
 [Prop. 10.9]. So, again, let it be contrived that as the number
 BA (is) to AC , so the (square) on FG (is) to the (square) on GH
 [Prop. 10.6 corr.]. Thus, the (square) on FG is commensurable
 with the (square) on GH [Prop. 10.6]. And FG (is) a rational
 (straight-line). Thus, GH (is) also a rational (straight-line).
 And since BA does not have to AC the ratio which (some)
 square number (has) to (some) square number, the (square)
 on FG does not have to the (square) on HG the ratio which
 (some) square number (has) to (some) square number either.
 Thus, FG is incommensurable in length with GH [Prop. 10.9].
 FG and GH are thus rational (straight-lines which are) com-
 mensurable in square only. Thus, FH is a binomial (straight-
 line) [Prop. 10.36]. So, I say that (it is) also a third (binomial
 straight-line).



Ἐπεὶ γάρ ἐστιν ὡς ὁ $Δ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς
 $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$, ὡς δὲ ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως

For since as D is to AB , so the (square) on E (is) to the
 (square) on FG , and as BA (is) to AC , so the (square) on FG

τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, δι' ἴσον ἄρα ἐστὶν ὡς ὁ Δ πρὸς τὸν $ΑΓ$, οὔτως τὸ ἀπὸ τῆς E πρὸς τὸ ἀπὸ τῆς $H\Theta$. ὁ δὲ Δ πρὸς τὸν $ΑΓ$ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὲ τὸ ἀπὸ τῆς E ἄρα πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ E τῇ $H\Theta$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ὁ BA πρὸς τὸν $ΑΓ$, οὔτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, μείζον ἄρα τὸ ἀπὸ τῆς ZH τοῦ ἀπὸ τῆς $H\Theta$. ἔστω οὖν τῶ ἀπὸ τῆς ZH ἴσα τὰ ἀπὸ τῶν $H\Theta$, K · ἀναστρέψαντι ἄρα [ἐστὶν] ὡς ὁ AB πρὸς τὸν $BΓ$, οὔτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ AB πρὸς τὸν $BΓ$ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὸ ἀπὸ τῆς ZH ἄρα πρὸς τὸ ἀπὸ τῆς K λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα [ἐστὶν] ἡ ZH τῇ K μήκει. ἡ ZH ἄρα τῆς $H\Theta$ μείζον δύναται τῶ ἀπὸ σύμμετρον ἑαντῇ. καὶ εἰσιν αἱ ZH , $H\Theta$ ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα αὐτῶν σύμμετρος ἐστὶ τῇ E μήκει.

Ἡ $Z\Theta$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη· ὅπερ ἔδει δεῖξαι.

(is) to the (square) on GH , thus, via equality, as D (is) to AC , so the (square) on E (is) to the (square) on GH [Prop. 5.22]. And D does not have to AC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on E does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, E is incommensurable in length with GH [Prop. 10.9]. And since as BA is to AC , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus greater than the (square) on GH [Prop. 5.14]. Therefore, let (the sum of) the (squares) on GH and K be equal to the (square) on FG . Thus, via conversion, as AB [is] to BC , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on K the ratio which (some) square number (has) to (some) square number. Thus, FG [is] commensurable in length with K [Prop. 10.9]. Thus, the square on FG is greater than (the square on) GH by the (square) on (some straight-line) commensurable (in length) with (FG) . And FG and GH are rational (straight-lines which are) commensurable in square only, and neither of them is commensurable in length with E .

Thus, FH is a third binomial (straight-line) [Def. 10.7].[†] (Which is) the very thing it was required to show.

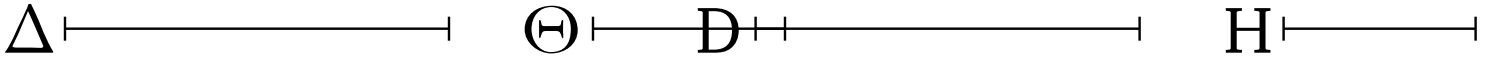
[†] If the rational straight-line has unit length then the length of a third binomial straight-line is $k^{1/2}(1 + \sqrt{1 - k'^2})$. This, and the third apotome, whose length is $k^{1/2}(1 - \sqrt{1 - k'^2})$ [Prop. 10.87], are the roots of $x^2 - 2k^{1/2}x + k'^2 = 0$.

51.

Proposition 51

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων τετάρτην.

To find a fourth binomial (straight-line).



Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν AB πρὸς τὸν $BΓ$ λόγον μὴ ἔχειν μήτε μὴν πρὸς τὸν $ΑΓ$, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ ἐκκεῖσθω ῥητὴ ἡ Δ , καὶ τῇ Δ σύμμετρος ἔστω μήκει ἡ EZ · ῥητὴ ἄρα ἐστὶ καὶ ἡ EZ . καὶ γεγονέντω ὡς ὁ BA ἀριθμὸς πρὸς τὸν $ΑΓ$, οὔτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH · σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ

Let the two numbers AC and CB be laid down such that AB does not have to BC , or to AC either, the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let the rational (straight-line) D be laid down. And let EF be commensurable in length with D . Thus, EF is also a rational (straight-line). And let it be

τῆς EZ τῷ ἀπὸ τῆς ZH ῥητῇ ἄρα ἐστὶ καὶ ἡ ZH . καὶ ἐπεὶ ὁ BA πρὸς τὸν AG λόγον οὐκ ἔθλει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔθλει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ EZ τῇ ZH μήκει. αἱ EZ , ZH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ὥστε ἡ EH ἐκ δύο ὀνομάτων ἐστίν. λέγω δὴ, ὅτι καὶ τετάρτη.

Ἐπεὶ γάρ ἐστιν ὡς ὁ BA πρὸς τὸν AG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH [μείζων δὲ ὁ BA τοῦ AG], μείζον ἄρα τὸ ἀπὸ τῆς EZ τοῦ ἀπὸ τῆς ZH . ἔστω οὖν τῷ ἀπὸ τῆς EZ ἴσα τὰ ἀπὸ τῶν ZH , Θ · ἀναστρέψαντι ἄρα ὡς ὁ AB ἀριθμὸς πρὸς τὸν BG , οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς Θ . ὁ δὲ AB πρὸς τὸν BG λόγον οὐκ ἔθλει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδ' ἄρα τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔθλει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ EZ τῇ Θ μήκει· ἡ EZ ἄρα τῆς HZ μείζον δύνανται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ. καὶ εἰσιν αἱ EZ , ZH ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ ἡ EZ τῇ Δ σύμμετρος ἐστὶ μήκει.

Ἡ EH ἄρα ἐκ δύο ὀνομάτων ἐστὶ τετάρτη· ὅπερ ἔδει δεῖν.

contrived that as the number BA (is) to AC , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. Thus, FG is also a rational (straight-line). And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop. 10.9]. Thus, EF and FG are rational (straight-lines which are) commensurable in square only. Hence, EG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fourth (binomial straight-line).

For since as BA is to AC , so the (square) on EF (is) to the (square) on FG [and BA (is) greater than AC], the (square) on EF (is) thus greater than the (square) on FG [Prop. 5.14]. Therefore, let (the sum of) the squares on FG and H be equal to the (square) on EF . Thus, via conversion, as the number AB (is) to BC , so the (square) on EF (is) to the (square) on H [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with H [Prop. 10.9]. Thus, the square on EF is greater than (the square on) GF by the (square) on (some straight-line) incommensurable (in length) with (EF). And EF and FG are rational (straight-lines which are) commensurable in square only. And EF is commensurable in length with D .

Thus, EG is a fourth binomial (straight-line) [Def. 10.8].[†] (Which is) the very thing it was required to show.

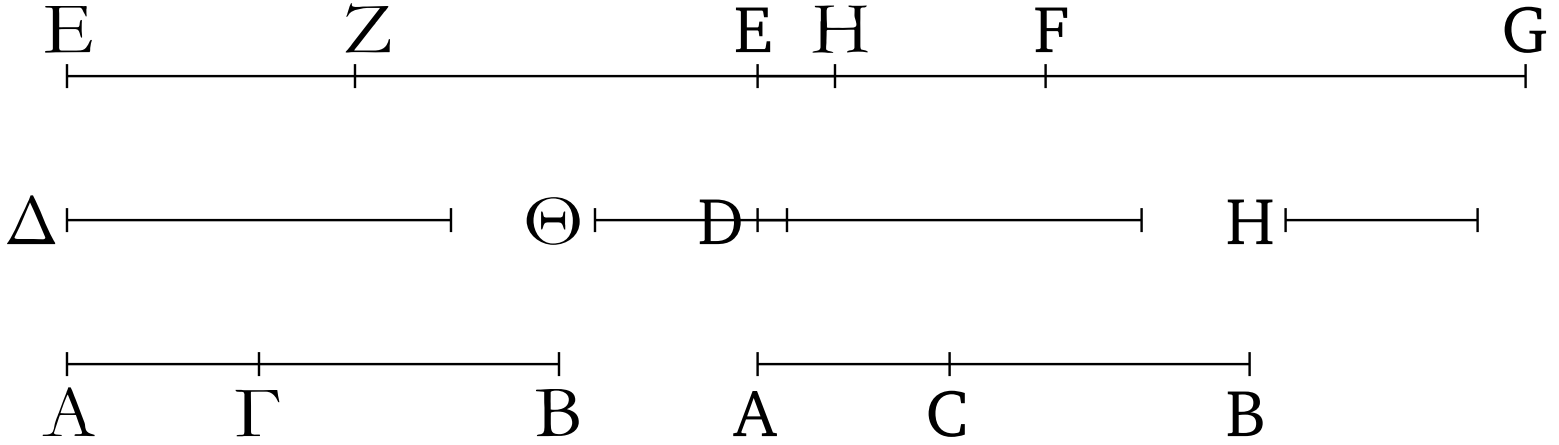
[†] If the rational straight-line has unit length then the length of a fourth binomial straight-line is $k(1 + 1/\sqrt{1+k'})$. This, and the fourth apotome, whose length is $k(1 - 1/\sqrt{1+k'})$ [Prop. 10.88], are the roots of $x^2 - 2kx + k^2k'/(1+k') = 0$.

52.

Εὕρεῖν τὴν ἐκ δύο ὀνομάτων πέμπτην.

Proposition 52

To find a fifth binomial straight-line.



Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν $ΑΒ$ πρὸς ἑκάτερον αὐτῶν λόγον μὴ ἔδειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ ἐκκεῖσθω ῥητὴ τις εὐθεῖα ἡ $Δ$, καὶ τῇ $Δ$ σύμμετρος ἔστω [μήκει] ἡ $ΕΖ$. ῥητὴ ἄρα ἡ $ΕΖ$. καὶ γεγονέτω ὡς ὁ $ΓΑ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $ΕΖ$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$. ὁ δὲ $ΓΑ$ πρὸς τὸν $ΑΒ$ λόγον οὐκ ἔδει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὲ τὸ ἀπὸ τῆς $ΕΖ$ ἄρα πρὸς τὸ ἀπὸ τῆς $ΖΗ$ λόγον ἔδει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. αἱ $ΕΖ$, $ΖΗ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $ΕΗ$. λέγω δὴ, ὅτι καὶ πέμπτη.

Ἐπεὶ γάρ ἐστιν ὡς ὁ $ΓΑ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $ΕΖ$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$, ἀνάπαλιν ὡς ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΖΕ$. μείζον ἄρα τὸ ἀπὸ τῆς $ΗΖ$ τοῦ ἀπὸ τῆς $ΖΕ$. ἔστω οὖν τῶ ἀπὸ τῆς $ΗΖ$ ἴσα τὰ ἀπὸ τῶν $ΕΖ$, $Θ$. ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ $ΑΒ$ ἀριθμὸς πρὸς τὸν $ΒΓ$, οὕτως τὸ ἀπὸ τῆς $ΗΖ$ πρὸς τὸ ἀπὸ τῆς $Θ$. ὁ δὲ $ΑΒ$ πρὸς τὸν $ΒΓ$ λόγον οὐκ ἔδει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδ' ἄρα τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔδει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ $ΖΗ$ τῇ $Θ$ μήκει· ὥστε ἡ $ΖΗ$ τῆς $ΖΕ$ μείζον δύναται τῶ ἀπὸ ἀσυνμέτρων ἑαυτῇ. καὶ εἰσιν αἱ $ΗΖ$, $ΖΕ$ ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ τὸ $ΕΖ$ ἕλαττον ὄνομα σύμμετρόν ἐστι τῇ ἐκκειμένῃ ῥητῇ τῇ $Δ$ μήκει.

Ἡ $ΕΗ$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ πέμπτη· ὅπερ ἔδει δεῖχαι.

Let the two numbers AC and CB be laid down such that AB does not have to either of them the ratio which (some) square number (has) to (some) square number [Prop. 10.38 lem.]. And let some rational straight-line D be laid down. And let EF be commensurable [in length] with D . Thus, EF (is) a rational (straight-line). And let it be contrived that as CA (is) to AB , so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. And CA does not have to AB the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF and FG are rational (straight-lines which are) commensurable in square only [Prop. 10.9]. Thus, EG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fifth (binomial straight-line).

For since as CA is to AB , so the (square) on EF (is) to the (square) on FG , inversely, as BA (is) to AC , so the (square) on FG (is) to the (square) on FE [Prop. 5.7 corr.]. Thus, the (square) on GF (is) greater than the (square) on FE [Prop. 5.14]. Therefore, let (the sum of) the (squares) on EF and H be equal to the (square) on GF . Thus, via conversion, as the number AB is to BC , so the (square) on GF (is) to the (square) on H [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with H [Prop. 10.9]. Hence, the square on FG is greater than (the square on) FE by the (square) on (some straight-line) incommensurable (in length) with (FG). And GF and FE are rational (straight-lines which are) commensurable in square only. And the lesser term EF is commensurable in length with the rational (straight-line previously) laid down, D .

Thus, EG is a fifth binomial (straight-line).[†] (Which is) the very thing it was required to show.

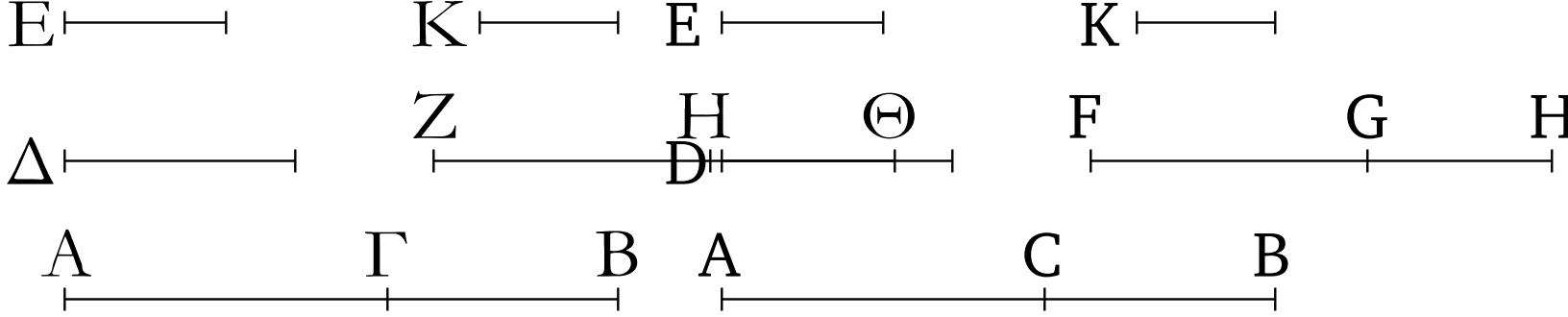
[†] If the rational straight-line has unit length then the length of a fifth binomial straight-line is $k(\sqrt{1+k'} + 1)$. This, and the fifth apotome, whose length is $k(\sqrt{1+k'} - 1)$ [Prop. 10.89], are the roots of $x^2 - 2k\sqrt{1+k'}x + k^2k' = 0$.

53.

Proposition 53

Εὐρεῖν τὴν ἐκ δύο ὀνομάτων ἑκτὴν.

To find a sixth binomial (straight-line).



Ἐκκεῖσθωσαν δύο ἀριθμοὶ οἱ $ΑΓ$, $ΓΒ$, ὥστε τὸν $ΑΒ$ πρὸς ἑκάτερον αὐτῶν λόγον μὴ ἔδειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἔστω δὲ καὶ ἕτερος ἀριθμὸς ὁ $Δ$ μὴ τετράγωνος ὡς μηδὲ πρὸς ἑκάτερον τῶν $ΒΑ$, $ΑΓ$ λόγον ἔθων, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· καὶ ἐκκεῖσθω τις ῥητὴ εὐθεῖα ἡ $Ε$, καὶ γερονέτω ὡς ὁ $Δ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$ · σύμμετρον ἄρα τὸ ἀπὸ τῆς $Ε$ τῷ ἀπὸ τῆς $ΖΗ$. καὶ ἐστὶ ῥητὴ ἡ $Ε$ · ῥητὴ ἄρα καὶ ἡ $ΖΗ$. καὶ ἐπεὶ οὐκ ἔθει ὁ $Δ$ πρὸς τὸν $ΑΒ$ λόγον, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς $Ε$ ἄρα πρὸς τὸ ἀπὸ τῆς $ΖΗ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἡ $Ε$ τῇ $ΖΗ$ μήκει. γερονέτω δὲ πάλιν ὡς ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$. σύμμετρον ἄρα τὸ ἀπὸ τῆς $ΖΗ$ τῷ ἀπὸ τῆς $ΗΘ$. ῥητὸν ἄρα τὸ ἀπὸ τῆς $ΗΘ$ · ῥητὴ ἄρα ἡ $ΗΘ$. καὶ ἐπεὶ ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδὲ τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἐστὶν ἡ $ΖΗ$ τῇ $ΗΘ$ μήκει. αἱ $ΖΗ$, $ΗΘ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $ΖΘ$. δεικτέον δὲ, ὅτι καὶ ἑκτὴ.

Ἐπεὶ γάρ ἐστιν ὡς ὁ $Δ$ πρὸς τὸν $ΑΒ$, οὕτως τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΖΗ$, ἐστὶ δὲ καὶ ὡς ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$, δι' ἴσον ἄρα ἐστὶν ὡς ὁ $Δ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $Ε$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$. ὁ δὲ $Δ$ πρὸς τὸν $ΑΓ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· οὐδὲ τὸ ἀπὸ τῆς $Ε$ ἄρα πρὸς τὸ ἀπὸ τῆς $ΗΘ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ἀσύμμετρος ἄρα ἐστὶν ἡ $Ε$ τῇ $ΗΘ$ μήκει. ἐδείχθη δὲ καὶ τῇ $ΖΗ$ ἀσύμμετρος· ἑκάτερα ἄρα τῶν $ΖΗ$, $ΗΘ$ ἀσύμμετρος ἐστὶ τῇ $Ε$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ὁ $ΒΑ$ πρὸς τὸν $ΑΓ$, οὕτως τὸ ἀπὸ τῆς $ΖΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΘ$, μείζον ἄρα τὸ ἀπὸ τῆς $ΖΗ$ τοῦ ἀπὸ τῆς $ΗΘ$. ἔστω οὖν

Let the two numbers AC and CB be laid down such that AB does not have to each of them the ratio which (some) square number (has) to (some) square number. And let D also be another number, which is not square, and does not have to each of BA and AC the ratio which (some) square number (has) to (some) square number either [Prop. 10.28 lem. I]. And let some rational straight-line E be laid down. And let it be contrived that as D (is) to AB , so the (square) on E (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on E (is) commensurable with the (square) on FG [Prop. 10.6]. And E is rational. Thus, FG (is) also rational. And since D does not have to AB the ratio which (some) square number (has) to (some) square number, the (square) on E thus does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, E (is) incommensurable in length with FG [Prop. 10.9]. So, again, let it have be contrived that as BA (is) to AC , so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.]. The (square) on FG (is) thus commensurable with the (square) on HG [Prop. 10.6]. The (square) on HG (is) thus rational. Thus, HG (is) rational. And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on FG does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. Thus, FG and GH are rational (straight-lines which are) commensurable in square only. Thus, FH is a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a sixth (binomial straight-line).

For since as D is to AB , so the (square) on E (is) to the (square) on FG , and also as BA is to AC , so the (square) on FG (is) to the (square) on GH , thus, via equality, as D is to AC , so the (square) on E (is) to the (square) on GH [Prop. 5.22]. And D does not have to AC the ratio which (some) square

τῷ ἀπὸ [τῆς] ZH ἴσα τὰ ἀπὸ τῶν $H\Theta$, K ἀναστρέψαντι ἄρα ὥς ὁ AB πρὸς $B\Gamma$, οὕτως τὸ ἀπὸ ZH πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ AB πρὸς τὸν $B\Gamma$ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν· ὥστε οὐδὲ τὸ ἀπὸ ZH πρὸς τὸ ἀπὸ τῆς K λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἐστὶν ἡ ZH τῇ K μήκει· ἡ ZH ἄρα τῆς $H\Theta$ μείζον δύναιται τῷ ἀπὸ ἀσυμμέτρον εἶναι. καὶ εἰσιν αἱ ZH , $H\Theta$ ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ οὐδετέρᾳ αὐτῶν σύμμετρος ἐστὶ μήκει τῇ ἐκκειμένη ῥῆτι τῇ E .

Ἡ $Z\Theta$ ἄρα ἐκ δύο ὀνομάτων ἐστὶν ἑκτα· ὅπερ ἔδει δεῖξαι.

number (has) to (some) square number. Thus, the (square) on E does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. E is thus incommensurable in length with GH [Prop. 10.9]. And (E) was also shown (to be) incommensurable (in length) with FG . Thus, FG and GH are each incommensurable in length with E . And since as BA is to AC , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus greater than the (square) on GH [Prop. 5.14]. Therefore, let (the sum of) the (squares) on GH and K be equal to the (square) on FG . Thus, via conversion, as AB (is) to BC , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Hence, the (square) on FG does not have to the (square) on K the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with K [Prop. 10.9]. The square on FG is thus greater than (the square on) GH by the (square) on (some straight-line which is) incommensurable (in length) with (FG). And FG and GH are rational (straight-lines which are) commensurable in square only, and neither of them is commensurable in length with the rational (straight-line) E (previously) laid down.

Thus, FH is a sixth binomial (straight-line) [Def. 10.10].[†] (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then the length of a sixth binomial straight-line is $\sqrt{k} + \sqrt{k'}$. This, and the sixth apotome, whose length is $\sqrt{k} - \sqrt{k'}$ [Prop. 10.90], are the roots of $x^2 - 2\sqrt{k}x + (k - k') = 0$.

Λήμμα.

Ἐστω δύο τετράγωνα τὰ AB , $B\Gamma$ καὶ κείσθωσαν ὥστε ἐπ' ἐνθείᾳ εἶναι τὴν ΔB τῇ BE · ἐπ' ἐνθείᾳ ἄρα ἐστὶ καὶ ἡ ZB τῇ BH . καὶ συμπεπληρώσθω τὸ AG παραλληλόγραμμον· λέγω, ὅτι τετράγωνόν ἐστι τὸ AG , καὶ ὅτι τῶν AB , $B\Gamma$ μέσον ἀνάλογόν ἐστι τὸ ΔH , καὶ ἔτι τῶν AG , ΓB μέσον ἀνάλογόν ἐστι τὸ $\Delta\Gamma$.

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ μὲν ΔB τῇ BZ , ἡ δὲ BE τῇ BH , ὅλη ἄρα ἡ ΔE ὅλη τῇ ZH ἐστὶν ἴση. ἀλλ' ἡ μὲν ΔE ἐκατέρᾳ τῶν $A\Theta$, $K\Gamma$ ἐστὶν ἴση, ἡ δὲ ZH ἐκατέρᾳ τῶν AK , $\Theta\Gamma$ ἐστὶν ἴση· καὶ ἐκατέρᾳ ἄρα τῶν $A\Theta$, $K\Gamma$ ἐκατέρᾳ τῶν AK , $\Theta\Gamma$ ἐστὶν ἴση. ἰσόπλευρον ἄρα ἐστὶ τὸ AG παραλληλόγραμμον· ἐστὶ δὲ καὶ ὀρθογώνιον· τετράγωνον ἄρα ἐστὶ τὸ AG .

Καὶ ἐπεὶ ἐστὶν ὥς ἡ ZB πρὸς τὴν BH , οὕτως ἡ ΔB πρὸς τὴν BE , ἀλλ' ὥς μὲν ἡ ZB πρὸς τὴν BH , οὕτως τὸ AB πρὸς τὸ ΔH , ὥς δὲ ἡ ΔB πρὸς τὴν BE , οὕτως τὸ ΔH πρὸς τὸ $B\Gamma$, καὶ ὥς ἄρα τὸ AB πρὸς τὸ ΔH , οὕτως τὸ ΔH πρὸς τὸ $B\Gamma$. τῶν AB , $B\Gamma$ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΔH .

Λέγω δὴ, ὅτι καὶ τῶν AG , ΓB μέσον ἀνάλογόν [ἐστὶ] τὸ $\Delta\Gamma$.

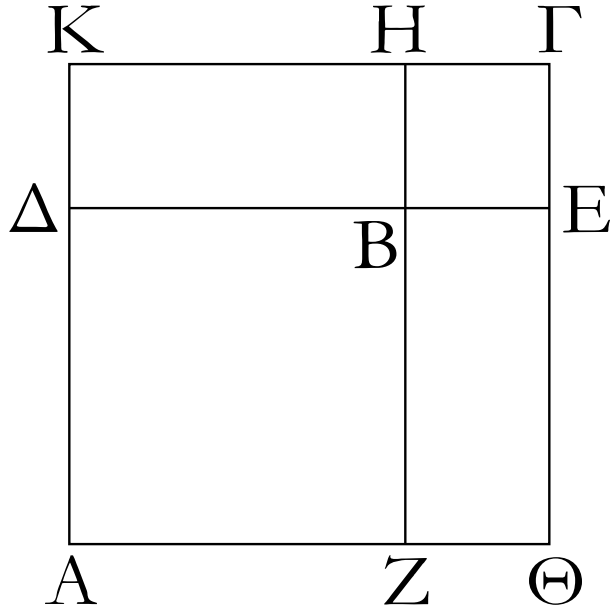
Lemma

Let AB and BC be two squares, and let them be laid down such that DB is straight-on to BE . FB is, thus, also straight-on to BG . And let the parallelogram AC be completed. I say that AC is a square, and that DG is the mean proportional to AB and BC , and, moreover, DC is the mean proportional to AC and CB .

For since DB is equal to BF , and BE to BG , the whole of DE is thus equal to the whole of FG . But DE is equal to each of AH and KC , and FG is equal to each of AK and HC [Prop. 1.34]. Thus, AH and KC are also equal to AK and HC , respectively. Thus, the parallelogram AC is equilateral. And (it is) also right-angled. Thus, AC is a square.

And since as FB is to BG , so DB (is) to BE , but as FB (is) to BG , so AB (is) to DG , and as DB (is) to BE , so DG (is) to BC [Prop. 6.1], thus also as AB (is) to DG , so DG (is) to BC [Prop. 5.11]. Thus, DG is the mean proportional to AB and BC .

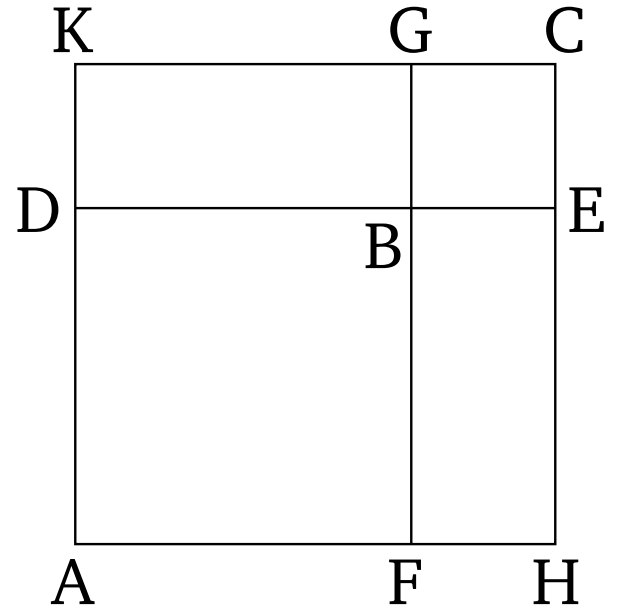
So I also say that DC [is] the mean proportional to AC and CB .



Ἐπεὶ γάρ ἐστιν ὡς ἡ AD πρὸς τὴν $ΔK$, οὕτως ἡ KH πρὸς τὴν $HΓ$. ἴση γάρ [ἐστίν] ἑκατέρω ἑκατέρω· καὶ συνθέντι ὡς ἡ AK πρὸς $KΔ$, οὕτως ἡ $KΓ$ πρὸς $ΓH$, ἀλλ' ὡς μὲν ἡ AK πρὸς $KΔ$, οὕτως τὸ $ΑΓ$ πρὸς τὸ $ΓΔ$, ὡς δὲ ἡ $KΓ$ πρὸς $ΓH$, οὕτως τὸ $ΔΓ$ πρὸς $ΓB$, καὶ ὡς ἄρα τὸ $ΑΓ$ πρὸς $ΔΓ$, οὕτως τὸ $ΔΓ$ πρὸς τὸ $BΓ$. τῶν $ΑΓ$, $ΓB$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $ΔΓ$ · ὃ προέκειτο δεῖξαι.

54.

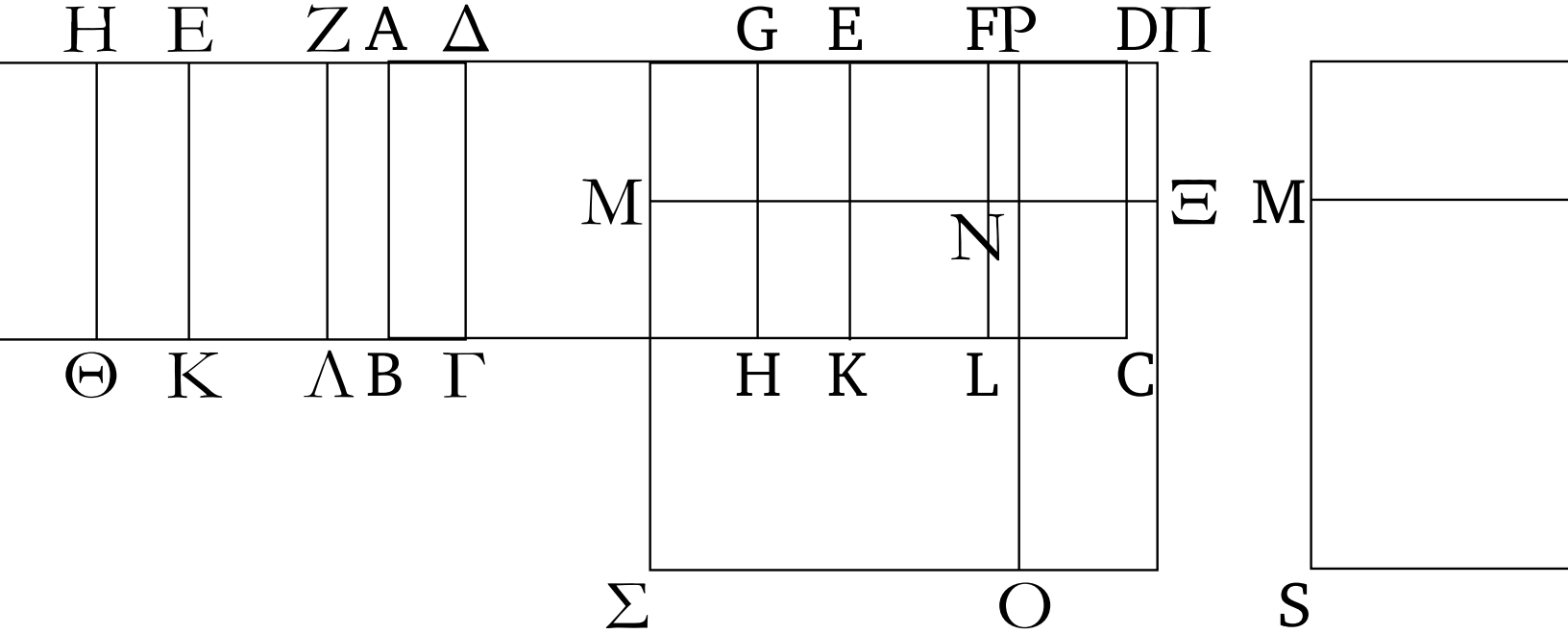
Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πρώτης, ἡ τὸ θωρίον δυναμένη ἀλογόξ ἐστίν ἡ καλονμένη ἐκ δύο ὀνομάτων.



For since as AD is to DK , so KG (is) to GC . For [they are] respectively equal. And, via composition, as AK (is) to KD , so KC (is) to CG [Prop. 5.18]. But as AK (is) to KD , so AC (is) to CD , and as KC (is) to CG , so DC (is) to CB [Prop. 6.1]. Thus, also, as AC (is) to DC , so DC (is) to BC [Prop. 5.11]. Thus, DC is the mean proportional to AC and CB . Which (is the very thing) it was prescribed to show.

Proposition 54

If an area is contained by a rational (straight-line) and a first binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called binomial.[†]



Θωρίον γάρ τὸ ΑΓ περιεθέσθω ὑπὸ ῥητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων πρώτης τῆς ΑΔ· λέγω, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἄλογός ἐστιν ἢ καλουμένη ἐκ δύο ὀνομάτων.

Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων ἐστὶ πρώτη ἡ ΑΔ, διηγήσθω εἰς τὰ ὀνόματα κατὰ τὸ Ε, καὶ ἔστω τὸ μείζον ὄνομα τὸ ΑΕ. φανερόν δὴ, ὅτι αἱ ΑΕ, ΕΔ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ, καὶ ἡ ΑΕ σύμμετρος ἐστὶ τῇ ἐκκειμένη ῥητῇ τῇ ΑΒ μήκει. τετμήσθω δὴ ἡ ΕΔ δίδια κατὰ τὸ Ζ σημείον. καὶ ἐπεὶ ἡ ΑΕ τῆς ΕΔ μείζον δύναται τῷ ἀπὸ συμέτρου ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος, τοντέστι τῷ ἀπὸ τῆς ΕΖ, ἴσον παρὰ τὴν μείζονα τὴν ΑΕ παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, εἰς σύμμετρα αὐτὴν διαιρεῖ. παραβελήσθω οὖν παρὰ τὴν ΑΕ τῷ ἀπὸ τῆς ΕΖ ἴσον τὸ ὑπὸ ΑΗ, ΗΕ· σύμμετρος ἄρα ἐστὶν ἡ ΑΗ τῇ ΕΗ μήκει. καὶ ἤθθωσαν ἀπὸ τῶν Η, Ε, Ζ ὁποτέρᾳ τῶν ΑΒ, ΓΔ παράλληλοι αἱ ΗΘ, ΕΚ, ΖΛ· καὶ τῷ μὲν ΑΘ παραλληλογράμῳ ἴσον τετράγωνον συνεστάτω τὸ ΣΝ, τῷ δὲ ΗΚ ἴσον τὸ ΝΠ, καὶ κείσθω ὥστε ἐπ' εὐθείᾳ εἶναι τὴν ΜΝ τῇ ΝΧ· ἐπ' εὐθείᾳ ἄρα ἐστὶ καὶ ἡ ΡΝ τῇ ΝΟ. καὶ συμπεπληρώσθω τὸ ΣΠ παραλληλόγραμμον· τετράγωνον ἄρα ἐστὶ τὸ ΣΠ. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΑΗ, ΗΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΖ, ἔστιν ἄρα ὡς ἡ ΑΗ πρὸς ΕΖ, οὕτως ἡ ΖΕ πρὸς ΕΗ· καὶ ὡς ἄρα τὸ ΑΘ πρὸς ΕΛ, τὸ ΕΛ πρὸς ΚΗ· τῶν ΑΘ, ΗΚ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΕΛ. ἀλλὰ τὸ μὲν ΑΘ ἴσον ἐστὶ τῷ ΣΝ, τὸ δὲ ΗΚ ἴσον τῷ ΝΠ· τῶν ΣΝ, ΝΠ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΕΛ. ἔστι δὲ τῶν αὐτῶν τῶν ΣΝ, ΝΠ μέσον ἀνάλογον καὶ τὸ ΜΡ· ἴσον ἄρα ἐστὶ τὸ ΕΛ τῷ ΜΡ· ὥστε καὶ τῷ ΟΧ ἴσον ἐστίν. ἔστι δὲ καὶ τὰ ΑΘ, ΗΚ τοῖς ΣΝ, ΝΠ ἴσα· ὅλον ἄρα τὸ ΑΓ ἴσον ἐστὶν ὅλῳ τῷ ΣΠ, τοντέστι τῷ ἀπὸ τῆς ΜΧ τετραγώνῳ· τὸ ΑΓ

For let the area AC be contained by the rational (straight-line) AB and by the first binomial (straight-line) AD. I say that square-root of area AC is the irrational (straight-line which is) called binomial.

For since AD is a first binomial (straight-line), let it be divided into its (component) terms at E, and let AE be the greater term. So, (it is) clear that AE and ED are rational (straight-lines which are) commensurable in square only, and that the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE), and that AE is commensurable (in length) with the rational (straight-line) AB (first) laid out [Def. 10.5]. So, let ED be cut in half at point F. And since the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE), thus if a (rectangle) equal to the fourth part of the (square) on the lesser (term)—that is to say, the (square) on EF—falling short by a square figure, is applied to the greater (term) AE, then it divides it into (terms which are) commensurable (in length) [Prop 10.17]. Therefore, let the (rectangle contained) by AG and GE, equal to the (square) on EF, be applied to AE. AG is thus commensurable in length with EG. And let GH, EK, and FL be drawn from (points) G, E, and F (respectively), parallel to either of AB or CD. And let the square SN, equal to the parallelogram AH, be constructed, and (the square) NQ, equal to (the parallelogram) GK [Prop. 2.14]. And let MN be laid down so as to be straight-on to NO. RN is thus also straight-on to NP. And let the parallelogram SQ be completed. SQ is thus a square [Prop. 10.53 lem.]. And since the (rectangle contained) by AG and GE is equal to the (square) on EF, thus

ἄρα δύναται ἡ MX . λέγω, ὅτι ἡ MX ἐκ δύο ὀνομάτων ἐστίν.

Ἐπεὶ γὰρ σύμμετρος ἐστὶν ἡ AH τῇ HE , σύμμετρος ἐστὶ καὶ ἡ AE ἑκατέρᾳ τῶν AH , HE . ὑπόκειται δὲ καὶ ἡ AE τῇ AB σύμμετρος· καὶ αἱ AH , HE ἄρα τῇ AB σύμμετροί εἰσιν. καὶ ἐστὶ ῥητὴ ἡ AB · ῥητὴ ἄρα ἐστὶ καὶ ἑκάτερα τῶν AH , HE · ῥητὸν ἄρα ἐστὶν ἑκάτερον τῶν $A\Theta$, HK , καὶ ἐστὶ σύμμετρον τὸ $A\Theta$ τῷ HK . ἀλλὰ τὸ μὲν $A\Theta$ τῷ ΣN ἴσον ἐστίν, τὸ δὲ HK τῷ $N\Pi$ · καὶ τὰ ΣN , $N\Pi$ ἄρα, τοντέσσι τὰ ἀπὸ τῶν MN , NX , ῥητά ἐστὶ καὶ σύμμετρα. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AE τῇ $E\Delta$ μήκει, ἀλλ' ἡ μὲν AE τῇ AH ἐστὶ σύμμετρος, ἡ δὲ ΔE τῇ EZ σύμμετρος, ἀσύμμετρος ἄρα καὶ ἡ AH τῇ EZ · ὥστε καὶ τὸ $A\Theta$ τῷ $E\Lambda$ ἀσύμμετρον ἐστίν. ἀλλὰ τὸ μὲν $A\Theta$ τῷ ΣN ἐστὶν ἴσον, τὸ δὲ $E\Lambda$ τῷ MP · καὶ τὸ ΣN ἄρα τῷ MP ἀσύμμετρον ἐστίν. ἀλλ' ὡς τὸ ΣN προξ MP , ἡ ON προξ τῇν NP · ἀσύμμετρος ἄρα ἐστὶν ἡ ON τῇ NP . ἴση δὲ ἡ μὲν ON τῇ MN , ἡ δὲ NP τῇ NX · ἀσύμμετρος ἄρα ἐστὶν ἡ MN τῇ NX . καὶ ἐστὶ τὸ ἀπὸ τῆς MN σύμμετρον τῷ ἀπὸ τῆς NX , καὶ ῥητὸν ἑκάτερον· αἱ MN , NX ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι.

Ἡ MX ἄρα ἐκ δύο ὀνομάτων ἐστὶ καὶ δύναται τὸ AG · ὅπερ ἔδει δεῖχαι.

as AG is to EF , so FE (is) to EG [Prop. 6.17]. And thus as AH (is) to EL , (so) EL (is) to KG [Prop. 6.1]. Thus, EL is the mean proportional to AH and GK . But, AH is equal to SN , and GK (is) equal to NQ . EL is thus the mean proportional to SN and NQ . And MR is also the mean proportional to the same—(namely), SN and NQ [Prop. 10.53 lem.]. EL is thus equal to MR . Hence, it is also equal to PO [Prop. 1.43]. And AH plus GK is equal to SN plus NQ . Thus, the whole of AC is equal to the whole of SQ —that is to say, to the square on MO . Thus, MO (is) the square-root of (area) AC . I say that MO is a binomial (straight-line).

For since AG is commensurable (in length) with GE , AE is also commensurable (in length) with each of AG and GE [Prop. 10.15]. And AE was also assumed (to be) commensurable (in length) with AB . Thus, AG and GE are also commensurable (in length) with AB [Prop. 10.12]. And AB is rational. AG and GE are thus each also rational. Thus, AH and GK are each rational (areas), and AH is commensurable with GK [Prop. 10.19]. But, AH is equal to SN , and GK to NQ . SN and NQ —that is to say, the (squares) on MN and NO (respectively)—are thus also rational and commensurable. And since AE is incommensurable in length with ED , but AE is commensurable (in length) with AG , and DE (is) commensurable (in length) with EF , AG (is) thus also incommensurable (in length) with EF [Prop. 10.13]. Hence, AH is also incommensurable with EL [Props. 6.1, 10.11]. But, AH is equal to SN , and EL to MR . Thus, SN is also incommensurable with MR . But, as SN (is) to MR , (so) PN (is) to NR [Prop. 6.1]. PN is thus incommensurable (in length) with NR [Prop. 10.11]. And PN (is) equal to MN , and NR to NO . Thus, MN is incommensurable (in length) with NO . And the (square) on MN is commensurable with the (square) on NO , and each (is) rational. MN and NO are thus rational (straight-lines which are) commensurable in square only.

Thus, MO is (both) a binomial (straight-line) [Prop. 10.36], and the square-root of AC . (Which is) the very thing it was required to show.

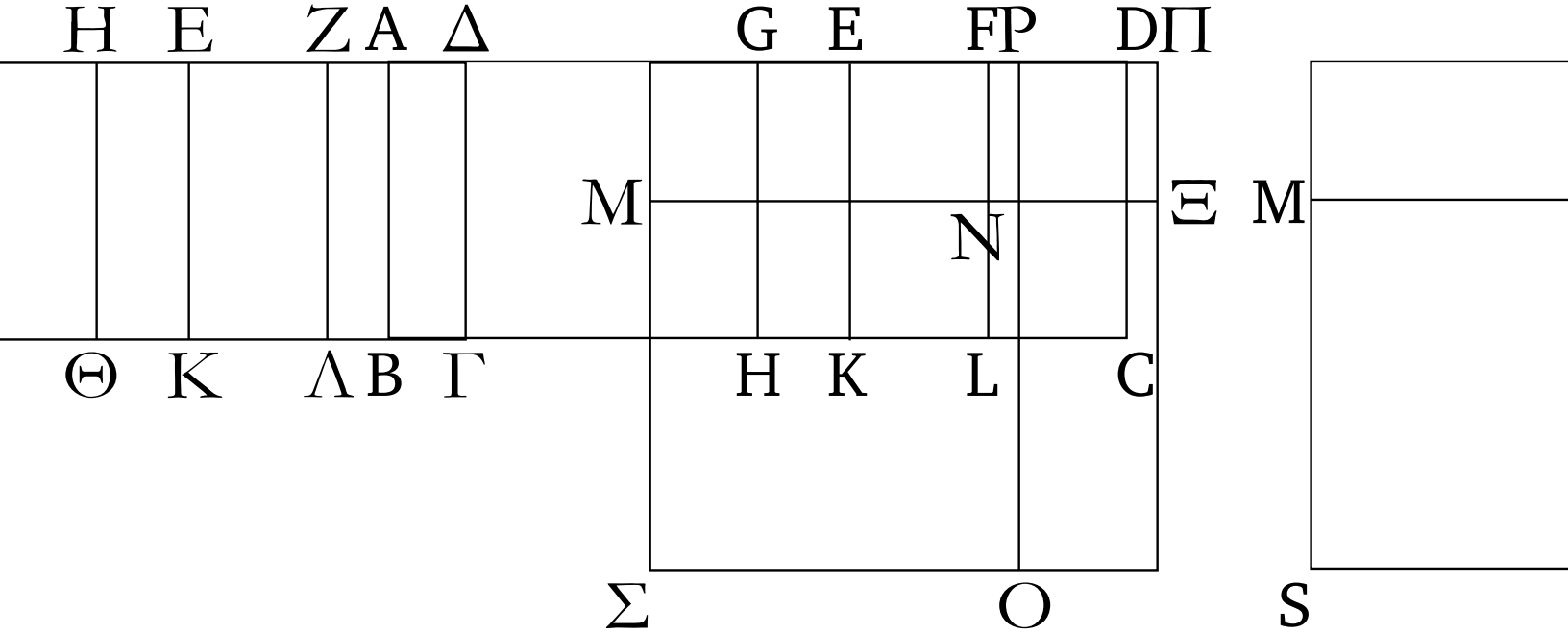
[†] If the rational straight-line has unit length then this proposition states that the square-root of a first binomial straight-line is a binomial straight-line: i.e., a first binomial straight-line has a length $k + k\sqrt{1 - k'^2}$ whose square-root can be written $\rho(1 + \sqrt{k''})$, where $\rho = \sqrt{k(1 + k')}/2$ and $k'' = (1 - k')/(1 + k')$. This is the length of a binomial straight-line (see Prop. 10.36), since ρ is rational.

55.

Proposition 55

Ἐάν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας, ἡ τὸ θωρίον δυνάμενη ἄλογός ἐστιν ἡ καλουμένη ἐκ δύο μέσων πρώτη.

If an area is contained by a rational (straight-line) and a second binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called first bimedial.[†]



Περιεθέσθω γάρ θωρίον τὸ $AB\Gamma\Delta$ ὑπὸ ῥητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας τῆς AD · λέγω, ὅτι ἡ τὸ AF θωρίον δυναμένη ἐκ δύο μέσων πρώτη ἐστίν.

Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων δευτέρα ἐστὶν ἡ AD , διηγήσθω εἰς τὰ ὀνόματα κατὰ τὸ E , ὥστε τὸ μείζον ὄνομα εἶναι τὸ AE · αἱ AE , ED ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AE τῆς ED μείζον δύναται τῷ ἀπὸ συμμετρῶν ἑαυτῇ, καὶ τὸ ἔλαττον ὄνομα ἡ ED σύμμετρόν ἐστι τῇ AB μήκει. τεμήσθω ἡ ED δίχα κατὰ τὸ Z , καὶ τῷ ἀπὸ τῆς EZ ἴσον παρὰ τὴν AE παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν AHE · σύμμετρος ἄρα ἡ AH τῇ HE μήκει. καὶ διὰ τῶν H , E , Z παράλληλοι ἤθθωσαν ταῖς AB , $\Gamma\Delta$ αἱ $H\Theta$, EK , $Z\Lambda$, καὶ τῷ μὲν $A\Theta$ παραλληλογράμμῳ ἴσον τετράγωνον συνεστάτω τὸ SN , τῷ δὲ HK ἴσον τετράγωνον τὸ $N\Pi$, καὶ κείσθω ὥστε ἐπ' εὐθείας εἶναι τὴν MN τῇ NX · ἐπ' εὐθείας ἄρα [ἐστὶ] καὶ ἡ PN τῇ NO . καὶ συμπληρώσθω τὸ $\Sigma\Pi$ τετράγωνον· φανερόν δὲ ἐκ τοῦ προδεδειγμένου, ὅτι τὸ MP μέσον ἀνάλογόν ἐστι τῶν SN , $N\Pi$, καὶ ἴσον τῷ EL , καὶ ὅτι τὸ AF θωρίον δύναται ἡ MX . δεικτέον δὲ, ὅτι ἡ MX ἐκ δύο μέσων ἐστὶ πρώτη.

Ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AE τῇ ED μήκει, σύμμετρος δὲ ἡ ED τῇ AB , ἀσύμμετρος ἄρα ἡ AE τῇ AB . καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ AH τῇ EH , σύμμετρος ἐστὶ καὶ ἡ AE ἑκάτερά τῶν AH , HE . ἀλλὰ ἡ AE ἀσύμμετρος τῇ AB μήκει· καὶ αἱ AH , HE ἄρα ἀσύμμετροί εἰσι τῇ AB . αἱ BA , AH , HE ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ὥστε μέσον ἐστὶν ἑκάτερον τῶν $A\Theta$, HK . ὥστε καὶ ἑκάτερον τῶν SN , $N\Pi$ μέσον ἐστίν. καὶ αἱ MN , NX ἄρα μέσαι εἰσίν. καὶ ἐπεὶ σύμμετρος ἡ AH τῇ HE μήκει, σύμμετρόν ἐστι καὶ τὸ $A\Theta$ τῷ HK , τοντέστι τὸ SN τῷ $N\Pi$, τοντέστι τὸ ἀπὸ τῆς MN τῷ

For let the area $ABCD$ be contained by the rational (straight-line) AB and by the second binomial (straight-line) AD . I say that the square-root of area AC is a first bimedial (straight-line).

For since AD is a second binomial (straight-line), let it be divided into its (component) terms at E , such that AE is the greater term. Thus, AE and ED are rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE), and the lesser term ED is commensurable in length with AB [Def. 10.6]. Let ED be cut in half at F . And let the (rectangle contained) by AGE , equal to the (square) on EF , be applied to AE , falling short by a square figure. AG (is) thus commensurable in length with GE [Prop. 10.17]. And let GH , EK , and FL be drawn through (points) G , E , and F (respectively), parallel to AB and CD . And let the square SN , equal to the parallelogram AH , be constructed, and the square NQ , equal to GK . And let MN be laid down so as to be straight-on to NO . Thus, RN [is] also straight-on to NP . And let the square SQ be completed. So, (it is) clear from what has been previously demonstrated [Prop. 10.53 lem.] that MR is the mean proportional to SN and NQ , and (is) equal to EL , and that MO is the square-root of the area AC . So, we must show that MO is a first bimedial (straight-line).

Since AE is incommensurable in length with ED , and ED (is) commensurable (in length) with AB , AE (is) thus incommensurable (in length) with AB [Prop. 10.13]. And since AG is commensurable (in length) with EG , AE is also commensurable (in length) with each of AG and GE [Prop. 10.15].

ἀπὸ τῆς NX [ὥστε δυνάμει εἰσὶ σύμμετροι αἱ MN , NX]. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AE τῇ $EΔ$ μήκει, ἀλλ' ἡ μὲν AE σύμμετρος ἐστὶ τῇ AH , ἡ δὲ $EΔ$ τῇ EZ σύμμετρος, ἀσύμμετρος ἄρα ἡ AH τῇ EZ . ὥστε καὶ τὸ $AΘ$ τῷ $ΕΛ$ ἀσύμμετρόν ἐστιν, τοντέστι τὸ $ΣΝ$ τῷ MP , τοντέστιν ὁ ON τῇ NP , τοντέστιν ἡ MN τῇ NX ἀσύμμετρος ἐστὶ μήκει. ἐδείκθησαν δὲ αἱ MN , NX καὶ μέσαι οὕσαι καὶ δυνάμει σύμμετροι· αἱ MN , NX ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δὴ, ὅτι καὶ ῥητὸν περιέθουσιν. ἐπεὶ γάρ ἡ $ΔΕ$ ὑπόκειται ἐκατέρῃ τῶν AB , EZ σύμμετρος, σύμμετρος ἄρα καὶ ἡ EZ τῇ $EΚ$. καὶ ῥητὴ ἐκατέρα αὐτῶν ῥητὸν ἄρα τὸ $ΕΛ$, τοντέστι τὸ MP . τὸ δὲ MP ἐστὶ τὸ ὑπὸ τῶν MNX . ἔαν δὲ δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι ῥητὸν περιέθουσαι, ἡ ὅλη ἄλογός ἐστιν, καλεῖται δὲ ἐκ δύο μέσων πρώτη.

Ἡ ἄρα MX ἐκ δύο μέσων ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

But, AE is incommensurable in length with AB . Thus, AG and GE are also (both) incommensurable (in length) with AB [Prop. 10.13]. Thus, BA , AG , and (BA , and) GE are (pairs of) rational (straight-lines which are) commensurable in square only. And, hence, each of AH and GK is a medial (area) [Prop. 10.21]. Hence, each of SN and NQ is also a medial (area). Thus, MN and NO are medial (straight-lines). And since AG (is) commensurable in length with GE , AH is also commensurable with GK —that is to say, SN with NQ —that is to say, the (square) on MN with the (square) on NO [hence, MN and NO are commensurable in square] [Props. 6.1, 10.11]. And since AE is incommensurable in length with ED , but AE is commensurable (in length) with AG , and ED commensurable (in length) with EF , AG (is) thus incommensurable (in length) with EF [Prop. 10.13]. Hence, AH is also incommensurable with EL —that is to say, SN with MR —that is to say, PN with NR —that is to say, MN is incommensurable in length with NO [Props. 6.1, 10.11]. But MN and NO have also been shown to be medial (straight-lines) which are commensurable in square. Thus, MN and NO are medial (straight-lines which are) commensurable in square only. So, I say that they also contain a rational (area). For since DE was assumed (to be) commensurable (in length) with each of AB and EF , EF (is) thus also commensurable with EK [Prop. 10.12]. And they (are) each rational. Thus, EL —that is to say, MR —(is) rational [Prop. 10.19]. And MR is the (rectangle contained) by MNO . And if two medial (straight-lines), commensurable in square only, which contain a rational (area), are added together, then the whole is (that) irrational (straight-line which is) called first bimedial [Prop. 10.37].

Thus, MO is a first bimedial (straight-line). (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a second binomial straight-line is a first bimedial straight-line: i.e., a second binomial straight-line has a length $k/\sqrt{1-k'^2} + k$ whose square-root can be written $\rho(k'^{1/4} + k'^{3/4})$, where $\rho = \sqrt{(k/2)(1+k')/(1-k')}$ and $k'' = (1-k')/(1+k')$. This is the length of a first bimedial straight-line (see Prop. 10.37), since ρ is rational.

56.

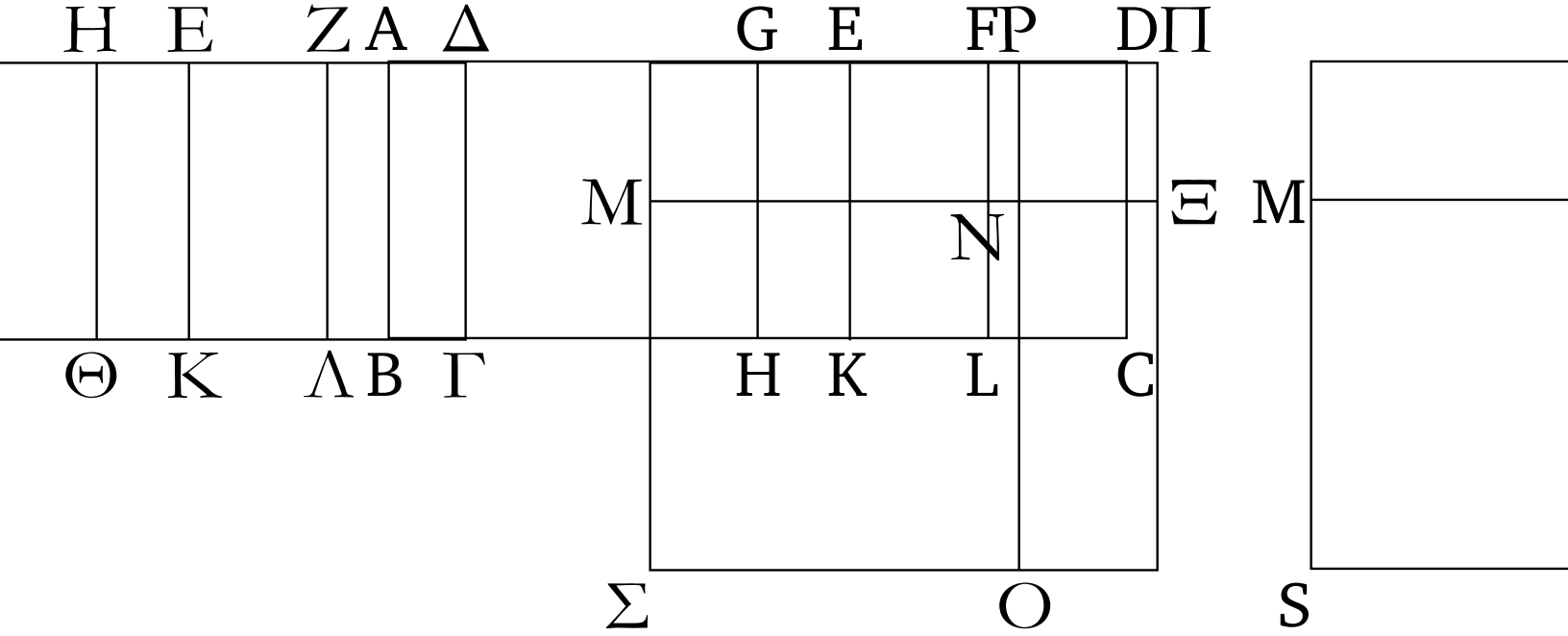
Proposition 56

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων τρίτης, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλουμένη ἐκ δύο μέσων δευτέρα.

Θωρίον γάρ τὸ $ABΓΔ$ περιέθῃται ὑπὸ ῥητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων τρίτης τῆς $ΑΔ$ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ E , ὧν μείζον ἐστὶ τὸ AE . λέγω, ὅτι ἡ τὸ $ΑΓ$ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλουμένη ἐκ δύο μέσων δευτέρα.

If an area is contained by a rational (straight-line) and a third binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called second bimedial.†

For let the area $ABCD$ be contained by the rational (straight-line) AB and by the third binomial (straight-line) AD , which has been divided into its (component) terms at E , of which AE is the greater. I say that the square-root of area AC is the irrational (straight-line which is) called second bimedial.



Κατεσκευάσθω γάρ τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶ τρίτη ἡ $ΑΔ$, αἱ $ΑΕ$, $ΕΔ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ $ΑΕ$ τῇ $ΕΔ$ μείζον δύνανται τῷ ἀπὸ συμέτρου ἐναντῇ, καὶ οὐδετέρω τῶν $ΑΕ$, $ΕΔ$ σύμμετρόξ [ἐστὶ] τῇ $ΑΒ$ μήκει. ὁμοίως δὲ τοῖς προοδεδειγμένοις δείχομεν, ὅτι ἡ $ΜΧ$ ἐστὶν ἡ τὸ $ΑΓ$ θωρίον δυναμένη, καὶ αἱ $ΜΝ$, $ΝΧ$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· ὥστε ἡ $ΜΧ$ ἐκ δύο μέσων ἐστίν. δεικτέον δὲ, ὅτι καὶ δευτέρα.

[Καί] ἐπεὶ ἀσύμμετρός ἐστὶν ἡ $ΔΕ$ τῇ $ΑΒ$ μήκει, τοντέστι τῇ $ΕΚ$, σύμμετρος δὲ ἡ $ΔΕ$ τῇ $ΕΖ$, ἀσύμμετρος ἄρα ἐστὶν ἡ $ΕΖ$ τῇ $ΕΚ$ μήκει. καὶ εἰσι ῥηταὶ· αἱ $ΖΕ$, $ΕΚ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. μέσον ἄρα [ἐστὶ] τὸ $ΕΛ$, τοντέστι τὸ $ΜΡ$ · καὶ περιέχεται ὑπὸ τῶν $ΜΝΞ$ · μέσον ἄρα ἐστὶ τὸ ὑπὸ τῶν $ΜΝΞ$.

Ἡ $ΜΧ$ ἄρα ἐκ δύο μέσων ἐστὶ δευτέρα· ὅπερ ἔδει δεῖξαι.

For let the same construction be made as previously. And since AD is a third binomial (straight-line), AE and ED are thus rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE) , and neither of AE and ED [is] commensurable in length with AB [Def. 10.7]. So, similarly to that which has been previously demonstrated, we can show that MO is the square-root of area AC , and MN and NO are medial (straight-lines which are) commensurable in square only. Hence, MO is bimedral. So, we must show that (it is) also second (bimedial).

[And] since DE is incommensurable in length with AB —that is to say, with EK —and DE (is) commensurable (in length) with EF , EF is thus incommensurable in length with EK [Prop. 10.13]. And they are (both) rational (straight-lines). Thus, FE and EK are rational (straight-lines which are) commensurable in square only. EL —that is to say, MR —[is] thus medial [Prop. 10.21]. And it is contained by MNO . Thus, the (rectangle contained) by MNO is medial.

Thus, MO is a second bimedral (straight-line) [Prop. 10.38]. (Which is) the very thing it was required to show.

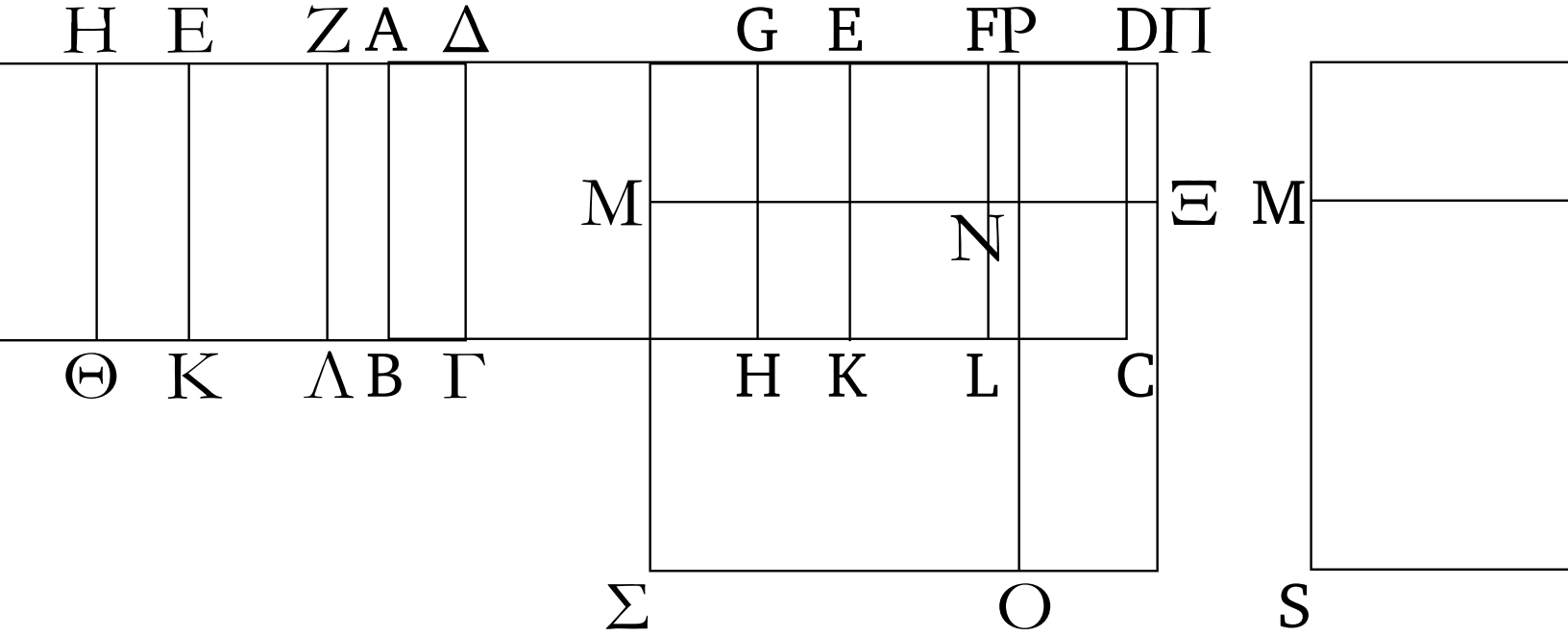
† If the rational straight-line has unit length then this proposition states that the square-root of a third binomial straight-line is a second bimedral straight-line: i.e., a third binomial straight-line has a length $k^{1/2}(1 + \sqrt{1 - k'^2})$ whose square-root can be written $\rho(k^{1/4} + k'^{1/2}/k^{1/4})$, where $\rho = \sqrt{(1 + k')/2}$ and $k' = k(1 - k)/(1 + k)$. This is the length of a second bimedral straight-line (see Prop. 10.38), since ρ is rational.

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῇς ἐκ δύο ὀνομάτων τετάρτης, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἡ κα-

If an area is contained by a rational (straight-line) and a fourth binomial (straight-line) then the square-root of the area

λογυμένη μείζων.

is the irrational (straight-line which is) called major.[†]



Θωρίον γάρ τὸ ΑΓ περιεθέσθω ὑπὸ ῥητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης τῆς ΑΔ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ Ε, ὧν μείζων ἔστω τὸ ΑΕ· λέγω, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλουμένη μείζων.

Ἐπεὶ γὰρ ἡ ΑΔ ἐκ δύο ὀνομάτων ἐστὶ τετάρτη, αἱ ΑΕ, ΕΔ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μείζων δύναται τῷ ἀπὸ ἀσύμμετρον ἐναντῆ, καὶ ἡ ΑΕ τῇ ΑΒ σύμμετρος [ἐστὶ] μήκει. τετμήσθω ἡ ΔΕ δίχα κατὰ τὸ Ζ, καὶ τῷ ἀπὸ τῆς ΕΖ ἴσον παρὰ τὴν ΑΕ παραβελθήσθω παραλληλόγραμμον τὸ ὑπὸ ΑΗ, ΗΕ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΗ τῇ ΗΕ μήκει. ἠθῶσαν παράλληλοι τῇ ΑΒ αἱ ΗΘ, ΕΚ, ΖΑ, καὶ τὰ λοιπὰ τὰ αὐτὰ τοῖς πρὸ τούτου γερονέτω· φανερόν δὴ, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἐστὶν ἡ ΜΧ. δεικτέον δὴ, ὅτι ἡ ΜΧ ἄλογός ἐστιν ἡ καλουμένη μείζων.

Ἐπεὶ ἀσύμμετρος ἐστὶν ἡ ΑΗ τῇ ΕΗ μήκει, ἀσύμμετρόν ἐστι καὶ τὸ ΑΘ τῷ ΗΚ, τοντέστι τὸ ΣΝ τῷ ΝΠ· αἱ ΜΝ, ΝΧ ἄρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ ΑΕ τῇ ΑΒ μήκει, ῥητόν ἐστι τὸ ΑΚ· καὶ ἐστὶν ἴσον τοῖς ἀπὸ τῶν ΜΝ, ΝΧ· ῥητόν ἄρα [ἐστὶ] καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ. καὶ ἐπεὶ ἀσύμμετρος [ἐστὶν] ἡ ΔΕ τῇ ΑΒ μήκει, τοντέστι τῇ ΕΚ, ἀλλὰ ἡ ΔΕ σύμμετρος ἐστὶ τῇ ΕΖ, ἀσύμμετρος ἄρα ἡ ΕΖ τῇ ΕΚ μήκει. αἱ ΕΚ, ΕΖ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα τὸ ΑΕ, τοντέστι τὸ ΜΡ. καὶ περιέχεται ὑπὸ τῶν ΜΝ, ΝΧ· μέσον ἄρα ἐστὶ τὸ ὑπὸ τῶν ΜΝ, ΝΧ. καὶ ῥητόν τὸ [συγκείμενον] ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ, καὶ εἰσὶν ἀσύμμετροι αἱ ΜΝ, ΝΧ δυνάμει. ἔαν δὲ δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον, ἡ ὅλη ἄλογός ἐστιν, καλεῖται δὲ μείζων.

For let the area AC be contained by the rational (straight-line) AB and the fourth binomial (straight-line) AD, which has been divided into its (component) terms at E, of which let AE be the greater. I say that the square-root of AC is the irrational (straight-line which is) called major.

For since AD is a fourth binomial (straight-line), AE and ED are thus rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) incommensurable (in length) with (AE), and AE [is] commensurable in length with AB [Def. 10.8]. Let DE be cut in half at F, and let the parallelogram (contained by) AG and GE, equal to the (square) on EF, (and falling short by a square figure) be applied to AE. AG is thus incommensurable in length with GE [Prop. 10.18]. Let GH, EK, and FL be drawn parallel to AB, and let the rest (of the construction) be made the same as the (proposition) before this. So, it is clear that MO is the square-root of area AC. So, we must show that MO is the irrational (straight-line which is) called major.

Since AG is incommensurable in length with EG, AH is also incommensurable with GK—that is to say, SN with NQ [Props. 6.1, 10.11]. Thus, MN and NO are incommensurable in square. And since AE is commensurable in length with AB, AK is rational [Prop. 10.19]. And it is equal to the (sum of the squares) on MN and NO. Thus, the sum of the (squares) on MN and NO [is] also rational. And since DE [is] incommensurable in length with AB [Prop. 10.13]—that is to say, with EK—but DE is commensurable (in length) with EF, EF (is) thus incommensurable in length with EK [Prop. 10.13].

Ἡ MX ἄρα ἄλογός ἐστιν ἡ καλονμένη μείζων, καὶ δύναιται τὸ AG θωρίον· ὅπερ ἔδει δεῖχαι.

Thus, EK and EF are rational (straight-lines which are) commensurable in square only. LE —that is to say, MR —(is) thus medial [Prop. 10.21]. And it is contained by MN and NO . The (rectangle contained) by MN and NO is thus medial. And the [sum] of the (squares) on MN and NO (is) rational, and MN and NO are incommensurable in square. And if two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial, are added together, then the whole is the irrational (straight-line which is) called major [Prop. 10.39].

Thus, MO is the irrational (straight-line which is) called major. And (it is) the square-root of area AC . (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then this proposition states that the square-root of a fourth binomial straight-line is a major straight-line: i.e., a fourth binomial straight-line has a length $k(1 + 1/\sqrt{1+k'})$ whose square-root can be written $\rho\sqrt{[1+k''/(1+k''^2)^{1/2}]/2} + \rho\sqrt{[1-k''/(1+k''^2)^{1/2}]/2}$, where $\rho = \sqrt{k}$ and $k''^2 = k'$. This is the length of a major straight-line (see Prop. 10.39), since ρ is rational.

58.

Proposition 58

Ἐάν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πέμπτῃς, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλονμένη ῥητὸν καὶ μέσον δυναμένη.

Θωρίον γὰρ τὸ AG περιεθέσθω ὑπὸ ῥητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων πέμπτῃς τῆς AD διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ E , ὥστε τὸ μείζον ὄνομα εἶναι τὸ AE · λέγω [δὴ], ὅτι ἡ τὸ AG θωρίον δυναμένη ἄλογός ἐστιν ἡ καλονμένη ῥητὸν καὶ μέσον δυναμένη.

Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρότερον δεδειγμένοις· φανερόν δὴ, ὅτι ἡ τὸ AG θωρίον δυναμένη ἐστὶν ἡ MX . δεῖκτέον δὴ, ὅτι ἡ MX ἐστὶν ἡ ῥητὸν καὶ μέσον δυναμένη.

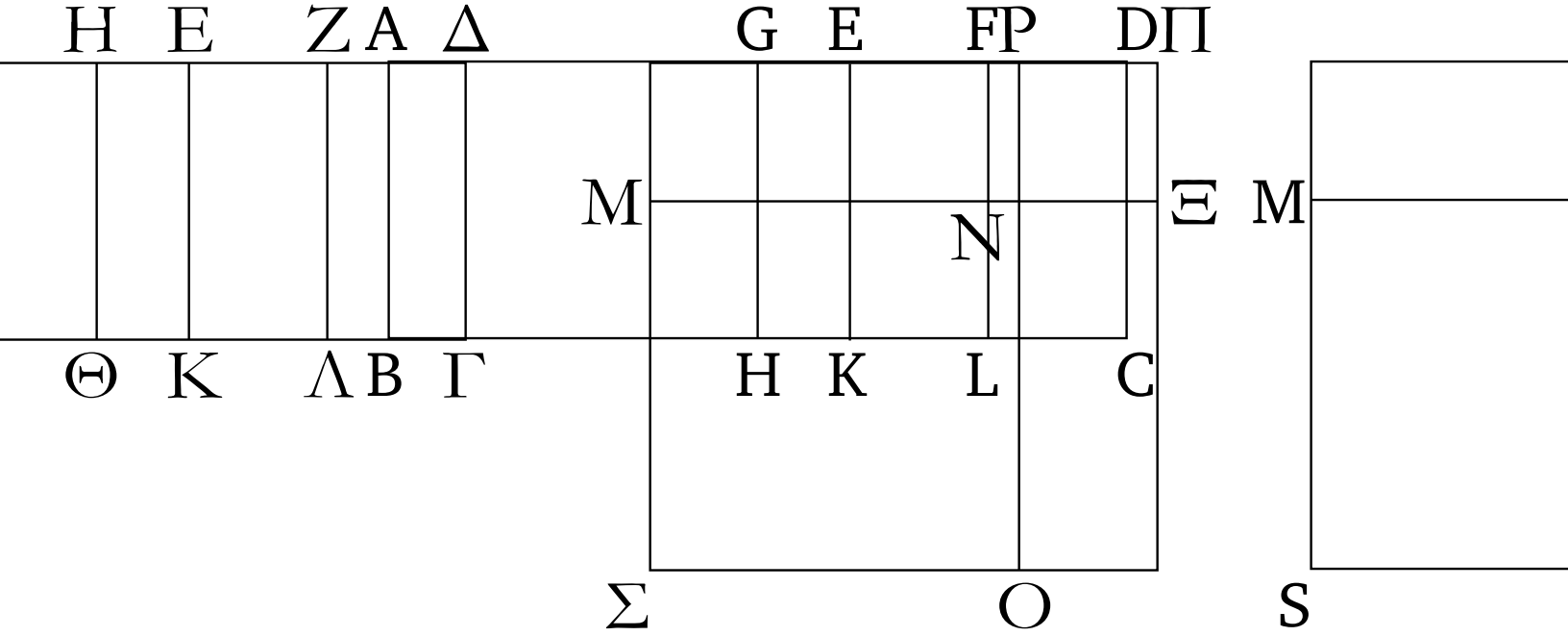
If an area is contained by a rational (straight-line) and a fifth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called the square-root of a rational plus a medial (area).[†]

For let the area AC be contained by the rational (straight-line) AB and the fifth binomial (straight-line) AD , which has been divided into its (component) terms at E , such that AE is the greater term. [So] I say that the square-root of area AC is the irrational (straight-line which is) called the square-root of a rational plus a medial (area).

For let the same construction be made as that shown previously. So, (it is) clear that MO is the square-root of area AC . So, we must show that MO is the square-root of a rational plus a medial (area).

Ἐάν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων ἑκτῆς, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἡ καλουμένη δύο μέσα δυναμένη.

If an area is contained by a rational (straight-line) and a sixth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called the square-root of (the sum of) two medial (areas).[†]



Θωρίον γάρ τὸ $AB\Gamma\Delta$ περιεθέσθω ὑπὸ ῥητῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων ἑκτῆς τῆς $A\Delta$ διηρημένης εἰς τὰ ὀνόματα κατὰ τὸ E , ὥστε τὸ μείζον ὄνομα εἶναι τὸ AE · λέγω, ὅτι ἡ τὸ $A\Gamma$ δυναμένη ἡ δύο μέσα δυναμένη ἐστίν.

Κατεσκευάσθω [γάρ] τὰ αὐτὰ τοῖς προοδεξιμένοις. φανερόν δὴ, ὅτι [ἡ] τὸ $A\Gamma$ δυναμένη ἐστίν ἡ MX , καὶ ὅτι ἀσύμμετρος ἐστὶν ἡ MN τῇ NX δυνάμει. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ EA τῇ AB μήκει, αἱ EA , AB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα ἐστὶ τὸ AK , τοντέστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν MN , NX . πάλιν, ἐπεὶ ἀσύμμετρος ἐστὶν ἡ ED τῇ AB μήκει, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ZE τῇ EK · αἱ ZE , EK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα ἐστὶ τὸ EL , τοντέστι τὸ MP , τοντέστι τὸ ὑπὸ τῶν MNX . καὶ ἐπεὶ ἀσύμμετρος ἡ AE τῇ EZ , καὶ τὸ AK τῷ EL ἀσύμμετρόν ἐστιν. ἀλλὰ τὸ μὲν AK ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν MN , NX , τὸ δὲ EL ἐστὶ τὸ ὑπὸ τῶν MNX · ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν MNX τῷ ὑπὸ τῶν MNX . καὶ ἐστὶ μέσον ἐκάτερον αὐτῶν, καὶ αἱ MN , NX δυνάμει εἰσὶν ἀσύμμετροι.

Ἡ MX ἄρα δύο μέσα δυναμένη ἐστὶ καὶ δύναται τὸ $A\Gamma$ · ὅπερ ἔδει δεῖξαι.

For let the area $ABCD$ be contained by the rational (straight-line) AB and the sixth binomial (straight-line) AD , which has been divided into its (component) terms at E , such that AE is the greater term. So, I say that the square-root of AC is the square-root of (the sum of) two medial (areas).

[For] let the same construction be made as that shown previously. So, (it is) clear that MO is the square-root of AC , and that MN is incommensurable in square with NO . And since EA is incommensurable in length with AB [Def. 10.10], EA and AB are thus rational (straight-lines which are) commensurable in square only. Thus, AK —that is to say, the sum of the (squares) on MN and NO —is medial [Prop. 10.21]. Again, since ED is incommensurable in length with AB [Def. 10.10], FE is thus also incommensurable (in length) with EK [Prop. 10.13]. Thus, FE and EK are rational (straight-lines which are) commensurable in square only. Thus, EL —that is to say, MR —that is to say, the (rectangle contained) by MNO —is medial [Prop. 10.21]. And since AE is incommensurable (in length) with EF , AK is also incommensurable with EL [Props. 6.1, 10.11]. But, AK is the sum of the (squares) on MN and NO , and EL is the (rectangle contained) by MNO . Thus, the sum of the (squares) on MNO is incommensurable with the (rectangle contained) by MNO . And each of them is medial. And MN and NO are incommensurable in square.

Thus, MO is the square-root of (the sum of) two me-

dial (areas) [Prop. 10.41]. And (it is) the square-root of AC . (Which is) the very thing it was required to show.

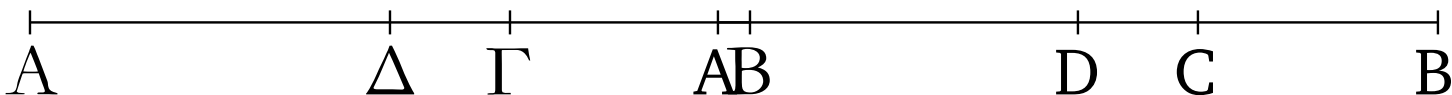
† If the rational straight-line has unit length then this proposition states that the square-root of a sixth binomial straight-line is the square root of the sum of two medial areas: i.e., a sixth binomial straight-line has a length $\sqrt{k} + \sqrt{k'}$ whose square-root can be written $k^{1/4} \left(\sqrt{[1 + k''/(1 + k''^2)^{1/2}]/2} + \sqrt{[1 - k''/(1 + k''^2)^{1/2}]/2} \right)$, where $k''^2 = (k - k')/k'$. This is the length of the square-root of the sum of two medial areas (see Prop. 10.41).

Λήμμα.

Lemma

Ἐάν εὐθεῖα γραμμὴ τμηθῇ εἰς ἄνισα, τὰ ἀπὸ τῶν ἀνίσων τετράγωνα μείζονά ἐστι τοῦ διζ ὑπὸ τῶν ἀνίσων περιεθρομένου ὀρθογωνίου.

If a straight-line is cut unequally then (the sum of) the squares on the unequal (parts) is greater than twice the rectangle contained by the unequal (parts).



Ἐστω εὐθεῖα ἡ AB καὶ τετμήσθω εἰς ἄνισα κατὰ τὸ Γ , καὶ ἔστω μείζων ἡ AG . λέγω, ὅτι τὰ ἀπὸ τῶν AG , GB μείζονά ἐστι τοῦ διζ ὑπὸ τῶν AG , GB .

Let AB be a straight-line, and let it be cut unequally at C , and let AC be greater (than CB). I say that (the sum of) the (squares) on AC and CB is greater than twice the (rectangle contained) by AC and CB .

Τετμήσθω γάρ ἡ AB δίχα κατὰ τὸ Δ . ἐπεὶ οὖν εὐθεῖα γραμμὴ τέτμηται εἰς μὲν ἴσα κατὰ τὸ Δ , εἰς δὲ ἄνισα κατὰ τὸ Γ , τὸ ἄρα ὑπὸ τῶν AG , GB μετὰ τοῦ ἀπὸ $\Gamma\Delta$ ἴσον ἐστὶ τῷ ἀπὸ $A\Delta$. ὥστε τὸ ὑπὸ τῶν AG , GB ἔλαττον ἐστι τοῦ ἀπὸ $A\Delta$. τὸ ἄρα διζ ὑπὸ τῶν AG , GB ἔλαττον ἢ διπλάσιόν ἐστι τοῦ ἀπὸ $A\Delta$. ἀλλὰ τὰ ἀπὸ τῶν AG , GB διπλάσιά [ἐστι] τῶν ἀπὸ τῶν $A\Delta$, $\Delta\Gamma$. τὰ ἄρα ἀπὸ τῶν AG , GB μείζονά ἐστι τοῦ διζ ὑπὸ τῶν AG , GB . ὅπερ ἔδει δεῖξαι.

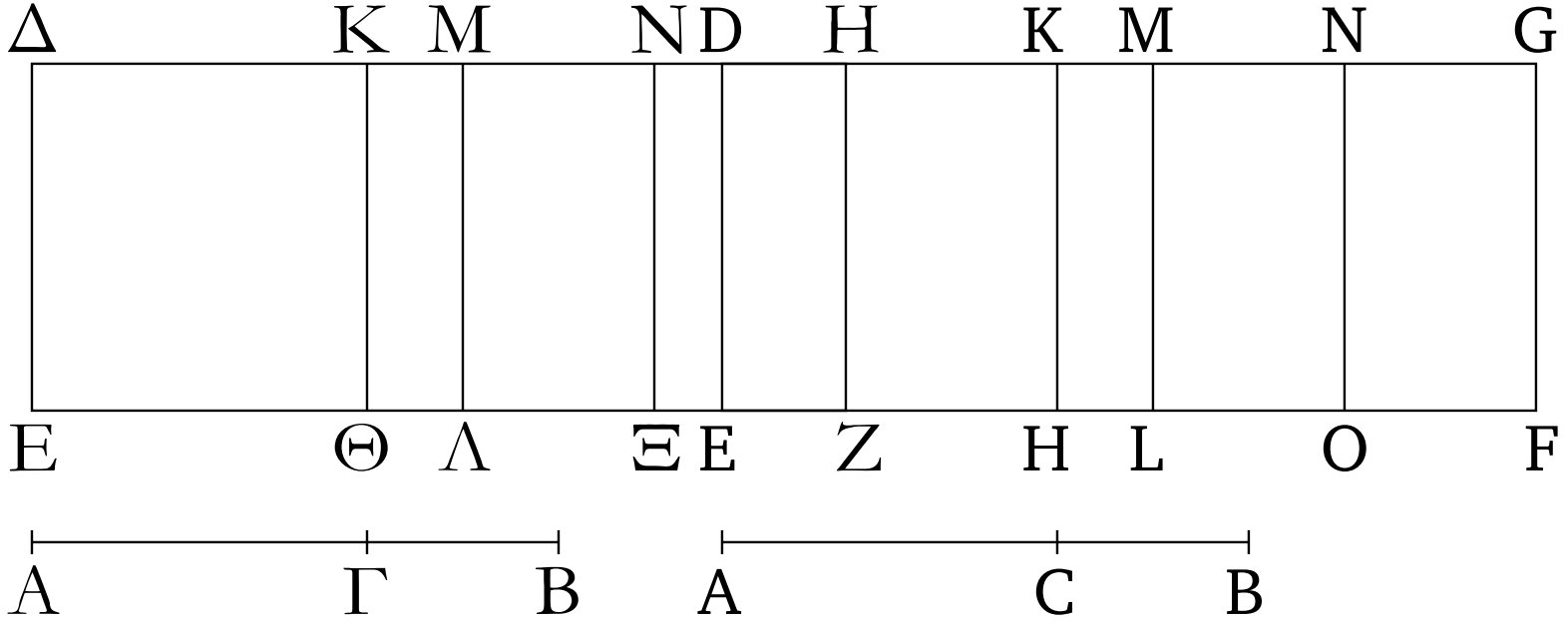
For let AB be cut in half at D . Therefore, since a straight-line has been cut into equal (parts) at D , and into unequal (parts) at C , the (rectangle contained) by AC and CB , plus the (square) on CD , is thus equal to the (square) on AD [Prop. 2.5]. Hence, the (rectangle contained) by AC and CB is less than the (square) on AD . Thus, twice the (rectangle contained) by AC and CB is less than double the (square) on AD . But, (the sum of) the (squares) on AC and CB [is] double (the sum of) the (squares) on AD and DC [Prop. 2.9]. Thus, (the sum of) the (squares) on AC and CB is greater than twice the (rectangle contained) by AC and CB . (Which is) the very thing it was required to show.

60.

Proposition 60

Τὸ ἀπὸ τῆς ἐκ δύο ὀνομάτων παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πρῶτην.

The square on a binomial (straight-line) applied to a rational (straight-line) produces as breadth a first binomial (straight-line).†



Ἐστω ἐκ δύο ὀνομάτων ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ , ὥστε τὸ μείζον ὄνομα εἶναι τὸ AG , καὶ ἐκκείσθω ῥητὴ ἡ DE , καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν DE παραβεβλήσθω τὸ $DEZH$ πλάτος ποιῶν τὴν DH . λέγω, ὅτι ἡ DH ἐκ δύο ὀνομάτων ἐστὶ πρώτη.

Παραβεβλήσθω γάρ παρὰ τὴν DE τῷ μὲν ἀπὸ τῆς AG ἴσον τὸ $\Delta\Theta$, τῷ δὲ ἀπὸ τῆς $B\Gamma$ ἴσον τὸ $ΚΛ$. λοιπὸν ἄρα τὸ διζ ὑπὸ τῶν AG , GB ἴσον ἐστὶ τῷ MZ . τετμήσθω ἡ MH διὰ κατὰ τὸ N , καὶ παράλληλος ἡθῇ ἡ NX [ἐκατέρω τῶν ML , HZ]. ἐκάτερον ἄρα τῶν MX , NZ ἴσον ἐστὶ τῷ ἅπασι ὑπὸ τῶν AGB . καὶ ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶν ἡ AB διηρημένη εἰς τὰ ὀνόματα κατὰ τὸ Γ , αἱ AG , GB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· τὰ ἄρα ἀπὸ τῶν AG , GB ῥητὰ ἐστὶ καὶ σύμμετρα ἀλλήλοισι· ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB . καὶ ἐστὶν ἴσον τῷ $\Delta\Lambda$. ῥητὸν ἄρα ἐστὶ τὸ $\Delta\Lambda$. καὶ παρὰ ῥητὴν τὴν DE παράκειται ῥητὴ ἄρα ἐστὶν ἡ ΔM καὶ σύμμετρος τῇ DE μήκει. πάλιν, ἐπεὶ αἱ AG , GB ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, μέσον ἄρα ἐστὶ τὸ διζ ὑπὸ τῶν AG , GB , τουτέστι τὸ MZ . καὶ παρὰ ῥητὴν τὴν ML παράκειται ῥητὴ ἄρα καὶ ἡ MH καὶ ἀσύμμετρος τῇ ML , τουτέστι τῇ DE , μήκει. ἐστὶ δὲ καὶ ἡ $M\Delta$ ῥητὴ καὶ τῇ DE μήκει σύμμετρος· ἀσύμμετρος ἄρα ἐστὶν ἡ ΔM τῇ MH μήκει. καὶ εἰσι ῥηταὶ αἱ ΔM , MH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ DH . δεικτέον δὴ, ὅτι καὶ πρώτη.

Ἐπεὶ τῶν ἀπὸ τῶν AG , GB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AGB , καὶ τῶν $\Delta\Theta$, $ΚΛ$ ἄρα μέσον ἀνάλογόν ἐστι τὸ MX . ἐστὶν ἄρα ὥς τὸ $\Delta\Theta$ πρὸς τὸ MX , οὕτως τὸ MX πρὸς τὸ $ΚΛ$, τουτέστιν ὥς ἡ ΔK πρὸς τὴν MN , ἡ MN πρὸς τὴν MK . τὸ ἄρα ὑπὸ τῶν ΔK , KM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN . καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AG τῷ ἀπὸ τῆς GB , σύμμετρόν ἐστι καὶ τὸ $\Delta\Theta$ τῷ $ΚΛ$. ὥστε καὶ ἡ ΔK τῇ KM

Let AB be a binomial (straight-line), having been divided into its (component) terms at C , such that AC is the greater term. And let the rational (straight-line) DE be laid down. And let the (rectangle) $DEFG$, equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a first binomial (straight-line).

For let DH , equal to the (square) on AC , and KL , equal to the (square) on BC , be applied to DE . Thus, the remaining twice the (rectangle contained) by AC and CB is equal to MF [Prop. 2.4]. Let MG be cut in half at N , and let NO be drawn parallel [to each of ML and GF]. MO and NF are thus each equal to once the (rectangle contained) by ACB . And since AB is a binomial (straight-line), having been divided into its (component) terms at C , AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. Thus, the (squares) on AC and CB are rational, and commensurable with one another. And hence the sum of the (squares) on AC and CB (is rational) [Prop. 10.15], and is equal to DL . Thus, DL is rational. And it is applied to the rational (straight-line) DE . DM is thus rational, and commensurable in length with DE [Prop. 10.20]. Again, since AC and CB are rational (straight-lines which are) commensurable in square only, twice the (rectangle contained) by AC and CB —that is to say, MF —is thus medial [Prop. 10.21]. And it is applied to the rational (straight-line) ML . MG is thus also rational, and incommensurable in length with ML —that is to say, with DE [Prop. 10.22]. And MD is also rational, and commensurable in length with DE . Thus, DM is incommensurable in length with MG [Prop. 10.13]. And they are rational. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line)

σύμμετρος ἔστιν. καὶ ἐπεὶ μείζονά ἐστι τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$ τοῦ διζ ὑπὸ τῶν $ΑΓ$, $ΓΒ$, μείζων ἄρα καὶ τὸ $ΔΑ$ τοῦ $ΜΖ$. ὥστε καὶ ἡ $ΔΜ$ τῆς $ΜΗ$ μείζων ἔστιν. καὶ ἔστιν ἴσον τὸ ὑπὸ τῶν $ΔΚ$, $ΚΜ$ τῷ ἀπὸ τῆς $ΜΝ$, τουτέστι τῷ τετάρτῳ τοῦ ἀπὸ τῆς $ΜΗ$ —.7πτ η, καὶ σύμμετρος ἡ $ΔΚ$ τῇ $ΚΜ$. ἔαν δὲ ὥσι δύο εὐθεῖαι ἄνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ καὶ εἷς σύμμετρα αὐτὴν διαιρῇ, ἡ μείζων τῆς ἐλάσσονος μείζων δύναται τῷ ἀπὸ συμμέτρου ἐκείνης· ἡ $ΔΜ$ ἄρα τῆς $ΜΗ$ μείζων δύναται τῷ ἀπὸ συμμέτρου ἐκείνης. καὶ εἰσι ῥηταὶ αἱ $ΔΜ$, $ΜΗ$, καὶ ἡ $ΔΜ$ μείζων ὄνομα ὅσα σύμμετρος ἔστι τῇ ἐκκειμένῃ ῥητῇ τῇ $ΔΕ$ μήκει.

Ἡ $ΔΗ$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

[Prop. 10.36]. So, we must show that (it is) also a first (binomial straight-line).

Since the (rectangle contained) by ACB is the mean proportional to the squares on AC and CB [Prop. 10.53 lem.], MO is thus also the mean proportional to DH and KL . Thus, as DH is to MO , so MO (is) to KL —that is to say, as DK (is) to MN , (so) MN (is) to MK [Prop. 6.1]. Thus, the (rectangle contained) by DK and KM is equal to the (square) on MN [Prop. 6.17]. And since the (square) on AC is commensurable with the (square) on CB , DH is also commensurable with KL . Hence, DK is also commensurable with KM [Props. 6.1, 10.11]. And since (the sum of) the squares on AC and CB is greater than twice the (rectangle contained) by AC and CB [Prop. 10.59 lem.], DL (is) thus also greater than MF . Hence, DM is also greater than MG [Props. 6.1, 5.14]. And the (rectangle contained) by DK and KM is equal to the (square) on MN —that is to say, to one quarter the (square) on MG . And DK (is) commensurable (in length) with KM . And if there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) commensurable (in length), then the square on the greater is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable (in length) with the greater [Prop. 10.17]. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM). And DM and MG are rational. And DM , which is the greater term, is commensurable in length with the (previously) laid down rational (straight-line) DE .

Thus, DG is a first binomial (straight-line) [Def. 10.5]. (Which is) the very thing it was required to show.

† In other words, the square of a binomial is a first binomial. See Prop. 10.54.

61.

Τὸ ἀπὸ τῆς ἐκ δύο μέσων πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων δευτέραν.

Ἐστω ἐκ δύο μέσων πρώτη ἡ $ΑΒ$ διηρημένη εἰς τὰς μέσας κατὰ τὸ $Γ$, ὧν μείζων ἡ $ΑΓ$, καὶ ἐκκείσθω ῥητὴ ἡ $ΔΕ$, καὶ παραβεβλήσθω παρὰ τὴν $ΔΕ$ τῷ ἀπὸ τῆς $ΑΒ$ ἴσον παραλληλόγραμμον τὸ $ΔΖ$ πλάτος ποιῶν τὴν $ΔΗ$. λέγω, ὅτι ἡ $ΔΗ$ ἐκ δύο ὀνομάτων ἐστὶ δευτέρα.

Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρὸ τούτου. καὶ ἐπεὶ ἡ $ΑΒ$ ἐκ δύο μέσων ἐστὶ πρώτη διηρημένη κατὰ τὸ $Γ$, αἱ $ΑΓ$, $ΓΒ$ ἄρα μέσαι εἰσι δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι· ὥστε καὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$ μέσα ἔστιν. μέσον ἄρα ἐστὶ τὸ $ΔΑ$. καὶ παρὰ ῥητὴν τὴν $ΔΕ$ παραβεβλήται· ῥητὴ ἄρα ἐστὶν ἡ $ΜΔ$ καὶ ἀσύμμετρος τῇ $ΔΕ$ μήκει. πάλιν, ἐπεὶ ῥητὸν ἐστὶ τὸ διζ ὑπὸ τῶν $ΑΓ$, $ΓΒ$, ῥητὸν ἐστὶ καὶ τὸ $ΜΖ$.

Proposition 61

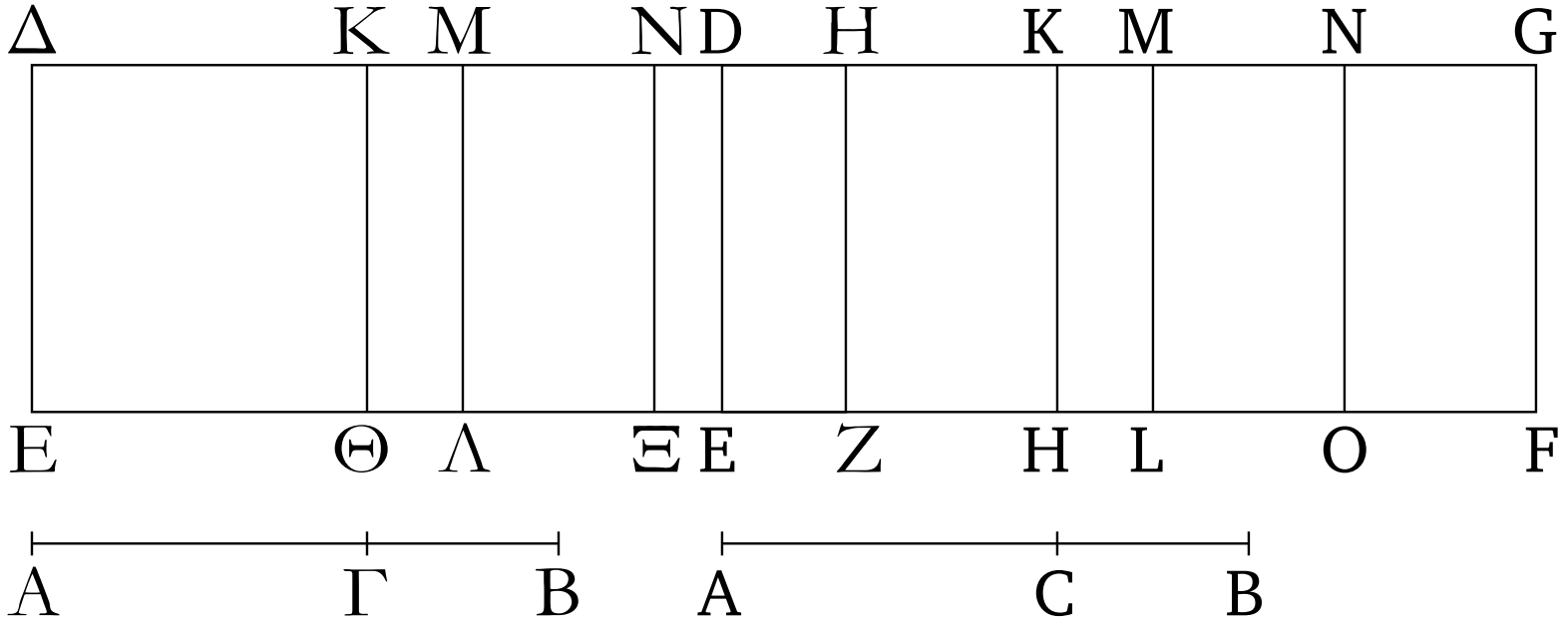
The square on a first binomial (straight-line) applied to a rational (straight-line) produces as breadth a second binomial (straight-line).[†]

Let AB be a first binomial (straight-line) having been divided into its (component) medial (straight-lines) at C , of which AC (is) the greater. And let the rational (straight-line) DE be laid down. And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a second binomial (straight-line).

For let the same construction be made as in the (proposition) before this. And since AB is a first binomial (straight-line), having been divided at C , AC and CB are thus medial (straight-lines) commensurable in square only, and containing a rational (area) [Prop. 10.37]. Hence, the (squares) on

καὶ παρὰ ῥητὴν τὴν $ΜΛ$ παράκειται ῥητὴ ἄρα [ἐστὶ] καὶ ἡ $ΜΗ$ καὶ μήκει σύμμετρος τῇ $ΜΛ$, τούτεστι τῇ $ΔΕ$. ἀσύμμετρος ἄρα ἐστὶν ἡ $ΔΜ$ τῇ $ΜΗ$ μήκει. καὶ εἰσι ῥηταί· αἱ $ΔΜ$, $ΜΗ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $ΔΗ$. δεικτέον δὴ, ὅτι καὶ δευτέρα.

AC and CB are also medial [Prop. 10.21]. Thus, DL is medial [Props. 10.15, 10.23 corr.]. And it has been applied to the rational (straight-line) DE . MD is thus rational, and incommensurable in length with DE [Prop. 10.22]. Again, since twice the (rectangle contained) by AC and CB is rational, MF is also rational. And it is applied to the rational (straight-line) ML . Thus, MG [is] also rational, and commensurable in length with ML —that is to say, with DE [Prop. 10.20]. DM is thus incommensurable in length with MG [Prop. 10.13]. And they are rational. DM and MG are thus rational, and commensurable in square only. DG is thus a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a second (binomial straight-line).



Ἐπεὶ γὰρ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$ μείζονά ἐστι τοῦ δι᾽ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ μείζον ἄρα καὶ τὸ $ΔΛ$ τοῦ $ΜΖ$. ὥστε καὶ ἡ $ΔΜ$ τῇς $ΜΗ$. καὶ ἐπεὶ σύμμετρον ἐστὶ τὸ ἀπὸ τῇς $ΑΓ$ τῷ ἀπὸ τῇς $ΓΒ$, σύμμετρον ἐστὶ καὶ τὸ $ΔΘ$ τῷ $ΚΛ$. ὥστε καὶ ἡ $ΔΚ$ τῇ $ΚΜ$ σύμμετρος ἐστίν. καὶ ἐστὶ τὸ ὑπὸ τῶν $ΔΚΜ$ ἴσον τῷ ἀπὸ τῇς $ΜΝ$. ἡ $ΔΜ$ ἄρα τῇς $ΜΗ$ μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντῇ. καὶ ἐστὶν ἡ $ΜΗ$ σύμμετρος τῇ $ΔΕ$ μήκει.

Ἡ $ΔΗ$ ἄρα ἐκ δύο ὀνομάτων ἐστὶ δευτέρα.

For since (the sum of) the squares on AC and CB is greater than twice the (rectangle contained) by AC and CB [Prop. 10.59], DL (is) thus also greater than MF . Hence, DM (is) also (greater) than MG [Prop. 6.1]. And since the (square) on AC is commensurable with the (square) on CB , DH is also commensurable with KL . Hence, DK is also commensurable (in length) with KM [Props. 6.1, 10.11]. And the (rectangle contained) by DKM is equal to the (square) on MN . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM) [Prop. 10.17]. And MG is commensurable in length with DE .

Thus, DG is a second binomial (straight-line) [Def. 10.6].

† In other words, the square of a first binomial is a second binomial. See Prop. 10.55.

Τὸ ἀπὸ τῇς ἐκ δύο μέσων δευτέρας παρὰ ῥητὴν παραβαλ-

The square on a second binomial (straight-line) applied to

λόμενον πλάτος ποιῇ τὴν ἐκ δύο ὀνομάτων τρίτην.

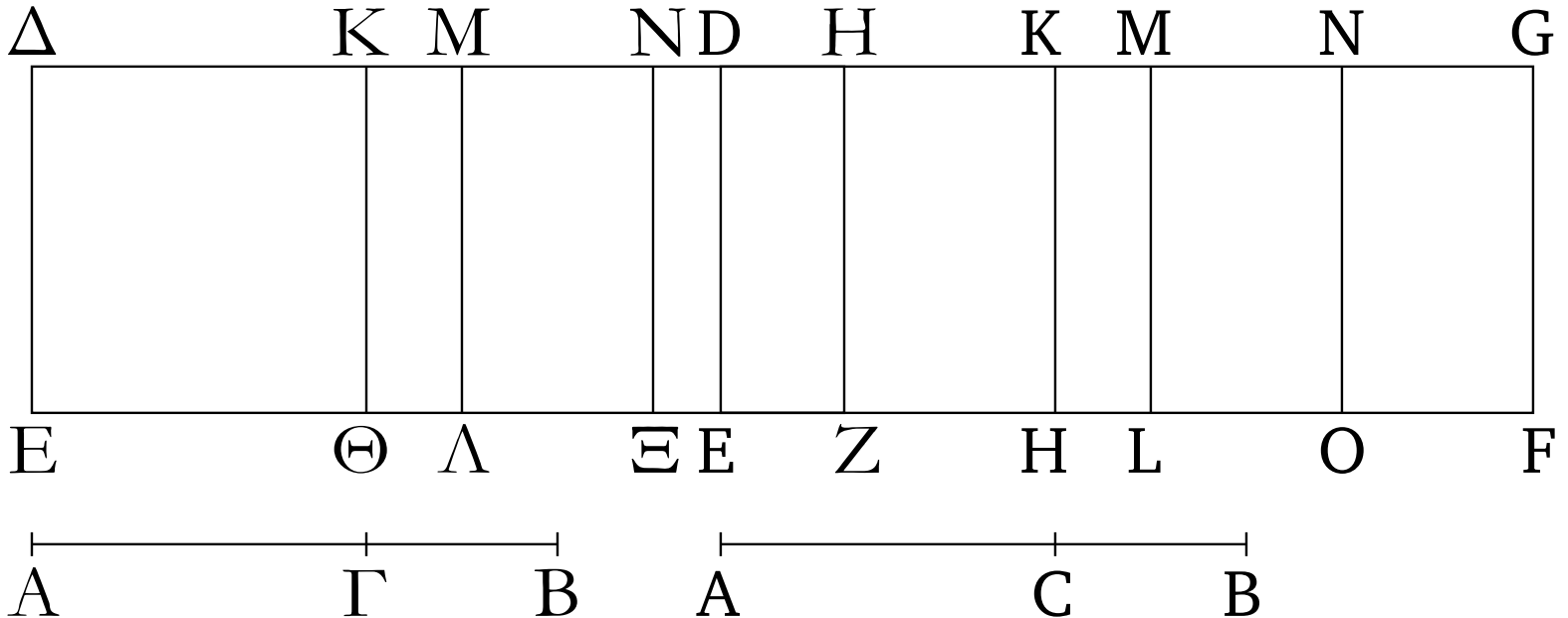
Ἐστω ἐκ δύο μέσων δευτέρα ἡ AB διηρημένη εἰς τὰς μέσας κατὰ τὸ Γ , ὥστε τὸ μείζον τμήμα εἶναι τὸ AG , ῥητὴ δὲ τις ἔστω ἡ ΔE , καὶ παρὰ τὴν ΔE τῷ ἀπὸ τῆς AB ἴσον παραλληλόγραμμον παραβεβλήσθω τὸ ΔZ πλάτος ποιῶν τὴν ΔH . λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶ τρίτη.

Κατεσκευάσθω τὰ αὐτὰ τοῖς προοδευμένοις. καὶ ἐπεὶ ἐκ δύο μέσων δευτέρα ἐστὶν ἡ AB διηρημένη κατὰ τὸ Γ , αἱ AG , GB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι. ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB μέσον ἐστίν. καὶ ἐστὶν ἴσον τῷ $\Delta\Lambda$. μέσον ἄρα καὶ τὸ $\Delta\Lambda$. καὶ παράκειται παρὰ ῥητὴν τὴν ΔE . ῥητὴ ἄρα ἐστὶ καὶ ἡ $M\Delta$ καὶ ἀσύμμετρος τῇ ΔE μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ MH ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ $M\Delta$, τοντέστι τῇ ΔE , μήκει. ῥητὴ ἄρα ἐστὶν ἑκατέρα τῶν ΔM , MH καὶ ἀσύμμετρος τῇ ΔE μήκει. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AG τῇ GB μήκει, ὡς δὲ ἡ AG πρὸς τὴν GB , οὕτως τὸ ἀπὸ τῆς AG πρὸς τὸ ὑπὸ τῶν AGB , ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς AG τῷ ὑπὸ τῶν AGB . ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB τῷ δι᾽ ὑπὸ τῶν AGB ἀσύμμετρόν ἐστιν, τοντέστι τὸ $\Delta\Lambda$ τῷ MZ . ὥστε καὶ ἡ ΔM τῷ MH ἀσύμμετρος ἐστίν. καὶ εἰσι ῥηταί. ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH . δεικτέον [δὴ], ὅτι καὶ τρίτη.

a rational (straight-line) produces as breadth a third binomial (straight-line).[†]

Let AB be a second binomial (straight-line) having been divided into its (component) medial (straight-lines) at C , such that AC is the greater segment. And let DE be some rational (straight-line). And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a third binomial (straight-line).

Let the same construction be made as that shown previously. And since AB is a second binomial (straight-line), having been divided at C , AC and CB are thus medial (straight-lines) commensurable in square only, and containing a medial (area) [Prop. 10.38]. Hence, the sum of the (squares) on AC and CB is also medial [Props. 10.15, 10.23 corr.]. And it is equal to DL . Thus, DL (is) also medial. And it is applied to the rational (straight-line) DE . MD is thus also rational, and incommensurable in length with DE [Prop. 10.22]. So, for the same (reasons), MG is also rational, and incommensurable in length with ML —that is to say, with DE . Thus, DM and MG are each rational, and incommensurable in length with DE . And since AC is incommensurable in length with CB , and as AC (is) to CB , so the (square) on AC (is) to the (rectangle contained) by ACB [Prop. 10.21 lem.], the (square) on AC (is) also incommensurable with the (rectangle contained) by ACB [Prop. 10.11]. And hence the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by ACB —that is to say, DL with MF [Props. 10.12, 10.13]. Hence, DM is also incommensurable (in length) with MG [Props. 6.1, 10.11]. And they are rational. DG is thus a binomial (straight-line) [Prop. 10.36]. [So] we must show that (it is) also a third (binomial straight-line).



Ὁμοίως δὴ τοῖς προτέροις ἐπιλογιόμεθα, ὅτι μείζων ἐστὶν ἡ ΔM τῆς MH , καὶ σύμμετρος ἡ ΔK τῇ KM . καὶ ἐστὶ τὸ ὑπὸ τῶν ΔKM ἴσον τῷ ἀπὸ τῆς MN . ἡ ΔM ἄρα τῆς MH μείζων δύνатаι τῷ ἀπὸ συμμέτρου ἐαυτῆς. καὶ οὐδετέρα τῶν ΔM , MH σύμμετρος ἐστὶ τῇ ΔE μήκει.

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη· ὅπερ ἔδει δεῖχαι.

So, similarly to the previous (propositions), we can conclude that DM is greater than MG , and DK (is) commensurable (in length) with KM . And the (rectangle contained) by DKM is equal to the (square) on MN . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM) [Prop. 10.17]. And neither of DM and MG is commensurable in length with DE .

Thus, DG is a third binomial (straight-line) [Def. 10.7]. (Which is) the very thing it was required to show.

† In other words, the square of a second binomial is a third binomial. See Prop. 10.56.

63.

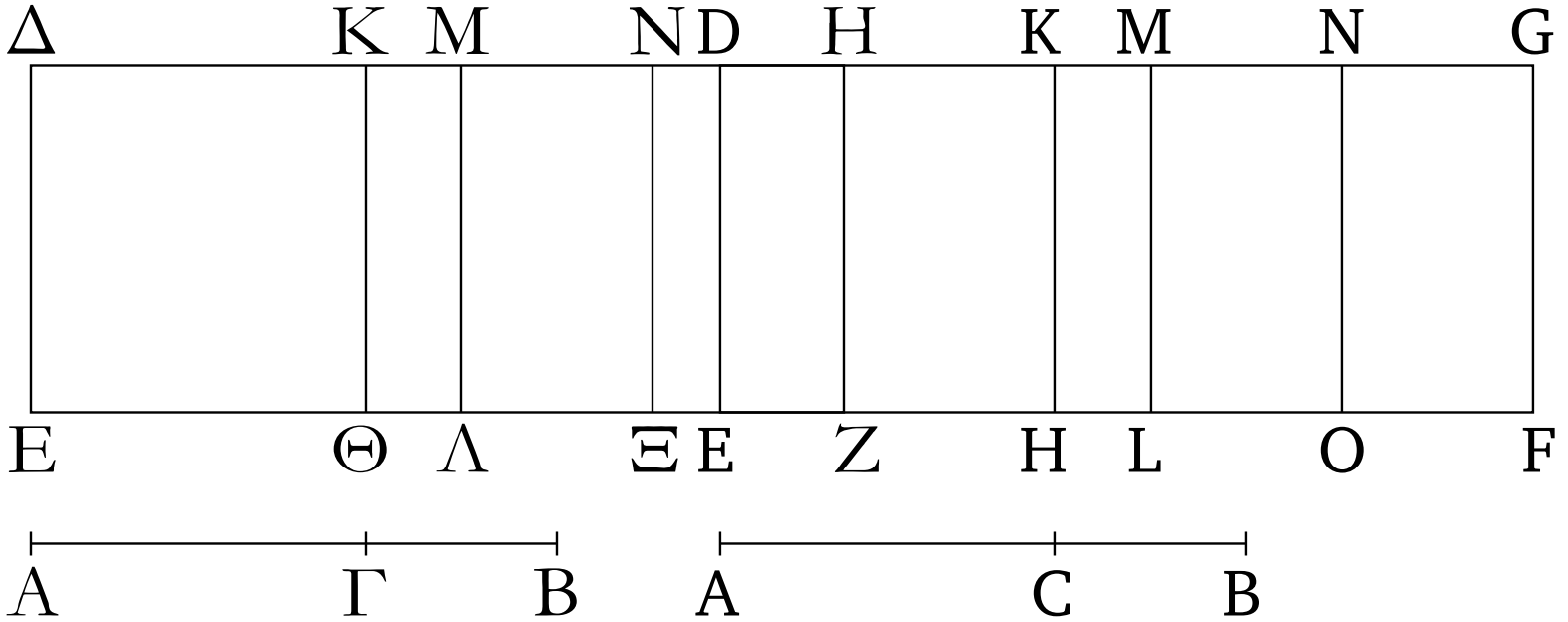
Proposition 63

Τὸ ἀπὸ τῆς μείζονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τετάρτην.

Ἐστω μείζων ἡ AB διηρημένη κατὰ τὸ Γ , ὥστε μείζονα εἶναι τὴν AG τῆς GB , ῥητὴ δὲ ἡ ΔE , καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν ΔE παραβεβλήσθω τὸ ΔZ παραλληλόγραμμον πλάτος ποιοῦν τὴν ΔH . λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶ τετάρτη.

The square on a major (straight-line) applied to a rational (straight-line) produces as breadth a fourth binomial (straight-line).†

Let AB be a major (straight-line) having been divided at C , such that AC is greater than CB , and (let) DE (be) a rational (straight-line). And let the parallelogram DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a fourth binomial (straight-line).



Κατεσκευάσθω τὰ αὐτὰ τοῖς προοδεγεγμένοις. καὶ ἐπεὶ μείζων ἐστὶν ἡ AB διηρημένη κατὰ τὸ Γ , αἱ AG , GB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δὲ ὑπ' αὐτῶν μέσον. ἐπεὶ οὖν ῥητόν ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB , ῥητόν ἄρα ἐστὶ τὸ ΔA . ῥητὴ ἄρα καὶ ἡ ΔM καὶ σύμμετρος τῇ ΔE μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ διξ ὑπὸ τῶν AG , GB , τουτέστι τὸ MZ , καὶ παρὰ ῥητὴν ἐστὶ τὴν MA , ῥητὴ ἄρα ἐστὶ καὶ ἡ MH καὶ ἀσύμμετρος τῇ ΔE μήκει· ἀσύμμετρος

Let the same construction be made as that shown previously. And since AB is a major (straight-line), having been divided at C , AC and CB are incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial [Prop. 10.39]. Therefore, since the sum of the (squares) on AC and CB is rational, DL is thus rational. Thus, DM (is) also rational, and commensurable in length with DE [Prop. 10.20]. Again, since twice the (rectangle contained) by AC and CB —that is to say,

ἄρα ἐστὶ καὶ ἡ ΔM τῇ MH μήκει. αἱ ΔM , MH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH . δεικτέον [δὴ], ὅτι καὶ τετάρτη.

Ὁμοίως δὴ δείχομεν τοῖς πρότερον, ὅτι μείζων ἐστὶν ἡ ΔM τῇ MH , καὶ ὅτι τὸ ὑπὸ ΔKM ἴσον ἐστὶ τῷ ἀπὸ τῇ MN . ἐπεὶ οὖν ἀσύμμετρόν ἐστι τὸ ἀπὸ τῇ AF τῷ ἀπὸ τῇ GB , ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ $\Delta \Theta$ τῷ KL . ὥστε ἀσύμμετρος καὶ ἡ ΔK τῇ KM ἐστίν. ἔαν δὲ ὥσι δύο εὐθεῖαι ἄνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῇ ἐλάσσονος ἴσον παραλληλόγραμμον παρὰ τὴν μείζονα παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ καὶ εἰς ἀσύμμετρα αὐτὴν διαιρῇ, ἡ μείζων τῇ ἐλάσσονος μείζον δυνήσεται τῷ ἀπὸ ἀσύμμετρον ἐαυτῇ μήκει· ἡ ΔM ἄρα τῇ MH μείζον δύνатаι τῷ ἀπὸ ἀσύμμετρον ἐαυτῇ. καὶ εἰσιν αἱ ΔM , MH ῥηταὶ δυνάμει μόνον σύμμετροι, καὶ ἡ ΔM σύμμετρος ἐστὶ τῇ ἐκκειμένῃ ῥητῇ τῇ ΔE .

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶ τετάρτη· ὅπερ ἔδει δεῖχαι.

MF —is medial, and is (applied to) the rational (straight-line) ML , MG is thus also rational, and incommensurable in length with DE [Prop. 10.22]. DM is thus also incommensurable in length with MG [Prop. 10.13]. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. [So] we must show that (it is) also a fourth (binomial straight-line).

So, similarly to the previous (propositions), we can show that DM is greater than MG , and that the (rectangle contained) by DKM is equal to the (square) on MN . Therefore, since the (square) on AC is incommensurable with the (square) on CB , DH is also incommensurable with KL . Hence, DK is also incommensurable with KM [Props. 6.1, 10.11]. And if there are two unequal straight-lines, and a parallelogram equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) incommensurable (in length), then the square on the greater will be larger than (the square on) the lesser by the (square) on (some straight-line) incommensurable in length with the greater [Prop. 10.18]. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) incommensurable (in length) with (DM). And DM and MG are rational (straight-lines which are) commensurable in square only. And DM is commensurable (in length) with the (previously) laid down rational (straight-line) DE .

Thus, DG is a fourth binomial (straight-line) [Def. 10.8]. (Which is) the very thing it was required to show.

† In other words, the square of a major is a fourth binomial. See Prop. 10.57.

64.

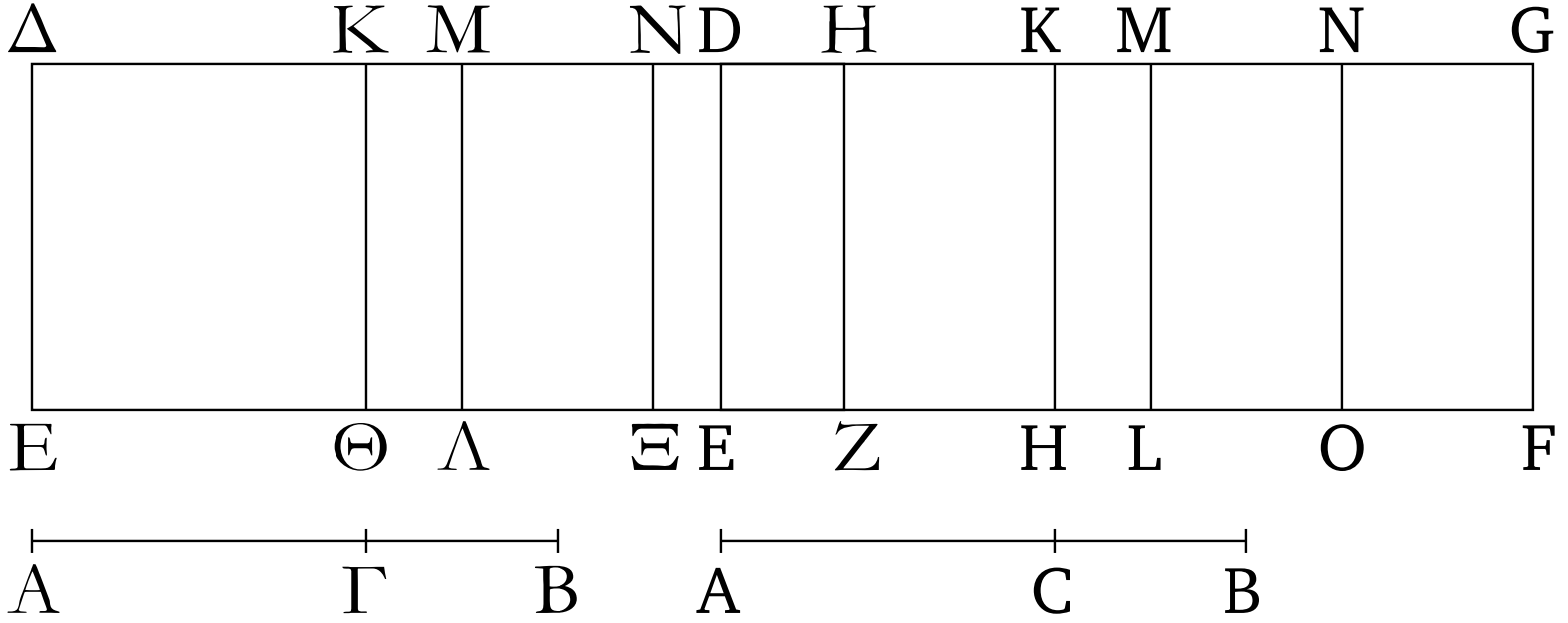
Τὸ ἀπὸ τῇ ῥητὸν καὶ μέσον δυναμένης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πέμπτην.

Ἐστω ῥητὸν καὶ μέσον δυναμένη ἡ AB διηρημένη εἰς τὰς εὐθείας κατὰ τὸ Γ , ὥστε μείζονα εἶναι τὴν AG , καὶ ἐκκείσθω ῥητὴ ἡ ΔE , καὶ τῷ ἀπὸ τῇ AB ἴσον παρὰ τὴν ΔE παραβλήσθω τὸ ΔZ πλάτος ποιοῦν τὴν ΔH . λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶ πέμπτη.

Proposition 64

The square on the square-root of a rational plus a medial (area) applied to a rational (straight-line) produces as breadth a fifth binomial (straight-line).[†]

Let AB be the square-root of a rational plus a medial (area) having been divided into its (component) straight-lines at C , such that AC is greater. And let the rational (straight-line) DE be laid down. And let the (parallelogram) DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a fifth binomial straight-line.



Κατεσκευάσθω τὰ αὐτὰ τοῖς πρὸ τούτων. ἐπεὶ οὖν ῥητόν καὶ μέσον δυναμένη ἐστὶν ἡ AB διηρημένη κατὰ τὸ Γ , αἱ AG , GB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν. ἐπεὶ οὖν μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB , μέσον ἄρα ἐστὶ τὸ $\Delta\Lambda$. ὥστε ῥητὴ ἐστὶν ἡ ΔM καὶ μήκει ἀσύμμετρος τῇ ΔE . πάλιν, ἐπεὶ ῥητόν ἐστὶ τὸ διξ ὑπὸ τῶν AGB , τοντέστι τὸ MZ , ῥητὴ ἄρα ἡ MH καὶ σύμμετρος τῇ ΔE . ἀσύμμετρος ἄρα ἡ ΔM τῇ MH . αἱ ΔM , MH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH . λέγω δὴ, ὅτι καὶ πέμπτη.

Ὅμοιως γὰρ διεθθήσεται, ὅτι τὸ ὑπὸ τῶν ΔKM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN , καὶ ἀσύμμετρος ἡ ΔK τῇ KM μήκει· ἡ ΔM ἄρα τῆς MH μείζον δύνανται τῷ ἀπὸ ἀσυμμέτρον ἑαντῇ. καὶ εἰσιν αἱ ΔM , MH [ῥηταὶ] δυνάμει μόνον σύμμετροι, καὶ ἡ ἐλάσσων ἡ MH σύμμετρος τῇ ΔE μήκει.

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶ πέμπτη· ὅπερ ἔδει δεῖχαι.

Let the same construction be made as in the (propositions) before this. Therefore, since AB is the square-root of a rational plus a medial (area), having been divided at C , AC and CB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational [Prop. 10.40]. Therefore, since the sum of the (squares) on AC and CB is medial, DL is thus medial. Hence, DM is rational and incommensurable in length with DE [Prop. 10.22]. Again, since twice the (rectangle contained) by ACB —that is to say, MF —is rational, MG (is) thus rational and commensurable (in length) with DE [Prop. 10.20]. DM (is) thus incommensurable (in length) with MG [Prop. 10.13]. Thus, DM and MG are rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fifth (binomial straight-line).

For, similarly (to the previous propositions), it can be shown that the (rectangle contained) by DKM is equal to the (square) on MN , and DK (is) incommensurable in length with KM . Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) incommensurable (in length) with (DM) [Prop. 10.18]. And DM and MG are [rational] (straight-lines which are) commensurable in square only, and the lesser MG is commensurable in length with DE .

Thus, DG is a fifth binomial (straight-line) [Def. 10.9]. (Which is) the very thing it was required to show.

† In other words, the square of the square-root of a rational plus medial is a fifth binomial. See Prop. 10.58.

Τὸ ἀπὸ τῆς δύο μέσα δυναμένης παρὰ ῥητὴν παραβαλ-

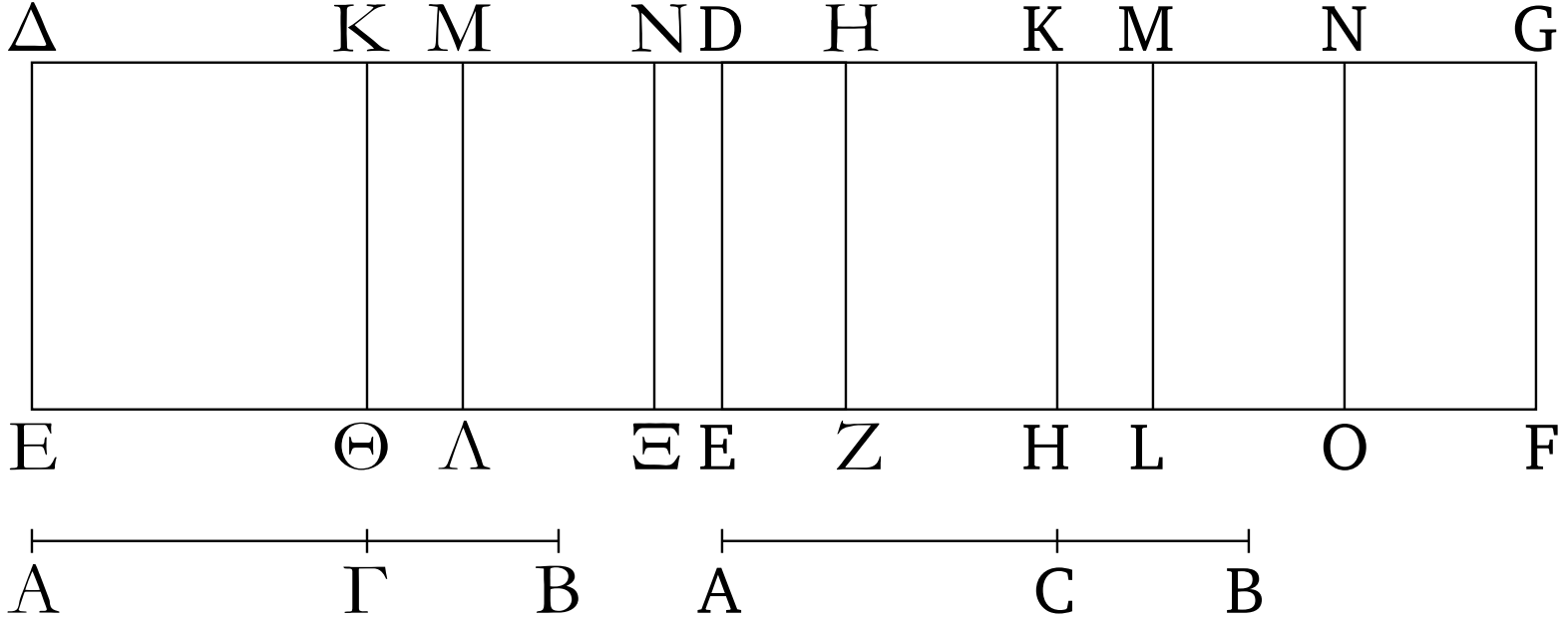
The square on the square-root of (the sum of) two medial

λόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων ἑκτὴν.

Ἐστω δύο μέσα δυναμένη ἡ AB διηρημένη κατὰ τὸ Γ , ῥητὴ δὲ ἔστω ἡ ΔE , καὶ παρὰ τὴν ΔE τῷ ἀπὸ τῆς AB ἴσον παραβεβλήσθω τὸ ΔZ πλάτος ποιοῦν τὴν ΔH . λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶν ἑκτὴν.

(areas) applied to a rational (straight-line) produces as breadth a sixth binomial (straight-line).[†]

Let AB be the square-root of (the sum of) two medial (areas), having been divided at C . And let DE be a rational (straight-line). And let the (parallelogram) DF , equal to the (square) on AB , be applied to DE , producing DG as breadth. I say that DG is a sixth binomial (straight-line).



Κατεσκευάσθω γάρ τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἡ AB δύο μέσα δυναμένη ἐστὶ διηρημένη κατὰ τὸ Γ , αἱ $ΑΓ$, $ΓΒ$ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τὸ ἐκ τῶν ἀπ' αὐτῶν τετραγώνων συγκείμενον τῷ ὑπ' αὐτῶν ὥστε κατὰ τὰ προδεδειγμένα μέσον ἐστὶν ἑκάτερον τῶν $\Delta\Lambda$, MZ . καὶ παρὰ ῥητὴν τὴν ΔE παράκειται ῥητὴ ἄρα ἐστὶν ἑκάτερον τῶν ΔM , MH καὶ ἀσύμμετρος τῇ ΔE μήκει. καὶ ἐπεὶ ἀσύμμετρον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΑΓ$, $ΓΒ$ τῷ διζ' ὑπὸ τῶν $ΑΓ$, $ΓΒ$, ἀσύμμετρον ἄρα ἐστὶ τὸ $\Delta\Lambda$ τῷ MZ . ἀσύμμετρος ἄρα καὶ ἡ ΔM τῇ MH . αἱ ΔM , MH ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH . λέγω δὴ, ὅτι καὶ ἑκτὴν.

Ὁμοίως δὴ πάλιν δεῖχμεν, ὅτι τὸ ὑπὸ τῶν ΔKM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN , καὶ ὅτι ἡ ΔK τῇ KM μήκει ἐστὶν ἀσύμμετρος· καὶ διὰ τὰ αὐτὰ δὴ ἡ ΔM τῇ MH μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἐναντὶ μήκει. καὶ οὐδετέρα τῶν ΔM , MH σύμμετρος ἐστὶ τῇ ἐκκεκλιμένη ῥητῇ τῇ ΔE μήκει.

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶν ἑκτὴν ὅπερ ἔδει δεῖξαι.

For let the same construction be made as in the previous (propositions). And since AB is the square-root of (the sum of) two medial (areas), having been divided at C , AC and CB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, the sum of the squares on them incommensurable with the (rectangle contained) by them [Prop. 10.41]. Hence, according to what has been previously demonstrated, DL and MF are each medial. And they are applied to the rational (straight-line) DE . Thus, DM and MG are each rational, and incommensurable in length with DE [Prop. 10.22]. And since the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , DL is thus incommensurable with MF . Thus, DM (is) also incommensurable (in length) with MG [Props. 6.1, 10.11]. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a sixth (binomial straight-line).

So, similarly (to the previous propositions), we can again show that the (rectangle contained) by DKM is equal to the (square) on MN , and that DK is incommensurable in length with KM . And so, for the same (reasons), the square on DM is greater than (the square on) MG by the (square) on

(some straight-line) incommensurable in length with (DM) [Prop. 10.18]. And neither of DM and MG is commensurable in length with the (previously) laid down rational (straight-line) DE .

Thus, DG is a sixth binomial (straight-line) [Def. 10.10]. (Which is) the very thing it was required to show.

† In other words, the square of the square-root of two medials is a sixth binomial. See Prop. 10.59.

66.

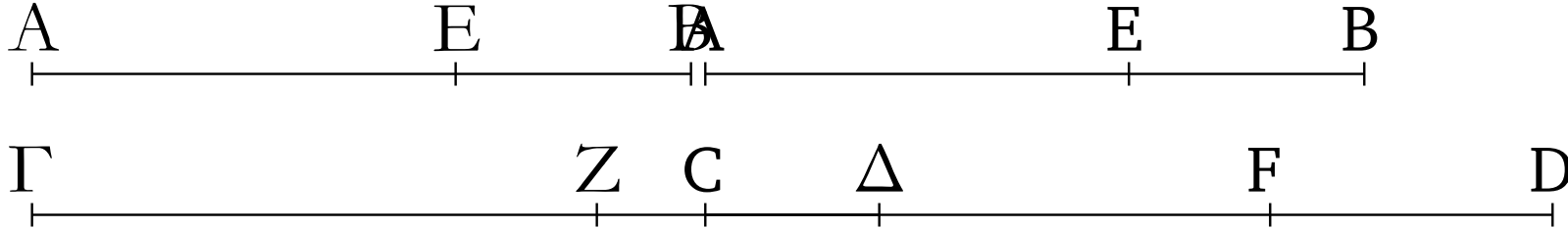
Proposition 66

Ἡ τῇ ἐκ δύο ὀνομάτων μήκει σύμμετρος καὶ αὐτῇ ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ τάχει ἡ αὐτή.

Ἐστω ἐκ δύο ὀνομάτων ἡ AB , καὶ τῇ AB μήκει σύμμετρος ἔστω ἡ $\Gamma\Delta$. λέγω, ὅτι ἡ $\Gamma\Delta$ ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ τάχει ἡ αὐτῇ τῇ AB .

A (straight-line) commensurable in length with a binomial (straight-line) is itself also binomial, and the same in order.

Let AB be a binomial (straight-line), and let CD be commensurable in length with AB . I say that CD is a binomial (straight-line), and (is) the same in order as AB .



Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων ἐστὶν ἡ AB , διηρήσθω εἰς τὰ ὀνόματα κατὰ τὸ E , καὶ ἔστω μείζων ὄνομα τὸ AE . αἱ AE , EB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. γερονέτω ὡς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ AE πρὸς τὴν ΓZ καὶ λοιπὴ ἄρα ἡ EB πρὸς λοιπὴν τὴν $Z\Delta$ ἐστίν, ὡς ἡ AB πρὸς τὴν $\Gamma\Delta$. σύμμετρος δὲ ἡ AB τῇ $\Gamma\Delta$ μήκει· σύμμετρος ἄρα ἐστὶ καὶ ἡ μὲν AE τῇ ΓZ , ἡ δὲ EB τῇ $Z\Delta$. καὶ εἰσι ῥηταὶ αἱ AE , EB . ῥηταὶ ἄρα εἰσὶ καὶ αἱ ΓZ , $Z\Delta$. καὶ ἐστὶν ὡς ἡ AE πρὸς ΓZ , ἡ EB πρὸς $Z\Delta$. ἐναλλάξ ἄρα ἐστὶν ὡς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$. αἱ δὲ AE , EB δυνάμει μόνον [εἰσὶ] σύμμετροι· καὶ αἱ ΓZ , $Z\Delta$ ἄρα δυνάμει μόνον εἰσὶ σύμμετροι. καὶ εἰσι ῥηταὶ· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $\Gamma\Delta$. λέγω δὴ, ὅτι τῇ τάχει ἐστὶν ἡ αὐτῇ τῇ AB .

Ἡ γὰρ AE τῇ EB μείζων δύναται ἥτοι τῷ ἀπὸ συμμέτρου ἐαντῇ ἢ τῷ ἀπὸ ἀσυνμέτρου. εἰ μὲν οὖν ἡ AE τῇ EB μείζων δύναται τῷ ἀπὸ συμμέτρου ἐαντῇ, καὶ ἡ ΓZ τῇ $Z\Delta$ μείζων δυνήσεται τῷ ἀπὸ συμμέτρου ἐαντῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ AE τῇ ἐκκεκμένη ῥητῇ, καὶ ἡ ΓZ σύμμετρος αὐτῇ ἔσται, καὶ διὰ τοῦτο ἑκάτερα τῶν AB , $\Gamma\Delta$ ἐκ δύο ὀνομάτων ἐστὶ πρώτη, τουτέστι τῇ τάχει ἡ αὐτή. εἰ δὲ ἡ EB σύμμετρος ἐστὶ τῇ ἐκκεκμένη ῥητῇ, καὶ ἡ $Z\Delta$ σύμμετρος αὐτῇ, καὶ διὰ τοῦτο πάλιν τῇ τάχει ἡ αὐτῇ ἔσται τῇ AB . ἑκάτερα γὰρ αὐτῶν ἔσται ἐκ δύο ὀνομάτων δευτέρα. εἰ δὲ οὐδετέρα τῶν AE , EB σύμμετρος ἐστὶ τῇ ἐκκεκμένη ῥητῇ, οὐδετέρα τῶν ΓZ , $Z\Delta$ σύμμετρος αὐτῇ ἔσται, καὶ ἐστὶν ἑκάτερα τρίτη. εἰ δὲ ἡ AE τῇ EB μείζων δύναται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ, καὶ ἡ ΓZ τῇ $Z\Delta$ μείζων δύναται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ.

For since AB is a binomial (straight-line), let it be divided into its (component) terms at E , and let AE be the greater term. AE and EB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. Let it be contrived that as AB (is) to CD , so AE (is) to CF [Prop. 6.12]. Thus, the remainder EB is also to the remainder FD , as AB (is) to CD [Props. 6.16, 5.19 corr.]. And AB (is) commensurable in length with CD . Thus, AE is also commensurable (in length) with CF , and EB with FD [Prop. 10.11]. And AE and EB are rational. Thus, CF and FD are also rational. And as AE is to CF , (so) EB (is) to FD [Prop. 5.11]. Thus, alternately, as AE is to EB , (so) CF (is) to FD [Prop. 5.16]. And AE and EB [are] commensurable in square only. Thus, CF and FD are also commensurable in square only [Prop. 10.11]. And they are rational. CD is thus a binomial (straight-line) [Prop. 10.36]. So, I say that it is the same in order as AB .

For the square on AE is greater than (the square on) EB by the (square) on (some straight-line) either commensurable or incommensurable (in length) with (AE). Therefore, if the square on AE is greater than (the square on) EB by the (square) on (some straight-line) commensurable (in length) with (AE) then the square on CF will also be greater than (the square on) FD by the (square) on (some straight-line) commensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable (in length) with (some previously) laid down rational (straight-line) then CF will also be commensurable (in length) with it [Prop. 10.12]. And, on account of this, AB and CD are each

καὶ εἰ μὲν ἡ AE σύμμετρος ἔστω τῇ ἐκκεκμημένη ῥήτῃ, καὶ ἡ ΓZ σύμμετρος ἔστω αὐτῇ, καὶ ἔστω ἑκατέρα τετάρτη. εἰ δὲ ἡ EB , καὶ ἡ $Z\Delta$, καὶ ἔστω ἑκατέρα πέμπτη. εἰ δὲ οὐδετέρα τῶν AE , EB , καὶ τῶν ΓZ , $Z\Delta$ οὐδετέρα σύμμετρος ἔστω τῇ ἐκκεκμημένη ῥήτῃ, καὶ ἔστω ἑκατέρα ἕκτη.

Ὡστε ἡ τῇ ἐκ δύο ὀνομάτων μήκει σύμμετρος ἐκ δύο ὀνομάτων ἔστι καὶ τῇ τάχει ἡ αὐτή· ὅπερ ἔδει δεῖξαι.

first binomial (straight-lines) [Def. 10.5]—that is to say, the same in order. And if EB is commensurable (in length) with the (previously) laid down rational (straight-line) then FD is also commensurable (in length) with it [Prop. 10.12], and, again, on account of this, (CD) will be the same in order as AB . For each of them will be second binomial (straight-lines) [Def. 10.6]. And if neither of AE and EB is commensurable (in length) with the (previously) laid down rational (straight-line) then neither of CF and FD will be commensurable (in length) with it [Prop. 10.13], and each (of AB and CD) is a third (binomial straight-line) [Def. 10.7]. And if the square on AE is greater than (the square on) EB by the (square) on (some straight-line) incommensurable (in length) with (AE) then the square on CF is also greater than (the square on) FD by the (square) on (some straight-line) incommensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable (in length) with the (previously) laid down rational (straight-line) then CF is also commensurable (in length) with it [Prop. 10.12], and each (of AB and CD) is a fourth (binomial straight-line) [Def. 10.8]. And if EB (is commensurable in length with the previously laid down rational straight-line) then FD (is) also (commensurable in length with it), and each (of AB and CD) will be a fifth (binomial straight-line) [Def. 10.9]. And if neither of AE and EB (is commensurable in length with the previously laid down rational straight-line) then also neither of CF and FD is commensurable (in length) with the laid down rational (straight-line), and each (of AB and CD) will be a sixth (binomial straight-line) [Def. 10.10].

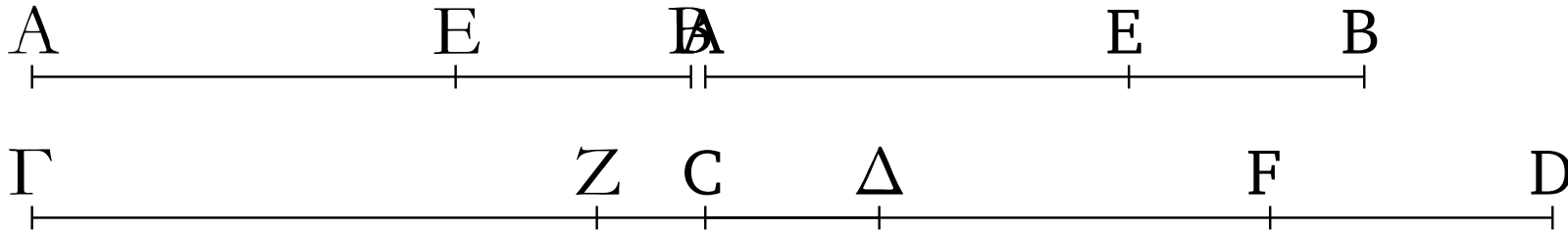
Hence, a (straight-line) commensurable in length with a binomial (straight-line) is a binomial (straight-line), and the same in order. (Which is) the very thing it was required to show.

67.

Proposition 67

Ἡ τῇ ἐκ δύο μέσων μήκει σύμμετρος καὶ αὐτὴ ἐκ δύο μέσων ἔστι καὶ τῇ τάχει ἡ αὐτή.

A (straight-line) commensurable in length with a bimedral (straight-line) is itself also bimedral, and the same in order.



Ἐστω ἐκ δύο μέσων ἡ AB , καὶ τῇ AB σύμμετρος ἔστω μήκει ἡ $\Gamma\Delta$. λέγω, ὅτι ἡ $\Gamma\Delta$ ἐκ δύο μέσων ἔστι καὶ τῇ τάχει ἡ αὐτὴ τῇ AB .

Ἐπεὶ γὰρ ἐκ δύο μέσων ἐστὶν ἡ AB , διηγήσθω εἰς τὰς μέσας κατὰ τὸ E αἱ AE , EB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ γεγονέντω ὥς ἡ AB πρὸς $\Gamma\Delta$, ἡ AE πρὸς

Let AB be a bimedral (straight-line), and let CD be commensurable in length with AB . I say that CD is bimedral, and the same in order as AB .

For since AB is a bimedral (straight-line), let it be divided into its (component) medial (straight-lines) at E . Thus, AE and EB are medial (straight-lines which are) commensurable

ΓZ · καὶ λοιπὴ ἄρα ἡ EB πρὸς λοιπὴν τὴν $Z\Delta$ ἐστίν, ὡς ἡ AB πρὸς $\Gamma\Delta$. σύμμετρος δὲ ἡ AB τῇ $\Gamma\Delta$ μήκει· σύμμετρος ἄρα καὶ ἑκατέρα τῶν AE , EB ἑκατέρῃ τῶν ΓZ , $Z\Delta$. μέσαι δὲ αἱ AE , EB · μέσαι ἄρα καὶ αἱ ΓZ , $Z\Delta$. καὶ ἐπεὶ ἐστὶν ὡς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$, αἱ δὲ AE , EB δυνάμει μόνον σύμμετροί εἰσιν, καὶ αἱ ΓZ , $Z\Delta$ [ἄρα] δυνάμει μόνον σύμμετροί εἰσιν, ἐδείκθησαν δὲ καὶ μέσαι· ἡ $\Gamma\Delta$ ἄρα ἐκ δύο μέσων ἐστίν. λέγω δὴ, ὅτι καὶ τῇ τάχει ἡ αὐτὴ ἐστὶ τῇ AB .

Ἐπεὶ γάρ ἐστιν ὡς ἡ AE πρὸς EB , ἡ ΓZ πρὸς $Z\Delta$, καὶ ὡς ἄρα τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AEB , οὕτως τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν $\Gamma Z\Delta$ · ἐναλλάξ ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς ΓZ , οὕτως τὸ ὑπὸ τῶν AEB πρὸς τὸ ὑπὸ τῶν $\Gamma Z\Delta$. σύμμετρον δὲ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς ΓZ · σύμμετρον ἄρα καὶ τὸ ὑπὸ τῶν AEB τῷ ὑπὸ τῶν $\Gamma Z\Delta$. εἴτε οὖν ῥητὸν ἐστὶ τὸ ὑπὸ τῶν AEB , καὶ τὸ ὑπὸ τῶν $\Gamma Z\Delta$ ῥητὸν ἐστὶν [καὶ διὰ τοῦτο ἐστὶν ἐκ δύο μέσων πρώτης]. εἴτε μέσον, καὶ ἐστὶν ἑκατέρα δευτέρα.

Καὶ διὰ τοῦτο ἔσται ἡ $\Gamma\Delta$ τῇ AB τῇ τάχει ἡ αὐτὴ· ὅπερ ἔδει δεῖχαι.

in square only [Props. 10.37, 10.38]. And let it be contrived that as AB (is) to CD , (so) AE (is) to CF [Prop. 6.12]. And thus as the remainder EB is to the remainder FD , so AB (is) to CD [Props. 5.19 corr., 6.16]. And AB (is) commensurable in length with CD . Thus, AE and EB are also commensurable (in length) with CF and FD , respectively [Prop. 10.11]. And AE and EB (are) medial. Thus, CF and FD (are) also medial [Prop. 10.23]. And since as AE is to EB , (so) CF (is) to FD , and AE and EB are commensurable in square only, CF and FD are [thus] also commensurable in square only [Prop. 10.11]. And they were also shown (to be) medial. Thus, CD is a bimedral (straight-line). So, I say that it is also the same in order as AB .

For since as AE is to EB , (so) CF (is) to FD , thus also as the (square) on AE (is) to the (rectangle contained) by AEB , so the (square) on CF (is) to the (rectangle contained) by CFD [Prop. 10.21 lem.]. Alternately, as the (square) on AE (is) to the (square) on CF , so the (rectangle contained) by AEB (is) to the (rectangle contained) by CFD [Prop. 5.16]. And the (square) on AE (is) commensurable with the (square) on CF . Thus, the (rectangle contained) by AEB (is) also commensurable with the (rectangle contained) by CFD [Prop. 10.11]. Therefore, either the (rectangle contained) by AEB is rational, and the (rectangle contained) by CFD is rational [and, on account of this, (AE and CD) are first bimedral (straight-lines)], or (the rectangle contained by AEB is) medial, and (the rectangle contained by CFD is) medial, and (AB and CD) are each second (bimedral straight-lines) [Props. 10.23, 10.37, 10.38].

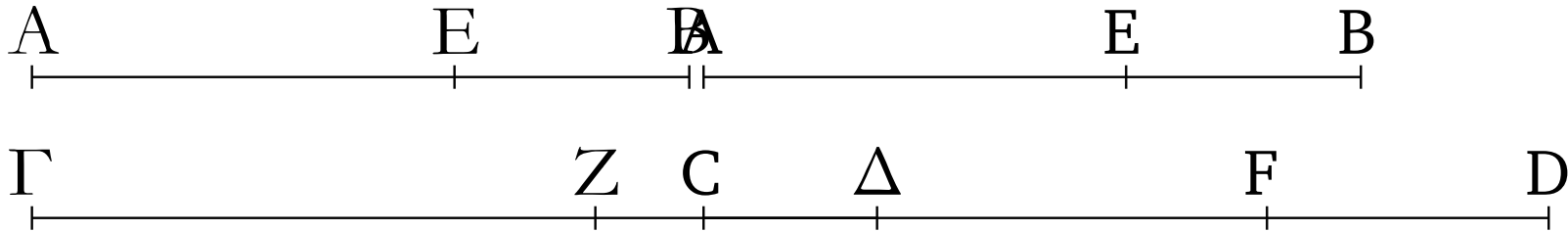
And, on account of this, CD will be the same in order as AB . (Which is) the very thing it was required to show.

68.

Ἡ τῇ μείζονι σύμμετρος καὶ αὐτὴ μείζων ἐστίν.

Proposition 68

A (straight-line) commensurable (in length) with a major (straight-line) is itself also major.



Ἐστω μείζων ἡ AB , καὶ τῇ AB σύμμετρος ἔστω ἡ $\Gamma\Delta$. λέγω, ὅτι ἡ $\Gamma\Delta$ μείζων ἐστίν.

Διηγήσθω ἡ AB κατὰ τὸ E · αἱ AE , EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητὸν, τὸ δ' ὑπ' αὐτῶν μέσον· καὶ γερονέτω τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἐστὶν ὡς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ AE πρὸς τὴν ΓZ καὶ ἡ EB πρὸς τὴν $Z\Delta$, καὶ ὡς ἄρα ἡ AE πρὸς τὴν ΓZ , οὕτως ἡ EB πρὸς τὴν $Z\Delta$.

Let AB be a major (straight-line), and let CD be commensurable (in length) with AB . I say that CD is a major (straight-line).

Let AB be divided (into its component terms) at E . AE and EB are thus incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial [Prop. 10.39]. And let (the) same (things) be contrived as in the previous (propositions). And since as

σύμμετρος δὲ ἡ AB τῇ $\Gamma\Delta$. σύμμετρος ἄρα καὶ ἑκάτερα τῶν AE , EB ἑκάτερα τῶν ΓZ , $Z\Delta$. καὶ ἐπεὶ ἐστὶν ὡς ἡ AE πρὸς τὴν ΓZ , οὕτως ἡ EB πρὸς τὴν $Z\Delta$, καὶ ἐναλλάχῃ ὡς ἡ AE πρὸς EB , οὕτως ἡ ΓZ πρὸς $Z\Delta$, καὶ συνθέντι ἄρα ἐστὶν ὡς ἡ AB πρὸς τὴν BE , οὕτως ἡ $\Gamma\Delta$ πρὸς τὴν ΔZ . καὶ ὡς ἄρα τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BE , οὕτως τὸ ἀπὸ τῆς $\Gamma\Delta$ πρὸς τὸ ἀπὸ τῆς ΔZ . ὁμοίως δὲ δείχομεν, ὅτι καὶ ὡς τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς AE , οὕτως τὸ ἀπὸ τῆς $\Gamma\Delta$ πρὸς τὸ ἀπὸ τῆς ΓZ . καὶ ὡς ἄρα τὸ ἀπὸ τῆς AB πρὸς τὰ ἀπὸ τῶν AE , EB , οὕτως τὸ ἀπὸ τῆς $\Gamma\Delta$ πρὸς τὰ ἀπὸ τῶν ΓZ , $Z\Delta$. καὶ ἐναλλάχῃ ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς $\Gamma\Delta$, οὕτως τὰ ἀπὸ τῶν AE , EB πρὸς τὰ ἀπὸ τῶν ΓZ , $Z\Delta$. σύμμετρον δὲ τὸ ἀπὸ τῆς AB τῷ ἀπὸ τῆς $\Gamma\Delta$. σύμμετρα ἄρα καὶ τὰ ἀπὸ τῶν AE , EB τοῖς ἀπὸ τῶν ΓZ , $Z\Delta$. καὶ ἐστὶ τὰ ἀπὸ τῶν AE , EB ἅμα ῥητόν, καὶ τὰ ἀπὸ τῶν ΓZ , $Z\Delta$ ἅμα ῥητόν ἐστὶν. ὁμοίως δὲ καὶ τὸ διξ ὑπὸ τῶν AE , EB σύμμετρόν ἐστὶ τῷ διξ ὑπὸ τῶν ΓZ , $Z\Delta$. καὶ ἐστὶ μέσον τὸ διξ ὑπὸ τῶν AE , EB . μέσον ἄρα καὶ τὸ διξ ὑπὸ τῶν ΓZ , $Z\Delta$. αἱ ΓZ , $Z\Delta$ ἄρα δυνάμει ἀσύμμετροί εἰσι ποιοῦσαι τὸ μὲν συγκεκλιμένον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ἅμα ῥητόν, τὸ δὲ διξ ὑπ' αὐτῶν μέσον· ὅλη ἄρα ἡ $\Gamma\Delta$ ἀλογόξ ἐστὶν ἡ καλουμένη μείζων.

Ἡ ἄρα τῇ μείζονι σύμμετρος μείζων ἐστίν· ὅπερ ἔδει δεῖξαι.

AB is to CD , so AE (is) to CF and EB to FD , thus also as AE (is) to CF , so EB (is) to FD [Prop. 5.11]. And AB (is) commensurable (in length) with CD . Thus, AE and EB (are) also commensurable (in length) with CF and FD , respectively [Prop. 10.11]. And since as AE is to CF , so EB (is) to FD , also, alternately, as AE (is) to EB , so CF (is) to FD [Prop. 5.16], and thus, via composition, as AB is to BE , so CD (is) to DF [Prop. 5.18]. And thus as the (square) on AB (is) to the (square) on BE , so the (square) on CD (is) to the (square) on DF [Prop. 6.20]. So, similarly, we can also show that as the (square) on AB (is) to the (square) on AE , so the (square) on CD (is) to the (square) on CF . And thus as the (square) on AB (is) to (the sum of) the (squares) on AE and EB , so the (square) on CD (is) to (the sum of) the (squares) on CF and FD . And thus, alternately, as the (square) on AB is to the (square) on CD , so (the sum of) the (squares) on AE and EB (is) to (the sum of) the (squares) on CF and FD [Prop. 5.16]. And the (square) on AB (is) commensurable with the (square) on CD . Thus, (the sum of) the (squares) on AE and EB (is) also commensurable with (the sum of) the (squares) on CF and FD [Prop. 10.11]. And the (squares) on AE and EB (added) together are rational. The (squares) on CF and FD (added) together (are) thus also rational. So, similarly, twice the (rectangle contained) by AE and EB is also commensurable with twice the (rectangle contained) by CF and FD . And twice the (rectangle contained) by AE and EB is medial. Therefore, twice the (rectangle contained) by CF and FD (is) also medial [Prop. 10.23 corr.]. CF and FD are thus (straight-lines which are) incommensurable in square [Prop 10.13], simultaneously making the sum of the squares on them rational, and twice the (rectangle contained) by them medial. The whole, CD , is thus that irrational (straight-line) called major [Prop. 10.39].

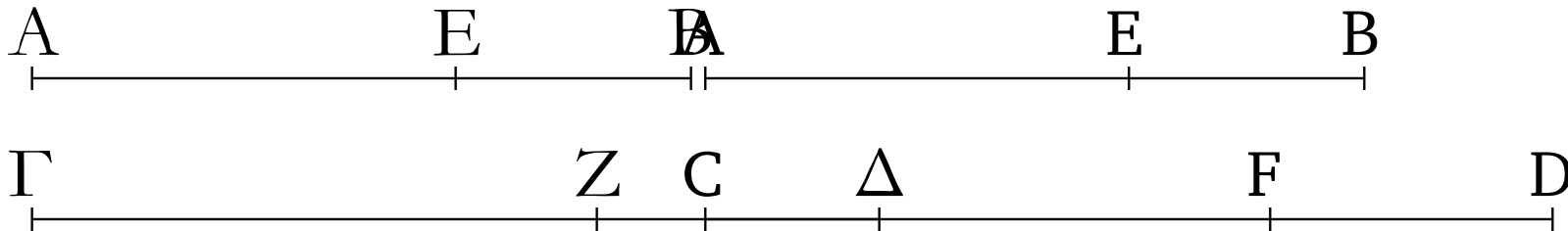
Thus, a (straight-line) commensurable (in length) with a major (straight-line) is major. (Which is) the very thing it was required to show.

69.

Proposition 69

Ἡ τῇ ῥητόν καὶ μέσον δυναμένη σύμμετρος [καὶ αὐτὴ] ῥητόν καὶ μέσον δυναμένη ἐστίν.

A (straight-line) commensurable (in length) with the square-root of a rational plus a medial (area) is [itself also] the square-root of a rational plus a medial (area).



Ἐστω ῥητόν καὶ μέσον δυναμένη ἡ AB , καὶ τῇ AB σύμμετρος ἔστω ἡ $\Gamma\Delta$. δεικτέον, ὅτι καὶ ἡ $\Gamma\Delta$ ῥητόν καὶ μέσον

Let AB be the square-root of a rational plus a medial (area), and let CD be commensurable (in length) with AB . We must

δυναμένη ἐστίν.

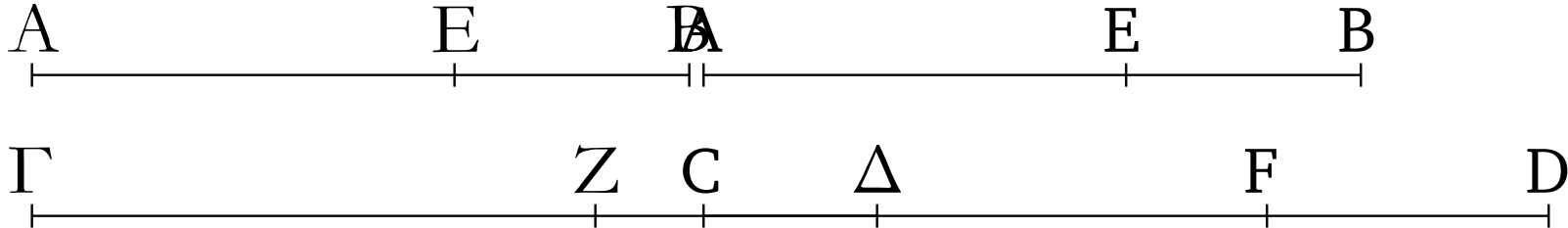
Διηγήσθω ἡ AB εἰς τὰς εὐθείας κατὰ τὸ E · αἱ AE , EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν· καὶ τὰ αὐτὰ κατεσκευάσθω τοῖς πρότερον. ὁμοίως δὲ δείχομεν, ὅτι καὶ αἱ ΓZ , $Z\Delta$ δυνάμει εἰσὶν ἀσύμμετροι, καὶ σύμμετρον τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τῶν συγκειμένων ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$, τὸ δὲ ὑπὸ AE , EB τῶν ὑπὸ ΓZ , $Z\Delta$ · ὥστε καὶ τὸ [μὲν] συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων ἐστὶ μέσον, τὸ δ' ὑπὸ τῶν ΓZ , $Z\Delta$ ῥητόν.

Ῥητόν ἄρα καὶ μέσον δυναμένη ἐστίν ἡ $\Gamma\Delta$ · ὅπερ ἔδει δεῖχαι.

70.

Ἡ τῇ δύο μέσα δυναμένη σύμμετρος δύο μέσα δυναμένη ἐστίν.

Ἐστω δύο μέσα δυναμένη ἡ AB , καὶ τῇ AB σύμμετρος ἡ $\Gamma\Delta$ · δεικτέον, ὅτι καὶ ἡ $\Gamma\Delta$ δύο μέσα δυναμένη ἐστίν.



Ἐπεὶ γὰρ δύο μέσα δυναμένη ἐστίν ἡ AB , διηγήσθω εἰς τὰς εὐθείας κατὰ τὸ E · αἱ AE , EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν [τετραγώνων] μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῶν ὑπὸ τῶν AE , EB · καὶ κατεσκευάσθω τὰ αὐτὰ τοῖς πρότερον. ὁμοίως δὲ δείχομεν, ὅτι καὶ αἱ ΓZ , $Z\Delta$ δυνάμει εἰσὶν ἀσύμμετροι καὶ σύμμετρον τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τῶν συγκειμένων ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$, τὸ δὲ ὑπὸ τῶν AE , EB τῶν ὑπὸ τῶν ΓZ , $Z\Delta$ · ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων μέσον ἐστὶ καὶ τὸ ὑπὸ τῶν ΓZ , $Z\Delta$ μέσον καὶ ἔτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΓZ , $Z\Delta$ τετραγώνων τῶν ὑπὸ τῶν ΓZ , $Z\Delta$.

Ἡ ἄρα $\Gamma\Delta$ δύο μέσα δυναμένη ἐστίν· ὅπερ ἔδει δεῖχαι.

show that CD is also the square-root of a rational plus a medial (area).

Let AB be divided into its (component) straight-lines at E . AE and EB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational [Prop. 10.40]. And let the same construction be made as in the previous (propositions). So, similarly, we can show that CF and FD are also incommensurable in square, and that the sum of the (squares) on AE and EB (is) commensurable with the sum of the (squares) on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . And hence the sum of the squares on CF and FD is medial, and the (rectangle contained) by CF and FD (is) rational.

Thus, CD is the square-root of a rational plus a medial (area) [Prop. 10.40]. (Which is) the very thing it was required to show.

Proposition 70

A (straight-line) commensurable (in length) with the square-root of (the sum of) two medial (areas) is (itself also) the square-root of (the sum of) two medial (areas).

Let AB be the square-root of (the sum of) two medial (areas), and (let) CD (be) commensurable (in length) with AB . We must show that CD is also the square-root of (the sum of) two medial (areas).

For since AB is the square-root of (the sum of) two medial (areas), let it be divided into its (component) straight-lines at E . Thus, AE and EB are incommensurable in square, making the sum of the [squares] on them medial, and the (rectangle contained) by them medial, and, moreover, the sum of the (squares) on AE and EB incommensurable with the (rectangle) contained by AE and EB [Prop. 10.41]. And let the same construction be made as in the previous (propositions). So, similarly, we can show that CF and FD are also incommensurable in square, and (that) the sum of the (squares) on AE and EB (is) commensurable with the sum of the (squares) on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Hence, the sum of the squares on CF and FD is also medial, and the (rectangle contained) by CF and FD (is) medial, and, moreover, the sum of the squares on CF and FD (is) incommensurable with the

71.

Ῥητοῦ καὶ μέσου συντιθεμένου τέσσαρες ἄλογοι γίνονται ἥτοι ἐκ δύο ὀνομάτων ἢ ἐκ δύο μέσων πρώτη ἢ μείζων ἢ ῥητόν καὶ μέσον δυναμένη.

Ἐστω ῥητόν μὲν τὸ AB , μέσον δὲ τὸ $\Gamma\Delta$. λέγω, ὅτι ἡ τὸ $A\Delta$ θωρίον δυναμένη ἥτοι ἐκ δύο ὀνομάτων ἐστὶν ἢ ἐκ δύο μέσων πρώτη ἢ μείζων ἢ ῥητόν καὶ μέσον δυναμένη.

Τὸ γὰρ AB τοῦ $\Gamma\Delta$ ἥτοι μείζον ἐστὶν ἢ ἔλασσον. ἔστω πρότερον μείζον· καὶ ἐκκείσθω ῥητὴ ἡ EZ , καὶ παραβελθήσθω παρὰ τὴν EZ τῷ AB ἴσον τὸ EH πλάτος ποιοῦν τὴν $E\Theta$. τῷ δὲ $\Delta\Gamma$ ἴσον παρὰ τὴν EZ παραβελθήσθω τὸ ΘI πλάτος ποιοῦν τὴν ΘK . καὶ ἐπεὶ ῥητόν ἐστι τὸ AB καὶ ἐστιν ἴσον τῷ EH , ῥητόν ἄρα καὶ τὸ EH . καὶ παρὰ [ῥητὴν] τὴν EZ παραβελθήσθω πλάτος ποιοῦν τὴν $E\Theta$. ἡ $E\Theta$ ἄρα ῥητὴ ἐστὶ καὶ σύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ $\Gamma\Delta$ καὶ ἐστὶν ἴσον τῷ ΘI , μέσον ἄρα ἐστὶ καὶ τὸ ΘI . καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν ΘK . ῥητὴ ἄρα ἐστὶν ἡ ΘK καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ μέσον ἐστὶ τὸ $\Gamma\Delta$, ῥητόν δὲ τὸ AB , ἀσύμμετρον ἄρα ἐστὶ τὸ AB τῷ $\Gamma\Delta$. ὥστε καὶ τὸ EH ἀσύμμετρον ἐστὶ τῷ ΘI . ὥς δὲ τὸ EH πρὸς τὸ ΘI , οὕτως ἐστὶν ἡ $E\Theta$ πρὸς τὴν ΘK . ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ $E\Theta$ τῇ ΘK μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ $E\Theta$, ΘK ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EK διηρημένη κατὰ τὸ Θ . καὶ ἐπεὶ μείζον ἐστὶ τὸ AB τοῦ $\Gamma\Delta$, ἴσον δὲ τὸ μὲν AB τῷ EH , τὸ δὲ $\Gamma\Delta$ τῷ ΘI , μείζον ἄρα καὶ τὸ EH τοῦ ΘI . καὶ ἡ $E\Theta$ ἄρα μείζων ἐστὶ τῆς ΘK . ἥτοι οὗν ἡ $E\Theta$ τῆς ΘK μείζον δύναται τῷ ἀπὸ συμμέτρου ἐαντὶ μήκει ἢ τῷ ἀπὸ ἀσυνμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμμέτρου ἐαντὶ· καὶ ἐστὶν ἡ μείζων ἡ ΘE σύμμετρος τῇ ἐκκευμένη ῥητὴ τῇ EZ . ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶ πρώτη. ῥητὴ δὲ ἡ EZ . ἔαν δὲ θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πρώτης, ἡ τὸ θωρίον δυναμένη ἐκ δύο ὀνομάτων ἐστὶν. ἡ ἄρα τὸ EI δυναμένη ἐκ δύο ὀνομάτων ἐστὶν. ὥστε καὶ ἡ τὸ $A\Delta$ δυναμένη ἐκ δύο ὀνομάτων ἐστὶν. ἀλλὰ δὴ δυνάσθω ἡ $E\Theta$ τῆς ΘK μείζον τῷ ἀπὸ ἀσυνμέτρου ἐαντὶ· καὶ ἐστὶν ἡ μείζων ἡ $E\Theta$ σύμμετρος τῇ ἐκκευμένη ῥητὴ τῇ EZ μήκει. ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶ τετάρτη. ῥητὴ δὲ ἡ EZ . ἔαν δὲ θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης, ἡ τὸ θωρίον δυναμένη ἄλογός ἐστιν ἢ καλουμένη μείζων. ἡ ἄρα τὸ EI θωρίον δυναμένη μείζων ἐστὶν. ὥστε καὶ ἡ τὸ $A\Delta$ δυναμένη μείζων ἐστὶν.

Ἀλλὰ δὴ ἔστω ἔλασσον τὸ AB τοῦ $\Gamma\Delta$. καὶ τὸ EH ἄρα ἔλασσόν ἐστι τοῦ ΘI . ὥστε καὶ ἡ $E\Theta$ ἐλάσσων ἐστὶ τῆς ΘK . ἥτοι δὲ ἡ ΘK τῆς $E\Theta$ μείζον δύναται τῷ ἀπὸ συμμέτρου ἐαντὶ ἢ τῷ ἀπὸ ἀσυνμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμμέτρου

(rectangle contained) by CF and FD .

Thus, CD is the square-root of (the sum of) two medial (areas) [Prop. 10.41]. (Which is) the very thing it was required to show.

Proposition 71

When a rational and a medial (area) are added together, four irrational (straight-lines) arise (as the square-roots of the total area)—either a binomial, or a first bimedial, or a major, or the square-root of a rational plus a medial (area).

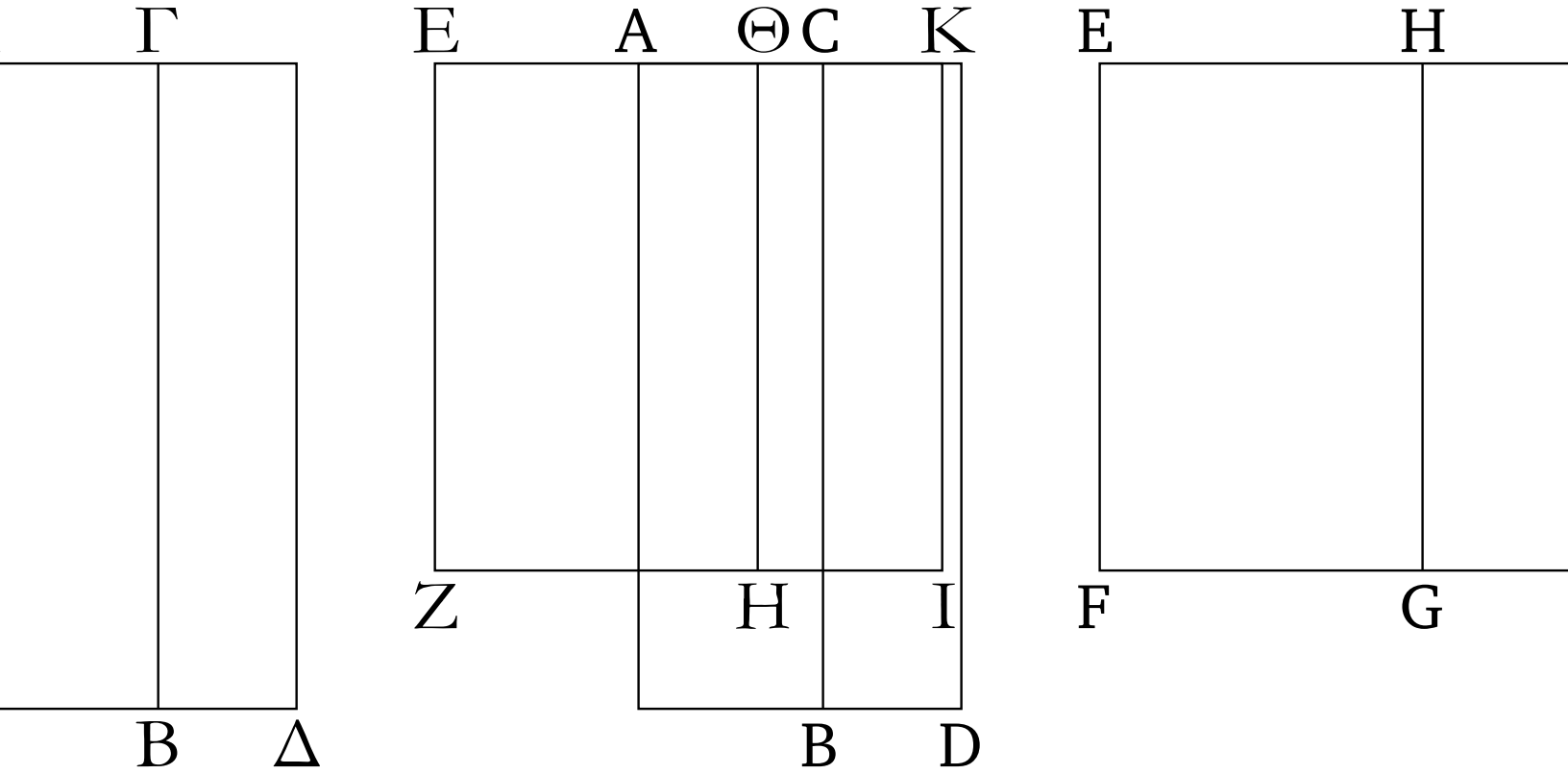
Let AB be a rational (area), and CD a medial (area). I say that the square-root of area AD is either binomial, or first bimedial, or major, or the square-root of a rational plus a medial (area).

For AB is either greater or less than CD . Let it, first of all, be greater. And let the rational (straight-line) EF be laid down. And let (the rectangle) EG , equal to AB , be applied to EF , producing EH as breadth. And let (the rectangle) HI , equal to DC , be applied to EF , producing HK as breadth. And since AB is rational, and is equal to EG , EG is thus also rational. And it has been applied to the [rational] (straight-line) EF , producing EH as breadth. EH is thus rational, and commensurable in length with EF [Prop. 10.20]. Again, since CD is medial, and is equal to HI , HI is thus also medial. And it is applied to the rational (straight-line) EF , producing HK as breadth. HK is thus rational, and incommensurable in length with EF [Prop. 10.22]. And since CD is medial, and AB rational, AB is thus incommensurable with CD . Hence, EG is also incommensurable with HI . And as EG (is) to HI , so EH is to HK [Prop. 6.1]. Thus, EH is also incommensurable in length with HK [Prop. 10.11]. And they are both rational. Thus, EH and HK are rational (straight-lines which are) commensurable in square only. EK is thus a binomial (straight-line), having been divided (into its component terms) at H [Prop. 10.36]. And since AB is greater than CD , and AB (is) equal to EG , and CD to HI , EG (is) thus also greater than HI . Thus, EH is also greater than HK [Prop. 5.14]. Therefore, the square on EH is greater than (the square on) HK either by the (square) on (some straight-line) commensurable in length with (EH) , or by the (square) on (some straight-line) incommensurable (in length with EH). Let it, first of all, be greater by the (square) on (some straight-line) commensurable (in length with EH). And the greater (of the two components of EK) HE is commensurable (in length) with the (previously) laid down (straight-line) EF . EK is thus a first binomial (straight-line) [Def. 10.5]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a first bimedial (straight-line) then the square-root of the area is a binomial (straight-line) [Prop. 10.54]. Thus, the square-root of EI is a binomial (straight-line). Hence the square-root of

ἐαντὴ μήκει· καὶ ἐστὶν ἡ ἐλάσσων ἢ $E\Theta$ σύμμετρος τῇ ἐκκειμένη ῥητῇ τῇ EZ μήκει· ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶ δευτέρα. ῥητὴ δὲ ἡ EZ · ἔαν δὲ θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας, ἡ τὸ θωρίον δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη· ὥστε καὶ ἡ τὸ $A\Delta$ δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη. ἀλλὰ δὴ ἡ ΘK τῆς ΘE μείζον δυνάσθω τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ. καὶ ἐστὶν ἡ ἐλάσσων ἢ $E\Theta$ σύμμετρος τῇ ἐκκειμένη ῥητῇ τῇ EZ · ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶ πέμπτη. ῥητὴ δὲ ἡ EZ · ἔαν δὲ θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πέμπτης, ἡ τὸ θωρίον δυναμένη ῥητόν καὶ μέσον δυναμένη ἐστίν· ὥστε καὶ ἡ τὸ $A\Delta$ θωρίον δυναμένη ῥητόν καὶ μέσον δυναμένη ἐστίν.

AD is also a binomial (straight-line). And, so, let the square on EH be greater than (the square on) HK by the (square) on (some straight-line) incommensurable (in length) with (EH). And the greater (of the two components of EK) EH is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a fourth binomial (straight-line) [Def. 10.8]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a fourth binomial (straight-line) then the square-root of the area is the irrational (straight-line) called major [Prop. 10.57]. Thus, the square-root of area EI is a major (straight-line). Hence, the square-root of AD is also major.

And so, let AB be less than CD . Thus, EG is also less than HI . Hence, EH is also less than HK [Props. 6.1, 5.14]. And the square on HK is greater than (the square on) EH either by the (square) on (some straight-line) commensurable (in length) with (HK), or by the (square) on (some straight-line) incommensurable (in length) with (HK). Let it, first of all, be greater by the square on (some straight-line) commensurable in length with (HK). And the lesser (of the two components of EK) EH is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a second binomial (straight-line) [Def. 10.6]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a second binomial (straight-line) then the square-root of the area is a first bimedral (straight-line) [Prop. 10.55]. Thus, the square-root of area EI is a first bimedral (straight-line). Hence, the square-root of AD is also a first bimedral (straight-line). And so, let the square on HK be greater than (the square on) HE by the (square) on (some straight-line) incommensurable (in length) with (HK). And the lesser (of the two components of EK) EH is commensurable (in length) with the (previously) laid down rational (straight-line) EF . Thus, EK is a fifth binomial (straight-line) [Def. 10.9]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a fifth binomial (straight-line) then the square-root of the area is the square-root of a rational plus a medial (area) [Prop. 10.58]. Thus, the square-root of area EI is the square-root of a rational plus a medial (area). Hence, the square-root of area AD is also the square-root of a rational plus a medial (area).



Πρὸ τοῦ ἄρα καὶ μέσον συντιθεμένων τέσσαρες ἄλογοι γίνονται ἥτοι ἐκ δύο ὀνομάτων ἢ ἐκ δύο μέσων πρώτη ἢ μείζων ἢ ῥητὸν καὶ μέσον δυναμένη· ὅπερ ἔδει δεῖξαι.

Thus, when a rational and a medial area are added together, four irrational (straight-lines) arise (as the square-roots of the total area)—either a binomial, or a first bimedral, or a major, or the square-root of a rational plus a medial (area). (Which is) the very thing it was required to show.

62.

Proposition 72

Δύο μέσων ἀσυμμέτρων ἀλλήλοις συντιθεμένων αἱ λοιπαὶ δύο ἄλογοι γίνονται ἥτοι ἐκ δύο μέσων δευτέρα ἢ [ἢ] δύο μέσα δυναμένη.

Συγκείσθω γάρ δύο μέσα ἀσύμμετρα ἀλλήλοις τὰ AB, ΓΔ· λέγω, ὅτι ἡ τὸ AD θωρίον δυναμένη ἥτοι ἐκ δύο μέσων ἐστὶ δευτέρα ἢ δύο μέσα δυναμένη.

Τὸ γὰρ AB τοῦ ΓΔ ἥτοι μείζον ἐστὶν ἢ ἔλασσον. ἔστω, εἰ τύχον, πρότερον μείζον τὸ AB τοῦ ΓΔ· καὶ ἐκκείσθω ῥητὴ ἢ EZ, καὶ τῷ μὲν AB ἴσον παρὰ τὴν EZ παραβελθήσθω τὸ EH πλάτος ποιῶν τὴν EΘ, τῷ δὲ ΓΔ ἴσον τὸ ΘΙ πλάτος ποιῶν τὴν ΘΚ. καὶ ἐπεὶ μέσον ἐστὶν ἑκάτερον τῶν AB, ΓΔ, μέσον ἄρα καὶ ἑκάτερον τῶν EH, ΘΙ. καὶ παρὰ ῥητὴν τὴν ZE παράκειται πλάτος ποιῶν τὰς EΘ, ΘΚ· ἑκάτερα ἄρα τῶν EΘ, ΘΚ ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρον ἐστὶ τὸ AB τῷ ΓΔ, καὶ ἐστὶν ἴσον τὸ μὲν AB τῷ EH, τὸ δὲ ΓΔ τῷ ΘΙ, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ EH τῷ ΘΙ. ὥς δὲ τὸ EH πρὸς τὸ ΘΙ, οὕτως ἐστὶν ἡ EΘ πρὸς ΘΚ· ἀσύμμετρος ἄρα ἐστὶν ἡ EΘ τῇ ΘΚ μήκει. αἱ EΘ, ΘΚ ἄρα

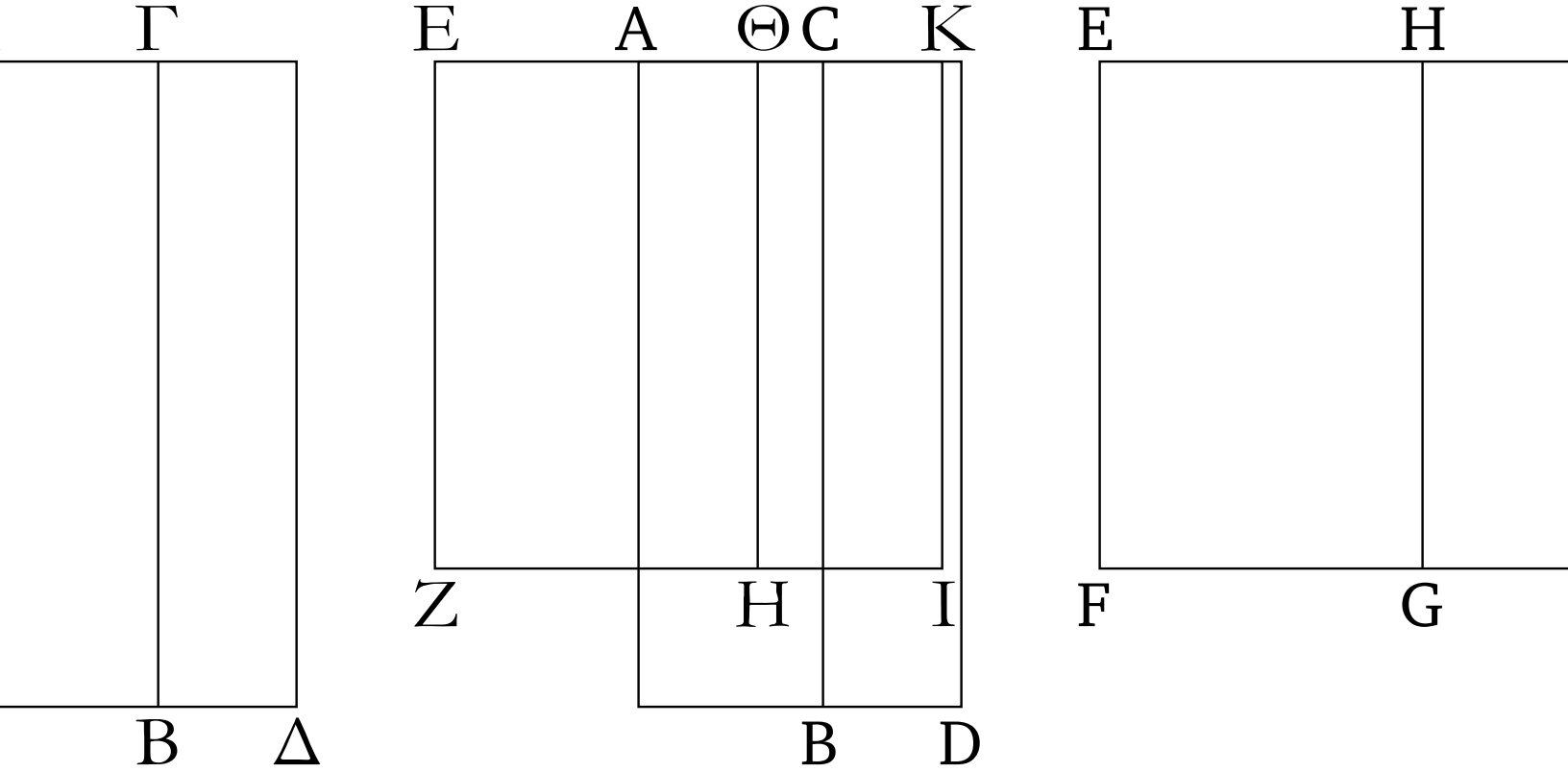
When two medial (areas which are) incommensurable with one another are added together, the remaining two irrational (straight-lines) arise (as the square-roots of the total area)—either a second bimedral, or the square-root of (the sum of) two medial (areas).

For let the two medial (areas) AB and CD, (which are) incommensurable with one another, be added together. I say that the square-root of area AD is either a second bimedral, or the square-root of (the sum of) two medial (areas).

For AB is either greater than or less than CD. By chance, let AB, first of all, be greater than CD. And let the rational (straight-line) EF be laid down. And let EG, equal to AB, be applied to EF, producing EH as breadth, and HI, equal to CD, producing HK as breadth. And since AB and CD are each medial, EG and HI (are) thus also each medial. And they are applied to the rational straight-line FE, producing EH and HK (respectively) as breadth. Thus, EH and HK are each rational (straight-lines which are) incommensurable in length with EF

ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EK . ἤτοι δὲ ἡ $E\Theta$ τῆς ΘK μείζον δύνανται τῷ ἀπὸ συμμέτρου ἐναντὶ ἢ τῷ ἀπὸ ἀσυμμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμμέτρου ἐναντὶ μήκει· καὶ οὐδετέρα τῶν $E\Theta$, ΘK σύμμετρόξ ἐστι τῇ ἐκκεκμένη ῥητῇ τῇ EZ μήκει· ἡ EK ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη. ῥητὴ δὲ ἡ EZ · ἔαν δὲ θεωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων τρίτης, ἡ τὸ θεωρίον δυναμένη ἐκ δύο μέσων ἐστὶ δευτέρα· ἡ ἄρα τὸ EI , τουτέστι τὸ $A\Delta$, δυναμένη ἐκ δύο μέσων ἐστὶ δευτέρα. ἀλλὰ δὴ ἡ $E\Theta$ τῆς ΘK μείζον δυνάσθω τῷ ἀπὸ ἀσυμμέτρου ἐναντὶ μήκει· καὶ ἀσύμμετρόξ ἐστὶν ἑκάτερα τῶν $E\Theta$, ΘK τῇ EZ μήκει· ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶν ἕκτη. ἔαν δὲ θεωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων ἕκτης, ἡ τὸ θεωρίον δυναμένη ἢ δύο μέσα δυναμένη ἐστὶν· ὥστε καὶ ἡ τὸ $A\Delta$ θεωρίον δυναμένη ἢ δύο μέσα δυναμένη ἐστὶν.

[Prop. 10.22]. And since AB is incommensurable with CD , and AB is equal to EG , and CD to HI , EG is thus also incommensurable with HI . And as EG (is) to HI , so EH is to HK [Prop. 6.1]. EH is thus incommensurable in length with HK [Prop. 10.11]. Thus, EH and HK are rational (straight-lines which are) commensurable in square only. EK is thus a binomial (straight-line) [Prop. 10.36]. And the square on EH is greater than (the square on) HK either by the (square) on (some straight-line) commensurable (in length) with (EH), or by the (square) on (some straight-line) incommensurable (in length with EH). Let it, first of all, be greater by the square on (some straight-line) commensurable in length with (EH). And neither of EH or HK is commensurable in length with the (previously) laid down rational (straight-line) EF . Thus, EK is a third binomial (straight-line) [Def. 10.7]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a third binomial (straight-line) then the square-root of the area is a second bimedral (straight-line) [Prop. 10.56]. Thus, the square-root of EI —that is to say, of AD —is a second bimedral. And so, let the square on EH be greater than (the square) on HK by the (square) on (some straight-line) incommensurable in length with (EH). And EH and HK are each incommensurable in length with EF . Thus, EK is a sixth binomial (straight-line) [Def. 10.10]. And if an area is contained by a rational (straight-line) and a sixth binomial (straight-line) then the square-root of the area is the square-root of (the sum of) two medial (areas) [Prop. 10.59]. Hence, the square-root of area AD is also the square-root of (the sum of) two medial (areas).



[Ὅμοιως δὴ δείξομεν, ὅτι κἂν ἔλαττον ᾖ τὸ AB τοῦ $\Gamma\Delta$, ἢ τὸ $A\Delta$ χωρίον δυναμένη ᾖ ἐκ δύο μέσων δευτέρα ἐστὶν ἥτοι δύο μέσα δυναμένη].

Δύο ἄρα μέσων ἀσυμμέτρων ἀλλήλοις συντιθεμένων αἰ λοιπαὶ δύο ἄλογοι γίνονται ἥτοι ἐκ δύο μέσων δευτέρα ἢ δύο μέσα δυναμένη.

Ἡ ἐκ δύο ὀνομάτων καὶ αἱ μετ' αὐτὴν ἄλογοι οὔτε τῇ μέσῃ οὔτε ἀλλήλαις εἰσὶν αἱ αὐταί. τὸ μὲν γὰρ ἀπὸ μέσῃ παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ῥητὴν καὶ ἀσύμμετρον τῇ παρ' ἣν παράκειται μήκει. τὸ δὲ ἀπὸ τῆς ἐκ δύο ὀνομάτων παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πρώτην. τὸ δὲ ἀπὸ τῆς ἐκ δύο μέσων πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων δευτέραν. τὸ δὲ ἀπὸ τῆς ἐκ δύο μέσων δευτέρας παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τρίτην. τὸ δὲ ἀπὸ τῆς μείζονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τετάρτην. τὸ δὲ ἀπὸ τῆς ῥητὸν καὶ μέσον δυναμένης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πέμπτην. τὸ δὲ ἀπὸ τῆς δύο μέσα δυναμένης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων ἕκτην. τὰ δ' εἰρημένα πλάτη διαφέρει τοῦ τε πρώτου καὶ ἀλλήλων, τοῦ μὲν πρώτου, ὅτι ῥητὴ ἐστίν, ἀλλήλων δέ, ὅτι τῇ τάχει οὐκ εἰσὶν αἱ αὐταί· ὥστε καὶ αὐταὶ αἱ ἄλογοι διαφέρουσιν ἀλλήλων.

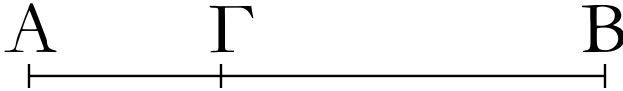
[So, similarly, we can show that, even if AB is less than CD , the square-root of area AD is either a second binomial or the square-root of (the sum of) two medial (areas).]

Thus, when two medial (areas which are) incommensurable with one another are added together, the remaining two irrational (straight-lines) arise (as the square-roots of the total area)—either a second binomial, or the square-root of (the sum of) two medial (areas).

A binomial (straight-line), and the (other) irrational (straight-lines) after it, are neither the same as a medial (straight-line) nor (the same) as one another. For the (square) on a medial (straight-line), applied to a rational (straight-line), produces as breadth a rational (straight-line which is) also incommensurable in length with (the straight-line) to which it is applied [Prop. 10.22]. And the (square) on a binomial (straight-line), applied to a rational (straight-line), produces as breadth a first binomial [Prop. 10.60]. And the (square) on a first binomial (straight-line), applied to a rational (straight-line), produces as breadth a second binomial [Prop. 10.61]. And the (square) on a second binomial (straight-line), applied to a rational (straight-line), produces as breadth a third binomial [Prop. 10.62]. And the (square) on a major (straight-line), applied to a rational (straight-line), produces as breadth a fourth binomial [Prop. 10.63]. And the (square) on the square-root of a rational plus a medial (area), applied to a rational (straight-line),

73.

Ἐάν ἀπὸ ῥητῆς ῥητῆ ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ, ἡ λοιπὴ ἄλογός ἐστιν· καλείσθω δὲ ἀποτομή.

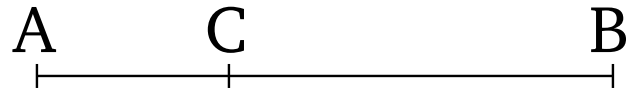


Ἀπὸ γὰρ ῥητῆς τῆς AB ῥητῆ ἀφαιρεθῶ ἡ $BΓ$ δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ· λέγω, ὅτι ἡ λοιπὴ ἡ $AΓ$ ἄλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐπεὶ γὰρ ἀσύμμετρος ἐστὶν ἡ AB τῇ $BΓ$ μήκει, καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ὑπὸ τῶν AB , $BΓ$, ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB , $BΓ$. ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρά ἐστι τὰ ἀπὸ τῶν AB , $BΓ$ τετράγωνα, τῷ δὲ ὑπὸ τῶν AB , $BΓ$ σύμμετρόν ἐστι τὸ δι᾽ ὑπὸ τῶν AB , $BΓ$. καὶ ἐπειδήπερ τὰ ἀπὸ τῶν AB , $BΓ$ ἴσα ἐστὶ τῷ δι᾽ ὑπὸ τῶν AB , $BΓ$ μετὰ τοῦ ἀπὸ $ΓΑ$, καὶ λοιπῷ ἄρα τῷ ἀπὸ τῆς $AΓ$ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB , $BΓ$. ῥητὰ δὲ τὰ ἀπὸ τῶν AB , $BΓ$ ἄλογοις ἄρα ἐστὶν ἡ $AΓ$ · καλείσθω δὲ ἀποτομή. ὅπερ ἔδει δεῖχαι.

Proposition 73

If a rational (straight-line), which is commensurable in square only with the whole, is subtracted from a(nother) rational (straight-line) then the remainder is an irrational (straight-line). Let it be called an apotome.



For let the rational (straight-line) BC , which commensurable in square only with the whole, be subtracted from the rational (straight-line) AB . I say that the remainder AC is that irrational (straight-line) called an apotome.

For since AB is incommensurable in length with BC , and as AB is to BC , so the (square) on AB (is) to the (rectangle contained) by AB and BC [Prop. 10.21 lem.], the (square) on AB is thus incommensurable with the (rectangle contained) by AB and BC [Prop. 10.11]. But, the (sum of the) squares on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. And, inasmuch as the (sum of the) squares on AB and BC is equal to twice the (rectangle contained) by AB and BC plus the (square) on CA [Prop. 2.7], the (sum of the) squares on AB and BC is thus also incommensurable with the remaining (square) on AC [Props. 10.13, 10.16]. And the (sum of the) squares on AB and BC is rational. AC is thus an irrational (straight-line) [Def. 10.4]. And let it be called an apotome.[†] (Which is) the very thing it was required to show.

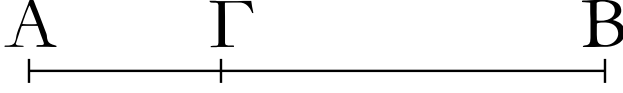
[†] See footnote to Prop. 10.36.

74.

Ἐάν ἀπὸ μέσης μέσης ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ῥητὸν περιέθουσα, ἡ λοιπὴ ἄλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομή πρώτη.

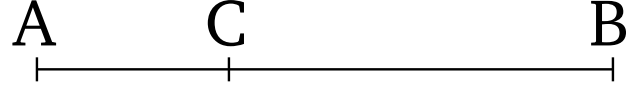
Proposition 74

If a medial (straight-line), which is commensurable in square only with the whole, and which contains a rational (area) with the whole, is subtracted from a(nother) medial (straight-line) then the remainder is an irrational (straight-line). Let it be called a first apotome of a medial (straight-line).



Ἀπὸ γὰρ μέσης τῆς AB μέση ἀφηγήσθω ἡ $BΓ$ δυνάμει μόνον σύμμετρος ὅσα τῇ AB , μετὰ δὲ τῆς AB ῥητὸν ποιῶσα τὸ ὑπὸ τῶν AB , $BΓ$ λέγω, ὅτι ἡ λοιπὴ ἡ $ΑΓ$ ἄλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ πρώτη.

Ἐπεὶ γὰρ αἱ AB , $BΓ$ μέσαι εἰσὶν, μέσα ἐστὶ καὶ τὰ ἀπὸ τῶν AB , $BΓ$. ῥητὸν δὲ τὸ διξ ὑπὸ τῶν AB , $BΓ$ · ἀσύμμετρα ἄρα τὰ ἀπὸ τῶν AB , $BΓ$ τῷ διξ ὑπὸ τῶν AB , $BΓ$ · καὶ λοιπῶν ἄρα τῷ ἀπὸ τῆς $ΑΓ$ ἀσύμμετρον ἐστὶ τὸ διξ ὑπὸ τῶν AB , $BΓ$, ἐπεὶ κ' ἂν τὸ ὅλον ἐν αὐτῶν ἀσύμμετρον ᾖ, καὶ τὰ ἐχ ἀρθῆς μεγέθη ἀσύμμετρα ἐσταί. ῥητὸν δὲ τὸ διξ ὑπὸ τῶν AB , $BΓ$ · ἄλογον ἄρα τὸ ἀπὸ τῆς $ΑΓ$ · ἄλογος ἄρα ἐστὶν ἡ $ΑΓ$ · καλείσθω δὲ μέσης ἀποτομὴ πρώτη.



For let the medial (straight-line) BC , which is commensurable in square only with AB , and which makes with AB the rational (rectangle contained) by AB and BC , be subtracted from the medial (straight-line) AB [Prop. 10.27]. I say that the remainder AC is an irrational (straight-line). Let it be called the first apotome of a medial (straight-line).

For since AB and BC are medial (straight-lines), the (sum of the squares) on AB and BC is also medial. And twice the (rectangle contained) by AB and BC (is) rational. The (sum of the squares) on AB and BC (is) thus incommensurable with twice the (rectangle contained) by AB and BC . Thus, twice the (rectangle contained) by AB and BC is also incommensurable with the remaining (square) on AC [Prop. 2.7], since if the whole is incommensurable with one of the (constituent magnitudes) then the original magnitudes will also be incommensurable (with one another) [Prop. 10.16]. And twice the (rectangle contained) by AB and BC (is) rational. Thus, the (square) on AC is irrational. Thus, AC is an irrational (straight-line) [Def. 10.4]. Let it be called a first apotome of a medial (straight-line).[†]

[†] See footnote to Prop. 10.37.

75.

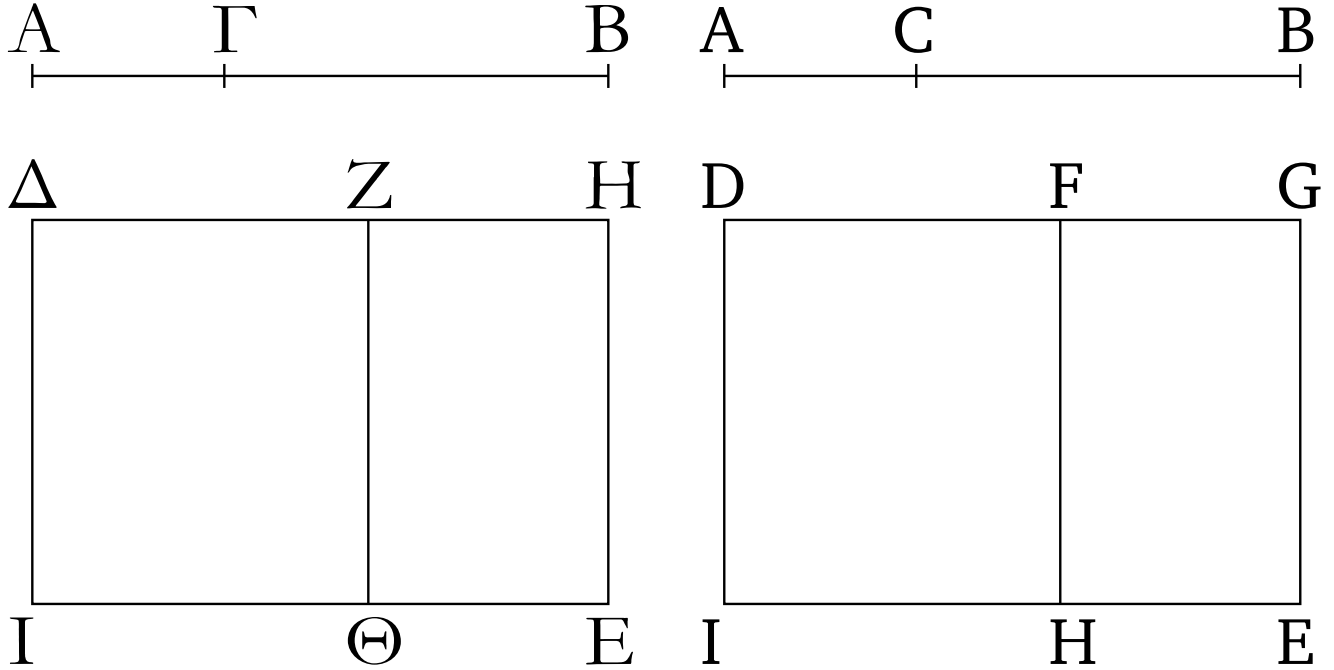
Ἐὰν ἀπὸ μέσης μέση ἀφαιρεθῇ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης μέσον περιέθουσα, ἡ λοιπὴ ἄλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ δευτέρα.

Ἀπὸ γὰρ μέσης τῆς AB μέση ἀφηγήσθω ἡ $ΓΒ$ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ τῇ AB , μετὰ δὲ τῆς ὅλης τῆς AB μέσον περιέθουσα τὸ ὑπὸ τῶν AB , $BΓ$ λέγω, ὅτι ἡ λοιπὴ ἡ $ΑΓ$ ἄλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ δευτέρα.

Proposition 75

If a medial (straight-line), which is commensurable in square only with the whole, and which contains a medial (area) with the whole, is subtracted from a(nother) medial (straight-line) then the remainder is an irrational (straight-line). Let it be called a second apotome of a medial (straight-line).

For let the medial (straight-line) CB , which is commensurable in square only with the whole, AB , and which contains with the whole, AB , the medial (rectangle contained) by AB and BC , be subtracted from the medial (straight-line) AB [Prop. 10.28]. I say that the remainder AC is an irrational (straight-line). Let it be called a second apotome of a medial (straight-line).



Ἐκκείσθω γάρ ῥητὴ ἡ DI , καὶ τοῖς μὲν ἀπὸ τῶν AB , BF ἴσον παρὰ τὴν DI παραβεβλήσθω τὸ ΔE πλάτος ποιῶν τὴν ΔH , τῷ δὲ διζῷ ὑπὸ τῶν AB , BF ἴσον παρὰ τὴν DI παραβεβλήσθω τὸ $\Delta \Theta$ πλάτος ποιῶν τὴν ΔZ . λοιπὸν ἄρα τὸ ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς AG . καὶ ἐπεὶ μέσα καὶ σύμμετρά ἐστι τὰ ἀπὸ τῶν AB , BF , μέσον ἄρα καὶ τὸ ΔE . καὶ παρὰ ῥητὴν τὴν DI παρὰκαίται πλάτος ποιῶν τὴν ΔH . ῥητὴ ἄρα ἐστὶν ἡ ΔH καὶ ἀσύμμετρος τῇ DI μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ὑπὸ τῶν AB , BF , καὶ τὸ διζῷ ἄρα ὑπὸ τῶν AB , BF μέσον ἐστίν. καὶ ἐστὶν ἴσον τῷ $\Delta \Theta$. καὶ τὸ $\Delta \Theta$ ἄρα μέσον ἐστίν. καὶ παρὰ ῥητὴν τὴν DI παραβεβλήται πλάτος ποιῶν τὴν ΔZ . ῥητὴ ἄρα ἐστὶν ἡ ΔZ καὶ ἀσύμμετρος τῇ DI μήκει. καὶ ἐπεὶ αἱ AB , BF δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶν ἡ AB τῇ BF μήκει. ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς AB τετράγωνον τῷ ὑπὸ τῶν AB , BF . ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρά ἐστι τὰ ἀπὸ τῶν AB , BF , τῷ δὲ ὑπὸ τῶν AB , BF σύμμετρον ἐστὶ τὸ διζῷ ὑπὸ τῶν AB , BF . ἀσύμμετρον ἄρα ἐστὶ τὸ διζῷ ὑπὸ τῶν AB , BF τοῖς ἀπὸ τῶν AB , BF . ἴσον δὲ τοῖς μὲν ἀπὸ τῶν AB , BF τὸ ΔE , τῷ δὲ διζῷ ὑπὸ τῶν AB , BF τὸ $\Delta \Theta$. ἀσύμμετρον ἄρα [ἐστὶ] τὸ ΔE τῷ $\Delta \Theta$. ὥς δὲ τὸ ΔE πρὸς τὸ $\Delta \Theta$, οὕτως ἡ $H\Delta$ πρὸς τὴν ΔZ . ἀσύμμετρος ἄρα ἐστὶν ἡ $H\Delta$ τῇ ΔZ . καὶ εἰσιν ἀμφοτέραι ῥηταί. αἱ ἄρα $H\Delta$, ΔZ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἡ ZH ἄρα ἀποτομή ἐστίν. ῥητὴ δὲ ἡ DI . τὸ δὲ ὑπὸ ῥητῆς καὶ ἀλόγου περιεχόμενον ἄλογόν ἐστιν, καὶ ἡ δυνάμεν αὐτὸ ἄλογός ἐστιν. καὶ δύναται τὸ ZE ἢ AG . ἡ AG ἄρα ἄλογός ἐστιν. καλεῖσθω δὲ μέσης ἀποτομὴ δευτέρα. ὅπερ ἔδει δεῖξαι.

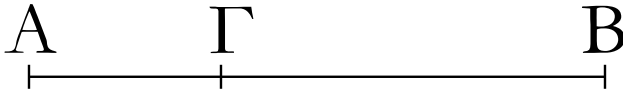
For let the rational (straight-line) DI be laid down. And let DE , equal to the (sum of the squares) on AB and BC , be applied to DI , producing DG as breadth. And let DH , equal to twice the (rectangle contained) by AB and BC , be applied to DI , producing DF as breadth. The remainder FE is thus equal to the (square) on AC [Prop. 2.7]. And since the (squares) on AB and BC are medial and commensurable (with one another), DE (is) thus also medial [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) DI , producing DG as breadth. Thus, DG is rational, and incommensurable in length with DI [Prop. 10.22]. Again, since the (rectangle contained) by AB and BC is medial, twice the (rectangle contained) by AB and BC is thus also medial [Prop. 10.23 corr.]. And it is equal to DH . Thus, DH is also medial. And it has been applied to the rational (straight-line) DI , producing DF as breadth. DF is thus rational, and incommensurable in length with DI [Prop. 10.22]. And since AB and BC are commensurable in square only, AB is thus incommensurable in length with BC . Thus, the square on AB (is) also incommensurable with the (rectangle contained) by AB and BC [Props. 10.21 lem., 10.11]. But, the (sum of the squares) on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. Thus, twice the (rectangle contained) by AB and BC is incommensurable with the (sum of the squares) on AB and BC [Prop. 10.13]. And DE is equal to the (sum of the squares) on AB and BC , and DH to twice the (rectangle contained) by AB and BC . Thus, DE [is] incommensurable with DH . And as DE (is) to DH , so GD (is) to DF [Prop. 6.1]. Thus, GD is in-

commensurable with DF [Prop. 10.11]. And they are both rational (straight-lines). Thus, GD and DF are rational (straight-lines which are) commensurable in square only. Thus, FG is an apotome [Prop. 10.73]. And DI (is) rational. And the (area) contained by a rational and an irrational (straight-line) is irrational [Prop. 10.20], and its square-root is irrational. And AC is the square-root of FE . Thus, AC is an irrational (straight-line) [Def. 10.4]. And let it be called the second apotome of a medial (straight-line).[†] (Which is) the very thing it was required to show.

[†] See footnote to Prop. 10.38.

76.

Ἐάν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιούσα τὰ μὲν ἀπ' αὐτῶν ἅμα ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον, ἡ λοιπὴ ἄλογός ἐστιν· καλείσθω δὲ ἐλάσσων.

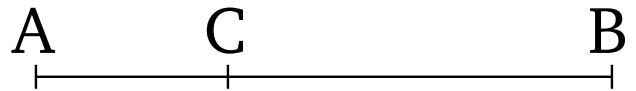


Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφαιρήσθω ἡ $BΓ$ δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ ποιούσα τὰ προκείμενα. λέγω, ὅτι ἡ λοιπὴ ἡ $ΑΓ$ ἄλογός ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ τετραγώνων ῥητόν ἐστιν, τὸ δὲ διξ ὑπὸ τῶν AB , $BΓ$ μέσον, ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$ τῷ διξ ὑπὸ τῶν AB , $BΓ$ · καὶ ἀναστρέψαντι λοιπῷ τῷ ἀπὸ τῆς $ΑΓ$ ἀσύμμετρά ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$. ῥητὰ δὲ τὰ ἀπὸ τῶν AB , $BΓ$ ἄλογον ἄρα τὸ ἀπὸ τῆς $ΑΓ$ ἄλογος ἄρα ἡ $ΑΓ$ · καλείσθω δὲ ἐλάσσων. ὅπερ ἔδει δεῖχαι.

Proposition 76

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the (squares) on them (added) together rational, and the (rectangle contained) by them medial, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called a minor (straight-line).



For let the straight-line BC , which is incommensurable in square with the whole, and fulfils the (other) prescribed (conditions), be subtracted from the straight-line AB [Prop. 10.33]. I say that the remainder AC is that irrational (straight-line) called minor.

For since the sum of the squares on AB and BC is rational, and twice the (rectangle contained) by AB and BC (is) medial, the (sum of the squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . And, via conversion, the (sum of the squares) on AB and BC is incommensurable with the remaining (square) on AC [Props. 2.7, 10.16]. And the (sum of the squares) on AB and BC (is) rational. The (square) on AC (is) thus irrational. Thus, AC (is) an irrational (straight-line) [Def. 10.4]. Let it be called a minor (straight-line).[†] (Which is) the very thing it was required to show.

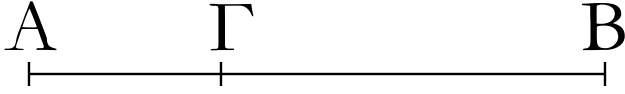
[†] See footnote to Prop. 10.39.

77.

Ἐάν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιούσα τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διξ ὑπ' αὐτῶν ῥητόν, ἡ λοιπὴ ἄλογός ἐστιν· καλείσθω δὲ ἡ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα.

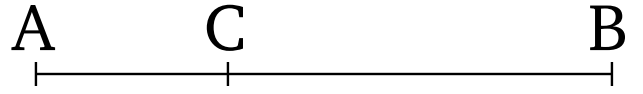
Proposition 77

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them rational, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called that which makes with a rational (area) a medial whole.



Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφηρήσθω ἡ $BΓ$ δυνάμει ἀσύμμετος ὅσα τῇ AB ποιοῦσα τὰ προκείμενα· λέγω, ὅτι ἡ λοιπὴ ἡ $AΓ$ ἄλογός ἐστιν ἡ προειρημένη.

Ἐπεὶ γὰρ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , $BΓ$ τετραγώνων μέσον ἐστίν, τὸ δὲ διξ ὑπὸ τῶν AB , $BΓ$ ῥητόν, ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AB , $BΓ$ τῷ διξ ὑπὸ τῶν AB , $BΓ$ · καὶ λοιπὸν ἄρα τὸ ἀπὸ τῆς $AΓ$ ἀσύμμετρόν ἐστι τῷ διξ ὑπὸ τῶν AB , $BΓ$. καὶ ἐστὶ τὸ διξ ὑπὸ τῶν AB , $BΓ$ ῥητόν· τὸ ἄρα ἀπὸ τῆς $AΓ$ ἄλογόν ἐστιν· ἄλογος ἄρα ἐστὶν ἡ $AΓ$ · καλεῖσθω δὲ ἡ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσα. ὅπερ ἔδει δεῖξαι.



For let the straight-line BC , which is incommensurable in square with AB , and fulfils the (other) prescribed (conditions), be subtracted from the straight-line AB [Prop. 10.34]. I say that the remainder AC is the aforementioned irrational (straight-line).

For since the sum of the squares on AB and BC is medial, and twice the (rectangle contained) by AB and BC rational, the (sum of the squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC . Thus, the remaining (square) on AC is also incommensurable with twice the (rectangle contained) by AB and BC [Props. 2.7, 10.16]. And twice the (rectangle contained) by AB and BC is rational. Thus, the (square) on AC is irrational. Thus, AC is an irrational (straight-line) [Def. 10.4]. And let it be called that which makes with a rational (area) a medial whole.[†] (Which is) the very thing it was required to show.

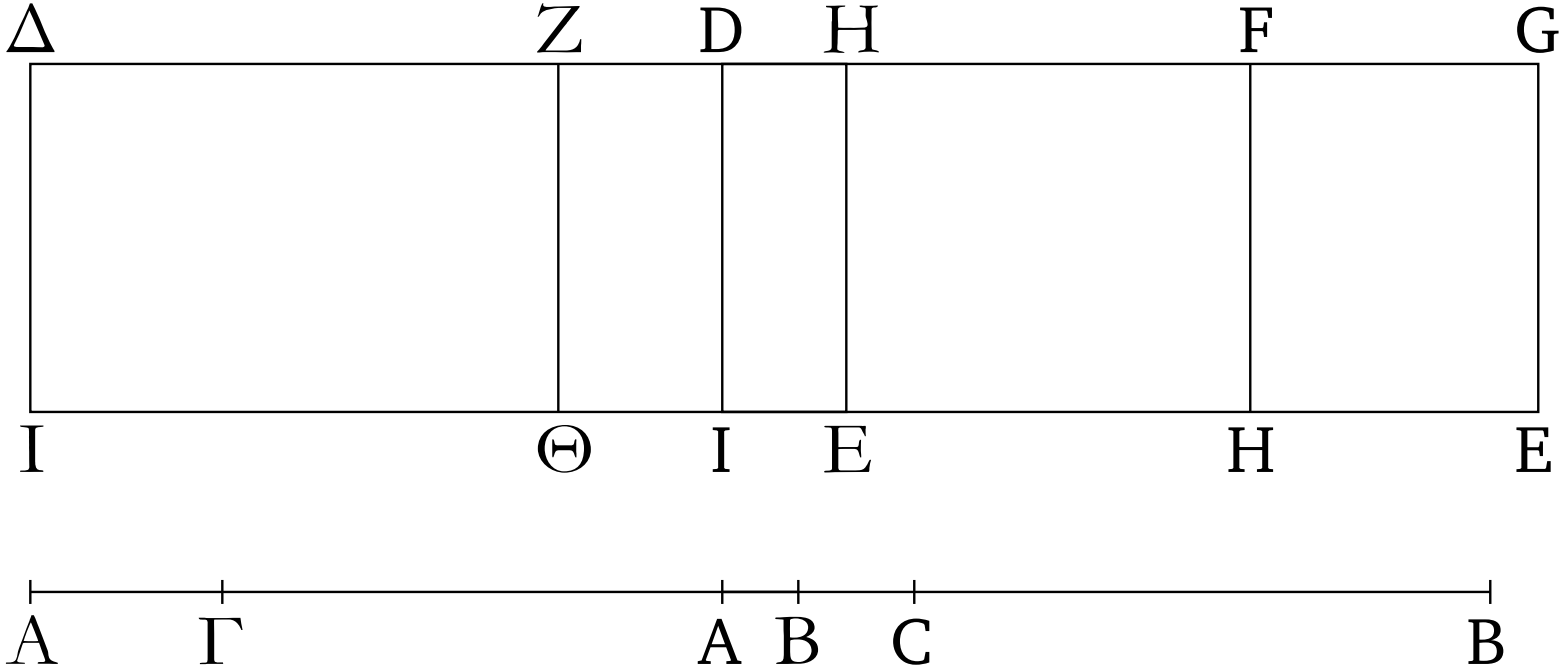
[†] See footnote to Prop. 10.40.

78.

Proposition 78

Ἐὰν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος ὅσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον τὸ τε διξ ὑπ' αὐτῶν μέσον καὶ ἔτι τὰ ἀπ' αὐτῶν τετράγωνα ἀσύμμετρα τῷ διξ ὑπ' αὐτῶν, ἡ λοιπὴ ἄλογός ἐστιν· καλεῖσθω δὲ ἡ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα.

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them medial, and, moreover, the (sum of the) squares on them incommensurable with twice the (rectangle contained) by them, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called that which makes with a medial (area) a medial whole.



Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφηρήσθω ἡ BI δυνάμει ἀσύμμετρος ὅσα τῇ AB ποιοῦσα τὰ προκείμενα· λέγω, ὅτι ἡ λοιπὴ ἡ AI ἄλογός ἐστιν ἡ καλουμένη ἡ μετὰ μέσον μέσον τὸ ὅλον ποιοῦσα.

Ἐκκείσθω γὰρ ῥητὴ ἡ DI , καὶ τοῖς μὲν ἀπὸ τῶν AB , BI ἴσον παρὰ τὴν DI παραβεβλήσθω τὸ DE πλάτος ποιοῦν τὴν DH , τῷ δὲ διζὺ ὑπὸ τῶν AB , BI ἴσον ἀφηρήσθω τὸ $ΔΘ$ [πλάτος ποιοῦν τὴν $ΔΖ$]. λοιπὸν ἄρα τὸ ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς AI · ὥστε ἡ AI δύνатаι τὸ ZE . καὶ ἐπεὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB , BI τετραγώνων μέσον ἐστὶ καὶ ἐστὶν ἴσον τῷ DE , μέσον ἄρα [ἐστὶ] τὸ DE . καὶ παρὰ ῥητὴν τὴν DI παράκειται πλάτος ποιοῦν τὴν DH · ῥητὴ ἄρα ἐστὶν ἡ DH καὶ ἀσύμμετρος τῇ DI μήκει. πάλιν, ἐπεὶ τὸ διζὺ ὑπὸ τῶν AB , BI μέσον ἐστὶ καὶ ἐστὶν ἴσον τῷ $ΔΘ$, τὸ ἄρα $ΔΘ$ μέσον ἐστὶν. καὶ παρὰ ῥητὴν τὴν DI παράκειται πλάτος ποιοῦν τὴν $ΔΖ$ · ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΔΖ$ καὶ ἀσύμμετρος τῇ DI μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB , BI τῷ διζὺ ὑπὸ τῶν AB , BI , ἀσύμμετρον ἄρα καὶ τὸ DE τῷ $ΔΘ$. ὥς δὲ τὸ DE πρὸς τὸ $ΔΘ$, οὕτως ἐστὶ καὶ ἡ DH πρὸς τὴν $ΔΖ$ · ἀσύμμετρος ἄρα ἡ DH τῇ $ΔΖ$. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ HD , $ΔΖ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἀποτομὴ ἄρα ἐστὶν ἡ ZH · ῥητὴ δὲ ἡ $ZΘ$. τὸ δὲ ὑπὸ ῥητῆς καὶ ἀποτομῆς περιεχόμενον [ὀρθογώνιον] ἄλογόν ἐστιν, καὶ ἡ δυνάμεν αὐτὸ ἄλογός ἐστιν· καὶ δύνатаι τὸ ZE ἡ AI · ἡ AI ἄρα ἄλογός ἐστιν· καλεῖσθω δὲ ἡ μετὰ μέσον μέσον τὸ ὅλον ποιοῦσα. ὅπερ ἔδει δεῖχαι.

For let the straight-line BC , which is incommensurable in square AB , and fulfils the (other) prescribed (conditions), be subtracted from the (straight-line) AB [Prop. 10.35]. I say that the remainder AC is the irrational (straight-line) called that which makes with a medial (area) a medial whole.

For let the rational (straight-line) DI be laid down. And let DE , equal to the (sum of the squares) on AB and BC , be applied to DI , producing DG as breadth. And let DH , equal to twice the (rectangle contained) by AB and BC , be subtracted (from DE) [producing DF as breadth]. Thus, the remainder FE is equal to the (square) on AC [Prop. 2.7]. Hence, AC is the square-root of FE . And since the sum of the squares on AB and BC is medial, and is equal to DE , DE [is] thus medial. And it is applied to the rational (straight-line) DI , producing DG as breadth. Thus, DG is rational, and incommensurable in length with DI [Prop 10.22]. Again, since twice the (rectangle contained) by AB and BC is medial, and is equal to DH , DH is thus medial. And it is applied to the rational (straight-line) DI , producing DF as breadth. Thus, DF is also rational, and incommensurable in length with DI [Prop. 10.22]. And since the (sum of the squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC , DE (is) also incommensurable with DH . And as DE (is) to DH , so DG also is to DF [Prop. 6.1]. Thus, DG (is) incommensurable (in length) with DF [Prop. 10.11]. And they are both rational. Thus, GD and DF are rational (straight-lines which are) commensurable in square only. Thus, FG is an apotome [Prop. 10.73]. And FH (is) rational. And the [rectangle] contained by a rational (straight-line) and an apotome is irrational [Prop. 10.20], and its square-root is irrational. And AC is the

square-root of FE . Thus, AC is irrational. Let it be called that which makes with a medial (area) a medial whole.[†] (Which is) the very thing it was required to show.

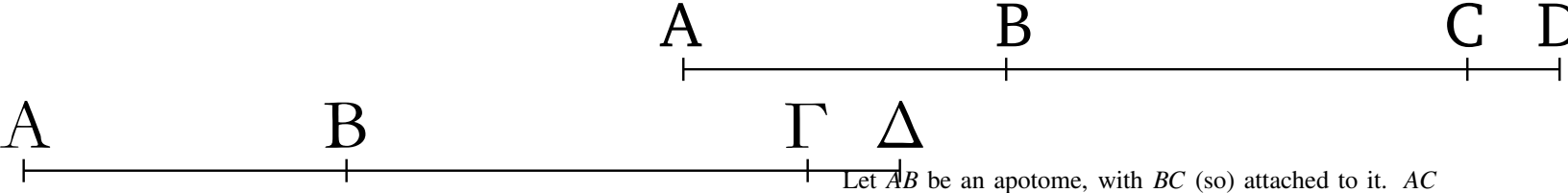
[†] See footnote to Prop. 10.41.

79.

Proposition 79

Τῇ ἀποτομῇ μία [μόνον] προσαρμόζει εὐθεῖα ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ.

[Only] one rational straight-line, which is commensurable in square only with the whole, can be attached to an apotome.[†]



Ἐστω ἀποτομή ἡ AB , προσαρμόζουσα δὲ αὐτῇ ἡ $BΓ$. αἱ $ΑΓ$, $ΓΒ$ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμόζει ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ.

Εἰ γὰρ δυνατόν, προσαρμόξέτω ἡ $ΒΔ$ · καὶ αἱ $ΑΔ$, $ΔΒ$ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ, ὥς ὑπερέθει τὰ ἀπὸ τῶν $ΑΔ$, $ΔΒ$ τοῦ διξ ὑπὸ τῶν $ΑΔ$, $ΔΒ$, τούτῳ ὑπερέθει καὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$ τοῦ διξ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ · τῷ γὰρ αὐτῷ τῷ ἀπὸ τῆς AB ἀμφοτέρω ὑπερέθει· ἐναλλάξ ἄρα, ὥς ὑπερέθει τὰ ἀπὸ τῶν $ΑΔ$, $ΔΒ$ τῶν ἀπὸ τῶν $ΑΓ$, $ΓΒ$, τούτῳ ὑπερέθει [καὶ] τὸ διξ ὑπὸ τῶν $ΑΔ$, $ΔΒ$ τοῦ διξ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. τὰ δὲ ἀπὸ τῶν $ΑΔ$, $ΔΒ$ τῶν ἀπὸ τῶν $ΑΓ$, $ΓΒ$ ὑπερέθει ῥητῶν· ῥητὰ γὰρ ἀμφοτέρω. καὶ τὸ διξ ἄρα ὑπὸ τῶν $ΑΔ$, $ΔΒ$ τοῦ διξ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ ὑπερέθει ῥητῶν· ὅπερ ἐστὶν ἀδύνατον· μέσα γὰρ ἀμφοτέρω, μέσον δὲ μέσον οὐδ' ὑπερέθει ῥητῶν. τῇ ἄρα AB ἑτέρα οὐ προσαρμόζει ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ.

Μία ἄρα μόνη τῇ ἀποτομῇ προσαρμόζει ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ· ὅπερ ἔδει δεῖξαι.

Let AB be an apotome, with BC (so) attached to it. AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.73]. I say that another rational (straight-line), which is commensurable in square only with the whole, cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds twice the (rectangle contained) by AD and DB , the (sum of the squares) on AC and CB also exceeds twice the (rectangle contained) by AC and CB by this (same area). For both exceed by the same (area)—(namely), the (square) on AB [Prop. 2.7]. Thus, alternately, by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB [also] exceeds twice the (rectangle contained) by AC and CB by this (same area). And the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB by a rational (area). For both (are) rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.21], and a medial (area) cannot exceed a(nother) medial (area) by a rational (area) [Prop. 10.26]. Thus, another rational (straight-line), which is commensurable in square only with the whole, cannot be attached to AB .

Thus, only one rational (straight-line), which is commensurable in square only with the whole, can be attached to an apotome. (Which is) the very thing it was required to show.

[†] This proposition is equivalent to Prop. 10.42, with minus signs instead of plus signs.

80.

Proposition 80

Τῇ μέσῃ ἀποτομῇ πρώτη μία μόνον προσαρμόζει εὐθεῖα

Only one medial straight-line, which is commensurable

μέση δυνάμει μόνον σύμμετροξ ὅσα τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃς ῥητὸν περιέθουσα.

in square only with the whole, and contains a rational (area) with the whole, can be attached to a first apotome of a medial (straight-line).[†]



Ἐστω γάρ μέσηξ ἀποτομή πρώτη ἡ AB , καὶ τῇ AB προσαρμύζετω ἡ BF . αἱ AF , FB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι τὸ ὑπὸ τῶν AF , FB . λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμύζει μέση δυνάμει μόνον σύμμετροξ ὅσα τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃς ῥητὸν περιέθουσα.

Εἰ γὰρ δυνατόν, προσαρμύζετω καὶ ἡ $ΔB$. αἱ ἄρα AD , $ΔB$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι τὸ ὑπὸ τῶν AD , $ΔB$. καὶ ἐπεὶ, ὥς ὑπερέθει τὰ ἀπὸ τῶν AD , $ΔB$ τοῦ διξ ὑπὸ τῶν AD , $ΔB$, τούτῳ ὑπερέθει καὶ τὰ ἀπὸ τῶν AF , FB τοῦ διξ ὑπὸ τῶν AF , FB . τῷ γὰρ αὐτῷ [πάλιν] ὑπερέθουσι τῷ ἀπὸ τῇξ AB . ἐναλλάχ ἄρα, ὥς ὑπερέθει τὰ ἀπὸ τῶν AD , $ΔB$ τῶν ἀπὸ τῶν AF , FB , τούτῳ ὑπερέθει καὶ τὸ διξ ὑπὸ τῶν AD , $ΔB$ τοῦ διξ ὑπὸ τῶν AF , FB . τὸ δὲ διξ ὑπὸ τῶν AD , $ΔB$ τοῦ διξ ὑπὸ τῶν AF , FB ὑπερέθει ῥητῷ ῥητὰ γὰρ ἀμφοτέρω. καὶ τὰ ἀπὸ τῶν AD , $ΔB$ ἄρα τῶν ἀπὸ τῶν AF , FB [τετραγώνων] ὑπερέθει ῥητῷ. ὅπερ ἐστὶν ἀδύνατον· μέσα γὰρ ἐστὶν ἀμφοτέρω, μέσον δὲ μέσον οὐδ' ὑπερέθει ῥητῷ.

Τῇ ἄρα μέσηξ ἀποτομῇ πρώτῃ μία μόνον προσαρμύζει ἐνθεῖα μέση δυνάμει μόνον σύμμετροξ ὅσα τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃς ῥητὸν περιέθουσα. ὅπερ ἔδει δεῖξαι.

For let AB be a first apotome of a medial (straight-line), and let BC be (so) attached to AB . Thus, AC and CB are medial (straight-lines which are) commensurable in square only, containing a rational (area)—(namely, that contained) by AC and CB [Prop. 10.74]. I say that a(nother) medial (straight-line), which is commensurable in square only with the whole, and contains a rational (area) with the whole, cannot be attached to AB .

For, if possible, let DB also be (so) attached to AB . Thus, AD and DB are medial (straight-lines which are) commensurable in square only, containing a rational (area)—(namely, that) contained by AD and DB [Prop. 10.74]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds twice the (rectangle contained) by AD and DB , the (sum of the squares) on AC and CB also exceeds twice the (rectangle contained) by AC and CB by this (same area). For [again] both exceed by the same (area)—(namely), the (square) on AB [Prop. 2.7]. Thus, alternately, by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area). And twice the (rectangle contained) by AD and DB exceeds twice the (rectangle contained) by AC and CB by a rational (area). For both (are) rational (areas). Thus, the (sum of the squares) on AD and DB also exceeds the (sum of the) [squares] on AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Props. 10.15, 10.23 corr.], and a medial (area) cannot exceed a(nother) medial (area) by a rational (area) [Prop. 10.26].

Thus, only one medial (straight-line), which is commensurable in square only with the whole, and contains a rational (area) with the whole, can be attached to a first apotome of a medial (straight-line). (Which is) the very thing it was required to show.

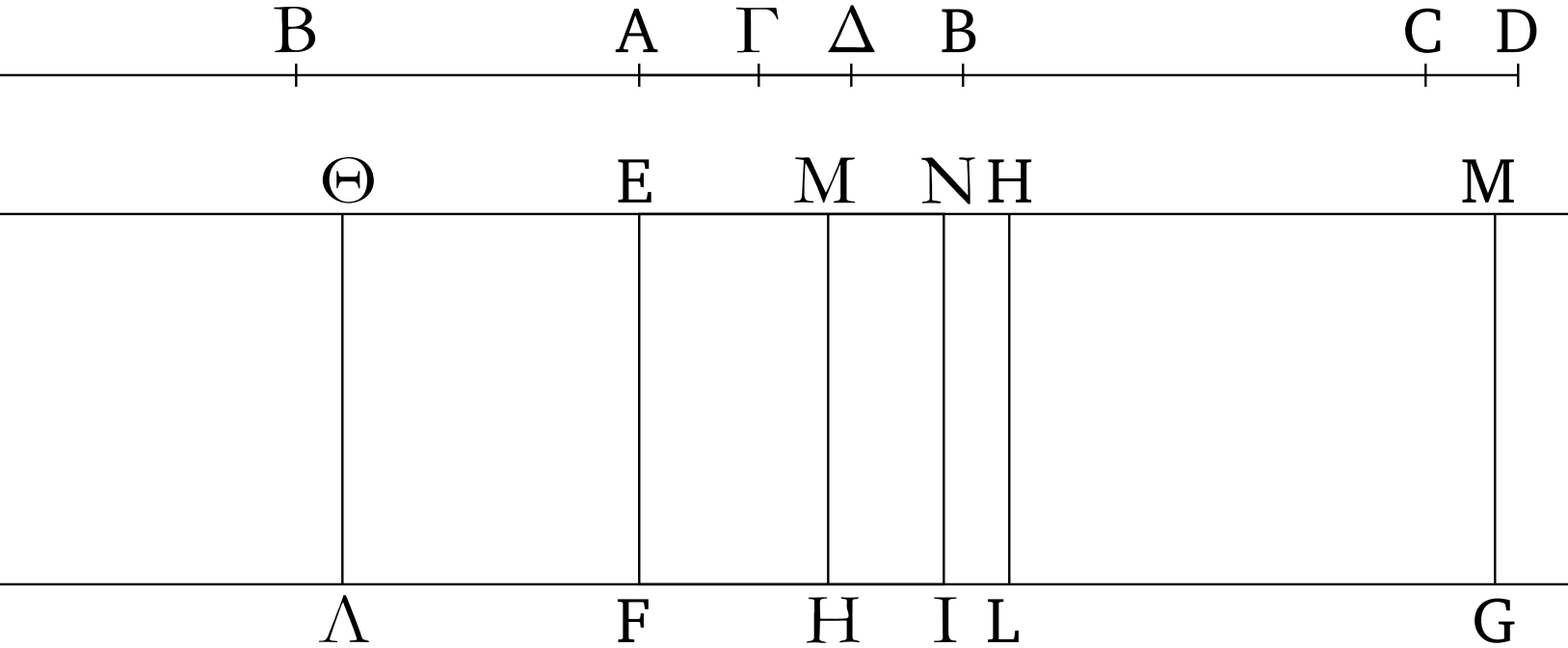
[†] This proposition is equivalent to Prop. 10.43, with minus signs instead of plus signs.

81.

Proposition 81

Τῇ μέσηξ ἀποτομῇ δευτέρῃ μία μόνον προσαρμύζει ἐνθεῖα μέση δυνάμει μόνον σύμμετροξ τῇ ὅλῃ, μετὰ δὲ τῇξ ὅλῃς μέσον περιέθουσα.

Only one medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, can be attached to a second apotome of a medial (straight-line).[†]



Ἐστω μέσης ἀποτομή δευτέρα ἡ AB καὶ τῇ AB προσαρμόζονσα ἡ $BΓ$. αἱ ἄρα $ΑΓ$, $ΓΒ$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι τὸ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμόσει εὐθεῖα μέση δυνάμει μόνον σύμμετρος οὕσα τῇ ὅλῃ, μετὰ δὲ τῇς ὅλης μέσον περιέθουσα.

Εἰ γὰρ δυνατόν, προσαρμόξέτω ἡ $BΔ$. καὶ αἱ $ΑΔ$, $ΔΒ$ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι τὸ ὑπὸ τῶν $ΑΔ$, $ΔΒ$. καὶ ἐκκείσθω ῥητὴ ἡ EZ , καὶ τοῖς μὲν ἀπὸ τῶν $ΑΓ$, $ΓΒ$ ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ $ΕΗ$ πλάτος ποιοῦν τὴν $ΕΜ$. τῷ δὲ διζῷ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ ἴσον ἀφηρήσθω τὸ $ΘΗ$ πλάτος ποιοῦν τὴν $ΘΜ$. λοιπὸν ἄρα τὸ $ΕΛ$ ἴσον ἐστὶ τῷ ἀπὸ τῇς $ΑΒ$. ὥστε ἡ $ΑΒ$ δύναται τὸ $ΕΛ$. πάλιν δὲ τοῖς ἀπὸ τῶν $ΑΔ$, $ΔΒ$ ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ $ΕΙ$ πλάτος ποιοῦν τὴν $ΕΝ$. ἔστι δὲ καὶ τὸ $ΕΛ$ ἴσον τῷ ἀπὸ τῇς $ΑΒ$ τετραγώνῳ. λοιπὸν ἄρα τὸ $ΘΙ$ ἴσον ἐστὶ τῷ διζῷ ὑπὸ τῶν $ΑΔ$, $ΔΒ$. καὶ ἐπεὶ μέσαι εἰσὶν αἱ $ΑΓ$, $ΓΒ$, μέσα ἄρα ἐστὶ καὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$. καὶ ἐστὶν ἴσα τῷ $ΕΗ$. μέσον ἄρα καὶ τὸ $ΕΗ$. καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν $ΕΜ$. ῥητὴ ἄρα ἐστὶν ἡ $ΕΜ$ καὶ ἀσύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ὑπὸ τῶν $ΑΓ$, $ΓΒ$, καὶ τὸ διζῷ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ μέσον ἐστίν. καὶ ἐστὶν ἴσον τῷ $ΘΗ$. καὶ τὸ $ΘΗ$ ἄρα μέσον ἐστίν. καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν $ΘΜ$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΘΜ$ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ αἱ $ΑΓ$, $ΓΒ$ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶν ἡ $ΑΓ$ τῇ $ΓΒ$ μήκει. ὥς δὲ ἡ $ΑΓ$ πρὸς τὴν $ΓΒ$, οὕτως ἐστὶ τὸ ἀπὸ τῇς $ΑΓ$ πρὸς τὸ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῇς $ΑΓ$ τῷ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. ἀλλὰ τῷ μὲν ἀπὸ τῇς $ΑΓ$ σύμμετρον ἐστὶ τὰ ἀπὸ τῶν $ΑΓ$, $ΓΒ$, τῷ δὲ ὑπὸ τῶν $ΑΓ$, $ΓΒ$ σύμμετρον ἐστὶ τὸ διζῷ ὑπὸ τῶν $ΑΓ$, $ΓΒ$. ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν $ΑΓ$,

Let AB be a second apotome of a medial (straight-line), with BC (so) attached to AB . Thus, AC and CB are medial (straight-lines which are) commensurable in square only, containing a medial (area)—(namely, that contained) by AC and CB [Prop. 10.75]. I say that a(nother) medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, cannot be attached to AB .

For, if possible, let BD be (so) attached. Thus, AD and DB are also medial (straight-lines which are) commensurable in square only, containing a medial (area)—(namely, that contained) by AD and DB [Prop. 10.75]. And let the rational (straight-line) EF be laid down. And let EG , equal to the (sum of the squares) on AC and CB , be applied to EF , producing EM as breadth. And let HG , equal to twice the (rectangle contained) by AC and CB , be subtracted (from EG), producing HM as breadth. The remainder EL is thus equal to the (square) on AB [Prop. 2.7]. Hence, AB is the square-root of EL . So, again, let EI , equal to the (sum of the squares) on AD and DB be applied to EF , producing EN as breadth. And EL is also equal to the square on AB . Thus, the remainder HI is equal to twice the (rectangle contained) by AD and DB [Prop. 2.7]. And since AC and CB are (both) medial (straight-lines), the (sum of the squares) on AC and CB is also medial. And it is equal to EG . Thus, EG is also medial [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) EF , producing EM as breadth. Thus, EM is rational, and incommensurable in length with EF [Prop. 10.22]. Again, since the (rectangle contained) by AC and CB is medial, twice the (rectangle contained) by AC and CB is also me-

GB τῷ διζ ὑπὸ τῶν AG , GB . καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν AG , GB ἴσον τὸ EH , τῷ δὲ διζ ὑπὸ τῶν AG , GB ἴσον τὸ $H\Theta$. ἀσύμμετρον ἄρα ἐστὶ τὸ EH τῷ ΘH . ὥς δὲ τὸ EH πρὸς τὸ ΘH , οὕτως ἐστὶν ἡ EM πρὸς τὴν ΘM . ἀσύμμετρος ἄρα ἐστὶν ἡ EM τῇ $M\Theta$ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ EM , $M\Theta$ ἄρα ῥηταὶ εἰςὶ δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ $E\Theta$, προσαρμόζουσα δὲ αὐτῇ ἡ ΘM . ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΘN αὐτῇ προσαρμόζει· τῇ ἄρα ἀποτομῇ ἄλλη καὶ ἄλλη προσαρμόζει εὐθεῖα δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ· ὅπερ ἐστὶν ἀδύνατον.

Τῇ ἄρα μέσῃ ἀποτομῇ δευτέρᾳ μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῇς ὅλης μέσον περιέθουσα· ὅπερ ἔδει δεῖχαι.

dial [Prop. 10.23 corr.]. And it is equal to HG . Thus, HG is also medial. And it is applied to the rational (straight-line) EF , producing HM as breadth. Thus, HM is also rational, and incommensurable in length with EF [Prop. 10.22]. And since AC and CB are commensurable in square only, AC is thus incommensurable in length with CB . And as AC (is) to CB , so the (square) on AC is to the (rectangle contained) by AC and CB [Prop. 10.21 corr.]. Thus, the (square) on AC is incommensurable with the (rectangle contained) by AC and CB [Prop. 10.11]. But, the (sum of the squares) on AC and CB is commensurable with the (square) on AC , and twice the (rectangle contained) by AC and CB is commensurable with the (rectangle contained) by AC and CB [Prop. 10.6]. Thus, the (sum of the squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB [Prop. 10.13]. And EG is equal to the (sum of the squares) on AC and CB . And GH is equal to twice the (rectangle contained) by AC and CB . Thus, EG is incommensurable with HG . And as EG (is) to HG , so EM is to HM [Prop. 6.1]. Thus, EM is incommensurable in length with MH [Prop. 10.11]. And they are both rational (straight-lines). Thus, EM and MH are rational (straight-lines which are) commensurable in square only. Thus, EH is an apotome [Prop. 10.73], and HM (is) attached to it. So, similarly, we can show that HN (is) also (commensurable in square only with EN and is) attached to (EH). Thus, different straight-lines, which are commensurable in square only with the whole, are attached to an apotome. The very thing is impossible [Prop. 10.79].

Thus, only one medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, can be attached to a second apotome of a medial (straight-line). (Which is) the very thing it was required to show.

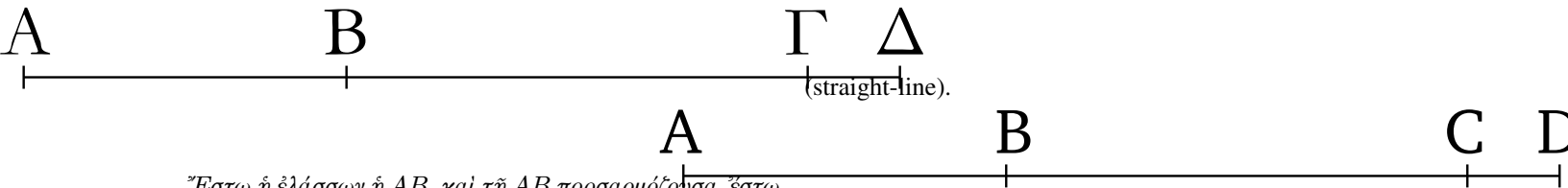
† This proposition is equivalent to Prop. 10.44, with minus signs instead of plus signs.

82.

Proposition 82

Τῇ ἐλάσσονι μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ ποιοῦσα μετὰ τῇς ὅλης τὸ μὲν ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δὲ διζ ὑπ' αὐτῶν μέσον.

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the (sum of the) squares on them rational, and twice the (rectangle contained) by them medial, can be attached to a minor



Ἐστω ἡ ἐλάσσων ἡ AB , καὶ τῇ AB προσαρμόζουσα ἔστω ἡ BF . αἱ ἄρα AG , GB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκεκλιμένον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δὲ διζ ὑπ' αὐτῶν μέσον· λέγω, ὅτι τῇ AB ἑτέρα εὐθεῖα οὐ

Let AB be a minor (straight-line), and let BC be (so) attached to AB . Thus, AC and CB are (straight-lines which are) incommensurable in square, making the sum of the squares

προσαρμόσει τὰ αὐτὰ ποιῶσα.

Εἰ γὰρ δυνατόν, προσαρμोजέτω ἡ BD · καὶ αἱ AD , DB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὰ προειρημένα. καὶ ἐπεὶ, ὥς ὑπερέθει τὰ ἀπὸ τῶν AD , DB τῶν ἀπὸ τῶν AG , GB , τούτῳ ὑπερέθει καὶ τὸ διξ ὑπὸ τῶν AD , DB τοῦ διξ ὑπὸ τῶν AG , GB , τὰ δὲ ἀπὸ τῶν AD , DB τετράγωνα τῶν ἀπὸ τῶν AG , GB τετραγώνων ὑπερέθει ῥητῶ· ῥητὰ γάρ ἐστιν ἀμφοτέρω· καὶ τὸ διξ ὑπὸ τῶν AD , DB ἄρα τοῦ διξ ὑπὸ τῶν AG , GB ὑπερέθει ῥητῶ· ὅπερ ἐστὶν ἀδύνατον· μέσα γάρ ἐστιν ἀμφοτέρω.

Τῇ ἄρα ἐλάσσονι μία μόνον προσαρμोजεῖ εὐθεῖα δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ καὶ ποιῶσα τὰ μὲν ἀπ' αὐτῶν τετράγωνα ἅμα ῥητόν, τὸ δὲ διξ ὑπ' αὐτῶν μέσον· ὅπερ ἔδει δεῖχαι.

on them rational, and twice the (rectangle contained) by them medial [Prop. 10.76]. I say that another another straight-line fulfilling the same (conditions) cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also (straight-lines which are) incommensurable in square, fulfilling the (other) aforementioned (conditions) [Prop. 10.76]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area) [Prop. 2.7]. And the (sum of the) squares on AD and DB exceeds the (sum of the) squares on AC and CB by a rational (area). For both are rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.26].

Thus, only one straight-line, which is incommensurable in square with the whole, and (with the whole) makes the squares on them (added) together rational, and twice the (rectangle contained) by them medial, can be attached to a minor (straight-line). (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.45, with minus signs instead of plus signs.

83.

Proposition 83

Τῇ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσῃ μία μόνον προσαρμोजεῖ εὐθεῖα δυνάμει ἀσύμμετρος οὔσα τῇ ὅλῃ, μετὰ δὲ τῇ ὅλῃ ποιῶσα τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διξ ὑπ' αὐτῶν ῥητόν.

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them rational, can be attached to that (straight-line) which with a rational (area) makes a medial whole.†



Ἐστω ἡ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα ἡ AB , καὶ τῇ AB προσαρμोजέτω ἡ $BΓ$ · αἱ ἄρα AG , GB δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὰ προκείμενα· λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμोजεῖ τὰ αὐτὰ ποιῶσα.

Εἰ γὰρ δυνατόν, προσαρμोजέτω ἡ BD · καὶ αἱ AD , DB ἄρα εὐθεῖαι δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὰ προκείμενα. ἐπεὶ οὖν, ὥς ὑπερέθει τὰ ἀπὸ τῶν AD , DB τῶν ἀπὸ τῶν AG , GB , τούτῳ ὑπερέθει καὶ τὸ διξ ὑπὸ τῶν AD , DB τοῦ διξ ὑπὸ τῶν AG , GB ἀκολούθως τοῖς πρὸ αὐτοῦ, τὸ δὲ διξ ὑπὸ τῶν AD , DB τοῦ διξ ὑπὸ τῶν AG , GB ὑπερέθει ῥητῶ· ῥητὰ γάρ ἐστιν ἀμφοτέρω· καὶ τὰ ἀπὸ τῶν AD , DB ἄρα τῶν ἀπὸ τῶν AG , GB ὑπερέθει ῥητῶ· ὅπερ ἐστὶν ἀδύνατον· μέσα γάρ ἐστιν ἀμφοτέρω.

Οὐκ ἄρα τῇ AB ἑτέρα προσαρμोजεῖ εὐθεῖα δυνάμει ἀσύμ-

Let AB be a (straight-line) which with a rational (area) makes a medial whole, and let BC be (so) attached to AB . Thus, AC and CB are (straight-lines which are) incommensurable in square, fulfilling the (other) proscribed (conditions) [Prop. 10.77]. I say that another (straight-line) fulfilling the same (conditions) cannot be attached to AB .

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also straight-lines (which are) incommensurable in square, fulfilling the (other) proscribed (conditions) [Prop. 10.77]. Therefore, analogously to the (propositions) before this, since by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB , twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this

μετροῦς οὕσα τῇ ὅλῃ, μετὰ δὲ τῇ ὅλῃ ποιοῦσα τὰ προειρη-
μένα· μία ἄρα μόνον προσαρμόσει· ὅπερ ἔδει δεῖχαι.

(same area). And twice the (rectangle contained) by AD and DB exceeds twice the (rectangle contained) by AC and CB by a rational (area). For they are (both) rational (areas). Thus, the (sum of the squares) on AD and DB also exceeds the (sum of the squares) on AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.26].

Thus, another straight-line cannot be attached to AB , which is incommensurable in square with the whole, and fulfills the (other) aforementioned (conditions) with the whole. Thus, only one (such straight-line) can be (so) attached. (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.46, with minus signs instead of plus signs.

84.

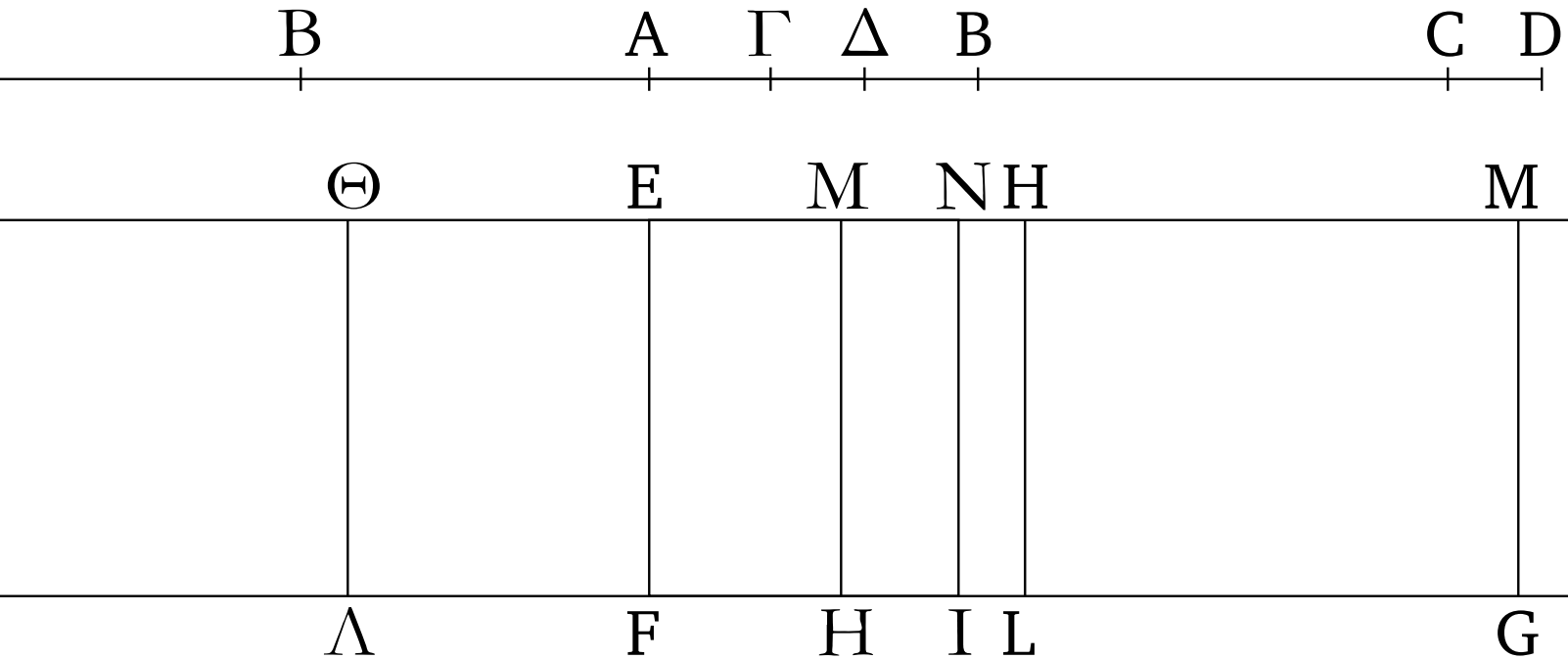
Proposition 84

Τῇ μετὰ μέσον μέσον τὸ ὅλον ποιούσῃ μία μόνῃ προσ-
αρμόζει εὐθεῖα δυνάμει ἀσύμμετρος οὕσα τῇ ὅλῃ, μετὰ δὲ
τῇ ὅλῃ ποιοῦσα τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετρα-
γώνων μέσον τὸ τε διξ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον
τῷ συγκειμένῳ ἐκ τῶν ἀπ' αὐτῶν.

Ἐστω ἡ μετὰ μέσον μέσον τὸ ὅλον ποιοῦσα ἡ AB , προσ-
αρμόζουσα δὲ αὐτῇ ἡ $B\Gamma$. αἱ ἄρα AG , GB δυνάμει εἰσὶν ἀσύμ-
μετροι ποιοῦσαι τὰ προειρημένα. λέγω, ὅτι τῇ AB ἑτέρα οὐ
προσαρμόσει ποιοῦσα προειρημένα.

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the (squares) on them, can be attached to that (straight-line) which with a medial (area) makes a medial whole.†

Let AB be a (straight-line) which with a medial (area) makes a medial whole, BC being (so) attached to it. Thus, AC and CB are incommensurable in square, fulfilling the (other) aforementioned (conditions) [Prop. 10.78]. I say that a(nother) (straight-line) fulfilling the aforementioned (conditions) cannot be attached to AB .



Εἰ γὰρ δυνατόν, προσαρμोजέτω ἡ $B\Delta$, ὥστε καὶ τὰς $A\Delta$,

For, if possible, let BD be (so) attached. Hence, AD and

ΔB δυνάμει ἀσύμμετρον εἶναι ποιούσαξ τά τε ἀπὸ τῶν ΔA , ΔB τετράγωνα ἅμα μέσον καὶ τὸ διζ ὑπὸ τῶν ΔA , ΔB μέσον καὶ ἔτι τὰ ἀπὸ τῶν ΔA , ΔB ἀσύμμετρα τῷ διζ ὑπὸ τῶν ΔA , ΔB · καὶ ἐκκείσθω ῥητὴ ἡ EZ , καὶ τοῖς μὲν ἀπὸ τῶν AG , GB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EH πλάτος ποιῶν τὴν EM , τῷ δὲ διζ ὑπὸ τῶν AG , GB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ ΘH πλάτος ποιῶν τὴν ΘM · λοιπὸν ἄρα τὸ ἀπὸ τῆς AB ἴσον ἐστὶ τῷ EL · ἡ ἄρα AB δύνανται τὸ EL . πάλιν τοῖς ἀπὸ τῶν ΔA , ΔB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EI πλάτος ποιῶν τὴν EN . ἔστι δὲ καὶ τὸ ἀπὸ τῆς AB ἴσον τῷ EL · λοιπὸν ἄρα τὸ διζ ὑπὸ τῶν ΔA , ΔB ἴσον [ἐστὶ] τῷ ΘI . καὶ ἐπεὶ μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AG , GB καὶ ἐστὶν ἴσον τῷ EH , μέσον ἄρα ἐστὶ καὶ τὸ EH . καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιῶν τὴν EM · ῥητὴ ἄρα ἐστὶν ἡ EM καὶ ἀσύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ διζ ὑπὸ τῶν AG , GB καὶ ἐστὶν ἴσον τῷ ΘH , μέσον ἄρα καὶ τὸ ΘH . καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιῶν τὴν ΘM · ῥητὴ ἄρα ἐστὶν ἡ ΘM καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὰ ἀπὸ τῶν AG , GB τῷ διζ ὑπὸ τῶν AG , GB , ἀσύμμετρόν ἐστι καὶ τὸ EH τῷ ΘH · ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ EM τῇ $M\Theta$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ἄρα EM , $M\Theta$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ $E\Theta$, προσαρμόζονσα δὲ αὐτῇ ἡ ΘM . ὁμοίως δὲ δείχμεν, ὅτι ἡ $E\Theta$ πάλιν ἀποτομὴ ἐστὶν, προσαρμόζονσα δὲ αὐτῇ ἡ ΘN . τῇ ἄρα ἀποτομῇ ἄλλη καὶ ἄλλη προσαρμόζει ῥητὴ δυνάμει μόνον σύμμετρος ὅσα τῇ ὅλῃ· ὅπερ εἰδείσθαι ἀδύνατον. οὐκ ἄρα τῇ AB ἑτέρα προσαρμόσει εὐθεΐα.

Τῇ ἄρα AB μία μόνον προσαρμόζει εὐθεΐα δυνάμει ἀσύμμετρος ὅσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιούσα τά τε ἀπ' αὐτῶν τετράγωνα ἅμα μέσον καὶ τὸ διζ ὑπ' αὐτῶν μέσον καὶ ἔτι τὰ ἀπ' αὐτῶν τετράγωνα ἀσύμμετρα τῷ διζ ὑπ' αὐτῶν· ὅπερ ἔδει δεῖξαι.

DB are also (straight-lines which are) incommensurable in square, making the squares on AD and DB (added) together medial, and twice the (rectangle contained) by AD and DB medial, and, moreover, the (sum of the squares) on AD and DB incommensurable with twice the (rectangle contained) by AD and DB [Prop. 10.78]. And let the rational (straight-line) EF be laid down. And let EG , equal to the (sum of the squares) on AC and CB , be applied to EF , producing EM as breadth. And let HG , equal to twice the (rectangle contained) by AC and CB , be applied to EF , producing HM as breadth. Thus, the remaining (square) on AB is equal to EL [Prop. 2.7]. Thus, AB is the square-root of EL . Again, let EI , equal to the (sum of the squares) on AD and DB , be applied to EF , producing EN as breadth. And the (square) on AB is also equal to EL . Thus, the remaining twice the (rectangle contained) by AD and DB [is] equal to HI [Prop. 2.7]. And since the sum of the (squares) on AC and CB is medial, and is equal to EG , EG is thus also medial. And it is applied to the rational (straight-line) EF , producing EM as breadth. EM is thus rational, and incommensurable in length with EF [Prop. 10.22]. Again, since twice the (rectangle contained) by AC and CB is medial, and is equal to HG , HG is thus also medial. And it is applied to the rational (straight-line) EF , producing HM as breadth. HM is thus rational, and incommensurable in length with EF [Prop. 10.22]. And since the (sum of the squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB , EG is also incommensurable with HG . Thus, EM is also incommensurable in length with MH [Props. 6.1, 10.11]. And they are both rational (straight-lines). Thus, EM and MH are rational (straight-lines which are) commensurable in square only. Thus, EH is an apotome [Prop. 10.73], with HM attached to it. So, similarly, we can show that EH is again an apotome, with HN attached to it. Thus, different rational (straight-lines), which are commensurable in square only with the whole, are attached to an apotome. The very thing was shown (to be) impossible [Prop. 10.79]. Thus, another straight-line cannot be (so) attached to AB .

Thus, only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the squares on them (added) together medial, and twice the (rectangle contained) by them medial, and, moreover, the (sum of the) squares on them incommensurable with the (rectangle contained) by them, can be attached to AB . (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.47, with minus signs instead of plus signs.

Ὅροι τρίτοι.

11. Ὑποκειμένης ῥητῆς καὶ ἀποτομῆς, ἔαν μὲν ἡ ὅλη τῆς προσαρμόζουσῆς μείζον δύνηται τῷ ἀπὸ συμμέτρον ἐαυτῇ

Definitions III

11. Given a rational (straight-line) and an apotome, if the square on the whole is greater than the (square on a straight-line) attached (to the apotome) by the (square) on (some

μήκει, καὶ ἡ ὅλη σύμμετρος ἢ τῇ ἐκκειμένῃ ῥητῇ μήκει, καλείσθω ἀποτομή πρώτη.

12. Ἐὰν δὲ ἡ προσαρμόζουσα σύμμετρος ἢ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ ὅλη τῆς προσαρμοζούσης μείζον δύνηται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καλείσθω ἀποτομή δευτέρα.

13. Ἐὰν δὲ μηδετέρα σύμμετρος ἢ τῇ ἐκκειμένῃ ῥητῇ μήκει, ἡ δὲ ὅλη τῆς προσαρμοζούσης μείζον δύνηται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καλείσθω ἀποτομή τρίτη.

14. Πάλιν, ἔαν ἡ ὅλη τῆς προσαρμοζούσης μείζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ [μήκει], ἔαν μὲν ἡ ὅλη σύμμετρος ἢ τῇ ἐκκειμένῃ ῥητῇ μήκει, καλείσθω ἀποτομή τετάρτη.

15. Ἐὰν δὲ ἡ προσαρμόζουσα, πέμπτη.

16. Ἐὰν δὲ μηδετέρα, ἕκτη.

straight-line) commensurable in length with (the whole), and the whole is commensurable in length with the (previously) laid down rational (straight-line), then let the (apotome) be called a first apotome.

12. And if the attached (straight-line) is commensurable in length with the (previously) laid down rational (straight-line), and the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) commensurable (in length) with (the whole), then let the (apotome) be called a second apotome.

13. And if neither of (the whole or the attached straight-line) is commensurable in length with the (previously) laid down rational (straight-line), and the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) commensurable (in length) with (the whole), then let the (apotome) be called a third apotome.

14. Again, if the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) incommensurable [in length] with (the whole), and the whole is commensurable in length with the (previously) laid down rational (straight-line), then let the (apotome) be called a fourth apotome.

15. And if the attached (straight-line is commensurable), a fifth (apotome).

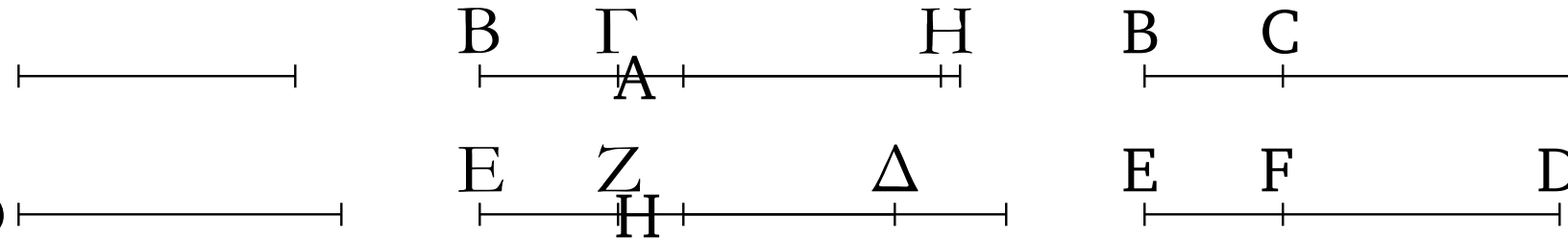
16. And if neither (the whole nor the attached straight-line is commensurable), a sixth (apotome).

85.

Proposition 85

Εὕρεῖν τὴν πρώτην ἀποτομήν.

To find a first apotome.



Ἐκκείσθω ῥητὴ ἡ A , καὶ τῇ A μήκει σύμμετρος ἔστω ἡ BH . ῥητὴ ἄρα ἐστὶ καὶ ἡ BH . καὶ ἐκκείσθωσαν δύο τετράγωνοι ἀριθμοὶ οἱ ΔE , EZ , ὧν ἡ ὑπεροχὴ ὁ $Z\Delta$ μὴ ἔστω τετράγωνος· οὐδ' ἄρα ὁ $E\Delta$ πρὸς τὸν ΔZ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ πεποιήσθω ὥς ὁ $E\Delta$ πρὸς τὸν ΔZ , οὕτως τὸ ἀπὸ τῆς BH τετράγωνον πρὸς τὸ ἀπὸ τῆς $H\Gamma$ τετράγωνον· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς BH τῷ ἀπὸ τῆς $H\Gamma$. ῥητὸν δὲ τὸ ἀπὸ τῆς BH . ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $H\Gamma$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $H\Gamma$. καὶ ἐπεὶ ὁ $E\Delta$ πρὸς τὸν ΔZ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' ἄρα τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $H\Gamma$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν.

Let the rational (straight-line) A be laid down. And let BG be commensurable in length with A . BG is thus also a rational (straight-line). And let two square numbers DE and EF be laid down, and let their difference FD be not square [Prop. 10.28 lem. I]. Thus, ED does not have to DF the ratio which (some) square number (has) to (some) square number. And let it be contrived that as ED (is) to DF , so the square on BG (is) to the square on GC [Prop. 10.6. corr.]. Thus, the (square) on BG is commensurable with the (square) on GC [Prop. 10.6]. And the (square) on BG (is) rational. Thus, the (square) on GC (is) also rational. Thus, GC is also rational. And since ED does not have to DF the ratio which (some)

νον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ BH τῇ $HΓ$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ BH , $HΓ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ ἄρα $ΒΓ$ ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ πρώτη.

Ἦ γὰρ μείζον ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $HΓ$, ἔστω τὸ ἀπὸ τῆς $Θ$. καὶ ἐπεὶ ἐστὶν ὡς ὁ $ΕΔ$ πρὸς τὸν $ΖΔ$, οὕτως τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$, καὶ ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ $ΔΕ$ πρὸς τὸν $ΕΖ$, οὕτως τὸ ἀπὸ τῆς HB πρὸς τὸ ἀπὸ τῆς $Θ$. ὁ δὲ $ΔΕ$ πρὸς τὸν $ΕΖ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἐκάτερος γὰρ τετράγωνός ἐστιν· καὶ τὸ ἀπὸ τῆς HB ἄρα πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα ἐστὶν ἡ BH τῇ $Θ$ μήκει. καὶ δύναιται ἡ BH τῆς $HΓ$ μείζον τῷ ἀπὸ τῆς $Θ$ · ἡ BH ἄρα τῆς $HΓ$ μείζον δύναιται τῷ ἀπὸ συμμετρὸν ἐαυτῇ μήκει. καὶ ἐστὶν ἡ ὅλη ἡ BH σύμμετρος τῇ ἐκκειμένη ῥητῇ μήκει τῇ A . ἡ $ΒΓ$ ἄρα ἀποτομή ἐστὶ πρώτη.

Εὐρεται ἄρα ἡ πρώτη ἀποτομή ἡ $ΒΓ$ · ὅπερ ἔδει εὐρεῖν.

square number (has) to (some) square number, the (square) on BH thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BH is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). Thus, BH and GC are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that (it is) also a first (apotome).

Let the (square) on H be that (area) by which the (square) on BH is greater than the (square) on GC [Prop. 10.13 lem.]. And since as ED is to FD , so the (square) on BH (is) to the (square) on GC , thus, via conversion, as DE is to EF , so the (square) on GB (is) to the (square) on H [Prop. 5.19 corr.]. And DE has to EF the ratio which (some) square-number (has) to (some) square-number. For each is a square (number). Thus, the (square) on GB also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, BH is commensurable in length with H [Prop. 10.9]. And the square on BH is greater than (the square on) GC by the (square) on H . Thus, the square on BH is greater than (the square on) GC by the (square) on (some straight-line) commensurable in length with (BH). And the whole, BH , is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is a first apotome [Def. 10.11].

Thus, the first apotome BC has been found. (Which is) the very thing it was required to find.

† See footnote to Prop. 10.48.

86.

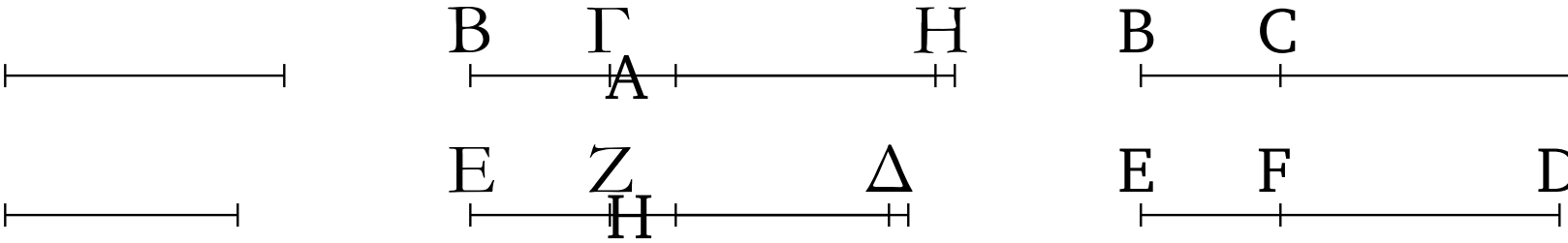
Εὐρεῖν τὴν δευτέραν ἀποτομήν.

Ἐκκείσθω ῥητὴ ἡ A καὶ τῇ A σύμμετρος μήκει ἡ $HΓ$. ῥητὴ ἄρα ἐστὶν ἡ $HΓ$. καὶ ἐκκείσθωσαν δύο τετράγωνοι ἀριθμοὶ οἱ $ΔΕ$, $ΕΖ$, ὧν ἡ ὑπεροβὴ ὁ $ΔΖ$ μὴ ἔστω τετράγωνος. καὶ πεποιήσθω ὡς ὁ $ΖΔ$ πρὸς τὸν $ΔΕ$, οὕτως τὸ ἀπὸ τῆς $ΓΗ$ τετράγωνον πρὸς τὸ ἀπὸ τῆς HB τετράγωνον. σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΓΗ$ τετράγωνον τῷ ἀπὸ τῆς HB τετράγωνῳ. ῥητὸν δὲ τὸ ἀπὸ τῆς $ΓΗ$. ῥητὸν ἄρα [ἐστὶ] καὶ τὸ ἀπὸ τῆς HB · ῥητὴ ἄρα ἐστὶν ἡ BH . καὶ ἐπεὶ τὸ ἀπὸ τῆς $HΓ$ τετράγωνον πρὸς τὸ ἀπὸ τῆς HB λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, ἀσύμμετρος ἐστὶν ἡ $ΓΗ$ τῇ HB μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ $ΓΗ$, HB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ $ΒΓ$ ἄρα ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ δευτέρα.

Proposition 86

To find a second apotome.

Let the rational (straight-line) A , and GC (which is) commensurable in length with A , be laid down. Thus, GC is a rational (straight-line). And let the two square numbers DE and EF be laid down, and let their difference DF be not square [Prop. 10.28 lem. I]. And let it be contrived that as FD (is) to DE , so the square on CG (is) to the square on GB [Prop. 10.6 corr.]. Thus, the square on CG is commensurable with the square on GB [Prop. 10.6]. And the (square) on CG (is) rational. Thus, the (square) on GB [is] also rational. Thus, BH is a rational (straight-line). And since the square on GC does not have to the (square) on GB the ratio which (some) square number (has) to (some) square number, CG is incommensurable in length with GB [Prop. 10.9]. And they are both rational (straight-lines). Thus, CG and GB are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that it is also a second (apotome).



ἵ γὰρ μείζον ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $HΓ$, ἔστω τὸ ἀπὸ τῆς $Θ$. ἐπεὶ οὖν ἐστὶν ὡς τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$, οὕτως ὁ $EΔ$ ἀριθμὸς πρὸς τὸν $ΔΖ$ ἀριθμόν, ἀναστρέψαντι ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $Θ$, οὕτως ὁ $ΔΕ$ πρὸς τὸν $ΕΖ$. καὶ ἐστὶν ἐκάτερος τῶν $ΔΕ$, $ΕΖ$ τετράγωνος· τὸ ἄρα ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα ἐστὶν ἡ BH τῇ $Θ$ μήκει. καὶ δύνатаι ἡ BH τῆς $HΓ$ μείζον τῷ ἀπὸ τῆς $Θ$ · ἡ BH ἄρα τῆς $HΓ$ μείζον δύνатаι τῷ ἀπὸ συμέτρου ἐναυτῇ μήκει. καὶ ἐστὶν ἡ προσαρμοζούσα ἡ $ΓΗ$ τῇ ἐκκειμένῃ ῥητῇ σύμμετρος τῇ $Α$. ἡ $ΒΓ$ ἄρα ἀποτομή ἐστὶ δευτέρα.

Εὐρίηται ἄρα δευτέρα ἀποτομή ἡ $ΒΓ$. ὅπερ ἔδει δεῖξαι.

For let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.1 lem.]. Therefore, since as the (square) on BG is to the (square) on GC , so the number ED (is) to the number DF , thus, also, via conversion, as the (square) on BG is to the (square) on H , so DE (is) to EF [Prop. 5.19 corr.]. And DE and EF are each square (numbers). Thus, the (square) on BG has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, BG is commensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on BG is greater than (the square on) GC by the (square) on (some straight-line) commensurable in length with (BG). And the attachment CG is commensurable (in length) with the (previously) laid down rational (straight-line) A . Thus, BC is a second apotome [Def. 10.12].[†]

Thus, the second apotome BC has been found. (Which is) the very thing it was required to show.

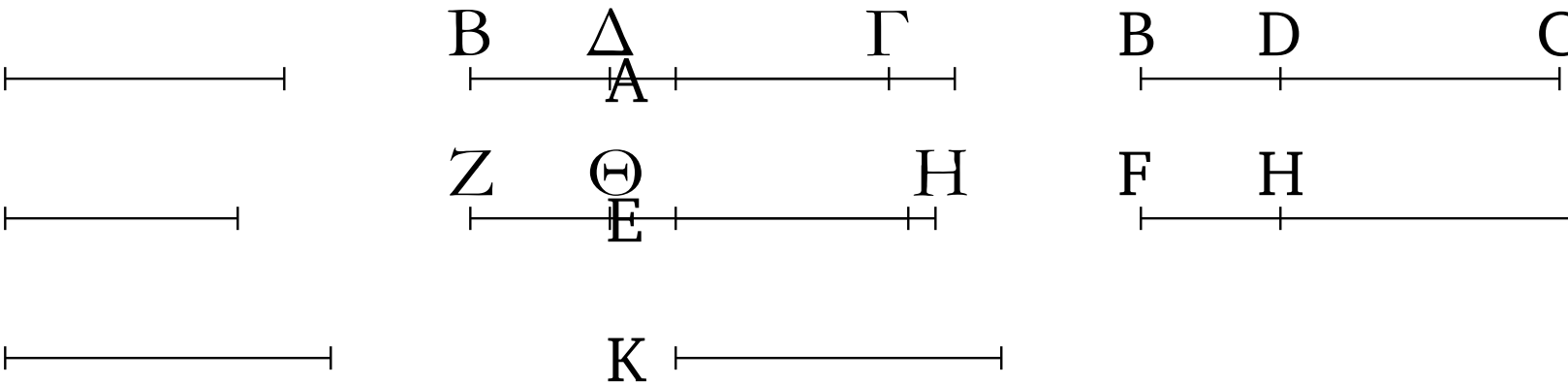
[†] See footnote to Prop. 10.49.

87.

Proposition 87

Εὐρεῖν τὴν τρίτην ἀποτομήν.

To find a third apotome.



Ἐκκείσθω ῥητὴ ἡ A , καὶ ἐκκείσθωσαν τρεῖς ἀριθμοὶ οἱ E , $BΓ$, $ΓΔ$ λόγον μὴ ἔχοντες πρὸς ἀλλήλους, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, ὃ δὲ $ΓΒ$ πρὸς τὸν $ΒΔ$ λόγον ἔθεται, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ πεποιήσθω ὡς μὲν ὁ E πρὸς τὸν $BΓ$, οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς ZH τετράγωνον, ὡς δὲ ὁ $BΓ$ πρὸς τὸν $ΓΔ$, οὕτως τὸ ἀπὸ τῆς ZH τετράγωνον πρὸς

Let the rational (straight-line) A be laid down. And let the three numbers, E , BC , and CD , not having to one another the ratio which (some) square number (has) to (some) square number, be laid down. And let CB have to BD the ratio which (some) square number (has) to (some) square number. And let it be contrived that as E (is) to BC , so the square on A (is) to the square on FG , and as BC (is) to CD , so the square on

τὸ ἀπὸ τῆς $H\Theta$. ἐπεὶ οὖν ἔστιν ὡς ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς ZH τετράγωνον, σύμμετρον ἄρα ἔστι τὸ ἀπὸ τῆς A τετράγωνον τῷ ἀπὸ τῆς ZH τετραγώνῳ. ῥητὸν δὲ τὸ ἀπὸ τῆς A τετράγωνον. ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς ZH . ῥητὴ ἄρα ἔστιν ἡ ZH . καὶ ἐπεὶ ὁ E πρὸς τὸν $B\Gamma$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ' ἄρα τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς ZH [τετράγωνον] λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἔστιν ἡ A τῇ ZH μήκει. πάλιν, ἐπεὶ ἔστιν ὡς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH τετράγωνον πρὸς τὸ ἀπὸ τῆς $H\Theta$, σύμμετρον ἄρα ἔστι τὸ ἀπὸ τῆς ZH τῷ ἀπὸ τῆς $H\Theta$. ῥητὸν δὲ τὸ ἀπὸ τῆς ZH . ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $H\Theta$. ῥητὴ ἄρα ἔστιν ἡ $H\Theta$. καὶ ἐπεὶ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ' ἄρα τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἔστιν ἡ ZH τῇ $H\Theta$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ZH , $H\Theta$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἔστιν ἡ $Z\Theta$. λέγω δὴ, ὅτι καὶ τρίτη.

Ἐπεὶ γάρ ἔστιν ὡς μὲν ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A τετράγωνον πρὸς τὸ ἀπὸ τῆς ZH , ὡς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς ΘH , δι' ἴσου ἄρα ἔστιν ὡς ὁ E πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ΘH . ὁ δὲ E πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. οὐδ' ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἡ A τῇ $H\Theta$ μήκει. οὐδετέρα ἄρα τῶν ZH , $H\Theta$ σύμμετρος ἔστι τῇ ἐκκειμένῃ ῥητῇ τῇ A μήκει. ὥ οὖν μείζον ἔστι τὸ ἀπὸ τῆς ZH τοῦ ἀπὸ τῆς $H\Theta$, ἔστω τὸ ἀπὸ τῆς K . ἐπεὶ οὖν ἔστιν ὡς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, ἀναστρέψαντι ἄρα ἔστιν ὡς ὁ $B\Gamma$ πρὸς τὸν $B\Delta$, οὕτως τὸ ἀπὸ τῆς ZH τετράγωνον πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ $B\Gamma$ πρὸς τὸν $B\Delta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. καὶ τὸ ἀπὸ τῆς ZH ἄρα πρὸς τὸ ἀπὸ τῆς K λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. σύμμετρος ἄρα ἔστιν ἡ ZH τῇ K μήκει, καὶ δύναται ἡ ZH τῆς $H\Theta$ μείζον τῷ ἀπὸ συμμέτρων ἑαυτῇ. καὶ οὐδετέρα τῶν ZH , $H\Theta$ σύμμετρος ἔστι τῇ ἐκκειμένῃ ῥητῇ τῇ A μήκει· ἡ $Z\Theta$ ἄρα ἀποτομή ἔστι τρίτη.

Ἐύρηται ἄρα ἡ τρίτη ἀποτομή ἡ $Z\Theta$. ὅπερ ἔδει δεῖχαι.

FG (is) to the (square) on GH [Prop. 10.6 corr.]. Therefore, since as E is to BC , so the square on A (is) to the square on FG , the square on A is thus commensurable with the square on FG [Prop. 10.6]. And the square on A (is) rational. Thus, the (square) on FG (is) also rational. Thus, FG is a rational (straight-line). And since E does not have to BC the ratio which (some) square number (has) to (some) square number, the square on A thus does not have to the [square] on FG the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with FG [Prop. 10.9]. Again, since as BC is to CD , so the square on FG is to the (square) on GH , the square on FG is thus commensurable with the (square) on GH [Prop. 10.6]. And the (square) on FG (is) rational. Thus, the (square) on GH (is) also rational. Thus, GH is a rational (straight-line). And since BC does not have to CD the ratio which (some) square number (has) to (some) square number, the (square) on FG thus does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. And both are rational (straight-lines). FG and GH are thus rational (straight-lines which are) commensurable in square only. Thus, FH is an apotome [Prop. 10.73]. So, I say that (it is) also a third (apotome).

For since as E is to BC , so the square on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on HG , thus, via equality, as E is to CD , so the (square) on A (is) to the (square) on HG [Prop. 5.22]. And E does not have to CD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on A does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. A (is) thus incommensurable in length with GH [Prop. 10.9]. Thus, neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Therefore, let the (square) on K be that (area) by which the (square) on FG is greater than the (square) on GH [Prop. 10.13 lem.]. Therefore, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , thus, via conversion, as BC is to BD , so the square on FG (is) to the square on K [Prop. 5.19 corr.]. And BC has to BD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on K the ratio which (some) square number (has) to (some) square number. FG is thus commensurable in length with K [Prop. 10.9]. And the square on FG is (thus) greater than (the square on) GH by the (square) on (some straight-line) commensurable (in length) with (FG). And neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, FH is a third apotome [Def. 10.13].

Thus, the third apotome FH has been found. (Which is)

very thing it was required to show.

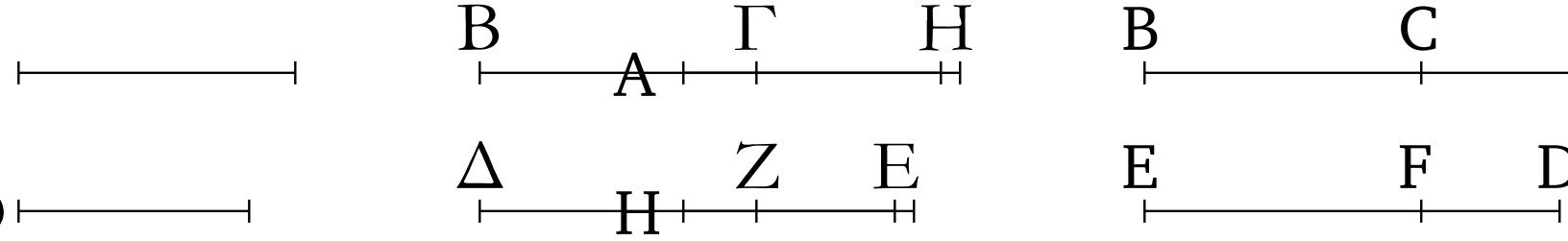
† See footnote to Prop. 10.50.

88.

Proposition 88

Εὐρεῖν τὴν τετάρτην ἀποτομήν.

To find a fourth apotome.



Ἐκκείσθω ῥητὴ ἡ A καὶ τῇ A μήκει σύμμετρος ἡ BH . ῥητὴ ἄρα ἐστὶ καὶ ἡ BH . καὶ ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΔZ , ZE , ὥστε τὸν ΔE ὅλον πρὸς ἑκάτερον τῶν ΔZ , EZ λόγον μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ πεποιήσθω ὥς ὁ ΔE πρὸς τὸν EZ , οὕτως τὸ ἀπὸ τῆς BH τετράγωνον πρὸς τὸ ἀπὸ τῆς $HΓ$. σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς BH τῷ ἀπὸ τῆς $HΓ$. ῥητὸν δὲ τὸ ἀπὸ τῆς BH . ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $HΓ$. ῥητὴ ἄρα ἐστὶν ἡ $HΓ$. καὶ ἐπεὶ ὁ ΔE πρὸς τὸν EZ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' ἄρα τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ BH τῇ $HΓ$ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ BH , $HΓ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἀποτομή ἄρα ἐστὶν ἡ $BΓ$. [λέγω δὴ, ὅτι καὶ τετάρτη.]

Ἴ οὖν μείζον ἐστὶ τὸ ἀπὸ τῆς BH τοῦ ἀπὸ τῆς $HΓ$, ἔστω τὸ ἀπὸ τῆς Θ . ἐπεὶ οὖν ἐστὶν ὥς ὁ ΔE πρὸς τὸν EZ , οὕτως τὸ ἀπὸ τῆς BH πρὸς τὸ ἀπὸ τῆς $HΓ$, καὶ ἀναστρέψαντι ἄρα ἐστὶν ὥς ὁ $E\Delta$ πρὸς τὸν ΔZ , οὕτως τὸ ἀπὸ τῆς HB πρὸς τὸ ἀπὸ τῆς Θ . ὁ δὲ $E\Delta$ πρὸς τὸν ΔZ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. οὐδ' ἄρα τὸ ἀπὸ τῆς HB πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ BH τῇ Θ μήκει. καὶ δύναται ἡ BH τῇ $HΓ$ μείζον τῷ ἀπὸ τῆς Θ . ἡ ἄρα BH τῇ $HΓ$ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἐαντῇ. καὶ ἐστὶν ὅλη ἡ BH σύμμετρος τῇ ἐκκεκμένη ῥητῇ μήκει τῇ A . ἡ ἄρα $BΓ$ ἀποτομή ἐστὶ τετάρτη.

Εὐρηται ἄρα ἡ τετάρτη ἀποτομή· ὅπερ ἔδει δεῖξαι.

Let the rational (straight-line) A , and BG (which is) commensurable in length with A , be laid down. Thus, BG is also a rational (straight-line). And let the two numbers DF and FE be laid down such that the whole, DE , does not have to each of DF and EF the ratio which (some) square number (has) to (some) square number. And let it be contrived that as DE (is) to EF , so the square on BG (is) to the (square) on GC [Prop. 10.6 corr.]. The (square) on BG is thus commensurable with the (square) on GC [Prop. 10.6]. And the (square) on BG (is) rational. Thus, the (square) on GC (is) also rational. Thus, GC (is) a rational (straight-line). And since DE does not have to EF the ratio which (some) square number (has) to (some) square number, the (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). Thus, BG and GC are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. [So, I say that (it is) also a fourth (apotome).]

Now, let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.]. Therefore, since as DE is to EF , so the (square) on BG (is) to the (square) on GC , thus, also, via conversion, as ED is to DF , so the (square) on GB (is) to the (square) on H [Prop. 5.19 corr.]. And ED does not have to DF the ratio which (some) square number (has) to (some) square number. Thus, the (square) on GB does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on BG is greater than (the square) on GC by the (square) on (some straight-line) incommensurable (in length) with (BG). And the whole, BG , is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is

a fourth apotome [Def. 10.14].[†]

Thus, a fourth apotome has been found. (Which is) the very thing it was required to show.

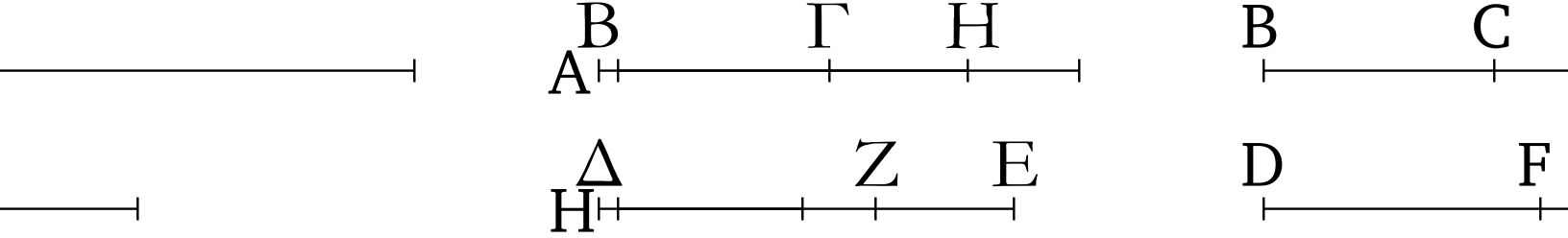
[†] See footnote to Prop. 10.51.

89.

Proposition 89

Εὕρεῖν τὴν πέμπτην ἀποτομήν.

To find a fifth apotome.



Ἐκκείσθω ῥητὴ ἡ A , καὶ τῇ A μήκει σύμμετρος ἔστω ἡ $ΓΗ$. ῥητὴ ἄρα [ἐστίν] ἡ $ΓΗ$. καὶ ἐκκείσθωσαν δύο ἀριθμοὶ οἱ $ΔΖ$, $ΖΕ$, ὥστε τὸν $ΔΕ$ πρὸς ἑκάτερον τῶν $ΔΖ$, $ΖΕ$ λόγον πάλιν μὴ ἔθειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ πεποιήσθω ὥς ὁ $ΖΕ$ πρὸς τὸν $ΕΔ$, οὕτως τὸ ἀπὸ τῆς $ΓΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΒ$. ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $ΗΒ$. ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΒΗ$. καὶ ἐπεὶ ἐστὶν ὥς ὁ $ΔΕ$ πρὸς τὸν $ΕΖ$, οὕτως τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΓ$, ὁ δὲ $ΔΕ$ πρὸς τὸν $ΕΖ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδ' ἄρα τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΓ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ $ΒΗ$ τῇ $ΗΓ$ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ $ΒΗ$, $ΗΓ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ $ΒΓ$ ἄρα ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ πέμπτη.

Ἦ γὰρ μείζων ἐστὶ τὸ ἀπὸ τῆς $ΒΗ$ τοῦ ἀπὸ τῆς $ΗΓ$, ἔστω τὸ ἀπὸ τῆς $Θ$. ἐπεὶ οὖν ἐστὶν ὥς τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $ΗΓ$, οὕτως ὁ $ΔΕ$ πρὸς τὸν $ΕΖ$, ἀναστρέψαντι ἄρα ἐστὶν ὥς ὁ $ΕΔ$ πρὸς τὸν $ΔΖ$, οὕτως τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $Θ$, ὁ δὲ $ΕΔ$ πρὸς τὸν $ΔΖ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδ' ἄρα τὸ ἀπὸ τῆς $ΒΗ$ πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ $ΒΗ$ τῇ $Θ$ μήκει. καὶ δύναται ἡ $ΒΗ$ τῆς $ΗΓ$ μείζον τῷ ἀπὸ τῆς $Θ$ · ἡ $ΗΒ$ ἄρα τῆς $ΗΓ$ μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἐαντὶ μήκει. καὶ ἐστὶν ἡ προσαρμόζουσα ἡ $ΓΗ$ σύμμετρος τῇ ἐκκεκλιμένη ῥητὶ τῇ A μήκει· ἡ ἄρα $ΒΓ$ ἀποτομή ἐστὶ πέμπτη.

Εὐρίηται ἄρα ἡ πέμπτη ἀποτομή ἡ $ΒΓ$. ὅπερ ἔδει δεῖχαι.

Let the rational (straight-line) A be laid down, and let CG be commensurable in length with A . Thus, CG [is] a rational (straight-line). And let the two numbers DF and FE be laid down such that DE again does not have to each of DF and FE the ratio which (some) square number (has) to (some) square number. And let it be contrived that as FE (is) to ED , so the (square) on CG (is) to the (square) on GB . Thus, the (square) on GB (is) also rational [Prop. 10.6]. Thus, BG is also rational. And since as DE is to EF , so the (square) on BG (is) to the (square) on GC . And DE does not have to EF the ratio which (some) square number (has) to (some) square number. The (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). BG and GC are thus rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that (it is) also a fifth (apotome).

For, let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.]. Therefore, since as the (square) on BG (is) to the (square) on GC , so DE (is) to EF , thus, via conversion, as ED is to DF , so the (square) on BG (is) to the (square) on H [Prop. 5.19 corr.]. And ED does not have to DF the ratio which (some) square number (has) to (some) square number. Thus, the (square) on BG does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H . Thus, the square on GB is greater than (the square on) GC by the (square) on (some straight-line) incommensurable in length with (GB). And the attachment CG is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, BC is a fifth

apotome [Def. 10.15].[†]

Thus, the fifth apotome BC has been found. (Which is) the very thing it was required to show.

[†] See footnote to Prop. 10.52.

90.

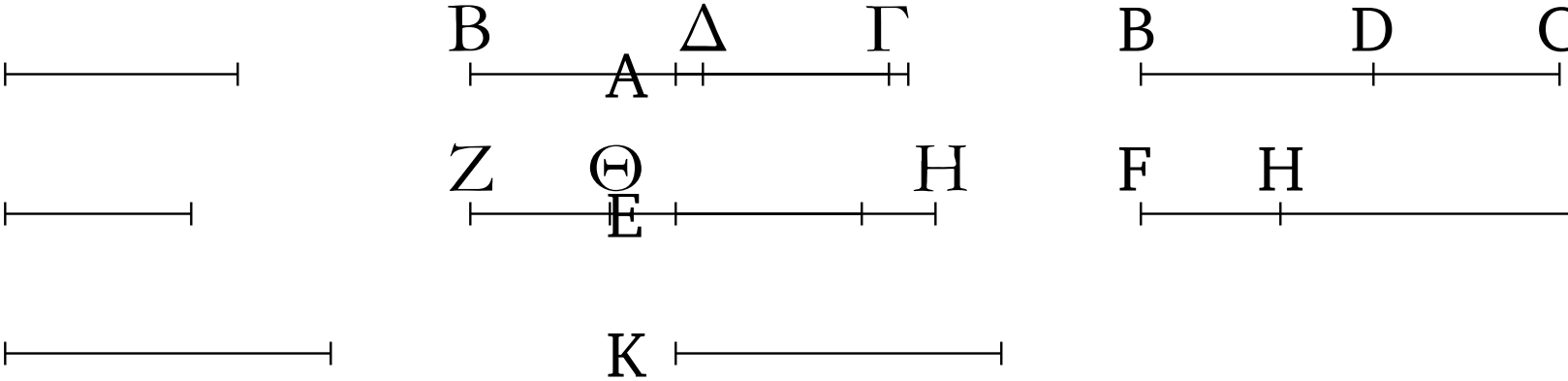
Εὐρεῖν τὴν ἕκτην ἀποτομήν.

Ἐκκείσθω ῥητὴ ἡ A καὶ τρεῖς ἀριθμοὶ οἱ E , $B\Gamma$, $\Gamma\Delta$ λόγον μὴ ἔχοντες πρὸς ἀλλήλους, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ἔτι δὲ καὶ ὁ ΓB πρὸς τὸν $B\Delta$ λόγον μὴ ἔθετώ, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν καὶ πεποιήσθω ὥς μὲν ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH , ὥς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$.

Proposition 90

To find a sixth apotome.

Let the rational (straight-line) A , and the three numbers E , BC , and CD , not having to one another the ratio which (some) square number (has) to (some) square number, be laid down. Furthermore, let CB also not have to BD the ratio which (some) square number (has) to (some) square number. And let it be contrived that as E (is) to BC , so the (square) on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.].



Ἐπεὶ οὖν ἔστιν ὥς ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH , σύμμετρον ἄρα τὸ ἀπὸ τῆς A τῷ ἀπὸ τῆς ZH . ῥητὸν δὲ τὸ ἀπὸ τῆς A ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς ZH . ῥητὴ ἄρα ἐστὶ καὶ ἡ ZH . καὶ ἐπεὶ ὁ E πρὸς τὸν $B\Gamma$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ' ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ ZH μήκει. πάλιν, ἐπεὶ ἔστιν ὥς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, σύμμετρον ἄρα τὸ ἀπὸ τῆς ZH τῷ ἀπὸ τῆς $H\Theta$. ῥητὸν δὲ τὸ ἀπὸ τῆς ZH . ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς $H\Theta$. ῥητὴ ἄρα καὶ ἡ $H\Theta$. καὶ ἐπεὶ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν, οὐδ' ἄρα τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν. ἀσύμμετρος ἄρα ἐστὶν ἡ ZH τῇ $H\Theta$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ZH , $H\Theta$ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ἄρα $Z\Theta$ ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ ἕκτη.

Ἐπεὶ γάρ ἐστιν ὥς μὲν ὁ E πρὸς τὸν $B\Gamma$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς ZH , ὥς δὲ ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, δι' ἴσον ἄρα ἐστὶν ὥς ὁ E πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $H\Theta$. ὁ δὲ E πρὸς τὸν $\Gamma\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος

Therefore, since as E is to BC , so the (square) on A (is) to the (square) on FG , the (square) on A (is) thus commensurable with the (square) on FG [Prop. 10.6]. And the (square) on A (is) rational. Thus, the (square) on FG (is) also rational. Thus, FG is also a rational (straight-line). And since E does not have to BC the ratio which (some) square number (has) to (some) square number, the (square) on A thus does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with FG [Prop. 10.9]. Again, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , the (square) on FG (is) thus commensurable with the (square) on GH [Prop. 10.6]. And the (square) on FG (is) rational. Thus, the (square) on GH (is) also rational. Thus, GH (is) also rational. And since BC does not have to CD the ratio which (some) square number (has) to (some) square number, the (square) on FG thus does not have to the (square) on GH the ratio which (some) square (number) has to (some) square (number) either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. And both are rational (straight-lines). Thus, FG and GH are rational (straight-lines which are) commensurable in square only. Thus, FH is an apotome [Prop. 10.73]. So, I say that (it is) also a sixth (apotome).

ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν οὐδ' ἄρα τὸ ἀπὸ τῆς A πρὸς τὸ ἀπὸ τῆς $H\Theta$ λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ἀσύμμετρος ἄρα ἐστὶν ἡ A τῇ $H\Theta$ μήκει· οὐδετέρα ἄρα τῶν ZH , $H\Theta$ σύμμετρος ἐστὶ τῇ A ῥητῇ μήκει. ὥ οὖν μείζον ἐστὶ τὸ ἀπὸ τῆς ZH τοῦ ἀπὸ τῆς $H\Theta$, ἔστω τὸ ἀπὸ τῆς K . ἐπεὶ οὖν ἐστὶν ὡς ὁ $B\Gamma$ πρὸς τὸν $\Gamma\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $H\Theta$, ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ ΓB πρὸς τὸν $B\Delta$, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς K . ὁ δὲ ΓB πρὸς τὸν $B\Delta$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν οὐδ' ἄρα τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς K λόγον ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμὸν ἀσύμμετρος ἄρα ἐστὶν ἡ ZH τῇ K μήκει. καὶ δύνανται ἡ ZH τῆς $H\Theta$ μείζον τῷ ἀπὸ τῆς K · ἡ ZH ἄρα τῆς $H\Theta$ μείζον δύνανται τῷ ἀπὸ ἀσυνμέτρου ἐαυτῇ μήκει. καὶ οὐδετέρα τῶν ZH , $H\Theta$ σύμμετρος ἐστὶ τῇ $\epsilon\kappa\kappa\epsilon\iota\mu\acute{\epsilon}\nu\eta$ ῥητῇ μήκει τῇ A . ἡ ἄρα $Z\Theta$ ἀποτομή ἐστὶν ἑκτῇ.

E ῥηται ἄρα ἡ ἑκτῇ ἀποτομή ἡ $Z\Theta$. ὅπερ ἔδει δεῖξαι.

For since as E is to BC , so the (square) on A (is) to the (square) on FG , and as BC (is) to CD , so the (square) on FG (is) to the (square) on GH , thus, via equality, as E is to CD , so the (square) on A (is) to the (square) on GH [Prop. 5.22]. And E does not have to CD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on A does not have to the (square) GH the ratio which (some) square number (has) to (some) square number either. A is thus incommensurable in length with GH [Prop. 10.9]. Thus, neither of FG and GH is commensurable in length with the rational (straight-line) A . Therefore, let the (square) on K be that (area) by which the (square) on FG is greater than the (square) on GH [Prop. 10.13 lem.]. Therefore, since as BC is to CD , so the (square) on FG (is) to the (square) on GH , thus, via conversion, as CB is to BD , so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And CB does not have to BD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG does not have to the (square) on K the ratio which (some) square number (has) to (some) square number either. FG is thus incommensurable in length with K [Prop. 10.9]. And the square on FG is greater than (the square on) GH by the (square) on K . Thus, the square on FG is greater than (the square on) GH by the (square) on (some straight-line) incommensurable in length with (FG). And neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A . Thus, FH is a sixth apotome [Def. 10.16].

Thus, the sixth apotome FH has been found. (Which is) the very thing it was required to show.

† See footnote to Prop. 10.53.

91.

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς πρώτης, ἡ τὸ θωρίον δυναμένη ἀπορομή ἐστὶν.

Περιεθέσθω γὰρ θωρίον τὸ AB ὑπὸ ῥητῆς τῆς AG καὶ ἀποτομῆς πρώτης τῆς AD . λέγω, ὅτι ἡ τὸ AB θωρίον δυναμένη ἀποτομή ἐστὶν.

Ἐπεὶ γὰρ ἀποτομή ἐστὶ πρώτη ἡ AD , ἔστω αὐτῇ προσαρμόζουσα ἡ ΔH . αἱ AH , $H\Delta$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ ὅλη ἡ AH σύμμετρος ἐστὶ τῇ ἑκκειμένῃ ῥητῇ τῇ AG , καὶ ἡ AH τῆς $H\Delta$ μείζον δύνανται τῷ ἀπὸ συμέτρου ἐαυτῇ μήκει· ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔH ἴσον παρὰ τὴν AH παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, εἷς σύμμετρα αὐτῇ διαιρεῖ. τετμήσθω ἡ ΔH δίῃ κατὰ τὸ E , καὶ τῷ ἀπὸ τῆς EH ἴσον παρὰ τὴν AH παραβεβλήσθω ἑλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν AZ , ZH · σύμμετρος ἄρα ἐστὶν ἡ AZ τῇ ZH . καὶ διὰ τῶν E , Z , H σημείων τῇ AG παράλληλοι ἡθῇσαν αἱ $E\Theta$, ZI , HK .

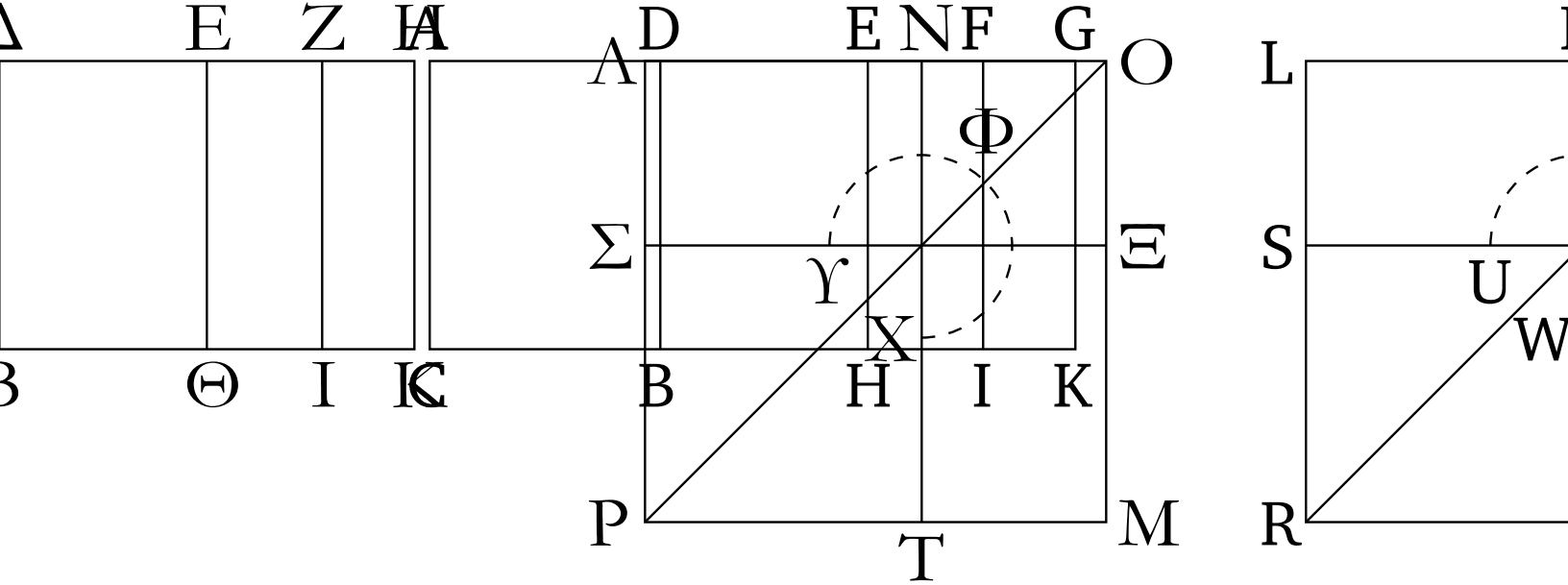
Proposition 91

If an area is contained by a rational (straight-line) and a first apotome then the square-root of the area is an apotome.

For let the area AB be contained by the rational (straight-line) AC and the first apotome AD . I say that the square-root of area AB is an apotome.

For since AD is a first apotome, let DG be its attachment. Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And the whole, AG , is commensurable (in length) with the (previously) laid down rational (straight-line) AC , and the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable in length with (AG) [Def. 10.11]. Thus, if (an area) equal to the fourth part of the (square) on DG is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short

by a square figure. And let it be the (rectangle contained) by AF and FG . AF is thus commensurable (in length) with FG . And let EH , FI , and GK be drawn through points E , F , and G (respectively), parallel to AC .



Καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ AZ τῇ ZH μήκει, καὶ ἡ AH ἄρα ἐκατέρω τῶν AZ , ZH σύμμετρος ἐστὶ μήκει. ἀλλὰ ἡ AH σύμμετρος ἐστὶ τῇ $ΑΓ$ · καὶ ἐκατέρα ἄρα τῶν AZ , ZH σύμμετρος ἐστὶ τῇ $ΑΓ$ μήκει. καὶ ἐστὶ ῥητὴ ἡ $ΑΓ$ · ῥητὴ ἄρα καὶ ἐκατέρα τῶν AZ , ZH · ὥστε καὶ ἐκάτερον τῶν $ΑΙ$, $ΖΚ$ ῥητόν ἐστιν. καὶ ἐπεὶ σύμμετρος ἐστὶν ἡ $ΔΕ$ τῇ $ΕΗ$ μήκει, καὶ ἡ $ΔΗ$ ἄρα ἐκατέρω τῶν $ΔΕ$, $ΕΗ$ σύμμετρος ἐστὶ μήκει. ῥητὴ δὲ ἡ $ΔΗ$ καὶ ἀσύμμετρος τῇ $ΑΓ$ μήκει· ῥητὴ ἄρα καὶ ἐκατέρα τῶν $ΔΕ$, $ΕΗ$ καὶ ἀσύμμετρος τῇ $ΑΓ$ μήκει· ἐκάτερον ἄρα τῶν $ΔΘ$, $ΕΚ$ μέσον ἐστίν.

Κεῖσθω δὴ τῷ μὲν $ΑΙ$ ἴσον τετράγωνον τὸ $ΑΜ$, τῷ δὲ $ΖΚ$ ἴσον τετράγωνον ἀφηρήσθω κοινὴν γωνίαν ἔθον αὐτῶ τὴν ὑπὸ $ΛΟΜ$ τὸ $ΝΧ$ · περὶ τὴν αὐτὴν ἄρα διάμετρον ἐστὶ τὰ $ΑΜ$, $ΝΧ$ τετράγωνα. ἔστω αὐτῶν διάμετρος ἡ $ΟΡ$, καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ οὖν ἴσον ἐστὶ τὸ ὑπὸ τῶν AZ , ZH περιεχόμενον ὀρθογώνιον τῷ ἀπὸ τῆς $ΕΗ$ τετραγώνῳ, ἔστιν ἄρα ὥς ἡ AZ πρὸς τὴν $ΕΗ$, οὕτως ἡ $ΕΗ$ πρὸς τὴν ZH . ἀλλ' ὥς μὲν ἡ AZ πρὸς τὴν $ΕΗ$, οὕτως τὸ $ΑΙ$ πρὸς τὸ $ΕΚ$, ὥς δὲ ἡ $ΕΗ$ πρὸς τὴν ZH , οὕτως ἐστὶ τὸ $ΕΚ$ πρὸς τὸ $ΚΖ$ · τῶν ἄρα $ΑΙ$, $ΚΖ$ μέσον ἀνάλογόν ἐστι τὸ $ΕΚ$. ἔστι δὲ καὶ τῶν $ΑΜ$, $ΝΧ$ μέσον ἀνάλογον τὸ $ΜΝ$, ὥς ἐν τοῖς ἔμπροσθεν ἐδείχθη, καὶ ἐστὶ τὸ [μὲν] $ΑΙ$ τῷ $ΑΜ$ τετραγώνῳ ἴσον, τὸ δὲ $ΚΖ$ τῷ $ΝΧ$ · καὶ τὸ $ΜΝ$ ἄρα τῷ $ΕΚ$ ἴσον ἐστίν. ἀλλὰ τὸ μὲν $ΕΚ$ τῷ $ΔΘ$ ἐστὶν ἴσον, τὸ δὲ $ΜΝ$ τῷ $ΑΧ$ · τὸ ἄρα $ΔΚ$ ἴσον ἐστὶ τῷ $ΥΦΘ$ γνώμονι καὶ τῷ $ΝΧ$. ἔστι δὲ καὶ τὸ $ΑΚ$ ἴσον τοῖς $ΑΜ$, $ΝΧ$ τετραγώνοις· λοιπὸν ἄρα τὸ $ΑΒ$ ἴσον ἐστὶ τῷ $ΣΤ$. τὸ δὲ $ΣΤ$ τὸ ἀπὸ τῆς $ΑΝ$ ἐστὶ τετράγωνον· τὸ ἄρα ἀπὸ

And since AF is commensurable in length with FG , AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. But AG is commensurable (in length) with AC . Thus, each of AF and FG is also commensurable in length with AC [Prop. 10.12]. And AC is a rational (straight-line). Thus, AF and FG (are) each also rational (straight-lines). Hence, AI and FK are also each rational (areas) [Prop. 10.19]. And since DE is commensurable in length with EG , DG is thus also commensurable in length with each of DE and EG [Prop. 10.15]. And DG (is) rational, and incommensurable in length with AC . DE and EG (are) thus each rational, and incommensurable in length with AC [Prop. 10.13]. Thus, DH and EK are each medial (areas) [Prop. 10.21].

So let the square LM , equal to AI , be laid down. And let the square NO , equal to FK , be subtracted (from LM), having with it the common angle LPM . Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the rectangle contained by AF and FG is equal to the square EG , thus as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG , so AI (is) to EK , and as EG (is) to FG , so EK is to KF [Prop. 6.1]. Thus, EK is the mean proportional to AI and KF [Prop. 5.11]. And MN is also the mean proportional to LM and NO , as shown before [Prop. 10.53 lem.]. And AI is equal to the square LM , and KF to NO . Thus, MN is also equal to EK . But, EK is equal to DH , and MN to LO [Prop. 1.43]. Thus, DK is equal to the

τῆς AN τετραγώνων ἴσον ἐστὶ τῷ AB . ἡ AN ἄρα δύναται τὸ AB . λέγω δὴ, ὅτι ἡ AN ἀποτομή ἐστίν.

Ἐπεὶ γὰρ ῥητόν ἐστιν ἑκάτερον τῶν AI , ZK , καὶ ἐστὶν ἴσον τοῖς AM , NX , καὶ ἑκάτερον ἄρα τῶν AM , NX ῥητόν ἐστιν, τουτέστι τὸ ἀπὸ ἑκατέρωξ τῶν AO , ON · καὶ ἑκατέρα ἄρα τῶν AO , ON ῥητὴ ἐστίν. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ $\Delta\Theta$ καὶ ἐστὶν ἴσον τῷ AX , μέσον ἄρα ἐστὶ καὶ τὸ AX . ἐπεὶ οὖν τὸ μὲν AX μέσον ἐστίν, τὸ δὲ NX ῥητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ AX τῷ NX . ὥς δὲ τὸ AX πρὸς τὸ NX , οὕτως ἐστὶν ἡ AO πρὸς τὴν ON . ἀσύμμετρος ἄρα ἐστὶν ἡ AO τῇ ON μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ AO , ON ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ AN . καὶ δύναται τὸ AB θωρίον ἡ ἄρα τὸ AB θωρίον δυνάμενη ἀποτομή ἐστίν.

Ἐὰν ἄρα θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ τὰ ἐχθῆς.

gnomon UVW and NO . And AK is also equal to (the sum of) the squares LM and NO . Thus, the remainder AB is equal to ST . And ST is the square on LN . Thus, the square on LN is equal to AB . Thus, LN is the square-root of AB . So, I say that LN is an apotome.

For since AI and FK are each rational (areas), and are equal to LM and NO (respectively), thus LM and NO —that is to say, the (squares) on each of LP and PN (respectively)—are also each rational (areas). Thus, LP and PN are also each rational (straight-lines). Again, since DH is a medial (area), and is equal to LO , LO is thus also a medial (area). Therefore, since LO is medial, and NO rational, LO is thus incommensurable with NO . And as LO (is) to NO , so LP is to PN [Prop. 6.1]. LP is thus incommensurable in length with PN [Prop. 10.11]. And they are both rational (straight-lines). Thus, LP and PN are rational (straight-lines which are) commensurable in square only. Thus, LN is an apotome [Prop. 10.73]. And it is the square-root of area AB . Thus, the square-root of area AB is an apotome.

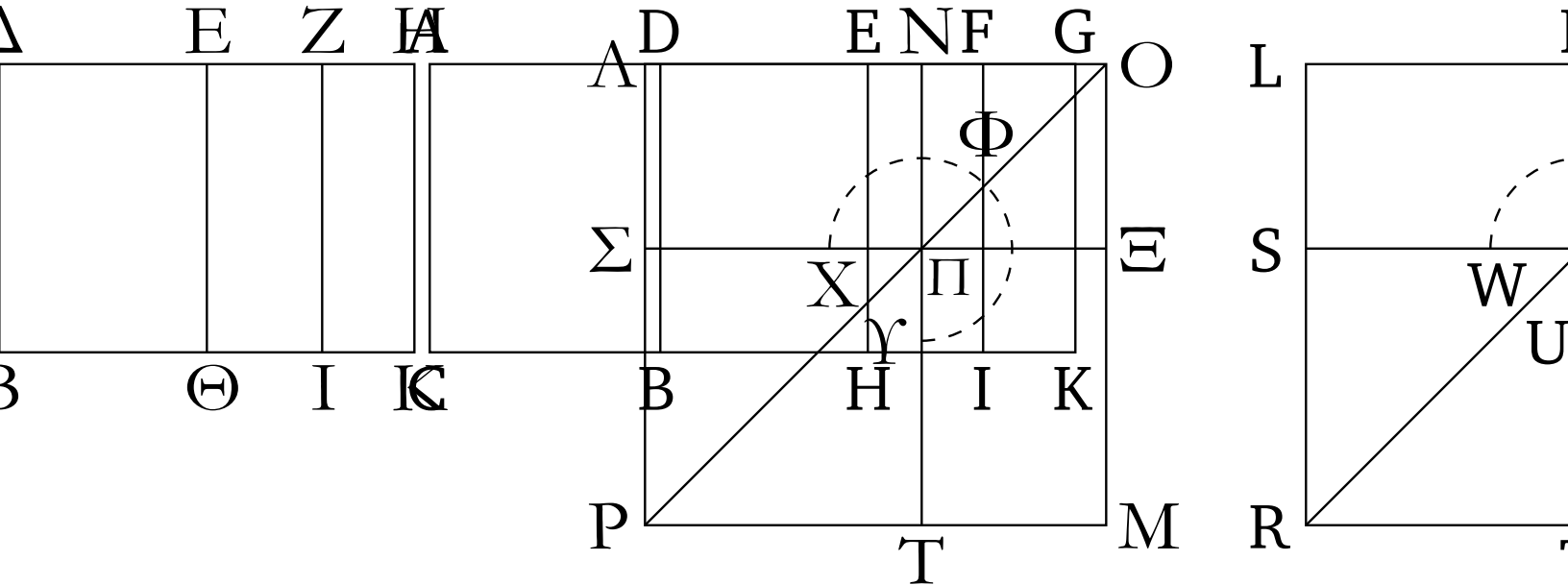
Thus, if an area is contained by a rational (straight-line), and so on . . .

92.

Proposition 92

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς δευτέρας, ἡ τὸ θωρίον δυνάμενη μέση ἀποτομή ἐστὶ πρώτη.

If an area is contained by a rational (straight-line) and a second apotome then the square-root of the area is a first apotome of a medial (straight-line).



Θωρίον γὰρ τὸ AB περιέθῃται ὑπὸ ῥητῆς τῆς AG καὶ ἀποτομῆς δευτέρας τῆς AD . λέγω, ὅτι ἡ τὸ AB θωρίον δυνάμενη μέση ἀποτομή ἐστὶ πρώτη.

Ἐστω γὰρ τῇ AD προσαρμόζονσα ἡ ΔH . αἱ ἄρα AH , $H\Delta$ ῥηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ προσαρμόζονσα

For let the area AB be contained by the rational (straight-line) AC and the second apotome AD . I say that the square-root of area AB is the first apotome of a medial (straight-line).

For let DG be an attachment to AD . Thus, AG and GD are rational (straight-lines which are) commensurable in

ἡ ΔH σύμμετρος ἐστὶ τῇ ἐκκεκμένη ῥητῇ τῇ AG , ἡ δὲ ὅλη ἡ AH τῇ προσαρμοζούσῃ τῇ $H\Delta$ μείζον δύναιται τῷ ἀπὸ συμμετρὸν ἑαυτῇ μήκει. ἐπεὶ οὖν ἡ AH τῇ $H\Delta$ μείζον δύναιται τῷ ἀπὸ συμμετρὸν ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῇ $H\Delta$ ἴσον παρὰ τὴν AH παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, εἷς σύμμετρος αὐτὴν διαιρεῖ. τετμήσθω οὖν ἡ ΔH διὰ κατὰ τὸ E . καὶ τῷ ἀπὸ τῇ EH ἴσον παρὰ τὴν AH παραβληθῇ ἑλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν AZ , ZH . σύμμετρος ἄρα ἐστὶν ἡ AZ τῇ ZH μήκει. καὶ ἡ AH ἄρα ἑκατέρᾳ τῶν AZ , ZH σύμμετρος ἐστὶ μήκει. ῥητὴ δὲ ἡ AH καὶ ἀσύμμετρος τῇ AG μήκει· καὶ ἑκατέρᾳ ἄρα τῶν AZ , ZH ῥητὴ ἐστὶ καὶ ἀσύμμετρος τῇ AG μήκει· ἑκάτερον ἄρα τῶν AI , ZK μέσον ἐστὶν. πάλιν, ἐπεὶ σύμμετρος ἐστὶν ἡ ΔE τῇ EH , καὶ ἡ ΔH ἄρα ἑκατέρᾳ τῶν ΔE , EH σύμμετρος ἐστὶν. ἀλλ' ἡ ΔH σύμμετρος ἐστὶ τῇ AG μήκει [ῥητὴ ἄρα καὶ ἑκατέρᾳ τῶν ΔE , EH καὶ σύμμετρος τῇ AG μήκει]. ἑκάτερον ἄρα τῶν $\Delta\Theta$, EK ῥητὸν ἐστὶν.

Συνεστάτω οὖν τῷ μὲν AI ἴσον τετραγώνον τὸ AM , τῷ δὲ ZK ἴσον ἀφῆρησθω τὸ NX περὶ τὴν αὐτὴν γωνίαν ὅν τῷ AM τὴν ὑπὸ τῶν $ΛΟΜ$. περὶ τὴν αὐτὴν ἄρα ἐστὶ διάμετρον τὰ AM , NX τετραγώνων. ἔστω αὐτῶν διάμετρος ἡ OP , καὶ καταγεγραμθῇ τὸ σθῆμα. ἐπεὶ οὖν τὰ AI , ZK μέσα ἐστὶ καὶ ἐστὶν ἴσα τοῖς ἀπὸ τῶν $ΛΟ$, $ΟΝ$, καὶ τὰ ἀπὸ τῶν $ΛΟ$, $ΟΝ$ [ἄρα] μέσα ἐστὶν· καὶ αἱ $ΛΟ$, $ΟΝ$ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ τὸ ὑπὸ τῶν AZ , ZH ἴσον ἐστὶ τῷ ἀπὸ τῇ EH , ἔστιν ἄρα ὡς ἡ AZ πρὸς τὴν EH , οὕτως ἡ EH πρὸς τὴν ZH . ἀλλ' ὡς μὲν ἡ AZ πρὸς τὴν EH , οὕτως [ἐστὶ] τὸ AI πρὸς τὸ EK . ὡς δὲ ἡ EH πρὸς τὴν ZH , οὕτως [ἐστὶ] τὸ EK πρὸς τὸ ZK . τῶν ἄρα AI , ZK μέσον ἀνάλογόν ἐστι τὸ EK . ἔστι δὲ καὶ τῶν AM , NX τετραγώνων μέσον ἀνάλογον τὸ MN . καὶ ἐστὶν ἴσον τὸ μὲν AI τῷ AM , τὸ δὲ ZK τῷ NX . καὶ τὸ MN ἄρα ἴσον ἐστὶ τῷ EK . ἀλλὰ τῷ μὲν EK ἴσον [ἐστὶ] τὸ $\Delta\Theta$, τῷ δὲ MN ἴσον τὸ AX . ὅλον ἄρα τὸ ΔK ἴσον ἐστὶ τῷ $Y\Phi\Theta$ γνώμονι καὶ τῷ NX . ἐπεὶ οὖν ὅλον τὸ AK ἴσον ἐστὶ τοῖς AM , NX , ὣν τὸ ΔK ἴσον ἐστὶ τῷ $Y\Phi\Theta$ γνώμονι καὶ τῷ NX , λοιπὸν ἄρα τὸ AB ἴσον ἐστὶ τῷ $T\Sigma$. τὸ δὲ $T\Sigma$ ἐστὶ τὸ ἀπὸ τῇ AN . τὸ ἀπὸ τῇ AN ἄρα ἴσον ἐστὶ τῷ AB θωρίῳ· ἡ AN ἄρα δύναιται τὸ AB θωρίον. λέγω [δή], ὅτι ἡ AN μέσης ἀποτομῆς ἐστὶ πρώτη.

Ἐπεὶ γὰρ ῥητὸν ἐστὶ τὸ EK καὶ ἐστὶν ἴσον τῷ AX , ῥητὸν ἄρα ἐστὶ τὸ AX , τουτέστι τὸ ὑπὸ τῶν $ΛΟ$, $ΟΝ$. μέσον δὲ εἰδείθῃ τὸ NX . ἀσύμμετρον ἄρα ἐστὶ τὸ AX τῷ NX . ὡς δὲ τὸ AX πρὸς τὸ NX , οὕτως ἐστὶν ἡ $ΛΟ$ πρὸς $ΟΝ$. αἱ $ΛΟ$, $ΟΝ$ ἄρα ἀσύμμετροί εἰσι μήκει. αἱ ἄρα $ΛΟ$, $ΟΝ$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι· ἡ AN ἄρα μέσης ἀποτομῆς ἐστὶ πρώτη καὶ δύναιται τὸ AB θωρίον.

Ἡ ἄρα τὸ AB θωρίον δυναμένη μέσης ἀποτομῆς ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

square only [Prop. 10.73], and the attachment DG is commensurable (in length) with the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, GD , by the (square) on (some straight-line) commensurable in length with (AG) [Def. 10.12]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable (in length) with (AG), thus if (an area) equal to the fourth part of the (square) on GD is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Therefore, let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short by a square figure. And let it be the (rectangle contained) by AF and FG . Thus, AF is commensurable in length with FG . AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. And AG (is) a rational (straight-line), and incommensurable in length with AC . AF and FG are thus also each rational (straight-lines), and incommensurable in length with AC [Prop. 10.13]. Thus, AI and FK are each medial (areas) [Prop. 10.21]. Again, since DE is commensurable (in length) with EG , thus DG is also commensurable (in length) with each of DE and EG [Prop. 10.15]. But, DG is commensurable in length with AC [thus, DE and EG are also each rational, and commensurable in length with AC]. Thus, DH and EK are each rational (areas) [Prop. 10.19].

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , which is about the same angle LPM as LM , be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since AI and FK are medial (areas), and are equal to the (squares) on LP and PN (respectively), [thus] the (squares) on LP and PN are also medial. Thus, LP and PN are also medial (straight-lines which are) commensurable in square only.[†] And since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus as AF is to EG , so EG (is) to FG [Prop. 10.17]. But, as AF (is) to EG , so AI (is) to EK . And as EG (is) to FG , so EK [is] to FK [Prop. 6.1]. Thus, EK is the mean proportional to AI and FK [Prop. 5.11]. And MN is also the mean proportional to the squares LM and NO [Prop. 10.53 lem.]. And AI is equal to LM , and FK to NO . Thus, MN is also equal to EK . But, DH [is] equal to EK , and LO equal to MN [Prop. 1.43]. Thus, the whole (of) DK is equal to the gnomon UVW and NO . Therefore, since the whole (of) AK is equal to LM and NO , of which DK is equal to the gnomon UVW and NO , the remainder AB is thus equal to TS . And TS is the (square) on LN . Thus, the (square) on LN is equal to the area AB . LN is thus the square-root of area AB . [So], I say that LN is the first apotome of a medial (straight-line).

For since EK is a rational (area), and is equal to LO , LO —that is to say, the (rectangle contained) by LP and PN —is thus a rational (area). And NO was shown (to be) a medial (area). Thus, LO is incommensurable with NO . And as LO (is) to NO , so LP is to PN [Prop. 6.1]. Thus, LP and PN are incommensurable in length [Prop. 10.11]. LP and PN are thus medial (straight-lines which are) commensurable in square only, and which contain a rational (area). Thus, LN is the first apotome of a medial (straight-line) [Prop. 10.74]. And it is the square-root of area AB .

Thus, the square root of area AB is the first apotome of a medial (straight-line). (Which is) the very thing it was required to show.

† There is an error in the argument here. It should just say that LP and PN are commensurable in square, rather than in square only, since LP and PN are only shown to be incommensurable in length later on.

93.

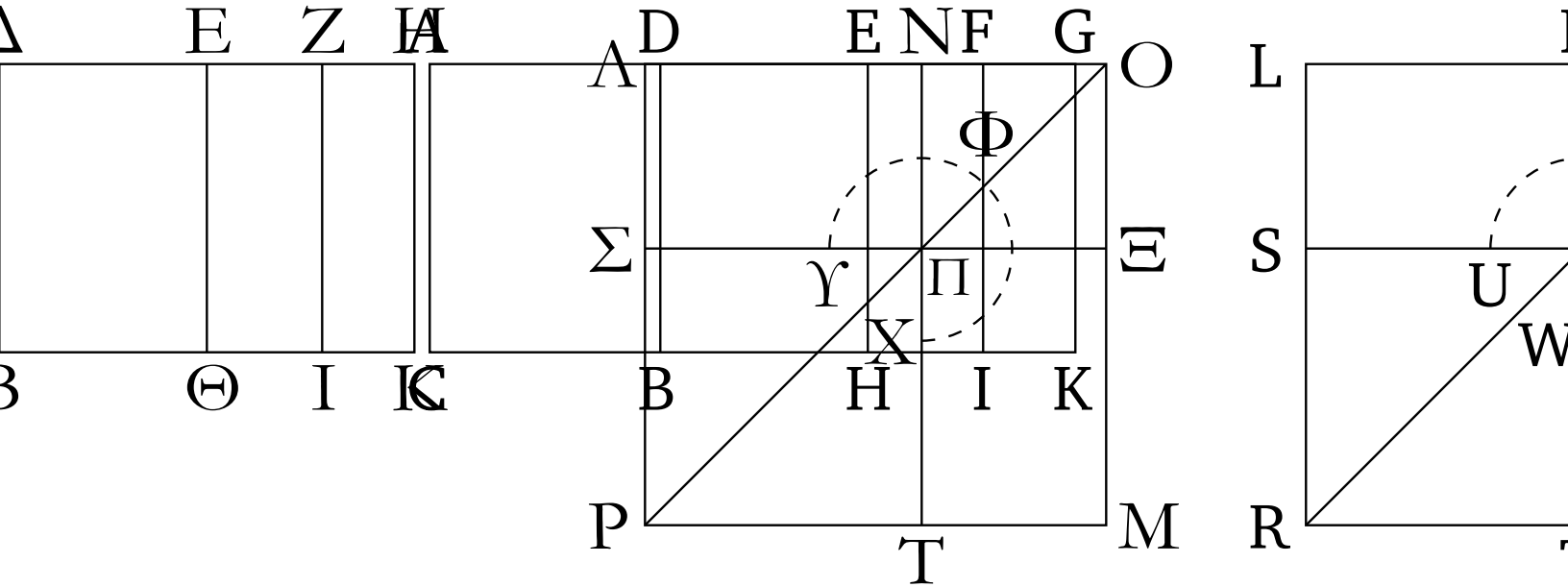
Ἐάν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς τρίτης, ἢ τὸ θωρίον δυνάμει μέσης ἀποτομῆς ἐστὶ δευτέρα.

Θωρίον γάρ τὸ AB περιέθῃται ὑπὸ ῥητῆς τῆς AG καὶ ἀποτομῆς τρίτης τῆς AD . λέγω, ὅτι ἡ τὸ AB θωρίον δυνάμει μέσης ἀποτομῆς ἐστὶ δευτέρα.

Proposition 93

If an area is contained by a rational (straight-line) and a third apotome then the square-root of the area is a second apotome of a medial (straight-line).

For let the area AB be contained by the rational (straight-line) AC and the third apotome AD . I say that the square-root of area AB is the second apotome of a medial (straight-line).



Ἐστω γάρ τῇ AD προσαρμόζονσα ἡ ΔH . αἱ AH , $H\Delta$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα τῶν AH , $H\Delta$ σύμμετρος ἐστὶ μήκει τῇ ἐκκεκμένη ῥητῇ τῇ AG , ἢ δὲ ὅλη ἡ AH τῆς προσαρμοζούσης τῆς ΔH μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. ἐπεὶ οὖν ἡ AH τῆς $H\Delta$ μείζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔH ἴσον παρὰ τὴν AH παραβληθῇ ἐλλείπον

For let DG be an attachment to AD . Thus, AG and GD are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and neither of AG and GD is commensurable in length with the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, DG , by the (square) on (some straight-line) commensurable (in length) with (AG)

εἶδει τετραγώνω, εἰς σύμμετρα αὐτὴν διελεί. τετμήσθω οὖν ἡ ΔΗ διὰ κατὰ τὸ Ε, καὶ τῷ ἀπὸ τῆς ΕΗ ἴσον παρὰ τὴν ΑΗ παραβεβλήσθω ἐλλείπον εἶδει τετραγώνω, καὶ ἔστω τὸ ὑπὸ τῶν ΑΖ, ΖΗ. καὶ ἡθῆσαν διὰ τῶν Ε, Ζ, Η σημείων τῇ ΑΓ παράλληλοι αἱ ΕΘ, ΖΙ, ΗΚ· σύμμετροι ἄρα εἰσὶν αἱ ΑΖ, ΖΗ· σύμμετρον ἄρα καὶ τὸ ΑΙ τῷ ΖΚ. καὶ ἐπεὶ αἱ ΑΖ, ΖΗ σύμμετροί εἰσι μήκει, καὶ ἡ ΑΗ ἄρα ἐκατέρω τῶν ΑΖ, ΖΗ σύμμετρος ἐστὶ μήκει. ῥητὴ δὲ ἡ ΑΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ὥστε καὶ αἱ ΑΖ, ΖΗ· ἐκότερον ἄρα τῶν ΑΙ, ΖΚ μέσον ἐστίν. πάλιν, ἐπεὶ σύμμετρος ἐστὶν ἡ ΔΕ τῇ ΕΗ μήκει, καὶ ἡ ΔΗ ἄρα ἐκατέρω τῶν ΔΕ, ΕΗ σύμμετρος ἐστὶ μήκει. ῥητὴ δὲ ἡ ΗΔ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ῥητὴ ἄρα καὶ ἐκατέρω τῶν ΔΕ, ΕΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ἐκότερον ἄρα τῶν ΔΘ, ΕΚ μέσον ἐστίν. καὶ ἐπεὶ αἱ ΑΗ, ΗΔ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶ μήκει ἡ ΑΗ τῇ ΗΔ. ἀλλ' ἡ μὲν ΑΗ τῇ ΑΖ σύμμετρος ἐστὶ μήκει ἡ δὲ ΔΗ τῇ ΕΗ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΖ τῇ ΕΗ μήκει. ὥς δὲ ἡ ΑΖ πρὸς τὴν ΕΗ, οὕτως ἐστὶ τὸ ΑΙ πρὸς τὸ ΕΚ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΑΙ τῷ ΕΚ.

Συνεστάτω οὖν τῷ μὲν ΑΙ ἴσον τετραγώνον τὸ ΑΜ, τῷ δὲ ΖΚ ἴσον ἀφῆρσθω τὸ ΝΧ περὶ τὴν αὐτὴν γωνίαν ὅν τῷ ΑΜ· περὶ τὴν αὐτὴν ἄρα διάμετρον ἐστὶ τὰ ΑΜ, ΝΧ. ἔστω αὐτῶν διάμετρος ἡ ΟΡ, καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ οὖν τὸ ὑπὸ τῶν ΑΖ, ΖΗ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΗ, ἔστιν ἄρα ὡς ἡ ΑΖ πρὸς τὴν ΕΗ, οὕτως ἡ ΕΗ πρὸς τὴν ΖΗ. ἀλλ' ὡς μὲν ἡ ΑΖ πρὸς τὴν ΕΗ, οὕτως ἐστὶ τὸ ΑΙ πρὸς τὸ ΕΚ· ὥς δὲ ἡ ΕΗ πρὸς τὴν ΖΗ, οὕτως ἐστὶ τὸ ΕΚ πρὸς τὸ ΖΚ· καὶ ὥς ἄρα τὸ ΑΙ πρὸς τὸ ΕΚ, οὕτως τὸ ΕΚ πρὸς τὸ ΖΚ· τῶν ἄρα ΑΙ, ΖΚ μέσον ἀνάλογον ἐστὶ τὸ ΕΚ. ἔστι δὲ καὶ τῶν ΑΜ, ΝΧ τετραγώνων μέσον ἀνάλογον τὸ ΜΝ· καὶ ἐστὶν ἴσον τὸ μὲν ΑΙ τῷ ΑΜ, τὸ δὲ ΖΚ τῷ ΝΧ· καὶ τὸ ΕΚ ἄρα ἴσον ἐστὶ τῷ ΜΝ. ἀλλὰ τὸ μὲν ΜΝ ἴσον ἐστὶ τῷ ΑΧ, τὸ δὲ ΕΚ ἴσον [ἐστὶ] τῷ ΔΘ· καὶ ὅλον ἄρα τὸ ΔΚ ἴσον ἐστὶ τῷ ΥΦΘ γνώμῃ καὶ τῷ ΝΧ. ἔστι δὲ καὶ τὸ ΑΚ ἴσον τοῖς ΑΜ, ΝΧ· λοιπὸν ἄρα τὸ ΑΒ ἴσον ἐστὶ τῷ ΣΤ, τοντέστι τῷ ἀπὸ τῆς ΑΝ τετραγώνω· ἡ ΑΝ ἄρα δύναιται τὸ ΑΒ θωρίον. λέγω, ὅτι ἡ ΑΝ μέσης ἀποτομή ἐστὶ δευτέρα.

Ἐπεὶ γὰρ μέσα ἐδείχθη τὰ ΑΙ, ΖΚ καὶ ἐστὶν ἴσα τοῖς ἀπὸ τῶν ΑΟ, ΟΝ, μέσον ἄρα καὶ ἐκότερον τῶν ἀπὸ τῶν ΑΟ, ΟΝ· μέση ἄρα ἐκατέρω τῶν ΑΟ, ΟΝ. καὶ ἐπεὶ σύμμετρον ἐστὶ τὸ ΑΙ τῷ ΖΚ, σύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς ΑΟ τῷ ἀπὸ τῆς ΟΝ. πάλιν, ἐπεὶ ἀσύμμετρον ἐδείχθη τὸ ΑΙ τῷ ΕΚ, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ΑΜ τῷ ΜΝ, τοντέστι τὸ ἀπὸ τῆς ΑΟ τῷ ὑπὸ τῶν ΑΟ, ΟΝ· ὥστε καὶ ἡ ΑΟ ἀσύμμετρος ἐστὶ μήκει τῇ ΟΝ· αἱ ΑΟ, ΟΝ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δὴ, ὅτι καὶ μέσον περιέθουσιν.

Ἐπεὶ γὰρ μέσον ἐδείχθη τὸ ΕΚ καὶ ἐστὶν ἴσον τῷ ὑπὸ τῶν ΑΟ, ΟΝ, μέσον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν ΑΟ, ΟΝ· ὥστε αἱ ΑΟ, ΟΝ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσιν. ἡ ΑΝ ἄρα μέσης ἀποτομή ἐστὶ δευτέρα· καὶ δύναιται τὸ ΑΒ θωρίον.

[Def. 10.13]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable (in length) with (AG) , thus if (an area) equal to the fourth part of the square on DG is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Therefore, let DG be cut in half at E . And let (an area) equal to the (square) on EG be applied to AG , falling short by a square figure. And let it be the (rectangle contained) by AF and FG . And let EH , FI , and GK be drawn through points E , F , and G (respectively), parallel to AC . Thus, AF and FG are commensurable (in length). AI (is) thus also commensurable with FK [Props. 6.1, 10.11]. And since AF and FG are commensurable in length, AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. And AG (is) rational, and incommensurable in length with AC . Hence, AF and FG (are) also (rational, and incommensurable in length with AC) [Prop. 10.13]. Thus, AI and FK are each medial (areas) [Prop. 10.21]. Again, since DE is commensurable in length with EG , DG is also commensurable in length with each of DE and EG [Prop. 10.15]. And GD (is) rational, and incommensurable in length with AC . Thus, DE and EG (are) each also rational, and incommensurable in length with AC [Prop. 10.13]. DH and EK are thus each medial (areas) [Prop. 10.21]. And since AG and GD are commensurable in square only, AG is thus incommensurable in length with GD . But, AG is commensurable in length with AF , and DG with EG . Thus, AF is incommensurable in length with EG [Prop. 10.13]. And as AF (is) to EG , so AI is to EK [Prop. 6.1]. Thus, AI is incommensurable with EK [Prop. 10.11].

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , which is about the same angle as LM , be subtracted (from LM). Thus, LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG , so AI is to EK [Prop. 6.1]. And as EG (is) to FG , so EK is to FK [Prop. 6.1]. And thus as AI (is) to EK , so EK (is) to FK [Prop. 5.11]. Thus, EK is the mean proportional to AI and FK . And MN is also the mean proportional to the squares LM and NO [Prop. 10.53 lem.]. And AI is equal to LM , and FK to NO . Thus, EK is also equal to MN . But, MN is equal to LO , and EK [is] equal to DH [Prop. 1.43]. And thus the whole of DK is equal to the gnomon UVW and NO . And AK (is) also equal to LM and NO . Thus, the remainder AB is equal to ST —that is to say, to the square on LN . Thus, LN is the square-root of area AB . I say that LN is the second apotome of a medial (straight-line).

For since AI and FK were shown (to be) medial (areas), and are equal to the (squares) on LP and PN (respectively),

Ἡ ἄρα τὸ AB θωρίον δυναμένη μέση ἀποτομή ἐστὶ δυνάτω· ὅπερ εἶδει δεῖχαι.

the (squares) on each of LP and PN (are) thus also medial. Thus, LP and PN (are) each medial (straight-lines). And since AI is commensurable with FK [Props. 6.1, 10.11], the (square) on LP (is) thus also commensurable with the (square) on PN . Again, since AI was shown (to be) incommensurable with EK , LM is thus also incommensurable with MN —that is to say, the (square) on LP with the (rectangle contained) by LP and PN . Hence, LP is also incommensurable in length with PN [Props. 6.1, 10.11]. Thus, LP and PN are medial (straight-lines which are) commensurable in square only. So, I say that they also contain a medial (area).

For since EK was shown (to be) a medial (area), and is equal to the (rectangle contained) by LP and PN , the (rectangle contained) by LP and PN is thus also medial. Hence, LP and PN are medial (straight-lines which are) commensurable in square only, and which contain a medial (area). Thus, LN is the second apotome of a medial (straight-line) [Prop. 10.75]. And it is the square-root of area AB .

Thus, the square-root of area AB is the second apotome of a medial (straight-line). (Which is) the very thing it was required to show.

94.

Proposition 94

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς τετάρτης, ἡ τὸ θωρίον δυναμένη ἐλάσσων ἐστίν.

Θωρίον γάρ τὸ AB περιέθῃται ὑπὸ ῥητῆς τῆς AG καὶ ἀποτομῆς τετάρτης τῆς AD . λέγω, ὅτι ἡ τὸ AB θωρίον δυναμένη ἐλάσσων ἐστίν.

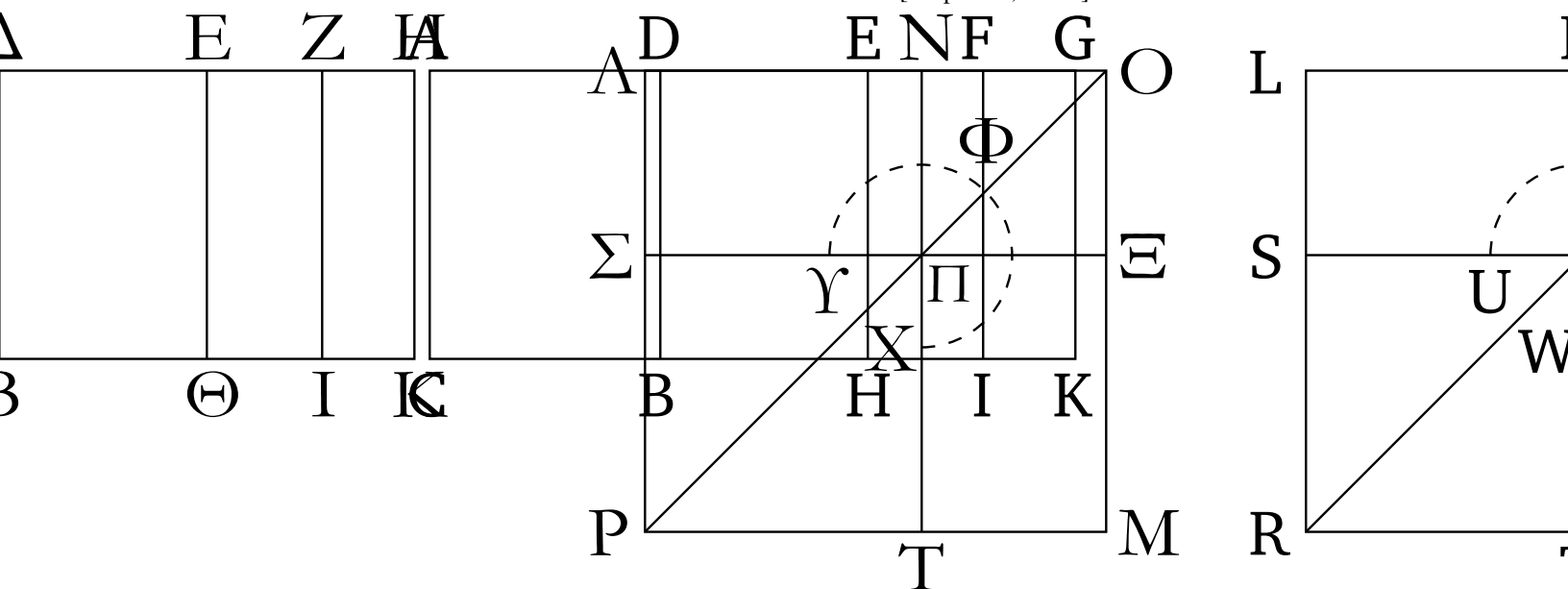
Ἐστω γάρ τῇ AD προσαρμόζουσα ἡ ΔH . αἱ ἄρα AH , $H\Delta$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AH σύμμετρος ἐστὶ τῇ ἐκκεκλιμένη ῥητῇ τῇ AG μήκει, ἡ δὲ ὅλη ἡ AH τῇς προσαρμόζουσας τῆς ΔH μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἐκκεκλιμένη μήκει. ἐπεὶ οὖν ἡ AH τῇς $H\Delta$ μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἐκκεκλιμένη μήκει, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔH ἴσον παρὰ τὴν AH παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ, εἷς ἀσύμμετρος αὐτὴν διελεί. τεμήσθω οὖν ἡ ΔH διὰ κατὰ τὸ E , καὶ τῷ ἀπὸ τῆς EH ἴσον παρὰ τὴν AH παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ, καὶ ἔστω τὸ ὑπὸ τῶν AZ , ZH . ἀσύμμετρος ἄρα ἐστὶ μήκει ἡ AZ τῇ ZH . ἡθροισαν οὖν διὰ τῶν E , Z , H παράλληλοι ταῖς AG , BD αἱ $E\Theta$, ZI , HK . ἐπεὶ οὖν ῥητὴ ἐστὶν ἡ AH καὶ σύμμετρος τῇ AG μήκει, ῥητὸν ἄρα ἐστὶν ὅλον τὸ AK . πάλιν, ἐπεὶ ἀσύμμετρος ἐστὶν ἡ ΔH τῇ AG μήκει, καὶ εἰσιν ἀμφοτέραι ῥηταὶ, μέσον ἄρα ἐστὶ τὸ ΔK . πάλιν, ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AZ τῇ ZH μήκει, ἀσύμμετρον ἄρα καὶ τὸ AI τῷ ZK .

If an area is contained by a rational (straight-line) and a fourth apotome then the square-root of the area is a minor (straight-line).

For let the area AB be contained by the rational (straight-line) AC and the fourth apotome AD . I say that the square-root of area AB is a minor (straight-line).

For let DG be an attachment to AD . Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and AG is commensurable in length with the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, DG , by the square on (some straight-line) incommensurable in length with (AG) [Def. 10.14]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) incommensurable in length with (AG) , thus if (some area), equal to the fourth part of the (square) on DG , is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let DG be cut in half at E , and let (some area), equal to the (square) on EG , be applied to AG , falling short by a square figure, and let it be the (rectangle contained) by AF and FG . Thus, AF is incommensurable in length with FG . Therefore, let EH , FI , and GK be drawn through E , F , and G (respectively), parallel to AC and BD . Therefore, since AG is rational, and commensurable in length with AC , the whole (area) AK is thus rational [Prop. 10.19]. Again, since DG is incommensurable in length

with AC , and both are rational (straight-lines), DK is thus a medial (area) [Prop. 10.21]. Again, since AF is incommensurable in length with FG , AI (is) thus also incommensurable with FK [Props. 6.1, 10.11].



Συνεοστάτω οὖν τῷ μὲν AI ἴσον τετραγώνων τὸ AM , τῷ δὲ ZK ἴσον ἀφηρηθῶ περὶ τὴν αὐτὴν γωνίαν τὴν ὑπὸ τῶν $ΛΟΜ$ τὸ NX . περὶ τὴν αὐτὴν ἄρα διάμετόν ἐστι τὰ AM , NX τετράγωνα. ἔστω αὐτῶν διάμετος ἡ OP , καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ οὖν τὸ ὑπὸ τῶν AZ , ZH ἴσον ἐστὶ τῷ ἀπὸ τῆς EH , ἀνάλογον ἄρα ἐστὶν ὥς ἡ AZ πρὸς τὴν EH , οὕτως ἡ EH πρὸς τὴν ZH . ἀλλ' ὥς μὲν ἡ AZ πρὸς τὴν EH , οὕτως ἐστὶ τὸ AI πρὸς τὸ EK , ὥς δὲ ἡ EH πρὸς τὴν ZH , οὕτως ἐστὶ τὸ EK πρὸς τὸ ZK . τῶν ἄρα AI , ZK μέσον ἀνάλογόν ἐστι τὸ EK . ἐστὶ δὲ καὶ τῶν AM , NX τετραγώνων μέσον ἀνάλογον τὸ MN , καὶ ἐστὶν ἴσον τὸ μὲν AI τῷ AM , τὸ δὲ ZK τῷ NX . καὶ τὸ EK ἄρα ἴσον ἐστὶ τῷ MN . ἀλλὰ τῷ μὲν EK ἴσον ἐστὶ τὸ $ΔΘ$, τῷ δὲ MN ἴσον ἐστὶ τὸ $ΛΧ$. ὅλον ἄρα τὸ $ΔΚ$ ἴσον ἐστὶ τῷ $ΥΦΘ$ γνώμονι καὶ τῷ NX . ἐπεὶ οὖν ὅλον τὸ AK ἴσον ἐστὶ τοῖς AM , NX τετραγώνοις, ὧν τὸ $ΔΚ$ ἴσον ἐστὶ τῷ $ΥΦΘ$ γνώμονι καὶ τῷ NX τετραγώνῳ, λοιπὸν ἄρα τὸ AB ἴσον ἐστὶ τῷ $ΣΤ$, τουτέστι τῷ ἀπὸ τῆς AN τετραγώνῳ. ἡ AN ἄρα δύναται τὸ AB θωρίον. λέγω, ὅτι ἡ AN ἄλογός ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ ῥητόν ἐστι τὸ AK καὶ ἔστιν ἴσον τοῖς ἀπὸ τῶν AO, ON τετράγωνοις, τὸ ἄρα συγκεκριμένον ἐκ τῶν ἀπὸ τῶν AO, ON ῥητόν ἐστιν. πάλιν, ἔπει τὸ $ΔK$ μέσον ἐστίν, καὶ ἔστιν ἴσον τὸ $ΔK$ τῷ διξ ὑπὸ τῶν AO, ON , τὸ ἄρα διξ ὑπὸ τῶν AO, ON μέσον ἐστίν. καὶ ἔπει ἀσύμμετρον ἐδεῖσθαι τὸ AI τῷ ZK , ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς AO τετράγωνον τῷ ἀπὸ τῆς ON τετραγώνῳ. αἱ AO, ON ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκεκριμένον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων

Therefore, let the square LM , equal to AI , be constructed. And let NO , equal to FK , (and) about the same angle, LPM , be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the (rectangle contained) by AF and FG is equal to the (square) on EG , thus, proportionally, as AF is to EG , so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG , so AI is to EK , and as EG (is) to FG , so EK is to FK [Prop. 6.1]. Thus, EK is the mean proportional to AI and FK [Prop. 5.11]. And MN is also the mean proportional to the squares LM and NO [Prop. 10.13 lem.], and AI is equal to LM , and FK to NO . EK is thus also equal to MN . But, DH is equal to EK , and LO is equal to MN [Prop. 1.43]. Thus, the whole of DK is equal to the gnomon UVW and NO . Therefore, since the whole of AK is equal to the (sum of the) squares LM and NO , of which DK is equal to the gnomon UVW and the square NO , the remainder AB is thus equal to ST —that is to say, to the square on LN . Thus, LN is the square-root of area AB . I say that LN is the irrational (straight-line which is) called minor.

For since AK is rational, and is equal to the (sum of the) squares LP and PN , the sum of the (squares) on LP and PN is thus rational. Again, since DK is medial, and DK is equal to twice the (rectangle contained) by LP and PN , thus twice the (rectangle contained) by LP and PN is medial. And since AI was shown (to be) incommensurable with FK , the square on LP (is) thus also incommensurable with the square on PN .

ῥητόν, τὸ δὲ διζ' ὑπ' αὐτῶν μέσον. ἡ AN ἄρα ἄλογός ἐστιν ἢ καλουμένη ἐλάσσων· καὶ δύναται τὸ AB θωρίον.

Ἡ ἄρα τὸ AB θωρίον δυναμένη ἐλάσσων ἐστίν· ὅπερ ἔδει δεῖξαι.

Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and twice the (rectangle contained) by them medial. LN is thus the irrational (straight-line) called minor [Prop. 10.76]. And it is the square-root of area AB .

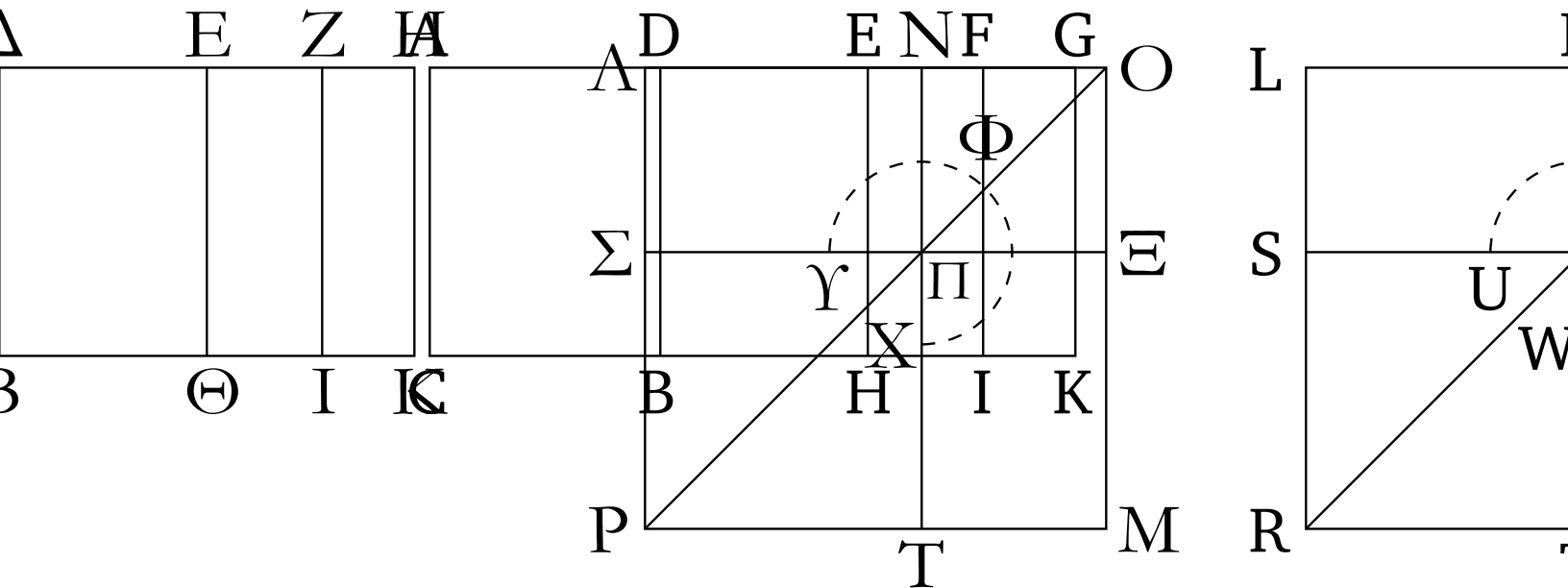
Thus, the square-root of area AB is a minor (straight-line). (Which is) the very thing it was required to show.

95.

Proposition 95

Ἐάν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς πέμπτης, ἢ τὸ θωρίον δυναμένη [ῥη] μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἐστίν.

If an area is contained by a rational (straight-line) and a fifth apotome then the square-root of the area is that (straight-line) which with a rational (area) makes a medial whole.



Θωρίον γάρ τὸ AB περιέθῃται ὑπὸ ῥητῆς τῆς AG καὶ ἀποτομῆς πέμπτης τῆς AD · λέγω, ὅτι ἡ τὸ AB θωρίον δυναμένη [ῥη] μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἐστίν.

Ἐστω γάρ τῇ AD προσαρμόζονσα ἡ ΔH · αἱ ἄρα AH , $H\Delta$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ προσαρμόζονσα ἡ $H\Delta$ σύμμετρος ἐστὶ μήκει τῇ ἐκκειμένῃ ῥητῇ τῇ AG , ἡ δὲ ὅλη ἡ AH τῆς προσαρμόζουσας τῆς ΔH μείζον δύναται τῷ ἀπὸ ἀσύμμετρον ἑαυτῇ. ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔH ἴσον παρὰ τὴν AH παραβληθῇ ἐλλείπον εἶδει τετραγώνῳ, εἷξ ἀσύμμετρα αὐτὴν διελεῖ. τετμήσθω οὖν ἡ ΔH δίῃα κατὰ τὸ E σημεῖον, καὶ τῷ ἀπὸ τῆς EH ἴσον παρὰ τὴν AH παραβελήσθω ἐλλείπον εἶδει τετραγώνῳ καὶ ἔστω τὸ ὑπὸ τῶν AZ , ZH · ἀσύμμετρος ἄρα ἐστὶν ἡ AZ τῇ ZH μήκει. καὶ ἐπεὶ ἀσύμμετρος ἐστὶν ἡ AH τῇ GA μήκει, καὶ εἰσιν ἀμφοτέραι ῥηταί, μέσον ἄρα ἐστὶ τὸ AK . πάλιν, ἐπεὶ ῥητὴ ἐστὶν ἡ ΔH καὶ σύμμετρος τῇ AG μήκει, ῥητόν ἐστι τὸ ΔK .

Συνεστάτω οὖν τῷ μὲν AI ἴσον τετράγωνον τὸ AM , τῷ δὲ ZK ἴσον τετράγωνον ἀφρηθήσθω τὸ NX περὶ τὴν αὐτὴν

For let the area AB be contained by the rational (straight-line) AC and the fifth apotome AD . I say that the square-root of area AB is that (straight-line) which with a rational (area) makes a medial whole.

For let DG be an attachment to AD . Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and the attachment GD is commensurable in length the (previously) laid down rational (straight-line) AC , and the square on the whole, AG , is greater than (the square on) the attachment, DG , by the (square) on (some straight-line) incommensurable (in length) with (AG) [Def. 10.15]. Thus, if (some area), equal to the fourth part of the (square) on DG , is applied to AG , falling short by a square figure, then it divides (AG) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let DG be divided in half at point E , and let (some area), equal to the (square) on EG , be applied to AG , falling short by a square figure, and let it be the (rectangle contained) by AF and FG . Thus, AF is incommensurable in length with FG . And since AG is incom-

γωνίαν τὴν ὑπὸ ΛOM · περὶ τὴν αὐτὴν ἄρα διάμετρον ἔστι τὰ ΛM , NX τετράγωνα. ἔστω αὐτῶν διάμετρος ἡ OP , καὶ καταγεγράφθω τὸ σθῆμα. ὁμοίως δὲ δείχομεν, ὅτι ἡ ΛN δύναται τὸ AB θωρίον. λέγω, ὅτι ἡ ΛN ἢ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα ἔστιν.

Ἐπεὶ γὰρ μέσον ἐδείχθη τὸ AK καὶ ἔστιν ἴσον τοῖς ἀπὸ τῶν ΛO , ON , τὸ ἄρα συγκείμενον ἐκ τῶν ἀπὸ τῶν ΛO , ON μέσον ἔστιν. πάλιν, ἐπεὶ ῥητόν ἐστι τὸ ΔK καὶ ἔστιν ἴσον τῷ διζ ὑπὸ τῶν ΛO , ON , καὶ αὐτὸ ῥητόν ἐστιν. καὶ ἐπεὶ ἀσύμμετρον ἔστι τὸ AI τῷ ZK , ἀσύμμετρον ἄρα ἔστι καὶ τὸ ἀπὸ τῆς ΛO τῷ ἀπὸ τῆς ON · αἱ ΛO , ON ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διζ ὑπ' αὐτῶν ῥητόν. ἡ λοιπὴ ἄρα ἡ ΛN ἀλογόξ ἐστιν ἢ καλουμένη μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα· καὶ δύναται τὸ AB θωρίον.

Ἡ τὸ AB ἄρα θωρίον δυναμένη μετὰ ῥητοῦ μέσον τὸ ὅλον ποιῶσα ἔστιν· ὅπερ ἔδει δεῖχαι.

mensurable in length with CA , and both are rational (straight-lines), AK is thus a medial (area) [Prop. 10.21]. Again, since DG is rational, and commensurable in length with AC , DK is a rational (area) [Prop. 10.19].

Therefore, let the square LM , equal to AI , be constructed. And let the square NO , equal to FK , (and) about the same angle, LPM , be subtracted (from NO). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let (the rest of) the figure be drawn. So, similarly (to the previous propositions), we can show that LN is the square-root of area AB . I say that LN is that (straight-line) which with a rational (area) makes a medial whole.

For since AK was shown (to be) a medial (area), and is equal to (the sum of) the squares on LP and PN , the sum of the (squares) on LP and PN is thus medial. Again, since DK is rational, and is equal to twice the (rectangle contained) by LP and PN , (the latter) is also rational. And since AI is incommensurable with FK , the (square) on LP is thus also incommensurable with the (square) on PN . Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by them rational. Thus, the remainder LN is the irrational (straight-line) called that which with a rational (area) makes a medial whole [Prop. 10.77]. And it is the square-root of area AB .

Thus, the square-root of area AB is that (straight-line) which with a rational (area) makes a medial whole. (Which is) the very thing it was required to show.

96.

Ἐὰν θωρίον περιέθῃται ὑπὸ ῥητῆς καὶ ἀποτομῆς ἑκτῆς, ἡ τὸ θωρίον δυναμένη μετὰ μέσον μέσον τὸ ὅλον ποιῶσα ἔστιν.

Proposition 96

If an area is contained by a rational (straight-line) and a sixth apotome then the square-root of the area is that (straight-line) which with a medial (area) makes a medial whole.

ἐστίν. καὶ ἐπεὶ ἀσύμμετρον ἐδείχθη τὸ AK τῷ $ΔK$, ἀσύμμετρα [ἄρα] ἐστὶ καὶ τὰ ἀπὸ τῶν $ΛO$, ON τετράγωνα τῷ διζυπὸ τῶν $ΛO$, ON . καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ AI τῷ ZK , ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς $ΛO$ τῷ ἀπὸ τῆς ON αἱ $ΛO$, ON ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ διζυπὸ αὐτῶν μέσον ἔτι τε τὰ ἀπ' αὐτῶν τετράγωνα ἀσύμμετρα τῷ διζυπὸ αὐτῶν. ἡ ἄρα $ΛN$ ἀλογόξ ἐστιν ἡ καλουμένη μετὰ μέσον μέσον τὸ ὅλον ποιοῦσα· καὶ δύναται τὸ AB θεωρίον.

Ἡ ἄρα τὸ θεωρίον δυναμένη μετὰ μέσον μέσον τὸ ὅλον ποιοῦσά ἐστιν· ὅπερ ἔδει δεῖχαι.

And let NO , equal to FK , (and) about the same angle, be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let (the rest of) the figure be drawn. So, similarly to the above, we can show that LN is the square-root of area AB . I say that LN is that (straight-line) which with a medial (area) makes a medial whole.

For since AK was shown (to be) a medial (area), and is equal to the (sum of the) squares on LP and PN , the sum of the (squares) on LP and PN is medial. Again, since DK was shown (to be) a medial (area), and is equal to twice the (rectangle contained) by LP and PN , twice the (rectangle contained) by LP and PN is also medial. And since AK was shown (to be) incommensurable with DK , [thus] the (sum of the) squares on LP and PN is also incommensurable with twice the (rectangle contained) by LP and PN . And since AI is incommensurable with FK , the (square) on LP (is) thus also incommensurable with the (square) on PN . Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by medial, and, furthermore, the (sum of the) squares on them incommensurable with twice the (rectangle contained) by them. Thus, LN is the irrational (straight-line) called that which with a medial (area) makes a medial whole [Prop. 10.78]. And it is the square-root of area AB .

Thus, the square-root of area (AB) is that (straight-line) which with a medial (area) makes a medial whole. (Which is) the very thing it was required to show.

97.

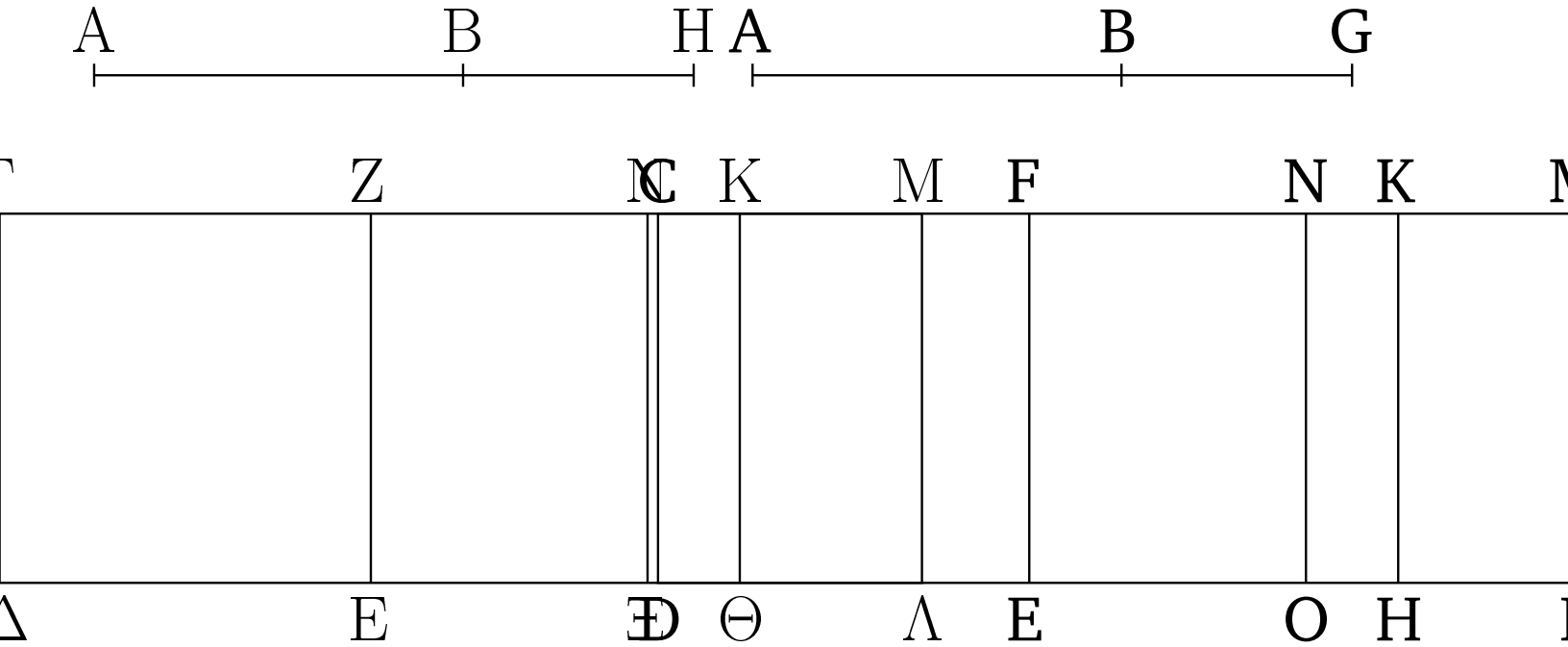
Proposition 97

Τὸ ἀπὸ ἀποτομῆς παρὰ ῥητὴν παραβαλλόμενον πλάτοξ ποιεῖ ἀποτομήν πρώτην.

Ἐστω ἀποτομή ἡ AB , ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $ΓΔ$ παραβελήσθω τὸ $ΓΕ$ πλάτοξ ποιῶν τὴν $ΓΖ$ · λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστὶ πρώτη.

The (square) on an apotome, applied to a rational (straight-line), produces a first apotome as breadth.

Let AB be an apotome, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a first apotome.



Ἐστω γάρ τῇ AB προσαρμόζονσα ἡ BH . αἱ ἄρα AH , HB ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΘ$, τῷ δὲ ἀπὸ τῆς BH τὸ $ΚΛ$. ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB . ὦν τὸ $ΓΕ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB . λοιπὸν ἄρα τὸ $ΖΛ$ ἴσον ἐστὶ τῷ διζ ὑπὸ τῶν AH , HB . τετμήσθω ἡ ZM διῖθα κατὰ τὸ N σημεῖον, καὶ ῥηθῶ διὰ τοῦ N τῇ $ΓΔ$ παράλληλος ἡ NX . ἐκάτερον ἄρα τῶν ZX , AN ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τὰ ἀπὸ τῶν AH , HB ῥητά ἐστιν, καὶ ἐστὶ τοῖς ἀπὸ τῶν AH , HB ἴσον τὸ $ΔΜ$, ῥητὸν ἄρα ἐστὶ τὸ $ΔΜ$. καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παραβέβληται πλάτος ποιοῦν τὴν $ΓΜ$. ῥητὴ ἄρα ἐστὶν ἡ $ΓΜ$ καὶ σύμμετρος τῇ $ΓΔ$ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ διζ ὑπὸ τῶν AH , HB , καὶ τῷ διζ ὑπὸ τῶν AH , HB ἴσον τὸ $ΖΛ$, μέσον ἄρα τὸ $ΖΛ$. καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παράκειται πλάτος ποιοῦν τὴν ZM . ῥητὴ ἄρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ τὰ μὲν ἀπὸ τῶν AH , HB ῥητά ἐστιν, τὸ δὲ διζ ὑπὸ τῶν AH , HB μέσον, ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AH , HB τῷ διζ ὑπὸ τῶν AH , HB . καὶ τοῖς μὲν ἀπὸ τῶν AH , HB ἴσον ἐστὶ τὸ $ΓΛ$, τῷ δὲ διζ ὑπὸ τῶν AH , HB τὸ $ΖΛ$. ἀσύμμετρον ἄρα ἐστὶ τὸ $ΔΜ$ τῷ $ΖΛ$. ὥς δὲ τὸ $ΔΜ$ πρόξ τὸ $ΖΛ$, οὕτως ἐστὶν ἡ $ΓΜ$ πρόξ τὴν ZM . ἀσύμμετρος ἄρα ἐστὶν ἡ $ΓΜ$ τῇ ZM μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ἄρα $ΓΜ$, MZ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ $ΓΖ$ ἄρα ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ πρώτη.

Ἐπεὶ γάρ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $ΓΘ$, τῷ δὲ ἀπὸ τῆς BH ἴσον τὸ $ΚΛ$, τῷ δὲ ὑπὸ τῶν AH , HB τὸ $ΝΛ$, καὶ τῶν $ΓΘ$, $ΚΛ$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $ΝΛ$. ἐστὶν ἄρα ὡς τὸ $ΓΘ$ πρόξ τὸ $ΝΛ$, οὕτως τὸ $ΝΛ$ πρόξ τὸ $ΚΛ$. ἀλλ’

For let BG be an attachment to AB . Thus, AG and GB are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let CH , equal to the (square) on AG , and KL , (equal) to the (square) on BG , be applied to CD . Thus, the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB . The remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Let FM be cut in half at point N . And let NO be drawn through N , parallel to CD . Thus, FO and LN are each equal to the (rectangle contained) by AG and GB . And since the (sum of the squares) on AG and GB is rational, and DM is equal to the (sum of the squares) on AG and GB , DM is thus rational. And it has been applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and commensurable in length with CD [Prop. 10.20]. Again, since twice the (rectangle contained) by AG and GB is medial, and FL (is) equal to twice the (rectangle contained) by AG and GB , FL (is) thus a medial (area). And it is applied to the rational (straight-line) CD , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since the (sum of the squares) on AG and GB is rational, and twice the (rectangle contained) by AG and GB medial, the (sum of the squares) on AG and GB is thus incommensurable with twice the (rectangle contained) by AG and GB . And CL is equal to the (sum of the squares) on AG and GB , and FL to twice the (rectangle contained) by AG and GB . DM is thus incommensurable with FL . And as DM (is) to FL , so CM is to FM [Prop. 6.1]. CM is thus incommensurable in length with FM [Prop. 10.11]. And both are rational (straight-lines). Thus, CM and MF are rational (straight-lines).

ὥς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $\Lambda\Delta$, οὕτως ἐστὶν ἡ $\Gamma\Lambda$ πρὸς τὴν NM . ὥς δὲ τὸ $\Lambda\Delta$ πρὸς τὸ $ΚΑ$, οὕτως ἐστὶν ἡ NM πρὸς τὴν KM . τὸ ἄρα ὑπὸ τῶν $\Gamma\Lambda$, KM ἴσον ἐστὶ τῷ ἀπὸ τῆς NM , τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . καὶ ἐπεὶ σύμμετρον ἐστὶ τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB , σύμμετρον [ἐστὶ] καὶ τὸ $\Gamma\Theta$ τῷ $ΚΑ$. ὥς δὲ τὸ $\Gamma\Theta$ πρὸς τὸ $ΚΑ$, οὕτως ἡ $\Gamma\Lambda$ πρὸς τὴν KM . σύμμετρος ἄρα ἐστὶν ἡ $\Gamma\Lambda$ τῇ KM . ἐπεὶ οὖν δύο εὐθεῖαι ἄκισοι εἰσιν αἱ ΓM , MZ , καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM ἴσον παρὰ τὴν ΓM παραβέβληται ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν $\Gamma\Lambda$, KM , καὶ ἐστὶ σύμμετρος ἡ $\Gamma\Lambda$ τῇ KM , ἡ ἄρα ΓM τῆς MZ μείζον δύνανται τῷ ἀπὸ συμέτρου ἐαυτῇ μήκει. καὶ ἐστὶν ἡ ΓM σύμμετρος τῇ ἐκκειμένῃ ῥητῇ τῇ $\Gamma\Delta$ μήκει. ἡ ἄρα ΓZ ἀποτομή ἐστὶ πρώτη.

Τὸ ἄρα ἀπὸ ἀποτομῆς παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην· ὅπερ ἔδει δεῖξαι.

98.

Τὸ ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν.

Ἐστω μέσης ἀποτομῆς πρώτη ἡ AB , ῥητὴ δὲ ἡ $\Gamma\Delta$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $\Gamma\Delta$ παραβέβλησθω τὸ ΓE πλάτος ποιοῦν τὴν ΓZ . λέγω, ὅτι ἡ ΓZ ἀποτομή ἐστὶ δευτέρα.

Ἐστω γάρ τῇ AB προσαρμόζουσα ἡ BH . αἱ ἄρα AH , HB μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ῥητὸν περιέθουσαι. καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $\Gamma\Delta$ παραβέβλησθω τὸ $\Gamma\Theta$ πλάτος ποιοῦν τὴν $\Gamma\Lambda$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $ΚΑ$ πλάτος ποιοῦν τὴν KM . ὅλον ἄρα τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB . μέσον ἄρα καὶ τὸ $\Gamma\Lambda$. καὶ παρὰ ῥητὴν τὴν $\Gamma\Delta$ παράκειται πλάτος ποιοῦν τὴν ΓM . ῥητὴ ἄρα ἐστὶν ἡ ΓM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , ὧν τὸ ἀπὸ τῆς AB ἴσον ἐστὶ τῷ ΓE , λοιπὸν ἄρα τὸ διζ ὑπὸ τῶν AH , HB ἴσον ἐστὶ τῷ $Z\Lambda$. ῥητὸν δὲ [ἐστὶ] τὸ διζ ὑπὸ τῶν AH , HB . ῥητὸν ἄρα τὸ $Z\Lambda$. καὶ παρὰ ῥητὴν τὴν ZE παράκειται πλάτος ποιοῦν τὴν ZM . ῥητὴ ἄρα ἐστὶ καὶ ἡ ZM καὶ σύμμετρος τῇ $\Gamma\Delta$ μήκει. ἐπεὶ οὖν τὰ μὲν ἀπὸ τῶν AH , HB , τουτέστι τὸ $\Gamma\Lambda$, μέσον ἐστὶν, τὸ δὲ διζ ὑπὸ τῶν AH , HB , τουτέστι τὸ $Z\Lambda$, ῥητὸν ἀσύμμετρον

which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a first (apotome).

For since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG , and KL equal to the (square) on GB , and NL to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to KM [Prop. 6.1]. Thus, the (rectangle contained) by CK and KM is equal to the (square) on NM —that is to say, to the fourth part of the (square) on FM [Prop. 6.17]. And since the (square) on AG is commensurable with the (square) on GB , CH [is] also commensurable with KL . And as CH (is) to KL , so CK (is) to KM [Prop. 6.1]. CK is thus commensurable (in length) with KM [Prop. 10.11]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on FM , has been applied to CM , falling short by a square figure, and CK is commensurable (in length) with KM , the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable in length with (CM) [Prop. 10.17]. And CM is commensurable in length with the (previously) laid down rational (straight-line) CD . Thus, CF is a first apotome [Def. 10.15].

Thus, the (square) on an apotome, applied to a rational (straight-line), produces a first apotome as breadth. (Which is) the very thing it was required to show.

Proposition 98

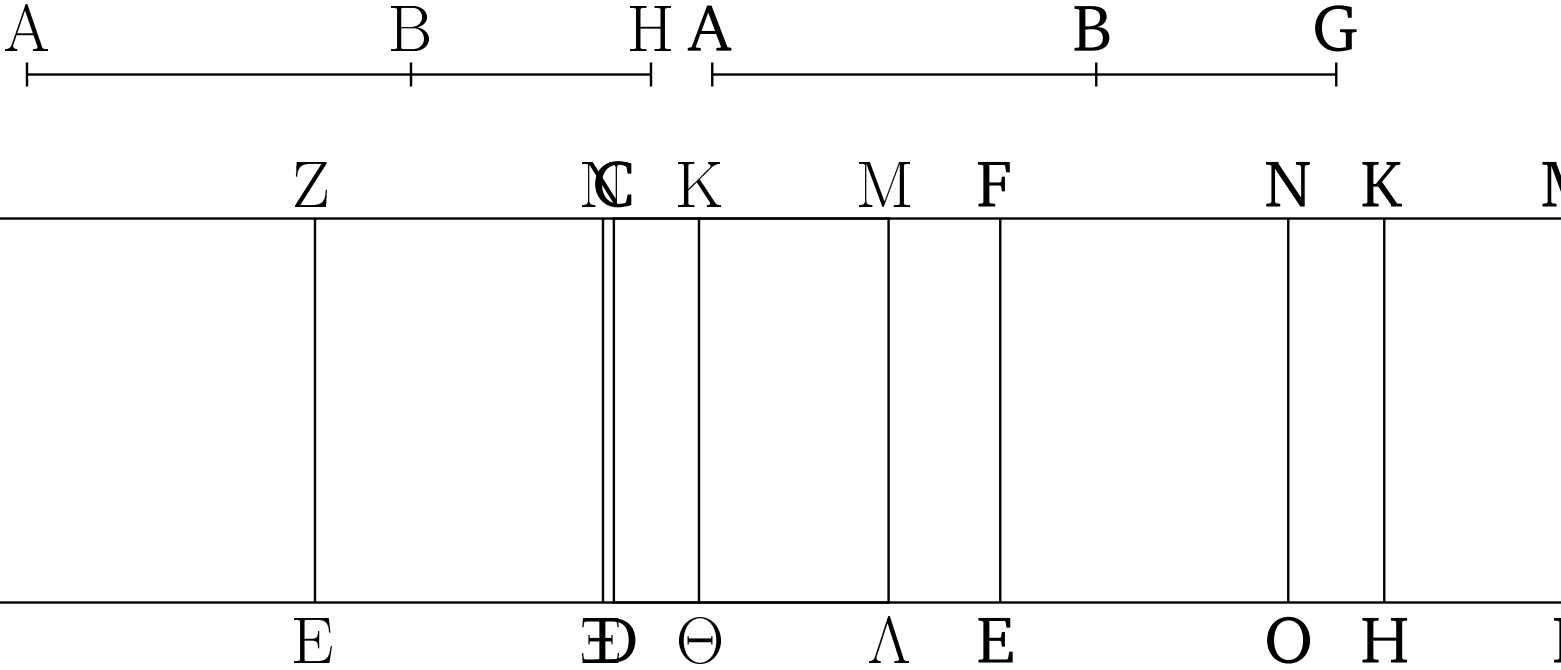
The (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces a second apotome as breadth.

Let AB be a first apotome of a medial (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a second apotome.

For let BG be an attachment to AB . Thus, AG and GB are medial (straight-lines which are) commensurable in square only, containing a rational (area) [Prop. 10.74]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on GB , producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . Thus, CL (is) also a medial (area) [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) CD , producing CM as breadth. CM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since CL is equal to the (sum of the squares) on AG and GB , of which the (square) on AB is equal to CE , the remainder, twice the (rectangle contained) by AG and GB , is thus equal to

ἄρα ἐστὶ τὸ $\Gamma\Lambda$ τῷ $Ζ\Lambda$. ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Ζ\Lambda$, οὕτως ἐστὶν ἡ $\Gamma\mathcal{M}$ πρὸς τὴν $Ζ\mathcal{M}$: ἀσύμμετρος ἄρα ἡ $\Gamma\mathcal{M}$ τῇ $Ζ\mathcal{M}$ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ἄρα $\Gamma\mathcal{M}$, $\mathcal{M}Ζ$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἡ $\Gamma Ζ$ ἄρα ἀποτομή ἐστίν. λέγω δὴ, ὅτι καὶ δευτέρα.

FL [Prop. 2.7]. And twice the (rectangle contained) by AG and GB [is] rational. Thus, FL (is) rational. And it is applied to the rational (straight-line) FE , producing FM as breadth. FM is thus also rational, and commensurable in length with CD [Prop. 10.20]. Therefore, since the (sum of the squares) on AG and GB —that is to say, CL —is medial, and twice the (rectangle contained) by AG and GB —that is to say, FL —(is) rational, CL is thus incommensurable with FL . And as CL (is) to FL , so CM is to FM [Prop. 6.1]. Thus, CM (is) incommensurable in length with FM [Prop. 10.11]. And they are both rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a second (apotome).



Τετμήσθω γὰρ ἡ $Ζ\mathcal{M}$ δίῃα κατὰ τὸ N , καὶ ἤθθω διὰ τοῦ N τῇ $\Gamma\Delta$ παράλληλοξ ἡ NX . ἐκάτερον ἄρα τῶν $ΖΧ$, $N\Lambda$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB τετραγώνων μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶν ἴσον τὸ μὲν ἀπὸ τῆς AH τῷ $\Gamma\Theta$, τὸ δὲ ὑπὸ τῶν AH , HB τῷ $N\Lambda$, τὸ δὲ ἀπὸ τῆς BH τῷ $K\Lambda$, καὶ τῶν $\Gamma\Theta$, $K\Lambda$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $N\Lambda$. ἔστιν ἄρα ὥς τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$. ἀλλ' ὥς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως ἐστὶν ἡ ΓK πρὸς τὴν NM , ὥς δὲ τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$, οὕτως ἐστὶν ἡ NM πρὸς τὴν MK . ὥς ἄρα ἡ ΓK πρὸς τὴν NM , οὕτως ἐστὶν ἡ NM πρὸς τὴν KM . τὸ ἄρα ὑπὸ τῶν ΓK , KM ἴσον ἐστὶ τῷ ἀπὸ τῆς NM , τοντέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $Ζ\mathcal{M}$ [καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς BH , σύμμετρόν ἐστι καὶ τὸ $\Gamma\Theta$ τῷ $K\Lambda$, τοντέστιν ἡ ΓK τῇ KM]. ἐπεὶ οὖν δύο εὐθεῖαι ἄνισοι εἰσιν αἱ $\Gamma\mathcal{M}$, $\mathcal{M}Ζ$, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $\mathcal{M}Ζ$ ἴσον παρὰ τὴν μείζονα

For let FM be cut in half at N . And let NO be drawn through (point) N , parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since the (rectangle contained) by AG and GB is the mean proportional to the squares on AG and GB [Prop. 10.21 lem.], and the (square) on AG is equal to CH , and the (rectangle contained) by AG and GB to NL , and the (square) on BG to KL , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL [Prop. 5.11]. But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to MK [Prop. 6.1]. Thus, as CK (is) to NM , so NM is to KM [Prop. 5.11]. The (rectangle contained) by CK and KM is thus equal to the (square) on NM [Prop. 6.17]—that is to say, to the fourth part of the (square) on FM [and since the (square) on AG is commensurable with the (square) on BG , CH is also commensurable with KL —that is to say, CK with

τὴν $ΓΜ$ παραβέβληται ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν $ΓΚ$, $ΚΜ$ καὶ εἰς σύμμετρα αὐτὴν διαιρεῖ, ἡ ἄρα $ΓΜ$ τῆς $ΜΖ$ μείζον δύνатаι τῷ ἀπὸ σύμμετρον ἐκείνῃ μήκει. καὶ ἐστὶν ἡ προσαρμόζουσα ἡ $ΖΜ$ σύμμετρος μήκει τῇ ἐκκειμένῃ ῥητῇ τῇ $ΓΔ$. ἡ ἄρα $ΓΖ$ ἀποτομή ἐστὶ δευτέρα.

Τὸ ἄρα ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν· ὅπερ ἔδει δεῖχαι.

KM]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on MF , has been applied to the greater CM , falling short by a square figure, and divides it into commensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable in length with (CM) [Prop. 10.17]. The attachment FM is also commensurable in length with the (previously) laid down rational (straight-line) CD . CF is thus a second apotome [Def. 10.16].

Thus, the (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces a second apotome as breadth. (Which is) the very thing it was required to show.

99.

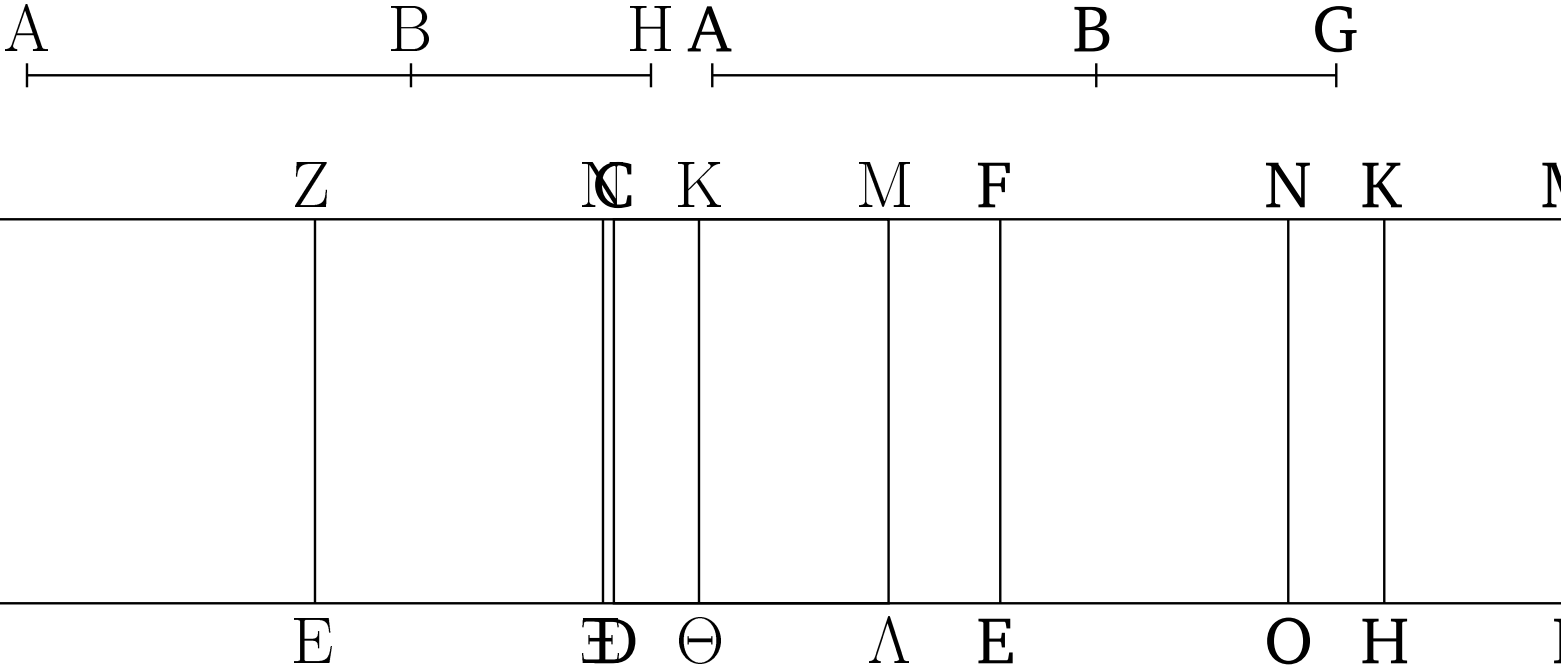
Proposition 99

Τὸ ἀπὸ μέσης ἀποτομῆς δευτέρας παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην.

Ἐστω μέσης ἀποτομῆς δευτέρας ἡ $ΑΒ$, ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς $ΑΒ$ ἴσον παρὰ τὴν $ΓΔ$ παραβελήσθω τὸ $ΓΕ$ πλάτος ποιοῦν τὴν $ΓΖ$. λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστὶ τρίτη.

The (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces a third apotome as breadth.

Let AB be the second apotome of a medial (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a third apotome.



Ἐστω γάρ τῇ $ΑΒ$ προσαρμόζουσα ἡ $ΒΗ$. αἱ ἄρα $ΑΗ$, $ΗΒ$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέχονσαι. καὶ τῷ μὲν ἀπὸ τῆς $ΑΗ$ ἴσον παρὰ τὴν $ΓΔ$ παραβελήσθω τὸ $ΓΘ$ πλάτος ποιοῦν τὴν $ΓΚ$, τῷ δὲ ἀπὸ τῆς $ΒΗ$ ἴσον παρὰ τὴν $ΚΘ$ παραβελήσθω τὸ $ΚΛ$ πλάτος ποιοῦν τὴν $ΚΜ$. ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΑΗ$, $ΗΒ$ [καὶ ἐστὶ μέσα τὰ ἀπὸ

For let BG be an attachment to AB . Thus, AG and GB are medial (straight-lines which are) commensurable in square only, containing a medial (area) [Prop. 10.75]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth. And let KL , equal to the (square) on BG , be applied to KH , producing KM as breadth. Thus, the whole of

τῶν AH, HB]· μέσον ἄρα καὶ τὸ $\Gamma\Lambda$. καὶ παρὰ ῥητὴν τὴν $\Gamma\Delta$ παραβέβληται πλάτος ποιοῦν τὴν ΓM · ῥητὴ ἄρα ἐστὶν ἡ ΓM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ ὅλον τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB , ὥν τὸ ΓE ἴσον ἐστὶ τῷ ἀπὸ τῆς AB , λοιπὸν ἄρα τὸ ΛZ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AH, HB . τετμήσθω οὖν ἡ ZM δίδια κατὰ τὸ N σημείον, καὶ τῇ $\Gamma\Delta$ παράλληλος ἡθῆτω ἡ NX · ἐκάτερον ἄρα τῶν $ZX, \Lambda\Lambda$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH, HB . μέσον δὲ τὸ ὑπὸ τῶν AH, HB · μέσον ἄρα ἐστὶ καὶ τὸ $Z\Lambda$. καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν ZM · ῥητὴ ἄρα καὶ ἡ ZM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ αἱ AH, HB δυνάμει μόνον εἰσὶ σύμμετροι, ἀσύμμετρος ἄρα [ἐστὶ] μήκει ἡ AH τῇ HB · ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς AH τῷ ὑπὸ τῶν AH, HB . ἀλλὰ τῷ μὲν ἀπὸ τῆς AH σύμμετρά ἐστὶ τὰ ἀπὸ τῶν AH, HB , τῷ δὲ ὑπὸ τῶν AH, HB τὸ διξ ὑπὸ τῶν AH, HB · ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AH, HB τῷ διξ ὑπὸ τῶν AH, HB . ἀλλὰ τοῖς μὲν ἀπὸ τῶν AH, HB ἴσον ἐστὶ τὸ $\Gamma\Lambda$, τῷ δὲ διξ ὑπὸ τῶν AH, HB ἴσον ἐστὶ τὸ $Z\Lambda$ · ἀσύμμετρον ἄρα ἐστὶ τὸ $\Gamma\Lambda$ τῷ $Z\Lambda$. ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Z\Lambda$, οὕτως ἐστὶν ἡ ΓM πρὸς τὴν ZM · ἀσύμμετρος ἄρα ἐστὶν ἡ ΓM τῇ ZM μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ ἄρα $\Gamma M, MZ$ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ ΓZ . λέγω δὴ, ὅτι καὶ τρίτη.

Ἐπεὶ γὰρ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB , σύμμετρον ἄρα καὶ τὸ $\Gamma\Theta$ τῷ $ΚΛ$ · ὥστε καὶ ἡ ΓK τῇ KM . καὶ ἐπεὶ τῶν ἀπὸ τῶν AH, HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH, HB , καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $ΚΛ$, τῷ δὲ ὑπὸ τῶν AH, HB ἴσον τὸ $\Lambda\Lambda$, καὶ τῶν $\Gamma\Theta, ΚΛ$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $\Lambda\Lambda$ · ἐστὶν ἄρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $\Lambda\Lambda$, οὕτως τὸ $\Lambda\Lambda$ πρὸς τὸ $ΚΛ$. ἀλλ' ὡς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $\Lambda\Lambda$, οὕτως ἐστὶν ἡ ΓK πρὸς τὴν NM , ὡς δὲ τὸ $\Lambda\Lambda$ πρὸς τὸ $ΚΛ$, οὕτως ἐστὶν ἡ NM πρὸς τὴν KM · ὡς ἄρα ἡ ΓK πρὸς τὴν MN , οὕτως ἐστὶν ἡ MN πρὸς τὴν KM · τὸ ἄρα ὑπὸ τῶν $\Gamma K, KM$ ἴσον ἐστὶ τῷ [ἀπὸ τῆς MN , τουτέστι τῷ] τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . ἐπεὶ οὖν δύο εὐθεῖαι ἄνισοι εἰσὶν αἱ $\Gamma M, MZ$, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM ἴσον παρὰ τὴν ΓM παραβέβληται ἐλλείπον εἶδει τετραγώνῳ καὶ εἰς σύμμετρα αὐτὴν διαιρεῖ, ἡ ΓM ἄρα τῆς MZ μείζον δύναται τῷ ἀπὸ συμέτρου ἐαντῇ. καὶ οὐδετέρα τῶν $\Gamma M, MZ$ σύμμετρος ἐστὶ μήκει τῇ ἐκκειμένη ῥητῇ τῇ $\Gamma\Delta$ · ἡ ἄρα ΓZ ἀποτομή ἐστὶ τρίτη.

Τὸ ἄρα ἀπὸ μέσης ἀποτομῆς δευτέραξ παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην· ὅπερ ἔδει δεῖξαι.

CL is equal to the (sum of the squares) on AG and GB [and the (sum of the squares) on AG and GB is medial]. CL (is) thus also medial [Props. 10.15, 10.23 corr.]. And it has been applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and incommensurable in length with CD [Prop. 10.22]. And since the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB , the remainder LF is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at point N . And let NO be drawn parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And the (rectangle contained) by AG and GB (is) medial. Thus, FL is also medial. And it is applied to the rational (straight-line) EF , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since AG and GB are commensurable in square only, AG [is] thus incommensurable in length with GB . Thus, the (square) on AG is also incommensurable with the (rectangle contained) by AG and GB [Props. 6.1, 10.11]. But, the (sum of the squares) on AG and GB is commensurable with the (square) on AG , and twice the (rectangle contained) by AG and GB with the (rectangle contained) by AG and GB . The (sum of the squares) on AG and GB is thus incommensurable with twice the (rectangle contained) by AG and GB [Prop. 10.13]. But, CL is equal to the (sum of the squares) on AG and GB , and FL is equal to twice the (rectangle contained) by AG and GB . Thus, CL is incommensurable with FL . And as CL (is) to FL , so CM is to FM [Prop. 6.1]. CM is thus incommensurable in length with FM [Prop. 10.11]. And they are both rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a third (apotome).

For since the (square) on AG is commensurable with the (square) on GB , CH (is) thus also commensurable with KL . Hence, CK (is) also (commensurable in length) with KM [Props. 6.1, 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG , and KL equal to the (square) on GB , and NL equal to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM (is) to KM [Prop. 6.1]. Thus, as CK (is) to MN , so MN is to KM [Prop. 5.11]. Thus, the (rectangle contained) by CK and KM is equal to the [(square) on MN —that is to say, to the] fourth part of the (square) on FM [Prop. 6.17]. Therefore, since CM and MF are two unequal straight-lines, and (some area), equal to the fourth part of the (square) on FM , has been applied to CM , falling short by a square figure, and divides it into commensurable (parts),

the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable (in length) with (CM) [Prop. 10.17]. And neither of CM and MF is commensurable in length with the (previously) laid down rational (straight-line) CD . CF is thus a third apotome [Def. 10.13].

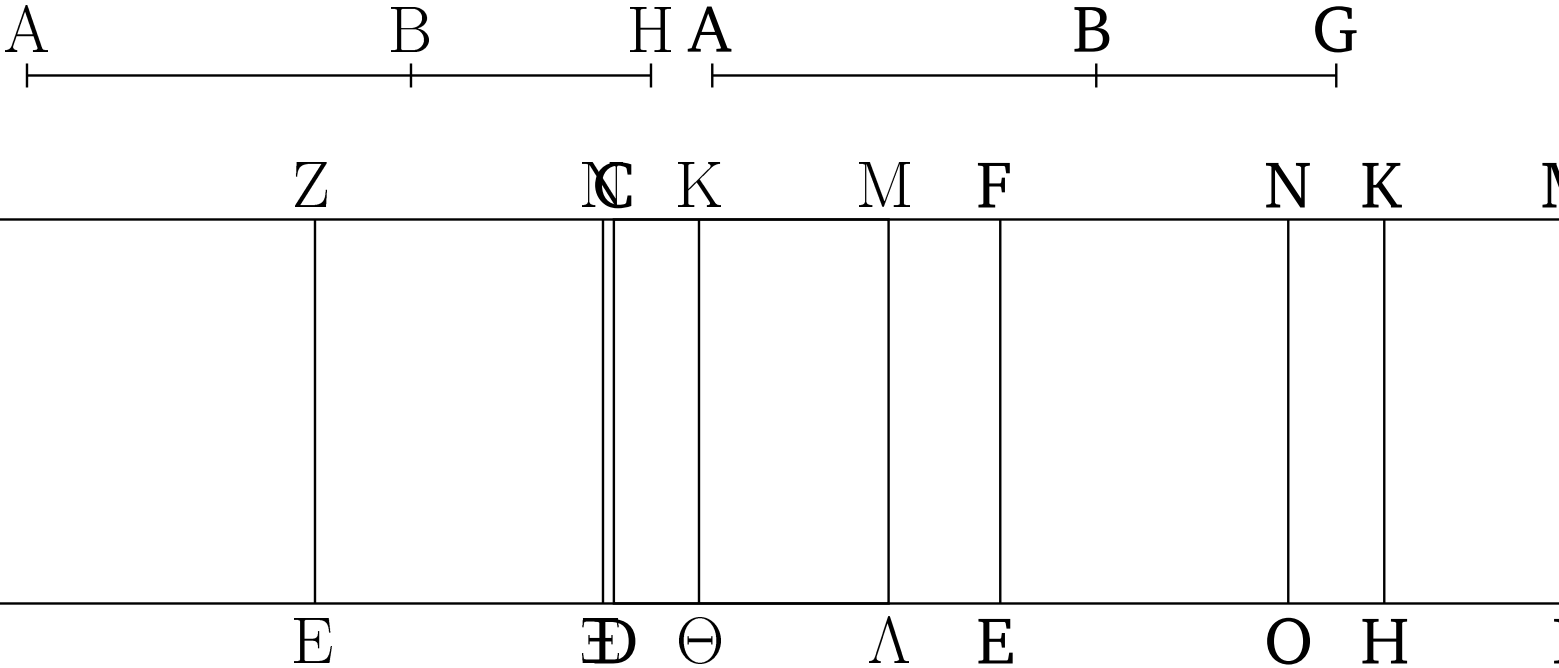
Thus, the (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces a third apotome as breadth. (Which is) the very thing it was required to show.

100.

Proposition 100

Τὸ ἀπὸ ἐλάσσονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν τετάρτην.

The (square) on a minor (straight-line), applied to a rational (straight-line), produces a fourth apotome as breadth.



Ἐστω ἐλάσσων ἡ AB , ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ ῥητὴν τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΕ$ πλάτος ποιοῦν τὴν $ΓΖ$ · λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστι τετάρτη.

Ἐστω γάρ τῇ AB προσαρμόζονσα ἡ BH · αἱ ἄρα AH , HB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB τετραγώνων ῥητόν, τὸ δὲ διξ ὑπὸ τῶν AH , HB μέσον. καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΘ$ πλάτος ποιοῦν τὴν $ΓΚ$, τῷ δὲ ἀπὸ τῆς BH ἴσον τὸ $ΚΛ$ πλάτος ποιοῦν τὴν $ΚΜ$ · ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB . καὶ ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB ῥητόν· ῥητόν ἄρα ἐστὶ καὶ τὸ $ΓΛ$. καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παράκειται πλάτος ποιοῦν τὴν $ΓΜ$ · ῥητὴ ἄρα καὶ ἡ $ΓΜ$ καὶ σύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ ὅλον τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , ὧν τὸ $ΓΕ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB , λοιπὸν ἄρα τὸ $ΖΛ$ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AH , HB . τετραγώνω ὃν ἡ $ΖΜ$ δίθω κατὰ τὸ $Ν$ σημείον, καὶ

Let AB be a minor (straight-line), and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to the rational (straight-line) CD , producing CF as breadth. I say that CF is a fourth apotome.

For let BG be an attachment to AB . Thus, AG and GB are incommensurable in square, making the sum of the squares on AG and GB rational, and twice the (rectangle contained) by AG and GB medial [Prop. 10.76]. And let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on BG , producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . And the sum of the (squares) on AG and GB is rational. CL is thus also rational. And it is applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM (is) also rational, and commensurable in length with CD [Prop. 10.20]. And since the whole of CL is

ἡθθω δια τοῦ N ὁποτέρᾳ τῶν $\Gamma\Delta$, $M\Lambda$ παράλληλος ἡ NX · ἐκάτερον ἄρα τῶν ZX , $N\Lambda$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB . καὶ ἐπεὶ τὸ διζ ὑπὸ τῶν AH , HB μέσον ἐστὶ καὶ ἐστὶν ἴσον τῷ $Z\Lambda$, καὶ τὸ $Z\Lambda$ ἄρα μέσον ἐστίν. καὶ παρὰ ῥητὴν τὴν ZE παράκειται πλάτος ποιῶν τὴν ZM · ῥητὴ ἄρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB ῥητόν ἐστιν, τὸ δὲ διζ ὑπὸ τῶν AH , HB μέσον, ἀσύμμετρον [ἄρα] ἐστὶ τὰ ἀπὸ τῶν AH , HB τῷ διζ ὑπὸ τῶν AH , HB . ἴσον δὲ [ἐστὶ] τὸ $\Gamma\Lambda$ τοῖς ἀπὸ τῶν AH , HB , τῷ δὲ διζ ὑπὸ τῶν AH , HB ἴσον τὸ $Z\Lambda$ · ἀσύμμετρον ἄρα [ἐστὶ] τὸ $\Gamma\Lambda$ τῷ $Z\Lambda$. ὥς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Z\Lambda$, οὕτως ἐστὶν ἡ ΓM πρὸς τὴν MZ · ἀσύμμετρος ἄρα ἐστὶν ἡ ΓM τῇ MZ μήκει. καὶ εἰσιν ἀμφοτέραι ῥηταί· αἱ ἄρα ΓM , MZ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ ΓZ . λέγω [δὴ], ὅτι καὶ τετάρτη.

Ἐπεὶ γὰρ αἱ AH , HB δυνάμει εἰσὶν ἀσύμμετροι, ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB . καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $K\Lambda$ · ἀσύμμετρον ἄρα ἐστὶ τὸ $\Gamma\Theta$ τῷ $K\Lambda$. ὥς δὲ τὸ $\Gamma\Theta$ πρὸς τὸ $K\Lambda$, οὕτως ἐστὶν ἡ ΓK πρὸς τὴν KM · ἀσύμμετρος ἄρα ἐστὶν ἡ ΓK τῇ KM μήκει. καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶν ἴσον τὸ μὲν ἀπὸ τῆς AH τῷ $\Gamma\Theta$, τὸ δὲ ἀπὸ τῆς HB τῷ $K\Lambda$, τὸ δὲ ὑπὸ τῶν AH , HB τῷ $N\Lambda$, τῶν ἄρα $\Gamma\Theta$, $K\Lambda$ μέσον ἀνάλογόν ἐστι τὸ $N\Lambda$ · ἴσον ἄρα ὥς τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$. ἀλλ' ὥς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως ἐστὶν ἡ ΓK πρὸς τὴν NM , ὥς δὲ τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$, οὕτως ἐστὶν ἡ NM πρὸς τὴν KM · ὥς ἄρα ἡ ΓK πρὸς τὴν MN , οὕτως ἐστὶν ἡ MN πρὸς τὴν KM · τὸ ἄρα ὑπὸ τῶν ΓK , KM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN , τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . ἐπεὶ οὖν δύο εὐθεῖαι ἄνισοι εἰσιν αἱ ΓM , MZ , καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς MZ ἴσον παρὰ τὴν ΓM παραβέβληται ἐλλείπον εἶδει τετραγώνῳ τὸ ὑπὸ τῶν ΓK , KM καὶ εἷς ἀσύμμετρος αὐτὴν διαιρεῖ, ἡ ἄρα ΓM τῆς MZ μείζον δύνатаι τῷ ἀπὸ ἀσυμμέτρου ἐαντῇ. καὶ ἐστὶν ὅλη ἡ ΓM σύμμετρος μήκει τῇ ἐκκειμένη ῥητῇ τῇ $\Gamma\Delta$ · ἡ ἄρα ΓZ ἀποτομή ἐστὶ τετάρτη.

Τὸ ἄρα ἀπὸ ἐλάσσονος καὶ τὰ ἐχθῆς.

equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB , the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at point N . And let NO be drawn through N , parallel to either of CD or ML . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since twice the (rectangle contained) by AG and GB is medial, and is equal to FL , FL is thus also medial. And it is applied to the rational (straight-line) FE , producing FM as breadth. Thus, FM is rational, and incommensurable in length with CD [Prop. 10.22]. And since the sum of the (squares) on AG and GB is rational, and twice the (rectangle contained) by AG and GB medial, the (sum of the squares) on AG and GB is [thus] incommensurable with twice the (rectangle contained) by AG and GB . And CL (is) equal to the (sum of the squares) on AG and GB , and FL equal to twice the (rectangle contained) by AG and GB . CL [is] thus incommensurable with FL . And as CL (is) to FL , so CM is to MF [Prop. 6.1]. CM is thus incommensurable in length with MF [Prop. 10.11]. And both are rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. [So], I say that (it is) also a fourth (apotome).

For since AG and GB are incommensurable in square, the (square) on AG (is) thus also incommensurable with the (square) on GB . And CH is equal to the (square) on AG , and KL equal to the (square) on GB . Thus, CH is incommensurable with KL . And as CH (is) to KL , so CK is to KM [Prop. 6.1]. CK is thus incommensurable in length with KM [Prop. 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and the (square) on AG is equal to CH , and the (square) on GB to KL , and the (rectangle contained) by AG and GB to NL , NL is thus the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . But, as CH (is) to NL , so CK is to NM , and as NL (is) to KL , so NM is to KM [Prop. 6.1]. Thus, as CK (is) to NM , so NM is to KM [Prop. 5.11]. The (rectangle contained) by CK and KM is thus equal to the (square) on MN —that is to say, to the fourth part of the (square) on FM [Prop. 6.17]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM , equal to the fourth part of the (square) on MF , has been applied to CM , falling short by a square figure, and divides it into incommensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And the whole of CM is commensurable in length with the (previously) laid down rational (straight-line) CD . Thus, CF is a fourth apotome [Def. 10.14].

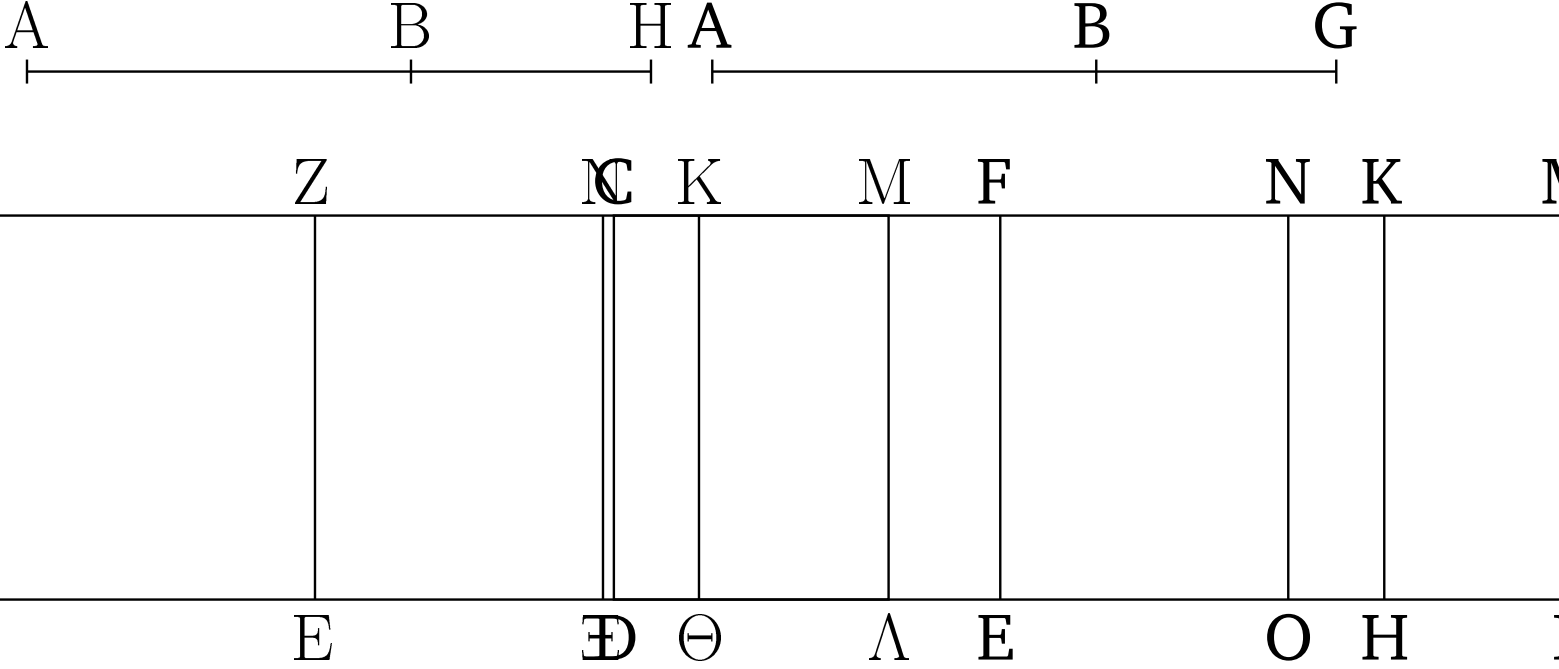
Thus, the (square) on a minor, and so on ...

101.

Proposition 101

Τὸ ἀπὸ τῆς μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσῃ παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν πέμπτην.

The (square) on that (straight-line) which with a rational (area) makes a medial whole, applied to a rational (straight-line), produces a fifth apotome as breadth.



Ἐστω ἡ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἡ AB , ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΕ$ πλάτος ποιοῦν τὴν $ΓΖ$ · λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστι πέμπτη.

Ἐστω γάρ τῃ AB προσαρμόζονσα ἡ BH · αἱ ἄρα AH , HB εὐθεῖαι δυνάμει εἰσὶν ἀσύμμετροι ποιούσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον, τὸ δὲ διξ ὑπ' αὐτῶν ῥητόν, καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΘ$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $ΚΛ$ · ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB . τὸ δὲ συγκείμενον ἐκ τῶν ἀπὸ τῶν AH , HB ἅμα μέσον ἐστὶν μέσον ἄρα ἐστὶ τὸ $ΓΛ$. καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παράκειται πλάτος ποιοῦν τὴν $ΓΜ$ · ῥητὴ ἄρα ἐστὶν ἡ $ΓΜ$ καὶ ἀσύμμετρος τῇ $ΓΔ$. καὶ ἐπεὶ ὅλον τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , ὧν τὸ $ΓΕ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB , λοιπὸν ἄρα τὸ $ΖΛ$ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AH , HB . τετμήσθω οὖν ἡ ZM διὰ κατὰ τὸ N , καὶ ἦθθω διὰ τοῦ N ὁποτέρῃ τῶν $ΓΔ$, $ΜΛ$ παράλληλος ἡ NX · ἐκάτερον ἄρα τῶν ZX , $NΛ$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB , καὶ ἐπεὶ τὸ διξ ὑπὸ τῶν AH , HB ῥητόν ἐστι καὶ [ἐστίν] ἴσον τῷ $ΖΛ$, ῥητόν ἄρα ἐστὶ τὸ $ΖΛ$. καὶ παρὰ ῥητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν ZM · ῥητὴ ἄρα ἐστὶν ἡ ZM καὶ σύμμετρος τῇ $ΓΔ$ μήκει. καὶ ἐπεὶ τὸ μὲν $ΓΛ$ μέσον ἐστὶν, τὸ δὲ $ΖΛ$ ῥητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ $ΓΛ$ τῷ $ΖΛ$. ὥς δὲ τὸ $ΓΛ$ πρόξ τὸ $ΖΛ$, οὕτως ἡ $ΓΜ$ πρόξ τὴν MZ · ἀσύμμετρος ἄρα ἐστὶν ἡ $ΓΜ$ τῇ MZ μήκει. καὶ εἰσὶν ἀμφοτέραι ῥηταί· αἱ ἄρα $ΓΜ$,

Let AB be that (straight-line) which with a rational (area) makes a medial whole, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a fifth apotome.

Let BG be an attachment to AB . Thus, the straight-lines AG and GB are incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by them rational [Prop. 10.77]. And let CH , equal to the (square) on AG , be applied to CD , and KL , equal to the (square) on GB . The whole of CL is thus equal to the (sum of the squares) on AG and GB . And the sum of the (squares) on AG and GB together is medial. Thus, CL is medial. And it has been applied to the rational (straight-line) CD , producing CM as breadth. CM is thus rational, and incommensurable (in length) with CD [Prop. 10.22]. And since the whole of CL is equal to the (sum of the squares) on AG and GB , of which CE is equal to the (square) on AB , the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at N . And let NO be drawn through N , parallel to either of CD or ML . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since twice the (rectangle contained) by AG and GB is rational, and [is] equal to FL , FL is thus rational. And it is applied to the rational (straight-line) EF , producing FM as breadth. Thus, FM is rational, and commensurable in length with CD .

MZ ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἢ FZ . λέγω δὴ, ὅτι καὶ πέμπτη.

Ὅμοίως γὰρ δείχουμεν, ὅτι τὸ ὑπὸ τῶν $ΓΚΜ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς NM , τοντέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM . καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB , ἴσον δὲ τὸ μὲν ἀπὸ τῆς AH τῷ $ΓΘ$, τὸ δὲ ἀπὸ τῆς HB τῷ $ΚΛ$, ἀσύμμετρον ἄρα τὸ $ΓΘ$ τῷ $ΚΛ$. ὥς δὲ τὸ $ΓΘ$ πρὸς τὸ $ΚΛ$, οὕτως ἢ $ΓΚ$ πρὸς τὴν KM · ἀσύμμετρος ἄρα ἢ $ΓΚ$ τῇ KM μήκει. ἐπεὶ οὖν δύο εὐθεῖαι ἄνισοί εἰσιν αἱ $ΓΜ$, MZ , καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM ἴσον παρὰ τὴν $ΓΜ$ παραβέβληται ἐλλείπον εἶδει τετραγώνῳ καὶ εἰς ἀσύμμετρα αὐτὴν διαιρεῖ, ἢ ἄρα $ΓΜ$ τῆς MZ μείζον δύναται τῷ ἀπὸ ἀσύμμετρον ἐαντῇ. καὶ ἐστὶν ἢ προσαρμόζουσα ἢ ZM σύμμετρος τῇ ἐκκειμένῃ ῥητῇ τῇ $ΓΔ$ · ἢ ἄρα $ΓΖ$ ἀποτομή ἐστὶ πέμπτη· ὅπερ ἔδει δεῖχαι.

[Prop. 10.20]. And since CL is medial, and FL rational, CL is thus incommensurable with FL . And as CL (is) to FL , so CM (is) to MF [Prop. 6.1]. CM is thus incommensurable in length with MF [Prop. 10.11]. And both are rational. Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a fifth (apotome).

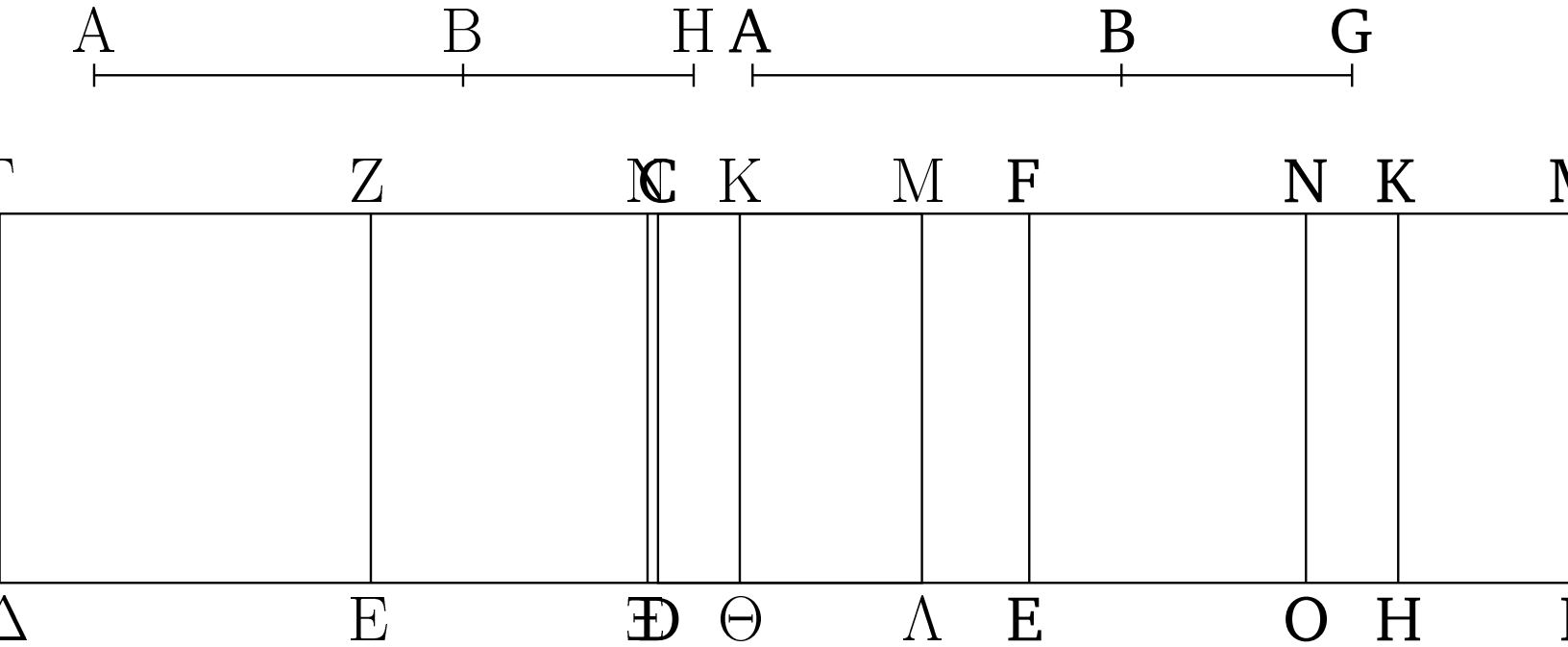
For, similarly (to the previous propositions), we can show that the (rectangle contained) by CKM is equal to the (square) on NM —that is to say, to the fourth part of the (square) on FM . And since the (square) on AG is incommensurable with the (square) on GB , and the (square) on AG (is) equal to CH , and the (square) on GB to KL , CH (is) thus incommensurable with KL . And as CH (is) to KL , so CK (is) to KM [Prop. 6.1]. Thus, CK (is) incommensurable in length with KM [Prop. 10.11]. Therefore, since CM and MF are two unequal straight-lines, and (some area), equal to the fourth part of the (square) on FM , has been applied to CM , falling short by a square figure, and divides it into incommensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And the attachment FM is commensurable with the (previously) laid down rational (straight-line) CD . Thus, CF is a fifth apotome [Def. 10.15]. (Which is) the very thing it was required to show.

102.

Τὸ ἀπὸ τῆς μετὰ μέσον μέσον τὸ ὅλον ποιούσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν ἕκτην.

Proposition 102

The (square) on that (straight-line) which with a medial (area) makes a medial whole, applied to a rational (straight-line), produces a sixth apotome as breadth.



Ἐστω ἡ μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἡ AB , ῥητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΕ$ πλάτος ποιῶν τὴν $ΓΖ$ · λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστὶν ἕκτη.

Ἐστω γάρ τῃ AB προσαρμόζονσα ἡ BH · αἱ ἄρα AH , HB δυνάμει εἰσὶν ἀσύμμετροι ποιῶσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ διξ ὑπὸ τῶν AH , HB μέσον καὶ ἀσύμμετρον τὰ ἀπὸ τῶν AH , HB τῷ διξ ὑπὸ τῶν AH , HB · παραβεβλήσθω οὖν παρὰ τὴν $ΓΔ$ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $ΓΘ$ πλάτος ποιῶν τὴν $ΓΚ$, τῷ δὲ ἀπὸ τῆς BH τὸ $ΚΑ$ · ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB · μέσον ἄρα [ἐστὶ] καὶ τὸ $ΓΛ$ · καὶ παρὰ ῥητὴν τὴν $ΓΔ$ παράκειται πλάτος ποιῶν τὴν $ΓΜ$ · ῥητὴ ἄρα ἐστὶν ἡ $ΓΜ$ καὶ ἀσύμμετρος τῇ $ΓΔ$ μήκει· ἐπεὶ οὖν τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH , HB , ὧν τὸ $ΓΕ$ ἴσον τῷ ἀπὸ τῆς AB , λοιπὸν ἄρα τὸ $ΖΛ$ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AH , HB · καὶ ἐστὶ τὸ διξ ὑπὸ τῶν AH , HB μέσον καὶ τὸ $ΖΛ$ ἄρα μέσον ἐστίν· καὶ παρὰ ῥητὴν τὴν $ΖΕ$ παράκειται πλάτος ποιῶν τὴν $ΖΜ$ · ῥητὴ ἄρα ἐστὶν ἡ $ΖΜ$ καὶ ἀσύμμετρος τῇ $ΓΔ$ μήκει· καὶ ἐπεὶ τὰ ἀπὸ τῶν AH , HB ἀσύμμετρά ἐστι τῷ διξ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τοῖς μὲν ἀπὸ τῶν AH , HB ἴσον τὸ $ΓΛ$, τῷ δὲ διξ ὑπὸ τῶν AH , HB ἴσον τὸ $ΖΛ$, ἀσύμμετρος ἄρα [ἐστὶ] τὸ $ΓΛ$ τῷ $ΖΛ$ · ὥς δὲ τὸ $ΓΛ$ πρὸς τὸ $ΖΛ$, οὕτως ἐστὶν ἡ $ΓΜ$ πρὸς τὴν $ΜΖ$ · ἀσύμμετρος ἄρα ἐστὶν ἡ $ΓΜ$ τῇ $ΜΖ$ μήκει· καὶ εἰσὶν ἀμφότεραι ῥηταί· αἱ $ΓΜ$, $ΜΖ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ $ΓΖ$ · λέγω δὴ, ὅτι καὶ ἕκτη.

Ἐπεὶ γάρ τὸ $ΖΛ$ ἴσον ἐστὶ τῷ διξ ὑπὸ τῶν AH , HB , τετμήσθω δίθῃ ἡ $ΖΜ$ κατὰ τὸ N , καὶ ἡθθω διὰ τοῦ N τῇ $ΓΔ$ παράλληλος ἡ NX · ἐκάτερον ἄρα τῶν ZX , $ΝΑ$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH , HB · καὶ ἐπεὶ αἱ AH , HB δυνάμει εἰσὶν ἀσύμμε-

Let AB be that (straight-line) which with a medial (area) makes a medial whole, and CD a rational (straight-line). And let CE , equal to the (square) on AB , be applied to CD , producing CF as breadth. I say that CF is a sixth apotome.

For let BG be an attachment to AB . Thus, AG and GB are incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by AG and GB medial, and the (sum of the squares) on AG and GB incommensurable with twice the (rectangle contained) by AG and GB [Prop. 10.78]. Therefore, let CH , equal to the (square) on AG , be applied to CD , producing CK as breadth, and KL , equal to the (square) on BG . Thus, the whole of CL is equal to the (sum of the squares) on AG and GB . CL [is] thus also medial. And it is applied to the rational (straight-line) CD , producing CM as breadth. Thus, CM is rational, and incommensurable in length with CD [Prop. 10.22]. Therefore, since CL is equal to the (sum of the squares) on AG and GB , of which CE (is) equal to the (square) on AB , the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. And twice the (rectangle contained) by AG and GB (is) medial. Thus, FL is also medial. And it is applied to the rational (straight-line) FE , producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since the (sum of the squares) on AG and GB is incommensurable with twice the (rectangle contained) by AG and GB , and CL equal to the (sum of the squares) on AG and GB , and FL equal to twice the (rectangle contained) by AG and GB , CL [is] thus incommensurable with FL . And as CL (is) to FL , so CM is to MF [Prop. 6.1]. Thus, CM is incommensurable in length with MF [Prop. 10.11]. And they

τροι, ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB . ἀλλὰ τῷ μὲν ἀπὸ τῆς AH ἴσον ἐστὶ τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον ἐστὶ τὸ $ΚΛ$. ἀσύμμετρον ἄρα ἐστὶ τὸ $\Gamma\Theta$ τῷ $ΚΛ$. ὥς δὲ τὸ $\Gamma\Theta$ πρὸς τὸ $ΚΛ$, οὕτως ἐστὶν ἡ $\GammaΚ$ πρὸς τὴν $ΚΜ$. ἀσύμμετρος ἄρα ἐστὶν ἡ $\GammaΚ$ τῇ $ΚΜ$. καὶ ἐπεὶ τῶν ἀπὸ τῶν AH , HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH , HB , καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $ΚΛ$, τῷ δὲ ὑπὸ τῶν AH , HB ἴσον τὸ $ΝΛ$, καὶ τῶν ἄρα $\Gamma\Theta$, $ΚΛ$ μέσον ἀνάλογόν ἐστι τὸ $ΝΛ$. ἐστὶν ἄρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $ΝΛ$, οὕτως τὸ $ΝΛ$ πρὸς τὸ $ΚΛ$. καὶ διὰ τὰ αὐτὰ ἡ $\GammaΜ$ τῆς $ΜΖ$ μείζον δύνανται τῷ ἀπὸ ἀσυνμέτρου ἐαντῇ. καὶ οὐδετέρω αὐτῶν σύμμετρος ἐστὶ τῇ ἐκκειμένη ῥητῇ τῇ $\GammaΔ$. ἡ $\GammaΖ$ ἄρα ἀποτομή ἐστὶν ἑκτῇ· ὅπερ εἶδει δεῖχαι.

are both rational. Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a sixth (apotome).

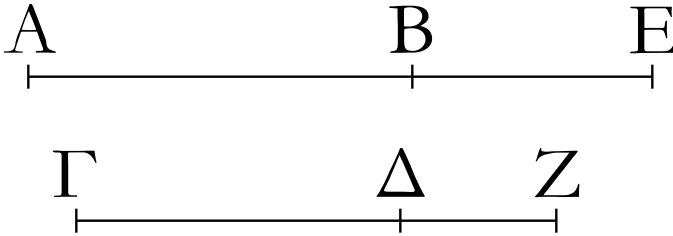
For since FL is equal to twice the (rectangle contained) by AG and GB , let FM be cut in half at N , and let NO be drawn through N , parallel to CD . Thus, FO and NL are each equal to the (rectangle contained) by AG and GB . And since AG and GB are incommensurable in square, the (square) on AG is thus incommensurable with the (square) on GB . But, CH is equal to the (square) on AG , and KL is equal to the (square) on GB . Thus, CH is incommensurable with KL . And as CH (is) to KL , so CK is to KM [Prop. 6.1]. Thus, CK is incommensurable (in length) with KM [Prop. 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG , and KL equal to the (square) on GB , and NL equal to the (rectangle contained) by AG and GB , NL is thus also the mean proportional to CH and KL . Thus, as CH is to NL , so NL (is) to KL . And for the same (reasons as the preceding propositions), the square on CM is greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And neither of them is commensurable with the (previously) laid down rational (straight-line) CD . Thus, CF is a sixth apotome [Def. 10.16]. (Which is) the very thing it was required to show.

103.

Proposition 103

Ἡ τῇ ἀποτομῇ μήκει σύμμετρος ἀποτομή ἐστὶ καὶ τῇ τάξει ἡ αὐτή.

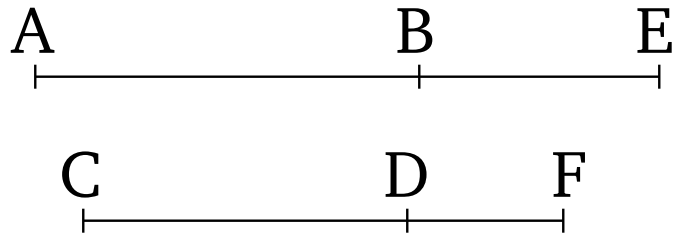
Ἐστω ἀποτομή ἡ AB , καὶ τῇ AB μήκει σύμμετρος ἔστω ἡ $\GammaΔ$. λέγω, ὅτι καὶ ἡ $\GammaΔ$ ἀποτομή ἐστὶ καὶ τῇ τάξει ἡ αὐτή τῇ AB .



Ἐπεὶ γὰρ ἀποτομή ἐστὶν ἡ AB , ἔστω αὐτῇ προσαρμόζουσα ἡ BE . αἱ AE , EB ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ τῆς AB πρὸς τὴν $\GammaΔ$ λόγῳ ὁ αὐτὸς γερονέτω ὁ τῆς BE πρὸς τὴν $\Delta Ζ$. καὶ ὥς ἐν ἄρα πρὸς ἓν, πάντα [ἐστὶ] πρὸς πάντα. ἐστὶν ἄρα καὶ ὥς ὅλη ἡ AE πρὸς ὅλην τὴν $\GammaΖ$, οὕτως ἡ AB πρὸς τὴν $\GammaΔ$. σύμμετρος δὲ ἡ AB τῇ $\GammaΔ$ μήκει. σύμμετρος ἄρα καὶ ἡ AE μὲν τῇ $\GammaΖ$, ἡ δὲ BE τῇ $\Delta Ζ$. καὶ αἱ AE , EB ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ αἱ $\GammaΖ$, $\Delta Ζ$

A (straight-line) commensurable in length with an apotome is an apotome, and (is) the same in order.

Let AB be an apotome, and let CD be commensurable in length with AB . I say that CD is also an apotome, and (is) the same in order as AB .



For since AB is an apotome, let BE be an attachment to it. Thus, AE and EB are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let it be contrived that the (ratio) of BE to DF is the same as the ratio of AB to CD [Prop. 6.12]. Thus, also, as one is to one, (so) all [are] to all [Prop. 5.12]. And thus as the whole AE is to the whole CF , so AB (is) to CD . And AB (is) commensurable in length with CD . AE (is) thus also commensurable (in

ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι [ἀποτομή ἄρα ἐστὶν ἡ ΓΔ. λέγω δὴ, ὅτι καὶ τῇ τάχει ἡ αὐτὴ τῇ ΑΒ].

Ἐπεὶ οὖν ἐστὶν ὥς ἡ ΑΕ πρὸς τὴν ΓΖ, οὕτως ἡ ΒΕ πρὸς τὴν ΔΖ, ἐναλλάξ ἄρα ἐστὶν ὥς ἡ ΑΕ πρὸς τὴν ΕΒ, οὕτως ἡ ΓΖ πρὸς τὴν ΖΔ. ἦτοι δὴ ἡ ΑΕ τῆς ΕΒ μείζον δύνανται τῷ ἀπὸ συμμέτρου ἐναντῇ ἢ τῷ ἀπὸ ἀσυνμέτρου. εἰ μὲν οὖν ἡ ΑΕ τῆς ΕΒ μείζον δύνανται τῷ ἀπὸ συμμέτρου ἐναντῇ, καὶ ἡ ΓΖ τῆς ΖΔ μείζον δυνήσεται τῷ ἀπὸ συμμέτρου ἐναντῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ ΑΕ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ ΓΖ, εἰ δὲ ἡ ΒΕ, καὶ ἡ ΔΖ, εἰ δὲ οὐδετέρα τῶν ΑΕ, ΕΒ, καὶ οὐδετέρα τῶν ΓΖ, ΖΔ. εἰ δὲ ἡ ΑΕ [τῆς ΕΒ] μείζον δύνανται τῷ ἀπὸ ἀσυνμέτρου ἐναντῇ, καὶ ἡ ΓΖ τῆς ΖΔ μείζον δυνήσεται τῷ ἀπὸ ἀσυνμέτρου ἐναντῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ ΑΕ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ ΓΖ, εἰ δὲ ἡ ΒΕ, καὶ ἡ ΔΖ, εἰ δὲ οὐδετέρα τῶν ΑΕ, ΕΒ, οὐδετέρα τῶν ΓΖ, ΖΔ.

Ἀποτομή ἄρα ἐστὶν ἡ ΓΔ καὶ τῇ τάχει ἡ αὐτὴ τῇ ΑΒ· ὅπερ ἔδει δεῖξαι.

length) with CF , and BE with DF [Prop. 10.11]. And AE and BE are rational (straight-lines which are) commensurable in square only. Thus, CF and FD are also rational (straight-lines which are) commensurable in square only [Prop. 10.13]. [CD is thus an apotome. So, I say that (it is) also the same in order as AB .]

Therefore, since as AE is to CF , so BE (is) to DF , thus, alternately, as AE is to EB , so CF (is) to FD [Prop. 5.16]. So, the square on AE is greater than (the square on) EB either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (AE) . Therefore, if the (square) on AE is greater than (the square on) EB by the (square) on (some straight-line) commensurable (in length) with (AE) then the square on CF will also be greater than (the square on) FD by the (square) on (some straight-line) commensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable in length with a (previously) laid down rational (straight-line) then so (is) CF [Prop. 10.12], and if BE (is commensurable), so (is) DF , and if neither of AE or EB (are commensurable), neither (are) either of CF or FD [Prop. 10.13]. And if the (square) on AE is greater [than (the square on) EB] by the (square) on (some straight-line) incommensurable (in length) with (AE) then the (square) on CF will also be greater than (the square on) FD by the (square) on (some straight-line) incommensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable in length with a (previously) laid down rational (straight-line), so (is) CF [Prop. 10.12], and if BE (is commensurable), so (is) DF , and if neither of AE or EB (are commensurable), neither (are) either of CF or FD [Prop. 10.13].

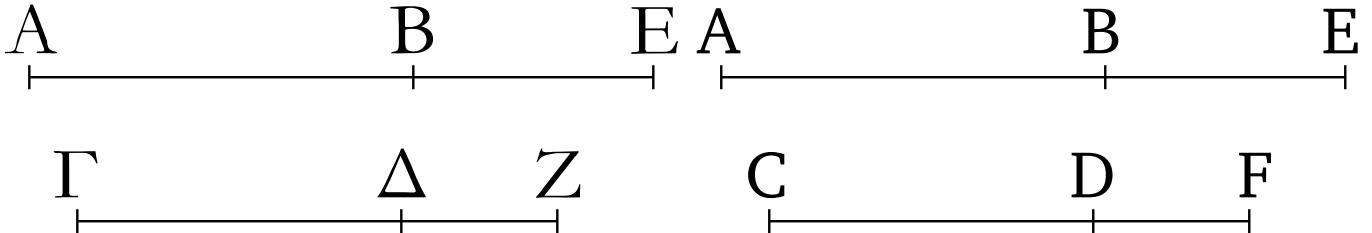
Thus, CD is an apotome, and (is) the same in order as AB [Defs. 10.11—10.16]. (Which is) the very thing it was required to show.

104.

Proposition 104

Ἡ τῇ μέσῃ ἀποτομῇ σύμμετρος μέσῃ ἀποτομῇ ἐστὶ καὶ τῇ τάχει ἡ αὐτή.

A (straight-line) commensurable (in length) with an apotome of a medial (straight-line) is an apotome of a medial (straight-line), and (is) the same in order.



Ἐστω μέσῃ ἀποτομῇ ἡ ΑΒ, καὶ τῇ ΑΒ μήκει σύμμετρος ἔστω ἡ ΓΔ· λέγω, ὅτι καὶ ἡ ΓΔ μέσῃ ἀποτομῇ ἐστὶ καὶ τῇ τάχει ἡ αὐτὴ τῇ ΑΒ.

Ἐπεὶ γὰρ μέσῃ ἀποτομῇ ἐστὶν ἡ ΑΒ, ἔστω αὐτῇ προσ-

Let AB be an apotome of a medial (straight-line), and let CD be commensurable in length with AB . I say that CD is also an apotome of a medial (straight-line), and (is) the same in order as AB .

αρμόζονσα ἡ EB . αἱ AE , EB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ γερονέτω ὡς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ BE πρὸς τὴν ΔZ · σύμμετρος ἄρα [ἐστὶ] καὶ ἡ AE τῇ ΓZ , ἡ δὲ BE τῇ ΔZ . αἱ δὲ AE , EB μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· καὶ αἱ ΓZ , $Z\Delta$ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· μέσης ἄρα ἀποτομή ἐστὶν ἡ $\Gamma\Delta$. λέγω δὴ, ὅτι καὶ τῇ τάχει ἐστὶν ἡ αὐτὴ τῇ AB .

Ἐπεὶ [γάρ] ἐστὶν ὡς ἡ AE πρὸς τὴν EB , οὕτως ἡ ΓZ πρὸς τὴν $Z\Delta$ [ἀλλ' ὡς μὲν ἡ AE πρὸς τὴν EB , οὕτως τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE , EB , ὡς δὲ ἡ ΓZ πρὸς τὴν $Z\Delta$, οὕτως τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν ΓZ , $Z\Delta$], ἐστὶν ἄρα καὶ ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE , EB , οὕτως τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν ΓZ , $Z\Delta$ [καὶ ἐναλλάχῃ ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς ΓZ , οὕτως τὸ ὑπὸ τῶν AE , EB πρὸς τὸ ὑπὸ τῶν ΓZ , $Z\Delta$]. σύμμετρον δὲ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς ΓZ · σύμμετρον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν AE , EB τῷ ὑπὸ τῶν ΓZ , $Z\Delta$. εἴτε οὖν ῥητόν ἐστι τὸ ὑπὸ τῶν AE , EB , ῥητόν ἐστὶ καὶ τὸ ὑπὸ τῶν ΓZ , $Z\Delta$, εἴτε μέσον [ἐστὶ] τὸ ὑπὸ τῶν AE , EB , μέσον [ἐστὶ] καὶ τὸ ὑπὸ τῶν ΓZ , $Z\Delta$.

Μέσης ἄρα ἀποτομή ἐστὶν ἡ $\Gamma\Delta$ καὶ τῇ τάχει ἡ αὐτὴ τῇ AB · ὅπερ ἔδει δεῖχαι.

For since AB is an apotome of a medial (straight-line), let EB be an attachment to it. Thus, AE and EB are medial (straight-lines which are) commensurable in square only [Props. 10.74, 10.75]. And let it be contrived that as AB is to CD , so BE (is) to DF [Prop. 6.12]. Thus, AE [is] also commensurable (in length) with CF , and BE with DF [Props. 5.12, 10.11]. And AE and EB are medial (straight-lines which are) commensurable in square only. CF and FD are thus also medial (straight-lines which are) commensurable in square only [Props. 10.23, 10.13]. Thus, CD is an apotome of a medial (straight-line) [Props. 10.74, 10.75]. So, I say that it is also the same in order as AB .

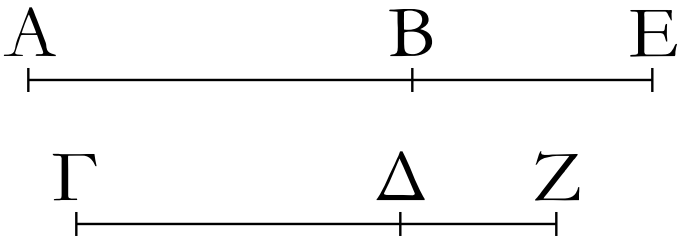
[For] since as AE is to EB , so CF (is) to FD [Props. 5.12, 5.16] [but as AE (is) to EB , so the (square) on AE (is) to the (rectangle contained) by AE and EB , and as CF (is) to FD , so the (square) on CF (is) to the (rectangle contained) by CF and FD], thus as the (square) on AE is to the (rectangle contained) by AE and EB , so the (square) on CF also (is) to the (rectangle contained) by CF and FD [Prop. 10.21 lem.] [and, alternately, as the (square) on AE (is) to the (square) on CF , so the (rectangle contained) by AE and EB (is) to the (rectangle contained) by CF and FD]. And the (square) on AE (is) commensurable with the (square) on CF . Thus, the (rectangle contained) by AE and EB is also commensurable with the (rectangle contained) by CF and FD [Props. 5.16, 10.11]. Therefore, either the (rectangle contained) by AE and EB is rational, and the (rectangle contained) by CF and FD will also be rational [Def. 10.4], or the (rectangle contained) by AE and EB [is] medial, and the (rectangle contained) by CF and FD [is] also medial [Prop. 10.23 corr.].

Therefore, CD is the apotome of a medial (straight-line), and is the same in order as AB [Props. 10.74, 10.75]. (Which is) the very thing it was required to show.

105.

Ἡ τῇ ἐλάσσονι σύμμετρος ἐλάσσων ἐστίν.

Ἐστω γὰρ ἐλάσσων ἡ AB καὶ τῇ AB σύμμετρος ἡ $\Gamma\Delta$. λέγω, ὅτι καὶ ἡ $\Gamma\Delta$ ἐλάσσων ἐστίν.

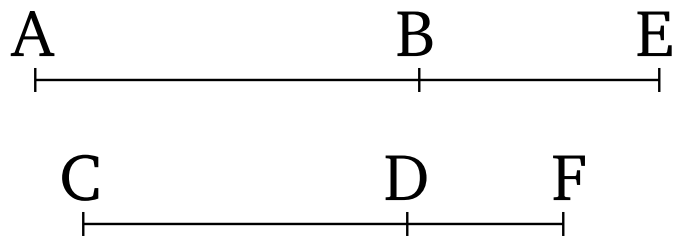


Γερονέτω γὰρ τὰ αὐτά· καὶ ἐπεὶ αἱ AE , EB δυνάμει εἰσὶν ἀσύμμετροι, καὶ αἱ ΓZ , $Z\Delta$ ἄρα δυνάμει εἰσὶν ἀσύμμετροι.

Proposition 105

A (straight-line) commensurable (in length) with a minor (straight-line) is a minor (straight-line).

For let AB be a minor (straight-line), and (let) CD (be) commensurable (in length) with AB . I say that CD is also a minor (straight-line).



For let the same things be contrived (as in the former proposition). And since AE and EB are (straight-lines which

ἐπεὶ οὖν ἔστιν ὥς ἡ AE πρὸς τὴν EB , οὕτως ἡ GZ πρὸς τὴν ZD , ἔστιν ἄρα καὶ ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς EB , οὕτως τὸ ἀπὸ τῆς GZ πρὸς τὸ ἀπὸ τῆς ZD . συνθέντι ἄρα ἔστιν ὥς τὰ ἀπὸ τῶν AE , EB πρὸς τὸ ἀπὸ τῆς EB , οὕτως τὰ ἀπὸ τῶν GZ , ZD πρὸς τὸ ἀπὸ τῆς ZD [καὶ ἐναλλάξ]. σύμμετρον δὲ ἔστι τὸ ἀπὸ τῆς BE τῷ ἀπὸ τῆς ΔZ · σύμμετρον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν GZ , ZD τετραγώνων. ῥητόν δὲ ἔστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων· ῥητόν ἄρα ἔστι καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν GZ , ZD τετραγώνων. πάλιν, ἐπεὶ ἔστιν ὥς τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE , EB , οὕτως τὸ ἀπὸ τῆς GZ πρὸς τὸ ὑπὸ τῶν GZ , ZD , σύμμετρον δὲ τὸ ἀπὸ τῆς AE τετραγώνων τῷ ἀπὸ τῆς GZ τετραγώνων, σύμμετρον ἄρα ἔστι καὶ τὸ ὑπὸ τῶν AE , EB τῷ ὑπὸ τῶν GZ , ZD . μέσον δὲ τὸ ὑπὸ τῶν AE , EB · μέσον ἄρα καὶ τὸ ὑπὸ τῶν GZ , ZD · αἱ GZ , ZD ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων ῥητόν, τὸ δ' ὑπ' αὐτῶν μέσον.

Ἐλάσσων ἄρα ἔστιν ἡ $\Gamma\Delta$ · ὅπερ ἔδει δεῖλαι.

106.

Ἡ τῇ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούση σύμμετρος μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἔστιν.

Ἐστω μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἡ AB καὶ τῇ AB σύμμετρος ἡ $\Gamma\Delta$ · λέγω, ὅτι καὶ ἡ $\Gamma\Delta$ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἔστιν.

Ἐστω γὰρ τῇ AB προσαρμόζουσα ἡ BE · αἱ AE , EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν. καὶ τὰ αὐτὰ κατεσκευάσθω. ὁμοίως δὲ δείχομεν τοῖς πρότερον, ὅτι αἱ GZ , ZD ἐν τῷ αὐτῷ λόγῳ εἰσὶ ταῖς AE , EB , καὶ σύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE , EB τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν GZ , ZD τετραγώνων, τὸ δὲ ὑπὸ τῶν AE , EB τῷ ὑπὸ τῶν GZ , ZD · ὥστε καὶ αἱ GZ , ZD δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν GZ , ZD τετραγώνων μέσον, τὸ δ' ὑπ' αὐτῶν ῥητόν.

are) incommensurable in square [Prop. 10.76], CF and FD are thus also (straight-lines which are) incommensurable in square [Prop. 10.13]. Therefore, since as AE is to EB , so CF (is) to FD [Props. 5.12, 5.16], thus also as the (square) on AE is to the (square) on EB , so the (square) on CF (is) to the (square) on FD [Prop. 6.22]. Thus, via composition, as the (sum of the squares) on AE and EB is to the (square) on EB , so the (sum of the squares) on CF and FD (is) to the (square) on FD [Prop. 5.18], [also alternately]. And the (square) on BE is commensurable with the (square) on DF [Prop. 10.104]. The sum of the squares on AE and EB (is) thus also commensurable with the sum of the squares on CF and FD [Prop. 5.16, 10.11]. And the sum of the (squares) on AE and EB is rational [Prop. 10.76]. Thus, the sum of the (squares) on CF and FD is also rational [Def. 10.4]. Again, since as the (square) on AE is to the (rectangle contained) by AE and EB , so the (square) on CF (is) to the (rectangle contained) by CF and FD [Prop. 10.21 lem.], and the square on AE (is) commensurable with the square on CF , the (rectangle contained) by AE and EB is thus also commensurable with the (rectangle contained) by CF and FD . And the (rectangle contained) by AE and EB (is) medial [Prop. 10.76]. Thus, the (rectangle contained) by CF and FD (is) also medial [Prop. 10.23 corr.]. CF and FD are thus (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial.

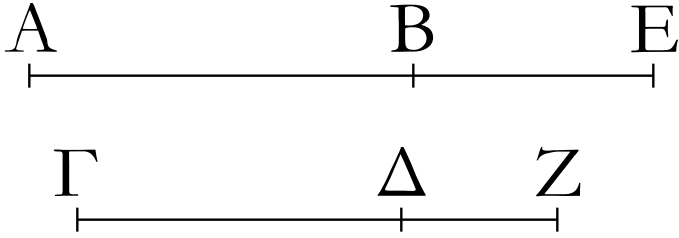
Thus, CD is a minor (straight-line) [Prop. 10.76]. (Which is) the very thing it was required to show.

Proposition 106

A (straight-line) commensurable (in length) with a (straight-line) which with a rational (area) makes a medial whole is a (straight-line) which with a rational (area) makes a medial whole.

Let AB be a (straight-line) which with a rational (area) makes a medial whole, and (let) CD (be) commensurable (in length) with AB . I say that CD is also a (straight-line) which with a rational (area) makes a medial (whole).

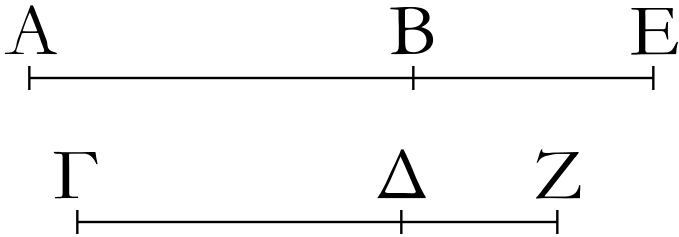
For let BE be an attachment to AB . Thus, AE and EB are (straight-lines which are) incommensurable in square, making the sum of the squares on AE and EB medial, and the (rectangle contained) by them rational [Prop. 10.77]. And let the same construction be made (as in the previous propositions). So, similarly to the previous (propositions), we can show that CF and FD are in the same ratio as AE and EB , and the sum of the squares on AE and EB is commensurable with the sum of the squares on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Hence, CF and FD are also (straight-lines which are) incommensurable in square, making the sum of the squares on CF



Ἡ ΓΔ ἄρα μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἐστίν· ὅπερ ἔδει δεῖχαι.

107.

Ἡ τῇ μετὰ μέσον μέσον τὸ ὅλον ποιούση σύμμετρος καὶ αὐτὴ μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἐστίν.

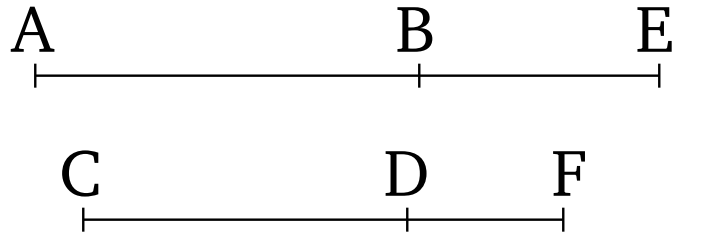


Ἐστω μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἡ ΑΒ, καὶ τῇ ΑΒ ἔστω σύμμετρος ἡ ΓΔ· λέγω, ὅτι καὶ ἡ ΓΔ μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἐστίν.

Ἐστω γάρ τῇ ΑΒ προσαρμόζονσα ἡ ΒΕ, καὶ τὰ αὐτὰ κατεσκευάσθω· αἱ ΑΕ, ΕΒ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιούσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων τῷ ὑπ' αὐτῶν. καὶ εἰσιν, ὥς ἐδείκθη, αἱ ΑΕ, ΕΒ σύμμετροι ταῖς ΓΖ, ΖΔ, καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΕ, ΕΒ τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν ΓΖ, ΖΔ, τὸ δὲ ὑπὸ τῶν ΑΕ, ΕΒ τῷ ὑπὸ τῶν ΓΖ, ΖΔ· καὶ αἱ ΓΖ, ΖΔ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιούσαι τὸ τε συγκείμενον ἐκ τῶν ἀπ' αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπ' αὐτῶν μέσον καὶ ἔτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπ' αὐτῶν [τετραγώνων] τῷ ὑπ' αὐτῶν.

Ἡ ΓΔ ἄρα μετὰ μέσον μέσον τὸ ὅλον ποιούσα ἐστίν· ὅπερ ἔδει δεῖχαι.

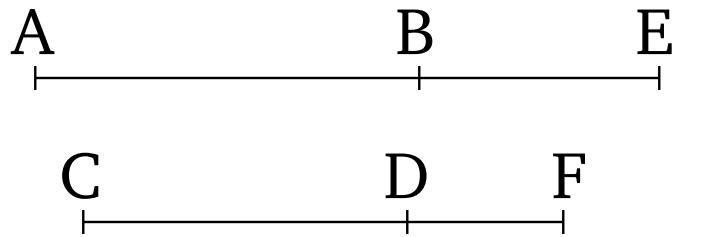
and FD medial, and the (rectangle contained) by them rational.



CD is thus a (straight-line) which with a rational (area) makes a medial whole [Prop. 10.77]. (Which is) the very thing it was required to show.

Proposition 107

A (straight-line) commensurable (in length) with a (straight-line) which with a medial (area) makes a medial whole is itself also a (straight-line) which with a medial (area) makes a medial whole.



Let AB be a (straight-line) which with a medial (area) makes a medial whole, and let CD be commensurable (in length) with AB . I say that CD is also a (straight-line) which with a medial (area) makes a medial whole.

For let BE be an attachment to AB . And let the same construction be made (as in the previous propositions). Thus, AE and EB are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, further, the sum of the squares on them incommensurable with the (rectangle contained) by them [Prop. 10.78]. And, as was shown (previously), AE and EB are commensurable (in length) with CF and FD (respectively), and the sum of the squares on AE and EB with the sum of the squares on CF and FD , and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD . Thus, CF and FD are also (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, further, the sum of the [squares] on them incommensurable with the (rectangle contained) by them.

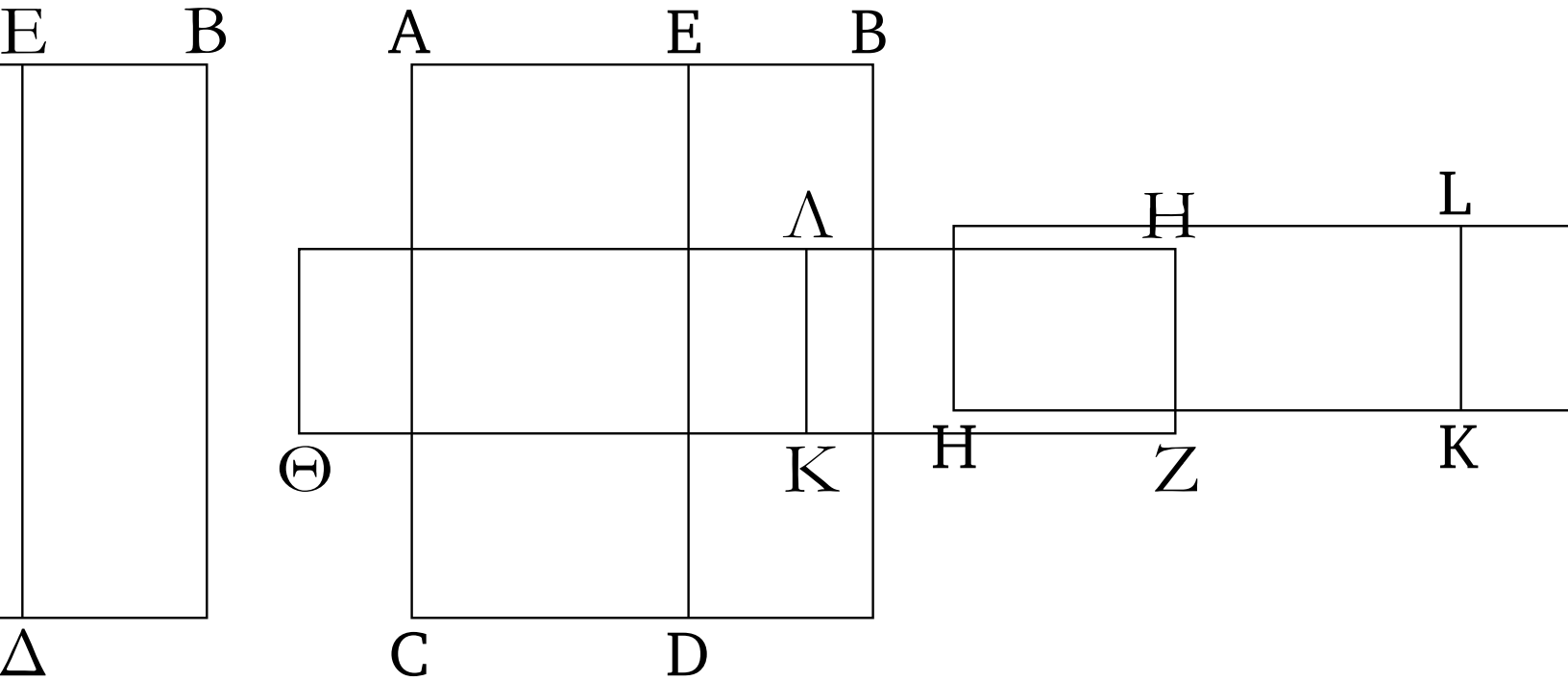
Thus, CD is a (straight-line) which with a medial (area) makes a medial whole [Prop. 10.78]. (Which is) the very thing it was required to show.

108.

Proposition 108

Ἀπὸ ῥητοῦ μέσον ἀφαιρουμένον ἢ τὸ λοιπὸν θεωρίον δυναμένη μία δύο ἀλόγων γίνεται ἥτοι ἀποτομή ἢ ἐλάσσων.

A medial (area) being subtracted from a rational (area), one of two irrational (straight-lines) arise (as) the square-root of the remaining area—either an apotome, or a minor (straight-line).



Ἀπὸ γάρ ῥητοῦ τοῦ BF μέσον ἀφηρήσθω τὸ $BΔ$ · λέγω, ὅτι ἢ τὸ λοιπὸν δυναμένη τὸ EF μία δύο ἀλόγων γίνεται ἥτοι ἀποτομή ἢ ἐλάσσων.

Ἐκκείσθω γάρ ῥητὴ ἡ ZH , καὶ τῷ μὲν BF ἴσον παρὰ τὴν ZH παραβελήσθω ὀρθογώνιον παραλληλόγραμμον τὸ $HΘ$, τῷ δὲ $ΔB$ ἴσον ἀφηρήσθω τὸ HK · λοιπὸν ἄρα τὸ EF ἴσον ἐστὶ τῷ $ΛΘ$. ἐπεὶ οὖν ῥητὸν μὲν ἐστὶ τὸ BF , μέσον δὲ τὸ $BΔ$, ἴσον δὲ τὸ μὲν BF τῷ $HΘ$, τὸ δὲ $BΔ$ τῷ HK , ῥητὸν μὲν ἄρα ἐστὶ τὸ $HΘ$, μέσον δὲ τὸ HK . καὶ παρὰ ῥητὴν τὴν ZH παράκειται· ῥητὴ μὲν ἄρα ἡ $ZΘ$ καὶ σύμμετρος τῇ ZH μήκει, ῥητὴ δὲ ἡ ZK καὶ ἀσύμμετρος τῇ ZH μήκει· ἀσύμμετρος ἄρα ἐστὶν ἡ $ZΘ$ τῇ ZK μήκει. αἱ $ZΘ$, ZK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ $KΘ$, προσαρμύζουσα δὲ αὐτῇ ἡ KZ . ἥτοι δὴ ἡ $ΘZ$ τῇ ZK μείζον δύναται τῷ ἀπὸ συμέτρον ἢ οὐ.

Δυνάσθω πρότερον τῷ ἀπὸ συμέτρον. καὶ ἐστὶν ὅλη ἡ $ΘZ$ σύμμετρος τῇ ἐκκειμένη ῥητῇ μήκει τῇ ZH · ἀποτομή ἄρα πρώτη ἐστὶν ἡ $KΘ$. τὸ δ' ὑπὸ ῥητῆς καὶ ἀποτομῆς πρώτης περιεχόμενον ἢ δυναμένη ἀποτομή ἐστὶν. ἢ ἄρα τὸ $ΛΘ$, τοντέστι τὸ EF , δυναμένη ἀποτομή ἐστὶν.

Εἰ δὲ ἡ $ΘZ$ τῇ ZK μείζον δύναται τῷ ἀπὸ ἀσυμέτρον ἐαντῇ, καὶ ἐστὶν ὅλη ἡ $ZΘ$ σύμμετρος τῇ ἐκκειμένη ῥητῇ μήκει τῇ ZH , ἀποτομή τετάρτη ἐστὶν ἡ $KΘ$. τὸ δ' ὑπὸ ῥητῆς καὶ

For let the medial (area) BD be subtracted from the rational (area) BC . I say that one of two irrational (straight-lines) arise (as) the square-root of the remaining (area), EC —either an apotome, or a minor (straight-line).

For let the rational (straight-line) FG be laid out, and let the right-angled parallelogram GH , equal to BC , be applied to FG , and let GK , equal to DB , be subtracted (from GH). Thus, the remainder EC is equal to LH . Therefore, since BC is a rational (area), and BD a medial (area), and BC (is) equal to GH , and BD to GK , GH is thus a rational (area), and GK a medial (area). And they are applied to the rational (straight-line) FG . Thus, FH (is) rational, and commensurable in length with FG [Prop. 10.20], and FK (is) also rational, and incommensurable in length with FG [Prop. 10.22]. Thus, FH is incommensurable in length with FK [Prop. 10.13]. FH and FK are thus rational (straight-lines which are) commensurable in square only. Thus, KH is an apotome [Prop. 10.73], and KF an attachment to it. So, the square on HF is greater than (the square on) FK by the (square) on (some straight-line which is) either commensurable, or not (commensurable), (in length with HF).

First, let the square (on it) be (greater) by the (square) on (some straight-line which is) commensurable (in length with

ἀποτομῇς τετάρτης περιεχόμενον ἢ δυναμένη ἐλάσσων ἐστίν· ὅπερ ἔδει δεῖχαι.

HF). And the whole of HF is commensurable in length with the (previously) laid down rational (straight-line) FG . Thus, KH is a first apotome [Def. 10.1]. And the square-root of an (area) contained by a rational (straight-line) and a first apotome is an apotome [Prop. 10.91]. Thus, the square-root of LH —that is to say, (of) EC —is an apotome.

And if the square on HF is greater than (the square on) FK by the (square) on (some straight-line which is) incommensurable (in length) with (HF) , and (since) the whole of FH is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a fourth apotome [Prop. 10.14]. And the square-root of an (area) contained by a rational (straight-line) and a fourth apotome is a minor (straight-line) [Prop. 10.94]. (Which is) the very thing it was required to show.

109.

Proposition 109

Ἀπὸ μέσον ῥητοῦ ἀφαιρουμένον ἄλλαι δύο ἄλογοι γίνονται ἥτοι μέση ἀποτομή πρώτη ἢ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσα.

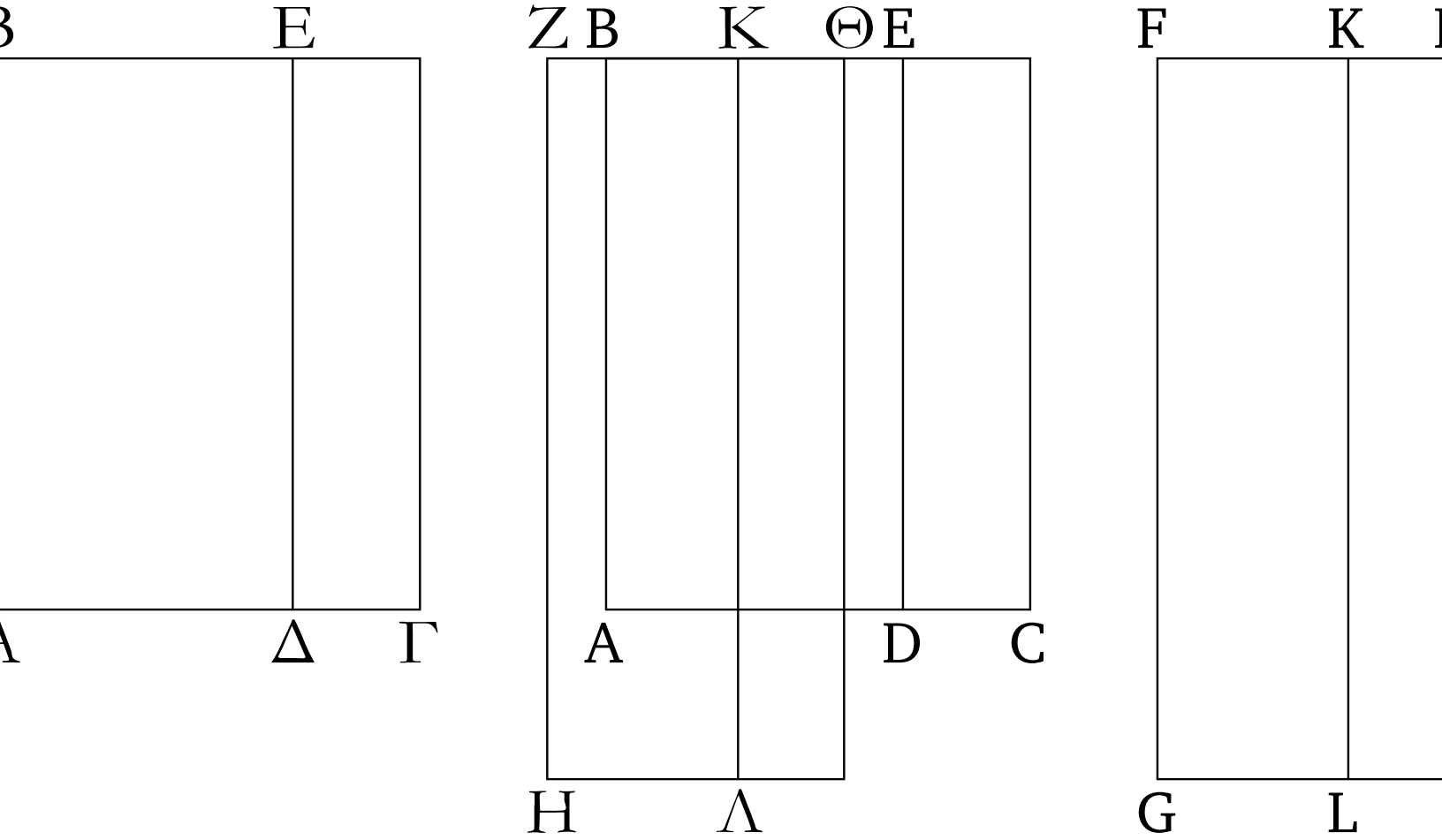
Ἀπὸ γὰρ μέσον τοῦ $BΓ$ ῥητὸν ἀφηρήσθω τὸ $ΒΔ$. λέγω, ὅτι ἡ τὸ λοιπὸν τὸ $ΕΓ$ δυναμένη μία δύο ἀλόγων γίνεται ἥτοι μέση ἀποτομή πρώτη ἢ μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσα.

Ἐκκείσθω γὰρ ῥητὴ ἡ ZH , καὶ παραβεβλήσθω ὁμοίως τὰ $θωρία$. ἔστι δὴ ἀκολουθῶς ῥητὴ μὲν ἡ $ZΘ$ καὶ ἀσύμμετρος τῇ ZH μήκει, ῥητὴ δὲ ἡ KZ καὶ σύμμετρος τῇ ZH μήκει· αἱ $ZΘ$, ZK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ $KΘ$, προσαρμόζουσα δὲ ταύτῃ ἡ ZK . ἥτοι δὴ ἡ $ΘΖ$ τῇ ZK μείζον δύνανται τῷ ἀπὸ συμμέτρον ἑαυτῇ ἢ τῷ ἀπὸ ἀσύμμετρον.

A rational (area) being subtracted from a medial (area), two other irrational (straight-lines) arise (as the square-root of the remaining area)—either a first apotome of a medial (straight-line), or that (straight-line) which with a rational (area) makes a medial whole.

For let the rational (area) BD be subtracted from the medial (area) BC . I say that one of two irrational (straight-lines) arise (as) the square-root of the remaining (area), EC —either a first apotome of a medial (straight-line), or that (straight-line) which with a rational (area) makes a medial whole.

For let the rational (straight-line) FG be laid down, and let similar areas (to the preceding proposition) have been applied (to it). So, accordingly, FH is rational, and incommensurable in length with FG , and KF (is) also rational, and commensurable in length with FG . Thus, FH and FK are rational (straight-lines which are) commensurable in square only [Prop. 10.13]. KH is thus an apotome [Prop. 10.73], and FK an attachment to it. So, the square on HF is greater than (the square on) FK either by the (square) on (some straight-line) commensurable (in length) with (HF) , or by the (square) on (some straight-line) incommensurable (in length with HF).



Εἰ μὲν οὖν ἡ ΘΖ τῆς ΖΚ μείζον δύναται τῷ ἀπὸ συμμετρων ἑαυτῇ, καὶ ἐστὶν ἡ προσαρμόζουσα ἡ ΖΚ σύμμετρος τῇ ἐκκειμένῃ ῥητῇ μήκει τῇ ΖΗ, ἀποτομή δευτέρα ἐστὶν ἡ ΚΘ. ῥητὴ δὲ ἡ ΖΗ· ὥστε ἡ τὸ ΛΘ, τουτέστι τὸ ΕΓ, δυναμένη μέσης ἀποτομῇ πρώτη ἐστίν.

Εἰ δὲ ἡ ΘΖ τῆς ΖΚ μείζον δύναται τῷ ἀπὸ ἀσυμμέτρων, καὶ ἐστὶν ἡ προσαρμόζουσα ἡ ΖΚ σύμμετρος τῇ ἐκκειμένῃ ῥητῇ μήκει τῇ ΖΗ, ἀποτομή πέμπτη ἐστὶν ἡ ΚΘ· ὥστε ἡ τὸ ΕΓ δυναμένη μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσα ἐστίν· ὅπερ ἔδει δεῖξαι.

Therefore, if the square on HZ is greater than (the square on) FK by the (square) on (some straight-line) commensurable (in length) with (HF) , and (since) the attachment FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a second apotome [Def. 10.12]. And FG (is) rational. Hence, the square-root of LH —that is to say, (of) EC —is a first apotome of a medial (straight-line) [Prop. 10.92].

And if the square on HZ is greater than (the square on) FK by the (square) on (some straight-line) incommensurable (in length with HF), and (since) the attachment FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a fifth apotome [Def. 10.15]. Hence, the square-root of EC is that (straight-line) which with a rational (area) makes a medial whole [Prop. 10.95]. (Which is) the very thing it was required to show.

110.

Proposition 110

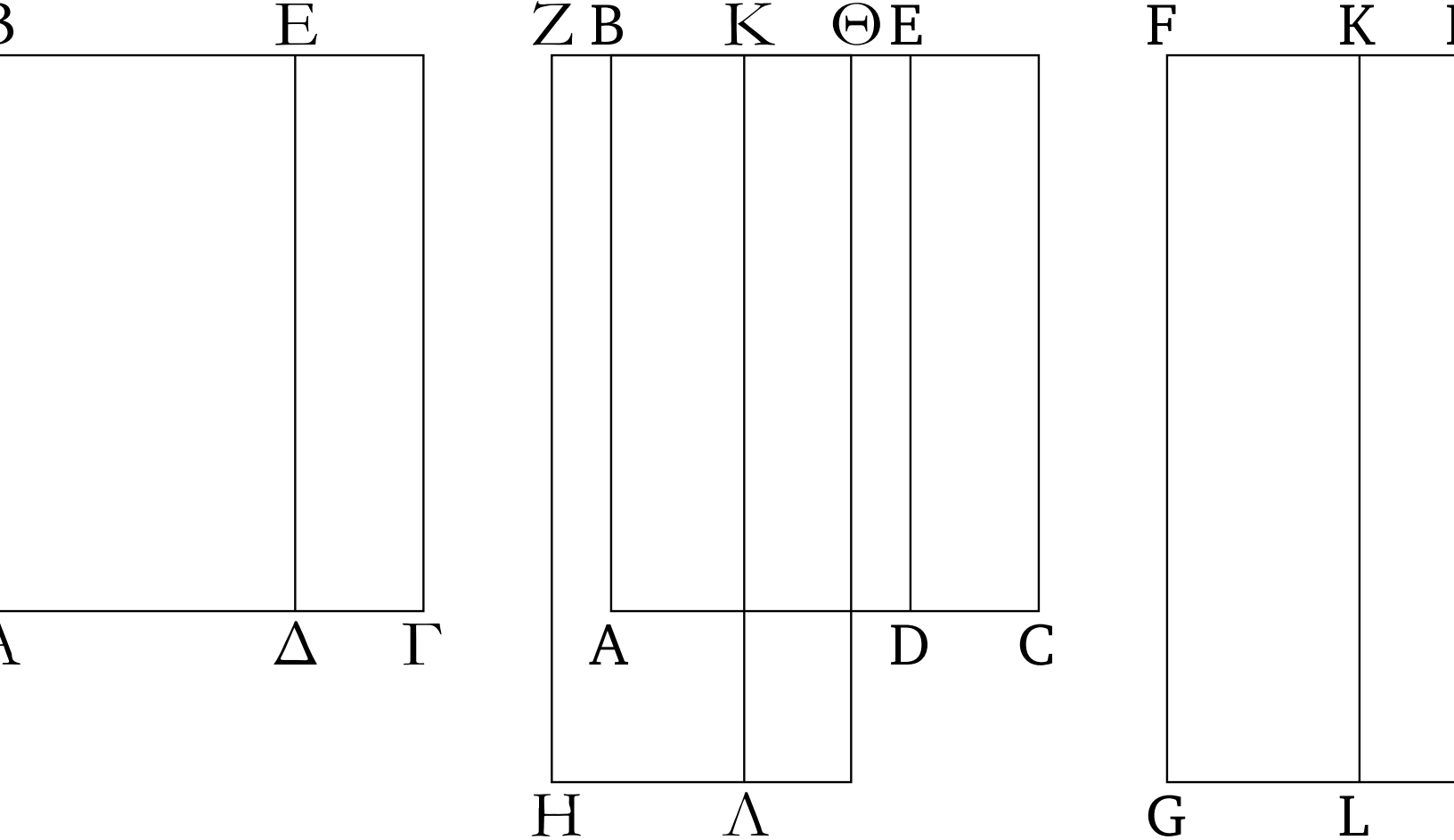
Ἀπὸ μέσον μέσον ἀφαιρουμένον ἀσυμμέτρον τῷ ὅλῳ αἰλοιπαὶ δύο ἄλογοι γίνονται ἥτοι μέσης ἀποτομῇ δευτέρα ἢ μετὰ μέσον μέσον τὸ ὅλον ποιούσα.

A medial (area), incommensurable with the whole, being subtracted from a medial (area), the two remaining irrational (straight-lines) arise (as) the (square-root of the area)—either

Ἀφρηρήσθω γάρ ὡς ἐπὶ τῶν προκειμένων καταγραφῶν ἀπὸ μέσον τοῦ $BΓ$ μέσον τὸ $BΔ$ ἀσύμμετρον τῷ ὅλῳ· λέγω, ὅτι ἡ τὸ $EΓ$ δυναμένη μία ἐστὶ δύο ἀλόγων ἥτοι μέσῃς ἀποτομῇ δευτέρᾳ ἢ μετὰ μέσον μέσον τὸ ὅλον ποιούσα.

a second apotome of a medial (straight-line), or that (straight-line) which with a medial (area) makes a medial whole.

For, as in the previous figures, let the medial (area) BD , incommensurable with the whole, be subtracted from the medial (area) BC . I say that the square-root of EC is one of two irrational (straight-lines)—either a second apotome of a medial (straight-line), or that (straight-line) which with a medial (area) makes a medial whole.



Ἐπεὶ γὰρ μέσον ἐστὶν ἑκάτερον τῶν $BΓ$, $BΔ$, καὶ ἀσύμμετρον τὸ $BΓ$ τῷ $BΔ$, ἔσται ἀκολούθως ῥητὴ ἑκάτερα τῶν $ZΘ$, ZK καὶ ἀσύμμετροξ τῇ ZH μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ $BΓ$ τῷ $BΔ$, τοντέστι τὸ $HΘ$ τῷ HK , ἀσύμμετροξ καὶ ἡ $ΘZ$ τῇ ZK . αἱ $ZΘ$, ZK ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομῇ ἄρα ἐστὶν ἡ $KΘ$ [προσαρμόζουσα δὲ ἡ ZK . ἥτοι δὴ ἡ $ZΘ$ τῇ ZK μείζον δύνανται τῷ ἀπὸ συμμέτρον ἢ τῷ ἀπὸ ἀσυμμέτρον ἑαυτῇ].

Εἰ μὲν δὴ ἡ $ZΘ$ τῇ ZK μείζον δύνανται τῷ ἀπὸ συμμέτρον ἑαυτῇ, καὶ οὐθετέρα τῶν $ZΘ$, ZK σύμμετροξ ἐστὶ τῇ ἑκκειμένη ῥητῇ μήκει τῇ ZH , ἀποτομῇ τρίτῃ ἐστὶν ἡ $KΘ$. ῥητὴ δὲ ἡ $ΚΛ$, τὸ δ' ὑπὸ ῥητῇ καὶ ἀποτομῇ τρίτῃ περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ ἄλογόξ ἐστιν,

For since BC and BD are each medial (areas), and BC (is) incommensurable with BD , accordingly, FH and FK will each be rational (straight-lines), and incommensurable in length with FG [Prop. 10.22]. And since BC is incommensurable with BD —that is to say, GH with GK — HF (is) also incommensurable (in length) with FK [Props. 6.1, 10.11]. Thus, FH and FK are rational (straight-lines which are) commensurable in square only. KH is thus as apotome [Prop. 10.73], [and FK an attachment (to it)]. So, the square on FH is greater than (the square on) FK either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (FH) .]

So, if the square on FH is greater than (the square on)

καλεῖται δὲ μέση ἀποτομή δευτέρα· ὥστε ἡ τὸ $\Lambda\Theta$, τοντέστι τὸ $ΕΓ$, δυναμένη μέση ἀποτομή ἐστὶ δευτέρα.

Εἰ δὲ ἡ $Z\Theta$ τῆς ZK μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἐαντὴ [μήκει], καὶ οὐδετέρα τῶν ΘZ , ZK σύμμετρος ἐστὶ τῇ ZH μήκει, ἀποτομή ἕκτη ἐστὶν ἡ $K\Theta$. τὸ δ' ὑπὸ ῥητῆς καὶ ἀποτομῆς ἕκτης ἡ δυναμένη ἐστὶ μετὰ μέσον μέσον τὸ ὅλον ποιῶσα. ἡ τὸ $\Lambda\Theta$ ἄρα, τοντέστι τὸ $ΕΓ$, δυναμένη μετὰ μέσον μέσον τὸ ὅλον ποιῶσα ἐστὶν· ὅπερ ἔδει δεῖχαι.

FK by the (square) on (some straight-line) commensurable (in length) with (FH), and (since) neither of FH and FK is commensurable in length with the (previously) laid down rational (straight-line) FG , KH is a third apotome [Def. 10.3]. And KL (is) rational. And the rectangle contained by a rational (straight-line) and a third apotome is irrational, and the square-root of it is that irrational (straight-line) called a second apotome of a medial (straight-line) [Prop. 10.93]. Hence, the square-root of LH —that is to say, (of) EC —is a second apotome of a medial (straight-line).

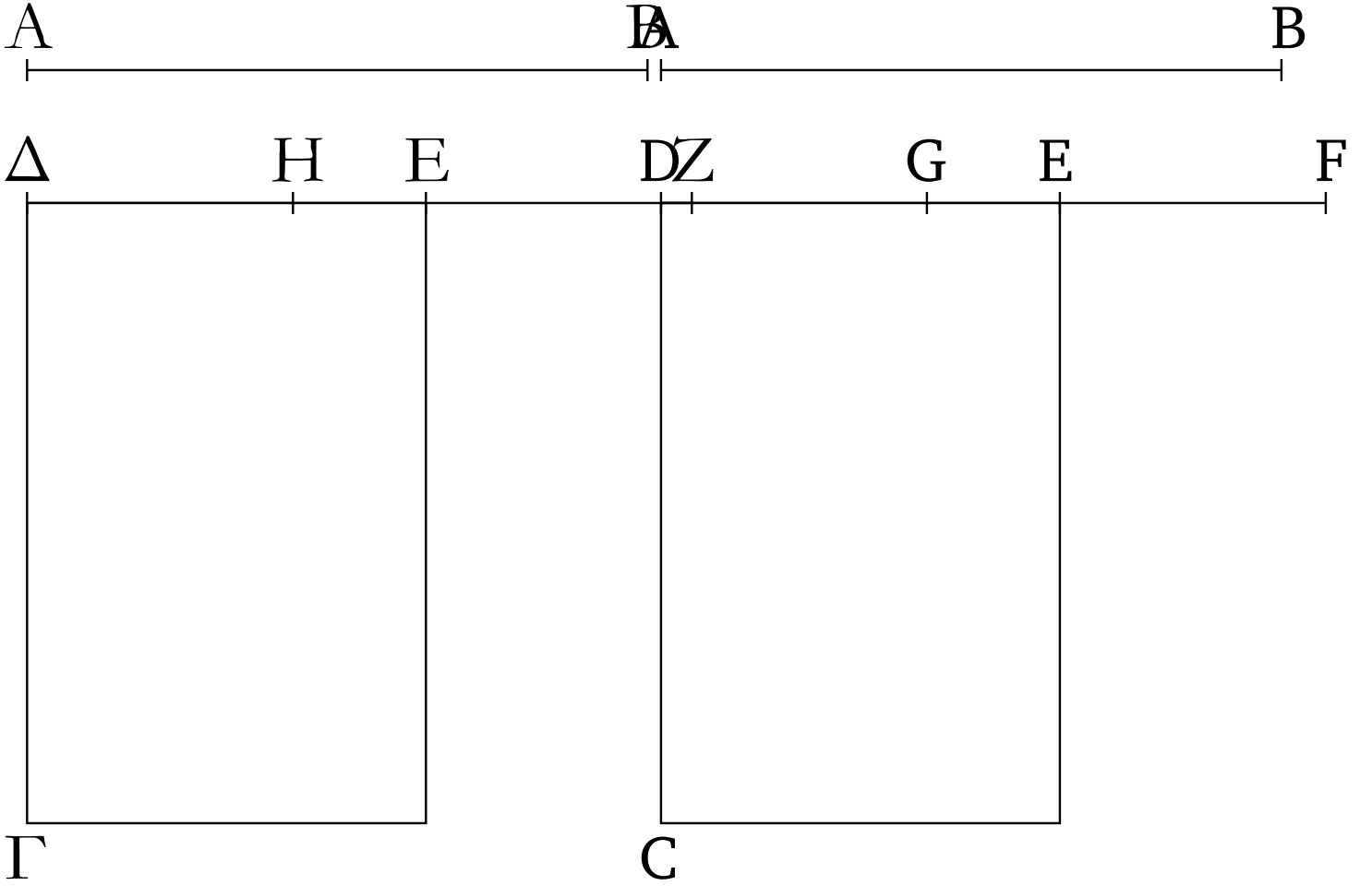
And if the square on FH is greater than (the square on) FK by the (square) on (some straight-line) incommensurable [in length] with (FH), and (since) neither of HF and FK is commensurable in length with FG , KH is a sixth apotome [Def. 10.16]. And the square-root of the (rectangle contained) by a rational (straight-line) and a sixth apotome is that (straight-line) which with a medial (area) makes a medial whole [Prop. 10.96]. Thus, the square-root of LH —that is to say, (of) EC —is that (straight-line) which with a medial (area) makes a medial whole. (Which is) the very thing it was required to show.

111.

Ἡ ἀποτομή οὐκ ἔστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων.

Proposition 111

An apotome is not the same as a binomial.



Ἐστω ἀποτομή ἡ AB . λέγω, ὅτι ἡ AB οὐκ ἔστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων.

Εἰ γὰρ δυνατόν, ἔστω· καὶ ἐκκείσθω ῥητὴ ἡ $\Delta\Gamma$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω ὀρθογώνιον τὸ ΓE πλάτος ποιῶν τὴν ΔE . ἔπει οὖν ἀποτομή ἐστὶν ἡ AB , ἀποτομή πρώτη ἐστὶν ἡ ΔE . ἔστω αὐτῇ προσαρμύζουσα ἡ EZ . αἱ ΔZ , ZE ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΔZ τῆς ZE μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντῆς, καὶ ἡ ΔZ σύμμετρος ἐστὶ τῇ ἐκκειμένῃ ῥητῇ μήκει τῇ $\Delta\Gamma$. πάλιν, ἔπει ἐκ δύο ὀνομάτων ἐστὶν ἡ AB , ἐκ δύο ἄρα ὀνομάτων πρώτη ἐστὶν ἡ ΔE . διηρήσθω εἰς τὰ ὀνόματα κατὰ τὸ H , καὶ ἔστω μείζον ὄνομα τὸ ΔH . αἱ ΔH , HE ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΔH τῆς HE μείζον δύναται τῷ ἀπὸ συμμέτρου ἐναντῆς, καὶ τὸ μείζον ἡ ΔH σύμμετρος ἐστὶ τῇ ἐκκειμένῃ ῥητῇ μήκει τῇ $\Delta\Gamma$. καὶ ἡ ΔZ ἄρα τῇ ΔH σύμμετρος ἐστὶ μήκει· καὶ λοιπὴ ἄρα ἡ HZ σύμμετρος ἐστὶ τῇ ΔZ μήκει. [ἔπει οὖν σύμμετρος ἐστὶν ἡ ΔZ τῇ HZ , ῥητὴ δὲ ἐστὶν ἡ ΔZ , ῥητὴ ἄρα ἐστὶ καὶ ἡ HZ . ἔπει οὖν σύμμετρος ἐστὶν ἡ ΔZ τῇ HZ μήκει] ἀσύμμετρος δὲ ἡ ΔZ τῇ EZ μήκει. ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ZH τῇ EZ μήκει. αἱ HZ , ZE ἄρα ῥηταὶ [εἰσι] δυνάμει μόνον σύμμετροι· ἀποτομή ἄρα ἐστὶν

Let AB be an apotome. I say that AB is not the same as a binomial.

For, if possible, let it be (the same). And let a rational (straight-line) DC be laid down. And let the rectangle CE , equal to the (square) on AB , be applied to CD , producing DE as breadth. Therefore, since AB is an apotome, DE is a first apotome [Prop. 10.97]. Let EF be an attachment to it. Thus, DF and FE are rational (straight-lines which are) commensurable in square only, and the square on DF is greater than (the square on) FE by the (square) on (some straight-line) commensurable (in length) with (DF), and DF is commensurable in length with the (previously) laid down rational (straight-line) DC [Def. 10.10]. Again, since AB is a binomial, DE is thus a first binomial [Prop. 10.60]. Let (DE) be divided into its (component) terms at G , and let DG be the greater term. Thus, DG and GE are rational (straight-lines which are) commensurable in square only, and the square on DG is greater than (the square on) GE by the (square) on (some straight-line) commensurable (in length) with (DG), and the greater (term) DG is commensurable in length with the (previously) laid down rational (straight-line) DC [Def. 10.5]. Thus, DF is also com-

ἡ EH . ἀλλὰ καὶ ῥητὴ· ὅπερ ἐστὶν ἀδύνατον.

Ἡ ἄρα ἀποτομή οὐκ ἔστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων· ὅπερ ἔδει δεῖξαι.

[Πόρισμα.]

Ἡ ἀποτομή καὶ αἱ μετ' αὐτὴν ἄλογοι οὔτε τῇ μέσῃ οὔτε ἀλλήλαις εἰσὶν αἱ αὐταί.

Τὸ μὲν γὰρ ἀπὸ μέσῃς παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ῥητὴν καὶ ἀσύμμετρον τῇ, παρ' ἣν παράκειται, μήκει, τὸ δὲ ἀπὸ ἀποτομῆς παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην, τὸ δὲ ἀπὸ μέσῃς ἀποτομῆς πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν, τὸ δὲ ἀπὸ μέσῃς ἀποτομῆς δευτέρας παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην, τὸ δὲ ἀπὸ ἐλάσσονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τετάρτην, τὸ δὲ ἀπὸ τῆς μετὰ ῥητοῦ μέσον τὸ ὅλον ποιούσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πέμπτην, τὸ δὲ ἀπὸ τῆς μετὰ μέσον μέσον τὸ ὅλον ποιούσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν ἕκτην. ἐπεὶ οὖν τὰ εἰρημένα πλάτη διαφέρει τοῦ τε πρώτου καὶ ἀλλήλων, τοῦ μὲν πρώτου, ὅτι ῥητὴ ἐστίν, ἀλλήλων δὲ, ἐπεὶ τῇ τάχει οὐκ εἰσὶν αἱ αὐταί, δῆλον, ὥς καὶ αὐταί αἱ ἄλογοι διαφέρουσιν ἀλλήλων. καὶ ἐπεὶ δέδεικται ἡ ἀποτομή οὐκ οὔσα ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων, ποιοῦσι δὲ πλάτη παρὰ ῥητὴν παραβαλλόμενα αἱ μετὰ τὴν ἀποτομὴν ἀποτομάς ἀκολουθῶς ἐκάστη τῇ τάχει τῇ καθ' αὐτήν, αἱ δὲ μετὰ τὴν ἐκ δύο ὀνομάτων τάξ ἐκ δύο ὀνομάτων καὶ αὐταί τῇ τάχει ἀκολουθῶς, ἕτεραι ἄρα εἰσὶν αἱ μετὰ τὴν ἀποτομὴν καὶ ἕτεραι αἱ μετὰ τὴν ἐκ δύο ὀνομάτων, ὥς εἶναι τῇ τάχει πάσας ἀλόγους γ,

measurable in length with DG [Prop. 10.12]. The remainder GF is thus commensurable in length with DF [Prop. 10.15]. [Therefore, since DF is commensurable with GF , and DF is rational, GF is thus also rational. Therefore, since DF is commensurable in length with GF ,] DF (is) incommensurable in length with EF . Thus, FG is also incommensurable in length with EF [Prop. 10.13]. GF and FE [are] thus rational (straight-lines which are) commensurable in square only. Thus, EG is an apotome [Prop. 10.73]. But, (it is) also rational. The very thing is impossible.

Thus, an apotome is not the same as a binomial. (Which is) the very thing it was required to show.

[Corollary]

The apotome and the irrational (straight-lines) after it are neither the same as a medial (straight-line) nor (the same) as one another.

For the (square) on a medial (straight-line), applied to a rational (straight-line), produces as breadth a rational (straight-line) which is) incommensurable in length with the (straight-line) to which (the area) is applied [Prop. 10.22]. And the (square) on an apotome, applied to a rational (straight-line), produces as breadth a first apotome [Prop. 10.97]. And the (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces as breadth a second apotome [Prop. 10.98]. And the (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces as breadth a third apotome [Prop. 10.99]. And (square) on a minor (straight-line), applied to a rational (straight-line), produces as breadth a fourth apotome [Prop. 10.100]. And (square) on that (straight-line) which with a rational (area) produces a medial whole, applied to a rational (straight-line), produces as breadth a fifth apotome [Prop. 10.101]. And (square) on that (straight-line) which with a medial (area) produces a medial whole, applied to a rational (straight-line), produces as breadth a sixth apotome [Prop. 10.102]. Therefore, since the aforementioned breadths differ from the first (breadth), and from one another—from the first, because it is rational, and from one another since they are not the same in order—clearly, the irrational (straight-lines) themselves also differ from one another. And since it has been shown that an apotome is not the same as a binomial [Prop. 10.111], and (that) the (irrational straight-lines) after the apotome, being applied to a rational (straight-line), produce as breadth, each according to its own (order), apotomes, and (that) the (irrational straight-lines) after the binomial themselves also (produce as breadth), according (to their) order, binomials, the (irrational straight-lines) after the apotome are thus different, and the (irrational straight-lines) after the binomial (are also) different, so that there are, in order, 13 irrational (straight-lines) in all:

Μέσσην,
 Ἐκ δύο ὀνομάτων,
 Ἐκ δύο μέσων πρώτην,
 Ἐκ δύο μέσων δευτέραν,
 Μείζονα,
 μὴ ῥητὸν καὶ μέσον δυναμένην,
 Δύο μέσα δυναμένην,
 Ἀποτομήν,
 Μέσῃς ἀποτομὴν πρώτην,
 Μέσῃς ἀποτομὴν δευτέραν,
 Ἐλάσσονα,
 Μετὰ ῥητοῦ μέσον τὸ ὅλον ποιοῦσαν,
 Μετὰ μέσου μέσον τὸ ὅλον ποιοῦσαν.

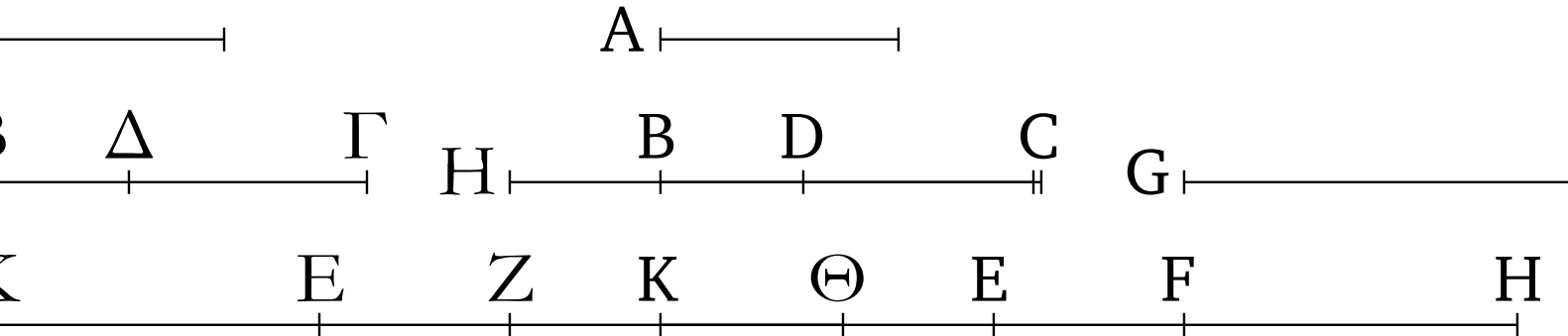
Medial,
 Binomial,
 First bimedral,
 Second bimedral,
 Major,
 Square-root of a rational plus a medial (area),
 Square-root of (the sum of) two medial (areas),
 Apotome,
 First apotome of a medial,
 Second apotome of a medial,
 Minor,
 That which with a rational (area) produces a medial whole,
 That which with a medial (area) produces a medial whole.

112.

Proposition 112[†]

Τὸ ἀπὸ ῥητῆς παρὰ τὴν ἐκ δύο ὀνομάτων παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν, ἥς τὰ ὀνόματα σύμμετρα ἔστι τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι καὶ ἔτι ἐν τῷ αὐτῷ λόγῳ, καὶ ἔτι ἡ γινόμενη ἀποτομή τὴν αὐτὴν ἔχει τάχιν τῇ ἐκ δύο ὀνομάτων.

The (square) on a rational (straight-line), applied to a binomial (straight-line), produces as breadth an apotome whose terms are commensurable (in length) with the terms of the binomial, and, furthermore, in the same ratio. Moreover, the created apotome will have the same order as the binomial.



Ἐστω ῥητὴ μὲν ἡ A , ἐκ δύο ὀνομάτων δὲ ἡ BF , ἥς μείζον ὄνομα ἔστω ἡ $\Delta\Gamma$, καὶ τῷ ἀπὸ τῆς A ἴσον ἔστω τὸ ὑπὸ τῶν $B\Gamma$, EZ · λέγω, ὅτι ἡ EZ ἀποτομή ἐστίν, ἥς τὰ ὀνόματα σύμμετρα ἔστι τοῖς $\Gamma\Delta$, ΔB , καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἔτι ἡ EZ τὴν αὐτὴν ἔχει τάχιν τῇ $B\Gamma$.

Ἐστω γάρ πάλιν τῷ ἀπὸ τῆς A ἴσον τὸ ὑπὸ τῶν $B\Delta$, H . ἐπεὶ οὖν τὸ ὑπὸ τῶν $B\Gamma$, EZ ἴσον ἐστὶ τῷ ὑπὸ τῶν $B\Delta$, H , ἔστιν ἄρα ὡς ἡ ΓB πρὸς τὴν $B\Delta$, οὕτως ἡ H πρὸς τὴν EZ . μείζων δὲ ἡ ΓB τῆς $B\Delta$ · μείζων ἄρα ἐστὶ καὶ ἡ H τῆς EZ . ἔστω τῇ H ἴση ἡ $E\Theta$ · ἔστιν ἄρα ὡς ἡ ΓB πρὸς τὴν $B\Delta$, οὕτως ἡ ΘE πρὸς τὴν EZ · διελόντι ἄρα ἐστὶν ὡς ἡ $\Gamma\Delta$ πρὸς τὴν $B\Delta$, οὕτως ἡ ΘZ πρὸς τὴν ZE . γεγονένω ὡς ἡ ΘZ πρὸς τὴν ZE , οὕτως ἡ ZE πρὸς τὴν KE · καὶ ὅλη ἄρα ἡ ΘK πρὸς ὅλην τὴν KZ ἐστίν, ὡς ἡ ZK πρὸς KE · ὡς γὰρ ἐν τῶν ἡγουμένων πρὸς ἐν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα. ὡς δὲ ἡ ZK πρὸς KE ,

Let A be a rational (straight-line), and BC a binomial (straight-line), of which let DC be the greater term. And let the (rectangle contained) by BC and EF be equal to the (square) on A . I say that EF is an apotome whose terms are commensurable (in length) with CD and DB , and in the same ratio, and, moreover, that EF will have the same order as BC .

For, again, let the (rectangle contained) by BD and G be equal to the (square) on A . Therefore, since the (rectangle contained) by BC and EF is equal to the (rectangle contained) by BD and G , thus as CB is to BD , so G (is) to EF [Prop. 6.16]. And CB (is) greater than BD . Thus, G is also greater than EF [Props. 5.16, 5.14]. Let EH be equal to G . Thus, as CB is to BD , so HE (is) to EF . Thus, via separation, as CD is to BD , so HF (is) to FE [Prop. 5.17]. Let it be contrived that as HF (is) to FE , so FK (is) to KE . And, thus, the whole HK is to the whole KF , as FK (is) to KE . For as one of the leading (pro-

οὕτως ἐστὶν ἡ $\Gamma\Delta$ πρὸς τὴν ΔB · καὶ ὥς ἄρα ἡ ΘK πρὸς KZ , οὕτως ἡ $\Gamma\Delta$ πρὸς τὴν ΔB . σύμμετρον δὲ τὸ ἀπὸ τῆς $\Gamma\Delta$ τῷ ἀπὸ τῆς ΔB · σύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΘK τῷ ἀπὸ τῆς KZ . καὶ ἐστὶν ὥς τὸ ἀπὸ τῆς ΘK πρὸς τὸ ἀπὸ τῆς KZ , οὕτως ἡ ΘK πρὸς τὴν KE , ἐπεὶ αἱ $\tau\rho\epsilon\acute{\iota}\varsigma$ αἱ ΘK , KZ , KE ἀνάλογόν εἰσιν. σύμμετρος ἄρα ἡ ΘK τῇ KE μήκει. ὥστε καὶ ἡ ΘE τῇ EK σύμμετρος ἐστὶ μήκει. καὶ ἐπεὶ τὸ ἀπὸ τῆς A ἴσον ἐστὶ τῷ ὑπὸ τῶν $E\Theta$, $B\Delta$, ῥητὸν δὲ ἐστὶ τὸ ἀπὸ τῆς A , ῥητὸν ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν $E\Theta$, $B\Delta$. καὶ παρὰ ῥητὴν τὴν $B\Delta$ παράκειται· ῥητὴ ἄρα ἐστὶν ἡ $E\Theta$ καὶ σύμμετρος τῇ $B\Delta$ μήκει· ὥστε καὶ ἡ σύμμετρος αὐτῇ ἡ EK ῥητὴ ἐστὶ καὶ σύμμετρος τῇ $B\Delta$ μήκει. ἐπεὶ οὖν ἐστὶν ὥς ἡ $\Gamma\Delta$ πρὸς ΔB , οὕτως ἡ ZK πρὸς KE , αἱ δὲ $\Gamma\Delta$, ΔB δυνάμει μόνον εἰσὶ σύμμετροι, καὶ αἱ ZK , KE δυνάμει μόνον εἰσὶ σύμμετροι. ῥητὴ δὲ ἐστὶν ἡ KE · ῥητὴ ἄρα ἐστὶ καὶ ἡ ZK . αἱ ZK , KE ἄρα ῥηταὶ δυνάμει μόνον εἰσὶ σύμμετροι· ἀποτομή ἄρα ἐστὶν ἡ EZ .

Ἦτοι δὲ ἡ $\Gamma\Delta$ τῆς ΔB μείζον δύνатаι τῷ ἀπὸ συμμέτρον ἑαντῇ ἢ τῷ ἀπὸ ἀσυνμέτρον.

Εἰ μὲν οὖν ἡ $\Gamma\Delta$ τῆς ΔB μείζον δύνатаι τῷ ἀπὸ συμμέτρον [ἑαντῇ], καὶ ἡ ZK τῆς KE μείζον δυνήσεται τῷ ἀπὸ συμμέτρον ἑαντῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ $\Gamma\Delta$ τῇ ἐκκεκλεμένη ῥητῇ μήκει, καὶ ἡ ZK · εἰ δὲ ἡ $B\Delta$, καὶ ἡ KE · εἰ δὲ οὐδετέρα τῶν $\Gamma\Delta$, ΔB , καὶ οὐδετέρα τῶν ZK , KE .

Εἰ δὲ ἡ $\Gamma\Delta$ τῆς ΔB μείζον δύνатаι τῷ ἀπὸ ἀσυνμέτρον ἑαντῇ, καὶ ἡ ZK τῆς KE μείζον δυνήσεται τῷ ἀπὸ ἀσυνμέτρον ἑαντῇ. καὶ εἰ μὲν ἡ $\Gamma\Delta$ σύμμετρος ἐστὶ τῇ ἐκκεκλεμένη ῥητῇ μήκει, καὶ ἡ ZK · εἰ δὲ ἡ $B\Delta$, καὶ ἡ KE · εἰ δὲ οὐδετέρα τῶν $\Gamma\Delta$, ΔB , καὶ οὐδετέρα τῶν ZK , KE · ὥστε ἀποτομή ἐστὶν ἡ ZE , ἥς τὰ ὀνόματα τὰ ZK , KE σύμμετρα ἐστὶ τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι τοῖς $\Gamma\Delta$, ΔB καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ τὴν αὐτὴν τάχιν ἔθει $B\Gamma$ · ὅπερ ἔδει δεῖξαι.

portional magnitudes is) to one of the following, so all of the leading (magnitudes) are to all of the following [Prop. 5.12]. And as FK (is) to KE , so CD is to DB [Prop. 5.11]. And, thus, as HK (is) to KF , so CD is to DB [Prop. 5.11]. And the (square) on CD (is) commensurable with the (square) on DB [Prop. 10.36]. The (square) on HK is thus also commensurable with the (square) on KF [Props. 6.22, 10.11]. And as the (square) on HK is to the (square) on KF , so HK (is) to KE , since the three (straight-lines) HK , KF , and KE are proportional [Def. 5.9]. HK is thus commensurable in length with KE [Prop. 10.11]. Hence, HE is also commensurable in length with EK [Prop. 10.15]. And since the (square) on A is equal to the (rectangle contained) by EH and BD , and the (square) on A is rational, the (rectangle contained) by EH and BD is thus also rational. And it is applied to the rational (straight-line) BD . Thus, EH is rational, and commensurable in length with BD [Prop. 10.20]. And, hence, the (straight-line) commensurable (in length) with it, EK , is also rational [Def. 10.3], and commensurable in length with BD [Prop. 10.12]. Therefore, since as CD is to DB , so FK (is) to KE , and CD and DB are (straight-lines which are) commensurable in square only, FK and KE are also commensurable in square only [Prop. 10.11]. And KE is rational. Thus, FK is also rational. FK and KE are thus rational (straight-lines which are) commensurable in square only. Thus, EF is an apotome [Prop. 10.73].

And the square on CD is greater than (the square on) DB either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (the square on) CD .

Therefore, if the square on CD is greater than (the square on) DB by the (square) on (some straight-line) commensurable (in length) with $[CD]$ then the square on FK will also be greater than (the square on) KE by the (square) on (some straight-line) commensurable (in length) with (FK) [Prop. 10.14]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FK [Props. 10.11, 10.12]. And if BD (is) commensurable, (so) also (is) KE [Prop. 10.12]. And if neither of CD or DB (is) commensurable, neither also (are) either of FK or KE .

And if the square on CD is greater than (the square on) DB by the (square) on (some straight-line) incommensurable (in length) with (CD) then the square on FK will also be greater than (the square on) KE by the (square) on (some straight-line) incommensurable (in length) with (FK) [Prop. 10.14]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FK [Props. 10.11, 10.12]. And if BD (is) commensurable, (so) also (is) KE [Prop. 10.12]. And if neither of CD or DB (is) commensurable, neither also (are) either of FK or KE . Hence, FE is an apotome whose terms, FK and KE , are commensurable (in length) with the terms, CD and DB , of the binomial, and in the same ratio. And

(FE) has the same order as BC [Defs. 10.5—10.10]. (Which is) the very thing it was required to show.

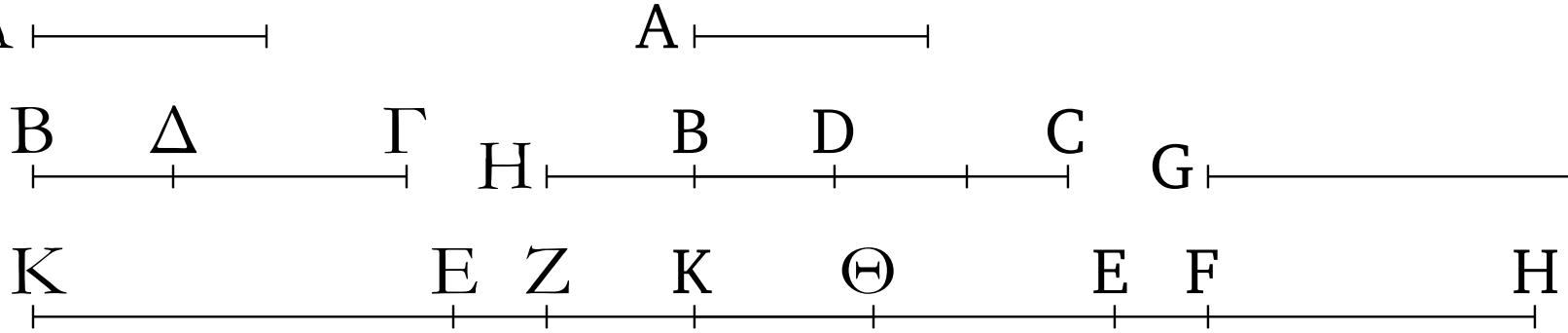
[†] Heiberg considers this proposition, and the succeeding ones, to be relatively early interpolations into the original text.

113.

Proposition 113

Τὸ ἀπὸ ῥητῆς παρὰ ἀποτομὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων, ἥξ τὰ ὀνόματα σύμμετρα ἔστι τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἔτι δὲ ἡ γινομένη ἐκ δύο ὀνομάτων τὴν αὐτὴν τάξιν ἔθει τῇ ἀποτομῇ.

The (square) on a rational (straight-line), applied to an apotome, produces as breadth a binomial whose terms are commensurable with the terms of the apotome, and in the same ratio. Moreover, the created binomial has the same order as the apotome.



Ἐστω ῥητὴ μὲν ἡ A , ἀποτομή δὲ ἡ BD , καὶ τῷ ἀπὸ τῆς A ἴσον ἔστω τὸ ὑπὸ τῶν BD , $KΘ$, ὥστε τὸ ἀπὸ τῆς A ῥητῆς παρὰ τὴν BD ἀποτομὴν παραβαλλόμενον πλάτος ποιεῖ τὴν $KΘ$. λέγω, ὅτι ἐκ δύο ὀνομάτων ἐστὶν ἡ $KΘ$, ἥξ τὰ ὀνόματα σύμμετρα ἔστι τοῖς τῆς BD ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἔτι ἡ $KΘ$ τὴν αὐτὴν ἔθει τάξιν τῇ BD .

Ἐστω γάρ τῇ BD προσαρμόζουσα ἡ $ΔΓ$. αἱ $ΒΓ$, $ΓΔ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ ἀπὸ τῆς A ἴσον ἔστω καὶ τὸ ὑπὸ τῶν $ΒΓ$, H . ῥητὸν δὲ τὸ ἀπὸ τῆς A ῥητὸν ἄρα καὶ τὸ ὑπὸ τῶν $ΒΓ$, H . καὶ παρὰ ῥητὴν τὴν $ΒΓ$ παραβέβληται ῥητὴ ἄρα ἐστὶν ἡ H καὶ σύμμετρος τῇ $ΒΓ$ μήκει. ἐπεὶ οὖν τὸ ὑπὸ τῶν $ΒΓ$, H ἴσον ἐστὶ τῷ ὑπὸ τῶν BD , $KΘ$, ἀνάλογον ἄρα ἐστὶν ὡς ἡ $ΓB$ πρὸς BD , οὕτως ἡ $KΘ$ πρὸς H . μείζων δὲ ἡ $ΒΓ$ τῆς BD . μείζων ἄρα καὶ ἡ $KΘ$ τῆς H . κείσθω τῇ H ἴση ἡ KE . σύμμετρος ἄρα ἐστὶν ἡ KE τῇ $ΒΓ$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ἡ $ΓB$ πρὸς BD , οὕτως ἡ $ΘK$ πρὸς KE , ἀναστρέψαντι ἄρα ἐστὶν ὡς ἡ $ΒΓ$ πρὸς τὴν $ΓΔ$, οὕτως ἡ $KΘ$ πρὸς $ΘE$. γεγονέτω ὡς ἡ $KΘ$ πρὸς $ΘE$, οὕτως ἡ $ΘZ$ πρὸς ZE . καὶ λοιπὴ ἄρα ἡ KZ πρὸς $ZΘ$ ἐστὶν, ὡς ἡ $KΘ$ πρὸς $ΘE$, τοντέστιν [ὡς] ἡ $ΒΓ$ πρὸς $ΓΔ$. αἱ δὲ $ΒΓ$, $ΓΔ$ δυνάμει μόνον [εἰσὶ] σύμμετροι. καὶ αἱ KZ , $ZΘ$ ἄρα δυνάμει μόνον εἰσὶ σύμμετροι. καὶ ἐπεὶ ἐστὶν ὡς ἡ $KΘ$ πρὸς $ΘE$, ἡ KZ πρὸς $ZΘ$, ἀλλ' ὡς ἡ $KΘ$ πρὸς $ΘE$, ἡ $ΘZ$ πρὸς ZE , καὶ ὡς ἄρα ἡ KZ πρὸς $ZΘ$, ἡ $ΘZ$ πρὸς ZE . ὥστε καὶ ὡς ἡ πρώτη πρὸς τὴν τρίτην, τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας. καὶ ὡς ἄρα ἡ KZ πρὸς ZE , οὕτως τὸ ἀπὸ τῆς KZ πρὸς τὸ ἀπὸ τῆς $ZΘ$. σύμμετρον δὲ ἐστὶ τὸ ἀπὸ τῆς KZ τῷ ἀπὸ τῆς $ZΘ$. αἱ γὰρ KZ , $ZΘ$ δυνάμει εἰσὶ σύμμετροι.

Let A be a rational (straight-line), and BD an apotome. And let the (rectangle contained) by BD and KH be equal to the (square) on A , such that the square on the rational (straight-line) A , applied to the apotome BD , produces KH as breadth. I say that KH is a binomial whose terms are commensurable with the terms of BD , and in the same ratio, and, moreover, that KH has the same order as BD .

For let DC be an attachment to BD . Thus, BC and CD are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let the (rectangle contained) by BC and G also be equal to the (square) on A . And the (square) on A (is) rational. The (rectangle contained) by BC and G (is) thus also rational. And it has been applied to the rational (straight-line) BC . Thus, G is rational, and commensurable in length with BC [Prop. 10.20]. Therefore, since the (rectangle contained) by BC and G is equal to the (rectangle contained) by BD and KH , thus, proportionally, as CB is to BD , so KH (is) to G [Prop. 6.16]. And BC (is) greater than BD . Thus, KH (is) also greater than G [Prop. 5.16, 5.14]. Let KE be made equal to G . KE is thus commensurable in length with BC . And since as CB is to BD , so HK (is) to KE , thus, via conversion, as BC (is) to CD , so KH (is) to HE [Prop. 5.19 corr.]. Let it be contrived that as KH (is) to HE , so HF (is) to FE . And thus the remainder KF is to FH , as KH (is) to HE —that is to say, [as] BC (is) to CD [Prop. 5.19]. And BC and CD [are] commensurable in square only. KF and FH are thus also commensurable in square only [Prop. 10.11]. And since as KH is to HE , (so) KF (is) to FH , but as KH (is) to HE ,

σύμμετρος ἄρα ἐστὶ καὶ ἡ KZ τῇ ZE μήκει· ὥστε ἡ KZ καὶ τῇ KE σύμμετρος [ἐστὶ] μήκει. ῥητὴ δὲ ἐστὶν ἡ KE καὶ σύμμετρος τῇ $BΓ$ μήκει. ῥητὴ ἄρα καὶ ἡ KZ καὶ σύμμετρος τῇ $BΓ$ μήκει. καὶ ἐπεὶ ἐστὶν ὡς ἡ $BΓ$ πρὸς $ΓΔ$, οὕτως ἡ KZ πρὸς $ZΘ$, ἐναλλάξ ὡς ἡ $BΓ$ πρὸς KZ , οὕτως ἡ $ΔΓ$ πρὸς $ZΘ$. σύμμετρος δὲ ἡ $BΓ$ τῇ KZ · σύμμετρος ἄρα καὶ ἡ $ZΘ$ τῇ $ΓΔ$ μήκει. αἱ $BΓ$, $ΓΔ$ δὲ ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· καὶ αἱ KZ , $ZΘ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ὀνομάτων ἐστὶν ἄρα ἡ $KΘ$.

Εἰ μὲν οὖν ἡ $BΓ$ τῇ $ΓΔ$ μείζον δύναιται τῷ ἀπὸ συμμέτρον ἑαυτῇ, καὶ ἡ KZ τῇ $ZΘ$ μείζον δυνήσεται τῷ ἀπὸ συμμέτρον ἑαυτῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ $BΓ$ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ KZ , εἰ δὲ ἡ $ΓΔ$ σύμμετρος ἐστὶ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ $ZΘ$, εἰ δὲ οὐδετέρα τῶν $BΓ$, $ΓΔ$, οὐδετέρα τῶν KZ , $ZΘ$.

Εἰ δὲ ἡ $BΓ$ τῇ $ΓΔ$ μείζον δύναιται τῷ ἀπὸ ἀσυνμέτρον ἑαυτῇ, καὶ ἡ KZ τῇ $ZΘ$ μείζον δυνήσεται τῷ ἀπὸ ἀσυνμέτρον ἑαυτῇ. καὶ εἰ μὲν σύμμετρος ἐστὶν ἡ $BΓ$ τῇ ἐκκειμένῃ ῥητῇ μήκει, καὶ ἡ KZ , εἰ δὲ ἡ $ΓΔ$, καὶ ἡ $ZΘ$, εἰ δὲ οὐδετέρα τῶν $BΓ$, $ΓΔ$, οὐδετέρα τῶν KZ , $ZΘ$.

Ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $KΘ$, ἥξ τὰ ὀνόματα τὰ KZ , $ZΘ$ σύμμετρα [ἐστὶ] τοῖς τῇς ἀποτομῆς ὀνόμασι τοῖς $BΓ$, $ΓΔ$ καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἔτι ἡ $KΘ$ τῇ $BΓ$ τὴν αὐτὴν ἔχει τάχιν· ὅπερ ἔδει δεῖξαι.

(so) HF (is) to FE , thus, also as KF (is) to FH , (so) HF (is) to FE [Prop. 5.11]. And hence as the first (is) to the third, so the (square) on the first (is) to the (square) on the second [Def. 5.9]. And thus as KF (is) to FE , so the (square) on KF (is) to the (square) on FH . And the (square) on KF is commensurable with the (square) on FH . For KF and FH are commensurable in square. Thus, KF is also commensurable in length with FE [Prop. 10.11]. Hence, KF [is] also commensurable in length with KE [Prop. 10.15]. And KE is rational, and commensurable in length with BC . Thus, KF (is) also rational, and commensurable in length with BC [Prop. 10.12]. And since as BC is to CD , (so) KF (is) to FH , alternately, as BC (is) to KF , so DC (is) to FH [Prop. 5.16]. And BC (is) commensurable (in length) with KF . Thus, FH (is) also commensurable in length with CD [Prop. 10.11]. And BC and CD are rational (straight-lines which are) commensurable in square only. KF and FH are thus also rational (straight-lines which are) commensurable in square only [Def. 10.3, Prop. 10.13]. Thus, KH is a binomial [Prop. 10.36].

Therefore, if the square on BC is greater than (the square on) CD by the (square) on (some straight-line) commensurable (in length) with (BC), then the square on KF will also be greater than (the square on) FH by the (square) on (some straight-line) commensurable (in length) with (KF) [Prop. 10.14]. And if BC is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) KF [Prop. 10.12]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FH [Prop. 10.12]. And if neither of BC or CD (are) commensurable, neither also (are) either of KF or FH [Prop. 10.13].

And if the square on BC is greater than (the square on) CD by the (square) on (some straight-line) incommensurable (in length) with (BC) then the square on KF will also be greater than (the square on) FH by the (square) on (some straight-line) incommensurable (in length) with (KF) [Prop. 10.14]. And if BC is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) KF [Prop. 10.12]. And if CD is commensurable, (so) also (is) FH [Prop. 10.12]. And if neither of BC or CD (are) commensurable, neither also (are) either of KF or FH [Prop. 10.13].

KH is thus a binomial whose terms, KF and FH , [are] commensurable (in length) with the terms, BC and CD , of the apotome, and in the same ratio. Moreover, KH will have the same order as BC [Defs. 10.5—10.10]. (Which is) the very thing it was required to show.

114.

Proposition 114

Ἐάν θωρίον περιέθῃται ὑπὸ ἀποτομῆς καὶ τῇς ἐκ δύο ὀνομάτων, ἥξ τὰ ὀνόματα σύμμετρα τέ ἐστι τοῖς τῇς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἡ τὸ θωρίον δυναμένη ῥητὴ

If an area is contained by an apotome, and a binomial whose terms are commensurable with, and in the same ratio as, the terms of the apotome then the square-root of the area is

ἔστιν.

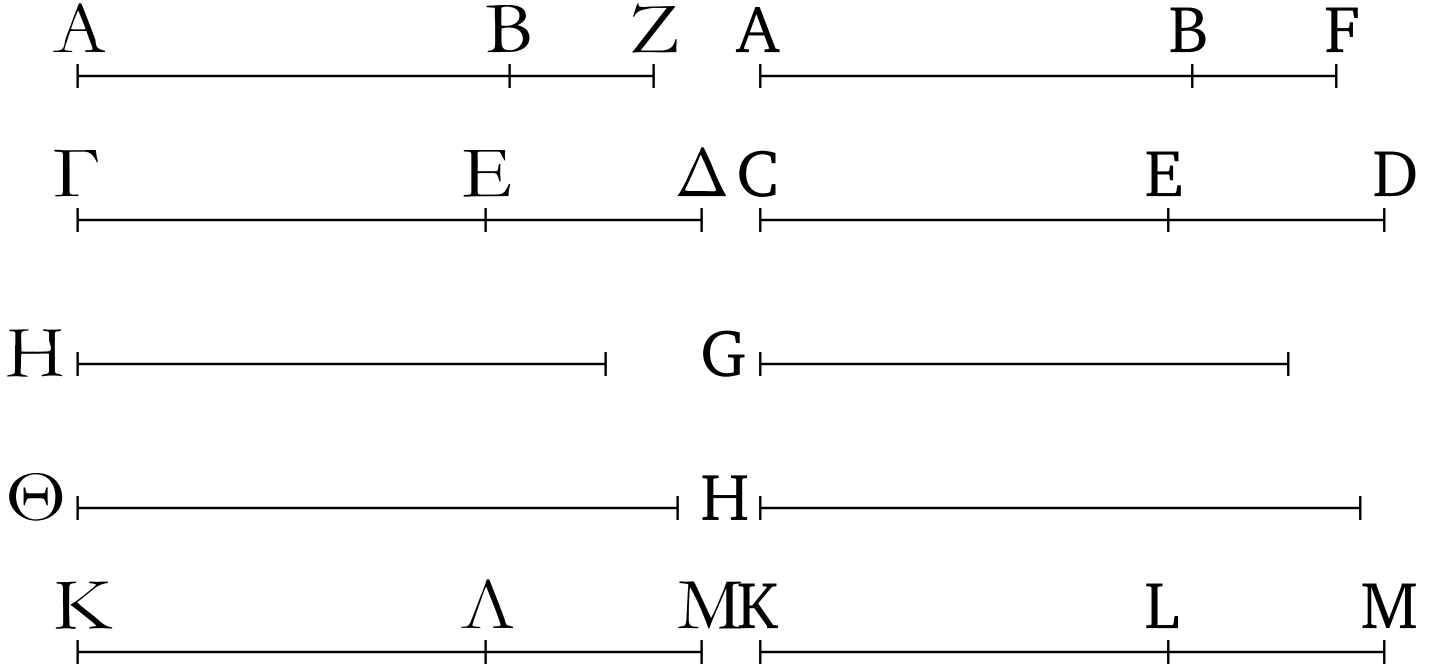
Περιεθέσθω γάρ θωρίον τὸ ὑπὸ τῶν AB , $\Gamma\Delta$ ὑπὸ ἀποτομῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων τῆς $\Gamma\Delta$, ἣς μείζον ὄνομα ἔστω τὸ ΓE , καὶ ἔστω τὰ ὀνόματα τῆς ἐκ δύο ὀνομάτων τὰ ΓE , $E\Delta$ σύμμετρά τε τοῖς τῆς ἀποτομῆς ὀνόμασι τοῖς AZ , ZB καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἔστω ἡ τὸ ὑπὸ τῶν AB , $\Gamma\Delta$ δυναμένη ἡ H · λέγω, ὅτι ῥητὴ ἔστιν ἡ H .

Ἐκκείσθω γάρ ῥητὴ ἡ Θ , καὶ τῷ ἀπὸ τῆς Θ ἴσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω πλάτος ποιῶν τὴν $ΚΛ$ · ἀποτομὴ ἄρα ἔστιν ἡ $ΚΛ$, ἣς τὰ ὀνόματα ἔστω τὰ KM , ML σύμμετρα τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι τοῖς ΓE , $E\Delta$ καὶ ἐν τῷ αὐτῷ λόγῳ. ἀλλὰ καὶ αἱ ΓE , $E\Delta$ σύμμετροί τε εἰσι ταῖς AZ , ZB καὶ ἐν τῷ αὐτῷ λόγῳ· ἔστιν ἄρα ὡς ἡ AZ πρὸς τὴν ZB , οὕτως ἡ KM πρὸς ML . ἐναλλάχ ἄρα ἔστιν ὡς ἡ AZ πρὸς τὴν KM , οὕτως ἡ BZ πρὸς τὴν $ΛΜ$ · καὶ λοιπὴ ἄρα ἡ AB πρὸς λοιπὴν τὴν $ΚΛ$ ἔστιν ὡς ἡ AZ πρὸς KM . σύμμετρος δὲ ἡ AZ τῇ KM · σύμμετρος ἄρα ἔστι καὶ ἡ AB τῇ $ΚΛ$. καὶ ἔστιν ὡς ἡ AB πρὸς $ΚΛ$, οὕτως τὸ ὑπὸ τῶν $\Gamma\Delta$, AB πρὸς τὸ ὑπὸ τῶν $\Gamma\Delta$, $ΚΛ$ · σύμμετρον ἄρα ἔστι καὶ τὸ ὑπὸ τῶν $\Gamma\Delta$, AB τῷ ὑπὸ τῶν $\Gamma\Delta$, $ΚΛ$. ἴσον δὲ τὸ ὑπὸ τῶν $\Gamma\Delta$, $ΚΛ$ τῷ ἀπὸ τῆς Θ · σύμμετρον ἄρα ἔστι τὸ ὑπὸ τῶν $\Gamma\Delta$, AB τῷ ἀπὸ τῆς Θ . τῷ δὲ ὑπὸ τῶν $\Gamma\Delta$, AB ἴσον ἔστι τὸ ἀπὸ τῆς H · σύμμετρον ἄρα ἔστι τὸ ἀπὸ τῆς H τῷ ἀπὸ τῆς Θ . ῥητὸν δὲ τὸ ἀπὸ τῆς Θ · ῥητὸν ἄρα ἔστι καὶ τὸ ἀπὸ τῆς H · ῥητὴ ἄρα ἔστιν ἡ H . καὶ δύναται τὸ ὑπὸ τῶν $\Gamma\Delta$, AB .

a rational (straight-line).

For let an area, the (rectangle contained) by AB and CD , be contained by the apotome AB , and the binomial CD , of which let the greater term be CE . And let the terms of the binomial, CE and ED , be commensurable with the terms of the apotome, AF and FB (respectively), and in the same ratio. And let the square-root of the (rectangle contained) by AB and CD be G . I say that G is a rational (straight-line).

For let the rational (straight-line) H be laid down. And let (some rectangle), equal to the (square) on H , be applied to CD , producing KL as breadth. Thus, KL is an apotome, of which let the terms, KM and ML , be commensurable with the terms of the binomial, CE and ED (respectively), and in the same ratio [Prop. 10.112]. But, CE and ED are also commensurable with AF and FB (respectively), and in the same ratio. Thus, as AF is to FB , so KM (is) to ML . Thus, alternately, as AF is to KM , so BF (is) to LM [Prop. 5.16]. Thus, the remainder AB is also to the remainder KL as AF (is) to KM [Prop. 5.19]. And AF (is) commensurable with KM [Prop. 10.12]. AB is thus also commensurable with KL [Prop. 10.11]. And as AB is to KL , so the (rectangle contained) by CD and AB (is) to the (rectangle contained) by CD and KL [Prop. 6.1]. Thus, the (rectangle contained) by CD and AB is also commensurable with the (rectangle contained) by CD and KL [Prop. 10.11]. And the (rectangle contained) by CD and KL (is) equal to the (square) on H . Thus, the (rectangle contained) by CD and AB is commensurable with the (square) on H . And the (square) on G is equal to the (rectangle contained) by CD and AB . The (square) on G is thus commensurable with the (square) on H . And the (square) on H (is) rational. Thus, the (square) on G is also rational. G is thus rational. And it is the square-root of the (rectangle contained) by CD and AB .



Ἐάν ἄρα θωρίον περιέθῃται ὑπὸ ἀποτομῆς καὶ τῆς ἐκ δύο ὀνομάτων, ἥξ τὰ ὀνόματα σύμμετρα ἔστι τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἡ τὸ θωρίον δυναμένη ῥητὴ ἔστιν.

Thus, if an area is contained by an apotome, and a binomial whose terms are commensurable with, and in the same ratio as, the terms of the apotome, then the square-root of the area is a rational (straight-line).

Πόρισμα.

Καὶ γέγονεν ἡμῖν καὶ διὰ τούτου φανερόν, ὅτι δυνατόν ἐστι ῥητὸν θωρίον ὑπὸ ἀλόγων εὐθειῶν περιέθεσθαι. ὅπερ ἔδει δεῖχαι.

Corollary

And it has also been made clear to us, through this, that it is possible for a rational area to be contained by irrational straight-lines. (Which is) the very thing it was required to show.

115.

Ἀπὸ μέσης ἄπειροι ἄλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾶ τῶν πρότερον ἢ αὐτῇ.

Ἐστω μέση ἡ A· λέγω, ὅτι ἀπὸ τῆς A ἄπειροι ἄλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾶ τῶν πρότερον ἢ αὐτῇ.

Proposition 115

An infinite (series) of irrational (straight-lines) can be created from a medial (straight-line), and none of them is the same as any of the preceding (straight-lines).

Let A be a medial (straight-line). I say that an infinite (series) of irrational (straight-lines) can be created from A, and that none of them is the same as any of the preceding (straight-lines).

A ————

B —————

Γ —————

Δ —————

A ————

B —————

C —————

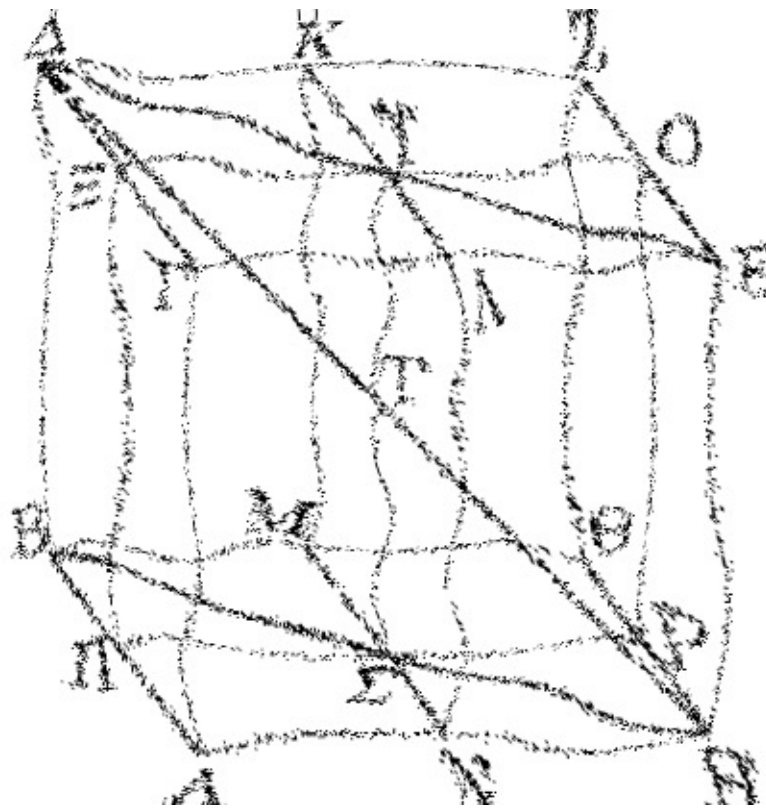
D —————

Ἐκκείσθω ῥητὴ ἡ B , καὶ τῷ ὑπὸ τῶν B , A ἴσον ἔστω τὸ ἀπὸ τῆς Γ · ἄλογος ἄρα ἐστὶν ἡ Γ · τὸ γὰρ ὑπὸ ἀλόγων καὶ ῥητῆς ἄλογόν ἐστιν. καὶ οὐδεμιᾶ τῶν πρότερον ἡ αὐτὴ· τὸ γὰρ ἀπ' οὐδεμιᾶς τῶν πρότερον παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ μέσην. πάλιν δὴ τῷ ὑπὸ τῶν B , Γ ἴσον ἔστω τὸ ἀπὸ τῆς Δ · ἄλογον ἄρα ἐστὶ τὸ ἀπὸ τῆς Δ · ἄλογος ἄρα ἐστὶν ἡ Δ · καὶ οὐδεμιᾶ τῶν πρότερον ἡ αὐτὴ· τὸ γὰρ ἀπ' οὐδεμιᾶς τῶν πρότερον παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν Γ . ὁμοίως δὴ τῆς τοιαύτης τάξεως ἐπ' ἄπειρον προβαينούσης φανερόν, ὅτι ἀπὸ τῆς μέσης ἄπειροι ἄλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾶ τῶν πρότερον ἡ αὐτὴ· ὅπερ ἔδει δεῖξαι.

Let the rational (straight-line) B be laid down. And let the (square) on C be equal to the (rectangle contained) by B and A . Thus, C is irrational [Def. 10.4]. For an (area contained) by an irrational and a rational (straight-line) is irrational [Prop. 10.20]. And (C is) not the same as any of the preceding (straight-lines). For the (square) on none of the preceding (straight-lines), applied to a rational (straight-line), produces a medial (straight-line) as breadth. So, again, let the (square) on D be equal to the (rectangle contained) by B and C . Thus, the (square) on D is irrational [Prop. 10.20]. D is thus irrational [Def. 10.4]. And (D is) not the same as any of the preceding (straight-lines). For the (square) on none of the preceding (straight-lines), applied to a rational (straight-line), produces C as breadth. So, similarly, this arrangement being advanced to infinity, it is clear that an infinite (series) of irrational (straight-lines) can be created from a medial (straight-line), and that none of them is the same as any of the preceding (straight-lines). (Which is) the very thing it was required to show.

ELEMENTS BOOK 11

Elementary Stereometry



Ὅροι.

1. Στερεόν ἐστι τὸ μήκοξ καὶ πλάτοξ καὶ βάθος ἔθον.
2. Στερεοῦ δὲ πέραξ ἐπιφάνεια.
3. Εὐθεΐα πρὸς ἐπίπεδον ὀρθή ἐστιν, ὅταν πρὸς πάσαξ τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὐσαξ ἐν τῷ [ὑποκειμένῳ] ἐπιπέδῳ ὀρθὰς ποιῇ γωνίαξ.
4. Ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, ὅταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰς ἀγόμεναι εὐθεΐαι ἐν ἐνὶ τῶν ἐπιπέδων τῷ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾧσιν.
5. Εὐθείαξ πρὸς ἐπίπεδον κλίσιξ ἐστίν, ὅταν ἀπὸ τοῦ μετεώρου πέρατοξ τῆς εὐθείας ἐπὶ τὸ ἐπίπεδον κάθετοξ ἀθθῇ, καὶ ἀπὸ τοῦ γενομένου σημείου ἐπὶ τὸ ἐν τῷ ἐπιπέδῳ πέραξ τῆς εὐθείας εὐθεΐα ἐπιξενθθῇ, ἡ περιεθιομένη γωνία ὑπὸ τῆς ἀθθείσης καὶ τῆς ἐφεστώσης.
6. Ἐπιπέδον πρὸς ἐπίπεδον κλίσιξ ἐστίν ἡ περιεθιομένη ὀχεΐα γωνία ὑπὸ τῶν πρὸς ὀρθὰς τῇ κοινῇ τομῇ ἀγομένων πρὸς τῷ αὐτῷ σημείῳ ἐν ἐκατέρῳ τῶν ἐπιπέδων.
7. Ἐπίπεδον πρὸς ἐπίπεδον ὁμοίωξ κεκλίσθαι λέγεται καὶ ἕτερον πρὸς ἕτερον, ὅταν αἱ εἰρημέναι τῶν κλίσεων γωναὶ ἴσαι ἀλλήλαιξ ᾧσιν.
8. Παράλληλα ἐπίπεδά ἐστι τὰ ἀσύμπτωτα.
9. Ὅμοια στερεὰ σθῆματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεθιόμενα ᾧσιν τὸ πλήθος.
10. ἴσα δὲ καὶ ὁμοια στερεὰ σθῆματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεθιόμενα ᾧσιν τῷ πλήθει καὶ τῷ μεγέθει.
11. Στερεὰ γωνία ἐστίν ἡ ὑπὸ πλειόνων ἢ δύο γραμμῶν ἀπτομένων ἀλλήλων καὶ μὴ ἐν τῇ αὐτῇ ἐπιφανείᾳ οὐσῶν πρὸς πάσαιξ ταῖξ γραμμαῖξ κλίσιξ. ἄλλωξ· στερεὰ γωνία ἐστίν ἡ ὑπὸ πλειόνων ἢ δύο γωνιῶν ἐπιπέδων περιεθιομένη μὴ οὐσῶν ἐν τῷ αὐτῷ ἐπιπέδῳ πρὸς ἐνὶ σημείῳ συνισταμένων.
12. Πυραμίξ ἐστὶ σθῆμα στερεὸν ἐπιπέδοιξ περιθιόμενον ἀπὸ ἐνὸς ἐπιπέδου πρὸς ἐνὶ σημείῳ συνεστῶξ.
13. Πρίσμα ἐστὶ σθῆμα στερεὸν ἐπιπέδοιξ περιεχόμενον, ὧν δύο τὰ ἀπεναντίον ἴσα τε καὶ ὁμοιά ἐστι καὶ παράλληλα, τὰ δὲ λοιπά παραλληλόγραμμα.
14. Σφαῖρά ἐστιν, ὅταν ἡμικυκλίον μενούσης τῆς διαμέτρου περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα.
15. Ἄχων δὲ τῆς σφαίραξ ἐστίν ἡ μένουσα εὐθεΐα, περιῖ ἢν τὸ ἡμικύκλιον στρέφεται.
16. Κέντρον δὲ τῆς σφαίραξ ἐστὶ τὸ αὐτό, ὃ καὶ τοῦ ἡμικυκλίου.
17. Διάμετροξ δὲ τῆς σφαίραξ ἐστίν εὐθεΐα τιξ διὰ τοῦ κέντρου ἡγμένη καὶ περατουμένη ἐφ' ἐκάτερα τὰ μέρη ὑπὸ τῆς ἐπιφανείας τῆς σφαίραξ.
18. Κωνός ἐστιν, ὅταν ὀρθογωνίου τριγώνου μενούσης μιᾶξ πλευρᾶξ τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθθὲν τὸ τρίγωνον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα. κ' ἂν μὲν ἡ μένουσα εὐθεΐα ἴση ᾧ τῇ λοιπῇ

[τῇ] περὶ τὴν ὀρθὴν περιφερομένη, ὀρθογώνιος ἔσται ὁ κῶνος, ἔαν δὲ ἐλάττων, ἀμβλυγώνιος, ἔαν δὲ μείζων, ὀχυγώνιος.

19. Ἄχων δὲ τοῦ κώνου ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ τρίγωνον στρέφεται.

20. Βάσις δὲ ὁ κύκλος ὁ ὑπὸ τῆς περιφερομένης εὐθείας γραφόμενος.

21. Κύλινδρος ἐστὶν, ὅταν ὀρθογωνίου παραλληλογράμμου μενούσης μιᾶς πλευρᾶς τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθ-
θὲν τὸ παραλληλόγραμμον εἰς τὸ αὐτὸ πάλιν ἀποκατασταθῇ,
ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα.

22. Ἄχων δὲ τοῦ κυλίνδρου ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ παραλληλόγραμμον στρέφεται.

23. Βάσεις δὲ οἱ κύκλοι οἱ ὑπὸ τῶν ἀπεναντίον περιγο-
μένων δύο πλευρῶν γραφόμενοι.

24. Ὅμοιοι κῶνοι καὶ κύλινδροί εἰσιν, ὧν οἱ τε ἄχονες καὶ αἱ διάμετροι τῶν βάσεων ἀνάλογόν εἰσιν.

25. Κύβοξ ἐστὶ σθῆμα στερεὸν ὑπὸ ἑχ τετραγώνων ἴσων περιεχόμενον.

26. Ὀκτάεδρόν ἐστὶ σθῆμα στερεὸν ὑπὸ ὀκτὼ τριγώνων ἴσων καὶ ἰσοπλεύρων περιεχόμενον.

27. Εἰκοσάεδρόν ἐστὶ σθῆμα στερεὸν ὑπὸ εἴκοσι τριγώνων ἴσων καὶ ἰσοπλεύρων περιεχόμενον.

28. Δωδεκάεδρόν ἐστὶ σθῆμα στερεὸν ὑπὸ δώδεκα πεντα-
γώνων ἴσων καὶ ἰσοπλεύρων καὶ ἰσογωνίων περιεχόμενον.

Definitions

1. A solid is a (figure) having length and breadth and depth.

2. The extremity of a solid (is) a surface.

3. A straight-line is at right-angles to a plane when it makes right-angles with all of the straight-lines joined to it which are also in the plane.

4. A plane is at right-angles to a(nother) plane when (all of) the straight-lines drawn in one of the planes, at right-angles to the common section of the planes, are at right-angles to the remaining plane.

5. The inclination of a straight-line to a plane is the angle contained by the drawn and standing (straight-lines), when a perpendicular is lead to the plane from the end of the (stand-
ing) straight-line raised (out of the plane), and a straight-line is (then) joined from the point (so) generated to the end of the (standing) straight-line (lying) in the plane.

6. The inclination of a plane to a(nother) plane is the acute angle contained by the (straight-lines), (one) in each of the planes, drawn at right-angles to the common segment (of the planes), at the same point.

7. A plane is said to be similarly inclined to a plane, as another to another, when the aforementioned angles of incli-
nation are equal to one another.

8. Parallel planes are those which do not meet (one an-
other).

9. Similar solid figures are those contained by equal num-
bers of similar planes (which are similarly arranged).

10. But equal and similar solid figures are those contained by similar planes equal in number and in magnitude (which are similarly arranged).

11. A solid angle is the inclination (constituted) by more than two lines joining one another (at the same point), and not being in the same surface, to all of the lines. Otherwise, a solid angle is that contained by more than two plane angles, not being in the same plane, and constructed at one point.

12. A pyramid is a solid figure, contained by planes, (which is) constructed from one plane to one point.

13. A prism is a solid figure, contained by planes, of which the two opposite (planes) are equal, similar, and parallel, and the remaining (planes are) parallelograms.

14. A sphere is the figure enclosed when, the diameter of a semicircle remaining (fixed), the semicircle is carried around, and again established at the same (position) from which it began to be moved.

15. And the axis of the sphere is the fixed straight-line about which the semicircle is turned.

16. And the center of the sphere is the same as that of the semicircle.

17. And the diameter of the sphere is any straight-line which is drawn through the center and terminated in both directions by the surface of the sphere.

18. A cone is the figure enclosed when, one of the sides of a right-angled triangle about the right-angle remaining (fixed), the triangle is carried around, and again established at the same (position) from which it began to be moved. And if the fixed straight-line is equal to the remaining (straight-line) about the right-angle, (which is) carried around, then the cone will be right-angled, and if less, obtuse-angled, and if greater, acute-angled.

19. And the axis of the cone is the fixed straight-line about which the triangle is turned.

20. And the base (of the cone is) the circle described by the (remaining) straight-line (about the right-angle which is) carried around (the axis).

21. A cylinder is the figure enclosed when, one of the sides of a right-angled parallelogram about the right-angle remaining (fixed), the parallelogram is carried around, and again established at the same (position) from which it began to be moved.

22. And the axis of the cylinder is the stationary straight-line about which the parallelogram is turned.

23. And the bases (of the cylinder are) the circles described by the two opposite sides (which are) carried around.

24. Similar cones and cylinders are those for which the axes and the diameters of the bases are proportional.

25. A cube is a solid figure contained by six equal squares.

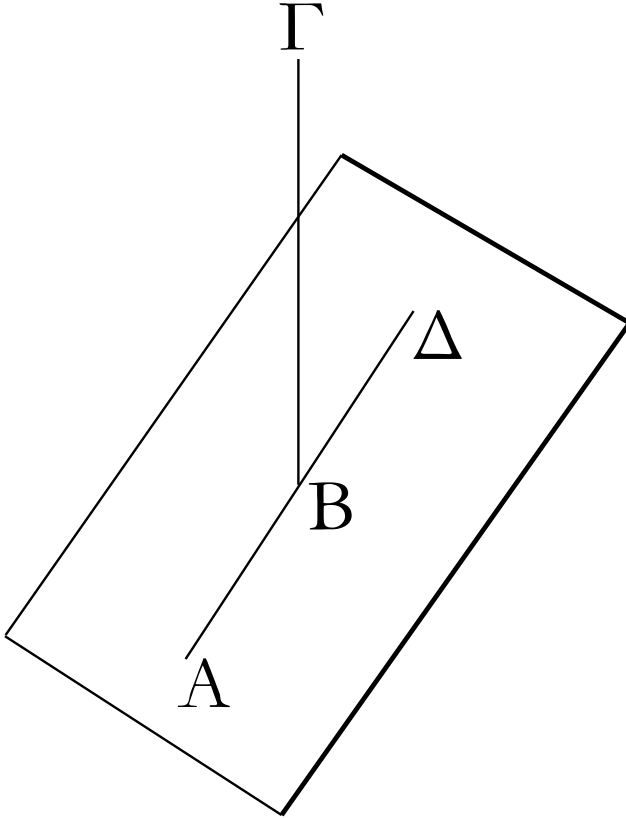
26. An octahedron is a solid figure contained by eight equal and equilateral triangles.

1.

Εὐθεία γραμμῆς μέρος μὲν τι οὐκ ἔστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι ἐν μετεωροτέρῳ.

Εἰ γὰρ δυνατόν, εὐθεία γραμμῆς τῆς $AB\Gamma$ μέρος μὲν τι τὸ AB ἔστω ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι τὸ $B\Gamma$ ἐν μετεωροτέρῳ.

Ἔσται δὴ τῆς AB συννεθῆς εὐθεῖα ἐπ' εὐθείας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ. ἔστω ἡ BD . δύο ἄρα εὐθειῶν τῶν $AB\Gamma$, ABD κοινὸν τμήμα ἔστιν ἡ AB . ὅπερ ἐστὶν ἀδύνατον, ἐπειδήπερ ἔαν κέντρῳ τῷ B καὶ διαστήματι τῷ AB κύκλον γράψωμεν, αἱ διάμετροι ἀνίσουσιν ἀπολήγονται τοῦ κύκλου περιφερείας.



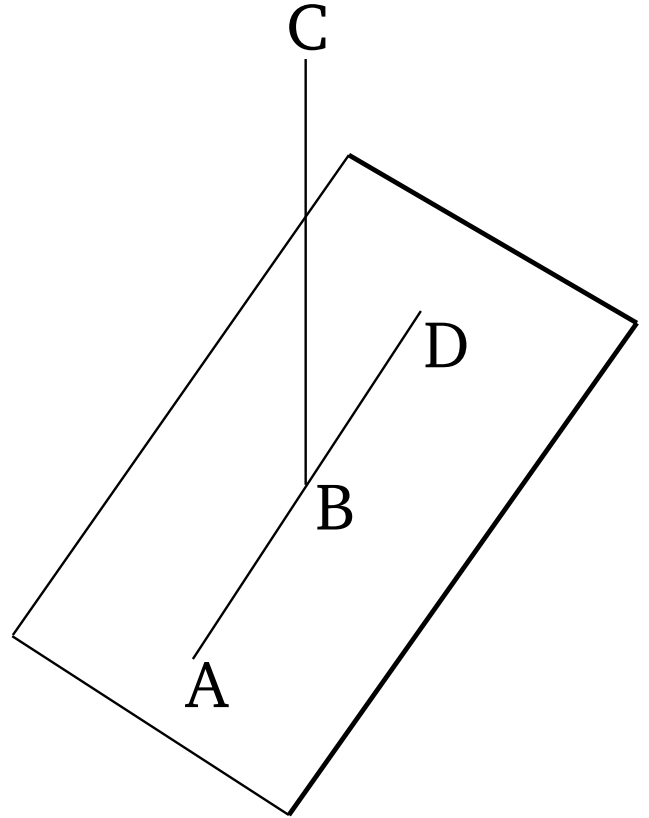
Εὐθεία ἄρα γραμμῆς μέρος μὲν τι οὐκ ἔστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δέ ἐν μετεωροτέρῳ. ὅπερ ἔδει δεῖξαι.

Proposition 1[†]

Some part of a straight-line cannot be in a reference plane, and some part in a more elevated (plane).

For, if possible, let some part, AB , of the straight-line ABC be in a reference plane, and some part, BC , in a more elevated (plane).

In the reference plane, there will be some straight-line continuous with, and straight-on to, AB .[‡] Let it be BD . Thus, AB is a common segment of the two (different) straight-lines ABC and ABD . The very thing is impossible, inasmuch as if we draw a circle with center B and radius AB then the diameters (ABD and ABC) will cut off unequal circumferences of the circle.



Thus, some part of a straight-line cannot be in a reference plane, and (some part) in a more elevated (plane). (Which is) the very thing it was required to show.

[†] The proofs of the first three propositions in this book are not at all rigorous. Hence, these three propositions should properly be regarded as additional axioms.

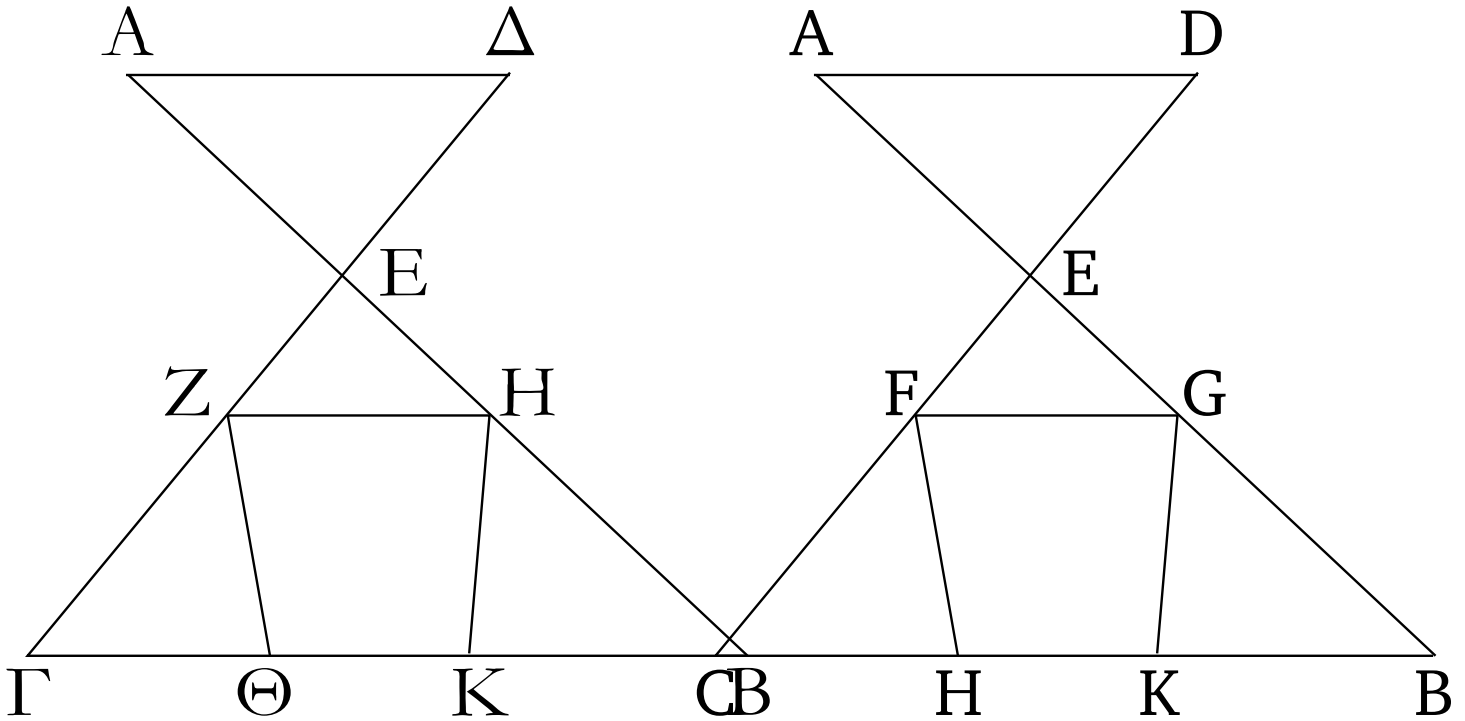
‡ This assumption essentially presupposes the validity of the proposition under discussion.

2.

Proposition 2

Ἐάν δύο εὐθεῖαι τέμνωσιν ἀλλήλας, ἐν ἐνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ.

If two straight-lines cut one another then they are in one plane, and every triangle (formed using segments of both lines) is in one plane.



Δύο γὰρ εὐθεῖαι αἱ AB , $ΓΔ$ τεμνέτωσαν ἀλλήλας κατὰ τὸ E σημεῖον. λέγω, ὅτι αἱ AB , $ΓΔ$ ἐν ἐνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ.

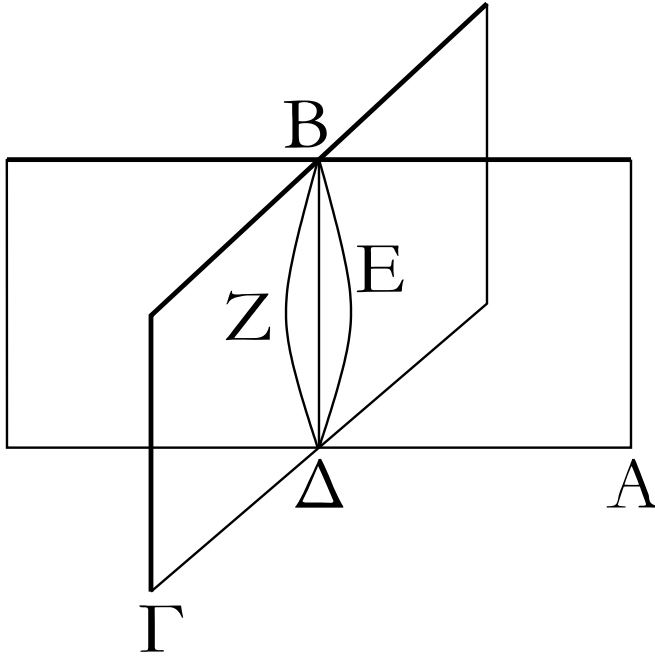
Εἰλήφθω γὰρ ἐπὶ τῶν $ΕΓ$, $ΕΒ$ τυθόντα σημεῖα τὰ Z , H , καὶ ἐπεξεύθθωσαν αἱ $ΓΒ$, ZH , καὶ διήθθωσαν αἱ $ZΘ$, HK . λέγω πρῶτον, ὅτι τὸ $ΕΓΒ$ τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ. εἰ γὰρ ἐστὶ τοῦ $ΕΓΒ$ τριγώνου μέρος ἥτοι τὸ $ZΘΓ$ ἢ τὸ $ΗΒΚ$ ἐν τῷ ὑποκειμένῳ [ἐπιπέδῳ], τὸ δὲ λοιπὸν ἐν ἄλλῳ, ἔσται καὶ μιᾷ τῶν $ΕΓ$, $ΕΒ$ εὐθειῶν μέρος μὲν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν ἄλλῳ. εἰ δὲ τοῦ $ΕΓΒ$ τριγώνου τὸ $ZΓΒΗ$ μέρος ἢ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ λοιπὸν ἐν ἄλλῳ, ἔσται καὶ ἀμφοτέρων τῶν $ΕΓ$, $ΕΒ$ εὐθειῶν μέρος μὲν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν ἄλλῳ· ὅπερ ἄτοπον ἐδείχθη. τὸ ἄρα $ΕΓΒ$ τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ. ἐν ᾧ δὲ ἐστὶ τὸ $ΕΓΒ$ τρίγωνον, ἐν τούτῳ καὶ ἑκατέρα τῶν $ΕΓ$, $ΕΒ$, ἐν ᾧ δὲ ἑκατέρα τῶν $ΕΓ$, $ΕΒ$, ἐν τούτῳ καὶ αἱ AB , $ΓΔ$. αἱ AB , $ΓΔ$ ἄρα εὐθεῖαι ἐν ἐνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ· ὅπερ ἔδει δεῖξαι.

For let the two straight-lines AB and CD have cut one another at point E . I say that AB and CD are in one plane, and that every triangle (formed using segments of both lines) is in one plane.

For let the random points F and G be taken on EC and EB (respectively). And let CB and FG be joined, and let FH and GK be drawn across. I say, first of all, that triangle ECB is in one (reference) plane. For if part of triangle ECB , either FHC or GBK , is in the reference [plane], and the remainder in a different (plane) then a part of one the straight-lines EC and EB will also be in the reference plane, and (a part) in a different (plane). And if the part $FCBG$ of triangle ECB is in the reference plane, and the remainder in a different (plane) then parts of both of the straight-lines EC and EB will also be in the reference plane, and (parts) in a different (plane). The very thing was shown to be absurd [Prop. 11.1]. Thus, triangle ECB is in one plane. And in whichever (plane) triangle ECB is (found), in that (plane) EC and EB (will) each also (be found). And in whichever (plane) EC and EB (are) each (found), in that (plane) AB and CD (will) also (be found) [Prop. 11.1]. Thus, the straight-lines AB and CD are in one plane, and every

3.

Ἐάν δύο ἐπίπεδα τεμνῇ ἄλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεΐά ἐστιν.



Δύο γάρ ἐπίπεδα τὰ AB , $BΓ$ τεμνέτω ἄλληλα, κοινὴ δὲ αὐτῶν τομὴ ἔστω ἡ $ΔΒ$ γραμμὴ· λέγω, ὅτι ἡ $ΔΒ$ γραμμὴ εὐθεΐά ἐστιν.

Εἰ γὰρ μή, ἐπεζεύθθω ἀπὸ τοῦ $Δ$ ἐπὶ τὸ B ἐν μὲν τῷ AB ἐπιπέδῳ εὐθεΐα ἡ $ΔΕΒ$, ἐν δὲ τῷ $BΓ$ ἐπιπέδῳ εὐθεΐα ἡ $ΔΖΒ$. ἔσται δὴ δύο εὐθειῶν τῶν $ΔΕΒ$, $ΔΖΒ$ τὰ αὐτὰ πέρατα, καὶ περιέχονσι δηλαδὴ θωρίον· ὅπερ ἄτοπον. οὐκ ἄρα αἱ $ΔΕΒ$, $ΔΖΒ$ εὐθεΐαι εἰσιν. ὁμοίως δὲ δείχμεν, ὅτι οὐδὲ ἄλλη τις ἀπὸ τοῦ $Δ$ ἐπὶ τὸ B ἐπιζευγνυμένη εὐθεΐα ἔσται πλὴν τῆς $ΔΒ$ κοινῆς τομῆς τῶν AB , $BΓ$ ἐπιπέδων.

Ἐάν ἄρα δύο ἐπίπεδα τέμνῃ ἄλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεΐά ἐστιν· ὅπερ ἔδει δεῖχαι.

4.

Ἐάν εὐθεΐα δύο εὐθείαις τεμνούσαις ἀλλήλας πρὸς ὀρθὰς ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ δι' αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰς ἔσται.

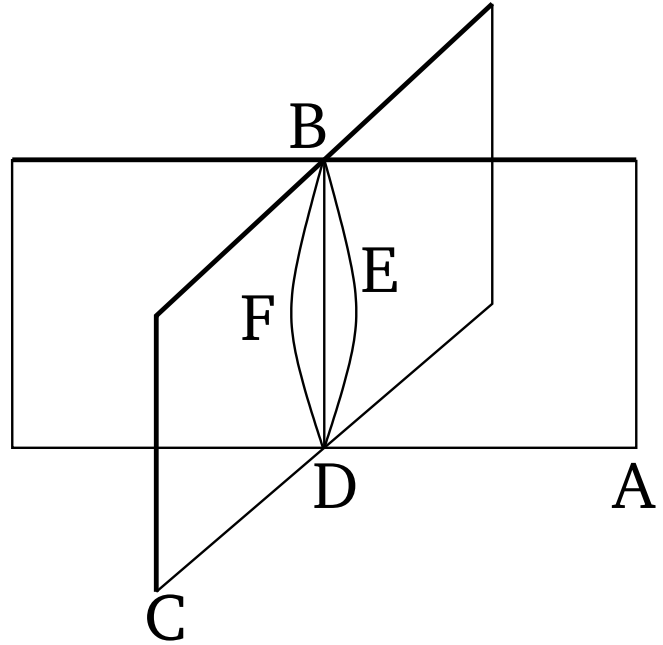
Εὐθεΐα γὰρ τις ἡ EZ δύο εὐθείαις ταῖς AB , $ΓΔ$ τεμνούσαις ἀλλήλας κατὰ τὸ E σημεῖον ἀπὸ τοῦ E πρὸς ὀρθὰς ἐφρεστάτω· λέγω, ὅτι ἡ EZ καὶ τῷ διὰ τῶν AB , $ΓΔ$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν.

Ἀπειλήφθωσαν γὰρ αἱ AE , EB , $ΓΕ$, $ΕΔ$ ἴσαι ἀλλήλαις,

triangle (formed using segments of both lines) is in one plane. (Which is) the very thing it was required to show.

Proposition 3

If two planes cut one another then their common section is a straight-line.



For let the two planes AB and BC cut one another, and let their common section be the line DB . I say that the line DB is straight.

For, if not, let the straight-line DEB be joined from D to B in the plane AB , and the straight-line DFB in the plane BC . So two straight-lines, DEB and DFB , will have the same ends, and they will clearly enclose an area. The very thing (is) absurd. Thus, DEB and DFB are not straight-lines. So, similarly, we can show that no other straight-line can be joined from D to B except DB , the common section of the planes AB and BC .

Thus, if two planes cut one another then their common section is a straight-line. (Which is) the very thing it was required to show.

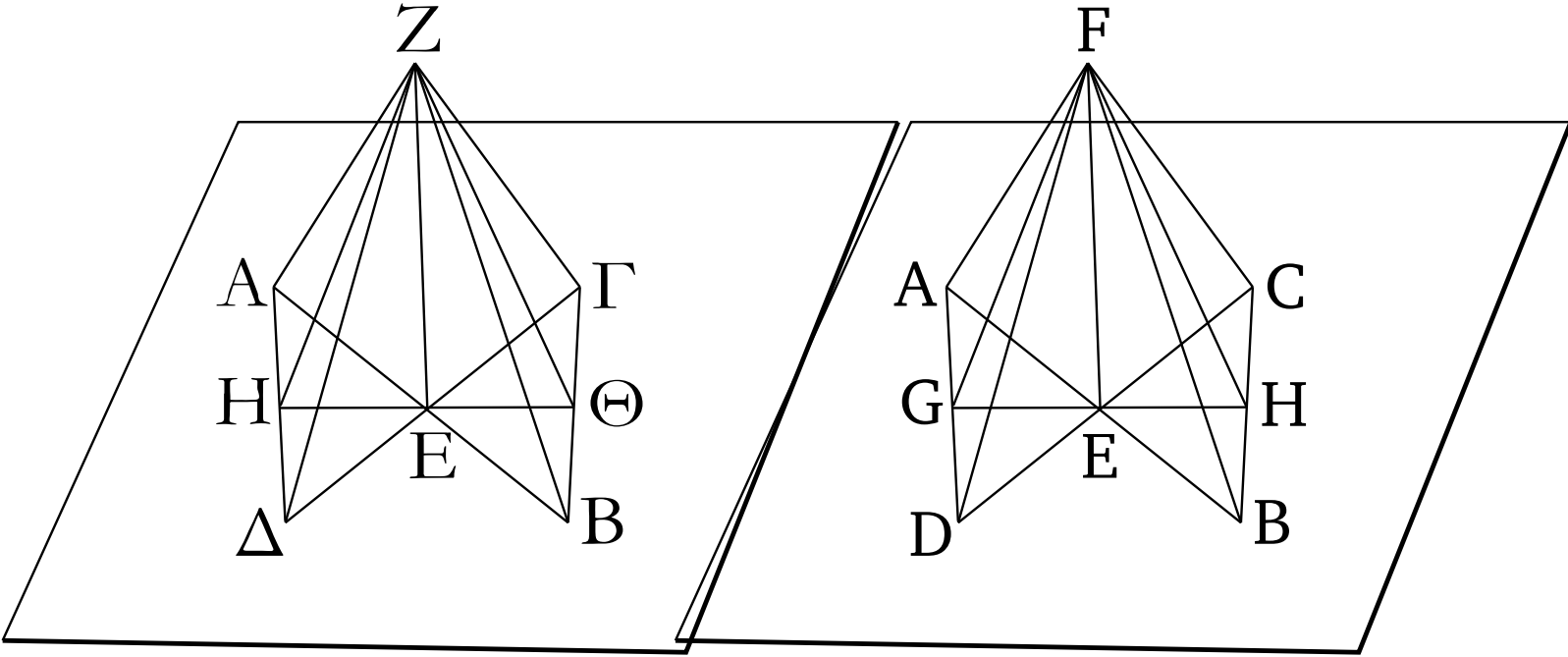
Proposition 4

If a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both).

For let some straight-line EF have (been) set up at right-angles to two straight-lines, AB and CD , cutting one another at point E , at E . I say that EF is also at right-angles to the plane (passing) through AB and CD .

καὶ διήθηται διὰ τοῦ E , ὥς ἔτυθεν, ἡ $HE\Theta$, καὶ ἐπεζεύθησαν αἱ $A\Delta$, ΓB , καὶ ἔτι ἀπὸ τυθόντος τοῦ Z ἐπεζεύθησαν αἱ ZA , ZH , $Z\Delta$, $Z\Gamma$, $Z\Theta$, ZB .

For let AE , EB , CE and ED be cut off from (the two straight-lines so as to be) equal to one another. And let GEH be drawn, at random, through E (in the plane passing through AB and CD). And let AD and CB be joined. And, furthermore, let FA , FG , FD , FC , FH , and FB be joined from the random (point) F (on EF).



Καὶ ἐπεὶ δύο αἱ AE , $E\Delta$ δυοὶ ταῖς $ΓE$, EB ἴσαι εἰσὶ καὶ γωνίαι ἴσαι περιέθουσιν, βάσεις ἄρα ἡ $A\Delta$ βάσει τῇ ΓB ἴση ἐστίν, καὶ τὸ AED τρίγωνον τῷ ΓEB τριγώνῳ ἴσον ἔσται· ὥστε καὶ γωνία ἡ ὑπὸ ΔAE γωνία τῇ ὑπὸ $EB\Gamma$ ἴση [ἐστίν]. ἔστι δὲ καὶ ἡ ὑπὸ AHE γωνία τῇ ὑπὸ $BE\Theta$ ἴση. δύο δὲ τρίγωνα ἐστὶ τὰ AHE , $BE\Theta$ τὰς δύο γωνίας δυοὶ γωνίαις ἴσαις ἔθοντα ἐκατέρωθεν ἐκατέρω καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν AE τῇ EB · καὶ τὰς λοιπὰς ἄρα πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαις ἔχουσιν. ἴση ἄρα ἡ μὲν HE τῇ $E\Theta$, ἡ δὲ AH τῇ $B\Theta$. καὶ ἐπεὶ ἴση ἐστὶν ἡ AE τῇ EB , κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ ZE , βάσεις ἄρα ἡ ZA βάσει τῇ ZB ἐστὶν ἴση. διὰ τὰ αὐτὰ δὲ καὶ ἡ $Z\Gamma$ τῇ $Z\Delta$ ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ $A\Delta$ τῇ ΓB , ἔστι δὲ καὶ ἡ ZA τῇ ZB ἴση, δύο δὲ αἱ ZA , $A\Delta$ δυοὶ ταῖς ZB , $B\Gamma$ ἴσαι εἰσὶν ἐκατέρωθεν· καὶ βάσεις ἡ $Z\Delta$ βάσει τῇ $Z\Gamma$ ἐδείχθη ἴση· καὶ γωνία ἄρα ἡ ὑπὸ $\Delta A\Delta$ γωνία τῇ ὑπὸ $ZB\Gamma$ ἴση ἐστίν. καὶ ἐπεὶ πάλιν ἐδείχθη ἡ AH τῇ $B\Theta$ ἴση, ἀλλὰ μὴν καὶ ἡ ZA τῇ ZB ἴση, δύο δὲ αἱ ZA , AH δυοὶ ταῖς ZB , $B\Theta$ ἴσαι εἰσὶν. καὶ γωνία ἡ ὑπὸ ZAH ἐδείχθη ἴση τῇ ὑπὸ $ZB\Theta$ · βάσεις ἄρα ἡ ZH βάσει τῇ $Z\Theta$ ἐστὶν ἴση. καὶ ἐπεὶ πάλιν ἴση ἐδείχθη ἡ HE τῇ $E\Theta$, κοινὴ δὲ ἡ EZ , δύο δὲ αἱ HE , EZ δυοὶ ταῖς ΘE , EZ ἴσαι εἰσὶν· καὶ βάσεις ἡ ZH βάσει τῇ $Z\Theta$ ἴση· γωνία ἄρα ἡ ὑπὸ HEZ γωνία τῇ ὑπὸ ΘEZ ἴση ἐστίν. ὀρθὴ ἄρα ἐκατέρα τῶν ὑπὸ

For since the two (straight-lines) AE and ED are equal to the two (straight-lines) CE and EB , and they enclose equal angles [Prop. 1.15], the base AD is thus equal to the base CB , and triangle AED will be equal to triangle CEB [Prop. 1.4]. Hence, the angle DAE [is] equal to the angle EBC . And the angle AEG (is) also equal to the angle BEH [Prop. 1.15]. So AGE and BEH are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), those by the equal angles, AE and EB . Thus, they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, GE (is) equal to EH , and AG to BH . And since AE is equal to EB , and FE is common and at right-angles, the base FA is thus equal to the base FB [Prop. 1.4]. So, for the same (reasons), FC is also equal to FD . And since AD is equal to CB , and FA is also equal to FB , the two (straight-lines) FA and AD are equal to the two (straight-lines) FB and BC , respectively. And the base FD was shown (to be) equal to the base FC . Thus, the angle FAD is also equal to the angle FBC [Prop. 1.8]. And, again, since AG was shown (to be) equal to BH , but FA (is) also equal to FB , the two (straight-lines) FA and AG are equal to the two (straight-lines) FB and BH (respectively). And the angle FAG was shown (to be) equal to the angle FBH . Thus, the base FG is equal to the

HEZ , ΘEZ γωνιών. ἡ ZE ἄρα πρόξ τὴν $H\Theta$ τυθόντως διὰ τοῦ E ἀθθεῖσαν ὀρθή ἐστιν. ὁμοίως δὲ δείχομεν, ὅτι ἡ ZE καὶ πρόξ πάσας τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὐσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. εὐθεῖα δὲ πρόξ ἐπίπεδον ὀρθή ἐστιν, ὅταν πρόξ πάσας τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὐσας ἐν τῷ αὐτῷ ἐπιπέδῳ ὀρθὰς ποιῇ γωνίας. ἡ ZE ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρόξ ὀρθὰς ἐστιν. τὸ δὲ ὑποκείμενον ἐπίπεδόν ἐστι τὸ διὰ τῶν AB , $\Gamma\Delta$ εὐθειῶν. ἡ ZE ἄρα πρόξ ὀρθὰς ἐστι τῷ διὰ τῶν AB , $\Gamma\Delta$ ἐπιπέδῳ.

Ἐάν ἄρα εὐθεῖα δύο εὐθείαις τεμνούσαις ἀλλήλας πρόξ ὀρθὰς ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ δι' αὐτῶν ἐπιπέδῳ πρόξ ὀρθὰς ἔσται. ὅπερ ἔδει δεῖχαι.

5.

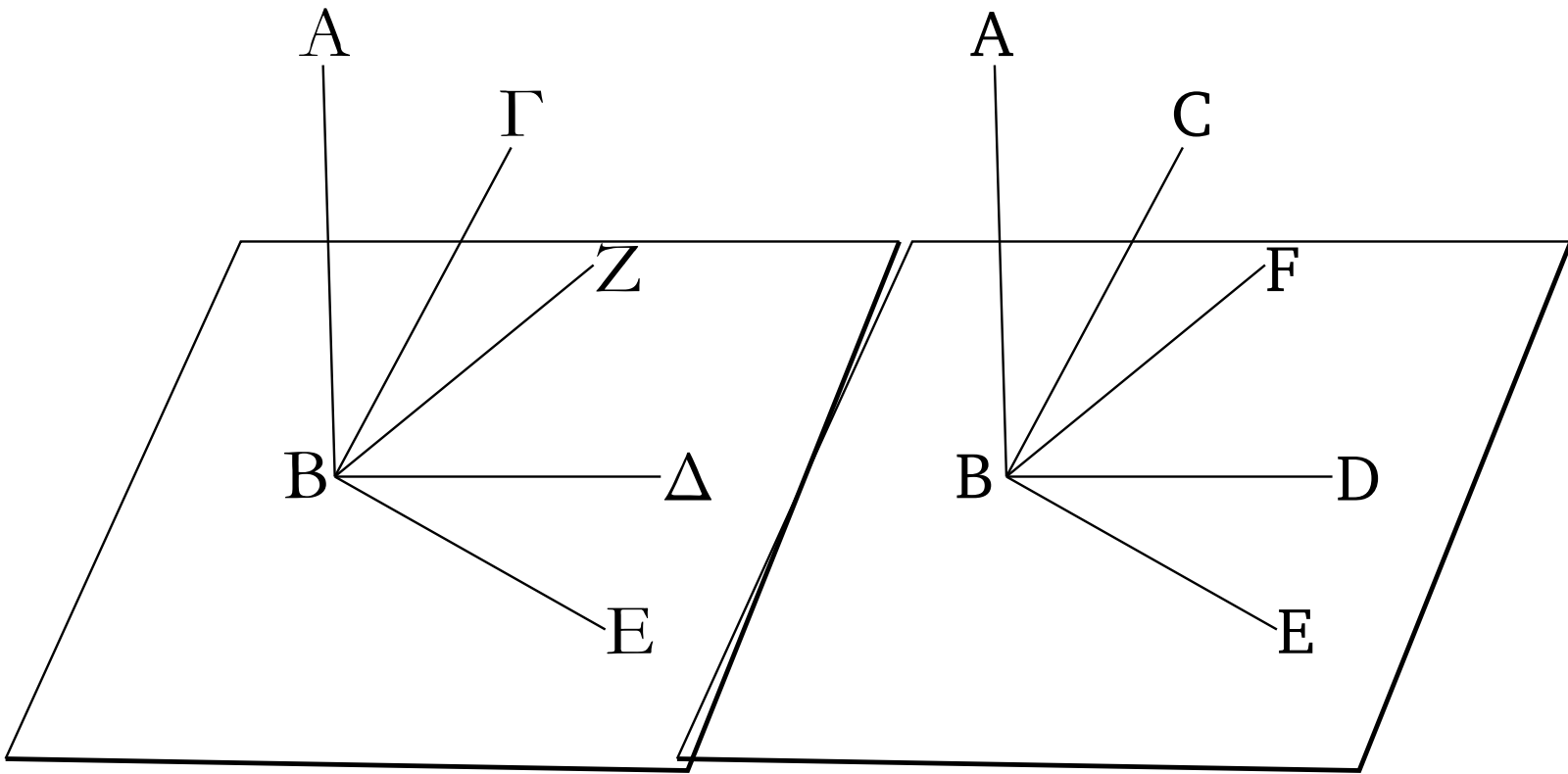
Ἐάν εὐθεῖα τρισὶν εὐθείαις ἀπτομέναις ἀλλήλων πρόξ ὀρθὰς ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἐνὶ εἰσὶν ἐπιπέδῳ.

base FH [Prop. 1.4]. And, again, since GE was shown (to be) equal to EH , and EF (is) common, the two (straight-lines) GE and EF are equal to the two (straight-lines) HE and EF (respectively). And the base FG (is) equal to the base FH . Thus, the angle GEF is equal to the angle HEF [Prop. 1.8]. Each of the angles GEF and HEF (are) thus right-angles [Def. 1.10]. Thus, FE is at right-angles to GH , which was drawn at random through E (in the reference plane passing through AB and AC). So, similarly, we can show that FE will make right-angles with all straight-lines joined to it which are in the reference plane. And a straight-line is at right-angles to a plane when it makes right-angles with all straight-lines joined to it which are in the plane [Def. 11.3]. Thus, FE is at right-angles to the reference plane. And the reference plane is that (passing) through the straight-lines AB and CD . Thus, FE is at right-angles to the plane (passing) through AB and CD .

Thus, if a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both). (Which is) the very thing it was required to show.

Proposition 5

If a straight-line is set up at right-angles to three straight-lines cutting one another, at the common point of section, then the three straight-lines are in one plane.



Εὐθεῖα γάρ τις ἡ AB τρισὶν εὐθείαις ταῖς $BΓ$, $BΔ$, BE πρὸς ὀρθὰς ἐπὶ τῆς κατὰ τὸ B ἀφ᾽ ἧς ἐφεστάτω· λέγω, ὅτι αἱ $BΓ$, $BΔ$, BE ἐν ἐνὶ εἰσὶν ἐπιπέδῳ.

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἔστωσαν αἱ μὲν $BΔ$, BE ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, ἡ δὲ $BΓ$ ἐν μετεωροτέρῳ, καὶ ἐκβεβλήσθω τὸ διὰ τῶν AB , $BΓ$ ἐπίπεδον· κοινὴν δὴ τομὴν ποιήσῃ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιεῖτω τὴν BZ . ἐν ἐνὶ ἄρα εἰσὶν ἐπιπέδῳ τῷ διηγμένῳ διὰ τῶν AB , $BΓ$ αἱ τρεῖς εὐθεῖαι αἱ AB , $BΓ$, BZ . καὶ ἐπεὶ ἡ AB ὀρθὴ ἐστὶ πρὸς ἑκατέραν τῶν $BΔ$, BE , καὶ τῷ διὰ τῶν $BΔ$, BE ἄρα ἐπιπέδῳ ὀρθὴ ἐστὶν ἡ AB . τὸ δὲ διὰ τῶν $BΔ$, BE ἐπίπεδον τὸ ὑποκείμενόν ἐστιν· ἡ AB ἄρα ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὥστε καὶ πρὸς πάσας τὰς ἀπομένους αὐτῆς εὐθείας καὶ οὕσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας ἡ AB . ἀπτεται δὲ αὐτῆς ἡ BZ ὅσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ἡ ἄρα ὑπὸ ABZ γωνία ὀρθὴ ἐστὶν. ὑπόκειται δὲ καὶ ἡ ὑπὸ $ABΓ$ ὀρθή· ἴση ἄρα ἡ ὑπὸ ABZ γωνία τῇ ὑπὸ $ABΓ$. καὶ εἰσὶν ἐν ἐνὶ ἐπιπέδῳ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ $BΓ$ εὐθεῖα ἐν μετεωροτέρῳ ἐστὶν ἐπιπέδῳ· αἱ τρεῖς ἄρα εὐθεῖαι αἱ $BΓ$, $BΔ$, BE ἐν ἐνὶ εἰσὶν ἐπιπέδῳ.

Ἐάν ἄρα εὐθεῖα τρισὶν εὐθείαις ἀπτομέναις ἀλλήλων ἐπὶ τῆς ἀφ᾽ ἧς πρὸς ὀρθὰς ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἐνὶ εἰσὶν ἐπιπέδῳ· ὅπερ ἔδει δεῖξαι.

For let some straight-line AB be set up at right-angles to three straight-lines BC , BD , and BE , at the (common) point of section B . I say that BC , BD , and BE are in one plane.

For (if) not, and if possible, let BD and BE be in the reference plane, and BC in a more elevated (plane). And let the plane through AB and BC be produced. So it will make a straight-line as a common section with the reference plane [Def. 11.3]. Let it make BF . Thus, the three straight-lines AB , BC , and BF are in one plane—(namely), that drawn through AB and BC . And since AB is at right-angles to each of BD and BE , AB is thus also at right-angles to the plane (passing) through BD and BE [Prop. 11.4]. And the plane (passing) through BD and BE is the reference plane. Thus, AB is at right-angles to the reference plane. Hence, AB will also make right-angles with all straight-lines joined to it which are also in the reference plane [Def. 11.3]. And BF , which is in the reference plane, is joined to it. Thus, the angle ABF is a right-angle. And ABC was also assumed to be a right-angle. Thus, angle ABF (is) equal to ABC . And they are in one plane. The very thing is impossible. Thus, BC is not in a more elevated plane. Thus, the three straight-lines BC , BD , and BE are in one plane.

Thus, if a straight-line is set up at right-angles to three straight-lines cutting one another, at the (common) point of section, then the three straight-lines are in one plane. (Which

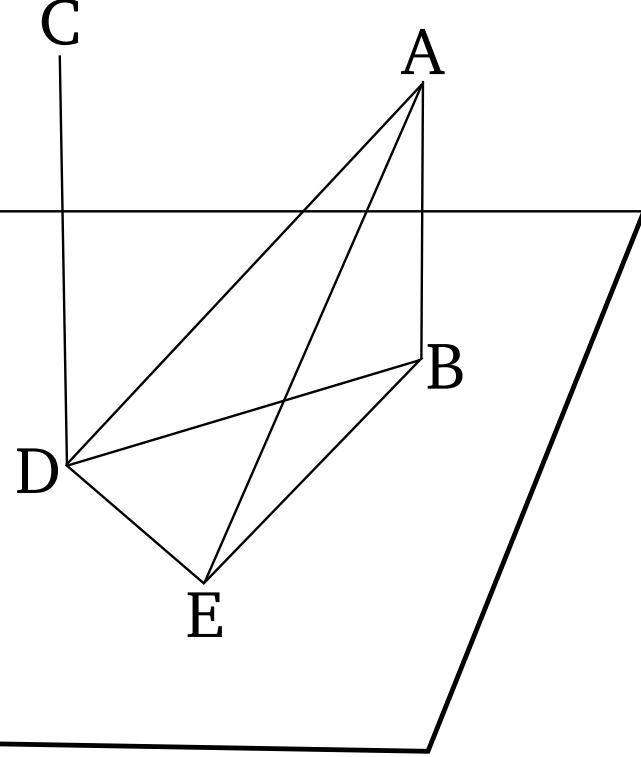
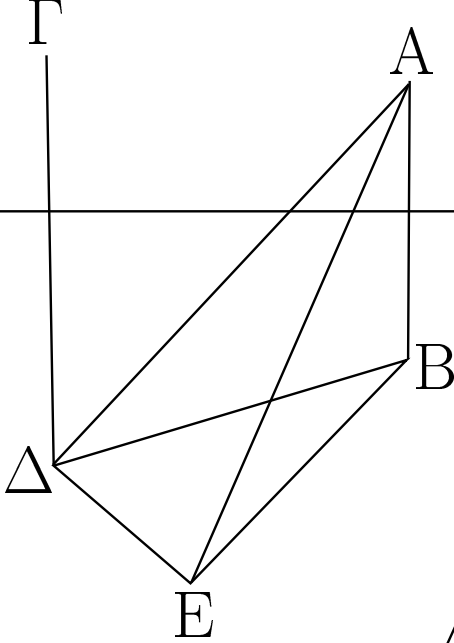
is) the very thing it was required to show.

6.

Proposition 6

Ἐάν δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾖσιν, παράλληλοι ἔσονται αἱ εὐθεῖαι.

If two straight-lines are at right-angles to the same plane then the straight-lines will be parallel.[†]



Δύο γὰρ εὐθεῖαι αἱ AB , $\Gamma\Delta$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἔστωσαν· λέγω, ὅτι παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$.

Συμβαλλέτωσαν γὰρ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ B , Δ σημεία, καὶ ἐπεζεύθω ἡ $B\Delta$ εὐθεῖα, καὶ ἤθθω τῇ $B\Delta$ πρὸς ὀρθὰς ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ ΔE , καὶ κείσθω τῇ AB ἴση ἡ ΔE , καὶ ἐπεζεύθωσαν αἱ BE , AE , AD .

Καὶ ἐπεὶ ἡ AB ὀρθὴ ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας [ἄρα] τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ὅσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίαξ. ἄπτεται δὲ τῇ AB ἑκάτερα τῶν $B\Delta$, BE ὅσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ὀρθὴ ἄρα ἐστὶν ἑκάτερα τῶν ὑπὸ $AB\Delta$, ABE γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερα τῶν ὑπὸ $\Gamma\Delta B$, $\Gamma\Delta E$ ὀρθὴ ἐστὶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ ΔE , κοινὴ δὲ ἡ $B\Delta$, δύο δὴ αἱ AB , $B\Delta$ δυοὶ ταῖς $E\Delta$, ΔB ἴσαι εἰσὶν· καὶ γωνίαξ ὀρθὰς περιέθουσιν· βάσειξ ἄρα ἡ AD βάσει τῇ BE ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ ΔE , ἀλλὰ καὶ ἡ AD τῇ BE , δύο δὴ αἱ AB , BE δυοὶ ταῖς $E\Delta$, ΔA ἴσαι εἰσὶν· καὶ βάσειξ αὐτῶν κοινὴ ἡ AE · γωνία ἄρα ἡ ὑπὸ ABE γωνιά τῇ ὑπὸ $E\Delta A$ ἐστὶν ἴση. ὀρθὴ δὲ ἡ ὑπὸ ABE · ὀρθὴ ἄρα καὶ ἡ ὑπὸ $E\Delta A$ · ἡ $E\Delta$ ἄρα πρὸς τὴν ΔA ὀρθὴ ἐστὶν. ἔστι δὲ καὶ πρὸς ἑκάτεραν τῶν $B\Delta$, $\Delta\Gamma$ ὀρθή. ἡ $E\Delta$ ἄρα τρισὶν εὐθείαις

For let the two straight-lines AB and CD be at right-angles to a reference plane. I say that AB is parallel to CD .

For let them meet the reference plane at points B and D (respectively). And let the straight-line BD be joined. And let DE be drawn at right-angles to BD in the reference plane. And let DE be made equal to AB . And let BE , AE , and AD be joined.

And since AB is at right-angles to the reference plane, it will [thus] also make right-angles with all straight-lines joined to it which are in the reference plane [Def. 11.3]. And BD and BE , which are in the reference plane, are each joined to AB . Thus, each of the angles ABD and ABE are right-angles. So, for the same (reasons), each of the angles CDB and CDE are also right-angles. And since AB is equal to DE , and BD (is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And they contain right-angles. Thus, the base AD is equal to the base BE [Prop. 1.4]. And since AB is equal to DE , and AD (is) also (equal) to BE , the two (straight-lines) AB and BE are thus equal to the two (straight-lines) ED and DA (respectively). And their base AE (is) common. Thus, angle ABE is equal

ταῖς $B\Delta$, ΔA , $\Delta\Gamma$ πρὸς ὀρθὰς ἐπὶ τῆς ἀφῆς ἐφέστηκεν· αἱ τρεῖς ἄρα εὐθεῖαι αἱ $B\Delta$, ΔA , $\Delta\Gamma$ ἐν ἐνὶ εἰσὶν ἐπιπέδῳ. ἐν ᾧ δὲ αἱ ΔB , ΔA , ἐν τούτῳ καὶ ἡ AB · πᾶν γὰρ τρίγωνον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ· αἱ ἄρα AB , $B\Delta$, $\Delta\Gamma$ εὐθεῖαι ἐν ἐνὶ εἰσὶν ἐπιπέδῳ. καὶ ἐστὶν ὀρθὴ ἑκατέρα τῶν ὑπὸ $AB\Delta$, $B\Delta\Gamma$ γωνιῶν· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

Ἐὰν ἄρα δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾤσιν, παράλληλοι ἔσονται αἱ εὐθεῖαι· ὅπερ ἔδει δεῖχαι.

to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. Thus, EDA (is) also a right-angle. ED is thus at right-angles to DA . And it is also at right-angles to each of BD and DC . Thus, ED is standing at right-angles to the three straight-lines BD , DA , and DC at the (common) point of section. Thus, the three straight-lines BD , DA , and DC are in one plane [Prop. 11.5]. And in which(ever) plane DB and DA (are found), in that (plane) AB (will) also (be found). For every triangle is in one plane [Prop. 11.2]. And each of the angles ABD and BDC is a right-angle. Thus, AB is parallel to CD [Prop. 1.28].

Thus, if two straight-lines are at right-angles to the same plane then the straight-lines will be parallel. (Which is) the very thing it was required to show.

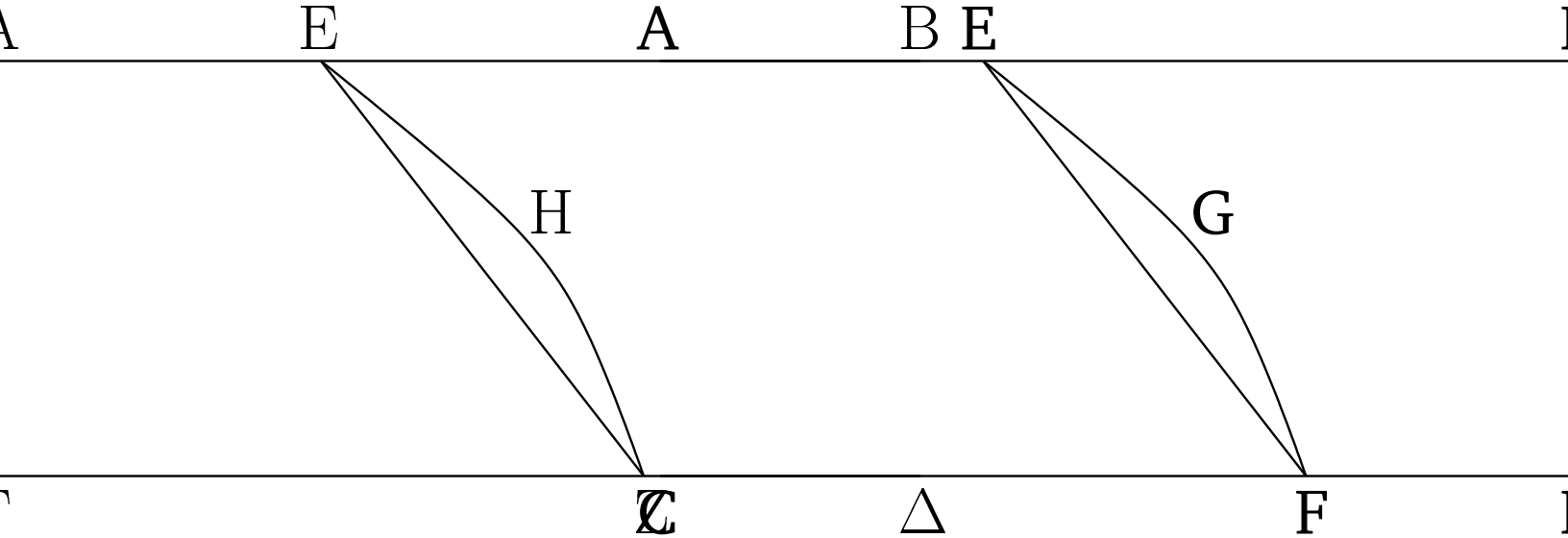
† In other words, the two straight-lines lie in the same plane, and never meet when produced in either direction.

7.

Proposition 7

Ἐὰν ᾧσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ' ἑκατέρας αὐτῶν τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

If there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines).



Ἐστωσαν δύο εὐθεῖαι παράλληλοι αἱ AB , $\Gamma\Delta$, καὶ ἐ-
κλήθω ἐφ' ἑκατέρας αὐτῶν τυθόντα σημεῖα τὰ E , Z · λέγω,
ὅτι ἡ ἐπὶ τὰ E , Z σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ
ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἔστω ἐν μετεωροτέρῳ ὥς ἡ EHZ ,
καὶ διήθω διὰ τῆς EHZ ἐπίπεδον· τομὴν δὴ ποιήσει ἐν τῷ
ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω ὥς τὴν EZ · δύο ἄρα
εὐθεῖαι αἱ EHZ , EZ θωρίον περιέχουσιν· ὅπερ ἐστὶν ἀδύνα-
τον. οὐκ ἄρα ἡ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖα ἐν
μετεωροτέρῳ ἐστὶν ἐπιπέδῳ· ἐν τῷ διὰ τῶν AB , $\Gamma\Delta$ ἄρα πα-
ραλλήλων ἐστὶν ἐπιπέδῳ ἡ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιζευγνυμένη
εὐθεῖα.

Ἐὰν ἄρα ᾧσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ' ἑκα-

Let AB and CD be two parallel straight-lines, and let the
random points E and F be taken on each of them (respec-
tively). I say that the straight-line joining points E and F is
in the same (reference) plane as the parallel (straight-lines).

For (if) not, and if possible, let it be in a more elevated
(plane), such as EGF . And let a plane be drawn through
 EGF . So it will make a straight cutting in the reference plane
[Prop. 11.3]. Let it make EF . Thus, two straight-lines (with
the same end-points), EGF and EF , will enclose an area. The
very thing is impossible. Thus, the straight-line joining E to F
is not in a more elevated plane. The straight-line joining E to
 F is thus in the plane through the parallel (straight-lines) AB
and CD .

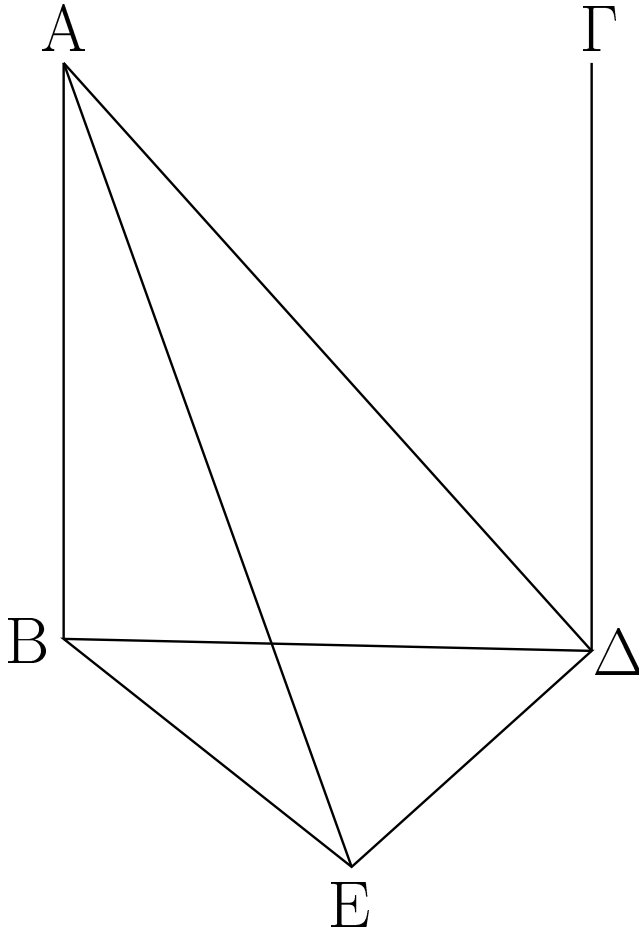
τέραξ αὐτῶν τυθόντα σημεία, ἡ ἐπὶ τὰ σημεία ἐπιζευγνυμένη εὐθεΐα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις· ὅπερ ἔδει δεῖξαι.

8.

Ἐάν ὥσι δύο εὐθεΐαι παράλληλοι, ἡ δὲ ἑτέρα αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾗ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται.

Ἐστωσαν δύο εὐθεΐαι παράλληλοι αἱ AB , $\Gamma\Delta$, ἡ δὲ ἑτέρα αὐτῶν ἡ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἔστω· λέγω, ὅτι καὶ ἡ λοιπὴ ἡ $\Gamma\Delta$ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται.

Συμβαλλέτωσαν γὰρ αἱ AB , $\Gamma\Delta$ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ B , Δ σημεία, καὶ ἐπεξέσθω ἡ $B\Delta$ · αἱ AB , $\Gamma\Delta$, $B\Delta$ ἄρα ἐν ἐνὶ εἰσιν ἐπιπέδῳ. ἦθθω τῇ BA πρὸς ὀρθὰς ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ ΔE , καὶ κείσθω τῇ AB ἴση ἡ ΔE , καὶ ἐπεξέσθωσαν αἱ BE , AE , $A\Delta$.



Καὶ ἐπεὶ ἡ AB ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῇ εὐθείας καὶ οὐσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστὶν ἡ AB · ὀρθὴ ἄρα

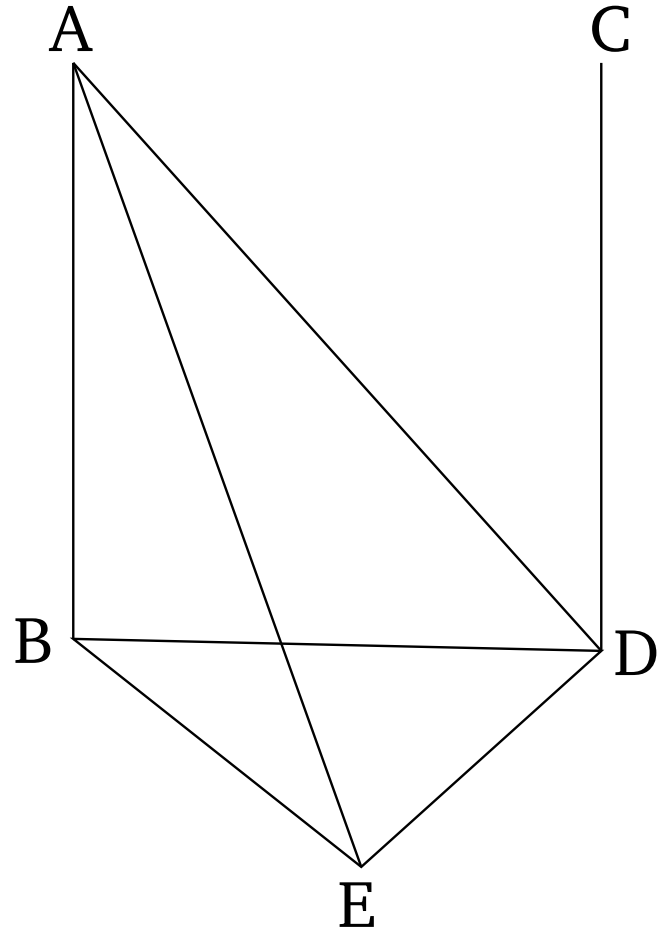
Thus, if there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines). (Which is) the very thing it was required to show.

Proposition 8

If two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane.

Let AB and CD be two parallel straight-lines, and let one of them, AB , be at right-angles to a reference plane. I say that the remaining (one), CD , will also be at right-angles to the same plane.

For let AB and CD meet the reference plane at points B and D (respectively). And let BD be joined. AB , CD , and BD are thus in one plane [Prop. 11.7]. Let DE be drawn at right-angles to BD in the reference plane, and let DE be made equal to AB , and let BE , AE , and AD be joined.



And since AB is at right-angles to the reference plane, AB is thus also at right-angles to all of the straight-lines joined to it which are in the reference plane [Def. 11.3]. Thus, the

[ἐστίν] ἑκατέρω τῶν ὑπὸ $AB\Delta$, ABE γωνιῶν. καὶ ἐπεὶ εἰς παραλλήλων τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπέπτωκεν ἡ $B\Delta$, αἱ ἄρα ὑπὸ $AB\Delta$, $\Gamma\Delta B$ γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσίν. ὀρθὴ δὲ ἡ ὑπὸ $AB\Delta$ · ὀρθὴ ἄρα καὶ ἡ ὑπὸ $\Gamma\Delta B$ · ἡ $\Gamma\Delta$ ἄρα πρὸς τὴν $B\Delta$ ὀρθὴ ἐστίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ ΔE , κοινὴ δὲ ἡ $B\Delta$, δύο δὲ αἱ AB , $B\Delta$ δυοῖν ταῖς $E\Delta$, ΔB ἴσαι εἰσίν· καὶ γωνία ἡ ὑπὸ $AB\Delta$ γωνία τῇ ὑπὸ $E\Delta B$ ἴση· ὀρθὴ γὰρ ἑκατέρω· βάσεις ἄρα ἡ $A\Delta$ βάσει τῇ BE ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν AB τῇ ΔE , ἡ δὲ BE τῇ $A\Delta$, δύο δὲ αἱ AB , BE δυοῖν ταῖς $E\Delta$, ΔA ἴσαι εἰσίν ἑκατέρω ἑκατέρω. καὶ βάσεις αὐτῶν κοινὴ ἡ AE · γωνία ἄρα ἡ ὑπὸ ABE γωνία τῇ ὑπὸ $E\Delta A$ ἐστὶν ἴση. ὀρθὴ δὲ ἡ ὑπὸ ABE · ὀρθὴ ἄρα καὶ ἡ ὑπὸ $E\Delta A$ · ἡ $E\Delta$ ἄρα πρὸς τὴν $A\Delta$ ὀρθὴ ἐστίν. ἔστι δὲ καὶ πρὸς τὴν ΔB ὀρθὴ· ἡ $E\Delta$ ἄρα καὶ τῷ διὰ τῶν $B\Delta$, ΔA ἐπιπέδῳ ὀρθὴ ἐστίν. καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὕσας ἐν τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας ἡ $E\Delta$. ἐν δὲ τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ἐστὶν ἡ $\Delta\Gamma$, ἐπειδήπερ ἐν τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ἐστὶν αἱ AB , $B\Delta$, ἐν ᾧ δὲ αἱ AB , $B\Delta$, ἐν τούτῳ ἐστὶ καὶ ἡ $\Delta\Gamma$. ἡ $E\Delta$ ἄρα τῇ $\Delta\Gamma$ πρὸς ὀρθὰς ἐστίν· ὥστε καὶ ἡ $\Gamma\Delta$ τῇ ΔE πρὸς ὀρθὰς ἐστίν. ἔστι δὲ καὶ ἡ $\Gamma\Delta$ τῇ $B\Delta$ πρὸς ὀρθὰς. ἡ $\Gamma\Delta$ ἄρα δύο εὐθείαις τεμνούσαις ἀλλήλαις ταῖς ΔE , ΔB ἀπὸ τῆς κατὰ τὸ Δ τομῆς πρὸς ὀρθὰς ἐφέστηκεν· ὥστε ἡ $\Gamma\Delta$ καὶ τῷ διὰ τῶν ΔE , ΔB ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. τὸ δὲ διὰ τῶν ΔE , ΔB ἐπίπεδον τὸ ὑποκείμενόν ἐστίν· ἡ $\Gamma\Delta$ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

Ἐὰν ἄρα ᾧσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾗ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· ὅπερ ἔδει δεῖχαι.

9.

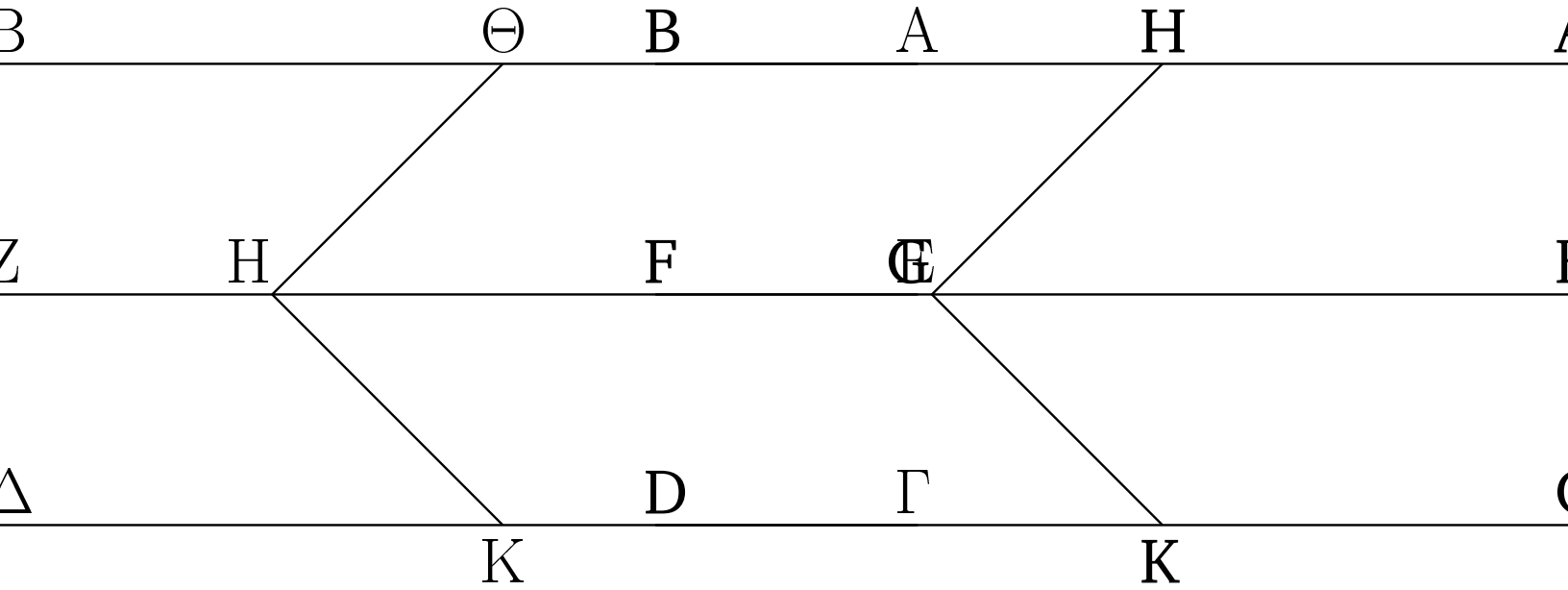
Αἱ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ μὴ οὕσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσι παράλληλοι.

angles ABD and ABE [are] each right-angles. And since the straight-line BD has met the parallel (straight-lines) AB and CD , the (sum of the) angles ABD and CDB is thus equal to two right-angles [Prop. 1.29]. And ABD (is) a right-angle. Thus, CDB (is) also a right-angle. CD is thus at right-angles to BD . And since AB is equal to DE , and BD (is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And angle ABD (is) equal to angle EDB . For each (is) a right-angle. Thus, the base AD (is) equal to the base BE [Prop. 1.4]. And since AB is equal to DE , and BE to AD , the two (sides) AB , BE are equal to the two (sides) ED , DA , respectively. And their base AE is common. Thus, angle ABE is equal to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. EDA (is) thus also a right-angle. Thus, ED is at right-angles to AD . And it is also at right-angles to DB . Thus, ED is also at right-angles to the plane through BD and DA [Prop. 11.4]. And ED will thus make right-angles with all of the straight-lines joined to it which are also in the plane through BDA . And DC is in the plane through BDA , inasmuch as AB and BD are in the plane through BDA [Prop. 11.2], and in which (ever plane) AB and BD (are found), DC is also (found). Thus, ED is at right-angles to DC . Hence, CD is also at right-angles to DE . And CD is also at right-angles to BD . Thus, CD is standing at right-angles to two straight-lines, DE and DB , which meet one another, at the (point) of section, D . Hence, CD is also at right-angles to the plane through DE and DB [Prop. 11.4]. And the plane through DE and DB is the reference (plane). CD is thus at right-angles to the reference plane.

Thus, if two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

Proposition 9

(Straight-lines) parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another.



Ἐστω γάρ ἑκατέρα τῶν AB , $\Gamma\Delta$ τῇ EZ παράλληλος μὴ οὔσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ· λέγω, ὅτι παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$.

Εἰλήφθω γάρ ἐπὶ τῆς EZ τυθὸν σημεῖον τὸ H , καὶ ἀπ' αὐτοῦ τῇ EZ ἐν μὲν τῷ διὰ τῶν EZ , AB ἐπιπέδῳ πρὸς ὀρθὰς ἦθθω ἡ $H\Theta$, ἐν δὲ τῷ διὰ τῶν ZE , $\Gamma\Delta$ τῇ EZ πάλιν πρὸς ὀρθὰς ἦθθω ἡ HK .

Καὶ ἐπεὶ ἡ EZ πρὸς ἑκατέραν τῶν $H\Theta$, HK ὀρθή ἐστιν, ἡ EZ ἄρα καὶ τῷ διὰ τῶν $H\Theta$, HK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. καὶ ἐστιν ἡ EZ τῇ AB παράλληλος· καὶ ἡ AB ἄρα τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ $\Gamma\Delta$ τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν· ἑκατέρα ἄρα τῶν AB , $\Gamma\Delta$ τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. ἔαν δὲ δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾤσιν, παράλληλοί εἰσιν αἱ εὐθεῖαι· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$ · ὅπερ ἔδει δεῖχαι.

10.

Ἐὰν δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων ᾧσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσαξ γωνίαξ περιέχουσιν.

Δύο γάρ εὐθεῖαι αἱ AB , BF ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας τὰς ΔE , EZ ἀπτομένας ἀλλήλων ἔστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ· λέγω, ὅτι ἴση ἐστὶν ἡ ὑπὸ ABF γωνία τῇ ὑπὸ ΔEZ .

Ἀπειλήφθωσαν γάρ αἱ BA , BF , $E\Delta$, EZ ἴσαι ἀλλήλαις, καὶ ἐπεξεὐθῆθωσαν αἱ $A\Delta$, ΓZ , BE , AF , ΔZ .

Καὶ ἐπεὶ ἡ BA τῇ $E\Delta$ ἴση ἐστὶ καὶ παράλληλος, καὶ ἡ $A\Delta$ ἄρα τῇ BE ἴση ἐστὶ καὶ παράλληλος. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΓZ τῇ BE ἴση ἐστὶ καὶ παράλληλος· ἑκατέρα ἄρα τῶν $A\Delta$, ΓZ τῇ BE ἴση ἐστὶ καὶ παράλληλος. αἱ δὲ τῇ αὐτῇ

For let AB and CD each be parallel to EF , not being in the same plane as it. I say that AB is parallel to CD .

For let some point G be taken at random on EF . And from it let GH be drawn at right-angles to EF in the plane through EF and AB . And let GK be drawn, again at right-angles to EF , in the plane through FE and CD .

And since EF is at right-angles to each of GH and GK , EF is thus also at right-angles to the plane through GH and GK [Prop. 11.4]. And EF is parallel to AB . Thus, AB is also at right-angles to the plane through HGK [Prop. 11.8]. So, for the same (reasons), CD is also at right-angles to the plane through HGK . Thus, AB and CD are each at right-angles to the plane through HGK . And if two straight-lines are at right-angles to the same plane then the straight-lines are parallel [Prop. 11.6]. Thus, AB is parallel to CD . (Which is) the very thing it was required to show.

Proposition 10

If two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles.

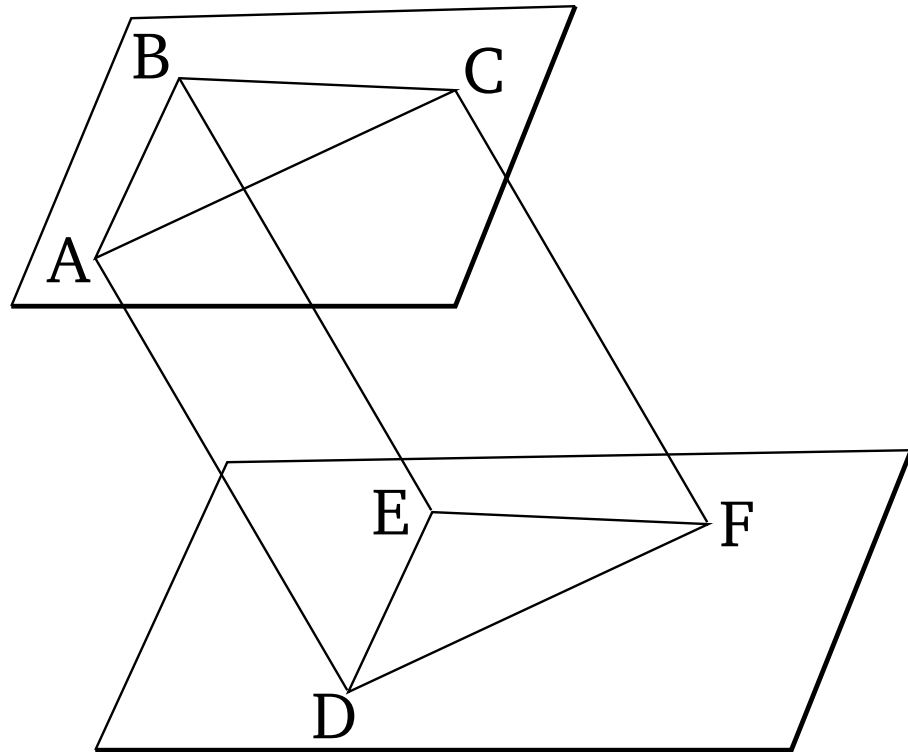
For let the two straight-lines joined to one another, AB and BC , be (respectively) parallel to the two straight-lines joined to one another, DE and EF , (but) not in the same plane. I say that angle ABC is equal to (angle) DEF .

For let BA , BC , ED , and EF be cut off (so as to be, respectively) equal to one another. And let AD , CF , BE , AC , and DF be joined.

And since BA is equal and parallel to ED , AD is thus also equal and parallel to BE [Prop. 1.33]. So, for the same reasons, CF is also equal and parallel to BE . Thus, AD and CF are

εὐθεία παράλληλοι καὶ μὴ οὔσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσι παράλληλοι· παράλληλος ἄρα ἐστὶν ἡ AD τῇ $ΓΖ$ καὶ ἴση. καὶ ἐπιξενγνύουσιν αὐτάς αἱ $ΑΓ$, $ΔΖ$ · καὶ ἡ $ΑΓ$ ἄρα τῇ $ΔΖ$ ἴση ἐστὶ καὶ παράλληλος. καὶ ἐπεὶ δύο αἱ AB , $ΒΓ$ δυοὶ ταῖς $ΔΕ$, $ΕΖ$ ἴσαι εἰσὶν, καὶ βάσις ἡ $ΑΓ$ βάσει τῇ $ΔΖ$ ἴση, γωνία ἄρα ἡ ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ ἐστὶν ἴση.

each equal and parallel to BE . And straight-lines parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another [Prop. 11.9]. Thus, AD is parallel and equal to CF . And AC and DF join them. Thus, AC is also equal and parallel to DF [Prop. 1.33]. And since the two (straight-lines) AB and BC are equal to the two (straight-lines) DE and EF (respectively), and the base AC (is) equal to the base DF , the angle ABC is thus equal to the (angle) DEF [Prop. 1.8].



Book11/fig10g.pdf

Ἐάν ἄρα δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθεῖας ἀπτομένας ἀλλήλων ὥσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσαί γωνίαί τις περιέχουσιν· ὅπερ ἔδει δεῖχαι.

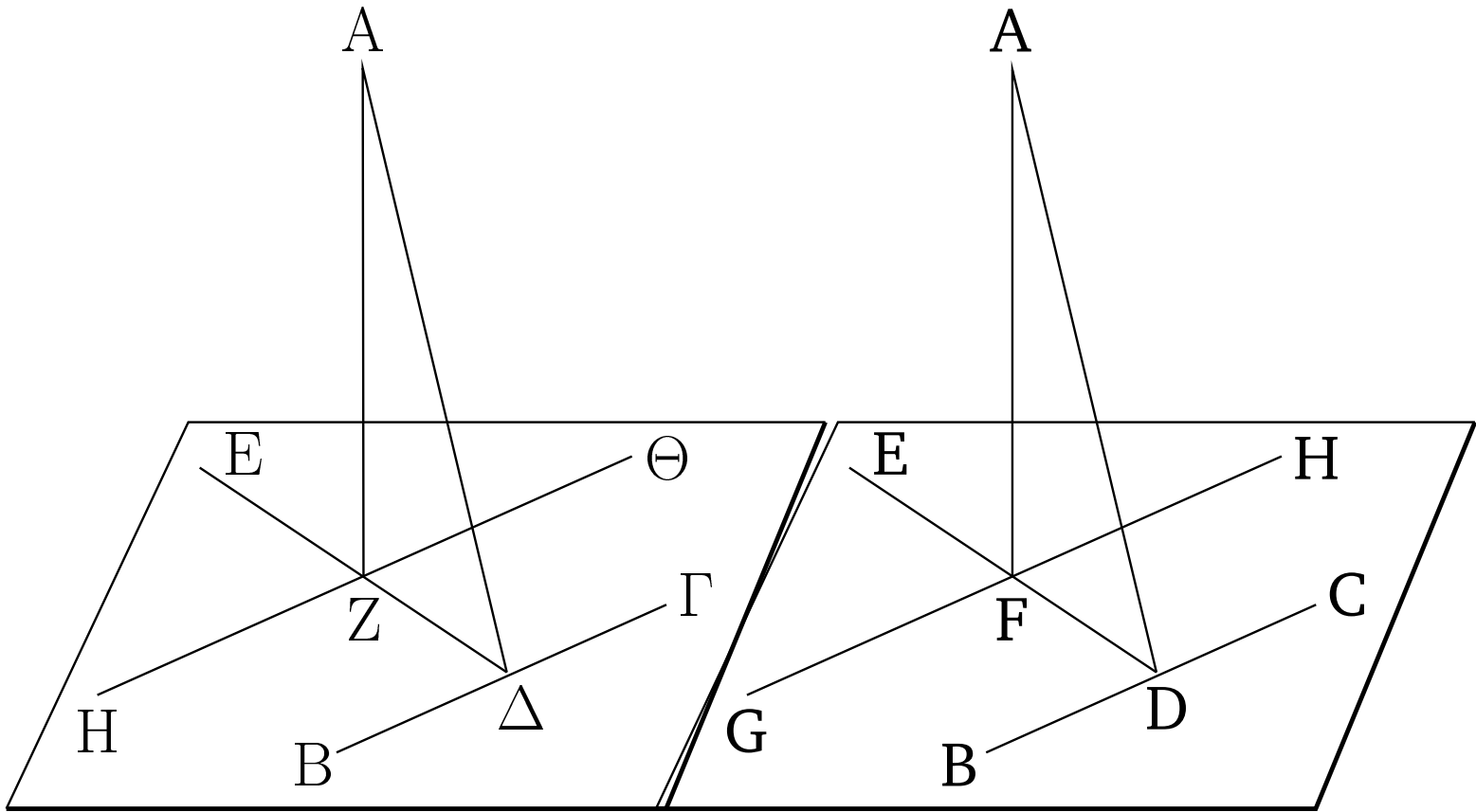
Thus, if two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles. (Which is) the very thing it was required to show.

11.

Ἀπὸ τοῦ δοθέντος σημείου μετεώρου ἐπὶ τὸ δοθὲν ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Proposition 11

To draw a perpendicular straight-line from a given raised point to a given plane.



Ἐστω τὸ μὲν δοθὲν σημεῖον μετέωρον τὸ A , τὸ δὲ δοθὲν ἐπίπεδον τὸ ὑποκείμενον· δεῖ δὴ ἀπὸ τοῦ A σημείου ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Διήθθω γάρ τις ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖα, ὡς ἔτυθεν, ἡ $BΓ$, καὶ ἤθθω ἀπὸ τοῦ A σημείου ἐπὶ τὴν $BΓ$ κάθετος ἡ $ΑΔ$. εἰ μὲν οὖν ἡ $ΑΔ$ κάθετός ἐστι καὶ ἐπὶ τὸ ὑποκείμενον ἐπίπεδον, γερονὸς ἂν εἴη τὸ ἐπιταθὲν. εἰ δὲ οὐ, ἤθθω ἀπὸ τοῦ $Δ$ σημείου τῇ $BΓ$ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἡ $ΔΕ$, καὶ ἤθθω ἀπὸ τοῦ A ἐπὶ τὴν $ΔΕ$ κάθετος ἡ $ΑΖ$, καὶ διὰ τοῦ Z σημείου τῇ $BΓ$ παράλληλος ἤθθω ἡ $HΘ$.

Καὶ ἐπεὶ ἡ $BΓ$ ἑκατέρω τῶν $ΔΑ$, $ΔΕ$ πρὸς ὀρθὰς ἐστίν, ἡ $BΓ$ ἄρα καὶ τῷ διὰ τῶν $ΕΔΑ$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. καὶ ἐστὶν αὐτῇ παράλληλος ἡ $HΘ$. ἔαν δὲ ὡσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾗ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· καὶ ἡ $HΘ$ ἄρα τῷ διὰ τῶν $ΕΔ$, $ΔΑ$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. καὶ πρὸς πάσαις ἄρα τὰς ἀποτομένας αὐτῆς εὐθείαις καὶ οὕσαι ἐν τῷ διὰ τῶν $ΕΔ$, $ΔΑ$ ἐπιπέδῳ ὀρθὴ ἐστὶν ἡ $HΘ$. ἄπτεται δὲ αὐτῆς ἡ $ΑΖ$ οὕσα ἐν τῷ διὰ τῶν $ΕΔ$, $ΔΑ$ ἐπιπέδῳ· ἡ $HΘ$ ἄρα ὀρθὴ ἐστὶ πρὸς τὴν $ΖΑ$. ὥστε καὶ ἡ $ΖΑ$ ὀρθὴ ἐστὶ πρὸς τὴν $ΘΗ$. ἔστι δὲ ἡ $ΑΖ$ καὶ πρὸς τὴν $ΔΕ$ ὀρθὴ· ἡ $ΑΖ$ ἄρα πρὸς ἑκατέραν τῶν $HΘ$, $ΔΕ$ ὀρθὴ ἐστίν. ἔαν δὲ εὐθεῖα δυνὸν εὐθείαις τεμνοῦσαι ἀλλήλας ἐπὶ τῇς τομῆς πρὸς ὀρθὰς ἐπισταθῇ, καὶ τῷ δι' αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· ἡ $ΖΑ$ ἄρα τῷ διὰ τῶν $ΕΔ$, $HΘ$

Let A be the given raised point, and the given plane the reference (plane). So, it is required to draw a perpendicular straight-line from point A to the reference plane.

Let some random straight-line BC be drawn across in the reference plane, and let the (straight-line) AD be drawn from point A perpendicular to BC [Prop. 1.12]. If, therefore, AD is also perpendicular to the reference plane then that which was prescribed will have occurred. And, if not, let DE be drawn in the reference plane from point D at right-angles to BC [Prop. 1.11], and let the (straight-line) AF be drawn from A perpendicular to DE [Prop. 1.12], and let GH be drawn through point F , parallel to BC [Prop. 1.31].

And since BC is at right-angles to each of DA and DE , BC is thus also at right-angles to the plane through EDA [Prop. 11.4]. And GH is parallel to it. And if two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (straight-line) will also be at right-angles to the same plane [Prop. 11.8]. Thus, GH is also at right-angles to the plane through ED and DA . And GH is thus at right-angles to all of the straight-lines joined to it which are also in the plane through ED and AD [Def. 11.3]. And AF , which is in the plane through ED and DA , is joined to it. Thus, GH is at right-angles to FA . Hence, FA is also at right-angles to HG . And AF is also at right-angles to DE . Thus, AF is at

ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. τὸ δὲ διὰ τῶν $ΕΔ$, $ΗΘ$ ἐπίπεδόν ἐστι τὸ ὑποκείμενον· ἡ AZ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

Ἀπὸ τοῦ ἄρα δοθέντος σημείου μετέωρον τοῦ A ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος εὐθεῖα γραμμὴ ἦκται ἡ AZ · ὅπερ ἔδει ποιῆσαι.

right-angles to each of GH and DE . And if a straight-line is set up at right-angles to two straight-lines cutting one another, at the point of section, then it will also be at right-angles to the plane through them [Prop. 11.4]. Thus, FA is at right-angles to the plane through ED and GH . And the plane through ED and GH is the reference (plane). Thus, AF is at right-angles to the reference plane.

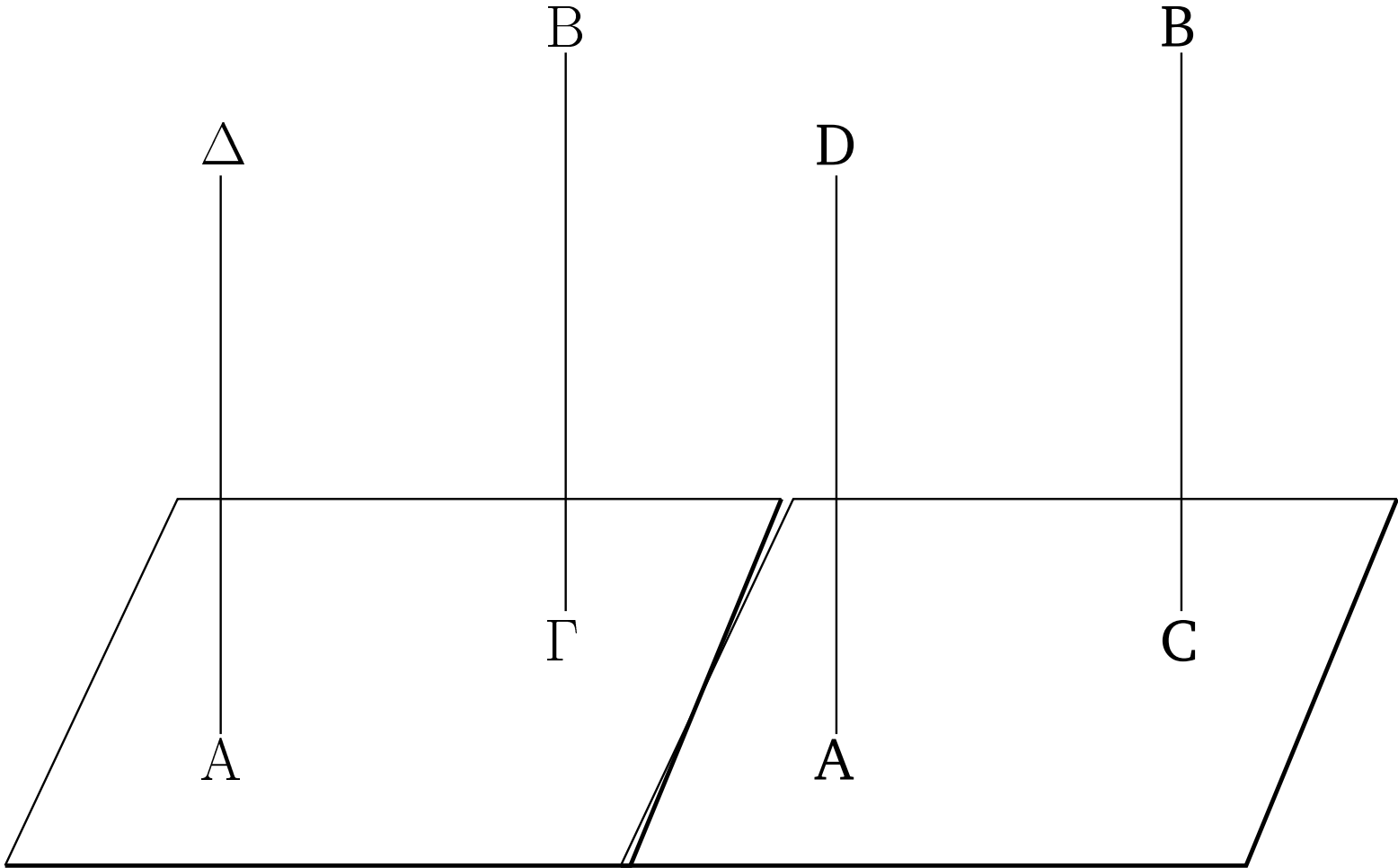
Thus, the straight-line AF has been drawn from the given raised point A perpendicular to the reference plane. (Which is) the very thing it was required to do.

12.

Τῷ δοθέντι ἐπιπέδῳ ἀπὸ τοῦ πρὸς αὐτῷ δοθέντος σημείου πρὸς ὀρθὰς εὐθεῖαν γραμμὴν ἀναστήσαι.

Proposition 12

To set up a straight-line at right-angles to a given plane from a given point in it.



Ἐστω τὸ μὲν δοθέν ἐπίπεδον τὸ ὑποκείμενον, τὸ δὲ πρὸς αὐτῷ σημείον τὸ A · δεῖ δὴ ἀπὸ τοῦ A σημείου τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς εὐθεῖαν γραμμὴν ἀναστήσαι.

Νενοήσθω τι σημείον μετέωρον τὸ B , καὶ ἀπὸ τοῦ B ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος ἦθθω ἡ $ΒΓ$, καὶ διὰ τοῦ A

Let the given plane be the reference (plane), and A a point in it. So, it is required to set up a straight-line at right-angles to the reference plane at point A .

Let some raised point B be assumed, and let the perpendicular (straight-line) BC be drawn from B to the reference plane

σημείον τῇ $B\Gamma$ παράλληλος ἦθθω ἡ $A\Delta$.

Ἐπεὶ οὖν δύο εὐθεῖαι παράλληλοί εἰσιν αἱ $A\Delta$, ΓB , ἡ δὲ μία αὐτῶν ἡ $B\Gamma$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν, καὶ ἡ λοιπὴ ἄρα ἡ $A\Delta$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

Τῷ ἄρα δοθέντι ἐπιπέδῳ ἀπὸ τοῦ πρὸς αὐτῷ σημείου τοῦ A πρὸς ὀρθὰς ἀνέσταται ἡ $A\Delta$. ὅπερ ἔδει ποιῆσαι.

[Prop. 11.11]. And let AD be drawn from point A parallel to BC [Prop. 1.31].

Therefore, since AD and CB are two parallel straight-lines, and one of them, BC , is at right-angles to the reference plane, the remaining (one) AD is thus also at right-angles to the reference plane [Prop. 11.8].

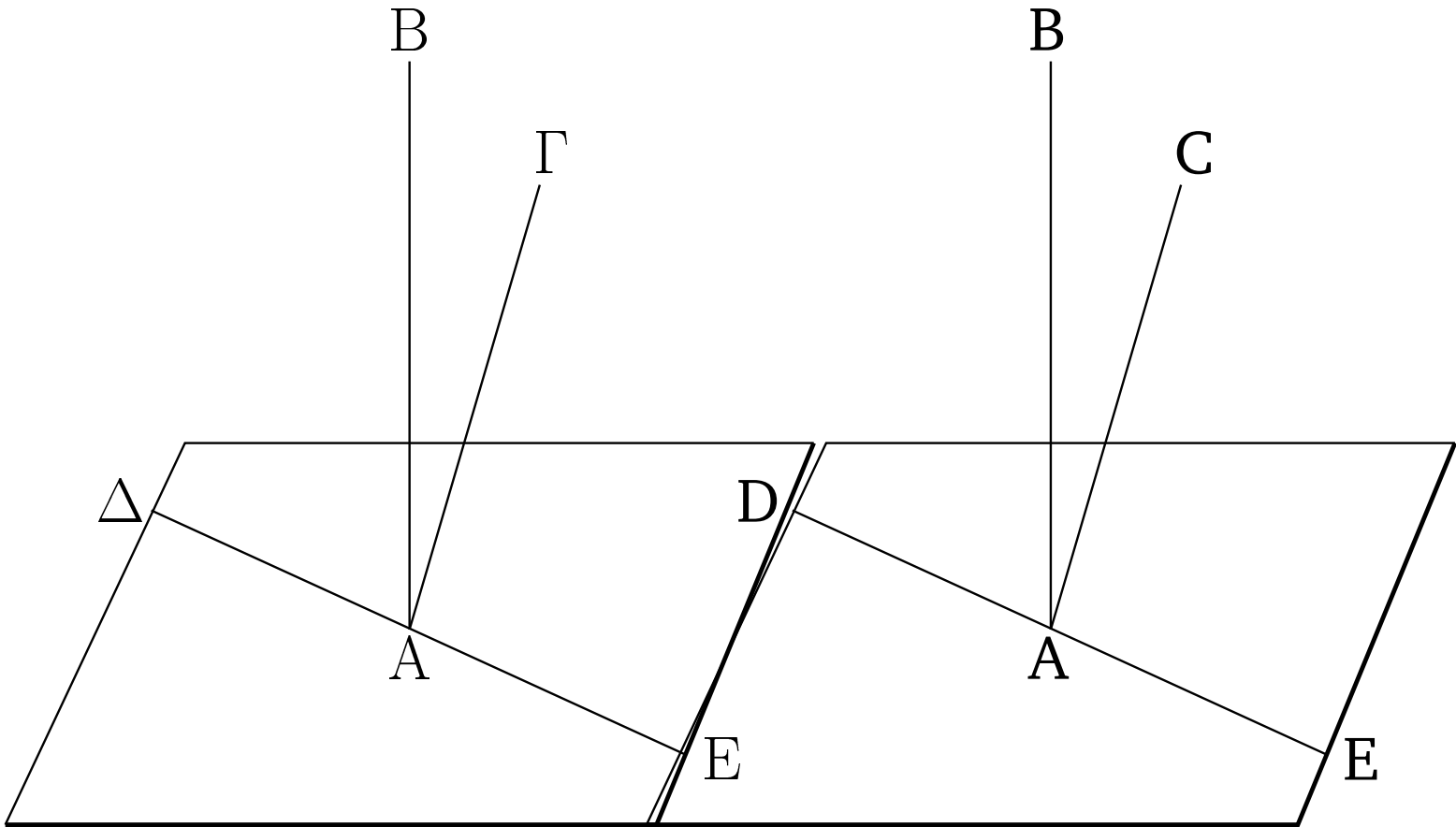
Thus, AD has been set up at right-angles to the given plane, from the point in it, A . (Which is) the very thing it was required to do.

13.

Ἀπὸ τοῦ αὐτοῦ σημείου τῷ αὐτῷ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰς οὐκ ἀναστήσονται ἐπὶ τὰ αὐτὰ μέρος.

Proposition 13

Two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side.



Εἰ γὰρ δυνατόν, ἀπὸ τοῦ αὐτοῦ σημείου τοῦ A τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖαι αἱ AB , $B\Gamma$ πρὸς ὀρθὰς ἀνεστάτωσαν ἐπὶ τὰ αὐτὰ μέρη, καὶ διήθθω τὸ διὰ τῶν BA , AG ἐπίπεδον· τομὴν δὴ ποιήσει διὰ τοῦ A ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω τὴν ΔAE . αἱ ἄρα AB , AG , ΔAE εὐθεῖαι ἐν ἑνὶ εἰσιν ἐπιπέδῳ. καὶ ἐπεὶ ἡ ΓA τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν, καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ὅσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. ἄπτεται δὲ αὐτῆς ἡ ΔAE ὅσα ἐν τῷ

For, if possible, let the two straight-lines AB and AC be set up at the same point A at right-angles to the reference plane, on the same side. And let the plane through BA and AC be drawn. So it will make a straight cutting (passing) through (point) A in the reference plane [Prop. 11.3]. Let it have made DAE . Thus, AB , AC , and DAE are straight-lines in one plane. And since CA is at right-angles to the reference plane, it will thus also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. And DAE ,

ὑποκειμένῳ ἐπιπέδῳ ἡ ἄρα ὑπὸ ΓΑΕ γωνία ὀρθή ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΒΑΕ ὀρθή ἐστιν· ἴση ἄρα ἡ ὑπὸ ΓΑΕ τῇ ὑπὸ ΒΑΕ καὶ εἰσιν ἐν ἐνὶ ἐπιπέδῳ· ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἀπὸ τοῦ αὐτοῦ σημείου τῶν αὐτῶν ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰς ἀνασταθήσονται ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἔδει δεῖχαι.

which is in the reference plane, is joined to it. Thus, angle CAE is a right-angle. So, for the same (reasons), BAE is also a right-angle. Thus, CAE (is) equal to BAE. And they are in one plane. The very thing is impossible.

Thus, two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side. (Which is) the very thing it was required to show.

14.

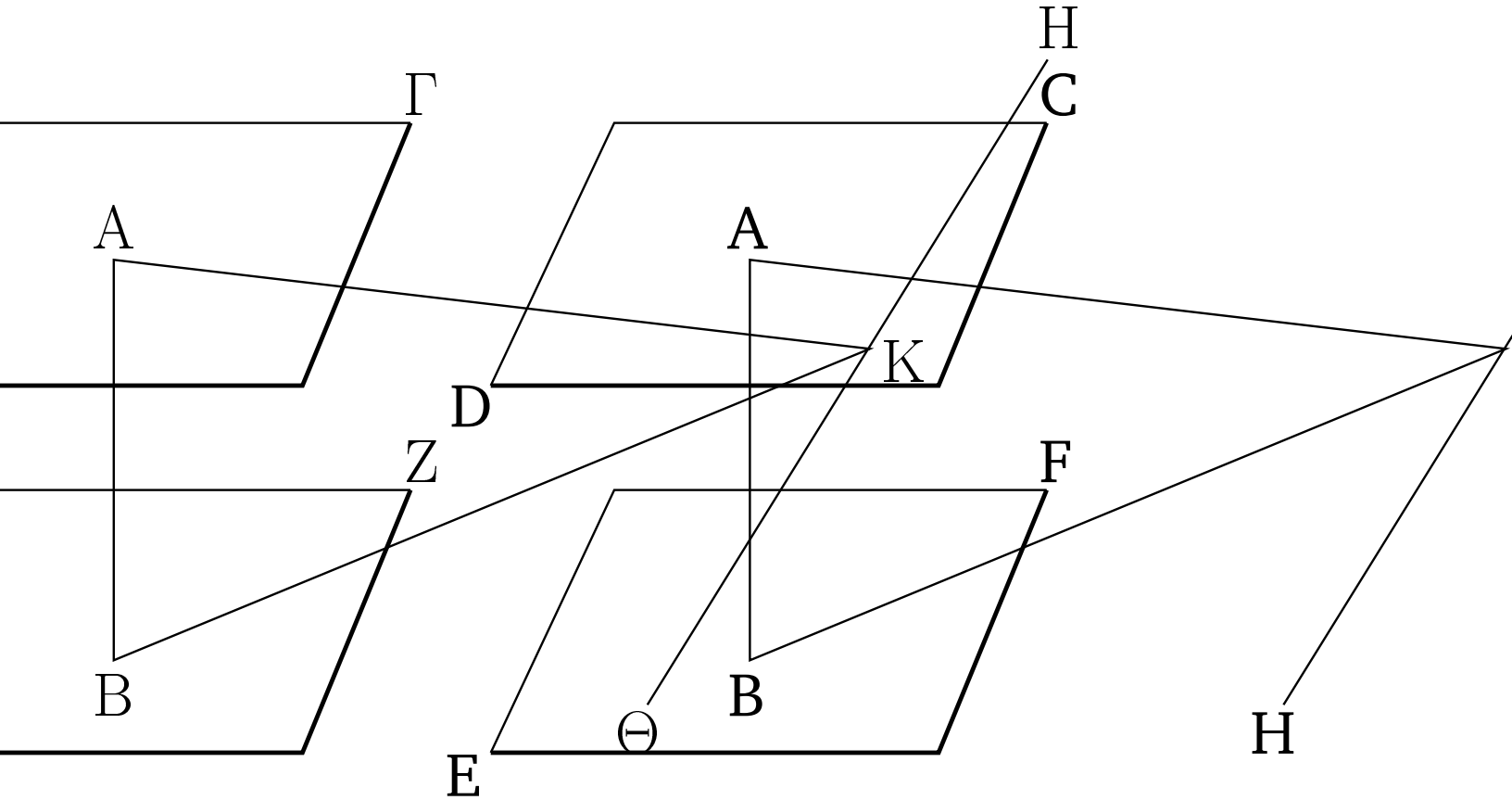
Proposition 14

Πρὸς ἃ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλα ἔσται τὰ ἐπίπεδα.

Εὐθεῖα γάρ τις ἡ ΑΒ πρὸς ἑκάτερον τῶν ΓΔ, ΕΖ ἐπιπέδων πρὸς ὀρθὰς ἔστω· λέγω, ὅτι παράλληλά ἐστι τὰ ἐπίπεδα.

Planes to which the same straight-line is at right-angles will be parallel planes.

For let some straight-line AB be at right-angles to each of the planes CD and EF. I say that the planes are parallel.



Εἰ γὰρ μὴ, ἐκβαλλόμενα συμπεσοῦνται. συμπιπτέτωσαν· ποιήσουσι δὴ κοινὴν τομὴν εὐθεῖαν. ποιείτωσαν τὴν ΗΘ, καὶ εἰλήφθω ἐπὶ τῇς ΗΘ τυθὸν σημεῖον τὸ Κ, καὶ ἐπεζεύθωσαν αἱ ΑΚ, ΒΚ.

Καὶ ἐπεὶ ἡ ΑΒ ὀρθή ἐστι πρὸς τὸ ΕΖ ἐπίπεδον, καὶ πρὸς τὴν ΒΚ ἄρα εὐθεῖαν ὅσον ἐν τῷ ΕΖ ἐκβληθέντι ἐπιπέδῳ ὀρθή ἐστιν ἡ ΑΒ· ἡ ἄρα ὑπὸ ΑΒΚ γωνία ὀρθή ἐστιν. διὰ τὰ

For, if not, being produced, they will meet. Let them have met. So they will make a straight-line as a common section [Prop. 11.3]. Let them have made GH. And let some random point K be taken on GH. And let AK and BK be joined.

And since AB is at right-angles to the plane EF, AB is thus also at right-angles to BK, which is a straight-line in the produced plane EF [Def. 11.3]. Thus, angle ABK is a right-

αὐτὰ δὴ καὶ ἡ ὑπὸ BAK ὀρθή ἐστιν. τριγώνον δὴ τοῦ ABK αἱ δύο γωνίαι αἱ ὑπὸ ABK , BAK δυσὶν ὀρθαῖς εἰσι ἴσαι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ $\Gamma\Delta$, EZ ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται· παράλληλα ἄρα ἐστὶ τὰ $\Gamma\Delta$, EZ ἐπίπεδα.

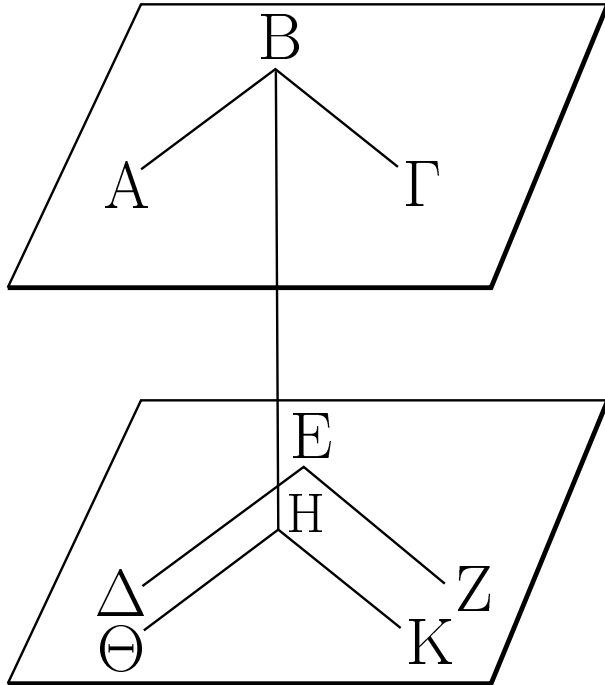
Πρὸς ἃ ἐπίπεδα ἄρα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλά ἐστι τὰ ἐπίπεδα· ὅπερ ἔδει δεῖχαι.

15.

Ἐάν δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων ὥσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ οὔσαι, παράλληλά ἐστι τὰ δι' αὐτῶν ἐπίπεδα.

Δύο γάρ εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ AB , $B\Gamma$ παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων τὰς ΔE , EZ ἔστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ οὔσαι· λέγω, ὅτι ἐκβαλλόμενα τὰ διὰ τῶν AB , $B\Gamma$, ΔE , EZ ἐπίπεδα οὐ συμπεσεῖται ἀλλήλοις.

Ἦθθω γάρ ἀπὸ τοῦ B σημείου ἐπὶ τὸ διὰ τῶν ΔE , EZ ἐπίπεδον κάθετος ἡ BH καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ H σημεῖον, καὶ διὰ τοῦ H τῇ μὲν $E\Delta$ παράλληλος ἡ $\Theta\Delta$ ἡ $H\Theta$, τῇ δὲ EZ ἡ HK .



Καὶ ἐπεὶ ἡ BH ὀρθή ἐστι πρὸς τὸ διὰ τῶν ΔE , EZ ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὔσαι ἐν τῷ διὰ τῶν ΔE , EZ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. ἄπτεται δὲ αὐτῆς ἑκατέρα τῶν $H\Theta$, HK οὔσα ἐν τῷ διὰ τῶν ΔE , EZ ἐπιπέδῳ· ὀρθή ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ $BH\Theta$,

angle. So, for the same (reasons), BAK is also a right-angle. So the (sum of the) two angles ABK and BAK in the triangle ABK is equal to two right-angles. The very thing is impossible [Prop. 1.17]. Thus, planes CD and EF , being produced, will not meet. Planes CD and EF are thus parallel [Def. 11.8].

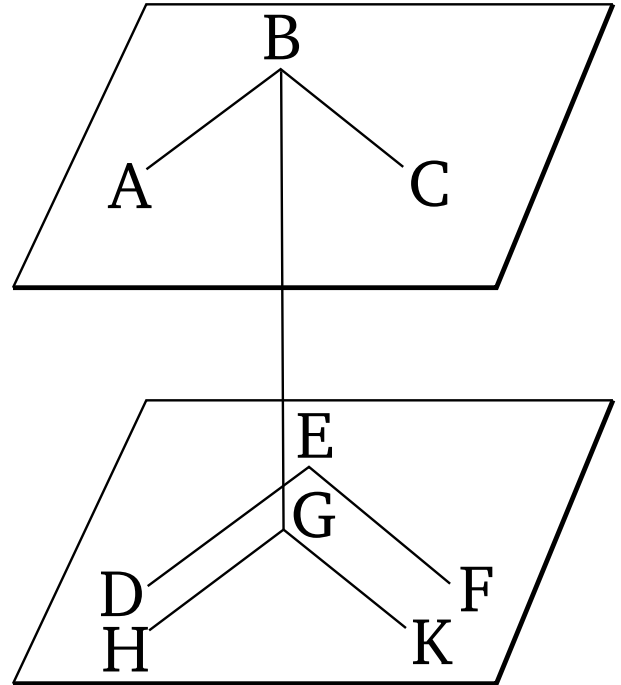
Thus, planes to which the same straight-line is at right-angles are parallel planes. (Which is) the very thing it was required to show.

Proposition 15

If two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another).

For let the two straight-lines joined to one another, AB and BC , be parallel to the two straight-lines joined to one another, DE and EF (respectively), not being in the same plane. I say that the planes through AB , BC and DE , EF will not meet one another (when) produced.

For let BG be drawn from point B perpendicular to the plane through DE and EF [Prop. 11.11], and let it meet the plane at point G . And let GH be drawn through G parallel to ED , and GK (parallel) to EF [Prop. 1.31].



And since BG is at right-angles to the plane through DE and EF , it will thus also make right-angles with all of the straight-lines joined to it, which are also in the plane through DE and EF [Def. 11.3]. And each of GH and GK , which are in the plane through DE and EF , are joined to it. Thus, each

BHK γωνιών. καὶ ἐπεὶ παράλληλός ἐστιν ἡ BA τῇ $HΘ$, αἱ ἄρα ὑπὸ HBA , $BHΘ$ γωνίαι δυοὶν ὀρθαῖς ἴσαι εἰσίν. ὀρθὴ δὲ ἡ ὑπὸ $BHΘ$ ὀρθὴ ἄρα καὶ ἡ ὑπὸ HBA . ἡ HB ἄρα τῇ BA πρὸς ὀρθάξ ἐστιν. διὰ τὰ αὐτὰ δὲ ἡ HB καὶ τῇ $ΒΓ$ ἐστι πρὸς ὀρθάξ. ἐπεὶ οὖν εὐθεῖα ἡ HB δυοὶν εὐθείαις ταῖς BA , $ΒΓ$ τεμνούσαις ἀλλήλας πρὸς ὀρθάξ ἐφέστηκεν, ἡ HB ἄρα καὶ τῷ διὰ τῶν BA , $ΒΓ$ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. [διὰ τὰ αὐτὰ δὲ ἡ BH καὶ τῷ διὰ τῶν $HΘ$, HK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. τὸ δὲ διὰ τῶν $HΘ$, HK ἐπίπεδόν ἐστι τὸ διὰ τῶν $ΔΕ$, $ΕΖ$. ἡ BH ἄρα τῷ διὰ τῶν $ΔΕ$, $ΕΖ$ ἐπιπέδῳ ἐστὶ πρὸς ὀρθάξ. ἐδείκθη δὲ ἡ HB καὶ τῷ διὰ τῶν AB , $ΒΓ$ ἐπιπέδῳ πρὸς ὀρθάξ]. πρὸς ἃ δὲ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθὴ ἐστιν, παράλληλά ἐστι τὰ ἐπίπεδα. παράλληλον ἄρα ἐστὶ τὸ διὰ τῶν AB , $ΒΓ$ ἐπίπεδον τῷ διὰ τῶν $ΔΕ$, $ΕΖ$.

Ἐάν ἄρα δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείαις ἀπτομένας ἀλλήλων ὥσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, παράλληλά ἐστι τὰ δι' αὐτῶν ἐπίπεδα. ὅπερ ἔδει δεῖχαι.

16.

Ἐάν δύο ἐπίπεδα παράλληλα ὑπὸ ἐπιπέδου τινὸς τέμνηται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν.

Δύο γάρ ἐπίπεδα παράλληλα τὰ AB , $ΓΔ$ ὑπὸ ἐπιπέδου τοῦ $EZHΘ$ τεμνέσθω, κοιναὶ δὲ αὐτῶν τομαὶ ἔστωσαν αἱ EZ , $HΘ$. λέγω, ὅτι παράλληλός ἐστιν ἡ EZ τῇ $HΘ$.

Εἰ γὰρ μὴ, ἐκβαλλόμεναι αἱ EZ , $HΘ$ ἤτοι ἐπὶ τὰ Z , $Θ$ μέρη ἢ ἐπὶ τὰ E , H συμπεσοῦνται. ἐκβεβλήσθωσαν ὥς ἐπὶ τὰ Z , $Θ$ μέρη καὶ συμπίπτωσαν πρότερον κατὰ τὸ K . καὶ ἐπεὶ ἡ EZK ἐν τῷ AB ἐστὶν ἐπιπέδῳ, καὶ πάντα ἄρα τὰ ἐπὶ τῆς EZK σημεία ἐν τῷ AB ἐστὶν ἐπιπέδῳ. ἐν δὲ τῶν ἐπὶ τῆς EZK εὐθείας σημείων ἐστὶ τὸ K . τὸ K ἄρα ἐν τῷ AB ἐστὶν ἐπιπέδῳ. διὰ τὰ αὐτὰ δὲ τὸ K καὶ ἐν τῷ $ΓΔ$ ἐστὶν ἐπιπέδῳ. τὰ AB , $ΓΔ$ ἄρα ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται. οὐ συμπίπτουσι δὲ διὰ τὸ παράλληλα ὑποκεῖσθαι. οὐκ ἄρα αἱ EZ , $HΘ$ εὐθεῖαι ἐκβαλλόμεναι ἐπὶ τὰ Z , $Θ$ μέρη συμπεσοῦνται. ὁμοίως δὲ δείχομεν, ὅτι αἱ EZ , $HΘ$ εὐθεῖαι οὐδὲ ἐπὶ τὰ E , H μέρη ἐκβαλλόμεναι συμπεσοῦνται. αἱ δὲ ἐπὶ μηδέτερα τὰ μέρη συμπίπτουσαι παράλληλοί εἰσιν. παράλληλός ἄρα ἐστὶν ἡ EZ τῇ $HΘ$.

of the angles BGH and BGK are right-angles. And since BA is parallel to GH [Prop. 11.9], the (sum of the) angles GBA and BGH is equal to two right-angles [Prop. 1.29]. And BGH (is) a right-angle. GBA (is) thus also a right-angle. Thus, GB is at right-angles to BA . So, for the same (reasons), GB is also at right-angles to BC . Therefore, since the straight-line GB has been set up at right-angles to two straight-lines, BA and BC , cutting one another, GB is thus at right-angles to the plane through BA and BC [Prop. 11.4]. [So, for the same (reasons), BG is also at right-angles to the plane through GH and GK . And the plane through GH and GK is the (plane) through DE and EF . And it was also shown that GB is at right-angles to the plane through AB and BC .] And planes to which the same straight-line is at right-angles are parallel planes [Prop. 11.14]. Thus, the plane through AB and BC is parallel to the (plane) through DE and EF .

Thus, if two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another). (Which is) the very thing it was required to show.

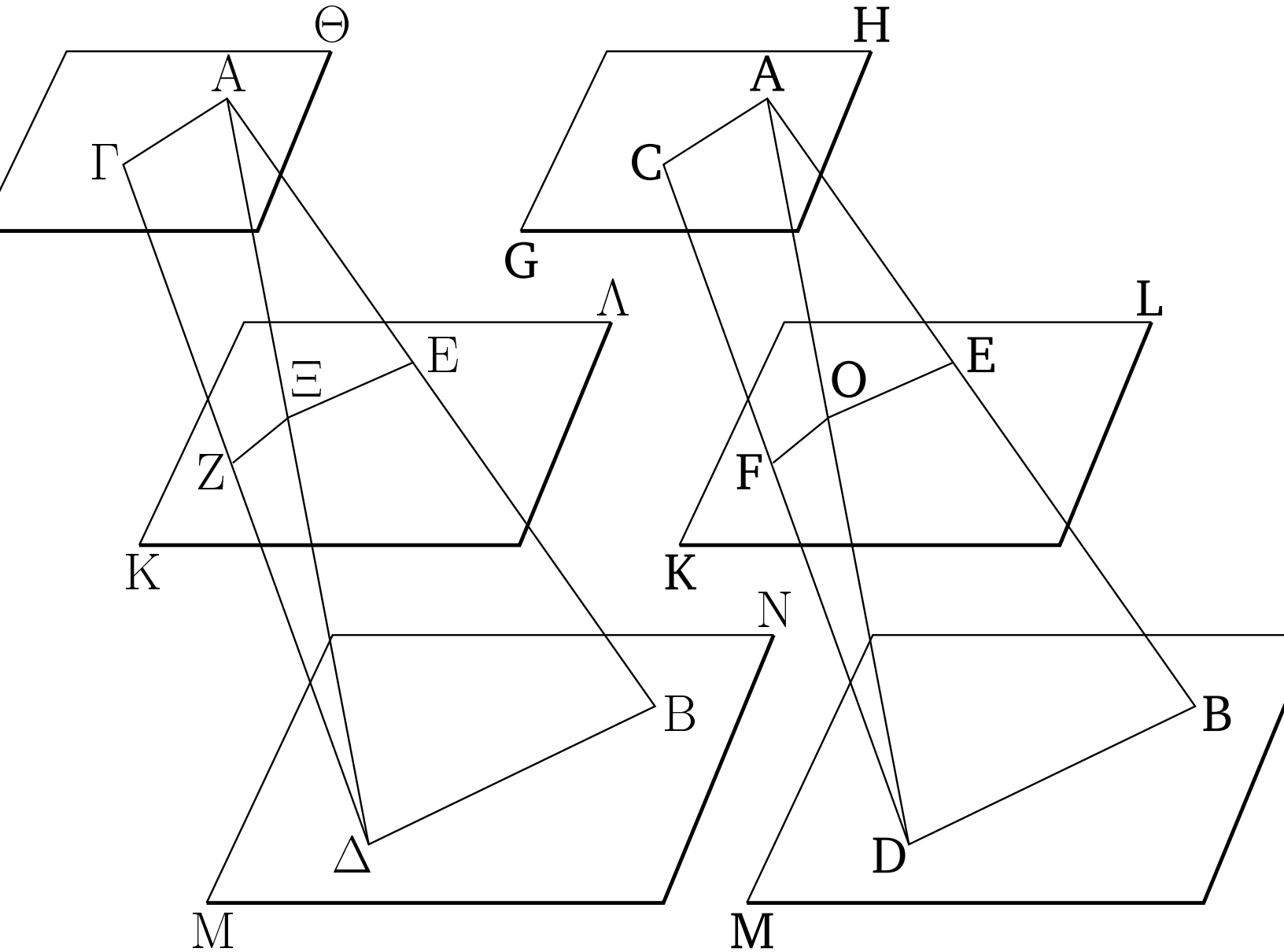
Proposition 16

If two parallel planes are cut by some plane then their common sections are parallel.

For let the two parallel planes AB and CD be cut by the plane $EFGH$. And let EF and GH be their common sections. I say that EF is parallel to GH .

For, if not, being produced, EF and GH will meet either in the direction of F , H , or of E , G . Let them be produced, as in the direction of F , H , and let them, first of all, have met at K . And since EFK is in the plane AB , all of the points on EFK are thus also in the plane AB [Prop. 11.1]. And K is one of the points on EFK . Thus, K is in the plane AB . So, for the same (reasons), K is also in the plane CD . Thus, the planes AB and CD , being produced, will meet. But they do not meet, on account of being (initially) assumed (to be mutually) parallel. Thus, the straight-lines EF and GH , being produced in the direction of F , H , will not meet. So, similarly, we can show that the straight-lines EF and GH , being produced in the direction of E , G , will not meet either. And (straight-lines in one plane which), being produced, do not meet in either direction are parallel [Def. 1.23]. EF is thus parallel to GH .

EB , οὕτως ἡ ΓZ πρὸς $Z\Delta$.



Ἐάν ἄρα δύο εὐθεῖαι ὑπὸ παραλλήλων ἐπιπέδων τέμνονται, εἰς τούξ αὐτὸνξ λόγονξ τμηθήσονται· ὅπερ ἔδει δεῖχαι.

Thus, if two straight-lines are cut by parallel planes then they will be cut in the same ratios. (Which is) the very thing it was required to show.

18.

Proposition 18

Ἐάν εὐθεῖα ἐπιπέδῳ τινὶ πρὸξ ὀρθάξ ᾗ, καὶ πάντα τὰ δι' αὐτῆξ ἐπίπεδα τῶ αὐτῶ ἐπιπέδῳ πρὸξ ὀρθάξ ἔσται.

Εὐθεῖα γάρ τις ἡ AB τῶ ὑποκειμένῳ ἐπιπέδῳ πρὸξ ὀρθάξ ἔστω· λέγω, ὅτι καὶ πάντα τὰ διὰ τῆξ AB ἐπίπεδα τῶ ὑποκειμένῳ ἐπιπέδῳ πρὸξ ὀρθάξ ἔστιν.

Ἐκβεβλήσθω γάρ διὰ τῆξ AB ἐπίπεδον τὸ ΔE , καὶ ἔστω

If a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane.

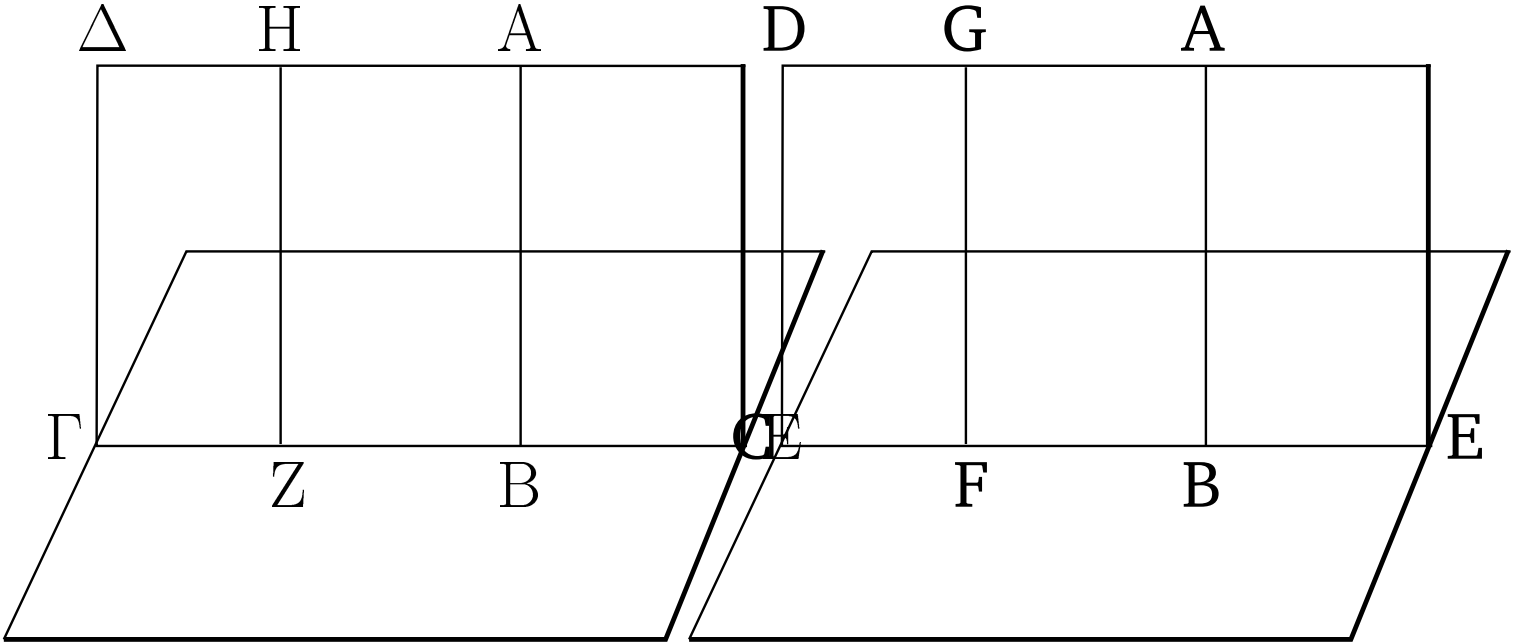
For let some straight-line AB be at right-angles to a reference plane. I say that all of the planes (passing) through AB are also at right-angles to the reference plane.

κοινή τομή τοῦ ΔE ἐπιπέδου καὶ τοῦ ὑποκειμένου ἢ GE , καὶ εἰλήφθω ἐπὶ τῇ GE τυθὸν σημεῖον τὸ Z , καὶ ἀπὸ τοῦ Z τῇ GE πρὸς ὀρθὰς ἤθθω ἐν τῷ ΔE ἐπιπέδῳ ἢ ZH .

Καὶ ἐπεὶ ἡ AB πρὸς τὸ ὑποκειμενον ἐπίπεδον ὀρθή ἐστιν, καὶ πρὸς πάσαξ ἄρα τὰς ἀπτομένας αὐτῇς εὐθείας καὶ ὅσαξ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθή ἐστιν ἡ AB . ὥστε καὶ πρὸς τὴν GE ὀρθή ἐστιν ἡ ἄρα ὑπὸ ABZ γωνία ὀρθή ἐστιν. ἔστι δὲ καὶ ἡ ὑπὸ HZB ὀρθή· παράλληλοξ ἄρα ἐστὶν ἡ AB τῇ ZH . ἡ δὲ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστὶν· καὶ ἡ ZH ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστὶν. καὶ ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, ὅταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰς ἀρόμεναι εὐθεῖαι ἐν ἐνὶ τῶν ἐπιπέδων τῷ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾤσιν. καὶ τῇ κοινῇ τομῇ τῶν ἐπιπέδων τῇ GE ἐν ἐνὶ τῶν ἐπιπέδων τῷ ΔE πρὸς ὀρθὰς ἀθθεῖσα ἢ ZH ἐδείχθη τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς· τὸ ἄρα ΔE ἐπίπεδον ὀρθόν ἐστὶ πρὸς τὸ ὑποκείμενον. ὁμοίωξ δὴ δευθθήσεται καὶ πάντα τὰ διὰ τῇ AB ἐπίπεδα ὀρθὰ τυγθανοντα πρὸς τὸ ὑποκείμενον ἐπίπεδον.

For let the plane DE be produced through AB . And let CE be the common section of the plane DE and the reference (plane). And let some random point F be taken on CE . And let FG be drawn from F , at right-angles to CE , in the plane DE [Prop. 1.11].

And since AB is at right-angles to the reference plane, AB is thus also at right-angles to all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Hence, it is also at right-angles to CE . Thus, angle ABF is a right-angle. And GFB is also a right-angle. Thus, AB is parallel to FG [Prop. 1.28]. And AB is at right-angles to the reference plane. Thus, FG is also at right-angles to the reference plane [Prop. 11.8]. And a plane is at right-angles to a(nother) plane when the straight-lines drawn at right-angles to the common section of the planes, (and lying) in one of the planes, are at right-angles to the remaining plane [Def. 11.4]. And FG , (which was) drawn at right-angles to the common section of the planes, CE , in one of the planes, DE , was shown to be at right-angles to the reference plane. Thus, plane DE is at right-angles to the reference (plane). So, similarly, it can be shown that all of the planes (passing) at random through AB (are) at right-angles to the reference plane.



Ἐὰν ἄρα εὐθεῖα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾤ, καὶ πάντα τὰ δι' αὐτῇς ἐπίπεδα τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· ὅπερ ἔδει δεῖξαι.

Thus, if a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

19.

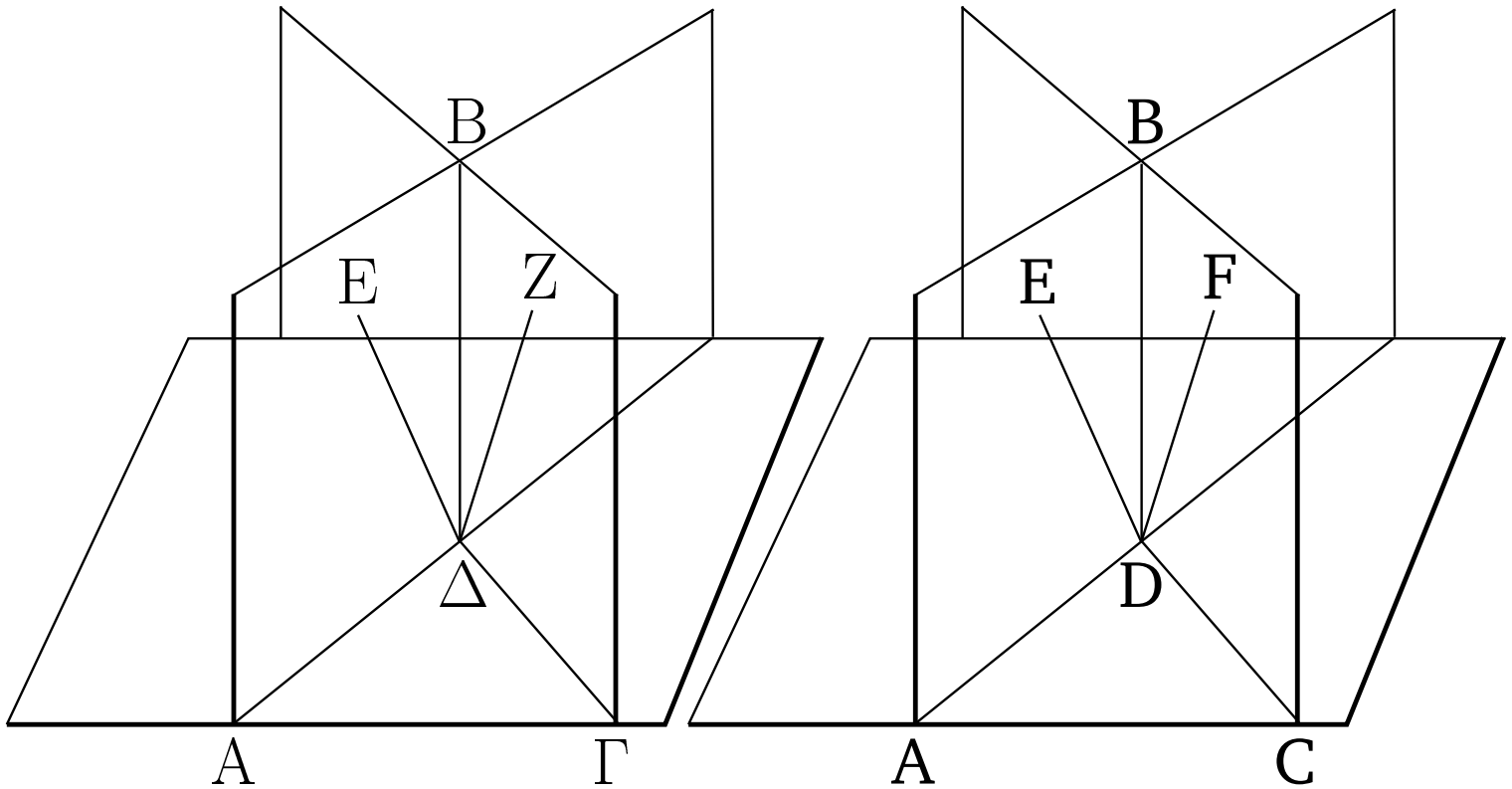
Proposition 19

Ἐὰν δύο ἐπίπεδα τέμνοντα ἄλληλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾤ, καὶ ἡ κοινὴ αὐτῶν τομή τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς

If two planes cutting one another are at right-angles to some plane then their common section will also be at right-

ἔσται.

angles to the same plane.



Δύο γάρ ἐπίπεδα τὰ AB , $BΓ$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἔστω, κοινὴ δὲ αὐτῶν τομὴ ἔστω ἡ BD . λέγω, ὅτι ἡ BD τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἔστιν.

μή γάρ, καὶ ἤθθωσαν ἀπὸ τοῦ Δ σημείου ἐν μὲν τῷ AB ἐπιπέδῳ τῇ AD εὐθείᾳ πρὸς ὀρθὰς ἡ ΔE , ἐν δὲ τῷ $BΓ$ ἐπιπέδῳ τῇ GD πρὸς ὀρθὰς ἡ ΔZ .

Καὶ ἐπεὶ τὸ AB ἐπίπεδον ὀρθὸν ἐστὶ πρὸς τὸ ὑποκείμενον, καὶ τῇ κοινῇ αὐτῶν τομῇ τῇ AD πρὸς ὀρθὰς ἐν τῷ AB ἐπιπέδῳ ἤκται ἡ ΔE , ἡ ΔE ἄρα ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΔZ ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον. ἀπὸ τοῦ αὐτοῦ ἄρα σημείου τοῦ Δ τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰς ἀνισταμέναι εἰσὶν ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ ἀπὸ τοῦ Δ σημείου ἀνασταθῆσεται πρὸς ὀρθὰς πλὴν τῆς ΔB κοινῆς τομῆς τῶν AB , $BΓ$ ἐπιπέδων.

Ἐάν ἄρα δύο ἐπίπεδα τέμνοντα ἄλληλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾗ, καὶ ἡ κοινὴ αὐτῶν τομὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· ὅπερ ἔδει δεῖξαι.

For let the two planes AB and BC be at right-angles to a reference plane, and let their common section be BD . I say that BD is at right-angles to the reference plane.

For (if) not, let DE also be drawn from point D , in the plane AB , at right-angles to the straight-line AD , and DF , in the plane BC , at right-angles to CD .

And since the plane AB is at right-angles to the reference (plane), and DE has been drawn at right-angles to their common section AD , in the plane AB , DE is thus at right-angles to the reference plane [Def. 11.4]. So, similarly, we can show that DF is also at right-angles to the reference plane. Thus, two (different) straight-lines are set up, at the same point D , at right-angles to the reference plane, on the same side. The very thing is impossible [Prop. 11.13]. Thus, no (other straight-line) except the common section DB of the planes AB and BC can be set up at point D , at right-angles to the reference plane.

Thus, if two planes cutting one another are at right-angles to some plane then their common section will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

20.

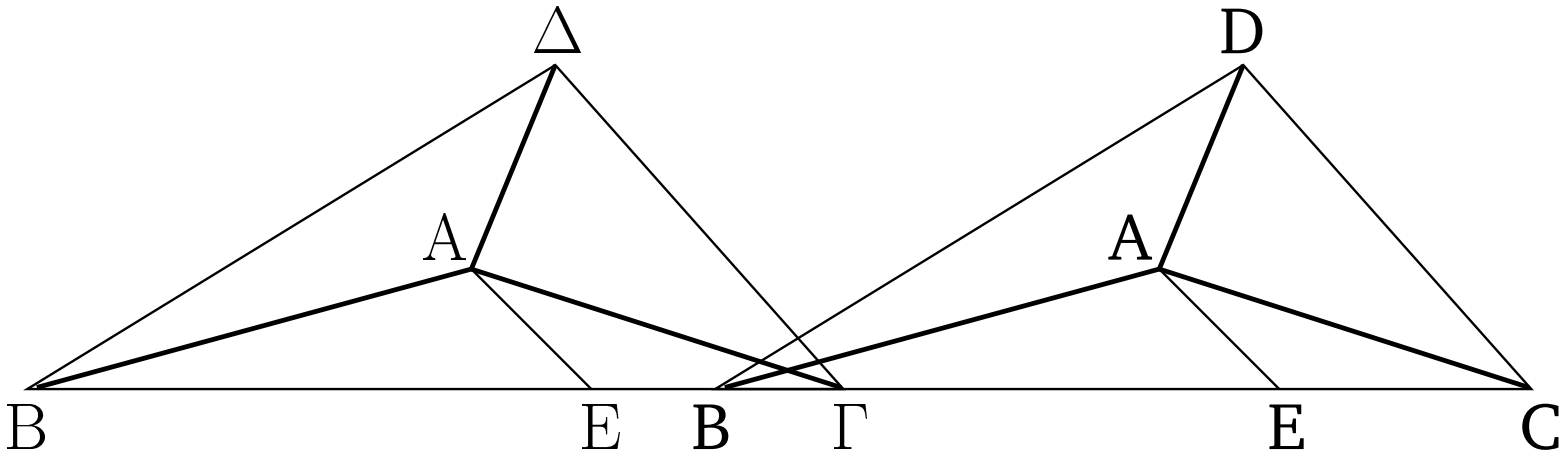
Proposition 20

Ἐάν στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθῃται, δύο ὁποιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβα-

If a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one),

νόμεναι.

(the angles) being taken up in any (possible way).



Στερεὰ γὰρ γωνία ἡ πρὸς τῷ A ὑπὸ τριῶν γωνιῶν ἐπιπέδων τῶν ὑπὸ BAG , $ΓΑΔ$, $ΔAB$ περιεθέσθω· λέγω, ὅτι τῶν ὑπὸ BAG , $ΓΑΔ$, $ΔAB$ γωνιῶν δύο ὅποιοι οὖν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

Εἰ μὲν οὖν αἱ ὑπὸ BAG , $ΓΑΔ$, $ΔAB$ γωνίαι ἴσαι ἀλλήλαις εἴσιν, φανερόν, ὅτι δύο ὅποιοι οὖν τῆς λοιπῆς μείζονές εἰσιν. εἰ δὲ οὐκ, ἔστω μείζων ἡ ὑπὸ BAG , καὶ συνεστίτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ ὑπὸ $ΔAB$ γωνίᾳ ἐν τῷ διὰ τῶν BAG ἐπιπέδῳ ἴση ἡ ὑπὸ BAE , καὶ κείσθω τῇ AD ἴση ἡ AE , καὶ διὰ τοῦ E σημείου διαθθεῖσα ἡ BEG τεμνέτω τὰς AB , AG εὐθείας κατὰ τὰ B , $Γ$ σημεῖα, καὶ ἐπεξεύθθωσαν αἱ $ΔB$, $ΔΓ$.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΔA$ τῇ AE , κοινὴ δὲ ἡ AB , δύο δυνάμει ἴσαι· καὶ γωνία ἡ ὑπὸ $ΔAB$ γωνία τῇ ὑπὸ BAE ἴση· βάσεις ἄρα ἡ $ΔB$ βάσει τῇ BE ἐστὶν ἴση. καὶ ἐπεὶ δύο αἱ BD , $ΔΓ$ τῆς $BΓ$ μείζονές εἰσιν, ὧν ἡ $ΔB$ τῇ BE ἐδείχθη ἴση, λοιπὴ ἄρα ἡ $ΔΓ$ λοιπῆς τῆς EG μείζων ἐστίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΔA$ τῇ AE , κοινὴ δὲ ἡ AG , καὶ βάσεις ἡ $ΔΓ$ βάσεως τῆς EG μείζων ἐστίν, γωνία ἄρα ἡ ὑπὸ $ΔAG$ γωνία τῆς ὑπὸ EAG μείζων ἐστίν. ἐδείχθη δὲ καὶ ἡ ὑπὸ $ΔAB$ τῇ ὑπὸ BAE ἴση· αἱ ἄρα ὑπὸ $ΔAB$, $ΔAG$ τῆς ὑπὸ BAG μείζονές εἰσιν. ὁμοίως δὲ δειχόμεν, ὅτι καὶ αἱ λοιπαὶ σύνδυο λαμβανόμεναι τῆς λοιπῆς μείζονές εἰσιν.

Ἐὰν ἄρα στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθῃται, δύο ὅποιοι οὖν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.

21.

Ἄπαντα στερεὰ γωνία ὑπὸ ἐλασσόνων [ἢ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέθεται.

For let the solid angle A be contained by the three plane angles BAC , CAD , and DAB . I say that (the sum of) any two of the angles BAC , CAD , and DAB is greater than the remaining (one), (the angles) being taken up in any (possible way).

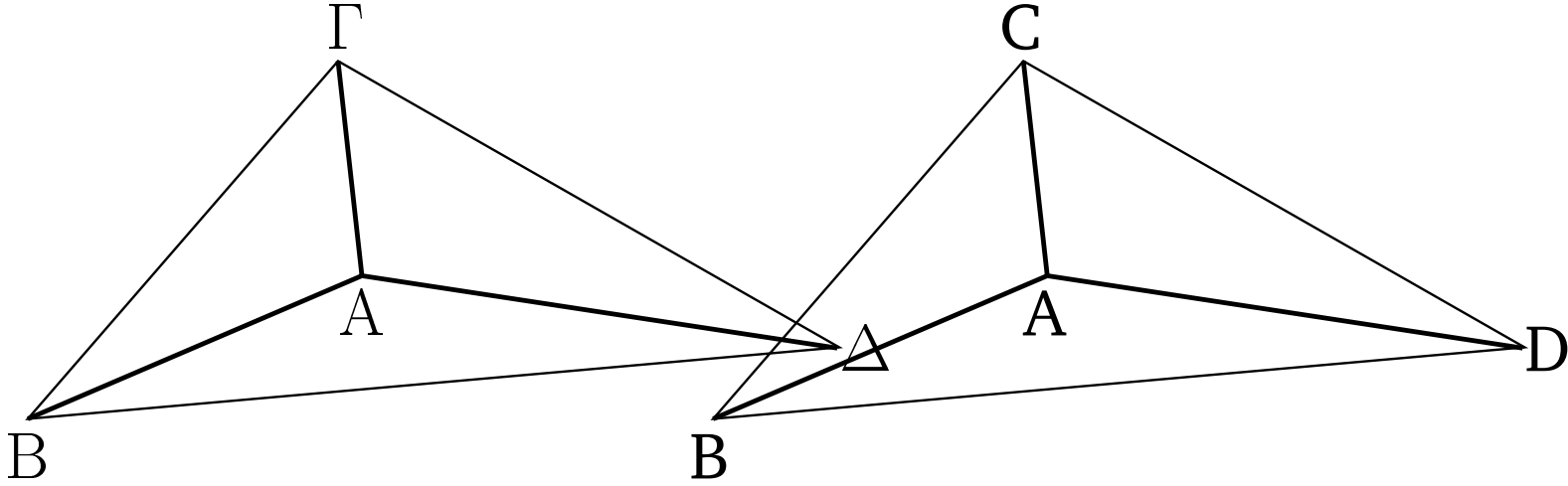
For if the angles BAC , CAD , and DAB are equal to one another then (it is) clear that (the sum of) any two is greater than the remaining (one). But, if not, let BAC be greater (than CAD or DAB). And let (angle) BAE , equal to the angle DAB , be constructed in the plane through BAC , on the straight-line AB , at the point A on it. And let AE be made equal to AD . And BEC being drawn across through point E , let it cut the straight-lines AB and AC at points B and C (respectively). And let DB and DC be joined.

And since DA is equal to AE , and AB (is) common, the two (straight-lines AD and AB are) equal to the two (straight-lines EA and AB , respectively). And angle DAB (is) equal to angle BAE . Thus, the base DB is equal to the base BE [Prop. 1.4]. And since the (sum of the) two (straight-lines) BD and DC is greater than BC [Prop. 1.20], of which DB was shown (to be) equal to BE , the remainder DC is thus greater than the remainder EC . And since DA is equal to AE , but AC (is) common, and the base DC is greater than the base EC , the angle DAC is thus greater than the angle EAC [Prop. 1.25]. And DAB was also shown (to be) equal to BAE . Thus, (the sum of) DAB and DAC is greater than BAC . So, similarly, we can also show that the remaining (angles), being taken in pairs, are greater than the remaining (one).

Thus, if a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one), (the angles) being taken up in any (possible way). (Which is) the very thing it was required to show.

Proposition 21

Any solid angle is contained by plane angles (whose sum is) less [than] four right-angles.[†]



Ἐστω στερεὰ γωνία ἡ πρὸς τῷ A περιεχομένη ὑπὸ ἐπιπέδων γωνιῶν τῶν ὑπὸ BAF , $ΓAF$, $ΔAB$ · λέγω, ὅτι αἱ ὑπὸ BAF , $ΓAF$, $ΔAB$ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

Εἰλήφθω γὰρ ἐφ' ἐκάστης τῶν AB , AF , AD τυθόντα σημεῖα τὰ B , $Γ$, $Δ$, καὶ ἐπεξεύθωσαν αἱ $BΓ$, $ΓΔ$, $ΔB$. καὶ ἐπεὶ στερεὰ γωνία ἡ πρὸς τῷ B ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέχεται τῶν ὑπὸ $ΓBA$, $ABΔ$, $ΓBΔ$, δύο ὁποιοῦν τῆς λοιπῆς μείζονές εἰσιν· αἱ ἄρα ὑπὸ $ΓBA$, $ABΔ$ τῆς ὑπὸ $ΓBΔ$ μείζονές εἰσιν. διὰ τὰ αὐτὰ δὴ καὶ αἱ μὲν ὑπὸ $BΓA$, $AFΔ$ τῆς ὑπὸ $BΓΔ$ μείζονές εἰσιν, αἱ δὲ ὑπὸ $ΓΔA$, ADB τῆς ὑπὸ $ΓΔB$ μείζονές εἰσιν· αἱ ἔχ' ἄρα γωνίαι αἱ ὑπὸ $ΓBA$, $ABΔ$, $BΓA$, $AFΔ$, $ΓΔA$, ADB τριῶν τῶν ὑπὸ $ΓBΔ$, $BΓA$, $ΓΔB$ μείζονές εἰσιν. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ $ΓBΔ$, $BΔΓ$, $BΓΔ$ δυοῖν ὀρθαῖς ἴσαι εἰσὶν· αἱ ἔχ' ἄρα αἱ ὑπὸ $ΓBA$, $ABΔ$, $BΓA$, $AFΔ$, $ΓΔA$, ADB δύο ὀρθῶν μείζονές εἰσιν. καὶ ἐπεὶ ἐκάστον τῶν $ABΓ$, $AFΔ$, ADB τριγώνων αἱ τρεῖς γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν, αἱ ἄρα τῶν τριῶν τριγώνων ἑννέα γωνίαι αἱ ὑπὸ $ΓBA$, AFB , BAF , $AFΔ$, $ΓΔA$, $ΓAΔ$, ADB , $ΔBA$, $BAΔ$ ἔχ' ὀρθαῖς ἴσαι εἰσὶν, ὧν αἱ ὑπὸ $ABΓ$, $BΓA$, $AFΔ$, $ΓΔA$, ADB , $ΔBA$ ἔχ' γωνίαι δύο ὀρθῶν εἰσι μείζονες· λοιπαὶ ἄρα αἱ ὑπὸ BAF , $ΓAΔ$, $ΔAB$ τρεῖς [γωνίαι] περιέχουσιν τὴν στερεάν γωνίαν τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

Ἄπανα ἄρα στερεὰ γωνία ὑπὸ ἐλασσόνων [ῆ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέχεται· ὅπερ ἔδει δεῖξαι.

Let the solid angle A be contained by the plane angles BAC , CAD , and DAB . I say that (the sum of) BAC , CAD , and DAB is less than four right-angles.

For let the random points B , C , and D be taken on each of (the straight-lines) AB , AC , and AD (respectively). And let BC , CD , and DB be joined. And since the solid angle at B is contained by the three plane angles CBA , ABD , and CBD , (the sum of) any two is greater than the remaining (one) [Prop. 11.20]. Thus, (the sum of) CBA and ABD is greater than CBD . So, for the same (reasons), (the sum of) BCA and ACD is also greater than BCD , and (the sum of) CDA and ADB is greater than CDB . Thus, the (sum of the) six angles CBA , ABD , BCA , ACD , CDA , and ADB is greater than the (sum of the) three (angles) CBD , BCD , and CDB . But, the (sum of the) three (angles) CBD , BDC , and BCD is equal to two right-angles [Prop. 1.32]. Thus, the (sum of the) six angles CBA , ABD , BCA , ACD , CDA , and ADB is greater than two right-angles. And since the (sum of the) three angles of each of the triangles ABC , ACD , and ADB is equal to two right-angles, the (sum of the) nine angles CBA , ACB , BAC , ACD , CDA , CAD , ADB , DBA , and BAD of the three triangles is equal to six right-angles, of which the (sum of the) six angles ABC , BCA , ACD , CDA , ADB , and DBA is greater than two right-angles. Thus, the (sum of the) remaining three [angles] BAC , CAD , and DAB , containing the solid angle, is less than four right-angles.

Thus, any solid angle is contained by plane angles (whose sum is) less [than] four right-angles. (Which is) the very thing it was required to show.

† This proposition is only proved for the case of a solid angle contained by three plane angles. However, the generalization to a solid angle contained by more than three plane angles is straightforward.

22.

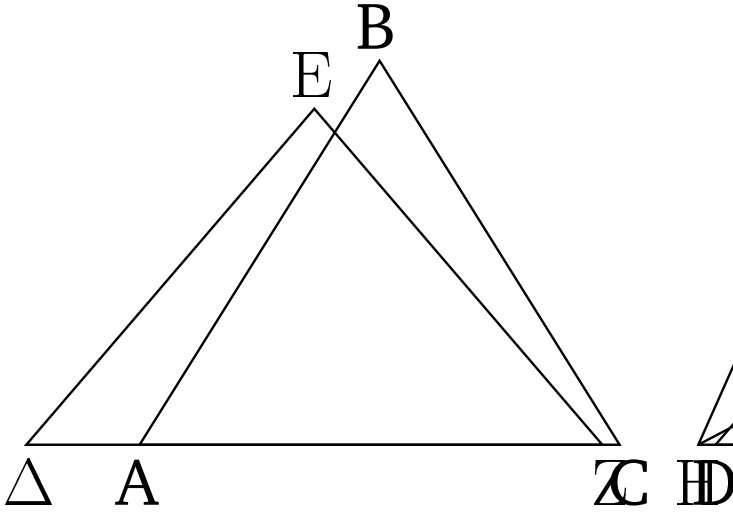
Proposition 22

Ἐάν ὡς τρεῖς γωνίαι ἐπίπεδοι, ὧν αἱ δύο τῆς λοιπῆς

If there are three plane angles, of which (the sum of any)

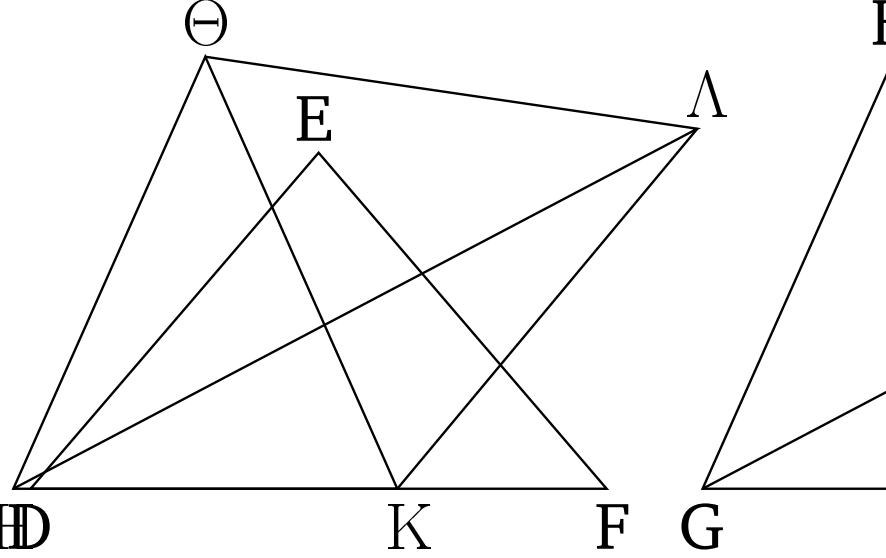
μειζονέξ εἰσι πάντα μεταλαμβανόμεναι, περιέθωσι δὲ αὐτάξ ἴσαι εὐθείαι, δυνατόν ἐστιν ἐκ τῶν ἐπιζευγνυσουσῶν τάξ ἴσαξ εὐθείαξ τρίγωνον συστήσασθαι.

two is greater than the remaining (one), (the angles) being taken up in any (possible way), and if equal straight-lines contain them, then it is possible to construct a triangle from (the straight-lines created by) joining the (ends of the) equal straight-lines.



Ἐστωσαν τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, ὧν αἱ δύο τῆξ λοιπῆξ μειζονέξ εἰσι πάντα μεταλαμβάνόμεναι, αἱ μὲν ὑπὸ ΑΒΓ, ΔΕΖ τῆξ ὑπὸ ΗΘΚ, αἱ δὲ ὑπὸ ΔΕΖ, ΗΘΚ τῆξ ὑπὸ ΑΒΓ, καὶ ἔτι αἱ ὑπὸ ΗΘΚ, ΑΒΓ τῆξ ὑπὸ ΔΕΖ, καὶ ἔστωσαν ἴσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ εὐθείαι, καὶ ἐπεξεύθθωσαν αἱ ΑΓ, ΔΖ, ΗΚ λέγω, ὅτι δυνατόν ἐστιν ἐκ τῶν ἴσων ταῖξ ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι, τουτέστιν ὅτι τῶν ΑΓ, ΔΖ, ΗΚ δύο ὁποιαοῦν τῆξ λοιπῆξ μειζονέξ εἰσιν.

Εἰ μὲν οὖν αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ γωνίαι ἴσαι ἀλλήλαιξ εἰσίν, φανερόν, ὅτι καὶ τῶν ΑΓ, ΔΖ, ΗΚ ἴσων γινόμενων δυνατόν ἐστιν ἐκ τῶν ἴσων ταῖξ ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. εἰ δὲ οὐ, ἔστωσαν ἄνωσοι, καὶ συνεστάτω πρόξ τῆξ ΘΚ εὐθεία καὶ τῶ πρόξ αὐτῆξ σημείω τῶ Θ τῆξ ὑπὸ ΑΒΓ γωνία ἴση ἢ ὑπὸ ΚΘΛ· καὶ κείσθω μιὰ τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ ἴση ἢ ΘΛ, καὶ ἐπεξεύθθωσαν αἱ ΚΛ, ΗΛ. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΓ δυοὶ ταῖξ ΚΘ, ΘΛ ἴσαι εἰσίν, καὶ γωνία ἢ πρόξ τῶ Β γωνία τῆξ ὑπὸ ΚΘΛ ἴση, βάσειξ ἄρα ἢ ΑΓ βάσει τῆξ ΚΛ ἴση. καὶ ἐπεὶ αἱ ὑπὸ ΑΒΓ, ΗΘΚ τῆξ ὑπὸ ΔΕΖ μειζονέξ εἰσιν, ἴση δὲ ἢ ὑπὸ ΑΒΓ τῆξ ὑπὸ ΚΘΛ, ἢ ἄρα ὑπὸ ΗΘΛ τῆξ ὑπὸ ΔΕΖ μειζων ἐστίν. καὶ ἐπεὶ δύο αἱ ΗΘ, ΘΛ δύο ταῖξ ΔΕ, ΕΖ ἴσαι εἰσίν, καὶ γωνία ἢ ὑπὸ ΗΘΛ γωνίαξ τῆξ ὑπὸ ΔΕΖ μειζων, βάσειξ ἄρα ἢ ΗΛ βάσεωξ τῆξ ΔΖ μειζων ἐστίν. ἀλλὰ αἱ ΗΚ, ΚΛ τῆξ ΗΛ μειζονέξ εἰσιν. πολλῶ ἄρα αἱ ΗΚ, ΚΛ τῆξ ΔΖ μειζονέξ εἰσιν. ἴση δὲ ἢ ΚΛ τῆξ ΑΓ· αἱ ΑΓ, ΗΚ ἄρα τῆξ λοιπῆξ τῆξ ΔΖ μειζονέξ εἰσιν. ὁμοίωξ δὴ δείχομεν, ὅτι καὶ αἱ μὲν ΑΓ, ΔΖ τῆξ ΗΚ μειζονέξ εἰσιν, καὶ ἔτι αἱ ΔΖ, ΗΚ τῆξ ΑΓ μειζονέξ εἰσιν. δυνατόν ἄρα ἐστίν ἐκ τῶν ἴσων



Let ABC , DEF , and GHK be three plane angles, of which the sum of any two is greater than the remaining (one), (the angles) being taken up in any (possible way)—(that is), ABC and DEF (greater) than GHK , DEF and GHK (greater) than ABC , and, further, GHK and ABC (greater) than DEF . And let AB , BC , DE , EF , GH , and HK be equal straight-lines. And let AC , DF , and GK be joined. I say that that it is possible to construct a triangle out of (straight-lines) equal to AC , DF , and GK —that is to say, that (the sum of) any two of AC , DF , and GK is greater than the remaining (one).

Now, if the angles ABC , DEF , and GHK are equal to one another then (it is) clear that, (with) AC , DF , and GK also becoming equal, it is possible to construct a triangle from (straight-lines) equal to AC , DF , and GK . And if not, let them be unequal, and let KHL , equal to angle ABC , be constructed on the straight-line HK , at the point H on it. And let HL be made equal to one of AB , BC , DE , EF , GH , and HK . And let KL and GL be joined. And since the two (straight-lines) AB and BC are equal to the two (straight-lines) KH and HL (respectively), and the angle at B (is) equal to KHL , the base AC is thus equal to the base KL [Prop. 1.4]. And since (the sum of) ABC and GHK is greater than DEF , and ABC equal to KHL , GHL is thus greater than DEF . And since the two (straight-lines) GH and HL are equal to the two (straight-lines) DE and EF (respectively), and angle GHL (is) greater than DEF , the base GL is thus greater than the base DF [Prop. 1.24]. But, (the sum of) GK and KL is greater than GL [Prop. 1.20]. Thus, (the sum of) GK and KL is much greater than DF . And KL

ταῖς $ΑΓ$, $ΔΖ$, $ΗΚ$ τρίγωνον συστήσασθαι· ὅπερ ἔδει δεῖξαι.

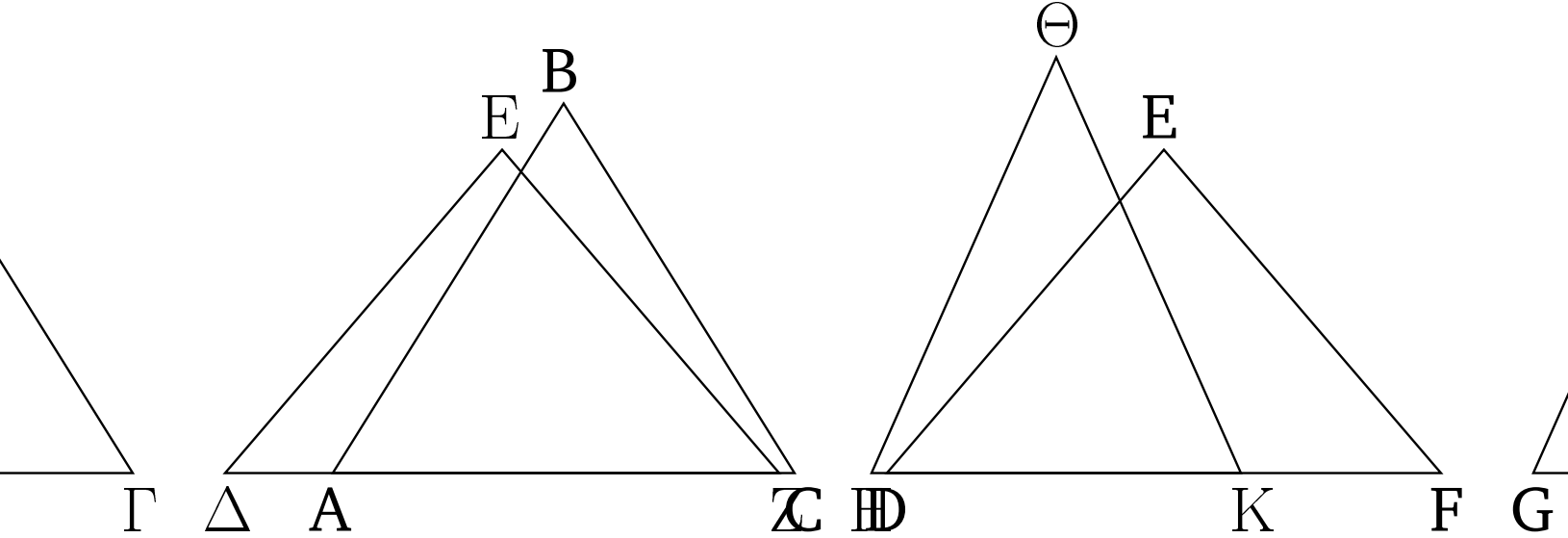
(is) equal to AC . Thus, (the sum of) AC and GK is greater than the remaining (straight-line) DF . So, similarly, we can show that (the sum of) AC and DF is greater than GK , and, further, that (the sum of) DF and GK is greater than AC . Thus, it is possible to construct a triangle from (straight-lines) equal to AC , DF , and GK . (Which is) the very thing it was required to show.

23.

Ἐκ τριῶν γωνιῶν ἐπιπέδων, ὧν αἱ δύο τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, στερεὰν γωνίαν συστήσασθαι· δεῖ δὴ τὰς τρεῖς τεσσάρων ὀρθῶν ἐλάσσοναξ εἶναι.

Proposition 23

To construct a solid angle from three (given) plane angles, (the sum of) two of which is greater than the remaining (one, the angles) being taken up in any (possible way). So, it is necessary for the (sum of the) three (angles) to be less than four right-angles [Prop. 11.21].

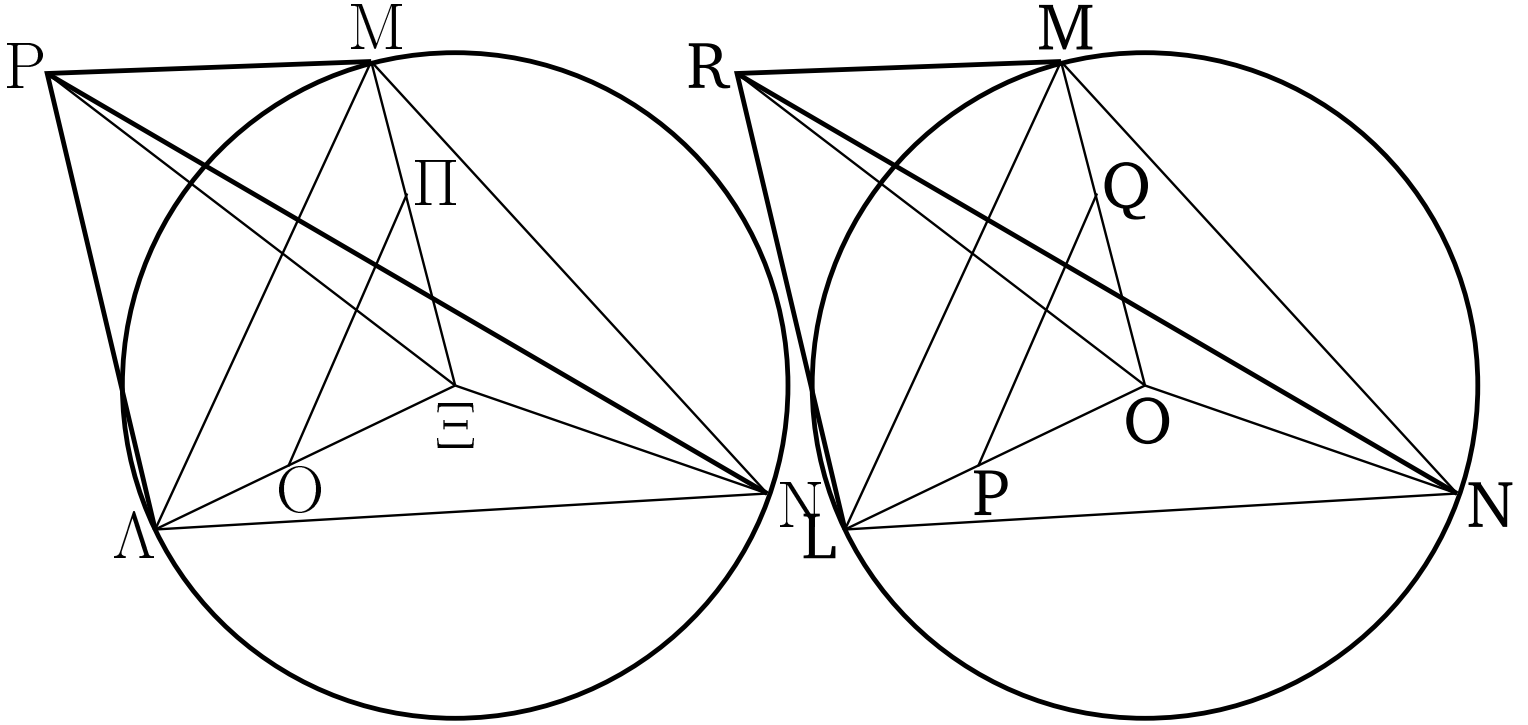


Ἐστωσαν αἱ δοθεῖσαι τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ $ΑΒΓ$, $ΔΕΖ$, $ΗΘΚ$, ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, ἔτι δὲ αἱ τρεῖς τεσσάρων ὀρθῶν ἐλάσσονες· δεῖ δὴ ἐκ τῶν ἴσων ταῖς ὑπὸ $ΑΒΓ$, $ΔΕΖ$, $ΗΘΚ$ στερεὰν γωνίαν συστήσασθαι.

Ἀπειλήφθωσαν ἴσαι αἱ $ΑΒ$, $ΒΓ$, $ΔΕ$, $ΕΖ$, $ΗΘ$, $ΘΚ$, καὶ ἐπεξεύθθωσαν αἱ $ΑΓ$, $ΔΖ$, $ΗΚ$ · δυνατὸν ἄρα ἐστὶν ἐκ τῶν ἴσων ταῖς $ΑΓ$, $ΔΖ$, $ΗΚ$ τρίγωνον συστήσασθαι. συνεστάτω τὸ $ΛΜΝ$, ὥστε ἴσῃν εἶναι τὴν μὲν $ΑΓ$ τῇ $ΛΜ$, τὴν δὲ $ΔΖ$ τῇ $ΜΝ$, καὶ ἔτι τὴν $ΗΚ$ τῇ $ΝΛ$, καὶ περιγεγράφθω περὶ τὸ $ΛΜΝ$ τρίγωνον κύκλος ὁ $ΛΜΝ$, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον καὶ ἔστω τὸ $Χ$, καὶ ἐπεξεύθθωσαν αἱ $ΑΧ$, $ΜΧ$, $ΝΧ$.

Let ABC , DEF , and GHK be the three given plane angles, of which let (the sum of) two be greater than the remaining (one, the angles) being taken up in any (possible way), and, further, (let) the (sum of the) three (be) less than four right-angles. So, it is necessary to construct a solid angle from (plane angles) equal to ABC , DEF , and GHK .

Let AB , BC , DE , EF , GH , and HK be cut off (so as to be) equal (to one another). And let AC , DF , and GK be joined. It is, thus, possible to construct a triangle from (straight-lines) equal to AC , DF , and GK [Prop. 11.22]. Let (such a triangle), LMN , have been constructed, such that AC is equal to LM , DF to MN , and, further, GK to NL . And let the circle LMN be circumscribed about triangle LMN [Prop. 4.5]. And let its center be found, and let it be (at) O . And let LO , MO , and NO be joined.



Λέγω, ὅτι ἡ AB μείζων ἐστὶ τῆς AX . εἰ γὰρ μή, ἦτοι ἴση ἐστὶν ἡ AB τῇ AX ἢ ἐλάττων. ἔστω πρότερον ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ AX , ἀλλὰ ἡ μὲν AB τῇ BF ἐστὶν ἴση, ἡ δὲ XA τῇ XM , δύο δὴ αἱ AB , BF δύο ταῖς AX , XM ἴσαι εἰσὶν ἑκατέρω ἑκατέρῳ· καὶ βάσεις ἡ AF βάσει τῇ AM ὑπόκειται ἴση· γωνία ἄρα ἡ ὑπὸ ABF γωνία τῇ ὑπὸ AXM ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ $ΔΕΖ$ τῇ ὑπὸ $ΜΧΝ$ ἐστὶν ἴση, καὶ ἔτι ἡ ὑπὸ $ΗΘΚ$ τῇ ὑπὸ $ΝΧΛ$ · αἱ ἄρα τρεῖς αἱ ὑπὸ ABF , $ΔΕΖ$, $ΗΘΚ$ γωνίαί τρισι ταῖς ὑπὸ AXM , MXN , $ΝΧΛ$ εἰσιν ἴσαι. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ AXM , MXN , $ΝΧΛ$ τέτταρσιν ὀρθαῖς εἰσιν ἴσαι· καὶ αἱ τρεῖς ἄρα αἱ ὑπὸ ABF , $ΔΕΖ$, $ΗΘΚ$ τέτταρσιν ὀρθαῖς ἴσαι εἰσὶν. ὑπόκεινται δὲ καὶ τεσσάρων ὀρθῶν ἐλάσσονες· ὅπερ ἄτοπον. οὐκ ἄρα ἡ AB τῇ AX ἴση ἐστὶν. λέγω δὴ, ὅτι οὐδὲ ἐλάττων ἐστὶν ἡ AB τῇς AX . εἰ γὰρ δυνατόν, ἔστω· καὶ κείσθω τῇ μὲν AB ἴση ἡ XO , τῇ δὲ BF ἴση ἡ $XΠ$, καὶ ἐπεξεύθθω ἡ $OΠ$. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ BF , ἴση ἐστὶ καὶ ἡ XO τῇ $XΠ$ · ὥστε καὶ λοιπὴ ἡ AO τῇ $ΠΜ$ ἐστὶν ἴση. παράλληλος ἄρα ἐστὶν ἡ AM τῇ $OΠ$, καὶ ἰσογώνιον τὸ AMX τῷ $OΠX$ · ἔστιν ἄρα ὡς ἡ XA πρὸς AM , οὕτως ἡ XO πρὸς $OΠ$ · ἐναλλάξ ὡς ἡ AX πρὸς XO , οὕτως ἡ AM πρὸς $OΠ$. μείζων δὲ ἡ AX τῆς XO · μείζων ἄρα καὶ ἡ AM τῆς $OΠ$. ἀλλὰ ἡ AM κείται τῇ AF ἴση· καὶ ἡ AF ἄρα τῆς $OΠ$ μείζων ἐστὶν. ἐπεὶ οὖν δύο αἱ AB , BF δυοὶ ταῖς OX , $XΠ$ ἴσαι εἰσὶν, καὶ βάσεις ἡ AF βάσεως τῆς $OΠ$ μείζων ἐστὶν, γωνία ἄρα ἡ ὑπὸ ABF γωνία τῆς ὑπὸ $OXΠ$ μείζων ἐστὶν. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ μὲν ὑπὸ $ΔΕΖ$ τῆς ὑπὸ $ΜΧΝ$ μείζων ἐστὶν, ἡ δὲ ὑπὸ $ΗΘΚ$ τῆς ὑπὸ $ΝΧΛ$. αἱ ἄρα τρεῖς γωνίαί αἱ ὑπὸ ABF , $ΔΕΖ$, $ΗΘΚ$ τριῶν τῶν ὑπὸ AXM ,

I say that AB is greater than LO . For, if not, AB is either equal to, or less than, LO . Let it, first of all, be equal. And since AB is equal to LO , but AB is equal to BC , and OL to OM , so the two (straight-lines) AB and BC are equal to the two (straight-lines) LO and OM , respectively. And the base AC was assumed (to be) equal to the base LM . Thus, angle ABC is equal to angle LOM [Prop. 1.8]. So, for the same (reasons), DEF is also equal to MON , and, further, GHK to NOL . Thus, the three angles ABC , DEF , and GHK are equal to the three angles LOM , MON , and NOL , respectively. But, the (sum of the) three angles LOM , MON , and NOL is equal to four right-angles. Thus, the (sum of the) three angles ABC , DEF , and GHK is also equal to four right-angles. And it was also assumed (to be) less than four right-angles. The very thing (is) absurd. Thus, AB is not equal to LO . So, I say that AB is not less than LO either. For, if possible, let it be (less). And let OP be made equal to AB , and OQ equal to BC , and let PQ be joined. And since AB is equal to BC , OP is also equal to OQ . Hence, the remainder LP is also equal to (the remainder) QM . LM is thus parallel to PQ [Prop. 6.2], and (triangle) LMO (is) equiangular with (triangle) PQO [Prop. 1.29]. Thus, as OL is to LM , so OP (is) to PQ [Prop. 6.4]. Alternately, as LO (is) to OP , so LM (is) to PQ [Prop. 5.16]. And LO (is) greater than OP . Thus, LM (is) also greater than PQ [Prop. 5.14]. But LM was made equal to AC . Thus, AC is also greater than PQ . Therefore, since the two (straight-lines) AB and BC are equal to the two (straight-lines) PO and OQ (respectively), and the base AC is greater than the base PQ , the angle ABC is

MXN , NXA μείζονές εἰσιν. ἀλλὰ αἱ ὑπὸ $ABΓ$, $ΔΕΖ$, $ΗΘΚ$ τεσσάρων ὀρθῶν ἐλάσσονες ὑπόκεινται· πολλῶν ἄρα αἱ ὑπὸ $ΛΧΜ$, $ΜΧΝ$, $ΝΧΑ$ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν. ἀλλὰ καὶ ἴσαι· ὅπερ ἐστὶν ἄτοπον. οὐκ ἄρα ἡ AB ἐλάσσων ἐστὶ τῆς $ΛΧ$. ἐδείχθη δέ, ὅτι οὐδὲ ἴση· μείζων ἄρα ἡ AB τῆς $ΛΧ$.

Ἀνεστάτω δὴ ἀπὸ τοῦ X σημεῖον τῷ τοῦ $ΑΜΝ$ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἡ XP , καὶ ὥς μείζον ἐστὶ τὸ ἀπὸ τῆς AB τετράγωνον τοῦ ἀπὸ τῆς $ΛΧ$, ἐκείνῳ ἴσον ἔστω τὸ ἀπὸ τῆς XP , καὶ ἐπεξεύθωσαν αἱ PA , PM , PN .

Καὶ ἐπεὶ ἡ PX ὀρθή ἐστι πρὸς τὸ τοῦ $ΑΜΝ$ κύκλου ἐπίπεδον, καὶ πρὸς ἐκάστην ἄρα τῶν $ΛΧ$, $ΜΧ$, $ΝΧ$ ὀρθή ἐστὶν ἡ PX . καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΛΧ$ τῇ $ΧΜ$, κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ XP , βάσει ἄρα ἡ PA βάσει τῇ PM ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ PN ἐκατέρᾳ τῶν PA , PM ἐστὶν ἴση· αἱ τρεῖς ἄρα αἱ PA , PM , PN ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ὥς μείζον ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς $ΛΧ$, ἐκείνῳ ἴσον ὑπόκειται τὸ ἀπὸ τῆς XP , τὸ ἄρα ἀπὸ τῆς AB ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΛΧ$, XP . τοῖς δὲ ἀπὸ τῶν $ΛΧ$, XP ἴσον ἐστὶ τὸ ἀπὸ τῆς AP ὀρθῇ γὰρ ἡ ὑπὸ $ΛXP$ · τὸ ἄρα ἀπὸ τῆς AB ἴσον ἐστὶ τῷ ἀπὸ τῆς PA · ἴση ἄρα ἡ AB τῇ PA . ἀλλὰ τῇ μὲν AB ἴση ἐστὶν ἐκάστη τῶν $ΒΓ$, $ΔΕ$, $ΕΖ$, $ΗΘ$, $ΘΚ$, τῇ δὲ PA ἴση ἐκατέρα τῶν PM , PN · ἐκάστη ἄρα τῶν AB , $ΒΓ$, $ΔΕ$, $ΕΖ$, $ΗΘ$, $ΘΚ$ ἐκάστη τῶν PA , PM , PN ἴση ἐστὶν. καὶ ἐπεὶ δύο αἱ AP , PM δυοὶ ταῖς AB , $ΒΓ$ ἴσαι εἰσίν, καὶ βάσει ἡ AM βάσει τῇ $ΑΓ$ ὑπόκειται ἴση, γωνία ἄρα ἡ ὑπὸ APM γωνία τῇ ὑπὸ $ΑΒΓ$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ MPN τῇ ὑπὸ $ΔΕΖ$ ἐστὶν ἴση, ἡ δὲ ὑπὸ APN τῇ ὑπὸ $ΗΘΚ$.

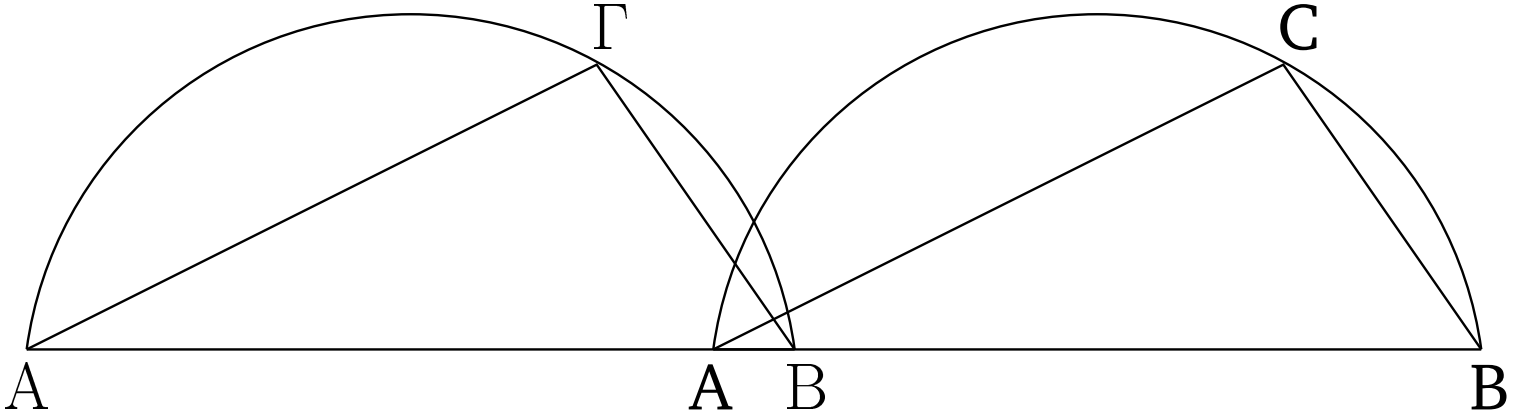
Ἐκ τριῶν ἄρα γωνιῶν ἐπιπέδων τῶν ὑπὸ APM , MPN , APN , αἱ εἰσὶν ἴσαι τρισὶ ταῖς δοθείσαις ταῖς ὑπὸ $ΑΒΓ$, $ΔΕΖ$, $ΗΘΚ$, στερεὰ γωνία συνέσταται ἡ πρὸς τῷ P περιεθρομένη ὑπὸ τῶν APM , MPN , APN γωνιῶν· ὅπερ ἔδει ποιῆσαι.

thus greater than the angle POQ [Prop. 1.25]. So, similarly, we can show that DEF is also greater than MON , and GHK than NOL . Thus, the (sum of the) three angles ABC , DEF , and GHK is greater than the (sum of the) three angles LOM , MON , and NOL . But, (the sum of) ABC , DEF , and GHK was assumed (to be) less than four right-angles. Thus, (the sum of) LOM , MON , and NOL is much less than four right-angles. But, (it is) also equal (to four right-angles). The very thing is absurd. Thus, AB is not less than LO . And it was shown (to be) not equal either. Thus, AB (is) greater than LO .

So let OR be set up at point O at right-angles to the plane of circle LMN [Prop. 11.12]. And let the (square) on OR be equal to that (area) by which the square on AB is greater than the (square) on LO [Prop. 11.23 lem.]. And let RL , RM , and RN be joined.

And since RO is at right-angles to the plane of circle LMN , RO is thus also at right-angles to each of LO , MO , and NO . And since LO is equal to OM , and OR is common and at right-angles, the base RL is thus equal to the base RM [Prop. 1.4]. So, for the same (reasons), RN is also equal to each of RL and RM . Thus, the three (straight-lines) RL , RM , and RN are equal to one another. And since the (square) on OR was assumed to be equal to that (area) by which the (square) on AB is greater than the (square) on LO , the (square) on AB is thus equal to the (sum of the squares) on LO and OR . And the (square) on LR is equal to the (sum of the squares) on LO and OR . For LOR (is) a right-angle [Prop. 1.47]. Thus, the (square) on AB is equal to the (square) on RL . Thus, AB (is) equal to RL . But, each of BC , DE , EF , GH , and HK is equal to AB , and each of RM and RN equal to RL . Thus, each of AB , BC , DE , EF , GH , and HK is equal to each of RL , RM , and RN . And since the two (straight-lines) LR and RM are equal to the two (straight-lines) AB and BC (respectively), and the base LM was assumed (to be) equal to the base AC , the angle LRM is thus equal to the angle ABC [Prop. 1.8]. So, for the same (reasons), MRN is also equal to DEF , and LRN to GHK .

Thus, the solid angle R , contained by the angles LRM , MRN , and LRN , has been constructed out of the three plane angles LRM , MRN , and LRN , which are equal to the three given (plane angles) ABC , DEF , and GHK (respectively). (Which is) the very thing it was required to do.



Λήμμα.

Lemma

Ὅν δὲ τρόπον, ὅῃ μείζον ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX, ἐκείνῳ ἴσον λαβεῖν ἐστὶ τὸ ἀπὸ τῆς XP, δείχομεν οὕτως. ἐκκείσθωσαν αἱ AB, AX εὐθεῖαι, καὶ ἔστω μείζων ἢ AB, καὶ γεγράφθω ἐπ' αὐτῆς ἡμικύκλιον τὸ ABΓ, καὶ εἷς τὸ ABΓ ἡμικύκλιον ἐνηρμόσθω τῇ AX εὐθείᾳ μὴ μείζονι οὕση τῆς AB διαμέτρον ἴση ἢ AG, καὶ ἐπεζεύθθω ἡ GB. ἐπεὶ οὖν ἐν ἡμικυκλίῳ τῷ AΓB γωνία ἐστὶν ἡ ὑπὸ AΓB, ὀρθή ἄρα ἐστὶν ἡ ὑπὸ AΓB. τὸ ἄρα ἀπὸ τῆς AB ἴσον ἐστὶ τοῖς ἀπὸ τῶν AG, GB. ὥστε τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AG μείζον ἐστὶ τῷ ἀπὸ τῆς GB. ἴση δὲ ἡ AG τῇ AX. τὸ ἄρα ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX μείζον ἐστὶ τῷ ἀπὸ τῆς GB. ἔαν οὖν τῇ BG ἴσην τὴν XP ἀπολάβωμεν, ἔσται τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX μείζον τῷ ἀπὸ τῆς XP· ὅπερ προέκειτο ποιῆσαι.

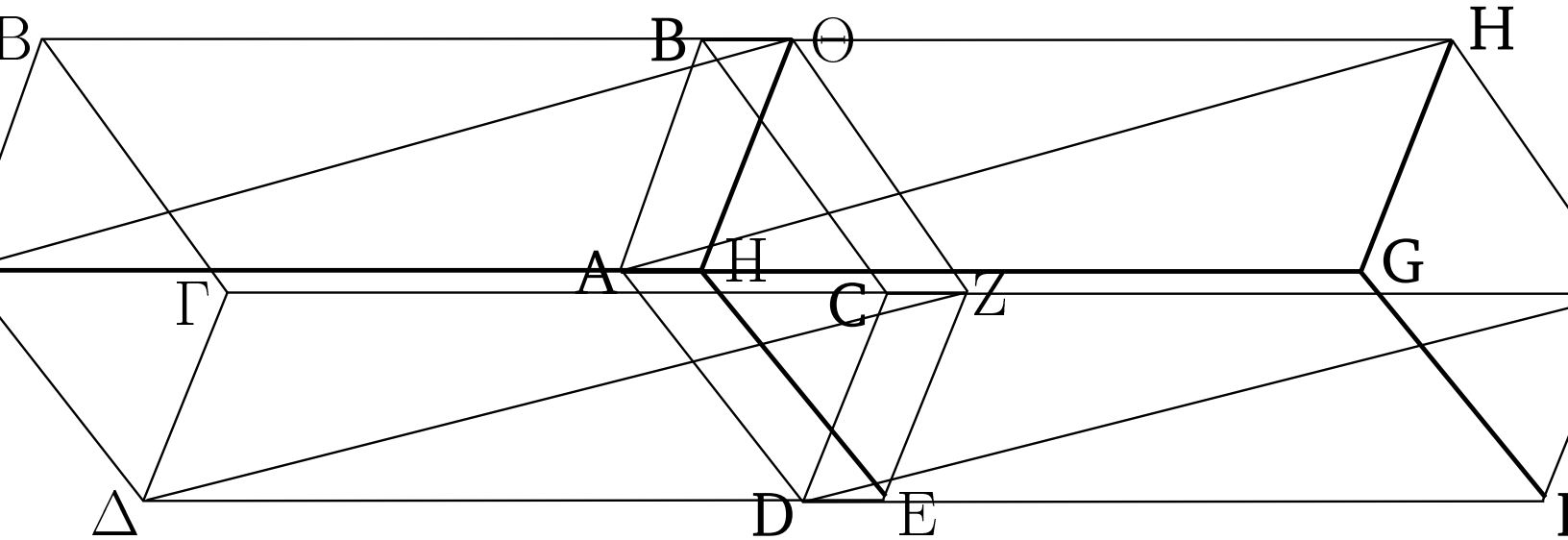
And we can demonstrate, thusly, in which manner to take the (square) on OR equal to that (area) by which the (square) on AB is greater than the (square) on LO . Let the straight-lines AB and LO be set out, and let AB be greater, and let the semicircle ABC be drawn around it. And let AC , equal to the straight-line LO , which is not greater than the diameter AB , be inserted into the semicircle ABC [Prop. 4.1]. And let CB be joined. Therefore, since the angle ACB is in the semicircle ACB , ACB is thus a right-angle [Prop. 3.31]. Thus, the (square) on AB is equal to the (sum of the) squares on AC and CB [Prop. 1.47]. Hence, the (square) on AB is greater than the (square) on AC by the (square) on CB . And AC (is) equal to LO . Thus, the (square) on AB is greater than the (square) on LO by the (square) on CB . Therefore, if we take OR equal to BC then the (square) on AB will be greater than the (square) on LO by the (square) on OR . (Which is) the very thing it was prescribed to do.

24.

Proposition 24

Ἐὰν στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθῃται, τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν.

If a solid (figure) is contained by (six) parallel planes then its opposite planes are both equal and parallelogrammic.



Στερεὸν γάρ τὸ ΓΔΘΗ ὑπὸ παραλλήλων ἐπιπέδων περι-
εθέσθω τῶν ΑΓ, ΗΖ, ΑΘ, ΔΖ, ΒΖ, ΑΕ· λέγω, ὅτι τὰ ἀπε-
ναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν.

Ἐπεὶ γὰρ δύο ἐπίπεδα παράλληλα τὰ ΒΗ, ΓΕ ὑπὸ ἐπι-
πέδον τοῦ ΑΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί
εἰσιν. παράλληλος ἄρα ἐστὶν ἡ ΑΒ τῇ ΔΓ. πάλιν, ἐπεὶ δύο
ἐπίπεδα παράλληλα τὰ ΒΖ, ΑΕ ὑπὸ ἐπιπέδον τοῦ ΑΓ τέμνε-
ται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν. παράλληλος ἄρα
ἐστὶν ἡ ΒΓ τῇ ΑΔ. ἐδείχθη δὲ καὶ ἡ ΑΒ τῇ ΔΓ παράλληλος·
παραλληλόγραμμον ἄρα ἐστὶ τὸ ΑΓ. ὁμοίως δὲ δείχομεν, ὅτι
καὶ ἕκαστον τῶν ΔΖ, ΖΗ, ΗΒ, ΒΖ, ΑΕ παραλληλόγραμμόν
ἐστίν.

Ἐπεξυῖθθωσαν αἱ ΑΘ, ΔΖ. καὶ ἐπεὶ παράλληλός ἐστιν ἡ
μὲν ΑΒ τῇ ΔΓ, ἡ δὲ ΒΘ τῇ ΓΖ, δύο δὲ αἱ ΑΒ, ΒΘ ἀπτόμεναι
ἀλλήλων παρὰ δύο εὐθείας τὰς ΔΓ, ΓΖ ἀπτομένας ἀλλήλων
εἰσὶν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ· ἴσαξ ἄρα γωνίαξ περιέχουσιν
ἴση ἄρα ἡ ὑπὸ ΑΒΘ γωνία τῇ ὑπὸ ΔΓΖ. καὶ ἐπεὶ δύο αἱ
ΑΒ, ΒΘ δυοὶ ταῖς ΔΓ, ΓΖ ἴσαι εἰσὶν, καὶ γωνία ἡ ὑπὸ ΑΒΘ
γωνία τῇ ὑπὸ ΔΓΖ ἐστὶν ἴση, βάσις ἄρα ἡ ΑΘ βάσει τῇ ΔΖ
ἐστὶν ἴση, καὶ τὸ ΑΒΘ τρίγωνον τῷ ΔΓΖ τριγόνῳ ἴσον ἐστίν.
καὶ ἐστὶ τοῦ μὲν ΑΒΘ διπλάσιον τὸ ΒΗ παραλληλόγραμμον,
τοῦ δὲ ΔΓΖ διπλάσιον τὸ ΓΕ παραλληλόγραμμον· ἴσον ἄρα
τὸ ΒΗ παραλληλόγραμμον τῷ ΓΕ παραλληλόγραμμῳ· ὁμοίως
δὲ δείχομεν, ὅτι καὶ τὸ μὲν ΑΓ τῷ ΗΖ ἐστὶν ἴσον, τὸ δὲ ΑΕ
τῷ ΒΖ.

Ἐὰν ἄρα στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθῃται,
τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά
ἐστίν· ὅπερ ἔδει δεῖξαι.

25.

Ἐὰν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμηθῇ παραλ-
λήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔσται ὡς ἡ βάσις πρὸς
τὴν βάσιν, οὕτως τὸ στερεὸν πρὸς τὸ στερεόν.

For let the solid (figure) $CDHG$ be contained by the par-
allel planes AC , GF , and AH , DF , and BF , AE . I say that its
opposite planes are both equal and parallelogrammic.

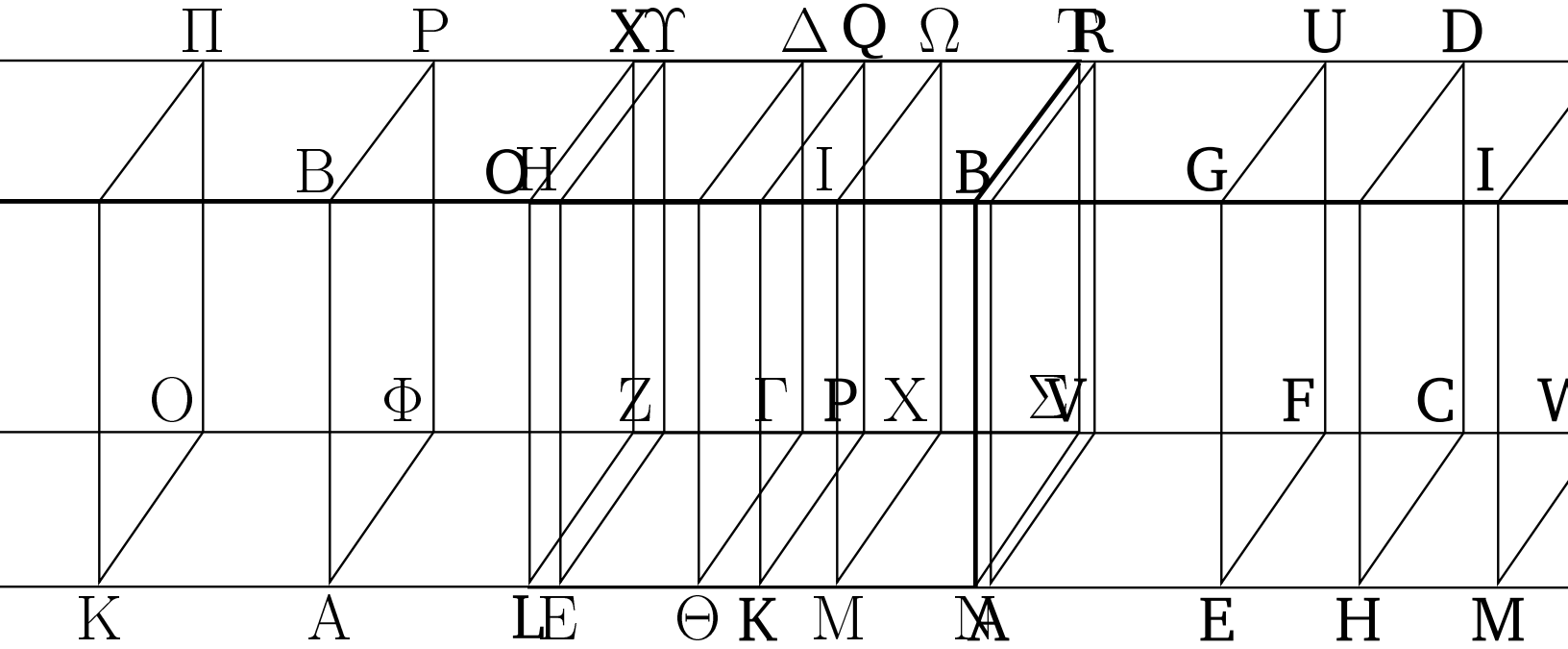
For since the two parallel planes BG and CE are cut by the
plane AC , their common sections are parallel [Prop. 11.16].
Thus, AB is parallel to DC . Again, since the two parallel
planes BF and AE are cut by the plane AC , their common
sections are parallel [Prop. 11.16]. Thus, BC is parallel to AD .
And AB was also shown (to be) parallel to DC . Thus, AC is a
parallelogram. So, similarly, we can also show that DF , FG ,
 GB , BF , and AE are each parallelograms.

Let AH and DF be joined. And since AB is parallel to DC ,
and BH to CF , so the two (straight-lines) joining one another,
 AB and BH , are parallel to the two straight-lines joining one
another, DC and CF (respectively), not (being) in the same
plane. Thus, they will contain equal angles [Prop. 11.10].
Thus, angle ABH (is) equal to (angle) DCF . And since the
two (straight-lines) AB and BH are equal to the two (straight-
lines) DC and CF (respectively) [Prop. 1.34], and angle ABH
is equal to angle DCF , the base AH is thus equal to the base
 DF , and triangle ABH is equal to triangle DCF [Prop. 1.4].
And parallelogram BG is double (triangle) ABH , and parallel-
ogram CE double (triangle) DCF [Prop. 1.34]. Thus, parallel-
ogram BG (is) equal to parallelogram CE . So, similarly, we
can show that AC is also equal to GF , and AE to BF .

Thus, if a solid (figure) is contained by (six) parallel planes
then its opposite planes are both equal and parallelogrammic.
(Which is) the very thing it was required to show.

Proposition 25

If a parallelepiped solid is cut by a plane which is parallel
to the opposite planes (of the parallelepiped) then as the base
(is) to the base, so the solid will be to the solid.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ $AB\Gamma\Delta$ ἐπιπέδῳ τῷ ZH τεμήσθω παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς PA , $\Delta\Theta$ · λέγω, ὅτι ἐστὶν ὡς ἡ $AEZ\Phi$ βάσις πρὸς τὴν $E\Theta\Gamma Z$ βάσιν, οὕτως τὸ $ABZY$ στερεὸν πρὸς τὸ $E\eta\Gamma\Delta$ στερεόν.

Ἐκβεβλήσθω γάρ ἡ $A\Theta$ ἐφ' ἐκάτερα τὰ μέρη, καὶ κείσθωσαν τῇ μὲν AE ἴσαι ὁσαιδηποῦν αἱ AK , KL , τῇ δὲ $E\Theta$ ἴσαι ὁσαιδηποῦν αἱ ΘM , MN , καὶ συμπεπληρώσθω τὰ AO , $K\Phi$, $\Theta\Theta$, $M\Sigma$ παραλληλόγραμμα καὶ τὰ $\Lambda\Pi$, KP , ΔM , MT στερεά.

Καὶ ἐπεὶ ἴσαι εἰσὶν αἱ ΛK , KA , AE εὐθεῖαι ἀλλήλαις, ἴσα ἐστὶ καὶ τὰ μὲν ΛO , $K\Phi$, AZ παραλληλόγραμμα ἀλλήλοις, τὰ δὲ KX , KB , AH ἀλλήλοις καὶ ἔτι τὰ $\Lambda\Psi$, $K\Pi$, AP ἀλλήλοις· ἀπεναντίον γάρ. διὰ τὰ αὐτὰ δὴ καὶ τὰ μὲν $E\Gamma$, $\Theta\Theta$, $M\Sigma$ παραλληλόγραμμα ἴσα εἰσὶν ἀλλήλοις, τὰ δὲ ΘH , ΘI , IN ἴσα εἰσὶν ἀλλήλοις, καὶ ἔτι τὰ $\Delta\Theta$, $M\Omega$, NT · τρία ἄρα ἐπίπεδα τῶν $\Lambda\Pi$, KP , AY στερεῶν τρισὶν ἐπιπέδοις ἐστὶν ἴσα· ἀλλὰ τὰ τρία τρισὶ τοῖς ἀπεναντίον ἐστὶν ἴσα· τὰ ἄρα τρία στερεὰ τὰ $\Lambda\Pi$, KP , AY ἴσα ἀλλήλοις ἐστὶν. διὰ τὰ αὐτὰ δὴ καὶ τὰ τρία στερεὰ τὰ $E\Delta$, ΔM , MT ἴσα ἀλλήλοις ἐστὶν· ὁσαπλασίον ἄρα ἐστὶν ἡ AZ βάσις τῇς AZ βάσεως, τοσανταπλάσιόν ἐστὶ καὶ τὸ AY στερεὸν τοῦ AY στερεοῦ. διὰ τὰ αὐτὰ δὴ ὁσαπλασίον ἐστὶν ἡ NZ βάσις τῇς $Z\Theta$ βάσεως, τοσανταπλάσιόν ἐστὶ καὶ τὸ NY στερεὸν τοῦ ΘY στερεοῦ. καὶ εἰ ἴση ἐστὶν ἡ AZ βάσις τῇ NZ βάσει, ἴσον ἐστὶ καὶ τὸ AY στερεὸν τῷ NY στερεῷ, καὶ εἰ ὑπερέθει ἡ AZ βάσις τῇς NZ βάσεως, ὑπερέθει καὶ τὸ AY στερεὸν τοῦ NY στερεοῦ, καὶ εἰ ἐλλείπει, ἐλλείπει. τεσσάρων δὴ ὄντων μεγεθῶν, δύο μὲν βάσεων τῶν AZ , $Z\Theta$, δύο δὲ στερεῶν τῶν AY , $Y\Theta$, εἴληπται ἰσάκως πολλαπλάσια

For let the parallelepiped solid $ABCD$ be cut by the plane FG which is parallel to the opposite planes RA and DH . I say that as the base $AEFV$ (is) to the base $EHCF$, so the solid $ABFU$ (is) to the solid $EGCD$.

For let AH be produced in each direction. And let any number whatsoever (of lengths), AK and KL , be made equal to AE , and any number whatsoever (of lengths), HM and MN , equal to EH . And let the parallelograms LP , KV , HW , and MS be completed, and the solids LQ , KR , DM , and MT .

And since the straight-lines LK , KA , and AE are equal to one another, the parallelograms LP , KV , and AF are also equal to one another, and KO , KB , and AG (are equal) to one another, and, further, LX , KQ , and AR (are equal) to one another. For (they are) opposite [Prop. 11.24]. So, for the same (reasons), the parallelograms EC , HW , and MS are also equal to one another, and HG , HI , and IN are equal to one another, and, further, DH , MY , and NT (are equal to one another). Thus, three planes of (one of) the solids LQ , KR , and AU are equal to the (corresponding) three planes (of the others). But, the three planes (in one of the solids) are equal to the three opposite planes [Prop. 11.24]. Thus, the three solids LQ , KR , and AU are equal to one another [Def. 11.10]. So, for the same (reasons), the three solids ED , DM , and MT are also equal to one another. Thus, as many multiples as the base LF is of the base AF , so many multiples is the solid LU also of the solid AU . So, for the same (reasons), as many multiples as the base NF is of the base FH , so many multiples is the solid NU also of the solid HU . And if the base LF is equal to the base

τῆς μὲν AZ βάσεως καὶ τοῦ AY στερεοῦ ἢ τε AZ βάσις καὶ τὸ AY στερεόν, τῆς δὲ ΘZ βάσεως καὶ τοῦ ΘY στερεοῦ ἢ τε NZ βάσις καὶ τὸ NY στερεόν, καὶ δέδεικται, ὅτι εἰ ὑπερέθει ἢ AZ βάσις τῆς ZN βάσεως, ὑπερέθει καὶ τὸ AY στερεόν τοῦ NY [στερεοῦ], καὶ εἰ ἴση, ἴσον, καὶ εἰ ἐλλείπει, ἐλλείπει. ἔστιν ἄρα ὡς ἡ AZ βάσις πρὸς τὴν $Z\Theta$ βάσιν, οὕτως τὸ AY στερεόν πρὸς τὸ $Y\Theta$ στερεόν· ὅπερ ἔδει δεῖξαι.

NF then the solid LU is also equal to the solid NU .[†] And if the base LF exceeds the base NF then the solid LU also exceeds the solid NU . And if (LF) is less than (NF) then (LU) is (also) less than (NU) . So, there are four magnitudes, the two bases AF and FH , and the two solids AU and UH , and equal multiples be taken of the base AF and the solid AU —(namely), the base LF and the solid LU —and of the base FH and the solid HU —(namely), the base NF and the solid NU . And it has been shown that if the base LF exceeds the base FN then the solid LU also exceeds the [solid] NU , and if (LF) is equal (to FN) then (LU) is equal (to NU), and if (LF) is less than (FN) then (LU) is less than (NU) . Thus, as the base AF is to the base FH , so the solid AU (is) to the solid UH [Def. 5.5]. (Which is) the very thing it was required to show.

[†] Here, Euclid assumes that $LF \cong NF$ implies $LU \cong NU$. This is easily demonstrated.

26.

Proposition 26

Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ στερεᾷ γωνίᾳ ἴσην στερεὰν γωνίαν συστήσασθαι.

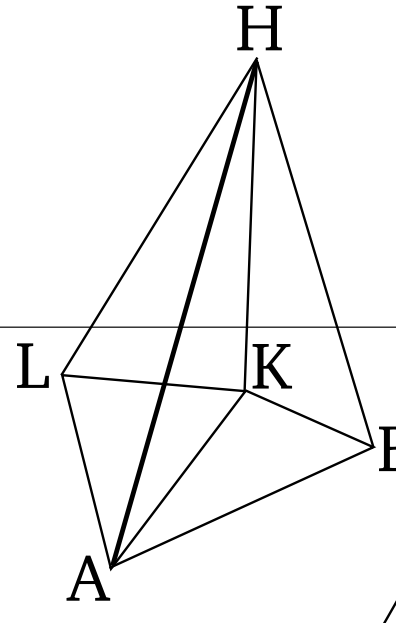
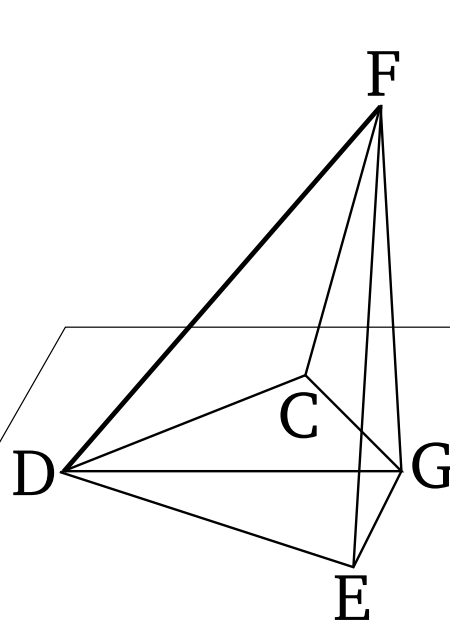
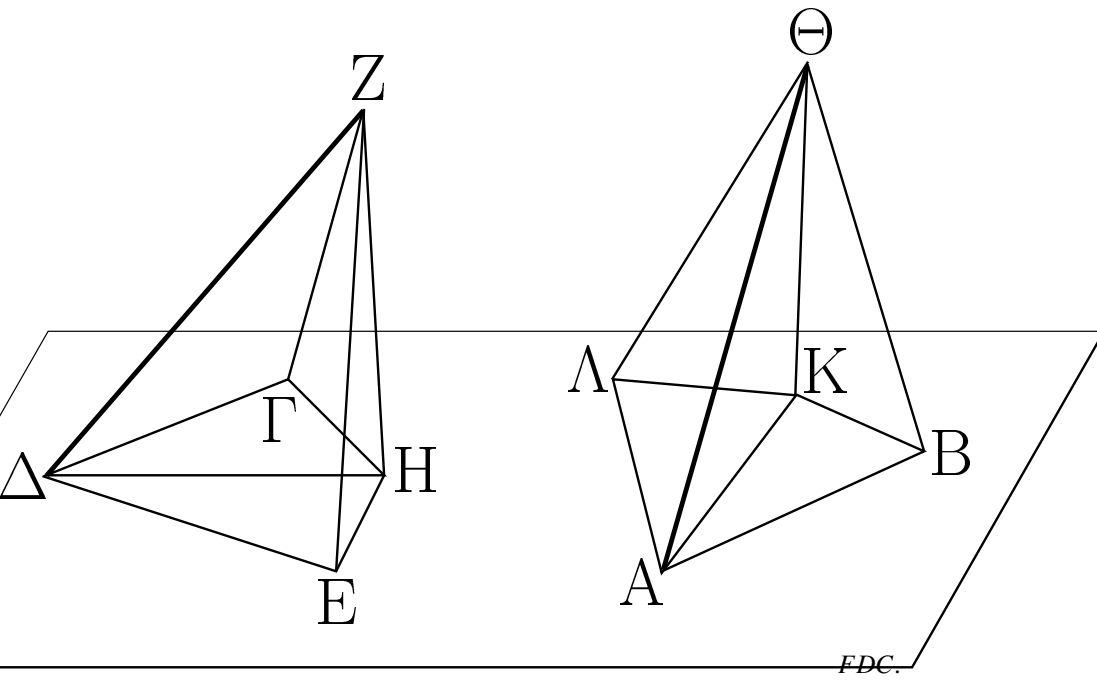
Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ πρὸς αὐτῇ δοθέν σημείον τὸ A , ἡ δὲ δοθεῖσα στερεὰ γωνία ἡ πρὸς τῷ Δ περιεθρομένη ὑπὸ τῶν ὑπὸ $E\Delta\Gamma$, $E\Delta Z$, $Z\Delta\Gamma$ γωνιῶν ἐπιπέδων· δεῖ δὲ πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ πρὸς τῷ Δ στερεᾷ γωνίᾳ ἴσην στερεὰν γωνίαν συστήσασθαι.

Εἰλήφθω γὰρ ἐπὶ τῆς ΔZ τυθὸν σημεῖον τὸ Z , καὶ ἦθθω ἀπὸ τοῦ Z ἐπὶ τὸ διὰ τῶν $E\Delta$, $\Delta\Gamma$ ἐπίπεδον κάθετος ἡ ZH , καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ H , καὶ ἐπεξέυθθω ἡ ΔH , καὶ συνεστέτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ μὲν ὑπὸ $E\Delta\Gamma$ γωνίᾳ ἴση ἡ ὑπὸ BAA , τῇ δὲ ὑπὸ $E\Delta H$ ἴση ἡ ὑπὸ BAK , καὶ κείσθω τῇ ΔH ἴση ἡ AK , καὶ ἀνεστέτω ἀπὸ τοῦ K σημείου τῷ διὰ τῶν BAA ἐπιπέδῳ πρὸς ὀρθὰς ἡ $K\Theta$, καὶ κείσθω ἴση τῇ HZ ἡ $K\Theta$, καὶ ἐπεξέυθθω ἡ ΘA . λέγω, ὅτι ἡ πρὸς τῷ A στερεὰ γωνία περιεθρομένη ὑπὸ τῶν BAA , $BA\Theta$, ΘAA γωνιῶν ἴση ἐστὶ τῇ πρὸς τῷ Δ στερεᾷ γωνίᾳ τῇ περιεθρομένη ὑπὸ τῶν $E\Delta\Gamma$, $E\Delta Z$, $Z\Delta\Gamma$ γωνιῶν.

To construct a solid angle equal to a given solid angle on a given straight-line, and at a given point on it.

Let AB be the given straight-line, and A the given point on it, and D the given solid angle, contained by the plane angles EDC , EDF , and FDC . So, it is necessary to construct a solid angle equal to the solid angle D on the straight-line AB , and at the point A on it.

For let some random point F be taken on DF , and let FG be drawn from F perpendicular to the plane through ED and DC [Prop. 11.11], and let it meet the plane at G , and let DG be joined. And let BAL , equal to the angle EDC , and BAK , equal to EDG , be constructed on the straight-line AB at the point A on it [Prop. 1.23]. And let AK be made equal to DG . And let KH be set up at the point K at right-angles to the plane through BAL [Prop. 11.12]. And let KH be made equal to GF . And let HA be joined. I say that the solid angle at A , contained by the (plane) angles BAL , BAH , and HAL , is equal to the solid angle at D , contained by the (plane) angles EDC , EDF , and



Ἀπειλήφθωσαν γὰρ ἴσαι αἱ AB , DE , καὶ ἐπεξέστησαν αἱ $ΘB$, KB , ZE , HE . καὶ ἐπεὶ ἡ ZH ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῆς ἐνθείας καὶ οὕσαξ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας· ὀρθὴ ἄρα ἐστὶν ἑκάτερα τῶν ὑπὸ $ZHΔ$, ZHE γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερα τῶν ὑπὸ $ΘKA$, $ΘKB$ γωνιῶν ὀρθὴ ἐστίν. καὶ ἐπεὶ δύο αἱ KA , AB δύο ταῖς $HΔ$, DE ἴσαι εἰσὶν ἑκάτερα ἑκάτερα, καὶ γωνίαξ ἴσαξ περιέθουσιν, βάσει ἄρα ἡ KB βάσει τῇ HE ἴση ἐστίν. ἔστι δὲ καὶ ἡ $KΘ$

For let AB and DE be cut off (so as to be) equal, and let HB , KB , FE , and GE be joined. And since FG is at right-angles to the reference plane (EDC), it will also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Thus, the angles FGD and FGE are right-angles. So, for the same (reasons), the angles HKA and HKB are also right-angles. And since the two (straight-lines) KA and AB are equal to the two (straight-lines)

τῇ HZ ἴση· καὶ γωνίαξ ὀρθὰς περιέθουσιν· ἴση ἄρα καὶ ἡ ΘB τῇ ZE . πάλιν ἐπεὶ δύο αἱ AK , $K\Theta$ δυοὶ ταῖς ΔH , HZ ἴσαι εἰσὶν, καὶ γωνίαξ ὀρθὰς περιέθουσιν, βάσις ἄρα ἡ $A\Theta$ βάσει τῇ $Z\Delta$ ἴση ἐστίν. ἔστι δὲ καὶ ἡ AB τῇ ΔE ἴση· δύο δὴ αἱ ΘA , AB δύο ταῖς ΔZ , ΔE ἴσαι εἰσὶν. καὶ βάσις ἡ ΘB βάσει τῇ ZE ἴση· γωνία ἄρα ἡ ὑπὸ $BA\Theta$ γωνία τῇ ὑπὸ $E\Delta Z$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $\Theta A\Lambda$ τῇ ὑπὸ $Z\Delta\Gamma$ ἐστὶν ἴση. ἔστι δὲ καὶ ἡ ὑπὸ BAL τῇ ὑπὸ $E\Delta\Gamma$ ἴση.

Πρὸς ἄρα τῇ δοθείσῃ εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ δοθείσῃ στερεᾷ γωνίᾳ τῇ πρὸς τῷ Δ ἴση συνέσταιται· ὅπερ ἔδει ποιῆσαι.

GD and DE , respectively, and they contain equal angles, the base KB is thus equal to the base GE [Prop. 1.4]. And KH is also equal to GF . And they contain right-angles (with the respective bases). Thus, HB (is) also equal to FE [Prop. 1.4]. Again, since the two (straight-lines) AK and KH are equal to the two (straight-lines) DG and GF (respectively), and they contain right-angles, the base AH is thus equal to the base FD [Prop. 1.4]. And AB (is) also equal to DE . So, the two (straight-lines) HA and AB are equal to the two (straight-lines) DF and DE (respectively). And the base HB (is) equal to the base FE . Thus, the angle BAH is equal to the angle EDF [Prop. 1.8]. So, for the same (reasons), HAL is also equal to FDC . And BAL is also equal to EDC .

Thus, (a solid angle) has been constructed, equal to the given solid angle at D , on the given straight-line AB , at the given point A on it. (Which is) the very thing it was required to do.

27.

Proposition 27

Ἀπὸ τῆς δοθείσης εὐθείας τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ ὁμοίον τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

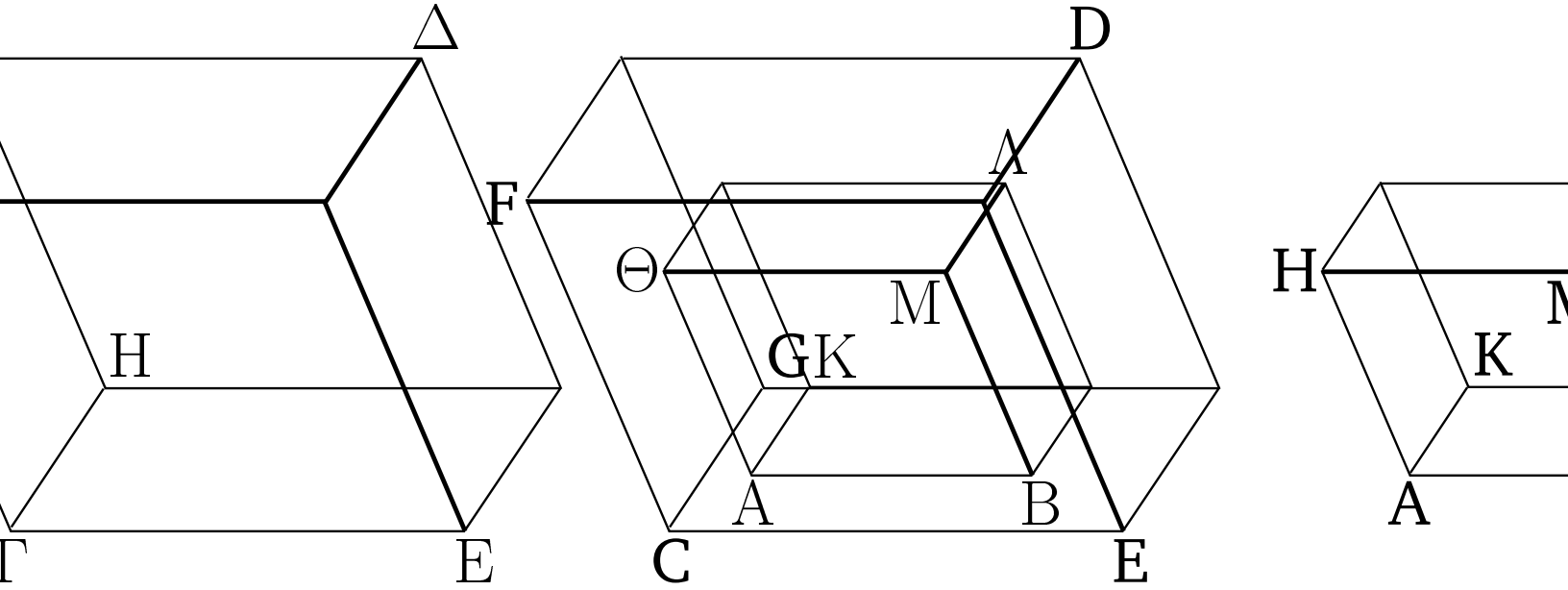
Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθέν στερεὸν παραλληλεπίπεδον τὸ $\Gamma\Delta$. δεῖ δὴ ἀπὸ τῆς δοθείσης εὐθείας τῆς AB τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ τῷ $\Gamma\Delta$ ὁμοίον τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

Συνεστάτω γάρ πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ πρὸς τῷ Γ στερεᾷ γωνίᾳ ἴση ἡ περιεθομένη ὑπὸ τῶν $BA\Theta$, ΘAK , KAB , ὥστε ἴσην εἶναι τὴν μὲν ὑπὸ $BA\Theta$ γωνίαν τῇ ὑπὸ $E\Gamma Z$, τὴν δὲ ὑπὸ BAK τῇ ὑπὸ $E\Gamma H$, τὴν δὲ ὑπὸ $KA\Theta$ τῇ ὑπὸ $H\Gamma Z$. καὶ γερονέτω ὥς μὲν ἡ $E\Gamma$ πρὸς τὴν ΓH , οὕτως ἡ BA πρὸς τὴν AK , ὥς δὲ ἡ $H\Gamma$ πρὸς τὴν ΓZ , οὕτως ἡ KA πρὸς τὴν $A\Theta$. καὶ δι' ἴσον ἄρα ἐστὶν ὥς ἡ $E\Gamma$ πρὸς τὴν ΓZ , οὕτως ἡ BA πρὸς τὴν $A\Theta$. καὶ συμπληρώσθω τὸ ΘB παραλληλόγραμμον καὶ τὸ AL στερεόν.

To describe a parallelepiped solid similar, and similarly laid out, to a given parallelepiped solid on a given straight-line.

Let the given straight-line be AB , and the given parallelepiped solid CD . So, it is necessary to describe a parallelepiped solid similar, and similarly laid out, to the given parallelepiped solid CD on the given straight-line AB .

For, let a (solid angle) contained by the (plane angles) BAH , HAK , and KAB be constructed, equal to solid angle at C , on the straight-line AB at the point A on it [Prop. 11.26], such that angle BAH is equal to ECF , and BAK to ECG , and KAH to GCF . And let it be contrived that as EC (is) to CG , so BA (is) to AK , and as GC (is) to CF , so KA (is) to AH [Prop. 6.12]. And thus, via equality, as EC is to CF , so BA (is) to AH [Prop. 5.22]. And let the parallelogram HB be completed, and the solid AL .



Καὶ ἐπεὶ ἐστὶν ὡς ἡ ΕΓ πρὸς τὴν ΓΗ, οὕτως ἡ ΒΑ πρὸς τὴν ΑΚ, καὶ περὶ ἴσαξ γωνίαξ τὰς ὑπὸ ΕΓΗ, ΒΑΚ αἱ πλεοναὶ ἀνάλογόν εἰσιν, ὁμοίων ἄρα ἐστὶ τὸ ΗΕ παραλληλόγραμμον τῷ ΚΒ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΚΘ παραλληλόγραμμον τῷ ΗΖ παραλληλογράμμῳ ὁμοίων ἐστὶ καὶ ἐτι τὸ ΖΕ τῷ ΘΒ· τρία ἄρα παραλληλόγραμμα τοῦ ΓΔ στερεοῦ τρισὶ παραλληλογράμμοις τοῦ ΑΛ στερεοῦ ὁμοία ἐστὶν. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὁμοία, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὁμοία· ὅλον ἄρα τὸ ΓΔ στερεὸν ὅλῳ τῷ ΑΛ στερεῷ ὁμοίων ἐστὶν.

Ἀπὸ τῆς δοθείσης ἄρα εὐθείας τῆς ΑΒ τῷ δοθέντι στερεῷ παραλληλεπίπεδῳ τῷ ΓΔ ὁμοίων τε καὶ ὁμοίωξ κείμενον ἀναγέγραπται τὸ ΑΛ· ὅπερ ἔδει ποιῆσαι.

28.

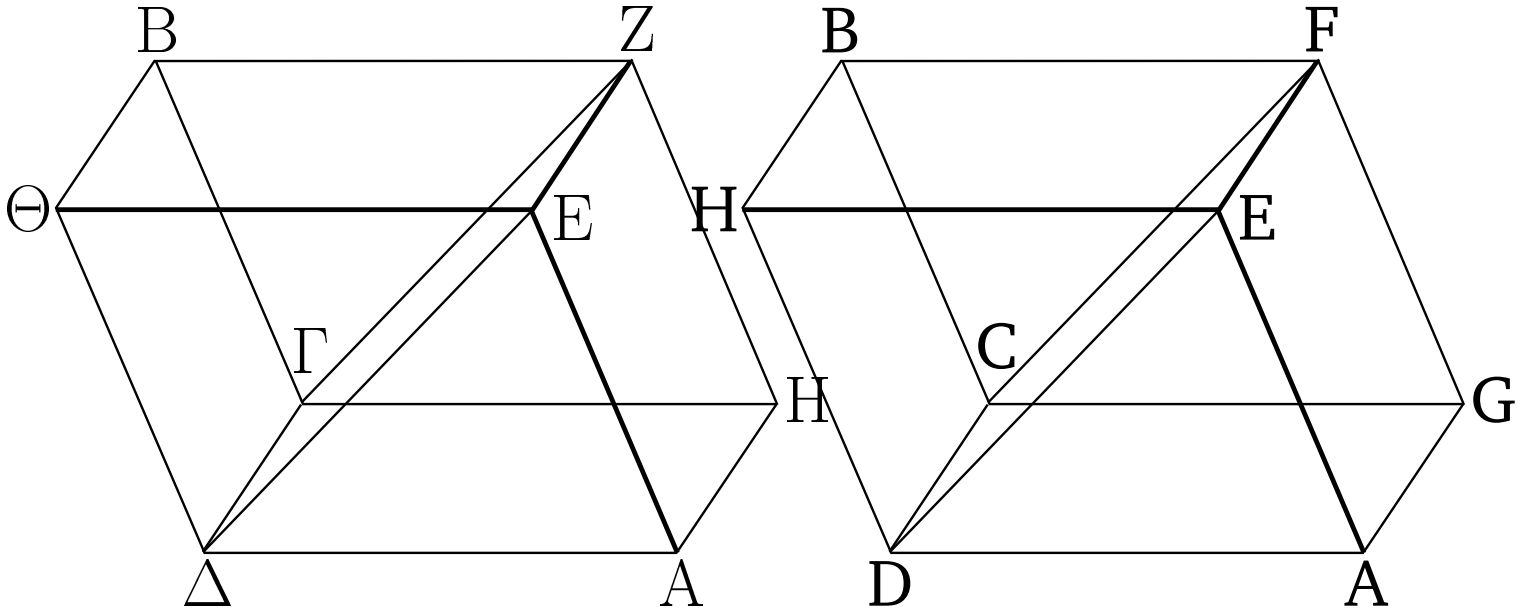
Ἐάν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμηθῇ κατὰ τὰς διαγωνίονξ τῶν ἀπεναντίον ἐπιπέδων, δίθῃ τμηθήσεται τὸ στερεὸν ὑπὸ τοῦ ἐπιπέδου.

And since as EC is to CG , so BA (is) to AK , and the sides about the equal angles ECG and BAK are (thus) proportional, the parallelogram GE is thus similar to the parallelogram KB . So, for the same (reasons), the parallelogram KH is also similar to the parallelogram GF , and, further, FE (is similar) to HB . Thus, three of the parallelograms of solid CD are similar to three of the parallelograms of solid AL . But, the (former) three are equal and similar to the three opposite, and the (latter) three are equal and similar to the three opposite. Thus, the whole solid CD is similar to the whole solid AL [Def. 11.9].

Thus, AL , similar, and similarly laid out, to the given parallelepiped solid CD , has been described on the given straight-lines AB . (Which is) the very thing it was required to do.

Proposition 28

If a parallelepiped solid is cut by a plane (passing) through the diagonals of (a pair of) opposite planes then the solid will be cut in half by the plane.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ AB ἐπιπέδῳ τῷ $ΓΔΕΖ$ τετμήσθω κατὰ τὰς διαγωνίους τῶν ἀπεναντίον ἐπιπέδων τὰς $ΓΖ$, $ΔΕ$ · λέγω, ὅτι διῖθα τμηθήσεται τὸ AB στερεὸν ὑπὸ τοῦ $ΓΔΕΖ$ ἐπιπέδου.

Ἐπεὶ γὰρ ἴσον ἐστὶ τὸ μὲν $ΓΗΖ$ τρίγωνον τῷ $ΓΖΒ$ τριγώνῳ, τὸ δὲ $ΑΔΕ$ τῷ $ΔΕΘ$, ἔστι δὲ καὶ τὸ μὲν $ΓΑ$ παραλληλογράμμον τῷ $ΕΒ$ ἴσον· ἀπεναντίον γάρ· τὸ δὲ $ΗΕ$ τῷ $ΓΘ$, καὶ τὸ πρίσμα ἄρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν $ΓΗΖ$, $ΑΔΕ$, τριῶν δὲ παραλληλογράμμων τῶν $ΗΕ$, $ΑΓ$, $ΓΕ$ ἴσον ἐστὶ τῷ πρίσματι τῷ περιεθρομένῳ ὑπὸ δύο μὲν τριγώνων τῶν $ΓΖΒ$, $ΔΕΘ$, τριῶν δὲ παραλληλογράμμων τῶν $ΓΘ$, $ΒΕ$, $ΓΕ$ · ὑπὸ γὰρ ἴσων ἐπιπέδων περιέθρονται τῷ τε πλήθει καὶ τῷ μεγέθει. ὥστε ὅλον τὸ AB στερεὸν διῖθα τέτμηται ὑπὸ τοῦ $ΓΔΕΖ$ ἐπιπέδου· ὅπερ ἔδει δεῖξαι.

For let the parallelepiped solid AB be cut by the plane $CDEF$ (passing) through the diagonals of the opposite planes CF and DE .[†] I say that the solid AB will be cut in half by the plane $CDEF$.

For since triangle CGF is equal to triangle CFB , and ADE (is equal) to DEH [Prop. 1.34], and parallelogram CA is also equal to EB —for (they are) opposite [Prop. 11.24]—and GE (equal) to CH , thus the prism contained by the two triangles CGF and ADE , and the three parallelograms GE , AC , and CE , is also equal to the prism contained by the two triangles CFB and DEH , and the three parallelograms CH , BE , and CE . For they are contained by planes (which are) equal in number and in magnitude [Def. 11.10].[‡] Thus, the whole of solid AB is cut in half by the plane $CDEF$. (Which is) the very thing it was required to show.

[†] Here, it is assumed that the two diagonals lie in the same plane. The proof is easily supplied.

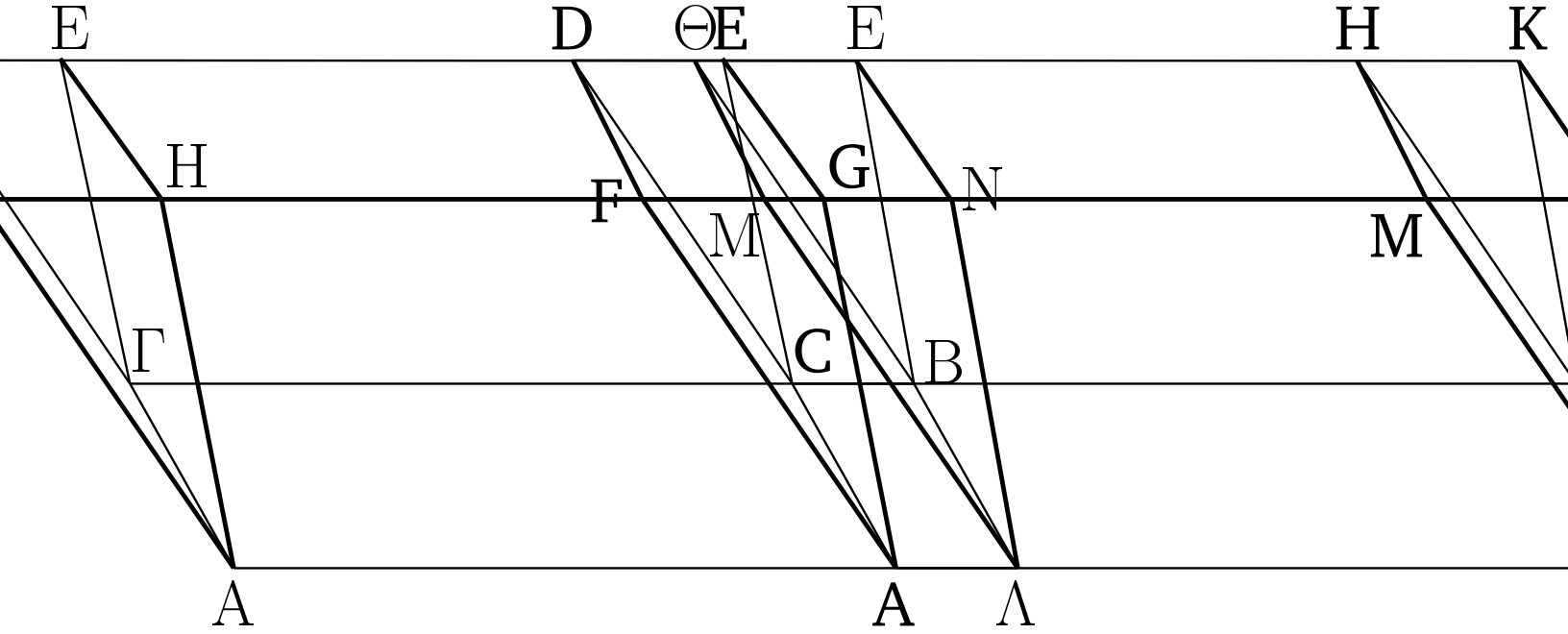
[‡] However, strictly speaking, the prisms are not similarly arranged, being mirror images of one another.

29.

Τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν.

Proposition 29

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are on the same straight-lines, are equal to one another.



Ἐστω ἐπὶ τῆς αὐτῆς βάσεως τῆς AB στερεὰ παραλληλεπίπεδα τὰ $ΓΜ$, $ΓΝ$ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AH , AZ , $ΑΜ$, $ΑΝ$, $ΓΔ$, $ΓΕ$, $BΘ$, BK ἐπὶ τῶν αὐτῶν εὐθειῶν ἕστωσαν τῶν ZN , $ΔΚ$ · λέγω, ὅτι ἴσον ἐστὶ τὸ $ΓΜ$ στερεὸν τῷ $ΓΝ$ στερεῷ.

Ἐπεὶ γὰρ παραλληλόγραμμον ἐστὶν ἑκάτερον τῶν $ΓΘ$, $ΓΚ$, ἴση ἐστὶν ἡ $ΓΒ$ ἑκατέρω τῶν $ΔΘ$, $ΕΚ$ · ὥστε καὶ ἡ $ΔΘ$ τῇ $ΕΚ$ ἐστὶν ἴση. κοινὴ ἀφηρήσθω ἡ $ΕΘ$ · λοιπὴ ἄρα ἡ $ΔΕ$ λοιπῇ τῇ $ΘΚ$ ἐστὶν ἴση. ὥστε καὶ τὸ μὲν $ΔΓΕ$ τρίγωνον τῷ $ΘΒΚ$ τριγώνῳ ἴσον ἐστίν, τὸ δὲ $ΔΗ$ παραλληλόγραμμον τῷ $ΘΝ$ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ AZH τριγώνον τῷ $ΜΑΝ$ τριγώνῳ ἴσον ἐστίν. ἐστὶ δὲ καὶ τὸ μὲν $ΓΖ$ παραλληλόγραμμον τῷ $ΒΜ$ παραλληλογράμμῳ ἴσον, τὸ δὲ $ΓΗ$ τῷ $ΒΝ$ · ἀπεναντίον γάρ· καὶ τὸ πρίσμα ἄρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν AZH , $ΔΓΕ$, τριῶν δὲ παραλληλογράμμων τῶν $ΑΔ$, $ΔΗ$, $ΓΗ$ ἴσον ἐστὶ τῷ πρίσματι τῷ περιεχομένῳ ὑπὸ δύο μὲν τριγώνων τῶν $ΜΑΝ$, $ΘΒΚ$, τριῶν δὲ παραλληλογράμμων τῶν $ΒΜ$, $ΘΝ$, $ΒΝ$. κοινὸν προσκείσθω τὸ στερεόν, οὗ βάσις μὲν τὸ AB παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $HEOM$ · ὅλον ἄρα τὸ $ΓΜ$ στερεὸν παραλληλεπίπεδον ὅλῳ τῷ $ΓΝ$ στερεῷ παραλληλεπίπεδῳ ἴσον ἐστίν.

Τὰ ἄρα ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

For let the parallelepiped solids CM and CN be on the same base AB , and (have) the same height, and let the (ends of the straight-lines) standing up in them, AG , AF , LM , LN , CD , CE , BH , and BK , be on the same straight-lines, FN and DK . I say that solid CM is equal to solid CN .

For since CH and CK are each parallelograms, CB is equal to each of DH and EK [Prop. 1.34]. Hence, DH is also equal to EK . Let EH be subtracted from both. Thus, the remainder DE is equal to the remainder HK . Hence, triangle DCE is also equal to triangle HBK [Props. 1.4, 1.8], and parallelogram DG to parallelogram HN [Prop. 1.36]. So, for the same (reasons), triangle AFG is also equal to triangle MLN . And parallelogram CF is also equal to parallelogram BM , and CG to BN [Prop. 11.24]. For they are opposite. Thus, the prism contained by the two triangles AFG and DCE , and the three parallelograms AD , DG , and CG , is equal to the prism contained by the two triangles MLN and HBK , and the three parallelograms BM , HN , and BN . Let the solid whose base (is) parallelogram AB , and (whose) opposite (face is) $GEHM$, be added to both (prisms). Thus, the whole parallelepiped solid CM is equal to the whole parallelepiped solid CN .

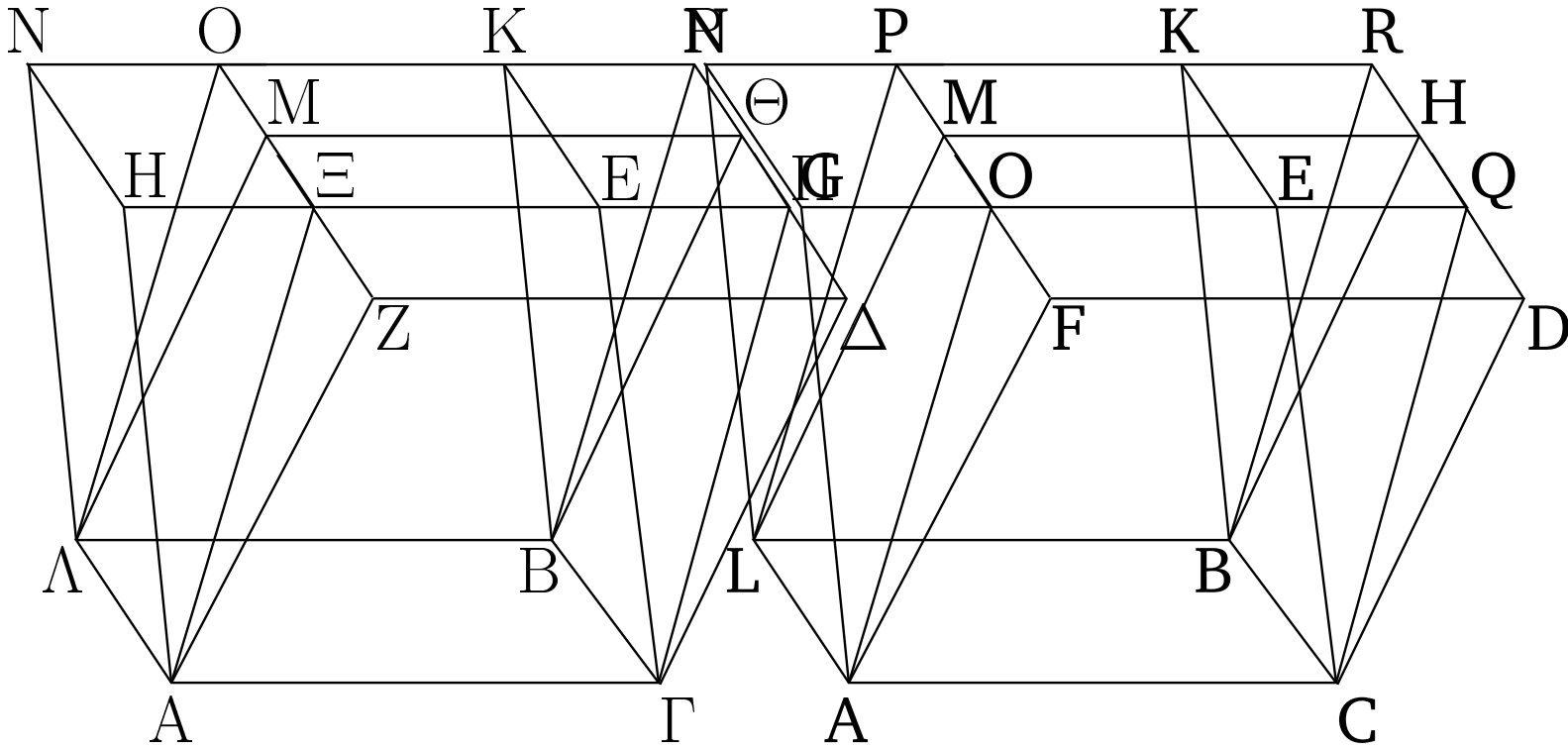
Thus, parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up (are) on the same straight-lines, are equal to one another. (Which is) the very thing it was required to show.

Τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσιν ἐπὶ τῶν

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-

αὐτῶν εὐθειῶν, ἴσα ἀλλήλοις ἔστί·

lines) standing up are not on the same straight-lines, are equal to one another.



Ἐστω ἐπὶ τῆς αὐτῆς βάσεως τῆς AB στερεὰ παραλληλεπίπεδα τὰ GM , GN ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AZ , AH , AM , AN , $ΓΔ$, $ΓΕ$, $ΒΘ$, $ΒΚ$ μὴ ἔστωσαν ἐπὶ τῶν αὐτῶν εὐθειῶν· λέγω, ὅτι ἴσον ἐστὶ τὸ GM στερεὸν τῷ GN στερεῷ.

Ἐκβεβλήσθωσαν γὰρ αἱ NK , $ΔΘ$ καὶ συμπιπτεύωσαν ἀλλήλαις κατὰ τὸ P , καὶ ἔτι ἐκβεβλήσθωσαν αἱ ZM , HE ἐπὶ τὰ O , $Π$, καὶ ἐπεξευθύνωσαν αἱ AX , AO , $ΓΠ$, BP . ἴσον δὲ ἐστὶ τὸ GM στερεόν, ὃ ὅ βασις μὲν τὸ $ΑΓΒΑ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ZΔΘΜ$, τῷ $ΓΟ$ στερεῷ, ὃ ὅ βασις μὲν τὸ $ΑΓΒΑ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ΧΠΡΟ$ · ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς $ΑΓΒΑ$ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AZ , AX , AM , AO , $ΓΔ$, $ΓΠ$, $ΒΘ$, BP ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ZO , $ΔΡ$. ἀλλὰ τὸ $ΓΟ$ στερεόν, ὃ ὅ βασις μὲν ἐστὶ τὸ $ΑΓΒΑ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ΧΠΡΟ$, ἴσον ἐστὶ τῷ GN στερεῷ, ὃ ὅ βασις μὲν τὸ $ΑΓΒΑ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $HEKN$ · ἐπὶ τε γὰρ πάλιν τῆς αὐτῆς βάσεως εἰσι τῆς $ΑΓΒΑ$ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AH , AX , $ΓΕ$, $ΓΠ$, AN , AO , $ΒΚ$, BP ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν $ΗΠ$, NP . ὥστε καὶ τὸ GM στερεὸν ἴσον ἐστὶ τῷ GN στερεῷ.

Τὰ ἄρα ἐπὶ τῆς αὐτῆς βάσεως στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσὶν ἐπὶ τῶν αὐτῶν εὐθειῶν, ἴσα ἀλλήλοις ἔστί· ὅπερ ἔδει δεῖξαι.

Let the parallelepiped solids CM and CN be on the same base, AB , and (have) the same height, and let the (ends of the straight-lines) standing up in them, AF , AG , LM , LN , CD , CE , BH , and BK , not be on the same straight-lines. I say that the solid CM is equal to the solid CN .

For let NK and DH be produced, and let them have joined one another at R . And, further, let FM and GE be produced to P and Q (respectively). And let AO , LP , CQ , and BR be joined. So, solid CM , whose base (is) parallelogram $ACBL$, and opposite (face) $FDHM$, is equal to solid CP , whose base (is) parallelogram $ACBL$, and opposite (face) $OQRP$. For they are on the same base, $ACBL$, and (have) the same height, and the (ends of the straight-lines) standing up in them, AF , AO , LM , LP , CD , CQ , BH , and BR , are on the same straight-lines, FP and DR [Prop. 11.29]. But, solid CP , whose base is parallelogram $ACBL$, and opposite (face) $OQRP$, is equal to solid CN , whose base (is) parallelogram $ACBL$, and opposite (face) $GEKN$. For, again, they are on the same base, $ACBL$, and (have) the same height, and the (ends of the straight-lines) standing up in them, AG , AO , CE , CQ , LN , LP , BK , and BR , are on the same straight-lines, GQ and NR [Prop. 11.29]. Hence, solid CM is also equal to solid CN .

Thus, parallelepiped solids (which are) on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are not on the same straight-lines,

are equal to one another. (Which is) the very thing it was required to show.

31.

Τὰ ἐπὶ ἴσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν.

Ἐστω ἐπὶ ἴσων βάσεων τῶν AB , $\Gamma\Delta$ στερεὰ παραλληλεπίπεδα τὰ AE , ΓZ ὑπὸ τὸ αὐτὸ ὕψος. λέγω, ὅτι ἴσον ἐστὶ τὸ AE στερεὸν τῷ ΓZ στερεῷ.

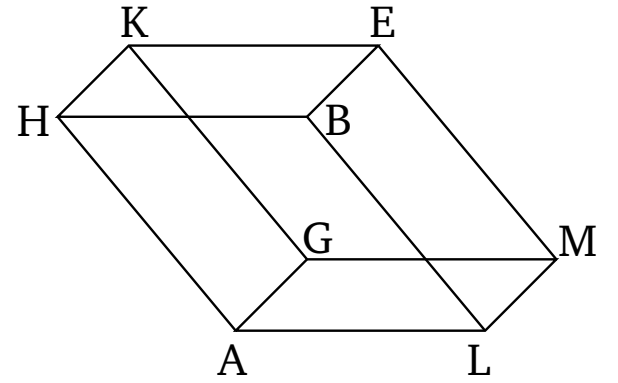
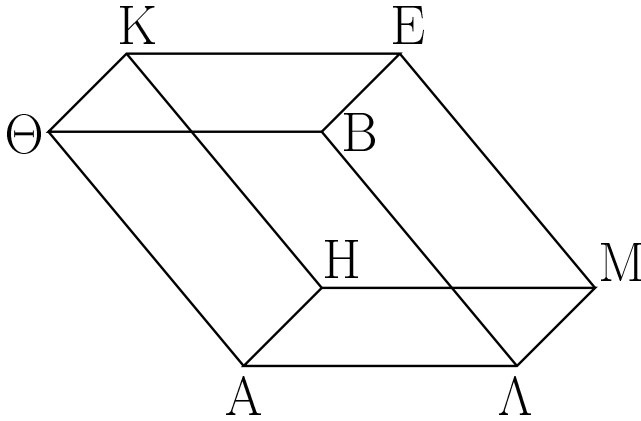
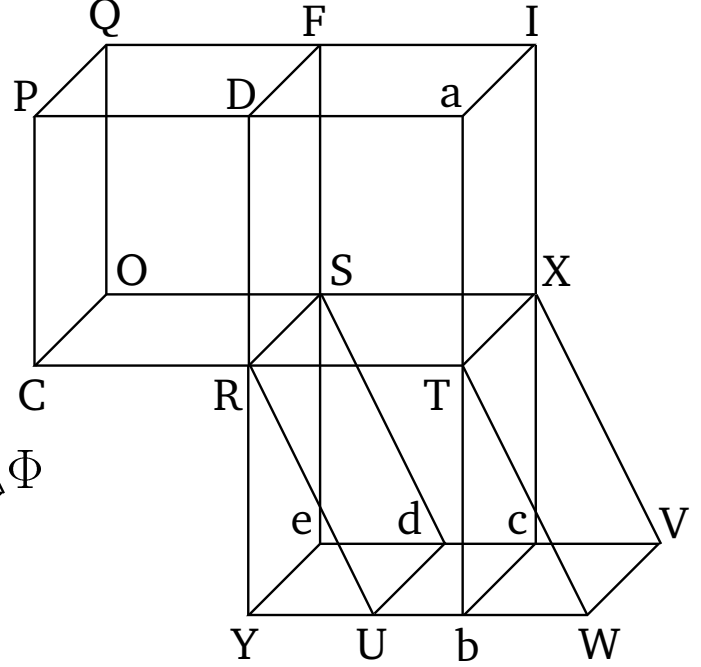
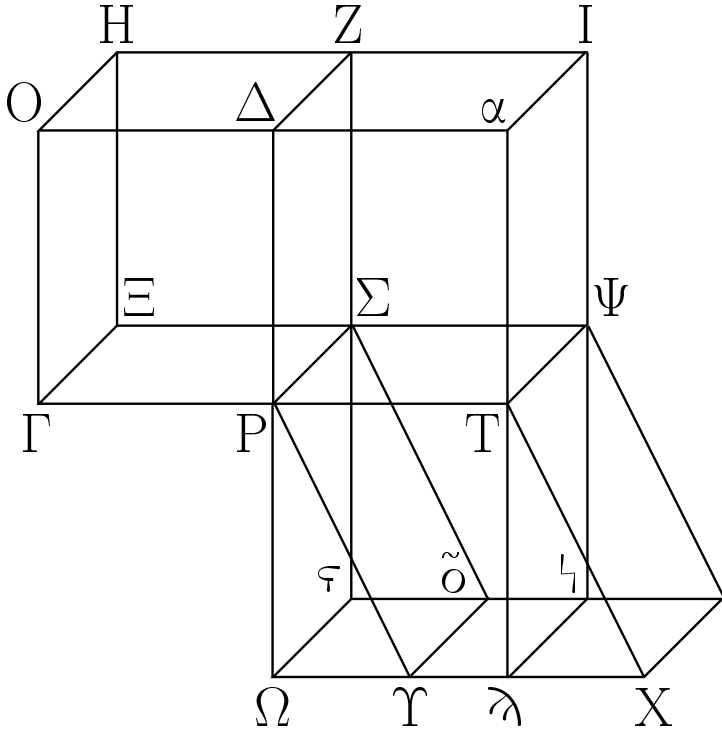
Ἐστωσαν δὴ πρότερον αἱ ἐφεστηκύναι αἱ ΘK , BE , AH , ΛM , $O\Pi$, ΔZ , ΓX , $P\Sigma$ πρὸς ὀρθὰς ταῖς AB , $\Gamma\Delta$ βάσεσιν, καὶ ἐκβεβλήσθω ἐπ' εὐθείᾳ τῇ ΓP εὐθεῖα ἡ PT , καὶ συνεστώτω πρὸς τῇ PT εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ P τῇ ὑπὸ $\Lambda\Lambda B$ γωνίᾳ ἴση ἢ ὑπὸ TPY , καὶ κείσθω τῇ μὲν $\Lambda\Lambda$ ἴση ἢ PT , τῇ δὲ ΛB ἴση ἢ PY , καὶ συμπληρώσθω ἡ τε $P\Theta$ βάσις καὶ τὸ ΨY στερεόν.

Proposition 31

Parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another.

Let the parallelepiped solids AE and CF be on the equal bases AB and CD (respectively), and (have) the same height. I say that solid AE is equal to solid CF .

So, let the (straight-lines) standing up, HK , BE , AG , LM , PQ , DF , CO , and RS , first of all, be at right-angles to the bases AB and CD . And let RT be produced in a straight-line with CR . And let (angle) TRU , equal to angle ALB , be constructed on the straight-line RT , at the point R on it [Prop. 1.23]. And let RT be made equal to AL , and RU to LB . And let the base RW , and the solid XU , be completed.

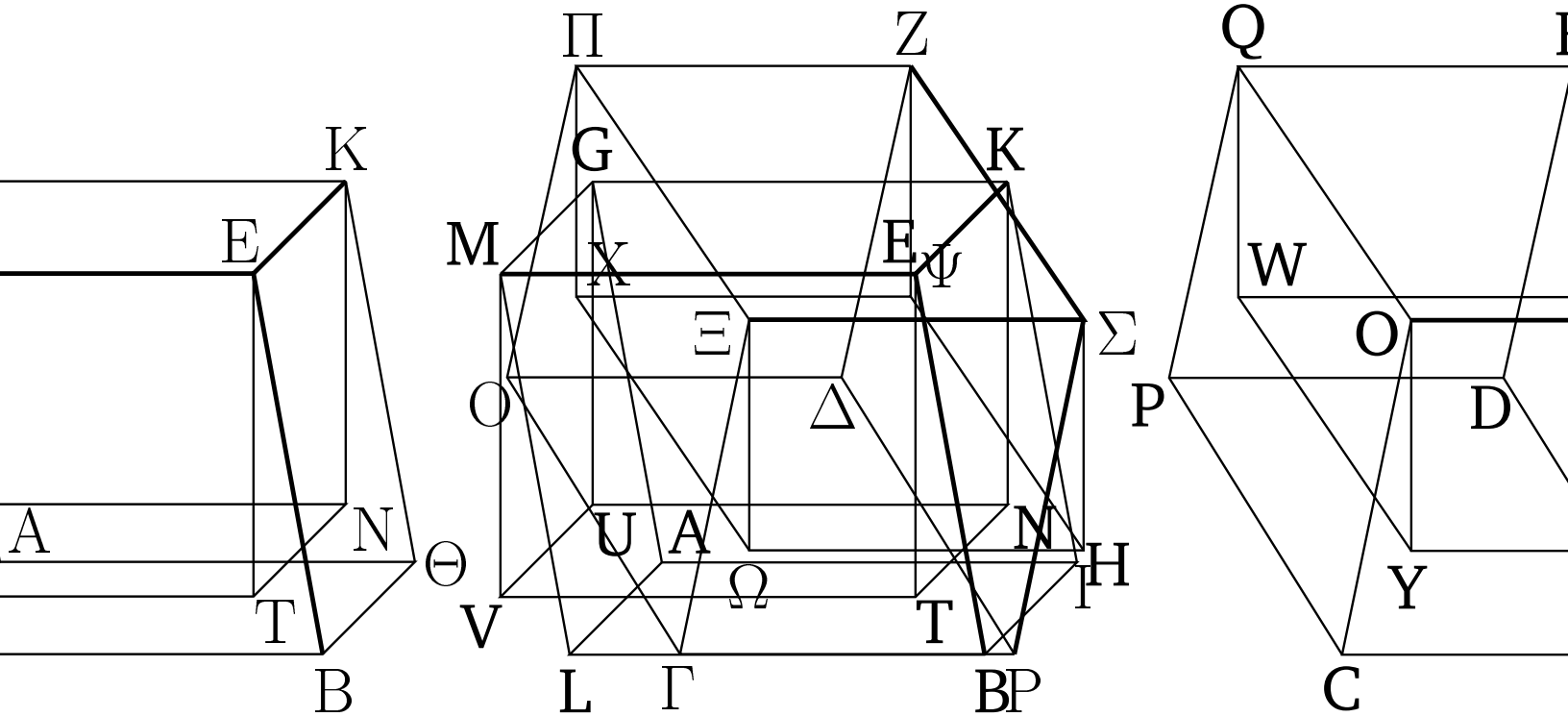


Καὶ ἐπεὶ δύο αἱ TP , PY ὀνοῖται AA , AB ἴσαι εἰσὶν, καὶ γωνίαξ ἴσαξ περιέθουσιν, ἴσον ἄρα καὶ ὅμοιον τὸ $P\Theta$ παραλληλόγραμμον τῷ $\Theta\Lambda$ παραλληλογράμμῳ. καὶ ἐπεὶ πάλιν ἴση μὲν ἡ AA τῇ PT , ἡ δὲ ΛM τῇ $P\Sigma$, καὶ γωνίαξ ὀρθάξ περιέθουσιν, ἴσον ἄρα καὶ ὅμοιον ἐστὶ τὸ $P\Psi$ παραλληλόγραμμον τῷ AM παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ AE τῷ ΣY ἴσον τέ ἐστι καὶ ὅμοιον· τρία ἄρα παραλληλόγραμμα τοῦ AE στερεοῦ τρισὶ παραλληλογράμμοιξ τοῦ ΨY στερεοῦ ἴσα τέ ἐστι καὶ ὅμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖξ ἀπεναντίον ἴσα τέ ἐστι καὶ ὅμοια, τὰ δὲ τρία τρισὶ τοῖξ ἀπεναντίον ὅλον ἄρα τὸ AE στερεὸν παραλληλεπίπεδον ὅλῳ τῷ ΨY στερεῷ παραλληλεπίπεδῳ ἴσον ἐστίν. διήθθωσαν αἱ ΔP , ΘY καὶ συμπιπτέτωσαν ἀλλήλαιξ κατὰ τὸ Ω , καὶ διὰ τοῦ T τῇ

And since the two (straight-lines) TR and RU are equal to the two (straight-lines) AL and LB (respectively), and they contain equal angles, parallelogram RW is thus equal and similar to parallelogram HL [Prop. 6.14]. And, again, since AL is equal to RT , and LM to RS , and they contain right-angles, parallelogram RX is thus equal and similar to parallelogram AM [Prop. 6.14]. So, for the same (reasons), LE is also equal and similar to SU . Thus, three parallelograms of solid AE are equal and similar to three parallelograms of solid XU . But, the three (faces of the former solid) are equal and similar to the three opposite (faces), and the three (faces of the latter solid) to the three opposite (faces) [Prop. 11.24]. Thus, the whole parallelepiped solid AE is equal to the whole parallelepiped solid

$\Delta\Omega$ παράλληλος ἦθθω ἡ aT , καὶ ἐκβεβλήσθω ἡ $O\Delta$ κατὰ τὸ a , καὶ συμπληρώσθω τὰ $\Omega\Psi$, PI στερεά. ἴσον δὲ ἔστι τὸ $\Psi\Omega$ στερεόν, οὗ βάσις μὲν ἔστι τὸ $P\Psi$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ Ω , τῷ ΨY στερεῶ, οὗ βάσις μὲν τὸ $P\Psi$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $Y\Phi$. ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς $P\Psi$ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ $P\Omega$, PY , T , $T\Theta$, Σ , $\Sigma\delta$, $\Psi\rho$, $\Psi\Phi$ ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ΩX , $\zeta\Phi$. ἀλλὰ τὸ ΨY στερεόν τῷ AE ἔστιν ἴσον· καὶ τὸ $\Psi\Omega$ ἄρα στερεόν τῷ AE στερεῶ ἔστιν ἴσον. καὶ ἐπεὶ ἴσον ἔστι τὸ $PYXT$ παραλληλόγραμμον τῷ ΩT παραλληλόγραμμῳ· ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς PT καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς PT , ΩX . ἀλλὰ τὸ $PYXT$ τῷ $\Gamma\Delta$ ἔστιν ἴσον, ἐπεὶ καὶ τῷ AB , καὶ τὸ ΩT ἄρα παραλληλόγραμμον τῷ $\Gamma\Delta$ ἔστιν ἴσον. ἄλλο δὲ τὸ ΔT · ἔστιν ἄρα ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT , οὕτως ἡ ΩT πρὸς τὴν ΔT . καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓI ἐπιπέδῳ τῷ PZ τέμνεται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔστιν ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT βᾶσιν, οὕτως τὸ ΓZ στερεὸν πρὸς τὸ PI στερεόν. διὰ τὰ αὐτὰ δὲ, ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΩI ἐπιπέδῳ τῷ $P\Psi$ τέμνεται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔστιν ὡς ἡ ΩT βάσις πρὸς τὴν $T\Lambda$ βᾶσιν, οὕτως τὸ $\Omega\Psi$ στερεὸν πρὸς τὸ PI . ἀλλ' ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT , οὕτως ἡ ΩT πρὸς τὴν ΔT . καὶ ὡς ἄρα τὸ ΓZ στερεὸν πρὸς τὸ PI στερεόν, οὕτως τὸ $\Omega\Psi$ στερεὸν πρὸς τὸ PI . ἐκότερον ἄρα τῶν ΓZ , $\Omega\Psi$ στερεῶν πρὸς τὸ PI τὸν αὐτὸν ἔχει λόγον· ἴσον ἄρα ἔστι τὸ ΓZ στερεὸν τῷ $\Omega\Psi$ στερεῶ. ἀλλὰ τὸ $\Omega\Psi$ τῷ AE ἐδείχθη ἴσον· καὶ τὸ AE ἄρα τῷ ΓZ ἔστιν ἴσον.

XU [Def. 11.10]. Let DR and WU be drawn across, and let them have met one another at Y . And let aTb be drawn through T parallel to DY . And let PD be produced to a . And let the solids YX and RI be completed. So, solid XY , whose base is parallelogram RX , and opposite (face) Yc , is equal to solid XU , whose base (is) parallelogram RX , and opposite (face) UV . For they are on the same base RX , and (have) the same height, and the (ends of the straight-lines) standing up in them, RY , RU , Tb , TW , Se , Sc , Xc and XV , are on the same straight-lines, YW and eV [Prop. 11.29]. But, solid XU is equal to AE . Thus, solid XY is also equal to solid AE . And since parallelogram $RUWT$ is equal to parallelogram YT . For they are on the same base RT , and between the same parallels RT and YW [Prop. 1.35]. But, $RUWT$ is equal to CD , since (it is) also (equal) to AB . Parallelogram YT is thus also equal to CD . And DT is another (parallelogram). Thus, as base CD is to DT , so YT (is) to DT [Prop. 5.7]. And since the parallelepiped solid CI has been cut by the plane RF , which is parallel to the opposite planes (of CI), as base CD is to base DT , so solid CF (is) to solid RI [Prop. 11.25]. So, for the same (reasons), since the parallelepiped solid YI has been cut by the plane RX , which is parallel to the opposite planes (of YI), as base YT is to base TD , so solid YX (is) to solid RI [Prop. 11.25]. But, as base CD (is) to DT , so YT (is) to DT . And, thus, as solid CF (is) to solid RI , so solid YX (is) to solid RI . Thus, solids CF and YX each have the same ratio to RI [Prop. 5.11]. Thus, solid CF is equal to solid YX [Prop. 5.9]. But, YX was show (to be) equal to AE . Thus, AE is also equal to CF .



μη' ἔστωσαν δὴ αἱ ἐφεστηκυῖαι αἱ AH , ΘK , BE , ΛM , $ΓX$, $ΟΠ$, ΔZ , $P\Sigma$ πρὸς ὀρθὰς ταῖς AB , $\Gamma\Delta$ βάσεσιν· λέγω πάλιν, ὅτι ἴσον τὸ AE στερεὸν τῷ ΓZ στερεῷ. ἤθθωσαν γὰρ ἀπὸ τῶν K , E , H , M , Π , Z , X , Σ σημείων ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετοι αἱ KN , ET , HY , $M\Phi$, $\Pi\Theta$, $Z\Psi$, $X\Omega$, ΣI , καὶ συμβαλλέτωσαν τῷ ἐπιπέδῳ κατὰ τὰ N , T , Y , Φ , Θ , Ψ , Ω , I σημεία, καὶ ἐπεξεύθθωσαν αἱ NT , NY , $Y\Phi$, $T\Phi$, $\Theta\Psi$, $\Theta\Omega$, ΩI , $I\Psi$. ἴσον δὴ ἐστὶ τὸ $K\Phi$ στερεὸν τῷ $\Pi\Psi$ στερεῷ· ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν KM , $\Pi\Psi$ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι πρὸς ὀρθὰς εἰσι ταῖς βάσεσιν. ἀλλὰ τὸ μὲν $K\Phi$ στερεὸν τῷ AE στερεῷ ἐστὶν ἴσον, τὸ δὲ $\Pi\Psi$ τῷ ΓZ · ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσιν ἐπὶ τῶν αὐτῶν εὐθειῶν. καὶ τὸ AE ἄρα στερεὸν τῷ ΓZ στερεῷ ἐστὶν ἴσον.

Τὰ ἄρα ἐπὶ ἴσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστὶν· ὅπερ ἔδει δεῖξαι.

And so let the (straight-lines) standing up, AG , HK , BE , LM , CO , PQ , DF , and RS , not be at right-angles to the bases AB and CD . Again, I say that solid AE (is) equal to solid CF . For let KN , ET , GU , MV , QW , FX , OY , and SI be drawn from points K , E , G , M , Q , F , O , and S (respectively) perpendicular to the reference plane (i.e., the plane of the bases AB and CD), and let them have met the plane at points N , T , U , V , W , X , Y , and I (respectively). And let NT , NU , UV , TV , WX , WY , YI , and IX be joined. So solid KV is equal to solid QI . For they are on the equal bases KM and QS , and (have) the same height, and the (straight-lines) standing up in them are at right-angles to their bases (see first part of proposition). But, solid KV is equal to solid AE , and QI to CF . For they are on the same base, and (have) the same height, and the (straight-lines) standing up in them are not on the same straight-lines [Prop. 11.30]. Thus, solid AE is also equal to solid CF .

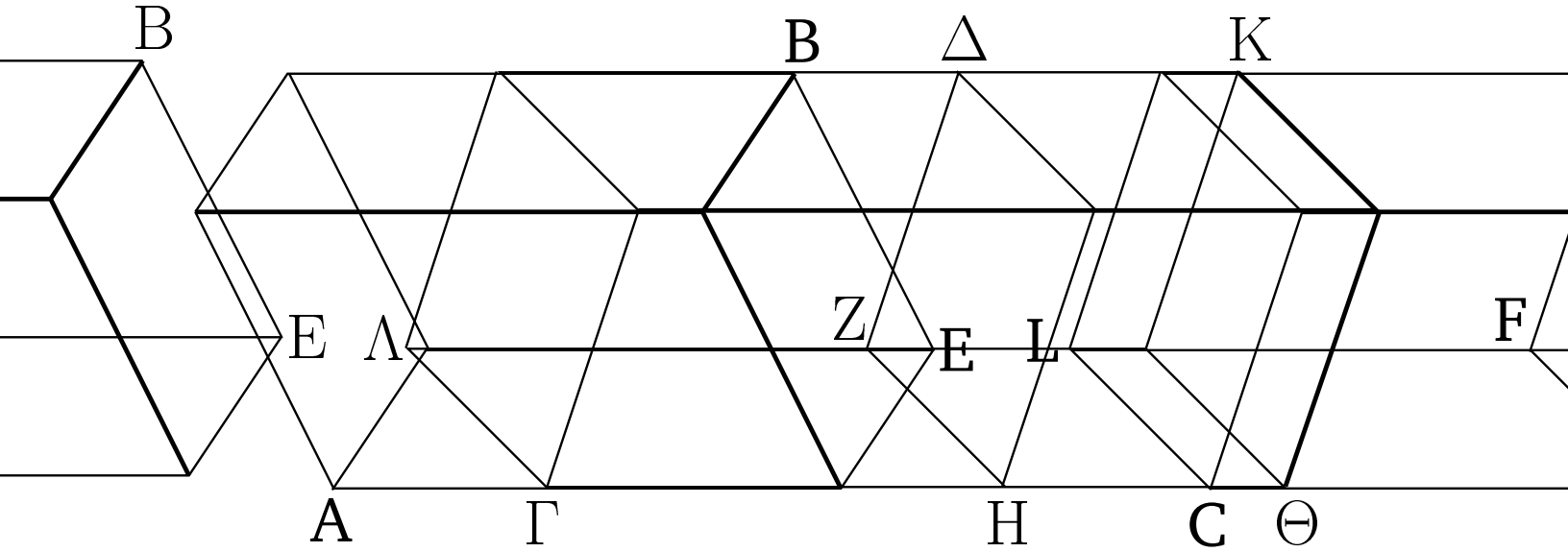
Thus, parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another. (Which is) the very thing it was required to show.

32.

Proposition 32

Τὰ ὑπὸ τὸ αὐτὸ ὕψος ὄντα στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐστὶν ὡς αἱ βάσεις.

Parallelepiped solids which (have) the same height are to one another as their bases.



Ἐστω ὑπὸ τὸ αὐτὸ ὕψος στερεὰ παραλληλεπίπεδα τὰ AB , $\Gamma\Delta$. λέγω, ὅτι τὰ AB , $\Gamma\Delta$ στερεὰ παραλληλεπίπεδα πρὸς ἄλληλά ἐστιν ὡς αἱ βάσεις, τοντέστιν ὅτι ἐστὶν ὡς ἡ AE βάσις πρὸς τὴν ΓZ βάσιν, οὕτως τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεόν.

Παραβελήσθω γὰρ παρὰ τὴν ZH τῷ AE ἴσον τὸ $Z\Theta$, καὶ ἀπὸ βάσεως μὲν τῆς $Z\Theta$, ὕψον δὲ τοῦ αὐτοῦ τῷ $\Gamma\Delta$ στερεὸν παραλληλεπίπεδον συμπληρώσθω τὸ HK . ἴσον δὴ ἐστὶ τὸ AB στερεὸν τῷ HK στερεῷ· ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν AE , $Z\Theta$ καὶ ὑπὸ τὸ αὐτὸ ὕψος. καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓK ἐπιπέδῳ τῷ ΔH τέμνεται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἐστὶν ἄρα ὡς ἡ ΓZ βάσις πρὸς τὴν $Z\Theta$ βάσιν, οὕτως τὸ $\Gamma\Delta$ στερεὸν πρὸς τὸ $\Delta\Theta$ στερεόν. ἴση δὲ ἡ μὲν $Z\Theta$ βάσις τῇ AE βάσει, τὸ δὲ HK στερεὸν τῷ AB στερεῷ· ἐστὶν ἄρα καὶ ὡς ἡ AE βάσις πρὸς τὴν ΓZ βάσιν, οὕτως τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεόν.

Τὰ ἄρα ὑπὸ τὸ αὐτὸ ὕψος ὄντα στερεὰ παραλληλεπίπεδα πρὸς ἄλληλά ἐστιν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖξαι.

33.

Τὰ ὅμοια στερεὰ παραλληλεπίπεδα πρὸς ἄλληλα ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.

Ἐστω ὅμοια στερεὰ παραλληλεπίπεδα τὰ AB , $\Gamma\Delta$, ὁμόλογος δὲ ἔστω ἡ AE τῇ ΓZ . λέγω, ὅτι τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεὸν τριπλασίονα λόγον ἔθει, ἥπερ ἡ AE πρὸς τὴν ΓZ .

Ἐκβεβλήσθωσαν γὰρ ἐπ' εὐθείας ταῖς AE , HE , ΘE αἱ EK , EL , EM , καὶ κείσθω τῇ μὲν ΓZ ἴση ἡ EK , τῇ δὲ ZN ἴση ἡ EL , καὶ ἔτι τῇ ZP ἴση ἡ EM , καὶ συμπληρώσθω τὸ KA παραλληλόγραμμον καὶ τὸ KO στερεόν.

Let AB and CD be parallelepiped solids (having) the same height. I say that the parallelepiped solids AB and CD are to one another as their bases. That is to say, as base AE is to base CF , so solid AB (is) to solid CD .

For let FH , equal to AE , be applied to FG (in the angle FGH equal to angle LCG) [Prop. 1.45]. And let the parallelepiped solid GK , (having) the same height as CD , be completed on the base FH . So solid AB is equal to solid GK . For they are on the equal bases AE and FH , and (have) the same height [Prop. 11.31]. And since the parallelepiped solid CK has been cut by the plane DG , which is parallel to the opposite planes (of CK), thus as the base CF is to the base FH , so the solid CD (is) to the solid DH [Prop. 11.25]. And base FH (is) equal to base AE , and solid GK to solid AB . And thus as base AE is to base CF , so solid AB (is) to solid CD .

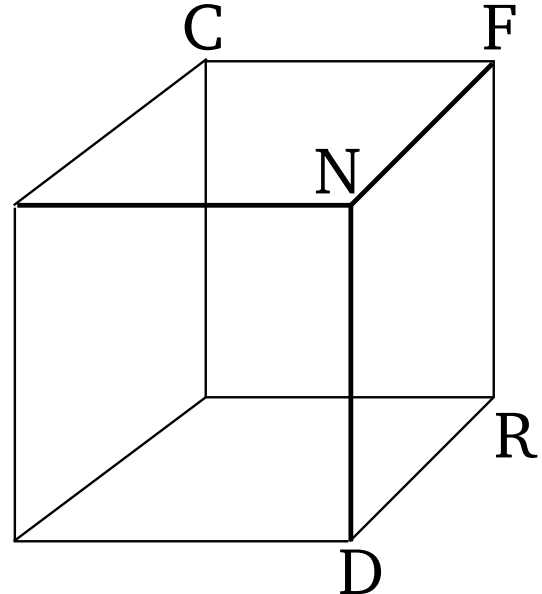
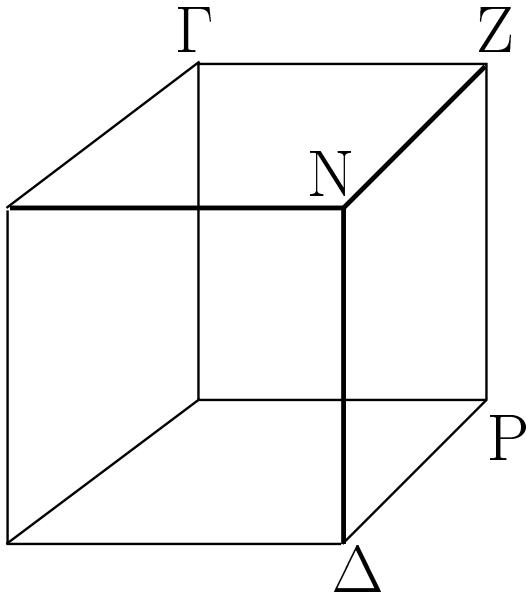
Thus, parallelepiped solids which (have) the same height are to one another as their bases. (Which is) the very thing it was required to show.

Proposition 33

Similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides.

Let AB and CD be similar parallelepiped solids, and let AE correspond to CF . I say that solid AB has to solid CD the cubed ratio that AE (has) to CF .

For let EK , EL , and EM be produced in a straight-line with AE , GE , and HE (respectively). And let EK be made equal to CF , and EL equal to FN , and, further, EM equal to FR . And let the parallelogram KL be completed, and the solid KP .

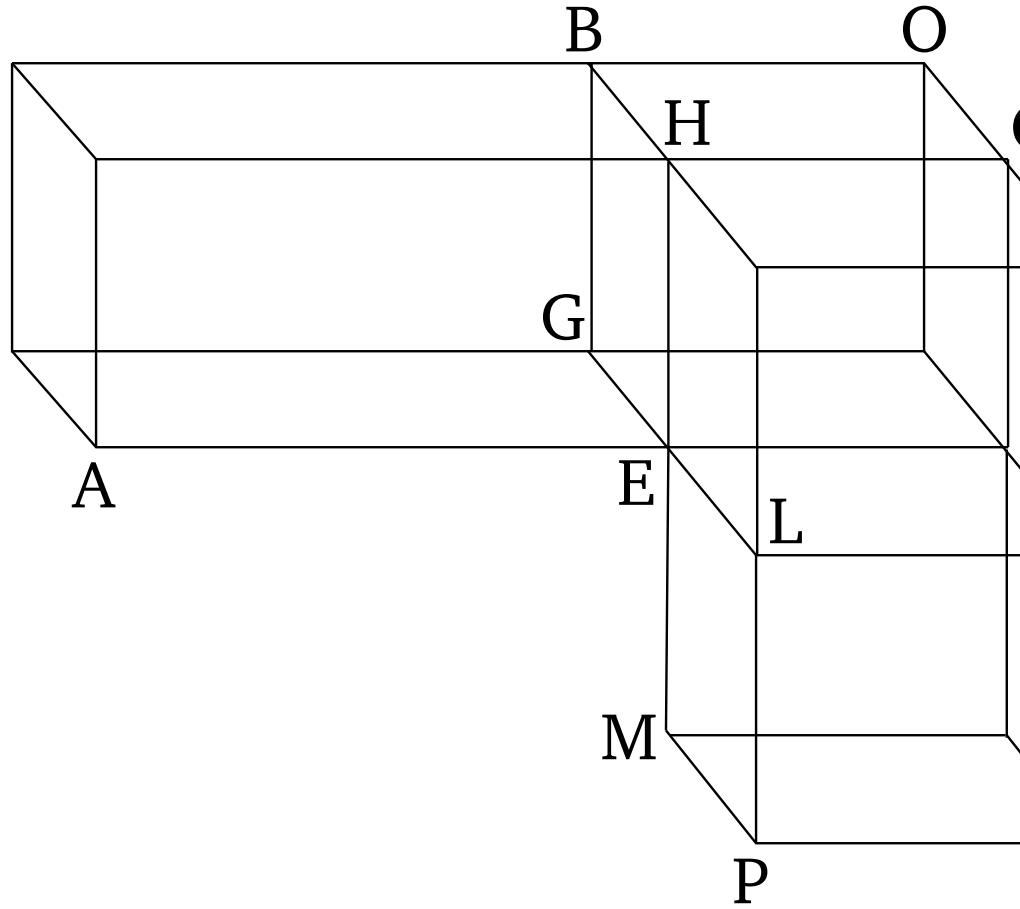


Καὶ ἐπεὶ δύο αἱ KE , EL δυοὶ ταῖς GZ , ZN ἴσαι εἰσίν, ἀλλὰ καὶ γωνία ἡ ὑπὸ KEA γωνία τῇ ὑπὸ GZN ἐστὶν ἴση, ἐπειδὴ περ καὶ ἡ ὑπὸ AEH τῇ ὑπὸ GZN ἐστὶν ἴση διὰ τὴν ὁμοιότητα τῶν AB , $ΓΔ$ στερεῶν, ἴσον ἄρα ἐστὶ [καὶ ὅμοιον] τὸ $ΚΛ$ παραλληλόγραμμον τῷ $ΓΝ$ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν KM παραλληλόγραμμον ἴσον ἐστὶ καὶ ὅμοιον τῷ $ΓΡ$ [παραλληλογράμμῳ] καὶ ἔτι τὸ EO τῷ $ΔΖ$ · τρία ἄρα παραλληλόγραμμα τοῦ KO στερεοῦ τρισὶ παραλληλογράμμοις τοῦ $ΓΔ$ στερεοῦ ἴσα ἐστὶ καὶ ὅμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα ἐστὶ καὶ ὅμοια, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ἴσα ἐστὶ καὶ ὅμοια· ὅλον ἄρα τὸ KO στερεὸν ὅλῳ τῷ $ΓΔ$ στερεῷ ἴσον ἐστὶ καὶ ὅμοιον. συμπληρώσθω τὸ HK παραλληλόγραμμον, καὶ ἀπὸ βάσεων μὲν τῶν HK , $ΚΛ$ παραλληλόγραμμων, ὕψονξ δὲ τοῦ αὐτοῦ τῷ AB στερεῷ συμπληρώσθω τὰ EX , $ΠΠ$. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν AB , $ΓΔ$ στερεῶν ἐστὶν ὡς ἡ AE πρὸς τὴν GZ , οὕτως ἡ EH πρὸς τὴν ZN , καὶ ἡ $EΘ$ πρὸς τὴν ZP , ἴση δὲ ἡ μὲν GZ τῇ EK , ἡ δὲ ZN τῇ $ΕΛ$, ἡ δὲ ZP τῇ $ΕΜ$, ἔστιν ἄρα ὡς ἡ AE πρὸς τὴν EK , οὕτως ἡ HE πρὸς τὴν $ΕΛ$ καὶ ἡ $ΘΕ$ πρὸς τὴν $ΕΜ$. ἀλλ' ὡς μὲν ἡ AE πρὸς τὴν EK , οὕτως τὸ AH [παραλληλόγραμμον] πρὸς τὸ HK παραλληλόγραμμον, ὡς δὲ ἡ HE πρὸς τὴν $ΕΛ$, οὕτως τὸ HK πρὸς τὸ $ΚΛ$, ὡς δὲ ἡ $ΘΕ$ πρὸς $ΕΜ$, οὕτως τὸ $ΠΕ$ πρὸς τὸ KM · καὶ ὡς ἄρα τὸ AH παραλληλόγραμμον πρὸς τὸ HK , οὕτως τὸ HK πρὸς τὸ $ΚΛ$ καὶ τὸ $ΠΕ$ πρὸς τὸ KM . ἀλλ' ὡς μὲν τὸ AH πρὸς τὸ HK , οὕτως τὸ AB στερεὸν πρὸς τὸ EX στερεὸν, ὡς δὲ τὸ HK πρὸς τὸ $ΚΛ$, οὕτως τὸ XE στερεὸν πρὸς τὸ $ΠΑ$ στερεὸν, ὡς δὲ τὸ $ΠΕ$ πρὸς τὸ KM , οὕτως τὸ $ΠΑ$ στερεὸν πρὸς τὸ KO στερεὸν καὶ ὡς ἄρα τὸ AB στερεὸν πρὸς τὸ EX , οὕτως τὸ EX πρὸς τὸ $ΠΑ$ καὶ τὸ $ΠΑ$ πρὸς τὸ KO . ἔαν δὲ τέσσαρα μεγέθη κατὰ τὸ συνθεθὲς ἀνάλογον ᾗ, τὸ πρῶτον πρὸς τὸ τέταρτον τριπλασίονα λόγον ἔχει ἥπερ πρὸς

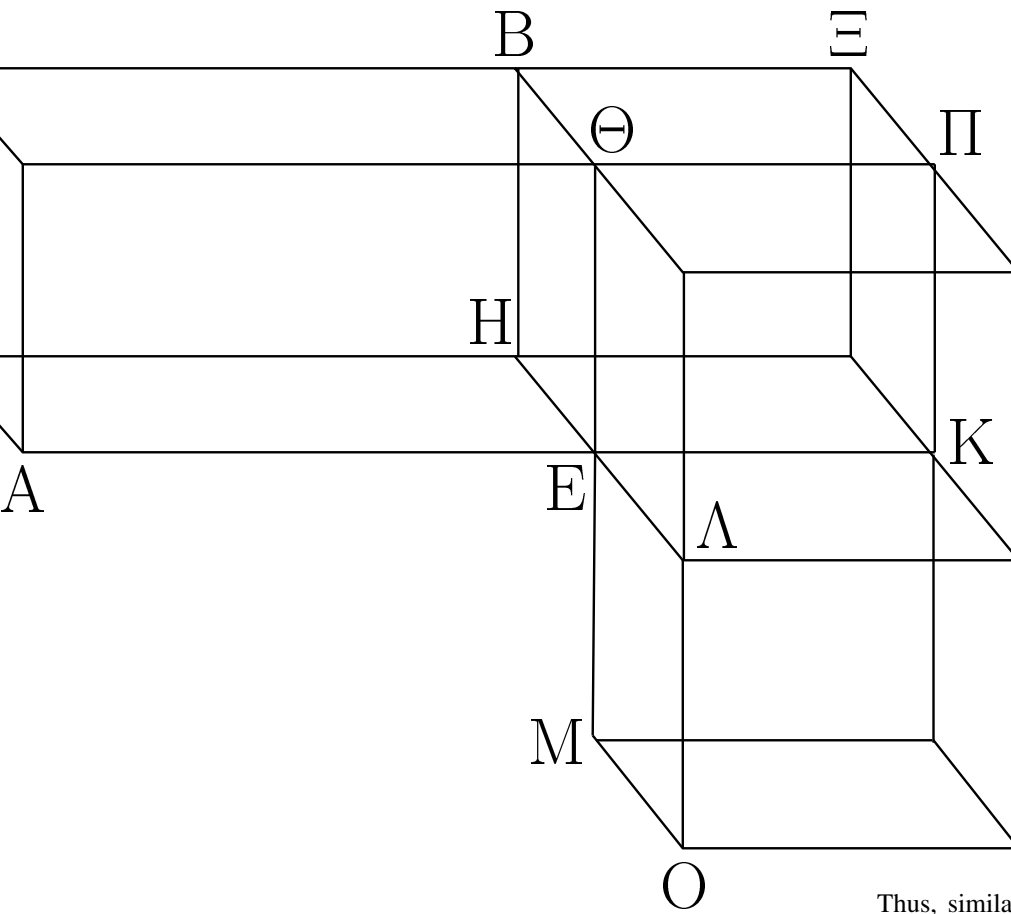
And since the two (straight-lines) KE and EL are equal to the two (straight-lines) CF and FN , but angle KEL is also equal to angle CFN , inasmuch as AEG is also equal to CFN , on account of the similarity of the solids AB and CD , parallelogram KL is thus equal [and similar] to parallelogram CN . So, for the same (reasons), parallelogram KM is also equal and similar to [parallelogram] CR , and, further, EP to DF . Thus, three parallelograms of solid KP are equal and similar to three parallelograms of solid CD . But the three (former parallelograms) are equal and similar to the three opposite (parallelograms), and the three (latter parallelograms) are equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the whole of solid KP is equal and similar to the whole of solid CD [Def. 11.10]. Let parallelogram GK be completed. And let the the solids EO and LQ , with bases the parallelograms GK and KL (respectively), and with the same height as AB , be completed. And since, on account of the similarity of solids AB and CD , as AE is to CF , so EG (is) to FN , and EH to FR [Defs. 6.1, 11.9], and CF (is) equal to EK , and FN to EL , and FR to $ΕΜ$, thus as AE is to EK , so GE (is) to EL , and HE to $ΕΜ$. But, as AE (is) to EK , so [parallelogram] AG (is) to parallelogram GK , and as GE (is) to EL , so GK (is) to KL , and as HE (is) to $ΕΜ$, so QE (is) to KM [Prop. 6.1]. And thus as parallelogram AG (is) to GK , so GK (is) to KL , and QE (is) to KM . But, as AG (is) to GK , so solid AB (is) to solid EO , and as GK (is) to KL , so solid OE (is) to solid QL , and as QE (is) to KM , so solid QL (is) to solid KP [Prop. 11.32]. And, thus, as solid AB is to EO , so EO (is) to QL , and QL to KP . And if four magnitudes are continuously proportional then the first has to the fourth the cubed ratio that (it has) to the second [Def. 5.10]. Thus, solid AB has to KP the cubed ratio which AB (has) to EO . But,

τὸ δεύτερον· τὸ AB ἄρα στερεὸν πρὸς τὸ KO τριπλασίονα λόγον ἔθει ἥπερ τὸ AB πρὸς τὸ EX . ἀλλ' ὥς τὸ AB πρὸς τὸ EX , οὕτως τὸ AH παραλληλόγραμμον πρὸς τὸ HK καὶ ἡ AE εὐθεῖα πρὸς τὴν EK · ὥστε καὶ τὸ AB στερεὸν πρὸς τὸ KO τριπλασίονα λόγον ἔθει ἥπερ ἡ AE πρὸς τὴν EK . ἴσον δὲ τὸ [μὲν] KO στερεὸν τῷ $\Gamma\Delta$ στερεῷ, ἡ δὲ EK εὐθεῖα τῇ ΓZ · καὶ τὸ AB ἄρα στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεὸν τριπλασίονα λόγον ἔθει ἥπερ ἡ ὁμόλογος αὐτοῦ πλευρὰ ἡ AE πρὸς τὴν

as AB (is) to EO , so parallelogram AG (is) to GK , and the straight-line AE to EK [Prop. 6.1]. Hence, solid AB also has to KP the cubed ratio that AE (has) to EK . And solid KP (is) equal to solid CD , and straight-line EK to CF . Thus, solid AB also has to solid CD the cubed ratio which its corresponding side AE (has) to the corresponding side CF .



ὁμόλογον πλευρὰν τὴν ΓZ .



Τὰ ἄρα ὅμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν· ὅπερ ἔδει δεῖξαι.

Thus, similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides. (Which is) the very thing it was required to show.

Πόρισμα.

Corollary

Ἐκ δὴ τούτου φανερόν, ὅτι ἂν τέσσαρες εὐθεῖαι ἀνάλογον ᾗσιν, ἔσται ὡς ἡ πρώτη πρὸς τὴν τετάρτην, οὕτω τὸ ἀπὸ τῆς πρώτης στερεὸν παραλληλεπίπεδον πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ ὅμοιον καὶ ὁμοίως ἀναγραφόμενον, ἐπεὶ καὶ ἡ πρώτη πρὸς τὴν τετάρτην τριπλασίονα λόγον ἔθει ἥπερ πρὸς τὴν δευτέραν.

So, (it is) clear, from this, that if four straight-lines are (continuously) proportional then as the first is to the fourth, so the parallelepiped solid on the first will be to the similar, and similarly described, parallelepiped solid on the second, since the first also has to the fourth the cubed ratio that (it has) to the second.

34.

Proposition 34[†]

Τῶν ἴσων στερεῶν παραλληλεπιπέδων ἀντιπεπόνθαι αἱ βάσεις τοῖς ὕψουσιν· καὶ ὧν στερεῶν παραλληλεπιπέδων ἀντιπεπόνθαι αἱ βάσεις τοῖς ὕψουσιν, ἴσα ἐστὶν ἐκεῖνα.

The bases of equal parallelepiped solids are reciprocally proportional to their heights. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal.

Ἐστω ἴσα στερεὰ παραλληλεπίπεδα τὰ AB, ΓΔ· λέγω, ὅτι τῶν AB, ΓΔ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθαι αἱ βάσεις τοῖς ὕψουσιν, καὶ ἐστὶν ὡς ἡ EΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως τὸ τοῦ ΓΔ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος.

Let AB and CD be equal parallelepiped solids. I say that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights, and (so) as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB.

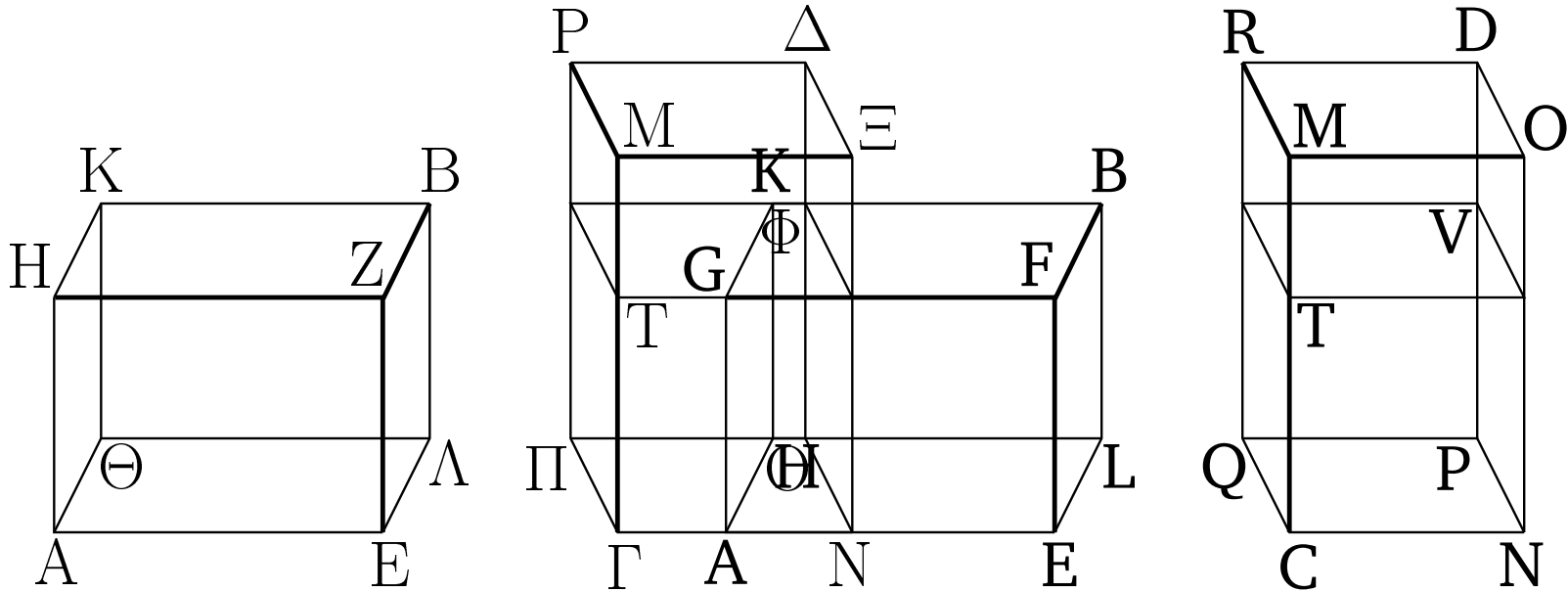
Ἐστωσαν γὰρ πρότερον αἱ ἐφεστηκυῖαι αἱ AH, EZ, AB,

ΘK , ΓM , $N X$, $O \Delta$, ΠP πρὸς ὀρθὰς ταῖς βάσεων αὐτῶν λέγω, ὅτι ἐστὶν ὡς ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν, οὕτως ἡ ΓM πρὸς τὴν $A H$.

Εἰ μὲν οὖν ἴση ἐστὶν ἡ $E \Theta$ βάσις τῇ $N \Pi$ βάσει, ἔστι δὲ καὶ τὸ AB στερεὸν τῷ $\Gamma \Delta$ στερεῷ ἴσον, ἔσται καὶ ἡ ΓM τῇ $A H$ ἴση. τὰ γὰρ ὑπὸ τὸ αὐτὸ ὕψος στερεὰ παραλληλεπίπεδα πρὸς ἄλληλά ἐστιν ὡς αἱ βάσεις. καὶ ἔσται ὡς ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$, οὕτως ἡ ΓM πρὸς τὴν $A H$, καὶ φανερόν, ὅτι τῶν AB , $\Gamma \Delta$ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

For, first of all, let the (straight-lines) standing up, AG , EF , LB , HK , CM , NO , PD , and QR , be at right-angles to their bases. I say that as base EH is to base NQ , so CM (is) to AG .

Therefore, if base EH is equal to base NQ , and solid AB is also equal to solid CD , CM will also be equal to AG . For parallelepiped solids of the same height are to one another as their bases [Prop. 11.32]. And as base EH (is) to NQ , so CM will be to AG . And (so it is) clear that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.



μὴ ἔστω δὲ ἴση ἡ $E \Theta$ βάσις τῇ $N \Pi$ βάσει, ἀλλ' ἔστω μείζων ἡ $E \Theta$. ἔστι δὲ καὶ τὸ AB στερεὸν τῷ $\Gamma \Delta$ στερεῷ ἴσον· μείζων ἄρα ἐστὶ καὶ ἡ ΓM τῇ $A H$. κείσθω οὖν τῇ $A H$ ἴση ἡ ΓT , καὶ συμπληρώσθω ἀπὸ βάσεως μὲν τῇς $N \Pi$, ὕψους δὲ τοῦ ΓT , στερεὸν παραλληλεπίπεδον τὸ $\Phi \Gamma$. καὶ ἐπεὶ ἴσον ἐστὶ τὸ AB στερεὸν τῷ $\Gamma \Delta$ στερεῷ, ἔχωνθεν δὲ τὸ $\Gamma \Phi$, τὰ δὲ ἴσα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὡς τὸ AB στερεὸν πρὸς τὸ $\Gamma \Phi$ στερεόν, οὕτως τὸ $\Gamma \Delta$ στερεὸν πρὸς τὸ $\Gamma \Phi$ στερεόν. ἀλλ' ὡς μὲν τὸ AB στερεὸν πρὸς τὸ $\Gamma \Phi$ στερεόν, οὕτως ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν· ἰσοῦψῃ γὰρ τὰ AB , $\Gamma \Phi$ στερεὰ· ὡς δὲ τὸ $\Gamma \Delta$ στερεὸν πρὸς τὸ $\Gamma \Phi$ στερεόν, οὕτως ἡ $M \Pi$ βάσις πρὸς τὴν $T \Pi$ βάσιν καὶ ἡ ΓM πρὸς τὴν ΓT . καὶ ὡς ἄρα ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν, οὕτως ἡ $M \Gamma$ πρὸς τὴν ΓT . ἴση δὲ ἡ ΓT τῇ $A H$. καὶ ὡς ἄρα ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν, οὕτως ἡ $M \Gamma$ πρὸς τὴν $A H$. τῶν AB , $\Gamma \Delta$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Πάλιν δὲ τῶν AB , $\Gamma \Delta$ στερεῶν παραλληλεπιπέδων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστω ὡς ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν, οὕτως τὸ τοῦ $\Gamma \Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος· λέγω, ὅτι ἴσον ἐστὶ τὸ AB στερεὸν τῷ $\Gamma \Delta$ στερεῷ.

So let base EH not be equal to base NQ , but let EH be greater. And solid AB is also equal to solid CD . Thus, CM is also greater than AG . Therefore, let CT be made equal to AG . And let the parallelepiped solid VC be completed on the base NQ , with height CT . And since solid AB is equal to solid CD , and CV (is) extrinsic (to them), and equal (magnitudes) have the same ratio to the same (magnitude) [Prop. 5.7], thus as solid AB is to solid CV , so solid CD (is) to solid CV . But, as solid AB (is) to solid CV , so base EH (is) to base NQ . For the solids AB and CV (are) of equal height [Prop. 11.32]. And as solid CD (is) to solid CV , so base MQ (is) to base TQ [Prop. 11.25], and CM to CT [Prop. 6.1]. And, thus, as base EH is to base NQ , so MC (is) to AG . And CT (is) equal to AG . And thus as base EH (is) to base NQ , so MC (is) to AG . Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

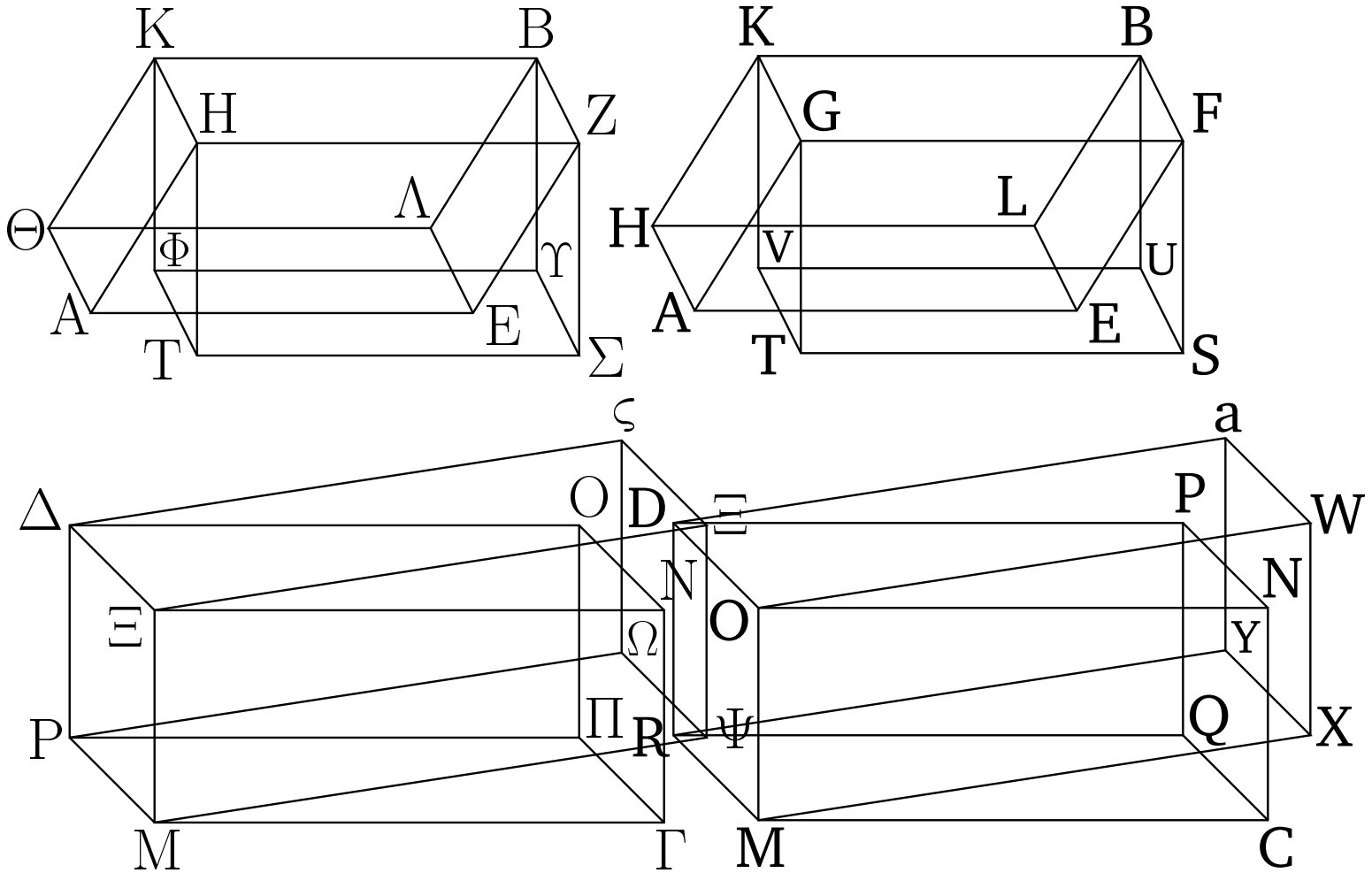
So, again, let the bases of the parallelepiped solids AB and CD be reciprocally proportional to their heights, and let base EH be to base NQ , as the height of solid CD (is) to the height of solid AB . I say that solid AB is equal to solid CD . [For] let the (straight-lines) standing up again be at right-angles to the bases. And if base EH is equal to base NQ , and as base EH is

Ἐστωσαν [γάρ] πάλιν αἱ ἐφεστηκυῖαι πρὸς ὀρθὰς ταῖς βάσεων. καὶ εἰ μὲν ἴση ἐστὶν ἡ $EΘ$ βάσις τῇ $ΝΠ$ βάσει, καὶ ἐστὶν ὡς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος, ἴσον ἄρα ἐστὶ καὶ τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος τῷ τοῦ $ΑΒ$ στερεοῦ ὕψει. τὰ δὲ ἐπὶ ἴσων βάσεων στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ $ΑΒ$ στερεὸν τῷ $ΓΔ$ στερεῷ.

μὴ ἔστω δὴ ἡ $EΘ$ βάσις τῇ $ΝΠ$ [βάσει] ἴση, ἀλλ' ἔστω μείζων ἡ $EΘ$ · μείζον ἄρα ἐστὶ καὶ τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος τοῦ τοῦ $ΑΒ$ στερεοῦ ὕψους, τοντέστιν ἡ $ΓΜ$ τῆς $ΑΗ$. κείσθω τῇ $ΑΗ$ ἴση πάλιν ἡ $ΓΤ$, καὶ συμπληρώσθω ὁμοίως τὸ $ΓΦ$ στερεόν. ἐπεὶ ἐστὶν ὡς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως ἡ $ΜΓ$ πρὸς τὴν $ΑΗ$, ἴση δὲ ἡ $ΑΗ$ τῇ $ΓΤ$, ἔστιν ἄρα ὡς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως ἡ $ΓΜ$ πρὸς τὴν $ΓΤ$. ἀλλ' ὡς μὲν ἡ $EΘ$ [βάσις] πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ $ΑΒ$ στερεὸν πρὸς τὸ $ΓΦ$ στερεόν· ἰσοῦσιν γάρ ἐστι τὰ $ΑΒ$, $ΓΦ$ στερεά· ὡς δὲ ἡ $ΓΜ$ πρὸς τὴν $ΓΤ$, οὕτως ἡ τε $ΜΠ$ βάσις πρὸς τὴν $ΠΤ$ βάσιν καὶ τὸ $ΓΔ$ στερεὸν πρὸς τὸ $ΓΦ$ στερεόν. καὶ ὡς ἄρα τὸ $ΑΒ$ στερεὸν πρὸς τὸ $ΓΦ$ στερεόν, οὕτως τὸ $ΓΔ$ στερεὸν πρὸς τὸ $ΓΦ$ στερεόν· ἐκότερον ἄρα τῶν $ΑΒ$, $ΓΔ$ πρὸς τὸ $ΓΦ$ τὸν αὐτὸν ἔθει λόγον. ἴσον ἄρα ἐστὶ τὸ $ΑΒ$ στερεὸν τῷ $ΓΔ$ στερεῷ.

to base NQ , so the height of solid CD (is) to the height of solid AB , the height of solid CD is thus also equal to the height of solid AB . And parallelepiped solids on equal bases, and also with the same height, are equal to one another [Prop. 11.31]. Thus, solid AB is equal to solid CD .

So, let base EH not be equal to [base] NQ , but let EH be greater. Thus, the height of solid CD is also greater than the height of solid AB , that is to say CM (greater) than AG . Let CT again be made equal to AG , and let the solid CV be similarly completed. Since as base EH is to base NQ , so MC (is) to AG , and AG (is) equal to CT , thus as base EH (is) to base NQ , so CM (is) to CT . But, as [base] EH (is) to base NQ , so solid AB (is) to solid CV . For solids AB and CV are of equal heights [Prop. 11.32]. And as CM (is) to CT , so (is) base MQ to base QT [Prop. 6.1], and solid CD to solid CV [Prop. 11.25]. And thus as solid AB (is) to solid CV , so solid CD (is) to solid CV . Thus, AB and CD each have the same ratio to CV . Thus, solid AB is equal to solid CD [Prop. 5.9].



μή ἔστωσαν δὴ αἱ ἐφεστηκνῦναι αἱ $ZE, BA, HA, K\Theta, XN, \Delta O, M\Gamma, P\Pi$ πρὸς ὀρθὰς ταῖς βάσεσιν αὐτῶν, καὶ ἤθθωσαν ἀπὸ τῶν $Z, H, B, K, X, M, P, \Delta$ σημείων ἐπὶ τὰ διὰ τῶν $E\Theta, N\Pi$ ἐπίπεδα κάθετοι καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ $\Sigma, T, Y, \Phi, \Theta, \Psi, \Omega, \xi$, καὶ συμπληρώσθω τὰ $Z\Phi, X\Omega$ στερεά· λέγω, ὅτι καὶ οὕτως ἴσων ὄντων τῶν $AB, \Gamma\Delta$ στερεῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστιν ὥς ἡ $E\Theta$ βάσιν πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Gamma\Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος.

Ἐπεὶ ἴσον ἐστὶ τὸ AB στερεὸν τῷ $\Gamma\Delta$ στερεῷ, ἀλλὰ τὸ μὲν AB τῷ BT ἔστιν ἴσον· ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς ZK καὶ ὑπὸ τὸ αὐτὸ ὕψος· τὸ δὲ $\Gamma\Delta$ στερεὸν τῷ $\Delta\Psi$ ἔστιν ἴσον· ἐπὶ τε γὰρ πάλιν τῆς αὐτῆς βάσεως εἰσι τῆς PX καὶ ὑπὸ τὸ αὐτὸ ὕψος· καὶ τὸ BT ἄρα στερεὸν τῷ $\Delta\Psi$ στερεῷ ἴσον ἐστίν. ἔστιν ἄρα ὥς ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. ἴση δὲ ἡ μὲν ZK βάσις τῇ $E\Theta$ βάσει, ἡ δὲ XP βάσις τῇ $N\Pi$ βάσει· ἔστιν ἄρα ὥς ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. τὰ δ' αὐτὰ ὕψη ἐστὶ τῶν $\Delta\Psi, BT$ στερεῶν καὶ τῶν $\Delta\Gamma, BA$.

So, let the (straight-lines) standing up, $FE, BL, GA, KH, ON, DP, MC$, and RQ , not be at right-angles to their bases. And let perpendiculars be drawn to the planes through EH and NQ from points F, G, B, K, O, M, R , and D , and let them have joined the planes at (points) S, T, U, V, W, X, Y , and a (respectively). And let the solids FV and OY be completed. In this case, also, I say that the solids AB and CD being equal, their bases are reciprocally proportional to their heights, and (so) as base EH is to base NQ , so the height of solid CD (is) to the height of solid AB .

Since solid AB is equal to solid CD , but AB is equal to BT . For they are on the same base FK , and (have) the same height [Props. 11.29, 11.30]. And solid CD is equal is equal to DX . For, again, they are on the same base RO , and (have) the same height [Props. 11.29, 11.30]. Solid BT is thus also equal to solid DX . Thus, as base FK (is) to base OR , so the height of solid DX (is) to the height of solid BT (see first part of proposition). And base FK (is) equal to base EH , and base OR to NQ . Thus, as base EH is to base NQ , so the height of solid DX (is) to the height of solid BT . And solids DX, BT

ἔστιν ἄρα ὥς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ τοῦ $ΔΓ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος. τῶν $ΑΒ$, $ΓΔ$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Πάλιν δὴ τῶν $ΑΒ$, $ΓΔ$ στερεῶν παραλληλεπιπέδων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστω ὥς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος· λέγω, ὅτι ἴσον ἐστὶ τὸ $ΑΒ$ στερεὸν τῷ $ΓΔ$ στερεῷ.

Τῶν γάρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἐστὶν ὥς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος, ἴση δὲ ἡ μὲν $EΘ$ βάσις τῇ ZK βάσει, ἡ δὲ $ΝΠ$ τῇ XP , ἔστιν ἄρα ὥς ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος. τὰ δ' αὐτὰ ὕψη ἐστὶ τῶν $ΑΒ$, $ΓΔ$ στερεῶν καὶ τῶν $ΒΤ$, $ΔΨ$ · ἔστιν ἄρα ὥς ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $ΔΨ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΒΤ$ στερεοῦ ὕψος. τῶν $ΒΤ$, $ΔΨ$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· ἴσον ἄρα ἐστὶ τὸ $ΒΤ$ στερεὸν τῷ $ΔΨ$ στερεῷ. ἀλλὰ τὸ μὲν $ΒΤ$ τῷ $ΒΑ$ ἴσον ἐστίν· ἐπὶ τε γάρ τῇ αὐτῇ βάσει [εἴσι] τῇ ZK καὶ ὑπὸ τὸ αὐτὸ ὕψος. τὸ δὲ $ΔΨ$ στερεὸν τῷ $ΔΓ$ στερεῷ ἴσον ἐστίν. καὶ τὸ $ΑΒ$ ἄρα στερεὸν τῷ $ΓΔ$ στερεῷ ἐστὶν ἴσον· ὅπερ εἶδει δεῖχαι.

are the same height as (solids) DC , BA (respectively). Thus, as base EH is to base NQ , so the height of solid DC (is) to the height of solid AB . Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

So, again, let the bases of the parallelepiped solids AB and CD be reciprocally proportional to their heights, and (so) let base EH be to base NQ , as the height of solid CD (is) to the height of solid AB . I say that solid AB is equal to solid CD .

For, with the same construction (as before), since as base EH is to base NQ , so the height of solid CD (is) to the height of solid AB , and base EH (is) equal to base FK , and NQ to OR , thus as base FK is to base OR , so the height of solid CD (is) to the height of solid AB . And solids AB , CD are the same height as (solids) BT , DX (respectively). Thus, as base FK is to base OR , so the height of solid DX (is) to the height of solid BT . Thus, the bases of the parallelepiped solids BT and DX are reciprocally proportional to their heights. Thus, solid BT is equal to solid DX (see first part of proposition). But, BT is equal to BA . For [they are] on the same base FK , and (have) the same height [Props. 11.29, 11.30]. And solid DX is equal to solid DC [Props. 11.29, 11.30]. Thus, solid AB is also equal to solid CD . (Which is) the very thing it was required to show.

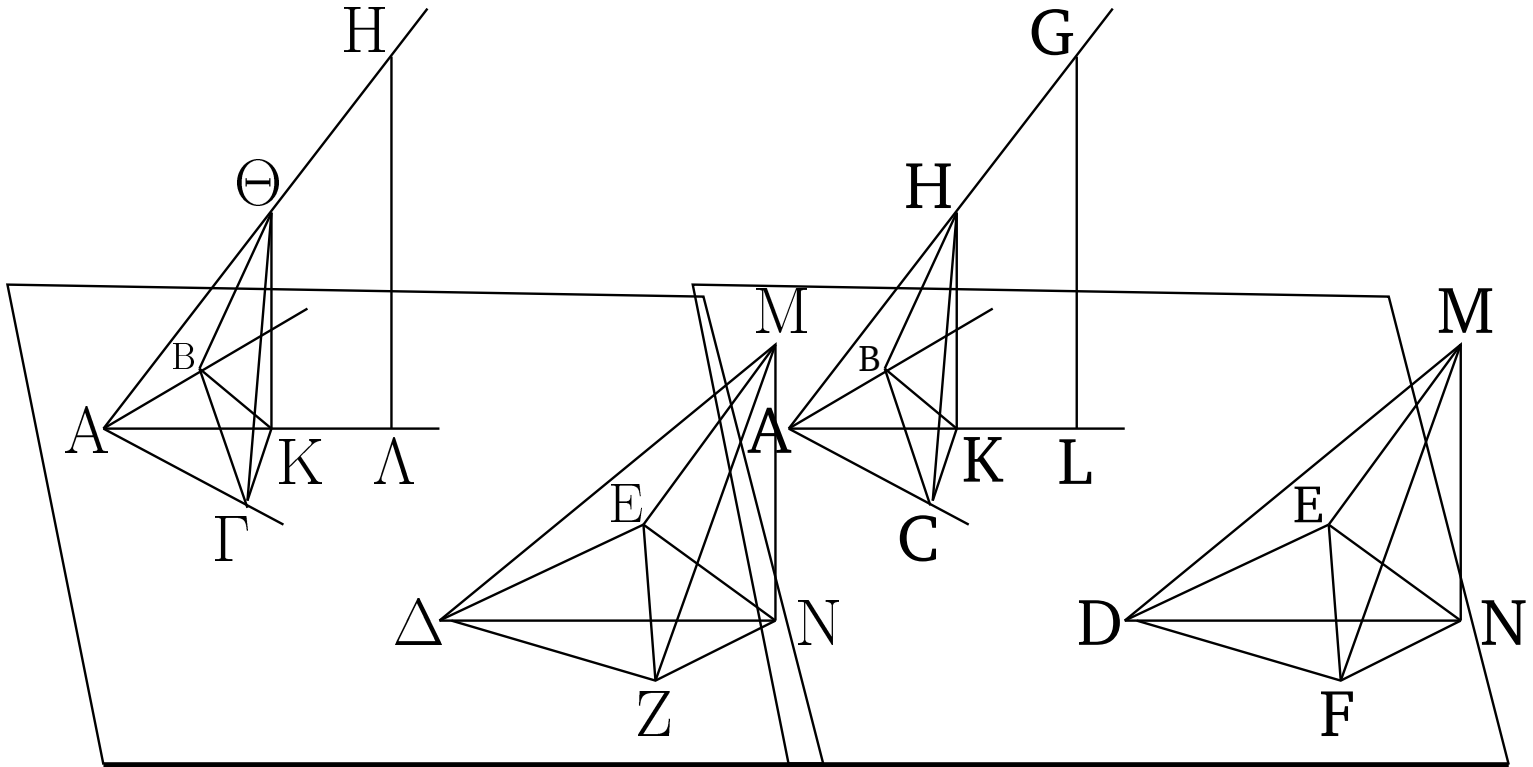
† This proposition assumes that (a) if two parallelepipeds are equal, and have equal bases, then their heights are equal, and (b) if the bases of two equal parallelepipeds are unequal, then that solid which has the lesser base has the greater height.

35.

Ἐὰν ὥσι δύο γωνίαι ἐπίπεδοι ἴσαι, ἐπὶ δὲ τῶν κορυφῶν αὐτῶν μετέωροι εὐθεῖαι ἐπισταθῶσιν ἴσαξ γωνίαξ περιέθουσαι μετὰ τῶν ἐξ ἀρθῆξ εὐθειῶν ἑκατέραν ἑκατέρῃ, ἐπὶ δὲ τῶν μετεώρων ληφθῇ τυθόντα σημεῖα, καὶ ἀπ' αὐτῶν ἐπὶ τὰ ἐπίπεδα, ἐν οἷς εἰσιν αἱ ἐξ ἀρθῆξ γωνίαι, κάθετοι ἀθθῶσιν, ἀπὸ δὲ τῶν γενομένων σημείων ἐν τοῖς ἐπιπέδοις ἐπὶ τὰς ἐξ ἀρθῆξ γωνίαξ ἐπιξενθῶσιν εὐθεῖαι, ἴσαξ γωνίαξ περιέχουσι μετὰ τῶν μετεώρων.

Proposition 35

If there are two equal plane angles, and raised straight-lines are stood on the apexes of them, containing equal angles respectively with the original straight-lines (forming the angles), and random points are taken on the raised (straight-lines), and perpendiculars are drawn from them to the planes in which the original angles are, and straight-lines are joined from the points created in the planes to the (vertices of the) original angles, then they will enclose equal angles with the raised (straight-lines).



Ἐστωσαν δύο γωνίαι εὐθύγραμμοι ἴσαι αἱ ὑπὸ BAG , $EΔZ$, ἀπὸ δὲ τῶν A , $Δ$ σημείων μετέωροι εὐθεῖαι ἐφεστώσας αἱ AH , $ΔM$ ἴσαξ γωνίας περιέθουσιν μετὰ τῶν ἐχ' ἀρθῆξ εὐθειῶν ἑκατέραν ἑκατέρῃ, τὴν μὲν ὑπὸ $MΔE$ τῇ ὑπὸ HAB , τὴν δὲ ὑπὸ $MΔZ$ τῇ ὑπὸ $HΔΓ$, καὶ εἰλήφθω ἐπὶ τῶν AH , $ΔM$ τυθόντα σημεία τὰ H , M , καὶ ἤρθωσαν ἀπὸ τῶν H , M σημείων ἐπὶ τὰ διὰ τῶν BAG , $EΔZ$ ἐπίπεδα κάθετοι αἱ HL , MN , καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ L , N , καὶ ἐπεξεύθωσαν αἱ LA , ND · λέγω, ὅτι ἴση ἐστὶν ἡ ὑπὸ HAA γωνία τῇ ὑπὸ $MΔN$ γωνίᾳ.

Κείσθω τῇ $ΔM$ ἴση ἡ $AΘ$, καὶ ἤρθω διὰ τοῦ $Θ$ σημείου τῇ HL παράλληλος ἡ $ΘK$. ἡ δὲ HL κάθετός ἐστιν ἐπὶ τὸ διὰ τῶν BAG ἐπίπεδον· καὶ ἡ $ΘK$ ἄρα κάθετός ἐστιν ἐπὶ τὸ διὰ τῶν BAG ἐπίπεδον. ἤρθωσαν ἀπὸ τῶν K , N σημείων ἐπὶ τὰς AG , $ΔZ$, AB , $ΔE$ εὐθείας κάθετοι αἱ $KΓ$, NZ , KB , NE , καὶ ἐπεξεύθωσαν αἱ $ΘΓ$, $ΓB$, MZ , ZE . ἐπεὶ τὸ ἀπὸ τῆς $ΘA$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΘK$, KA , τῷ δὲ ἀπὸ τῆς KA ἴσα ἐστὶ τὰ ἀπὸ τῶν $KΓ$, $ΓA$, καὶ τὸ ἀπὸ τῆς $ΘA$ ἄρα ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΘK$, $KΓ$, $ΓA$. τοῖς δὲ ἀπὸ τῶν $ΘK$, $KΓ$ ἴσον ἐστὶ τὸ ἀπὸ τῆς $ΘΓ$ · τὸ ἄρα ἀπὸ τῆς $ΘA$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΘΓ$, $ΓA$. ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ $ΘΓA$ γωνία. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $ΔZM$ γωνία ὀρθὴ ἐστίν. ἴση ἄρα ἐστὶν ἡ ὑπὸ $AGΘ$ γωνία τῇ ὑπὸ $ΔZM$. ἐστὶ δὲ καὶ ἡ ὑπὸ $ΘAG$ τῇ ὑπὸ $MΔZ$ ἴση. δύο δὲ τρίγωνά ἐστι τὰ $MΔZ$, $ΘAG$ δύο γωνίας δυοὶ γωνίαιξ ἴσαξ ἔχοντα ἑκατέραν ἑκατέρῃ καὶ μίαν πλευράν μιᾷ πλευρᾷ ἴσην τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν τὴν $ΘA$ τῇ $MΔ$ · καὶ τὰς λοιπάς ἄρα πλευράς ταῖς λοιπαῖς πλευραῖς

Let BAC and EDF be two equal rectilinear angles. And let the raised straight-lines AG and DM be stood on points A and D , containing equal angles respectively with the original straight-lines. (That is) MDE (equal) to GAB , and MDF (to) GAC . And let the random points G and M be taken on AG and DM (respectively). And let the GL and MN be drawn from points G and M perpendicular to the planes through BAC and EDF (respectively). And let them have joined the planes at points L and N (respectively). And let LA and ND be joined. I say that angle GAL is equal to angle MDN .

Let AH be made equal to DM . And let HK be drawn through point H parallel to GL . And GL is perpendicular to the plane through BAC . Thus, HK is also perpendicular to the plane through BAC [Prop. 11.8]. And let KC , NF , KB , and NE be drawn from points K and N perpendicular to the straight-lines AC , DF , AB , and DE . And let HC , CB , MF , and FE be joined. Since the (square) on HA is equal to the (sum of the squares) on HK and KA [Prop. 1.47], and the (sum of the squares) on KC and CA is equal to the (square) on KA [Prop. 1.47], thus the (square) on HA is equal to the (sum of the squares) on HK , KC , and CA . And the (square) on HC is equal to the (sum of the squares) on HK and KC [Prop. 1.47]. Thus, the (square) on HA is equal to the (sum of the squares) on HC and CA . Thus, angle HCA is a right-angle [Prop. 1.48]. So, for the same (reasons), angle DFM is also a right-angle. Thus, angle ACH is equal to (angle) DFM . And HAC is also equal to MDF . So, MDF and HAC are two triangles hav-

ἴσαξ ἔχει ἑκατέραν ἑκατέρω. ἴση ἄρα ἐστὶν ἡ $ΑΓ$ τῇ $ΔΖ$. ὁμοίωξ δὴ δείχομεν, ὅτι καὶ ἡ $ΑΒ$ τῇ $ΔΕ$ ἐστὶν ἴση. ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν $ΑΓ$ τῇ $ΔΖ$, ἡ δὲ $ΑΒ$ τῇ $ΔΕ$, δύο δὴ αἱ $ΓΑ$, $ΑΒ$ δυοὶ ταῖς $ΖΔ$, $ΔΕ$ ἴσαι εἰσὶν. ἀλλὰ καὶ γωνία ἡ ὑπὸ $ΓΑΒ$ γωνία τῇ ὑπὸ $ΖΔΕ$ ἐστὶν ἴση· βάσεις ἄρα ἡ $ΒΓ$ βάσει τῇ $ΕΖ$ ἴση ἐστὶ καὶ τὸ τρίγωνον τῷ τριγώνῳ καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις· ἴση ἄρα ἡ ὑπὸ $ΑΓΒ$ γωνία τῇ ὑπὸ $ΔΖΕ$. ἐστὶ δὲ καὶ ὀρθή ἡ ὑπὸ $ΑΓΚ$ ὀρθή τῇ ὑπὸ $ΔΖΝ$ ἴση· καὶ λοιπὴ ἄρα ἡ ὑπὸ $ΒΓΚ$ λοιπὴ τῇ ὑπὸ $ΕΖΝ$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $ΓΒΚ$ τῇ ὑπὸ $ΖΕΝ$ ἐστὶν ἴση. δύο δὴ τριγώνω ἐστὶ τὰ $ΒΓΚ$, $ΕΖΝ$ [τάξ] δύο γωνίαξ δυοὶ γωνίαιξ ἴσαξ ἔθοντα ἑκατέραν ἑκατέρω καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν $ΒΓ$ τῇ $ΕΖ$ · καὶ τὰς λοιπάς ἄρα πλευράς ταῖς λοιπαῖς πλευραῖς ἴσαξ ἔχουσιν. ἴση ἄρα ἐστὶν ἡ $ΓΚ$ τῇ $ΖΝ$. ἐστὶ δὲ καὶ ἡ $ΑΓ$ τῇ $ΔΖ$ ἴση· δύο δὴ αἱ $ΑΓ$, $ΓΚ$ δυοὶ ταῖς $ΔΖ$, $ΖΝ$ ἴσαι εἰσὶν καὶ ὀρθὰς γωνίαξ περιέθουσιν. βάσεις ἄρα ἡ $ΑΚ$ βάσει τῇ $ΔΝ$ ἴση ἐστὶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΑΘ$ τῇ $ΔΜ$, ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς $ΑΘ$ τῷ ἀπὸ τῆς $ΔΜ$. ἀλλὰ τῷ μὲν ἀπὸ τῆς $ΑΘ$ ἴσα ἐστὶ τὰ ἀπὸ τῶν $ΑΚ$, $ΚΘ$ · ὀρθή γάρ ἡ ὑπὸ $ΑΚΘ$ · τῷ δὲ ἀπὸ τῆς $ΔΜ$ ἴσα τὰ ἀπὸ τῶν $ΔΝ$, $ΝΜ$ · ὀρθή γάρ ἡ ὑπὸ $ΔΝΜ$ · τὰ ἄρα ἀπὸ τῶν $ΑΚ$, $ΚΘ$ ἴσα ἐστὶ τοῖς ἀπὸ τῶν $ΔΝ$, $ΝΜ$, ὧν τὸ ἀπὸ τῆς $ΑΚ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΔΝ$ · λοιπὸν ἄρα τὸ ἀπὸ τῆς $ΚΘ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΝΜ$ · ἴση ἄρα ἡ $ΘΚ$ τῇ $ΜΝ$. καὶ ἐπεὶ δύο αἱ $ΘΑ$, $ΑΚ$ δυοὶ ταῖς $ΜΔ$, $ΔΝ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρω, καὶ βάσεις ἡ $ΘΚ$ βάσει τῇ $ΜΝ$ ἐδείχθη ἴση, γωνία ἄρα ἡ ὑπὸ $ΘΑΚ$ γωνία τῇ ὑπὸ $ΜΔΝ$ ἐστὶν ἴση.

Ἐὰν ἄρα ὦσι δύο γωνίαι ἐπίπεδοι ἴσαι καὶ τὰ ἐχθῆς τῆς προτάσεως [ὅπερ ἔδει δεῖχαι].

ing two angles equal to two angles, respectively, and one side equal to one side—(namely), that subtending one of the equal angles —(that is), HA (equal) to MD . Thus, they will also have the remaining sides equal to the remaining sides, respectively [Prop. 1.26]. Thus, AC is equal to DF . So, similarly, we can show that AB is also equal to DE . Therefore, since AC is equal to DF , and AB to DE , so the two (straight-lines) CA and AB are equal to the two (straight-lines) FD and DE (respectively). But, angle CAB is also equal to angle FDE . Thus, base BC is equal to base EF , and triangle (ACB) to triangle (DFE), and the remaining angles to the remaining angles (respectively) [Prop. 1.4]. Thus, angle ACB (is) equal to DFE . And the right-angle ACK is also equal to the right-angle DFN . Thus, the remainder BCK is equal to the remainder EFN . So, for the same (reasons), CBK is also equal to FEN . So, BCK and EFN are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), that by the equal angles—(that is), BC (equal) to EF . Thus, they will also have the remaining sides equal to the remaining sides (respectively) [Prop. 1.26]. Thus, CK is equal to FN . And AC (is) also equal to DF . So, the two (straight-lines) AC and CK are equal to the two (straight-lines) DF and FN (respectively). And they enclose right-angles. Thus, base AK is equal to base DN [Prop. 1.4]. And since AH is equal to DM , the (square) on AH is also equal to the (square) on DM . But, the the (sum of the squares) on AK and KH is equal to the (square) on AH . For angle AKH (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on DN and NM (is) equal to the square on DM . For angle DNM (is) a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AK and KH is equal to the (sum of the squares) on DN and NM , of which the (square) on AK is equal to the (square) on DN . Thus, the remaining (square) on KH is equal to the (square) on NM . Thus, HK (is) equal to MN . And since the two (straight-lines) HA and AK are equal to the two (straight-lines) MD and DN , respectively, and base HK was shown (to be) equal to base MN , angle HAK is thus equal to angle MDN [Prop. 1.8].

Thus, if there are two equal plane angles, and so on of the proposition. [(Which is) the very thing it was required to show].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἔαν ὦσι δύο γωνίαι ἐπίπεδοι ἴσαι, ἐπισταθῶσι δὲ ἐπ' αὐτῶν μετέωροι εὐθεῖαι ἴσαι ἴσαξ γωνίαξ περιέθουσαι μετὰ τῶν ἐχ ἄρθῆς εὐθειῶν ἑκατέραν ἑκατέρω, αἱ ἀπ' αὐτῶν κάθετοι ἀρόμεναι ἐπὶ τὰ ἐπίπεδα, ἐν οἷς εἰσὶν αἱ ἐχ ἄρθῆς γωνίαι, ἴσαι ἀλλήλαις εἰσὶν. ὅπερ ἔδει δεῖχαι.

Corollary

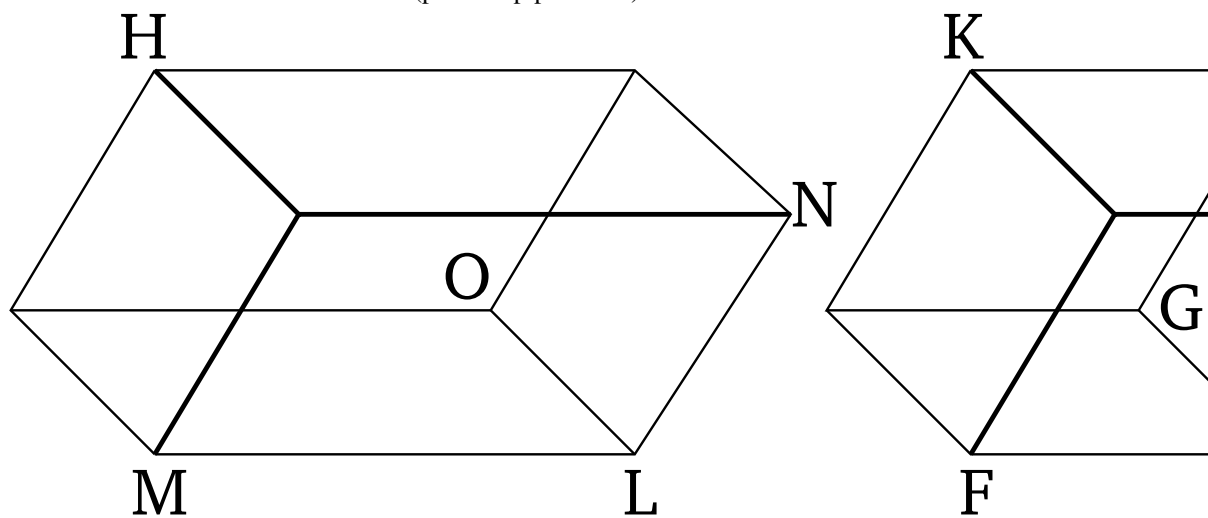
So, it is clear, from this, that if there are two equal plane angles, and equal raised straight-lines are stood on them (at their apexes), containing equal angles respectively with the original straight-lines (forming the angles), then the perpendiculars drawn from (the raised ends of) them to the planes in which the original angles lie are equal to one another. (Which is) the very thing it was required to show.

36.

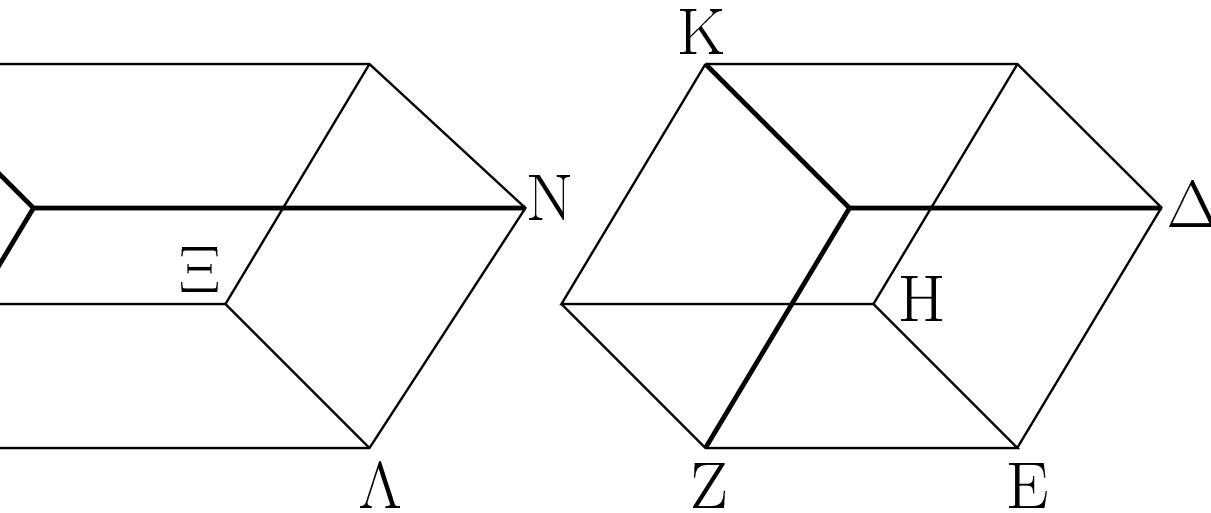
Proposition 36

Ἐάν τρεῖς εὐθεῖαι ἀνάλογον ᾗσιν, τὸ ἐκ τῶν τριῶν στερεὸν παραλληλεπίπεδον ἴσον ἐστὶ τῷ ἀπὸ τῆς μέσης στερεῷ παραλληλεπίπεδῳ ἰσοπλεύρῳ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

If three straight-lines are (continuously) proportional then the parallelepiped solid (formed) from the three (straight-lines) is equal to the equilateral parallelepiped solid on the middle (straight-line which is) equiangular to the aforementioned (parallelepiped solid).



- A _____
- B _____
- C _____



Ἐστωσαν τρεῖς εὐθεῖαι ἀνάλογον αἱ A, B, Γ , ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ . λέγω, ὅτι τὸ ἐκ τῶν A, B, Γ στερεὸν ἴσον ἐστὶ τῷ ἀπὸ τῆς B στερεῷ ἰσοπλεύρῳ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

Ἐκκείσθω στερεὰ γωνία ἡ πρὸς τῷ E περιεθομένη ὑπὸ τῶν ὑπὸ $\Delta E H, H E Z, Z E \Delta$, καὶ κείσθω τῇ μὲν B ἴση ἐκάστη τῶν $\Delta E, H E, E Z$, καὶ συμπληρώσθω τὸ $E K$ στερεὸν παραλληλεπίπεδον, τῇ δὲ A ἴση ἡ ΛM , καὶ συνεστάτω πρὸς τῇ ΛM εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Λ τῇ πρὸς τῷ E στερεᾷ γωνία ἴση στερεὰ γωνία ἡ περιεθομένη ὑπὸ τῶν $N \Lambda X, X \Lambda M, M \Lambda N$, καὶ κείσθω τῇ μὲν B ἴση ἡ ΛX , τῇ δὲ Γ ἴση ἡ ΛN . καὶ ἐπεὶ ἐστὶν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ , ἴση δὲ ἡ μὲν A τῇ ΛM , ἡ δὲ B ἐκατέρα τῶν $\Lambda X, E \Delta$, ἡ δὲ Γ τῇ ΛN , ἔστιν ἄρα ὥς ἡ ΛM πρὸς τὴν $E Z$, οὕτως ἡ ΔE πρὸς τὴν ΛN . καὶ περὶ ἴσαξ γωνίας τὰς ὑπὸ $N \Lambda M, \Delta E Z$ αἱ πλευраὶ ἀντιπεπόνθασιν· ἴσον ἄρα ἐστὶ τὸ $M N$ παραλληλόγραμμον τῷ ΔZ παραλληλογραμμίῳ. καὶ ἐπεὶ δύο γωνίαι ἐπίπεδοι εὐθύγραμμοι ἴσαι εἰσὶν αἱ ὑπὸ $\Delta E Z, N \Lambda M$, καὶ ἐπ' αὐτῶν μετέωροι εὐθεῖαι ἐφεστᾶσιν αἱ $\Lambda X, E H$ ἴσαι τε ἀλλήλαις καὶ ἴσαξ γωνίαξ περιέθουσαι μετὰ τῶν ἐχ' ἀρθῆξ εὐθειῶν ἐκατέραν ἐκατέρα, αἱ ἄρα ἀπὸ τῶν H, X σημείων κάθετοι ἀγόμεναι ἐπὶ τὰ διὰ τῶν $N \Lambda M, \Delta E Z$ ἐπίπεδα ἴσαι ἀλλήλαις εἰσὶν· ὥστε τὰ $\Lambda \Theta, E K$ στερεὰ ὑπὸ τὸ αὐτὸ ὕψος ἐστίν. τὰ δὲ ἐπὶ ἴσων βάσεων στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοισι ἐστίν· ἴσον ἄρα ἐστὶ τὸ $\Theta \Lambda$

Let A, B , and C be three (continuously) proportional straight-lines, (such that) as A (is) to B , so B (is) to C . I say that the (parallelepiped) solid (formed) from A, B , and C is equal to the equilateral solid on B (which is) equiangular with the aforementioned (solid).

Let the solid angle at E , contained by DEG, GEF , and FED , be set out. And let DE, GE , and EF each be made equal to B . And let the parallelepiped solid $E K$ be completed. And (let) LM (be made) equal to A . And let the solid angle contained by NLO, OLM , and MLN be constructed on the straight-line LM , and at the point L on it, (so as to be) equal to the solid angle E [Prop. 11.23]. And let LO be made equal to B , and LN equal to C . And since as A (is) to B , so B (is) to C , and A (is) equal to LM , and B to each of LO and ED , and C to LN , thus as LM (is) to EF , so DE (is) to LN . And (so) the sides around the equal angles NLM and DEF are reciprocally proportional. Thus, parallelogram MN is equal to parallelogram DF [Prop. 6.14]. And since the two plane rectilinear angles DEF and NLM are equal, and the raised straight-lines stood on them (at their apexes), LO and EG , are equal to one another, and contain equal angles respectively with the original straight-lines (forming the angles), the perpendiculars drawn from points G and O to the planes through NLM and DEF (respectively) are thus equal to one another [Prop. 11.35 corr.]. Thus, the solids LH and $E K$ (have) the same height. And parallelepiped solids on equal bases, and with the same height, are

στερεὸν τῷ EK στερεῷ. καὶ ἐστὶ τὸ μὲν $\Lambda\Theta$ τὸ ἐκ τῶν A, B, Γ στερεόν, τὸ δὲ EK τὸ ἀπὸ τῆς B στερεόν· τὸ ἄρα ἐκ τῶν A, B, Γ στερεὸν παραλληλεπίπεδον ἴσον ἐστὶ τῷ ἀπὸ τῆς B στερεῷ ἰσοπλεύρῳ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ· ὅπερ ἔδει δεῖξαι.

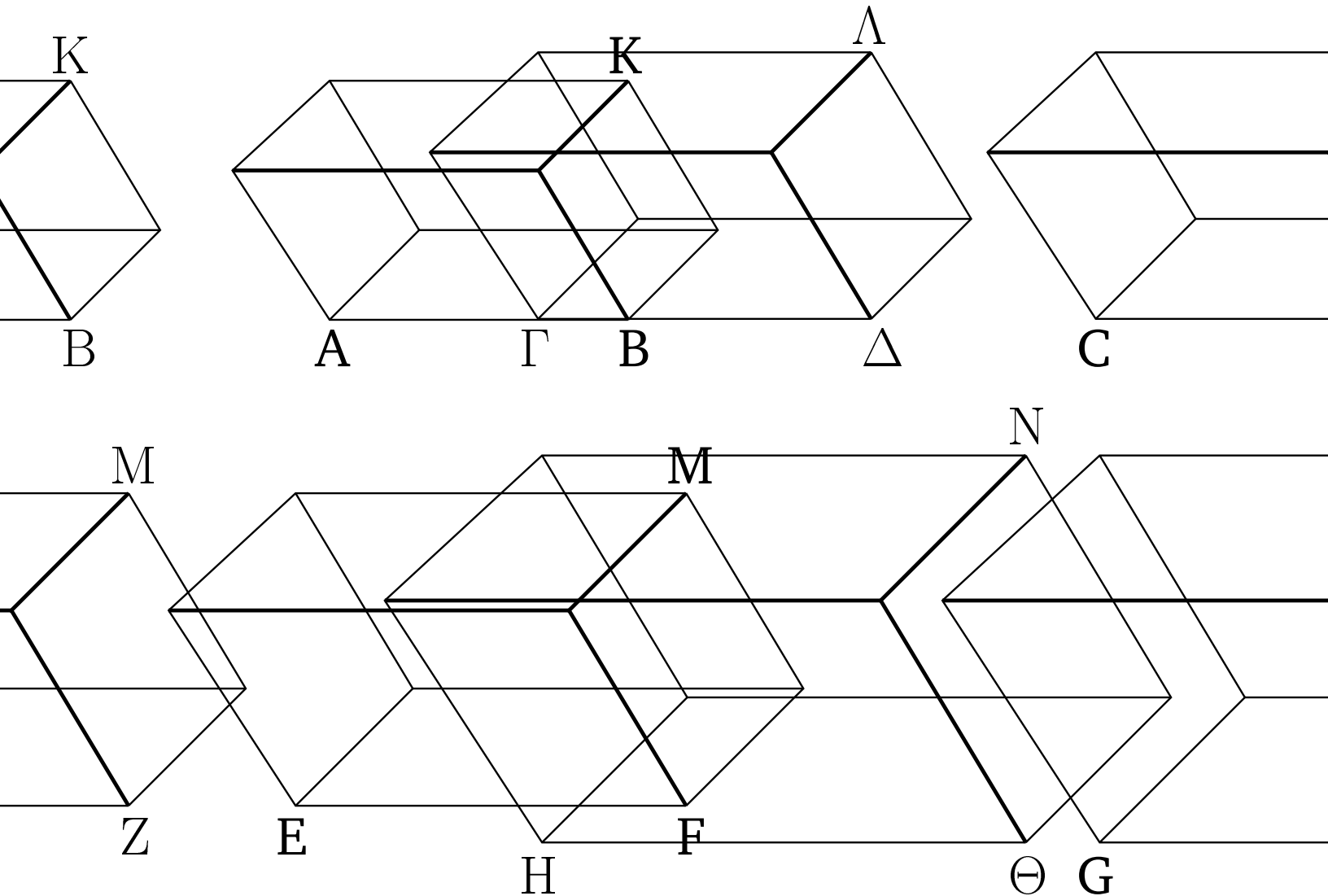
equal to one another [Prop. 11.31]. Thus, solid HL is equal to solid EK . And LH is the solid (formed) from A, B , and C , and EK the solid on B . Thus, the parallelepiped solid (formed) from A, B , and C is equal to the equilateral solid on B (which is) equiangular with the aforementioned (solid). (Which is) the very thing it was required to show.

37.

Proposition 37[†]

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ᾧσιν, καὶ τὰ ἀπ' αὐτῶν στερεὰ παραλληλεπίπεδα ὁμοία τε καὶ ὁμοίῳξ ἀναγραφόμενα ἀνάλογον ἔσται· καὶ ἔαν τὰ ἀπ' αὐτῶν στερεὰ παραλληλεπίπεδα ὁμοία τε καὶ ὁμοίῳξ ἀναγραφόμενα ἀνάλογον ᾧ, καὶ αὐταὶ αἱ εὐθεῖαι ἀνάλογον ἔσονται.

If four straight-lines are proportional then the similar, and similarly described, parallelepiped solids on them will also be proportional. And if the similar, and similarly described, parallelepiped solids on them are proportional then the straight-lines themselves will be proportional.



Ἐστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ $AB, \Gamma\Delta, EZ,$

Let AB, CD, EF , and GH , be four proportional straight-

$H\Theta$, ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$, καὶ ἀναγεγράφθωσαν ἀπὸ τῶν AB , $\Gamma\Delta$, EZ , $H\Theta$ ὁμοιά τε καὶ ὁμοίως κείμενα στερεὰ παραλληλεπίπεδα τὰ KA , $\Lambda\Gamma$, ME , NH . λέγω, ὅτι ἐστὶν ὥς τὸ KA πρὸς τὸ $\Lambda\Gamma$, οὕτως τὸ ME πρὸς τὸ NH .

Ἐπεὶ γὰρ ὁμοίων ἐστι τὸ KA στερεὸν παραλληλεπίπεδον τῷ $\Lambda\Gamma$, τὸ KA ἄρα πρὸς τὸ $\Lambda\Gamma$ τριπλασίονα λόγον ἔθει ἥπερ ἡ AB πρὸς τὴν $\Gamma\Delta$. διὰ τὰ αὐτὰ δὴ καὶ τὸ ME πρὸς τὸ NH τριπλασίονα λόγον ἔθει ἥπερ ἡ EZ πρὸς τὴν $H\Theta$. καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$. καὶ ὥς ἄρα τὸ AK πρὸς τὸ $\Lambda\Gamma$, οὕτως τὸ ME πρὸς τὸ NH .

Ἀλλὰ δὴ ἔστω ὥς τὸ AK στερεὸν πρὸς τὸ $\Lambda\Gamma$ στερεόν, οὕτως τὸ ME στερεὸν πρὸς τὸ NH . λέγω, ὅτι ἐστὶν ὥς ἡ AB εὐθεῖα πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$.

Ἐπεὶ γὰρ πάλιν τὸ KA πρὸς τὸ $\Lambda\Gamma$ τριπλασίονα λόγον ἔθει ἥπερ ἡ AB πρὸς τὴν $\Gamma\Delta$, ἔθει δὲ καὶ τὸ ME πρὸς τὸ NH τριπλασίονα λόγον ἥπερ ἡ EZ πρὸς τὴν $H\Theta$, καὶ ἐστὶν ὥς τὸ KA πρὸς τὸ $\Lambda\Gamma$, οὕτως τὸ ME πρὸς τὸ NH , καὶ ὥς ἄρα ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$.

Ἐάν ἄρα τέσσαρες εὐθεῖαι ἀνάλογον ᾧσι καὶ τὰ ἐχῇς τῆς προτάσεως· ὅπερ ἔδει δεῖχαι.

lines, (such that) as AB (is) to CD , so EF (is) to GH . And let the similar, and similarly laid out, parallelepiped solids KA , LC , ME and NG be described on AB , CD , EF , and GH (respectively). I say that as KA is to LC , so ME (is) to NG .

For since the parallelepiped solid KA is similar to LC , KA thus has to LC the cubed ratio that AB (has) to CD [Prop. 11.33]. So, for the same (reasons), ME also has to NG the cubed ratio that EF (has) to GH [Prop. 11.33]. And since as AB is to CD , so EF (is) to GH , thus, also, as AK (is) to LC , so ME (is) to NG .

And so let solid AK be to solid LC , as solid ME (is) to NG . I say that as straight-line AB is to CD , so EF (is) to GH .

For, again, since KA has to LC the cubed ratio that AB (has) to CD [Prop. 11.33], and ME also has to NG the cubed ratio that EF (has) to GH [Prop. 11.33], and as KA is to LC , so ME (is) to NG , thus, also, as AB (is) to CD , so EF (is) to GH .

Thus, if four straight-lines are proportional, and so on of the proposition. (Which is) the very thing it was required to show.

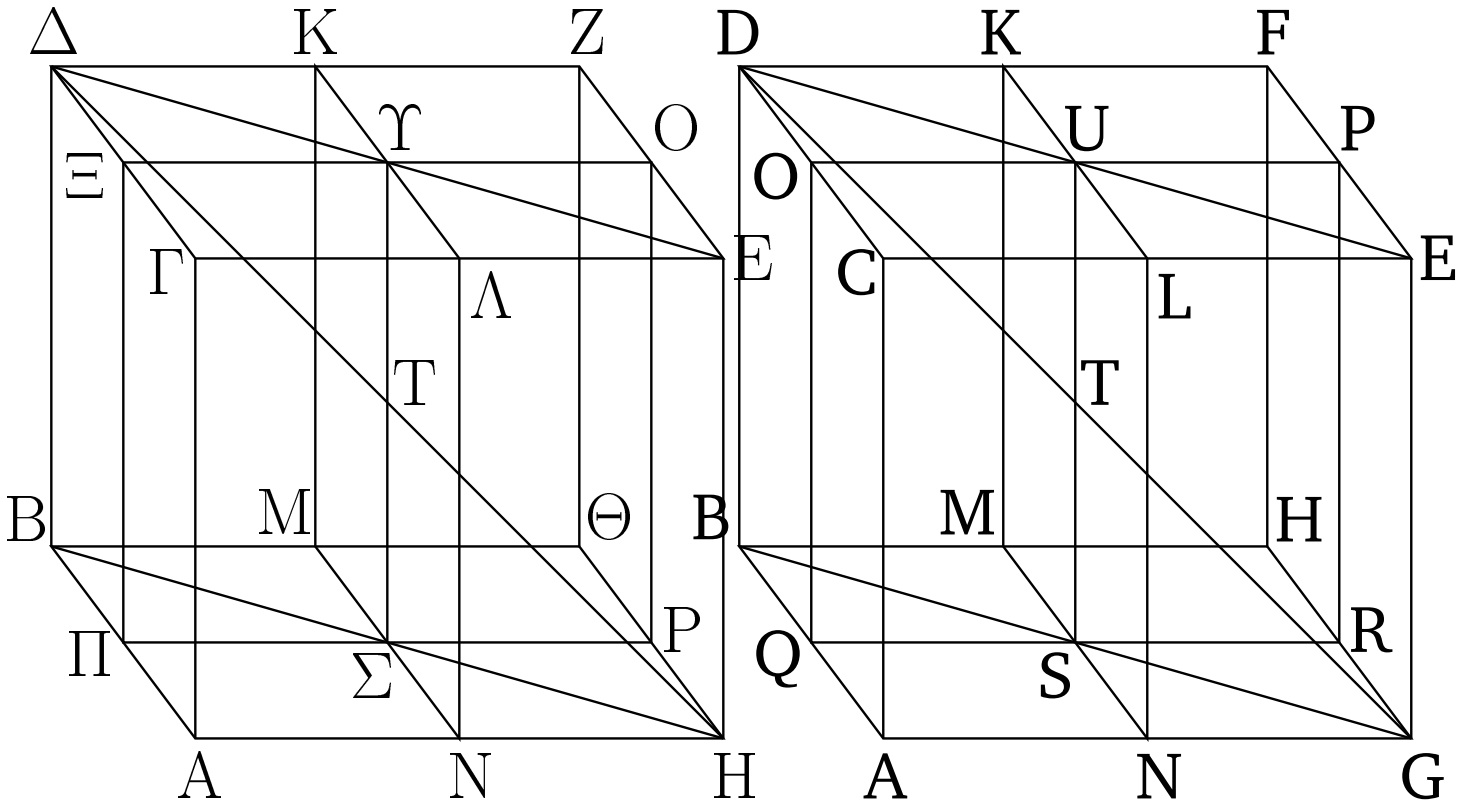
[†] This proposition assumes that if two ratios are equal then the cube of the former is also equal to the cube of the latter, and vice versa.

38.

Ἐάν κύβον τῶν ἀπεναντίον ἐπιπέδων αἱ πλευραὶ δίθῃ τμηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος δίθῃ τέμνουσιν ἀλλήλας.

Proposition 38

If the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut one another in half.



Κύβον γὰρ τοῦ ΑΖ τῶν ἀπεναντίον ἐπιπέδων τῶν ΓΖ, ΑΘ αἱ πλευραὶ διῖθα τετμήσθωσαν κατὰ τὰ Κ, Λ, Μ, Ν, Χ, Π, Ο, Ρ σημεῖα, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβεβλήσθω τὰ ΚΝ, ΧΡ, κοινὴ δὲ τομὴ τῶν ἐπιπέδων ἔστω ἡ ΥΣ, τοῦ δὲ ΑΖ κύβου διαγώνιος ἡ ΔΗ. λέγω, ὅτι ἴση ἐστὶν ἡ μὲν ΥΤ τῇ ΤΣ, ἡ δὲ ΔΤ τῇ ΤΗ.

Ἐπεξενύθωσαν γὰρ αἱ ΔΥ, ΥΕ, ΒΣ, ΣΗ. καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΔΧ τῇ ΟΕ, αἱ ἐναλλάξ γωνίαι αἱ ὑπὸ ΔΧΥ, ΥΟΕ ἴσαι ἀλλήλαις εἰσὶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν ΔΧ τῇ ΟΕ, ἡ δὲ ΧΥ τῇ ΥΟ, καὶ γωνίαξ ἴσαξ περιέθουσιν, βάσιξ ἴσα ἡ ΔΥ τῇ ΥΕ ἐστὶν ἴση, καὶ τὸ ΔΧΥ τρίγωνον τῷ ΟΥΕ τρίγωνῳ ἐστὶν ἴσον καὶ αἱ λοιπαὶ γωνίαι ταῖξ λοιπαῖξ γωνίαξ ἴσαι· ἴση ἄρα ἡ ὑπὸ ΧΥΔ γωνία τῇ ὑπὸ ΟΥΕ γωνία. διὰ δὲ τοῦτο εὐθεῖά ἐστιν ἡ ΔΥΕ. διὰ τὰ αὐτὰ δὴ καὶ ΒΣΗ εὐθεῖά ἐστιν, καὶ ἴση ἡ ΒΣ τῇ ΣΗ. καὶ ἐπεὶ ἡ ΓΑ τῇ ΔΒ ἴση ἐστὶ καὶ παράλληλος, ἀλλὰ ἡ ΓΑ καὶ τῇ ΕΗ ἴση τέ ἐστι καὶ παράλληλος, καὶ ἡ ΔΒ ἴσα τῇ ΕΗ ἴση τέ ἐστι καὶ παράλληλος. καὶ ἐπιξενυνύουσιν αὐτάξ εὐθεῖαι αἱ ΔΕ, ΒΗ· παράλληλός ἄρα ἐστὶν ἡ ΔΕ τῇ ΒΗ. ἴση ἄρα ἡ μὲν ὑπὸ ΕΔΤ γωνία τῇ ὑπὸ ΒΗΤ· ἐναλλάξ γάρ· ἡ δὲ ὑπὸ ΔΤΥ τῇ ὑπὸ ΗΤΣ. δύο δὴ τρίγωνά ἐστι τὰ ΔΤΥ, ΗΤΣ τὰξ δύο γωνίαξ ταῖξ δυοὶ γωνίαξ ἴσαξ ἔθοντα καὶ μίαν πλευράν μιᾷ πλευρᾷ ἴσην τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν τὴν ΔΥ τῇ ΗΣ· ἡμίσειαι γὰρ εἰσὶ τῶν ΔΕ, ΒΗ· καὶ τὰξ λοιπάξ πλευράξ ταῖξ λοιπαῖξ πλευραῖξ ἴσαξ ἔχει. ἴση ἄρα ἡ μὲν ΔΤ τῇ ΤΗ, ἡ δὲ ΥΤ τῇ ΤΣ.

For let the opposite planes CF and AH of the cube AF be cut in half at the points K, L, M, N, O, Q, P , and R . And let the planes KN and OR be produced through the pieces. And let US be the common section of the planes, and DG the diameter of cube AF . I say that UT is equal to TS , and DT to TG .

For let DU, UE, BS , and SG be joined. And since DO is parallel to PE , the alternate angles DOU and UPE are equal to one another [Prop. 1.29]. And since DO is equal to PE , and OU to UP , and they contain equal angles, base DU is thus equal to base UE , and triangle DOU is equal to triangle PUE , and the remaining angles (are) equal to the remaining angles [Prop. 1.4]. Thus, angle ODU (is) equal to angle PUE . So, for this (reason), DUE is a straight-line [Prop. 1.14]. So, for the same (reason), BSG is also a straight-line, and BS equal to SG . And since CA is equal and parallel to DB , but CA is also equal and parallel to EG , DB is thus also equal and parallel to EG [Prop. 11.9]. And the straight-lines DE and BG join them. DE is thus parallel to BG [Prop. 1.33]. Thus, angle EDT (is) equal to BGT . For (they are) alternate [Prop. 1.29]. And (angle) DTU (is equal) to GTS [Prop. 1.15]. So, DTU and GTS are two triangles having two angles equal to two angles, and one side equal to one side—(namely), that subtended by one of the equal angles—(that is), DU (equal) to GS . For they are halves of DE and BG (respectively). (Thus), they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, DT (is) equal to TG , and UT to TS .

Ἐάν ἄρα κύβον τῶν ἀπεναντίον ἐπιπέδων αἱ πλευραὶ διῖτα τμηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος διῖτα τέμνουσιν ἀλλήλας· ὅπερ ἔδει δεῖξαι.

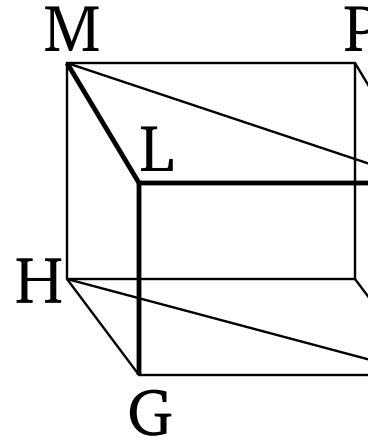
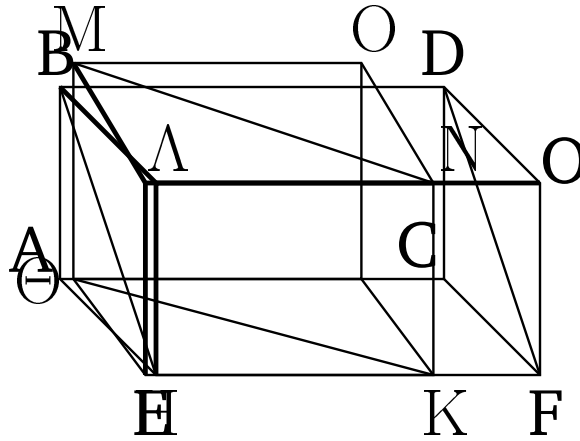
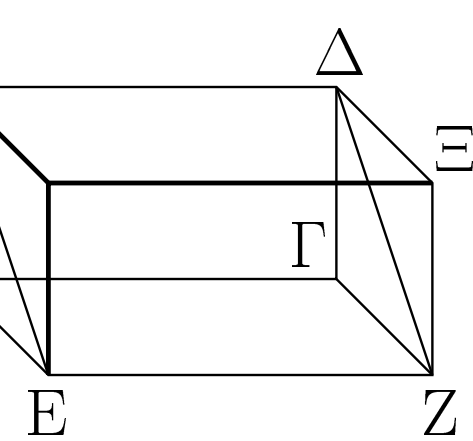
Thus, if the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut one another in half. (Which is) the very thing it was required to show.

39.

Proposition 39

Ἐάν ἡ δύο πρίσματα ἰσοῦψῃ, καὶ τὸ μὲν ἔθῃ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ἢ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσα ἔσται τὰ πρίσματα.

If there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms will be equal.



Ἐστω δύο πρίσματα ἰσοῦψῃ τὰ $ABΓΔΕΖ$, $ΗΘΚΛΜΝ$, καὶ τὸ μὲν ἔθῃ βάσιν τὸ AZ παραλληλόγραμμον, τὸ δὲ τὸ $ΗΘΚ$ τρίγωνον, διπλάσιον δὲ ἔστω τὸ AZ παραλληλόγραμμον τοῦ $ΗΘΚ$ τριγώνου· λέγω, ὅτι ἴσον ἐστὶ τὸ $ABΓΔΕΖ$ πρίσμα τῷ $ΗΘΚΛΜΝ$ πρίσματι.

Let $ABCDEF$ and $GHKLMN$ be two equal height prisms, and let the former have the parallelogram AF , and the latter the triangle GHK , as a base. And let parallelogram AF be twice triangle GHK . I say that prism $ABCDEF$ is equal to prism $GHKLMN$.

Συμπεπληρώσθω γάρ τὰ AX , HO στερεά. ἐπεὶ διπλάσιόν ἐστι τὸ AZ παραλληλόγραμμον τοῦ $ΗΘΚ$ τριγώνου, ἔστι δὲ καὶ τὸ $ΘΚ$ παραλληλόγραμμον διπλάσιον τοῦ $ΗΘΚ$ τριγώνου, ἴσον ἄρα ἐστὶ τὸ AZ παραλληλόγραμμον τῷ $ΘΚ$ παραλληλόγραμμῳ. τὰ δὲ ἐπὶ ἴσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ AX στερεὸν τῷ HO στερεῷ. καὶ ἐστὶ τοῦ μὲν AX στερεοῦ ἡμισὺν τὸ $ABΓΔΕΖ$ πρίσμα, τοῦ δὲ HO στερεοῦ ἡμισὺν τὸ $ΗΘΚΛΜΝ$ πρίσμα· ἴσον ἄρα ἐστὶ τὸ $ABΓΔΕΖ$ πρίσμα τῷ $ΗΘΚΛΜΝ$ πρίσματι.

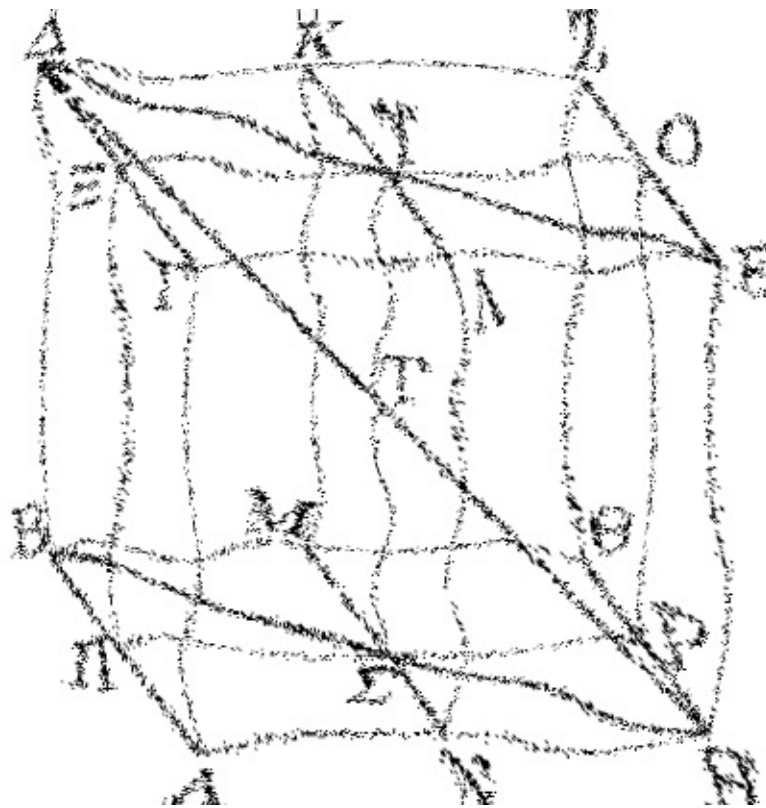
For let the solids AO and GP be completed. Since parallelogram AF is double triangle GHK , and parallelogram HK is also double triangle GHK [Prop. 1.34], parallelogram AF is thus equal to parallelogram HK . And parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another [Prop. 11.31]. Thus, solid AO is equal to solid GP . And prism $ABCDEF$ is half of solid AO , and prism $GHKLMN$ half of solid GP [Prop. 11.28]. Prism $ABCDEF$ is thus equal to prism $GHKLMN$.

Ἐάν ἄρα ἡ δύο πρίσματα ἰσοῦψῃ, καὶ τὸ μὲν ἔθῃ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ἢ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσα ἔσται τὰ πρίσματα· ὅπερ ἔδει δεῖξαι.

Thus, if there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms are equal. (Which is) the very thing it was required to show.

ELEMENTS BOOK 11

Elementary Stereometry



Ὅροι.

1. Στερεόν ἐστι τὸ μήκοξ καὶ πλάτοξ καὶ βάθοξ ἔθον.
2. Στερεοῦ δὲ πέραξ ἐπιφάνεια.
3. Εὐθεῖα πρὸς ἐπίπεδον ὀρθή ἐστιν, ὅταν πρὸς πάσαξ τὰς ἀπτομέναξ αὐτῆξ εὐθείαξ καὶ οὐσαξ ἐν τῷ [ὑποκειμένῳ] ἐπιπέδῳ ὀρθὰξ ποιῇ γωνίαξ.
4. Ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, ὅταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰξ ἀγόμεναι εὐθεῖαι ἐν ἐνὶ τῶν ἐπιπέδων τῷ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰξ ᾧσιν.
5. Εὐθείαξ πρὸς ἐπίπεδον κλίσιξ ἐστίν, ὅταν ἀπὸ τοῦ μετεώρου πέρατοξ τῆξ εὐθείαξ ἐπὶ τὸ ἐπίπεδον κάθετοξ ἀθθῇ, καὶ ἀπὸ τοῦ γενομένου σημείου ἐπὶ τὸ ἐν τῷ ἐπιπέδῳ πέραξ τῆξ εὐθείαξ εὐθεῖα ἐπιξενθθῇ, ἡ περιεθιομένη γωνία ὑπὸ τῆξ ἀθθείσηξ καὶ τῆξ ἐφεστώσηξ.
6. Ἐπιπέδον πρὸς ἐπίπεδον κλίσιξ ἐστίν ἡ περιεθιομένη ὀχεῖα γωνία ὑπὸ τῶν πρὸς ὀρθὰξ τῇ κοινῇ τομῇ ἀγομένων πρὸς τῷ αὐτῷ σημείῳ ἐν ἐκατέρῳ τῶν ἐπιπέδων.
7. Ἐπίπεδον πρὸς ἐπίπεδον ὁμοίωξ κεκλίσθαι λέγεται καὶ ἕτερον πρὸς ἕτερον, ὅταν αἱ εἰρημέναι τῶν κλίσεων γωνίαι ἴσαι ἀλλήλαιξ ᾧσιν.
8. Παράλληλα ἐπίπεδά ἐστι τὰ ἀσύμπτωτα.
9. Ὅμοια στερεὰ σθῆματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεθιόμενα ᾧσιν τὸ πλήθοξ.
10. ἴσα δὲ καὶ ὅμοια στερεὰ σθῆματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεθιόμενα ᾧσιν τῷ πλήθει καὶ τῷ μεγέθει.
11. Στερεὰ γωνία ἐστίν ἡ ὑπὸ πλειόνων ἢ δύο γραμμῶν ἀπτομένων ἀλλήλων καὶ μὴ ἐν τῇ αὐτῇ ἐπιφανείᾳ οὐσῶν πρὸς πάσαιξ ταῖξ γραμμαῖξ κλίσιξ. ἄλλωξ· στερεὰ γωνία ἐστίν ἡ ὑπὸ πλειόνων ἢ δύο γωνιῶν ἐπιπέδων περιεθιομένη μὴ οὐσῶν ἐν τῷ αὐτῷ ἐπιπέδῳ πρὸς ἐνὶ σημείῳ συνισταμένων.
12. Πυραμὶξ ἐστὶ σθῆμα στερεὸν ἐπιπέδοιξ περιθιόμενον ἀπὸ ἐνὸξ ἐπιπέδου πρὸς ἐνὶ σημείῳ συνεστώξ.
13. Πρίσμα ἐστὶ σθῆμα στερεὸν ἐπιπέδοιξ περιεχόμενον, ὧν δύο τὰ ἀπεναντίον ἴσα τε καὶ ὅμοιά ἐστι καὶ παράλληλα, τὰ δὲ λοιπά παραλληλόγραμμα.
14. Σφαῖρά ἐστιν, ὅταν ἡμικυκλίον μενούσηξ τῆξ διαμέτρου περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα.
15. Ἄχων δὲ τῆξ σφαίραξ ἐστίν ἡ μένουσα εὐθεῖα, περιῆν τὸ ἡμικύκλιον στρέφεται.
16. Κέντρον δὲ τῆξ σφαίραξ ἐστὶ τὸ αὐτό, ὃ καὶ τοῦ ἡμικυκλίου.
17. Διάμετροξ δὲ τῆξ σφαίραξ ἐστίν εὐθεῖα τιξ διὰ τοῦ κέντρου ἡγμένη καὶ περατουμένη ἐφ' ἐκάτερα τὰ μέρη ὑπὸ τῆξ ἐπιφανείαξ τῆξ σφαίραξ.
18. Κωνός ἐστιν, ὅταν ὀρθογωνίου τριγώνου μενούσηξ μιᾷξ πλευρᾷξ τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθθὲν τὸ τρίγωνον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα. καὶ μὲν ἡ μένουσα εὐθεῖα ἴση ἢ τῇ λοιπῇ

[τῇ] περί τὴν ὀρθὴν περιφερομένη, ὀρθογώνιος ἔσται ὁ κῶνος, ἔαν δὲ ἐλάττων, ἀμβλυγώνιος, ἔαν δὲ μείζων, ὀχυγώνιος.

19. Ἄχων δὲ τοῦ κώνου ἐστὶν ἡ μένουσα εὐθεΐα, περί ἣν τὸ τρίγωνον στρέφεται.

20. Βάσις δὲ ὁ κύκλος ὁ ὑπὸ τῆς περιφερομένης εὐθείας γραφόμενος.

21. Κύλινδρος ἐστὶν, ὅταν ὀρθογωνίου παραλληλογράμμου μενούσης μιᾶς πλευρᾶς τῶν περί τὴν ὀρθὴν γωνίαν περιενεθ-
θὲν τὸ παραλληλόγραμμον εἰς τὸ αὐτὸ πάλιν ἀποκατασταθῇ,
ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σθῆμα.

22. Ἄχων δὲ τοῦ κυλίνδρου ἐστὶν ἡ μένουσα εὐθεΐα, περί ἣν τὸ παραλληλόγραμμον στρέφεται.

23. Βάσεις δὲ οἱ κύκλοι οἱ ὑπὸ τῶν ἀπεναντίον περιγο-
μένων δύο πλευρῶν γραφόμενοι.

24. Ὅμοιοι κῶνοι καὶ κύλινδροι εἰσιν, ὧν οἱ τε ἄχονες καὶ αἱ διάμετροι τῶν βάσεων ἀνάλογόν εἰσιν.

25. Κύβοξ ἐστὶ σθῆμα στερεόν ὑπὸ ἑξ τετραγώνων ἴσων περιεχόμενον.

26. Ὀκτάεδρόν ἐστὶ σθῆμα στερεόν ὑπὸ ὀκτὼ τριγώνων ἴσων καὶ ἰσοπλεύρων περιεχόμενον.

27. Εἰκοσάεδρόν ἐστὶ σθῆμα στερεόν ὑπὸ εἴκοσι τριγώνων ἴσων καὶ ἰσοπλεύρων περιεχόμενον.

28. Δωδεκάεδρόν ἐστὶ σθῆμα στερεόν ὑπὸ δώδεκα πεντα-
γώνων ἴσων καὶ ἰσοπλεύρων καὶ ἰσογωνίων περιεχόμενον.

Definitions

1. A solid is a (figure) having length and breadth and depth.

2. The extremity of a solid (is) a surface.

3. A straight-line is at right-angles to a plane when it makes right-angles with all of the straight-lines joined to it which are also in the plane.

4. A plane is at right-angles to a(nother) plane when (all of) the straight-lines drawn in one of the planes, at right-angles to the common section of the planes, are at right-angles to the remaining plane.

5. The inclination of a straight-line to a plane is the angle contained by the drawn and standing (straight-lines), when a perpendicular is lead to the plane from the end of the (stand-
ing) straight-line raised (out of the plane), and a straight-line is (then) joined from the point (so) generated to the end of the (standing) straight-line (lying) in the plane.

6. The inclination of a plane to a(nother) plane is the acute angle contained by the (straight-lines), (one) in each of the planes, drawn at right-angles to the common segment (of the planes), at the same point.

7. A plane is said to be similarly inclined to a plane, as another to another, when the aforementioned angles of incli-
nation are equal to one another.

8. Parallel planes are those which do not meet (one an-
other).

9. Similar solid figures are those contained by equal num-
bers of similar planes (which are similarly arranged).

10. But equal and similar solid figures are those contained by similar planes equal in number and in magnitude (which are similarly arranged).

11. A solid angle is the inclination (constituted) by more than two lines joining one another (at the same point), and not being in the same surface, to all of the lines. Otherwise, a solid angle is that contained by more than two plane angles, not being in the same plane, and constructed at one point.

12. A pyramid is a solid figure, contained by planes, (which is) constructed from one plane to one point.

13. A prism is a solid figure, contained by planes, of which the two opposite (planes) are equal, similar, and parallel, and the remaining (planes are) parallelograms.

14. A sphere is the figure enclosed when, the diameter of a semicircle remaining (fixed), the semicircle is carried around, and again established at the same (position) from which it began to be moved.

15. And the axis of the sphere is the fixed straight-line about which the semicircle is turned.

16. And the center of the sphere is the same as that of the semicircle.

17. And the diameter of the sphere is any straight-line which is drawn through the center and terminated in both directions by the surface of the sphere.

18. A cone is the figure enclosed when, one of the sides of a right-angled triangle about the right-angle remaining (fixed), the triangle is carried around, and again established at the same (position) from which it began to be moved. And if the fixed straight-line is equal to the remaining (straight-line) about the right-angle, (which is) carried around, then the cone will be right-angled, and if less, obtuse-angled, and if greater, acute-angled.

19. And the axis of the cone is the fixed straight-line about which the triangle is turned.

20. And the base (of the cone is) the circle described by the (remaining) straight-line (about the right-angle which is) carried around (the axis).

21. A cylinder is the figure enclosed when, one of the sides of a right-angled parallelogram about the right-angle remaining (fixed), the parallelogram is carried around, and again established at the same (position) from which it began to be moved.

22. And the axis of the cylinder is the stationary straight-line about which the parallelogram is turned.

23. And the bases (of the cylinder are) the circles described by the two opposite sides (which are) carried around.

24. Similar cones and cylinders are those for which the axes and the diameters of the bases are proportional.

25. A cube is a solid figure contained by six equal squares.

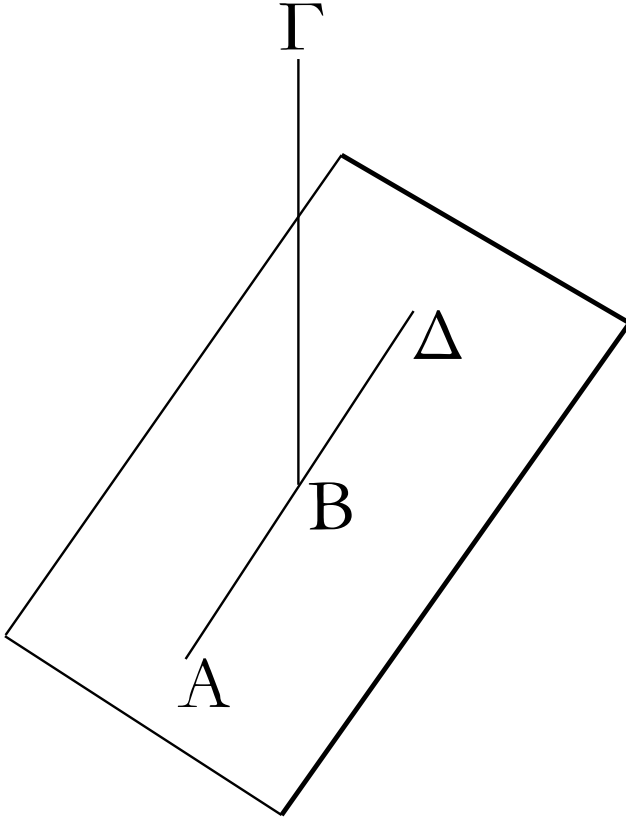
26. An octahedron is a solid figure contained by eight equal and equilateral triangles.

1.

Εὐθεία γραμμῆς μέρος μὲν τι οὐκ ἔστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι ἐν μετεωροτέρῳ.

Εἰ γὰρ δυνατόν, εὐθεία γραμμῆς τῆς $AB\Gamma$ μέρος μὲν τι τὸ AB ἔστω ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι τὸ $B\Gamma$ ἐν μετεωροτέρῳ.

Ἔσται δὴ τῆς AB συννεθῆς εὐθεῖα ἐπ' εὐθείας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ. ἔστω ἡ BD . δύο ἄρα εὐθειῶν τῶν $AB\Gamma$, ABD κοινὸν τμήμα ἔστιν ἡ AB . ὅπερ ἐστὶν ἀδύνατον, ἐπειδήπερ ἔαν κέντρῳ τῷ B καὶ διαστήματι τῷ AB κύκλον γράψωμεν, αἱ διάμετροι ἀνίσουσιν ἀπολήγονται τοῦ κύκλου περιφερείας.



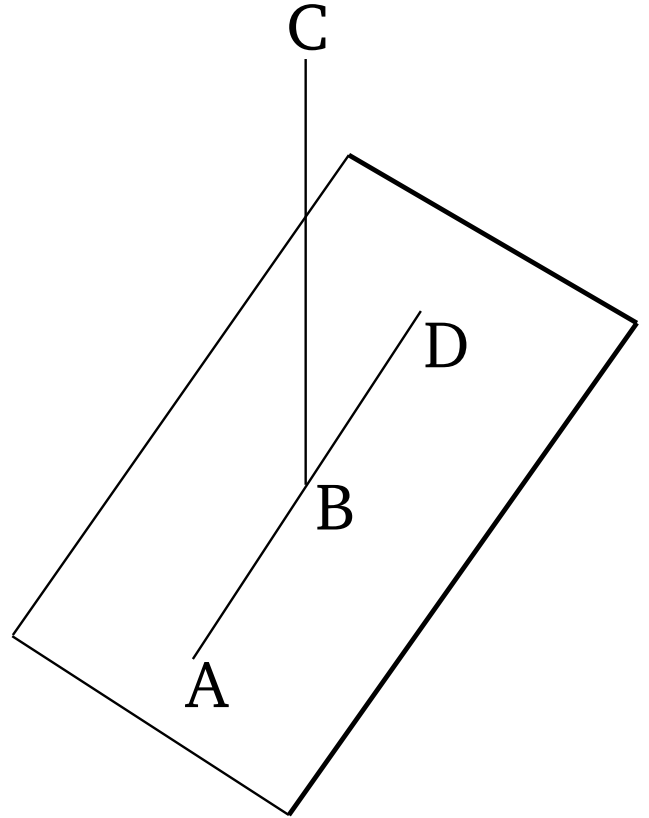
Εὐθεία ἄρα γραμμῆς μέρος μὲν τι οὐκ ἔστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δέ ἐν μετεωροτέρῳ. ὅπερ ἔδει δεῖχαι.

Proposition 1[†]

Some part of a straight-line cannot be in a reference plane, and some part in a more elevated (plane).

For, if possible, let some part, AB , of the straight-line ABC be in a reference plane, and some part, BC , in a more elevated (plane).

In the reference plane, there will be some straight-line continuous with, and straight-on to, AB .[‡] Let it be BD . Thus, AB is a common segment of the two (different) straight-lines ABC and ABD . The very thing is impossible, inasmuch as if we draw a circle with center B and radius AB then the diameters (ABD and ABC) will cut off unequal circumferences of the circle.



Thus, some part of a straight-line cannot be in a reference plane, and (some part) in a more elevated (plane). (Which is) the very thing it was required to show.

[†] The proofs of the first three propositions in this book are not at all rigorous. Hence, these three propositions should properly be regarded as additional axioms.

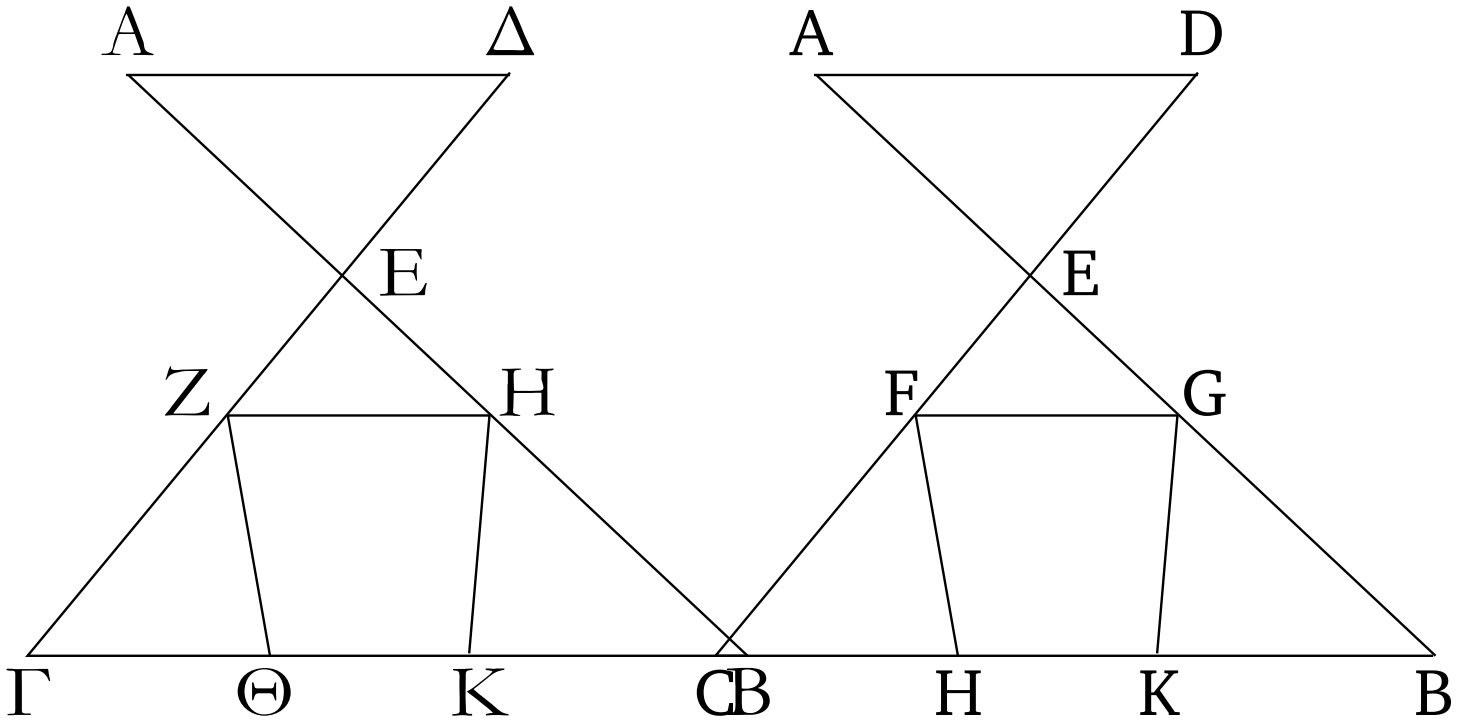
‡ This assumption essentially presupposes the validity of the proposition under discussion.

2.

Proposition 2

Ἐάν δύο εὐθεῖαι τέμνωσιν ἀλλήλας, ἐν ἐνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ.

If two straight-lines cut one another then they are in one plane, and every triangle (formed using segments of both lines) is in one plane.



Δύο γὰρ εὐθεῖαι αἱ AB , $ΓΔ$ τεμνέτωσαν ἀλλήλας κατὰ τὸ E σημεῖον. λέγω, ὅτι αἱ AB , $ΓΔ$ ἐν ἐνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ.

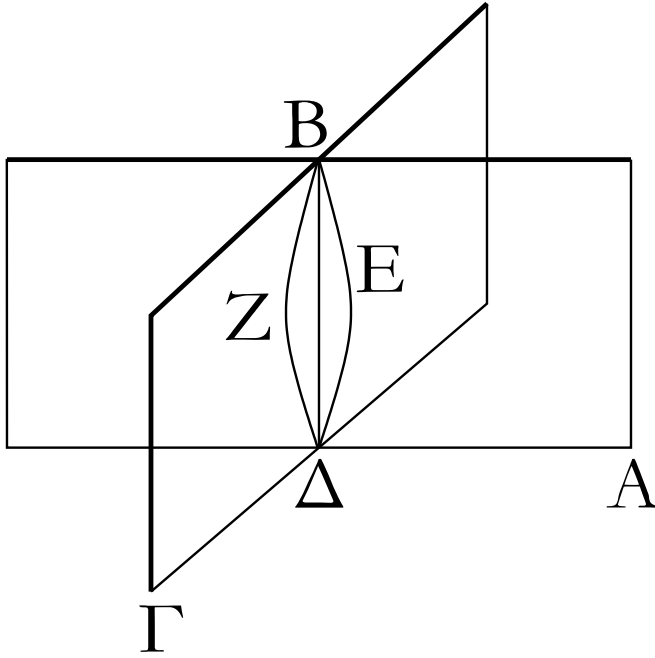
Εἰλήφθω γὰρ ἐπὶ τῶν $ΕΓ$, $ΕΒ$ τυθόντα σημεῖα τὰ Z , H , καὶ ἐπεξεύθθωσαν αἱ $ΓΒ$, ZH , καὶ διήθθωσαν αἱ $ZΘ$, HK . λέγω πρῶτον, ὅτι τὸ $ΕΓΒ$ τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ. εἰ γὰρ ἐστὶ τοῦ $ΕΓΒ$ τριγώνου μέρος ἥτοι τὸ $ZΘΓ$ ἢ τὸ HBK ἐν τῷ ὑποκειμένῳ [ἐπιπέδῳ], τὸ δὲ λοιπὸν ἐν ἄλλῳ, ἔσται καὶ μιᾶς τῶν $ΕΓ$, $ΕΒ$ εὐθειῶν μέρος μὲν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν ἄλλῳ. εἰ δὲ τοῦ $ΕΓΒ$ τριγώνου τὸ $ZΓΒH$ μέρος ἢ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ λοιπὸν ἐν ἄλλῳ, ἔσται καὶ ἀμφοτέρων τῶν $ΕΓ$, $ΕΒ$ εὐθειῶν μέρος μὲν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν ἄλλῳ. ὅπερ ἄτοπον ἐδείχθη. τὸ ἄρα $ΕΓΒ$ τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ. ἐν ᾧ δὲ ἐστὶ τὸ $ΕΓΒ$ τρίγωνον, ἐν τούτῳ καὶ ἑκατέρα τῶν $ΕΓ$, $ΕΒ$, ἐν ᾧ δὲ ἑκατέρα τῶν $ΕΓ$, $ΕΒ$, ἐν τούτῳ καὶ αἱ AB , $ΓΔ$. αἱ AB , $ΓΔ$ ἄρα εὐθεῖαι ἐν ἐνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἐνί ἐστὶν ἐπιπέδῳ. ὅπερ ἔδει δεῖξαι.

For let the two straight-lines AB and CD have cut one another at point E . I say that AB and CD are in one plane, and that every triangle (formed using segments of both lines) is in one plane.

For let the random points F and G be taken on EC and EB (respectively). And let CB and FG be joined, and let FH and GK be drawn across. I say, first of all, that triangle ECB is in one (reference) plane. For if part of triangle ECB , either FHC or GBK , is in the reference [plane], and the remainder in a different (plane) then a part of one the straight-lines EC and EB will also be in the reference plane, and (a part) in a different (plane). And if the part $FCBG$ of triangle ECB is in the reference plane, and the remainder in a different (plane) then parts of both of the straight-lines EC and EB will also be in the reference plane, and (parts) in a different (plane). The very thing was shown to be absurd [Prop. 11.1]. Thus, triangle ECB is in one plane. And in whichever (plane) triangle ECB is (found), in that (plane) EC and EB (will) each also (be found). And in whichever (plane) EC and EB (are) each (found), in that (plane) AB and CD (will) also (be found) [Prop. 11.1]. Thus, the straight-lines AB and CD are in one plane, and every

3.

Ἐάν δύο ἐπίπεδα τεμνῇ ἄλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεΐά ἐστιν.



Δύο γάρ ἐπίπεδα τὰ AB , $BΓ$ τεμνέτω ἄλληλα, κοινὴ δὲ αὐτῶν τομὴ ἔστω ἡ $ΔΒ$ γραμμὴ· λέγω, ὅτι ἡ $ΔΒ$ γραμμὴ εὐθεΐά ἐστιν.

Εἰ γὰρ μή, ἐπεζεύθθω ἀπὸ τοῦ $Δ$ ἐπὶ τὸ B ἐν μὲν τῷ AB ἐπιπέδῳ εὐθεΐα ἡ $ΔΕΒ$, ἐν δὲ τῷ $BΓ$ ἐπιπέδῳ εὐθεΐα ἡ $ΔΖΒ$. ἔσται δὴ δύο εὐθειῶν τῶν $ΔΕΒ$, $ΔΖΒ$ τὰ αὐτὰ πέρατα, καὶ περιέχονσι δηλαδὴ θωρίον· ὅπερ ἄτοπον. οὐκ ἄρα αἱ $ΔΕΒ$, $ΔΖΒ$ εὐθεΐαι εἰσιν. ὁμοίως δὲ δείχμεν, ὅτι οὐδὲ ἄλλη τις ἀπὸ τοῦ $Δ$ ἐπὶ τὸ B ἐπιζευγνυμένη εὐθεΐα ἔσται πλὴν τῆς $ΔΒ$ κοινῆς τομῆς τῶν AB , $BΓ$ ἐπιπέδων.

Ἐάν ἄρα δύο ἐπίπεδα τέμνῃ ἄλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεΐά ἐστιν· ὅπερ ἔδει δεῖχαι.

4.

Ἐάν εὐθεΐα δύο εὐθείαις τεμνούσαις ἀλλήλας πρὸς ὀρθὰς ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ δι' αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰς ἔσται.

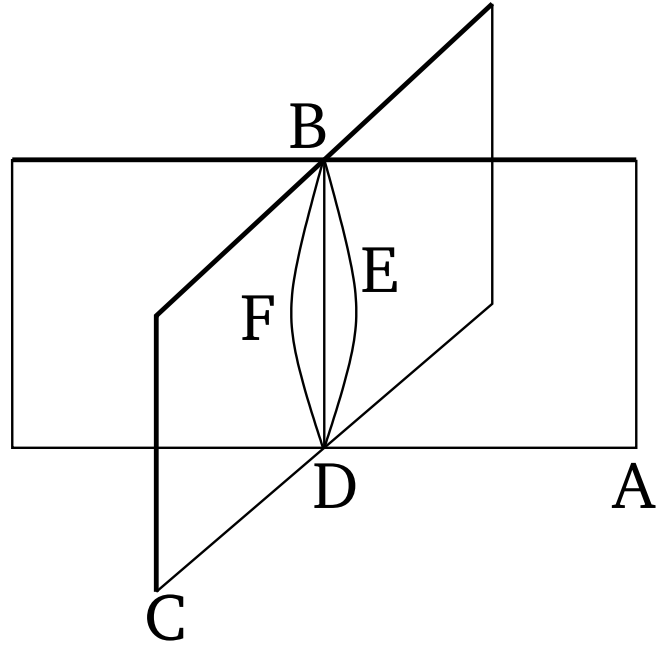
Εὐθεΐα γὰρ τις ἡ EZ δύο εὐθείαις ταῖς AB , $ΓΔ$ τεμνούσαις ἀλλήλας κατὰ τὸ E σημεῖον ἀπὸ τοῦ E πρὸς ὀρθὰς ἐφρεστώτω· λέγω, ὅτι ἡ EZ καὶ τῷ διὰ τῶν AB , $ΓΔ$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν.

Ἀπειλήφθωσαν γὰρ αἱ AE , EB , $ΓΕ$, $ΕΔ$ ἴσαι ἀλλήλαις,

triangle (formed using segments of both lines) is in one plane. (Which is) the very thing it was required to show.

Proposition 3

If two planes cut one another then their common section is a straight-line.



For let the two planes AB and BC cut one another, and let their common section be the line DB . I say that the line DB is straight.

For, if not, let the straight-line DEB be joined from D to B in the plane AB , and the straight-line DFB in the plane BC . So two straight-lines, DEB and DFB , will have the same ends, and they will clearly enclose an area. The very thing (is) absurd. Thus, DEB and DFB are not straight-lines. So, similarly, we can show that no other straight-line can be joined from D to B except DB , the common section of the planes AB and BC .

Thus, if two planes cut one another then their common section is a straight-line. (Which is) the very thing it was required to show.

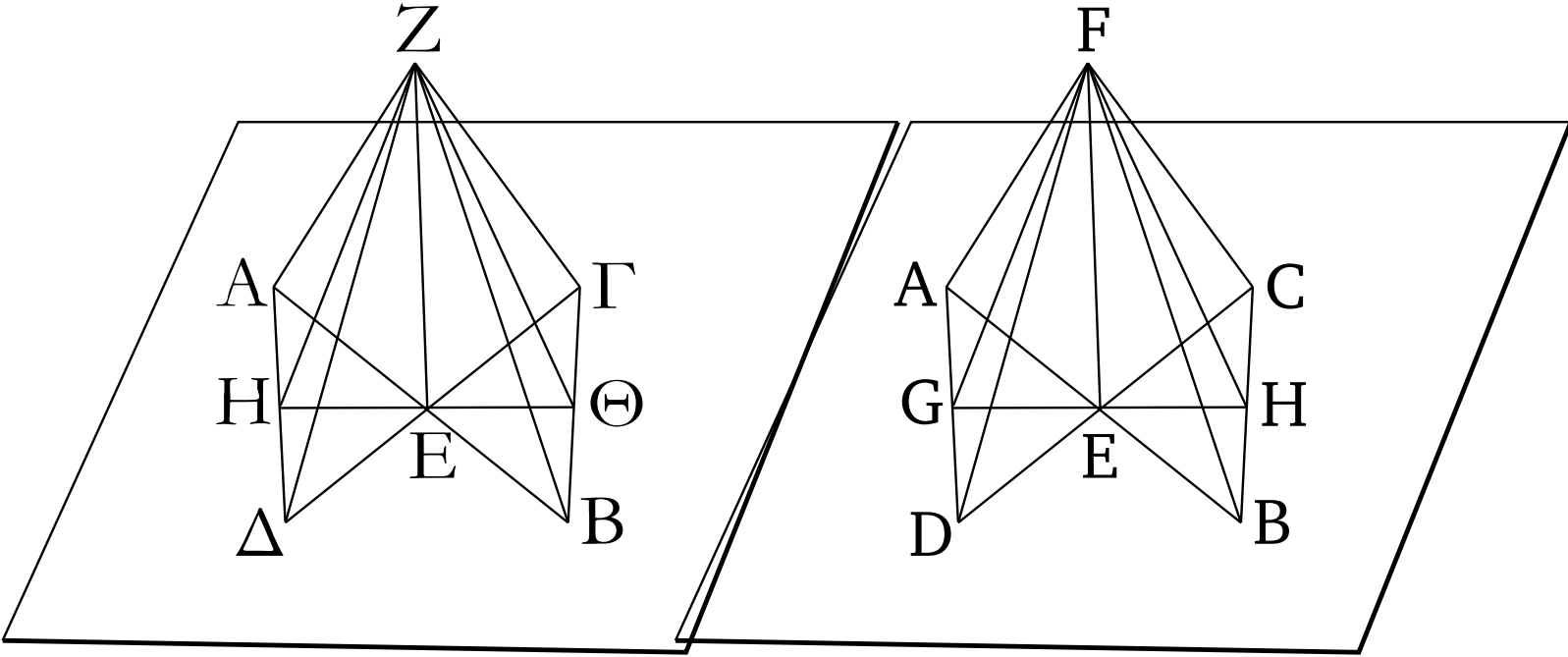
Proposition 4

If a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both).

For let some straight-line EF have (been) set up at right-angles to two straight-lines, AB and CD , cutting one another at point E , at E . I say that EF is also at right-angles to the plane (passing) through AB and CD .

καὶ διήθηται διὰ τοῦ E , ὥς ἔτυθεν, ἡ $HE\Theta$, καὶ ἐπεξεύθη-
θωσαν αἱ $A\Delta$, ΓB , καὶ ἔτι ἀπὸ τυθόντοξ τοῦ Z ἐπεξεύθηθωσαν
αἱ ZA , ZH , $Z\Delta$, $Z\Gamma$, $Z\Theta$, ZB .

For let AE , EB , CE and ED be cut off from (the two
straight-lines so as to be) equal to one another. And let GEH
be drawn, at random, through E (in the plane passing through
 AB and CD). And let AD and CB be joined. And, furthermore,
let FA , FG , FD , FC , FH , and FB be joined from the random
(point) F (on EF).



Καὶ ἐπεὶ δύο αἱ AE , $E\Delta$ δυοὶ ταῖς ΓE , EB ἴσαι εἰσὶ καὶ
γωνίαις ἴσαις περιέθουσιν, βάσεις ἄρα ἡ $A\Delta$ βάσει τῇ ΓB ἴση
ἐστίν, καὶ τὸ AED τρίγωνον τῷ ΓEB τριγώνῳ ἴσον ἔσται·
ὥστε καὶ γωνία ἡ ὑπὸ ΔAE γωνία τῇ ὑπὸ $EB\Gamma$ ἴση [ἐστίν].
ἔστι δὲ καὶ ἡ ὑπὸ AHE γωνία τῇ ὑπὸ $BE\Theta$ ἴση. δύο δὲ
τρίγωνα ἐστὶ τὰ AHE , $BE\Theta$ τὰς δύο γωνίας δυοὶ γωνίαις
ἴσαις ἔθοντα ἐκατέρωθεν ἐκατέρω καὶ μίαν πλευρὰν μιᾷ πλευρᾷ
ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν AE τῇ EB · καὶ τὰς
λοιπὰς ἄρα πλευρὰς ταῖς λοιπαῖς πλευραῖς ἴσαις ἔχουσιν. ἴση
ἄρα ἡ μὲν HE τῇ $E\Theta$, ἡ δὲ AH τῇ $B\Theta$. καὶ ἐπεὶ ἴση ἐστὶν ἡ
 AE τῇ EB , κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ ZE , βάσεις ἄρα ἡ ZA
βάσει τῇ ZB ἐστὶν ἴση. διὰ τὰ αὐτὰ δὲ καὶ ἡ $Z\Gamma$ τῇ $Z\Delta$ ἐστὶν
ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ $A\Delta$ τῇ ΓB , ἔστι δὲ καὶ ἡ ZA τῇ ZB
ἴση, δύο δὲ αἱ ZA , $A\Delta$ δυοὶ ταῖς ZB , $B\Gamma$ ἴσαι εἰσὶν ἐκατέρω
ἐκατέρω· καὶ βάσεις ἡ $Z\Delta$ βάσει τῇ $Z\Gamma$ ἐδείχθη ἴση· καὶ γωνία
ἄρα ἡ ὑπὸ ZAD γωνία τῇ ὑπὸ $ZB\Gamma$ ἴση ἐστίν. καὶ ἐπεὶ πάλιν
ἐδείχθη ἡ AH τῇ $B\Theta$ ἴση, ἀλλὰ μὴν καὶ ἡ ZA τῇ ZB ἴση,
δύο δὲ αἱ ZA , AH δυοὶ ταῖς ZB , $B\Theta$ ἴσαι εἰσὶν. καὶ γωνία ἡ
ὑπὸ ZAH ἐδείχθη ἴση τῇ ὑπὸ $ZB\Theta$ · βάσεις ἄρα ἡ ZH βάσει
τῇ $Z\Theta$ ἐστὶν ἴση. καὶ ἐπεὶ πάλιν ἴση ἐδείχθη ἡ HE τῇ $E\Theta$,
κοινὴ δὲ ἡ EZ , δύο δὲ αἱ HE , EZ δυοὶ ταῖς ΘE , EZ ἴσαι εἰσὶν·
καὶ βάσεις ἡ ZH βάσει τῇ $Z\Theta$ ἴση· γωνία ἄρα ἡ ὑπὸ HEZ
γωνία τῇ ὑπὸ ΘEZ ἴση ἐστίν. ὀρθὴ ἄρα ἐκατέρα τῶν ὑπὸ

For since the two (straight-lines) AE and ED are equal to
the two (straight-lines) CE and EB , and they enclose equal an-
gles [Prop. 1.15], the base AD is thus equal to the base CB ,
and triangle AED will be equal to triangle CEB [Prop. 1.4].
Hence, the angle DAE [is] equal to the angle EBC . And the
angle AEG (is) also equal to the angle BEH [Prop. 1.15]. So
 AGE and BEH are two triangles having two angles equal to
two angles, respectively, and one side equal to one side—
(namely), those by the equal angles, AE and EB . Thus, they
will also have the remaining sides equal to the remaining sides
[Prop. 1.26]. Thus, GE (is) equal to EH , and AG to BH . And
since AE is equal to EB , and FE is common and at right-
angles, the base FA is thus equal to the base FB [Prop. 1.4].
So, for the same (reasons), FC is also equal to FD . And
since AD is equal to CB , and FA is also equal to FB , the two
(straight-lines) FA and AD are equal to the two (straight-lines)
 FB and BC , respectively. And the base FD was shown (to be)
equal to the base FC . Thus, the angle FAD is also equal to
the angle FBC [Prop. 1.8]. And, again, since AG was shown
(to be) equal to BH , but FA (is) also equal to FB , the two
(straight-lines) FA and AG are equal to the two (straight-lines)
 FB and BH (respectively). And the angle FAG was shown (to
be) equal to the angle FBH . Thus, the base FG is equal to the

HEZ , ΘEZ γωνιών. ἡ ZE ἄρα πρὸς τὴν $H\Theta$ τυθόντως διὰ τοῦ E ἀθθεῖσαν ὀρθή ἐστιν. ὁμοίως δὲ δείχουμεν, ὅτι ἡ ZE καὶ πρὸς πάσας τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὐσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. εὐθεῖα δὲ πρὸς ἐπίπεδον ὀρθή ἐστιν, ὅταν πρὸς πάσας τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὐσας ἐν τῷ αὐτῷ ἐπιπέδῳ ὀρθὰς ποιῇ γωνίας. ἡ ZE ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. τὸ δὲ ὑποκείμενον ἐπίπεδόν ἐστι τὸ διὰ τῶν AB , $\Gamma\Delta$ εὐθειῶν. ἡ ZE ἄρα πρὸς ὀρθὰς ἐστι τῷ διὰ τῶν AB , $\Gamma\Delta$ ἐπιπέδῳ.

Ἐάν ἄρα εὐθεῖα δύο εὐθείαις τεμνούσαις ἀλλήλας πρὸς ὀρθὰς ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ δι' αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰς ἔσται. ὅπερ ἔδει δεῖχαι.

5.

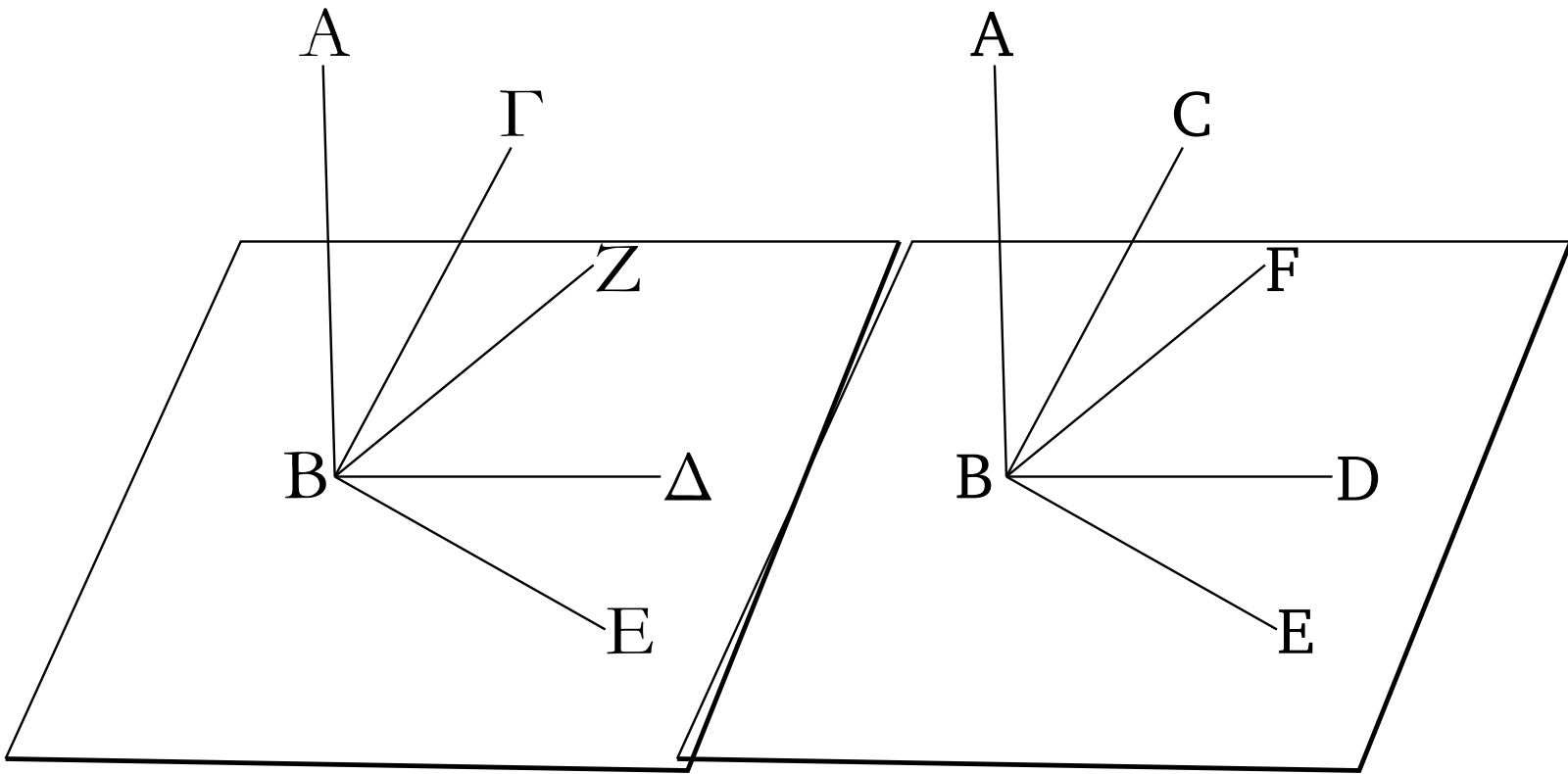
Ἐάν εὐθεῖα τρισὶν εὐθείαις ἀπτομέναις ἀλλήλων πρὸς ὀρθὰς ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἐνὶ εἰσὶν ἐπιπέδῳ.

base FH [Prop. 1.4]. And, again, since GE was shown (to be) equal to EH , and EF (is) common, the two (straight-lines) GE and EF are equal to the two (straight-lines) HE and EF (respectively). And the base FG (is) equal to the base FH . Thus, the angle GEF is equal to the angle HEF [Prop. 1.8]. Each of the angles GEF and HEF (are) thus right-angles [Def. 1.10]. Thus, FE is at right-angles to GH , which was drawn at random through E (in the reference plane passing through AB and AC). So, similarly, we can show that FE will make right-angles with all straight-lines joined to it which are in the reference plane. And a straight-line is at right-angles to a plane when it makes right-angles with all straight-lines joined to it which are in the plane [Def. 11.3]. Thus, FE is at right-angles to the reference plane. And the reference plane is that (passing) through the straight-lines AB and CD . Thus, FE is at right-angles to the plane (passing) through AB and CD .

Thus, if a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both). (Which is) the very thing it was required to show.

Proposition 5

If a straight-line is set up at right-angles to three straight-lines cutting one another, at the common point of section, then the three straight-lines are in one plane.



Εὐθεῖα γάρ τις ἡ AB τρισὶν εὐθείαις ταῖς $BΓ$, $BΔ$, BE πρὸς ὀρθὰς ἐπὶ τῆς κατὰ τὸ B ἀφ᾽ ἧς ἐφεστάτω· λέγω, ὅτι αἱ $BΓ$, $BΔ$, BE ἐν ἐνὶ εἰσὶν ἐπιπέδῳ.

μη γάρ, ἀλλ' εἰ δυνατόν, ἔστωσαν αἱ μὲν $BΔ$, BE ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, ἡ δὲ $BΓ$ ἐν μετεωροτέρῳ, καὶ ἐκβεβλήσθω τὸ διὰ τῶν AB , $BΓ$ ἐπίπεδον· κοινὴν δὴ τομὴν ποιήσῃ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιεῖτω τὴν BZ . ἐν ἐνὶ ἄρα εἰσὶν ἐπιπέδῳ τῷ διηγμένῳ διὰ τῶν AB , $BΓ$ αἱ τρεῖς εὐθεῖαι αἱ AB , $BΓ$, BZ . καὶ ἐπεὶ ἡ AB ὀρθή ἐστι πρὸς ἑκατέραν τῶν $BΔ$, BE , καὶ τῷ διὰ τῶν $BΔ$, BE ἄρα ἐπιπέδῳ ὀρθή ἐστὶν ἡ AB . τὸ δὲ διὰ τῶν $BΔ$, BE ἐπίπεδον τὸ ὑποκείμενόν ἐστιν· ἡ AB ἄρα ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὥστε καὶ πρὸς πάσας τὰς ἀπομένους αὐτῆς εὐθείας καὶ οὕσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας ἡ AB . ἀπτεται δὲ αὐτῆς ἡ BZ ὅσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ἡ ἄρα ὑπὸ ABZ γωνία ὀρθή ἐστὶν. ὑπόκειται δὲ καὶ ἡ ὑπὸ $ABΓ$ ὀρθή· ἴση ἄρα ἡ ὑπὸ ABZ γωνία τῇ ὑπὸ $ABΓ$. καὶ εἰσὶν ἐν ἐνὶ ἐπιπέδῳ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ $BΓ$ εὐθεῖα ἐν μετεωροτέρῳ ἐστὶν ἐπιπέδῳ· αἱ τρεῖς ἄρα εὐθεῖαι αἱ $BΓ$, $BΔ$, BE ἐν ἐνὶ εἰσὶν ἐπιπέδῳ.

Ἐάν ἄρα εὐθεῖα τρισὶν εὐθείαις ἀπομέναις ἀλλήλων ἐπὶ τῆς ἀφ᾽ ἧς πρὸς ὀρθὰς ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἐνὶ εἰσὶν ἐπιπέδῳ· ὅπερ ἔδει δεῖξαι.

For let some straight-line AB be set up at right-angles to three straight-lines BC , BD , and BE , at the (common) point of section B . I say that BC , BD , and BE are in one plane.

For (if) not, and if possible, let BD and BE be in the reference plane, and BC in a more elevated (plane). And let the plane through AB and BC be produced. So it will make a straight-line as a common section with the reference plane [Def. 11.3]. Let it make BF . Thus, the three straight-lines AB , BC , and BF are in one plane—(namely), that drawn through AB and BC . And since AB is at right-angles to each of BD and BE , AB is thus also at right-angles to the plane (passing) through BD and BE [Prop. 11.4]. And the plane (passing) through BD and BE is the reference plane. Thus, AB is at right-angles to the reference plane. Hence, AB will also make right-angles with all straight-lines joined to it which are also in the reference plane [Def. 11.3]. And BF , which is in the reference plane, is joined to it. Thus, the angle ABF is a right-angle. And ABC was also assumed to be a right-angle. Thus, angle ABF (is) equal to ABC . And they are in one plane. The very thing is impossible. Thus, BC is not in a more elevated plane. Thus, the three straight-lines BC , BD , and BE are in one plane.

Thus, if a straight-line is set up at right-angles to three straight-lines cutting one another, at the (common) point of section, then the three straight-lines are in one plane. (Which

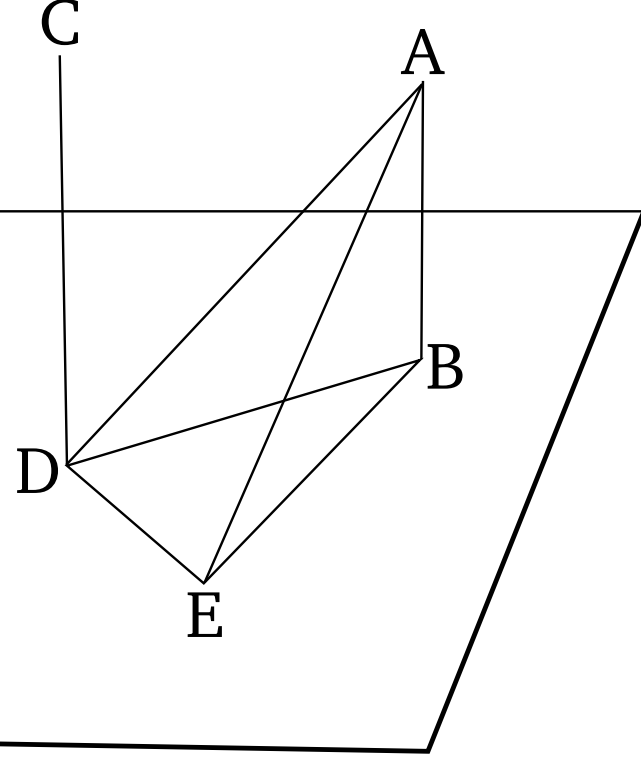
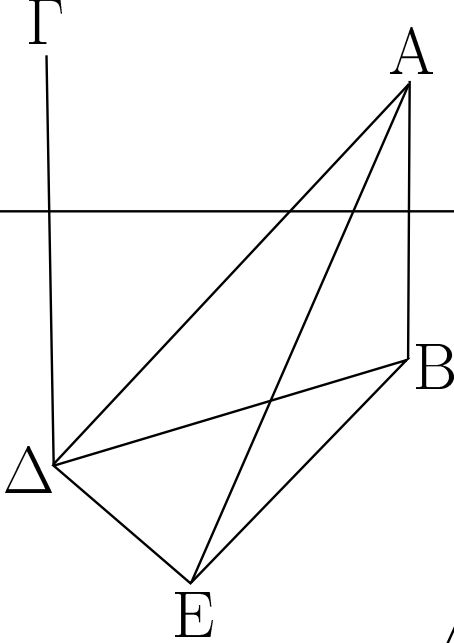
is) the very thing it was required to show.

6.

Proposition 6

Ἐάν δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾤων, παράλληλοι ἔσονται αἱ εὐθεῖαι.

If two straight-lines are at right-angles to the same plane then the straight-lines will be parallel.[†]



Δύο γὰρ εὐθεῖαι αἱ AB , $\Gamma\Delta$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἔστωσαν· λέγω, ὅτι παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$.

Συμβαλλέτωσαν γὰρ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ B , Δ σημεία, καὶ ἐπεζεύθω ἡ $B\Delta$ εὐθεῖα, καὶ ἤθθω τῇ $B\Delta$ πρὸς ὀρθὰς ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ ΔE , καὶ κείσθω τῇ AB ἴση ἡ ΔE , καὶ ἐπεζεύθωσαν αἱ BE , AE , AD .

Καὶ ἐπεὶ ἡ AB ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας [ἄρα] τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ὅσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίαξ. ἄπτεται δὲ τῇς AB ἑκάτερα τῶν $B\Delta$, BE ὅσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ὀρθὴ ἄρα ἐστὶν ἑκάτερα τῶν ὑπὸ $AB\Delta$, ABE γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερα τῶν ὑπὸ $\Gamma\Delta B$, $\Gamma\Delta E$ ὀρθὴ ἐστὶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ ΔE , κοινὴ δὲ ἡ $B\Delta$, δύο δὴ αἱ AB , $B\Delta$ δυοὶ ταῖς $E\Delta$, ΔB ἴσαι εἰσὶν· καὶ γωνίαξ ὀρθὰς περιέθουσιν· βάσεις ἄρα ἡ AD βάσει τῇ BE ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ ΔE , ἀλλὰ καὶ ἡ AD τῇ BE , δύο δὴ αἱ AB , BE δυοὶ ταῖς $E\Delta$, ΔA ἴσαι εἰσὶν· καὶ βάσεις αὐτῶν κοινὴ ἡ AE · γωνία ἄρα ἡ ὑπὸ ABE γωνιά τῇ ὑπὸ $E\Delta A$ ἐστὶν ἴση. ὀρθὴ δὲ ἡ ὑπὸ ABE · ὀρθὴ ἄρα καὶ ἡ ὑπὸ $E\Delta A$ · ἡ $E\Delta$ ἄρα πρὸς τὴν ΔA ὀρθὴ ἐστὶν. ἔστι δὲ καὶ πρὸς ἑκάτεραν τῶν $B\Delta$, $\Delta\Gamma$ ὀρθή. ἡ $E\Delta$ ἄρα τρισὶν εὐθείαις

For let the two straight-lines AB and CD be at right-angles to a reference plane. I say that AB is parallel to CD .

For let them meet the reference plane at points B and D (respectively). And let the straight-line BD be joined. And let DE be drawn at right-angles to BD in the reference plane. And let DE be made equal to AB . And let BE , AE , and AD be joined.

And since AB is at right-angles to the reference plane, it will [thus] also make right-angles with all straight-lines joined to it which are in the reference plane [Def. 11.3]. And BD and BE , which are in the reference plane, are each joined to AB . Thus, each of the angles ABD and ABE are right-angles. So, for the same (reasons), each of the angles CDB and CDE are also right-angles. And since AB is equal to DE , and BD (is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And they contain right-angles. Thus, the base AD is equal to the base BE [Prop. 1.4]. And since AB is equal to DE , and AD (is) also (equal) to BE , the two (straight-lines) AB and BE are thus equal to the two (straight-lines) ED and DA (respectively). And their base AE (is) common. Thus, angle ABE is equal

ταῖς $B\Delta$, ΔA , $\Delta\Gamma$ πρὸς ὀρθὰς ἐπὶ τῆς ἀφ᾽ ἑφ᾽ ἐφέστηκεν· αἱ τρεῖς ἄρα εὐθεῖαι αἱ $B\Delta$, ΔA , $\Delta\Gamma$ ἐν ἐνὶ εἰσιν ἐπιπέδῳ. ἐν ᾧ δὲ αἱ ΔB , ΔA , ἐν τούτῳ καὶ ἡ AB · πᾶν γὰρ τρίγωνον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ· αἱ ἄρα AB , $B\Delta$, $\Delta\Gamma$ εὐθεῖαι ἐν ἐνὶ εἰσιν ἐπιπέδῳ. καὶ ἐστὶν ὀρθὴ ἑκατέρα τῶν ὑπὸ $AB\Delta$, $B\Delta\Gamma$ γωνιῶν· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

Ἐὰν ἄρα δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾤσιν, παράλληλοι ἔσονται αἱ εὐθεῖαι· ὅπερ ἔδει δεῖχαι.

to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. Thus, EDA (is) also a right-angle. ED is thus at right-angles to DA . And it is also at right-angles to each of BD and DC . Thus, ED is standing at right-angles to the three straight-lines BD , DA , and DC at the (common) point of section. Thus, the three straight-lines BD , DA , and DC are in one plane [Prop. 11.5]. And in which(ever) plane DB and DA (are found), in that (plane) AB (will) also (be found). For every triangle is in one plane [Prop. 11.2]. And each of the angles ABD and BDC is a right-angle. Thus, AB is parallel to CD [Prop. 1.28].

Thus, if two straight-lines are at right-angles to the same plane then the straight-lines will be parallel. (Which is) the very thing it was required to show.

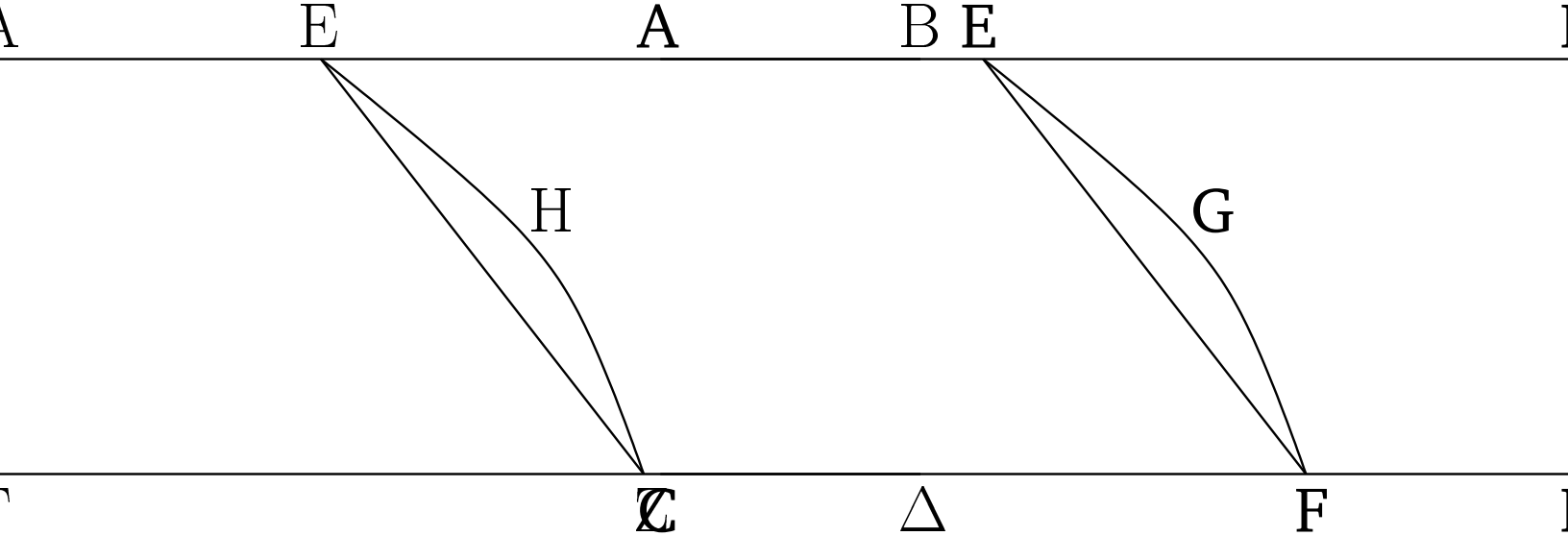
† In other words, the two straight-lines lie in the same plane, and never meet when produced in either direction.

7.

Proposition 7

Ἐὰν ᾧσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ' ἑκατέρας αὐτῶν τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιξεννυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

If there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines).



Ἐστωσαν δύο εὐθεῖαι παράλληλοι αἱ AB , $\Gamma\Delta$, καὶ ἐ-
κλήθω ἐφ' ἑκατέρας αὐτῶν τυθόντα σημεῖα τὰ E , Z · λέγω,
ὅτι ἡ ἐπὶ τὰ E , Z σημεῖα ἐπιξεννυμένη εὐθεῖα ἐν τῷ αὐτῷ
ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

μὴ γάρ, ἀλλ' εἰ δυνατόν, ἔστω ἐν μετεωροτέρῳ ὥς ἡ EHZ ,
καὶ διήθω διὰ τῆς EHZ ἐπίπεδον· τομὴν δὴ ποιήσει ἐν τῷ
ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω ὥς τὴν EZ · δύο ἄρα
εὐθεῖαι αἱ EHZ , EZ θωρίον περιέχουσιν· ὅπερ ἐστὶν ἀδύνα-
τον. οὐκ ἄρα ἡ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιξεννυμένη εὐθεῖα ἐν
μετεωροτέρῳ ἐστὶν ἐπιπέδῳ· ἐν τῷ διὰ τῶν AB , ΓB ἄρα πα-
ραλλήλων ἐστὶν ἐπιπέδῳ ἡ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιξεννυμένη
εὐθεῖα.

Ἐὰν ἄρα ᾧσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ' ἑκα-

Let AB and CD be two parallel straight-lines, and let the
random points E and F be taken on each of them (respec-
tively). I say that the straight-line joining points E and F is
in the same (reference) plane as the parallel (straight-lines).

For (if) not, and if possible, let it be in a more elevated
(plane), such as EGF . And let a plane be drawn through
 EGF . So it will make a straight cutting in the reference plane
[Prop. 11.3]. Let it make EF . Thus, two straight-lines (with
the same end-points), EGF and EF , will enclose an area. The
very thing is impossible. Thus, the straight-line joining E to F
is not in a more elevated plane. The straight-line joining E to
 F is thus in the plane through the parallel (straight-lines) AB
and CD .

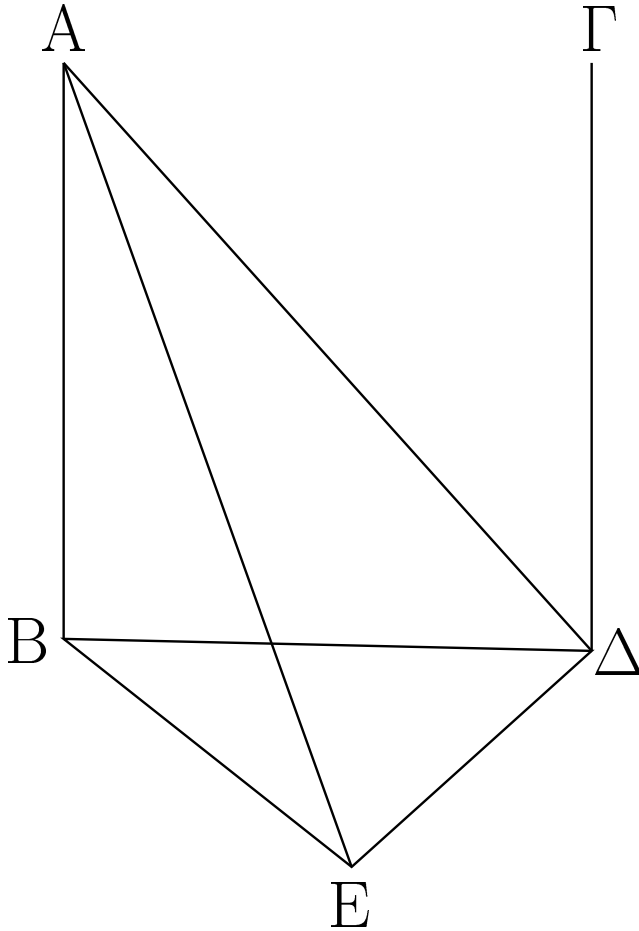
τέραξ αὐτῶν τυθόντα σημεία, ἡ ἐπὶ τὰ σημεία ἐπιζευγνυμένη εὐθεΐα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις· ὅπερ ἔδει δεῖχαι.

8.

Ἐάν ὥσι δύο εὐθεΐαι παράλληλοι, ἡ δὲ ἑτέρα αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾗ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται.

Ἐστωσαν δύο εὐθεΐαι παράλληλοι αἱ AB , $\Gamma\Delta$, ἡ δὲ ἑτέρα αὐτῶν ἡ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἔστω· λέγω, ὅτι καὶ ἡ λοιπὴ ἡ $\Gamma\Delta$ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται.

Συμβαλλέτωσαν γὰρ αἱ AB , $\Gamma\Delta$ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ B , Δ σημεία, καὶ ἐπεξέσθω ἡ $B\Delta$ · αἱ AB , $\Gamma\Delta$, $B\Delta$ ἄρα ἐν ἐνὶ εἰσιν ἐπιπέδῳ. ἦθθω τῇ BA πρὸς ὀρθὰς ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ ΔE , καὶ κείσθω τῇ AB ἴση ἡ ΔE , καὶ ἐπεξέσθωσαν αἱ BE , AE , $A\Delta$.



Καὶ ἐπεὶ ἡ AB ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῇ εὐθείας καὶ οὐσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστὶν ἡ AB · ὀρθὴ ἄρα

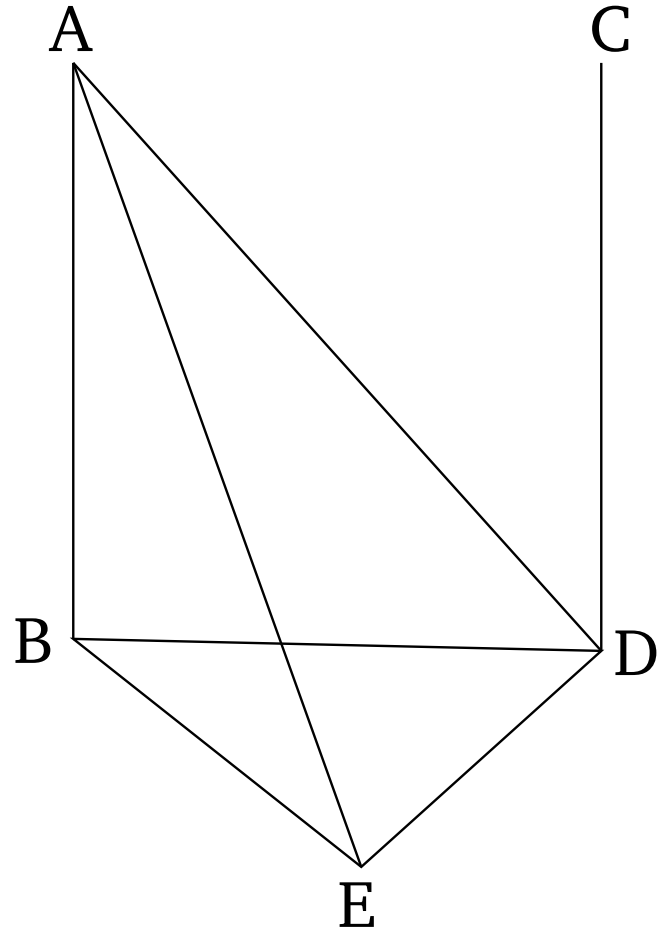
Thus, if there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines). (Which is) the very thing it was required to show.

Proposition 8

If two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane.

Let AB and CD be two parallel straight-lines, and let one of them, AB , be at right-angles to a reference plane. I say that the remaining (one), CD , will also be at right-angles to the same plane.

For let AB and CD meet the reference plane at points B and D (respectively). And let BD be joined. AB , CD , and BD are thus in one plane [Prop. 11.7]. Let DE be drawn at right-angles to BD in the reference plane, and let DE be made equal to AB , and let BE , AE , and AD be joined.



And since AB is at right-angles to the reference plane, AB is thus also at right-angles to all of the straight-lines joined to it which are in the reference plane [Def. 11.3]. Thus, the

[ἐστίν] ἑκατέρω τῶν ὑπὸ $AB\Delta$, ABE γωνιῶν. καὶ ἐπεὶ εἰς παραλλήλων τὰς AB , $\Gamma\Delta$ εὐθεῖα ἐμπέπτωκεν ἡ $B\Delta$, αἱ ἄρα ὑπὸ $AB\Delta$, $\Gamma\Delta B$ γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσίν. ὀρθή δὲ ἡ ὑπὸ $AB\Delta$ · ὀρθή ἄρα καὶ ἡ ὑπὸ $\Gamma\Delta B$ · ἡ $\Gamma\Delta$ ἄρα πρὸς τὴν $B\Delta$ ὀρθή ἐστίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ ΔE , κοινὴ δὲ ἡ $B\Delta$, δύο δὴ αἱ AB , $B\Delta$ δυοῖν ταῖς $E\Delta$, ΔB ἴσαι εἰσίν· καὶ γωνία ἡ ὑπὸ $AB\Delta$ γωνία τῇ ὑπὸ $E\Delta B$ ἴση· ὀρθή γὰρ ἑκατέρω· βάσεις ἄρα ἡ $A\Delta$ βάσει τῇ BE ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν AB τῇ ΔE , ἡ δὲ BE τῇ $A\Delta$, δύο δὴ αἱ AB , BE δυοῖν ταῖς $E\Delta$, ΔA ἴσαι εἰσίν ἑκατέρω ἑκατέρω. καὶ βάσεις αὐτῶν κοινὴ ἡ AE · γωνία ἄρα ἡ ὑπὸ ABE γωνία τῇ ὑπὸ $E\Delta A$ ἐστὶν ἴση. ὀρθή δὲ ἡ ὑπὸ ABE · ὀρθή ἄρα καὶ ἡ ὑπὸ $E\Delta A$ · ἡ $E\Delta$ ἄρα πρὸς τὴν $A\Delta$ ὀρθή ἐστίν. ἔστι δὲ καὶ πρὸς τὴν ΔB ὀρθή· ἡ $E\Delta$ ἄρα καὶ τῷ διὰ τῶν $B\Delta$, ΔA ἐπιπέδῳ ὀρθή ἐστίν. καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὕσας ἐν τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας ἡ $E\Delta$. ἐν δὲ τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ἐστὶν ἡ $\Delta\Gamma$, ἐπειδήπερ ἐν τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ἐστὶν αἱ AB , $B\Delta$, ἐν ᾧ δὲ αἱ AB , $B\Delta$, ἐν τούτῳ ἐστὶ καὶ ἡ $\Delta\Gamma$. ἡ $E\Delta$ ἄρα τῇ $\Delta\Gamma$ πρὸς ὀρθὰς ἐστίν· ὥστε καὶ ἡ $\Gamma\Delta$ τῇ ΔE πρὸς ὀρθὰς ἐστίν. ἔστι δὲ καὶ ἡ $\Gamma\Delta$ τῇ $B\Delta$ πρὸς ὀρθὰς. ἡ $\Gamma\Delta$ ἄρα δύο εὐθείαις τεμνούσαις ἀλλήλαις ταῖς ΔE , ΔB ἀπὸ τῆς κατὰ τὸ Δ τομῆς πρὸς ὀρθὰς ἐφέστηκεν· ὥστε ἡ $\Gamma\Delta$ καὶ τῷ διὰ τῶν ΔE , ΔB ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. τὸ δὲ διὰ τῶν ΔE , ΔB ἐπίπεδον τὸ ὑποκείμενόν ἐστιν ἡ $\Gamma\Delta$ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

Ἐὰν ἄρα ᾧσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾗ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· ὅπερ ἔδει δεῖχαι.

9.

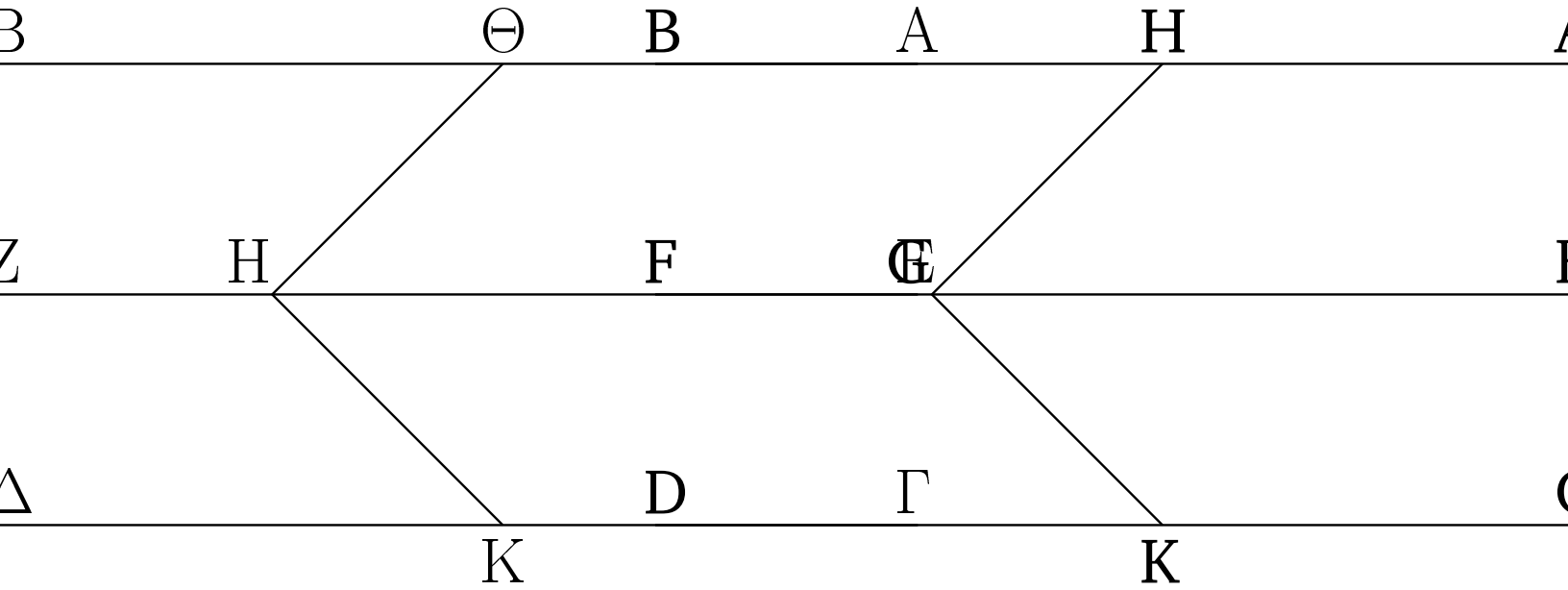
Αἱ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ μὴ οὕσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσι παράλληλοι.

angles ABD and ABE [are] each right-angles. And since the straight-line BD has met the parallel (straight-lines) AB and CD , the (sum of the) angles ABD and CDB is thus equal to two right-angles [Prop. 1.29]. And ABD (is) a right-angle. Thus, CDB (is) also a right-angle. CD is thus at right-angles to BD . And since AB is equal to DE , and BD (is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And angle ABD (is) equal to angle EDB . For each (is) a right-angle. Thus, the base AD (is) equal to the base BE [Prop. 1.4]. And since AB is equal to DE , and BE to AD , the two (sides) AB , BE are equal to the two (sides) ED , DA , respectively. And their base AE is common. Thus, angle ABE is equal to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. EDA (is) thus also a right-angle. Thus, ED is at right-angles to AD . And it is also at right-angles to DB . Thus, ED is also at right-angles to the plane through BD and DA [Prop. 11.4]. And ED will thus make right-angles with all of the straight-lines joined to it which are also in the plane through BDA . And DC is in the plane through BDA , inasmuch as AB and BD are in the plane through BDA [Prop. 11.2], and in which (ever plane) AB and BD (are found), DC is also (found). Thus, ED is at right-angles to DC . Hence, CD is also at right-angles to DE . And CD is also at right-angles to BD . Thus, CD is standing at right-angles to two straight-lines, DE and DB , which meet one another, at the (point) of section, D . Hence, CD is also at right-angles to the plane through DE and DB [Prop. 11.4]. And the plane through DE and DB is the reference (plane). CD is thus at right-angles to the reference plane.

Thus, if two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

Proposition 9

(Straight-lines) parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another.



Ἐστω γάρ ἑκατέρα τῶν AB , $\Gamma\Delta$ τῇ EZ παράλληλος μὴ οὔσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ· λέγω, ὅτι παράλληλός ἐστιν ἡ AB τῇ $\Gamma\Delta$.

Εἰλήφθω γάρ ἐπὶ τῆς EZ τυθὸν σημεῖον τὸ H , καὶ ἀπ' αὐτοῦ τῇ EZ ἐν μὲν τῷ διὰ τῶν EZ , AB ἐπιπέδῳ πρὸς ὀρθὰς ἦθθω ἡ $H\Theta$, ἐν δὲ τῷ διὰ τῶν ZE , $\Gamma\Delta$ τῇ EZ πάλιν πρὸς ὀρθὰς ἦθθω ἡ HK .

Καὶ ἐπεὶ ἡ EZ πρὸς ἑκατέραν τῶν $H\Theta$, HK ὀρθή ἐστιν, ἡ EZ ἄρα καὶ τῷ διὰ τῶν $H\Theta$, HK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. καὶ ἐστιν ἡ EZ τῇ AB παράλληλος· καὶ ἡ AB ἄρα τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ $\Gamma\Delta$ τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν· ἑκατέρα ἄρα τῶν AB , $\Gamma\Delta$ τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθὰς ἐστιν. ἔαν δὲ δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾤσιν, παράλληλοί εἰσιν αἱ εὐθεῖαι· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$ · ὅπερ ἔδει δεῖχαι.

10.

Ἐὰν δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων ᾧσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσαξ γωνίαξ περιέχουσιν.

Δύο γάρ εὐθεῖαι αἱ AB , $B\Gamma$ ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας τὰς ΔE , EZ ἀπτομένας ἀλλήλων ἔστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ· λέγω, ὅτι ἴση ἐστὶν ἡ ὑπὸ $AB\Gamma$ γωνία τῇ ὑπὸ ΔEZ .

Ἀπειλήφθωσαν γάρ αἱ BA , $B\Gamma$, $E\Delta$, EZ ἴσαι ἀλλήλαις, καὶ ἐπεξεὐθῆθωσαν αἱ $A\Delta$, ΓZ , BE , $A\Gamma$, ΔZ .

Καὶ ἐπεὶ ἡ BA τῇ $E\Delta$ ἴση ἐστὶ καὶ παράλληλος, καὶ ἡ $A\Delta$ ἄρα τῇ BE ἴση ἐστὶ καὶ παράλληλος. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΓZ τῇ BE ἴση ἐστὶ καὶ παράλληλος· ἑκατέρα ἄρα τῶν $A\Delta$, ΓZ τῇ BE ἴση ἐστὶ καὶ παράλληλος. αἱ δὲ τῇ αὐτῇ

For let AB and CD each be parallel to EF , not being in the same plane as it. I say that AB is parallel to CD .

For let some point G be taken at random on EF . And from it let GH be drawn at right-angles to EF in the plane through EF and AB . And let GK be drawn, again at right-angles to EF , in the plane through FE and CD .

And since EF is at right-angles to each of GH and GK , EF is thus also at right-angles to the plane through GH and GK [Prop. 11.4]. And EF is parallel to AB . Thus, AB is also at right-angles to the plane through HGK [Prop. 11.8]. So, for the same (reasons), CD is also at right-angles to the plane through HGK . Thus, AB and CD are each at right-angles to the plane through HGK . And if two straight-lines are at right-angles to the same plane then the straight-lines are parallel [Prop. 11.6]. Thus, AB is parallel to CD . (Which is) the very thing it was required to show.

Proposition 10

If two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles.

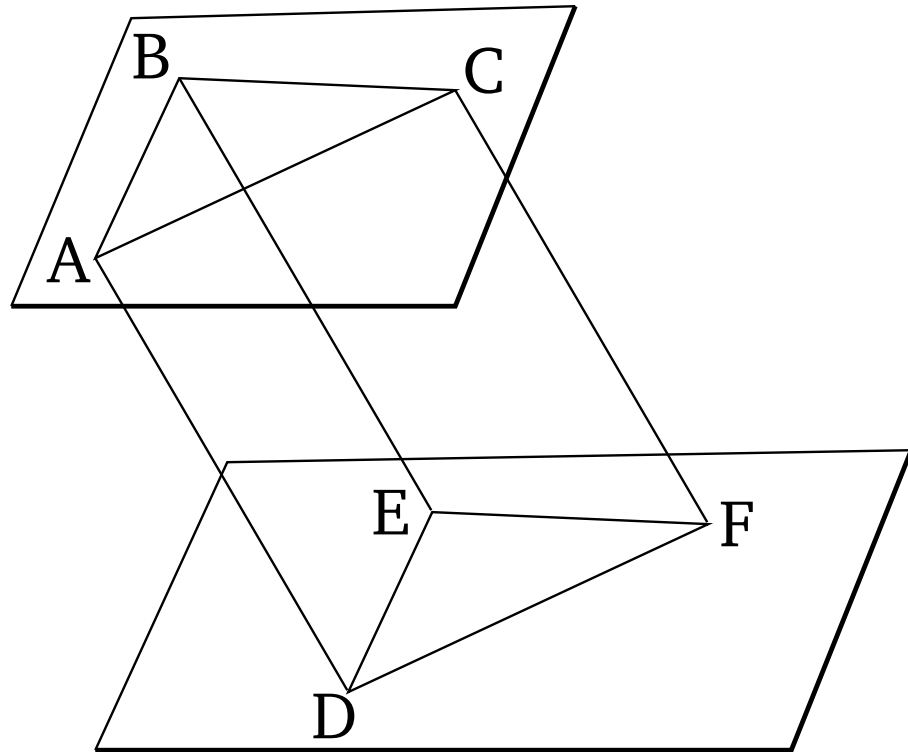
For let the two straight-lines joined to one another, AB and BC , be (respectively) parallel to the two straight-lines joined to one another, DE and EF , (but) not in the same plane. I say that angle ABC is equal to (angle) DEF .

For let BA , BC , ED , and EF be cut off (so as to be, respectively) equal to one another. And let AD , CF , BE , AC , and DF be joined.

And since BA is equal and parallel to ED , AD is thus also equal and parallel to BE [Prop. 1.33]. So, for the same reasons, CF is also equal and parallel to BE . Thus, AD and CF are

εὐθεία παράλληλοι καὶ μὴ οὔσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσι παράλληλοι· παράλληλος ἄρα ἐστὶν ἡ AD τῇ $ΓΖ$ καὶ ἴση. καὶ ἐπιξενγνύουσιν αὐτάς αἱ $ΑΓ$, $ΔΖ$ · καὶ ἡ $ΑΓ$ ἄρα τῇ $ΔΖ$ ἴση ἐστὶ καὶ παράλληλος. καὶ ἐπεὶ δύο αἱ AB , $ΒΓ$ δυοὶ ταῖς $ΔΕ$, $ΕΖ$ ἴσαι εἰσὶν, καὶ βάσις ἡ $ΑΓ$ βάσει τῇ $ΔΖ$ ἴση, γωνία ἄρα ἡ ὑπὸ $ΑΒΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ ἐστὶν ἴση.

each equal and parallel to BE . And straight-lines parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another [Prop. 11.9]. Thus, AD is parallel and equal to CF . And AC and DF join them. Thus, AC is also equal and parallel to DF [Prop. 1.33]. And since the two (straight-lines) AB and BC are equal to the two (straight-lines) DE and EF (respectively), and the base AC (is) equal to the base DF , the angle ABC is thus equal to the (angle) DEF [Prop. 1.8].



Book11/fig10g.pdf

Ἐάν ἄρα δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθεῖας ἀπτομένας ἀλλήλων ὥσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσαξ γωνίαξ περιέχουσιν· ὅπερ ἔδει δεῖχαι.

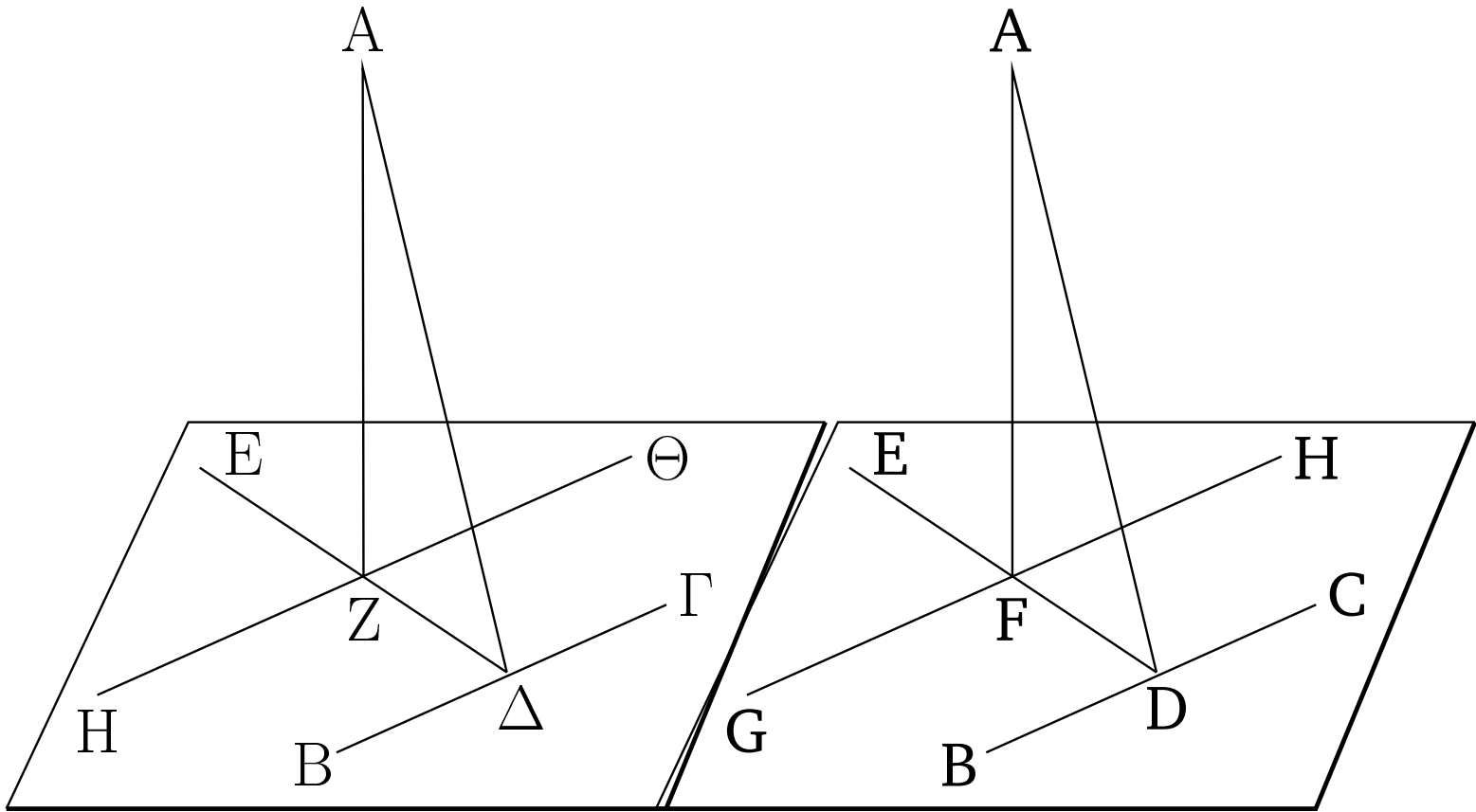
Thus, if two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles. (Which is) the very thing it was required to show.

11.

Ἀπὸ τοῦ δοθέντος σημείου μετεώρου ἐπὶ τὸ δοθὲν ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Proposition 11

To draw a perpendicular straight-line from a given raised point to a given plane.



Ἐστω τὸ μὲν δοθὲν σημεῖον μετέωρον τὸ A , τὸ δὲ δοθὲν ἐπίπεδον τὸ ὑποκείμενον· δεῖ δὴ ἀπὸ τοῦ A σημείου ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Διήθθω γάρ τις ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖα, ὡς ἔτυθεν, ἡ $BΓ$, καὶ ἤθθω ἀπὸ τοῦ A σημείου ἐπὶ τὴν $BΓ$ κάθετος ἡ $ΑΔ$. εἰ μὲν οὖν ἡ $ΑΔ$ κάθετός ἐστι καὶ ἐπὶ τὸ ὑποκείμενον ἐπίπεδον, γερονὸς ἂν εἴη τὸ ἐπιταθὲν. εἰ δὲ οὐ, ἤθθω ἀπὸ τοῦ $Δ$ σημείου τῇ $BΓ$ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἡ $ΔΕ$, καὶ ἤθθω ἀπὸ τοῦ A ἐπὶ τὴν $ΔΕ$ κάθετος ἡ $ΑΖ$, καὶ διὰ τοῦ Z σημείου τῇ $BΓ$ παράλληλος ἤθθω ἡ $ΗΘ$.

Καὶ ἐπεὶ ἡ $BΓ$ ἑκατέρω τῶν $ΔΑ$, $ΔΕ$ πρὸς ὀρθὰς ἐστίν, ἡ $BΓ$ ἄρα καὶ τῷ διὰ τῶν $ΕΔΑ$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. καὶ ἐστὶν αὐτῇ παράλληλος ἡ $ΗΘ$. ἔαν δὲ ὧσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾗ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· καὶ ἡ $ΗΘ$ ἄρα τῷ διὰ τῶν $ΕΔ$, $ΔΑ$ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. καὶ πρὸς πάσαις ἄρα τὰς ἀποτομένας αὐτῆς εὐθείαις καὶ οὕσαι ἐν τῷ διὰ τῶν $ΕΔ$, $ΔΑ$ ἐπιπέδῳ ὀρθαὶ ἐστὶν ἡ $ΗΘ$. ἄπτεται δὲ αὐτῆς ἡ $ΑΖ$ οὕσα ἐν τῷ διὰ τῶν $ΕΔ$, $ΔΑ$ ἐπιπέδῳ· ἡ $ΗΘ$ ἄρα ὀρθαὶ ἐστὶ πρὸς τὴν $ΖΑ$. ὥστε καὶ ἡ $ΖΑ$ ὀρθαὶ ἐστὶ πρὸς τὴν $ΘΗ$. ἔστι δὲ ἡ $ΑΖ$ καὶ πρὸς τὴν $ΔΕ$ ὀρθαὶ· ἡ $ΑΖ$ ἄρα πρὸς ἑκατέραν τῶν $ΗΘ$, $ΔΕ$ ὀρθαὶ ἐστίν. ἔαν δὲ εὐθεῖα δυνὸν εὐθείαις τεμνοῦσαις ἀλλήλας ἐπὶ τῇς τομῆς πρὸς ὀρθὰς ἐπισταθῇ, καὶ τῷ δι' αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· ἡ $ΖΑ$ ἄρα τῷ διὰ τῶν $ΕΔ$, $ΗΘ$

Let A be the given raised point, and the given plane the reference (plane). So, it is required to draw a perpendicular straight-line from point A to the reference plane.

Let some random straight-line BC be drawn across in the reference plane, and let the (straight-line) AD be drawn from point A perpendicular to BC [Prop. 1.12]. If, therefore, AD is also perpendicular to the reference plane then that which was prescribed will have occurred. And, if not, let DE be drawn in the reference plane from point D at right-angles to BC [Prop. 1.11], and let the (straight-line) AF be drawn from A perpendicular to DE [Prop. 1.12], and let GH be drawn through point F , parallel to BC [Prop. 1.31].

And since BC is at right-angles to each of DA and DE , BC is thus also at right-angles to the plane through EDA [Prop. 11.4]. And GH is parallel to it. And if two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (straight-line) will also be at right-angles to the same plane [Prop. 11.8]. Thus, GH is also at right-angles to the plane through ED and DA . And GH is thus at right-angles to all of the straight-lines joined to it which are also in the plane through ED and AD [Def. 11.3]. And AF , which is in the plane through ED and DA , is joined to it. Thus, GH is at right-angles to FA . Hence, FA is also at right-angles to HG . And AF is also at right-angles to DE . Thus, AF is at

ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν. τὸ δὲ διὰ τῶν $ΕΔ$, $ΗΘ$ ἐπίπεδόν ἐστι τὸ ὑποκείμενον· ἡ AZ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

Ἀπὸ τοῦ ἄρα δοθέντος σημείου μετέωρον τοῦ A ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος εὐθεῖα γραμμὴ ἦκται ἡ AZ · ὅπερ ἔδει ποιῆσαι.

right-angles to each of GH and DE . And if a straight-line is set up at right-angles to two straight-lines cutting one another, at the point of section, then it will also be at right-angles to the plane through them [Prop. 11.4]. Thus, FA is at right-angles to the plane through ED and GH . And the plane through ED and GH is the reference (plane). Thus, AF is at right-angles to the reference plane.

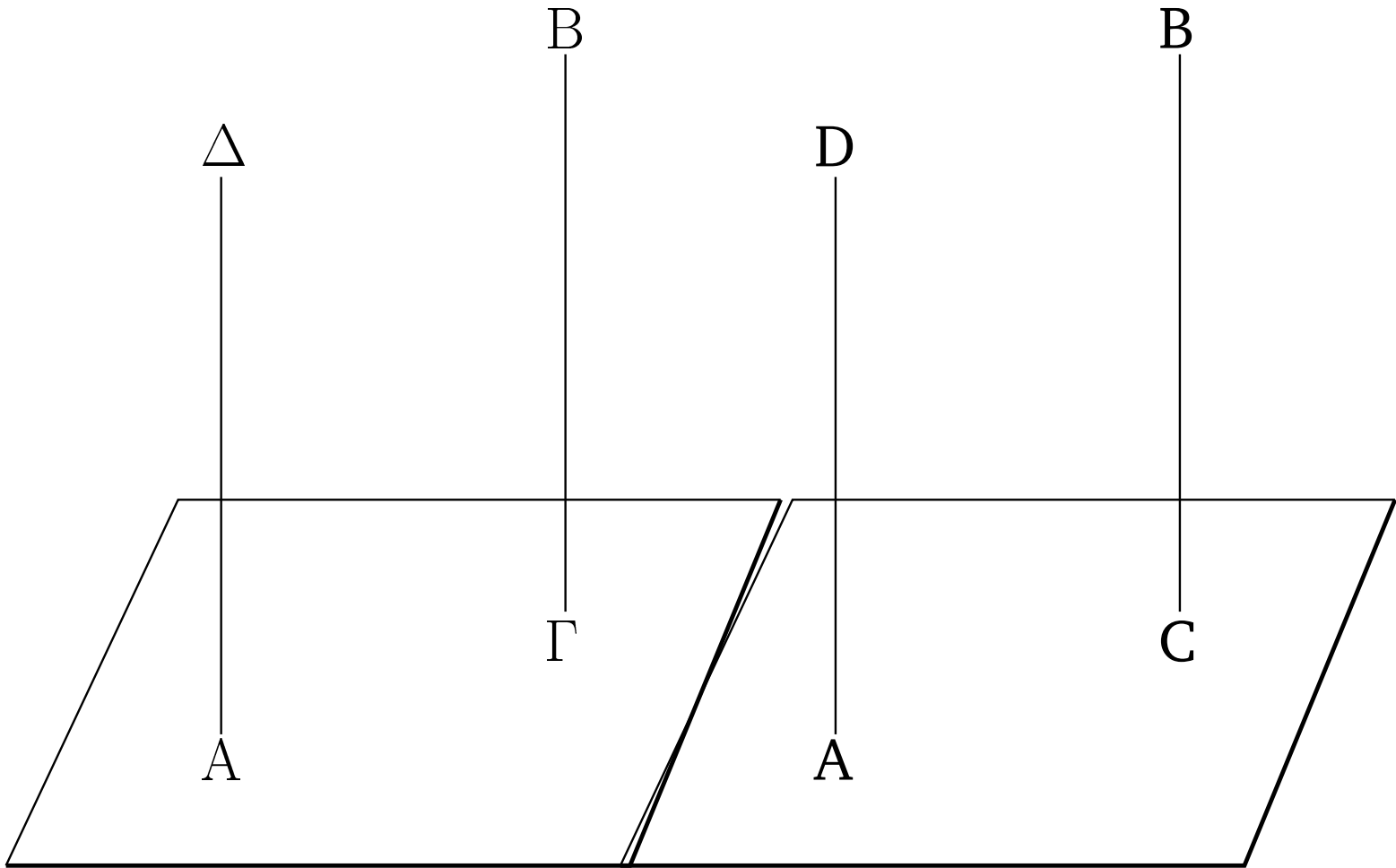
Thus, the straight-line AF has been drawn from the given raised point A perpendicular to the reference plane. (Which is) the very thing it was required to do.

12.

Τῷ δοθέντι ἐπιπέδῳ ἀπὸ τοῦ πρὸς αὐτῷ δοθέντος σημείου πρὸς ὀρθὰς εὐθεῖαν γραμμὴν ἀναστήσαι.

Proposition 12

To set up a straight-line at right-angles to a given plane from a given point in it.



Ἐστω τὸ μὲν δοθέν ἐπίπεδον τὸ ὑποκείμενον, τὸ δὲ πρὸς αὐτῷ σημείον τὸ A · δεῖ δὴ ἀπὸ τοῦ A σημείου τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς εὐθεῖαν γραμμὴν ἀναστήσαι.

Νενοήσθω τι σημείον μετέωρον τὸ B , καὶ ἀπὸ τοῦ B ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος ἦθθω ἡ $ΒΓ$, καὶ διὰ τοῦ A

Let the given plane be the reference (plane), and A a point in it. So, it is required to set up a straight-line at right-angles to the reference plane at point A .

Let some raised point B be assumed, and let the perpendicular (straight-line) BC be drawn from B to the reference plane

σημείον τῇ $B\Gamma$ παράλληλος ἦθθω ἡ $A\Delta$.

Ἐπεὶ οὖν δύο εὐθεῖαι παράλληλοί εἰσιν αἱ $A\Delta$, ΓB , ἡ δὲ μία αὐτῶν ἡ $B\Gamma$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν, καὶ ἡ λοιπὴ ἄρα ἡ $A\Delta$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

Τῷ ἄρα δοθέντι ἐπιπέδῳ ἀπὸ τοῦ πρὸς αὐτῷ σημείου τοῦ A πρὸς ὀρθὰς ἀνέσταται ἡ $A\Delta$. ὅπερ ἔδει ποιῆσαι.

[Prop. 11.11]. And let AD be drawn from point A parallel to BC [Prop. 1.31].

Therefore, since AD and CB are two parallel straight-lines, and one of them, BC , is at right-angles to the reference plane, the remaining (one) AD is thus also at right-angles to the reference plane [Prop. 11.8].

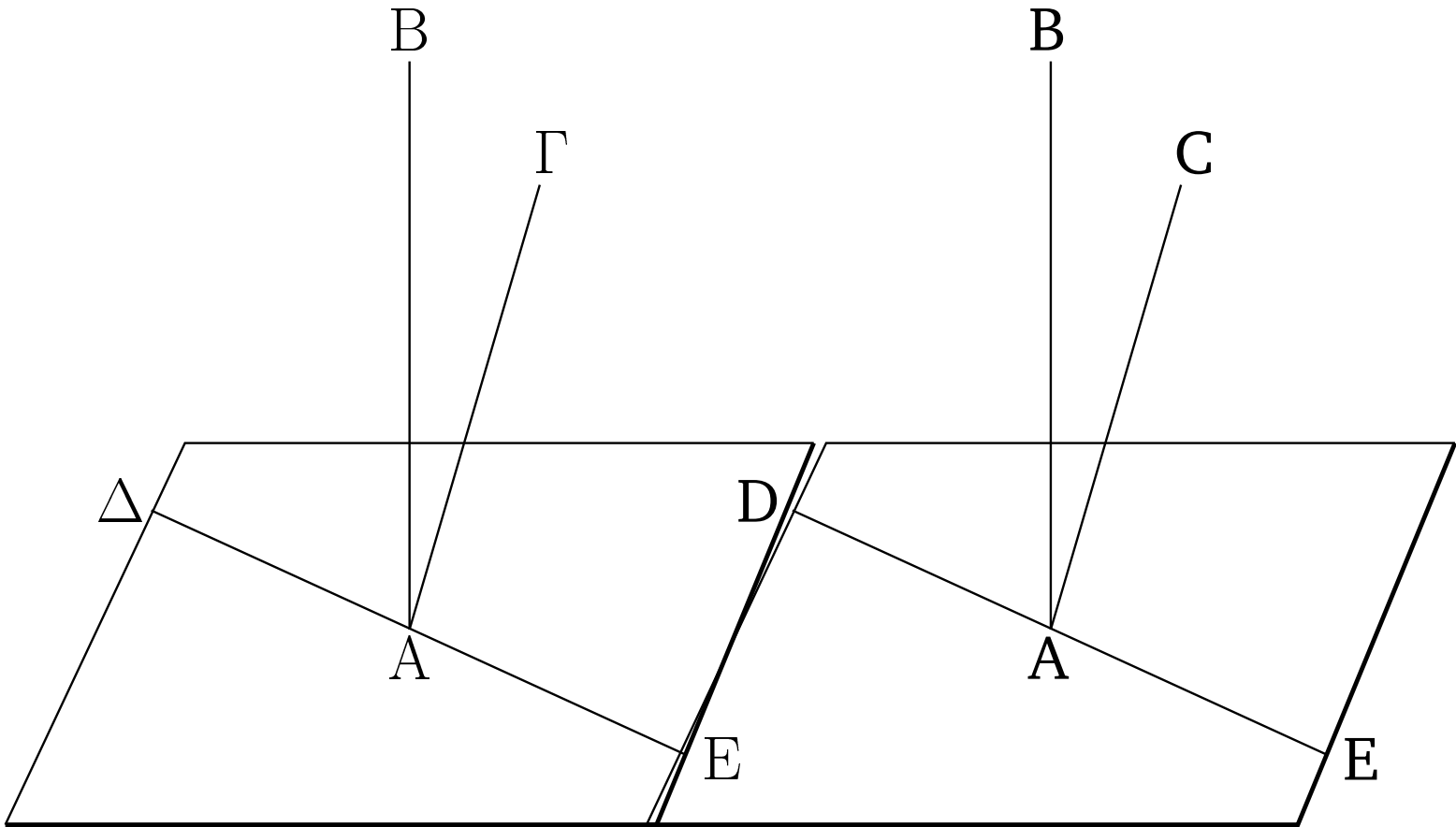
Thus, AD has been set up at right-angles to the given plane, from the point in it, A . (Which is) the very thing it was required to do.

13.

Ἀπὸ τοῦ αὐτοῦ σημείου τῷ αὐτῷ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰς οὐκ ἀναστήσονται ἐπὶ τὰ αὐτὰ μέρος.

Proposition 13

Two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side.



Εἰ γὰρ δυνατόν, ἀπὸ τοῦ αὐτοῦ σημείου τοῦ A τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖαι αἱ AB , $B\Gamma$ πρὸς ὀρθὰς ἀνεστάτωσαν ἐπὶ τὰ αὐτὰ μέρη, καὶ διήθθω τὸ διὰ τῶν BA , AG ἐπίπεδον· τομὴν δὲ ποιήσει διὰ τοῦ A ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω τὴν ΔAE . αἱ ἄρα AB , AG , ΔAE εὐθεῖαι ἐν ἐν εἰσιν ἐπιπέδῳ. καὶ ἐπεὶ ἡ ΓA τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν, καὶ πρὸς πάσαις ἄρα τὰς ἀπτομένης αὐτῆς εὐθείαις καὶ ὅσαις ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. ἄπτεται δὲ αὐτῆς ἡ ΔAE ὅσα ἐν τῷ

For, if possible, let the two straight-lines AB and AC be set up at the same point A at right-angles to the reference plane, on the same side. And let the plane through BA and AC be drawn. So it will make a straight cutting (passing) through (point) A in the reference plane [Prop. 11.3]. Let it have made DAE . Thus, AB , AC , and DAE are straight-lines in one plane. And since CA is at right-angles to the reference plane, it will thus also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. And DAE ,

ὑποκειμένῳ ἐπιπέδῳ ἡ ἄρα ὑπὸ ΓΑΕ γωνία ὀρθή ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΒΑΕ ὀρθή ἐστιν· ἴση ἄρα ἡ ὑπὸ ΓΑΕ τῇ ὑπὸ ΒΑΕ καὶ εἰσιν ἐν ἐνὶ ἐπιπέδῳ· ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἀπὸ τοῦ αὐτοῦ σημείου τῶν αὐτῶν ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰς ἀνασταθήσονται ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἔδει δεῖχαι.

which is in the reference plane, is joined to it. Thus, angle CAE is a right-angle. So, for the same (reasons), BAE is also a right-angle. Thus, CAE (is) equal to BAE . And they are in one plane. The very thing is impossible.

Thus, two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side. (Which is) the very thing it was required to show.

14.

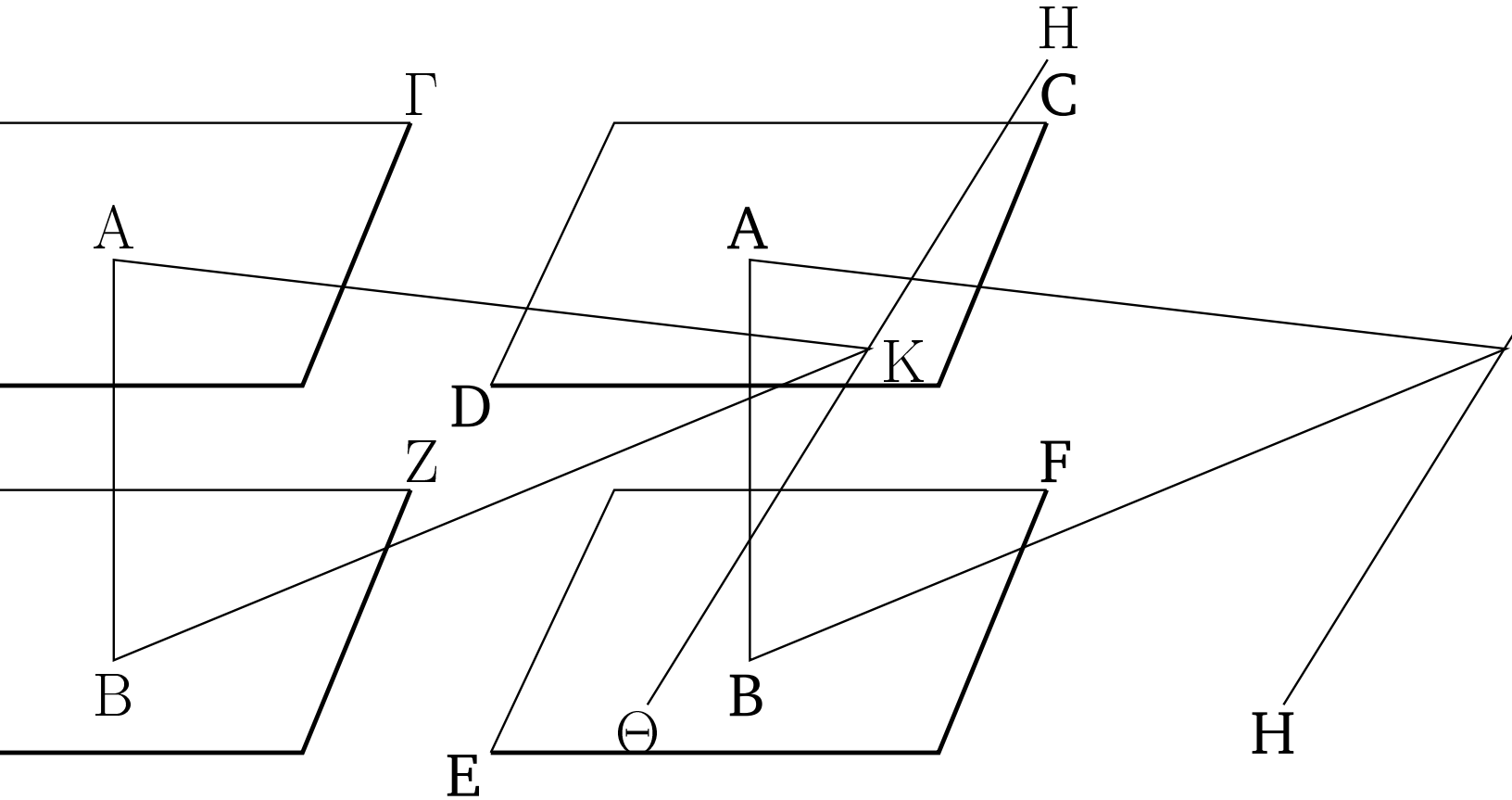
Proposition 14

Πρὸς ᾧ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθὴ ἐστίν, παράλληλα ἔσται τὰ ἐπίπεδα.

Εὐθεῖα γάρ τις ἡ AB πρὸς ἑκάτερον τῶν $\Gamma\Delta$, EZ ἐπιπέδων πρὸς ὀρθὰς ἔστω· λέγω, ὅτι παράλληλά ἐστι τὰ ἐπίπεδα.

Planes to which the same straight-line is at right-angles will be parallel planes.

For let some straight-line AB be at right-angles to each of the planes CD and EF . I say that the planes are parallel.



Εἰ γὰρ μή, ἐκβαλλόμενα συμπεσοῦνται. συμπιπτέτωσαν· ποιήσουσι δὴ κοινὴν τομὴν εὐθεῖαν. ποιείτωσαν τὴν $H\Theta$, καὶ εἰλήφθω ἐπὶ τῆς $H\Theta$ τυθὸν σημεῖον τὸ K , καὶ ἐπεζεύθωσαν αἱ AK , BK .

Καὶ ἐπεὶ ἡ AB ὀρθὴ ἐστὶ πρὸς τὸ EZ ἐπίπεδον, καὶ πρὸς τὴν BK ἄρα εὐθεῖαν ὅσον ἐν τῷ EZ ἐκβληθέντι ἐπιπέδῳ ὀρθὴ ἐστὶν ἡ AB · ἡ ἄρα ὑπὸ ABK γωνία ὀρθή ἐστιν. διὰ τὰ

For, if not, being produced, they will meet. Let them have met. So they will make a straight-line as a common section [Prop. 11.3]. Let them have made GH . And let some random point K be taken on GH . And let AK and BK be joined.

And since AB is at right-angles to the plane EF , AB is thus also at right-angles to BK , which is a straight-line in the produced plane EF [Def. 11.3]. Thus, angle ABK is a right-

αὐτὰ δὴ καὶ ἡ ὑπὸ BAK ὀρθή ἐστιν. τριγώνον δὴ τοῦ ABK αἱ δύο γωνίαι αἱ ὑπὸ ABK , BAK δυσὶν ὀρθαῖς εἰσι ἴσαι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ $\Gamma\Delta$, EZ ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται· παράλληλα ἄρα ἐστὶ τὰ $\Gamma\Delta$, EZ ἐπίπεδα.

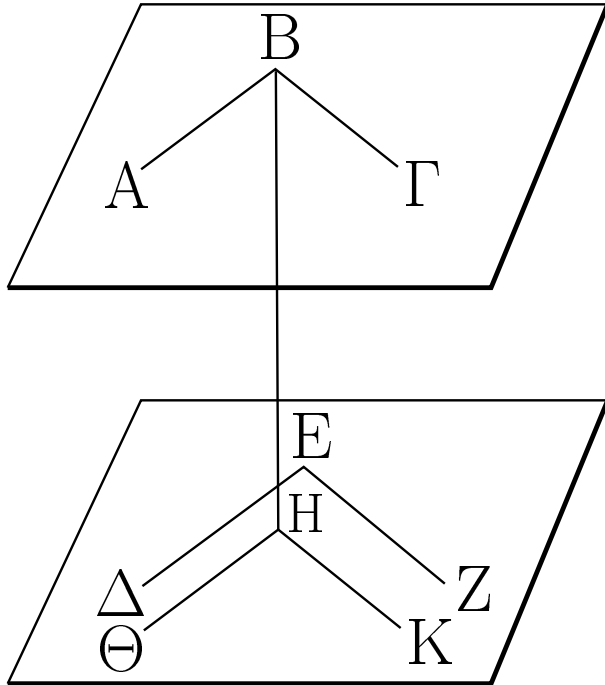
Πρὸς ἃ ἐπίπεδα ἄρα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλά ἐστι τὰ ἐπίπεδα· ὅπερ ἔδει δεῖχαι.

15.

Ἐάν δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων ὥσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ οὔσαι, παράλληλά ἐστι τὰ δι' αὐτῶν ἐπίπεδα.

Δύο γάρ εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ AB , $B\Gamma$ παρὰ δύο εὐθείας ἀπτομένας ἀλλήλων τὰς ΔE , EZ ἔστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ οὔσαι· λέγω, ὅτι ἐκβαλλόμενα τὰ διὰ τῶν AB , $B\Gamma$, ΔE , EZ ἐπίπεδα οὐ συμπεσεῖται ἀλλήλοις.

Ἦθθω γάρ ἀπὸ τοῦ B σημείου ἐπὶ τὸ διὰ τῶν ΔE , EZ ἐπίπεδον κάθετος ἡ BH καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ H σημεῖον, καὶ διὰ τοῦ H τῇ μὲν $E\Delta$ παράλληλος ἡ $\Theta\Delta$ ἢ $H\Theta$, τῇ δὲ EZ ἡ HK .



Καὶ ἐπεὶ ἡ BH ὀρθή ἐστι πρὸς τὸ διὰ τῶν ΔE , EZ ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ οὔσαι ἐν τῷ διὰ τῶν ΔE , EZ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. ἄπτεται δὲ αὐτῆς ἑκατέρα τῶν $H\Theta$, HK οὔσα ἐν τῷ διὰ τῶν ΔE , EZ ἐπιπέδῳ· ὀρθή ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ $BH\Theta$,

angle. So, for the same (reasons), BAK is also a right-angle. So the (sum of the) two angles ABK and BAK in the triangle ABK is equal to two right-angles. The very thing is impossible [Prop. 1.17]. Thus, planes CD and EF , being produced, will not meet. Planes CD and EF are thus parallel [Def. 11.8].

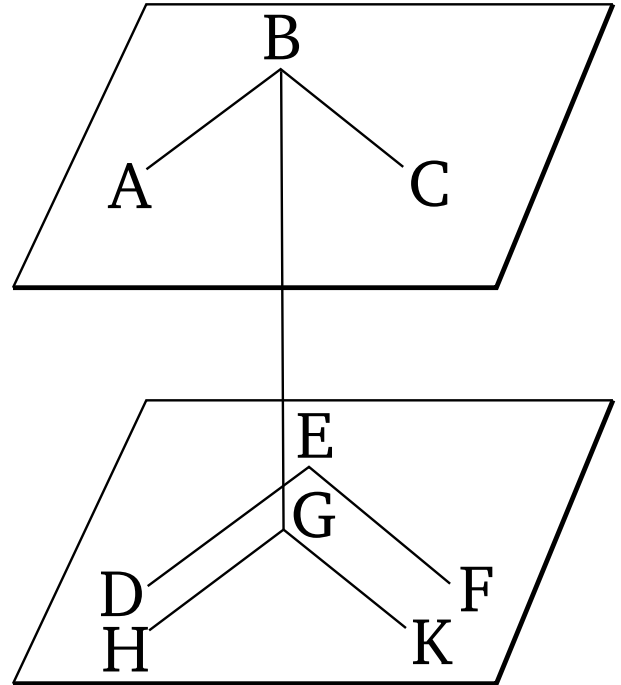
Thus, planes to which the same straight-line is at right-angles are parallel planes. (Which is) the very thing it was required to show.

Proposition 15

If two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another).

For let the two straight-lines joined to one another, AB and BC , be parallel to the two straight-lines joined to one another, DE and EF (respectively), not being in the same plane. I say that the planes through AB , BC and DE , EF will not meet one another (when) produced.

For let BG be drawn from point B perpendicular to the plane through DE and EF [Prop. 11.11], and let it meet the plane at point G . And let GH be drawn through G parallel to ED , and GK (parallel) to EF [Prop. 1.31].



And since BG is at right-angles to the plane through DE and EF , it will thus also make right-angles with all of the straight-lines joined to it, which are also in the plane through DE and EF [Def. 11.3]. And each of GH and GK , which are in the plane through DE and EF , are joined to it. Thus, each

BHK γωνιών. καὶ ἐπεὶ παράλληλός ἐστιν ἡ BA τῇ $HΘ$, αἱ ἄρα ὑπὸ HBA , $BHΘ$ γωνίαι δυοὶν ὀρθαῖς ἴσαι εἰσὶν. ὀρθὴ δὲ ἡ ὑπὸ $BHΘ$. ὀρθὴ ἄρα καὶ ἡ ὑπὸ HBA . ἡ HB ἄρα τῇ BA πρὸς ὀρθάξ ἐστιν. διὰ τὰ αὐτὰ δὲ ἡ HB καὶ τῇ $ΒΓ$ ἐστι πρὸς ὀρθάξ. ἐπεὶ οὖν εὐθεῖα ἡ HB δυοὶν εὐθείαις ταῖς BA , $ΒΓ$ τεμνούσαις ἀλλήλας πρὸς ὀρθάξ ἐφέστηκεν, ἡ HB ἄρα καὶ τῷ διὰ τῶν BA , $ΒΓ$ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. [διὰ τὰ αὐτὰ δὲ ἡ BH καὶ τῷ διὰ τῶν $HΘ$, HK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. τὸ δὲ διὰ τῶν $HΘ$, HK ἐπίπεδόν ἐστι τὸ διὰ τῶν $ΔΕ$, $ΕΖ$. ἡ BH ἄρα τῷ διὰ τῶν $ΔΕ$, $ΕΖ$ ἐπιπέδῳ ἐστὶ πρὸς ὀρθάξ. ἐδείχθη δὲ ἡ HB καὶ τῷ διὰ τῶν AB , $ΒΓ$ ἐπιπέδῳ πρὸς ὀρθάξ]. πρὸς ἃ δὲ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθὴ ἐστιν, παράλληλά ἐστι τὰ ἐπίπεδα. παράλληλον ἄρα ἐστὶ τὸ διὰ τῶν AB , $ΒΓ$ ἐπίπεδον τῷ διὰ τῶν $ΔΕ$, $ΕΖ$.

Ἐάν ἄρα δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων παρὰ δύο εὐθείαις ἀπτομένας ἀλλήλων ὥσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, παράλληλά ἐστι τὰ δι' αὐτῶν ἐπίπεδα. ὅπερ ἔδει δεῖχαι.

16.

Ἐάν δύο ἐπίπεδα παράλληλα ὑπὸ ἐπιπέδου τινὸς τέμνηται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν.

Δύο γάρ ἐπίπεδα παράλληλα τὰ AB , $ΓΔ$ ὑπὸ ἐπιπέδου τοῦ $EZHΘ$ τεμνέσθω, κοιναὶ δὲ αὐτῶν τομαὶ ἔστωσαν αἱ EZ , $HΘ$. λέγω, ὅτι παράλληλός ἐστιν ἡ EZ τῇ $HΘ$.

Εἰ γὰρ μὴ, ἐκβαλλόμεναι αἱ EZ , $HΘ$ ἤτοι ἐπὶ τὰ Z , $Θ$ μέρη ἢ ἐπὶ τὰ E , H συμπεσοῦνται. ἐκβεβλήσθωσαν ὥς ἐπὶ τὰ Z , $Θ$ μέρη καὶ συμπίπτωσαν πρότερον κατὰ τὸ K . καὶ ἐπεὶ ἡ EZK ἐν τῷ AB ἐστὶν ἐπιπέδῳ, καὶ πάντα ἄρα τὰ ἐπὶ τῇ EZK σημεία ἐν τῷ AB ἐστὶν ἐπιπέδῳ. ἐν δὲ τῶν ἐπὶ τῇ EZK εὐθείας σημείων ἐστὶ τὸ K . τὸ K ἄρα ἐν τῷ AB ἐστὶν ἐπιπέδῳ. διὰ τὰ αὐτὰ δὲ τὸ K καὶ ἐν τῷ $ΓΔ$ ἐστὶν ἐπιπέδῳ. τὰ AB , $ΓΔ$ ἄρα ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται. οὐ συμπίπτουσι δὲ διὰ τὸ παράλληλα ὑποκεῖσθαι. οὐκ ἄρα αἱ EZ , $HΘ$ εὐθεῖαι ἐκβαλλόμεναι ἐπὶ τὰ Z , $Θ$ μέρη συμπεσοῦνται. ὁμοίως δὲ δείχομεν, ὅτι αἱ EZ , $HΘ$ εὐθεῖαι οὐδὲ ἐπὶ τὰ E , H μέρη ἐκβαλλόμεναι συμπεσοῦνται. αἱ δὲ ἐπὶ μηδέτερα τὰ μέρη συμπίπτουσαι παράλληλοί εἰσιν. παράλληλός ἄρα ἐστὶν ἡ EZ τῇ $HΘ$.

of the angles BGH and BGK are right-angles. And since BA is parallel to GH [Prop. 11.9], the (sum of the) angles GBA and BGH is equal to two right-angles [Prop. 1.29]. And BGH (is) a right-angle. GBA (is) thus also a right-angle. Thus, GB is at right-angles to BA . So, for the same (reasons), GB is also at right-angles to BC . Therefore, since the straight-line GB has been set up at right-angles to two straight-lines, BA and BC , cutting one another, GB is thus at right-angles to the plane through BA and BC [Prop. 11.4]. [So, for the same (reasons), BG is also at right-angles to the plane through GH and GK . And the plane through GH and GK is the (plane) through DE and EF . And it was also shown that GB is at right-angles to the plane through AB and BC .] And planes to which the same straight-line is at right-angles are parallel planes [Prop. 11.14]. Thus, the plane through AB and BC is parallel to the (plane) through DE and EF .

Thus, if two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another). (Which is) the very thing it was required to show.

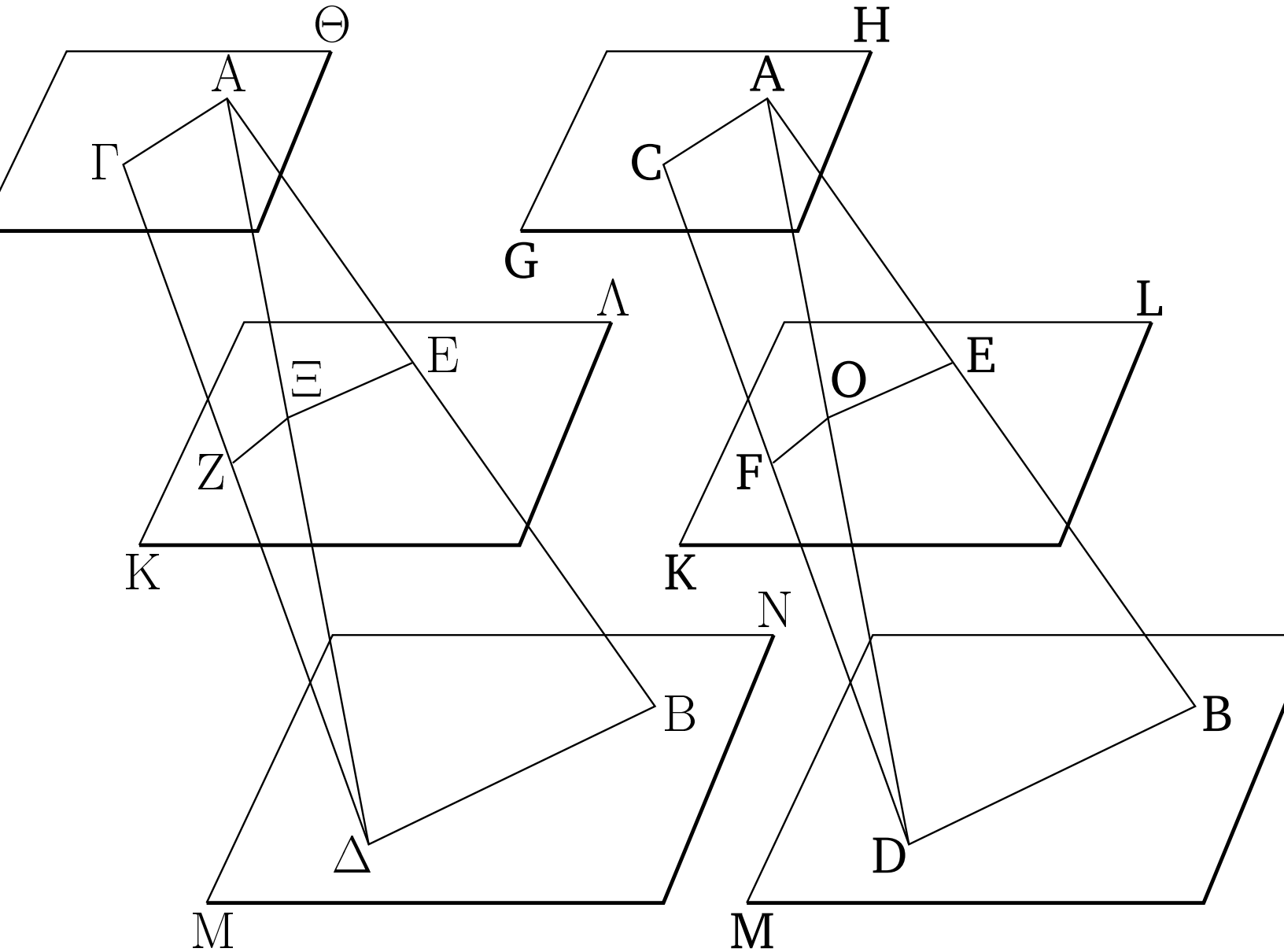
Proposition 16

If two parallel planes are cut by some plane then their common sections are parallel.

For let the two parallel planes AB and CD be cut by the plane $EFGH$. And let EF and GH be their common sections. I say that EF is parallel to GH .

For, if not, being produced, EF and GH will meet either in the direction of F , H , or of E , G . Let them be produced, as in the direction of F , H , and let them, first of all, have met at K . And since EFK is in the plane AB , all of the points on EFK are thus also in the plane AB [Prop. 11.1]. And K is one of the points on EFK . Thus, K is in the plane AB . So, for the same (reasons), K is also in the plane CD . Thus, the planes AB and CD , being produced, will meet. But they do not meet, on account of being (initially) assumed (to be mutually) parallel. Thus, the straight-lines EF and GH , being produced in the direction of F , H , will not meet. So, similarly, we can show that the straight-lines EF and GH , being produced in the direction of E , G , will not meet either. And (straight-lines in one plane which), being produced, do not meet in either direction are parallel [Def. 1.23]. EF is thus parallel to GH .

EB , οὕτως ἡ ΓZ πρὸς $Z\Delta$.



Ἐάν ἄρα δύο εὐθεῖαι ὑπὸ παραλλήλων ἐπιπέδων τέμνονται, εἰς τούξ αὐτοῦξ λόγονξ τμηθήσονται· ὅπερ ἔδει δεῖχαι.

Thus, if two straight-lines are cut by parallel planes then they will be cut in the same ratios. (Which is) the very thing it was required to show.

18.

Proposition 18

Ἐάν εὐθεῖα ἐπιπέδῳ τινὶ πρὸξ ὀρθάξ ᾗ, καὶ πάντα τὰ δι' αὐτῆξ ἐπίπεδα τῷ αὐτῷ ἐπιπέδῳ πρὸξ ὀρθάξ ἔσται.

Εὐθεῖα γάρ τις ἡ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸξ ὀρθάξ ἔστω· λέγω, ὅτι καὶ πάντα τὰ διὰ τῆξ AB ἐπίπεδα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸξ ὀρθάξ ἔστιν.

Ἐκβεβλήσθω γάρ διὰ τῆξ AB ἐπίπεδον τὸ ΔE , καὶ ἔστω

If a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane.

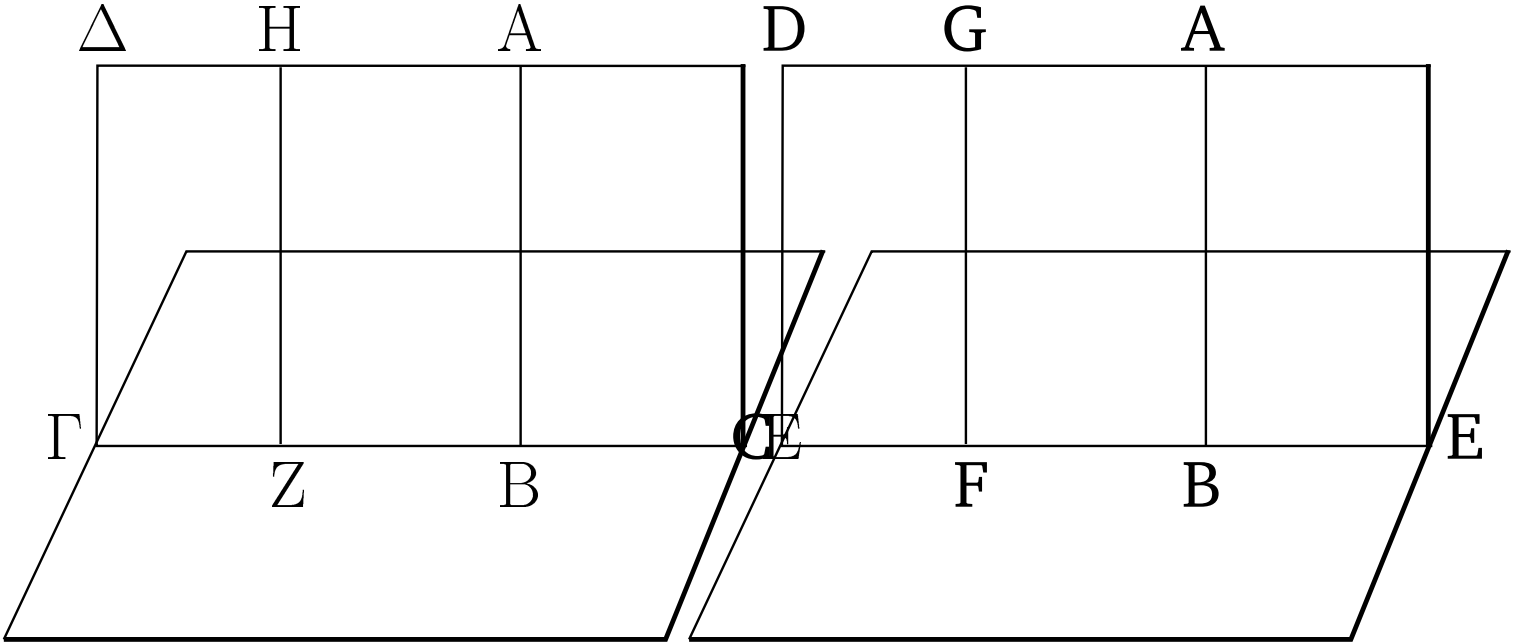
For let some straight-line AB be at right-angles to a reference plane. I say that all of the planes (passing) through AB are also at right-angles to the reference plane.

κοινή τομή τοῦ ΔE ἐπιπέδου καὶ τοῦ ὑποκειμένου ἢ GE , καὶ εἰλήφθω ἐπὶ τῇ GE τυθὸν σημεῖον τὸ Z , καὶ ἀπὸ τοῦ Z τῇ GE πρὸς ὀρθὰς ἤθθω ἐν τῷ ΔE ἐπιπέδῳ ἢ ZH .

Καὶ ἐπεὶ ἡ AB πρὸς τὸ ὑποκειμενον ἐπίπεδον ὀρθή ἐστιν, καὶ πρὸς πάσαξ ἄρα τὰς ἀπτομένας αὐτῇς εὐθείαις καὶ ὅσαξ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθή ἐστιν ἡ AB . ὥστε καὶ πρὸς τὴν GE ὀρθή ἐστιν ἡ ἄρα ὑπὸ ABZ γωνία ὀρθή ἐστιν. ἔστι δὲ καὶ ἡ ὑπὸ HZB ὀρθή· παράλληλοξ ἄρα ἐστὶν ἡ AB τῇ ZH . ἡ δὲ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστὶν· καὶ ἡ ZH ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστὶν. καὶ ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, ὅταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰς ἀγόμεναι εὐθεῖαι ἐν ἐνὶ τῶν ἐπιπέδων τῷ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰς ᾤσιν. καὶ τῇ κοινῇ τομῇ τῶν ἐπιπέδων τῇ GE ἐν ἐνὶ τῶν ἐπιπέδων τῷ ΔE πρὸς ὀρθὰς ἀθθεῖσα ἢ ZH ἐδείχθη τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς· τὸ ἄρα ΔE ἐπίπεδον ὀρθόν ἐστὶ πρὸς τὸ ὑποκείμενον. ὁμοίωξ δὴ δευθθήσεται καὶ πάντα τὰ διὰ τῇ AB ἐπίπεδα ὀρθὰ τυγθανοντα πρὸς τὸ ὑποκείμενον ἐπίπεδον.

For let the plane DE be produced through AB . And let CE be the common section of the plane DE and the reference (plane). And let some random point F be taken on CE . And let FG be drawn from F , at right-angles to CE , in the plane DE [Prop. 1.11].

And since AB is at right-angles to the reference plane, AB is thus also at right-angles to all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Hence, it is also at right-angles to CE . Thus, angle ABF is a right-angle. And GFB is also a right-angle. Thus, AB is parallel to FG [Prop. 1.28]. And AB is at right-angles to the reference plane. Thus, FG is also at right-angles to the reference plane [Prop. 11.8]. And a plane is at right-angles to a(nother) plane when the straight-lines drawn at right-angles to the common section of the planes, (and lying) in one of the planes, are at right-angles to the remaining plane [Def. 11.4]. And FG , (which was) drawn at right-angles to the common section of the planes, CE , in one of the planes, DE , was shown to be at right-angles to the reference plane. Thus, plane DE is at right-angles to the reference (plane). So, similarly, it can be shown that all of the planes (passing) at random through AB (are) at right-angles to the reference plane.



Ἐὰν ἄρα εὐθεῖα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾤ, καὶ πάντα τὰ δι' αὐτῇς ἐπίπεδα τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· ὅπερ ἔδει δεῖξαι.

Thus, if a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

19.

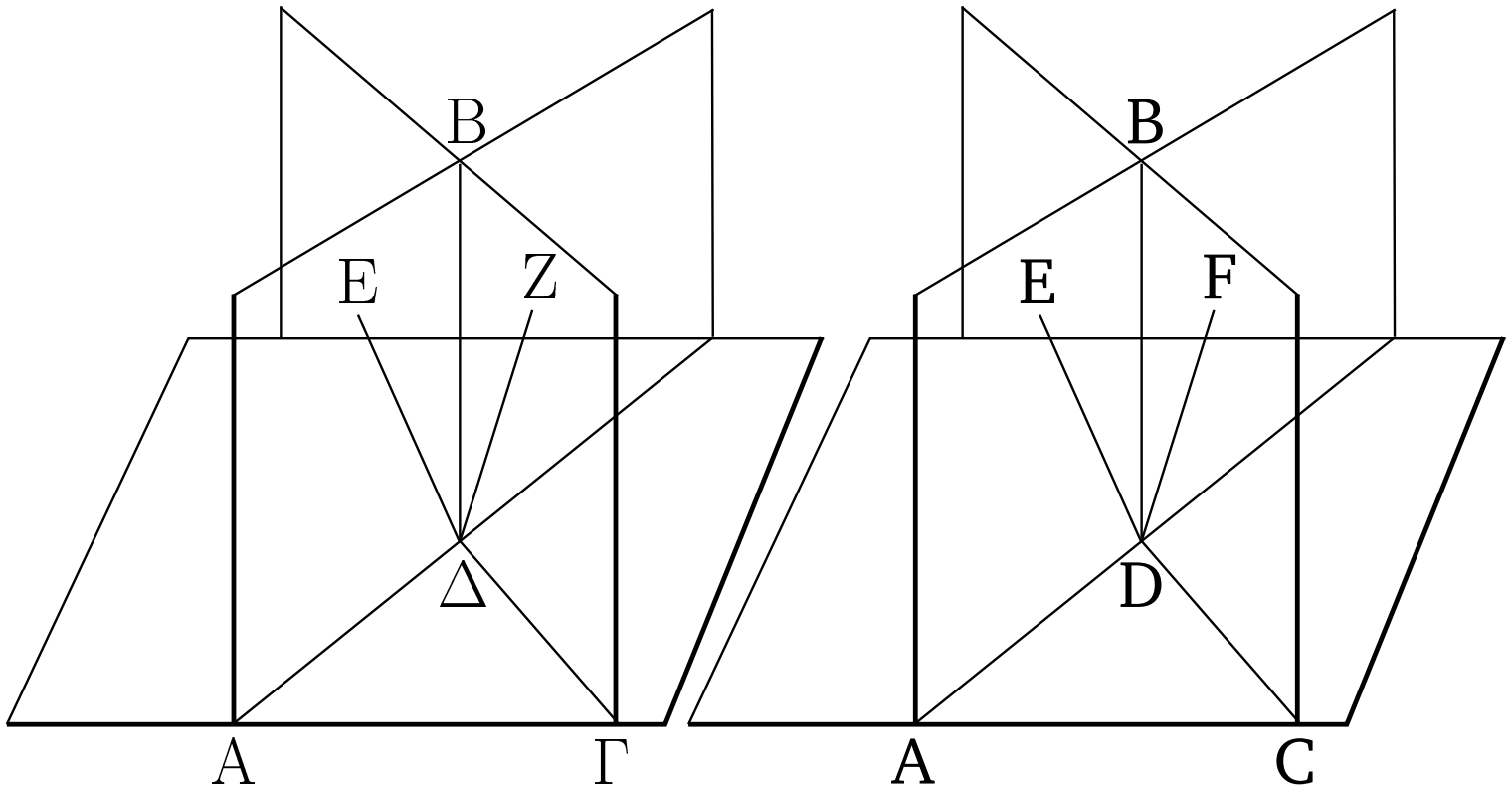
Proposition 19

Ἐὰν δύο ἐπίπεδα τέμνοντα ἄλληλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾤ, καὶ ἡ κοινὴ αὐτῶν τομή τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς

If two planes cutting one another are at right-angles to some plane then their common section will also be at right-

ἔσται.

angles to the same plane.



Δύο γάρ ἐπίπεδα τὰ AB , $BΓ$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἔστω, κοινὴ δὲ αὐτῶν τομὴ ἔστω ἡ BD . λέγω, ὅτι ἡ BD τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν.

μή γάρ, καὶ ἤθθωσαν ἀπὸ τοῦ Δ σημείου ἐν μὲν τῷ AB ἐπιπέδῳ τῇ AD εὐθείᾳ πρὸς ὀρθὰς ἡ ΔE , ἐν δὲ τῷ $BΓ$ ἐπιπέδῳ τῇ GD πρὸς ὀρθὰς ἡ ΔZ .

Καὶ ἐπεὶ τὸ AB ἐπίπεδον ὀρθὸν ἐστὶ πρὸς τὸ ὑποκείμενον, καὶ τῇ κοινῇ αὐτῶν τομῇ τῇ AD πρὸς ὀρθὰς ἐν τῷ AB ἐπιπέδῳ ἤκται ἡ ΔE , ἡ ΔE ἄρα ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ΔZ ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον. ἀπὸ τοῦ αὐτοῦ ἄρα σημείου τοῦ Δ τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰς ἀνισταμέναι εἰσὶν ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ ἀπὸ τοῦ Δ σημείου ἀνασταθῆσεται πρὸς ὀρθὰς πλὴν τῆς ΔB κοινῆς τομῆς τῶν AB , $BΓ$ ἐπιπέδων.

Ἐάν ἄρα δύο ἐπίπεδα τέμνοντα ἄλληλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰς ᾗ, καὶ ἡ κοινὴ αὐτῶν τομὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἔσται· ὅπερ ἔδει δεῖξαι.

For let the two planes AB and BC be at right-angles to a reference plane, and let their common section be BD . I say that BD is at right-angles to the reference plane.

For (if) not, let DE also be drawn from point D , in the plane AB , at right-angles to the straight-line AD , and DF , in the plane BC , at right-angles to CD .

And since the plane AB is at right-angles to the reference (plane), and DE has been drawn at right-angles to their common section AD , in the plane AB , DE is thus at right-angles to the reference plane [Def. 11.4]. So, similarly, we can show that DF is also at right-angles to the reference plane. Thus, two (different) straight-lines are set up, at the same point D , at right-angles to the reference plane, on the same side. The very thing is impossible [Prop. 11.13]. Thus, no (other straight-line) except the common section DB of the planes AB and BC can be set up at point D , at right-angles to the reference plane.

Thus, if two planes cutting one another are at right-angles to some plane then their common section will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

20.

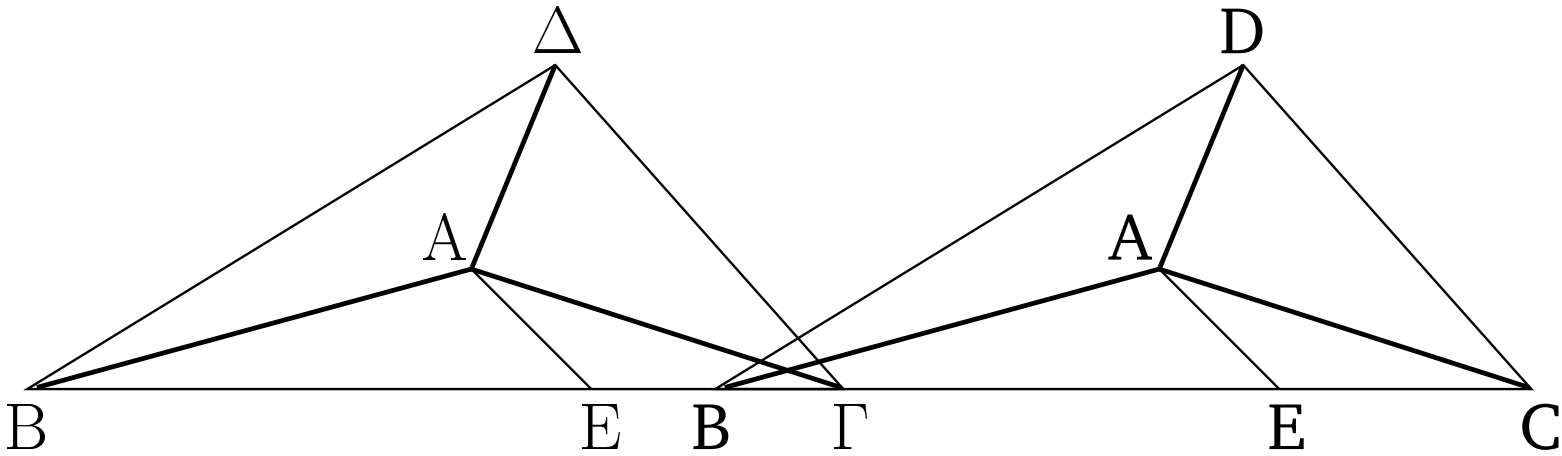
Proposition 20

Ἐάν στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθῃται, δύο ὁποιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβα-

If a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one),

νόμεναι.

(the angles) being taken up in any (possible way).



Στερεὰ γὰρ γωνία ἡ πρὸς τῷ A ὑπὸ τριῶν γωνιῶν ἐπιπέδων τῶν ὑπὸ BAG , $ΓΑΔ$, $ΔAB$ περιεθέσθω· λέγω, ὅτι τῶν ὑπὸ BAG , $ΓΑΔ$, $ΔAB$ γωνιῶν δύο ὁποιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

Εἰ μὲν οὖν αἱ ὑπὸ BAG , $ΓΑΔ$, $ΔAB$ γωνίαι ἴσαι ἀλλήλαις εἴσιν, φανερόν, ὅτι δύο ὁποιοῦν τῆς λοιπῆς μείζονές εἰσιν. εἰ δὲ οὐ, ἔστω μείζων ἡ ὑπὸ BAG , καὶ συνεστίτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ ὑπὸ $ΔAB$ γωνίᾳ ἐν τῷ διὰ τῶν BAG ἐπιπέδῳ ἴση ἡ ὑπὸ BAE , καὶ κείσθω τῇ AD ἴση ἡ AE , καὶ διὰ τοῦ E σημείου διαθθεῖσα ἡ BEG τεμνέτω τὰς AB , AG εὐθείας κατὰ τὰ B , $Γ$ σημεῖα, καὶ ἐπεξεύθθωσαν αἱ $ΔB$, $ΔΓ$.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΔA$ τῇ AE , κοινὴ δὲ ἡ AB , δύο δυνάμει ἴσαι· καὶ γωνία ἡ ὑπὸ $ΔAB$ γωνία τῇ ὑπὸ BAE ἴση· βάσεις ἄρα ἡ $ΔB$ βάσει τῇ BE ἐστὶν ἴση. καὶ ἐπεὶ δύο αἱ BD , $ΔΓ$ τῆς $BΓ$ μείζονές εἰσιν, ὣν ἡ $ΔB$ τῇ BE ἐδείχθη ἴση, λοιπὴ ἄρα ἡ $ΔΓ$ λοιπῆς τῆς EG μείζων ἐστίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΔA$ τῇ AE , κοινὴ δὲ ἡ AG , καὶ βάσεις ἡ $ΔΓ$ βάσεως τῆς EG μείζων ἐστίν, γωνία ἄρα ἡ ὑπὸ $ΔAG$ γωνία τῆς ὑπὸ EAG μείζων ἐστίν. ἐδείχθη δὲ καὶ ἡ ὑπὸ $ΔAB$ τῇ ὑπὸ BAE ἴση· αἱ ἄρα ὑπὸ $ΔAB$, $ΔAG$ τῆς ὑπὸ BAG μείζονές εἰσιν. ὁμοίως δὲ δειχόμεν, ὅτι καὶ αἱ λοιπαὶ σύνδυο λαμβανόμεναι τῆς λοιπῆς μείζονές εἰσιν.

Ἐὰν ἄρα στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθῃται, δύο ὁποιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.

21.

Ἄπανα στερεὰ γωνία ὑπὸ ἐλασσόνων [ἢ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέθεται.

For let the solid angle A be contained by the three plane angles BAC , CAD , and DAB . I say that (the sum of) any two of the angles BAC , CAD , and DAB is greater than the remaining (one), (the angles) being taken up in any (possible way).

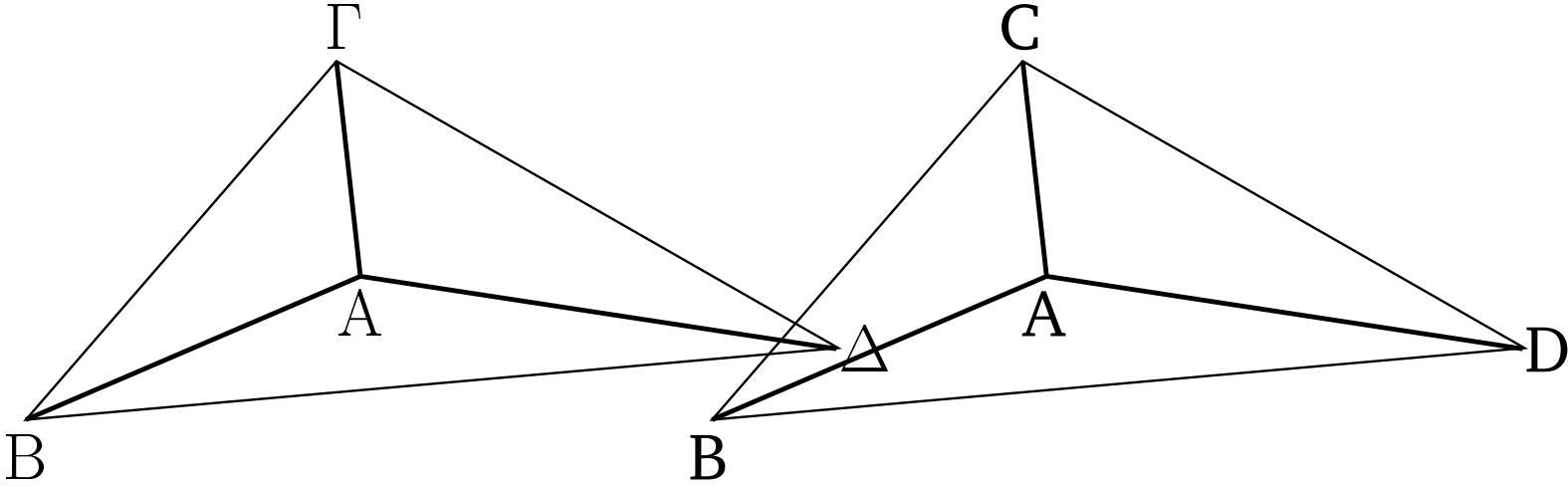
For if the angles BAC , CAD , and DAB are equal to one another then (it is) clear that (the sum of) any two is greater than the remaining (one). But, if not, let BAC be greater (than CAD or DAB). And let (angle) BAE , equal to the angle DAB , be constructed in the plane through BAC , on the straight-line AB , at the point A on it. And let AE be made equal to AD . And BEC being drawn across through point E , let it cut the straight-lines AB and AC at points B and C (respectively). And let DB and DC be joined.

And since DA is equal to AE , and AB (is) common, the two (straight-lines AD and AE) are equal to the two (straight-lines EA and AB , respectively). And angle DAB (is) equal to angle BAE . Thus, the base DB is equal to the base BE [Prop. 1.4]. And since the (sum of the) two (straight-lines) BD and DC is greater than BC [Prop. 1.20], of which DB was shown (to be) equal to BE , the remainder DC is thus greater than the remainder EC . And since DA is equal to AE , but AC (is) common, and the base DC is greater than the base EC , the angle DAC is thus greater than the angle EAC [Prop. 1.25]. And DAB was also shown (to be) equal to BAE . Thus, (the sum of) DAB and DAC is greater than BAC . So, similarly, we can also show that the remaining (angles), being taken in pairs, are greater than the remaining (one).

Thus, if a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one), (the angles) being taken up in any (possible way). (Which is) the very thing it was required to show.

Proposition 21

Any solid angle is contained by plane angles (whose sum is) less [than] four right-angles.[†]



Ἐστω στερεὰ γωνία ἡ πρὸς τῷ A περιεχομένη ὑπὸ ἐπιπέδων γωνιῶν τῶν ὑπὸ BAF , $ΓΑΔ$, $ΔΑΒ$ · λέγω, ὅτι αἱ ὑπὸ BAF , $ΓΑΔ$, $ΔΑΒ$ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

Εἰλήφθω γάρ ἐφ' ἐκάστης τῶν AB , $ΑΓ$, $ΑΔ$ τυθόντα σημεῖα τὰ B , $Γ$, $Δ$, καὶ ἐπεξεύθωσαν αἱ $BΓ$, $ΓΔ$, $ΔΒ$. καὶ ἐπεὶ στερεὰ γωνία ἡ πρὸς τῷ B ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέχεται τῶν ὑπὸ $ΓΒΑ$, $ΑΒΔ$, $ΓΒΔ$, δύο ὁποιοῦν τῆς λοιπῆς μείζονές εἰσιν· αἱ ἄρα ὑπὸ $ΓΒΑ$, $ΑΒΔ$ τῆς ὑπὸ $ΓΒΔ$ μείζονές εἰσιν. διὰ τὰ αὐτὰ δὴ καὶ αἱ μὲν ὑπὸ $ΒΓΑ$, $ΑΓΔ$ τῆς ὑπὸ $ΒΓΔ$ μείζονές εἰσιν, αἱ δὲ ὑπὸ $ΓΔΑ$, $ΑΔΒ$ τῆς ὑπὸ $ΓΔΒ$ μείζονές εἰσιν· αἱ ἔχ' ἄρα γωνίαι αἱ ὑπὸ $ΓΒΑ$, $ΑΒΔ$, $ΒΓΑ$, $ΑΓΔ$, $ΓΔΑ$, $ΑΔΒ$ τριῶν τῶν ὑπὸ $ΓΒΔ$, $ΒΓΑ$, $ΓΔΒ$ μείζονές εἰσιν. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ $ΓΒΔ$, $ΒΔΓ$, $ΒΓΔ$ δυοῖν ὀρθαῖς ἴσαι εἰσὶν· αἱ ἔχ' ἄρα αἱ ὑπὸ $ΓΒΑ$, $ΑΒΔ$, $ΒΓΑ$, $ΑΓΔ$, $ΓΔΑ$, $ΑΔΒ$ δύο ὀρθῶν μείζονές εἰσιν. καὶ ἐπεὶ ἐκάστον τῶν $ΑΒΓ$, $ΑΓΔ$, $ΑΔΒ$ τριγώνων αἱ τρεῖς γωνίαι δυοῖν ὀρθαῖς ἴσαι εἰσὶν, αἱ ἄρα τῶν τριῶν τριγώνων ἑννέα γωνίαι αἱ ὑπὸ $ΓΒΑ$, $ΑΓΒ$, $ΒΑΓ$, $ΑΓΔ$, $ΓΔΑ$, $ΓΑΔ$, $ΑΔΒ$, $ΔΒΑ$, $ΒΑΔ$ ἔχ' ὀρθαῖς ἴσαι εἰσὶν, ὧν αἱ ὑπὸ $ΑΒΓ$, $ΒΓΑ$, $ΑΓΔ$, $ΓΔΑ$, $ΑΔΒ$, $ΔΒΑ$ ἔχ' γωνίαι δύο ὀρθῶν εἰσι μείζονες· λοιπαὶ ἄρα αἱ ὑπὸ $ΒΑΓ$, $ΓΑΔ$, $ΔΑΒ$ τρεῖς [γωνίαι] περιέχουσιν τὴν στερεάν γωνίαν τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

Ἄπανα ἄρα στερεὰ γωνία ὑπὸ ἐλασσόνων [ῆ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέχεται· ὅπερ ἔδει δεῖξαι.

Let the solid angle A be contained by the plane angles BAC , CAD , and DAB . I say that (the sum of) BAC , CAD , and DAB is less than four right-angles.

For let the random points B , C , and D be taken on each of (the straight-lines) AB , AC , and AD (respectively). And let BC , CD , and DB be joined. And since the solid angle at B is contained by the three plane angles CBA , ABD , and CBD , (the sum of) any two is greater than the remaining (one) [Prop. 11.20]. Thus, (the sum of) CBA and ABD is greater than CBD . So, for the same (reasons), (the sum of) BCA and ACD is also greater than BCD , and (the sum of) CDA and ADB is greater than CDB . Thus, the (sum of the) six angles CBA , ABD , BCA , ACD , CDA , and ADB is greater than the (sum of the) three (angles) CBD , BCD , and CDB . But, the (sum of the) three (angles) CBD , BDC , and BCD is equal to two right-angles [Prop. 1.32]. Thus, the (sum of the) six angles CBA , ABD , BCA , ACD , CDA , and ADB is greater than two right-angles. And since the (sum of the) three angles of each of the triangles ABC , ACD , and ADB is equal to two right-angles, the (sum of the) nine angles CBA , ACB , BAC , ACD , CDA , CAD , ADB , DBA , and BAD of the three triangles is equal to six right-angles, of which the (sum of the) six angles ABC , BCA , ACD , CDA , ADB , and DBA is greater than two right-angles. Thus, the (sum of the) remaining three [angles] BAC , CAD , and DAB , containing the solid angle, is less than four right-angles.

Thus, any solid angle is contained by plane angles (whose sum is) less [than] four right-angles. (Which is) the very thing it was required to show.

† This proposition is only proved for the case of a solid angle contained by three plane angles. However, the generalization to a solid angle contained by more than three plane angles is straightforward.

22.

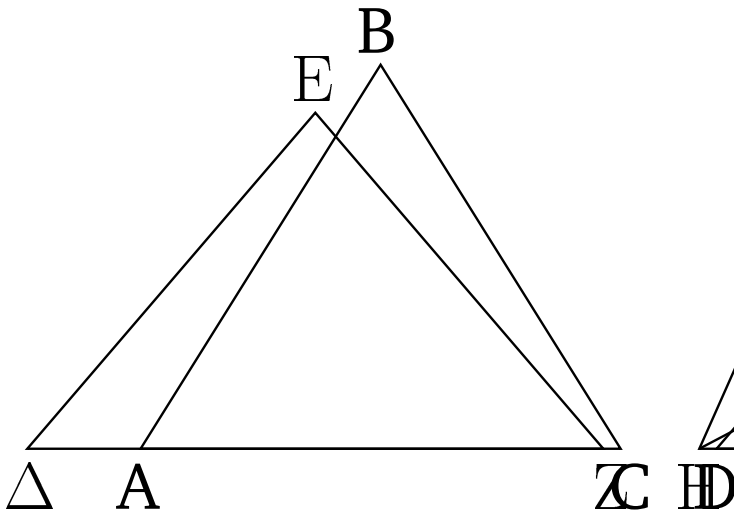
Proposition 22

Ἐάν ὡς τρεῖς γωνίαι ἐπίπεδοι, ὧν αἱ δύο τῆς λοιπῆς

If there are three plane angles, of which (the sum of any)

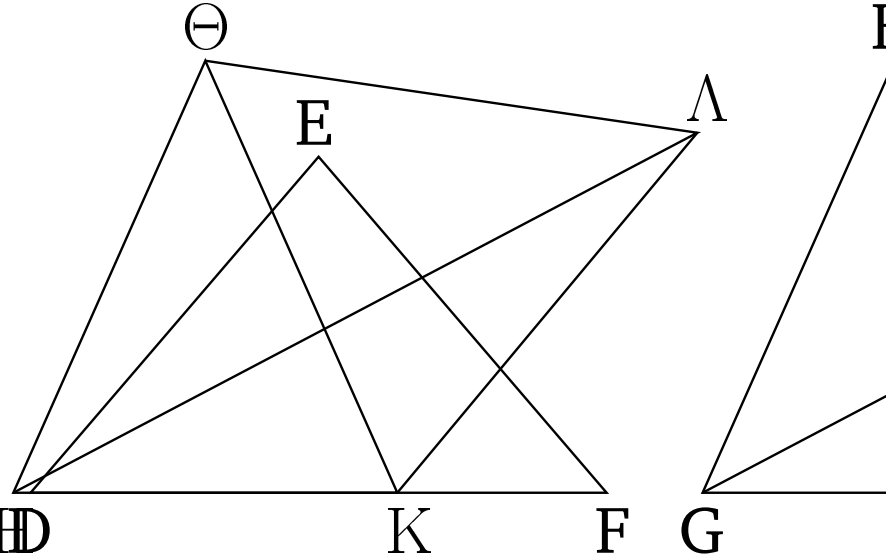
μειζονές εἰσι πάντα μεταλαμβανόμεναι, περιέθωσι δὲ αὐτάξ ἴσαι εὐθείαι, δυνατόν ἐστιν ἐκ τῶν ἐπιζευγνυσουσῶν τὰς ἴσας εὐθείας τρίγωνον συστήσασθαι.

two is greater than the remaining (one), (the angles) being taken up in any (possible way), and if equal straight-lines contain them, then it is possible to construct a triangle from (the straight-lines created by) joining the (ends of the) equal straight-lines.



Ἐστωσαν τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, ὧν αἱ δύο τῆς λοιπῆς μειζονές εἰσι πάντα μεταλαμβάνόμεναι, αἱ μὲν ὑπὸ ΑΒΓ, ΔΕΖ τῆς ὑπὸ ΗΘΚ, αἱ δὲ ὑπὸ ΔΕΖ, ΗΘΚ τῆς ὑπὸ ΑΒΓ, καὶ ἔτι αἱ ὑπὸ ΗΘΚ, ΑΒΓ τῆς ὑπὸ ΔΕΖ, καὶ ἔστωσαν ἴσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ εὐθείαι, καὶ ἐπεξεύθθωσαν αἱ ΑΓ, ΔΖ, ΗΚ λέγω, ὅτι δυνατόν ἐστιν ἐκ τῶν ἴσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι, τουτέστιν ὅτι τῶν ΑΓ, ΔΖ, ΗΚ δύο ὁποιαοῦν τῆς λοιπῆς μειζονές εἰσιν.

Εἰ μὲν οὖν αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ γωνίαι ἴσαι ἀλλήλαις εἰσίν, φανερόν, ὅτι καὶ τῶν ΑΓ, ΔΖ, ΗΚ ἴσων γινόμενων δυνατόν ἐστιν ἐκ τῶν ἴσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. εἰ δὲ οὐ, ἔστωσαν ἄνωσοι, καὶ συνεστάτω πρὸς τῇ ΘΚ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Θ τῇ ὑπὸ ΑΒΓ γωνίᾳ ἴση ἡ ὑπὸ ΚΘΛ· καὶ κείσθω μὲν τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ ἴση ἡ ΘΛ, καὶ ἐπεξεύθθωσαν αἱ ΚΛ, ΗΛ. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΓ δυοὶ ταῖς ΚΘ, ΘΛ ἴσαι εἰσίν, καὶ γωνία ἡ πρὸς τῷ Β γωνία τῇ ὑπὸ ΚΘΛ ἴση, βάσεις ἄρα ἡ ΑΓ βάσει τῇ ΚΛ ἴση. καὶ ἐπεὶ αἱ ὑπὸ ΑΒΓ, ΗΘΚ τῆς ὑπὸ ΔΕΖ μειζονές εἰσιν, ἴση δὲ ἡ ὑπὸ ΑΒΓ τῇ ὑπὸ ΚΘΛ, ἡ ἄρα ὑπὸ ΗΘΛ τῆς ὑπὸ ΔΕΖ μείζων ἐστίν. καὶ ἐπεὶ δύο αἱ ΗΘ, ΘΛ δύο ταῖς ΔΕ, ΕΖ ἴσαι εἰσίν, καὶ γωνία ἡ ὑπὸ ΗΘΛ γωνία τῆς ὑπὸ ΔΕΖ μείζων, βάσεις ἄρα ἡ ΗΛ βάσει τῆς ΔΖ μείζων ἐστίν. ἀλλὰ αἱ ΗΚ, ΚΛ τῆς ΗΛ μείζονές εἰσιν. πολλῶν ἄρα αἱ ΗΚ, ΚΛ τῆς ΔΖ μείζονές εἰσιν. ἴση δὲ ἡ ΚΛ τῇ ΑΓ· αἱ ΑΓ, ΗΚ ἄρα τῆς λοιπῆς τῆς ΔΖ μείζονές εἰσιν. ὁμοίως δὲ δείχουμεν, ὅτι καὶ αἱ μὲν ΑΓ, ΔΖ τῆς ΗΚ μείζονές εἰσιν, καὶ ἔτι αἱ ΔΖ, ΗΚ τῆς ΑΓ μείζονές εἰσιν. δυνατόν ἄρα ἐστίν ἐκ τῶν ἴσων



Let ABC , DEF , and GHK be three plane angles, of which the sum of any two is greater than the remaining (one), (the angles) being taken up in any (possible way)—(that is), ABC and DEF (greater) than GHK , DEF and GHK (greater) than ABC , and, further, GHK and ABC (greater) than DEF . And let AB , BC , DE , EF , GH , and HK be equal straight-lines. And let AC , DF , and GK be joined. I say that that it is possible to construct a triangle out of (straight-lines) equal to AC , DF , and GK —that is to say, that (the sum of) any two of AC , DF , and GK is greater than the remaining (one).

Now, if the angles ABC , DEF , and GHK are equal to one another then (it is) clear that, (with) AC , DF , and GK also becoming equal, it is possible to construct a triangle from (straight-lines) equal to AC , DF , and GK . And if not, let them be unequal, and let KHL , equal to angle ABC , be constructed on the straight-line HK , at the point H on it. And let HL be made equal to one of AB , BC , DE , EF , GH , and HK . And let KL and GL be joined. And since the two (straight-lines) AB and BC are equal to the two (straight-lines) KH and HL (respectively), and the angle at B (is) equal to KHL , the base AC is thus equal to the base KL [Prop. 1.4]. And since (the sum of) ABC and GHK is greater than DEF , and ABC equal to KHL , GHL is thus greater than DEF . And since the two (straight-lines) GH and HL are equal to the two (straight-lines) DE and EF (respectively), and angle GHL (is) greater than DEF , the base GL is thus greater than the base DF [Prop. 1.24]. But, (the sum of) GK and KL is greater than GL [Prop. 1.20]. Thus, (the sum of) GK and KL is much greater than DF . And KL

ταῖς $ΑΓ$, $ΔΖ$, $ΗΚ$ τρίγωνον συστήσασθαι· ὅπερ ἔδει δεῖξαι.

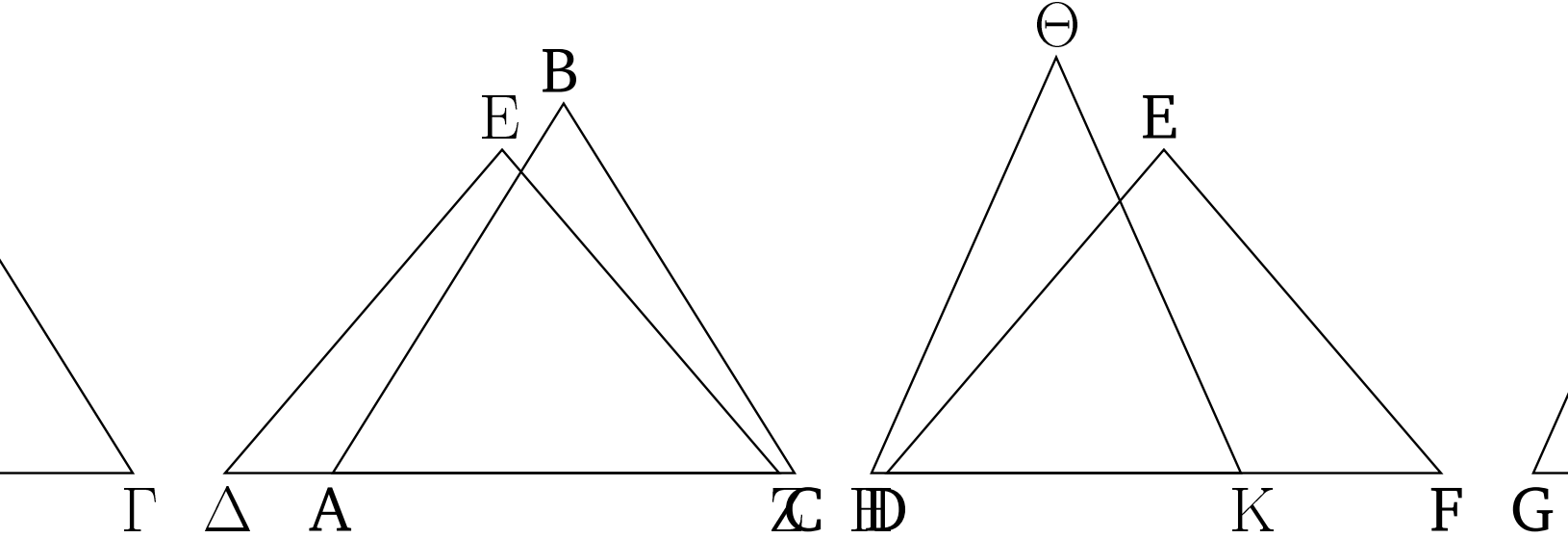
(is) equal to AC . Thus, (the sum of) AC and GK is greater than the remaining (straight-line) DF . So, similarly, we can show that (the sum of) AC and DF is greater than GK , and, further, that (the sum of) DF and GK is greater than AC . Thus, it is possible to construct a triangle from (straight-lines) equal to AC , DF , and GK . (Which is) the very thing it was required to show.

23.

Ἐκ τριῶν γωνιῶν ἐπιπέδων, ὧν αἱ δύο τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, στερεὰν γωνίαν συστήσασθαι· δεῖ δὴ τὰς τρεῖς τεσσάρων ὀρθῶν ἐλάσσοναξ εἶναι.

Proposition 23

To construct a solid angle from three (given) plane angles, (the sum of) two of which is greater than the remaining (one, the angles) being taken up in any (possible way). So, it is necessary for the (sum of the) three (angles) to be less than four right-angles [Prop. 11.21].

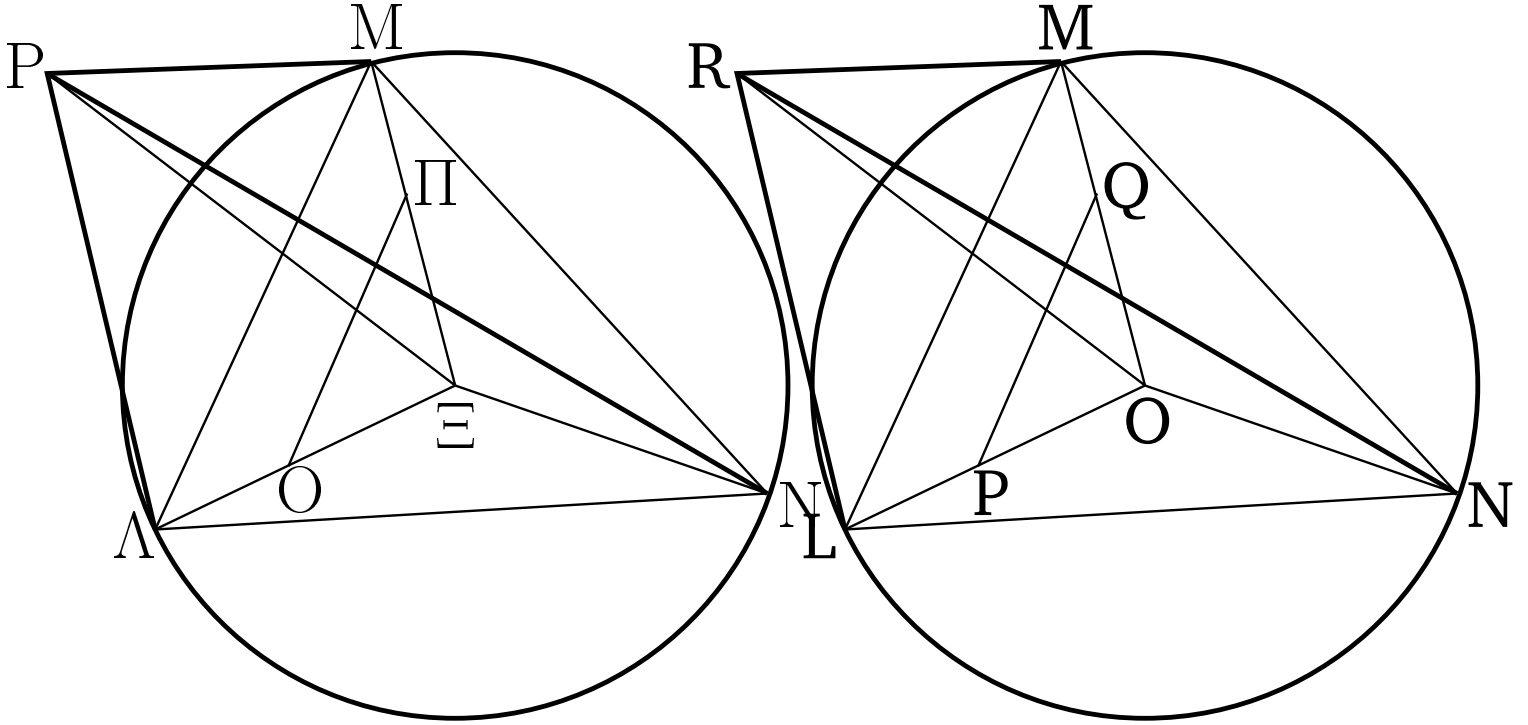


Ἐστωσαν αἱ δοθεῖσαι τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ $ΑΒΓ$, $ΔΕΖ$, $ΗΘΚ$, ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, ἔτι δὲ αἱ τρεῖς τεσσάρων ὀρθῶν ἐλάσσονες· δεῖ δὴ ἐκ τῶν ἴσων ταῖς ὑπὸ $ΑΒΓ$, $ΔΕΖ$, $ΗΘΚ$ στερεὰν γωνίαν συστήσασθαι.

Ἀπειλήφθωσαν ἴσαι αἱ $ΑΒ$, $ΒΓ$, $ΔΕ$, $ΕΖ$, $ΗΘ$, $ΘΚ$, καὶ ἐπεξεύθθωσαν αἱ $ΑΓ$, $ΔΖ$, $ΗΚ$ · δυνατὸν ἄρα ἐστὶν ἐκ τῶν ἴσων ταῖς $ΑΓ$, $ΔΖ$, $ΗΚ$ τρίγωνον συστήσασθαι. συνεστάτω τὸ $ΛΜΝ$, ὥστε ἴσῃ εἶναι τὴν μὲν $ΑΓ$ τῇ $ΛΜ$, τὴν δὲ $ΔΖ$ τῇ $ΜΝ$, καὶ ἔτι τὴν $ΗΚ$ τῇ $ΝΛ$, καὶ περιγεγράφθω περὶ τὸ $ΛΜΝ$ τρίγωνον κύκλος ὁ $ΛΜΝ$, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον καὶ ἔστω τὸ $Χ$, καὶ ἐπεξεύθθωσαν αἱ $ΑΧ$, $ΜΧ$, $ΝΧ$.

Let ABC , DEF , and GHK be the three given plane angles, of which let (the sum of) two be greater than the remaining (one, the angles) being taken up in any (possible way), and, further, (let) the (sum of the) three (be) less than four right-angles. So, it is necessary to construct a solid angle from (plane angles) equal to ABC , DEF , and GHK .

Let AB , BC , DE , EF , GH , and HK be cut off (so as to be) equal (to one another). And let AC , DF , and GK be joined. It is, thus, possible to construct a triangle from (straight-lines) equal to AC , DF , and GK [Prop. 11.22]. Let (such a triangle), LMN , have been constructed, such that AC is equal to LM , DF to MN , and, further, GK to NL . And let the circle LMN be circumscribed about triangle LMN [Prop. 4.5]. And let its center be found, and let it be (at) O . And let LO , MO , and NO be joined.



Λέγω, ὅτι ἡ AB μείζων ἐστὶ τῆς AX . εἰ γὰρ μή, ἦτοι ἴση ἐστὶν ἡ AB τῇ AX ἢ ἐλάττων. ἔστω πρότερον ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ AX , ἀλλὰ ἡ μὲν AB τῇ BF ἐστὶν ἴση, ἡ δὲ XA τῇ XM , δύο δὴ αἱ AB , BF δύο ταῖς AX , XM ἴσαι εἰσὶν ἑκατέρα ἑκατέρῃ· καὶ βάσεις ἡ AF βάσει τῇ AM ὑπόκειται ἴση· γωνία ἄρα ἡ ὑπὸ ABF γωνία τῇ ὑπὸ AXM ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ $ΔΕΖ$ τῇ ὑπὸ $ΜΧΝ$ ἐστὶν ἴση, καὶ ἔτι ἡ ὑπὸ $ΗΘΚ$ τῇ ὑπὸ $ΝΧΛ$ · αἱ ἄρα τρεῖς αἱ ὑπὸ ABF , $ΔΕΖ$, $ΗΘΚ$ γωνίαί τρισι ταῖς ὑπὸ AXM , MXN , $ΝΧΛ$ εἰσιν ἴσαι. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ AXM , MXN , $ΝΧΛ$ τέτταρσιν ὀρθαῖς εἰσιν ἴσαι· καὶ αἱ τρεῖς ἄρα αἱ ὑπὸ ABF , $ΔΕΖ$, $ΗΘΚ$ τέτταρσιν ὀρθαῖς ἴσαι εἰσὶν. ὑπόκεινται δὲ καὶ τεσσάρων ὀρθῶν ἐλάσσονες· ὅπερ ἄτοπον. οὐκ ἄρα ἡ AB τῇ AX ἴση ἐστὶν. λέγω δὴ, ὅτι οὐδὲ ἐλάττων ἐστὶν ἡ AB τῇς AX . εἰ γὰρ δυνατόν, ἔστω· καὶ κείσθω τῇ μὲν AB ἴση ἡ XO , τῇ δὲ BF ἴση ἡ $XΠ$, καὶ ἐπεξεύθθω ἡ $OΠ$. καὶ ἐπεὶ ἴση ἐστὶν ἡ AB τῇ BF , ἴση ἐστὶ καὶ ἡ XO τῇ $XΠ$ · ὥστε καὶ λοιπὴ ἡ AO τῇ $ΠΜ$ ἐστὶν ἴση. παρὰλληλοξ ἄρα ἐστὶν ἡ AM τῇ $OΠ$, καὶ ἰσογώνιον τὸ AMX τῷ $OΠX$ · ἔστιν ἄρα ὡς ἡ XA πρὸς AM , οὕτως ἡ XO πρὸς $OΠ$ · ἐναλλάξ ὡς ἡ AX πρὸς XO , οὕτως ἡ AM πρὸς $OΠ$. μείζων δὲ ἡ AX τῇς XO · μείζων ἄρα καὶ ἡ AM τῇς $OΠ$. ἀλλὰ ἡ AM κείται τῇ AF ἴση· καὶ ἡ AF ἄρα τῇς $OΠ$ μείζων ἐστὶν. ἐπεὶ οὖν δύο αἱ AB , BF δυοὶ ταῖς OX , $XΠ$ ἴσαι εἰσὶν, καὶ βάσεις ἡ AF βάσεως τῇς $OΠ$ μείζων ἐστὶν, γωνία ἄρα ἡ ὑπὸ ABF γωνία τῇς ὑπὸ $OXΠ$ μείζων ἐστὶν. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ μὲν ὑπὸ $ΔΕΖ$ τῇς ὑπὸ $ΜΧΝ$ μείζων ἐστὶν, ἡ δὲ ὑπὸ $ΗΘΚ$ τῇς ὑπὸ $ΝΧΛ$. αἱ ἄρα τρεῖς γωνίαί αἱ ὑπὸ ABF , $ΔΕΖ$, $ΗΘΚ$ τριῶν τῶν ὑπὸ AXM ,

I say that AB is greater than LO . For, if not, AB is either equal to, or less than, LO . Let it, first of all, be equal. And since AB is equal to LO , but AB is equal to BC , and OL to OM , so the two (straight-lines) AB and BC are equal to the two (straight-lines) LO and OM , respectively. And the base AC was assumed (to be) equal to the base LM . Thus, angle ABC is equal to angle LOM [Prop. 1.8]. So, for the same (reasons), DEF is also equal to MON , and, further, GHK to NOL . Thus, the three angles ABC , DEF , and GHK are equal to the three angles LOM , MON , and NOL , respectively. But, the (sum of the) three angles LOM , MON , and NOL is equal to four right-angles. Thus, the (sum of the) three angles ABC , DEF , and GHK is also equal to four right-angles. And it was also assumed (to be) less than four right-angles. The very thing (is) absurd. Thus, AB is not equal to LO . So, I say that AB is not less than LO either. For, if possible, let it be (less). And let OP be made equal to AB , and OQ equal to BC , and let PQ be joined. And since AB is equal to BC , OP is also equal to OQ . Hence, the remainder LP is also equal to (the remainder) QM . LM is thus parallel to PQ [Prop. 6.2], and (triangle) LMO (is) equiangular with (triangle) PQO [Prop. 1.29]. Thus, as OL is to LM , so OP (is) to PQ [Prop. 6.4]. Alternately, as LO (is) to OP , so LM (is) to PQ [Prop. 5.16]. And LO (is) greater than OP . Thus, LM (is) also greater than PQ [Prop. 5.14]. But LM was made equal to AC . Thus, AC is also greater than PQ . Therefore, since the two (straight-lines) AB and BC are equal to the two (straight-lines) PO and OQ (respectively), and the base AC is greater than the base PQ , the angle ABC is

MXN , NXA μείζονές εἰσιν. ἀλλὰ αἱ ὑπὸ $ABΓ$, $ΔΕΖ$, $ΗΘΚ$ τεσσάρων ὀρθῶν ἐλάσσονες ὑπόκεινται· πολλῶν ἄρα αἱ ὑπὸ $ΛΧΜ$, $ΜΧΝ$, $ΝΧΑ$ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν. ἀλλὰ καὶ ἴσαι· ὅπερ ἐστὶν ἄτοπον. οὐκ ἄρα ἡ AB ἐλάσσων ἐστὶ τῆς $ΛΧ$. ἐδείχθη δέ, ὅτι οὐδὲ ἴση· μείζων ἄρα ἡ AB τῆς $ΛΧ$.

Ἀνεστάτω δὴ ἀπὸ τοῦ X σημεῖον τῷ τοῦ $ΑΜΝ$ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἡ XP , καὶ ὥς μείζον ἐστὶ τὸ ἀπὸ τῆς AB τετραγώνου τοῦ ἀπὸ τῆς $ΛΧ$, ἐκείνῳ ἴσον ἔστω τὸ ἀπὸ τῆς XP , καὶ ἐπεξεύχθωσαν αἱ PA , PM , PN .

Καὶ ἐπεὶ ἡ PX ὀρθή ἐστι πρὸς τὸ τοῦ $ΑΜΝ$ κύκλου ἐπίπεδον, καὶ πρὸς ἐκάστην ἄρα τῶν $ΛΧ$, $ΜΧ$, $ΝΧ$ ὀρθή ἐστὶν ἡ PX . καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΛΧ$ τῇ $ΧΜ$, κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ XP , βάσει ἄρα ἡ PA βάσει τῇ PM ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ PN ἐκατέρᾳ τῶν PA , PM ἐστὶν ἴση· αἱ τρεῖς ἄρα αἱ PA , PM , PN ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ὥς μείζον ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς $ΛΧ$, ἐκείνῳ ἴσον ὑπόκειται τὸ ἀπὸ τῆς XP , τὸ ἄρα ἀπὸ τῆς AB ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΛΧ$, XP . τοῖς δὲ ἀπὸ τῶν $ΛΧ$, XP ἴσον ἐστὶ τὸ ἀπὸ τῆς AP · ὀρθὴ γάρ ἡ ὑπὸ $ΛXP$. τὸ ἄρα ἀπὸ τῆς AB ἴσον ἐστὶ τῷ ἀπὸ τῆς PA . ἴση ἄρα ἡ AB τῇ PA . ἀλλὰ τῇ μὲν AB ἴση ἐστὶν ἐκάστη τῶν $ΒΓ$, $ΔΕ$, $ΕΖ$, $ΗΘ$, $ΘΚ$, τῇ δὲ PA ἴση ἐκατέρᾳ τῶν PM , PN · ἐκάστη ἄρα τῶν AB , $ΒΓ$, $ΔΕ$, $ΕΖ$, $ΗΘ$, $ΘΚ$ ἐκάστη τῶν PA , PM , PN ἴση ἐστὶν. καὶ ἐπεὶ δύο αἱ AP , PM δυοὶ ταῖς AB , $ΒΓ$ ἴσαι εἰσίν, καὶ βάσει ἡ AM βάσει τῇ $ΑΓ$ ὑπόκειται ἴση, γωνία ἄρα ἡ ὑπὸ APM γωνία τῇ ὑπὸ $ΑΒΓ$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ MPN τῇ ὑπὸ $ΔΕΖ$ ἐστὶν ἴση, ἡ δὲ ὑπὸ APN τῇ ὑπὸ $ΗΘΚ$.

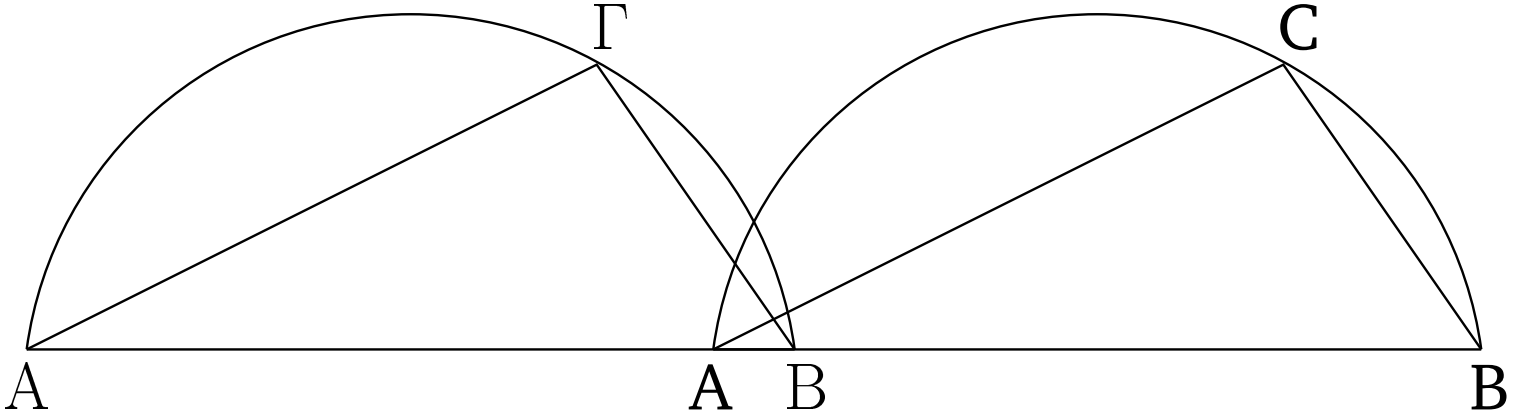
Ἐκ τριῶν ἄρα γωνιῶν ἐπιπέδων τῶν ὑπὸ APM , MPN , APN , αἱ εἰσὶν ἴσαι τρισὶ ταῖς δοθείσαις ταῖς ὑπὸ $ΑΒΓ$, $ΔΕΖ$, $ΗΘΚ$, στερεὰ γωνία συνίσταται ἡ πρὸς τῷ P περιεθρομένη ὑπὸ τῶν APM , MPN , APN γωνιῶν· ὅπερ ἔδει ποιῆσαι.

thus greater than the angle POQ [Prop. 1.25]. So, similarly, we can show that DEF is also greater than MON , and GHK than NOL . Thus, the (sum of the) three angles ABC , DEF , and GHK is greater than the (sum of the) three angles LOM , MON , and NOL . But, (the sum of) ABC , DEF , and GHK was assumed (to be) less than four right-angles. Thus, (the sum of) LOM , MON , and NOL is much less than four right-angles. But, (it is) also equal (to four right-angles). The very thing is absurd. Thus, AB is not less than LO . And it was shown (to be) not equal either. Thus, AB (is) greater than LO .

So let OR be set up at point O at right-angles to the plane of circle LMN [Prop. 11.12]. And let the (square) on OR be equal to that (area) by which the square on AB is greater than the (square) on LO [Prop. 11.23 lem.]. And let RL , RM , and RN be joined.

And since RO is at right-angles to the plane of circle LMN , RO is thus also at right-angles to each of LO , MO , and NO . And since LO is equal to OM , and OR is common and at right-angles, the base RL is thus equal to the base RM [Prop. 1.4]. So, for the same (reasons), RN is also equal to each of RL and RM . Thus, the three (straight-lines) RL , RM , and RN are equal to one another. And since the (square) on OR was assumed to be equal to that (area) by which the (square) on AB is greater than the (square) on LO , the (square) on AB is thus equal to the (sum of the squares) on LO and OR . And the (square) on LR is equal to the (sum of the squares) on LO and OR . For LOR (is) a right-angle [Prop. 1.47]. Thus, the (square) on AB is equal to the (square) on RL . Thus, AB (is) equal to RL . But, each of BC , DE , EF , GH , and HK is equal to AB , and each of RM and RN equal to RL . Thus, each of AB , BC , DE , EF , GH , and HK is equal to each of RL , RM , and RN . And since the two (straight-lines) LR and RM are equal to the two (straight-lines) AB and BC (respectively), and the base LM was assumed (to be) equal to the base AC , the angle LRM is thus equal to the angle ABC [Prop. 1.8]. So, for the same (reasons), MRN is also equal to DEF , and LRN to GHK .

Thus, the solid angle R , contained by the angles LRM , MRN , and LRN , has been constructed out of the three plane angles LRM , MRN , and LRN , which are equal to the three given (plane angles) ABC , DEF , and GHK (respectively). (Which is) the very thing it was required to do.



Λήμμα.

Lemma

Ὅν δὲ τρόπον, ὅῃ μείζον ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX , ἐκείνῳ ἴσον λαβεῖν ἐστὶ τὸ ἀπὸ τῆς XP , δείχομεν οὕτως. ἐκκείσθωσαν αἱ AB , AX εὐθεῖαι, καὶ ἔστω μείζων ἢ AB , καὶ γεγράφθω ἐπ' αὐτῆς ἡμικύκλιον τὸ $ABΓ$, καὶ εἷς τὸ $ABΓ$ ἡμικύκλιον ἐνηρμόσθω τῇ AX εὐθείᾳ μὴ μείζον οὕση τῆς AB διαμέτρον ἴση ἢ AG , καὶ ἐπεζεύθθω ἡ GB . ἐπεὶ οὖν ἐν ἡμικυκλίῳ τῷ AGB γωνία ἐστὶν ἡ ὑπὸ AGB , ὀρθή ἄρα ἐστὶν ἡ ὑπὸ AGB . τὸ ἄρα ἀπὸ τῆς AB ἴσον ἐστὶ τοῖς ἀπὸ τῶν AG , GB . ὥστε τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AG μείζον ἐστὶ τῷ ἀπὸ τῆς GB . ἴση δὲ ἡ AG τῇ AX . τὸ ἄρα ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX μείζον ἐστὶ τῷ ἀπὸ τῆς GB . ἔαν οὖν τῇ $BΓ$ ἴσην τὴν XP ἀπολάβωμεν, ἔσται τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AX μείζον τῷ ἀπὸ τῆς XP . ὅπερ προέκειτο ποιῆσαι.

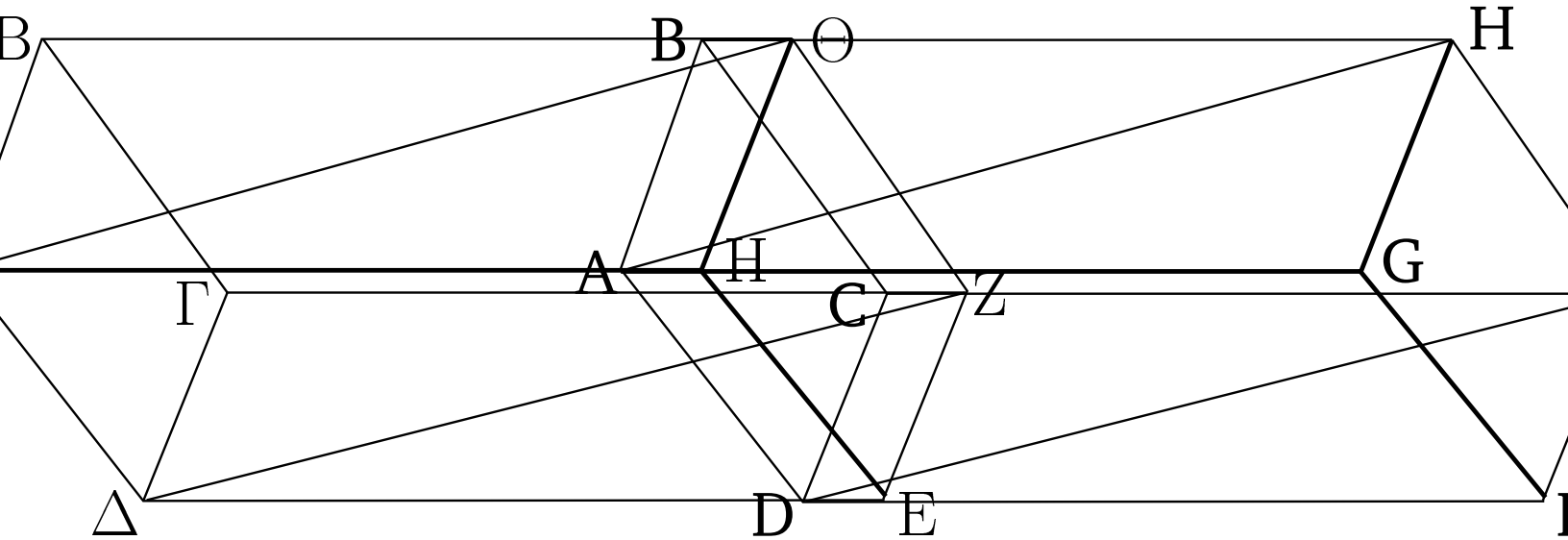
And we can demonstrate, thusly, in which manner to take the (square) on OR equal to that (area) by which the (square) on AB is greater than the (square) on LO . Let the straight-lines AB and LO be set out, and let AB be greater, and let the semicircle ABC be drawn around it. And let AC , equal to the straight-line LO , which is not greater than the diameter AB , be inserted into the semicircle ABC [Prop. 4.1]. And let CB be joined. Therefore, since the angle ACB is in the semicircle ACB , ACB is thus a right-angle [Prop. 3.31]. Thus, the (square) on AB is equal to the (sum of the) squares on AC and CB [Prop. 1.47]. Hence, the (square) on AB is greater than the (square) on AC by the (square) on CB . And AC (is) equal to LO . Thus, the (square) on AB is greater than the (square) on LO by the (square) on CB . Therefore, if we take OR equal to BC then the (square) on AB will be greater than the (square) on LO by the (square) on OR . (Which is) the very thing it was prescribed to do.

24.

Proposition 24

Ἐὰν στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθῃται, τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν.

If a solid (figure) is contained by (six) parallel planes then its opposite planes are both equal and parallelogrammic.



Στερεὸν γάρ τὸ ΓΔΘΗ ὑπὸ παραλλήλων ἐπιπέδων περι-
εθέσθω τῶν ΑΓ, ΗΖ, ΑΘ, ΔΖ, ΒΖ, ΑΕ· λέγω, ὅτι τὰ ἀπε-
ναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν.

Ἐπεὶ γὰρ δύο ἐπίπεδα παράλληλα τὰ ΒΗ, ΓΕ ὑπὸ ἐπι-
πέδον τοῦ ΑΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί
εἰσιν. παράλληλος ἄρα ἐστὶν ἡ ΑΒ τῇ ΔΓ. πάλιν, ἐπεὶ δύο
ἐπίπεδα παράλληλα τὰ ΒΖ, ΑΕ ὑπὸ ἐπιπέδον τοῦ ΑΓ τέμνε-
ται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν. παράλληλος ἄρα
ἐστὶν ἡ ΒΓ τῇ ΑΔ. ἐδείχθη δὲ καὶ ἡ ΑΒ τῇ ΔΓ παράλληλος·
παραλληλόγραμμον ἄρα ἐστὶ τὸ ΑΓ. ὁμοίως δὲ δείχομεν, ὅτι
καὶ ἕκαστον τῶν ΔΖ, ΖΗ, ΗΒ, ΒΖ, ΑΕ παραλληλόγραμμόν
ἐστίν.

Ἐπεξεύχθωσαν αἱ ΑΘ, ΔΖ. καὶ ἐπεὶ παράλληλός ἐστιν ἡ
μὲν ΑΒ τῇ ΔΓ, ἡ δὲ ΒΘ τῇ ΓΖ, δύο δὲ αἱ ΑΒ, ΒΘ ἀπτόμεναι
ἀλλήλων παρὰ δύο εὐθείας τὰς ΔΓ, ΓΖ ἀπτομένας ἀλλήλων
εἰσὶν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ· ἴσαξ ἄρα γωνίαξ περιέχουσιν
ἴση ἄρα ἡ ὑπὸ ΑΒΘ γωνία τῇ ὑπὸ ΔΓΖ. καὶ ἐπεὶ δύο αἱ
ΑΒ, ΒΘ δυοὶ ταῖς ΔΓ, ΓΖ ἴσαι εἰσὶν, καὶ γωνία ἡ ὑπὸ ΑΒΘ
γωνία τῇ ὑπὸ ΔΓΖ ἐστὶν ἴση, βάσις ἄρα ἡ ΑΘ βάσει τῇ ΔΖ
ἐστὶν ἴση, καὶ τὸ ΑΒΘ τρίγωνον τῷ ΔΓΖ τριγόνῳ ἴσον ἐστίν.
καὶ ἐστὶ τοῦ μὲν ΑΒΘ διπλάσιον τὸ ΒΗ παραλληλόγραμμον,
τοῦ δὲ ΔΓΖ διπλάσιον τὸ ΓΕ παραλληλόγραμμον· ἴσον ἄρα
τὸ ΒΗ παραλληλόγραμμον τῷ ΓΕ παραλληλόγραμμῳ· ὁμοίως
δὲ δείχομεν, ὅτι καὶ τὸ μὲν ΑΓ τῷ ΗΖ ἐστὶν ἴσον, τὸ δὲ ΑΕ
τῷ ΒΖ.

Ἐὰν ἄρα στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθῃται,
τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά
ἐστιν· ὅπερ ἔδει δεῖξαι.

25.

Ἐὰν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμηθῇ παραλ-
λήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔσται ὡς ἡ βάσις πρὸς
τὴν βάσιν, οὕτως τὸ στερεὸν πρὸς τὸ στερεόν.

For let the solid (figure) $CDHG$ be contained by the par-
allel planes AC , GF , and AH , DF , and BF , AE . I say that its
opposite planes are both equal and parallelogrammic.

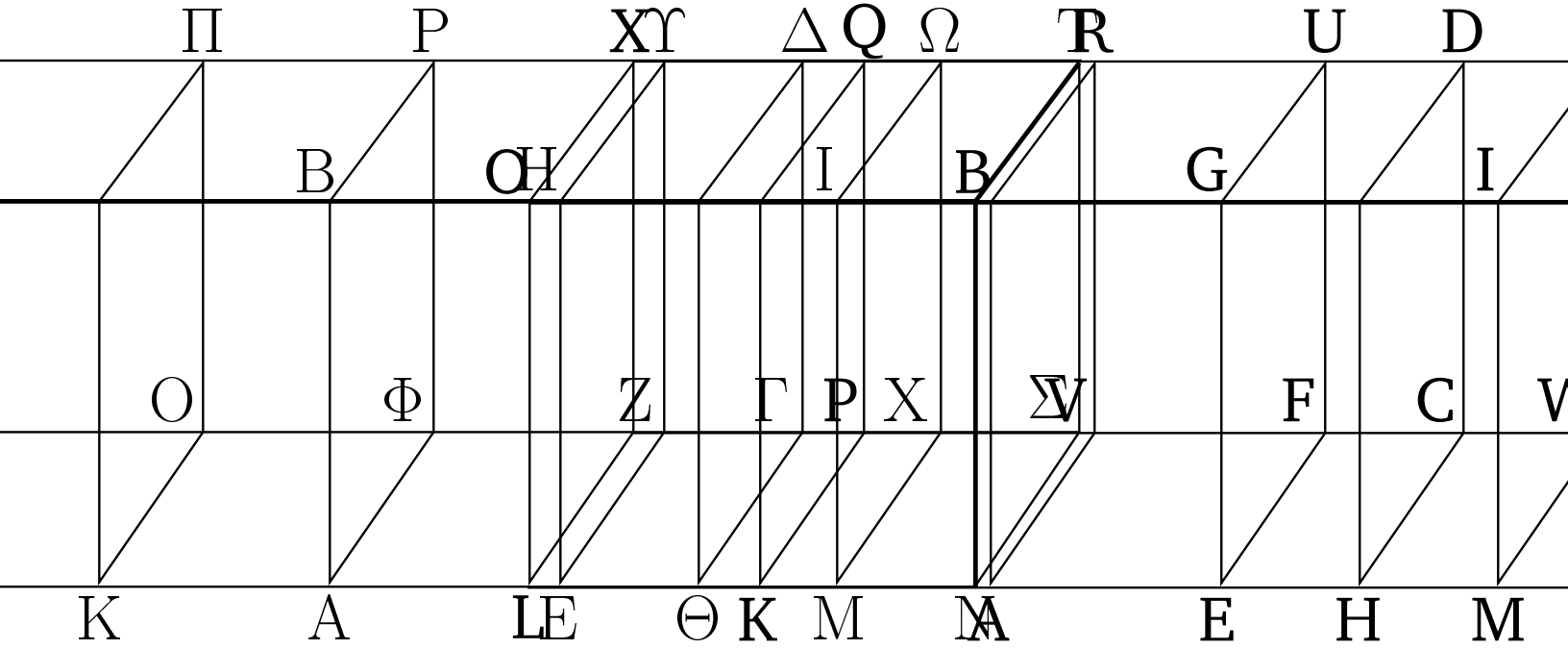
For since the two parallel planes BG and CE are cut by the
plane AC , their common sections are parallel [Prop. 11.16].
Thus, AB is parallel to DC . Again, since the two parallel
planes BF and AE are cut by the plane AC , their common
sections are parallel [Prop. 11.16]. Thus, BC is parallel to AD .
And AB was also shown (to be) parallel to DC . Thus, AC is a
parallelogram. So, similarly, we can also show that DF , FG ,
 GB , BF , and AE are each parallelograms.

Let AH and DF be joined. And since AB is parallel to DC ,
and BH to CF , so the two (straight-lines) joining one another,
 AB and BH , are parallel to the two straight-lines joining one
another, DC and CF (respectively), not (being) in the same
plane. Thus, they will contain equal angles [Prop. 11.10].
Thus, angle ABH (is) equal to (angle) DCF . And since the
two (straight-lines) AB and BH are equal to the two (straight-
lines) DC and CF (respectively) [Prop. 1.34], and angle ABH
is equal to angle DCF , the base AH is thus equal to the base
 DF , and triangle ABH is equal to triangle DCF [Prop. 1.4].
And parallelogram BG is double (triangle) ABH , and parallel-
ogram CE double (triangle) DCF [Prop. 1.34]. Thus, parallel-
ogram BG (is) equal to parallelogram CE . So, similarly, we
can show that AC is also equal to GF , and AE to BF .

Thus, if a solid (figure) is contained by (six) parallel planes
then its opposite planes are both equal and parallelogrammic.
(Which is) the very thing it was required to show.

Proposition 25

If a parallelepiped solid is cut by a plane which is parallel
to the opposite planes (of the parallelepiped) then as the base
(is) to the base, so the solid will be to the solid.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ $AB\Gamma\Delta$ ἐπιπέδῳ τῷ ZH τεμήσθω παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς PA , $\Delta\Theta$ · λέγω, ὅτι ἐστὶν ὡς ἡ $AEZ\Phi$ βάσις πρὸς τὴν $E\Theta\Gamma Z$ βάσιν, οὕτως τὸ $ABZY$ στερεὸν πρὸς τὸ $E\Theta\Gamma\Delta$ στερεόν.

Ἐκβεβλήσθω γάρ ἡ $A\Theta$ ἐφ' ἐκάτερα τὰ μέρη, καὶ κείσθωσαν τῇ μὲν AE ἴσαι ὁσαιδηποῦν αἱ AK , KL , τῇ δὲ $E\Theta$ ἴσαι ὁσαιδηποῦν αἱ ΘM , MN , καὶ συμπληρώσθω τὰ AO , $K\Phi$, $\Theta\Theta$, $M\Sigma$ παραλληλόγραμμα καὶ τὰ $\Lambda\Pi$, KP , ΔM , MT στερεά.

Καὶ ἐπεὶ ἴσαι εἰσὶν αἱ ΛK , KA , AE εὐθεῖαι ἀλλήλαις, ἴσα ἐστὶ καὶ τὰ μὲν ΛO , $K\Phi$, AZ παραλληλόγραμμα ἀλλήλοις, τὰ δὲ KX , KB , AH ἀλλήλοις καὶ ἔτι τὰ $\Lambda\Psi$, $K\Pi$, AP ἀλλήλοις· ἀπεναντίον γάρ. διὰ τὰ αὐτὰ δὴ καὶ τὰ μὲν $E\Gamma$, $\Theta\Theta$, $M\Sigma$ παραλληλόγραμμα ἴσα εἰσὶν ἀλλήλοις, τὰ δὲ ΘH , ΘI , IN ἴσα εἰσὶν ἀλλήλοις, καὶ ἔτι τὰ $\Delta\Theta$, $M\Omega$, NT · τρία ἄρα ἐπίπεδα τῶν $\Lambda\Pi$, KP , AY στερεῶν τρισὶν ἐπιπέδοις ἐστὶν ἴσα· ἀλλὰ τὰ τρία τρισὶ τοῖς ἀπεναντίον ἐστὶν ἴσα· τὰ ἄρα τρία στερεὰ τὰ $\Lambda\Pi$, KP , AY ἴσα ἀλλήλοις ἐστὶν. διὰ τὰ αὐτὰ δὴ καὶ τὰ τρία στερεὰ τὰ $E\Delta$, ΔM , MT ἴσα ἀλλήλοις ἐστὶν· ὁσαπλασίον ἄρα ἐστὶν ἡ AZ βάσις τῇ AZ βάσει, τοσανταπλάσιόν ἐστὶ καὶ τὸ AY στερεὸν τοῦ AY στερεοῦ. διὰ τὰ αὐτὰ δὴ ὁσαπλασίον ἐστὶν ἡ NZ βάσις τῇ $Z\Theta$ βάσει, τοσανταπλάσιόν ἐστὶ καὶ τὸ NY στερεὸν τοῦ ΘY στερεοῦ. καὶ εἰ ἴση ἐστὶν ἡ AZ βάσις τῇ NZ βάσει, ἴσον ἐστὶ καὶ τὸ AY στερεὸν τῷ NY στερεῷ, καὶ εἰ ὑπερέθει ἡ AZ βάσις τῇ NZ βάσει, ὑπερέθει καὶ τὸ AY στερεὸν τοῦ NY στερεοῦ, καὶ εἰ ἐλλείπει, ἐλλείπει. τεσσάρων δὲ ὄντων μεγεθῶν, δύο μὲν βάσεων τῶν AZ , $Z\Theta$, δύο δὲ στερεῶν τῶν AY , $Y\Theta$, εἴληπται ἰσάκως πολλαπλάσια

For let the parallelepiped solid $ABCD$ be cut by the plane FG which is parallel to the opposite planes RA and DH . I say that as the base $AEFV$ (is) to the base $EHCF$, so the solid $ABFU$ (is) to the solid $EGCD$.

For let AH be produced in each direction. And let any number whatsoever (of lengths), AK and KL , be made equal to AE , and any number whatsoever (of lengths), HM and MN , equal to EH . And let the parallelograms LP , KV , HW , and MS be completed, and the solids LQ , KR , DM , and MT .

And since the straight-lines LK , KA , and AE are equal to one another, the parallelograms LP , KV , and AF are also equal to one another, and KO , KB , and AG (are equal) to one another, and, further, LX , KQ , and AR (are equal) to one another. For (they are) opposite [Prop. 11.24]. So, for the same (reasons), the parallelograms EC , HW , and MS are also equal to one another, and HG , HI , and IN are equal to one another, and, further, DH , MY , and NT (are equal to one another). Thus, three planes of (one of) the solids LQ , KR , and AU are equal to the (corresponding) three planes (of the others). But, the three planes (in one of the solids) are equal to the three opposite planes [Prop. 11.24]. Thus, the three solids LQ , KR , and AU are equal to one another [Def. 11.10]. So, for the same (reasons), the three solids ED , DM , and MT are also equal to one another. Thus, as many multiples as the base LF is of the base AF , so many multiples is the solid LU also of the solid AU . So, for the same (reasons), as many multiples as the base NF is of the base FH , so many multiples is the solid NU also of the solid HU . And if the base LF is equal to the base

τῆς μὲν AZ βάσεως καὶ τοῦ AY στερεοῦ ἢ τε AZ βάσις καὶ τὸ AY στερεόν, τῆς δὲ ΘZ βάσεως καὶ τοῦ ΘY στερεοῦ ἢ τε NZ βάσις καὶ τὸ NY στερεόν, καὶ δέδεικται, ὅτι εἰ ὑπερέθει ἢ AZ βάσις τῆς ZN βάσεως, ὑπερέθει καὶ τὸ AY στερεόν τοῦ NY [στερεοῦ], καὶ εἰ ἴση, ἴσον, καὶ εἰ ἐλλείπει, ἐλλείπει. ἔστιν ἄρα ὥς ἢ AZ βάσις πρὸς τὴν $Z\Theta$ βάσιν, οὕτως τὸ AY στερεόν πρὸς τὸ $Y\Theta$ στερεόν· ὅπερ ἔδει δεῖξαι.

NF then the solid LU is also equal to the solid NU .[†] And if the base LF exceeds the base NF then the solid LU also exceeds the solid NU . And if (LF) is less than (NF) then (LU) is (also) less than (NU) . So, there are four magnitudes, the two bases AF and FH , and the two solids AU and UH , and equal multiples be taken of the base AF and the solid AU —(namely), the base LF and the solid LU —and of the base FH and the solid HU —(namely), the base NF and the solid NU . And it has been shown that if the base LF exceeds the base FN then the solid LU also exceeds the [solid] NU , and if (LF) is equal (to FN) then (LU) is equal (to NU), and if (LF) is less than (FN) then (LU) is less than (NU) . Thus, as the base AF is to the base FH , so the solid AU (is) to the solid UH [Def. 5.5]. (Which is) the very thing it was required to show.

[†] Here, Euclid assumes that $LF \cong NF$ implies $LU \cong NU$. This is easily demonstrated.

26.

Proposition 26

Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ στερεᾷ γωνίᾳ ἴσην στερεὰν γωνίαν συστήσασθαι.

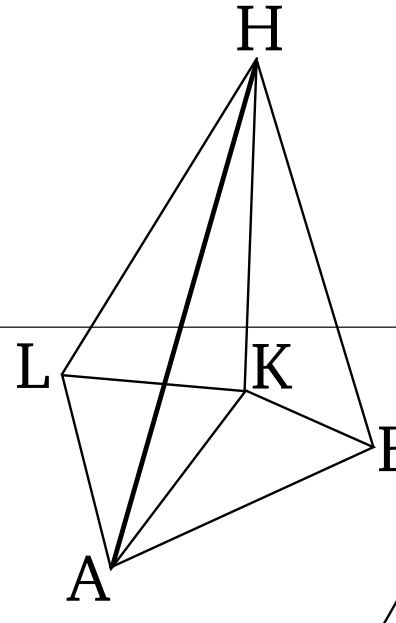
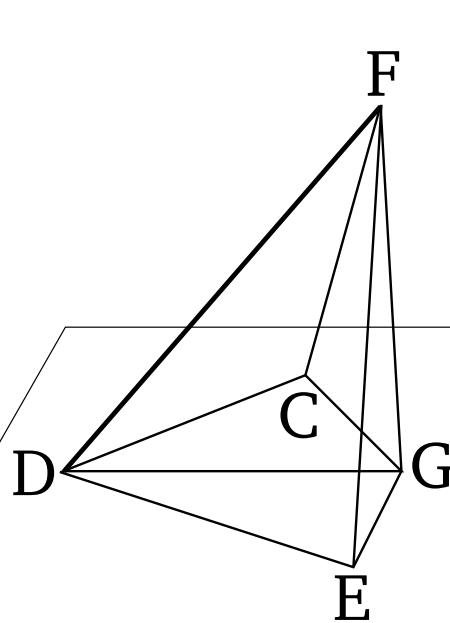
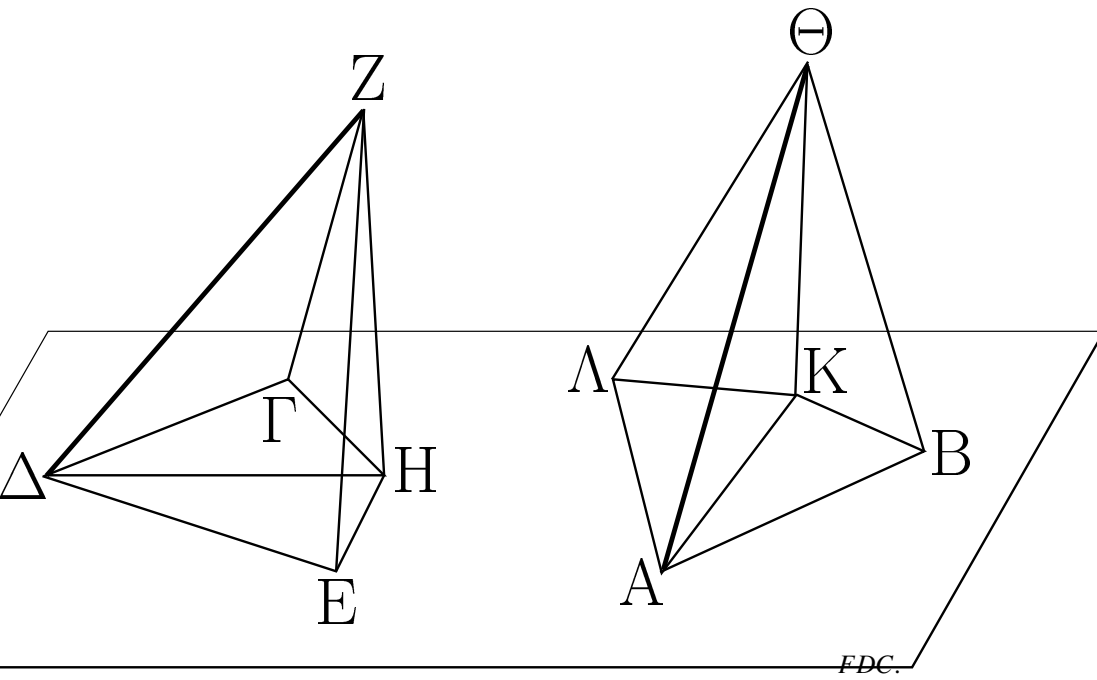
Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ πρὸς αὐτῇ δοθέν σημείον τὸ A , ἡ δὲ δοθεῖσα στερεὰ γωνία ἡ πρὸς τῷ Δ περιεθρομένη ὑπὸ τῶν ὑπὸ $E\Delta\Gamma$, $E\Delta Z$, $Z\Delta\Gamma$ γωνιῶν ἐπιπέδων· δεῖ δὲ πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ πρὸς τῷ Δ στερεᾷ γωνίᾳ ἴσην στερεὰν γωνίαν συστήσασθαι.

Εἰλήφθω γάρ ἐπὶ τῆς ΔZ τυθὸν σημείον τὸ Z , καὶ ἦθθω ἀπὸ τοῦ Z ἐπὶ τὸ διὰ τῶν $E\Delta$, $\Delta\Gamma$ ἐπίπεδον κάθετος ἡ ZH , καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ H , καὶ ἐπεξέυθθω ἡ ΔH , καὶ συνεστέτω πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ μὲν ὑπὸ $E\Delta\Gamma$ γωνίᾳ ἴση ἢ ὑπὸ BAA , τῇ δὲ ὑπὸ $E\Delta H$ ἴση ἢ ὑπὸ BAK , καὶ κείσθω τῇ ΔH ἴση ἢ AK , καὶ ἀνεστέτω ἀπὸ τοῦ K σημείου τῷ διὰ τῶν BAA ἐπιπέδῳ πρὸς ὀρθὰς ἡ $K\Theta$, καὶ κείσθω ἴση τῇ HZ ἢ $K\Theta$, καὶ ἐπεξέυθθω ἡ ΘA · λέγω, ὅτι ἡ πρὸς τῷ A στερεὰ γωνία περιεθρομένη ὑπὸ τῶν BAA , $BA\Theta$, ΘAA γωνιῶν ἴση ἐστὶ τῇ πρὸς τῷ Δ στερεᾷ γωνίᾳ τῇ περιεθρομένη ὑπὸ τῶν $E\Delta\Gamma$, $E\Delta Z$, $Z\Delta\Gamma$ γωνιῶν.

To construct a solid angle equal to a given solid angle on a given straight-line, and at a given point on it.

Let AB be the given straight-line, and A the given point on it, and D the given solid angle, contained by the plane angles EDC , EDF , and FDC . So, it is necessary to construct a solid angle equal to the solid angle D on the straight-line AB , and at the point A on it.

For let some random point F be taken on DF , and let FG be drawn from F perpendicular to the plane through ED and DC [Prop. 11.11], and let it meet the plane at G , and let DG be joined. And let BAL , equal to the angle EDC , and BAK , equal to EDG , be constructed on the straight-line AB at the point A on it [Prop. 1.23]. And let AK be made equal to DG . And let KH be set up at the point K at right-angles to the plane through BAL [Prop. 11.12]. And let KH be made equal to GF . And let HA be joined. I say that the solid angle at A , contained by the (plane) angles BAL , BAH , and HAL , is equal to the solid angle at D , contained by the (plane) angles EDC , EDF , and



Ἀπειλήφθωσαν γὰρ ἴσαι αἱ AB , DE , καὶ ἐπεξέσθωσαν αἱ $ΘB$, KB , ZE , HE . καὶ ἐπεὶ ἡ ZH ὀρθὴ ἐστὶ πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσαις ἄρα τὰς ἀπτομένας αὐτῆς ἐνθεῖαις καὶ οὐσαὶ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας· ὀρθὴ ἄρα ἐστὶν ἑκάτερα τῶν ὑπὸ $ZHΔ$, ZHE γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερα τῶν ὑπὸ $ΘKA$, $ΘKB$ γωνιῶν ὀρθὴ ἐστὶν. καὶ ἐπεὶ δύο αἱ KA , AB δύο ταῖς $HΔ$, DE ἴσαι εἰσὶν ἑκάτερα ἑκάτερα, καὶ γωνίαξ ἴσαξ περιέθουσιν, βάσει ἄρα ἡ KB βάσει τῇ HE ἴση ἐστίν. ἔστι δὲ καὶ ἡ $KΘ$

For let AB and DE be cut off (so as to be) equal, and let HB , KB , FE , and GE be joined. And since FG is at right-angles to the reference plane (EDC), it will also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Thus, the angles FGD and FGE are right-angles. So, for the same (reasons), the angles HKA and HKB are also right-angles. And since the two (straight-lines) KA and AB are equal to the two (straight-lines)

τῇ HZ ἴση· καὶ γωνίαξ ὀρθὰς περιέθουσιν· ἴση ἄρα καὶ ἡ ΘB τῇ ZE . πάλιν ἐπεὶ δύο αἱ AK , $K\Theta$ δυοὶ ταῖς ΔH , HZ ἴσαι εἰσὶν, καὶ γωνίαξ ὀρθὰς περιέθουσιν, βάσις ἄρα ἡ $A\Theta$ βάσει τῇ $Z\Delta$ ἴση ἐστίν. ἔστι δὲ καὶ ἡ AB τῇ ΔE ἴση· δύο δὴ αἱ ΘA , AB δύο ταῖς ΔZ , ΔE ἴσαι εἰσὶν. καὶ βάσις ἡ ΘB βάσει τῇ ZE ἴση· γωνία ἄρα ἡ ὑπὸ $BA\Theta$ γωνία τῇ ὑπὸ $E\Delta Z$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $\Theta A\Lambda$ τῇ ὑπὸ $Z\Delta\Gamma$ ἐστὶν ἴση. ἔστι δὲ καὶ ἡ ὑπὸ BAL τῇ ὑπὸ $E\Delta\Gamma$ ἴση.

Πρὸς ἄρα τῇ δοθείσῃ εὐθείᾳ τῇ AB καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ δοθείσῃ στερεᾷ γωνίᾳ τῇ πρὸς τῷ Δ ἴση συνέσταιται· ὅπερ ἔδει ποιῆσαι.

GD and DE , respectively, and they contain equal angles, the base KB is thus equal to the base GE [Prop. 1.4]. And KH is also equal to GF . And they contain right-angles (with the respective bases). Thus, HB (is) also equal to FE [Prop. 1.4]. Again, since the two (straight-lines) AK and KH are equal to the two (straight-lines) DG and GF (respectively), and they contain right-angles, the base AH is thus equal to the base FD [Prop. 1.4]. And AB (is) also equal to DE . So, the two (straight-lines) HA and AB are equal to the two (straight-lines) DF and DE (respectively). And the base HB (is) equal to the base FE . Thus, the angle BAH is equal to the angle EDF [Prop. 1.8]. So, for the same (reasons), HAL is also equal to FDC . And BAL is also equal to EDC .

Thus, (a solid angle) has been constructed, equal to the given solid angle at D , on the given straight-line AB , at the given point A on it. (Which is) the very thing it was required to do.

27.

Proposition 27

Ἀπὸ τῆς δοθείσης εὐθείας τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ ὁμοίον τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

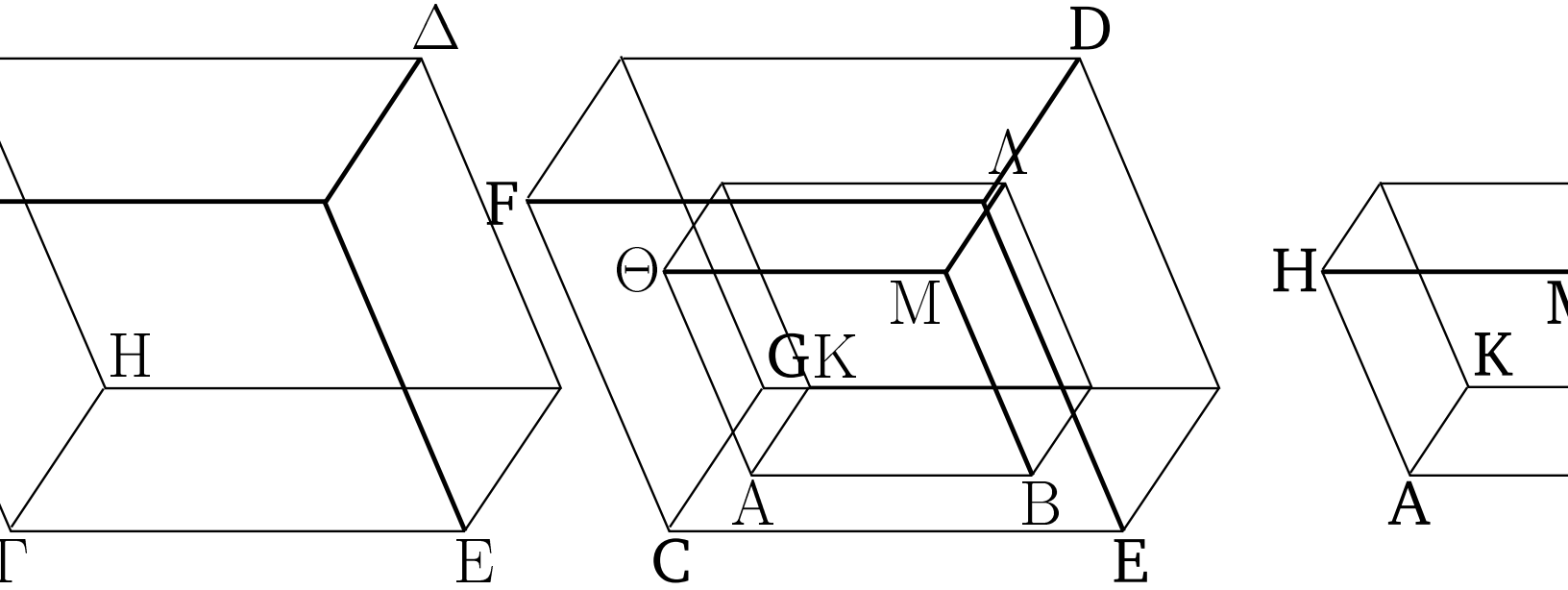
Ἐστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB , τὸ δὲ δοθέν στερεὸν παραλληλεπίπεδον τὸ $\Gamma\Delta$. δεῖ δὴ ἀπὸ τῆς δοθείσης εὐθείας τῆς AB τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ τῷ $\Gamma\Delta$ ὁμοίον τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

Συνεστάτω γάρ πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ πρὸς τῷ Γ στερεᾷ γωνίᾳ ἴση ἡ περιεθομένη ὑπὸ τῶν $BA\Theta$, ΘAK , KAB , ὥστε ἴσην εἶναι τὴν μὲν ὑπὸ $BA\Theta$ γωνίαν τῇ ὑπὸ $E\Gamma Z$, τὴν δὲ ὑπὸ BAK τῇ ὑπὸ $E\Gamma H$, τὴν δὲ ὑπὸ $KA\Theta$ τῇ ὑπὸ $H\Gamma Z$. καὶ γερονέτω ὥς μὲν ἡ $E\Gamma$ πρὸς τὴν ΓH , οὕτως ἡ BA πρὸς τὴν AK , ὥς δὲ ἡ $H\Gamma$ πρὸς τὴν ΓZ , οὕτως ἡ KA πρὸς τὴν $A\Theta$. καὶ δι' ἴσον ἄρα ἐστὶν ὥς ἡ $E\Gamma$ πρὸς τὴν ΓZ , οὕτως ἡ BA πρὸς τὴν $A\Theta$. καὶ συμπληρώσθω τὸ ΘB παραλληλόγραμμον καὶ τὸ AL στερεόν.

To describe a parallelepiped solid similar, and similarly laid out, to a given parallelepiped solid on a given straight-line.

Let the given straight-line be AB , and the given parallelepiped solid CD . So, it is necessary to describe a parallelepiped solid similar, and similarly laid out, to the given parallelepiped solid CD on the given straight-line AB .

For, let a (solid angle) contained by the (plane angles) BAH , HAK , and KAB be constructed, equal to solid angle at C , on the straight-line AB at the point A on it [Prop. 11.26], such that angle BAH is equal to ECF , and BAK to ECG , and KAH to GCF . And let it be contrived that as EC (is) to CG , so BA (is) to AK , and as GC (is) to CF , so KA (is) to AH [Prop. 6.12]. And thus, via equality, as EC is to CF , so BA (is) to AH [Prop. 5.22]. And let the parallelogram HB be completed, and the solid AL .



Καὶ ἐπεὶ ἐστὶν ὡς ἡ ΕΓ πρὸς τὴν ΓΗ, οὕτως ἡ ΒΑ πρὸς τὴν ΑΚ, καὶ περὶ ἴσαξ γωνίαξ τὰς ὑπὸ ΕΓΗ, ΒΑΚ αἱ πλεοναὶ ἀνάλογόν εἰσιν, ὁμοίων ἄρα ἐστὶ τὸ ΗΕ παραλληλόγραμμον τῷ ΚΒ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΚΘ παραλληλόγραμμον τῷ ΗΖ παραλληλογράμμῳ ὁμοίων ἐστὶ καὶ ἐτι τὸ ΖΕ τῷ ΘΒ· τρία ἄρα παραλληλόγραμμα τοῦ ΓΔ στερεοῦ τρισὶ παραλληλογράμμοις τοῦ ΑΛ στερεοῦ ὁμοία ἐστὶν. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὁμοία, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὁμοία· ὅλον ἄρα τὸ ΓΔ στερεὸν ὅλῳ τῷ ΑΛ στερεῷ ὁμοίων ἐστὶν.

Ἀπὸ τῆς δοθείσης ἄρα εὐθείας τῆς ΑΒ τῷ δοθέντι στερεῷ παραλληλεπίπεδῳ τῷ ΓΔ ὁμοίων τε καὶ ὁμοίωξ κείμενον ἀναγέγραπται τὸ ΑΛ· ὅπερ ἔδει ποιῆσαι.

28.

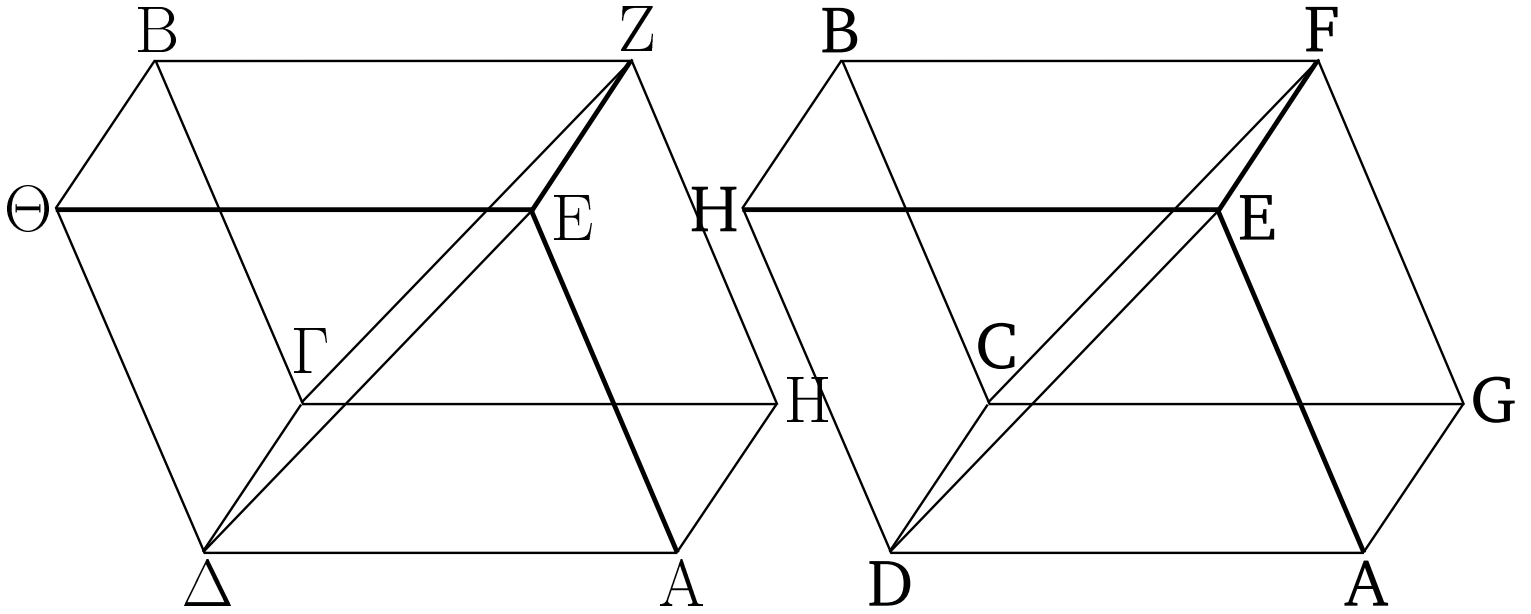
Ἐάν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμηθῇ κατὰ τὰς διαγωνίονξ τῶν ἀπεναντίον ἐπιπέδων, δίθῃ τμηθήσεται τὸ στερεὸν ὑπὸ τοῦ ἐπιπέδου.

And since as EC is to CG , so BA (is) to AK , and the sides about the equal angles ECG and BAK are (thus) proportional, the parallelogram GE is thus similar to the parallelogram KB . So, for the same (reasons), the parallelogram KH is also similar to the parallelogram GF , and, further, FE (is similar) to HB . Thus, three of the parallelograms of solid CD are similar to three of the parallelograms of solid AL . But, the (former) three are equal and similar to the three opposite, and the (latter) three are equal and similar to the three opposite. Thus, the whole solid CD is similar to the whole solid AL [Def. 11.9].

Thus, AL , similar, and similarly laid out, to the given parallelepiped solid CD , has been described on the given straight-lines AB . (Which is) the very thing it was required to do.

Proposition 28

If a parallelepiped solid is cut by a plane (passing) through the diagonals of (a pair of) opposite planes then the solid will be cut in half by the plane.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ AB ἐπιπέδῳ τῷ $ΓΔΕΖ$ τετμήσθω κατὰ τὰς διαγωνίους τῶν ἀπεναντίον ἐπιπέδων τὰς $ΓΖ$, $ΔΕ$ · λέγω, ὅτι διῖθα τμηθήσεται τὸ AB στερεὸν ὑπὸ τοῦ $ΓΔΕΖ$ ἐπιπέδου.

Ἐπεὶ γὰρ ἴσον ἐστὶ τὸ μὲν $ΓΗΖ$ τρίγωνον τῷ $ΓΖΒ$ τριγώνῳ, τὸ δὲ $ΑΔΕ$ τῷ $ΔΕΘ$, ἔστι δὲ καὶ τὸ μὲν $ΓΑ$ παραλληλογράμμον τῷ $ΕΒ$ ἴσον· ἀπεναντίον γάρ· τὸ δὲ $ΗΕ$ τῷ $ΓΘ$, καὶ τὸ πρίσμα ἄρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν $ΓΗΖ$, $ΑΔΕ$, τριῶν δὲ παραλληλογράμμων τῶν $ΗΕ$, $ΑΓ$, $ΓΕ$ ἴσον ἐστὶ τῷ πρίσματι τῷ περιεθρομένῳ ὑπὸ δύο μὲν τριγώνων τῶν $ΓΖΒ$, $ΔΕΘ$, τριῶν δὲ παραλληλογράμμων τῶν $ΓΘ$, $ΒΕ$, $ΓΕ$ · ὑπὸ γὰρ ἴσων ἐπιπέδων περιέθρονται τῷ τε πλήθει καὶ τῷ μεγέθει. ὥστε ὅλον τὸ AB στερεὸν διῖθα τέτμηται ὑπὸ τοῦ $ΓΔΕΖ$ ἐπιπέδου· ὅπερ ἔδει δεῖξαι.

For let the parallelepiped solid AB be cut by the plane $CDEF$ (passing) through the diagonals of the opposite planes CF and DE .[†] I say that the solid AB will be cut in half by the plane $CDEF$.

For since triangle CGF is equal to triangle CFB , and ADE (is equal) to DEH [Prop. 1.34], and parallelogram CA is also equal to EB —for (they are) opposite [Prop. 11.24]—and GE (equal) to CH , thus the prism contained by the two triangles CGF and ADE , and the three parallelograms GE , AC , and CE , is also equal to the prism contained by the two triangles CFB and DEH , and the three parallelograms CH , BE , and CE . For they are contained by planes (which are) equal in number and in magnitude [Def. 11.10].[‡] Thus, the whole of solid AB is cut in half by the plane $CDEF$. (Which is) the very thing it was required to show.

[†] Here, it is assumed that the two diagonals lie in the same plane. The proof is easily supplied.

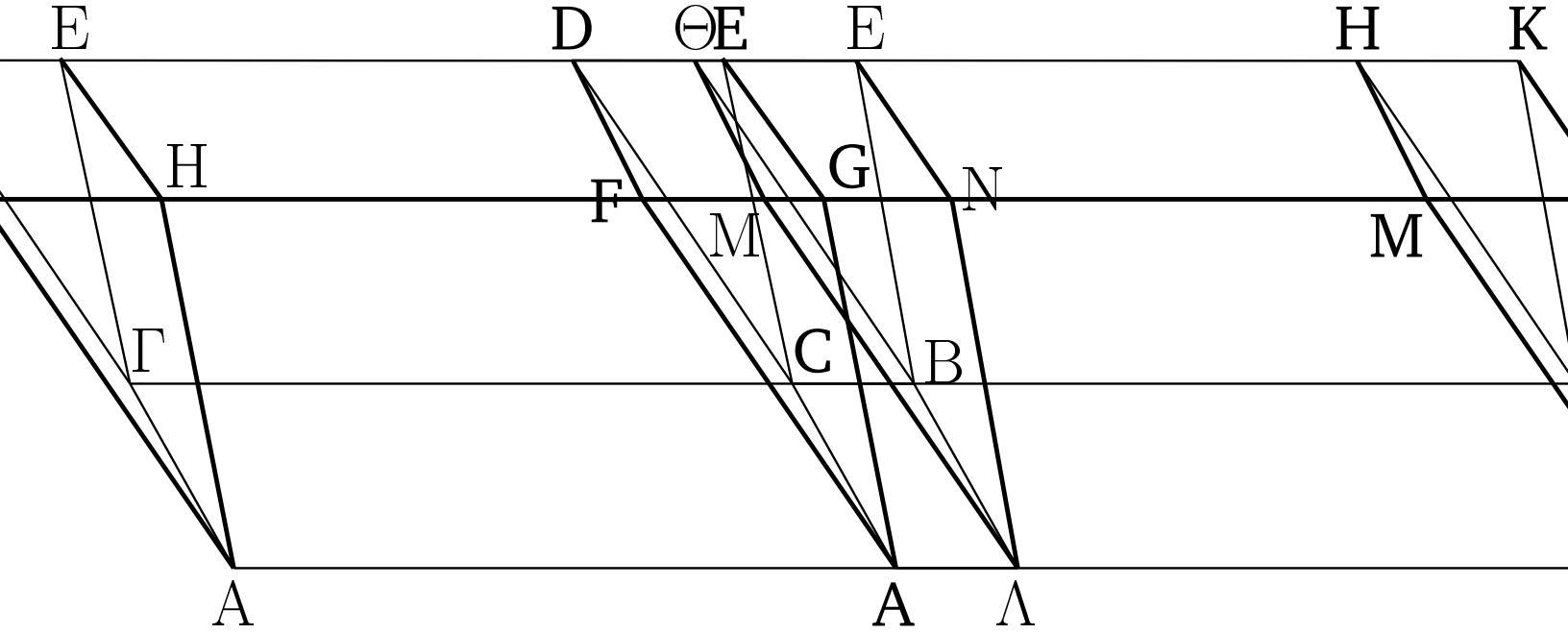
[‡] However, strictly speaking, the prisms are not similarly arranged, being mirror images of one another.

29.

Τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν.

Proposition 29

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are on the same straight-lines, are equal to one another.



Ἐστω ἐπὶ τῆς αὐτῆς βάσεως τῆς AB στερεὰ παραλληλεπίπεδα τὰ $ΓΜ$, $ΓΝ$ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AH , AZ , $ΑΜ$, $ΑΝ$, $ΓΔ$, $ΓΕ$, $BΘ$, BK ἐπὶ τῶν αὐτῶν εὐθειῶν ἕστωσαν τῶν ZN , $ΔΚ$ · λέγω, ὅτι ἴσον ἐστὶ τὸ $ΓΜ$ στερεὸν τῷ $ΓΝ$ στερεῷ.

Ἐπεὶ γὰρ παραλληλόγραμμον ἐστὶν ἑκάτερον τῶν $ΓΘ$, $ΓΚ$, ἴση ἐστὶν ἡ $ΓΒ$ ἑκατέρω τῶν $ΔΘ$, $ΕΚ$ · ὥστε καὶ ἡ $ΔΘ$ τῇ $ΕΚ$ ἐστὶν ἴση. κοινὴ ἀφηρήσθω ἡ $ΕΘ$ · λοιπὴ ἄρα ἡ $ΔΕ$ λοιπῇ τῇ $ΘΚ$ ἐστὶν ἴση. ὥστε καὶ τὸ μὲν $ΔΓΕ$ τρίγωνον τῷ $ΘΒΚ$ τριγώνῳ ἴσον ἐστίν, τὸ δὲ $ΔΗ$ παραλληλόγραμμον τῷ $ΘΝ$ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ AZH τριγώνον τῷ $ΜΑΝ$ τριγώνῳ ἴσον ἐστίν. ἔστι δὲ καὶ τὸ μὲν $ΓΖ$ παραλληλόγραμμον τῷ $ΒΜ$ παραλληλογράμμῳ ἴσον, τὸ δὲ $ΓΗ$ τῷ $ΒΝ$ · ἀπεναντίον γάρ· καὶ τὸ πρίσμα ἄρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν AZH , $ΔΓΕ$, τριῶν δὲ παραλληλογράμμων τῶν $ΑΔ$, $ΔΗ$, $ΓΗ$ ἴσον ἐστὶ τῷ πρίσματι τῷ περιεχομένῳ ὑπὸ δύο μὲν τριγώνων τῶν $ΜΑΝ$, $ΘΒΚ$, τριῶν δὲ παραλληλογράμμων τῶν $ΒΜ$, $ΘΝ$, $ΒΝ$. κοινὸν προσκείσθω τὸ στερεόν, οὗ βάσις μὲν τὸ AB παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $HEOM$ · ὅλον ἄρα τὸ $ΓΜ$ στερεὸν παραλληλεπίπεδον ὅλῳ τῷ $ΓΝ$ στερεῷ παραλληλεπίπεδῳ ἴσον ἐστίν.

Τὰ ἄρα ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

30.

Τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσιν ἐπὶ τῶν

For let the parallelepiped solids CM and CN be on the same base AB , and (have) the same height, and let the (ends of the straight-lines) standing up in them, AG , AF , LM , LN , CD , CE , BH , and BK , be on the same straight-lines, FN and DK . I say that solid CM is equal to solid CN .

For since CH and CK are each parallelograms, CB is equal to each of DH and EK [Prop. 1.34]. Hence, DH is also equal to EK . Let EH be subtracted from both. Thus, the remainder DE is equal to the remainder HK . Hence, triangle DCE is also equal to triangle HBK [Props. 1.4, 1.8], and parallelogram DG to parallelogram HN [Prop. 1.36]. So, for the same (reasons), triangle AFG is also equal to triangle MLN . And parallelogram CF is also equal to parallelogram BM , and CG to BN [Prop. 11.24]. For they are opposite. Thus, the prism contained by the two triangles AFG and DCE , and the three parallelograms AD , DG , and CG , is equal to the prism contained by the two triangles MLN and HBK , and the three parallelograms BM , HN , and BN . Let the solid whose base (is) parallelogram AB , and (whose) opposite (face is) $GEHM$, be added to both (prisms). Thus, the whole parallelepiped solid CM is equal to the whole parallelepiped solid CN .

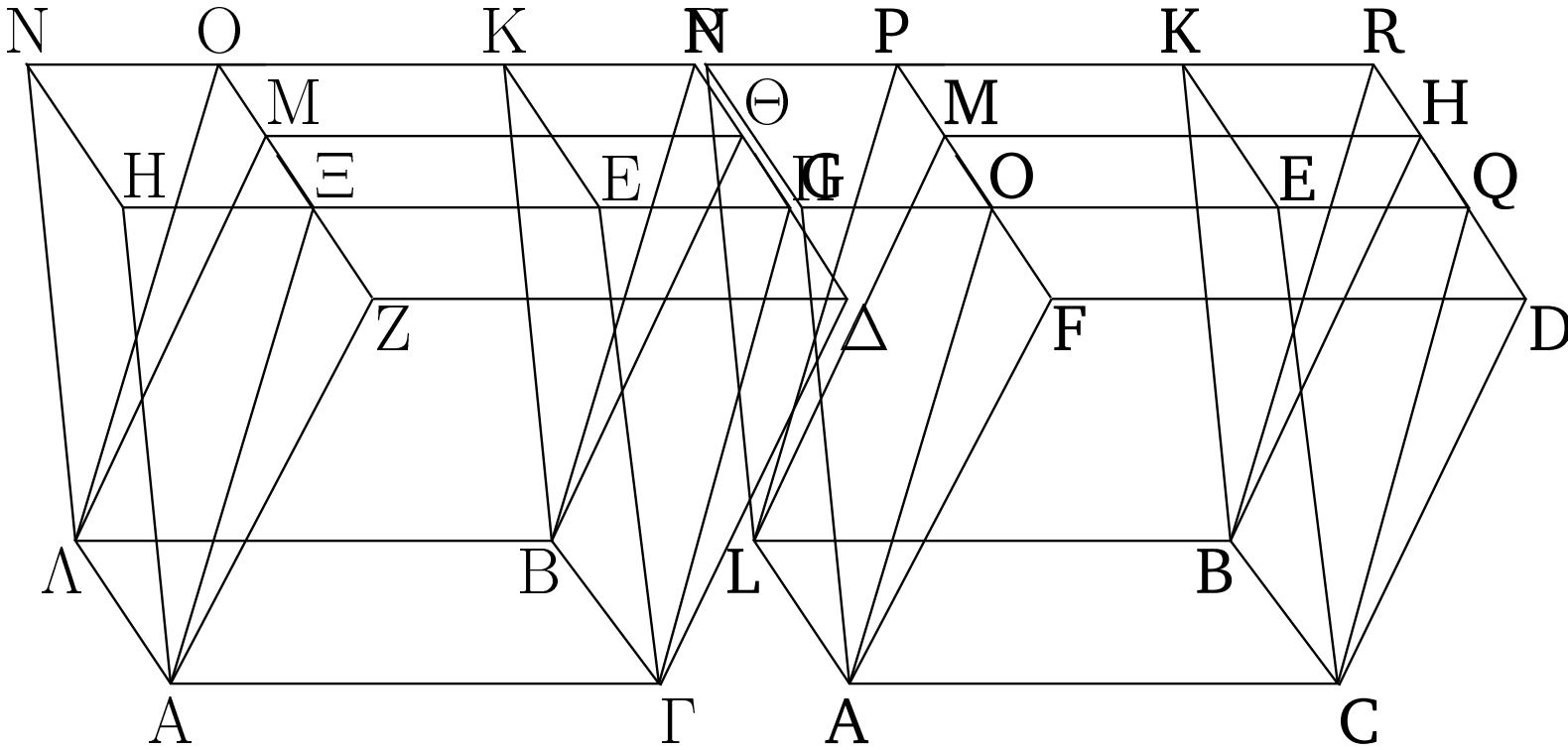
Thus, parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up (are) on the same straight-lines, are equal to one another. (Which is) the very thing it was required to show.

Proposition 30

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-

αὐτῶν εὐθειῶν, ἴσα ἀλλήλοις ἔστί·

lines) standing up are not on the same straight-lines, are equal to one another.



Ἐστω ἐπὶ τῆς αὐτῆς βάσεως τῆς AB στερεὰ παραλληλεπίπεδα τὰ GM , GN ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AZ , AH , AM , AN , $ΓΔ$, $ΓΕ$, $BΘ$, BK μὴ ἔστωσαν ἐπὶ τῶν αὐτῶν εὐθειῶν· λέγω, ὅτι ἴσον ἐστὶ τὸ GM στερεὸν τῷ GN στερεῷ.

Ἐκβεβλήσθωσαν γὰρ αἱ NK , $ΔΘ$ καὶ συμπιπτεύωσαν ἀλλήλαις κατὰ τὸ P , καὶ ἔτι ἐκβεβλήσθωσαν αἱ ZM , HE ἐπὶ τὰ O , $Π$, καὶ ἐπεξευθύνωσαν αἱ AX , AO , $ΓΠ$, BP . ἴσον δὲ ἐστὶ τὸ GM στερεόν, ὃ ὅ βάσις μὲν τὸ $AGBL$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ZΔΘM$, τῷ $ΓO$ στερεῷ, ὃ ὅ βάσις μὲν τὸ $AGBL$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $XΠPO$ · ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς $AGBL$ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AZ , AX , AM , AO , $ΓΔ$, $ΓΠ$, $BΘ$, BP ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ZO , $ΔP$. ἀλλὰ τὸ $ΓO$ στερεόν, ὃ ὅ βάσις μὲν ἐστὶ τὸ $AGBL$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $XΠPO$, ἴσον ἐστὶ τῷ GN στερεῷ, ὃ ὅ βάσις μὲν τὸ $AGBL$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $HEKN$ · ἐπὶ τε γὰρ πάλιν τῆς αὐτῆς βάσεως εἰσι τῆς $AGBL$ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AH , AX , $ΓΕ$, $ΓΠ$, AN , AO , BK , BP ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν $HΠ$, NP . ὥστε καὶ τὸ GM στερεὸν ἴσον ἐστὶ τῷ GN στερεῷ.

Τὰ ἄρα ἐπὶ τῆς αὐτῆς βάσεως στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσὶν ἐπὶ τῶν αὐτῶν εὐθειῶν, ἴσα ἀλλήλοις ἔστί· ὅπερ ἔδει δεῖξαι.

Let the parallelepiped solids CM and CN be on the same base, AB , and (have) the same height, and let the (ends of the straight-lines) standing up in them, AF , AG , LM , LN , CD , CE , BH , and BK , not be on the same straight-lines. I say that the solid CM is equal to the solid CN .

For let NK and DH be produced, and let them have joined one another at R . And, further, let FM and GE be produced to P and Q (respectively). And let AO , LP , CQ , and BR be joined. So, solid CM , whose base (is) parallelogram $ACBL$, and opposite (face) $FDHM$, is equal to solid CP , whose base (is) parallelogram $ACBL$, and opposite (face) $OQRP$. For they are on the same base, $ACBL$, and (have) the same height, and the (ends of the straight-lines) standing up in them, AF , AO , LM , LP , CD , CQ , BH , and BR , are on the same straight-lines, FP and DR [Prop. 11.29]. But, solid CP , whose base is parallelogram $ACBL$, and opposite (face) $OQRP$, is equal to solid CN , whose base (is) parallelogram $ACBL$, and opposite (face) $GEKN$. For, again, they are on the same base, $ACBL$, and (have) the same height, and the (ends of the straight-lines) standing up in them, AG , AO , CE , CQ , LN , LP , BK , and BR , are on the same straight-lines, GQ and NR [Prop. 11.29]. Hence, solid CM is also equal to solid CN .

Thus, parallelepiped solids (which are) on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are not on the same straight-lines,

are equal to one another. (Which is) the very thing it was required to show.

31.

Τὰ ἐπὶ ἴσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν.

Ἐστω ἐπὶ ἴσων βάσεων τῶν AB , $\Gamma\Delta$ στερεὰ παραλληλεπίπεδα τὰ AE , ΓZ ὑπὸ τὸ αὐτὸ ὕψος. λέγω, ὅτι ἴσον ἐστὶ τὸ AE στερεὸν τῷ ΓZ στερεῷ.

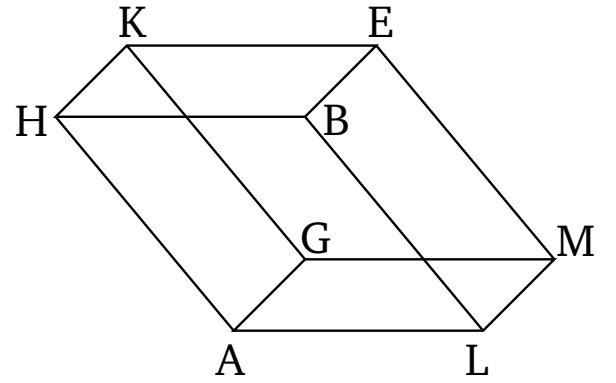
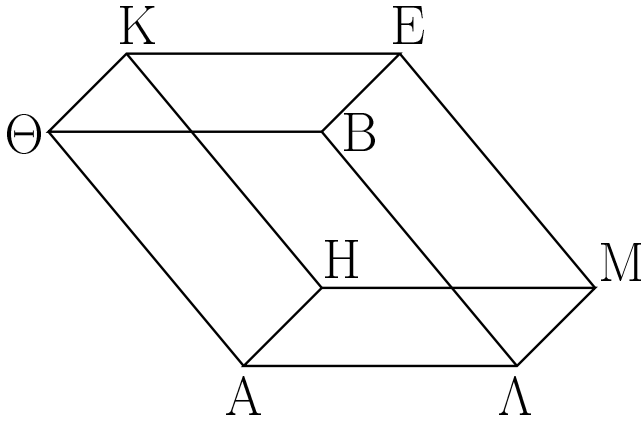
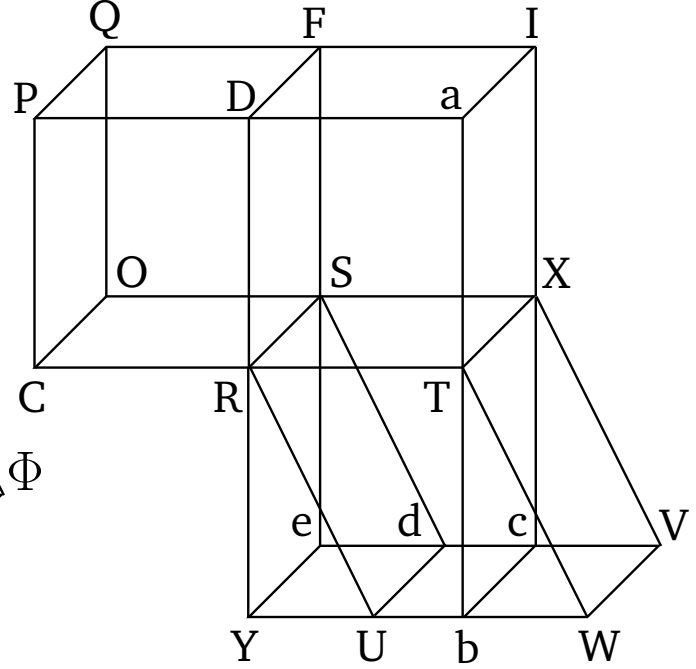
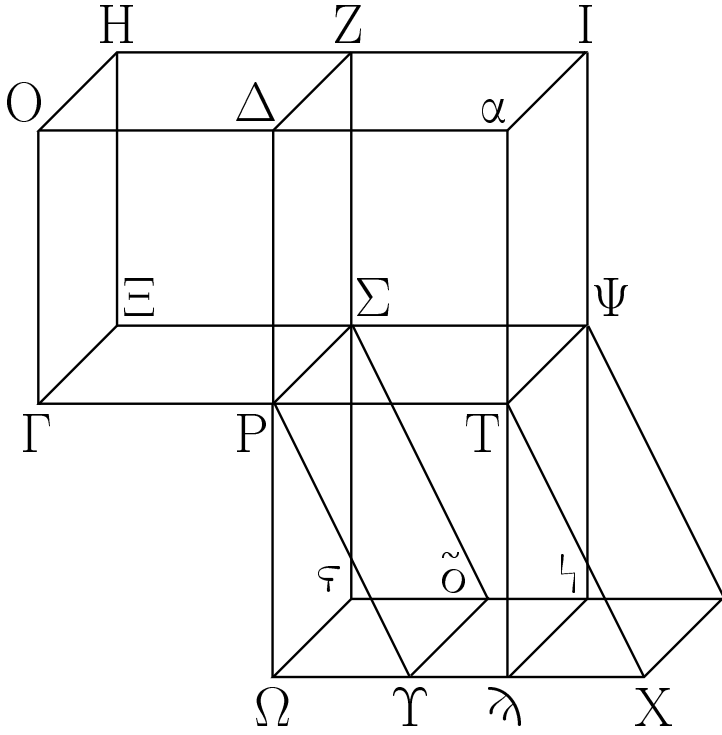
Ἐστωσαν δὴ πρότερον αἱ ἐφεστηκύναι αἱ ΘK , BE , AH , ΛM , $O\Pi$, ΔZ , ΓX , $P\Sigma$ πρὸς ὀρθὰς ταῖς AB , $\Gamma\Delta$ βάσεσιν, καὶ ἐκβεβλήσθω ἐπ' εὐθείᾳ τῇ ΓP εὐθεῖα ἡ PT , καὶ συνεστώτω πρὸς τῇ PT εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ P τῇ ὑπὸ $\Lambda\Lambda B$ γωνίᾳ ἴση ἢ ὑπὸ TPY , καὶ κείσθω τῇ μὲν $\Lambda\Lambda$ ἴση ἢ PT , τῇ δὲ ΛB ἴση ἢ PY , καὶ συμπληρώσθω ἡ τε $P\Theta$ βάσις καὶ τὸ ΨY στερεόν.

Proposition 31

Parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another.

Let the parallelepiped solids AE and CF be on the equal bases AB and CD (respectively), and (have) the same height. I say that solid AE is equal to solid CF .

So, let the (straight-lines) standing up, HK , BE , AG , LM , PQ , DF , CO , and RS , first of all, be at right-angles to the bases AB and CD . And let RT be produced in a straight-line with CR . And let (angle) TRU , equal to angle ALB , be constructed on the straight-line RT , at the point R on it [Prop. 1.23]. And let RT be made equal to AL , and RU to LB . And let the base RW , and the solid XU , be completed.

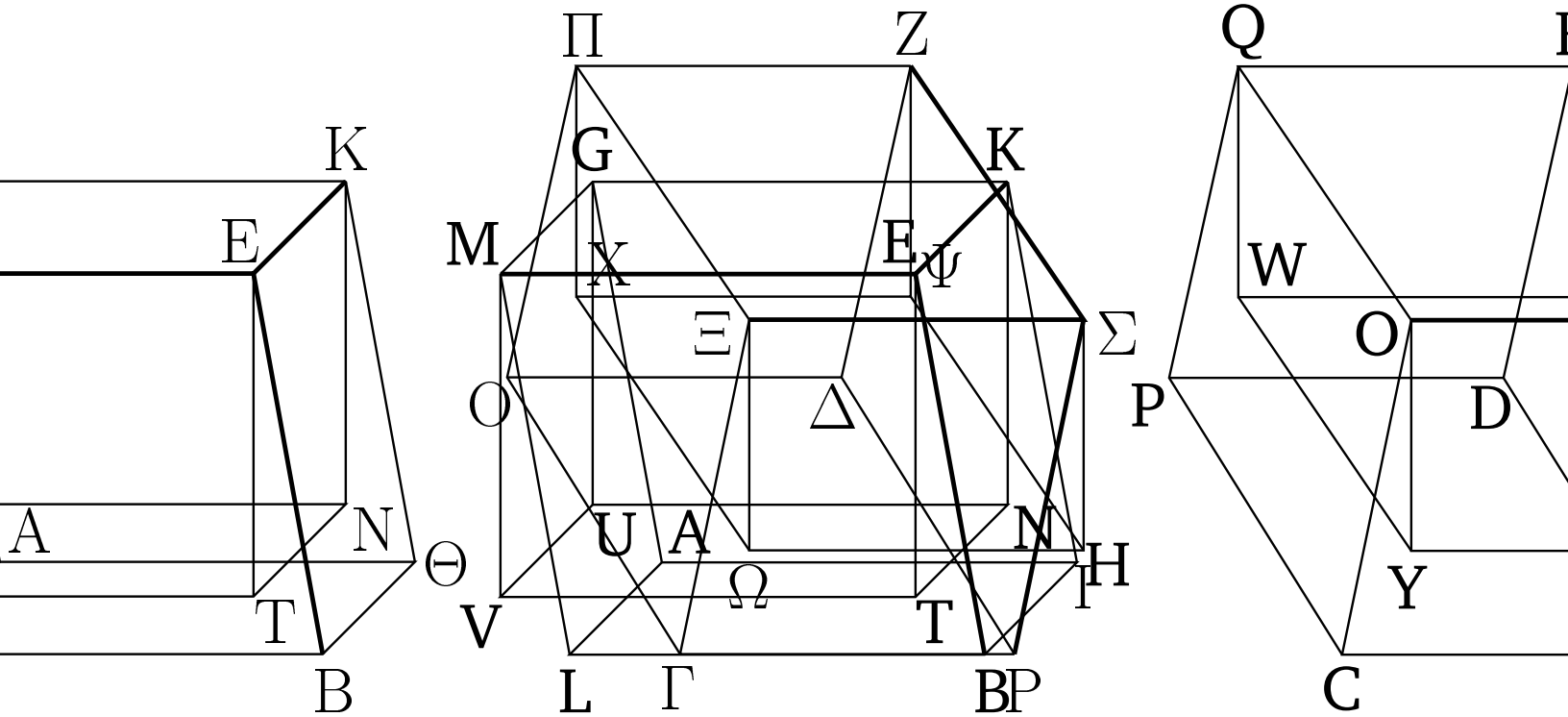


Καὶ ἐπεὶ δύο αἱ TP , PY ὁνοῖται AA , AB ἴσαι εἰσὶν, καὶ γωνίαξ ἴσαξ περιέθουσιν, ἴσον ἄρα καὶ ὅμοιον τὸ $P\Theta$ παραλληλόγραμμον τῷ $\Theta\Lambda$ παραλληλογράμμῳ. καὶ ἐπεὶ πάλιν ἴση μὲν ἡ AA τῇ PT , ἡ δὲ AM τῇ $P\Sigma$, καὶ γωνίαξ ὀρθάξ περιέθουσιν, ἴσον ἄρα καὶ ὅμοιον ἐστὶ τὸ $P\Psi$ παραλληλόγραμμον τῷ AM παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ AE τῷ ΣY ἴσον τέ ἐστι καὶ ὅμοιον· τρία ἄρα παραλληλόγραμμα τοῦ AE στερεοῦ τρισὶ παραλληλογράμμοιξ τοῦ ΨY στερεοῦ ἴσα τέ ἐστι καὶ ὅμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖξ ἀπεναντίον ἴσα τέ ἐστι καὶ ὅμοια, τὰ δὲ τρία τρισὶ τοῖξ ἀπεναντίον ὅλον ἄρα τὸ AE στερεὸν παραλληλεπίπεδον ὅλῳ τῷ ΨY στερεῷ παραλληλεπίπεδῳ ἴσον ἐστίν. διήθθωσαν αἱ ΔP , ΘY καὶ συμπιπτέτωσαν ἀλλήλαιξ κατὰ τὸ Ω , καὶ διὰ τοῦ T τῇ

And since the two (straight-lines) TR and RU are equal to the two (straight-lines) AL and LB (respectively), and they contain equal angles, parallelogram RW is thus equal and similar to parallelogram HL [Prop. 6.14]. And, again, since AL is equal to RT , and LM to RS , and they contain right-angles, parallelogram RX is thus equal and similar to parallelogram AM [Prop. 6.14]. So, for the same (reasons), LE is also equal and similar to SU . Thus, three parallelograms of solid AE are equal and similar to three parallelograms of solid XU . But, the three (faces of the former solid) are equal and similar to the three opposite (faces), and the three (faces of the latter solid) to the three opposite (faces) [Prop. 11.24]. Thus, the whole parallelepiped solid AE is equal to the whole parallelepiped solid

$\Delta\Omega$ παράλληλος ἦθθω ἡ aT , καὶ ἐκβεβλήσθω ἡ $O\Delta$ κατὰ τὸ a , καὶ συμπληρώσθω τὰ $\Omega\Psi$, PI στερεά. ἴσον δὲ ἔστι τὸ $\Psi\Omega$ στερεόν, οὗ βάσις μὲν ἔστι τὸ $P\Psi$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ Ω , τῷ ΨY στερεῶ, οὗ βάσις μὲν τὸ $P\Psi$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $Y\Phi$. ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς $P\Psi$ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφραστῶσαι αἱ $P\Omega$, PY , T , $T\Theta$, Σ , $\Sigma\delta$, $\Psi\rho$, $\Psi\Phi$ ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ΩX , $\zeta\Phi$. ἀλλὰ τὸ ΨY στερεόν τῷ AE ἔστιν ἴσον· καὶ τὸ $\Psi\Omega$ ἄρα στερεόν τῷ AE στερεῶ ἔστιν ἴσον. καὶ ἐπεὶ ἴσον ἔστι τὸ $PYXT$ παραλληλόγραμμον τῷ ΩT παραλληλόγραμμῳ· ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς PT καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς PT , ΩX . ἀλλὰ τὸ $PYXT$ τῷ $\Gamma\Delta$ ἔστιν ἴσον, ἐπεὶ καὶ τῷ AB , καὶ τὸ ΩT ἄρα παραλληλόγραμμον τῷ $\Gamma\Delta$ ἔστιν ἴσον. ἄλλο δὲ τὸ ΔT · ἔστιν ἄρα ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT , οὕτως ἡ ΩT πρὸς τὴν ΔT . καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓI ἐπιπέδῳ τῷ PZ τέμνεται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔστιν ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT βάσιν, οὕτως τὸ ΓZ στερεὸν πρὸς τὸ PI στερεόν. διὰ τὰ αὐτὰ δὲ, ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΩI ἐπιπέδῳ τῷ $P\Psi$ τέμνεται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔστιν ὡς ἡ ΩT βάσις πρὸς τὴν $T\Lambda$ βάσιν, οὕτως τὸ $\Omega\Psi$ στερεὸν πρὸς τὸ PI . ἀλλ' ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT , οὕτως ἡ ΩT πρὸς τὴν ΔT . καὶ ὡς ἄρα τὸ ΓZ στερεὸν πρὸς τὸ PI στερεόν, οὕτως τὸ $\Omega\Psi$ στερεὸν πρὸς τὸ PI . ἐκότερον ἄρα τῶν ΓZ , $\Omega\Psi$ στερεῶν πρὸς τὸ PI τὸν αὐτὸν ἔχει λόγον· ἴσον ἄρα ἔστι τὸ ΓZ στερεὸν τῷ $\Omega\Psi$ στερεῶ. ἀλλὰ τὸ $\Omega\Psi$ τῷ AE ἐδείχθη ἴσον· καὶ τὸ AE ἄρα τῷ ΓZ ἔστιν ἴσον.

XU [Def. 11.10]. Let DR and WU be drawn across, and let them have met one another at Y . And let aTb be drawn through T parallel to DY . And let PD be produced to a . And let the solids YX and RI be completed. So, solid XY , whose base is parallelogram RX , and opposite (face) Yc , is equal to solid XU , whose base (is) parallelogram RX , and opposite (face) UV . For they are on the same base RX , and (have) the same height, and the (ends of the straight-lines) standing up in them, RY , RU , Tb , TW , Se , Sd , Xc and XV , are on the same straight-lines, YW and eV [Prop. 11.29]. But, solid XU is equal to AE . Thus, solid XY is also equal to solid AE . And since parallelogram $RUWT$ is equal to parallelogram YT . For they are on the same base RT , and between the same parallels RT and YW [Prop. 1.35]. But, $RUWT$ is equal to CD , since (it is) also (equal) to AB . Parallelogram YT is thus also equal to CD . And DT is another (parallelogram). Thus, as base CD is to DT , so YT (is) to DT [Prop. 5.7]. And since the parallelepiped solid CI has been cut by the plane RF , which is parallel to the opposite planes (of CI), as base CD is to base DT , so solid CF (is) to solid RI [Prop. 11.25]. So, for the same (reasons), since the parallelepiped solid YI has been cut by the plane RX , which is parallel to the opposite planes (of YI), as base YT is to base TD , so solid YX (is) to solid RI [Prop. 11.25]. But, as base CD (is) to DT , so YT (is) to DT . And, thus, as solid CF (is) to solid RI , so solid YX (is) to solid RI . Thus, solids CF and YX each have the same ratio to RI [Prop. 5.11]. Thus, solid CF is equal to solid YX [Prop. 5.9]. But, YX was show (to be) equal to AE . Thus, AE is also equal to CF .



μή ἔστωσαν δὴ αἱ ἐφεστηκυῖαι αἱ AH , ΘK , BE , ΛM , $ΓX$, $ΟΠ$, ΔZ , $P\Sigma$ πρὸς ὀρθὰς ταῖς AB , $\Gamma\Delta$ βάσεσιν· λέγω πάλιν, ὅτι ἴσον τὸ AE στερεὸν τῷ ΓZ στερεῷ. ἤθθωσαν γὰρ ἀπὸ τῶν K , E , H , M , Π , Z , X , Σ σημείων ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετοι αἱ KN , ET , HY , $M\Phi$, $\Pi\Theta$, $Z\Psi$, $X\Omega$, ΣI , καὶ συμβαλλέτωσαν τῷ ἐπιπέδῳ κατὰ τὰ N , T , Y , Φ , Θ , Ψ , Ω , I σημεία, καὶ ἐπεξεύθθωσαν αἱ NT , NY , $Y\Phi$, $T\Phi$, $\Theta\Psi$, $\Theta\Omega$, ΩI , $I\Psi$. ἴσον δὴ ἐστὶ τὸ $K\Phi$ στερεὸν τῷ $\Pi\Psi$ στερεῷ· ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν KM , $\Pi\Psi$ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι πρὸς ὀρθὰς εἰσι ταῖς βάσεσιν. ἀλλὰ τὸ μὲν $K\Phi$ στερεὸν τῷ AE στερεῷ ἐστὶν ἴσον, τὸ δὲ $\Pi\Psi$ τῷ ΓZ · ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσιν ἐπὶ τῶν αὐτῶν εὐθειῶν. καὶ τὸ AE ἄρα στερεὸν τῷ ΓZ στερεῷ ἐστὶν ἴσον.

Τὰ ἄρα ἐπὶ ἴσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστὶν· ὅπερ ἔδει δεῖξαι.

And so let the (straight-lines) standing up, AG , HK , BE , LM , CO , PQ , DF , and RS , not be at right-angles to the bases AB and CD . Again, I say that solid AE (is) equal to solid CF . For let KN , ET , GU , MV , QW , FX , OY , and SI be drawn from points K , E , G , M , Q , F , O , and S (respectively) perpendicular to the reference plane (i.e., the plane of the bases AB and CD), and let them have met the plane at points N , T , U , V , W , X , Y , and I (respectively). And let NT , NU , UV , TV , WX , WY , YI , and IX be joined. So solid KV is equal to solid QI . For they are on the equal bases KM and QS , and (have) the same height, and the (straight-lines) standing up in them are at right-angles to their bases (see first part of proposition). But, solid KV is equal to solid AE , and QI to CF . For they are on the same base, and (have) the same height, and the (straight-lines) standing up in them are not on the same straight-lines [Prop. 11.30]. Thus, solid AE is also equal to solid CF .

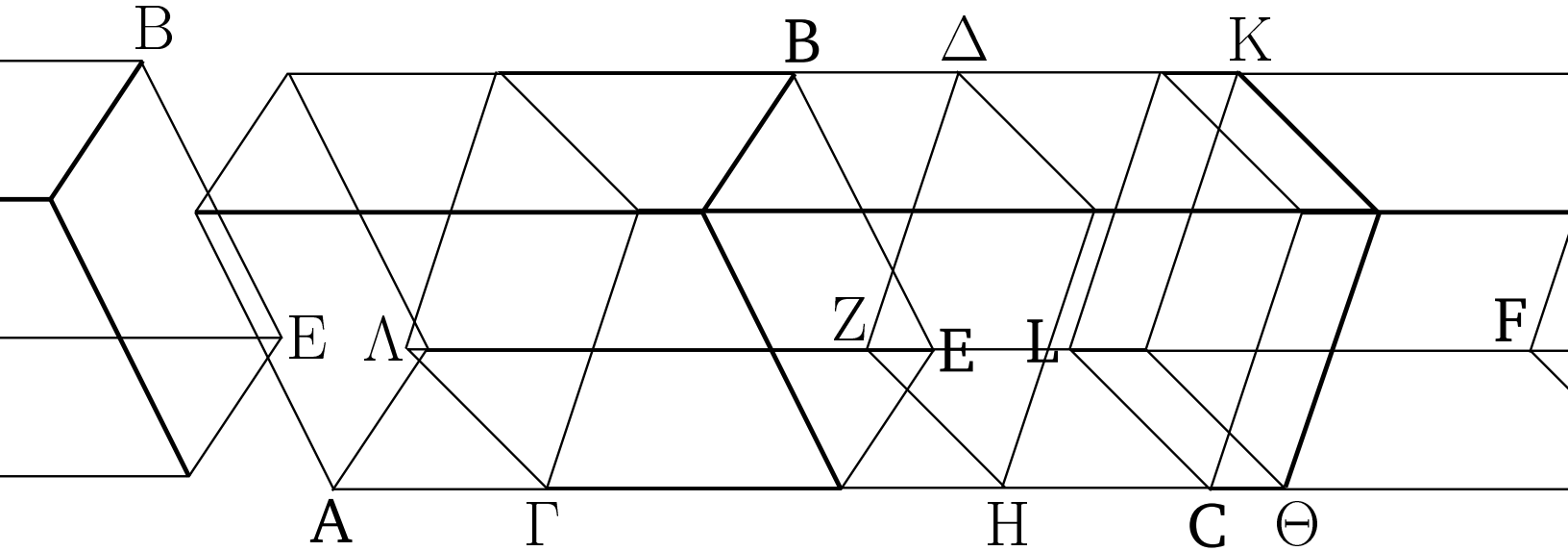
Thus, parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another. (Which is) the very thing it was required to show.

32.

Proposition 32

Τὰ ὑπὸ τὸ αὐτὸ ὕψος ὄντα στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐστὶν ὡς αἱ βάσεις.

Parallelepiped solids which (have) the same height are to one another as their bases.



Ἐστω ὑπὸ τὸ αὐτὸ ὕψος στερεὰ παραλληλεπίπεδα τὰ AB , $\Gamma\Delta$. λέγω, ὅτι τὰ AB , $\Gamma\Delta$ στερεὰ παραλληλεπίπεδα πρὸς ἄλληλά ἐστιν ὡς αἱ βάσεις, τοντέστιν ὅτι ἐστὶν ὡς ἡ AE βάσις πρὸς τὴν ΓZ βάσιν, οὕτως τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεόν.

Παραβελήσθω γὰρ παρὰ τὴν ZH τῷ AE ἴσον τὸ $Z\Theta$, καὶ ἀπὸ βάσεως μὲν τῆς $Z\Theta$, ὕψον δὲ τοῦ αὐτοῦ τῷ $\Gamma\Delta$ στερεὸν παραλληλεπίπεδον συμπληρώσθω τὸ HK . ἴσον δὴ ἐστὶ τὸ AB στερεὸν τῷ HK στερεῷ· ἐπὶ τε γὰρ ἴσων βάσεων εἰσι τῶν AE , $Z\Theta$ καὶ ὑπὸ τὸ αὐτὸ ὕψος. καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓK ἐπιπέδῳ τῷ ΔH τέμνεται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἐστὶν ἄρα ὡς ἡ ΓZ βάσις πρὸς τὴν $Z\Theta$ βάσιν, οὕτως τὸ $\Gamma\Delta$ στερεὸν πρὸς τὸ $\Delta\Theta$ στερεόν. ἴση δὲ ἡ μὲν $Z\Theta$ βάσις τῇ AE βάσει, τὸ δὲ HK στερεὸν τῷ AB στερεῷ· ἐστὶν ἄρα καὶ ὡς ἡ AE βάσις πρὸς τὴν ΓZ βάσιν, οὕτως τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεόν.

Τὰ ἄρα ὑπὸ τὸ αὐτὸ ὕψος ὄντα στερεὰ παραλληλεπίπεδα πρὸς ἄλληλά ἐστιν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖξαι.

33.

Τὰ ὅμοια στερεὰ παραλληλεπίπεδα πρὸς ἄλληλα ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.

Ἐστω ὅμοια στερεὰ παραλληλεπίπεδα τὰ AB , $\Gamma\Delta$, ὁμόλογος δὲ ἔστω ἡ AE τῇ ΓZ . λέγω, ὅτι τὸ AB στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεὸν τριπλασίονα λόγον ἔθει, ἥπερ ἡ AE πρὸς τὴν ΓZ .

Ἐκβεβλήσθωσαν γὰρ ἐπ' εὐθείας ταῖς AE , HE , ΘE αἱ EK , EL , EM , καὶ κείσθω τῇ μὲν ΓZ ἴση ἡ EK , τῇ δὲ ZN ἴση ἡ EL , καὶ ἔτι τῇ ZP ἴση ἡ EM , καὶ συμπληρώσθω τὸ KA παραλληλόγραμμον καὶ τὸ KO στερεόν.

Let AB and CD be parallelepiped solids (having) the same height. I say that the parallelepiped solids AB and CD are to one another as their bases. That is to say, as base AE is to base CF , so solid AB (is) to solid CD .

For let FH , equal to AE , be applied to FG (in the angle FGH equal to angle LCG) [Prop. 1.45]. And let the parallelepiped solid GK , (having) the same height as CD , be completed on the base FH . So solid AB is equal to solid GK . For they are on the equal bases AE and FH , and (have) the same height [Prop. 11.31]. And since the parallelepiped solid CK has been cut by the plane DG , which is parallel to the opposite planes (of CK), thus as the base CF is to the base FH , so the solid CD (is) to the solid DH [Prop. 11.25]. And base FH (is) equal to base AE , and solid GK to solid AB . And thus as base AE is to base CF , so solid AB (is) to solid CD .

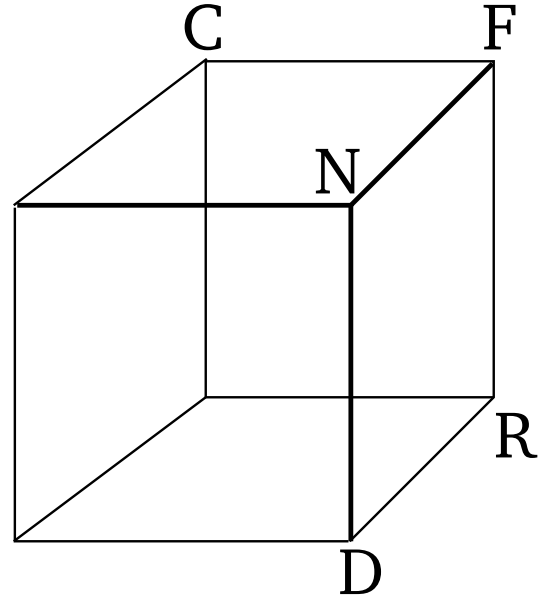
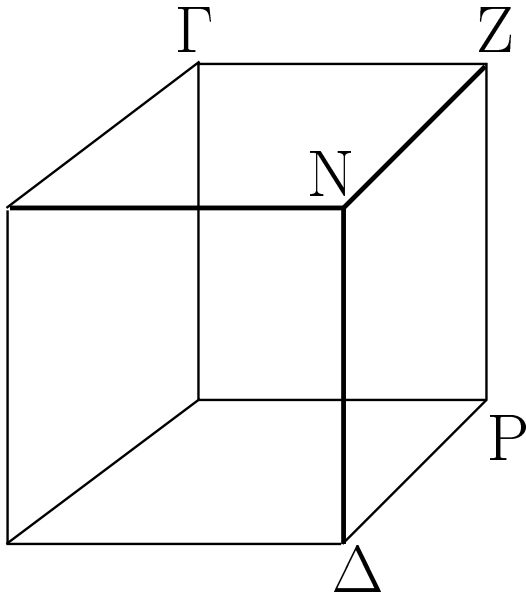
Thus, parallelepiped solids which (have) the same height are to one another as their bases. (Which is) the very thing it was required to show.

Proposition 33

Similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides.

Let AB and CD be similar parallelepiped solids, and let AE correspond to CF . I say that solid AB has to solid CD the cubed ratio that AE (has) to CF .

For let EK , EL , and EM be produced in a straight-line with AE , GE , and HE (respectively). And let EK be made equal to CF , and EL equal to FN , and, further, EM equal to FR . And let the parallelogram KL be completed, and the solid KP .

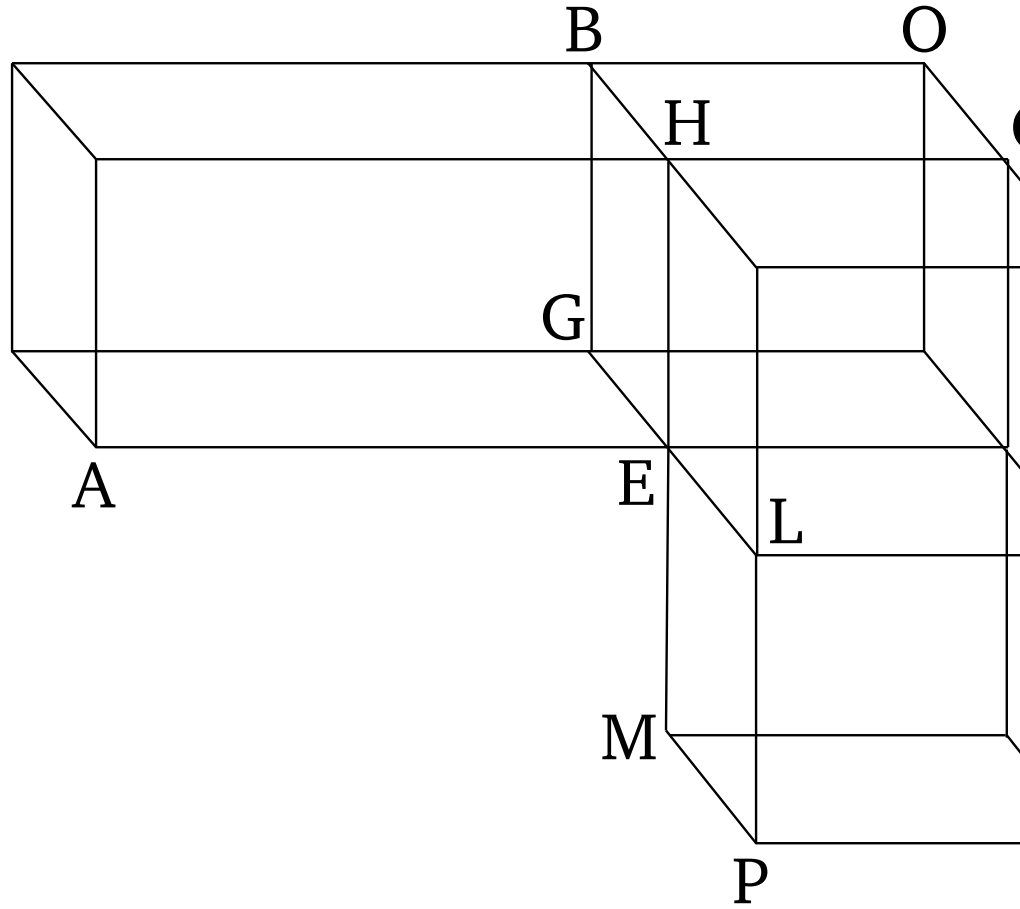


Καὶ ἐπεὶ δύο αἱ KE , EA δυοὶ ταῖς GZ , ZN ἴσαι εἰσίν, ἀλλὰ καὶ γωνία ἡ ὑπὸ KEA γωνία τῇ ὑπὸ GZN ἐστὶν ἴση, ἐπειδὴ περ καὶ ἡ ὑπὸ AEH τῇ ὑπὸ GZN ἐστὶν ἴση διὰ τὴν ὁμοιότητα τῶν AB , $ΓΔ$ στερεῶν, ἴσον ἄρα ἐστὶ [καὶ ὅμοιον] τὸ $ΚΛ$ παραλληλόγραμμον τῷ $ΓΝ$ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν KM παραλληλόγραμμον ἴσον ἐστὶ καὶ ὅμοιον τῷ $ΓΡ$ [παραλληλογράμμῳ] καὶ ἔτι τὸ EO τῷ $ΔΖ$ · τρία ἄρα παραλληλόγραμμα τοῦ KO στερεοῦ τρισὶ παραλληλογράμμοις τοῦ $ΓΔ$ στερεοῦ ἴσα ἐστὶ καὶ ὅμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα ἐστὶ καὶ ὅμοια, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ἴσα ἐστὶ καὶ ὅμοια· ὅλον ἄρα τὸ KO στερεὸν ὅλῳ τῷ $ΓΔ$ στερεῷ ἴσον ἐστὶ καὶ ὅμοιον. συμπληρώσθω τὸ HK παραλληλόγραμμον, καὶ ἀπὸ βάσεων μὲν τῶν HK , $ΚΛ$ παραλληλόγραμμων, ὕψονξ δὲ τοῦ αὐτοῦ τῷ AB στερεῷ συμπληρώσθω τὰ EX , $ΠΠ$. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν AB , $ΓΔ$ στερεῶν ἐστὶν ὡς ἡ AE πρὸς τὴν GZ , οὕτως ἡ EH πρὸς τὴν ZN , καὶ ἡ $EΘ$ πρὸς τὴν ZP , ἴση δὲ ἡ μὲν GZ τῇ EK , ἡ δὲ ZN τῇ EA , ἡ δὲ ZP τῇ EM , ἔστιν ἄρα ὡς ἡ AE πρὸς τὴν EK , οὕτως ἡ HE πρὸς τὴν EA καὶ ἡ $ΘE$ πρὸς τὴν EM . ἀλλ' ὡς μὲν ἡ AE πρὸς τὴν EK , οὕτως τὸ AH [παραλληλόγραμμον] πρὸς τὸ HK παραλληλόγραμμον, ὡς δὲ ἡ HE πρὸς τὴν EA , οὕτως τὸ HK πρὸς τὸ $ΚΛ$, ὡς δὲ ἡ $ΘE$ πρὸς EM , οὕτως τὸ $ΠE$ πρὸς τὸ KM · καὶ ὡς ἄρα τὸ AH παραλληλόγραμμον πρὸς τὸ HK , οὕτως τὸ HK πρὸς τὸ $ΚΛ$ καὶ τὸ $ΠE$ πρὸς τὸ KM . ἀλλ' ὡς μὲν τὸ AH πρὸς τὸ HK , οὕτως τὸ AB στερεὸν πρὸς τὸ EX στερεὸν, ὡς δὲ τὸ HK πρὸς τὸ $ΚΛ$, οὕτως τὸ XE στερεὸν πρὸς τὸ $ΠA$ στερεὸν, ὡς δὲ τὸ $ΠE$ πρὸς τὸ KM , οὕτως τὸ $ΠA$ στερεὸν πρὸς τὸ KO στερεὸν καὶ ὡς ἄρα τὸ AB στερεὸν πρὸς τὸ EX , οὕτως τὸ EX πρὸς τὸ $ΠA$ καὶ τὸ $ΠA$ πρὸς τὸ KO . ἔαν δὲ τέσσαρα μεγέθη κατὰ τὸ συνθεθὲς ἀνάλογον ᾗ, τὸ πρῶτον πρὸς τὸ τέταρτον τριπλασίονα λόγον ἔθει ἥπερ πρὸς

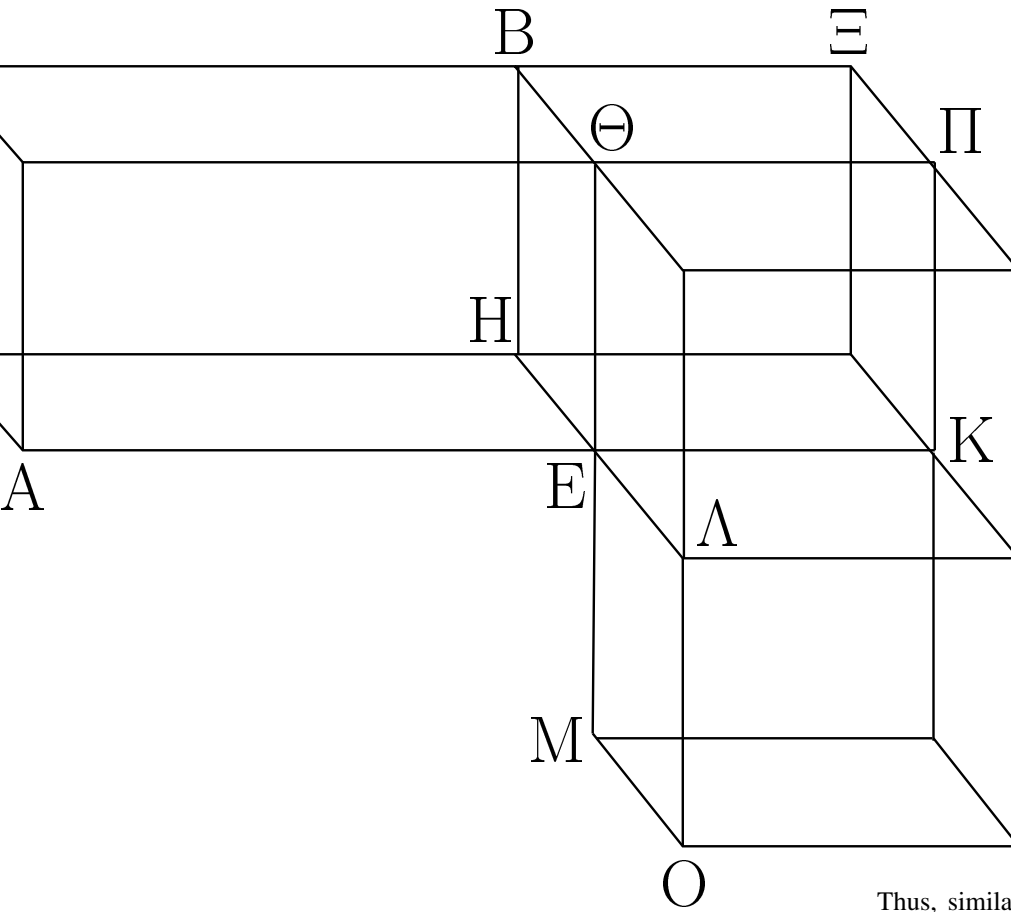
And since the two (straight-lines) KE and EA are equal to the two (straight-lines) CF and FN , but angle KEL is also equal to angle CFN , inasmuch as AEG is also equal to CFN , on account of the similarity of the solids AB and CD , parallelogram KL is thus equal [and similar] to parallelogram CN . So, for the same (reasons), parallelogram KM is also equal and similar to [parallelogram] CR , and, further, EP to DF . Thus, three parallelograms of solid KP are equal and similar to three parallelograms of solid CD . But the three (former parallelograms) are equal and similar to the three opposite (parallelograms), and the three (latter parallelograms) are equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the whole of solid KP is equal and similar to the whole of solid CD [Def. 11.10]. Let parallelogram GK be completed. And let the the solids EO and LQ , with bases the parallelograms GK and KL (respectively), and with the same height as AB , be completed. And since, on account of the similarity of solids AB and CD , as AE is to CF , so EG (is) to FN , and EH to FR [Defs. 6.1, 11.9], and CF (is) equal to EK , and FN to EL , and FR to EM , thus as AE is to EK , so GE (is) to EL , and HE to EM . But, as AE (is) to EK , so [parallelogram] AG (is) to parallelogram GK , and as GE (is) to EL , so GK (is) to KL , and as HE (is) to EM , so QE (is) to KM [Prop. 6.1]. And thus as parallelogram AG (is) to GK , so GK (is) to KL , and QE (is) to KM . But, as AG (is) to GK , so solid AB (is) to solid EO , and as GK (is) to KL , so solid OE (is) to solid QL , and as QE (is) to KM , so solid QL (is) to solid KP [Prop. 11.32]. And, thus, as solid AB is to EO , so EO (is) to QL , and QL to KP . And if four magnitudes are continuously proportional then the first has to the fourth the cubed ratio that (it has) to the second [Def. 5.10]. Thus, solid AB has to KP the cubed ratio which AB (has) to EO . But,

τὸ δεύτερον· τὸ AB ἄρα στερεὸν πρὸς τὸ KO τριπλασίονα λόγον ἔθει ἥπερ τὸ AB πρὸς τὸ EX . ἀλλ' ὥς τὸ AB πρὸς τὸ EX , οὕτως τὸ AH παραλληλόγραμμον πρὸς τὸ HK καὶ ἡ AE εὐθεῖα πρὸς τὴν EK · ὥστε καὶ τὸ AB στερεὸν πρὸς τὸ KO τριπλασίονα λόγον ἔθει ἥπερ ἡ AE πρὸς τὴν EK . ἴσον δὲ τὸ [μὲν] KO στερεὸν τῷ $\Gamma\Delta$ στερεῷ, ἡ δὲ EK εὐθεῖα τῇ ΓZ · καὶ τὸ AB ἄρα στερεὸν πρὸς τὸ $\Gamma\Delta$ στερεὸν τριπλασίονα λόγον ἔθει ἥπερ ἡ ὁμόλογος αὐτοῦ πλευρὰ ἡ AE πρὸς τὴν

as AB (is) to EO , so parallelogram AG (is) to GK , and the straight-line AE to EK [Prop. 6.1]. Hence, solid AB also has to KP the cubed ratio that AE (has) to EK . And solid KP (is) equal to solid CD , and straight-line EK to CF . Thus, solid AB also has to solid CD the cubed ratio which its corresponding side AE (has) to the corresponding side CF .



ὁμόλογον πλευρὰν τὴν ΓZ .



Τὰ ἄρα ὅμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν· ὅπερ ἔδει δεῖξαι.

Thus, similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides. (Which is) the very thing it was required to show.

Πόρισμα.

Corollary

Ἐκ δὴ τούτου φανερόν, ὅτι ἂν τέσσαρες εὐθεῖαι ἀνάλογον ᾖσιν, ἔσται ὡς ἡ πρώτη πρὸς τὴν τετάρτην, οὕτω τὸ ἀπὸ τῆς πρώτης στερεὸν παραλληλεπίπεδον πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ ὅμοιον καὶ ὁμοίως ἀναγραφόμενον, ἐπεὶ καὶ ἡ πρώτη πρὸς τὴν τετάρτην τριπλασίονα λόγον ἔθει ἢ περ πρὸς τὴν δευτέραν.

So, (it is) clear, from this, that if four straight-lines are (continuously) proportional then as the first is to the fourth, so the parallelepiped solid on the first will be to the similar, and similarly described, parallelepiped solid on the second, since the first also has to the fourth the cubed ratio that (it has) to the second.

34.

Proposition 34[†]

Τῶν ἴσων στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· καὶ ὡν στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσα ἐστὶν ἐκεῖνα.

The bases of equal parallelepiped solids are reciprocally proportional to their heights. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal.

Ἐστω ἴσα στερεὰ παραλληλεπίπεδα τὰ AB, ΓΔ· λέγω, ὅτι τῶν AB, ΓΔ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἐστὶν ὡς ἡ EΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως τὸ τοῦ ΓΔ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος.

Let AB and CD be equal parallelepiped solids. I say that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights, and (so) as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB.

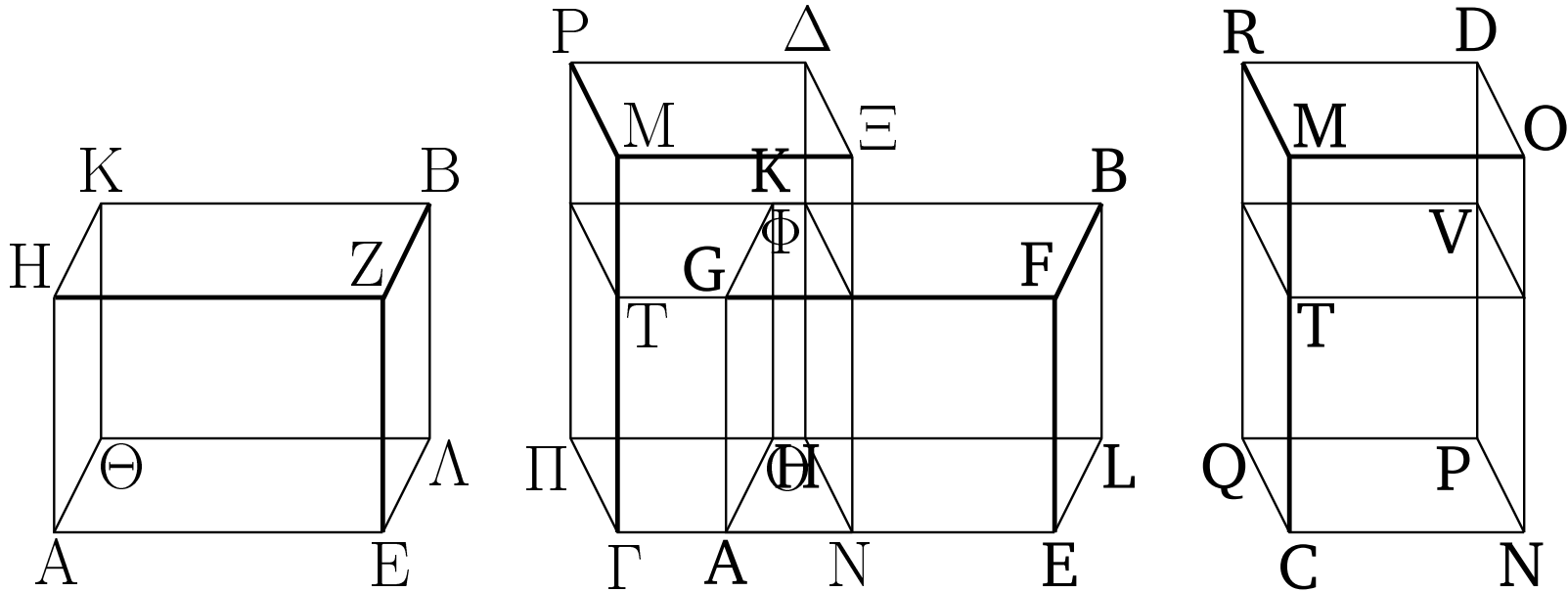
Ἐστωσαν γὰρ πρότερον αἱ ἐφεστηκυῖαι αἱ AH, EZ, AB,

ΘK , ΓM , $N X$, $O \Delta$, ΠP πρὸς ὀρθὰς ταῖς βάσεων αὐτῶν λέγω, ὅτι ἐστὶν ὡς ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν, οὕτως ἡ ΓM πρὸς τὴν $A H$.

Εἰ μὲν οὖν ἴση ἐστὶν ἡ $E \Theta$ βάσις τῇ $N \Pi$ βάσει, ἔστι δὲ καὶ τὸ AB στερεὸν τῷ $\Gamma \Delta$ στερεῷ ἴσον, ἔσται καὶ ἡ ΓM τῇ $A H$ ἴση. τὰ γὰρ ὑπὸ τὸ αὐτὸ ὕψος στερεὰ παραλληλεπίπεδα πρὸς ἄλληλά ἐστιν ὡς αἱ βάσεις. καὶ ἔσται ὡς ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$, οὕτως ἡ ΓM πρὸς τὴν $A H$, καὶ φανερόν, ὅτι τῶν AB , $\Gamma \Delta$ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

For, first of all, let the (straight-lines) standing up, AG , EF , LB , HK , CM , NO , PD , and QR , be at right-angles to their bases. I say that as base EH is to base NQ , so CM (is) to AG .

Therefore, if base EH is equal to base NQ , and solid AB is also equal to solid CD , CM will also be equal to AG . For parallelepiped solids of the same height are to one another as their bases [Prop. 11.32]. And as base EH (is) to NQ , so CM will be to AG . And (so it is) clear that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.



μὴ ἔστω δὲ ἴση ἡ $E \Theta$ βάσις τῇ $N \Pi$ βάσει, ἀλλ' ἔστω μείζων ἡ $E \Theta$. ἔστι δὲ καὶ τὸ AB στερεὸν τῷ $\Gamma \Delta$ στερεῷ ἴσον· μείζων ἄρα ἐστὶ καὶ ἡ ΓM τῇ $A H$. κείσθω οὖν τῇ $A H$ ἴση ἡ ΓT , καὶ συμπληρώσθω ἀπὸ βάσεως μὲν τῇς $N \Pi$, ὕψους δὲ τοῦ ΓT , στερεὸν παραλληλεπίπεδον τὸ $\Phi \Gamma$. καὶ ἐπεὶ ἴσον ἐστὶ τὸ AB στερεὸν τῷ $\Gamma \Delta$ στερεῷ, ἔχωνθεν δὲ τὸ $\Gamma \Phi$, τὰ δὲ ἴσα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔθει λόγον, ἔστιν ἄρα ὡς τὸ AB στερεὸν πρὸς τὸ $\Gamma \Phi$ στερεόν, οὕτως τὸ $\Gamma \Delta$ στερεὸν πρὸς τὸ $\Gamma \Phi$ στερεόν. ἀλλ' ὡς μὲν τὸ AB στερεὸν πρὸς τὸ $\Gamma \Phi$ στερεόν, οὕτως ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν· ἰσοῦψῃ γὰρ τὰ AB , $\Gamma \Phi$ στερεὰ· ὡς δὲ τὸ $\Gamma \Delta$ στερεὸν πρὸς τὸ $\Gamma \Phi$ στερεόν, οὕτως ἡ $M \Pi$ βάσις πρὸς τὴν $T \Pi$ βάσιν καὶ ἡ ΓM πρὸς τὴν ΓT . καὶ ὡς ἄρα ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν, οὕτως ἡ $M \Gamma$ πρὸς τὴν ΓT . ἴση δὲ ἡ ΓT τῇ $A H$. καὶ ὡς ἄρα ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν, οὕτως ἡ $M \Gamma$ πρὸς τὴν $A H$. τῶν AB , $\Gamma \Delta$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Πάλιν δὲ τῶν AB , $\Gamma \Delta$ στερεῶν παραλληλεπιπέδων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστω ὡς ἡ $E \Theta$ βάσις πρὸς τὴν $N \Pi$ βάσιν, οὕτως τὸ τοῦ $\Gamma \Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος· λέγω, ὅτι ἴσον ἐστὶ τὸ AB στερεὸν τῷ $\Gamma \Delta$ στερεῷ.

So let base EH not be equal to base NQ , but let EH be greater. And solid AB is also equal to solid CD . Thus, CM is also greater than AG . Therefore, let CT be made equal to AG . And let the parallelepiped solid VC be completed on the base NQ , with height CT . And since solid AB is equal to solid CD , and CV (is) extrinsic (to them), and equal (magnitudes) have the same ratio to the same (magnitude) [Prop. 5.7], thus as solid AB is to solid CV , so solid CD (is) to solid CV . But, as solid AB (is) to solid CV , so base EH (is) to base NQ . For the solids AB and CV (are) of equal height [Prop. 11.32]. And as solid CD (is) to solid CV , so base MQ (is) to base TQ [Prop. 11.25], and CM to CT [Prop. 6.1]. And, thus, as base EH is to base NQ , so MC (is) to AG . And CT (is) equal to AG . And thus as base EH (is) to base NQ , so MC (is) to AG . Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

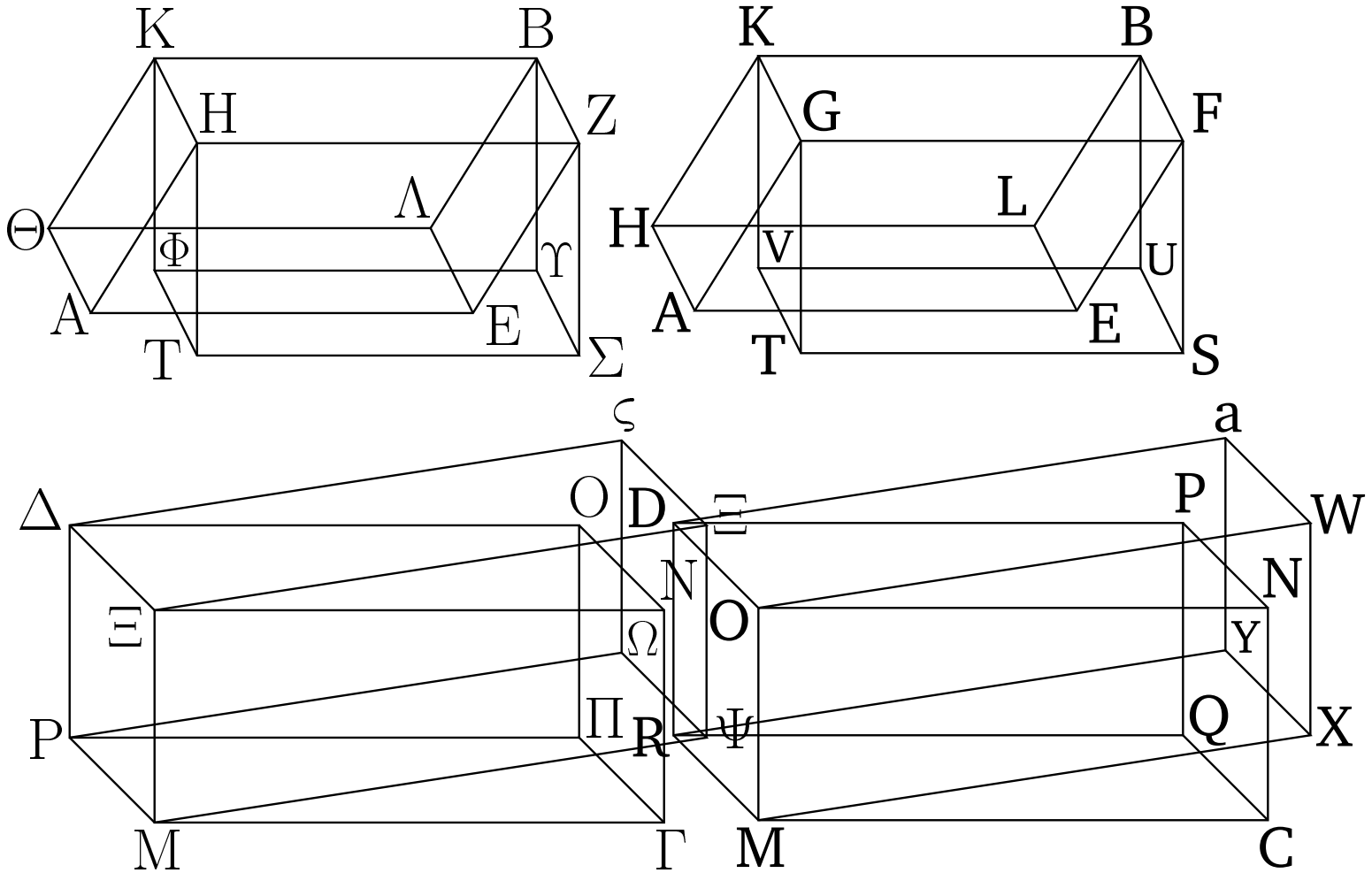
So, again, let the bases of the parallelepiped solids AB and CD be reciprocally proportional to their heights, and let base EH be to base NQ , as the height of solid CD (is) to the height of solid AB . I say that solid AB is equal to solid CD . [For] let the (straight-lines) standing up again be at right-angles to the bases. And if base EH is equal to base NQ , and as base EH is

Ἐστωσαν [γάρ] πάλιν αἱ ἐφεστηκυῖαι πρὸς ὀρθὰς ταῖς βάσεων. καὶ εἰ μὲν ἴση ἐστὶν ἡ $EΘ$ βάσις τῇ $ΝΠ$ βάσει, καὶ ἐστὶν ὡς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος, ἴσον ἄρα ἐστὶ καὶ τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος τῷ τοῦ $ΑΒ$ στερεοῦ ὕψει. τὰ δὲ ἐπὶ ἴσων βάσεων στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ $ΑΒ$ στερεὸν τῷ $ΓΔ$ στερεῷ.

μὴ ἔστω δὴ ἡ $EΘ$ βάσις τῇ $ΝΠ$ [βάσει] ἴση, ἀλλ' ἔστω μείζων ἡ $EΘ$ · μείζον ἄρα ἐστὶ καὶ τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος τοῦ τοῦ $ΑΒ$ στερεοῦ ὕψους, τοντέστιν ἡ $ΓΜ$ τῆς $ΑΗ$. κείσθω τῇ $ΑΗ$ ἴση πάλιν ἡ $ΓΤ$, καὶ συμπληρώσθω ὁμοίως τὸ $ΓΦ$ στερεόν. ἐπεὶ ἐστὶν ὡς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως ἡ $ΜΓ$ πρὸς τὴν $ΑΗ$, ἴση δὲ ἡ $ΑΗ$ τῇ $ΓΤ$, ἔστιν ἄρα ὡς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως ἡ $ΓΜ$ πρὸς τὴν $ΓΤ$. ἀλλ' ὡς μὲν ἡ $EΘ$ [βάσις] πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ $ΑΒ$ στερεὸν πρὸς τὸ $ΓΦ$ στερεόν· ἰσοῦσιν γάρ ἐστι τὰ $ΑΒ$, $ΓΦ$ στερεά· ὡς δὲ ἡ $ΓΜ$ πρὸς τὴν $ΓΤ$, οὕτως ἡ τε $ΜΠ$ βάσις πρὸς τὴν $ΠΤ$ βάσιν καὶ τὸ $ΓΔ$ στερεὸν πρὸς τὸ $ΓΦ$ στερεόν. καὶ ὡς ἄρα τὸ $ΑΒ$ στερεὸν πρὸς τὸ $ΓΦ$ στερεόν, οὕτως τὸ $ΓΔ$ στερεὸν πρὸς τὸ $ΓΦ$ στερεόν· ἐκότερον ἄρα τῶν $ΑΒ$, $ΓΔ$ πρὸς τὸ $ΓΦ$ τὸν αὐτὸν ἔθει λόγον. ἴσον ἄρα ἐστὶ τὸ $ΑΒ$ στερεὸν τῷ $ΓΔ$ στερεῷ.

to base NQ , so the height of solid CD (is) to the height of solid AB , the height of solid CD is thus also equal to the height of solid AB . And parallelepiped solids on equal bases, and also with the same height, are equal to one another [Prop. 11.31]. Thus, solid AB is equal to solid CD .

So, let base EH not be equal to [base] NQ , but let EH be greater. Thus, the height of solid CD is also greater than the height of solid AB , that is to say CM (greater) than AG . Let CT again be made equal to AG , and let the solid CV be similarly completed. Since as base EH is to base NQ , so MC (is) to AG , and AG (is) equal to CT , thus as base EH (is) to base NQ , so CM (is) to CT . But, as [base] EH (is) to base NQ , so solid AB (is) to solid CV . For solids AB and CV are of equal heights [Prop. 11.32]. And as CM (is) to CT , so (is) base MQ to base QT [Prop. 6.1], and solid CD to solid CV [Prop. 11.25]. And thus as solid AB (is) to solid CV , so solid CD (is) to solid CV . Thus, AB and CD each have the same ratio to CV . Thus, solid AB is equal to solid CD [Prop. 5.9].



μή ἔστωσαν δὴ αἱ ἐφεστηκνῦναι αἱ $ZE, BA, HA, K\Theta, XN, \Delta O, M\Gamma, P\Pi$ πρὸς ὀρθὰς ταῖς βάσεσιν αὐτῶν, καὶ ἤθθωσαν ἀπὸ τῶν $Z, H, B, K, X, M, P, \Delta$ σημείων ἐπὶ τὰ διὰ τῶν $E\Theta, N\Pi$ ἐπίπεδα κάθετοι καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ $\Sigma, T, Y, \Phi, \Theta, \Psi, \Omega, \xi$, καὶ συμπληρώσθω τὰ $Z\Phi, X\Omega$ στερεά· λέγω, ὅτι καὶ οὕτως ἴσων ὄντων τῶν $AB, \Gamma\Delta$ στερεῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστιν ὥς ἡ $E\Theta$ βάσιν πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Gamma\Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος.

Ἐπεὶ ἴσον ἐστὶ τὸ AB στερεὸν τῷ $\Gamma\Delta$ στερεῷ, ἀλλὰ τὸ μὲν AB τῷ BT ἔστιν ἴσον· ἐπὶ τε γὰρ τῆς αὐτῆς βάσεως εἰσι τῆς ZK καὶ ὑπὸ τὸ αὐτὸ ὕψος· τὸ δὲ $\Gamma\Delta$ στερεὸν τῷ $\Delta\Psi$ ἔστιν ἴσον· ἐπὶ τε γὰρ πάλιν τῆς αὐτῆς βάσεως εἰσι τῆς PX καὶ ὑπὸ τὸ αὐτὸ ὕψος· καὶ τὸ BT ἄρα στερεὸν τῷ $\Delta\Psi$ στερεῷ ἴσον ἐστίν. ἔστιν ἄρα ὥς ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. ἴση δὲ ἡ μὲν ZK βάσις τῇ $E\Theta$ βάσει, ἡ δὲ XP βάσις τῇ $N\Pi$ βάσει· ἔστιν ἄρα ὥς ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. τὰ δ' αὐτὰ ὕψη ἐστὶ τῶν $\Delta\Psi, BT$ στερεῶν καὶ τῶν $\Delta\Gamma, BA$.

So, let the (straight-lines) standing up, $FE, BL, GA, KH, ON, DP, MC$, and RQ , not be at right-angles to their bases. And let perpendiculars be drawn to the planes through EH and NQ from points F, G, B, K, O, M, R , and D , and let them have joined the planes at (points) S, T, U, V, W, X, Y , and a (respectively). And let the solids FV and OY be completed. In this case, also, I say that the solids AB and CD being equal, their bases are reciprocally proportional to their heights, and (so) as base EH is to base NQ , so the height of solid CD (is) to the height of solid AB .

Since solid AB is equal to solid CD , but AB is equal to BT . For they are on the same base FK , and (have) the same height [Props. 11.29, 11.30]. And solid CD is equal is equal to DX . For, again, they are on the same base RO , and (have) the same height [Props. 11.29, 11.30]. Solid BT is thus also equal to solid DX . Thus, as base FK (is) to base OR , so the height of solid DX (is) to the height of solid BT (see first part of proposition). And base FK (is) equal to base EH , and base OR to NQ . Thus, as base EH is to base NQ , so the height of solid DX (is) to the height of solid BT . And solids DX, BT

ἔστιν ἄρα ὥς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ τοῦ $ΔΓ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος. τῶν $ΑΒ$, $ΓΔ$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Πάλιν δὴ τῶν $ΑΒ$, $ΓΔ$ στερεῶν παραλληλεπιπέδων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστω ὥς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος· λέγω, ὅτι ἴσον ἐστὶ τὸ $ΑΒ$ στερεὸν τῷ $ΓΔ$ στερεῷ.

Τῶν γάρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἐστὶν ὥς ἡ $EΘ$ βάσις πρὸς τὴν $ΝΠ$ βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος, ἴση δὲ ἡ μὲν $EΘ$ βάσις τῇ ZK βάσει, ἡ δὲ $ΝΠ$ τῇ XP , ἔστιν ἄρα ὥς ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΑΒ$ στερεοῦ ὕψος. τὰ δ' αὐτὰ ὕψη ἐστὶ τῶν $ΑΒ$, $ΓΔ$ στερεῶν καὶ τῶν $ΒΤ$, $ΔΨ$ · ἔστιν ἄρα ὥς ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $ΔΨ$ στερεοῦ ὕψος πρὸς τὸ τοῦ $ΒΤ$ στερεοῦ ὕψος. τῶν $ΒΤ$, $ΔΨ$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· ἴσον ἄρα ἐστὶ τὸ $ΒΤ$ στερεὸν τῷ $ΔΨ$ στερεῷ. ἀλλὰ τὸ μὲν $ΒΤ$ τῷ $ΒΑ$ ἴσον ἐστίν· ἐπὶ τε γάρ τῇ αὐτῇ βάσει [εἶσι] τῇ ZK καὶ ὑπὸ τὸ αὐτὸ ὕψος. τὸ δὲ $ΔΨ$ στερεὸν τῷ $ΔΓ$ στερεῷ ἴσον ἐστίν. καὶ τὸ $ΑΒ$ ἄρα στερεὸν τῷ $ΓΔ$ στερεῷ ἐστὶν ἴσον· ὅπερ εἶδει δεῖχαι.

are the same height as (solids) DC , BA (respectively). Thus, as base EH is to base NQ , so the height of solid DC (is) to the height of solid AB . Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

So, again, let the bases of the parallelepiped solids AB and CD be reciprocally proportional to their heights, and (so) let base EH be to base NQ , as the height of solid CD (is) to the height of solid AB . I say that solid AB is equal to solid CD .

For, with the same construction (as before), since as base EH is to base NQ , so the height of solid CD (is) to the height of solid AB , and base EH (is) equal to base FK , and NQ to OR , thus as base FK is to base OR , so the height of solid CD (is) to the height of solid AB . And solids AB , CD are the same height as (solids) BT , DX (respectively). Thus, as base FK is to base OR , so the height of solid DX (is) to the height of solid BT . Thus, the bases of the parallelepiped solids BT and DX are reciprocally proportional to their heights. Thus, solid BT is equal to solid DX (see first part of proposition). But, BT is equal to BA . For [they are] on the same base FK , and (have) the same height [Props. 11.29, 11.30]. And solid DX is equal to solid DC [Props. 11.29, 11.30]. Thus, solid AB is also equal to solid CD . (Which is) the very thing it was required to show.

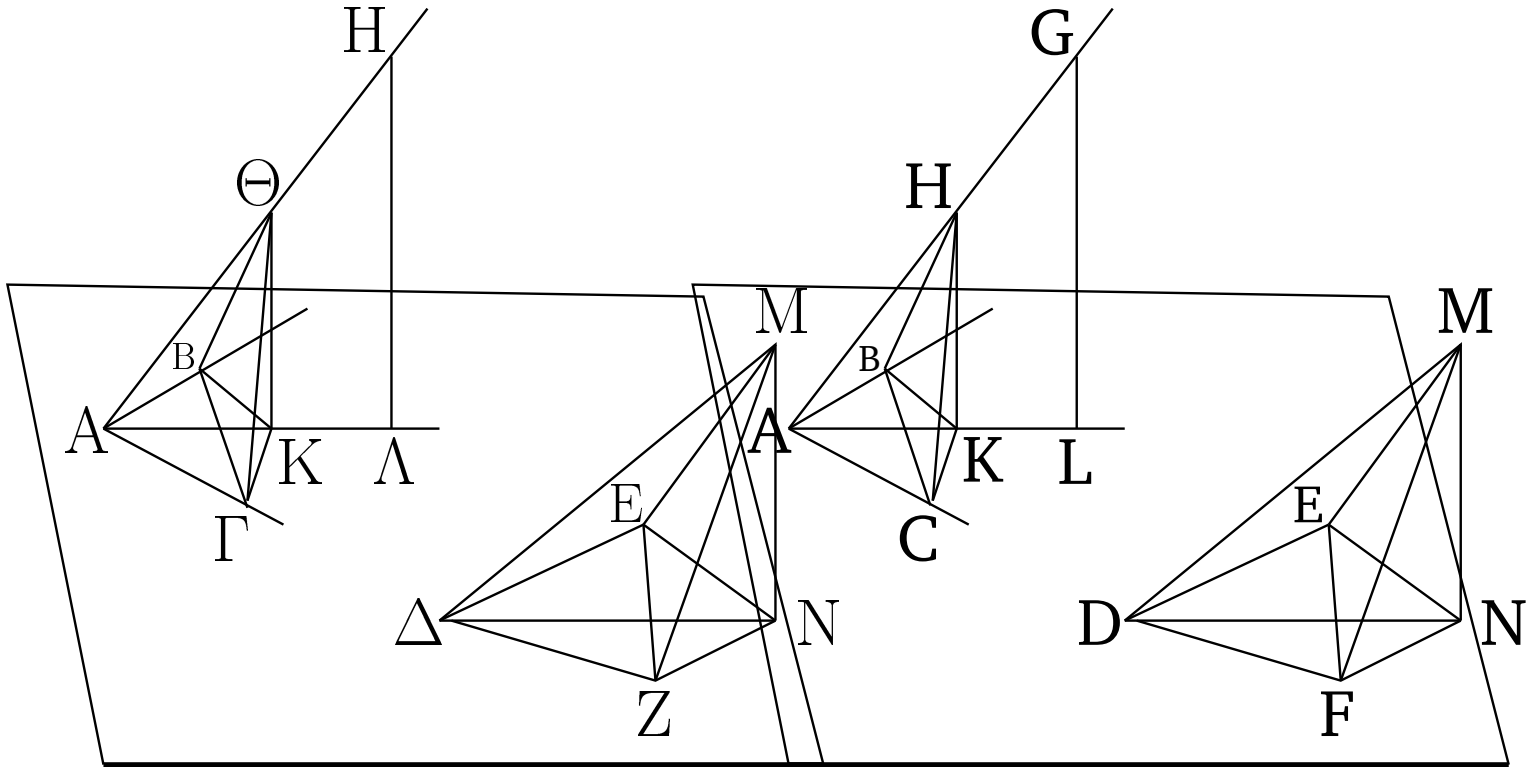
† This proposition assumes that (a) if two parallelepipeds are equal, and have equal bases, then their heights are equal, and (b) if the bases of two equal parallelepipeds are unequal, then that solid which has the lesser base has the greater height.

35.

Ἐὰν ὥσι δύο γωνίαι ἐπίπεδοι ἴσαι, ἐπὶ δὲ τῶν κορυφῶν αὐτῶν μετέωροι εὐθεῖαι ἐπισταθῶσιν ἴσαξ γωνίαξ περιέθουσαι μετὰ τῶν ἐξ ἀρθῆξ εὐθειῶν ἑκατέραν ἑκατέρῃ, ἐπὶ δὲ τῶν μετεώρων ληφθῇ τυθόντα σημεῖα, καὶ ἀπ' αὐτῶν ἐπὶ τὰ ἐπίπεδα, ἐν οἷς εἰσιν αἱ ἐξ ἀρθῆξ γωνίαι, κάθετοι ἀθθῶσιν, ἀπὸ δὲ τῶν γενομένων σημείων ἐν τοῖς ἐπιπέδοις ἐπὶ τὰς ἐξ ἀρθῆξ γωνίαξ ἐπιξενθῶσιν εὐθεῖαι, ἴσαξ γωνίαξ περιέχουσι μετὰ τῶν μετεώρων.

Proposition 35

If there are two equal plane angles, and raised straight-lines are stood on the apexes of them, containing equal angles respectively with the original straight-lines (forming the angles), and random points are taken on the raised (straight-lines), and perpendiculars are drawn from them to the planes in which the original angles are, and straight-lines are joined from the points created in the planes to the (vertices of the) original angles, then they will enclose equal angles with the raised (straight-lines).



Ἐστωσαν δύο γωνίαι εὐθύγραμμοι ἴσαι αἱ ὑπὸ BAG , $EΔZ$, ἀπὸ δὲ τῶν A , $Δ$ σημείων μετέωροι εὐθεῖαι ἐφεσταύωσαν αἱ AH , $ΔM$ ἴσαξ γωνίας περιέθουσιν μετὰ τῶν ἐχ' ἀρθῆξ εὐθειῶν ἑκατέραν ἑκατέρῳ, τὴν μὲν ὑπὸ $MΔE$ τῇ ὑπὸ HAB , τὴν δὲ ὑπὸ $MΔZ$ τῇ ὑπὸ HAG , καὶ εἰλήφθω ἐπὶ τῶν AH , $ΔM$ τυθόντα σημεία τὰ H , M , καὶ ἤρθωσαν ἀπὸ τῶν H , M σημείων ἐπὶ τὰ διὰ τῶν BAG , $EΔZ$ ἐπίπεδα κάθετοι αἱ HL , MN , καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ L , N , καὶ ἐπεξεύθωσαν αἱ LA , ND · λέγω, ὅτι ἴση ἐστὶν ἡ ὑπὸ HAA γωνία τῇ ὑπὸ $MΔN$ γωνίᾳ.

Κείσθω τῇ $ΔM$ ἴση ἡ $AΘ$, καὶ ἤρθω διὰ τοῦ $Θ$ σημείου τῇ HL παράλληλος ἡ $ΘK$. ἡ δὲ HL κάθετός ἐστιν ἐπὶ τὸ διὰ τῶν BAG ἐπίπεδον· καὶ ἡ $ΘK$ ἄρα κάθετός ἐστιν ἐπὶ τὸ διὰ τῶν BAG ἐπίπεδον. ἤρθωσαν ἀπὸ τῶν K , N σημείων ἐπὶ τὰς AG , $ΔZ$, AB , $ΔE$ εὐθείας κάθετοι αἱ $KΓ$, NZ , KB , NE , καὶ ἐπεξεύθωσαν αἱ $ΘΓ$, $ΓB$, MZ , ZE . ἐπεὶ τὸ ἀπὸ τῆς $ΘA$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΘK$, KA , τῷ δὲ ἀπὸ τῆς KA ἴσα ἐστὶ τὰ ἀπὸ τῶν $KΓ$, $ΓA$, καὶ τὸ ἀπὸ τῆς $ΘA$ ἄρα ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΘK$, $KΓ$, $ΓA$. τοῖς δὲ ἀπὸ τῶν $ΘK$, $KΓ$ ἴσον ἐστὶ τὸ ἀπὸ τῆς $ΘΓ$ · τὸ ἄρα ἀπὸ τῆς $ΘA$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν $ΘΓ$, $ΓA$. ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ $ΘΓA$ γωνία. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $ΔZM$ γωνία ὀρθή ἐστιν. ἴση ἄρα ἐστὶν ἡ ὑπὸ $AGΘ$ γωνία τῇ ὑπὸ $ΔZM$. ἐστὶ δὲ καὶ ἡ ὑπὸ $ΘAG$ τῇ ὑπὸ $MΔZ$ ἴση. δύο δὴ τρίγωνά ἐστι τὰ $MΔZ$, $ΘAG$ δύο γωνίας δυσὶ γωνίαις ἴσαξ ἔχοντα ἑκατέραν ἑκατέρῳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν τὴν $ΘA$ τῇ $MΔ$ · καὶ τὰς λοιπάς ἄρα πλευράς ταῖς λοιπαῖς πλευραῖς

Let BAC and EDF be two equal rectilinear angles. And let the raised straight-lines AG and DM be stood on points A and D , containing equal angles respectively with the original straight-lines. (That is) MDE (equal) to GAB , and MDF (to) GAC . And let the random points G and M be taken on AG and DM (respectively). And let the GL and MN be drawn from points G and M perpendicular to the planes through BAC and EDF (respectively). And let them have joined the planes at points L and N (respectively). And let LA and ND be joined. I say that angle GAL is equal to angle MDN .

Let AH be made equal to DM . And let HK be drawn through point H parallel to GL . And GL is perpendicular to the plane through BAC . Thus, HK is also perpendicular to the plane through BAC [Prop. 11.8]. And let KC , NF , KB , and NE be drawn from points K and N perpendicular to the straight-lines AC , DF , AB , and DE . And let HC , CB , MF , and FE be joined. Since the (square) on HA is equal to the (sum of the squares) on HK and KA [Prop. 1.47], and the (sum of the squares) on KC and CA is equal to the (square) on KA [Prop. 1.47], thus the (square) on HA is equal to the (sum of the squares) on HK , KC , and CA . And the (square) on HC is equal to the (sum of the squares) on HK and KC [Prop. 1.47]. Thus, the (square) on HA is equal to the (sum of the squares) on HC and CA . Thus, angle HCA is a right-angle [Prop. 1.48]. So, for the same (reasons), angle DFM is also a right-angle. Thus, angle ACH is equal to (angle) DFM . And HAC is also equal to MDF . So, MDF and HAC are two triangles hav-

ἴσαξ ἔχει ἑκατέραν ἑκατέρω. ἴση ἄρα ἐστὶν ἡ $ΑΓ$ τῇ $ΔΖ$. ὁμοίωξ δὴ δείχομεν, ὅτι καὶ ἡ $ΑΒ$ τῇ $ΔΕ$ ἐστὶν ἴση. ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν $ΑΓ$ τῇ $ΔΖ$, ἡ δὲ $ΑΒ$ τῇ $ΔΕ$, δύο δὴ αἱ $ΓΑ$, $ΑΒ$ δυοὶ ταῖς $ΖΔ$, $ΔΕ$ ἴσαι εἰσὶν. ἀλλὰ καὶ γωνία ἡ ὑπὸ $ΓΑΒ$ γωνία τῇ ὑπὸ $ΖΔΕ$ ἐστὶν ἴση· βάσεις ἄρα ἡ $ΒΓ$ βάσει τῇ $ΕΖ$ ἴση ἐστὶ καὶ τὸ τρίγωνον τῷ τριγώνῳ καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις· ἴση ἄρα ἡ ὑπὸ $ΑΓΒ$ γωνία τῇ ὑπὸ $ΔΖΕ$. ἐστὶ δὲ καὶ ὀρθή ἡ ὑπὸ $ΑΓΚ$ ὀρθή τῇ ὑπὸ $ΔΖΝ$ ἴση· καὶ λοιπὴ ἄρα ἡ ὑπὸ $ΒΓΚ$ λοιπὴ τῇ ὑπὸ $ΕΖΝ$ ἐστὶν ἴση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ $ΓΒΚ$ τῇ ὑπὸ $ΖΕΝ$ ἐστὶν ἴση. δύο δὴ τριγώνω ἐστὶ τὰ $ΒΓΚ$, $ΕΖΝ$ [τάξ] δύο γωνίαξ δυοὶ γωνίαιξ ἴσαξ ἔθοντα ἑκατέραν ἑκατέρω καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν $ΒΓ$ τῇ $ΕΖ$ · καὶ τὰς λοιπάς ἄρα πλευράς ταῖς λοιπαῖς πλευραῖς ἴσαξ ἔχουσιν. ἴση ἄρα ἐστὶν ἡ $ΓΚ$ τῇ $ΖΝ$. ἐστὶ δὲ καὶ ἡ $ΑΓ$ τῇ $ΔΖ$ ἴση· δύο δὴ αἱ $ΑΓ$, $ΓΚ$ δυοὶ ταῖς $ΔΖ$, $ΖΝ$ ἴσαι εἰσὶν καὶ ὀρθὰς γωνίαξ περιέθουσιν. βάσεις ἄρα ἡ $ΑΚ$ βάσει τῇ $ΔΝ$ ἴση ἐστὶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΑΘ$ τῇ $ΔΜ$, ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς $ΑΘ$ τῷ ἀπὸ τῆς $ΔΜ$. ἀλλὰ τῷ μὲν ἀπὸ τῆς $ΑΘ$ ἴσα ἐστὶ τὰ ἀπὸ τῶν $ΑΚ$, $ΚΘ$ · ὀρθή γάρ ἡ ὑπὸ $ΑΚΘ$ · τῷ δὲ ἀπὸ τῆς $ΔΜ$ ἴσα τὰ ἀπὸ τῶν $ΔΝ$, $ΝΜ$ · ὀρθή γάρ ἡ ὑπὸ $ΔΝΜ$ · τὰ ἄρα ἀπὸ τῶν $ΑΚ$, $ΚΘ$ ἴσα ἐστὶ τοῖς ἀπὸ τῶν $ΔΝ$, $ΝΜ$, ὣν τὸ ἀπὸ τῆς $ΑΚ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΔΝ$ · λοιπὸν ἄρα τὸ ἀπὸ τῆς $ΚΘ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΝΜ$ · ἴση ἄρα ἡ $ΘΚ$ τῇ $ΜΝ$. καὶ ἐπεὶ δύο αἱ $ΘΑ$, $ΑΚ$ δυοὶ ταῖς $ΜΔ$, $ΔΝ$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρω, καὶ βάσεις ἡ $ΘΚ$ βάσει τῇ $ΜΝ$ ἐδείχθη ἴση, γωνία ἄρα ἡ ὑπὸ $ΘΑΚ$ γωνία τῇ ὑπὸ $ΜΔΝ$ ἐστὶν ἴση.

Ἐὰν ἄρα ὦσι δύο γωνίαι ἐπίπεδοι ἴσαι καὶ τὰ ἐχθῆς τῆς προτάσεως [ὅπερ ἔδει δεῖχαι].

ing two angles equal to two angles, respectively, and one side equal to one side—(namely), that subtending one of the equal angles —(that is), HA (equal) to MD . Thus, they will also have the remaining sides equal to the remaining sides, respectively [Prop. 1.26]. Thus, AC is equal to DF . So, similarly, we can show that AB is also equal to DE . Therefore, since AC is equal to DF , and AB to DE , so the two (straight-lines) CA and AB are equal to the two (straight-lines) FD and DE (respectively). But, angle CAB is also equal to angle FDE . Thus, base BC is equal to base EF , and triangle (ACB) to triangle (DFE), and the remaining angles to the remaining angles (respectively) [Prop. 1.4]. Thus, angle ACB (is) equal to DFE . And the right-angle ACK is also equal to the right-angle DFN . Thus, the remainder BCK is equal to the remainder EFN . So, for the same (reasons), CBK is also equal to FEN . So, BCK and EFN are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), that by the equal angles—(that is), BC (equal) to EF . Thus, they will also have the remaining sides equal to the remaining sides (respectively) [Prop. 1.26]. Thus, CK is equal to FN . And AC (is) also equal to DF . So, the two (straight-lines) AC and CK are equal to the two (straight-lines) DF and FN (respectively). And they enclose right-angles. Thus, base AK is equal to base DN [Prop. 1.4]. And since AH is equal to DM , the (square) on AH is also equal to the (square) on DM . But, the the (sum of the squares) on AK and KH is equal to the (square) on AH . For angle AKH (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on DN and NM (is) equal to the square on DM . For angle DNM (is) a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AK and KH is equal to the (sum of the squares) on DN and NM , of which the (square) on AK is equal to the (square) on DN . Thus, the remaining (square) on KH is equal to the (square) on NM . Thus, HK (is) equal to MN . And since the two (straight-lines) HA and AK are equal to the two (straight-lines) MD and DN , respectively, and base HK was shown (to be) equal to base MN , angle HAK is thus equal to angle MDN [Prop. 1.8].

Thus, if there are two equal plane angles, and so on of the proposition. [(Which is) the very thing it was required to show].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἔαν ὦσι δύο γωνίαι ἐπίπεδοι ἴσαι, ἐπισταθῶσι δὲ ἐπ' αὐτῶν μετέωροι εὐθεῖαι ἴσαι ἴσαξ γωνίαξ περιέθουσαι μετὰ τῶν ἐχ ἄρθῆς εὐθειῶν ἑκατέραν ἑκατέρω, αἱ ἀπ' αὐτῶν κάθετοι ἀρόμεναι ἐπὶ τὰ ἐπίπεδα, ἐν οἷς εἰσὶν αἱ ἐχ ἄρθῆς γωνίαι, ἴσαι ἀλλήλαις εἰσὶν. ὅπερ ἔδει δεῖχαι.

Corollary

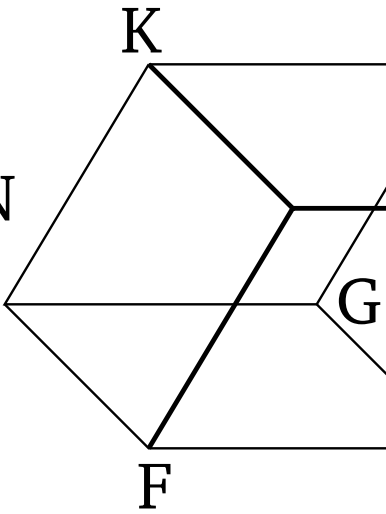
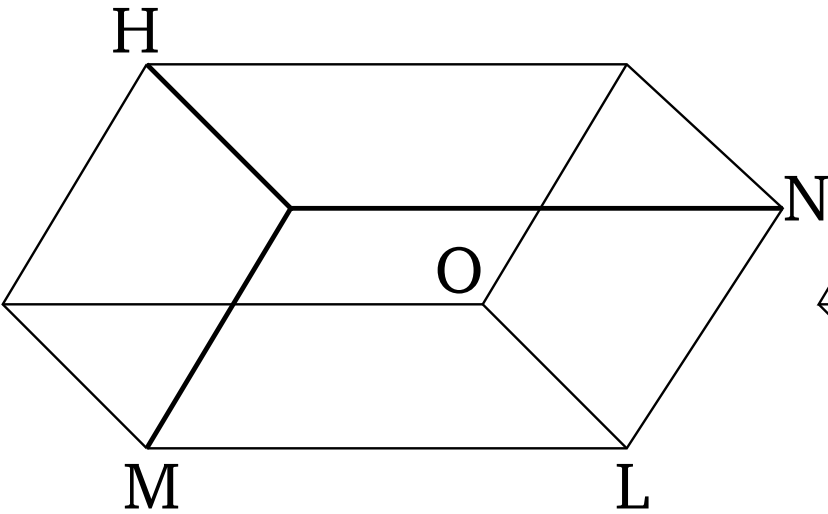
So, it is clear, from this, that if there are two equal plane angles, and equal raised straight-lines are stood on them (at their apexes), containing equal angles respectively with the original straight-lines (forming the angles), then the perpendiculars drawn from (the raised ends of) them to the planes in which the original angles lie are equal to one another. (Which is) the very thing it was required to show.

36.

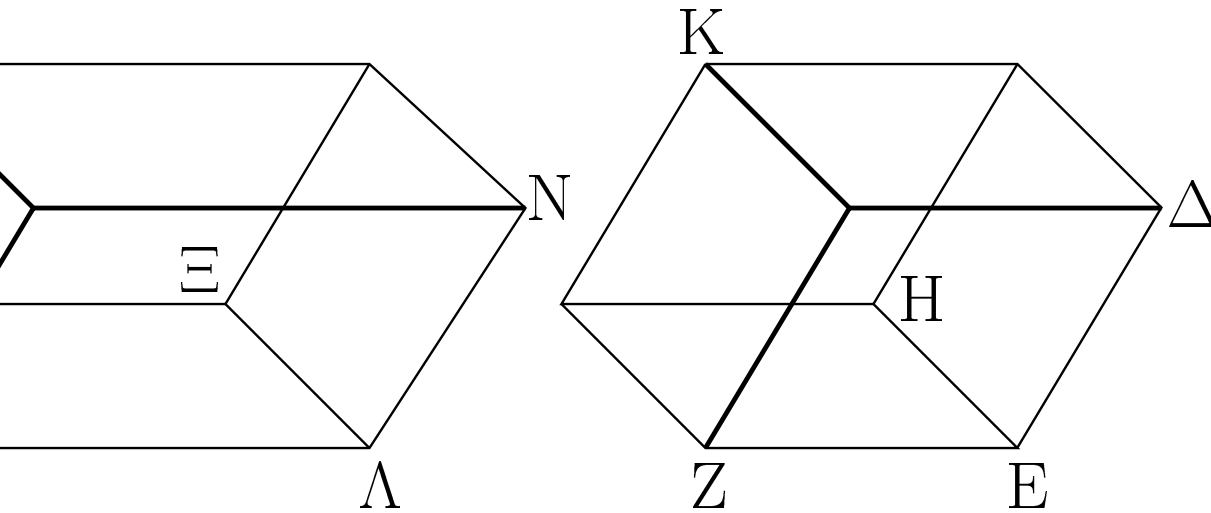
Proposition 36

Ἐάν τρεῖς εὐθεῖαι ἀνάλογον ᾗσιν, τὸ ἐκ τῶν τριῶν στερεὸν παραλληλεπίπεδον ἴσον ἐστὶ τῷ ἀπὸ τῆς μέσης στερεῷ παραλληλεπίπεδῳ ἰσοπλεύρῳ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

If three straight-lines are (continuously) proportional then the parallelepiped solid (formed) from the three (straight-lines) is equal to the equilateral parallelepiped solid on the middle (straight-line which is) equiangular to the aforementioned (parallelepiped solid).



- A _____
- B _____
- C _____



Ἐστωσαν τρεῖς εὐθεῖαι ἀνάλογον αἱ A, B, Γ , ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ . λέγω, ὅτι τὸ ἐκ τῶν A, B, Γ στερεὸν ἴσον ἐστὶ τῷ ἀπὸ τῆς B στερεῷ ἰσοπλεύρῳ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

Ἐκκείσθω στερεὰ γωνία ἡ πρὸς τῷ E περιεθομένη ὑπὸ τῶν ὑπὸ $\Delta E H, H E Z, Z E \Delta$, καὶ κείσθω τῇ μὲν B ἴση ἐκάστη τῶν $\Delta E, H E, E Z$, καὶ συμπληρώσθω τὸ $E K$ στερεὸν παραλληλεπίπεδον, τῇ δὲ A ἴση ἡ ΛM , καὶ συνεστάτω πρὸς τῇ ΛM εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Λ τῇ πρὸς τῷ E στερεᾷ γωνία ἴση στερεὰ γωνία ἡ περιεθομένη ὑπὸ τῶν $N \Lambda X, X \Lambda M, M \Lambda N$, καὶ κείσθω τῇ μὲν B ἴση ἡ ΛX , τῇ δὲ Γ ἴση ἡ ΛN . καὶ ἐπεὶ ἐστὶν ὥς ἡ A πρὸς τὴν B , οὕτως ἡ B πρὸς τὴν Γ , ἴση δὲ ἡ μὲν A τῇ ΛM , ἡ δὲ B ἐκατέρᾳ τῶν $\Lambda X, E \Delta$, ἡ δὲ Γ τῇ ΛN , ἔστιν ἄρα ὥς ἡ ΛM πρὸς τὴν $E Z$, οὕτως ἡ ΔE πρὸς τὴν ΛN . καὶ περὶ ἴσαξ γωνίας τὰς ὑπὸ $N \Lambda M, \Delta E Z$ αἱ πλευраὶ ἀντιπεπόνθασιν· ἴσον ἄρα ἐστὶ τὸ $M N$ παραλληλόγραμμον τῷ ΔZ παραλληλογραμμίῳ. καὶ ἐπεὶ δύο γωνίαι ἐπίπεδοι εὐθύγραμμοι ἴσαι εἰσὶν αἱ ὑπὸ $\Delta E Z, N \Lambda M$, καὶ ἐπ' αὐτῶν μετέωροι εὐθεῖαι ἐφεστᾶσιν αἱ $\Lambda X, E H$ ἴσαι τε ἀλλήλαις καὶ ἴσαξ γωνίαξ περιέθουσαι μετὰ τῶν ἐχ' ἀρθῆξ εὐθειῶν ἐκατέραν ἐκατέρᾳ, αἱ ἄρα ἀπὸ τῶν H, X σημείων κάθετοι ἀγόμεναι ἐπὶ τὰ διὰ τῶν $N \Lambda M, \Delta E Z$ ἐπίπεδα ἴσαι ἀλλήλαις εἰσὶν· ὥστε τὰ $\Lambda \Theta, E K$ στερεὰ ὑπὸ τὸ αὐτὸ ὕψος ἐστίν. τὰ δὲ ἐπὶ ἴσων βάσεων στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ $\Theta \Lambda$

Let A, B , and C be three (continuously) proportional straight-lines, (such that) as A (is) to B , so B (is) to C . I say that the (parallelepiped) solid (formed) from A, B , and C is equal to the equilateral solid on B (which is) equiangular with the aforementioned (solid).

Let the solid angle at E , contained by DEG, GEF , and FED , be set out. And let DE, GE , and EF each be made equal to B . And let the parallelepiped solid $E K$ be completed. And (let) LM (be made) equal to A . And let the solid angle contained by NLO, OLM , and MLN be constructed on the straight-line LM , and at the point L on it, (so as to be) equal to the solid angle E [Prop. 11.23]. And let LO be made equal to B , and LN equal to C . And since as A (is) to B , so B (is) to C , and A (is) equal to LM , and B to each of LO and ED , and C to LN , thus as LM (is) to EF , so DE (is) to LN . And (so) the sides around the equal angles NLM and DEF are reciprocally proportional. Thus, parallelogram $M N$ is equal to parallelogram $D F$ [Prop. 6.14]. And since the two plane rectilinear angles DEF and NLM are equal, and the raised straight-lines stood on them (at their apexes), LO and EG , are equal to one another, and contain equal angles respectively with the original straight-lines (forming the angles), the perpendiculars drawn from points G and O to the planes through NLM and DEF (respectively) are thus equal to one another [Prop. 11.35 corr.]. Thus, the solids LH and $E K$ (have) the same height. And parallelepiped solids on equal bases, and with the same height, are

στερεὸν τῷ EK στερεῷ. καὶ ἐστὶ τὸ μὲν $\Lambda\Theta$ τὸ ἐκ τῶν A, B, Γ στερεόν, τὸ δὲ EK τὸ ἀπὸ τῆς B στερεόν· τὸ ἄρα ἐκ τῶν A, B, Γ στερεὸν παραλληλεπίπεδον ἴσον ἐστὶ τῷ ἀπὸ τῆς B στερεῷ ἰσοπλεύρῳ μὲν, ἰσογωνίῳ δὲ τῷ προειρημένῳ· ὅπερ ἔδει δεῖξαι.

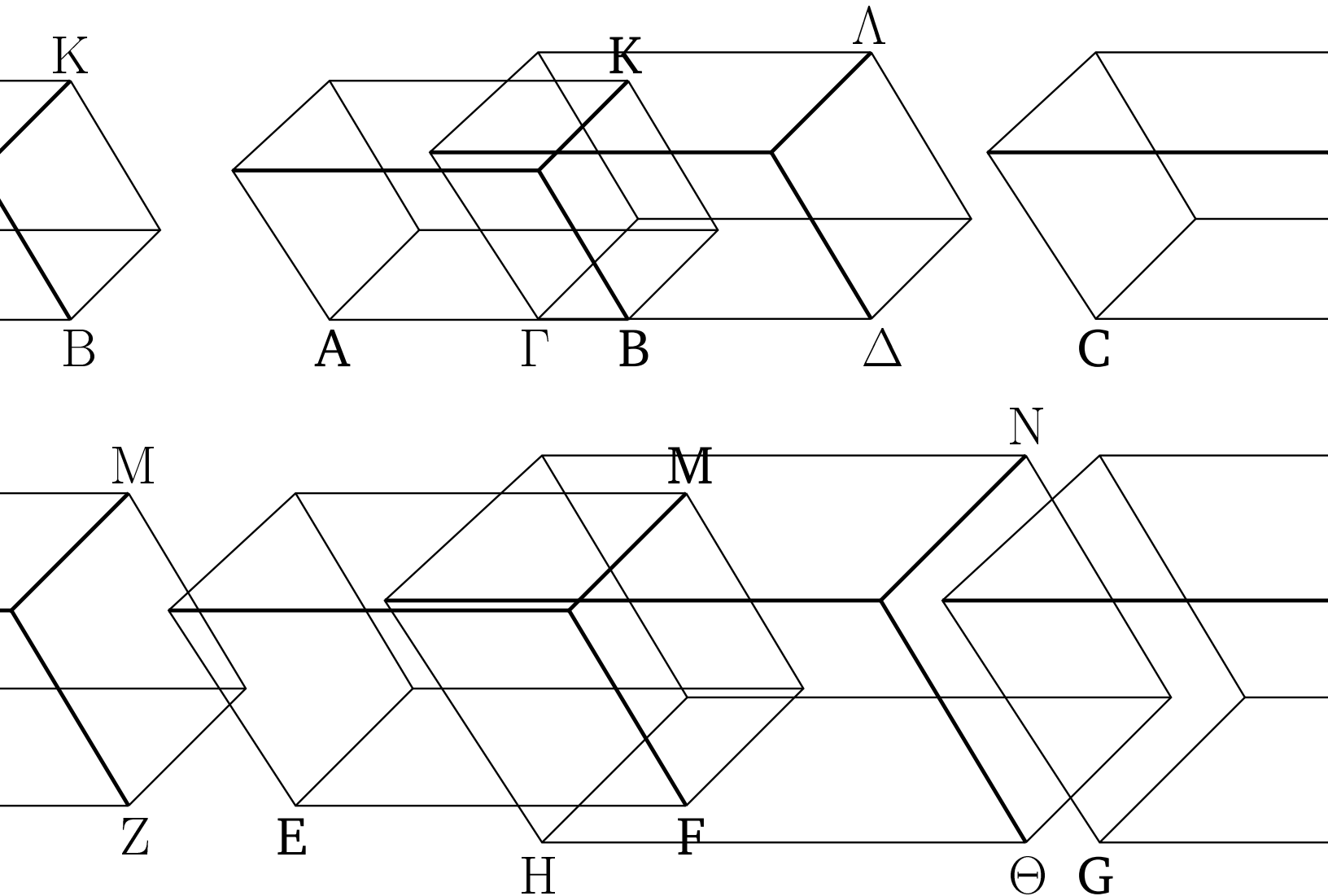
equal to one another [Prop. 11.31]. Thus, solid HL is equal to solid EK . And LH is the solid (formed) from A, B , and C , and EK the solid on B . Thus, the parallelepiped solid (formed) from A, B , and C is equal to the equilateral solid on B (which is) equiangular with the aforementioned (solid). (Which is) the very thing it was required to show.

37.

Proposition 37[†]

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ᾧσιν, καὶ τὰ ἀπ' αὐτῶν στερεὰ παραλληλεπίπεδα ὁμοία τε καὶ ὁμοίῳξ ἀναγραφόμενα ἀνάλογον ἔσται· καὶ ἔαν τὰ ἀπ' αὐτῶν στερεὰ παραλληλεπίπεδα ὁμοία τε καὶ ὁμοίῳξ ἀναγραφόμενα ἀνάλογον ᾧ, καὶ αὐταὶ αἱ εὐθεῖαι ἀνάλογον ἔσονται.

If four straight-lines are proportional then the similar, and similarly described, parallelepiped solids on them will also be proportional. And if the similar, and similarly described, parallelepiped solids on them are proportional then the straight-lines themselves will be proportional.



Ἐστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ $AB, \Gamma\Delta, EZ,$

Let $AB, CD, EF,$ and $GH,$ be four proportional straight-

$H\Theta$, ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$, καὶ ἀναγεγράφθωσαν ἀπὸ τῶν AB , $\Gamma\Delta$, EZ , $H\Theta$ ὁμοιά τε καὶ ὁμοίως κείμενα στερεὰ παραλληλεπίπεδα τὰ KA , $\Lambda\Gamma$, ME , NH . λέγω, ὅτι ἐστὶν ὥς τὸ KA πρὸς τὸ $\Lambda\Gamma$, οὕτως τὸ ME πρὸς τὸ NH .

Ἐπεὶ γὰρ ὁμοίον ἐστὶ τὸ KA στερεὸν παραλληλεπίπεδον τῷ $\Lambda\Gamma$, τὸ KA ἄρα πρὸς τὸ $\Lambda\Gamma$ τριπλασίονα λόγον ἔθει ἥπερ ἡ AB πρὸς τὴν $\Gamma\Delta$. διὰ τὰ αὐτὰ δὴ καὶ τὸ ME πρὸς τὸ NH τριπλασίονα λόγον ἔθει ἥπερ ἡ EZ πρὸς τὴν $H\Theta$. καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$. καὶ ὥς ἄρα τὸ AK πρὸς τὸ $\Lambda\Gamma$, οὕτως τὸ ME πρὸς τὸ NH .

Ἀλλὰ δὴ ἔστω ὥς τὸ AK στερεὸν πρὸς τὸ $\Lambda\Gamma$ στερεόν, οὕτως τὸ ME στερεὸν πρὸς τὸ NH . λέγω, ὅτι ἐστὶν ὥς ἡ AB εὐθεῖα πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$.

Ἐπεὶ γὰρ πάλιν τὸ KA πρὸς τὸ $\Lambda\Gamma$ τριπλασίονα λόγον ἔθει ἥπερ ἡ AB πρὸς τὴν $\Gamma\Delta$, ἔθει δὲ καὶ τὸ ME πρὸς τὸ NH τριπλασίονα λόγον ἥπερ ἡ EZ πρὸς τὴν $H\Theta$, καὶ ἐστὶν ὥς τὸ KA πρὸς τὸ $\Lambda\Gamma$, οὕτως τὸ ME πρὸς τὸ NH , καὶ ὥς ἄρα ἡ AB πρὸς τὴν $\Gamma\Delta$, οὕτως ἡ EZ πρὸς τὴν $H\Theta$.

Ἐὰν ἄρα τέσσαρες εὐθεῖαι ἀνάλογον ᾧσι καὶ τὰ ἐχῇς τῆς προτάσεως· ὅπερ ἔδει δεῖχαι.

lines, (such that) as AB (is) to CD , so EF (is) to GH . And let the similar, and similarly laid out, parallelepiped solids KA , LC , ME and NG be described on AB , CD , EF , and GH (respectively). I say that as KA is to LC , so ME (is) to NG .

For since the parallelepiped solid KA is similar to LC , KA thus has to LC the cubed ratio that AB (has) to CD [Prop. 11.33]. So, for the same (reasons), ME also has to NG the cubed ratio that EF (has) to GH [Prop. 11.33]. And since as AB is to CD , so EF (is) to GH , thus, also, as AK (is) to LC , so ME (is) to NG .

And so let solid AK be to solid LC , as solid ME (is) to NG . I say that as straight-line AB is to CD , so EF (is) to GH .

For, again, since KA has to LC the cubed ratio that AB (has) to CD [Prop. 11.33], and ME also has to NG the cubed ratio that EF (has) to GH [Prop. 11.33], and as KA is to LC , so ME (is) to NG , thus, also, as AB (is) to CD , so EF (is) to GH .

Thus, if four straight-lines are proportional, and so on of the proposition. (Which is) the very thing it was required to show.

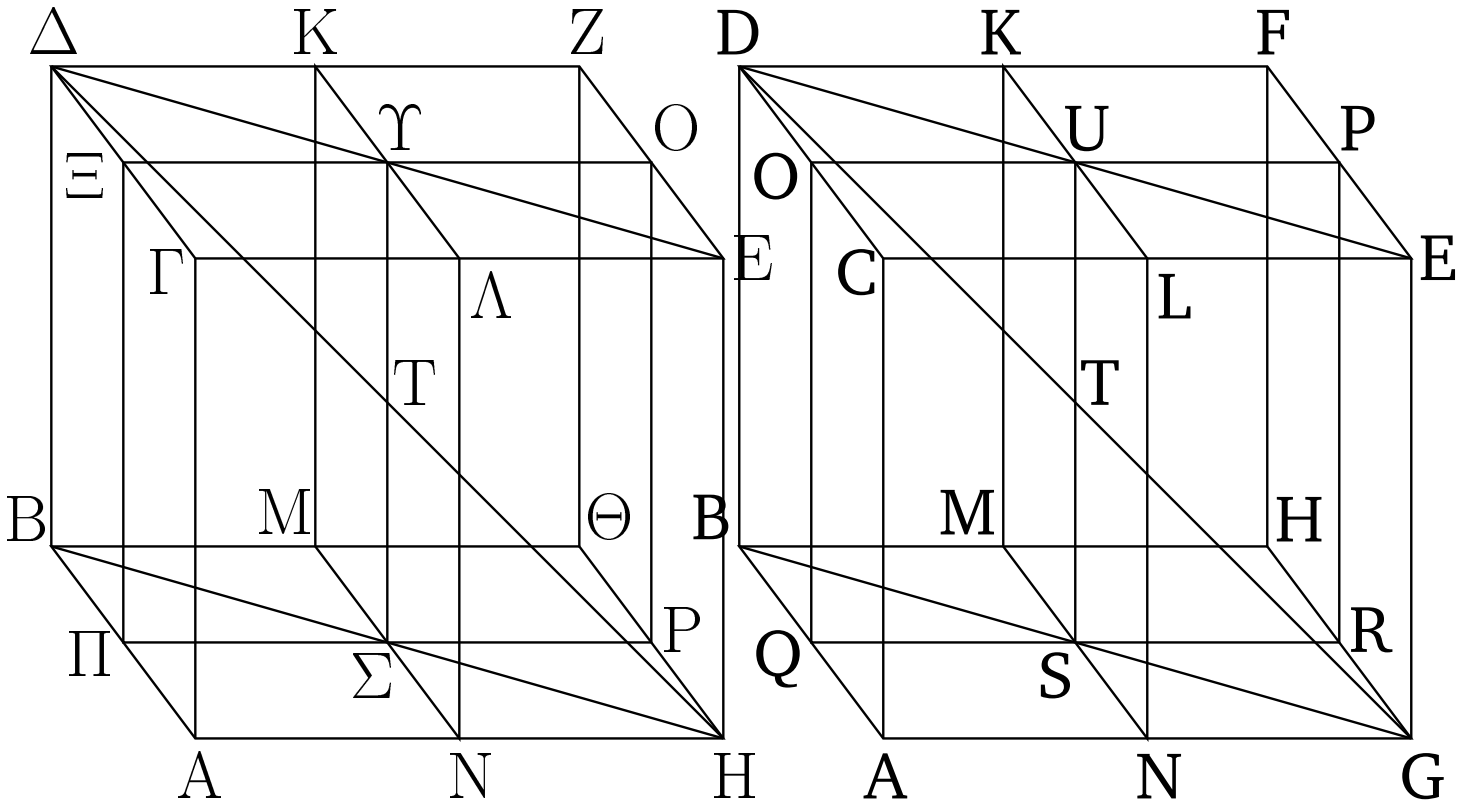
[†] This proposition assumes that if two ratios are equal then the cube of the former is also equal to the cube of the latter, and vice versa.

38.

Ἐὰν κύβον τῶν ἀπεναντίον ἐπιπέδων αἱ πλευραὶ δίθῃ τμηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος δίθῃ τέμνουσιν ἀλλήλας.

Proposition 38

If the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut one another in half.



Κύβον γάρ τοῦ ΑΖ τῶν ἀπεναντίον ἐπιπέδων τῶν ΓΖ, ΑΘ αἱ πλευραὶ διῖθα τετμήσθωσαν κατὰ τὰ Κ, Λ, Μ, Ν, Χ, Π, Ο, Ρ σημεῖα, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβεβλήσθω τὰ ΚΝ, ΧΡ, κοινὴ δὲ τομὴ τῶν ἐπιπέδων ἔστω ἡ ΥΣ, τοῦ δὲ ΑΖ κύβου διαγώνιος ἡ ΔΗ. λέγω, ὅτι ἴση ἐστὶν ἡ μὲν ΥΤ τῇ ΤΣ, ἡ δὲ ΔΤ τῇ ΤΗ.

Ἐπεξενύθωσαν γάρ αἱ ΔΥ, ΥΕ, ΒΣ, ΣΗ. καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΔΧ τῇ ΟΕ, αἱ ἐναλλάξ γωνίαι αἱ ὑπὸ ΔΧΥ, ΥΟΕ ἴσαι ἀλλήλαις εἰσὶν. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν ΔΧ τῇ ΟΕ, ἡ δὲ ΧΥ τῇ ΥΟ, καὶ γωνίαξ ἴσαξ περιέθουσιν, βάσιξ ἄρα ἡ ΔΥ τῇ ΥΕ ἐστὶν ἴση, καὶ τὸ ΔΧΥ τρίγωνον τῷ ΟΥΕ τρίγωνῳ ἐστὶν ἴσον καὶ αἱ λοιπαὶ γωνίαι ταῖξ λοιπαῖξ γωνίαξ ἴσαι· ἴση ἄρα ἡ ὑπὸ ΧΥΔ γωνία τῇ ὑπὸ ΟΥΕ γωνία. διὰ δὲ τοῦτο εὐθεῖά ἐστιν ἡ ΔΥΕ. διὰ τὰ αὐτὰ δὴ καὶ ΒΣΗ εὐθεῖά ἐστιν, καὶ ἴση ἡ ΒΣ τῇ ΣΗ. καὶ ἐπεὶ ἡ ΓΑ τῇ ΔΒ ἴση ἐστὶ καὶ παράλληλος, ἀλλὰ ἡ ΓΑ καὶ τῇ ΕΗ ἴση τέ ἐστὶ καὶ παράλληλος, καὶ ἡ ΔΒ ἄρα τῇ ΕΗ ἴση τέ ἐστὶ καὶ παράλληλος. καὶ ἐπιξενυνύουσιν αὐτάξ εὐθεῖαι αἱ ΔΕ, ΒΗ· παράλληλός ἄρα ἐστὶν ἡ ΔΕ τῇ ΒΗ. ἴση ἄρα ἡ μὲν ὑπὸ ΕΔΤ γωνία τῇ ὑπὸ ΒΗΤ· ἐναλλάξ γάρ· ἡ δὲ ὑπὸ ΔΤΥ τῇ ὑπὸ ΗΤΣ. δύο δὴ τρίγωνα ἐστὶ τὰ ΔΤΥ, ΗΤΣ τὰξ δύο γωνίαξ ταῖξ δυοὶ γωνίαξ ἴσαξ ἔθοντα καὶ μίαν πλευράν μιᾷ πλευρᾷ ἴσην τὴν ὑποτείνουσιν ὑπὸ μίαν τῶν ἴσων γωνιῶν τὴν ΔΥ τῇ ΗΣ· ἡμίσειαι γάρ εἰσι τῶν ΔΕ, ΒΗ· καὶ τὰξ λοιπάξ πλευράξ ταῖξ λοιπαῖξ πλευραῖξ ἴσαξ ἔχει. ἴση ἄρα ἡ μὲν ΔΤ τῇ ΤΗ, ἡ δὲ ΥΤ τῇ ΤΣ.

For let the opposite planes CF and AH of the cube AF be cut in half at the points K, L, M, N, O, Q, P , and R . And let the planes KN and OR be produced through the pieces. And let US be the common section of the planes, and DG the diameter of cube AF . I say that UT is equal to TS , and DT to TG .

For let DU, UE, BS , and SG be joined. And since DO is parallel to PE , the alternate angles DOU and UPE are equal to one another [Prop. 1.29]. And since DO is equal to PE , and OU to UP , and they contain equal angles, base DU is thus equal to base UE , and triangle DOU is equal to triangle PUE , and the remaining angles (are) equal to the remaining angles [Prop. 1.4]. Thus, angle ODU (is) equal to angle PUE . So, for this (reason), DUE is a straight-line [Prop. 1.14]. So, for the same (reason), BSG is also a straight-line, and BS equal to SG . And since CA is equal and parallel to DB , but CA is also equal and parallel to EG , DB is thus also equal and parallel to EG [Prop. 11.9]. And the straight-lines DE and BG join them. DE is thus parallel to BG [Prop. 1.33]. Thus, angle EDT (is) equal to BGT . For (they are) alternate [Prop. 1.29]. And (angle) DTU (is equal) to GTS [Prop. 1.15]. So, DTU and GTS are two triangles having two angles equal to two angles, and one side equal to one side—(namely), that subtended by one of the equal angles—(that is), DU (equal) to GS . For they are halves of DE and BG (respectively). (Thus), they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, DT (is) equal to TG , and UT to TS .

Ἐάν ἄρα κύβον τῶν ἀπεναντίον ἐπιπέδων αἱ πλευραὶ διῆται μὴθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος διῆται τέμνονσιν ἀλλήλας· ὅπερ ἔδει δεῖξαι.

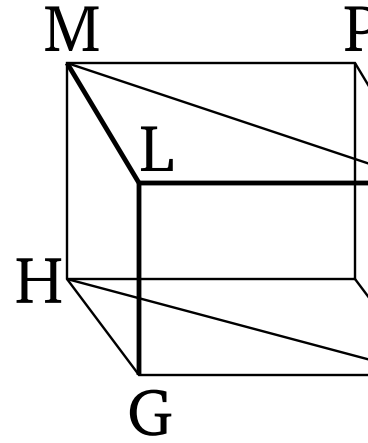
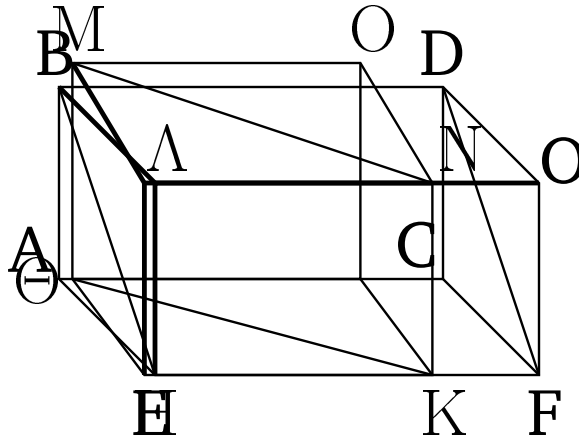
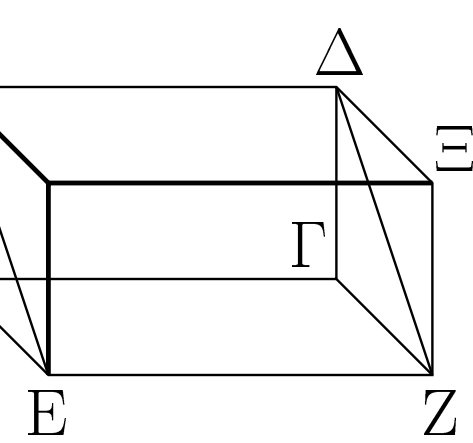
Thus, if the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut one another in half. (Which is) the very thing it was required to show.

39.

Proposition 39

Ἐάν ἡ δύο πρίσματα ἰσοῦσῃ, καὶ τὸ μὲν ἔθῃ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ἢ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσα ἔσται τὰ πρίσματα.

If there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms will be equal.



Ἐστω δύο πρίσματα ἰσοῦσῃ τὰ $ABΓΔΕΖ$, $HΘΚΛΜΝ$, καὶ τὸ μὲν ἔθῃ βάσιν τὸ AZ παραλληλόγραμμον, τὸ δὲ τὸ $HΘΚ$ τρίγωνον, διπλάσιον δὲ ἔστω τὸ AZ παραλληλόγραμμον τοῦ $HΘΚ$ τριγώνου· λέγω, ὅτι ἴσον ἐστὶ τὸ $ABΓΔΕΖ$ πρίσμα τῷ $HΘΚΛΜΝ$ πρίσματι.

Let $ABCDEF$ and $GHKLMN$ be two equal height prisms, and let the former have the parallelogram AF , and the latter the triangle GHK , as a base. And let parallelogram AF be twice triangle GHK . I say that prism $ABCDEF$ is equal to prism $GHKLMN$.

Συμπεπληρώσθω γάρ τὰ AX , HO στερεά. ἐπεὶ διπλάσιόν ἐστι τὸ AZ παραλληλόγραμμον τοῦ $HΘΚ$ τριγώνου, ἔστι δὲ καὶ τὸ $ΘΚ$ παραλληλόγραμμον διπλάσιον τοῦ $HΘΚ$ τριγώνου, ἴσον ἄρα ἐστὶ τὸ AZ παραλληλόγραμμον τῷ $ΘΚ$ παραλληλόγραμμῳ. τὰ δὲ ἐπὶ ἴσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ AX στερεὸν τῷ HO στερεῷ. καὶ ἐστὶ τοῦ μὲν AX στερεοῦ ἡμισὺν τὸ $ABΓΔΕΖ$ πρίσμα, τοῦ δὲ HO στερεοῦ ἡμισὺν τὸ $HΘΚΛΜΝ$ πρίσμα· ἴσον ἄρα ἐστὶ τὸ $ABΓΔΕΖ$ πρίσμα τῷ $HΘΚΛΜΝ$ πρίσματι.

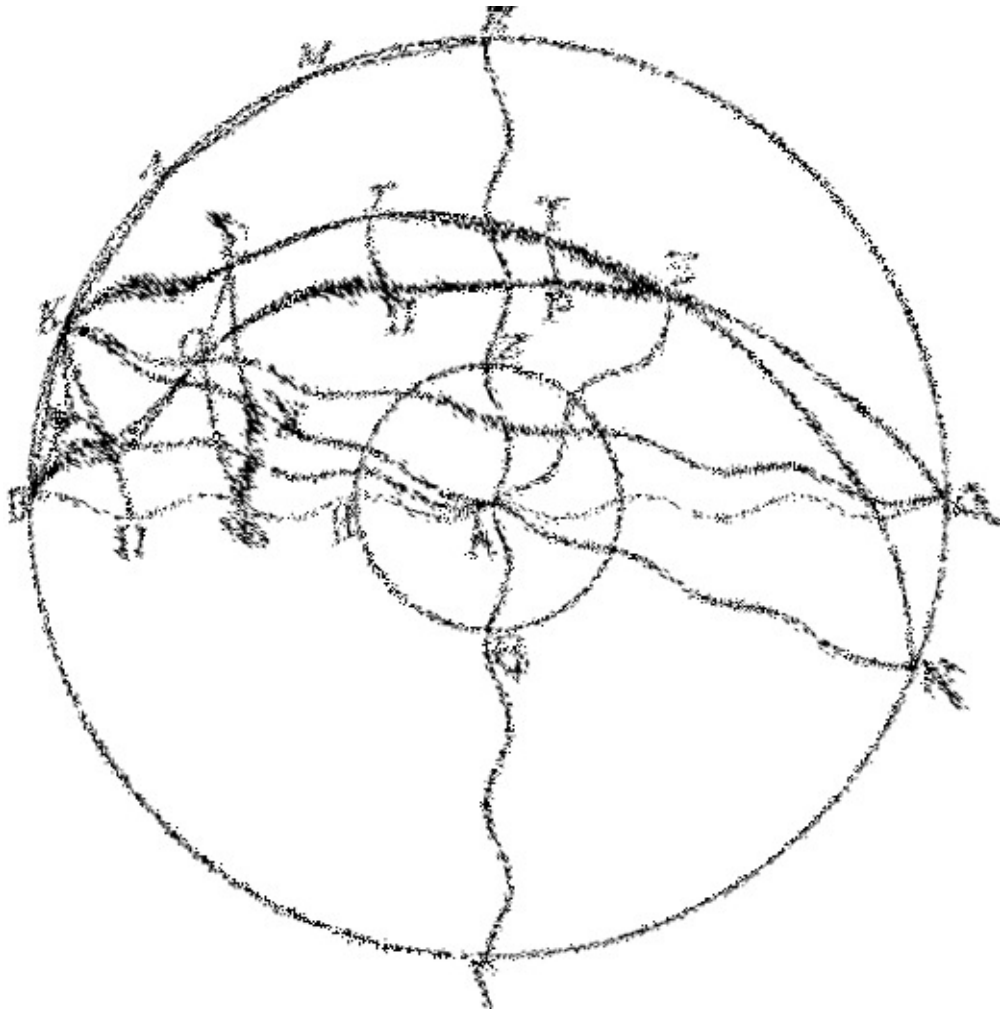
For let the solids AO and GP be completed. Since parallelogram AF is double triangle GHK , and parallelogram HK is also double triangle GHK [Prop. 1.34], parallelogram AF is thus equal to parallelogram HK . And parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another [Prop. 11.31]. Thus, solid AO is equal to solid GP . And prism $ABCDEF$ is half of solid AO , and prism $GHKLMN$ half of solid GP [Prop. 11.28]. Prism $ABCDEF$ is thus equal to prism $GHKLMN$.

Ἐάν ἄρα ἡ δύο πρίσματα ἰσοῦσῃ, καὶ τὸ μὲν ἔθῃ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ἢ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσα ἔσται τὰ πρίσματα· ὅπερ ἔδει δεῖξαι.

Thus, if there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms are equal. (Which is) the very thing it was required to show.

ELEMENTS BOOK 12

Proportional Stereometry[†]



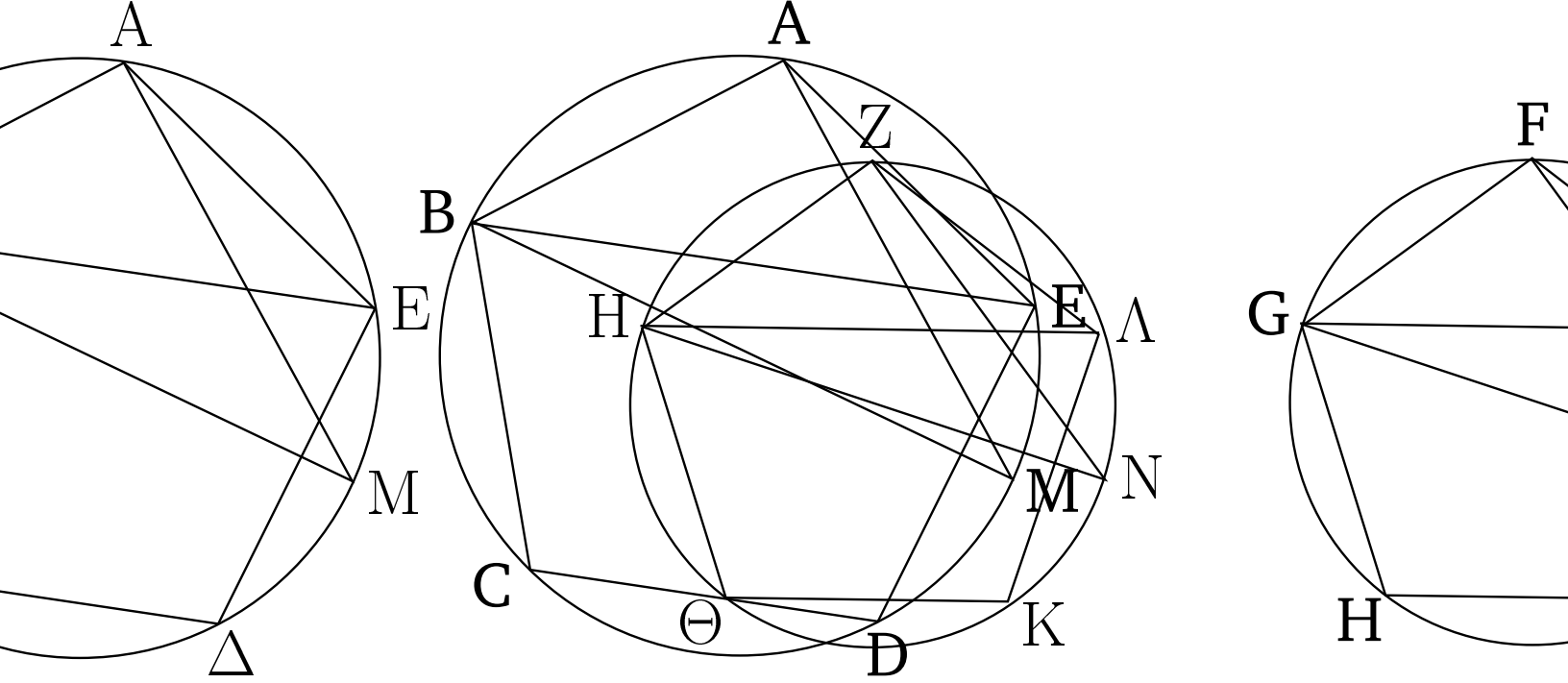
[†]The novel feature of this book is the use of the so-called method of exhaustion (see Prop. 10.1), a precursor to integration which is generally attributed to Eudoxus of Cnidus.

1.

Proposition 1

Τὰ ἐν τοῖς κύκλοις ὅμοια πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.

Similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles).



Ἐστωσαν κύκλοι οἱ $ABΓ$, $ZHΘ$, καὶ ἐν αὐτοῖς ὅμοια πολύγωνα ἔστω τὰ $ABΓΔΕ$, $ZHΘΚΛ$, διαμέτροι δὲ τῶν κύκλων ἔστωσαν BM , HN . λέγω, ὅτι ἐστὶν ὡς τὸ ἀπὸ τῆς BM τετράγωνον πρὸς τὸ ἀπὸ τῆς HN τετράγωνον, οὕτως τὸ $ABΓΔΕ$ πολύγωνον πρὸς τὸ $ZHΘΚΛ$ πολύγωνον.

Ἐπεξεύθωσαν γὰρ αἱ BE , AM , HA , ZN . καὶ ἐπεὶ ὅμοιον τὸ $ABΓΔΕ$ πολύγωνον τῷ $ZHΘΚΛ$ πολυγώνῳ, ἴση ἐστὶ καὶ ἡ ὑπὸ BAE γωνία τῇ ὑπὸ HZA , καὶ ἐστὶν ὡς ἡ BA πρὸς τὴν AE , οὕτως ἡ HZ πρὸς τὴν ZA . δύο δὲ τρίγωνα ἐστὶ τὰ BAE , HZA μίαν γωνίαν μὲν γωνία ἴσην ἔχοντα τὴν ὑπὸ BAE τῇ ὑπὸ HZA , περὶ δὲ τὰς ἴσας γωνίας τὰς πλευρὰς ἀνάλογον· ἰσογώνιον ἄρα ἐστὶ τὸ ABE τρίγωνον τῷ ZHA τριγώνῳ. ἴση ἄρα ἐστὶν ἡ ὑπὸ AEB γωνία τῇ ὑπὸ ZAH . ἀλλ' ἡ μὲν ὑπὸ AEB τῇ ὑπὸ AMB ἐστὶν ἴση· ἐπὶ γὰρ τῆς αὐτῆς περιφερείας βεβήκασιν· ἡ δὲ ὑπὸ ZAH τῇ ὑπὸ ZNH καὶ ἡ ὑπὸ AMB ἄρα τῇ ὑπὸ ZNH ἐστὶν ἴση. ἔστι δὲ καὶ ὀρθή ἡ ὑπὸ BAM ὀρθή τῇ ὑπὸ HZN ἴση· καὶ ἡ λοιπὴ ἄρα τῇ λοιπῇ ἐστὶν ἴση. ἰσογώνιον ἄρα ἐστὶ τὸ ABM τρίγωνον τῷ ZHN τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ BM πρὸς τὴν HN , οὕτως ἡ BA πρὸς τὴν HZ . ἀλλὰ τοῦ μὲν τῆς BM πρὸς τὴν HN λόγον διπλασίων ἐστὶν ὁ τοῦ ἀπὸ τῆς BM τετραγώνου πρὸς τὸ ἀπὸ τῆς HN τετράγωνον, τοῦ δὲ τῆς BA πρὸς τὴν HZ διπλασίων ἐστὶν ὁ τοῦ $ABΓΔΕ$ πολυγώνου πρὸς τὸ $ZHΘΚΛ$ πολύγωνον· καὶ ὡς ἄρα τὸ ἀπὸ τῆς BM τετράγωνον πρὸς τὸ

Let ABC and FGH be circles, and let $ABCDE$ and $FGHKL$ be similar polygons (inscribed) in them (respectively), and let BM and GN be the diameters of the circles (respectively). I say that as the square on BM is to the square on GN , so polygon $ABCDE$ (is) to polygon $FGHKL$.

For let BE , AM , GL , and FN be joined. And since polygon $ABCDE$ (is) similar to polygon $FGHKL$, angle BAE is also equal to (angle) GFL , and as BA is to AE , so GF (is) to FL [Def. 6.1]. So, BAE and GFL are two triangles having one angle equal to one angle, (namely), BAE (equal) to GFL , and the sides around the equal angles proportional. Triangle ABE is thus equiangular with triangle FGL [Prop. 6.6]. Thus, angle AEB is equal to (angle) FLG . But, AEB is equal to AMB , and FLG to FNG , for they stand on the same circumference [Prop. 3.27]. Thus, AMB is also equal to FNG . And the right-angle BAM is also equal to the right-angle GFN [Prop. 3.31]. Thus, the remaining (angle) is also equal to the remaining (angle) [Prop. 1.32]. Thus, triangle ABM is equiangular with triangle FGN . Thus, proportionally, as BM is to GN , so BA (is) to GF [Prop. 6.4]. But, the (ratio) of the square on BM to the square on GN is the square of the ratio of BM to GN , and the (ratio) of polygon $ABCDE$ to polygon $FGHKL$ is the square of the (ratio) of BA to GF [Prop. 6.20]. And, thus, as the square on BM (is) to the square on GN , so polygon $ABCDE$

ἀπὸ τῆς HN τετραγώνου, οὕτως τὸ $ABΓΔΕ$ πολύγωνον πρὸς τὸ $ZHΘΚΛ$ πολύγωνον.

Τὰ ἄρα ἐν τοῖς κύκλοις ὅμοια πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα· ὅπερ εἶδει δεῖχαι.

2.

Οἱ κύκλοι πρὸς ἀλλήλους εἶσιν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.

Ἐστωσαν κύκλοι οἱ $ABΓΔ$, $EZHΘ$, διάμετροι δὲ αὐτῶν $[ἐστῶσαν]$ αἱ $BΔ$, $ZΘ$. λέγω, ὅτι ἐστὶν ὡς ὁ $ABΓΔ$ κύκλος πρὸς τὸν $EZHΘ$ κύκλον, οὕτως τὸ ἀπὸ τῆς $BΔ$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $ZΘ$ τετράγωνον.

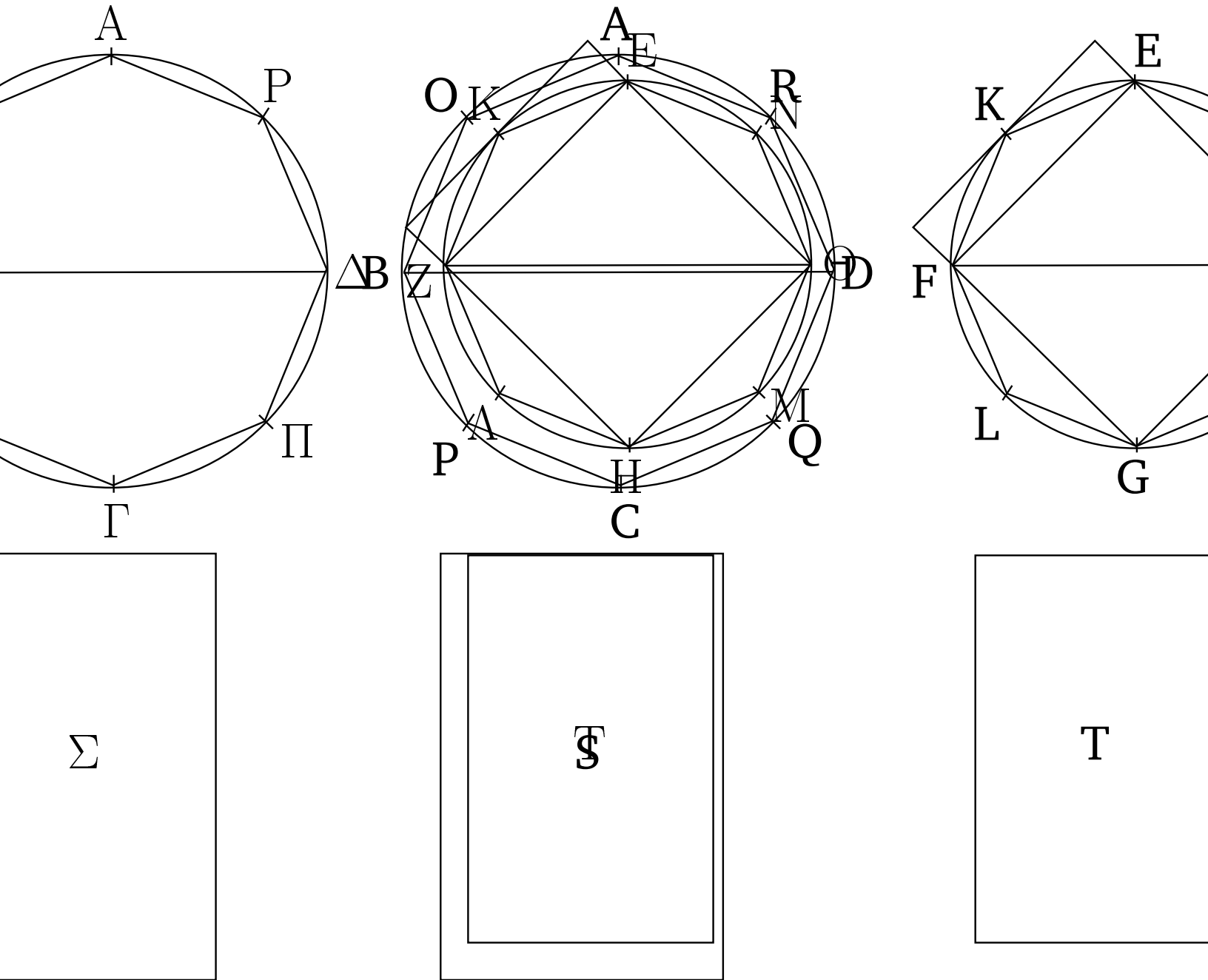
(is) to polygon $FGHKL$.

Thus, similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles). (Which is) the very thing it was required to show.

Proposition 2

Circles are to one another as the squares on (their) diameters.

Let $ABCD$ and $EFGH$ be circles, and [let] BD and FH [be] their diameters. I say that as circle $ABCD$ is to circle $EFGH$, so the square on BD (is) to the square on FH .



Εἰ γὰρ μὴ ἔστιν ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$, οὕτως τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, ἔσται ὡς τὸ ἀπὸ τῆς $B\Delta$ πρὸς τὸ ἀπὸ τῆς $Z\Theta$, οὕτως ὁ $AB\Gamma\Delta$ κύκλος ἦτοι πρὸς ἑλασσόν τι τοῦ $EZH\Theta$ κύκλου θωρίον ἢ πρὸς μείζον. ἔστω πρότερον πρὸς ἑλασσόν τὸ Σ . καὶ ἐγγεγράφῃω εἰς τὸν $EZH\Theta$ κύκλον τετράγωνον τὸ $EZH\Theta$. τὸ δὴ ἐγγεγραμμένον τετράγωνον μείζον ἔστιν ἢ τὸ ἥμισυ τοῦ $EZH\Theta$ κύκλου, ἐπειδήπερ ἔαν διὰ τῶν E, Z, H, Θ σημείων ἐφαπτομένας [εὐθείας] τοῦ κύκλου ἀγάγωμεν, τοῦ περιγρα-

For if the circle $ABCD$ is not to the (circle) $EFGH$, as the square on BD (is) to the (square) on FH , then as the (square) on BD (is) to the (square) on FH , so circle $ABCD$ will be to some area either less than, or greater than, circle $EFGH$. Let it, first of all, be (in that ratio) to (some) lesser (area), S . And let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. So the inscribed square is greater than half of circle $EFGH$, inasmuch as if we draw tangents to the circle through the points E, F, G , and H , then square $EFGH$ is half

φομένον περι τὸν κύκλον τετραγώνον ἥμισυ ἐστὶ τὸ $EZH\Theta$ τετράγωνον, τοῦ δὲ περιγραφέντος τετραγώνου ἐλάττων ἐστὶν ὁ κύκλος· ὥστε τὸ $EZH\Theta$ ἐγγεγραμμένον μὲν τετράγωνον μεῖζόν ἐστι τοῦ ἡμίσεως τοῦ $EZH\Theta$ κύκλου. τεμήσθωσαν δὶθὰ αἱ EZ , ZH , $H\Theta$, ΘE περιφέρειαι κατὰ τὰ K , Λ , M , N σημεία, καὶ ἐπεξεύθωσαν αἱ EK , KZ , $Z\Lambda$, ΛH , HM , $M\Theta$, ΘN , NE · καὶ ἕκαστον ἄρα τῶν EKZ , $Z\Lambda H$, $HM\Theta$, ΘNE τριγώνων μεῖζόν ἐστιν ἢ τὸ ἥμισυ τοῦ καθ' ἑαυτὸ τμήματος τοῦ κύκλου, ἐπειδήπερ ἔαν διὰ τῶν K , Λ , M , N σημείων ἐφαπτομένας τοῦ κύκλου ἀγάγωμεν καὶ ἀναπληρώσωμεν τὰ ἐπὶ τῶν EZ , ZH , $H\Theta$, ΘE εὐθειῶν παραλληλόγραμμα, ἕκαστον τῶν EKZ , $Z\Lambda H$, $HM\Theta$, ΘNE τριγώνων ἥμισυ ἔσται τοῦ καθ' ἑαυτὸ παραλληλογράμμου, ἀλλὰ τὸ καθ' ἑαυτὸ τμήμα ἐλάττον ἐστὶ τοῦ παραλληλογράμμου· ὥστε ἕκαστον τῶν EKZ , $Z\Lambda H$, $HM\Theta$, ΘNE τριγώνων μεῖζόν ἐστι τοῦ ἡμίσεως τοῦ καθ' ἑαυτὸ τμήματος τοῦ κύκλου. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας δὶθὰ καὶ ἐπιξυγγνόντες εὐθείας καὶ τοῦτο ἀεὶ ποιῶντες καταλείβομεν τινα ἀποτμήματα τοῦ κύκλου, ἃ ἔσται ἐλάσσονα τῆς ὑπεροθῆς, ἢ ὑπερέθει ὁ $EZH\Theta$ κύκλος τοῦ Σ θωρίον. ἐδείχθη γάρ ἐν τῷ πρώτῳ θεωρήματι τοῦ δεκάτου βιβλίου, ὅτι δύο μεγεθῶν ἀνίσων ἐκκειμένων, ἔαν ἀπὸ τοῦ μεῖζονος ἀφαιρεθῇ μεῖζον ἢ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μεῖζον ἢ τὸ ἥμισυ, καὶ τοῦτο ἀεὶ γίγνηται, λειφθήσεται τι μέγεθος, ὃ ἔσται ἐλάσσον τοῦ ἐκκειμένου ἐλάσσονος μεγέθους. λελείφθω οὖν, καὶ ἔστω τὰ ἐπὶ τῶν EK , KZ , $Z\Lambda$, ΛH , HM , $M\Theta$, ΘN , NE τμήματα τοῦ $EZH\Theta$ κύκλου ἐλάττονα τῆς ὑπεροθῆς, ἢ ὑπερέθει ὁ $EZH\Theta$ κύκλος τοῦ Σ θωρίον. λοιπὸν ἄρα τὸ $EKZ\Lambda HM\Theta N$ πολύγωνον μεῖζόν ἐστι τοῦ Σ θωρίον. ἐγγεγράφθω καὶ εἰς τὸν $AB\Gamma\Delta$ κύκλον τῷ $EKZ\Lambda HM\Theta N$ πολυγώνῳ ὅμοιον πολύγωνον τὸ $AXBO\Gamma\Pi\Delta P$ · ἔστιν ἄρα ὡς τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$ τετράγωνον, οὕτως τὸ $AXBO\Gamma\Pi\Delta P$ πολύγωνον πρὸς τὸ $EKZ\Lambda HM\Theta N$ πολύγωνον. ἀλλὰ καὶ ὡς τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, οὕτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ Σ θωρίον· καὶ ὡς ἄρα ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ Σ θωρίον, οὕτως τὸ $AXBO\Gamma\Pi\Delta P$ πολύγωνον πρὸς τὸ $EKZ\Lambda HM\Theta N$ πολύγωνον· ἐναλλάξ ἄρα ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ ἐν αὐτῷ πολύγωνον, οὕτως τὸ Σ θωρίον πρὸς τὸ $EKZ\Lambda HM\Theta N$ πολύγωνον. μεῖζον δὲ ὁ $AB\Gamma\Delta$ κύκλος τοῦ ἐν αὐτῷ πολυγώνου· μεῖζον ἄρα καὶ τὸ Σ θωρίον τοῦ $EKZ\Lambda HM\Theta N$ πολυγώνου. ἀλλὰ καὶ ἐλάττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, οὕτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς ἑλασσόν τι τοῦ $EZH\Theta$ κύκλου θωρίον. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὡς τὸ ἀπὸ $Z\Theta$ πρὸς τὸ ἀπὸ $B\Delta$, οὕτως ὁ $EZH\Theta$ κύκλος πρὸς ἑλασσόν τι τοῦ $AB\Gamma\Delta$ κύκλου θωρίον.

Λέγω δὴ, ὅτι οὐδὲ ὡς τὸ ἀπὸ τῆς $B\Delta$ πρὸς τὸ ἀπὸ τῆς $Z\Theta$, οὕτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς μεῖζόν τι τοῦ $EZH\Theta$ κύκλου θωρίον.

Εἰ γὰρ δυνατόν, ἔστω πρὸς μεῖζον τὸ Σ . ἀνάπαλιν ἄρα [ἐστίν] ὡς τὸ ἀπὸ τῆς $Z\Theta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $B\Delta$, οὕτως τὸ Σ θωρίον πρὸς τὸν $AB\Gamma\Delta$ κύκλον. ἀλλ' ὡς τὸ

of the square circumscribed about the circle [Prop. 1.47], and the circle is less than the circumscribed square. Hence, the inscribed square $EFGH$ is greater than half of circle $EFGH$. Let the circumferences EF , FG , GH , and HE be cut in half at points K , L , M , and N (respectively), and let EK , KF , FL , LG , GM , MH , HN , and NE be joined. And, thus, each of the triangles EKF , FLG , GMH , and HNE is greater than half of the segment of the circle about it, inasmuch as if we draw tangents to the circle through points K , L , M , and N , and complete the parallelograms on the straight-lines EF , FG , GH , and HE , then each of the triangles EKF , FLG , GMH , and HNE will be half of the parallelogram about it, but the segment about it is less than the parallelogram. Hence, each of the triangles EKF , FLG , GMH , and HNE is greater than half of the segment of the circle about it. So, by cutting the circumferences remaining behind in half, and joining straight-lines, and doing this continually, we will (eventually) leave behind some segments of the circle whose (sum) will be less than the excess by which circle $EFGH$ exceeds the area S . For we showed in the first theorem of the tenth book that if two unequal magnitudes are laid out, and if (a part) greater than a half is subtracted from the greater, and (if from) the remainder (a part) greater than a half (is subtracted), and this happens continually, then some magnitude will (eventually) be left which will be less than the lesser laid out magnitude [Prop. 10.1]. Therefore, let the (segments) be left, and let the (sum of the) segments of the circle $EFGH$ on EK , KF , FL , LG , GM , MH , HN , and NE be less than the excess by which circle $EFGH$ exceeds area S . Thus, the remaining polygon $EKFLGMHN$ is greater than area S . And let the polygon $AOBPCQDR$, similar to the polygon $EKFLGMHN$, be inscribed in circle $ABCD$. Thus, as the square on BD is to the square on FH , so polygon $AOBPCQDR$ (is) to polygon $EKFLGMHN$ [Prop. 12.1]. But, also, as the square on BD (is) to the square on FH , so circle $ABCD$ (is) to area S . And, thus, as circle $ABCD$ (is) to area S , so polygon $AOBPGQDR$ (is) to polygon $EKFLGMHN$ [Prop. 5.11]. Thus, alternately, as circle $ABCD$ (is) to the polygon (inscribed) within it, so area S (is) to polygon $EKFLGMHN$ [Prop. 5.16]. And circle $ABCD$ (is) greater than the polygon (inscribed) within it. Thus, area S is also greater than polygon $EKFLGMHN$. But, (it is) also less. The very thing is impossible. Thus, the square on BD is not to the (square) on FH , as circle $ABCD$ (is) to some area less than circle $EFGH$. So, similarly, we can show that the (square) on FH (is) not to the (square) on BD as circle $EFGH$ (is) to some area less than circle $ABCD$ either.

So, I say that neither (is) the (square) on BD to the (square) on FH , as circle $ABCD$ (is) to some area greater than circle $EFGH$.

For, if possible, let it be (in that ratio) to (some) greater (area), S . Thus, inversely, as the square on FH [is] to the (square) on DB , so area S (is) to circle $ABCD$ [Prop. 5.7 corr.].

Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ ἑλαττόν τι τοῦ $ABΓΔ$ κύκλου θωρίον· καὶ ὡς ἄρα τὸ ἀπὸ τῆς $ZΘ$ πρόξ τὸ ἀπὸ τῆς $BΔ$, οὕτως ὁ $EZHΘ$ κύκλος πρόξ ἑλασσόν τι τοῦ $ABΓΔ$ κύκλου θωρίον· ὅπερ ἀδύνατον ἐδείχθη. οὐκ ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς $BΔ$ τετράγωνον πρόξ τὸ ἀπὸ τῆς $ZΘ$, οὕτως ὁ $ABΓΔ$ κύκλος πρόξ μείζον τι τοῦ $EZHΘ$ κύκλου θωρίον. ἐδείχθη δέ, ὅτι οὐδὲ πρόξ ἑλασσον· ἔστιν ἄρα ὡς τὸ ἀπὸ τῆς $BΔ$ τετράγωνον πρόξ τὸ ἀπὸ τῆς $ZΘ$, οὕτως ὁ $ABΓΔ$ κύκλος πρόξ τὸν $EZHΘ$ κύκλον.

Οἱ ἄρα κύκλοι πρόξ ἀλλήλους εἰσὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα· ὅπερ ἔδει δεῖχαι.

Λήμμα.

Λέγω δὴ, ὅτι τοῦ Σ θωρίον μείζονος ὄντος τοῦ $EZHΘ$ κύκλου ἐστὶν ὡς τὸ Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ ἑλαττόν τι τοῦ $ABΓΔ$ κύκλου θωρίον.

Γεγονέτω γάρ ὡς τὸ Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ τὸ T θωρίον. λέγω, ὅτι ἑλαττόν ἐστι τὸ T θωρίον τοῦ $ABΓΔ$ κύκλου. ἐπεὶ γάρ ἐστιν ὡς τὸ Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ τὸ T θωρίον, ἐναλλάξ ἐστὶν ὡς τὸ Σ θωρίον πρόξ τὸν $EZHΘ$ κύκλον, οὕτως ὁ $ABΓΔ$ κύκλος πρόξ τὸ T θωρίον. μείζον δὲ τὸ Σ θωρίον τοῦ $EZHΘ$ κύκλου· μείζων ἄρα καὶ ὁ $ABΓΔ$ κύκλος τοῦ T θωρίον. ὥστε ἐστὶν ὡς τὸ Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ ἑλαττόν τι τοῦ $ABΓΔ$ κύκλου θωρίον· ὅπερ ἔδει δεῖχαι.

3.

Πᾶσα πυραμὶς τριγώνων ἔθουσα βάσιν διαιρεῖται εἰς δύο πυραμίδας ἴσας τε καὶ ὁμοίας ἀλλήλαις καὶ [ὁμοίας] τῇ ὅλῃ τριγώνονος ἔθουσαξ βάσεις καὶ εἰς δύο πρίσματα ἴσα· καὶ τὰ δύο πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος.

But, as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$ (see lemma). And, thus, as the (square) on FH (is) to the (square) on BD , so circle $EFGH$ (is) to some area less than circle $ABCD$ [Prop. 5.11]. The very thing was shown (to be) impossible. Thus, as the square on BD is to the (square) on FH , so circle $ABCD$ (is) not to some area greater than circle $EFGH$. And it was shown that neither (is it in that ratio) to (some) lesser (area). Thus, as the square on BD is to the (square) on FH , so circle $ABCD$ (is) to circle $EFGH$.

Thus, circles are to one another as the squares on (their) diameters. (Which is) the very thing it was required to show.

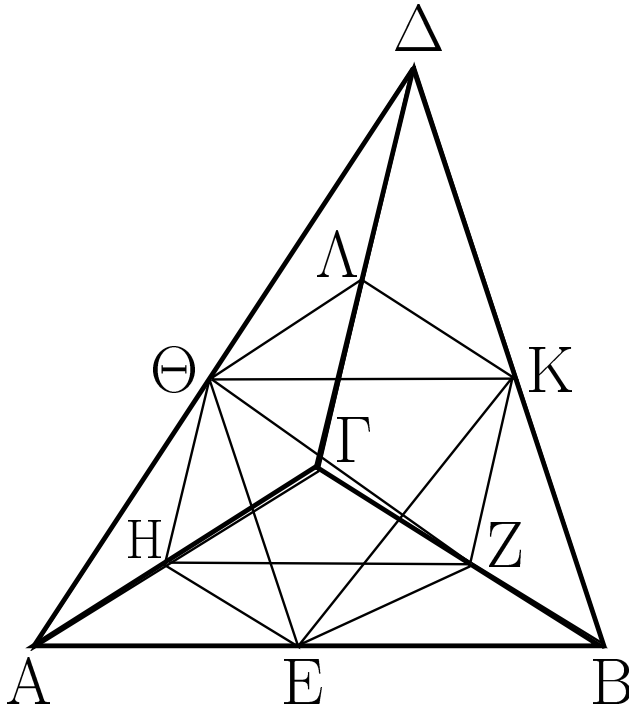
Lemma

So, I say that, area S being greater than circle $EFGH$, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$.

For let it be contrived that as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to area T . I say that area T is less than circle $ABCD$. For since as area S is to circle $ABCD$, so circle $EFGH$ (is) to area T , alternately, as area S is to circle $EFGH$, so circle $ABCD$ (is) to area T [Prop. 5.16]. And area S (is) greater than circle $EFGH$. Thus, circle $ABCD$ (is) also greater than area T [Prop. 5.14]. Hence, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$. (Which is) the very thing it was required to show.

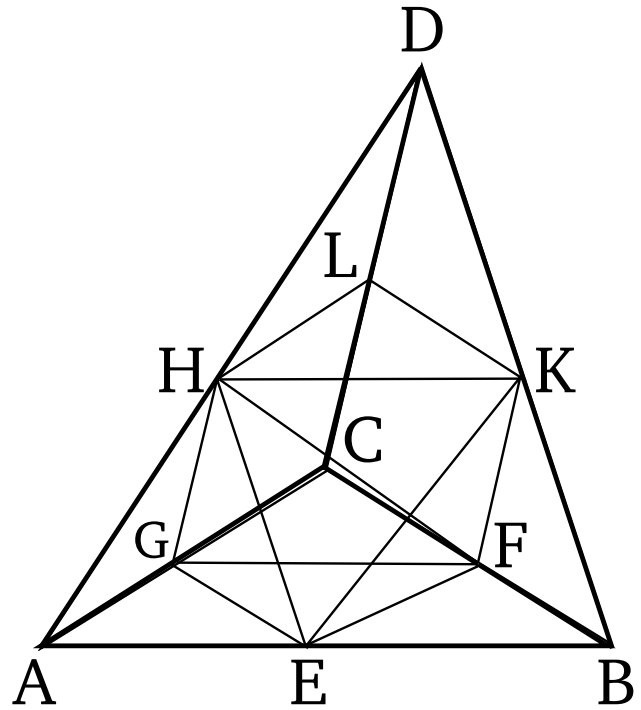
Proposition 3

Any pyramid having a triangular base is divided into two pyramids having triangular bases (which are) equal, similar to one another, and [similar] to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.



Ἐστω πυραμὶς, ἧς βάσις μὲν ἐστὶ τὸ $AB\Gamma$ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον· λέγω, ὅτι ἡ $AB\Gamma\Delta$ πυραμὶς διαιρεῖται εἰς δύο πυραμίδας ἴσας ἀλλήλαις τριγώνου βάσεις ἐθούσας καὶ ὁμοίας τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα· καὶ τὰ δύο πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος.

Τετμήσθωσαν γὰρ αἱ AB , $B\Gamma$, ΓA , AD , ΔB , $\Delta \Gamma$ διὰ κατὰ τὰ E , Z , H , Θ , K , Λ σημεία, καὶ ἐπεζεύθωσαν αἱ ΘE , $E H$, $H \Theta$, ΘK , $K \Lambda$, $\Lambda \Theta$, $K Z$, $Z H$. Ἐπεὶ ἴση ἐστὶν ἡ μὲν $A E$ τῇ $E B$, ἡ δὲ $A \Theta$ τῇ $\Delta \Theta$, παράλληλος ἄρα ἐστὶν ἡ $E \Theta$ τῇ ΔB . διὰ τὰ αὐτὰ δὴ καὶ ἡ ΘK τῇ $A B$ παράλληλός ἐστιν. παραλληλόγραμμον ἄρα ἐστὶ τὸ $\Theta E B K$. ἴση ἄρα ἐστὶν ἡ ΘK τῇ $E B$. ἀλλὰ ἡ $E B$ τῇ $E A$ ἐστὶν ἴση· καὶ ἡ $A E$ ἄρα τῇ ΘK ἐστὶν ἴση. ἔστι δὲ καὶ ἡ $A \Theta$ τῇ $\Theta \Delta$ ἴση· δύο δὴ αἱ $E A$, $A \Theta$ δυοὶ ταῖς $K \Theta$, $\Theta \Delta$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ $E A \Theta$ γωνία τῇ ὑπὸ $K \Theta \Delta$ ἴση· βάσις ἄρα ἡ $E \Theta$ βάσει τῇ $K \Delta$ ἐστὶν ἴση. ἴσον ἄρα καὶ ὁμοῖόν ἐστι τὸ $A E \Theta$ τρίγωνον τῷ $\Theta K \Delta$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $A \Theta H$ τρίγωνον τῷ $\Theta \Lambda \Delta$ τριγώνῳ ἴσον τε καὶ ὁμοῖον. καὶ ἐπεὶ δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ $E \Theta$, ΘH παρὰ δύο εὐθείας ἀπτόμενας ἀλλήλων τὰς $K \Delta$, $\Delta \Lambda$ εἰσὶν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ οὔσαι, ἴσας γωνίας περιέχουσιν. ἴση ἄρα ἐστὶν ἡ ὑπὸ $E \Theta H$ γωνία τῇ ὑπὸ $K \Delta \Lambda$ γωνίᾳ. καὶ ἐπεὶ δύο εὐθεῖαι αἱ $E \Theta$, ΘH δυοὶ ταῖς $K \Delta$, $\Delta \Lambda$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ γωνία ἡ ὑπὸ $E \Theta H$ γωνία τῇ ὑπὸ $K \Delta \Lambda$ ἐστὶν ἴση, βάσις ἄρα ἡ $E H$ βάσει τῇ $K \Lambda$ [ἐστὶν] ἴση· ἴσον ἄρα καὶ ὁμοῖόν ἐστι τὸ $E \Theta H$ τρίγωνον τῷ $K \Delta \Lambda$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $A E H$ τρίγωνον τῷ $\Theta K \Lambda$ τριγώνῳ ἴσον τε καὶ ὁμοῖόν ἐστιν. ἡ ἄρα πυραμὶς, ἧς βάσις μὲν ἐστὶ τὸ $A E H$ τρίγωνον, κορυφή δὲ τὸ Θ σημείον, ἴση καὶ ὁμοία ἐστὶ πυραμίδι, ἧς



Let there be a pyramid whose base is triangle ABC , and (whose) apex (is) point D . I say that pyramid $ABCD$ is divided into two pyramids having triangular bases (which are) equal to one another, and similar to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.

For let AB , BC , CA , AD , DB , and DC be cut in half at points E , F , G , H , K , and L (respectively). And let HE , EG , GH , HK , KL , LH , KF , and FG be joined. Since AE is equal to EB , and AH to HD , EH is thus parallel to DB [Prop. 6.2]. So, for the same (reasons), HK is also parallel to AB . Thus, $HEBK$ is a parallelogram. Thus, HK is equal to EB [Prop. 1.34]. But, EB is equal to EA . Thus, AE is also equal to HK . And AH is also equal to HD . So the two (straight-lines) EA and AH are equal to the two (straight-lines) KH and HD , respectively. And angle EAH (is) equal to angle KHD [Prop. 1.29]. Thus, base EH is equal to base KD [Prop. 1.4]. Thus, triangle AEH is equal and similar to triangle HKD [Prop. 1.4]. So, for the same (reasons), triangle AHG is also equal and similar to triangle HLD . And since EH and HG are two straight-lines joining one another (which are respectively) parallel to two straight-lines joining one another, KD and DL , not being in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle EHG is equal to angle KDL . And since the two straight-lines EH and HG are equal to the two straight-lines KD and DL , respectively, and angle EHG is equal to angle KDL , base EG [is] thus equal to base KL [Prop. 1.4]. Thus, triangle EHG is equal and similar to triangle KDL . So, for the same (reasons), triangle AEG

βάσις μὲν ἐστὶ τὸ $\Theta\text{ΚΛ}$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. καὶ ἐπεὶ τρίγωνον τοῦ ΑΔΒ παρὰ μίαν τῶν πλευρῶν τὴν ΑΒ ἦκται ἡ $\Theta\text{Κ}$, ἰσογώνιον ἐστὶ τὸ ΑΔΒ τρίγωνον τῷ $\Delta\text{ΘΚ}$ τριγώνῳ, καὶ τὰς πλευράς ἀνάλογον ἔθουσιν· ὅμοιον ἄρα ἐστὶ τὸ ΑΔΒ τρίγωνον τῷ $\Delta\text{ΘΚ}$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν $\Delta\text{ΒΓ}$ τρίγωνον τῷ $\Delta\text{ΚΛ}$ τριγώνῳ ὅμοιον ἐστίν, τὸ δὲ ΑΔΓ τῷ $\Delta\text{ΛΘ}$. καὶ ἐπεὶ δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ ΒΑ , ΑΓ παρὰ δύο εὐθείας ἀπτόμενάς ἀλλήλων τὰς ΚΘ , ΘΛ εἰσιν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσας γωνίας περιέχουσιν. ἴση ἄρα ἐστὶν ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΚΘΛ . καὶ ἐστὶν ὥς ἡ ΒΑ πρὸς τὴν ΑΓ , οὕτως ἡ ΚΘ πρὸς τὴν ΘΛ . ὅμοιον ἄρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ $\Theta\text{ΚΛ}$ τριγώνῳ. καὶ πυραμὶς ἄρα, ἥς βάσις μὲν ἐστὶ τὸ ΑΒΓ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ὁμοία ἐστὶ πυραμίδι, ἥς βάσις μὲν ἐστὶ τὸ $\Theta\text{ΚΛ}$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. ἀλλὰ πυραμὶς, ἥς βάσις μὲν [ἐστὶ] τὸ $\Theta\text{ΚΛ}$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ὁμοία ἐδείχθη πυραμίδι, ἥς βάσις μὲν ἐστὶ τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημείον. ἑκατέρα ἄρα τῶν ΑΕΗΘ , $\Theta\text{ΚΛΔ}$ πυραμίδων ὁμοία ἐστὶ τῇ ὅλῃ τῇ ΑΒΓΔ πυραμίδι.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ ΒΖ τῇ ΖΓ , διπλάσιον ἐστὶ τὸ ΕΒΖΗ παραλληλόγραμμον τοῦ ΗΖΓ τριγώνου. καὶ ἐπεὶ, ἔαν ἦ δύο πρίσματα ἰσοῦψῃ, καὶ τὸ μὲν ἔθῃ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ἦ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσα ἐστὶ τὰ πρίσματα, ἴσον ἄρα ἐστὶ τὸ πρίσμα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΒΚΖ , ΕΘΗ , τριῶν δὲ παραλληλογράμμων τῶν ΕΒΖΗ , ΕΒΚΘ , $\Theta\text{ΚΖΗ}$ τῷ πρισματι τῷ περιεχομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΗΖΓ , $\Theta\text{ΚΛ}$, τριῶν δὲ παραλληλογράμμων τῶν ΚΖΓΛ , ΛΓΗΘ , $\Theta\text{ΚΖΗ}$. καὶ φανερόν, ὅτι ἐκάτρων τῶν πρισμάτων, οὗ τε βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $\Theta\text{Κ}$ εὐθεῖα, καὶ οὗ βάσις τὸ ΗΖΓ τρίγωνον, ἀπεναντίον δὲ τὸ $\Theta\text{ΚΛ}$ τρίγωνον, μεῖζον ἐστὶν ἑκατέρως τῶν πυραμίδων, ὧν βάσεις μὲν τὰ ΑΕΗ , $\Theta\text{ΚΛ}$ τρίγωνα, κορυφαί, δὲ τὰ Θ , Δ σημεία, ἐπειδὴ περ [καὶ] ἔαν ἐπιτείνωμεν τὰς ΕΖ , ΕΚ εὐθείας, τὸ μὲν πρίσμα, οὗ βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $\Theta\text{Κ}$ εὐθεῖα, μεῖζον ἐστὶ τῆς πυραμίδος, ἥς βάσις τὸ ΕΒΖ τρίγωνον, κορυφή δὲ τὸ Κ σημείον. ἀλλ' ἡ πυραμὶς, ἥς βάσις τὸ ΕΒΖ τρίγωνον, κορυφή δὲ τὸ Κ σημείον, ἴση ἐστὶ πυραμίδι, ἥς βάσις τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημείον· ὑπὸ γὰρ ἴσων καὶ ὁμοίων ἐπιπέδων περιέχονται. ὥστε καὶ τὸ πρίσμα, οὗ βάσις μὲν τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $\Theta\text{Κ}$ εὐθεῖα, μεῖζον ἐστὶ πυραμίδος, ἥς βάσις μὲν τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημείον. ἴσον δὲ τὸ μὲν πρίσμα, οὗ βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $\Theta\text{Κ}$ εὐθεῖα, τῷ πρισματι, οὗ βάσις μὲν τὸ ΗΖΓ τρίγωνον, ἀπεναντίον δὲ τὸ $\Theta\text{ΚΛ}$ τρίγωνον· ἡ δὲ πυραμὶς, ἥς βάσις τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημείον, ἴση ἐστὶ πυραμίδι, ἥς βάσις τὸ $\Theta\text{ΚΛ}$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. τὰ ἄρα εἰρημένα δύο πρίσματα μεῖζονά ἐστὶ τῶν εἰρημένων δύο πυραμίδων, ὧν βάσεις μὲν τὰ ΑΕΗ , $\Theta\text{ΚΛ}$ τρίγωνα, κορυφαί δὲ τὰ Θ , Δ σημεία.

is also equal and similar to triangle ΗΚΛ . Thus, the pyramid whose base is triangle ΑΕΓ , and apex the point Η , is equal and similar to the pyramid whose base is triangle ΗΚΛ , and apex the point Δ [Def. 11.10]. And since ΗΚ has been drawn parallel to one of the sides, ΑΒ , of triangle ΑΔΒ , triangle ΑΔΒ is equiangular to triangle ΔΗΚ [Prop. 1.29], and they have proportional sides. Thus, triangle ΑΔΒ is similar to triangle ΔΗΚ [Def. 6.1]. So, for the same (reasons), triangle ΔΒC is also similar to triangle ΔΚΛ , and ΑΔC to ΔΛΗ . And since two straight-lines joining one another, ΒΑ and ΑC , are parallel to two straight-lines joining one another, ΚΗ and ΗΛ , not in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle ΒΑC is equal to (angle) ΚΗΛ . And as ΒΑ is to ΑC , so ΚΗ (is) to ΗΛ . Thus, triangle ΑΒC is similar to triangle ΗΚΛ [Prop. 6.6]. And, thus, the pyramid whose base is triangle ΑΒC , and apex the point Δ , is similar to the pyramid whose base is triangle ΗΚΛ , and apex the point Δ [Def. 11.9]. But, the pyramid whose base [is] triangle ΗΚΛ , and apex the point Δ , was shown (to be) similar to the pyramid whose base is triangle ΑΕΓ , and apex the point Η . Thus, each of the pyramids ΑΕΓΗ and ΗΚΛΔ is similar to the whole pyramid ΑΒCΔ .

And since ΒF is equal to FC , parallelogram ΕΒFG is double triangle GFC [Prop. 1.41]. And since, if two prisms (have) equal heights, and the former has a parallelogram as a base, and the latter a triangle, and the parallelogram (is) double the triangle, then the prisms are equal [Prop. 11.39], the prism contained by the two triangles ΒΚF and ΕΗG , and the three parallelograms ΕΒFG , ΕΒΚΗ , and ΗΚFG , is thus equal to the prism contained by the two triangles GFC and ΗΚΛ , and the three parallelograms KFCL , LCGH , and ΗΚFG . And (it is) clear that each of the prisms whose base (is) parallelogram ΕΒFG , and opposite (side) straight-line ΗΚ , and whose base (is) triangle GFC , and opposite (plane) triangle ΗΚΛ , is greater than each of the pyramids whose bases are triangles ΑΕΓ and ΗΚΛ , and apexes the points Η and Δ (respectively), inasmuch as, if we [also] join the straight-lines ΕF and ΕΚ then the prism whose base (is) parallelogram ΕΒFG , and opposite (side) straight-line ΗΚ , is greater than the pyramid whose base (is) triangle ΕΒF , and apex the point Κ . But the pyramid whose base (is) triangle ΕΒF , and apex the point Κ , is equal to the pyramid whose base is triangle ΑΕΓ , and apex point Η . For they are contained by equal and similar planes. And, hence, the prism whose base (is) parallelogram ΕΒFG , and opposite (side) straight-line ΗΚ , is greater than the pyramid whose base (is) triangle ΑΕΓ , and apex the point Η . And the prism whose base is parallelogram ΕΒFG , and opposite (side) straight-line ΗΚ , (is) equal to the prism whose base (is) triangle GFC , and opposite (plane) triangle ΗΚΛ . And the pyramid whose base (is) triangle ΑΕΓ , and apex the point Η , is equal to the pyramid whose base (is) triangle ΗΚΛ , and apex the point Δ . Thus, the (sum of the) aforementioned two prisms

Ἡ ἄρα ὅλη πυραμὶς, ἣς βάσις τὸ $ABΓ$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, διήρηται εἰς τε δύο πυραμίδαξ ἴσαξ ἀλλήλαις [καὶ ὁμοίαξ τῇ ὅλῃ] καὶ εἰς δύο πρίσματα ἴσα, καὶ τὰ δύο πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος· ὅπερ ἔδει δεῖχαι.

4.

Ἐάν ᾧσι δύο πυραμίδες ὑπὸ τὸ αὐτὸ ὕψος τριγώνουξ ἔθουσαι βάσεις, διαιρεθῇ δὲ ἑκάτερα αὐτῶν εἰς τε δύο πυραμίδαξ ἴσαξ ἀλλήλαις καὶ ὁμοίαξ τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα, ἔσται ὥς ἡ τῆς μιᾶς πυραμίδος βάσις πρὸς τὴν τῆς ἐτέρας πυραμίδος βάσιν, οὕτως τὰ ἐν τῇ μιᾷ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ ἐτέρᾳ πυραμίδι πρίσματα πάντα ἰσοπληθῇ.

Ἔστωσαν δύο πυραμίδες ὑπὸ τὸ αὐτὸ ὕψος τριγώνουξ ἔθουσαι βάσεις τὰς $ABΓ$, ΔEZ , κορυφὰς δὲ τὰ H , Θ σημεία, καὶ διηρησθῶ ἑκάτερα αὐτῶν εἰς τε δύο πυραμίδαξ ἴσαξ ἀλλήλαις καὶ ὁμοίαξ τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα· λέγω, ὅτι ἐστὶν ὥς ἡ $ABΓ$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὰ ἐν τῇ $ABΓH$ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα ἰσοπληθῇ.

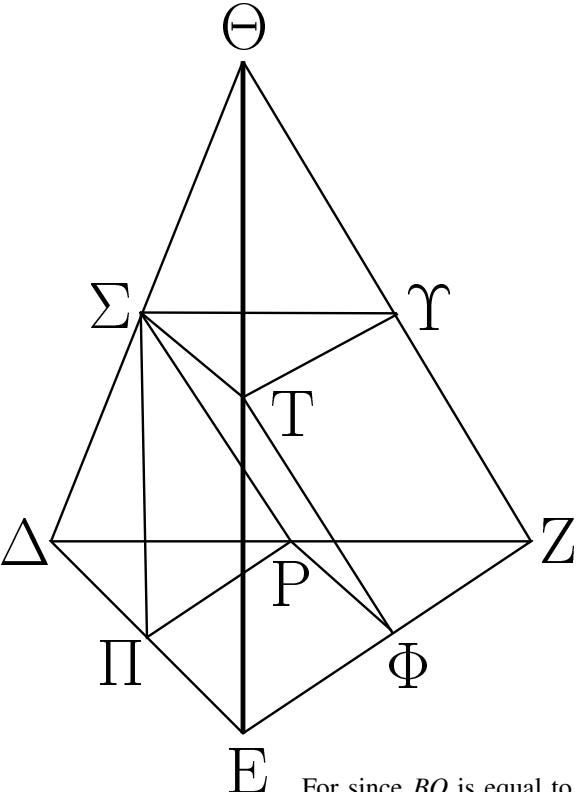
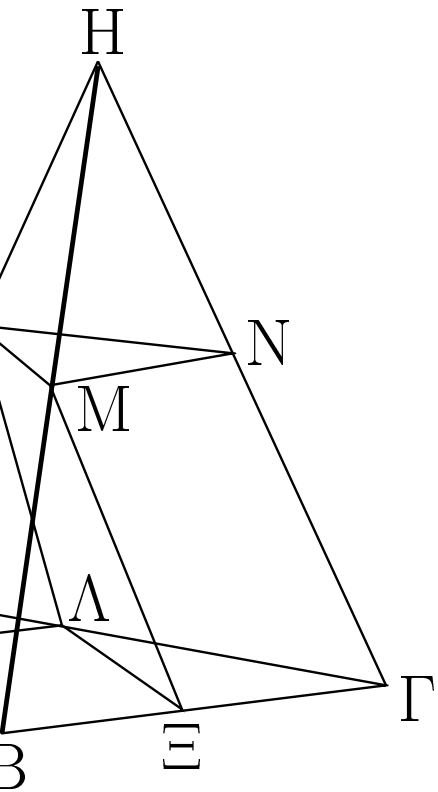
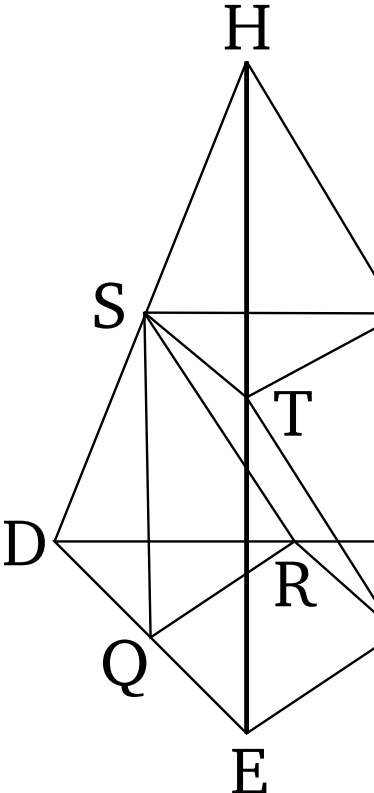
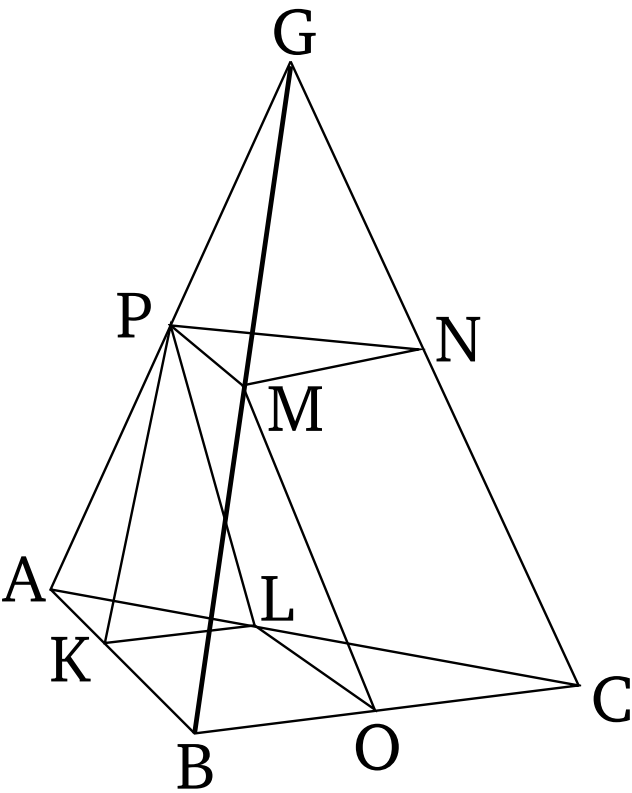
is greater than the (sum of the) aforementioned two pyramids, whose bases (are) triangles AEG and HKL , and apexes the points H and D (respectively).

Thus, the whole pyramid, whose base (is) triangle ABC , and apex the point D , has been divided into two pyramids (which are) equal to one another [and similar to the whole], and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid. (Which is) the very thing it was required to show.

Proposition 4

If there are two pyramids with the same height, having triangular bases, and each of them is divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms then as the base of one pyramid (is) to the base of the other pyramid, so (the sum of) all the prisms in one pyramid will be to (the sum of all) the equal number of prisms in the other pyramid.

Let there be two pyramids with the same height, having the triangular bases ABC and DEF , (with) apexes the points G and H (respectively). And let each of them be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms [Prop. 12.3]. I say that as base ABC is to base DEF , so (the sum of) all the prisms in pyramid $ABCG$ (is) to (the sum of) all the equal number of prisms in pyramid $DEFH$.



E For since BO is equal to OC , and AL to LC , LO is thus

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ μὲν BX τῇ $XΓ$, ἡ δὲ $ΑΛ$ τῇ $ΛΓ$, παράλληλος ἄρα ἐστὶν ἡ $ΑΧ$ τῇ $ΑΒ$ καὶ ὅμοιον τὸ $ΑΒΓ$ τρίγωνον τῷ $ΑΧΓ$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $ΔΕΖ$ τρίγωνον τῷ $ΡΦΖ$ τριγώνῳ ὅμοιον ἐστίν. καὶ ἐπεὶ διπλασίῳ ἐστὶν ἡ μὲν $ΒΓ$ τῇ $ΓΧ$, ἡ δὲ $ΕΖ$ τῇ $ΖΦ$, ἔστιν ἄρα ὡς ἡ $ΒΓ$ πρὸς τὴν $ΓΧ$, οὕτως ἡ $ΕΖ$ πρὸς τὴν $ΖΦ$. καὶ ἀναγέγραπται ἀπὸ μὲν τῶν $ΒΓ$, $ΓΧ$ ὁμοία τε καὶ ὁμοίως κείμενα εὐθύγραμμα τὰ $ΑΒΓ$, $ΑΧΓ$, ἀπὸ δὲ τῶν $ΕΖ$, $ΖΦ$ ὁμοία τε καὶ ὁμοίως κείμενα [εὐθύγραμμα] τὰ $ΔΕΖ$, $ΡΦΖ$. ἔστιν ἄρα ὡς τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΑΧΓ$ τρίγωνον, οὕτως τὸ $ΔΕΖ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον· ἐναλλάξ ἄρα ἐστὶν ὡς τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΔΕΖ$ [τρίγωνον], οὕτως τὸ $ΑΧΓ$ [τρίγωνον] πρὸς τὸ $ΡΦΖ$ τρίγωνον. ἀλλ' ὡς τὸ $ΑΧΓ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις μὲν [ἐστὶ] τὸ $ΑΧΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΟΜΝ$, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$. καὶ ὡς ἄρα τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΔΕΖ$ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΑΧΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΟΜΝ$, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$. ὡς δὲ τὰ εἰρημένα πρίσματα πρὸς ἄλληλα, οὕτως τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΚΒΧΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $ΟΜ$ εὐθεῖα, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΠΕΦΡ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $ΣΤ$ εὐθεῖα. καὶ τὰ δύο ἄρα πρίσματα, οὗ τε βάσις μὲν τὸ $ΚΒΧΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $ΟΜ$, καὶ οὗ βάσις μὲν τὸ $ΑΧΓ$, ἀπεναντίον δὲ τὸ $ΟΜΝ$, πρὸς τὰ πρίσματα, οὗ τε βάσις μὲν τὸ $ΠΕΦΡ$, ἀπεναντίον δὲ ἡ $ΣΤ$ εὐθεῖα, καὶ οὗ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$. καὶ ὡς ἄρα ἡ $ΑΒΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ εἰρημένα δύο πρίσματα πρὸς τὰ εἰρημένα δύο πρίσματα.

Καὶ ὁμοίως, εἰαν διαιρεθῶσιν αἱ $ΟΜΝΗ$, $ΣΤΥΘ$ πυραμίδες εἴς τε δύο πρίσματα καὶ δύο πυραμίδας, ἔσται ὡς ἡ $ΟΜΝ$ βάσις πρὸς τὴν $ΣΤΥ$ βάσιν, οὕτως τὰ ἐν τῇ $ΟΜΝΗ$ πυραμίδι δύο πρίσματα πρὸς τὰ ἐν τῇ $ΣΤΥΘ$ πυραμίδι δύο πρίσματα. ἀλλ' ὡς ἡ $ΟΜΝ$ βάσις πρὸς τὴν $ΣΤΥ$ βάσιν, οὕτως ἡ $ΑΒΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν· ἴσον γὰρ ἑκάτερον τῶν $ΟΜΝ$, $ΣΤΥ$ τριγώνων ἑκατέρῳ τῶν $ΑΧΓ$, $ΡΦΖ$. καὶ ὡς ἄρα ἡ $ΑΒΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ τέσσαρα πρίσματα πρὸς τὰ τέσσαρα πρίσματα. ὁμοίως δὲ κ' ἂν τὰς ὑπολειπομένας πυραμίδας διέλωμεν εἴς τε δύο πυραμίδας καὶ εἴς δύο πρίσματα, ἔσται ὡς ἡ $ΑΒΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ ἐν τῇ $ΑΒΓΗ$ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ $ΔΕΖΘ$ πυραμίδι πρίσματα πάντα ἰσοπληθῆ· ὅπερ ἔδει δεῖχαι.

Λήμμα.

Ὅτι δὲ ἐστὶν ὡς τὸ $ΑΧΓ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις τὸ $ΑΧΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΟΜΝ$, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΡΦΖ$ [τρίγωνον], ἀπεναντίον δὲ τὸ $ΣΤΥ$, οὕτω δεικτέον.

parallel to AB , and triangle ABC similar to triangle LOC [Prop. 12.3]. So, for the same (reasons), triangle DEF is also similar to triangle RVF . And since BC is double CO , and EF (double) FV , thus as BC (is) to CO , so EF (is) to FV . And the similar, and similarly laid out, rectilinear (figures) ABC and LOC be described on BC and CO (respectively), and the similar, and similarly laid out, [rectilinear] (figures) DEF and RVF on EF and FV (respectively). Thus, as triangle ABC is to triangle LOC , so triangle DEF (is) to triangle RVF [Prop. 6.22]. Thus, alternately, as triangle ABC is to [triangle] DEF , so [triangle] LOC (is) to triangle RVF [Prop. 5.16]. But, as triangle LOC (is) to triangle RVF , so the prism whose base [is] triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) triangle RVF , and opposite (plane) STU (see lemma). And, thus, as triangle ABC (is) to triangle DEF , so the prism whose base (is) triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) triangle RVF , and opposite (plane) STU . And as the aforementioned prisms (are) to one another, so the prism whose base (is) parallelogram $KBOL$, and opposite (side) straight-line PM , (is) to the prism whose base (is) parallelogram $QEVN$, and opposite (side) straight-line ST [Props. 11.39, 12.3]. Thus, also, (is) the (sum of the) two prisms—that whose base (is) parallelogram $KBOL$, and opposite (side) PM , and that whose base (is) LOC , and opposite (plane) PMN —to (the sum of) the (two) prisms—that whose base (is) $QEVN$, and opposite (side) straight-line ST , and that whose base (is) triangle RVF , and opposite (plane) STU [Prop. 5.12]. And, thus, as base ABC (is) to base DEF , so the (sum of the first) aforementioned two prisms (is) to the (sum of the second) aforementioned two prisms.

And, similarly, if pyramids $PMNG$ and $STUH$ are divided into two prisms, and two pyramids, as base PMN (is) to base STU , so (the sum of) the two prisms in pyramid $PMNG$ will be to (the sum of) the two prisms in pyramid $STUH$. But, as base PMN (is) to base STU , so base ABC (is) to base DEF . For the triangles PMN and STU (are) equal to LOC and RVF , respectively. And, thus, as base ABC (is) to base DEF , so (the sum of) the four prisms (is) to (the sum of) the four prisms [Prop. 5.12]. So, similarly, even if we divide the pyramids left behind into two pyramids and into two prisms, as base ABC (is) to base DEF , so (the sum of) all the prisms in pyramid $ABCG$ will be to (the sum of) all the equal number of prisms in pyramid $DEFH$. (Which is) the very thing it was required to show.

Lemma

And one may show, as follows, that as triangle LOC is to triangle RVF , so the prism whose base (is) triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) [triangle] RVF , and opposite (plane) STU .

Ἐπὶ γὰρ τῆς αὐτῆς καταγραφῆς νενοήσθωσαν ἀπὸ τῶν H , Θ κάθετοι ἐπὶ τὰ $AB\Gamma$, ΔEZ ἐπίπεδα, ἴσαι δηλαδὴ τυγθάνουσαι διὰ τὸ ἰσοῦν εἶναι ὑποκείσθαι τὰς πυραμίδας. καὶ ἐπεὶ δύο εὐθεῖαι ἢ τε $H\Gamma$ καὶ ἢ ἀπὸ τοῦ H κάθετος ὑπὸ παραλλήλων ἐπιπέδων τῶν $AB\Gamma$, OMN τέμνονται, εἷς τούξ αὐτοῦξ λόγους τμηθήσονται. καὶ τέμνεται ἡ $H\Gamma$ διὰ τοῦ OMN ἐπιπέδου κατὰ τὸ N · καὶ ἡ ἀπὸ τοῦ H ἄρα κάθετος ἐπὶ τὸ $AB\Gamma$ ἐπίπεδον διὰ τμηθήσεται ὑπὸ τοῦ OMN ἐπιπέδου. διὰ τὰ αὐτὰ δὴ καὶ ἡ ἀπὸ τοῦ Θ κάθετος ἐπὶ τὸ ΔEZ ἐπίπεδον διὰ τμηθήσεται ὑπὸ τοῦ ΣTY ἐπιπέδου. καὶ εἰσιν ἴσαι αἱ ἀπὸ τῶν H , Θ κάθετοι ἐπὶ τὰ $AB\Gamma$, ΔEZ ἐπίπεδα· ἴσαι ἄρα καὶ αἱ ἀπὸ τῶν OMN , ΣTY τριγόνων ἐπὶ τὰ $AB\Gamma$, ΔEZ κάθετοι. ἰσοῦν γὰρ [ἐστὶ] τὰ πρίσματα, ὧν βάσεις μὲν εἰσὶ τὰ $\Lambda X\Gamma$, $P\Phi Z$ τρίγωνα, ἀπεναντίον δὲ τὰ OMN , ΣTY . ὥστε καὶ τὰ στερεὰ παραλληλεπίπεδα τὰ ἀπὸ τῶν εἰρημένων πρισμάτων ἀναγραφόμενα ἰσοῦν καὶ πρὸς ἄλληλά [εἰσιν] ὡς αἱ βάσεις· καὶ τὰ ἡμίση ἄρα ἐστὶν ὡς ἡ $\Lambda X\Gamma$ βάσις πρὸς τὴν $P\Phi Z$ βάσιν, οὕτως τὰ εἰρημένα πρίσματα πρὸς ἄλληλα· ὅπερ ἔδει δεῖχαι.

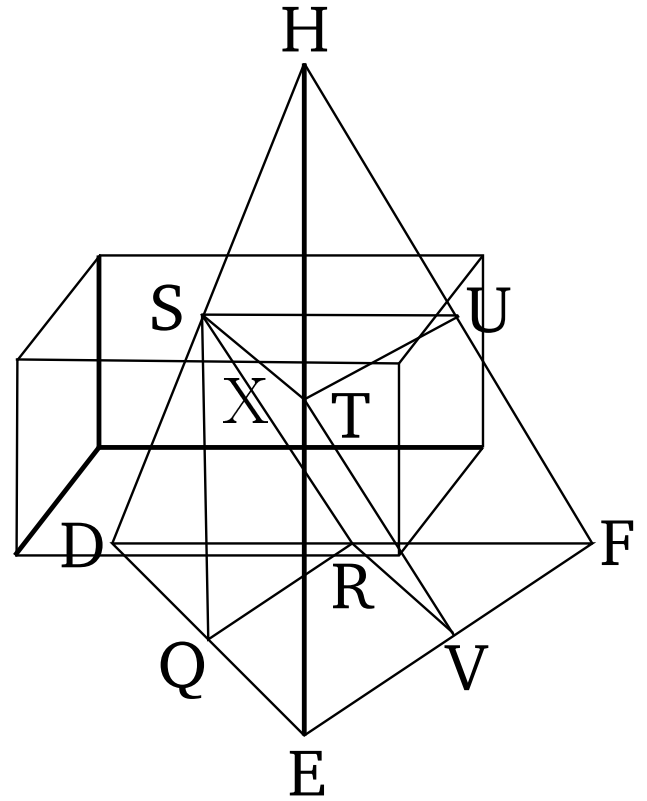
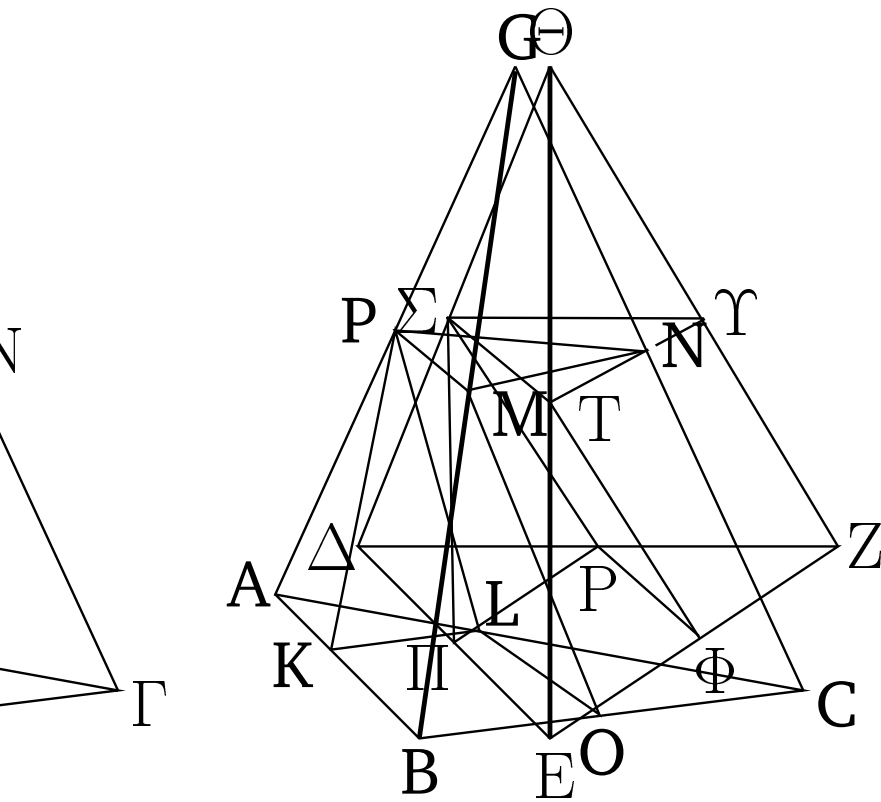
5.

Αἱ ὑπὸ τὸ αὐτὸ ὕψος οὔσαι πυραμίδες καὶ τριγώνουξ ἔθουσαι βάσεις πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις.

For, in the same figure, let perpendiculars be conceived (drawn) from (points) G and H to the planes ABC and DEF (respectively). These clearly turn out to be equal, on account of the pyramids being assumed (to be) of equal height. And since two straight-lines, GC and the perpendicular from G , are cut by the parallel planes ABC and PMN they will be cut in the same ratios [Prop. 11.17]. And GC was cut in half by the plane PMN at N . Thus, the perpendicular from G to the plane ABC will also be cut in half by the plane PMN . So, for the same (reasons), the perpendicular from H to the plane DEF will also be cut in half by the plane STU . And the perpendiculars from G and H to the planes ABC and DEF (respectively) are equal. Thus, the perpendiculars from the triangles PMN and STU to ABC and DEF (respectively, are) also equal. Thus, the prisms whose bases are triangles LOC and RVF , and opposite (sides) PMN and STU (respectively), [are] of equal height. And, hence, the parallelepiped solids described on the aforementioned prisms [are] of equal height and (are) to one another as their bases [Prop. 11.32]. Likewise, the halves (of the solids) [Prop. 11.28]. Thus, as base LOC is to base RVF , so the aforementioned prisms (are) to one another. (Which is) the very thing it was required to show.

Proposition 5

Pyramids which are of the same height, and have triangular bases, are to one another as their bases.



Ἐστωσαν ὑπὸ τὸ αὐτὸ ὕψος πυραμίδες, ὧν βάσεις μὲν τὰ $AB\Gamma$, ΔEZ τρίγωνα, κορυφαὶ δὲ τὰ H , Θ σημεία· λέγω, ὅτι ἐστὶν ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως ἡ $AB\Gamma H$ πυραμὶς πρὸς τὴν $\Delta EZ\Theta$ πυραμίδα.

Εἰ γὰρ μὴ ἐστὶν ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως ἡ $AB\Gamma H$ πυραμὶς πρὸς τὴν $\Delta EZ\Theta$ πυραμίδα, ἔσται ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως ἡ $AB\Gamma H$ πυραμὶς ἥτοι πρὸς ἑλασσόν τι τῆς $\Delta EZ\Theta$ πυραμίδος στερεὸν ἢ πρὸς μείζον. ἔστω πρότερον πρὸς ἑλασσόν τὸ Θ , καὶ διηρῆσθω ἡ $\Delta EZ\Theta$ πυραμὶς εἰς τε δύο πυραμίδας ἴσας ἀλλήλαις καὶ ὁμοίαις τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα· τὰ δὴ δύο πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος. καὶ πάλιν αἱ ἐκ τῆς διαιρέσεως γινόμεναι πυραμίδες ὁμοίως διηρῆσθωσαν, καὶ τοῦτο ἀεὶ γινέσθω, ἕως ὅτ' ὑλειφθῶσί τινας πυραμίδες ἀπὸ τῆς $\Delta EZ\Theta$ πυραμίδος, αἱ εἰσὶν ἐλάττωες τῆς ὑπεροθῆς, ἣν ὑπερέθει ἡ $\Delta EZ\Theta$ πυραμὶς τοῦ Θ στερεοῦ. λελειφθωσαν καὶ ἔστωσαν λόγον ἕνεκεν αἱ $\Delta\text{Π}\text{Ρ}\Sigma$, $\Sigma\text{Τ}\Upsilon\Theta$ · λοιπὰ ἄρα τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα μείζονά ἐστι τοῦ Θ στερεοῦ. διηρῆσθω καὶ ἡ $AB\Gamma H$ πυραμὶς ὁμοίως καὶ ἱσοπληθῶς τῇ $\Delta EZ\Theta$ πυραμίδι· ἐστὶν ἄρα ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὰ ἐν τῇ $AB\Gamma H$ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα, ἀλλὰ καὶ ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως ἡ $AB\Gamma H$ πυραμὶς πρὸς τὸ Θ στερεόν· καὶ ὡς ἄρα ἡ $AB\Gamma H$ πυραμὶς πρὸς τὸ Θ στερεόν, οὕτως τὰ ἐν τῇ $AB\Gamma H$ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα· ἐναλλάχ' ἄρα ὡς ἡ AB

Let there be pyramids of the same height whose bases (are) the triangles ABC and DEF , and apexes the points G and H (respectively). I say that as base ABC is to base DEF , so pyramid $ABCG$ (is) to pyramid $DEFH$.

For if base ABC is not to base DEF , as pyramid $ABCG$ (is) to pyramid $DEFH$, then base ABC will be to base DEF , as pyramid $ABCG$ (is) to some solid either less than, or greater than, pyramid $DEFH$. Let it, first of all, be (in this ratio) to (some) lesser (solid), W . And let pyramid $DEFH$ be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms. So, the (sum of the) two prisms is greater than half of the whole pyramid [Prop. 12.3]. And, again, let the pyramids generated by the division be similarly divided, and let this be done continually until some pyramids are left from pyramid $DEFH$ which (when added together) are less than the excess by which pyramid $DEFH$ exceeds the solid W [Prop. 10.1]. Let them be left, and, for the sake of argument, let them be $DQRS$ and $STUH$. Thus, the (sum of the) remaining prisms within pyramid $DEFH$ is greater than solid W . Let pyramid $ABCG$ also be divided similarly, and a similar number of times, as pyramid $DEFH$. Thus, as base ABC is to base DEF , so the (sum of the) prisms within pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 12.4]. But, also, as base ABC (is) to base DEF , so pyramid $ABCG$ (is) to solid W . And, thus, as pyramid $ABCG$ (is) to solid W , so the (sum of the) prisms within

ΓΗ πυραμίδες πρὸς τὰ ἐν αὐτῇ πρίσματα, οὕτως τὸ Θ στερεὸν πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα. μείζων δὲ ἡ ΑΒΓΗ πυραμὶς τῶν ἐν αὐτῇ πρισμάτων· μείζον ἄρα καὶ τὸ Θ στερεὸν τῶν ἐν τῇ ΔΕΖΘ πυραμίδι πρισμάτων. ἀλλὰ καὶ ἔλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς ἑλασσόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ὁμοίως δὲ δευτέρησεται, ὅτι οὐδὲ ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως ἡ ΔΕΖΘ πυραμὶς πρὸς ἑλαττόν τι τῆς ΑΒΓΗ πυραμίδος στερεόν.

Λέγω δὴ, ὅτι οὐκ ἐστὶν οὐδὲ ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς μείζον τι τῆς ΔΕΖΘ πυραμίδος στερεόν.

Εἰ γὰρ δυνατόν, ἔστω πρὸς μείζον τὸ Θ· ἀνάπαλιν ἄρα ἐστὶν ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα. ὡς δὲ τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα, οὕτως ἡ ΔΕΖΘ πυραμὶς πρὸς ἑλασσόν τι τῆς ΑΒΓΗ πυραμίδος, ὡς ἔμπροσθεν ἐδείχθη· καὶ ὡς ἄρα ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως ἡ ΔΕΖΘ πυραμὶς πρὸς ἑλασσόν τι τῆς ΑΒΓΗ πυραμίδος· ὅπερ ἄτοπον ἐδείχθη. οὐκ ἄρα ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς μείζον τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ἐδείχθη δέ, ὅτι οὐδὲ πρὸς ἑλασσόν. ἔστιν ἄρα ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα· ὅπερ ἔδει δεῖχαι.

6.

Αἱ ὑπὸ τὸ αὐτὸ ὕψος οὔσαι πυραμίδες καὶ πολυγώνου ἐθνουσι βάσεις πρὸς ἀλλήλας εἶσιν ὡς αἱ βάσεις.

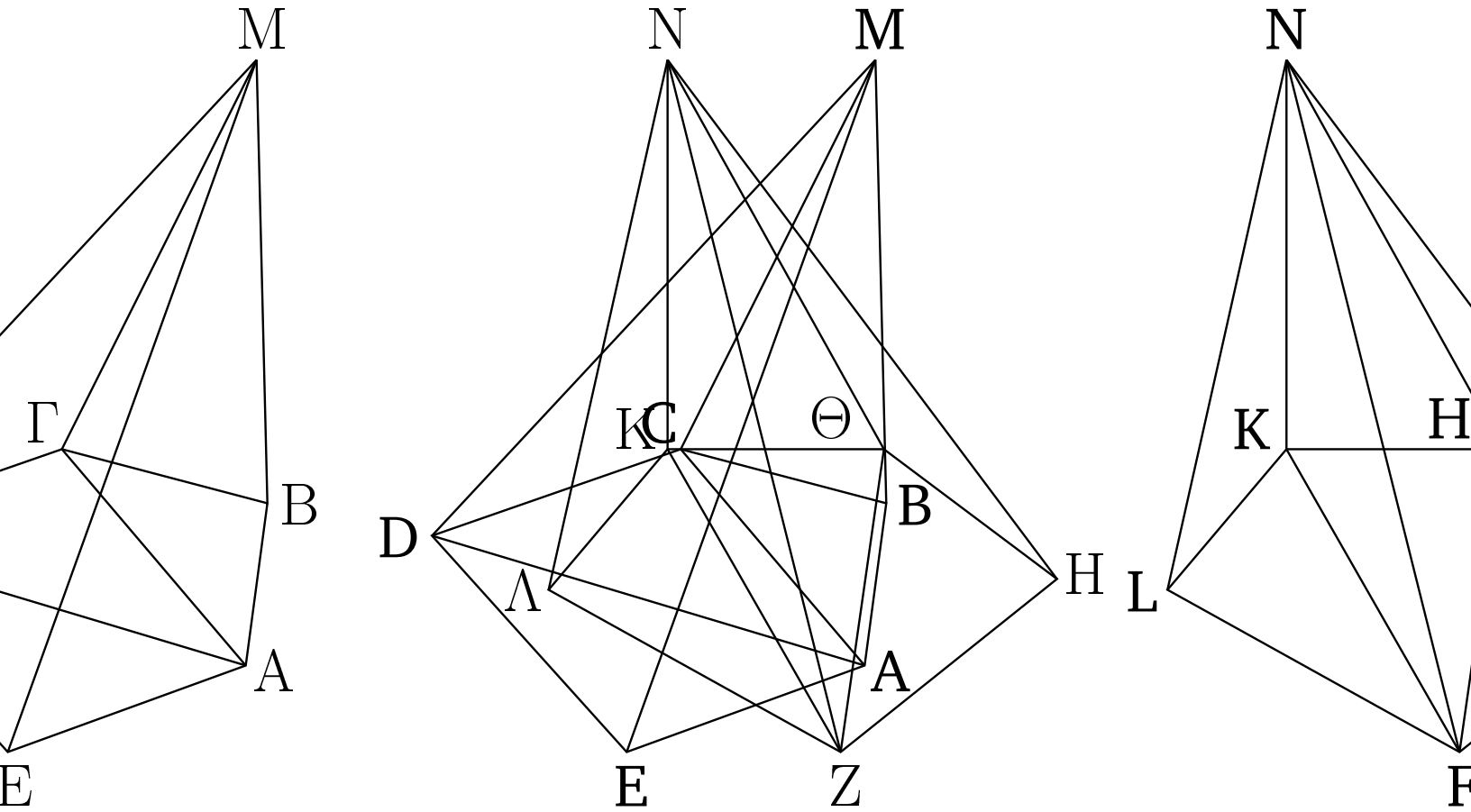
pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.11]. Thus, alternately, as pyramid $ABCG$ (is) to the (sum of the) prisms within it, so solid W (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.16]. And pyramid $ABCG$ (is) greater than the (sum of the) prisms within it. Thus, solid W (is) also greater than the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.14]. But, (it is) also less. This very thing is impossible. Thus, as base ABC is to base DEF , so pyramid $ABCG$ (is) not to some solid less than pyramid $DEFH$. So, similarly, we can show that base DEF is not to base ABC , as pyramid $DEFH$ (is) to some solid less than pyramid $ABCG$ either.

So, I say that neither is base ABC to base DEF , as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$.

For, if possible, let it be (in this ratio) to some greater (solid), W . Thus, inversely, as base DEF (is) to base ABC , so solid W (is) to pyramid $ABCG$ [Prop. 5.7. corr.]. And as solid W (is) to pyramid $ABCG$, so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$, as shown before [Prop. 12.2 lem.]. And, thus, as base DEF (is) to base ABC , so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$ [Prop. 5.11]. The very thing was shown (to be) absurd. Thus, base ABC is not to base DEF , as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$. And, it was shown that neither (is it in this ratio) to a lesser (solid). Thus, as base ABC is to base DEF , so pyramid $ABCG$ (is) to pyramid $DEFH$. (Which is) the very thing it was required to show.

Proposition 6

Pyramids which are of the same height, and have polygonal bases, are to one another as their bases.



Ἐστωσαν ὑπὸ τὸ αὐτὸ ὕψος πυραμίδες, ὧν [αἱ] βάσεις μὲν τὰ $ABΓΔΕ$, $ZHΘΚΛ$ πολύγωνα, κορυφαὶ δὲ τὰ M , N σημεῖα· λέγω, ὅτι ἐστὶν ὡς ἡ $ABΓΔΕ$ βάσις πρὸς τὴν $ZHΘΚΛ$ βάσιν, οὕτως ἡ $ABΓΔΕΜ$ πυραμὶς πρὸς τὴν $ZHΘΚΛΝ$ πυραμίδα.

Ἐπεξεύθθωσαν γάρ αἱ $ΑΓ$, $ΑΔ$, $ΖΘ$, $ΖΚ$. ἐπεὶ οὖν δύο πυραμίδες εἰσὶν αἱ $ABΓΜ$, $ΑΓΔΜ$ τριγώνων ἐξ ἑνὸς ὕψους καὶ ὅμοιαι, πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις· ἐστὶν ἄρα ὡς ἡ $ABΓ$ βάσις πρὸς τὴν $ΑΓΔ$ βάσιν, οὕτως ἡ $ABΓΜ$ πυραμὶς πρὸς τὴν $ΑΓΔΜ$ πυραμίδα. καὶ συνθέντι ὡς ἡ $ABΓΔ$ βάσις πρὸς τὴν $ΑΓΔ$ βάσιν, οὕτως ἡ $ABΓΔΜ$ πυραμὶς πρὸς τὴν $ΑΓΔΜ$ πυραμίδα. ἀλλὰ καὶ ὡς ἡ $ΑΓΔ$ βάσις πρὸς τὴν $ΑΔΕ$ βάσιν, οὕτως ἡ $ΑΓΔΜ$ πυραμὶς πρὸς τὴν $ΑΔΕΜ$ πυραμίδα. δι' ἴσον ἄρα ὡς ἡ $ABΓΔ$ βάσις πρὸς τὴν $ΑΔΕ$ βάσιν, οὕτως ἡ $ABΓΔΜ$ πυραμὶς πρὸς τὴν $ΑΔΕΜ$ πυραμίδα. καὶ συνθέντι πάλιν, ὡς ἡ $ABΓΔΕ$ βάσις πρὸς τὴν $ΑΔΕ$ βάσιν, οὕτως ἡ $ABΓΔΕΜ$ πυραμὶς πρὸς τὴν $ΑΔΕΜ$ πυραμίδα. ὁμοίως δὲ δευτέρως, ὅτι καὶ ὡς ἡ $ZHΘΚΛ$ βάσις πρὸς τὴν $ZHΘ$ βάσιν, οὕτως καὶ ἡ $ZHΘΚΛΝ$ πυραμὶς πρὸς τὴν $ZHΘΝ$ πυραμίδα. καὶ ἐπεὶ δύο πυραμίδες εἰσὶν αἱ $ΑΔΕΜ$, $ZHΘΝ$ τριγώνων ἐξ ἑνὸς ὕψους καὶ ὅμοιαι, ἐστὶν ἄρα ὡς ἡ $ΑΔΕ$ βάσις πρὸς τὴν $ZHΘ$ βάσιν, οὕτως ἡ $ΑΔΕΜ$ πυραμὶς πρὸς τὴν $ZHΘΝ$ πυραμίδα. ἀλλ' ὡς ἡ

Let there be pyramids of the same height whose bases (are) the polygons $ABCDE$ and $FGHKL$, and apexes the points M and N (respectively). I say that as base $ABCDE$ is to base $FGHKL$, so pyramid $ABCDEM$ (is) to pyramid $FGHKLN$.

For let AC , AD , FH , and FK be joined. Therefore, since $ABCM$ and $ACDM$ are two pyramids having triangular bases and equal height, they are to one another as their bases [Prop. 12.5]. Thus, as base ABC is to base ACD , so pyramid $ABCM$ (is) to pyramid $ACDM$. And, via composition, as base $ABCD$ (is) to base ACD , so pyramid $ABCDM$ (is) to pyramid $ACDM$ [Prop. 5.18]. But, as base ACD (is) to base ADE , so pyramid $ACDM$ (is) also to pyramid $ADEM$ [Prop. 12.5]. Thus, via equality, as base $ABCD$ (is) to base ADE , so pyramid $ABCDM$ (is) to pyramid $ADEM$ [Prop. 5.22]. And, again, via composition, as base $ABCDE$ (is) to base ADE , so pyramid $ABCDEM$ (is) to pyramid $ADEM$ [Prop. 5.18]. So, similarly, it can also be shown that as base $FGHKL$ (is) to base FGH , so pyramid $FGHKLN$ (is) also to pyramid $FGHN$. And since $ADEM$ and $FGHN$ are two pyramids having triangular bases and equal height, thus as base ADE (is) to base FGH , so pyramid $ADEM$ (is) to pyramid $FGHN$ [Prop. 12.5]. But, as base ADE (is) to base $ABCDE$, so pyramid $ADEM$ (was) to pyramid $ABCDEM$. Thus, via equality, as base $ABCDE$

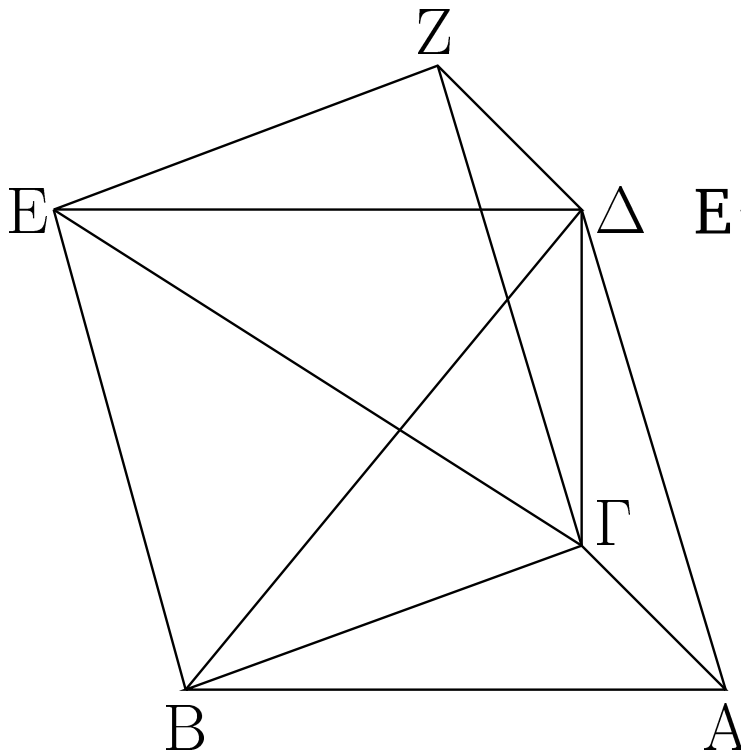
$A\Delta E$ βάσις πρὸς τὴν $AB\Gamma\Delta E$ βάσιν, οὕτως ἢ ἡ $A\Delta E M$ πυραμὶς πρὸς τὴν $AB\Gamma\Delta E M$ πυραμίδα. καὶ δι' ἴσον ἄρα ὡς ἡ $AB\Gamma\Delta E$ βάσις πρὸς τὴν $ZH\Theta$ βάσιν, οὕτως ἡ $AB\Gamma\Delta E M$ πυραμὶς πρὸς τὴν $ZH\Theta N$ πυραμίδα. ἀλλὰ μὴν καὶ ὡς ἡ $ZH\Theta$ βάσις πρὸς τὴν $ZH\Theta K\Lambda$ βάσιν, οὕτως ἢ καὶ ἡ $ZH\Theta N$ πυραμὶς πρὸς τὴν $ZH\Theta K\Lambda N$ πυραμίδα, καὶ δι' ἴσον ἄρα ὡς ἡ $AB\Gamma\Delta E$ βάσις πρὸς τὴν $ZH\Theta K\Lambda$ βάσιν, οὕτως ἡ $AB\Gamma\Delta E M$ πυραμὶς πρὸς τὴν $ZH\Theta K\Lambda N$ πυραμίδα· ὅπερ ἔδει δεῖξαι.

7.

Πᾶν πρίσμα τρίγωνον ἔθον βάσιν διαιρεῖται εἰς τρεῖς πυραμίδας ἴσαξ ἀλλήλαις τριγώνον βάσεις ἔθούσας.

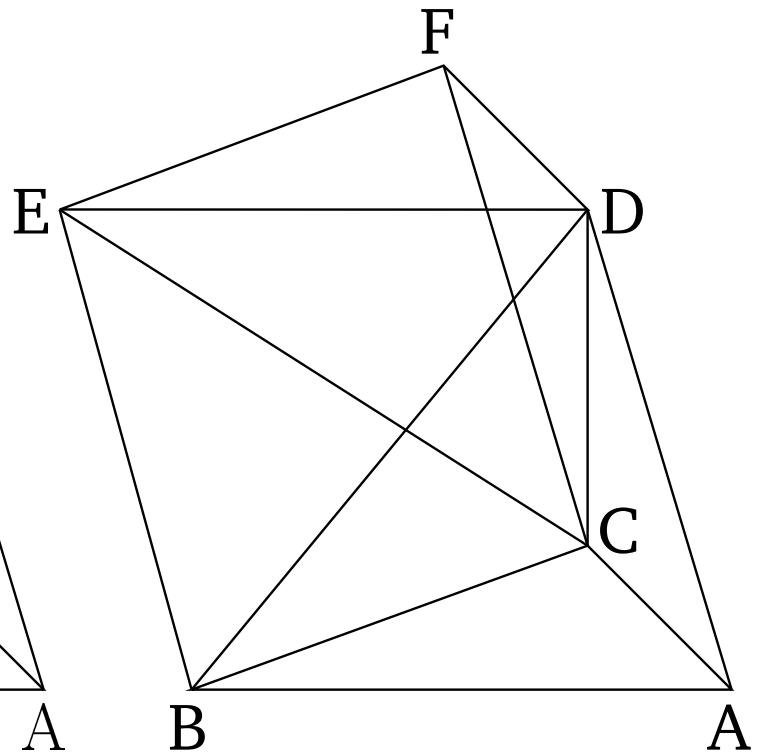
Proposition 7

Any prism having a triangular base is divided into three pyramids having triangular bases (which are) equal to one another.



Ἐστω πρίσμα, οὗ βάσις μὲν τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ ΔEZ . λέγω, ὅτι τὸ $AB\Gamma\Delta EZ$ πρίσμα διαιρεῖται εἰς τρεῖς πυραμίδας ἴσαξ ἀλλήλαις τριγώνον βάσεις ἔθούσας.

Ἐπεξεύθωσαν γὰρ αἱ $B\Delta$, $E\Gamma$, $\Gamma\Delta$. ἐπεὶ παραλληλόγραμμον ἔστι τὸ $ABE\Delta$, διάμετρος δὲ αὐτοῦ ἔστιν ἡ $B\Delta$, ἴσον ἄρα ἔστι τὸ $AB\Delta$ τρίγωνον τῷ $EB\Delta$ τριγώνῳ· καὶ ἡ πυραμὶς ἄρα, ἥτις βάσις μὲν τὸ $AB\Delta$ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἴση ἐστὶ πυραμίδι, ἥτις βάσις μὲν ἔστι τὸ ΔEB τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον. ἀλλὰ ἡ πυραμὶς, ἥτις βάσις μὲν ἔστι τὸ ΔEB τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἡ αὐτὴ ἔστι πυραμίδι, ἥτις βάσις μὲν ἔστι τὸ $EB\Gamma$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον· ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέχεται. καὶ πυραμὶς ἄρα, ἥτις βάσις μὲν ἔστι τὸ $AB\Delta$ τρίγωνον, κο-



Let there be a prism whose base (is) triangle ABC , and opposite (plane) DEF . I say that prism $ABCDEF$ is divided into three pyramids having triangular bases (which are) equal to one another.

For let BD , EC , and CD be joined. Since $ABED$ is a parallelogram, and BD is its diagonal, triangle ABD is thus equal to triangle EBD [Prop. 1.34]. And, thus, the pyramid whose base (is) triangle ABD , and apex the point C , is equal to the pyramid whose base is triangle DEB , and apex the point C [Prop. 12.5]. But, the pyramid whose base is triangle DEB , and apex the point C , is the same as the pyramid whose base is triangle EBC , and apex the point D . For they are contained by the same planes. And, thus, the pyramid whose base is ABD ,

ρυφή δὲ τὸ Γ σημείον, ἴση ἐστὶ πυραμίδι, ἥξ βάσις μὲν ἐστὶ τὸ $EB\Gamma$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. πάλιν, ἐπεὶ παραλληλόγραμμον ἐστὶ τὸ $Z\Gamma BE$, διάμετρος δὲ ἐστὶν αὐτοῦ ἡ GE , ἴσον ἐστὶ τὸ GEZ τρίγωνον τῷ ΓBE τριγώνῳ. καὶ πυραμὶς ἄρα, ἥξ βάσις μὲν ἐστὶ τὸ $B\Gamma E$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ἴση ἐστὶ πυραμίδι, ἥξ βάσις μὲν ἐστὶ τὸ EGZ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. ἡ δὲ πυραμὶς, ἥξ βάσις μὲν ἐστὶ τὸ $B\Gamma E$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ἴση ἐδείχθη πυραμίδι, ἥξ βάσις μὲν ἐστὶ τὸ $AB\Delta$ τρίγωνον, κορυφή δὲ τὸ Γ σημείον καὶ πυραμὶς ἄρα, ἥξ βάσις μὲν ἐστὶ τὸ GEZ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ἴση ἐστὶ πυραμίδι, ἥξ βάσις μὲν [ἐστὶ] τὸ $AB\Delta$ τρίγωνον, κορυφή δὲ τὸ Γ σημείον· διήρηται ἄρα τὸ $AB\Gamma\Delta EZ$ πρίσμα εἰς τρεῖς πυραμίδας ἴσας ἀλλήλαις τριγώνους ἐθνόσας βάσεις.

Καὶ ἐπεὶ πυραμὶς, ἥξ βάσις μὲν ἐστὶ τὸ $AB\Delta$ τρίγωνον, κορυφή δὲ τὸ Γ σημείον, ἡ αὐτὴ ἐστὶ πυραμίδι, ἥξ βάσις τὸ ΓAB τρίγωνον, κορυφή δὲ τὸ Δ σημείον· ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέχονται· ἡ δὲ πυραμὶς, ἥξ βάσις τὸ $AB\Delta$ τρίγωνον, κορυφή δὲ τὸ Γ σημείον, τρίτον ἐδείχθη τοῦ πρίσματος, ὅς ἡ βάσις τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ ΔEZ , καὶ ἡ πυραμὶς ἄρα, ἥξ βάσις τὸ $AB\Gamma$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, τρίτον ἐστὶ τοῦ πρίσματος τοῦ ἐθοντος βάσις τὴν αὐτὴν τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ ΔEZ .

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι πᾶσα πυραμὶς τρίτον μέρος ἐστὶ τοῦ πρίσματος τοῦ τὴν αὐτὴν βάσιν ἐθοντος αὐτῇ καὶ ὕψος ἴσον· ὅπερ ἔδει δεῖχαι.

8.

Αἱ ὅμοιαι πυραμίδες καὶ τριγώνους ἐθονσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.

Ἐστωσαν ὅμοιαι καὶ ὁμοίως κείμεναι πυραμίδες, ὧν βάσεις μὲν εἰσὶ τὰ $AB\Gamma$, ΔEZ τρίγωνα, κορυφαὶ δὲ τὰ H , Θ σημεία· λέγω, ὅτι ἡ $AB\Gamma H$ πυραμὶς πρὸς τὴν $\Delta EZ\Theta$ πυραμίδα τριπλασίονα λόγον ἔθει ἢ περὶ ἡ $B\Gamma$ πρὸς τὴν EZ .

and apex the point C , is equal to the pyramid whose base is EBC and apex the point D . Again, since $FCBE$ is a parallelogram, and CE is its diagonal, triangle CEF is equal to triangle CBE [Prop. 1.34]. And, thus, the pyramid whose base is triangle BCE , and apex the point D , is equal to the pyramid whose base is triangle ECF , and apex the point D [Prop. 12.5]. And the pyramid whose base is triangle BCE , and apex the point D , was shown (to be) equal to the pyramid whose base is triangle ABD , and apex the point C . Thus, the pyramid whose base is triangle CEF , and apex the point D , is also equal to the pyramid whose base [is] triangle ABD , and apex the point C . Thus, the prism $ABCDEF$ has been divided into three pyramids having triangular bases (which are) equal to one another.

And since the pyramid whose base is triangle ABD , and apex the point C , is the same as the pyramid whose base is triangle CAB , and apex the point D . For they are contained by the same planes. And the pyramid whose base (is) triangle ABD , and apex the point C , was shown (to be) a third of the prism whose base is triangle ABC , and opposite (plane) DEF , thus the pyramid whose base is triangle ABC , and apex the point D , is also a third of the pyramid having the same base, triangle ABC , and opposite (plane) DEF .

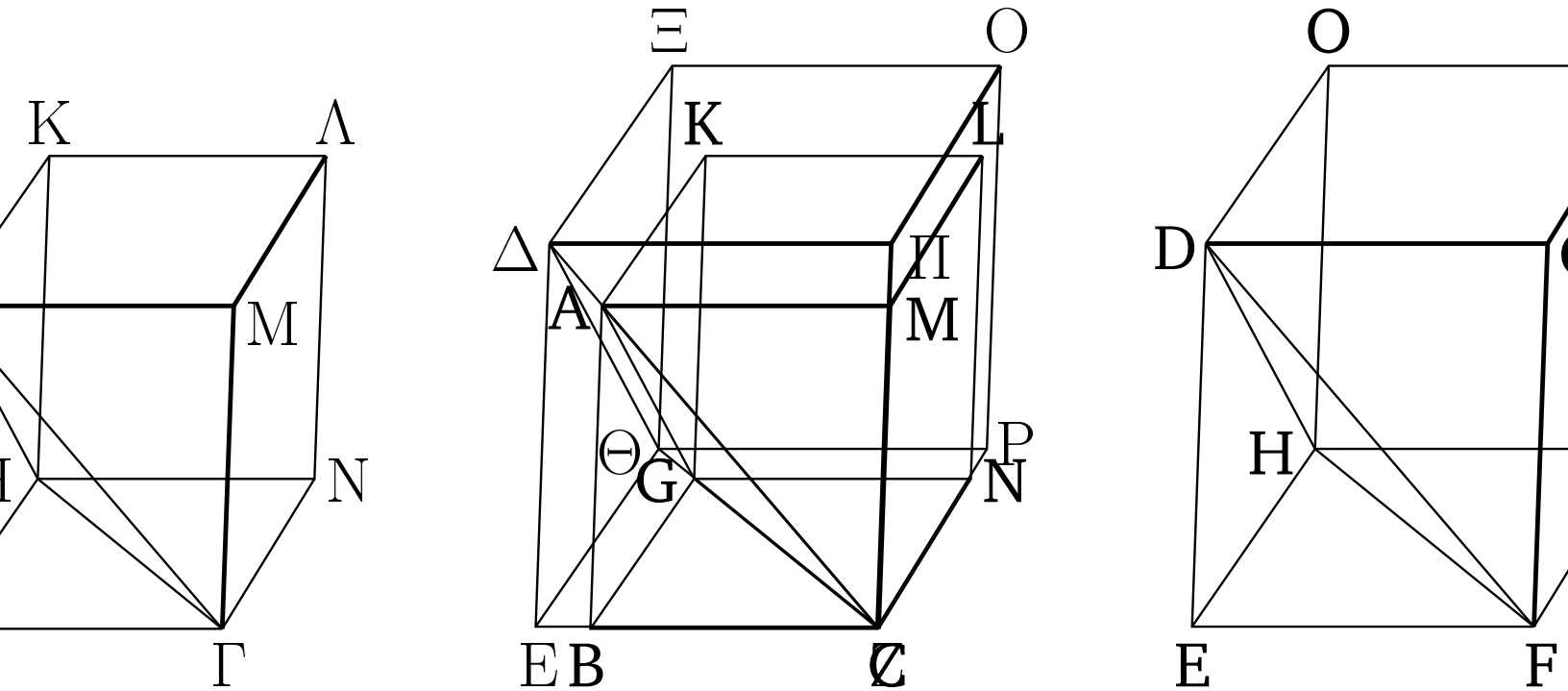
Corollary

And, from this, (it is) clear that any pyramid is the third part of the prism having the same base as it, and an equal height. (Which is) the very thing it was required to show.

Proposition 8

Similar pyramids which also have triangular bases are in the cubed ratio of their corresponding sides.

Let there be similar, and similarly laid out, pyramids whose bases are triangles ABC and DEF , and apexes the points G and H (respectively). I say that pyramid $ABCG$ has to pyramid $DEFH$ the cubed ratio of that BC (has) to EF .



Συμπληρώσω γάρ τὰ $BHML$, $EΘΠΟ$ στερεὰ παραλληλεπίπεδα. καὶ ἐπεὶ ὁμοία ἐστὶν ἡ $ABΓH$ πυραμὶς τῇ $ΔΕΖΘ$ πυραμίδι, ἴση ἄρα ἐστὶν ἡ μὲν ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ γωνίᾳ, ἡ δὲ ὑπὸ HBF τῇ ὑπὸ $ΘΕΖ$, ἡ δὲ ὑπὸ ABH τῇ ὑπὸ $ΔΕΘ$, καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $ΔΕ$, οὕτως ἡ BF πρὸς τὴν $ΕΖ$, καὶ ἡ BH πρὸς τὴν $ΕΘ$. καὶ ἐπεὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $ΔΕ$, οὕτως ἡ BF πρὸς τὴν $ΕΖ$, καὶ περὶ ἴσα ἑσὶν γωνίας αἱ πλευραὶ ἀνάλογον εἰσιν, ὅμοιον ἄρα ἐστὶ τὸ BM παραλληλόγραμμον τῷ $ΕΠ$ παραλληλογράμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν BN τῷ $ΕΡ$ ὅμοιον ἐστὶ, τὸ δὲ BK τῷ $ΕΧ$. τὰ τρία ἄρα τὰ MB , BK , BN τρισὶ τοῖς $ΕΠ$, $ΕΧ$, $ΕΡ$ ὅμοια ἐστὶν. ἀλλὰ τὰ μὲν τρία τὰ MB , BK , BN τρισὶ τοῖς ἀπεναντίον ἴσα τε καὶ ὅμοια ἐστὶν, τὰ δὲ τρία τὰ $ΕΠ$, $ΕΧ$, $ΕΡ$ τρισὶ τοῖς ἀπεναντίον ἴσα τε καὶ ὅμοια ἐστὶν. τὰ $BHML$, $EΘΠΟ$ ἄρα στερεὰ ὑπὸ ὁμοίων ἐπιπέδων ἴσων τὸ πλῆθος περιέχεται. ὅμοιον ἄρα ἐστὶ τὸ $BHML$ στερεὸν τῷ $EΘΠΟ$ στερεῷ. τὰ δὲ ὅμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν. τὸ $BHML$ ἄρα στερεὸν πρὸς τὸ $EΘΠΟ$ στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ὁμόλογος πλευρὰ ἡ BF πρὸς τὴν ὁμόλογον πλευρὰν τὴν $ΕΖ$. ὥς δὲ τὸ $BHML$ στερεὸν πρὸς τὸ $EΘΠΟ$ στερεὸν, οὕτως ἡ $ABΓH$ πυραμὶς πρὸς τὴν $ΔΕΖΘ$ πυραμίδα, ἐπειδὴ περ ἡ πυραμὶς ἕκτον μέρος ἐστὶ τοῦ στερεοῦ διὰ τὸ καὶ τὸ πρίσμα ἡμισυ ὂν τοῦ στερεοῦ παραλληλεπιπέδον τριπλασίον εἶναι τῆς πυραμίδος. καὶ ἡ $ABΓH$ ἄρα πυραμὶς πρὸς τὴν $ΔΕΖΘ$ πυραμίδα τριπλασίονα λόγον ἔχει ἥπερ ἡ BF πρὸς τὴν $ΕΖ$. ὅπερ ἔδει δεῖξαι.

For let the parallelepiped solids $BGML$ and $EHQP$ be completed. And since pyramid $ABCH$ is similar to pyramid $DEFH$, angle ABC is thus equal to angle DEF , and GBC to HEF , and ABG to DEH . And as AB is to DE , so BC (is) to EF , and BG to EH [Def. 11.9]. And since as AB is to DE , so BC (is) to EF , and (so) the sides around equal angles are proportional, parallelogram BM is thus similar to parallelogram EQ . So, for the same (reasons), BN is also similar to ER , and BK to EO . Thus, the three (parallelograms) MB , BK , and BN are similar to the three (parallelograms) EQ , EO , ER (respectively). But, the three (parallelograms) MB , BK , and BN are (both) equal and similar to the three opposite (parallelograms), and the three (parallelograms) EQ , EO , and ER are (both) equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the solids $BGML$ and $EHQP$ are contained by equal numbers of similar (and similarly laid out) planes. Thus, solid $BGML$ is similar to solid $EHQP$ [Def. 11.9]. And similar parallelepiped solids are in the cubed ratio of corresponding sides [Prop. 11.33]. Thus, solid $BGML$ has to solid $EHQP$ the cubed ratio that the corresponding side BC (has) to the corresponding side EF . And as solid $BGML$ (is) to solid $EHQP$, so pyramid $ABCH$ (is) to pyramid $DEFH$, inasmuch as the pyramid is the sixth part of the solid, on account of the prism, being half of the parallelepiped solid [Prop. 11.28], also being three times the pyramid [Prop. 12.7]. Thus, pyramid $ABCH$ also has to pyramid $DEFH$ the cubed ratio that BC (has) to EF . (Which is) the very thing it was required to show.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι καὶ αἱ πολυγώνουζ ἔθουσαι βάσειζ ὅμοιαι πυραμίδεζ πρὸς ἀλλήλαζ ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. διαιρεθεισῶν γάρ αὐτῶν εἰς τὰς ἐν αὐταῖς πυραμίδαζ τριγώνουζ βάσειζ ἔθουσαζ τῷ καὶ τὰ ὅμοια πολύγωνα τῶν βάσεων εἰς ὅμοια τρίγωνα διαιρεῖσθαι καὶ ἴσα τῷ πλήθει καὶ ὁμόλογα τοῖς ὅλοις ἔσται ὥς [ή] ἐν τῇ ἐτέρῳ μία πυραμίς τριγώνον ἔθουσα βάσιν πρὸς τὴν ἐν τῇ ἐτέρῳ μίαν πυραμίδα τριγώνον ἔθουσαν βάσιν, οὕτως καὶ ἅπασαι αἱ ἐν τῇ ἐτέρῳ πυραμίδι πυραμίδεζ τριγώνουζ ἔθουσαι βάσειζ πρὸς τὰς ἐν τῇ ἐτέρῳ πυραμίδι πυραμίδαζ τριγώνουζ βάσειζ ἔθουσαζ, τοντέστιν αὐτὴ ἡ πολύγωνον βάσιν ἔθουσα πυραμίς πρὸς τὴν πολύγωνον βάσιν ἔθουσαν πυραμίδα. ἡ δὲ τριγώνον βάσιν ἔθουσα πυραμίς πρὸς τὴν τριγώνον βάσιν ἔθουσαν ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν· καὶ ἡ πολύγωνον ἄρα βάσιν ἔθουσα πρὸς τὴν ὁμοίαν βάσιν ἔθουσαν τριπλασίονα λόγον ἔθει ἢ περ ἢ πλευρὰ πρὸς τὴν πλευράν.

9.

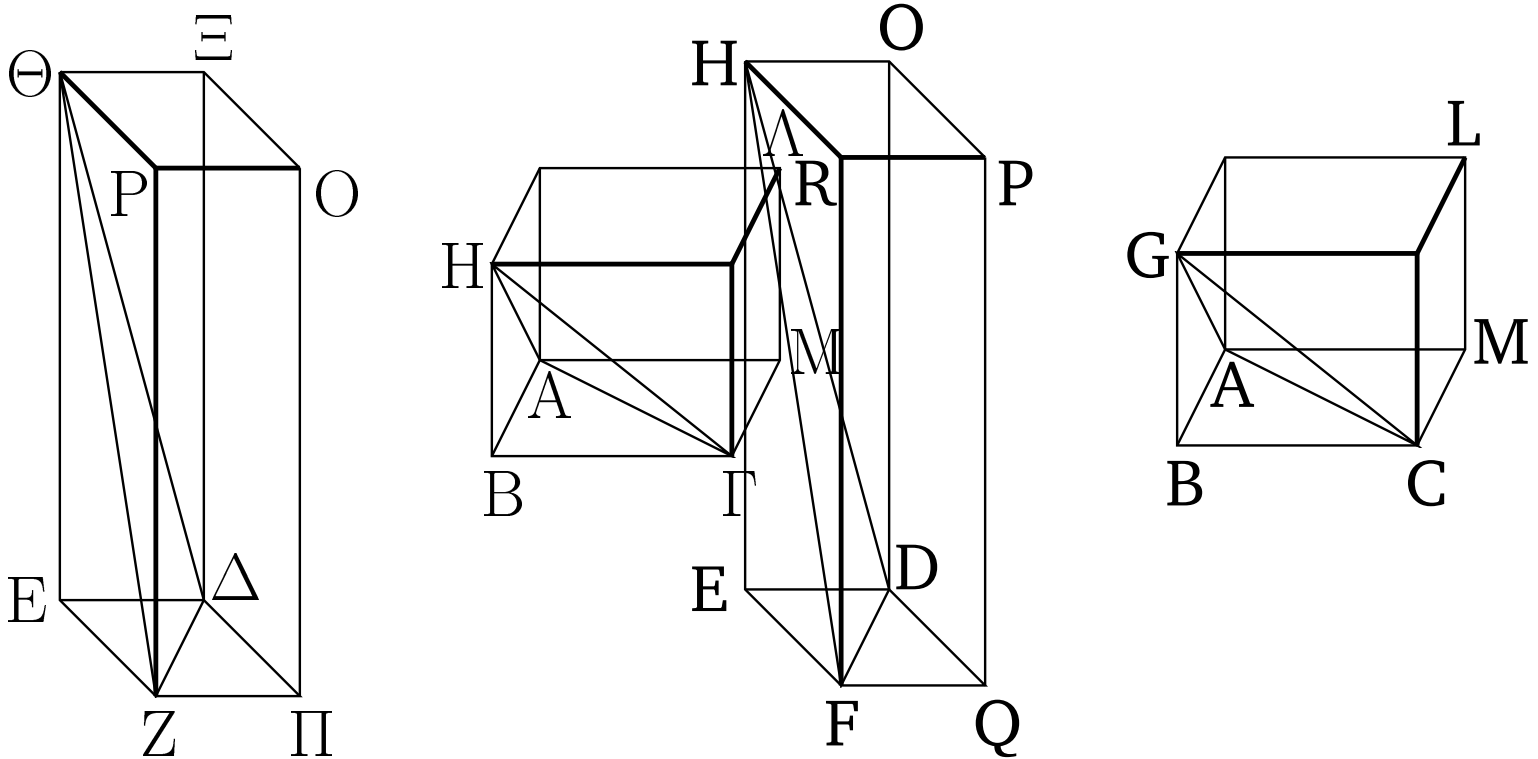
Τῶν ἴσων πυραμίδων καὶ τριγώνουζ βάσειζ ἔθουσῶν ἀντιπεπύονθαι αἱ βάσειζ τοῖς ὕψουσιν· καὶ ὧν πυραμίδων τριγώνουζ βάσειζ ἔθουσῶν ἀντιπεπύονθαι αἱ βάσειζ τοῖς ὕψουσιν, ἴσαι εἶσιν ἐκεῖναι.

Corollary

So, from this, (it is) also clear that similar pyramids having polygonal bases (are) to one another as the cubed ratio of their corresponding sides. For, dividing them into the pyramids (contained) within them which have triangular bases, with the similar polygons of the bases also being divided into similar triangles (which are) both equal in number, and corresponding, to the wholes [Prop. 6.20]. As one pyramid having a triangular base in the former (pyramid having a polygonal base is) to one pyramid having a triangular base in the latter (pyramid having a polygonal base), so (the sum of) all the pyramids having triangular bases in the former pyramid will also be to (the sum of) all the pyramids having triangular bases in the latter pyramid [Prop. 5.12]—that is to say, the (former) pyramid itself having a polygonal base to the (latter) pyramid having a polygonal base. And a pyramid having a triangular base is to a (pyramid) having a triangular base in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, a (pyramid) having a polygonal base also has to to a (pyramid) having a similar base the cubed ratio of a (corresponding) side to a (corresponding) side.

Proposition 9

The bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids which have triangular bases whose bases are reciprocally proportional to their heights are equal.



Ἐστωσαν γάρ ἴσαι πυραμίδες τριγώνου βάσεις ἔθουσai τὰς $AB\Gamma$, ΔEZ , κορυφὰς δὲ τὰ H , Θ σημεία· λέγω, ὅτι τῶν $AB\Gamma H$, $\Delta EZ\Theta$ πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν, καὶ ἐστὶν ὥς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος.

Συμπεληρώσω γάρ τὰ $BHMA$, $E\Theta\Pi O$ στερεὰ παραλληλεπίπεδα. καὶ ἐπεὶ ἴση ἐστὶν ἡ $AB\Gamma H$ πυραμὶς τῇ $\Delta EZ\Theta$ πυραμίδι, καὶ ἐστὶ τῆς μὲν $AB\Gamma H$ πυραμίδος ἑξαπλάσιον τὸ $BHMA$ στερεόν, τῆς δὲ $\Delta EZ\Theta$ πυραμίδος ἑξαπλάσιον τὸ $E\Theta\Pi O$ στερεόν, ἴσον ἄρα ἐστὶ τὸ $BHMA$ στερεόν τῷ $E\Theta\Pi O$ στερεῷ. τῶν δὲ ἴσων στερεῶν παραλληλεπιπύδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν· ἐστὶν ἄρα ὥς ἡ BM βάσις πρὸς τὴν $E\Pi$ βάσιν, οὕτως τὸ τοῦ $E\Theta\Pi O$ στερεοῦ ὕψος πρὸς τὸ τοῦ $BHMA$ στερεοῦ ὕψος. ἀλλ' ὥς ἡ BM βάσις πρὸς τὴν $E\Pi$, οὕτως τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ ΔEZ τρίγωνον. καὶ ὥς ἄρα τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ ΔEZ τρίγωνον, οὕτως τὸ τοῦ $E\Theta\Pi O$ στερεοῦ ὕψος πρὸς τὸ τοῦ $BHMA$ στερεοῦ ὕψος. ἀλλὰ τὸ μὲν τοῦ $E\Theta\Pi O$ στερεοῦ ὕψος τὸ αὐτὸ ἐστὶ τῷ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψει, τὸ δὲ τοῦ $BHMA$ στερεοῦ ὕψος τὸ αὐτὸ ἐστὶ τῷ τῆς $AB\Gamma H$ πυραμίδος ὕψει· ἐστὶν ἄρα ὥς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος. τῶν $AB\Gamma H$, $\Delta EZ\Theta$ ἄρα πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν.

Ἀλλὰ δὴ τῶν $AB\Gamma H$, $\Delta EZ\Theta$ πυραμίδων ἀντιπεπονθέντων αἱ βάσεις τοῖς ὕψουσιν, καὶ ἔστω ὥς ἡ $AB\Gamma$ βάσις πρὸς

For let there be (two) equal pyramids having the triangular bases ABC and DEF , and apexes the points G and H (respectively). I say that the bases of the pyramids $ABCG$ and $DEFH$ are reciprocally proportional to their heights, and (so) that as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$.

For let the parallelepiped solids $BGML$ and $EHQP$ be completed. And since pyramid $ABCG$ is equal to pyramid $DEFH$, and solid $BGML$ is six times pyramid $ABCG$ (see previous proposition), and solid $EHQP$ (is) six times pyramid $DEFH$, solid $BGML$ is thus equal to solid $EHQP$. And the bases of equal parallelepiped solids are reciprocally proportional to their heights [Prop. 11.34]. Thus, as base BM is to base EQ , so the height of solid $EHQP$ (is) to the height of solid $BGML$. But, as base BM (is) to base EQ , so triangle ABC (is) to triangle DEF [Prop. 1.34]. And, thus, as triangle ABC (is) to triangle DEF , so the height of solid $EHQP$ (is) to the height of solid $BGML$ [Prop. 5.11]. But, the height of solid $EHQP$ is the same as the height of pyramid $DEFH$, and the height of solid $BGML$ is the same as the height of pyramid $ABCG$. Thus, as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$. Thus, the bases of pyramids $ABCG$ and $DEFH$ are reciprocally proportional to their heights.

And so, let the bases of pyramids $ABCG$ and $DEFH$ be reciprocally proportional to their heights, and (thus) let base ABC be to base DEF , as the height of pyramid $DEFH$ (is)

τὴν ΔEZ βάσιν, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος· λέγω, ὅτι ἴση ἐστὶν ἡ $AB\Gamma H$ πυραμὶς τῇ $\Delta EZ\Theta$ πυραμίδι.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἐστὶν ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος, ἀλλ' ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὸ BM παραλληλόγραμμον πρὸς τὸ $E\Pi$ παραλληλόγραμμον, καὶ ὡς ἄρα τὸ BM παραλληλόγραμμον πρὸς τὸ $E\Pi$ παραλληλόγραμμον, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος. ἀλλὰ τὸ [μὲν] τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος τὸ αὐτὸ ἐστὶ τῷ τοῦ $E\Theta\Pi O$ παραλληλεπιπέδου ὕψει, τὸ δὲ τῆς $AB\Gamma H$ πυραμίδος ὕψος τὸ αὐτὸ ἐστὶ τῷ τοῦ $BHMA$ παραλληλεπιπέδου ὕψει· ἐστὶν ἄρα ὡς ἡ BM βάσις πρὸς τὴν $E\Pi$ βάσιν, οὕτως τὸ τοῦ $E\Theta\Pi O$ παραλληλεπιπέδου ὕψος πρὸς τὸ τοῦ $BHMA$ παραλληλεπιπέδου ὕψος. ὧν δὲ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσα ἐστὶν ἐκείνα· ἴσον ἄρα ἐστὶ τὸ $BHMA$ στερεὸν παραλληλεπιπέδον τῷ $E\Theta\Pi O$ στερεῷ παραλληλεπιπέδῳ. καὶ ἐστὶ τοῦ μὲν $BHMA$ ἕκτον μέρος ἡ $AB\Gamma H$ πυραμὶς, τοῦ δὲ $E\Theta\Pi O$ παραλληλεπιπέδου ἕκτον μέρος ἡ $\Delta EZ\Theta$ πυραμὶς· ἴση ἄρα ἡ $AB\Gamma H$ πυραμὶς τῇ $\Delta EZ\Theta$ πυραμίδι.

Τῶν ἄρα ἴσων πυραμίδων καὶ τριγώνου βάσεις ἐθνουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν καὶ ὧν πυραμίδων τριγώνου βάσεις ἐθνουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσαι εἰσὶν ἐκείναι· ὅπερ εἶδει δεῖχαι.

10.

Πᾶς κῶνος κυλίνδρον τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον.

Ἐθέτω γὰρ κῶνος κυλίνδρῳ βάσιν τε τὴν αὐτὴν τὸν $AB\Gamma\Delta$ κύκλον καὶ ὕψος ἴσον· λέγω, ὅτι ὁ κῶνος τοῦ κυλίνδρου τρίτον ἐστὶ μέρος, τουτέστιν ὅτι ὁ κύλινδρος τοῦ κώνου τριπλασίῳ ἐστίν.

to the height of pyramid $ABCG$. I say that pyramid $ABCG$ is equal to pyramid $DEFH$.

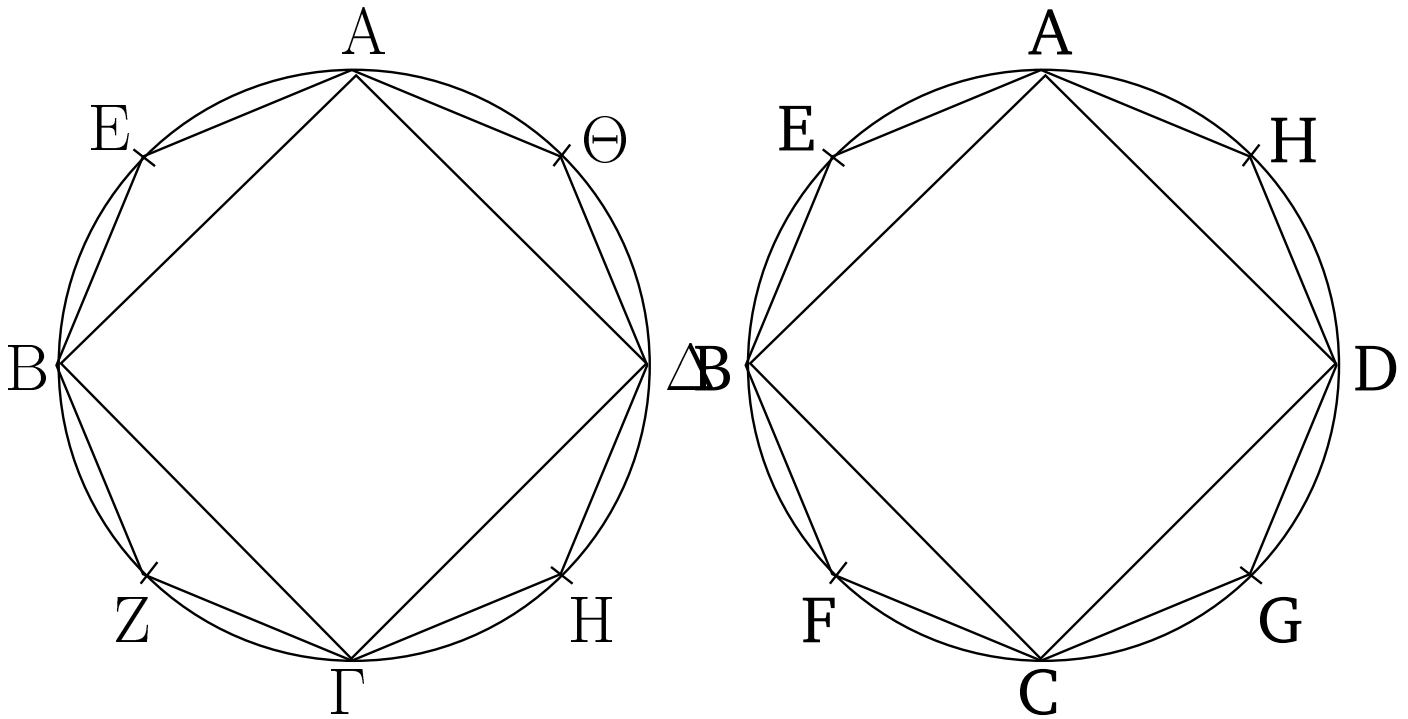
For, with the same construction, since as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$, but as base ABC (is) to base DEF , so parallelogram BM (is) to parallelogram EQ [Prop. 1.34], thus as parallelogram BM (is) to parallelogram EQ , so the height of pyramid $DEFH$ (is) also to the height of pyramid $ABCG$ [Prop. 5.11]. But, the height of pyramid $DEFH$ is the same as the height of parallelepiped $EHQP$, and the height of pyramid $ABCG$ is the same as the height of parallelepiped $BGML$. Thus, as base BM is to base EQ , so the height of parallelepiped $EHQP$ (is) to the height of parallelepiped $BGML$. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal [Prop. 11.34]. Thus, the parallelepiped solid $BGML$ is equal to the parallelepiped solid $EHQP$. And pyramid $ABCG$ is a sixth part of $BGML$, and pyramid $DEFH$ a sixth part of parallelepiped $EHQP$. Thus, pyramid $ABCG$ is equal to pyramid $DEFH$.

Thus, the bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids having triangular bases whose bases are reciprocally proportional to their heights are equal. (Which is) the very thing it was required to show.

Proposition 10

Every cone is the third part of the cylinder which has the same base as it, and an equal height.

For let there be a cone (with) the same base as a cylinder, (namely) the circle $ABCD$, and an equal height. I say that the cone is the third part of the cylinder—that is to say, that the cylinder is three times the cone.



Εἰ γὰρ μὴ ἐστὶν ὁ κύλινδρος τοῦ κώνου τριπλασίων, ἔσται ὁ κύλινδρος τοῦ κώνου ἢ τοῦ μείζων ἢ τριπλασίων ἢ ἐλάσσων ἢ τριπλασίων. ἔστω πρότερον μείζων ἢ τριπλασίων, καὶ ἐγγεγράφθω εἰς τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον τὸ $AB\Gamma\Delta$. τὸ δὴ $AB\Gamma\Delta$ τετράγωνον μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ $AB\Gamma\Delta$ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ $AB\Gamma\Delta$ τετραγώνου πρίσμα ἰσοῦν πρὸς τῷ κυλίνδρῳ. τὸ δὴ ἀνιστάμενον πρίσμα μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ κυλίνδρου, ἐπειδὴ περὶ τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον περιγράφωμεν, τὸ ἐγγεγραμμένον εἰς τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον ἥμισυ ἐστὶ τοῦ περιγεγραμμένου· καὶ ἐστὶ τὰ ἀπ' αὐτῶν ἀνιστάμενα στερεὰ παραλληλεπίπεδα πρίσματα ἰσοῦν· τὰ δὲ ὑπὸ τὸ αὐτὸ ὕψος ὄντα στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐστὶν ὡς αἱ βάσεις· καὶ τὸ ἐπὶ τοῦ $AB\Gamma\Delta$ ἄρα τετραγώνου ἀνασταθέν πρίσμα ἥμισυ ἐστὶ τοῦ ἀνασταθέντος πρίματος ἀπὸ τοῦ περὶ τὸν $AB\Gamma\Delta$ κύκλον περιγραφέντος τετραγώνου· καὶ ἐστὶν ὁ κύλινδρος ἐλάττω τοῦ πρίματος τοῦ ἀνασταθέντος ἀπὸ τοῦ περὶ τὸν $AB\Gamma\Delta$ κύκλον περιγραφέντος τετραγώνου· τὸ ἄρα πρίσμα τὸ ἀνασταθέν ἀπὸ τοῦ $AB\Gamma\Delta$ τετραγώνου ἰσοῦν πρὸς τῷ κυλίνδρῳ μείζον ἐστὶ τοῦ ἡμίσεως τοῦ κυλίνδρου. τεμήσθωσαν αἱ AB , $B\Gamma$, $\Gamma\Delta$, ΔA περιφέρειαι διῖθα κατὰ τὰ E , Z , H , Θ σημεία, καὶ ἐπεζεύθωσαν αἱ AE , EB , BZ , $Z\Gamma$, ΓH , $H\Delta$, $\Delta\Theta$, ΘA · καὶ ἕκαστον ἄρα τῶν AEB , $BZ\Gamma$, $\Gamma H\Delta$, $\Delta\Theta A$ τριγώνων μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ καθ' ἑαυτὸ τμήματος τοῦ $AB\Gamma\Delta$ κύκλου, ὡς ἔμπροσθεν ἐδείκνυμεν. ἀνεστάτω ἐφ' ἑκάστου τῶν AEB , $BZ\Gamma$, $\Gamma H\Delta$, $\Delta\Theta A$ τριγώνων πρίσματα ἰσοῦν πρὸς τῷ κυλίνδρῳ· καὶ ἕκαστον ἄρα τῶν ἀνασταθέντων πρισμάτων μείζον ἐστὶν ἢ τὸ ἥμισυ μέρους τοῦ καθ' ἑαυτὸ τμήματος τοῦ

For if the cylinder is not three times the cone then the cylinder will be either more than three times, or less than three times, (the cone). Let it, first of all, be more than three times (the cone). And let the square $ABCD$ be inscribed in circle $ABCD$ [Prop. 4.6]. So, square $ABCD$ is more than half of circle $ABCD$ [Prop. 12.2]. And let a prism of equal height to the cylinder be set up on square $ABCD$. So, the prism set up is more than half of the cylinder, inasmuch as if we also circumscribe a square around circle $ABCD$ [Prop. 4.7] then the square inscribed in circle $ABCD$ is half of the circumscribed (square). And the solids set up on them are parallelepiped prisms of equal height. And parallelepiped solids having the same height are to one another as their bases [Prop. 11.32]. And, thus, the prism set up on square $ABCD$ is half of the prism set up on the square circumscribed about circle $ABCD$. And the cylinder is less than the prism set up on the square circumscribed about circle $ABCD$. Thus, the prism set up on square $ABCD$ of the same height as the cylinder is more than half of the cylinder. Let the circumferences AB , BC , CD , and DA be cut in half at points E , F , G , and H . And let AE , EB , BF , FC , CG , GD , DH , and HA be joined. And thus each of the triangles AEB , BFC , CGD , and DHA is more than half of the segment of circle $ABCD$ about it, as was shown previously [Prop. 12.2]. Let prisms of equal height to the cylinder be set up on each of the triangles AEB , BFC , CGD , and DHA . And each of the prisms set up is greater than the half part of the segment of the cylinder about it—inasmuch as if we draw (straight-lines) parallel to AB , BC , CD , and DA through

κυλίνδρον, ἐπειδήπερ ἔαν διὰ τῶν E, Z, H, Θ σημείων παραλλήλων ταῖς $AB, BF, \Gamma\Delta, \Delta A$ ἀγάγωμεν, καὶ συμπληρώσωμεν τὰ ἐπὶ τῶν $AB, BF, \Gamma\Delta, \Delta A$ παραλληλόγραμμα, καὶ ἀπ' αὐτῶν ἀναστήσωμεν στερεὰ παραλληλεπίπεδα ἰσοῦσῃ τῷ κυλίνδρῳ, ἑκάστων τῶν ἀνασταθέντων ἡμίση ἐστὶ τὰ πρίσματα τὰ ἐπὶ τῶν $AEB, BZF, \Gamma H\Delta, \Delta\Theta A$ τριγώνων· καὶ ἐστὶ τὰ τοῦ κυλίνδρου τμήματα ἐλάττωνα τῶν ἀνασταθέντων στερεῶν παραλληλεπιπέδων· ὥστε καὶ τὰ ἐπὶ τῶν $AEB, BZF, \Gamma H\Delta, \Delta\Theta A$ τριγώνων πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῶν καθ' ἑαυτὰ τοῦ κυλίνδρου τμημάτων. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας διῖθα καὶ ἐπιξενυγνόντες εὐθείας καὶ ἀνοστάντες ἑφ' ἑκάστων τῶν τριγώνων πρίσματα ἰσοῦσῃ τῷ κυλίνδρῳ καὶ τοῦτο αἰ ποιοῦντες καταλείβομεν τινα ἀποτμήματα τοῦ κυλίνδρου, ἃ ἔσται ἐλάττωνα τῆς ὑπεροθῆς, ἣ ὑπερέθει ὁ κυλίνδρος τοῦ τριπλασίου τοῦ κώνου. λελείφθω, καὶ ἔστω τὰ $AE, EB, BZ, ZF, \Gamma H, H\Delta, \Delta\Theta, \Theta A$ · λοιπὸν ἄρα τὸ πρίσμα, οὗ βάσις μὲν τὸ $AEBZ\Gamma H\Delta\Theta$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, μείζον ἐστὶν ἢ τριπλάσιον τοῦ κώνου. ἀλλὰ τὸ πρίσμα, οὗ βάσις μὲν ἐστὶ τὸ $AEBZ\Gamma H\Delta\Theta$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, τριπλάσιόν ἐστι τῆς πυραμίδος, ἥς βάσις μὲν ἐστὶ τὸ $AEBZ\Gamma H\Delta\Theta$ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ· καὶ ἡ πυραμὶς ἄρα, ἥς βάσις μὲν [ἐστὶ] τὸ $AEBZ\Gamma H\Delta\Theta$ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶ τοῦ κώνου τοῦ βάσιν ἔθοντες τὸν $AB\Gamma\Delta$ κύκλον. ἀλλὰ καὶ ἐλάττων ἐμπεριέθεται γὰρ ὑπ' αὐτοῦ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὁ κύλινδρος τοῦ κώνου μείζων ἢ τριπλάσιος.

Λέγω δὴ, ὅτι οὐδὲ ἐλάττων ἐστὶν ἢ τριπλάσιος ὁ κύλινδρος τοῦ κώνου.

Εἰ γὰρ δυνατόν, ἔστω ἐλάττων ἢ τριπλάσιος ὁ κύλινδρος τοῦ κώνου· ἀνάπαλιν ἄρα ὁ κώνος τοῦ κυλίνδρου μείζων ἐστὶν ἢ τρίτον μέρος. ἐγγεγράφθω δὴ εἰς τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον τὸ $AB\Gamma\Delta$ · τὸ $AB\Gamma\Delta$ ἄρα τετράγωνον μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ $AB\Gamma\Delta$ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ $AB\Gamma\Delta$ τετραγώνου πυραμὶς τὴν αὐτὴν κορυφὴν ἔθονσα τῷ κώνῳ· ἡ ἄρα ἀνασταθεῖσα πυραμὶς μείζων ἐστὶν ἢ τὸ ἥμισυ μέρος τοῦ κώνου, ἐπειδήπερ, ὡς ἐμπροσθεν ἐδείκνυμεν, ὅτι ἔαν περὶ τὸν κύκλον τετράγωνον περιγράψωμεν, ἔσται τὸ $AB\Gamma\Delta$ τετράγωνον ἥμισυ τοῦ περὶ τὸν κύκλον περιγεγραμμένου τετραγώνου· καὶ ἔαν ἀπὸ τῶν τετραγώνων στερεὰ παραλληλεπίπεδα ἀναστήσωμεν ἰσοῦσῃ τῷ κώνῳ, ἃ καὶ καλεῖται πρίσματα, ἔσται τὸ ἀνασταθέν ἀπὸ τοῦ $AB\Gamma\Delta$ τετραγώνου ἥμισυ τοῦ ἀνασταθέντος ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου· πρὸς ἄλληλα γὰρ εἰσιν ὡς αἱ βάσεις. ὥστε καὶ τὰ τρίτα· καὶ πυραμὶς ἄρα, ἥς βάσις τὸ $AB\Gamma\Delta$ τετράγωνον, ἥμισυ ἐστὶ τῆς πυραμίδος τῆς ἀνασταθείσης ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου. καὶ ἐστὶ μείζων ἢ πυραμὶς ἢ ἀνασταθεῖσα ἀπὸ τοῦ περὶ τὸν κύκλον τετραγώνου τοῦ κώνου· ἐμπεριέθει γὰρ αὐτόν. ἡ ἄρα πυραμὶς, ἥς βάσις τὸ $AB\Gamma\Delta$ τετράγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶν ἢ τὸ ἥμισυ τοῦ κώνου. τεμήσθωσαν αἱ $AB, BF, \Gamma\Delta, \Delta A$ περιφέρειαι διῖθα κατὰ τὰ E, Z, H, Θ σημεία, καὶ ἐπεξέυθωσαν αἱ

points E, F, G , and H (respectively), and complete the parallelograms on AB, BC, CD , and DA , and set up parallelepiped solids of equal height to the cylinder on them, then the prisms on triangles AEB, BFC, CGD , and DHA are each half of the set up (parallelepipeds). And the segments of the cylinder are less than the set up parallelepiped solids. Hence, the prisms on triangles AEB, BFC, CGD , and DHA are also greater than half of the segments of the cylinder about them. So (if) the remaining circumferences are cut in half, and straight-lines are joined, and prisms of equal height to the cylinder are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cylinder whose (sum) is less than the excess by which the cylinder exceeds three times the cone [Prop. 10.1]. Let them be left, and let them be $AE, EB, BF, FC, CG, GD, DH$, and HA . Thus, the remaining prism whose base (is) polygon $AEBFCGDH$, and height the same as the cylinder, is greater than three times the cone. But, the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder, is three times the pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone [Prop. 12.7 corr.]. And thus the pyramid whose base [is] polygon $AEBFCGDH$, and apex the same as the cone, is greater than the cone having (as) base circle $ABCD$. But (it is) also less. For it is encompassed by it. The very thing (is) impossible. Thus, the cylinder is not more than three times the cone.

So, I say that neither (is) the cylinder less than three times the cone.

For, if possible, let the cylinder be less than three times the cone. Thus, inversely, the cone is greater than the third part of the cylinder. So, let the square $ABCD$ be inscribed in circle $ABCD$ [Prop. 4.6]. Thus, square $ABCD$ is greater than half of circle $ABCD$. And let a pyramid having the same apex as the cone be set up on square $ABCD$. Thus, the pyramid set up is greater than the half part of the cone, inasmuch as we showed previously that if we circumscribe a square about the circle [Prop. 4.7] then the square $ABCD$ will be half of the square circumscribed about the circle [Prop. 12.2]. And if we set up on the squares parallelepiped solids—which are also called prisms—of the same height as the cone, then the (prism) set up on square $ABCD$ will be half of the (prism) set up on the square circumscribed about the circle. For they are to one another as their bases [Prop. 11.32]. Hence, (the same) also (goes for) the thirds. Thus, the pyramid whose base is square $ABCD$ is half of the pyramid set up on the square circumscribed about the circle [Prop. 12.7 corr.]. And the pyramid set up on the square circumscribed about the circle is greater than the cone. For it encompasses it. Thus, the pyramid whose base is square $ABCD$, and apex the same as the cone, is greater than half of the cone. Let the circumferences AB, BC, CD , and DA be cut in half at points E, F, G , and H (respectively). And let $AE, EB,$

$AE, EB, BZ, ZΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ$ · καὶ ἕκαστον ἄρα τῶν $AEB, BZΓ, ΓΗΔ, ΔΘΑ$ τριγώνων μείζον ἐστὶν ἢ τὸ ἥμισυ μέρος του καθ' ἑαυτὸ τμήματος τοῦ $ABΓΔ$ κύκλου. καὶ ἀνεστάτωσαν ἐφ' ἕκαστον τῶν $AEB, BZΓ, ΓΗΔ, ΔΘΑ$ τριγώνων πυραμίδες τὴν αὐτὴν κορυφὴν ἔθουσαι τῷ κώνῳ· καὶ ἕκαστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων κατὰ τὸν αὐτὸν τρόπον μείζων ἐστὶν ἢ τὸ ἥμισυ μέρος τοῦ καθ' ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὲ τὰς ὑπολειπομένας περιφερείας διῆθα καὶ ἐπιξενυγνύντες εὐθείας καὶ ἀνιστάντες ἐφ' ἕκαστον τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴν ἔθουσας τῷ κώνῳ καὶ τοῦτο αἰ ποιοῦντες καταλείβομεν τινα ἀποτμήματα τοῦ κώνου, ἃ ἔσται ἐλάττωνα τῆς ὑπεροθῆς, ἢ ὑπερέθει ὁ κώνος τοῦ τρίτου μέρους τοῦ κυλίνδρου. λελείφθω, καὶ ἔστω τὰ ἐπὶ τῶν $AE, EB, BZ, ZΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ$ · λοιπὴ ἄρα ἡ πυραμὶς, ἥς βάσις μὲν ἐστὶ τὸ $AEBZΓΗΔΘ$ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶν ἢ τρίτον μέρος τοῦ κυλίνδρου. ἀλλ' ἡ πυραμὶς, ἥς βάσις μὲν ἐστὶ τὸ $AEBZΓΗΔΘ$ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, τρίτον ἐστὶ μέρος τοῦ πρίσματος, ὃς βάσις μὲν ἐστὶ τὸ $AEBZΓΗΔΘ$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ· τὸ ἄρα πρίσμα, ὃς βάσις μὲν ἐστὶ τὸ $AEBZΓΗΔΘ$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, μείζον ἐστὶ τοῦ κυλίνδρου, ὃς βάσις ἐστὶν ὁ $ABΓΔ$ κύκλος. ἀλλὰ καὶ ἐλάττων ἐμπεριέχεται γὰρ ὑπ' αὐτοῦ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ κύλινδρος τοῦ κώνου ἐλάττων ἐστὶν ἢ τριπλάσιος. ἐδείχθη δέ, ὅτι οὐδὲ μείζων ἢ τριπλάσιος· τριπλάσιος ἄρα ὁ κύλινδρος τοῦ κώνου· ὥστε ὁ κώνος τρίτον ἐστὶ μέρος τοῦ κυλίνδρου.

Πᾶς ἄρα κώνος κυλίνδρου τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον· ὅπερ ἔδει δεῖξαι.

11.

Οἱ ὑπο τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρόξ ἀλλήλων εἰσὶν ὥς αἱ βάσεις.

Ἐστωσαν ὑπὸ τὸ αὐτὸ ὕψος κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν [εἰσιν] οἱ $ABΓΔ, EZHΘ$ κύκλοι, ἄχονες δὲ οἱ KL, MN , διαμέτροι δὲ τῶν βάσεων αἱ AG, EH · λέγω, ὅτι ἐστὶν ὥς ὁ $ABΓΔ$ κύκλος πρόξ τὸν $EZHΘ$ κύκλον, οὕτως ὁ AL κώνος πρόξ τὸν EN κώνον.

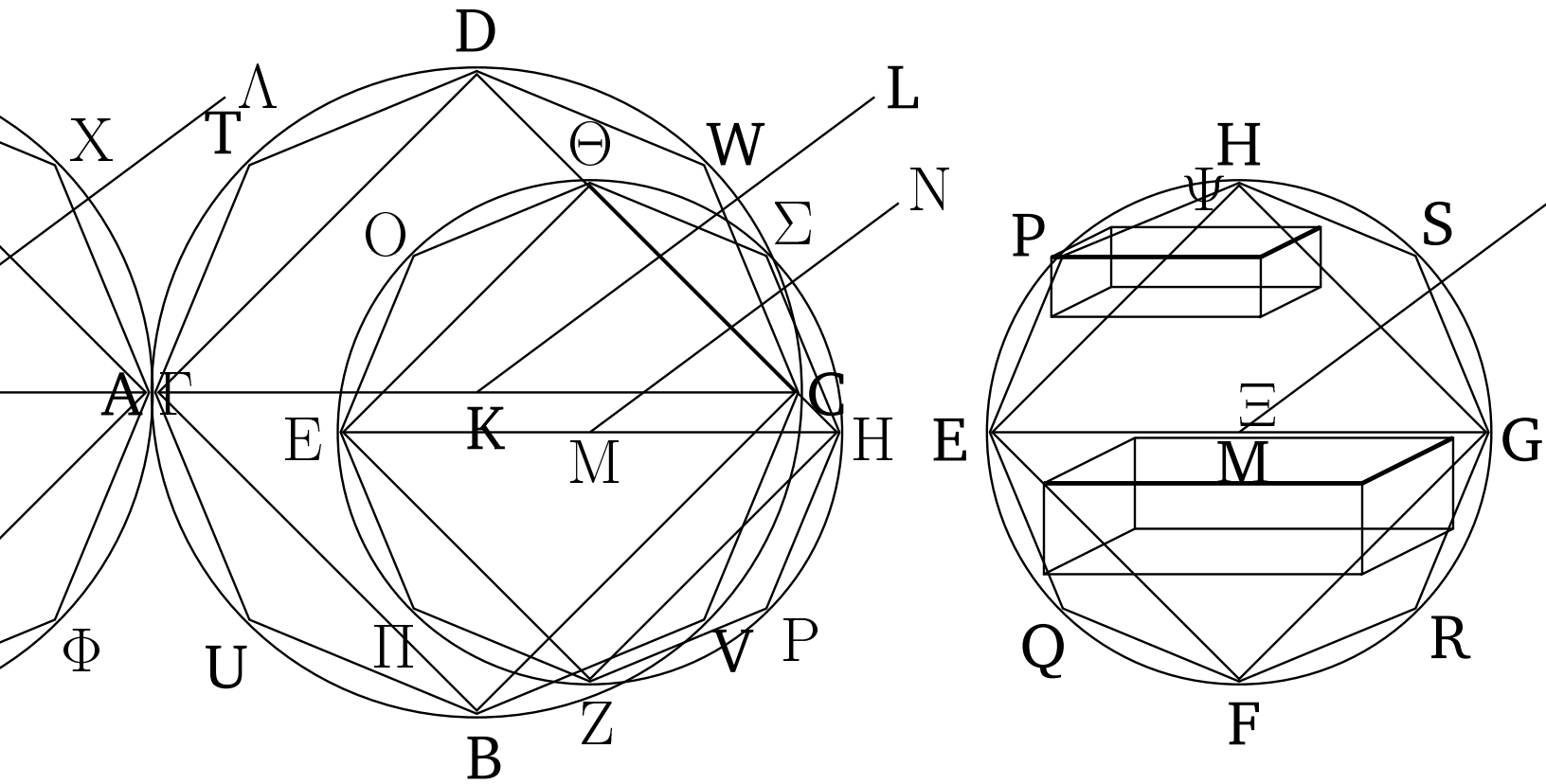
BF, FC, CG, GD, DH , and HA be joined. And, thus, each of the triangles AEB, BFC, CGD , and DHA is greater than the half part of the segment of circle $ABCD$ about it [Prop. 12.2]. And let pyramids having the same apex as the cone be set up on each of the triangles AEB, BFC, CGD , and DHA . And, thus, in the same way, each of the pyramids set up is more than the half part of the segment of the cone about it. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which the cone exceeds the third part of the cylinder [Prop. 10.1]. Let them be left, and let them be the (segments) on $AE, EB, BF, FC, CG, GD, DH$, and HA . Thus, the remaining pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is greater than the third part of the cylinder. But, the pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is the third part of the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder [Prop. 12.7 corr.]. Thus, the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder, is greater than the cylinder whose base is circle $ABCD$. But, (it is) also less. For it is encompassed by it. The very thing is impossible. Thus, the cylinder is not less than three times the cone. And it was shown that neither (is it) greater than three times (the cone). Thus, the cylinder (is) three times the cone. Hence, the cone is the third part of the cylinder.

Thus, every cone is the third part of the cylinder which has the same base as it, and an equal height. (Which is) the very thing it was required to show.

Proposition 11

Cones and cylinders having the same height are to one another as their bases.

Let there be cones and cylinders of the same height whose bases [are] the circles $ABCD$ and $EFGH$, axes KL and MN , and diameters of the bases AC and EG (respectively). I say that as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to cone EN .



Εἰ γὰρ μή, ἔσται ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως ὁ AL κώνοξ ἦτοι πρὸς ἑλασσόν τι τοῦ EN κώνου στερεόν ἢ πρὸς μείζον. ἔστω πρότερον πρὸς ἑλασσον τὸ X , καὶ ὥς ἑλασσόν ἐστι τὸ X στερεόν τοῦ EN κώνου, ἐκεῖνῳ ἴσον ἔστω τὸ Ψ στερεόν· ὁ EN κώνοξ ἄρα ἴσος ἐστὶ τοῖς X , Ψ στερεοῖς. ἐγγεγράφω εἰς τὸν $EZH\Theta$ κύκλον τετράγωνον τὸ $EZH\Theta$ · τὸ ἄρα τετράγωνον μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ κύκλου. ἀνεστίτω ἀπὸ τοῦ $EZH\Theta$ τετραγώνου πυραμὶς ἰσοῦψῆς τῷ κώνῳ· ἡ ἄρα ἀνασταθεῖσα πυραμὶς μείζων ἐστὶν ἢ τὸ ἥμισυ τοῦ κώνου, ἐπειδὴ περ ἔαν περιγράψωμεν περὶ τὸν κύκλον τετράγωνον, καὶ ἀπ' αὐτοῦ ἀναστήσωμεν πυραμίδα ἰσοῦψῆ τῷ κώνῳ, ἡ ἐγγραφεῖσα πυραμὶς ἥμισυ ἐστὶ τῆς περιγραφείσης· πρὸς ἀλλήλας γὰρ εἰσιν ὡς αἱ βάσεις· ἐλάττων δὲ ὁ κώνοξ τῆς περιγραφείσης πυραμίδος. τετμήσθωσαν αἱ EZ , ZH , $H\Theta$, ΘE περιφέρειαι δίδυα κατὰ τὰ O , Π , P , Σ σημεία, καὶ ἐπεζεύθωσαν αἱ ΘO , $O E$, $E\Pi$, ΠZ , $Z P$, $P H$, $H\Sigma$, $\Sigma\Theta$. ἕκαστον ἄρα τῶν $\Theta O E$, $E\Pi Z$, $Z P H$, $H Z \Theta$ τριγώνων μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ καθ' ἑαυτὸ τμήματος τοῦ κύκλου. ἀνεστίτω ἐφ' ἑκάστον τῶν $\Theta O E$, $E\Pi Z$, $Z P H$, $H\Sigma\Theta$ τριγώνων πυραμὶς ἰσοῦψῆς τῷ κώνῳ· καὶ ἑκάστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων μείζων ἐστὶν ἢ τὸ ἥμισυ τοῦ καθ' ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας δίδυα καὶ ἐπιζευγνόντες εὐθείας καὶ ἀνιστάντες ἐπὶ ἑκάστον τῶν τριγώνων πυραμίδας ἰσοῦψεῖς τῷ κώνῳ καὶ ἀεὶ τοῦτο ποιοῦντες καταλείψομεν τινα ἀποτμήματα τοῦ κώνου,

For if not, then as circle $ABCD$ (is) to circle $EFGH$, so cone AL will be to some solid either less than, or greater than, cone EN . Let it, first of all, be (in this ratio) to (some) lesser (solid), O . And let solid X be equal to that (magnitude) by which solid O is less than cone EN . Thus, cone EN is equal to (the sum of) solids O and X . Let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. Thus, the square is greater than half of the circle [Prop. 12.2]. Let a pyramid of the same height as the cone be set up on square $EFGH$. Thus, the pyramid set up is greater than half of the cone, inasmuch as, if we circumscribe a square about the circle [Prop. 4.7], and set up on it a pyramid of the same height as the cone, then the inscribed pyramid is half of the circumscribed pyramid. For they are to one another as their bases [Prop. 12.6]. And the cone (is) less than the circumscribed pyramid. Let the circumferences EF , FG , GH , and HE be cut in half at points P , Q , R , and S . And let HP , PE , EQ , QF , FR , RG , GS , and SH be joined. Thus, each of the triangles HPE , EQF , FRG , and GSH is greater than half of the segment of the circle about it [Prop. 12.2]. Let pyramids of the same height as the cone be set up on each of the triangles HPE , EQF , FRG , and GSH . And, thus, each of the pyramids set up is greater than half of the segment of the cone about it [Prop. 12.10]. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids of equal height to the cone are

ἂ ἔσται ἐλάσσονα τοῦ Ψ στερεοῦ. λελείφθω, καὶ ἔστω τὰ ἐπὶ τῶν $\Theta\Omega\epsilon$, $\epsilon\Pi\zeta$, $\zeta\rho\eta$, $\eta\varsigma\theta$: λοιπὴ ἄρα ἡ πυραμὶς, ἣς βάσις τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\varsigma$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κώνῳ, μείζων ἐστὶ τοῦ X στερεοῦ. ἐγγεγράφθω καὶ εἰς τὸν $AB\Gamma\Delta$ κύκλον τῷ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\varsigma$ πολυγώνῳ ὁμοίον τε καὶ ὁμοίως κείμενον πολύγωνον τὸ $\Delta TAYB\Phi\Gamma\Theta$, καὶ ἀνεστιάτω ἐπ' αὐτοῦ πυραμὶς ἰσοῦν ἡς τῷ AA κώνῳ. ἐπεὶ οὖν ἐστὶν ὡς τὸ ἀπὸ τῆς AG πρὸς τὸ ἀπὸ τῆς EH , οὕτως τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον πρὸς τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\varsigma$ πολύγωνον, ὡς δὲ τὸ ἀπὸ τῆς AG πρὸς τὸ ἀπὸ τῆς EH , οὕτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\theta$ κύκλον, καὶ ὡς ἄρα ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\theta$ κύκλον, οὕτως τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον πρὸς τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\varsigma$ πολύγωνον. ὡς δὲ ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς τὸ X στερεόν, ὡς δὲ τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον πρὸς τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\varsigma$ πολύγωνον, οὕτως ἡ πυραμὶς, ἣς βάσις μὲν τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον, κορυφὴ δὲ τὸ A σημεῖον, πρὸς τὴν πυραμίδα, ἣς βάσις μὲν τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\varsigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον. καὶ ὡς ἄρα ὁ AA κώνος πρὸς τὸ X στερεόν, οὕτως ἡ πυραμὶς, ἣς βάσις μὲν τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον, κορυφὴ δὲ τὸ A σημεῖον, πρὸς τὴν πυραμίδα, ἣς βάσις μὲν τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\varsigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον ἐναλλάξ ἄρα ἐστὶν ὡς ὁ AA κώνος πρὸς τὴν ἐν αὐτῷ πυραμίδα, οὕτως τὸ X στερεόν πρὸς τὴν ἐν τῷ EN κώνῳ πυραμίδα. μείζων δὲ ὁ AA κώνος τῆς ἐν αὐτῷ πυραμίδος· μείζων ἄρα καὶ τὸ X στερεόν τῆς ἐν τῷ EN κώνῳ πυραμίδος. ἀλλὰ καὶ ἔλασσον· ὅπερ ἄτοπον. οὐκ ἄρα ἐστὶν ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς ἔλασσόν τι τοῦ EN κώνου στερεόν. ὁμοίως δὲ δείχομεν, ὅτι οὐδὲ ἐστὶν ὡς ὁ $EZH\theta$ κύκλος πρὸς τὸν $AB\Gamma\Delta$ κύκλον, οὕτως ὁ EN κώνος πρὸς ἔλασσόν τι τοῦ AA κώνου στερεόν.

Λέγω δὴ, ὅτι οὐδὲ ἐστὶν ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς μείζον τι τοῦ EN κώνου στερεόν.

Εἰ γὰρ δυνατόν, ἔστω πρὸς μείζον τὸ X : ἀνάπαλιν ἄρα ἐστὶν ὡς ὁ $EZH\theta$ κύκλος πρὸς τὸν $AB\Gamma\Delta$ κύκλον, οὕτως τὸ X στερεόν πρὸς τὸν AA κώνον. ἀλλ' ὡς τὸ X στερεόν πρὸς τὸν AA κώνον, οὕτως ὁ EN κώνος πρὸς ἔλασσόν τι τοῦ AA κώνου στερεόν· καὶ ὡς ἄρα ὁ $EZH\theta$ κύκλος πρὸς τὸν $AB\Gamma\Delta$ κύκλον, οὕτως ὁ EN κώνος πρὸς ἔλασσόν τι τοῦ AA κώνου στερεόν· ὅπερ ἀδύνατον ἐδείχθη. οὐκ ἄρα ἐστὶν ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς μείζον τι τοῦ EN κώνου στερεόν. ἐδείχθη δέ, ὅτι οὐδὲ πρὸς ἔλασσον· ἐστὶν ἄρα ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς τὸν EN κώνον.

Ἀλλ' ὡς ὁ κώνος πρὸς τὸν κώνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον τριπλασίῳ γὰρ ἐκάτερος ἐκατέρων. καὶ ὡς ἄρα ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\theta$ κύκλον, οὕτως οἱ ἐπ' αὐτῶν ἰσοῦν εἶξ.

Οἱ ἄρα ὑπὸ τὸ αὐτὸ ὕψος ὄντες κώνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖξαι.

set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone (the sum of) which is less than solid X [Prop. 10.1]. Let them be left, and let them be the (segments) on HPE , EQF , FRG , and GSH . Thus, the remaining pyramid whose base is polygon $HPEQFRGS$, and height the same as the cone, is greater than solid O [Prop. 6.18]. And let the polygon $DTAUBVCW$, similar, and similarly laid out, to polygon $HPEQFRGS$, be inscribed in circle $ABCD$. And on it let a pyramid of the same height as cone AL be set up. Therefore, since as the (square) on AC is to the (square) on EG , so polygon $DTAUBVCW$ (is) to polygon $HPEQFRGS$ [Prop. 12.1], and as the (square) on AC (is) to the (square) on EG , so circle $ABCD$ (is) to circle $EFGH$ [Prop. 12.2], thus as circle $ABCD$ (is) to circle $EFGH$, so polygon $DTAUBVCW$ also (is) to polygon $HPEQFRGS$. And as circle $ABCD$ (is) to circle $EFGH$, so cone AL (is) to solid O . And as polygon $DTAUBVCW$ (is) to polygon $HPEQFRGS$, so the pyramid whose base is polygon $DTAUBVCW$, and apex the point L , (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 12.6]. And, thus, as cone AL (is) to solid O , so the pyramid whose base is $DTAUBVCW$, and apex the point L , (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 5.11]. Thus, alternately, as cone AL is to the pyramid within it, so solid O (is) to the pyramid within cone EN [Prop. 5.16]. But, cone AL (is) greater than the pyramid within it. Thus, solid O (is) also greater than the pyramid within cone EN [Prop. 5.14]. But, (it is) also less. The very thing (is) absurd. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid less than cone EN . So, similarly, we can show that neither is circle $EFGH$ to circle $ABCD$, as cone EN (is) to some solid less than cone AL .

So, I say that neither is circle $ABCD$ to circle $EFGH$, as cone AL (is) to some solid greater than cone EN .

For, if possible, let it be (in this ratio) to (some) greater (solid), O . Thus, inversely, as circle $EFGH$ is to circle $ABCD$, so solid O (is) to cone AL [Prop. 5.7 corr.]. But, as solid O (is) to cone AL , so cone EN (is) to some solid less than cone AL [Prop. 12.2 lem.]. And, thus, as circle $EFGH$ (is) to circle $ABCD$, so cone EN (is) to some solid less than cone AL . The very thing was shown (to be) impossible. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid greater than cone EN . And, it was shown that neither (is it in this ratio) to (some) lesser (solid). Thus, as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to cone EN .

But, as the cone (is) to the cone, (so) the cylinder (is) to the cylinder. For each (is) three times each [Prop. 12.10]. Thus, circle $ABCD$ (is) also to circle $EFGH$, as (the ratio of the cylinders) on them (having) the same height.

Thus, cones and cylinders having the same height are to one another as their bases. (Which is) the very thing it was

required to show.

12.

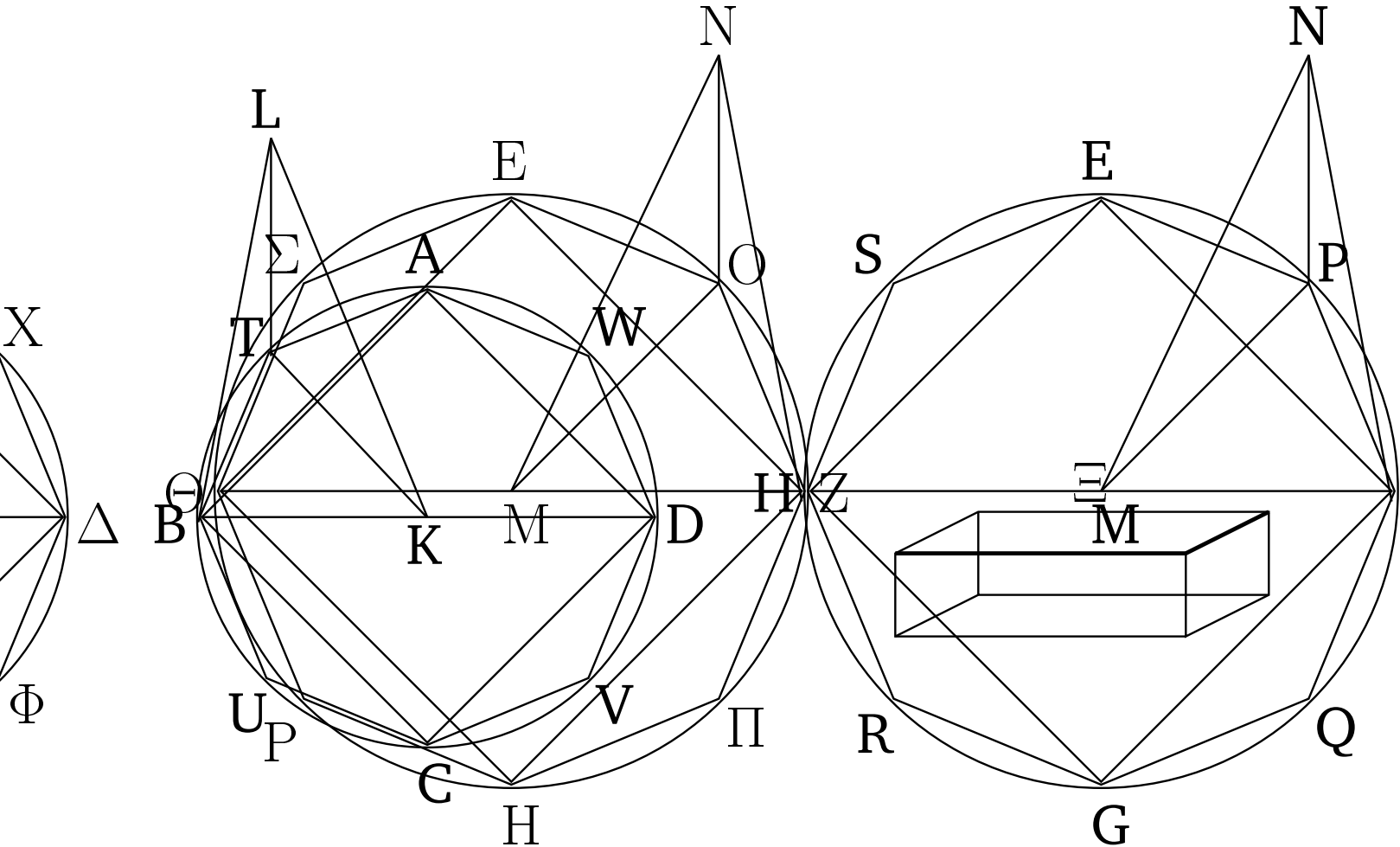
Οἱ ὅμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἐν ταῖς βάσεσι διαμέτρων.

Ἐστωσαν ὅμοιοι κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν οἱ $ABΓΔ$, $EZHΘ$ κύκλοι, διαμέτροι δὲ τῶν βάσεων αἱ $BΔ$, $ZΘ$, ἄχονες δὲ τῶν κῶνων καὶ κυλίνδρων οἱ KL , MN . λέγω, ὅτι ὁ κῶνος, ὃν βάσις μὲν [ἐστίν] ὁ $ABΓΔ$ κύκλος, κορυφή δὲ τὸ $Λ$ σημεῖον, πρὸς τὸν κῶνον, ὃν βάσις μὲν [ἐστίν] ὁ $EZHΘ$ κύκλος, κορυφή δὲ τὸ N σημεῖον, τριπλασίονα λόγον ἔθχει ἢ $BΔ$ πρὸς τὴν $ZΘ$.

Proposition 12

Similar cones and cylinders are to one another in the cubed ratio of the diameters of their bases.

Let there be similar cones and cylinders of which the bases (are) the circles $ABCD$ and $EFGH$, the diameters of the bases (are) BD and FH , and the axes of the cones and cylinders (are) KL and MN (respectively). I say that the cone whose base [is] circle $ABCD$, and apex the point L , has to the cone whose base [is] circle $EFGH$, and apex the point N , the cubed ratio that BD (has) to FH .



Εἰ γὰρ μὴ ἔθχει ὁ $ABΓΔΛ$ κῶνος πρὸς τὸν $EZHΘN$ κῶνον τριπλασίονα λόγον ἢ περὶ ἢ $BΔ$ πρὸς τὴν $ZΘ$, ἔχει ὁ $ABΓΔΛ$ κῶνος ἢ πρὸς ἔλασσόν τι τοῦ $EZHΘN$ κῶνον στερεὸν τριπλασίονα λόγον ἢ πρὸς μείζον. ἐθέτω πρότερον πρὸς ἔλασσον τὸ X , καὶ ἐγγεγράφθω εἰς τὸν $EZHΘ$ κύκλον τετράγωνον τὸ $EZHΘ$. τὸ ἄρα $EZHΘ$ τετράγωνον μείζον ἐστίν ἢ τὸ ἥμισυ τοῦ $EZHΘ$ κύκλου. καὶ ἀνεστάτω ἐπὶ τοῦ $EZHΘ$ τετραγώνου

For if cone $ABCDL$ does not have to cone $EFGHN$ the cubed ratio that BD (has) to FH then cone $ABCDL$ will have the cubed ratio to some solid either less than, or greater than, cone $EFGHN$. Let it, first of all, have (such a ratio) to (some) lesser (solid), O . And let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. Thus, square $EFGH$ is greater than half of circle $EFGH$ [Prop. 12.2]. And let a pyramid hav-

πυραμὶς τὴν αὐτὴν κορυφὴν ἔθουσα τῷ κώνῳ· ἡ ἄρα ἀνασταθεῖσα πυραμὶς μείζων ἐστὶν ἢ τὸ ἥμισυ μέρους τοῦ κώνου. τετμήσθωσαν δὴ αἱ EZ , ZH , $H\Theta$, ΘE περιφέρειαι διὰ κατὰ τὰ O , Π , P , Σ σημεία, καὶ ἐπεξεύθωσαν αἱ EO , OZ , $Z\Pi$, ΠH , HP , $P\Theta$, $\Theta\Sigma$, ΣE . καὶ ἕκαστον ἄρα τῶν EOZ , $Z\Pi H$, $HP\Theta$, $\Theta\Sigma E$ τριγώνων μείζον ἐστὶν ἢ τὸ ἥμισυ μέρους τοῦ καθ' ἑαυτὸ τμήματος τοῦ $EZH\Theta$ κύκλου. καὶ ἀνεστιάτω ἐφ' ἕκαστον τῶν EOZ , $Z\Pi H$, $HP\Theta$, $\Theta\Sigma E$ τριγώνων πυραμὶς τὴν αὐτὴν κορυφὴν ἔθουσα τῷ κώνῳ· καὶ ἕκαστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων μείζων ἐστὶν ἢ τὸ ἥμισυ μέρους τοῦ καθ' ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὲ τὰς ὑπολειπομένας περιφερείας διὰ καὶ ἐπιξενγνόντες εὐθείας καὶ ἀνοστήσαντες ἐφ' ἕκαστον τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴν ἔθουσας τῷ κώνῳ καὶ τοῦτο αἰεὶ ποιοῦντες καταλείπομέναι τινα ἀποτμήματα τοῦ κώνου, ἃ ἔσται ἐλάσσονα τῆς ὑπεροπῆς, ἣ ὑπερέθει ὁ $EZH\Theta N$ κώνος τοῦ X στερεοῦ. λελεῖφθω, καὶ ἔστω τὰ ἐπὶ τῶν EO , OZ , $Z\Pi$, ΠH , HP , $P\Theta$, $\Theta\Sigma$, ΣE λοιπὴ ἄρα ἡ πυραμὶς, ἥς βάσις μὲν ἐστὶ τὸ $EOZ\Pi HP\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον, μείζων ἐστὶ τοῦ X στερεοῦ. ἐγγεγράφθω καὶ εἰς τὸν $AB\Gamma\Delta$ κύκλον τῷ $EOZ\Pi HP\Theta\Sigma$ πολυγώνῳ ὁμοίον τε καὶ ὁμοίως κείμενον πολυγώνον τὸ $ATBYT\Phi\Delta\Theta$, καὶ ἀνεστιάτω ἐπὶ τοῦ $ATBYT\Phi\Delta\Theta$ πολυγώνου πυραμὶς τὴν αὐτὴν κορυφὴν ἔθουσα τῷ κώνῳ, καὶ τῶν μὲν περιεθόντων τὴν πυραμίδα, ἥς βάσις μὲν ἐστὶ τὸ $ATBYT\Phi\Delta\Theta$ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, ἐν τριγώνον ἔστω τὸ ΛBT , τῶν δὲ περιεθόντων τὴν πυραμίδα, ἥς βάσις μὲν ἐστὶ τὸ $EOZ\Pi HP\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον, ἐν τριγώνον ἔστω τὸ NZO , καὶ ἐπεξεύθωσαν αἱ KT , MO . καὶ ἐπεὶ ὁμοίός ἐστιν ὁ $AB\Gamma\Delta$ κώνος τῷ $EZH\Theta N$ κώνῳ, ἔστιν ἄρα ὥς ἡ $B\Delta$ πρὸς τὴν $Z\Theta$, οὕτως ὁ $K\Lambda$ ἄκων πρὸς τὸν MN ἄκωνα. ὥς δὲ ἡ $B\Delta$ πρὸς τὴν $Z\Theta$, οὕτως ἡ BK πρὸς τὴν ZM · καὶ ὥς ἄρα ἡ BK πρὸς τὴν ZM , οὕτως ἡ $K\Lambda$ πρὸς τὴν MN . καὶ ἐναλλάξ ὥς ἡ BK πρὸς τὴν $K\Lambda$, οὕτως ἡ ZM πρὸς τὴν MN . καὶ περὶ ἴσας γωνίας τὰς ὑπὸ BKA , ZMN αἱ πλευραὶ ἀνάλογόν εἰσιν· ὁμοίον ἄρα ἐστὶ τὸ BKA τρίγωνον τῷ ZMN τριγώνῳ. πάλιν, ἐπεὶ ἐστὶν ὥς ἡ BK πρὸς τὴν KT , οὕτως ἡ ZM πρὸς τὴν MO , καὶ περὶ ἴσας γωνίας τὰς ὑπὸ BKT , ZMO , ἐπειδήπερ, ὁ μέρους ἐστὶν ἡ ὑπὸ BKT γωνία τῶν πρὸς τῷ K κέντρῳ τεσσάρων ὀρθῶν, τὸ αὐτὸ μέρος ἐστὶ καὶ ἡ ὑπὸ ZMO γωνία τῶν πρὸς τῷ M κέντρῳ τεσσάρων ὀρθῶν· ἐπεὶ οὖν περὶ ἴσας γωνίας αἱ πλευραὶ ἀνάλογόν εἰσιν, ὁμοίον ἄρα ἐστὶ τὸ BKT τρίγωνον τῷ ZMO τριγώνῳ. πάλιν, ἐπεὶ ἐδείχθη ὥς ἡ BK πρὸς τὴν $K\Lambda$, οὕτως ἡ ZM πρὸς τὴν MN , ἴση δὲ ἡ μὲν BK τῇ KT , ἡ δὲ ZM τῇ OM , ἔστιν ἄρα ὥς ἡ TK πρὸς τὴν $K\Lambda$, οὕτως ἡ OM πρὸς τὴν MN . καὶ περὶ ἴσας γωνίας τὰς ὑπὸ TKA , OMN · ὀρθαὶ γάρ· αἱ πλευραὶ ἀνάλογόν εἰσιν· ὁμοίον ἄρα ἐστὶ τὸ AKT τρίγωνον τῷ NMO τριγώνῳ. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν AKB , NMZ τριγώνων ἐστὶν ὥς ἡ AB πρὸς τὴν BK , οὕτως ἡ NZ πρὸς τὴν ZM , διὰ δὲ τὴν ὁμοιότητα τῶν BKT , ZMO τριγώνων ἐστὶν ὥς ἡ KB πρὸς τὴν BT , οὕτως ἡ MZ πρὸς

ing the same apex as the cone be set up on square $EFGH$. Thus, the pyramid set up is greater than the half part of the cone [Prop. 12.10]. So, let the circumferences EF , FG , GH , and HE be cut in half at points P , Q , R , and S (respectively). And let EP , PF , FQ , QG , GR , RH , HS , and SE be joined. And, thus, each of the triangles EPF , FQG , GRH , and HSE is greater than the half part of the segment of circle $EFGH$ about it [Prop. 12.2]. And let a pyramid having the same apex as the cone be set up on each of the triangles EPF , FQG , GRH , and HSE . And thus each of the pyramids set up is greater than the half part of the segment of the cone about it [Prop. 12.10]. So, (if) the the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which cone $EFGHN$ exceeds solid O [Prop. 10.1]. Let them be left, and let them be the (segments) on EP , PF , FQ , QG , GR , RH , HS , and SE . Thus, the remaining pyramid whose base is polygon $EPFQGRHS$, and apex the point N , is greater than solid O . And let the polygon $ATBUCVDW$, similar, and similarly laid out, to polygon $EPFQGRHS$, be inscribed in circle $ABCD$ [Prop. 6.18]. And let a pyramid having the same apex as the cone be set up on polygon $ATBUCVDW$. And let LBT be one of the triangles containing the pyramid whose base is polygon $ATBUCVDW$, and apex the point L . And let NFP be one of the triangles containing the pyramid whose base is triangle $EPFQGRHS$, and apex the point N . And let KT and MP be joined. And since cone $ABCDL$ is similar to cone $EFGHN$, thus as BD is to FH , so axis KL (is) to axis MN [Def. 11.24]. And as BD (is) to FH , so BK (is) to FM . And, thus, as BK (is) to FM , so KL (is) to MN . And, alternately, as BK (is) to KL , so FM (is) to MN [Prop. 5.16]. And the sides around the equal angles BKL and FMN are proportional. Thus, triangle BKL is similar to triangle FMN [Prop. 6.6]. Again, since as BK (is) to KT , so FM (is) to MP , and (they are) about the equal angles BKT and FMP , inasmuch as whatever part angle BKT is of the four right-angles at the center K , angle FMP is also the same part of the four right-angles at the center M . Therefore, since the sides about equal angles are proportional, triangle BKT is thus similar to triangle FMP [Prop. 6.6]. Again, since it was shown that as BK (is) to KL , so FM (is) to MN , and BK (is) equal to KT , and FM to PM , thus as TK (is) to KL , so PM (is) to MN . And the sides about the equal angles TKL and PMN —for (they are both) right-angles—are proportional. Thus, triangle LKT (is) similar to triangle NMP [Prop. 6.6]. And since, on account of the similarity of triangles LKB and NMF , as LB (is) to BK , so NF (is) to FM , and, on account of the similarity of triangles BKT and FMP , as KB (is) to BT , so MF (is) to FP [Def. 6.1], thus, via equality, as LB (is) to BT , so NF (is) to FP [Prop. 5.22]. Again, since, on

τὴν ZO , δι' ἴσον ἄρα ὡς ἡ AB πρὸς τὴν BT , οὕτως ἡ NZ πρὸς τὴν ZO . πάλιν, ἐπεὶ διὰ τὴν ομοιότητα τῶν ATK , NOM τριγώνων ἐστὶν ὡς ἡ AT πρὸς τὴν TK , οὕτως ἡ NO πρὸς τὴν OM , διὰ δὲ τὴν ομοιότητα τῶν TKB , OMZ τριγώνων ἐστὶν ὡς ἡ KT πρὸς τὴν TB , οὕτως ἡ MO πρὸς τὴν OZ , δι' ἴσον ἄρα ὡς ἡ AT πρὸς τὴν TB , οὕτως ἡ NO πρὸς τὴν OZ . ἐδείκθη δὲ καὶ ὡς ἡ TB πρὸς τὴν BA , οὕτως ἡ OZ πρὸς τὴν ZN . δι' ἴσον ἄρα ὡς ἡ TA πρὸς τὴν AB , οὕτως ἡ ON πρὸς τὴν NZ . τῶν ATB , NOZ ἄρα τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ ἰσογώνια ἄρα ἐστὶ τὰ ATB , NOZ τρίγωνα· ὥστε καὶ ὅμοια. καὶ πυραμῖς ἄρα, ἥς βάσις μὲν τὸ BKT τρίγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, ὁμοία ἐστὶ πυραμίδι, ἥς βάσις μὲν τὸ ZMO τρίγωνον, κορυφὴ δὲ τὸ N σημεῖον· ὑπὸ γάρ ὁμοίων ἐπιπέδων περιέχονται ἴσων τὸ πλῆθος. αἱ δὲ ὅμοιαι πυραμίδες καὶ τριγώνουζ ἔθουσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. ἡ ἄρα $BKTA$ πυραμῖς πρὸς τὴν $ZMON$ πυραμίδα τριπλασίονα λόγον ἔθει ἥπερ ἡ BK πρὸς τὴν ZM . ὁμοίως δὲ ἐπιξενυνόντες ἀπὸ τῶν A , Θ , Δ , Φ , Γ , Y ἐπὶ τὸ K εὐθείας καὶ ἀπὸ τῶν E , Σ , Θ , P , H , Π ἐπὶ τὸ M καὶ ἀνιστάντες ἐφ' ἐκάστον τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴν ἐθούσας τοῖς κῶνοις δείχομεν, ὅτι καὶ ἐκάστη τῶν ὁμοταγῶν πυραμίδων πρὸς ἐκάστην ὁμοταγῇ πυραμίδα τριπλασίονα λόγον ἔχει ἥπερ ἡ BK ὁμόλογος πλευρὰ πρὸς τὴν ZM ὁμόλογον πλευράν, τουτέστιν ἥπερ ἡ BD πρὸς τὴν $Z\Theta$. καὶ ὡς ἐν τῶν ἡγουμένων πρὸς ἐν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα· ἐστὶν ἄρα καὶ ὡς ἡ $BKTA$ πυραμῖς πρὸς τὴν $ZMON$ πυραμίδα, οὕτως ἡ ὅλη πυραμῖς, ἥς βάσις τὸ $ATBYT\Phi\Delta\Theta$ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὴν ὅλην πυραμίδα, ἥς βάσις μὲν τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον· ὥστε καὶ πυραμῖς, ἥς βάσις μὲν τὸ $ATBYT\Phi\Delta\Theta$, κορυφὴ δὲ τὸ Λ , πρὸς τὴν πυραμίδα, ἥς βάσις [μὲν] τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον, τριπλασίονα λόγον ἔθει ἥπερ ἡ BD πρὸς τὴν $Z\Theta$. ὑπόκειται δὲ καὶ ὁ κῶνος, ὃς βάσις [μὲν] ὁ $AB\Gamma\Delta$ κύκλος, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὸ X στερεὸν τριπλασίονα λόγον ἔθων ἥπερ ἡ BD πρὸς τὴν $Z\Theta$. ἐστὶν ἄρα ὡς ὁ κῶνος, ὃς βάσις μὲν ἐστὶν ὁ $AB\Gamma\Delta$ κύκλος, κορυφὴ δὲ τὸ Λ , πρὸς τὸ X στερεόν, οὕτως ἡ πυραμῖς, ἥς βάσις μὲν τὸ $ATBYT\Phi\Delta\Theta$ [πολύγωνον], κορυφὴ δὲ τὸ Λ , πρὸς τὴν πυραμίδα, ἥς βάσις μὲν ἐστὶ τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N . ἐναλλάξ ἄρα, ὡς ὁ κῶνος, ὃς βάσις μὲν ὁ $AB\Gamma\Delta$ κύκλος, κορυφὴ δὲ τὸ Λ , πρὸς τὴν ἐν αὐτῷ πυραμίδα, ἥς βάσις μὲν τὸ $ATBYT\Phi\Delta\Theta$ πολύγωνον, κορυφὴ δὲ τὸ Λ , οὕτως τὸ X [στερεόν] πρὸς τὴν πυραμίδα, ἥς βάσις μὲν ἐστὶ τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N . μείζων δὲ ὁ εἰρημένος κῶνος τῆς ἐν αὐτῷ πυραμίδος· ἐμπεριέθει γάρ αὐτήν. μείζον ἄρα καὶ τὸ X στερεόν τῆς πυραμίδος, ἥς βάσις μὲν ἐστὶ τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N . ἀλλὰ καὶ ἔλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ κῶνος, ὃς βάσις ὁ $AB\Gamma\Delta$ κύκλος, κορυφὴ δὲ τὸ Λ [σημεῖον], πρὸς ἔλαττον τι τοῦ κῶνον στερεόν, ὃς βάσις μὲν ὁ $EZH\Theta$

account of the similarity of triangles LTK and NPM , as LT (is) to TK , so NP (is) to PM , and, on account of the similarity of triangles TKB and PMF , as KT (is) to TB , so MP (is) to PF , thus, via equality, as LT (is) to TB , so NP (is) to PF [Prop. 5.22]. And it was shown that as TB (is) to BL , so PF (is) to FN . Thus, via equality, as TL (is) to LB , so PN (is) to NF [Prop. 5.22]. Thus, the sides of triangles LTB and NPF are proportional. Thus, triangles LTB and NPF are equiangular [Prop. 6.5]. And, hence, (they are) similar [Def. 6.1]. And, thus, the pyramid whose base is triangle BKT , and apex the point L , is similar to the pyramid whose base is triangle FMP , and apex the point N . For they are contained by equal numbers of similar planes [Def. 11.9]. And similar pyramids which also have triangular bases are in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, pyramid $BKTL$ has to pyramid $FMPN$ the cubed ratio that BK (has) to FM . So, similarly, joining straight-lines from (points) A , W , D , V , C , and U to (center) K , and from (points) E , S , H , R , G , and Q to (center) M , and setting up pyramids having the same apexes as the cones on each of the triangles (so formed), we can also show that each of the pyramids (on base $ABCD$ taken) in order will have to each of the pyramids (on base $EFGH$ taken) in order the cubed ratio that the corresponding side BK (has) to the corresponding side FM —that is to say, that BD (has) to FH . And (for two sets of proportional magnitudes) as one of the leading (magnitudes is) to one of the following, so (the sum of) all of the leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. And, thus, as pyramid $BKTL$ (is) to pyramid $FMPN$, so the whole pyramid whose base is polygon $ATBUCVDW$, and apex the point L , (is) to the whole pyramid whose base is polygon $EPFQGRHS$, and apex the point N . And, hence, the pyramid whose base is polygon $ATBUCVDW$, and apex the point L , has to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N , the cubed ratio that BD (has) to FH . And it was also assumed that the cone whose base is circle $ABCD$, and apex the point L , has to solid O the cubed ratio that BD (has) to FH . Thus, as the cone whose base is circle $ABCD$, and apex the point L , is to solid O , so the pyramid whose base (is) [polygon] $ATBUCVDW$, and apex the point L , (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N . Thus, alternately, as the cone whose base (is) circle $ABCD$, and apex the point L , (is) to the pyramid within it whose base (is) the polygon $ATBUCVDW$, and apex the point L , so the [solid] O (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N [Prop. 5.16]. And the aforementioned cone (is) greater than the pyramid within it. For it encompasses it. Thus, solid O (is) also greater than the pyramid whose base is polygon $EPFQGRHS$, and apex the point N . But, (it is) also less. The very thing is impossible. Thus, the cone whose base (is) circle $ABCD$, and apex the [point] L , does not have

κύκλος, κορυφή δὲ τὸ N σημεῖον, τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Delta$ πρὸς τὴν $Z\Theta$. ὁμοίως δὴ δείχουμεν, ὅτι οὐδὲ ὁ $EZH\Theta N$ κῶνος πρὸς ἑλαττόν τι τοῦ $AB\Gamma\Delta\Lambda$ κώνου στερεὸν τριπλασίονα λόγον ἔθει ἥπερ ἡ $Z\Theta$ πρὸς τὴν $B\Delta$.

Λέγω δὴ, ὅτι οὐδὲ ὁ ΑΒΓΔΛ κῶνος πρὸς μείζον τι τοῦ ΕΖΗΘΝ κώνου στερεὸν τριπλασίονα λόγον ἔθει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

Εἰ γὰρ θνατόν, ἐθέτω πρὸς μείζον τὸ *X*. ἀνάπαλιν ἄρα τὸ *X* στερεὸν πρὸς τὸν *ΑΒΓΔΛ* κῶνον τριπλασίονα λόγον ἔθει ἥπερ ἡ *ΖΘ* πρὸς τὴν *ΒΔ*. ὥς δὲ τὸ *X* στερεὸν πρὸς τὸν *ΑΒΓΔΛ* κῶνον, οὕτως ὁ *ΕΖΗΘΝ* κῶνος πρὸς ἑλαττόν τι τοῦ *ΑΒΓΔΛ* κῶνου στερεόν. καὶ ὁ *ΕΖΗΘΝ* ἄρα κῶνος πρὸς ἑλαττόν τι τοῦ *ΑΒΓΔΛ* κῶνου στερεόν τριπλασίονα λόγον ἔθει ἥπερ ἡ *ΖΘ* πρὸς τὴν *ΒΔ*. ὅπερ ἀδύνατον ἐδείκθῃ. οὐκ ἄρα ὁ *ΑΒΓΔΛ* κῶνος πρὸς μείζον τι τοῦ *ΕΖΗΘΝ* κῶνου στερεόν τριπλασίονα λόγον ἔθει ἥπερ ἡ *ΒΔ* πρὸς τὴν *ΖΘ*. ἐδείκθῃ δέ, ὅτι οὐδὲ πρὸς ἑλαττον. ὁ *ΑΒΓΔΛ* ἄρα κῶνος πρὸς τὸν *ΕΖΗΘΝ* κῶνον τριπλασίονα λόγον ἔθει ἥπερ ἡ *ΒΔ* πρὸς τὴν *ΖΘ*.

Ἰὼς δὲ ὁ κῶνος πρὸς τὸν κῶνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον· τριπλασίους γάρ ὁ κύλινδρος τοῦ κῶνον ὁ ἐπὶ τῆς αὐτῆς βάσεως τῷ κῶνῳ καὶ ἰσοῦνῃς αὐτῷ. καὶ ὁ κύλινδρος ἄρα πρὸς τὸν κύλινδρον τριπλασίονα λόγον ἔθει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

Οἱ ἄρα ὅμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριπλασίονι λόγῳ εἰς τῶν ἐν ταῖς βάσεσι διαμέτρων· ὅπερ ἔδει δεῖξαι.

to some solid less than the cone whose base (is) circle $EFGH$, and apex the point N , the cubed ratio that BD (has) to EH . So, similarly, we can show that neither does cone $EFGHN$ have to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD .

So, I say that neither does cone $ABCDL$ have to some solid greater than cone $EFGHN$ the cubed ratio that BD (has) to FH .

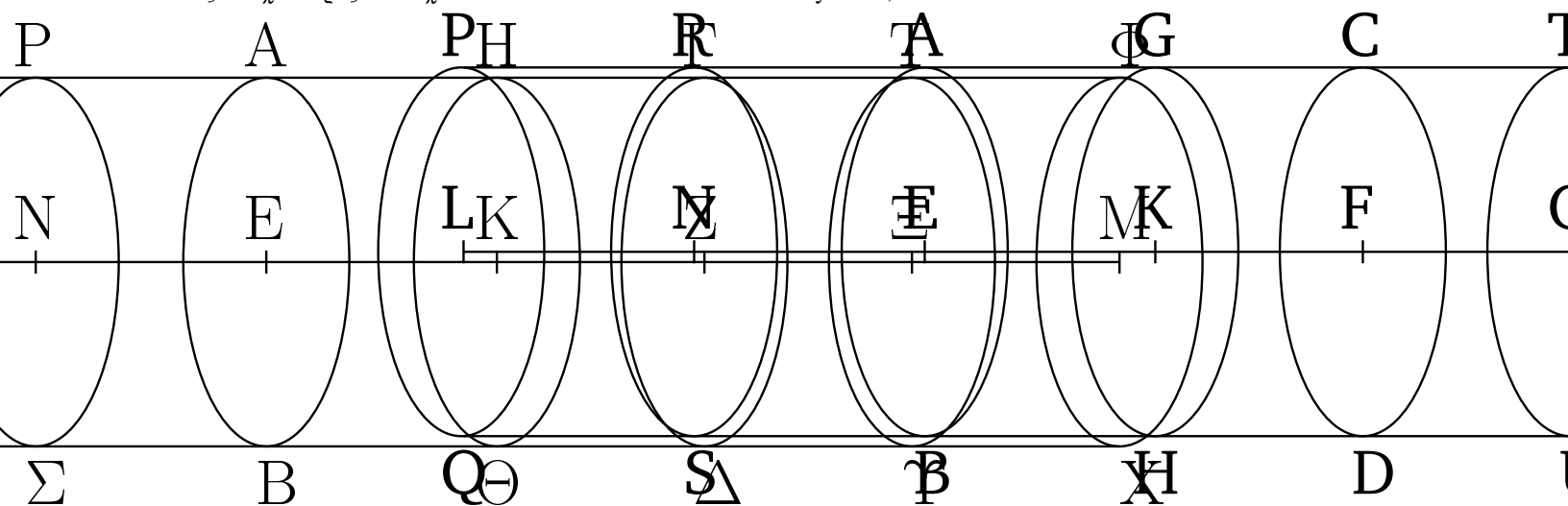
For, if possible, let it have (such a ratio) to a greater (solid), O . Thus, inversely, solid O has to cone $ABCDL$ the cubed ratio that FH (has) to BD [Prop. 5.7 corr.]. And as solid O (is) to cone $ABCDL$, so cone $EFGHN$ (is) to some solid less than cone $ABCDL$ [12.2 lem.]. Thus, cone $EFGHN$ also has to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD . The very thing was shown (to be) impossible. Thus, cone $ABCDL$ does not have to some solid greater than cone $EFGHN$ the cubed ratio than BD (has) to FH . And it was shown that neither (does it have such a ratio) to a lesser (solid). Thus, cone $ABCDL$ has to cone $EFGHN$ the cubed ratio that BD (has) to FG .

And as the cone (is) to the cone, so the cylinder (is) to the cylinder. For a cylinder is three times a cone on the same base as the cone, and of the same height as it [Prop. 12.10]. Thus, the cylinder also has to the cylinder the cubed ratio that BD (has) to FH .

Thus, similar cones and cylinders are in the cubed ratio of the diameters of their bases. (Which is) the very thing it was required to show.

13.

Ἐὰν κύλινδρος ἐπιπέδῳ τμηθῇ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔσται ὡς ὁ κύλινδρος πρὸς τὸν κύλινδρον, οὔτως ὁ ἄγων πρὸς τὸν ἄγονα.



Κύλινδροξ γάρ ὁ $A\Delta$ ἐπιπέδω τῷ $H\Theta$ τετμήσθω παραλ-

Proposition 13

If a cylinder is cut by a plane which is parallel to the opposite planes (of the cylinder) then as the cylinder (is) to the cylinder, so the axis will be to the axis.

For let the cylinder AD be cut by the plane GH which is

λήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς AB , $\Gamma\Delta$, καὶ συμβαλλέτω τῷ ἄχονι τὸ $H\Theta$ ἐπίπεδον κατὰ τὸ K σημείον λέγω, ὅτι ἐστὶν ὡς ὁ BH κύλινδρος πρὸς τὸν $H\Delta$ κύλινδρον, οὕτως ὁ EK ἄχων πρὸς τὸν KZ ἄχονα.

Ἐκβεβλήσθω γὰρ ὁ EZ ἄχων ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ A , M σημεία, καὶ ἐκκείσθωσαν τῷ EK ἄχονι ἴσοι ὁσοιδηποτοῦν οἱ EN , NA , τῷ δὲ ZK ἴσοι ὁσοιδηποτοῦν οἱ ZX , XM , καὶ νοείσθω ὁ ἐπὶ τοῦ AM ἄχονος κύλινδρος ὁ $O\Theta$, οὗ βάσεις οἱ $O\Pi$, $\Phi\Theta$ κύκλοι. καὶ ἐκβεβλήσθω διὰ τῶν N , X σημείων ἐπίπεδα παράλληλα τοῖς AB , $\Gamma\Delta$ καὶ ταῖς βάσεσι τοῦ $O\Theta$ κύλινδρον καὶ ποιέτωσαν τοὺς $P\Sigma$, TY κύκλους περὶ τὰ N , X κέντρα. καὶ ἐπεὶ οἱ AN , NE , EK ἄχονες ἴσοι εἰσὶν ἀλλήλοις, οἱ ἄρα ΠP , PB , BH κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις. ἴσοι δὲ εἰσὶν αἱ βάσεις· ἴσοι ἄρα καὶ οἱ ΠP , PB , BH κύλινδροι ἀλλήλοις. ἐπεὶ οὖν οἱ AN , NE , EK ἄχονες ἴσοι εἰσὶν ἀλλήλοις, εἰσὶ δὲ καὶ οἱ ΠP , PB , BH κύλινδροι ἴσοι ἀλλήλοις, καὶ ἐστὶν ἴσον τὸ πλῆθος τῷ πλῆθει, ὅσαπλασίων ἄρα ὁ $ΚΑ$ ἄχων τοῦ EK ἄχονος, τοσανταπλασίων ἔσται καὶ ὁ ΠH κύλινδρος τοῦ $H\Delta$ κύλινδρον. διὰ τὰ αὐτὰ δὴ καὶ ὅσαπλασίων ἐστὶν ὁ MK ἄχων τοῦ KZ ἄχονος, τοσανταπλασίων ἐστὶ καὶ ὁ ΘH κύλινδρος τοῦ $H\Delta$ κύλινδρον. καὶ εἰ μὲν ἴσος ἐστὶν ὁ $ΚΑ$ ἄχων τῷ KM ἄχονι, ἴσος ἔσται καὶ ὁ ΠH κύλινδρος τῷ $H\Theta$ κύλινδρῳ, εἰ δὲ μείζων ὁ ἄχων τοῦ ἄχονος, μείζων καὶ ὁ κύλινδρος τοῦ κύλινδρον, καὶ εἰ ἐλάσσων, ἐλάσσων. τεσσάρων δὴ μεγεθῶν ὄντων, ἁχόνων μὲν τῶν EK , KZ , κύλινδρων δὲ τῶν BH , $H\Delta$, εἴληπται ἰσάκις πολλαπλάσια, τοῦ μὲν EK ἄχονος καὶ τοῦ BH κύλινδρον ὅ τε AK ἄχων καὶ ὁ ΠH κύλινδρος, τοῦ δὲ KZ ἄχονος καὶ τοῦ $H\Delta$ κύλινδρον ὅ τε KM ἄχων καὶ ὁ $H\Theta$ κύλινδρος, καὶ δέδεικται, ὅτι εἰ ὑπερέθει ὁ $ΚΑ$ ἄχων τοῦ KM ἄχονος, ὑπερέθει καὶ ὁ ΠH κύλινδρος τοῦ $H\Theta$ κύλινδρον, καὶ εἰ ἴσος, ἴσος, καὶ εἰ ἐλάσσων, ἐλάσσων. ἔστιν ἄρα ὡς ὁ EK ἄχων πρὸς τὸν KZ ἄχονα, οὕτως ὁ BH κύλινδρος πρὸς τὸν $H\Delta$ κύλινδρον· ὅπερ ἔδει δεῖχαι.

14.

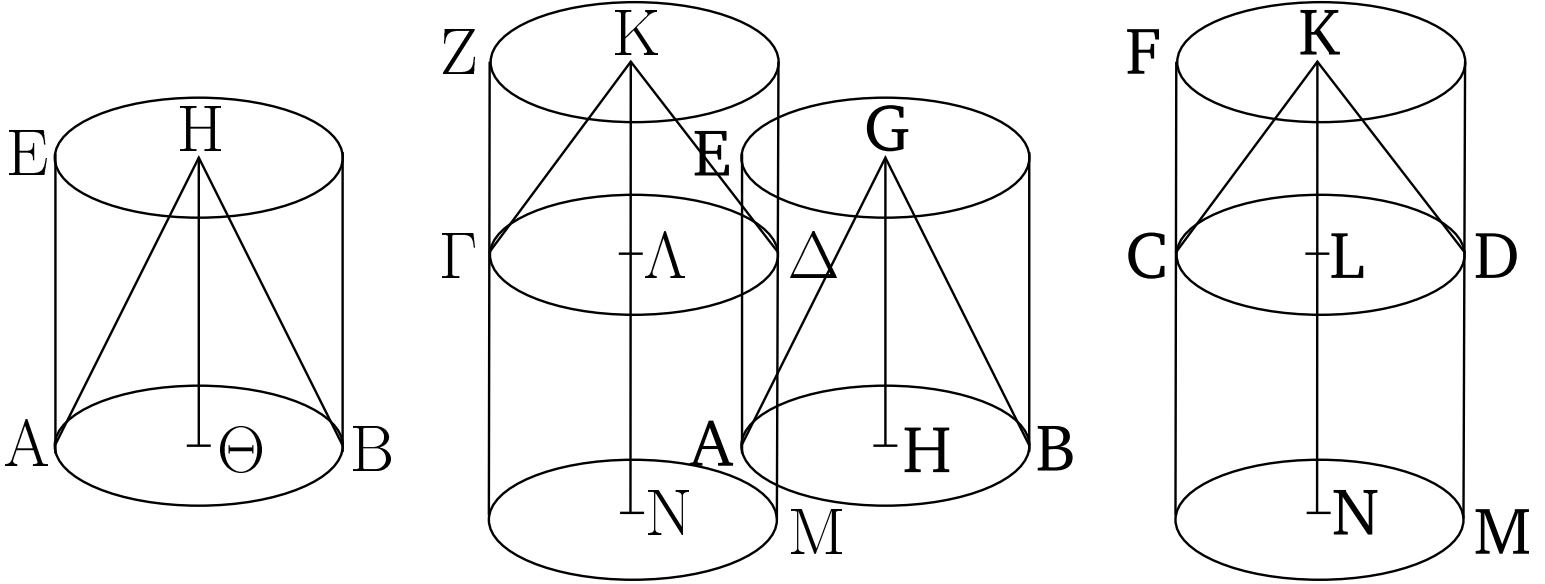
Οἱ ἐπὶ ἴσων βάσεων ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς τὰ ὕψη.

parallel to the opposite planes (of the cylinder), AB and CD . And let the plane GH have met the axis at point K . I say that as cylinder BG is to cylinder GD , so axis EK (is) to axis KF .

For let axis EF be produced in each direction to points L and M . And let any number whatsoever (of lengths), EN and NL , equal to axis EK , be set out (on the axis EL), and any number whatsoever (of lengths), FO and OM , equal to (axis) FK , (on the axis KM). And let the cylinder PW , whose bases (are) the circles PQ and VW , be conceived on axis LM . And let planes parallel to AB , CD , and the bases of cylinder PW , be produced through points N and O , and let them have made the circles RS and TU around the centers N and O (respectively). And since axes LN , NE , and EK are equal to one another, the cylinders QR , RB , and BG are to one another as their bases [Prop. 12.11]. But the bases are equal. Thus, the cylinders QR , RB , and BG (are) also equal to one another. Therefore, since the axes LN , NE , and EK are equal to one another, and the cylinders QR , RB , and BG are also equal to one another, and the number (of the former) is equal to the number (of the latter), thus as many multiples as axis KL is of axis EK , so many multiples is cylinder QG also of cylinder GB . And so, for the same (reasons), as many multiples as axis MK is of axis KF , so many multiples is cylinder WG also of cylinder GD . And if axis KL is equal to axis KM then cylinder QG will also be equal to cylinder GW , and if the axis (is) greater than the axis then the cylinder (will also be) greater than the cylinder, and if (the axis is) less then (the cylinder will also be) less. So, there are four magnitudes—the axes EK and KF , and the cylinders BG and GD —and equal multiples be taken of axis EK and cylinder BG —(namely), axis LK and cylinder QG —and of axis KF and cylinder GD —(namely), axis KM and cylinder GW . And it has been shown that if axis KL exceeds axis KM then cylinder QG also exceeds cylinder GW , and if (the axes are) equal then (the cylinders are) equal, and if (KL is) less then (QG is) less. Thus, as axis EK is to axis KF , so cylinder BG (is) to cylinder GD [Def. 5.5]. (Which is) the very thing it was required to show.

Proposition 14

Cones and cylinders which are on equal bases are to one another as their heights.



Ἐστωσαν γάρ ἐπὶ ἴσων βάσεων τῶν AB , $\Gamma\Delta$ κύκλων κύλινδροι οἱ EB , $Z\Delta$. λέγω, ὅτι ἐστὶν ὡς ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, οὕτως ὁ $H\Theta$ ἄξων πρὸς τὸν ΚΛ ἄξωνα.

Ἐκβεβλήσθω γάρ ὁ ΚΛ ἄξων ἐπὶ τὸ N σημεῖον, καὶ κείσθω τῷ $H\Theta$ ἄξωνι ἴσος ὁ ΛΝ , καὶ περὶ ἄξωνα τὸν ΛΝ κύλινδρος γενοήσθω ὁ $\Gamma\text{Μ}$. ἐπεὶ οὖν οἱ EB , $\Gamma\text{Μ}$ κύλινδροι ὑπὸ τὸ αὐτὸ ὕψος εἰσὶν, πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις. ἴσαι δὲ εἰσὶν αἱ βάσεις ἀλλήλαις· ἴσοι ἄρα εἰσὶ καὶ οἱ EB , $\Gamma\text{Μ}$ κύλινδροι. καὶ ἐπεὶ κύλινδρος ὁ $Z\text{Μ}$ ἐπιπέδῳ τέτμηται τῷ $\Gamma\Delta$ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἐστὶν ἄρα ὡς ὁ $\Gamma\text{Μ}$ κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, οὕτως ὁ ΛΝ ἄξων πρὸς τὸν ΚΛ ἄξωνα. ἴσος δὲ ἐστὶν ὁ μὲν $\Gamma\text{Μ}$ κύλινδρος τῷ EB κύλινδρῳ, ὁ δὲ ΛΝ ἄξων τῷ $H\Theta$ ἄξωνι· ἐστὶν ἄρα ὡς ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, οὕτως ὁ ABH κῶνος πρὸς τὸν $\Gamma\Delta\text{Κ}$ κῶνον. καὶ ὡς ἄρα ὁ $H\Theta$ ἄξων πρὸς τὸν ΚΛ ἄξωνα, οὕτως ὁ ABH κῶνος πρὸς τὸν $\Gamma\Delta\text{Κ}$ κῶνον καὶ ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον· ὅπερ ἔδει δεῖξαι.

15.

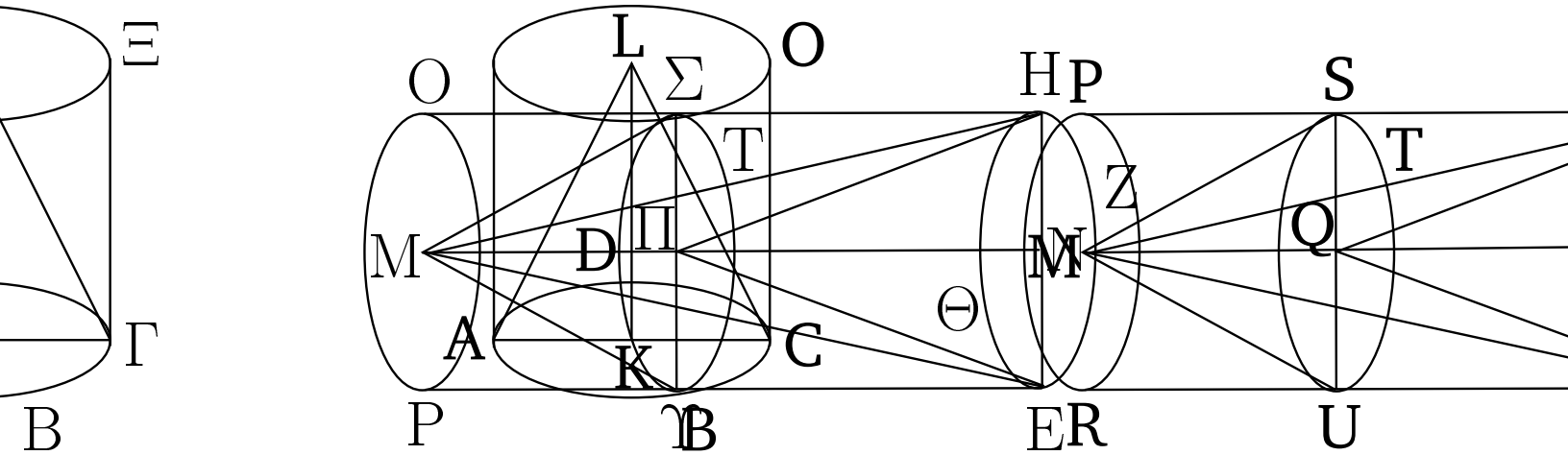
Τῶν ἴσων κόνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν καὶ τῶν κόνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν, ἴσοι εἰσὶν ἐκείνοι.

For let EB and FD be cylinders on equal bases, (namely) the circles AB and CD (respectively). I say that as cylinder EB is to cylinder FD , so axis GH (is) to axis KL .

For let the axis KL be produced to point N . And let LN be made equal to axis GH . And let the cylinder CM be conceived about axis LN . Therefore, since cylinders EB and CM have the same height they are to one another as their bases [Prop. 12.11]. And the bases are equal to one another. Thus, cylinders EB and CM are also equal to one another. And since cylinder FM has been cut by the plane CD , which is parallel to its opposite planes, thus as cylinder CM is to cylinder FD , so axis LN (is) to axis KL [Prop. 12.13]. And cylinder CM is equal to cylinder EB , and axis LN to axis GH . Thus, as cylinder EB is to cylinder FD , so axis GH (is) to axis KL . And as cylinder EB (is) to cylinder FD , so cone ABG (is) to cone CDK [Prop. 12.10]. Thus, also, as axis GH (is) to axis KL , so cone ABG (is) to cone CDK , and cylinder EB to cylinder FD . (Which is) the very thing it was required to show.

Proposition 15

The bases of equal cones and cylinders are reciprocally proportional to their heights. And, those cones and cylinders whose bases (are) reciprocally proportional to their heights are equal.



Ἐστωσαν ἴσοι κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν οἱ $AB\Gamma\Delta$, $EZH\Theta$ κύκλοι, διάμετροι δὲ αὐτῶν αἱ AF , EH , ἄχονες δὲ οἱ KA , MN , οὔτινες καὶ ὕψη εἰσὶ τῶν κῶνων ἢ κυλίνδρων, καὶ συμπεπληρώσθωσαν οἱ AX , EO κύλινδροι. λέγω, ὅτι τῶν AX , EO κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστιν ὥς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ KA ὕψος.

Τὸ γὰρ AK ὕψος τῷ MN ὕψει ἴσον ἔστιν ἢ οὐ. ἔστω πρότερον ἴσον. ἔστι δὲ καὶ ὁ AX κύλινδρος τῷ EO κυλίνδρῳ ἴσος. οἱ δὲ ὑπὸ τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὥς αἱ βάσεις· ἴση ἄρα καὶ ἡ $AB\Gamma\Delta$ βάσις τῇ $EZH\Theta$ βάσει. ὥστε καὶ ἀντιπέπονθεν, ὥς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ KA ὕψος. ἀλλὰ δὴ μὴ ἔστω τὸ AK ὕψος τῷ MN ἴσον, ἀλλ' ἔστω μείζον τὸ MN , καὶ ἀφηρήσθω ἀπὸ τοῦ MN ὕψους τῷ KA ἴσον τὸ ΠN , καὶ διὰ τοῦ Π σημείου τετμήσθω ὁ EO κύλινδρος ἐπιπέδῳ τῷ $TY\S$ παραλλήλῳ τοῖς τῶν $EZH\Theta$, PO κύκλων ἐπιπέδοις, καὶ ἀπὸ βάσεως μὲν τοῦ $EZH\Theta$ κύκλου, ὕψους δὲ τοῦ $N\Pi$ κυλίνδρου γενησθῶ ὁ $E\S$. καὶ ἐπεὶ ἴσος ἔστιν ὁ AX κύλινδρος τῷ EO κυλίνδρῳ, ἔστιν ἄρα ὥς ὁ AX κύλινδρος πρὸς τὸν $E\S$ κύλινδρον, οὕτως ὁ EO κύλινδρος πρὸς τὸν $E\S$ κύλινδρον. ἀλλ' ὥς μὲν ὁ AX κύλινδρος πρὸς τὸν $E\S$ κύλινδρον, οὕτως ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ · ὑπὸ γὰρ τὸ αὐτὸ ὕψος εἰσὶν οἱ AX , $E\S$ κύλινδροι· ὥς δὲ ὁ EO κύλινδρος πρὸς τὸν $E\S$, οὕτως τὸ MN ὕψος πρὸς τὸ ΠN ὕψος· ὁ γὰρ EO κύλινδρος ἐπιπέδῳ τέτμηται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις. ἔστιν ἄρα καὶ ὥς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ ΠN ὕψος. ἴσον δὲ τὸ ΠN ὕψος τῷ KA ὕψει· ἔστιν ἄρα ὥς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ KA ὕψος. τῶν ἄρα AX , EO κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Ἀλλὰ δὴ τῶν AX , EO κυλίνδρων ἀντιπεπονητόσθωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστω ὥς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ KA ὕψος· λέγω, ὅτι ἴσος ἔστιν ὁ AX κύλινδρος τῷ EO κυλίνδρῳ.

Let there be equal cones and cylinders whose bases are the circles $ABCD$ and $EFGH$, and the diameters of (the bases) AC and EG , and (whose) axes (are) KL and MN , which are also the heights of the cones and cylinders (respectively). And let the cylinders AO and EP be completed. I say that the bases of cylinders AO and EP are reciprocally proportional to their heights, and (so) as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL .

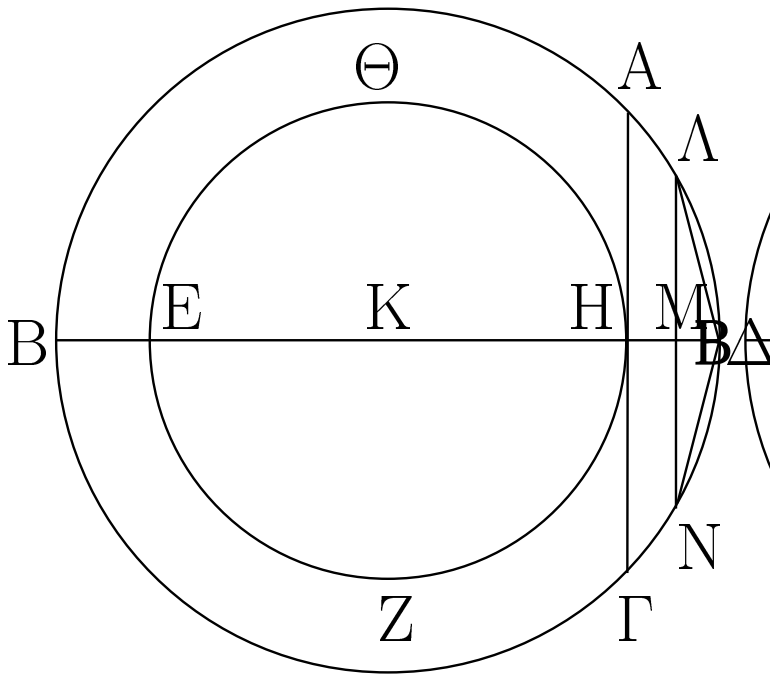
For height LK is either equal to height MN , or not. Let it, first of all, be equal. And cylinder AO is also equal to cylinder EP . And cones and cylinders having the same height are to one another as their bases [Prop. 12.11]. Thus, base $ABCD$ (is) also equal to base $EFGH$. And, hence, reciprocally, as base $ABCD$ (is) to base $EFGH$, so height MN (is) to height KL . And so, let height LK not be equal to MN , but let MN be greater. And let QN , equal to KL , be cut off from height MN . And let the cylinder EP be cut, through point Q , by the plane TUS (which is) parallel to the planes of the circles $EFGH$ and RP . And let cylinder ES be conceived, with base the circle $EFGH$, and height NQ . And since cylinder AO is equal to cylinder EP , thus, as cylinder AO (is) to cylinder ES , so cylinder EP (is) to cylinder ES [Prop. 5.7]. But, as cylinder AO (is) to cylinder ES , so base $ABCD$ (is) to base $EFGH$. For cylinders AO and ES (have) the same height [Prop. 12.11]. And as cylinder EP (is) to (cylinder) ES , so height MN (is) to height QN . For cylinder EP has been cut by a plane which is parallel to its opposite planes [Prop. 12.13]. And, thus, as base $ABCD$ is to base $EFGH$, so height MN (is) to height QN [Prop. 5.11]. And height QN (is) equal to height KL . Thus, as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL . Thus, the bases of cylinders AO and EP are reciprocally proportional to their heights.

And, so, let the bases of cylinders AO and EP be reciprocally proportional to their heights, and (thus) let base $ABCD$ be to base $EFGH$, as height MN (is) to height KL . I say that cylinder AO is equal to cylinder EP .

Τῶν γάρ αὐτῶν κατασκευασθέντων ἐπεὶ ἐστὶν ὡς ἡ $ABΓΔ$ βάσις πρὸς τὴν $EZHΘ$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ $ΚΛ$ ὕψος, ἴσον δὲ τὸ $ΚΛ$ ὕψος τῷ $ΠΝ$ ὕψει, ἔσται ἄρα ὡς ἡ $ABΓΔ$ βάσις πρὸς τὴν $EZHΘ$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ $ΠΝ$ ὕψος. ἀλλ' ὡς μὲν ἡ $ABΓΔ$ βάσις πρὸς τὴν $EZHΘ$ βάσιν, οὕτως ὁ $ΑΧ$ κύλινδρος πρὸς τὸν $ΕΣ$ κύλινδρον· ὑπὸ γάρ τὸ αὐτὸ ὕψος εἰσὶν· ὡς δὲ τὸ MN ὕψος πρὸς τὸ $ΠΝ$ [ὕψος], οὕτως ὁ $ΕΟ$ κύλινδρος πρὸς τὸν $ΕΣ$ κύλινδρον· ἔστιν ἄρα ὡς ὁ $ΑΧ$ κύλινδρος πρὸς τὸν $ΕΣ$ κύλινδρον, οὕτως ὁ $ΕΟ$ κύλινδρος πρὸς τὸν $ΕΣ$. ἴσοις ἄρα ὁ $ΑΧ$ κύλινδρος τῷ $ΕΟ$ κύλινδρῳ. ὡσαύτως δὲ καὶ ἐπὶ τῶν κώνων· ὅπερ ἔδει δεῖχαι.

16.

Δύο κύκλων περὶ τὸ αὐτὸ κέντρον ὄντων εἷς τὸν μείζονα κύκλον πολύγωνον ἰσόπλευρόν τε καὶ ἀρτιόπλευρον ἐγγράφει μὴ φανόν τοῦ ἐλάσσονος κύκλου.



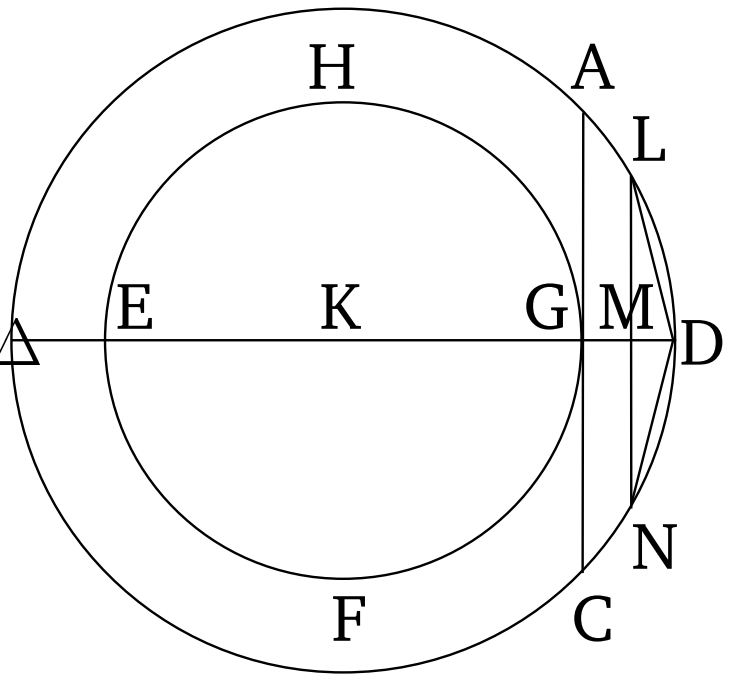
Ἐστωσαν οἱ δοθέντες δύο κύκλοι οἱ $ABΓΔ$, $EZHΘ$ περὶ τὸ αὐτὸ κέντρον τὸ K . δεῖ δὲ εἷς τὸν μείζονα κύκλον τὸν $ABΓΔ$ πολύγωνον ἰσόπλευρόν τε καὶ ἀρτιόπλευρον ἐγγράφει μὴ φανόν τοῦ $EZHΘ$ κύκλου.

Ἦθθω γάρ διὰ τοῦ K κέντρον εὐθεῖα ἡ $BKΔ$, καὶ ἀπὸ τοῦ H σημείου τῇ $BΔ$ εὐθείᾳ πρὸς ὀρθὰς ἡθθῶ ἡ HA καὶ διήθθω ἐπὶ τὸ $Γ$ · ἡ $ΑΓ$ ἄρα ἐφάπτεται τοῦ $EZHΘ$ κύκλου. τέμνοντες δὲ τὴν $BAΔ$ περιφέρειαν διῖθα καὶ τὴν ἡμίσειαν αὐτῆς διῖθα καὶ τοῦτο αἰ ποιοῦντες καταλείβομεν περιφέρειαν ἐλάσσονα τῆς $ΑΔ$. λελείφθω, καὶ ἔστω ἡ $ΛΔ$, καὶ ἀπὸ τοῦ $Λ$ ἐπὶ τὴν $BΔ$ κάθετος ἡθθῶ ἡ $ΛΜ$ καὶ διήθθω ἐπὶ τὸ N , καὶ ἐπεξεύθθωσαν αἱ $ΛΔ$, $ΔΝ$ · ἴση ἄρα ἐστὶν ἡ $ΛΔ$ τῇ $ΔΝ$.

For, with the same construction, since as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL , and height KL (is) equal to height QN , thus, as base $ABCD$ (is) to base $EFGH$, so height MN will be to height QN . But, as base $ABCD$ (is) to base $EFGH$, so cylinder AO (is) to cylinder ES . For they are the same height [Prop. 12.11]. And as height MN (is) to [height] QN , so cylinder EP (is) to cylinder ES [Prop. 12.13]. Thus, as cylinder AO is to cylinder ES , so cylinder EP (is) to (cylinder) ES [Prop. 5.11]. Thus, cylinder AO (is) equal to cylinder EP [Prop. 5.9]. In the same manner, (the proposition can) also (be demonstrated) for the cones. (Which is) the very thing it was required to show.

Proposition 16

There being two circles about the same center, to inscribe an equilateral and even-sided polygon in the greater circle, not touching the lesser circle.



Let $ABCD$ and $EFGH$ be the given two circles, about the same center, K . So, it is necessary to inscribe an equilateral and even-sided polygon in the greater circle $ABCD$, not touching circle $EFGH$.

Let the straight-line BKD be drawn through the center K . And let GA be drawn, at right-angles to the straight-line BD , through point G , and let it be drawn through to C . Thus, AC touches circle $EFGH$ [Prop. 3.16 corr.]. So, (by) cutting circumference BAD in half, and the half of it in half, and doing this continually, we will (eventually) leave a circumference less than AD [Prop. 10.1]. Let it be left, and let it be LD . And let LM be drawn, from L , perpendicular to BD , and let it be

καὶ ἔπει παράλληλός ἐστιν ἡ ΛN τῇ AG , ἡ δὲ AG ἐφάπτεται τοῦ $\text{EZH}\Theta$ κύκλου, ἡ ΛN ἄρα οὐκ ἐφάπτεται τοῦ $\text{EZH}\Theta$ κύκλου· πολλῶν ἄρα αἱ $\Lambda\Delta$, ΔN οὐκ ἐφάπτονται τοῦ $\text{EZH}\Theta$ κύκλου. ἔαν δὴ τῇ $\Lambda\Delta$ εὐθείᾳ ἴσας κατὰ τὸ συνθεῆς ἑναρμόσωμεν εἰς τὸν $\text{AB}\Gamma\Delta$ κύκλον, ἐγγραφήσεται εἰς τὸν $\text{AB}\Gamma\Delta$ κύκλον πολύγωνον ἰσοπλευρόν τε καὶ ἀρτιόπλευρον μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ $\text{EZH}\Theta$ · ὅπερ ἔδει ποιῆσαι.

drawn through to N . And let LD and DN be joined. Thus, LD is equal to DN [Props. 3.3, 1.4]. And since LN is parallel to AC [Prop. 1.28], and AC touches circle $EFGH$, LN thus does not touch circle $EFGH$. Thus, even more so, LD and DN do not touch circle $EFGH$. And if we continuously insert (straight-lines) equal to straight-line LD into circle $ABCD$ [Prop. 4.1] then an equilateral and even-sided polygon, not touching the lesser circle $EFGH$, will be inscribed in circle $ABCD$.[†] (Which is) the very thing it was required to do.

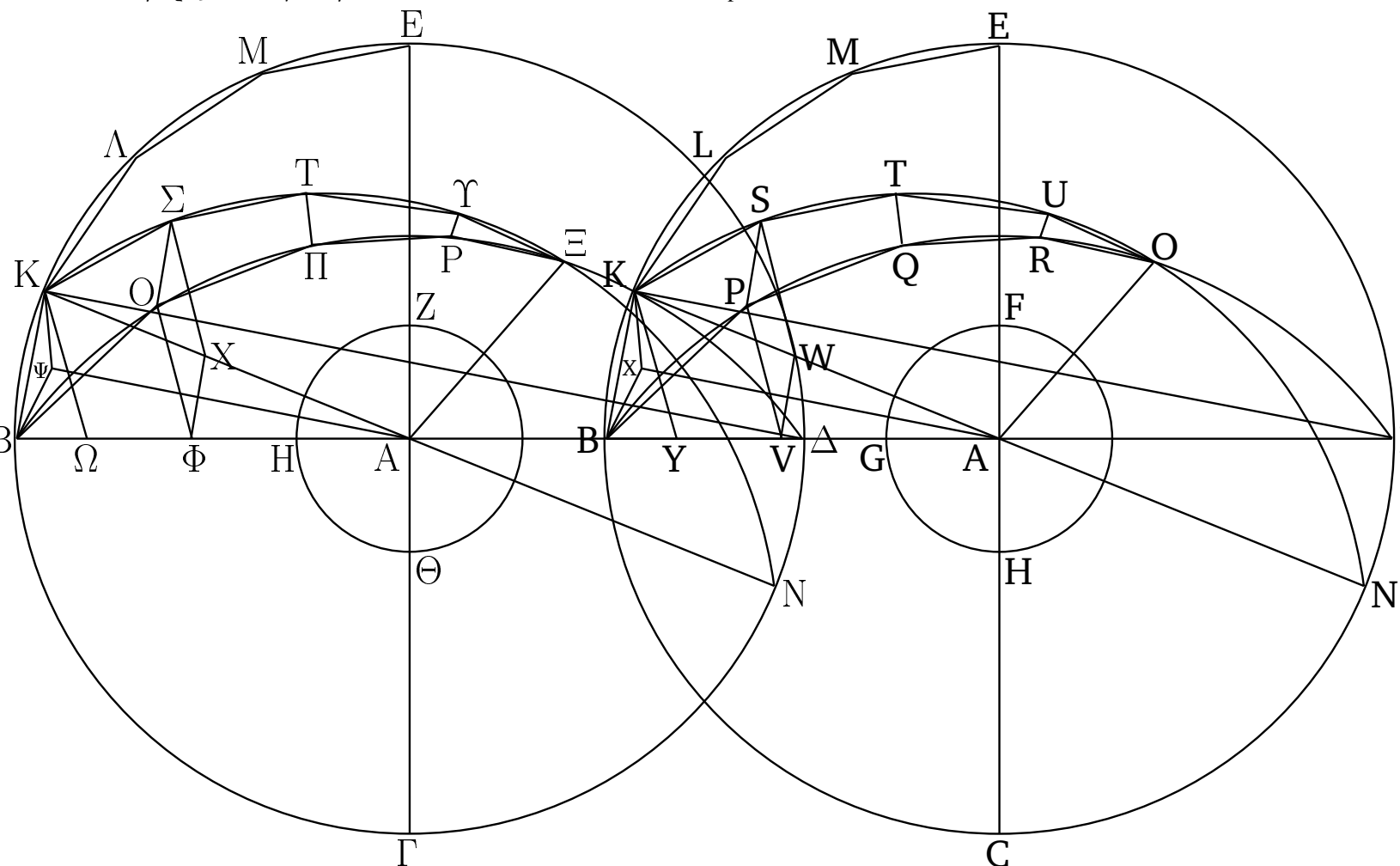
[†] Note that the chord of the polygon, LN , does not touch the inner circle either.

17.

Δύο σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἰς τὴν μείζονα σφαῖραν στερεὸν πολυέδρον ἐγγράψαι μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Proposition 17

There being two spheres about the same center, to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.



Νενοήσθωσαν δύο σφαίραι περί τὸ αὐτὸ κέντρον τὸ Α·
 δεῖ δὴ εἶξ τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγράφαι
 μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Let two spheres be conceived about the same center, A. So, it is necessary to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.

Τετμήσθωσαν αἱ σφαῖραι ἐπιπέδῳ τινὶ διὰ τοῦ κέντρον· ἔσονται δὴ αἱ τομαὶ κύκλοι, ἐπειδήπερ μενούσης τῆς διαμέτρου καὶ περιφερομένου τοῦ ἡμικυκλίου ἐγίγνετο ἡ σφαῖρα· ὥστε καὶ καθ' οἷαζ' ἂν θέσεως ἐπινοήσωμεν τὸ ἡμικύκλιον, τὸ δι' αὐτοῦ ἐκβαλλόμενον ἐπίπεδον ποιήσει ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας κύκλον. καὶ φανερόν, ὅτι καὶ μέγιστον, ἐπειδήπερ ἡ διάμετρος τῆς σφαίρας, ἥτις ἐστὶ καὶ τοῦ ἡμικυκλίου διάμετρος δηλαδὴ καὶ τοῦ κύκλου, μείζων ἐστὶ πασῶν τῶν εἰς τὸν κύκλον ἢ τὴν σφαῖραν διαγομένων [εὐθειῶν]. ἔστω οὖν ἐν μὲν τῇ μείζονι σφαίρᾳ κύκλος ὁ ΒΓΔΕ, ἐν δὲ τῇ ἐλάσσονι σφαίρᾳ κύκλος ὁ ΖΗΘ, καὶ ἡθῶσαν αὐτῶν δύο διαμέτροι πρὸς ὀρθὰς ἀλλήλαις αἱ ΒΔ, ΓΕ, καὶ δύο κύκλων περὶ τὸ αὐτὸ κέντρον ὄντων τῶν ΒΓΔΕ, ΖΗΘ εἰς τὸν μείζονα κύκλον τὸν ΒΓΔΕ πολὺγωνον ἰσοπλευρον καὶ ἀρτιόπλευρον ἐγγεγράφθω μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ ΖΗΘ, οὗ πλευραὶ ἔστωσαν ἐν τῷ ΒΕ τεταρτημορίῳ αἱ ΒΚ, ΚΛ, ΛΜ, ΜΕ, καὶ ἐπιξευθθεῖσα ἡ ΚΑ διήθῃ ἐπὶ τὸ Ν, καὶ ἀνεσάτω ἀπὸ τοῦ Α σημείον τῷ τοῦ ΒΓΔΕ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἡ ΑΧ καὶ συμβαλλέτω τῇ ἐπιφανείᾳ τῆς σφαίρας κατὰ τὸ Χ, καὶ διὰ τῆς ΑΧ καὶ ἑκατέρᾳ τῶν ΒΔ, ΚΝ ἐπίπεδα ἐκβεβλήσθω· ποιήσουσι δὴ διὰ τὰ εἰρημένα ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας μεγίστους κύκλους. ποιείτωσαν, ὧν ἡμικύκλια ἔστω ἐπὶ τῶν ΒΔ, ΚΝ διαμέτρων τὰ ΒΧΔ, ΚΧΝ. καὶ ἐπεὶ ἡ ΧΑ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, καὶ πάντα ἄρα τὰ διὰ τῆς ΧΑ ἐπίπεδά ἐστιν ὀρθὰ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον· ὥστε καὶ τὰ ΒΧΔ, ΚΧΝ ἡμικύκλια ὀρθὰ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. καὶ ἐπεὶ ἴσα ἐστὶ τὰ ΒΕΔ, ΒΧΔ, ΚΧΝ ἡμικύκλια· ἐπὶ γὰρ ἴσων εἰσὶ διαμέτρων τῶν ΒΔ, ΚΝ· ἴσα ἐστὶ καὶ τὰ ΒΕ, ΒΧ, ΚΧ τεταρτημόρια ἀλλήλοισι. ὅσαι ἄρα εἰσὶν ἐν τῷ ΒΕ τεταρτημορίῳ πλευραὶ τοῦ πολυγώνου, τοσαῦταί εἰσι καὶ ἐν τοῖς ΒΧ, ΚΧ τεταρτημορίοις ἴσαι ταῖς ΒΚ, ΚΛ, ΛΜ, ΜΕ εὐθείαις. ἐγγεγράφθωσαν καὶ ἔστωσαν αἱ ΒΟ, ΟΠ, ΠΡ, ΡΧ, ΚΣ, ΣΤ, ΤΥ, ΥΧ, καὶ ἐπεξευθθωσαν αἱ ΣΟ, ΤΠ, ΥΡ, καὶ ἀπὸ τῶν Ο, Σ ἐπὶ τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον κάθετοι ἡθῶσαν· πεσοῦνται δὴ ἐπὶ τὰς κοινὰς τομὰς τῶν ἐπιπέδων τὰς ΒΔ, ΚΝ, ἐπειδήπερ καὶ τὰ τῶν ΒΧΔ, ΚΧΝ ἐπίπεδα ὀρθὰ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. πιπτέτωσαν, καὶ ἔστωσαν αἱ ΟΦ, ΣΘ, καὶ ἐπεξευθθῶ ἡ ΘΦ. καὶ ἐπεὶ ἐν ἴσοις ἡμικυκλίοις τοῖς ΒΧΔ, ΚΧΝ ἴσαι ἀπειλημέναι εἰσὶν αἱ ΒΟ, ΚΣ, καὶ κάθετοι ἡγμέναι εἰσὶν αἱ ΟΦ, ΣΘ, ἴση [ἄρα] ἐστὶν ἡ μὲν ΟΦ τῇ ΣΘ, ἡ δὲ ΒΦ τῇ ΚΘ. ἐστὶ δὲ καὶ ὅλη ἡ ΒΑ ὅλη τῇ ΚΑ ἴση· καὶ λοιπὴ ἄρα ἡ ΦΑ λοιπὴ τῇ ΘΑ ἐστὶν ἴση· ἐστὶν ἄρα ὥς ἡ ΒΦ πρὸς τὴν ΦΑ, οὕτως ἡ ΚΘ πρὸς τὴν ΘΑ· παράλληλος ἄρα ἐστὶν ἡ ΘΦ τῇ ΚΒ. καὶ ἐπεὶ ἑκατέρα τῶν ΟΦ, ΣΘ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, παράλληλος ἄρα ἐστὶν ἡ ΟΦ τῇ ΣΘ. εἰδείθη δὲ αὐτῇ καὶ ἴση καὶ αἱ ΘΦ, ΣΟ ἄρα ἴσαι εἰσὶ καὶ παράλληλοι. καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΘΦ τῇ ΣΟ, ἀλλὰ ἡ ΘΦ τῇ ΚΒ ἐστὶ παράλληλος, καὶ ἡ ΣΟ ἄρα τῇ ΚΒ ἐστὶ παράλληλος. καὶ ἐπιξευγνόνουσιν αὐτὰς αἱ ΒΟ, ΚΣ· τὸ ΚΒΟΣ

Let the spheres be cut by some plane through the center. So, the sections will be circles, inasmuch as a sphere is generated by the diameter remaining behind, and a semi-circle being carried around [Def. 11.14]. And, hence, whatever position we conceive (of for) the semi-circle, the plane produced through it will make a circle on the surface of the sphere. And (it is) clear that (it is) also a great (circle), inasmuch as the diameter of the sphere, which is also manifestly the diameter of the semi-circle and the circle, is greater than all of the (other) [straight-lines] drawn across in the circle or the sphere [Prop. 3.15]. Therefore, let $BCDE$ be the circle in the greater sphere, and FGH the circle in the lesser sphere. And let two diameters of them be drawn at right-angles to one another, (namely), BD and CE . And there being two circles about the same center—(namely), $BCDE$ and FGH —let an equilateral and even-sided polygon be inscribed in the greater circle, $BCDE$, not touching the lesser circle, FGH [Prop. 12.16], of which let the sides in the quadrant BE be BK , KL , LM , and ME . And, KA being joined, let it be drawn across to N . And let AO be set up at point A , at right-angles to the plane of circle $BCDE$. And let it meet the surface of the (greater) sphere at O . And let planes be produced through AO and each of BD and KN . So, according to the aforementioned (discussion), they will make great circles on the surface of the (greater) sphere. Let them make (great circles), of which let BOD and KON be semi-circles on the diameters BD and KN (respectively). And since OA is at right-angles to the plane of circle $BCDE$, all of the planes through OA are thus also at right-angles to the plane of circle $BCDE$ [Prop. 11.18]. And, hence, the semi-circles BOD and KON are also at right-angles to the plane of circle $BCDE$. And since semi-circles BED , BOD , and KON are equal—for (they are) on the equal diameters BD and KN [Def. 3.1]—the quadrants BE , BO , and KO are also equal to one another. Thus, as many sides of the polygon as are in quadrant BE , so many are also in quadrants BO and KO equal to the straight-lines BK , KL , LM , and ME . Let them be inscribed, and let them be BP , PQ , QR , RO , KS , ST , TU , and UO . And let SP , TQ , and UR be joined. And let perpendiculars be drawn from P and S to the plane of circle $BCDE$ [Prop. 11.11]. So, they will fall on the common sections of the planes BD and KN (with $BCDE$), inasmuch as the planes of BOD and KON are also at right-angles to the plane of circle $BCDE$ [Def. 11.4]. Let them have fallen, and let them be PV and SW . And let VW be joined. And since BP and KS are equal (circumferences) having been cut off in the equal semi-circles BOD and KON [Def. 3.28], and PV and SW are perpendiculars having been drawn (from them), PV is [thus] equal to SW , and BV to KW [Props. 3.27, 1.26]. And the whole of BA is also equal to the whole of KA . And, thus, as BV is to VA , so KW (is) to WA . VW is thus parallel to KB [Prop. 6.2]. And since PV and SW are each at right-angles to the plane of circle $BCDE$, PV is thus parallel

ἄρα τετράπλευρον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ, ἐπειδήπερ, ἔαν ὥσι δύο εὐθείαι παράλληλοι, καὶ ἐφ' ἑκατέρᾳ αὐτῶν ληφθῇ τυθόντα σημεῖα, ἢ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεία ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παράλληλοις. διὰ τὰ αὐτὰ δὴ καὶ ἐκάτερον τῶν ΣΟΠΤ, ΤΠΡΥ τετραπλεύρων ἐν ἐνὶ ἐστὶν ἐπιπέδῳ. ἔστι δὲ καὶ τὸ ΥΡΧ τρίγωνον ἐν ἐνὶ ἐπιπέδῳ. ἔαν δὴ νοήσωμεν ἀπὸ τῶν Ο, Σ, Π, Τ, Ρ, Υ σημείων ἐπὶ τὸ Α ἐπιζευγνυμένας εὐθείας, συσταθήσεται τι σῆμα στερεὸν πολύεδρον ματαχὺ τῶν ΒΧ, ΚΧ περιφερειῶν ἐκ πυραμίδων συγκείμενον, ὧν βάσεις μὲν τὰ ΚΒΟΣ, ΣΟΠΤ, ΤΠΡΥ τετράπλευρα καὶ τὸ ΥΡΧ τρίγωνον, κορυφὴ δὲ τὸ Α σημεῖον. ἔαν δὲ καὶ ἐπὶ ἐκάστης τῶν ΚΛ, ΛΜ, ΜΕ πλευρῶν καθάπερ ἐπὶ τῆς ΒΚ τὰ αὐτὰ κατασκευάσωμεν καὶ ἔτι τῶν λοιπῶν τριῶν τεταρτημορίων, συσταθήσεται τι σῆμα πολύεδρον ἐγγεγραμμένον εἰς τὴν σφαῖραν πυραμίσι περιεχόμενον, ὧν βάσεις [μὲν] τὰ εἰρημένα τετράπλευρα καὶ τὸ ΥΡΧ τρίγωνον καὶ τὰ ὁμοταγῇ αὐτοῖς, κορυφὴ δὲ τὸ Α σημεῖον.

Λέγω ὅτι τὸ εἰρημένον πολύεδρον οὐκ ἐφάπεται τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν, ἐφ' ἣς ἐστὶν ὁ ΖΗΘ κύκλος.

Ἦθθω ἀπὸ τοῦ Α σημεῖον ἐπὶ τὸ τοῦ ΚΒΟΣ τετραπλεύρου ἐπίπεδον κάθετος ἡ ΑΨ καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ Ψ σημεῖον, καὶ ἐπεξενῆθωσαν αἱ ΨΒ, ΨΚ. καὶ ἐπεὶ ἡ ΑΨ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΚΒΟΣ τετραπλεύρου ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰς ἀπομένους αὐτῆς εὐθείας καὶ οὕτως ἐν τῷ τοῦ τετραπλεύρου ἐπιπέδῳ ὀρθὴ ἐστὶν. ἡ ΑΨ ἄρα ὀρθὴ ἐστὶ πρὸς ἑκατέραν τῶν ΒΨ, ΨΚ. καὶ ἐπεὶ ἴση ἐστὶν ἡ ΑΒ τῇ ΑΚ, ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς ΑΒ τῷ ἀπὸ τῆς ΑΚ. καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς ΑΒ ἴσα τὰ ἀπὸ τῶν ΑΨ, ΨΒ· ὀρθὴ γάρ ἡ πρὸς τῷ Ψ· τῷ δὲ ἀπὸ τῆς ΑΚ ἴσα τὰ ἀπὸ τῶν ΑΨ, ΨΚ. τὰ ἄρα ἀπὸ τῶν ΑΨ, ΨΒ ἴσα ἐστὶ τοῖς ἀπὸ τῶν ΑΨ, ΨΚ. κοινὸν ἀφηρήσθω τὸ ἀπὸ τῆς ΑΨ· λοιπὸν ἄρα τὸ ἀπὸ τῆς ΒΨ λοιπὸν τῷ ἀπὸ τῆς ΨΚ ἴσον ἐστίν· ἴση ἄρα ἡ ΒΨ τῇ ΨΚ. ὁμοίως δὲ δείχμεν, ὅτι καὶ αἱ ἀπὸ τοῦ Ψ ἐπὶ τὰ Ο, Σ ἐπιζευγνυμένα εὐθείαι ἴσαι εἰσὶν ἑκατέρᾳ τῶν ΒΨ, ΨΚ. ὁ ἄρα κέντρον τῷ Ψ καὶ διαστήματι ἐνὶ τῶν ΨΒ, ΨΚ γραφόμενος κύκλος ἔχει καὶ διὰ τῶν Ο, Σ, καὶ ἔσται ἐν κύκλῳ τὸ ΚΒΟΣ τετράπλευρον.

Καὶ ἐπεὶ μείζων ἐστὶν ἡ ΚΒ τῆς ΘΦ, ἴση δὲ ἡ ΘΦ τῇ ΣΟ, μείζων ἄρα ἡ ΚΒ τῆς ΣΟ. ἴση δὲ ἡ ΚΒ ἑκατέρᾳ τῶν ΚΣ, ΒΟ· καὶ ἑκάτερα ἄρα τῶν ΚΣ, ΒΟ τῆς ΣΟ μείζων ἐστίν. καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρον ἐστὶ τὸ ΚΒΟΣ, καὶ ἴσαι αἱ ΚΒ, ΒΟ, ΚΣ, καὶ ἐλάττων ἡ ΟΣ, καὶ ἐκ τοῦ κέντρον τοῦ κύκλου ἐστὶν ἡ ΒΨ, τὸ ἄρα ἀπὸ τῆς ΚΒ τοῦ ἀπὸ τῆς ΒΨ μείζον ἐστὶν ἢ διπλάσιον. ἦθθω ἀπὸ τοῦ Κ ἐπὶ τὴν ΒΦ κάθετος ἡ ΚΩ. καὶ ἐπεὶ ἡ ΒΔ τῆς ΔΩ ἐλάττων ἐστὶν ἢ διπλῇ, καὶ ἐστὶν ὡς ἡ ΒΔ πρὸς τὴν ΔΩ, οὕτως τὸ ὑπὸ τῶν ΔΒ, ΒΩ πρὸς τὸ ὑπὸ [τῶν] ΔΩ, ΩΒ, ἀναγραφόμενον ἀπὸ τῆς ΒΩ τετραγώνον καὶ συμπληρουμένον τοῦ ἐπὶ τῆς ΩΔ παραλληλογράμμου καὶ τὸ ὑπὸ ΔΒ, ΒΩ ἄρα τοῦ ὑπὸ ΔΩ, ΩΒ ἐλαττόν ἐστὶν ἢ διπλάσιον. καὶ ἐστὶ τῆς ΚΔ ἐπιζευγνυμένης τὸ μὲν ὑπὸ ΔΒ,

to SW [Prop. 11.6]. And it was also shown (to be) equal to it. And, thus, WV and SP are equal and parallel [Prop. 1.33]. And since WV is parallel to SP, but WV is parallel to KB, SP is thus also parallel to KB [Prop. 11.1]. And BP and KS join them. Thus, the quadrilateral KBPS is in one plane, inasmuch as if there are two parallel straight-lines, and a random point is taken on each of them, then the straight-line joining the points is in the same plane as the parallel (straight-lines) [Prop. 11.7]. So, for the same (reasons), each of the quadrilaterals SPQT and TQRU is also in one plane. And triangle URO is also in one plane [Prop. 11.2]. So, if we conceive straight-lines joining points P, S, Q, T, R, and U to A then some solid polyhedral figure will be constructed between the circumferences BO and KO, being composed of pyramids whose bases (are) the quadrilaterals KBPS, SPQT, TQRU, and the triangle URO, and apex the point A. And if we also make the same construction on each of the sides KL, LM, and ME, just as on BK, and, further, (repeat the construction) in the remaining three quadrants, then some polyhedral figure which has been inscribed in the sphere will be constructed, being contained by pyramids whose bases (are) the aforementioned quadrilaterals, and triangle URO, and the (quadrilaterals and triangles) similarly arranged to them, and apex the point A.

So, I say that the aforementioned polyhedron will not touch the lesser sphere on the surface on which the circle FGH is (situated).

Let the perpendicular (straight-line) AX be drawn from point A to the plane KBPS, and let it meet the plane at point X [Prop. 11.11]. And let XB and XK be joined. And since AX is at right-angles to the plane of quadrilateral KBPS, it is thus also at right-angles to all of the straight-lines joined to it which are also in the plane of the quadrilateral [Def. 11.3]. Thus, AX is at right-angles to each of BX and XK. And since AB is equal to AK, the (square) on AB is also equal to the (square) on AK. And the (sum of the squares) on AX and XB is equal to the (square) on AB. For the angle at X (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on AX and XK is equal to the (square) on AK [Prop. 1.47]. Thus, the (sum of the squares) on AX and XB is equal to the (sum of the squares) on AX and XK. Let the (square) on AX be subtracted from both. Thus, the remaining (square) on BX is equal to the remaining (square) on XK. Thus, BX (is) equal to XK. So, similarly, we can show that the straight-lines joined from X to P and S are equal to each of BX and XK. Thus, a circle drawn (in the plane of the quadrilateral) with center X, and radius one of XB or XK, will also pass through P and S, and the quadrilateral KBPS will be inside the circle.

And since KB is greater than WV, and WV (is) equal to SP, KB (is) thus greater than SP. And KB (is) equal to each of KS and BP. Thus, KS and BP are each greater than SP. And since quadrilateral KBPS is in a circle, and KB, BP, and

$B\Omega$ ἴσον τῷ ἀπὸ τῆς BK , τὸ δὲ ὑπὸ τῶν $\Delta\Omega$, ΩB ἴσον τῷ ἀπὸ τῆς $K\Omega$. τὸ ἄρα ἀπὸ τῆς KB τοῦ ἀπὸ τῆς $K\Omega$ ἔλασσόν ἐστιν ἢ διπλάσιον. ἀλλὰ τὸ ἀπὸ τῆς KB τοῦ ἀπὸ τῆς $B\psi$ μείζον ἐστιν ἢ διπλάσιον· μείζον ἄρα τὸ ἀπὸ τῆς $K\Omega$ τοῦ ἀπὸ τῆς $B\psi$. καὶ ἐπεὶ ἴση ἐστὶν ἡ BA τῇ KA , ἴσον ἐστὶ τὸ ἀπὸ τῆς BA τῷ ἀπὸ τῆς AK . καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς BA ἴσα τὰ ἀπὸ τῶν $B\psi$, ψA , τῷ δὲ ἀπὸ τῆς KA ἴσα τὰ ἀπὸ τῶν $K\Omega$, ΩA . τὰ ἄρα ἀπὸ τῶν $B\psi$, ψA ἴσα ἐστὶ τοῖς ἀπὸ τῶν $K\Omega$, ΩA , ὧν τὸ ἀπὸ τῆς $K\Omega$ μείζον τοῦ ἀπὸ τῆς $B\psi$. λοιπὸν ἄρα τὸ ἀπὸ τῆς ΩA ἔλασσόν ἐστι τοῦ ἀπὸ τῆς ψA . μείζον ἄρα ἡ $A\psi$ τῆς $A\Omega$. πολλῷ ἄρα ἡ $A\psi$ μείζον ἐστὶ τῆς AH . καὶ ἐστὶν ἡ μὲν $A\psi$ ἐπὶ μίαν τοῦ πολυέδρου βάσιν, ἡ δὲ AH ἐπὶ τὴν τῆς ἐλάσσονος σφαίρας ἐπιφάνειαν· ὥστε τὸ πολυέδρον οὐ ψαύσει τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Δύο ἄρα σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἷς τὴν μείζονα σφαῖραν στερεὸν πολυέδρον ἐγγέγραπται μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν· ὅπερ ἔδει ποιῆσαι.

KS are equal (to one another), and PS (is) less (than them), and BX is the radius of the circle, the (square) on KB is thus greater than double the (square) on BX .[†] Let the perpendicular KY be drawn from K to BV .[‡] And since BD is less than double DY , and as BD is to DY , so the (rectangle contained) by DB and BY (is) to the (rectangle contained) by DY and YB —a square being described on BY , and a (rectangular) parallelogram (with short side equal to BY) completed on YD —the (rectangle contained) by DB and BY is thus also less than double the (rectangle contained) by DY and YB . And, KD being joined, the (rectangle contained) by DB and BY is equal to the (square) on BK , and the (rectangle contained) by DY and YB equal to the (square) on KY [Props. 3.31, 6.8 corr.]. Thus, the (square) on KB is less than double the (square) on KY . But, the (square) on KB is greater than double the (square) on BX . Thus, the (square) on KY (is) greater than the (square) on BX . And since BA is equal to KA , the (square) on BA is equal to the (square) on AK . And the (sum of the squares) on BX and XA is equal to the (square) on BA , and the (sum of the squares) on KY and YA (is) equal to the (square) on KA [Prop. 1.47]. Thus, the (sum of the squares) on BX and XA is equal to the (sum of the squares) on KY and YA , of which the (square) on KY (is) greater than the (square) on BX . Thus, the remaining (square) on YA is less than the (square) on XA . Thus, AX (is) greater than AY . Thus, AX is much greater than AG .[§] And AX is (a perpendicular) on one of the bases of the polyhedron, and AG (is a perpendicular) on the surface of the lesser sphere. Hence, the polyhedron will not touch the lesser sphere on its surface.

Thus, there being two spheres about the same center, a polyhedral solid has been inscribed in the greater sphere which does not touch the lesser sphere on its surface. (Which is) the very thing it was required to do.

[†] Since KB , BP , and KS are greater than the sides of an inscribed square, which are each of length $\sqrt{2}BX$.

[‡] Note that points Y and V are actually identical.

[§] This conclusion depends on the fact that the chord of the polygon in proposition 12.16 does not touch the inner circle.

Πόρισμα.

Ἐάν δὲ καὶ εἷς ἐτάραν σφαῖραν τῷ ἐν τῇ $B\Gamma\Delta E$ σφαίρᾳ στερεῷ πολυέδρῳ ὅμοιον στερεὸν πολυέδρον ἐγγραφῇ, τὸ ἐν τῇ $B\Gamma\Delta E$ σφαίρᾳ στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ἐτέρᾳ σφαίρᾳ στερεὸν πολυέδρον τριπλασίονα λόγον ἔθει, ἥπερ ἡ τῆς $B\Gamma\Delta E$ σφαίρας διάμετρος πρὸς τὴν τῆς ἐτέρας σφαίρας διάμετρον. διαιρεθέντων γὰρ τῶν στερεῶν εἷς τὰς ὁμοιοπληθεῖς καὶ ὁμοιοταγεῖς πυραμίδας ἔσονται αἱ πυραμίδες ὅμοιαι. αἱ δὲ ὅμοιαι πυραμίδες πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν· ἡ ἄρα πυραμὶς, ἥξ βασις μὲν ἐστὶ τὸ $K\theta\sigma$ τετράπλευρον, κορυφή δὲ τὸ A σημεῖον, πρὸς τὴν

Corollary

And, also, if a similar polyhedral solid to that in sphere $BCDE$ is inscribed in another sphere then the polyhedral solid in sphere $BCDE$ has to the polyhedral solid in the other sphere the cubed ratio that the diameter of sphere $BCDE$ has to the diameter of the other sphere. For if the solids are divided into similarly numbered, and similarly situated, pyramids, then the pyramids will be similar. And similar pyramids are in the cubed ratio of corresponding sides [Prop. 12.8 corr.]. Thus, the pyramid whose base is quadrilateral $K\theta PS$, and apex the point A , will have to the similarly situated pyramid in the other

ἐν τῇ ἐτέρᾳ σφαίρᾳ ὁμοιοταγῇ πυραμίδα τριπλασίονα λόγον ἔθει, ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἥπερ ἡ AB ἐκ τοῦ κέντρον τῆς σφαίρας τῆς περὶ κέντρον τὸ A πρὸς τὴν ἐκ τοῦ κέντρον τῆς ἐτέρας σφαίρας. ὁμοίως καὶ ἐκάστη πυραμίς τῶν ἐν τῇ περὶ κέντρον τὸ A σφαίρᾳ πρὸς ἐκάστην ὁμοιοταγῇ πυραμίδα τῶν ἐν τῇ ἐτέρᾳ σφαίρᾳ τριπλασίονα λόγον ἔχει, ἥπερ ἡ AB πρὸς τὴν ἐκ τοῦ κέντρον τῆς ἐτέρας σφαίρας. καὶ ὥς ἐν τῶν ἡγουμένων πρὸς ἐν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα· ὥστε ὅλον τὸ ἐν τῇ περὶ κέντρον τὸ A σφαίρᾳ στερεὸν πολυέδρον πρὸς ὅλον τὸ ἐν τῇ ἐτέρᾳ [σφαίρᾳ] στερεὸν πολυέδρον τριπλασίονα λόγον ἔχει, ἥπερ ἡ AB πρὸς τὴν ἐκ τοῦ κέντρον τῆς ἐτέρας σφαίρας, τουτέστιν ἥπερ ἡ BD διάμετρος πρὸς τὴν τῆς ἐτέρας σφαίρας διάμετρον· ὅπερ ἔδει δεῖξαι.

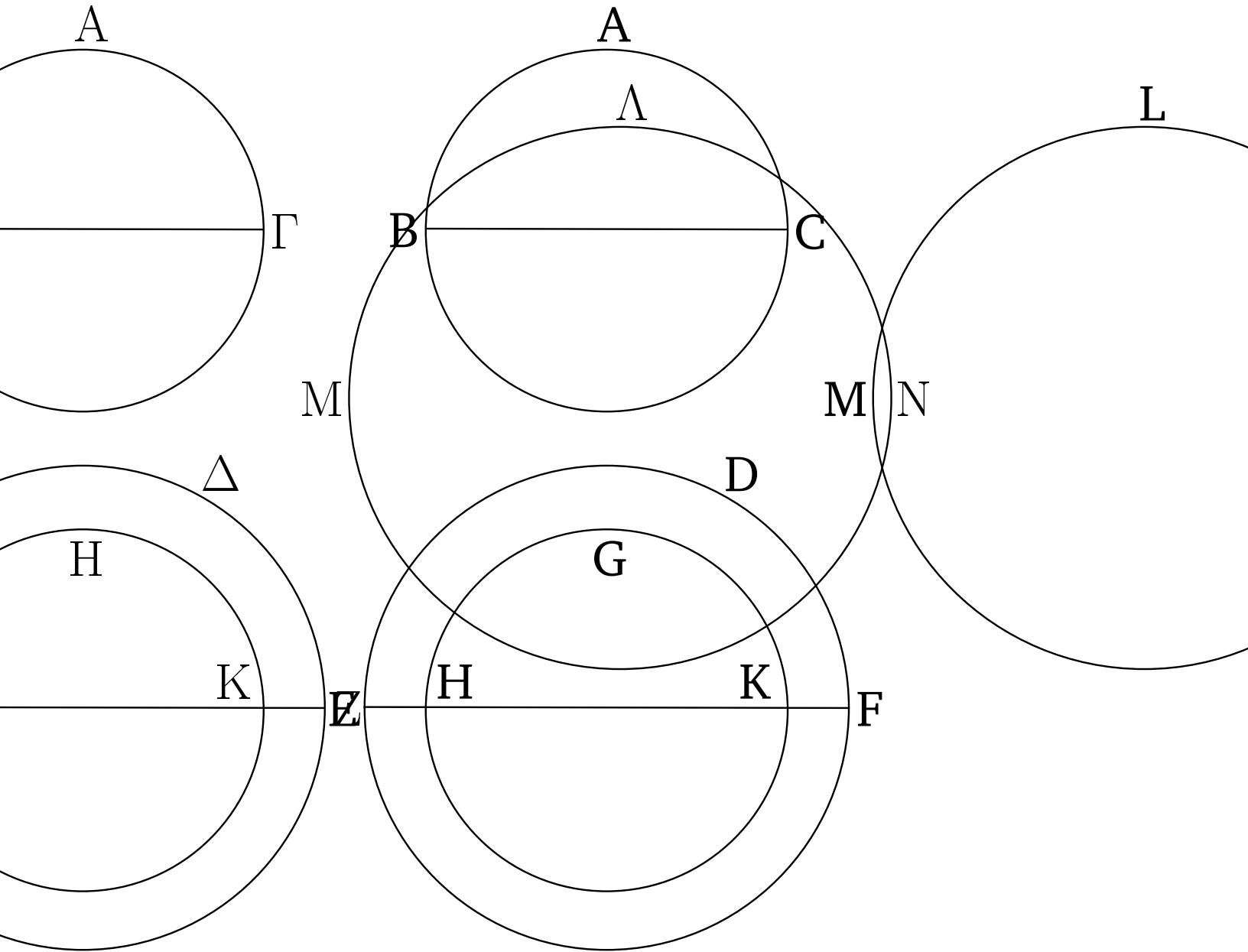
18.

Αἱ σφαῖραι πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἰδίων διαμέτρων.

sphere the cubed ratio that a corresponding side (has) to a corresponding side. That is to say, that of radius AB of the sphere about center A to the radius of the other sphere. And, similarly, each pyramid in the sphere about center A will have to each similarly situated pyramid in the other sphere the cubed ratio that AB (has) to the radius of the other sphere. And as one of the leading (magnitudes is) to one of the following (in two sets of proportional magnitudes), so (the sum of) all the leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. Hence, the whole polyhedral solid in the sphere about center A will have to the whole polyhedral solid in the other [sphere] the cubed ratio that (radius) AB (has) to the radius of the other sphere. That is to say, that diameter BD (has) to the diameter of the other sphere. (Which is) the very thing it was required to show.

Proposition 18

Spheres are to one another in the cubed ratio of their respective diameters.



Νενοήσθωσαν σφαῖραι αἱ $ABΓ$, $ΔΕΖ$, διάμετροι δὲ αὐτῶν αἱ $BΓ$, $ΕΖ$: λέγω, ὅτι ἡ $ABΓ$ σφαῖρα πρὸς τὴν $ΔΕΖ$ σφαῖραν τριπλασίονα λόγον ἔχει ἢ περὶ ἡ $BΓ$ πρὸς τὴν $ΕΖ$.

Εἰ γάρ μὴ ἡ $ABΓ$ σφαῖρα πρὸς τὴν $ΔΕΖ$ σφαῖραν τριπλασίονα λόγον ἔχει ἢ περὶ ἡ $BΓ$ πρὸς τὴν $ΕΖ$, ἔχει ἄρα ἡ $ABΓ$ σφαῖρα πρὸς ἐλάσσονά τινα τῆς $ΔΕΖ$ σφαίρας τριπλασίονα λόγον ἢ πρὸς μείζονα ἢ περὶ ἡ $BΓ$ πρὸς τὴν $ΕΖ$. ἐθέτω πρότερον πρὸς ἐλάσσονα τὴν $ΗΘΚ$, καὶ νενοήσθω ἡ $ΔΕΖ$ τῇ $ΗΘΚ$ περὶ τὸ αὐτὸ κέντρον, καὶ ἐγγεγράφθω εἰς τὴν μείζονα σφαῖραν τὴν $ΔΕΖ$ στερεὸν πολύεδρον μὴ ψαῖον τῆς ἐλάσσονος σφαίρας τῆς $ΗΘΚ$ κατὰ τὴν ἐπιφάνειαν, ἐγγεγράφθω δὲ καὶ εἰς τὴν $ABΓ$ σφαῖραν τῷ ἐν τῇ $ΔΕΖ$ σφαίρᾳ στερεῷ πολυ-

Let the spheres ABC and DEF be conceived, and (let) their diameters (be) BC and EF (respectively). I say that sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF .

For if sphere ABC does not have to sphere DEF the cubed ratio that BC (has) to EF then sphere ABC will have to some (sphere) either less than, or greater than, sphere DEF the cubed ratio that BC (has) to EF . Let it, first of all, have (such a ratio) to a lesser (sphere), GHK . And let DEF be conceived about the same center as GHK . And let a polyhedral solid be inscribed in the greater sphere DEF , not touching the lesser sphere GHK on its surface [Prop. 12.17]. And let a polyhedral solid, similar to the polyhedral solid in sphere DEF ,

ἐδρῳ ὅμοιον στερεὸν πολυέδρον· τὸ ἄρα ἐν τῇ $AB\Gamma$ στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ΔEZ στερεὸν πολυέδρον τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ἔθει δὲ καὶ ἡ $AB\Gamma$ σφαῖρα πρὸς τὴν $H\Theta K$ σφαῖραν τριπλασίονα λόγον ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ἔστιν ἄρα ὥς ἡ $AB\Gamma$ σφαῖρα πρὸς τὴν $H\Theta K$ σφαῖραν, οὕτως τὸ ἐν τῇ $AB\Gamma$ σφαίρα στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ΔEZ σφαίρα στερεὸν πολυέδρον· ἐναλλάξ [ἄρα] ὥς ἡ $AB\Gamma$ σφαῖρα πρὸς τὸ ἐν αὐτῇ πολυέδρον, οὕτως ἡ $H\Theta K$ σφαῖρα πρὸς τὸ ἐν τῇ ΔEZ σφαίρα στερεὸν πολυέδρον. μείζων δὲ ἡ $AB\Gamma$ σφαῖρα τοῦ ἐν αὐτῇ πολυέδρου· μείζων ἄρα καὶ ἡ $H\Theta K$ σφαῖρα τοῦ ἐν τῇ ΔEZ σφαίρα πολυέδρου. ἀλλὰ καὶ ἐλάττων· ἐμπεριέθεται γάρ ὑπ' αὐτοῦ. οὐκ ἄρα ἡ $AB\Gamma$ σφαῖρα πρὸς ἐλάσσονα τῆς ΔEZ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ διάμετρος πρὸς τὴν EZ . ὁμοίως δὲ δείχομεν, ὅτι οὐδὲ ἡ ΔEZ σφαῖρα πρὸς ἐλάσσονα τῆς $AB\Gamma$ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ EZ πρὸς τὴν $B\Gamma$.

Λέγω δὴ, ὅτι οὐδὲ ἡ $AB\Gamma$ σφαῖρα πρὸς μείζονά τινα τῆς ΔEZ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ .

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μείζονα τὴν ΛMN . ἀνάπαλιν ἄρα ἡ ΛMN σφαῖρα πρὸς τὴν $AB\Gamma$ σφαῖραν τριπλασίονα λόγον ἔθει ἥπερ ἡ EZ διάμετρος πρὸς τὴν $B\Gamma$ διάμετρον. ὥς δὲ ἡ ΛMN σφαῖρα πρὸς τὴν $AB\Gamma$ σφαῖραν, οὕτως ἡ ΔEZ σφαῖρα πρὸς ἐλάσσονά τινα τῆς $AB\Gamma$ σφαίρας, ἐπειδήπερ μείζων ἐστὶν ἡ ΛMN τῆς ΔEZ , ὥς ἔμπροσθεν ἐδείχθη. καὶ ἡ ΔEZ ἄρα σφαῖρα πρὸς ἐλάσσονά τινα τῆς $AB\Gamma$ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ EZ πρὸς τὴν $B\Gamma$. ὅπερ ἀδύνατον ἐδείχθη. οὐκ ἄρα ἡ $AB\Gamma$ σφαῖρα πρὸς μείζονά τινα τῆς ΔEZ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ἐδείχθη δέ, ὅτι οὐδὲ πρὸς ἐλάσσονα. ἡ ἄρα $AB\Gamma$ σφαῖρα πρὸς τὴν ΔEZ σφαῖραν τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ὅπερ ἔδει δεῖχαι.

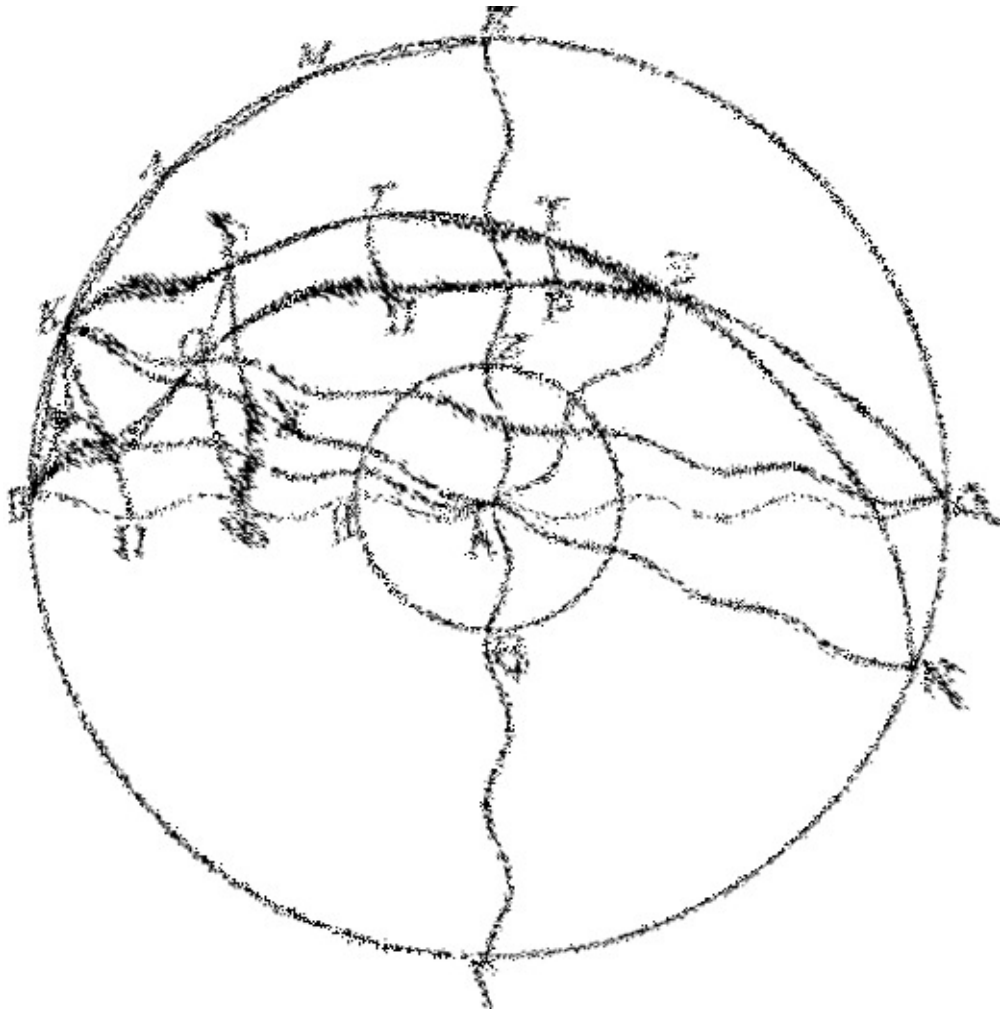
have also been inscribed in sphere ABC . Thus, the polyhedral solid in sphere ABC has to the polyhedral solid in sphere DEF the cubed ratio that BC (has) to EF [Prop. 12.17 corr.]. And sphere ABC also has to sphere GHK the cubed ratio that BC (has) to EF . Thus, as sphere ABC is to sphere GHK , so the polyhedral solid in sphere ABC (is) to the polyhedral solid in sphere DEF . [Thus], alternately, as sphere ABC (is) to the polygon within it, so sphere GHK (is) to the polyhedral solid within sphere DEF [Prop. 5.16]. And sphere ABC (is) greater than the polyhedron within it. Thus, sphere GHK (is) also greater than the polyhedron within sphere DEF [Prop. 5.14]. But, (it is) also less. For it is encompassed by it. Thus, sphere ABC does not have to (a sphere) less than sphere DEF the cubed ratio that diameter BC (has) to EF . So, similarly, we can show that sphere DEF does not have to (a sphere) less than sphere ABC the cubed ratio that EF (has) to BC either.

So, I say that sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF either.

For, if possible, let it have (the cubed ratio) to a greater (sphere), LMN . Thus, inversely, sphere LMN (has) to sphere ABC the cubed ratio that diameter EF (has) to diameter BC [Prop. 5.7 corr.]. And as sphere LMN (is) to sphere ABC , so sphere DEF (is) to some (sphere) less than sphere ABC , inasmuch as LMN is greater than DEF , as was shown before [Prop. 12.2 lem.]. And, thus, sphere DEF has to some (sphere) less than sphere ABC the cubed ratio that EF (has) to BC . The very thing was shown (to be) impossible. Thus, sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF . And it was shown that neither (does it have such a ratio) to a lesser (sphere). Thus, sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF . (Which is) the very thing it was required to show.

ELEMENTS BOOK 12

Proportional Stereometry[†]



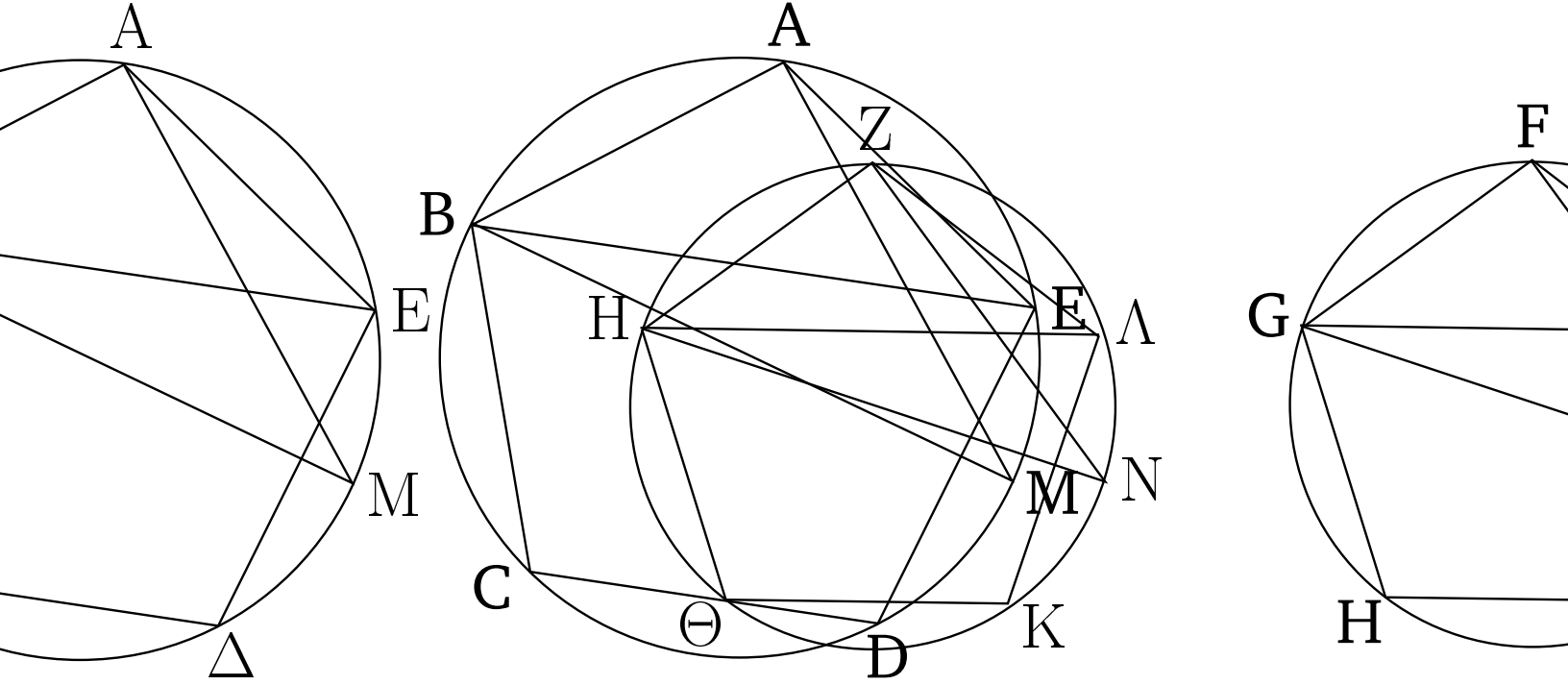
[†]The novel feature of this book is the use of the so-called method of exhaustion (see Prop. 10.1), a precursor to integration which is generally attributed to Eudoxus of Cnidus.

1.

Proposition 1

Τὰ ἐν τοῖς κύκλοις ὅμοια πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.

Similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles).



Ἐστωσαν κύκλοι οἱ $ABΓ$, $ZHΘ$, καὶ ἐν αὐτοῖς ὅμοια πολύγωνα ἔστω τὰ $ABΓΔΕ$, $ZHΘΚΛ$, διαμέτροι δὲ τῶν κύκλων ἔστωσαν BM , HN . λέγω, ὅτι ἐστὶν ὡς τὸ ἀπὸ τῆς BM τετράγωνον πρὸς τὸ ἀπὸ τῆς HN τετράγωνον, οὕτως τὸ $ABΓΔΕ$ πολύγωνον πρὸς τὸ $ZHΘΚΛ$ πολύγωνον.

Ἐπεξεύθωσαν γάρ αἱ BE , AM , HA , ZN . καὶ ἐπεὶ ὅμοιον τὸ $ABΓΔΕ$ πολύγωνον τῷ $ZHΘΚΛ$ πολυγώνῳ, ἴση ἐστὶ καὶ ἡ ὑπὸ BAE γωνία τῇ ὑπὸ HZA , καὶ ἐστὶν ὡς ἡ BA πρὸς τὴν AE , οὕτως ἡ HZ πρὸς τὴν ZA . δύο δὲ τρίγωνα ἐστὶ τὰ BAE , HZA μίαν γωνίαν μὲν γωνία ἴσην ἔχοντα τὴν ὑπὸ BAE τῇ ὑπὸ HZA , περὶ δὲ τὰς ἴσας γωνίας τὰς πλευρὰς ἀνάλογον· ἰσογώνιον ἄρα ἐστὶ τὸ ABE τρίγωνον τῷ ZHA τριγώνῳ. ἴση ἄρα ἐστὶν ἡ ὑπὸ AEB γωνία τῇ ὑπὸ ZAH . ἀλλ' ἡ μὲν ὑπὸ AEB τῇ ὑπὸ AMB ἐστὶν ἴση· ἐπὶ γὰρ τῆς αὐτῆς περιφερείας βεβήκασιν· ἡ δὲ ὑπὸ ZAH τῇ ὑπὸ ZNH καὶ ἡ ὑπὸ AMB ἄρα τῇ ὑπὸ ZNH ἐστὶν ἴση. ἔστι δὲ καὶ ὀρθή ἡ ὑπὸ BAM ὀρθή τῇ ὑπὸ HZN ἴση· καὶ ἡ λοιπὴ ἄρα τῇ λοιπῇ ἐστὶν ἴση. ἰσογώνιον ἄρα ἐστὶ τὸ ABM τρίγωνον τῷ ZHN τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ BM πρὸς τὴν HN , οὕτως ἡ BA πρὸς τὴν HZ . ἀλλὰ τοῦ μὲν τῆς BM πρὸς τὴν HN λόγον διπλασίων ἐστὶν ὁ τοῦ ἀπὸ τῆς BM τετραγώνου πρὸς τὸ ἀπὸ τῆς HN τετράγωνον, τοῦ δὲ τῆς BA πρὸς τὴν HZ διπλασίων ἐστὶν ὁ τοῦ $ABΓΔΕ$ πολυγώνου πρὸς τὸ $ZHΘΚΛ$ πολύγωνον· καὶ ὡς ἄρα τὸ ἀπὸ τῆς BM τετράγωνον πρὸς τὸ

Let ABC and FGH be circles, and let $ABCDE$ and $FGHKL$ be similar polygons (inscribed) in them (respectively), and let BM and GN be the diameters of the circles (respectively). I say that as the square on BM is to the square on GN , so polygon $ABCDE$ (is) to polygon $FGHKL$.

For let BE , AM , GL , and FN be joined. And since polygon $ABCDE$ (is) similar to polygon $FGHKL$, angle BAE is also equal to (angle) GFL , and as BA is to AE , so GF (is) to FL [Def. 6.1]. So, BAE and GFL are two triangles having one angle equal to one angle, (namely), BAE (equal) to GFL , and the sides around the equal angles proportional. Triangle ABE is thus equiangular with triangle FGL [Prop. 6.6]. Thus, angle AEB is equal to (angle) FLG . But, AEB is equal to AMB , and FLG to FNG , for they stand on the same circumference [Prop. 3.27]. Thus, AMB is also equal to FNG . And the right-angle BAM is also equal to the right-angle GFN [Prop. 3.31]. Thus, the remaining (angle) is also equal to the remaining (angle) [Prop. 1.32]. Thus, triangle ABM is equiangular with triangle FNG . Thus, proportionally, as BM is to GN , so BA (is) to GF [Prop. 6.4]. But, the (ratio) of the square on BM to the square on GN is the square of the ratio of BM to GN , and the (ratio) of polygon $ABCDE$ to polygon $FGHKL$ is the square of the (ratio) of BA to GF [Prop. 6.20]. And, thus, as the square on BM (is) to the square on GN , so polygon $ABCDE$

ἀπὸ τῆς HN τετραγώνου, οὕτως τὸ $ABΓΔΕ$ πολύγωνον πρὸς τὸ $ZHΘΚΛ$ πολύγωνον.

Τὰ ἄρα ἐν τοῖς κύκλοις ὅμοια πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα· ὅπερ εἶδει δεῖχαι.

2.

Οἱ κύκλοι πρὸς ἀλλήλους εἶσιν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.

Ἐστωσαν κύκλοι οἱ $ABΓΔ$, $EZHΘ$, διάμετροι δὲ αὐτῶν $[ἐστῶσαν]$ αἱ $BΔ$, $ZΘ$. λέγω, ὅτι ἐστὶν ὡς ὁ $ABΓΔ$ κύκλος πρὸς τὸν $EZHΘ$ κύκλον, οὕτως τὸ ἀπὸ τῆς $BΔ$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $ZΘ$ τετράγωνον.

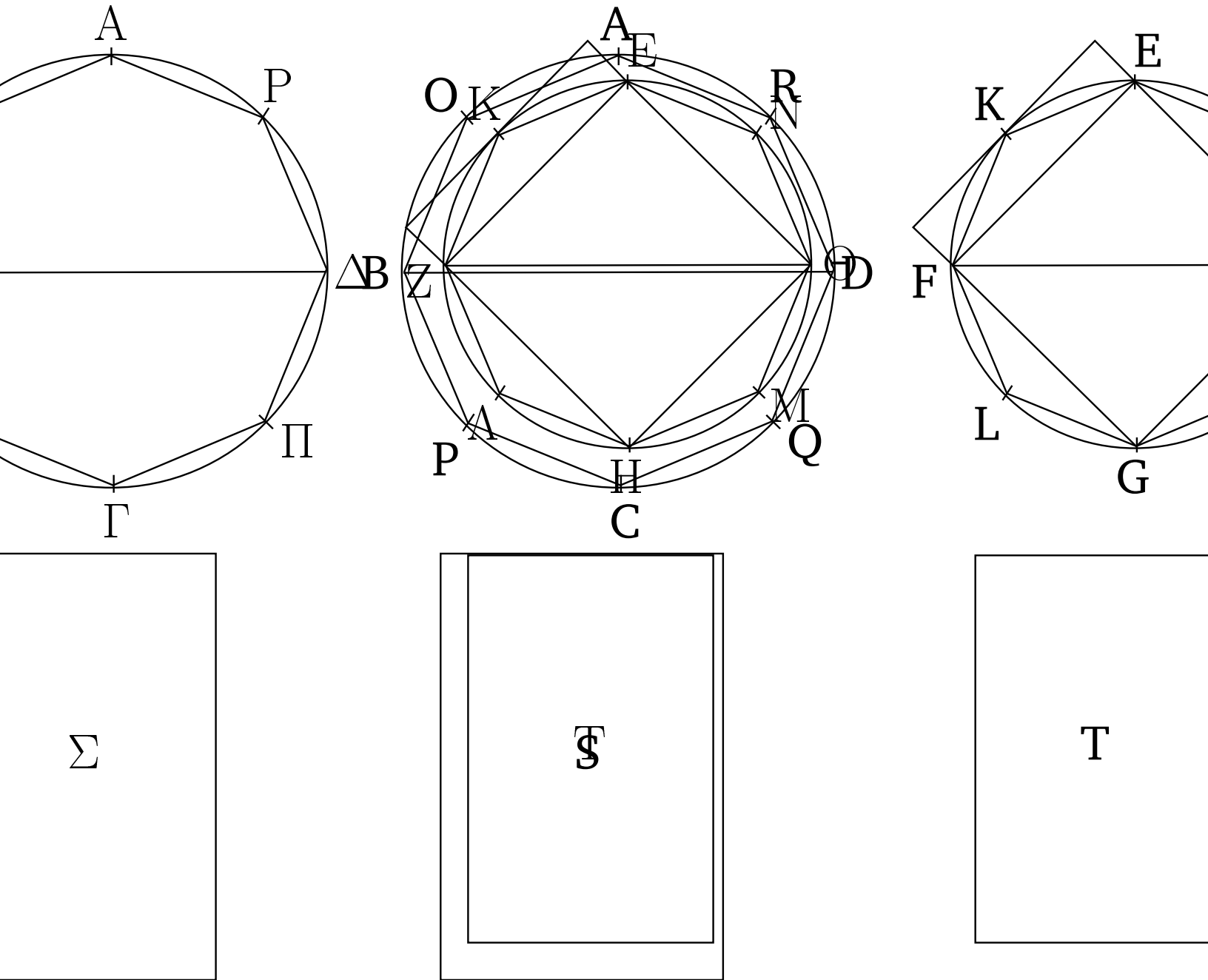
(is) to polygon $FGHKL$.

Thus, similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles). (Which is) the very thing it was required to show.

Proposition 2

Circles are to one another as the squares on (their) diameters.

Let $ABCD$ and $EFGH$ be circles, and [let] BD and FH [be] their diameters. I say that as circle $ABCD$ is to circle $EFGH$, so the square on BD (is) to the square on FH .



Εἰ γὰρ μὴ ἔστιν ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$, οὕτως τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, ἔσται ὡς τὸ ἀπὸ τῆς $B\Delta$ πρὸς τὸ ἀπὸ τῆς $Z\Theta$, οὕτως ὁ $AB\Gamma\Delta$ κύκλος ἦτοι πρὸς ἑλασσόν τι τοῦ $EZH\Theta$ κύκλου θωρίον ἢ πρὸς μείζον. ἔστω πρότερον πρὸς ἑλασσόν τὸ Σ . καὶ ἐγγεγράφω εἰς τὸν $EZH\Theta$ κύκλον τετράγωνον τὸ $EZH\Theta$. τὸ δὴ ἐγγεγραμμένον τετράγωνον μείζον ἔστιν ἢ τὸ ἥμισυ τοῦ $EZH\Theta$ κύκλου, ἐπειδήπερ ἔαν διὰ τῶν E, Z, H, Θ σημείων ἐφαπτομένας [εὐθείας] τοῦ κύκλου ἀγάγωμεν, τοῦ περιγρα-

For if the circle $ABCD$ is not to the (circle) $EFGH$, as the square on BD (is) to the (square) on FH , then as the (square) on BD (is) to the (square) on FH , so circle $ABCD$ will be to some area either less than, or greater than, circle $EFGH$. Let it, first of all, be (in that ratio) to (some) lesser (area), S . And let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. So the inscribed square is greater than half of circle $EFGH$, inasmuch as if we draw tangents to the circle through the points E, F, G , and H , then square $EFGH$ is half

φομένον περὶ τὸν κύκλον τετραγώνον ἥμισυ ἐστὶ τὸ $EZH\Theta$ τετράγωνον, τοῦ δὲ περιγραφέντος τετραγώνου ἐλάττων ἐστὶν ὁ κύκλος· ὥστε τὸ $EZH\Theta$ ἐγγεγραμμένον μὴν τετράγωνον μείζον ἐστὶ τοῦ ἡμίσεως τοῦ $EZH\Theta$ κύκλου. τεμήσθωσαν δὶθα αἱ EZ , ZH , $H\Theta$, ΘE περιφέρειαι κατὰ τὰ K , Λ , M , N σημεία, καὶ ἐπεξεύθθωσαν αἱ EK , KZ , $Z\Lambda$, ΛH , HM , $M\Theta$, ΘN , NE · καὶ ἕκαστον ἄρα τῶν EKZ , $Z\Lambda H$, $HM\Theta$, ΘNE τριγώνων μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ καθ' ἑαυτὸ τμήματος τοῦ κύκλου, ἐπειδήπερ ἔαν διὰ τῶν K , Λ , M , N σημείων ἐφαπτομένας τοῦ κύκλου ἀγάγωμεν καὶ ἀναπληρώσωμεν τὰ ἐπὶ τῶν EZ , ZH , $H\Theta$, ΘE εὐθειῶν παραλληλόγραμμα, ἕκαστον τῶν EKZ , $Z\Lambda H$, $HM\Theta$, ΘNE τριγώνων ἥμισυ ἔσται τοῦ καθ' ἑαυτὸ παραλληλογράμμου, ἀλλὰ τὸ καθ' ἑαυτὸ τμήμα ἐλάττον ἐστὶ τοῦ παραλληλογράμμου· ὥστε ἕκαστον τῶν EKZ , $Z\Lambda H$, $HM\Theta$, ΘNE τριγώνων μείζον ἐστὶ τοῦ ἡμίσεως τοῦ καθ' ἑαυτὸ τμήματος τοῦ κύκλου. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας δὶθα καὶ ἐπιξυρνήντες εὐθείας καὶ τοῦτο αἰεὶ ποιῶντες καταλείβομεν τινα ἀποτμήματα τοῦ κύκλου, ἃ ἔσται ἐλάσσονα τῆς ὑπεροθῆς, ἢ ὑπερέθει ὁ $EZH\Theta$ κύκλος τοῦ Σ θωρίον. ἐδείχθη γάρ ἐν τῷ πρώτῳ θεωρήματι τοῦ δεκάτου βιβλίου, ὅτι δύο μεγεθῶν ἀνίσων ἐκκειμένων, ἔαν ἀπὸ τοῦ μείζονος ἀφαιρεθῇ μείζον ἢ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μείζον ἢ τὸ ἥμισυ, καὶ τοῦτο αἰεὶ γίγνηται, λειφθήσεται τι μέγεθος, ὃ ἔσται ἐλάσσον τοῦ ἐκκειμένου ἐλάσσονος μεγέθους. λελείφθω οὖν, καὶ ἔστω τὰ ἐπὶ τῶν EK , KZ , $Z\Lambda$, ΛH , HM , $M\Theta$, ΘN , NE τμήματα τοῦ $EZH\Theta$ κύκλου ἐλάττονα τῆς ὑπεροθῆς, ἢ ὑπερέθει ὁ $EZH\Theta$ κύκλος τοῦ Σ θωρίον. λοιπὸν ἄρα τὸ $EKZ\Lambda HM\Theta N$ πολύγωνον μείζον ἐστὶ τοῦ Σ θωρίον. ἐγγεγράφθω καὶ εἰς τὸν $AB\Gamma\Delta$ κύκλον τῷ $EKZ\Lambda HM\Theta N$ πολυγώνῳ ὅμοιον πολύγωνον τὸ $AXBO\Gamma\Pi\Delta P$ · ἐστὶν ἄρα ὡς τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$ τετράγωνον, οὕτως τὸ $AXBO\Gamma\Pi\Delta P$ πολύγωνον πρὸς τὸ $EKZ\Lambda HM\Theta N$ πολύγωνον. ἀλλὰ καὶ ὡς τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, οὕτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ Σ θωρίον· καὶ ὡς ἄρα ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ Σ θωρίον, οὕτως τὸ $AXBO\Gamma\Pi\Delta P$ πολύγωνον πρὸς τὸ $EKZ\Lambda HM\Theta N$ πολύγωνον· ἐναλλάξ ἄρα ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸ ἐν αὐτῷ πολύγωνον, οὕτως τὸ Σ θωρίον πρὸς τὸ $EKZ\Lambda HM\Theta N$ πολύγωνον. μείζων δὲ ὁ $AB\Gamma\Delta$ κύκλος τοῦ ἐν αὐτῷ πολυγώνου· μείζων ἄρα καὶ τὸ Σ θωρίον τοῦ $EKZ\Lambda HM\Theta N$ πολυγώνου. ἀλλὰ καὶ ἐλάττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς $B\Delta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $Z\Theta$, οὕτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς ἑλασσόν τι τοῦ $EZH\Theta$ κύκλου θωρίον. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὡς τὸ ἀπὸ $Z\Theta$ πρὸς τὸ ἀπὸ $B\Delta$, οὕτως ὁ $EZH\Theta$ κύκλος πρὸς ἑλασσόν τι τοῦ $AB\Gamma\Delta$ κύκλου θωρίον.

Λέγω δὴ, ὅτι οὐδὲ ὡς τὸ ἀπὸ τῆς $B\Delta$ πρὸς τὸ ἀπὸ τῆς $Z\Theta$, οὕτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς μείζον τι τοῦ $EZH\Theta$ κύκλου θωρίον.

Εἰ γὰρ δυνατόν, ἔστω πρὸς μείζον τὸ Σ . ἀνάπαλιν ἄρα [ἐστίν] ὡς τὸ ἀπὸ τῆς $Z\Theta$ τετράγωνον πρὸς τὸ ἀπὸ τῆς $B\Delta$, οὕτως τὸ Σ θωρίον πρὸς τὸν $AB\Gamma\Delta$ κύκλον. ἀλλ' ὡς τὸ

of the square circumscribed about the circle [Prop. 1.47], and the circle is less than the circumscribed square. Hence, the inscribed square $EFGH$ is greater than half of circle $EFGH$. Let the circumferences EF , FG , GH , and HE be cut in half at points K , L , M , and N (respectively), and let EK , KF , FL , LG , GM , MH , HN , and NE be joined. And, thus, each of the triangles EKF , FLG , GMH , and HNE is greater than half of the segment of the circle about it, inasmuch as if we draw tangents to the circle through points K , L , M , and N , and complete the parallelograms on the straight-lines EF , FG , GH , and HE , then each of the triangles EKF , FLG , GMH , and HNE will be half of the parallelogram about it, but the segment about it is less than the parallelogram. Hence, each of the triangles EKF , FLG , GMH , and HNE is greater than half of the segment of the circle about it. So, by cutting the circumferences remaining behind in half, and joining straight-lines, and doing this continually, we will (eventually) leave behind some segments of the circle whose (sum) will be less than the excess by which circle $EFGH$ exceeds the area S . For we showed in the first theorem of the tenth book that if two unequal magnitudes are laid out, and if (a part) greater than a half is subtracted from the greater, and (if from) the remainder (a part) greater than a half (is subtracted), and this happens continually, then some magnitude will (eventually) be left which will be less than the lesser laid out magnitude [Prop. 10.1]. Therefore, let the (segments) be left, and let the (sum of the) segments of the circle $EFGH$ on EK , KF , FL , LG , GM , MH , HN , and NE be less than the excess by which circle $EFGH$ exceeds area S . Thus, the remaining polygon $EKFLGMHN$ is greater than area S . And let the polygon $AOBPCQDR$, similar to the polygon $EKFLGMHN$, be inscribed in circle $ABCD$. Thus, as the square on BD is to the square on FH , so polygon $AOBPCQDR$ (is) to polygon $EKFLGMHN$ [Prop. 12.1]. But, also, as the square on BD (is) to the square on FH , so circle $ABCD$ (is) to area S . And, thus, as circle $ABCD$ (is) to area S , so polygon $AOBPGQDR$ (is) to polygon $EKFLGMHN$ [Prop. 5.11]. Thus, alternately, as circle $ABCD$ (is) to the polygon (inscribed) within it, so area S (is) to polygon $EKFLGMHN$ [Prop. 5.16]. And circle $ABCD$ (is) greater than the polygon (inscribed) within it. Thus, area S is also greater than polygon $EKFLGMHN$. But, (it is) also less. The very thing is impossible. Thus, the square on BD is not to the (square) on FH , as circle $ABCD$ (is) to some area less than circle $EFGH$. So, similarly, we can show that the (square) on FH (is) not to the (square) on BD as circle $EFGH$ (is) to some area less than circle $ABCD$ either.

So, I say that neither (is) the (square) on BD to the (square) on FH , as circle $ABCD$ (is) to some area greater than circle $EFGH$.

For, if possible, let it be (in that ratio) to (some) greater (area), S . Thus, inversely, as the square on FH [is] to the (square) on DB , so area S (is) to circle $ABCD$ [Prop. 5.7 corr.].

Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ ἑλαττόν τι τοῦ $ABΓΔ$ κύκλου θωρίον· καὶ ὡς ἄρα τὸ ἀπὸ τῆς $ZΘ$ πρόξ τὸ ἀπὸ τῆς $BΔ$, οὕτως ὁ $EZHΘ$ κύκλος πρόξ ἑλασσόν τι τοῦ $ABΓΔ$ κύκλου θωρίον· ὅπερ ἀδύνατον ἐδείχθη. οὐκ ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς $BΔ$ τετράγωνον πρόξ τὸ ἀπὸ τῆς $ZΘ$, οὕτως ὁ $ABΓΔ$ κύκλος πρόξ μείζον τι τοῦ $EZHΘ$ κύκλου θωρίον. ἐδείχθη δέ, ὅτι οὐδὲ πρόξ ἑλασσον· ἔστιν ἄρα ὡς τὸ ἀπὸ τῆς $BΔ$ τετράγωνον πρόξ τὸ ἀπὸ τῆς $ZΘ$, οὕτως ὁ $ABΓΔ$ κύκλος πρόξ τὸν $EZHΘ$ κύκλον.

Οἱ ἄρα κύκλοι πρόξ ἀλλήλους εἰσὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα· ὅπερ ἔδει δεῖχαι.

Λήμμα.

Λέγω δὴ, ὅτι τοῦ Σ θωρίον μείζονος ὄντος τοῦ $EZHΘ$ κύκλου ἐστὶν ὡς τὸ Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ ἑλαττόν τι τοῦ $ABΓΔ$ κύκλου θωρίον.

Γεγονέτω γάρ ὡς τὸ Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ τὸ T θωρίον. λέγω, ὅτι ἑλαττόν ἐστι τὸ T θωρίον τοῦ $ABΓΔ$ κύκλου. ἐπεὶ γάρ ἐστιν ὡς τὸ Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ τὸ T θωρίον, ἐναλλάξ ἐστὶν ὡς τὸ Σ θωρίον πρόξ τὸν $EZHΘ$ κύκλον, οὕτως ὁ $ABΓΔ$ κύκλος πρόξ τὸ T θωρίον. μείζον δὲ τὸ Σ θωρίον τοῦ $EZHΘ$ κύκλου· μείζων ἄρα καὶ ὁ $ABΓΔ$ κύκλος τοῦ T θωρίον. ὥστε ἐστὶν ὡς τὸ Σ θωρίον πρόξ τὸν $ABΓΔ$ κύκλον, οὕτως ὁ $EZHΘ$ κύκλος πρόξ ἑλαττόν τι τοῦ $ABΓΔ$ κύκλου θωρίον· ὅπερ ἔδει δεῖχαι.

3.

Πᾶσα πυραμὶς τριγώνων ἔθουσα βάσιν διαιρεῖται εἰς δύο πυραμίδας ἴσας τε καὶ ὁμοίας ἀλλήλαις καὶ [ὁμοίας] τῇ ὅλῃ τριγώνον· ἔθουσαξ βάσεις καὶ εἰς δύο πρίσματα ἴσα· καὶ τὰ δύο πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος.

But, as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$ (see lemma). And, thus, as the (square) on FH (is) to the (square) on BD , so circle $EFGH$ (is) to some area less than circle $ABCD$ [Prop. 5.11]. The very thing was shown (to be) impossible. Thus, as the square on BD is to the (square) on FH , so circle $ABCD$ (is) not to some area greater than circle $EFGH$. And it was shown that neither (is it in that ratio) to (some) lesser (area). Thus, as the square on BD is to the (square) on FH , so circle $ABCD$ (is) to circle $EFGH$.

Thus, circles are to one another as the squares on (their) diameters. (Which is) the very thing it was required to show.

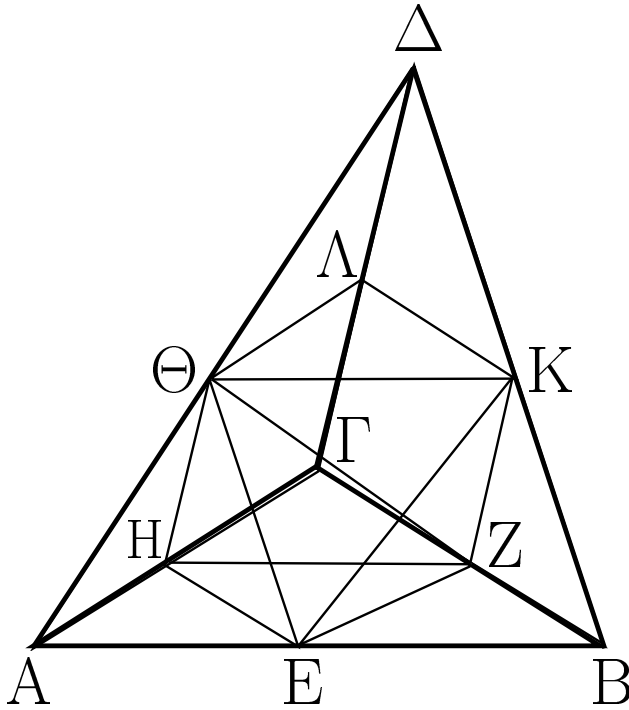
Lemma

So, I say that, area S being greater than circle $EFGH$, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$.

For let it be contrived that as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to area T . I say that area T is less than circle $ABCD$. For since as area S is to circle $ABCD$, so circle $EFGH$ (is) to area T , alternately, as area S is to circle $EFGH$, so circle $ABCD$ (is) to area T [Prop. 5.16]. And area S (is) greater than circle $EFGH$. Thus, circle $ABCD$ (is) also greater than area T [Prop. 5.14]. Hence, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$. (Which is) the very thing it was required to show.

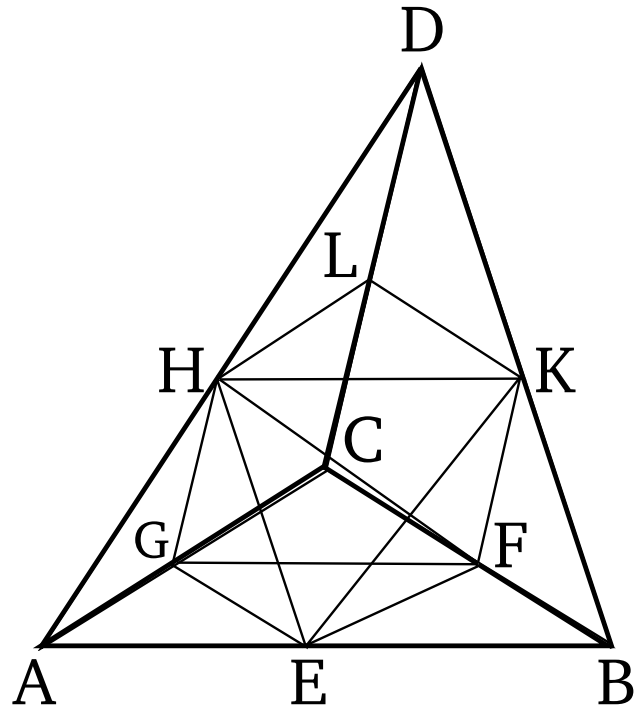
Proposition 3

Any pyramid having a triangular base is divided into two pyramids having triangular bases (which are) equal, similar to one another, and [similar] to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.



Ἐστω πυραμὶς, ἧς βάσις μὲν ἐστὶ τὸ $AB\Gamma$ τρίγωνον, κορυφή δὲ τὸ Δ σημεῖον· λέγω, ὅτι ἡ $AB\Gamma\Delta$ πυραμὶς διαιρεῖται εἰς δύο πυραμίδας ἴσας ἀλλήλαις τριγώνου βάσεις ἐθούσας καὶ ὁμοίας τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα· καὶ τὰ δύο πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος.

Τετμήσθωσαν γὰρ αἱ AB , $B\Gamma$, ΓA , AD , ΔB , $\Delta \Gamma$ διὰ κατὰ τὰ E , Z , H , Θ , K , Λ σημεία, καὶ ἐπεζεύθωσαν αἱ ΘE , $E H$, $H \Theta$, ΘK , $K \Lambda$, $\Lambda \Theta$, $K Z$, $Z H$. Ἐπεὶ ἴση ἐστὶν ἡ μὲν $A E$ τῇ $E B$, ἡ δὲ $A \Theta$ τῇ $\Delta \Theta$, παράλληλός ἔστιν ἡ $E \Theta$ τῇ ΔB . διὰ τὰ αὐτὰ δὴ καὶ ἡ ΘK τῇ $A B$ παράλληλός ἐστιν. παραλληλόγραμμον ἔστι τὸ $\Theta E B K$. ἴση ἔστιν ἡ ΘK τῇ $E B$. ἀλλὰ ἡ $E B$ τῇ $E A$ ἐστὶν ἴση· καὶ ἡ $A E$ ἔστιν ἴση τῇ ΘK ἐστὶν ἴση. ἔστι δὲ καὶ ἡ $A \Theta$ τῇ $\Theta \Delta$ ἴση· δύο δὴ αἱ $E A$, $A \Theta$ δυοὶ ταῖς $K \Theta$, $\Theta \Delta$ ἴσαι εἰσὶν ἑκατέρω ἐκατέρω· καὶ γωνία ἡ ὑπὸ $E A \Theta$ γωνία τῇ ὑπὸ $K \Theta \Delta$ ἴση· βάσις ἔστιν ἡ $E \Theta$ βάσει τῇ $K \Delta$ ἐστὶν ἴση. ἴσον ἔστιν ἔτι καὶ ὁμοῖόν ἐστι τὸ $A E \Theta$ τρίγωνον τῷ $\Theta K \Delta$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $A \Theta H$ τρίγωνον τῷ $\Theta \Lambda \Delta$ τριγώνῳ ἴσον τε καὶ ὁμοῖον. καὶ ἐπεὶ δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ $E \Theta$, ΘH παρὰ δύο εὐθείας ἀπτόμενας ἀλλήλων τὰς $K \Delta$, $\Delta \Lambda$ εἰσὶν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ οὔσαι, ἴσας γωνίας περιέχουσιν. ἴση ἔστιν ἡ ὑπὸ $E \Theta H$ γωνία τῇ ὑπὸ $K \Delta \Lambda$ γωνίᾳ. καὶ ἐπεὶ δύο εὐθεῖαι αἱ $E \Theta$, ΘH δυοὶ ταῖς $K \Delta$, $\Delta \Lambda$ ἴσαι εἰσὶν ἑκατέρω ἐκατέρω, καὶ γωνία ἡ ὑπὸ $E \Theta H$ γωνία τῇ ὑπὸ $K \Delta \Lambda$ ἐστὶν ἴση, βάσις ἔστιν ἡ $E H$ βάσει τῇ $K \Lambda$ [ἐστὶν] ἴση· ἴσον ἔστιν ἔτι καὶ ὁμοῖόν ἐστι τὸ $E \Theta H$ τρίγωνον τῷ $K \Delta \Lambda$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $A E H$ τρίγωνον τῷ $\Theta K \Lambda$ τριγώνῳ ἴσον τε καὶ ὁμοῖον ἐστὶν. ἡ ἔστιν πυραμὶς, ἧς βάσις μὲν ἐστὶ τὸ $A E H$ τρίγωνον, κορυφή δὲ τὸ Θ σημείον, ἴση καὶ ὁμοία ἐστὶ πυραμίδι, ἧς



Let there be a pyramid whose base is triangle ABC , and (whose) apex (is) point D . I say that pyramid $ABCD$ is divided into two pyramids having triangular bases (which are) equal to one another, and similar to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.

For let AB , BC , CA , AD , DB , and DC be cut in half at points E , F , G , H , K , and L (respectively). And let HE , EG , GH , HK , KL , LH , KF , and FG be joined. Since AE is equal to EB , and AH to HD , EH is thus parallel to DB [Prop. 6.2]. So, for the same (reasons), HK is also parallel to AB . Thus, $HEBK$ is a parallelogram. Thus, HK is equal to EB [Prop. 1.34]. But, EB is equal to EA . Thus, AE is also equal to HK . And AH is also equal to HD . So the two (straight-lines) EA and AH are equal to the two (straight-lines) KH and HD , respectively. And angle EAH (is) equal to angle KHD [Prop. 1.29]. Thus, base EH is equal to base KD [Prop. 1.4]. Thus, triangle AEH is equal and similar to triangle HKD [Prop. 1.4]. So, for the same (reasons), triangle AHG is also equal and similar to triangle HLD . And since EH and HG are two straight-lines joining one another (which are respectively) parallel to two straight-lines joining one another, KD and DL , not being in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle EHG is equal to angle KDL . And since the two straight-lines EH and HG are equal to the two straight-lines KD and DL , respectively, and angle EHG is equal to angle KDL , base EG [is] thus equal to base KL [Prop. 1.4]. Thus, triangle EHG is equal and similar to triangle KDL . So, for the same (reasons), triangle AEG

βάσις μὲν ἐστὶ τὸ $\Theta\text{ΚΛ}$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. καὶ ἐπεὶ τρίγωνον τοῦ ΑΔΒ παρὰ μίαν τῶν πλευρῶν τὴν ΑΒ ἥκται ἡ $\Theta\text{Κ}$, ἰσογώνιον ἐστὶ τὸ ΑΔΒ τρίγωνον τῷ $\Delta\Theta\text{Κ}$ τριγώνῳ, καὶ τὰς πλευράς ἀνάλογον ἔθουσιν· ὅμοιον ἄρα ἐστὶ τὸ ΑΔΒ τρίγωνον τῷ $\Delta\Theta\text{Κ}$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν $\Delta\text{ΒΓ}$ τρίγωνον τῷ $\Delta\text{ΚΛ}$ τριγώνῳ ὅμοιον ἐστίν, τὸ δὲ ΑΔΓ τῷ $\Delta\text{ΛΘ}$. καὶ ἐπεὶ δύο εὐθεῖαι ἀπτόμεναι ἀλλήλων αἱ ΒΑ , ΑΓ παρὰ δύο εὐθείας ἀπτόμενάς ἀλλήλων τὰς ΚΘ , $\Theta\text{Λ}$ εἰσιν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσας γωνίας περιέχουσιν. ἴση ἄρα ἐστὶν ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΚΘΛ . καὶ ἐστὶν ὥς ἡ ΒΑ πρὸς τὴν ΑΓ , οὕτως ἡ ΚΘ πρὸς τὴν $\Theta\text{Λ}$. ὅμοιον ἄρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ $\Theta\text{ΚΛ}$ τριγώνῳ. καὶ πυραμὶς ἄρα, ἥς βάσις μὲν ἐστὶ τὸ ΑΒΓ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ὁμοία ἐστὶ πυραμίδι, ἥς βάσις μὲν ἐστὶ τὸ $\Theta\text{ΚΛ}$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. ἀλλὰ πυραμὶς, ἥς βάσις μὲν [ἐστὶ] τὸ $\Theta\text{ΚΛ}$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ὁμοία ἐδείχθη πυραμίδι, ἥς βάσις μὲν ἐστὶ τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημείον. ἑκατέρα ἄρα τῶν ΑΕΗΘ , $\Theta\text{ΚΛΔ}$ πυραμίδων ὁμοία ἐστὶ τῇ ὅλῃ τῇ ΑΒΓΔ πυραμίδι.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ ΒΖ τῇ ΖΓ , διπλάσιον ἐστὶ τὸ ΕΒΖΗ παραλληλόγραμμον τοῦ ΗΖΓ τριγώνου. καὶ ἐπεὶ, ἔαν ἦ δύο πρίσματα ἰσοῦπῃ, καὶ τὸ μὲν ἔθῃ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ἦ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσα ἐστὶ τὰ πρίσματα, ἴσον ἄρα ἐστὶ τὸ πρίσμα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΒΚΖ , ΕΘΗ , τριῶν δὲ παραλληλογράμμων τῶν ΕΒΖΗ , ΕΒΚΘ , $\Theta\text{ΚΖΗ}$ τῷ πρισματι τῷ περιεχομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΗΖΓ , $\Theta\text{ΚΛ}$, τριῶν δὲ παραλληλογράμμων τῶν ΚΖΓΛ , ΛΓΗΘ , $\Theta\text{ΚΖΗ}$. καὶ φανερόν, ὅτι ἐκάτρων τῶν πρισμάτων, οὗ τε βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $\Theta\text{Κ}$ εὐθεῖα, καὶ οὗ βάσις τὸ ΗΖΓ τρίγωνον, ἀπεναντίον δὲ τὸ $\Theta\text{ΚΛ}$ τρίγωνον, μεῖζόν ἐστὶν ἑκατέρως τῶν πυραμίδων, ὧν βάσεις μὲν τὰ ΑΕΗ , $\Theta\text{ΚΛ}$ τρίγωνα, κορυφαί, δὲ τὰ Θ , Δ σημεία, ἐπειδὴ περ [καὶ] ἔαν ἐπιζεύχωμεν τὰς ΕΖ , ΕΚ εὐθείας, τὸ μὲν πρίσμα, οὗ βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $\Theta\text{Κ}$ εὐθεῖα, μεῖζόν ἐστὶ τῆς πυραμίδος, ἥς βάσις τὸ ΕΒΖ τρίγωνον, κορυφή δὲ τὸ Κ σημείον. ἀλλ' ἡ πυραμὶς, ἥς βάσις τὸ ΕΒΖ τρίγωνον, κορυφή δὲ τὸ Κ σημείον, ἴση ἐστὶ πυραμίδι, ἥς βάσις τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημείον· ὑπὸ γὰρ ἴσων καὶ ὁμοίων ἐπιπέδων περιέθονται. ὥστε καὶ τὸ πρίσμα, οὗ βάσις μὲν τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $\Theta\text{Κ}$ εὐθεῖα, μεῖζόν ἐστὶ πυραμίδος, ἥς βάσις μὲν τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημείον. ἴσον δὲ τὸ μὲν πρίσμα, οὗ βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $\Theta\text{Κ}$ εὐθεῖα, τῷ πρισματι, οὗ βάσις μὲν τὸ ΗΖΓ τρίγωνον, ἀπεναντίον δὲ τὸ $\Theta\text{ΚΛ}$ τρίγωνον· ἡ δὲ πυραμὶς, ἥς βάσις τὸ ΑΕΗ τρίγωνον, κορυφή δὲ τὸ Θ σημείον, ἴση ἐστὶ πυραμίδι, ἥς βάσις τὸ $\Theta\text{ΚΛ}$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. τὰ ἄρα εἰρημένα δύο πρίσματα μεῖζονά ἐστὶ τῶν εἰρημένων δύο πυραμίδων, ὧν βάσεις μὲν τὰ ΑΕΗ , $\Theta\text{ΚΛ}$ τρίγωνα, κορυφαί δὲ τὰ Θ , Δ σημεία.

is also equal and similar to triangle ΗΚΛ . Thus, the pyramid whose base is triangle ΑΕΓ , and apex the point Η , is equal and similar to the pyramid whose base is triangle ΗΚΛ , and apex the point Δ [Def. 11.10]. And since ΗΚ has been drawn parallel to one of the sides, ΑΒ , of triangle ΑΔΒ , triangle ΑΔΒ is equiangular to triangle ΔΗΚ [Prop. 1.29], and they have proportional sides. Thus, triangle ΑΔΒ is similar to triangle ΔΗΚ [Def. 6.1]. So, for the same (reasons), triangle ΔΒC is also similar to triangle ΔΚΛ , and ΑΔC to ΔΛΗ . And since two straight-lines joining one another, ΒΑ and ΑC , are parallel to two straight-lines joining one another, ΚΗ and ΗΛ , not in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle ΒΑC is equal to (angle) ΚΗΛ . And as ΒΑ is to ΑC , so ΚΗ (is) to ΗΛ . Thus, triangle ΑΒC is similar to triangle ΗΚΛ [Prop. 6.6]. And, thus, the pyramid whose base is triangle ΑΒC , and apex the point Δ , is similar to the pyramid whose base is triangle ΗΚΛ , and apex the point Δ [Def. 11.9]. But, the pyramid whose base [is] triangle ΗΚΛ , and apex the point Δ , was shown (to be) similar to the pyramid whose base is triangle ΑΕΓ , and apex the point Η . Thus, each of the pyramids ΑΕΓΗ and ΗΚΛΔ is similar to the whole pyramid ΑΒCΔ .

And since ΒF is equal to FC , parallelogram ΕΒFG is double triangle GFC [Prop. 1.41]. And since, if two prisms (have) equal heights, and the former has a parallelogram as a base, and the latter a triangle, and the parallelogram (is) double the triangle, then the prisms are equal [Prop. 11.39], the prism contained by the two triangles ΒΚF and ΕΗG , and the three parallelograms ΕΒFG , ΕΒΚΗ , and ΗΚFG , is thus equal to the prism contained by the two triangles GFC and ΗΚΛ , and the three parallelograms KFCL , LCGH , and ΗΚFG . And (it is) clear that each of the prisms whose base (is) parallelogram ΕΒFG , and opposite (side) straight-line ΗΚ , and whose base (is) triangle GFC , and opposite (plane) triangle ΗΚΛ , is greater than each of the pyramids whose bases are triangles ΑΕΓ and ΗΚΛ , and apexes the points Η and Δ (respectively), inasmuch as, if we [also] join the straight-lines ΕF and ΕΚ then the prism whose base (is) parallelogram ΕΒFG , and opposite (side) straight-line ΗΚ , is greater than the pyramid whose base (is) triangle ΕΒF , and apex the point Κ . But the pyramid whose base (is) triangle ΕΒF , and apex the point Κ , is equal to the pyramid whose base is triangle ΑΕΓ , and apex point Η . For they are contained by equal and similar planes. And, hence, the prism whose base (is) parallelogram ΕΒFG , and opposite (side) straight-line ΗΚ , is greater than the pyramid whose base (is) triangle ΑΕΓ , and apex the point Η . And the prism whose base is parallelogram ΕΒFG , and opposite (side) straight-line ΗΚ , (is) equal to the prism whose base (is) triangle GFC , and opposite (plane) triangle ΗΚΛ . And the pyramid whose base (is) triangle ΑΕΓ , and apex the point Η , is equal to the pyramid whose base (is) triangle ΗΚΛ , and apex the point Δ . Thus, the (sum of the) aforementioned two prisms

Ἡ ἄρα ὅλη πυραμὶς, ἣς βάσις τὸ $ABΓ$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, διήρηται εἰς τε δύο πυραμίδαξ ἴσαξ ἀλλήλαις [καὶ ὁμοίαξ τῇ ὅλῃ] καὶ εἰς δύο πρίσματα ἴσα, καὶ τὰ δύο πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος· ὅπερ ἔδει δεῖχαι.

4.

Ἐάν ᾧσι δύο πυραμίδες ὑπὸ τὸ αὐτὸ ὕψος τριγώνουξ ἔθουσαι βάσεις, διαιρεθῇ δὲ ἑκάτερα αὐτῶν εἰς τε δύο πυραμίδαξ ἴσαξ ἀλλήλαις καὶ ὁμοίαξ τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα, ἔσται ὥς ἡ τῆς μιᾶς πυραμίδος βάσις πρὸς τὴν τῆς ἐτέρας πυραμίδος βάσιν, οὕτως τὰ ἐν τῇ μιᾷ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ ἐτέρᾳ πυραμίδι πρίσματα πάντα ἰσοπληθῇ.

Ἔστωσαν δύο πυραμίδες ὑπὸ τὸ αὐτὸ ὕψος τριγώνουξ ἔθουσαι βάσεις τὰς $ABΓ$, ΔEZ , κορυφαί δὲ τὰ H , Θ σημεία, καὶ διηρησθῶ ἑκάτερα αὐτῶν εἰς τε δύο πυραμίδαξ ἴσαξ ἀλλήλαις καὶ ὁμοίαξ τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα· λέγω, ὅτι ἐστὶν ὥς ἡ $ABΓ$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὰ ἐν τῇ $ABΓH$ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα ἰσοπληθῇ.

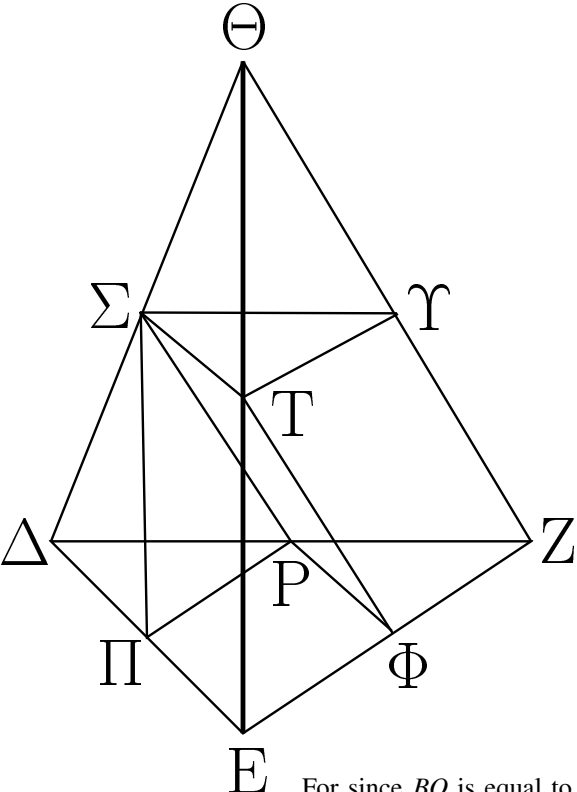
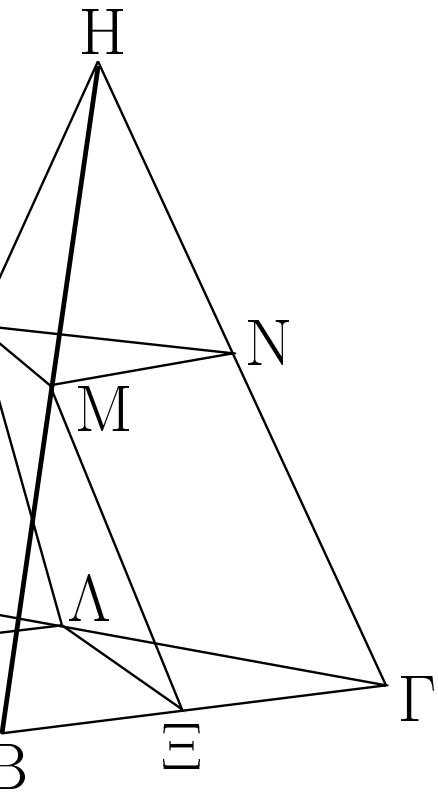
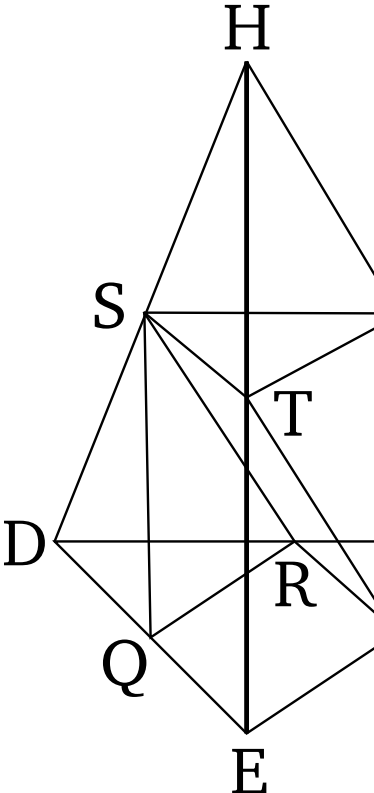
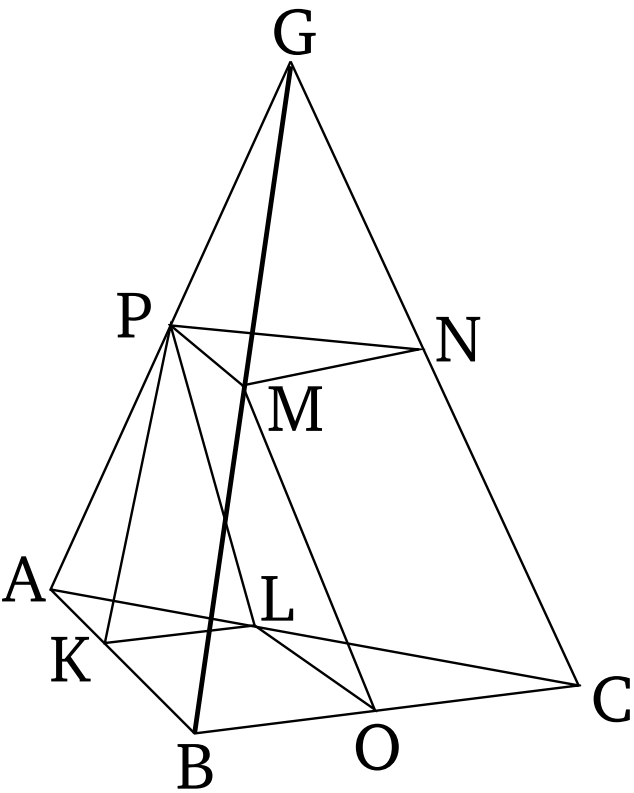
is greater than the (sum of the) aforementioned two pyramids, whose bases (are) triangles AEG and HKL , and apexes the points H and D (respectively).

Thus, the whole pyramid, whose base (is) triangle ABC , and apex the point D , has been divided into two pyramids (which are) equal to one another [and similar to the whole], and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid. (Which is) the very thing it was required to show.

Proposition 4

If there are two pyramids with the same height, having triangular bases, and each of them is divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms then as the base of one pyramid (is) to the base of the other pyramid, so (the sum of) all the prisms in one pyramid will be to (the sum of all) the equal number of prisms in the other pyramid.

Let there be two pyramids with the same height, having the triangular bases ABC and DEF , (with) apexes the points G and H (respectively). And let each of them be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms [Prop. 12.3]. I say that as base ABC is to base DEF , so (the sum of) all the prisms in pyramid $ABCG$ (is) to (the sum of) all the equal number of prisms in pyramid $DEFH$.



For since BO is equal to OC , and AL to LC , LO is thus

Ἐπεὶ γὰρ ἴση ἐστὶν ἡ μὲν BX τῇ $XΓ$, ἡ δὲ $ΑΛ$ τῇ $ΛΓ$, παράλληλος ἄρα ἐστὶν ἡ $ΑΧ$ τῇ $ΑΒ$ καὶ ὅμοιον τὸ $ΑΒΓ$ τρίγωνον τῷ $ΑΧΓ$ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $ΔΕΖ$ τρίγωνον τῷ $ΡΦΖ$ τριγώνῳ ὅμοιον ἐστίν. καὶ ἐπεὶ διπλασίῳ ἐστὶν ἡ μὲν $ΒΓ$ τῇ $ΓΧ$, ἡ δὲ $ΕΖ$ τῇ $ΖΦ$, ἔστιν ἄρα ὡς ἡ $ΒΓ$ πρὸς τὴν $ΓΧ$, οὕτως ἡ $ΕΖ$ πρὸς τὴν $ΖΦ$. καὶ ἀναγέγραπται ἀπὸ μὲν τῶν $ΒΓ$, $ΓΧ$ ὁμοία τε καὶ ὁμοίως κείμενα εὐθύγραμμα τὰ $ΑΒΓ$, $ΑΧΓ$, ἀπὸ δὲ τῶν $ΕΖ$, $ΖΦ$ ὁμοία τε καὶ ὁμοίως κείμενα [εὐθύγραμμα] τὰ $ΔΕΖ$, $ΡΦΖ$. ἔστιν ἄρα ὡς τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΑΧΓ$ τρίγωνον, οὕτως τὸ $ΔΕΖ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον· ἐναλλάξ ἄρα ἐστὶν ὡς τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΔΕΖ$ [τρίγωνον], οὕτως τὸ $ΑΧΓ$ [τρίγωνον] πρὸς τὸ $ΡΦΖ$ τρίγωνον. ἀλλ' ὡς τὸ $ΑΧΓ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις μὲν [ἐστὶ] τὸ $ΑΧΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΟΜΝ$, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$. καὶ ὡς ἄρα τὸ $ΑΒΓ$ τρίγωνον πρὸς τὸ $ΔΕΖ$ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΑΧΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΟΜΝ$, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$. ὡς δὲ τὰ εἰρημένα πρίσματα πρὸς ἄλληλα, οὕτως τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΚΒΧΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $ΟΜ$ εὐθεῖα, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΠΕΦΡ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $ΣΤ$ εὐθεῖα. καὶ τὰ δύο ἄρα πρίσματα, οὗ τε βάσις μὲν τὸ $ΚΒΧΛ$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ $ΟΜ$, καὶ οὗ βάσις μὲν τὸ $ΑΧΓ$, ἀπεναντίον δὲ τὸ $ΟΜΝ$, πρὸς τὰ πρίσματα, οὗ τε βάσις μὲν τὸ $ΠΕΦΡ$, ἀπεναντίον δὲ ἡ $ΣΤ$ εὐθεῖα, καὶ οὗ βάσις μὲν τὸ $ΡΦΖ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΣΤΥ$. καὶ ὡς ἄρα ἡ $ΑΒΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ εἰρημένα δύο πρίσματα πρὸς τὰ εἰρημένα δύο πρίσματα.

Καὶ ὁμοίως, εἰαν διαιρεθῶσιν αἱ $ΟΜΝΗ$, $ΣΤΥΘ$ πυραμίδες εἴς τε δύο πρίσματα καὶ δύο πυραμίδας, ἔσται ὡς ἡ $ΟΜΝ$ βάσις πρὸς τὴν $ΣΤΥ$ βάσιν, οὕτως τὰ ἐν τῇ $ΟΜΝΗ$ πυραμίδι δύο πρίσματα πρὸς τὰ ἐν τῇ $ΣΤΥΘ$ πυραμίδι δύο πρίσματα. ἀλλ' ὡς ἡ $ΟΜΝ$ βάσις πρὸς τὴν $ΣΤΥ$ βάσιν, οὕτως ἡ $ΑΒΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν· ἴσον γὰρ ἑκάτερον τῶν $ΟΜΝ$, $ΣΤΥ$ τριγώνων ἑκατέρῳ τῶν $ΑΧΓ$, $ΡΦΖ$. καὶ ὡς ἄρα ἡ $ΑΒΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ τέσσαρα πρίσματα πρὸς τὰ τέσσαρα πρίσματα. ὁμοίως δὲ κ' ἂν τὰς ὑπολειπομένας πυραμίδας διέλωμεν εἴς τε δύο πυραμίδας καὶ εἴς δύο πρίσματα, ἔσται ὡς ἡ $ΑΒΓ$ βάσις πρὸς τὴν $ΔΕΖ$ βάσιν, οὕτως τὰ ἐν τῇ $ΑΒΓΗ$ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ $ΔΕΖΘ$ πυραμίδι πρίσματα πάντα ἰσοπληθῆ· ὅπερ ἔδει δεῖχαι.

Λήμμα.

Ὅτι δὲ ἐστὶν ὡς τὸ $ΑΧΓ$ τρίγωνον πρὸς τὸ $ΡΦΖ$ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις τὸ $ΑΧΓ$ τρίγωνον, ἀπεναντίον δὲ τὸ $ΟΜΝ$, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ $ΡΦΖ$ [τρίγωνον], ἀπεναντίον δὲ τὸ $ΣΤΥ$, οὕτω δεικτέον.

parallel to AB , and triangle ABC similar to triangle LOC [Prop. 12.3]. So, for the same (reasons), triangle DEF is also similar to triangle RVF . And since BC is double CO , and EF (double) FV , thus as BC (is) to CO , so EF (is) to FV . And the similar, and similarly laid out, rectilinear (figures) ABC and LOC be described on BC and CO (respectively), and the similar, and similarly laid out, [rectilinear] (figures) DEF and RVF on EF and FV (respectively). Thus, as triangle ABC is to triangle LOC , so triangle DEF (is) to triangle RVF [Prop. 6.22]. Thus, alternately, as triangle ABC is to [triangle] DEF , so [triangle] LOC (is) to triangle RVF [Prop. 5.16]. But, as triangle LOC (is) to triangle RVF , so the prism whose base [is] triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) triangle RVF , and opposite (plane) STU (see lemma). And, thus, as triangle ABC (is) to triangle DEF , so the prism whose base (is) triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) triangle RVF , and opposite (plane) STU . And as the aforementioned prisms (are) to one another, so the prism whose base (is) parallelogram $KBOL$, and opposite (side) straight-line PM , (is) to the prism whose base (is) parallelogram $QEVF$, and opposite (side) straight-line ST [Props. 11.39, 12.3]. Thus, also, (is) the (sum of the) two prisms—that whose base (is) parallelogram $KBOL$, and opposite (side) PM , and that whose base (is) LOC , and opposite (plane) PMN —to (the sum of) the (two) prisms—that whose base (is) $QEVF$, and opposite (side) straight-line ST , and that whose base (is) triangle RVF , and opposite (plane) STU [Prop. 5.12]. And, thus, as base ABC (is) to base DEF , so the (sum of the first) aforementioned two prisms (is) to the (sum of the second) aforementioned two prisms.

And, similarly, if pyramids $PMNG$ and $STUH$ are divided into two prisms, and two pyramids, as base PMN (is) to base STU , so (the sum of) the two prisms in pyramid $PMNG$ will be to (the sum of) the two prisms in pyramid $STUH$. But, as base PMN (is) to base STU , so base ABC (is) to base DEF . For the triangles PMN and STU (are) equal to LOC and RVF , respectively. And, thus, as base ABC (is) to base DEF , so (the sum of) the four prisms (is) to (the sum of) the four prisms [Prop. 5.12]. So, similarly, even if we divide the pyramids left behind into two pyramids and into two prisms, as base ABC (is) to base DEF , so (the sum of) all the prisms in pyramid $ABCG$ will be to (the sum of) all the equal number of prisms in pyramid $DEFH$. (Which is) the very thing it was required to show.

Lemma

And one may show, as follows, that as triangle LOC is to triangle RVF , so the prism whose base (is) triangle LOC , and opposite (plane) PMN , (is) to the prism whose base (is) [triangle] RVF , and opposite (plane) STU .

Ἐπὶ γάρ τῃς αὐτῆς καταγραφῆς νενοήσθωσαν ἀπὸ τῶν H , Θ κάθετοι ἐπὶ τὰ $AB\Gamma$, ΔEZ ἐπίπεδα, ἴσαι δηλαδὴ τυγθάνουσαι διὰ τὸ ἰσοῦν εἶναι ὑποκείσθαι τὰς πυραμίδας. καὶ ἐπεὶ δύο εὐθεῖαι ἢ τε $H\Gamma$ καὶ ἢ ἀπὸ τοῦ H κάθετος ὑπὸ παραλλήλων ἐπιπέδων τῶν $AB\Gamma$, OMN τέμνονται, εἷς τούξ αὐτούξ λόγους τμηθήσονται. καὶ τέμνεται ἡ $H\Gamma$ διὰ ὑπὸ τοῦ OMN ἐπιπέδου κατὰ τὸ N · καὶ ἡ ἀπὸ τοῦ H ἄρα κάθετος ἐπὶ τὸ $AB\Gamma$ ἐπίπεδον διὰ τμηθήσεται ὑπὸ τοῦ OMN ἐπιπέδου. διὰ τὰ αὐτὰ δὴ καὶ ἡ ἀπὸ τοῦ Θ κάθετος ἐπὶ τὸ ΔEZ ἐπίπεδον διὰ τμηθήσεται ὑπὸ τοῦ ΣTY ἐπιπέδου. καὶ εἰσιν ἴσαι αἱ ἀπὸ τῶν H , Θ κάθετοι ἐπὶ τὰ $AB\Gamma$, ΔEZ ἐπίπεδα· ἴσαι ἄρα καὶ αἱ ἀπὸ τῶν OMN , ΣTY τριγώνων ἐπὶ τὰ $AB\Gamma$, ΔEZ κάθετοι. ἰσοῦν γὰρ [ἐστὶ] τὰ πρίσματα, ὧν βάσεις μὲν εἰσὶ τὰ $\Lambda X\Gamma$, $P\Phi Z$ τρίγωνα, ἀπεναντίον δὲ τὰ OMN , ΣTY . ὥστε καὶ τὰ στερεὰ παραλληλεπίπεδα τὰ ἀπὸ τῶν εἰρημένων πρισμάτων ἀναγραφόμενα ἰσοῦν καὶ πρὸς ἄλληλά [εἰσιν] ὡς αἱ βάσεις· καὶ τὰ ἡμίση ἄρα ἐστὶν ὡς ἡ $\Lambda X\Gamma$ βάσις πρὸς τὴν $P\Phi Z$ βάσιν, οὕτως τὰ εἰρημένα πρίσματα πρὸς ἄλληλα· ὅπερ ἔδει δεῖχαι.

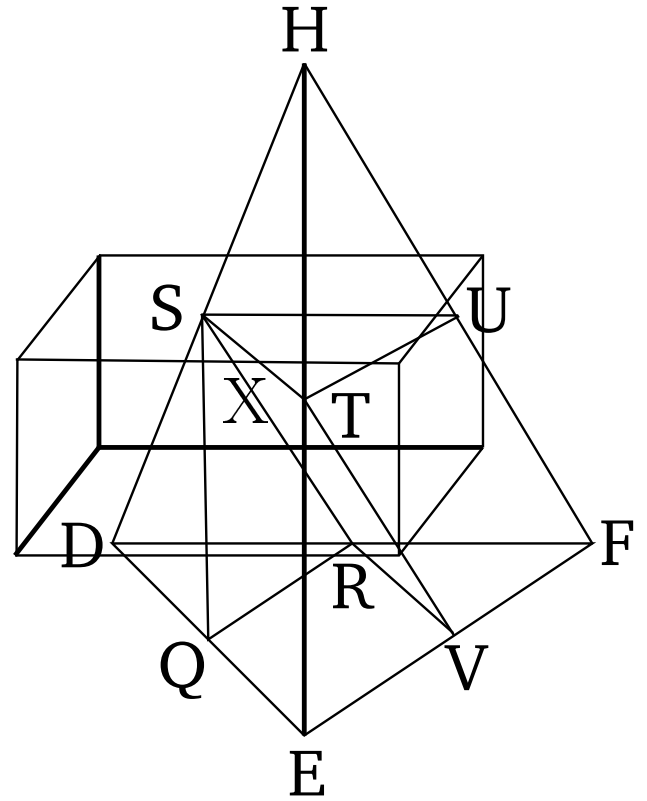
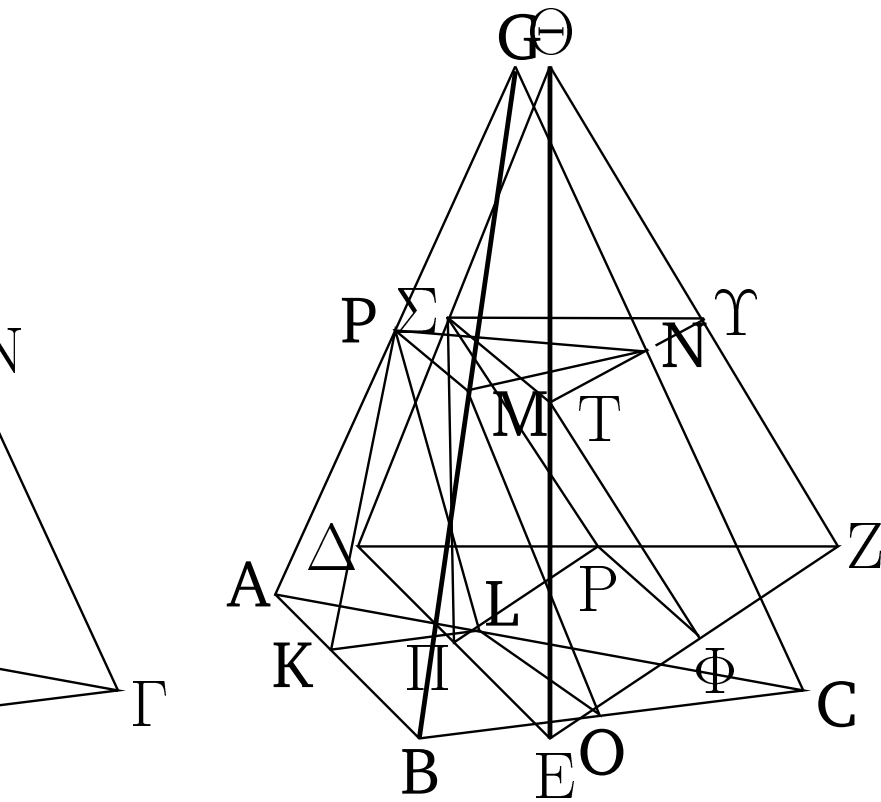
5.

Αἱ ὑπὸ τὸ αὐτὸ ὕψος οὔσαι πυραμίδες καὶ τριγώνουξ ἔθουσαι βάσεις πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις.

For, in the same figure, let perpendiculars be conceived (drawn) from (points) G and H to the planes ABC and DEF (respectively). These clearly turn out to be equal, on account of the pyramids being assumed (to be) of equal height. And since two straight-lines, GC and the perpendicular from G , are cut by the parallel planes ABC and PMN they will be cut in the same ratios [Prop. 11.17]. And GC was cut in half by the plane PMN at N . Thus, the perpendicular from G to the plane ABC will also be cut in half by the plane PMN . So, for the same (reasons), the perpendicular from H to the plane DEF will also be cut in half by the plane STU . And the perpendiculars from G and H to the planes ABC and DEF (respectively) are equal. Thus, the perpendiculars from the triangles PMN and STU to ABC and DEF (respectively, are) also equal. Thus, the prisms whose bases are triangles LOC and RVF , and opposite (sides) PMN and STU (respectively), [are] of equal height. And, hence, the parallelepiped solids described on the aforementioned prisms [are] of equal height and (are) to one another as their bases [Prop. 11.32]. Likewise, the halves (of the solids) [Prop. 11.28]. Thus, as base LOC is to base RVF , so the aforementioned prisms (are) to one another. (Which is) the very thing it was required to show.

Proposition 5

Pyramids which are of the same height, and have triangular bases, are to one another as their bases.



Ἐστωσαν ὑπὸ τὸ αὐτὸ ὕψος πυραμίδες, ὧν βάσεις μὲν τὰ $AB\Gamma$, ΔEZ τρίγωνα, κορυφαὶ δὲ τὰ H , Θ σημεία· λέγω, ὅτι ἐστὶν ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως ἡ $AB\Gamma H$ πυραμὶς πρὸς τὴν $\Delta EZ\Theta$ πυραμίδα.

Εἰ γὰρ μὴ ἐστὶν ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως ἡ $AB\Gamma H$ πυραμὶς πρὸς τὴν $\Delta EZ\Theta$ πυραμίδα, ἔσται ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως ἡ $AB\Gamma H$ πυραμὶς ἥτοι πρὸς ἑλασσόν τι τῆς $\Delta EZ\Theta$ πυραμίδος στερεὸν ἢ πρὸς μείζον. ἔστω πρότερον πρὸς ἑλασσόν τὸ Θ , καὶ διηγήσθω ἡ $\Delta EZ\Theta$ πυραμὶς εἰς τε δύο πυραμίδας ἴσας ἀλλήλαις καὶ ὁμοίας τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα· τὰ δὴ δύο πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος. καὶ πάλιν αἱ ἐκ τῆς διαιρέσεως γινόμεναι πυραμίδες ὁμοίως διηγήσθωσαν, καὶ τοῦτο ἀεὶ γινέσθω, ἕως ὅτ' ὑλειφθῶσί τινας πυραμίδες ἀπὸ τῆς $\Delta EZ\Theta$ πυραμίδος, αἱ εἰσὶν ἐλάττωες τῆς ὑπεροθῆς, ἣν ὑπερέθει ἡ $\Delta EZ\Theta$ πυραμὶς τοῦ Θ στερεοῦ. λελειφθῶσαν καὶ ἔστωσαν λόγον ἕνεκεν αἱ $\Delta\text{I}\text{P}\Sigma$, $\Sigma\text{T}\Upsilon\Theta$ · λοιπὰ ἄρα τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα μείζονά ἐστι τοῦ Θ στερεοῦ. διηγήσθω καὶ ἡ $AB\Gamma H$ πυραμὶς ὁμοίως καὶ ἰσοπληθῶς τῇ $\Delta EZ\Theta$ πυραμίδι· ἐστὶν ἄρα ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὰ ἐν τῇ $AB\Gamma H$ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα, ἀλλὰ καὶ ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως ἡ $AB\Gamma H$ πυραμὶς πρὸς τὸ Θ στερεόν· καὶ ὡς ἄρα ἡ $AB\Gamma H$ πυραμὶς πρὸς τὸ Θ στερεόν, οὕτως τὰ ἐν τῇ $AB\Gamma H$ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα· ἐναλλάχ' ἄρα ὡς ἡ AB

Let there be pyramids of the same height whose bases (are) the triangles ABC and DEF , and apexes the points G and H (respectively). I say that as base ABC is to base DEF , so pyramid $ABCG$ (is) to pyramid $DEFH$.

For if base ABC is not to base DEF , as pyramid $ABCG$ (is) to pyramid $DEFH$, then base ABC will be to base DEF , as pyramid $ABCG$ (is) to some solid either less than, or greater than, pyramid $DEFH$. Let it, first of all, be (in this ratio) to (some) lesser (solid), W . And let pyramid $DEFH$ be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms. So, the (sum of the) two prisms is greater than half of the whole pyramid [Prop. 12.3]. And, again, let the pyramids generated by the division be similarly divided, and let this be done continually until some pyramids are left from pyramid $DEFH$ which (when added together) are less than the excess by which pyramid $DEFH$ exceeds the solid W [Prop. 10.1]. Let them be left, and, for the sake of argument, let them be $DQRS$ and $STUH$. Thus, the (sum of the) remaining prisms within pyramid $DEFH$ is greater than solid W . Let pyramid $ABCG$ also be divided similarly, and a similar number of times, as pyramid $DEFH$. Thus, as base ABC is to base DEF , so the (sum of the) prisms within pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 12.4]. But, also, as base ABC (is) to base DEF , so pyramid $ABCG$ (is) to solid W . And, thus, as pyramid $ABCG$ (is) to solid W , so the (sum of the) prisms within

ΓΗ πυραμίδι πρὸς τὰ ἐν αὐτῇ πρίσματα, οὕτως τὸ Θ στερεὸν πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα. μείζων δὲ ἡ ΑΒΓΗ πυραμὶς τῶν ἐν αὐτῇ πρισμάτων· μείζον ἄρα καὶ τὸ Θ στερεὸν τῶν ἐν τῇ ΔΕΖΘ πυραμίδι πρισμάτων. ἀλλὰ καὶ ἔλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς ἑλασσόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ὁμοίως δὲ δευτέρησεται, ὅτι οὐδὲ ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως ἡ ΔΕΖΘ πυραμὶς πρὸς ἑλαττόν τι τῆς ΑΒΓΗ πυραμίδος στερεόν.

Λέγω δὴ, ὅτι οὐκ ἐστὶν οὐδὲ ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς μείζον τι τῆς ΔΕΖΘ πυραμίδος στερεόν.

Εἰ γὰρ δυνατὸν, ἔστω πρὸς μείζον τὸ Θ· ἀνάπαλιν ἄρα ἐστὶν ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα. ὡς δὲ τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα, οὕτως ἡ ΔΕΖΘ πυραμὶς πρὸς ἑλασσόν τι τῆς ΑΒΓΗ πυραμίδος, ὡς ἔμπροσθεν ἐδείχθη· καὶ ὡς ἄρα ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως ἡ ΔΕΖΘ πυραμὶς πρὸς ἑλασσόν τι τῆς ΑΒΓΗ πυραμίδος· ὅπερ ἄτοπον ἐδείχθη. οὐκ ἄρα ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς μείζον τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ἐδείχθη δέ, ὅτι οὐδὲ πρὸς ἑλασσόν. ἔστιν ἄρα ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα· ὅπερ ἔδει δεῖχαι.

6.

Αἱ ὑπὸ τὸ αὐτὸ ὕψος οὔσαι πυραμίδες καὶ πολυγώνουζ ἔθουσαι βάσεις πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις.

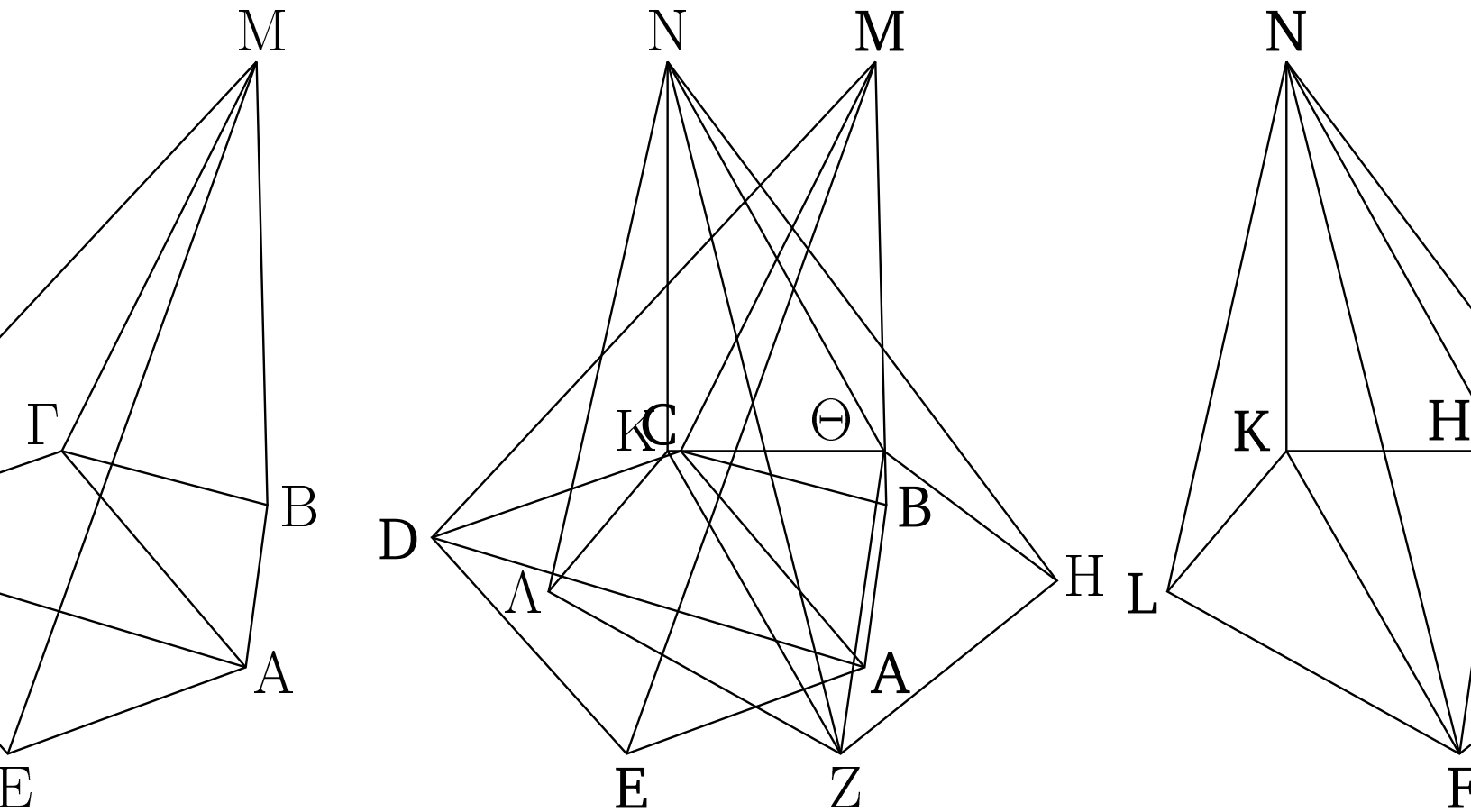
pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.11]. Thus, alternately, as pyramid $ABCG$ (is) to the (sum of the) prisms within it, so solid W (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.16]. And pyramid $ABCG$ (is) greater than the (sum of the) prisms within it. Thus, solid W (is) also greater than the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.14]. But, (it is) also less. This very thing is impossible. Thus, as base ABC is to base DEF , so pyramid $ABCG$ (is) not to some solid less than pyramid $DEFH$. So, similarly, we can show that base DEF is not to base ABC , as pyramid $DEFH$ (is) to some solid less than pyramid $ABCG$ either.

So, I say that neither is base ABC to base DEF , as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$.

For, if possible, let it be (in this ratio) to some greater (solid), W . Thus, inversely, as base DEF (is) to base ABC , so solid W (is) to pyramid $ABCG$ [Prop. 5.7. corr.]. And as solid W (is) to pyramid $ABCG$, so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$, as shown before [Prop. 12.2 lem.]. And, thus, as base DEF (is) to base ABC , so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$ [Prop. 5.11]. The very thing was shown (to be) absurd. Thus, base ABC is not to base DEF , as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$. And, it was shown that neither (is it in this ratio) to a lesser (solid). Thus, as base ABC is to base DEF , so pyramid $ABCG$ (is) to pyramid $DEFH$. (Which is) the very thing it was required to show.

Proposition 6

Pyramids which are of the same height, and have polygonal bases, are to one another as their bases.



Ἐστωσαν ὑπὸ τὸ αὐτὸ ὕψος πυραμίδες, ὧν [αἱ] βάσεις μὲν τὰ $ABΓΔΕ$, $ZHΘΚΛ$ πολύγωνα, κορυφαὶ δὲ τὰ M , N σημεῖα· λέγω, ὅτι ἐστὶν ὡς ἡ $ABΓΔΕ$ βάσις πρὸς τὴν $ZHΘΚΛ$ βάσιν, οὕτως ἡ $ABΓΔΕΜ$ πυραμὶς πρὸς τὴν $ZHΘΚΛΝ$ πυραμίδα.

Ἐπεξεύθθωσαν γάρ αἱ $ΑΓ$, $ΑΔ$, $ΖΘ$, $ΖΚ$. ἐπεὶ οὖν δύο πυραμίδες εἰσὶν αἱ $ABΓΜ$, $ΑΓΔΜ$ τριγώνων ἐθονσαι βάσεις καὶ ὕψος ἴσον, πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις· ἐστὶν ἄρα ὡς ἡ $ABΓ$ βάσις πρὸς τὴν $ΑΓΔ$ βάσιν, οὕτως ἡ $ABΓΜ$ πυραμὶς πρὸς τὴν $ΑΓΔΜ$ πυραμίδα. καὶ συνθέντι ὡς ἡ $ABΓΔ$ βάσις πρὸς τὴν $ΑΓΔ$ βάσιν, οὕτως ἡ $ABΓΔΜ$ πυραμὶς πρὸς τὴν $ΑΓΔΜ$ πυραμίδα. ἀλλὰ καὶ ὡς ἡ $ΑΓΔ$ βάσις πρὸς τὴν $ΑΔΕ$ βάσιν, οὕτως ἡ $ΑΓΔΜ$ πυραμὶς πρὸς τὴν $ΑΔΕΜ$ πυραμίδα. δι' ἴσον ἄρα ὡς ἡ $ABΓΔ$ βάσις πρὸς τὴν $ΑΔΕ$ βάσιν, οὕτως ἡ $ABΓΔΜ$ πυραμὶς πρὸς τὴν $ΑΔΕΜ$ πυραμίδα. καὶ συνθέντι πάλιν, ὡς ἡ $ABΓΔΕ$ βάσις πρὸς τὴν $ΑΔΕ$ βάσιν, οὕτως ἡ $ABΓΔΕΜ$ πυραμὶς πρὸς τὴν $ΑΔΕΜ$ πυραμίδα. ὁμοίως δὲ δευτέρησεται, ὅτι καὶ ὡς ἡ $ZHΘΚΛ$ βάσις πρὸς τὴν $ZHΘ$ βάσιν, οὕτως καὶ ἡ $ZHΘΚΛΝ$ πυραμὶς πρὸς τὴν $ZHΘΝ$ πυραμίδα. καὶ ἐπεὶ δύο πυραμίδες εἰσὶν αἱ $ΑΔΕΜ$, $ZHΘΝ$ τριγώνων ἐθονσαι βάσεις καὶ ὕψος ἴσον, ἐστὶν ἄρα ὡς ἡ $ΑΔΕ$ βάσις πρὸς τὴν $ZHΘ$ βάσιν, οὕτως ἡ $ΑΔΕΜ$ πυραμὶς πρὸς τὴν $ZHΘΝ$ πυραμίδα. ἀλλ' ὡς ἡ

Let there be pyramids of the same height whose bases (are) the polygons $ABCDE$ and $FGHKL$, and apexes the points M and N (respectively). I say that as base $ABCDE$ is to base $FGHKL$, so pyramid $ABCDEM$ (is) to pyramid $FGHKLN$.

For let AC , AD , FH , and FK be joined. Therefore, since $ABCM$ and $ACDM$ are two pyramids having triangular bases and equal height, they are to one another as their bases [Prop. 12.5]. Thus, as base ABC is to base ACD , so pyramid $ABCM$ (is) to pyramid $ACDM$. And, via composition, as base $ABCD$ (is) to base ACD , so pyramid $ABCDM$ (is) to pyramid $ACDM$ [Prop. 5.18]. But, as base ACD (is) to base ADE , so pyramid $ACDM$ (is) also to pyramid $ADEM$ [Prop. 12.5]. Thus, via equality, as base $ABCD$ (is) to base ADE , so pyramid $ABCDM$ (is) to pyramid $ADEM$ [Prop. 5.22]. And, again, via composition, as base $ABCDE$ (is) to base ADE , so pyramid $ABCDEM$ (is) to pyramid $ADEM$ [Prop. 5.18]. So, similarly, it can also be shown that as base $FGHKL$ (is) to base FGH , so pyramid $FGHKLN$ (is) also to pyramid $FGHN$. And since $ADEM$ and $FGHN$ are two pyramids having triangular bases and equal height, thus as base ADE (is) to base FGH , so pyramid $ADEM$ (is) to pyramid $FGHN$ [Prop. 12.5]. But, as base ADE (is) to base $ABCDE$, so pyramid $ADEM$ (was) to pyramid $ABCDEM$. Thus, via equality, as base $ABCDE$

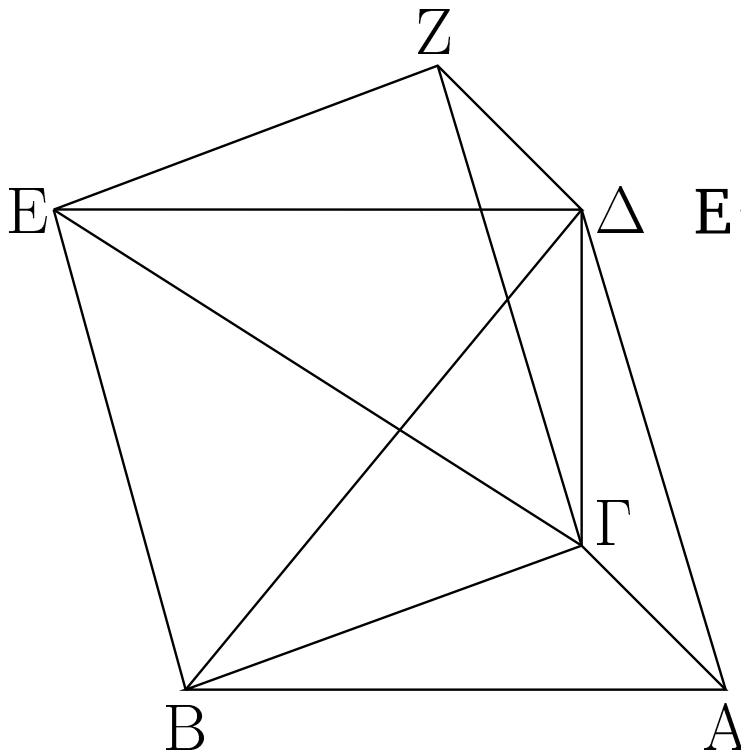
$A\Delta E$ βάσις πρὸς τὴν $AB\Gamma\Delta E$ βάσιν, οὕτως ἢν ἡ $A\Delta EM$ πυραμὶς πρὸς τὴν $AB\Gamma\Delta EM$ πυραμίδα. καὶ δι' ἴσον ἄρα ὡς ἡ $AB\Gamma\Delta E$ βάσις πρὸς τὴν $ZH\Theta$ βάσιν, οὕτως ἡ $AB\Gamma\Delta EM$ πυραμὶς πρὸς τὴν $ZH\Theta N$ πυραμίδα. ἀλλὰ μὴν καὶ ὡς ἡ $ZH\Theta$ βάσις πρὸς τὴν $ZH\Theta K\Lambda$ βάσιν, οὕτως ἢν καὶ ἡ $ZH\Theta N$ πυραμὶς πρὸς τὴν $ZH\Theta K\Lambda N$ πυραμίδα, καὶ δι' ἴσον ἄρα ὡς ἡ $AB\Gamma\Delta E$ βάσις πρὸς τὴν $ZH\Theta K\Lambda$ βάσιν, οὕτως ἡ $AB\Gamma\Delta EM$ πυραμὶς πρὸς τὴν $ZH\Theta K\Lambda N$ πυραμίδα· ὅπερ ἔδει δεῖξαι.

7.

Πᾶν πρίσμα τρίγωνον ἔθον βάσιν διαιρεῖται εἰς τρεῖς πυραμίδας ἴσαξ ἀλλήλαις τριγώνονξ βάσειξ ἐθούσαξ.

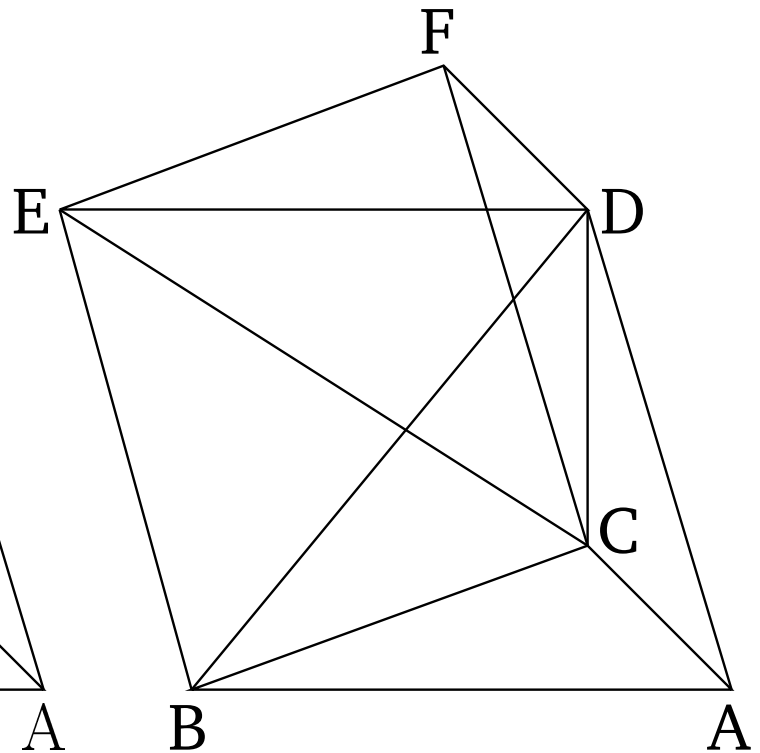
Proposition 7

Any prism having a triangular base is divided into three pyramids having triangular bases (which are) equal to one another.



Ἐστω πρίσμα, οὗ βάσις μὲν τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ ΔEZ . λέγω, ὅτι τὸ $AB\Gamma\Delta EZ$ πρίσμα διαιρεῖται εἰς τρεῖς πυραμίδας ἴσαξ ἀλλήλαις τριγώνονξ ἐθούσαξ βάσειξ.

Ἐπεξεύθωσαν γὰρ αἱ $B\Delta$, $E\Gamma$, $\Gamma\Delta$. ἐπεὶ παραλληλόγραμμον ἔστι τὸ $ABE\Delta$, διάμετροξ δὲ αὐτοῦ ἔστιν ἡ $B\Delta$, ἴσον ἄρα ἔστι τὸ $AB\Delta$ τρίγωνον τῷ $EB\Delta$ τριγώνῳ· καὶ ἡ πυραμὶς ἄρα, ἥξ βάσις μὲν τὸ $AB\Delta$ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἴση ἔστι πυραμίδι, ἥξ βάσις μὲν ἔστι τὸ ΔEB τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον. ἀλλὰ ἡ πυραμὶς, ἥξ βάσις μὲν ἔστι τὸ ΔEB τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἡ αὐτὴ ἔστι πυραμίδι, ἥξ βάσις μὲν ἔστι τὸ $EB\Gamma$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον· ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέχεται. καὶ πυραμὶς ἄρα, ἥξ βάσις μὲν ἔστι τὸ $AB\Delta$ τρίγωνον, κο-



Let there be a prism whose base (is) triangle ABC , and opposite (plane) DEF . I say that prism $ABCDEF$ is divided into three pyramids having triangular bases (which are) equal to one another.

For let BD , EC , and CD be joined. Since $ABED$ is a parallelogram, and BD is its diagonal, triangle ABD is thus equal to triangle EBD [Prop. 1.34]. And, thus, the pyramid whose base (is) triangle ABD , and apex the point C , is equal to the pyramid whose base is triangle DEB , and apex the point C [Prop. 12.5]. But, the pyramid whose base is triangle DEB , and apex the point C , is the same as the pyramid whose base is triangle EBC , and apex the point D . For they are contained by the same planes. And, thus, the pyramid whose base is ABD ,

ρυνφή δὲ τὸ Γ σημείον, ἴση ἐστὶ πυραμίδι, ἥξ βάσις μὲν ἐστὶ τὸ $EB\Gamma$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. πάλιν, ἐπεὶ παραλληλόγραμμον ἐστὶ τὸ $Z\Gamma BE$, διάμετρος δὲ ἐστὶν αὐτοῦ ἡ GE , ἴσον ἐστὶ τὸ GEZ τρίγωνον τῷ ΓBE τριγώνῳ. καὶ πυραμὶς ἄρα, ἥξ βάσις μὲν ἐστὶ τὸ $B\Gamma E$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ἴση ἐστὶ πυραμίδι, ἥξ βάσις μὲν ἐστὶ τὸ EGZ τρίγωνον, κορυφή δὲ τὸ Δ σημείον. ἡ δὲ πυραμὶς, ἥξ βάσις μὲν ἐστὶ τὸ $B\Gamma E$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ἴση ἐδείχθη πυραμίδι, ἥξ βάσις μὲν ἐστὶ τὸ $AB\Delta$ τρίγωνον, κορυφή δὲ τὸ Γ σημείον καὶ πυραμὶς ἄρα, ἥξ βάσις μὲν ἐστὶ τὸ GEZ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, ἴση ἐστὶ πυραμίδι, ἥξ βάσις μὲν [ἐστὶ] τὸ $AB\Delta$ τρίγωνον, κορυφή δὲ τὸ Γ σημείον· διήρηται ἄρα τὸ $AB\Gamma\Delta EZ$ πρίσμα εἰς τρεῖς πυραμίδας ἴσας ἀλλήλαις τριγώνους ἐθνόσας βάσεις.

Καὶ ἐπεὶ πυραμὶς, ἥξ βάσις μὲν ἐστὶ τὸ $AB\Delta$ τρίγωνον, κορυφή δὲ τὸ Γ σημείον, ἡ αὐτὴ ἐστὶ πυραμίδι, ἥξ βάσις τὸ ΓAB τρίγωνον, κορυφή δὲ τὸ Δ σημείον· ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέχονται· ἡ δὲ πυραμὶς, ἥξ βάσις τὸ $AB\Delta$ τρίγωνον, κορυφή δὲ τὸ Γ σημείον, τρίτον ἐδείχθη τοῦ πρίσματος, ὅς ἡ βάσις τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ ΔEZ , καὶ ἡ πυραμὶς ἄρα, ἥξ βάσις τὸ $AB\Gamma$ τρίγωνον, κορυφή δὲ τὸ Δ σημείον, τρίτον ἐστὶ τοῦ πρίσματος τοῦ ἐθοντος βάσις τὴν αὐτὴν τὸ $AB\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ ΔEZ .

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι πᾶσα πυραμὶς τρίτον μέρος ἐστὶ τοῦ πρίσματος τοῦ τὴν αὐτὴν βάσιν ἐθοντος αὐτῇ καὶ ὕψος ἴσον· ὅπερ ἔδει δεῖχαι.

8.

Αἱ ὅμοιαι πυραμίδες καὶ τριγώνους ἐθονσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.

Ἐστωσαν ὅμοιαι καὶ ὁμοίως κείμεναι πυραμίδες, ὧν βάσεις μὲν εἰσὶ τὰ $AB\Gamma$, ΔEZ τρίγωνα, κορυφαὶ δὲ τὰ H , Θ σημεία· λέγω, ὅτι ἡ $AB\Gamma H$ πυραμὶς πρὸς τὴν $\Delta EZ\Theta$ πυραμίδα τριπλασίονα λόγον ἔθει ἢ περὶ ἡ $B\Gamma$ πρὸς τὴν EZ .

and apex the point C , is equal to the pyramid whose base is EBC and apex the point D . Again, since $FCBE$ is a parallelogram, and CE is its diagonal, triangle CEF is equal to triangle CBE [Prop. 1.34]. And, thus, the pyramid whose base is triangle BCE , and apex the point D , is equal to the pyramid whose base is triangle ECF , and apex the point D [Prop. 12.5]. And the pyramid whose base is triangle BCE , and apex the point D , was shown (to be) equal to the pyramid whose base is triangle ABD , and apex the point C . Thus, the pyramid whose base is triangle CEF , and apex the point D , is also equal to the pyramid whose base [is] triangle ABD , and apex the point C . Thus, the prism $ABCDEF$ has been divided into three pyramids having triangular bases (which are) equal to one another.

And since the pyramid whose base is triangle ABD , and apex the point C , is the same as the pyramid whose base is triangle CAB , and apex the point D . For they are contained by the same planes. And the pyramid whose base (is) triangle ABD , and apex the point C , was shown (to be) a third of the prism whose base is triangle ABC , and opposite (plane) DEF , thus the pyramid whose base is triangle ABC , and apex the point D , is also a third of the pyramid having the same base, triangle ABC , and opposite (plane) DEF .

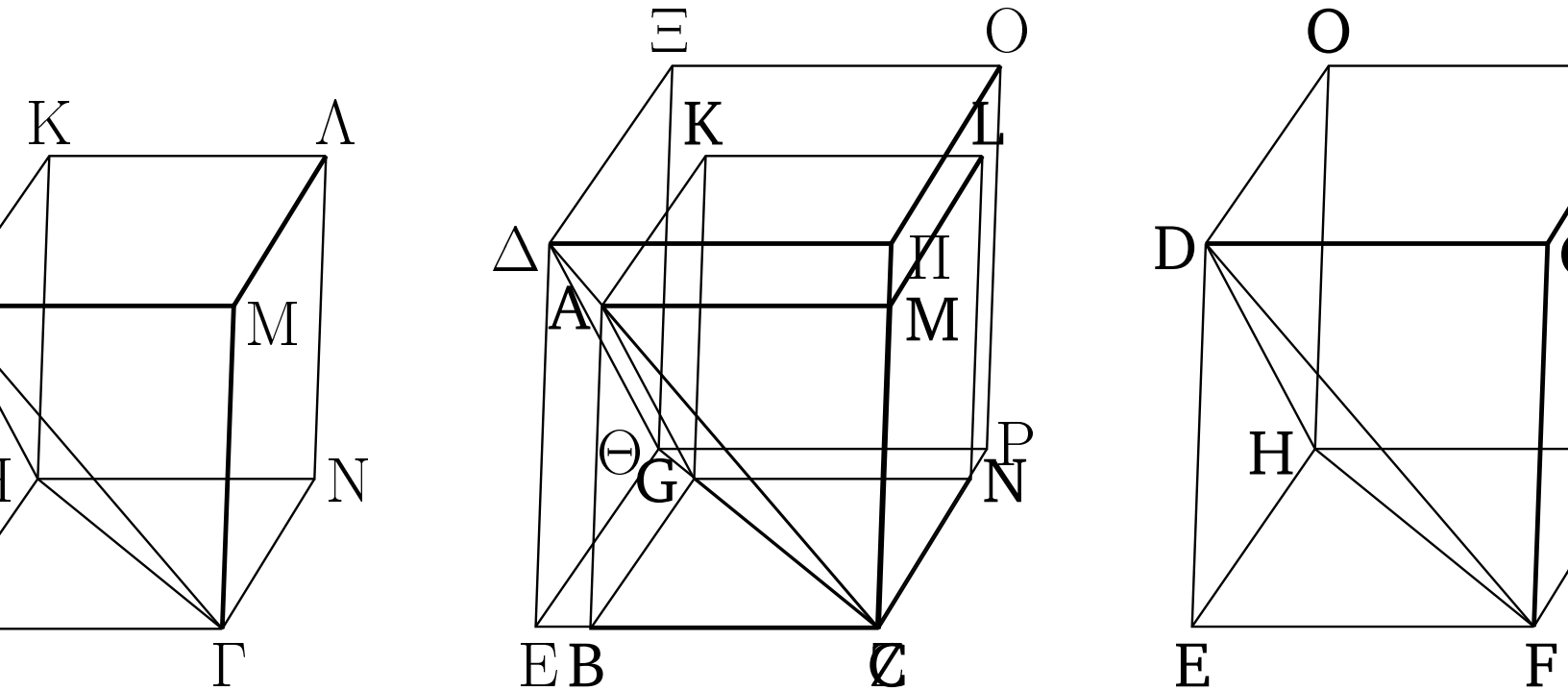
Corollary

And, from this, (it is) clear that any pyramid is the third part of the prism having the same base as it, and an equal height. (Which is) the very thing it was required to show.

Proposition 8

Similar pyramids which also have triangular bases are in the cubed ratio of their corresponding sides.

Let there be similar, and similarly laid out, pyramids whose bases are triangles ABC and DEF , and apexes the points G and H (respectively). I say that pyramid $ABCG$ has to pyramid $DEFH$ the cubed ratio of that BC (has) to EF .



Συμπληρώσθω γάρ τὰ $BHML$, $EΘΠΟ$ στερεὰ παραλληλεπίπεδα. καὶ ἐπεὶ ὁμοία ἐστὶν ἡ $ABΓH$ πυραμὶς τῇ $ΔΕΖΘ$ πυραμίδι, ἴση ἄρα ἐστὶν ἡ μὲν ὑπὸ $ABΓ$ γωνία τῇ ὑπὸ $ΔΕΖ$ γωνίᾳ, ἡ δὲ ὑπὸ $HΒΓ$ τῇ ὑπὸ $ΘΕΖ$, ἡ δὲ ὑπὸ ABH τῇ ὑπὸ $ΔΕΘ$, καὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $ΔΕ$, οὕτως ἡ $BΓ$ πρὸς τὴν $ΕΖ$, καὶ ἡ BH πρὸς τὴν $ΕΘ$. καὶ ἐπεὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $ΔΕ$, οὕτως ἡ $BΓ$ πρὸς τὴν $ΕΖ$, καὶ περὶ ἴσα γωνίας αἱ πλευραὶ ἀνάλογον εἰσιν, ὅμοιον ἄρα ἐστὶ τὸ BM παραλληλόγραμμον τῷ $ΕΠ$ παραλληλογράμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν BN τῷ $ΕΡ$ ὅμοιον ἐστὶ, τὸ δὲ BK τῷ $ΕΧ$. τὰ τρία ἄρα τὰ MB , BK , BN τρισὶ τοῖς $ΕΠ$, $ΕΧ$, $ΕΡ$ ὅμοία ἐστὶν. ἀλλὰ τὰ μὲν τρία τὰ MB , BK , BN τρισὶ τοῖς ἀπεναντίον ἴσα τε καὶ ὅμοία ἐστὶν, τὰ δὲ τρία τὰ $ΕΠ$, $ΕΧ$, $ΕΡ$ τρισὶ τοῖς ἀπεναντίον ἴσα τε καὶ ὅμοία ἐστὶν. τὰ $BHML$, $EΘΠΟ$ ἄρα στερεὰ ὑπὸ ὁμοίων ἐπιπέδων ἴσων τὸ πλῆθος περιέχεται. ὅμοιον ἄρα ἐστὶ τὸ $BHML$ στερεὸν τῷ $EΘΠΟ$ στερεῷ. τὰ δὲ ὅμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν. τὸ $BHML$ ἄρα στερεὸν πρὸς τὸ $EΘΠΟ$ στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ὁμόλογος πλευρὰ ἡ $BΓ$ πρὸς τὴν ὁμόλογον πλευρὰν τὴν $ΕΖ$. ὥς δὲ τὸ $BHML$ στερεὸν πρὸς τὸ $EΘΠΟ$ στερεὸν, οὕτως ἡ $ABΓH$ πυραμὶς πρὸς τὴν $ΔΕΖΘ$ πυραμίδα, ἐπειδὴ περ ἡ πυραμὶς ἕκτον μέρος ἐστὶ τοῦ στερεοῦ διὰ τὸ καὶ τὸ πρίσμα ἥμισυ ὂν τοῦ στερεοῦ παραλληλεπιπέδον τριπλασίον εἶναι τῆς πυραμίδος. καὶ ἡ $ABΓH$ ἄρα πυραμὶς πρὸς τὴν $ΔΕΖΘ$ πυραμίδα τριπλασίονα λόγον ἔχει ἥπερ ἡ $BΓ$ πρὸς τὴν $ΕΖ$. ὅπερ ἔδει δεῖξαι.

For let the parallelepiped solids $BGML$ and $EHQP$ be completed. And since pyramid $ABCG$ is similar to pyramid $DEFH$, angle ABC is thus equal to angle DEF , and GBC to HEF , and ABG to DEH . And as AB is to DE , so BC (is) to EF , and BG to EH [Def. 11.9]. And since as AB is to DE , so BC (is) to EF , and (so) the sides around equal angles are proportional, parallelogram BM is thus similar to parallelogram EQ . So, for the same (reasons), BN is also similar to ER , and BK to EO . Thus, the three (parallelograms) MB , BK , and BN are similar to the three (parallelograms) EQ , EO , ER (respectively). But, the three (parallelograms) MB , BK , and BN are (both) equal and similar to the three opposite (parallelograms), and the three (parallelograms) EQ , EO , and ER are (both) equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the solids $BGML$ and $EHQP$ are contained by equal numbers of similar (and similarly laid out) planes. Thus, solid $BGML$ is similar to solid $EHQP$ [Def. 11.9]. And similar parallelepiped solids are in the cubed ratio of corresponding sides [Prop. 11.33]. Thus, solid $BGML$ has to solid $EHQP$ the cubed ratio that the corresponding side BC (has) to the corresponding side EF . And as solid $BGML$ (is) to solid $EHQP$, so pyramid $ABCG$ (is) to pyramid $DEFH$, inasmuch as the pyramid is the sixth part of the solid, on account of the prism, being half of the parallelepiped solid [Prop. 11.28], also being three times the pyramid [Prop. 12.7]. Thus, pyramid $ABCG$ also has to pyramid $DEFH$ the cubed ratio that BC (has) to EF . (Which is) the very thing it was required to show.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι καὶ αἱ πολυγώνουζ ἔθουσαι βάσειζ ὅμοιαι πυραμίδεζ πρὸς ἀλλήλαζ ἐν τριπλασίῳ λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. διαιρεθεισῶν γάρ αὐτῶν εἰς τὰς ἐν αὐταῖς πυραμίδαζ τριγώνουζ βάσειζ ἔθουσαζ τῷ καὶ τὰ ὅμοια πολύγωνα τῶν βάσεων εἰς ὅμοια τρίγωνα διαιρεῖσθαι καὶ ἴσα τῷ πλήθει καὶ ὁμόλογα τοῖς ὅλοις ἔσται ὡς [ή] ἐν τῇ ἐτέρῳ μία πυραμίς τριγώνον ἔθουσα βάσιν πρὸς τὴν ἐν τῇ ἐτέρῳ μίαν πυραμίδα τριγώνον ἔθουσαν βάσιν, οὕτως καὶ ἅπασαι αἱ ἐν τῇ ἐτέρῳ πυραμίδι πυραμίδεζ τριγώνουζ ἔθουσαι βάσειζ πρὸς τὰς ἐν τῇ ἐτέρῳ πυραμίδι πυραμίδαζ τριγώνουζ βάσειζ ἔθουσαζ, τοντέστιν αὐτὴ ἡ πολύγωνον βάσιν ἔθουσα πυραμίς πρὸς τὴν πολύγωνον βάσιν ἔθουσαν πυραμίδα. ἡ δὲ τριγώνον βάσιν ἔθουσα πυραμίς πρὸς τὴν τριγώνον βάσιν ἔθουσαν ἐν τριπλασίῳ λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν· καὶ ἡ πολύγωνον ἄρα βάσιν ἔθουσα πρὸς τὴν ὁμοίαν βάσιν ἔθουσαν τριπλασίονα λόγον ἔθει ἢ περ ἢ πλευρὰ πρὸς τὴν πλευράν.

9.

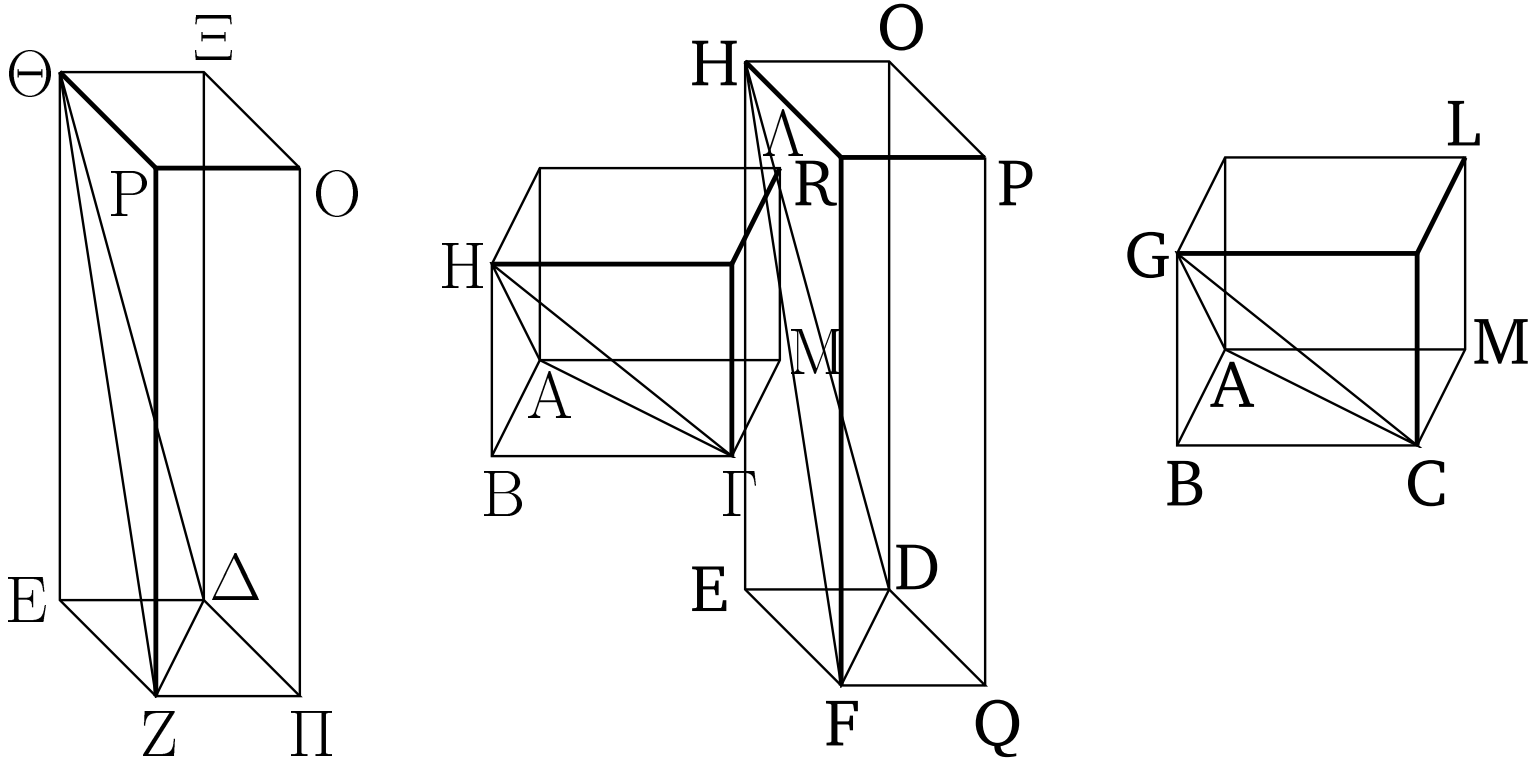
Τῶν ἴσων πυραμίδων καὶ τριγώνουζ βάσειζ ἔθουσῶν ἀντιπεπύονθαι αἱ βάσειζ τοῖς ὕψουσιν· καὶ ὧν πυραμίδων τριγώνουζ βάσειζ ἔθουσῶν ἀντιπεπύονθαι αἱ βάσειζ τοῖς ὕψουσιν, ἴσαι εἶσιν ἐκεῖναι.

Corollary

So, from this, (it is) also clear that similar pyramids having polygonal bases (are) to one another as the cubed ratio of their corresponding sides. For, dividing them into the pyramids (contained) within them which have triangular bases, with the similar polygons of the bases also being divided into similar triangles (which are) both equal in number, and corresponding, to the wholes [Prop. 6.20]. As one pyramid having a triangular base in the former (pyramid having a polygonal base is) to one pyramid having a triangular base in the latter (pyramid having a polygonal base), so (the sum of) all the pyramids having triangular bases in the former pyramid will also be to (the sum of) all the pyramids having triangular bases in the latter pyramid [Prop. 5.12]—that is to say, the (former) pyramid itself having a polygonal base to the (latter) pyramid having a polygonal base. And a pyramid having a triangular base is to a (pyramid) having a triangular base in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, a (pyramid) having a polygonal base also has to to a (pyramid) having a similar base the cubed ratio of a (corresponding) side to a (corresponding) side.

Proposition 9

The bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids which have triangular bases whose bases are reciprocally proportional to their heights are equal.



Ἐστωσαν γάρ ἴσαι πυραμίδες τριγώνου βάσεις ἔθουσai τὰς $AB\Gamma$, ΔEZ , κορυφὰς δὲ τὰ H , Θ σημεία· λέγω, ὅτι τῶν $AB\Gamma H$, $\Delta EZ\Theta$ πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν, καὶ ἐστὶν ὥς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος.

Συμπεληρώσω γὰρ τὰ $BHMA$, $E\Theta\Pi O$ στερεὰ παραλληλεπίπεδα. καὶ ἐπεὶ ἴση ἐστὶν ἡ $AB\Gamma H$ πυραμὶς τῇ $\Delta EZ\Theta$ πυραμίδι, καὶ ἐστὶ τῆς μὲν $AB\Gamma H$ πυραμίδος ἑξαπλάσιον τὸ $BHMA$ στερεόν, τῆς δὲ $\Delta EZ\Theta$ πυραμίδος ἑξαπλάσιον τὸ $E\Theta\Pi O$ στερεόν, ἴσον ἄρα ἐστὶ τὸ $BHMA$ στερεόν τῷ $E\Theta\Pi O$ στερεῷ. τῶν δὲ ἴσων στερεῶν παραλληλεπιπύδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν· ἐστὶν ἄρα ὥς ἡ BM βάσις πρὸς τὴν $E\Pi$ βάσιν, οὕτως τὸ τοῦ $E\Theta\Pi O$ στερεοῦ ὕψος πρὸς τὸ τοῦ $BHMA$ στερεοῦ ὕψος. ἀλλ' ὥς ἡ BM βάσις πρὸς τὴν $E\Pi$, οὕτως τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ ΔEZ τρίγωνον. καὶ ὥς ἄρα τὸ $AB\Gamma$ τρίγωνον πρὸς τὸ ΔEZ τρίγωνον, οὕτως τὸ τοῦ $E\Theta\Pi O$ στερεοῦ ὕψος πρὸς τὸ τοῦ $BHMA$ στερεοῦ ὕψος. ἀλλὰ τὸ μὲν τοῦ $E\Theta\Pi O$ στερεοῦ ὕψος τὸ αὐτὸ ἐστὶ τῷ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψει, τὸ δὲ τοῦ $BHMA$ στερεοῦ ὕψος τὸ αὐτὸ ἐστὶ τῷ τῆς $AB\Gamma H$ πυραμίδος ὕψει· ἐστὶν ἄρα ὥς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος. τῶν $AB\Gamma H$, $\Delta EZ\Theta$ ἄρα πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν.

Ἀλλὰ δὴ τῶν $AB\Gamma H$, $\Delta EZ\Theta$ πυραμίδων ἀντιπεπονθέντων αἱ βάσεις τοῖς ὕψουσιν, καὶ ἔστω ὥς ἡ $AB\Gamma$ βάσις πρὸς

For let there be (two) equal pyramids having the triangular bases ABC and DEF , and apexes the points G and H (respectively). I say that the bases of the pyramids $ABCG$ and $DEFH$ are reciprocally proportional to their heights, and (so) that as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$.

For let the parallelepiped solids $BGML$ and $EHQP$ be completed. And since pyramid $ABCG$ is equal to pyramid $DEFH$, and solid $BGML$ is six times pyramid $ABCG$ (see previous proposition), and solid $EHQP$ (is) six times pyramid $DEFH$, solid $BGML$ is thus equal to solid $EHQP$. And the bases of equal parallelepiped solids are reciprocally proportional to their heights [Prop. 11.34]. Thus, as base BM is to base EQ , so the height of solid $EHQP$ (is) to the height of solid $BGML$. But, as base BM (is) to base EQ , so triangle ABC (is) to triangle DEF [Prop. 1.34]. And, thus, as triangle ABC (is) to triangle DEF , so the height of solid $EHQP$ (is) to the height of solid $BGML$ [Prop. 5.11]. But, the height of solid $EHQP$ is the same as the height of pyramid $DEFH$, and the height of solid $BGML$ is the same as the height of pyramid $ABCG$. Thus, as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$. Thus, the bases of pyramids $ABCG$ and $DEFH$ are reciprocally proportional to their heights.

And so, let the bases of pyramids $ABCG$ and $DEFH$ be reciprocally proportional to their heights, and (thus) let base ABC be to base DEF , as the height of pyramid $DEFH$ (is)

τὴν ΔEZ βάσιν, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος· λέγω, ὅτι ἴση ἐστὶν ἡ $AB\Gamma H$ πυραμὶς τῇ $\Delta EZ\Theta$ πυραμίδι.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεὶ ἐστὶν ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος, ἀλλ' ὡς ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὕτως τὸ BM παραλληλόγραμμον πρὸς τὸ $E\Pi$ παραλληλόγραμμον, καὶ ὡς ἄρα τὸ BM παραλληλόγραμμον πρὸς τὸ $E\Pi$ παραλληλόγραμμον, οὕτως τὸ τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος πρὸς τὸ τῆς $AB\Gamma H$ πυραμίδος ὕψος. ἀλλὰ τὸ [μὲν] τῆς $\Delta EZ\Theta$ πυραμίδος ὕψος τὸ αὐτὸ ἐστὶ τῷ τοῦ $E\Theta\Pi O$ παραλληλεπιπέδου ὕψει, τὸ δὲ τῆς $AB\Gamma H$ πυραμίδος ὕψος τὸ αὐτὸ ἐστὶ τῷ τοῦ $BHMA$ παραλληλεπιπέδου ὕψει· ἐστὶν ἄρα ὡς ἡ BM βάσις πρὸς τὴν $E\Pi$ βάσιν, οὕτως τὸ τοῦ $E\Theta\Pi O$ παραλληλεπιπέδου ὕψος πρὸς τὸ τοῦ $BHMA$ παραλληλεπιπέδου ὕψος. ὧν δὲ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσα ἐστὶν ἐκείνα· ἴσον ἄρα ἐστὶ τὸ $BHMA$ στερεὸν παραλληλεπιπέδον τῷ $E\Theta\Pi O$ στερεῷ παραλληλεπιπέδῳ. καὶ ἐστὶ τοῦ μὲν $BHMA$ ἕκτον μέρος ἡ $AB\Gamma H$ πυραμὶς, τοῦ δὲ $E\Theta\Pi O$ παραλληλεπιπέδου ἕκτον μέρος ἡ $\Delta EZ\Theta$ πυραμὶς· ἴση ἄρα ἡ $AB\Gamma H$ πυραμὶς τῇ $\Delta EZ\Theta$ πυραμίδι.

Τῶν ἄρα ἴσων πυραμίδων καὶ τριγώνου βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν καὶ ὧν πυραμίδων τριγώνου βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσαι εἰσὶν ἐκείναι· ὅπερ εἶδει δεῖχαι.

10.

Πᾶς κῶνος κυλίνδρον τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον.

Ἐθέτω γὰρ κῶνος κυλίνδρῳ βάσιν τε τὴν αὐτὴν τὸν $AB\Gamma\Delta$ κύκλον καὶ ὕψος ἴσον· λέγω, ὅτι ὁ κῶνος τοῦ κυλίνδρου τρίτον ἐστὶ μέρος, τουτέστιν ὅτι ὁ κύλινδρος τοῦ κώνου τριπλασίῳ ἐστίν.

to the height of pyramid $ABCG$. I say that pyramid $ABCG$ is equal to pyramid $DEFH$.

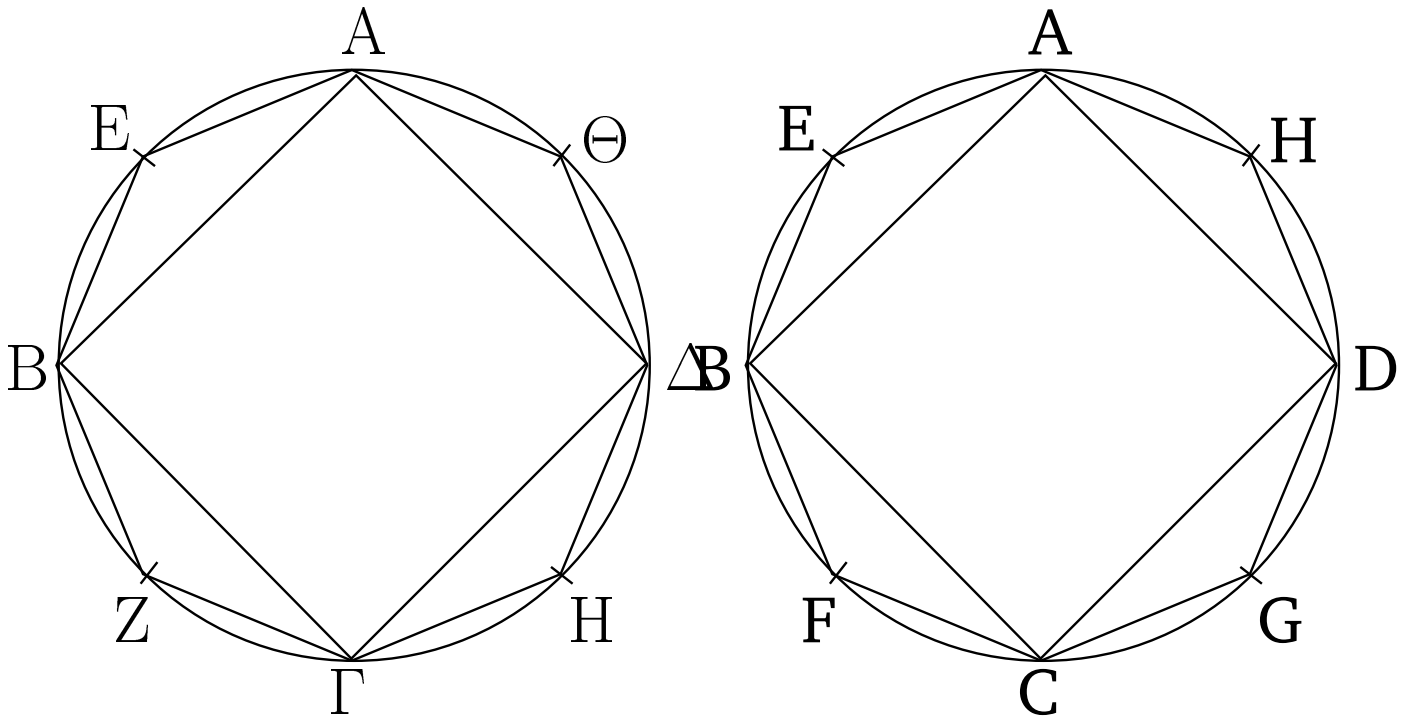
For, with the same construction, since as base ABC is to base DEF , so the height of pyramid $DEFH$ (is) to the height of pyramid $ABCG$, but as base ABC (is) to base DEF , so parallelogram BM (is) to parallelogram EQ [Prop. 1.34], thus as parallelogram BM (is) to parallelogram EQ , so the height of pyramid $DEFH$ (is) also to the height of pyramid $ABCG$ [Prop. 5.11]. But, the height of pyramid $DEFH$ is the same as the height of parallelepiped $EHQP$, and the height of pyramid $ABCG$ is the same as the height of parallelepiped $BGML$. Thus, as base BM is to base EQ , so the height of parallelepiped $EHQP$ (is) to the height of parallelepiped $BGML$. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal [Prop. 11.34]. Thus, the parallelepiped solid $BGML$ is equal to the parallelepiped solid $EHQP$. And pyramid $ABCG$ is a sixth part of $BGML$, and pyramid $DEFH$ a sixth part of parallelepiped $EHQP$. Thus, pyramid $ABCG$ is equal to pyramid $DEFH$.

Thus, the bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids having triangular bases whose bases are reciprocally proportional to their heights are equal. (Which is) the very thing it was required to show.

Proposition 10

Every cone is the third part of the cylinder which has the same base as it, and an equal height.

For let there be a cone (with) the same base as a cylinder, (namely) the circle $ABCD$, and an equal height. I say that the cone is the third part of the cylinder—that is to say, that the cylinder is three times the cone.



Εἰ γὰρ μὴ ἔστιν ὁ κύλινδρος τοῦ κώνου τριπλασίων, ἔσται ὁ κύλινδρος τοῦ κώνου ἢ τοῦ μείζων ἢ τριπλασίων ἢ ἐλάσσων ἢ τριπλασίων. ἔστω πρότερον μείζων ἢ τριπλασίων, καὶ ἐγγεγράφθω εἰς τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον τὸ $AB\Gamma\Delta$. τὸ δὴ $AB\Gamma\Delta$ τετράγωνον μείζον ἔστιν ἢ τὸ ἥμισυ τοῦ $AB\Gamma\Delta$ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ $AB\Gamma\Delta$ τετραγώνου πρίσμα ἰσοῦν πρὸς τῷ κυλίνδρῳ. τὸ δὴ ἀνιστάμενον πρίσμα μείζον ἔστιν ἢ τὸ ἥμισυ τοῦ κυλίνδρου, ἐπειδὴ περὶ τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον περιγράφωμεν, τὸ ἐγγεγραμμένον εἰς τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον ἥμισυ ἔστι τοῦ περιγεγραμμένου· καὶ ἔστι τὰ ἀπ' αὐτῶν ἀνιστάμενα στερεὰ παραλληλεπίπεδα πρίσματα ἰσοῦν· τὰ δὲ ὑπὸ τὸ αὐτὸ ὕψος ὄντα στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἔστιν ὡς αἱ βάσεις· καὶ τὸ ἐπὶ τοῦ $AB\Gamma\Delta$ ἄρα τετραγώνου ἀνασταθέν πρίσμα ἥμισυ ἔστι τοῦ ἀνασταθέντος πρίσματος ἀπὸ τοῦ περὶ τὸν $AB\Gamma\Delta$ κύκλον περιγραφέντος τετραγώνου· καὶ ἔστιν ὁ κύλινδρος ἐλάττω τοῦ πρίσματος τοῦ ἀνασταθέντος ἀπὸ τοῦ περὶ τὸν $AB\Gamma\Delta$ κύκλον περιγραφέντος τετραγώνου· τὸ ἄρα πρίσμα τὸ ἀνασταθέν ἀπὸ τοῦ $AB\Gamma\Delta$ τετραγώνου ἰσοῦν πρὸς τῷ κυλίνδρῳ μείζον ἔστι τοῦ ἡμίσεως τοῦ κυλίνδρου. τετμήσθωσαν αἱ AB , $B\Gamma$, $\Gamma\Delta$, ΔA περιφέρειαι διῖθα κατὰ τὰ E , Z , H , Θ σημεία, καὶ ἐπεζεύθωσαν αἱ AE , EB , BZ , $Z\Gamma$, ΓH , $H\Delta$, $\Delta\Theta$, ΘA . καὶ ἕκαστον ἄρα τῶν AEB , $BZ\Gamma$, $\Gamma H\Delta$, $\Delta\Theta A$ τριγώνων μείζον ἔστιν ἢ τὸ ἥμισυ τοῦ καθ' ἑαυτὸ τμήματος τοῦ $AB\Gamma\Delta$ κύκλου, ὡς ἔμπροσθεν ἐδείκνυμεν. ἀνεστάτω ἐφ' ἑκάστου τῶν AEB , $BZ\Gamma$, $\Gamma H\Delta$, $\Delta\Theta A$ τριγώνων πρίσματα ἰσοῦν πρὸς τῷ κυλίνδρῳ· καὶ ἕκαστον ἄρα τῶν ἀνασταθέντων πρισμάτων μείζον ἔστιν ἢ τὸ ἥμισυ μέρους τοῦ καθ' ἑαυτὸ τμήματος τοῦ

For if the cylinder is not three times the cone then the cylinder will be either more than three times, or less than three times, (the cone). Let it, first of all, be more than three times (the cone). And let the square $ABCD$ be inscribed in circle $ABCD$ [Prop. 4.6]. So, square $ABCD$ is more than half of circle $ABCD$ [Prop. 12.2]. And let a prism of equal height to the cylinder be set up on square $ABCD$. So, the prism set up is more than half of the cylinder, inasmuch as if we also circumscribe a square around circle $ABCD$ [Prop. 4.7] then the square inscribed in circle $ABCD$ is half of the circumscribed (square). And the solids set up on them are parallelepiped prisms of equal height. And parallelepiped solids having the same height are to one another as their bases [Prop. 11.32]. And, thus, the prism set up on square $ABCD$ is half of the prism set up on the square circumscribed about circle $ABCD$. And the cylinder is less than the prism set up on the square circumscribed about circle $ABCD$. Thus, the prism set up on square $ABCD$ of the same height as the cylinder is more than half of the cylinder. Let the circumferences AB , BC , CD , and DA be cut in half at points E , F , G , and H . And let AE , EB , BF , FC , CG , GD , DH , and HA be joined. And thus each of the triangles AEB , BFC , CGD , and DHA is more than half of the segment of circle $ABCD$ about it, as was shown previously [Prop. 12.2]. Let prisms of equal height to the cylinder be set up on each of the triangles AEB , BFC , CGD , and DHA . And each of the prisms set up is greater than the half part of the segment of the cylinder about it—inasmuch as if we draw (straight-lines) parallel to AB , BC , CD , and DA through

κυλίνδρον, ἐπειδήπερ ἔαν διὰ τῶν E, Z, H, Θ σημείων παραλλήλων ταῖς $AB, BF, \Gamma\Delta, \Delta A$ ἀγάγωμεν, καὶ συμπληρώσωμεν τὰ ἐπὶ τῶν $AB, BF, \Gamma\Delta, \Delta A$ παραλληλόγραμμα, καὶ ἀπ' αὐτῶν ἀναστήσωμεν στερεὰ παραλληλεπίπεδα ἰσοῦσῃ τῷ κυλίνδρῳ, ἑκάστων τῶν ἀνασταθέντων ἡμίση ἐστὶ τὰ πρίσματα τὰ ἐπὶ τῶν $AEB, BZF, \Gamma H\Delta, \Delta\Theta A$ τριγώνων καὶ ἐστὶ τὰ τοῦ κυλίνδρου τμήματα ἐλάττωνα τῶν ἀνασταθέντων στερεῶν παραλληλεπιπέδων· ὥστε καὶ τὰ ἐπὶ τῶν $AEB, BZF, \Gamma H\Delta, \Delta\Theta A$ τριγώνων πρίσματα μείζονά ἐστιν ἢ τὸ ἥμισυ τῶν καθ' ἑαυτὰ τοῦ κυλίνδρου τμημάτων. τέμνοντες δὲ τὰς ὑπολειπομένας περιφερείας διῖθα καὶ ἐπιξενυγνόντες εὐθείας καὶ ἀνιστάντες ἑφ' ἑκάστων τῶν τριγώνων πρίσματα ἰσοῦσῃ τῷ κυλίνδρῳ καὶ τοῦτο αἰ ποιοῦντες καταλείβομεν τινα ἀποτμήματα τοῦ κυλίνδρου, ἃ ἔσται ἐλάττωνα τῆς ὑπεροθῆς, ἣ ὑπερέθει ὁ κύλινδρος τοῦ τριπλασίου τοῦ κώνου. λελείφθω, καὶ ἔστω τὰ $AE, EB, BZ, ZF, \Gamma H, H\Delta, \Delta\Theta, \Theta A$ · λοιπὸν ἄρα τὸ πρίσμα, οὗ βάσις μὲν τὸ $AEBZ\Gamma H\Delta\Theta$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, μείζον ἐστὶν ἢ τριπλάσιον τοῦ κώνου. ἀλλὰ τὸ πρίσμα, οὗ βάσις μὲν ἐστὶ τὸ $AEBZ\Gamma H\Delta\Theta$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, τριπλάσιόν ἐστι τῆς πυραμίδος, ἥς βάσις μὲν ἐστὶ τὸ $AEBZ\Gamma H\Delta\Theta$ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ· καὶ ἡ πυραμὶς ἄρα, ἥς βάσις μὲν [ἐστὶ] τὸ $AEBZ\Gamma H\Delta\Theta$ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶ τοῦ κώνου τοῦ βάσιν ἔχοντος τὸν $AB\Gamma\Delta$ κύκλον. ἀλλὰ καὶ ἐλάττων ἐμπεριέθεται γὰρ ὑπ' αὐτοῦ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὁ κύλινδρος τοῦ κώνου μείζων ἢ τριπλάσιος.

Λέγω δὴ, ὅτι οὐδὲ ἐλάττων ἐστὶν ἢ τριπλάσιος ὁ κύλινδρος τοῦ κώνου.

Εἰ γὰρ δυνατόν, ἔστω ἐλάττων ἢ τριπλάσιος ὁ κύλινδρος τοῦ κώνου· ἀνάπαλιν ἄρα ὁ κώνος τοῦ κυλίνδρου μείζων ἐστὶν ἢ τρίτον μέρος. ἐγγεγράφθω δὲ εἰς τὸν $AB\Gamma\Delta$ κύκλον τετράγωνον τὸ $AB\Gamma\Delta$ · τὸ $AB\Gamma\Delta$ ἄρα τετράγωνον μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ $AB\Gamma\Delta$ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ $AB\Gamma\Delta$ τετραγώνου πυραμὶς τὴν αὐτὴν κορυφὴν ἔχουσα τῷ κώνῳ· ἡ ἄρα ἀνασταθεῖσα πυραμὶς μείζων ἐστὶν ἢ τὸ ἥμισυ μέρος τοῦ κώνου, ἐπειδήπερ, ὡς ἐμπροσθεν ἐδείκνυμεν, ὅτι ἔαν περὶ τὸν κύκλον τετράγωνον περιγράψωμεν, ἔσται τὸ $AB\Gamma\Delta$ τετράγωνον ἥμισυ τοῦ περὶ τὸν κύκλον περιγεγραμμένου τετραγώνου· καὶ ἔαν ἀπὸ τῶν τετραγώνων στερεὰ παραλληλεπίπεδα ἀναστήσωμεν ἰσοῦσῃ τῷ κώνῳ, ἃ καὶ καλεῖται πρίσματα, ἔσται τὸ ἀνασταθέν ἀπὸ τοῦ $AB\Gamma\Delta$ τετραγώνου ἥμισυ τοῦ ἀνασταθέντος ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου· πρὸς ἄλληλα γὰρ εἰσιν ὡς αἱ βάσεις. ὥστε καὶ τὰ τρίτα· καὶ πυραμὶς ἄρα, ἥς βάσις τὸ $AB\Gamma\Delta$ τετράγωνον, ἥμισυ ἐστὶ τῆς πυραμίδος τῆς ἀνασταθείσης ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου. καὶ ἐστὶ μείζων ἢ πυραμὶς ἢ ἀνασταθεῖσα ἀπὸ τοῦ περὶ τὸν κύκλον τετραγώνου τοῦ κώνου· ἐμπεριέθει γὰρ αὐτόν. ἡ ἄρα πυραμὶς, ἥς βάσις τὸ $AB\Gamma\Delta$ τετράγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶν ἢ τὸ ἥμισυ τοῦ κώνου. τεμήσθωσαν αἱ $AB, BF, \Gamma\Delta, \Delta A$ περιφέρειαι διῖθα κατὰ τὰ E, Z, H, Θ σημεία, καὶ ἐπεξέυθωσαν αἱ

points E, F, G , and H (respectively), and complete the parallelograms on AB, BC, CD , and DA , and set up parallelepiped solids of equal height to the cylinder on them, then the prisms on triangles AEB, BFC, CGD , and DHA are each half of the set up (parallelepipeds). And the segments of the cylinder are less than the set up parallelepiped solids. Hence, the prisms on triangles AEB, BFC, CGD , and DHA are also greater than half of the segments of the cylinder about them. So (if) the remaining circumferences are cut in half, and straight-lines are joined, and prisms of equal height to the cylinder are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cylinder whose (sum) is less than the excess by which the cylinder exceeds three times the cone [Prop. 10.1]. Let them be left, and let them be $AE, EB, BF, FC, CG, GD, DH$, and HA . Thus, the remaining prism whose base (is) polygon $AEBFCGDH$, and height the same as the cylinder, is greater than three times the cone. But, the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder, is three times the pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone [Prop. 12.7 corr.]. And thus the pyramid whose base [is] polygon $AEBFCGDH$, and apex the same as the cone, is greater than the cone having (as) base circle $ABCD$. But (it is) also less. For it is encompassed by it. The very thing (is) impossible. Thus, the cylinder is not more than three times the cone.

So, I say that neither (is) the cylinder less than three times the cone.

For, if possible, let the cylinder be less than three times the cone. Thus, inversely, the cone is greater than the third part of the cylinder. So, let the square $ABCD$ be inscribed in circle $ABCD$ [Prop. 4.6]. Thus, square $ABCD$ is greater than half of circle $ABCD$. And let a pyramid having the same apex as the cone be set up on square $ABCD$. Thus, the pyramid set up is greater than the half part of the cone, inasmuch as we showed previously that if we circumscribe a square about the circle [Prop. 4.7] then the square $ABCD$ will be half of the square circumscribed about the circle [Prop. 12.2]. And if we set up on the squares parallelepiped solids—which are also called prisms—of the same height as the cone, then the (prism) set up on square $ABCD$ will be half of the (prism) set up on the square circumscribed about the circle. For they are to one another as their bases [Prop. 11.32]. Hence, (the same) also (goes for) the thirds. Thus, the pyramid whose base is square $ABCD$ is half of the pyramid set up on the square circumscribed about the circle [Prop. 12.7 corr.]. And the pyramid set up on the square circumscribed about the circle is greater than the cone. For it encompasses it. Thus, the pyramid whose base is square $ABCD$, and apex the same as the cone, is greater than half of the cone. Let the circumferences AB, BC, CD , and DA be cut in half at points E, F, G , and H (respectively). And let $AE, EB,$

$AE, EB, BZ, ZΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ$ · και ἕκαστον ἄρα τῶν $AEB, BZΓ, ΓΗΔ, ΔΘΑ$ τριγώνων μείζον ἐστὶν ἢ τὸ ἥμισυ μέρος του καθ' ἑαυτὸ τμήματος τοῦ $ABΓΔ$ κύκλου. καὶ ἀνεστάτωσαν ἐφ' ἕκαστον τῶν $AEB, BZΓ, ΓΗΔ, ΔΘΑ$ τριγώνων πυραμίδες τὴν αὐτὴν κορυφὴν ἔθουσαι τῷ κώνῳ· καὶ ἕκαστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων κατὰ τὸν αὐτὸν τρόπον μείζων ἐστὶν ἢ τὸ ἥμισυ μέρος τοῦ καθ' ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰς ὑπολειπομένας περιφερείας διῆθα καὶ ἐπιξενυγνύντες εὐθείας καὶ ἀνιστάντες ἐφ' ἕκαστον τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴν ἔθουσας τῷ κώνῳ καὶ τοῦτο αἰ ποιοῦντες καταλείβομεν τινα ἀποτμήματα τοῦ κώνου, ἃ ἔσται ἐλάττωνα τῆς ὑπεροθῆς, ἢ ὑπερέθει ὁ κώνος τοῦ τρίτου μέρους τοῦ κυλίνδρου. λελείφθω, καὶ ἔστω τὰ ἐπὶ τῶν $AE, EB, BZ, ZΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ$ · λοιπὴ ἄρα ἡ πυραμὶς, ἥς βάσις μὲν ἐστὶ τὸ $AEBZΓΗΔΘ$ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶν ἢ τρίτον μέρος τοῦ κυλίνδρου. ἀλλ' ἡ πυραμὶς, ἥς βάσις μὲν ἐστὶ τὸ $AEBZΓΗΔΘ$ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, τρίτον ἐστὶ μέρος τοῦ πρίσματος, ὃς βάσις μὲν ἐστὶ τὸ $AEBZΓΗΔΘ$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ· τὸ ἄρα πρίσμα, ὃς βάσις μὲν ἐστὶ τὸ $AEBZΓΗΔΘ$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, μείζον ἐστὶ τοῦ κυλίνδρου, ὃς βάσις ἐστὶν ὁ $ABΓΔ$ κύκλος. ἀλλὰ καὶ ἐλάττων ἐμπεριέθεται γὰρ ὑπ' αὐτοῦ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ κύλινδρος τοῦ κώνου ἐλάττων ἐστὶν ἢ τριπλάσιος. ἐδείχθη δέ, ὅτι οὐδὲ μείζων ἢ τριπλάσιος· τριπλάσιος ἄρα ὁ κύλινδρος τοῦ κώνου· ὥστε ὁ κώνος τρίτον ἐστὶ μέρος τοῦ κυλίνδρου.

Πᾶς ἄρα κώνος κυλίνδρου τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον· ὅπερ ἔδει δεῖξαι.

11.

Οἱ ὑπο τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρόξ ἀλλήλων εἰσὶν ὥς αἱ βάσεις.

Ἐστωσαν ὑπὸ τὸ αὐτὸ ὕψος κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν [εἰσιν] οἱ $ABΓΔ, EZHΘ$ κύκλοι, ἄχονες δὲ οἱ KL, MN , διαμέτροι δὲ τῶν βάσεων αἱ AG, EH · λέγω, ὅτι ἐστὶν ὥς ὁ $ABΓΔ$ κύκλος πρόξ τὸν $EZHΘ$ κύκλον, οὕτως ὁ AL κώνος πρόξ τὸν EN κώνον.

BF, FC, CG, GD, DH , and HA be joined. And, thus, each of the triangles AEB, BFC, CGD , and DHA is greater than the half part of the segment of circle $ABCD$ about it [Prop. 12.2]. And let pyramids having the same apex as the cone be set up on each of the triangles AEB, BFC, CGD , and DHA . And, thus, in the same way, each of the pyramids set up is more than the half part of the segment of the cone about it. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which the cone exceeds the third part of the cylinder [Prop. 10.1]. Let them be left, and let them be the (segments) on $AE, EB, BF, FC, CG, GD, DH$, and HA . Thus, the remaining pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is greater than the third part of the cylinder. But, the pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is the third part of the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder [Prop. 12.7 corr.]. Thus, the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder, is greater than the cylinder whose base is circle $ABCD$. But, (it is) also less. For it is encompassed by it. The very thing is impossible. Thus, the cylinder is not less than three times the cone. And it was shown that neither (is it) greater than three times (the cone). Thus, the cylinder (is) three times the cone. Hence, the cone is the third part of the cylinder.

Thus, every cone is the third part of the cylinder which has the same base as it, and an equal height. (Which is) the very thing it was required to show.

Proposition 11

Cones and cylinders having the same height are to one another as their bases.

Let there be cones and cylinders of the same height whose bases [are] the circles $ABCD$ and $EFGH$, axes KL and MN , and diameters of the bases AC and EG (respectively). I say that as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to cone EN .

ἂ ἔσται ἐλάσσονα τοῦ Ψ στερεοῦ. λελείφθω, καὶ ἔστω τὰ ἐπὶ τῶν $\Theta\Omega\epsilon$, $\epsilon\Pi\zeta$, $\zeta\rho\eta$, $\eta\sigma\theta$: λοιπὴ ἄρα ἡ πυραμὶς, ἣς βάσις τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\sigma$ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κώνῳ, μείζων ἐστὶ τοῦ X στερεοῦ. ἐγγεγράφθω καὶ εἰς τὸν $AB\Gamma\Delta$ κύκλον τῷ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\sigma$ πολυγώνῳ ὁμοίον τε καὶ ὁμοίως κείμενον πολύγωνον τὸ $\Delta TAYB\Phi\Gamma\Theta$, καὶ ἀνεστιάτω ἐπ' αὐτοῦ πυραμὶς ἰσοῦν ἡς τῷ AA κώνῳ. ἐπεὶ οὖν ἐστὶν ὡς τὸ ἀπὸ τῆς AG πρὸς τὸ ἀπὸ τῆς EH , οὕτως τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον πρὸς τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\sigma$ πολύγωνον, ὡς δὲ τὸ ἀπὸ τῆς AG πρὸς τὸ ἀπὸ τῆς EH , οὕτως ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, καὶ ὡς ἄρα ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον πρὸς τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\sigma$ πολύγωνον. ὡς δὲ ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς τὸ X στερεόν, ὡς δὲ τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον πρὸς τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\sigma$ πολύγωνον, οὕτως ἡ πυραμὶς, ἣς βάσις μὲν τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον, κορυφὴ δὲ τὸ A σημεῖον, πρὸς τὴν πυραμίδα, ἣς βάσις μὲν τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον. καὶ ὡς ἄρα ὁ AA κώνος πρὸς τὸ X στερεόν, οὕτως ἡ πυραμὶς, ἣς βάσις μὲν τὸ $\Delta TAYB\Phi\Gamma\Theta$ πολύγωνον, κορυφὴ δὲ τὸ A σημεῖον, πρὸς τὴν πυραμίδα, ἣς βάσις μὲν τὸ $\Theta\Omega\epsilon\Pi\zeta\rho\eta\sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον ἐναλλάξ ἄρα ἐστὶν ὡς ὁ AA κώνος πρὸς τὴν ἐν αὐτῷ πυραμίδα, οὕτως τὸ X στερεόν πρὸς τὴν ἐν τῷ EN κώνῳ πυραμίδα. μείζων δὲ ὁ AA κώνος τῆς ἐν αὐτῷ πυραμίδος· μείζων ἄρα καὶ τὸ X στερεόν τῆς ἐν τῷ EN κώνῳ πυραμίδος. ἀλλὰ καὶ ἔλασσον· ὅπερ ἄτοπον. οὐκ ἄρα ἐστὶν ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς ἔλασσόν τι τοῦ EN κώνου στερεόν. ὁμοίως δὲ δείχομεν, ὅτι οὐδὲ ἐστὶν ὡς ὁ $EZH\Theta$ κύκλος πρὸς τὸν $AB\Gamma\Delta$ κύκλον, οὕτως ὁ EN κώνος πρὸς ἔλασσόν τι τοῦ AA κώνου στερεόν.

Λέγω δὴ, ὅτι οὐδὲ ἐστὶν ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς μείζον τι τοῦ EN κώνου στερεόν.

Εἰ γὰρ δυνατόν, ἔστω πρὸς μείζον τὸ X : ἀνάπαλιν ἄρα ἐστὶν ὡς ὁ $EZH\Theta$ κύκλος πρὸς τὸν $AB\Gamma\Delta$ κύκλον, οὕτως τὸ X στερεόν πρὸς τὸν AA κώνον. ἀλλ' ὡς τὸ X στερεόν πρὸς τὸν AA κώνον, οὕτως ὁ EN κώνος πρὸς ἔλασσόν τι τοῦ AA κώνου στερεόν· καὶ ὡς ἄρα ὁ $EZH\Theta$ κύκλος πρὸς τὸν $AB\Gamma\Delta$ κύκλον, οὕτως ὁ EN κώνος πρὸς ἔλασσόν τι τοῦ AA κώνου στερεόν· ὅπερ ἀδύνατον ἐδείχθη. οὐκ ἄρα ἐστὶν ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς μείζον τι τοῦ EN κώνου στερεόν. ἐδείχθη δέ, ὅτι οὐδὲ πρὸς ἔλασσον· ἐστὶν ἄρα ὡς ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως ὁ AA κώνος πρὸς τὸν EN κώνον.

Ἀλλ' ὡς ὁ κώνος πρὸς τὸν κώνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον τριπλασίῳ γὰρ ἐκάτερος ἐκατέρων. καὶ ὡς ἄρα ὁ $AB\Gamma\Delta$ κύκλος πρὸς τὸν $EZH\Theta$ κύκλον, οὕτως οἱ ἐπ' αὐτῶν ἰσοῦνται.

Οἱ ἄρα ὑπὸ τὸ αὐτὸ ὕψος ὄντες κώνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖξαι.

set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone (the sum of) which is less than solid X [Prop. 10.1]. Let them be left, and let them be the (segments) on HPE , EQF , FRG , and GSH . Thus, the remaining pyramid whose base is polygon $HPEQFRGS$, and height the same as the cone, is greater than solid O [Prop. 6.18]. And let the polygon $DTAUBVCW$, similar, and similarly laid out, to polygon $HPEQFRGS$, be inscribed in circle $ABCD$. And on it let a pyramid of the same height as cone AL be set up. Therefore, since as the (square) on AC is to the (square) on EG , so polygon $DTAUBVCW$ (is) to polygon $HPEQFRGS$ [Prop. 12.1], and as the (square) on AC (is) to the (square) on EG , so circle $ABCD$ (is) to circle $EFGH$ [Prop. 12.2], thus as circle $ABCD$ (is) to circle $EFGH$, so polygon $DTAUBVCW$ also (is) to polygon $HPEQFRGS$. And as circle $ABCD$ (is) to circle $EFGH$, so cone AL (is) to solid O . And as polygon $DTAUBVCW$ (is) to polygon $HPEQFRGS$, so the pyramid whose base is polygon $DTAUBVCW$, and apex the point L , (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 12.6]. And, thus, as cone AL (is) to solid O , so the pyramid whose base is $DTAUBVCW$, and apex the point L , (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 5.11]. Thus, alternately, as cone AL is to the pyramid within it, so solid O (is) to the pyramid within cone EN [Prop. 5.16]. But, cone AL (is) greater than the pyramid within it. Thus, solid O (is) also greater than the pyramid within cone EN [Prop. 5.14]. But, (it is) also less. The very thing (is) absurd. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid less than cone EN . So, similarly, we can show that neither is circle $EFGH$ to circle $ABCD$, as cone EN (is) to some solid less than cone AL .

So, I say that neither is circle $ABCD$ to circle $EFGH$, as cone AL (is) to some solid greater than cone EN .

For, if possible, let it be (in this ratio) to (some) greater (solid), O . Thus, inversely, as circle $EFGH$ is to circle $ABCD$, so solid O (is) to cone AL [Prop. 5.7 corr.]. But, as solid O (is) to cone AL , so cone EN (is) to some solid less than cone AL [Prop. 12.2 lem.]. And, thus, as circle $EFGH$ (is) to circle $ABCD$, so cone EN (is) to some solid less than cone AL . The very thing was shown (to be) impossible. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid greater than cone EN . And, it was shown that neither (is it in this ratio) to (some) lesser (solid). Thus, as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to cone EN .

But, as the cone (is) to the cone, (so) the cylinder (is) to the cylinder. For each (is) three times each [Prop. 12.10]. Thus, circle $ABCD$ (is) also to circle $EFGH$, as (the ratio of the cylinders) on them (having) the same height.

Thus, cones and cylinders having the same height are to one another as their bases. (Which is) the very thing it was

required to show.

12.

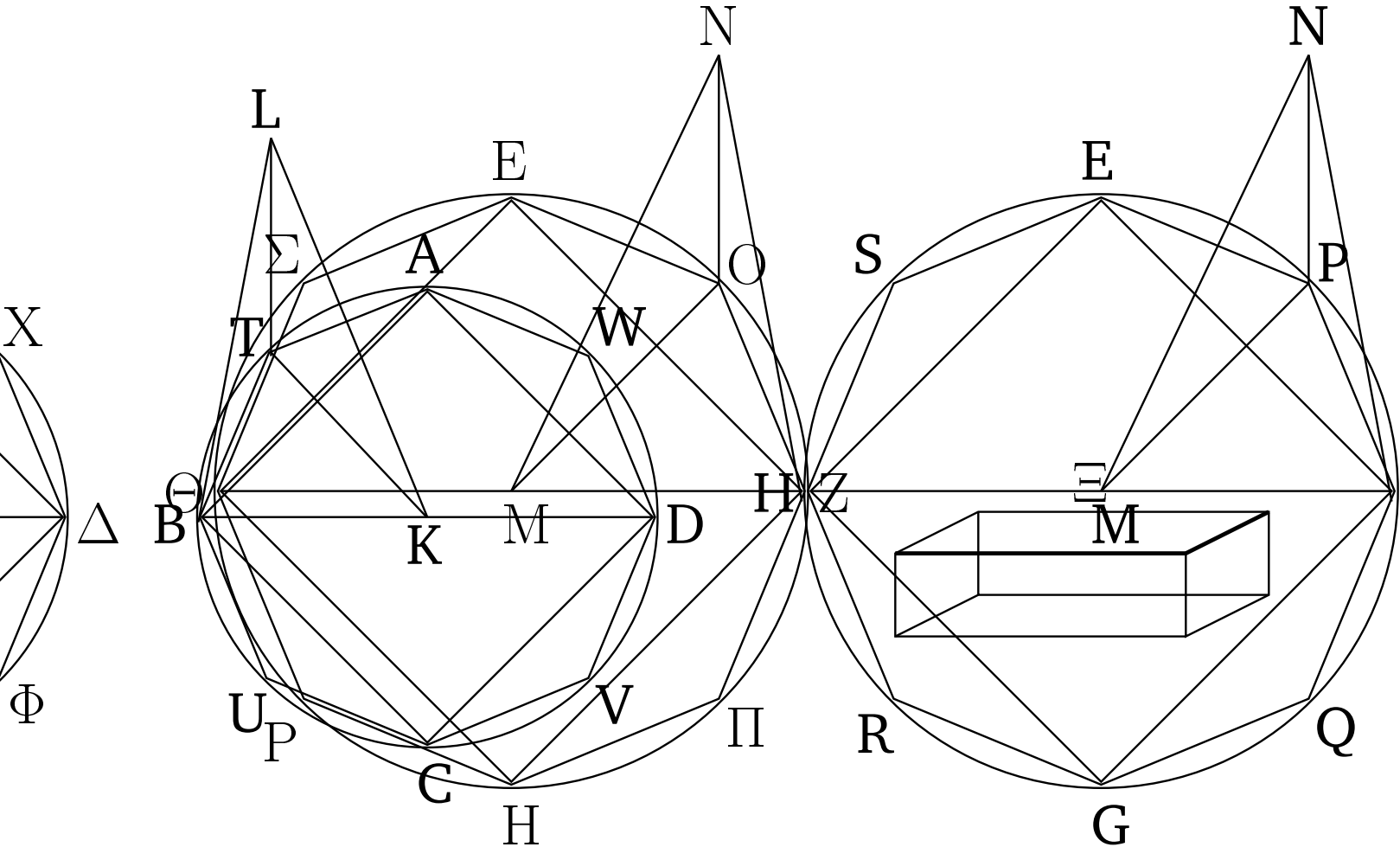
Οἱ ὅμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἐν ταῖς βάσεσι διαμέτρων.

Ἐστωσαν ὅμοιοι κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν οἱ $ABΓΔ$, $EZHΘ$ κύκλοι, διαμέτροι δὲ τῶν βάσεων αἱ $BΔ$, $ZΘ$, ἄχονες δὲ τῶν κῶνων καὶ κυλίνδρων οἱ KL , MN . λέγω, ὅτι ὁ κῶνος, ὃς ἔχει μὲν [ἔστιν] ὁ $ABΓΔ$ κύκλος, κορυφή δὲ τὸ $Λ$ σημεῖον, πρὸς τὸν κῶνον, ὃς ἔχει μὲν [ἔστιν] ὁ $EZHΘ$ κύκλος, κορυφή δὲ τὸ N σημεῖον, τριπλασίονα λόγον ἔχει ἢπερ ἡ $BΔ$ πρὸς τὴν $ZΘ$.

Proposition 12

Similar cones and cylinders are to one another in the cubed ratio of the diameters of their bases.

Let there be similar cones and cylinders of which the bases (are) the circles $ABCD$ and $EFGH$, the diameters of the bases (are) BD and FH , and the axes of the cones and cylinders (are) KL and MN (respectively). I say that the cone whose base [is] circle $ABCD$, and apex the point L , has to the cone whose base [is] circle $EFGH$, and apex the point N , the cubed ratio that BD (has) to FH .



Εἰ γὰρ μὴ ἔθῃ ὁ $ABΓΔΛ$ κῶνος πρὸς τὸν $EZHΘN$ κῶνον τριπλασίονα λόγον ἢπερ ἡ $BΔ$ πρὸς τὴν $ZΘ$, ἔχει ὁ $ABΓΔΛ$ κῶνος ἢ πρὸς ἑλασσόν τι τοῦ $EZHΘN$ κῶνον στερεὸν τριπλασίονα λόγον ἢ πρὸς μείζον. ἐθέτω πρότερον πρὸς ἑλασσόν τὸ X , καὶ ἐγγεγράφθω εἰς τὸν $EZHΘ$ κύκλον τετράγωνον τὸ $EZHΘ$. τὸ ἄρα $EZHΘ$ τετράγωνον μείζον ἐστὶν ἢ τὸ ἥμισυ τοῦ $EZHΘ$ κύκλου. καὶ ἀνεστάτω ἐπὶ τοῦ $EZHΘ$ τετραγώνου

For if cone $ABCDL$ does not have to cone $EFGHN$ the cubed ratio that BD (has) to FH then cone $ABCDL$ will have the cubed ratio to some solid either less than, or greater than, cone $EFGHN$. Let it, first of all, have (such a ratio) to (some) lesser (solid), O . And let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. Thus, square $EFGH$ is greater than half of circle $EFGH$ [Prop. 12.2]. And let a pyramid hav-

πυραμὶς τὴν αὐτὴν κορυφὴν ἔθουσα τῷ κώνῳ· ἡ ἄρα ἀνασταθεῖσα πυραμὶς μείζων ἐστὶν ἢ τὸ ἥμισυ μέρους τοῦ κώνου. τετμήσθωσαν δὴ αἱ EZ , ZH , $H\Theta$, ΘE περιφέρειαι διὰ κατὰ τὰ O , Π , P , Σ σημεία, καὶ ἐπεξεύθωσαν αἱ EO , OZ , $Z\Pi$, ΠH , HP , $P\Theta$, $\Theta\Sigma$, ΣE . καὶ ἕκαστον ἄρα τῶν EOZ , $Z\Pi H$, $HP\Theta$, $\Theta\Sigma E$ τριγώνων μείζον ἐστὶν ἢ τὸ ἥμισυ μέρους τοῦ καθ' ἑαυτὸ τμήματος τοῦ $EZH\Theta$ κύκλου. καὶ ἀνεστιάτω ἐφ' ἕκαστον τῶν EOZ , $Z\Pi H$, $HP\Theta$, $\Theta\Sigma E$ τριγώνων πυραμὶς τὴν αὐτὴν κορυφὴν ἔθουσα τῷ κώνῳ· καὶ ἕκαστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων μείζων ἐστὶν ἢ τὸ ἥμισυ μέρους τοῦ καθ' ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὲ τὰς ὑπολειπομένας περιφερείας διὰ καὶ ἐπιξενγνόντες εὐθείας καὶ ἀνοστήσαντες ἐφ' ἕκαστον τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴν ἔθουσας τῷ κώνῳ καὶ τοῦτο αἶε ποιοῦντες καταλείπομέναι ἀποτμήματα τοῦ κώνου, ἃ ἔσται ἐλάσσονα τῆς ὑπεροπῆς, ἣ ὑπερέθει ὁ $EZH\Theta$ κώνος τοῦ X στερεοῦ. λελεῖφθω, καὶ ἔστω τὰ ἐπὶ τῶν EO , OZ , $Z\Pi$, ΠH , HP , $P\Theta$, $\Theta\Sigma$, ΣE λοιπὴ ἄρα ἡ πυραμὶς, ἥς βάσις μὲν ἐστὶ τὸ $EOZ\Pi HP\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον, μείζων ἐστὶ τοῦ X στερεοῦ. ἐγγεγράφθω καὶ εἰς τὸν $AB\Gamma\Delta$ κύκλον τῷ $EOZ\Pi HP\Theta\Sigma$ πολυγώνῳ ὁμοίον τε καὶ ὁμοίως κείμενον πολυγώνον τὸ $ATBY\Gamma\Phi\Delta\Theta$, καὶ ἀνεστιάτω ἐπὶ τοῦ $ATBY\Gamma\Phi\Delta\Theta$ πολυγώνου πυραμὶς τὴν αὐτὴν κορυφὴν ἔθουσα τῷ κώνῳ, καὶ τῶν μὲν περιεθόντων τὴν πυραμίδα, ἥς βάσις μὲν ἐστὶ τὸ $ATBY\Gamma\Phi\Delta\Theta$ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, ἐν τριγώνον ἔστω τὸ ΛBT , τῶν δὲ περιεθόντων τὴν πυραμίδα, ἥς βάσις μὲν ἐστὶ τὸ $EOZ\Pi HP\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημεῖον, ἐν τριγώνον ἔστω τὸ NZO , καὶ ἐπεξεύθωσαν αἱ KT , MO . καὶ ἐπεὶ ὁμοίός ἐστιν ὁ $AB\Gamma\Delta$ κώνος τῷ $EZH\Theta$ κώνῳ, ἔστιν ἄρα ὥς ἡ $B\Delta$ πρὸς τὴν $Z\Theta$, οὕτως ὁ $K\Lambda$ ἄκων πρὸς τὸν MN ἄκωνα. ὥς δὲ ἡ $B\Delta$ πρὸς τὴν $Z\Theta$, οὕτως ἡ BK πρὸς τὴν ZM · καὶ ὥς ἄρα ἡ BK πρὸς τὴν ZM , οὕτως ἡ $K\Lambda$ πρὸς τὴν MN . καὶ ἐναλλάξ ὥς ἡ BK πρὸς τὴν $K\Lambda$, οὕτως ἡ ZM πρὸς τὴν MN . καὶ περὶ ἴσας γωνίας τὰς ὑπὸ $BK\Lambda$, ZMN αἱ πλευραὶ ἀνάλογόν εἰσιν· ὁμοίον ἄρα ἐστὶ τὸ $BK\Lambda$ τρίγωνον τῷ ZMN τριγώνῳ. πάλιν, ἐπεὶ ἐστὶν ὥς ἡ BK πρὸς τὴν KT , οὕτως ἡ ZM πρὸς τὴν MO , καὶ περὶ ἴσας γωνίας τὰς ὑπὸ BKT , ZMO , ἐπειδήπερ, ὁ μέρους ἐστὶν ἡ ὑπὸ BKT γωνία τῶν πρὸς τῷ K κέντρῳ τεσσάρων ὀρθῶν, τὸ αὐτὸ μέρος ἐστὶ καὶ ἡ ὑπὸ ZMO γωνία τῶν πρὸς τῷ M κέντρῳ τεσσάρων ὀρθῶν· ἐπεὶ οὖν περὶ ἴσας γωνίας αἱ πλευραὶ ἀνάλογόν εἰσιν, ὁμοίον ἄρα ἐστὶ τὸ BKT τρίγωνον τῷ ZMO τριγώνῳ. πάλιν, ἐπεὶ ἐδείχθη ὥς ἡ BK πρὸς τὴν $K\Lambda$, οὕτως ἡ ZM πρὸς τὴν MN , ἴση δὲ ἡ μὲν BK τῇ KT , ἡ δὲ ZM τῇ OM , ἔστιν ἄρα ὥς ἡ TK πρὸς τὴν $K\Lambda$, οὕτως ἡ OM πρὸς τὴν MN . καὶ περὶ ἴσας γωνίας τὰς ὑπὸ $TK\Lambda$, OMN · ὀρθαὶ γάρ· αἱ πλευραὶ ἀνάλογόν εἰσιν· ὁμοίον ἄρα ἐστὶ τὸ ΛKT τρίγωνον τῷ NMO τριγώνῳ. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ΛKB , NMZ τριγώνων ἐστὶν ὥς ἡ ΛB πρὸς τὴν BK , οὕτως ἡ NZ πρὸς τὴν ZM , διὰ δὲ τὴν ὁμοιότητα τῶν BKT , ZMO τριγώνων ἐστὶν ὥς ἡ KB πρὸς τὴν BT , οὕτως ἡ MZ πρὸς

ing the same apex as the cone be set up on square $EFGH$. Thus, the pyramid set up is greater than the half part of the cone [Prop. 12.10]. So, let the circumferences EF , FG , GH , and HE be cut in half at points P , Q , R , and S (respectively). And let EP , PF , FQ , QG , GR , RH , HS , and SE be joined. And, thus, each of the triangles EPF , FQG , GRH , and HSE is greater than the half part of the segment of circle $EFGH$ about it [Prop. 12.2]. And let a pyramid having the same apex as the cone be set up on each of the triangles EPF , FQG , GRH , and HSE . And thus each of the pyramids set up is greater than the half part of the segment of the cone about it [Prop. 12.10]. So, (if) the the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which cone $EFGHN$ exceeds solid O [Prop. 10.1]. Let them be left, and let them be the (segments) on EP , PF , FQ , QG , GR , RH , HS , and SE . Thus, the remaining pyramid whose base is polygon $EPFQGRHS$, and apex the point N , is greater than solid O . And let the polygon $ATBUCVDW$, similar, and similarly laid out, to polygon $EPFQGRHS$, be inscribed in circle $ABCD$ [Prop. 6.18]. And let a pyramid having the same apex as the cone be set up on polygon $ATBUCVDW$. And let LBT be one of the triangles containing the pyramid whose base is polygon $ATBUCVDW$, and apex the point L . And let NFP be one of the triangles containing the pyramid whose base is triangle $EPFQGRHS$, and apex the point N . And let KT and MP be joined. And since cone $ABCDL$ is similar to cone $EFGHN$, thus as BD is to FH , so axis KL (is) to axis MN [Def. 11.24]. And as BD (is) to FH , so BK (is) to FM . And, thus, as BK (is) to FM , so KL (is) to MN . And, alternately, as BK (is) to KL , so FM (is) to MN [Prop. 5.16]. And the sides around the equal angles BKL and FMN are proportional. Thus, triangle BKL is similar to triangle FMN [Prop. 6.6]. Again, since as BK (is) to KT , so FM (is) to MP , and (they are) about the equal angles BKT and FMP , inasmuch as whatever part angle BKT is of the four right-angles at the center K , angle FMP is also the same part of the four right-angles at the center M . Therefore, since the sides about equal angles are proportional, triangle BKT is thus similar to triangle FMP [Prop. 6.6]. Again, since it was shown that as BK (is) to KL , so FM (is) to MN , and BK (is) equal to KT , and FM to PM , thus as TK (is) to KL , so PM (is) to MN . And the sides about the equal angles TKL and PMN —for (they are both) right-angles—are proportional. Thus, triangle LKT (is) similar to triangle NMP [Prop. 6.6]. And since, on account of the similarity of triangles LKB and NMF , as LB (is) to BK , so NF (is) to FM , and, on account of the similarity of triangles BKT and FMP , as KB (is) to BT , so MF (is) to FP [Def. 6.1], thus, via equality, as LB (is) to BT , so NF (is) to FP [Prop. 5.22]. Again, since, on

τὴν ZO , δι' ἴσων ἄρα ὡς ἡ AB πρὸς τὴν BT , οὕτως ἡ NZ πρὸς τὴν ZO . πάλιν, ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ATK , NOM τριγώνων ἐστὶν ὡς ἡ AT πρὸς τὴν TK , οὕτως ἡ NO πρὸς τὴν OM , διὰ δὲ τὴν ὁμοιότητα τῶν TKB , OMZ τριγώνων ἐστὶν ὡς ἡ KT πρὸς τὴν TB , οὕτως ἡ MO πρὸς τὴν OZ , δι' ἴσων ἄρα ὡς ἡ AT πρὸς τὴν TB , οὕτως ἡ NO πρὸς τὴν OZ . ἐδείχθη δὲ καὶ ὡς ἡ TB πρὸς τὴν BA , οὕτως ἡ OZ πρὸς τὴν ZN . δι' ἴσων ἄρα ὡς ἡ TA πρὸς τὴν AB , οὕτως ἡ ON πρὸς τὴν NZ . τῶν ATB , NOZ ἄρα τριγώνων ἀνάλογόν εἰσιν αἱ πλευραὶ ἰσογώνια ἄρα ἐστὶ τὰ ATB , NOZ τρίγωνα· ὥστε καὶ ὅμοια. καὶ πυραμῖς ἄρα, ἥς βάσεις μὲν τὸ BKT τρίγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, ὁμοία ἐστὶ πυραμίδι, ἥς βάσεις μὲν τὸ ZMO τρίγωνον, κορυφὴ δὲ τὸ N σημείον· ὑπὸ γὰρ ὁμοίων ἐπιπέδων περιέχονται ἴσων τὸ πλῆθος. αἱ δὲ ὅμοιαι πυραμίδες καὶ τριγώνουζ ἔθουσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. ἡ ἄρα $BKTA$ πυραμῖς πρὸς τὴν $ZMON$ πυραμίδα τριπλασίονα λόγον ἔθει ἥπερ ἡ BK πρὸς τὴν ZM . ὁμοίως δὲ ἐπιξενυνόντες ἀπὸ τῶν A , Θ , Δ , Φ , Γ , Y ἐπὶ τὸ K εὐθείας καὶ ἀπὸ τῶν E , Σ , Θ , P , H , Π ἐπὶ τὸ M καὶ ἀνιστάντες ἐφ' ἐκάστον τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴν ἐθούσας τοῖς κῶνοις δείχομεν, ὅτι καὶ ἐκάστη τῶν ὁμοταγῶν πυραμίδων πρὸς ἐκάστην ὁμοταγῇ πυραμίδα τριπλασίονα λόγον ἔχει ἥπερ ἡ BK ὁμόλογος πλευρὰ πρὸς τὴν ZM ὁμόλογον πλευράν, τουτέστιν ἥπερ ἡ BD πρὸς τὴν $Z\Theta$. καὶ ὡς ἐν τῶν ἡγουμένων πρὸς ἐν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα· ἐστὶν ἄρα καὶ ὡς ἡ $BKTA$ πυραμῖς πρὸς τὴν $ZMON$ πυραμίδα, οὕτως ἡ ὅλη πυραμῖς, ἥς βάσεις τὸ $ATBYT\Phi\Delta\Theta$ πολύγωνον, κορυφὴ δὲ τὸ Λ σημείον, πρὸς τὴν ὅλην πυραμίδα, ἥς βάσεις μὲν τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημείον· ὥστε καὶ πυραμῖς, ἥς βάσεις μὲν τὸ $ATBYT\Phi\Delta\Theta$, κορυφὴ δὲ τὸ Λ , πρὸς τὴν πυραμίδα, ἥς βάσεις [μὲν] τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N σημείον, τριπλασίονα λόγον ἔθει ἥπερ ἡ BD πρὸς τὴν $Z\Theta$. ὑπόκειται δὲ καὶ ὁ κῶνος, ὃς βάσεις [μὲν] ὁ $AB\Gamma\Delta$ κύκλος, κορυφὴ δὲ τὸ Λ σημείον, πρὸς τὸ X στερεὸν τριπλασίονα λόγον ἔθων ἥπερ ἡ BD πρὸς τὴν $Z\Theta$ · ἐστὶν ἄρα ὡς ὁ κῶνος, ὃς βάσεις μὲν ἐστὶν ὁ $AB\Gamma\Delta$ κύκλος, κορυφὴ δὲ τὸ Λ , πρὸς τὸ X στερεόν, οὕτως ἡ πυραμῖς, ἥς βάσεις μὲν τὸ $ATBYT\Phi\Delta\Theta$ [πολύγωνον], κορυφὴ δὲ τὸ Λ , πρὸς τὴν πυραμίδα, ἥς βάσεις μὲν ἐστὶ τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N · ἐναλλάξ ἄρα, ὡς ὁ κῶνος, ὃς βάσεις μὲν ὁ $AB\Gamma\Delta$ κύκλος, κορυφὴ δὲ τὸ Λ , πρὸς τὴν ἐν αὐτῷ πυραμίδα, ἥς βάσεις μὲν τὸ $ATBYT\Phi\Delta\Theta$ πολύγωνον, κορυφὴ δὲ τὸ Λ , οὕτως τὸ X [στερεόν] πρὸς τὴν πυραμίδα, ἥς βάσεις μὲν ἐστὶ τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N . μείζων δὲ ὁ εἰρημένος κῶνος τῆς ἐν αὐτῷ πυραμίδος· ἐμπεριέθει γὰρ αὐτήν. μείζων ἄρα καὶ τὸ X στερεόν τῆς πυραμίδος, ἥς βάσεις μὲν ἐστὶ τὸ $EOZ\Pi\text{HP}\Theta\Sigma$ πολύγωνον, κορυφὴ δὲ τὸ N . ἀλλὰ καὶ ἔλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ κῶνος, ὃς βάσεις ὁ $AB\Gamma\Delta$ κύκλος, κορυφὴ δὲ τὸ Λ [σημεῖον], πρὸς ἔλαττον τι τοῦ κῶνον στερεόν, ὃς βάσεις μὲν ὁ $EZH\Theta$

account of the similarity of triangles LTK and NPM , as LT (is) to TK , so NP (is) to PM , and, on account of the similarity of triangles TKB and PMF , as KT (is) to TB , so MP (is) to PF , thus, via equality, as LT (is) to TB , so NP (is) to PF [Prop. 5.22]. And it was shown that as TB (is) to BL , so PF (is) to FN . Thus, via equality, as TL (is) to LB , so PN (is) to NF [Prop. 5.22]. Thus, the sides of triangles LTB and NPF are proportional. Thus, triangles LTB and NPF are equiangular [Prop. 6.5]. And, hence, (they are) similar [Def. 6.1]. And, thus, the pyramid whose base is triangle BKT , and apex the point L , is similar to the pyramid whose base is triangle FMP , and apex the point N . For they are contained by equal numbers of similar planes [Def. 11.9]. And similar pyramids which also have triangular bases are in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, pyramid $BKTL$ has to pyramid $FMPN$ the cubed ratio that BK (has) to FM . So, similarly, joining straight-lines from (points) A , W , D , V , C , and U to (center) K , and from (points) E , S , H , R , G , and Q to (center) M , and setting up pyramids having the same apexes as the cones on each of the triangles (so formed), we can also show that each of the pyramids (on base $ABCD$ taken) in order will have to each of the pyramids (on base $EFGH$ taken) in order the cubed ratio that the corresponding side BK (has) to the corresponding side FM —that is to say, that BD (has) to FH . And (for two sets of proportional magnitudes) as one of the leading (magnitudes is) to one of the following, so (the sum of) all of the leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. And, thus, as pyramid $BKTL$ (is) to pyramid $FMPN$, so the whole pyramid whose base is polygon $ATBUCVDW$, and apex the point L , (is) to the whole pyramid whose base is polygon $EPFQGRHS$, and apex the point N . And, hence, the pyramid whose base is polygon $ATBUCVDW$, and apex the point L , has to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N , the cubed ratio that BD (has) to FH . And it was also assumed that the cone whose base is circle $ABCD$, and apex the point L , has to solid O the cubed ratio that BD (has) to FH . Thus, as the cone whose base is circle $ABCD$, and apex the point L , is to solid O , so the pyramid whose base (is) [polygon] $ATBUCVDW$, and apex the point L , (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N . Thus, alternately, as the cone whose base (is) circle $ABCD$, and apex the point L , (is) to the pyramid within it whose base (is) the polygon $ATBUCVDW$, and apex the point L , so the [solid] O (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N [Prop. 5.16]. And the aforementioned cone (is) greater than the pyramid within it. For it encompasses it. Thus, solid O (is) also greater than the pyramid whose base is polygon $EPFQGRHS$, and apex the point N . But, (it is) also less. The very thing is impossible. Thus, the cone whose base (is) circle $ABCD$, and apex the [point] L , does not have

κύκλος, κορυφή δὲ τὸ N σημεῖον, τριπλασίονα λόγον ἔθει ἥπερ ἡ BD πρὸς τὴν $Z\Theta$. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὁ $EZH\Theta$ κῶνος πρὸς ἑλαττόν τι τοῦ $AB\Gamma\Delta$ κώνου στερεὸν τριπλασίονα λόγον ἔθει ἥπερ ἡ $Z\Theta$ πρὸς τὴν BD .

Λέγω δὴ, ὅτι οὐδὲ ὁ $AB\Gamma\Delta$ κῶνος πρὸς μείζον τι τοῦ $EZH\Theta$ κώνου στερεὸν τριπλασίονα λόγον ἔθει ἥπερ ἡ BD πρὸς τὴν $Z\Theta$.

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μείζον τὸ X . ἀνάπαλιν ἄρα τὸ X στερεὸν πρὸς τὸν $AB\Gamma\Delta$ κώνον τριπλασίονα λόγον ἔθει ἥπερ ἡ $Z\Theta$ πρὸς τὴν BD . ὥς δὲ τὸ X στερεὸν πρὸς τὸν $AB\Gamma\Delta$ κώνον, οὕτως ὁ $EZH\Theta$ κῶνος πρὸς ἑλαττόν τι τοῦ $AB\Gamma\Delta$ κώνου στερεόν. καὶ ὁ $EZH\Theta$ ἄρα κῶνος πρὸς ἑλαττόν τι τοῦ $AB\Gamma\Delta$ κώνου στερεὸν τριπλασίονα λόγον ἔθει ἥπερ ἡ $Z\Theta$ πρὸς τὴν BD . ὅπερ ἀδύνατον ἐδείκθη. οὐκ ἄρα ὁ $AB\Gamma\Delta$ κῶνος πρὸς μείζον τι τοῦ $EZH\Theta$ κώνου στερεὸν τριπλασίονα λόγον ἔθει ἥπερ ἡ BD πρὸς τὴν $Z\Theta$. ἐδείκθη δέ, ὅτι οὐδὲ πρὸς ἑλαττόν. ὁ $AB\Gamma\Delta$ ἄρα κῶνος πρὸς τὸν $EZH\Theta$ κώνον τριπλασίονα λόγον ἔθει ἥπερ ἡ BD πρὸς τὴν $Z\Theta$.

Ὡς δὲ ὁ κῶνος πρὸς τὸν κώνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον· τριπλασίος γὰρ ὁ κύλινδρος τοῦ κώνου ὁ ἐπὶ τῇ αὐτῇ βάσει καὶ ἰσοῦψής αὐτῷ. καὶ ὁ κύλινδρος ἄρα πρὸς τὸν κύλινδρον τριπλασίονα λόγον ἔθει ἥπερ ἡ BD πρὸς τὴν $Z\Theta$.

Οἱ ἄρα ὅμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριπλασίονι λόγῳ εἰσι τῶν ἐν ταῖς βάσεσι διαμέτρων· ὅπερ ἔδει δεῖξαι.

to some solid less than the cone whose base (is) circle $EFGH$, and apex the point N , the cubed ratio that BD (has) to EH . So, similarly, we can show that neither does cone $EFGHN$ have to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD .

So, I say that neither does cone $ABCDL$ have to some solid greater than cone $EFGHN$ the cubed ratio that BD (has) to FH .

For, if possible, let it have (such a ratio) to a greater (solid), O . Thus, inversely, solid O has to cone $ABCDL$ the cubed ratio that FH (has) to BD [Prop. 5.7 corr.]. And as solid O (is) to cone $ABCDL$, so cone $EFGHN$ (is) to some solid less than cone $ABCDL$ [12.2 lem.]. Thus, cone $EFGHN$ also has to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD . The very thing was shown (to be) impossible. Thus, cone $ABCDL$ does not have to some solid greater than cone $EFGHN$ the cubed ratio than BD (has) to FH . And it was shown that neither (does it have such a ratio) to a lesser (solid). Thus, cone $ABCDL$ has to cone $EFGHN$ the cubed ratio that BD (has) to FG .

And as the cone (is) to the cone, so the cylinder (is) to the cylinder. For a cylinder is three times a cone on the same base as the cone, and of the same height as it [Prop. 12.10]. Thus, the cylinder also has to the cylinder the cubed ratio that BD (has) to FH .

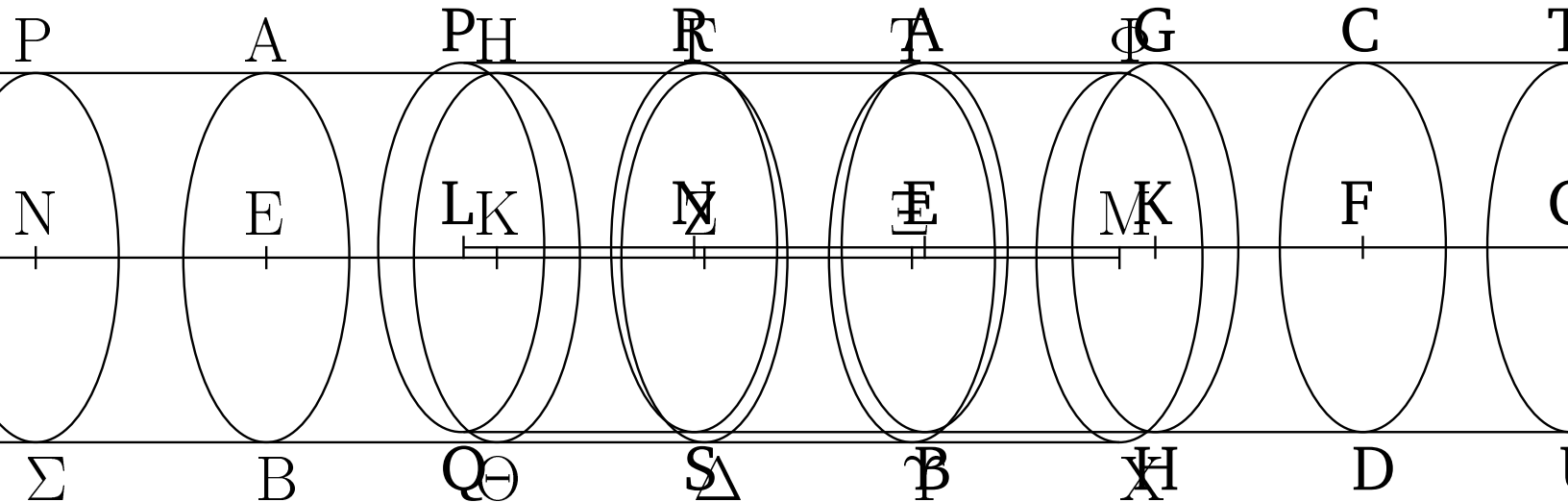
Thus, similar cones and cylinders are in the cubed ratio of the diameters of their bases. (Which is) the very thing it was required to show.

13.

Proposition 13

Ἐάν κύλινδρος ἐπιπέδῳ τμηθῇ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔσται ὥς ὁ κύλινδρος πρὸς τὸν κύλινδρον, οὕτως ὁ ἄκων πρὸς τὸν ἄκων.

If a cylinder is cut by a plane which is parallel to the opposite planes (of the cylinder) then as the cylinder (is) to the cylinder, so the axis will be to the axis.



Κύλινδρος γὰρ ὁ AD ἐπιπέδῳ τῷ $H\Theta$ τετμήσθω παραλ-

For let the cylinder AD be cut by the plane GH which is

λήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς AB , $\Gamma\Delta$, καὶ συμβαλλέτω τῷ ἄχονι τὸ $H\Theta$ ἐπίπεδον κατὰ τὸ K σημείον λέγω, ὅτι ἐστὶν ὡς ὁ BH κύλινδρος πρὸς τὸν $H\Delta$ κύλινδρον, οὕτως ὁ EK ἄχων πρὸς τὸν KZ ἄχονα.

Ἐκβεβλήσθω γὰρ ὁ EZ ἄχων ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ A , M σημεία, καὶ ἐκκείσθωσαν τῷ EK ἄχονι ἴσοι ὁσοιδηποτοῦν οἱ EN , NA , τῷ δὲ ZK ἴσοι ὁσοιδηποτοῦν οἱ ZX , XM , καὶ νοείσθω ὁ ἐπὶ τοῦ AM ἄχονος κύλινδρος ὁ $O\Theta$, οὗ βάσεις οἱ $O\Pi$, $\Phi\Theta$ κύκλοι. καὶ ἐκβεβλήσθω διὰ τῶν N , X σημείων ἐπίπεδα παράλληλα τοῖς AB , $\Gamma\Delta$ καὶ ταῖς βάσεσι τοῦ $O\Theta$ κύλινδρον καὶ ποιέτωσαν τοὺς $P\Sigma$, $T\Upsilon$ κύκλους περὶ τὰ N , X κέντρα. καὶ ἐπεὶ οἱ AN , NE , EK ἄχονες ἴσοι εἰσὶν ἀλλήλοις, οἱ ἄρα ΠP , PB , BH κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις. ἴσοι δὲ εἰσὶν αἱ βάσεις· ἴσοι ἄρα καὶ οἱ ΠP , PB , BH κύλινδροι ἀλλήλοις. ἐπεὶ οὖν οἱ AN , NE , EK ἄχονες ἴσοι εἰσὶν ἀλλήλοις, εἰσὶ δὲ καὶ οἱ ΠP , PB , BH κύλινδροι ἴσοι ἀλλήλοις, καὶ ἐστὶν ἴσον τὸ πλῆθος τῷ πλῆθει, ὅσαπλασίων ἄρα ὁ $ΚΑ$ ἄχων τοῦ EK ἄχονος, τοσανταπλασίων ἔσται καὶ ὁ ΠH κύλινδρος τοῦ $H\Delta$ κύλινδρον. διὰ τὰ αὐτὰ δὴ καὶ ὅσαπλασίων ἐστὶν ὁ MK ἄχων τοῦ KZ ἄχονος, τοσανταπλασίων ἐστὶ καὶ ὁ ΘH κύλινδρος τοῦ $H\Delta$ κύλινδρον. καὶ εἰ μὲν ἴσος ἐστὶν ὁ $ΚΑ$ ἄχων τῷ KM ἄχονι, ἴσος ἔσται καὶ ὁ ΠH κύλινδρος τῷ $H\Theta$ κύλινδρῳ, εἰ δὲ μείζων ὁ ἄχων τοῦ ἄχονος, μείζων καὶ ὁ κύλινδρος τοῦ κύλινδρον, καὶ εἰ ἐλάσσων, ἐλάσσων. τεσσάρων δὴ μεγεθῶν ὄντων, ἁχόνων μὲν τῶν EK , KZ , κύλινδρων δὲ τῶν BH , $H\Delta$, εἴληπται ἰσάκις πολλαπλάσια, τοῦ μὲν EK ἄχονος καὶ τοῦ BH κύλινδρον ὅ τε AK ἄχων καὶ ὁ ΠH κύλινδρος, τοῦ δὲ KZ ἄχονος καὶ τοῦ $H\Delta$ κύλινδρον ὅ τε KM ἄχων καὶ ὁ $H\Theta$ κύλινδρος, καὶ δέδεικται, ὅτι εἰ ὑπερέθει ὁ $ΚΑ$ ἄχων τοῦ KM ἄχονος, ὑπερέθει καὶ ὁ ΠH κύλινδρος τοῦ $H\Theta$ κύλινδρον, καὶ εἰ ἴσος, ἴσος, καὶ εἰ ἐλάσσων, ἐλάσσων. ἔστιν ἄρα ὡς ὁ EK ἄχων πρὸς τὸν KZ ἄχονα, οὕτως ὁ BH κύλινδρος πρὸς τὸν $H\Delta$ κύλινδρον· ὅπερ ἔδει δεῖχαι.

14.

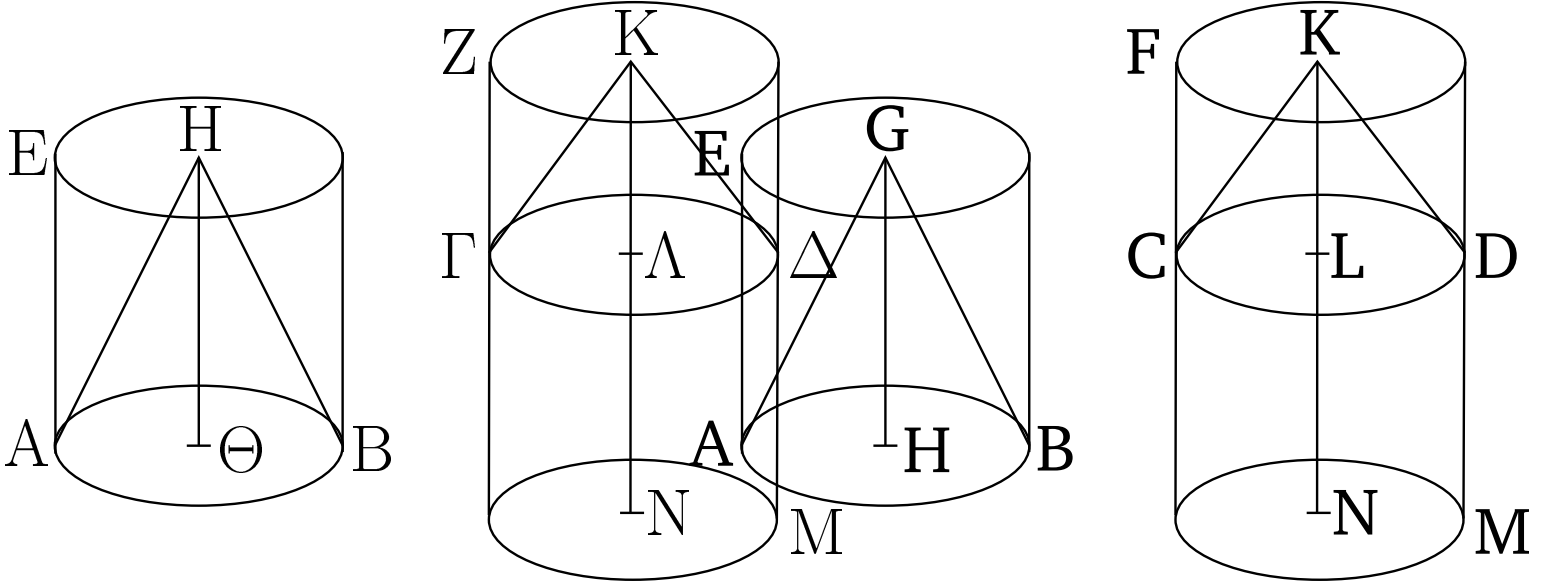
Οἱ ἐπὶ ἴσων βάσεων ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς τὰ ὕψη.

parallel to the opposite planes (of the cylinder), AB and CD . And let the plane GH have met the axis at point K . I say that as cylinder BG is to cylinder GD , so axis EK (is) to axis KF .

For let axis EF be produced in each direction to points L and M . And let any number whatsoever (of lengths), EN and NL , equal to axis EK , be set out (on the axis EL), and any number whatsoever (of lengths), FO and OM , equal to (axis) FK , (on the axis KM). And let the cylinder PW , whose bases (are) the circles PQ and VW , be conceived on axis LM . And let planes parallel to AB , CD , and the bases of cylinder PW , be produced through points N and O , and let them have made the circles RS and TU around the centers N and O (respectively). And since axes LN , NE , and EK are equal to one another, the cylinders QR , RB , and BG are to one another as their bases [Prop. 12.11]. But the bases are equal. Thus, the cylinders QR , RB , and BG (are) also equal to one another. Therefore, since the axes LN , NE , and EK are equal to one another, and the cylinders QR , RB , and BG are also equal to one another, and the number (of the former) is equal to the number (of the latter), thus as many multiples as axis KL is of axis EK , so many multiples is cylinder QG also of cylinder GB . And so, for the same (reasons), as many multiples as axis MK is of axis KF , so many multiples is cylinder WG also of cylinder GD . And if axis KL is equal to axis KM then cylinder QG will also be equal to cylinder GW , and if the axis (is) greater than the axis then the cylinder (will also be) greater than the cylinder, and if (the axis is) less then (the cylinder will also be) less. So, there are four magnitudes—the axes EK and KF , and the cylinders BG and GD —and equal multiples be taken of axis EK and cylinder BG —(namely), axis LK and cylinder QG —and of axis KF and cylinder GD —(namely), axis KM and cylinder GW . And it has been shown that if axis KL exceeds axis KM then cylinder QG also exceeds cylinder GW , and if (the axes are) equal then (the cylinders are) equal, and if (KL is) less then (QG is) less. Thus, as axis EK is to axis KF , so cylinder BG (is) to cylinder GD [Def. 5.5]. (Which is) the very thing it was required to show.

Proposition 14

Cones and cylinders which are on equal bases are to one another as their heights.



Ἐστωσαν γὰρ ἐπὶ ἴσων βάσεων τῶν AB , $\Gamma\Delta$ κύκλων κύλινδροι οἱ EB , $Z\Delta$. λέγω, ὅτι ἐστὶν ὡς ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, οὕτως ὁ $H\Theta$ ἄξων πρὸς τὸν ΚΛ ἄξωνα.

Ἐκβεβλήσθω γὰρ ὁ ΚΛ ἄξων ἐπὶ τὸ N σημεῖον, καὶ κείσθω τῷ $H\Theta$ ἄξωνι ἴσος ὁ ΛN , καὶ περὶ ἄξωνα τὸν ΛN κύλινδρος γενοήσθω ὁ ΓM . ἐπεὶ οὖν οἱ EB , ΓM κύλινδροι ὑπὸ τὸ αὐτὸ ὕψος εἰσὶν, πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις. ἴσαι δὲ εἰσὶν αἱ βάσεις ἀλλήλαις· ἴσοι ἄρα εἰσὶ καὶ οἱ EB , ΓM κύλινδροι. καὶ ἐπεὶ κύλινδρος ὁ ZM ἐπιπέδῳ τέτμηται τῷ $\Gamma\Delta$ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἐστὶν ἄρα ὡς ὁ ΓM κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, οὕτως ὁ ΛN ἄξων πρὸς τὸν ΚΛ ἄξωνα. ἴσος δὲ ἐστὶν ὁ μὲν ΓM κύλινδρος τῷ EB κύλινδρῳ, ὁ δὲ ΛN ἄξων τῷ $H\Theta$ ἄξωνι· ἐστὶν ἄρα ὡς ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον, οὕτως ὁ ABH κῶνος πρὸς τὸν $\Gamma\Delta K$ κῶνον. καὶ ὡς ἄρα ὁ $H\Theta$ ἄξων πρὸς τὸν ΚΛ ἄξωνα, οὕτως ὁ ABH κῶνος πρὸς τὸν $\Gamma\Delta K$ κῶνον καὶ ὁ EB κύλινδρος πρὸς τὸν $Z\Delta$ κύλινδρον· ὅπερ ἔδει δεῖξαι.

15.

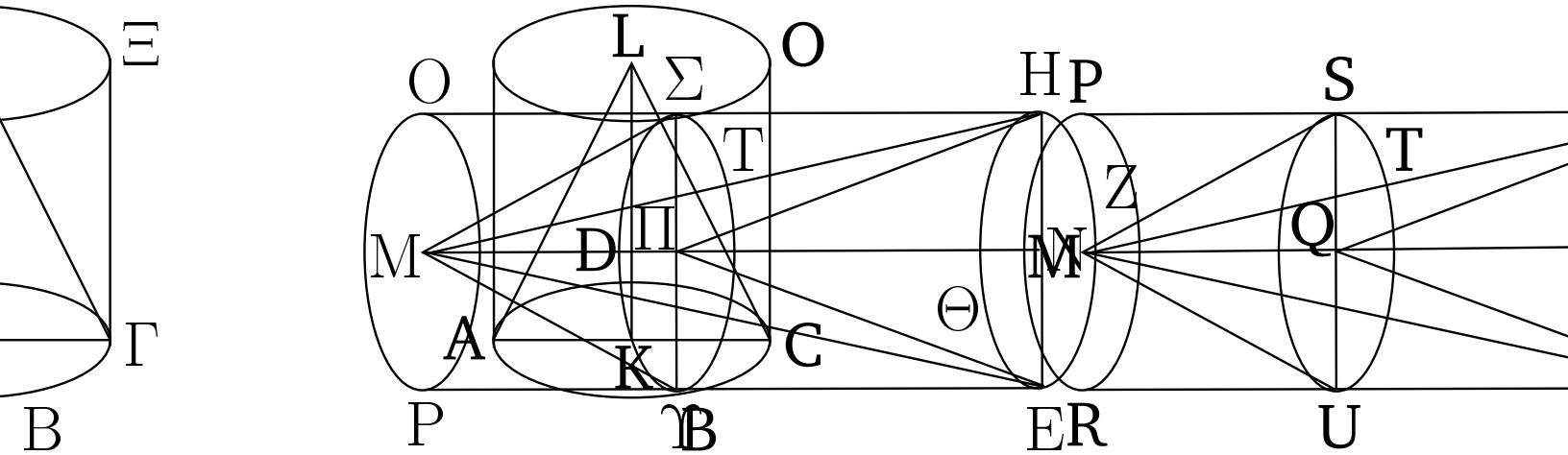
Τῶν ἴσων κόνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν καὶ τῶν κόνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψουσιν, ἴσοι εἰσὶν ἐκείνοι.

For let EB and FD be cylinders on equal bases, (namely) the circles AB and CD (respectively). I say that as cylinder EB is to cylinder FD , so axis GH (is) to axis KL .

For let the axis KL be produced to point N . And let LN be made equal to axis GH . And let the cylinder CM be conceived about axis LN . Therefore, since cylinders EB and CM have the same height they are to one another as their bases [Prop. 12.11]. And the bases are equal to one another. Thus, cylinders EB and CM are also equal to one another. And since cylinder FM has been cut by the plane CD , which is parallel to its opposite planes, thus as cylinder CM is to cylinder FD , so axis LN (is) to axis KL [Prop. 12.13]. And cylinder CM is equal to cylinder EB , and axis LN to axis GH . Thus, as cylinder EB is to cylinder FD , so axis GH (is) to axis KL . And as cylinder EB (is) to cylinder FD , so cone ABG (is) to cone CDK [Prop. 12.10]. Thus, also, as axis GH (is) to axis KL , so cone ABG (is) to cone CDK , and cylinder EB to cylinder FD . (Which is) the very thing it was required to show.

Proposition 15

The bases of equal cones and cylinders are reciprocally proportional to their heights. And, those cones and cylinders whose bases (are) reciprocally proportional to their heights are equal.



Ἐστωσαν ἴσοι κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν οἱ $AB\Gamma\Delta$, $EZH\Theta$ κύκλοι, διάμετροι δὲ αὐτῶν αἱ AF , EH , ἄχονες δὲ οἱ KA , MN , οὔτινες καὶ ὕψη εἰσὶ τῶν κῶνων ἢ κυλίνδρων, καὶ συμπεπληρώσθωσαν οἱ AX , EO κύλινδροι. λέγω, ὅτι τῶν AX , EO κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστιν ὡς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ KA ὕψος.

Τὸ γὰρ AK ὕψος τῷ MN ὕψει ἴσος ἔστιν ἢ οὐ. ἔστω πρότερον ἴσον. ἔστι δὲ καὶ ὁ AX κύλινδρος τῷ EO κυλίνδρῳ ἴσος. οἱ δὲ ὑπὸ τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις· ἴση ἄρα καὶ ἡ $AB\Gamma\Delta$ βάσις τῇ $EZH\Theta$ βάσει. ὥστε καὶ ἀντιπέπονθεν, ὡς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ KA ὕψος. ἀλλὰ δὴ μὴ ἔστω τὸ AK ὕψος τῷ MN ἴσον, ἀλλ' ἔστω μείζον τὸ MN , καὶ ἀφηρήσθω ἀπὸ τοῦ MN ὕψους τῷ KA ἴσον τὸ ΠN , καὶ διὰ τοῦ Π σημείου τετμήσθω ὁ EO κύλινδρος ἐπιπέδῳ τῷ $TY\S$ παραλλήλῳ τοῖς τῶν $EZH\Theta$, PO κύκλων ἐπιπέδοις, καὶ ἀπὸ βάσεως μὲν τοῦ $EZH\Theta$ κύκλου, ὕψους δὲ τοῦ $N\Pi$ κυλίνδρου γενησθῶ ὁ $E\Sigma$. καὶ ἐπεὶ ἴσος ἔστιν ὁ AX κύλινδρος τῷ EO κυλίνδρῳ, ἔστιν ἄρα ὡς ὁ AX κύλινδρος πρὸς τὸν $E\Sigma$ κύλινδρον, οὕτως ὁ EO κύλινδρος πρὸς τὸν $E\Sigma$ κύλινδρον. ἀλλ' ὡς μὲν ὁ AX κύλινδρος πρὸς τὸν $E\Sigma$ κύλινδρον, οὕτως ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ · ὑπὸ γὰρ τὸ αὐτὸ ὕψος εἰσὶν οἱ AX , $E\Sigma$ κύλινδροι· ὡς δὲ ὁ EO κύλινδρος πρὸς τὸν $E\Sigma$, οὕτως τὸ MN ὕψος πρὸς τὸ ΠN ὕψος· ὁ γὰρ EO κύλινδρος ἐπιπέδῳ τέτμηται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις. ἔστιν ἄρα καὶ ὡς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ ΠN ὕψος. ἴσον δὲ τὸ ΠN ὕψος τῷ KA ὕψει· ἔστιν ἄρα ὡς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ KA ὕψος. τῶν ἄρα AX , EO κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Ἀλλὰ δὴ τῶν AX , EO κυλίνδρων ἀντιπεπονητέωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ἔστω ὡς ἡ $AB\Gamma\Delta$ βάσις πρὸς τὴν $EZH\Theta$ βάσιν, οὕτως τὸ MN ὕψος πρὸς τὸ KA ὕψος· λέγω, ὅτι ἴσος ἔστιν ὁ AX κύλινδρος τῷ EO κυλίνδρῳ.

Let there be equal cones and cylinders whose bases are the circles $ABCD$ and $EFGH$, and the diameters of (the bases) AC and EG , and (whose) axes (are) KL and MN , which are also the heights of the cones and cylinders (respectively). And let the cylinders AO and EP be completed. I say that the bases of cylinders AO and EP are reciprocally proportional to their heights, and (so) as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL .

For height LK is either equal to height MN , or not. Let it, first of all, be equal. And cylinder AO is also equal to cylinder EP . And cones and cylinders having the same height are to one another as their bases [Prop. 12.11]. Thus, base $ABCD$ (is) also equal to base $EFGH$. And, hence, reciprocally, as base $ABCD$ (is) to base $EFGH$, so height MN (is) to height KL . And so, let height LK not be equal to MN , but let MN be greater. And let QN , equal to KL , be cut off from height MN . And let the cylinder EP be cut, through point Q , by the plane TUS (which is) parallel to the planes of the circles $EFGH$ and RP . And let cylinder ES be conceived, with base the circle $EFGH$, and height NQ . And since cylinder AO is equal to cylinder EP , thus, as cylinder AO (is) to cylinder ES , so cylinder EP (is) to cylinder ES [Prop. 5.7]. But, as cylinder AO (is) to cylinder ES , so base $ABCD$ (is) to base $EFGH$. For cylinders AO and ES (have) the same height [Prop. 12.11]. And as cylinder EP (is) to (cylinder) ES , so height MN (is) to height QN . For cylinder EP has been cut by a plane which is parallel to its opposite planes [Prop. 12.13]. And, thus, as base $ABCD$ is to base $EFGH$, so height MN (is) to height QN [Prop. 5.11]. And height QN (is) equal to height KL . Thus, as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL . Thus, the bases of cylinders AO and EP are reciprocally proportional to their heights.

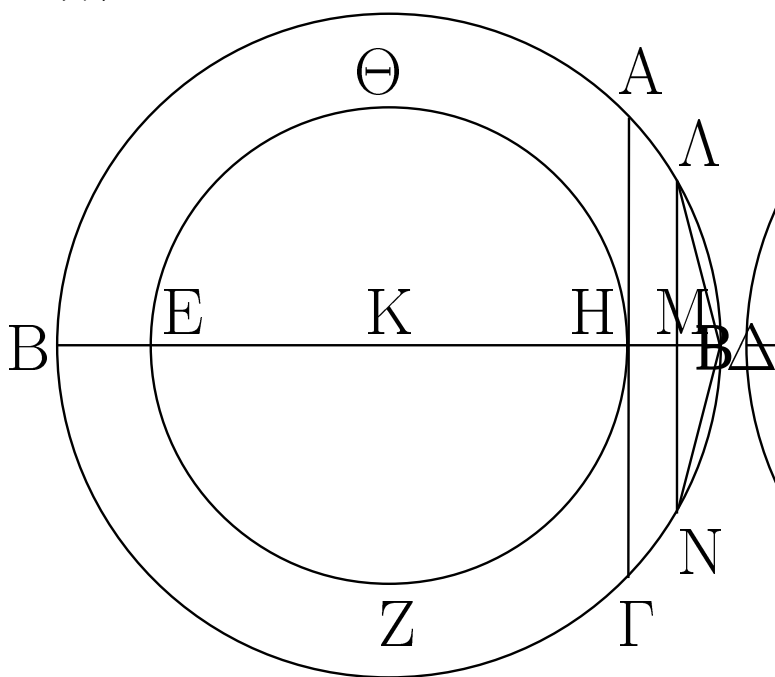
And, so, let the bases of cylinders AO and EP be reciprocally proportional to their heights, and (thus) let base $ABCD$ be to base $EFGH$, as height MN (is) to height KL . I say that cylinder AO is equal to cylinder EP .

Τῶν γὰρ αὐτῶν κατασκευασθέντων ἐπεὶ ἐστὶν ὡς ἡ ΑΒΓΔ
βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ
ΚΑ ὕψος, ἴσον δὲ τὸ ΚΑ ὕψος τῷ ΠΝ ὕψει, ἔσται ἄρα ὡς
ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὕτως τὸ ΜΝ ὕψος
πρὸς τὸ ΠΝ ὕψος. ἀλλ' ὡς μὲν ἡ ΑΒΓΔ βάσις πρὸς τὴν
ΕΖΗΘ βάσιν, οὕτως ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον·
ὑπὸ γὰρ τὸ αὐτὸ ὕψος εἰσὶν ὡς δὲ τὸ ΜΝ ὕψος πρὸς τὸ ΠΝ
[ὕψος], οὕτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον· ἔστιν
ἄρα ὡς ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον, οὕτως ὁ
ΕΟ κύλινδρος πρὸς τὸν ΕΣ. ἴσος ἄρα ὁ ΑΧ κύλινδρος τῷ
ΕΟ κυλίνδρῳ. ὠσαύτως δὲ καὶ ἐπὶ τῶν κώνων· ὅπερ ἔδει
δείξαι.

For, with the same construction, since as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL , and height KL (is) equal to height QN , thus, as base $ABCD$ (is) to base $EFGH$, so height MN will be to height QN . But, as base $ABCD$ (is) to base $EFGH$, so cylinder AO (is) to cylinder ES . For they are the same height [Prop. 12.11]. And as height MN (is) to [height] QN , so cylinder EP (is) to cylinder ES [Prop. 12.13]. Thus, as cylinder AO is to cylinder ES , so cylinder EP (is) to (cylinder) ES [Prop. 5.11]. Thus, cylinder AO (is) equal to cylinder EP [Prop. 5.9]. In the same manner, (the proposition can) also (be demonstrated) for the cones. (Which is) the very thing it was required to show.

16.

Δύο κύκλων περι τὸ αὐτὸ κέντρον ὄντων εἷς τὸν μείζονα κύκλον πολύγωνον ἰσοπλευρόν τε καὶ ἀρτιόπλευρον ἐγγράφαι μὴ ψαῦον τοῦ ἐλάσσονος κύκλου.

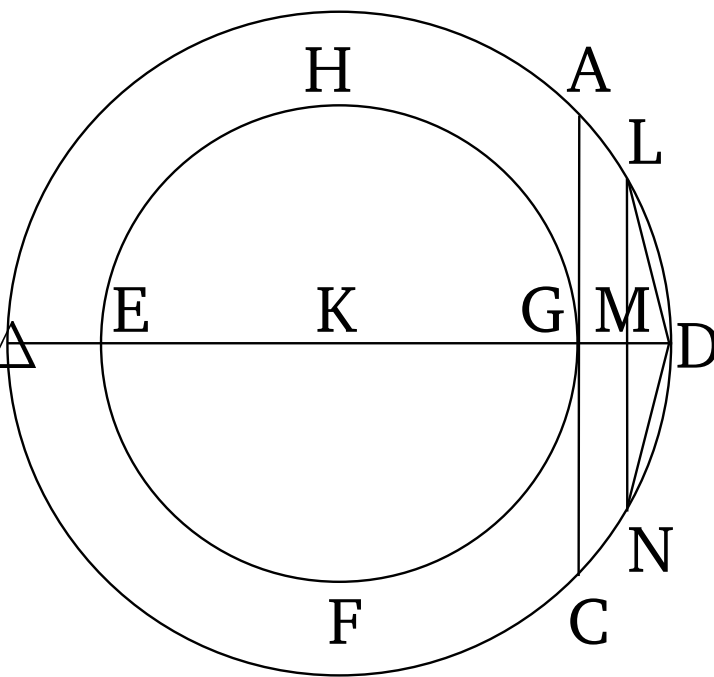


Ἐστωσαν οἱ δοθέντες δύο κύκλοι οἱ $ΑΒΓΔ$, $ΕΖΗΘ$ περὶ τὸ αὐτὸ κέντρον τὸ $Κ$. δεῖ δὴ εἰς τὸν μείζονα κύκλον τὸν $ΑΒΓΔ$ πολύγωνον ἰσοπλευρόν τε καὶ ἀρτιόπλευρον ἐγγράφαι μὴ ψαῶν τοῦ $ΕΖΗΘ$ κύκλου.

Ἦθηθω γὰρ διὰ τοῦ K κέντρον εὐθεία ἡ $BK\Delta$, καὶ ἀπὸ τοῦ H σημείου τῇ $B\Delta$ εὐθείᾳ πρὸς ὀρθὰς ἤθηθω ἡ HA καὶ διήθηθω ἐπὶ τὸ Γ ἡ AG ἄρα ἐφάπτεται τοῦ $EZH\Theta$ κύκλου. τέμνοντες δὴ τὴν $BA\Delta$ περιφέρειαν διθα καὶ τὴν ἡμίσειαν αὐτῆς διθα καὶ τοῦτο αἶ ποιῶντες καταλείπομεν περιφέρειαν ἐλάσσονα τῆς $A\Delta$. λελείφθω, καὶ ἔστω ἡ $\Lambda\Delta$, καὶ ἀπὸ τοῦ Λ ἐπὶ τὴν $B\Delta$ κάθετος ἤθηθω ἡ AM καὶ διήθηθω ἐπὶ τὸ N , καὶ ἐπεξεύθησαν αἱ $\Lambda\Delta$, ΔN . ἴση ἄρα ἐστὶν ἡ $\Lambda\Delta$ τῇ ΔN .

Proposition 16

There being two circles about the same center, to inscribe an equilateral and even-sided polygon in the greater circle, not touching the lesser circle.



Let $ABCD$ and $EFGH$ be the given two circles, about the same center, K . So, it is necessary to inscribe an equilateral and even-sided polygon in the greater circle $ABCD$, not touching circle $EFGH$.

Let the straight-line BKD be drawn through the center K . And let GA be drawn, at right-angles to the straight-line BD , through point G , and let it be drawn through to C . Thus, AC touches circle $EFGH$ [Prop. 3.16 corr.]. So, (by) cutting circumference BAD in half, and the half of it in half, and doing this continually, we will (eventually) leave a circumference less than AD [Prop. 10.1]. Let it be left, and let it be LD . And let LM be drawn, from L , perpendicular to BD , and let it be

καὶ ἔπει παρὰλληλὸς ἐστὶν ἡ ΛN τῇ $\text{A}\Gamma$, ἡ δὲ $\text{A}\Gamma$ ἐφάπτεται τοῦ $\text{EZH}\Theta$ κύκλου, ἡ ΛN ἄρα οὐκ ἐφάπτεται τοῦ $\text{EZH}\Theta$ κύκλου· πολλῶν ἄρα αἱ $\Lambda\Delta$, ΔN οὐκ ἐφάπτονται τοῦ $\text{EZH}\Theta$ κύκλου. ἔαν δὴ τῇ $\Lambda\Delta$ εὐθείᾳ ἴσας κατὰ τὸ συνθεῖς ἑναρμόσωμεν εἰς τὸν $\text{A}\text{B}\Gamma\Delta$ κύκλον, ἐγγραφήσεται εἰς τὸν $\text{A}\text{B}\Gamma\Delta$ κύκλον πολύγωνον ἰσοπλευρὸν τε καὶ ἀρτιόπλευρον μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ $\text{EZH}\Theta$ · ὅπερ ἔδει ποιῆσαι.

drawn through to N . And let LD and DN be joined. Thus, LD is equal to DN [Props. 3.3, 1.4]. And since LN is parallel to AC [Prop. 1.28], and AC touches circle $EFGH$, LN thus does not touch circle $EFGH$. Thus, even more so, LD and DN do not touch circle $EFGH$. And if we continuously insert (straight-lines) equal to straight-line LD into circle $ABCD$ [Prop. 4.1] then an equilateral and even-sided polygon, not touching the lesser circle $EFGH$, will be inscribed in circle $ABCD$.[†] (Which is) the very thing it was required to do.

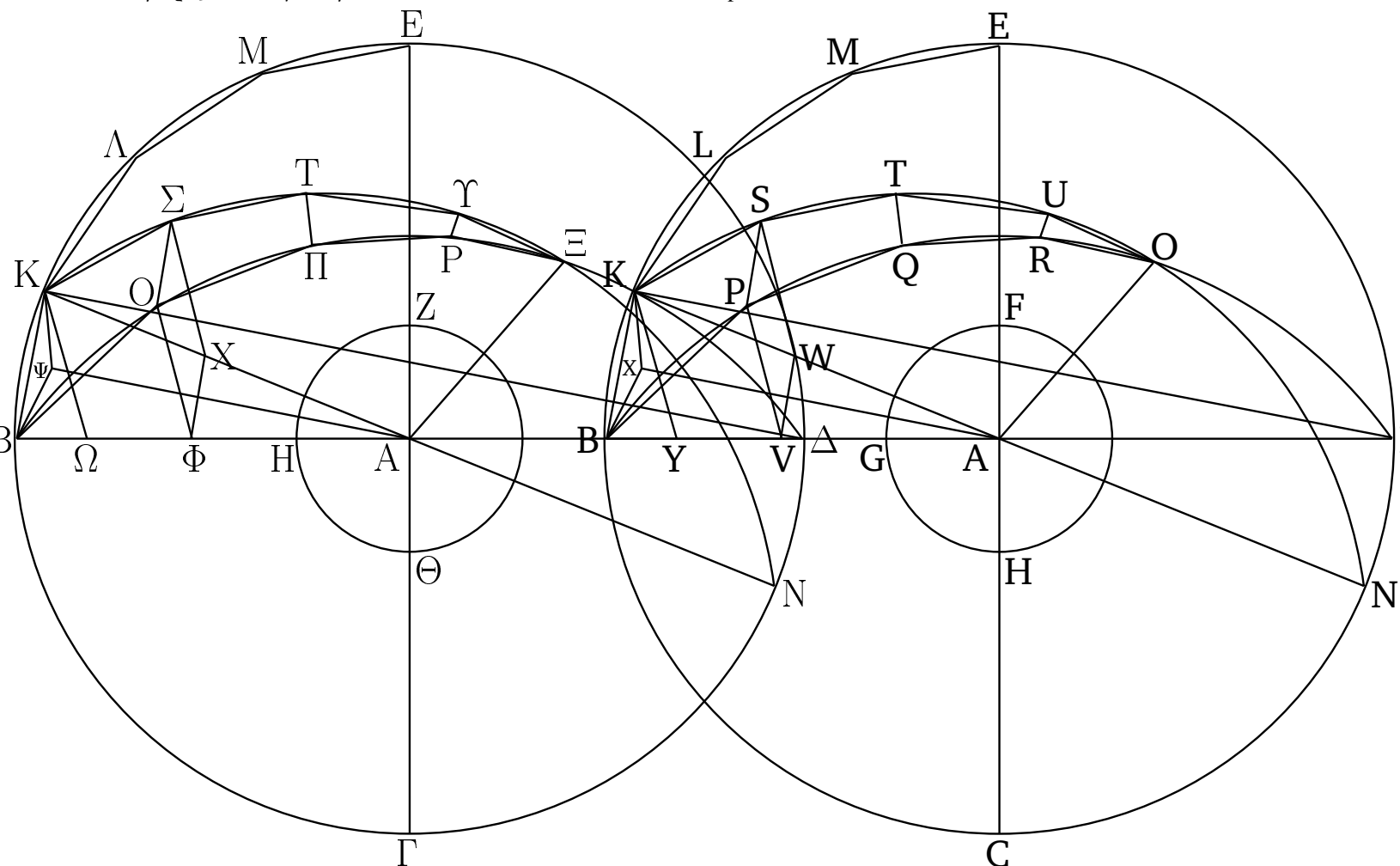
[†] Note that the chord of the polygon, LN , does not touch the inner circle either.

17.

Δύο σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἰς τὴν μείζονα σφαῖραν στερεὸν πολυέδρον ἐγγράψαι μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Proposition 17

There being two spheres about the same center, to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.



Νενοήσθωσαν δύο σφαῖραι περὶ τὸ αὐτὸ κέντρον τὸ Α·
 δεῖ δὴ εἶξ τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγράφαι
 μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Let two spheres be conceived about the same center, *A*. So, it is necessary to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.

Τετμήσθωσαν αἱ σφαῖραι ἐπιπέδῳ τινὶ διὰ τοῦ κέντρον· ἔσονται δὴ αἱ τομαὶ κύκλοι, ἐπειδὴ περ μενούσης τῆς διαμέτρου καὶ περιφερομένου τοῦ ἡμικυκλίου ἐγίγνετο ἡ σφαῖρα· ὥστε καὶ καθ' οἷαζ' ἂν θέσεως ἐπινοήσωμεν τὸ ἡμικύκλιον, τὸ δι' αὐτοῦ ἐκβαλλόμενον ἐπίπεδον ποιήσει ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας κύκλον. καὶ φανερόν, ὅτι καὶ μέγιστον, ἐπειδὴ περ ἡ διάμετρος τῆς σφαίρας, ἥτις ἐστὶ καὶ τοῦ ἡμικυκλίου διάμετρος δηλαδὴ καὶ τοῦ κύκλου, μείζων ἐστὶ πασῶν τῶν εἰς τὸν κύκλον ἢ τὴν σφαῖραν διαγομένων [εὐθειῶν]. ἔστω οὖν ἐν μὲν τῇ μείζονι σφαίρᾳ κύκλος ὁ ΒΓΔΕ, ἐν δὲ τῇ ἐλάσσονι σφαίρᾳ κύκλος ὁ ΖΗΘ, καὶ ἡθῶσαν αὐτῶν δύο διαμέτροι πρὸς ὀρθὰς ἀλλήλαις αἱ ΒΔ, ΓΕ, καὶ δύο κύκλων περὶ τὸ αὐτὸ κέντρον ὄντων τῶν ΒΓΔΕ, ΖΗΘ εἰς τὸν μείζονα κύκλον τὸν ΒΓΔΕ πολὺγωνον ἰσοπλευρον καὶ ἀρτιόπλευρον ἐγγεγράφθω μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ ΖΗΘ, οὗ πλευραὶ ἔστωσαν ἐν τῷ ΒΕ τεταρτημορίῳ αἱ ΒΚ, ΚΛ, ΛΜ, ΜΕ, καὶ ἐπιξευθθεῖσα ἡ ΚΑ διήθῃ ἐπὶ τὸ Ν, καὶ ἀνεστάτω ἀπὸ τοῦ Α σημείον τῷ τοῦ ΒΓΔΕ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἡ ΑΧ καὶ συμβαλλέτω τῇ ἐπιφανείᾳ τῆς σφαίρας κατὰ τὸ Χ, καὶ διὰ τῆς ΑΧ καὶ ἑκατέρᾳ τῶν ΒΔ, ΚΝ ἐπίπεδα ἐκβεβλήσθω· ποιήσουσι δὴ διὰ τὰ εἰρημένα ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας μεγίστους κύκλους. ποιείτωσαν, ὧν ἡμικύκλια ἔστω ἐπὶ τῶν ΒΔ, ΚΝ διαμέτρων τὰ ΒΧΔ, ΚΧΝ. καὶ ἐπεὶ ἡ ΧΑ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, καὶ πάντα ἄρα τὰ διὰ τῆς ΧΑ ἐπίπεδά ἐστιν ὀρθὰ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον· ὥστε καὶ τὰ ΒΧΔ, ΚΧΝ ἡμικύκλια ὀρθὰ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. καὶ ἐπεὶ ἴσα ἐστὶ τὰ ΒΕΔ, ΒΧΔ, ΚΧΝ ἡμικύκλια· ἐπὶ γὰρ ἴσων εἰσὶ διαμέτρων τῶν ΒΔ, ΚΝ· ἴσα ἐστὶ καὶ τὰ ΒΕ, ΒΧ, ΚΧ τεταρτημόρια ἀλλήλοισι. ὅσαι ἄρα εἰσὶν ἐν τῷ ΒΕ τεταρτημορίῳ πλευραὶ τοῦ πολυγώνου, τοσαῦταί εἰσι καὶ ἐν τοῖς ΒΧ, ΚΧ τεταρτημορίοις ἴσαι ταῖς ΒΚ, ΚΛ, ΛΜ, ΜΕ εὐθείαις. ἐγγεγράφθωσαν καὶ ἔστωσαν αἱ ΒΟ, ΟΠ, ΠΡ, ΡΧ, ΚΣ, ΣΤ, ΤΥ, ΥΧ, καὶ ἐπεξευθθωσαν αἱ ΣΟ, ΤΠ, ΥΡ, καὶ ἀπὸ τῶν Ο, Σ ἐπὶ τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον κάθετοι ἡθῶσαν· πεσοῦνται δὴ ἐπὶ τὰς κοινὰς τομὰς τῶν ἐπιπέδων τὰς ΒΔ, ΚΝ, ἐπειδὴ περ καὶ τὰ τῶν ΒΧΔ, ΚΧΝ ἐπίπεδα ὀρθὰ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. πιπτέτωσαν, καὶ ἔστωσαν αἱ ΟΦ, ΣΘ, καὶ ἐπεξευθθῶ ἡ ΘΦ. καὶ ἐπεὶ ἐν ἴσοις ἡμικυκλίοις τοῖς ΒΧΔ, ΚΧΝ ἴσαι ἀπειλημέναι εἰσὶν αἱ ΒΟ, ΚΣ, καὶ κάθετοι ἡγμέναι εἰσὶν αἱ ΟΦ, ΣΘ, ἴση [ἄρα] ἐστὶν ἡ μὲν ΟΦ τῇ ΣΘ, ἡ δὲ ΒΦ τῇ ΚΘ. ἐστὶ δὲ καὶ ὅλη ἡ ΒΑ ὅλη τῇ ΚΑ ἴση· καὶ λοιπὴ ἄρα ἡ ΦΑ λοιπὴ τῇ ΘΑ ἐστὶν ἴση· ἐστὶν ἄρα ὡς ἡ ΒΦ πρὸς τὴν ΦΑ, οὕτως ἡ ΚΘ πρὸς τὴν ΘΑ· παράλληλος ἄρα ἐστὶν ἡ ΘΦ τῇ ΚΒ. καὶ ἐπεὶ ἑκατέρα τῶν ΟΦ, ΣΘ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, παράλληλος ἄρα ἐστὶν ἡ ΟΦ τῇ ΣΘ. ἐδείχθη δὲ αὐτῇ καὶ ἴση καὶ αἱ ΘΦ, ΣΟ ἄρα ἴσαι εἰσὶ καὶ παράλληλοι. καὶ ἐπεὶ παράλληλός ἐστιν ἡ ΘΦ τῇ ΣΟ, ἀλλὰ ἡ ΘΦ τῇ ΚΒ ἐστὶ παράλληλος, καὶ ἡ ΣΟ ἄρα τῇ ΚΒ ἐστὶ παράλληλος. καὶ ἐπιξευγνόνουσιν αὐτὰς αἱ ΒΟ, ΚΣ· τὸ ΚΒΟΣ

Let the spheres be cut by some plane through the center. So, the sections will be circles, inasmuch as a sphere is generated by the diameter remaining behind, and a semi-circle being carried around [Def. 11.14]. And, hence, whatever position we conceive (of for) the semi-circle, the plane produced through it will make a circle on the surface of the sphere. And (it is) clear that (it is) also a great (circle), inasmuch as the diameter of the sphere, which is also manifestly the diameter of the semi-circle and the circle, is greater than all of the (other) [straight-lines] drawn across in the circle or the sphere [Prop. 3.15]. Therefore, let $BCDE$ be the circle in the greater sphere, and FGH the circle in the lesser sphere. And let two diameters of them be drawn at right-angles to one another, (namely), BD and CE . And there being two circles about the same center—(namely), $BCDE$ and FGH —let an equilateral and even-sided polygon be inscribed in the greater circle, $BCDE$, not touching the lesser circle, FGH [Prop. 12.16], of which let the sides in the quadrant BE be BK , KL , LM , and ME . And, KA being joined, let it be drawn across to N . And let AO be set up at point A , at right-angles to the plane of circle $BCDE$. And let it meet the surface of the (greater) sphere at O . And let planes be produced through AO and each of BD and KN . So, according to the aforementioned (discussion), they will make great circles on the surface of the (greater) sphere. Let them make (great circles), of which let BOD and KON be semi-circles on the diameters BD and KN (respectively). And since OA is at right-angles to the plane of circle $BCDE$, all of the planes through OA are thus also at right-angles to the plane of circle $BCDE$ [Prop. 11.18]. And, hence, the semi-circles BOD and KON are also at right-angles to the plane of circle $BCDE$. And since semi-circles BED , BOD , and KON are equal—for (they are) on the equal diameters BD and KN [Def. 3.1]—the quadrants BE , BO , and KO are also equal to one another. Thus, as many sides of the polygon as are in quadrant BE , so many are also in quadrants BO and KO equal to the straight-lines BK , KL , LM , and ME . Let them be inscribed, and let them be BP , PQ , QR , RO , KS , ST , TU , and UO . And let SP , TQ , and UR be joined. And let perpendiculars be drawn from P and S to the plane of circle $BCDE$ [Prop. 11.11]. So, they will fall on the common sections of the planes BD and KN (with $BCDE$), inasmuch as the planes of BOD and KON are also at right-angles to the plane of circle $BCDE$ [Def. 11.4]. Let them have fallen, and let them be PV and SW . And let VW be joined. And since BP and KS are equal (circumferences) having been cut off in the equal semi-circles BOD and KON [Def. 3.28], and PV and SW are perpendiculars having been drawn (from them), PV is [thus] equal to SW , and BV to KW [Props. 3.27, 1.26]. And the whole of BA is also equal to the whole of KA . And, thus, as BV is to VA , so KW (is) to WA . VW is thus parallel to KB [Prop. 6.2]. And since PV and SW are each at right-angles to the plane of circle $BCDE$, PV is thus parallel

ἄρα τετράπλευρον ἐν ἐνὶ ἐστὶν ἐπιπέδῳ, ἐπειδήπερ, ἔαν ὥσι δύο εὐθείαι παράλληλοι, καὶ ἐφ' ἑκατέρᾳ αὐτῶν ληφθῇ τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεία ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παράλληλοις. διὰ τὰ αὐτὰ δὴ καὶ ἐκάτερον τῶν ΣΟΠΤ, ΤΠΡΥ τετραπλεύρων ἐν ἐνὶ ἐστὶν ἐπιπέδῳ. ἔστι δὲ καὶ τὸ ΥΡΧ τρίγωνον ἐν ἐνὶ ἐπιπέδῳ. ἔαν δὴ νοήσωμεν ἀπὸ τῶν Ο, Σ, Π, Τ, Ρ, Υ σημείων ἐπὶ τὸ Α ἐπιζευγνυμένας εὐθείας, συσταθήσεται τι σῆμα στερεὸν πολυέδρον ματαχὺ τῶν ΒΧ, ΚΧ περιφερειῶν ἐκ πυραμίδων συγκείμενον, ὧν βάσεις μὲν τὰ ΚΒΟΣ, ΣΟΠΤ, ΤΠΡΥ τετράπλευρα καὶ τὸ ΥΡΧ τρίγωνον, κορυφὴ δὲ τὸ Α σημεῖον. ἔαν δὲ καὶ ἐπὶ ἐκάστης τῶν ΚΛ, ΛΜ, ΜΕ πλευρῶν καθάπερ ἐπὶ τῆς ΒΚ τὰ αὐτὰ κατασκευάσωμεν καὶ ἔτι τῶν λοιπῶν τριῶν τεταρτημορίων, συσταθήσεται τι σῆμα πολυέδρον ἐγγεγραμμένον εἰς τὴν σφαῖραν πυραμίσι περιεχόμενον, ὧν βάσεις [μὲν] τὰ εἰρημένα τετράπλευρα καὶ τὸ ΥΡΧ τρίγωνον καὶ τὰ ὁμοταγῇ αὐτοῖς, κορυφὴ δὲ τὸ Α σημεῖον.

Λέγω ὅτι τὸ εἰρημένον πολυέδρον οὐκ ἐφάπεται τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν, ἐφ' ἣς ἐστὶν ὁ ΖΗΘ κύκλος.

Ἦθθω ἀπὸ τοῦ Α σημεῖον ἐπὶ τὸ τοῦ ΚΒΟΣ τετραπλεύρου ἐπίπεδον κάθετος ἡ ΑΨ καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ Ψ σημεῖον, καὶ ἐπεξενύθωσαν αἱ ΨΒ, ΨΚ. καὶ ἐπεὶ ἡ ΑΨ ὀρθὴ ἐστὶ πρὸς τὸ τοῦ ΚΒΟΣ τετραπλεύρου ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰς ἀπομένους αὐτῆς εὐθείας καὶ οὕτως ἐν τῷ τοῦ τετραπλεύρου ἐπιπέδῳ ὀρθὴ ἐστὶν. ἡ ΑΨ ἄρα ὀρθὴ ἐστὶ πρὸς ἑκατέραν τῶν ΒΨ, ΨΚ. καὶ ἐπεὶ ἴση ἐστὶν ἡ ΑΒ τῇ ΑΚ, ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς ΑΒ τῷ ἀπὸ τῆς ΑΚ. καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς ΑΒ ἴσα τὰ ἀπὸ τῶν ΑΨ, ΨΒ. ὀρθὴ γάρ ἡ πρὸς τῷ Ψ. τῷ δὲ ἀπὸ τῆς ΑΚ ἴσα τὰ ἀπὸ τῶν ΑΨ, ΨΚ. τὰ ἄρα ἀπὸ τῶν ΑΨ, ΨΒ ἴσα ἐστὶ τοῖς ἀπὸ τῶν ΑΨ, ΨΚ. κοινὸν ἀφηρήσθω τὸ ἀπὸ τῆς ΑΨ. λοιπὸν ἄρα τὸ ἀπὸ τῆς ΒΨ λοιπὸν τῷ ἀπὸ τῆς ΨΚ ἴσον ἐστίν. ἴση ἄρα ἡ ΒΨ τῇ ΨΚ. ὁμοίως δὲ δείχμεν, ὅτι καὶ αἱ ἀπὸ τοῦ Ψ ἐπὶ τὰ Ο, Σ ἐπιζευγνυμένα εὐθείαι ἴσαι εἰσὶν ἑκατέρᾳ τῶν ΒΨ, ΨΚ. ὁ ἄρα κέντρον τῷ Ψ καὶ διαστήματι ἐνὶ τῶν ΨΒ, ΨΚ γραφόμενος κύκλος ἔχει καὶ διὰ τῶν Ο, Σ, καὶ ἔσται ἐν κύκλῳ τὸ ΚΒΟΣ τετράπλευρον.

Καὶ ἐπεὶ μείζων ἐστὶν ἡ ΚΒ τῆς ΘΦ, ἴση δὲ ἡ ΘΦ τῇ ΣΟ, μείζων ἄρα ἡ ΚΒ τῆς ΣΟ. ἴση δὲ ἡ ΚΒ ἑκατέρᾳ τῶν ΚΣ, ΒΟ. καὶ ἑκάτερα ἄρα τῶν ΚΣ, ΒΟ τῆς ΣΟ μείζων ἐστίν. καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρον ἐστὶ τὸ ΚΒΟΣ, καὶ ἴσαι αἱ ΚΒ, ΒΟ, ΚΣ, καὶ ἐλάττων ἡ ΟΣ, καὶ ἐκ τοῦ κέντρον τοῦ κύκλου ἐστὶν ἡ ΒΨ, τὸ ἄρα ἀπὸ τῆς ΚΒ τοῦ ἀπὸ τῆς ΒΨ μείζον ἐστὶν ἢ διπλάσιον. ἦθθω ἀπὸ τοῦ Κ ἐπὶ τὴν ΒΦ κάθετος ἡ ΚΩ. καὶ ἐπεὶ ἡ ΒΔ τῆς ΔΩ ἐλάττων ἐστὶν ἢ διπλῇ, καὶ ἐστὶν ὡς ἡ ΒΔ πρὸς τὴν ΔΩ, οὕτως τὸ ὑπὸ τῶν ΔΒ, ΒΩ πρὸς τὸ ὑπὸ [τῶν] ΔΩ, ΩΒ, ἀναγραφόμενον ἀπὸ τῆς ΒΩ τετραγώνου καὶ συμπληρουμένου τοῦ ἐπὶ τῆς ΩΔ παραλληλογράμμου καὶ τὸ ὑπὸ ΔΒ, ΒΩ ἄρα τοῦ ὑπὸ ΔΩ, ΩΒ ἐλαττόν ἐστὶν ἢ διπλάσιον. καὶ ἐστὶ τῆς ΚΔ ἐπιζευγνυμένης τὸ μὲν ὑπὸ ΔΒ,

to SW [Prop. 11.6]. And it was also shown (to be) equal to it. And, thus, WV and SP are equal and parallel [Prop. 1.33]. And since WV is parallel to SP, but WV is parallel to KB, SP is thus also parallel to KB [Prop. 11.1]. And BP and KS join them. Thus, the quadrilateral KBPS is in one plane, inasmuch as if there are two parallel straight-lines, and a random point is taken on each of them, then the straight-line joining the points is in the same plane as the parallel (straight-lines) [Prop. 11.7]. So, for the same (reasons), each of the quadrilaterals SPQT and TQRU is also in one plane. And triangle URO is also in one plane [Prop. 11.2]. So, if we conceive straight-lines joining points P, S, Q, T, R, and U to A then some solid polyhedral figure will be constructed between the circumferences BO and KO, being composed of pyramids whose bases (are) the quadrilaterals KBPS, SPQT, TQRU, and the triangle URO, and apex the point A. And if we also make the same construction on each of the sides KL, LM, and ME, just as on BK, and, further, (repeat the construction) in the remaining three quadrants, then some polyhedral figure which has been inscribed in the sphere will be constructed, being contained by pyramids whose bases (are) the aforementioned quadrilaterals, and triangle URO, and the (quadrilaterals and triangles) similarly arranged to them, and apex the point A.

So, I say that the aforementioned polyhedron will not touch the lesser sphere on the surface on which the circle FGH is (situated).

Let the perpendicular (straight-line) AX be drawn from point A to the plane KBPS, and let it meet the plane at point X [Prop. 11.11]. And let XB and XK be joined. And since AX is at right-angles to the plane of quadrilateral KBPS, it is thus also at right-angles to all of the straight-lines joined to it which are also in the plane of the quadrilateral [Def. 11.3]. Thus, AX is at right-angles to each of BX and XK. And since AB is equal to AK, the (square) on AB is also equal to the (square) on AK. And the (sum of the squares) on AX and XB is equal to the (square) on AB. For the angle at X (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on AX and XK is equal to the (square) on AK [Prop. 1.47]. Thus, the (sum of the squares) on AX and XB is equal to the (sum of the squares) on AX and XK. Let the (square) on AX be subtracted from both. Thus, the remaining (square) on BX is equal to the remaining (square) on XK. Thus, BX (is) equal to XK. So, similarly, we can show that the straight-lines joined from X to P and S are equal to each of BX and XK. Thus, a circle drawn (in the plane of the quadrilateral) with center X, and radius one of XB or XK, will also pass through P and S, and the quadrilateral KBPS will be inside the circle.

And since KB is greater than WV, and WV (is) equal to SP, KB (is) thus greater than SP. And KB (is) equal to each of KS and BP. Thus, KS and BP are each greater than SP. And since quadrilateral KBPS is in a circle, and KB, BP, and

$B\Omega$ ἴσον τῷ ἀπὸ τῆς BK , τὸ δὲ ὑπὸ τῶν $\Delta\Omega$, ΩB ἴσον τῷ ἀπὸ τῆς $K\Omega$. τὸ ἄρα ἀπὸ τῆς KB τοῦ ἀπὸ τῆς $K\Omega$ ἔλασσόν ἐστιν ἢ διπλάσιον. ἀλλὰ τὸ ἀπὸ τῆς KB τοῦ ἀπὸ τῆς $B\Psi$ μείζον ἐστιν ἢ διπλάσιον· μείζον ἄρα τὸ ἀπὸ τῆς $K\Omega$ τοῦ ἀπὸ τῆς $B\Psi$. καὶ ἐπεὶ ἴση ἐστὶν ἡ BA τῇ KA , ἴσον ἐστὶ τὸ ἀπὸ τῆς BA τῷ ἀπὸ τῆς AK . καὶ ἐστὶ τῷ μὲν ἀπὸ τῆς BA ἴσα τὰ ἀπὸ τῶν $B\Psi$, ΨA , τῷ δὲ ἀπὸ τῆς KA ἴσα τὰ ἀπὸ τῶν $K\Omega$, ΩA . τὰ ἄρα ἀπὸ τῶν $B\Psi$, ΨA ἴσα ἐστὶ τοῖς ἀπὸ τῶν $K\Omega$, ΩA , ὧν τὸ ἀπὸ τῆς $K\Omega$ μείζον τοῦ ἀπὸ τῆς $B\Psi$. λοιπὸν ἄρα τὸ ἀπὸ τῆς ΩA ἔλασσόν ἐστι τοῦ ἀπὸ τῆς ΨA . μείζον ἄρα ἡ $A\Psi$ τῆς $A\Omega$. πολλῷ ἄρα ἡ $A\Psi$ μείζον ἐστὶ τῆς AH . καὶ ἐστὶν ἡ μὲν $A\Psi$ ἐπὶ μίαν τοῦ πολυέδρου βάσιν, ἡ δὲ AH ἐπὶ τὴν τῆς ἐλάσσονος σφαίρας ἐπιφάνειαν· ὥστε τὸ πολυέδρον οὐ ψαύσει τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Δύο ἄρα σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἷς τὴν μείζονα σφαῖραν στερεὸν πολυέδρον ἐγγέγραπται μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν· ὅπερ ἔδει ποιῆσαι.

KS are equal (to one another), and PS (is) less (than them), and BX is the radius of the circle, the (square) on KB is thus greater than double the (square) on BX .[†] Let the perpendicular KY be drawn from K to BV .[‡] And since BD is less than double DY , and as BD is to DY , so the (rectangle contained) by DB and BY (is) to the (rectangle contained) by DY and YB —a square being described on BY , and a (rectangular) parallelogram (with short side equal to BY) completed on YD —the (rectangle contained) by DB and BY is thus also less than double the (rectangle contained) by DY and YB . And, KD being joined, the (rectangle contained) by DB and BY is equal to the (square) on BK , and the (rectangle contained) by DY and YB equal to the (square) on KY [Props. 3.31, 6.8 corr.]. Thus, the (square) on KB is less than double the (square) on KY . But, the (square) on KB is greater than double the (square) on BX . Thus, the (square) on KY (is) greater than the (square) on BX . And since BA is equal to KA , the (square) on BA is equal to the (square) on AK . And the (sum of the squares) on BX and XA is equal to the (square) on BA , and the (sum of the squares) on KY and YA (is) equal to the (square) on KA [Prop. 1.47]. Thus, the (sum of the squares) on BX and XA is equal to the (sum of the squares) on KY and YA , of which the (square) on KY (is) greater than the (square) on BX . Thus, the remaining (square) on YA is less than the (square) on XA . Thus, AX (is) greater than AY . Thus, AX is much greater than AG .[§] And AX is (a perpendicular) on one of the bases of the polyhedron, and AG (is a perpendicular) on the surface of the lesser sphere. Hence, the polyhedron will not touch the lesser sphere on its surface.

Thus, there being two spheres about the same center, a polyhedral solid has been inscribed in the greater sphere which does not touch the lesser sphere on its surface. (Which is) the very thing it was required to do.

[†] Since KB , BP , and KS are greater than the sides of an inscribed square, which are each of length $\sqrt{2}BX$.

[‡] Note that points Y and V are actually identical.

[§] This conclusion depends on the fact that the chord of the polygon in proposition 12.16 does not touch the inner circle.

Πόρισμα.

Ἐάν δὲ καὶ εἷς ἐτάραν σφαῖραν τῷ ἐν τῇ $B\Gamma\Delta E$ σφαίρᾳ στερεῷ πολυέδρῳ ὅμοιον στερεὸν πολυέδρον ἐγγραφῇ, τὸ ἐν τῇ $B\Gamma\Delta E$ σφαίρᾳ στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ἐτέρᾳ σφαίρᾳ στερεὸν πολυέδρον τριπλασίονα λόγον ἔθει, ἥπερ ἡ τῆς $B\Gamma\Delta E$ σφαίρας διάμετρος πρὸς τὴν τῆς ἐτέρας σφαίρας διάμετρον. διαιρεθέντων γὰρ τῶν στερεῶν εἷς τὰς ὁμοιοπληθεῖς καὶ ὁμοιοταγεῖς πυραμίδας ἔσονται αἱ πυραμίδες ὅμοιαι. αἱ δὲ ὅμοιαι πυραμίδες πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν· ἡ ἄρα πυραμὶς, ἥξ βασις μὲν ἐστὶ τὸ $K\Theta\Omega$ τετράπλευρον, κορυφή δὲ τὸ A σημεῖον, πρὸς τὴν

Corollary

And, also, if a similar polyhedral solid to that in sphere $BCDE$ is inscribed in another sphere then the polyhedral solid in sphere $BCDE$ has to the polyhedral solid in the other sphere the cubed ratio that the diameter of sphere $BCDE$ has to the diameter of the other sphere. For if the solids are divided into similarly numbered, and similarly situated, pyramids, then the pyramids will be similar. And similar pyramids are in the cubed ratio of corresponding sides [Prop. 12.8 corr.]. Thus, the pyramid whose base is quadrilateral $K\Theta\Omega$, and apex the point A , will have to the similarly situated pyramid in the other

ἐν τῇ ἐτέρᾳ σφαίρᾳ ὁμοιοταγῇ πυραμίδα τριπλασίονα λόγον ἔθει, ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἥπερ ἡ AB ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περὶ κέντρον τὸ A πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἐτέρας σφαίρας. ὁμοίως καὶ ἐκάστη πυραμίς τῶν ἐν τῇ περὶ κέντρον τὸ A σφαίρᾳ πρὸς ἐκάστην ὁμοιοταγῇ πυραμίδα τῶν ἐν τῇ ἐτέρᾳ σφαίρᾳ τριπλασίονα λόγον ἔχει, ἥπερ ἡ AB πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἐτέρας σφαίρας. καὶ ὥς ἐν τῶν ἡγουμένων πρὸς ἐν τῶν ἐπομένων, οὕτως ἅπαντα τὰ ἡγούμενα πρὸς ἅπαντα τὰ ἐπόμενα· ὥστε ὅλον τὸ ἐν τῇ περὶ κέντρον τὸ A σφαίρᾳ στερεὸν πολυέδρον πρὸς ὅλον τὸ ἐν τῇ ἐτέρᾳ [σφαίρᾳ] στερεὸν πολυέδρον τριπλασίονα λόγον ἔχει, ἥπερ ἡ AB πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἐτέρας σφαίρας, τουτέστιν ἥπερ ἡ BD διάμετρος πρὸς τὴν τῆς ἐτέρας σφαίρας διάμετρον· ὅπερ ἔδει δεῖξαι.

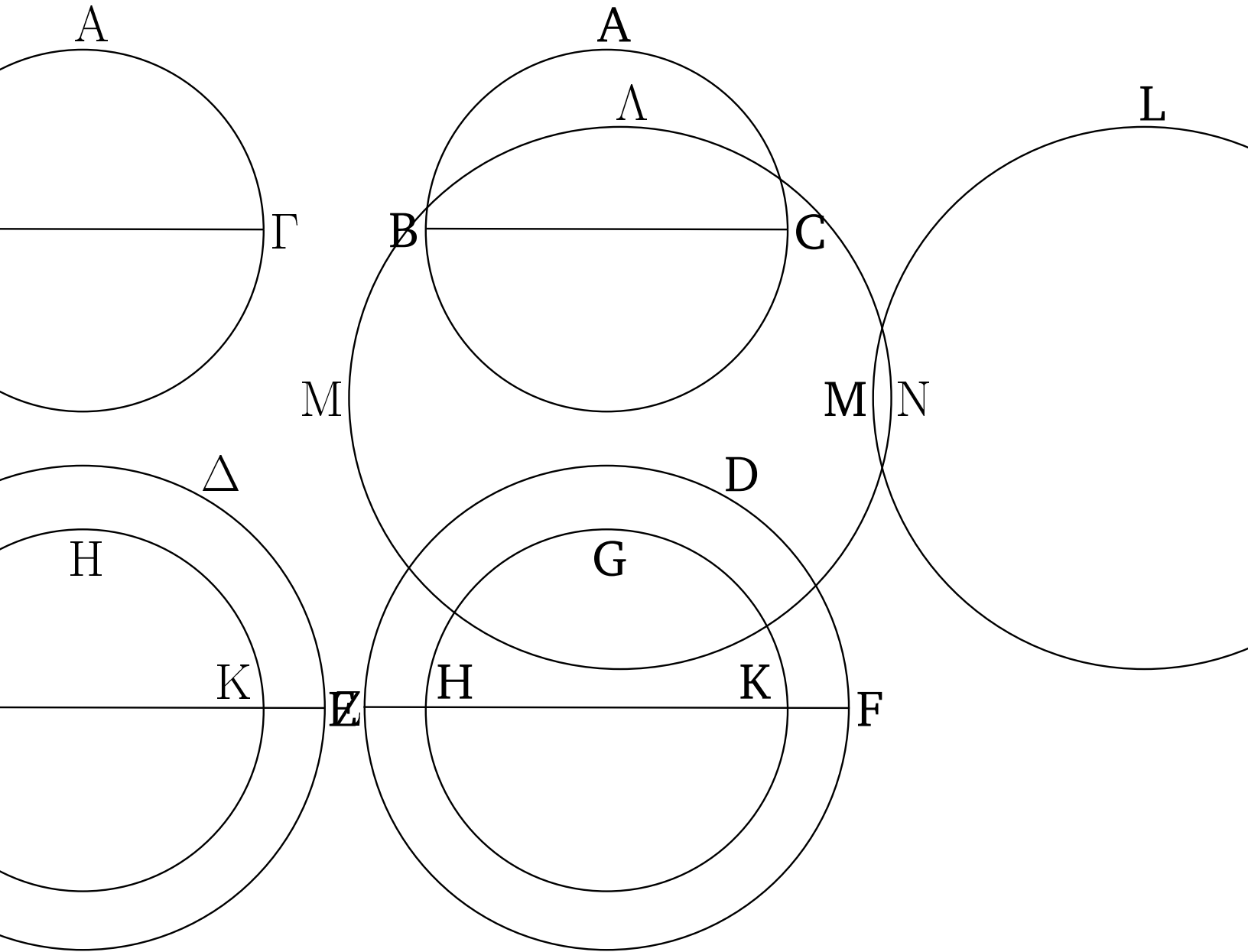
18.

Αἱ σφαῖραι πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἰδίων διαμέτρων.

sphere the cubed ratio that a corresponding side (has) to a corresponding side. That is to say, that of radius AB of the sphere about center A to the radius of the other sphere. And, similarly, each pyramid in the sphere about center A will have to each similarly situated pyramid in the other sphere the cubed ratio that AB (has) to the radius of the other sphere. And as one of the leading (magnitudes is) to one of the following (in two sets of proportional magnitudes), so (the sum of) all the leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. Hence, the whole polyhedral solid in the sphere about center A will have to the whole polyhedral solid in the other [sphere] the cubed ratio that (radius) AB (has) to the radius of the other sphere. That is to say, that diameter BD (has) to the diameter of the other sphere. (Which is) the very thing it was required to show.

Proposition 18

Spheres are to one another in the cubed ratio of their respective diameters.



Νενοήσθωσαν σφαῖραι αἱ $ABΓ$, $ΔΕΖ$, διάμετροι δὲ αὐτῶν αἱ $BΓ$, $EΖ$: λέγω, ὅτι ἡ $ABΓ$ σφαῖρα πρὸς τὴν $ΔΕΖ$ σφαῖραν τριπλασίονα λόγον ἔχει ἢ περὶ ἡ $BΓ$ πρὸς τὴν $EΖ$.

Εἰ γάρ μὴ ἡ $ABΓ$ σφαῖρα πρὸς τὴν $ΔΕΖ$ σφαῖραν τριπλασίονα λόγον ἔχει ἢ περὶ ἡ $BΓ$ πρὸς τὴν $EΖ$, ἔχει ἄρα ἡ $ABΓ$ σφαῖρα πρὸς ἐλάσσονά τινα τῆς $ΔΕΖ$ σφαίρας τριπλασίονα λόγον ἢ πρὸς μείζονα ἢ περὶ ἡ $BΓ$ πρὸς τὴν $EΖ$. ἐθέτω πρότερον πρὸς ἐλάσσονα τὴν $ΗΘΚ$, καὶ νενοήσθω ἡ $ΔΕΖ$ τῇ $ΗΘΚ$ περὶ τὸ αὐτὸ κέντρον, καὶ ἐγγεγράφθω εἰς τὴν μείζονα σφαῖραν τὴν $ΔΕΖ$ στερεὸν πολύεδρον μὴ ψαῖον τῆς ἐλάσσονος σφαίρας τῆς $ΗΘΚ$ κατὰ τὴν ἐπιφάνειαν, ἐγγεγράφθω δὲ καὶ εἰς τὴν $ABΓ$ σφαῖραν τῷ ἐν τῇ $ΔΕΖ$ σφαίρᾳ στερεῷ πολυ-

Let the spheres ABC and DEF be conceived, and (let) their diameters (be) BC and EF (respectively). I say that sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF .

For if sphere ABC does not have to sphere DEF the cubed ratio that BC (has) to EF then sphere ABC will have to some (sphere) either less than, or greater than, sphere DEF the cubed ratio that BC (has) to EF . Let it, first of all, have (such a ratio) to a lesser (sphere), GHK . And let DEF be conceived about the same center as GHK . And let a polyhedral solid be inscribed in the greater sphere DEF , not touching the lesser sphere GHK on its surface [Prop. 12.17]. And let a polyhedral solid, similar to the polyhedral solid in sphere DEF ,

ἐδρῳ ὅμοιον στερεὸν πολυέδρον· τὸ ἄρα ἐν τῇ $AB\Gamma$ στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ΔEZ στερεὸν πολυέδρον τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ἔθει δὲ καὶ ἡ $AB\Gamma$ σφαῖρα πρὸς τὴν $H\Theta K$ σφαῖραν τριπλασίονα λόγον ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ἔστιν ἄρα ὡς ἡ $AB\Gamma$ σφαῖρα πρὸς τὴν $H\Theta K$ σφαῖραν, οὕτως τὸ ἐν τῇ $AB\Gamma$ σφαίρα στερεὸν πολυέδρον πρὸς τὸ ἐν τῇ ΔEZ σφαίρα στερεὸν πολυέδρον· ἐναλλάξ [ἄρα] ὡς ἡ $AB\Gamma$ σφαῖρα πρὸς τὸ ἐν αὐτῇ πολυέδρον, οὕτως ἡ $H\Theta K$ σφαῖρα πρὸς τὸ ἐν τῇ ΔEZ σφαίρα στερεὸν πολυέδρον. μείζων δὲ ἡ $AB\Gamma$ σφαῖρα τοῦ ἐν αὐτῇ πολυέδρου· μείζων ἄρα καὶ ἡ $H\Theta K$ σφαῖρα τοῦ ἐν τῇ ΔEZ σφαίρα πολυέδρου. ἀλλὰ καὶ ἐλάττων· ἐμπεριέθεται γάρ ὑπ' αὐτοῦ. οὐκ ἄρα ἡ $AB\Gamma$ σφαῖρα πρὸς ἐλάσσονα τῆς ΔEZ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ διάμετρος πρὸς τὴν EZ . ὁμοίως δὲ δείχομεν, ὅτι οὐδὲ ἡ ΔEZ σφαῖρα πρὸς ἐλάσσονα τῆς $AB\Gamma$ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ EZ πρὸς τὴν $B\Gamma$.

Λέγω δὴ, ὅτι οὐδὲ ἡ $AB\Gamma$ σφαῖρα πρὸς μείζονά τινα τῆς ΔEZ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ .

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μείζονα τὴν ΛMN . ἀνάπαλιν ἄρα ἡ ΛMN σφαῖρα πρὸς τὴν $AB\Gamma$ σφαῖραν τριπλασίονα λόγον ἔθει ἥπερ ἡ EZ διάμετρος πρὸς τὴν $B\Gamma$ διάμετρον. ὡς δὲ ἡ ΛMN σφαῖρα πρὸς τὴν $AB\Gamma$ σφαῖραν, οὕτως ἡ ΔEZ σφαῖρα πρὸς ἐλάσσονά τινα τῆς $AB\Gamma$ σφαίρας, ἐπειδὴ περ μείζων ἐστὶν ἡ ΛMN τῆς ΔEZ , ὡς ἔμπροσθεν ἐδείχθη. καὶ ἡ ΔEZ ἄρα σφαῖρα πρὸς ἐλάσσονά τινα τῆς $AB\Gamma$ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ EZ πρὸς τὴν $B\Gamma$. ὅπερ ἀδύνατον ἐδείχθη. οὐκ ἄρα ἡ $AB\Gamma$ σφαῖρα πρὸς μείζονά τινα τῆς ΔEZ σφαίρας τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ἐδείχθη δέ, ὅτι οὐδὲ πρὸς ἐλάσσονα. ἡ ἄρα $AB\Gamma$ σφαῖρα πρὸς τὴν ΔEZ σφαῖραν τριπλασίονα λόγον ἔθει ἥπερ ἡ $B\Gamma$ πρὸς τὴν EZ . ὅπερ ἔδει δεῖχαι.

have also been inscribed in sphere ABC . Thus, the polyhedral solid in sphere ABC has to the polyhedral solid in sphere DEF the cubed ratio that BC (has) to EF [Prop. 12.17 corr.]. And sphere ABC also has to sphere GHK the cubed ratio that BC (has) to EF . Thus, as sphere ABC is to sphere GHK , so the polyhedral solid in sphere ABC (is) to the polyhedral solid in sphere DEF . [Thus], alternately, as sphere ABC (is) to the polygon within it, so sphere GHK (is) to the polyhedral solid within sphere DEF [Prop. 5.16]. And sphere ABC (is) greater than the polyhedron within it. Thus, sphere GHK (is) also greater than the polyhedron within sphere DEF [Prop. 5.14]. But, (it is) also less. For it is encompassed by it. Thus, sphere ABC does not have to (a sphere) less than sphere DEF the cubed ratio that diameter BC (has) to EF . So, similarly, we can show that sphere DEF does not have to (a sphere) less than sphere ABC the cubed ratio that EF (has) to BC either.

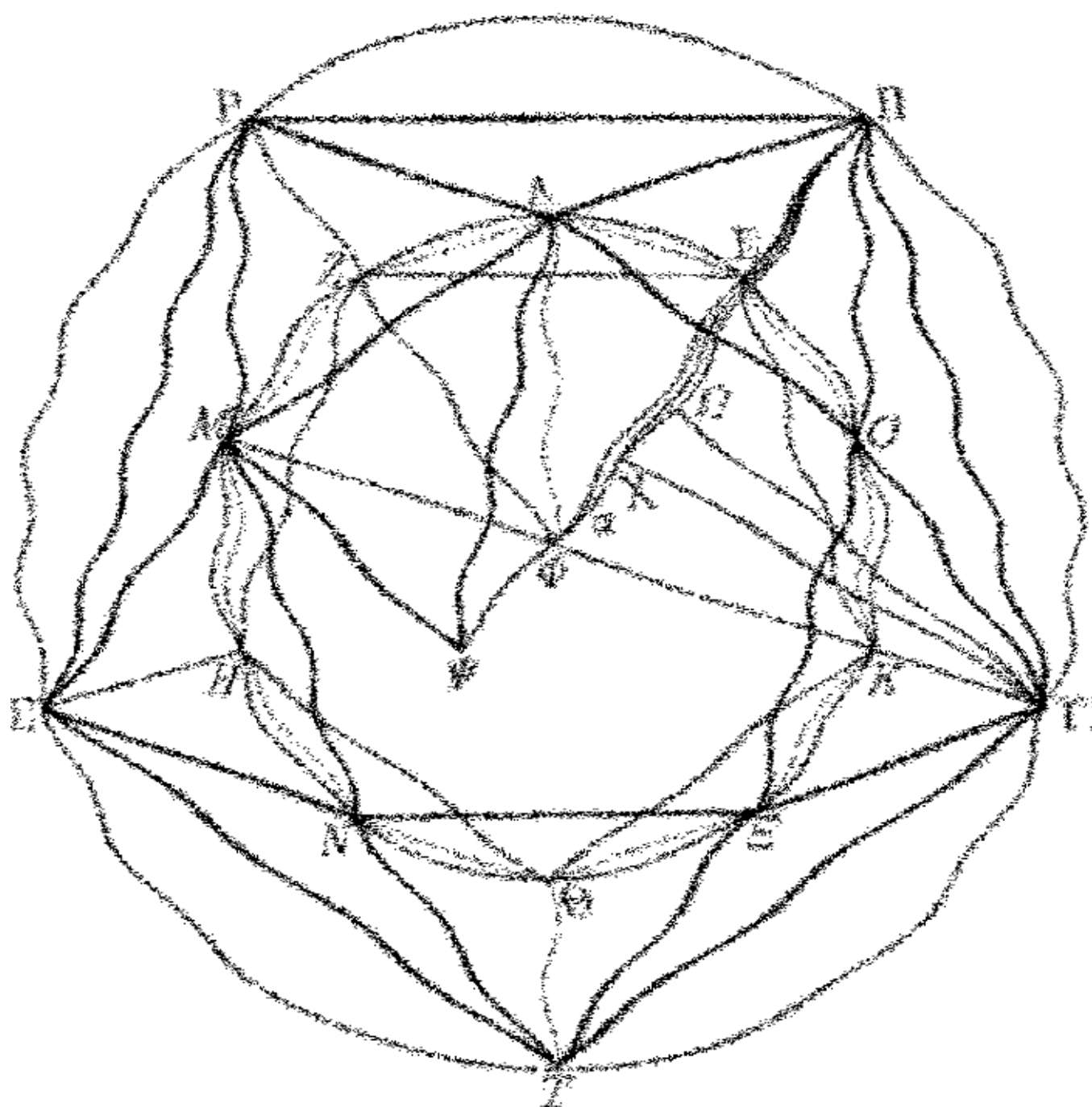
So, I say that sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF either.

For, if possible, let it have (the cubed ratio) to a greater (sphere), LMN . Thus, inversely, sphere LMN (has) to sphere ABC the cubed ratio that diameter EF (has) to diameter BC [Prop. 5.7 corr.]. And as sphere LMN (is) to sphere ABC , so sphere DEF (is) to some (sphere) less than sphere ABC , inasmuch as LMN is greater than DEF , as was shown before [Prop. 12.2 lem.]. And, thus, sphere DEF has to some (sphere) less than sphere ABC the cubed ratio that EF (has) to BC . The very thing was shown (to be) impossible. Thus, sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF . And it was shown that neither (does it have such a ratio) to a lesser (sphere). Thus, sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF . (Which is) the very thing it was required to show.

ELEMENTS BOOK 13

The Platonic Solids[†]

[†]The five regular solids—the cube, tetrahedron (i.e., pyramid), octahedron, icosahedron, and dodecahedron—were probably discovered by the school of Pythagoras. They are generally termed “Platonic” solids because they feature prominently in Plato’s famous dialogue *Timaeus*. Many of the theorems contained in this book—particularly those which pertain to the last two solids—are ascribed to Theaetetus of Athens.

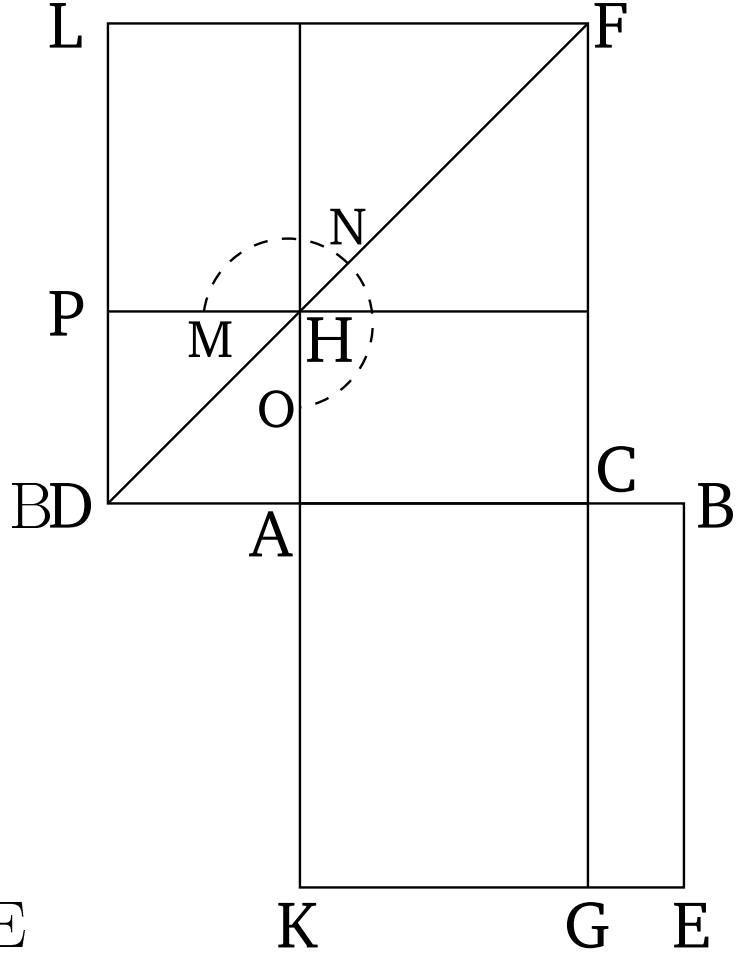
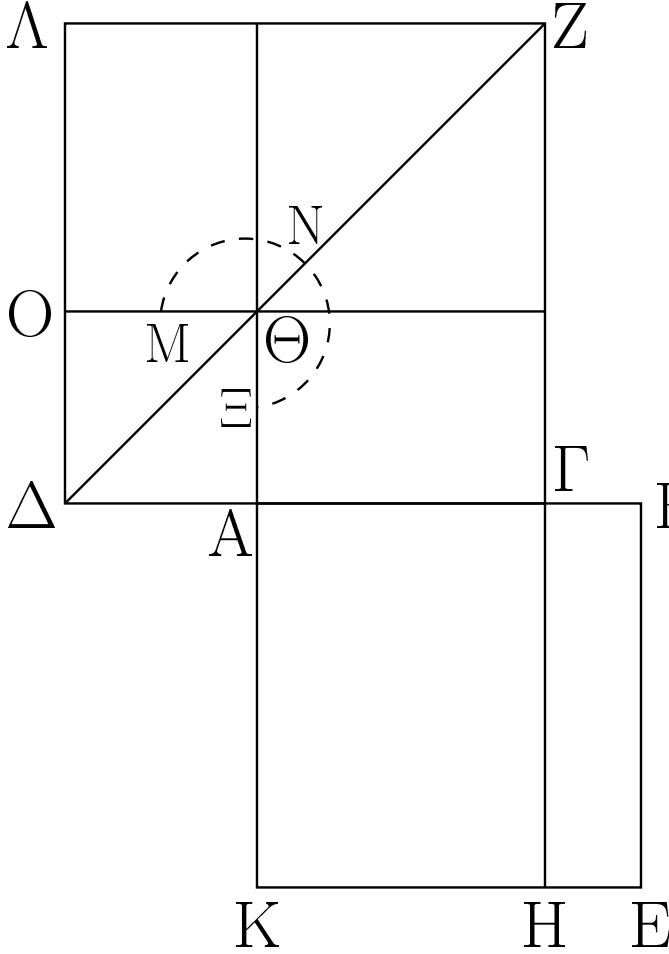


1.

Proposition 1

Ἐάν εὐθεῖα γραμμὴ ἄκρον καὶ μέσον λόγον τμηθῇ, τὸ μείζον τμήμα προσλαβὼν τὴν ἡμίσειαν τῆς ὅλης πενταπλάσιον δύνανται τοῦ ἀπὸ τῆς ἡμισείας τετραγώνον.

If a straight-line is cut in extreme and mean ratio then the square on the greater piece, added to half of the whole, is five times the square on the half.



Εὐθεῖα γὰρ γραμμὴ ἡ ΑΒ ἄκρον καὶ μέσον λόγον τε-
τμήσθω κατὰ τὸ Γ σημεῖον, καὶ ἔστω μείζον τμήμα τὸ ΑΓ,
καὶ ἐκβεβλήσθω ἐπ' εὐθείας τῇ ΓΑ εὐθεῖα ἡ ΑΔ, καὶ κείσθω
τῆς ΑΒ ἡμίσεια ἡ ΑΔ· λέγω, ὅτι πενταπλάσιόν ἐστι τὸ ἀπὸ
τῆς ΓΔ τοῦ ἀπὸ τῆς ΔΑ.

Ἀναγεγράφθωσαν γὰρ ἀπὸ τῶν ΑΒ, ΔΓ τετράγωνα τὰ
ΑΕ, ΔΖ, καὶ καταγεγράφθω ἐν τῷ ΔΖ τὸ σθῆμα, καὶ διήθθω
ἡ ΖΓ ἐπὶ τὸ Η. καὶ ἐπεὶ ἡ ΑΒ ἄκρον καὶ μέσον λόγον τέτμηται
κατὰ τὸ Γ, τὸ ἄρα ὑπὸ τῶν ΑΒΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΓ.
καὶ ἐστὶ τὸ μὲν ὑπὸ τῶν ΑΒΓ τὸ ΓΕ, τὸ δὲ ἀπὸ τῆς ΑΓ τὸ
ΖΘ· ἴσον ἄρα τὸ ΓΕ τῷ ΖΘ. καὶ ἐπεὶ διπλῇ ἐστὶν ἡ ΒΑ τῇ
ΑΔ, ἴση δὲ ἡ μὲν ΒΑ τῇ ΚΑ, ἡ δὲ ΑΔ τῇ ΑΘ, διπλῇ ἄρα
καὶ ἡ ΚΑ τῇ ΑΘ. ὥς δὲ ἡ ΚΑ πρὸς τὴν ΑΘ, οὕτως τὸ
ΓΚ πρὸς τὸ ΓΘ· διπλάσιον ἄρα τὸ ΓΚ τοῦ ΓΘ. εἰσὶ δὲ καὶ
τὰ ΛΘ, ΘΓ διπλάσια τοῦ ΓΘ. ἴσον ἄρα τὸ ΚΓ τοῖς ΛΘ,

For let the straight-line AB be cut in extreme and mean ratio at point C , and let AC be the greater piece. And let the straight-line AD be produced in a straight-line with CA . And let AD be made (equal to) half of AB . I say that the (square) on CD is five times the (square) on DA .

For let the squares AE and DF be described on AB and DC (respectively). And let the figure in DF be drawn. And let FC be drawn across to G . And since AB has been cut in extreme and mean ratio at C , the (rectangle contained) by ABC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. And CE is the (rectangle contained) by ABC , and FH the (square) on AC . Thus, CE (is) equal to FH . And since BA is double AD , and BA (is) equal to KA , and AD to AH , KA (is) thus also double AH . And as KA (is) to AH , so CK (is) to CH [Prop. 6.1]. Thus, CK (is) double CH . And LH plus HC is also double

ΘΓ. ἐδείκθη δὲ καὶ τὸ ΓΕ τῷ ΘΖ ἴσον· ὅλον ἄρα τὸ ΑΕ τετραγώνον ἴσον ἐστὶ τῷ ΜΝΧ γνώμονι. καὶ ἐπεὶ διπλὴ ἐστὶν ἡ ΒΑ τῇς ΑΔ, τετραπλάσιόν ἐστι τὸ ἀπὸ τῇς ΒΑ τοῦ ἀπὸ τῇς ΑΔ, τοντέστι τὸ ΑΕ τοῦ ΔΘ. ἴσον δὲ τὸ ΑΕ τῷ ΜΝΧ γνώμονι· καὶ ὁ ΜΝΧ ἄρα γνώμων τετραπλάσιός ἐστι τοῦ ΑΟ· ὅλον ἄρα τὸ ΔΖ πενταπλάσιόν ἐστι τοῦ ΑΟ. καὶ ἐστὶ τὸ μὲν ΔΖ τὸ ἀπὸ τῇς ΔΓ, τὸ δὲ ΑΟ τὸ ἀπὸ τῇς ΔΑ· τὸ ἄρα ἀπὸ τῇς ΓΔ πενταπλάσιόν ἐστι τοῦ ἀπὸ τῇς ΔΑ.

Ἐάν ἄρα εὐθεῖα ἄκρον καὶ μέσον λόγον τμηθῇ, τὸ μείζον τμήμα προσλαβὸν τὴν ἡμίσειαν τῇς ὅλης πενταπλάσιον δύναται τοῦ ἀπὸ τῇς ἡμισείας τετραγώνου· ὅπερ ἔδει δεῖξαι.

CH [Prop. 1.43]. Thus, *KC* (is) equal to *LH* plus *HC*. And *CE* was also shown (to be) equal to *HF*. Thus, the whole square *AE* is equal to the gnomon *MNO*. And since *BA* is double *AD*, the (square) on *BA* is four times the (square) on *AD*—that is to say, *AE* (is four times) *DH*. And *AE* (is) equal to gnomon *MNO*. And, thus, gnomon *MNO* is also four times *AP*. Thus, the whole of *DF* is five times *AP*. And *DF* is the (square) on *DC*, and *AP* the (square) on *DA*. Thus, the (square) on *CD* is five times the (square) on *DA*.

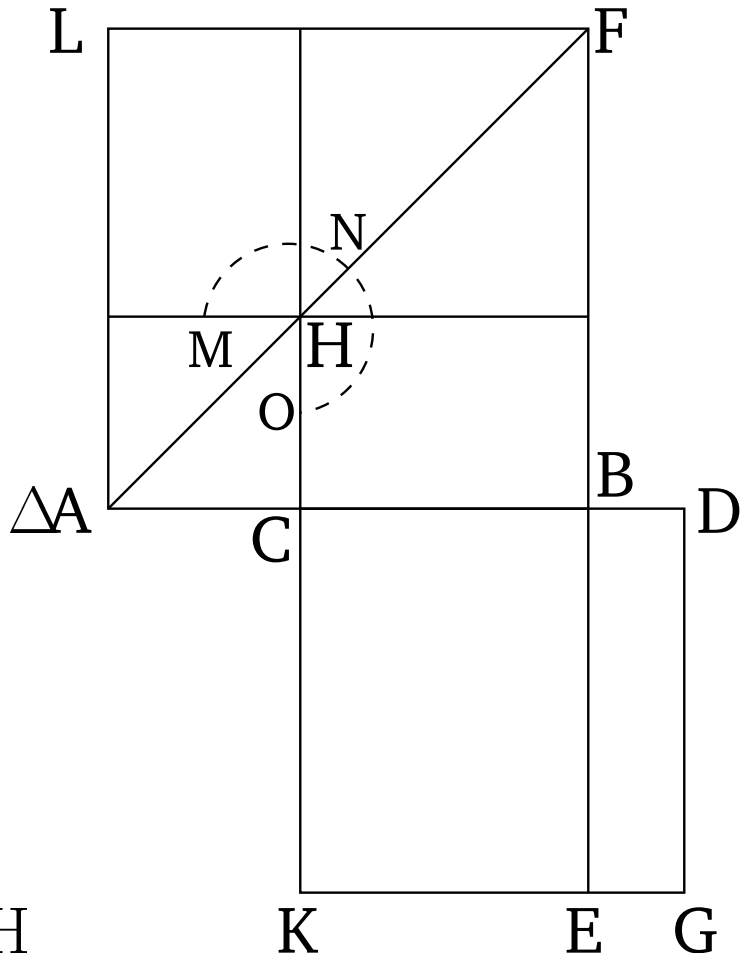
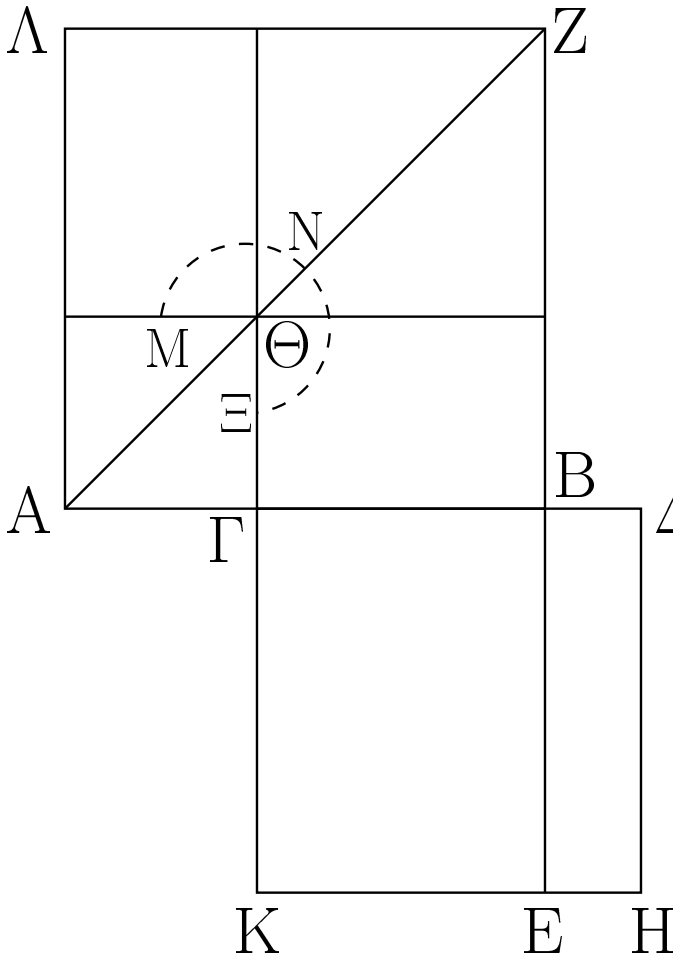
Thus, if a straight-line is cut in extreme and mean ratio then the square on the greater piece, added to half of the whole, is five times the square on the half. (Which is) the very thing it was required to show.

2.

Proposition 2

Ἐάν εὐθεῖα γραμμὴ τμήματοξ ἑαντῇς πενταπλάσιον δύνῃται, τῇς διπλασίαξ τοῦ εἰρημένον τμήματοξ ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα τὸ λοιπὸν μέρος ἐστὶ τῇς ἐχάρθης εὐθείας.

If the square on a straight-line is five times the (square) on a piece of it, and double the aforementioned piece is cut in extreme and mean ratio, then the greater piece is the remaining part of the original straight-line.



Εὐθεΐα γὰρ γραμμὴ ἡ AB τμήματος ἑαυτῆς τοῦ AG πενταπλάσιον δυνάσθω, τῆς δὲ AG διπλῆ ἔστω ἡ $ΓΔ$. λέγω, ὅτι τῆς $ΓΔ$ ἄκρον καὶ μέσον λόγον τεμνομένου τὸ μείζον τμήμα ἐστὶν ἡ GB .

Ἀναγεγράφθω γὰρ ἀφ' ἑκατέρας τῶν AB , $ΓΔ$ τετραγώνων τὰ AZ , $ΓH$, καὶ καταγεγράφθω ἐν τῷ AZ τὸ σθῆμα, καὶ διήθθω ἡ BE . καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς BA τοῦ ἀπὸ τῆς AG , πενταπλάσιόν ἐστι τὸ AZ τοῦ $AΘ$. τετραπλάσιος ἄρα ὁ MNX γνώμων τοῦ $AΘ$. καὶ ἐπεὶ διπλῆ ἐστὶν ἡ $ΔΓ$ τῆς $ΓΑ$, τετραπλάσιον ἄρα ἐστὶ τὸ ἀπὸ $ΔΓ$ τοῦ ἀπὸ $ΓΑ$, τοντέστι τὸ $ΓH$ τοῦ $AΘ$. ἐδείκθη δὲ καὶ ὁ MNX γνώμων τετραπλάσιος τοῦ $AΘ$. ἴσος ἄρα ὁ MNX γνώμων τῷ $ΓH$. καὶ ἐπεὶ διπλῆ ἐστὶν ἡ $ΔΓ$ τῆς $ΓΑ$, ἴση δὲ ἡ μὲν $ΔΓ$ τῇ $ΓΚ$, ἡ δὲ AG τῇ $ΓΘ$, [διπλῆ ἄρα καὶ ἡ $ΚΓ$ τῆς $ΓΘ$], διπλάσιον ἄρα καὶ τὸ KB τοῦ $BΘ$. εἰσὶ δὲ καὶ τὰ $ΛΘ$, $ΘB$ τοῦ $ΘB$ διπλάσια· ἴσον ἄρα τὸ KB τοῖς $ΛΘ$, $ΘB$. ἐδείκθη δὲ καὶ ὅλος ὁ MNX γνώμων ὅλῳ τῷ $ΓH$ ἴσος· καὶ λοιπὸν ἄρα τὸ $ΘZ$ τῷ BH ἐστὶν ἴσον. καὶ ἐστὶ τὸ μὲν BH τὸ ὑπὸ τῶν $ΓΔB$ · ἴση γὰρ ἡ $ΓΔ$ τῇ $ΔH$ · τὸ δὲ $ΘZ$ τὸ ἀπὸ τῆς $ΓB$ · τὸ ἄρα ὑπὸ τῶν $ΓΔB$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΓB$. ἐστὶν ἄρα ὥς ἡ $ΔΓ$ πρὸς τὴν $ΓB$, οὕτως ἡ $ΓB$ πρὸς τὴν $BΔ$. μείζων δὲ ἡ $ΔΓ$ τῆς $ΓB$ · μείζων ἄρα καὶ ἡ $ΓB$ τῆς $BΔ$. τῆς $ΓΔ$ ἄρα εὐθείας ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα ἐστὶν ἡ GB .

Ἐὰν ἄρα εὐθεΐα γραμμὴ τμήματος ἑαυτῆς πενταπλάσιον δύνῃται, τῆς διπλασίας τοῦ εἰρημένου τμήματος ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα τὸ λοιπὸν μέρος ἐστὶ τῆς ἐκ ἀρθῆς εὐθείας· ὅπερ ἔδει δεῖχαι.

Λήμμα.

Ὅτι δὲ ἡ διπλῆ τῆς AG μείζων ἐστὶ τῆς $BΓ$, οὕτως δεῖκνται.

Εἰ γὰρ μή, ἔστω, εἰ δυνατόν, ἡ $BΓ$ διπλῆ τῆς $ΓΑ$. τετραπλάσιον ἄρα τὸ ἀπὸ τῆς $BΓ$ τοῦ ἀπὸ τῆς $ΓΑ$ · πενταπλάσια ἄρα τὰ ἀπὸ τῶν $BΓ$, $ΓΑ$ τοῦ ἀπὸ τῆς $ΓΑ$. ὑπόκειται δὲ καὶ τὸ ἀπὸ τῆς BA πενταπλάσιον τοῦ ἀπὸ τῆς $ΓΑ$ · τὸ ἄρα ἀπὸ τῆς BA ἴσον ἐστὶ τοῖς ἀπὸ τῶν $BΓ$, $ΓΑ$ · ὅπερ ἀδύνατον. οὐκ ἄρα ἡ $ΓB$ διπλασία ἐστὶ τῆς AG . ὁμοίως δὲ δείχουμεν, ὅτι οὐδὲ ἡ ἐλάττων τῆς $ΓB$ διπλασίον ἐστὶ τῆς $ΓΑ$ · πολλῶ γὰρ [μείζον] τὸ ἄτοπον.

Ἡ ἄρα τῆς AG διπλῆ μείζων ἐστὶ τῆς $ΓB$ · ὅπερ ἔδει δεῖχαι.

3.

Ἐὰν εὐθεΐα γραμμὴ ἄκρον καὶ μέσον λόγον τμηθῇ, τὸ ἔλασσον τμήμα προσλαβὼν τὴν ἡμίσειαν τοῦ μείζονος τμήματος

For let the square on the straight-line AB be five times the (square) on the piece of it, AC . And let CD be double AC . I say that if CD is cut in extreme and mean ratio then the greater piece is CB .

For let the squares AF and CG be described on each of AB and CD (respectively). And let the figure in AF be drawn. And let BE be drawn across. And since the (square) on BA is five times the (square) on AC , AF is five times AH . Thus, gnomon MNO (is) four times AH . And since DC is double CA , the (square) on DC is thus four times the (square) on CA —that is to say, CG (is four times) AH . And the gnomon MNO was also shown (to be) four times AH . Thus, gnomon MNO (is) equal to CG . And since DC is double CA , and DC (is) equal to CK , and AC to CH , [KC (is) thus also double CH], (and) KB (is) also double BH [Prop. 6.1]. And LH plus HB is also double HB [Prop. 1.43]. Thus, KB (is) equal to LH plus HB . And the whole gnomon MNO was also shown (to be) equal to the whole of CG . Thus, the remainder HF is also equal to (the remainder) BG . And BG is the (rectangle contained) by CDB . For CD (is) equal to DG . And HF (is) the square on CB . Thus, the (rectangle contained) by CDB is equal to the (square) on CB . Thus, as DC is to CB , so CB (is) to BD [Prop. 6.17]. And DC (is) greater than CB (see lemma). Thus, CB (is) also greater than BD [Prop. 5.14]. Thus, if the straight-line CD is cut in extreme and mean ratio then the greater piece is CB .

Thus, if the square on a straight-line is five times the (square) on a piece of itself, and double the aforementioned piece is cut in extreme and mean ratio, then the greater piece is the remaining part of the original straight-line. (Which is) the very thing it was required to show.

Lemma

And it can be shown that double AC (i.e., DC) is greater than BC , as follows.

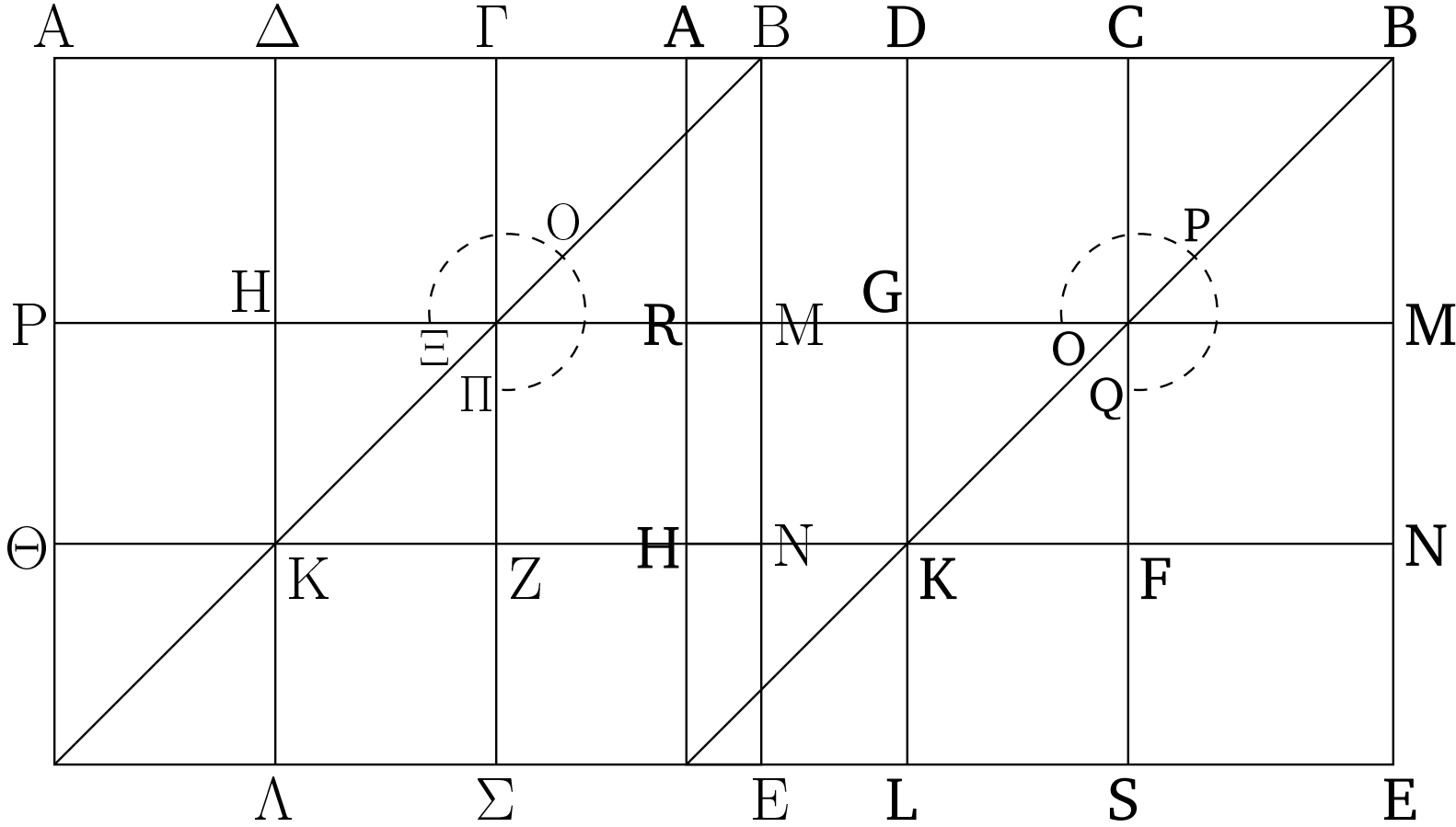
For if (double AC is) not (greater than BC), if possible, let BC be double CA . Thus, the (square) on BC (is) four times the (square) on CA . Thus, the (sum of) the (squares) on BC and CA (is) five times the (square) on CA . And the (square) on BA was assumed (to be) five times the (square) on CA . Thus, the (square) on BA is equal to the (sum of) the (squares) on BC and CA . The very thing (is) impossible [Prop. 2.4]. Thus, CB is not double AC . So, similarly, we can show that a (straight-line) less than CB is not double AC either. For (in this case) the absurdity is much [greater].

Thus, double AC is greater than CB . (Which is) the very thing it was required to show.

Proposition 3

If a straight-line is cut in extreme and mean ratio then the square on the lesser piece added to half of the greater piece is

πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τοῦ μείζονος five times the square on half of the greater piece.
 τμήματος τετραγώνου.



Εὐθεΐα γάρ τις ἡ AB ἄκρον καὶ μέσον λόγον τετμήσθω κατὰ τὸ Γ σημείον, καὶ ἔστω μείζον τμήμα τὸ AG , καὶ τετμήσθω ἡ AG δίδια κατὰ τὸ Δ · λέγω, ὅτι πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς BD τοῦ ἀπὸ τῆς $\Delta\Gamma$.

Ἀναγεγράφθω γάρ ἀπὸ τῆς AB τετράγωνον τὸ AE , καὶ καταγεγράφθω διπλοῦν τὸ σθῆμα. ἐπεὶ διπλῇ ἐστὶν ἡ AG τῆς $\Delta\Gamma$, τετραπλάσιον ἄρα τὸ ἀπὸ τῆς AG τοῦ ἀπὸ τῆς $\Delta\Gamma$, τουτέστι τὸ $P\Sigma$ τοῦ ZH . καὶ ἐπεὶ τὸ ὑπὸ τῶν $AB\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AG , καὶ ἐστὶ τὸ ὑπὸ τῶν $AB\Gamma$ τὸ ΓE , τὸ ἄρα ΓE ἴσον ἐστὶ τῷ $P\Sigma$. τετραπλάσιον δὲ τὸ $P\Sigma$ τοῦ ZH · τετραπλάσιον ἄρα καὶ τὸ ΓE τοῦ ZH . πάλιν ἐπεὶ ἴση ἐστὶν ἡ $A\Delta$ τῇ $\Delta\Gamma$, ἴση ἐστὶ καὶ ἡ ΘK τῇ KZ . ὥστε καὶ τὸ HZ τετράγωνον ἴσον ἐστὶ τῷ $\Theta\Lambda$ τετραγώνῳ. ἴση ἄρα ἡ HK τῇ KL , τουτέστιν ἡ MN τῇ NE . ὥστε καὶ τὸ MZ τῷ ZE ἐστὶν ἴσον. ἀλλὰ τὸ MZ τῷ ΓH ἐστὶν ἴσον· καὶ τὸ ΓH ἄρα τῷ ZE ἐστὶν ἴσον. κοινὸν προσκείσθω τὸ ΓN · ὁ ἄρα $XO\Pi$ γνῶμων ἴσος ἐστὶ τῷ ΓE . ἀλλὰ τὸ ΓE τετραπλάσιον ἐδείκθη τοῦ HZ · καὶ ὁ $XO\Pi$ ἄρα γνῶμων τετραπλάσιός ἐστι τοῦ ZH τετραγώνου. ὁ $XO\Pi$ ἄρα γνῶμων καὶ τὸ ZH τετράγωνον πενταπλάσιός ἐστι τοῦ ZH . ἀλλὰ ὁ $XO\Pi$ γνῶμων καὶ τὸ ZH τετράγωνόν ἐστι τὸ ΔN . καὶ ἐστὶ τὸ μὲν ΔN τὸ ἀπὸ τῆς ΔB ,

For let some straight-line AB be cut in extreme and mean ratio at point C . And let AC be the greater piece. And let AC be cut in half at D . I say that the (square) on BD is five times the (square) on DC .

For let the square AE be described on AB . And let the figure be drawn double. Since AC is double DC , the (square) on AC (is) thus four times the (square) on DC —that is to say, RS (is four times) FG . And since the (rectangle contained) by ABC is equal to the (square) on AC [Def. 6.3, Prop. 6.17], and CE is the (rectangle contained) by ABC , CE is thus equal to RS . And RS (is) four times FG . Thus, CE (is) also four times FG . Again, since AD is equal to DC , HK is also equal to KF . Hence, square GF is also equal to square HL . Thus, GK (is) equal to KL —that is to say, MN to NE . Hence, MF is also equal to FE . But, MF is equal to CG . Thus, CG is also equal to FE . Let CN be added to both. Thus, gnomon OPQ is equal to CE . But, CE was shown (to be) equal to four times GF . Thus, gnomon OPQ is also four times square FG . Thus, gnomon OPQ plus square FG is five times FG . But, gnomon OPQ plus square FG is (square) DN . And DN is the (square) on DB , and GF the (square) on DC . Thus, the (square) on DB

τὸ δὲ HZ τὸ ἀπὸ τῆς $\Delta\Gamma$. τὸ ἄρα ἀπὸ τῆς ΔB πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς $\Delta\Gamma$. ὅπερ ἔδει δεῖχαι.

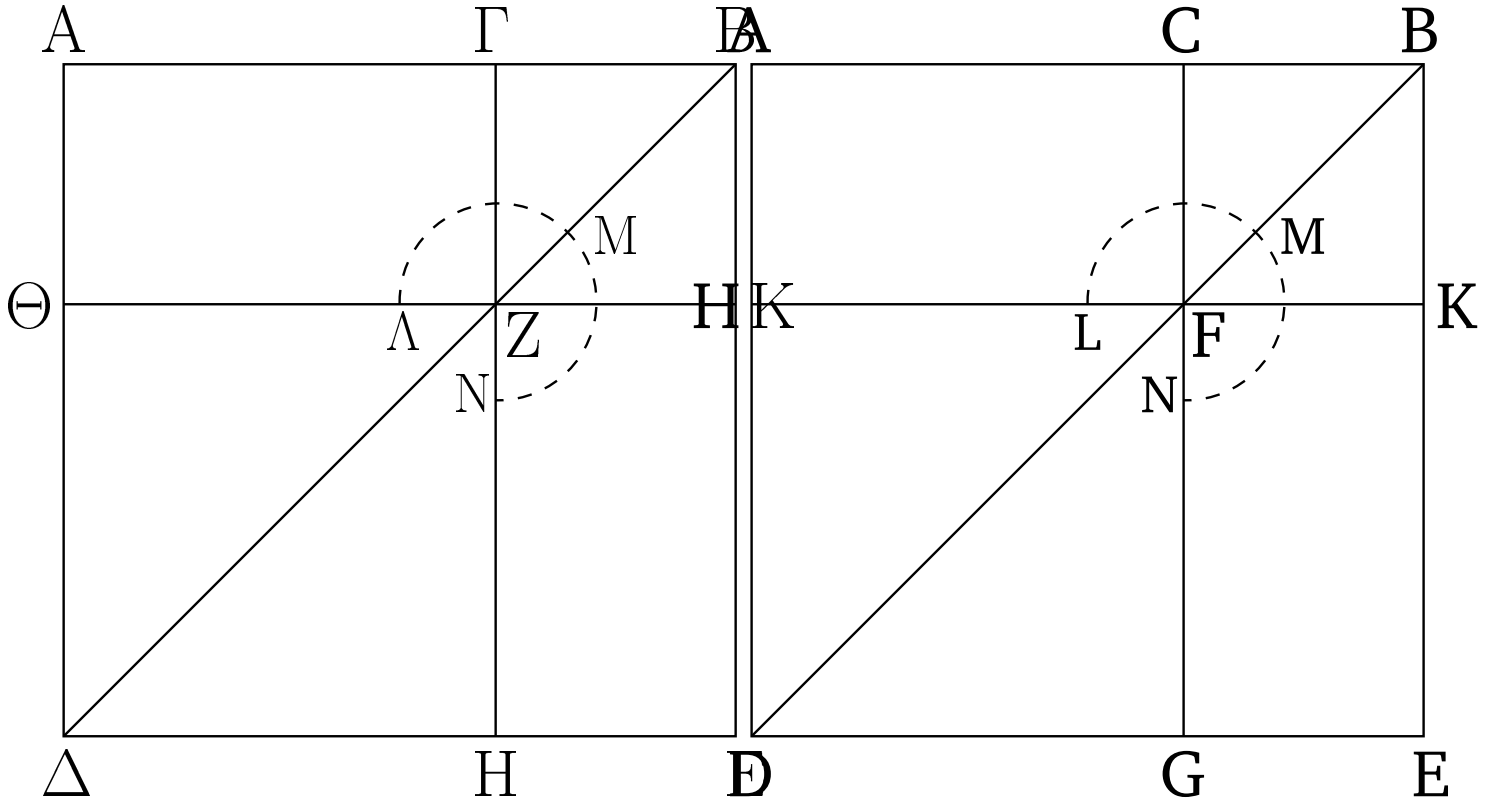
is five times the (square) on DC . (Which is) the very thing it was required to show.

4.

Proposition 4

Ἐὰν εὐθεῖα γραμμὴ ἄκρον καὶ μέσον λόγον τμηθῇ, τὸ ἀπὸ τῆς ὅλης καὶ τοῦ ἐλάσσονος τμήματος, τὰ συναμφότερα τετράγωνα, τριπλάσιά ἐστι τοῦ ἀπὸ τοῦ μείζονος τμήματος τετραγώνου.

If a straight-line is cut in extreme and mean ratio then the sum of the squares on the whole and the lesser piece is three times the square on the greater piece.



Ἐστω εὐθεῖα ἡ AB , καὶ τετμήσθω ἄκρον καὶ μέσον λόγον κατὰ τὸ Γ , καὶ ἔστω μείζον τμήμα τὸ AG . λέγω, ὅτι τὰ ἀπὸ τῶν AB , $B\Gamma$ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς GA .

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $ADEB$, καὶ καταγεγράφθω τὸ στήμα. ἐπεὶ οὖν ἡ AB ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Γ , καὶ τὸ μείζον τμήμα ἐστὶν ἡ AG , τὸ ἄρα ὑπὸ τῶν $AB\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AG . καὶ ἐστὶ τὸ μὲν ὑπὸ τῶν $AB\Gamma$ τὸ AK , τὸ δὲ ἀπὸ τῆς AG τὸ ΘH . ἴσον ἄρα ἐστὶ τὸ AK τῷ ΘH . καὶ ἐπεὶ ἴσον ἐστὶ τὸ AZ τῷ ZE , κοινὸν προσκείσθω τὸ $ΓΚ$. ὅλον ἄρα τὸ AK ὅλῳ τῷ $ΓΕ$ ἐστὶν ἴσον· τὰ ἄρα AK , $ΓΕ$ τοῦ AK ἐστὶ διπλάσια. ἀλλὰ τὰ AK , $ΓΕ$ ὁ ΛMN γνώμων ἐστὶ καὶ τὸ $ΓΚ$ τετράγωνον· ὁ ἄρα ΛMN γνώμων καὶ τὸ $ΓΚ$ τετράγωνον διπλάσιά ἐστι τοῦ AK . ἀλλὰ μὴν καὶ τὸ AK τῷ ΘH ἐδείκθη ἴσον· ὁ ἄρα ΛMN γνώμων καὶ [τὸ $ΓΚ$ τετράγωνον διπλάσιά ἐστι τοῦ ΘH · ὥστε ὁ ΛMN γνώμων καὶ] τὰ $ΓΚ$, ΘH τετράγωνα τριπλάσιά ἐστι τοῦ ΘH τετραγώνου. καὶ ἐστὶν ὁ [μὲν] ΛMN γνώμων καὶ

Let AB be a straight-line, and let it be cut in extreme and mean ratio at C , and let AC be the greater piece. I say that the (sum of the squares) on AB and BC is three times the (square) on CA .

For let the square $ADEB$ be described on AB , and let the (remainder of the) figure be drawn. Therefore, since AB has been cut in extreme and mean ratio at C , and AC is the greater piece, the (rectangle contained) by ABC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. And AK is the (rectangle contained) by ABC , and HG the (square) on AC . Thus, AK is equal to HG . And since AF is equal to FE [Prop. 1.43], let CK be added to both. Thus, the whole of AK is equal to the whole of CE . Thus, AK plus CE is double AK . But, AK plus CE is the gnomon LMN plus the square CK . Thus, gnomon LMN plus square CK is double AK . But, indeed, AK was also shown (to be) equal to HG . Thus, gnomon LMN plus [square CK is double HG . Hence, gnomon LMN plus] the squares CK

τά $ΓΚ$, $ΘΗ$ τετράγωνα ὅλον τὸ $ΑΕ$ καὶ τὸ $ΓΚ$, ἕπερ ἐστὶ τὰ ἀπὸ τῶν $ΑΒ$, $ΒΓ$ τετράγωνα, τὸ δὲ $ΗΘ$ τὸ ἀπὸ τῆς $ΑΓ$ τετράγωνον. τὰ ἄρα ἀπὸ τῶν $ΑΒ$, $ΒΓ$ τετράγωνα τριπλάσιά ἐστι τοῦ ἀπὸ τῆς $ΑΓ$ τετραγώνου· ὅπερ ἔδει δεῖχαι.

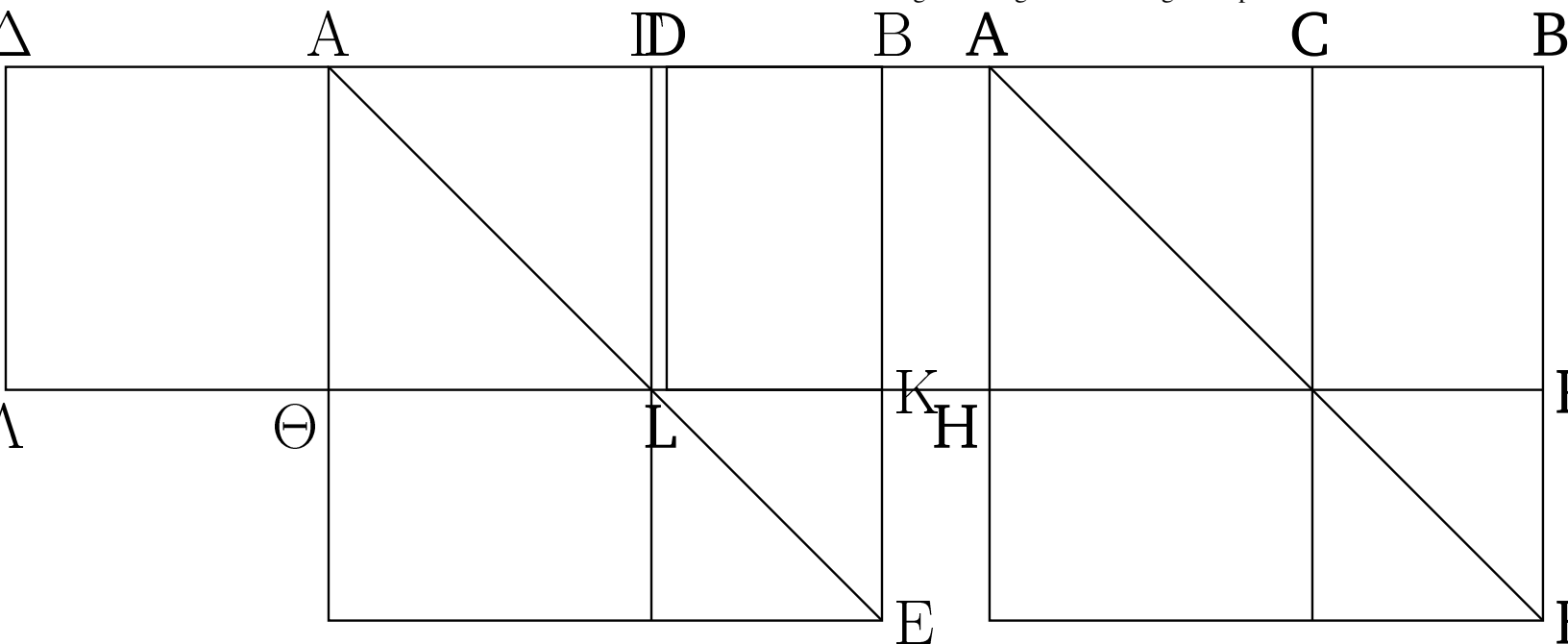
and HG is three times the square HG . And gnomon LMN plus the squares CK and HG is the whole of AE plus CK —which are the squares on AB and BC (respectively)—and GH (is) the square on AC . Thus, the (sum of the) squares on AB and BC is three times the square on AC . (Which is) the very thing it was required to show.

5.

Ἐάν εὐθεία γραμμὴ ἄκρον καὶ μέσον λόγον τηθῇ, καὶ προστεθῇ αὐτῇ ἴση τῷ μείζονι τμήματι, ἡ ὅλη εὐθεία ἄκρον καὶ μέσον λόγον τέμνεται, καὶ τὸ μείζον τμήμα ἐστὶν ἡ ἐκ ἀρθῆξ εὐθεία.

Proposition 5

If a straight-line is cut in extreme and mean ratio, and a (straight-line) equal to the greater piece is added to it, then the whole straight-line has been cut in extreme and mean ratio, and the original straight-line is the greater piece.



Εὐθεία γὰρ γραμμὴ ἡ AB ἄκρον καὶ μέσον λόγον τε-
τμήσθω κατὰ τὸ Γ σημεῖον, καὶ ἔστω μείζον τμήμα ἡ $ΑΓ$, καὶ
τῇ $ΑΓ$ ἴση [κείσθω] ἡ $ΑΔ$. λέγω, ὅτι ἡ $ΔΒ$ εὐθεία ἄκρον
καὶ μέσον λόγον τέτμηται κατὰ τὸ A , καὶ τὸ μείζον τμήμά
ἐστιν ἡ ἐκ ἀρθῆς εὐθεία ἡ AB .

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετραγώνον τὸ AE , καὶ καταγεγράφθω τὸ σθῆμα. ἐπεὶ ἡ AB ἄκρον καὶ μέσον λόγον τέμνεται κατὰ τὸ Γ , τὸ ἄρα ὑπὸ $AB\Gamma$ ἴσον ἐστὶ τῷ ἀπὸ AI . καὶ ἐστὶ τὸ μὲν ὑπὸ $AB\Gamma$ τὸ ΓE , τὸ δὲ ἀπὸ τῆς $A\Gamma$ τὸ $\Gamma\Theta$. ἴσον ἄρα τὸ ΓE τῷ $\Theta\Gamma$. ἀλλὰ τῷ μὲν ΓE ἴσον ἐστὶ τὸ ΘE , τῷ δὲ $\Theta\Gamma$ ἴσον τὸ $\Delta\Theta$. καὶ τὸ $\Delta\Theta$ ἄρα ἴσον ἐστὶ τῷ ΘE [κοινὸν προσκείσθω τὸ ΘB]. ὅλον ἄρα τὸ ΔK ὅλῳ τῷ AE ἴσον ἐστὶν. καὶ ἐστὶ τὸ μὲν ΔK τὸ ὑπὸ τῶν $B\Delta$, ΔA . ἴση γὰρ ἡ $A\Delta$ τῇ ΔA . τὸ δὲ AE τὸ ἀπὸ τῆς AB . τὸ ἄρα ὑπὸ τῶν $B\Delta A$ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB . ἔστιν ἄρα ὥς ἡ ΔB πρὸς τὴν BA , οὕτως ἡ BA πρὸς τὴν AD . μείζων δὲ ἡ ΔB τῆς BA . μείζων ἄρα καὶ ἡ BA τῆς AD .

For let the straight-line AB be cut in extreme and mean ratio at point C . And let AC be the greater piece. And let AD be [made] equal to AC . I say that the straight-line DB has been cut in extreme and mean ratio at A , and that the original straight-line AB is the greater piece.

For let the square AE be described on AB , and let the (remainder of the) figure be drawn. And since AB has been cut in extreme and mean ratio at C , the (rectangle contained) by ABC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. And CE is the (rectangle contained) by ABC , and CH the (square) on AC . But, HE is equal to CE [Prop. 1.43], and DH equal to HC . Thus, DH is also equal to HE . [Let HB be added to both.] Thus, the whole of DK is equal to the whole of AE . And DK is the (rectangle contained) by BD and DA . For AD (is) equal to DL . And AE (is) the (square) on AB . Thus, the (rectangle contained) by BDA is equal to the (square) on AB . Thus, as DB (is) to BA , so BA (is) to AD [Prop. 6.17]. And

Ἡ ἄρα $ΔΒ$ ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ $Α$, καὶ τὸ μείζον τμήμα ἐστὶν ἡ $ΑΒ$. ὅπερ ἔδει δεῖχαι.

DB (is) greater than BA . Thus, BA (is) also greater than AD [Prop. 5.14].

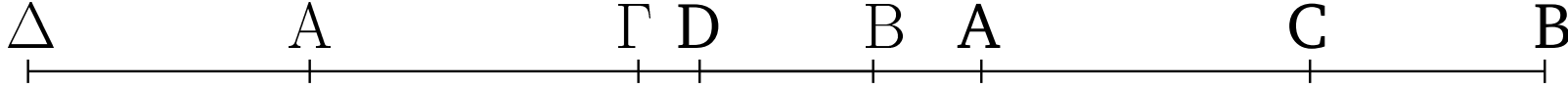
Thus, DB has been cut in extreme and mean ratio at A , and the greater piece is AB . (Which is) the very thing it was required to show.

6.

Proposition 6

Ἐὰν εὐθεῖα ῥητὴ ἄκρον καὶ μέσον λόγον τμηθῇ, ἐκάτερον τῶν τμημάτων ἄλογός ἐστιν ἡ καλουμένη ἀποτομή.

If a rational straight-line is cut in extreme and mean ratio then each of the pieces is that irrational (straight-line) called an apotome.



Ἐστω εὐθεῖα ῥητὴ ἡ $ΑΒ$ καὶ τετμήσθω ἄκρον καὶ μέσον λόγον κατὰ τὸ $Γ$, καὶ ἔστω μείζον τμήμα ἡ $ΑΓ$. λέγω, ὅτι ἐκάτερα τῶν $ΑΓ$, $ΓΒ$ ἄλογός ἐστιν ἡ καλουμένη ἀποτομή.

Let AB be a rational straight-line cut in extreme and mean ratio at C , and let AC be the greater piece. I say that AC and CB is each that irrational (straight-line) called an apotome.

Ἐκβεβλήσθω γάρ ἡ $ΒΑ$, καὶ κείσθω τῆς $ΒΑ$ ἡμίσεια ἡ $ΑΔ$. ἐπεὶ οὖν εὐθεῖα ἡ $ΑΒ$ τέτμηται ἄκρον καὶ μέσον λόγον κατὰ τὸ $Γ$, καὶ τῷ μείζονι τμήματι τῷ $ΑΓ$ πρόσκειται ἡ $ΑΔ$ ἡμίσεια οὕσα τῆς $ΑΒ$, τὸ ἄρα ἀπὸ $ΓΔ$ τοῦ ἀπὸ $ΔΑ$ πενταπλάσιόν ἐστιν. τὸ ἄρα ἀπὸ $ΓΔ$ πρὸς τὸ ἀπὸ $ΔΑ$ λόγον ἔθει, ὃν ἀριθμὸς πρὸς ἀριθμόν· σύμμετρον ἄρα τὸ ἀπὸ $ΓΔ$ τῷ ἀπὸ $ΔΑ$. ῥητὸν δὲ τὸ ἀπὸ $ΔΑ$ · ῥητὴ γάρ [ἐστὶν] ἡ $ΔΑ$ ἡμίσεια οὕσα τῆς $ΑΒ$ ῥητῆς οὕσης· ῥητὸν ἄρα καὶ τὸ ἀπὸ $ΓΔ$ · ῥητὴ ἄρα ἐστὶ καὶ ἡ $ΓΔ$. καὶ ἐπεὶ τὸ ἀπὸ $ΓΔ$ πρὸς τὸ ἀπὸ $ΔΑ$ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, ἀσύμμετρος ἄρα μήκει ἡ $ΓΔ$ τῇ $ΔΑ$. αἱ $ΓΔ$, $ΔΑ$ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ $ΑΓ$. πάλιν, ἐπεὶ ἡ $ΑΒ$ ἄκρον καὶ μέσον λόγον τέτμηται, καὶ τὸ μείζον τμήμα ἐστὶν ἡ $ΑΓ$, τὸ ἄρα ὑπὸ $ΑΒ$, $ΒΓ$ τῷ ἀπὸ $ΑΓ$ ἴσον ἐστίν. τὸ ἄρα ἀπὸ τῆς $ΑΓ$ ἀποτομῆς παρὰ τὴν $ΑΒ$ ῥητὴν παραβληθὲν πλάτοξ ποιεῖ τὴν $ΒΓ$. τὸ δὲ ἀπὸ ἀποτομῆς παρὰ ῥητὴν παραβαλλόμενον πλάτοξ ποιεῖ ἀποτομήν· ἀποτομὴ ἄρα πρώτη ἐστὶν ἡ $ΓΒ$. ἐδείχθη δὲ καὶ ἡ $ΓΑ$ ἀποτομή.

For let BA be produced, and let AD be made (equal) to half of BA . Therefore, since the straight-line AB has been cut in extreme and mean ratio at C , and AD , which is half of AB , has been added to the greater piece AC , the (square) on CD is thus five times the (square) on DA [Prop. 13.1]. Thus, the (square) on CD has to the (square) on DA the ratio which a number (has) to a number. The (square) on CD (is) thus commensurable with the (square) on DA [Prop. 10.6]. And the (square) on DA (is) rational. For DA [is] rational, being half of AB , which is rational. Thus, the (square) on CD (is) also rational [Def. 10.4]. Thus, CD is also rational. And since the (square) on CD does not have to the (square) on DA the ratio which a square number (has) to a square number, CD (is) thus incommensurable in length with DA [Prop. 10.9]. Thus, CD and DA are rational (straight-lines which are) commensurable in square only. Thus, AC is an apotome [Prop. 10.73]. Again, since AB has been cut in extreme and mean ratio, and AC is the greater piece, the (rectangle contained) by AB and BC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. Thus, the (square) on the apotome AC , applied to the rational (straight-line) AB , makes BC as width. And the (square) on an apotome, applied to a rational (straight-line), makes a first apotome as width [Prop. 10.97]. Thus, CB is a first apotome. And CA was also shown (to be) an apotome.

Ἐὰν ἄρα εὐθεῖα ῥητὴ ἄκρον καὶ μέσον λόγον τμηθῇ, ἐκάτερον τῶν τμημάτων ἄλογός ἐστιν ἡ καλουμένη ἀποτομή· ὅπερ ἔδει δεῖχαι.

Thus, if a rational straight-line is cut in extreme and mean ratio then each of the pieces is that irrational (straight-line) called an apotome.

7.

Proposition 7

Ἐὰν πενταγώνον ἰσοπλεύρον αἱ τρεῖς γωνίαι ᾗτοι αἱ κατὰ τὸ ἐχθὲς ἢ αἱ μὴ κατὰ τὸ ἐχθὲς ἴσαι ᾖσιν, ἰσογώνιον ἔσται τὸ πεντάγωνον.

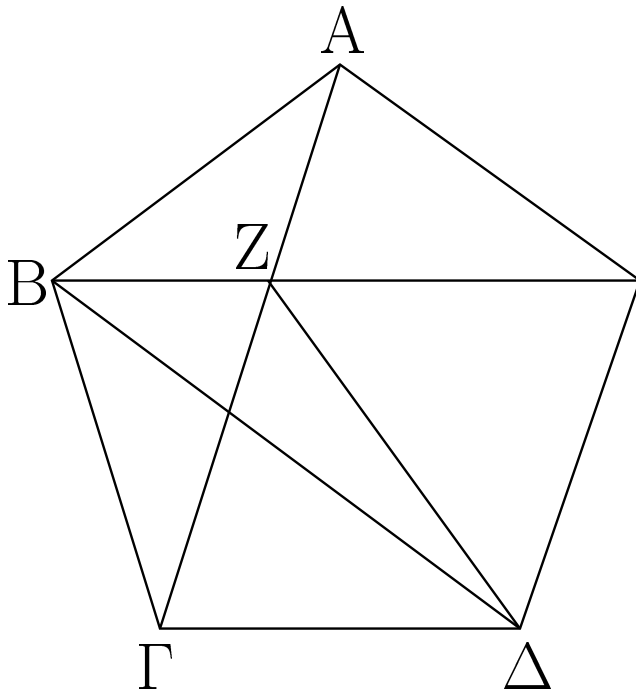
If three angles, either consecutive or not consecutive, of an equilateral pentagon are equal then the pentagon will be equiangular.

Πενταγώνον γάρ ἰσοπλεύρον τοῦ $ΑΒΓΔΕ$ αἱ τρεῖς γωνίαι

For let three angles of the equilateral pentagon $ABCDE$ —

πρότερον αἱ κατὰ τὸ ἐχθῆς αἱ πρὸς τοῖς A, B, Γ ἴσαι ἀλλήλαις ἔστωσαν· λέγω, ὅτι ἰσογώνιον ἐστὶ τὸ $AB\Gamma\Delta E$ πεντάγωνον.

Ἐπεξεύθθωσαν γὰρ αἱ $AF, BE, Z\Delta$. καὶ ἐπεὶ δύο αἱ $\Gamma B, BA$ δυοὶ ταῖς BA, AE ἴσαι εἰσὶν ἑκατέρα ἑκατέρῃ, καὶ γωνία ἡ ὑπὸ ΓBA γωνία τῇ ὑπὸ BAE ἐστὶν ἴση, βάσεις ἄρα ἡ AF βάσει τῇ BE ἐστὶν ἴση, καὶ τὸ $AB\Gamma$ τρίγωνον τῷ ABE τριγώνῳ ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ ΓBA τῇ ὑπὸ BEA , ἡ δὲ ὑπὸ ABE τῇ ὑπὸ ΓAB · ὥστε καὶ πλευρὰ ἡ AZ πλευρᾷ τῇ BZ ἐστὶν ἴση. ἐδείκθη δὲ καὶ ὅλη ἡ AF ὅλη τῇ BE ἴση· καὶ λοιπὴ ἄρα ἡ $Z\Gamma$ λοιπῇ τῇ ZE ἐστὶν ἴση. ἐστὶ δὲ καὶ ἡ $\Gamma\Delta$ τῇ ΔE ἴση. δύο δὲ αἱ $Z\Gamma, \Gamma\Delta$ δυοὶ ταῖς $ZE, E\Delta$ ἴσαι εἰσὶν καὶ βάσεις αὐτῶν κοινὴ ἡ $Z\Delta$ · γωνία ἄρα ἡ ὑπὸ $Z\Gamma\Delta$ γωνία τῇ ὑπὸ $ZE\Delta$ ἐστὶν ἴση. ἐδείκθη δὲ καὶ ἡ ὑπὸ ΓBA τῇ ὑπὸ AEB ἴση· καὶ ὅλη ἄρα ἡ ὑπὸ $\Gamma\Delta E$ ὅλη τῇ ὑπὸ $AE\Delta$ ἴση. ἀλλ' ἡ ὑπὸ $\Gamma\Delta E$ ἴση ὑπόκειται ταῖς πρὸς τοῖς A, B γωνίαις· καὶ ἡ ὑπὸ $AE\Delta$ ἄρα ταῖς πρὸς τοῖς A, B γωνίαις ἴση ἐστίν. ὁμοίως δὲ δείκνυσθαι, ὅτι καὶ ἡ ὑπὸ $\Gamma\Delta E$ γωνία ἴση ἐστὶ ταῖς πρὸς τοῖς A, B, Γ γωνίαις· ἰσογώνιον ἄρα ἐστὶ τὸ $AB\Gamma\Delta E$ πεντάγωνον.

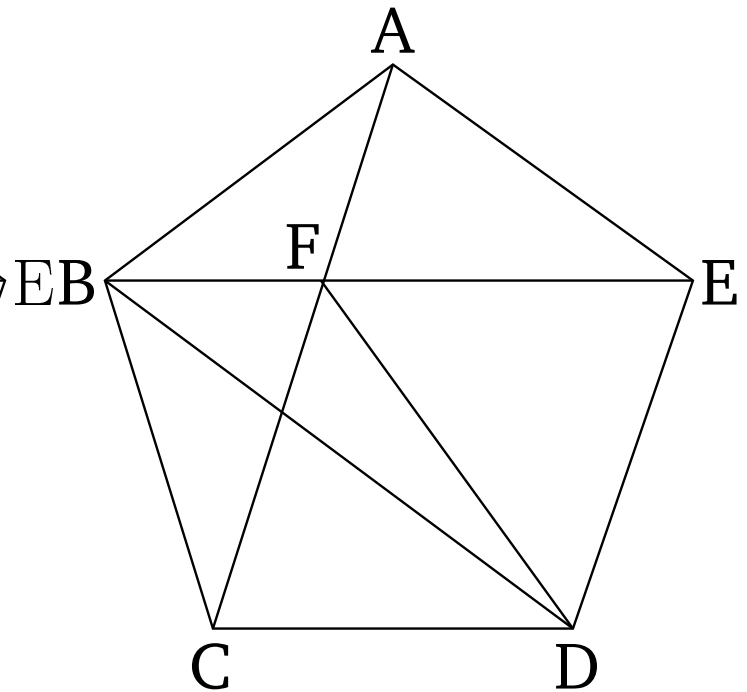


Ἀλλὰ δὴ μὴ ἔστωσαν ἴσαι αἱ κατὰ τὸ ἐχθῆς γωνίαι, ἀλλ' ἔστωσαν ἴσαι αἱ πρὸς τοῖς A, Γ, Δ σημείοις· λέγω, ὅτι καὶ οὕτως ἰσογώνιον ἐστὶ τὸ $AB\Gamma\Delta E$ πεντάγωνον.

Ἐπεξεύθθω γὰρ ἡ $B\Delta$. καὶ ἐπεὶ δύο αἱ BA, AE δυοὶ ταῖς $B\Gamma, \Gamma\Delta$ ἴσαι εἰσὶ καὶ γωνίαξ ἴσαξ περιέθουσιν, βάσεις ἄρα ἡ BE βάσει τῇ $B\Delta$ ἴση ἐστίν, καὶ τὸ ABE τρίγωνον τῷ $B\Gamma\Delta$ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις

first of all, the consecutive (angles) at A, B , and C —be equal to one another. I say that pentagon $ABCDE$ is equiangular.

For let AC, BE , and FD be joined. And since the two (straight-lines) CB and BA are equal to the two (straight-lines) BA and AE , respectively, and angle CBA is equal to angle BAE , base AC is thus equal to base BE , and triangle ABC equal to triangle ABE , and the remaining angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4], (that is), BCA (equal) to BEA , and ABE to CAB . And hence side AF is also equal to side BF [Prop. 1.6]. And the whole of AC was also shown (to be) equal to the whole of BE . Thus, the remainder FC is also equal to the remainder FE . And CD is also equal to DE . So, the two (straight-lines) FC and CD are equal to the two (straight-lines) FE and ED (respectively). And FD is their common base. Thus, angle FCD is equal to angle FED [Prop. 1.8]. And BCA was also shown (to be) equal to AEB . And thus the whole of BCD (is) equal to the whole of AED . But, (angle) BCD was assumed (to be) equal to the angles at A and B . Thus, (angle) AED is also equal to the angles at A and B . So, similarly, we can show that angle CDE is also equal to the angles at A, B, C . Thus, pentagon $ABCDE$ is equiangular.



And so let consecutive angles not be equal, but let the (angles) at points A, C , and D be equal. I say that pentagon $ABCDE$ is also equiangular in this case.

For let BD be joined. And since the two (straight-lines) BA and AE are equal to the (straight-lines) BC and CD , and they contain equal angles, base BE is thus equal to base BD , and triangle ABE is equal to triangle BCD , and the remaining

ἴσαι ἔσονται, ὅφ' ἂν αἱ ἴσαι πλευраὶ ὑποτείνουσιν· ἴση ἄρα ἔστιν ἡ ὑπὸ AEB γωνία τῇ ὑπὸ $ΓΔΒ$. ἔστι δὲ καὶ ἡ ὑπὸ $ΒΕΔ$ γωνία τῇ ὑπὸ $ΒΔΕ$ ἴση, ἐπεὶ καὶ πλευρὰ ἡ $ΒΕ$ πλευρᾷ τῇ $ΒΔ$ ἔστιν ἴση. καὶ ὅλη ἄρα ἡ ὑπὸ $ΑΕΔ$ γωνία ὅλη τῇ ὑπὸ $ΓΔΕ$ ἔστιν ἴση. ἀλλὰ ἡ ὑπὸ $ΓΔΕ$ ταῖς πρὸς τοῖς $A, Γ$ γωνίαις ὑπόκειται ἴση καὶ ἡ ὑπὸ $ΑΕΔ$ ἄρα γωνία ταῖς πρὸς τοῖς $A, Γ$ ἴση ἔστιν. διὰ τὰ αὐτὰ δὲ καὶ ἡ ὑπὸ $ΑΒΓ$ ἴση ἔστι ταῖς πρὸς τοῖς $A, Γ, Δ$ γωνίαις. ἰσογώνιον ἄρα ἔστι τὸ $ΑΒΓΔΕ$ πεντάγωνον· ὅπερ ἔδει δεῖξαι.

angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4]. Thus, angle AEB is equal to (angle) CDB . And angle BED is also equal to (angle) BDE , since side BE is also equal to side BD [Prop. 1.5]. Thus, the whole angle AED is also equal to the whole (angle) CDE . But, (angle) CDE was assumed (to be) equal to the angles at A and C . Thus, angle AED is also equal to the (angles) at A and C . So, for the same (reasons), (angle) ABC is also equal to the angles at A, C , and D . Thus, pentagon $ABCDE$ is equiangular. (Which is) the very thing it was required to show.

8.

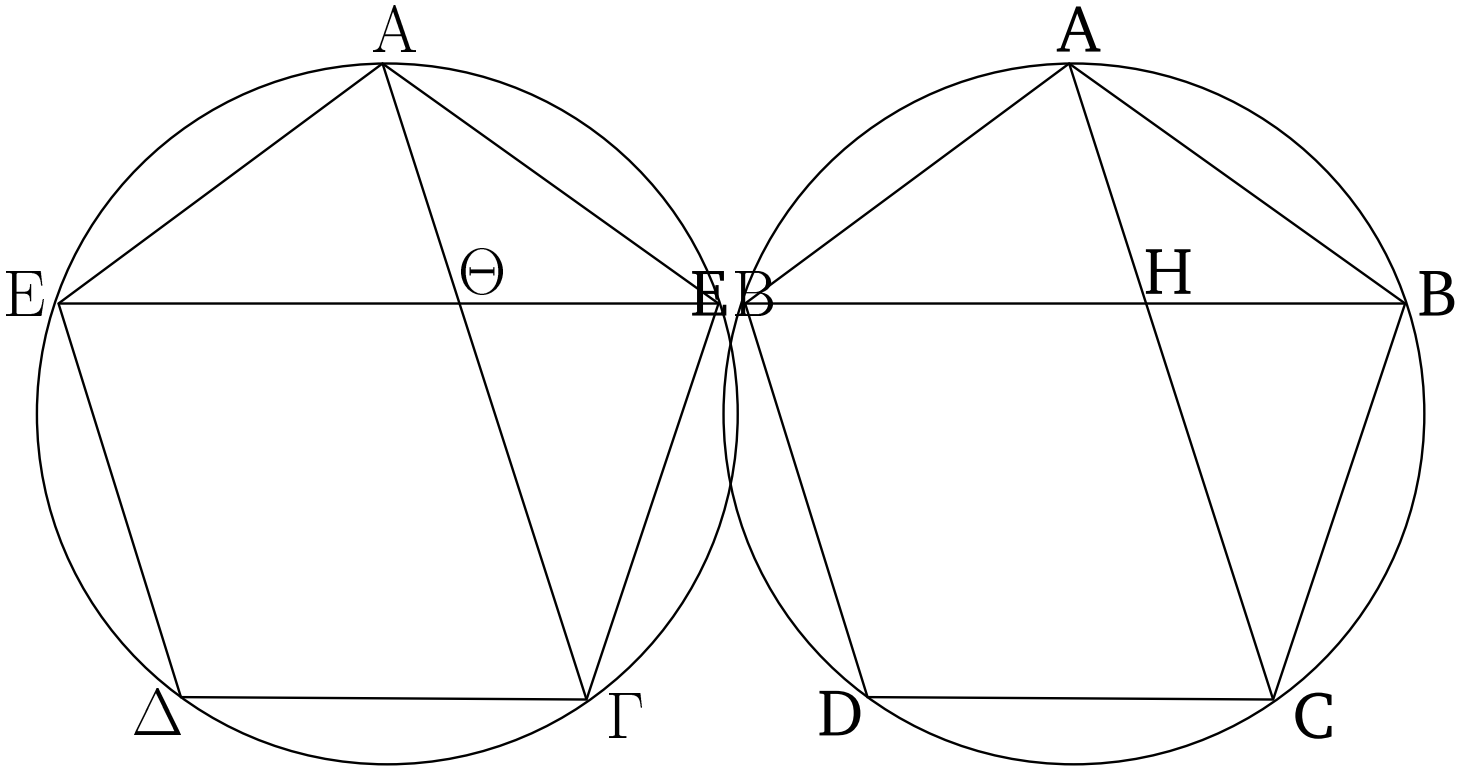
Proposition 8

Ἐάν πενταγώνον ἰσοπλεύρον καὶ ἰσογώνιον τὰς κατὰ τὸ ἐχθὲς δύο γωνίας ὑποτείνωσιν εὐθεῖαι, ἄκρον καὶ μέσον λόγον τέμνουσιν ἀλλήλας, καὶ τὰ μείζονα αὐτῶν τμήματα ἴσα ἔσθι τῇ τοῦ πενταγώνου πλευρᾷ.

If straight-lines subtend two consecutive angles of an equilateral and equiangular pentagon then they cut one another in extreme and mean ratio, and their greater pieces are equal to the sides of the pentagon.

Πενταγώνον γὰρ ἰσοπλεύρον καὶ ἰσογώνιον τοῦ $ΑΒΓΔΕ$ δύο γωνίας τὰς κατὰ τὸ ἐχθὲς τὰς πρὸς τοῖς A, B ὑποτείνεωσαν εὐθεῖαι αἱ $ΑΓ, ΒΕ$ τέμνουσαι ἀλλήλας κατὰ τὸ $Θ$ σημεῖον· λέγω, ὅτι ἑκατέρα αὐτῶν ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ $Θ$ σημεῖον, καὶ τὰ μείζονα αὐτῶν τμήματα ἴσα ἔσθι τῇ τοῦ πενταγώνου πλευρᾷ.

For let the two straight-lines, AC and BE , cutting one another at point H , have subtended two consecutive angles, at A and B (respectively), of the equilateral and equiangular pentagon $ABCDE$. I say that each of them has been cut in extreme and mean ratio at point H , and that their greater pieces are equal to the sides of the pentagon.



Περιγεγράφθω γὰρ περὶ τὸ $ΑΒΓΔΕ$ πεντάγωνον κύκλος ὁ $ΑΒΓΔΕ$. καὶ ἐπεὶ δύο εὐθεῖαι αἱ $ΕΑ, ΑΒ$ δυοὶ ταῖς $ΑΒ, ΒΓ$ ἴσαι εἰσὶ καὶ γωνίαξ ἴσαξ περιέθουσιν, βάσειξ ἄρα ἡ $ΒΕ$ βάσει τῇ $ΑΓ$ ἴση ἔστιν, καὶ τὸ $ΑΒΕ$ τρίγωνον τῷ $ΑΒΓ$ τριγώνῳ ἴσον

For let the circle $ABCDE$ be circumscribed about pentagon $ABCDE$ [Prop. 4.14]. And since the two straight-lines EA and AB are equal to the two (straight-lines) AB and BC (respectively), and they contain equal angles, the base BE is

ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρω ἑκατέρω, ὅφ' ἂν αἱ ἴσαι πλευραὶ ὑποτείνουσιν. ἴση ἄρα ἐστὶν ἡ ὑπὸ $BA\Gamma$ γωνία τῇ ὑπὸ ABE . διπλὴ ἄρα ἡ ὑπὸ $A\Theta E$ τῆς ὑπὸ $BA\Theta$. ἔστι δὲ καὶ ἡ ὑπὸ EAF τῆς ὑπὸ $BA\Gamma$ διπλὴ, ἐπειδὴ περ καὶ περιφέρεια ἡ $E\Delta\Gamma$ περιφερείας τῆς ΓB ἐστὶ διπλὴ· ἴση ἄρα ἡ ὑπὸ $\Theta A E$ γωνία τῇ ὑπὸ $A\Theta E$. ὥστε καὶ ἡ ΘE εὐθεῖα τῇ EA , τοντέστι τῇ AB ἐστὶν ἴση. καὶ ἐπεὶ ἴση ἐστὶν ἡ BA εὐθεῖα τῇ AE , ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ ABE τῇ ὑπὸ AEB . ἀλλὰ ἡ ὑπὸ ABE τῇ ὑπὸ $BA\Theta$ ἐδείχθη ἴση· καὶ ἡ ὑπὸ BEA ἄρα τῇ ὑπὸ $BA\Theta$ ἐστὶν ἴση. καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε ABE καὶ τοῦ $AB\Theta$ ἐστὶν ἡ ὑπὸ ABE . λοιπὴ ἄρα ἡ ὑπὸ BAE γωνία λοιπὴ τῇ ὑπὸ $A\Theta B$ ἐστὶν ἴση· ἰσογώνιον ἄρα ἐστὶ τὸ ABE τρίγωνον τῷ $AB\Theta$ τριγώνῳ· ἀνάλογον ἄρα ἐστὶν ὡς ἡ EB πρὸς τὴν BA , οὕτως ἡ AB πρὸς τὴν $B\Theta$. ἴση δὲ ἡ BA τῇ $E\Theta$. ὡς ἄρα ἡ BE πρὸς τὴν $E\Theta$, οὕτως ἡ $E\Theta$ πρὸς τὴν ΘB . μείζων δὲ ἡ BE τῆς $E\Theta$. μείζων ἄρα καὶ ἡ $E\Theta$ τῆς ΘB . ἡ BE ἄρα ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Θ , καὶ τὸ μείζον τμήμα τὸ ΘE ἴσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ. ὁμοίως δὲ δεῖξαι, ὅτι καὶ ἡ AF ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Θ , καὶ τὸ μείζον αὐτῆς τμήμα ἡ $\Gamma\Theta$ ἴσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ· ὅπερ ἔδει δεῖξαι.

9.

Ἐάν ἡ τοῦ ἑξαγώνου πλευρὰ καὶ ἡ τοῦ δεκαγώνου τῶν εἰς τὸν αὐτὸν κύκλον ἐγγραφόμενων συντεθῶσιν, ἡ ὅλη εὐθεῖα ἄκρον καὶ μέσον λόγον τέτμηται, καὶ τὸ μείζον αὐτῆς τμήμα ἐστὶν ἡ τοῦ ἑξαγώνου πλευρὰ.

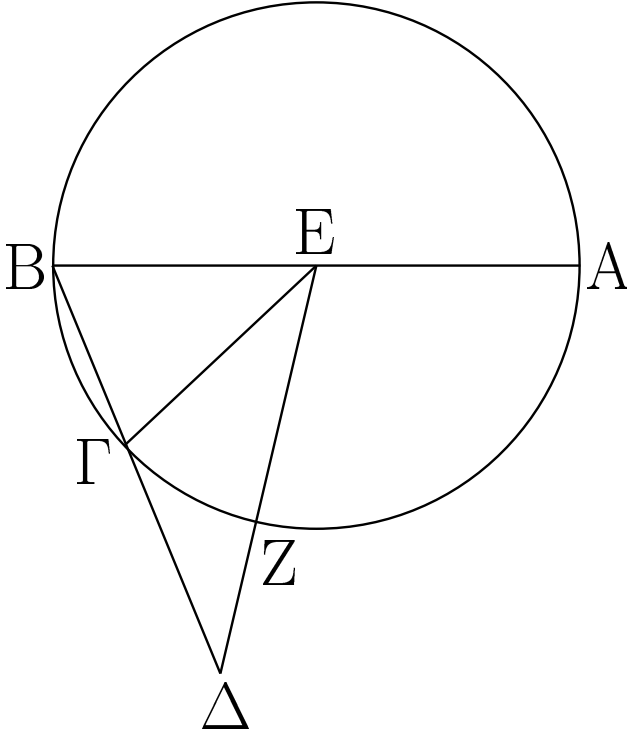
Ἐστω κύκλος ὁ $AB\Gamma$, καὶ τῶν εἰς τὸν $AB\Gamma$ κύκλον ἐγγραφόμενων σθημάτων, δεκαγώνον μὲν ἔστω πλευρὰ ἡ $B\Gamma$, ἑξαγώνον δὲ ἡ $\Gamma\Delta$, καὶ ἔστωσαν ἐπ' εὐθείας· λέγω, ὅτι ἡ ὅλη εὐθεῖα ἡ $B\Delta$ ἄκρον καὶ μέσον λόγον τέτμηται, καὶ τὸ μείζον αὐτῆς τμήμα ἐστὶν ἡ $\Gamma\Delta$.

thus equal to the base AC , and triangle ABE is equal to triangle ABC , and the remaining angles will be equal to the remaining angles, respectively, which the equal sides subtend [Prop. 1.4]. Thus, angle BAC is equal to (angle) ABE . Thus, (angle) AHE (is) double (angle) BAH [Prop. 1.32]. And EAC is also double BAC , inasmuch as circumference EDC is also double circumference CB [Props. 3.28, 6.33]. Thus, angle HAE (is) equal to (angle) AHE . Hence, straight-line HE is also equal to (straight-line) EA —that is to say, to (straight-line) AB [Prop. 1.6]. And since straight-line BA is equal to AE , angle ABE is also equal to AEB [Prop. 1.5]. But, ABE was shown (to be) equal to BAH . Thus, BEA is also equal to BAH . And (angle) ABE is common to the two triangles ABE and ABH . Thus, the remaining angle BAE is equal to the remaining (angle) AHB [Prop. 1.32]. Thus, triangle ABE is equiangular to triangle ABH . Thus, proportionally, as EB is to BA , so AB (is) to BH [Prop. 6.4]. And BA (is) equal to EH . Thus, as BE (is) to EH , so EH (is) to HB . And BE (is) greater than EH . EH (is) thus also greater than HB [Prop. 5.14]. Thus, BE has been cut in extreme and mean ratio at H , and the greater piece HE is equal to the side of the pentagon. So, similarly, we can show that AC has also been cut in extreme and mean ratio at H , and that its greater piece CH is equal to the side of the pentagon. (Which is) the very thing it was required to show.

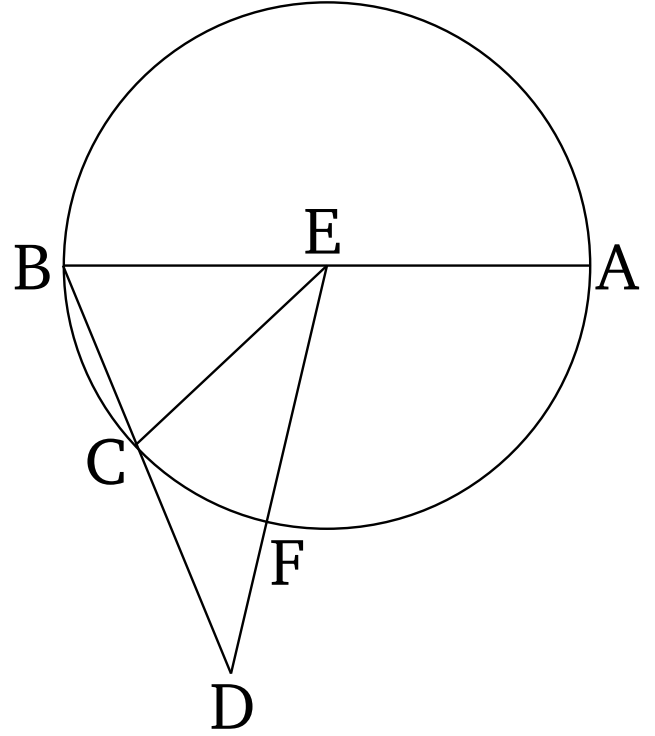
Proposition 9

If the side of a hexagon and of a decagon inscribed in the same circle are added together then the whole straight-line has been cut in extreme and mean ratio (at the junction point), and its greater piece is the side of the hexagon.[†]

Let ABC be a circle. And of the figures inscribed in circle ABC , let BC be the side of a decagon, and CD (the side) of a hexagon. And let them be (laid down) straight-on (to one another). I say that the whole straight-line BD has been cut in extreme and mean ratio (at C), and that CD is its greater piece.



Εἰλήφθω γάρ τὸ κέντρον τοῦ κύκλου τὸ E σημείον, καὶ ἐπεξεύθθωσαν αἱ EB , EG , ED , καὶ διήθθω ἡ BE ἐπὶ τὸ A . ἐπεὶ δεκαγώνου ἰσοπλευρόν πλευρά ἐστὶν ἡ $BΓ$, πενταπλασίον ἄρα ἡ $AΓB$ περιφέρεια τῆς $BΓ$ περιφερείας· τετραπλασίον ἄρα ἡ $AΓ$ περιφέρεια τῆς $ΓB$. ὥς δὲ ἡ $AΓ$ περιφέρεια πρὸς τὴν $ΓB$, οὕτως ἡ ὑπὸ AEG γωνία πρὸς τὴν ὑπὸ $ΓEB$ · τετραπλασίον ἄρα ἡ ὑπὸ AEG τῆς ὑπὸ $ΓEB$. καὶ ἐπεὶ ἴση ἡ ὑπὸ $EBΓ$ γωνία τῇ ὑπὸ $ΕΓB$, ἡ ἄρα ὑπὸ AEG γωνία διπλασία ἐστὶ τῆς ὑπὸ $ΕΓB$. καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΕΓ$ εὐθεῖα τῇ $ΓΔ$ · ἑκατέρα γὰρ αὐτῶν ἴση ἐστὶ τῇ τοῦ ἑξαγώνου πλευρᾷ τοῦ εἰς τὸν $ABΓ$ κύκλον [ἐγγραφομένου]· ἴση ἐστὶ καὶ ἡ ὑπὸ $ΓΕΔ$ γωνία τῇ ὑπὸ $ΓΔΕ$ γωνίᾳ· διπλασία ἄρα ἡ ὑπὸ $ΕΓB$ γωνία τῆς ὑπὸ $ΕΔΓ$. ἀλλὰ τῆς ὑπὸ $ΕΓB$ διπλασία ἐδείκθη ἡ ὑπὸ AEG · τετραπλασία ἄρα ἡ ὑπὸ AEG τῆς ὑπὸ $ΕΔΓ$. ἐδείκθη δὲ καὶ τῆς ὑπὸ $ΒΕΓ$ τετραπλασία ἡ ὑπὸ AEG · ἴση ἄρα ἡ ὑπὸ $ΕΔΓ$ τῇ ὑπὸ $ΒΕΓ$. κοινὴ δὲ τῶν δύο τριγώνων, τοῦ τε $ΒΕΓ$ καὶ τοῦ $ΒΕΔ$, ἡ ὑπὸ $ΕΒΔ$ γωνία· καὶ λοιπὴ ἄρα ἡ ὑπὸ $ΒΕΔ$ τῇ ὑπὸ $ΕΓB$ ἐστὶν ἴση· ἰσογώνιον ἄρα ἐστὶ τὸ $ΕΒΔ$ τρίγωνον τῷ $ΕΒΓ$ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὥς ἡ $ΔB$ πρὸς τὴν BE , οὕτως ἡ EB πρὸς τὴν $BΓ$. ἴση δὲ ἡ EB τῇ $ΓΔ$. ἐστὶν ἄρα ὥς ἡ $BΔ$ πρὸς τὴν $ΔΓ$, οὕτως ἡ $ΔΓ$ πρὸς τὴν $ΓB$. μείζων δὲ ἡ $BΔ$ τῆς $ΔΓ$ · μείζων ἄρα καὶ ἡ $ΔΓ$ τῆς $ΓB$. ἡ $BΔ$ ἄρα εὐθεῖα ἄκρον καὶ μέσον λόγον τέτμηται [κατὰ τὸ $Γ$], καὶ τὸ μείζον τμήμα αὐτῆς ἐστὶν ἡ $ΔΓ$ · ὅπερ ἔδει δεῖξαι.



For let the center of the circle, point E , be found [Prop. 3.1], and let EB , EC , and ED be joined, and let BE be drawn across to A . Since BC is a side on an equilateral decagon, circumference ACB (is) thus five times circumference BC . Thus, circumference AC (is) four times CB . And as circumference AC (is) to CB , so angle AEC (is) to CEB [Prop. 6.33]. Thus, (angle) AEC (is) four times CEB . And since angle EBC (is) equal to ECB [Prop. 1.5], angle AEC is thus double ECB [Prop. 1.32]. And since straight-line EC is equal to CD —for each of them is equal to the side of the hexagon [inscribed] in circle ABC [Prop. 4.15 corr.]—angle CED is also equal to angle CDE [Prop. 1.5]. Thus, angle ECB (is) double EDC [Prop. 1.32]. But, AEC was shown (to be) double ECB . Thus, AEC (is) four times EDC . And AEC was also shown (to be) four times BEC . Thus, EDC (is) equal to BEC . And angle EBD (is) common to the two triangles BEC and BED . Thus, the remaining (angle) BED is equal to the (remaining angle) ECB [Prop. 1.32]. Thus, triangle EBD is equiangular to triangle EBC . Thus, proportionally, as DB is to BE , so EB (is) to BC [Prop. 6.4]. And EB (is) equal to CD . Thus, as BD is to DC , so DC (is) to CB . And BD (is) greater than DC . Thus, DC (is) also greater than CB [Prop. 5.14]. Thus, the straight-line BD has been cut in extreme and mean ratio [at C], and DC is its greater piece. (Which is), the very thing it was required to show.

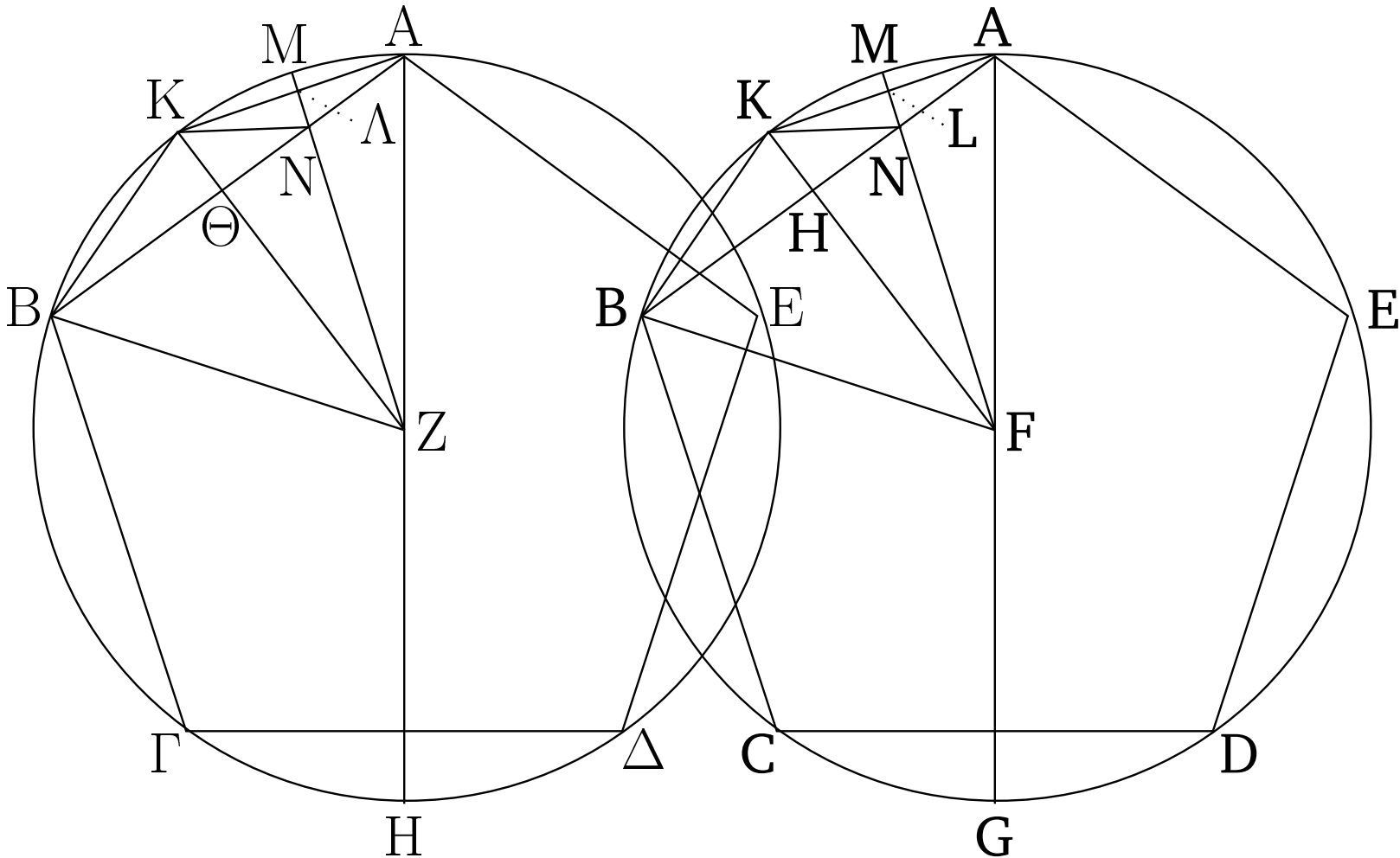
[†] If the circle is of unit radius then the side of the hexagon is 1, whereas the side of the decagon is $(1/2)(\sqrt{5} - 1)$.

10.

Proposition 10

Ἐάν εἷς κύκλον πεντάγωνον ἰσόπλευρον ἐγγράφῃ, ἡ τοῦ πενταγώνου πλευρὰ δύναται τὴν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἷς τὸν αὐτὸν κύκλον ἐγγραφομένων.

If an equilateral pentagon is inscribed in a circle then the square on the side of the pentagon is (equal to) the (sum of the squares) on the (sides) of the hexagon and of the decagon inscribed in the same circle.[†]



Ἐστω κύκλος ὁ ABΓΔE, καὶ εἷς τὸ ABΓΔE κύκλον πεντάγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ABΓΔE. λέγω, ὅτι ἡ τοῦ ABΓΔE πενταγώνου πλευρὰ δύναται τὴν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου πλευρὰν τῶν εἷς τὸν ABΓΔE κύκλον ἐγγραφομένων.

Εἰλήφθω γάρ τὸ κέντρον τοῦ κύκλου τὸ Z σημεῖον, καὶ ἐπιξενυθεῖσα ἡ AZ διήθθω ἐπὶ τὸ H σημεῖον, καὶ ἐπεξεύθθω ἡ ZB, καὶ ἀπὸ τοῦ Z ἐπὶ τὴν AB κάθετοξ ἤθθω ἡ ZΘ, καὶ διήθθω ἐπὶ τὸ K, καὶ ἐπεξεύθθωσαν αἱ AK, KB, καὶ πάλιν ἀπὸ τοῦ Z ἐπὶ τὴν AK κάθετοξ ἤθθω ἡ ZΛ, καὶ διήθθω ἐπὶ τὸ M, καὶ ἐπεξεύθθω ἡ KN.

Ἐπεὶ ἴση ἐστὶν ἡ ABΓH περιφέρεια τῇ AEΔH περιφερείᾳ, ὧν ἡ ABΓ τῇ AEΔ ἐστὶν ἴση, λοιπὴ ἄρα ἡ ΓH περιφέρει-α λοιπῇ τῇ HΔ ἐστὶν ἴση. πενταγώνου δὲ ἡ ΓΔ· δεκαγώνου

Let $ABCDE$ be a circle. And let the equilateral pentagon $ABCDE$ be inscribed in circle $ABCDE$. I say that the square on the side of pentagon $ABCDE$ is the (sum of the squares) on the sides of the hexagon and of the decagon inscribed in circle $ABCDE$.

For let the center of the circle, point F , be found [Prop. 3.1]. And, AF being joined, let it be drawn across to point G . And let FB be joined. And let FH be drawn from F perpendicular to AB . And let it be drawn across to K . And let AK and KB be joined. And, again, let FL have been drawn from F perpendicular to AK . And let it be drawn across to M . And let KN be joined.

Since circumference $ABCG$ is equal to circumference $AEDG$, of which ABC is equal to AED , the remaining cir-

ἄρα ἡ GH . καὶ ἐπεὶ ἴση ἐστὶν ἡ ZA τῇ ZB , καὶ κάθετοξ ἡ $ZΘ$, ἴση ἄρα καὶ ἡ ὑπὸ AZK γωνία τῇ ὑπὸ KZB . ὥστε καὶ περιφέρεια ἡ AK τῇ KB ἐστὶν ἴση· διπλῇ ἄρα ἡ AB περιφέρεια τῇ BK περιφερείᾳ· δεκαγώνον ἄρα πλευρὰ ἐστὶν ἡ AK εὐθεία. διὰ τὰ αὐτὰ δὴ καὶ ἡ AK τῇ KM ἐστὶ διπλῇ. καὶ ἐπεὶ διπλῇ ἐστὶν ἡ AB περιφέρεια τῇ BK περιφερείᾳ, ἴση δὲ ἡ $ΓΔ$ περιφέρεια τῇ AB περιφερείᾳ, διπλῇ ἄρα καὶ ἡ $ΓΔ$ περιφέρεια τῇ BK περιφερείᾳ. ἐστὶ δὲ ἡ $ΓΔ$ περιφέρεια καὶ τῇ GH διπλῇ· ἴση ἄρα ἡ GH περιφέρεια τῇ BK περιφερείᾳ. ἀλλὰ ἡ BK τῇ KM ἐστὶ διπλῇ, ἐπεὶ καὶ ἡ KA καὶ ἡ GH ἄρα τῇ KM ἐστὶ διπλῇ. ἀλλὰ μὴν καὶ ἡ GB περιφέρεια τῇ BK περιφερείᾳ ἐστὶ διπλῇ· ἴση γάρ ἡ GB περιφέρεια τῇ BA . καὶ ὅλη ἄρα ἡ HB περιφέρεια τῇ BM ἐστὶ διπλῇ· ὥστε καὶ γωνία ἡ ὑπὸ HZB γωνίαξ τῇ ὑπὸ BZM [ἐστὶ] διπλῇ. ἐστὶ δὲ ἡ ὑπὸ HZB καὶ τῇ ὑπὸ ZAB διπλῇ· ἴση γάρ ἡ ὑπὸ ZAB τῇ ὑπὸ ABZ . καὶ ἡ ὑπὸ BZN ἄρα τῇ ὑπὸ ZAB ἐστὶν ἴση. κοινὴ δὲ τῶν δύο τριγώνων, τοῦ τε ABZ καὶ τοῦ BZN , ἡ ὑπὸ ABZ γωνία· λοιπὴ ἄρα ἡ ὑπὸ AZB λοιπῇ τῇ ὑπὸ BNZ ἐστὶν ἴση· ἰσογώνον ἄρα ἐστὶ τὸ ABZ τρίγωνον τῷ BZN τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ AB εὐθεία πρὸς τὴν BZ , οὕτως ἡ ZB πρὸς τὴν BN · τὸ ἄρα ὑπὸ τῶν ABN ἴσον ἐστὶ τῷ ἀπὸ BZ . πάλιν ἐπεὶ ἴση ἐστὶν ἡ AA τῇ AK , κοινὴ δὲ καὶ πρὸς ὀρθὰς ἡ AN , βάσειξ ἄρα ἡ KN βάσει τῇ AN ἐστὶν ἴση· καὶ γωνία ἄρα ἡ ὑπὸ AKN γωνίαξ τῇ ὑπὸ LAN ἐστὶν ἴση. ἀλλὰ ἡ ὑπὸ LAN τῇ ὑπὸ KBN ἐστὶν ἴση· καὶ ἡ ὑπὸ AKN ἄρα τῇ ὑπὸ KBN ἐστὶν ἴση. καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε AKB καὶ τοῦ AKN ἡ πρὸς τῷ A . λοιπὴ ἄρα ἡ ὑπὸ AKB λοιπῇ τῇ ὑπὸ KNA ἐστὶν ἴση· ἰσογώνον ἄρα ἐστὶ τὸ KBA τρίγωνον τῷ KNA τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ BA εὐθεία πρὸς τὴν AK , οὕτως ἡ KA πρὸς τὴν AN · τὸ ἄρα ὑπὸ τῶν BAN ἴσον ἐστὶ τῷ ἀπὸ τῇ AK . ἐδείκθη δὲ καὶ τὸ ὑπὸ τῶν ABN ἴσον τῷ ἀπὸ τῇ BZ · τὸ ἄρα ὑπὸ τῶν ABN μετὰ τοῦ ὑπὸ BAN , ὅπερ ἐστὶ τὸ ἀπὸ τῇ BA , ἴσον ἐστὶ τῷ ἀπὸ τῇ BZ μετὰ τοῦ ἀπὸ τῇ AK . καὶ ἐστὶν ἡ μὲν BA πενταγώνου πλευρὰ, ἡ δὲ BZ ἑξαγώνου, ἡ δὲ AK δεκαγώνου.

Ἡ ἄρα τοῦ πενταγώνου πλευρὰ δύναται τὴν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰς τὸν αὐτὸν κύκλον ἐγγραφομένων· ὅπερ εἶδει δεῖχαι.

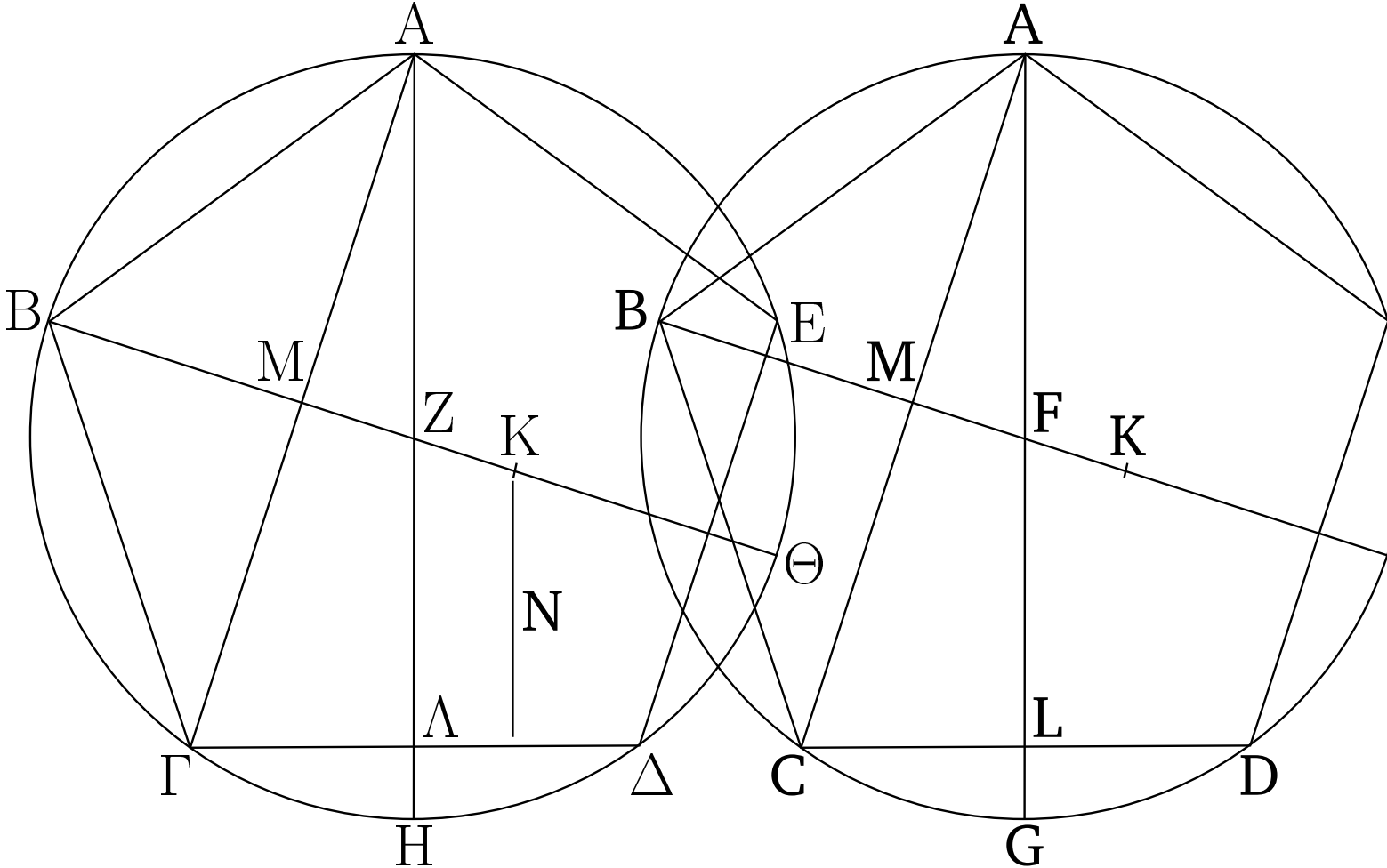
circumference CG is thus equal to the remaining (circumference) GD . And CD (is the side) of the pentagon. CG (is) thus (the side) of the decagon. And since FA is equal to FB , and FH is perpendicular (to AB), angle AFK (is) thus also equal to KFB [Props. 1.5, 1.26]. Hence, circumference AK is also equal to KB [Prop. 3.26]. Thus, circumference AB (is) double circumference BK . Thus, straight-line AK is the side of the decagon. So, for the same (reasons, circumference) AK is also double KM . And since circumference AB is double circumference BK , and circumference CD (is) equal to circumference AB , circumference CD (is) thus also double circumference BK . And circumference CD is also double CG . Thus, circumference CG (is) equal to circumference BK . But, BK is double KM , since KA (is) also (double KM). Thus, (circumference) CG is also double KM . But, indeed, circumference CB is also double circumference BK . For circumference CB (is) equal to BA . Thus, the whole circumference GB is also double BM . Hence, angle GFB [is] also double angle BFM [Prop. 6.33]. And GFB (is) also double FAB . For FAB (is) equal to ABF . Thus, BFN is also equal to FAB . And angle ABF (is) common to the two triangles ABF and BFN . Thus, the remaining (angle) AFB is equal to the remaining (angle) BNF [Prop. 1.32]. Thus, triangle ABF is equiangular to triangle BFN . Thus, proportionally, as straight-line AB (is) to BF , so FB (is) to BN [Prop. 6.4]. Thus, the (rectangle contained) by ABN is equal to the (square) on BF [Prop. 6.17]. Again, since AL is equal to LK , and LN is common and at right-angles (to KA), base KN is thus equal to base AN [Prop. 1.4]. And, thus, angle LKN is equal to angle LAN . But, LAN is equal to KBN [Props. 3.29, 1.5]. Thus, LKN is also equal to KBN . And the (angle) at A (is) common to the two triangles AKB and AKN . Thus, the remaining (angle) AKB is equal to the remaining (angle) KNA [Prop. 1.32]. Thus, triangle KBA is equiangular to triangle KNA . Thus, proportionally, as straight-line BA is to AK , so KA (is) to AN [Prop. 6.4]. Thus, the (rectangle contained) by BAN is equal to the (square) on AK [Prop. 6.17]. And the (rectangle contained) by ABN was also shown (to be) equal to the (square) on BF . Thus, the (rectangle contained) by ABN plus the (rectangle contained) by BAN , which is the (square) on BA [Prop. 2.2], is equal to the (square) on BF plus the (square) on AK . And BA is the side of the pentagon, and BF (the side) of the hexagon [Prop. 4.15 corr.], and AK (the side) of the decagon.

Thus, the square on the side of the pentagon (inscribed in a circle) is (equal to) the (sum of the squares) on the (sides) of the hexagon and of the decagon inscribed in the same circle.

† If the circle is of unit radius then the side of the pentagon is $(1/2) \sqrt{10 - 2\sqrt{5}}$.

Ἐάν εἷς κύκλον ῥητὴν ἔθοντα τὴν διάμετρον πεντάγωνον
ἰσόπλευρον ἐγγράφῃ, ἡ τοῦ πενταγώνου πλευρὰ ἄλογός ἐστιν
ἢ καλουμένη ἐλάσσων.

If an equilateral pentagon is inscribed in a circle which
has a rational diameter then the side of the pentagon is that
irrational (straight-line) called minor.



Εἷς γάρ κύκλον τὸν $ABΓΔE$ ῥητὴν ἔθοντα τὴν διάμετρον
πεντάγωνον ἰσόπλευρον ἐγγεγράφθω τὸ $ABΓΔE$ · λέγω, ὅτι ἡ
τοῦ $[ABΓΔE]$ πενταγώνου πλευρὰ ἄλογός ἐστιν ἢ καλουμένη
ἐλάσσων.

Εἰλήφθω γάρ τὸ κέντρον τοῦ κύκλου τὸ Z σημεῖον, καὶ
ἐπεζεύθωσαν αἱ AZ , ZB καὶ διήθωσαν ἐπὶ τὰ H , Θ σημεία,
καὶ ἐπεζεύθω ἡ AG , καὶ κείσθω τῆς AZ τέταρτον μέρος
ἡ ZK . ῥητὴ δὲ ἡ AZ · ῥητὴ ἄρα καὶ ἡ ZK . ἔστι δὲ καὶ ἡ
 BZ ῥητὴ· ὅλη ἄρα ἡ BK ῥητὴ ἐστίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ
 AGH περιφέρεια τῇ $AΔH$ περιφερείᾳ, ὥν ἡ $ABΓ$ τῇ $AEΔ$
ἐστὶν ἴση, λοιπὴ ἄρα ἡ $ΓH$ λοιπὴ τῇ $HΔ$ ἐστὶν ἴση. καὶ ἔαν
ἐπιζεύχωμεν τὴν AD , συνάγονται ὀρθαὶ αἱ πρὸς τῷ Λ γωνίαι,
καὶ διπλὴ ἡ $ΓΔ$ τῆς $ΓΛ$. διὰ τὰ αὐτὰ δὲ καὶ αἱ πρὸς τῷ M
ὀρθαὶ εἰσιν, καὶ διπλὴ ἡ AG τῆς $ΓM$. ἐπεὶ οὖν ἴση ἐστὶν ἡ
ὑπὸ $AAΓ$ γωνία τῇ ὑπὸ AMZ , κοινὴ δὲ τῶν δύο τριγώνων
τοῦ τε $AGΛ$ καὶ τοῦ AMZ ἡ ὑπὸ $AAΓ$, λοιπὴ ἄρα ἡ ὑπὸ

For let the equilateral pentagon $ABCDE$ be inscribed in
the circle $ABCDE$ which has a rational diameter. I say that
the side of pentagon $[ABCDE]$ is that irrational (straight-line)
called minor.

For let the center of the circle, point F , be found [Prop. 3.1].
And let AF and FB be joined. And let them be drawn across
to points G and H (respectively). And let AC be joined. And
let FK made (equal) to the fourth part of AF . And AF (is)
rational. FK (is) thus also rational. And BF is also rational.
Thus, the whole of BK is rational. And since circumference
 ACG is equal to circumference ADG , of which ABC is equal
to AED , the remainder CG is thus equal to the remainder GD .
And if we join AD then the angles at L are inferred (to be)
right-angles, and CD (is inferred to be) double CL [Prop. 1.4].
So, for the same (reasons), the (angles) at M are also right-
angles, and AC (is) double CM . Therefore, since angle ALC

ΑΓΑ λοιπὴ τῇ ὑπὸ ΜΖΑ ἔστιν ἴση· ἰσογώνιον ἄρα ἔστι τὸ ΑΓΑ τρίγωνον τῷ ΑΜΖ τριγώνῳ· ἀνάλογον ἄρα ἔστιν ὡς ἡ ΑΓ πρὸς ΓΑ, οὕτως ἡ ΜΖ πρὸς ΖΑ· καὶ τῶν ἡγουμένων τὰ διπλάσια· ὡς ἄρα ἡ τῆς ΑΓ διπλὴ πρὸς τὴν ΓΑ, οὕτως ἡ τῆς ΜΖ διπλὴ πρὸς τὴν ΖΑ. ὡς δὲ ἡ τῆς ΜΖ διπλὴ πρὸς τὴν ΖΑ, οὕτως ἡ ΜΖ πρὸς τὴν ἡμίσειαν τῆς ΖΑ· καὶ ὡς ἄρα ἡ τῆς ΑΓ διπλὴ πρὸς τὴν ΓΑ, οὕτως ἡ ΜΖ πρὸς τὴν ἡμίσειαν τῆς ΖΑ· καὶ τῶν ἐπομένων τὰ ἡμίσεια· ὡς ἄρα ἡ τῆς ΑΓ διπλὴ πρὸς τὴν ἡμίσειαν τῆς ΓΑ, οὕτως ἡ ΜΖ πρὸς τὸ τέτατρον τῆς ΖΑ. καὶ ἔστι τῆς μὲν ΑΓ διπλὴ ἡ ΔΓ, τῆς δὲ ΓΑ ἡμίσεια ἡ ΓΜ, τῆς δὲ ΖΑ τέτατρον μέρος ἡ ΖΚ· ἔστιν ἄρα ὡς ἡ ΔΓ πρὸς τὴν ΓΜ, οὕτως ἡ ΜΖ πρὸς τὴν ΖΚ. συνθέντι καὶ ὡς συναμφοτέρως ἡ ΔΓΜ πρὸς τὴν ΓΜ, οὕτως ἡ ΜΚ πρὸς ΚΖ· καὶ ὡς ἄρα τὸ ἀπὸ συναμφοτέρων τῆς ΔΓΜ πρὸς τὸ ἀπὸ ΓΜ, οὕτως τὸ ἀπὸ ΜΚ πρὸς τὸ ἀπὸ ΚΖ. καὶ ἐπεὶ τῆς ὑπὸ δύο πλευρὰς τοῦ πενταγώνου ὑποτείνουσῃς, οἷον τῆς ΑΓ, ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα ἴσον ἔστι τῇ τοῦ πενταγώνου πλευρᾷ, τουτέστι τῇ ΔΓ, τὸ δὲ μείζον τμήμα προσλαβὼν τὴν ἡμίσειαν τῆς ὅλης πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τῆς ὅλης, καὶ ἔστιν ὅλης τῆς ΑΓ ἡμίσεια ἡ ΓΜ, τὸ ἄρα ἀπὸ τῆς ΔΓΜ ὡς μιᾶς πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΓΜ. ὡς δὲ τὸ ἀπὸ τῆς ΔΓΜ ὡς μιᾶς πρὸς τὸ ἀπὸ τῆς ΓΜ, οὕτως ἐδείκθη τὸ ἀπὸ τῆς ΜΚ πρὸς τὸ ἀπὸ τῆς ΚΖ· πενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΜΚ τοῦ ἀπὸ τῆς ΚΖ. ῥητόν δὲ τὸ ἀπὸ τῆς ΚΖ· ῥητὴ γάρ ἡ διάμετρος· ῥητόν ἄρα καὶ τὸ ἀπὸ τῆς ΜΚ· ῥητὴ ἄρα ἔστιν ἡ ΜΚ [δυνάμει μόνον]. καὶ ἐπεὶ τετραπλασία ἔστιν ἡ ΒΖ τῆς ΖΚ, πενταπλασία ἄρα ἔστιν ἡ ΒΚ τῆς ΚΖ· εἰκοσιπενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΖ. πενταπλάσιον δὲ τὸ ἀπὸ τῆς ΜΚ τοῦ ἀπὸ τῆς ΚΖ· πενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ· τὸ ἄρα ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ ΚΜ λόγον οὐκ ἔθει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἔστιν ἡ ΒΚ τῇ ΚΜ μήκει. καὶ ἔστι ῥητὴ ἑκατέρα αὐτῶν. αἱ ΒΚ, ΚΜ ἄρα ῥηταὶ εἰσι δυνάμει μόνον σύμμετροι. ἔαν δὲ ἀπὸ ῥητῆς ῥητὴ ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὔσα τῇ ὅλῃ, ἡ λοιπὴ ἄλογός ἐστιν ἀποτομή· ἀποτομή ἄρα ἔστιν ἡ ΜΒ, προσαρμόζονσα δὲ αὐτῇ ἡ ΜΚ. λέγω δὴ, ὅτι καὶ τετάρτη. ὥ δὴ μείζον ἐστι τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ, ἐκείνῳ ἴσον ἔστω τὸ ἀπὸ τῆς Ν· ἡ ΒΚ ἄρα τῆς ΚΜ μείζον δύναται τῇ Ν. καὶ ἐπεὶ σύμμετρος ἐστιν ἡ ΚΖ τῇ ΖΒ, καὶ συνθέντι σύμμετρος ἐστι ἡ ΚΒ τῇ ΖΒ. ἀλλὰ ἡ ΒΖ τῇ ΒΘ σύμμετρος ἐστιν· καὶ ἡ ΒΚ ἄρα τῇ ΒΘ σύμμετρος ἐστιν. καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ, τὸ ἄρα ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ τῆς ΚΜ λόγον ἔθει, ὃν εἰς πρὸς ἕν. ἀναστρέψαντι ἄρα τὸ ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ τῆς Ν λόγον ἔχει, ὃν εἰς πρὸς δ, οὐκ ὃν τετράγωνος πρὸς τετράγωνον· ἀσύμμετρος ἄρα ἔστιν ἡ ΒΚ τῇ Ν· ἡ ΒΚ ἄρα τῆς ΚΜ μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἑαντῇ. ἐπεὶ οὖν ὅλη ἡ ΒΚ τῆς προσαρμοζούσης τῆς ΚΜ μείζον δύναται τῷ ἀπὸ ἀσυνμέτρου ἑαντῇ, καὶ ὅλη ἡ ΒΚ σύμμετρος ἐστι τῇ ἐκκεκμένη ῥητῇ τῇ ΒΘ, ἀποτομή ἄρα τετάρτη ἐστὶν ἡ ΜΒ. τὸ δὲ ὑπὸ ῥητῆς καὶ

(is) equal to AMF , and (angle) LAC (is) common to the two triangles ACL and AMF , the remaining (angle) ACL is thus equal to the remaining (angle) MFA [Prop. 1.32]. Thus, triangle ACL is equiangular to triangle AMF . Thus, proportionally, as LC (is) to CA , so MF (is) to FA [Prop. 6.4]. And (we can take) the doubles of the leading (magnitudes). Thus, as double LC (is) to CA , so double MF (is) to FA . And as double MF (is) to FA , so MF (is) to half of FA . And, thus, as double LC (is) to CA , so MF (is) to half of FA . And (we can take) the halves of the following (magnitudes). Thus, as double LC (is) to half of CA , so MF (is) to the fourth of FA . And DC is double LC , and CM half of CA , and FK the fourth part of FA . Thus, as DC is to CM , so MF (is) to FK . Via composition, as the sum of DCM (i.e., DC and CM) (is) to CM , so MK (is) to KF [Prop. 5.18]. And, thus, as the (square) on the sum of DCM (is) to the (square) on CM , so the (square) on MK (is) to the (square) on KF . And since the greater piece of a (straight-line) subtending two sides of a pentagon, such as AC , (which is) cut in extreme and mean ratio is equal to the side of the pentagon [Prop. 13.8]—that is to say, to DC —and the square on the greater piece added to half of the whole is five times the (square) on half of the whole [Prop. 13.1], and CM (is) half of the whole, AC , thus the (square) on DCM , (taken) as one, is five times the (square) on CM . And the (square) on DCM , (taken) as one, (is) to the (square) on CM , so the (square) on MK was shown (to be) to the (square) on KF . Thus, the (square) on MK (is) five times the (square) on KF . And the square on KF (is) rational. For the diameter (is) rational. Thus, the (square) on MK (is) also rational. Thus, MK is rational [in square only]. And since BF is four times FK , BK is thus five times KF . Thus, the (square) on BK (is) twenty-five times the (square) on KF . And the (square) on MK (is) five times the square on KF . Thus, the (square) on BK (is) five times the (square) on KM . Thus, the (square) on BK does not have to the (square) on KM the ratio which a square number (has) to a square number. Thus, BK is incommensurable in length with KM [Prop. 10.9]. And each of them is a rational (straight-line). Thus, BK and KM are rational (straight-lines) which are) commensurable in square only. And if from a rational (straight-line) a rational (straight-line) is subtracted, which is commensurable in square only with the whole, then the remainder is that irrational (straight-line called) an apotome [Prop. 10.73]. Thus, MB is an apotome, and MK its attachment. So, I say that (it is) also a fourth (apotome). So, let the (square) on N be (made) equal to that (magnitude) by which the (square) on BK is greater than the (square) on KM . Thus, the square on BK is greater than the (square) on KM by the (square) on N . And since KF is commensurable (in length) with FB then, via composition, KB is also commensurable (in length) with FB [Prop. 10.15]. But, BF is commensurable (in length) with BH . Thus, BK is also commensurable

ἀποτομῆς τετάρτης περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ ἄλογός ἐστιν, καλεῖται δὲ ἐλάττων. δύνανται δὲ τὸ ὑπὸ τῶν ΘBM ἢ AB διὰ τὸ ἐπιξεννυμένης τῆς $A\Theta$ ἰσογώνιον γίνεσθαι τὸ $AB\Theta$ τρίγωνον τῷ ABM τριγώνῳ καὶ εἶναι ὡς τὴν ΘB πρὸς τὴν BA , οὕτως τὴν AB πρὸς τὴν BM .

Ἡ ἄρα AB τοῦ πενταγώνου πλευρὰ ἄλογός ἐστιν ἡ καλουμένη ἐλάττων· ὅπερ ἔδει δεῖχαι.

(in length) with BH [Prop. 10.12]. And since the (square) on BK is five times the (square) on KM , the (square) on BK thus has to the (square) on KM the ratio which 5 (has) to one. Thus, via conversion, the (square) on BK has to the (square) on N the ratio which 5 (has) to 4 [Prop. 5.19 corr.], which is not (that) of a square (number) to a square (number). BK is thus incommensurable (in length) with N [Prop. 10.9]. Thus, the square on BK is greater than the (square) on KM by the (square) on (some straight-line which is) incommensurable (in length) with (BK) . Therefore, since the square on the whole, BK , is greater than the (square) on the attachment, KM , by the (square) on (some straight-line which is) incommensurable (in length) with (BK) , and the whole, BK , is commensurable (in length) with the (previously) laid down rational (straight-line) BH , MB is thus a fourth apotome [Def. 10.14]. And the rectangle contained by a rational (straight-line) and a fourth apotome is irrational, and its square-root is that irrational (straight-line) called minor [Prop. 10.94]. And the square on AB is the rectangle contained by HBM , on account of joining AH , (so that) triangle ABH becomes equiangular with triangle ABM [Prop. 6.8], and (proportionally) as HB is to BA , so AB (is) to BM .

Thus, the side AB of the pentagon is that irrational (straight-line) called minor.[†] (Which is) the very thing it was required to show.

[†] If the circle has unit radius then the side of the pentagon is $(1/2)\sqrt{10-2\sqrt{5}}$. However, this length can be written in the “minor” form (see Prop. 10.94) $(\rho/\sqrt{2})\sqrt{1+k/\sqrt{1+k^2}} - (\rho/\sqrt{2})\sqrt{1-k/\sqrt{1+k^2}}$, with $\rho = \sqrt{5}/2$ and $k = 2$.

12.

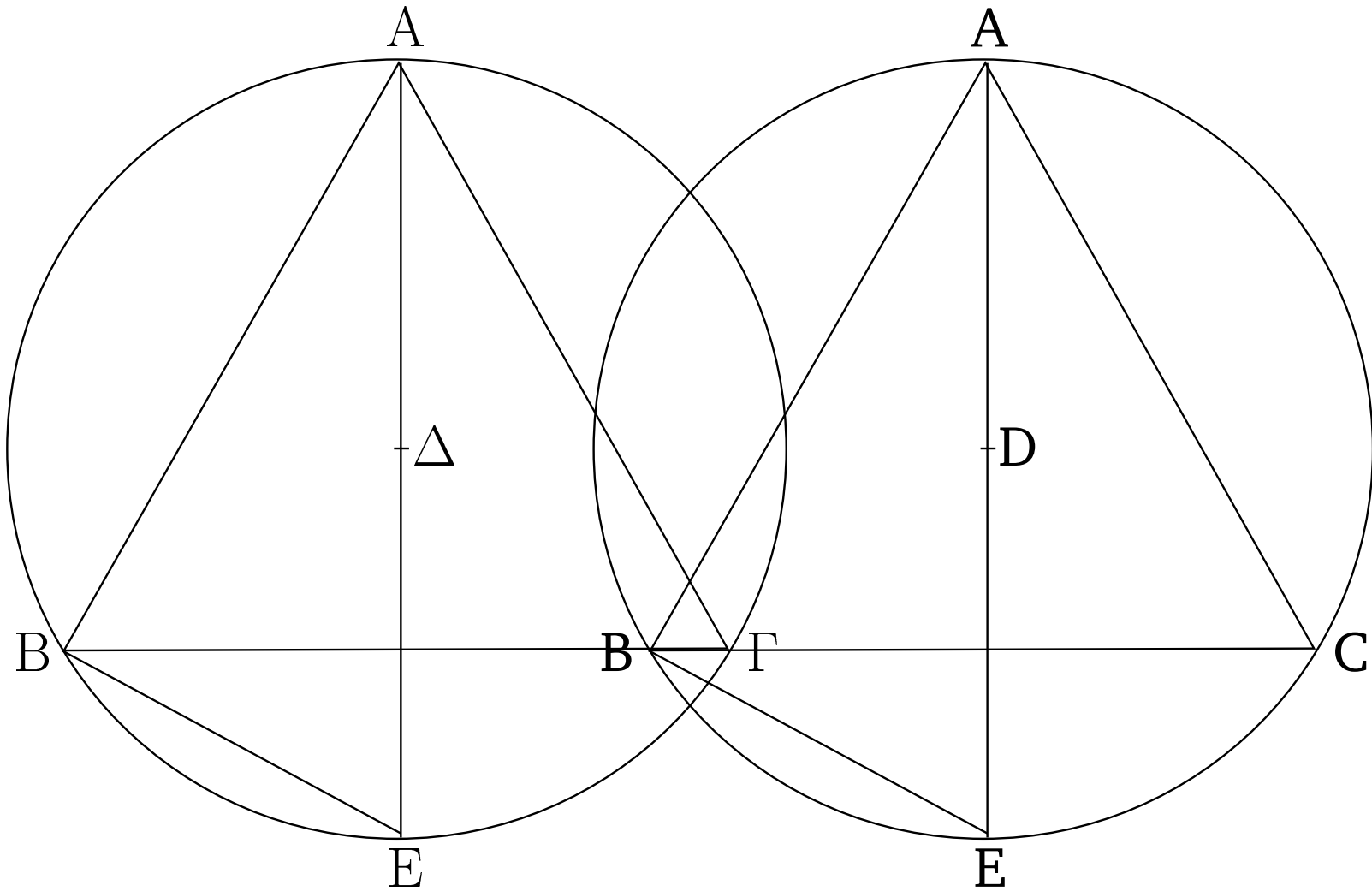
Proposition 12

Ἐάν εἷς κύκλον τρίγωνον ἰσόπλευρον ἐγγραφήῃ, ἡ τοῦ τριγώνου πλευρὰ δυνάμει τριπλασίῳ ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ κύκλου.

Ἐστω κύκλος ὁ $AB\Gamma$, καὶ εἷς αὐτὸν τρίγωνον ἰσόπλευρον ἐγγεγράφθω τὸ $AB\Gamma$. λέγω, ὅτι τοῦ $AB\Gamma$ τριγώνου μία πλευρὰ δυνάμει τριπλασίῳ ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ $AB\Gamma$ κύκλου.

If an equilateral triangle is inscribed in a circle then the square on the side of the triangle is three times the (square) on the radius of the circle.

Let there be a circle ABC , and let the equilateral triangle ABC have been inscribed in it [Prop. 4.2]. I say that the square on one side of triangle ABC is three times the (square) on the radius of circle ABC .



Εἰλήφθω γάρ τὸ κέντρον τοῦ $AB\Gamma$ κύκλου τὸ Δ , καὶ ἐπιξενυθῆῖσα ἡ $A\Delta$ διήθθω ἐπὶ τὸ E , καὶ ἐπεζεύθθω ἡ BE .

Καὶ ἐπεὶ ἰσόπλευρόν ἐστι τὸ $AB\Gamma$ τρίγωνον, ἡ $BE\Gamma$ ἄρα περιφέρεια τρίτον μέρος ἐστὶ τῆς τοῦ $AB\Gamma$ κύκλου περιφερείας. ἡ ἄρα BE περιφέρεια ἕκτον ἐστὶ μέρος τῆς τοῦ κύκλου περιφερείας· ἔχαλόνον ἄρα ἐστὶν ἡ BE εὐθεΐα· ἴση ἄρα ἐστὶ τῇ ἐκ τοῦ κέντρον τῇ ΔE . καὶ ἐπεὶ διπλὴ ἐστὶν ἡ AE τῆς ΔE , τετραπλάσιον ἐστὶ τὸ ἀπὸ τῆς AE τοῦ ἀπὸ τῆς ΔE , τοντέστι τοῦ ἀπὸ τῆς BE . ἴσον δὲ τὸ ἀπὸ τῆς AE τοῖς ἀπὸ τῶν AB , BE · τὰ ἄρα ἀπὸ τῶν AB , BE τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς BE . διελόντι ἄρα τὸ ἀπὸ τῆς AB τριπλάσιόν ἐστι τοῦ ἀπὸ BE . ἴση δὲ ἡ BE τῇ ΔE · τὸ ἄρα ἀπὸ τῆς AB τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΔE .

Ἡ ἄρα τοῦ τριγώνου πλευρὰ δυνάμει τριπλασία ἐστὶ τῆς ἐκ τοῦ κέντρον [τοῦ κύκλου]· ὅπερ ἔδει δεῖξαι.

For let the center, D , of circle ABC be found [Prop. 3.1]. And AD (being) joined, let it be drawn across to E . And let BE be joined.

And since triangle ABC is equilateral, circumference BEC is thus the third part of the circumference of circle ABC . Thus, circumference BE is the sixth part of the circumference of the circle. Thus, straight-line BE is (the side) of a hexagon. Thus, it is equal to the radius DE [Prop. 4.15 corr.]. And since AE is double DE , the (square) on AE is four times the (square) on ED —that is to say, of the (square) on BE . And the (square) on AE (is) equal to the (sum of the squares) on AB and BE [Props. 3.31, 1.47]. Thus, the (sum of the squares) on AB and BE is four times the (square) on BE . Thus, via separation, the (square) on AB is three times the (square) on BE . And BE (is) equal to DE . Thus, the (square) on AB is three times the (square) on DE .

Thus, the square on the side of the triangle is three times the (square) on the radius [of the circle]. (Which is) the very

13.

Πυραμίδα συστήσασθαι καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει ἡμιολία ἐστὶ τῆς πλευρᾶς τῆς πυραμίδος.

Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB , καὶ τετμήσθω κατὰ τὸ Γ σημείον, ὥστε διπλασίαν εἶναι τὴν AG τῆς GB · καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $A\Delta B$, καὶ ἡθθῶ ἀπὸ τοῦ Γ σημείου τῇ AB πρὸς ὀρθὰς ἡ $\Gamma\Delta$, καὶ ἐπεζεύθῃ ἡ ΔA · καὶ ἐκκείσθω κύκλος ὁ EZH ἴσην ἔχων τὴν ἐκ τοῦ κέντρου τῇ $\Delta\Gamma$, καὶ ἐγγεγράφθω εἰς τὸν EZH κύκλον τρίγωνον ἰσόπλευρον τὸ EZH · καὶ εἰλήφθω τὸ κέντρον τοῦ κύκλου τὸ Θ σημείον, καὶ ἐπεζεύθωσαν αἱ $E\Theta$, ΘZ , ΘH · καὶ ἀνεσπάτω ἀπὸ τοῦ Θ σημείου τῷ τοῦ EZH κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἡ ΘK , καὶ ἀφρηθήσθω ἀπὸ τῆς ΘK τῇ AG εὐθείᾳ ἴση ἡ ΘK , καὶ ἐπεζεύθωσαν αἱ KE , KZ , KH . καὶ ἐπεὶ ἡ $K\Theta$ ὀρθή ἐστι πρὸς τὸ τοῦ EZH κύκλου ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰς ἀπτομένας αὐτῆς εὐθείας καὶ ὅσας ἐν τῷ τοῦ EZH κύκλου ἐπιπέδῳ ὀρθὰς ποιήσει γωνίας. ἄπτεται δὲ αὐτῆς ἐκάστη τῶν ΘE , ΘZ , ΘH · ἡ ΘK ἄρα πρὸς ἐκάστη τῶν ΘE , ΘZ , ΘH ὀρθή ἐστιν. καὶ ἐπεὶ ἴση ἐστὶν ἡ μὲν AG τῇ ΘK , ἡ δὲ $\Gamma\Delta$ τῇ ΘE , καὶ ὀρθὰς γωνίας περιέθουσιν, βάσεις ἄρα ἡ ΔA βάσει τῇ KE ἐστὶν ἴση. διὰ τὰ αὐτὰ δὲ καὶ ἑκάτερα τῶν KZ , KH τῇ ΔA ἐστὶν ἴση· αἱ τρεῖς ἄρα αἱ KE , KZ , KH ἴσαι ἀλλήλαις εἰσὶν. καὶ ἐπεὶ διπλὴ ἐστὶν ἡ AG τῆς GB , τριπλὴ ἄρα ἡ AB τῆς BI . ὥς δὲ ἡ AB πρὸς τὴν BI , οὕτως τὸ ἀπὸ τῆς AD πρὸς τὸ ἀπὸ τῆς $\Delta\Gamma$, ὥς ἐχῆς δευθῆσεται. τριπλάσιον ἄρα τὸ ἀπὸ τῆς AD τοῦ ἀπὸ τῆς $\Delta\Gamma$. ἐστὶ δὲ καὶ τὸ ἀπὸ τῆς ZE τοῦ ἀπὸ τῆς $E\Theta$ τριπλάσιον, καὶ ἐστὶν ἴση ἡ $\Delta\Gamma$ τῇ $E\Theta$ · ἴση ἄρα καὶ ἡ ΔA τῇ EZ . ἀλλὰ ἡ ΔA ἐκάστη τῶν KE , KZ , KH ἐδείκθη ἴση· καὶ ἐκάστη ἄρα τῶν EZ , ZH , HE ἐκάστη τῶν KE , KZ , KH ἐστὶν ἴση· ἰσόπλευρα ἄρα ἐστὶ τὰ τέσσαρα τρίγωνα τὰ EZH , KEZ , KZH , KEH . πυραμὶς ἄρα συνέσταται ἐκ τεσσάρων τριγώνων ἰσοπλύνων, ἥς βάσις μὲν ἐστὶ τὸ EZH τρίγωνον, κορυφή δὲ τὸ K σημείον.

thing it was required to show.

Proposition 13

To construct a (regular) pyramid (*i.e.*, a tetrahedron), and to enclose (it) in a given sphere, and to show that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

Let the diameter AB of the given sphere be laid out, and let it be cut at point C such that AC is double CB [Prop. 6.10]. And let the semi-circle ADB be drawn on AB . And let CD be drawn from point C at right-angles to AB . And let DA be joined. And let the circle EFG be laid down having a radius equal to DC , and let the equilateral triangle EFG be inscribed in circle EFG [Prop. 4.2]. And let the center of the circle, point H , be found [Prop. 3.1]. And let EH , HF , and HG be joined. And let HK be set up, at point H , at right-angles to the plane of circle EFG [Prop. 11.12]. And let HK , equal to the straight-line AC , be cut off from HK . And let KE , KF , and KG be joined. And since KH is at right-angles to the plane of circle EFG , it will thus also make right-angles with all of the straight-lines joining it (which are) also in the plane of circle EFG [Def. 11.3]. And HE , HF , and HG each join it. Thus, HK is at right-angles to each of HE , HF , and HG . And since AC is equal to HK , and CD to HE , and they contain right-angles, the base DA is thus equal to the base KE [Prop. 1.4]. So, for the same (reasons), KF and KG is each equal to DA . Thus, the three (straight-lines) KE , KF , and KG are equal to one another. And since AC is double CB , AB (is) thus triple BC . And as AB (is) to BC , so the (square) on AD (is) to the (square) on DC , as will be shown later [see lemma]. Thus, the (square) on AD (is) three times the (square) on DC . And the (square) on FE is also three times the (square) on EH [Prop. 13.12], and DC is equal to EH . Thus, DA (is) also equal to EF . But, DA was shown (to be) equal to each of KE , KF , and KG . Thus, EF , FG , and GE are equal to KE , KF , and KG , respectively. Thus, the four triangles EFG , KEF , KFG , and KEG are equilateral. Thus, a pyramid, whose base is triangle EFG , and apex the point K , has been constructed from four equilateral triangles.

πρὸς τὴν ΘE , οὕτως ἡ $E\Theta$ πρὸς τὴν ΘA . τὸ ἄρα ὑπὸ τῶν $K\Theta$, ΘA ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Theta$. καὶ ἐστὶν ὀρθὴ ἑκατέρα τῶν ὑπὸ $K\Theta E$, $E\Theta A$ γωνιῶν· τὸ ἄρα ἐπὶ τῆς $K\Lambda$ γραφόμενον ἡμικύκλιον ἔχει καὶ διὰ τοῦ E [ἐπειδὴ περὶ εἶν ἐπιζεύχουμεν τὴν $E\Lambda$, ὀρθὴ γίνεται ἡ ὑπὸ $\Lambda E K$ γωνία διὰ τὸ ἰσογώνιον γίνεσθαι τὸ $E\Lambda K$ τρίγωνον ἑκατέρω τῶν $E\Lambda\Theta$, $E\Theta K$ τριγώνων]. εἰ δὲ ἡ μενούσης τῆς $K\Lambda$ περιεγεθῆν τὸ ἡμικύκλιον εἰς τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἔχει καὶ διὰ τῶν Z , H σημείων ἐπιζευγνυμένων τῶν $Z\Lambda$, ΛH καὶ ὀρθῶν ὁμοίως γινόμενων τῶν πρὸς τοῖς Z , H γωνιῶν· καὶ ἔσται ἡ πυραμὶς σφαῖρα περιελημμένη τῇ δοθείσῃ. ἡ γὰρ $K\Lambda$ τῆς σφαίρας διάμετρος ἴση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ τῇ AB , ἐπειδὴ περὶ τῇ μὲν AG ἴση κεῖται ἡ $K\Theta$, τῇ δὲ GB ἡ ΘA .

Λέγω δὴ, ὅτι ἡ τῆς σφαίρας διάμετρος ἡμολία ἐστὶ δυνάμει τῆς πλευρᾶς τῆς πυραμίδος.

Ἐπεὶ γὰρ διπλὴ ἐστὶν ἡ AG τῆς GB , τριπλὴ ἄρα ἐστὶν ἡ AB τῆς BF . ἀναστρέψαντι ἡμολία ἄρα ἐστὶν ἡ BA τῆς AG . ὥς δὲ ἡ BA πρὸς τὴν AG , οὕτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AD [ἐπειδὴ περὶ ἐπιζευγνυμένης τῆς DB ἐστὶν ὥς ἡ BA πρὸς τὴν AD , οὕτως ἡ DA πρὸς τὴν AG διὰ τὴν ὁμοιότητα τῶν ΔAB , ΔAG τριγώνων, καὶ εἶναι ὥς τὴν πρώτην πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς τρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας]. ἡμολιον ἄρα καὶ τὸ ἀπὸ τῆς BA τοῦ ἀπὸ τῆς AD . καὶ ἐστὶν ἡ μὲν BA ἡ τῆς δοθείσης σφαίρας διάμετρος, ἡ δὲ AD ἴση τῇ πλευρᾷ τῆς πυραμίδος.

Ἡ ἄρα τῆς σφαίρας διάμετρος ἡμολία ἐστὶ τῆς πλευρᾶς τῆς πυραμίδος· ὅπερ ἔδει δεῖξαι.

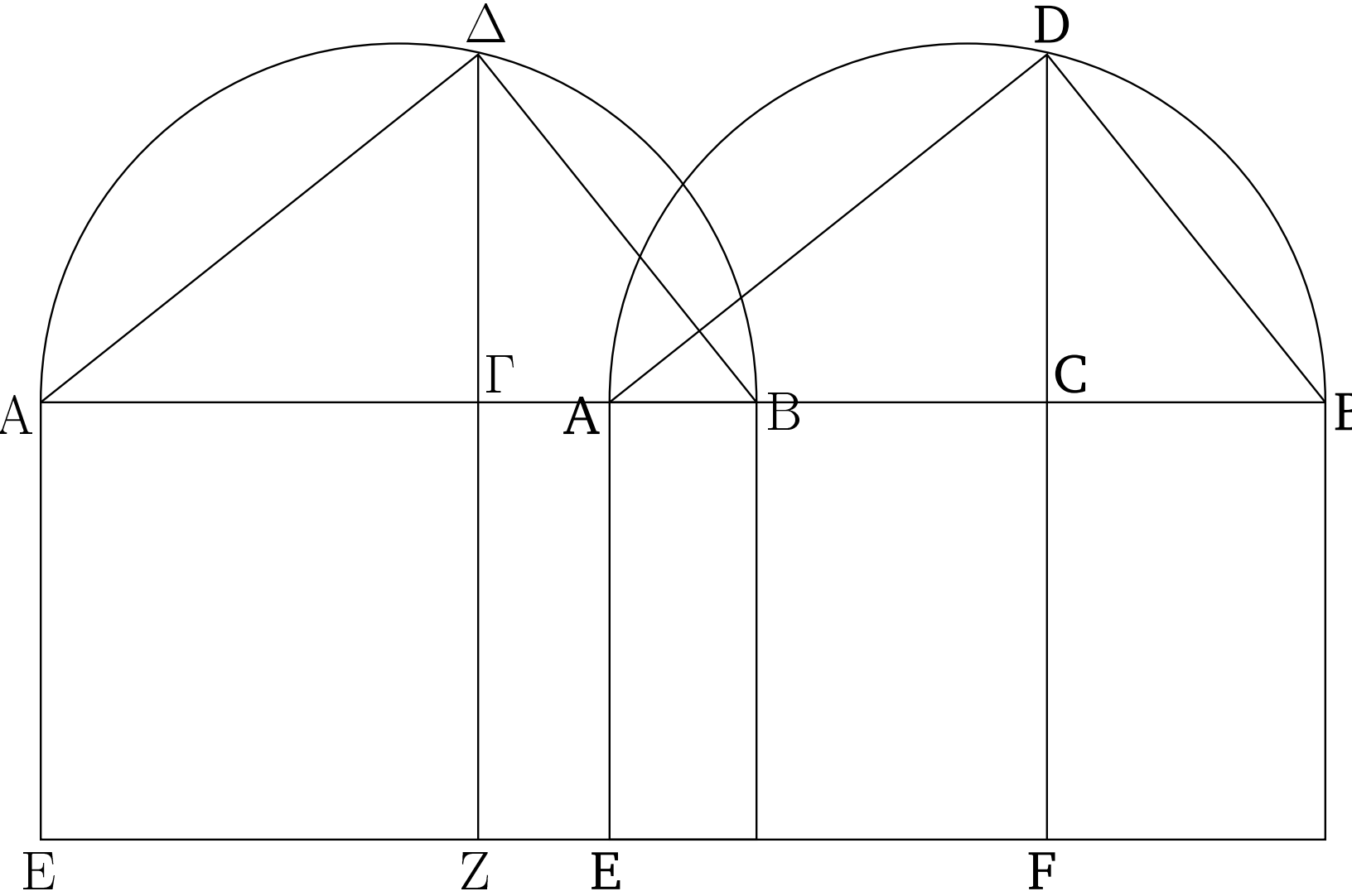
EH (is) to HL . Thus, the (rectangle contained) by KH and HL is equal to the (square) on EH [Prop. 6.17]. And each of the angles KHE and EHL is a right-angle. Thus, the semi-circle drawn on KL will also pass through E [inasmuch as if we join EL then the angle LEK becomes a right-angle, on account of triangle ELK becoming equiangular to each of the triangles ELH and EHK [Props. 6.8, 3.31]]. So, if KL remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, it will also pass through points F and G , (because) if FL and LG are joined, the angles at F and G will similarly become right-angles. And the pyramid will be enclosed by the given sphere. For the diameter, KL , of the sphere is equal to the diameter, AB , of the given sphere—inasmuch as KH was made equal to AC , and HL to CB .

So, I say that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

For since AC is double CB , AB is thus triple BC . Thus, via conversion, BA is one and a half times AC . And as BA (is) to AC , so the (square) on BA (is) to the (square) on AD [inasmuch as if DB is joined then as BA is to AD , so DA (is) to AC , on account of the similarity of triangles DAB and DAC . And as the first is to the third (of four proportional magnitudes), so the (square) on the first (is) to the (square) on the second.] Thus, the (square) on BA (is) also one and a half times the (square) on AD . And BA is the diameter of the given sphere, and AD (is) equal to the side of the pyramid.

Thus, the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.[†] (Which is) the very thing it was required to show.

[†] If the radius of the sphere is unity then the side of the pyramid (i.e., tetrahedron) is $\sqrt{8/3}$.



Λήμμα.

Lemma

Δεικτέον, ὅτι ἐστὶν ὥς ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ἀπὸ τῆς $AΔ$ πρὸς τὸ ἀπὸ τῆς $ΔΓ$.

Ἐκκείσθω γάρ ἡ τοῦ ἡμικυκλίου καταγραφή, καὶ ἐπεζεύθω ἡ $ΔB$, καὶ ἀναγεγράφθω ἀπὸ τῆς $AΓ$ τετράγωνον τὸ $EΓ$, καὶ συμπληρώσθω τὸ ZB παραλληλόγραμμον. ἐπεὶ οὖν διὰ τὸ ἰσογώνιον εἶναι τὸ $ΔAB$ τρίγωνον τῷ $ΔAΓ$ τριγώνῳ ἐστὶν ὥς ἡ BA πρὸς τὴν $AΔ$, οὕτως ἡ $ΔA$ πρὸς τὴν $AΓ$, τὸ ἄρα ὑπὸ τῶν BA , $AΓ$ ἴσον ἐστὶ τῷ ἀπὸ τῆς $AΔ$. καὶ ἐπεὶ ἐστὶν ὥς ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ EB πρὸς τὸ BZ , καὶ ἐστὶ τὸ μὲν EB τὸ ὑπὸ τῶν BA , $AΓ$ ἴση γάρ ἡ EA τῇ $AΓ$. τὸ δὲ BZ τὸ ὑπὸ τῶν $AΓ$, $ΓB$, ὥς ἄρα ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ὑπὸ τῶν BA , $AΓ$ πρὸν τὸ ὑπὸ τῶν $AΓ$, $ΓB$. καὶ ἐστὶ τὸ μὲν ὑπὸ τῶν BA , $AΓ$ ἴσον τῷ ἀπὸ τῆς $AΔ$, τὸ δὲ ὑπὸ τῶν $AΓB$ ἴσον τῷ ἀπὸ τῆς $ΔΓ$. ἡ γὰρ $ΔΓ$ κάθετος τῶν τῆς βάσεως τμημάτων τῶν $AΓ$, $ΓB$ μέση ἀνάλογόν ἐστι διὰ τὸ ὀρθὴν εἶναι

It must be shown that as AB is to BC , so the (square) on AD (is) to the (square) on DC .

For, let the figure of the semi-circle be set out, and let DB have been joined. And let the square EC be described on AC . And let the parallelogram FB be completed. Therefore, since, on account of triangle DAB being equiangular to triangle DAC [Props. 6.8, 6.4], (proportionally) as BA is to AD , so DA (is) to AC , the (rectangle contained) by BA and AC is thus equal to the (square) on AD [Prop. 6.17]. And since as AB is to BC , so EB (is) to BF [Prop. 6.1]. And EB is the (rectangle contained) by BA and AC —for EA (is) equal to AC . And BF the (rectangle contained) by AC and CB . Thus, as AB (is) to BC , so the (rectangle contained) by BA and AC (is) to the (rectangle contained) by AC and CB . And the (rectangle contained) by BA and AC is equal to the (square) on AD , and the (rectan-

τὴν ὑπὸ ADB . ὥς ἄρα ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ἀπὸ
τῆς AD πρὸς τὸ ἀπὸ τῆς $ΔΓ$. ὅπερ ἔδει δεῖξαι.

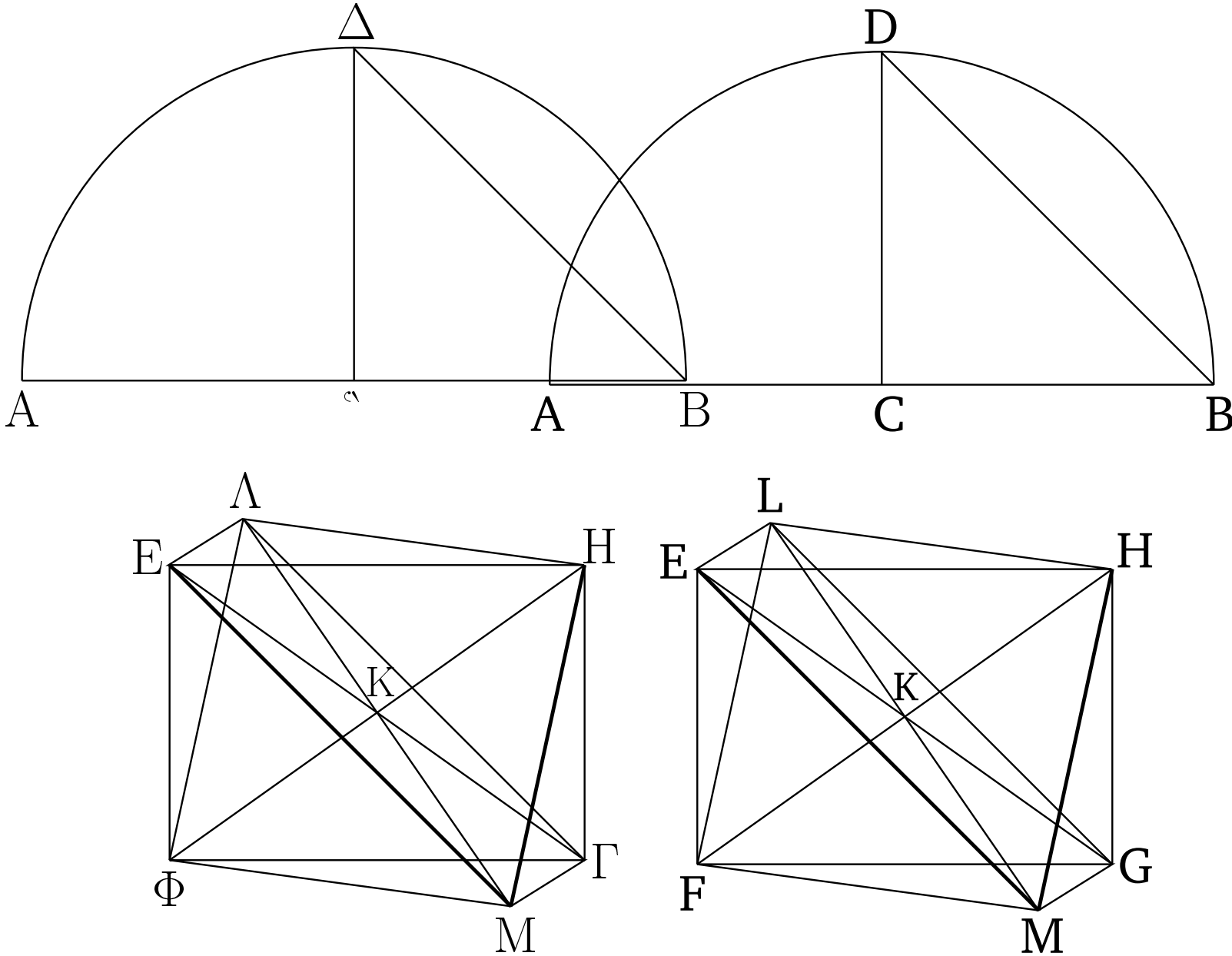
gle contained) by ACB (is) equal to the (square) on DC . For the perpendicular DC is the mean proportional to the pieces of the base, AC and CB , on account of ADB being a right-angle [Prop. 6.8 corr.]. Thus, as AB (is) to BC , so the (square) on AD (is) to the (square) on DC . (Which is) the very thing it was required to show.

14.

Ὀκτάεδρον συστήσασθαι καὶ σφαῖρα περιλαβεῖν, ἣ καὶ
τὰ πρότερα, καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει
διπλασία ἐστὶ τῆς πλευρᾶς τοῦ ὀκταέδρου.

Proposition 14

To construct an octahedron, and to enclose (it) in a (given) sphere, like in the preceding (proposition), and to show that the square on the diameter of the sphere is double the (square) on the side of the octahedron.



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB , καὶ τετμήσθω δὶνα κατὰ τὸ Γ , καὶ γεγράψθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $A\Delta B$, καὶ ἦθθω ἀπὸ τοῦ Γ τῇ AB πρὸς ὀρθὰς ἡ $\Gamma\Delta$, καὶ ἐπεξεύθθω ἡ ΔB , καὶ ἐκκείσθω τετράγωνον τὸ $EZH\Theta$ ἴσην ἔχον ἐκάστην τῶν πλευρῶν τῇ ΔB , καὶ ἐπεξεύθθωσαν αἱ ΘZ , EH , καὶ ἀνεστάτω ἀπὸ τοῦ K σημεῖον τῷ τοῦ $EZH\Theta$ τετραγώνου ἐπιπέδῳ πρὸς ὀρθὰς εὐθεῖα ἡ KL καὶ διήθθω ἐπὶ τὰ ἑτέρα μέρη τοῦ ἐπιπέδου ὡς ἡ KM , καὶ ἀφηρήσθω ἀφ' ἐκατέρᾳ τῶν KL , KM μιᾷ τῶν EK , ZK , HK , ΘK ἴση ἐκατέρα τῶν KL , KM , καὶ ἐπεξεύθθωσαν αἱ LE , AZ , AH , $\Lambda\Theta$, ME , MZ , MH , $M\Theta$.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ KE τῇ $K\Theta$, καὶ ἐστὶν ὀρθὴ ἡ ὑπὸ

Let the diameter AB of the given sphere be laid out, and let it have been cut in half at C . And let the semi-circle ADB be drawn on AB . And let CD be drawn from C at right-angles to AB . And let DB be joined. And let the square $EFGH$, having each of its sides equal to DB , be laid out. And let HF and EG be joined. And let the straight-line KL be set up, at point K , at right-angles to the plane of square $EFGH$ [Prop. 11.12]. And let it be drawn across on the other side of the plane, like KM . And let KL and KM , equal to one of EK , FK , GK , and HK , be cut off from KL and KM , respectively. And let LE , LF , LG , LH , ME , MF , MG , and MH be joined.

And since KE is equal to KH , and angle EKH is a right-

$EK\Theta$ γωνία, τὸ ἄρα ἀπὸ τῆς ΘE διπλάσιόν ἐστι τοῦ ἀπὸ τῆς EK . πάλιν, ἐπεὶ ἴση ἐστὶν ἡ AK τῇ KE , καὶ ἐστὶν ὀρθή ἡ ὑπὸ AKE γωνία, τὸ ἄρα ἀπὸ τῆς EA διπλάσιόν ἐστι τοῦ ἀπὸ EK . ἐδείκθη δὲ καὶ τὸ ἀπὸ τῆς ΘE διπλάσιον τοῦ ἀπὸ τῆς EK . τὸ ἄρα ἀπὸ τῆς AE ἴσον ἐστὶ τῷ ἀπὸ τῆς $E\Theta$. ἴση ἄρα ἐστὶν ἡ AE τῇ $E\Theta$. διὰ τὰ αὐτὰ δὲ καὶ ἡ AO τῇ ΘE ἐστὶν ἴση· ἰσοπλευρόν ἄρα ἐστὶ τὸ $AE\Theta$ τρίγωνον. ὁμοίως δὲ δέχομεν, ὅτι καὶ ἕκαστον τῶν λοιπῶν τριγώνων, ὧν βάσεις μὲν εἰσιν αἱ τοῦ $EZH\Theta$ τετραγώνου πλευραί, κορυφαὶ δὲ τὰ A, M σημεία, ἰσοπλευρόν ἐστιν· ὁκτάεδρον ἄρα συνέσταται ὑπὸ ὁκτώ τριγώνων ἰσοπλευρῶν περιεχόμενον.

Δεῖ δὲ αὐτὸ καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ καὶ δεῖχαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίον ἐστὶ τῆς τοῦ ὁκταέδρου πλευρᾶς.

Ἐπεὶ γὰρ αἱ τρεῖς αἱ AK, KM, KE ἴσαι ἀλλήλαις εἰσίν, τὸ ἄρα ἐπὶ τῆς AM γραφόμενον ἡμικύκλιον ἔχει καὶ διὰ τοῦ E . καὶ διὰ τὰ αὐτά, ἔαν μενούσης τῆς AM περιεγεσθὲν τὸ ἡμικύκλιον εἰς τὸ αὐτὸ ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἔχει καὶ διὰ τῶν Z, H, Θ σημείων, καὶ ἔσται σφαῖρα περιελημμένη τὸ ὁκτάεδρον. λέγω δὲ, ὅτι καὶ τῇ δοθείσῃ. ἐπεὶ γὰρ ἴση ἐστὶν ἡ AK τῇ KM , κοινὴ δὲ ἡ KE , καὶ γωνίαξ ὀρθὰς περιέθουσιν, βάσις ἄρα ἡ AE βάσει τῇ EM ἐστὶν ἴση. καὶ ἐπεὶ ὀρθή ἐστὶν ἡ ὑπὸ AEM γωνία· ἐν ἡμικυκλίῳ γάρ· τὸ ἄρα ἀπὸ τῆς AM διπλάσιόν ἐστι τοῦ ἀπὸ τῆς AE . πάλιν, ἐπεὶ ἴση ἐστὶν ἡ AG τῇ GB , διπλασία ἐστὶν ἡ AB τῆς BG . ὥς δὲ ἡ AB πρὸς τὴν BG , οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BD διπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς BD . ἐδείκθη δὲ καὶ τὸ ἀπὸ τῆς AM διπλάσιον τοῦ ἀπὸ τῆς AE . καὶ ἐστὶν ἴσον τὸ ἀπὸ τῆς AB τῷ ἀπὸ τῆς AE . ἴση γὰρ κεῖται ἡ $E\Theta$ τῇ ΔB . ἴσον ἄρα καὶ τὸ ἀπὸ τῆς AB τῷ ἀπὸ τῆς AM . ἴση ἄρα ἡ AB τῇ AM . καὶ ἐστὶν ἡ AB ἡ τῆς δοθείσης σφαίρας διάμετρος· ἡ AM ἄρα ἴση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ.

Περιεληπτὰ ἄρα τὸ ὁκτάεδρον τῇ δοθείσῃ σφαίρᾳ. καὶ συναποδέδεικται, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίον ἐστὶ τῆς τοῦ ὁκταέδρου πλευρᾶς· ὅπερ ἔδει δεῖχαι.

angle, the (square) on the HE is thus double the (square) on EK [Prop. 1.47]. Again, since LK is equal to KE , and angle LKE is a right-angle, the (square) on EL is thus double the (square) on EK [Prop. 1.47]. And the (square) on HE was also shown (to be) double the (square) on EK . Thus, the (square) on LE is equal to the (square) on EH . Thus, LE is equal to EH . So, for the same (reasons), LH is also equal to HE . Triangle LEH is thus equilateral. So, similarly, we can show that each of the remaining triangles, whose bases are the sides of the square $EFGH$, and apexes the points L and M , are equilateral. Thus, an octahedron contained by eight equilateral triangles has been constructed.

So, it is also necessary to enclose it by the given sphere, and to show that the square on the diameter of the sphere is double the (square) on the side of the octahedron.

For since the three (straight-lines) LK, KM , and KE are equal to one another, the semi-circle drawn on LM will thus also pass through E . And, for the same (reasons), if LM remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then it will also pass through points F, G , and H , and the octahedron will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For since LK is equal to KM , and KE (is) common, and they contain right-angles, the base LE is thus equal to the base EM [Prop. 1.4]. And since angle LEM is a right-angle—for (it is) in a semi-circle [Prop. 3.31]—the (square) on LM is thus double the (square) on LE [Prop. 1.47]. Again, since AC is equal to CB , AB is double BC . And as AB (is) to BC , so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is double the (square) on BD . And the (square) on LM was also shown (to be) double the (square) on LE . And the (square) on DB is equal to the (square) on LE . For EH was made equal to DB . Thus, the (square) on AB (is) also equal to the (square) on LM . Thus, AB (is) equal to LM . And AB is the diameter of the given sphere. Thus, LM is equal to the diameter of the given sphere.

Thus, the octahedron has been enclosed by the given sphere, and it has been simultaneously proved that the square on the diameter of the sphere is double the (square) on the side of the octahedron.[†] (Which is) the very thing it was required to show.

[†] If the radius of the sphere is unity then the side of octahedron is $\sqrt{2}$.

15.

Κύβον συστήσασθαι καὶ σφαῖρα περιλαβεῖν, ἥ καὶ τὴν πυραμίδα, καὶ δεῖχαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίον ἐστὶ τῆς τοῦ κύβου πλευρᾶς.

Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB καὶ

Proposition 15

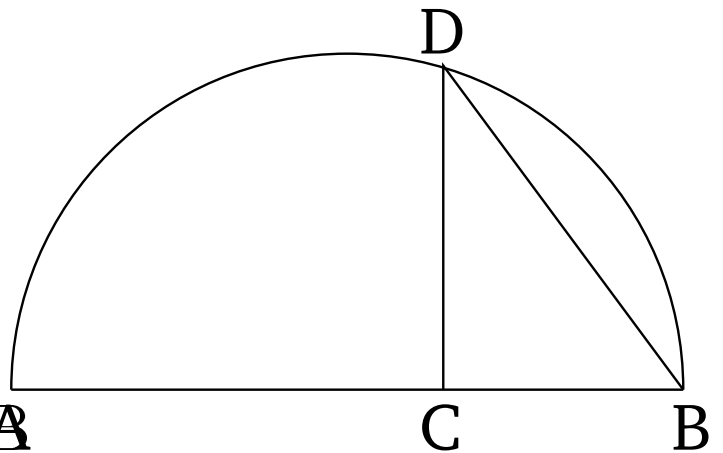
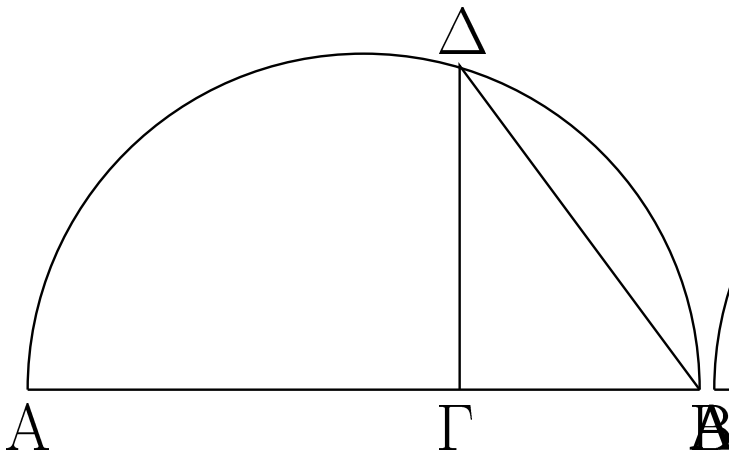
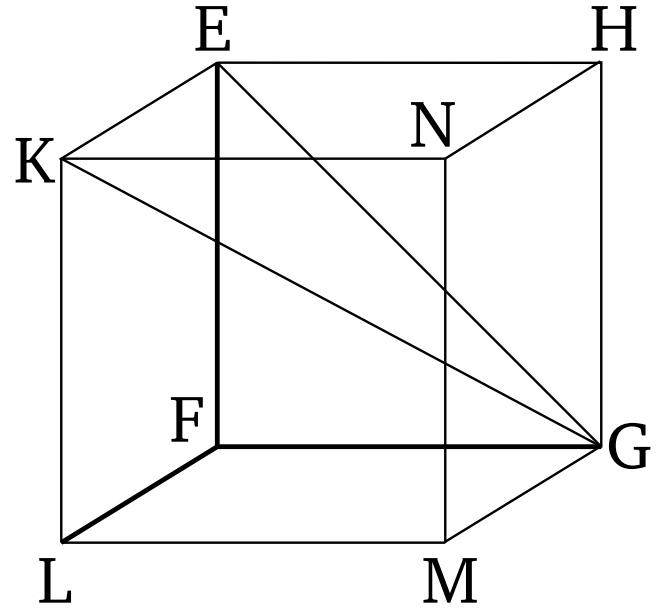
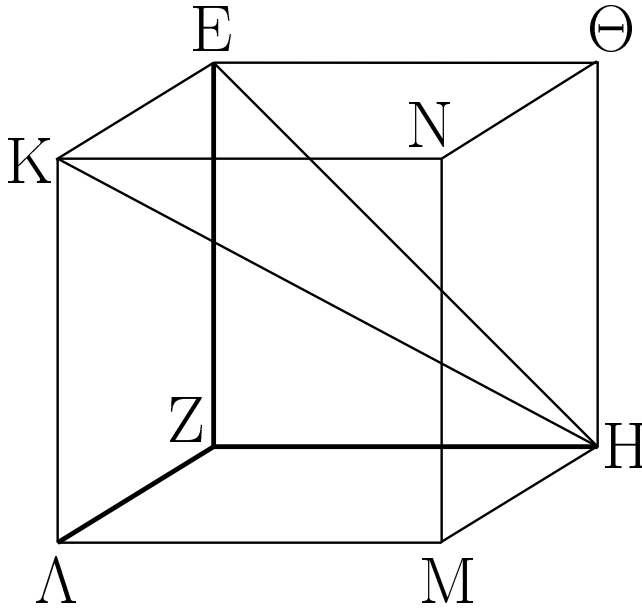
To construct a cube, and to enclose (it) in a sphere, like in the (case of the) pyramid, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube.

τετμήσθω κατὰ τὸ Γ ὥστε διπλὴν εἶναι τὴν $ΑΓ$ τῇς $ΓΒ$, καὶ γεγράφθω ἐπὶ τῇς $ΑΒ$ ἡμικύκλιον τὸ $ΑΔΒ$, καὶ ἀπὸ τοῦ Γ τῇ $ΑΒ$ πρὸς ὀρθὰς ῥθθῶ ἡ $ΓΔ$, καὶ ἐπεζεύθῃ ἡ $ΔΒ$, καὶ ἐκκεῖσθω τετράγωνον τὸ $ΕΖΗΘ$ ἴσην ἔχον τὴν πλευρὰν τῇ $ΔΒ$, καὶ ἀπὸ τῶν $Ε, Ζ, Η, Θ$ τῷ τοῦ $ΕΖΗΘ$ τετραγώνου ἐπιπέδῳ πρὸς ὀρθὰς ῥθθῶσαν αἱ $ΕΚ, ΖΑ, ΗΜ, ΘΝ$, καὶ ἀφηρήσθω ἀπὸ ἐκάστης τῶν $ΕΚ, ΖΑ, ΗΜ, ΘΝ$ μὲν τῶν $ΕΖ, ΖΗ, ΗΘ$, $ΘΕ$ ἴση ἐκάστη τῶν $ΕΚ, ΖΑ, ΗΜ, ΘΝ$, καὶ ἐπεζεύθῃσαν αἱ $ΚΛ, ΛΜ, ΜΝ, ΝΚ$. κύβος ἄρα συνέσταται ὁ $ΖΝ$ ὑπὸ ἑξ τετραγώνων ἴσων περιεθόμενος.

Δεῖ δὴ αὐτὸν καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τῇς σφαίρας διάμετρος δυνάμει τριπλασία ἐστὶ τῇς πλευρᾶς τοῦ κύβου.

Let the diameter AB of the given sphere be laid out, and let it be cut at C such that AC is double CB . And let the semi-circle ADB be drawn on AB . And let CD be drawn from C at right-angles to AB . And let DB be joined. And let the square $EFGH$, having (its) side equal to DB , be laid out. And let EK, FL, GM , and HN be drawn from (points) E, F, G , and H , (respectively), at right-angles to the plane of square $EFGH$. And let EK, FL, GM , and HN , equal to one of EF, FG, GH , and HE , be cut off from EK, FL, GM , and HN , respectively. And let KL, LM, MN , and NK be joined. Thus, a cube contained by six equal squares has been constructed.

So, it is also necessary to enclose it by the given sphere, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube.



Ἐπεζεύθῃσαν γὰρ αἱ $ΚΗ, ΕΗ$. καὶ ἐπεὶ ὀρθὴ ἐστὶν ἡ ὑπὸ $ΚΕΗ$ γωνία διὰ τὸ καὶ τὴν $ΚΕ$ ὀρθὴν εἶναι πρὸς τὸ $ΕΗ$ ἐπίπεδον δηλαδὴ καὶ πρὸς τὴν $ΕΗ$ εὐθεῖαν, τὸ ἄρα ἐπὶ

For let KG and EG be joined. And since angle KEG is a right-angle—on account of KE also being at right-angles to the plane EG , and manifestly also to the straight-line EG

τῆς KH γραφόμενον ἡμικύκλιον ἥξει καὶ διὰ τοῦ E σημείου. πάλιν, ἐπεὶ ἡ HZ ὀρθή ἐστι πρὸς ἑκατέραν τῶν ZA , ZE , καὶ πρὸς τὸ ZK ἄρα ἐπίπεδον ὀρθή ἐστιν ἡ HZ . ὥστε καὶ ἔαν ἐπιζεύχωμεν τὴν ZK , ἡ HZ ὀρθή ἐσται καὶ πρὸς τὴν ZK . καὶ διὰ τοῦτο πάλιν τὸ ἐπὶ τῆς HK γραφόμενον ἡμικύκλιον ἥξει καὶ διὰ τοῦ Z . ὁμοίως καὶ διὰ τῶν λοιπῶν τοῦ κύβου σημείων ἥξει. ἔαν δὲ μενούσῃ τῆς KH περιγενηθῇ τὸ ἡμικύκλιον εἰς τὸ αὐτὸ ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἐσται σφαῖρα περιελημμένη ὁ κύβος. λέγω δὴ, ὅτι καὶ τῇ δοθείσῃ. ἐπεὶ γὰρ ἴση ἐστὶν ἡ HZ τῇ ZE , καὶ ἐστὶν ὀρθή ἡ πρὸς τῷ Z γωνία, τὸ ἄρα ἀπὸ τῆς EH διπλάσιόν ἐστι τοῦ ἀπὸ τῆς EZ . ἴση δὲ ἡ EZ τῇ EK . τὸ ἄρα ἀπὸ τῆς EH διπλάσιόν ἐστι τοῦ ἀπὸ τῆς EK . ὥστε τὰ ἀπὸ τῶν HE , EK , τουτέστι τὸ ἀπὸ τῆς HK , τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς EK . καὶ ἐπεὶ τριπλασίον ἐστὶν ἡ AB τῆς BF , ὡς δὲ ἡ AB πρὸς τὴν BF , οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BD , τριπλάσιον ἄρα τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς BD . ἐδείχθη δὲ καὶ τὸ ἀπὸ τῆς HK τοῦ ἀπὸ τῆς KE τριπλάσιον. καὶ κεῖται ἴση ἡ KE τῇ DB . ἴση ἄρα καὶ ἡ KH τῇ AB . καὶ ἐστὶν ἡ AB τῆς δοθείσης σφαίρας διάμετρος· καὶ ἡ KH ἄρα ἴση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ.

Τῇ δοθείσῃ ἄρα σφαῖρα περιέληπται ὁ κύβος· καὶ συναποδέδεικται, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίον ἐστὶ τῆς τοῦ κύβου πλευρᾶς· ὅπερ ἔδει δεῖξαι.

[Def. 11.3]—the semi-circle drawn on KG will thus also pass through point E . Again, since GF is at right-angles to each of FL and FE , GF is thus also at right-angles to the plane FK [Prop. 11.4]. Hence, if we also join FK then GF will also be at right-angles to FK . And, again, on account of this, the semi-circle drawn on GK will also pass through point F . Similarly, it will also pass through the remaining (angular) points of the cube. So, if KG remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then the cube will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For since GF is equal to FE , and the angle at F is a right-angle, the (square) on EG is thus double the (square) on EF [Prop. 1.47]. And EF (is) equal to EK . Thus, the (square) on EG is double the (square) on EK . Hence, the (sum of the squares) on GE and EK —that is to say, the (square) on GK [Prop. 1.47]—is three times the (square) on EK . And since AB is three times BC , and as AB (is) to BC , so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9], the (square) on AB (is) thus three times the (square) on BD . And the (square) on GK was also shown (to be) three times the (square) on KE . And KE was made equal to DB . Thus, KG (is) also equal to AB . And AB is the radius of the given sphere. Thus, KG is also equal to the diameter of the given sphere.

Thus, the cube has been enclosed by the given sphere. And it has simultaneously been shown that the square on the diameter of the sphere is three times the (square) on the side of the cube.[†] (Which is) the very thing it was required to show.

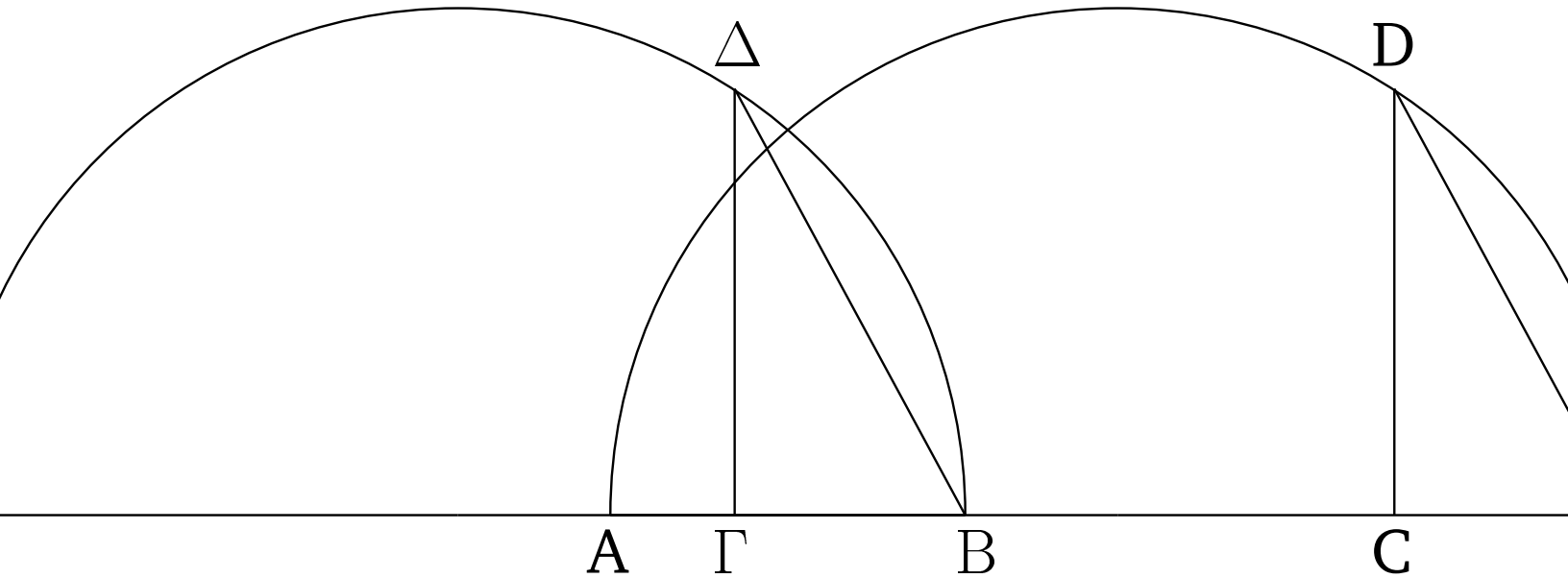
[†] If the radius of the sphere is unity then the side of the cube is $\sqrt{4/3}$.

16.

Εἰκοσαέδρον συστήσασθαι καὶ σφαῖρα περιλαβεῖν, ἥ καὶ τὰ προειρημένα σήματα, καὶ δεῖξαι, ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ ἄλογός ἐστιν ἡ καλουμένη ἐλάττων.

Proposition 16

To construct an icosahedron, and to enclose (it) in a sphere, like the aforementioned figures, and to show that the side of the icosahedron is that irrational (straight-line) called minor.



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB καὶ τετμήσθω κατὰ τὸ Γ ὥστε τετραπλὴν εἶναι τὴν $A\Gamma$ τῆς GB , καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ ADB , καὶ ᾗθθω ἀπὸ τοῦ Γ τῇ AB πρὸς ὀρθὰς γωνίᾳ εὐθεῖα γραμμὴ ἡ $\Gamma\Delta$, καὶ ἐπεξεύθθω ἡ ΔB , καὶ ἐκκείσθω κύκλος ὁ $EZH\Theta K$, ὃς ἢ ἐν τοῦ κέντρον ἴση ἔστω τῇ ΔB , καὶ ἐγγεγράφθω εἰς τὸν $EZH\Theta K$ κύκλον πεντάγωνον ἰσοπλευρόν τε καὶ ἰσογώνιον τὸ $EZH\Theta K$, καὶ τετμήσθωσαν αἱ EZ , ZH , $H\Theta$, ΘK , KE περιφέρειαι δίῃ κατὰ τὸ Λ , M , N , X , O σημεία, καὶ ἐπεξεύθθωσαν αἱ ΛM , MN , NX , XO , $O\Lambda$, EO . ἰσοπλευρον ἄρα ἐστὶ καὶ τὸ $\Lambda MNXO$ πεντάγωνον, καὶ δεκαγώνον ἡ EO εὐθεῖα. καὶ ἀνεστάτωσαν ἀπὸ τῶν E , Z , H , Θ , K σημείων τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς γωνίᾳ εὐθεῖαι αἱ $E\Pi$, ZP , $H\Sigma$, ΘT , KY ἴσαι ὅσαι τῇ ἐκ τοῦ κέντρον τοῦ $EZH\Theta K$ κύκλου, καὶ ἐπεξεύθθωσαν αἱ ΠP , $P\Sigma$, ΣT , TY , $Y\Pi$, $\Pi\Lambda$, ΛP , $P M$, $M\Sigma$, ΣN , $N T$, $T X$, $X Y$, $Y O$, $O\Pi$.

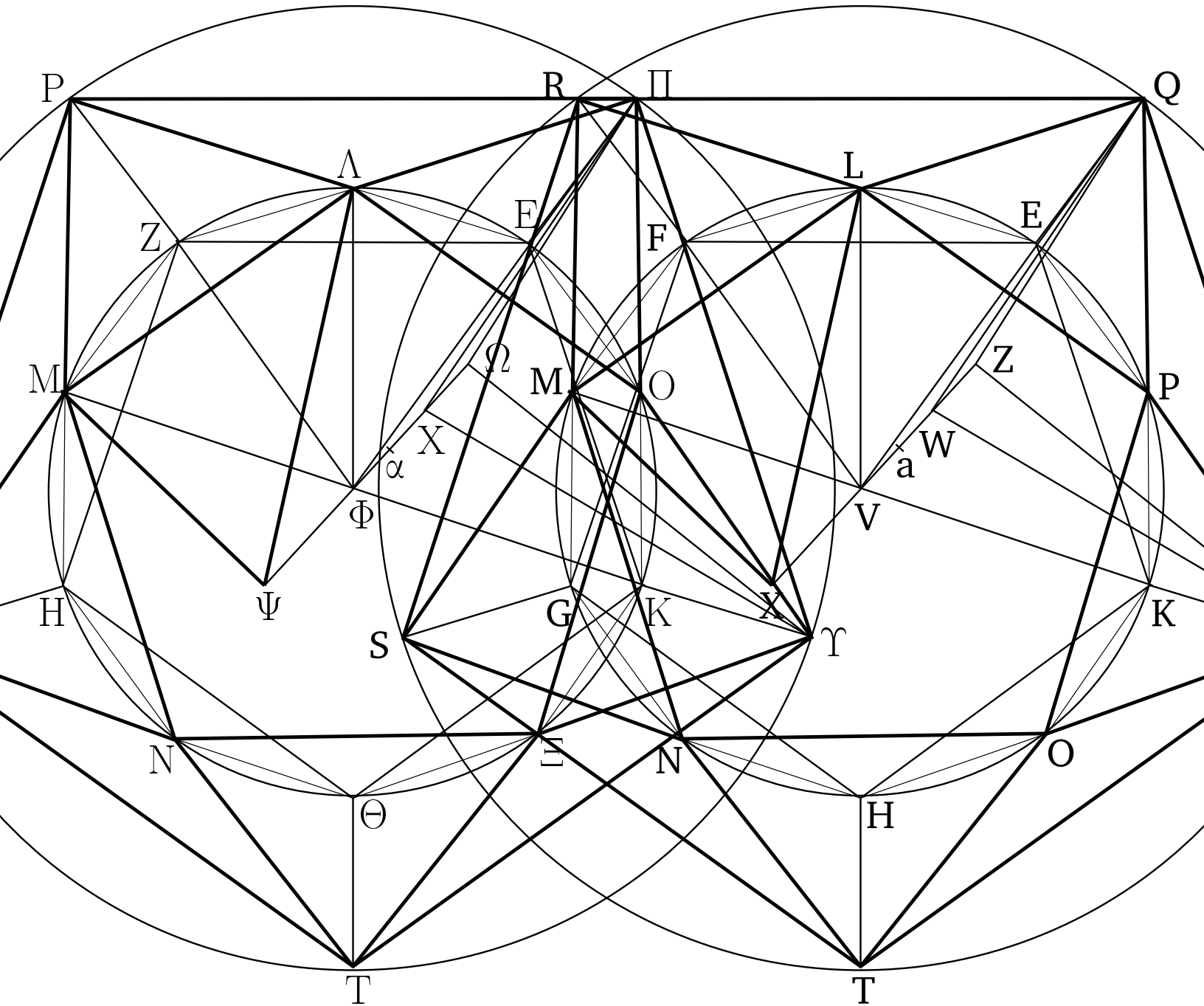
Καὶ ἐπεὶ ἑκάτερα τῶν $E\Pi$, KY τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰς ἐστίν, παράλληλος ἄρα ἐστὶν ἡ $E\Pi$ τῇ KY . ἔστι δὲ αὐτῇ καὶ ἴση· αἱ δὲ τὰς ἴσας τε καὶ παραλλήλους ἐπιξενγνύουσαι ἐπὶ τὰ αὐτὰ μέρη εὐθεῖαι ἴσαι τε καὶ παράλληλοί εἰσιν. ἡ ΠY ἄρα τῇ $E K$ ἴση τε καὶ παράλληλός ἐστιν. πενταγώνον δὲ ἰσοπλεύρον ἡ $E K$ · πενταγώνον ἄρα ἰσοπλεύρον καὶ ἡ ΠY τοῦ εἰς τὸν $EZH\Theta K$ κύκλον ἐγγραφομένου. διὰ τὰ αὐτὰ δὴ καὶ ἑκάστη τῶν ΠP , $P\Sigma$, ΣT , $T Y$ πενταγώνον ἐστὶν ἰσοπλεύρου τοῦ εἰς τὸν $EZH\Theta K$ κύκλον ἐγγραφομένου· ἰσοπλευρον ἄρα τὸ $\Pi P\Sigma T Y$ πεντάγωνον. καὶ ἐπεὶ ἑχαγώνον μὲν ἐστὶν ἡ ΠE , δεκαγώνον δὲ ἡ EO , καὶ ἐστὶν ὀρθὴ ἡ ὑπὸ $\Pi E O$, πενταγώνον ἄρα ἐστὶν ἡ ΠO · ἡ γὰρ τοῦ πενταγώνου πλευρὰ δύνатаι τὴν τε τοῦ ἑχαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰς τὸν αὐτὸν κύκλον ἐγγραφομένων. διὰ τὰ αὐτὰ δὴ καὶ ἡ $O Y$ πενταγώνον ἐστὶ πλευρά. ἔστι δὲ καὶ ἡ ΠY πενταγώνον· ἰσοπλευρον ἄρα ἐστὶ τὸ $\Pi O Y$ τρίγωνον. διὰ τὰ αὐτὰ

Let the diameter AB of the given sphere be laid out, and let it be cut at C such that AC is four times CB [Prop. 6.10]. And let the semi-circle ADB be drawn on AB . And let the straight-line CD be drawn from C at right-angles to AB . And let DB be joined. And let the circle $EFGHK$ be set down, and let its radius be equal to DB . And let the equilateral and equiangular pentagon $EFGHK$ be inscribed in circle $EFGHK$ [Prop. 4.11]. And let the circumferences EF , FG , GH , HK , and KE be cut in half at points L , M , N , O , and P (respectively). And let LM , MN , NO , OP , PL , and EP be joined. Thus, pentagon $LMNOP$ is also equilateral, and EP (is) the side of the decagon (inscribed in the circle). And let the straight-lines EQ , FR , GS , HT , and KU , which are equal to the radius of circle $EFGHK$, have been set up at right-angles to the plane of the circle, at points E , F , G , H , and K (respectively). And let QR , RS , ST , TU , UQ , QL , LR , RM , MS , SN , NT , TO , OU , UP , and PQ be joined.

And since EQ and KU are each at right-angles to the same plane, EQ is thus parallel to KU [Prop. 11.6]. And it is also equal to it. And straight-lines joining equal and parallel (straight-lines) on the same side are (themselves) equal and parallel [Prop. 1.33]. Thus, QU is equal and parallel to $E K$. And $E K$ (is the side) of an equilateral pentagon (inscribed in circle $EFGHK$). Thus, QU (is) also the side of an equilateral pentagon inscribed in circle $EFGHK$. So, for the same (reasons), QR , RS , ST , and TU are also the sides of an equilateral pentagon inscribed in circle $EFGHK$. Pentagon $QRSTU$ (is) thus equilateral. And side QE is (the side) of a hexagon (inscribed in circle $EFGHK$), and EP (the side) of a decagon, and (angle) QEP is a right-angle, thus QP is (the side) of a pentagon (inscribed in the same circle). For the square on the side of a pentagon is (equal to the sum of) the (squares) on (the

δὴ καὶ ἕκαστον τῶν $ΠΑΡ$, $ΡΜΣ$, $ΣΝΤ$, $ΤΧΥ$ ἰσόπλευρόν ἐστιν. καὶ ἐπεὶ πενταγώνον ἐδείκθη ἑκατέρα τῶν $ΠΑ$, $ΠΟ$, ἔστι δὲ καὶ ἡ $ΛΟ$ πενταγώνον, ἰσόπλευρον ἄρα ἐστὶ τὸ $ΠΛΟ$ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ ἕκαστον τῶν $ΛΡΜ$, $ΜΣΝ$, $ΝΤΧ$, $ΧΥΟ$ τριγώνων ἰσόπλευρόν ἐστιν.

sides of) a hexagon and a decagon inscribed in the same circle [Prop. 13.10]. So, for the same (reasons), PU is also the side of a pentagon. And QU is also (the side) of a pentagon. Thus, triangle QPU is equilateral. So, for the same (reasons), (triangles) QLR , RMS , $ΣΝΤ$, and $ΤΟΥ$ are each also equilateral. And since QL and QP were each shown (to be the sides) of a pentagon, and LP is also (the side) of a pentagon, triangle QLP is thus equilateral. So, for the same (reasons), triangles LRM , MSN , $ΝΤΟ$, and $ΟUP$ are each also equilateral.



Εἰλήφθω τὸ κέντρον τοῦ ΕΖΗΘΚ κύκλου τὸ Φ σημεῖον·
καὶ ἀπὸ τοῦ Φ τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθὰς ἀνεστάτω
ἡ ΦΩ, καὶ ἐκβεβλήσθω ἐπὶ τὰ ἕτερα μέρη ὡς ἡ ΦΨ, καὶ
ἀφηρήσθω ἑκαγώνου μὲν ἡ ΦΘ, δεκαγώνου δὲ ἑκατέρω τῶν
ΦΨ, ΘΩ, καὶ ἐπεζεύθωσαν αἱ ΠΩ, ΠΘ, ΥΩ, ΕΦ, ΛΦ, ΛΨ,
ΨΜ.

Καὶ ἐπεὶ ἑκατέρω τῶν ΦΘ, ΠΕ τῷ τοῦ κύκλου ἐπιπέδῳ
πρὸς ὀρθὰς ἔστιν, παράλληλοξ ἄρα ἔστιν ἡ ΦΘ τῇ ΠΕ. εἰσὶ

Let the center, point V, of circle $EFGHK$ be found
[Prop. 3.1]. And let VZ be set up, at (point) V, at right-angles
to the plane of the circle. And let it be produced on the other
side (of the circle), like VX . And let VW be cut off (from XZ
so as to be equal to the side) of a hexagon, and each of VX and
 WZ (so as to be equal to the side) of a decagon. And let QZ ,
 QW , UZ , EV , LV , LX , and XM be joined.

And since VW and QE are each at right-angles to the plane

δὲ καὶ ἴσαι· καὶ αἱ $E\Phi$, $\Pi\Theta$ ἄρα ἴσαι τε καὶ παράλληλοι εἰσιν. ἔχαρῶνον δὲ ἡ $E\Phi$ · ἔχαρῶνον ἄρα καὶ ἡ $\Pi\Theta$. καὶ ἐπεὶ ἔχαρῶνον μὲν ἔστιν ἡ $\Pi\Theta$, δεκαγῶνον δὲ ἡ $\Theta\Omega$, καὶ ὀρθή ἐστιν ἡ ὑπὸ $\Pi\Theta\Omega$ γωνία, πενταγῶνον ἄρα ἔστιν ἡ $\Pi\Omega$. διὰ τὰ αὐτὰ δὴ καὶ ἡ $ΥΩ$ πενταγῶνον ἔστιν, ἐπειδήπερ, ἔαν ἐπιζεύχωμεν τὰς ΦK , ΘY , ἴσαι καὶ ἀπεναντίον ἔσονται, καὶ ἔστιν ἡ ΦK ἐκ τοῦ κέντρον οὔσα ἔχαρῶνον. ἔχαρῶνον ἄρα καὶ ἡ ΘY . δεκαγῶνον δὲ ἡ $\Theta\Omega$, καὶ ὀρθή ἡ ὑπὸ $Y\Theta\Omega$ · πενταγῶνον ἄρα ἡ $ΥΩ$. ἔστι δὲ καὶ ἡ ΠY πενταγῶνον· ἰσόπλευρον ἄρα ἔστι τὸ $\Pi Y\Omega$ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ ἕκαστον τῶν λοιπῶν τριγῶνων, ὧν βάσεις μὲν εἰσιν αἱ ΠP , $P\Sigma$, ΣT , $T Y$ εὐθεῖαι, κορυφή δὲ τὸ Ω σημεῖον, ἰσόπλευρόν ἐστιν. πάλιν, ἐπεὶ ἔχαρῶνον μὲν ἡ $\Phi\Lambda$, δεκαγῶνον δὲ ἡ $\Phi\Psi$, καὶ ὀρθή ἐστιν ἡ ὑπὸ $\Lambda\Phi\Psi$ γωνία, πενταγῶνον ἄρα ἔστιν ἡ $\Lambda\Psi$. διὰ τὰ αὐτὰ δὴ ἔαν ἐπιζεύχωμεν τὴν $M\Phi$ οὔσαν ἔχαρῶνον, συνάγεται καὶ ἡ $M\Psi$ πενταγῶνον. ἔστι δὲ καὶ ἡ ΛM πενταγῶνον· ἰσόπλευρον ἄρα ἔστι τὸ $\Lambda M\Psi$ τρίγωνον. ὁμοίως δὴ δευτέρησεται, ὅτι καὶ ἕκαστον τῶν λοιπῶν τριγῶνων, ὧν βάσεις μὲν εἰσιν αἱ MN , NX , XO , $ΟΛ$, κορυφή δὲ τὸ Ψ σημεῖον, ἰσόπλευρόν ἐστιν. συνέσταται ἄρα εἰκοσάεδρον ὑπὸ εἴκοσι τριγῶνων ἰσοπλεύρων περιεχόμενον.

Δεῖ δὴ αὐτὸ καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ ἀλογόξ ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ ἔχαρῶνον ἔστιν ἡ $\Phi\Theta$, δεκαγῶνον δὲ ἡ $\Theta\Omega$, ἡ $\Phi\Omega$ ἄρα ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Θ , καὶ τὸ μείζον αὐτῆς τμήμα ἐστιν ἡ $\Phi\Theta$ · ἔστιν ἄρα ὡς ἡ $\Omega\Phi$ πρὸς τὴν $\Phi\Theta$, οὕτως ἡ $\Phi\Theta$ πρὸς τὴν $\Theta\Omega$. ἴση δὲ ἡ μὲν $\Phi\Theta$ τῇ ΦE , ἡ δὲ $\Theta\Omega$ τῇ $\Phi\Psi$ · ἔστιν ἄρα ὡς ἡ $\Omega\Phi$ πρὸς τὴν ΦE , οὕτως ἡ $E\Phi$ πρὸς τὴν $\Phi\Psi$. καὶ εἰσιν ὀρθαὶ αἱ ὑπὸ $\Omega\Phi E$, $E\Phi\Psi$ γωνίαι· ἔαν ἄρα ἐπιζεύχωμεν τὴν $E\Omega$ εὐθεῖαν, ὀρθή ἔσται ἡ ὑπὸ $\Psi E\Omega$ γωνία διὰ τὴν ὁμοιότητα τῶν $\Psi E\Omega$, $\Phi E\Omega$ τριγῶνων. διὰ τὰ αὐτὰ δὴ ἐπεὶ ἔστιν ὡς ἡ $\Omega\Phi$ πρὸς τὴν $\Phi\Theta$, οὕτως ἡ $\Phi\Theta$ πρὸς τὴν $\Theta\Omega$, ἴση δὲ ἡ μὲν $\Omega\Phi$ τῇ $\Psi\Theta$, ἡ δὲ $\Phi\Theta$ τῇ $\Theta\Pi$, ἔστιν ἄρα ὡς ἡ $\Psi\Theta$ πρὸς τὴν $\Theta\Pi$, οὕτως ἡ $\Pi\Theta$ πρὸς τὴν $\Theta\Omega$. καὶ διὰ τοῦτο πάλιν ἔαν ἐπιζεύχωμεν τὴν $\Pi\Psi$, ὀρθή ἔσται ἡ πρὸς τῷ Π γωνία· τὸ ἄρα ἐπὶ τῆς $\Psi\Omega$ γραφόμενον ἡμικύκλιον ἔχει καὶ διὰ τοῦ Π . καὶ ἔαν μενούσῃ τῇς $\Psi\Omega$ περιγενηθὲν τὸ ἡμικύκλιον εἴξῃ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἔχει καὶ διὰ τοῦ Π καὶ τῶν λοιπῶν σημείων τοῦ εἰκοσαέδρου, καὶ ἔσται σφαῖρα περιειλημμένη τὸ εἰκοσάεδρον. λέγω δὴ, ὅτι καὶ τῇ δοθείσῃ. τετμήσθω γὰρ ἡ $\Phi\Theta$ διὰ κατὰ τὸ α . καὶ ἐπεὶ εὐθεῖα γραμμὴ ἡ $\Phi\Omega$ ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Θ , καὶ τὸ ἐλάσσον αὐτῆς τμήμα ἐστιν ἡ $\Omega\Theta$, ἡ ἄρα $\Omega\Theta$ προσλαβοῦσα τὴν ἡμίσειαν τοῦ μείζονος τμήματος τὴν $\Theta\alpha$ πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τοῦ μείζονος τμήματος· πενταπλάσιον ἄρα ἔστι τὸ ἀπὸ τῆς $\Omega\alpha$ τοῦ ἀπὸ τῆς $\alpha\Theta$. καὶ ἔστι τῆς μὲν $\Omega\alpha$ διπλὴ ἡ $\Omega\Psi$, τῆς δὲ $\alpha\Theta$ διπλὴ ἡ $\Phi\Theta$ · πενταπλάσιον ἄρα ἔστι τὸ ἀπὸ τῆς $\Omega\Psi$ τοῦ ἀπὸ τῆς $\Theta\Phi$. καὶ ἐπεὶ τετραπλὴ ἐστιν ἡ $\Lambda\Gamma$ τῆς ΓB , πενταπλὴ ἄρα ἔστιν ἡ AB τῆς $B\Gamma$. ὡς δὲ ἡ

of the circle, VW is thus parallel to QE [Prop. 11.6]. And they are also equal. EV and QW are thus equal and parallel (to one another) [Prop. 1.33]. And EV (is the side) of a hexagon. Thus, QW (is) also (the side) of a hexagon. And since QW is (the side) of a hexagon, and WZ (the side) of a decagon, and angle QWZ is a right-angle [Def. 11.3, Prop. 1.29], QZ is thus (the side) of a pentagon [Prop. 13.10]. So, for the same (reasons), UZ is also (the side) of a pentagon—inasmuch as, if we join VK and WU then they will be equal and opposite. And VK , being (equal) to the radius (of the circle), is (the side) of a hexagon [Prop. 4.15 corr.]. Thus, WU (is) also the side of a hexagon. And WZ (is the side) of a decagon, and (angle) UWZ (is) a right-angle. Thus, UZ (is the side) of a pentagon [Prop. 13.10]. And QU is also (the side) of a pentagon. Triangle QUZ is thus equilateral. So, for the same (reasons), each of the remaining triangles, whose bases are the straight-lines QR , RS , ST , and TU , and apexes the point Z , are also equilateral. Again, since VL (is the side) of a hexagon, and VX (the side) of a decagon, and angle LVX is a right-angle, LX is thus (the side) of a pentagon [Prop. 13.10]. So, for the same (reasons), if we join MV , which is (the side) of a hexagon, MX is also inferred (to be the side) of a pentagon. And LM is also (the side) of a pentagon. Thus, triangle LMX is equilateral. So, similarly, it can be shown that each of the remaining triangles, whose bases are the (straight-lines) MN , NO , OP , and PL , and apexes the point X , are also equilateral. Thus, an icosahedron contained by twenty equilateral triangles has been constructed.

So, it is also necessary to enclose it in the given sphere, and to show that the side of the icosahedron is that irrational (straight-line) called minor.

For, since VW is (the side) of a hexagon, and WZ (the side) of a decagon, VZ has thus been cut in extreme and mean ratio at W , and VW is its greater piece [Prop. 13.9]. Thus, as ZV is to VW , so VW (is) to WZ . And VW (is) equal to VE , and WZ to VX . Thus, as ZV is to VE , so EV (is) to VX . And angles ZVE and EVX are right-angles. Thus, if we join straight-line EZ then angle XEZ will be a right-angle, on account of the similarity of triangles XEZ and VEZ . [Prop. 6.8]. So, for the same (reasons), since as ZV is to VW , so VW (is) to WZ , and ZV (is) equal to XW , and VW to WQ , thus as XW is to WQ , so QW (is) to WZ . And, again, on account of this, if we join QX then the angle at Q will be a right-angle [Prop. 6.8]. Thus, the semi-circle drawn on XZ will also pass through Q [Prop. 3.31]. And if XZ remains fixed, and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then it will also pass through (point) Q , and (through) the remaining (angular) points of the icosahedron. And the icosahedron will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For let VW be cut in half at α . And since the straight-line VZ has been cut in extreme and mean ratio at W , and ZW is its lesser piece, then

AB πρὸς τὴν $BΓ$, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς $BΔ$ · πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς $BΔ$. ἐδείκθη δὲ καὶ τὸ ἀπὸ τῆς $ΩΨ$ πενταπλάσιον τοῦ ἀπὸ τῆς $ΦΘ$. καὶ ἐστὶν ἴση ἡ $ΔB$ τῇ $ΦΘ$ · ἑκατέρα γὰρ αὐτῶν ἴση ἐστὶ τῇ ἐκ τοῦ κέντρου τοῦ $EZHΘK$ κύκλου· ἴση ἄρα καὶ ἡ AB τῇ $ΨΩ$. καὶ ἐστὶν ἡ AB ἡ τῆς δοθείσης σφαίρας διάμετρος· καὶ ἡ $ΨΩ$ ἄρα ἴση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ· τῇ ἄρα δοθείσῃ σφαίρᾳ περιελλίπται τὸ εἰκοσάεδρον.

Λέγω δὴ, ὅτι ἡ τοῦ εἰκοσάεδρου πλευρὰ ἄλογός ἐστιν ἡ καλουμένη ἐλάττων. ἐπεὶ γὰρ ῥητὴ ἐστὶν ἡ τῆς σφαίρας διάμετρος, καὶ ἐστὶ δυνάμει πενταπλασίῳ τῆς ἐκ τοῦ κέντρου τοῦ $EZHΘK$ κύκλου, ῥητὴ ἄρα ἐστὶ καὶ ἡ ἐκ τοῦ κέντρου τοῦ $EZHΘK$ κύκλου· ὥστε καὶ ἡ διάμετρος αὐτοῦ ῥητὴ ἐστὶν. ἔαν δὲ εἷς κύκλον ῥητὴν ἔθοντα τὴν διάμετρον πεντάγωνον ἰσόπλευρον ἐγγράφῃ, ἡ τοῦ πενταγώνου πλευρὰ ἄλογός ἐστιν ἡ καλουμένη ἐλάττων. ἡ δὲ τοῦ $EZHΘK$ πενταγώνου πλευρὰ ἡ τοῦ εἰκοσάεδρου ἐστίν. ἡ ἄρα τοῦ εἰκοσάεδρου πλευρὰ ἄλογός ἐστιν ἡ καλουμένη ἐλάττων.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει πενταπλασίῳ ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ κύκλου, ἀφ' οὗ τὸ εἰκοσάεδρον ἀναγέγραπται, καὶ ὅτι ἡ τῆς σφαίρας διάμετρος σύγκειται ἐκ τε τῆς τοῦ ἑξαγώνου καὶ δύο τῶν τοῦ δεκαγώνου τῶν εἷς τὸν αὐτὸν κύκλον ἐγγραφομένων. ὅπερ ἔδει δεῖξαι.

† If the radius of the sphere is unity then the radius of the circle is $2/\sqrt{5}$, and the sides of the hexagon, decagon, and pentagon/icosahedron are $2/\sqrt{5}$, $1 - 1/\sqrt{5}$, and $(1/\sqrt{5})\sqrt{10 - 2\sqrt{5}}$, respectively.

17.

Δωδεκάεδρον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν, ἥ καὶ τὰ προειρημένα στήματα, καὶ δεῖξαι, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ἄλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐκκείσθωσαν τοῦ προειρημένου κύβου δύο ἐπίπεδα πρὸς ὀρθὰς ἀλλήλοις τὰ $ABΓΔ$, $ΓΒΕΖ$, καὶ τετμήσθω ἑκάστη τῶν AB , $BΓ$, $ΓΔ$, $ΔΑ$, EZ , EB , $ZΓ$ πλευρῶν διὰ κατὰ τὰ H , $Θ$, K , $Λ$, M , N , X , καὶ ἐπεξέσθωσαν αἱ HK , $ΘΛ$, $MΘ$, NX , καὶ τετμήσθω ἑκάστη τῶν NO , OX , $ΘΠ$ ἄκρον καὶ μέσον λόγον κατὰ τὰ P , $Σ$, T σημεία, καὶ ἔστω αὐτῶν μείζονα τμήματα τὰ PO , $ΟΣ$, $ΤΠ$, καὶ ἀνεσπάτωσαν ἀπὸ τῶν P , $Σ$, T σημείων τοῖς τοῦ κύβου ἐπιπέδοις πρὸς ὀρθὰς ἐπὶ τὰ ἐκτὸς μέρη τοῦ κύβου αἱ PY , $ΣΦ$, $TΘ$, καὶ κείσθωσαν ἴσαι ταῖς PO , $ΟΣ$, $ΤΠ$, καὶ ἐπεξέσθωσαν αἱ YB , $BΘ$, $ΘΓ$, $ΓΦ$, $ΦΥ$.

the square on ZW added to half of the greater piece, Wa , is five times the (square) on half of the greater piece [Prop. 13.3]. Thus, the (square) on Za is five times the (square) on aW . And ZX is double Za , and VW double aW . Thus, the (square) on ZX is five times the (square) on WV . And since AC is four times CB , AB is thus five times BC . And as AB (is) to BC , so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is five times the (square) on BD . And the (square) on ZX was also shown (to be) five times the (square) on VW . And DB is equal to VW . For each of them is equal to the radius of circle $EFGHK$. Thus, AB (is) also equal to XZ . And AB is the diameter of the given sphere. Thus, XZ is equal to the diameter of the given sphere. Thus, the icosahedron has been enclosed by the given sphere.

So, I say that the side of the icosahedron is that irrational (straight-line) called minor. For since the diameter of the sphere is rational, and the square on it is five times the (square) on the radius of circle $EFGHK$, the radius of circle $EFGHK$ is thus also rational. Hence, its diameter is also rational. And if an equilateral pentagon is inscribed in a circle having a rational diameter then the side of the pentagon is that irrational (straight-line) called minor [Prop. 13.11]. And the side of pentagon $EFGHK$ is (the side) of the icosahedron. Thus, the side of the icosahedron is that irrational (straight-line) called minor.

Corollary

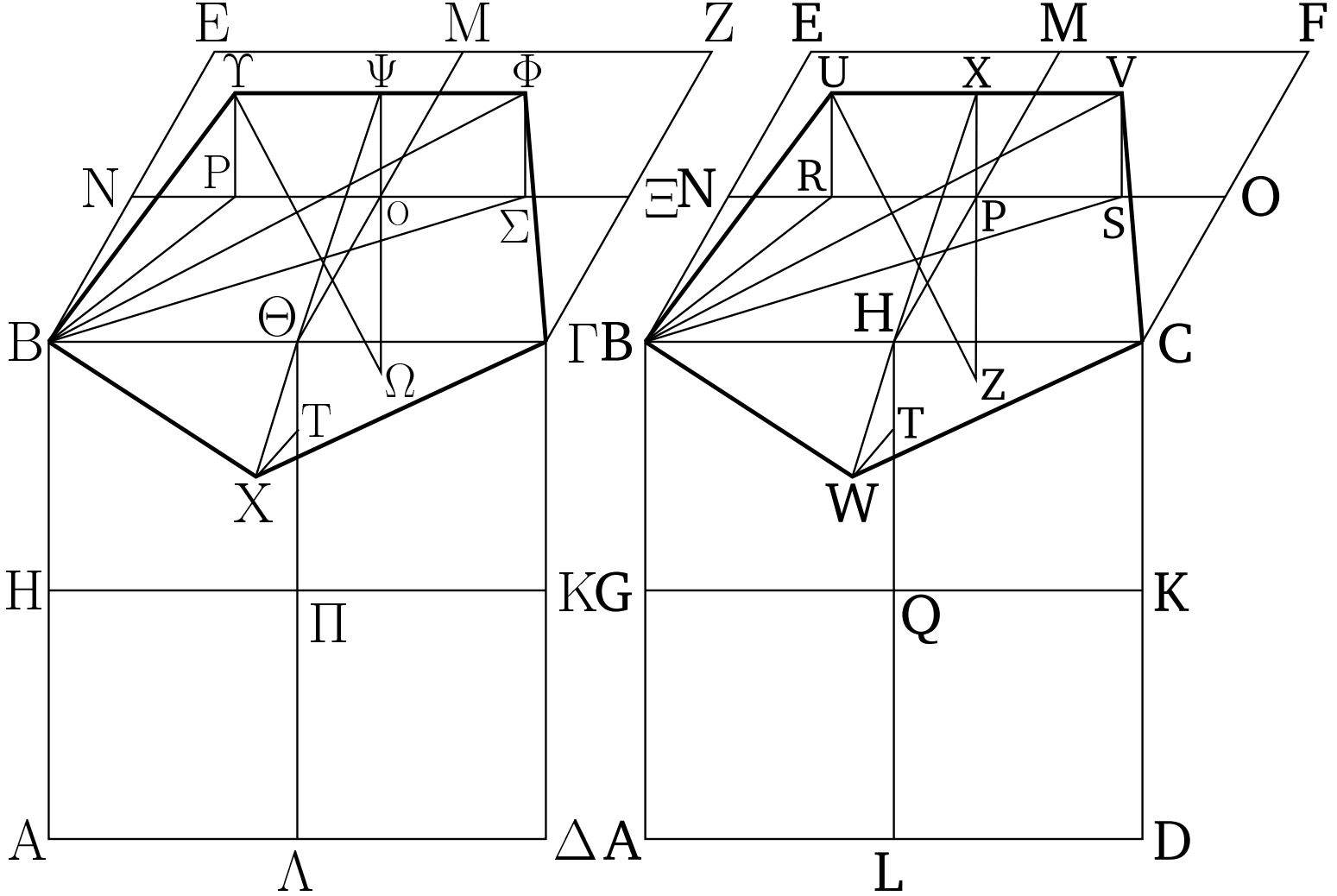
So, (it is) clear, from this, that the square on the diameter of the sphere is five times the square on the radius of the circle from which the icosahedron has been described, and that the diameter of the sphere is the sum of (the side) of the hexagon, and two of (the sides) of the decagon, inscribed in the same circle.†

Proposition 17

To construct a dodecahedron, and to enclose (it) in a sphere, like the aforementioned figures, and to show that the side of the dodecahedron is that irrational (straight-line) called an apotome.

Let two planes of the aforementioned cube [Prop. 13.15], $ABCD$ and $CBEF$, (which are) at right-angles to one another, be laid out. And let the sides AB , BC , CD , DA , EF , EB , and FC have each been cut in half at points G , H , K , L , M , N , and O (respectively). And let GK , HL , MH , and NO be joined. And let NP , PO , and HQ have each been cut in extreme and mean ratio at points R , S , and T (respectively). And let their greater pieces be RP , PS , and TQ (respectively). And let RU , SV , and TW be set up on the exterior side of the cube, at points

R , S , and T (respectively), at right-angles to the planes of the cube. And let them be made equal to RP , PS , and TQ . And let UB , BW , WC , CV , and VU be joined.



Λέγω, ὅτι τὸ $YB\Theta\Gamma\Phi$ πεντάγωνον ἰσοπλευρόν τε καὶ ἐν ἐνὶ ἐπιπέδῳ καὶ ἔτι ἰσογώνιον ἔστιν. ἐπεξεύθυνθωσαν γὰρ αἱ PB , ΣB , ΦB . καὶ ἐπεὶ εὐθεῖα ἡ NO ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ P , καὶ τὸ μείζον τμήμα ἔστιν ἡ PO , τὰ ἄρα ἀπὸ τῶν ON , NP τριπλάσιά ἐστι τοῦ ἀπὸ τῆς PO . ἴση δὲ ἡ μὲν ON τῇ NB , ἡ δὲ OP τῇ PY . τὰ ἄρα ἀπὸ τῶν BN , NP τριπλάσιά ἐστι τοῦ ἀπὸ τῆς PY . τοῖς δὲ ἀπὸ τῶν BN , NP τὸ ἀπὸ τῆς BP ἔστιν ἴσον· τὸ ἄρα ἀπὸ τῆς BP τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς PY . ὥστε τὰ ἀπὸ τῶν BP , PY τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς PY . τοῖς δὲ ἀπὸ τῶν BP , PY ἴσον ἔστι τὸ ἀπὸ τῆς BY . τὸ ἄρα ἀπὸ τῆς BY τετραπλάσιόν ἐστι τοῦ ἀπὸ τῆς YP . διπλῇ ἄρα ἔστιν ἡ BY τῇ PY . ἔστι δὲ καὶ ἡ ΦY τῇ YP διπλῇ, ἐπειδήπερ καὶ ἡ ΣP τῇ OP , τουτέστι τῇ PY , ἔστι διπλῇ· ἴση ἄρα ἡ BY τῇ $Y\Phi$. ὁμοίως δὲ δευθθήσεται, ὅτι καὶ ἐκάστη τῶν $B\Theta$, $\Theta\Gamma$, $\Gamma\Phi$ ἑκατέρω

I say that the pentagon $UBWCV$ is equilateral, and in one plane, and, further, equiangular. For let RB , SB , and VB be joined. And since the straight-line NP has been cut in extreme and mean ratio at R , and RP is the greater piece, the (sum of the squares) on PN and NR is thus three times the (square) on RP [Prop. 13.4]. And PN (is) equal to NB , and PR to RU . Thus, the (sum of the squares) on BN and NR is three times the (square) on RU . And the (square) on BR is equal to the (sum of the squares) on BN and NR [Prop. 1.47]. Thus, the (square) on BR is three times the (square) on RU . Hence, the (sum of the squares) on BR and RU is four times the (square) on RU . And the (square) on BU is equal to the (sum of the squares) on BR and RU [Prop. 1.47]. Thus, the (square) on BU is four times the (square) on RU . Thus, BU is double RU . And VU is also double UR , inasmuch as SR is also double PR —that is

τῶν BY , $Y\Phi$ ἔστιν ἴση. ἰσόπλευρον ἄρα ἔστι τὸ $BY\Phi\Gamma\Theta$ πεντάγωνον. λέγω δὴ, ὅτι καὶ ἐν ἐνὶ ἔστιν ἐπιπέδω. ἦθθω γὰρ ἀπὸ τοῦ O ἑκατέρω τῶν PY , $\Sigma\Phi$ παράλληλος ἐπὶ τὰ ἑκτὸς τοῦ κύβου μέρη ἢ $O\Psi$, καὶ ἐπεξεύθθωσαν αἱ $\Psi\Theta$, $\Theta\Theta$. λέγω, ὅτι ἡ $\Psi\Theta\Theta$ εὐθεῖα ἔστιν. ἐπεὶ γὰρ ἡ $\Theta\Pi$ ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ T , καὶ τὸ μείζον αὐτῆς τμήμα ἔστιν ἡ ΠT , ἔστιν ἄρα ὡς ἡ $\Theta\Pi$ πρὸς τὴν ΠT , οὕτως ἡ ΠT πρὸς τὴν $T\Theta$. ἴση δὲ ἡ μὲν $\Theta\Pi$ τῇ ΘO , ἡ δὲ ΠT ἑκατέρω τῶν $T\Theta$, $O\Psi$. ἔστιν ἄρα ὡς ἡ ΘO πρὸς τὴν $O\Psi$, οὕτως ἡ ΘT πρὸς τὴν $T\Theta$. καὶ ἔστι παράλληλος ἡ μὲν ΘO τῇ $T\Theta$. ἑκατέρω γὰρ αὐτῶν τῷ $B\Delta$ ἐπιπέδω πρὸς ὀρθὰς ἔστιν. ἡ δὲ $T\Theta$ τῇ $O\Psi$. ἑκατέρω γὰρ αὐτῶν τῷ BZ ἐπιπέδω πρὸς ὀρθὰς ἔστιν. ἔαν δὲ δύο τρίγωνα συντεθῇ κατὰ μίαν γωνίαν, ὡς τὰ $\Psi O\Theta$, $\Theta T\Theta$, τὰς δύο πλευρὰς ταῖς δυνὶ ἀνάλογον ἔθοντα, ὥστε τὰς ὁμολόγους αὐτῶν πλευρὰς καὶ παραλλήλους εἶναι, αἱ λοιπαὶ εὐθεῖαι ἐπ' εὐθείας ἔσονται. ἐπ' εὐθείας ἄρα ἔστιν ἡ $\Psi\Theta$ τῇ $\Theta\Theta$. πᾶσα δὲ εὐθεῖα ἐν ἐνὶ ἔστιν ἐπιπέδω. ἐν ἐνὶ ἄρα ἐπιπέδω ἔστι τὸ $YB\Theta\Gamma\Phi$ πεντάγωνον.

Λέγω δὴ, ὅτι καὶ ἰσογώνιον ἔστιν.

Ἐπεὶ γὰρ εὐθεῖα γραμμὴ ἡ NO ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ P , καὶ τὸ μείζον τμήμα ἔστιν ἡ OP [ἔστιν ἄρα ὡς συναμφοτέρωξ ἡ NO , OP πρὸς τὴν ON , οὕτως ἡ NO πρὸς τὴν OP], ἴση δὲ ἡ OP τῇ OS [ἔστιν ἄρα ὡς ἡ SN πρὸς τὴν NO , οὕτως ἡ NO πρὸς τὴν OS], ἡ NS ἄρα ἄκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ O , καὶ τὸ μείζον τμήμα ἔστιν ἡ NO . τὰ ἄρα ἀπὸ τῶν NS , SO τριπλάσιά ἐστι τοῦ ἀπὸ τῆς NO . ἴση δὲ ἡ μὲν NO τῇ NB , ἡ δὲ OS τῇ $\Sigma\Phi$. τὰ ἄρα ἀπὸ τῶν NS , $\Sigma\Phi$ τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς NB . ὥστε τὰ ἀπὸ τῶν $\Phi\Sigma$, ΣN , NB τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς NB . ταῖς δὲ ἀπὸ τῶν ΣN , NB ἴσον ἔστι τὸ ἀπὸ τῆς ΣB . τὰ ἄρα ἀπὸ τῶν $B\Sigma$, $\Sigma\Phi$, τουτέστι τὸ ἀπὸ τῆς $B\Phi$ [ὀρθὴ γὰρ ἡ ὑπὸ $\Phi\Sigma B$ γωνία], τετραπλάσιόν ἐστι τοῦ ἀπὸ τῆς NB . διπλῇ ἄρα ἔστιν ἡ ΦB τῇ BN . ἔστι δὲ καὶ ἡ $B\Gamma$ τῇ BN διπλῇ. ἴση ἄρα ἔστιν ἡ $B\Phi$ τῇ $B\Gamma$. καὶ ἐπεὶ δύο αἱ BY , $Y\Phi$ δυοὶ ταῖς $B\Theta$, $\Theta\Gamma$ ἴσαι εἰσίν, καὶ βάσις ἡ $B\Phi$ βάσει τῇ $B\Gamma$ ἴση, γωνία ἄρα ἡ ὑπὸ $BY\Phi$ γωνία τῇ ὑπὸ $B\Theta\Gamma$ ἔστιν ἴση. ὁμοίως δὲ δείχομεν, ὅτι καὶ ἡ ὑπὸ $Y\Phi\Gamma$ γωνία ἴση ἔστι τῇ ὑπὸ $B\Theta\Gamma$. αἱ ἄρα ὑπὸ $B\Theta\Gamma$, $BY\Phi$, $Y\Phi\Gamma$ τρεῖς γωνίαι ἴσαι ἀλλήλαις εἰσίν. ἔαν δὲ πεντάγωνον ἰσοπλεύρον αἱ τρεῖς γωνίαι ἴσαι ἀλλήλαις ᾖσιν, ἰσογώνιον ἔσται τὸ πεντάγωνον. ἰσογώνιον ἄρα ἔστι τὸ $BY\Phi\Gamma\Theta$ πεντάγωνον. ἐδείχθη δὲ καὶ ἰσόπλευρον τὸ ἄρα $BY\Phi\Gamma\Theta$ πεντάγωνον ἰσόπλευρόν ἐστι καὶ ἰσογώνιον, καὶ ἔστιν ἐπὶ μιᾷ τοῦ κύβου πλευρᾷ τῆς $B\Gamma$. ἔαν ἄρα ἐφ' ἑκάστης τῶν τοῦ κύβου δώδεκα πλευρῶν τὰ αὐτὰ κατασκευάσωμεν, συσταθήσεται τι σῆμα στερεόν ὑπὸ δώδεκα πενταγώνων ἰσοπλεύρων τε καὶ ἰσογώνιων περιεχόμενον, ὃ καλεῖται δωδεκάεδρον.

Δεῖ δὲ αὐτὸ καὶ σφαῖρα περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ἄλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐκβεβλήσθω γὰρ ἡ ΨO , καὶ ἔστω ἡ $\Psi\Omega$. συμβάλλει ἄρα

to say, RU . Thus, BU (is) equal to UV . So, similarly, it can be shown that each of BW , WC , CV is equal to each of BU and UV . Thus, pentagon $BUVCW$ is equilateral. So, I say that it is also in one plane. For let PX be drawn from P , parallel to each of RU and SV , on the exterior side of the cube. And let XH and HW be joined. I say that XHW is a straight-line. For since HQ has been cut in extreme and mean ratio at T , and QT is its greater piece, thus as HQ is to QT , so QT (is) to TH . And HQ (is) equal to HP , and QT to each of TW and PX . Thus, as HP is to PX , so WT (is) to TH . And HP is parallel to TW . For of each of them is at right-angles to the plane BD [Prop. 11.6]. And TH (is parallel) to PX . For each of them is at right-angles to the plane BF [Prop. 11.6]. And if two triangles, like XPH and HTW , having two sides proportional to two sides, are placed together at a single angle such that their corresponding sides are also parallel then the remaining sides will be straight-on (to one another) [Prop. 6.32]. Thus, XH is straight-on to HW . And every straight-line is in one plane [Prop. 11.1]. Thus, pentagon $UBWCV$ is in one plane.

So, I say that it is also equiangular.

For since the straight-line NP has been cut in extreme and mean ratio at R , and PR is the greater piece [thus as the sum of NP and PR is to PN , so NP (is) to PR], and PR (is) equal to PS [thus as SN is to NP , so NP (is) to PS], NS has thus also been cut in extreme and mean ratio at P , and NP is the greater piece [Prop. 13.5]. Thus, the (sum of the squares) on NS and SP is three times the (square) on NP [Prop. 13.4]. And NP (is) equal to NB , and PS to SV . Thus, the (sum of the) squares on NS and SV is three times the (square) on NB . Hence, the (sum of the squares) on VS , SN , and NB is four times the (square) on NB . And the (square) on SB is equal to the (sum of the squares) on SN and NB [Prop. 1.47]. Thus, the (sum of the squares) on BS and SV —that is to say, the (square) on BV [for angle $VS B$ (is) a right-angle]—is four times the (square) on NB [Def. 11.3, Prop. 1.47]. Thus, VB is double BN . And BC (is) also double BN . Thus, BV is equal to BC . And since the two (straight-lines) BU and UV are equal to the two (straight-lines) BW and WC (respectively), and the base BV (is) equal to the base BC , angle BUV is thus equal to angle BWC [Prop. 1.8]. So, similarly, we can show that angle UVC is equal to angle BWC . Thus, the three angles BWC , BUV , and UVC are equal to one another. And if three angles of an equilateral pentagon are equal to one another then the pentagon is equiangular [Prop. 13.7]. Thus, pentagon $BUVCW$ is equiangular. And it was also shown (to be) equilateral. Thus, pentagon $BUVCW$ is equilateral and equiangular, and it is on one of the sides, BC , of the cube. Thus, if we make the same construction on each of the twelve sides of the cube then some solid figure contained by twelve equilateral and equiangular pentagons will be constructed, which is called a dodecahedron.

So, it is necessary to enclose it in the given sphere, and

ἡ $O\Omega$ τῇ τοῦ κύβου διαμέτρῳ, καὶ διῖα τέμνονσιν ἀλλήλας· τοῦτο γὰρ δέδεικται ἐν τῷ παρατελευτῶ θεωρήματι τοῦ ἐνδεκάτου βιβλίου. τεμνέτωσαν κατὰ τὸ Ω · τὸ Ω ἄρα κέντρον ἐστὶ τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον, καὶ ἡ $O\Omega$ ἡμίσεια τῆς πλευρᾶς τοῦ κύβου. ἐπεξεύθθω δὴ ἡ $Y\Omega$. καὶ ἐπεὶ εὐθεῖα γραμμὴ ἡ $N\Sigma$ ἄκρον καὶ μέσον λόγον τέμνεται κατὰ τὸ O , καὶ τὸ μείζον αὐτῆς τμημᾶ ἐστὶν ἡ NO , τὰ ἄρα ἀπὸ τῶν $N\Sigma$, ΣO τριπλασιά ἐστι τοῦ ἀπὸ τῆς NO . ἴση δὲ ἡ μὲν $N\Sigma$ τῇ $\Psi\Omega$, ἐπειδήπερ καὶ ἡ μὲν NO τῇ $O\Omega$ ἐστὶν ἴση, ἡ δὲ ΨO τῇ $O\Sigma$. ἀλλὰ μὴν καὶ ἡ $O\Sigma$ τῇ ΨY , ἐπεὶ καὶ τῇ PO · τὰ ἄρα ἀπὸ τῶν $\Omega\Psi$, ΨY τριπλασιά ἐστὶ τοῦ ἀπὸ τῆς NO . τοῖς δὲ ἀπὸ τῶν $\Omega\Psi$, ΨY ἴσον ἐστὶ τὸ ἀπὸ τῆς $Y\Omega$ · τὸ ἄρα ἀπὸ τῆς $Y\Omega$ τριπλασίον ἐστὶ τοῦ ἀπὸ τῆς NO . ἐστὶ δὲ καὶ ἡ ἐκ τοῦ κέντρον τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον δυνάμει τριπλασίῳ τῆς ἡμισείας τῆς τοῦ κύβου πλευρᾶς· προδεδεικται γὰρ κύβον συστήσασθαι καὶ σφαῖρα περιλαβεῖν καὶ δεῖχαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίῳ ἐστὶ τῆς πλευρᾶς τοῦ κύβου. εἰ δὲ ὅλη τῆς ὅλης, καὶ [ἡ] ἡμίσεια τῆς ἡμισείας· καὶ ἐστὶν ἡ NO ἡμίσεια τῆς τοῦ κύβου πλευρᾶς· ἡ ἄρα $Y\Omega$ ἴση ἐστὶ τῇ ἐκ τοῦ κέντρον τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον. καὶ ἐστὶ τὸ Ω κέντρον τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον· τὸ Y ἄρα σημεῖον πρὸς τῇ ἐπιφανείᾳ ἐστὶ τῆς σφαίρας. ὁμοίως δὲ δεῖχμεν, ὅτι καὶ ἐκάστη τῶν λοιπῶν γωνιῶν τοῦ δωδεκαέδρου πρὸς τῇ ἐπιφανείᾳ ἐστὶ τῆς σφαίρας· περιεῖληπται ἄρα τὸ δωδεκαέδρον τῇ δοθείσῃ σφαίρᾳ.

Λέγω δὴ, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ἄλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐπεὶ γὰρ τῆς NO ἄκρον καὶ μέσον λόγον τετμημένης τὸ μείζον τμημᾶ ἐστὶν ὁ PO , τῆς δὲ OX ἄκρον καὶ μέσον λόγον τετμημένης τὸ μείζον τμημᾶ ἐστὶν ἡ $O\Sigma$, ὅλης ἄρα τῆς NX ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμημᾶ ἐστὶν ἡ $P\Sigma$. [οὔτον ἐπεὶ ἐστὶν ὡς ἡ NO πρὸς τὴν OP , ἡ OP πρὸς τὴν PN , καὶ τὰ διπλάσια· τὰ γὰρ μέρη τοῖς ἰσάκας πολλαπλασίοις τὸν αὐτὸν ἔθει λόγον· ὡς ἄρα ἡ NX πρὸς τὴν $P\Sigma$, οὔτως ἡ $P\Sigma$ πρὸς συναμφοτέρον τὴν NP , ΣX . μείζων δὲ ἡ NX τῆς $P\Sigma$ · μείζων ἄρα καὶ ἡ $P\Sigma$ συναμφοτέρον τῆς NP , ΣX · ἡ NX ἄρα ἄκρον καὶ μέσον λόγον τέμνεται, καὶ τὸ μείζον αὐτῆς τμημᾶ ἐστὶν ἡ $P\Sigma$.] ἴση δὲ ἡ $P\Sigma$ τῇ $Y\Phi$ · τῆς ἄρα NX ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμημᾶ ἐστὶν ἡ $Y\Phi$. καὶ ἐπεὶ ῥητὴ ἐστὶν τῆς σφαίρας διάμετρος καὶ ἐστὶ δυνάμει τριπλασίῳ τῆς τοῦ κύβου πλευρᾶς, ῥητὴ ἄρα ἐστὶν ἡ NX πλευρὰ οὔσα τοῦ κύβου. ἔαν δὲ ῥητὴ γραμμὴ ἄκρον καὶ μέσον λόγον τμηθῇ, ἐκάτερον τῶν τμημάτων ἄλογός ἐστιν ἀποτομή.

Ἡ $Y\Phi$ ἄρα πλευρὰ οὔσα τοῦ δωδεκαέδρου ἄλογός ἐστιν ἀποτομή.

to show that the side of the dodecahedron is that irrational (straight-line) called an apotome.

For let XP be produced, and let (the produced straight-line) be XZ . Thus, PZ meets the diameter of the cube, and they cut one another in half. For, this has been proved in the penultimate theorem of the eleventh book [Prop. 11.38]. Let them cut (one another) at Z . Thus, Z is the center of the sphere enclosing the cube, and ZP (is) half the side of the cube. So, let UZ be joined. And since the straight-line NS has been cut in extreme and mean ratio at P , and its greater piece is NP , the (sum of the squares) on NS and SP is thus three times the (square) on NP [Prop. 13.4]. And NS (is) equal to XZ , inasmuch as NP is also equal to PZ , and XP to PS . But, indeed, PS (is) also (equal) to XU , since (it is) also (equal) to RP . Thus, the (sum of the squares) on ZX and XU is three times the (square) on NP . And the (square) on UZ is equal to the (sum of the squares) on ZX and XU [Prop. 1.47]. Thus, the (square) on UZ is three times the (square) on NP . And the square on the radius of the sphere enclosing the cube is also three times the (square) on half the side of the cube. For it has previously been demonstrated (how to) construct the cube, and to enclose (it) in a sphere, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube [Prop. 13.15]. And if the (square on the) whole (is three times) the (square on the) whole, then the (square on the) half (is) also (three times) the (square on the) half. And NP is half of the side of the cube. Thus, UZ is equal to the radius of the sphere enclosing the cube. And Z is the center of the sphere enclosing the cube. Thus, point U is on the surface of the sphere. So, similarly, we can show that each of the remaining angles of the dodecahedron is also on the surface of the sphere. Thus, the dodecahedron has been enclosed by the given sphere.

So, I say that the side of the dodecahedron is that irrational straight-line called an apotome.

For since RP is the greater piece of NP , which has been cut in extreme and mean ratio, and PS is the greater piece of PO , which has been cut in extreme and mean ratio, RS is thus the greater piece of the whole of NO , which has been cut in extreme and mean ratio. [Thus, since as NP is to PR , (so) PR (is) to RN , and (the same is also true) of the doubles. For parts have the same ratio as similar multiples (taken in corresponding order) [Prop. 5.15]. Thus, as NO (is) to RS , so RS (is) to the sum of NR and SO . And NO (is) greater than RS . Thus, RS (is) also greater than the sum of NR and SO [Prop. 5.14]. Thus, NO has been cut in extreme and mean ratio, and RS is its greater piece.] And RS (is) equal to UV . Thus, UV is the greater piece of NO , which has been cut in extreme and mean ratio. And since the diameter of the sphere is rational, and the square on it is three times the (square) on the side of the cube, NO , which is the side of the cube, is thus rational. And if a rational (straight)-line is cut in extreme and

mean ratio then each of the pieces is the irrational (straight-line called) an apotome.

Thus, UV , which is the side of the dodecahedron, is the irrational (straight-line called) an apotome [Prop. 13.6].

Πόρισμα.

Ἐκ δὴ τούτων φανερόν, ὅτι τῆς τοῦ κύβου πλευρᾶς ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα ἐστὶν ἡ τοῦ δωδεκαέδρου πλευρά. ὅπερ ἔδει δεῖξαι.

Corollary

So, (it is) clear, from this, that the side of the dodecahedron is the greater piece of the side of the cube, when it is cut in extreme and mean ratio.[†] (Which is) the very thing it was required to show.

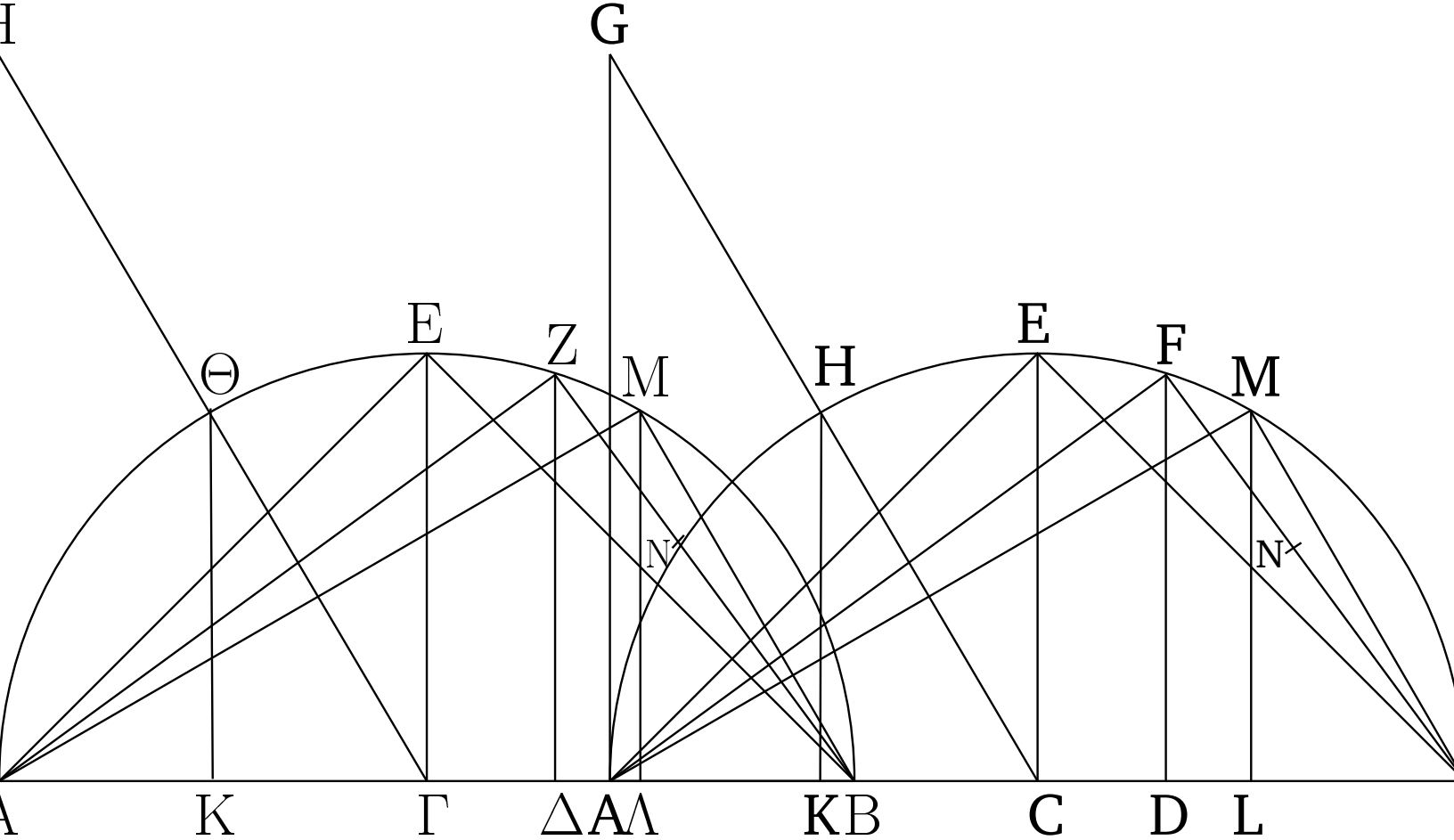
[†] If the radius of the circumscribed sphere is unity then the side of the cube is $\sqrt{4/3}$, and the side of the dodecahedron is $(1/3)(\sqrt{15} - \sqrt{3})$.

18.

Τὰς πλευρὰς τῶν πέντε σθημάτων ἐκθέσθαι καὶ συγκρῖναι πρὸς ἀλλήλας.

Proposition 18

To set out the sides of the five (aforementioned) figures, and to compare (them) with one another.[†]



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB , καὶ τετμήσθω κατὰ τὸ Γ ὥστε ἴσῃ εἶναι τὴν $A\Gamma$ τῇ ΓB , κατὰ δὲ τὸ Δ ὥστε διπλασίονα εἶναι τὴν $A\Delta$ τῆς ΔB , καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AEB , καὶ ἀπὸ τῶν Γ , Δ τῇ AB

Let the diameter, AB , of the given sphere be laid out. And let it be cut at C , such that AC is equal to CB , and at D , such that AD is double DB . And let the semi-circle AEB be drawn on AB . And let CE and DF be drawn from C and D (respec-

πρὸς ὀρθὰς ἡθῆσαν αἱ $ΓΕ$, $ΔΖ$, καὶ ἐπεξεύθθωσαν αἱ AZ , ZB , EB . καὶ ἐπεὶ διπλὴ ἐστὶν ἡ $ΑΔ$ τῆς $ΔΒ$, τριπλὴ ἄρα ἐστὶν ἡ $ΑΒ$ τῆς $ΒΔ$. ἀναστρέφαντι ἡμιολία ἄρα ἐστὶν ἡ $ΒΑ$ τῆς $ΑΔ$. ὥς δὲ ἡ $ΒΑ$ πρὸς τὴν $ΑΔ$, οὕτως τὸ ἀπὸ τῆς $ΒΑ$ πρὸς τὸ ἀπὸ τῆς AZ ἰσογώνιον γάρ ἐστι τὸ AZB τριγώνον τῷ $AZΔ$ τριγώνῳ· ἡμιόλιον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΒΑ$ τοῦ ἀπὸ τῆς AZ . ἔστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει ἡμιολία τῆς πλευρᾶς τῆς πυραμίδος. καὶ ἐστὶν ἡ $ΑΒ$ ἡ τῆς σφαίρας διάμετρος· ἡ AZ ἄρα ἴση ἐστὶ τῇ πλευρᾷ τῆς πυραμίδος.

Πάλιν, ἐπεὶ διπλασίον ἐστὶν ἡ $ΑΔ$ τῆς $ΔΒ$, τριπλὴ ἄρα ἐστὶν ἡ $ΑΒ$ τῆς $ΒΔ$. ὥς δὲ ἡ $ΑΒ$ πρὸς τὴν $ΒΔ$, οὕτως τὸ ἀπὸ τῆς $ΑΒ$ πρὸς τὸ ἀπὸ τῆς BZ · τριπλασίον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΑΒ$ τοῦ ἀπὸ τῆς BZ . ἔστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίον τῆς τοῦ κύβου πλευρᾶς. καὶ ἐστὶν ἡ $ΑΒ$ ἡ τῆς σφαίρας διάμετρος· ἡ BZ ἄρα τοῦ κύβου ἐστὶ πλευρά.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ $ΑΓ$ τῇ $ΓΒ$, διπλὴ ἄρα ἐστὶν ἡ $ΑΒ$ τῆς $ΒΓ$. ὥς δὲ ἡ $ΑΒ$ πρὸς τὴν $ΒΓ$, οὕτως τὸ ἀπὸ τῆς $ΑΒ$ πρὸς τὸ ἀπὸ τῆς $ΒΕ$ · διπλασίον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΑΒ$ τοῦ ἀπὸ τῆς $ΒΕ$. ἔστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίον τῆς τοῦ ὀκταέδρου πλευρᾶς. καὶ ἐστὶν ἡ $ΑΒ$ ἡ τῆς δοθείσης σφαίρας διάμετρος· ἡ $ΒΕ$ ἄρα τοῦ ὀκταέδρου ἐστὶ πλευρά.

Ἦθῃ δὴ ἀπὸ τοῦ A σημείου τῇ $ΑΒ$ εὐθείᾳ πρὸς ὀρθὰς ἡ AH , καὶ κείσθω ἡ AH ἴση τῇ $ΑΒ$, καὶ ἐπεξεύθθω ἡ $ΗΓ$, καὶ ἀπὸ τοῦ $Θ$ ἐπὶ τὴν $ΑΒ$ κάθετος ἡθῇ ἡ $ΘΚ$. καὶ ἐπεὶ διπλὴ ἐστὶν ἡ $ΗΑ$ τῆς $ΑΓ$ · ἴση γάρ ἡ $ΗΑ$ τῇ $ΑΒ$ · ὥς δὲ ἡ $ΗΑ$ πρὸς τὴν $ΑΓ$, οὕτως ἡ $ΘΚ$ πρὸς τὴν $ΚΓ$, διπλὴ ἄρα καὶ ἡ $ΘΚ$ τῆς $ΚΓ$. τετραπλασίον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΘΚ$ τοῦ ἀπὸ τῆς $ΚΓ$ · τὰ ἄρα ἀπὸ τῶν $ΘΚ$, $ΚΓ$, ὅπερ ἐστὶ τὸ ἀπὸ τῆς $ΘΓ$, πενταπλασίον ἐστὶ τοῦ ἀπὸ τῆς $ΚΓ$. ἴση δὲ ἡ $ΘΓ$ τῇ $ΓΒ$ · πενταπλασίον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΒΓ$ τοῦ ἀπὸ τῆς $ΓΚ$. καὶ ἐπεὶ διπλὴ ἐστὶν ἡ $ΑΒ$ τῆς $ΓΒ$, ὥν ἡ $ΑΔ$ τῆς $ΔΒ$ ἐστὶ διπλὴ, λοιπὴ ἄρα ἡ $ΒΔ$ λοιπῆς τῆς $ΔΓ$ ἐστὶ διπλὴ. τριπλὴ ἄρα ἡ $ΒΓ$ τῆς $ΓΔ$ · ἐνναπλασίον ἄρα τὸ ἀπὸ τῆς $ΒΓ$ τοῦ ἀπὸ τῆς $ΓΔ$. πενταπλασίον δὲ τὸ ἀπὸ τῆς $ΒΓ$ τοῦ ἀπὸ τῆς $ΓΚ$ · μείζον ἄρα τὸ ἀπὸ τῆς $ΓΚ$ τοῦ ἀπὸ τῆς $ΓΔ$. μείζων ἄρα ἐστὶν ἡ $ΓΚ$ τῆς $ΓΔ$. κείσθω τῇ $ΓΚ$ ἴση ἡ $ΓΛ$, καὶ ἀπὸ τοῦ $Λ$ τῇ $ΑΒ$ πρὸς ὀρθὰς ἡθῇ ἡ $ΛΜ$, καὶ ἐπεξεύθθω ἡ $ΜΒ$. καὶ ἐπεὶ πενταπλασίον ἐστὶ τὸ ἀπὸ τῆς $ΒΓ$ τοῦ ἀπὸ τῆς $ΓΚ$, καὶ ἐστὶ τῆς μὲν $ΒΓ$ διπλὴ ἡ $ΑΒ$, τῆς δὲ $ΓΚ$ διπλὴ ἡ $ΚΛ$, πενταπλασίον ἄρα ἐστὶ τὸ ἀπὸ τῆς $ΑΒ$ τοῦ ἀπὸ τῆς $ΚΛ$. ἔστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει πενταπλασίον τῆς ἐκ τοῦ κέντρου τοῦ κύκλου, ἀφ' οὗ τὸ εἰκοσάεδρον ἀναγέγραπται. καὶ ἐστὶν ἡ $ΑΒ$ ἡ τῆς σφαίρας διάμετρος· ἡ $ΚΛ$ ἄρα ἐκ τοῦ κέντρου ἐστὶ τοῦ κύκλου, ἀφ' οὗ τὸ εἰκοσάεδρον ἀναγέγραπται· ἡ $ΚΛ$ ἄρα ἐχαγώνον ἐστὶ πλευρά τοῦ εἰρημένου κύκλου. καὶ ἐπεὶ ἡ τῆς σφαίρας διάμετρος σύγκειται ἐκ τε τῆς τοῦ ἐχαγώνου καὶ δύο τῶν τοῦ δεκαγώνου τῶν εἰς τὸν εἰρημένον κύκλον ἐγγραφομένων, καὶ ἐστὶν ἡ μὲν $ΑΒ$ ἡ τῆς σφαίρας διάμετρος, ἡ δὲ $ΚΛ$ ἐχαγώνον πλευρά, καὶ ἴση ἡ $ΑΚ$ τῇ $ΛΒ$, ἐκατέρα ἄρα

tively), at right-angles to AB . And let AF , FB , and EB be joined. And since AD is double DB , AB is thus triple BD . Thus, via conversion, BA is one and a half times AD . And as BA (is) to AD , so the (square) on BA (is) to the (square) on AF [Def. 5.9]. For triangle AFB is equiangular to triangle AFD [Prop. 6.8]. Thus, the (square) on BA is one and a half times the (square) on AF . And the square on the diameter of the sphere is also one and a half times the (square) on the side of the pyramid [Prop. 13.13]. And AB is the diameter of the sphere. Thus, AF is equal to the side of the pyramid.

Again, since AD is double DB , AB is thus triple BD . And as AB (is) to BD , so the (square) on AB (is) to the (square) on BF [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is three times the (square) on BF . And the square on the diameter of the sphere is also three times the (square) on the side of the cube [Prop. 13.15]. And AB is the diameter of the sphere. Thus, BF is the side of the cube.

And since AC is equal to CB , AB is thus double BC . And as AB (is) to BC , so the (square) on AB (is) to the (square) on BE [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is double the (square) on BE . And the square on the diameter of the sphere is also double the (square) on the side of the octagon [Prop. 13.14]. And AB is the diameter of the given sphere. Thus, BE is the side of the octagon.

So let AG be drawn from point A at right-angles to the straight-line AB . And let AG be made equal to AB . And let GC be joined. And let HK be drawn from H , perpendicular to AB . And since GA is double AC . For GA (is) equal to AB . And as GA (is) to AC , so HK (is) to KC [Prop. 6.4]. HK (is) thus also double KC . Thus, the (square) on HK is four times the (square) on KC . Thus, the (sum of the squares) on HK and KC , which is the (square) on HC [Prop. 1.47], is five times the (square) on KC . And HC (is) equal to CB . Thus, the (square) on BC (is) five times the (square) on CK . And since AB is double CB , of which AD is double DB , the remainder BD is thus double the remainder DC . BC (is) thus triple CD . The (square) on BC (is) thus nine times the (square) on CD . And the (square) on BC (is) five times the (square) on CK . Thus, the (square) on CK (is) greater than the (square) on CD . CK is thus greater than CD . Let CL be made equal to CK . And let LM be drawn from L at right-angles to AB . And let MB be joined. And since the (square) on BC is five times the (square) on CK , and AB is double BC , and KL double CK , the (square) on AB is thus five times the (square) on KL . And the square on the diameter of the sphere is also five times the (square) on the radius of the circle from which the icosahedron has been described [Prop. 13.16 corr.]. And AB is the diameter of the sphere. Thus, KL is the radius of the circle from which the icosahedron has been described. Thus, KL is (the side) of the hexagon (inscribed) in the aforementioned circle [Prop. 4.15 corr.]. And since the diameter of the

τῶν AK , AB δεκαγώνου ἐστὶ πλευρὰ τοῦ ἐγγραφομένου εἰς τὸν κύκλον, ἀφ' οὗ τὸ εἰκοσαέδρον ἀναγράφεται. καὶ ἐπεὶ δεκαγώνου μὲν ἡ AB , ἑξαγώνου δὲ ἡ ML . ἴση γάρ ἐστι τῇ KL , ἐπεὶ καὶ τῇ OK . ἴσον γὰρ ἀπέθουσιν ἀπὸ τοῦ κέντρου· καὶ ἐστὶν ἑκατέρω τῶν OK , KL διπλασίων τῆς $KΓ$. πενταγώνου ἄρα ἐστὶν ἡ MB . ἡ δὲ τοῦ πενταγώνου ἐστὶν ἡ τοῦ εἰκοσαέδρου· εἰκοσαέδρου ἄρα ἐστὶν ἡ MB .

Καὶ ἐπεὶ ἡ ZB κύβου ἐστὶ πλευρά, τετμήσθω ἄκρον καὶ μέσον λόγον κατὰ τὸ N , καὶ ἔστω μείζον τμήμα τὸ NB . ἡ NB ἄρα δωδεκαέδρου ἐστὶ πλευρά.

Καὶ ἐπεὶ ἡ τῆς σφαίρας διάμετρος ἐδείχθη τῆς μὲν AZ πλευρᾶς τῆς πυραμίδος δυνάμει ἡμιολία, τῆς δὲ τοῦ ὀκταέδρου τῆς BE δυνάμει διπλασίον, τῆς δὲ τοῦ κύβου τῆς ZB δυνάμει τριπλασίον, οἷον ἄρα ἡ τῆς σφαίρας διάμετρος δυνάμει ἔχ, τοιούτων ἡ μὲν τῆς πυραμίδος τεσσάρων, ἡ δὲ τοῦ ὀκταέδρου τριῶν, ἡ δὲ τοῦ κύβου δύο. ἡ μὲν ἄρα τῆς πυραμίδος πλευρὰ τῆς μὲν τοῦ ὀκταέδρου πλευρᾶς δυνάμει ἐστὶν ἐπίτριτος, τῆς δὲ τοῦ κύβου δυνάμει διπλή, ἡ δὲ τοῦ ὀκταέδρου τῆς τοῦ κύβου δυνάμει ἡμιολία. αἱ μὲν οὖν εἰρημέναι τῶν τριῶν σθημάτων πλευραὶ, λέγω δὴ πυραμίδος καὶ ὀκταέδρου καὶ κύβου, πρὸς ἀλλήλας εἰσὶν ἐν λόγοις ῥητοῖς. αἱ δὲ λοιπαὶ δύο, λέγω δὴ ἡ τε τοῦ εἰκοσαέδρου καὶ ἡ τοῦ δωδεκαέδρου, οὔτε πρὸς ἀλλήλας οὔτε πρὸς τὰς προειρημένας εἰσὶν ἐν λόγοις ῥητοῖς· ἄλογοι γάρ εἰσιν, ἡ μὲν ἐλάττων, ἡ δὲ ἀποτομή.

Ὅτι μείζων ἐστὶν ἡ τοῦ εἰκοσαέδρου πλευρὰ ἡ MB τῆς τοῦ δωδεκαέδρου τῆς NB , δείχομεν οὕτως.

Ἐπεὶ γὰρ ἰσογώνιον ἐστὶ τὸ $ZΔB$ τρίγωνον τῷ ZAB τριγώνῳ, ἀνάλογόν ἐστιν ὡς ἡ $ΔB$ πρὸς τὴν BZ , οὕτως ἡ BZ πρὸς τὴν BA . καὶ ἐπεὶ τρεῖς εὐθεῖαι ἀνάλογόν εἰσιν, ἔστιν ὡς ἡ πρώτη πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας· ἔστιν ἄρα ὡς ἡ $ΔB$ πρὸς τὴν BA , οὕτως τὸ ἀπὸ τῆς $ΔB$ πρὸς τὸ ἀπὸ τῆς BZ . ἀνάπαλιν ἄρα ὡς ἡ AB πρὸς τὴν BD , οὕτως τὸ ἀπὸ τῆς ZB πρὸς τὸ ἀπὸ τῆς BD . τριπλὴ δὲ ἡ AB τῆς BD . τριπλασίον ἄρα τὸ ἀπὸ τῆς ZB τοῦ ἀπὸ τῆς BD . ἔστι δὲ καὶ τὸ ἀπὸ τῆς AD τοῦ ἀπὸ τῆς $ΔB$ τετραπλάσιον· διπλὴ γάρ ἡ AD τῆς $ΔB$. μείζον ἄρα τὸ ἀπὸ τῆς AD τοῦ ἀπὸ τῆς ZB . μείζων ἄρα ἡ AD τῆς ZB . πολλῶν ἄρα ἡ AD τῆς ZB μείζων ἐστίν. καὶ τῆς μὲν AD ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα ἐστὶν ἡ KA , ἐπειδήπερ ἡ μὲν AK ἑξαγώνου ἐστίν, ἡ δὲ KA δεκαγώνου· τῆς δὲ ZB ἄκρον καὶ μέσον λόγον τεμνομένης τὸ μείζον τμήμα ἐστὶν ἡ NB . μείζων ἄρα ἡ KA τῆς NB . ἴση δὲ ἡ KA τῇ AM . μείζων ἄρα ἡ AM τῆς NB [τῆς δὲ AM μείζων ἐστὶν ἡ MB]. πολλῶν ἄρα ἡ MB πλευρὰ οὕσα τοῦ εἰκοσαέδρου μείζων ἐστὶ τῆς NB πλευρᾶς οὕσης τοῦ δωδεκαέδρου· ὅπερ ἔδει δεῖχαι.

sphere is composed of (the side) of the hexagon, and two of (the sides) of the decagon, inscribed in the aforementioned circle, and AB is the diameter of the sphere, and KL the side of the hexagon, and AK (is) equal to LB , thus AK and LB are each sides of the decagon inscribed in the circle from which the icosahedron has been described. And since LB is (the side) of the decagon. And ML (is the side) of the hexagon—for (it is) equal to KL , since (it is) also (equal) to HK , for they are equally far from the center. And HK and KL are each double KC . MB is thus (the side) of the pentagon (inscribed in the circle) [Props. 13.10, 1.47]. And (the side) of the pentagon is (the side) of the icosahedron [Prop. 13.16]. Thus, MB is (the side) of the icosahedron.

And since FB is the side of the cube, let it be cut in extreme and mean ratio at N , and let NB be the greater piece. Thus, NB is the side of the dodecahedron [Prop. 13.17 corr.].

And since the (square) on the diameter of the sphere was shown (to be) one and a half times the square on the side, AF , of the pyramid, and twice the square on (the side), BE , of the octagon, and three times the square on (the side), FB , of the cube, thus, of whatever (parts) the (square) on the diameter of the sphere (makes) six, of such (parts) the (square) on (the side) of the pyramid (makes) four, and (the square) on (the side) of the octagon three, and (the square) on (the side) of the cube two. Thus, the (square) on the side of the pyramid is one and a third times the square on the side of the octagon, and double the square on (the side) of the cube. And the (square) on (the side) of the octahedron is one and a half times the square on (the side) of the cube. Therefore, the aforementioned sides of the three figures—I mean, of the pyramid, and of the octahedron, and of the cube—are in rational ratios to one another. And (the sides of) the remaining two (figures)—I mean, of the icosahedron, and of the dodecahedron—are neither in rational ratios to one another, nor to the (sides) of the aforementioned (three figures). For they are irrational (straight-lines): (namely), a minor [Prop. 13.16], and an apotome [Prop. 13.17].

(And), we can show that the side, MB , of the icosahedron is greater than the (side), NB , or the dodecahedron, as follows.

For, since triangle FDB is equiangular to triangle FAB [Prop. 6.8], proportionally, as DB is to BF , so BF (is) to BA [Prop. 6.4]. And since three straight-lines are (continually) proportional, as the first (is) to the third, so the (square) on the first (is) to the (square) on the second [Def. 5.9, Prop. 6.20 corr.]. Thus, as DB is to BA , so the (square) on DB (is) to the (square) on BF . Thus, inversely, as AB (is) to BD , so the (square) on FB (is) to the (square) on BD . And AB (is) triple BD . Thus, the (square) on FB (is) three times the (square) on BD . And the (square) on AD is also four times the (square) on DB . For AD (is) double DB . Thus, the (square) on AD (is) greater than the (square) on FB . Thus, AD (is)

greater than FB . Thus, AL is much greater than FB . And KL is the greater piece of AL , which is cut in extreme and mean ratio—inasmuch as LK is (the side) of the hexagon, and KA (the side) of the decagon [Prop. 13.9]. And NB is the greater piece of FB , which is cut in extreme and mean ratio. Thus, KL (is) greater than NB . And KL (is) equal to LM . Thus, LM (is) greater than NB [and MB is greater than LM]. Thus, MB , which is (the side) of the icosahedron, is much greater than NB , which is (the side) of the dodecahedron. (Which is) the very thing it was required to show.

† If the radius of the given sphere is unity then the sides of the pyramid (i.e., tetrahedron), octahedron, cube, icosahedron, and dodecahedron, respectively, satisfy the following inequality: $\sqrt{8/3} > \sqrt{2} > \sqrt{4/3} > (1/\sqrt{5}) \sqrt{10-2\sqrt{5}} > (1/3)(\sqrt{15}-\sqrt{3})$.

Λέγω δὴ, ὅτι παρὰ τὰ εἰρημένα πέντε σήματα οὐ συνσταθήσεται ἕτερον σῆμα περιεχόμενον ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίων ἴσων ἀλλήλοις.

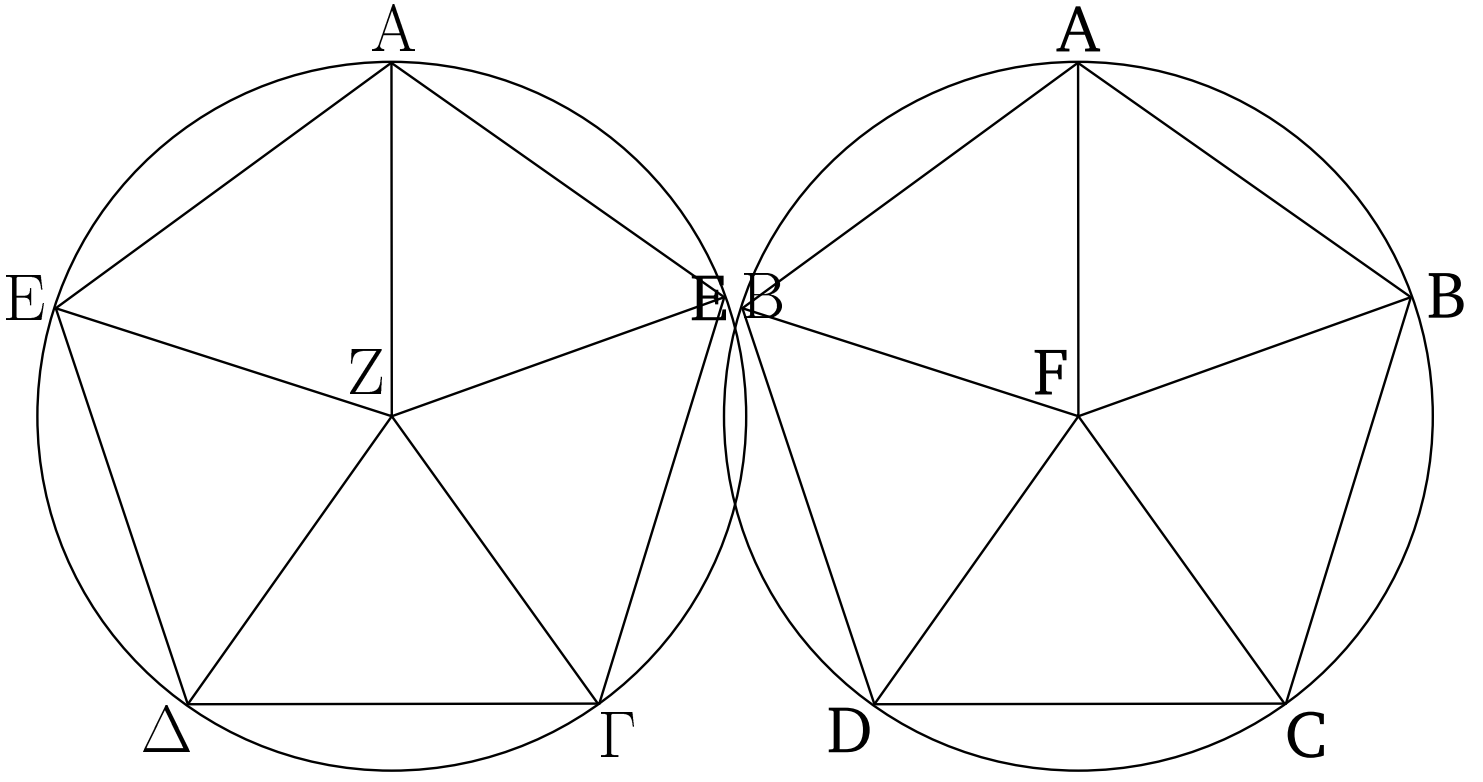
Ὑπὸ μὲν γὰρ δύο τριγώνων ἢ ὅλῳ ἐπιπέδων στερεὰ γωνία οὐ συνίσταται. ὑπὸ δὲ τριῶν τριγώνων ἢ τῆς πυραμίδος, ὑπὸ δὲ τεσσάρων ἢ τοῦ ὀκταέδρου, ὑπὸ δὲ πέντε ἢ τοῦ εἰκοσαέδρου· ὑπὸ δὲ ἔχ τριγώνων ἰσοπλεύρων τε καὶ ἰσογωνίων πρόξ ἐν σημείῳ συνισταμένων οὐκ ἔσται στερεὰ γωνία· οὕσης γὰρ τῆς τοῦ ἰσοπλεύρου τριγώνου γωνίας ὁμοίρου ὀρθῆς ἔσονται αἱ ἔχ τέσσαρσιν ὀρθαῖς ἴσαι· ὅπερ ἀδύνατον· ἅπαντα γὰρ στερεὰ γωνία ὑπὸ ἐλασσόνων ἢ τεσσάρων ὀρθῶν περιέθεται. διὰ τὰ αὐτὰ δὴ οὐδὲ ὑπὸ πλειόνων ἢ ἔχ γωνιῶν ἐπιπέδων στερεὰ γωνία συνίσταται. ὑπὸ δὲ τετραγώνων τριῶν ἢ τοῦ κύβου γωνία περιέθεται· ὑπὸ δὲ τεσσάρων ἀδύνατον· ἔσονται γὰρ πάλιν τέσσαρες ὀρθαί. ὑπὸ δὲ πενταγώνων ἰσοπλεύρων καὶ ἰσογωνίων, ὑπὸ μὲν τριῶν ἢ τοῦ δωδεκαέδρου· ὑπὸ δὲ τεσσάρων ἀδύνατον· οὕσης γὰρ τῆς τοῦ πενταγώνου ἰσοπλεύρου γωνίας ὀρθῆς καὶ πέμπτου, ἔσονται αἱ τέσσαρες γωνίαι τεσσάρων ὀρθῶν μείζονες· ὅπερ ἀδύνατον. οὐδὲ μὴν ὑπὸ πολυγώνων ἑτέρων σημάτων περισθεθήσεται στερεὰ γωνία διὰ τὸ αὐτὸ ἄτοπον.

Οὐκ ἄρα παρὰ τὰ εἰρημένα πέντε σήματα ἕτερον σῆμα στερεὸν συνσταθήσεται ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίων περιεχόμενον· ὅπερ ἔδει δεῖχαι.

So, I say that, beside the five aforementioned figures, no other (solid) figure can be constructed (which is) contained by equilateral and equiangular (planes), equal to one another.

For a solid angle cannot be constructed from two triangles, or indeed (two) planes (of any sort) [Def. 11.11]. And (the solid angle) of the pyramid (is constructed) from three (equiangular) triangles, and (that) of the octahedron from four (triangles), and (that) of the icosahedron from (five) triangles. And a solid angle cannot be (made) from six equilateral and equiangular triangles set up together at one point. For, since the angles of a equilateral triangle are (each) two-thirds of a right-angle, the (sum of the) six (plane) angles (containing the solid angle) will be four right-angles. The very thing (is) impossible. For every solid angle is contained by (plane angles whose sum is) less than four right-angles [Prop. 11.21]. So, for the same (reasons), a solid angle cannot be constructed from more than six plane angles (equal to two-thirds of a right-angle) either. And the (solid) angle of a cube is contained by three squares. And (a solid angle contained) by four (squares is) impossible. For, again, the (sum of the plane angles containing the solid angle) will be four right-angles. And (the solid angle) of a dodecahedron (is contained) by three equilateral and equiangular pentagons. And (a solid angle contained) by four (equiangular pentagons is) impossible. For, the angle of an equilateral pentagon being one and one-fifth of right-angle, four (such) angles will be greater (in sum) than four right-angles. The very thing (is) impossible. And, on account of the same absurdity, a solid angle cannot be constructed from any other (equiangular) polygonal figures either.

Thus, beside the five aforementioned figures, no other solid figure can be constructed (which is) contained by equilateral and equiangular (planes). (Which is) the very thing it was required to show.



Λήμμα.

Lemma

Ὅτι δὲ ἡ τοῦ ἰσοπλεύρου καὶ ἰσογωνίου πενταγώνου γωνία ὀρθή ἐστι καὶ πέμπτων, οὕτω δεικτέον.

Ἐστω γάρ πεντάγωνον ἰσόπλευρον καὶ ἰσογώνιον τὸ $AB\Gamma\Delta E$, καὶ περιγεγράφθω περὶ αὐτὸ κύκλος ὁ $AB\Gamma\Delta E$, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον τὸ Z , καὶ ἐπεξέυθθωσαν αἱ $ZA, ZB, Z\Gamma, Z\Delta, ZE$. διὶα ἄρα τέμνουσι τὰς πρὸς τοῖς A, B, Γ, Δ, E τοῦ πενταγώνου γωνίας. καὶ ἐπεὶ αἱ πρὸς τῷ Z πέντε γωνίαι τέσσαρσιν ὀρθαῖς ἴσαι εἰσὶ καὶ εἰσιν ἴσαι, μία ἄρα αὐτῶν, ὥς ἡ ὑπὸ AZB , μιᾶς ὀρθῆς ἐστὶ παρὰ πέμπτων· λοιπαὶ ἄρα αἱ ὑπὸ ZAB, ABZ μιᾶς εἰσιν ὀρθῆς καὶ πέμπτων. ἴση δὲ ἡ ὑπὸ ZAB τῇ ὑπὸ $ZB\Gamma$ · καὶ ὅλη ἄρα ἡ ὑπὸ $AB\Gamma$ τοῦ πενταγώνου γωνία μιᾶς ἐστὶν ὀρθῆς καὶ πέμπτων· ὅπερ ἔδει δεῖξαι.

It can be shown that the angle of an equilateral and equiangular pentagon is one and one-fifth of a right-angle, as follows.

For let $ABCDE$ be an equilateral and equiangular pentagon, and let the circle $ABCDE$ be circumscribed about it [Prop. 4.14]. And let its center, F , be found [Prop. 3.1]. And let FA, FB, FC, FD , and FE be joined. Thus, they cut the angles of the pentagon in half at (points) A, B, C, D , and E [Prop. 1.4]. And since the five angles at F are equal (in sum) to four right-angles, and are also equal (to one another), (any) one of them, like AFB , is thus one less a fifth of a right-angle. Thus, the (sum of the) remaining (angles in triangle ABF), FAB and ABF , is one plus a fifth of a right-angle [Prop. 1.32]. And FAB (is) equal to FBC . Thus, the whole angle, ABC , of the pentagon is also one and one-fifth of a right-angle. (Which is) the very thing it was required to show.

GREEK-ENGLISH LEXICON

ABBREVIATIONS: *act* - active; *adj* - adjective; *adv* - adverb; *conj* - conjunction; *fut* - future; *gen* - genitive; *imperat* - imperative; *imperf* - imperfect; *ind* - indeclinable; *indic* - indicative; *intr* - intransitive; *mid* - middle; *neut* - neuter; *no* - noun; *par* - particle; *part* - participle; *pass* - passive; *perf* - perfect; *pre* - preposition; *pres* - present; *pro* - pronoun; *sg* - singular; *tr* - transitive; *vb* - verb.

ἄγω, ἄχω, ἡγαγον, -ῆθα, ἡγμαι, ἡθθην : *vb*, lead, draw (a line).

ἀδύνατος -ον : *adj*, impossible.

ἀεί : *adv*, always, for ever.

αἰρέω, αἰρήσω, εἴ[~]λον, ἤρῃκα, ἤρημαι, ἡρέθην : *vb*, grasp.

αἰτέω, αἰτήσω, ἤτησα, ἤτηκα, ἤτημαι, ἡτήθη : *vb*, postulate.

ἀΐτημα -ατος, τό : *no*, postulate.

ἀκόλουθος -ον : *adj*, analogous, consequent on, in conformity with.

ἄκρος -α -ον : *adj*, outermost, end, extreme.

ἀλλά : *conj*, but, otherwise.

ἄλογος -ον : *adj*, irrational.

ἅμα : *adv*, at once, at the same time, together.

ἀμβλυγώνιος -ον : *adj*, obtuse-angled; τό ἀμβλυγώνιον, *no*, obtuse angle.

ἀμβλύς -εῖα -ύ : *adj*, obtuse.

ἀμφότερος -α -ον : *pro*, both.

ἀναγράφω : *vb*, describe (a figure); see γράφω.

ἀναλογία, ἡ : *no*, proportion, (geometric) progression.

ἀνάλογος -ον : *adj*, proportional.

ἀνάπαλιν : *adv*, inverse(ly).

αναπληρώω : *vb*, fill up.

ἀναστρέφω : *vb*, turn upside down, convert (ratio); see στρέφω.

ἀναστροφή, ἡ : *no*, turning upside down, conversion (of ratio).

ἀνθρῳαίρεω : *vb*, take away in turn; see αἰρέω.

ἀνίστημι : *vb*, set up; see ἵστημι.

ἄνισος -ον : *adj*, unequal, uneven.

ἀντιπάσθω : *vb*, be reciprocally proportional; see πάσθω.

ἄχων -ονος, ὁ : *vb*, axis.

ἅπαχ : *adv*, once.

ἅπαξ, ἅπασα, ἅπαν : *adj*, quite all, the whole.

ἄπειρος -ον : *adj*, infinite.

ἀπεναντίον : *ind*, opposite.

ἀπέθω : *vb*, be far from, be away from; see ἔθω.

ἀπλατήξ -έξ : *adj*, without breadth.

ἀπόδειξις -εως, ἡ : *no*, proof.

ἀποκαθίστημι : *vb*, re-establish, restore; see ἵστημι.

ἀπολαμβάνω : *vb*, take from, subtract from, cut off from; see λαμβάνω.

ἀποτέμνω : *vb*, cut off, subtend.

ἀπότμημα -ατος, τό : *no*, piece cut off, segment.

ἀποτομή, ἡ : *vb*, piece cut off, apotome.

ἅπτω, ἅψω, ἥψα, —, ἥμμαι, — : *vb*, touch, join, meet.

ἀπώτερος -α -ον : *adj*, further off.

ἄρα : *par*, thus, as it seems (inferential).

ἀριθμός, ὁ : *no*, number.

ἀρτιάκις : *adv*, an even number of times.

ἀρτιόπλευρος -ον : *adj*, having an even number of sides.

ἄρθω, ἄρχω, ἤρξα, ἤρθα, ἤρχμαι, ἤρθθην : *vb*, rule; *mid.*, begin.

ἀσύμμετρος -ον : *adj*, incommensurable.

ἀσύνπτωτος -ον : *adj*, not touching, not meeting.

ἄρτιος -α -ον : *adj*, even, perfect.

ἄτμητος -ον : *adj*, uncut.

ἀτόπος -ον : *adj*, absurd, paradoxical.

αὐτόθεν : *adv*, immediately, obviously.

ἀφαίρεω : *vb*, take from, subtract from, cut off from; see αἰρέω.

ἄφή, ἡ : *no*, point of contact.

βάθος -εος, τό : *no*, depth, height.

βαίνω, -βήσομαι, -έβην, βέβηκα, —, — : *vb*, walk; *perf*, stand (of angle).

βάλλω, βαλῶ, ἔβαλον, βέβληκα, βέβλημαι, ἐβλήθην : *vb*, throw.

βάσις -εως, ἡ : *no*, base (of a triangle).

γάρ : *conj*, for (explanatory).

γί[γ]νομαι, γενήσομαι, ἐγενόμην, γέγονα, γεγένημαι, — : *vb*, happen, become.

γνώμων -ονος, ἡ : *no*, gnomon.

γραμμή, ἡ : *no*, line.

γράφω, γράψω, ἔγρα[ψ]α, γέγραφα, γέγραμμαι, ἐραψάμην : *vb*, draw (a figure).

γωνία, ἡ : *no*, angle.

δεῖ : *vb*, be necessary; δεῖ, it is necessary; ἔδει, it was necessary; δεόν, being necessary.

δείκνυμι, δείχω, ἔδειχα, δέδειθα, δέδειγμαι, ἐδείθην : *vb*, show, demonstrate.

δεικτέον : *ind*, one must show.

δείξις -εως, ἡ : *no*, proof.

δεκαγώνος -ον : *adj*, ten-sided; τό δεκαγώνον, *no*, decagon.

δέδομαι, δέχομαι, ἐδεχάμην, —, δέδεγμαι, ἐδέθην : *vb*, receive, accept.

δή : *conj*, so (explanatory).

δηλαδὴ : *ind*, quite clear, manifest.

δῆλος -η -ον : *adj*, clear.

δηλονότι : *adv*, manifestly.

διάγω : *vb*, carry over, draw through, draw across; see ἄγω.

διαγώνιος -ον : *adj*, diagonal.

διαλείπω : *vb*, leave an interval between.

διάμετρος -ον : *adj*, diametrical; ἡ διάμετρος, *no*, diameter, diagonal.

διαίρεσις -εως, ἡ : *no*, division, separation.

διαίρεω : *vb*, divide (in two); διαρεθέντος -η -ον, *adj*, separated (ratio); see αἰρέω.

διάστημα -ατος, τό : *no*, radius.

διαφέρω : *vb*, differ; see φέρω.

δίδομι, δώσω, ἔδωκα, δέδωκα, δέδομαι, ἐδόθην : *vb*, give.

δυοίρος -ον : *adj*, two-thirds.

διπλασιάζω : *vb*, double.

διπλάσιος -α -ον : *adj*, double, twofold.

διπλασίων -ον : *adj*, double, twofold.

διπλοῦς -ῆ -οῦν : *adj*, double.

δίξ : *adv*, twice.

δίθια : *adv*, in two, in half.

διθορομία, ἡ : *no*, point of bisection.

δυνάξ -άδοξ, ἡ : *no*, the number two, dyad.

δύναμαι : *vb*, be able, be capable, generate, square, be when squared; *δυναμένη*, ἡ, *no*, square-root (of area)—*i.e.*, straight-line whose square is equal to a given area.

δύναμις -εωξ, ἡ : *no*, power (usually 2nd power when used in mathematical sence, hence), square.

δυνατός -ή -όν : *adj*, possible.

δωδεκάεδροξ -ον : *adj*, twelve-sided.

ἐαυτοῦ -ῆξ -οῦ : *adj*, of him/her/it/self, his/her/its/own.

ἐγγίων -ον : *adj*, nearer, nearest.

ἐγγράφω : *vb*, inscribe; see *γράφω*.

εἶδος -εοξ, τό : *no*, figure, form, shape.

εἰκοσάεδροξ -ον : *adj*, twenty-sided.

εἶρω/λέγω, ἐρῶ/ερέω, εἶπον, εἶρηκα, εἶρημαι, ἐρρήθην : *vb*, say, speak; *per pass part*, *ειρημένος* -η -ον, *adj*, said, aforementioned.

εἴτε ... εἴτε : *ind*, either ... or.

ἐκαστος -η -ον : *pro*, each, every one.

ἐκατέροξ -α -ον : *pro*, each (of two).

ἐκβάλλω : *vb*, produce (a line); see *βάλλω*.

ἐκθέω : *vb*, set out.

ἐκκεμαι : *vb*, be set out, be taken; see *κείμαι*.

ἐκτίθην : *vb*, set out; see *τίθην*.

ἐκτός : *pre + gen*, outside, external.

ἐλάσσων/ἐλάττων -ον : *adj*, less, lesser.

ἐλάχιστος -η -ον : *adj*, least.

ἐλλείπω : *vb*, be less than, fall short of.

ἐμπίπτω : *vb*, meet (of lines), fall on; see *πίπτω*.

ἐμπροσθεν : *adv*, in front.

ἐναλλάξ : *adv*, alternate(ly).

ἐναρμόζω : *vb*, insert; *perf indic pass 3rd sg*, ἐνήρμοσται.

ἐνδέθωμαι : *vb*, admit, allow.

ἐνεκεν : *ind*, on account of, for the sake of.

ἐνναπλάσιος -α -ον : *adj*, nine-fold, nine-times.

ἐννοια, ἡ : *no*, notion.

ενπεριέθω : *vb*, encompass; see *έθω*.

ἐνπίπτω : see *ἐμπίπτω*.

ἐντός : *pre + gen*, inside, interior, within, internal.

ἐξάγωνος -ον : *adj*, hexagonal; τό *ἐξάγωνον*, *no*, hexagon.

ἐξαπλάσιος -α -ον : *adj*, sixfold.

ἐχῆξ : *adv*, in order, successively, consecutively.

ἐχωθεν : *adv*, outside, extrinsic.

ἐπάνω : *adv*, above.

ἐπαφή, ἡ : *no*, point of contact.

ἐπεί : *conj*, since (causal).

ἐπειδήπερ : *ind*, inasmuch as, seeing that.

ἐπιξύνγνυμι, ἐπιξύνγω, ἐπέξενυχα, —, ἐπέξενυγμαι, ἐπέξενύθην : *vb*, join (by a line).

ἐπιλογίζομαι : *vb*, conclude.

ἐπινοέω : *vb*, think of, contrive.

ἐπιπέδοξ -ον : *adj*, level, flat, plane; τό *ἐπιπέδον*, *no*, plane.

ἐπισκέπτομαι : *vb*, investigate.

ἐπίσκεψις -εωξ, ἡ : *no*, inspection, investigation.

ἐπιτάσσω : *vb*, put upon, enjoin; τό *ἐπιταθέν*, *no*, the (thing) prescribed; see *τάσσω*.

ἐπίτριτος -ον : *adj*, one and a third times.

ἐπιφάνεια, ἡ : *no*, surface.

ἔπομαι : *vb*, follow.

ἐρθομαι, ἐλεύσομαι, ἦλθον, ἐλήλυθα, —, — : *vb*, come, go.

ἔσθαιτος -η -ον : *adj*, outermost, uttermost, last.

ἐτερόμηκεξ -εξ : *adj*, oblong; τό *ἐτερόμηκεξ*, *no*, rectangle.

ἕτεροξ -α -ον : *adj*, other (of two).

ἔτι : *par*, yet, still, besides.

εὐθύγραμμος -ον : *adj*, rectilinear; τό *εὐθύγραμμον*, *no*, rectilinear figure.

εὐθύς -εῖα -ύ : *adj*, straight; ἡ *εὐθεῖα*, *no*, straight-line; ἐπ' *εὐθείαις*, in a straight-line, straight-on.

εὐρίσκω, εὐρήσκω, ἦῤρον, εὔρεκα, εὔρημαι, εὔρέθην : *vb*, find.

ἐφάπτω : *vb*, bind to; *mid*, touch; ἡ *ἐφαπτομένη*, *no*, tangent; see *άπτω*.

ἐφαρμόζω, ἐφαρμόσω, ἐφήρμοσα, ἐφήμοκα, ἐφήμοσμαι, ἐφήμόσθην : *vb*, coincide; *pass*, be applied.

ἐφεχῆξ : *adv*, in order, adjacent.

ἐφίστημι : *vb*, set, stand, place upon; see *ἵστημι*.

έθω, έχω, έσθον, έσθηκα, -έσθημαι, — : *vb*, have.

ηγέομαι, ήγήσομαι, ήγησάμην, ήγηναι, —, ήγήθην : *vb*, lead.

ήδη : *ind*, already, now.

ήκω, ήχω, —, —, —, — : *vb*, have come, be present.

ήμικύκλιον, τό : *no*, semi-circle.

ήμιόλιος -α -ον : *adj*, containing one and a half, one and a half times.

ήμισυξ -εια -υ : *adj*, half.

ήπερ = ή + περ : *conj*, than, than indeed.

ήτοι ... ή : *par*, surely, either ... or; in fact, either ... or.

θέσις -εωξ, ἡ : *no*, placing, setting, position.

θεωρημα -ατοξ, τό : *no*, theorem.

ἴδιος -α -ον : *adj*, one's own.

ισάκιξ : *adv*, the same number of times; *ισάκις πολλαπλάσια*, the same multiples, equal multiples.

ισογώνιος -ον : *adj*, equiangular.

ισόπλευροξ -ον : *adj*, equilateral.

ισοπληθής -έξ : *adj*, equal in number.

ἴσοξ -η -ον : *adj*, equal; έχ *ἴσον*, equally, evenly.

ἰσοσκελής -έξ : *adj*, isosceles.

ἵστημι, στήσω, ἕστησα, —, —, ἐστάθην : *vb tr*, stand (something).

ἵστημι, στήσω, ἕστην, ἕστηκα, ἑσταμαι, ἐσταθην : *vb intr*, stand up (oneself); Note: perfect *I have stood up* can be taken to mean present *I am standing*.

ἰσοῦψής -έξ : *adj*, of equal height.

καθάπερ : *ind*, according as, just as.

κάθετος -ον : *adj*, perpendicular.

καθόλον : *adv*, on the whole, in general.

καλέω : *vb*, call.

καῖκινοξ = καὶ ἐκεῖνοξ .

κ'άν = καὶ ἂν : *ind*, even if, and if.

καταγραφή, ἥ : *no*, diagram, figure.

καταγράφω : *vb*, describe/draw, inscribe (a figure); see γράφω.

κατακολουθέω : *vb*, follow after.

καταλείπω : *vb*, leave behind; see λείπω; τὰ καταλειπόμενα, *no*, remainder.

κατάλληλος -ον : *adj*, in succession, in corresponding order.

καταμετρέω : *vb*, measure (exactly).

καταντάω : *vb*, come to, arrive at.

κατασκενάζω : *vb*, furnish, construct.

κεῖμαι, κεῖσομαι, —, —, —, — : *vb*, have been placed, lie, be made; see τίθημι.

κέντρον, τό : *no*, center.

κλάω : *vb*, break off, inflect.

κλίνω, κλίνω, ἔκλινα, κέκλικα, κέκλιναι, ἐκλίθην : *vb*, lean, incline.

κλίσις -εωξ, ἥ : *no*, inclination, bending.

κοῖλος -η -ον : *adj*, hollow, concave.

κορυφή, ἥ : *no*, top, summit, apex; κατὰ κορυφήν, vertically opposite (of angles).

κρίνω, κρίνω, ἔκρινα, κέκρικα, κέκριμαι, ἐκρίθην : *vb*, judge.

κύβοξ, ὁ : *no*, cube.

κύκλος, ὁ : *no*, circle.

κύλινδρος, ὁ : *no*, cylinder.

κυρτός -ή -όν : *adj*, convex.

κῶνος, ὁ : *no*, cone.

λαμβάνω, λήψομαι, ἔλαβον, εἴληφα εἴλημμαι, ἐλήφθην : *vb*, take.

λέγω : *vb*, say; *pres pass part*, λεγόμενος -η -ον, *adj*, so-called; see ἔιρω.

λείπω, λείπω, ἔλιπον, λέλοιπα, λέλειμμαι, ἐλείφθην : *vb*, leave, leave behind.

λημμάτιον, τό : *no*, diminutive of λήμμα.

λήμμα -ατος, τό : *no*, lemma.

λήψις -εωξ, ἥ : *no*, taking, catching.

λόγος, ὁ : *no*, ratio, proportion, argument.

λοιπός -ή -όν : *adj*, remaining.

μανθάνω, μαθήσομαι, ἔμαθον, μεμάθηκα, —, — : *vb*, learn.

μέγεθος -εοξ, τό : *no*, magnitude, size.

μεῖζων -ον : *adj*, greater.

μένω, μενῶ, ἔμεινα, μεμένηκα, —, — : *vb*, stay, remain.

μέρος -ονξ, τό : *no*, part, direction, side.

μέσος -η -ον : *adj*, middle, mean, medial; ἐκ δύο μέσων, bimedral.

μεταλαμβάνω : *vb*, take up.

μεταχύ : *adv*, between.

μετέωρος -ον : *adj*, raised off the ground.

μετρέω : *vb*, measure.

μέτρον, τό : *no*, measure.

μηδεῖς, μηδεμία, μηδέν : *adj*, not even one, (neut.) nothing.

μηδέποτε : *adv*, never.

μηδέτερος -α -ον : *pro*, neither (of two).

μηκοξ -εοξ, τό : *no*, length.

μήν : *par*, truly, indeed.

μονάξ -άδοξ, ἥ : *no*, unit, unity.

μοναθός -ή -όν : *adj*, unique.

μοναθῶξ : *adv*, uniquely.

μόνος -η -ον : *adj*, alone.

νοέω, —, νόησα, νενόηκα, νενόημαι, ἐνόηθην : *vb*, apprehend, conceive.

οἷος -α -ον : *pre*, such as, of what sort.

ὀκτάεδρος -ον : *adj*, eight-sided.

ὅλος -η -ον : *adj*, whole.

ὁμογενής -έξ : *adj*, of the same kind.

ὁμοιοξ -α -ον : *adj*, similar.

ὁμοιοπληθής -έξ : *adj*, similar in number.

ὁμοιοταγής -έξ : *adj*, similarly arranged.

ὁμοιότης -ητος, ἥ : *no* similarity.

ὁμοίως : *adv*, similarly.

ὁμόλογος -ον : *adj*, corresponding, homologous.

ὁμοταγής -έξ : *adj*, ranged in the same row or line.

ὁμώνυμος -ον : *adj*, having the same name.

ὄνομα -ατος, τό : *no*, name; ἐκ δύο ὀνομάτων, binomial.

ὀχρῶνος -ον : *adj*, acute-angled; τὸ ὀχρῶνον, *no*, acute angle.

ὀξύς -εῖα -ύ : *adj*, acute.

ὅποιοςσὺν = ὅποῖος -α -ον + οὔν : *adj*, of whatever kind, any kind whatsoever.

ὅποσοξ -η -ον : *pro*, as many, as many as.

ὅποσοδοποτοῦν = ὅπόσοξ -η -ον + δὴ + ποτέ + οὔν : *adj*, of whatever number, any number whatsoever.

ὅποσοσὺν = ὅπόσοξ -η -ον + οὔν : *adj*, of whatever number, any number whatsoever.

ὅποτερος -α -ον : *pro*, either (of two), which (of two).

ὀρθογώνιον, τό : *no*, rectangle, right-angle.

ὀρθός -ή -όν : *adj*, straight, right-angled, perpendicular; πρὸς ὀρθὰς γωνίας, at right-angles.

ὄρος, ὁ : *no*, boundary, definition, term (of a ratio).

ὅσαδηποτοῦν = ὅσα + δὴ + ποτέ + οὔν : *ind*, any number whatsoever.

ὁσάκιξ : *ind*, as many times as, as often as.

ὅσαπλάσιοξ -ον : *pro*, as many times as.

ὅσοξ -η -ον : *pro*, as many as.

ὅσπερ, ἥπερ, ὅπερ : *pro*, the very man who, the very thing which.

ὅστις, ἥτις, ὅ τι : *pro*, anyone who, anything which.

ὅταν : *adv*, when, whenever.

ὅτιοῦν : *ind*, whatsoever.

οὐδεὶς, οὐδεμία, οὐδέν : *pro*, not one, nothing.

οὐδέτερος -α -ον : *pro*, not either.

οὐθέτερος : see οὐδέτερος.

οὐθέν : *ind*, nothing.

οὕν : *adv*, therefore, in fact.

οὕτωξ : *adv*, thusly, in this case.

πάλιν : *adv*, back, again.

πάντωξ : *adv*, in all ways.

παρά : *prep* + *acc*, parallel to.

παραβάλλω : *vb*, apply (a figure); see βάλλω.

παραβολή, ἡ : *no*, application.

παράκειμαι : *vb*, lie beside, apply (a figure); see κείμει.

παραλλάσσω, παραλλάχω, —, παρήλλαθα, —, — : *vb*, miss, fall awry.

παραλληλεπίπεδος, -ον : *adj*, with parallel surfaces; τὸ παραλληλεπίπεδον, *no*, parallelepiped.

παραλληλόγραμμος -ον : *adj*, bounded by parallel lines; τὸ παραλληλόγραμμον, *no*, parallelogram.

παράλληλος -ον : *adj*, parallel; τὸ παράλληλον, *no*, parallel, parallel-line.

παραπλήρωμα -ατοξ, τό : *no*, complement (of a parallelogram).

παρατέλνετοξ -ον : *adj*, penultimate.

παρέκ : *prep* + *gen*, except.

παραεμπίπτω : *vb*, insert; see πίπτω.

πάσθω, πείσσομαι, ἔπαθον, πέπονθα, —, — : *vb*, suffer.

πεντάγωνος -ον : *adj*, pentagonal; τὸ πεντάγωνον, *no*, pentagon.

πενταπλάσιος -α -ον : *adj*, five-fold, five-times.

πεντεκαϊδεκάγωνον, τό : *no*, fifteen-sided figure.

πεπερασμένοξ -η -ον : *adj*, finite, limited; see περαίνω.

περαίνω, περανῶ, ἐπέρανα, —, πεπεράσμαι, ἐπερανάνθην : *vb*, bring to end, finish, complete; *pass*, be finite.

πέρας -ατοξ, τό : *no*, end, extremity.

περατόω, —, —, —, — : *vb*, bring to an end.

περιγράφω : *vb*, circumscribe; see γράφω.

περιέθω : *vb*, encompass, surround, contain, comprise; see ἔθω.

περιλαμβάνω : *vb*, enclose; see λαμβάνω.

περισσάκιξ : *adv*, an odd number of times.

περισσόξ -ή -όν : *adj*, odd.

περιφέρεια, ἡ : *no*, circumference.

περιφέρω : *vb*, carry round; see φέρω.

πηλικότης -ητοξ, ἡ : *no*, magnitude, size.

πίπτω, πεσοῦμαι, ἔπεσον, πέπτωκα, —, — : *vb*, fall.

πλάτος -εοξ, τό : *no*, breadth, width.

πλείων -ον : *adj*, more, several.

πλευρά, ἡ : *no*, side.

πληθος -εοξ, τό : *no*, great number, multitude, number.

πλήν : *adv* & *prep* + *gen*, more than.

ποιός -ά -όν : *adj*, of a certain nature, kind, quality, type.

πολλαπλασιάζω : *vb*, multiply.

πολλαπλασιασμός, ὁ : *no*, multiplication.

πολλαπλάσιον, τό : *no*, multiple.

πολύεδρος -ον : *adj*, polyhedral; τό πολυέδρον, *no*, polyhedron.

πολύγωνος -ον : *adj*, polygonal; τό πολύγωνον, *no*, polygon.

πολύπλευρος -ον : *adj*, multilateral.

πόρισμα -ατοξ, τό : *no*, corollary.

ποτέ : *ind*, at some time.

πρῶμα -ατοξ, τό : *no*, prism.

προβαίνω : *vb*, step forward, advance.

προδείκνυμι : *vb*, show previously; see δείκνυμι.

προεκτίθην : *vb*, set forth beforehand; see τίθην.

προερέω : *vb*, say beforehand; *perf pass part*, προεξηρημένοξ -η -ον, *adj*, aforementioned; see ἔζρω.

προσαναπληρόω : *vb*, fill up, complete.

προσαναγράφω : *vb*, complete (tracing of); see γράφω.

προσαρμόζω : *vb*, fit to, attach to.

προσεκβάλλω : *vb*, produce (a line); see ἐκβάλλω.

προσευρίσκω : *vb*, find besides, find; see εὐρίσκω.

προσλαμβάνω : *vb*, add.

πρόκειμαι : *vb*, set before, prescribe; see κείμει.

πρόσκειμαι : *vb*, be laid on, have been added to; see κείμει.

προσπίπτω : *vb*, fall on, fall toward, meet; see πίπτω.

προτασις -εωξ, ἡ : *no*, proposition.

προστάσσω : *vb*, prescribe, enjoin; τὸ τροσταθθέν, *no*, the thing prescribed; see τάσσω.

προστίθην : *vb*, add; see τίθην.

πρότερος -α -ον : *adj*, first (comparative), before, former.

προτίθην : *vb*, assign; see τίθην.

προθωρέω : *vb*, go/come forward, advance.

πρῶτος -α -ον : *adj*, first, prime.

πυραμίδς -ίδος, ἡ : *no*, pyramid.

ῥητός -ή -όν : *adj*, expressible, rational.

ῥομβοειδής -έξ : *adj*, rhomboidal; τὸ ῥομβοειδέξ, *no*, rhomboid.

ῥόμβος, ὁ : *no*, rhombus.

σημεῖον, τό : *no*, point.

σκαληνός -ή -όν : *adj*, scalene.

στερεός -ά -όν : *adj*, solid; τὸ στερεόν, *no*, solid, solid body.

στοιθεῖον, τό : *no*, element.

στρέφω, -στρέψω, ἔστρεψα, —, ἐσταμμαι, ἐστάφην : *vb*, turn.

σύνκειμαι : *vb*, lie together, be the sum of, be composed; συγκείμενος -η -ον, *adj*, composed (ratio), compounded; see κείμει.

σύνκρινω : *vb*, compare; see κρίνω.

συμβαίνω : *vb*, come to pass, happen, follow; see βαίνω.

συμβάλλω : *vb*, throw together, meet; see βάλλω.

σύμμετροξ -ον : *adj*, commensurable.

σύμπαξ -αντοξ, ὁ : *no*, sum, whole.

συνπίπτω : *vb*, meet together (of lines); see *πίπτω*.

συνπληρόω : *vb*, complete (a figure), fill in.

συνάγω : *vb*, conclude, infer; see *ἄγω*.

συναμφοτέροι -αι -α : *adj*, both together; ὁ *συναμφοτέροξ*, *no*, sum (of two things).

συναποδείκνυμι : *no*, demonstrate together; see *δείκνυμι*.

συναφή, ἡ : *no*, point of junction.

σύνδυο, οἱ, αἱ, τά : *no*, two together, in pairs.

συνετής -έξ : *adj*, continuous; *κατὰ τὸ συνεθέξ*, continuously.

σύνθεσις -εωξ, ἡ : *no*, putting together, composition.

σύνθετοξ -ον : *adj*, composite.

συ[γ]ίστημι : *vb*, construct (a figure), set up together; *perf imperat pass 3rd sg*, *συνεσάτω*; see *ἵστημι*.

συντίθημι : *vb*, put together, add together, compound (ratio); see *τίθημι*.

σθέσις -εωξ, ἡ : *no*, state, condition.

σθῆμα -ατοξ, τό : *no*, figure.

σφαῖρα -αξ, ἡ : *no*, sphere.

τάξις -εωξ, ἡ : *no*, arrangement, order.

ταράσσω, ταράχω, —, —, τετάραγμα, ἐταράθην : *vb*, stir, trouble, disturb; τεταραγμένοξ -η -ον, *adj*, disturbed, perturbed.

τάσσω, τάχω, ἔταχα, τέταθα, τέταγμα, ἐτάθην : *vb*, arrange, draw up.

τέλειος -α -ον : *adj*, perfect.

τέμνω, τεμνῶ, ἔτεμον, -τέμνηκα, τέτμημαι, ἐτμήθην : *vb*, cut; *pres/fut indic act 3rd sg*, τέμει.

τεταρτημοριον, τό : *no*, quadrant.

τετράγωνος -ον : *adj*, square; τὸ *τετράγωνον*, *no*, square.

τετράκις : *adv*, four times.

τετραπλάσιος -α -ον : *adj*, quadruple.

τετράπλευρος -ον : *adj*, quadrilateral.

τετραπλόος -η -ον : *adj*, fourfold.

τίθημι, θήσω, ἔθηκα, τέθηκα, κείμαι, ἐτέθην : *vb*, place, put.

τμήμα -ατοξ, τό : *no*, part cut off, piece, segment.

τοίνυν : *par*, accordingly.

τοιούτοξ -αύτη -οὔτο : *pro*, such as this.

τομεύξ -έωξ, ὁ : *no*, sector (of circle).

τομή, ἡ : *no*, cutting, stump, piece.

τόπος, ὁ : *no*, place, space.

τοσαντάκις : *adv*, so many times.

τοσανταπλάσιος -α -ον : *pro*, so many times.

τοσοῦτοξ -αύτη -οὔτο : *pro*, so many.

τουτέστι = τοῦτ' ἔστι : *par*, that is to say.

τραπέζιον, τό : *no*, trapezium.

τρίγωνος -ον : *adj*, triangular; τὸ *τρίγωνον*, *no*, triangle.

τριπλάσιος -α -ον : *adj*, triple, threefold.

τρίπλευρος -ον : *adj*, trilateral.

τριπλ -όος -η -ον : *adj*, triple.

τρόπος, ὁ : *no*, way.

τυγθάνω, τεύχομαι, ἔτυθον, τετύθηκα, τέτευγμαι, ἐτεύθην : *vb*, hit, happen to be at (a place).

ὑπάρθω : *vb*, begin, be, exist; see *ἄρθω*.

ὑπεχαίρεσις -εωξ, ἡ : *no*, removal.

ὑπερβάλλω : *vb*, overshoot, exceed; see *βάλλω*.

ὑπεροθή, ἡ : *no*, excess, difference.

ὑπερέθω : *vb*, exceed; see *ἔθω*.

ὑπόθεσις -εωξ, ἡ : *no*, hypothesis.

ὑποκείμαι : *vb*, underlie, be assumed (as hypothesis); see *κείμαι*.

ὑπολείπω : *vb*, leave remaining.

ὑποτείνω, ὑποτενῶ, ὑπέτενα, ὑποτέτακα, ὑποτετάμαι, ὑπετάθην : *vb*, subtend.

ῥυψος -εοξ, τό : *no*, height.

φανερὸξ -ά -όν : *adj*, visible, manifest.

φημί, φήσω, ἔφην, —, —, — : *vb*, say; ἔφαμεν, we said.

φέρω, οἴσω, ἡνεγκον, ἐνήνοθα, ἐνήνεγμα, ἡνέθην : *vb*, carry.

θώριον, τό : *no*, place, spot, area, figure.

θωρίξ : *pre + gen*, apart from.

ψάω : *vb*, touch.

ὥξ : *par*, as, like, for instance.

ὥξ ἔτυθεν : *par*, at random.

ὡσαύτωξ : *adv*, in the same manner, just so.

ὥστε : *conj*, so that (causal), hence.

GREEK-ENGLISH LEXICON

ABBREVIATIONS: *act* - active; *adj* - adjective; *adv* - adverb; *conj* - conjunction; *fut* - future; *gen* - genitive; *imperat* - imperative; *imperf* - imperfect; *ind* - indeclinable; *indic* - indicative; *intr* - intransitive; *mid* - middle; *neut* - neuter; *no* - noun; *par* - particle; *part* - participle; *pass* - passive; *perf* - perfect; *pre* - preposition; *pres* - present; *pro* - pronoun; *sg* - singular; *tr* - transitive; *vb* - verb.

ἄγω, ἄχω, ἡγαγον, -ῆθα, ἡγμαι, ἡθην : *vb*, lead, draw (a line).

ἀδύνατος -ον : *adj*, impossible.

ἀεί : *adv*, always, for ever.

αἰρέω, αἰρήσω, εἴ[~]λον, ἤρῃκα, ἤρημαι, ἡρέθην : *vb*, grasp.

αἰτέω, αἰτήσω, ἤτησα, ἤτηκα, ἤτημαι, ἡτήθη : *vb*, postulate.

αἵτημα -ατος, τό : *no*, postulate.

ἀκόλουθος -ον : *adj*, analogous, consequent on, in conformity with.

ἄκρος -α -ον : *adj*, outermost, end, extreme.

ἀλλά : *conj*, but, otherwise.

ἄλογος -ον : *adj*, irrational.

ἅμα : *adv*, at once, at the same time, together.

ἀμβλυγώνιος -ον : *adj*, obtuse-angled; τό ἀμβλυγώνιον, *no*, obtuse angle.

ἀμβλύς -εῖα -ύ : *adj*, obtuse.

ἀμφότερος -α -ον : *pro*, both.

ἀναγράφω : *vb*, describe (a figure); see γράφω.

ἀναλογία, ἡ : *no*, proportion, (geometric) progression.

ἀνάλογος -ον : *adj*, proportional.

ἀνάπαλιν : *adv*, inverse(ly).

αναπληρώω : *vb*, fill up.

ἀναστρέφω : *vb*, turn upside down, convert (ratio); see στρέφω.

ἀναστροφή, ἡ : *no*, turning upside down, conversion (of ratio).

ἀνθρῳαίρεω : *vb*, take away in turn; see αἰρέω.

ἀνίστημι : *vb*, set up; see ἵστημι.

ἄνισος -ον : *adj*, unequal, uneven.

ἀντιπάσθω : *vb*, be reciprocally proportional; see πάσθω.

ἄχων -ονος, ὁ : *vb*, axis.

ἅπαχ : *adv*, once.

ἅπαξ, ἅπασα, ἅπαν : *adj*, quite all, the whole.

ἄπειρος -ον : *adj*, infinite.

ἀπεναντίον : *ind*, opposite.

ἀπέθω : *vb*, be far from, be away from; see ἔθω.

ἀπλατήξ -έξ : *adj*, without breadth.

ἀπόδειξις -εως, ἡ : *no*, proof.

ἀποκαθίστημι : *vb*, re-establish, restore; see ἵστημι.

ἀπολαμβάνω : *vb*, take from, subtract from, cut off from; see λαμβάνω.

ἀποτέμνω : *vb*, cut off, subtend.

ἀπότμημα -ατος, τό : *no*, piece cut off, segment.

ἀποτομή, ἡ : *vb*, piece cut off, apotome.

ἅπτω, ἅψω, ἥψα, —, ἥμμαι, — : *vb*, touch, join, meet.

ἀπώτερος -α -ον : *adj*, further off.

ἄρα : *par*, thus, as it seems (inferential).

ἀριθμός, ὁ : *no*, number.

ἀρτιάκις : *adv*, an even number of times.

ἀρτιόπλευρος -ον : *adj*, having an even number of sides.

ἄρθω, ἄρχω, ἤρξα, ἤρθα, ἤρχμαι, ἤρθην : *vb*, rule; *mid.*, begin.

ἀσύμμετρος -ον : *adj*, incommensurable.

ἀσύνπτωτος -ον : *adj*, not touching, not meeting.

ἄρτιος -α -ον : *adj*, even, perfect.

ἄτμητος -ον : *adj*, uncut.

ἀτόπος -ον : *adj*, absurd, paradoxical.

αὐτόθεν : *adv*, immediately, obviously.

ἀφαίρεω : *vb*, take from, subtract from, cut off from; see αἰρέω.

ἄφή, ἡ : *no*, point of contact.

βάθος -εος, τό : *no*, depth, height.

βαίνω, -βήσομαι, -έβην, βέβηκα, —, — : *vb*, walk; *perf*, stand (of angle).

βάλλω, βαλῶ, ἔβαλον, βέβληκα, βέβλημαι, ἐβλήθην : *vb*, throw.

βάσις -εως, ἡ : *no*, base (of a triangle).

γάρ : *conj*, for (explanatory).

γί[γ]νομαι, γενήσομαι, ἐγενόμην, γέγονα, γεγένημαι, — : *vb*, happen, become.

γνώμων -ονος, ἡ : *no*, gnomon.

γραμμή, ἡ : *no*, line.

γράφω, γράψω, ἔγρα[ψ]α, γέγραφα, γέγραμμαι, ἐραψάμην : *vb*, draw (a figure).

γωνία, ἡ : *no*, angle.

δεῖ : *vb*, be necessary; δεῖ, it is necessary; ἔδει, it was necessary; δεόν, being necessary.

δείκνυμι, δείχω, ἔδειχα, δέδειθα, δέδειγμαι, ἐδείθην : *vb*, show, demonstrate.

δεικτέον : *ind*, one must show.

δείξις -εως, ἡ : *no*, proof.

δεκαγώνος -ον : *adj*, ten-sided; τό δεκαγώνον, *no*, decagon.

δέδομαι, δέχομαι, ἐδεχάμην, —, δέδεγμαι, ἐδέθην : *vb*, receive, accept.

δή : *conj*, so (explanatory).

δηλαδὴ : *ind*, quite clear, manifest.

δῆλος -η -ον : *adj*, clear.

δηλονότι : *adv*, manifestly.

διάγω : *vb*, carry over, draw through, draw across; see ἄγω.

διαγώνιος -ον : *adj*, diagonal.

διαλείπω : *vb*, leave an interval between.

διάμετρος -ον : *adj*, diametrical; ἡ διάμετρος, *no*, diameter, diagonal.

διαίρεσις -εως, ἡ : *no*, division, separation.

διαίρεω : *vb*, divide (in two); διαρεθέντος -η -ον, *adj*, separated (ratio); see αἰρέω.

διάστημα -ατος, τό : *no*, radius.

διαφέρω : *vb*, differ; see φέρω.

δίδομι, δώσω, ἔδωκα, δέδωκα, δέδομαι, ἐδόθην : *vb*, give.

δυοῖρος -ον : *adj*, two-thirds.

διπλασιάζω : *vb*, double.

διπλάσιος -α -ον : *adj*, double, twofold.

διπλασίων -ον : *adj*, double, twofold.

διπλοῦς -ῆ -οῦν : *adj*, double.

δίξ : *adv*, twice.

δίθια : *adv*, in two, in half.

διθορομία, ἡ : *no*, point of bisection.

δυνάξ -άδοξ, ἡ : *no*, the number two, dyad.

δύναμαι : *vb*, be able, be capable, generate, square, be when squared; δυναμένη, ἡ, *no*, square-root (of area)—i.e., straight-line whose square is equal to a given area.

δύναμις -εως, ἡ : *no*, power (usually 2nd power when used in mathematical sense, hence), square.

δυνατός -ή -όν : *adj*, possible.

δωδεκάεδροξ -ον : *adj*, twelve-sided.

ἐαυτοῦ -ῆς -οῦ : *adj*, of him/her/it/self, his/her/its/own.

ἐγγίων -ον : *adj*, nearer, nearest.

ἐγγράφω : *vb*, inscribe; see γράφω.

εἶδος -εος, τό : *no*, figure, form, shape.

εἰκοσάεδροξ -ον : *adj*, twenty-sided.

εἶρω/λέγω, ἐρῶ/ερέω, εἶπον, εἶρηκα, εἶρημαι, ἐρρήθην : *vb*, say, speak; *per pass part*, ειρημένος -η -ον, *adj*, said, aforementioned.

εἴτε ... εἴτε : *ind*, either ... or.

ἐκαστος -η -ον : *pro*, each, every one.

ἐκατέρωξ -α -ον : *pro*, each (of two).

ἐκβάλλω : *vb*, produce (a line); see βάλλω.

ἐκθέω : *vb*, set out.

ἐκκεμαι : *vb*, be set out, be taken; see κείμαι.

ἐκτίθην : *vb*, set out; see τίθην.

ἐκτός : *pre + gen*, outside, external.

ἐλάσσων/ἐλάττων -ον : *adj*, less, lesser.

ἐλάχιστος -η -ον : *adj*, least.

ἐλλείπω : *vb*, be less than, fall short of.

ἐμπίπτω : *vb*, meet (of lines), fall on; see πίπτω.

ἐμπροσθεν : *adv*, in front.

ἐναλλάξ : *adv*, alternate(ly).

ἐναρμόζω : *vb*, insert; *perf indic pass 3rd sg*, ἐνήρμοσται.

ἐνδέθωμαι : *vb*, admit, allow.

ἐνεκεν : *ind*, on account of, for the sake of.

ἐνναπλάσιος -α -ον : *adj*, nine-fold, nine-times.

ἐννοια, ἡ : *no*, notion.

ἐνπεριέθω : *vb*, encompass; see ἔθω.

ἐνπίπτω : see ἐμπίπτω.

ἐντός : *pre + gen*, inside, interior, within, internal.

ἐξάγωνος -ον : *adj*, hexagonal; τό ἐξάγωνον, *no*, hexagon.

ἐξαπλάσιος -α -ον : *adj*, sixfold.

ἐχῆς : *adv*, in order, successively, consecutively.

ἔχωθεν : *adv*, outside, extrinsic.

ἐπάνω : *adv*, above.

ἐπαφή, ἡ : *no*, point of contact.

ἐπεί : *conj*, since (causal).

ἐπειδήπερ : *ind*, inasmuch as, seeing that.

ἐπιξύνγνυμι, ἐπιξύνγω, ἐπέξενυχα, —, ἐπέξενυγμαι, ἐπέξενύθην : *vb*, join (by a line).

ἐπιλογίζομαι : *vb*, conclude.

ἐπινοέω : *vb*, think of, contrive.

ἐπιπέδοξ -ον : *adj*, level, flat, plane; τό ἐπιπέδον, *no*, plane.

ἐπισκέπτομαι : *vb*, investigate.

ἐπίσκεψις -εως, ἡ : *no*, inspection, investigation.

ἐπιτάσσω : *vb*, put upon, enjoin; τό ἐπιταθέν, *no*, the (thing) prescribed; see τάσσω.

ἐπίτριτος -ον : *adj*, one and a third times.

ἐπιφάνεια, ἡ : *no*, surface.

ἔπομαι : *vb*, follow.

ἔρθομαι, ἐλεύσομαι, ἦλθον, ἐλήλυθα, —, — : *vb*, come, go.

ἔσθαιτος -η -ον : *adj*, outermost, uttermost, last.

ἑτερόμηκεξ -εξ : *adj*, oblong; τό ἑτερόμηκεξ, *no*, rectangle.

ἕτερος -α -ον : *adj*, other (of two).

ἔτι : *par*, yet, still, besides.

εὐθύγραμμος -ον : *adj*, rectilinear; τό εὐθύγραμμον, *no*, rectilinear figure.

εὐθύς -εῖα -ύ : *adj*, straight; ἡ εὐθεῖα, *no*, straight-line; ἐπ' εὐθείᾱ, in a straight-line, straight-on.

εὐρίσκω, εὐρήσκω, ἦϋρον, εὔρεκα, εὔρημαι, εὔρέθην : *vb*, find.

ἐφάπτω : *vb*, bind to; mid, touch; ἡ ἐφαπτομένη, *no*, tangent; see ἄπτω.

ἐφαρμόζω, ἐφαρμόσω, ἐφήρμοσα, ἐφήμοκα, ἐφήμοσμαι, ἐφήμόσθην : *vb*, coincide; *pass*, be applied.

ἐφεχῆς : *adv*, in order, adjacent.

ἐφίστημι : *vb*, set, stand, place upon; see ἵστημι.

ἔθω, ἔχω, ἔσθον, ἔσθηκα, -έσθημαι, — : *vb*, have.

ἡγέομαι, ἡγήσομαι, ἡγησάμην, ἡγηναι, —, ἡγήθην : *vb*, lead.

ἤδη : *ind*, already, now.

ἦκω, ἦχω, —, —, —, — : *vb*, have come, be present.

ἡμικύκλιον, τό : *no*, semi-circle.

ἡμιόλιος -α -ον : *adj*, containing one and a half, one and a half times.

ἡμισυς -εῖα -υ : *adj*, half.

ἥπερ = ἥ + περ : *conj*, than, than indeed.

ἦτοι ... ἦ : *par*, surely, either ... or; in fact, either ... or.

θέσις -εως, ἡ : *no*, placing, setting, position.

θεωρημα -ατος, τό : *no*, theorem.

ἴδιος -α -ον : *adj*, one's own.

ισάκις : *adv*, the same number of times; ισάκις πολλαπλάσια, the same multiples, equal multiples.

ισογώνιος -ον : *adj*, equiangular.

ισόπλευρος -ον : *adj*, equilateral.

ισοπληθής -ές : *adj*, equal in number.

ἴσος -η -ον : *adj*, equal; ἐχ ἴσον, equally, evenly.

ἰσοσκελής -έξ : *adj*, isosceles.

ἵστημι, στήσω, ἕστησα, —, —, ἐστάθην : *vb tr*, stand (something).

ἵστημι, στήσω, ἕστην, ἕστηκα, ἑσταμαι, ἐσταθην : *vb intr*, stand up (oneself); Note: perfect *I have stood up* can be taken to mean present *I am standing*.

ἰσοῦψής -έξ : *adj*, of equal height.

καθάπερ : *ind*, according as, just as.

κάθετος -ον : *adj*, perpendicular.

καθόλον : *adv*, on the whole, in general.

καλέω : *vb*, call.

κάκεινος = καὶ ἐκεῖνος .

κ'άν = καὶ ἂν : *ind*, even if, and if.

καταγραφή, ἡ : *no*, diagram, figure.

καταγράφω : *vb*, describe/draw, inscribe (a figure); see γράφω.

κατακολουθέω : *vb*, follow after.

καταλείπω : *vb*, leave behind; see λείπω; τὰ καταλειπόμενα, *no*, remainder.

κατάλληλος -ον : *adj*, in succession, in corresponding order.

καταμετρέω : *vb*, measure (exactly).

καταντάω : *vb*, come to, arrive at.

κατασκενάζω : *vb*, furnish, construct.

κείμει, κείμεμαι, —, —, —, — : *vb*, have been placed, lie, be made; see τίθημι.

κέντρον, τό : *no*, center.

κλάω : *vb*, break off, inflect.

κλίνω, κλίνω, ἔκλινα, κέκλικα, κέκλιναι, ἐκλίθην : *vb*, lean, incline.

κλίσις -εως, ἡ : *no*, inclination, bending.

κοῖλος -η -ον : *adj*, hollow, concave.

κορυφή, ἡ : *no*, top, summit, apex; κατὰ κορυφήν, vertically opposite (of angles).

κρίνω, κρίνω, ἔκρινα, κέκρικα, κέκριμαι, ἐκρίθην : *vb*, judge.

κύβοξ, ὁ : *no*, cube.

κύκλος, ὁ : *no*, circle.

κύλινδρος, ὁ : *no*, cylinder.

κυρτός -ή -όν : *adj*, convex.

κῶνος, ὁ : *no*, cone.

λαμβάνω, λήψομαι, ἔλαβον, εἴληφα εἴλημμαι, ἐλήφθην : *vb*, take.

λέγω : *vb*, say; *pres pass part*, λεγόμενος -η -ον, *adj*, so-called; see ἔιρω.

λείπω, λείπω, ἔλιπον, λέλοιπα, λέλειμμαι, ἐλείφθην : *vb*, leave, leave behind.

λημμάτιον, τό : *no*, diminutive of λήμμα.

λήμμα -ατος, τό : *no*, lemma.

λήψις -εως, ἡ : *no*, taking, catching.

λόγος, ὁ : *no*, ratio, proportion, argument.

λοιπός -ή -όν : *adj*, remaining.

μανθάνω, μαθήσομαι, ἔμαθον, μεμάθηκα, —, — : *vb*, learn.

μέγεθος -εος, τό : *no*, magnitude, size.

μεῖζων -ον : *adj*, greater.

μένω, μενῶ, ἔμεινα, μεμένηκα, —, — : *vb*, stay, remain.

μέρος -ονξ, τό : *no*, part, direction, side.

μέσος -η -ον : *adj*, middle, mean, medial; ἐκ δύο μέσων, bimedral.

μεταλαμβάνω : *vb*, take up.

μεταχύ : *adv*, between.

μετέωρος -ον : *adj*, raised off the ground.

μετρέω : *vb*, measure.

μέτρον, τό : *no*, measure.

μηδεῖς, μηδεμία, μηδέν : *adj*, not even one, (neut.) nothing.

μηδέποτε : *adv*, never.

μηδέτερος -α -ον : *pro*, neither (of two).

μηκοξ -εος, τό : *no*, length.

μήν : *par*, truly, indeed.

μονάξ -άδοξ, ἡ : *no*, unit, unity.

μοναθός -ή -όν : *adj*, unique.

μοναθῶξ : *adv*, uniquely.

μόνος -η -ον : *adj*, alone.

νοέω, —, νόησα, νενόηκα, νενόημαι, ἐνόηθην : *vb*, apprehend, conceive.

οἷος -α -ον : *pre*, such as, of what sort.

ὀκτάεδρος -ον : *adj*, eight-sided.

ὅλος -η -ον : *adj*, whole.

ὁμογενής -έξ : *adj*, of the same kind.

ὁμοιοξ -α -ον : *adj*, similar.

ὁμοιοπληθής -έξ : *adj*, similar in number.

ὁμοιοταγής -έξ : *adj*, similarly arranged.

ὁμοιότης -ητος, ἡ : *no* similarity.

ὁμοίως : *adv*, similarly.

ὁμόλογος -ον : *adj*, corresponding, homologous.

ὁμοταγής -έξ : *adj*, ranged in the same row or line.

ὁμώνυμος -ον : *adj*, having the same name.

ὄνομα -ατος, τό : *no*, name; ἐκ δύο ὀνομάτων, binomial.

ὀχυνώνοξ -ον : *adj*, acute-angled; τὸ ὀχυνώνον, *no*, acute angle.

ὀχύς -εῖα -ύ : *adj*, acute.

ὅποιοςσὺν = ὅποῖος -α -ον + οὔν : *adj*, of whatever kind, any kind whatsoever.

ὅποσος -η -ον : *pro*, as many, as many as.

ὅποσοσδηποτοῦν = ὅπόσος -η -ον + δὴ + ποτέ + οὔν : *adj*, of whatever number, any number whatsoever.

ὅποσοσὺν = ὅπόσος -η -ον + οὔν : *adj*, of whatever number, any number whatsoever.

ὁπότερος -α -ον : *pro*, either (of two), which (of two).

ὀρθογώνιον, τό : *no*, rectangle, right-angle.

ὀρθός -ή -όν : *adj*, straight, right-angled, perpendicular; πρὸς ὀρθὰς γωνίας, at right-angles.

ὄρος, ὁ : *no*, boundary, definition, term (of a ratio).

ὅσαδηποτοῦν = ὅσα + δὴ + ποτέ + οὔν : *ind*, any number whatsoever.

ὁσάκις : *ind*, as many times as, as often as.

ὅσαπλάσιος -ον : *pro*, as many times as.

ὅσοξ -η -ον : *pro*, as many as.

ὅσπερ, ἥπερ, ὅπερ : *pro*, the very man who, the very thing which.

ὅστις, ἥτις, ὅ τι : *pro*, anyone who, anything which.

ὅταν : *adv*, when, whenever.

ὅτιοῦν : *ind*, whatsoever.

οὐδεὶς, οὐδεμία, οὐδέν : *pro*, not one, nothing.

οὐδέτερος -α -ον : *pro*, not either.

οὐθέτερος : see οὐδέτερος.

οὐθέν : *ind*, nothing.

οὕν : *adv*, therefore, in fact.

οὕτωξ : *adv*, thusly, in this case.

πάλιν : *adv*, back, again.

πάντωξ : *adv*, in all ways.

παρά : *prep* + *acc*, parallel to.

παραβάλλω : *vb*, apply (a figure); see βάλλω.

παραβολή, ἡ : *no*, application.

παράκειμαι : *vb*, lie beside, apply (a figure); see κείμει.

παραλλάσσω, παραλλάχω, —, παρήλλαθα, —, — : *vb*, miss, fall awry.

παραλληλεπίπεδος, -ον : *adj*, with parallel surfaces; τὸ παραλληλεπίπεδον, *no*, parallelepiped.

παραλληλόγραμμος -ον : *adj*, bounded by parallel lines; τὸ παραλληλόγραμμον, *no*, parallelogram.

παράλληλος -ον : *adj*, parallel; τὸ παράλληλον, *no*, parallel, parallel-line.

παραπλήρωμα -ατοξ, τό : *no*, complement (of a parallelogram).

παρατέλνετοξ -ον : *adj*, penultimate.

παρέκ : *prep* + *gen*, except.

παρεμπίπτω : *vb*, insert; see πίπτω.

πάσθω, πείσσομαι, ἔπαθον, πέπονθα, —, — : *vb*, suffer.

πεντάγωνος -ον : *adj*, pentagonal; τὸ πεντάγωνον, *no*, pentagon.

πενταπλάσιος -α -ον : *adj*, five-fold, five-times.

πεντεκαϊδεκάγωνον, τό : *no*, fifteen-sided figure.

πεπερασμένοξ -η -ον : *adj*, finite, limited; see περαίνω.

περαίνω, περανῶ, ἐπέρανα, —, πεπεράσμαι, ἐπερανάνθην : *vb*, bring to end, finish, complete; *pass*, be finite.

πέρας -ατοξ, τό : *no*, end, extremity.

περατόω, —, —, —, — : *vb*, bring to an end.

περιγράφω : *vb*, circumscribe; see γράφω.

περιέθω : *vb*, encompass, surround, contain, comprise; see ἔθω.

περιλαμβάνω : *vb*, enclose; see λαμβάνω.

περισσάκιξ : *adv*, an odd number of times.

περισσόξ -ή -όν : *adj*, odd.

περιφέρεια, ἡ : *no*, circumference.

περιφέρω : *vb*, carry round; see φέρω.

πηλικότης -ητοξ, ἡ : *no*, magnitude, size.

πίπτω, πεσοῦμαι, ἔπεσον, πέπτωκα, —, — : *vb*, fall.

πλάτος -εοξ, τό : *no*, breadth, width.

πλείων -ον : *adj*, more, several.

πλευρά, ἡ : *no*, side.

πλήθος -εοξ, τό : *no*, great number, multitude, number.

πλήν : *adv* & *prep* + *gen*, more than.

ποιός -ά -όν : *adj*, of a certain nature, kind, quality, type.

πολλαπλασιάζω : *vb*, multiply.

πολλαπλασιασμός, ὁ : *no*, multiplication.

πολλαπλάσιον, τό : *no*, multiple.

πολύεδρος -ον : *adj*, polyhedral; τό πολυέδρον, *no*, polyhedron.

πολύγωνος -ον : *adj*, polygonal; τό πολύγωνον, *no*, polygon.

πολύπλευρος -ον : *adj*, multilateral.

πόρισμα -ατοξ, τό : *no*, corollary.

ποτέ : *ind*, at some time.

πρῶμα -ατοξ, τό : *no*, prism.

προβαίνω : *vb*, step forward, advance.

προδείκνυμι : *vb*, show previously; see δείκνυμι.

προεκτίθην : *vb*, set forth beforehand; see τίθην.

προερέω : *vb*, say beforehand; *perf pass part*, προεξηρημένοξ -η -ον, *adj*, aforementioned; see ἔζρω.

προσαναπληρόω : *vb*, fill up, complete.

προσαναγράφω : *vb*, complete (tracing of); see γράφω.

προσαρμόζω : *vb*, fit to, attach to.

προσεκβάλλω : *vb*, produce (a line); see ἐκβάλλω.

προσευρίσκω : *vb*, find besides, find; see εὐρίσκω.

προσλαμβάνω : *vb*, add.

πρόκειμαι : *vb*, set before, prescribe; see κείμει.

πρόσκειμαι : *vb*, be laid on, have been added to; see κείμει.

προσπίπτω : *vb*, fall on, fall toward, meet; see πίπτω.

προτασις -εωξ, ἡ : *no*, proposition.

προστάσσω : *vb*, prescribe, enjoin; τὸ τροσταθθέν, *no*, the thing prescribed; see τάσσω.

προστίθην : *vb*, add; see τίθην.

πρότερος -α -ον : *adj*, first (comparative), before, former.

προτίθην : *vb*, assign; see τίθην.

προθωρέω : *vb*, go/come forward, advance.

πρῶτος -α -ον : *adj*, first, prime.

πυραμίδς -ίδος, ἡ : *no*, pyramid.

ῥητός -ή -όν : *adj*, expressible, rational.

ῥομβοειδής -έξ : *adj*, rhomboidal; τὸ ῥομβοειδέξ, *no*, rhomboid.

ῥόμβος, ὁ : *no*, rhombus.

σημεῖον, τό : *no*, point.

σκαληνός -ή -όν : *adj*, scalene.

στερεός -ά -όν : *adj*, solid; τὸ στερεόν, *no*, solid, solid body.

στοιθεῖον, τό : *no*, element.

στρέφω, -στρέψω, ἔστρεψα, —, ἐσταμμαι, ἐστάφην : *vb*, turn.

σύνκειμαι : *vb*, lie together, be the sum of, be composed; συγκείμενος -η -ον, *adj*, composed (ratio), compounded; see κείμει.

σύνκρινω : *vb*, compare; see κρίνω.

συμβαίνω : *vb*, come to pass, happen, follow; see βαίνω.

συμβάλλω : *vb*, throw together, meet; see βάλλω.

σύμμετροξ -ον : *adj*, commensurable.

σύμπαξ -αντοξ, ὁ : *no*, sum, whole.

συνπίπτω : *vb*, meet together (of lines); see *πίπτω*.

συνπληρόω : *vb*, complete (a figure), fill in.

συνάγω : *vb*, conclude, infer; see *ἄγω*.

συναμφοτέροι -αι -α : *adj*, both together; ὁ συναμφοτέροξ, *no*, sum (of two things).

συναποδείκνυμι : *no*, demonstrate together; see *δείκνυμι*.

συναφή, ἡ : *no*, point of junction.

σύνδυο, οἱ, αἱ, τά : *no*, two together, in pairs.

συνετής -έξ : *adj*, continuous; κατὰ τὸ συνεθέξ, continuously.

σύνθεσις -εωξ, ἡ : *no*, putting together, composition.

σύνθετοξ -ον : *adj*, composite.

συ[γ]ίστημι : *vb*, construct (a figure), set up together; *perf imperat pass 3rd sg*, συνεστάτω; see *ἵστημι*.

συντίθημι : *vb*, put together, add together, compound (ratio); see *τίθημι*.

σθέσις -εωξ, ἡ : *no*, state, condition.

σθῆμα -ατοξ, τό : *no*, figure.

σφαῖρα -αξ, ἡ : *no*, sphere.

τάξις -εωξ, ἡ : *no*, arrangement, order.

ταράσσω, ταράχω, —, —, τετάραγμα, ἐταράθην : *vb*, stir, trouble, disturb; τεταραγμένοξ -η -ον, *adj*, disturbed, perturbed.

τάσσω, τάχω, ἔταχα, τέταθα, τέταγμα, ἐτάθην : *vb*, arrange, draw up.

τέλειος -α -ον : *adj*, perfect.

τέμνω, τεμνῶ, ἔτεμον, -τέμνηκα, τέτμημαι, ἐτμήθην : *vb*, cut; *pres/fut indic act 3rd sg*, τέμει.

τεταρτημοριον, τό : *no*, quadrant.

τετράγωνος -ον : *adj*, square; τὸ τετράγωνον, *no*, square.

τετράκις : *adv*, four times.

τετραπλάσιος -α -ον : *adj*, quadruple.

τετράπλευρος -ον : *adj*, quadrilateral.

τετραπλόος -η -ον : *adj*, fourfold.

τίθημι, θήσω, ἔθηκα, τέθηκα, κείμαι, ἐτέθην : *vb*, place, put.

τμήμα -ατοξ, τό : *no*, part cut off, piece, segment.

τοίνυν : *par*, accordingly.

τοιούτοξ -αύτη -οὔτο : *pro*, such as this.

τομεύξ -έωξ, ὁ : *no*, sector (of circle).

τομή, ἡ : *no*, cutting, stump, piece.

τόπος, ὁ : *no*, place, space.

τοσαντάκις : *adv*, so many times.

τοσανταπλάσιος -α -ον : *pro*, so many times.

τοσοῦτοξ -αύτη -οὔτο : *pro*, so many.

τουτέστι = τοῦτ' ἔστι : *par*, that is to say.

τραπέζιον, τό : *no*, trapezium.

τρίγωνος -ον : *adj*, triangular; τὸ τρίγωνον, *no*, triangle.

τριπλάσιος -α -ον : *adj*, triple, threefold.

τρίπλευρος -ον : *adj*, trilateral.

τριπλ -όος -η -ον : *adj*, triple.

τρόπος, ὁ : *no*, way.

τυγθάνω, τεύχομαι, ἔτυθον, τετύθηκα, τέτευγμαι, ἐτεύθην : *vb*, hit, happen to be at (a place).

ὑπάρθω : *vb*, begin, be, exist; see *ἄρθω*.

ὑπεχαίρεσις -εωξ, ἡ : *no*, removal.

ὑπερβάλλω : *vb*, overshoot, exceed; see *βάλλω*.

ὑπεροθή, ἡ : *no*, excess, difference.

ὑπερέθω : *vb*, exceed; see *ἔθω*.

ὑπόθεσις -εωξ, ἡ : *no*, hypothesis.

ὑποκείμαι : *vb*, underlie, be assumed (as hypothesis); see *κείμαι*.

ὑπολείπω : *vb*, leave remaining.

ὑποτείνω, ὑποτενῶ, ὑπέτενα, ὑποτέτακα, ὑποτετάμαι, ὑπετάθην : *vb*, subtend.

ὑψοξ -εοξ, τό : *no*, height.

φανερὸξ -ά -όν : *adj*, visible, manifest.

φημί, φήσω, ἔφην, —, —, — : *vb*, say; ἔφαμεν, we said.

φέρω, οἴσω, ἡνεγκον, ἐνήνοθα, ἐνήνεγμα, ἡνέθην : *vb*, carry.

θώριον, τό : *no*, place, spot, area, figure.

θωρίξ : *pre + gen*, apart from.

ψάω : *vb*, touch.

ὥξ : *par*, as, like, for instance.

ὥξ ἔτυθεν : *par*, at random.

ὡσαύτωξ : *adv*, in the same manner, just so.

ὥστε : *conj*, so that (causal), hence.